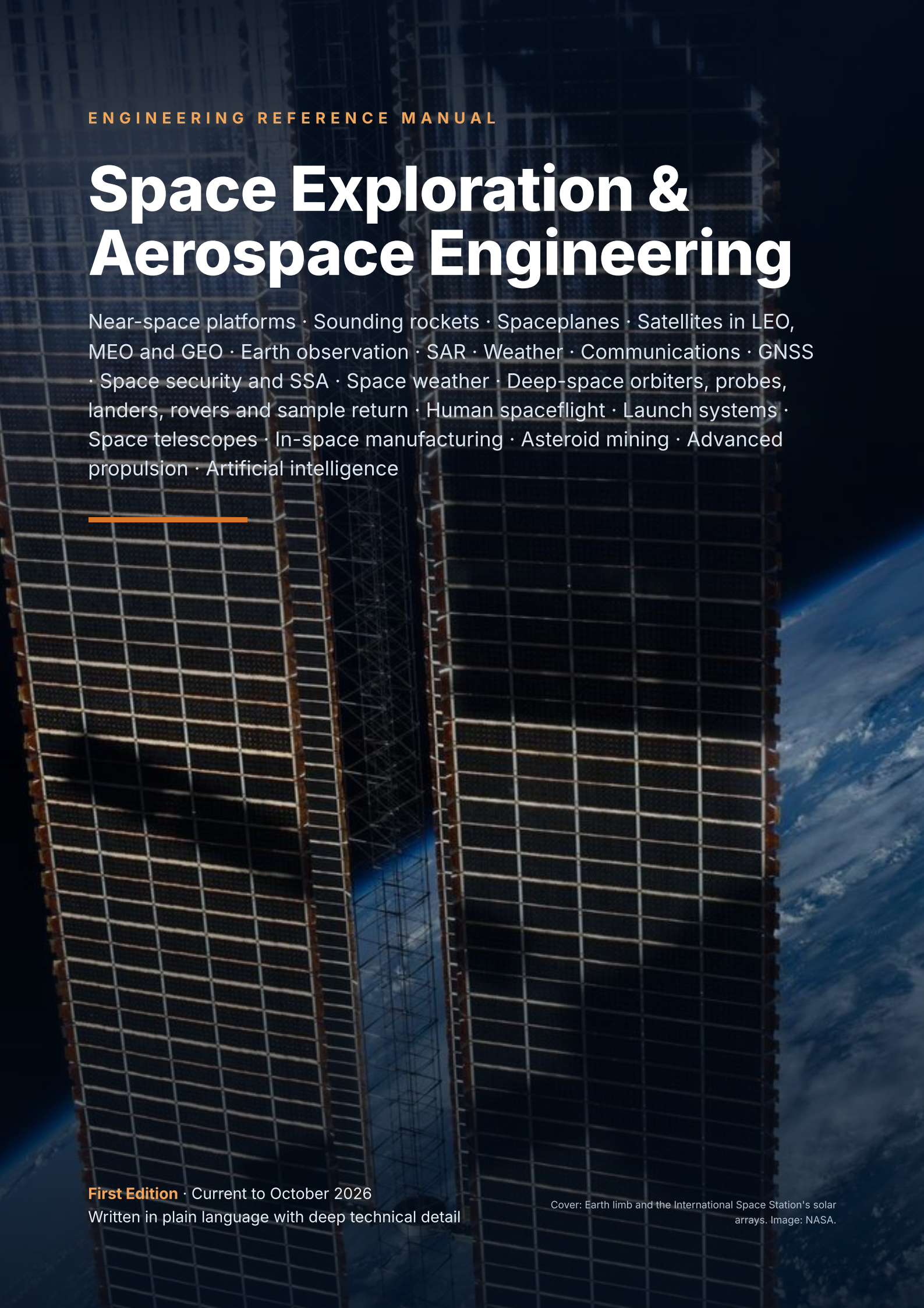
ENGINEERING REFERENCE MANUAL

# Space Exploration & Aerospace Engineering

Near-space platforms · Sounding rockets · Spaceplanes · Satellites in LEO, MEO and GEO · Earth observation · SAR · Weather · Communications · GNSS · Space security and SSA · Space weather · Deep-space orbiters, probes, landers, rovers and sample return · Human spaceflight · Launch systems · Space telescopes · In-space manufacturing · Asteroid mining · Advanced propulsion · Artificial intelligence



**First Edition** · Current to October 2026

Written in plain language with deep technical detail

Cover: Earth limb and the International Space Station's solar arrays. Image: NASA.

ENGINEERING REFERENCE MANUAL

# Space Exploration & Aerospace Engineering

A complete, illustrated guide to the space ecosystem — from stratospheric balloons to interstellar probes — explaining how every major class of space system works, how it is engineered, and where artificial intelligence is changing it.

First Edition · Information current to October 2026
A4 format · Fourteen Parts · Fifty-three chapters · Ten appendices
Version 1.0 · October 2026

## About this edition

**Space Exploration & Aerospace Engineering — An Engineering Reference Manual.** First Edition, Version 1.0, October 2026.

This manual was written to be read by students, engineers, analysts, managers and curious readers. Each topic starts with plain-language explanations and builds up to the equations, numbers and design trade-offs that practising engineers use. Sentences are kept short and every technical term is explained in plain words when it first appears; the glossary (Appendix H) and the list of acronyms (Appendix G) collect them. Facts, mission status and programme milestones are current to October 2026; space programmes change quickly, so dates of future events should be read as plans, not certainties.

## Figures, photographs and data

Diagrams, charts and graphs were drawn specifically for this manual from published physical relationships and public data. Photographs are from the NASA Image and Video Library and are used under NASA's media usage guidelines (NASA material is generally not subject to copyright in the United States); individual credits appear with each image and in Appendix J. NASA does not endorse this publication. Numerical values (masses, power levels, performance) are representative published figures and may differ slightly between sources and mission phases.

## Safety and responsibility

This is an educational reference. It is not a design standard, a flight rule, or a substitute for the requirements, analyses and verification processes of a qualified engineering organisation. Chapters dealing with military and intelligence systems use only openly published information and describe principles, not operational or classified details.

## How numbers are written

SI units are used throughout, with common aerospace units (nautical miles, feet, pounds-force) shown where the industry uses them. Large numbers use thin spacing (e.g., 35 786 km). "≈" means approximately, "~" means order-of-magnitude. Dates follow the ISO format (YYYY-MM-DD) in tables.

## Typography and production

Set in Source Serif 4, Inter and JetBrains Mono. Equations typeset with MathJax. Produced from Markdown sources with an automated pipeline (pandoc, Chromium, pypdf); the sources and build scripts accompany the PDF.

# Preface and How to Use This Manual

Space is no longer a niche. In 2025 the world made about 330 orbital launch attempts — more than in any previous year — and by late 2026 about sixteen thousand active satellites were in orbit, two-thirds of them Starlinks. A single company now lands and re-flies orbital boosters every few days, governments are building lunar landers with private partners, and the same smartphone that streams a video can receive a signal from four navigation constellations and, increasingly, from satellites directly. Behind all of this is a large, interconnected *ecosystem* of vehicles, instruments, ground systems, science, regulation and business.

This manual describes that ecosystem from end to end. It is organised in fourteen Parts that follow the journey of space systems from the edge of space to the edge of the Solar System:

- **Part I — Foundations** gives the physics and engineering every other Part relies on: the space environment, orbital mechanics, systems engineering, spacecraft subsystems and communications.
- **Parts II–III** climb from near-space balloons, sounding rockets and spaceplanes to satellites in Earth orbit.
- **Parts IV–VII** cover what satellites *do*: observe the Earth, carry communications and navigation signals, support national security and space situational awareness, and study space weather.
- **Parts VIII–IX** leave Earth orbit: robotic deep-space exploration and human spaceflight.
- **Parts X–XII** cover how we get to space (launch systems and spaceports), how we look at the universe (space telescopes), and the emerging frontiers of in-space manufacturing, space resources and advanced propulsion.
- **Part XIII** is an in-depth, eleven-chapter treatment of **artificial intelligence** across the whole space ecosystem: foundations (Chapters 43–45), applications (Chapters 46–52) and trustworthy AI (Chapter 53).
- **Part XIV** contains reference data: constants, equations, planetary data, frequency bands, mission databases, a timeline, acronyms, a glossary, sources and image credits.

## Reading paths

The manual can be read cover to cover, but it is designed as a reference. Each chapter opens with a short summary and a list of its sections, and cross-references (shown as blue links in the PDF) connect related figures, tables and sections. Suggested paths:

| If you are… | Start with | Then read |
|---|---|---|
| New to space engineering | Chapters 1, 2, 3 | Any application chapter; return to Chapters 4–7 as needed |
| A satellite or payload engineer | Chapters 4–7, 11–12 | Part IV (EO), Part V (comms/nav), Chapter 45 (edge AI in orbit) |
| Working in Earth observation or climate | Chapters 13–16 | Chapters 46–47 (AI for EO, weather and climate) |

| If you are... | Start with | Then read |
|---|---|---|
| Working in defence or space security | Chapters 19–21 | Chapters 22 (space weather), 50 (AI for SSA and defence) |
| A planetary or deep-space engineer | Chapters 3, 24 | Chapters 25–29, 42 (propulsion), 48 (autonomy) |
| Interested in human spaceflight | Chapters 30–33 | Chapters 10 (reentry), 31 (crewed spacecraft), 36 (launchers) |
| A launch or propulsion engineer | Chapters 34–37 | Chapters 9–10, 42 |
| An AI or data scientist entering space | Chapter 1, Chapters 43–45 | Chapters 13, 21, 24 for domain context, then Chapters 46–53 |

## Special boxes

**KEY CONCEPT**

The one idea in a section you should remember.

**WORKED EXAMPLE**

A calculation with real numbers, step by step.

**ENGINEERING NOTE**

Practical detail, rule of thumb, or design lesson.

**FAILURE LESSON**

What went wrong in a real mission and what engineers learned.

**DATA BOX**

Key numbers for a system or mission.

**AI IN PRACTICE**

How artificial intelligence is used for the topic at hand.

## A note on accuracy and currency

Space programmes evolve weekly: launch dates slip, missions end, and new vehicles fly. The information here reflects publicly available sources as of October 2026. Where a programme's future was uncertain at that date (for example, proposed cancellations in government budgets), the manual says so. Performance figures are representative values from official fact sheets, peer-reviewed literature and agency documentation; readers designing real systems should always go to the primary source listed in Appendix I.

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PART I

# Foundations of Space Engineering

Every space system — a weather balloon, a GPS satellite or a Mars rover — obeys the same physics and is built with the same engineering discipline. Part I explains the ecosystem, the environment a spacecraft must survive, how orbits work, how missions are engineered, the subsystems inside every spacecraft, and how data travels between space and the ground.


1 The Space Ecosystem: An Overview
2 The Space Environment
3 Orbital Mechanics and Astrodynamics
4 Spacecraft Systems Engineering
5 Spacecraft Subsystems I: Structures, Thermal Control, Power and Mechanisms
6 Spacecraft Subsystems II: Attitude Control, Propulsion, Data Handling and Flight Software
7 Space Communications and the Ground Segment


Image: NASA/JSC

CHAPTER 1

# The Space Ecosystem: An Overview

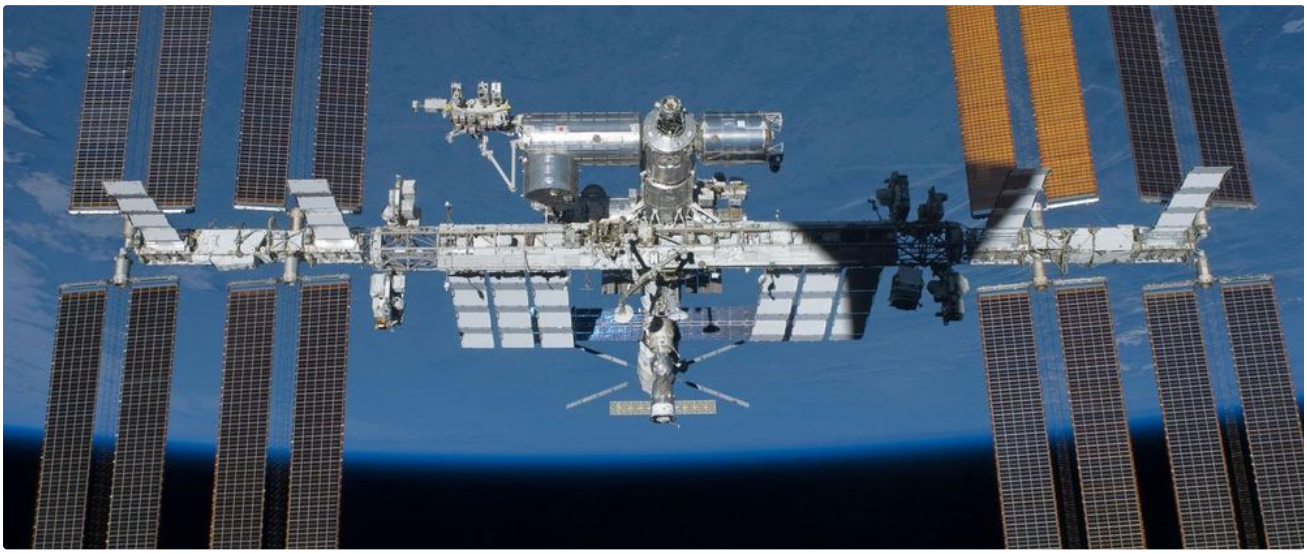


Image: NASA/JSC

*Space is not one place or one industry. It is a layered environment that begins a few tens of kilometres above our heads and stretches to interstellar space, and an interconnected ecosystem of vehicles, ground systems, users, companies, governments and rules. This chapter maps that ecosystem, defines the regimes in which space systems operate, describes the segments of every space system, and summarises the history, economics, statistics and governance that shape it today.*

**IN THIS CHAPTER**


1.1 Where Does Space Begin?
1.2 The Segments of a Space System
1.3 A Map of the Space Ecosystem
1.4 A Short History of Spaceflight
1.5 Space Activity in Numbers
1.6 Why Space Is Hard
1.7 Governance: The Rules of Space
1.8 The Engineering Disciplines of the Space Ecosystem
1.9 How This Manual Is Organised


## 1.1 Where Does Space Begin?

There is no physical surface where the atmosphere stops and space starts. Air simply becomes thinner with height: at 5.5 km half of the atmosphere's mass is below you, at 16 km about 90 %, and at 50 km more than 99.9 %. Engineers therefore use *functional* boundaries — heights at which the physics that governs a vehicle changes.

The best-known boundary is the **Kármán line** at **100 km**. It is named after Theodore von Kármán. He noticed that an aircraft trying to fly at about this height would need to travel very fast to generate enough lift. In fact, it would already be close to orbital speed. Above it, wings are useless; a vehicle must either follow a ballistic path or be in orbit. The Fédération Aéronautique

Internationale (FAI) uses 100 km for astronautics records. The United States Air Force, NASA and the US Federal Aviation Administration award astronaut wings above **50 miles (80.5 km)**. Some scientists argue that 80 km is physically better justified. This is because satellites with very low perigees can briefly dip to about 80–90 km without immediately re-entering.

Neither number is a legal border. International law has never defined where national airspace ends and outer space begins. In practice, no satellite can stay in orbit for long below about 160 km, because atmospheric drag removes its energy within days.

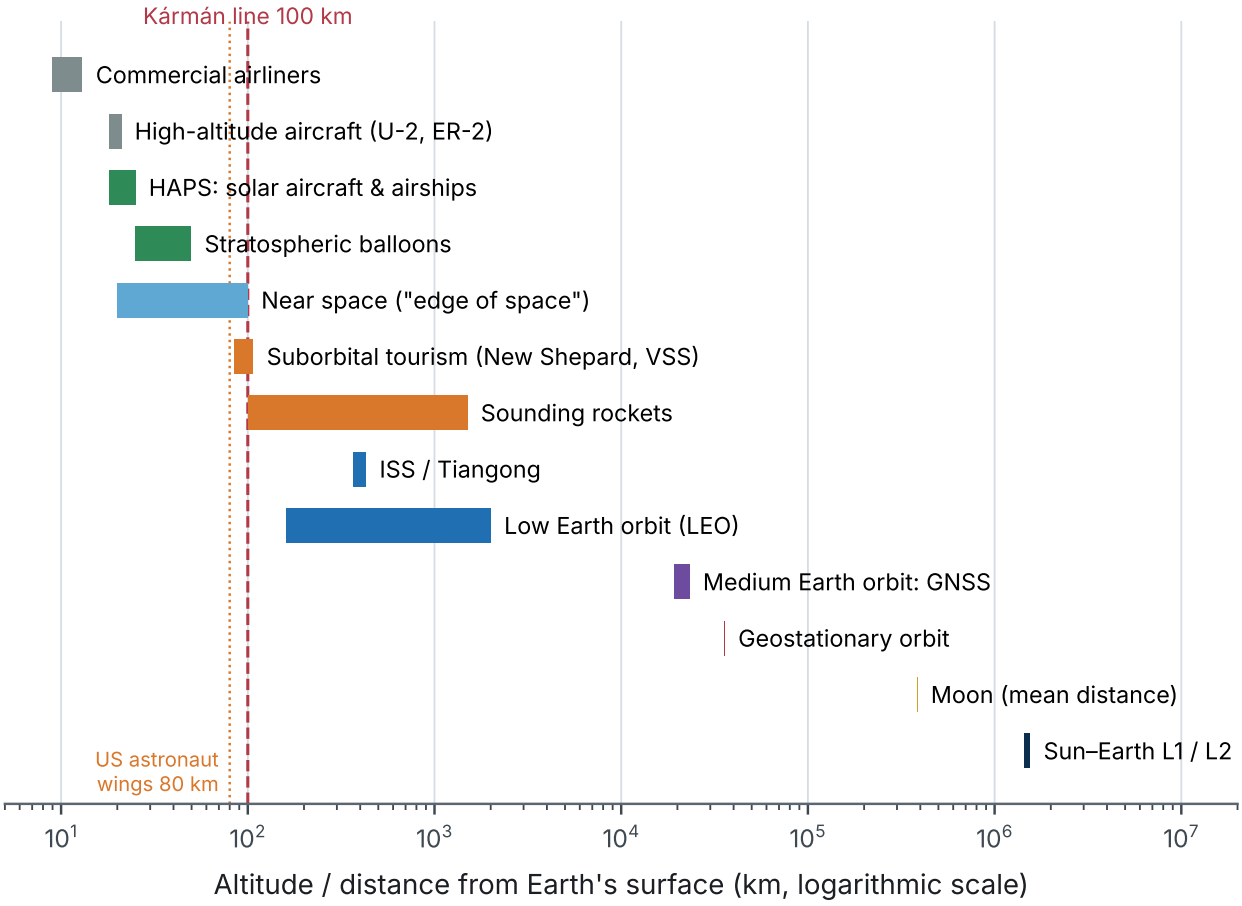



**FIGURE 1.1** The regimes of near-Earth and cislunar space shown on a logarithmic distance scale. Each order of magnitude in altitude corresponds to a different class of vehicle and a different engineering problem: aerodynamics and buoyancy below 50 km, ballistic flight up to a few hundred kilometres, orbital mechanics beyond, and interplanetary trajectories past the Moon.

Figure 1.1 and Table 1.1 summarise the regimes used throughout this manual. Moving outward, each regime adds new challenges. Lower air density first removes lift and cooling, then oxygen for air-breathing engines, then protection from radiation and debris. Finally, distance itself becomes the problem, delaying radio signals by seconds, minutes or hours.

**TABLE 1.1** Altitude regimes and the vehicles that operate in them.

| Regime | Typical altitude or distance | Main vehicles | Dominant physics / engineering problem |
|---|---|---|---|
| Troposphere | 0–12 km (0–18 km at the equator) | Aircraft, drones, weather balloons | Weather, lift, air-breathing propulsion |
| Stratosphere | 12–50 km | HAPS (solar aircraft, airships), stratospheric balloons, U-2/ER-2 | Low density, extreme cold (−56 °C), ozone, UV, long endurance |

| Regime | Typical altitude or distance | Main vehicles | Dominant physics / engineering problem |
|---|---|---|---|
| Mesosphere / lower thermosphere (“near space” / “ignorosphere”) | 50–100 km | Sounding rockets, suborbital vehicles, re-entering spacecraft | Too thin for aircraft, too dense for satellites; meteors burn up here |
| Suborbital | apogee 80–1 500 km | New Shepard, sounding rockets, ballistic missiles, test vehicles | Ballistic flight, minutes of microgravity, re-entry heating |
| Low Earth orbit (LEO) | 160–2 000 km | ISS, Tiangong, Starlink, Earth observation | Drag, atomic oxygen, debris, short ground contacts |
| Medium Earth orbit (MEO) | 2 000–35 786 km | GPS, Galileo, GLONASS, BeiDou, O3b mPOWER | Van Allen radiation belts, longer links |
| Geosynchronous orbit (GEO) | 35 786 km altitude (42 164 km radius) | Communications, weather, missile warning | Station-keeping, 0.24 s round-trip delay, crowded arc |
| Highly elliptical orbit (HEO) | perigee ~500 km, apogee ~40 000 km | Molniya/Tundra comms, early warning | Repeated belt crossings, high-latitude coverage |
| Cislunar space | to 384 400 km (Moon) and Earth–Moon L1/L2 | Orion, lunar orbiters, landers, CAPSTONE | Three-body dynamics, deep-space communications |
| Sun–Earth Lagrange points | 1.5 million km | SOHO, DSCOVR, JWST, Euclid, Roman | Halo orbits, thermal stability, station-keeping |
| Interplanetary / deep space | 0.4–100+ AU | Planetary probes, orbiters, landers, Voyager | Light-time delay, power far from the Sun, autonomy |

**KEY CONCEPT — SPACE IS ABOUT SPEED, NOT HEIGHT**

Reaching 100 km altitude needs only about 1.4 km/s of velocity change if you go straight up and come straight down. Staying in space at 400 km needs a horizontal speed of about **7.7 km/s** — roughly 28 000 km/h. Getting to orbit is therefore mostly a problem of going *sideways fast enough* that the ground curves away beneath you as quickly as you fall towards it. This is why an orbital rocket is roughly ten to twenty times larger than a suborbital rocket carrying the same payload.

## 1.2 The Segments of a Space System

Every space mission, whether a CubeSat built by students or a constellation of thousands of broadband satellites, can be described as a set of cooperating **segments** (Figure 1.2).

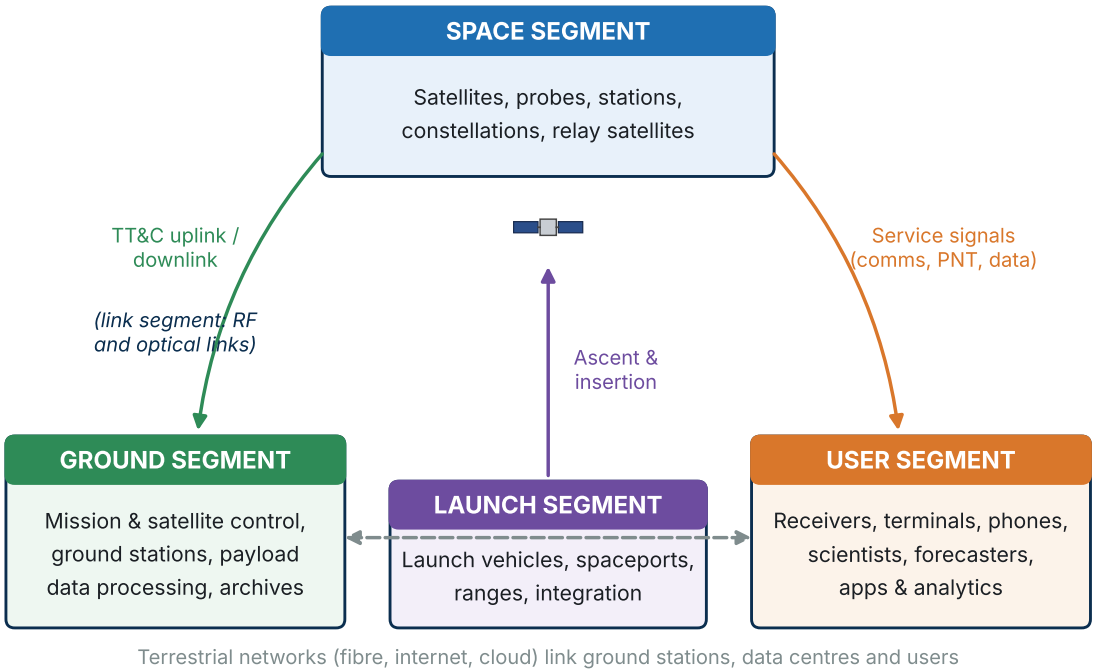



**FIGURE 1.2** The four segments of a space system and the links that join them. The space segment is usually the most visible, but the ground and user segments often cost as much over a mission's life.

**Space segment.** The spacecraft themselves: satellites, probes, stations and their payloads. A spacecraft is normally divided into two parts. The **payload** is the part that performs the mission — a camera, a transponder, a telescope. The **bus** or **platform** is the part that keeps the payload alive — power, thermal control, attitude control, propulsion, computing and communications. Part I explains the bus in detail; the application Parts explain the payloads.

**Ground segment.** Everything on the ground that controls the spacecraft and handles its data: mission control centres, ground stations with antennas, networks, flight dynamics software, payload data processing, archives and distribution systems. Large ground networks — NASA's Deep Space Network, ESA's ESTRACK, commercial "ground-station-as-a-service" networks such as KSAT, AWS Ground Station and Leaf Space — are shared by many missions.

**Launch segment.** The launch vehicle, the spaceport, the range that tracks the rocket and protects the public, and the facilities that integrate the payload into the fairing. For a single satellite the launch is a one-time event, but its cost and risk shape the entire mission design.

**User segment.** The people and equipment that use the service. Examples are a GPS receiver in a phone, a satellite TV dish and a farmer's crop-monitoring app. Others are a forecaster using satellite data in a weather model, or a scientist analysing telescope images. For commercial systems the user segment is where revenue is earned; for navigation and broadband systems the user equipment is produced in hundreds of millions of units.

The **link segment** — radio-frequency (RF) and, increasingly, optical (laser) links — joins all of them. Links include telemetry, tracking and command (TT&C) for spacecraft control, high-rate payload data downlinks, user service links, and inter-satellite links that relay data through space. Chapter 7 explains how links are designed.

**ENGINEERING NOTE — THINK IN SEGMENTS EARLY**

A frequent cause of mission failure or cost overrun is a space segment designed in isolation. Examples include satellites that produce more data than the ground stations can download, spacecraft whose radio frequencies were never coordinated with regulators, and payloads whose data products nobody was funded to process. Systems engineering (Chapter 4) exists largely to prevent these interface failures.

## 1.3 A Map of the Space Ecosystem

The word *ecosystem* is used deliberately: space activities depend on each other like species in an environment. Cheap launch enables mega-constellations. Mega-constellations create demand for ground terminals, spectrum and collision avoidance, and collision avoidance needs space situational awareness. All of them depend on scientific understanding of space weather. Figure 1.3 maps the major domains covered in this manual.

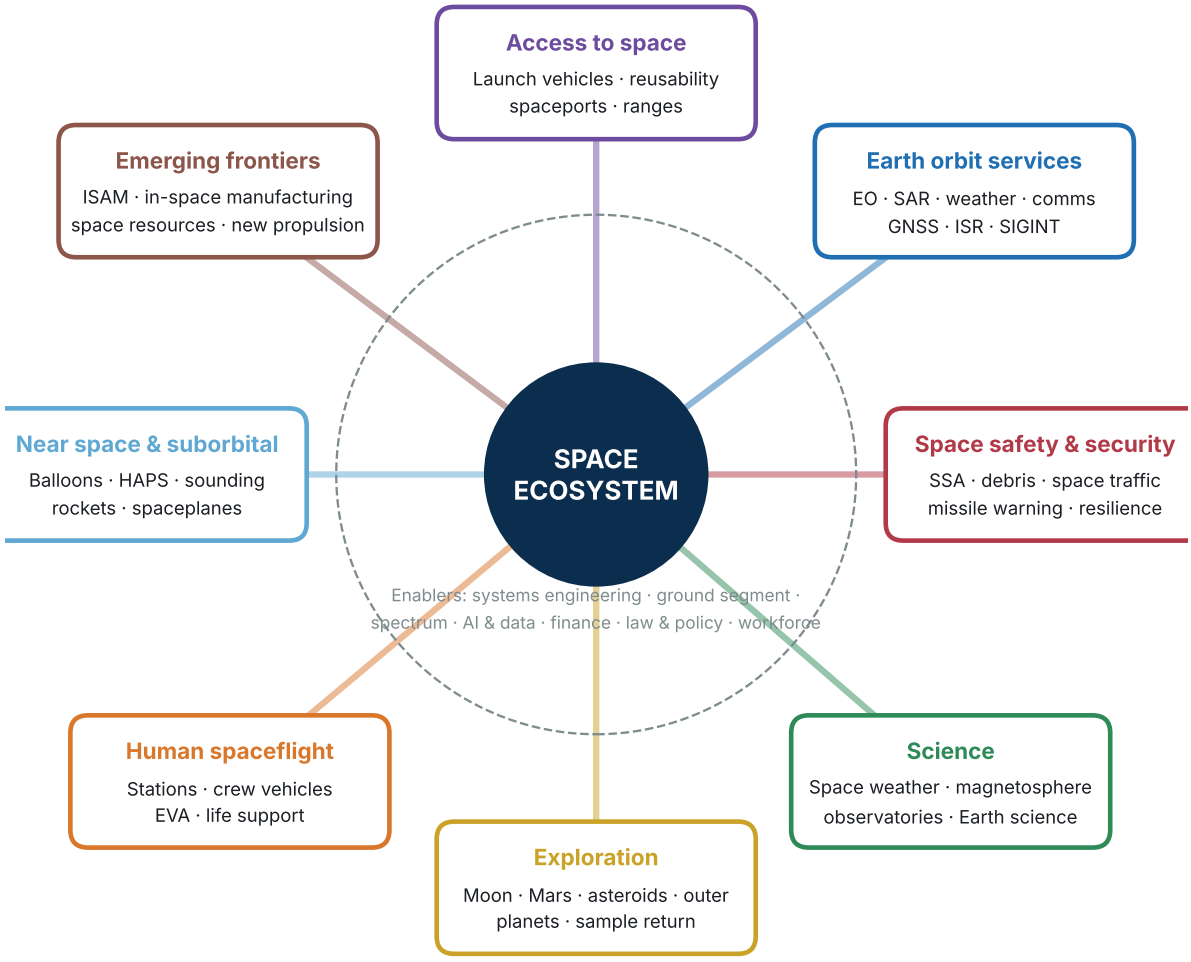



**FIGURE 1.3** The space ecosystem as eight interlocking domains, each covered by one or more Parts of this manual, supported by common enablers such as systems engineering, ground infrastructure, spectrum management, artificial intelligence and law.

### 1.3.1 Classification by Mission Purpose

Spacecraft are commonly grouped by what they do. The classification below follows the structure of this manual.

- **Near-space and suborbital platforms** — stratospheric balloons, high-altitude platform stations (HAPS), sounding rockets, suborbital crewed vehicles and spaceplanes (Part II).
- **Earth observation (EO) and remote sensing** — optical, infrared, radar, atmospheric and oceanographic satellites (Part IV).
- **Communications** — fixed and mobile satellite services, broadcasting, broadband, data relay, direct-to-device (Part V).
- **Positioning, navigation and timing (PNT)** — GPS, GLONASS, Galileo, BeiDou and regional systems (Part V).
- **National security** — intelligence, surveillance and reconnaissance (ISR), signals intelligence, missile warning, military communications (Part VI).
- **Space safety** — space situational awareness, debris tracking and removal (Part VI).
- **Space science** — heliophysics, magnetospheric physics, astronomy and astrophysics (Parts VII and XI).
- **Exploration** — robotic orbiters, probes, landers, rovers and sample-return missions; human spaceflight (Parts VIII and IX).
- **Technology and in-space economy** — servicing, assembly, manufacturing, resource extraction, propulsion demonstrations (Part XII).

### 1.3.2 Classification by Size

Satellites range from a few grams to more than 400 tonnes (the International Space Station). The industry uses an informal mass scale (Table 1.2). The boundaries vary slightly between organisations, but the categories are widely understood.

**TABLE 1.2** Informal spacecraft mass classes and examples.

| Class | Wet mass | Typical examples | Typical cost range (spacecraft only) |
|---|---|---|---|
| Femto/chip satellite | < 100 g | KickSat "Sprites", ChipSats | < US$1 000 each |
| Picosatellite | 0.1–1 kg | PocketQube, 1U CubeSat (≈1.3 kg is borderline) | US$10k–100k |
| Nanosatellite | 1–10 kg | 1U–6U CubeSats, Planet "Dove" (≈5 kg) | US$50k–2M |
| Microsatellite | 10–100 kg | 12U/16U CubeSats, ICEYE SAR (≈90 kg), Spire | US$1M–20M |
| Minisatellite / small satellite | 100–500 kg | OneWeb (≈150 kg), Starlink V1.5 (≈300 kg), NovaSAR-1 (≈450 kg) | US$5M–100M |
| Medium satellite | 500–2 500 kg | Starlink V2 Mini (≈800 kg), Iridium NEXT (≈860 kg), Sentinel-2 (≈1.1 t) | US$50M–500M |
| Large satellite | > 2 500 kg | Landsat 9 (≈2.7 t), GPS III (≈3.9 t), GEO communications satellites (3–7 t), JWST (6.2 t) | US$150M–10B |

| Class | Wet mass | Typical examples | Typical cost range (spacecraft only) |
|---|---|---|---|
| Space station | > 50 t | Tiangong (≈100 t), ISS (≈450 t) | US$10B–150B |

The CubeSat standard (Chapter 11) deserves special mention: by defining a standard 10 cm cube "unit" (1U) and a standard deployer, it made space accessible to universities and start-ups. More than 2 500 CubeSats have been launched since 2003.

## 1.4 A Short History of Spaceflight

The modern space age began on 4 October 1957, when the Soviet Union launched **Sputnik 1**, an 84 kg sphere with four antennas that transmitted a simple beep. Within twelve years humans had orbited the Earth (Yuri Gagarin, 12 April 1961) and walked in space (Alexei Leonov, 1965). They had also walked on the Moon (Neil Armstrong and Buzz Aldrin, 20 July 1969). Figure 1.4 marks major milestones.

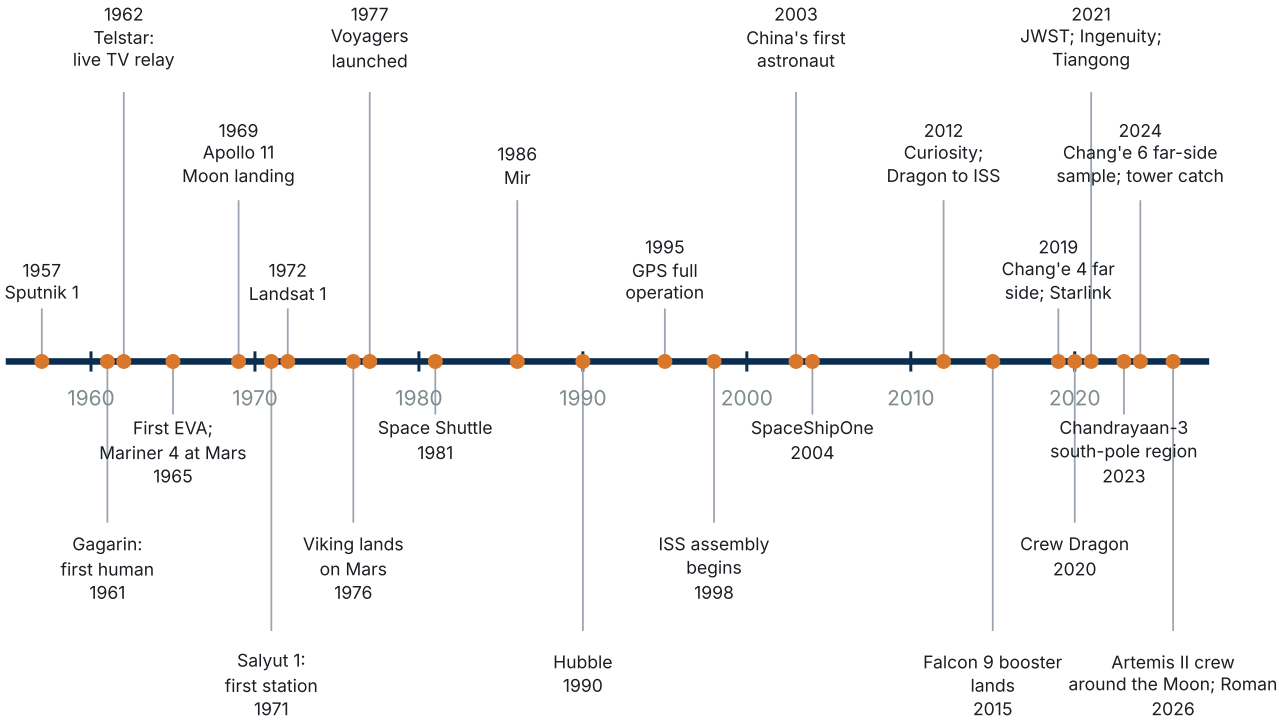



**FIGURE 1.4** Selected milestones of the space age, 1957–2026. The pace of "firsts" was highest in the 1960s; since 2015 a second wave has been driven by reusable rockets, small satellites and new national programmes.

The history can be divided into five broad eras:

1. **The Space Race (1957–1975).** A competition between the USA and the USSR produced the first satellites, human flights, planetary probes, weather and communications satellites, reconnaissance satellites (CORONA, Zenit), and the Apollo lunar landings. Launch rates peaked at about 140 orbital attempts in 1967.

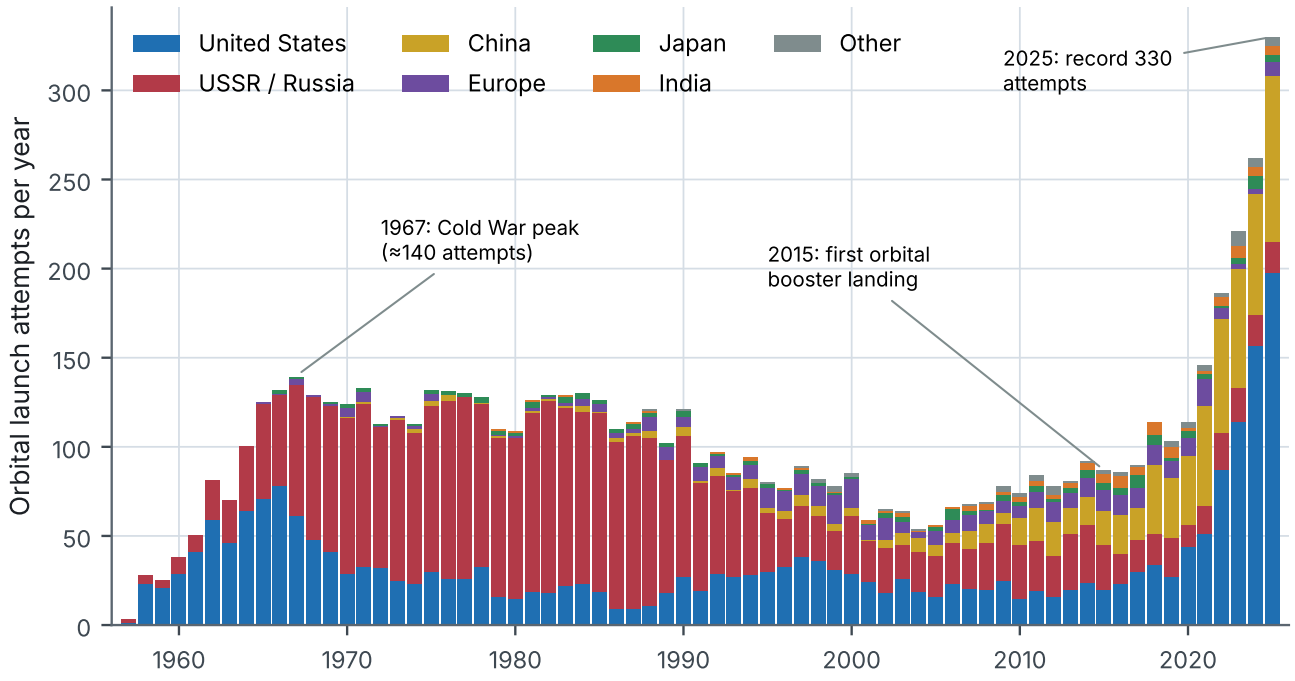
2. **Stations and shuttles (1971–2000).** The USSR operated Salyut stations and Mir; the USA flew Skylab and the reusable Space Shuttle. Satellite applications matured: Landsat (1972), GPS (first launch 1978, fully operational 1995), INTELSAT global communications, and the great observatories (Hubble 1990, Compton 1991, Chandra 1999).
3. **International cooperation and globalisation (1998–2010).** Assembly of the International Space Station began in 1998, permanently crewed since November 2000. China flew its first astronaut (Yang Liwei, 2003). Europe, Japan and India developed independent launch and exploration capabilities. Mars exploration intensified with orbiters and rovers.
4. **"New Space" (2010–2019).** Private capital entered the industry. SpaceX's Falcon 9 began flying in 2010, delivered cargo to the ISS in 2012 and landed an orbital-class booster in December 2015. CubeSats and small-satellite constellations (Planet, Spire) appeared. China landed on the far side of the Moon (Chang'e 4, 2019) and the first Starlink satellites launched in 2019.
5. **The industrial space age (2020–present).** This era brought commercial crew transport (Crew Dragon, 2020), mega-constellations with thousands of satellites, routine booster reuse and China's Tiangong station (2021–2022). It also saw the James Webb Space Telescope (2021), the first far-side sample return (Chang'e 6, 2024) and the first returning booster caught by its launch tower (Starship, October 2024). Then came the first crewed flight around the Moon since 1972 (Artemis II, April 2026) and the launch of the Nancy Grace Roman Space Telescope (August 2026).

> **HISTORY — THE SECOND SPACE RACE IS COMMERCIAL AND MULTIPOLAR**
>
> In 1967, two nations conducted nearly all space launches. In 2025, rockets from more than ten countries reached orbit, and private companies launched most of the payloads. The busiest single launch provider (SpaceX) flew more often than the whole world had in any year before 2021.

## 1.5 Space Activity in Numbers

### 1.5.1 Launch Rate

Figure 1.5 shows every orbital launch attempt since 1957, grouped by the nationality of the launching organisation. After the Cold War peak, launch rates fell for two decades as satellites became longer-lived and the Soviet programme contracted. Since 2016 the rate has risen steeply, driven first by reusable Falcon 9 boosters and Chinese commercial and government launches.




Data: J. McDowell, GCAT (planet4589.org), updated 13 Oct 2026. Includes failures.

**FIGURE 1.5** Orbital launch attempts per year, 1957–2025, by country of the launch provider. The 2025 total of 330 attempts (counts vary slightly between sources, from 323 to 330, depending on how test flights are treated) is the highest in history. "United States" includes Rocket Lab launches from New Zealand.

**TABLE 1.3** Orbital launch attempts by country of launch provider, 2025 and 2026 to mid-October (GCAT).

| Country / region | 2025 attempts | Share 2025 | 2026 (to 13 Oct) | Main vehicles |
|---|---|---|---|---|
| United States | 198 | 60 % | 146 | Falcon 9, Falcon Heavy, Electron, Starship, Atlas V, Vulcan, New Glenn |
| China | 93 | 28 % | 71 | Long March family, Zhuque, Kinetica-1, Ceres-1, Kuaizhou, Jielong, Gravity-1 |
| Russia | 17 | 5 % | 13 | Soyuz-2, Angara, Proton |
| Europe | 8 | 2 % | 7 | Ariane 6, Vega-C |
| India | 5 | 2 % | 3 | PSLV, GSLV, LVM3 |
| Japan | 4 | 1 % | 3 | H3, H-IIA (retired 2025) |
| Others (Iran, North Korea, South Korea, …) | 5 | 2 % | 1 | Simorgh, Qaem-100, Nuri |
| **World** | **330** | 100 % | **244** | |

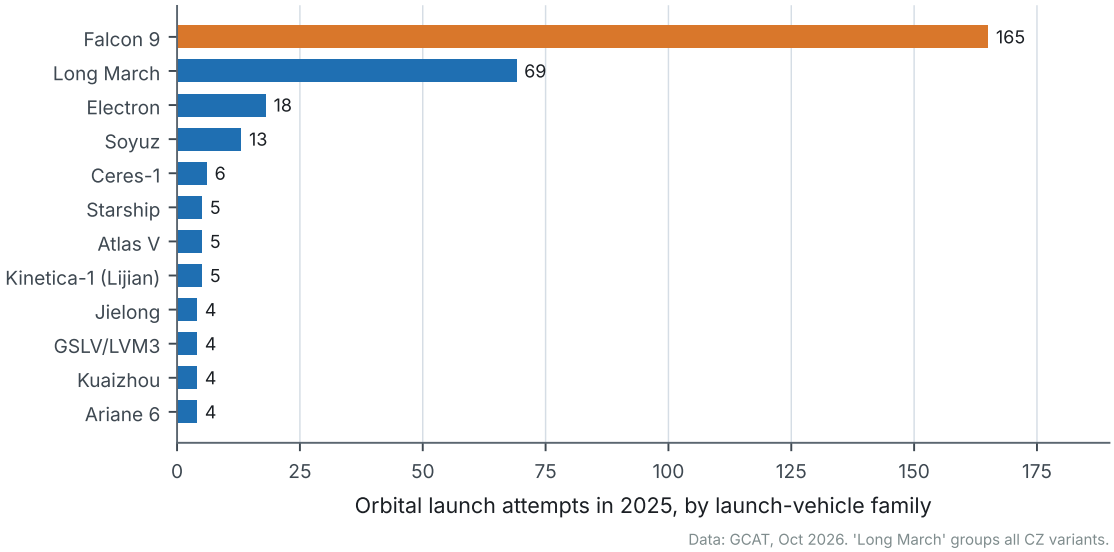



**FIGURE 1.6** Orbital launch attempts in 2025 by launch-vehicle family. A single vehicle family, Falcon 9, accounted for half of all orbital launches worldwide.

## 1.5.2 Satellites in Orbit

The number of satellites has grown even faster than the number of launches, because one launch can now deploy dozens or hundreds of satellites. In 2025 alone about **4 500 payloads** were placed in orbit — more than in the first fifty years of spaceflight combined (Figure 1.7). Most of the growth comes from broadband constellations, especially SpaceX's Starlink, which had more than 10 600 satellites in orbit by October 2026. Next come Eutelsat OneWeb (≈650), China's Guowang and Qianfan constellations, and Amazon Leo (formerly Project Kuiper, about 400 satellites by mid-2026).

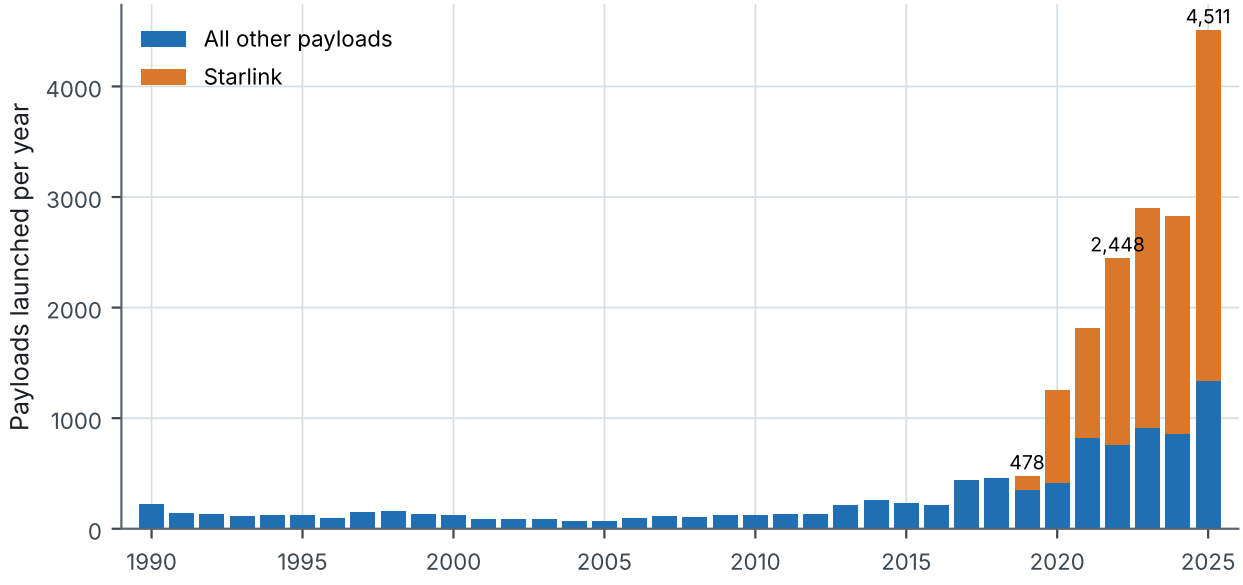
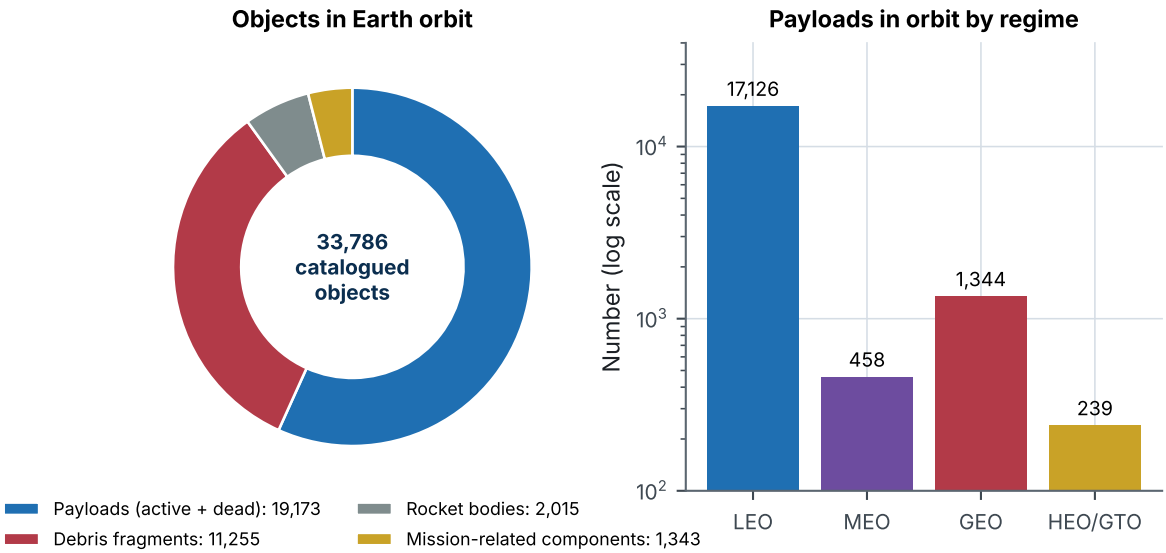



**FIGURE 1.7** Payloads launched per year, 1990–2025, separating Starlink from all other payloads. The step change after 2019 is the arrival of mega-constellations.




Data: GCAT, 13 Oct 2026. Tracked objects ≥10 cm; ~1 million objects 1–10 cm are untracked.

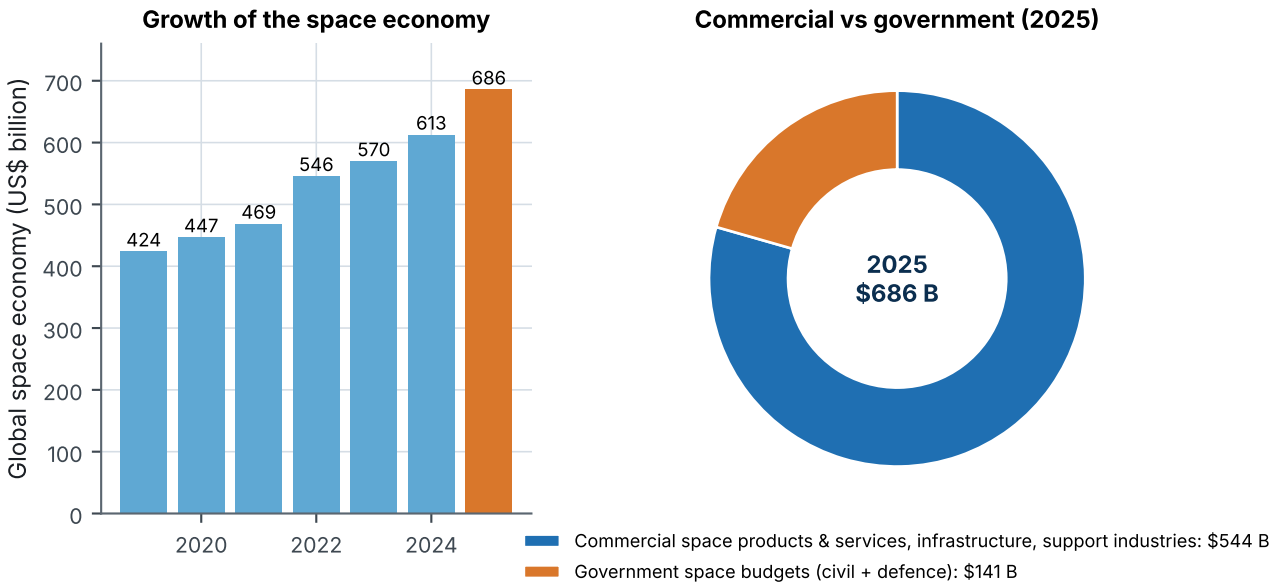
**FIGURE 1.8** Left: the ~33 800 catalogued objects in Earth orbit by type. Right: payloads in orbit by orbital regime. Low Earth orbit dominates in numbers; GEO dominates in value per satellite.

**DATA BOX — EARTH ORBIT POPULATION, OCTOBER 2026 (GCAT)**

- Catalogued objects in orbit: **≈33 800**, of which payloads ≈19 200, debris fragments ≈11 300, rocket bodies ≈2 000, other mission objects ≈1 300.
- Payloads believed to be operating: **≈16 000–17 000** (GCAT estimate ≈16 000; CelesTrak lists ≈17 200 as operational; definitions of "active" differ).
- Starlink satellites in orbit: **≈10 650**.
- Payloads by regime: LEO ≈17 100, GEO ≈1 340, MEO ≈460, highly elliptical ≈240.
- Estimated objects too small to track (1–10 cm): **about one million**; objects 1 mm–1 cm: over 100 million (ESA debris model).

## 1.5.3 The Space Economy

The **space economy** is the total of all revenues and budgets generated by space activities. Estimates differ in what they include (for example, whether satellite TV subscriptions or GNSS-enabled phones count), but the Space Foundation's series is widely quoted: the global space economy reached about **US$686 billion in 2025**, up 12 % from US$613 billion in 2024 (Figure 1.9). About 79 % was commercial; government space budgets totalled about US$141 billion, of which the United States accounted for 57 %.




Source: Space Foundation, The Space Report (2019–2025 editions; 2025 data released July 2026). Values in nominal US$.

**FIGURE 1.9** Left: growth of the global space economy, 2019–2025. Right: split between commercial activity and government budgets in 2025.

Three observations help interpret these numbers:

1. **Most money is made on the ground.** Satellite communications services, user terminals, GNSS chipsets and downstream data products earn far more than rockets or satellites themselves. The "upstream" manufacturing and launch sector is only roughly a tenth of the total.
2. **Governments remain the anchor customers.** Even in commercial markets, government contracts (for example NASA's Commercial Crew and Cargo programmes, the US Space Development Agency's constellation, or ESA's Galileo) fund the first systems and reduce investor risk.
3. **Defence spending is rising fastest.** Since 2022, military and intelligence space budgets have grown more quickly than civil budgets in the United States, Europe, Japan and elsewhere, reflecting the role of satellites in modern conflicts.

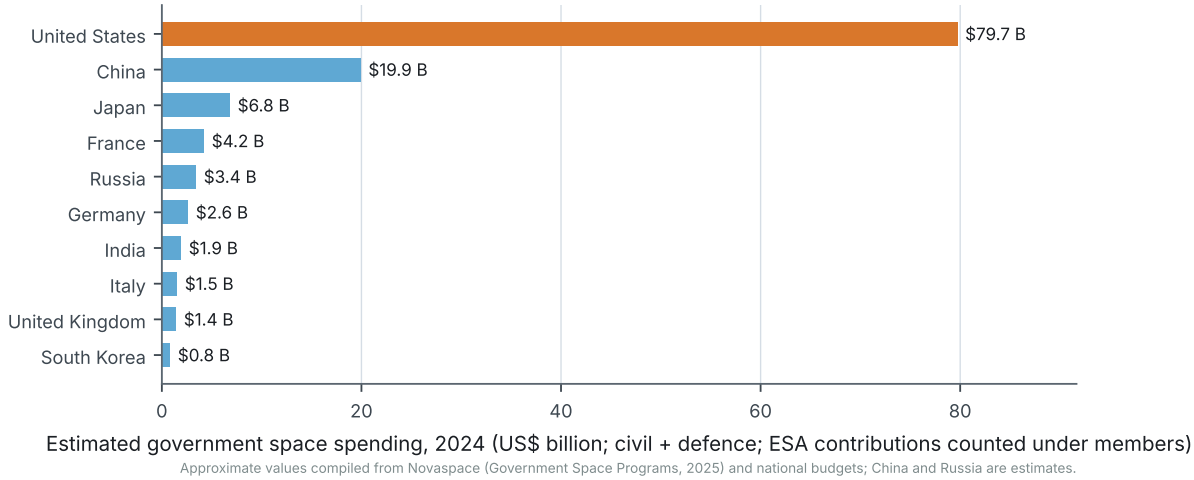



**FIGURE 1.10** Estimated government space spending by country in 2024 (civil and defence combined). The United States spends roughly four times more than China and more than all other governments together.

**TABLE 1.4** Major space agencies and organisations (selected).

| Organisation | Country / members | Founded | Notable capabilities (2026) |
|---|---|---|---|
| NASA | USA | 1958 | Artemis (SLS/Orion), ISS, deep-space science, JWST, Roman, Deep Space Network |
| US Space Force / NRO / SDA | USA | 2019 / 1961 / 2019 | GPS, missile warning, military comms, reconnaissance, proliferated LEO architecture |
| NOAA (NESDIS) | USA | 1970 | GOES, JPSS weather satellites, space weather (SWFO-L1), commercial remote-sensing licensing |
| Roscosmos | Russia | 1992 (successor to Soviet programme) | Soyuz crew and cargo, GLONASS, Angara, ISS Russian segment |
| CNSA / CMSA | China | 1993 / 1992 | Tiangong station, Chang'e lunar, Tianwen planetary, BeiDou, Long March |
| ESA | 23 member states | 1975 | Ariane 6, Vega-C, Copernicus (with EU), Galileo (with EU), science missions |
| EUSPA / European Commission | EU-27 | 2021 | Galileo, EGNOS, Copernicus programme management, IRIS² secure connectivity |
| JAXA | Japan | 2003 (merger) | H3, Hayabusa missions, SLIM lunar lander, MMX, HTV-X cargo |
| ISRO | India | 1969 | PSLV, LVM3, Chandrayaan-3, Aditya-L1, NavIC, Gaganyaan |
| CNES / DLR / ASI / UKSA | France / Germany / Italy / UK | 1961 / 1969 (as DFVLR) / 1988 / 2010 | National programmes and major ESA contributors |
| CSA | Canada | 1989 | Canadarm robotics, RADARSAT, Artemis II astronaut |
| KASA | South Korea | 2024 | Nuri launcher, Danuri lunar orbiter |
| UAE Space Agency / MBRSC | United Arab Emirates | 2014 / 2006 | Hope Mars orbiter, astronaut programme, Gateway airlock (planned) |

## 1.6 Why Space Is Hard

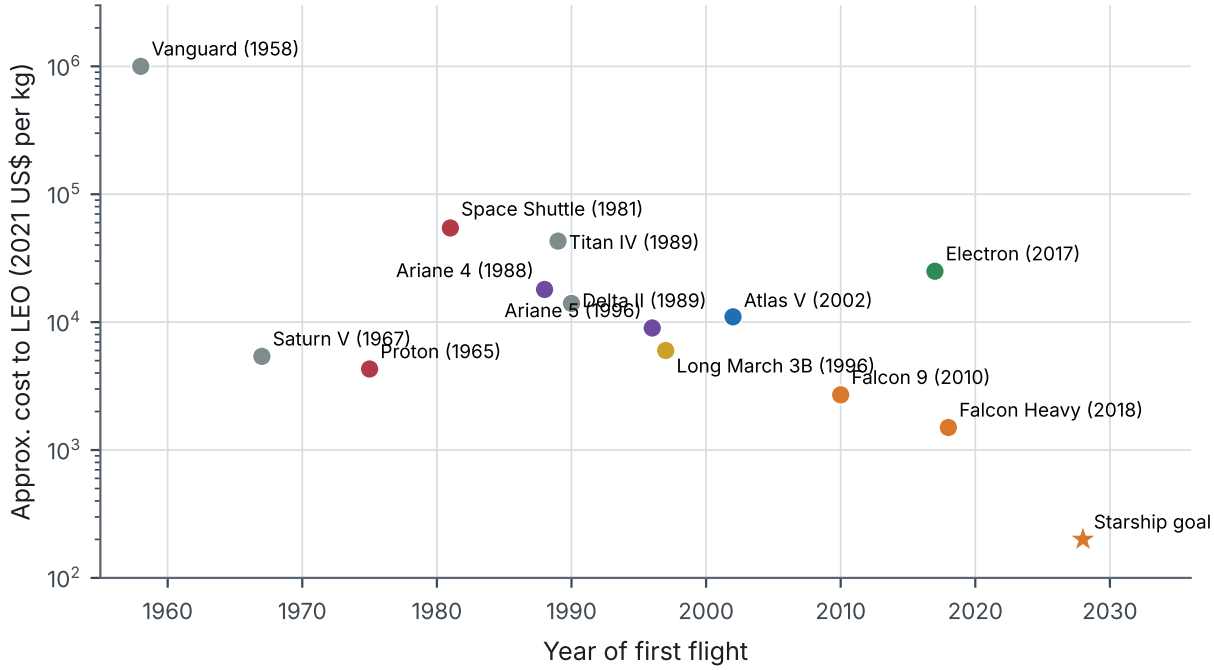
Space engineering combines three difficulties that are rarely found together in other industries.

**1. The energy problem.** To reach low Earth orbit a vehicle must gain about 9.3–9.6 km/s of velocity once gravity and drag losses are included. The rocket equation (Chapter 34) shows that this requires a vehicle that is typically 85–95 % propellant at lift-off; only 2–4 % of the lift-off mass reaches orbit as payload. Every kilogram of spacecraft therefore costs many kilograms of rocket.

**2. The environment problem.** In orbit, a spacecraft must survive vacuum, temperature swings of more than 200 °C and radiation that damages electronics and solar cells. It must also survive atomic oxygen, micrometeoroids and debris travelling at 10 km/s, and launch vibrations and shocks of tens of g (Chapter 2).

**3. The no-repair problem.** With very few exceptions (Hubble servicing, the ISS, the new generation of servicing vehicles), a spacecraft cannot be repaired after launch. It must work first time and keep working for 5–15 years. This drives redundancy, extensive testing, conservative design margins and long development schedules.

These difficulties explain why spacecraft traditionally cost tens of thousands of dollars per kilogram. Two trends are now lowering the cost. The first is reusable launch vehicles. The second is mass-produced small satellites, which accept higher individual failure rates because the constellation as a whole is resilient. Figure 1.11 shows how much the price of reaching orbit has fallen.




Approximate: list price ÷ maximum LEO payload (reusable vehicles at commercial price). After NASA/CSIS compilations; real costs vary by mission.

**FIGURE 1.11** Approximate cost per kilogram to low Earth orbit for representative launch vehicles, in inflation-adjusted dollars. Reusable boosters reduced commercial launch prices by about an order of magnitude; fully reusable vehicles aim for another order of magnitude.

### WORKED EXAMPLE — HOW MUCH DOES IT COST TO PUT A 6U CUBESAT IN ORBIT?

A 6U CubeSat has a mass of about 10 kg. On a rideshare mission (for example SpaceX's Transporter series), the list price in 2025 was about US$6 500 per kilogram. The minimum was about US$325 000 for 50 kg to sun-synchronous orbit. So a single 6U satellite would typically be launched through an integrator for roughly US$150 000–300 000, including the deployer. A dedicated small launcher (e.g., Rocket Lab Electron, about US$8.5 million per launch) is far more expensive per kilogram. However, it lets the customer choose the orbit and the launch date. The choice between *rideshare* and *dedicated* launch is one of the first trade-offs in a small-satellite mission (Chapter 11).

## 1.7 Governance: The Rules of Space

Space activities are regulated by a combination of international treaties, UN guidelines, international technical bodies and national laws.

**TABLE 1.5** The core instruments of international space law.

| Instrument | Year | Key provisions |
|---|---|---|
| Outer Space Treaty | 1967 | Space free for exploration by all states; no national appropriation of celestial bodies; no weapons of mass destruction in orbit; states responsible for national activities, including private ones ("authorisation and continuing supervision"); liability for damage |
| Rescue Agreement | 1968 | Assistance to astronauts in distress; return of space objects |
| Liability Convention | 1972 | Absolute liability for damage on Earth's surface or to aircraft; fault-based liability in space |
| Registration Convention | 1975 | States register launched objects with the UN (UNOOSA register) |
| Moon Agreement | 1979 | Moon resources as "common heritage of mankind" — ratified by few states, not by major space powers |
| ITU Constitution and Radio Regulations | updated at each World Radiocommunication Conference (latest WRC-23; next WRC-27) | Allocation of radio frequencies; coordination and registration of satellite networks and orbital positions |
| UN COPUOS Space Debris Mitigation Guidelines | 2007 | Limit debris release, avoid break-ups, post-mission disposal |
| UN Long-Term Sustainability Guidelines | 2019 | 21 voluntary guidelines on safety, SSA, debris, regulation |
| Artemis Accords | 2020 | Non-binding principles for civil lunar exploration: transparency, interoperability, emergency assistance, registration, release of scientific data, preserving heritage sites, deconfliction "safety zones", space-resource use consistent with the OST. 76 signatories by late September 2026 |

National regulators turn these principles into licences. In the United States, for example, the FAA licenses launches and re-entries. The Federal Communications Commission (FCC) licenses spectrum and imposes debris rules, including a five-year post-mission disposal rule for LEO satellites adopted in 2022. NOAA licenses commercial remote-sensing systems. Other countries, including the UK (Space Industry Act 2018), France (French Space Operations Act 2008), Japan (Space Activities Act 2016), New Zealand, Luxembourg and the UAE, have comparable laws. China's International Lunar Research Station (ILRS) programme forms a second international lunar partnership alongside the Artemis Accords.

**LESSON — DEBRIS IS A SHARED RISK**

In 2007 China destroyed its Fengyun-1C weather satellite with a ground-launched missile, creating more than 3 500 trackable fragments, many still in orbit. In 2009 the active Iridium 33 and the defunct Kosmos 2251 collided at 11.7 km/s, creating about 2 300 more. In November 2021 a Russian anti-satellite test destroyed Kosmos 1408 and forced the ISS crew to shelter. Because debris does not respect national boundaries, debris mitigation and space traffic coordination (Chapter 21) have become central policy issues.

## 1.8 The Engineering Disciplines of the Space Ecosystem

A space programme employs almost every branch of engineering and science. Table 1.6 lists the main disciplines and where they appear in this manual.

**TABLE 1.6** Engineering and science disciplines in the space ecosystem.

| Discipline | What it does in space programmes | Main chapters |
|---|---|---|
| Systems engineering | Requirements, architecture, interfaces, budgets, verification, risk | 4, 11, 24 |
| Astrodynamics / flight dynamics | Orbit design, manoeuvres, navigation, orbit determination | 3, 11, 21, 24 |
| Structural and mechanical engineering | Primary structure, mechanisms, deployables, launch loads | 5, 34 |
| Thermal engineering | Radiators, heaters, insulation, heat pipes, cryogenics | 5, 38 |
| Electrical power engineering | Solar arrays, batteries, power conditioning, RTGs | 5, 24 |
| Guidance, navigation and control (GNC) | Attitude determination and control, autonomous navigation, landing | 6, 27, 45 |
| Propulsion engineering | Rocket engines, electric thrusters, propellant systems | 6, 35, 42 |
| Avionics and embedded software | Flight computers, data handling, fault protection, flight software | 6, 45 |
| RF and optical communications | Antennas, transponders, link budgets, lasercom | 7, 17 |
| Payload and instrument engineering | Optics, detectors, radars, spectrometers, particle instruments | 12–16, 38–39 |
| Aerothermodynamics | Re-entry heating, heat shields, atmospheric entry | 10, 27 |
| Human factors and life sciences | Life support, radiation protection, ergonomics, medicine | 30–33 |
| Ground systems and operations | Mission control, scheduling, data processing | 7, 45 |
| Data science and artificial intelligence | Onboard processing, analytics, autonomy, forecasting | 43–53 |
| Product assurance and safety | Reliability, quality, parts, materials, safety analysis | 4 |
| Law, policy and economics | Licensing, spectrum, insurance, finance, export control | 1, 21, 41 |

## 1.9 How This Manual Is Organised

The remaining chapters follow a logical sequence. Part I builds the foundations: the environment (Chapter 2), orbits (Chapter 3), systems engineering (Chapter 4), spacecraft subsystems (Chapters 5 and 6) and communications (Chapter 7). Each later Part applies these foundations to a family of systems. Part XIII then shows how artificial intelligence is changing every one of these domains, from onboard image classification to AI weather models and autonomous rovers. Part XIV collects the reference data an engineer needs at hand.

### KEY TAKEAWAYS

- Space has no physical boundary; 100 km (Kármán line) and 80 km are conventional limits. Orbit requires horizontal speed (≈7.8 km/s), not just height.
- Every mission consists of space, ground, launch and user segments joined by links; interfaces between segments are where many problems occur.
- In 2025 the world made a record 330 orbital launch attempts and placed about 4 500 payloads in orbit; about 16 000 satellites were operating in late 2026, two-thirds of them Starlink.
- The space economy was about US$686 billion in 2025, roughly four-fifths commercial; government budgets (≈US$141 billion) remain the anchor of the industry.
- Space is hard because of the energy needed to reach orbit, the hostile environment, and the near-impossibility of repair; reusability and mass production are changing the economics.
- International treaties set principles, while national licences, ITU spectrum rules and debris guidelines govern day-to-day activities.

### REVIEW QUESTIONS

1. Why is reaching an altitude of 100 km not enough to stay in orbit? What horizontal speed is needed in low Earth orbit?
2. Name the four segments of a space system and describe one interface between two of them that could cause a mission to fail.
3. Classify each of the following by mission purpose and by size class: a 3U CubeSat that images crops, a 6-tonne geostationary communication satellite, and a Mars rover.
4. Describe how the number of orbital launch attempts per year has changed since 2015, and identify the main reasons for the change.
5. Roughly what fraction of the US$686 billion space economy in 2025 was commercial? Give two examples each of commercial and government space activities.
6. Give three reasons why space is “hard”, and for each describe an engineering approach used to overcome it.
7. What principles does the Outer Space Treaty establish? Which other rules and bodies govern spectrum use and debris mitigation in practice?
8. Choose one engineering discipline described in this chapter and explain how it contributes to a satellite mission from design to operations.

# 2 CHAPTER The Space Environment

Image: NASA/JSC

> *A spacecraft must survive two very different environments - the short, violent environment of launch and the long, quiet but hostile environment of space. This chapter explains each hazard in turn - the upper atmosphere and drag, vacuum, the thermal environment, atomic oxygen, plasma and charging, ionising radiation, micrometeoroids and debris, and the mechanical loads of launch - shows how each one varies with orbit, and gives the design rules engineers use to cope with them.*

IN THIS CHAPTER


2.1 Why the Environment Drives Design
2.2 The Neutral Atmosphere
2.3 Vacuum
2.4 The Thermal Environment
2.5 Atomic Oxygen
2.6 Plasma and Spacecraft Charging
2.7 Ionising Radiation
2.8 Micrometeoroids and Orbital Debris
2.9 Mechanical Environments: Launch and Microgravity
2.10 Environments Beyond Earth Orbit
2.11 Designing for the Environment: Standards and Tools


## 2.1 Why the Environment Drives Design

A spacecraft engineer's first question about any mission is *where* the spacecraft will fly, because the environment determines most design decisions. It sets the thickness of shielding, the type of electronic parts, the thermal coatings and the solar array degradation margin. It also sets the propellant needed to fight drag, and even the shape of the vehicle. Many famous failures were environmental failures. An early example is the Telstar 1 satellite, killed in 1962 by radiation from a high-altitude nuclear test. A recent one is the loss of 38 newly launched Starlink satellites to a minor geomagnetic storm in February 2022.

The space environment can be divided into the following components, each discussed in this chapter:

1. **The neutral atmosphere** — drag, atomic oxygen, and their dependence on solar activity.
2. **Vacuum** — outgassing, contamination, cold welding, discharges.
3. **The thermal environment** — sunlight, Earth albedo and infrared, eclipses, deep-space cold.
4. **Plasma** — charged particles that charge spacecraft surfaces and cause discharges.
5. **Ionising radiation** — trapped belts, solar particle events and galactic cosmic rays.

6. **Micrometeoroids and orbital debris (MMOD)** — hypervelocity impacts.
7. **Mechanical environments** — launch loads, vibration, acoustics, shock, and microgravity.

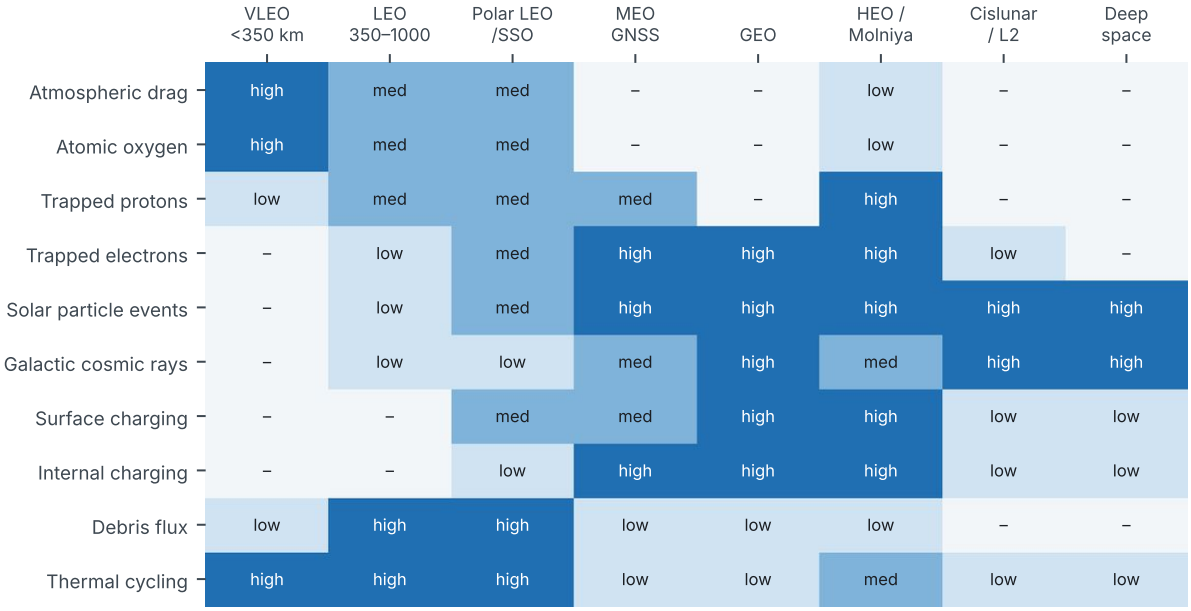



**FIGURE 2.1** Relative severity of the main environmental hazards in different orbital regimes. No orbit is benign: low orbits suffer drag, atomic oxygen and debris; medium and high orbits suffer radiation and charging; deep space adds cosmic rays and solar particle events with no magnetospheric protection.

## 2.2 The Neutral Atmosphere

### 2.2.1 Structure of the Atmosphere

The atmosphere is divided into layers by the way temperature changes with height (Figure 2.2). In the **troposphere** temperature falls about 6.5 K per kilometre. In the **stratosphere** it rises again because ozone absorbs ultraviolet sunlight. It falls through the **mesosphere** to the coldest point on Earth, the **mesopause** at 85–90 km (about −90 °C), where noctilucent clouds form and meteors burn up. Above that, in the **thermosphere**, the very thin gas absorbs extreme-ultraviolet (EUV) radiation from the Sun and is heated to 700–1 400 K or more. Above about 500 km lies the **exosphere**, where atoms are so sparse that they follow ballistic trajectories and can escape to space.

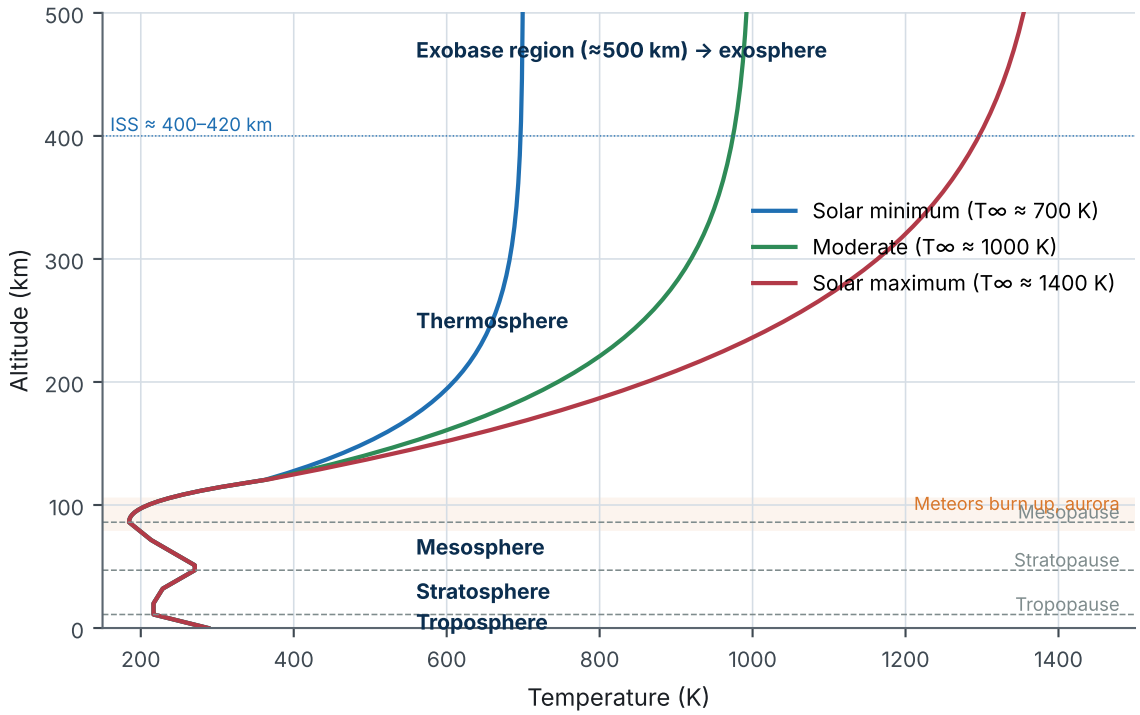



**FIGURE 2.2** Temperature profile of the atmosphere. Below 86 km the US Standard Atmosphere 1976 is shown; above, the thermosphere approaches an "exospheric temperature" $T_\infty$ that depends strongly on solar EUV output.

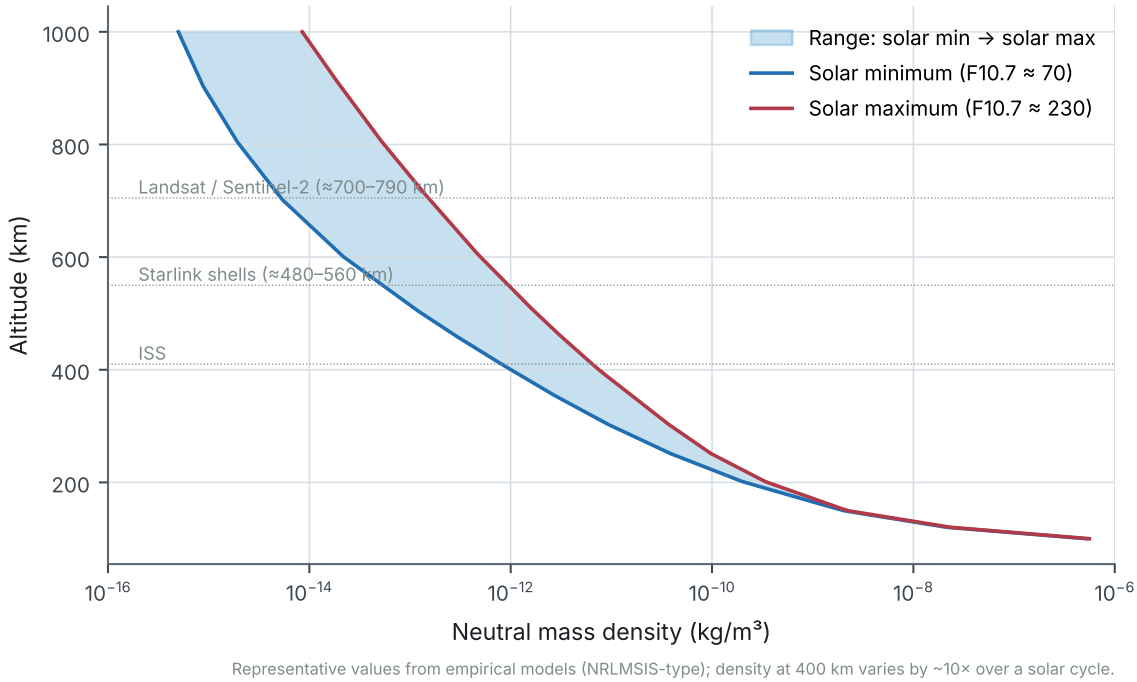
The composition also changes. Below about 100 km the atmosphere is well mixed (78 % $N_2$, 21 % $O_2$). Above, gases separate by molecular weight, and solar ultraviolet splits $O_2$ into **atomic oxygen (O)**, which becomes the dominant species between about 200 and 600 km. Higher still, helium and then hydrogen dominate.

### 2.2.2 Density and the Scale Height

In a column of gas in hydrostatic equilibrium, the pressure falls exponentially with altitude:

$$\rho(h) = \rho_0 \exp\left(-\frac{h - h_0}{H}\right), \qquad H = \frac{k_B T}{m g}$$

where $H$ is the **scale height** — the height over which density falls by a factor of $e \approx 2.72$. Near the ground $H \approx 8.5$ km; at 400 km, where the gas is hot and light, $H$ is about 50–60 km. The value of $H$ depends on temperature, and thermospheric temperature depends on the Sun. So **density at a given LEO altitude can change by a factor of ten or more** between solar minimum and solar maximum. It can also change by a factor of two to five within hours during a geomagnetic storm (Figure 2.3).




Representative values from empirical models (NRLMSIS-type); density at 400 km varies by ~10× over a solar cycle.

**FIGURE 2.3** Neutral density of the upper atmosphere for low and high solar activity. At typical LEO altitudes, the density — and therefore the drag on a satellite — varies by about an order of magnitude over the 11-year solar cycle.

Solar activity is tracked by two indices that drive the atmospheric models (NRLMSIS 2.0, JB2008, DTM-2020). The **F10.7 index** is the solar radio flux at 10.7 cm wavelength (a proxy for EUV output), typically 70 at solar minimum and 150–250 at maximum. The geomagnetic **Ap / Kp indices** measure the heating of the upper atmosphere by the solar wind (Chapter 22).

### 2.2.3 Atmospheric Drag

A satellite moving at velocity $v$ through gas of density $\rho$ experiences a drag force

$$F_D = \tfrac{1}{2}\,\rho\, v^2\, C_D\, A$$

where $A$ is the cross-sectional area and $C_D$ is the drag coefficient (typically 2.0–2.4 for satellites in free-molecular flow). The ratio $B = m/(C_D A)$ is the **ballistic coefficient**: a dense, compact satellite (large $B$) decays slowly; a satellite with large solar arrays (small $B$) decays quickly. For a circular orbit of radius $a$, drag reduces the orbit radius at the rate

$$\frac{da}{dt} = -\frac{\rho\sqrt{\mu\, a}}{B}$$

Counter-intuitively, drag makes a satellite *speed up*: as it loses energy it sinks to a lower orbit, where the orbital speed is higher. The decay accelerates as the satellite descends into denser air. So most of a satellite's orbital life is spent at the higher altitudes, and the final descent from 200 km takes only days. Figure 2.4 shows typical orbital lifetimes.

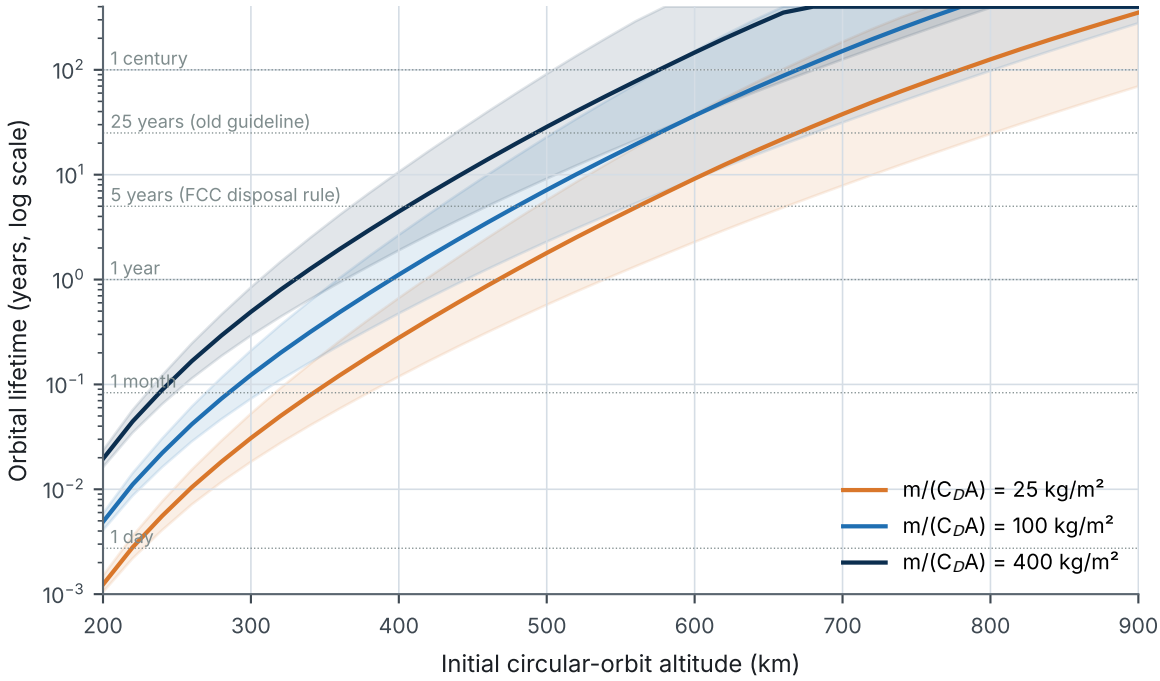



**FIGURE 2.4** Approximate orbital lifetime versus initial altitude for three ballistic coefficients. A satellite at 400 km with m/(C_D A) = 100 kg/m² lasts about one to three years; at 600 km it lasts decades; at 800 km it lasts more than a century. The 5-year line marks the US FCC post-mission disposal rule (2022) for LEO satellites.

**WORKED EXAMPLE — HOW FAST DOES THE ISS LOSE HEIGHT?**

Take $a = 6\,778$ km (400 km altitude), $\rho = 3 \times 10^{-12}$ kg/m³ (moderate activity), mass 420 t, effective $C_D A \approx 2.2 \times 1\,900$ m² ≈ 4 200 m², so $B \approx 100$ kg/m².

$$\frac{da}{dt} = -\frac{3 \times 10^{-12} \times \sqrt{3.986 \times 10^{14} \times 6.778 \times 10^{6}}}{100} \approx -1.6 \times 10^{-3}\ \text{m/s}$$

That is about 140 m per day, or 4 km per month. The real ISS loses 1–5 km per month depending on solar activity and attitude, and is periodically re-boosted by the engines of Progress or Cygnus cargo vehicles. Without re-boosts it would re-enter within one to two years.

**FAILURE LESSON — THE FEBRUARY 2022 STARLINK LOSS**

On 3 February 2022, 49 Starlink satellites were released into a low 210 × 340 km parking orbit, a deliberately low orbit that ensures failed satellites re-enter quickly. A moderate geomagnetic storm the next day increased atmospheric density at that altitude by up to 50 %. The satellites, placed in a low-drag "knife-edge" attitude, could not climb out against the extra drag, and 38 re-entered. The lesson: in very low orbits, space-weather forecasts are as important as weather forecasts are for aircraft.

### 2.2.4 Very Low Earth Orbit (VLEO)

Orbits below about 350 km are attracting interest for high-resolution imaging (resolution improves in proportion to distance), lower latency and natural debris removal. Satellites such as ESA's GOCE, Japan's SLATS/Tsubame and a new generation of commercial VLEO imagers use streamlined shapes, electric propulsion and atmosphere-resistant materials. GOCE operated at 255 km from 2009 to 2013, using an ion engine to cancel drag. SLATS/Tsubame flew as low as 167 km (2017–2019). Some concepts even scoop up atmospheric gas to feed an electric thruster ("air-breathing electric propulsion").

## 2.3 Vacuum

At 400 km the pressure is about $10^{-6}$ Pa, and in GEO about $10^{-11}$ Pa — far better vacuum than most laboratory chambers. Vacuum causes several problems:

- **Outgassing.** Polymers, adhesives, paints and composites release trapped water and volatile compounds. The vapour can condense on cold optics, star trackers and solar cells, reducing their performance. Materials are screened with the ASTM E595 test: total mass loss (TML) below 1.0 % and collected volatile condensable material (CVCM) below 0.1 %. Sensitive instruments use "bake-out" before launch, contamination budgets and heated covers.
- **Cold welding and friction.** Clean metal surfaces in contact can bond (cold weld) in vacuum because there is no oxide layer or adsorbed gas to separate them. Mechanisms use dissimilar materials, dry lubricants ($MoS_2$, lead), or space-qualified oils with very low vapour pressure (perfluoropolyethers, multiply-alkylated cyclopentanes).
- **Sublimation.** Some materials (cadmium, zinc, uncoated magnesium) sublimate at elevated temperatures; tin whiskers grow on pure-tin finishes. Space-grade parts forbid pure tin, cadmium and zinc plating.
- **Electrical discharge.** Between about 1 and 1 000 Pa — the pressure inside a spacecraft just after launch — the breakdown voltage of a gap reaches a minimum (Paschen's law). High-voltage equipment must be vented or switched off during ascent. In RF equipment, high-power signals can cause **multipaction**, a resonant electron avalanche between surfaces in vacuum.
- **No convection.** Heat moves only by conduction and radiation, so electronics cannot be cooled by fans and air (except inside pressurised modules).

## 2.4 The Thermal Environment

### 2.4.1 Heat Sources

A spacecraft in Earth orbit receives heat from three external sources and loses heat only by radiation (Figure 2.5):

1. **Direct sunlight.** The **solar constant** at 1 AU is about $S = 1\,361$ W/m² (it varies ±3.4 % between January perihelion and July aphelion, and only ±0.1 % over the solar cycle). The spectrum peaks in the visible.
2. **Albedo.** The Earth reflects on average about 30 % of incident sunlight (0.25–0.40 depending on clouds, ice and ocean).
3. **Earth infrared.** The Earth radiates about 240 W/m² at its effective temperature of 255 K, with a peak wavelength near 10 μm.

In addition, internal electronics dissipate heat. Deep space appears as a sink at 2.7 K.

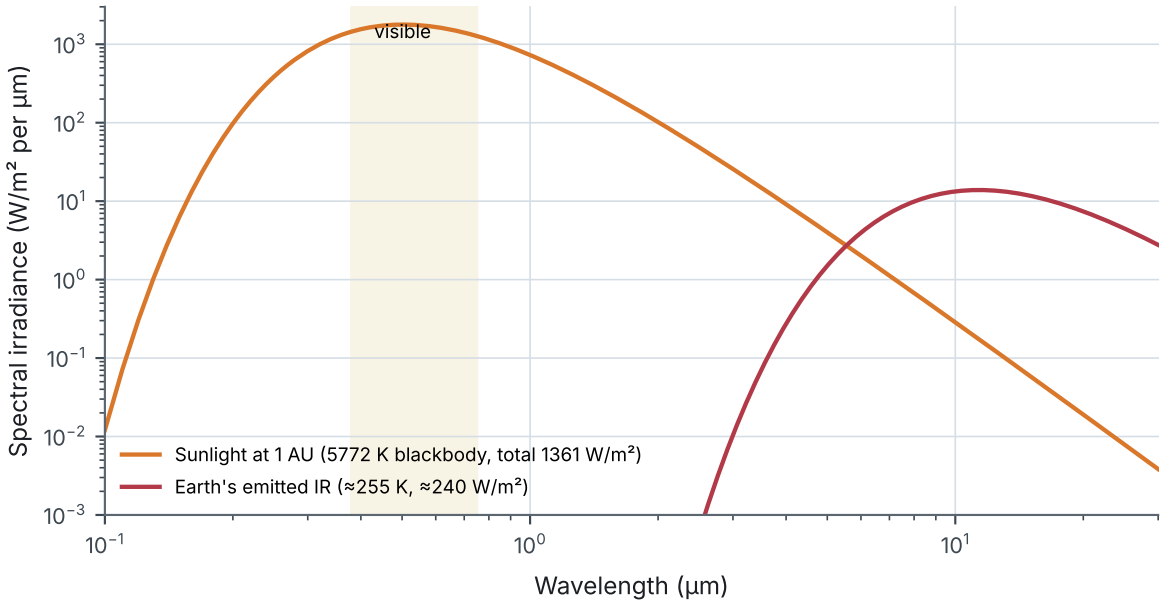



**FIGURE 2.5** Spectra of sunlight and of Earth's thermal emission. Because the two hardly overlap, a surface can be designed to absorb little sunlight (low α) while radiating strongly in the infrared (high ε) — the principle behind white paints and mirrors used on radiators.

### 2.4.2 Radiative Equilibrium

The temperature of a surface is set by the balance between absorbed and emitted power. For a flat plate facing the Sun with an insulated back:

$$\alpha S A = \varepsilon \sigma A T^4 \quad \Rightarrow \quad T = \left( \frac{\alpha}{\varepsilon} \frac{S}{\sigma} \right)^{1/4}$$

where $\alpha$ is the **solar absorptance**, $\varepsilon$ the **infrared emittance** and $\sigma = 5.67 \times 10^{-8}$ W m$^{-2}$ K$^{-4}$ the Stefan–Boltzmann constant. The ratio $\alpha/\varepsilon$ controls temperature. For an isolated sphere (absorbing on its cross-section $\pi r^2$, radiating from its surface $4\pi r^2$) the result is $T = (\alpha S / 4\varepsilon\sigma)^{1/4}$. Figure 2.6 shows how surface coatings span a range of more than 300 °C in equilibrium temperature.

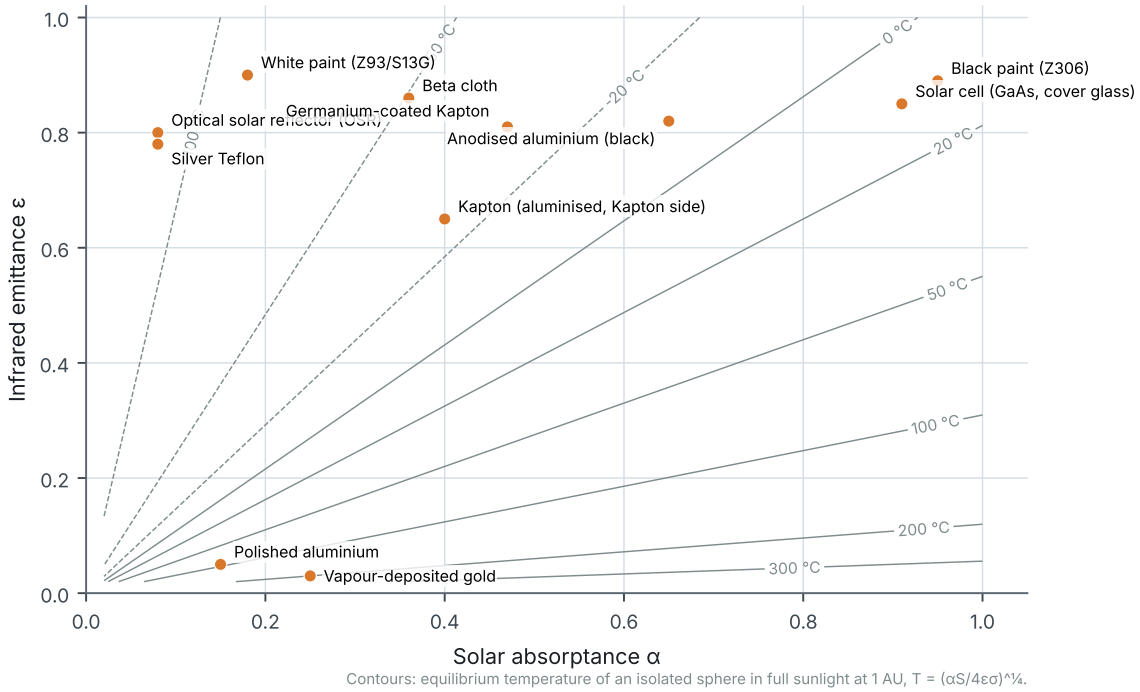



**FIGURE 2.6** Absorptance–emittance map of common spacecraft surface finishes with lines of equal equilibrium temperature. Radiators use low-α/high-ε finishes (white paint, optical solar reflectors); blankets use low-ε outer layers to isolate; black paint maximises heat exchange inside instruments.

**WORKED EXAMPLE — TEMPERATURE OF A CUBESAT IN SUNLIGHT AND ECLIPSE**

A 1U CubeSat (10 cm cube) is covered in solar cells (α = 0.91, ε = 0.85). In a 500 km orbit it absorbs sunlight on (on average) about 1.5 faces, plus albedo and Earth IR on the Earth-facing side. A simple estimate of absorbed power in sunlight is

$$P_{in} \approx \alpha S A_{sun} + \alpha a S F A_{earth} + \varepsilon q_{IR} F A_{earth} + P_{int}$$
$$\approx 0.91 \times 1361 \times 0.015 + 0.91 \times 0.3 \times 1361 \times 0.5 \times 0.01 + 0.85 \times 240 \times 0.5 \times 0.01 + 1 \approx 18.6 + 1.9 + 1.0 + 1 \approx 22.5 \text{ W}$$

The satellite radiates from 0.06 m², so $T = \left(22.5/(0.85 \times 5.67 \times 10^{-8} \times 0.06)\right)^{1/4} \approx 290$ K (17 °C). In eclipse (35 minutes of a 95-minute orbit) only Earth IR and internal power remain (≈2 W), and the equilibrium would be about 150 K. Thermal mass slows the cooling, so the real minimum is typically −10 to −30 °C. Batteries (operating range typically 0–40 °C for charging) usually need heaters or careful placement.

### 2.4.3 Eclipses and Thermal Cycling

A LEO satellite passes through Earth's shadow on most orbits — up to about 36 minutes of a 92-minute orbit at 400 km. It therefore experiences about 15–16 thermal cycles per day, or more than 50 000 cycles in ten years. These cycles fatigue solder joints, composite bonds and solar-array interconnects. Sun-synchronous dawn–dusk orbits avoid most eclipses. GEO satellites experience eclipses only during two **eclipse seasons** of about 45 days around the equinoxes, with a maximum eclipse of 72 minutes per day.

The thermal control techniques — insulation blankets, radiators, heat pipes, louvres and heaters — are described in Chapter 5.

## 2.5 Atomic Oxygen

Between about 200 and 700 km, atomic oxygen (AO) is the most abundant gas. A satellite moving at 7.7 km/s hits AO atoms with an energy of about 5 eV, enough to break chemical bonds. AO **erodes** organic materials such as Kapton, Mylar and carbon-fibre epoxy, oxidises silver, and degrades some paints. Erosion is characterised by the **erosion yield** — the volume of material removed per incident atom:

$$\Delta x = E_y \, \Phi$$

where $\Phi$ is the fluence (atoms/cm$^2$). Kapton H has $E_y \approx 3.0 \times 10^{-24}$ cm$^3$/atom. At 400 km during solar maximum, the ram-facing fluence can exceed $10^{22}$ atoms/cm$^2$ per year, removing about 0.3 mm of Kapton per year — enough to destroy an unprotected 25 µm film in weeks. Protective measures include silicon-dioxide or aluminium-oxide coatings, AO-resistant polymers (e.g., polysiloxane-based coatings, "CP1"), silicone-based paints and metal foils. NASA's Materials International Space Station Experiment (MISSE) series has exposed thousands of material samples on the ISS exterior since 2001 to measure these effects.

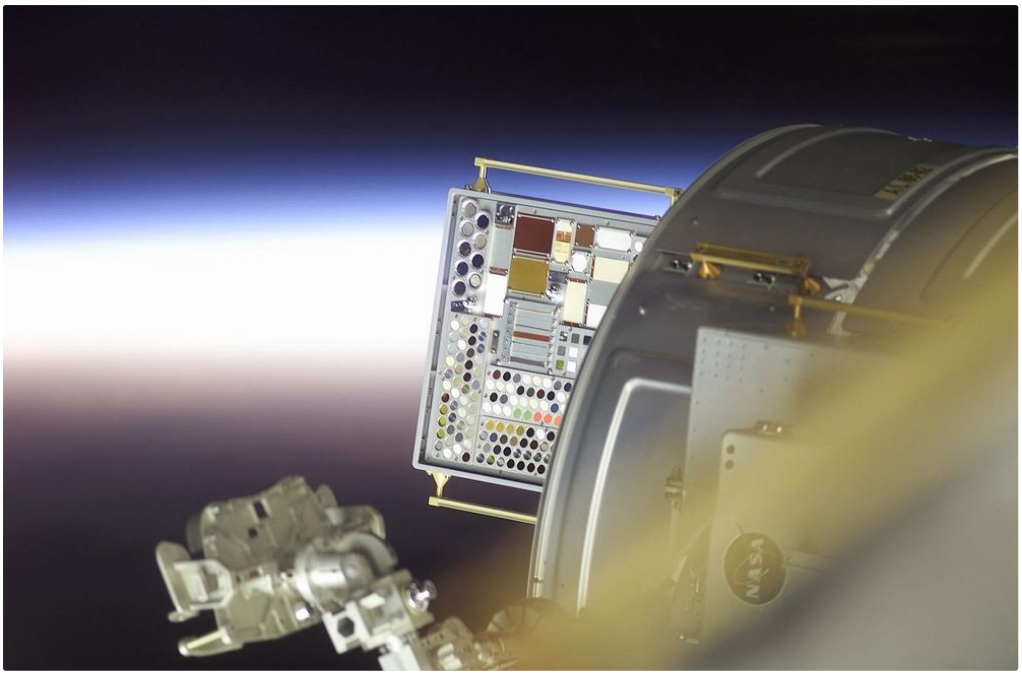


**FIGURE 2.7** The first Materials International Space Station Experiment (MISSE-1) mounted on the outside of the ISS airlock in 2001, backdropped by an orbital sunrise. Each suitcase-like carrier exposed hundreds of material samples to atomic oxygen, ultraviolet light, radiation and thermal cycling for about 18 months before being returned to Earth for analysis. Image: NASA/MSFC.

AO also produces a faint orange **"spacecraft glow"** on ram-facing surfaces and can affect the calibration of sensitive optical instruments.

## 2.6 Plasma and Spacecraft Charging

Space is filled with **plasma** — ionised gas containing free electrons and ions. In LEO the plasma is cold (≈0.1 eV) and dense ($10^{10}$–$10^{12}$ m$^{-3}$) — this is the ionosphere (Chapter 23). In GEO it is hot (keV energies) and tenuous. A spacecraft floating in plasma collects charge until the currents of electrons and ions to its surface balance, which sets its **floating potential**.

**Surface charging** is most severe in GEO and in high-inclination LEO orbits during geomagnetic substorms, when hot electrons (1–50 keV) are injected from the magnetotail. A sunlit surface emits photoelectrons and stays near zero potential, but shadowed dielectric surfaces may charge to thousands of volts negative. If neighbouring surfaces reach different potentials, an **electrostatic discharge (ESD)** can occur, injecting current pulses into cables, causing anomalies ("phantom commands"), or damaging solar arrays. Mitigations include conductive coatings (indium-tin oxide, Germanium-coated Kapton, carbon-loaded black Kapton), grounding all conductive parts, and limiting the voltage between adjacent solar cells.

**Internal (deep-dielectric) charging** occurs when energetic electrons (above about 0.5 MeV) penetrate shielding and accumulate inside cable insulation and circuit boards, eventually discharging. It is the dominant anomaly mechanism in GEO and MEO during periods of high outer-belt electron flux — for example after high-speed solar-wind streams. Guidelines (NASA-HDBK-4002B, ECSS-E-ST-20-06C) limit floating metal and require bleed paths for charge.

**High-voltage solar arrays** in LEO (above about 100 V) can arc into the surrounding plasma. The ISS uses 160 V arrays and a **plasma contactor unit** that emits xenon plasma to keep the structure near plasma potential, protecting astronauts during spacewalks.

## 2.7 Ionising Radiation

### 2.7.1 Sources

Three populations of energetic particles reach spacecraft:

1. **Trapped radiation (Van Allen belts).** Earth's magnetic field traps electrons and protons in two toroidal belts (Figure 2.8). The **inner belt** (1.2–2.5 Earth radii, $R_E$) contains protons of tens to hundreds of MeV, produced mainly by cosmic-ray neutron decay; it is very stable. The **outer belt** (3–7 $R_E$) contains electrons of 0.1–10 MeV; it varies dramatically with solar-wind conditions. Between them lies the **slot region**. GNSS satellites (≈4.2 $R_E$) fly in the heart of the outer belt; GEO satellites (6.6 $R_E$) on its outer edge.
2. **Solar particle events (SPEs).** Large solar flares and coronal mass ejections accelerate protons (and heavier ions) to tens or hundreds of MeV. Events last hours to days and can increase radiation doses by orders of magnitude. They are the dominant hazard for astronauts beyond LEO (Chapter 30). The largest events of the space age include August 1972 (between Apollo 16 and 17) and October 1989.
3. **Galactic cosmic rays (GCRs).** Fully ionised nuclei (87 % protons, 12 % helium, 1 % heavier ions up to iron and beyond) arriving from outside the Solar System with energies up to $10^{20}$ eV. Their flux is low but they are nearly impossible to shield, and heavy ions cause single-event effects in electronics. GCR flux is about 2–3 times higher at solar minimum, when the weaker solar wind lets more of them in.

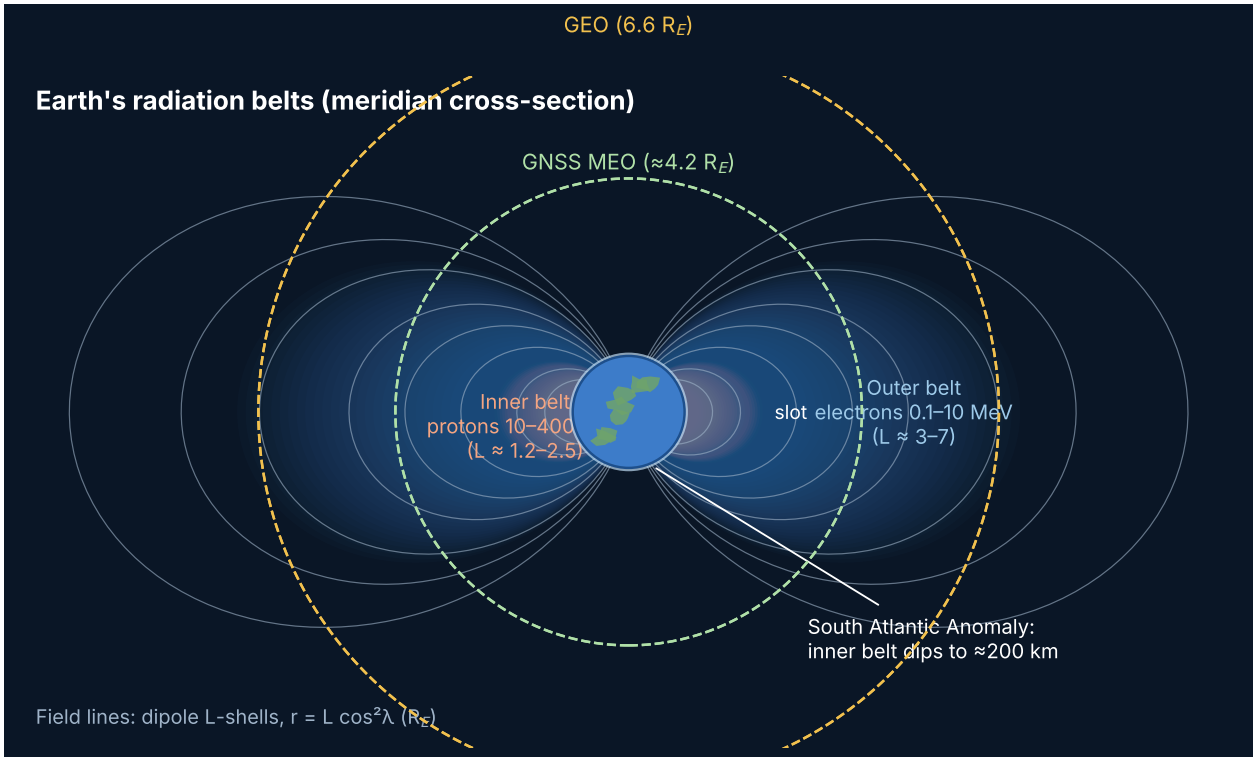



**FIGURE 2.8** Cross-section of Earth's magnetic field and radiation belts. Low Earth orbit lies below the belts except over the South Atlantic Anomaly; navigation satellites fly through the outer electron belt; geostationary satellites sit at its outer edge.

Because Earth's magnetic axis is offset and tilted relative to its rotation axis, the inner belt comes closest to the surface over South America and the South Atlantic — the **South Atlantic Anomaly (SAA)**. LEO satellites, including the ISS and Hubble, receive most of their trapped-proton dose during SAA passes, and many

instruments switch off or flag their data there (Figure 2.9). In polar orbits, satellites also cross the "horns" of the outer belt at high latitudes and are exposed to solar particles, which reach low altitudes near the magnetic poles.

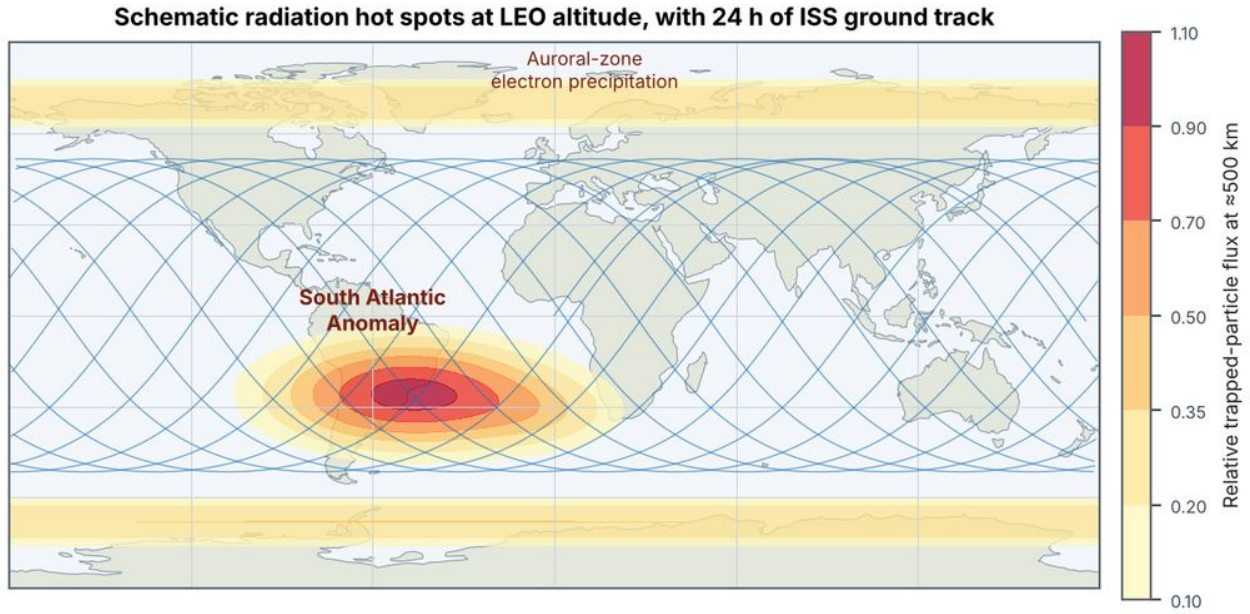



**FIGURE 2.9** Schematic map of the regions where LEO satellites receive most of their radiation: the South Atlantic Anomaly and the high-latitude auroral zones, with one day of International Space Station ground track overlaid.

### 2.7.2 Radiation Effects on Electronics and Materials

Radiation affects spacecraft in three main ways:

- **Total ionising dose (TID).** Energy deposited by ionisation accumulates over the mission, trapping charge in oxides of transistors and shifting their threshold voltages and leakage currents until circuits fail. Dose is measured in gray (Gy = 1 J/kg) or, traditionally, **rad** (1 rad = 0.01 Gy), quoted for silicon as krad(Si).
- **Displacement damage dose (DDD).** Protons and neutrons knock atoms out of the crystal lattice, degrading solar cells, optocouplers, CCD and CMOS image sensors (more "hot pixels", worse charge-transfer efficiency) and bipolar transistors.
- **Single-event effects (SEEs).** A single heavy ion or energetic proton deposits a dense track of charge. This can flip a memory bit (**single-event upset, SEU**) or disturb an analogue output (**single-event transient, SET**). It can also latch a CMOS circuit into a high-current state that can burn it out (**single-event latch-up, SEL**), or destroy power MOSFETs (**single-event burnout, SEB; single-event gate rupture, SEGR**).

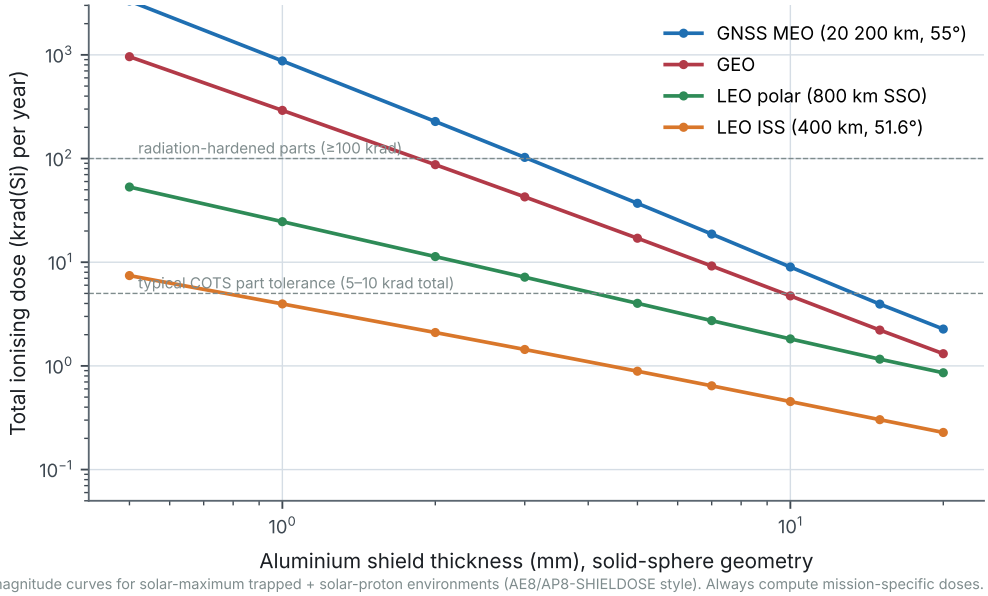
Table 2.1 summarises effects and mitigations.

**TABLE 2.1** Radiation effects, typical causes and mitigations.

| Effect | Main cause | Typical consequence | Mitigations |
|---|---|---|---|
| Total ionising dose (TID) | Trapped electrons and protons, SPE protons | Parameter drift, increased power, eventual failure | Shielding, rad-hard parts, derating, spot shielding, part screening (lot acceptance testing) |
| Displacement damage (DDD) | Protons, neutrons | Solar array power loss, detector dark current, CCD charge transfer loss | Cover glass, annealing, operating detectors cold, design margins (typically 15–30 % end-of-life array margin) |
| Single-event upset (SEU) | Heavy ions (GCR, SPE), protons in SAA | Bit flips in memories and registers | Error detection and correction (EDAC), triple modular redundancy (TMR), memory scrubbing, watchdogs |

| Effect | Main cause | Typical consequence | Mitigations |
|---|---|---|---|
| Single-event latch-up (SEL) | Heavy ions, protons | High current; destructive if not cleared | Latch-up-immune technologies (SOI), current limiters with automatic power cycling |
| Single-event burnout / gate rupture | Heavy ions | Permanent destruction of power devices | Voltage derating (e.g., operate at < 50–75 % of rated $V_{DS}$), rad-hard MOSFETs |
| Single-event functional interrupt (SEFI) | Heavy ions | Processor or FPGA hangs | Watchdog timers, reset logic, configuration scrubbing |

Electronic parts are classified by their tolerance. Commercial off-the-shelf (COTS) parts often survive 5–30 krad(Si) and can be susceptible to latch-up. By contrast, **radiation-tolerant** parts are characterised to 30–100 krad, while **radiation-hardened** parts are designed for 100 krad to more than 1 Mrad and are latch-up immune. Rad-hard parts are expensive and lag commercial technology by one to two decades in performance — a major reason why AI processors in space are a challenge (Chapter 45).




Illustrative order-of-magnitude curves for solar-maximum trapped + solar-proton environments (AE8/AP8-SHIELDOSE style). Always compute mission-specific doses.

**FIGURE 2.10** Illustrative annual total ionising dose behind spherical aluminium shielding for four orbits. MEO is the harshest common orbit; LEO at ISS altitude is comparatively benign. Shielding is effective against electrons (MEO, GEO) but less effective against high-energy protons.

### 2.7.3 Estimating Upset Rates

The susceptibility of a device to single-event effects is measured in particle accelerators. The result is a curve of **cross-section** $\sigma$ (the effective sensitive area per bit or per device) versus **linear energy transfer (LET)**, the energy deposited per unit path length, usually given in MeV·cm²/mg (Figure 2.11). A device with a threshold LET above about 37 MeV·cm²/mg is considered effectively immune to heavy-ion upsets in most environments, because heavier, more ionising particles are very rare. The upset rate is found by folding the cross-section curve with the LET spectrum of the orbit (tools: CREME96, OMERE, SPENVIS).

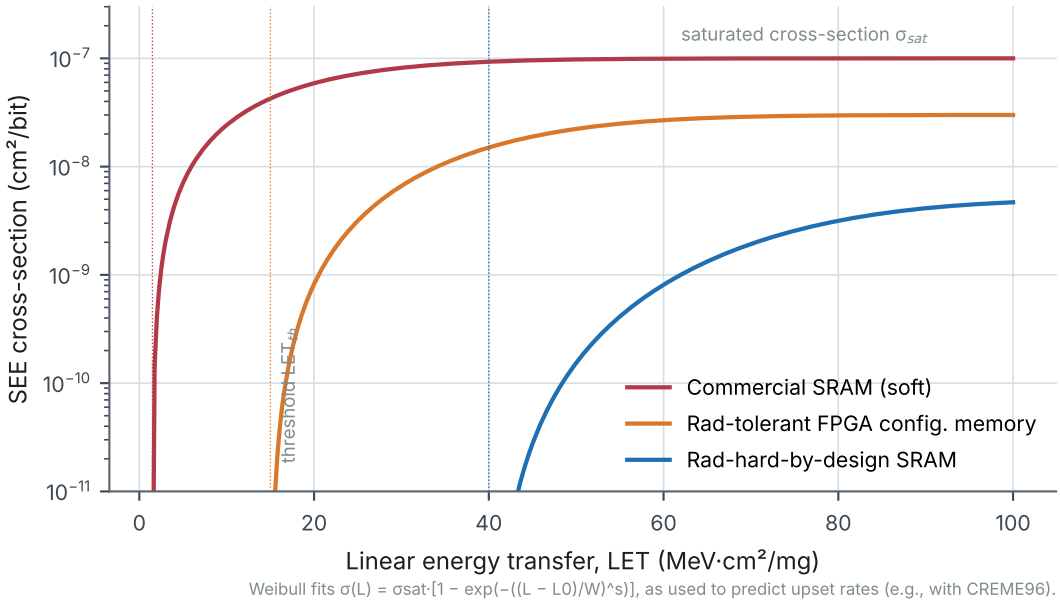



**FIGURE 2.11** Single-event upset cross-section versus LET for three classes of memory, fitted with Weibull functions. Hardening raises the threshold LET and lowers the saturated cross-section, reducing upset rates by orders of magnitude.

**ENGINEERING NOTE — RADIATION DESIGN MARGIN**

Agencies require a **radiation design margin (RDM)** — the ratio of a part's tested capability to the predicted mission dose at its location. NASA and ESA practice is typically RDM ≥ 2 for TID (with lot-specific test data), rising for parts with large lot-to-lot variability. Dose predictions use standard models: AE8/AP8 or the newer **AE9/AP9-IRENE** for trapped particles, ESP/PSYCHIC or SAPPHIRE for solar protons, and ISO 15390 or Badhwar–O'Neill for cosmic rays. These are combined with ray-tracing through a 3-D model of the spacecraft (sector analysis) to find the shielding at each part.

## 2.8 Micrometeoroids and Orbital Debris

### 2.8.1 The Particle Population

Two kinds of small particles strike spacecraft at hypervelocity:

- **Micrometeoroids** — natural dust from comets and asteroids. They arrive from all directions at 11–72 km/s (average about 20 km/s), with annual showers such as the Perseids and Leonids adding short peaks. About 30–40 tonnes of meteoroid material enter Earth's atmosphere every day.
- **Orbital debris** — artificial objects: fragments from explosions and collisions, paint flakes, solid-rocket-motor slag and aluminium-oxide dust, and dead satellites. In LEO, debris impacts occur at an average of about 10 km/s, mainly from directions near the velocity vector.

For particles smaller than about 0.1 mm, meteoroids dominate; for larger particles in LEO, debris dominates. Objects larger than about 10 cm in LEO (1 m in GEO) are tracked and catalogued (Chapter 21). The roughly one million objects between 1 and 10 cm are not tracked and cannot be avoided. Yet they carry enough energy to destroy a satellite. A 1 cm aluminium sphere at 10 km/s has a kinetic energy of about 70 kJ — similar to a small car at 50 km/h.

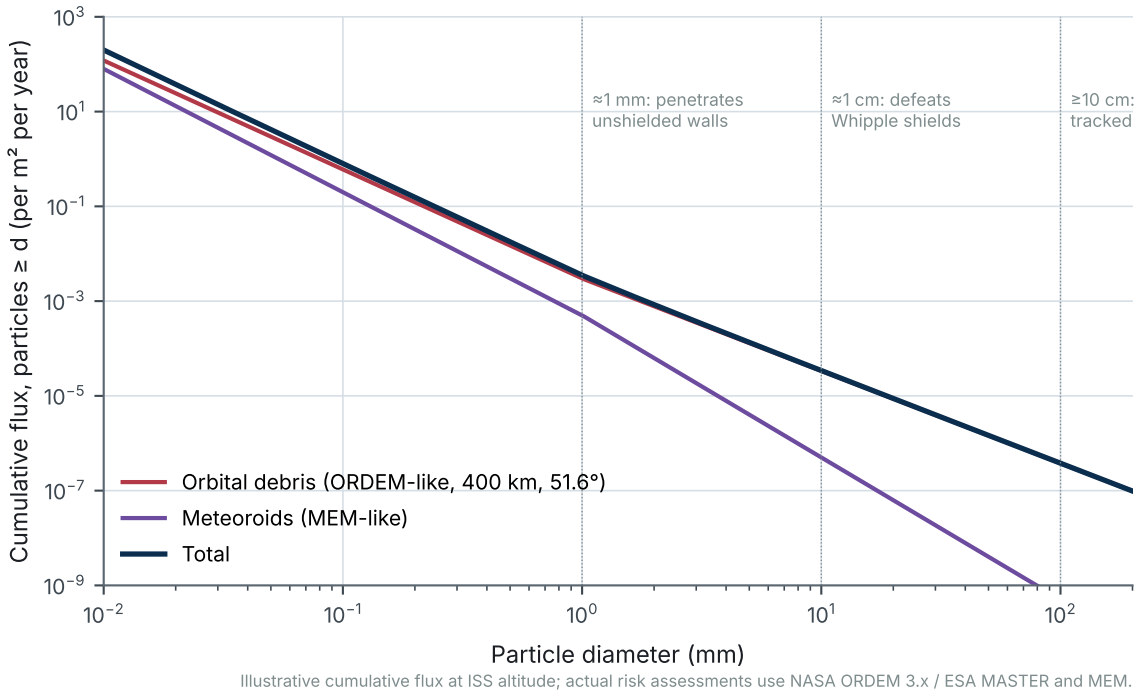



**FIGURE 2.12** Illustrative cumulative flux of meteoroids and orbital debris at ISS altitude. The steep size dependence means that small impacts are frequent but survivable, while the rare large impacts are catastrophic.

## 2.8.2 Shielding

The standard protection is the **Whipple shield**, invented by astronomer Fred Whipple in 1947 (Figure 2.13). A thin sacrificial **bumper** placed some distance in front of the main wall shatters the incoming particle into a cloud of fragments and vapour, which spreads out and is stopped by the rear wall. A Whipple shield is far more mass-efficient than a single thick wall. Variants include the **stuffed Whipple shield** (with layers of Nextel ceramic fabric and Kevlar between bumper and wall, used on the ISS) and **multi-shock shields**. The ISS has more than 100 types of shield and is the most heavily protected spacecraft ever flown; the Bigelow-built BEAM and future inflatable habitats use multiple layers of fabric.

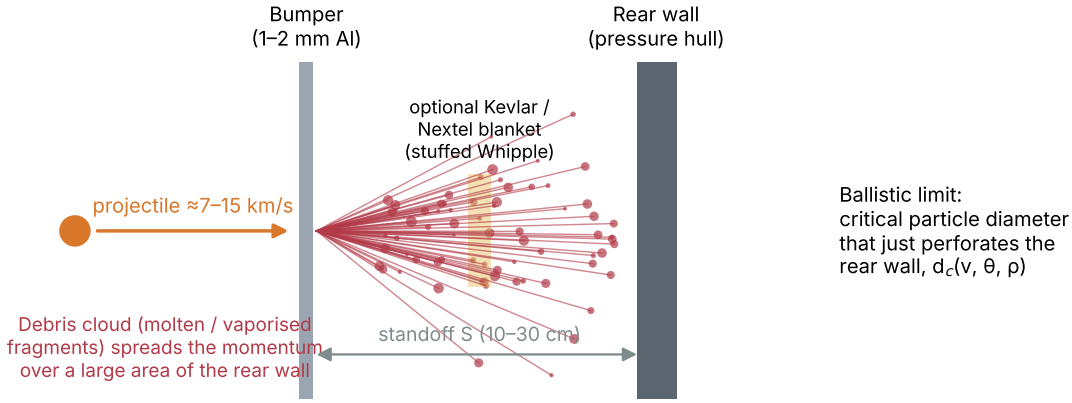



**FIGURE 2.13** Principle of the Whipple shield. The bumper breaks up and partly vaporises the projectile; the resulting cloud spreads its momentum over a large area of the rear wall.

**FIGURE 2.14** The Long Duration Exposure Facility (LDEF), a 9.7-tonne bus-sized cylinder carrying 57 experiments, being retrieved by Space Shuttle Columbia in January 1990 after almost six years in orbit. LDEF's surfaces recorded more than 30 000 impact craters visible to the naked eye and remain a cornerstone data set for meteoroid, debris and atomic-oxygen models. Image: NASA/JSC.

Shield performance is described by **ballistic limit equations** (BLEs). These are empirical fits to hypervelocity gun tests (light-gas guns reach about 7 km/s) and hydrocode simulations. They give the critical particle diameter $d_c$ that just perforates the rear wall, as a function of velocity, impact angle, particle density and shield geometry. Combining the BLE with the flux model gives the **probability of no penetration (PNP)** over the mission, typically required to be above 0.95–0.99 for critical elements. Tools include NASA's BUMPER and ESA's ESABASE2/Debris.

> **DATA BOX — MMOD ON THE ISS AND HUBBLE**
>
> - The ISS has been struck by many thousands of small particles; impact pits are regularly found on windows (the Cupola windows have external shutters for protection), radiators and handrails.
> - In 2021 an untracked object punched a 5 mm hole through the thermal blanket of the Canadarm2 boom — the arm continued to work.
> - Solar arrays returned from Hubble servicing missions (1993, 2002) showed thousands of impact features, used to calibrate debris models.
> - The ISS performs a debris-avoidance manoeuvre when a tracked object's collision probability exceeds about 1 in 10 000 (or about 1 in 100 000 if the manoeuvre is easy). It has manoeuvred more than 40 times since 1999.

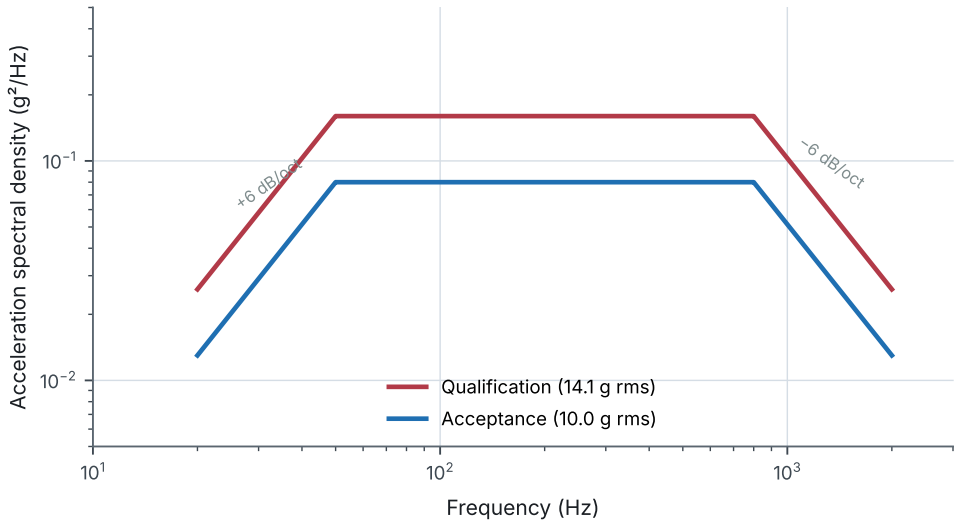
## 2.9 Mechanical Environments: Launch and Microgravity

### 2.9.1 Launch Loads

The most mechanically severe minutes of a spacecraft's life are during launch. Loads arise from:

- **Quasi-static acceleration.** Steady thrust produces axial accelerations of 3–6 g at burnout (Falcon 9 ≈ 4–5 g, Ariane 6 ≈ 4.5 g, Soyuz ≈ 4.3 g); low-frequency dynamic transients add lateral accelerations of 1–2 g.
- **Sine vibration.** Low-frequency (5–100 Hz) oscillations from engine ignition, pogo, wind gusts and staging. Spacecraft must have their primary structural natural frequencies above values specified by the launcher (typically > 20–25 Hz axial and > 10–15 Hz lateral) to avoid dynamic coupling.

- **Random vibration.** Broadband vibration (20–2 000 Hz) transmitted through the structure from engines and aerodynamic turbulence. Components are tested to spectra such as NASA's General Environmental Verification Standard (GEVS) shown in Figure 2.15.
- **Acoustic noise.** Sound pressure levels inside the fairing reach 135–145 dB overall at lift-off and in the transonic region, exciting light, large-area structures such as solar panels and antennas.
- **Shock.** Pyrotechnic separation events (fairing jettison, stage and spacecraft separation) produce short, high-frequency accelerations of thousands of g, characterised by a **shock response spectrum (SRS)**. Shock attenuates rapidly with distance, so sensitive units are placed away from separation interfaces.




Generalized random-vibration test levels for components ≤ 22.7 kg (NASA GSFC-STD-7000, GEVS). Duration 2 min/axis (qual), 1 min/axis (acceptance).

**FIGURE 2.15** Random-vibration qualification and acceptance spectra from NASA's GEVS for small components. The plateau from 50 to 800 Hz at 0.16 g²/Hz corresponds to an overall level of 14.1 g rms.

### 2.9.2 Microgravity

Once in orbit, the spacecraft and everything in it are in free fall. Residual accelerations of $10^{-6}$ to $10^{-3}$ g arise from drag, gravity gradient, rotation, crew motion and machinery. Microgravity matters for fluid management, because propellant does not settle at the bottom of a tank (see Chapter 6). It matters for heat transfer: there is no buoyant convection, so boiling and combustion behave differently. It also matters for deployable structures, which must be tested on the ground with gravity-offloading rigs, and for the human body (Chapter 30). It is also a resource: in-space manufacturing (Chapter 40) exploits it to make materials impossible to make on Earth.

## 2.10 Environments Beyond Earth Orbit

Deep-space missions face additional environments. Table 2.2 and Figure 2.16 summarise some of the most demanding.

**TABLE 2.2** Representative environmental challenges for planetary missions.

| Destination | Key environmental challenges | Engineering responses (examples) |
|---|---|---|
| Moon | Surface −170 °C to +120 °C at the equator; 14-day nights; permanently shadowed craters near 25–40 K; abrasive, electrostatically charged dust; no magnetic field or atmosphere | Radioisotope heater units, lunar-night survival modes, dust-tolerant seals and connectors, radiation shielding for crews |
| Mars | Thin $CO_2$ atmosphere (≈0.6 % of Earth's pressure); dust storms reducing sunlight; −125 °C to +20 °C; perchlorate-laden dust; no global magnetic field | Entry heat shields and parachutes (Chapter 27), RTG or dust-tolerant solar power, heaters, dust-clearing strategies |
| Venus | Surface 465 °C and 92 bar, sulfuric-acid clouds; super-rotating winds | Pressure vessels and insulation (Venera landers survived up to 127 minutes); silicon-carbide electronics research |

| Destination | Key environmental challenges | Engineering responses (examples) |
|---|---|---|
| Mercury / near-Sun | Solar flux up to 11 times Earth's (Mercury) or ≈500 times (Parker Solar Probe at closest approach); intense heating from planet's surface | Heat shields (Parker's 11.4 cm carbon-composite shield), high-temperature solar arrays, sunshades, special orbit design |
| Jupiter system | The most intense radiation belts in the Solar System — Europa Clipper receives ≈2.8 Mrad behind 0.1 inch of aluminium | Radiation vaults (Juno's 1 cm titanium vault, Clipper's aluminium-zinc vault), orbits that minimise belt passages |
| Outer Solar System | Sunlight 1/100 to 1/1000 of Earth's; temperatures 40–100 K; light-hour communication delays | RTGs, low-temperature materials, high-autonomy operations |
| Interplanetary cruise | Solar particle events, GCRs, micrometeoroid streams near comets | Shielding, fault protection, safe modes, trajectory selection |

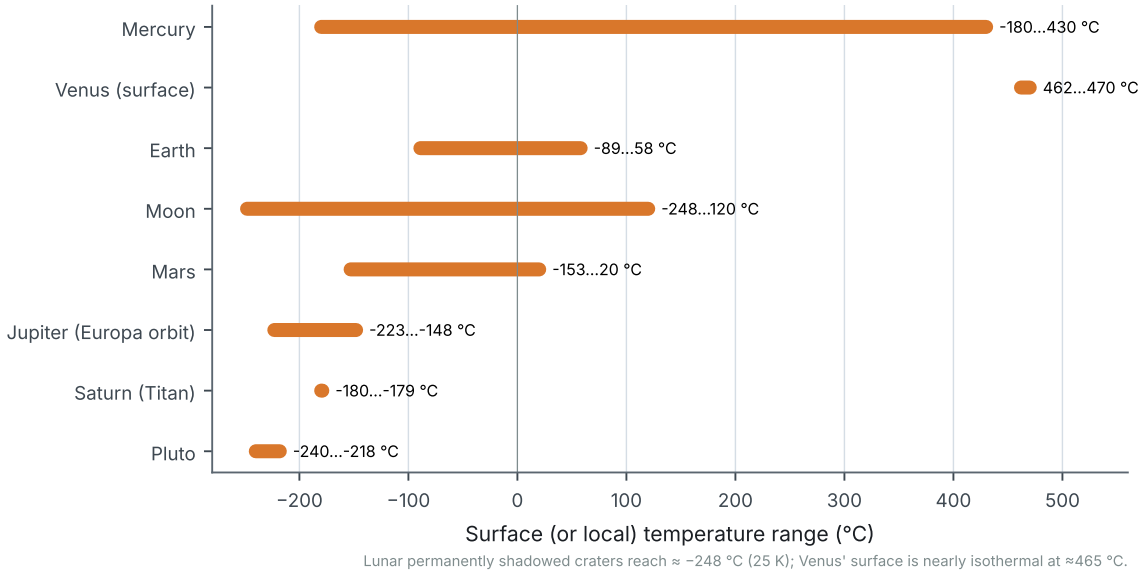



**FIGURE 2.16** Temperature ranges at the surfaces (or local environments) of selected Solar System bodies. Thermal design must cover both extremes — and, for landers, the transition from cruise vacuum to surface conditions.

## 2.11 Designing for the Environment: Standards and Tools

Engineers capture the environment in a project **environmental specification**. It is derived from standards such as **ECSS-E-ST-10-04C** (space environment), **NASA-STD-7009/NASA-HDBK-1001** (terrestrial and natural space environments) and **ISO 15390** (cosmic rays). Other sources include the **AE9/AP9** models, **ORDEM 3.2** and **MASTER-8** (debris), **MEM 3** (meteoroids) and **NRLMSIS 2.0** (atmosphere). ESA's online **SPENVIS** system and NASA's **Space Environments and Effects (SEE)** program provide integrated tools.

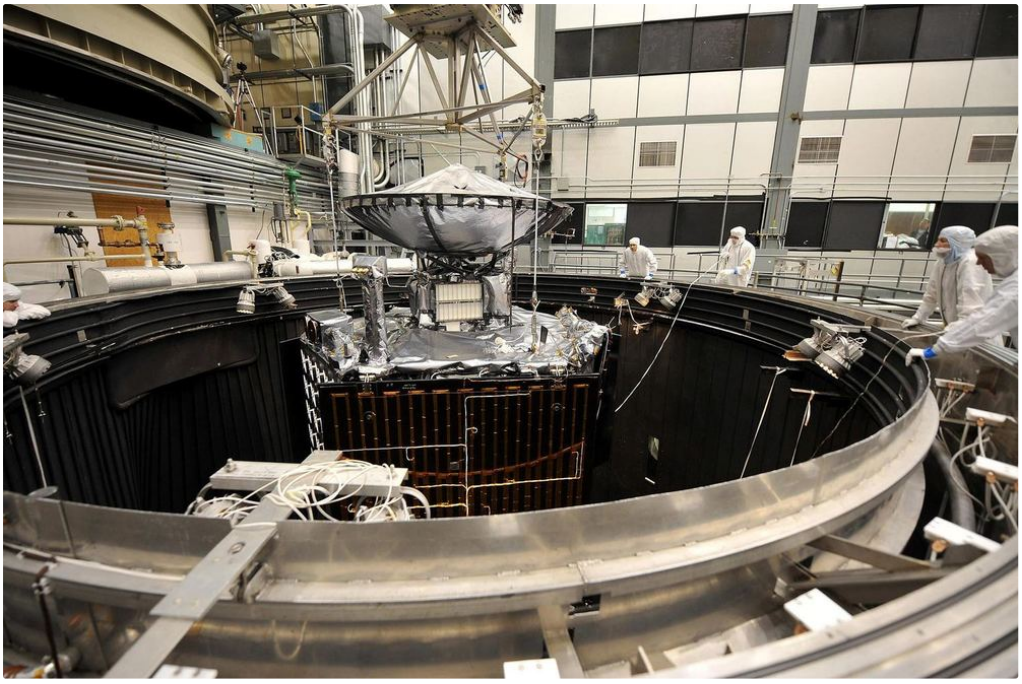


**FIGURE 2.17** NASA's Juno spacecraft emerging from thermal-vacuum testing at Lockheed Martin in 2011. In the chamber the spacecraft was cycled between the hot and cold extremes predicted for its mission while operating, verifying both the thermal design and the electronics. Image: NASA/JPL (NASA/JPL-Caltech/Lockheed Martin).

The environmental specification feeds the verification programme: thermal-vacuum and thermal-balance tests, vibration, acoustic and shock tests, EMC tests, radiation tests of parts, and material screening (Chapter 4). The general principle is **"test as you fly, fly as you test"**: every environment the spacecraft will meet in flight should, as far as practical, be reproduced or bounded on the ground.

### KEY TAKEAWAYS

- Atmospheric density at a given LEO altitude varies by ~10× over the solar cycle; drag sets satellite lifetime (about 1–3 years at 400 km, decades at 600 km, centuries at 800 km).
- Vacuum causes outgassing, cold welding and discharges; the thermal environment is governed by radiation balance and the $\alpha/\varepsilon$ ratio of surfaces.
- Atomic oxygen erodes polymers in LEO; plasma causes surface and internal charging, most severely in GEO and MEO.
- Radiation causes total dose, displacement damage and single-event effects; MEO is the harshest common orbit, and Jupiter the harshest planetary environment.
- Micrometeoroids and debris strike at 10–20 km/s; Whipple shields defeat particles up to about 1 cm, and larger objects must be tracked and avoided.
- Launch imposes quasi-static, sine, random, acoustic and shock loads that often size the structure; environmental testing reproduces these on the ground.

### REVIEW QUESTIONS

1. Why does atmospheric density at 400 km change by about a factor of ten over a solar cycle, and what does this mean for satellite lifetime?
2. With a scale height of 60 km, by what factor does density fall between 400 km and 450 km altitude?
3. A plate at 1 au faces the Sun with $\alpha = 0.2$ and $\varepsilon = 0.8$ on its sunlit face and loses heat only from that face. Estimate its equilibrium temperature.
4. What is atomic oxygen, at which altitudes does it matter most, and how are exposed materials protected?
5. Distinguish surface charging from internal (deep dielectric) charging. In which orbits is each most severe?
6. Explain the differences between total ionising dose, displacement damage and single-event effects, and give one mitigation for each.
7. How does a Whipple shield work, and why must objects larger than about 1 cm be tracked and avoided instead?

8. List the main mechanical loads during launch and explain why they, rather than the space environment, often size a spacecraft's structure.

CHAPTER

# 3 Orbital Mechanics and Astrodynamics

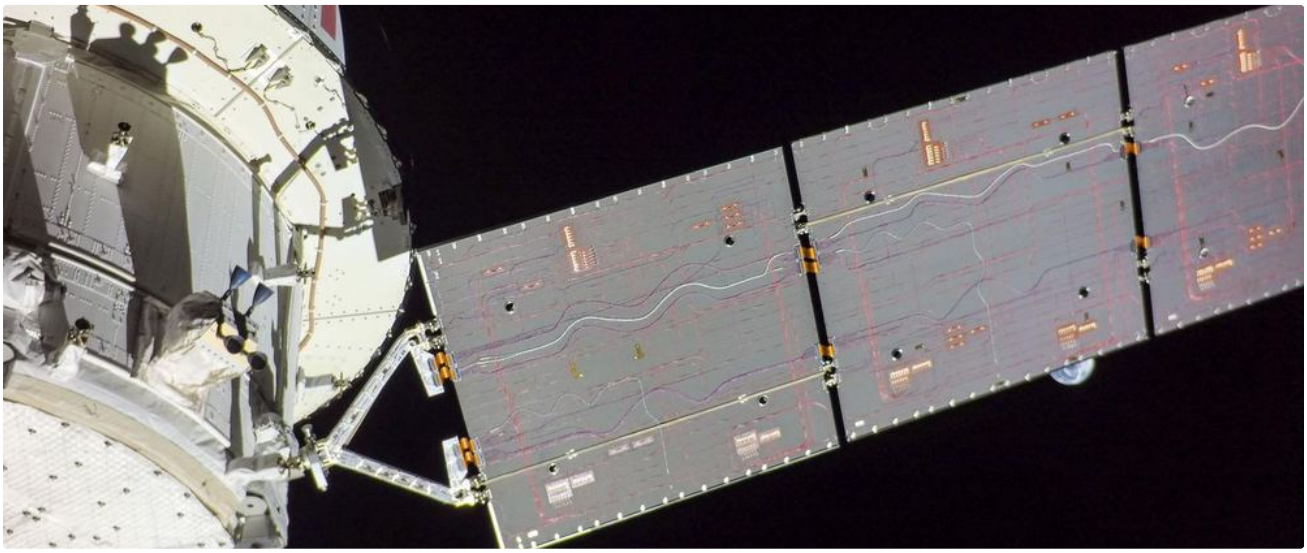


Image: NASA/JSC

> *Orbital mechanics explains how spacecraft move under gravity. It tells us why satellites stay up, how long an orbit takes, how much velocity change a mission needs, how orbits drift because of Earth's shape and the Sun and Moon, how to rendezvous with another spacecraft, and how to travel between planets. This chapter builds the subject from Newton's and Kepler's laws to the practical tools of mission design - Hohmann transfers, plane changes, ground tracks, Lambert's problem, gravity assists and the Lagrange points - with worked examples throughout.*

**IN THIS CHAPTER**


3.1 From Newton to Kepler
3.2 Describing an Orbit: Orbital Elements and Reference Frames
3.3 Where Is the Satellite? Kepler's Equation
3.4 Orbit Perturbations
3.5 Orbital Manoeuvres
3.6 Rendezvous and Relative Motion
3.7 Orbit Determination
3.8 Ground Tracks and Coverage
3.9 Interplanetary Trajectories
3.10 Three-Body Dynamics and the Lagrange Points
3.11 The Δv Budget


## 3.1 From Newton to Kepler

### 3.1.1 Gravity and the Two-Body Problem

Isaac Newton's law of universal gravitation states that two masses attract each other with a force proportional to the product of their masses and inversely proportional to the square of the distance between them:

$$F = \frac{GMm}{r^2}, \qquad G = 6.674 \times 10^{-11}\ \mathrm{N\,m^2\,kg^{-2}}$$

For a satellite of negligible mass orbiting a planet of mass $M$, it is convenient to use the **gravitational parameter** $\mu = GM$, which is known far more precisely than $G$ or $M$ separately. For Earth, $\mu_\oplus = 398\,600.4418$ km³/s²; for the Sun, $\mu_\odot = 1.327\,124 \times 10^{11}$ km³/s². The equation of motion of the satellite relative to the planet's centre is

$$\ddot{\mathbf{r}} = -\frac{\mu}{r^3}\,\mathbf{r}$$

This is the **two-body problem**. It has an exact solution: the satellite moves on a **conic section** — a circle, ellipse, parabola or hyperbola — with the planet at one focus (Figure 3.1). All the complexity of real orbits comes from small departures from this ideal, called **perturbations**.

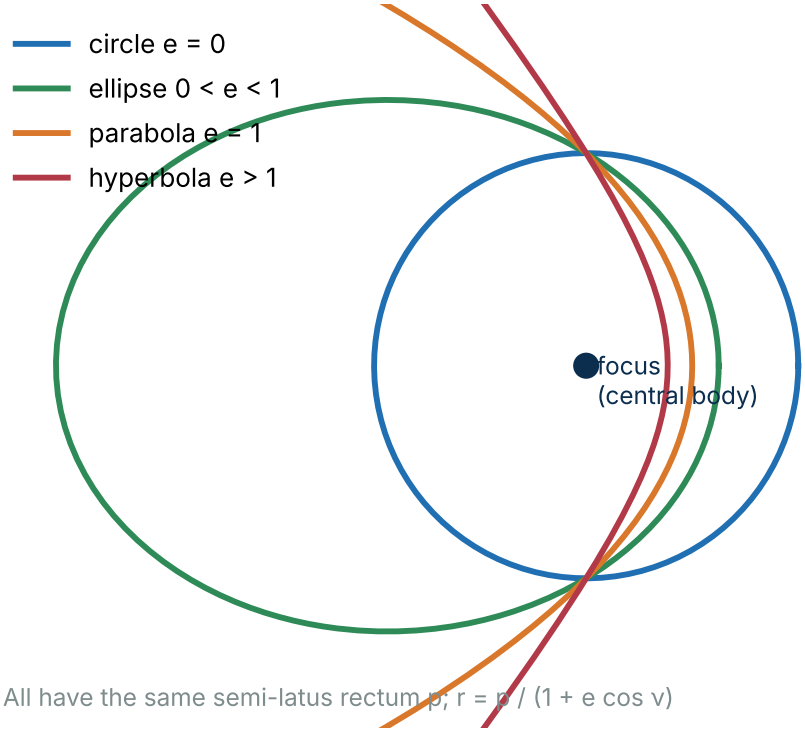



**FIGURE 3.1** The four conic sections. Bound orbits are circles and ellipses; parabolas are the boundary case of escape; hyperbolas describe flybys and escape trajectories. All share the polar equation r = p/(1 + e cos ν) with the attracting body at a focus.

### 3.1.2 Kepler's Three Laws

Johannes Kepler deduced three laws from Tycho Brahe's observations of Mars, decades before Newton explained them:

1. **The orbit of a planet is an ellipse with the Sun at one focus.**
2. **A line joining the planet and the Sun sweeps out equal areas in equal times** (Figure 3.2). Physically, this is conservation of angular momentum: the satellite moves fastest at its closest point (**periapsis**) and slowest at its farthest point (**apoapsis**).
3. **The square of the orbital period is proportional to the cube of the semi-major axis:**

$$T = 2\pi\sqrt{\frac{a^3}{\mu}}$$

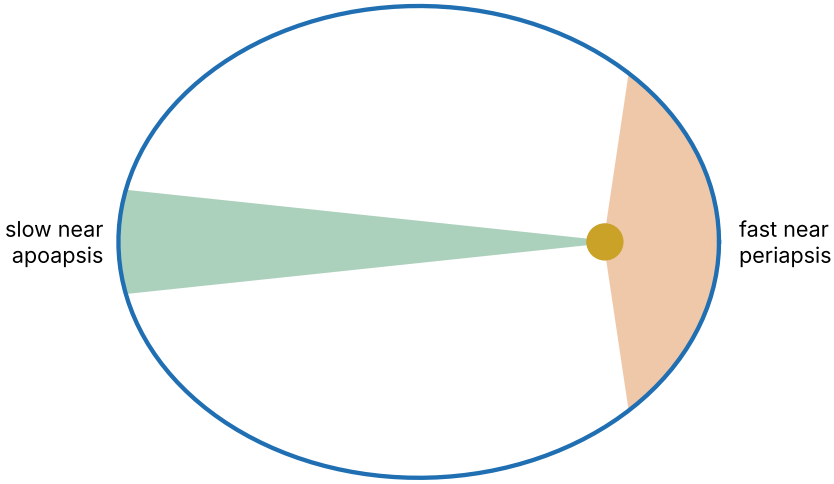



**FIGURE 3.2** Kepler's second law: the two shaded sectors have equal area and are swept in equal times, so the satellite moves much faster near periapsis.

For Earth orbits, the words perigee and apogee are used; around the Sun, perihelion and aphelion; around the Moon, perilune (periselene) and apolune.

### 3.1.3 The Geometry of the Ellipse

An ellipse is described by its **semi-major axis** $a$ (half the longest diameter) and **eccentricity** $e$ (how elongated it is; 0 for a circle). The orbit equation in polar form is

$$r = \frac{a(1 - e^2)}{1 + e\cos\nu}$$

where $\nu$ is the **true anomaly**, the angle from periapsis to the satellite measured at the focus. The periapsis and apoapsis radii are $r_p = a(1 - e)$ and $r_a = a(1 + e)$ (Figure 3.3).

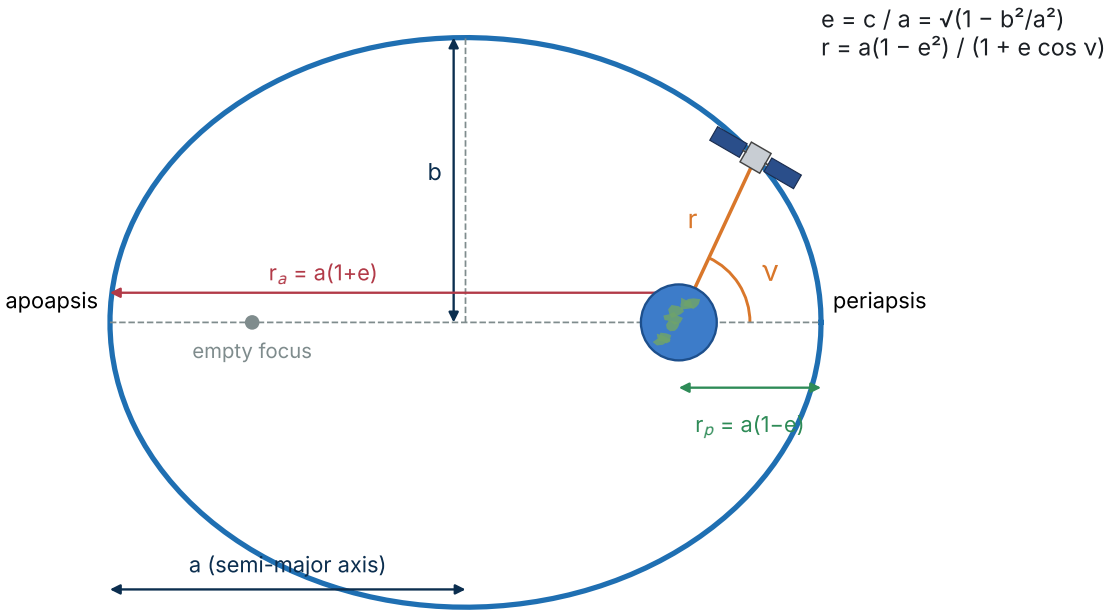



**FIGURE 3.3** Geometry of an elliptical orbit. The central body sits at one focus; the other focus is empty. Periapsis and apoapsis lie at the ends of the major axis.

### 3.1.4 Energy and Speed: the Vis-Viva Equation

Two quantities are conserved in the two-body problem: the **specific mechanical energy** (energy per unit mass) and the **specific angular momentum**:

$$\varepsilon = \frac{v^2}{2} - \frac{\mu}{r} = -\frac{\mu}{2a}, \qquad h = |\mathbf{r} \times \mathbf{v}| = \sqrt{\mu a (1 - e^2)}$$

Rearranging the energy equation gives the most useful single equation in orbital mechanics, the **vis-viva equation**:

$$v^2 = \mu \left( \frac{2}{r} - \frac{1}{a} \right)$$

It gives the speed anywhere on an orbit from the current radius and the semi-major axis alone. Special cases:

- **Circular orbit** ($r = a$): $v_c = \sqrt{\mu / r}$.
- **Escape** (parabola, $a \to \infty$): $v_{esc} = \sqrt{2\mu / r} = \sqrt{2}\, v_c$.
- **Hyperbolic excess speed** $v_\infty$ — the speed left over far from the planet: $v^2 = v_{esc}^2 + v_\infty^2$. Mission designers quote launch energy as $C_3 = v_\infty^2$ (km²/s²).

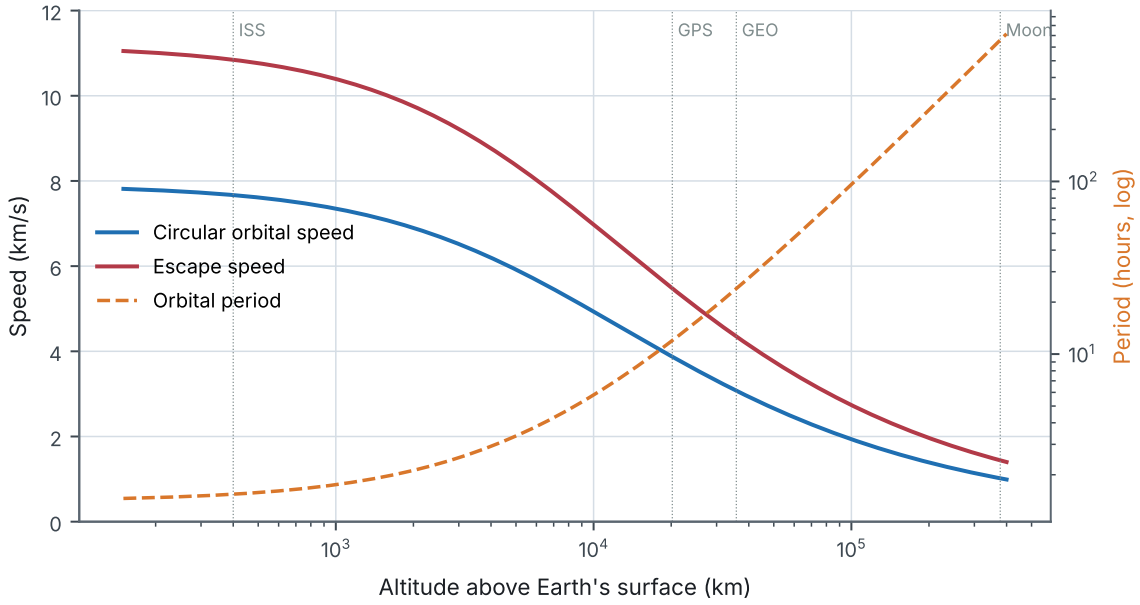



**FIGURE 3.4** Circular orbital speed, escape speed and orbital period as functions of altitude above Earth. At the ISS altitude a satellite circles the Earth every 92 minutes at 7.66 km/s; at GEO the period equals one sidereal day (23 h 56 min).

**WORKED EXAMPLE — THE INTERNATIONAL SPACE STATION'S ORBIT**

The ISS flies at about 420 km, so $r = 6\,378 + 420 = 6\,798$ km.

- Circular speed: $v = \sqrt{398\,600 / 6\,798} = 7.657$ km/s (27 570 km/h).
- Period: $T = 2\pi \sqrt{6\,798^3 / 398\,600} = 5\,578$ s = 93.0 min, so about 15.5 orbits per day.
- Escape speed from this altitude: $\sqrt{2} \times 7.657 = 10.83$ km/s, so leaving Earth from the ISS orbit would need $\Delta v = 3.17$ km/s.

# 3.2 Describing an Orbit: Orbital Elements and Reference Frames

## 3.2.1 The Six Classical Orbital Elements

A satellite's position and velocity at an instant (six numbers: $\mathbf{r}$, $\mathbf{v}$) define its two-body orbit completely. An equivalent and more intuitive description uses six **Keplerian (classical) orbital elements** (Figure 3.5):

| Element | Symbol | Describes |
|---|---|---|
| Semi-major axis | $a$ | Size of the orbit (and the period) |
| Eccentricity | $e$ | Shape |
| Inclination | $i$ | Tilt of the orbit plane relative to the equator (0°–180°; > 90° = retrograde) |
| Right ascension of the ascending node (RAAN) | $\Omega$ | Angle in the equatorial plane from the vernal equinox to the point where the satellite crosses the equator northward |
| Argument of periapsis | $\omega$ | Angle in the orbit plane from the ascending node to periapsis |
| True anomaly (or mean anomaly, or time of periapsis) | $\nu$ ($M$, $t_p$) | Position of the satellite on the orbit |

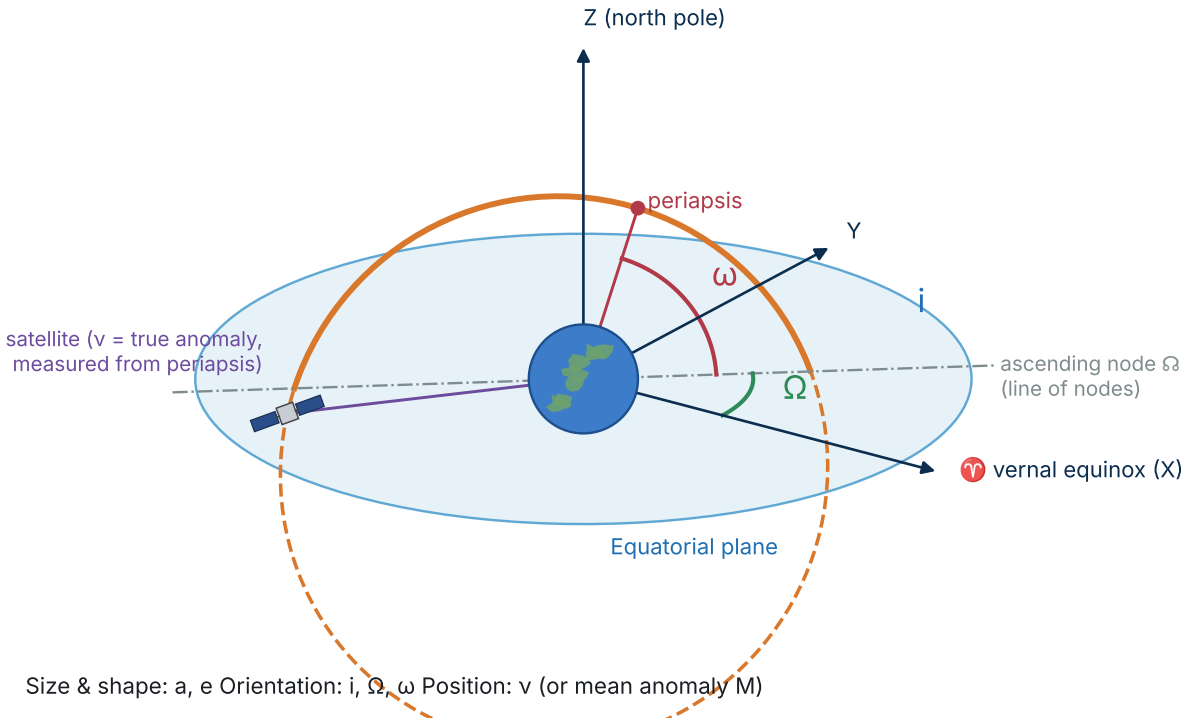



**FIGURE 3.5** The classical orbital elements. Two define the size and shape of the orbit (a, e), three its orientation in space (i, Ω, ω), and one the position of the satellite along it (ν).

Some elements are undefined for special orbits: for circular orbits there is no periapsis (ω is replaced by the **argument of latitude** $u = \omega + \nu$), and for equatorial orbits there is no ascending node. For such orbits, engineers use **equinoctial elements**, which have no singularities and are preferred in numerical software and for electric-propulsion trajectory optimisation.

### 3.2.2 Reference Frames and Time

Orbits are computed in **inertial** (non-rotating) frames, but ground stations, maps and GNSS work in **Earth-fixed** frames, and spacecraft attitude and rendezvous are described in **orbit-fixed** frames (Figure 3.6):

- **Earth-centred inertial (ECI)** frames: today the **Geocentric Celestial Reference Frame (GCRF)**, closely aligned with the older **J2000** (EME2000) frame. The X-axis points to the vernal equinox, Z to the celestial north pole.
- **Earth-centred, Earth-fixed (ECEF)**: the **International Terrestrial Reference Frame (ITRF)** and its GNSS realisations (WGS 84 for GPS). It rotates with the Earth; converting to ECI requires precession, nutation, Earth rotation angle and polar motion, published by the International Earth Rotation and Reference Systems Service (IERS).
- **Orbit frames**: **RTN/RSW** (radial, transverse/along-track, normal) and **LVLH** (local vertical, local horizontal), used for relative motion, attitude pointing and orbit-determination covariance.
- **Body frame**: fixed to the spacecraft structure.

Precise work also needs precise **time scales**. These include International Atomic Time (TAI) and Coordinated Universal Time (UTC), which differs from TAI by an integer number of leap seconds (37 s since 2017). There are also GPS time (TAI − 19 s, no leap seconds), Terrestrial Time (TT = TAI + 32.184 s) and Barycentric Dynamical Time (TDB) for interplanetary ephemerides. Confusing these has caused real errors.

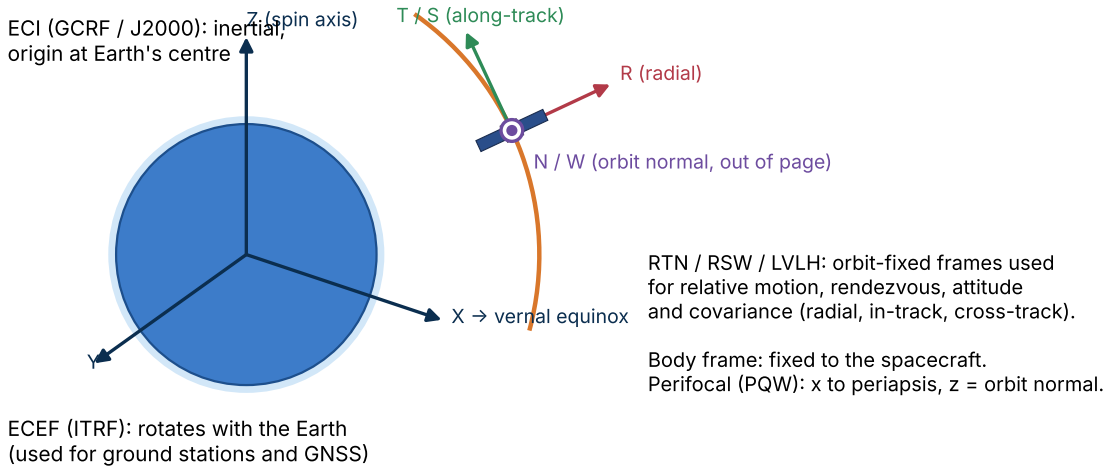
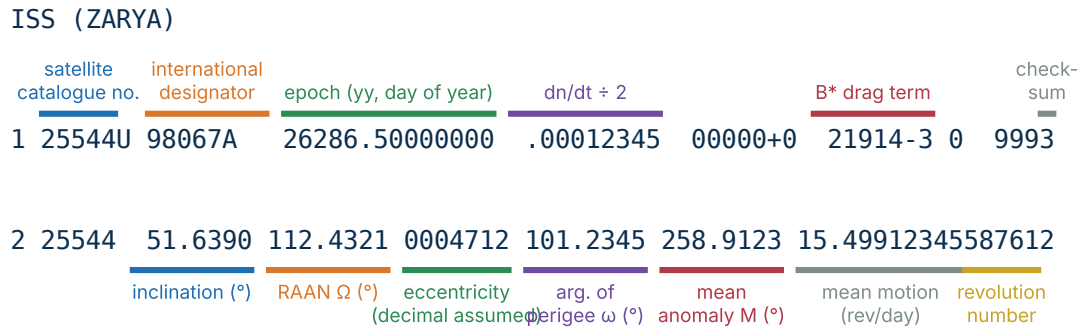



**FIGURE 3.6** Common reference frames: Earth-centred inertial and Earth-fixed frames, and the orbit-fixed radial / along-track / cross-track (RTN) frame attached to the satellite.

### 3.2.3 Two-Line Element Sets

The most widely distributed orbit format is the **two-line element set (TLE)** (Figure 3.7), produced by the US Space Force for the public catalogue. TLEs are *mean* elements fitted for the **SGP4** analytical propagator. They are adequate for antenna pointing and general awareness (errors of about 1 km at epoch, growing by kilometres per day). But they are not adequate for precise conjunction assessment, which uses covariance-carrying ephemerides (Chapter 21). The fixed five-character catalogue-number field is close to exhaustion as tens of thousands of new objects

are catalogued. An interim "Alpha-5" scheme replaces the first digit with a letter. So users are moving to the **CCSDS Orbit Mean-elements Message (OMM)** in XML, KVN or JSON, which has no such limit.




TLEs are mean elements for the SGP4/SDP4 propagator, not osculating Keplerian elements. Illustrative values; current sets are published by CelesTrak and Space-Track.org (also in the newer OMM / CCSDS format to allow catalogue numbers > 99 999).

**FIGURE 3.7** Anatomy of a two-line element set (illustrative values). Each field occupies fixed columns; the last digit of each line is a checksum.

## 3.3 Where Is the Satellite? Kepler's Equation

Knowing the orbit is not the same as knowing where the satellite will be at a given time. Because the speed varies around an elliptical orbit, the true anomaly does not increase uniformly. Kepler solved this with two auxiliary angles (Figure 3.8): the **mean anomaly** $M$, which does increase uniformly, $M = n(t - t_p)$ with **mean motion** $n = \sqrt{\mu/a^3}$; and the **eccentric anomaly** $E$, related to $M$ by **Kepler's equation**:

$$M = E - e \sin E$$

Given $M$, this transcendental equation is solved for $E$ by iteration (Newton's method converges in a few steps), and then

$$\tan\frac{\nu}{2} = \sqrt{\frac{1+e}{1-e}}\,\tan\frac{E}{2}, \qquad r = a(1 - e\cos E)$$

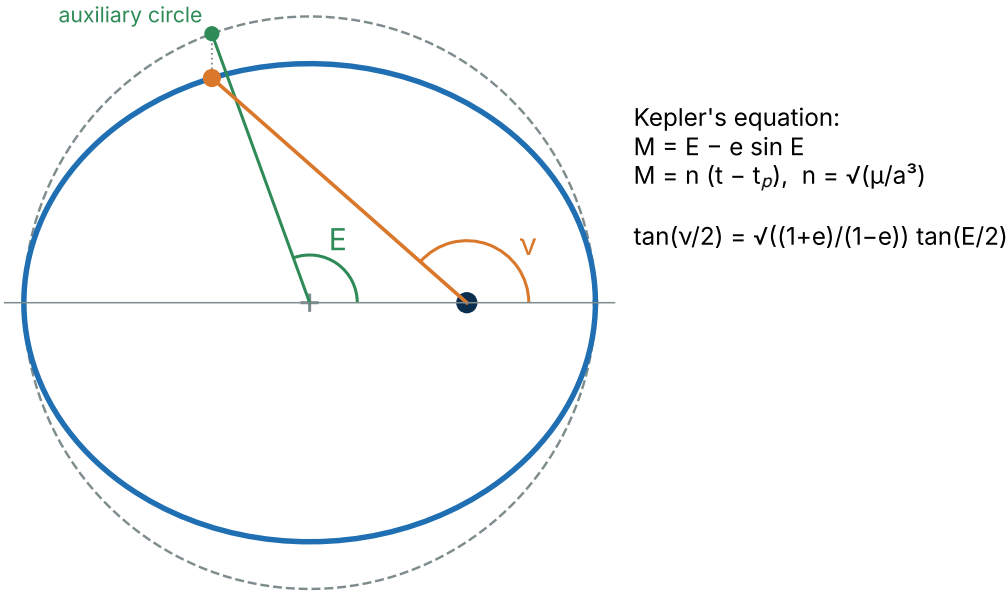
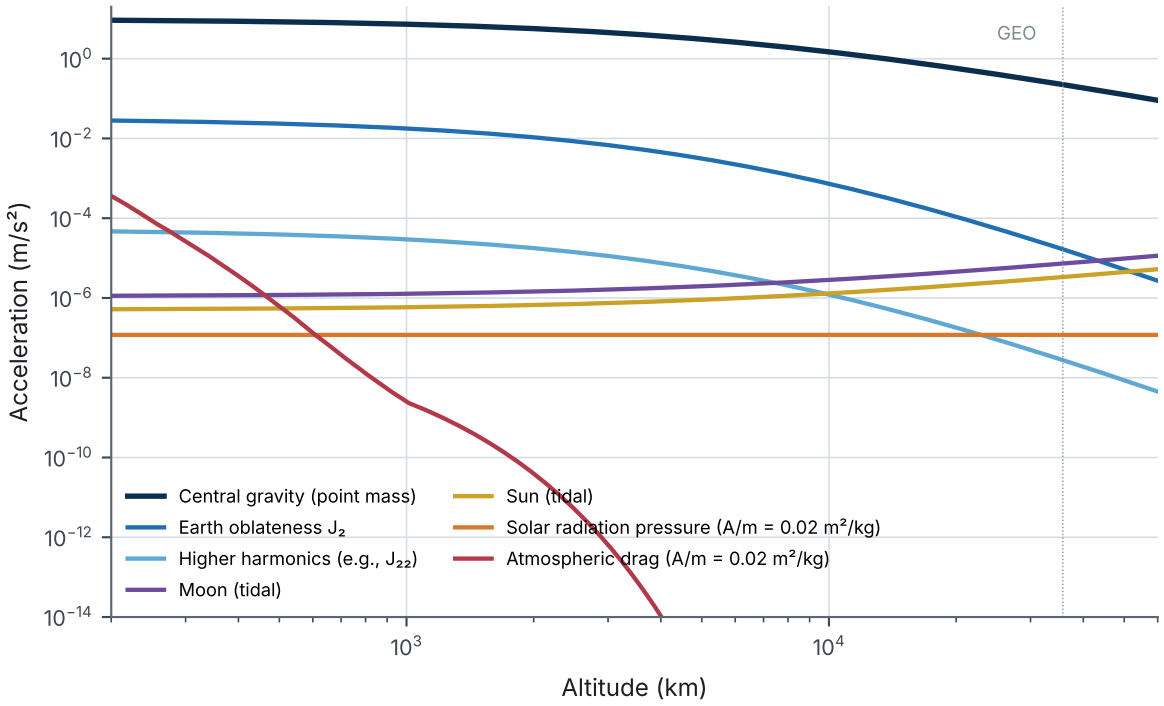



**FIGURE 3.8** The eccentric anomaly E is measured at the centre of the ellipse to the point on an auxiliary circle; the true anomaly v is measured at the focus. Kepler's equation links E to the uniformly increasing mean anomaly M.

Modern software usually avoids the anomalies altogether by integrating the equations of motion numerically (e.g., with Runge–Kutta or Gauss–Jackson integrators), which also allows any perturbing forces to be included. Analytical methods remain important for quick estimates, for onboard software with limited computing power, and for physical insight.

## 3.4 Orbit Perturbations

Real orbits deviate from Keplerian ellipses because of additional forces. Figure 3.9 compares their sizes.




Orders of magnitude after Montenbruck & Gill (2000). Drag uses moderate solar activity.

**FIGURE 3.9** Magnitude of perturbing accelerations as a function of altitude for a typical satellite. Earth's oblateness ($J_2$) dominates in LEO after central gravity; drag dominates the decay of low orbits; the Sun and Moon dominate high orbits such as GEO.

### 3.4.1 Earth's Oblateness ($J_2$)

The Earth is not a sphere: its equatorial radius (6 378.137 km) exceeds its polar radius (6 356.752 km) by 21 km. Its gravity field is expressed as a series of **spherical harmonics**; the largest term after the central field is the **$J_2$ zonal harmonic** ($J_2 = 1.082\,63 \times 10^{-3}$), about a thousand times larger than any other. The equatorial bulge exerts a torque on an inclined orbit, causing two secular drifts:

**Nodal regression** — the orbit plane precesses about Earth's axis:

$$\dot{\Omega} = -\frac{3}{2}\, n\, J_2 \left(\frac{R_\oplus}{p}\right)^2 \cos i$$

**Apsidal rotation** — the line of apsides rotates within the orbit plane:

$$\dot{\omega} = \frac{3}{4}\, n\, J_2 \left(\frac{R_\oplus}{p}\right)^2 \left(5\cos^2 i - 1\right)$$

where $p = a(1 - e^2)$. For the ISS ($i = 51.6^\circ$, 420 km), the node regresses by about 5° per day. This is why its ground track shifts each day, and why launch windows to the ISS move earlier by roughly 22 minutes per day.

These "perturbations" are put to good use in three special orbit families (Figure 3.10):

- **Sun-synchronous orbits (SSO).** Choosing a slightly retrograde inclination (about 97–99° for altitudes of 500–1 000 km) makes $\dot{\Omega} = +0.9856^\circ$/day, exactly matching the Earth's motion around the Sun. The orbit plane then keeps a constant angle to the Sun, so the satellite crosses each latitude at the same **local solar time** every day (e.g., Landsat at 10:00 descending node,

Sentinel-2 at 10:30). This gives consistent lighting for imaging and stable thermal conditions. Dawn–dusk SSOs (06:00/18:00) are almost always in sunlight — ideal for radar satellites and solar-powered missions.

- **Critically inclined orbits.** At $i = 63.4^\circ$ or $116.6^\circ$, $5\cos^2 i - 1 = 0$, so the apsides do not rotate. The **Molniya** (12 h period) and **Tundra** (24 h) orbits use this to keep their apogee permanently over high northern latitudes.
- **Frozen orbits.** A suitable choice of $e$ and $\omega = 90^\circ$ (using the $J_3$ harmonic) keeps the altitude profile over each latitude fixed — useful for altimetry missions such as Sentinel-3 and Jason/Sentinel-6.

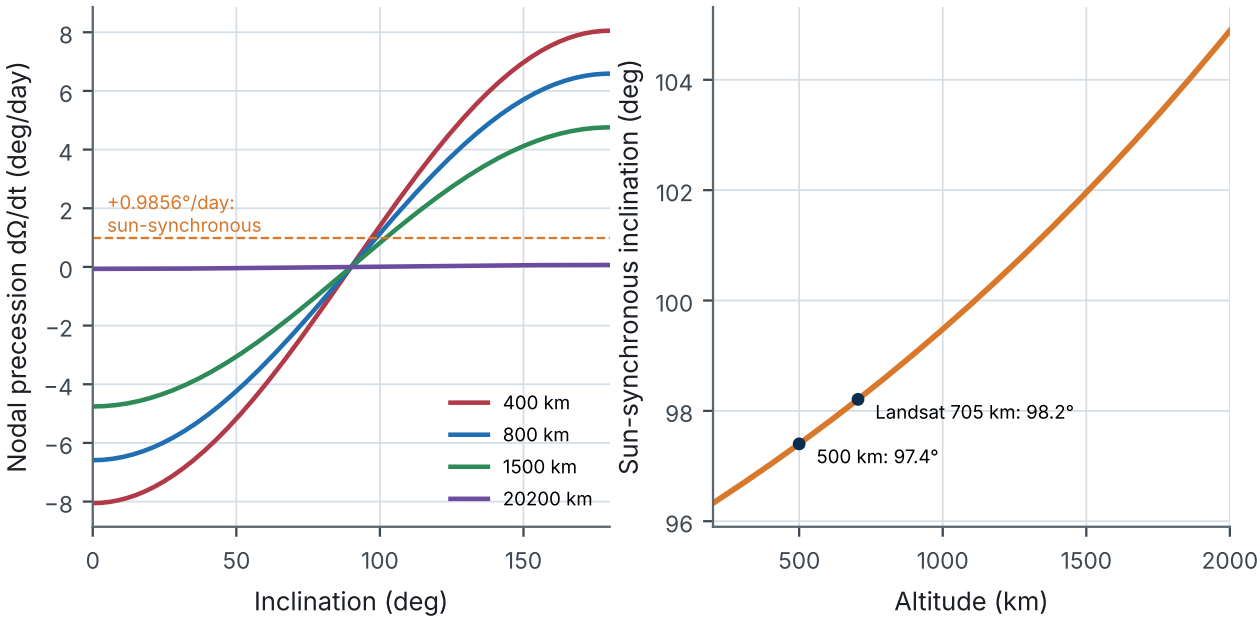



**FIGURE 3.10** Left: nodal precession rate versus inclination for several altitudes; the dashed line marks the sun-synchronous rate. Right: inclination required for a sun-synchronous circular orbit as a function of altitude.

## 3.4.2 Other Perturbations

- **Higher-order gravity.** Tesseral and sectoral harmonics include $J_{22}$, which is related to the ellipticity of the equator. They cause GEO satellites to drift east or west towards two stable points near 75° E and 105° W. They also cause resonances for orbits with periods that are simple fractions of a day. Precise orbit determination uses models such as EGM2008 or GGM05 to degree and order 100 or more, derived from the GRACE and GOCE missions.
- **Third-body gravity** from the Moon and Sun changes the inclination of GEO satellites by about 0.75–0.95° per year. This must be corrected by north–south station-keeping (about 45–50 m/s per year — the largest item in a GEO propellant budget).
- **Solar radiation pressure (SRP)** — sunlight carries momentum, producing a pressure of about 4.56 μN/m² on an absorbing surface at 1 AU (twice that on a perfect mirror). SRP perturbs eccentricity in GEO and is the main error source in GNSS orbit modelling; it is exploited by solar sails (Chapter 42).
- **Atmospheric drag** — dominant below about 600 km (Chapter 2).
- **Relativity, Earth tides, albedo and thermal re-radiation** — small effects that matter for centimetre-level orbits of geodetic and GNSS satellites.

## 3.5 Orbital Manoeuvres

A manoeuvre changes the satellite's velocity by firing a thruster. The **velocity change** $\Delta v$ is the universal currency of mission design, because the rocket equation (Chapter 34) converts it directly into propellant mass:

$$\frac{m_{prop}}{m_0} = 1 - \exp\left(-\frac{\Delta v}{I_{sp}\, g_0}\right)$$

For chemical engines, burns are short compared with the orbital period and are treated as **impulsive** (instantaneous) changes.

### 3.5.1 The Hohmann Transfer

The minimum-energy two-impulse transfer between two coplanar circular orbits, found by Walter Hohmann in 1925, uses an ellipse tangent to both (Figure 3.11). The first burn at periapsis raises the apoapsis to the target radius; the second burn at apoapsis circularises:

$$\Delta v_1 = \sqrt{\frac{\mu}{r_1}}\left(\sqrt{\frac{2r_2}{r_1 + r_2}} - 1\right), \qquad \Delta v_2 = \sqrt{\frac{\mu}{r_2}}\left(1 - \sqrt{\frac{2r_1}{r_1 + r_2}}\right)$$

The transfer takes half the period of the transfer ellipse: $t = \pi\sqrt{a_t^3/\mu}$ with $a_t = (r_1 + r_2)/2$.

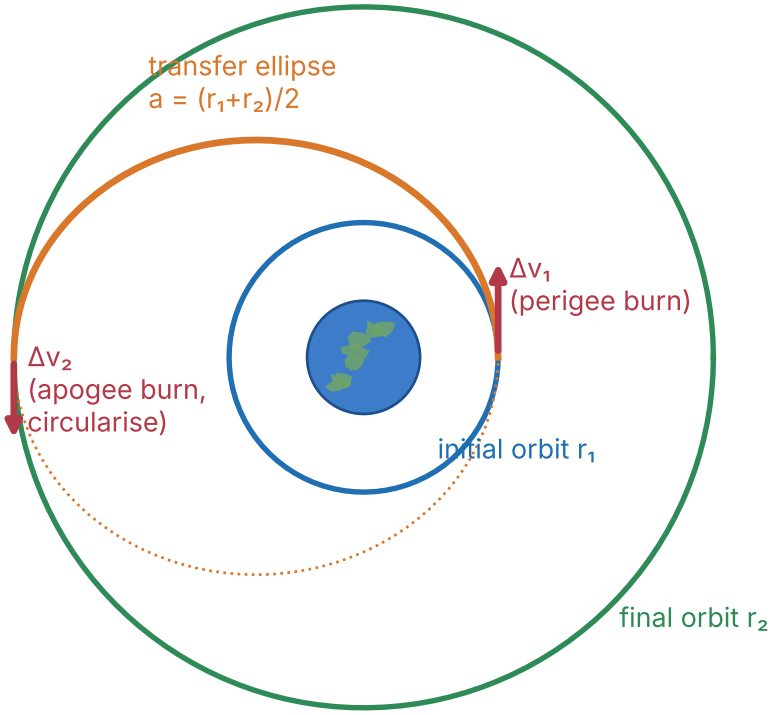



**FIGURE 3.11** The Hohmann transfer between circular orbits: a tangential burn at periapsis, a half-revolution coast on the transfer ellipse, and a tangential circularising burn at apoapsis.

**WORKED EXAMPLE — FROM A 300 KM PARKING ORBIT TO GEO**

$r_1 = 6\,678$ km, $r_2 = 42\,164$ km, so $a_t = 24\,421$ km.

- $v_{c1} = \sqrt{398\,600/6\,678} = 7.726$ km/s; speed at transfer perigee $= \sqrt{398\,600\,(2/6\,678 - 1/24\,421)} = 10.152$ km/s, so $\Delta v_1 = 2.426$ km/s.
- Speed at transfer apogee $= \sqrt{398\,600\,(2/42\,164 - 1/24\,421)} = 1.608$ km/s; $v_{GEO} = 3.075$ km/s, so $\Delta v_2 = 1.467$ km/s.
- Total: **3.89 km/s**; transfer time: $\pi\sqrt{24\,421^3/398\,600} = 18\,990$ s = **5.3 h**.

From Cape Canaveral (28.5°), the satellite must also remove its inclination. Doing this at apogee, where the speed is lowest, and combining it with the circularisation burn: $\Delta v_2 = \sqrt{1.608^2 + 3.075^2 - 2(1.608)(3.075)\cos 28.5^\circ} = 1.83$ km/s. In practice the launcher performs the first burn (injecting into GTO) and the satellite's own engine performs the second, often split into several apogee burns. Launching from Kourou (5.2° N) saves about 0.3 km/s, which is why Kourou is favoured for GEO missions.

## 3.5.2 Bi-elliptic Transfers

A three-burn **bi-elliptic transfer** first goes beyond the target orbit. It can be cheaper than a Hohmann transfer when the radius ratio exceeds 11.94, and always is when it exceeds 15.58. But it takes much longer (Figure 3.12). It is rarely used for Earth orbits but illustrates that "minimum energy" depends on the constraints.

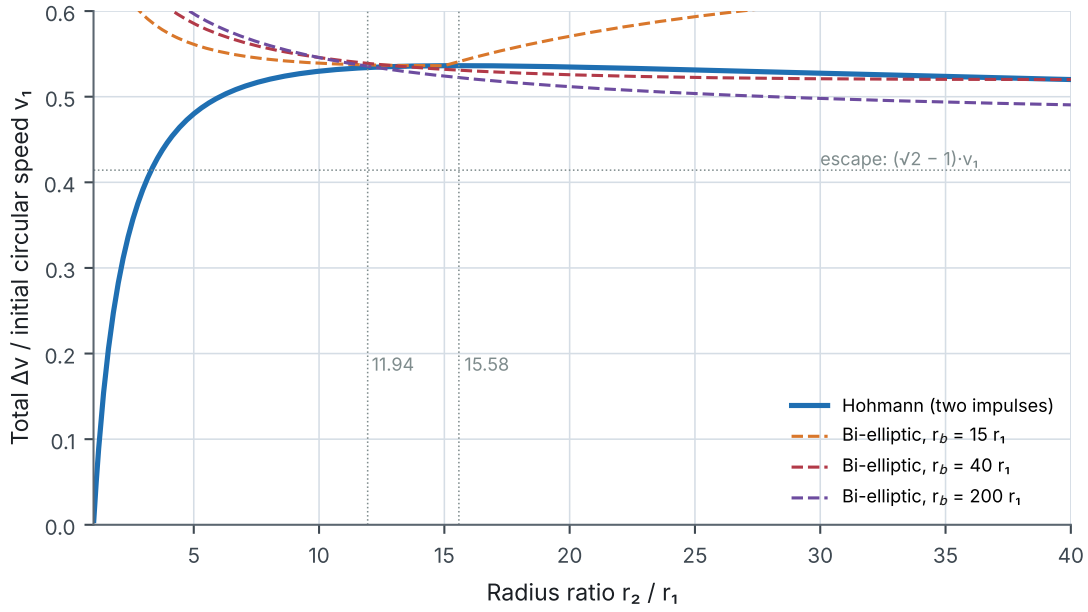



**FIGURE 3.12** Total Δv (normalised by initial circular speed) for Hohmann and bi-elliptic transfers. The Hohmann cost peaks at a radius ratio of 15.58; above that, a bi-elliptic transfer with a distant intermediate apoapsis is cheaper.

## 3.5.3 Plane Changes

Changing the orbit plane is expensive because the whole velocity vector must be rotated. A pure plane change by angle $\Delta i$ at speed $v$ costs

$$\Delta v = 2v \sin\frac{\Delta i}{2}$$

so it is best performed where the speed is lowest — at apoapsis, or at high altitude — and combined with other burns (Figure 3.13). Changing the plane of a LEO satellite by just 10° costs 1.3 km/s, more than most satellites carry. This is why the launch site latitude and launch azimuth, which set the initial inclination, are so important, and why constellations deploy separate planes from separate launches. Instead of plane changes, some constellations exploit differential nodal precession: satellites placed at slightly different altitudes drift apart in RAAN for free over months (Starlink, Planet).

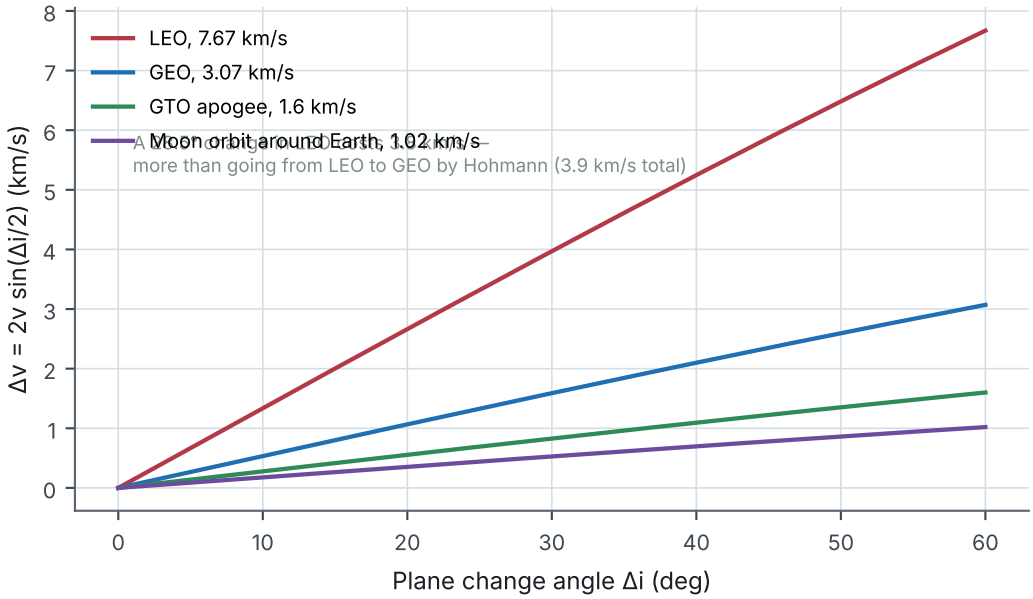



**FIGURE 3.13** Δv needed for a plane change as a function of angle, for four orbital speeds. Plane changes are cheapest where the spacecraft moves slowest.

### 3.5.4 Low-Thrust Transfers

Electric propulsion (Chapter 6) produces tiny thrust (millinewtons to a few newtons) for months. The orbit then spirals gradually; the impulsive approximation no longer applies. For a slow spiral between circular orbits, the Δv approaches the *difference in circular speeds*: $\Delta v \approx v_{c1} - v_{c2}$. This is larger than for a Hohmann transfer (4.6 km/s from LEO to GEO instead of 3.9 km/s). But the propellant efficiency is ten times higher, so far less propellant is used. The **Edelbaum equation** gives the optimal low-thrust transfer with a simultaneous plane change:

$$\Delta v = \sqrt{v_1^2 + v_2^2 - 2 v_1 v_2 \cos\left(\tfrac{\pi}{2}\Delta i\right)}$$

All-electric GEO satellites (Boeing 702SP, Airbus Eurostar Neo, Thales Spacebus Neo) spend four to six months raising their orbits from GTO.

## 3.6 Rendezvous and Relative Motion

To dock with the ISS, a visiting vehicle must arrive at the same place at the same time with the same velocity. Rendezvous begins with **phasing**: because a lower orbit has a shorter period, a chaser in a slightly lower orbit gains on the target each revolution. When close (tens of kilometres), relative motion is described in the target's RTN frame by the linearised **Hill–Clohessy–Wiltshire (CW) equations**, valid for a circular target orbit with mean motion $n$:

$$\ddot{x} - 3n^2x - 2n\dot{y} = 0, \qquad \ddot{y} + 2n\dot{x} = 0, \qquad \ddot{z} + n^2z = 0$$

where $x$ is radial, $y$ along-track and $z$ cross-track. Their solutions produce counter-intuitive behaviour (Figure 3.14): a thrust *forward* along the velocity raises the chaser, so it slows down and drops *behind*; a radial burn produces a closed loop with no net drift. Approaches are designed along the **V-bar** (target velocity vector) or **R-bar** (radial line). They include "hold points", corridors and abort trajectories that are passively safe. That is, they never collide with the target if thrusters fail.

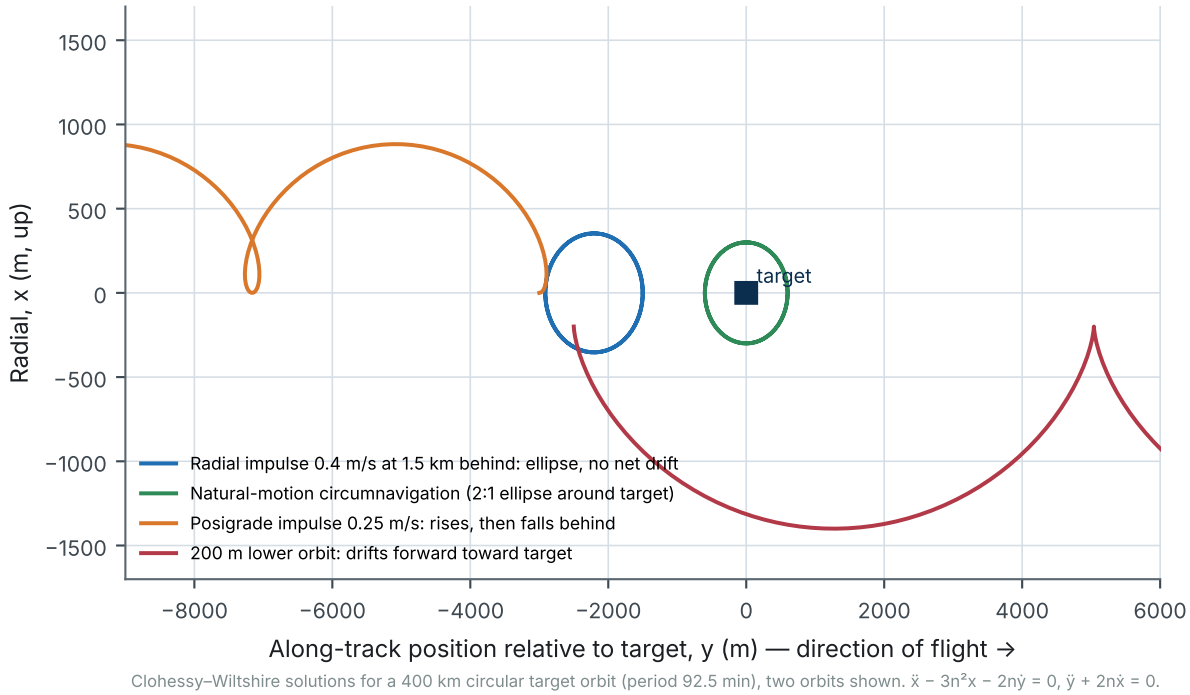



**FIGURE 3.14** Relative motion of a chaser near a target in a 400 km orbit, from the Clohessy–Wiltshire solutions. Each trajectory starts from a simple initial condition and is shown for two orbits.

Fast rendezvous profiles now let crews reach the ISS in about 3 hours (Soyuz and Progress since 2020) instead of the traditional 2 days; Crew Dragon typically takes 15–30 hours. Autonomous rendezvous uses relative GNSS, laser rangefinders (lidar) and cameras (Chapter 48).

## 3.7 Orbit Determination

**Orbit determination (OD)** estimates a satellite's orbit from measurements. Common measurements are:

- **Range** (distance), from the two-way travel time of a radio signal or laser pulse (satellite laser ranging achieves millimetre precision);
- **Range-rate**, from the Doppler shift of the carrier signal;
- **Angles**, from antennas, optical telescopes or interferometers (e.g., Delta-DOR for deep space, Chapter 24);
- **Onboard GNSS**, which gives LEO satellites their own position to better than 10 m in real time and to centimetres after processing.

Because each measurement is noisy, OD combines many of them with a dynamical model. The **batch least-squares** method adjusts the initial state to minimise the weighted sum of squared residuals over a data arc; the **(extended) Kalman filter** updates the estimate sequentially as each measurement arrives. Both produce a **covariance matrix** — a statistical description of the uncertainty — which is essential for conjunction assessment and for planning manoeuvres. The quality of OD depends as much on the force models (drag, SRP) as on the measurements.

## 3.8 Ground Tracks and Coverage

The **ground track** is the path traced on the Earth's surface by the point directly beneath the satellite. It combines the satellite's orbital motion with the Earth's rotation beneath it (Figure 3.15). Each successive pass of a LEO satellite crosses the equator about $360^\circ \times T/T_{day}$ further west — about 23° for a 92-minute orbit. The maximum latitude reached equals the inclination (or 180° − i for retrograde orbits).

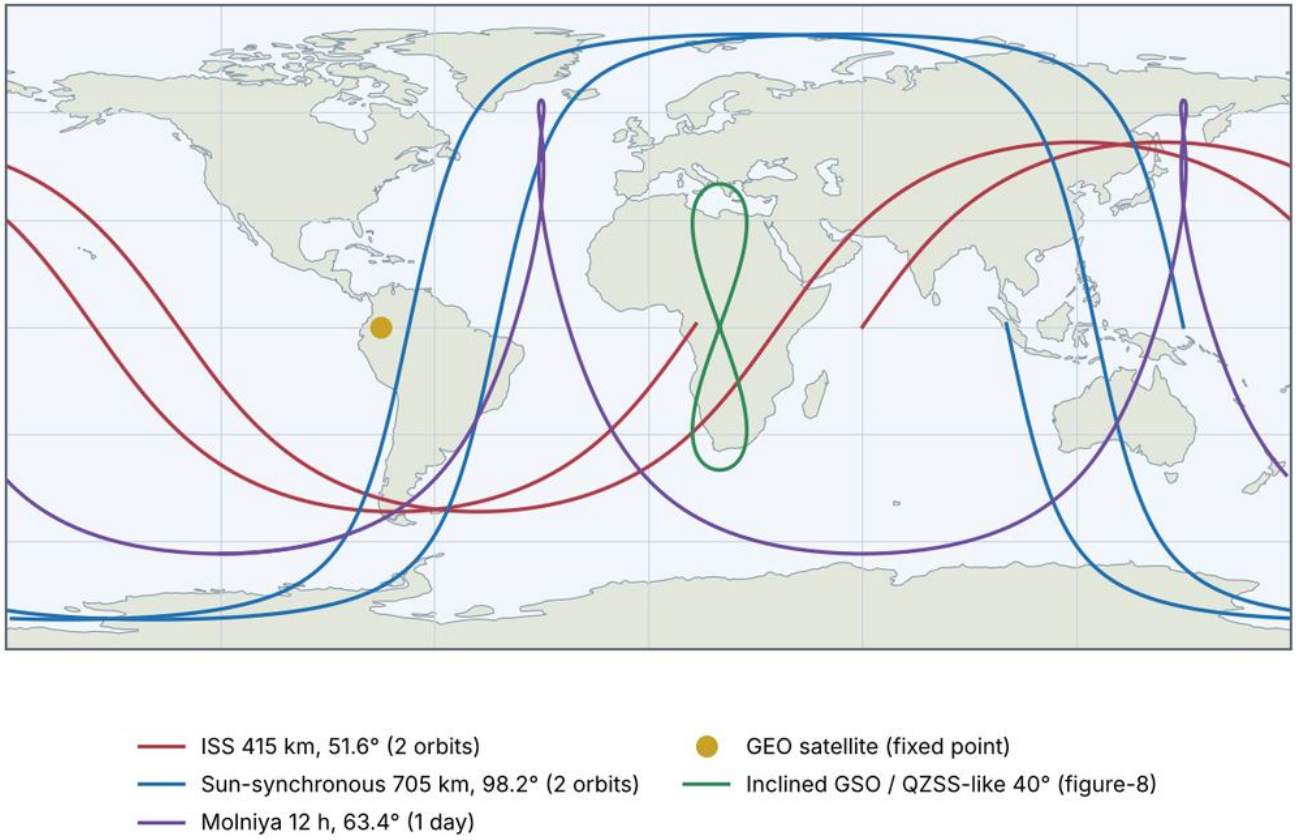



**FIGURE 3.15** Ground tracks of representative orbits: the ISS (51.6°), a sun-synchronous imaging orbit (98.2°), a 12-hour Molniya orbit that dwells over the northern hemisphere, a geostationary satellite (a fixed point) and an inclined geosynchronous orbit tracing a figure-eight.

A **repeat ground track** orbit returns over exactly the same points after an integer number of revolutions $N$ in an integer number of days $D$ ($NT = D \times T_{day,nodal}$). Landsat repeats after 233 orbits in 16 days; Sentinel-2 after 143 orbits in 10 days; Jason-3 after 127 orbits in about 9.9 days. Repeat orbits allow consistent comparison of images over time and accurate altimetry along fixed tracks.

The area a satellite can see — its **coverage** — depends on altitude and the minimum elevation angle at which a ground terminal can communicate. Coverage geometry and constellation design are covered in Chapter 11.

Table 3.1 summarises the main Earth-orbit types.

**TABLE 3.1** Principal Earth orbit types and their uses.

| Orbit | Typical altitude / shape | Inclination | Period | Main uses |
|---|---|---|---|---|
| Very low Earth orbit | 160–350 km, circular | any | 87–91 min | High-resolution imaging, technology demos, short-lived missions |
| Low Earth orbit (LEO) | 350–2 000 km | 0–90°+ | 91–127 min | Stations, EO, broadband constellations, science |
| Sun-synchronous (SSO) | 500–1 000 km | 97–99.5° | 95–105 min | Optical and radar EO, weather (polar), climate |
| Polar (non-SSO) | 700–1 400 km | ≈ 85–90° | 100–115 min | Iridium (86.4°), global coverage |
| Medium Earth orbit (MEO) | 8 000–23 300 km | 0–56° | 4.8–14 h | GNSS (≈20 000–23 200 km), O3b mPOWER (8 062 km, equatorial) |
| Geostationary (GEO) | 35 786 km, circular | 0° | 23 h 56 min | Communications, weather, missile warning, data relay |
| Inclined geosynchronous (IGSO) | 35 786 km | 30–60° | 23 h 56 min | QZSS, BeiDou IGSO, NavIC (regional navigation) |
| Molniya | 500 × 39 900 km | 63.4° | 12 h | High-latitude communications, early warning |
| Tundra | ≈ 25 000 × 46 000 km | 63.4° | 24 h | Sirius XM (formerly), QZSS (variant), Arctic comms |
| Geosynchronous transfer (GTO) | 200 × 35 786 km | launch-site dependent | 10.5 h | Transfer to GEO |
| Graveyard (super-synchronous) | GEO + ≥ 235 km | ≈ 0° | > 24 h | Disposal of GEO satellites |

## 3.9 Interplanetary Trajectories

### 3.9.1 Patched Conics and the Sphere of Influence

A spacecraft travelling to Mars is pulled by the Earth, Mars and the Sun at the same time. The **patched-conic approximation** splits the journey into segments, in each of which only one body matters, and "patches" them together at the boundary of each planet's **sphere of influence (SOI)**:

$$r_{SOI} \approx a_{planet}\left(\frac{m_{planet}}{m_{Sun}}\right)^{2/5}$$

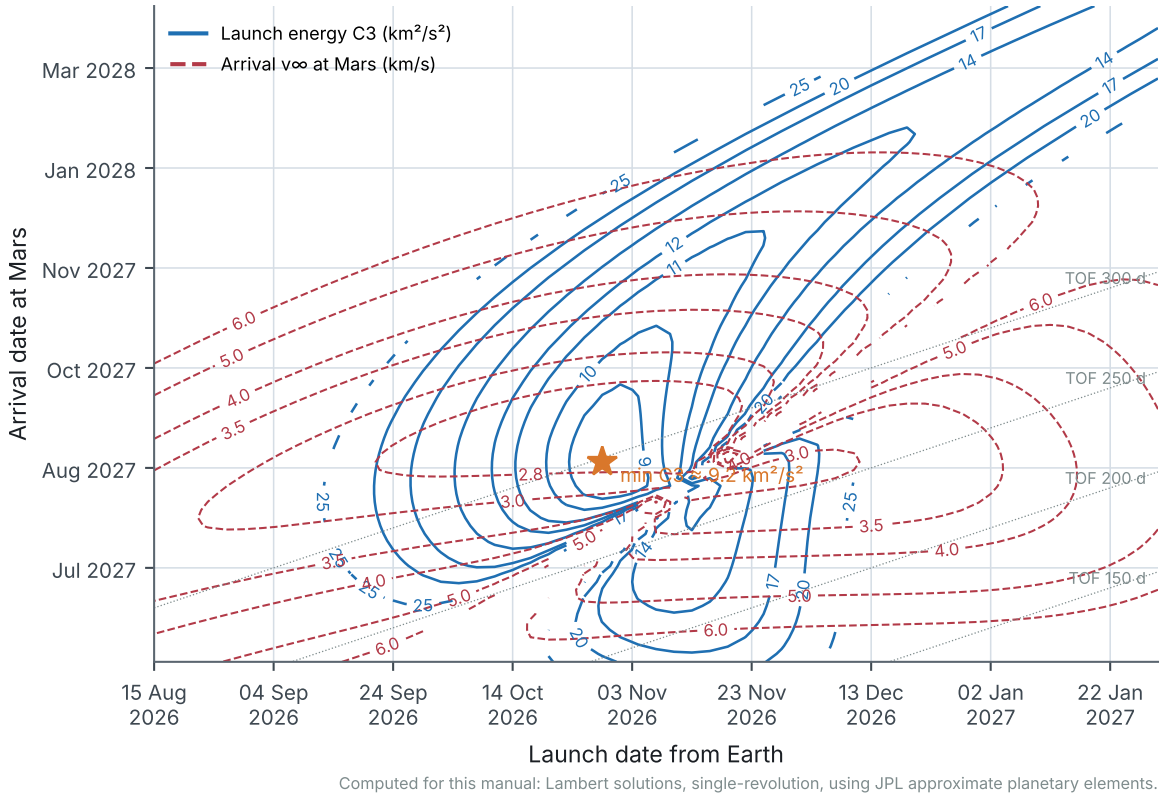
Earth's SOI has a radius of about 925 000 km, Mars' about 577 000 km, and the Moon's (relative to Earth) about 66 000 km. A mission thus consists of a **hyperbolic departure** from Earth, a **heliocentric transfer** ellipse, and a **hyperbolic arrival** at Mars, where the spacecraft either flies by, brakes into orbit, or enters the atmosphere.

**WORKED EXAMPLE — A HOHMANN TRANSFER FROM EARTH TO MARS**

Treating both orbits as circular and coplanar (1.000 and 1.524 AU): the transfer ellipse has $a = 1.262$ AU, so the flight time is half its period, $0.5 \times 1.262^{1.5}$ years = **259 days**. Earth's orbital speed is 29.78 km/s and the transfer perihelion speed is $29.78\sqrt{2 \times 1.524/2.524} = 32.73$ km/s, so the departure $v_\infty = 2.94$ km/s ($C_3 = 8.7$ km²/s²). From a 300 km parking orbit the injection burn is $\sqrt{8.7 + 2 \times 398\,600/6\,678} - 7.73 = 3.6$ km/s. At Mars the spacecraft arrives with $v_\infty = 2.65$ km/s. Because the planets must be correctly aligned at departure (Mars about 44° ahead of Earth), such opportunities recur only once every **synodic period** — 780 days, or about 26 months.

## 3.9.2 Lambert's Problem and Porkchop Plots

Real planetary orbits are elliptical and inclined, so mission designers solve **Lambert's problem**: find the orbit that connects position $\mathbf{r}_1$ at time $t_1$ to position $\mathbf{r}_2$ at time $t_2$. Solving it for a grid of launch and arrival dates gives a **porkchop plot** (Figure 3.16) — contours of departure energy $C_3$ and arrival $v_\infty$ — the basic tool for choosing launch periods. The plot for the late-2026 Mars window, computed for this manual, shows a minimum $C_3$ of about 9.2 km²/s² for launch near 29 October 2026 and arrival in late August 2027. The two lobes are "Type I" (transfer angle less than 180°) and "Type II" (more than 180°) trajectories; the ridge between them is a near-180° transfer that requires a large plane change.




Computed for this manual: Lambert solutions, single-revolution, using JPL approximate planetary elements.

**FIGURE 3.16** Porkchop plot for Earth-to-Mars transfers in the 2026 launch opportunity. Solid blue contours: launch energy C3 (km²/s²); dashed red: arrival hyperbolic speed at Mars (km/s); dotted: time of flight.

### 3.9.3 Gravity Assists

A **gravity assist** (or swing-by) uses a close flyby of a planet to change a spacecraft's heliocentric velocity without propellant (Figure 3.17). In the planet's frame, the hyperbolic flyby only rotates the velocity vector by a turning angle $\delta$, with

$$\sin\frac{\delta}{2} = \frac{1}{1 + r_p v_\infty^2 / \mu}$$

but in the Sun's frame the planet's orbital velocity is added before and after, so the spacecraft's speed can increase (passing behind the planet) or decrease (passing in front). Gravity assists made the Voyager "Grand Tour" possible and sent Cassini to Saturn via Venus–Venus–Earth–Jupiter. They slowed MESSENGER and BepiColombo enough to orbit Mercury. Parker Solar Probe uses them (seven Venus flybys) to approach within 6.2 million km of the Sun. Flybys can be "powered" (with a burn at periapsis, exploiting the **Oberth effect**: a burn is more effective where the speed is highest).

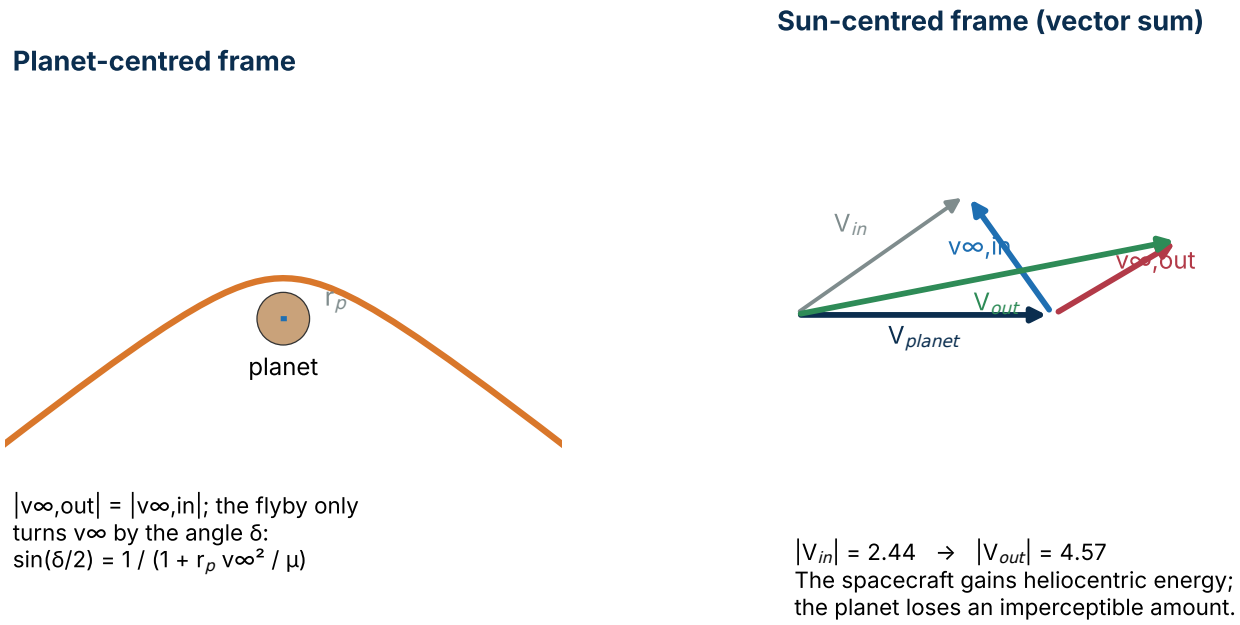
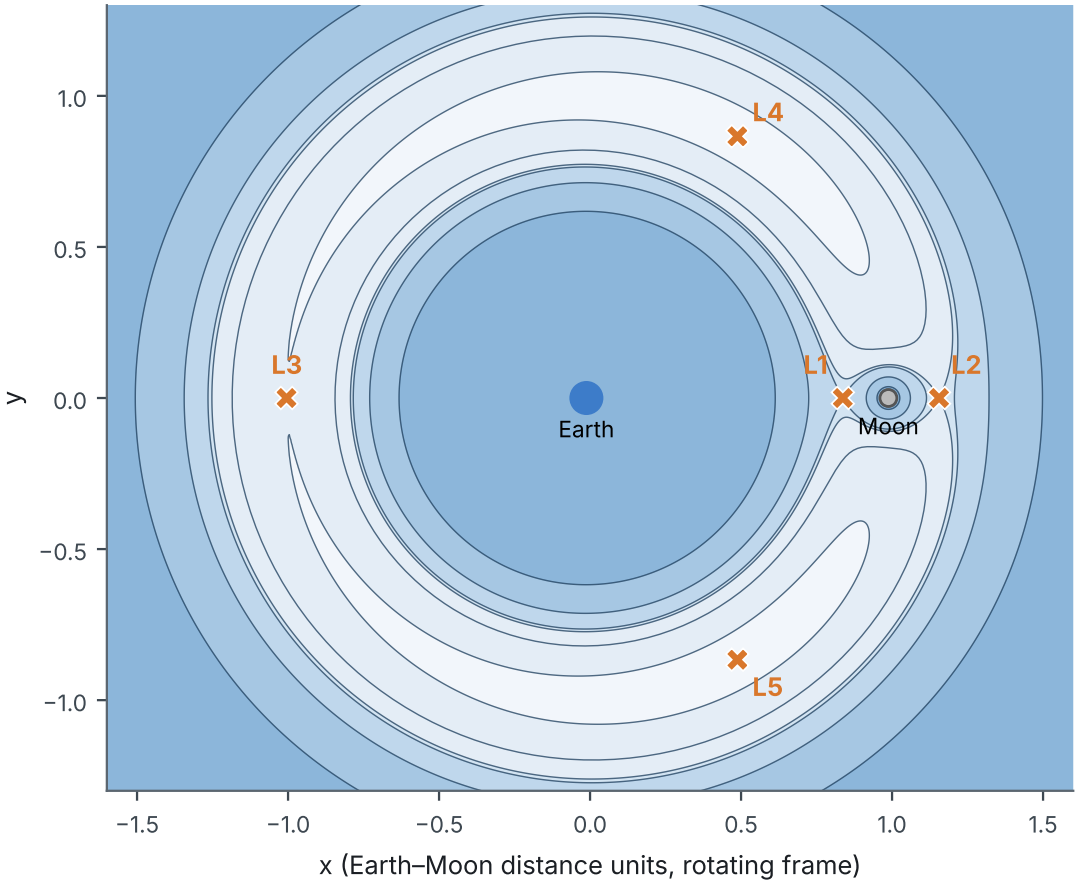



**FIGURE 3.17** Geometry of a gravity assist. Left: in the planet's frame the hyperbolic flyby rotates the excess velocity without changing its magnitude. Right: in the Sun's frame, adding the planet's orbital velocity shows a net gain in speed.

## 3.10 Three-Body Dynamics and the Lagrange Points

When two massive bodies (such as Sun–Earth or Earth–Moon) orbit each other, a third small body moves in the **circular restricted three-body problem (CR3BP)**. In a frame rotating with the two primaries there are five equilibrium points — the **Lagrange points** L1 to L5 (Figure 3.18). L1, L2 and L3 lie on the line through the primaries and are unstable (like a ball on a saddle). L4 and L5 form equilateral triangles with the primaries and are stable when the mass ratio is small. Jupiter's Trojan asteroids gather there.




Contours of the Jacobi constant (zero-velocity curves) for μ = 0.01215. L1 at 326,382 km and L2 at 448,914 km from Earth.

**FIGURE 3.18** The Earth–Moon Lagrange points and contours of the Jacobi constant ("zero-velocity curves"), which bound the regions a spacecraft with a given energy can reach in the rotating frame.

Spacecraft cannot sit exactly at an unstable point, but they can fly periodic **halo** or **Lissajous orbits** around it with small station-keeping burns (a few m/s per year):

- **Sun–Earth L1** (1.5 million km sunward) gives an uninterrupted view of the Sun and of the daylit Earth: SOHO, ACE, DSCOVR, Aditya-L1, SWFO-L1 and IMAP.
- **Sun–Earth L2** (1.5 million km anti-sunward) keeps the Sun, Earth and Moon on one side. This is ideal for cold infrared and survey telescopes: WMAP, Planck, Herschel, Gaia, JWST, Euclid and the Nancy Grace Roman Space Telescope.
- **Earth–Moon L1/L2** (about 61 000 km from the Moon) serve as relay and staging locations: China's Queqiao and Queqiao-2 relay satellites orbit near Earth–Moon L2 to support far-side landers.
- **Near-rectilinear halo orbits (NRHO)** are a family of L2 halo orbits that pass close over the lunar poles. They were tested by NASA's CAPSTONE CubeSat (2022) and were chosen for the planned Gateway station. Orion flew **distant retrograde orbits (DROs)**, very stable orbits around the Moon, on Artemis I.

The invariant manifolds of these orbits form low-energy "tubes" that connect them, enabling fuel-efficient transfers known as the **interplanetary superhighway**. The Genesis solar-wind sample-return mission and the GRAIL and CAPSTONE lunar missions used such low-energy trajectories.

## 3.11 The Δv Budget

Mission designers summarise trajectories as a **Δv budget**: the list of all velocity changes the vehicle must provide, plus margins. Figure 3.19 gives approximate values for the most common legs in the Earth–Moon–Mars system.

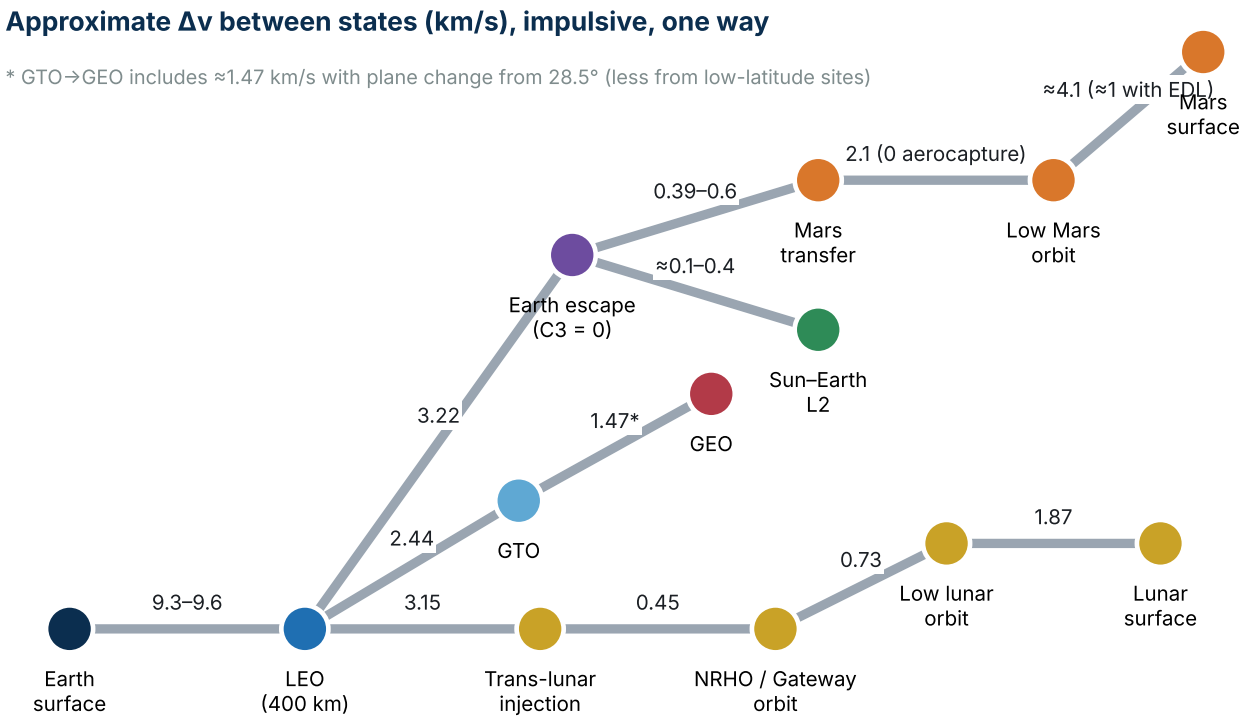



**FIGURE 3.19** An approximate "Δv subway map" of the Earth–Moon–Mars system. Values are impulsive, one-way and indicative; real missions add gravity and steering losses, navigation corrections and margins. Aerobraking or aerocapture can replace most of the braking at planets with atmospheres.

**TABLE 3.2** Example Δv budget for a geostationary communications satellite (15-year life, launched to standard GTO).

| Item | Δv (m/s) | Notes |
|---|---|---|
| GTO → GEO (apogee raising, inclination removal from 27°) | 1 500–1 800 | Several apogee-engine burns; less from supersynchronous or low-inclination GTO |
| Station acquisition and longitude relocation | 10–30 | Drift to assigned orbital slot |
| North–south station-keeping | 45–50 per year (≈ 700 total) | Counteracts luni-solar inclination drift; often done with electric thrusters |
| East–west station-keeping | 1–5 per year | Counteracts $J_{22}$ longitude drift and SRP eccentricity |
| Momentum management | 5–20 | Depends on ADCS design |
| End-of-life disposal to graveyard orbit | 7–11 | ≥ 235 km above GEO (IADC guideline) |
| Margin | 3–5 % | Dispersions, contingencies |

### KEY TAKEAWAYS

- Two-body orbits are conic sections; the vis-viva equation $v^2 = \mu(2/r - 1/a)$ gives the speed anywhere on an orbit, and Kepler's third law gives the period.
- Six orbital elements (a, e, i, Ω, ω, ν) describe an orbit; TLEs are mean elements for SGP4, not precise ephemerides.
- $J_2$ makes orbit planes precess; this enables sun-synchronous, critically inclined (Molniya/Tundra) and frozen orbits.
- Δv is the currency of mission design: LEO→GEO by Hohmann costs about 3.9 km/s; plane changes are expensive and done where speed is lowest.
- Rendezvous is governed by the counter-intuitive Clohessy–Wiltshire dynamics; orbit determination combines measurements and force models to produce states with covariance.
- Interplanetary missions are designed with patched conics, Lambert solutions and porkchop plots; gravity assists and Lagrange-point orbits extend what is possible with limited propellant.

### REVIEW QUESTIONS

1. Use the vis-viva equation to find the speeds at perigee and apogee of a transfer orbit with perigee altitude 250 km and apogee altitude 35 786 km.
2. Calculate the period of a circular orbit at 700 km altitude.
3. List the six classical orbital elements. Which describe the orbit's size and shape, which its orientation, and which the satellite's position?
4. Why are two-line element sets unsuitable for precise collision assessment or precise pointing?
5. Explain how the Earth's $J_2$ term makes sun-synchronous orbits possible, and why such orbits are slightly retrograde.
6. Calculate the Hohmann-transfer Δv from a 300 km circular orbit to geostationary radius, ignoring the plane change. Why is the plane change usually combined with the apogee burn?
7. In the Clohessy–Wiltshire frame, what happens when a chaser spacecraft makes a small burn forward along its velocity? Why is this counter-intuitive?
8. What is a porkchop plot, and what does a mission designer read from it?
9. Describe the five Lagrange points of the Sun–Earth system and name a mission that uses L1 and one that uses L2.

# 4 CHAPTER Spacecraft Systems Engineering

Image: NASA/GSFC

*A spacecraft is a system of systems that must work first time, in an environment it was never fully tested in, for many years without repair. Systems engineering is the discipline that makes this possible. This chapter explains the project life cycle and its reviews, requirements and architecture, technical budgets and margins, technology readiness, reliability and redundancy, risk management, model-based systems engineering, the verification and test campaign, product assurance, cost and schedule estimation, and the standards that codify decades of hard-won lessons.*

**IN THIS CHAPTER**


4.1 What Systems Engineering Does
4.2 The Project Life Cycle
4.3 Requirements Engineering
4.4 Architecture and Trade Studies
4.5 Technical Budgets and Margins
4.6 Technology Readiness
4.7 Reliability and Redundancy
4.8 Risk Management
4.9 Model-Based Systems Engineering
4.10 Verification and Validation
4.11 Product Assurance and Safety
4.12 Cost and Schedule
4.13 Standards and Bodies of Knowledge


## 4.1 What Systems Engineering Does

The International Council on Systems Engineering (INCOSE) defines systems engineering as "a transdisciplinary and integrative approach to enable the successful realisation, use and retirement of engineered systems." In plainer words: the systems engineer makes sure that the right system is built, that all its parts work together, and that it is shown to work before it is launched.

In a space project the systems engineering team is responsible for:

- **Defining the problem.** Understanding stakeholder needs and turning them into a concept of operations (ConOps) and verifiable requirements.
- **Designing the architecture.** Choosing the orbit, the segments, the spacecraft configuration and the split of functions between subsystems, hardware and software, space and ground.
- **Managing interfaces.** Every mechanical, electrical, thermal, data and RF interface between subsystems and segments is documented and controlled (interface control documents, ICDs).
- **Allocating and tracking budgets.** Mass, power, propellant, pointing, data, link and cost budgets, each with margins.
- **Managing technical risk.** Identifying what could go wrong and reducing likelihood or consequence.
- **Planning and closing verification.** Proving, by test, analysis, inspection or demonstration, that every requirement is met.
- **Making trade-offs.** Comparing alternatives using clear criteria (performance, cost, risk, schedule).

> **KEY CONCEPT — OPTIMISE THE WHOLE, NOT THE PARTS**
>
> A subsystem engineer naturally wants the best solar array, the lightest structure or the fastest computer. The systems engineer asks a different question: what combination of choices gives the best *mission* within cost, schedule and risk? A slightly heavier structure that simplifies the thermal design and the test campaign may be the better system choice.

## 4.2 The Project Life Cycle

Large space projects follow a phased life cycle with formal reviews and decision points. NASA's version (NPR 7120.5 and NPR 7123.1) is shown in Figure 4.1; ESA's ECSS-M-ST-10C uses equivalent phases 0, A, B, C, D, E and F.

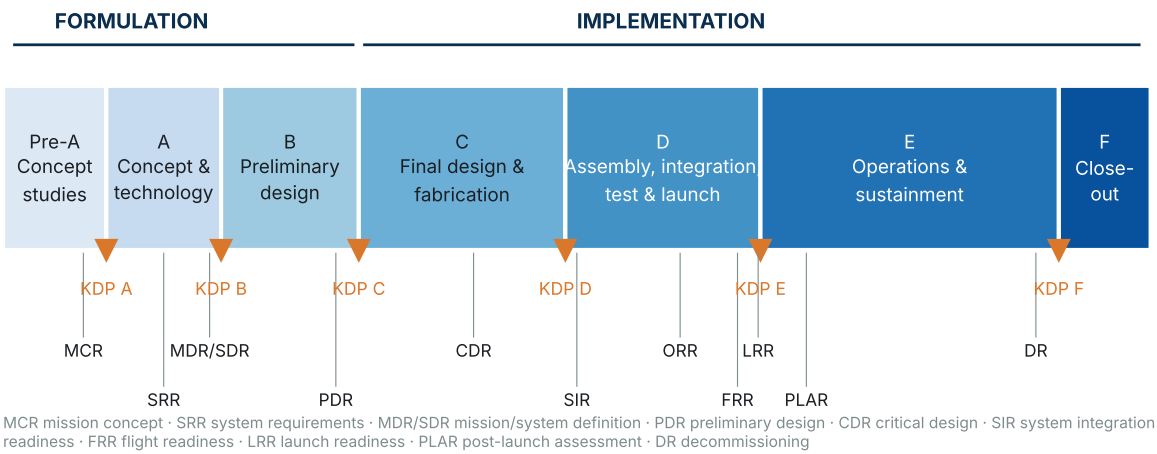



**FIGURE 4.1** NASA's project life cycle. Formulation (Pre-Phase A to Phase B) defines and plans the mission; Implementation (Phases C to F) builds, launches, operates and closes it out. Key Decision Points (KDPs) are management gates; technical reviews assess maturity before each gate.

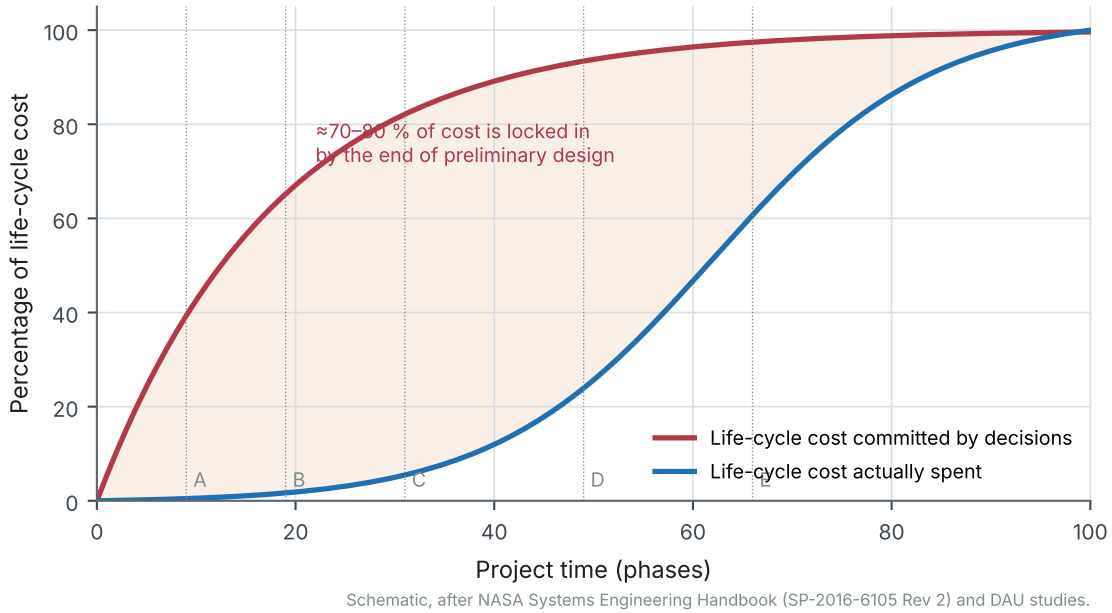
**TABLE 4.1** Phases of a space project and their main products.

| Phase | Purpose | Typical products | Exit review |
|---|---|---|---|
| Pre-Phase A — Concept studies | Explore the mission idea, identify needs and alternative concepts | Mission concept, feasibility study, rough cost | Mission Concept Review (MCR) |
| A — Concept and technology development | Define the system baseline, mature critical technologies | ConOps, system requirements, technology development plan, preliminary budgets | System Requirements Review (SRR), Mission/System Definition Review (MDR/SDR) |
| B — Preliminary design | Complete the preliminary design of all elements; establish cost and schedule baseline | Subsystem specifications, ICDs, preliminary design, verification plan, risk list | Preliminary Design Review (PDR) |
| C — Final design and fabrication | Complete the detailed design; build and test engineering and qualification models; start flight hardware | Build-to drawings, software code, qualification test results | Critical Design Review (CDR), System Integration Review (SIR) |
| D — System assembly, integration, test and launch | Integrate and verify the flight system; prepare operations; launch; commission | Verified flight system, operations procedures, trained team | Operational Readiness Review (ORR), Flight Readiness Review (FRR), Launch Readiness Review (LRR), Post-Launch Assessment Review (PLAR) |
| E — Operations and sustainment | Conduct the mission | Science and service data, operations reports | Critical events reviews, Decommissioning Review (DR) |
| F — Closeout | Dispose of the spacecraft and close the project | Disposal manoeuvres, data archiving, lessons learned | Disposal readiness |

Commercial and “New Space” projects compress these phases dramatically, often merging reviews and replacing paper documents with continuously updated models and agile software practices. A Starlink satellite design iterates in weeks; a flagship science mission such as Europa Clipper took about a decade from Phase A to launch. The principles — know what you need, design to it, prove it works — remain the same.

### 4.2.1 Why Early Decisions Matter

Most of a project’s life-cycle cost is *committed* by decisions made early, long before most of the money is *spent* (Figure 4.2). Choosing the orbit, launch vehicle, propulsion type or instrument technology in Phase A fixes most of what follows. A change that costs a meeting in Phase A may cost a redesign in Phase C and a mission failure if discovered in orbit. This is why systems engineering effort should be heaviest at the start of a project.



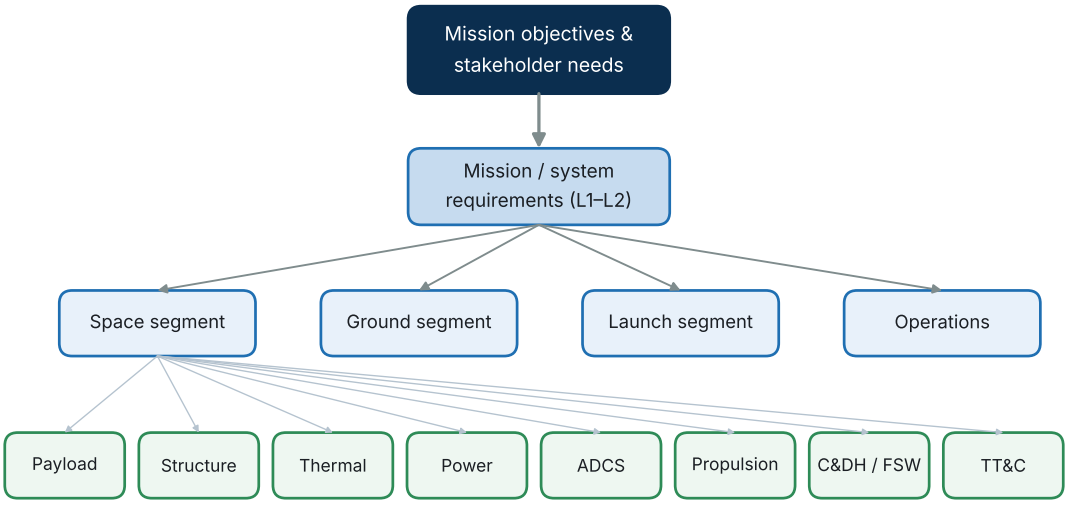

Schematic, after NASA Systems Engineering Handbook (SP-2016-6105 Rev 2) and DAU studies.

**FIGURE 4.2** Cumulative life-cycle cost committed by design decisions versus cost actually spent. By the end of the preliminary design phase, roughly three-quarters of the eventual cost is already determined.

# 4.3 Requirements Engineering

## 4.3.1 Good Requirements

A requirement states what the system *shall* do or be. Requirements form a hierarchy (Figure 4.3): Level 1 mission requirements (owned by the sponsor), Level 2 system requirements, Level 3 segment and subsystem requirements, and Level 4 component requirements. Every lower-level requirement should trace to a parent; requirements with no parent are "gold plating", and parents with no children are gaps.




Each requirement: unique ID · 'shall' statement · rationale · parent trace · verification method (Test / Analysis / Inspection / Demonstration) · owner

**FIGURE 4.3** The requirements hierarchy, from mission objectives through system and segment requirements to subsystem specifications. Each requirement carries a verification method and a trace to its parent.

A good requirement is:

- **Necessary** — traceable to a real need;

- **Unambiguous** — only one interpretation ("the spacecraft shall be lightweight" fails; "the spacecraft wet mass shall not exceed 450 kg" passes);
- **Verifiable** — there is a test, analysis, inspection or demonstration that proves it;
- **Achievable** — technically feasible within the budget;
- **Singular** — one requirement per statement;
- **Solution-independent** where possible — states *what*, not *how*.

> **WORKED EXAMPLE — FROM A MISSION OBJECTIVE TO A SUBSYSTEM REQUIREMENT**
>
> *Mission objective:* "Map crop health across Europe every five days."
>
> *System requirement:* "The system shall acquire multispectral imagery of all land between 35° N and 72° N. The revisit time shall be no more than 5 days, and the ground sampling distance shall be 10 m or better in the red, green, blue and near-infrared bands."
>
> *Derived requirements:* the constellation shall consist of two satellites in a 786 km sun-synchronous orbit phased 180° apart (orbit analysis). The instrument swath shall be ≥ 290 km (coverage analysis), and the instrument focal length and detector pitch shall give 10 m GSD (optics design). The data-handling subsystem shall store ≥ 2.4 Tbit and the X-band downlink shall transmit ≥ 520 Mbit/s (data budget). These numbers are, in fact, those of ESA's Sentinel-2 mission (Chapter 14).

### 4.3.2 Concept of Operations

A **concept of operations (ConOps)** describes how the system will be used from the user's viewpoint. It covers the mission phases (launch, early orbit, commissioning, nominal operations, contingencies, disposal) and what the spacecraft does on a typical day. It also covers who commands it and how often, how data flows to users, and how anomalies are handled. Writing the ConOps early exposes missing requirements — for example, a need for autonomy because ground contacts are only a few minutes per orbit.

## 4.4 Architecture and Trade Studies

The **architecture** is the high-level structure of the system: what elements exist and how functions are allocated among them. Typical architectural decisions for an Earth-observation mission include:

- One large satellite or a constellation of small ones?
- Optical or radar payload, or both?
- Onboard processing or raw-data downlink?
- Own ground stations, a commercial ground network or optical relay through GEO?
- Chemical, electric or no propulsion?

Alternatives are compared in **trade studies**. A transparent method is to define weighted criteria (e.g., performance 30 %, cost 25 %, risk 20 %, schedule 15 %, growth potential 10 %). The method then scores each option and tests how sensitive the result is to the weights. A trade study is only as good as its assumptions, so the assumptions must be documented.

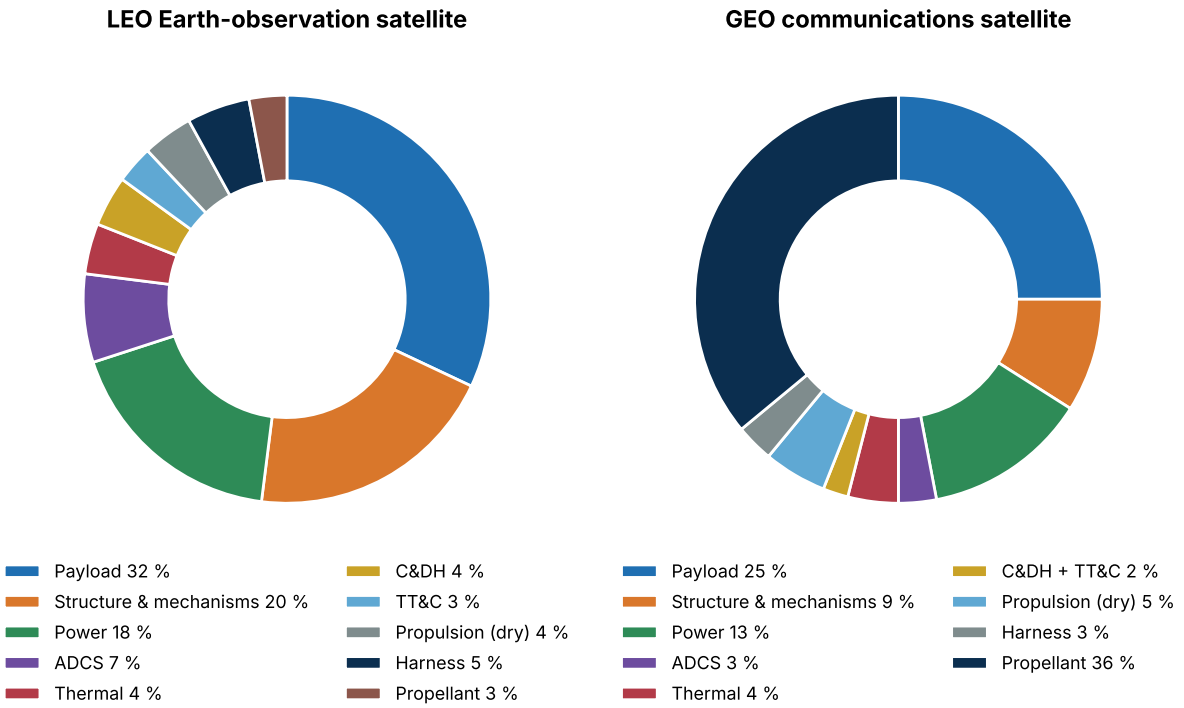
# 4.5 Technical Budgets and Margins

## 4.5.1 The Main Budgets

The systems engineer maintains a set of **budgets** — tables that allocate a limited resource among subsystems and track the current best estimate against the allocation:

| Budget | Resource tracked | Typical drivers |
|---|---|---|
| Mass | kg, including propellant | Launch vehicle capability, structure, propellant for Δv |
| Power | W (average and peak), Wh per orbit | Payload duty cycle, eclipse length, heaters, end-of-life degradation |
| Propellant / Δv | m/s and kg | Orbit raising, station-keeping, attitude control, disposal |
| Data | bits generated, stored, downlinked per day | Payload data rate, compression, ground contact time |
| Link | dB of margin | Distance, antenna gains, transmitter power, rain fade |
| Pointing | arcseconds of error and stability | Payload field of view, jitter, alignment, thermal distortion |
| Thermal | °C ranges for each unit | Dissipation, radiator area, orbit environment |
| Cost and schedule | money and time | All of the above |

Figure 4.4 shows how the launch mass of two typical spacecraft is divided.

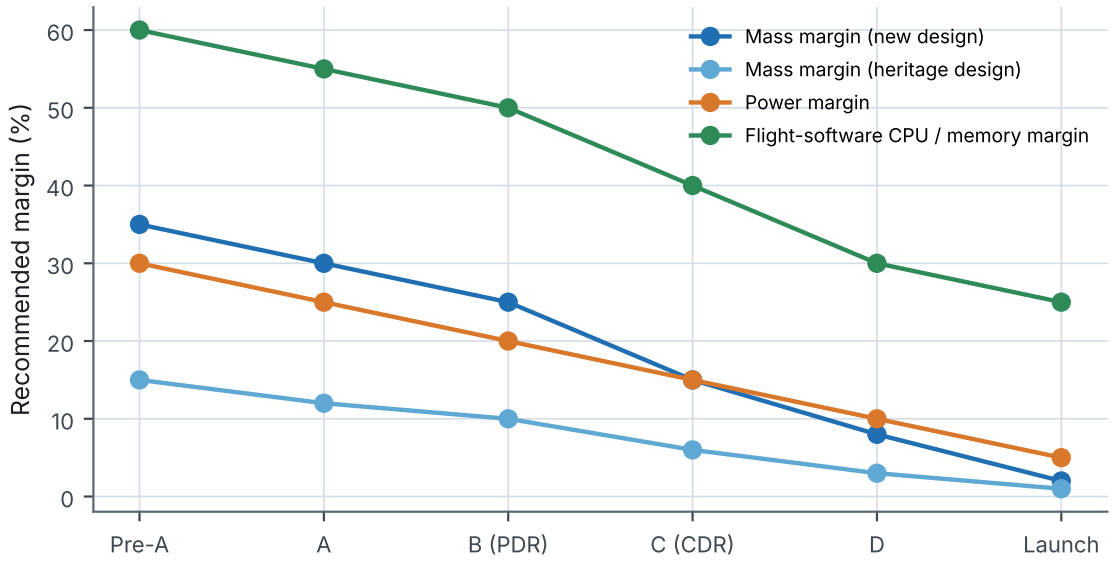



Representative fractions of launch (wet) mass, after SMAD (Wertz et al., 2011) and published spacecraft data.

**FIGURE 4.4** Typical distribution of launch mass for a LEO Earth-observation satellite and a GEO communications satellite. The GEO satellite carries a large propellant fraction for orbit raising and 15 years of station-keeping.

### 4.5.2 Margins and Contingency

Early in a project the estimates are uncertain, and history shows that mass and power almost always *grow* as the design matures. Projects therefore hold **contingency** and **margin**. Contingency is an allowance for growth in the estimate of each item, larger for new designs than for heritage hardware. Margin is a system-level reserve between the total estimate and the hard limit. Figure 4.5 shows typical guidance.




Typical contingency guidance, after AIAA S-120A-2015, ECSS-E-ST-10C and NASA GSFC Gold Rules; organisations differ.

**FIGURE 4.5** Typical recommended margins as a function of project phase. Margins are largest when uncertainty is largest and are progressively released as the design matures.

> **FAILURE LESSON — RUNNING OUT OF MARGIN**
>
> The James Webb Space Telescope grew in cost from an initial estimate of about US$1–3.5 billion to about US$10 billion, and its launch slipped from 2007 to 2021. This was in part because the technical complexity of the deployable sunshield and mirror had been underestimated, and early reserves were too small. The telescope ultimately performed better than its requirements. But the experience led NASA to require cost and schedule reserves for major projects to be budgeted at the 70 % confidence level ("joint confidence level", JCL).

## 4.6 Technology Readiness

**Technology Readiness Levels (TRLs)** — invented at NASA in the 1970s and now an international standard (ISO 16290) — measure how mature a technology is (Figure 4.6). Most agencies require all critical technologies to reach **TRL 6** (prototype demonstrated in a relevant environment) by PDR. Technologies at TRL 3–5 cross the "valley of death", where research funding ends but projects are not yet willing to adopt them. Related scales include Manufacturing Readiness Levels (MRL) and Software/AI readiness frameworks (Chapter 53).

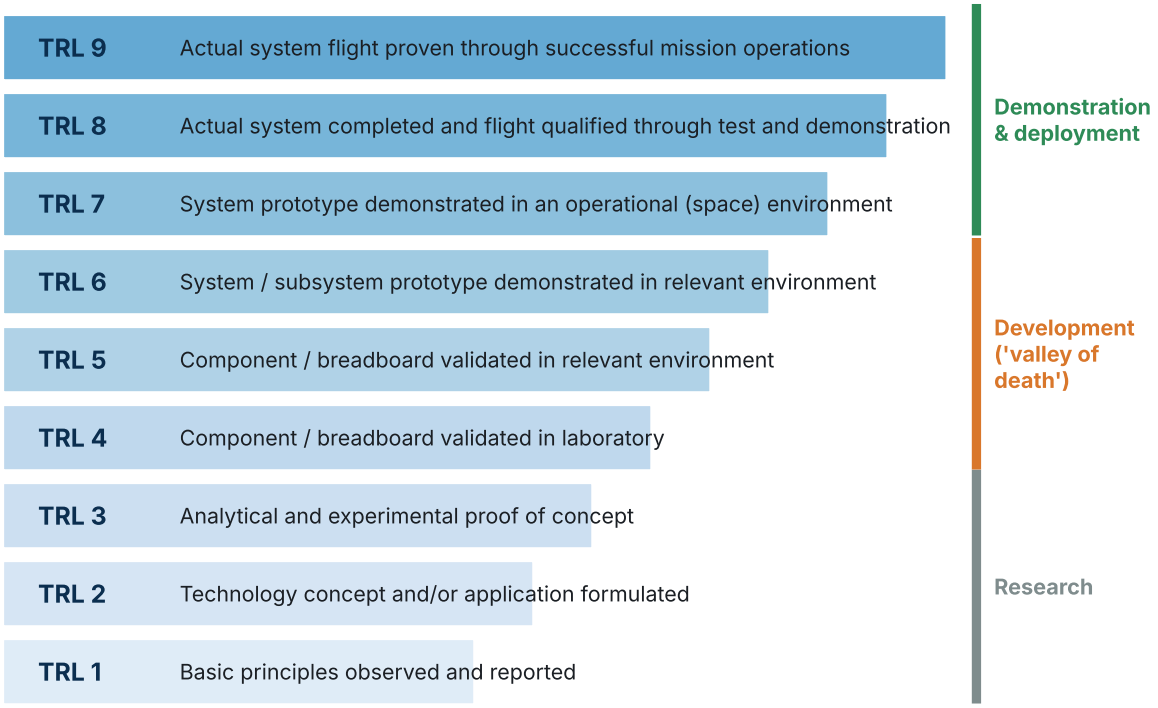



**FIGURE 4.6** The nine Technology Readiness Levels. Projects typically require TRL 6 for critical technologies by the Preliminary Design Review.

## 4.7 Reliability and Redundancy

### 4.7.1 Reliability Basics

**Reliability** $R(t)$ is the probability that an item performs its function for time $t$ under stated conditions. During the useful-life period of electronics, failures occur at a roughly constant rate $\lambda$, so

$$R(t) = e^{-\lambda t}, \qquad \text{MTTF} = 1/\lambda$$

Failure rates are often given in **FITs** (failures in $10^9$ device-hours). Before and after this period, the "bathtub curve" rises (Figure 4.7). Early *infant mortality* failures are weeded out by burn-in, screening and environmental testing. At the end, *wear-out* (battery capacity fade, bearing wear, propellant depletion, solar cell degradation) limits life.

For components in series (all must work), reliabilities multiply: $R_{sys} = \prod R_i$. For parallel redundancy (any one suffices), unreliabilities multiply: $R_{sys} = 1 - \prod (1 - R_i)$.

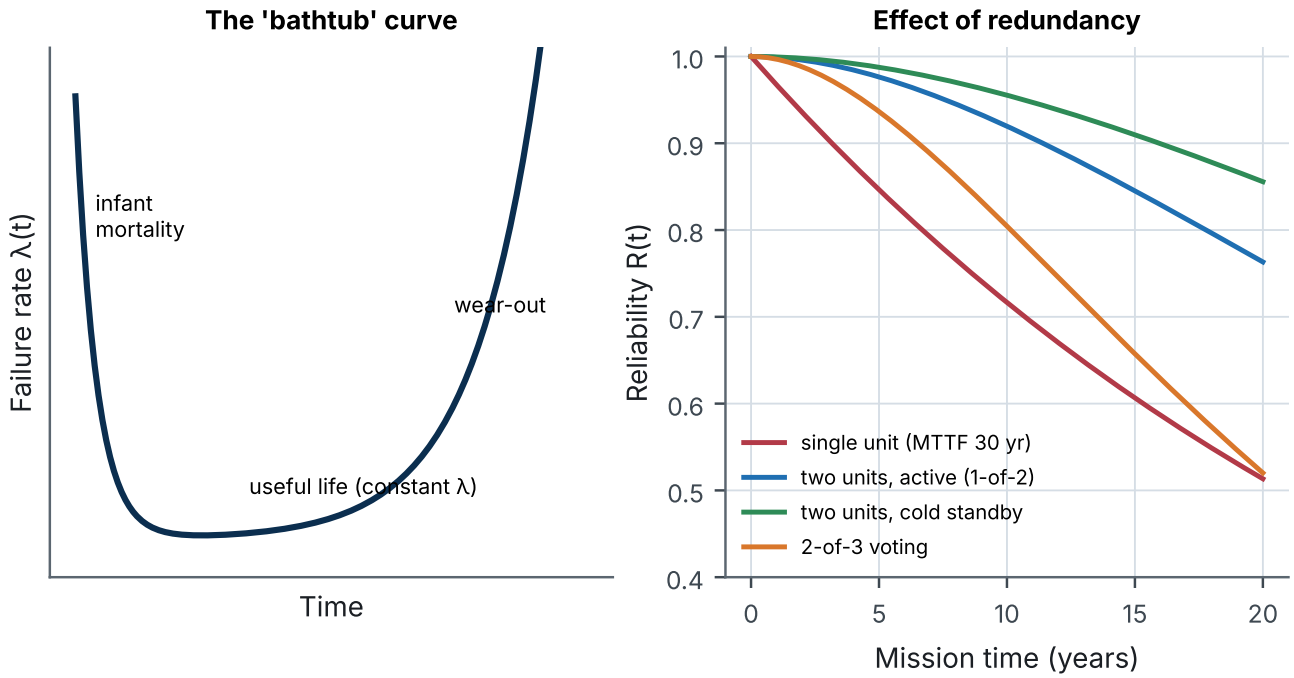



**FIGURE 4.7** Left: the bathtub curve of failure rate over life. Right: reliability over a 20-year mission for a single unit and for three redundancy schemes, assuming an MTTF of 30 years per unit.

### 4.7.2 Redundancy Strategies

- **Hot (active) redundancy** — two units operate together; either can carry the load.
- **Cold (standby) redundancy** — the backup is switched on only when the primary fails; it suffers less wear but needs reliable failure detection and switching.
- **Cross-strapping** — each of two redundant units of subsystem A can work with either unit of subsystem B, avoiding a single failure taking out a whole chain.
- **Voting (k-out-of-n)** — three computers compare outputs and outvote a faulty one (triple modular redundancy); used for launch vehicle avionics and safety-critical systems.
- **Functional redundancy** — a different method achieves the same function (e.g., attitude from star trackers or from Sun sensors and magnetometers).
- **Constellation-level redundancy** — spare satellites in orbit or on the ground, or graceful degradation by design (common in mega-constellations).

Single points of failure (SPFs) are identified by **failure modes, effects and criticality analysis (FMECA)** and either removed or justified. Typical SPFs that are accepted include primary structure, propellant tanks and some deployment mechanisms, which are designed with large margins instead.

**TABLE 4.2** Extract of a failure modes, effects and criticality analysis (FMECA) for a small-satellite power subsystem (illustrative).

| Item | Failure mode | Cause | Local effect | System effect | Severity | Detection | Mitigation |
|---|---|---|---|---|---|---|---|
| Solar array string | Open circuit | Interconnect fatigue, MMOD impact | Loss of one string (≈ 4 % power) | Reduced power margin | 3 (minor) | String current telemetry | 15 % end-of-life array margin; string-level blocking diodes |

| Item | Failure mode | Cause | Local effect | System effect | Severity | Detection | Mitigation |
|---|---|---|---|---|---|---|---|
| Battery cell | Internal short | Manufacturing defect, dendrite growth | Cell voltage drop, heating | Loss of battery module if not isolated | 2 (major) | Cell voltage and temperature monitoring | Cell screening, cell bypass, n+1 strings |
| Battery charge regulator | Fails “on” | Transistor short | Overcharge | Battery damage, possible thermal runaway | 1 (catastrophic) | Hardware over-voltage detector | Independent hardware cut-off, redundant regulator |
| Power distribution switch | Fails open | Latch-up, wear | Load unpowered | Loss of unit until switched to redundant | 2 | Current telemetry, watchdog | Cross-strapped redundant switches |
| Main bus | Short circuit | Harness chafing, debris | Bus voltage collapse | Loss of mission | 1 | Undervoltage monitor | Fault-current limiters, harness protection, design reviews |

## 4.8 Risk Management

A **risk** is a potential future event that would harm the project, characterised by its **likelihood** and **consequence**. Risks are recorded in a risk register, scored on a **5 × 5 matrix** (Figure 4.8) and assigned an owner. Each is handled by one of four strategies: *mitigate* (reduce likelihood or consequence), *accept*, *transfer* (e.g., insurance or contract terms), or *avoid* (change the design). Mitigation plans include “burn-down” steps tied to milestones, so that progress is visible.

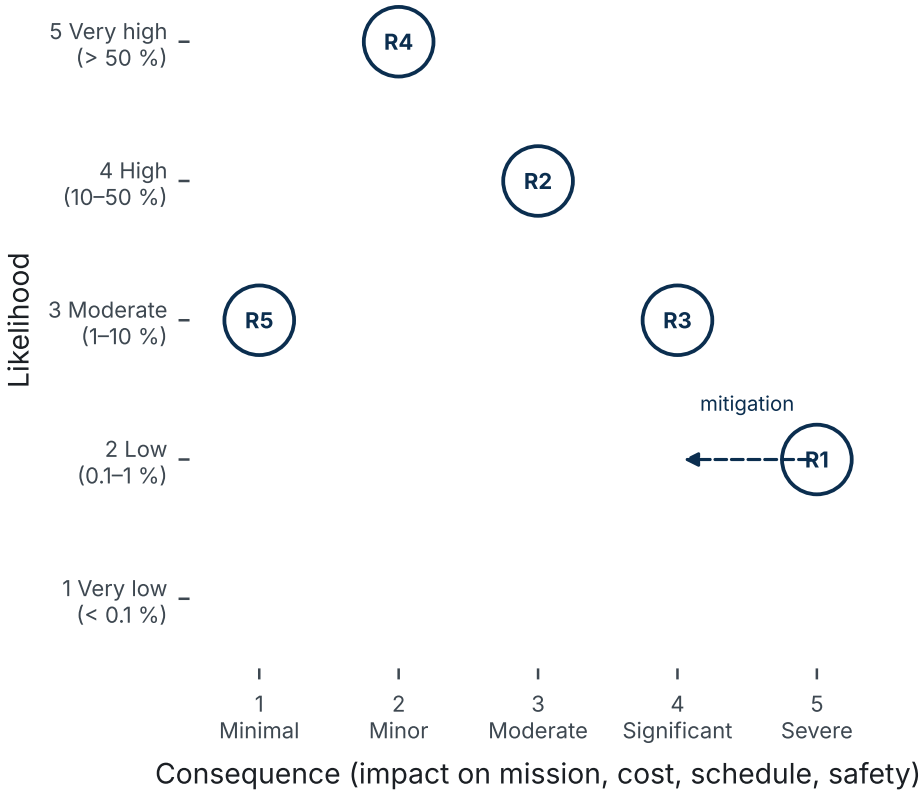



**FIGURE 4.8** A 5 × 5 risk matrix. Each risk (R1–R5) is plotted by likelihood and consequence; mitigation actions aim to move risks out of the red zone before key reviews.

Programmes also distinguish **risk classes**. NASA's NPR 8705.4 defines Class A to Class D, plus a "do no harm" category for very low-cost CubeSats. Class A has the highest priority and lowest risk tolerance (e.g., JWST, crewed systems). Class D accepts higher risk for lower cost (many small science missions). The class determines parts quality, redundancy, review depth and test requirements. ESA and other agencies use similar tailoring.

## 4.9 Model-Based Systems Engineering

Traditionally, the system design lived in hundreds of documents, spreadsheets and drawings, with inconsistencies discovered late. **Model-based systems engineering (MBSE)** replaces documents with a single, connected digital model of requirements, structure, behaviour and parameters (Figure 4.9). The standard language is **SysML** (Systems Modeling Language); **SysML v2**, formally adopted by the Object Management Group in 2025, adds a precise textual notation and an API that lets engineering tools exchange data automatically. MBSE tools link to CAD, finite-element, thermal, orbit and cost models, so that a change in one place propagates everywhere — the foundation of the "digital thread" and **digital twin** concepts (Chapter 52).

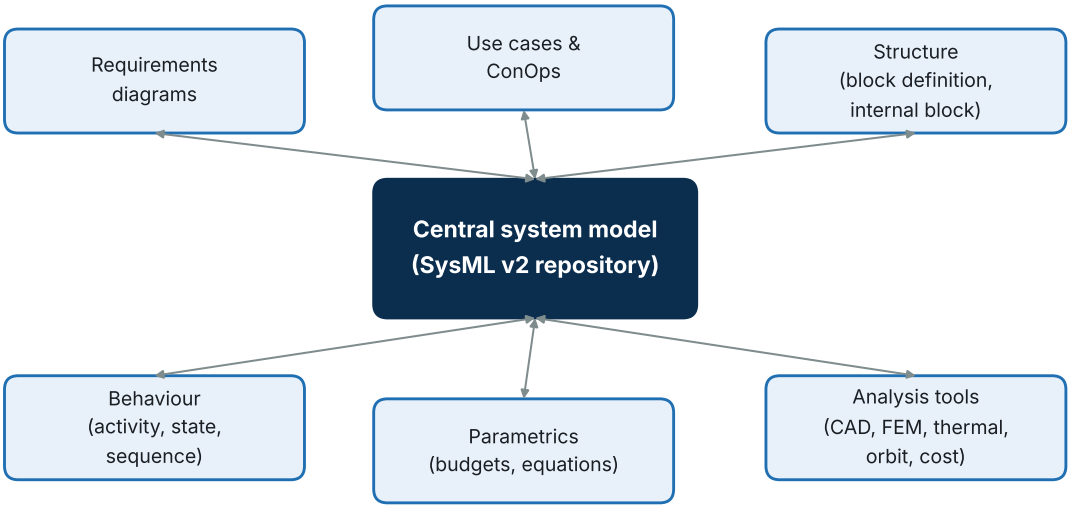



**FIGURE 4.9** Model-based systems engineering links requirements, operations, structure, behaviour and parametric analyses in one authoritative system model, connected to specialist engineering tools.

**AI IN PRACTICE — ASSISTANTS FOR SYSTEMS ENGINEERS**

Large language models are increasingly used to check requirement quality (ambiguity, missing "shall", non-verifiable wording), to trace requirements to design elements, to search lessons-learned databases and to draft SysML models from text. Agencies including NASA and ESA are evaluating such tools under strict data-security rules. Chapter 52 discusses how AI outputs must themselves be verified before they enter a controlled baseline.

## 4.10 Verification and Validation

**Verification** asks "did we build the system right?" — does it meet its requirements? **Validation** asks "did we build the right system?" — does it meet the stakeholders' needs in its real environment? The V-model (Figure 4.10) shows how each level of design on the left is matched by a level of verification on the right.

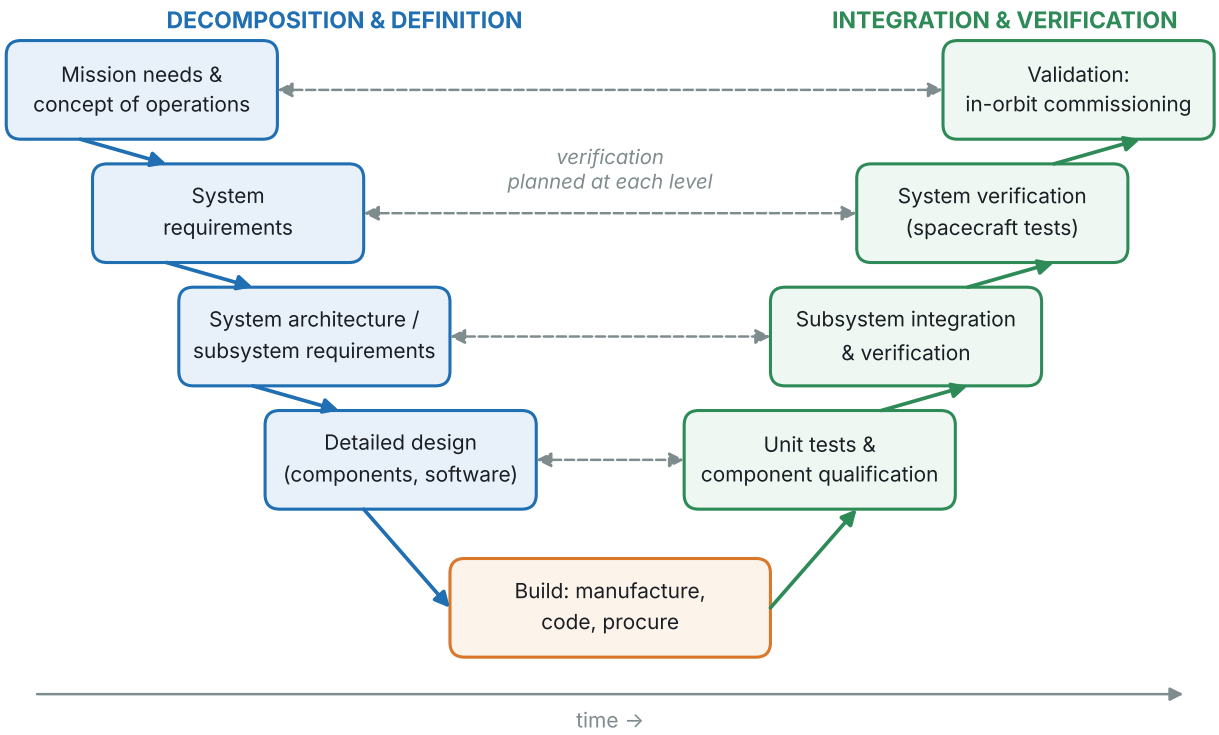



**FIGURE 4.10** The systems engineering "V". Requirements are decomposed down the left side, the system is built at the bottom, and verification proceeds up the right side, each level verified against the requirements defined at the same level.

### 4.10.1 Verification Methods

Every requirement is assigned one or more verification methods:

- **Test** — measurement under controlled conditions (e.g., thermal-vacuum test of the battery).
- **Analysis** — calculation or simulation, used when testing is impossible or too costly (e.g., MMOD survival probability, 15-year radiation dose).
- **Inspection** — visual or dimensional check (e.g., connector labelling, workmanship).
- **Demonstration** — operation showing a function without detailed measurement (e.g., deploying an antenna with gravity offloading).

The **verification matrix** lists every requirement, its method, the verification event and the closure evidence. Closing it is a condition of flight readiness.

### 4.10.2 Model Philosophy

Projects choose how many physical models to build (Figure 4.11). A classic approach builds a **structural-thermal model (STM)** to verify mechanical and thermal design, and an **engineering model (EM)** to verify functions and interfaces. It also builds a **qualification model (QM)** tested to qualification levels (typically flight levels plus 3–6 dB for vibration and ±10 °C beyond predicted temperatures). Finally, it builds a **flight model (FM)** tested to acceptance levels. The cheaper **protoflight** approach tests the single flight model to qualification levels for acceptance durations, accepting a slightly higher risk of fatigue damage.

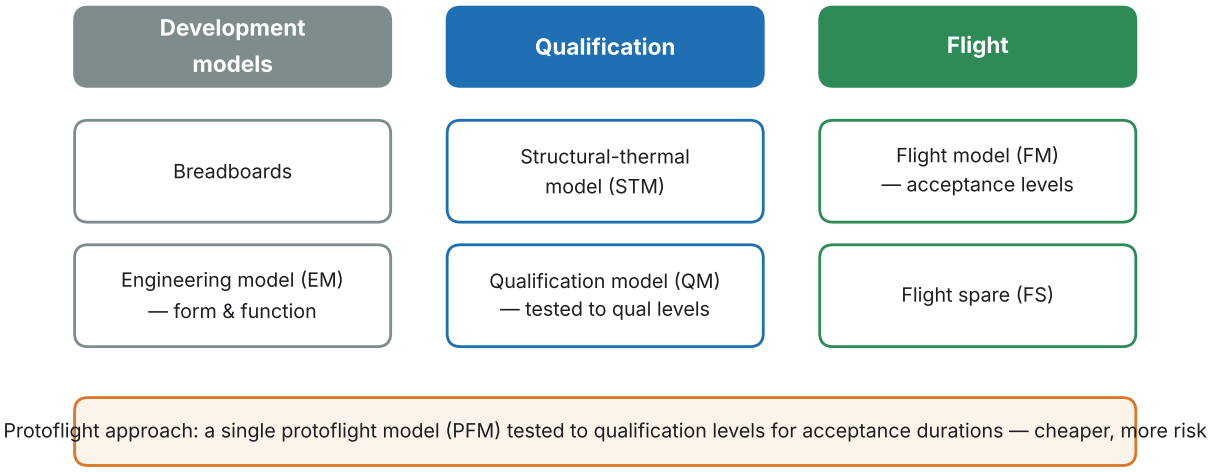
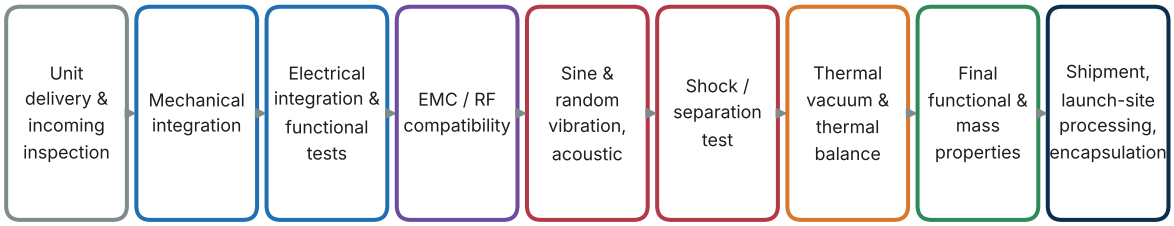



**FIGURE 4.11** Model philosophies. Classic approaches qualify a dedicated model before building the flight unit; protoflight approaches qualify and accept the flight unit itself.

### 4.10.3 The Integration and Test Campaign

Assembly, integration and test (AIT) — called I&T in the US — brings the spacecraft together and exposes it to every environment it will see (Figure 4.12). Key tests include:

- **Functional and performance tests** at unit, subsystem and system levels, repeated after each environmental test (comprehensive performance tests, CPTs, and shorter "aliveness" tests).
- **Mechanical tests**: sine vibration, random vibration (for smaller spacecraft), acoustic (for large spacecraft with light panels), shock and separation tests, modal surveys, and static load tests on structural models.
- **Thermal tests**: thermal balance (to validate the thermal model) and thermal-vacuum cycling (to show the hardware works at the hot and cold extremes; typically 4–8 cycles for units).
- **Electromagnetic compatibility (EMC)**: conducted and radiated emissions and susceptibility (e.g., MIL-STD-461), plus RF compatibility with ground stations and the launcher.
- **Deployment tests** with gravity off-loading (zero-g rigs, balloons, air-bearing tables).
- **End-to-end tests** with the real ground segment and mission operations team, and **mission simulations** that rehearse launch and early operations.
- **Mass properties** (mass, centre of gravity, moments of inertia) and **alignment** measurements.




Functional ("aliveness") tests are repeated after every environmental test to detect damage; a complete spacecraft test campaign typically lasts 6–18 months.

**FIGURE 4.12** A typical spacecraft assembly, integration and test (AIT) flow. Environmental tests are separated by functional tests that check for damage; the sequence is arranged so that the most severe mechanical tests precede the final thermal and functional verification.

**FIGURE 4.13** The High Bay 1 cleanroom at NASA's Jet Propulsion Laboratory during assembly of the Mars 2020 Perseverance rover and its cruise stage. Cleanrooms control particulate and molecular contamination and, for Mars missions, biological contamination. Image: NASA/JPL (NASA/JPL-Caltech).

**FIGURE 4.14** Europa Clipper in JPL's High Bay 1 after installation of its 3 m high-gain antenna in August 2023. The spacecraft was launched in October 2024 and will arrive at Jupiter in 2030. Image: NASA/JPL (NASA/JPL-Caltech).

**FIGURE 4.15** A technician inside Sierra Space's Dream Chaser spaceplane preparing for vibration testing at NASA's Neil Armstrong Test Facility in Ohio, home to the world's most powerful spacecraft shaker system and a large vacuum chamber. Image: NASA/GRC (NASA/Jordan Salkin).

**FAILURE LESSON — TEST AS YOU FLY**

In 1999 NASA's Mars Polar Lander was probably lost because its software interpreted vibrations from the deployment of its landing legs as touchdown, shutting off the descent engines about 40 m above the surface. The fault could have been found by a system-level test of the leg deployment with the flight software — a test that was planned but, after a wiring error was corrected, not repeated. The same year, Mars Climate Orbiter was lost because one team used pound-force seconds and another newton-seconds for thruster impulse data. Both losses led to stricter end-to-end testing and interface verification across NASA.

## 4.11 Product Assurance and Safety

**Product assurance (PA)** — called mission assurance in the US — ensures that the hardware and software are built as designed and are reliable. It includes:

- **Quality assurance**: configuration management, nonconformance reporting, material review boards, inspections, workmanship standards (e.g., soldering to ECSS-Q-ST-70-08 or NASA-STD-8739).
- **Electrical, electronic and electromechanical (EEE) parts**: selection from approved lists, screening (burn-in, X-ray), radiation testing, counterfeit avoidance, derating (operating parts well below their maximum ratings).
- **Materials and processes**: outgassing, flammability, stress corrosion, atomic-oxygen resistance.
- **Software product assurance**: coding standards, independent verification and validation (IV&V), software criticality classes (e.g., NASA Class A–E; ECSS criticality A–D).

- **Safety**: hazard analyses for launch-site operations (pressurised systems, pyrotechnics, batteries, propellants), launch-range safety, crew safety for human-rated systems, and re-entry casualty risk. The probability of injuring someone on the ground from re-entering debris must normally be below 1 in 10 000.

## 4.12 Cost and Schedule

### 4.12.1 Estimating Cost

Space projects are estimated by three methods: **analogy** (scaling from a similar past project), **parametric** and **engineering build-up** (summing labour and materials for every work-package). The parametric method uses cost-estimating relationships (CERs) that relate cost to mass, power, complexity and heritage. Examples are NASA's Project Cost Estimating Capability (PCEC), the USAF Unmanned Space Vehicle Cost Model and the Aerospace Corporation's Small Satellite Cost Model. Mature projects use all three and reconcile the results.

Typical cost elements for a science mission are the spacecraft bus (25–35 %), instruments (20–35 %) and project management/systems engineering/mission assurance (10–15 %). The others are integration and test (5–10 %), launch (10–25 %), and operations and data analysis (10–20 %). Reserves of 25–50 % are typical before PDR.

### 4.12.2 Schedule

Space schedules are dominated by long-lead items (detectors, star trackers, reaction wheels, radiation-hardened processors, propellant tanks). They are also dominated by integration and test, which cannot be easily parallelised, and by launch windows. For planetary missions, launch windows occur only every one to two years. A slip of a few weeks in delivering a Mars mission can delay launch by 26 months. This happened to ESA's ExoMars rover, delayed from 2020 to 2022 and then to 2028 after cooperation with Russia ended.

## 4.13 Standards and Bodies of Knowledge

Systems engineering practice is codified in standards that capture lessons from decades of successes and failures:

**TABLE 4.3** Key standards and handbooks for space systems engineering.

| Standard / handbook | Publisher | Scope |
|---|---|---|
| NASA Systems Engineering Handbook (SP-2016-6105 Rev 2) | NASA | Processes, life cycle, technical management |
| NPR 7120.5 / NPR 7123.1 | NASA | Programme/project management and systems engineering requirements |
| ECSS-E-ST-10C, ECSS-M-ST-10C, ECSS-Q-ST series | European Cooperation for Space Standardization | Systems engineering, project phasing, product assurance |
| ISO/IEC/IEEE 15288 | ISO / IEEE | System life-cycle processes (all industries) |
| ISO 24113 | ISO | Space debris mitigation requirements |

| Standard / handbook | Publisher | Scope |
|---|---|---|
| GSFC-STD-7000 (GEVS) | NASA Goddard | Environmental verification requirements for payloads and spacecraft |
| SMC-S-016 / SMC-S-010 | US Space Force | Test requirements for launch, upper-stage and space vehicles; systems engineering |
| AIAA S-120A | AIAA | Mass properties control and growth allowance |
| CCSDS Blue Books | Consultative Committee for Space Data Systems | Communication and data-system standards (Chapter 7) |
| *Space Mission Engineering: The New SMAD* (Wertz, Everett & Puschell, 2011) | Microcosm Press | Classic engineering reference for mission design |
| INCOSE Systems Engineering Handbook (5th ed., 2023) | INCOSE | General systems engineering body of knowledge |

### KEY TAKEAWAYS

- Systems engineering optimises the whole mission across segments, subsystems, cost, schedule and risk; most cost is committed in early phases.
- Life cycles move from concept (Pre-A, A) through design (B, C) to integration, launch and operations (D, E) and closeout (F), gated by reviews such as SRR, PDR and CDR.
- Good requirements are necessary, unambiguous, verifiable and traceable; budgets track mass, power, Δv, data, links and pointing with phase-dependent margins.
- Reliability is improved by part quality, derating, redundancy and testing; FMECA identifies single points of failure; risks are managed on a likelihood–consequence matrix.
- MBSE (SysML v2) connects requirements, design and analysis in a single model; verification closes every requirement by test, analysis, inspection or demonstration.
- "Test as you fly": many famous failures (Mars Polar Lander, Mars Climate Orbiter) were interface or end-to-end test gaps, not exotic physics.

### REVIEW QUESTIONS

1. Why is most of a project's life-cycle cost committed during the early phases, even though little money has been spent by then?
2. Outline the project phases from Pre-A to F and name the major reviews that close Phase B and Phase C.
3. Rewrite the requirement "The satellite shall have a good camera" so that it is necessary, unambiguous and verifiable.
4. Why do mass and power budgets carry larger margins early in a project? How do the margins change as the design matures?
5. Two identical units are flown in active parallel redundancy, each with a reliability of 0.90 over the mission. What is the reliability of the pair? What assumption does this answer rely on?
6. What is a technology readiness level? Why is TRL 6 normally expected by the preliminary design review?
7. Distinguish verification from validation and list the four verification methods, with an example of each.
8. Explain the principle "test as you fly" using one of the mission failures discussed in this chapter.

CHAPTER

# 5 Spacecraft Subsystems I: Structures, Thermal Control, Power and Mechanisms

Image: NASA/JPL (NASA/JPL-Caltech/Lockheed Martin)

*Every spacecraft, whatever its mission, needs a skeleton to carry it through launch, a way to keep its parts at safe temperatures, a source of electrical power, and mechanisms that release, deploy and point. This chapter explains how structures are designed and built, how thermal control works with insulation, radiators, heat pipes and heaters, how solar arrays, batteries and radioisotope generators produce and store power, and how mechanisms are made reliable in vacuum.*

IN THIS CHAPTER

5.1 The Spacecraft Bus
5.2 Structures
5.3 Thermal Control
5.4 Electrical Power
5.5 Mechanisms
5.6 Integrated Design: Example Bus

## 5.1 The Spacecraft Bus

A spacecraft consists of a **payload**, which performs the mission, and a **bus** (or **platform**), which supports the payload. The bus is divided into **subsystems** (Table 5.1). This chapter covers the "physical" subsystems — structures and mechanisms, thermal control, and electrical power. Chapter 6 covers attitude control, propulsion, data handling and flight software, and Chapter 7 covers communications.

**TABLE 5.1** The subsystems of a spacecraft bus and their main functions.

| Subsystem | Main functions | Key components |
|---|---|---|
| Structure and mechanisms | Carry loads, provide mounting, maintain alignment; release, deploy and point | Primary/secondary structure, adapters, hinges, release devices, drive mechanisms |
| Thermal control (TCS) | Keep every unit within its allowed temperature range | MLI blankets, coatings, radiators, heat pipes, heaters, louvres, thermostats |
| Electrical power (EPS) | Generate, store, regulate and distribute power | Solar arrays or RTGs, batteries, regulators, power distribution |
| Attitude determination and control (ADCS / GNC) | Know and control orientation (and position) | Sensors (star trackers, gyros, Sun sensors), actuators (wheels, thrusters, magnetorquers) |
| Propulsion | Change orbit, maintain orbit, control attitude, de-orbit | Thrusters, tanks, valves, feed system, electric thrusters |
| Command and data handling (C&DH / OBDH) | Process commands, run software, collect and store data | Onboard computer, mass memory, data buses, remote interface units |
| Telemetry, tracking and command (TT&C) and payload data | Link to ground for control and data | Transponders, antennas, high-rate transmitters, optical terminals |
| Flight software (FSW) | Autonomy, control laws, fault protection, payload operations | Operating system, application software, FDIR |

## 5.2 Structures

### 5.2.1 Functions and Requirements

The structure carries all the spacecraft's equipment through launch, holds instruments in precise alignment in orbit, and often contributes to thermal control and radiation shielding. The **primary structure** carries the main loads from the launch-vehicle interface; **secondary structures** (brackets, panels, booms) carry individual units; **tertiary structures** include cable trays and small supports.

Structural design is driven by:

- **Strength** — the structure must not yield or break under launch loads multiplied by **factors of safety**. For flight hardware that is tested, these are typically 1.1–1.25 against yield and 1.25–1.4 against ultimate. They are higher for untested designs or pressure vessels.
- **Stiffness** — the fundamental natural frequencies must exceed launcher requirements (e.g., > 20 Hz axial and > 10 Hz lateral for many launchers) to avoid dynamic coupling. Stiffness, not strength, usually sizes the structure.
- **Stability** — dimensional stability under temperature changes and moisture release, which is critical for telescopes and antennas (alignment tolerances of micrometres).
- **Mass** — every kilogram costs launch capacity; structures are typically 15–25 % of dry mass.
- **Interfaces** — standard launch-vehicle adapters (e.g., 937 mm, 1 194 mm, 1 666 mm clamp-band diameters) and CubeSat deployer rails.

### 5.2.2 Configurations

Figure 5.1 shows the three most common configurations. Large GEO platforms use a **central thrust cylinder** (often carbon-fibre) that carries the propellant tanks and transfers loads to the adapter, with honeycomb **shear panels** and equipment panels. Small and medium satellites use

**box structures** of bolted honeycomb panels. Stations and very large antennas use **trusses**. Mega-constellation satellites such as Starlink use flat-packed designs that stack dozens of satellites inside a fairing, with the stack itself carrying launch loads.

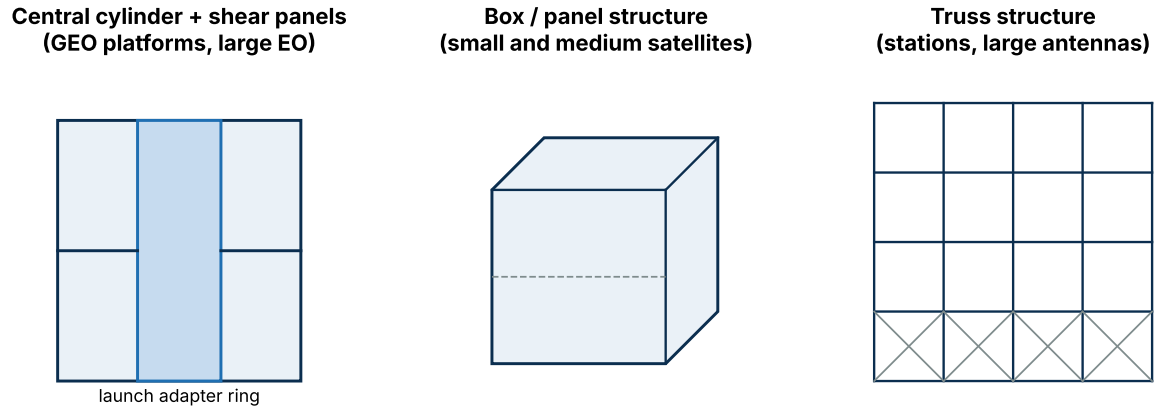



**FIGURE 5.1** Common spacecraft structural configurations: central cylinder with shear panels, box/panel structure, and truss.

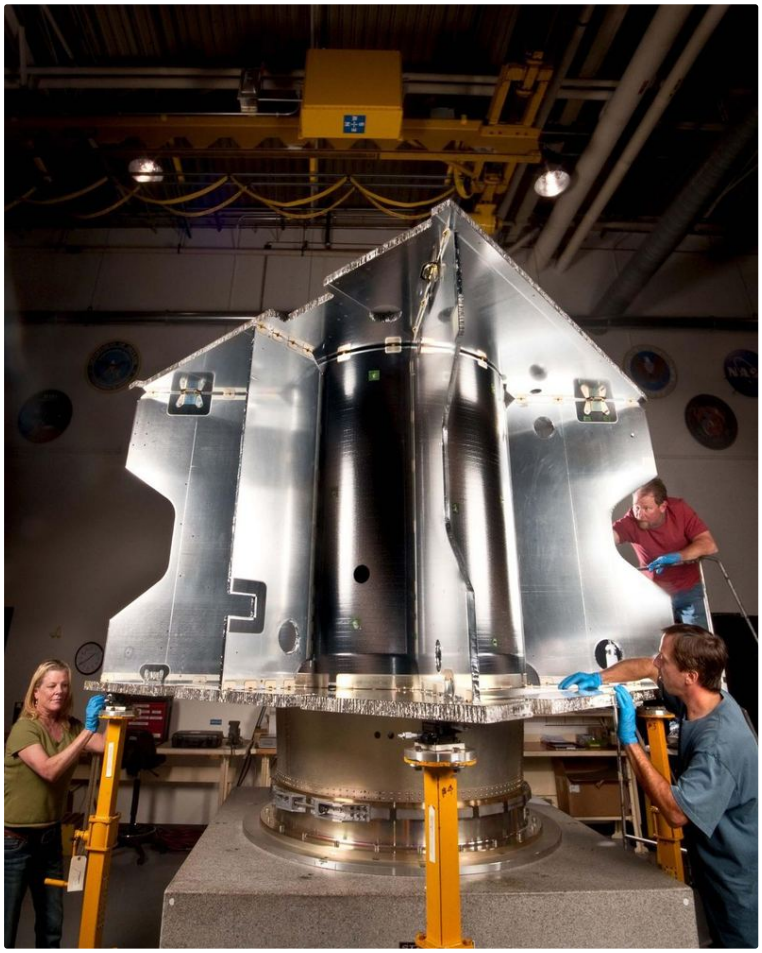
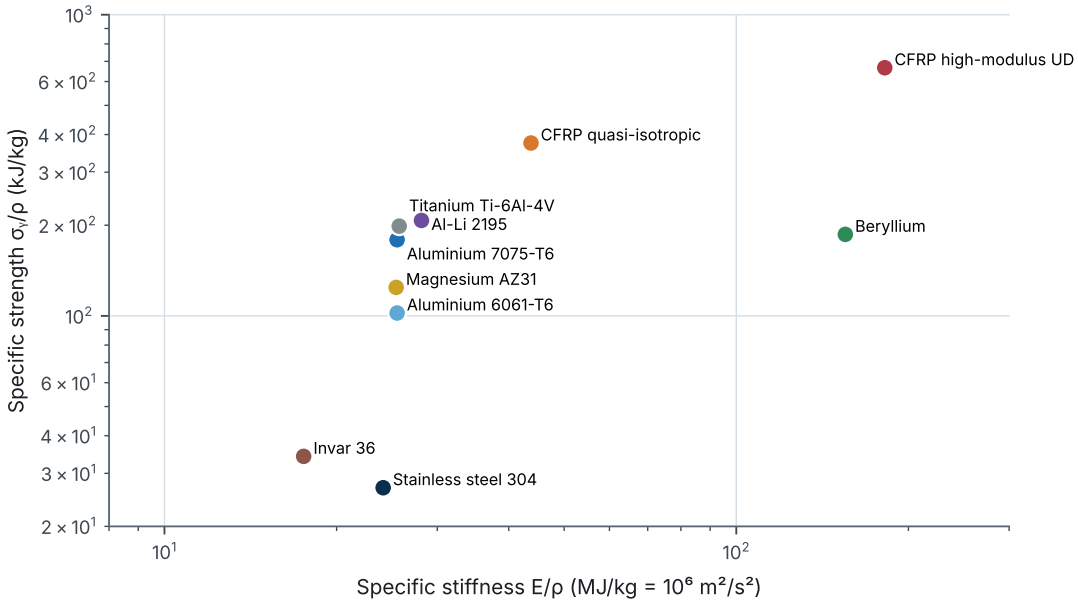


**FIGURE 5.2** The primary structure of NASA's MAVEN Mars orbiter during assembly at Lockheed Martin. The central core carries the propellant tanks and transmits launch loads to the launch-vehicle adapter. Image: NASA/GSFC.

### 5.2.3 Materials

Aluminium alloys remain the workhorse of spacecraft structures — cheap, easy to machine and well characterised. Titanium is used for highly loaded fittings and for its low thermal conductivity. Carbon-fibre reinforced polymers (CFRP) give the highest stiffness per unit mass

and near-zero thermal expansion, ideal for optical benches and antenna reflectors. Beryllium offers extreme stiffness for lightweight mirrors (JWST), and Invar and Super-Invar give dimensional stability (Figure 5.3). Additive manufacturing (3-D printing) of titanium, aluminium and polymer parts is now routine, allowing topology-optimised brackets that are 30–60 % lighter than machined parts (Chapter 52).




Representative room-temperature properties; CFRP values are laminate-dependent. Beryllium: high stiffness for optics (JWST mirrors) but toxic to machine.

**FIGURE 5.3** Specific stiffness and specific strength of common spacecraft structural materials. Composites and beryllium dominate where stiffness per kilogram matters most.

### 5.2.4 Sandwich Panels

The most common structural element is the **honeycomb sandwich panel** (Figure 5.4): two thin face sheets bonded to a lightweight honeycomb core. Like an I-beam, the faces carry bending stresses while the core keeps them apart and carries shear. Equipment is attached with **potted inserts**. Embedded heat pipes can be bonded inside panels to spread heat.

Bending stiffness grows with the square of the core thickness: doubling the core thickness makes the panel ~4× stiffer for a few % more mass.

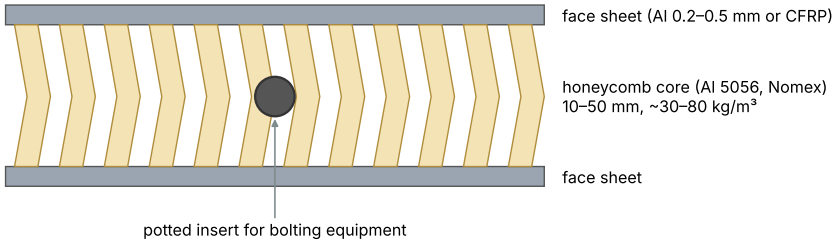



**FIGURE 5.4** Construction of a honeycomb sandwich panel. Separating the face sheets with a light core increases bending stiffness roughly with the square of the core thickness.

### 5.2.5 Structural Analysis

Structures are analysed with the **finite-element method (FEM)**, using codes such as NASTRAN, Abaqus or ANSYS. The main analyses are:

1. **Modal analysis** — natural frequencies and mode shapes.

2. **Coupled loads analysis (CLA)** — the launcher provider combines the spacecraft model with the launcher model and simulates flight events (lift-off, transonic buffet, engine cut-off, stage separation) to predict the real loads at each interface.
3. **Stress analysis** — margins of safety for every part:

$$MS = \frac{\text{allowable stress}}{\text{FS} \times \text{applied stress}} - 1 \geq 0$$

4. **Thermo-elastic analysis** — distortion caused by temperature changes, critical for pointing and optical alignment.
5. **Fracture control** — for pressure vessels and fracture-critical parts, demonstration that a crack of detectable size will not grow to failure within four lifetimes.

## 5.3 Thermal Control

### 5.3.1 Why Temperature Matters

Each component has an **operational temperature range** and a wider **survival range** (Table 5.2). The thermal control subsystem (TCS) must keep every unit within these limits during all mission phases, despite the large external variations described in Chapter 2. These phases are launch, eclipse, hot and cold attitudes, safe mode, and end of life.

**TABLE 5.2** Typical allowed temperature ranges of spacecraft components.

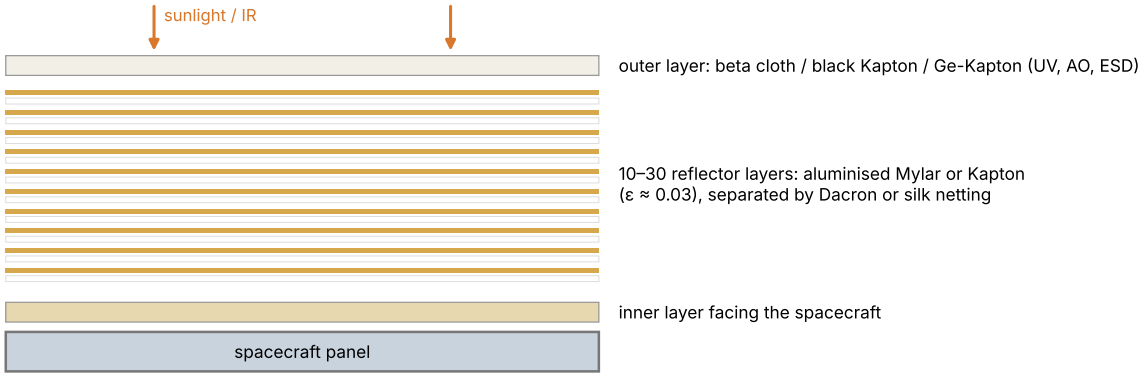
| Component | Operating range (°C) | Notes |
|---|---|---|
| Li-ion batteries | 0 to +30 (charging), −20 to +40 (discharge) | Narrow range; usually the TCS design driver |
| General electronics | −20 to +50 | Reliability falls ≈ 2× per 10 °C rise |
| Solar arrays | −150 to +110 | Efficiency falls as temperature rises |
| Hydrazine propellant and lines | +7 to +50 | Must not freeze (+2 °C) |
| Reaction wheels / bearings | −10 to +40 | Lubricant viscosity |
| Infrared detectors (HgCdTe) | −233 to −173 (40–100 K) | Cryocoolers or passive radiators |
| Optical structures | ± 0.1–1 °C stability | Thermal distortion budget |
| CCD/CMOS visible detectors | −40 to +20 | Cooler = lower dark current |

### 5.3.2 Passive Thermal Control

Most thermal control is **passive** — it uses no moving parts or power:

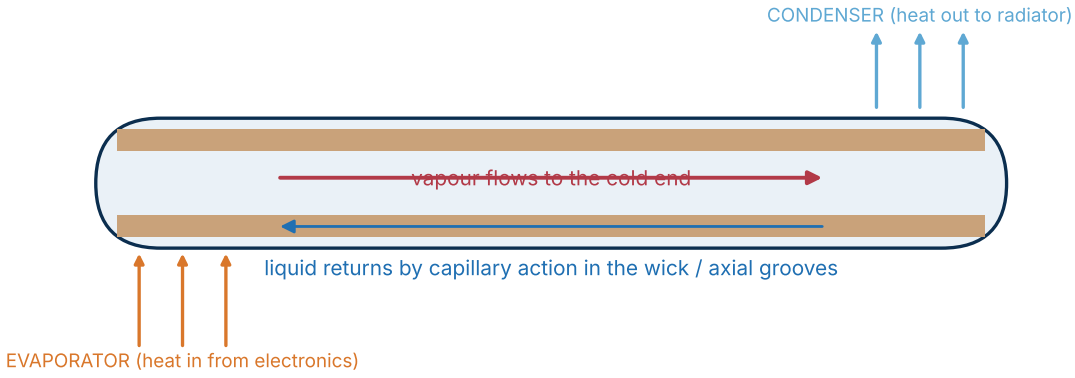
- **Surface finishes** set the balance between absorbed sunlight and emitted infrared (the α/ε ratio, Chapter 2). Radiators use white paints and optical solar reflectors (OSRs — silvered quartz tiles, α ≈ 0.08, ε ≈ 0.8). Low-emittance gold or aluminium surfaces are used where heat must be retained.
- **Multi-layer insulation (MLI)** blankets (Figure 5.5) are the "thermos flask" of spacecraft: many thin layers of aluminised polymer film separated by netting, each layer radiating only to its neighbours. MLI reduces radiative heat flow by a factor of 30–100 compared with a bare surface. The gold-coloured "foil" seen on most spacecraft is the outer Kapton layer of MLI.
- **Radiators** — surfaces with high ε and low α, oriented away from the Sun (and often Earth), that reject waste heat to space (Figure 5.7). The required area is $A = Q/(\varepsilon\sigma(T^4 - T_{sink}^4))$.

- **Conductive paths** — doublers, thermal straps, interface fillers and isolators control where heat flows.
- **Heat pipes** (Figure 5.6) — sealed tubes containing a working fluid (usually ammonia) that evaporates at the hot end and condenses at the cold end, with capillary wicks or grooves returning the liquid. They transport hundreds of watts across a spacecraft with only a few degrees of temperature drop. **Loop heat pipes** extend this over metres with flexible lines; **variable-conductance heat pipes** regulate temperature automatically using a non-condensable gas reservoir.




Effective emittance ε* ≈ 0.01–0.05 in practice (seams, vents and edges dominate); each layer radiates only to its neighbours.

**FIGURE 5.5** Construction of a multi-layer insulation blanket. Effective emittances of 0.01–0.05 are typical for real installations, limited by seams, penetrations and compression.




Working fluid: ammonia (−60 to +100 °C), also methanol, water, propylene; conductance ~1000× a solid aluminium bar of the same mass

**FIGURE 5.6** Principle of a heat pipe. Vapour carries heat from the evaporator to the condenser; capillary forces in the wick return the liquid without pumps.

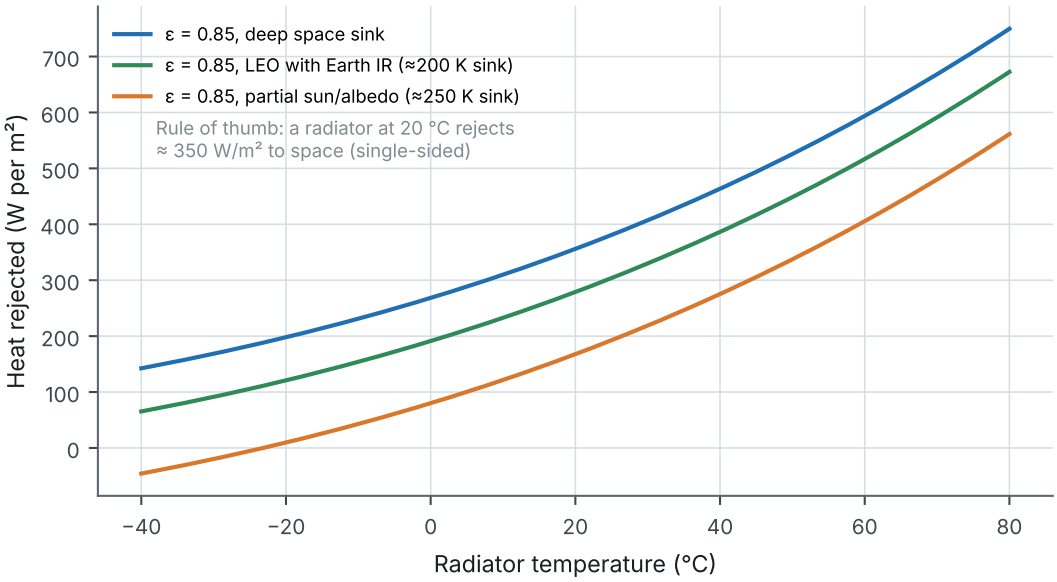



**FIGURE 5.7** Heat rejected per square metre by a radiator with ε = 0.85 as a function of its temperature, for three effective sink temperatures. Warmer radiators are smaller, which is why electronics are allowed to run warm when possible.

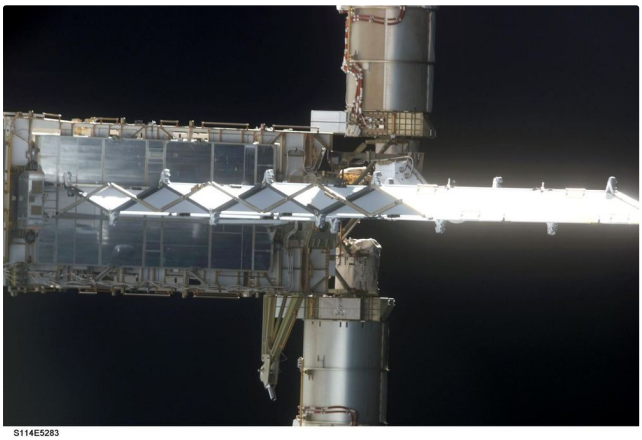


**FIGURE 5.8** Radiator panels of the ISS P6 truss photographed during STS-114 in 2005. The station's external active thermal control system pumps ammonia through radiators totalling more than 1 000 m². Image: NASA/JSC.

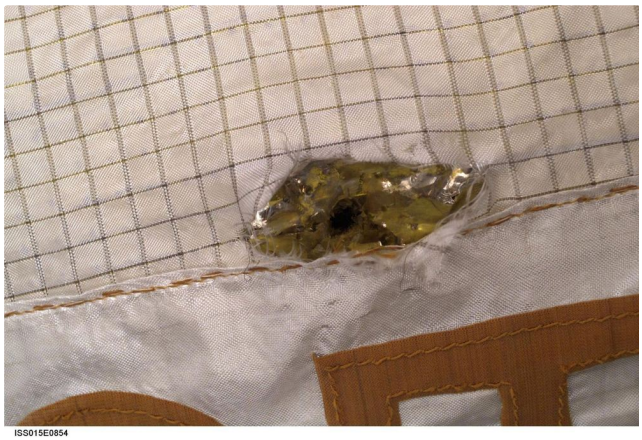


**FIGURE 5.9** Damage to a multi-layer insulation blanket on the exterior of the ISS Zarya module, photographed during an Expedition 15 spacewalk in 2007. MLI is lightweight but fragile; outer layers are chosen to resist ultraviolet light, atomic oxygen and handling. Image: NASA/JSC.

### 5.3.3 Active Thermal Control

Where passive methods are not enough, **active** methods add control:

- **Electric heaters** (Kapton film heaters with thermostats or software control) keep batteries, propellant lines and electronics above minimum temperatures during eclipses and cold attitudes. Heater power is a significant item in the power budget, particularly in safe mode.
- **Louvres** — blinds driven by bimetallic springs that open over radiators when the spacecraft is hot and close when cold (used on Rosetta and many science missions).
- **Mechanically pumped fluid loops** — pumps circulate a coolant (e.g., ammonia, Freon, water in pressurised modules) to carry large heat loads to radiators. The ISS rejects up to 70 kW; Mars rovers use fluid loops to carry heat from their RTG.

- **Cryocoolers** — Stirling, pulse-tube, Joule–Thomson or turbo-Brayton coolers that cool infrared detectors to 4–150 K. JWST's MIRI instrument uses a pulse-tube/Joule–Thomson cryocooler to reach 6.7 K.
- **Thermoelectric coolers** (Peltier devices) for small detectors.

### 5.3.4 Thermal Analysis

The TCS is designed with a **thermal mathematical model**: the spacecraft is divided into hundreds or thousands of "nodes" connected by conductive and radiative couplings. Tools such as ESATAN-TMS, Thermal Desktop/SINDA and Systema compute the radiative view factors and orbital heat inputs, then solve the energy balance

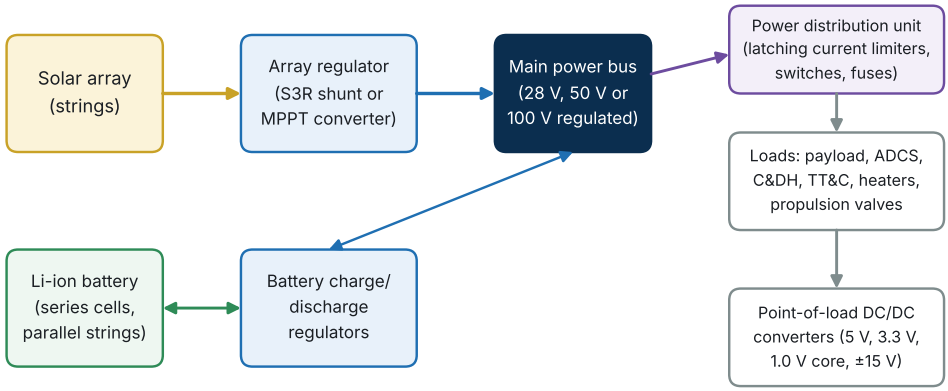
$$m_i c_i \frac{dT_i}{dt} = Q_{ext,i} + Q_{int,i} + \sum_j G_{ij}(T_j - T_i) + \sum_j \sigma R_{ij}(T_j^4 - T_i^4)$$

for each node $i$, over hot-case and cold-case scenarios (combinations of end-of-life coatings, maximum dissipation, solstice/equinox, attitudes). Predictions carry **uncertainty margins** (typically ±10–15 °C before thermal balance test correlation, ±5 °C after) so that the hardware is qualified beyond the predicted extremes.

## 5.4 Electrical Power

### 5.4.1 Power System Architecture

The electrical power subsystem (EPS) generates, stores, conditions and distributes power (Figure 5.10). A typical small satellite needs 50–500 W, and a large Earth-observation satellite 2–5 kW. A modern high-throughput GEO communications satellite needs 15–25 kW. The ISS needs about 120 kW (after the iROSA upgrade, up to 215 kW of array capacity).




Direct-energy-transfer (DET) systems pass array power straight to a regulated bus; MPPT systems extract maximum array power at the cost of converter losses.

**FIGURE 5.10** Architecture of a spacecraft electrical power system. Solar array power is regulated onto a main bus that charges the battery and supplies the loads through a protected distribution unit and point-of-load converters.

Two families of regulation exist:

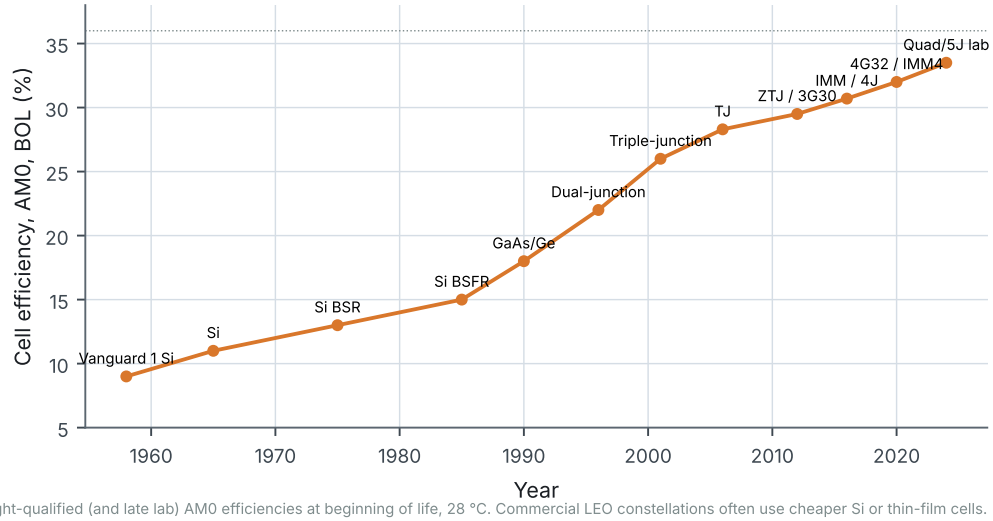
- **Direct energy transfer (DET)** — the array is connected directly (through shunt regulators) to the bus. Simple, efficient and common on GEO platforms (sequential switching shunt regulators, S3R) with regulated 50 V or 100 V buses.

- **Peak power tracking (MPPT)** — a DC/DC converter continuously adjusts the array operating point to its maximum power point (Figure 5.12). This extracts 10–25 % more power when the array is cold or illuminated at varying angles — common in LEO and on CubeSats.

Buses may be **regulated** (fixed voltage) or **unregulated** (voltage follows battery state of charge, typically 22–36 V). Larger spacecraft use higher voltages (100 V on GEO platforms, 120–160 V on the ISS) to reduce current and harness mass. The Lunar Gateway's Power and Propulsion Element was designed for 120 V to feed 50 kW-class Hall thrusters.

### 5.4.2 Solar Arrays

Photovoltaic **solar cells** convert sunlight into electricity. Early satellites used silicon cells of about 10 % efficiency. Modern space cells are **multi-junction** gallium-arsenide-based devices that stack three or four semiconductor layers, each absorbing a different part of the spectrum. They reach **30–33 %** efficiency at beginning of life under the space (AM0) spectrum (Figure 5.11). Cost-sensitive mega-constellations often use silicon or other lower-cost cells instead.




Approximate flight-qualified (and late lab) AM0 efficiencies at beginning of life, 28 °C. Commercial LEO constellations often use cheaper Si or thin-film cells.

**FIGURE 5.11** Evolution of the efficiency of flight-qualified space solar cells. Multi-junction III–V cells now exceed 30 %.

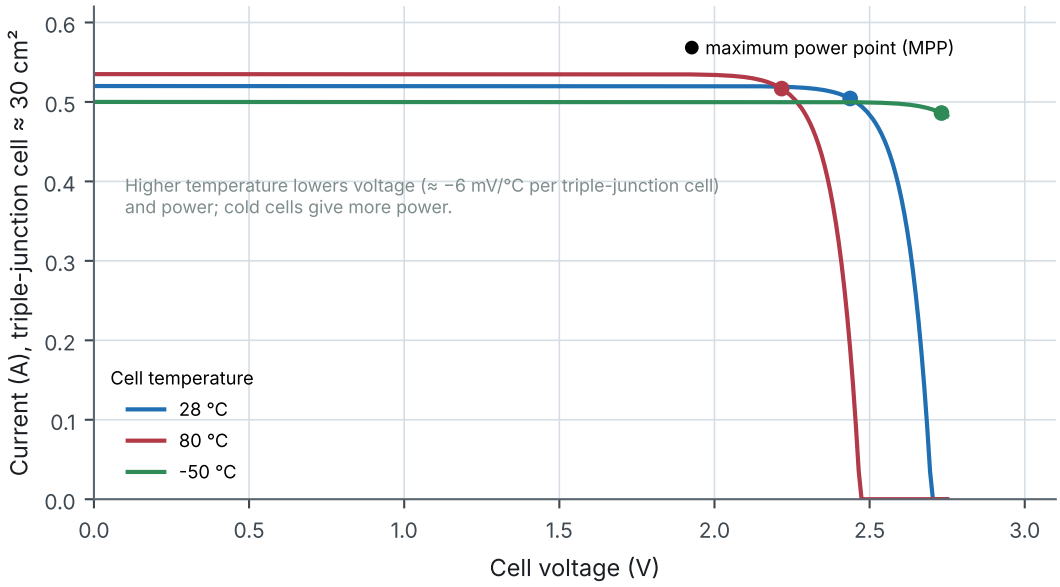



**FIGURE 5.12** Current–voltage characteristics of a triple-junction cell at three temperatures. The maximum power point lies at the "knee" of the curve; heating reduces the voltage and the power.

Array output at end of life (EOL) is estimated as

$$P_{EOL} = S\,\eta_{cell}\,A_{cells}\,\cos\theta \;\cdot\; F_{packing}\,F_{temp}\,F_{assembly}\,(1-d)^{L}$$

where $\theta$ is the Sun incidence angle, and $F$ terms account for packing factor (≈ 0.85–0.9), temperature and assembly losses. The term $d$ is the annual degradation from radiation, UV and micrometeoroids (typically 1–3 % per year in LEO; more in MEO). Arrays are sized for EOL conditions at the worst season (aphelion in July: 1 316 W/m$^2$).

**Array types:**

- **Body-mounted** cells on the spacecraft surface — simple, used on spinning and very small satellites.
- **Rigid deployable panels** — honeycomb panels hinged together, rotated to track the Sun by a **solar array drive assembly (SADA)**.
- **Flexible / roll-out arrays** — thin blankets on booms. The **Roll-Out Solar Array (ROSA)** and its ISS version **iROSA** (≈ 20 kW each, six installed 2021–2023, more delivered later) are lighter and more compact than rigid panels.
- **Concentrator arrays** — lenses or mirrors focus sunlight on fewer cells (e.g., Deep Space 1).

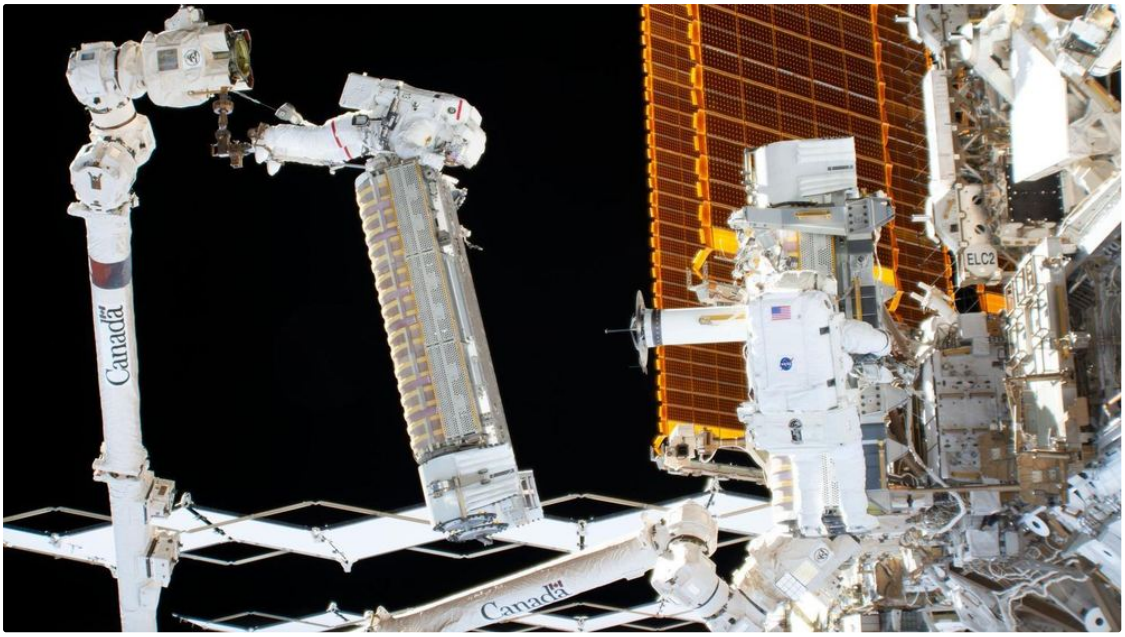



**FIGURE 5.13** Spacewalkers install an ISS Roll-Out Solar Array (iROSA) in December 2022. Each iROSA unrolls from a compact canister into a 6 × 19 m flexible wing over the station's original arrays. Image: NASA/JSC.

Solar power falls with the square of the distance from the Sun (Figure 5.14). NASA's Juno is the farthest solar-powered spacecraft (Jupiter, 5.2 AU), using three 9-metre wings that produce about 14 kW at Earth but only about 500 W at Jupiter. ESA's JUICE and NASA's Europa Clipper are also solar-powered at Jupiter, with arrays of 85 m² and about 100 m² respectively. Lucy uses two 7.3 m circular "UltraFlex" arrays to reach Jupiter's Trojan asteroids.

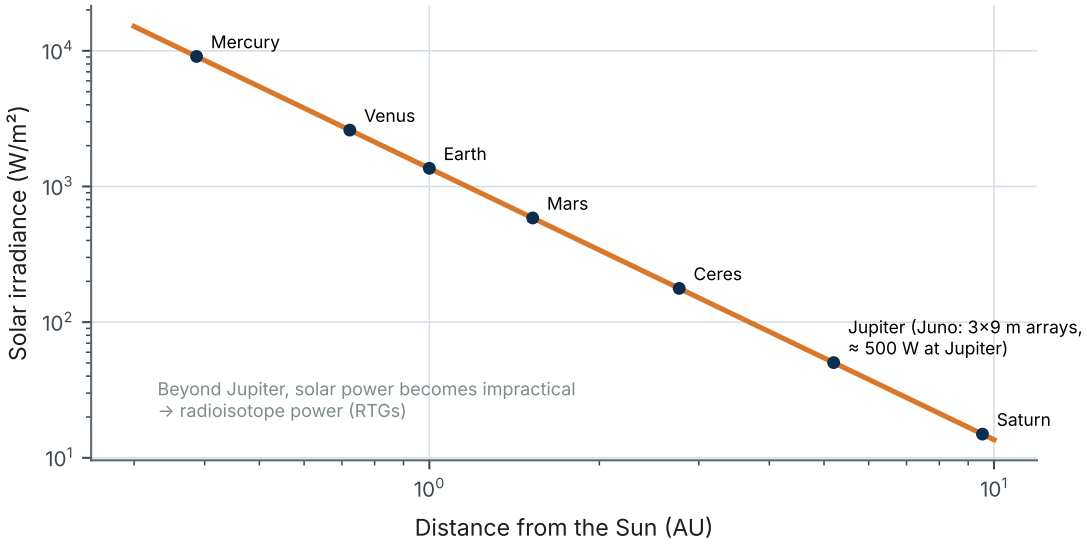



**FIGURE 5.14** Solar irradiance versus distance from the Sun. At Jupiter sunlight is 27 times weaker than at Earth; at Saturn about 90 times.

## 5.4.3 Batteries

Batteries supply power during eclipses, launch, peak loads and emergencies. **Lithium-ion** cells have replaced older nickel-cadmium (Ni-Cd) and nickel-hydrogen (Ni-$H_2$) technologies since the early 2000s because of their two to four times higher specific energy (Figure 5.15). Battery packs

are built from many small commercial-format cells (e.g., 18650 cylindrical cells) or larger prismatic cells, arranged in series strings (for voltage) and parallel strings (for capacity and redundancy).

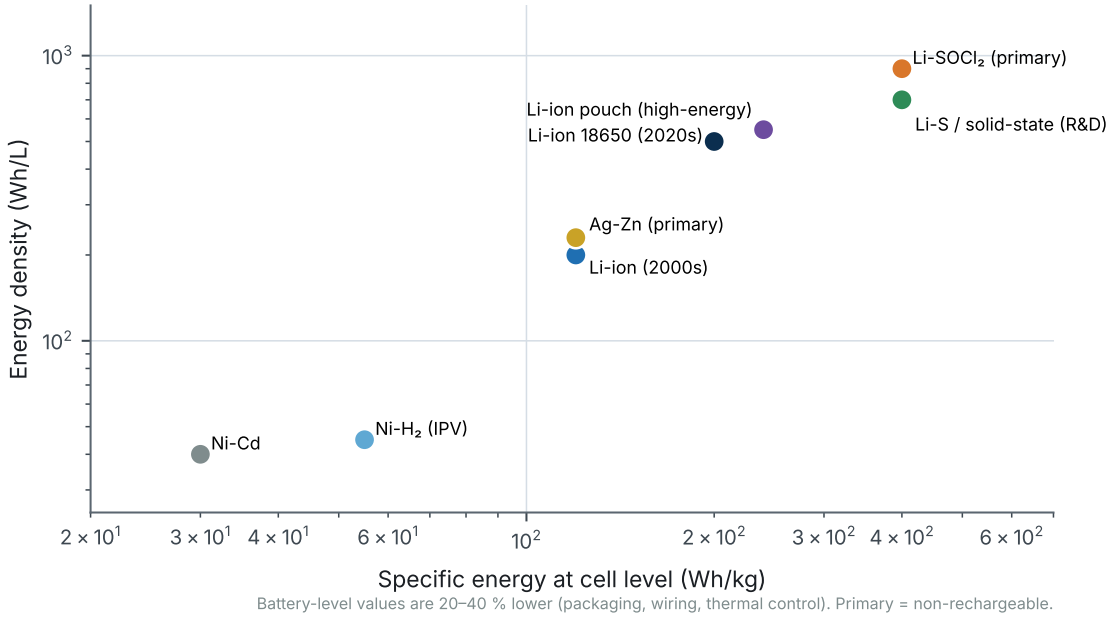



**FIGURE 5.15** Specific energy and energy density of space battery chemistries. Li-ion cells made rechargeable storage several times lighter than nickel-based batteries.

Battery life depends strongly on the **depth of discharge (DoD)** in each cycle (Figure 5.16). A LEO satellite cycles about 5 500 times per year, so its batteries are typically limited to 15–30 % DoD. A GEO satellite cycles only 90 times per year (during two eclipse seasons) and can use 60–80 % DoD. Battery capacity is sized as

$$C = \frac{P_{eclipse}\, t_{eclipse}}{\mathrm{DoD}\, \eta_{discharge}\, N_{strings,\,usable}/N_{strings}}$$

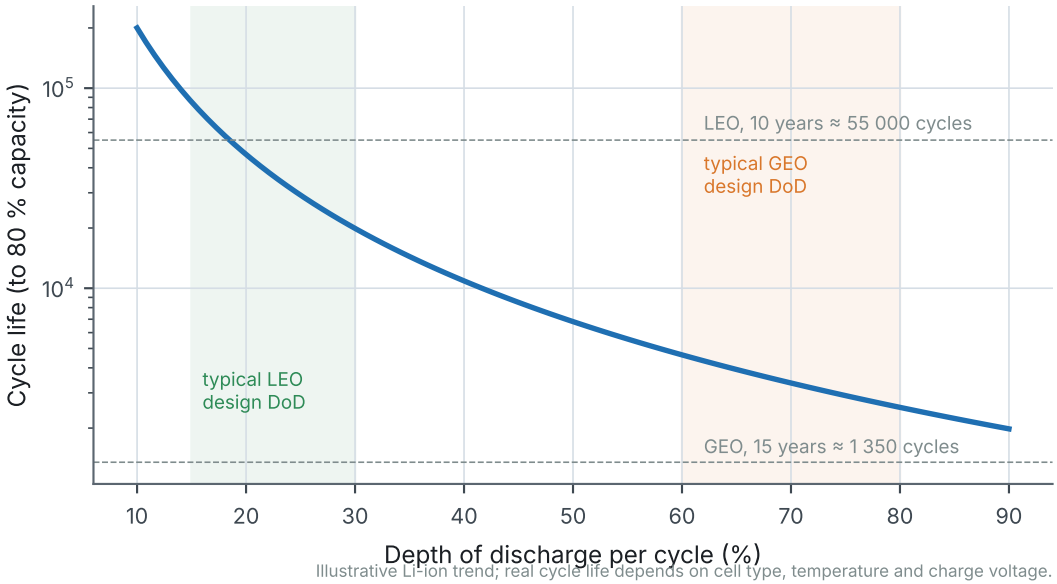



**FIGURE 5.16** Illustrative cycle life of Li-ion cells versus depth of discharge. Low-orbit satellites use shallow discharges because they cycle many thousands of times.

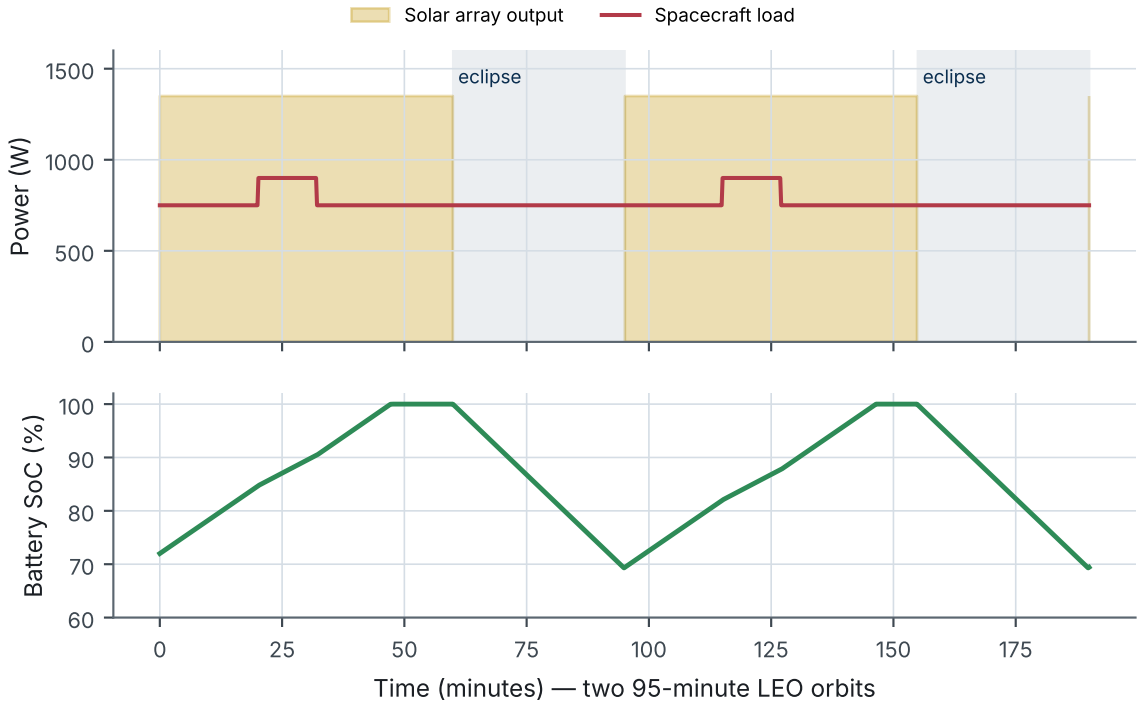



**FIGURE 5.17** Power generation, load and battery state of charge over two orbits for a LEO satellite. The battery charges in sunlight and supplies the load in eclipse; the array must be sized to recharge it within the sunlit part of each orbit.

**WORKED EXAMPLE — SIZING A LEO POWER SYSTEM**

A satellite in a 600 km orbit (period 96.7 min, maximum eclipse 35.5 min) needs 400 W continuously.

*Battery:* energy in eclipse = 400 W × 35.5/60 h = 237 Wh. With 25 % DoD and 95 % discharge efficiency, plus one failed string out of five: $C = 237/(0.25 \times 0.95 \times 0.8) = 1\,250$ Wh — about 7 kg of Li-ion cells at 180 Wh/kg cell level.

*Array:* in sunlight (61.2 min) the array must supply the load and recharge 237 Wh / (charge efficiency 0.9) = 263 Wh, so it needs $400 + 263/(61.2/60) = 658$ W at EOL. Assume 30 % cells, a packing factor of 0.88, 0.85 for temperature and other losses, 10 % degradation over 7 years and an average cosine loss of 0.9. Then each square metre gives $1\,361 \times 0.30 \times 0.88 \times 0.85 \times 0.9 \times 0.9 = 247$ W. So about **2.7 m²** of array is needed, plus margin.

### 5.4.4 Radioisotope Power

Beyond Jupiter, in permanent shadow, or on planetary surfaces with long nights and dust, spacecraft use **radioisotope power systems (RPS)**. A **radioisotope thermoelectric generator (RTG)** converts the heat of plutonium-238 decay into electricity using thermocouples (Figure 5.18). Pu-238 is used because it emits mainly alpha particles (easily shielded) and has an 87.7-year half-life. RTGs have powered Pioneer 10/11, Voyager 1/2 (still operating after 49 years, though at reduced power), Galileo, Cassini, New Horizons, Curiosity and Perseverance; NASA's Dragonfly rotorcraft (launch 2028) will use one on Titan.

**MMRTG — Curiosity, Perseverance, Dragonfly**

- 8 GPHS modules, 4.8 kg $PuO_2$
- ≈ 2 000 W thermal
- ≈ 110 W electric at launch
- conversion efficiency ≈ 6 %
- output falls ≈ 4 % per year (Pu-238 half-life 87.7 years plus thermocouple ageing)
- mass 45 kg, length 0.66 m

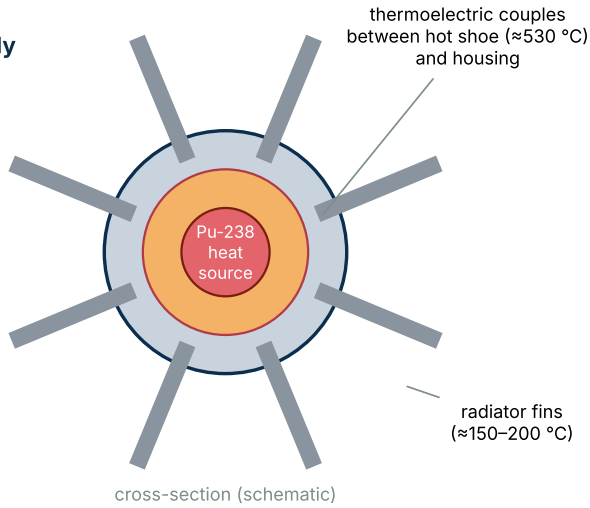



**FIGURE 5.18** Schematic of a Multi-Mission RTG. Heat from plutonium dioxide fuel flows through thermoelectric couples to radiator fins; about 6 % of it becomes electricity.

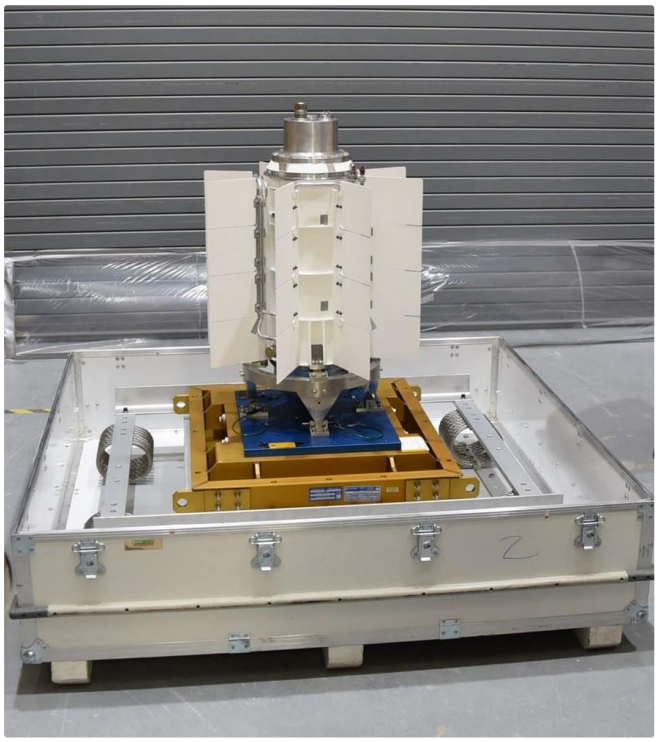


**FIGURE 5.19** The Multi-Mission Radioisotope Thermoelectric Generator (MMRTG) for the Mars 2020 Perseverance rover during testing at Idaho National Laboratory. Image: NASA/JPL (NASA/JPL-Caltech).

Smaller **radioisotope heater units (RHUs)**, each producing about 1 W of heat from 2.7 g of $PuO_2$, keep components warm on many missions (e.g., Mars rovers Sojourner–Opportunity, Chang'e 3/4 landers, which used Chinese-made RHUs). Russia and China use Pu-238 and Po-210 heaters. Future systems aim at higher efficiency: Stirling convertors (≈ 25–30 %), the next-generation RTG with skutterudite couples, and americium-241 RTGs under development in Europe. For high power, **fission reactors** are being developed for lunar bases and nuclear electric propulsion (Chapter 42). Examples are NASA's Kilopower/KRUSTY demonstration (2018) and the Fission Surface Power project. That project targets a lunar reactor of at least 40 kW (and, under 2025 direction, 100 kW-class) by about 2030.

### 5.4.5 Power Distribution and Protection

The **power distribution unit (PDU)** switches power to each load through **latching current limiters (LCLs)** or fuses, so that a short circuit in one unit cannot collapse the main bus. Harnesses are designed with derated wire gauges, twisted pairs and shielding for electromagnetic compatibility, and separated from RF and pyrotechnic lines. **Point-of-load converters** provide the low voltages (3.3 V, 1.8 V, 1.0 V core) that modern processors need. Fault protection includes undervoltage load shedding: when the battery falls below a threshold, non-essential loads are switched off and the spacecraft enters a power-safe mode.

## 5.5 Mechanisms

Mechanisms are among the most failure-prone items on spacecraft. They must work after launch vibration, in vacuum and over temperature extremes, often after years of storage. And many can be tested only a few times on the ground. The main families are summarised in Figure 5.20.

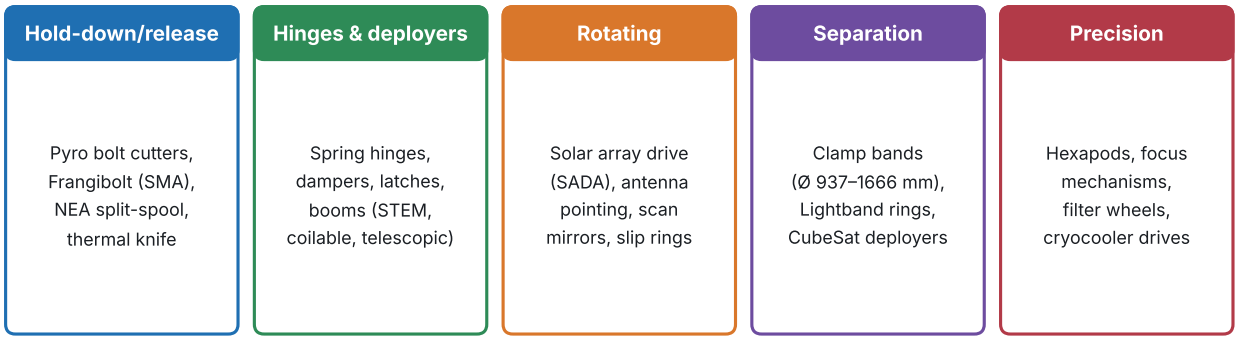



**FIGURE 5.20** Families of spacecraft mechanisms and typical examples.

- **Hold-down and release mechanisms (HDRMs)** lock deployables during launch and release them in orbit. Pyrotechnic devices (bolt cutters, separation nuts) are fast and reliable but produce shock; non-explosive actuators (NEA split-spool devices, Frangibolt shape-memory-alloy actuators, thermal knives that cut Dyneema cords on CubeSats) produce little shock.
- **Hinges and deployment devices** — spring-driven hinges with dampers or latches; booms such as coilable lattice masts, STEM/bi-STEM tubes and telescopic masts for magnetometers, antennas and solar sails.
- **Rotating mechanisms** — solar array drives, antenna pointing mechanisms, scan mirrors, reaction wheel bearings; they need space-qualified lubricants and transfer power and signals across rotating joints through slip rings or flexible cables.
- **Separation systems** — clamp bands, motorised Lightband rings and CubeSat deployers (e.g., the standard 3U/6U/12U deployers) separate the spacecraft from the launcher.
- **Precision mechanisms** — hexapods, focus and filter mechanisms, mirror actuators (JWST has 132 actuators to align its 18 primary mirror segments to tens of nanometres).

Design rules for reliable mechanisms include a **torque margin** of at least 2×. ECSS-E-ST-33-01C requires the motorisation torque to exceed twice the sum of resistive torques plus inertia terms, with higher factors early in design. Other rules are redundant springs or motor windings, and

avoidance of cold welding by dissimilar materials and coatings. The rules also call for life testing to several times the required number of cycles, and careful “test as you fly” deployment testing with gravity compensation.

**FAILURE LESSON — GALILEO’S HIGH-GAIN ANTENNA (1991)**

NASA’s Galileo Jupiter probe carried a 4.8 m umbrella-like high-gain antenna. When commanded to open in April 1991, three of its 18 ribs stuck. This was probably because the lubricant on the rib restraint pins had been lost during repeated truck transports, while the launch was delayed after the Challenger accident. Engineers could never free it. Through heroic software and ground-station upgrades — new data compression and coding, arraying of DSN antennas — the mission still returned about 70 % of its science objectives. It used the low-gain antenna at about 160 bit/s instead of 134 kbit/s. Lessons include lubricant life, the effect of ground handling, and the value of in-flight reprogrammability.

## 5.6 Integrated Design: Example Bus

To see how these subsystems combine, Table 5.3 lists the main characteristics of a representative 500 kg-class Earth-observation bus.

**TABLE 5.3** Representative characteristics of a 500 kg-class LEO Earth-observation satellite bus (illustrative).

| Parameter | Value |
|---|---|
| Launch mass / payload mass | 500 kg / 150 kg |
| Structure | Aluminium honeycomb box, CFRP payload bench, 1 194 mm interface ring |
| Orbit | 600 km sun-synchronous, 10:30 local time |
| Power | Two deployable GaAs triple-junction wings, 1.2 kW BOL / 0.95 kW EOL; 28 V unregulated bus; 60 Ah Li-ion battery |
| Thermal | MLI, white-paint and OSR radiators, heat pipes on payload panel, 120 W of thermostatic heaters |
| ADCS | 3-axis stabilised, 4 reaction wheels, 3 magnetorquers, 2 star trackers, fibre-optic gyro, GNSS receiver; pointing accuracy 0.02° |
| Propulsion | Hydrazine monopropellant, 4 × 1 N thrusters, 40 kg propellant (or Hall thruster, 15 kg xenon/krypton) |
| Data handling | Radiation-tolerant processor, 4 Tbit mass memory, SpaceWire payload interface |
| Communications | S-band TT&C (2 kbit/s up, 64 kbit/s down); X-band payload downlink 800 Mbit/s; optional optical terminal |
| Design life | 7 years |

### KEY TAKEAWAYS

- Structures are usually sized by stiffness, not strength; honeycomb sandwich panels, aluminium, titanium and CFRP dominate; FEM and coupled-loads analysis predict the real launch loads.
- Thermal control is mostly passive (coatings, MLI, radiators, heat pipes), supplemented by heaters, louvres, pumped loops and cryocoolers; models are correlated in thermal-balance tests.
- Multi-junction solar cells reach 30–33 % efficiency; arrays are sized for end-of-life, worst-case conditions; Li-ion batteries are sized by eclipse energy and allowable depth of discharge.
- Beyond Jupiter or on dark, dusty surfaces, RTGs (≈ 6 % efficient) provide power for decades; fission power is being developed for the Moon.
- Mechanisms are high-risk items: use torque margins, redundancy, proper lubrication and realistic deployment testing.

### REVIEW QUESTIONS

1. Why are spacecraft structures usually sized by stiffness rather than strength? Why do launch vehicles specify minimum fundamental frequencies?
2. Explain why honeycomb sandwich panels are so widely used in spacecraft structures.
3. A radiator with emittance 0.85 views deep space and must reject 500 W at 290 K. Estimate its area, ignoring absorbed environmental heat.
4. Describe four passive and two active thermal control techniques, and give a typical application of each.
5. A LEO spacecraft needs 800 W during a 35-minute eclipse. Size the battery capacity for a 30 % depth of discharge, ignoring losses.
6. Why are solar arrays sized for end-of-life conditions? List three factors that reduce array output during a mission.
7. Why do missions beyond Jupiter use radioisotope power, and what are its main drawbacks?
8. Why are mechanisms considered high-risk items, and how is that risk reduced?

CHAPTER

# 6 Spacecraft Subsystems II: Attitude Control, Propulsion, Data Handling and Flight Software

Image: NASA/GRC

*A spacecraft must know which way it is pointing and turn when needed, change its orbit, and think for itself between ground contacts. This chapter explains attitude determination and control - sensors, actuators, disturbance torques and control laws; spacecraft propulsion - cold gas, monopropellant, bipropellant and electric thrusters, feed systems and propellant management; command and data handling - onboard computers, memories, buses and radiation-tolerant electronics; and flight software, autonomy and fault protection.*

IN THIS CHAPTER

6.1 Attitude Determination and Control
6.2 Spacecraft Propulsion
6.3 Command and Data Handling
6.4 Flight Software

## 6.1 Attitude Determination and Control

### 6.1.1 What the ADCS Does

**Attitude** is the orientation of the spacecraft in space. Almost every mission needs attitude control: cameras must point at their targets, antennas at the Earth, solar arrays at the Sun, radiators away from it, and thrusters along the desired Δv direction. The **attitude determination**

**and control subsystem (ADCS)** is often combined with orbit control as **guidance, navigation and control (GNC)**. It measures the attitude with sensors and compares it with the desired attitude. It then applies torques with actuators in a feedback loop (Figure 6.1).

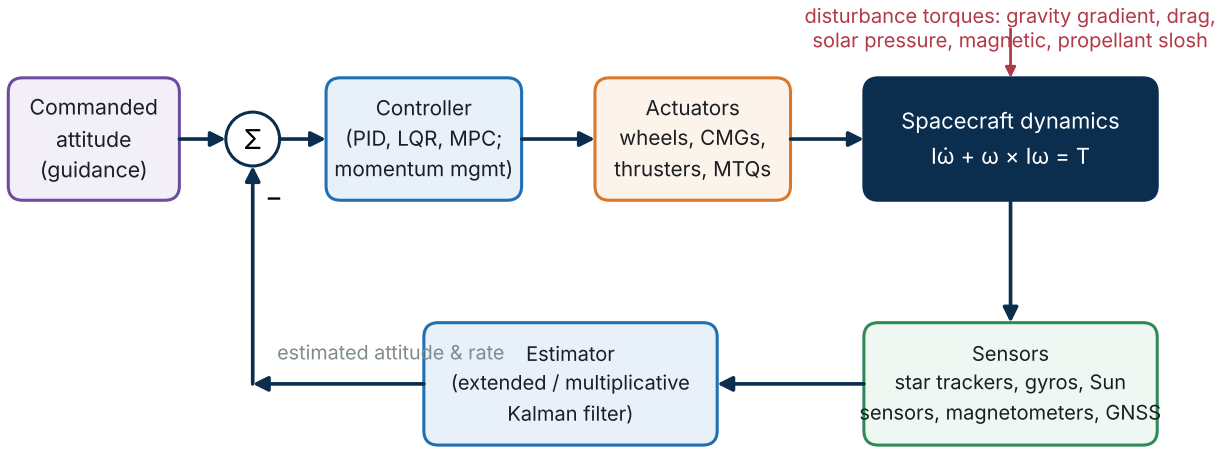



**FIGURE 6.1** The attitude control loop. Sensors feed an estimator (usually a Kalman filter); the controller compares estimated and commanded attitude and commands actuators to produce torques that act, together with disturbances, on the spacecraft dynamics.

Pointing requirements range enormously:

- **Coarse**: Sun-pointing for power, ±5–10° (safe modes, simple CubeSats).
- **Moderate**: Earth-pointing communications satellites, ±0.05–0.1°.
- **Fine**: Earth-observation imagers, 0.01–0.02° (tens of arcseconds) with stability of a few µrad/s during imaging.
- **Very fine**: space telescopes — Hubble holds its pointing to 0.007 arcsec (7 milliarcseconds) for hours; JWST about the same with a fine-steering mirror.

Requirements distinguish **accuracy** (how close to the target direction) and **knowledge** (how well the attitude is known after the fact). They also distinguish **stability** or **jitter** (how much it moves over a short time), and **agility** (how fast it can slew from target to target).

### 6.1.2 Stabilisation Methods

- **Spin stabilisation** — the spacecraft spins (typically 5–60 rpm) and behaves like a gyroscope, keeping its spin axis fixed in inertial space. Simple and robust; used by early satellites, many sounding rockets, the Juno spacecraft (spinning at 2–5 rpm) and some upper stages. **Dual-spin** satellites combine a spinning rotor with a de-spun platform for antennas.
- **Gravity-gradient stabilisation** — an elongated spacecraft tends to align its long axis with the local vertical because the near end feels slightly more gravity than the far end. Accuracy is a few degrees; used on early navigation satellites and some small satellites with deployable booms.
- **Passive magnetic stabilisation** — permanent magnets align a CubeSat with Earth's field, with hysteresis rods for damping.

- **Three-axis stabilisation** — active control of all three axes with wheels or thrusters. This is the standard for modern spacecraft.

### 6.1.3 Rigid-Body Dynamics

The rotational motion of a rigid spacecraft is governed by **Euler's equation**:

$$\mathbf{I}\dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times (\mathbf{I}\boldsymbol{\omega}) = \mathbf{T}_{control} + \mathbf{T}_{disturbance} - \dot{\mathbf{h}}_{wheels} - \boldsymbol{\omega} \times \mathbf{h}_{wheels}$$

where $\mathbf{I}$ is the inertia tensor, $\boldsymbol{\omega}$ the angular velocity, $\mathbf{T}$ the external torques and $\mathbf{h}$ the angular momentum stored in internal wheels. Attitude itself is represented with **direction-cosine matrices**, **Euler angles** (intuitive but singular at certain angles — "gimbal lock"), or **quaternions** (four parameters, no singularities, the standard in flight software).

### 6.1.4 Disturbance Torques

Small external torques continually push the spacecraft away from its desired attitude (Figure 6.2):

- **Gravity gradient** — $T = \frac{3\mu}{2r^3}\,|I_z - I_y|\sin 2\theta$, important for large, elongated spacecraft in LEO.
- **Aerodynamic** — drag acting at a centre of pressure offset from the centre of mass; dominant below about 500 km.
- **Solar radiation pressure** — the same offset effect for sunlight; dominant in high orbits and deep space.
- **Magnetic** — interaction of residual magnetic dipoles (from currents and magnetised materials) with Earth's field.
- **Internal** — propellant slosh, moving mechanisms, flexible appendages, outgassing and thruster misalignment.

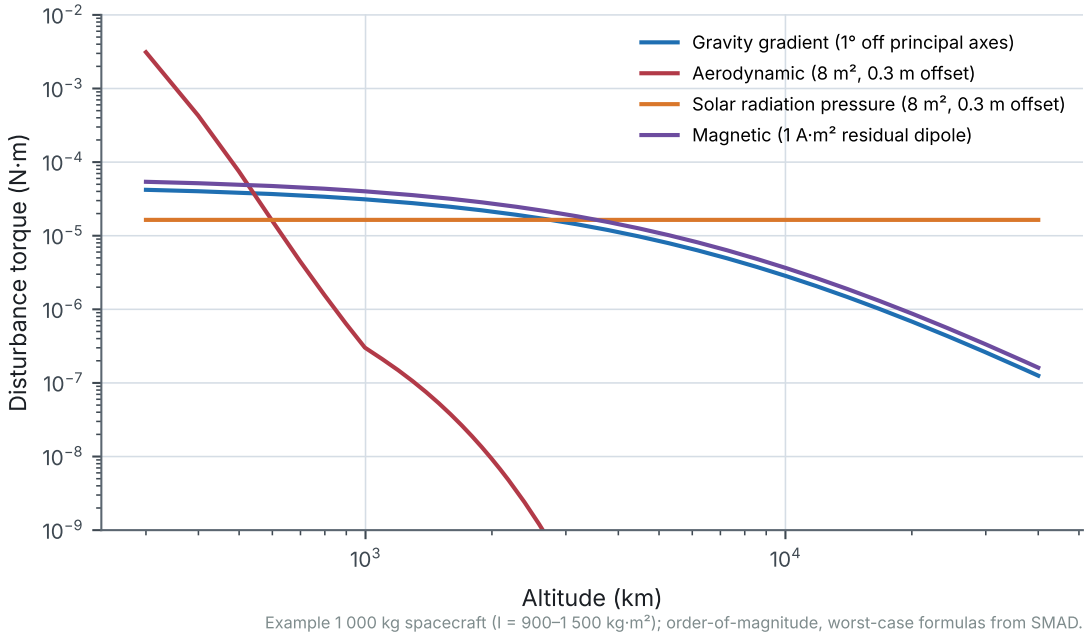



**FIGURE 6.2** Order-of-magnitude disturbance torques on a 1 000 kg spacecraft as a function of altitude. Aerodynamic torque dominates in low orbits; gravity-gradient and magnetic torques in mid-LEO; solar pressure in high orbits.

Because these torques act continuously in one direction, momentum accumulates in the reaction wheels until they approach their maximum speed; it must periodically be "dumped" using magnetorquers or thrusters (**momentum management**).

### 6.1.5 Attitude Sensors

**TABLE 6.1** Common attitude sensors.

| Sensor | Measures | Typical accuracy | Notes |
|---|---|---|---|
| Coarse Sun sensor | Sun direction (photodiode currents) | 1–5° | Cheap, robust; used for safe mode |
| Fine (digital) Sun sensor | Sun direction (slit or pinhole on detector array) | 0.01–0.1° | Field of view up to ±64° |
| Earth (horizon) sensor | Direction of Earth's IR limb (14–16 µm $CO_2$ band) | 0.03–0.3° | Classic for GEO and LEO Earth-pointing |
| Magnetometer | Local magnetic field vector | 0.5–5° (after comparison with field model) | Also used for magnetorquer control |
| Star tracker | Full 3-axis attitude by matching stars to a catalogue | 1–10 arcsec (cross-boresight); ≈ 5–40 arcsec (roll) | The precision sensor of modern spacecraft; blinded by Sun/Moon/Earth |
| Gyroscope / IMU | Angular rate (integrates to attitude change) | Drift from 0.001°/h (HRG, FOG) to 10°/h (MEMS) | Bridges star-tracker outages; ring-laser, fibre-optic, hemispherical resonator, MEMS |
| GNSS receiver | Position, velocity, time (and attitude with several antennas) | Position 1–10 m real-time | Standard in LEO; works up to GEO and beyond with weak-signal receivers |
| Fine guidance sensor | Offset of a guide star in the science telescope focal plane | Milliarcseconds | Hubble FGS, JWST FGS, Roman |

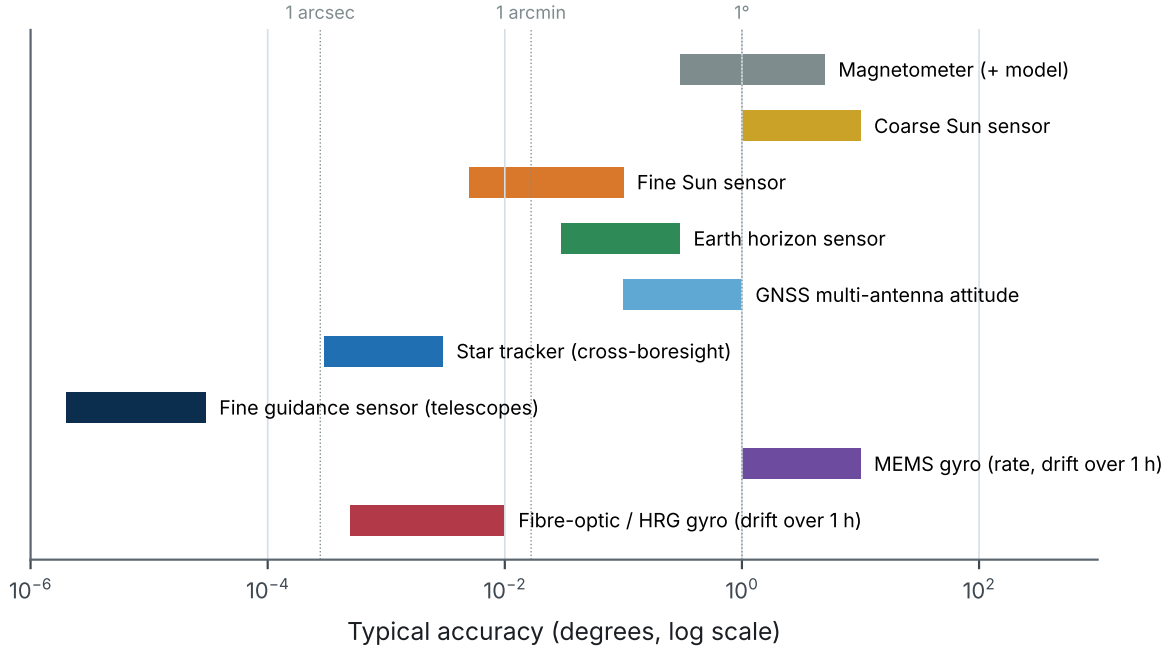



**FIGURE 6.3** Typical accuracy ranges of attitude sensors on a logarithmic scale. Star trackers improved attitude knowledge by two to three orders of magnitude over Sun and Earth sensors.

A **star tracker** is a small digital camera with a wide-angle lens (field of view 10–25°) and an onboard catalogue of thousands of stars. A "lost-in-space" algorithm identifies star patterns (triangles or pyramids of stars) without prior knowledge; once locked, it tracks the stars at 1–10 Hz. Modern trackers for small satellites weigh under 300 g; high-end units (e.g., Leonardo's AA-STR, Sodern's Hydra, Ball's CT-2020) serve large missions. Measurements from all sensors are fused by an **extended Kalman filter** (usually a multiplicative quaternion formulation) that also estimates gyro biases.

**FIGURE 6.4** Engineers install the star-tracker camera heads on NASA's Europa Clipper before launch (image released by JPL in 2025, when the trackers returned their first in-flight star-field images). Star trackers are the primary attitude sensors of almost every modern three-axis-stabilised spacecraft. Image: NASA/JPL (NASA/JPL-Caltech).

### 6.1.6 Attitude Actuators

- **Reaction wheels** — flywheels driven by brushless motors; accelerating a wheel in one direction rotates the spacecraft in the other. They give smooth, precise, propellant-free control. Typical wheels provide 0.01–0.3 N·m of torque and store 0.01 N·m·s (CubeSat wheels) to 200 N·m·s. Four wheels in a skewed pyramid (Figure 6.5) provide full control with one redundant wheel.
- **Momentum wheels** — biased wheels spinning at constant high speed for gyroscopic stiffness about one axis (classic GEO "momentum-bias" designs).
- **Control moment gyroscopes (CMGs)** — constantly spinning wheels whose spin axis is tilted by a gimbal motor. The gyroscopic reaction produces torques hundreds of times larger than a reaction wheel of similar mass. This enables very agile spacecraft (WorldView, Pléiades Neo) and the attitude control of the ISS (four 4 760 N·m·s double-gimbal CMGs).
- **Magnetorquers** — electromagnetic coils or rods that interact with Earth's field ($\mathbf{T} = \mathbf{m} \times \mathbf{B}$); used for momentum dumping and for CubeSat attitude control. They can only produce torque perpendicular to the local field.
- **Thrusters** — produce large torques for slews, momentum dumping and control during orbit manoeuvres; necessary beyond Earth's magnetic field.

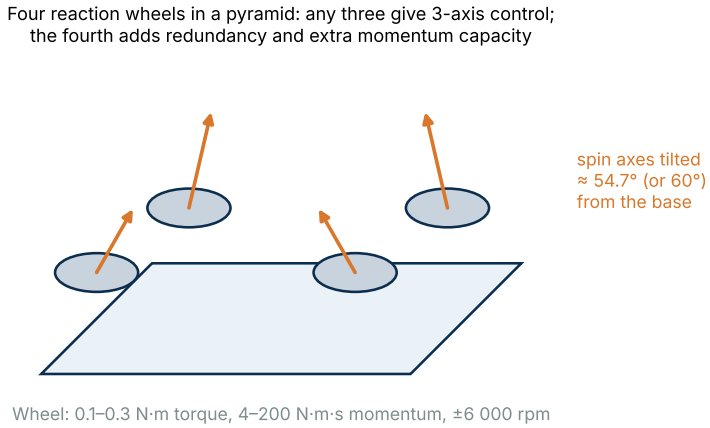



**FIGURE 6.5** Four reaction wheels in a pyramid arrangement. The configuration tolerates the failure of any one wheel.

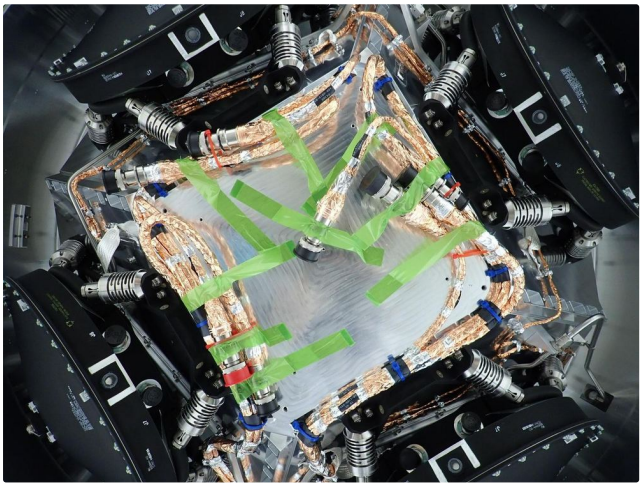


**FIGURE 6.6** The four reaction wheels of Europa Clipper installed on the spacecraft's propulsion module in 2022. Image: NASA/JPL (NASA/JPL-Caltech).

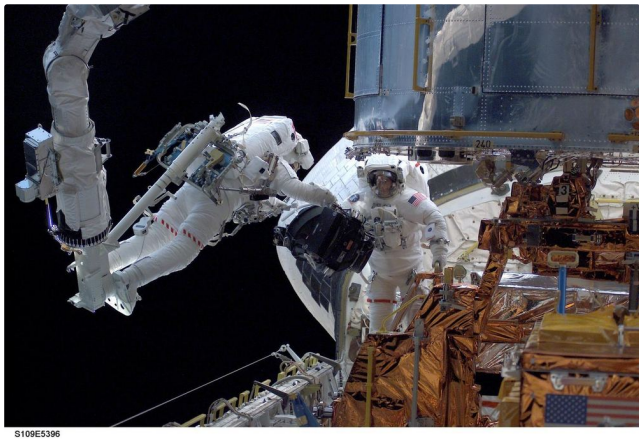


**FIGURE 6.7** Astronauts Mike Massimino and James Newman replace a reaction wheel assembly on the Hubble Space Telescope during Servicing Mission 3B in March 2002. Image: NASA/JSC.

**FAILURE LESSON — WHEN REACTION WHEELS FAIL**

Reaction wheels are a classic life-limiting item. NASA's Kepler planet-hunting telescope lost its second of four wheels in 2013 and could no longer point precisely. Engineers invented the "K2" mission, balancing the spacecraft against solar radiation pressure. This mission continued to discover exoplanets for five more years. The Dawn asteroid mission lost two wheels and completed its mission using hydrazine thrusters and careful propellant management. Hubble has had several gyro failures and since 2024 operates in a one-gyro mode to preserve its remaining gyros. Lessons include bearing lubrication, cage instability, conservative speed limits, and designing control modes that work with fewer actuators.

### 6.1.7 Control Laws and Modes

Most attitude controllers are variants of **proportional–integral–derivative (PID)** control around each axis. They are designed with classical frequency-domain margins (typically ≥ 6 dB gain margin and ≥ 30° phase margin). They also use filters to avoid exciting flexible modes of solar

arrays and antennas. More advanced methods include **linear-quadratic regulators (LQR)**, **model predictive control (MPC)** for constrained slews, and **eigen-axis slews** that rotate about a single axis for time-optimal reorientation.

A spacecraft has several **ADCS modes**: rate damping after separation (often magnetic "B-dot" control on small satellites), Sun acquisition, safe mode, nominal pointing (nadir, target or inertial), slew, orbit-control (thrusting) mode, and momentum-dumping mode. Mode transitions are managed by the flight software and by the fault-protection system.

## 6.2 Spacecraft Propulsion

### 6.2.1 Why Spacecraft Need Propulsion

On-board propulsion provides:

- **Orbit insertion and transfer** — e.g., GTO→GEO, lunar or planetary orbit insertion.
- **Orbit maintenance** — compensating drag in LEO, station-keeping in GEO, constellation phasing.
- **Attitude control and momentum dumping**.
- **Rendezvous, proximity operations and docking**.
- **Collision avoidance**.
- **End-of-life disposal** — de-orbiting within regulatory limits or raising to a graveyard orbit.

Every propulsion system is characterised by two numbers: **thrust** $F$ (newtons) and **specific impulse** $I_{sp}$ (seconds), which measures propellant efficiency:

$$I_{sp} = \frac{F}{\dot{m}\, g_0} = \frac{v_e}{g_0}$$

where $v_e$ is the effective exhaust velocity. Through the rocket equation, $I_{sp}$ determines how much propellant a given Δv requires (Figure 6.8). The landscape of thruster technologies spans eight orders of magnitude in thrust and two in specific impulse (Figure 6.9).

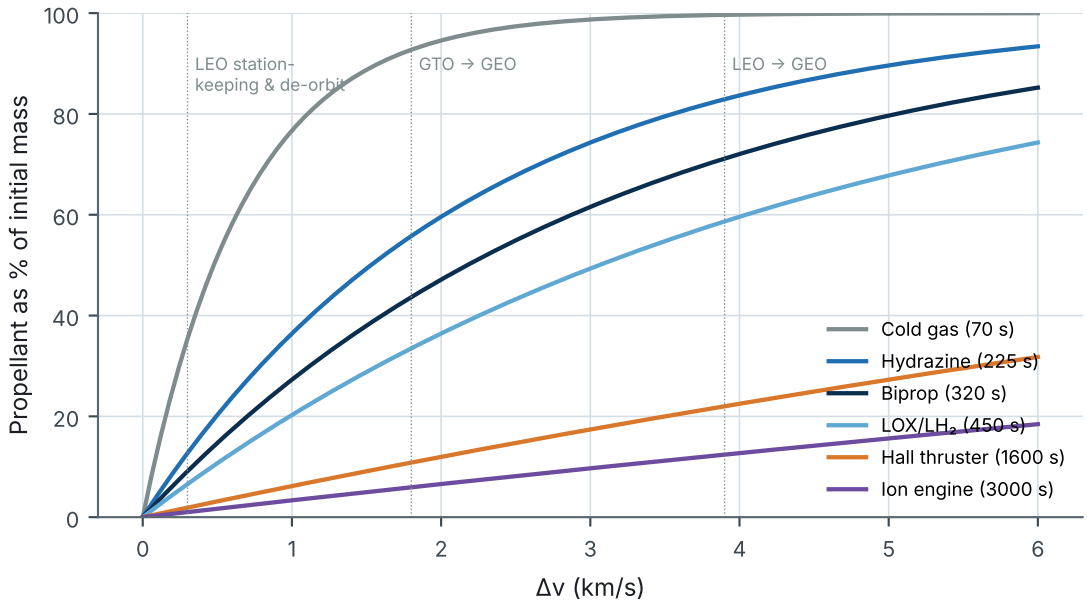



**FIGURE 6.8** Propellant required as a fraction of initial mass versus Δv for different propulsion technologies. High-Isp electric propulsion makes large Δv missions possible with modest propellant mass, at the cost of very long thrusting times.

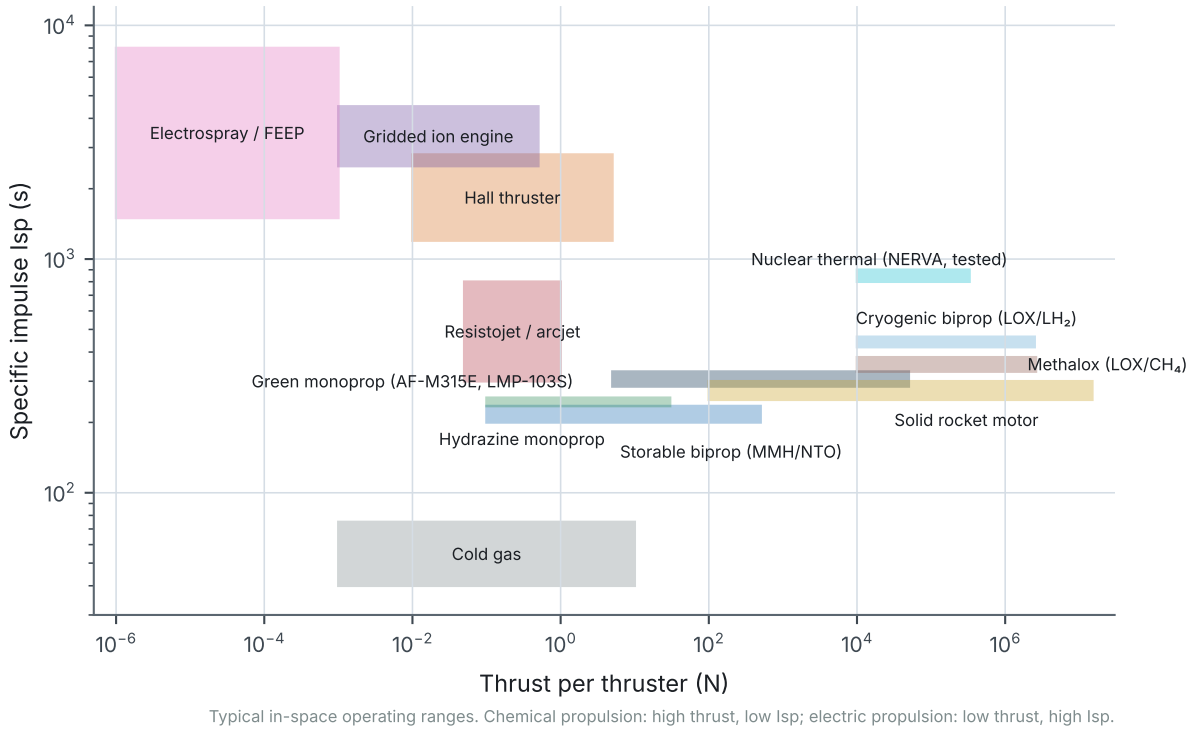



**FIGURE 6.9** Operating ranges of in-space propulsion technologies in thrust and specific impulse.

## 6.2.2 Chemical Propulsion

**Cold gas** thrusters simply release compressed gas (nitrogen, xenon, butane) through a nozzle. Isp is only 50–75 s, but they are simple, clean and produce precise small impulses — used on CubeSats and for fine control (Gaia used cold gas micro-thrusters).

**Monopropellant hydrazine** ($N_2H_4$) has been the standard for satellite attitude and orbit control since the 1970s. Hydrazine decomposes exothermically when it passes over a heated iridium-on-alumina catalyst bed, producing hot ammonia, nitrogen and hydrogen (Figure 6.10). Thrusters range from 0.5 N to several hundred newtons (Isp 220–235 s). Hydrazine is toxic and

carcinogenic, so handling requires protective suits (SCAPE suits) and is expensive. **"Green" monopropellants** — AF-M315E (ASCENT, flown on NASA's Green Propellant Infusion Mission, 2019) and LMP-103S (ammonium dinitramide-based, flown on PRISMA and Skybox/SkySat) — offer higher density-Isp and much safer handling.

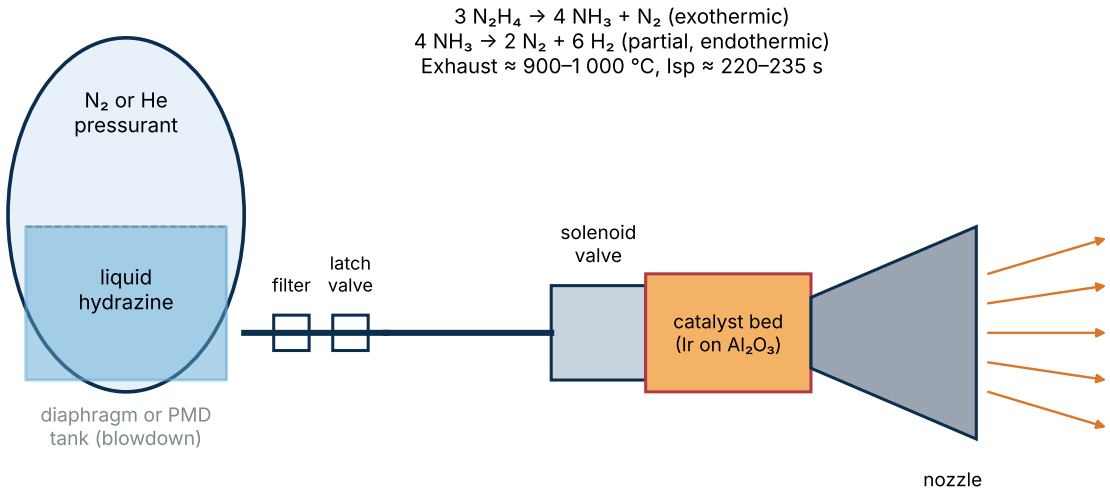



**FIGURE 6.10** A hydrazine monopropellant thruster system. Propellant flows from a blowdown tank through filters and valves to a catalyst bed where it decomposes; the hot gases expand through a nozzle.

**Bipropellant** systems burn a fuel (monomethylhydrazine, MMH, or hydrazine) with an oxidiser (nitrogen tetroxide, NTO, or mixed oxides of nitrogen, MON) that ignite spontaneously on contact (**hypergolic**). Isp is 300–330 s. A typical GEO satellite has a 400–500 N **liquid apogee engine** for orbit raising and 10–22 N thrusters for attitude and station-keeping, fed from regulated tanks (Figure 6.11). Planetary orbiters use biprop engines for orbit insertion (e.g., Juno's 645 N Leros-1b). **Dual-mode** systems use hydrazine both as monopropellant for small thrusters and as fuel for a biprop engine.

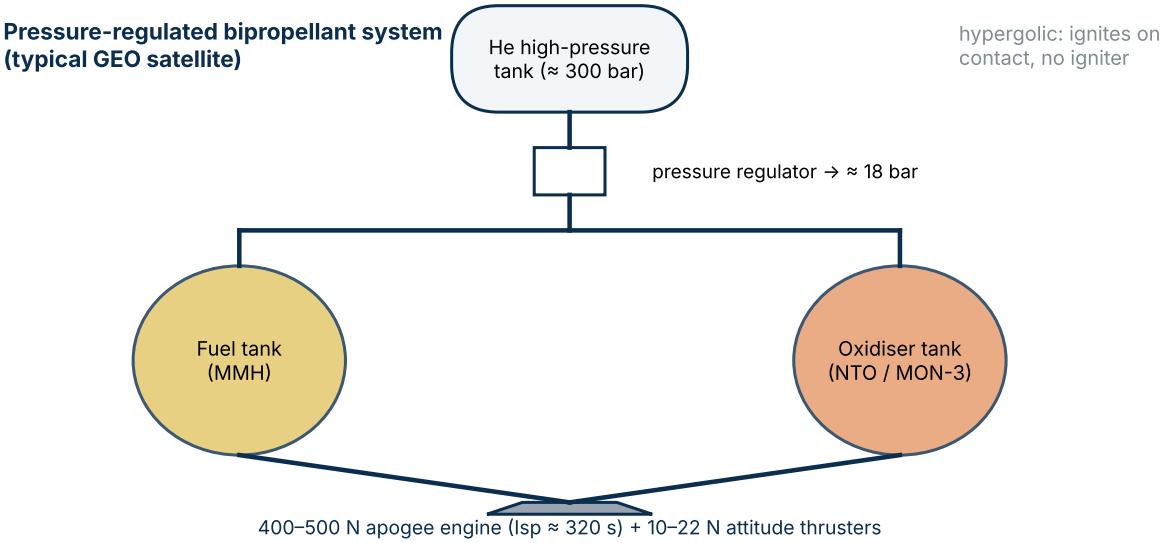



**FIGURE 6.11** A pressure-regulated bipropellant system. Helium from a high-pressure tank is regulated to constant pressure and pushes fuel and oxidiser to the engines.

**Solid rocket motors** provide a single, large impulse — historically used as apogee kick motors and for planetary orbit insertion, and still common in upper stages.

### 6.2.3 Feed Systems and Propellant Management

Propellant can be fed by **blowdown** or **pressure regulation**. With blowdown, the pressurant is in the same tank, and pressure and thrust fall as the tank empties; this is simple and typical of LEO satellites. Pressure regulation gives constant pressure from a separate high-pressure tank — needed for biprop engines with tight mixture-ratio control. Large launch-vehicle stages use pumps (Chapter 35).

In microgravity, liquid does not settle at the tank outlet. **Propellant management devices (PMDs)** — surface-tension vanes, sponges, galleries and screens — or **elastomeric diaphragms and bladders** keep gas-free liquid at the outlet. Alternatively, a small "ullage" burn settles the propellant before a main burn. **Propellant gauging** is difficult in weightlessness and uses bookkeeping (counting thruster pulses), pressure–volume–temperature methods, or the thermal response of the tank to heating.

### 6.2.4 Electric Propulsion

**Electric propulsion (EP)** uses electrical energy to accelerate propellant to much higher exhaust speeds than chemical reactions allow — 15–50 km/s instead of 3–4.5 km/s. Thrust is tiny (millinewtons to a few newtons) because it is limited by available power:

$$F = \frac{2\,\eta\,P}{v_e} = \frac{2\eta P}{g_0 I_{sp}}$$

so a 5 kW Hall thruster at 60 % efficiency and 1 800 s produces about 0.34 N. The main types (Figure 6.12) are:

- **Electrothermal** — resistojets and arcjets heat propellant electrically (Isp 300–800 s); used for GEO station-keeping in the 1990s.
- **Gridded ion engines** — xenon is ionised in a discharge chamber and accelerated through electrically charged grids; a neutraliser emits electrons to keep the spacecraft neutral. Isp 2 500–4 500 s. Flown on Deep Space 1, Hayabusa 1 and 2, Dawn (three NSTAR engines, total Δv ≈ 11 km/s — a record), BepiColombo (four T6 engines), DART (NEXT-C) and Boeing 702 GEO satellites (XIPS).
- **Hall-effect thrusters** — a radial magnetic field traps electrons in an annular channel, creating an axial electric field that accelerates ions. Isp 1 200–2 800 s, higher thrust per kW than ion engines. They are by far the most widely flown EP technology. Starlink uses krypton and (V2 Mini onwards) argon Hall thrusters on every satellite. The 6 kW-class units power Psyche (SPT-140) and all-electric GEO satellites, and 12 kW AEPS thrusters were developed for the Gateway Power and Propulsion Element.
- **Electrospray and field-emission (FEEP)** thrusters accelerate ions or charged droplets from liquid metals or ionic liquids, giving micronewton thrust for precise control (LISA Pathfinder) and compact CubeSat propulsion.
- **Pulsed plasma thrusters** ablate a Teflon block with an electric arc — simple, used on small satellites.

- **Advanced concepts** (VASIMR, magnetoplasmadynamic thrusters, nuclear electric propulsion) are covered in Chapter 42.

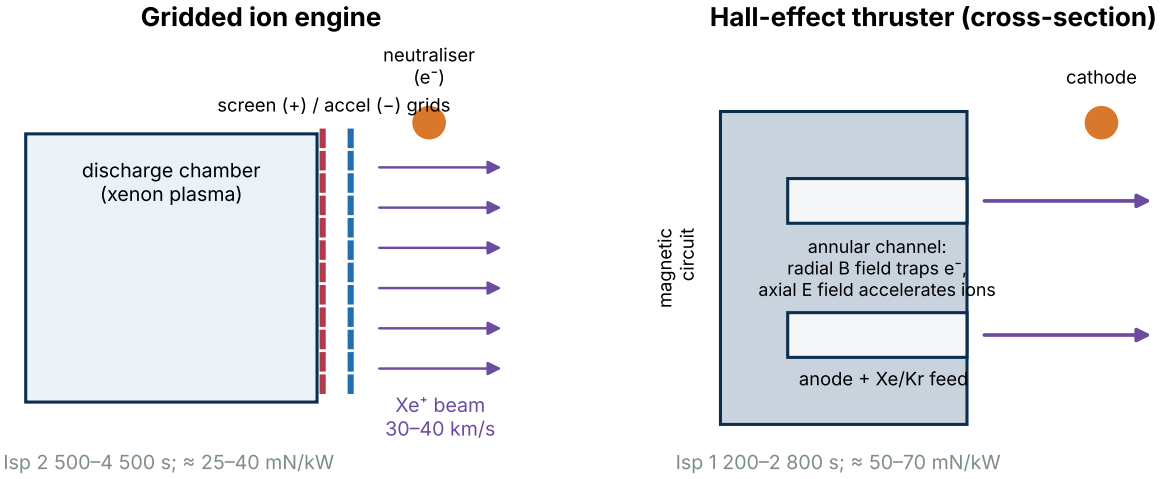



**FIGURE 6.12** Principles of gridded ion and Hall-effect thrusters. Both ionise a noble gas and accelerate the ions electrostatically; a cathode neutralises the beam.

DATA BOX — PROPELLANTS FOR ELECTRIC PROPULSION

- **Xenon**: heavy, easily ionised, inert, stores densely at high pressure (≈ 1.5 g/cm$^3$ at 80–150 bar). Expensive (US$ 3 000–20 000/kg depending on market; prices spiked in 2022).
- **Krypton**: about 10× cheaper than xenon, slightly lower thrust efficiency; used by Starlink V1.
- **Argon**: cheaper still; adopted by Starlink V2 Mini thrusters (≈ 2.4× more thrust and 1.5× more Isp than the earlier krypton thrusters, according to SpaceX).
- **Iodine**: stored as a solid at low pressure — no high-pressure tanks; first flown by ThrustMe on Beihangkongshi-1 in 2020.
- **Water** and other condensables for electrothermal and microwave thrusters on small satellites.

### 6.2.5 Choosing a Propulsion System

The choice depends on the Δv required, the time available, the power available, cost and safety. Rules of thumb: small LEO satellites needing < 100 m/s use cold gas, green monopropellant or electric propulsion, and GEO satellites increasingly use all-electric or hybrid architectures. Deep-space missions with large Δv use ion or Hall thrusters if solar power is sufficient. Planetary orbit insertion and landing need chemical thrust. Mega-constellations have made Hall thrusters a mass-produced commodity: SpaceX alone has built more electric thrusters than the rest of the world combined.

## 6.3 Command and Data Handling

### 6.3.1 Functions and Architecture

The **command and data handling (C&DH)** subsystem — **on-board data handling (OBDH)** in European terminology — is the spacecraft's nervous system. It receives and validates commands, distributes them to subsystems, runs the flight software and keeps time. It collects housekeeping

telemetry and payload data, stores data in mass memory and formats it for downlink. It also supervises the health of the spacecraft (Figure 6.13).

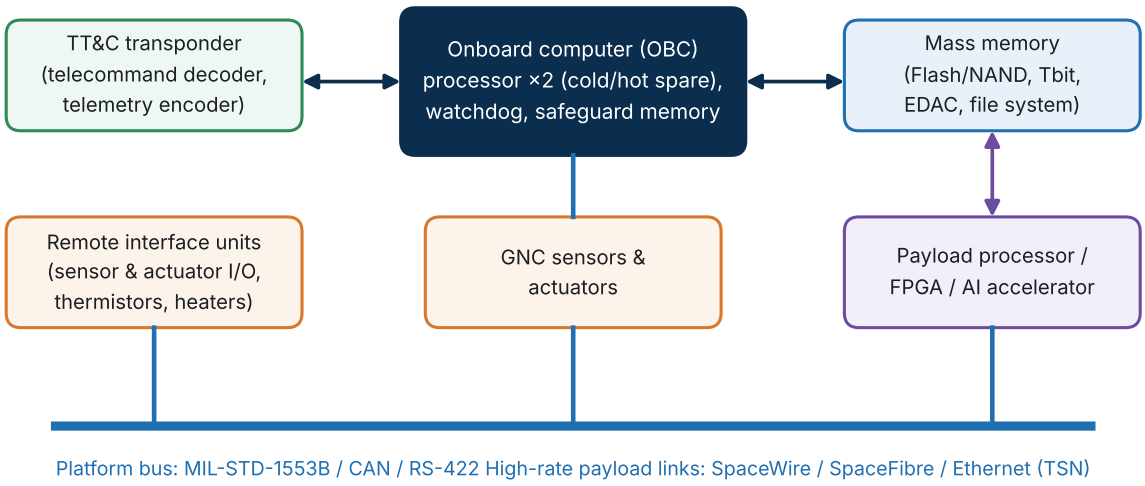



**FIGURE 6.13** A typical C&DH architecture: a redundant onboard computer connected through a platform bus to remote interface units and GNC equipment, a high-rate network to the payload and mass memory, and a link to the TT&C transponders.

Architectures may be **centralised** (one computer does everything — common on small satellites) or **federated** (separate computers for GNC, payload and platform). They may also be **distributed/integrated modular avionics**, with partitioned software on shared processors connected by deterministic networks. This is now common on human spacecraft and large platforms.

## 6.3.2 Data Buses and Networks

**TABLE 6.2** Common spacecraft data bus and network standards.

| Standard | Data rate | Typical use |
|---|---|---|
| MIL-STD-1553B | 1 Mbit/s, dual-redundant, command/response | Military and NASA platforms, ISS, Orion |
| CAN (and CANopen / CubeSat Space Protocol) | up to 1 Mbit/s | Small satellites, ESA platforms (ECSS-E-ST-50-15C) |
| RS-422 / RS-485, I²C, SPI | kbit/s–tens of Mbit/s | Simple point-to-point and board-level links |
| SpaceWire (ECSS-E-ST-50-12C) | 2–400 Mbit/s, packet-switched | Instrument and mass-memory networks (JWST, Gaia, many ESA/NASA/JAXA missions) |
| SpaceFibre | 2.5–6.25 Gbit/s per lane, multi-lane | High-rate payloads (SAR, hyperspectral) |
| Time-Triggered Ethernet / TSN | 100 Mbit/s–10 Gbit/s, deterministic | Orion, Gateway, launch vehicles, next-generation platforms |
| SpaceVPX / VITA standards | Backplane | Modular payload processors |

### 6.3.3 Processors and Memory

Spacecraft computers lag their terrestrial counterparts by 10–20 years in performance because they must tolerate radiation (Chapter 2) and are qualified slowly (Figure 6.14). The **BAE Systems RAD750**, a radiation-hardened PowerPC first flown in 2005, has powered more than 150 missions including Curiosity, Perseverance, JWST and Juno, at about 200–260 MIPS. European missions use **LEON** SPARC processors (GR712RC dual-core, GR740 quad-core). NASA's **High-Performance Spaceflight Computing (HPSC)** project with Microchip is producing a radiation-hardened multicore RISC-V processor (PIC64-HPSC) with roughly 100 times the performance of the RAD750, available to missions from about 2025–2026.

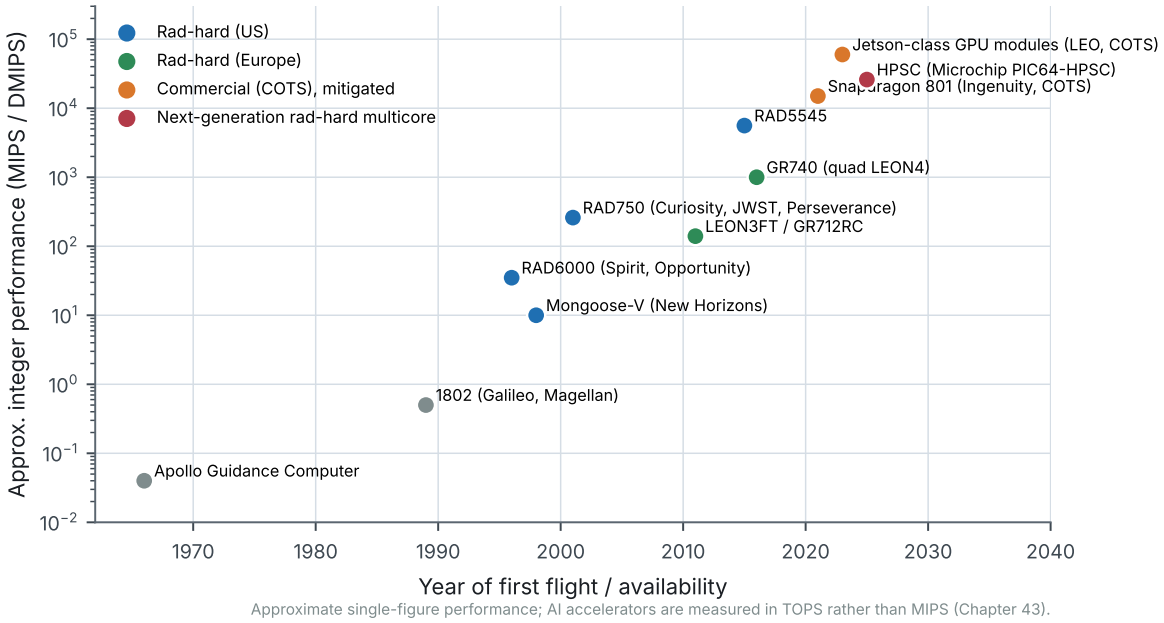



**FIGURE 6.14** Approximate integer performance of spaceflight processors over time. Commercial processors with radiation mitigation (e.g., the Snapdragon on the Ingenuity Mars helicopter) offer orders-of-magnitude more performance than rad-hard parts but less proven reliability.

A growing alternative is to fly **commercial (COTS)** processors, FPGAs and GPUs with **mitigation**. This includes screening and radiation testing, triple modular redundancy in FPGA logic, watchdogs and error-correcting memory. It also includes current-limited power with automatic latch-up recovery, and frequent memory scrubbing. Short-duration LEO missions, mega-constellations and technology demonstrations increasingly use this approach — the Ingenuity helicopter on Mars ran on a Qualcomm Snapdragon 801 smartphone processor for nearly three years. Reconfigurable **FPGAs** (e.g., AMD/Xilinx Kintex UltraScale and Versal space-grade parts, Microchip RTG4 and RT PolarFire) process high-rate payload data and run neural networks (Chapter 45).

**Memory** must cope with single-event upsets. Program memory is often in radiation-tolerant PROM, MRAM or protected Flash; working memory in SDRAM with **error detection and correction (EDAC)** that corrects single-bit errors and detects double-bit errors, continuously **scrubbed** in the background. Mass memories use NAND Flash with file systems (CCSDS File Delivery Protocol for ground transfer) and capacities from gigabits on CubeSats to tens of terabits on Earth-observation satellites.

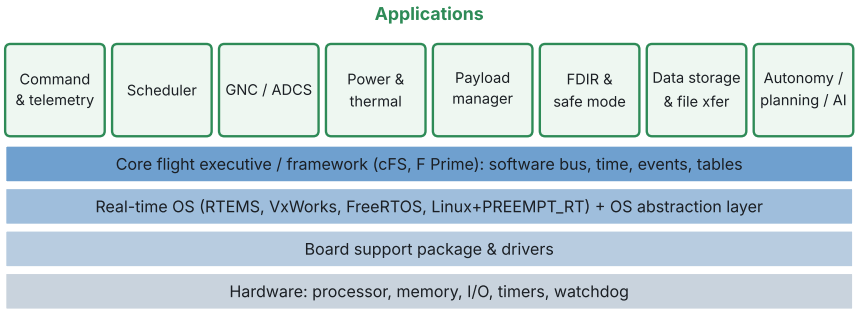
### 6.3.4 Time

The C&DH maintains **onboard time**, synchronised to UTC or GPS time via GNSS or ground correlation, to time-tag telemetry and to execute **time-tagged commands** stored in an onboard schedule. Earth-observation and science missions need time knowledge to microseconds; navigation satellites carry atomic clocks with nanosecond precision (Chapter 18).

## 6.4 Flight Software

### 6.4.1 Architecture

Flight software (FSW) has grown from a few kilobytes on early spacecraft to millions of lines of code (Orion ≈ 2 million; the F-35 fighter about 8 million; Starlink satellites run Linux). Good FSW is layered (Figure 6.15). It has a hardware abstraction layer and drivers, a **real-time operating system** (RTEMS, VxWorks, FreeRTOS, or Linux with real-time extensions), a reusable framework, and mission applications on top. The framework provides a software bus, time, event logging and table services. Examples are NASA's **core Flight System (cFS)**, JPL's **F Prime** (used on Ingenuity and many CubeSats) and ESA's component-based frameworks.




Layering isolates applications from hardware so that they can be reused across missions (NASA cFS has flown on dozens of spacecraft; JPL's F Prime flew on Ingenuity).

**FIGURE 6.15** Layered flight-software architecture. A framework such as NASA's cFS or JPL's F Prime separates mission applications from the operating system and hardware.

### 6.4.2 Development and Assurance

Flight software controls the whole spacecraft and can be the cause of mission loss, so it is developed under strict processes. These include coding standards such as MISRA C and JPL's "Power of Ten" rules (no recursion, no dynamic memory allocation after initialisation, bounded loops, assertions). They also include static analysis, unit and integration tests, hardware-in-the-loop testing on flight-like testbeds ("flatsats"), and **independent verification and validation (IV&V)** for high-criticality software. Most modern spacecraft can be **reprogrammed in flight** — a capability that has rescued many missions.

**FAILURE LESSON — SOFTWARE CAN END A MISSION IN SECONDS**

On its maiden flight in 1996, Ariane 5 broke up 37 seconds after lift-off. The cause was inertial-reference software reused from Ariane 4, which converted a 64-bit floating-point horizontal velocity into a 16-bit integer. This caused an overflow that shut down both (identical) inertial units. In 2016 the Japanese X-ray observatory Hitomi was destroyed when an incorrect parameter uploaded to its attitude-control software, combined with a star-tracker anomaly, caused it to spin up until it broke apart. In 2023 the ispace HAKUTO-R lander crashed on the Moon after its software rejected valid altimeter data as faulty while passing over a crater rim. Each case shows the importance of verifying software against realistic, end-to-end scenarios — including "off-nominal" ones.

### 6.4.3 Autonomy and Fault Protection

Spacecraft are out of contact much of the time — a LEO satellite sees a given ground station for only about 10 minutes per pass, and a Mars rover's commands take 5–22 minutes to arrive. They therefore need **autonomy**: the ability to execute plans, react to events and protect themselves.

**Fault detection, isolation and recovery (FDIR)** — called fault protection at NASA — is organised hierarchically (Figure 6.16). Faults are handled at the lowest level possible. A memory bit-flip is corrected by EDAC, a failed unit is replaced by its redundant twin, and a subsystem anomaly triggers a mode change. If all else fails, the spacecraft enters **safe mode** — pointing its solar arrays at the Sun, switching off non-essential loads, and waiting for instructions from the ground. Safe modes must be simple, robust, and independent of the failed hardware (many spacecraft have a separate "survival" computer or hardware-implemented safe mode).

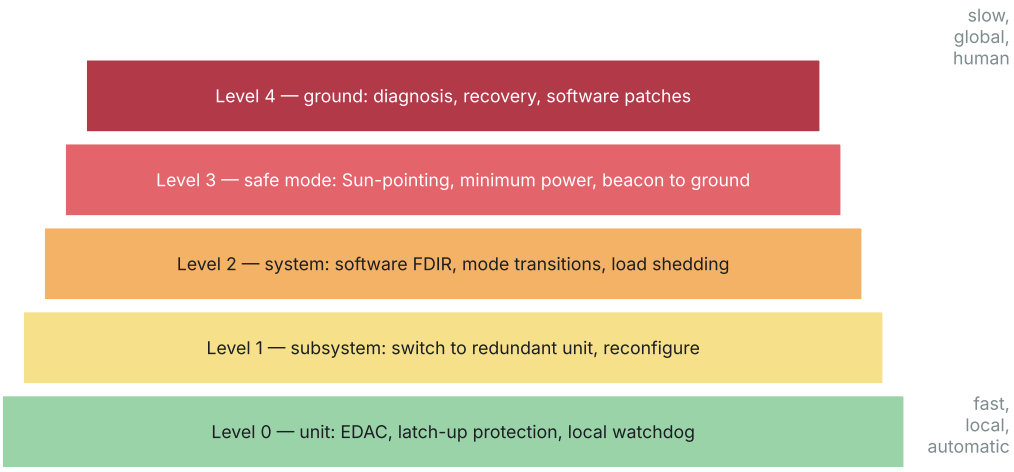



**FIGURE 6.16** The hierarchy of fault detection, isolation and recovery. Fast local responses handle simple faults; system-level safe mode and ground intervention handle complex ones.

Higher levels of autonomy — onboard planning and scheduling, autonomous navigation, opportunistic science and model-based diagnosis — are increasingly implemented with artificial intelligence techniques, discussed in depth in Chapters 48 and 49.

### KEY TAKEAWAYS

- The ADCS determines and controls attitude with sensors (star trackers, gyros, Sun sensors, magnetometers, GNSS) and actuators (reaction wheels, CMGs, magnetorquers, thrusters) in a Kalman-filter-based feedback loop. Momentum must be managed against persistent disturbance torques.
- Propulsion is characterised by thrust and Isp. Chemical systems (cold gas 70 s, hydrazine 225 s, biprop 320 s) give high thrust. Electric propulsion (Hall 1 200–2 800 s, ion 2 500–4 500 s) gives high efficiency at low thrust and is now standard on constellations and many GEO and deep-space missions.
- Microgravity requires propellant management devices; feed systems are blowdown or pressure-regulated.
- C&DH uses radiation-hardened or mitigated commercial processors, EDAC-protected memory and deterministic buses (1553, CAN, SpaceWire, TSN); rad-hard processors lag commercial ones by a decade or more.
- Flight software is layered (RTOS + framework + applications), rigorously verified and reprogrammable; hierarchical FDIR and safe modes protect the spacecraft between ground contacts.

### REVIEW QUESTIONS

1. Compare spin stabilisation with three-axis stabilisation and give one advantage of each.
2. List four attitude sensors and rank them by typical accuracy.
3. Why do reaction wheels need momentum management, and which actuators are used to unload them?
4. A reaction wheel can store 2 N m s. A constant disturbance torque of 10 μN m acts on the spacecraft. How long does the wheel take to saturate if it starts at zero speed?
5. Compare cold gas, hydrazine monopropellant, bipropellant and Hall-effect propulsion in specific impulse and thrust, and give a typical application of each.
6. A 200 kg satellite (initial mass) needs a Δv of 300 m/s. Compare the propellant mass needed with hydrazine ($I_{sp}$ = 225 s) and with a Hall thruster ($I_{sp}$ = 1 500 s).
7. Why do space-qualified processors lag commercial processors by a decade or more, and how can commercial parts be used safely?
8. Describe the layers of a typical flight-software architecture and the role of safe mode in fault protection.

CHAPTER

# 7 Space Communications and the Ground Segment

Image: NASA/JPL (NASA/JPL-Caltech)

> *Without communications a spacecraft is useless - it cannot be controlled, and its data cannot be used. This chapter explains how radio and laser links work, from frequency bands and antennas to the link budget that decides whether a signal can be received; how modulation and error-correcting codes approach the theoretical limits of communication; the ground stations, networks and mission control centres that make up the ground segment; deep-space communications; and the international standards that let spacecraft and ground systems from different organisations work together.*

IN THIS CHAPTER


7.1 The Role of Communications
7.2 Frequency Bands and Spectrum
7.3 Antennas
7.4 The Link Budget
7.5 Modulation and Coding
7.6 Optical (Laser) Communications
7.7 The Ground Segment
7.8 Space Data Standards: CCSDS


## 7.1 The Role of Communications

Every spacecraft has at least one radio link. Communications serve four purposes:

1. **Telecommand (TC)** — instructions from the ground to the spacecraft (uplink), typically at a few kbit/s with strong error protection and authentication.
2. **Telemetry (TM)** — health and status data from the spacecraft (housekeeping), typically kbit/s to tens of kbit/s.
3. **Tracking** — measuring range, range-rate and angles to determine the orbit (Chapter 3).

4. **Payload (mission) data** — the product of the mission: images, radar data, science measurements, or, for communications satellites, the user traffic itself. Rates range from bits per second (deep space) to tens of gigabits per second (laser crosslinks).

The first three are grouped as **TT&C** (telemetry, tracking and command). Payload data usually uses a separate, higher-frequency, higher-rate link.

## 7.2 Frequency Bands and Spectrum

### 7.2.1 The Radio Spectrum for Space

Radio frequencies are a limited, shared natural resource. The **International Telecommunication Union (ITU)** allocates frequency bands to services (e.g., Earth exploration-satellite, space research, fixed-satellite, mobile-satellite, radionavigation-satellite) in its **Radio Regulations**, revised at World Radiocommunication Conferences every three to four years. Satellite operators must file their networks with the ITU through national administrations and coordinate with others to avoid interference. This process can take years and is a central part of any commercial constellation business plan.

Space engineers use the IEEE radar letter designations for bands (Figure 7.1, Table 7.1).

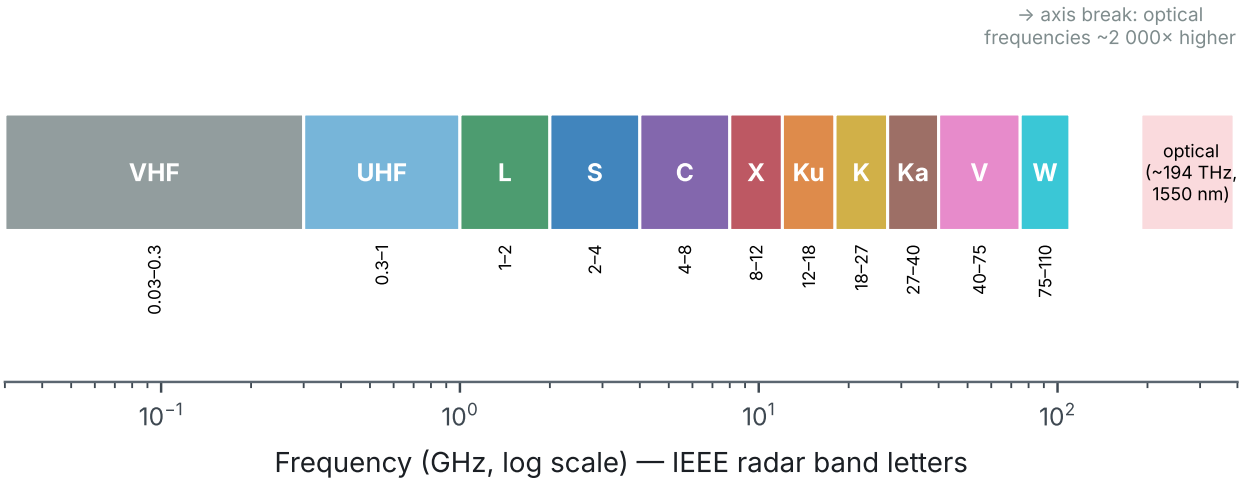



**FIGURE 7.1** Microwave frequency bands used for space communications, with their approximate frequency ranges in GHz. Optical communication uses frequencies about 2 000 times higher than Ka-band.

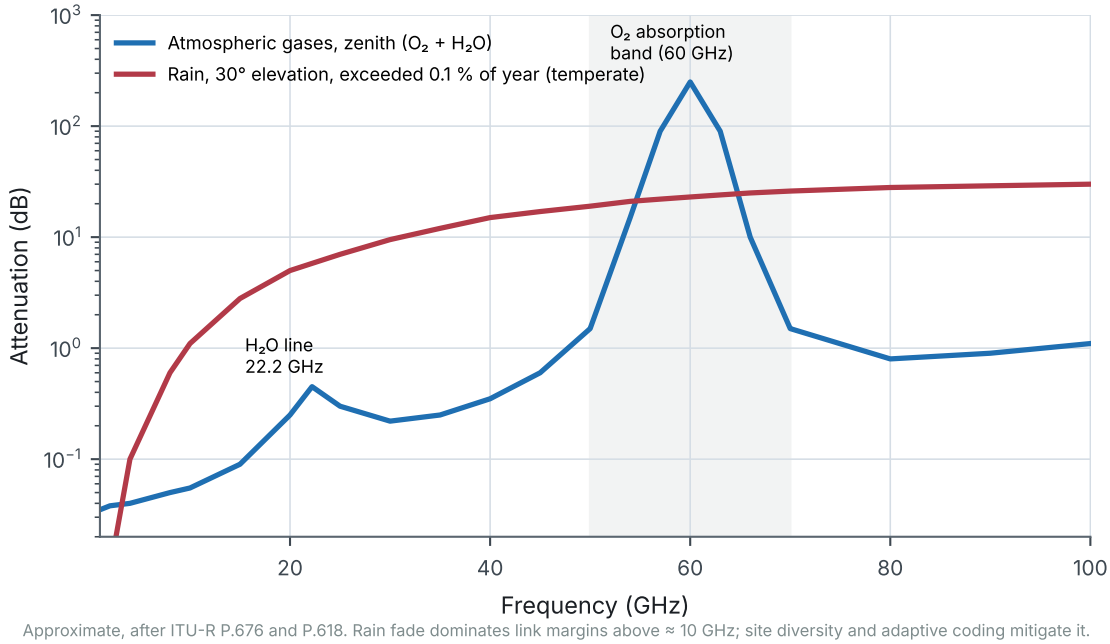
**TABLE 7.1** Principal frequency bands for space communications and their uses.

| Band | Frequency range | Typical space uses | Notes |
|---|---|---|---|
| VHF / UHF | 30–300 MHz / 300 MHz–1 GHz | Amateur and CubeSat TT&C, military UHF SATCOM (MUOS), AIS ship tracking, IoT | Low gain, crowded, ionospheric effects |
| L-band | 1–2 GHz | GNSS (1 164–1 610 MHz), Iridium, Inmarsat, Globalstar uplink, direct-to-device | Penetrates foliage and rain; small antennas |
| S-band | 2–4 GHz | TT&C (2 025–2 110 MHz up, 2 200–2 290 MHz down), Globalstar, some EO, radar | The classic TT&C band |

| Band | Frequency range | Typical space uses | Notes |
|---|---|---|---|
| C-band | 4–8 GHz | GEO TV distribution, rain-robust trunking, SAR (Sentinel-1, RADARSAT) | Little rain fade; large ground dishes |
| X-band | 8–12 GHz | EO payload downlinks (8 025–8 400 MHz), deep space (7.1–7.2 GHz up, 8.4–8.45 GHz down), military SATCOM, SAR | Workhorse for science downlinks |
| Ku-band | 12–18 GHz | Direct-to-home TV, VSAT, Starlink/OneWeb user links, data relay | Rain fade becomes significant |
| K / Ka-band | 18–40 GHz | High-throughput satellites, gateways, deep-space Ka (31.8–32.3 GHz), EO high-rate (25.5–27 GHz) | Wide bandwidths, strong rain fade |
| Q / V-band | 33–75 GHz | Feeder links for next-generation HTS and constellations | Experimental-to-early operational |
| W-band | 75–110 GHz | Experimental links and radiometry | Research |
| Optical | ≈ 200 THz (1 064 / 1 550 nm) | Laser crosslinks, data relay, deep-space optical | Unregulated by ITU (so far); blocked by cloud |

## 7.2.2 Why Higher Frequencies?

Higher frequencies offer more **bandwidth** (and therefore higher data rates) and allow small antennas to form narrow, high-gain beams. Their disadvantages are greater attenuation by rain and atmospheric gases (Figure 7.2), the need for more precise antenna pointing, and more expensive components. The trend over 60 years has been steadily upward: from VHF and S-band to X and Ka for downlinks, and now to optical.




Approximate, after ITU-R P.676 and P.618. Rain fade dominates link margins above ≈ 10 GHz; site diversity and adaptive coding mitigate it.

**FIGURE 7.2** Approximate atmospheric attenuation through the whole atmosphere as a function of frequency: gaseous absorption at zenith and rain attenuation exceeded 0.1 % of the time at 30° elevation in a temperate climate. Above 10 GHz, rain fade dominates link design.

## 7.3 Antennas

An antenna converts electrical signals into electromagnetic waves and focuses them in a direction. Its key property is **gain** $G$ — how much it concentrates power compared with an isotropic radiator, expressed in dBi. For a parabolic dish of diameter $D$ at wavelength $\lambda$ and aperture efficiency $\eta$ (typically 0.55–0.70):

$$G = \eta\left(\frac{\pi D}{\lambda}\right)^2, \qquad \theta_{3dB} \approx 70\,\frac{\lambda}{D}\ \text{degrees}$$

Gain rises with the square of frequency for a fixed size, while beamwidth shrinks (Figure 7.3). A 34 m Deep Space Network antenna has a gain of about 68 dBi at X-band and 80 dBi at Ka-band. Its Ka beam is only 0.017° wide, so pointing must be accurate to a few millidegrees.

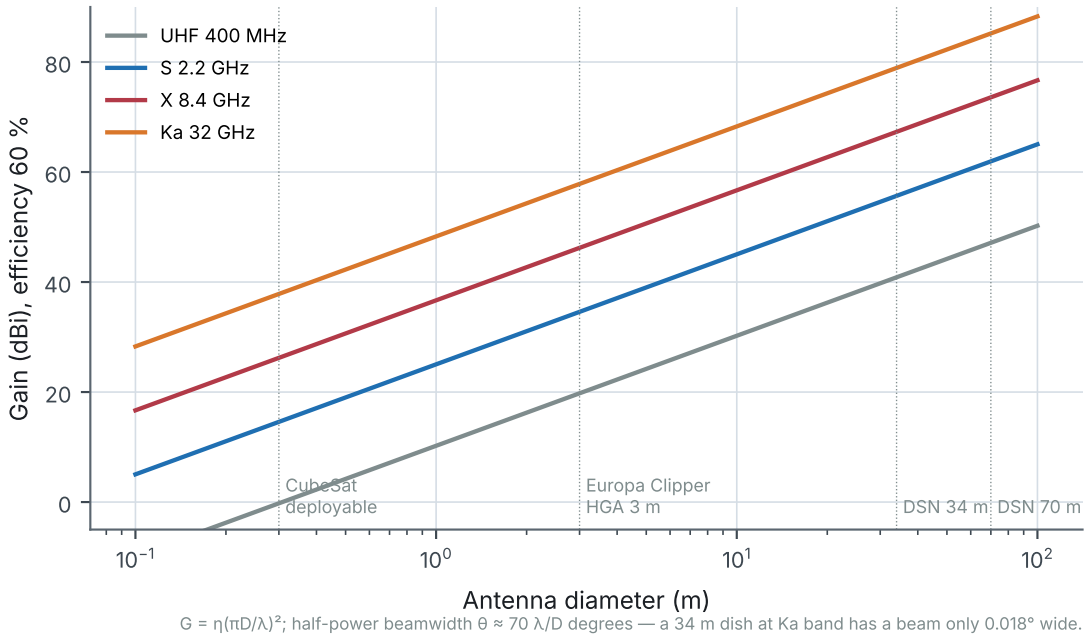



**FIGURE 7.3** Gain of parabolic antennas as a function of diameter for four frequency bands.

Spacecraft use several antenna types (Figure 7.4):

- **Low-gain antennas (LGAs)** — monopoles, patches, helices with almost hemispherical coverage, used for TT&C in any attitude and in safe mode.
- **Medium-gain antennas** — horns, helices and small arrays with tens of degrees of beamwidth.
- **High-gain antennas (HGAs)** — fixed or steerable parabolic reflectors; deployable mesh reflectors up to 22 m (e.g., on mobile-satellite GEO spacecraft such as TerreStar and AST SpaceMobile's large phased arrays in LEO).
- **Phased arrays** — many small elements whose signals are shifted in phase to steer the beam electronically, with no moving parts. They are now standard on LEO broadband satellites and user terminals (the Starlink user terminal is a phased array of more than a thousand elements) and on SAR satellites.

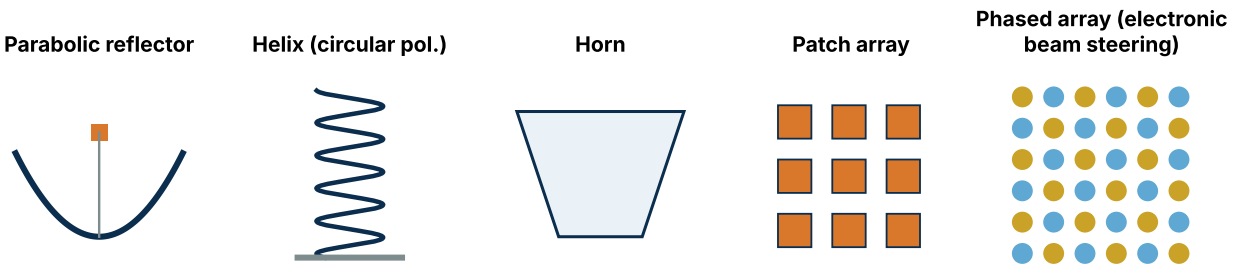



**FIGURE 7.4** Common antenna types used on spacecraft and ground terminals.

## 7.4 The Link Budget

### 7.4.1 The Link Equation

The **link budget** is the accounting that shows whether a link closes — whether the received signal is strong enough compared with noise to deliver the required data rate at the required error rate. In decibels, the received carrier power is:

$$C = P_t + G_t - L_t - L_{fs} - L_a - L_{pol} - L_{point} + G_r - L_r \quad [\text{dBW}]$$

where $P_t$ is transmitter power, $G_t$ and $G_r$ are antenna gains, $L_t$ and $L_r$ line losses, and $L_a$ atmospheric losses. The term $L_{fs}$ is the **free-space path loss**, which describes how power spreads over the surface of an ever-larger sphere:

$$L_{fs} = \left(\frac{4\pi d f}{c}\right)^2 \quad \Rightarrow \quad L_{fs}[\text{dB}] = 92.45 + 20\log_{10} d[\text{km}] + 20\log_{10} f[\text{GHz}]$$

The product $P_t + G_t - L_t$ is the **effective isotropic radiated power (EIRP)**. Path loss grows enormously with distance (Figure 7.5): about 177 dB for a 2 000 km LEO link at X-band, 205 dB to GEO at Ku-band, and about 280 dB to Mars at X-band.

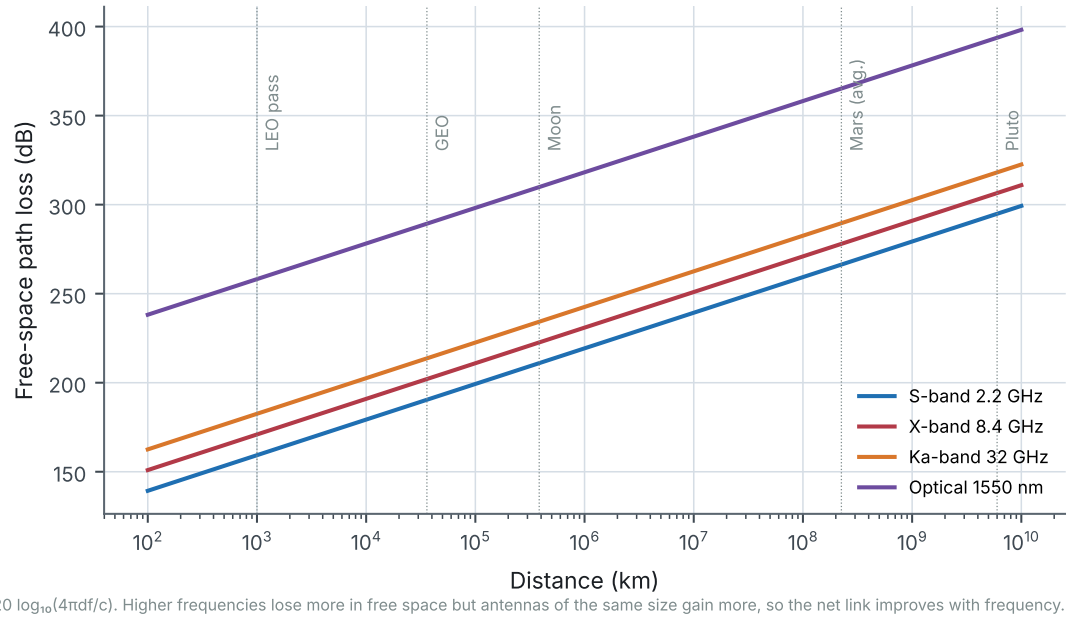



**FIGURE 7.5** Free-space path loss versus distance for four frequency bands. Each tenfold increase in distance costs 20 dB.

The noise that competes with the signal is characterised by the **noise power spectral density** $N_0 = kT_{sys}$, where $k = 1.38 \times 10^{-23}$ J/K (−228.6 dBW/K/Hz) is Boltzmann's constant and $T_{sys}$ is the system noise temperature (antenna, sky, receiver). A receiving system's quality is summarised by its **figure of merit** $G/T$ in dB/K. The ratio that matters is

$$\frac{C}{N_0} = \text{EIRP} - L + \frac{G}{T} + 228.6 \quad \text{[dB-Hz]}, \qquad \frac{E_b}{N_0} = \frac{C}{N_0} - 10\log_{10} R_b$$

where $R_b$ is the bit rate and $E_b/N_0$ the energy per bit relative to noise density. The link **margin** is the difference between the achieved and the required $E_b/N_0$; designs typically require 3 dB of margin, more for rain-affected links.

**WORKED EXAMPLE — AN X-BAND EARTH-OBSERVATION DOWNLINK**

A LEO satellite transmits 10 W (10 dBW) through 1.5 dB of losses into a 6 dBi patch antenna (EIRP = 14.5 dBW) at 8.2 GHz. At 5° elevation from a 600 km orbit, the slant range is about 2 000 km, so $L_{fs}$ = 176.7 dB; atmosphere and rain cost 1.5 dB, pointing and polarisation 1 dB. The ground station has a 5.4 m dish (51 dBi) and $T_{sys}$ = 150 K, so $G/T = 51 - 21.8 = 29.2$ dB/K. Then (Figure 7.6)

- $C = 14.5 - 176.7 - 2.5 + 51 = -113.7$ dBW
- $C/N_0 = -113.7 - 10\log(1.38 \times 10^{-23} \times 150) = -113.7 + 206.8 = 93.1$ dB-Hz
- For $R_b$ = 500 Mbit/s (87.0 dB-Hz): $E_b/N_0 = 6.1$ dB.

With LDPC-coded QPSK requiring about 3 dB, the margin is about 3 dB — adequate. At higher elevations the range shrinks to 600 km and the margin rises by 10 dB, which modern systems exploit with **variable coding and modulation (VCM)**, increasing the data rate during the pass.

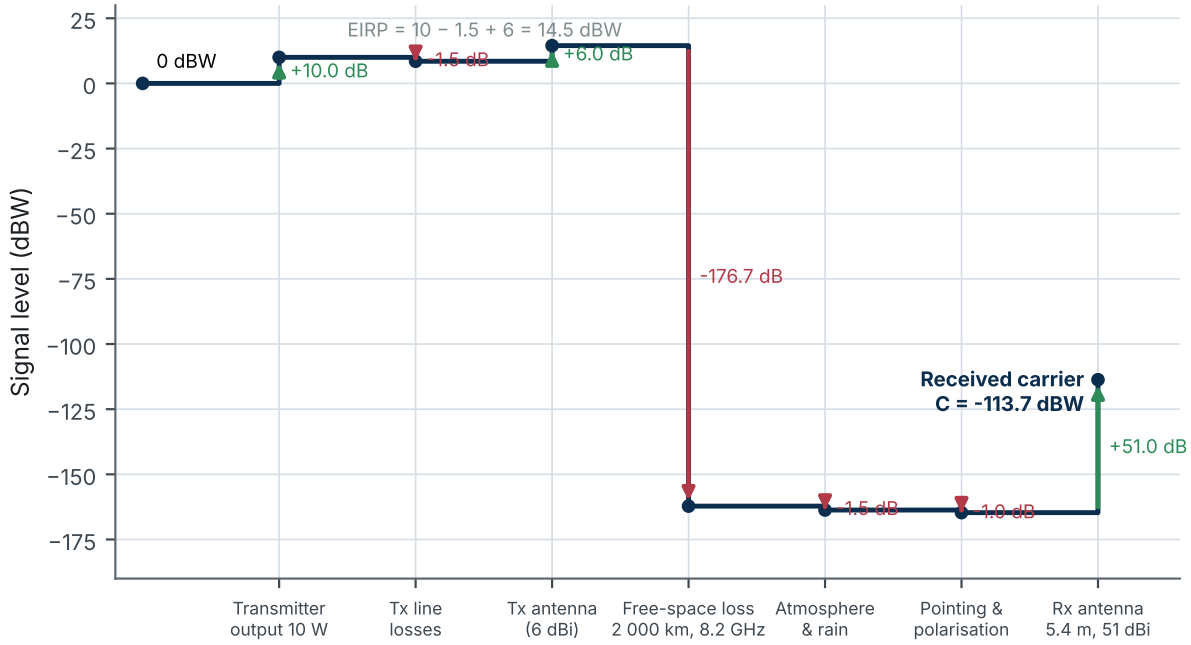



**FIGURE 7.6** Signal level through the example X-band downlink: free-space loss dominates, and is recovered mainly by the gain of the ground antenna.

# 7.5 Modulation and Coding

## 7.5.1 Modulation

**Modulation** imprints digital data on the carrier wave by changing its phase, amplitude or frequency. Phase-shift keying is dominant in space (Figure 7.7):

- **BPSK** — two phases, 1 bit per symbol, very robust; used for TT&C and deep space.
- **QPSK / OQPSK** — four phases, 2 bits per symbol, same power efficiency as BPSK with twice the spectral efficiency; the workhorse of EO downlinks.
- **8PSK, 16APSK, 32APSK** — more bits per symbol, at the cost of needing more $E_b/N_0$. Amplitude-phase shift keying (APSK) arranges points on concentric rings. This lets it tolerate the non-linear power amplifiers (travelling-wave tube amplifiers, TWTAs) used in space. DVB-S2/S2X and CCSDS 131.2 support them with adaptive switching.
- **GMSK and filtered modulations** — constant-envelope, spectrally compact schemes.

Ranging signals (tones or pseudo-noise codes) can share the carrier with data to measure distance.

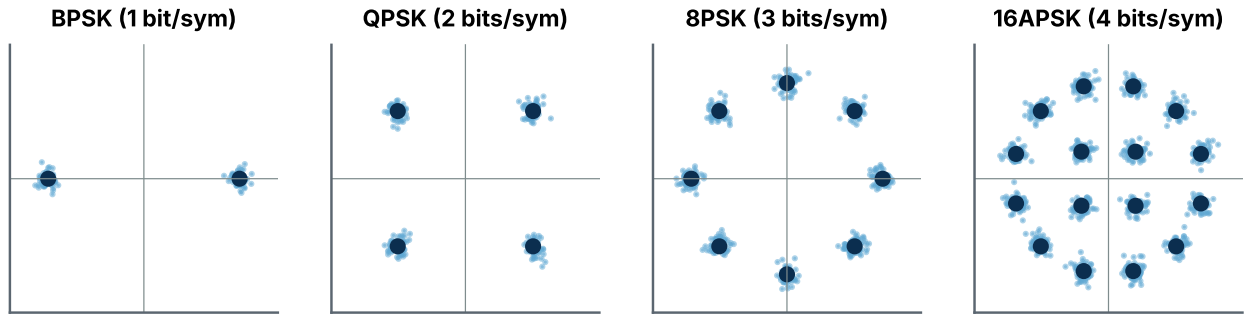



**FIGURE 7.7** Constellation diagrams of common modulation schemes with noise. Denser constellations carry more bits per symbol but need a stronger signal to separate the points.

## 7.5.2 Error-Correcting Codes

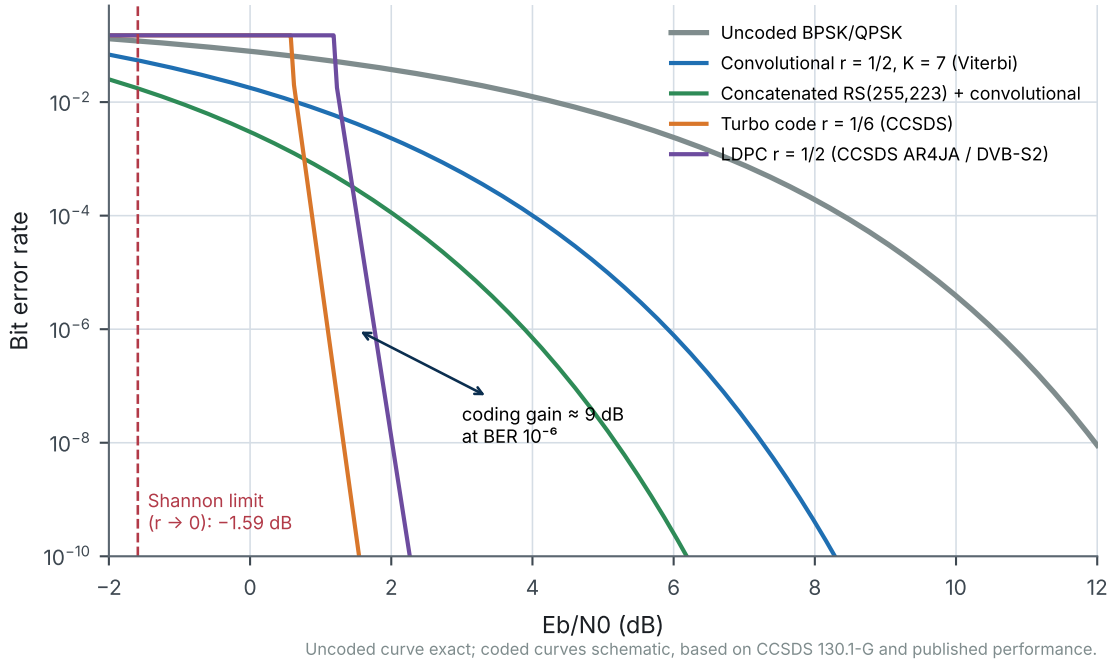
Claude Shannon showed in 1948 that the maximum error-free data rate (capacity) of a channel of bandwidth $B$ is

$$C = B \log_2 \left(1 + \frac{S}{N}\right)$$

and that, for any rate below capacity, a code exists that makes errors arbitrarily rare. The ultimate limit is $E_b/N_0 = -1.59$ dB. **Forward error correction (FEC)** codes add redundant bits that let the receiver correct errors; the improvement is called the **coding gain** (Figure 7.8). Space communications have driven coding theory for 60 years:

- **Reed–Muller codes** on Mariner 6/7/9 (1969–1971).
- **Convolutional codes with Viterbi decoding** (rate 1/2, constraint length 7) — the standard from Voyager onwards.
- **Concatenated Reed–Solomon + convolutional codes** — Voyager at Uranus and Neptune, Galileo, and many EO missions.
- **Turbo codes** (invented 1993; CCSDS standard 1999) — within 1 dB of the Shannon limit; used on MRO, New Horizons, MESSENGER and many deep-space missions.

- **Low-density parity-check (LDPC) codes** — near-Shannon performance with efficient parallel decoding; standardised in CCSDS 131.0 (AR4JA codes for deep space, rate 7/8 for near-Earth) and DVB-S2 for commercial satellite broadband.




Uncoded curve exact; coded curves schematic, based on CCSDS 130.1-G and published performance.

**FIGURE 7.8** Bit error rate versus Eb/N0 for uncoded and coded transmission. Modern turbo and LDPC codes operate within about 1–2 dB of the Shannon limit, a gain of roughly 9 dB over uncoded transmission — equivalent to an eightfold increase in transmitter power or antenna area.

Data is also **compressed** before transmission. CCSDS standards include lossless compression (CCSDS 121.0, based on Rice coding), image compression (CCSDS 122.0, wavelet-based) and multispectral/hyperspectral compression (CCSDS 123.0). Onboard AI that discards cloudy images or extracts only events of interest is an increasingly powerful form of "compression" (Chapter 45).

## 7.6 Optical (Laser) Communications

**Free-space optical communications** use lasers at near-infrared wavelengths (1 064 or 1 550 nm). The wavelength is about 10 000 times shorter than X-band. So a 10 cm telescope produces a beam about 10 µrad wide — far narrower than any radio beam — which concentrates the power on the receiver. Optical links offer data rates of gigabits to hundreds of gigabits per second, use spectrum that is not (yet) regulated, and are very hard to intercept or jam. Their challenges are clouds (which block the beam entirely), precise pointing, acquisition and tracking (to microradians while both terminals move), and atmospheric turbulence.

Milestones include:

- **LLCD** (NASA, 2013) — 622 Mbit/s from lunar orbit on LADEE.
- **EDRS** (ESA/Airbus "SpaceDataHighway", 2016 onwards) — GEO relay nodes receiving 1.8 Gbit/s laser links from Sentinel-1 and Sentinel-2 satellites and relaying the data to Europe in near-real time.
- **LCRD** (NASA, 2021) — a GEO laser relay, linked with the **ILLUMA-T** terminal on the ISS in 2023.

- **TBIRD** (NASA/MIT Lincoln Laboratory, 2022) — a CubeSat terminal that downlinked at 200 Gbit/s.
- **DSOC** (NASA/JPL, on the Psyche mission, 2023–2025) — the first deep-space laser link. It achieved 267 Mbit/s from 33 million km and continued to link from beyond Mars' distance. It used a superconducting nanowire photon-counting detector at Palomar Observatory.
- **Optical inter-satellite links in mega-constellations** — Starlink satellites carry three or more laser terminals, forming a mesh network in space that carries many petabytes per day. The US Space Development Agency's Transport Layer also uses optical crosslinks, and China's constellations are adopting them.
- **Artemis II's O2O terminal** (2026) — a laser link returning high-definition video from Orion during the crewed lunar flight.

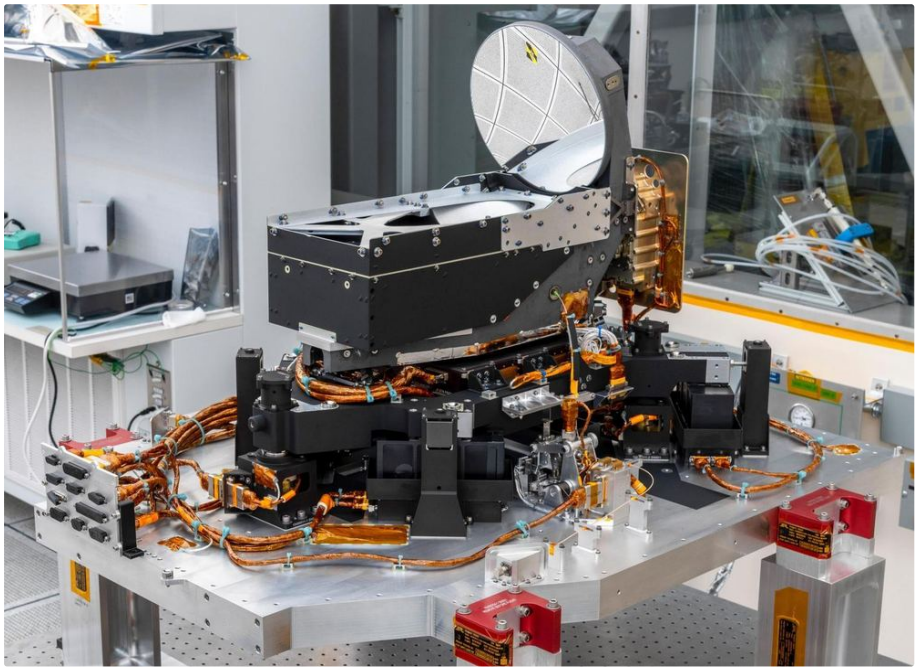


**FIGURE 7.9** The Deep Space Optical Communications (DSOC) flight laser transceiver before integration with NASA's Psyche spacecraft. Its 22 cm telescope sent the first high-rate laser signals from deep space. Image: NASA/JPL (NASA/JPL-Caltech).

**FIGURE 7.10** Artist's illustration of the ILLUMA-T terminal on the International Space Station relaying data by laser through the LCRD relay satellite in geostationary orbit. Image: NASA/JSC.

Figure 7.11 shows how the best achievable data rates have grown by more than ten orders of magnitude since the first planetary missions.

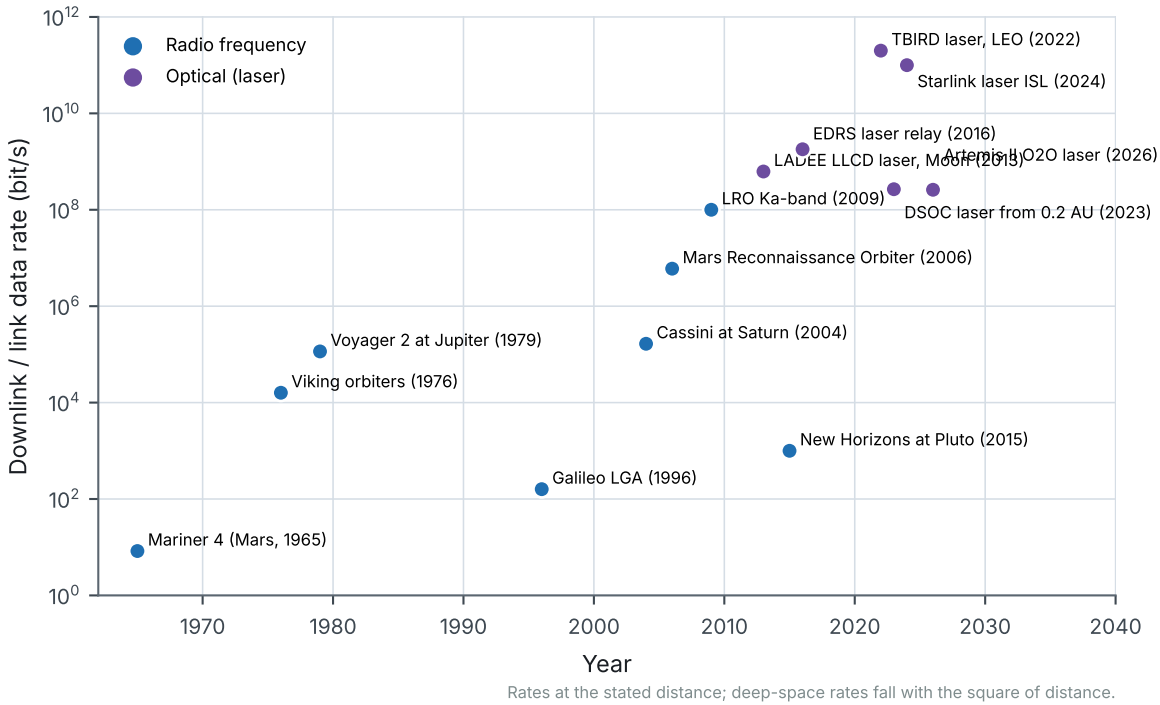



**FIGURE 7.11** Representative space communication data rates since 1965. Laser links have pushed rates into the tens and hundreds of gigabits per second.

## 7.7 The Ground Segment

### 7.7.1 Architecture

The **ground segment** comprises all the ground facilities needed to operate the spacecraft and exploit its data (Figure 7.12):

- **Ground stations** with antennas, RF equipment (low-noise amplifiers, high-power amplifiers, up/down-converters), modems/baseband units, timing, and network connections.
- **Mission operations centre (MOC)** — the control room where flight controllers monitor telemetry, send commands, and respond to anomalies. They use a **mission control system** (e.g., ESA's SCOS-2000 and its successor EGS-CC, NASA's ITOS and AMMOS, or open-source frameworks such as Yamcs and OpenMCT).
- **Flight dynamics** — orbit determination, manoeuvre planning, attitude reconstruction, collision avoidance, and product generation (ephemerides, pointing predictions).
- **Mission planning and scheduling** — turning user requests into conflict-free activity timelines that respect power, thermal, data-volume and visibility constraints (Chapter 49 discusses AI planners).
- **Payload data ground segment (PDGS)** — receiving, processing, calibrating, archiving and distributing payload data. Processing levels follow a standard ladder: Level 0 (raw, time-ordered), Level 1 (calibrated, geolocated), Level 2 (geophysical variables), Level 3 (gridded, composited), Level 4 (model output) (Chapter 13).
- **Security** — encryption and authentication of telecommands (e.g., CCSDS Space Data Link Security), network segmentation, intrusion detection, and protection of ground systems, which are the most frequently attacked part of space systems.

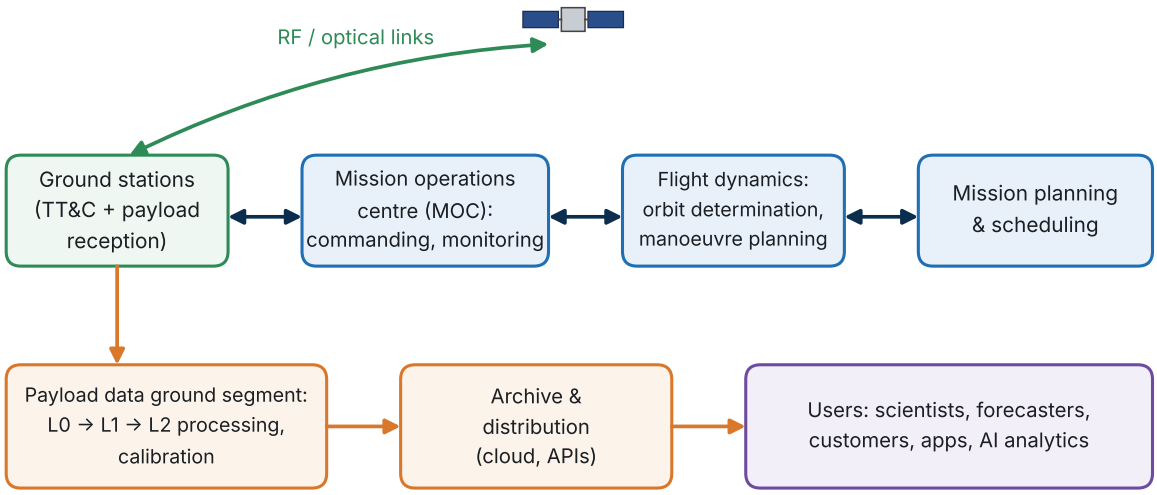



**FIGURE 7.12** Elements of a typical ground segment and their data flows.

## 7.7.2 Ground Station Networks

A LEO satellite is visible from a given station for only 5–15 minutes per pass, and from a mid-latitude station for only a few passes a day. Polar-orbiting satellites, however, pass over the poles on every orbit, so stations at high latitudes see every orbit (Figure 7.13). This is why **Svalbard** (78° N, KSAT's SvalSat with more than 100 antennas), **Inuvik**, **Kiruna**, **Fairbanks** and Antarctic stations such as **Troll** (72° S) and **McMurdo** host so many antennas.

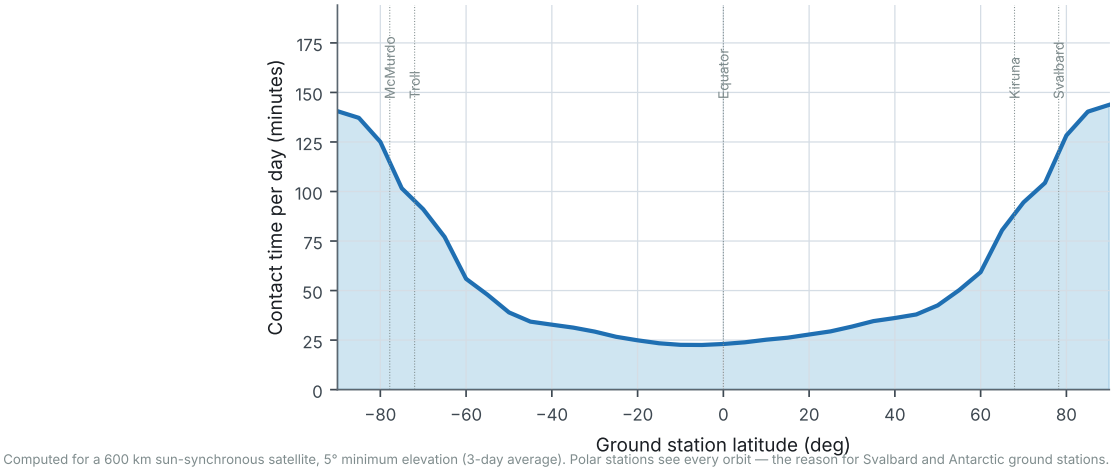



**FIGURE 7.13** Daily contact time between a 600 km sun-synchronous satellite and a ground station, as a function of station latitude, computed for this manual. A polar station sees the satellite for several times longer than an equatorial one.

Operators increasingly buy **ground-station-as-a-service**, paying per pass or per minute on shared networks: KSAT, SSC (Swedish Space Corporation), AWS Ground Station, Microsoft Azure Orbital (partnered networks), Leaf Space, Atlas Space Operations, RBC Signals and others. Electronically steered **phased-array ground stations** and software-defined radios allow one site to track several satellites at once. Data relay satellites in GEO — NASA's **TDRS** constellation, ESA's **EDRS**, China's **Tianlian**, Russia's **Luch** — provide nearly continuous contact for the ISS, Hubble, Tiangong and Earth-observation satellites.

### 7.7.3 Deep Space Communications

Deep-space missions require the largest antennas, the most sensitive receivers and the most powerful transmitters on Earth. NASA's **Deep Space Network (DSN)** has three complexes spaced about 120° apart in longitude: **Goldstone** (California), **Madrid** (Spain) and **Canberra** (Australia). This spacing means that any spacecraft beyond a few thousand kilometres is always visible from at least one of them (Figure 7.14). Each complex has a 70 m antenna and several 34 m beam-waveguide antennas; new 34 m antennas were added under the DSN Aperture Enhancement Project (DSS-53 in Madrid in 2022, DSS-56 and others). Antennas can be **arrayed** to combine their signals, as at Madrid in 2024, where six antennas were arrayed together for the first time to receive Voyager-class signals.

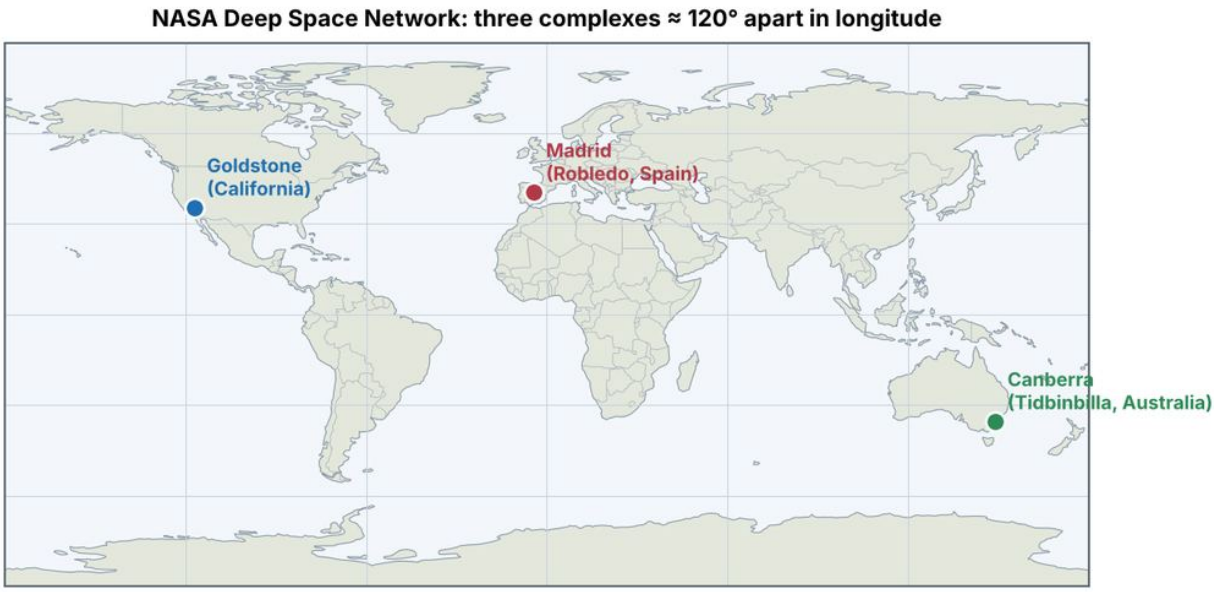



**FIGURE 7.14** Locations of the three Deep Space Network complexes.

**FIGURE 7.15** Six DSN antennas at the Madrid complex arrayed for the first time in April 2024 to receive a single spacecraft's signal. Image: NASA/JPL (MDSCC/INTA, Francisco "Paco" Moreno).

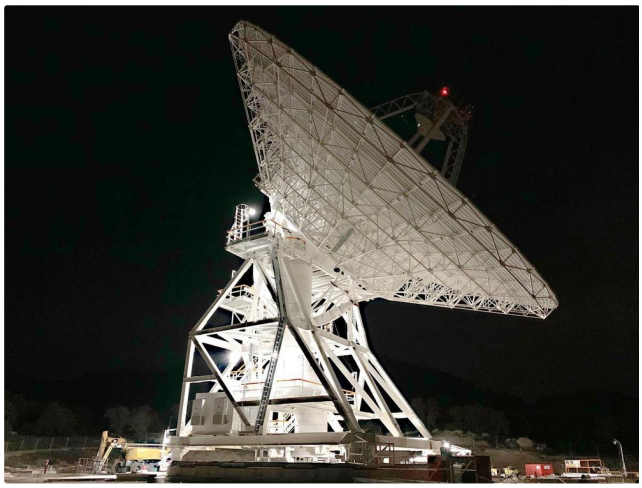


**FIGURE 7.16** The 34 m beam-waveguide antenna DSS-53, added to the Madrid complex in 2022. Image: NASA/JPL (NASA/JPL-Caltech).

Other agencies operate complementary networks, such as ESA's **ESTRACK** with 35 m deep-space antennas at New Norcia (Australia), Cebreros (Spain) and Malargüe (Argentina). China's deep-space network has 35 m and 66 m antennas at Kashgar, Jiamusi and Zapala (Argentina). There are also India's IDSN at Byalalu, JAXA's 54 m Misasa antenna and Russia's network. **Cross-support** agreements using CCSDS standards let agencies track each other's missions.

The DSN is heavily oversubscribed: in the late 2020s it supports more than 40 missions, and demand from lunar missions (Artemis, CLPS landers, international orbiters) is growing. NASA is complementing it with commercial lunar communication services and with optical ground stations.

DATA BOX — TALKING TO VOYAGER 1

- Distance (October 2026): about 171 AU, or 25.6 billion km; one-way light time ≈ 23.7 hours (it reaches one light-day from Earth in late 2026).
- Transmitter: about 22 W at 8.4 GHz through a 3.7 m dish (gain ≈ 48 dBi, EIRP ≈ 61 dBW).
- Free-space loss ≈ 319 dB; received power at a 70 m DSN antenna ≈ $4 \times 10^{-19}$ W (−184 dBW).
- Downlink rate: 160 bit/s (science), reduced over the years as power and distance changed.
- Commands are sent with transmitters of 20 kW or more; a round-trip command-and-response takes almost two days.

### 7.7.4 Delay- and Disruption-Tolerant Networking

The Internet assumes continuous end-to-end connectivity and round-trip times of milliseconds. In space, links are intermittent and delays can be minutes or hours. **Delay/Disruption-Tolerant Networking (DTN)** uses the **Bundle Protocol** to store data at each node until the next link is available ("store-and-forward"), forming an interplanetary internet. NASA has used DTN on the ISS since 2016 and in the PACE Earth-observation mission (2024). It is a core element of **LunaNet**, the planned interoperable network of lunar relays and services, and of ESA's **Moonlight** initiative.

## 7.8 Space Data Standards: CCSDS

The **Consultative Committee for Space Data Systems (CCSDS)**, founded in 1982 by the major space agencies, develops open standards ("Blue Books") for space communications and data handling. Using CCSDS standards lets spacecraft use any compliant ground station and lets agencies share hardware and software. Figure 7.17 shows the main layers.

| | |
|---|---|
| **Application** | File delivery (CFDP), image/data compression (CCSDS 121/122/123), Mission Operations services |
| **Transport / network** | Space Packet Protocol, Encapsulation, IP over CCSDS, Delay/Disruption-Tolerant Networking (Bundle Protocol) |
| **Data link — protocol** | TM / TC / AOS / Proximity-1 / USLP transfer frames; Space Data Link Security (SDLS) |
| **Data link — coding & sync** | Reed–Solomon, convolutional, turbo, LDPC codes; randomisation; frame sync |
| **Physical** | RF modulation (BPSK, QPSK, GMSK, APSK), ranging; optical (CCSDS 141/142 O3K, HPE) |

**FIGURE 7.17** The CCSDS protocol stack used by most civil spacecraft, from physical-layer modulation up to file delivery and mission operations services.

Key CCSDS standards include Space Packets (133.0), TM/TC/AOS frames (132.0/232.0/732.0), the Unified Space Data Link Protocol (USLP, 732.1) and channel coding (131.0, 231.0). Others cover the CCSDS File Delivery Protocol (CFDP, 727.0), Space Data Link Security (355.0), the Bundle Protocol (734.2) and optical communications (141.0, 142.0). Commercial communications

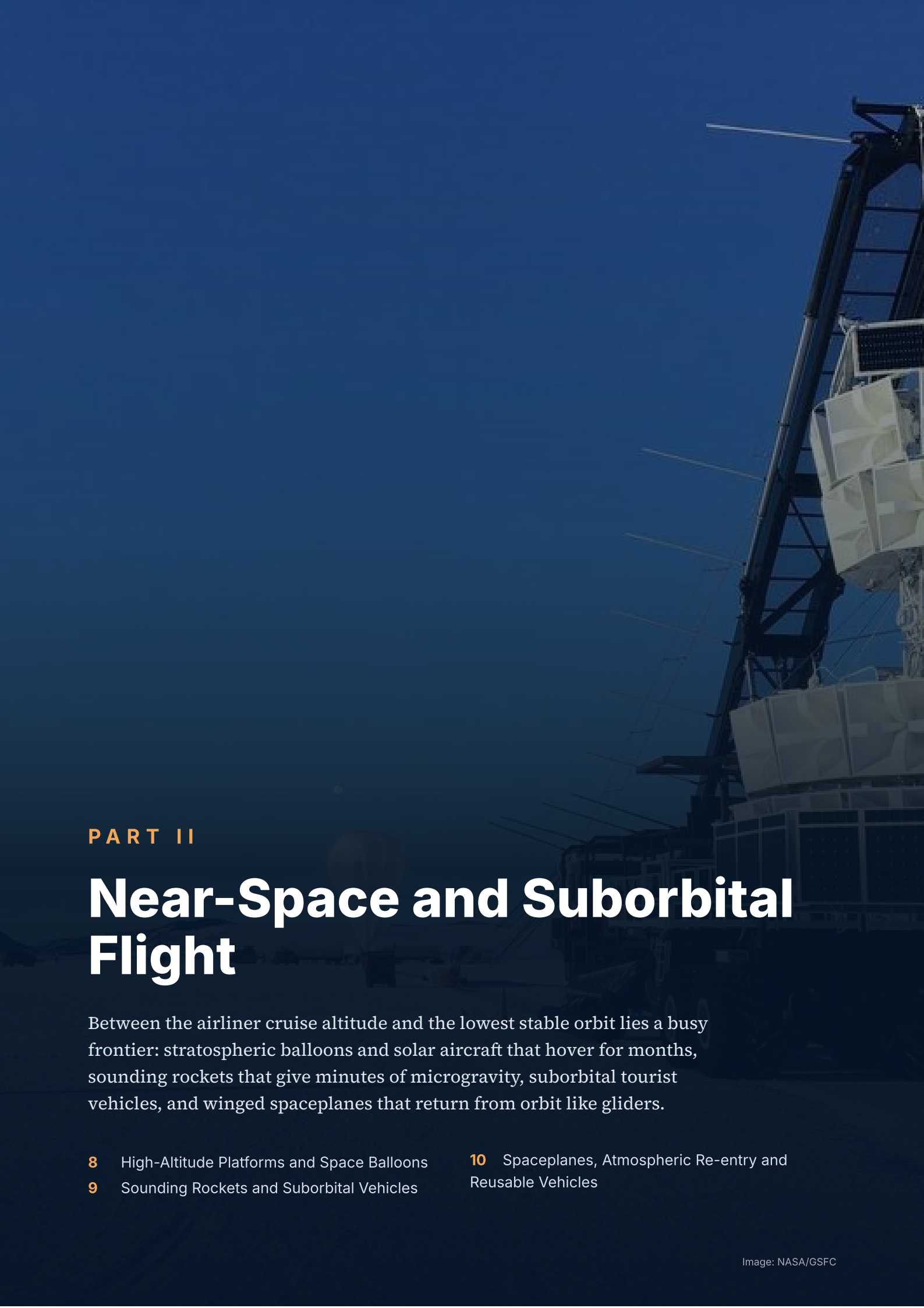
satellites use the DVB-S2/S2X family; LEO broadband constellations use proprietary protocols; and **3GPP non-terrestrial network (NTN)** standards (from Release 17 onwards) integrate satellites into 5G mobile networks (Chapter 17).

### KEY TAKEAWAYS

- TT&C links control the spacecraft; payload links deliver its product. Frequencies are allocated by the ITU; higher bands give more bandwidth but suffer rain fade.
- The link budget — EIRP minus losses plus G/T — determines $C/N_0$ and $E_b/N_0$; free-space loss grows 20 dB per decade of distance.
- Modern LDPC and turbo codes operate within 1–2 dB of the Shannon limit; adaptive coding and modulation use every available dB.
- Laser links deliver gigabit-to-hundreds-of-gigabit rates and now connect satellites in constellations, relays in GEO and even deep-space probes.
- The ground segment includes stations, mission control, flight dynamics, planning and payload data processing; polar stations and relay satellites maximise contact time; the DSN's three complexes provide continuous deep-space coverage.
- CCSDS standards enable interoperability; DTN extends the Internet to intermittent space links.

### REVIEW QUESTIONS

1. Distinguish TT&C links from payload data links and give typical frequency bands for each.
2. Calculate the gain of a 3.7 m ground antenna with 60 % efficiency at 8.2 GHz.
3. Calculate the free-space path loss at 2.2 GHz over a slant range of 2 000 km.
4. Write the link equation for $C/N_0$ and explain each term. Why does −228.6 dBW/(K Hz) appear?
5. Why do modern links use LDPC or turbo codes? What does "within 1–2 dB of the Shannon limit" mean?
6. Give two advantages and two disadvantages of laser communication compared with radio.
7. Why are many ground stations for polar-orbiting satellites located at high latitudes?
8. Why does deep-space communication need very large antennas and three stations spaced around the world?
9. What problem does delay- and disruption-tolerant networking solve, and how?



PART II

# Near-Space and Suborbital Flight

Between the airliner cruise altitude and the lowest stable orbit lies a busy frontier: stratospheric balloons and solar aircraft that hover for months, sounding rockets that give minutes of microgravity, suborbital tourist vehicles, and winged spaceplanes that return from orbit like gliders.


8 High-Altitude Platforms and Space Balloons

9 Sounding Rockets and Suborbital Vehicles

10 Spaceplanes, Atmospheric Re-entry and Reusable Vehicles


Image: NASA/GSFC

CHAPTER

# 8 High-Altitude Platforms and Space Balloons

Image: NASA/GSFC

*The stratosphere, 18 to 50 kilometres above the ground, is too thin for ordinary aircraft and too dense for satellites. Balloons have explored it for more than a century, and a new generation of solar-powered aircraft and airships - high-altitude platform stations, or HAPS - aims to hover there for months as "pseudo-satellites". This chapter explains the physics of buoyancy and solar-powered flight, the types of balloons and HAPS, their engineering challenges, their applications in science, telecommunications, Earth observation and defence, and the current state of the industry.*

IN THIS CHAPTER

8.1 Near Space: The Edge of the Atmosphere
8.2 The Physics of Buoyancy
8.3 Types of Balloons
8.4 High-Altitude Platform Stations (HAPS)
8.5 Other Near-Space Vehicles
8.6 Case Studies
8.7 Safety and Regulation
8.8 Balloons on Other Worlds
8.9 Near Space as a Test Bed for Space Systems

## 8.1 Near Space: The Edge of the Atmosphere

"Near space" usually means the region between the highest altitude of conventional aircraft (about 18–20 km) and the lowest altitude of satellites (about 100–160 km). Its lower part, the stratosphere, has properties that make it uniquely useful:

- **Above the weather.** Almost all clouds, storms and turbulence lie in the troposphere below 12–18 km.

- **Above most of the atmosphere.** At 20 km, 95 % of the atmosphere's mass lies below; at 40 km, 99.7 %. Telescopes see in infrared, ultraviolet and X-ray bands that never reach the ground, and cosmic-ray detectors meet primary particles before they shower.
- **Close to the ground.** A platform at 20 km is 25–30 times closer than a low-orbit satellite and 1 800 times closer than a geostationary one. It can see the ground in fine detail with modest optics and communicate with ordinary phones.
- **Calm winds.** Average winds reach a minimum between about 17 and 23 km (Figure 8.1), between the tropospheric jet stream and the stronger stratospheric winds above.
- **Above controlled airspace.** In most countries, airspace above about 18 km (FL600 in Europe and the US Class E airspace above FL600) is lightly used, which simplifies operations — although regulators are still developing rules for long-duration HAPS.

Its challenges are equally distinctive. Air density at 20 km is only 7 % of sea level (and 0.4 % at 40 km). Temperatures fall to −56 °C or lower (−80 °C near the tropical tropopause), ultraviolet radiation and ozone attack materials, and cooling by convection is weak.

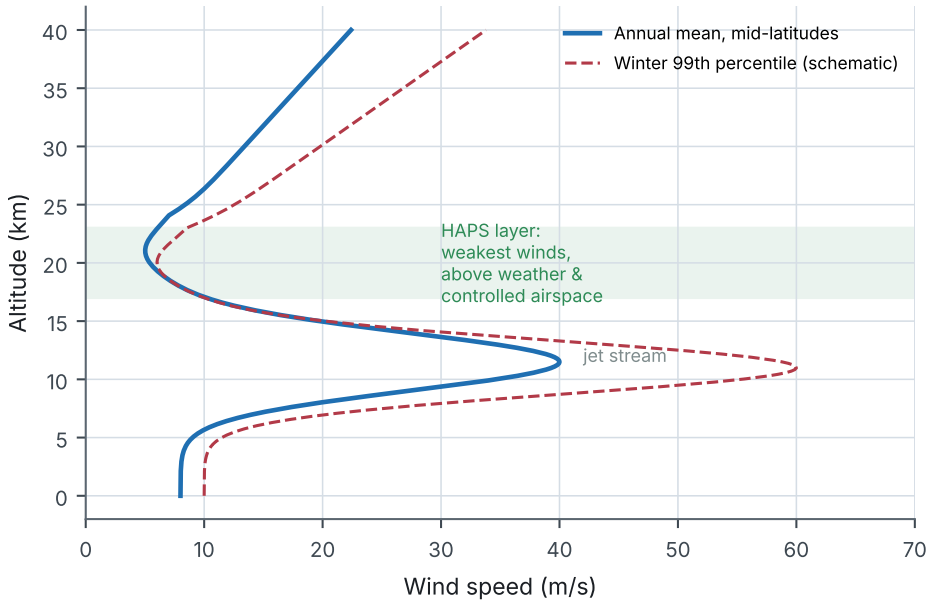



**FIGURE 8.1** Typical wind speed versus altitude at mid-latitudes. The "HAPS layer" between about 17 and 23 km has the weakest winds, making station-keeping practical.

## 8.2 The Physics of Buoyancy

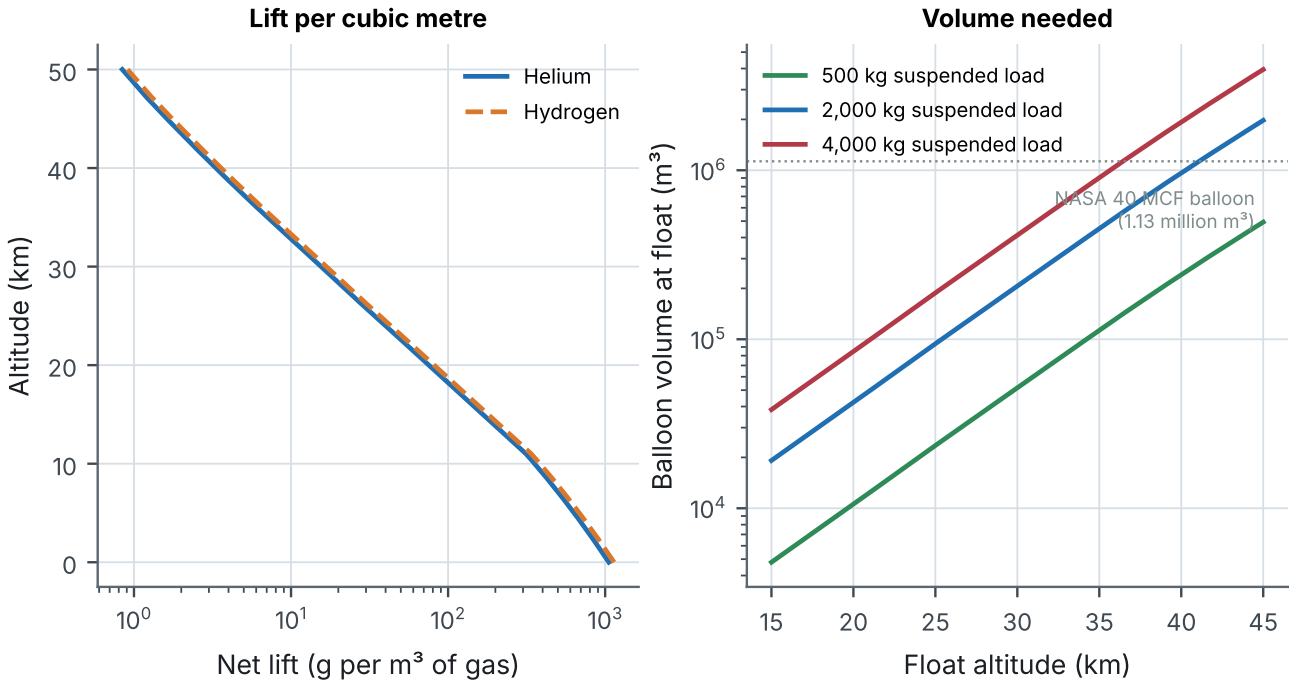
### 8.2.1 Archimedes in the Stratosphere

A balloon floats when the weight of the air it displaces equals the total weight of the balloon, its gas and its payload. The net (free) lift of a volume $V$ of lifting gas is

$$L = V\,(\rho_{air} - \rho_{gas})\,g = V\,\rho_{air}\left(1 - \frac{M_{gas}}{M_{air}}\right)g$$

assuming the gas and surrounding air have the same pressure and temperature, where $M$ is the molar mass. Helium ($M$ = 4.0 g/mol) gives 86 % of the lift of a vacuum; hydrogen ($M$ = 2.0 g/mol) 93 %. Hydrogen is cheaper and lifts more but is flammable; helium is inert but scarce and expensive, and some programmes (including Loon and many weather services) use hydrogen.

Because air density falls exponentially with height, the lift of each cubic metre of gas falls by a factor of about 160 between sea level and 40 km (Figure 8.2). A balloon must therefore expand enormously as it rises. A NASA 40 MCF (million cubic feet) zero-pressure balloon has a volume of 1.13 million m³ at float — large enough to hold a football stadium. Yet at launch only a small bubble at its top is filled with helium. The rest of the polyethylene film hangs limp below.




Gross mass ≈ 1.6 × load (balloon envelope, ballast, rigging). US Standard Atmosphere 1976.

**FIGURE 8.2** Left: net lift per cubic metre of helium and hydrogen versus altitude. Right: balloon volume required at float to carry a given payload, assuming the total system mass is about 1.6 times the suspended load.

**WORKED EXAMPLE — HOW BIG A BALLOON FOR A 2-TONNE TELESCOPE AT 38 KM?**

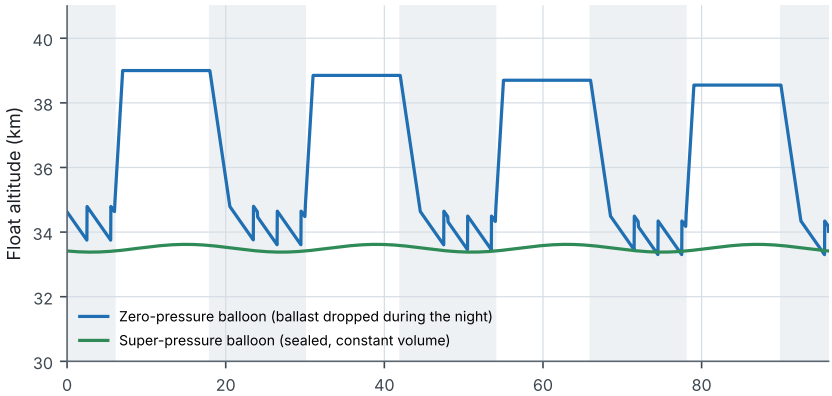
At 38 km the US Standard Atmosphere gives $\rho \approx 0.0058$ kg/m³. Each cubic metre of helium lifts $0.0058 \times (1 - 4.00/28.96) = 0.0050$ kg = 5.0 g. The suspended load is 2 000 kg. Adding the balloon film (≈ 1 800 kg for a 1 million m³ balloon), ballast, parachute and rigging gives a gross mass of about 3 200–4 000 kg. So the balloon needs about **650 000–800 000 m³** at float. NASA would fly this on a 29–40 MCF balloon (0.8–1.13 million m³). The film is polyethylene only about 20 µm thick — thinner than a sandwich bag — so that the balloon itself does not become too heavy.

## 8.2.2 Thermal Effects and Super-Pressure

The lift of a balloon depends on the gas temperature. During the day, sunlight and upwelling infrared warm the gas above the surrounding air ("superheat"), increasing lift; at sunset the gas cools by 20–40 K and lift falls sharply. This drives the main design distinction between balloon types:

- A **zero-pressure balloon** is open at the bottom through vent ducts, so internal and external pressures are equal. When the gas cools at night it contracts, the balloon loses lift and sinks; to stop the descent, ballast (fine steel shot or glass beads) is dropped. Each night costs typically 5–10 % of the gross mass in ballast, limiting flights to days or a few weeks — except in polar summer, when the Sun never sets (Figure 8.3).

- A **super-pressure balloon (SPB)** is sealed. It is designed so that, even when the gas is coldest at night, it is still fully inflated with a slight overpressure. At noon the overpressure rises, but the volume (and therefore the altitude) barely changes. Super-pressure balloons can fly for months without ballast. The challenge is building a hull that withstands the pressure difference: NASA uses a "pumpkin" design of polyethylene gores with high-strength load tendons, while Loon used a smaller, simpler polyethylene envelope.




Schematic. At sunset the lifting gas cools and a zero-pressure balloon sinks until ballast is released; at sunrise it warms and rises, venting gas. Each night costs ballast, limiting flight duration.

**FIGURE 8.3** Schematic altitude history over four days for zero-pressure and super-pressure balloons. Night-time descent forces a zero-pressure balloon to drop ballast; a super-pressure balloon holds altitude.

## 8.3 Types of Balloons

Figure 8.4 and Table 8.1 summarise the main families.

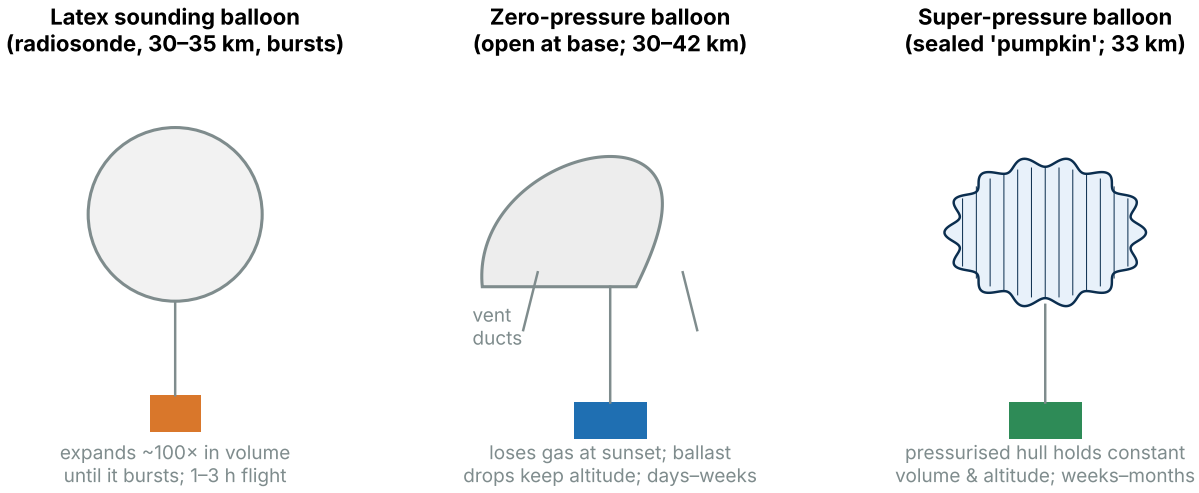



**FIGURE 8.4** The three main families of free balloons: latex sounding balloons that burst, zero-pressure balloons that vent gas, and sealed super-pressure balloons.

**TABLE 8.1** Main types of free balloons.

| Type | Typical volume | Payload | Float altitude | Duration | Main users |
|---|---|---|---|---|---|
| Latex sounding (weather) balloon | 1–3 m³ at launch (bursts at ≈ 10 m diameter) | 0.1–3 kg (radiosonde) | Rises to 30–35 km, bursts | 1.5–3 h | Weather services: ≈ 800 stations launch twice daily (≈ 1 500 per day worldwide) |

| Type | Typical volume | Payload | Float altitude | Duration | Main users |
|---|---|---|---|---|---|
| Small zero-pressure | 1 000–100 000 m³ | 5–500 kg | 25–35 km | Hours–days | Universities, atmospheric chemistry, technology tests |
| Large zero-pressure ("heavy lift") | 0.3–1.13 million m³ | 1 000–3 600 kg | 35–42 km | 1 day (conus), 10–57 days (polar) | NASA, CNES, JAXA, ISRO: astrophysics, cosmic rays |
| Super-pressure (scientific) | ≈ 530 000 m³ (18.8 MCF) | up to ≈ 1 t | ≈ 33 km | Weeks–months | NASA SPB programme (Wanaka, New Zealand) |
| Small super-pressure | 10–500 m³ | 1–50 kg | 15–20 km | Weeks–months | CNES Stratéole-2, Loon (ended 2021), Aerostar, World View |
| Montgolfière / infrared balloon | variable | small | 20–30 km | Weeks | CNES (historic MIR balloons), Venus concepts |
| Tethered aerostat | 100–10 000 m³ | 50–1 000 kg | 0.5–5 km | Weeks | Border surveillance, telecom relays |

### 8.3.1 Scientific Ballooning

Balloons have been scientific platforms since Victor Hess discovered cosmic rays on a manned balloon flight in 1912 (Nobel Prize 1936). In the space age they remain the cheapest way to reach near-space conditions with heavy instruments, and they are an important training ground for instrument scientists. Notable achievements include:

- **BOOMERanG** (1998) and **MAXIMA** — balloon-borne measurements of the cosmic microwave background that established that the universe is geometrically flat.
- **BLAST** — sub-millimetre surveys of star-forming galaxies; **SPIDER** — CMB polarisation from Antarctica.
- **CREAM** and **SuperTIGER** — direct measurements of heavy cosmic-ray nuclei; SuperTIGER flew 55 days over Antarctica in 2012–13.
- **GUSTO** (Galactic/Extragalactic ULDB Spectroscopic Terahertz Observatory) — a terahertz telescope that flew about 57 days over Antarctica from December 2023, a record for a NASA heavy-lift balloon.
- **COSI** — a Compton gamma-ray telescope whose 2016 super-pressure flight paved the way for the COSI satellite (launch planned 2027).
- **EUSO-SPB2** (2023) — ultra-high-energy cosmic-ray and neutrino detection from a super-pressure balloon.
- **Stratéole-2** (CNES/LMD) — fleets of small super-pressure balloons drifting for months in the tropical lower stratosphere, measuring winds, temperature, water vapour and waves; they also provided validation data for the Aeolus wind lidar satellite.

NASA's **Columbia Scientific Balloon Facility (CSBF)**, operated from Palestine, Texas, launches from Fort Sumner (New Mexico), Palestine, McMurdo Station (Antarctica), Kiruna (Sweden), Wanaka (New Zealand) and Alice Springs (Australia). Antarctic **long-duration balloon (LDB)**

flights exploit the austral summer — continuous daylight and a circumpolar wind pattern that returns the balloon close to McMurdo after one to three circuits of the continent. CNES operates from Aire-sur-l'Adour, Kiruna, Timmins (Canada) and the Seychelles; JAXA from Taiki (Hokkaido); ISRO from Hyderabad.

**FIGURE 8.5** A NASA super-pressure balloon carrying the COSI gamma-ray telescope ready for launch from McMurdo Station, Antarctica, in the 2014–15 campaign. Image: NASA/GSFC.

**FIGURE 8.6** The launch vehicle holds the ANITA radio-antenna payload, which searches for ultra-high-energy neutrinos interacting in the Antarctic ice, moments before a long-duration balloon launch. Image: NASA/GSFC.

### 8.3.2 Balloon Operations

A heavy-lift launch begins before dawn in calm winds (typically below 3 m/s at the surface and in the first few hundred metres). The payload hangs from a launch vehicle — a large mobile crane — while the balloon is laid out on a ground cloth and its top bubble is inflated with helium through fill tubes. On release, the bubble rises and lifts the rest of the balloon off the ground; the launch vehicle drives underneath to release the payload smoothly. Ascent takes 2–3 hours at about 5 m/s.

At float, the payload is controlled from the ground through line-of-sight telemetry, Iridium, TDRS or Starlink links. The flight is terminated by command: a pyrotechnic cutter releases the payload, which descends on a large parachute, while a rip panel opens the balloon so that it falls separately. Recovery teams then retrieve payloads, sometimes from remote Antarctic plateaus by aircraft. Safety rules forbid termination over populated areas, which limits flights from continental sites.

Engineers' main challenges include balloon film reliability (cold embrittlement at the tropopause) and thermal control of payloads (convective cooling is weak but not absent). Another is pointing of telescopes from a swinging gondola (arcsecond-level pointing is achieved with momentum wheels, reaction against the flight train, and star cameras). The others are power (solar panels) and communications.

### 8.3.3 Balloon and Gondola Design

**Envelope design.** A zero-pressure balloon at float takes up a "natural shape" — an inverted teardrop in which the film carries tension only along the meridians (Figure 8.7). The shape is computed by solving the equilibrium between internal gas pressure, film weight and the tension

carried by **load tapes**. These tapes run from the apex to the base along every seam. The gas pressure increases with height inside the balloon, because the gas column is lighter than the surrounding air. NASA's large balloons are made from dozens to more than 200 **gores** — long, narrow panels of linear low-density polyethylene film 20–38 µm thick, heat-sealed together. The film must stay flexible at the −80 °C tropopause, resist ultraviolet light, and have no pinholes. Manufacturing is done by hand on tables more than 100 m long; a 40 MCF balloon contains about 24 hectares of film.

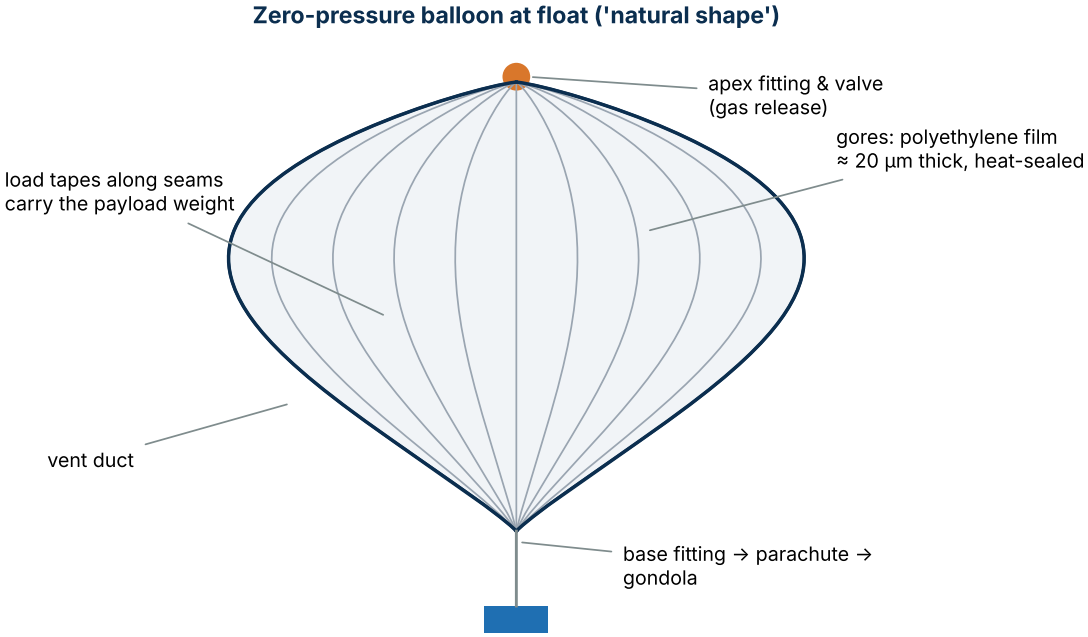



**FIGURE 8.7** Construction of a zero-pressure balloon at float. Load tapes along the gore seams carry the payload weight to the apex; vent ducts release gas when the balloon is fully inflated.

**Super-pressure hulls.** NASA's 18.8 MCF super-pressure balloon uses a "pumpkin" shape: the film bulges between high-strength tendons so that the film itself carries only a small local stress while the tendons carry the global load. Developing a reliable pumpkin balloon took more than a decade because of a deployment instability ("clefting") in which the lobes fold incorrectly during inflation. Flights from Wanaka, New Zealand, have lasted up to about 55 days at mid-latitudes. A 2025 campaign carried the HIWIND thermospheric wind instrument on a flight of about 17.5 days. It also launched a second balloon intended for up to 100 days.

**Gondolas.** The science payload hangs in a **gondola** — an aluminium or carbon-fibre frame carrying the instrument, solar panels, batteries, flight computers, telemetry, and crush pads for landing (Figure 8.8). Astrophysics gondolas point telescopes to a few arcseconds. Azimuth is controlled by a motorised pivot that torques against the balloon flight train and by a reaction wheel, while elevation is controlled by driving the telescope itself. Attitude is sensed with daytime star cameras, gyros, GPS compasses and sun sensors. Gondolas must survive a parachute landing at 5–8 m/s and, for the heaviest payloads, the "opening shock" of the parachute (up to 8–10 g).

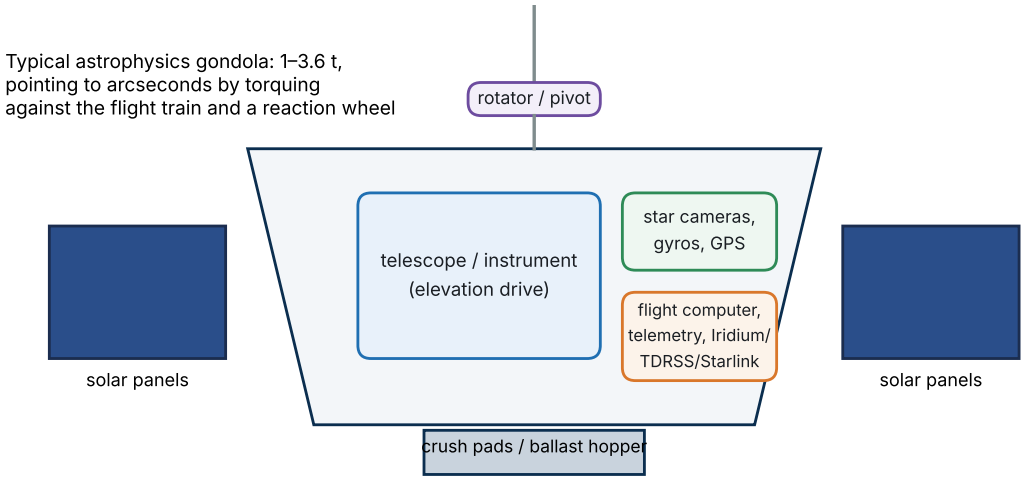



**FIGURE 8.8** A typical balloon-borne astrophysics gondola. Solar panels hang outside; the instrument points by torquing against the flight train.

### 8.3.4 Radiosondes: Balloons for Every Weather Forecast

The most numerous balloons by far are the small latex balloons that carry **radiosondes** — the backbone of upper-air meteorology since the 1930s. About 800 stations in the WMO Global Observing System launch them, mostly at 00 and 12 UTC, adding up to roughly 1 500 soundings per day worldwide. A modern radiosonde (for example the Vaisala RS41, Graw DFM-17 or Meteomodem M20) weighs 80–110 g. It measures temperature (with a small platinum or thermistor sensor), relative humidity (thin-film capacitive sensor) and pressure (or derives it from GNSS height). By tracking its GNSS position, it also measures winds. It transmits at around 400 MHz to a ground receiver for the 1.5–2-hour ascent, reaching 30–35 km before the balloon bursts (Figure 8.9).

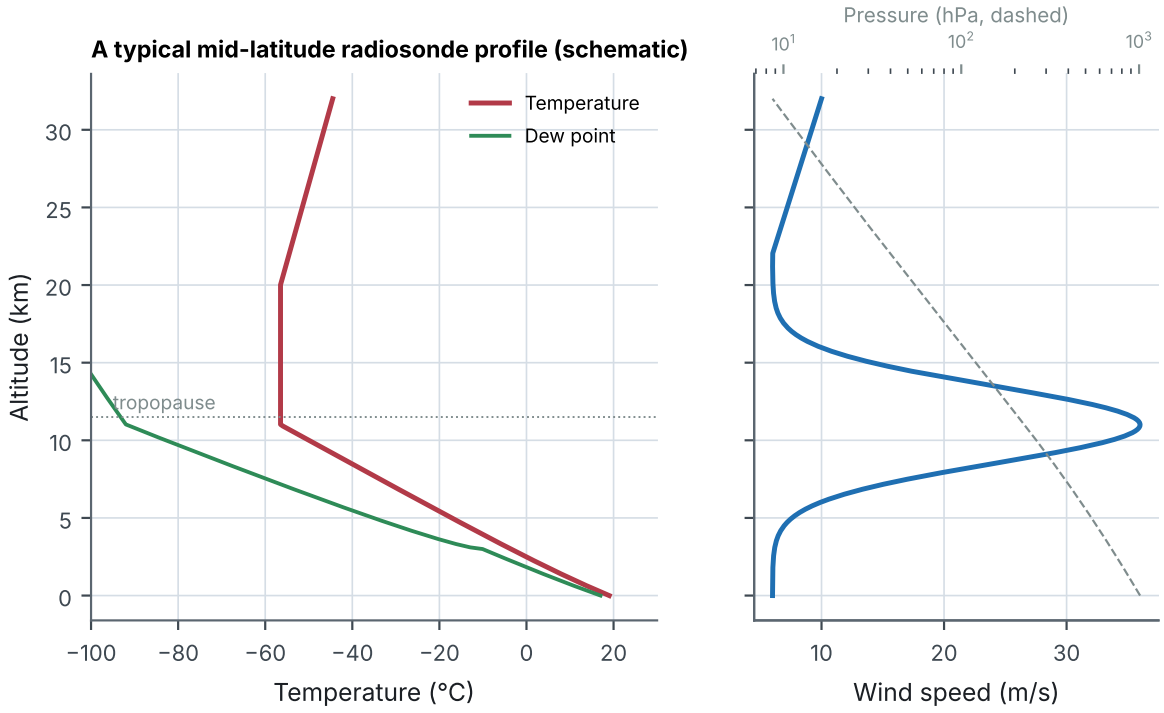



**FIGURE 8.9** A schematic mid-latitude radiosonde profile: temperature falls through the troposphere to the tropopause, the dew point shows how moist the air is, and winds peak in the jet stream near 10–12 km.

Radiosonde data are few compared with satellite observations, but they are among the most valuable inputs to numerical weather prediction. This is because they measure vertical profiles directly and serve as an absolute reference to calibrate satellite sounders (Chapter 16). **Dropsondes**, released from aircraft into hurricanes and storms, use the same technology in reverse. Weather-balloon launch stations are increasingly automated with "autolaunchers" that inflate and release balloons without staff.

## 8.4 High-Altitude Platform Stations (HAPS)

### 8.4.1 The Pseudo-Satellite Concept

A **high-altitude platform station** (the ITU term) or **high-altitude pseudo-satellite** is an aircraft, airship or balloon that stays in the stratosphere for weeks to months, typically at 18–25 km. It holds position over a region like a geostationary satellite, but much closer to the ground. HAPS promise:

- **Telecommunications** directly to unmodified phones (4G/5G "cell tower in the sky") over an area 100–200 km across, with latency below a millisecond, for rural connectivity, disaster recovery and events.
- **Earth observation** with persistence: continuous imaging of one area at 10–20 cm resolution, rather than a satellite snapshot once or twice a day. Uses include wildfire monitoring, maritime surveillance, border security, crop monitoring and environmental science.
- **Navigation augmentation** and **defence** applications such as communications relays and persistent intelligence, surveillance and reconnaissance (ISR).
- **Rapid deployment** without launch: a HAPS can be repositioned or recovered and upgraded.

The ITU allocated spectrum for HAPS (in bands near 2 GHz, 6 GHz, 21–31 GHz and 38–48 GHz) at WRC-19. At WRC-23, it approved the use of HAPS as **IMT base stations (HIBS)** in mobile bands below 2.7 GHz. This allows them to serve ordinary phones.

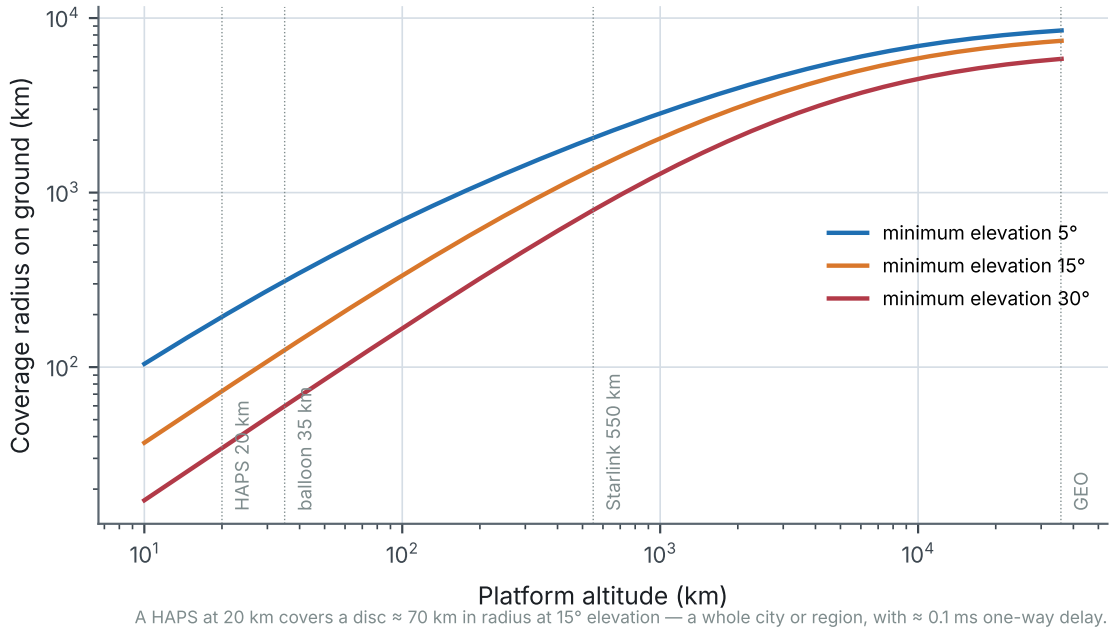



**FIGURE 8.10** Coverage radius on the ground versus platform altitude for three minimum elevation angles. A stratospheric platform covers a city-sized region; a LEO satellite covers a much larger area but moves away within minutes.

## 8.4.2 Three Architectures

HAPS designs fall into three families (Figure 8.11), each with distinct trade-offs:

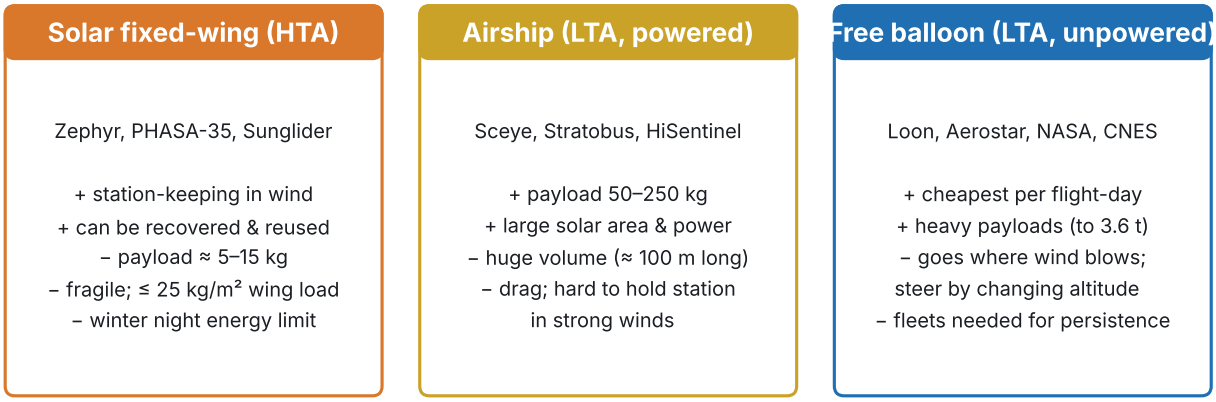



**FIGURE 8.11** The three HAPS architectures and their main advantages and limitations.

**Solar-powered fixed-wing aircraft (heavier than air).** Extremely light, high-aspect-ratio wings covered in solar cells power electric propellers by day and charge batteries for the night. Airbus' **Zephyr** — now operated by **AALTO HAPS** — has a 25 m wingspan and a mass of about 75 kg. In 2022 it flew 64 days continuously, and AALTO reported a new record of about 67 days in 2025. Commercial service is targeted from 2026 (including an operating base in Kenya). BAE Systems' **PHASA-35** (35 m span, 150 kg) completed stratospheric trials in 2023–2024; AeroVironment/SoftBank's **Sunglider** (78 m) flew to the stratosphere in 2020. NASA's **Pathfinder**, **Centurion** and **Helios** prototypes pioneered the concept: Helios reached a record 29.5 km in 2001 before breaking up in turbulence in 2003.

**FIGURE 8.12** NASA's Helios prototype, a 75 m solar-electric flying wing, taking off from the Pacific Missile Range Facility on Kauai, Hawaii, for its 2001 record flight to an altitude of 29 524 m — a record for a non-rocket-powered aircraft in sustained horizontal flight. Image: NASA/AFRC (NASA/Nick Galante).

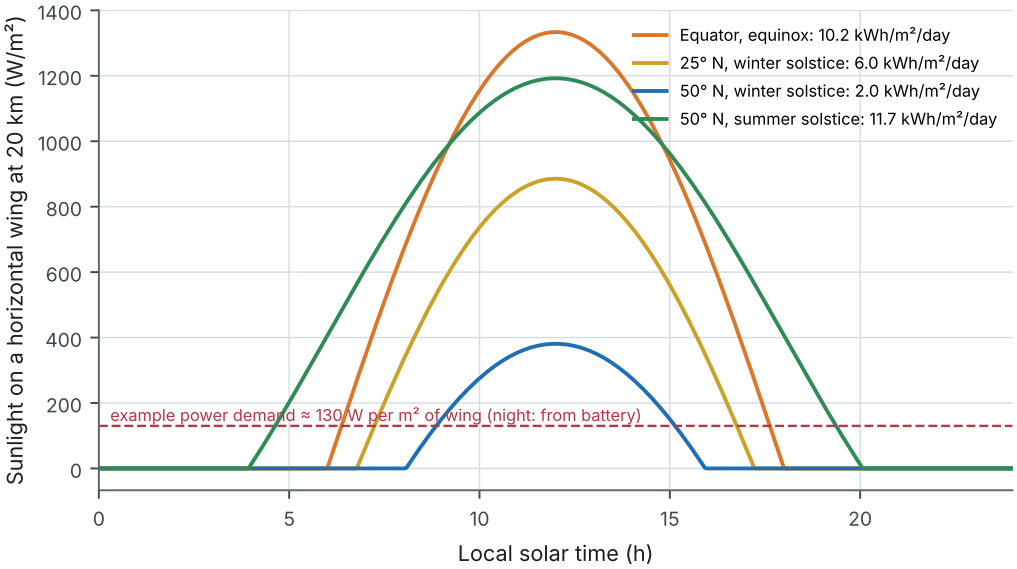
**Airships (lighter than air, powered).** Stratospheric airships use helium for lift and solar-powered propellers to hold position. Their large volume (typically 50–100 m long) gives large solar arrays and payloads of 100 kg or more. **Sceye** (USA) has flown a series of stratospheric airship tests from New Mexico since 2021, including a long-distance mission, and announced plans with SoftBank to start pre-commercial services in Japan in 2026. Thales Alenia Space's **Stratobus** and Lockheed Martin's **HiSentinel** are other examples. The difficulty is drag: a large envelope must push against winds that can exceed 20–30 m/s at times.

**Free balloons with altitude control (lighter than air, unpowered).** Google's **Loon** (2013–2021) showed that super-pressure balloons can be "steered" by changing altitude to catch winds blowing in different directions, using an air-ballast pump (a "ballonet") and machine-learning navigation (Chapter 48). Loon flew more than 40 million km in total and set a record of 312 days aloft for one balloon. It provided emergency LTE service in Puerto Rico after Hurricane Maria (2017) and commercial service in Kenya (2020). It shut down in 2021 because the business case did not close. Its flight data were published for research. **Aerostar** (Thunderhead balloons) supplies similar balloons for US defence experiments; **World View**, **Urban Sky**, **Near Space Labs** and others offer stratospheric imaging and ride services.

### 8.4.3 Engineering Challenges

**Energy closure.** The defining problem of solar HAPS is surviving the night. The platform must collect enough energy during the day to fly all day and night. Figure 8.13 shows that at the equator a horizontal wing collects about 10 kWh/m$^2$/day. At 50° latitude in winter it collects only about 2 kWh/m$^2$/day — with 16-hour nights. With present batteries and solar cells, most fixed-wing HAPS can operate year-round only within about ±25–35° of the equator. The batteries are

lithium-ion at 250–350 Wh/kg, with lithium-sulphur and silicon-anode cells promising 400–500 Wh/kg. The solar cells are thin-film or lightweight multi-junction types (20–30 %). Batteries often make up 30–40 % of the aircraft mass.




Computed clear-sky irradiance above most of the atmosphere. Winter nights at high latitude are the design case: little energy to collect, long nights to survive.

**FIGURE 8.13** Clear-sky solar irradiance on a horizontal wing at 20 km over one day for four latitude/season combinations. The energy collected per day ranges from about 12 kWh/m² in midsummer at 50° N to 2 kWh/m² in midwinter — the central design challenge for solar HAPS.

The power needed for level flight follows from aerodynamics. For an aircraft of mass $m$, wing area $S$, lift coefficient $C_L$ and drag coefficient $C_D$ flying at air density $\rho$:

$$P_{level} = \frac{D\,v}{\eta_{prop}} = \frac{mg}{\eta_{prop}}\,\frac{C_D}{C_L}\sqrt{\frac{2mg}{\rho S C_L}}$$

Because $\rho$ at 20 km is 14 times smaller than at sea level, the flight speed — and power — is $\sqrt{14} \approx 3.7$ times greater for the same aircraft. HAPS aircraft therefore have very low wing loadings (2–4 kg/m², a hundred times less than an airliner), very high lift-to-drag ratios (25–40), and must be extraordinarily light.

**Structural fragility and weather.** The lightest structures are vulnerable during ascent and descent through the troposphere, where turbulence and gusts are strongest. Most HAPS losses have occurred in these phases: Helios broke up in turbulence in 2003; a Zephyr crashed after 64 days in 2022; Facebook's Aquila programme ended in 2018 after a landing accident. Launch windows require calm surface winds.

**Station-keeping.** To act as a pseudo-satellite, a platform must stay within a station-keeping circle of perhaps 5–20 km radius. Aircraft fly loiter patterns; airships push against the wind; balloons use altitude control or fleets.

**Thermal and materials.** Cold temperatures (−60 to −80 °C) reduce battery capacity and make films brittle; ultraviolet radiation and ozone degrade polymers; solar cells run hot in weak convection.

**Regulation.** Long-endurance HAPS require certification and airspace rules that do not yet exist in many countries; flights through lower airspace must be coordinated with air-traffic control.

### 8.4.4 HAPS Payloads

**Telecommunications payloads.** A HAPS base station uses a phased-array or multi-beam antenna that projects tens of "cells" onto the ground, each acting like a terrestrial cell sector. Because the platform moves within its station-keeping circle and changes attitude, beams are steered electronically to stay fixed on the ground. The feeder link to a ground gateway (or to a satellite) uses Ka- or E-band, or free-space optics. Japan's SoftBank/HAPSMobile demonstrated LTE connectivity from Sunglider in 2020 and 5G-class links from Sceye in later tests; Loon's balloons relayed traffic between balloons over hundreds of kilometres using millimetre-wave mesh links. Capacity per HAPS is typically hundreds of Mbit/s to a few Gbit/s.

**Earth-observation payloads.** At 20 km, a camera with a 10 cm aperture achieves sub-metre resolution; 20–30 cm resolution is achievable with payloads of only a few kilograms. HAPS EO payloads include visible/near-infrared cameras, thermal infrared cameras for wildfire detection, hyperspectral imagers, and small synthetic aperture radars. Their key advantage is **persistence** — the same area can be watched continuously for hours or days, enabling detection of change and movement that satellites revisiting once a day would miss.

**Science payloads.** Stratospheric aircraft and airships can carry in-situ sensors for ozone, water vapour and aerosols, and long-duration balloons carry instruments for gravity-wave, atmospheric-chemistry and climate studies.

DATA BOX — REPRESENTATIVE HAPS (2026)

| Platform | Type | Span / length | Mass | Payload | Endurance (demonstrated) |
|---|---|---|---|---|---|
| AALTO Zephyr S | Solar aircraft | 25 m | ≈ 75 kg | ≈ 5 kg | ≈ 64–67 days |
| BAE PHASA-35 | Solar aircraft | 35 m | ≈ 150 kg | ≈ 15 kg | Stratospheric trials (hours–days) |
| Sceye | Solar airship | ≈ 65 m | ≈ 1 t-class | 100–250 kg (planned) | Multi-day missions |
| Loon (retired) | Super-pressure balloon | 15 m diameter | ≈ 75 kg (+ payload) | ≈ 15 kg | 312 days (record) |
| NASA super-pressure | Balloon | 0.53 million $m^3$ | ≈ 2.5 t system | ≈ 1 t | ≈ 55 days (2016 record flights) |

### 8.4.5 Applications and Business Case

The HAPS industry has experienced several boom-and-bust cycles: Google (Loon, Titan), Facebook (Aquila) and others invested heavily in the 2010s and withdrew. The renewed wave in the mid-2020s rests on better batteries, regulatory progress (WRC-23 HIBS decisions) and defence demand for persistent ISR and communications. It also rests on partnerships with mobile network operators (e.g., SoftBank, NTT Docomo, Deutsche Telekom, Vodafone in trials). HAPS compete with — and complement — LEO direct-to-device satellites (Chapter 17), drones, and terrestrial towers. Their advantages are persistence over one area, low latency, high capacity density and low cost per unit area served. Their disadvantages are weather sensitivity during ascent/descent, latitude limits and the need for fleets to provide continuous service.

## 8.5 Other Near-Space Vehicles

- **High-altitude aircraft.** The Lockheed U-2 and NASA's **ER-2** fly at up to about 21 km with a pilot in a pressure suit, carrying Earth-science instruments that test future satellite sensors. NASA's WB-57 (18 km) supports eclipse observations and launch tracking.
- **High-altitude drones.** RQ-4 Global Hawk (18 km, 34+ h endurance) and Chinese, European and Israeli equivalents.
- **Stratospheric tourism.** Several companies planned pressurised-capsule balloon flights to about 30 km. Space Perspective (Neptune capsule) flew an uncrewed test in 2024 but ran into financial difficulty. It was acquired by the Spanish company **EOS X Space** in 2025. Zephalto (France) and Halo Space (Spain) pursue similar concepts. Such flights are not spaceflight by any definition but offer views of Earth's curvature and the black sky.
- **Record jumps.** Joseph Kittinger (31.3 km, 1960), Felix Baumgartner (39.0 km, 2012, supersonic freefall) and Alan Eustace (41.4 km, 2014) used balloons to test pressure suits and parachute systems relevant to crew escape.

**FIGURE 8.14** NASA's ER-2 high-altitude research aircraft in flight. Flying at up to 21 km, ER-2s carry prototypes of satellite instruments and collect validation data for Earth-observation missions. Image: NASA/AFRC (NASA/Lori Losey).

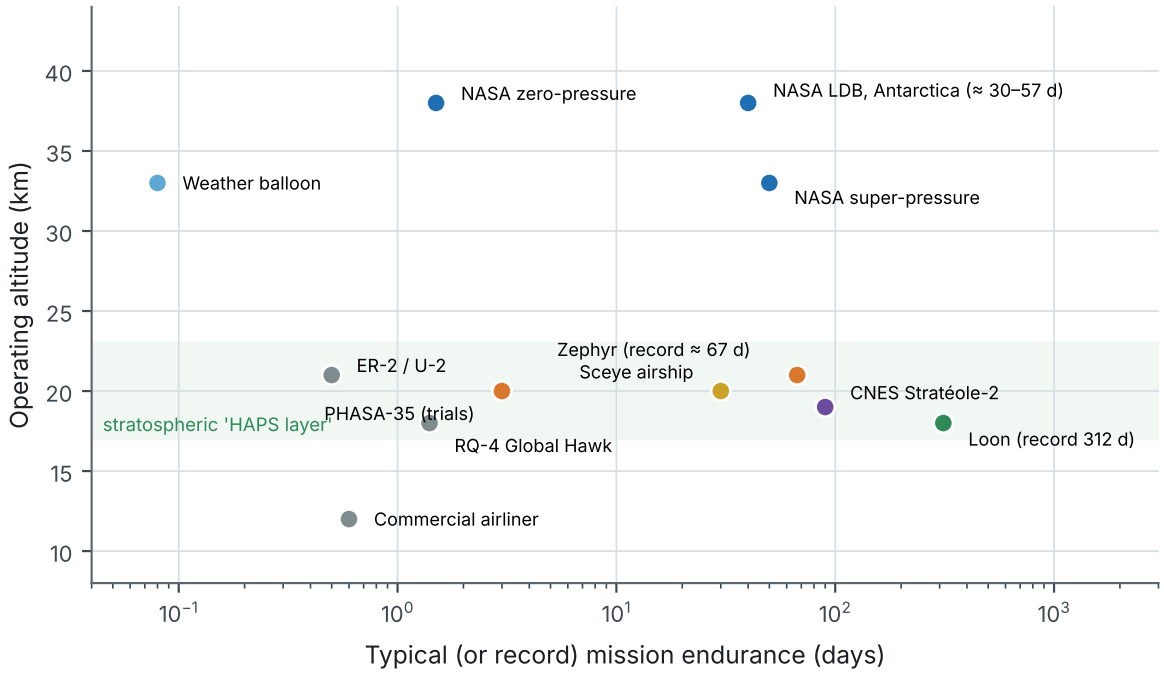



**FIGURE 8.15** Operating altitude versus typical or record endurance for atmospheric and stratospheric platforms. HAPS occupy a niche between aircraft (hours) and satellites (years).

## 8.6 Case Studies

### 8.6.1 Loon: Navigating by the Wind

Loon's balloons were 15 m-diameter super-pressure envelopes carrying about 15 kg of electronics powered by solar panels. Each balloon could change its altitude between about 15 and 20 km by pumping air into or out of an internal ballonet. Because winds at different altitudes blow in different directions, choosing an altitude was equivalent to choosing a direction. A fleet-management system combined global weather forecasts with real-time wind measurements from the balloons themselves to plan altitude changes. In 2020 Loon published results showing that a **deep reinforcement-learning controller**, trained in simulation, kept balloons near their target station better than the hand-engineered controller. This was one of the first operational uses of reinforcement learning in aerospace (Chapter 48). Loon's lessons for the industry included the value of fleet operations, the importance of regulatory partnerships in each country, and the difficulty of competing economically with terrestrial towers and LEO satellites.

### 8.6.2 Zephyr: Surviving the Night

Zephyr's flights illustrate the energy balance of solar HAPS. The aircraft climbs through the troposphere at dawn, reaches the stratosphere by mid-morning, and continues climbing to 21–23 km while its batteries charge. At night it slowly descends — storing energy as altitude in a glider-like "potential-energy battery" — to about 15–18 km by dawn, when the cycle repeats. Zephyr's record flights were made from Arizona in northern summer, when nights are short; year-round operation is planned from low-latitude bases such as Kenya.

### 8.6.3 Stratéole-2: Science in the Tropical Stratosphere

The French–American Stratéole-2 programme launches fleets of small super-pressure balloons from the Seychelles that drift around the Earth in the tropical lower stratosphere at 18–20 km for up to three months. They carry instruments for winds, temperature, water vapour, aerosols and clouds, and lidar and GNSS radio-occultation profilers. The data improve understanding of the quasi-biennial oscillation, gravity waves and stratospheric water vapour, and validated the Aeolus Doppler wind lidar satellite.

## 8.7 Safety and Regulation

Free balloons are regulated as aircraft. In the United States, 14 CFR Part 101 sets rules for unmanned free balloons. Heavy balloons (payloads above 6 lb with certain density limits, or above 12 lb) require notification of air-traffic control, position reporting, lighting, and a reliable termination system. ICAO Annex 2 provides the international framework, and balloons crossing borders need overflight permissions. Scientific balloon programmes conduct casualty-risk analyses for descent and terminate flights only over unpopulated areas. HAPS aircraft are being certified under special conditions developed by the European Union Aviation Safety Agency (EASA), the US FAA and national authorities, and ICAO is developing provisions for operations above FL600 (“higher airspace operations”).

## 8.8 Balloons on Other Worlds

Balloons can also explore other planets. In 1985 the Soviet **Vega 1** and **Vega 2** missions released two 3.5 m helium balloons into the atmosphere of **Venus** at about 54 km altitude. They floated for about 46 hours each, travelling more than 11 000 km in the super-rotating winds. They transmitted data to an international network of radio telescopes on Earth. NASA and JPL have studied variable-altitude “aerobots” for Venus (tested in the Nevada desert in 2022). Titan’s thick, cold atmosphere would make balloon (or rotorcraft) flight easy — the basis of NASA’s Dragonfly mission (Chapter 28). Mars’ thin atmosphere makes balloons very large; concepts use super-pressure or solar-heated Montgolfière designs.

## 8.9 Near Space as a Test Bed for Space Systems

The stratosphere above 30 km has less than 1 % of sea-level pressure and conditions resembling Mars’ surface pressure (≈ 600 Pa at about 35 km). It is therefore used to test:

- **Mars entry, descent and landing systems** — NASA’s Viking parachutes (1972) and the **Low-Density Supersonic Decelerator (LDSD)** tests (2014–2015) were dropped from balloons at 37 km and accelerated by rockets to Mars-relevant supersonic speeds.
- **Space instruments** — detectors, star trackers and telescopes are flown on balloons before satellite missions.
- **Mars aircraft and balloons** — prototypes of planetary aerial vehicles.
- **Astrobiology** — sampling of microbes carried to the stratosphere and survival studies.

### KEY TAKEAWAYS

- The stratosphere offers a unique vantage point: above weather and most of the atmosphere, close to the ground, with a wind minimum near 20 km.
- Balloon lift comes from displaced air; lift per cubic metre falls about 160-fold from sea level to 40 km, so stratospheric balloons reach volumes of up to 1.13 million $m^3$.
- Zero-pressure balloons vent gas and must drop ballast at night; sealed super-pressure balloons hold altitude for months.
- HAPS — solar aircraft, airships and steerable balloons — aim to be persistent "pseudo-satellites" for telecom and Earth observation; the energy balance of the night limits fixed-wing HAPS to lower latitudes with current batteries.
- After earlier boom-and-bust cycles, HAPS are approaching commercial service in the mid-2020s, supported by ITU spectrum decisions and defence demand; balloons remain the cheapest route to near-space science.

### REVIEW QUESTIONS

1. Why is the stratosphere near 20 km attractive for persistent platforms? Mention winds, weather and distance to the ground.
2. At 36 km the air density is about 0.007 2 $kg/m^3$. Estimate the helium volume needed to float a total mass of 3 500 kg (gondola, ballast and envelope), assuming the gas and the air are at the same temperature and pressure.
3. Explain the difference between zero-pressure and super-pressure balloons. Why must a zero-pressure balloon drop ballast at night?
4. Describe the three main HAPS architectures and give one advantage and one limitation of each.
5. Why does the energy balance of the night restrict solar-powered fixed-wing HAPS to lower latitudes and summer seasons?
6. How did Loon steer its balloons without propulsion? What data and algorithms made this possible?
7. What does a radiosonde measure, and why do weather services still launch hundreds every day in the satellite era?
8. Why is near space a useful test bed for space hardware and instruments? Give two examples.

CHAPTER

# 9 Sounding Rockets and Suborbital Vehicles

Image: NASA/GSFC

*A suborbital flight goes up into space and comes straight back down, without reaching orbital speed. Sounding rockets have used such flights since the 1940s to measure the upper atmosphere, observe the Sun and stars, test spacecraft technology and provide minutes of high-quality weightlessness. Since 2021, commercial vehicles have carried paying passengers on similar trajectories. This chapter explains suborbital trajectories, sounding-rocket design and operations, the science and technology they enable, microgravity research platforms, crewed suborbital vehicles, hypersonic test flights and the infrastructure and regulation that support them.*

IN THIS CHAPTER

9.1 What "Suborbital" Means
9.2 Suborbital Trajectories
9.3 Sounding Rockets
9.4 Microgravity Research Platforms
9.5 Crewed Suborbital Spaceflight
9.6 Hypersonic and Re-entry Test Flights
9.7 New Suborbital Vehicles and Reusable Sounding Rockets
9.8 Milestones of Suborbital Flight
9.9 Suborbital Point-to-Point Transport
9.10 Operations, Safety and Regulation

## 9.1 What "Suborbital" Means

A **suborbital** trajectory is one whose path intersects the Earth's surface: the vehicle does not have enough horizontal speed to stay in orbit. A suborbital flight may still go very high — sounding rockets have reached more than 1 500 km — and spend many minutes above the atmosphere. Suborbital flights include:

- **Sounding rockets** — unguided or minimally guided research rockets that carry instruments on a nearly vertical trajectory;
- **Crewed suborbital spacecraft** — vehicles such as Blue Origin's New Shepard and Virgin Galactic's SpaceShipTwo, carrying tourists and researchers;
- **Rocket-powered research aircraft and spaceplanes** — the X-15 of the 1960s, SpaceShipOne in 2004, and future hypersonic vehicles (Chapter 10);
- **Test flights** of orbital vehicles (for example Starship's early flights and launch-abort tests) and **hypersonic test vehicles**;
- **Ballistic missiles**, which follow suborbital trajectories (discussed only as context in Chapter 20).

The energy needed is much smaller than for orbit. Rising vertically to 100 km requires a burnout speed of about 1.4 km/s (ignoring drag and the height gained during the burn). Reaching 300 km requires about 2.4 km/s, and 1 000 km about 4 km/s. By comparison, reaching low Earth orbit needs roughly 9.4 km/s, including losses.

> **KEY CONCEPT — SUBORBITAL FLIGHT IS A VERTICAL CANNON SHOT**
>
> After the motor burns out, a sounding rocket coasts upward like a stone thrown in the air, slowing under gravity until it stops at apogee and falls back. Above about 80–100 km, air drag is negligible, so everything aboard — instruments, fluids, passengers — falls together and is weightless. The weightless time grows roughly with the square root of the height above the atmosphere: about 3 minutes for a 105 km apogee, 6 minutes for 260 km and 13 minutes for 700 km (Figure 9.2).

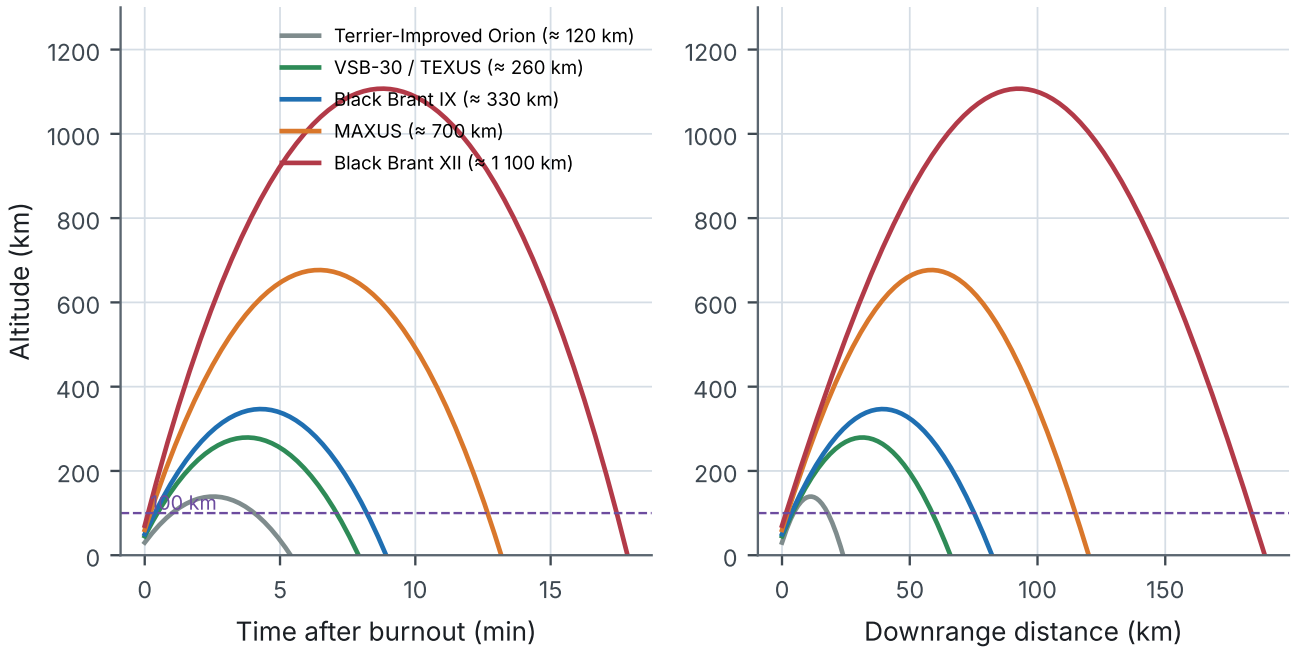
## 9.2 Suborbital Trajectories

### 9.2.1 Ballistic Flight

Neglecting drag above the atmosphere, a vertically launched rocket with burnout speed $v_b$ at altitude $h_b$ reaches an apogee given by energy conservation:

$$\frac{v_b^2}{2} - \frac{\mu}{R + h_b} = -\frac{\mu}{R + h_{apo}} \quad \Rightarrow \quad h_{apo} = \frac{R + h_b}{1 - \dfrac{v_b^2 (R + h_b)}{2\mu}} - R$$

For low apogees this reduces to the flat-Earth result $h_{apo} \approx h_b + v_b^2 / 2g$. Sounding rockets are not launched exactly vertically but typically at 80–88° elevation, so that spent stages and the payload fall into a designated impact area downrange (Figure 9.1). The launch elevation and azimuth are adjusted for winds before each launch (**wind weighting**). This is because unguided rockets weathercock into the wind while their fins are effective at low speed.

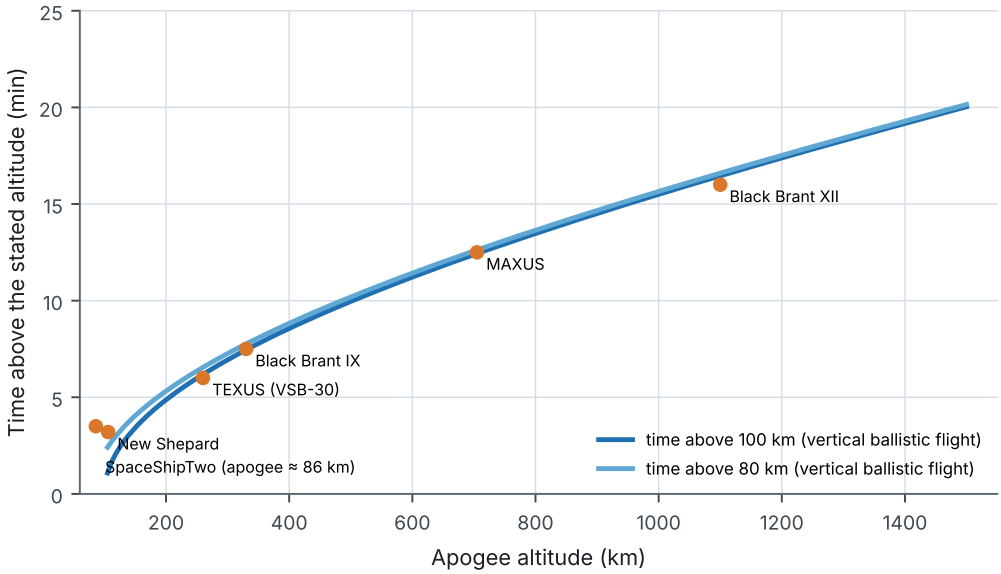



Drag-free trajectories from burnout, computed for this manual; real launches add a few km of drag losses.

**FIGURE 9.1** Computed altitude–time and altitude–downrange profiles for five representative sounding rockets after motor burnout. Higher apogees mean longer flights and longer impact ranges.

## 9.2.2 Microgravity Time

The time spent above a given altitude depends only on the apogee (Figure 9.2). Because drag is negligible above about 100 km, a sounding rocket provides residual accelerations of only $10^{-5}$–$10^{-6}$ g once its attitude control system has stopped firing. This is better than the International Space Station, where crew motion, pumps and fans create disturbances.




Orange points: typical quoted weightless times for each vehicle (they start after engine shutdown and despin, so are shorter than the time above 100 km).

**FIGURE 9.2** Time spent above 100 km and above 80 km as a function of apogee for a vertical ballistic flight, with typical quoted weightless times for several vehicles.

### 9.2.3 Re-entry

On the way down, the payload re-enters at roughly the speed it had at burnout. For a 300 km apogee this is about 2.3 km/s — enough to produce significant aerodynamic heating and decelerations of 20–50 g, but far milder than orbital re-entry. Payloads that are to be recovered carry a parachute system, deployed at about 5 km altitude after the payload has slowed to subsonic speed. Recovery is by helicopter over land (White Sands, Esrange) or by ship at sea (Wallops, Andøya).

## 9.3 Sounding Rockets

### 9.3.1 History

The first scientific rockets were captured German V-2 missiles, launched from White Sands, New Mexico, from 1946. They carried cosmic-ray counters, spectrographs and cameras, and in October 1946 returned the first photographs of Earth from above 100 km. Purpose-built sounding rockets followed: the US Aerobee (from 1947), Viking and Nike-based vehicles, and the British Skylark (1957–2005, 441 launches). Others were the Canadian Black Brant (from 1961) and many more in France, Japan, India, the USSR and China.

Sounding rockets made fundamental discoveries:

- **X-ray astronomy** was born on 18 June 1962. That day, an Aerobee rocket launched by Riccardo Giacconi's team detected the first X-ray source outside the Solar System, Scorpius X-1. This work was recognised with the 2002 Nobel Prize in Physics.
- The **solar extreme-ultraviolet and X-ray spectrum** was first measured from rockets in the late 1940s and 1950s.
- Rocket measurements mapped the **ionosphere**, the **mesosphere** and the **thermosphere**, and discovered phenomena such as the airglow layers and the auroral electrojet.
- India's space programme began with a Nike-Apache launched from a church at **Thumba** in November 1963.

### 9.3.2 Anatomy of a Sounding Rocket

A typical modern sounding rocket (Figure 9.3) has one to four solid-propellant motor stages, fins for aerodynamic stability, and a payload made of modular cylindrical sections:

- **Motors.** Military-surplus motors (Terrier, Talos, Nike) and purpose-built motors (Black Brant, Nihka, Oriole, Brazil's S-30/S-31, Germany's improved Orion and ESA/DLR's Rheinmetall-built motors) give total burn times of 30–90 seconds.
- **Ignition and despin module.** Rockets are spun at a few revolutions per second during powered flight for stability (by canted fins and spin motors). They are then despun with **yo-yo weights** — masses on cables that unwind and carry away angular momentum.
- **Telemetry and power.** S-band transmitters send data at up to tens of Mbit/s; batteries provide power for 10–20 minutes.
- **Attitude control system (ACS).** Cold-gas thrusters and gyros, often with a Sun sensor or star tracker, point instruments to arcseconds (for solar and astronomical payloads) or hold the payload still (for microgravity).

- **Experiment modules.** The scientific instruments, often with vacuum doors that open above the atmosphere.
- **Recovery system.** A parachute and, for sea recovery, flotation and beacons.
- **Nose cone.** Protects the payload during ascent and is ejected in space.

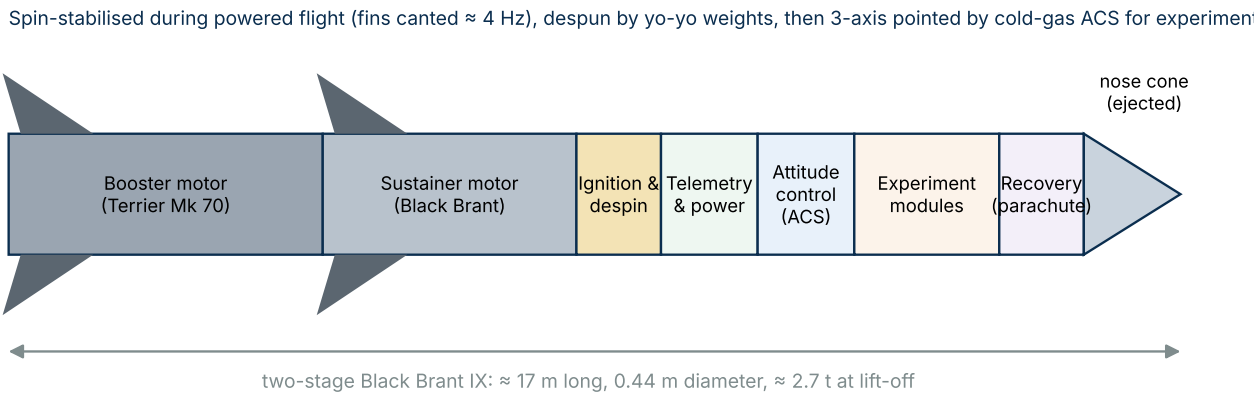



**FIGURE 9.3** Anatomy of a two-stage sounding rocket, based on the NASA Black Brant IX configuration.

### 9.3.3 Vehicles Around the World

NASA's **Sounding Rocket Program**, managed from Wallops Flight Facility and operated with contractors, flies about 20 missions a year using a family of vehicles assembled from a small set of motors (Figure 9.4, Table 9.1). Magellan Aerospace's Black Brant has flown more than 1 000 times, with a reported success rate of 99.4 % over its last 500 launches.

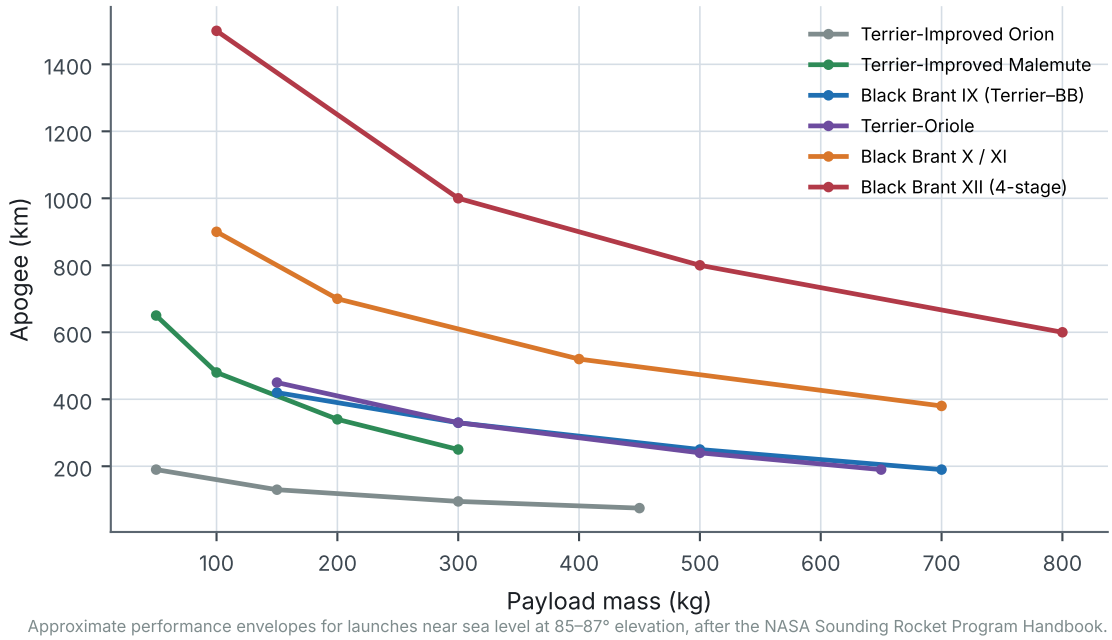



**FIGURE 9.4** Approximate payload–apogee performance of NASA sounding-rocket vehicles.

**TABLE 9.1** Selected sounding rockets in current or recent use.

| Vehicle | Country / operator | Stages | Typical payload | Typical apogee | Notes |
|---|---|---|---|---|---|
| Terrier-Improved Orion | USA (NASA) | 2 | 50–400 kg | 70–200 km | Student and technology flights (RockSat-X, RockOn) |

| Vehicle | Country / operator | Stages | Typical payload | Typical apogee | Notes |
|---|---|---|---|---|---|
| Terrier-Improved Malemute | USA (NASA) | 2 | 50–300 kg | 250–650 km | Atmospheric and technology missions |
| Black Brant IX | USA / Canada | 2 (Terrier–BB) | 150–700 kg | 200–420 km | Workhorse for solar and astrophysics |
| Black Brant X, XI, XII | USA / Canada | 3–4 | 100–800 kg | 380–1 500 km | Long-duration ionospheric and auroral missions |
| Terrier-Oriole | USA | 2 | 150–650 kg | 190–450 km | Higher-performance option |
| VSB-30 | Brazil (DCTA/IAE) with DLR | 2 | ≈ 400 kg | ≈ 260 km | Used for TEXUS and MAPHEUS microgravity missions |
| VS-30 / VS-31 | Brazil | 1–2 | 260–300 kg | ≈ 120–160 km | Atmospheric and hypersonic research |
| MAXUS | ESA / Sweden / Germany | 1 (Castor 4B) | ≈ 800 kg | ≈ 700 km | 12–13 minutes of microgravity |
| Improved Orion, REXUS | DLR / SSC | 1–2 | 30–100 kg | 80–100 km | European student programme REXUS/BEXUS |
| S-310, S-520, SS-520 | Japan (JAXA) | 1–3 | 50–150 kg | 150–800 km | SS-520-5 orbited a CubeSat in 2018 (smallest orbital rocket) |
| RH-200, RH-300, RH-560 | India (ISRO) | 1–2 | 10–100 kg | 70–500 km | Rohini series since 1967 |
| Various solid rockets | China (CAS, CASIC) | 1–2 | varied | 100–200+ km | Atmospheric research, technology and tourism demonstrators |

### 9.3.4 Launch Ranges

Sounding rockets need ranges with large unpopulated impact areas, tracking radars and telemetry stations, and safety infrastructure (Figure 9.5). Major ranges include NASA's **Wallops Flight Facility** (Virginia), **White Sands Missile Range** (New Mexico) and **Poker Flat Research Range** (Alaska, operated by the University of Alaska Fairbanks for auroral science). Others are the **Andøya Space** range and **SvalRak** (Norway), **Esrange** (Sweden), **Woomera** (Australia) and **Kwajalein** (Marshall Islands, for equatorial ionosphere studies). There are also **Alcântara** and **Barreira do Inferno** (Brazil), **Thumba** (India), **Uchinoura** (Japan) and **El Arenosillo** (Spain). High-latitude ranges let rockets fly directly into the aurora.

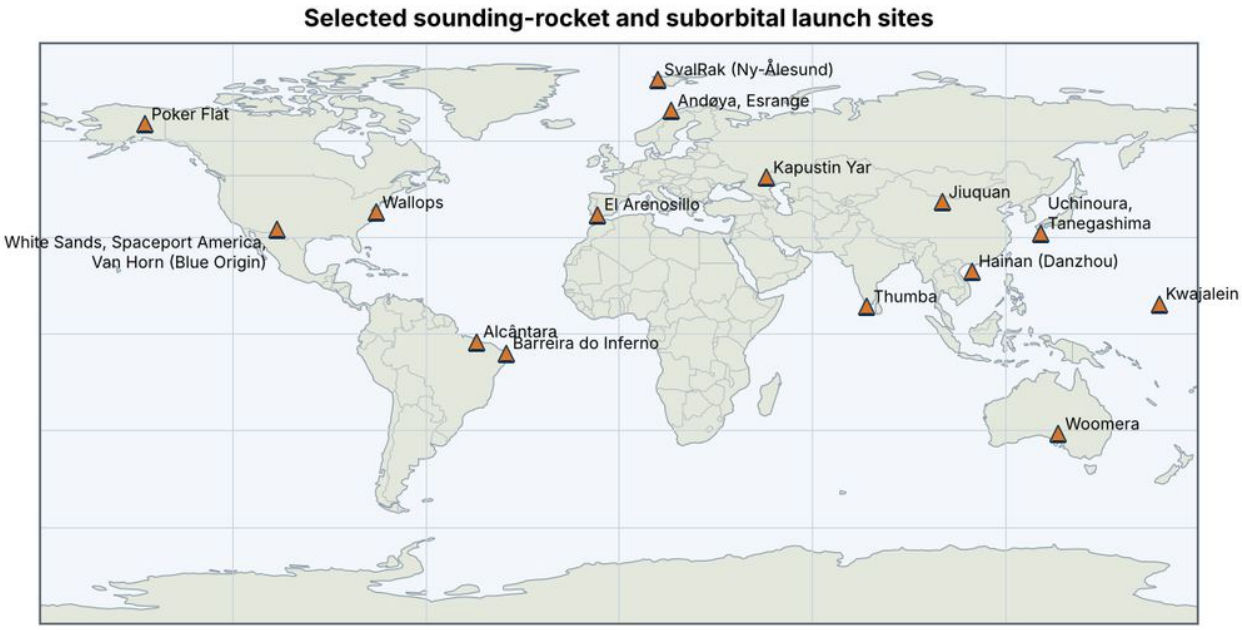



**FIGURE 9.5** Selected sounding-rocket and suborbital launch sites.

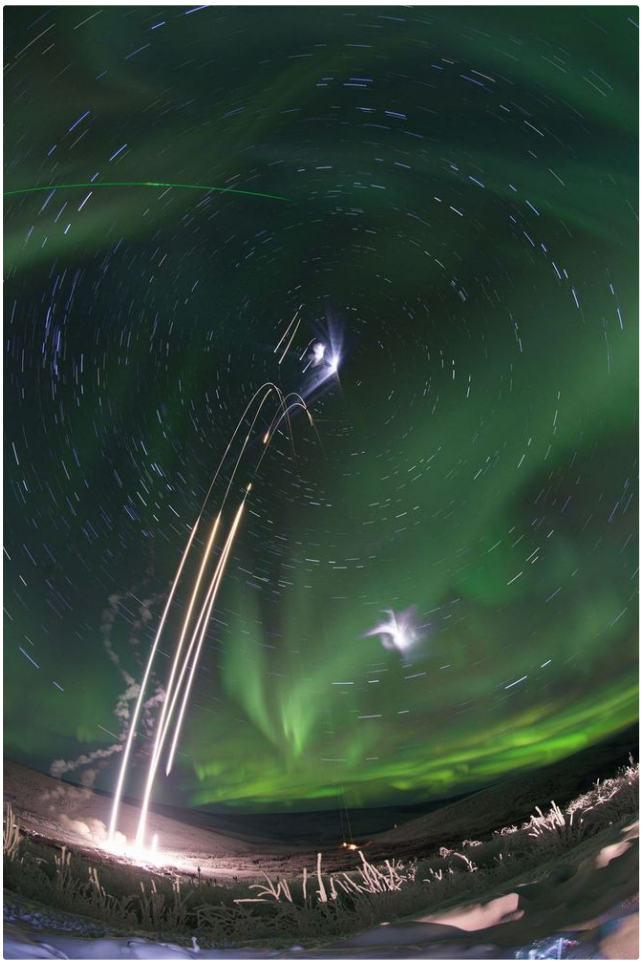


**FIGURE 9.6** A NASA rocket launches from Poker Flat Research Range in Alaska into the aurora during the M-TeX and MIST campaign. Long-exposure photographs record the rocket trails. Image: NASA/GSFC.

**FIGURE 9.7** Red vapour tracer released by the EVEX mission over Kwajalein Atoll (2013) to measure winds in the equatorial ionosphere. Image: NASA/GSFC.

### 9.3.5 Scientific Uses

Sounding rockets remain irreplaceable for several kinds of science:

- **In-situ measurement of the mesosphere and lower thermosphere (50–150 km)** — the "ignorosphere", too high for balloons and too low for satellites. Rockets carry neutral and ion mass spectrometers, Langmuir probes, electric-field booms and particle detectors directly through the region. They also release **chemical tracers** — trimethylaluminium, barium, sodium, lithium — that glow or scatter sunlight so that ground cameras can measure winds and electric fields (Figure 9.8).
- **Auroral and ionospheric physics** — rockets fly through auroral arcs to measure the particles and fields that create them. Multi-rocket campaigns (e.g., NASA's Grand Challenge Initiative – Cusp, 2019) and rocket–satellite conjunctions resolve spatial and temporal structure.
- **Solar physics** — high-resolution ultraviolet and X-ray instruments that test concepts for future satellites (Hi-C, CLASP, FOXSI, MaGIXS) observe the Sun for about 5 minutes per flight.
- **Astrophysics** — ultraviolet and X-ray spectroscopy of hot gas, galaxies and the diffuse background (e.g., CIBER, FIRE, the Micro-X microcalorimeter).
- **Planetary science** — rockets observed Comet Halley, Comet Hale-Bopp and planetary atmospheres in the ultraviolet.
- **Calibration** — under-flights of satellites with calibrated instruments (e.g., SDO's EVE instrument, calibrated by rocket flights).

**FIGURE 9.8** Artificial clouds formed by vapour tracers released from a NASA sounding rocket over the US East Coast. Ground cameras track their motion to measure upper-atmosphere winds. Image: NASA/GSFC.

### 9.3.6 Anatomy of a Mission: From Proposal to Flight

A NASA sounding-rocket mission typically takes two to three years from selection to launch — fast by space standards. A principal investigator proposes an experiment through NASA's Research Opportunities in Space and Earth Sciences (ROSES). Once the experiment is selected, the Wallops team assigns a mission manager. The team works with the scientists through a

requirements definition meeting, design review and mission readiness review. Payload integration and testing at Wallops (vibration, spin balance, bend tests, mass properties, horizontal and vertical sequence tests with the real telemetry and attitude systems) takes a few weeks. The payload is then shipped to the range, where a launch window of one to three weeks is held for the right geophysical or astronomical conditions. Examples are an active aurora overhead, a solar target at the right elevation, or dark skies for an astronomical observation.

Figure 9.9 shows the sequence of events for a typical Black Brant IX solar or astronomy mission.

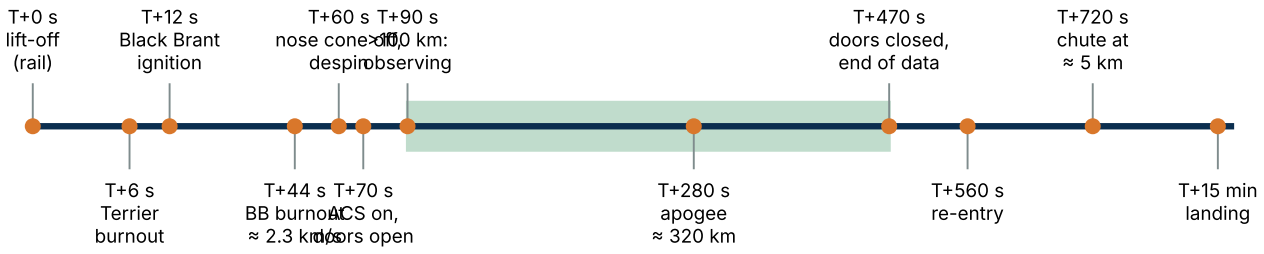



**FIGURE 9.9** Sequence of events for a typical Black Brant IX science mission, from rail launch to parachute landing about 15 minutes later.

**TABLE 9.2** A simplified sequence of events for a Black Brant IX solar mission (representative timings).

| Time | Event | Notes |
|---|---|---|
| T−6 h to T−30 min | Final payload checks, ACS gas loading, vacuum pump-down of instruments | Science team monitors detector temperatures |
| T−15 min | Wind measurement and launcher setting | Elevation ≈ 85–87°, azimuth adjusted for wind weighting |
| T+0 | Lift-off from the rail launcher; Terrier first stage burns ≈ 6 s | Spin-up by canted fins |
| T+12 s | Black Brant second-stage ignition after coast | Burn ≈ 32 s |
| T+44 s | Burnout at ≈ 2.3 km/s, ≈ 80 km altitude | Accelerations up to ≈ 12 g |
| T+60 s | Nose cone ejection, despin with yo-yos, motor separation | Payload rates < 1°/s |
| T+70–90 s | ACS acquires the Sun; shutter doors open above ≈ 100 km | Pointing accuracy ≈ 1 arcsec with fine Sun sensor |
| T+90–470 s | Science observations | Data telemetered live at up to ≈ 20–40 Mbit/s; images also stored onboard |
| T+280 s | Apogee ≈ 320 km | |
| T+470 s | Doors close, payload put in re-entry attitude | |
| T+720 s | Main parachute deploys at ≈ 5 km | Payload descends at ≈ 8 m/s |
| ≈ T+15 min | Landing ≈ 80 km downrange | Helicopter recovery; reflight of instrument possible |

### 9.3.7 Solar and Astrophysics Instruments on Rockets

Because a rocket flight lasts only minutes, instruments are designed to make one measurement extremely well. Recent examples include:

- **Hi-C (High-resolution Coronal Imager)** — an extreme-ultraviolet telescope that imaged the solar corona at about 0.3 arcsec resolution (five times better than SDO's AIA), revealing fine magnetic "braids" (flights in 2012, 2018 and 2024).
- **CLASP / CLASP2** — a Japan–US–Europe ultraviolet spectropolarimeter that measured magnetic fields in the solar chromosphere for the first time using the Hanle effect (2015, 2019, 2021).
- **FOXSI** — a focusing hard X-ray telescope that observes faint solar flares and nanoflares (flights 2012–2024), a pathfinder for future satellites.
- **MaGIXS** — a soft X-ray spectrometer that measures plasma composition in active regions.
- **Micro-X** — the first X-ray microcalorimeter flown with a transition-edge-sensor array, observing supernova remnants.
- **CIBER and CIBER-2** — measurements of the near-infrared extragalactic background light.

Many of these instruments are recovered and reflown, and their technologies feed into satellite missions such as Solar Orbiter, MUSE (Multi-slit Solar Explorer) and future X-ray observatories.

### 9.3.8 Technology Demonstration and Training

Sounding rockets also serve as cheap, fast test beds — "a space mission in a year" — for new technologies before they fly on satellites. These include star trackers, detectors, deployable booms, atmospheric re-entry experiments, telemetry systems and attitude-control software. NASA's **Flight Opportunities** programme buys flights on commercial suborbital vehicles for the same purpose. Equally important, sounding rockets train scientists and engineers: a graduate student can design, build, launch and analyse an instrument within a PhD. Student programmes include NASA's RockOn and RockSat-X, ESA/DLR/SNSA's REXUS/BEXUS, and many university rocketry teams.

## 9.4 Microgravity Research Platforms

Gravity affects many processes in subtle ways: it drives buoyant convection in fluids, sedimentation, hydrostatic pressure, and the structure of flames. Removing it lets researchers study fundamental physics (fluid dynamics, combustion, crystal growth, soft matter), biology (cell behaviour, plant growth, physiology) and materials processing. Several platforms provide reduced gravity, trading duration against quality and cost (Figure 9.10, Table 9.3).

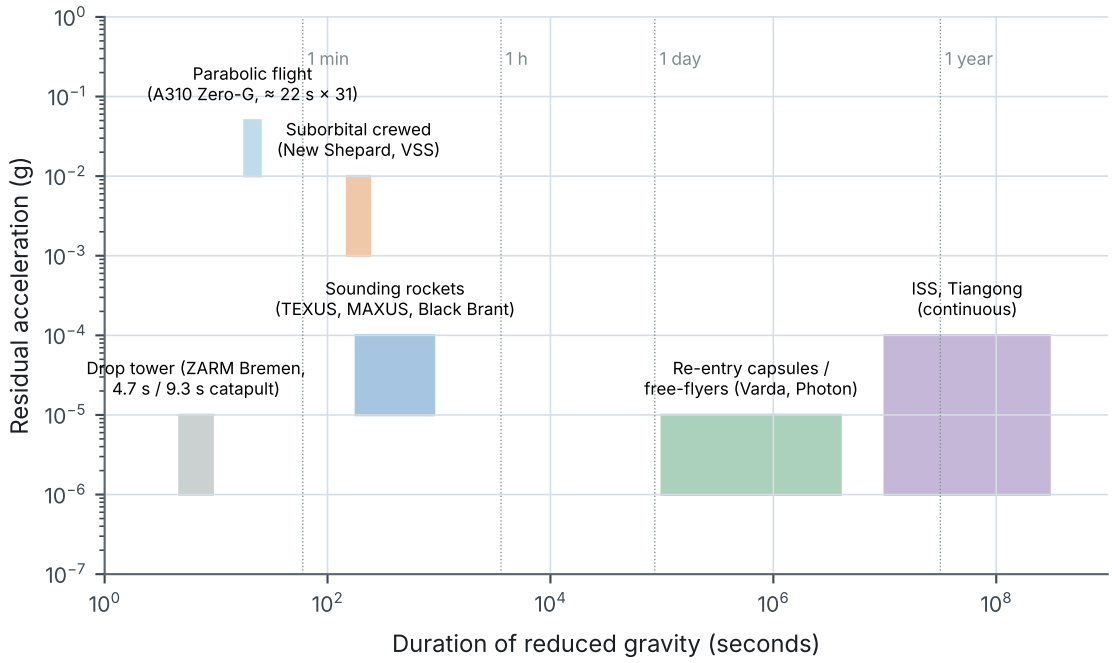



**FIGURE 9.10** Duration and quality of reduced gravity provided by different research platforms.

**TABLE 9.3** Microgravity research platforms.

| Platform | Duration | Residual acceleration | Typical cost per experiment | Examples |
|---|---|---|---|---|
| Drop towers and tubes | 2–10 s | $10^{-6}$ g | Low (thousands of €) | ZARM Bremen (146 m tower, 4.7 s; 9.3 s with catapult); NASA Glenn Zero Gravity Research Facility (5.2 s); Beijing NMLC |
| Parabolic flights | ≈ 20–25 s × 30+ parabolas | $10^{-2}$ g | Moderate | Novespace Airbus A310 Zero-G (France); Zero Gravity Corporation (US); national programmes in Russia, China |
| Sounding rockets | 3–15 min | $10^{-5}$–$10^{-6}$ g | Moderate–high | TEXUS, MAPHEUS, MASER (≈ 6 min), MAXUS (≈ 12.5 min), Black Brant |
| Crewed suborbital vehicles | 3–4 min | $10^{-2}$–$10^{-3}$ g (vibration, crew) | Moderate | New Shepard (paused 2026), SpaceShipTwo/Delta |
| Orbital free-flyers and capsules | days–months | $10^{-6}$ g | High | Chinese SJ-10, Varda W-series, Photon/Bion (Russia), Space Rider (planned) |
| Space stations | months–years | $10^{-6}$–$10^{-4}$ g | High | ISS (US, Europe, Japan, Russia), Tiangong |

**WORKED EXAMPLE — HOW LONG A MICROGRAVITY EXPERIMENT FOR AN APOGEE OF 260 KM?**

The TEXUS missions launched on VSB-30 rockets from Esrange reach about 260 km. Using the vertical-flight formula with an average gravity $g \approx 9.3$ m/s² between 100 and 260 km, the time above 100 km is $2\sqrt{2 \times 160\,000/9.3} \approx 371$ s ≈ 6.2 min. After motor separation, despin and settling of residual rates, about 6 minutes of high-quality microgravity are available — enough for experiments on crystal growth, solidification of metal alloys, and biological processes. The payload then re-enters, deploys its parachute at about 5 km and lands about 70 km downrange, where helicopters recover it within an hour.

## 9.5 Crewed Suborbital Spaceflight

### 9.5.1 From the X-15 to SpaceShipOne

The first humans to make suborbital spaceflights were the Mercury astronauts **Alan Shepard** (Freedom 7, 5 May 1961, 187 km apogee) and **Gus Grissom** (Liberty Bell 7), launched on Redstone rockets. Between 1959 and 1968 the **North American X-15** rocket plane, air-launched from a B-52 at about 13.7 km, made 199 flights. It reached Mach 6.7 and, on 13 flights, altitudes above 80 km. Two flights by Joe Walker in 1963 exceeded 100 km (maximum 107.8 km). The X-15 programme provided much of the knowledge of hypersonic flight, reaction controls and pressure suits used later in the Space Shuttle.

In 2004 **SpaceShipOne**, built by Scaled Composites with funding from Paul Allen, became the first privately developed crewed vehicle to reach space. It won the US$10 million **Ansari X Prize** with two flights above 100 km within two weeks. Its innovations — a hybrid rocket motor (rubber fuel with nitrous oxide oxidiser) and a “feathering” wing that folds upward to stabilise re-entry like a shuttlecock — were carried into Virgin Galactic’s SpaceShipTwo.

### 9.5.2 Blue Origin New Shepard

**New Shepard** is a fully reusable, vertically launched suborbital system named after Alan Shepard (Figure 9.11). A single-stage booster powered by one BE-3PM liquid-hydrogen/liquid-oxygen engine (≈ 490 kN) lifts a pressurised capsule with six large windows and six seats. After main engine cut-off at about 55–60 km, the capsule separates and coasts to an apogee of about 105–107 km, giving 3–4 minutes of weightlessness. The booster returns and lands vertically on a pad near the launch site using its engine and drag brakes. The capsule descends under three main parachutes, firing a retro-thrust system just before touchdown in the West Texas desert. The capsule has a launch escape motor that can pull it away from a failing booster — used successfully in an uncrewed in-flight test in 2016 and in the NS-23 booster failure in September 2022.

New Shepard began crewed flights on 20 July 2021 with Jeff Bezos aboard. By January 2026 it had flown 38 times, carried 98 people above the Kármán line and more than 200 research payloads. On 30 January 2026 Blue Origin announced that it would **pause New Shepard flights for at least two years** to redirect resources to its lunar lander programme.

### 9.5.3 Virgin Galactic SpaceShipTwo and the Delta Class

Virgin Galactic's **SpaceShipTwo** is air-launched from the carrier aircraft **VMS Eve** at about 14 km. Its hybrid rocket motor fires for about 60 seconds, accelerating the spaceplane to about Mach 3. The spaceplane reaches an apogee of about 85–90 km — above the US 80 km astronaut boundary but below the Kármán line. The twin tail booms then feather upward for a stable, self-correcting re-entry, after which the vehicle glides to a runway landing at Spaceport America, New Mexico. The programme suffered a fatal accident in 2014, when the first vehicle (VSS Enterprise) broke up after its feather was unlocked prematurely during a powered test flight, killing co-pilot Michael Alsbury. Its successor, **VSS Unity**, flew commercial missions from June 2023 (Galactic 01) to June 2024 (Galactic 07) and was then retired. Virgin Galactic is building a new **Delta-class** fleet designed to fly twice a week with six passengers each. As of mid-2026, the company planned flight tests of the first Delta ship in the third quarter of 2026 and rocket-powered spaceflight in the fourth quarter. Commercial service was planned to ramp up in 2027.

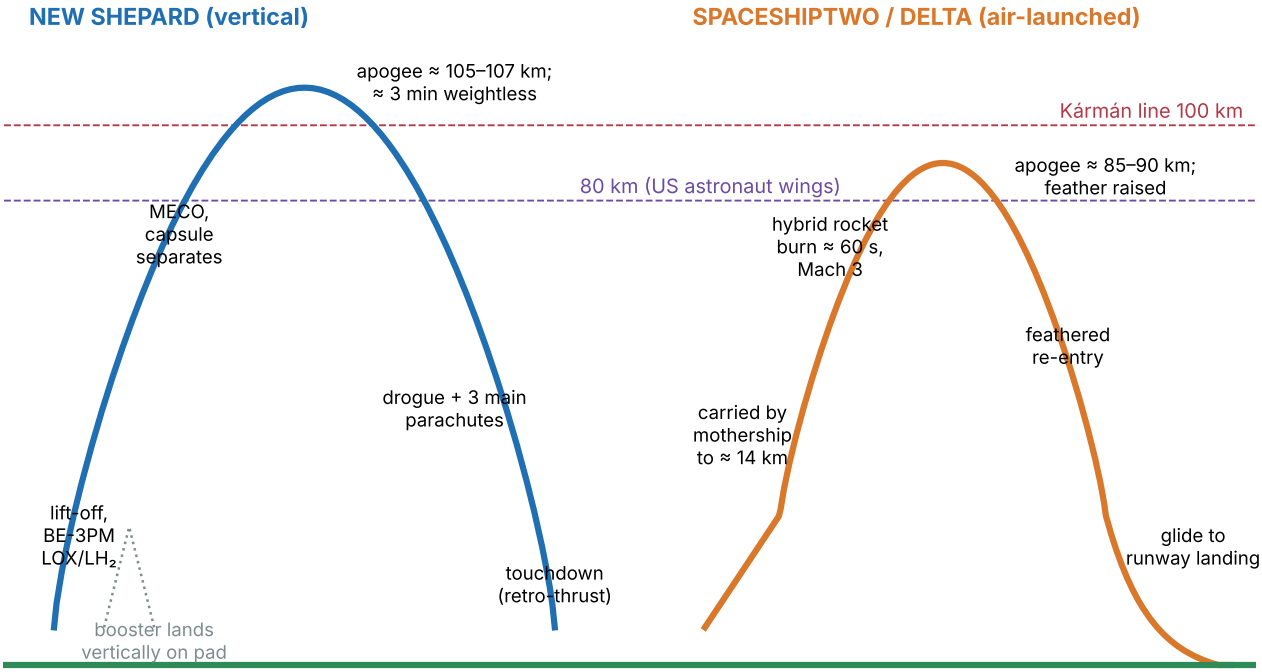



**FIGURE 9.11** Comparison of the New Shepard vertical-launch profile and the SpaceShipTwo air-launched profile.

**TABLE 9.4** Crewed suborbital vehicles.

| Vehicle | Operator | Launch mode | Propulsion | Apogee | Passengers | Status (Oct 2026) |
|---|---|---|---|---|---|---|
| Mercury-Redstone | NASA | Vertical, expendable | Alcohol/LOX | 187–190 km | 1 | 2 flights (1961) |
| X-15 | NASA / USAF | Air launch from B-52 | XLR99 ($NH_3$/LOX) | up to 107.8 km | 1 pilot | 199 flights (1959–68) |
| SpaceShipOne | Scaled Composites | Air launch from White Knight | Hybrid (HTPB/$N_2O$) | 112 km | 1 pilot (+ ballast) | 3 spaceflights (2004) |
| New Shepard | Blue Origin | Vertical, reusable booster and capsule | BE-3PM ($LH_2$/LOX) | ≈ 105–107 km | 6 | 38 flights, 98 people; paused 2026 |

| Vehicle | Operator | Launch mode | Propulsion | Apogee | Passengers | Status (Oct 2026) |
|---|---|---|---|---|---|---|
| SpaceShipTwo (VSS Unity) | Virgin Galactic | Air launch from VMS Eve | Hybrid (HTPB/$N_2O$) | ≈ 85–90 km | 2 pilots + 4 | Retired 2024 after 11 spaceflights |
| Delta class | Virgin Galactic | Air launch | Hybrid | ≈ 85–90 km | 2 pilots + 6 | First spaceflight planned late 2026 |
| Chinese suborbital tourism vehicles | CAS Space, Deep Blue Aerospace and others | Vertical | Liquid | > 100 km planned | 6–7 planned | In development (first flights targeted for late 2020s) |

## 9.5.4 Engineering of Crewed Suborbital Vehicles

**Hybrid propulsion.** SpaceShipOne and SpaceShipTwo use **hybrid rocket motors**, which combine a liquid or gaseous oxidiser with a solid fuel (Figure 9.12). Because the fuel and oxidiser are stored separately and only mix in the combustion port, a hybrid cannot detonate like a mixed solid propellant and can be shut down by closing the oxidiser valve. Nitrous oxide is attractive because it is self-pressurising (its vapour pressure is about 50 bar at room temperature), avoiding pumps; HTPB rubber or polyamide fuels are cheap and inert. Hybrids have lower regression rates (the speed at which the fuel surface burns back) and lower Isp (≈ 250 s) than liquid engines. Their combustion can also be rough. Paraffin-based fuels, which form a liquid layer that sheds droplets into the flow, burn three to four times faster.

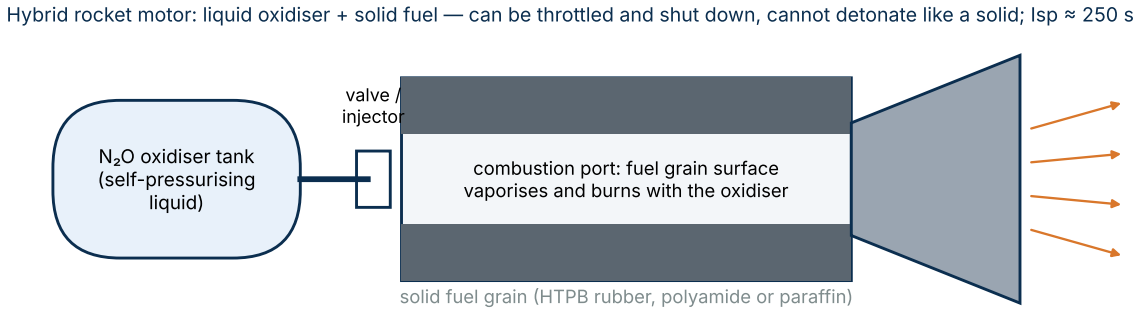



**FIGURE 9.12** Principle of a hybrid rocket motor as used on SpaceShipOne and SpaceShipTwo.

**Liquid-hydrogen propulsion and reusability.** New Shepard's BE-3PM engine burns liquid hydrogen and liquid oxygen at high efficiency and can be deeply throttled (to about 20 % of full thrust) for the vertical landing. The booster uses a ring fin and wedge fins for aerodynamic stability on descent, drag brakes to slow down, and the engine for a final landing burn. This approach was later scaled up for orbital boosters (Chapter 34).

**Escape systems.** A crew capsule escape system must pull the crew away from a failing booster at any point in flight, faster than an explosion can propagate. New Shepard's solid escape motor in the capsule base delivers a short, high-acceleration impulse (about 10 g). SpaceShipTwo is a

winged vehicle released from a carrier aircraft, and it has no escape system. Its safety relies on the reliability of the hybrid motor and on the feather mechanism. The feather mechanism was redesigned with an inhibit after the 2014 accident.

**Cabin and environment.** Suborbital cabins are pressurised to about sea-level conditions, with oxygen masks for emergencies (SpaceShipTwo pilots wear pressure suits; New Shepard passengers do not). Seats are designed for re-entry decelerations of 5–6 g in the "eyeballs-in" direction, which healthy adults tolerate well. Medical screening is lighter than for orbital flight: people in their 80s and 90s have flown, including Wally Funk (82) in 2021 and William Shatner (90) in 2021.

### 9.5.5 Passenger Experience and Safety

A suborbital passenger experiences 3–4 g during ascent (New Shepard), or up to about 3.5 g along the axis and 6 g during the pull-out on re-entry (SpaceShipTwo). The passenger also gets 3–4 minutes of weightlessness, and a view of the Earth's curvature against a black sky. Training takes one to a few days. In the United States, the **Commercial Space Launch Amendments Act of 2004** established an "informed consent" regime. Passengers ("spaceflight participants") must be told the risks, and the FAA regulates public safety rather than occupant safety. Congress has repeatedly extended a "learning period" moratorium on occupant-safety regulation. Suborbital flights are therefore not certified like airliners.

> DATA BOX — WHO COUNTS AS AN ASTRONAUT?
>
> - **FAI (international records body):** 100 km (Kármán line).
> - **US Air Force, NASA, FAA:** 50 statute miles (80.5 km).
> - **FAA Commercial Space Astronaut Wings:** awarded 2004–2021 to crew (not passengers) who reached 50 miles. The programme ended in January 2022. Since then, the names of all who reach the altitude are listed on the FAA website.
> - **Virgin Galactic flights** (≈ 86–89 km) cross the US boundary but not the FAI boundary; **New Shepard flights** cross both.

## 9.6 Hypersonic and Re-entry Test Flights

Suborbital trajectories are ideal for testing vehicles and materials at hypersonic speeds (above Mach 5):

- **Re-entry experiments** — heat-shield materials and capsule shapes, e.g., NASA's IRVE inflatable re-entry experiments on Black Brant rockets (2009–2012), ESA's EXPERT and the Intermediate eXperimental Vehicle (IXV, 2015, launched on Vega to 412 km).
- **Air-breathing hypersonic propulsion** — NASA's X-43A scramjet (2004, Mach 9.6, launched on a Pegasus booster from a B-52) and the USAF/DARPA X-51A Waverider (2013, Mach 5.1 for 210 s). Other examples are Australia's HIFiRE programme (sounding rockets from Woomera) and many more recent national programmes.
- **Commercial hypersonic test services** — Rocket Lab's **HASTE** (a suborbital variant of Electron, flying from Wallops since 2023) and several start-ups provide flights for defence hypersonic testing. This reflects a large increase in US Department of Defense test cadence.

**FIGURE 9.13** NASA's B-52B carrier aircraft takes off from Edwards Air Force Base in 2004 carrying the third X-43A scramjet research vehicle mounted on a Pegasus booster. The X-43A reached Mach 9.6 — the fastest flight of an air-breathing vehicle. Image: NASA/AFRC (NASA/Tom Tschida).

## 9.7 New Suborbital Vehicles and Reusable Sounding Rockets

A new generation of reusable suborbital vehicles aims to reduce cost and increase flight rates for research, testing and training:

- **Dawn Aerospace Mk-II Aurora** (New Zealand) — a small rocket-powered spaceplane that takes off from a runway, intended to fly to 100 km several times per day. It became the first civil aircraft to fly supersonic since Concorde when it exceeded Mach 1 in November 2024, using a liquid bipropellant engine.
- **PLD Space MIURA 1** (Spain) — a liquid-propellant suborbital rocket that flew in October 2023 from El Arenosillo as a pathfinder for the orbital MIURA 5.
- **Chinese suborbital vehicles** — several companies (CAS Space, Deep Blue Aerospace and others) are developing vertical-takeoff, vertical-landing suborbital rockets and capsules for tourism and research, with first crewed flights targeted for the late 2020s.
- **Reusable sounding rockets** — concepts in Europe and Japan (e.g., JAXA's RV-X demonstrator for reusable rocket technology) test throttleable engines, landing guidance and rapid turnaround.
- **Hypersonic test beds** — Rocket Lab's HASTE, Stratolaunch's Talon-A reusable hypersonic vehicle (first powered hypersonic flight in December 2024, launched from the Roc aircraft) and others serve the growing demand for hypersonic testing.

## 9.8 Milestones of Suborbital Flight

**TABLE 9.5** Milestones of suborbital and sounding-rocket flight.

| Year | Milestone |
|---|---|
| 1944 | A German V-2 test rocket (MW 18014) becomes the first human-made object to cross 100 km |
| 1946 | First scientific V-2 flights from White Sands; first photograph of Earth from above 100 km |
| 1947 | First animals (fruit flies) launched to space on a V-2 and recovered alive |
| 1949 | Bumper-WAC two-stage rocket reaches 393 km |
| 1957 | First British Skylark launch (Woomera) |
| 1961 | Alan Shepard's Freedom 7 — first American in space (suborbital) |
| 1962 | Aerobee rocket discovers Scorpius X-1, founding X-ray astronomy |
| 1963 | X-15 exceeds 100 km (Joe Walker); India's first sounding rocket from Thumba |
| 1980s | TEXUS (1977–) and MASER microgravity programmes establish European microgravity research |
| 1992 | First MAXUS flight (12+ minutes of microgravity) |
| 2004 | SpaceShipOne wins the Ansari X Prize |
| 2015 | New Shepard becomes the first booster to fly to space and land vertically |
| 2018 | Japan's SS-520-5, a modified sounding rocket, orbits a 3U CubeSat |
| 2021 | First commercial crewed suborbital flights: VSS Unity (Branson, 11 July) and New Shepard (Bezos, 20 July) |
| 2023 | Virgin Galactic begins commercial service; Rocket Lab's HASTE first flight |
| 2026 | New Shepard paused after 38 flights; Virgin Galactic prepares Delta-class flights |

## 9.9 Suborbital Point-to-Point Transport

Rocket-powered point-to-point travel would reach anywhere on Earth in under an hour on a suborbital trajectory. It has been proposed for passengers (SpaceX's "Earth to Earth" Starship concept) and for rapid military cargo delivery. The US Air Force/Space Force "Rocket Cargo" Vanguard programme is studying delivery of up to 100 t anywhere in about 90 minutes. A flight from New York to Shanghai would require a burnout speed of roughly 6–7 km/s, accelerations of about 3–4 g, and a re-entry comparable to an orbital vehicle. The challenges are cost per flight, noise and sonic booms near cities, safety certification, and the need for spaceports at sea or far from populated areas.

## 9.10 Operations, Safety and Regulation

A sounding-rocket or suborbital launch is planned against strict **range safety** criteria. Analysts compute the impact dispersion of each stage and payload from Monte Carlo simulations including motor performance, winds and misalignments. Launch is allowed only if the expected casualty risk to the public ($E_c$) is below limits such as $10^{-4}$ per mission (US FAA Part 450 and

range commanders' standards). Unguided vehicles may carry a thrust termination or flight termination system. Winds are measured with balloons shortly before launch to set the launcher elevation and azimuth. Aircraft and shipping are kept clear of hazard areas by NOTAMs (notices to air missions) and navigational warnings.

Commercial suborbital vehicles in the United States are licensed by the FAA Office of Commercial Space Transportation under **14 CFR Part 450** (streamlined launch and re-entry licensing, fully in force since 2021). In Europe, national laws (France, the UK, Norway, Sweden) govern sounding-rocket ranges, and EASA is developing a framework for suborbital vehicles that take off and land like aircraft.

### FAILURE LESSON — NS-23 (SEPTEMBER 2022)

On its 23rd flight, an uncrewed New Shepard mission carrying research payloads, the booster suffered a thermal-structural failure of its engine nozzle about one minute after lift-off. The capsule's escape system fired automatically and carried it safely to a parachute landing; the booster was destroyed on impact. The investigation led to a redesign of the nozzle and operating changes, and flights resumed in December 2023. The event demonstrated the value of a capsule escape system — the same principle protects crews on Soyuz, Crew Dragon, Orion and Shenzhou.

### KEY TAKEAWAYS

- Suborbital flights reach space without orbital speed. Weightless time scales roughly with the square root of the height above 100 km (≈ 3 min at 105 km, ≈ 6 min at 260 km, ≈ 13 min at 700 km).
- Sounding rockets, flown since 1946, remain unique for in-situ study of the 50–150 km region, auroral physics, solar and astrophysical observations, technology tests and training; NASA flies about 20 per year.
- Microgravity platforms range from drop towers (seconds) to sounding rockets (minutes) to orbital stations (months), each trading duration, quality and cost.
- Commercial crewed suborbital flights began in 2021; New Shepard flew 98 people before pausing in 2026, and Virgin Galactic plans to resume with its Delta class in late 2026–2027.
- Suborbital trajectories are also used for hypersonic and re-entry testing; range safety, wind weighting and impact-dispersion analysis are central to all operations.

### REVIEW QUESTIONS

1. What distinguishes a suborbital flight from an orbital one? Can a suborbital flight reach higher than the International Space Station?
2. Ignoring drag and taking $g = 9.5\ \text{m/s}^2$, estimate the time a sounding rocket with an apogee of 300 km spends above 100 km.
3. Why do sounding rockets remain the only way to make in-situ measurements in the 50–150 km region?
4. Compare drop towers, parabolic flights, sounding rockets and orbital stations as microgravity platforms in duration, quality of microgravity and cost.
5. Describe the main stages of a sounding-rocket mission, from proposal to recovery.
6. Compare the vehicle architectures and flight profiles of New Shepard and SpaceShipTwo.
7. What are wind weighting and impact-dispersion analysis, and why are they central to range safety?

8. What are the main technical and economic obstacles to suborbital point-to-point passenger transport?

CHAPTER

# 10 Spaceplanes, Atmospheric Re-entry and Reusable Vehicles

Image: NASA/JSC

> *Coming back from orbit is as hard as getting there. A returning spacecraft must shed the enormous kinetic energy it gained during launch - about 30 megajoules for every kilogram - as heat in the upper atmosphere, without burning up or crushing its crew. This chapter explains the physics of atmospheric entry and heating, the design of heat shields and thermal protection systems, capsules versus winged spaceplanes, the history and lessons of the Space Shuttle, Buran and the X-37B, today's reusable spaceplanes and lifting bodies, hypersonic air-breathing concepts, and the economics of reusability.*

IN THIS CHAPTER

10.1 The Re-entry Problem
10.2 Thermal Protection Systems
10.3 Capsules
10.4 Winged Spaceplanes and Lifting Bodies
10.5 Entry at Other Worlds
10.6 Uncontrolled Re-entries and Design for Demise
10.7 Hypersonic Air-Breathing Flight and Single-Stage-to-Orbit
10.8 Reusability Beyond Spaceplanes

## 10.1 The Re-entry Problem

### 10.1.1 Energy to Dissipate

A satellite in low Earth orbit moves at about 7.8 km/s. Its kinetic energy per kilogram is

$$\frac{v^2}{2} = \frac{(7\,800)^2}{2} \approx 3.0 \times 10^7 \ \text{J/kg} = 30 \ \text{MJ/kg}$$

— about seven times the chemical energy in a kilogram of TNT, and enough to vaporise the spacecraft several times over if it were all absorbed. A vehicle returning from the Moon arrives at about 11 km/s (60 MJ/kg); a Mars sample capsule at 12–13 km/s; the Stardust comet-sample capsule entered at 12.9 km/s, the fastest human-made re-entry to date.

Re-entry works because almost all this energy goes into heating the *air*, not the vehicle. A strong **bow shock wave** forms ahead of the vehicle. The air passing through it is compressed and heated to temperatures of 6 000–11 000 K (hotter than the surface of the Sun). It is also dissociated into atoms and partly ionised into a plasma. Most of this hot gas flows around the vehicle and is left behind. Only about 1–5 % of the energy reaches the vehicle surface — and the job of the **thermal protection system (TPS)** is to deal with that fraction.

### 10.1.2 The Blunt-Body Insight

In 1951, H. Julian Allen and Alfred Eggers at NACA Ames made a counter-intuitive discovery: a re-entry vehicle should be **blunt**, not sharp (Figure 10.1). A blunt body creates a detached shock that stands off from the surface, so most of the heat stays in the shock layer. A sharp body has an attached shock close to its nose and receives far more heating. The convective heat flux at the stagnation point scales approximately as

$$\dot{q}_s \approx k \sqrt{\frac{\rho}{r_n}}\, V^3 \qquad \text{(Sutton–Graves; } k \approx 1.74 \times 10^{-4} \text{ for Earth's atmosphere in SI units)}$$

where $r_n$ is the nose radius. Larger nose radii mean lower heat flux. This insight shaped every capsule since Mercury and the rounded noses of ICBM warheads and the Space Shuttle.

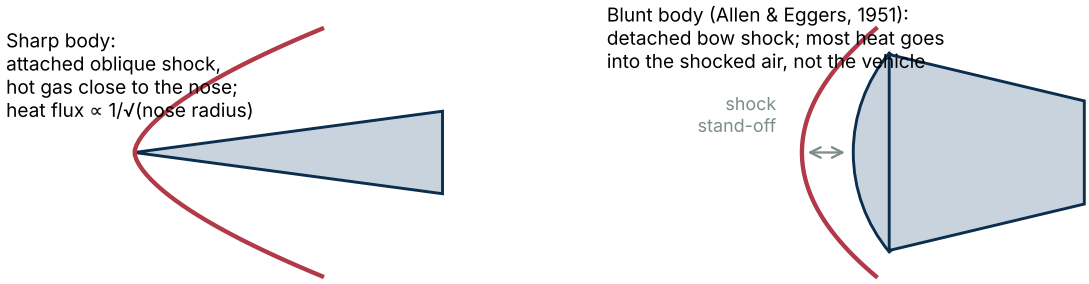



**FIGURE 10.1** Sharp and blunt bodies at hypersonic speed. A blunt nose pushes the bow shock away from the surface so that most of the heat remains in the air.

At speeds above about 10 km/s, the hot shock layer also emits strong **radiative heating** (from excited nitrogen and oxygen atoms and molecules), which, unlike convective heating, increases with nose radius. Lunar-return capsules such as Orion and Apollo experience significant radiative heating, and Mars-return or outer-planet probes even more.

### 10.1.3 The Entry Corridor

A returning vehicle must enter within a narrow range of **flight-path angles** — the **entry corridor** (Figure 10.2). If it enters too shallowly, it may skip back out of the atmosphere (for a lifting vehicle) or spend too long in the heating region; if too steeply, deceleration and peak heating become excessive. For a ballistic capsule, Allen and Eggers showed that the peak deceleration depends only on the entry speed $V_E$ and angle $\gamma_E$:

$$n_{max} = \frac{V_E^2 \sin|\gamma_E|}{2\,e\,H\,g_0}$$

where $H \approx 7.2$ km is the atmospheric scale height (Figure 10.3). Returning from LEO at 7.8 km/s, a ballistic capsule entering at −2° experiences about 8–9 g; at −6° it would experience over 20 g. Lunar returns must be precisely targeted — Apollo's corridor was only about 2° wide.

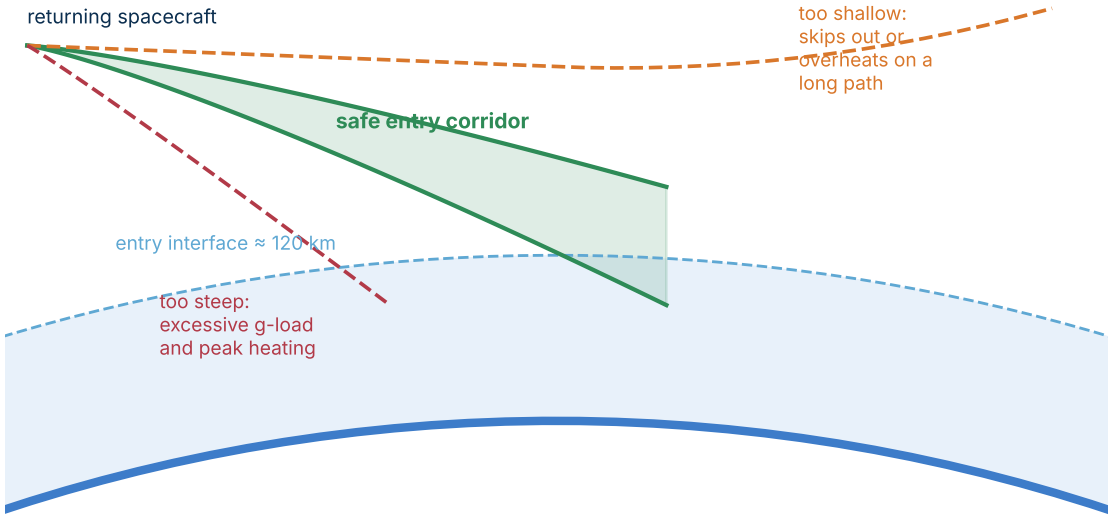



**FIGURE 10.2** The entry corridor. A returning vehicle must enter steeply enough not to skip out or overheat over a long path, but shallowly enough to keep deceleration and peak heating within limits.

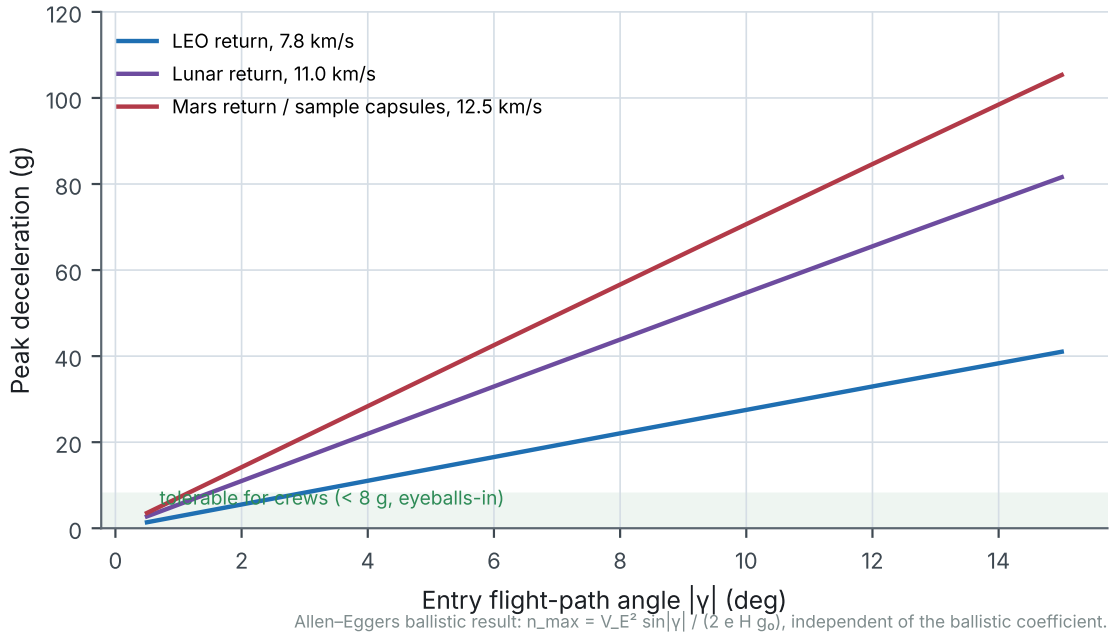


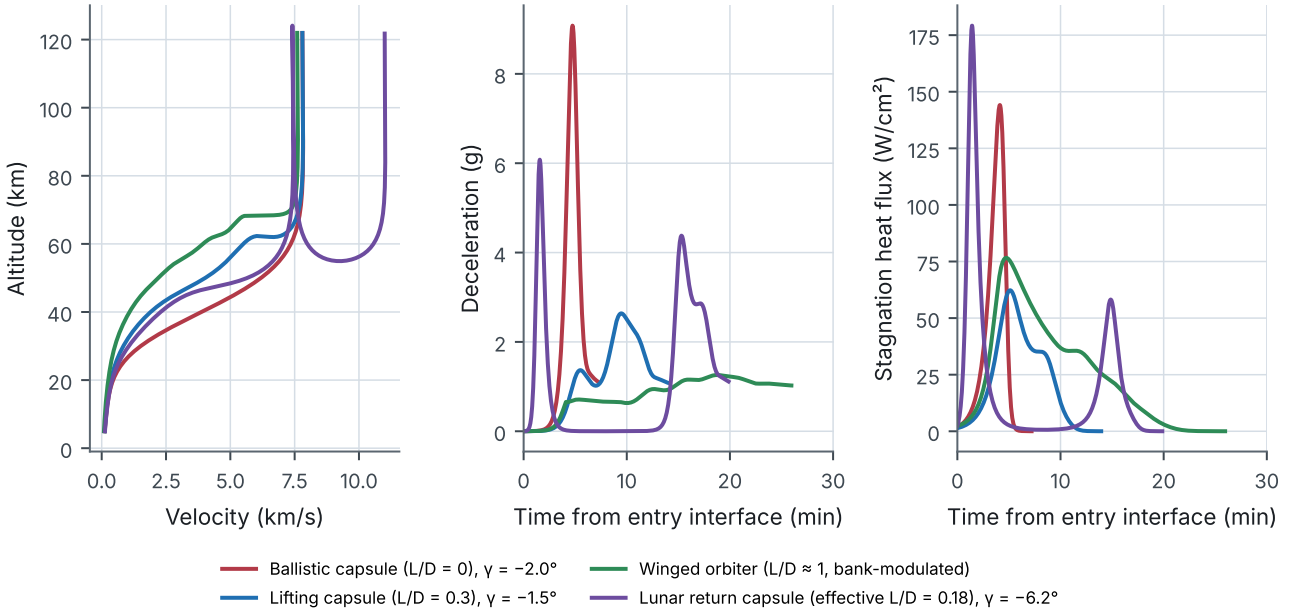

**FIGURE 10.3** Peak deceleration of a ballistic entry vehicle as a function of entry angle and speed (Allen–Eggers result). Crews can tolerate about 8 g in the eyeballs-in direction for short periods; lifting entries reduce the load dramatically.

### 10.1.4 Lift Changes Everything

A purely ballistic entry is simple but harsh. A capsule can generate a small amount of **lift** by flying at an angle of attack, with its centre of mass offset from the centreline. Even this small lift lets the vehicle stay higher where the air is thinner, spreading deceleration over a longer time. A lift-to-drag ratio of only 0.3 (Gemini, Apollo, Soyuz, Shenzhou, Crew Dragon, Orion) reduces peak

deceleration from LEO from about 8 g to 3–4 g. It also allows the vehicle to steer by rolling its lift vector ("bank-angle modulation"), giving landing accuracy of a few kilometres. Winged vehicles with L/D ≈ 1–1.5 at hypersonic speeds (the Space Shuttle, Buran, X-37B, Dream Chaser) can keep decelerations below 1.5–2 g. They can glide thousands of kilometres, with cross-range capability of hundreds to 2 000 km.

Figure 10.4 compares simulated entries for four classes of vehicle.




Computed for this manual: 2-D point-mass entry from 122 km, exponential atmosphere, Sutton–Graves stagnation heating; illustrative parameters.

**FIGURE 10.4** Simulated entries (computed for this manual) for a ballistic capsule, a lifting capsule, a winged orbiter and a lunar-return capsule: altitude–velocity profiles, deceleration and stagnation-point heat flux versus time. Lift reduces peak deceleration and heating rate but lengthens the heating pulse.

**WORKED EXAMPLE — READING THE SIMULATED ENTRIES**

In the simulation in Figure 10.4, a ballistic capsule from LEO entering at −2.0° peaks at about 9 g and 140 W/cm². A lifting capsule (L/D = 0.3) at −1.5° peaks at about 2.6 g and 60 W/cm². A winged orbiter flying a bank-modulated equilibrium glide peaks at about 1.3 g and 80 W/cm², but travels some 6 000 km during entry. A lunar-return capsule at 11 km/s entering at −6.2° peaks at about 6 g and 180 W/cm² (convective only — real lunar returns add radiative heating). These numbers are close to flight experience: Soyuz ballistic (backup) entries reach 8–9 g, nominal lifting entries about 4 g; Shuttle entries stayed below 1.5 g; Apollo returns peaked at 6–7 g.

## 10.1.5 Communications Blackout

The plasma around an entering vehicle reflects and absorbs radio waves, causing a **communications blackout** of several minutes (about 4 minutes for Apollo, 6 minutes for Orion on Artemis II). From about 1988, the Space Shuttle avoided blackout by transmitting upward through the plasma "wake" to TDRS relay satellites. Crew Dragon and Starship similarly maintain contact through relay links (Starship uses Starlink) for much of entry. Research continues on techniques such as magnetic "windows" in the plasma and higher frequencies.

### 10.1.6 Beyond the Basics: Real-Gas Aerothermodynamics

At entry speeds the air behind the shock does not behave as an ideal gas. Oxygen molecules begin to dissociate above about 2 500 K and nitrogen above about 4 000 K; above about 9 000 K the gas ionises. These **real-gas effects** lower the temperature behind the shock (energy goes into breaking bonds rather than raising temperature). They also change the shock stand-off distance and pressure distribution. For example, on the Shuttle's first flight this caused the body-flap deflection to be twice the predicted value, because the pitching moment had been underestimated. Because the flow moves so fast, chemical reactions may not have time to reach equilibrium (**thermochemical non-equilibrium**). Whether atoms recombine at the vehicle surface depends on the surface material's **catalytic efficiency**. Non-catalytic coatings (like the Shuttle's black glass) can reduce heating substantially compared with catalytic metals.

Other important phenomena include:

- **Boundary-layer transition** — at lower altitudes the boundary layer becomes turbulent, increasing heating by a factor of 3–5. Transition is triggered by surface roughness. Protruding gap fillers between Shuttle tiles were a concern after Columbia. On STS-114 (2005), astronaut Steve Robinson pulled out two protruding fillers during an unprecedented spacewalk under the orbiter.
- **Shock–shock and shock–boundary-layer interactions** — where one shock impinges on another surface (e.g., a wing leading edge behind the bow shock), local heating can be many times higher. The X-15 suffered severe burn-through on a pylon during a 1967 Mach 6.7 flight with a dummy ramjet because of such interactions.
- **Heat load versus heat rate** — two quantities size a heat shield. The *peak heat flux* (W/cm$^2$) determines which material can survive; the *integrated heat load* (J/cm$^2$, the heat flux integrated over time) determines how thick the insulation or ablator must be. Lifting entries reduce the peak but lengthen the pulse, so they may increase the total load.

### 10.1.7 Entry Guidance

A lifting vehicle steers by **rolling its lift vector**: flying with lift "up" keeps it high and extends range; rolling the lift "down" or sideways makes it descend faster and turn. Entry guidance algorithms compute the bank angle needed to reach the landing site within the allowed deceleration and heating:

- **Apollo guidance** predicted the remaining range from the current state and commanded bank angles, including a "skip" phase for long-range returns.
- **Shuttle entry guidance** followed a reference **drag-acceleration-versus-velocity profile** designed to respect heating and load limits. Deviations were corrected by bank angle. Periodic **roll reversals** kept the cross-range error within bounds as the orbiter weaved left and right across its ground track. The **Terminal Area Energy Management (TAEM)** phase then managed energy below Mach 2.5 to align with the runway using a "heading alignment cone".
- **Orion skip-entry guidance** can dip into the atmosphere, climb out, and re-enter to extend range to more than 8 000 km. This lets a lunar return always reach a landing zone off California. For Artemis II, NASA used a less aggressive profile.

- **Mars entry guidance** (MSL, Mars 2020) used Apollo-derived bank control to reduce the landing ellipse from about 100 km to 20 km × 7 km, enabling landing in Gale and Jezero craters (Chapter 27).

Modern research applies numerical predictor–corrector guidance and convex optimisation, increasingly with onboard computing power that would have been unthinkable in the Apollo era (Chapter 48).

### 10.1.8 Descent and Landing

After hypersonic deceleration, vehicles need a final descent and landing system (Figure 10.5). Capsules deploy **drogue parachutes** at about 7–10 km, often with reefing lines that limit initial inflation to reduce opening shock. They deploy **main parachutes** at 2–3 km. Orion's three 35 m main parachutes slow it to about 7.6 m/s at splashdown. Soyuz fires **soft-landing solid motors** about 0.7 m above the ground triggered by a gamma-ray altimeter, reducing touchdown speed from about 6 to 1.5 m/s. Boeing's Starliner lands on land using **airbags**. Winged vehicles glide to **runway** landings, and rocket stages and some landers land **propulsively**.

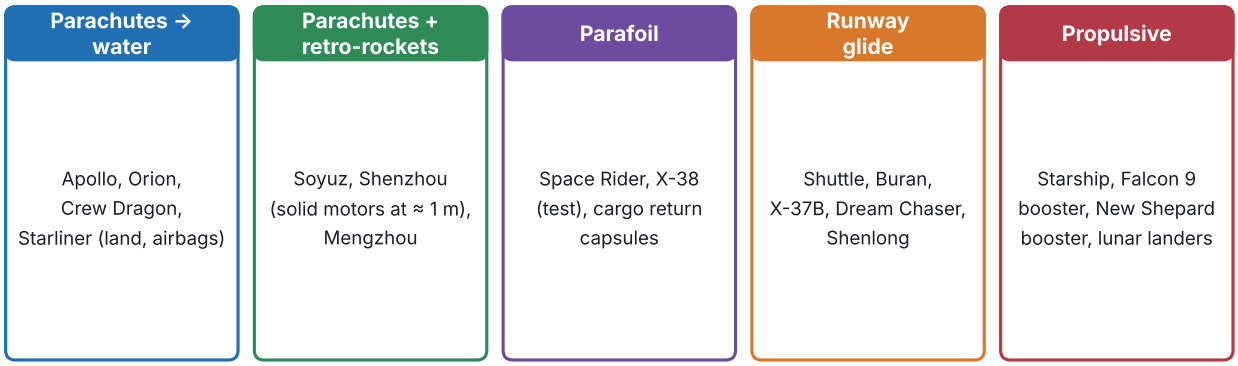
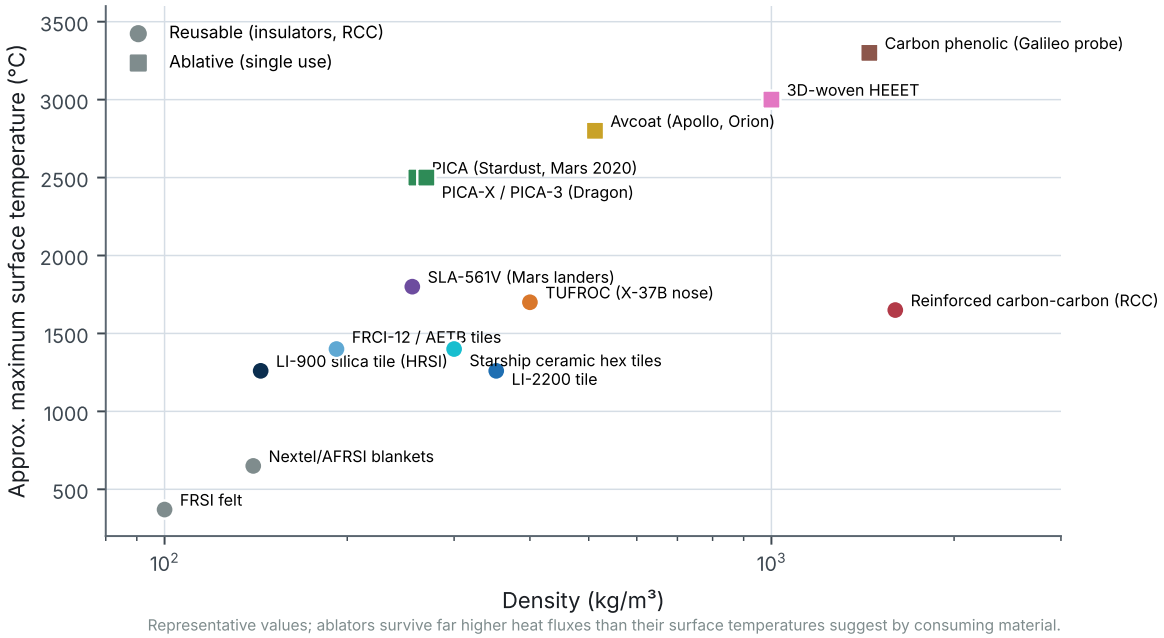



**FIGURE 10.5** Families of landing systems used by returning spacecraft.

## 10.2 Thermal Protection Systems

The TPS keeps the vehicle structure (typically aluminium or composite, limited to about 175 °C, or titanium to about 400 °C) cool enough during entry. There are two fundamentally different approaches: **ablative** and **reusable (insulating or radiative)**. Figure 10.6 compares materials.




Representative values; ablators survive far higher heat fluxes than their surface temperatures suggest by consuming material.

**FIGURE 10.6** Approximate maximum surface temperature and density of thermal protection materials. Reusable insulators such as silica tiles are extremely light; ablators tolerate far more severe conditions by sacrificing material.

### 10.2.1 Ablative Heat Shields

An **ablator** is a material that absorbs heat by decomposing (pyrolysing), charring and eroding. Pyrolysis gases flow out through the char and form a cooler boundary layer that blocks some of the incoming heat ("blowing"), while the char surface radiates heat away at temperatures of 2 000–3 000 °C. Ablators can handle very high heat fluxes but are consumed and are normally used once. Important ablators include:

- **Avcoat** — an epoxy-novolac resin in a fibreglass-phenolic honeycomb, used on Apollo and, in moulded block form, on Orion.
- **PICA (Phenolic-Impregnated Carbon Ablator)** — very light (≈ 0.27 g/cm$^3$) and efficient; used on Stardust (1999–2006), Mars Science Laboratory and Mars 2020 heat shields. SpaceX's **PICA-X** (and later versions) protects Crew and Cargo Dragon; its shield can be reused a limited number of times with refurbishment.
- **SLA-561V** — a cork-silicone ablator used on Viking and many Mars landers.
- **Carbon-phenolic** — dense and robust; used for ICBM warheads and the Galileo Jupiter probe, which entered at 47.4 km/s and lost about half its heat-shield mass.
- **HEEET (Heatshield for Extreme Entry Environment Technology)** — a 3-D woven carbon/phenolic system developed by NASA for Venus, Saturn and high-speed entries.
- **Cork- and silicone-based ablators** used by ESA (Norcoat Liège on ARD, IXV, Huygens) and in Chinese capsules.

**FIGURE 10.7** The Artemis I Orion heat shield after its 2022 lunar-return entry. Engineers found unexpected loss of charred Avcoat material in many locations; the investigation led NASA to fly Artemis II on a modified, shallower-skip entry trajectory rather than change the heat shield. Image: NASA/KSC (NASA).

**LESSON — ARTEMIS I HEAT-SHIELD CHAR LOSS**

After the uncrewed Artemis I flight in 2022, engineers found that pieces of charred Avcoat had broken away from Orion's heat shield in more than 100 places, unlike Apollo-era experience. Investigators traced the cause to gases generated inside the Avcoat during the "skip" entry, in which Orion dipped into the atmosphere, climbed back out and re-entered. The gases could not escape because the material was not permeable enough. They built up pressure that cracked the char. For Artemis II (April 2026), NASA kept the existing heat shield but flew a modified entry trajectory. This reduced the time spent in the conditions that caused the problem. The crew returned safely, and later Orion heat shields use a more permeable Avcoat formulation.

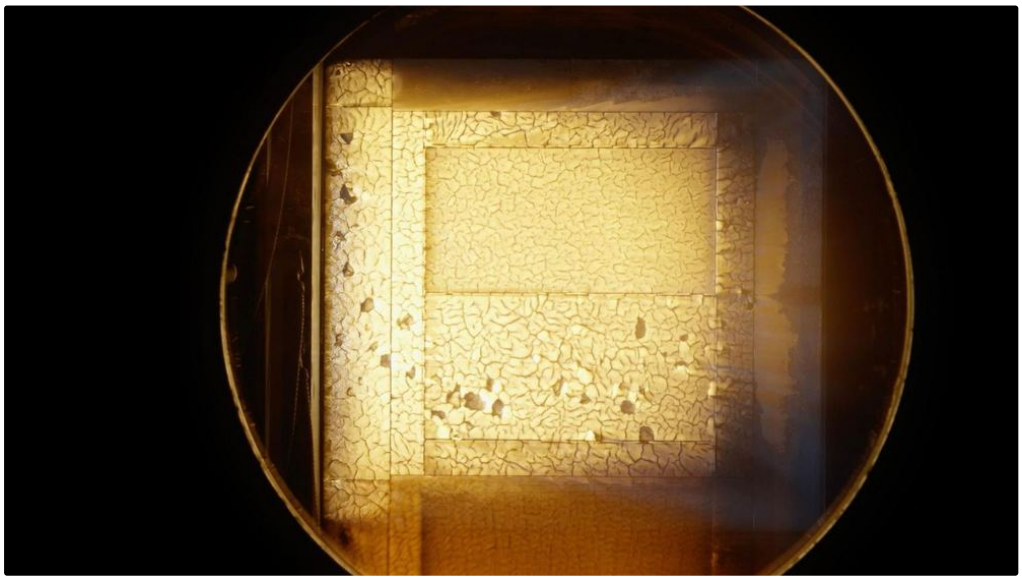


**FIGURE 10.8** Arc-jet testing of Orion's Avcoat heat-shield material at NASA Ames Research Center during the investigation. Arc jets heat air to plasma temperatures with an electric arc and accelerate it over material samples for minutes, reproducing entry heating on the ground. Image: NASA/ARC (NASA).

### 10.2.2 Reusable Thermal Protection

A **reusable** TPS must survive many flights without degradation. Approaches include:

- **Insulating ceramic tiles** — the Space Shuttle's tiles were made of 90 % empty space. They were very pure silica fibres sintered into blocks (LI-900 at 0.14 g/cm$^3$). The blocks were coated with a black, high-emittance borosilicate glass (on the hot underside) or white coating (upper surfaces). A tile can be held by its edges seconds after being removed from a 1 260 °C oven, because it conducts heat so poorly. Stronger toughened tiles (FRCI, AETB, TUFROC) followed.
- **Reinforced carbon-carbon (RCC)** — carbon fibres in a carbon matrix, protected from oxidation by a silicon-carbide coating; used on the Shuttle's nose cap and wing leading edges, where temperatures exceeded 1 260 °C.
- **Flexible blankets** — quilted silica-fabric blankets (AFRSI) and felt (FRSI) for cooler areas.
- **Metallic TPS** — superalloy or titanium shingles over insulation; studied for X-33 and other vehicles.
- **Hot structures** — load-bearing structures made of materials that can operate hot (carbon-silicon carbide, titanium aluminides), avoiding a separate TPS; used on X-37B control surfaces and hypersonic vehicles.
- **Transpiration and active cooling** — pumping a coolant through a porous surface (studied for Starship's leeward side early in development) or circulating fuel through panels (scramjets).

**Starship** uses about 18 000 hexagonal ceramic tiles attached to its stainless-steel windward side by pins, with the steel itself tolerating higher temperatures than aluminium on the leeward side. Its flight tests since 2023 have been as much about maturing a fully reusable TPS as about the vehicle itself.

### 10.2.3 Designing and Testing a Heat Shield

Heat-shield design couples **computational fluid dynamics** (CFD with thermochemical non-equilibrium models) with **material response** codes that compute pyrolysis, char and recession (e.g., NASA's FIAT and PATO). It also includes margins for uncertainty — typically 30–50 % on heating for new designs. Ground testing uses **arc-jet** facilities (NASA Ames, NASA Johnson, CIRA Scirocco in Italy, DLR) that expose samples to supersonic plasma for minutes. It also uses **shock tunnels** and **expansion tubes** that reproduce flow conditions for milliseconds, and **ballistic ranges**. Because no ground facility reproduces all flight conditions simultaneously, flight tests and instrumented flights (e.g., MEDLI sensors on Mars landers) remain essential.

## 10.3 Capsules

Most crewed and cargo vehicles that return from orbit are **capsules** — blunt, axisymmetric bodies with an ablative heat shield on the base, descending finally under parachutes to land or water (Chapter 31). Their advantages are robustness, simplicity, light weight and good packaging; their disadvantages are modest cross-range, limited reusability and relatively high landing loads. Recent trends include **propulsive landing**: Crew Dragon was designed for it but uses parachutes, while the Chinese Mengzhou and Russia's planned vehicles use retro-rockets and airbags.

Another trend is **reusable capsules** (Dragon capsules have flown up to six or more times). A third is **small sample-return and manufacturing capsules** (e.g., Varda Space's W-series capsules returning materials made in orbit; Chapter 40).

## 10.4 Winged Spaceplanes and Lifting Bodies

### 10.4.1 Why Wings?

Wings and lifting-body shapes provide high L/D, giving gentle decelerations, large cross-range (the ability to land at a site far from the orbital ground track), runway landings and — in principle — rapid reuse. The costs are mass (wings, landing gear and large TPS areas) and complexity. A winged orbital vehicle typically has 2–3 times the dry mass of a capsule carrying the same crew or cargo.

### 10.4.2 The X-15 and Early Spaceplanes

The X-15 (Chapter 9) explored hypersonic flight to Mach 6.7 and the edge of space. The US Air Force's **X-20 Dyna-Soar** (1957–1963), a small delta-winged orbital spaceplane, was cancelled before flight, but its research informed the Shuttle. Lifting-body research aircraft such as the M2-F2, HL-10 and X-24 (1960s–1970s) demonstrated that wingless shapes could glide to runway landings; the HL-10 shape later became the basis of Dream Chaser.

### 10.4.3 The Space Shuttle (1981–2011)

The **Space Transportation System** was the world's first reusable orbital spacecraft. The winged **orbiter** (37 m long, 24 m span, about 78 t dry) launched vertically with three reusable RS-25 main engines fed from an expendable external tank and boosted by two recoverable solid rocket boosters. It carried up to seven crew and 24 t of payload to LEO, deployed and retrieved satellites, serviced Hubble five times and assembled most of the ISS. Five orbiters flew 135 missions, carrying 355 different people.

**FIGURE 10.9** STS-1: Space Shuttle Columbia lifts off on the first Shuttle flight, 12 April 1981 — the first time a crewed vehicle flew to orbit on its first launch. Image: NASA/MSFC.

The orbiter's thermal protection system (Figure 10.10) was one of its most innovative and most troublesome elements. It was made of tens of thousands of individually shaped tiles, each different. They had to be inspected, and many repaired or replaced, between flights.

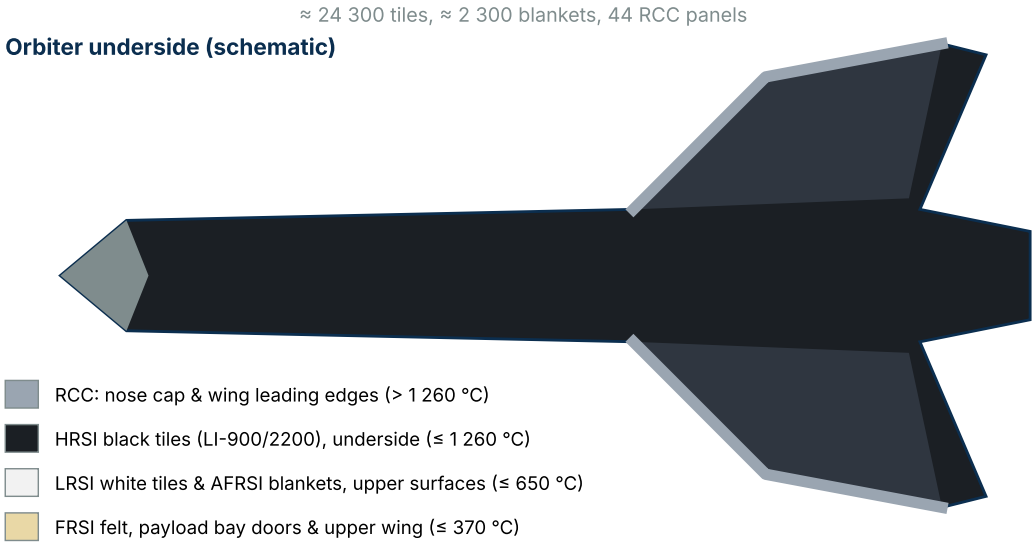



**FIGURE 10.10** Schematic of the orbiter's thermal protection zones: reinforced carbon-carbon on the hottest areas, black high-temperature tiles on the underside, white low-temperature tiles and blankets on upper surfaces.

**FIGURE 10.11** A technician installs a thermal protection tile on the underside of Space Shuttle Atlantis in the Orbiter Processing Facility at Kennedy Space Center (2003). Turnaround between flights required months of inspection and repair. Image: NASA/KSC.

**FAILURE LESSONS — CHALLENGER (1986) AND COLUMBIA (2003)**

On 28 January 1986, Challenger broke apart 73 seconds after launch when hot gas escaped past an O-ring seal in a solid rocket booster joint that had been weakened by cold weather, killing seven astronauts. On 1 February 2003, Columbia broke up during re-entry, killing seven. The cause was a piece of foam insulation from the external tank, which had struck and breached the RCC panels of the left wing's leading edge during launch. The breach let superheated plasma enter the wing. Both accidents had been preceded by earlier incidents that had been rationalised as acceptable — what sociologist Diane Vaughan called the **normalisation of deviance**. The investigations (Rogers Commission; Columbia Accident Investigation Board) led to safety-culture reforms across NASA. They also influenced the decision to retire the Shuttle, and to design the next crewed vehicles as capsules placed on top of their rockets, with launch escape systems.

The Shuttle demonstrated reusable spaceflight but not cheap spaceflight. Each flight cost on the order of US$450 million in marginal cost, and roughly US$1.5 billion when programme costs are included. This was because refurbishment was labour-intensive and flight rates (at most 9 per year) were far below the 50 per year originally envisioned. Figure 10.12 shows why refurbishment cost determines whether reuse pays.

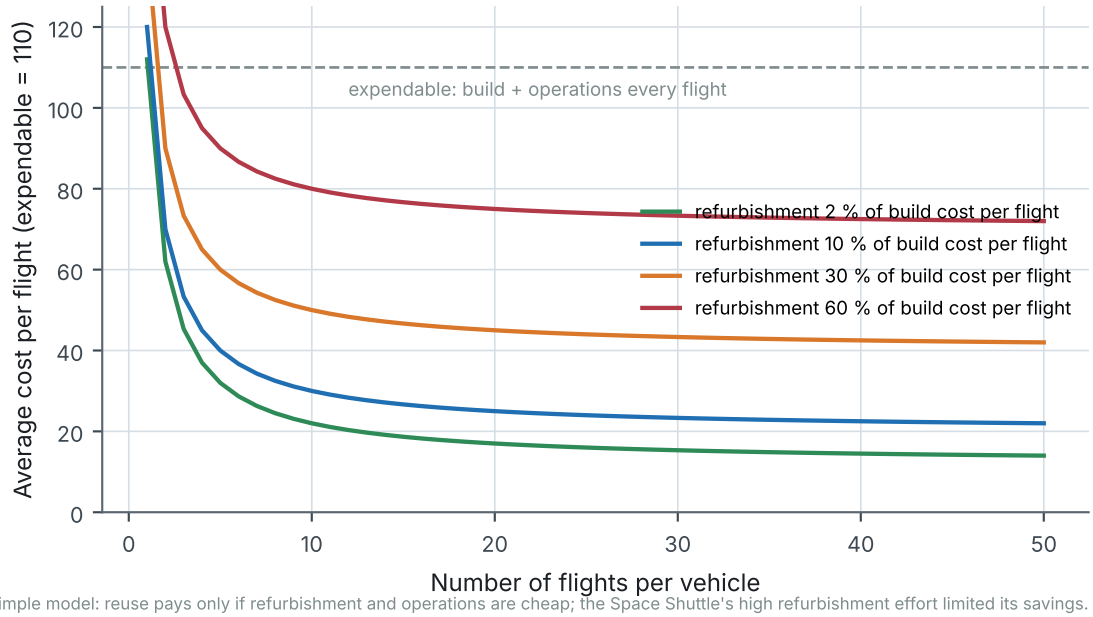



**FIGURE 10.12** A simple model of average cost per flight versus number of reuses for different refurbishment costs. Reuse saves money only when refurbishment and operations are a small fraction of the cost of building a new vehicle.

### 10.4.4 Buran

The Soviet **Buran** orbiter was outwardly similar to the Shuttle but was launched by the powerful Energia rocket. Energia's core stage carried the main engines, so Buran itself had only orbital manoeuvring engines. On 15 November 1988 it made a single **fully automatic uncrewed orbital flight**, completing two orbits. It landed autonomously on a runway at Baikonur in a crosswind of 17 m/s — a remarkable achievement in autonomous flight control. The programme was cancelled in 1993 for lack of funds; the flown vehicle was destroyed in 2002 when its hangar roof collapsed.

### 10.4.5 Today's Spaceplanes

**Boeing X-37B Orbital Test Vehicle (USA).** A 9 m-long, 5 t uncrewed spaceplane operated by the US Space Force that launches inside a fairing and lands on a runway. Two vehicles have flown eight missions since 2010, testing technologies in orbit for up to 908 days (OTV-6, 2020–2022). OTV-7 (December 2023 – March 2025) flew in a highly elliptical orbit and demonstrated **aerobraking** to lower its orbit before landing. OTV-8, launched in August 2025 on a Falcon 9, is testing laser communications and a quantum inertial sensor for navigation without GPS.

**China's reusable experimental spacecraft.** China has flown an uncrewed reusable spacecraft on three missions (2020, 2022–2023, and a third from December 2023 to September 2024). It was launched on Long March 2F rockets from Jiuquan and landed on a runway. Few details have been released.

**Sierra Space Dream Chaser (USA).** A 9 m lifting-body spaceplane derived from NASA's HL-20, designed to carry 5 t of cargo to the ISS and land on a runway with gentle (< 1.5 g) loads suitable for delicate experiments. A crewed version is planned. After years of delays, NASA and Sierra Space modified the cargo contract in September 2025. The first flight became a **free-flying demonstration** without ISS docking, targeted for late 2026 on a ULA Vulcan rocket. After it, NASA may order cargo missions.

**FIGURE 10.13** The Dream Chaser engineering test article arriving at NASA's Armstrong Flight Research Center in 2017 for approach-and-landing tests, in which it was dropped from a helicopter and glided to a runway landing. Image: NASA/AFRC (Ken Ulbrich).

**ESA Space Rider (Europe).** An uncrewed reusable lifting-body spacecraft, derived from the IXV demonstrator that flew in 2015, designed to carry 600 kg of experiments for two months in orbit and land under a steerable parafoil. It will be launched by Vega-C; its first flight has slipped to no earlier than 2028.

**ISRO RLV (India).** India's Reusable Launch Vehicle Technology Demonstrator flew a hypersonic test in 2016 (splashing down at sea). Later, the "Pushpak" vehicle completed three autonomous landing experiments (RLV-LEX) in 2023–2024. In these, it was dropped from a helicopter and landed on a runway at Chitradurga. An orbital re-entry experiment is planned.

**FIGURE 10.14** Space Shuttle Discovery lands at Edwards Air Force Base with its drag chute deployed at the end of STS-53 (December 1992). The Shuttle landed like a heavy glider with no engine power, at about 340 km/h. Image: NASA/JSC.

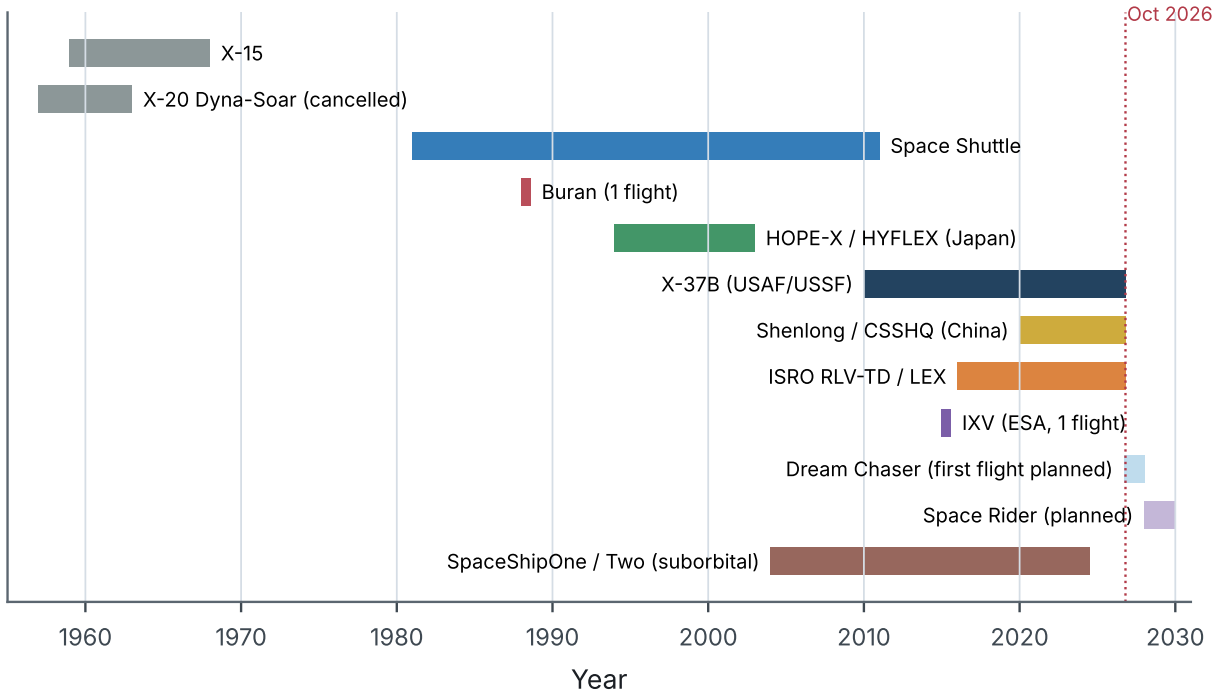



**FIGURE 10.15** Timeline of winged and lifting-body spaceplane programmes.

**TABLE 10.1** Orbital spaceplanes and lifting bodies.

| Vehicle | Country | Length / mass | Crew | Launch | Landing | Flights / status (Oct 2026) |
|---|---|---|---|---|---|---|
| Space Shuttle orbiter | USA | 37 m / ≈ 100 t at landing | 2–8 | Vertical (own engines + SRBs + ET) | Runway | 135 flights 1981–2011 (retired) |

| Vehicle | Country | Length / mass | Crew | Launch | Landing | Flights / status (Oct 2026) |
|---|---|---|---|---|---|---|
| Buran | USSR | 36 m / ≈ 80 t | 0 (designed for 2–10) | Energia | Runway (automatic) | 1 flight 1988 (cancelled) |
| X-37B OTV | USA | 8.9 m / ≈ 5 t | 0 | Atlas V, Falcon 9, Falcon Heavy | Runway | 8 missions (OTV-8 in orbit since Aug 2025) |
| Chinese reusable experimental spacecraft | China | not disclosed | 0 | Long March 2F | Runway | 3 missions 2020–2024 |
| Dream Chaser (Tenacity) | USA | 9 m / ≈ 9 t | 0 (crewed version planned) | Vulcan | Runway | First flight (free flyer) planned late 2026 |
| Space Rider | ESA | ≈ 4.6 m / ≈ 2.4 t | 0 | Vega-C | Parafoil on runway/ground | First flight NET 2028 |
| IXV | ESA | 5 m / ≈ 2 t | 0 | Vega | Parachute at sea | 1 suborbital flight 2015 |
| RLV-TD / Pushpak | India | 6.5 m / 1.75 t | 0 | Rohini booster / helicopter drops | Sea / runway | Hypersonic test 2016; landing tests 2023–24 |

## 10.5 Entry at Other Worlds

Atmospheric entry is not unique to Earth. Planetary missions enter atmospheres with very different compositions and speeds (Figure 10.16). Mars' thin $CO_2$ atmosphere provides little drag, so landers must also use parachutes and rockets (Chapter 27). Venus' dense $CO_2$ atmosphere and high entry speed produce severe heating, while Titan's thick, cold nitrogen atmosphere allowed Huygens a gentle 2.5-hour parachute descent. The Galileo probe's entry into Jupiter at 47.4 km/s remains the most extreme ever survived, consuming about half of its carbon-phenolic heat shield. **Aerocapture** and **aerobraking** exploit atmospheres to save propellant. Aerocapture uses a single pass through a planet's atmosphere to brake into orbit instead of a propulsive burn. Aerobraking uses many shallow passes to gradually lower an orbit, and has been used by Magellan, Mars Global Surveyor, MRO, TGO and X-37B.

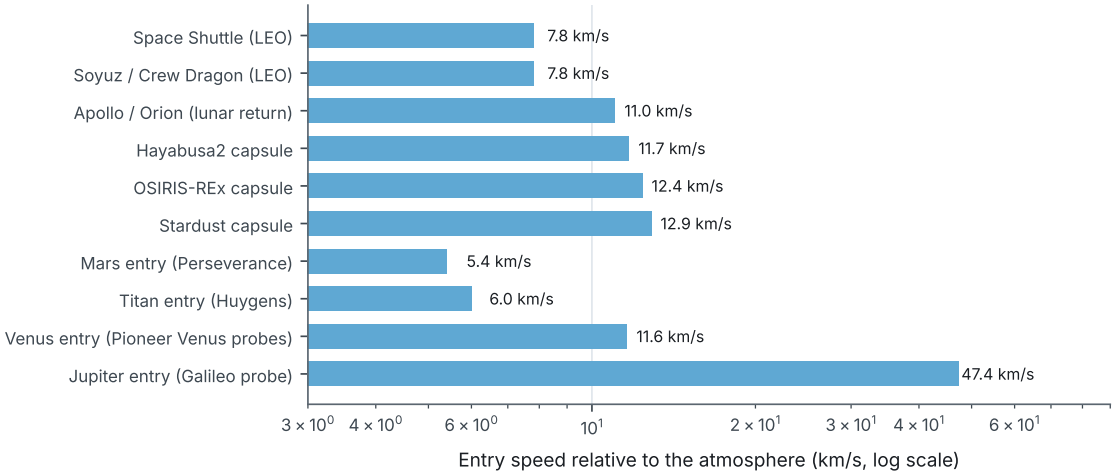



**FIGURE 10.16** Entry speeds of representative missions relative to the atmosphere they entered. The Galileo probe's Jupiter entry was nearly four times faster than any Earth return.

## 10.6 Uncontrolled Re-entries and Design for Demise

Not every re-entry is planned. Thousands of satellites, rocket stages and fragments re-enter each year; most burn up, but dense or high-melting-point components (titanium and stainless-steel tanks, optical benches, reaction-wheel hubs, carbon-fibre-overwrapped pressure vessels) can survive to the ground. About 10–40 % of the mass of large objects may survive.

- **Skylab** (77 t) re-entered over Western Australia in 1979, scattering debris over the town of Esperance.
- **Tiangong-1** (2018) and **Long March 5B core stages** (≈ 21 t, re-entering uncontrolled in 2020–2022) drew international criticism.
- In March 2024 a 0.7 kg fragment from a pallet of old ISS batteries, jettisoned in 2021, crashed through the roof of a house in Florida. This prompted legal claims against NASA under the Liability Convention principles.
- Fragments of SpaceX Dragon trunks have been found on farms in Australia (2022) and North Carolina (2024), and debris from a Falcon 9 upper stage fell in western Poland (2025).

International guidelines require that the **casualty risk** from an uncontrolled re-entry be below **1 in 10 000** (NASA-STD-8719.14, ISO 27875 and many national laws). Otherwise the object must perform a **controlled re-entry** over the South Pacific Ocean Uninhabited Area ("Point Nemo"). This is done for cargo vehicles, was done for the Mir station (2001) and is planned for the ISS (≈ 2030, using the SpaceX-built US Deorbit Vehicle). Engineers increasingly practise **design for demise**. They choose materials and configurations (aluminium rather than titanium tanks, components that separate early) so that nothing survives re-entry. Tools such as NASA's DAS and ORSAT and ESA's DRAMA predict surviving fragments. Re-entry is also a growing environmental question: aluminium oxide particles from thousands of burning satellites each year may affect stratospheric chemistry, an active research topic.

## 10.7 Hypersonic Air-Breathing Flight and Single-Stage-to-Orbit

A long-standing dream is a vehicle that takes off from a runway, breathes air for much of its ascent (avoiding the need to carry oxidiser), and flies to orbit in a single stage. Air-breathing engines offer specific impulses of 1 000–4 000 s, compared with about 450 s for the best rockets. But their thrust-to-weight ratio is low. They also work only within the atmosphere, up to about Mach 5 (turbojets, ramjets) or Mach 10–15 (scramjets, in theory).

- **Scramjets** (supersonic-combustion ramjets) burn fuel in a supersonic airstream. NASA's X-43A reached Mach 9.6 for about 10 seconds in 2004; the X-51A sustained Mach 5.1 for 210 seconds in 2013. Many countries now fly scramjet-powered hypersonic missiles.
- **Combined-cycle engines** switch between modes. The British **SABRE** (Synergetic Air-Breathing Rocket Engine) concept used a pre-cooler to chill incoming air from 1 000 °C to −150 °C in milliseconds. This allowed an engine to run as an air-breathing rocket up to Mach 5.5 and then switch to pure rocket mode. The engine was intended for the proposed **Skylon** single-stage-to-orbit spaceplane. Its developer, Reaction Engines, demonstrated the pre-cooler at Mach 5 conditions in 2019 but entered administration in October 2024.
- **Single-stage-to-orbit (SSTO)** remains elusive: the required propellant fraction leaves almost no mass for structure, TPS, engines and payload with current technology. The US NASP/X-30 (1986–1993), the X-33/VentureStar (cancelled 2001) and others were abandoned. Two-stage, partially or fully reusable rockets (Falcon 9, New Glenn, Starship) have instead become the route to cheap access (Chapter 34).

## 10.8 Reusability Beyond Spaceplanes

The reusability revolution of the 2010s came not from spaceplanes but from **vertically landing rocket stages** and **reusable capsules**. Falcon 9 first stages land on drone ships or landing pads and have flown more than 30 times each. New Glenn's booster landed on its second flight (November 2025), and China's commercial and state rockets have begun landing tests. Starship's Super Heavy boosters are caught by the launch tower's mechanical arms. These are discussed in Chapter 34 and Chapter 36. The lessons of the Shuttle — design for rapid inspection and minimal refurbishment, and fly often — have been central to their success.

### KEY TAKEAWAYS

- Returning from LEO means dissipating ≈ 30 MJ/kg (≈ 60 MJ/kg from the Moon); a blunt shape puts most of the heat into the air through a detached bow shock.
- Peak deceleration for a ballistic entry depends only on speed and angle. Lift (even L/D ≈ 0.3) cuts loads from ≈ 8–9 g to 3–4 g and allows steering. Winged vehicles keep loads near 1.5 g with large cross-range.
- Ablative heat shields (Avcoat, PICA, carbon-phenolic) handle the most severe entries; reusable TPS (silica tiles, RCC, ceramic hex tiles) must survive many flights with minimal refurbishment.
- The Shuttle proved reusable spaceflight but was expensive because of refurbishment; Columbia and Challenger taught lasting safety lessons.
- Today's spaceplanes are uncrewed or cargo vehicles (X-37B, Chinese spaceplane, Dream Chaser, Space Rider, ISRO RLV); air-breathing SSTO remains a research goal, while reusable rocket stages dominate in practice.

### REVIEW QUESTIONS

1. Calculate the kinetic energy per kilogram of a vehicle entering at 7.8 km/s and at 11 km/s. Why does only a small fraction of this energy reach the vehicle as heat?
2. Explain the “blunt-body insight” of Allen and Eggers. Why are re-entry capsules blunt rather than pointed?
3. Use the Allen–Eggers result to estimate the peak deceleration of a ballistic entry at 7.8 km/s with a 3° entry angle, taking a scale height of 7 km. How does a lift-to-drag ratio of 0.3 change the result?
4. What is the entry corridor? What happens if the entry angle is too shallow or too steep?
5. Why does communications blackout occur during re-entry, and how has it been overcome for some vehicles?
6. Compare ablative and reusable thermal protection systems, giving two examples of each and the missions they suit.
7. What made the Space Shuttle expensive to operate, and what safety lessons did the Challenger and Columbia accidents teach?
8. What is “design for demise”? Why does it matter for satellites that re-enter uncontrolled?

PART III

# Satellites in Earth Orbit

About sixteen thousand active satellites now circle the Earth. This Part explains where they fly (LEO, MEO, GEO and special orbits), how constellations are designed, how a satellite bus is built, and the families of payload instruments they carry.

11 Earth Orbits, Constellations and Small Satellites

12 Satellite Buses and Payload Instruments

Image: NASA/KSC (NASA/Randy Beaudoin)

CHAPTER

# 11 Earth Orbits, Constellations and Small Satellites

Image: NASA/JSC (Peggy Whitson)

*A satellite's orbit decides what it can see, how often, with what delay and at what cost. This chapter turns the orbital mechanics of Chapter 3 into practical mission design - choosing an orbit, calculating coverage and revisit, and designing constellations of many satellites. It then surveys the crowded reality of Earth orbit in 2026, the mega-constellations that dominate it, the geostationary belt, the small-satellite and CubeSat revolution, rideshare and in-orbit deployment, formation flying, and the sustainability questions raised by tens of thousands of satellites.*

IN THIS CHAPTER


11.1 Choosing an Orbit
11.2 Coverage Geometry
11.3 Constellation Design
11.4 The Population of Earth Orbit
11.5 Mega-Constellations
11.6 The Geostationary Orbit
11.7 Small Satellites and the CubeSat Revolution
11.8 Launch and Early Operations
11.9 Orbit Maintenance
11.10 A Worked Small-Satellite Design
11.11 Getting Small Satellites to Orbit
11.12 Formation Flying and Swarms
11.13 Operating Large Constellations
11.14 Sustainability, Regulation and the Night Sky


## 11.1 Choosing an Orbit

Orbit selection is usually the first and most important decision in mission design, because it sets the environment (Chapter 2), the launch cost, the communication geometry and the performance of the payload. The main drivers are:

- **What must be seen or served** — the whole globe, high latitudes, a specific region, or a point in deep space.
- **How often** — continuously (communications, early warning), several times a day (weather, maritime), daily (crop monitoring), or occasionally (mapping).
- **How closely** — resolution improves in proportion to distance for optical and radar sensors, and link budgets improve with the square of distance.
- **Lighting** — optical imagers want consistent Sun angles; thermal and power design prefer predictable eclipses.
- **Latency** — interactive services (voice, gaming, finance) want short round-trip delays.
- **Environment and lifetime** — drag, radiation, debris density and disposal rules.
- **Launch and cost** — higher-energy orbits need bigger rockets or longer transfers.

Table 11.1 gives typical choices.

**TABLE 11.1** Typical orbit choices by mission type.

| Mission | Typical orbit | Reason |
|---|---|---|
| High-resolution optical imaging | 400–700 km sun-synchronous (10:00–10:30 LTDN), increasingly VLEO 250–400 km | Short range for resolution; constant lighting |
| Radar imaging (SAR) | 500–700 km dawn–dusk sun-synchronous | Near-continuous sunlight for power; radar works day/night |
| Global weather (polar) | 800–850 km sun-synchronous, morning and afternoon planes | Full global coverage twice daily |
| Weather (regional, continuous) | GEO | Images every 30 s–10 min over a hemisphere |
| Ocean altimetry | 1 336 km, 66° (Jason, Sentinel-6); 890 km, 77.6° (SWOT) | Avoid sun-synchronous tidal aliasing; long-lived repeat tracks |
| Gravity mapping | 450–500 km polar, low-low formation (GRACE-FO) | Sensitivity to gravity anomalies |
| Broadband internet | 340–1 200 km LEO (constellations); GEO; MEO (O3b) | Low latency (LEO) or wide coverage per satellite (GEO) |
| Direct-to-device mobile | 500–700 km LEO; GEO for narrowband | Short range to tiny phone antennas |
| Navigation | 19 000–23 300 km MEO, GEO/IGSO for regional | Many satellites visible at once; stable orbits |
| Missile warning | GEO and HEO (Molniya-type), increasingly LEO/MEO layers | Continuous view of launch regions; polar coverage |
| Space science (astronomy) | LEO (Hubble), HEO (Chandra), Sun–Earth L2 (JWST, Euclid, Roman) | Cold, stable, unobstructed sky |
| Solar wind monitoring | Sun–Earth L1 | Upstream of Earth, ≈ 1 hour warning |
| Technology demos, education | 400–600 km (rideshare or ISS deployment) | Cheap launch, natural decay |

# 11.2 Coverage Geometry

## 11.2.1 Footprint and Elevation

What a satellite can see from altitude $h$ is described by four angles (Figure 11.1). The **elevation angle** $\varepsilon$ is the angle of the satellite above the local horizon of a ground point. The **nadir angle** $\eta$ is the angle at the satellite between the nadir and the ground point; the others are the **Earth central angle** $\lambda$ and the slant range $\rho$. They are linked by

$$\sin\eta = \frac{R_\oplus}{R_\oplus + h}\cos\varepsilon, \qquad \lambda = 90^\circ - \eta - \varepsilon, \qquad \rho = R_\oplus \frac{\sin\lambda}{\sin\eta}$$

and the area of the coverage circle is $A = 2\pi R_\oplus^2 (1 - \cos\lambda)$.

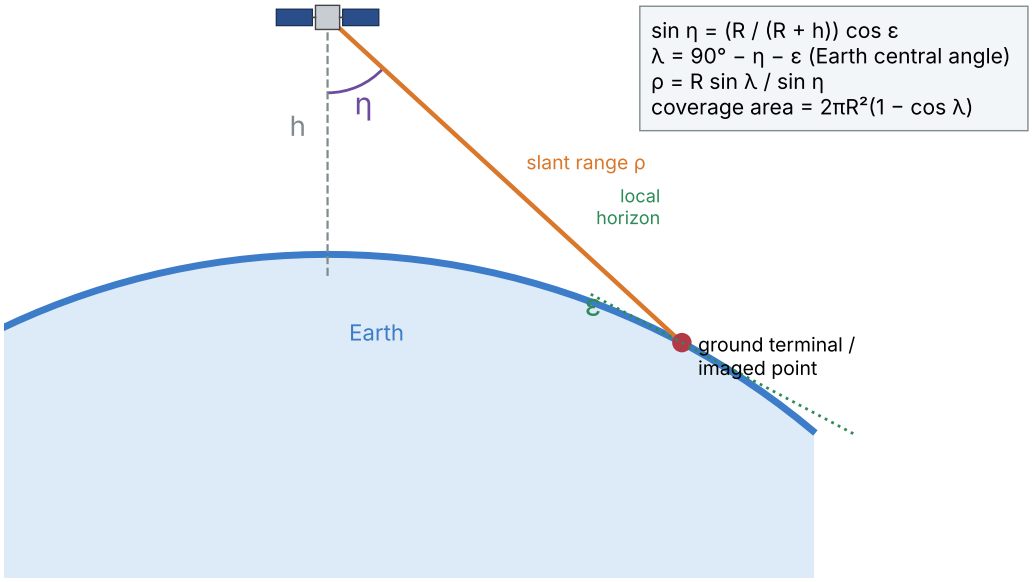



**FIGURE 11.1** Coverage geometry of a satellite. The minimum elevation angle at which a terminal can use the satellite — limited by terrain, buildings, atmosphere and antenna design — sets the coverage radius.

The **minimum elevation angle** is a key design parameter. It is 5–10° for ground stations with clear horizons. For broadband user terminals it is 25–40° (to avoid blockage and long atmospheric paths and to share spectrum with GEO systems); for navigation it is 10–20°. A higher minimum elevation shrinks the footprint, so more satellites are needed (Figure 11.2).

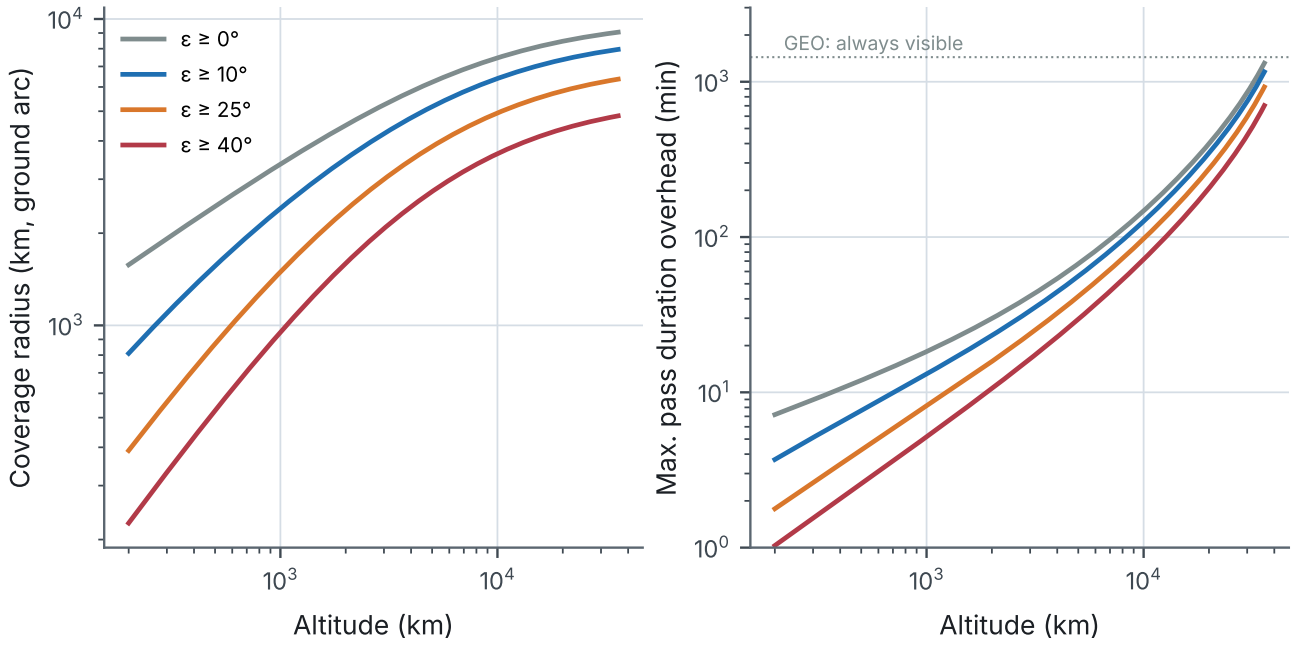



**FIGURE 11.2** Left: coverage radius on the ground versus altitude for four minimum elevation angles. Right: maximum duration of an overhead pass. LEO satellites are visible for minutes, MEO for hours, GEO continuously.

**WORKED EXAMPLE — HOW MANY MINUTES CAN A GROUND STATION TALK TO A 550 KM SATELLITE?**

For $h$ = 550 km and $\varepsilon$ = 10°: $\sin\eta = (6\,378/6\,928)\cos 10^\circ = 0.9066$, so $\eta = 65.0^\circ$ and $\lambda = 90 - 65.0 - 10 = 15.0^\circ$. The coverage radius is $6\,378 \times 0.262 = 1\,670$ km. The satellite's angular rate is its mean motion, $n = 360^\circ/95.6$ min = 3.77°/min, so an overhead pass lasts at most $2 \times 15.0/3.77 \approx 8.0$ minutes; passes that do not go directly overhead are shorter. A station sees such a satellite on 4–6 passes a day at mid-latitudes, but on almost every orbit from Svalbard if the orbit is polar.

## 11.2.2 Swath, Revisit and Repeat

For an Earth-observation satellite, the relevant footprint is the **swath** — the width of ground imaged as the satellite moves. It is set by the instrument field of view or, for agile satellites, by how far off-nadir it can point. Two different time measures are used:

- **Revisit time** — how often a given place *can* be observed (possibly at different angles). Agile pointing and constellations reduce revisit to hours or less.
- **Repeat cycle** — how long until the satellite passes over exactly the same ground track (Chapter 3), giving identical viewing geometry. Landsat 8 and 9 each repeat every 16 days; together they give 8-day coverage.

For a single sun-synchronous imager with swath $W$ at the equator, global coverage takes about $2\pi R_\oplus/(W \times N_{orbits/day})$ days. For example, a 290 km swath (Sentinel-2) with 14.3 orbits per day covers the equator in about 10 days. Two satellites halve this to 5 days. Revisit improves at higher latitudes, where ground tracks converge.

## 11.3 Constellation Design

When one satellite is not enough — for continuous coverage, frequent revisit, more capacity or redundancy — engineers design **constellations**.

### 11.3.1 Walker Constellations

The most widely used patterns were described by J. G. Walker in 1970–1984. A **Walker delta** constellation is written **i: T/P/F** — inclination, total number of satellites $T$, number of equally spaced orbital planes $P$ (with RAAN spaced 360°/P), and a phasing parameter $F$ (0 to P−1) that sets the relative position of satellites in adjacent planes (in units of 360°/T). GPS's original design is close to a 55°: 24/6/1 Walker; Galileo is a 56°: 24/3/1 Walker (plus spares). A **Walker star** (or polar, "street-of-coverage") constellation places planes over 180° of RAAN with near-polar inclination, so that adjacent planes overlap like streets; Iridium uses 66 active satellites in six planes at 86.4° (Figure 11.3).

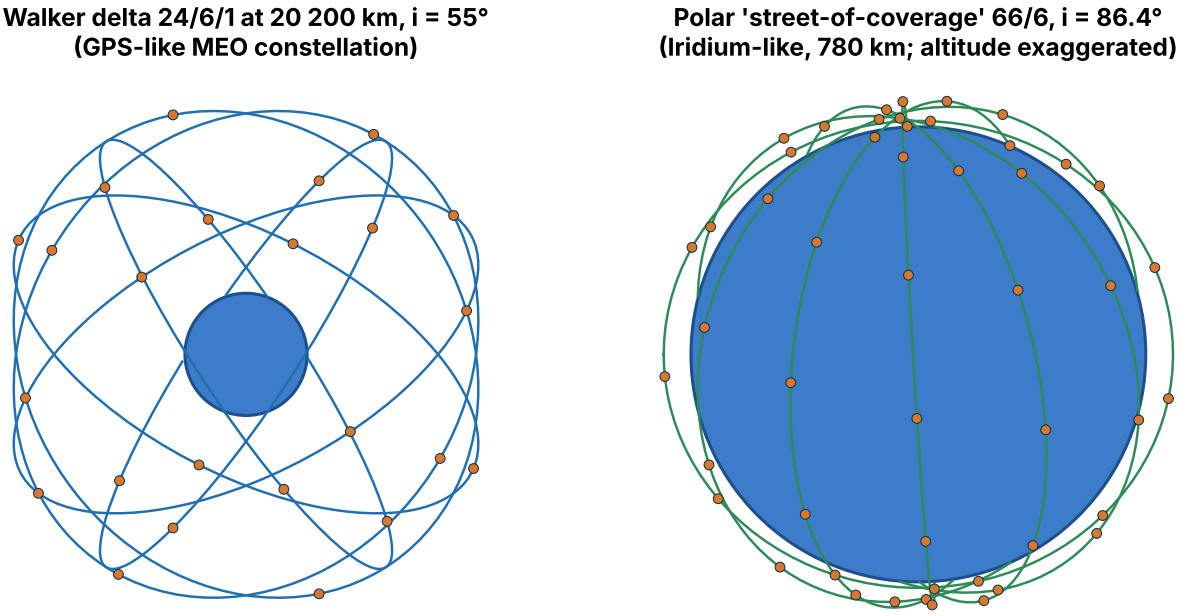


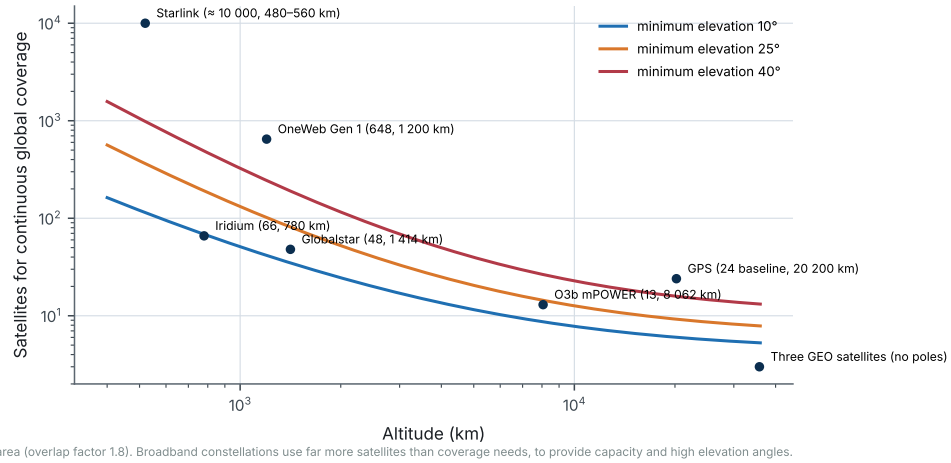

**FIGURE 11.3** Two classic constellation geometries: a Walker delta MEO constellation like GPS, and a near-polar "street-of-coverage" constellation like Iridium.

Other families include **Flower constellations** (satellites on identical repeating ground tracks) and **elliptical constellations** that concentrate coverage over high latitudes (Molniya, Ellipso). There are also hybrid multi-orbit systems combining LEO, MEO and GEO (e.g., Eutelsat OneWeb with Eutelsat's GEO fleet; SES with O3b mPOWER and GEO; IRIS$^2$).

### 11.3.2 How Many Satellites?

The number of satellites needed for continuous global coverage falls rapidly with altitude because each satellite sees more of the Earth (Figure 11.4). Three GEO satellites cover everything except the polar caps; about 24 MEO satellites give global navigation (which needs at least four in view everywhere); 66 satellites at 780 km give Iridium global voice coverage. Broadband mega-

constellations use far more satellites than coverage alone requires — thousands. This is because what limits them is **capacity**. Each satellite can serve only so many users with its spectrum and beams, and users want high elevation angles and low latency.




Approximate: N ≈ 1.8 × 4π / cap area (overlap factor 1.8). Broadband constellations use far more satellites than coverage needs, to provide capacity and high elevation angles.

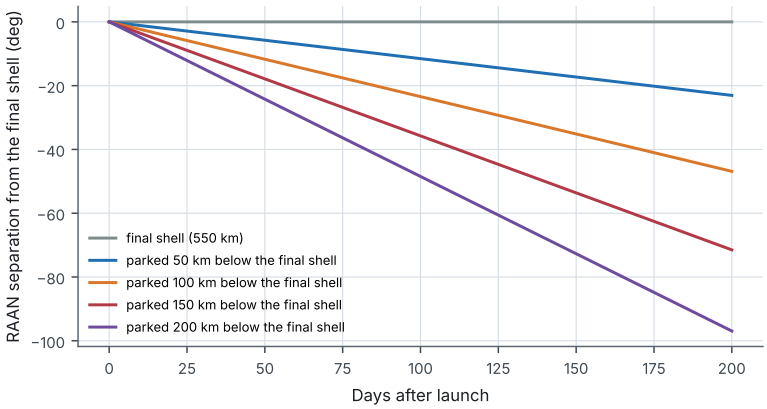
**FIGURE 11.4** Approximate number of satellites needed for continuous single global coverage versus altitude, compared with real systems. Broadband constellations exceed the coverage requirement many times over to provide capacity.

The main constellation trade-offs are:

| Choice | Lower altitude | Higher altitude |
|---|---|---|
| Number of satellites | More | Fewer |
| Latency (round trip, user–satellite–gateway) | ≈ 20–40 ms in LEO | ≈ 125 ms in MEO, ≈ 500+ ms in GEO |
| Link budget, user terminal size | Better / smaller | Worse / larger |
| Satellite lifetime vs drag | Shorter (natural disposal) | Longer (must actively dispose) |
| Radiation | Lower | Higher (especially MEO) |
| Handover and tracking | Frequent (minutes) | Rare or none |
| Launch cost per satellite | Lower | Higher |
| Debris risk if satellites fail | Self-cleaning below ≈ 550 km | Persistent above ≈ 600 km |

### 11.3.3 Deploying a Constellation: Plane Phasing

Placing satellites in many orbital planes is expensive if each plane needs its own launch. Constellation operators therefore exploit **differential nodal precession** (Chapter 3). Because the $J_2$ precession rate depends on altitude, satellites from a single launch can wait at different "parking" altitudes, where their planes drift apart in RAAN at different rates. They then raise themselves to the operational shell at the right moment (Figure 11.5). A satellite parked 100 km below a 550 km shell drifts about 0.18° per day relative to it, so three months of waiting separates planes by about 15°. The cost is time rather than propellant — a trade that suits electric propulsion and mass-produced satellites.




$J_2$ nodal drift at 53° inclination. Satellites from one launch wait at different altitudes and drift apart in RAAN for free before climbing to the operational shell — 'differential precession' plane phasing.

**FIGURE 11.5** Relative RAAN drift between satellites parked below a 550 km, 53° shell and satellites already in it. Waiting at lower altitudes spreads orbital planes using Earth's oblateness instead of propellant.

**TABLE 11.2** Key parameters of major satellite constellations (October 2026).

| Constellation | Operator | Service | Altitude / inclination | Satellites (in orbit / planned) | Mass per satellite | Notable features |
|---|---|---|---|---|---|---|
| Starlink | SpaceX (USA) | Broadband, D2C, government | ≈ 480–570 km; 33°–97.6° | ≈ 10 650 / 12 000 (Gen1) + Gen2 | ≈ 300 kg (V1.5), ≈ 800 kg (V2 Mini), ≈ 2 t (V3) | Laser mesh, argon Hall thrusters, Starship-launched V3 |
| Eutelsat OneWeb | Eutelsat (France/UK) | Enterprise broadband | 1 200 km, 87.9° | ≈ 650 / 648 + Gen2 | ≈ 150 kg | Bent-pipe Ku/Ka; GEO + LEO offering |
| Amazon Leo | Amazon (USA) | Broadband | 590–630 km, 33°–52° | ≈ 400 / 3 232 | ≈ 600 kg | Launches on five vehicle types |
| Qianfan | SSST (China) | Broadband | ≈ 800–1 000 km (first shells) | ≈ 220–240 / > 14 000 | ≈ 250–300 kg | Launched in batches of 18 on Long March 6A |
| Guowang | China SatNet | Broadband, government | ≈ 500–1 145 km | ≈ 200 / ≈ 13 000 | various | State-owned |
| Iridium NEXT | Iridium (USA) | Voice, data, IoT, PNT | 780 km, 86.4° | 66 active + spares | 860 kg | Cross-links; hosted ADS-B (Aireon) |
| Globalstar | Globalstar (USA) | Voice, data, Apple emergency SOS | 1 414 km, 52° | ≈ 25 active + new generation | ≈ 700 kg | Bent-pipe; new satellites from MDA |
| O3b mPOWER | SES (Luxembourg) | High-throughput MEO | 8 062 km, 0° | 13 | ≈ 1 700 kg | Thousands of steerable beams |
| AST SpaceMobile | AST (USA) | Direct-to-device | ≈ 520–700 km | 13 / ≈ 45 (2027) | ≈ 1.5–6 t | Giant phased arrays |
| SDA PWSA | US Space Development Agency | Military data transport, missile tracking | ≈ 1 000 km | 42 / ≈ 1 000 (all tranches) | ≈ 500–1 000 kg | Optical crosslinks, Link 16 |
| Planet (Dove/Pelican) | Planet Labs (USA) | Earth observation | ≈ 475–525 km SSO | ≈ 200 | 5 kg (Dove), ≈ 100–200 kg (Pelican) | Daily global imaging |

| Constellation | Operator | Service | Altitude / inclination | Satellites (in orbit / planned) | Mass per satellite | Notable features |
|---|---|---|---|---|---|---|
| Spire | Spire Global (USA) | RF data (AIS, ADS-B, GNSS-RO) | ≈ 400–600 km | ≈ 80–100 | ≈ 5 kg (3U) | GNSS radio occultation for weather |
| ICEYE | ICEYE (Finland) | SAR imaging | ≈ 550 km SSO | ≈ 50 | ≈ 90–130 kg | Largest SAR constellation |

## 11.4 The Population of Earth Orbit

The orbital population in 2026 is dominated by a few altitude bands (Figure 11.6). CelesTrak's public catalogue lists about **35 000 objects** in Earth orbit — 20 100 payloads (of which about 17 200 are operational), 12 500 debris fragments and 2 300 rocket bodies. More than half of all objects now lie between 450 and 600 km, where Starlink operates. A second, older concentration between 750 and 1 000 km contains sun-synchronous Earth-observation satellites and Iridium. It also holds the long-lived debris from the 2007 Fengyun-1C test and the 2009 Iridium–Cosmos collision.

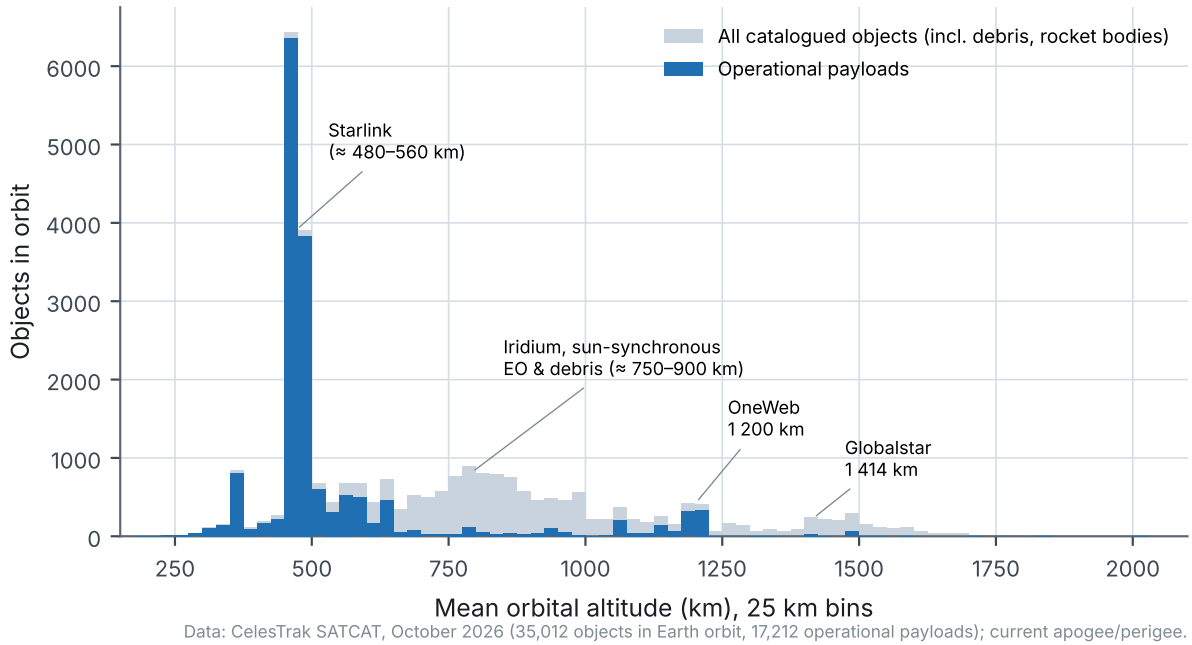



**FIGURE 11.6** Distribution of catalogued objects in Earth orbit by mean altitude, October 2026. Operational payloads (blue) cluster in constellation shells; debris (grey) dominates the 700–1 000 km region.

## 11.5 Mega-Constellations

### 11.5.1 Starlink

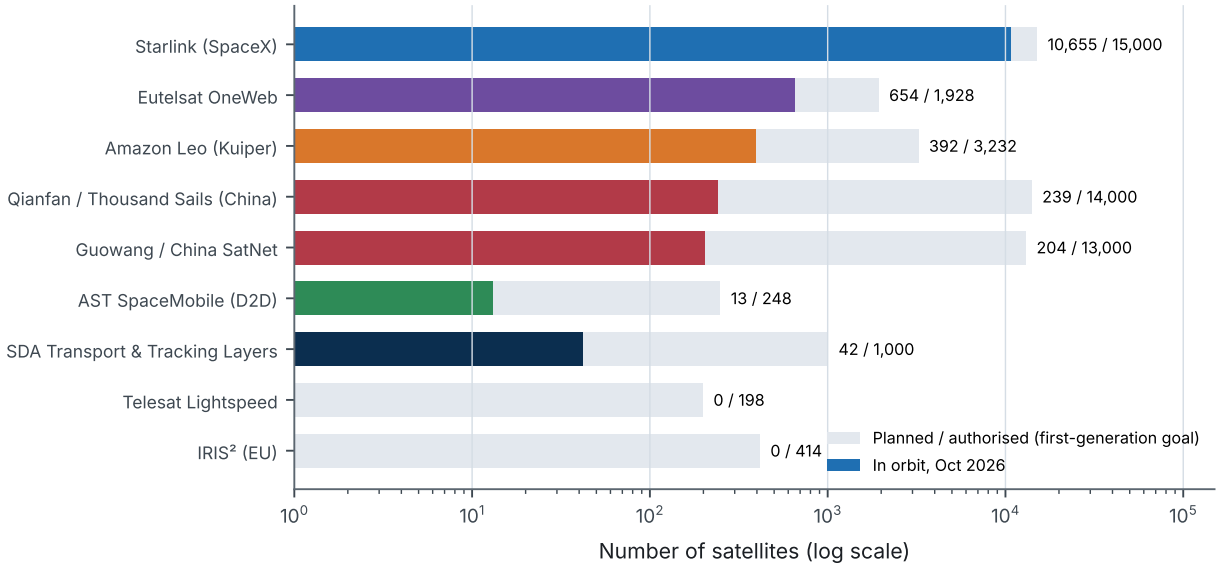
SpaceX's **Starlink** is the largest satellite system ever built. Its first operational satellites launched in May 2019. By October 2026 more than 10 600 were in orbit, serving more than 8 million subscribers in over 140 countries and territories. They provide broadband, maritime, aviation and mobile ("Direct to Cell") services. Key engineering features include:

- **Mass production** — satellites are built at rates of several per day in Redmond, Washington. They are launched in batches of 21–29 (V2 Mini) on Falcon 9 or, from September 2026, in larger batches of the much bigger **V3** satellites on Starship.
- **Flat-pack design** — satellites stack in the fairing without a dispenser; a single solar array per satellite on V1/V1.5, two on V2 Mini.
- **Electric propulsion** — krypton (V1) and argon (V2 Mini) Hall thrusters for orbit raising, station-keeping, collision avoidance and de-orbit.
- **Phased-array antennas** for Ku- and Ka-band user and gateway links and E-band backhaul; **laser inter-satellite links** (three or more per satellite) forming a mesh network that routes traffic across oceans and polar regions.
- **Autonomous collision avoidance** — satellites manoeuvre automatically using conjunction data; SpaceX reports tens of thousands of avoidance manoeuvres per half-year (Chapter 21).
- **Low orbits** for natural disposal: SpaceX began lowering about 4 400 satellites from about 550 km to about 480 km during 2026, citing lower debris density and much faster decay of failed satellites.

The US government uses a derivative, **Starshield**, for national-security applications, including a proliferated reconnaissance constellation for the National Reconnaissance Office.

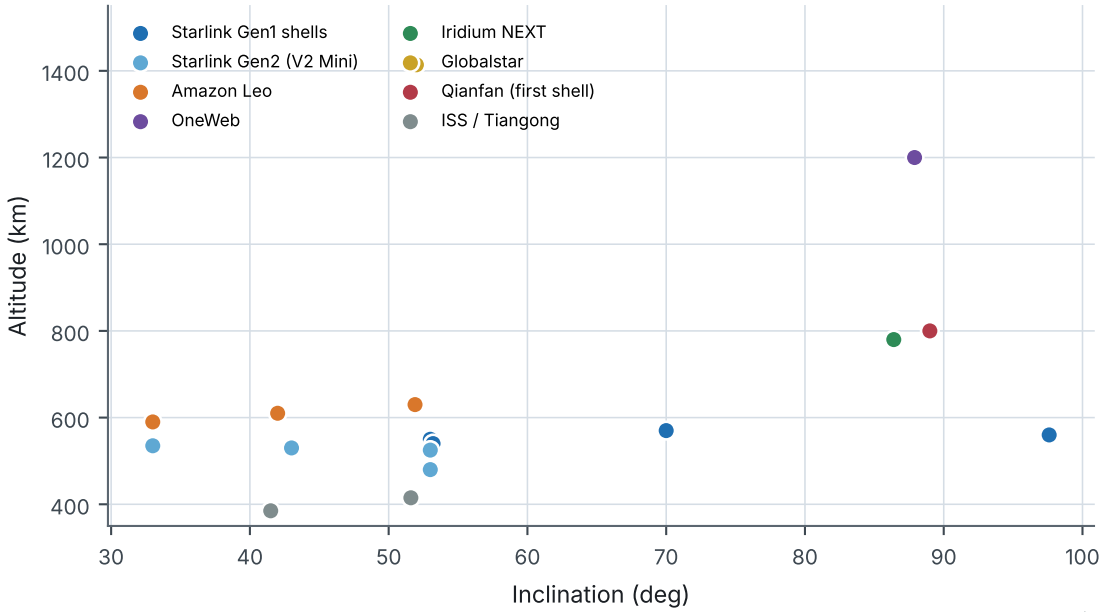
### 11.5.2 Other Broadband Constellations

- **Eutelsat OneWeb** — about 650 satellites at 1 200 km in 12 near-polar planes, serving enterprise, government, aviation and maritime customers through partners; second-generation satellites are being procured.
- **Amazon Leo** (formerly Project Kuiper) — authorised for 3 232 satellites in three shells at 590–630 km. About 400 were in orbit by mid-2026 after launches on Atlas V, Falcon 9, Vulcan, Ariane 6 and New Glenn. Beta service was planned for autumn 2026. The FCC waived its July 2026 half-constellation milestone.
- **Qianfan / "Thousand Sails"** (Shanghai Spacecom Satellite Technology) — planned to exceed 14 000 satellites; about 220–240 launched by mid/late 2026.
- **Guowang** (China Satellite Network Group, "SatNet") — a state constellation of about 13 000 satellites; about 200 launched by September 2026.
- **Telesat Lightspeed** — 198 advanced satellites with optical links planned for enterprise and government service from the late 2020s.
- **IRIS²** — the EU's secure multi-orbit constellation (SpaceRISE consortium of SES, Eutelsat and Hispasat). Under an August 2026 agreement, it comprises 330 LEO and 18 MEO satellites plus 66 dedicated government satellites, with first launches planned for 2029.
- **AST SpaceMobile** — very large (up to ≈ 220 $m^2$ phased-array) "BlueBird" satellites that connect directly to unmodified phones; 13 in orbit by August 2026, with about 45 targeted by early 2027 (Chapter 17).




In-orbit counts: GCAT (Oct 2026) and company/agency reports; planned numbers are first-generation authorisations or declared targets and change frequently.

**FIGURE 11.7** Mega-constellations: satellites in orbit (coloured) compared with first-generation plans or authorisations (grey), October 2026.




Representative published shells; constellations are reconfigured over time (e.g., Starlink lowering shells from ≈ 550 to ≈ 480 km during 2026).

**FIGURE 11.8** Altitudes and inclinations of representative constellation shells. Shells at different inclinations concentrate capacity where most users live (mid-latitudes) while polar shells cover high latitudes.

### 11.5.3 Military Proliferated Constellations

The US **Space Development Agency (SDA)** is deploying the **Proliferated Warfighter Space Architecture** — hundreds of relatively low-cost satellites in LEO in successive "tranches" every two years. It has a **Transport Layer** of data-relay satellites with optical crosslinks and Link 16 tactical radios. It also has a **Tracking Layer** of infrared satellites that detect and track missiles (Chapter 20). Tranche 1 launches began in September 2025. China, Europe and others are pursuing similar concepts, reflecting a shift from a few exquisite satellites to resilient proliferated architectures.

# 11.6 The Geostationary Orbit

## 11.6.1 A Unique Resource

A satellite at 35 786 km altitude above the equator, moving eastward with a period of one sidereal day, appears fixed in the sky. Antennas on the ground need not track it, and three satellites cover most of the populated Earth. The **geostationary arc** is therefore a limited natural resource, shared through ITU coordination. Satellites are assigned **orbital slots** (longitudes), typically spaced 2–3° for the same frequency bands. They must stay within a **station-keeping box** of about ±0.05–0.1° in longitude and latitude. About 1 340 payloads are in GEO-type orbits, of which roughly 600–650 are operational.

Natural forces push GEO satellites out of their boxes. The Sun and Moon tilt the orbit by about 0.8° per year (north–south drift). The ellipticity of Earth's equator causes east–west drift towards stable points near 75° E and 105° W, and solar radiation pressure changes the eccentricity. Station-keeping requires about 50 m/s per year (Chapter 3). At end of life, operators must raise their satellites at least 235 km plus a margin depending on area-to-mass ratio above GEO into a **graveyard orbit** (IADC guideline). Some operators extend life by allowing inclination to grow ("inclined-orbit operation") or by using life-extension servicers (Chapter 40).

## 11.6.2 Eclipses and Sun Interference

GEO satellites experience eclipses only around the equinoxes (up to 72 minutes per day for about 45 days in each season). Around the equinoxes, the Sun also passes directly behind satellites as seen from ground antennas, causing **Sun outages** of a few minutes per day due to solar noise.

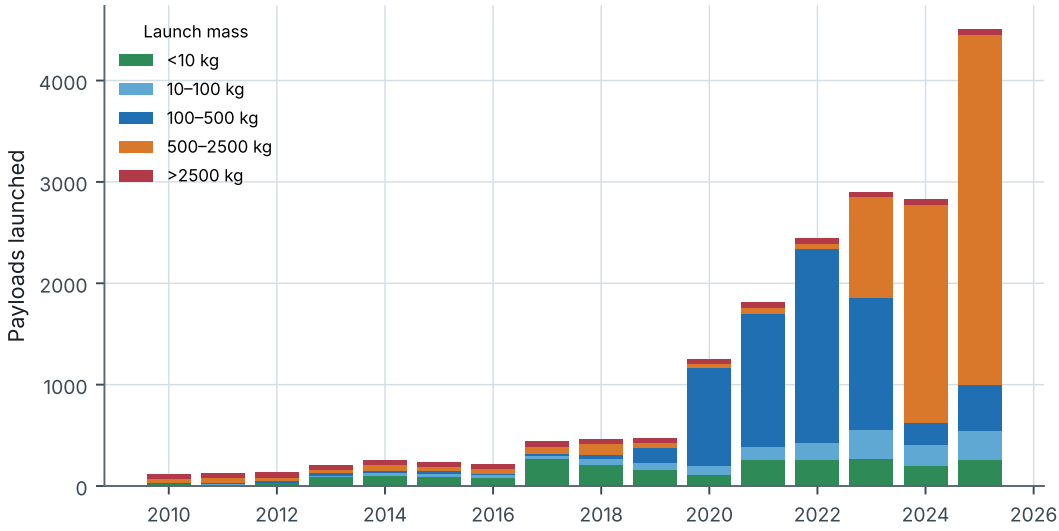
## 11.6.3 Trends in GEO

GEO remains the workhorse for broadcasting and many government services but faces competition from LEO broadband. Trends include **software-defined, reconfigurable payloads** that can change coverage and frequency plans in orbit (Airbus OneSat, Thales Inspire, Boeing 702X). Other trends are **all-electric propulsion**, **smaller GEO satellites** (Astranis MicroGEO, about 400 kg), and **in-orbit servicing** to extend life.

# 11.7 Small Satellites and the CubeSat Revolution

## 11.7.1 Why Small?

"Small satellites" (below about 500 kg) have transformed the industry by trading individual capability for speed, cost and numbers. Their enablers were: commercial electronics (processors, cameras, radios) that became small, capable and cheap; standardised interfaces and deployers; inexpensive rideshare launch; and a culture of rapid iteration ("fail fast"). Figure 11.9 shows how launches have shifted over the past 15 years — first towards very small satellites, then towards mass-produced constellation satellites in the 100–2 500 kg range.




Data: GCAT. Starlink V1/V1.5 (≈ 260–310 kg) fill the 100–500 kg class to 2022; V2 Mini (≈ 800 kg) and Amazon Leo (≈ 600 kg) fill the 500–2 500 kg class from 2023.

**FIGURE 11.9** Payloads launched per year by launch mass, 2010–2025 (GCAT). The explosion after 2019 is driven by mass-produced constellation satellites; CubeSats (< 10 kg) continue at several hundred per year.

## 11.7.2 The CubeSat Standard

The **CubeSat** was proposed in 1999 by Jordi Puig-Suari (California Polytechnic State University) and Bob Twiggs (Stanford) as a standard for student satellites. It is a 10 cm cube ("1U") of up to about 2 kg. It is launched from a standard spring-loaded box deployer (the P-POD and its successors) that protects the launch vehicle and primary payload. Larger formats stack units: 3U, 6U, 12U and 16U are now the most common (Figure 11.10). The standard's genius is that it decouples the satellite from the launcher: any CubeSat fits any compatible deployer on any rocket or on the ISS.

CubeSat Design Specification (Cal Poly, rev. 14):
1U = 10 cm cube; rails slide in a
spring-loaded deployer

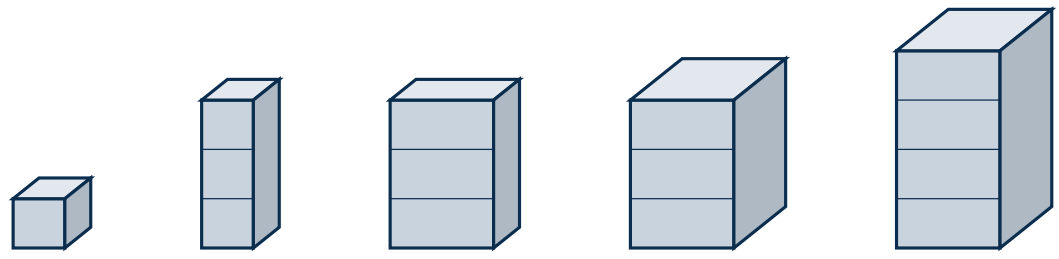



**FIGURE 11.10** Common CubeSat form factors and approximate maximum masses.

A thriving commercial supply chain now sells CubeSat buses and components "off the shelf". These include structures, solar panels, batteries and power units, reaction wheels and star trackers. There are also flight computers running Linux or FreeRTOS, S- and X-band radios, and even electric thrusters. A 6U Earth-observation CubeSat can be built and launched for a few million dollars in under two years.

### 11.7.3 What Small Satellites Do

- **Earth observation** — Planet's fleet of about 200 "Dove" and "SuperDove" 3U CubeSats images the entire land surface daily at 3–5 m resolution; larger SkySats and Pelican satellites offer 50 cm imaging.
- **Radio-frequency data** — Spire's Lemur CubeSats collect GNSS radio-occultation weather data, ship AIS and aircraft ADS-B signals; HawkEye 360 geolocates RF emitters (Chapter 19).
- **Radar** — ICEYE, Capella and Umbra fly 50–150 kg SAR microsatellites (Chapter 15).
- **Science** — NASA's TROPICS CubeSats measure tropical-cyclone microwave signatures every hour; GRACE-like and space-weather missions use CubeSat swarms.
- **Deep space** — NASA's twin **MarCO** CubeSats relayed InSight's landing data from Mars in 2018 — the first CubeSats beyond Earth orbit. Ten CubeSats rode on Artemis I to the Moon (2022), and CAPSTONE pioneered the Gateway's NRHO.
- **Technology demonstration and education** — thousands of students worldwide have built CubeSats; the ISS deploys many of them.

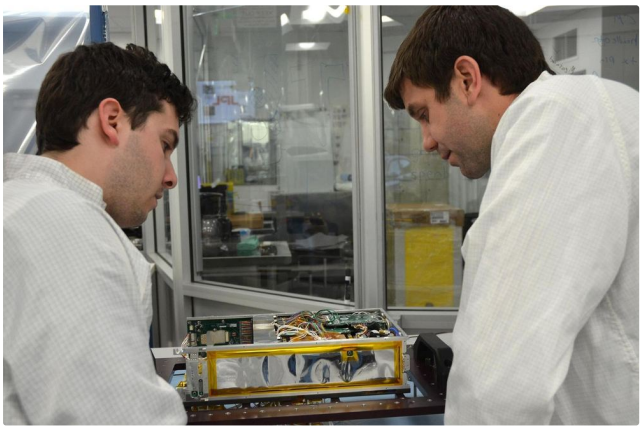


**FIGURE 11.11** JPL engineers with one of the twin MarCO 6U CubeSats, which flew alongside InSight to Mars in 2018 and relayed its landing data in real time. Image: NASA/JPL (Elizabeth Landau).

**FIGURE 11.12** NASA astronaut Frank Rubio installs a NanoRacks CubeSat Deployer in the ISS Kibo airlock (2023). Deployers are passed outside through the airlock and the station's robotic arm releases the CubeSats. Image: NASA/JSC.

### 11.7.4 Reliability of Small Satellites

Early CubeSats had high failure rates. Studies of missions launched up to the mid-2010s found that roughly 20–40 % failed early in their missions, often due to power, communications or deployment failures. University "first missions" fared worst. Reliability has improved greatly with commercial heritage hardware, more thorough testing ("flatsat" functional testing, thermal-vacuum and vibration) and better operations. Commercial constellations accept some individual failures because the system degrades gracefully; science and defence small satellites adopt selected high-reliability practices ("Class D" risk posture, Chapter 4).

## 11.8 Launch and Early Operations

The first days after launch — the **launch and early orbit phase (LEOP)** — are the riskiest of a satellite's life. A typical LEO satellite's LEOP sequence is:

1. **Separation** from the launch vehicle; the spacecraft wakes on a separation switch, powers essential units and starts timers.
2. **Detumbling** — rates from separation (up to a few degrees per second) are damped using magnetorquers ("B-dot" control) or thrusters.
3. **Sun acquisition** — solar arrays are deployed (if folded) and pointed at the Sun to make the spacecraft power-positive.
4. **First contact** — ground stations around the world, often including commercial network stations, search for the beacon. Orbit determination from the launcher's injection data and early ranging refines the orbit; for rideshare launches, operators must first identify which of dozens of newly catalogued objects is theirs.
5. **Deployments** — antennas, booms, radiator covers and instrument doors are released.
6. **Commissioning** — each subsystem and the payload are switched on and verified; propulsion is tested with small burns; the satellite is manoeuvred to its operational orbit.
7. **Calibration and validation** of the payload, typically lasting weeks (communications) to months (science instruments), before the satellite enters routine operations.

## 11.9 Orbit Maintenance

Satellites in LEO must counter drag to stay in their slots; constellations must also maintain their relative phasing. The velocity lost to drag each year is the integral of the drag deceleration over the year:

$$\Delta v_{year} = \int_{1\,\mathrm{yr}} \frac{\rho v^2}{2B}\, dt \;\approx\; \frac{\rho v^2}{2B} \times 3.16 \times 10^{7}\ \mathrm{s}$$

At 550 km with moderate solar activity ($\rho \approx 5 \times 10^{-13}$ kg/m$^3$, $v$ = 7.6 km/s) and $B$ = 50 kg/m$^2$, the drag deceleration is $5 \times 10^{-13} \times (7\,600)^2/100 \approx 2.9 \times 10^{-7}$ m/s$^2$. This is about **9 m/s per year**. At 400 km and solar maximum it can exceed 100 m/s per year.

## 11.10 A Worked Small-Satellite Design

To see how the material of Part I comes together, consider the preliminary design of a 12U CubeSat for agricultural monitoring.

**Requirements.** Multispectral images (blue, green, red, red-edge, near-infrared) at 5 m ground sampling distance (GSD) over Europe, revisit ≤ 5 days with a constellation, 3-year life, launch on a rideshare to SSO.

**Orbit.** 525 km sun-synchronous (97.5°), 10:30 local time. Lifetime analysis (Chapter 2) gives 6–15 years for a 12U CubeSat with deployed panels, so a small thruster for collision avoidance and end-of-life lowering is advisable to meet the 5-year disposal rule with margin.

**Payload optics.** GSD = altitude × pixel pitch / focal length. With 5.5 µm pixels: $f = 525\,000 \times 5.5 \times 10^{-6}/5 = 0.58$ m. A 9 cm-aperture compact catadioptric telescope with a 580 mm focal length fits in about 4U. At f/6.4 the diffraction limit at 800 nm ($2.44\lambda N \approx 12.5$ µm spot diameter) is acceptable for 5 m GSD, with some MTF loss. A 4 096-pixel line array gives a 20 km swath.

**Data budget.** Imaging Europe's croplands ≈ 2 million km² per 5 days per satellite with 5 bands × 12 bits at 5 m: $2 \times 10^6\,\text{km}^2/5\,\text{days} \times (10^6/25)\,\text{pixels/km}^2 \times 60\,\text{bits} \approx 1 \times 10^{12}$ bits/day ≈ 1 Tbit/day before compression. With 3:1 compression and onboard cloud screening (Chapter 45) removing about half, the downlink requirement is ≈ 170 Gbit/day. An X-band transmitter at 200 Mbit/s needs about 14 minutes of contact per day — two or three passes over a high-latitude station.

**Power budget.**

| Mode | Power (W) | Duty / orbit | Energy per orbit (Wh) |
|---|---|---|---|
| Platform (OBC, ADCS, receivers, heaters) | 12 | 100 % | 19.1 |
| Imaging (payload + ADCS fine pointing) | 25 | 10 % | 4.0 |
| X-band downlink | 30 | 7 % | 3.3 |
| Total | — | — | ≈ 26.4 per 95.3-min orbit |

Deployable panels with triple-junction cells (≈ 80 W peak, ≈ 45 W orbit-average after pointing losses) give a positive balance with 30 % margin. A 90 Wh Li-ion battery supports eclipse at < 20 % depth of discharge.

**Mass budget.**

| Item | Mass (kg) |
|---|---|
| Structure and deployers | 3.2 |
| Payload telescope and focal-plane electronics | 5.5 |
| Power (panels, battery, EPS) | 3.0 |
| ADCS (3 wheels, star tracker, magnetorquers, sensors) | 1.6 |
| OBC and payload processor (with AI accelerator) | 0.6 |
| S-band TT&C and X-band transmitter with patch array | 0.9 |
| Propulsion (electrospray or cold gas) | 1.2 |
| Harness, thermal hardware | 0.8 |
| **Subtotal** | **16.8** |
| Margin (25 %) | 4.2 |
| **Total (12U limit ≈ 24 kg)** | **21.0** |

**Constellation.** With a 20 km swath, one satellite cannot meet a 5-day revisit over Europe by nadir imaging alone. Pointing ±20° off-nadir gives an accessible swath of about 400 km. So a constellation of four to six satellites in one or two planes meets the requirement. This example illustrates how requirements cascade through optics, data, power, mass and constellation size — the essence of systems engineering (Chapter 4).

## 11.11 Getting Small Satellites to Orbit

### 11.11.1 Rideshare

Most small satellites fly as **secondary payloads** sharing a launch with others. SpaceX's **Transporter** (sun-synchronous) and **Bandwagon** (mid-inclination) missions carry 50–120 satellites per flight from many customers; ISRO's PSLV, Rocket Lab's Electron, Arianespace's Vega-C (SSMS dispenser) and China's Long March and commercial rockets offer similar services. Rideshare is cheap (≈ US$6 000–7 000 per kg on Transporter in 2025) but the customer does not choose the orbit or the date. **Dedicated small launchers** (Electron, Firefly Alpha, Kinetica-1, Ceres-1, Zhuque-2, Vega-C, PSLV, SSLV) provide orbit and schedule control at higher cost per kilogram (Chapter 36).

### 11.11.2 Orbital Transfer Vehicles

**Orbital transfer vehicles (OTVs)** or "space tugs" bridge the gap: they ride a rideshare launch, then carry their satellite passengers to individual orbits or planes. Examples include D-Orbit's ION Satellite Carrier (more than 15 missions since 2020), Exotrail's spacevan, Momentus' Vigoride, Impulse Space's Mira, Rocket Lab's Photon and Firefly's Elytra. Some OTVs also host experiments and are evolving into servicing vehicles.

### 11.11.3 Deployment from the ISS

The ISS has deployed hundreds of CubeSats through the Japanese **J-SSOD** and commercial NanoRacks deployers in the Kibo module airlock. ISS-deployed satellites start at about 410 km and decay within one to three years — an advantage for debris mitigation.

## 11.12 Formation Flying and Swarms

Some missions require two or more satellites to fly in a controlled geometry:

- **GRACE and GRACE-FO** (US–Germany) — two satellites 220 km apart in the same orbit measure their separation to micrometres (GRACE-FO adds a laser ranging interferometer at nanometre precision). These measurements map changes in Earth's gravity from melting ice, groundwater and ocean mass.
- **TanDEM-X** (Germany) — two radar satellites flying in a close "helix" formation a few hundred metres apart acted as a single-pass interferometer to produce a global digital elevation model (Chapter 15).
- **PROBA-3** (ESA, launched December 2024) — two satellites flying about 150 m apart in a highly elliptical orbit. One acts as an occulting disc for the other's coronagraph to create artificial solar eclipses, with millimetre-level relative positioning. It produced its first artificial-eclipse images of the solar corona in 2025.
- **MMS** (NASA) — four spacecraft in a tetrahedral formation measuring magnetic reconnection (Chapter 23).
- **Starling** (NASA, 2023) — four CubeSats demonstrating autonomous swarm manoeuvre planning and relative navigation.

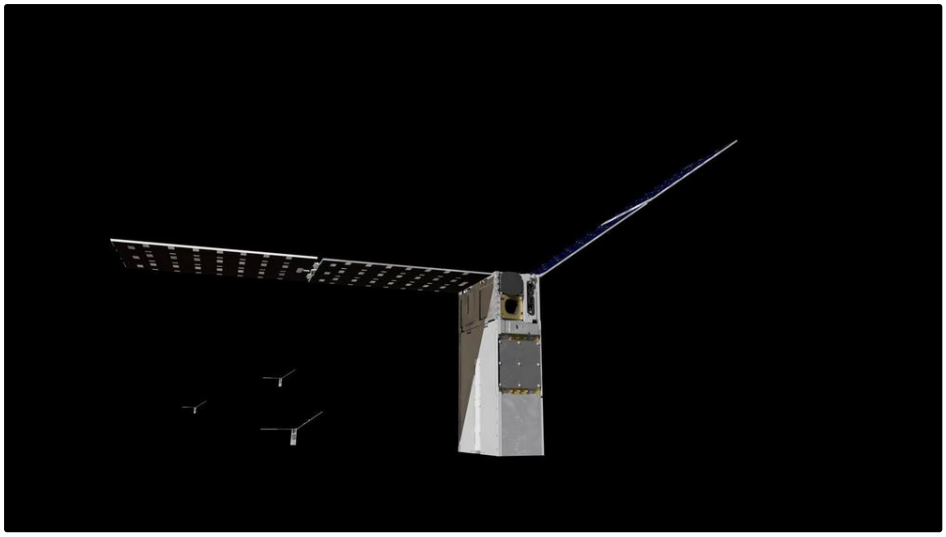


**FIGURE 11.13** Illustration of NASA's Starling mission: four CubeSats flying as an autonomous swarm in low Earth orbit (2023–2024). Image: NASA/ARC (NASA).

Formation flying requires precise relative navigation (carrier-phase differential GNSS, inter-satellite radio or optical ranging, vision sensors) and control using small thrusters or differential drag. Future concepts include distributed-aperture telescopes and interferometers, synthetic-aperture radar clusters, and gravitational-wave observatories (LISA, Chapter 39).

## 11.13 Operating Large Constellations

Operating thousands of satellites is impossible with traditional mission control, where a team watches each spacecraft. Constellation operators rely on:

- **Automation** — automated planning and scheduling of ground contacts, autonomous orbit maintenance and collision avoidance, automated anomaly detection and recovery; humans manage by exception.
- **Fleet management** — configuration management of thousands of identical vehicles, software updates over the air, health trending with machine learning (Chapter 49).
- **Ground networks** — hundreds of gateways (Starlink operates more than 150 gateway sites) and cloud-based processing.
- **Replenishment** — continuous launches to replace satellites that reach end of life (typically after five years in LEO) and to upgrade technology.
- **Space traffic coordination** — sharing ephemerides and manoeuvre plans with other operators and with space-surveillance organisations (Chapter 21).

## 11.14 Sustainability, Regulation and the Night Sky

The rapid growth of constellations raises four sustainability issues:

1. **Orbital debris and collision risk.** Large numbers of satellites increase conjunctions. Mitigation measures include low operating altitudes, reliable propulsion and prompt disposal. The US FCC requires LEO satellites ending their missions to deorbit within **5 years**, a

rule adopted in 2022 that replaced the older 25-year guideline. Other measures are high post-mission disposal reliability (> 90–99 %) and design for demise (Chapter 21).

2. **Spectrum.** The ITU requires non-geostationary systems to deploy **10 % of their filed satellites within 2 years**, **50 % within 5 years** and **100 % within 7 years** after the end of the regulatory period. These rules were adopted at WRC-19. Systems must meet them to keep their priority — driving the "race" to launch. Interference with GEO systems is limited by equivalent power-flux-density (EPFD) rules.
3. **Astronomy.** Sunlit satellites appear as moving streaks in telescope images, particularly in twilight and in wide-field surveys such as the Vera C. Rubin Observatory's Legacy Survey of Space and Time. Operators have reduced brightness with dark coatings, visors and attitude changes, and the International Astronomical Union's Centre for the Protection of the Dark and Quiet Sky coordinates mitigation. Unintended radio emissions from satellite electronics also affect radio astronomy.
4. **Atmosphere.** Re-entering satellites (several per day from large constellations) deposit aluminium oxide and other materials in the upper atmosphere, with possible effects on ozone chemistry that are under study.

### KEY TAKEAWAYS

- Orbit choice sets coverage, revisit, resolution, latency, environment and cost; coverage geometry links altitude, minimum elevation and footprint.
- Walker delta and polar "street-of-coverage" constellations are the classic designs; broadband constellations need thousands of satellites for capacity, not just coverage.
- In October 2026 about 35 000 objects were catalogued in Earth orbit (≈ 17 200 operational payloads). More than half of all objects orbit between 450 and 600 km, dominated by Starlink (> 10 600 satellites).
- Mega-constellations (Starlink, OneWeb, Amazon Leo, Qianfan, Guowang, IRIS$^2$, SDA layers) are reshaping the industry and the orbital environment.
- GEO is a limited resource managed through ITU slots and station-keeping boxes, now evolving towards software-defined and smaller satellites.
- CubeSats and small satellites, rideshare launches and orbital transfer vehicles made space accessible to universities, start-ups and new nations; sustainability rules (5-year disposal, ITU milestones, brightness mitigation) are evolving with them.

### REVIEW QUESTIONS

1. List six mission characteristics that depend on the choice of orbit.
2. A satellite at 550 km altitude serves users down to 25° elevation. Calculate the Earth central angle and the ground radius of its coverage circle.
3. Explain Walker delta notation $i$: $T/P/F$, using a GNSS constellation as an example.
4. Why do broadband constellations need thousands of satellites when a few dozen could cover the whole Earth?
5. How are satellites launched into one plane redistributed into several planes without large plane-change manoeuvres?
6. When do geostationary satellites experience eclipses, and how long can an eclipse last?
7. What features of the CubeSat standard made small satellites affordable? What are the main reliability concerns?

8. Describe three rules or practices that aim to keep low Earth orbit sustainable and limit the effect of constellations on astronomy.

CHAPTER

# 12 Satellite Buses and Payload Instruments

Image: NASA/KSC (Michael Starobin NOAA NASA)

*Every satellite is a payload carried by a bus. This chapter surveys the commercial and government satellite platforms on which payloads fly - from standard geostationary buses to mass-produced constellation platforms and CubeSat kits - and the interfaces between bus and payload. It then introduces the families of payload instruments that later chapters apply: optical and infrared imagers and spectrometers, microwave radiometers, radars, lidars, particle and field instruments and service payloads, with the physics of detectors, telescopes, scanning methods, calibration and the payload data chain.*

**IN THIS CHAPTER**


12.1 The Bus as a Product
12.2 A Taxonomy of Payload Instruments
12.3 Optical and Infrared Instruments
12.4 Microwave Instruments
12.5 Lidar Instruments
12.6 In-Situ Instruments: Fields and Particles
12.7 Service Payloads
12.8 Performance, Calibration and Data
12.9 Designing an Optical Payload: Radiometry and SNR
12.10 Instrument Case Studies
12.11 Large Communications Platforms: Power as the Measure


## 12.1 The Bus as a Product

### 12.1.1 From Custom Spacecraft to Product Lines

Early satellites were built one at a time around their payloads. Today most satellites use a **standard bus** (platform) from a manufacturer's product line, adapted to the payload — much as a truck chassis carries different bodies. Product lines lower cost and risk through heritage (flight-proven designs), reuse of qualification data, and streamlined production. Table 12.1 lists representative product lines.

**TABLE 12.1** Representative satellite bus product lines (2026).

| Class | Bus | Manufacturer | Typical mass / power | Typical missions |
|---|---|---|---|---|
| Large GEO | Eurostar Neo; OneSat (software-defined) | Airbus (Europe) | 3–6.5 t / up to 25 kW | Broadband and broadcast comsats |

| Class | Bus | Manufacturer | Typical mass / power | Typical missions |
|---|---|---|---|---|
| Large GEO | Spacebus Neo; Inspire | Thales Alenia Space (Europe) | 3–6 t / up to 20 kW | Comsats, Galileo second generation (with Airbus) |
| Large GEO | 702 family (702SP, 702X) | Boeing (USA) | 2–6 t / 3–25 kW | Comsats, WGS, O3b mPOWER (MEO) |
| Large GEO | LM 2100 / LM 400 | Lockheed Martin (USA) | 2–6 t | GOES-R, GPS III (A2100-derived), SBIRS, comsats |
| Large GEO | 1300 series | Maxar Space Systems (USA) | 3–7 t / up to 25 kW | Comsats, Psyche (derived bus) |
| Large GEO | DFH-4 / DFH-4E / DFH-5 | CAST (China) | 5–9 t / up to 28 kW | Chinese comsats, Tianlian relays |
| Large GEO | I-3K / I-4K / I-6K | ISRO (India) | 2–6 t | GSAT, INSAT |
| Small GEO | MicroGEO | Astranis (USA) | ≈ 400 kg | Regional broadband |
| LEO / MEO medium | Eurostar / AstroBus / Arrow | Airbus | 0.3–2 t | Sentinel, EO, OneWeb (Arrow) |
| LEO medium | SSTL-150 / -300 / -S1 | Surrey Satellite Technology (UK) | 100–500 kg | EO, Galileo FOC payloads, technology |
| LEO medium | LM 400 / LM 50 | Lockheed Martin | 50–2 000 kg | SDA, military |
| LEO productised | Aries / Nova / Comet | Apex (USA) | 100–500 kg | Rapid-delivery buses for many customers |
| LEO productised | X-Bus, MicroSat | York Space / Blue Canyon (RTX) | 50–500 kg | SDA Transport Layer, NASA science (TROPICS) |
| Constellation-specific | Starlink V2 Mini / V3 | SpaceX | 0.8–2 t | Vertical integration, in-house only |
| CubeSat kits | 3U–16U buses | Blue Canyon, EnduroSat, GomSpace, NanoAvionics, AAC Clyde, Open Cosmos | 4–30 kg | EO, IoT, science, education |

Three major trends are reshaping bus design:

1. **Productisation and speed.** Companies such as Apex and York aim to deliver standard buses in months rather than years, with large factories building dozens per year. Governments (e.g., the US Space Development Agency) buy satellites in “tranches” of tens to hundreds.
2. **Software-defined platforms and payloads.** Digital processors and flexible antennas let a satellite’s coverage, frequency plan and power allocation be changed in orbit — and let the bus’s own behaviour be updated like a phone’s operating system.
3. **Vertical integration.** The largest constellation operators build their own buses, payloads, ground terminals and launches, optimising the whole system rather than buying components.

### 12.1.2 What the Bus Provides

From the payload’s point of view, the bus is a service provider. It must deliver:

- **A place to live**: mechanical mounting with the required alignment and stiffness, protection during launch.
- **Power**: average and peak power at the right voltage, with protected switching.
- **Thermal environment**: heat removal for electronics, stable temperatures for optics, cooling for detectors.
- **Pointing**: accuracy, knowledge, stability (jitter), agility and keep-out zones (e.g., never point a sensitive telescope near the Sun).
- **Data handling**: command distribution, data collection, storage and downlink capacity.
- **Time and position**: precise time (often a GNSS-derived pulse per second) and orbit/attitude data for geolocation.

- **A clean environment**: low contamination, low electromagnetic interference and low microvibration.

These services are defined in a **payload–bus interface control document** (Figure 12.1). Mismatches at interfaces — a payload drawing more peak power than expected, or a bus reaction wheel exciting a resonance of the telescope — are among the commonest causes of problems in integration.

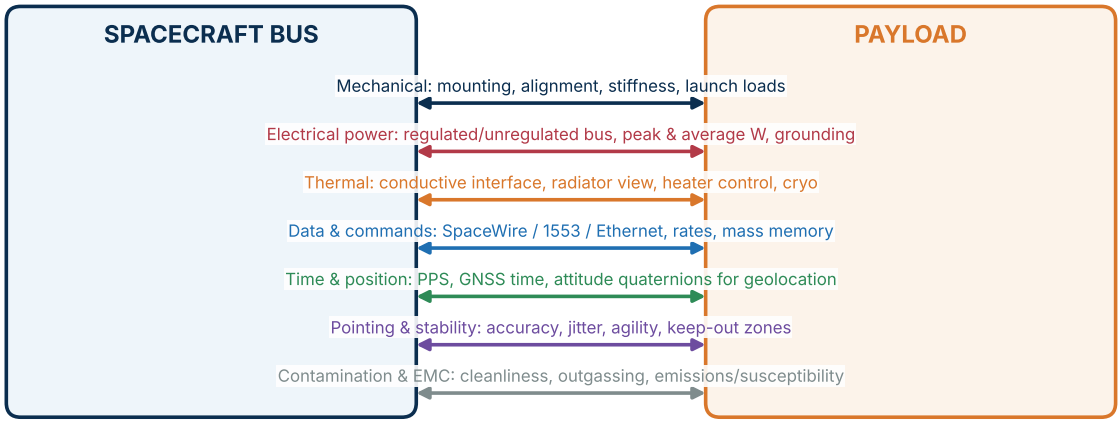



**FIGURE 12.1** The payload–bus interfaces that must be specified and verified for every mission.

### 12.1.3 Hosted Payloads

A **hosted payload** is an instrument or transponder that flies on someone else's satellite, using spare mass, power and data capacity — typically on a commercial GEO communications satellite. Hosting can cut costs by a factor of several and speed access to orbit. Examples include NASA's GOLD ultraviolet imager (on SES-14, 2018) and TEMPO air-quality spectrometer (on Intelsat 40e, 2023). Others are the Aireon ADS-B receivers on all 66 Iridium NEXT satellites (giving global air-traffic surveillance), and several US and European government payloads on commercial satellites. The challenges are aligning schedules, defining interfaces and ensuring that the host's operations do not compromise the hosted mission.

## 12.2 A Taxonomy of Payload Instruments

Payloads can be grouped in four families (Figure 12.2):

- **Passive remote sensing** — instruments that measure radiation emitted or reflected by the target: cameras, imagers, spectrometers, radiometers, polarimeters, and receivers of GNSS signals (radio occultation and reflectometry).
- **Active remote sensing** — instruments that transmit their own signal and measure its echo: radars (synthetic aperture radar, altimeters, scatterometers, precipitation and cloud radars) and lidars.
- **In-situ instruments** — instruments that measure their immediate environment: magnetometers, plasma and particle detectors, electric-field and wave instruments, dust detectors, mass spectrometers and accelerometers.
- **Service payloads** — communications transponders, navigation payloads (clocks and signal generators), data relay terminals, search-and-rescue repeaters.

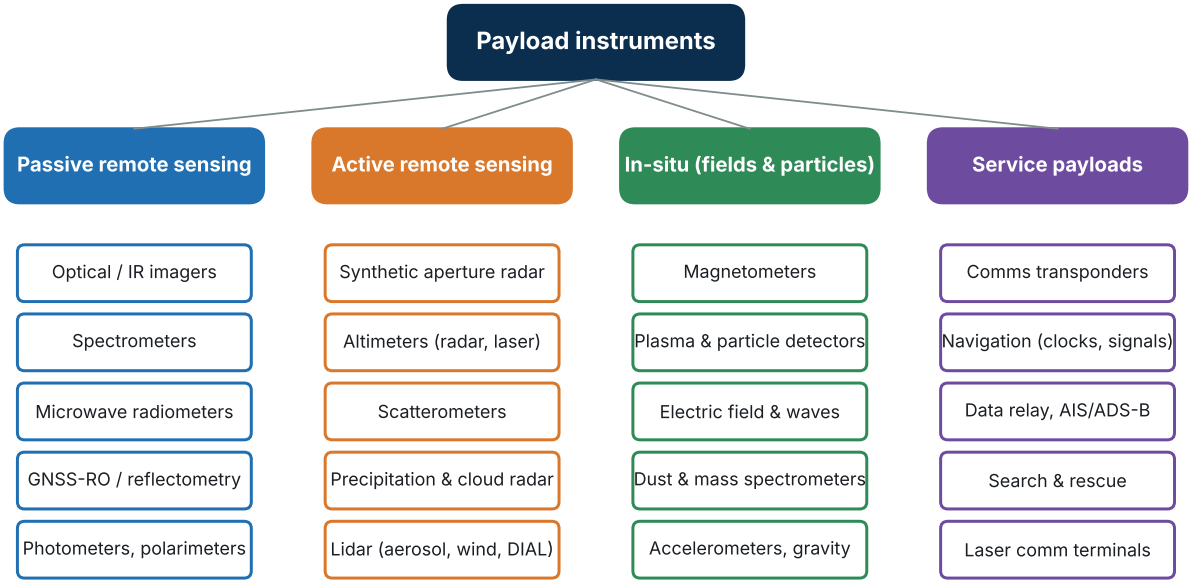



**FIGURE 12.2** A taxonomy of satellite payloads.

Each part of the electromagnetic spectrum carries different information and needs different technology (Figure 12.3). Short wavelengths (X-rays, gamma rays) reveal high-energy processes, and ultraviolet senses ozone, the upper atmosphere and hot plasmas. Visible and near-infrared light shows surface reflectance (vegetation, minerals, water), and thermal infrared shows temperature and composition. Microwaves penetrate clouds and sense moisture, temperature profiles and surface roughness; radio wavelengths are used for radar, navigation and communications.

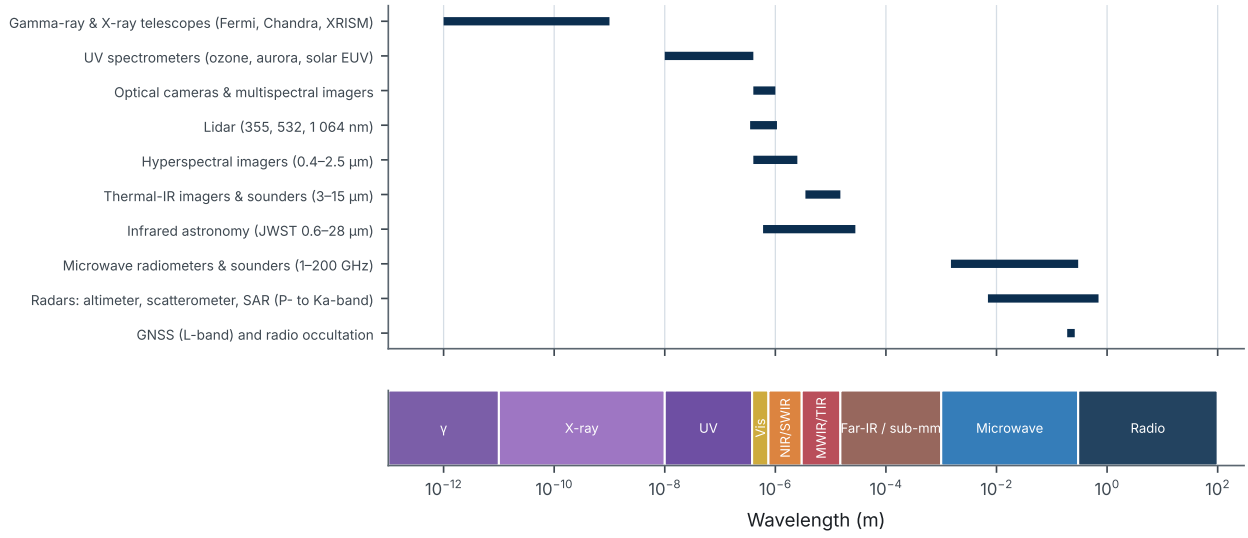
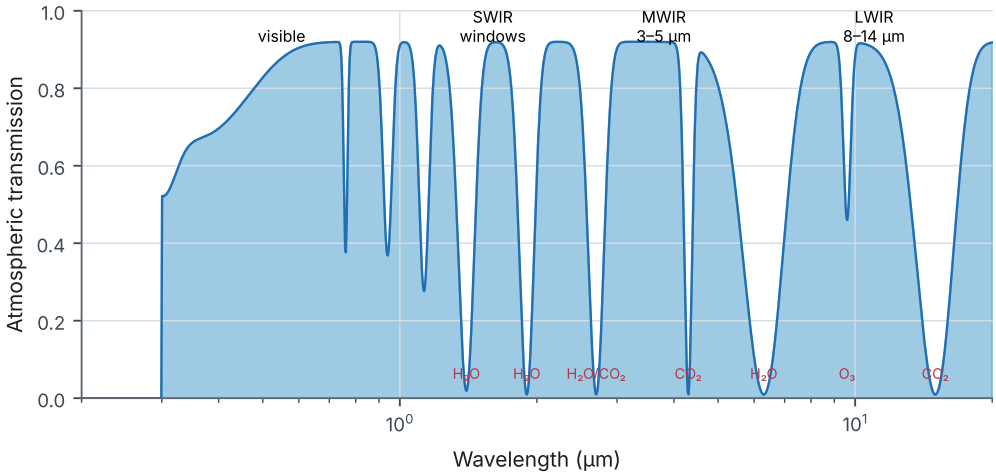



**FIGURE 12.3** The electromagnetic spectrum and the wavelength ranges used by different classes of space instruments.

Not all wavelengths reach the ground from space: the atmosphere absorbs strongly in certain bands (Figure 12.4). Imagers that want to see the surface use **atmospheric windows**, while **sounders** deliberately use absorption bands. At different distances from the centre of a $CO_2$ or $O_2$ band, the radiation comes from different heights. This lets sounders retrieve vertical profiles of temperature and humidity (Chapter 16).




Schematic vertical transmission of a mid-latitude atmosphere. Imagers use the windows; sounders deliberately observe within absorption bands to probe different heights.

**FIGURE 12.4** Schematic atmospheric transmission from 0.2 to 20 µm. Windows are used for surface imaging; absorption bands of water vapour, carbon dioxide and ozone are used for atmospheric sounding.

## 12.3 Optical and Infrared Instruments

### 12.3.1 Telescopes

An optical instrument collects light with a **telescope** and focuses it onto a detector. The main designs (Figure 12.5) are:

- **Refractors** — lenses; simple but heavy and limited in aperture; used in star trackers, small cameras and wide-field instruments.
- **Cassegrain-type reflectors** — a concave primary and convex secondary mirror; the **Ritchey–Chrétien** variant (Hubble, many Earth-imaging telescopes) corrects coma for a wider field.
- **Three-mirror anastigmats (TMA)** — three curved mirrors, often off-axis, give wide, well-corrected fields with no central obstruction; used in pushbroom imagers (Landsat OLI, Sentinel-2 MSI, WorldView) and JWST.
- **Segmented and deployable primaries** — large telescopes that fold for launch (JWST's 6.5 m, 18-segment mirror; Chapter 38).
- **Catadioptric** designs combining lenses and mirrors, compact enough for small satellites.

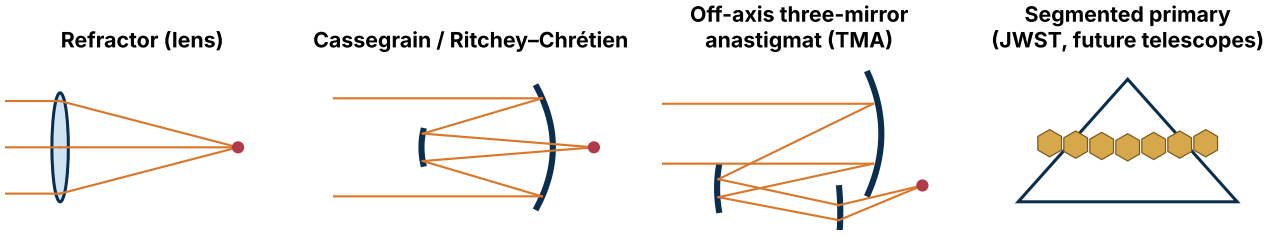



**FIGURE 12.5** Telescope configurations used in space instruments.

The angular resolution of a telescope is limited by diffraction:

$$\theta \approx 1.22 \, \frac{\lambda}{D}$$

For a 1.1 m telescope (WorldView-3) at 0.5 µm, $\theta = 0.55$ µrad, so from 617 km it can resolve about 34 cm on the ground. Real resolution also depends on the detector pixel size, optical aberrations, satellite motion and jitter, and the atmosphere — summarised in the **modulation transfer function (MTF)** (Chapter 13).

### 12.3.2 Detectors

**Detectors** convert photons into electrical signals. Visible and near-infrared imagers use **silicon** detectors — CCDs (charge-coupled devices) and, increasingly, CMOS image sensors, which integrate readout electronics on the chip, consume less power and are more tolerant of radiation. **Time-delay integration (TDI)** sensors shift charge along columns in step with the image motion, adding up the signal from tens of rows to increase sensitivity for fast-moving LEO imagers.

Infrared detectors use semiconductors with smaller band gaps: **InGaAs** (short-wave IR to 1.7 µm, usable at near room temperature) and **InSb** (to 5.5 µm, cooled to about 80 K). **HgCdTe** ("MCT") is tunable from 1 to more than 14 µm by changing its composition; it is the workhorse of space IR instruments, including JWST's NIRCam and NIRSpec. Newer types are **type-II superlattice** (T2SL) and **barrier** detectors. **Microbolometers** detect the heating of tiny suspended membranes and need no cooling, making thermal cameras possible on small satellites, although with lower sensitivity (Figure 12.6).

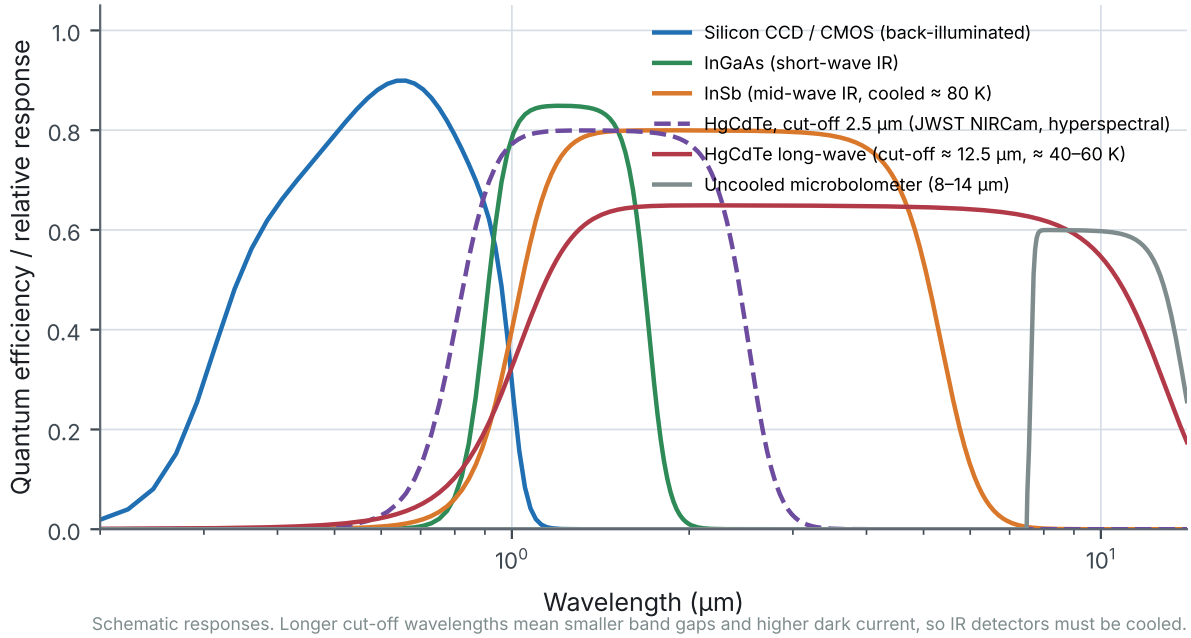



**FIGURE 12.6** Schematic quantum efficiency or relative response of common detector materials. Each material covers a spectral range set by its band gap; infrared detectors must be cooled to suppress dark current.

The dark current of a detector grows exponentially with temperature and inversely with band gap. So long-wave infrared detectors must be cooled — to 40–80 K for HgCdTe and InSb, and to below 10 K for some astronomical detectors. Cooling uses **passive radiators** (for missions with cold views, such as polar orbiters and L2 telescopes) or **cryocoolers** (Chapter 5).

### 12.3.3 Imaging Architectures

Imagers build a 2-D picture of the ground in three ways (Figure 12.7):

- **Whiskbroom** scanners use a rotating or oscillating mirror to sweep a few detectors across the swath, building the image line by line as the satellite moves (Landsat 1–7 MSS/TM/ETM+, AVHRR, MODIS). They need few detectors but have short dwell times and moving parts.
- **Pushbroom** imagers use long linear arrays (thousands to tens of thousands of pixels) perpendicular to the ground track; the satellite's motion sweeps the array along. They have long dwell time, no scanning mirror and excellent geometry — the standard for modern EO (Landsat 8/9 OLI, Sentinel-2 MSI, WorldView, Pléiades).
- **Framing** (staring) cameras use 2-D arrays to capture whole scenes at once, like a normal camera; used on SkySat, Planet Doves, many GEO imagers in step-stare mode, and astronomical telescopes. **Video** from orbit is possible with framing cameras on agile satellites.

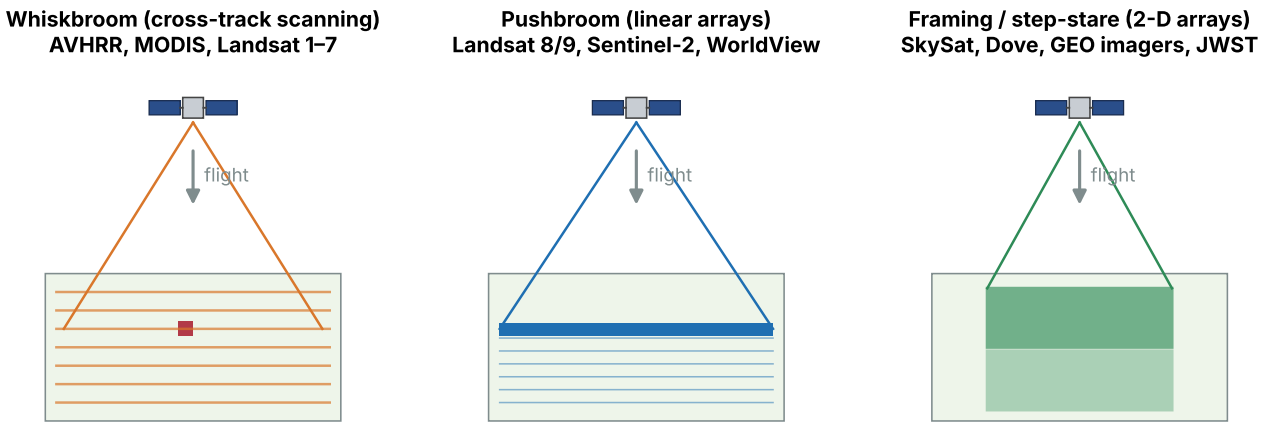



**FIGURE 12.7** The three basic imaging architectures.

### 12.3.4 Spectrometers

A **spectrometer** splits light into its component wavelengths to measure composition. Main types:

- **Dispersive** spectrometers use a **prism** or **diffraction grating**. An imaging spectrometer forms a 2-D detector image with one axis spatial and one axis spectral (hyperspectral pushbroom imagers such as EMIT, PRISMA and EnMAP; Chapter 14).
- **Fourier-transform spectrometers (FTS)** measure an interferogram with a moving mirror (Michelson interferometer) and transform it into a spectrum with high spectral resolution; used by infrared sounders (IASI, CrIS) and greenhouse-gas instruments (GOSAT).
- **Filter-based** spectral imagers use fixed or tunable filters (multispectral imagers; Fabry–Pérot and acousto-optic tunable filters).
- **Spatial heterodyne** and **grating-based high-resolution** spectrometers for specific gases (OCO-2/3 measure $CO_2$ with three grating spectrometers at 0.76, 1.61 and 2.06 μm).

## 12.4 Microwave Instruments

### 12.4.1 Passive Microwave Radiometers

Every object emits microwave radiation in proportion to its temperature and **emissivity**. In the microwave region the Rayleigh–Jeans approximation holds, so the measured radiance is expressed as a **brightness temperature** $T_B = \varepsilon T_{physical}$. Radiometers measure $T_B$ at several frequencies and polarisations to retrieve:

- **Sea-surface temperature and wind speed** (6–37 GHz; AMSR2, GMI);
- **Soil moisture and ocean salinity** at L-band (1.4 GHz; SMOS, SMAP), where emissivity depends strongly on water content;
- **Sea ice, snow and rain** (10–89 GHz);
- **Atmospheric temperature profiles** using the 50–60 GHz oxygen band and **humidity profiles** using the 183 GHz water-vapour line (ATMS, AMSU, MWS on MetOp-SG); see Chapter 16.

Because microwave wavelengths are long, antennas must be large for fine resolution. At 1.4 GHz a 6 m reflector (SMAP) gives about 40 km resolution. SMOS used a 2-D **interferometric** array of 69 small antennas to synthesise a large aperture. Radiometers are calibrated on every scan by viewing a warm blackbody target and cold space.

### 12.4.2 Radars

Active microwave instruments transmit pulses and analyse the echoes:

- **Synthetic aperture radar (SAR)** — high-resolution imaging day and night through cloud, using the satellite's motion to synthesise a long antenna (Chapter 15).
- **Radar altimeters** — measure the height of the ocean, ice or land surface to centimetre precision from the round-trip time of nadir pulses (Jason, Sentinel-3, Sentinel-6 Michael Freilich). **SWOT** (2022) uses Ka-band radar interferometry to map water heights in 2-D swaths.
- **Scatterometers** — measure backscatter from the ocean surface from several directions to infer wind speed and direction (ASCAT on MetOp, HY-2 series).
- **Precipitation radars** (Ku/Ka-band; GPM's Dual-frequency Precipitation Radar) and **cloud radars** (94 GHz; CloudSat, EarthCARE's Doppler cloud profiling radar).
- **Sounding radars** — low-frequency radars that penetrate ice and dry ground (MARSIS and SHARAD at Mars, RIME on JUICE, REASON on Europa Clipper).

### 12.4.3 GNSS Remote Sensing

Navigation signals are a free, global source of microwave illumination. In **radio occultation (RO)**, a LEO receiver tracks a GNSS signal as it sets behind the Earth's atmosphere. The bending of the signal gives profiles of refractivity, hence temperature and humidity, with high vertical resolution and no calibration drift. These profiles are among the most valuable inputs to weather forecasting (COSMIC-2, MetOp GRAS, Spire, PlanetiQ). In **GNSS reflectometry (GNSS-R)**, receivers measure signals reflected from the ocean or land to infer wind speed, soil moisture and flooding (CYGNSS, 2016).

## 12.5 Lidar Instruments

A **lidar** (light detection and ranging) fires laser pulses and measures the return. Space lidars include:

- **Altimetry lidars** — ICESat (2003–2009) and **ICESat-2** (2018–). ICESat-2's ATLAS instrument fires 10 000 pulses per second split into six beams, and counts individual returning photons to measure ice-sheet and sea-ice elevation to centimetres. **GEDI** on the ISS measures forest canopy height.
- **Backscatter lidars** — CALIPSO (2006–2023) and EarthCARE's ATLID (2024) profile aerosols and clouds, separating particle types by depolarisation.
- **Doppler wind lidar** — ESA's **Aeolus** (2018–2023) measured wind profiles globally for the first time from the Doppler shift of molecular and aerosol backscatter at 355 nm, improving weather forecasts.
- **Differential absorption lidar (DIAL)** — measures gases such as methane (France–Germany's planned MERLIN) by comparing absorption at two nearby wavelengths.
- **Planetary lidars** — the Mars Orbiter Laser Altimeter (MOLA) and Lunar Orbiter Laser Altimeter (LOLA) mapped topography; lidars guide landers (Chapter 27).

Space lidars need powerful, efficient, long-life lasers (diode-pumped Nd:YAG at 1 064 nm and its harmonics at 532 and 355 nm; fibre lasers at 1.5–2 μm). They also need large receiving telescopes and sensitive detectors (photomultipliers, avalanche photodiodes, single-photon counters). Laser-induced damage of optics in vacuum and laser lifetime were major challenges for Aeolus.

## 12.6 In-Situ Instruments: Fields and Particles

Space-physics missions (Part VII) and planetary missions carry instruments that sample their surroundings directly:

- **Magnetometers** — **fluxgate** vector magnetometers (accurate to ≈ 0.01–0.1 nT) and **scalar** magnetometers (optically pumped helium, Overhauser) that measure field magnitude absolutely. They are mounted on long booms (5–10 m) to escape the spacecraft's own magnetic field, and missions follow strict **magnetic cleanliness** programmes. The ESA **Swarm** constellation maps Earth's field; Juno's magnetometers revealed Jupiter's complex dynamo.

- **Plasma analysers** — electrostatic analysers that measure the energy, direction and (with time-of-flight sections) mass of ions and electrons of eV–tens of keV.
- **Energetic particle detectors** — solid-state silicon telescopes and scintillators for MeV particles in radiation belts and solar events.
- **Electric-field and plasma-wave instruments** — long wire booms (tens of metres) and search-coil magnetometers.
- **Langmuir probes** — measure plasma density and temperature.
- **Dust detectors and mass spectrometers** — analyse cometary dust, planetary atmospheres and exospheres (e.g., Rosetta's COSIMA and ROSINA, Europa Clipper's SUDA and MASPEX).
- **Accelerometers** — measure non-gravitational accelerations for gravity missions (GOCE, GRACE-FO) and atmospheric density.

**FIGURE 12.8** The Parker Solar Probe magnetometer boom being tested for deployment before launch (2018). Booms separate sensitive magnetometers from the spacecraft's own magnetic fields. Image: NASA/KSC (NASA/Leif Heimbold).

**FIGURE 12.9** Europa Clipper's 8.5 m magnetometer boom, a coilable lattice mast, during deployment testing at JPL (2021). Image: NASA/JPL (NASA/JPL-Caltech).

## 12.7 Service Payloads

**Communications payloads** — transponders that receive, amplify, frequency-convert and retransmit signals, and modern regenerative payloads with digital processors and beam-forming networks — are described in Chapter 17. **Navigation payloads** — atomic clocks, signal generators and high-stability transmitters — are described in Chapter 18. Other service payloads include **search-and-rescue repeaters** (Cospas-Sarsat MEOSAR on GPS, Galileo, GLONASS and BeiDou satellites). These relay 406 MHz distress beacons and, on Galileo, send a "return link" acknowledgement. Further examples are **AIS and ADS-B receivers** for ship and aircraft tracking, **data collection systems** (Argos for animal tracking and buoys), and **laser communication terminals**.

## 12.8 Performance, Calibration and Data

### 12.8.1 Key Performance Measures

Instrument performance is captured by a small set of figures of merit:

| Measure | Meaning | Typical use |
|---|---|---|
| Ground sampling distance (GSD) / spatial resolution | Size of a pixel on the ground; ability to separate nearby objects | Imaging (0.3 m–1 km) |

| Measure | Meaning | Typical use |
|---|---|---|
| Modulation transfer function (MTF) | Contrast preserved at a given spatial frequency (sharpness) | Optical imagers (MTF at Nyquist ≈ 0.1–0.3) |
| Signal-to-noise ratio (SNR) | Signal relative to noise at a reference radiance | Optical/IR imagers (100–500) |
| Noise-equivalent temperature difference (NEΔT) | Smallest detectable temperature change | Thermal IR (0.05–0.3 K) |
| Noise-equivalent differential reflectance | Smallest detectable change in reflectance | Ocean colour, hyperspectral |
| Spectral resolution / sampling | Width of spectral channels | Hyperspectral (5–10 nm), sounders (< 0.5 $cm^{-1}$) |
| Radiometric accuracy and stability | Agreement with SI-traceable truth over time | Climate records (≤ 1–3 %; ≤ 0.1 K) |
| Geolocation accuracy | Error in where a pixel is placed on Earth | EO (≤ 1 pixel; < 5 m for high-res) |
| Revisit / temporal resolution | Time between observations | EO, weather |

## 12.8.2 Calibration

A measurement is only useful if it is accurate and comparable over time and between instruments. Calibration (Figure 12.10) starts with **pre-launch laboratory calibration** against standards traceable to national metrology institutes (NIST, NPL, PTB). It continues with **onboard references** — sunlight scattered from diffuser panels, internal lamps, blackbodies at known temperatures, views of cold space. It also uses natural targets: the **Moon** (whose reflectance is extremely stable and is modelled by the ROLO and LIME models) and stable desert and ice sites (**vicarious calibration**, e.g., the RadCalNet network). Finally, it uses **inter-calibration** against reference instruments during simultaneous observations (the WMO **Global Space-based Inter-Calibration System, GSICS**). Future "climate benchmark" missions such as NASA's CLARREO Pathfinder and ESA's TRUTHS aim to fly SI-traceable references in orbit with uncertainties ten times smaller than today's instruments.

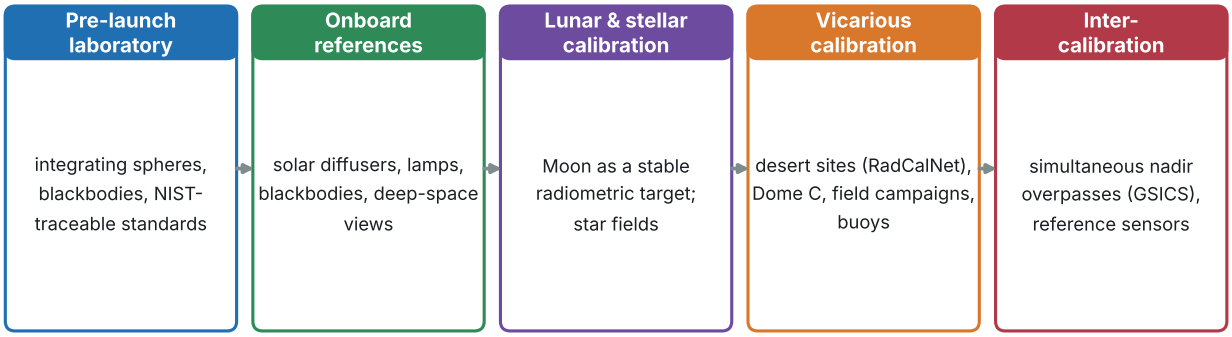



**FIGURE 12.10** The calibration chain for Earth-observation instruments, from laboratory standards to in-orbit inter-calibration.

## 12.8.3 The Payload Data Chain

Inside the instrument, photons or radio signals become information through a chain of steps. First come the detector, analogue signal conditioning and **analogue-to-digital conversion** (typically 12–16 bits for imagers). Then come **non-uniformity correction** and binning, **compression** (lossless or near-lossless, CCSDS 121/122/123), formatting into packets with time and attitude tags, **mass memory** and downlink. Increasingly, an onboard processor or AI accelerator analyses the data before downlink — detecting clouds, ships, fires or scientific events — and sends only what matters (Chapters 45–46). On the ground, processing turns raw data (Level 0) into calibrated, geolocated products (Level 1) and geophysical variables (Level 2 and above), described in Chapter 13.

## 12.9 Designing an Optical Payload: Radiometry and SNR

### 12.9.1 How Much Light Reaches a Pixel?

The signal in an Earth-imaging detector can be estimated from the scene radiance $L$ (W m$^{-2}$ sr$^{-1}$ µm$^{-1}$), the telescope aperture diameter $D$ and focal length $f$, and the optical transmission $\tau$. The other inputs are the spectral bandwidth $\Delta\lambda$, the detector pixel pitch $p$, the quantum efficiency $\eta$ and the integration time $t_{int}$. The number of photoelectrons collected is

$$N_e = \frac{\pi}{4}\,\frac{L\,\tau\,\eta\,\Delta\lambda\,p^2\,t_{int}}{F^2}\,\frac{\lambda}{hc}, \qquad F = f/D$$

where $F$ is the focal ratio (f-number). The noise has contributions from **photon shot noise** ($\sqrt{N_e}$), **dark current**, **read noise** and **quantisation**, so

$$\mathrm{SNR} = \frac{N_e}{\sqrt{N_e + N_{dark} + \sigma_{read}^2 + \sigma_{quant}^2}}$$

For a pushbroom imager, the integration time is limited by the time the ground image takes to move one pixel: $t_{int} = \mathrm{GSD}/v_{ground}$. This is about 1 ms for a 7 m GSD at a ground speed of 7 km/s, and only about 50 µs for a 30 cm GSD. This is why very-high-resolution imagers need large apertures, **time-delay integration** with dozens of stages, or satellite "slow-down" manoeuvres that rotate the spacecraft to slow the apparent ground motion.

> **WORKED EXAMPLE — SNR OF A 10 M MULTISPECTRAL BAND**
>
> Consider a red band (centre 665 nm, $\Delta\lambda$ = 30 nm) viewing vegetation of radiance $L$ = 50 W m$^{-2}$ sr$^{-1}$ µm$^{-1}$ from 786 km with a 15 cm aperture, f/6.5 telescope ($\tau$ = 0.8), 7.5 µm pixels ($\eta$ = 0.8), and $t_{int}$ = 1.5 ms (10 m pixel at 6.7 km/s ground speed).
>
> $N_e = (\pi/4) \times 50 \times 0.8 \times 0.8 \times 0.030 \times (7.5\times10^{-6})^2 \times 1.5\times10^{-3} / 6.5^2 \times (0.665\times10^{-6}/(6.63\times10^{-34}\times 3\times10^{8}))$
>
> $= 0.785 \times 50 \times 0.64 \times 0.030 \times 5.6\times10^{-11} \times 1.5\times10^{-3}/42.3 \times 3.34\times10^{18} \approx 2.6\times10^{5}$ electrons.
>
> With shot noise dominating, SNR $\approx \sqrt{2.6\times10^{5}} \approx 500$ (before other noise sources), comfortably above typical requirements of 100–200. Real designs check SNR at a dark reference radiance (e.g., over water) where the margin is much smaller — the reason ocean-colour sensors use large apertures and wider bands.

### 12.9.2 Pointing, Jitter and Microvibration

A high-resolution imager magnifies every tiny motion of the spacecraft. If the line of sight moves by more than a fraction of a pixel during integration, the image blurs. For a 30 cm GSD at 600 km, one pixel subtends 0.5 µrad (0.1 arcsec) — so **jitter** during the 50–100 µs line time must be held to a fraction of that. Sources include reaction-wheel imbalance (at the wheel spin frequency and harmonics), cryocooler compressors, solar-array drive steps and thermal "snaps". Mitigations include isolating wheels on vibration-damping mounts, balancing wheels to very tight tolerances, using control-moment gyroscopes with high rotor speeds above structural modes, quiet (counter-balanced) cryocoolers, **fine-steering mirrors** in the optical path, and line-of-sight stabilisation. JWST and Hubble use dedicated fine guidance sensors; high-resolution EO satellites use extremely stiff, thermally stable optical benches made of silicon carbide, Zerodur or carbon-fibre composites.

### 12.9.3 Field-of-View and Keep-Out Analysis

Every instrument has a field of view that must not be blocked by solar arrays, antennas or thruster plumes. It also has keep-out zones that must avoid the Sun (to prevent detector damage or stray light), the Earth limb or the Moon. Accommodation studies use 3-D models to compute fields of view, stray-light paths (with tools

such as FRED or ASAP), thruster-plume impingement, and contamination deposition. Thrusters are placed so that plumes do not reach cold optics, and sensitive instruments carry covers that open only after outgassing has subsided in orbit.

## 12.10 Instrument Case Studies

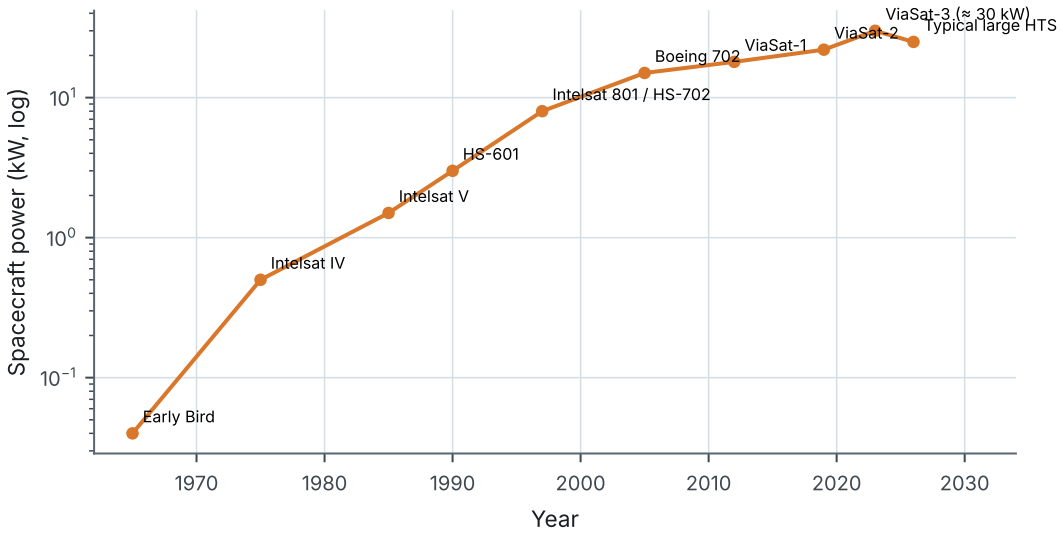
**TABLE 12.2** Representative modern instruments and their key parameters.

| Instrument (mission, launch) | Type | Key parameters | What it measures |
|---|---|---|---|
| OLI-2 (Landsat 9, 2021) | Pushbroom multispectral imager, 4-mirror anastigmat | 9 bands 0.43–2.29 µm; 15 m pan, 30 m multispectral; 185 km swath; 14-bit | Land cover, agriculture, water quality, long-term record since 1972 |
| TIRS-2 (Landsat 9) | Pushbroom thermal imager, QWIP detectors at ≈ 40 K | 2 bands 10.9 and 12.0 µm; 100 m | Land-surface temperature, evapotranspiration |
| MSI (Sentinel-2A/B/C, 2015–2024) | Pushbroom TMA imager | 13 bands 0.44–2.19 µm; 10/20/60 m; 290 km swath | Vegetation, coastal and inland waters |
| ABI (GOES-R series, 2016–2024) | Scanning GEO imager | 16 bands 0.47–13.3 µm; 0.5–2 km; full disk every 10 min, mesoscale every 30–60 s | Weather, fire, volcanic ash, cloud properties |
| VIIRS (Suomi NPP, NOAA-20/21) | Whiskbroom scanning radiometer | 22 bands 0.41–12.5 µm; 375 m / 750 m; day–night band | Weather, fires, night lights, ocean colour, aerosols |
| IASI-NG (MetOp-SG A, 2025) | Fourier-transform IR sounder | 3.62–15.5 µm, 16 921 channels, 0.25 $cm^{-1}$ | Temperature, humidity and trace-gas profiles |
| ATMS (JPSS) | Microwave sounder | 22 channels 23–183 GHz | All-weather temperature and humidity profiles |
| SMAP radiometer (2015) | L-band radiometer, 6 m rotating mesh antenna | 1.41 GHz, ≈ 40 km, 1 000 km swath | Soil moisture, freeze/thaw |
| ATLAS (ICESat-2, 2018) | Photon-counting lidar | 532 nm, 10 kHz, 6 beams, ≈ 0.7 m along-track sampling | Ice-sheet, sea-ice and vegetation height |
| C-SAR (Sentinel-1, 2014–2025) | C-band SAR, active phased array 12.3 × 0.8 m | 5.405 GHz; 5 × 20 m (IW mode), 250 km swath | Deformation, floods, ice, ships |
| TROPOMI (Sentinel-5P, 2017) | UV–SWIR imaging spectrometer | 270–2 385 nm; 5.5 × 3.5 km; 2 600 km swath | $NO_2$, $O_3$, $SO_2$, CO, $CH_4$, aerosols |
| MAG (Swarm, 2013) | Vector fluxgate + absolute scalar magnetometers on boom | 0.1 nT resolution | Geomagnetic field and its variations |

These instruments show recurring design patterns: pushbroom imagers with TMA optics for land imaging, scanning imagers in GEO for weather, and FTS and grating spectrometers for atmospheric composition. Others are large deployable antennas for long-wavelength microwave measurements, and booms for magnetic measurements.

## 12.11 Large Communications Platforms: Power as the Measure

For communications satellites, available power largely determines capacity. Over six decades GEO communications satellites grew from the 40 W *Early Bird* (1965) to platforms of 20–30 kW (Figure 12.11). This growth was enabled by multi-junction solar cells, lithium-ion batteries, deployable radiators with loop heat pipes, and electric propulsion that frees mass for payload. Very-high-throughput satellites such as **ViaSat-3** (each about 6.4 t and about 30 kW, deploying a very large reflector) push the limits of what a single platform can do. However, the deployment of ViaSat-3 F1's reflector partially failed in 2023, showing that large deployables remain a risk.




Approximate end-of-life power of representative geostationary communications satellites.

**FIGURE 12.11** Approximate power of representative geostationary communications satellites since 1965.

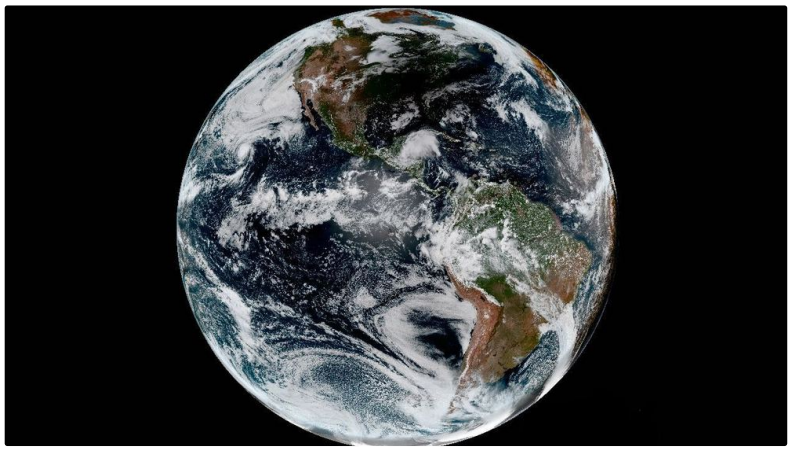


**FIGURE 12.12** A GOES-R series full-disk image of the Western Hemisphere from the Advanced Baseline Imager in geostationary orbit — the kind of product made possible by a capable payload carried on a stable bus. Image: NASA/GSFC.

### KEY TAKEAWAYS

- Most satellites use standard buses from product lines; trends are productisation, software-defined platforms and vertical integration; hosted payloads share buses to reduce cost.
- Payload–bus interfaces (mechanical, power, thermal, data, time, pointing, contamination/EMC) must be specified and verified carefully.
- Payloads are passive or active remote-sensing instruments, in-situ field and particle instruments, or service payloads; each spectral region reveals different physics, and atmospheric windows and absorption bands are both useful.
- Optical systems combine telescopes (refractors, Ritchey–Chrétien, TMA, segmented mirrors) with detectors chosen for wavelength (Si, InGaAs, InSb, HgCdTe, microbolometers) and cooled as needed; pushbroom imaging dominates modern EO.
- Microwave radiometers, radars, GNSS radio occultation and lidars measure the atmosphere and surface in all weather; magnetometers and particle instruments sample space plasmas.
- Performance is described by GSD, MTF, SNR, NE$\Delta$T, spectral resolution, radiometric accuracy and geolocation; calibration from the laboratory to the Moon and vicarious sites makes measurements trustworthy.

### REVIEW QUESTIONS

1. Why have most satellite manufacturers moved to standard bus product lines? What are hosted payloads?
2. List the interfaces between a payload and its bus that must be specified and verified.

3. Which detector materials are used for visible, short-wave infrared and thermal-infrared imaging, and which require cooling?
4. Compare framing, whisk-broom and push-broom imaging architectures.
5. A push-broom imager at 500 km altitude has 8 μm pixels and a 4 m focal length. What is its ground sample distance at nadir? What aperture is needed for its diffraction-limited resolution at 0.6 μm to match that GSD?
6. How does GNSS radio occultation measure atmospheric temperature and humidity profiles?
7. Explain why calibration is needed before launch, in orbit and with vicarious sites, and give an example of each.
8. Why is a communication satellite's payload capability usually described by its power?

PART IV

# Earth Observation and Remote Sensing

Satellites that look down at our planet measure crops, cities, ice, oceans, storms and greenhouse gases. Part IV covers the physics of remote sensing, high-resolution optical and thermal imagers, synthetic aperture radar, and the weather and environmental satellites behind every forecast.


13 Remote Sensing Fundamentals
14 Electro-Optical and Infrared Imaging Satellites
15 Synthetic Aperture Radar
16 Weather and Environmental Satellites


Image: NASA/GSFC

# 13 CHAPTER Remote Sensing Fundamentals

Image: NASA/GSFC

*Remote sensing is the science of measuring things without touching them. From orbit, satellites measure the light, heat and radio waves leaving the Earth and turn them into maps of crops, forests, cities, oceans, ice and air. This chapter explains the physics that makes this possible - electromagnetic radiation, its interaction with the atmosphere and the surface, and the spectral signatures that identify materials - and the engineering concepts used to describe sensors: spatial, spectral, radiometric and temporal resolution, image quality, geometric correction, processing levels and analysis-ready data. It ends with the methods used to turn images into information and the open-data ecosystem that has made satellite data a global public good.*

IN THIS CHAPTER

13.1 What Remote Sensing Is
13.2 Electromagnetic Radiation
13.3 Interaction with the Atmosphere
13.4 Interaction with the Surface: Spectral Signatures
13.5 The Four Resolutions
13.6 Image Quality
13.7 Geometry: Putting Pixels on the Map
13.8 Processing Levels and Analysis-Ready Data
13.9 Turning Images into Information
13.10 The Open-Data Revolution and the Users

## 13.1 What Remote Sensing Is

**Remote sensing** means acquiring information about an object from a distance, usually by measuring electromagnetic radiation that the object reflects or emits. Satellites have made it possible to observe the whole planet repeatedly, consistently and objectively. The first weather satellite, TIROS-1, returned its first television images of clouds in April 1960. The US CORONA reconnaissance programme returned film capsules from 1960. **Landsat 1**, launched in July 1972,

began the longest continuous record of Earth's land surface — a record that continues today with Landsat 8 and 9. Since 2008, all Landsat data have been free. Since 2014 the European Union's **Copernicus** programme has provided free and open data from its Sentinel satellites. Together they have transformed Earth observation (EO) into a global public good used by millions.

Remote sensing systems are classified by the source of the energy they measure:

- **Passive optical** systems measure reflected sunlight (visible, near-infrared and short-wave infrared, about 0.4–2.5 µm). They only work in daylight and are blocked by clouds.
- **Passive thermal** systems measure radiation emitted by the surface and atmosphere (mid-wave and long-wave infrared, about 3–14 µm). They work day and night but are blocked by clouds.
- **Passive microwave** systems measure natural microwave emission. They see through most clouds but have coarse resolution.
- **Active** systems — radar and lidar — provide their own illumination. Radar works day and night through clouds (Chapter 15); lidar measures heights and atmospheric profiles precisely.

The complete remote-sensing chain (Figure 13.1) links a source of energy, its passage through the atmosphere and its interaction with the surface. It continues with the return trip through the atmosphere, the sensor, and the processing and analysis that turn measurements into decisions.

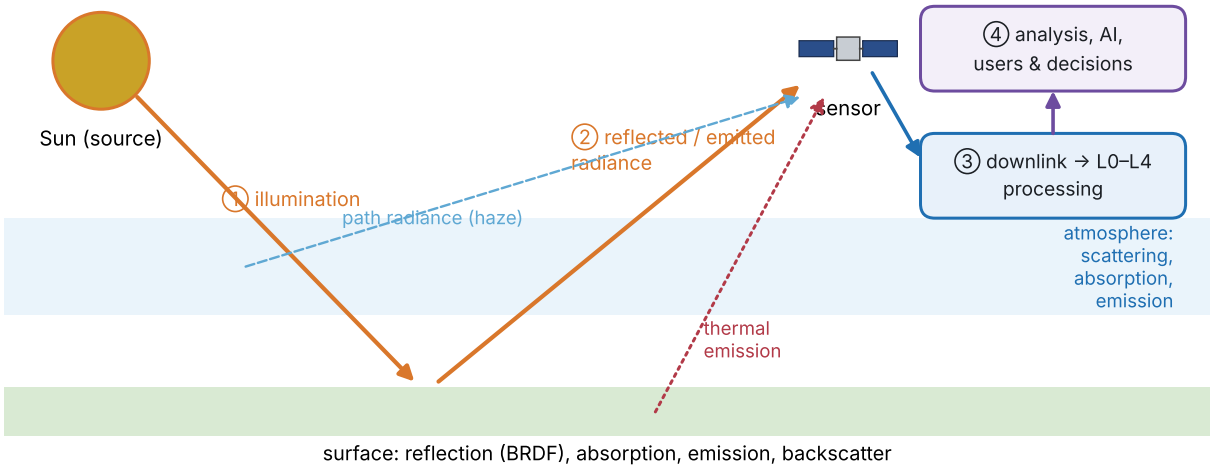



**FIGURE 13.1** The remote-sensing chain: illumination, interaction with the atmosphere and surface, measurement by the sensor, and processing into information for users.

## 13.2 Electromagnetic Radiation

### 13.2.1 Waves, Photons and the Spectrum

Electromagnetic radiation travels at the speed of light $c = 299\,792\,458$ m/s. Its wavelength $\lambda$ and frequency $\nu$ are related by $c = \lambda\nu$, and each photon carries an energy $E = h\nu = hc/\lambda$, where $h = 6.626 \times 10^{-34}$ J s is Planck's constant. Remote-sensing wavelengths span from about 0.3 µm (ultraviolet) to tens of centimetres (microwave radar) — a range of a million to one (Chapter 12, Figure 12.3).

### 13.2.2 Thermal Emission: Planck's Law

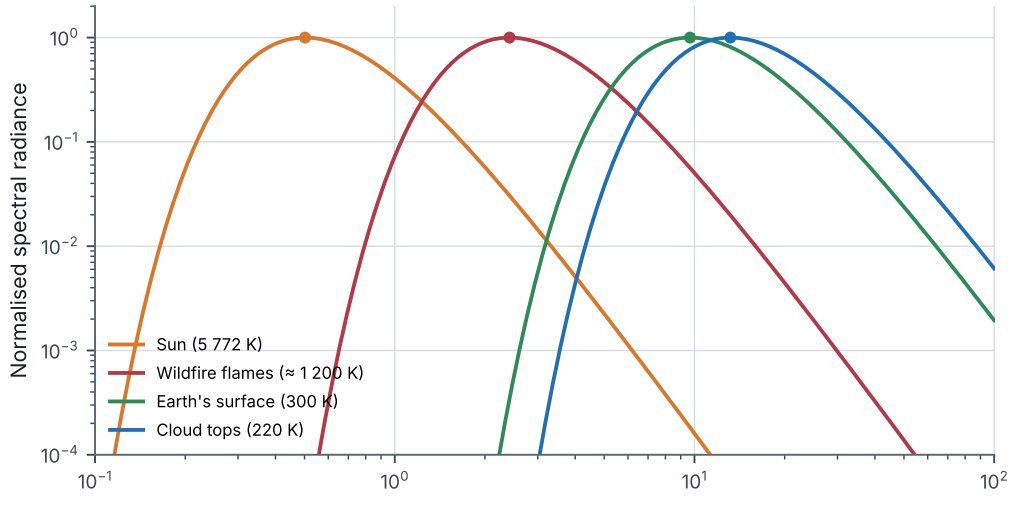
Every object above absolute zero emits radiation. An ideal emitter — a **blackbody** — emits a spectral radiance given by **Planck's law**:

$$B_\lambda(T) = \frac{2hc^2}{\lambda^5}\,\frac{1}{e^{hc/\lambda k_B T} - 1}$$

Two consequences are central to remote sensing (Figure 13.2):

- **Wien's displacement law**: the peak wavelength is $\lambda_{max} = 2\,898\ \mu\text{m·K}/T$. The Sun (≈ 5 800 K) peaks near 0.5 µm — which is why our eyes evolved to see there. The Earth's surface (≈ 300 K) peaks near 10 µm. A forest fire (≈ 1 000–1 200 K) peaks at 2.4–2.9 µm, which is why fires are detected in the mid-wave infrared around 4 µm.
- **Stefan–Boltzmann law**: the total emitted power per unit area is $M = \varepsilon\sigma T^4$; hot objects radiate enormously more energy than cool ones.

Real surfaces emit less than a blackbody by a factor called the **emissivity** $\varepsilon(\lambda)$ (0–1). Water and vegetation have thermal-infrared emissivities of 0.97–0.99; bare rocks and soils vary from 0.85 to 0.98, and their emissivity spectra identify minerals.




Planck curves, each normalised to its peak. Wien's law: λ_max = 2 898 µm·K / T — sunlight peaks at 0.5 µm, Earth's emission at ≈ 10 µm, fires at ≈ 2.4 µm.

**FIGURE 13.2** Planck curves for sunlight, a wildfire, the Earth's surface and cold cloud tops, each normalised to its peak.

### 13.2.3 Radiometric Quantities

Engineers use precise quantities to describe radiation:

**TABLE 13.1** Radiometric quantities used in remote sensing.

| Quantity | Symbol | Unit | Meaning |
|---|---|---|---|
| Radiant flux | $\Phi$ | W | Power |
| Irradiance | $E$ | W m$^{-2}$ | Power arriving per unit area |
| Radiant exitance | $M$ | W m$^{-2}$ | Power leaving per unit area |
| Radiance | $L$ | W m$^{-2}$ sr$^{-1}$ | Power per unit area per unit solid angle — what a sensor pixel measures |

| Quantity | Symbol | Unit | Meaning |
|---|---|---|---|
| Spectral radiance | $L_\lambda$ | W m$^{-2}$ sr$^{-1}$ µm$^{-1}$ | Radiance per unit wavelength |
| Reflectance | $\rho$ | dimensionless | Fraction of incident radiation reflected |
| Brightness temperature | $T_B$ | K | Temperature of a blackbody producing the measured radiance |

A sensor measures **radiance** at the top of the atmosphere. Converting it to **top-of-atmosphere (TOA) reflectance** removes the effect of solar illumination:

$$\rho_{TOA} = \frac{\pi \, L_\lambda \, d^2}{E_{sun,\lambda} \cos\theta_s}$$

where $d$ is the Earth–Sun distance in AU, $E_{sun,\lambda}$ the exo-atmospheric solar irradiance and $\theta_s$ the solar zenith angle. Removing the effects of the atmosphere then gives **surface reflectance** — the property of the ground itself.

## 13.3 Interaction with the Atmosphere

Between the Sun, the ground and the sensor, the atmosphere **scatters** and **absorbs** radiation:

- **Rayleigh scattering** by gas molecules (much smaller than the wavelength) is proportional to $\lambda^{-4}$. Blue light (450 nm) scatters roughly four to five times more than red light (650 nm). This is why the sky is blue and why images in the blue band look hazy (Figure 13.3).
- **Mie scattering** by aerosols (dust, smoke, sea salt, pollution particles of similar size to the wavelength) depends more weakly on wavelength and produces the whitish haze over cities and deserts. The **aerosol optical depth (AOD)** at 550 nm ranges from about 0.05 in clean air to more than 2 in heavy smoke or dust.
- **Absorption** by water vapour, carbon dioxide, ozone and oxygen removes radiation in specific bands (Chapter 12, Figure 12.4). Sensors observe the surface through the **atmospheric windows**.
- **Clouds** block optical and infrared radiation completely; on average about 67 % of the Earth is cloud-covered at any time — a fundamental limitation that makes all-weather radar so valuable.

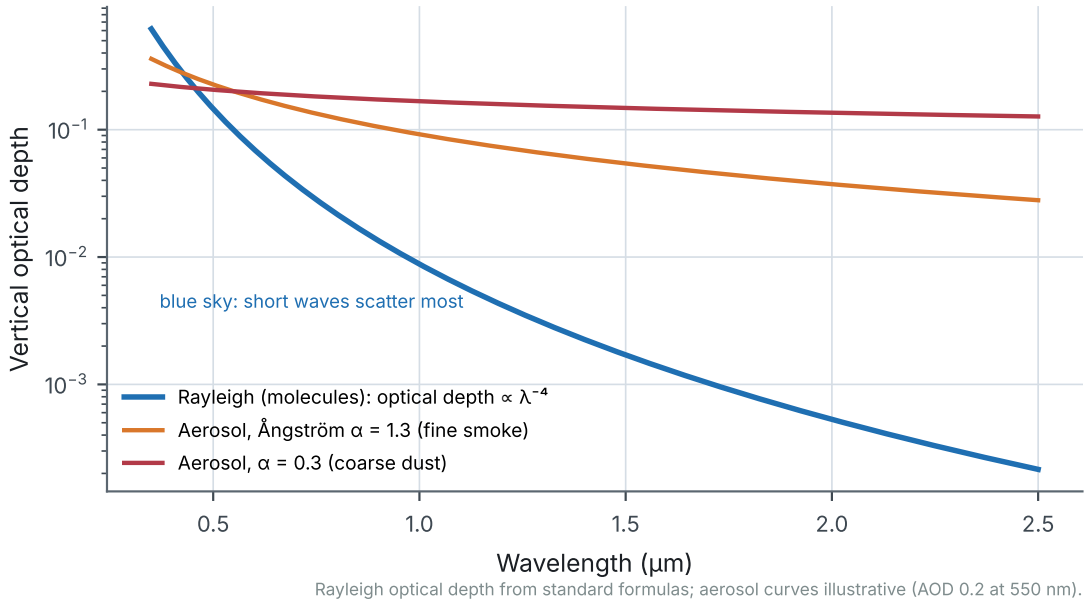



**FIGURE 13.3** Vertical optical depth for Rayleigh scattering by air molecules and for two aerosol types as a function of wavelength. Short wavelengths are attenuated far more.

The radiance reaching a sensor over a dark surface may be more than half **path radiance** — light scattered into the sensor by the atmosphere without ever touching the ground. **Atmospheric correction** removes these effects using radiative-transfer models (e.g., 6S, MODTRAN, libRadtran) and information on aerosols, water vapour and ozone, often retrieved from the image itself (e.g., using dark-target or "dense dark vegetation" methods). Operational processors such as ESA's Sen2Cor and NASA's LaSRC/LEDAPS produce surface-reflectance products for Sentinel-2 and Landsat.

## 13.4 Interaction with the Surface: Spectral Signatures

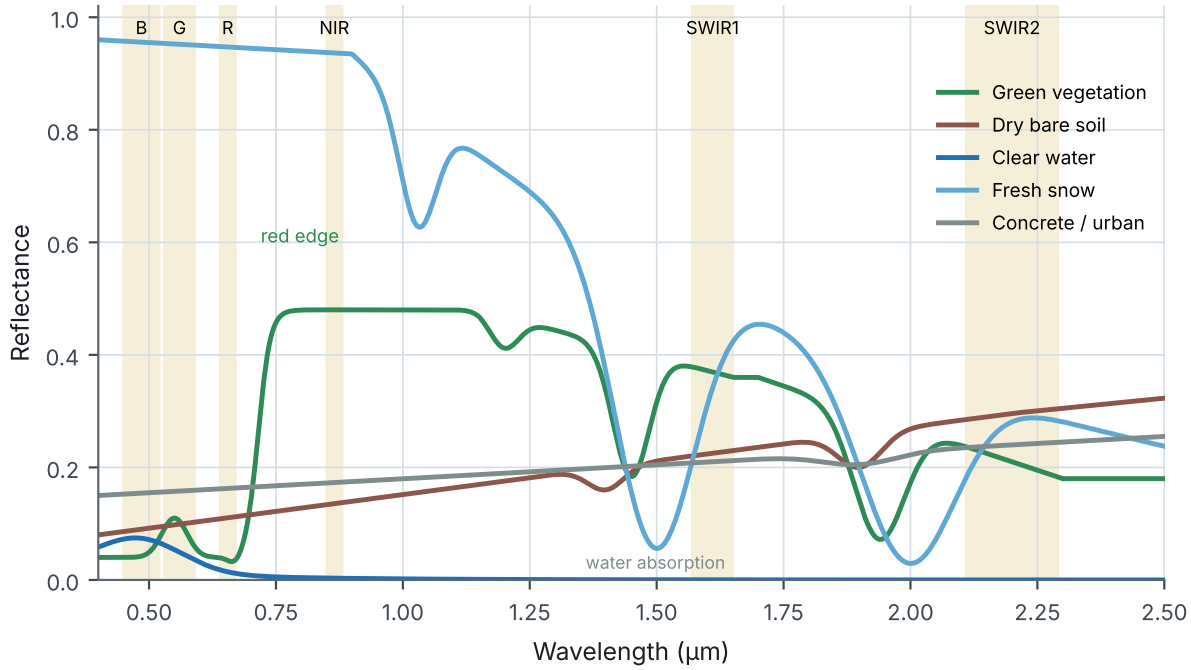
### 13.4.1 Why Things Look Different

When radiation strikes a surface it may be reflected, absorbed or transmitted. The fraction reflected depends on wavelength in ways that are characteristic of each material — its **spectral signature** (Figure 13.4):

- **Healthy vegetation** absorbs strongly in the blue and red (by chlorophyll, for photosynthesis) and reflects a little more in the green (which is why leaves look green). It reflects strongly in the near-infrared (40–50 %) because of scattering in the internal cell structure of leaves. The sharp rise between red and near-infrared at about 0.70–0.75 µm is the **red edge**; its position shifts with chlorophyll content and plant stress. In the short-wave infrared, reflectance falls in the water absorption bands at 1.45 and 1.94 µm; drier leaves reflect more.
- **Soils and rocks** generally show gently increasing reflectance with wavelength, modified by moisture (wetter soils are darker) and organic matter. Iron oxides (absorption near 0.9 µm) and clay and carbonate minerals (absorption features near 2.2 and 2.3 µm) also modify it — the basis of geological mapping.
- **Water** reflects little except in the blue and green, and absorbs almost everything beyond about 0.75 µm — so water appears black in near-infrared images, making shorelines and floods easy to map. Suspended sediment and phytoplankton change the visible reflectance of

water, the basis of **ocean-colour** remote sensing.

- **Snow** is highly reflective in the visible but much darker in the short-wave infrared, whereas clouds stay bright — a property used to distinguish snow from cloud.
- **Built surfaces** (concrete, asphalt, roofs) have relatively flat, moderate spectra.




Schematic spectra after USGS/ECOSTRESS spectral libraries; shaded: Landsat 8/9 OLI bands.

**FIGURE 13.4** Schematic spectral reflectance signatures of common surface materials, with Landsat 8/9 OLI bands shaded. Differences in these curves allow materials to be distinguished from orbit.

### 13.4.2 Bidirectional Reflectance

Real surfaces are not perfect diffuse (Lambertian) reflectors: their brightness depends on the directions of illumination and viewing, described by the **bidirectional reflectance distribution function (BRDF)**. Forests appear brightest when viewed with the Sun behind the observer (the "hot spot", where shadows are hidden); water shows sharp **sun glint** in the specular direction. Wide-swath sensors and agile satellites that view at different angles must correct for BRDF effects to compare observations.

### 13.4.3 Spectral Indices

Simple combinations of bands — **spectral indices** — capture key surface properties and are widely used because they are easy to compute and interpret:

**TABLE 13.2** Widely used spectral indices. NIR = near-infrared; SWIR1 ≈ 1.6 µm; SWIR2 ≈ 2.2 µm.

| Index | Formula | Use |
|---|---|---|
| Normalised Difference Vegetation Index (NDVI) | $(\rho_{NIR} - \rho_{red})/(\rho_{NIR} + \rho_{red})$ | Vegetation greenness and density |
| Enhanced Vegetation Index (EVI) | $2.5(\rho_{NIR} - \rho_{red})/(\rho_{NIR} + 6\rho_{red} - 7.5\rho_{blue} + 1)$ | Vegetation in dense canopies, less atmospheric effect |
| Normalised Difference Water Index (NDWI, McFeeters) | $(\rho_{green} - \rho_{NIR})/(\rho_{green} + \rho_{NIR})$ | Open water |
| Normalised Difference Moisture Index (NDMI) | $(\rho_{NIR} - \rho_{SWIR1})/(\rho_{NIR} + \rho_{SWIR1})$ | Vegetation water content |

| Index | Formula | Use |
|---|---|---|
| Normalised Burn Ratio (NBR) | $(\rho_{NIR} - \rho_{SWIR2})/(\rho_{NIR} + \rho_{SWIR2})$ | Burn severity (difference before/after fire, dNBR) |
| Normalised Difference Snow Index (NDSI) | $(\rho_{green} - \rho_{SWIR1})/(\rho_{green} + \rho_{SWIR1})$ | Snow cover |
| Normalised Difference Built-up Index (NDBI) | $(\rho_{SWIR1} - \rho_{NIR})/(\rho_{SWIR1} + \rho_{NIR})$ | Urban areas |

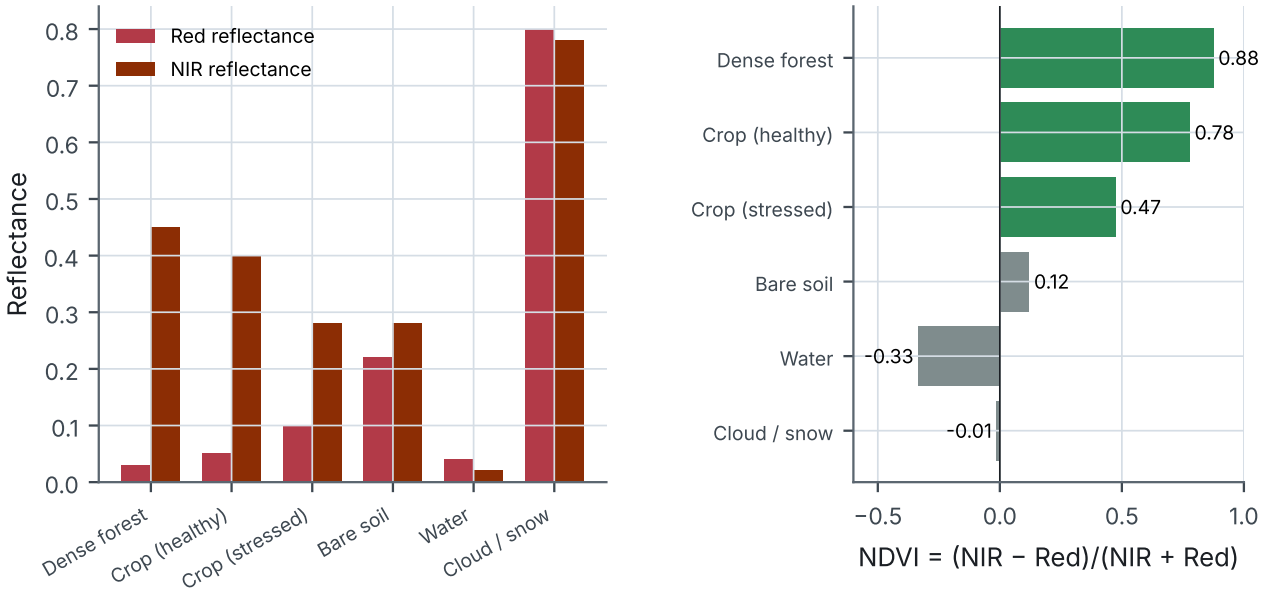



**FIGURE 13.5** Red and near-infrared reflectance of six surface types (left) and the resulting NDVI (right). Healthy vegetation gives NDVI values of 0.6–0.9; bare soil 0.1–0.2; water and clouds near zero or negative.

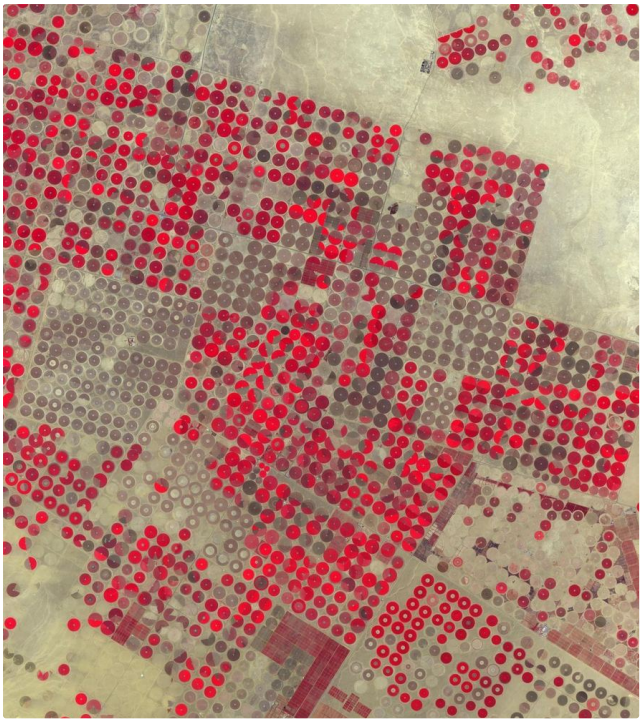


**FIGURE 13.6** Centre-pivot irrigated fields in the desert of Al Jowf, Saudi Arabia, imaged by NASA's ASTER instrument in false colour, where vigorous vegetation appears red because it reflects strongly in the near-infrared. Image: NASA/JPL.

**FIGURE 13.7** The San Francisco Bay Area in an ASTER false-colour composite. Vegetation is red; urban areas appear blue-grey; water is dark. Image: NASA/JPL.

## 13.5 The Four Resolutions

A remote-sensing system is characterised by four kinds of resolution (Figure 13.8):

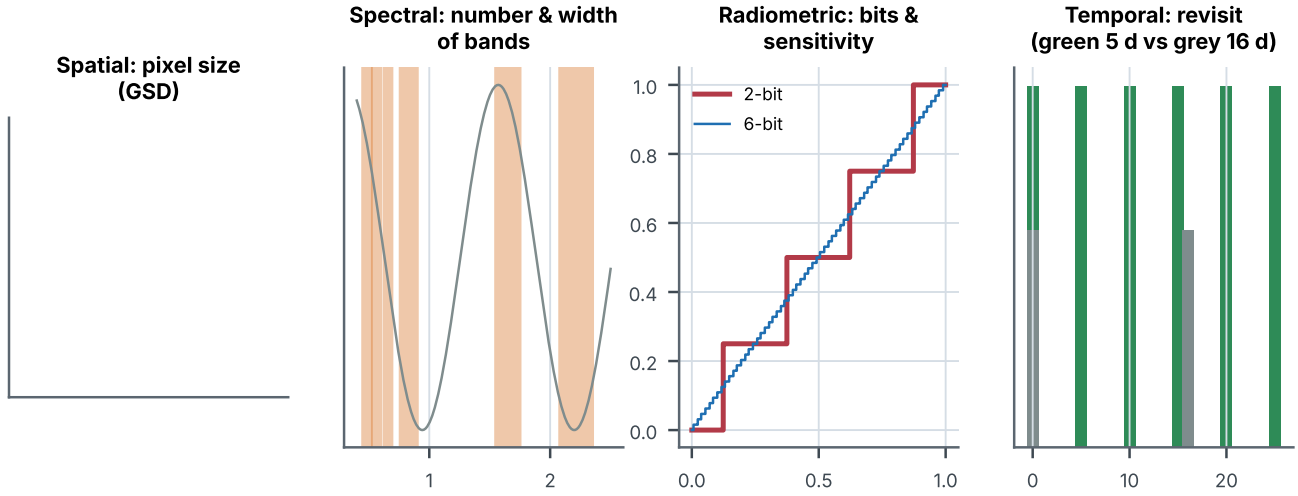



**FIGURE 13.8** The four resolutions of a remote-sensing system: spatial, spectral, radiometric and temporal.

**Spatial resolution** is the size of the smallest feature that can be distinguished. It is commonly expressed by the **ground sampling distance (GSD)** — the distance between pixel centres on the ground — although true resolving power also depends on optical quality and motion (see MTF

below). EO systems range from about 30 cm (very-high-resolution commercial imagers) to 1–2 km (global weather imagers).

**Spectral resolution** is the number, position and width of the spectral bands. **Panchromatic** sensors have one broad band; **multispectral** sensors have 4–20 bands; **hyperspectral** (imaging spectroscopy) sensors have 100–400 contiguous narrow bands (5–10 nm wide) that record a near-continuous spectrum for every pixel (Chapter 14).

**Radiometric resolution** is the sensor's ability to distinguish small differences in brightness. It is set by the number of bits used to record each pixel (8-bit = 256 levels; Landsat 9 uses 14 bits = 16 384 levels) and, more fundamentally, by its signal-to-noise ratio.

**Temporal resolution** is how often the same place is observed (revisit time), from 30 seconds (geostationary "rapid scan" over storms) to 16 days (a single Landsat) or longer.

These resolutions trade against one another and against swath width, data volume and cost. A satellite cannot image a wide area at very fine resolution in hundreds of bands every day, all at the same time. Figure 13.9 shows how today's systems span the trade space.

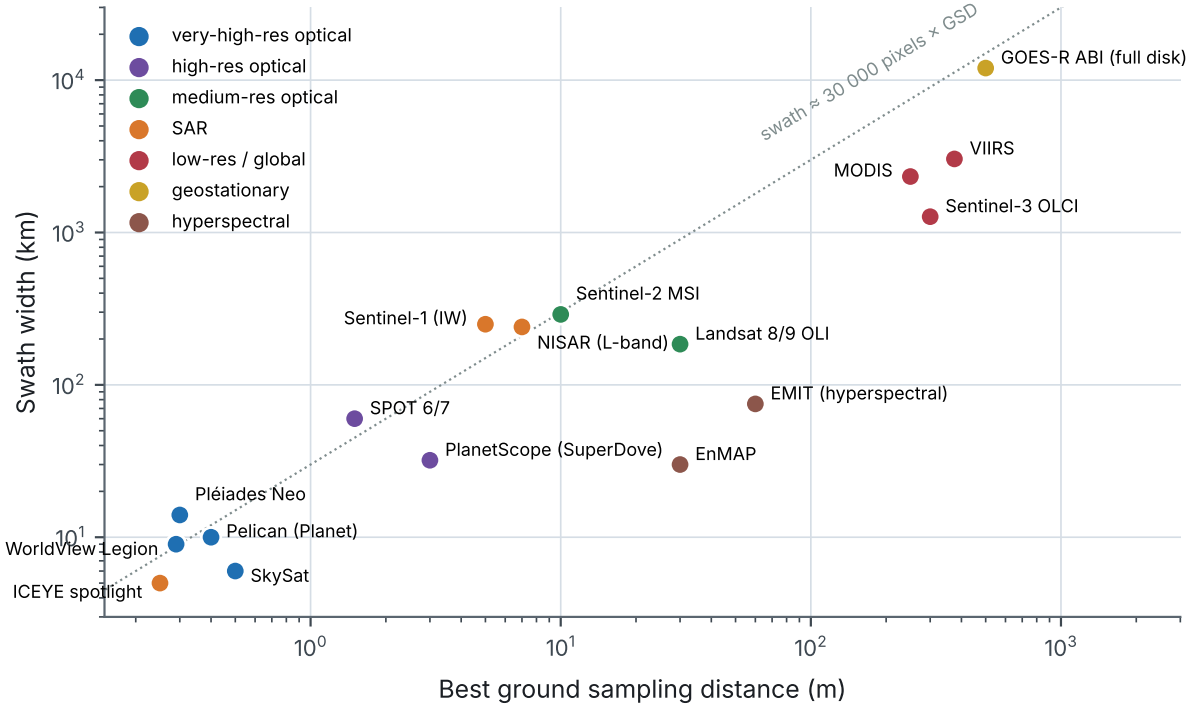



**FIGURE 13.9** Ground sampling distance versus swath width for representative EO systems. Most sensors lie near a line corresponding to a few tens of thousands of pixels across the swath, the practical limit of detector arrays and data rates.

## 13.6 Image Quality

### 13.6.1 Modulation Transfer Function

GSD alone does not tell how sharp an image is. A sensor blurs the scene through optical diffraction and aberrations, the finite size of each detector pixel, motion of the satellite and pointing jitter during the exposure, and the atmosphere. The **modulation transfer function (MTF)** describes how much contrast the system preserves as a function of spatial frequency; the

system MTF is the product of the MTFs of each contributor (Figure 13.10). A common specification is the MTF at the **Nyquist frequency** (one cycle per two pixels): values of 0.1–0.3 are typical. Too low and images look soft; too high and fine patterns produce **aliasing** artefacts.

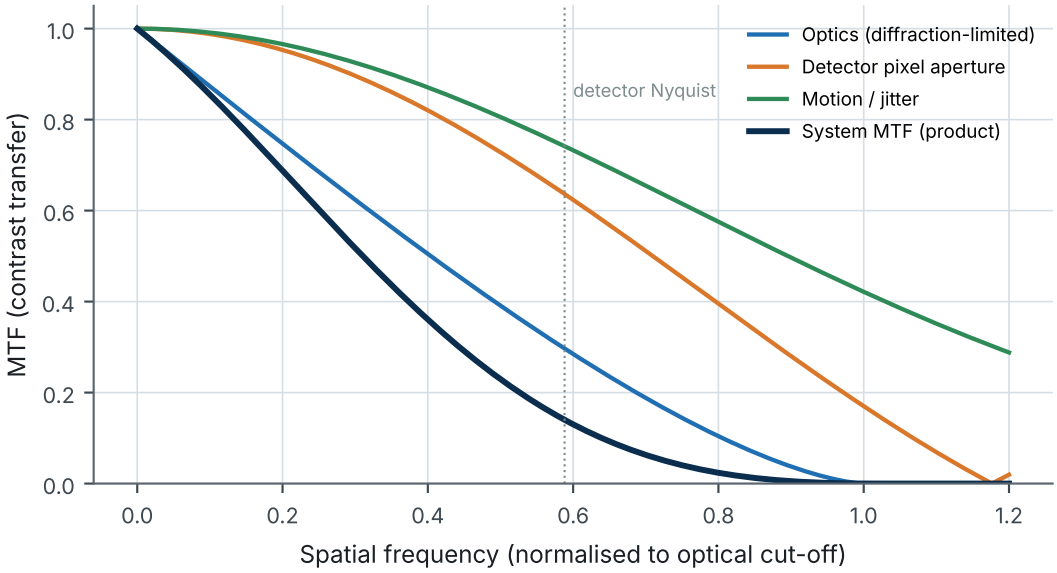



**FIGURE 13.10** Contributions to the modulation transfer function of a typical EO imager. The system MTF is the product of optics, detector and motion terms; its value at the detector Nyquist frequency is a key image-quality specification.

### 13.6.2 Signal-to-Noise Ratio

The **signal-to-noise ratio (SNR)** expresses how well the signal stands above random noise from photon statistics (shot noise, $\sqrt{N}$ for $N$ photo-electrons), detector dark current, read-out electronics and quantisation. A related figure of merit is the **noise-equivalent change in reflectance (NEΔρ)** or, for thermal sensors, the **noise-equivalent temperature difference (NEΔT)** — typically 0.05–0.1 K for modern thermal imagers. This figure states the smallest change that can be detected. SNR improves with larger apertures, wider bands, longer integration time (using time-delay integration, TDI) and cooler detectors (Chapter 12).

### 13.6.3 Interpretability: NIIRS and GIQE

For high-resolution imagery, users care about what tasks an image supports. The US **National Imagery Interpretability Rating Scale (NIIRS)** rates images from 0 to 9 by the objects that can be identified. NIIRS 5 allows individual rail cars to be identified by type, and NIIRS 6 distinguishes between car types (sedans and station wagons). NIIRS 7 allows individual rail ties to be identified. The **General Image Quality Equation (GIQE)** estimates NIIRS from GSD, the relative edge response (a measure of sharpness), image processing gain and SNR. Halving the GSD raises NIIRS by about one level.

## 13.7 Geometry: Putting Pixels on the Map

### 13.7.1 Sources of Geometric Distortion

A raw image is not a map. Its geometry is distorted by:

- the **viewing geometry** — off-nadir viewing stretches pixels and makes them oblique;

- **Earth's curvature and rotation** during a push-broom scan (the image is skewed westward as the Earth turns beneath it);
- **terrain relief** — elevated points appear displaced away from the nadir point (Figure 13.11);
- **platform position and attitude errors** from GNSS and star trackers; and
- **sensor geometry** — lens distortion and focal-plane alignment.

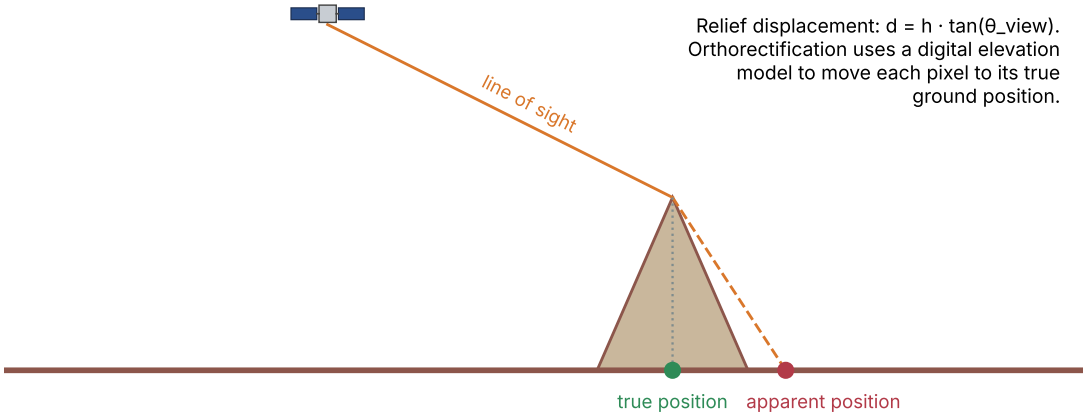



**FIGURE 13.11** Relief displacement: the top of a tall object (or mountain) appears shifted away from nadir by d = h·tan θ. Without correction, a 1 000 m mountain imaged 20° off nadir would be displaced by 364 m.

### 13.7.2 Geolocation and Orthorectification

**Geolocation** computes the ground position of each pixel by projecting its line of sight from the satellite's position and attitude onto the Earth's surface model. Before ground control, modern satellites achieve geolocation accuracy of better than 5 m CE90 for very-high-resolution systems and about 10–15 m for Landsat-class sensors. With **ground control points (GCPs)** and precision processing, the Landsat and Sentinel-2 archives are co-registered to better than one pixel.

**Orthorectification** removes the effects of terrain and viewing geometry using a **digital elevation model (DEM)** — a gridded map of surface heights. The result is an **orthoimage** in which every pixel is at its true map position, like a map. Global DEMs widely used for this purpose include the SRTM (Shuttle Radar Topography Mission, 2000, 30 m) and the Copernicus DEM (derived from TanDEM-X, 30 m and 90 m global). Others are ALOS World 3D (30 m) and NASADEM.

For commercial satellites, the camera model is often distributed as **rational polynomial coefficients (RPCs)**. These are ratios of cubic polynomials that map ground coordinates (latitude, longitude, height) to image coordinates. They let users orthorectify images without knowing the proprietary sensor model.

### 13.7.3 Map Projections and Coordinate Systems

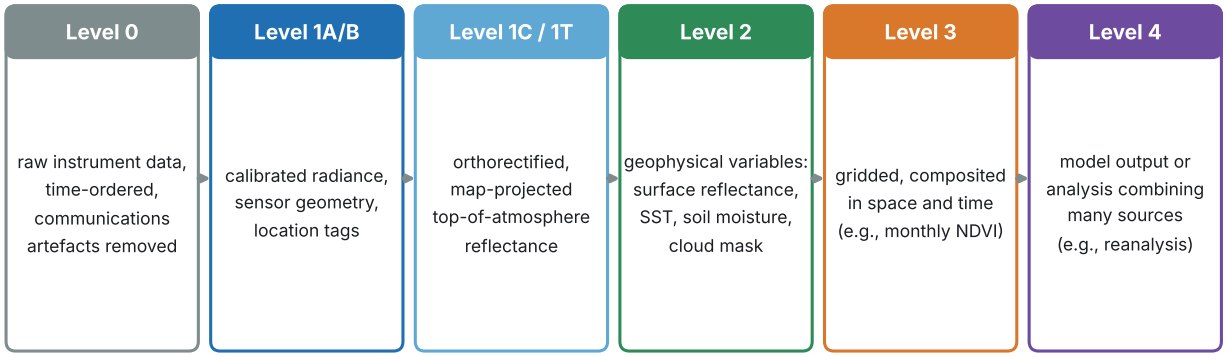
Orthoimages are delivered in a **map projection** — a mathematical transformation of the curved Earth onto a flat plane. The most common for EO is the **Universal Transverse Mercator (UTM)** system, which divides the Earth into 60 zones 6° wide; Sentinel-2 and Landsat products are delivered in UTM tiles. Global products often use a geographic latitude–longitude grid, sinusoidal (MODIS) or equal-area grids (EASE-Grid). All coordinates refer to a **datum**, today almost always WGS 84 (Chapter 18).

# 13.8 Processing Levels and Analysis-Ready Data

## 13.8.1 From Raw Bits to Information

Space agencies classify data products by **processing level** (Figure 13.12):

- **Level 0** — raw instrument data at full resolution, time-ordered, with communications artefacts removed.
- **Level 1A/1B** — data with radiometric calibration applied (Level 1B in radiance units) and geolocation information attached.
- **Level 1C/1T** — orthorectified, map-projected TOA reflectance or radiance (e.g., Sentinel-2 L1C, Landsat L1TP).
- **Level 2** — derived geophysical variables at the same resolution: surface reflectance, land-surface temperature, sea-surface temperature, cloud masks.
- **Level 3** — variables mapped onto uniform space–time grids, often composited over days or months (e.g., monthly global NDVI).
- **Level 4** — model output or results from analysis of lower-level data (e.g., gross primary productivity, crop yield, carbon flux).




CEOS Analysis Ready Data (CARD4L) defines Level-2-type products that users can analyse directly.

**FIGURE 13.12** Data processing levels from raw telemetry (Level 0) to model-based information products (Level 4).

## 13.8.2 Calibration and Validation

**Calibration** establishes the link between the sensor's digital numbers and physical units. It begins with pre-flight laboratory calibration against standards traceable to national metrology institutes. In orbit, it continues with on-board calibrators (solar diffusers, lamps, blackbodies) and observations of the Moon (whose reflectance is extremely stable). It also uses **vicarious calibration** sites with stable bright surfaces (such as Railroad Valley in Nevada, La Crau in France and Gobabeb in Namibia, part of the **RadCalNet** network). Finally, there is **cross-calibration** between satellites viewing the same place at nearly the same time. The radiometric accuracy of good modern sensors is about 3–5 % absolute and better than 1 % relative stability. Upcoming climate-benchmark missions, such as the UK-led ESA mission **TRUTHS**, aim for SI-traceable accuracy of 0.3 % or better so that they can serve as "standards in space" for other sensors.

**Validation** checks derived products against independent reference data — field measurements, airborne campaigns, in-situ networks (such as AERONET for aerosols, FLUXNET for carbon fluxes and Argo floats for the ocean) — and reports their uncertainty. The international **Committee on Earth Observation Satellites (CEOS)** coordinates calibration and validation among agencies.

### 13.8.3 Analysis-Ready Data

Most users do not want to correct each image themselves. **Analysis-ready data (ARD)** are satellite data processed to a minimum set of requirements — calibrated, geometrically corrected, atmospherically corrected to surface reflectance, cloud-masked, with per-pixel quality information — so that users can analyse them directly. CEOS has defined the **CEOS Analysis Ready Data (CEOS-ARD)** framework (formerly CARD4L) for optical, thermal and radar data; Landsat Collection 2, Sentinel-2 L2A and the harmonised Landsat–Sentinel (HLS) dataset are examples. HLS combines Landsat 8/9 and the Sentinel-2 satellites into a single 30 m surface-reflectance record with a revisit of 2–3 days.

### 13.8.4 Cloud Platforms, Data Cubes and Standards

Earth-observation archives now grow by many petabytes per year. The Copernicus Sentinel satellites alone produce more than 15 TB of data per day, and a single radar mission such as NISAR adds tens of terabytes per day. Instead of downloading data, users increasingly bring their algorithms to the data on **cloud computing platforms**:

- **Google Earth Engine** — a planetary-scale analysis platform with a multi-petabyte catalogue of Landsat, Sentinel, MODIS and many other datasets, free for research and non-commercial use;
- the **Copernicus Data Space Ecosystem** — the EU's platform providing free access to all Sentinel data with processing services;
- **Microsoft Planetary Computer**, **Amazon Web Services Open Data** and the **NASA Earthdata Cloud**, which hosts NASA's archive in AWS;
- national **data cubes** such as Digital Earth Australia and Digital Earth Africa, which organise ARD into time series for every pixel of a country or continent.

Common open standards make this possible. **Cloud-Optimized GeoTIFF (COG)** files are organised so that a program can read just the part of an image it needs over the internet. **Zarr** stores multidimensional arrays in chunks. The **SpatioTemporal Asset Catalog (STAC)** specification provides a common way to describe and search EO data across providers.

## 13.9 Turning Images into Information

### 13.9.1 Visual Interpretation and Colour Composites

Humans interpret images using tone, colour, texture, pattern, shape, size, shadow, context and association. Multispectral data are displayed by assigning three bands to the red, green and blue channels of a display. A **true-colour** composite (red, green, blue bands) looks like a photograph. A **false-colour infrared** composite (near-infrared, red, green displayed as red, green, blue) makes healthy vegetation bright red, as in Figure 13.6 and Figure 13.13, and is the traditional way to assess vegetation. A **SWIR composite** (SWIR2, NIR, red) highlights burned areas and active fires while seeing through smoke, as in Figure 13.14.

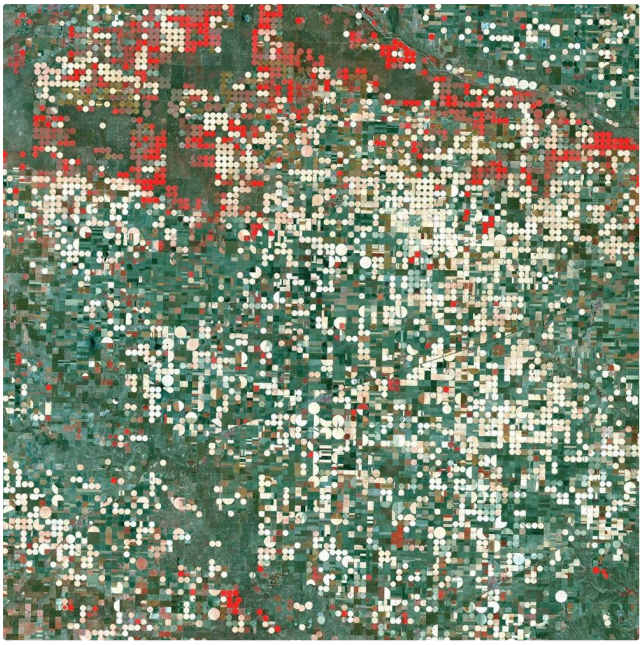


**FIGURE 13.13** Centre-pivot irrigated fields near Garden City, Kansas, in a Landsat 7 false-colour composite. Each circle is typically 800 m or 1.6 km in diameter; colour shows crop vigour and growth stage. Image: NASA/GSFC.

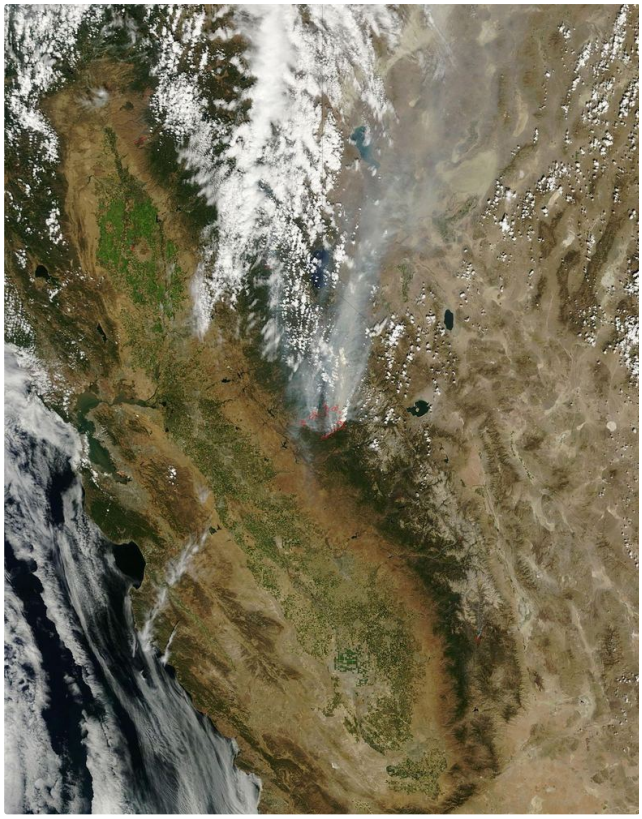


**FIGURE 13.14** The 2013 Rim Fire near Yosemite, California, imaged by MODIS on NASA's Terra satellite. The smoke plume streams north-east; thermal and short-wave infrared bands map the active fire front through the smoke. Image: NASA/GSFC.

### 13.9.2 Classification

**Image classification** assigns each pixel (or object) to a class such as forest, cropland, water or urban:

- In **unsupervised classification** (e.g., k-means, ISODATA), an algorithm groups pixels into spectral clusters, which the analyst then labels.
- In **supervised classification**, the analyst provides labelled training samples, and a classifier learns to recognise each class. Traditional classifiers include maximum likelihood; modern ones include **random forests**, support vector machines, gradient-boosted trees and **deep learning** (convolutional neural networks and transformers — see Part XIII).
- **Object-based image analysis (OBIA)** first segments the image into homogeneous objects (fields, buildings) and then classifies the objects using their spectral, shape and context attributes — useful for high-resolution imagery.

Global land-cover maps are now produced at 10 m resolution: ESA WorldCover (2020, 2021), Esri/Impact Observatory's annual land cover and Google's **Dynamic World**, which classifies every Sentinel-2 image in near-real time using deep learning.

### 13.9.3 Accuracy Assessment

A map is only useful if its accuracy is known. Accuracy is assessed by comparing the map with independent **reference data** collected by a statistically valid sample and tabulating the result in a **confusion matrix** (also called an error matrix):

**TABLE 13.3** Example confusion matrix for a four-class land-cover map with 550 validation samples (illustrative values).

| Map class ↓ / Reference → | Forest | Crop | Water | Urban | Total | User's accuracy |
|---|---|---|---|---|---|---|
| Forest | 182 | 12 | 1 | 2 | 197 | 92.4 % |
| Crop | 15 | 141 | 0 | 9 | 165 | 85.5 % |
| Water | 1 | 0 | 88 | 1 | 90 | 97.8 % |
| Urban | 2 | 11 | 1 | 84 | 98 | 85.7 % |
| Total | 200 | 164 | 90 | 96 | 550 | |
| Producer's accuracy | 91.0 % | 86.0 % | 97.8 % | 87.5 % | | **Overall 90.0 %** |

From the confusion matrix (Table 13.3), **overall accuracy** is the fraction of samples correctly classified (here 495/550 = 90.0 %). **User's accuracy** (the complement of commission error) tells a map user how often a pixel labelled "crop" really is crop. By contrast, **producer's accuracy** (the complement of omission error) tells the map producer how much of the real crop was found. Best practice (Olofsson et al., 2014) also uses the matrix to correct area estimates for map errors and to report confidence intervals.

### 13.9.4 Change Detection and Time Series

Many of the most important applications are about **change**: deforestation, urban growth, flood extent, crop development, glacier retreat. Methods include:

- **image differencing and ratioing** of bands or indices between two dates (e.g., differenced NBR for burn severity);
- **post-classification comparison** of two classified maps;
- **time-series analysis** of every observation of a pixel over years or decades, using algorithms such as LandTrendr, CCDC (Continuous Change Detection and Classification) and BFAST, which model seasonal cycles and detect breaks;
- **deep-learning change detection**, which learns to recognise meaningful changes (new buildings, damage) while ignoring irrelevant ones (shadows, season).

Dense time series are possible because of free data and combined constellations. The Global Forest Watch annual tree-cover-loss maps (University of Maryland) are based on the full Landsat archive since 2000. These maps, and the GLAD and RADD near-real-time deforestation alerts (using Landsat, Sentinel-2 and Sentinel-1 radar), are examples used by governments and companies to enforce laws and supply-chain commitments.

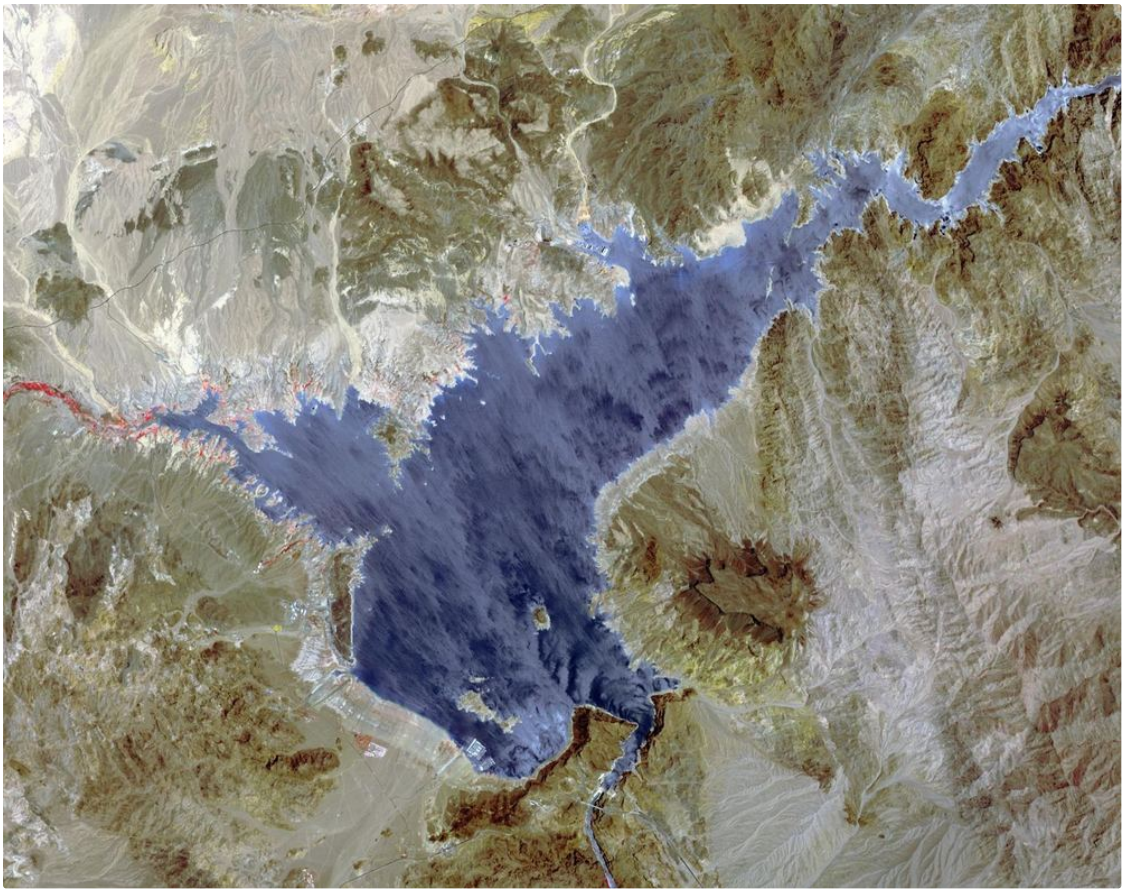


**FIGURE 13.15** Lake Mead on the Colorado River seen by ASTER on NASA's Terra satellite. Comparison of images from 2000 and 2015 reveals the light-coloured "bathtub ring" of exposed shoreline left as the water level fell by about 40 m to a record low. Image: NASA/JPL.

### 13.9.5 From Pixels to Physical Variables

Beyond classification, remote sensing retrieves quantitative **geophysical variables** by inverting physical models: leaf-area index, soil moisture, land-surface temperature, sea-surface height, ocean chlorophyll concentration, aerosol optical depth, trace-gas columns. The Global Climate Observing System (GCOS) has defined 55 **Essential Climate Variables (ECVs)**, more than half of which can be observed globally only from space. ESA's Climate Change Initiative and the Copernicus Climate Change Service produce long, consistent satellite records of many of them.

## 13.10 The Open-Data Revolution and the Users

### 13.10.1 Data Policy

For its first decades, satellite data were expensive: a Landsat Thematic Mapper scene cost about US $4 400 in the 1990s, and most data were never used. In 2008 the US Geological Survey made the entire Landsat archive free to all users, and downloads increased from about 25 000 scenes per year to millions. In 2014 the EU adopted a **full, free and open** data policy for Copernicus. Studies have estimated the annual economic benefit of open Landsat data at several billion US dollars. Other agencies, including JAXA, INPE (Brazil), CSA and ISRO, have followed with open policies for many of their datasets.

Commercial data, by contrast, are licensed. Very-high-resolution optical and SAR imagery from companies such as Maxar (Vantor), Airbus, Planet, BlackSky, ICEYE, Capella and Umbra is sold under licences that restrict redistribution. Governments are often anchor customers through programmes such as the US National Reconnaissance Office's Electro-Optical Commercial Layer. Another is NASA's Commercial SmallSat Data Acquisition (CSDA) programme, which buys data for use by NASA-funded scientists.

National regulations also govern remote sensing. In the USA, commercial remote-sensing systems are licensed by NOAA's Commercial Remote Sensing Regulatory Affairs office under a tiered system introduced in 2020; Germany has the SatDSiG data security law; other countries have similar rules. Internationally, the **UN Principles Relating to Remote Sensing of the Earth from Outer Space** (1986) state that remote sensing shall be carried out for the benefit of all countries. They also state that a sensed state shall have access to data concerning its territory on a non-discriminatory basis. Privacy concerns grow as resolution improves, although at 30 cm individuals cannot be identified.

### 13.10.2 Applications

**TABLE 13.4** Major application areas of Earth observation.

| Sector | Typical applications | Typical data |
|---|---|---|
| Agriculture | Crop type and area, crop health, yield forecasting, irrigation, insurance | Sentinel-2, Landsat, Planet, Sentinel-1, MODIS/VIIRS |
| Forestry and carbon | Deforestation alerts, forest biomass and height, carbon credits verification | Landsat, Sentinel-1/2, GEDI lidar, BIOMASS radar, ICESat-2 |
| Water resources | Reservoir levels, irrigation water use, snow cover, groundwater change | SWOT, Landsat thermal, GRACE-FO, MODIS snow |
| Disasters | Flood, fire, earthquake and storm mapping, damage assessment | SAR constellations, VHR optical, VIIRS fire, GOES/Himawari |
| Urban and infrastructure | Urban growth, informal settlements, subsidence, building footprints | VHR optical, InSAR (Sentinel-1, ICEYE), night lights |
| Oceans and coasts | Sea-surface temperature, ocean colour, oil spills, ship detection, coastal erosion | Sentinel-3, PACE, SAR, AIS from space |
| Climate and atmosphere | Greenhouse gases, aerosols, ozone, ice sheets, sea-level rise | OCO-2/3, Sentinel-5P, Copernicus Sentinel-6, ICESat-2, CryoSat-2 |
| Defence and security | Monitoring of facilities, borders, maritime domain awareness | VHR optical, SAR, RF geolocation (Chapter 19) |
| Finance and insurance | Commodity monitoring, parametric insurance, supply-chain verification | Optical and SAR constellations, analytics platforms |
| Public health | Disease vector habitats, heat stress, air quality | Thermal, aerosol and land-cover data |

The **International Charter "Space and Major Disasters"** was founded in 2000 by ESA, CNES and CSA. It now unites some 17 space agencies and operators who provide satellite data free of charge to countries affected by major disasters. It has been activated more than 900 times. The Copernicus Emergency Management Service and UN-SPIDER/UNOSAT provide rapid mapping services.

### EXAMPLE: ESTIMATING THE IRRIGATED AREA OF A DISTRICT

An analyst wants the area of irrigated cropland in a 5 000 km² district during the dry season.

1. Select all Sentinel-2 L2A (surface reflectance) images for the dry season and mask clouds using the scene classification layer.
2. Compute NDVI for each image and take the maximum value composite per pixel (to capture peak green-up).
3. Collect 400 training samples (irrigated, rain-fed, natural vegetation, bare, water, urban) from field visits and very-high-resolution images.
4. Train a random-forest classifier using the dry-season NDVI time series plus selected bands.
5. Classify the district and assess accuracy with 600 independent stratified-random samples; suppose the overall accuracy is 88 % and the user's and producer's accuracies for "irrigated" are 84 % and 90 %.
6. Use the error matrix to adjust the mapped area: if 610 km² is mapped as irrigated, the bias-corrected estimate (Olofsson method) might be 640 ± 45 km² at 95 % confidence.

The entire workflow can run in a few minutes on a cloud platform at zero data cost — a task that would have taken months and thousands of dollars in data purchases in the 1990s.

### AI IN REMOTE SENSING (PREVIEW)

Machine learning now underpins most large-scale EO mapping. Deep convolutional networks and vision transformers segment buildings, roads and fields; detect ships, aircraft and vehicles; and classify land cover. **Geospatial foundation models** pre-trained on millions of unlabelled images — such as NASA–IBM's Prithvi, Clay, SatMAE and Google's AlphaEarth Foundations embeddings — can be fine-tuned for new tasks with few labels. Part XIII (Chapters 43–53) covers these methods in depth, including on-board AI that decides which images to keep and downlink.

### KEY TAKEAWAYS

- Remote sensing measures electromagnetic radiation reflected or emitted from the Earth. Planck's law and Wien's law explain why reflected sunlight dominates at 0.4–2.5 µm and thermal emission at 3–14 µm.
- The atmosphere scatters (strongly at short wavelengths) and absorbs (in gas bands) radiation; sensors observe through atmospheric windows and use atmospheric correction to retrieve surface reflectance.
- Each material has a characteristic spectral signature. Vegetation's red-edge contrast underlies NDVI and many other indices.
- Spatial, spectral, radiometric and temporal resolution trade against each other and against swath, data volume and cost.
- Image quality depends on MTF and SNR, not just GSD; NIIRS and GIQE relate them to interpretability.
- Geometric correction (geolocation and orthorectification with a DEM) turns images into maps; calibration and validation make measurements trustworthy.
- Data progress from Level 0 to Level 4; analysis-ready data, open standards (COG, STAC, Zarr) and cloud platforms let users analyse petabytes without downloading them.
- Classification, change detection and time-series analysis, with rigorous accuracy assessment, turn images into information. Free and open Landsat and Copernicus data have made EO a global public good.

### REVIEW QUESTIONS

1. Using Wien's law, find the wavelength of peak emission for the Sun (5 772 K) and for the Earth's surface (300 K). What does this imply for optical and thermal sensors?
2. Why does the sky look blue, and why do Earth-observation sensors avoid the strong water-vapour and $CO_2$ absorption bands for surface imaging?
3. Calculate the NDVI for a vegetated pixel (red reflectance 0.05, near-infrared 0.45) and for bare soil (red 0.25, near-infrared 0.30).
4. Describe the four resolutions of a remote-sensing system and give an example of a trade-off between two of them.
5. Why can two imagers with the same ground sample distance produce images of very different quality?
6. Why does orthorectification need a digital elevation model? What happens to geolocation accuracy in mountainous terrain without one?
7. A classification is checked at 500 points: 180 forest correctly mapped, 20 forest mapped as non-forest, 30 non-forest mapped as forest and 270 non-forest correct. Calculate the overall accuracy and the user's and producer's accuracies for forest.
8. What are analysis-ready data, and how do cloud platforms and standards such as COG and STAC change the way users work with satellite archives?

CHAPTER

# 14 Electro-Optical and Infrared Imaging Satellites

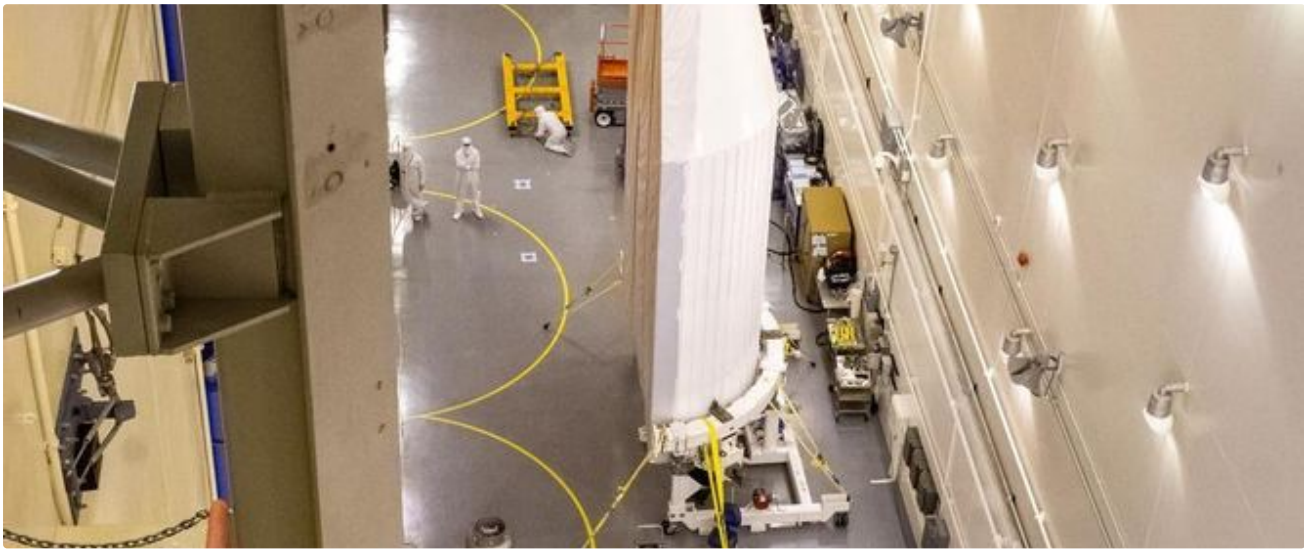


Image: NASA/KSC (NASA/Randy Beaudoin)

*Electro-optical and infrared (EO/IR) satellites are cameras in orbit. They range from government "workhorse" missions such as Landsat and Sentinel-2, which map the whole land surface every few days at 10–30 m, through commercial constellations of hundreds of shoebox-sized satellites that image the planet daily, to very-high-resolution satellites that can see objects 30 cm across. This chapter explains how these systems are designed — telescopes, focal planes, time-delay integration, agility and data handling — and why the laws of optics and photon statistics set their limits. It then covers thermal infrared imaging for temperature and fire, imaging spectroscopy (hyperspectral) for minerals, vegetation chemistry and methane, global moderate-resolution imagers, and the state of the industry in 2026.*

IN THIS CHAPTER

14.1 The Family of EO/IR Imagers
14.2 Landsat and Sentinel-2: The Workhorses
14.3 Very-High-Resolution Imaging
14.4 Constellations: Trading Resolution for Revisit
14.5 Thermal Infrared Imaging
14.6 Imaging Spectroscopy (Hyperspectral)
14.7 Global Imagers: MODIS, VIIRS and Their Successors
14.8 Design Example: A Small VHR Imaging Satellite

## 14.1 The Family of EO/IR Imagers

**Electro-optical (EO)** sensors convert visible and near-infrared light into electrical signals; **infrared (IR)** sensors extend this to the short-wave, mid-wave and long-wave infrared. Together they form the largest and most familiar class of Earth-observation satellites. Their principal

parameters — resolution, spectral bands, swath and revisit — were introduced in Chapter 13; their instrument building blocks (telescopes, detectors, scanning methods) in Chapter 12. This chapter looks at complete systems.

EO/IR satellites are usually grouped by their spatial resolution, which largely determines their applications, size and cost (Table 14.1).

**TABLE 14.1** Classes of EO/IR imaging satellites.

| Class | GSD | Swath | Typical missions | Typical uses |
|---|---|---|---|---|
| Low resolution | 250 m – 5 km | 1 000 – 3 000 km | MODIS, VIIRS, Sentinel-3, geostationary imagers | Weather, climate, ocean colour, fires, global vegetation |
| Medium resolution | 10 – 100 m | 100 – 300 km | Landsat 8/9, Sentinel-2, CBERS, Resourcesat | Land cover, agriculture, forestry, water, change monitoring |
| High resolution | 1 – 10 m | 10 – 60 km | PlanetScope, SPOT 6/7, Gaofen-1, RapidEye (retired) | Field-level agriculture, mapping, monitoring |
| Very high resolution (VHR) | 0.25 – 1 m | 5 – 20 km | WorldView, Pléiades Neo, SkySat, BlackSky, Jilin-1, Cartosat-3 | Urban mapping, infrastructure, defence and intelligence, disaster damage |
| Extremely high resolution | < 0.25 m | < 5 km | Military reconnaissance; very-low-orbit commercial concepts | Detailed object identification |

There is also a sharp division between **government science and operational missions**, which emphasise calibration accuracy, long-term continuity and free data, and **commercial missions**, which emphasise resolution, revisit, responsiveness and fast delivery to paying customers. The two are increasingly complementary: scientists use commercial data under licence, and companies build value-added products on top of free government data.

## 14.2 Landsat and Sentinel-2: The Workhorses

### 14.2.1 Landsat — Fifty Years of Land Observation

The Landsat programme is managed jointly by NASA (which builds and launches the satellites) and the US Geological Survey (which operates them and archives and distributes the data). It has observed the Earth's land surface continuously since 1972 (Figure 14.1). Its value comes from **continuity**: the same places observed in consistent, carefully cross-calibrated ways for more than five decades, providing the longest record of human and natural change on the land surface.

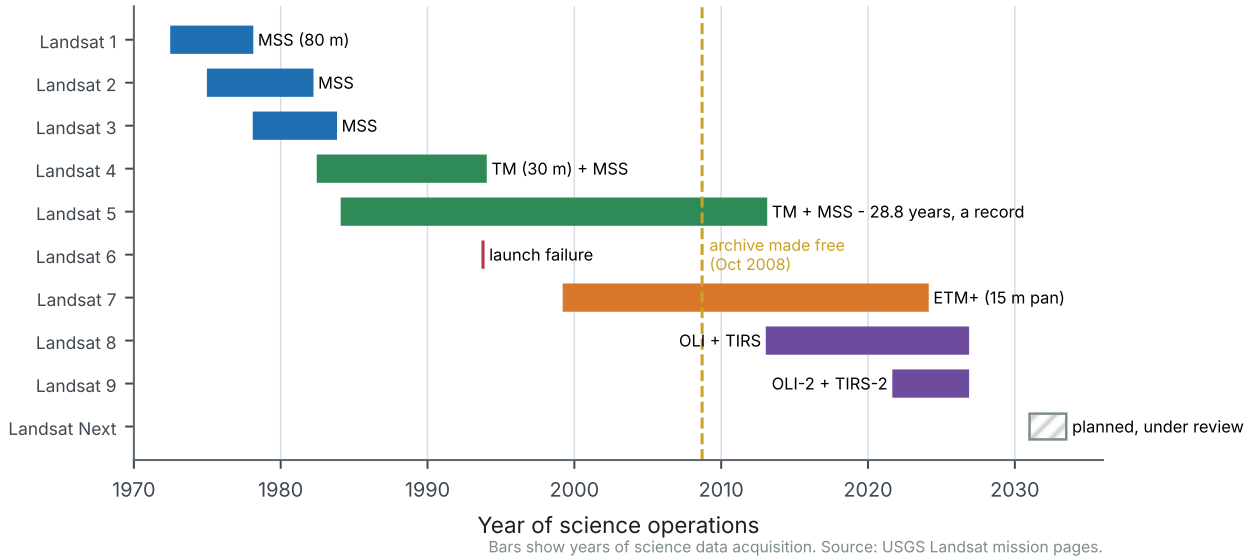



**FIGURE 14.1** The Landsat missions. Landsat 5 holds the record for the longest-operating Earth-observation satellite (28.8 years). The free-data decision of 2008 transformed the use of the archive.

Landsat 8 (launched 2013) and Landsat 9 (2021) fly in the same 705 km sun-synchronous orbit with a 10:00 a.m. descending-node crossing. Each repeats its ground track every 16 days. They are spaced half a cycle apart, so together they revisit every point every **8 days**. Each carries two instruments:

- The **Operational Land Imager (OLI / OLI-2)**, built by Ball Aerospace, is a **push-broom** imager with a four-mirror telescope. Its focal plane of 14 modules carries more than 6 500 detectors per spectral band across the 185 km swath (twice as many for the panchromatic band). It has nine bands — including a coastal-aerosol band (0.43–0.45 µm) and a cirrus band (1.36–1.39 µm) for detecting thin high clouds — at 30 m (15 m panchromatic). Landsat 9's OLI-2 records data with 14-bit quantisation, giving more brightness levels than Landsat 8's 12 bits.
- The **Thermal Infrared Sensor (TIRS / TIRS-2)**, built by NASA Goddard, images in two long-wave bands (10.6–11.2 and 11.5–12.5 µm) at 100 m. It uses **quantum-well infrared photodetectors (QWIPs)** cooled to about 43 K by a cryocooler. TIRS-2 corrected a stray-light problem discovered in TIRS on Landsat 8.

Landsat 7, launched in 1999, suffered a failure of its scan-line corrector in 2003 that left wedge-shaped gaps in its images. Even so, it continued to provide useful data until January 2024 and was decommissioned in June 2025. The planned successor, **Landsat Next**, was originally conceived as a three-satellite "superspectral" constellation with 26 bands, 10–20 m resolution and a 6-day revisit, launching around 2030–31. It was restructured in the US administration's 2026 budget proposal, and its final architecture was still under review in 2026.

### 14.2.2 Sentinel-2 — The Copernicus High-Resolution Imager

The European Union's **Sentinel-2** mission, developed by ESA with Airbus as prime contractor, has become the most widely used medium-resolution optical mission. Each satellite carries the **Multispectral Instrument (MSI)**, a push-broom imager with a 15 cm three-mirror anastigmat telescope and two focal planes (visible/near-infrared silicon and short-wave infrared HgCdTe). It

records **13 spectral bands**: four at 10 m, six at 20 m and three atmospheric bands at 60 m, across a **290 km swath**. Its three **red-edge** bands at 705, 740 and 783 nm are especially valuable for vegetation monitoring (Figure 14.2).

The satellites fly at 786 km with a 10:30 a.m. descending node; each repeats every 10 days and two together give a **5-day revisit** at the equator — 2–3 days at mid-latitudes, where swaths overlap. Sentinel-2A was launched in 2015, 2B in 2017 and **2C in September 2024** on the last flight of Europe's Vega rocket. Sentinel-2C took over from 2A in early 2025, and 2A continues in an extended campaign. Sentinel-2D is planned for about 2028, and a **Sentinel-2 Next Generation** with finer resolution and more bands is planned for the 2030s.

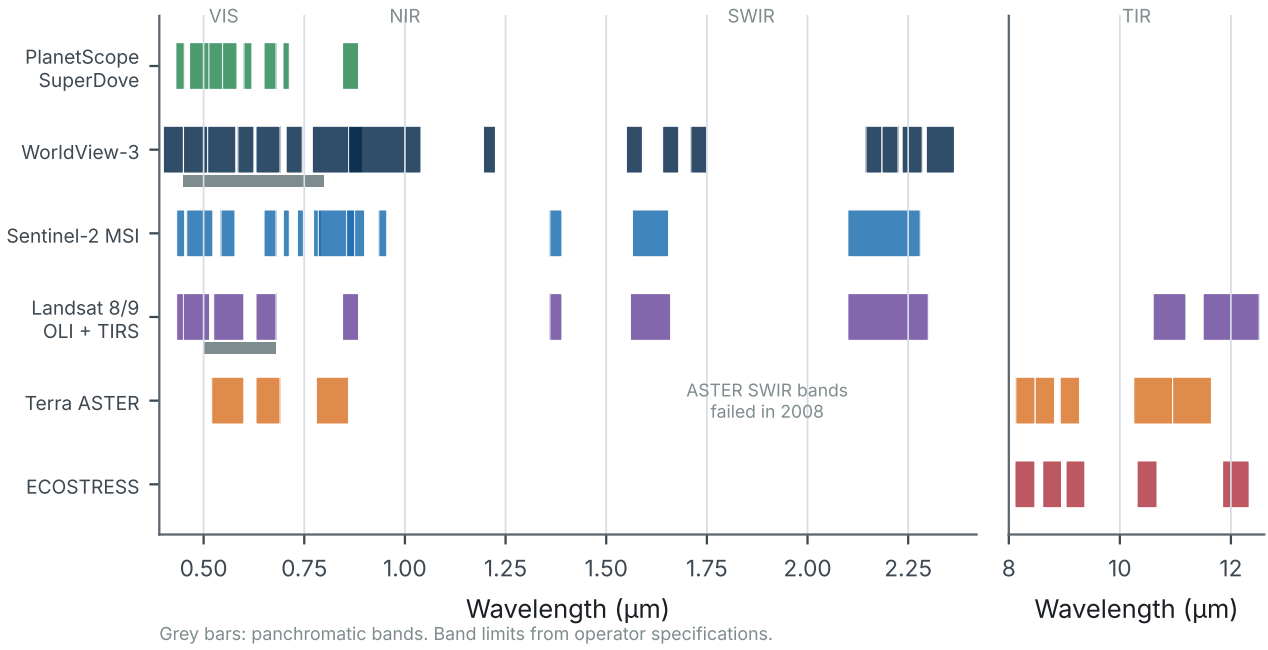



**FIGURE 14.2** Spectral bands of representative EO/IR sensors. Commercial imagers concentrate bands in the visible and near-infrared; science missions add short-wave and thermal infrared bands for atmospheric correction, geology, fire and temperature.

**TABLE 14.2** Comparison of the two leading medium-resolution land-imaging missions.

| Parameter | Landsat 8/9 | Sentinel-2A/B/C |
|---|---|---|
| Operators | NASA (development) / USGS (operations) | EU Copernicus programme; developed and operated by ESA |
| Orbit | 705 km SSO, 10:00 descending | 786 km SSO, 10:30 descending |
| Revisit (one / two satellites) | 16 / 8 days | 10 / 5 days |
| Swath | 185 km | 290 km |
| Bands | 9 reflective + 2 thermal | 13 reflective |
| Resolution | 15 m pan, 30 m MS, 100 m thermal | 10, 20 and 60 m |
| Quantisation | 12 bit (L8), 14 bit (L9) | 12 bit |
| Data volume | ≈ 1 500 scenes per day (two satellites) | ≈ 1.6 TB of compressed data per satellite per day |
| Data policy | Free and open (since 2008) | Free, full and open (since 2014) |

The two missions are complementary. Landsat has the longer record, thermal bands and stringent calibration; Sentinel-2 has finer resolution, red-edge bands and more frequent revisit. NASA's **Harmonized Landsat and Sentinel-2 (HLS)** product combines them into a single 30 m surface-reflectance record with observations every 2–3 days.

Other national programmes contribute medium-resolution data: the China–Brazil **CBERS** series, India's **Resourcesat** (LISS-III and AWiFS) and China's **Gaofen-1/6** wide-field cameras. Commercial wide-area constellations also contribute, such as **EarthDaily**, which is building a constellation designed for scientific-grade daily global coverage.

## 14.3 Very-High-Resolution Imaging

### 14.3.1 The Optics of Seeing Small Things

For a push-broom imager, the ground sampling distance is set by the detector pixel pitch $p$, the telescope focal length $f$ and the altitude $h$:

$$\mathrm{GSD} = \frac{p\,h}{f}$$

A satellite at 600 km with 5 µm detectors therefore needs a **10 m focal length** to achieve 30 cm GSD. Such long focal lengths are folded into compact telescopes with three or four curved mirrors (Chapter 12).

A long focal length alone is not enough: the aperture must be large enough that diffraction does not blur the image. The smallest resolvable separation on the ground is roughly given by the Rayleigh criterion:

$$r \approx 1.22\,\frac{\lambda\,h}{D}$$

For λ = 0.6 µm and h = 600 km, resolving 30 cm requires an aperture of at least 1.5 m by this criterion. In practice, however, systems sample somewhat finer than the Rayleigh limit and recover detail through processing (Figure 14.3). This is the fundamental reason why very-high-resolution satellites are large. WorldView-3 has a 1.1 m telescope and weighs about 2 800 kg, while a 3U CubeSat with a 9 cm aperture cannot do better than about 3 m. It also explains the military reconnaissance satellites believed to have 2.4 m mirrors, and the commercial ventures aiming for 10 cm. These either use very large telescopes or fly much lower (see "Very low Earth orbit" below).

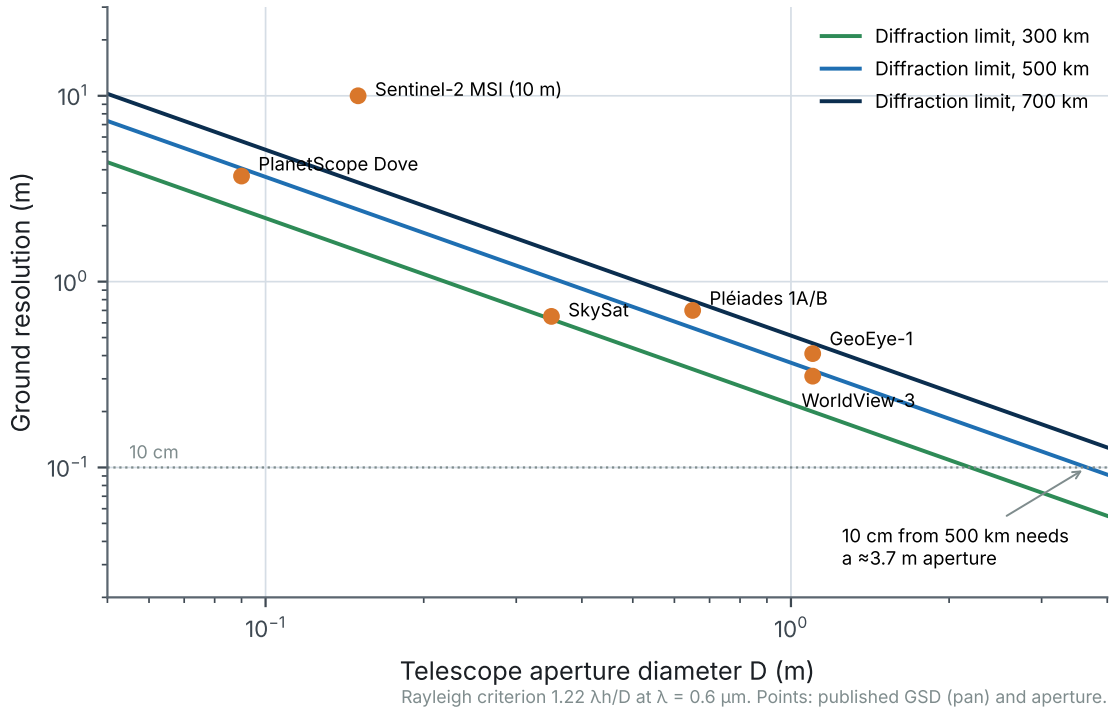
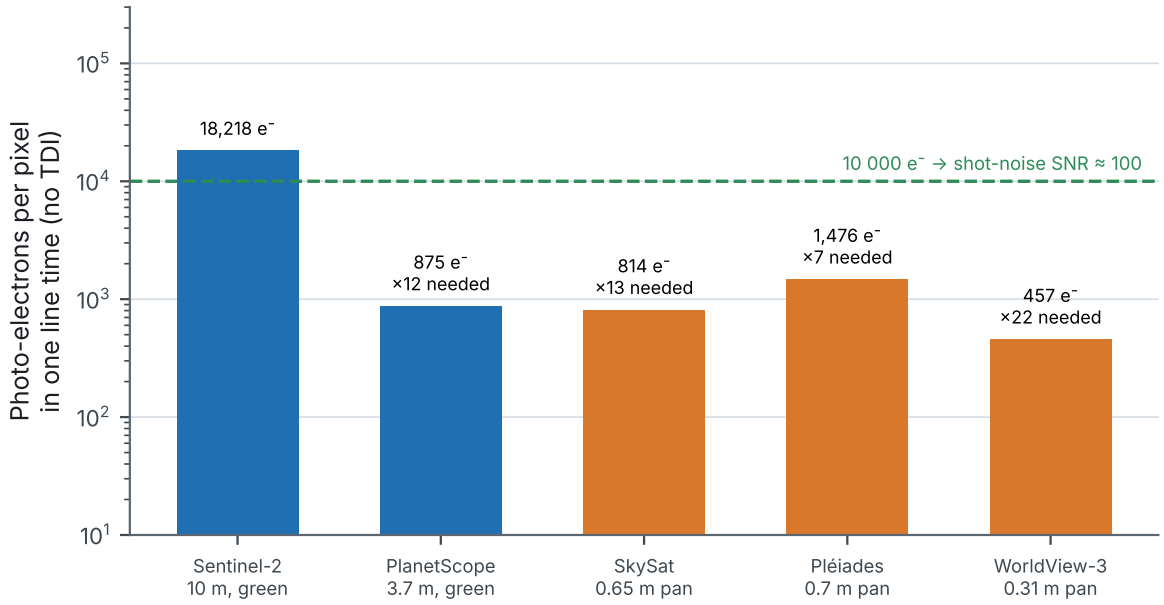



**FIGURE 14.3** Diffraction-limited ground resolution versus telescope aperture at three altitudes, with the published panchromatic GSD of representative systems. Small apertures (Sentinel-2) are deliberately not diffraction-limited; VHR systems sample at or slightly beyond the diffraction limit.

### 14.3.2 Not Enough Photons: Time-Delay Integration

The second limit is light. As the GSD shrinks, each pixel collects light from a smaller area, for a shorter time. The satellite moves 7 km/s over the ground, so a 31 cm pixel is crossed in about 45 µs. Figure 14.4 estimates the number of photo-electrons collected per pixel in one line time for several systems. A 10 m Sentinel-2 pixel collects tens of thousands of electrons, but a 31 cm WorldView-3 pixel collects only a few hundred. This gives a shot-noise-limited signal-to-noise ratio of about 20, far too low for good imagery.




Assumes L = 100 W m⁻² sr⁻¹ µm⁻¹, optics × detector efficiency 0.4, dwell = GSD / ground speed.
Push-broom imagers make up the shortfall with TDI; frame cameras (Dove, SkySat) stack several frames.

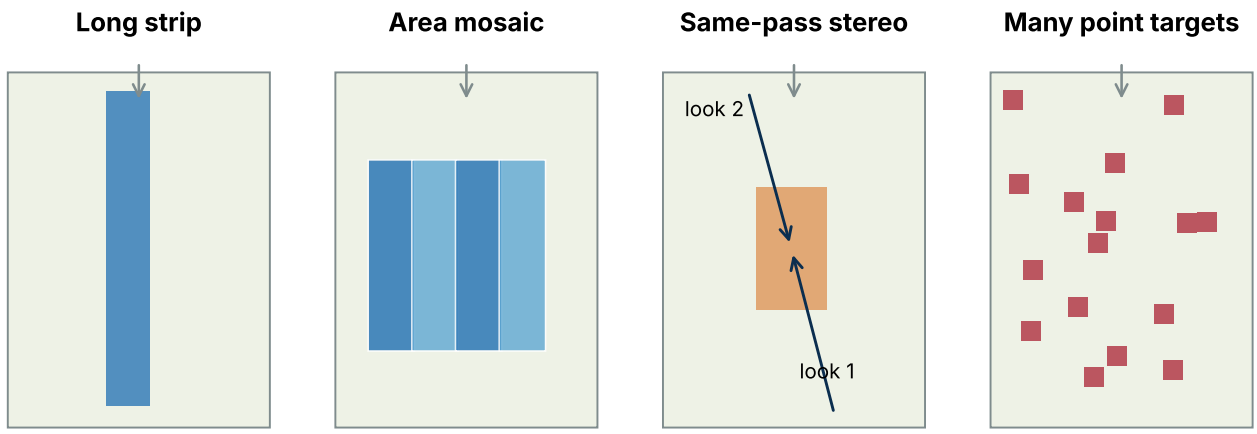
**FIGURE 14.4** Estimated photo-electrons per pixel collected in one line time for representative imagers. VHR systems collect far fewer photons per line and rely on time-delay integration (TDI) or frame stacking to reach useful signal-to-noise ratios.

The solution is **time-delay integration (TDI)**. A TDI detector has many rows (stages) of pixels aligned along the direction of motion. As the image of a ground point moves across the detector, the charge collected in one row is shifted to the next row in exact synchrony. In this way the same ground point is integrated by 8, 16, 32, 64 or more rows in succession. The signal grows in proportion to the number of stages $N$ while the shot noise grows only as $\sqrt{N}$. Modern VHR focal planes use CCD or CMOS TDI detectors with up to 64–128 stages, selectable to adapt to illumination — fewer stages for bright desert, more for dark forest or low sun. TDI requires extremely precise synchronisation between the charge transfer and the ground image motion, which depends on altitude, satellite speed, viewing angle and Earth rotation; errors smear the image.

Frame cameras, such as those on PlanetScope Doves and SkySats, use another approach. They take many overlapping short exposures and combine ("stack") them on the ground. This also allows **super-resolution** processing that recovers detail finer than a single frame.

### 14.3.3 Agility, Pointing and Collection Modes

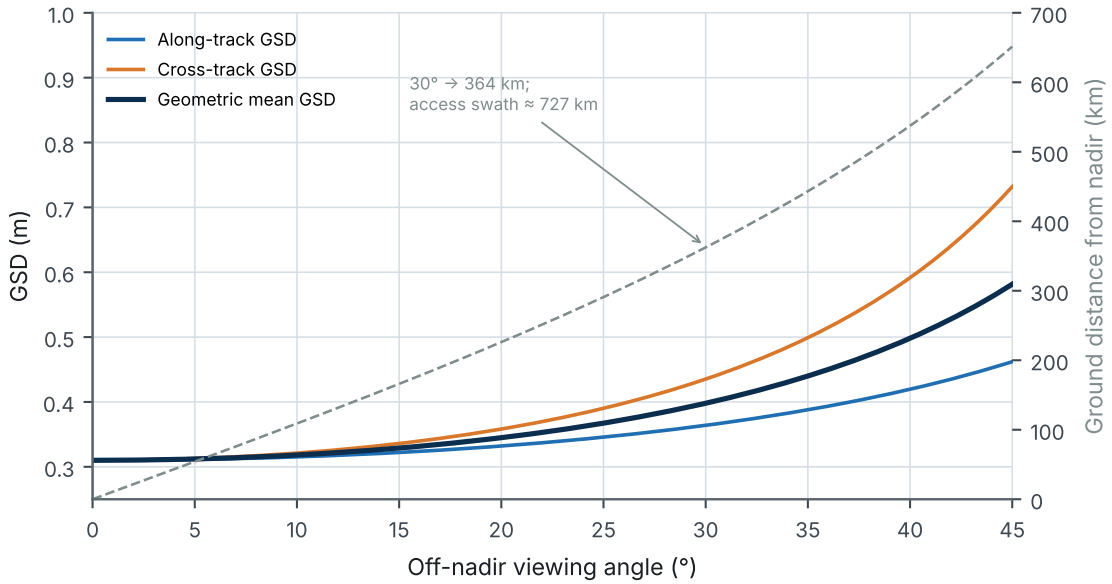
A VHR satellite with a swath of only 10–15 km would take years to map the whole Earth. Instead, VHR satellites are **agile**: they rotate the whole spacecraft to point the telescope at targets up to 30–45° away from the ground track, using control-moment gyroscopes (CMGs) or large reaction wheels (Chapter 6). An agile satellite can typically retarget across hundreds of kilometres in about ten seconds, collecting many separate targets on a single pass or sweeping its view to build an area larger than its swath (Figure 14.5).




Arrows show the direction of flight. Agile satellites slew between targets in seconds.

**FIGURE 14.5** Collection modes of an agile VHR satellite: a long strip along the ground track, an area mosaic built from adjacent strips on one pass, a stereo pair from fore and aft views, and many point targets collected in rapid succession.

Off-nadir viewing has a cost: the distance to the ground increases and the ground is viewed obliquely, so the GSD grows (Figure 14.6). At 30° off nadir, a satellite with 31 cm nadir GSD produces imagery with about 40 cm (geometric mean) GSD; at 45°, about 60 cm. Oblique views also suffer more atmospheric haze, longer building shadows and greater layover of tall structures — so customers specify a maximum off-nadir angle when ordering.




WorldView-3-like case: 617 km altitude, 0.31 m GSD at nadir, spherical Earth.

**FIGURE 14.6** Ground sampling distance versus off-nadir angle for a WorldView-3-like satellite. The dashed curve gives the distance from the ground track: viewing up to 30° off nadir gives access to a strip about 725 km wide.

### 14.3.4 Stereo Imaging and 3-D Mapping

By imaging the same area from two different directions on the same pass — looking forward and then backward — an agile satellite collects a **stereo pair**. Tall features appear shifted (parallax) between the two images in proportion to their height. Photogrammetric software measures the parallax by automatically matching features and computes a **digital surface model (DSM)** (Figure 14.7). The precision of height measurement depends on the ratio of the stereo base to the height (B/H) and the precision of image matching. VHR stereo pairs routinely produce DSMs with

1–2 m vertical accuracy. They are used to build 3-D city models, measure volumes of mines and stockpiles, and produce terrain data for defence and aviation. Some missions, such as Japan's ALOS PRISM (2006–2011) and the Chinese ZY-3 series, carry two or three fixed cameras looking forward, nadir and backward to collect stereo continuously.

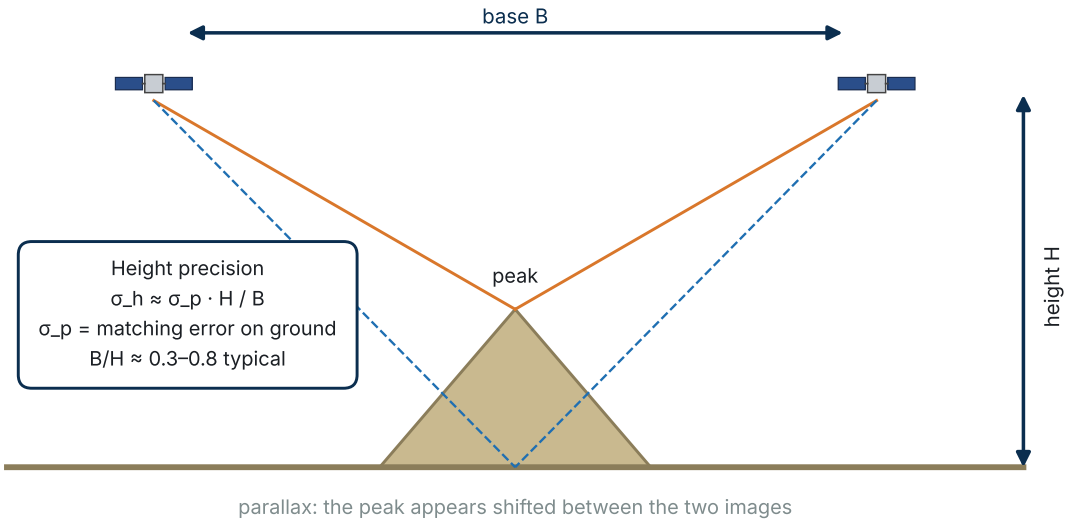



**FIGURE 14.7** Stereo imaging geometry. Two views separated by base B from height H see a tall feature displaced by parallax; height precision is approximately the matching precision multiplied by H/B.

### 14.3.5 Pan-Sharpening

Most VHR satellites collect a high-resolution **panchromatic** band and lower-resolution (typically 4×) **multispectral** bands. The panchromatic band collects more light because it is wide, so its pixels can be smaller with the same SNR. **Pan-sharpening** combines the two to create a colour image with the panchromatic resolution (Figure 14.8). Methods range from simple ratio (Brovey) and intensity–hue–saturation substitution to Gram–Schmidt, wavelet and, increasingly, deep-learning fusion. Pan-sharpening is cosmetically excellent but can distort the spectral values, so scientists usually prefer the original multispectral data for quantitative analysis.

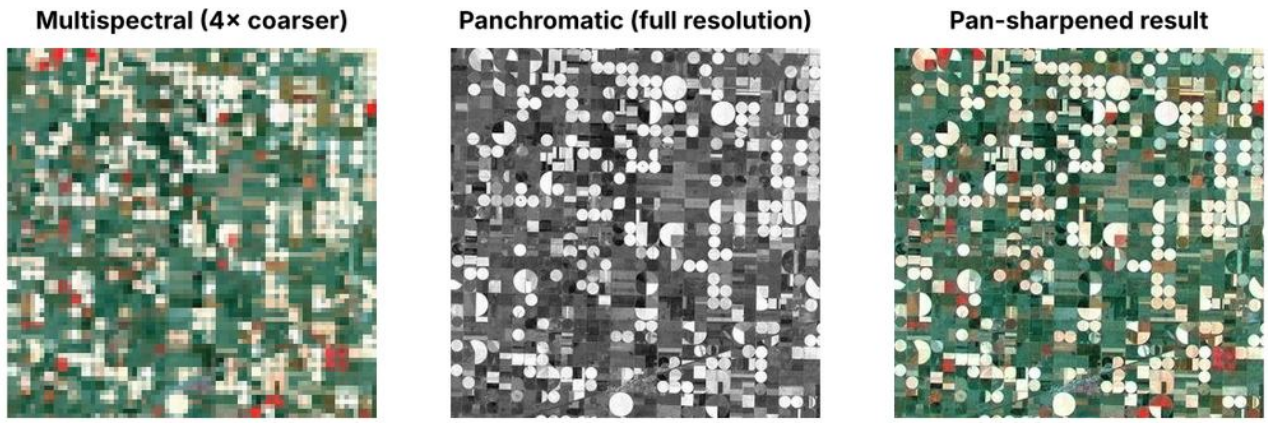



**FIGURE 14.8** Pan-sharpening simulated with part of a Landsat scene: coarse multispectral data (left) and a sharp panchromatic image (centre) combine into a sharp colour image (right).

### 14.3.6 The Commercial VHR Landscape

The commercial VHR industry began in 1999, when **IKONOS** became the first commercial satellite to sell 1 m imagery. Since then, resolution has improved steadily, limited as much by regulation as by technology (Figure 14.9). In the USA, commercial imaging satellites are licensed by NOAA. The resolution permitted for sale was relaxed from 0.5 m to 0.25 m in 2014. In 2020 fixed limits were replaced by a tiered system that considers what is already available on the world market.

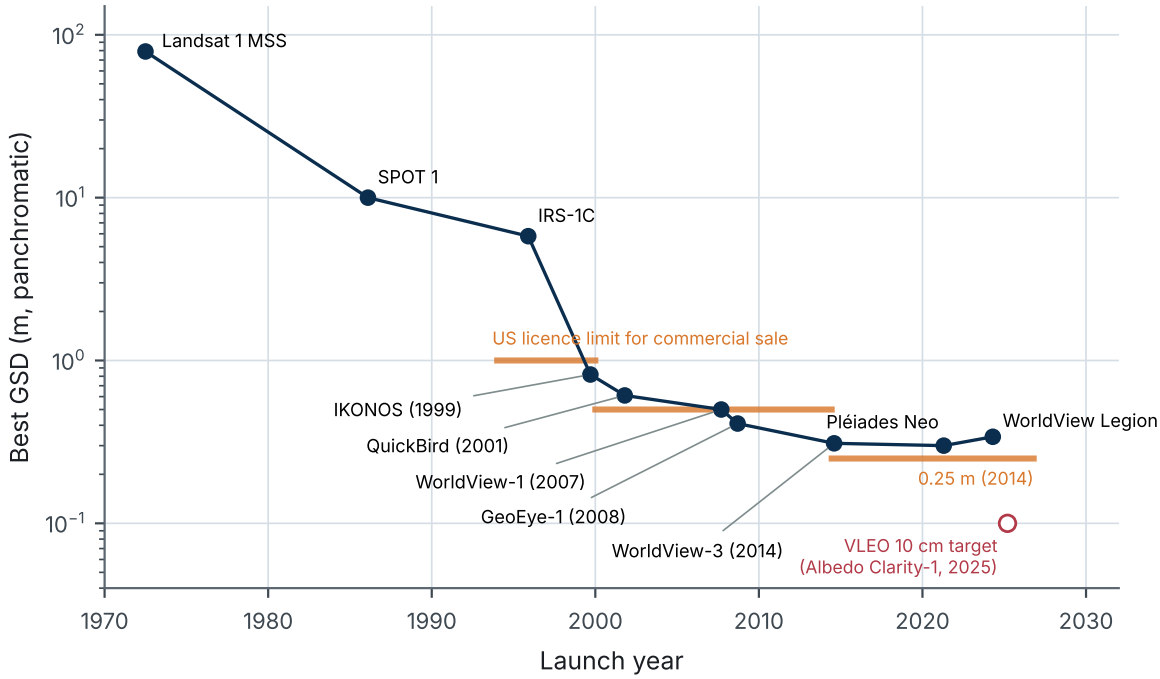



**FIGURE 14.9** Best panchromatic GSD of civil and commercial imaging satellites by launch year, and the US licensing limit for commercial sale.

**TABLE 14.3** Representative very-high-resolution imaging satellites in 2026 (approximate values from operators' published specifications).

| System (operator, country) | First launch | Altitude | Pan / MS GSD | Swath | Notes |
|---|---|---|---|---|---|
| WorldView-3 (Vantor, USA) | 2014 | 617 km | 0.31 / 1.24 m | 13 km | 8 VNIR + 8 SWIR bands, CAVIS atmospheric sensor |
| WorldView Legion (Vantor, USA) | 2024 | 518 km | 0.34 / 1.36 m (0.29 m best) | 10 km | Six ≈ 630 kg satellites (two SSO, four mid-inclination); rapid revisit |
| Pléiades Neo (Airbus, France) | 2021 | 620 km | 0.30 / 1.2 m | 14 km | Two in orbit; two lost in a 2022 Vega C failure |
| Cartosat-3 (ISRO, India) | 2019 | ≈ 510 km | ≈ 0.25 / 1.1 m | 16 km | Government system; data sold commercially |
| SuperView Neo-1 (Siwei/CASC, China) | 2022 | ≈ 530 km | 0.3 / 1.2 m | 12 km | Part of a larger Chinese commercial fleet |

| System (operator, country) | First launch | Altitude | Pan / MS GSD | Swath | Notes |
|---|---|---|---|---|---|
| Jilin-1 (Chang Guang, China) | 2015 | 500–650 km | 0.5–1 m (several types) | 10–150 km | Largest Chinese commercial constellation; video and wide-field types |
| KOMPSAT-3A (KARI, Korea) | 2015 | 528 km | 0.55 / 2.2 m | 12 km | Also a 5.5 m mid-wave infrared camera |
| SkySat (Planet, USA) | 2013 | ≈ 450 km | 0.5 m (ortho) | ≈ 6 km | Frame camera; up to 90 s video; many images per day |
| BlackSky Gen-3 (BlackSky, USA) | 2025 | ≈ 450 km | 0.35 m | ≈ 5 km | Small agile satellites; rapid tasking |
| Pelican (Planet, USA) | 2023 | ≈ 500 km | ≈ 0.5 m (Gen 1); 0.3 m class (Gen 2) | — | On-board AI processing; up to 32 planned |
| NewSat Mark IV (Satellogic, Argentina/USA) | 2020 | ≈ 470 km | ≈ 0.7 m / ≈ 25 m hyperspectral | 5 km | Low-cost microsatellites |

The market is dominated by government customers. In the USA, the National Reconnaissance Office's **Electro-Optical Commercial Layer** contracts (awarded in 2022 to Maxar — now **Vantor** — BlackSky and Planet) buy commercial imagery for defence and intelligence users. Many other governments buy imagery or entire satellites. Since Russia's invasion of Ukraine in 2022, commercial imagery has been used openly by governments, news media and researchers on an unprecedented scale, demonstrating the role of commercial EO as an independent source of information.

### 14.3.7 Very Low Earth Orbit

Because resolution improves in proportion to altitude for a given aperture, some companies are pursuing **very low Earth orbit (VLEO)** — below about 400 km. Halving the altitude from 600 to 300 km halves the GSD for the same telescope, increases the light received per pixel area, and reduces the power needed for radar or optical communications. The penalties are atmospheric drag and atomic-oxygen erosion (Chapter 2): a satellite at 250–300 km must thrust almost continuously to stay in orbit. Precedents include ESA's GOCE gravity mission (2009–2013), which flew at about 255 km using an ion engine, and JAXA's **SLATS** (Tsubame), which in 2019 set a record by imaging from 167 km. The US company **Albedo** launched its first VLEO satellite, **Clarity-1**, in March 2025 to demonstrate 10 cm visible and 2 m thermal imaging. It operated for several months, but contact was later lost and it re-entered in June 2026. Air-breathing electric propulsion, which collects the thin atmosphere itself as propellant, is being studied as a long-term enabler (Chapter 42).

# 14.4 Constellations: Trading Resolution for Revisit

## 14.4.1 The PlanetScope Revolution

In 2013 the start-up **Planet** (then Planet Labs) began launching "Doves" — 3U CubeSats with a 9 cm telescope and a frame camera — in large "Flocks". The company's insight was that many cheap satellites in sun-synchronous orbits, each with a modest 24 km swath, could image **almost the entire landmass of the Earth every day** at about 3–4 m resolution. The current **SuperDove** generation records eight bands, including coastal blue, yellow and red-edge bands designed to be interoperable with Sentinel-2. Each satellite has a lifetime of only a few years, so the fleet is constantly replenished; in October 2026 about 90 Doves were active (Figure 14.10).

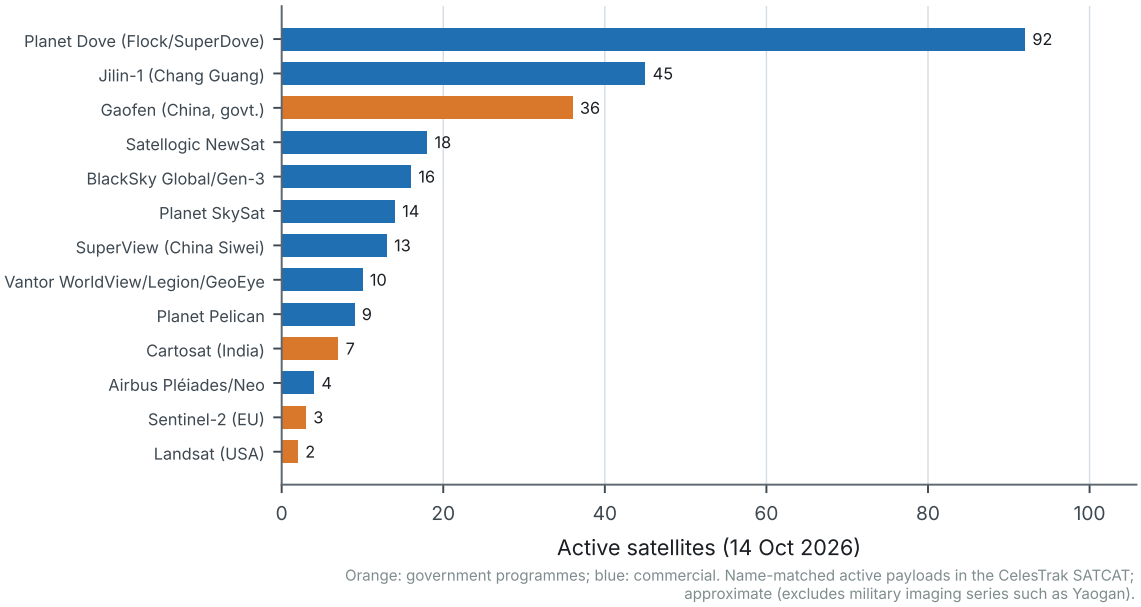



**FIGURE 14.10** Active commercial and government optical imaging satellites by operator or series in October 2026 (name-matched in the CelesTrak SATCAT, approximate).

Daily imaging changed the questions that could be asked. Instead of a few cloud-free images per year, analysts can follow a construction site, a crop field, a mine or a refugee camp day by day. The trade-off is radiometric and geometric consistency: hundreds of small satellites with slightly different sensors require careful cross-calibration, often against Sentinel-2 and Landsat.

## 14.4.2 How Many Satellites for Daily Coverage?

The time to cover the whole equator with nadir-looking imagers can be estimated simply. Each satellite makes about 14.6 orbits per day, and each ascending (or descending) pass covers a strip as wide as its swath. Covering the 40 075 km equator therefore takes about

$$T \approx \frac{40\,075\ \text{km}}{14.6 \times \text{swath} \times N}\ \text{days}$$

for $N$ satellites (Figure 14.11). Two Landsats with 185 km swaths need about 7–8 days, and two Sentinel-2s with 290 km swaths about 5. MODIS, with a 2 330 km swath, covers the Earth every 1–2 days with a single satellite. About 130 Doves with 24 km swaths achieve near-daily coverage. Agile VHR satellites do not cover the Earth systematically: their "revisit" is the time between

opportunities to point at a given target. This is typically less than a day for a constellation of a few satellites at different local times. For the mid-inclination WorldView Legion satellites over populated latitudes, it is down to about an hour.

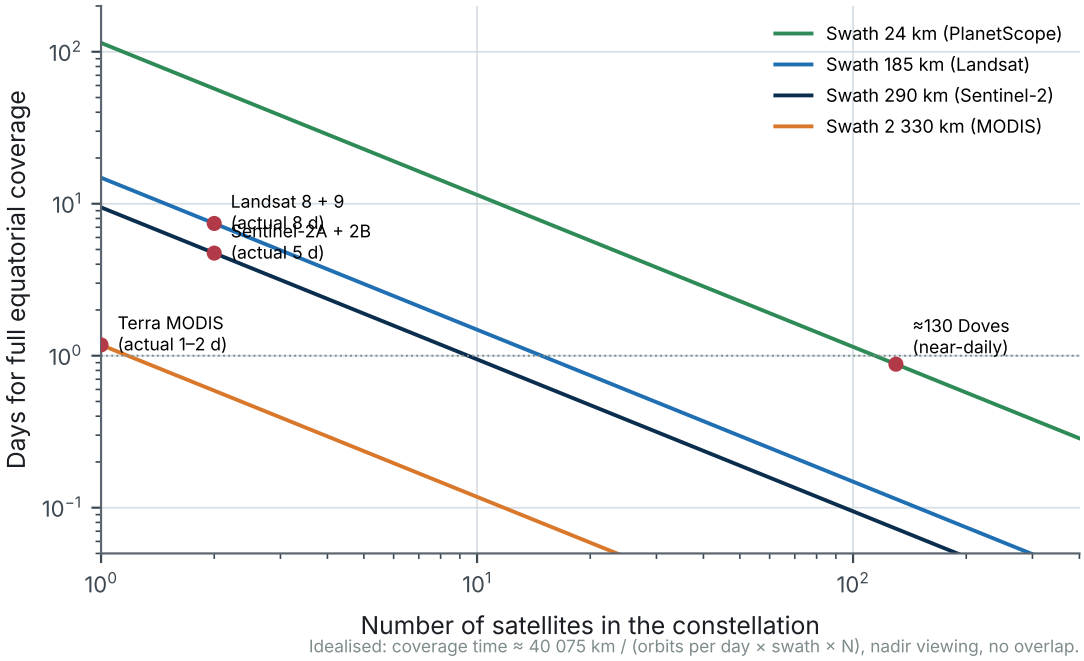



**FIGURE 14.11** Idealised days needed for full equatorial coverage versus the number of satellites, for four swath widths, compared with real systems.

### 14.4.3 Data Volume and Delivery

Imaging satellites produce enormous amounts of data. A push-broom VHR imager with 0.3 m pixels and a 13 km swath generates about 23 000 lines per second, each of about 43 000 pixels — about 1 gigapixel per second before multispectral bands. On-board **compression** (JPEG 2000 or the CCSDS 122 image-compression standard, typically 2:1 to 4:1 with near-lossless quality) and large solid-state recorders (terabits) buffer the data until it can be downlinked. Downlink uses **X-band** (typically 0.5–3 Gbit/s on modern satellites) or **Ka-band** links to networks of ground stations at high latitudes (Svalbard, Inuvik, Troll in Antarctica) and around the world (Chapter 7). Some operators now relay data through **optical inter-satellite links** or data-relay satellites such as the European Data Relay System (EDRS), which relays Sentinel data from LEO via GEO at 1.8 Gbit/s.

For customers, the key metric is the time from **tasking request to image delivery**. Modern operators accept orders through web portals and application programming interfaces (APIs), plan collections automatically, and deliver orthorectified images within hours — sometimes minutes — of collection. Planet's Pelican and BlackSky's Gen-3 satellites include on-board processors that can run AI models to detect objects and send alerts before the full image is downlinked (Part XIII).

# 14.5 Thermal Infrared Imaging

## 14.5.1 Why Measure Temperature from Space?

Thermal infrared (TIR) sensors measure radiation emitted by the surface, from which **land-surface temperature (LST)** and **emissivity** are derived (Chapter 13). Surface temperature is one of the most informative variables about the land. It responds immediately to water stress (plants that cannot transpire heat up), drives urban heat islands, and reveals volcanic activity, geothermal resources and industrial heat sources. It is also the key input for mapping **evapotranspiration** — the water used by crops — which is essential for water management in arid regions.

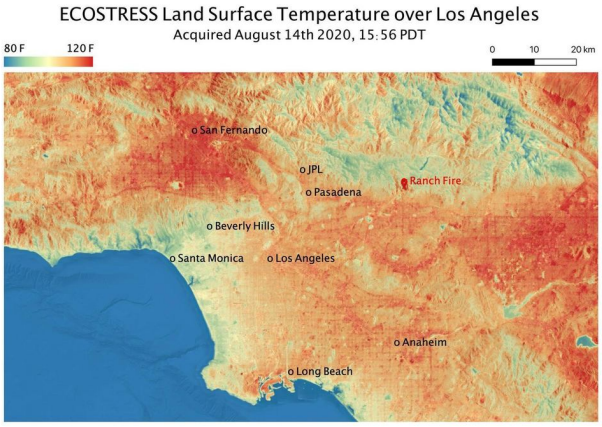



**FIGURE 14.12** Land-surface temperature over Los Angeles during an August 2020 heat wave, measured by ECOSTRESS on the International Space Station. Streets and roofs exceeded 50 °C, while parks and the coast stayed far cooler. Image: NASA/JPL (NASA/JPL-Caltech).

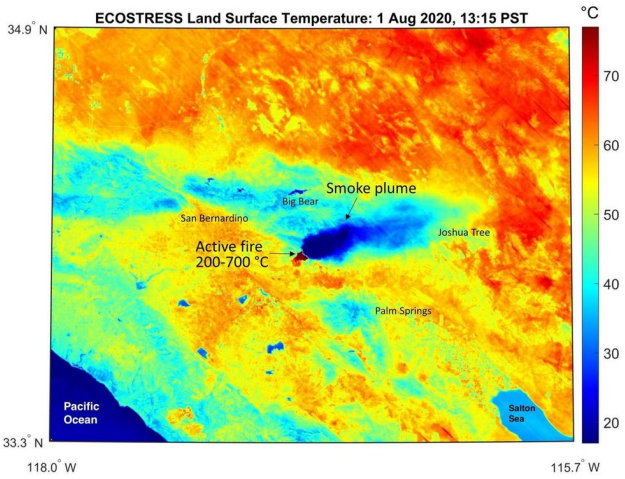



**FIGURE 14.13** ECOSTRESS land-surface temperature during the Apple Fire in California, 1 August 2020. The active fire front exceeded 200 °C, and the smoke plume appears cooler than the surrounding ground. Image: NASA/JPL (NASA/JPL-Caltech).

## 14.5.2 Thermal Instrument Design

Thermal imaging is harder than visible imaging for three reasons:

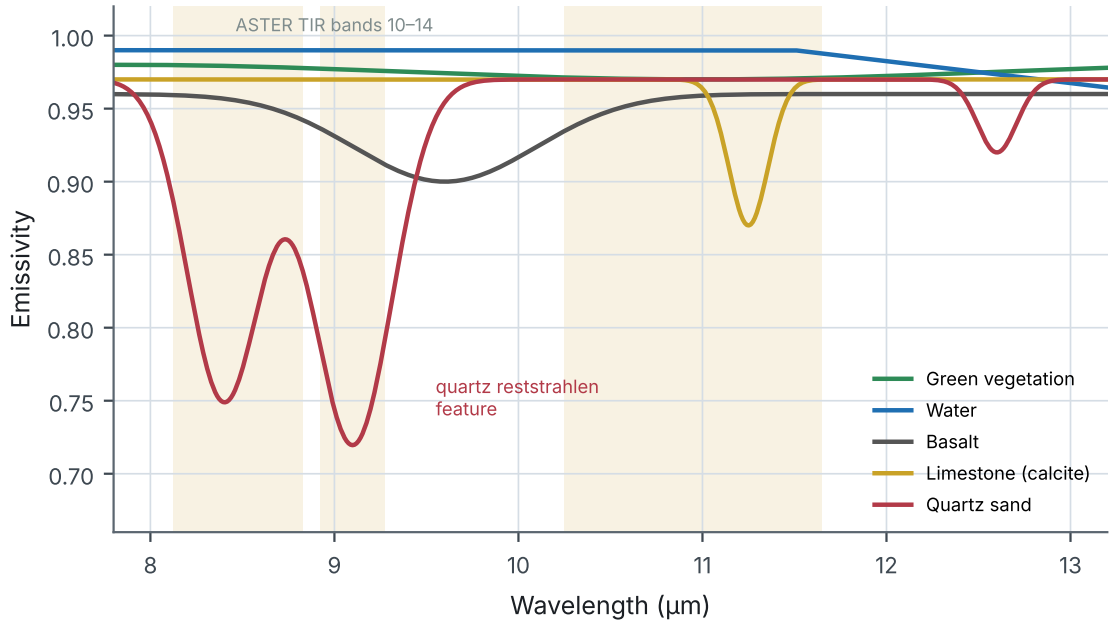
1. **Diffraction:** at 10 µm, the diffraction limit is about 17 times coarser than at 0.6 µm for the same aperture. So thermal pixels are larger (typically 60–100 m from LEO for science missions, 3–10 m from VHR systems with large apertures).
2. **Detectors:** sensitive photon detectors for the long-wave infrared (mercury cadmium telluride, HgCdTe; QWIPs; strained-layer superlattices) must be cooled to 40–80 K by mechanical cryocoolers. Uncooled **microbolometers**, which sense the heating of a tiny suspended membrane, are cheaper and smaller and are used by many small satellites, but are less sensitive and slower.
3. **Self-emission:** at room temperature, the telescope and instrument themselves glow in the thermal infrared. Thermal instruments therefore cool their optics or baffles, use cold stops, and calibrate frequently by viewing an on-board **blackbody** of known temperature and a cold reference (deep space).

The sensitivity of a thermal sensor is expressed as the **noise-equivalent temperature difference (NEΔT)** — 0.03–0.1 K for science instruments at 300 K.

### 14.5.3 Retrieving Temperature and Emissivity

The radiance measured by a thermal sensor depends on both the temperature and the emissivity of the surface, and on the atmosphere's emission and absorption — mainly by water vapour. Several methods untangle these:

- **Split-window algorithms** use two adjacent bands (around 11 and 12 µm) in which water vapour absorbs differently; the brightness-temperature difference between them corrects for the atmosphere. They are widely used for sea-surface temperature, where emissivity is known.
- **Temperature–emissivity separation (TES)** was developed for ASTER and is used for ECOSTRESS. It uses five or more bands and an empirical relationship between the spectral contrast and minimum emissivity to solve for both temperature and the emissivity spectrum.
- **Emissivity spectra** in the 8–12 µm range identify minerals (Figure 14.14): silicate rocks show strong "reststrahlen" features near 8–10 µm whose positions shift with silica content, while carbonates show features near 11.3 µm. Multispectral thermal data from ASTER, with five TIR bands, have been used extensively for geological mapping.




Schematic after the ECOSTRESS (ASTER) spectral library.

**FIGURE 14.14** Thermal-infrared emissivity spectra of common surface materials (schematic) with the ASTER thermal bands. Quartz shows strong reststrahlen features; vegetation and water are nearly black bodies.

### 14.5.4 Detecting Fires

Active fires are among the most important things to detect from space. Their detection relies on Planck's law (Figure 14.15). A 1 000 K fire emits thousands of times more radiance than a 300 K background at **3.9 µm** (mid-wave infrared), but only about 30 times more at 11 µm. A fire covering just 0.01 % of a 1 km pixel (100 m$^2$) raises the pixel's 3.9 µm brightness temperature by about 10 K but its 11 µm temperature by only 0.2 K. Fire-detection algorithms therefore look for

pixels that are much hotter at 3.9 µm than at 11 µm and than their neighbours. They use the excess radiance to estimate the **fire radiative power (FRP)**, which is related to the rate at which biomass is burned.

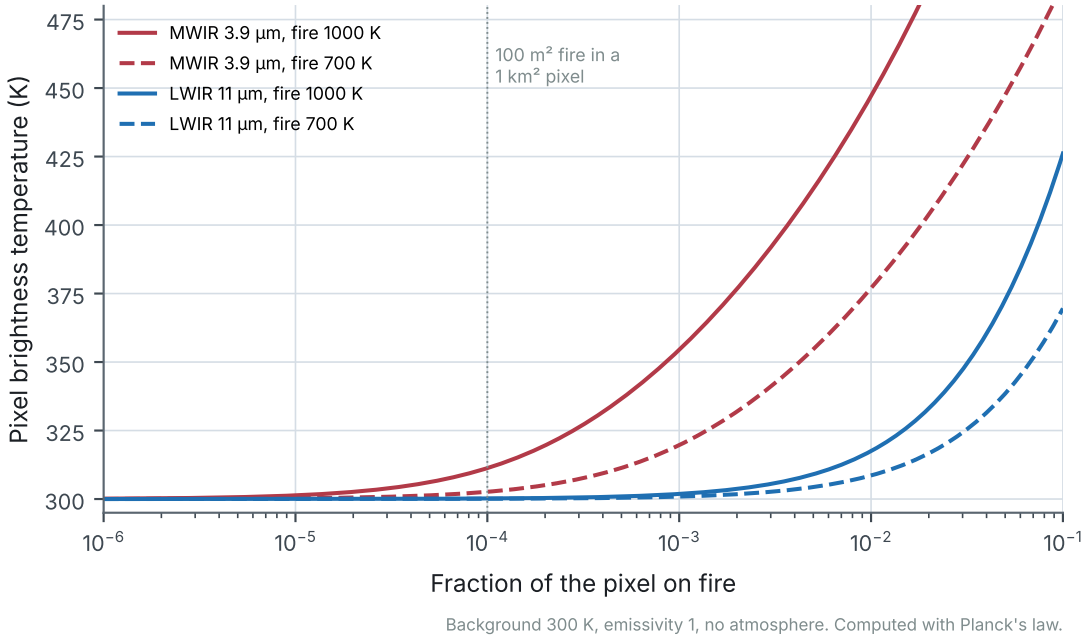



**FIGURE 14.15** Brightness temperature of a pixel containing a small fire versus the fraction of the pixel burning. The 3.9 µm band is far more sensitive to small hot fires than the 11 µm band.

NASA's **Fire Information for Resource Management System (FIRMS)** distributes active fire detections from MODIS (1 km) and VIIRS (375 m) within about three hours of observation worldwide. Geostationary imagers (GOES-R, Himawari, Meteosat Third Generation) detect fires every 5–10 minutes (Chapter 16). Dedicated commercial constellations are now adding faster, finer fire detection. The German company **OroraTech** had 20 thermal satellites of its FOREST series in orbit by October 2026, aiming at a global revisit of about 30 minutes. The Earth Fire Alliance's **FireSat** constellation, developed with Muon Space and Google, aims to detect fires as small as 5 m × 5 m.

**TABLE 14.4** Thermal infrared imaging missions (resolution and dates approximate).

| Mission | Operator | Launch | TIR resolution | TIR bands | Status (2026) |
|---|---|---|---|---|---|
| ASTER (Terra) | NASA / Japan | 1999 | 90 m | 5 | Operating; Terra nearing end of mission |
| Landsat 8 / 9 TIRS | NASA / USGS | 2013 / 2021 | 100 m | 2 | Operating |
| ECOSTRESS (ISS) | NASA JPL | 2018 | ≈ 70 m | 5 (3 used since 2019) | Operating |
| VIIRS | NOAA / NASA | 2011, 2017, 2022 | 375–750 m | 7 MWIR/LWIR | Operating (Suomi NPP, NOAA-20, NOAA-21) |
| FOREST series | OroraTech (Germany) | 2022– | ≈ 200 m | MWIR + LWIR | 20 satellites in orbit |
| HiVE (SkyBee) | constellr (Germany) | 2025– | ≈ 30 m class | 4 | First satellites in orbit |
| VanZyl | Hydrosat (USA) | 2024– | ≈ 50–70 m | Multi-band | Early constellation |

| Mission | Operator | Launch | TIR resolution | TIR bands | Status (2026) |
|---|---|---|---|---|---|
| TRISHNA | CNES / ISRO | late 2020s | 57 m | 4 | In development |
| LSTM | EU Copernicus / ESA | ≈ 2029 | ≈ 50 m | 5 | In development |
| SBG-TIR | NASA / ASI | ≈ 2029 | ≈ 60 m | 8 | In development |

### 14.5.5 Night-Time Lights

The **Day/Night Band (DNB)** of VIIRS is sensitive enough to image city lights, gas flares, fishing boats and moonlit clouds at night (Figure 14.16). NASA's **Black Marble** products correct these images for moonlight, atmosphere and seasonal effects to produce daily, monthly and annual maps of night-time lights. These maps are used to estimate economic activity and population, monitor power outages after disasters and conflicts, and track light pollution.

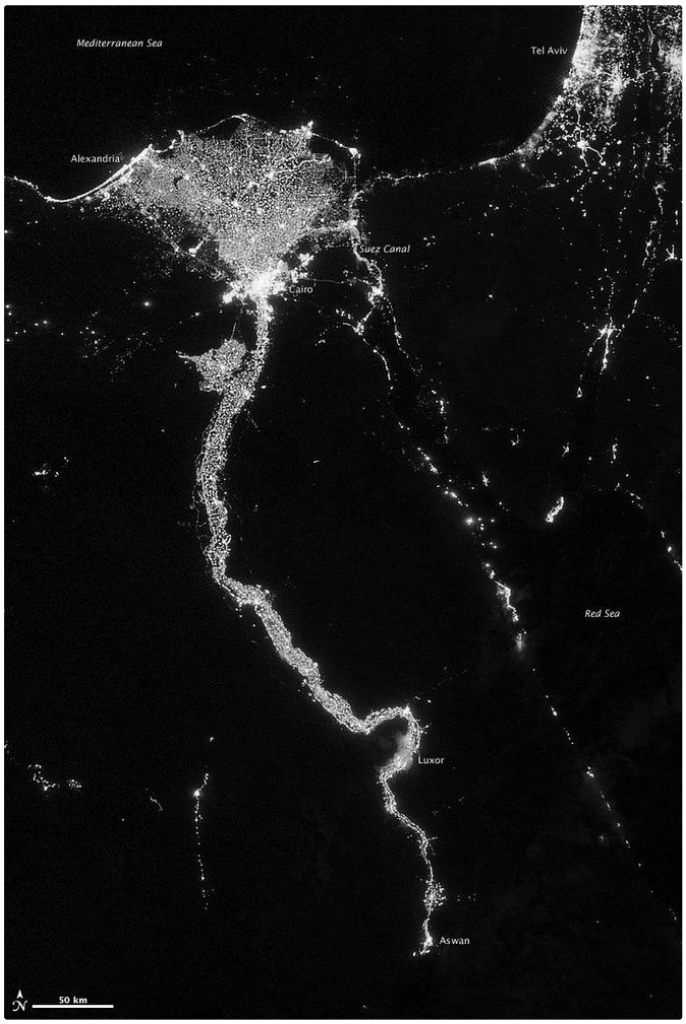



**FIGURE 14.16** City lights along the Nile Valley and Delta imaged by the VIIRS Day/Night Band on 13 October 2012. About 97 % of Egypt's population lives on less than 5 % of its land, along the river, which stands out as a ribbon of light through the dark desert. Image: NASA/GSFC.

# 14.6 Imaging Spectroscopy (Hyperspectral)

## 14.6.1 The Hypercube

A **hyperspectral imager**, or **imaging spectrometer**, records a continuous spectrum for every pixel — typically 200–300 contiguous bands 5–10 nm wide covering 0.4–2.5 µm. The result is a three-dimensional **hypercube** with two spatial dimensions and one spectral dimension (Figure 14.17). Where a multispectral sensor can tell vegetation from soil, an imaging spectrometer can identify specific minerals (by their narrow absorption features) and estimate leaf chemistry (chlorophyll, water, nitrogen, lignin). It can also map phytoplankton types in water and detect gases such as methane and carbon dioxide by their absorption.

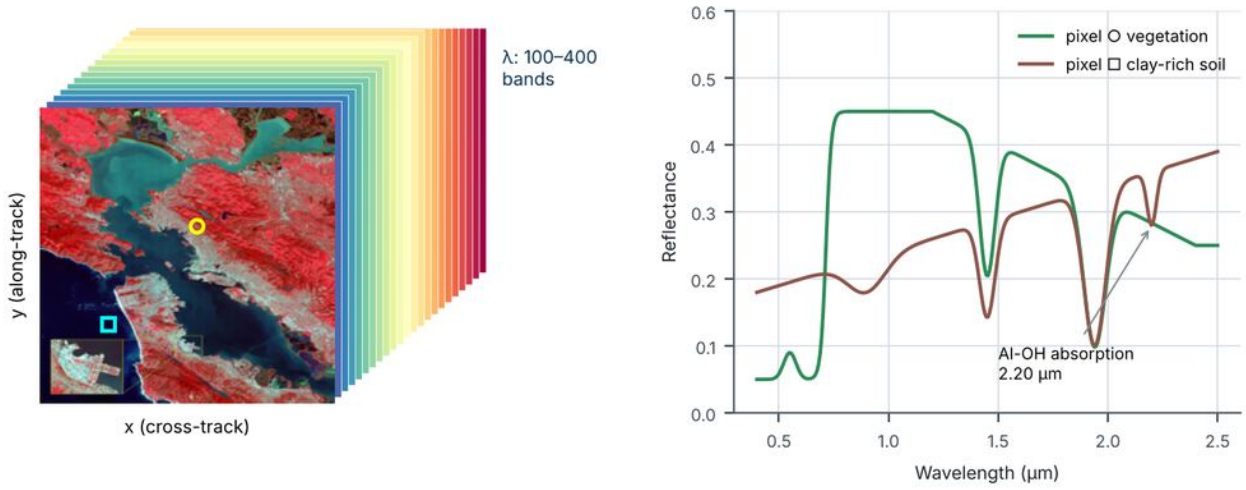



**FIGURE 14.17** A hyperspectral data cube (left) records a full spectrum for every pixel (right). Narrow absorption features — here the 2.20 µm aluminium–hydroxyl feature of clay minerals — identify materials that broad-band sensors cannot distinguish.

Most imaging spectrometers are push-broom instruments in which a slit selects one line on the ground. A dispersing element then spreads the light from each point along the slit into a spectrum across a two-dimensional detector. This element is a prism or a diffraction grating, often in compact **Offner** or **Dyson** designs. Each detector frame thus contains one spatial line and its full spectrum. Achieving high SNR in hundreds of narrow bands requires large apertures and long dwell times, which is why space hyperspectral systems usually have 30–60 m pixels and narrow swaths (30–80 km). Calibration is demanding: errors of a fraction of a nanometre in the spectral position of bands ("smile" and "keystone" distortions) can corrupt the retrieval of narrow absorption features.

## 14.6.2 Missions

NASA's **Hyperion** on EO-1 (2000–2017) was the first civil spaceborne imaging spectrometer, with 220 bands at 30 m. Today, several missions are operating and more are coming (Table 14.5).

**TABLE 14.5** Spaceborne imaging spectrometers (approximate characteristics).

| Mission | Operator | Launch | GSD | Spectral range / bands | Notes |
|---|---|---|---|---|---|
| PRISMA | ASI (Italy) | 2019 | 30 m (5 m pan) | 0.4–2.5 µm, ≈ 240 | Second generation in development |

| Mission | Operator | Launch | GSD | Spectral range / bands | Notes |
|---|---|---|---|---|---|
| DESIS (ISS) | DLR / Teledyne | 2018 | 30 m | 0.4–1.0 µm, 235 | Commercial access |
| HISUI (ISS) | Japan METI | 2019 | 20 × 31 m | 0.4–2.5 µm, 185 | |
| EnMAP | DLR (Germany) | 2022 | 30 m | 0.42–2.45 µm, > 220 | Science mission, open data |
| EMIT (ISS) | NASA JPL | 2022 | 60 m | 0.38–2.5 µm, 285 | Mineral dust sources; methane plumes |
| Tanager-1 / -2 | Carbon Mapper / Planet / JPL | 2024 / 2026 | ≈ 30 m | 0.4–2.5 µm | Methane and $CO_2$ point sources |
| Firefly | Pixxel (India) | 2025 | ≈ 5 m | VNIR, > 150 | Commercial constellation |
| CHIME | EU Copernicus / ESA | ≈ 2029 | 30 m | 0.4–2.5 µm | Copernicus Expansion mission |
| SBG-VSWIR | NASA | 2030s | ≈ 30 m | 0.4–2.5 µm | Recommended by the 2017 Decadal Survey |

### 14.6.3 Seeing Methane

One of the most consequential new uses of imaging spectroscopy is mapping **methane plumes** from individual facilities — leaking oil and gas wells, pipelines, coal mines and landfills. Methane absorbs sunlight in narrow bands near 1.65 and 2.3 µm; a plume of excess methane slightly deepens these absorptions in the pixels it covers, and matched-filter algorithms detect this signature (Figure 14.18). Because methane is a powerful greenhouse gas with a short atmospheric lifetime (about a decade), finding and fixing large leaks — "super-emitters" — is one of the fastest ways to slow global warming. EMIT detected hundreds of methane plumes in its first years; Carbon Mapper's Tanager satellites and the commercial **GHGSat** constellation, which uses Fabry–Pérot interferometers with about 25 m resolution, specialise in this task. Chapter 16 describes broader mapping of methane and carbon dioxide over regions and the globe. This is done by spectrometers such as TROPOMI on Sentinel-5P, NASA's OCO-2/3, Japan's GOSAT series (including GOSAT-GW, launched in June 2025) and Europe's future CO2M mission. The Environmental Defense Fund's MethaneSAT, launched in 2024 to quantify emissions from whole oil and gas basins, was lost in June 2025 after about a year in orbit.

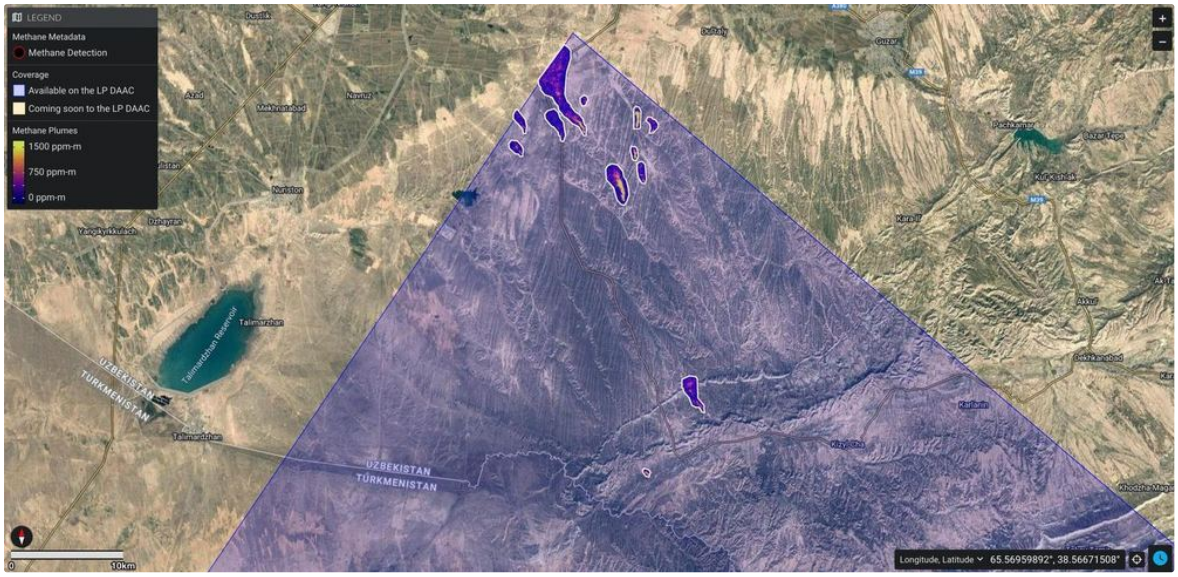



**FIGURE 14.18** A cluster of 12 methane plumes detected by NASA's EMIT imaging spectrometer over a 400 km² oil and gas field in southern Uzbekistan on 1 September 2022. Each purple plume marks a source releasing methane at a high rate. Image: NASA/JPL (NASA/JPL-Caltech).

## 14.7 Global Imagers: MODIS, VIIRS and Their Successors

The **Moderate Resolution Imaging Spectroradiometer (MODIS)**, on NASA's Terra (1999) and Aqua (2002) satellites, has been one of the most productive scientific instruments ever flown. It has 36 spectral bands from 0.41 to 14.4 µm, at 250 m (two bands), 500 m (five bands) and 1 km. Its 2 330 km swath covers the whole Earth every one to two days. Its standard products — surface reflectance, vegetation indices, land cover, snow and ice, fires, aerosols, clouds, ocean colour and sea-surface temperature — have been used in tens of thousands of scientific papers. Terra and Aqua are well beyond their design lives and have run out of the fuel needed to maintain their overpass times, so their observations are drifting and will end in the next few years.

The **Visible Infrared Imaging Radiometer Suite (VIIRS)** continues the MODIS record operationally on the NOAA–NASA Suomi NPP (2011), NOAA-20 (2017) and NOAA-21 (2022) satellites. It has 22 bands at 375 m and 750 m, and a 3 040 km swath that leaves no gaps between orbits even at the equator. In Europe, **Sentinel-3**'s OLCI (ocean and land colour) and SLSTR (sea and land surface temperature) instruments provide similar global observations with an emphasis on climate-quality accuracy. These sensors are described further in Chapter 16.

**FIGURE 14.19** The Hyperion imaging spectrometer on NASA's EO-1 satellite views Iceland's Eyjafjallajökull volcano on 2 May 2010. Visible bands (left) show only ash plume and snow; combinations using short-wave infrared bands (centre and right) reveal the hot lava flow extending north from the vent. Image: NASA/JPL (NASA/JPL/EO-1 Mission/GSFC/Ashley Davies).

## 14.8 Design Example: A Small VHR Imaging Satellite

**EXAMPLE: SIZING A 0.5 M SMALL-SATELLITE IMAGER**

*Requirement:* 0.5 m panchromatic GSD from a 500 km orbit, 6 km swath.

1. **Focal length.** With a CMOS TDI detector of 5.5 µm pixels: $f = p\,h/\mathrm{GSD} = 5.5 \times 10^{-6} \times 500\,000/0.5 = 5.5$ m.
2. **Aperture.** The Rayleigh limit at 0.6 µm for 0.5 m is $D = 1.22\lambda h/r = 0.73$ m. Choosing a 0.5 m aperture (f/11) gives an optical sampling ratio $Q = \lambda f/(D\,p) = 1.2$ — slightly under-sampled relative to the optics, a common compromise that keeps the telescope compact. The system MTF at Nyquist will be about 0.1.
3. **Swath and detector.** 6 km / 0.5 m = 12 000 pixels across the swath, for example three staggered 4 096-pixel TDI arrays.
4. **Line rate.** Ground speed at 500 km ≈ 7.06 km/s, so the line rate is 7 060 / 0.5 ≈ 14 100 lines per second.
5. **Signal.** Using the method of Figure 14.4, a 0.5 m aperture with 0.5 m pixels collects roughly 1 000 photo-electrons per line time in the panchromatic band. With 16–32 TDI stages, this rises to 16 000–32 000 electrons and a shot-noise SNR above 100 for bright targets.
6. **Data rate.** 12 000 pixels × 14 100 lines/s × 12 bits ≈ 2.0 Gbit/s for the panchromatic band, plus about 25 % for four multispectral bands at 2 m. With 3:1 compression this is about 0.85 Gbit/s; a 10-second, 70 km long image produces about 1 GB.
7. **Downlink.** An X-band link at 800 Mbit/s over a 6-minute ground-station pass can return about 36 GB per pass — some 36 such images.
8. **Agility.** To collect 20 targets per pass, the satellite must slew roughly 1–2° per second and settle within a few seconds — requiring CMGs or high-torque reaction wheels and a rigid structure with good jitter isolation.

The result is a satellite of roughly 150–250 kg — similar to commercial systems such as BlackSky Gen-3 or Planet's Pelicans.

**AI ON EO/IR SATELLITES**

Optical imaging is where AI in space is most mature. On-board neural networks screen out cloudy images before downlink (ESA's Φ-sat-1 first demonstrated this in 2020), detect ships, aircraft and vehicles, and generate alerts within minutes of collection. On the ground, deep learning performs super-resolution, pan-sharpening, cloud removal and automatic change detection over entire countries. Chapters 45 and 46 discuss these applications in detail.

### KEY TAKEAWAYS

- EO/IR satellites range from global imagers (≈ 1 km) through medium-resolution workhorses (10–30 m) to very-high-resolution systems (≈ 0.3 m).
- Landsat (since 1972) and Sentinel-2 (since 2015) provide free, well-calibrated imagery of the whole land surface every few days and are combined in harmonised products.
- Resolution is limited by diffraction ($r \approx 1.22\lambda h/D$) and by the number of photons per pixel; VHR systems need large apertures, long focal lengths and time-delay integration.
- Agile satellites point off nadir to collect many targets, area mosaics and stereo pairs, at the cost of coarser GSD at large angles.
- Constellations of small satellites trade resolution for revisit: about 90 Doves image the land surface almost daily at 3–4 m.
- Thermal infrared sensors measure surface temperature and emissivity, essential for water use, urban heat, geology and fire detection; fire detection exploits the extreme sensitivity of the 3.9 µm band to small hot sources.
- Imaging spectrometers record hundreds of narrow bands, identifying minerals, plant chemistry and methane plumes from individual facilities.
- MODIS is ending after more than two decades; VIIRS and Sentinel-3 continue the global record.

### REVIEW QUESTIONS

1. Calculate the diffraction-limited ground resolution of a telescope with a 1.1 m aperture at 0.55 µm from 500 km altitude.
2. What is time-delay integration, and why do very-high-resolution imagers need it?
3. Why does ground sample distance degrade when an agile satellite images far off nadir?
4. How are stereo pairs collected, and what products are made from them?
5. Explain the trade between resolution and revisit using the PlanetScope constellation as an example.
6. Why is the 3.9 µm band so sensitive to small fires? Use Planck's law in your answer.
7. What is an imaging spectrometer's "hypercube", and how can imaging spectroscopy detect methane from individual facilities?
8. Why are Landsat and Sentinel-2 data often combined, and what must be done to harmonise them?

# 15 CHAPTER Synthetic Aperture Radar

Image: NASA/JPL (NASA/JPL-Caltech)

*Synthetic aperture radar (SAR) satellites carry their own illumination — microwave pulses — and so can image the Earth day and night, through clouds, smoke and rain. By recording the echoes of thousands of pulses as the satellite moves and processing them together, a SAR synthesises an antenna kilometres long and achieves resolutions as fine as 16 cm from hundreds of kilometres away. This chapter explains how SAR works — pulse compression, aperture synthesis, imaging geometry and modes — how radar waves interact with the surface, and why SAR images look so different from photographs. It then covers SAR interferometry, which measures ground movements of millimetres and maps topography, the main applications, and the rapidly growing fleet of government and commercial radar satellites, from Sentinel-1 and NISAR to constellations of 100 kg smallsats.*

IN THIS CHAPTER

15.1 Why Radar?
15.2 Radar Fundamentals
15.3 Frequency and Polarisation
15.4 How the Surface Looks to a Radar
15.5 Imaging Modes
15.6 SAR Interferometry
15.7 Applications
15.8 SAR Missions in 2026
15.9 Design Example: An X-band SAR Smallsat

## 15.1 Why Radar?

Optical and infrared sensors are passive: they need sunlight or thermal emission and they cannot see through clouds, which cover about two-thirds of the Earth at any moment. In the humid tropics, weeks or months can pass without a single clear optical image. A **radar** (radio detection and ranging) sends its own microwave pulses and measures the echoes. Because microwaves

with wavelengths of centimetres to decimetres pass almost unaffected through clouds, rain (except at the shortest wavelengths), smoke, dust and darkness, an imaging radar can observe any place on Earth at any time. This makes SAR indispensable for disaster response, maritime surveillance, sea-ice monitoring, defence and any application that requires guaranteed acquisitions.

Radar also measures things that optical sensors cannot. The echo depends on the **roughness**, **geometry** and **moisture** of the surface rather than its colour, and longer wavelengths **penetrate** vegetation, dry soil and ice. Because the radar records the exact **phase** of the echo, two radar images can be combined to measure distance changes of a few millimetres. This is the basis of **SAR interferometry (InSAR)**.

### 15.1.1 A Short History

Side-looking airborne radars were developed for military reconnaissance in the 1950s, and the synthetic-aperture principle was invented by Carl Wiley at Goodyear in 1951. The first civilian imaging radar in space was carried by NASA's **Seasat** in 1978, which operated for only 106 days but proved the value of SAR for oceanography and land studies (Figure 15.1). The Shuttle Imaging Radar experiments (SIR-A 1981, SIR-B 1984, and the multi-frequency, multi-polarisation **SIR-C/X-SAR** in 1994, Figure 15.2) and the Soviet **Almaz** missions followed. The era of operational SAR began with ESA's **ERS-1** (1991), Japan's **JERS-1** (1992) and Canada's **RADARSAT-1** (1995). In 2000 the **Shuttle Radar Topography Mission (SRTM)** mapped the topography of 80 % of the Earth's land in 11 days. Today, dozens of radar satellites are in orbit (Figure 15.3).

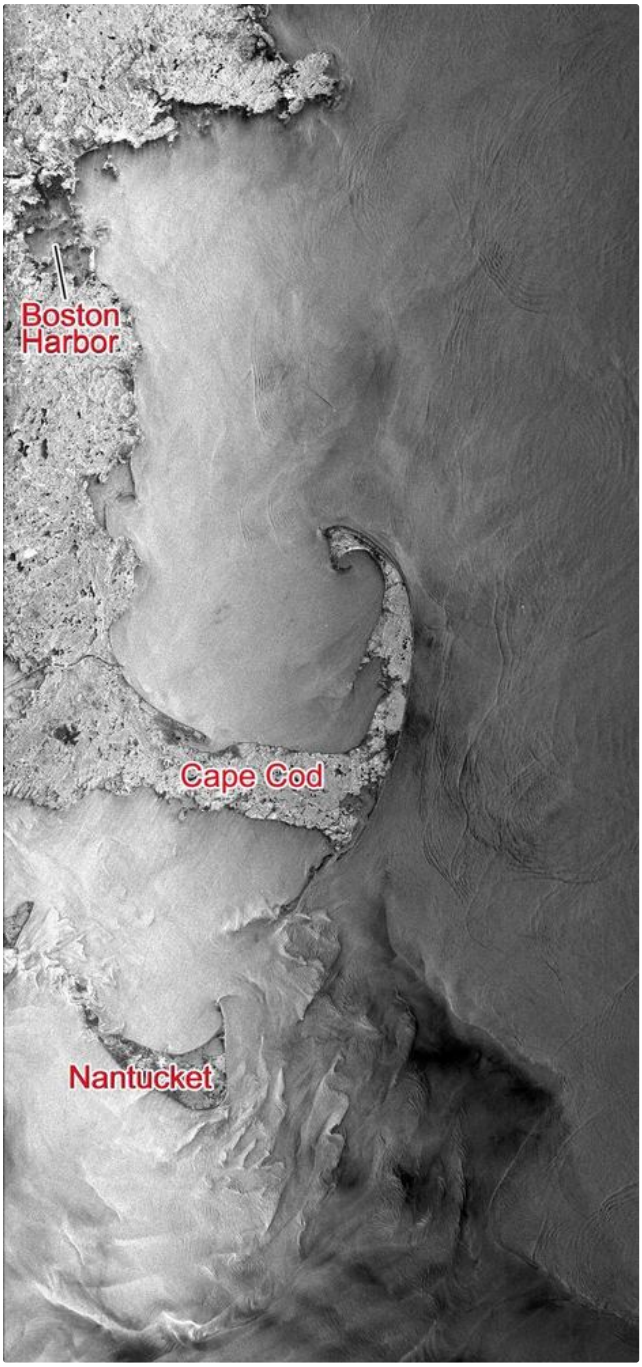



**FIGURE 15.1** Seasat's L-band SAR image of the Massachusetts coast and Cape Cod, acquired in 1978. Seasat was the first civilian satellite to carry an imaging radar. Image: NASA/JPL (NASA/JPL-Caltech/Alaska Satellite Facility).

**FIGURE 15.2** The SIR-C/X-SAR antenna — 12 m long and weighing 11 tonnes — being prepared for its first Space Shuttle flight in 1994. It was the first spaceborne SAR to image in three frequencies (L, C and X) and multiple polarisations. Image: NASA/JSC.

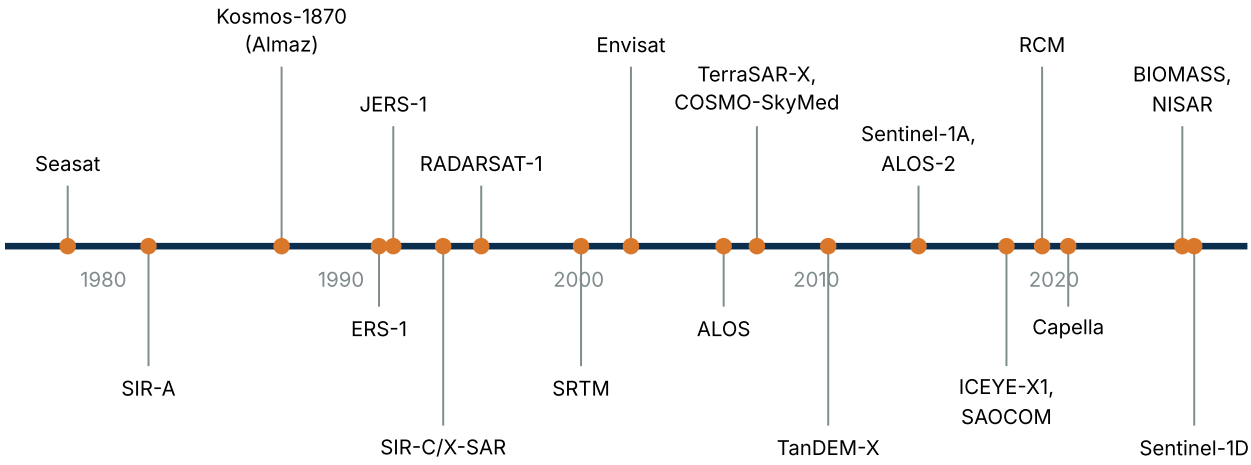



**FIGURE 15.3** Milestones in spaceborne SAR, from Seasat to the 2025 launches of NISAR, BIOMASS and Sentinel-1D.

## 15.2 Radar Fundamentals

### 15.2.1 Ranging and the Radar Equation

A radar transmits a short pulse and measures the delay $\tau$ of its echo; the distance (range) to the target is $R = c\tau/2$. The power of the echo is given by the **radar equation**:

$$P_r = \frac{P_t\, G^2\, \lambda^2\, \sigma}{(4\pi)^3\, R^4\, L}$$

where $P_t$ is the transmitted power, $G$ the antenna gain, $\lambda$ the wavelength, $\sigma$ the **radar cross-section** of the target (m$^2$) and $L$ the system losses. The $R^4$ term reflects that the signal spreads out on the way to the target and again on the way back. It makes spaceborne radar demanding: an echo from 800 km is $10^{12}$ times weaker than from 800 m. For extended natural surfaces, the radar cross-section is expressed per unit area as the dimensionless **backscatter coefficient** $\sigma^0$ (sigma-nought), usually quoted in decibels. It ranges from about −25 dB for calm water to +5 dB or more for urban areas.

### 15.2.2 Imaging Geometry

An imaging radar looks **sideways**, perpendicular to its direction of flight (Figure 15.4). Looking straight down would be useless: points equally distant on both sides of the ground track would return their echoes at the same time and could not be separated. The direction of flight is called **azimuth** (or along-track); the direction of the radar beam is called **range**. The angle between the beam and the vertical at the satellite is the **look angle**. The angle at the ground between the beam and the local vertical is the **incidence angle**, which is slightly larger because of the Earth's curvature. The strip of ground illuminated between **near range** and **far range** is the **swath**.

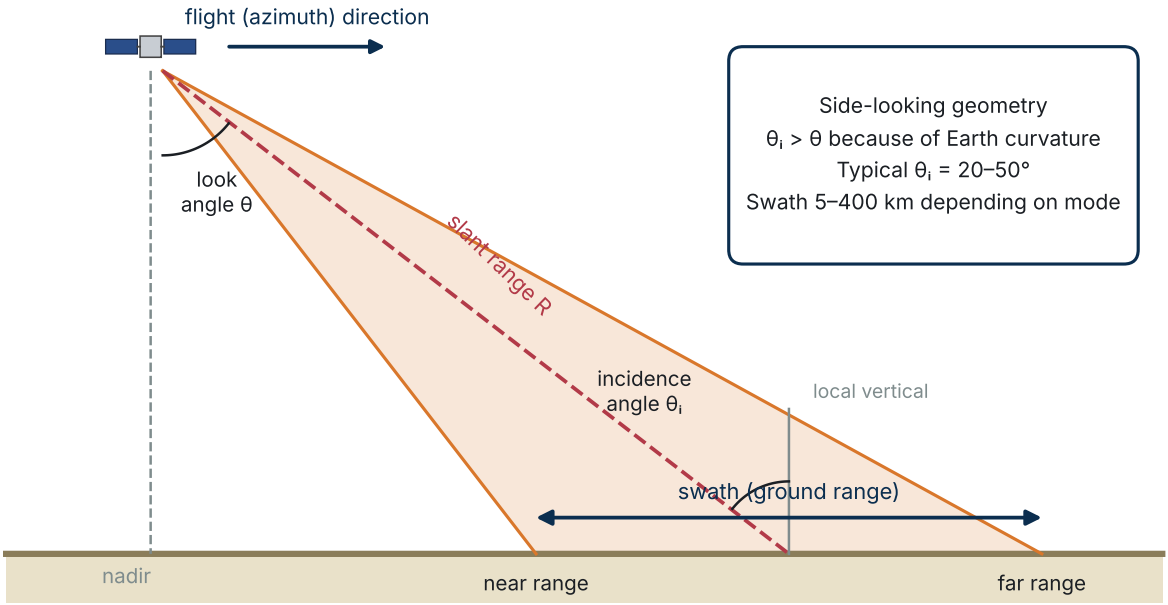



**FIGURE 15.4** Side-looking SAR imaging geometry. The radar measures distance along the slant range; images are later projected onto the ground.

### 15.2.3 Range Resolution and Pulse Compression

Two targets can be separated in range if their echoes do not overlap. For a pulse of duration $\tau_p$, the slant-range resolution is $c\tau_p/2$; a 10 µs pulse would give 1.5 km. To achieve metre-scale resolution with enough energy, radars transmit a long pulse whose frequency changes linearly with time — a **chirp** — and compress it on reception with a **matched filter** (Figure 15.5). The resolution then depends only on the bandwidth $B$ of the chirp:

$$\delta_r = \frac{c}{2B}, \qquad \delta_g = \frac{c}{2B\sin\theta_i}$$

where $\delta_g$ is the ground-range resolution. A bandwidth of 50 MHz gives 3 m slant-range resolution; modern X-band systems use up to 600–1 200 MHz for resolutions of 25 cm or better. Pulse compression is a beautiful example of signal processing replacing raw power. The transmitter can operate at modest peak power for a long pulse, while the receiver recovers the resolution of a pulse hundreds of times shorter.

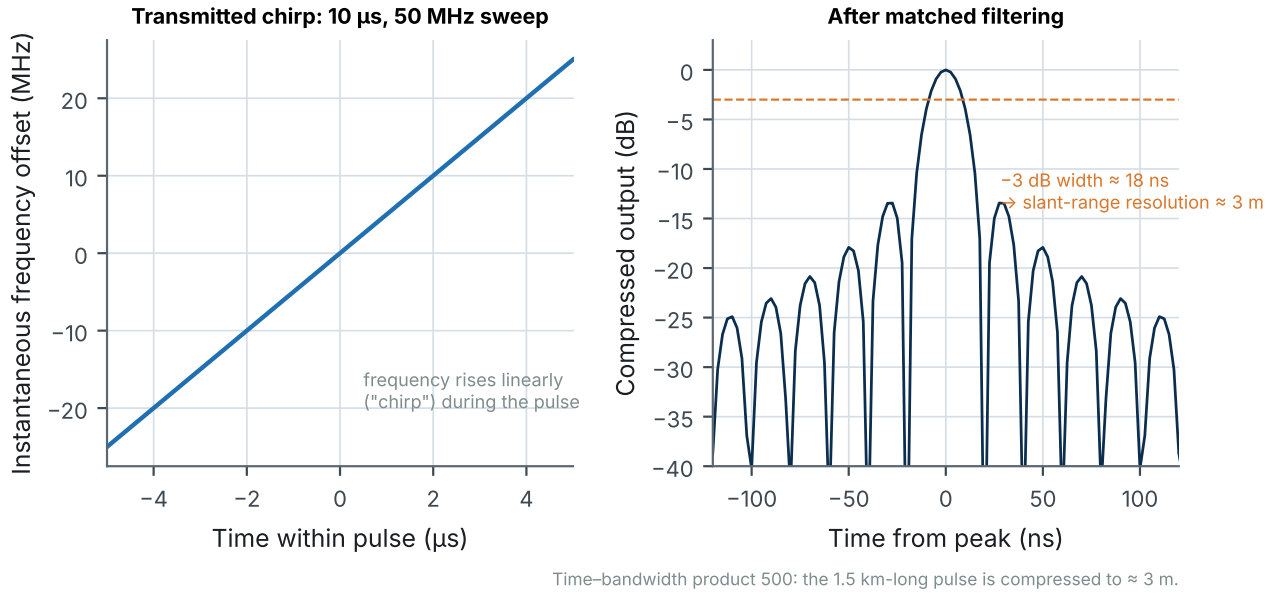



**FIGURE 15.5** Pulse compression. A 10 µs chirp sweeping 50 MHz (left) is compressed by matched filtering into a peak about 18 ns wide (right), giving 3 m slant-range resolution. The sidelobes at −13 dB are usually reduced further by weighting.

### 15.2.4 The Synthetic Aperture

In the azimuth direction, a real antenna of length $L$ forms a beam of angular width about $\lambda/L$. At range $R$ the beam footprint — the real-aperture azimuth resolution — is $\lambda R/L$: for a C-band radar ($\lambda$ = 5.6 cm) with a 10 m antenna at 850 km range, about 4.7 km. To achieve 5 m resolution with a real aperture would require an antenna 10 km long.

The **synthetic aperture** principle solves this (Figure 15.6). As the satellite moves, each point on the ground stays inside the beam for a time during which the satellite travels a distance $L_{sa} = \lambda R/L$ — the length of the **synthetic aperture**. During this time the radar transmits thousands of pulses and records the amplitude and phase of each echo. Because the distance to the target first decreases and then increases, the echo's phase follows a characteristic quadratic history (equivalently, its **Doppler frequency** sweeps from positive to negative). A processor matches this phase history — just as pulse compression matches the chirp in range — and focuses all the echoes from each target into a single sharp point. The resulting azimuth resolution for stripmap imaging is

$$\delta_a \approx \frac{L}{2}$$

a remarkable result: the resolution is **independent of range and wavelength**, and *smaller* antennas give *finer* resolution (because their wider beams illuminate each target for longer). A 10 m antenna gives 5 m resolution whether the satellite is at 500 or 800 km.

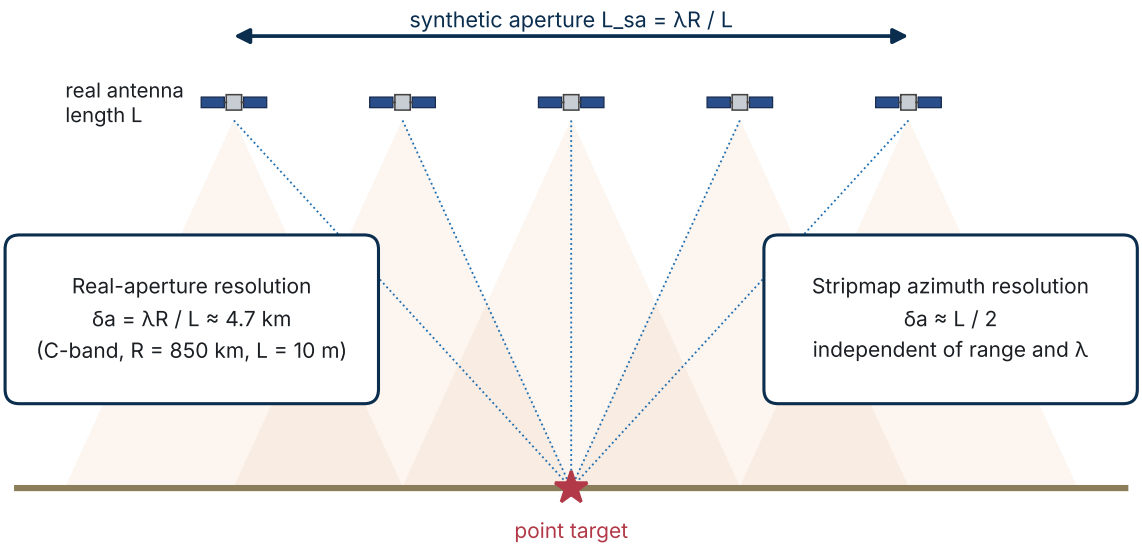



**FIGURE 15.6** Aperture synthesis. Each target is illuminated by many successive pulses as the satellite passes; coherent processing combines them into a virtual antenna kilometres long.

Turning raw echoes into an image — **SAR focusing** — requires the processing of enormous amounts of data with precise knowledge of the satellite's orbit. Classical algorithms include the **range–Doppler**, **chirp scaling** and **omega-k** algorithms, which work efficiently in the frequency domain using fast Fourier transforms. The **back-projection** algorithm is computationally heavier but exact for any geometry, and has become practical with GPUs. Focusing a single Sentinel-1 scene takes minutes on a modern server; some new satellites focus images on board.

### 15.2.5 Constraints: PRF, Ambiguities and Antenna Size

The radar must transmit pulses at a **pulse repetition frequency (PRF)** high enough to sample the Doppler bandwidth of the echoes. This means at least one pulse each time the satellite moves half an antenna length, $\mathrm{PRF} > 2v/L$. But the PRF must be low enough that the echo from the whole swath arrives before the next pulse is sent. Violating these conditions creates **ambiguities**: ghost images displaced in azimuth or range. Together they imply a **minimum antenna area**:

$$A = L\,W > \frac{4\,v\,\lambda\,R\,\tan\theta_i}{c}$$

which grows with wavelength (Figure 15.7). C-band and L-band satellites therefore need antennas of 10–30 m², while X-band smallsats can use antennas of about 1–2 m². Wide swaths, fine resolution and low ambiguities cannot all be achieved simultaneously with a conventional single-beam radar — the central trade-off of SAR design. New techniques overcome it. One is **digital beamforming** with multiple receive channels (as in Germany's planned HRWS mission and Sentinel-1 Next Generation). Another is the **SweepSAR** technique of NASA–ISRO's NISAR, which illuminates a wide swath and uses a large reflector with an array feed to "sweep" a narrow receive beam across it, following the echoes.

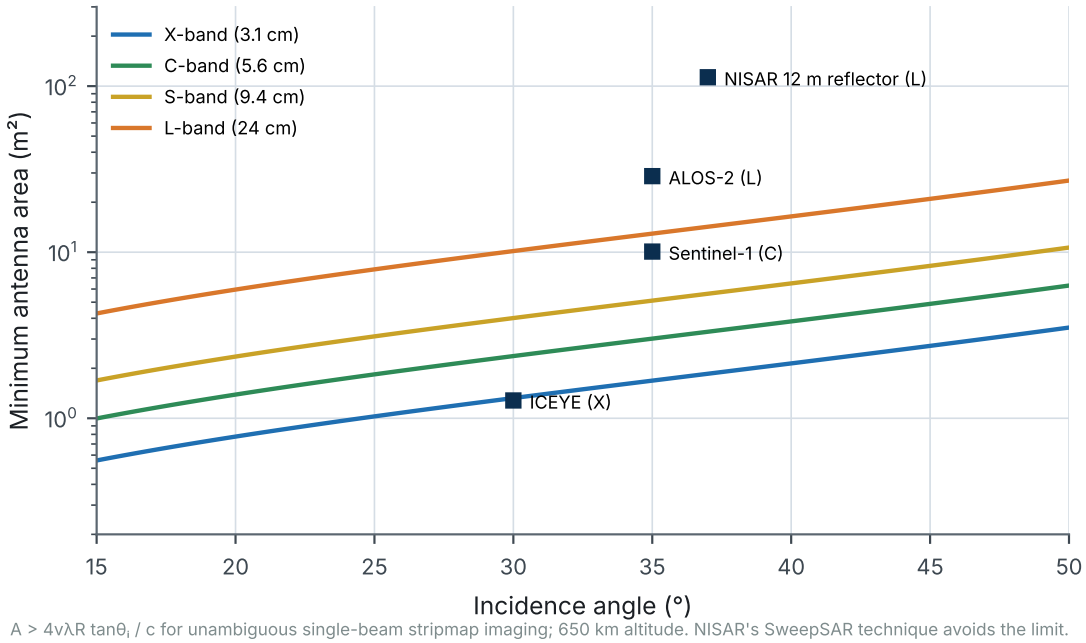



**FIGURE 15.7** Minimum antenna area for unambiguous single-beam imaging versus incidence angle, compared with real systems.

The sensitivity of a SAR is described by its **noise-equivalent sigma-zero (NESZ)** — the backscatter coefficient that gives a signal equal to the noise. Values of −20 to −25 dB are typical of large government SARs; smallsats achieve −15 to −20 dB. Surfaces darker than the NESZ, such as calm water, appear as noise.

## 15.3 Frequency and Polarisation

### 15.3.1 Radar Bands

The choice of wavelength is the most fundamental decision in SAR design (Table 15.1). Shorter wavelengths give finer resolution with small antennas and respond to small-scale roughness and the tops of vegetation canopies. Longer wavelengths penetrate deeper into vegetation, soils and ice and remain coherent over longer periods for interferometry (Figure 15.8).

**TABLE 15.1** Radar frequency bands used for Earth observation (IEEE letter designations).

| Band | Frequency | Wavelength | Main characteristics | Example missions |
|---|---|---|---|---|
| Ka | 26.5–40 GHz | 0.75–1.1 cm | Very fine resolution; affected by rain; altimetry | SWOT (KaRIn interferometer) |
| Ku | 12–18 GHz | 1.7–2.5 cm | Altimeters, scatterometers, precipitation radar | CryoSat-2, Sentinel-3/6 altimeters |
| X | 8–12 GHz | 2.4–3.8 cm | High resolution with small antennas; canopy-top scattering | TerraSAR-X, COSMO-SkyMed, ICEYE, Capella, Umbra |
| C | 4–8 GHz | 3.8–7.5 cm | Good all-round compromise; oceans, ice, land | ERS, Envisat, RADARSAT, Sentinel-1, Gaofen-3 |
| S | 2–4 GHz | 7.5–15 cm | Intermediate penetration | NISAR (S-band), NovaSAR-1, Kondor-FKA |

| Band | Frequency | Wavelength | Main characteristics | Example missions |
|---|---|---|---|---|
| L | 1–2 GHz | 15–30 cm | Penetrates vegetation; long-term coherence for InSAR | ALOS-2/4, SAOCOM, NISAR (L-band) |
| P | 420–450 MHz (space allocation) | ≈ 69 cm | Penetrates forest canopy to trunks; biomass | BIOMASS |

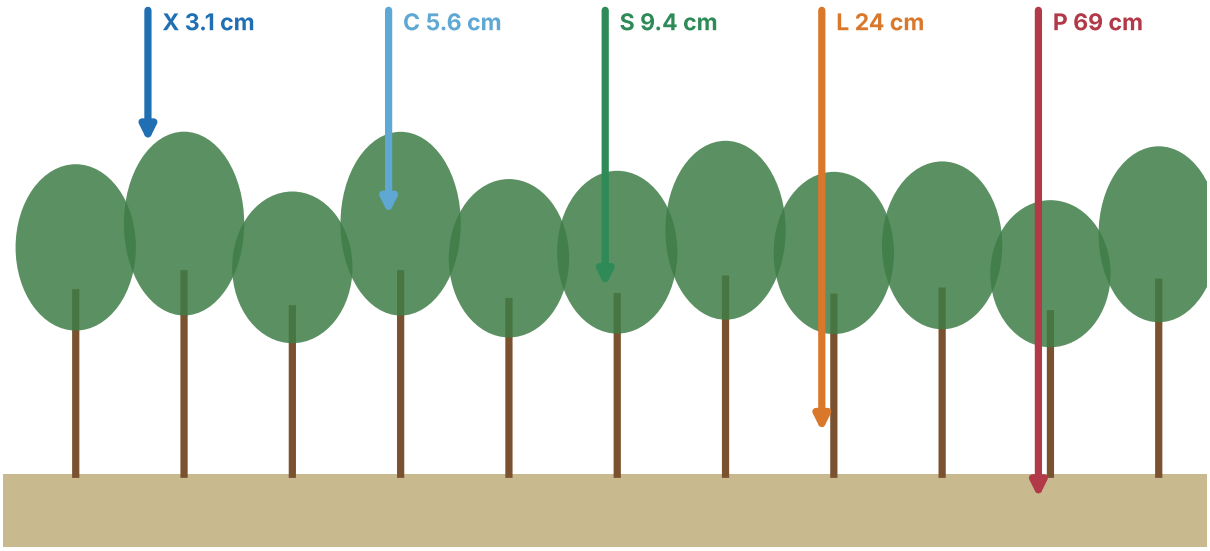



**FIGURE 15.8** Schematic penetration of radar waves of different wavelengths into a forest. X-band scatters mainly from leaves at the top of the canopy, while L- and P-band reach the branches, trunks and ground.

### 15.3.2 Polarisation

Radar waves are polarised: the electric field oscillates in a particular plane. Spaceborne SARs transmit and receive horizontally (H) or vertically (V) polarised waves. An image is described by its transmit and receive polarisations: **HH**, **VV** (co-polarised) and **HV**, **VH** (cross-polarised). Single-polarisation radars record one channel, **dual-pol** radars two (for example VV + VH, Sentinel-1's standard over land), and **quad-pol** (fully polarimetric) radars all four with their relative phases. Some missions use **compact polarimetry**, transmitting circular polarisation and receiving two linear polarisations, as an efficient compromise (RADARSAT Constellation Mission).

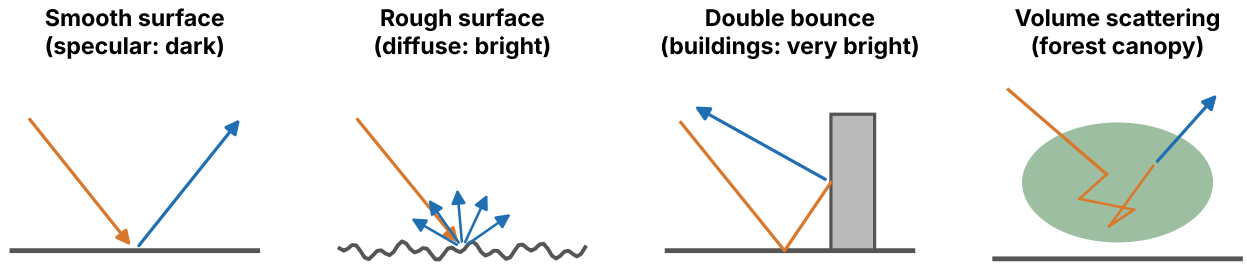
Polarisation reveals the scattering mechanism. Smooth surfaces and the ocean return mostly co-polarised energy; volume scattering by randomly oriented branches and leaves **depolarises** the wave, producing strong cross-polarised (HV) returns — making HV an excellent indicator of vegetation and biomass. **Polarimetric decompositions** (such as the Pauli, Freeman–Durden and Cloude–Pottier entropy/alpha decompositions) separate each pixel's response into surface, double-bounce and volume components, often displayed as red–green–blue colour images.

# 15.4 How the Surface Looks to a Radar

## 15.4.1 Brightness: Roughness, Geometry and Moisture

The brightness of a SAR image depends on how much energy is scattered back towards the radar (Figure 15.9):

- **Smooth surfaces** (calm water, roads, runways, smooth sand) reflect the wave away like a mirror and appear **dark**. A surface is "smooth" if its height variations are small compared with the wavelength (the Rayleigh criterion), so the same surface can be smooth at L-band and rough at X-band.
- **Rough surfaces** (ploughed fields, wind-roughened water, rocky terrain) scatter in all directions and appear **brighter**.
- **Double-bounce** reflection — wave bouncing off the ground and then a vertical wall back to the radar — makes buildings, ships, bridges and flooded forests appear **very bright**. Corner reflectors used for calibration exploit this effect.
- **Volume scattering** within vegetation canopies, dry snow or sea ice produces moderate, depolarised returns.
- **Moisture** increases the dielectric constant of soils and vegetation and therefore their reflectivity, so wet soil is brighter than dry soil. This allows SAR to measure soil moisture and to detect flooding under vegetation.




Orange: incident wave from the radar (upper left); blue: scattered energy.

**FIGURE 15.9** The main scattering mechanisms that determine the brightness of SAR images.

Backscatter also depends strongly on the incidence angle (Figure 15.10). Calm water reflects strongly only near vertical incidence and is black at normal SAR angles. This is why oil slicks, which damp the small capillary waves on the sea, appear as dark patches and can be detected. Forests are relatively insensitive to angle, while soils and the wind-roughened sea darken steadily with increasing incidence angle. The ocean's backscatter depends on wind speed and direction, which is how SAR and scatterometers measure **ocean winds**.

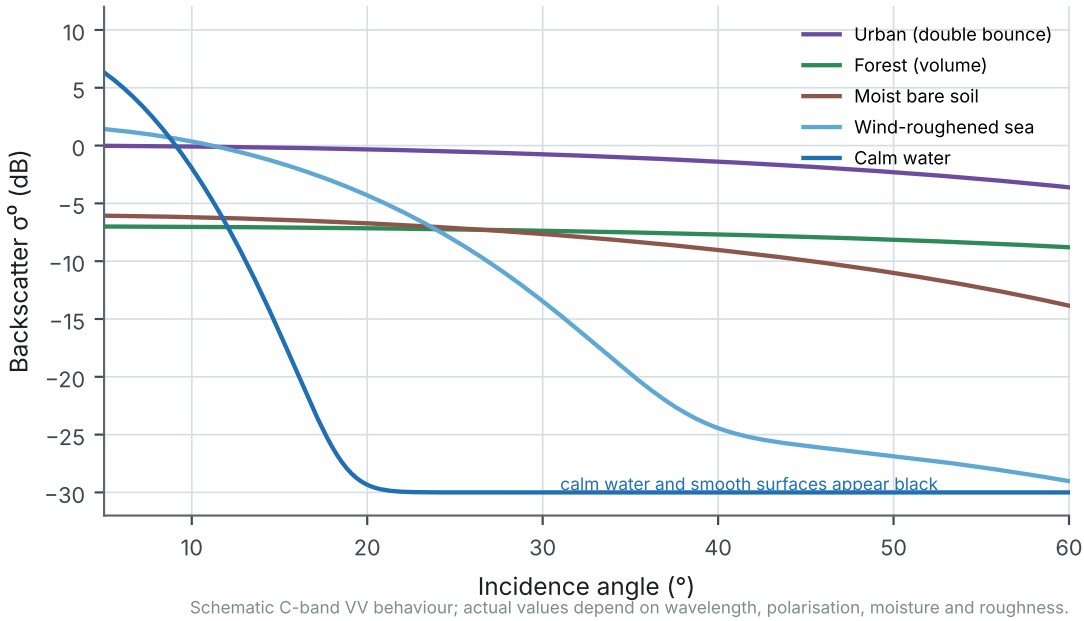



**FIGURE 15.10** Schematic variation of backscatter with incidence angle for different surfaces at C-band.

## 15.4.2 Geometric Distortions

Because a radar measures distances rather than angles, terrain is distorted in characteristic ways (Figure 15.11):

- **Foreshortening**: slopes facing the radar appear compressed and bright.
- **Layover**: when a slope facing the radar is steeper than the incidence angle, its top is closer to the radar than its bottom and is imaged *before* it. Mountains appear to lean towards the radar, and tall buildings lay over onto the street in front of them.
- **Shadow**: slopes facing away from the radar that are steeper than the grazing angle receive no energy and appear black.

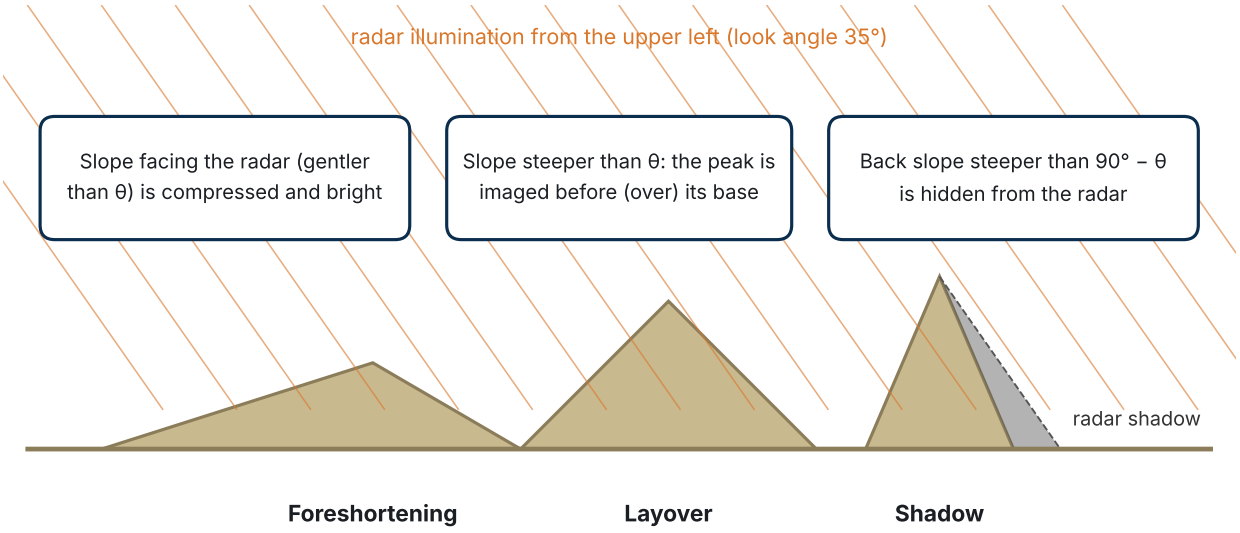



**FIGURE 15.11** Radar geometric distortions in mountainous terrain: foreshortening, layover and shadow.

Distortions are corrected by **terrain-corrected geocoding** using a DEM. Brightness variations due to slope can also be corrected — **radiometric terrain correction** converts backscatter to the "terrain-flattened" coefficient $\gamma^0$ — but layover and shadow areas contain no recoverable

information. Combining images from ascending and descending passes, which look from opposite directions, fills many gaps.

### 15.4.3 Speckle

SAR images have a characteristic grainy appearance called **speckle** (Figure 15.12). Each resolution cell contains many individual scatterers whose echoes add coherently with random phases; the sum varies randomly from pixel to pixel even over a uniform surface. Speckle is not noise in the usual sense. It is a real, deterministic consequence of coherent imaging that repeats if the same scene is imaged under identical conditions. But it makes single-look images hard to interpret. Speckle is reduced by **multi-looking** (averaging several independent looks, at the cost of resolution), by averaging images over time, and by adaptive filters (Lee, Frost, non-local means, and deep-learning despeckling).

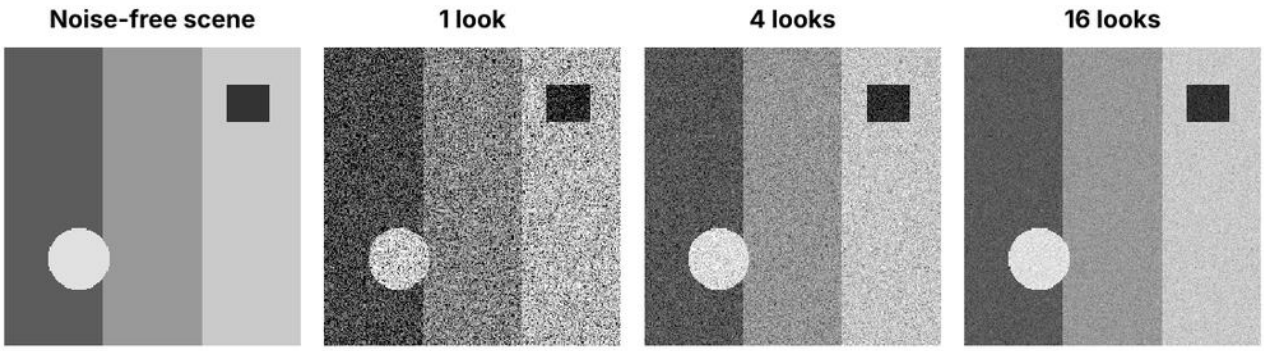



**FIGURE 15.12** Speckle in a simulated SAR scene. Averaging 4 or 16 independent looks reduces the speckle but also reduces resolution.

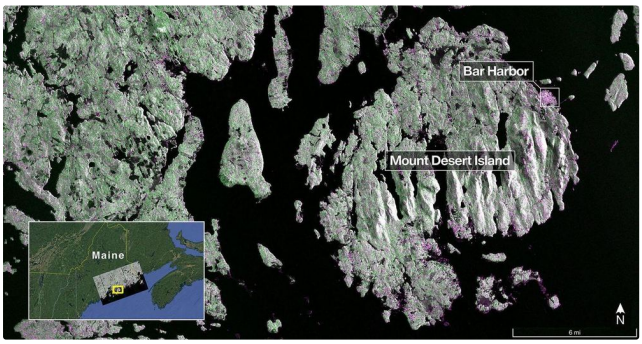



**FIGURE 15.13** The first L-band image from NISAR, acquired on 21 August 2025, showing Mount Desert Island on the coast of Maine. Forests appear bright, water dark and urban areas very bright. Image: NASA/JPL (NASA/JPL-Caltech).

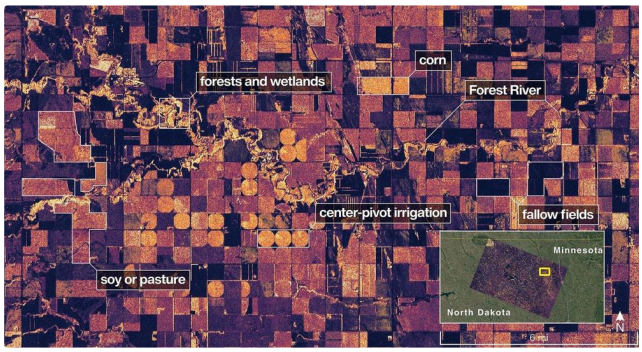
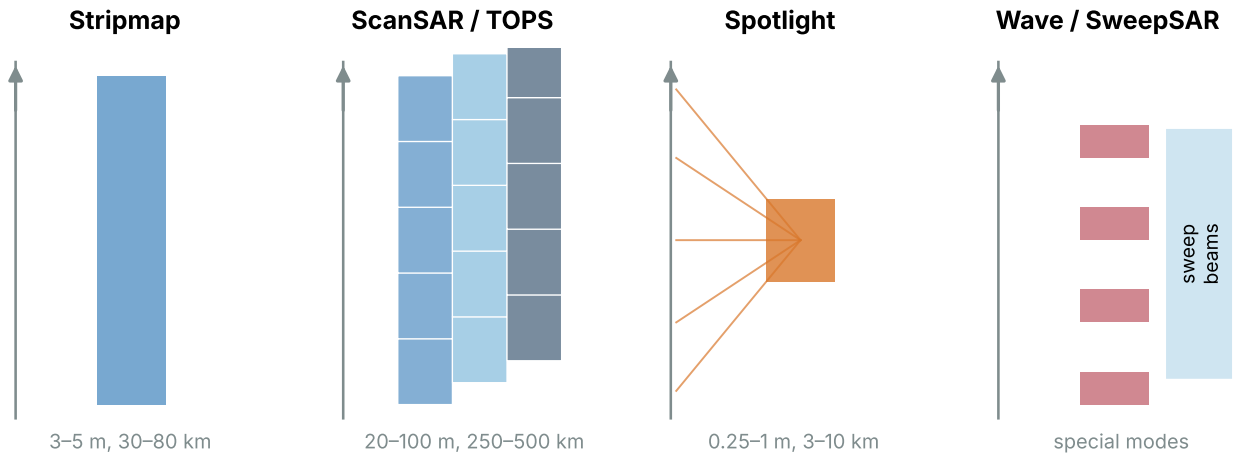



**FIGURE 15.14** An early NISAR image of farmland and wetlands in North Dakota. Fields with different crops, moisture and roughness show distinct brightness; centre-pivot fields appear as circles. Image: NASA/JPL (NASA/JPL-Caltech).

## 15.5 Imaging Modes

By steering its beam electronically (with a phased-array antenna) or mechanically (by rotating the spacecraft), a SAR can trade resolution against swath width in several **imaging modes** (Figure 15.15):

- **Stripmap**: the beam points at a fixed angle and sweeps a continuous strip; resolution about half the antenna length (typically 3–5 m), swath 30–80 km.
- **ScanSAR**: the beam switches periodically between several adjacent sub-swaths, illuminating each in short bursts. The swath becomes several times wider (250–500 km) at the cost of coarser azimuth resolution. ESA's **TOPS** (Terrain Observation by Progressive Scans) mode, used by Sentinel-1's Interferometric Wide-swath (IW) mode, also steers the beam forward-to-backward during each burst, giving uniform image quality across the swath — important for interferometry.
- **Spotlight**: the beam is steered backwards during the pass to keep illuminating the same small area for longer, synthesising a longer aperture and achieving resolutions of 0.25–1 m over a few kilometres. **Staring spotlight** and **dwell** modes extend this further.
- **Wave mode** collects small vignettes over the open ocean to measure wave spectra, and specialised modes such as NISAR's **SweepSAR** combine wide swaths with fine resolution.




Grey arrow: satellite ground track and direction of flight. The radar looks to one side.

**FIGURE 15.15** Schematic SAR imaging modes. Grey arrows show the ground track; the radar looks to the right.

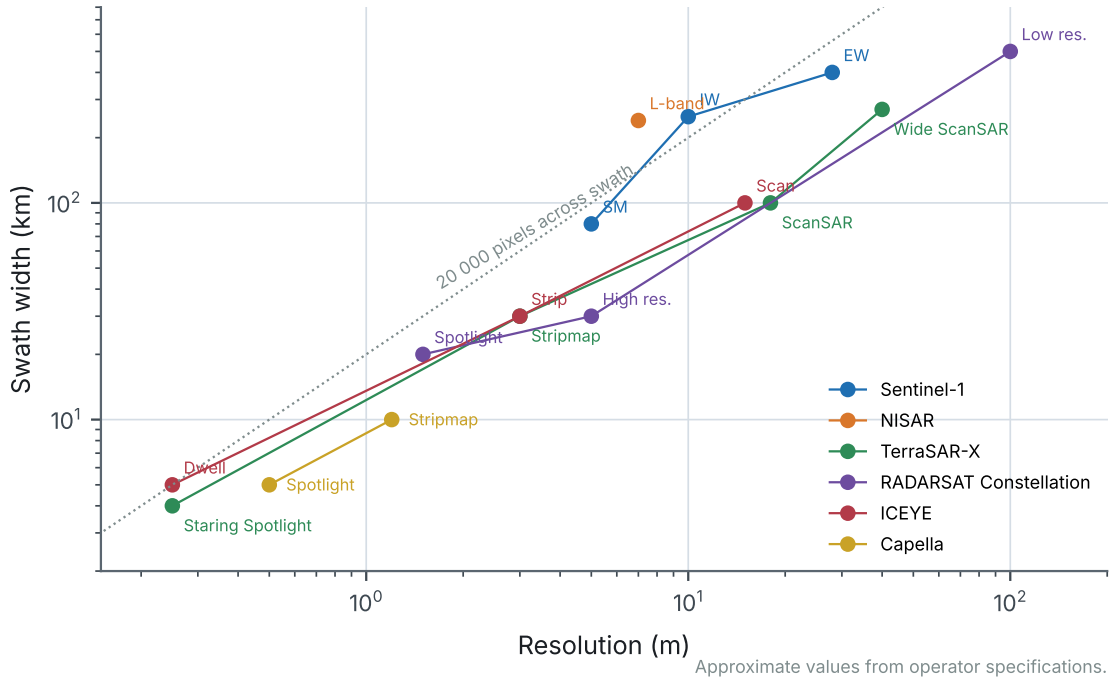



**FIGURE 15.16** Resolution versus swath width for the imaging modes of representative SAR missions. Each system offers a family of modes along a line of roughly constant number of pixels across the swath.

## 15.6 SAR Interferometry

### 15.6.1 The Principle

A SAR image is a matrix of complex numbers. Each pixel has an **amplitude** (brightness) and a **phase**. The phase depends on the exact two-way distance between the antenna and the scatterers in the pixel, measured in fractions of a wavelength. The phase of a single image looks random, because it includes the unknown arrangement of scatterers within each pixel. But if two images are acquired from nearly the same position, this random part is almost identical in both. The **phase difference** between them — the **interferogram** — then depends only on the difference in distance (Figure 15.17):

$$\Delta\phi = \frac{4\pi}{\lambda}\,\Delta R$$

One complete cycle of phase (a **fringe**) corresponds to a change in distance of half a wavelength — 2.8 cm at C-band and 12 cm at L-band. Since phase can be measured to a small fraction of a cycle, InSAR measures distance changes of **millimetres** from hundreds of kilometres away.

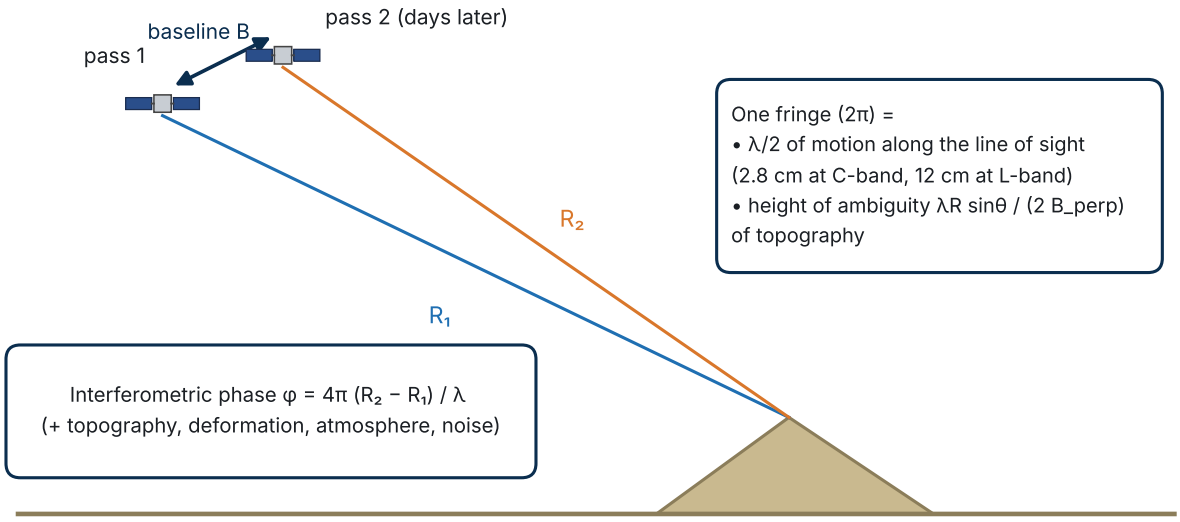



**FIGURE 15.17** Geometry of SAR interferometry. The phase difference between two acquisitions depends on topography (through the baseline geometry) and on any movement of the ground between them.

The interferometric phase has several contributions. One is the **topography** (because the two antennas view the terrain from slightly different positions separated by the **baseline**). Others are any **deformation** of the ground along the line of sight between acquisitions, differences in **atmospheric delay** (mainly water vapour in the troposphere and electrons in the ionosphere), orbit errors, and noise. Different applications isolate different components.

### 15.6.2 Measuring Topography

The two images can also be acquired at the same time from two antennas separated by a known baseline — **single-pass** or **bistatic** interferometry. Then there is no deformation or atmospheric change, and the phase directly measures height. The **height of ambiguity**, the height change that produces one fringe, is $h_a = \lambda R \sin\theta / (2B_\perp)$, where $B_\perp$ is the perpendicular baseline; longer baselines give greater height sensitivity.

The **Shuttle Radar Topography Mission** (February 2000) used a C-band antenna in the Shuttle's payload bay and a second receive antenna on the end of a 60 m mast. It mapped 80 % of the Earth's land surface between 60° N and 56° S, producing the first near-global high-resolution DEM (Figure 15.18). Germany's **TanDEM-X** mission (2010) flew a second satellite in close formation, a few hundred metres from TerraSAR-X. It produced the global **WorldDEM** / Copernicus DEM with 12 m posting and a vertical accuracy of a few metres — the most accurate consistent global elevation model.

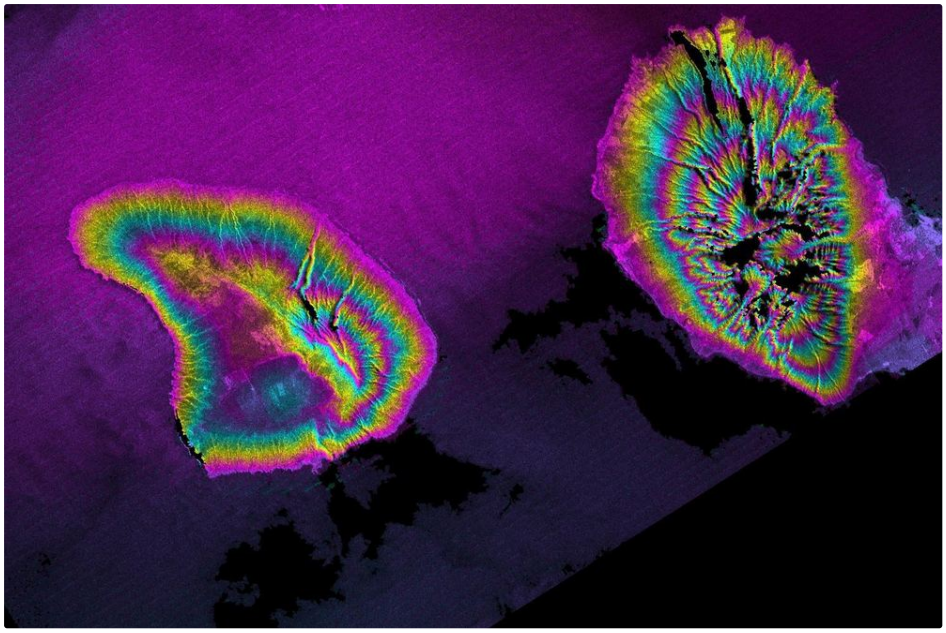


**FIGURE 15.18** SRTM interferometric height of Lanai and West Maui, Hawaii. Each cycle of colours represents a fixed increment of elevation. Image: NASA/JPL (NASA/JPL).

### 15.6.3 Measuring Deformation

**Differential InSAR (DInSAR)** uses two images taken days to years apart by the same satellite from nearly the same orbit. After the topographic phase is removed using a DEM, the remaining fringes show how the ground moved between the acquisitions. The first spectacular demonstration was the 1992 Landers, California, earthquake, imaged by ERS-1 (published in *Nature* in 1993). Since then InSAR has mapped the deformation of almost every major earthquake (Figure 15.19), volcanic inflation and deflation (Figure 15.20), landslides, and glacier and ice-sheet flow. It has also mapped subsidence caused by groundwater pumping, mining, oil and gas extraction and tunnelling.

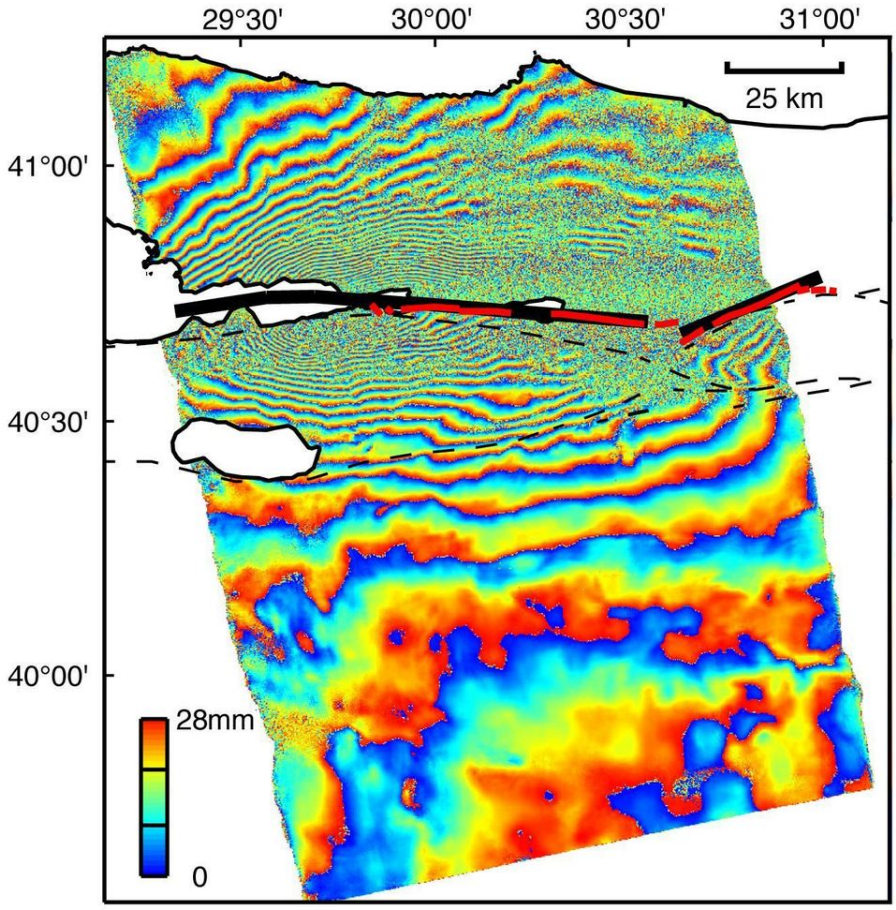



**FIGURE 15.19** Wrapped interferogram from ERS-1 and ERS-2 data showing ground displacement from the magnitude 7.6 Izmit, Turkey, earthquake of 17 August 1999. Each colour cycle represents 28 mm of motion towards or away from the satellite. Image: NASA/JPL (NASA/JPL).

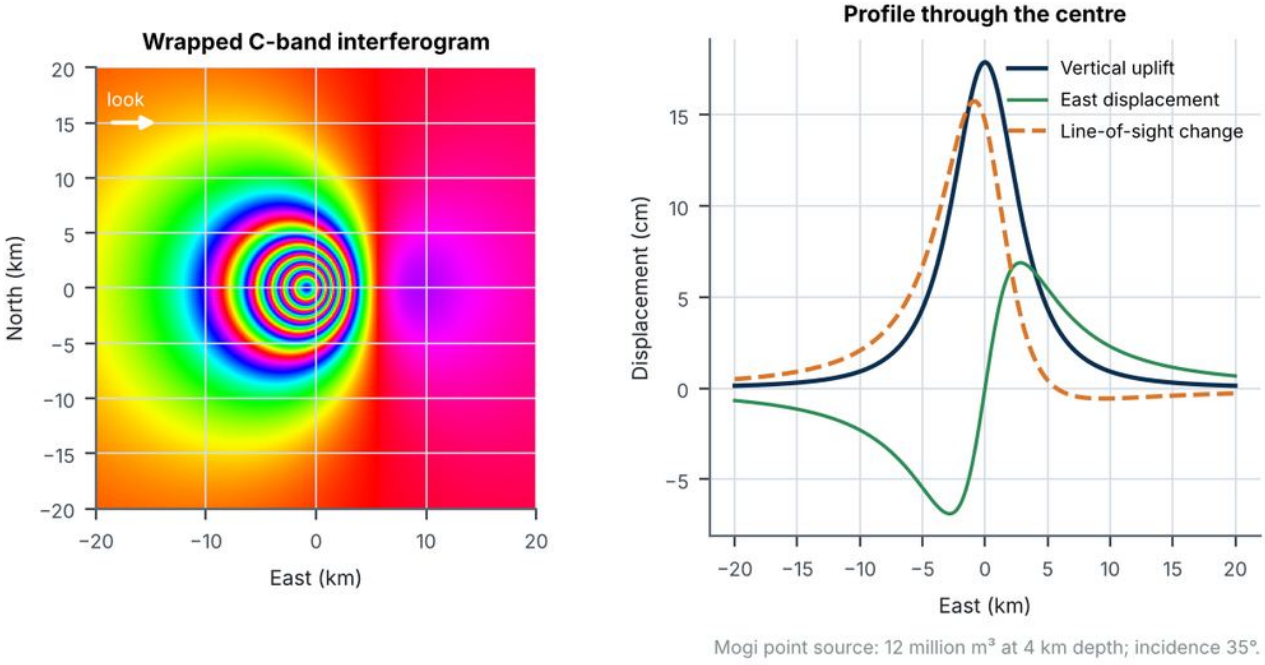



**FIGURE 15.20** Synthetic interferogram (left) and displacement profile (right) for a volcano inflating because of 12 million m³ of magma injected at 4 km depth (Mogi model). The radar measures only the component of motion along its line of sight, so the fringe pattern is asymmetric.

### 15.6.4 Coherence, Time Series and Accuracy

InSAR works only if the scatterers within each pixel remain arranged in the same way between acquisitions — that is, if the images remain **coherent**. **Coherence** (0 to 1) measures this. It is high over cities and bare rock, but decays over vegetation as leaves move and plants grow, and is destroyed by snowfall, ploughing or flooding (Figure 15.21). Longer wavelengths stay coherent longer, which is why L-band missions such as ALOS-2 and NISAR are preferred over vegetated land, and why short revisit intervals (6–12 days) are valuable. Loss of coherence is itself useful: **coherence change detection** reveals damaged buildings after earthquakes and disturbed ground.

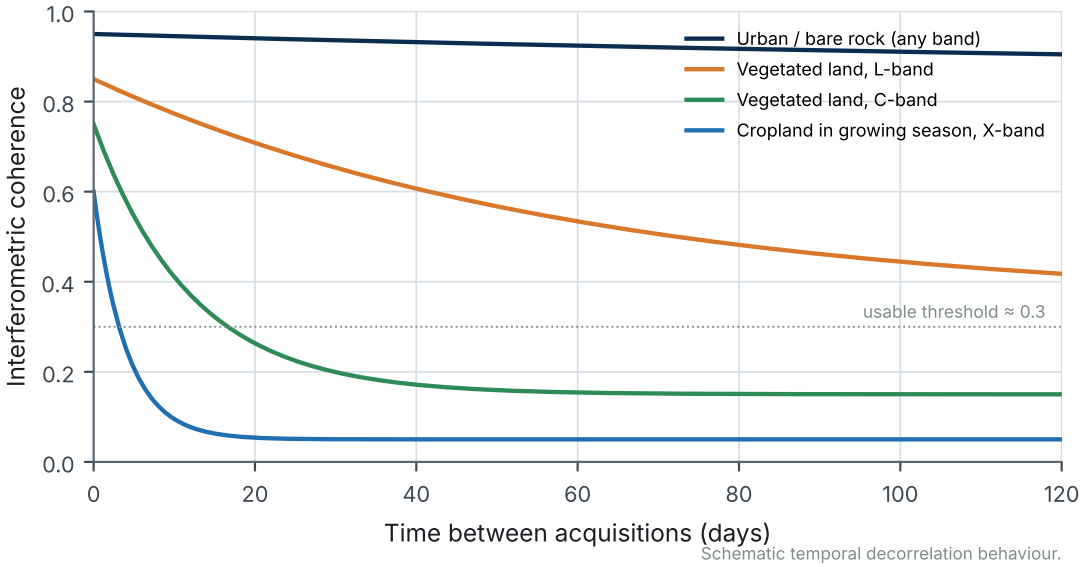



**FIGURE 15.21** Schematic decay of interferometric coherence with time between acquisitions for different surfaces and wavelengths.

The biggest error source for deformation measurements is the **atmosphere**: variations in water vapour delay the radar signal by amounts equivalent to several centimetres. **Multi-temporal InSAR** techniques overcome this by analysing stacks of dozens to hundreds of images. One method, **persistent scatterer interferometry (PSI)**, identifies stable point-like targets (buildings, rocks, infrastructure) and estimates their motion history with millimetre-per-year precision. By contrast, **small baseline subset (SBAS)** methods analyse distributed targets. The **European Ground Motion Service (EGMS)**, launched by Copernicus in 2022, provides Sentinel-1 PSI ground-motion measurements for almost all of Europe, updated annually. Its billions of measurement points show subsidence of cities, motion of landslides and the stability of dams, bridges and railways.

Other interferometric techniques include **along-track interferometry**, which uses two antennas separated in the flight direction. It measures the velocity of moving targets such as ocean currents and vehicles (ground moving target indication, Chapter 19). Another is **SAR tomography**, which combines many baselines to resolve the vertical structure of forests and buildings — a key technique of ESA's BIOMASS mission. Where coherence is low, **offset tracking** measures large displacements (metres) by correlating image patches, for example to map glacier velocities.

## 15.7 Applications

SAR has become essential in many fields (Table 15.2).

**TABLE 15.2** Major applications of spaceborne SAR.

| Field | Applications | Preferred band and technique |
|---|---|---|
| Disasters | Flood mapping (water appears dark), earthquake and landslide deformation, building damage, volcanic monitoring | All bands; amplitude change, coherence change, DInSAR |
| Oceans | Ship detection, oil-spill detection, ocean wind and waves, currents, internal waves | C and X; wave mode; along-track interferometry |
| Sea ice and polar regions | Ice type and concentration charts, iceberg tracking, ice-sheet velocity and grounding lines | C and L; ScanSAR; offset tracking |
| Geohazards and infrastructure | Subsidence of cities, mines and oilfields; stability of dams, bridges, railways, pipelines | C, L, X; PSI, SBAS time series |
| Forestry and carbon | Deforestation alerts in cloudy tropics, forest biomass and height | L, P; cross-polarisation, tomography |
| Agriculture | Crop type mapping, crop growth, soil moisture, rice paddies (flooded fields) | C, L; time series of polarimetric backscatter |
| Hydrology | Soil moisture, wetlands, snow water equivalent, river and lake levels | L, C; Ka-band interferometry (SWOT) |
| Defence and security | All-weather imaging, change detection, moving targets, maritime surveillance | X (high resolution), all modes |

**EXAMPLE: MAPPING A FLOOD WITH SENTINEL-1**

A major river floods during the monsoon, under persistent cloud.

1. Select a Sentinel-1 IW image acquired during the flood and a reference image from the same orbit before the flood.
2. Calibrate both to $\sigma^0$, apply terrain correction with the Copernicus DEM, and reduce speckle with a multi-temporal filter.
3. Open floodwater appears very dark (below about −18 dB in VV). Pixels that were bright before and dark now are classified as newly flooded; a threshold chosen automatically from the image histogram (Otsu's method) or a trained classifier separates water from land.
4. Flooded vegetation and flooded urban areas can appear *brighter* because of double bounce between water and trunks or walls; these are detected as increases in backscatter or drops in coherence.
5. Remove areas of radar shadow and permanent water using a height-above-nearest-drainage map and a reference water mask.

Free Sentinel-1 data are typically available within a few hours of acquisition. With these data and services such as the Copernicus Emergency Management Service and the Global Flood Monitoring system, flood maps can be produced the same day — regardless of clouds.

# 15.8 SAR Missions in 2026

## 15.8.1 Government Missions

**TABLE 15.3** Major government SAR missions (selected).

| Mission | Agency / country | Band | Launch | Key features |
|---|---|---|---|---|
| Sentinel-1A, 1C, 1D | EU Copernicus / ESA | C | 2014, 2024, 2025 | 693 km SSO, 12-day repeat per satellite; IW mode 250 km at 5 × 20 m; systematic global acquisitions, free data |
| RADARSAT Constellation Mission | Canada (CSA) | C | 2019 | Three satellites; 4-day repeat; compact polarimetry; maritime surveillance |
| TerraSAR-X / TanDEM-X / PAZ | Germany (DLR, Airbus) / Spain | X | 2007, 2010, 2018 | High resolution; bistatic interferometry (WorldDEM) |
| COSMO-SkyMed (1st and 2nd gen.) | Italy (ASI, Defence) | X | 2007– | Dual-use constellation; spotlight to 0.35 m class |
| ALOS-2 / ALOS-4 | Japan (JAXA) | L | 2014, 2024 | PALSAR-3 on ALOS-4 images a 200 km swath at 3 m |
| SAOCOM 1A/1B | Argentina (CONAE) | L | 2018, 2020 | Full polarimetry; soil moisture; cooperation with Italy |
| Gaofen-3 series | China | C | 2016– | Multi-mode, up to 1 m |
| EOS-04 (RISAT-1A) | India (ISRO) | C | 2022 | Agriculture, flood, forestry |
| Kondor-FKA | Russia | S | 2023, 2024 | Up to 1 m class |
| NISAR | NASA / ISRO | L + S | 2025 | 12 m reflector, SweepSAR, 242 km swath, 12-day repeat; global systematic coverage, free data |
| BIOMASS | ESA | P | 2025 | First spaceborne P-band SAR; forest biomass and height by tomography |
| ROSE-L | EU Copernicus / ESA | L | ≈ 2029 | Copernicus Expansion mission |
| Harmony | ESA | C (receive-only) | ≈ 2029 | Two satellites flying with a Sentinel-1 for multi-static measurements |

Two missions launched in 2025 stand out. **NISAR** is a partnership between NASA (L-band radar, 12 m deployable mesh reflector, data systems) and ISRO (S-band radar, spacecraft bus, GSLV launch). It is the first radar mission to map nearly all of the Earth's land and ice surfaces twice every 12 days at two frequencies. Launched on 30 July 2025, it began its science phase in November 2025; its free data — tens of terabytes per day — will measure ground deformation, ice-

sheet dynamics, forest biomass and agriculture globally. ESA's **BIOMASS**, launched on 29 April 2025, carries the first spaceborne P-band SAR. At 70 cm wavelength the radar penetrates to tree trunks, allowing the mass of forests — and hence their carbon content — to be estimated. Because P-band frequencies are shared with powerful military space-surveillance radars, BIOMASS is not permitted to operate over Europe and North America.

### 15.8.2 The Commercial SAR Revolution

Until 2018, all SAR satellites were large government systems weighing 1–3 tonnes. In January 2018 the Finnish start-up **ICEYE** launched ICEYE-X1, the first SAR satellite under 100 kg. Its innovations were a compact X-band phased array, high-power gallium-nitride amplifiers, batteries that supply short bursts of several kilowatts, and agile spacecraft that rotate to perform spotlight imaging. They showed that useful SAR imaging was possible from smallsats costing a few million dollars. By July 2026 ICEYE had launched 76 satellites, offered resolutions down to 16 cm with its fourth-generation design, and was producing about 50 satellites per year. Other commercial operators include **Capella Space** (USA, acquired in July 2025 by the quantum-computing company IonQ), **Umbra** (USA), **Synspective** (StriX, Japan), **iQPS** (QPS-SAR, Japan) and a growing number of Chinese commercial SAR companies (Figure 15.22).

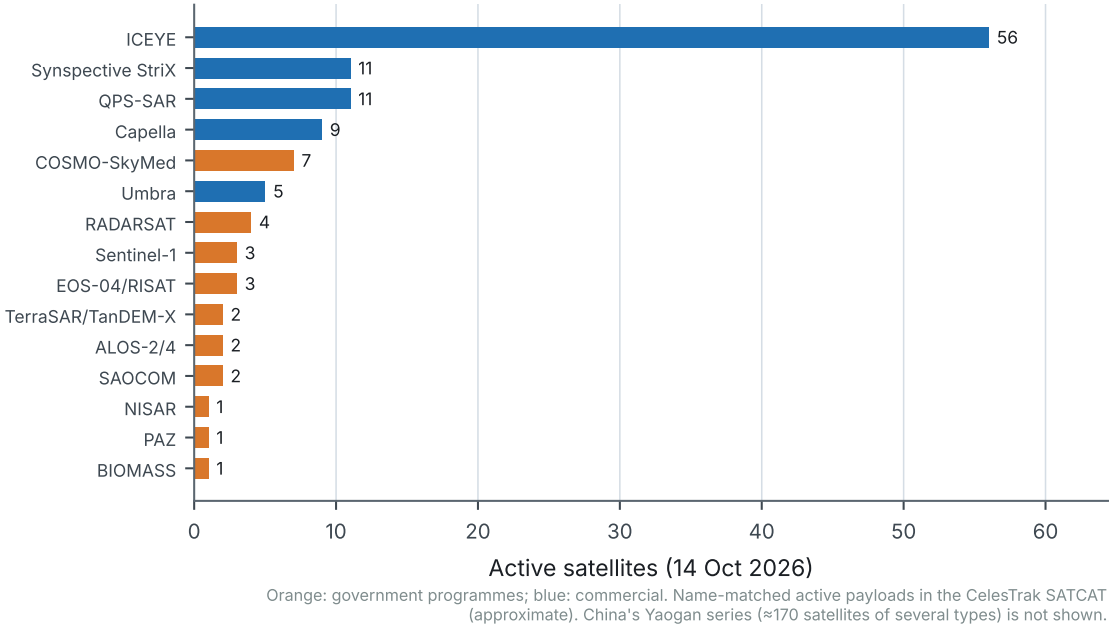



**FIGURE 15.22** Active SAR satellites by operator or series in October 2026 (name-matched in the CelesTrak SATCAT, approximate).

Commercial SAR constellations offer frequent revisit (several images per day of a target), fast tasking and delivery, and very high resolution. They have found strong markets in defence and intelligence (the war in Ukraine was a turning point), insurance (rapid flood extent after events), maritime monitoring and infrastructure. Government missions provide the systematic, well-calibrated, free global coverage needed for science and public services; the two are increasingly complementary.

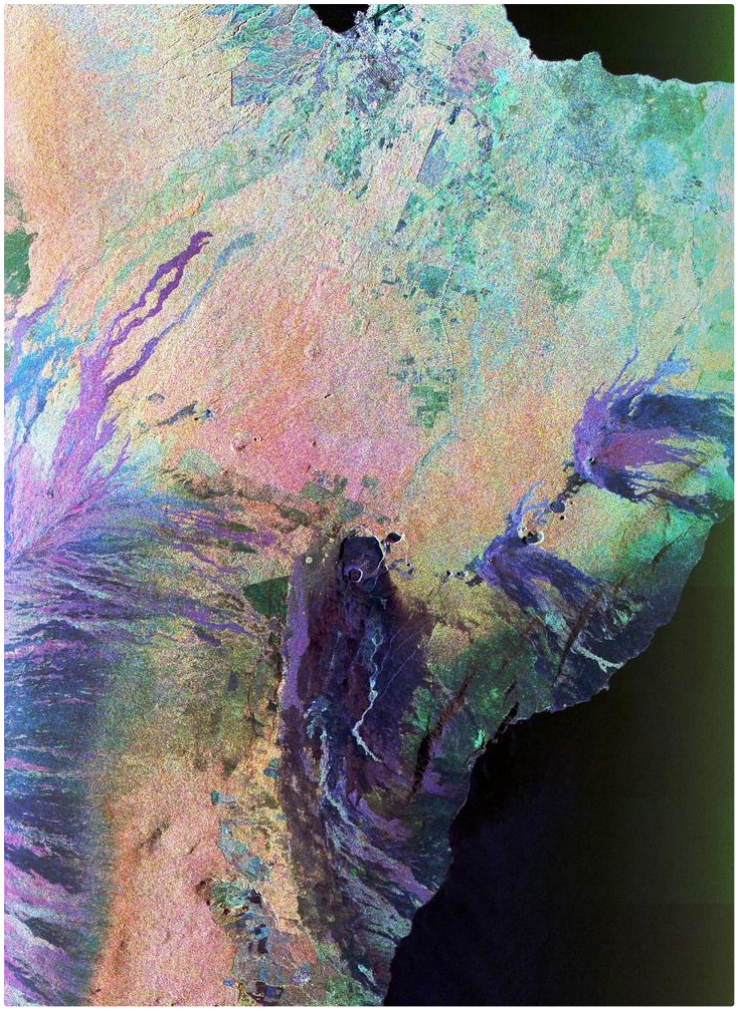


**FIGURE 15.23** SIR-C/X-SAR multi-frequency image of the Kilauea volcano, Hawaii, in 1994. Colours combine L-band and C-band channels in different polarisations: rough young lava flows appear bright, smooth older surfaces darker. Image: NASA/JPL (NASA/JPL/NGA).

## 15.9 Design Example: An X-band SAR Smallsat

**EXAMPLE: SIZING A 1 M-CLASS SAR MICROSATELLITE**

*Requirements:* ≈ 1 m stripmap resolution, NESZ better than −15 dB, 550 km orbit, 30° incidence.

1. **Wavelength.** X-band (9.65 GHz, λ = 3.1 cm) allows a small antenna.
2. **Antenna.** A 3.2 m × 0.4 m phased array (1.28 $m^2$) exceeds the minimum area of about 1.2 $m^2$ for these conditions (Figure 15.7). Stripmap azimuth resolution ≈ L/2 = 1.6 m; spotlight mode can do better.
3. **Bandwidth.** 300 MHz gives 0.5 m slant-range and 1.0 m ground-range resolution at 30° incidence.
4. **Gain.** With 60 % aperture efficiency, $G = 4\pi A\eta/\lambda^2 \approx 10\,000$ (40 dBi).
5. **Sensitivity.** Assume 800 W average radiated power (4 kW peak at 20 % duty cycle), slant range 650 km, 3 dB noise figure and 3 dB losses. The radar equation for distributed targets then gives NESZ ≈ −15.6 dB — just meeting the requirement.
6. **Power.** 4 kW peak cannot come from a smallsat's solar arrays (a few hundred watts); a high-rate battery supplies imaging bursts of one to two minutes per orbit, recharged between collections.
7. **Data.** Sampling 300 MHz of bandwidth with 8-bit complex samples (reduced to about 4 bits by block-adaptive quantisation) produces a raw data rate of roughly 2–3 Gbit/s during imaging. A 20-second collection produces about 6 GB, downlinked over X-band or Ka-band.

The result — a satellite of about 100 kg with a deployable antenna — is close to the design of the first commercial SAR microsatellites.

**AI AND SAR**

SAR images are difficult for humans to interpret but well suited to machine learning. Machine learning is now used for ship and aircraft detection, sea-ice classification, despeckling, flood and building-damage mapping, and translating SAR into optical-like images. Deep learning also helps unwrap interferometric phase, separate atmospheric noise from deformation in InSAR time series, and detect anomalies in millions of persistent scatterers — for example, early warning of landslides or structural failures. Chapter 46 discusses these methods.

### KEY TAKEAWAYS

- SAR provides its own microwave illumination and images day and night through clouds, smoke and rain.
- Range resolution comes from pulse bandwidth ($c/2B$) through pulse compression; azimuth resolution comes from aperture synthesis and is about half the antenna length, independent of range.
- The PRF constraints imply a minimum antenna area; resolution and swath trade off in stripmap, ScanSAR/TOPS and spotlight modes, and digital beamforming and SweepSAR relax the trade.
- Wavelength determines penetration and interferometric coherence: X-band for high resolution, C-band for all-round use, L- and P-band for forests and long-term interferometry.
- SAR brightness depends on roughness, geometry and moisture; images contain speckle and geometric distortions (foreshortening, layover, shadow).
- Interferometry measures topography (SRTM, TanDEM-X) and ground deformation with millimetre precision; persistent-scatterer time series monitor cities and infrastructure across whole continents.
- The SAR fleet now combines systematic government missions (Sentinel-1, NISAR, BIOMASS, ALOS-4) with fast-growing commercial constellations of 100 kg smallsats.

### REVIEW QUESTIONS

1. Why can a SAR image through clouds, smoke and darkness while an optical imager cannot?
2. A SAR transmits a chirp of 300 MHz bandwidth. What is its slant-range resolution, and what is its ground-range resolution at an incidence angle of 35°?
3. A stripmap SAR has a 3.2 m long antenna. What is its approximate azimuth resolution, and why does it not depend on range?
4. Explain how the pulse repetition frequency constrains the minimum antenna area of a SAR.
5. Compare X-, C- and L-band SAR for forest mapping, sea-ice monitoring and high-resolution imaging of ports.
6. Describe foreshortening, layover and shadow, and explain what causes speckle.
7. In C-band differential interferometry ($\lambda = 5.55$ cm), how much line-of-sight ground motion does one full interferometric fringe represent?
8. How have commercial SAR constellations of 100 kg satellites changed what SAR can deliver, compared with large government missions?

CHAPTER

# 16 Weather and Environmental Satellites

Image: NASA/JSC

*Every weather forecast you see depends on satellites. Geostationary satellites watch storms develop minute by minute over a third of the planet, while polar-orbiting satellites carry sounders that measure the temperature and humidity of the atmosphere layer by layer across the whole globe twice a day. Their observations are fed into numerical weather prediction models that now forecast as well five days ahead as they did three days ahead in the 1980s. This chapter explains the global satellite observing system, the instruments that make it work and the physics of atmospheric sounding, and then turns to environmental and climate monitoring — atmospheric composition and air quality, clouds, precipitation and radiation, oceans, sea level, sea ice and ice sheets — illustrated with real satellite records. It ends with the arrival of AI weather models.*

**IN THIS CHAPTER**

16.1 Satellites and the Weather
16.2 Geostationary Weather Satellites
16.3 Polar-Orbiting Weather Satellites
16.4 From Observations to Forecasts
16.5 Atmospheric Composition and Air Quality
16.6 Clouds, Aerosols, Precipitation and the Energy Budget
16.7 Oceans
16.8 The Cryosphere and Land

## 16.1 Satellites and the Weather

On 1 April 1960, NASA's **TIROS-1** returned the first television picture of the Earth's clouds from orbit. Forecasters immediately saw storms and weather systems that had been invisible to the scattered network of ground stations, especially over the oceans. Within a few years, the **Nimbus**

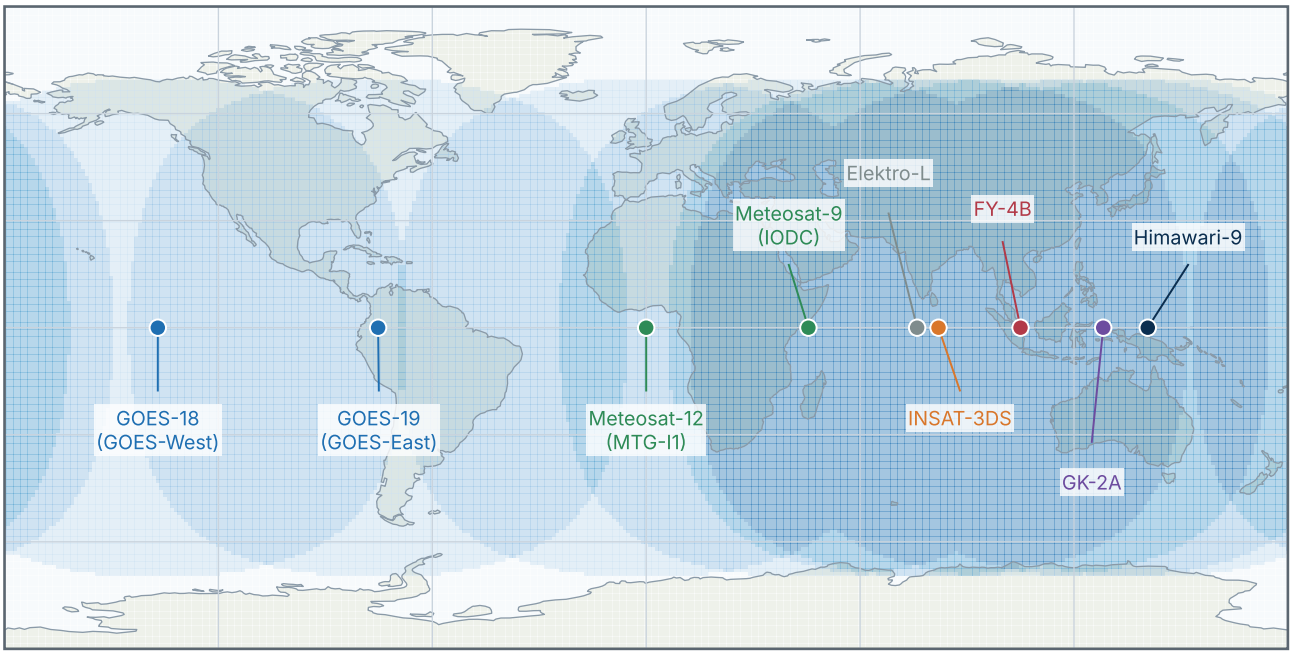
research satellites (from 1964) had demonstrated the first instruments to measure atmospheric temperature profiles from space. **ATS-1** (1966) then returned the first images of the full Earth disc from geostationary orbit. Operational geostationary satellites followed — the US **SMS/GOES** series (1974–75), Europe's **Meteosat** and Japan's **GMS** (both 1977) — and polar-orbiting satellites became the backbone of global forecasting.

Today, meteorological satellites are coordinated internationally through the World Meteorological Organization (WMO) and the **Coordination Group for Meteorological Satellites (CGMS)**. Their data are exchanged freely between nations under WMO resolutions — one of the most successful examples of international cooperation in space. More than 90 % of the observations assimilated into modern global weather forecasts come from satellites, and they account for most of the improvement in forecast skill over the past four decades.

### 16.1.1 The Space-Based Global Observing System

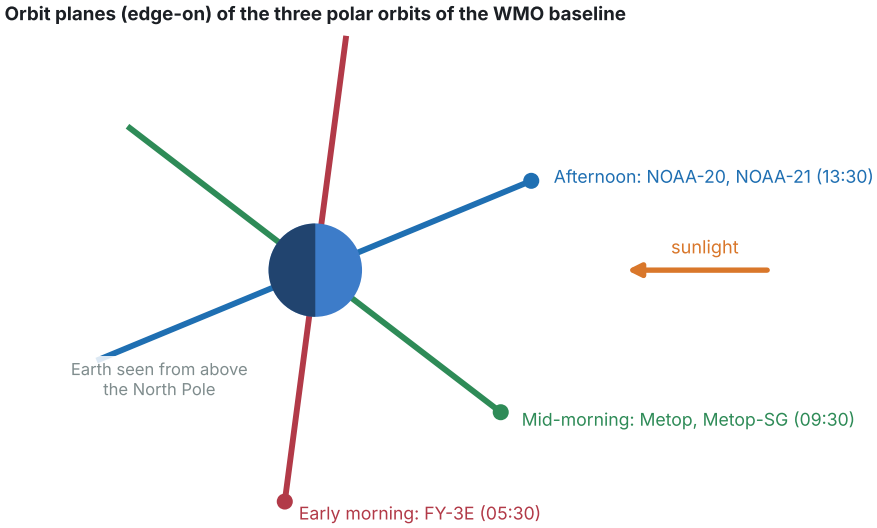
The WMO's baseline architecture combines two orbital regimes (Chapter 11):

- **Geostationary (GEO) satellites** at 35 786 km provide continuous observation of about a third of the Earth from a fixed position over the equator. A ring of five or six satellites covers the globe between about 60° N and 60° S (Figure 16.1). They are ideal for watching rapidly developing weather — thunderstorms, hurricanes, fog, fires, volcanic ash — and for deriving winds by tracking cloud and water-vapour features between successive images (**atmospheric motion vectors**). Their weakness is poor coverage of polar regions and the coarse resolution imposed by their great distance.
- **Low-Earth-orbit (LEO) polar satellites** in sun-synchronous orbits at 800–850 km cover the entire globe twice a day from close range, carrying the **sounders** that are the most important data source for numerical weather prediction. The WMO baseline calls for three orbital planes — early morning, mid-morning and afternoon — so that every place on Earth is observed at least every four hours (Figure 16.2). Europe (Metop) provides the mid-morning orbit, the USA (JPSS) the afternoon orbit, and China (FY-3E) has filled the previously empty early-morning orbit since 2021.




Approximate 2026 positions of primary meteorological geostationary satellites. Shading: number of satellites viewing each point within 70° Earth-central angle (the practical limit for quantitative imagery).

**FIGURE 16.1** Primary meteorological geostationary satellites in 2026. Shading shows how many satellites view each point at a usable angle; overlap provides backup and stereo views of clouds.




The plane stays fixed relative to the Sun, so each satellite crosses the equator at the same local times every day

**FIGURE 16.2** The three sun-synchronous orbit planes of the WMO polar-orbiting baseline, viewed from above the North Pole. Each plane keeps a fixed orientation relative to the Sun, so a satellite always crosses the equator at the same local solar times.

These are complemented by a growing number of other missions: **GNSS radio-occultation** constellations, **scatterometers** and **microwave imagers**, research missions with radars and lidars, and, increasingly, commercial providers selling weather data to governments.

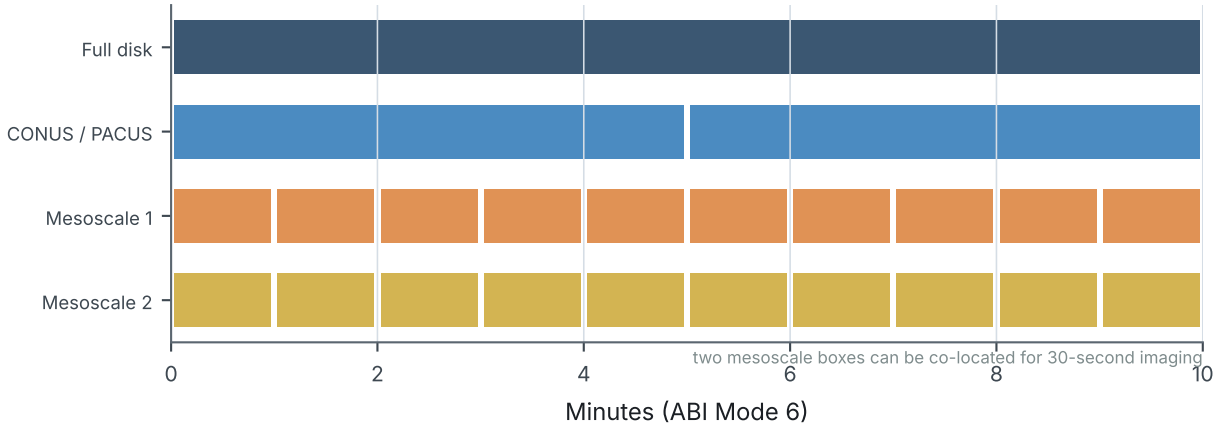
# 16.2 Geostationary Weather Satellites

## 16.2.1 The Current Generation

Since 2015 every geostationary weather operator has introduced a new generation of satellites with dramatically improved imagers (Table 16.1). The common design is represented by the US **Advanced Baseline Imager (ABI)**, Japan's **Advanced Himawari Imager (AHI)** and Europe's **Flexible Combined Imager (FCI)**. It scans the full disc in about 10 minutes in 16 spectral channels, with 0.5–1 km resolution in visible channels and 2 km in the infrared. At the same time it scans smaller regions every few minutes or even every 30 seconds (Figure 16.3).

**TABLE 16.1** Operational geostationary meteorological satellites in 2026 (selected).

| Operator | Series (current satellites) | Positions | Imager | Other instruments |
|---|---|---|---|---|
| USA (NOAA/NASA) | GOES-R: GOES-18 (West), GOES-19 (East) | 137° W, 75° W | ABI, 16 channels | GLM lightning mapper; space weather sensors; CCOR-1 coronagraph on GOES-19 |
| Europe (EUMETSAT/ESA) | Meteosat Third Generation: MTG-I1 (Meteosat-12), MTG-I2, MTG-S1; plus MSG | 0°, 9.5° E, 45.5° E | FCI, 16 channels | LI lightning imager; IRS hyperspectral sounder and Sentinel-4 UVN on MTG-S1 |
| Japan (JMA) | Himawari-9 (Himawari-8 backup) | 140.7° E | AHI, 16 channels | Himawari-10 (≈ 2029) to add a hyperspectral sounder |
| China (CMA) | Fengyun-4A, 4B | ≈ 86.5° E, 105° E | AGRI | GIIRS hyperspectral sounder; LMI lightning imager |
| South Korea (KMA) | GEO-KOMPSAT-2A | 128.2° E | AMI, 16 channels | Space weather; GEMS air quality on GK-2B |
| India (IMD/ISRO) | INSAT-3D, 3DR, 3DS | 74°–82° E | 6-channel imager | 19-channel infrared sounder |
| Russia (Roshydromet) | Elektro-L N2–N4 | 76° E, 165.8° E, etc. | MSU-GS | — |




The scans are interleaved; each bar shows how often a complete image of that domain is produced.

**FIGURE 16.3** Interleaved scanning of the GOES Advanced Baseline Imager in its standard mode: the full disc every 10 minutes, the continental USA every 5 minutes, and two 1 000 km mesoscale boxes every minute.

The GOES-R series illustrates the state of the art. Each satellite weighs about 5 200 kg at launch and carries the ABI, the **Geostationary Lightning Mapper (GLM)**, solar and space-environment instruments and a data-relay payload. ABI's 16 channels include two visible, four near-infrared and ten infrared bands; three water-vapour channels at 6.2, 6.9 and 7.3 µm map moisture at different heights, and the 3.9 µm channel detects fires and fog. GOES-16 was launched in 2016, GOES-17 in 2018, GOES-18 in 2022 and GOES-19 in June 2024. The ABI on GOES-17 suffered a cooling-system problem that degraded some infrared channels for several hours per night, and GOES-17 was replaced by GOES-18 in 2023. GOES-19 became the operational GOES-East satellite in April 2025. The follow-on **GeoXO** programme plans its first launch in 2032, adding a hyperspectral infrared sounder and an atmospheric-composition instrument.

**FIGURE 16.4** GOES-U (now GOES-19) on its Falcon Heavy rocket before launch in June 2024. The satellite became the operational GOES-East in April 2025. Image: NASA/KSC (SpaceX).

### 16.2.2 Lightning from Orbit

Lightning mappers fly on the newest geostationary satellites: GLM on GOES, the **Lightning Imager (LI)** on MTG and the **Lightning Mapping Imager (LMI)** on Fengyun-4. They detect the brief optical flashes of lightning at the oxygen emission line of 777.4 nm, day and night. They use staring detector arrays that take about 500 frames per second, and on-board processing that subtracts the bright background of sunlit clouds. Rapid increases in total lightning ("lightning jumps") often precede severe weather such as hail and tornadoes, and lightning data help pilots and air-traffic controllers avoid storms over the oceans, where radar coverage is absent.

### 16.2.3 Geostationary Sounding

In 2016 China's **Fengyun-4A** carried the first hyperspectral infrared sounder in geostationary orbit (GIIRS). In July 2025 Europe launched **MTG-S1** with the **Infrared Sounder (IRS)**, which measures about 1 700 spectral channels over Europe every 30 minutes. Geostationary sounders track the evolution of temperature and humidity in the hours before storms form — precisely the information needed for forecasting severe convection. MTG-S1 also carries the Copernicus **Sentinel-4** ultraviolet–visible–near-infrared spectrometer for hourly air-quality monitoring over Europe.

# 16.3 Polar-Orbiting Weather Satellites

## 16.3.1 The Main Systems

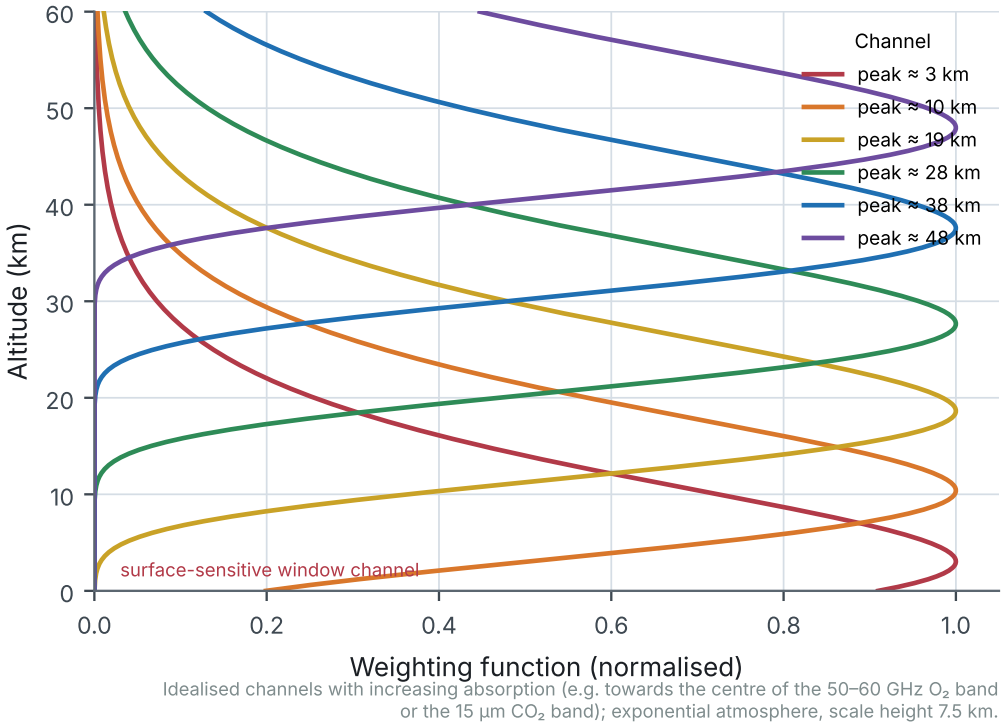
**TABLE 16.2** Operational polar-orbiting meteorological satellites (selected).

| System | Operator | Equator crossing | Satellites (2026) | Key instruments |
|---|---|---|---|---|
| JPSS | NOAA / NASA (USA) | 13:30 (afternoon) | Suomi NPP (2011), NOAA-20 (2017), NOAA-21 (2022); JPSS-4 from 2027 | VIIRS, CrIS, ATMS, OMPS, CERES |
| EPS / EPS-SG | EUMETSAT / ESA (Europe) | 09:30 (mid-morning) | Metop-B (2012), Metop-C (2018), Metop-SG A1 (2025); Metop-SG B1 (late 2026) | IASI and IASI-NG, AMSU-A/MHS and MWS, ASCAT and SCA, METimage, radio occultation, Sentinel-5 |
| Fengyun-3 | CMA (China) | early morning, morning, afternoon | FY-3C to FY-3G (FY-3E in the 05:30 orbit, FY-3G for precipitation radar) | MERSI, HIRAS, MWTS, MWHS, GNOS |
| Meteor-M | Roshydromet (Russia) | morning | Meteor-M N2-2 to N2-4 | MSU-MR, IKFS-2, MTVZA |
| DMSP | US Space Force | various | Last satellites near end of life | SSMIS microwave imager/sounder |

The European **Metop Second Generation (EPS-SG)** programme illustrates the trend to specialisation. Instead of one large satellite, it uses two series flying together. **Metop-SG A** (sounding and imaging) carries METimage, IASI-NG, MWS, the 3MI multi-viewing polarimeter, radio occultation and the Copernicus Sentinel-5 atmospheric-composition spectrometer. **Metop-SG B** (microwave imaging and scatterometry) carries SCA, MWI, the ICI ice-cloud imager, radio occultation and Argos. The first A satellite was launched on 13 August 2025 on an Ariane 6.

## 16.3.2 Sounders: Measuring the Atmosphere Layer by Layer

The most valuable instruments for weather forecasting are **sounders**, which measure the vertical profiles of temperature and humidity. They exploit a simple idea (Figure 16.5). In a spectral band where an atmospheric gas of known concentration absorbs strongly, the radiation that reaches space comes only from high in the atmosphere, because radiation from below is absorbed. Where it absorbs weakly, radiation from lower down escapes. A sounder measures many channels from the centre to the edge of an absorption band, such as the 15 µm band of $CO_2$ or the 50–60 GHz band of oxygen. The concentrations of these gases are well known. In this way the sounder samples the temperature at a range of heights. Each channel's sensitivity to different heights is described by its **weighting function**. Channels in water-vapour bands (6.3 µm and 183 GHz) similarly sample humidity.




Idealised channels with increasing absorption (e.g. towards the centre of the 50–60 GHz $O_2$ band or the 15 µm $CO_2$ band); exponential atmosphere, scale height 7.5 km.

**FIGURE 16.5** Idealised weighting functions of six sounding channels. Each channel senses a layer of the atmosphere a few kilometres thick; together they provide a temperature profile.

**Infrared sounders** offer fine vertical resolution. The hyperspectral sounders now flying are NASA's **AIRS** (2002, 2 378 channels), Europe's **IASI** (2006, 8 461 channels), the US **CrIS** (2011, 2 211 channels) and China's HIRAS. They measure the infrared spectrum in thousands of narrow channels and provide temperature profiles with about 1 K accuracy in 1 km layers. **IASI-NG** on Metop-SG doubles IASI's spectral resolution and halves its noise. Infrared sounders cannot see through clouds, however.

**Microwave sounders** (AMSU-A, MHS, ATMS, MWHS, and the new MWS) have coarser resolution but see through all but the thickest precipitating clouds. So they observe the atmosphere in exactly the stormy regions that matter most for forecasts. Figure 16.6 shows the absorption lines they use. Microwave sounders have consistently been found to have the largest single impact on global forecast skill.

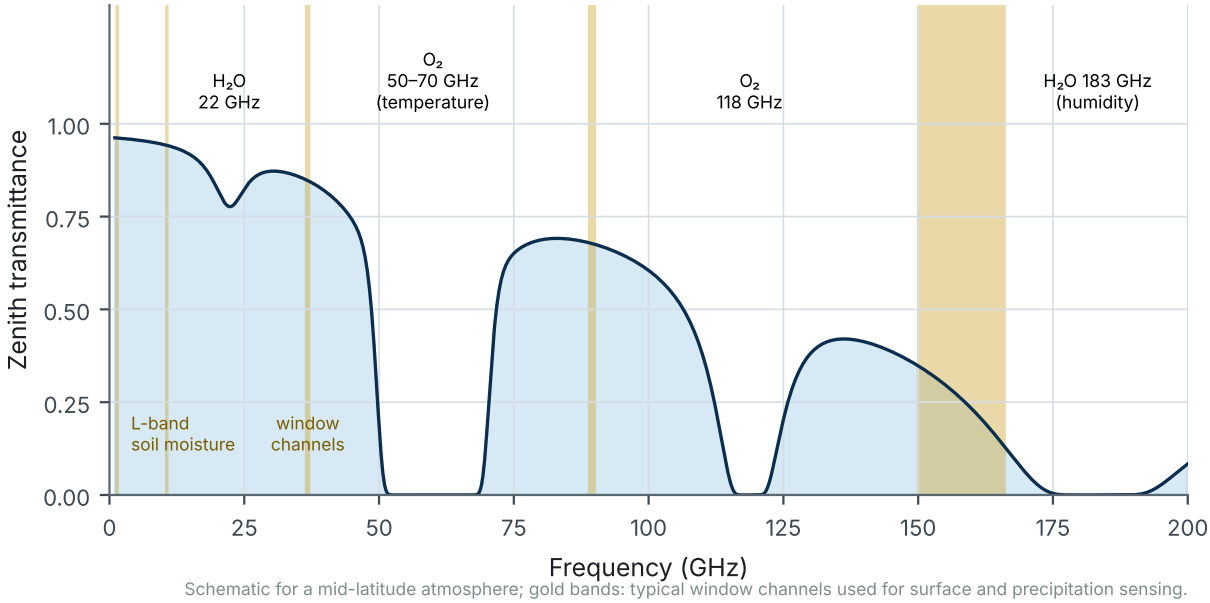



**FIGURE 16.6** Schematic transmittance of the atmosphere at microwave frequencies. Sounders use the oxygen bands near 60 and 118 GHz for temperature and the water-vapour line at 183 GHz for humidity; window channels see the surface and precipitation.

**Microwave imagers** (SSMIS, AMSR2, GMI, MWI) measure emission from the surface and the atmosphere at window frequencies, from which they derive precipitation, total water vapour, cloud liquid water, sea-surface wind speed, sea ice and snow. The SSMIS instruments on the ageing US military DMSP satellites provided about half of the world's microwave imagery. The planned end of these satellites — and NOAA's decision in 2025 to stop processing their data publicly — raised concerns among hurricane forecasters about gaps in observations of storm structure. This also increased the importance of newer instruments and commercial microwave small satellites.

### 16.3.3 GNSS Radio Occultation

A remarkably powerful technique uses signals from navigation satellites (Figure 16.7). As a GPS, Galileo, GLONASS or BeiDou satellite sets or rises behind the Earth's limb as seen from a LEO satellite, its signals pass through the atmosphere and are bent by refraction. A receiver on the LEO satellite measures the extra delay with millimetre precision. From this, the **bending angle** — and hence the refractivity, temperature, pressure and humidity of the atmosphere — can be computed as a function of height. **Radio occultation (RO)** measurements are self-calibrating (based on time measurements traceable to atomic clocks), unaffected by clouds and highly precise in the upper troposphere and stratosphere. They are used to "anchor" the bias corrections of other satellite instruments.

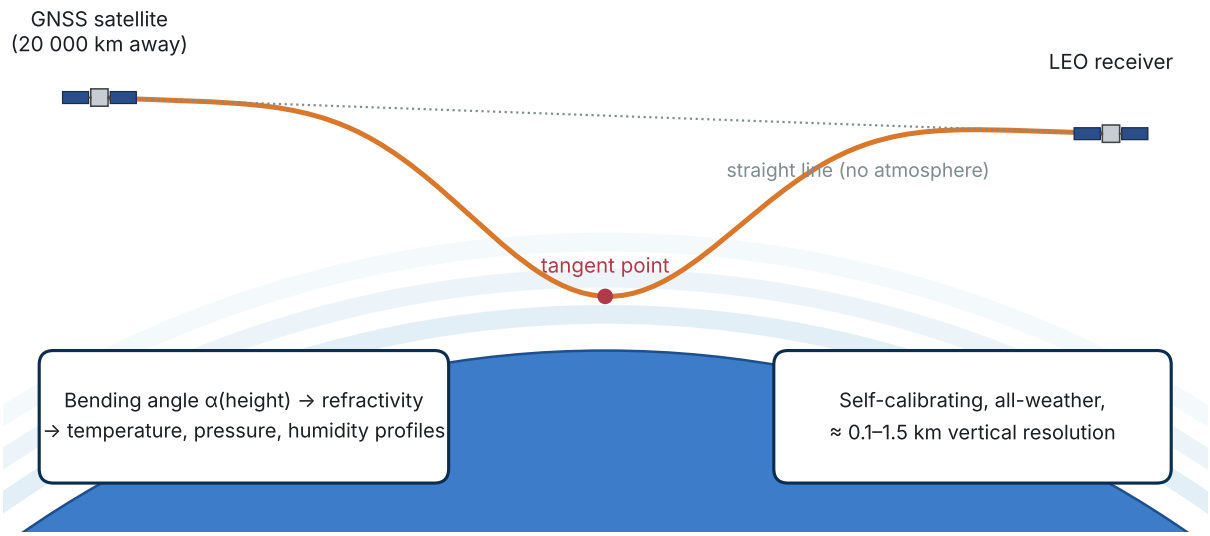



**FIGURE 16.7** GNSS radio occultation. A signal from a navigation satellite is bent as it passes through the atmosphere on its way to a receiver in low Earth orbit.

The US–Taiwan **COSMIC-2** constellation (2019) provides about 5 000 tropical occultations per day, and Metop carries the GRAS receiver. Since about 2020, commercial companies such as **Spire** and **PlanetiQ** have sold radio-occultation data from constellations of small satellites to NOAA, EUMETSAT and other agencies — one of the first successful commercial weather-data markets. Spire also measures reflected GNSS signals (GNSS-R) for ocean winds and soil moisture.

### 16.3.4 Winds

Wind is the hardest weather variable to observe from space. Most wind information comes from **atmospheric motion vectors**, found by tracking clouds and water-vapour features in successive geostationary images. It also comes from **scatterometers** such as ASCAT on Metop. These measure ocean-surface wind speed and direction from the radar backscatter of wind-driven ripples observed from several directions. ESA's **Aeolus** (2018–2023) was the first satellite to carry a Doppler wind lidar, measuring wind profiles directly by the Doppler shift of laser light backscattered by molecules and aerosols. Despite technical difficulties with the laser, Aeolus clearly improved forecasts, particularly in the tropics, and an operational follow-on (EPS-Aeolus) is planned for the 2030s.

## 16.4 From Observations to Forecasts

### 16.4.1 Numerical Weather Prediction and Data Assimilation

A weather forecast is produced by a **numerical weather prediction (NWP)** model that solves the equations of fluid dynamics and thermodynamics for the atmosphere on a three-dimensional grid. For the global model of the European Centre for Medium-Range Weather Forecasts (ECMWF), this grid is today about 9 km horizontally and 137 levels vertically. Because the atmosphere is chaotic, a forecast can only be as good as its **initial conditions** — the best estimate of the current state of the whole atmosphere, called the **analysis**.

Producing the analysis is the job of **data assimilation** (Figure 16.8). Every 6 or 12 hours, tens of millions of observations — over 90 % of them from satellites — are compared with a short-range forecast (the "first guess"). Modern systems assimilate satellite **radiances** directly, rather than retrieved temperatures. A fast **radiative-transfer model** (such as RTTOV or CRTM) computes what each instrument channel would measure given the model atmosphere. An optimisation algorithm (**4D-Var**, ensemble Kalman filters or hybrids) then adjusts the model state to fit all the observations over a time window as well as possible, weighted by their expected errors. Instrument biases are corrected continuously within the system, using unbiased "anchor" observations such as radiosondes and GNSS radio occultation.

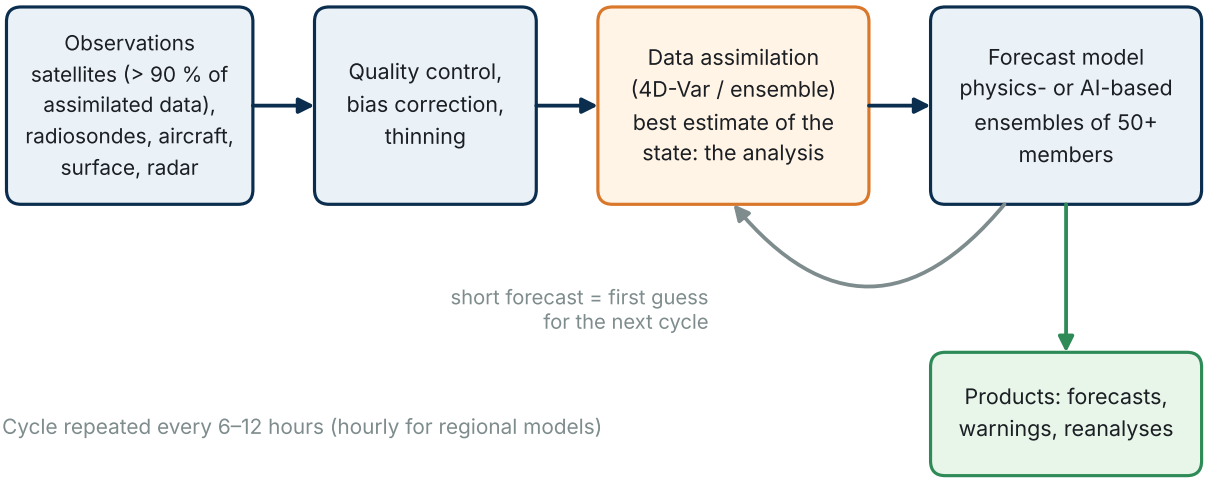
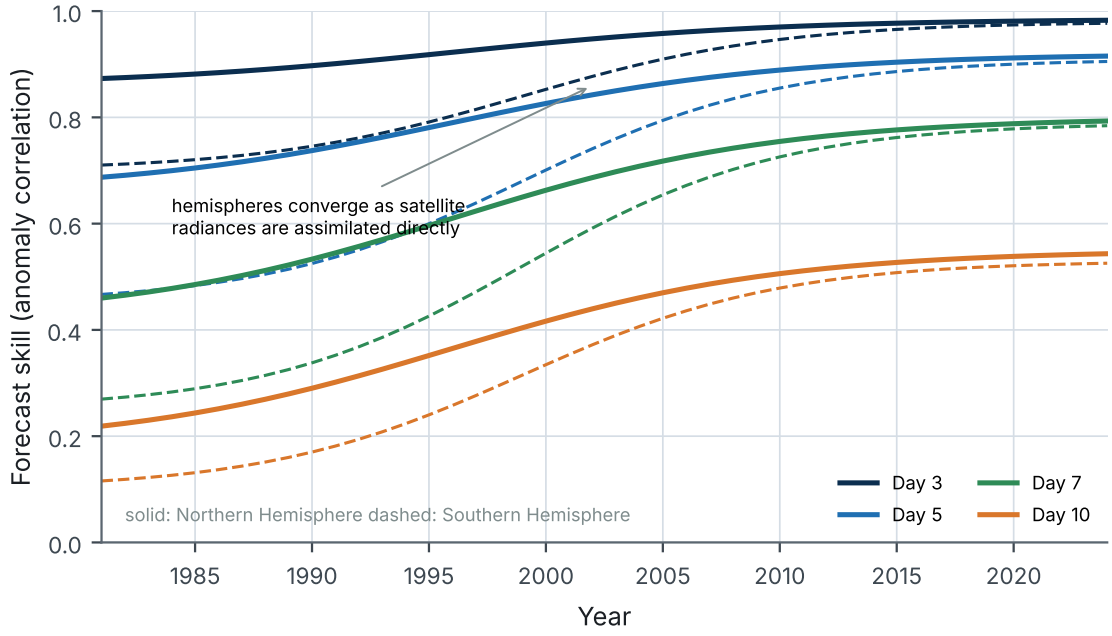



**FIGURE 16.8** The data-assimilation and forecasting cycle used by operational weather centres.

The result has been called a "quiet revolution": forecast skill has improved by about one day per decade, so that a five-day forecast today is as accurate as a three-day forecast in the 1980s (Figure 16.9). One of the clearest signs of the satellites' contribution is the closing of the gap between the Northern and Southern Hemispheres around 2000. The Southern Hemisphere has few ground observations; once satellite radiances were assimilated directly, its forecasts became nearly as good as those of the well-observed north.




Schematic redrawn after published ECMWF 500 hPa height anomaly-correlation records (Bauer et al., 2015).

**FIGURE 16.9** Schematic evolution of medium-range forecast skill. Skill has risen steadily, and the Southern Hemisphere caught up with the Northern Hemisphere as satellite data assimilation matured.

Studies measure each observing system's contribution, either by denying data in experiments or by adjoint-based "forecast sensitivity to observation impact" methods. They consistently find that microwave sounders, hyperspectral infrared sounders, GNSS radio occultation and atmospheric motion vectors are among the most valuable observations, together with conventional radiosondes and aircraft reports.

### 16.4.2 AI Weather Models

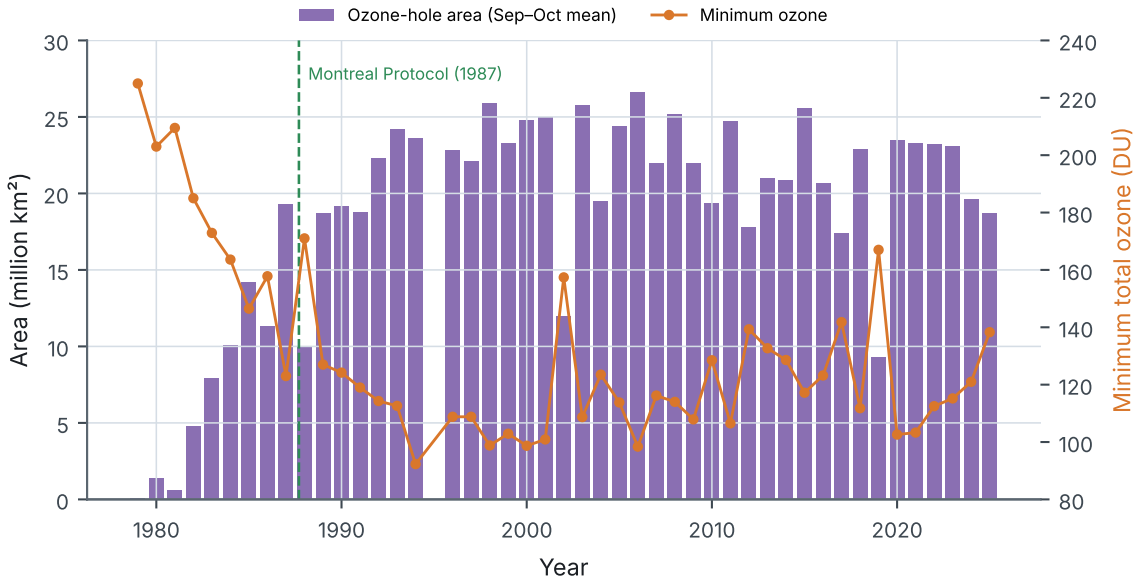
Machine-learning models can be trained on decades of ECMWF's **ERA5 reanalysis** — itself a product of assimilating satellite data. Since 2022, such models have matched or beaten physics-based models on many forecast scores, while running thousands of times faster. Notable models include NVIDIA's FourCastNet (2022), Huawei's Pangu-Weather and Google DeepMind's GraphCast (2023) and GenCast (2024). In February 2025 ECMWF made its own **Artificial Intelligence Forecasting System (AIFS)** operational. It was the first data-driven global model in an operational service at a major international centre, and was followed by an AI ensemble (AIFS ENS) in July 2025. These models still depend on conventional data assimilation for their initial conditions, and therefore on satellite observations; research models that learn to forecast directly from raw observations are under development. Part XIII discusses these advances in detail.

## 16.5 Atmospheric Composition and Air Quality

### 16.5.1 Ozone

Satellites played a central role in one of the great environmental success stories. The ozone layer, at 15–35 km, absorbs harmful solar ultraviolet radiation. After British Antarctic Survey scientists reported the **Antarctic ozone hole** in 1985 from ground measurements, NASA's Total Ozone

Mapping Spectrometer (TOMS) on Nimbus-7 revealed its full extent. The **Montreal Protocol** (1987) and its amendments phased out the chlorofluorocarbons responsible, and satellites have monitored the result ever since (Figure 16.10). The ozone hole still forms every southern spring but has begun to shrink; full recovery is expected around 2066 over the Antarctic.




Data: NASA Ozone Watch (TOMS, OMI, OMPS; gaps filled with MERRA-2). No satellite data in 1995.

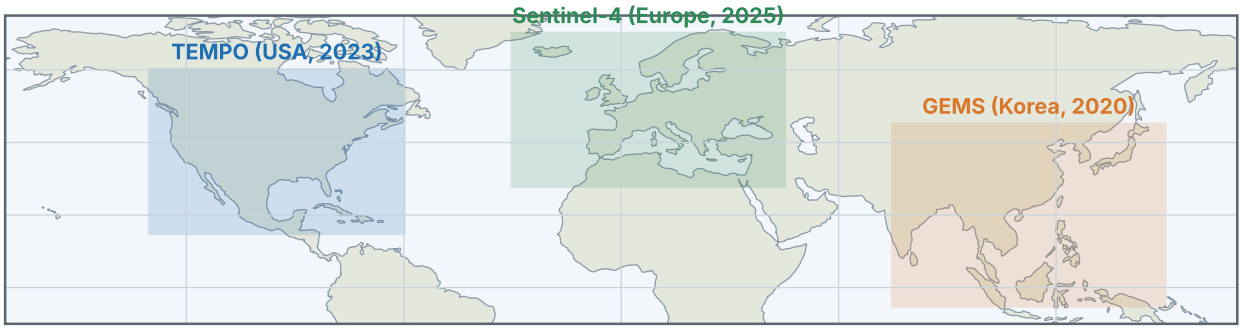
**FIGURE 16.10** Size of the Antarctic ozone hole and minimum total ozone, 1979–2025, from satellite measurements.

**FIGURE 16.11** The Antarctic ozone hole on 1 October 2010 seen by the Ozone Monitoring Instrument on NASA's Aura satellite. Blue and purple show the lowest ozone. Image: NASA/GSFC.

Ozone and other trace gases are measured by **ultraviolet–visible spectrometers** that observe sunlight scattered by the atmosphere and identify gases by their absorption features. Examples are TOMS (1978–2005), the Dutch–Finnish **OMI** on Aura (2004), **OMPS** on Suomi NPP and NOAA-20/21, and GOME-2 on Metop. Since 2017, **TROPOMI** on the Copernicus **Sentinel-5 Precursor** has mapped nitrogen dioxide, ozone, sulphur dioxide, formaldehyde, carbon monoxide and methane daily, with pixels of 5.5 × 3.5 km.

### 16.5.2 Air Quality from Geostationary Orbit

Air pollution changes hour by hour with traffic, industry and weather. A **geostationary air-quality constellation** now observes the world's three most populated northern regions every daylight hour (Figure 16.12). Korea's **GEMS** on GEO-KOMPSAT-2B (2020) covers Asia, and NASA's **TEMPO** (2023, hosted on a commercial communications satellite) covers North America. Copernicus **Sentinel-4** on MTG-S1 (2025) covers Europe and North Africa. These instruments map pollutants such as nitrogen dioxide at neighbourhood scales and show their daily cycles — rush-hour peaks, ship tracks, wildfire smoke — for the first time.




Approximate fields of regard of the geostationary air-quality constellation (hourly daytime $NO_2$, $O_3$, HCHO, aerosols).

**FIGURE 16.12** Fields of regard of the geostationary air-quality constellation.

### 16.5.3 Greenhouse Gases

Measuring carbon dioxide from space is extremely demanding. Adding all of humanity's emissions for a year changes the atmospheric concentration by only about 0.5 %. A large power plant raises it locally by a fraction of a percent. Dedicated missions use high-resolution spectrometers in the near-infrared and short-wave infrared absorption bands of $CO_2$ (1.6 and 2.06 µm) and of oxygen (0.76 µm, to measure the total air column). They aim for a precision of better than 0.25 %. Missions include Japan's **GOSAT** (2009), **GOSAT-2** (2018) and **GOSAT-GW** (2025), NASA's **OCO-2** (2014) and **OCO-3** on the ISS (2019), China's TanSat (2016) and the French **MicroCarb** (2025). Europe's **CO2M** constellation, the first Copernicus Expansion mission, is designed to monitor anthropogenic $CO_2$ emissions at the scale of countries and large cities for verification of the Paris Agreement. Methane — whose super-emitters are mapped by imaging spectrometers (Chapter 14) — is measured globally by TROPOMI and GOSAT.

## 16.6 Clouds, Aerosols, Precipitation and the Energy Budget

### 16.6.1 Profiling Radars and Lidars

Passive sensors see mostly the tops of clouds. Active instruments reveal their vertical structure. From 2006 until 2023, NASA's **CloudSat** (94 GHz cloud radar) and the NASA–CNES **CALIPSO** (lidar) flew in the "A-Train" formation with Aqua and Aura, revolutionising understanding of clouds and aerosols. Their successor, the ESA–JAXA **EarthCARE** (launched May 2024), combines an atmospheric lidar (ATLID), the first **Doppler cloud-profiling radar** in space (provided by Japan), a multispectral imager and a broadband radiometer. Its aim is to understand how clouds and aerosols affect the Earth's energy balance — the largest uncertainty in climate projections.

### 16.6.2 Precipitation

The US–Japan **Tropical Rainfall Measuring Mission (TRMM, 1997–2015)** carried the first precipitation radar in space. Its successor, the **Global Precipitation Measurement (GPM)** Core Observatory (2014, Figure 16.13), carries a dual-frequency precipitation radar (Ku- and Ka-band) and a 13-channel microwave imager. GPM serves as a calibration reference for a constellation of about ten microwave imagers and sounders from several nations. Together with geostationary infrared data, they feed the **IMERG** product, which maps rainfall worldwide every 30 minutes on a 0.1° grid. Small satellites are joining: NASA's **TROPICS** CubeSats (2023) observe tropical cyclones with microwave sounders roughly every hour.

**FIGURE 16.13** The GPM Core Observatory during testing at NASA Goddard. Its dual-frequency precipitation radar and microwave imager calibrate a whole constellation of precipitation-sensing satellites. Image: NASA/GSFC.

### 16.6.3 The Earth's Energy Budget

The climate is governed by the balance between absorbed sunlight and emitted thermal radiation. The **CERES** instruments, flying since 1997 on TRMM, Terra, Aqua, Suomi NPP and NOAA-20, measure reflected solar and emitted longwave radiation. Combined with ocean heat-content measurements, they show that the Earth's **energy imbalance** roughly doubled between 2005 and 2019, to about 1 W/m$^2$ — energy that is warming the oceans. Total solar irradiance has been measured from space since 1978, most recently by NASA's TSIS-1 on the ISS, showing that changes in the Sun are far too small to explain recent warming. Future missions such as the UK–ESA TRUTHS aim to set SI-traceable climate benchmarks (Chapter 13).

# 16.7 Oceans

## 16.7.1 Sea-Surface Temperature and Ocean Colour

Sea-surface temperature (SST) has been measured from space since the 1970s with infrared radiometers (AVHRR, and today VIIRS and Sentinel-3's dual-view SLSTR). SLSTR uses two viewing angles to correct for the atmosphere. Microwave radiometers (AMSR2) are also used, and these see through clouds. SST drives the weather and climate — it fuels hurricanes, defines El Niño, and is a key input to forecasts.

**Ocean colour** sensors measure the subtle colour of the sea to derive chlorophyll concentration, sediments and harmful algal blooms. Chlorophyll concentration shows the abundance of phytoplankton, which produce about half of the world's oxygen and form the base of the marine food web. After SeaWiFS (1997–2010), MODIS and Sentinel-3's OLCI, NASA's **PACE** mission (launched February 2024, Figure 16.14) carries the first hyperspectral ocean-colour instrument (OCI). OCI measures the full spectrum from ultraviolet to near-infrared in 5 nm steps, so that different types of phytoplankton can be distinguished. PACE also carries two polarimeters for aerosols and clouds.

**FIGURE 16.14** NASA's PACE spacecraft being prepared for launch in December 2023. Its Ocean Color Instrument measures more than 200 wavelengths. Image: NASA/KSC (NASA/Kim Shiflett).

## 16.7.2 Satellite Altimetry and Sea Level

A radar **altimeter** measures the height of the sea surface by timing radar pulses reflected vertically from the ocean (Figure 16.15). The satellite's orbit must be known with centimetre accuracy, which is achieved with GNSS receivers, the French DORIS system and laser ranging. Corrections are also applied for the ionosphere, water vapour, sea state and tides. The sea-surface height can then be measured to about 2–3 cm along each ground track. Ocean currents, eddies and El Niño show up as bumps and dips of tens of centimetres in the sea surface.

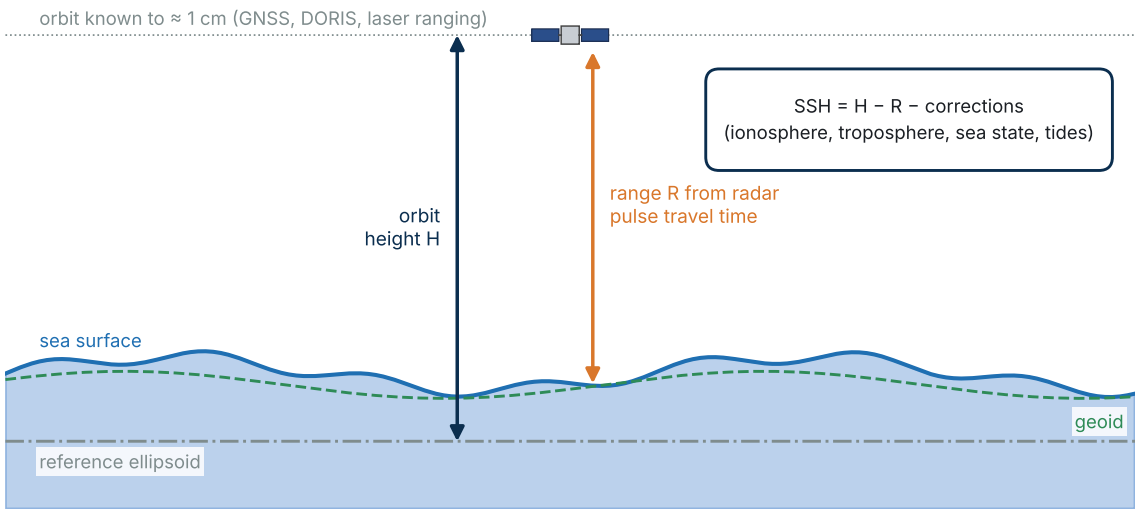
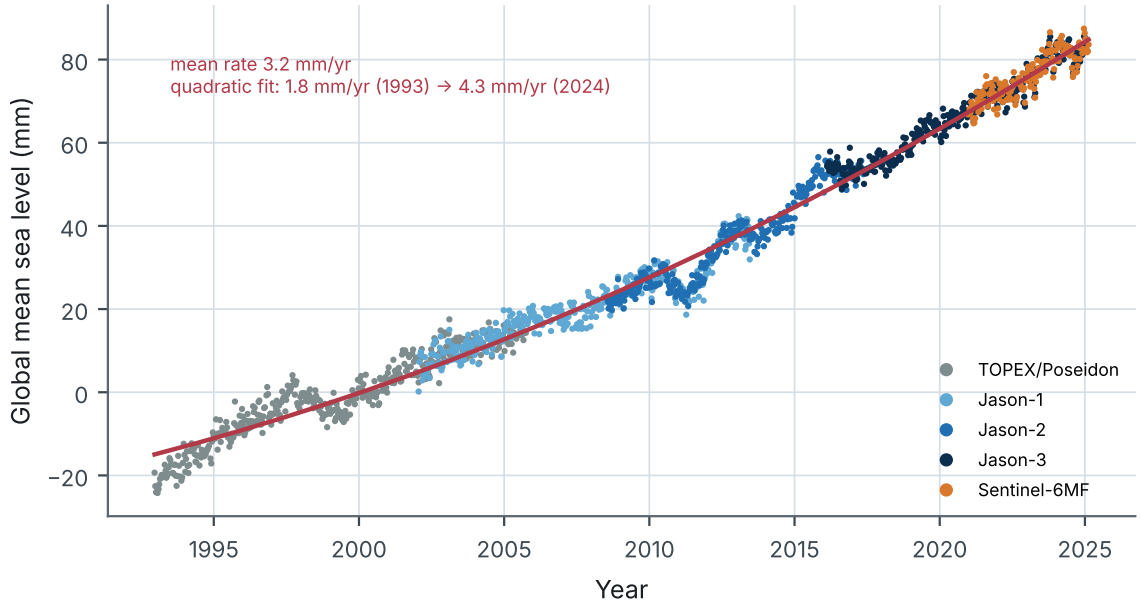



**FIGURE 16.15** Principle of satellite radar altimetry. Sea-surface height is the orbit height minus the measured range, corrected for atmospheric and sea-state effects.

Since the US–French **TOPEX/Poseidon** mission (1992), a series of reference altimetry missions has measured global mean sea level with continuous, cross-calibrated precision (Figure 16.16). These are Jason-1 (2001), Jason-2 (2008), Jason-3 (2016), **Sentinel-6 Michael Freilich** (2020) and **Sentinel-6B** (launched November 2025). The record shows a rise of about 10 cm since 1993, at a rate that has roughly doubled, driven by ocean warming (thermal expansion) and by melting glaciers and ice sheets.




Altimetry data are provided by NOAA Laboratory for Satellite Altimetry (seasonal signals removed, no GIA correction).

**FIGURE 16.16** Global mean sea level from the reference altimetry missions, 1993–2025, with a quadratic fit showing acceleration.

The NASA–CNES **SWOT** mission (2022) introduced a new technique: a Ka-band radar interferometer (KaRIn) with two antennas on a 10 m boom. It measures water-surface height across two 50 km swaths rather than along a single line. SWOT resolves ocean eddies ten times smaller than before and measures the height and slope of rivers and lakes worldwide — the first global survey of fresh water from space (Figure 16.17).

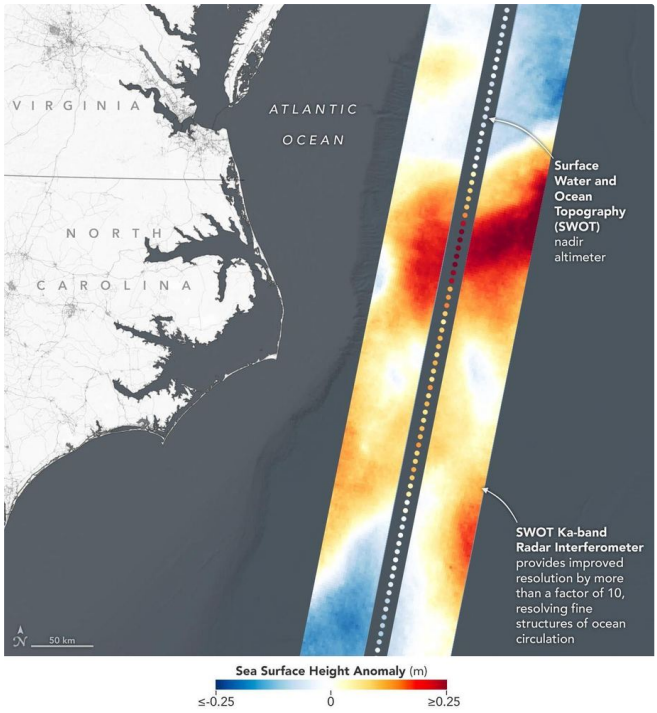
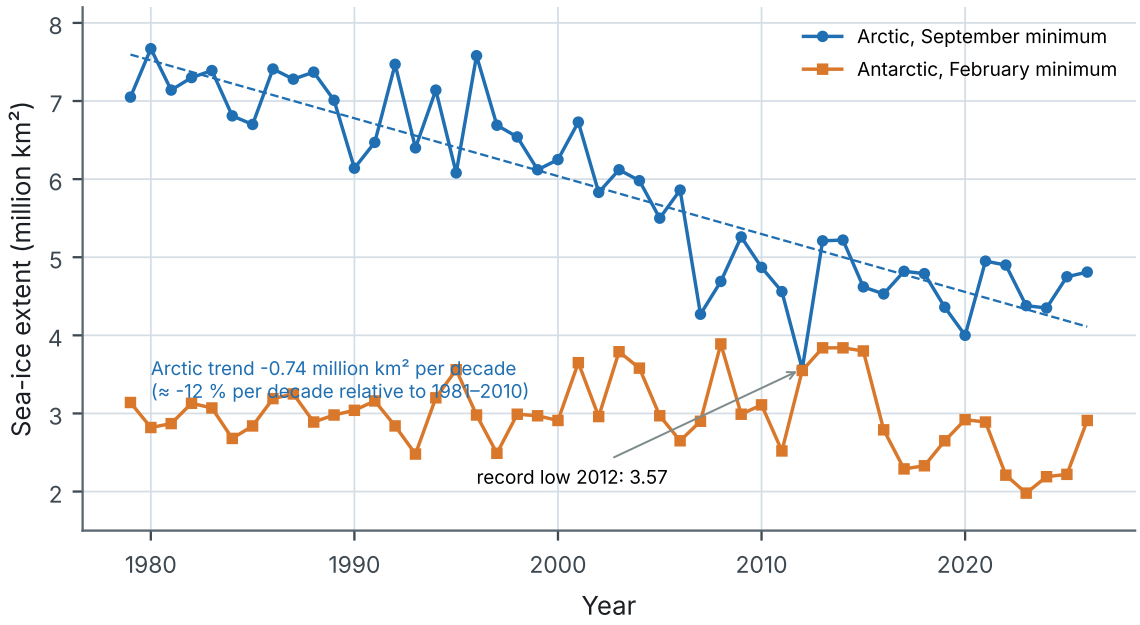



**FIGURE 16.17** SWOT's first-light sea-surface height measurements off the US East Coast (2023), showing fine-scale ocean features within the wide swath compared with conventional altimetry. Image: NASA/JPL (NASA/JPL-Caltech).

# 16.8 The Cryosphere and Land

## 16.8.1 Sea Ice and Ice Sheets

Passive microwave radiometers have mapped polar sea ice every day since 1978, regardless of clouds and polar darkness, because ice and open water emit microwaves very differently. The record shows a dramatic decline in Arctic sea ice, especially at the end of summer (Figure 16.18): the September minimum extent has fallen by about 12 % per decade, with a record low in 2012. Antarctic sea ice changed little until 2016, then fell sharply, with record lows in 2023.




Data: NSIDC Sea Ice Index (passive microwave, SMMR, SSM/I, SSMIS, AMSR2).

**FIGURE 16.18** Minimum sea-ice extent in the Arctic (September) and Antarctic (February) from the passive-microwave record, 1979–2026.

Radar altimeters (ESA's **CryoSat-2**, 2010) and laser altimeters (NASA's **ICESat-2**, 2018) measure the thickness of sea ice and the changing height of ice sheets and glaciers. The US–German **GRACE** (2002–2017) and **GRACE Follow-On** (2018) missions measure tiny changes in the Earth's gravity field to "weigh" ice sheets and groundwater. They do this by tracking the distance between two satellites 220 km apart to within a fraction of a micrometre, with microwave and laser ranging. They show Greenland losing about 270 billion tonnes of ice per year and Antarctica about 150 billion tonnes, and reveal groundwater depletion in regions such as northern India and California's Central Valley.

## 16.8.2 Soil Moisture and Land

ESA's **SMOS** (2009) and NASA's **SMAP** (2015) measure soil moisture and ocean salinity with L-band (1.4 GHz) radiometers, exploiting a protected frequency band where the emission of the surface depends strongly on its water content. SMOS uses a Y-shaped interferometric array of 69 small antennas to synthesise a large aperture; SMAP uses a 6 m rotating mesh antenna. Land-surface monitoring by optical, thermal and radar missions is described in Chapters 13–15.

### DATA BOX: ESSENTIAL CLIMATE VARIABLES FROM SPACE

Of the 55 Essential Climate Variables defined by the Global Climate Observing System, more than half can only be observed globally from space. Long, stable, cross-calibrated satellite records are a foundation of the assessments of the Intergovernmental Panel on Climate Change (IPCC). Examples are those produced by ESA's Climate Change Initiative, the EUMETSAT Climate Monitoring Satellite Application Facility and NOAA's climate data records. Producing them requires overlap between successive satellites, careful recalibration of old instruments and the reprocessing of decades of data.

### EXAMPLE: WHY A SOUNDER CHANNEL "SEES" A PARTICULAR HEIGHT

Consider a channel in which the absorption optical depth of the whole atmosphere is $\tau_0 = 40$, in an atmosphere where the absorbing gas density falls with a scale height $H$ = 7.5 km. The optical depth above height $z$ is $\tau(z) = \tau_0 e^{-z/H}$, and the radiation from height $z$ reaching space is attenuated by $e^{-\tau(z)}$. The weighting function $W(z) = d(e^{-\tau(z)})/dz$ peaks where $\tau(z) = 1$:

$$z_{peak} = H \ln \tau_0 = 7.5 \times \ln 40 \approx 28 \text{ km}$$

A channel with $\tau_0 = 4$ peaks at about 10 km; one with $\tau_0 = 600$ at about 48 km (Figure 16.5). The weighting functions are each about 10–15 km wide, which is why sounders need many overlapping channels and why the retrieved profiles are smooth. Hyperspectral sounders with thousands of channels sharpen the vertical resolution considerably.

### AI IN WEATHER AND ENVIRONMENTAL MONITORING

Artificial intelligence is transforming every stage of the chain. Uses include on-board cloud detection and data compression, fast neural-network radiative-transfer models for data assimilation, and nowcasting of thunderstorms from geostationary imagery. Others are retrieval of precipitation and atmospheric composition, and global AI forecast models such as ECMWF's AIFS. Chapter 47 discusses AI for weather, climate and Earth-system science in depth.

### KEY TAKEAWAYS

- Weather satellites are coordinated globally through WMO and CGMS; satellites supply more than 90 % of the observations used in global forecasts.
- Geostationary imagers (ABI, AHI, FCI) scan the full disc every 10 minutes in 16 channels and small regions every 30–60 seconds; lightning mappers and geostationary sounders are the newest additions.
- Polar orbiters in early-morning, mid-morning and afternoon orbits carry infrared and microwave sounders, the most valuable data for numerical weather prediction; GNSS radio occultation provides unbiased anchor measurements.
- Sounding works because channels with different absorption strengths sense different heights (weighting functions).
- Data assimilation combines observations with models; forecast skill has improved by about a day per decade, and AI forecast models became operational in 2025.
- Satellites monitor the ozone layer, air quality (now hourly from geostationary orbit), greenhouse gases, clouds, precipitation and the Earth's energy budget.
- Altimetry shows global sea level rising about 10 cm since 1993 at an accelerating rate. Passive microwave records show Arctic summer sea ice declining about 12 % per decade, and gravity missions weigh the ice sheets' losses.

### REVIEW QUESTIONS

1. Why are both geostationary and polar-orbiting satellites needed for weather forecasting? What does each do best?
2. Explain how infrared and microwave sounders measure temperature at different heights using weighting functions.
3. Why are GNSS radio-occultation measurements described as "unbiased", and why are they valuable for anchoring other observations?
4. What is data assimilation? Why has forecast skill improved by about one day per decade?

5. How do AI weather-forecast models differ from physics-based numerical models, and what are their present limitations?
6. Why is air-quality monitoring from geostationary orbit an important advance over polar-orbiting measurements?
7. Global mean sea level has risen about 10 cm since 1993. What is the average rate, and why is the present rate higher than the average?
8. Which satellite techniques measure sea ice, ice-sheet mass and soil moisture? Give one mission for each.

PART V

# Communication and Navigation Satellites

Satellites carry television, internet, phone calls and the timing signals that keep banks, power grids and phones synchronised. Part V explains communication payloads, broadband mega-constellations, direct-to-device service and the four global navigation systems: GPS, GLONASS, Galileo and BeiDou.

17 Communication Satellites

18 Global Navigation Satellite Systems

Image: NASA/KSC

# 17 CHAPTER Communication Satellites

Image: NASA/KSC (NASA/Kim Shiflett)

*Communication satellites carry television to hundreds of millions of homes, connect ships, aircraft and remote communities to the internet, relay data from other spacecraft, and now connect ordinary smartphones directly from space. This chapter traces their evolution from passive balloons to geostationary "high-throughput" satellites delivering a terabit per second and to constellations of thousands of satellites in low orbit linked by lasers. It explains the engineering choices that shape every system — orbit, frequency band, propagation, link budget, payload architecture, frequency reuse and terminals — and the regulatory framework that shares the radio spectrum and orbits among nations. It closes with the current state of the industry in 2026: Starlink, OneWeb, Amazon Leo and the Chinese constellations, direct-to-device services, optical communications and data relay.*

**IN THIS CHAPTER**

17.1 From Science Fiction to Global Infrastructure
17.2 Services and Markets
17.3 Choosing an Orbit
17.4 Frequencies, Propagation and Spectrum Sharing
17.5 Link Budgets and Terminals
17.6 The Communication Payload
17.7 LEO Broadband Constellations
17.8 Mobile Satellite Services and Direct-to-Device
17.9 Gateways, Teleports and Networks
17.10 Optical Communications and Data Relay

## 17.1 From Science Fiction to Global Infrastructure

In October 1945 the engineer and writer Arthur C. Clarke proposed in *Wireless World* that three satellites in a 24-hour orbit above the equator could relay radio signals around the whole world. Fifteen years later the idea began to come true. NASA's **Echo 1** (1960), a 30 m aluminised balloon, reflected radio signals passively between ground stations (Figure 17.1). AT&T's **Telstar 1** (1962)

was the first active communication satellite. It relayed the first live television pictures across the Atlantic. However, its low elliptical orbit meant it was visible from both continents for only minutes at a time. In 1963 Hughes' **Syncom 2** reached the first geosynchronous orbit, and in 1964 **Syncom 3** — in a true geostationary orbit — broadcast the Tokyo Olympic Games to the USA. In 1965 **Intelsat I** ("Early Bird") began commercial transatlantic telephone and television service with 240 voice circuits. The Soviet Union, whose territory lies at high latitudes, chose highly elliptical **Molniya** orbits for its own system from 1965 (Figure 17.3).

**FIGURE 17.1** Inflation test of an Echo balloon satellite in 1958. Echo 1, launched in 1960, was a 30 m passive reflector — the first communication satellite experiment to relay voice between continents. Image: NASA/LRC (NACA).

**FIGURE 17.2** A Syncom IV communication satellite after deployment from the Space Shuttle Discovery in 1984. The Syncom name dates back to the first geosynchronous satellites of 1963–64. Image: NASA/JSC.

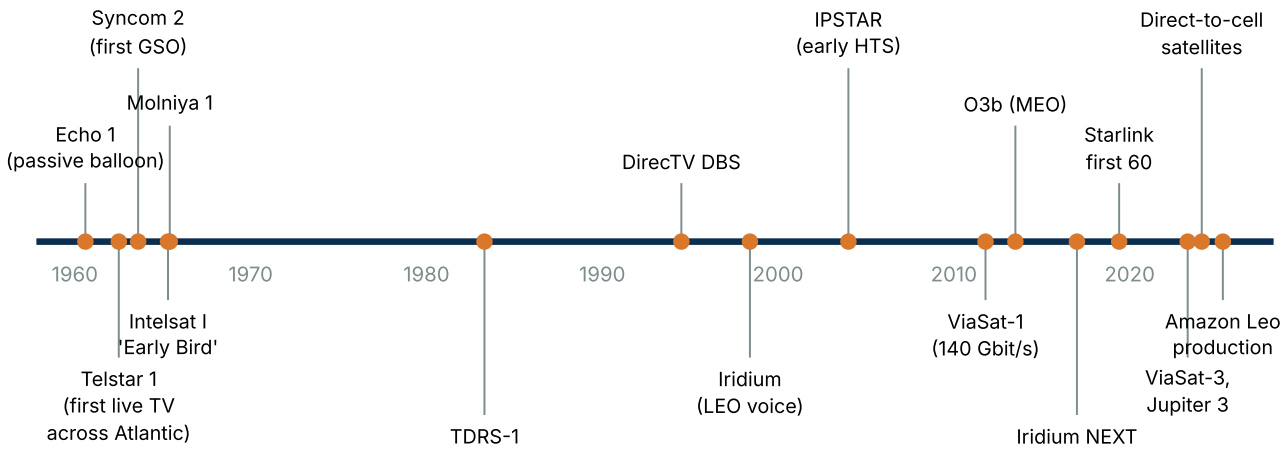



**FIGURE 17.3** Milestones in satellite communications.

For half a century, communication satellites were dominated by a few hundred large geostationary satellites operated by international organisations (Intelsat, Inmarsat, Eutelsat — later privatised) and national and commercial operators. They carried international telephone trunks (until undersea fibre took over from the 1990s), distributed television to cable head-ends and directly to homes, and provided mobile communications to ships and aircraft. In the 2010s two revolutions transformed the industry: **high-throughput satellites (HTS)** in GEO, which multiplied capacity a hundredfold using many small spot beams; and **low-Earth-orbit broadband constellations** of hundreds to thousands of mass-produced satellites. In 2026 satellite communications remain the largest commercial space market, generating more than $100 billion a year in services and equipment.

**FIGURE 17.4** The Space Shuttle crew of STS-49 captured the stranded Intelsat VI satellite in 1992 during the first three-person spacewalk, attached a new perigee motor and sent it on to geostationary orbit. Communication satellites were among the Shuttle's main commercial payloads. Image: NASA/MSFC.

## 17.2 Services and Markets

Satellite communications serve several distinct markets, each defined in the ITU Radio Regulations as a "service" with its own frequency allocations (Table 17.1).

**TABLE 17.1** Satellite communication services.

| Service | Typical users | Main bands | Example systems |
|---|---|---|---|
| Fixed Satellite Service (FSS): trunking, backhaul, enterprise networks, video contribution | Telecom operators, broadcasters, businesses | C, Ku, Ka | SES, Intelsat, Eutelsat, APT, Telesat |
| Broadcasting Satellite Service (BSS): direct-to-home TV and radio | Households | Ku (TV), S/L (radio) | SES Astra, DirecTV, Dish, Sky, SiriusXM |
| Satellite broadband | Homes, businesses, communities, governments | Ku, Ka, Q/V, E | Starlink, OneWeb, Amazon Leo, Viasat, Hughes |
| Mobility: maritime and aeronautical connectivity | Ships, oil rigs, airlines, business jets | L, Ku, Ka | Inmarsat (Viasat), Panasonic, Starlink, OneWeb |

| Service | Typical users | Main bands | Example systems |
|---|---|---|---|
| Mobile Satellite Service (MSS): handheld voice and data, IoT | Remote workers, emergency services, militaries | L, S | Iridium, Globalstar, Inmarsat, Thuraya |
| Direct-to-device (D2D) | Ordinary smartphones | Terrestrial cellular bands, S, L | Starlink Direct to Cell, AST SpaceMobile, Lynk, Apple–Globalstar |
| Government and military satellite communications | Armed forces, civil protection | UHF, X, Ka, EHF | WGS, AEHF/ESS, Skynet, Syracuse (Chapter 19) |
| Data relay | Other spacecraft (ISS, EO satellites) | S, Ku, Ka, optical | TDRS, EDRS, Tianlian, Luch |

Broadcasting was long the most profitable segment, but linear satellite TV is declining as viewers move to internet streaming, while satellite broadband — especially LEO broadband — is growing rapidly. Starlink alone reported about 12 million customers in more than 160 countries by June 2026.

## 17.3 Choosing an Orbit

The orbit determines coverage, latency, the number of satellites needed and the cost of terminals (Chapter 11).

- **Geostationary orbit (GEO)** allows a satellite to cover about a third of the Earth continuously, and simple fixed dishes to point at it. Three satellites cover the world except the polar regions. The penalty is distance: radio signals take about a quarter of a second to travel up and down. So a round trip through a GEO satellite (user → satellite → gateway → satellite → user) takes at least 477 ms, plus processing. This delay is noticeable in conversations and online gaming. The large distance also requires big antennas and high power.
- **Medium Earth orbit (MEO)** offers a compromise. SES's **O3b** and **O3b mPOWER** satellites, in an 8 062 km equatorial orbit, provide fibre-like links with round-trip latency of about 110–150 ms to cruise ships, mobile operators and governments.
- **Low Earth orbit (LEO)** at 500–1 200 km gives latencies comparable with terrestrial networks (Figure 17.5) and allows smaller terminals. But each satellite is visible from a given place for only a few minutes. So hundreds to thousands of satellites, steerable user antennas and handovers between satellites are required.
- **Highly elliptical orbits (HEO)** such as Molniya and Tundra serve high latitudes. Space Norway's two Arctic Satellite Broadband Mission satellites (2024), carrying military and commercial payloads, use such orbits to provide continuous coverage of the Arctic.

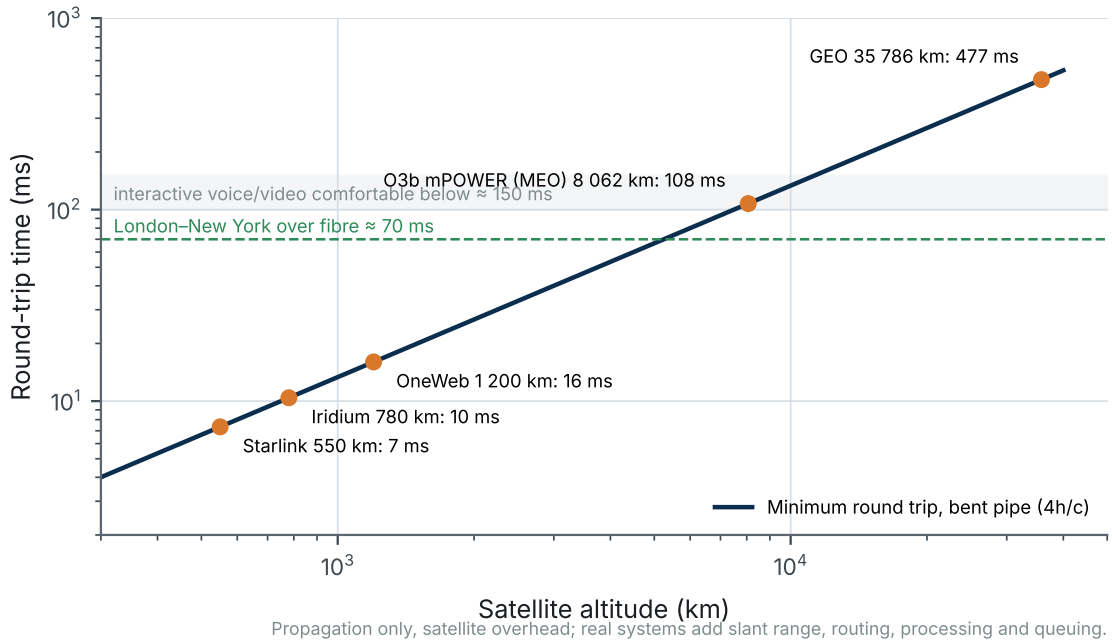



**FIGURE 17.5** Minimum round-trip propagation time through a bent-pipe satellite versus altitude. LEO systems approach the latency of terrestrial fibre.

## 17.4 Frequencies, Propagation and Spectrum Sharing

### 17.4.1 Frequency Bands

Satellite communications use bands from VHF to millimetre waves and, increasingly, optical wavelengths (Chapter 7). Lower bands (L, S) propagate well, penetrate foliage and work with small, non-directional antennas — ideal for mobile services — but offer little bandwidth. Higher bands (Ku, Ka, Q/V, E) offer gigahertz of bandwidth and allow small, high-gain antennas, but suffer from rain. The industry has migrated steadily upwards: C-band for early trunking, Ku-band for television from the 1980s, Ka-band for broadband from the 2000s, and Q/V- and E-band for gateway links in the newest systems.

### 17.4.2 Rain Fade

Above about 10 GHz, raindrops absorb and scatter radio waves. The **specific attenuation** (dB/km) grows rapidly with frequency and rain rate (Figure 17.6). A heavy tropical downpour of 100 mm/h can attenuate a Ka-band signal by 30–50 dB — enough to break any link. The same rain costs only a few dB at Ku-band and almost nothing at C-band. Because heavy rain is rare and localised, systems are designed for a specified **availability** (for example 99.7 % of the average year, about 26 hours of outage). They use statistical rain models from ITU-R Recommendations, and they combat fades with:

- **adaptive coding and modulation (ACM)**, which switches to more robust waveforms when the link degrades (standardised in DVB-S2 and DVB-S2X);
- **uplink power control**, which raises the transmitted power during a fade;
- **site diversity**, using two gateways several tens of kilometres apart, since intense rain cells rarely cover both.

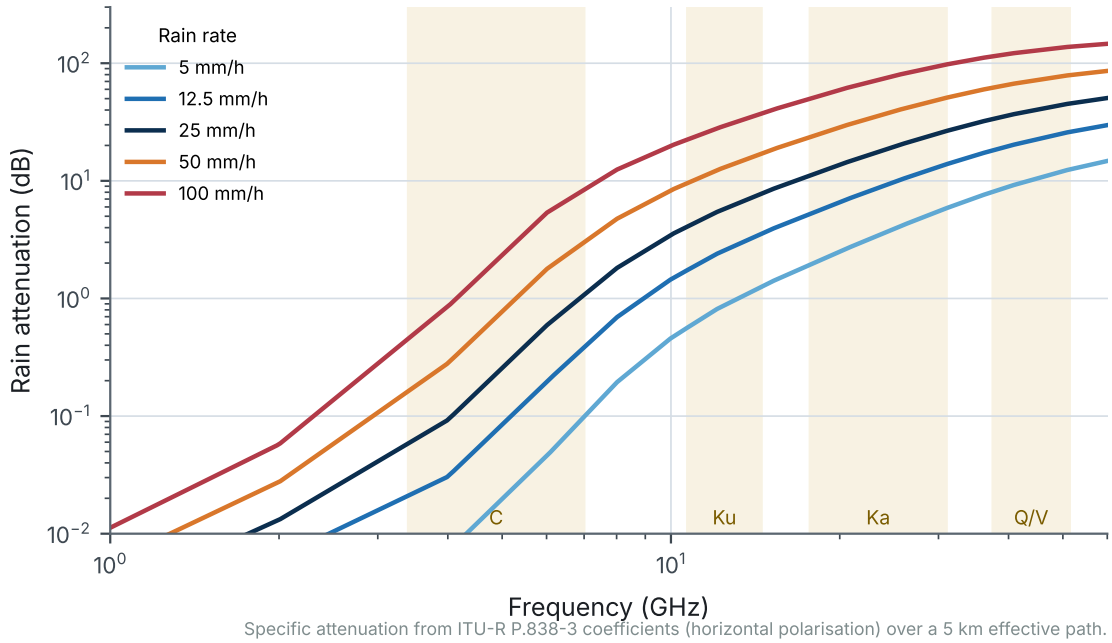



**FIGURE 17.6** Rain attenuation versus frequency for different rain rates. Attenuation increases dramatically above about 10 GHz.

### 17.4.3 Sharing the Spectrum and the Orbits

Radio spectrum and orbital positions are limited natural resources shared by all nations. The **International Telecommunication Union (ITU)**, a United Nations agency, maintains the **Radio Regulations** — an international treaty that allocates frequency bands to services — and the **Master International Frequency Register**. A satellite operator, through its national administration, must file its planned network with the ITU, **coordinate** it with earlier networks that might suffer interference, and bring it into use within seven years. Geostationary positions are typically spaced 2–3° apart, and coordination between neighbouring operators is a complex negotiation. Every three to four years a **World Radiocommunication Conference** (WRC-23 in Dubai; WRC-27 next) revises the Regulations — for example to allocate spectrum for direct-to-device services or to set rules for lunar communications.

Non-geostationary constellations must protect GEO networks. ITU **equivalent power-flux-density (EPFD)** limits cap the interference that their combined satellites may cause to GEO terminals. In practice, this means that a LEO system must not transmit along the line of sight between a ground terminal and the geostationary arc (Figure 17.7). Large constellations must also meet **milestones**. Since 2019, ITU rules require 10 % of a constellation to be deployed within two years of the end of the seven-year bring-into-use period. They also require 50 % within five years and 100 % within seven years. National licences add their own conditions. In the USA, for example, the FCC required Amazon's constellation to have half of its satellites (about 1 600) in orbit by July 2026. With roughly 300–400 in orbit, Amazon obtained a conditional waiver in 2026.

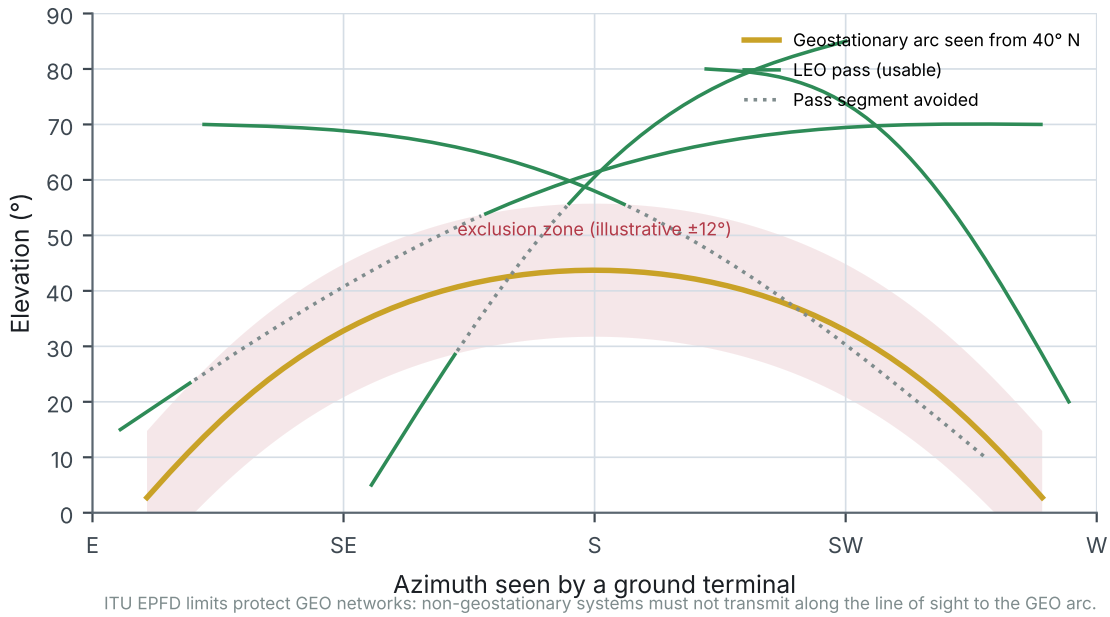



**FIGURE 17.7** A ground terminal's view of the sky. To protect geostationary networks, LEO systems avoid using satellites that appear near the geostationary arc (illustrative exclusion zone).

## 17.5 Link Budgets and Terminals

The link-budget method of Chapter 7 determines whether a link closes. Figure 17.8 shows the budget of a typical Ku-band direct-to-home television downlink as a "waterfall". A satellite EIRP of 52 dBW is attenuated by a free-space loss of about 206 dB over 38 000 km. A 60 cm dish with a low-noise block converter recovers about 36 dB of gain against a system noise temperature of about 120 K. In a 27 MHz channel, the resulting carrier-to-noise ratio is about 15 dB. This is enough for the DVB-S2 8PSK rate-3/4 waveform, with a margin of more than 6 dB for rain.

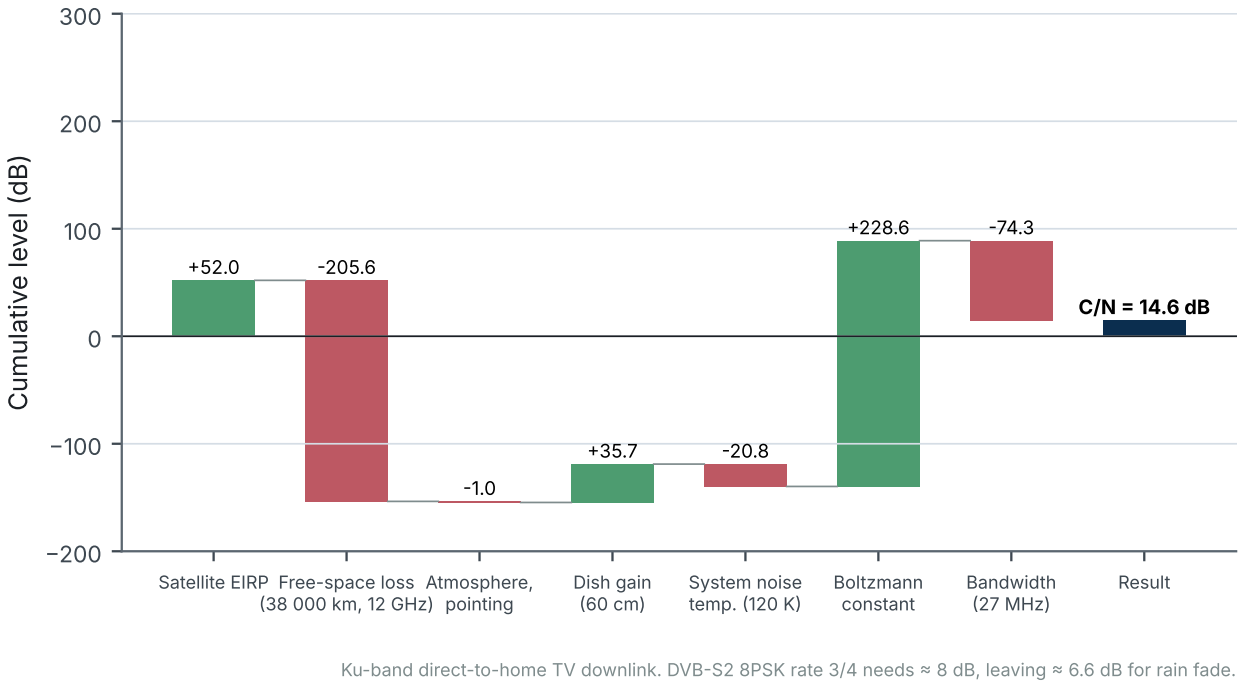



**FIGURE 17.8** Link budget of a Ku-band direct-to-home television downlink, shown as a waterfall of gains and losses.

Modern systems adapt their waveforms continuously. The **DVB-S2X** standard offers more than 100 combinations of modulation (from QPSK to 256APSK) and coding, achieving spectral efficiencies from below 0.5 to more than 5 bit/s/Hz, within about 1 dB of the Shannon limit. Since 2022 the 3GPP mobile standards (Release 17 onward) have included **non-terrestrial networks (NTN)**, allowing satellites to be integrated into 5G — the basis for direct-to-device services.

### 17.5.1 User Terminals

Terminals have changed as much as satellites (Figure 17.9). The 60 cm dish for satellite TV, fixed to a wall and pointed at a GEO satellite, costs a few tens of dollars. A LEO broadband system needs an antenna that follows satellites across the sky and switches between them every few minutes. The breakthrough was the low-cost **electronically steered phased array**. Starlink's flat terminal contains about a thousand or more antenna elements. Their signals are phase-shifted by custom chips to form and steer beams with no moving parts. Through mass production, the terminal's cost fell from thousands of dollars to a few hundred. Aircraft and ships use larger flat-panel or mechanically steered terminals.

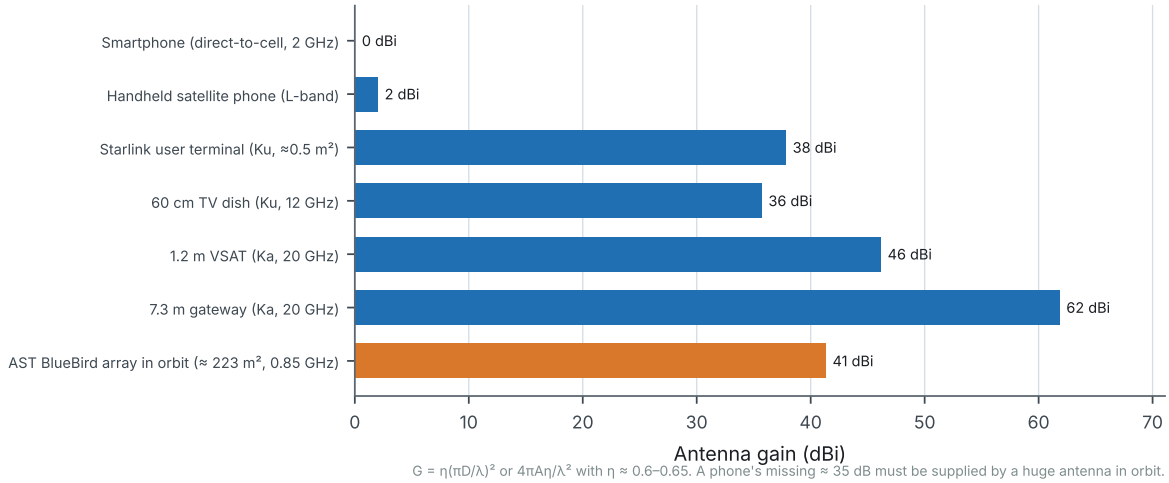



**FIGURE 17.9** Antenna gain of typical terminals. Direct-to-device service must close the link with a smartphone's near-isotropic antenna, so the satellite antenna must be enormous.

## 17.6 The Communication Payload

### 17.6.1 Transponders: Bent Pipe and Regenerative

The heart of a classic communication satellite is the **transponder** (Figure 17.10). In a **bent-pipe** (transparent) design, the receive antenna collects the uplink signals and a low-noise amplifier boosts them. Filters and frequency converters then shift them from the uplink to the downlink band (for example from 14 GHz to 12 GHz at Ku-band). An **input multiplexer** divides the band into channels, typically 36–72 MHz wide. Each channel is amplified by a **travelling-wave-tube amplifier (TWTA)** or a solid-state amplifier. Finally, an **output multiplexer** recombines the channels for the transmit antenna. A bent pipe does not care what signals it carries, which makes it flexible and long-lived — a satellite launched for analogue television can carry digital broadband 15 years later.

A **regenerative** payload demodulates and decodes the uplink signals, switches or routes the data, and re-modulates it for the downlink. Regeneration improves link quality (uplink and downlink noise no longer add), enables on-board routing between beams and inter-satellite links, and is essential for networks such as Iridium and Starlink. **Digital transparent processors** sit between the two: they digitise the band and route and filter channels flexibly without decoding them.

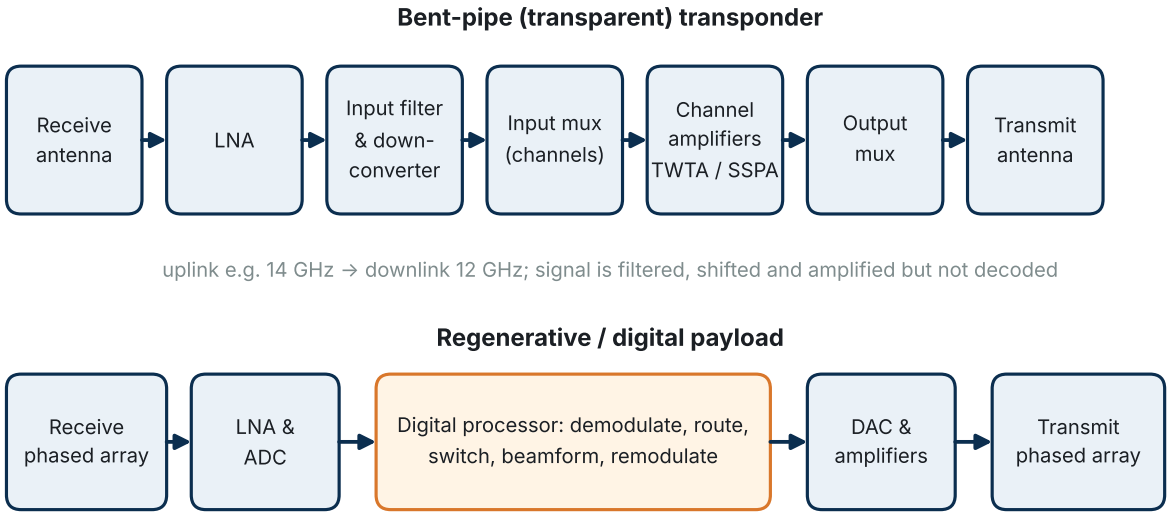



**FIGURE 17.10** Architecture of bent-pipe and regenerative communication payloads.

TWTAs remain the workhorse high-power amplifiers in space because of their efficiency (60–70 %) and power (tens to hundreds of watts per tube). Gallium-nitride solid-state amplifiers are increasingly used, especially in phased arrays where many small amplifiers are distributed over the antenna.

## 17.6.2 Antennas, Spot Beams and Frequency Reuse

Early satellites used a single **global beam** or a few regional beams. Capacity is limited by bandwidth, so the key to more capacity is to **reuse** the same frequencies many times. **Multi-beam antennas** — large reflectors fed by clusters of horns, or phased arrays — form tens to hundreds of narrow **spot beams**, each a few hundred kilometres wide. Adjacent beams use different frequency sub-bands or polarisations; beams that are far enough apart reuse the same ones. In the classic **four-colour** scheme (Figure 17.11), the available spectrum is split into two sub-bands and two polarisations. So each beam uses half the spectrum, and the whole band is reused about a quarter as many times as there are beams. A satellite with 100 beams thus reuses its spectrum about 25 times. Smaller beams also have higher antenna gain, so users can have smaller terminals and higher data rates.

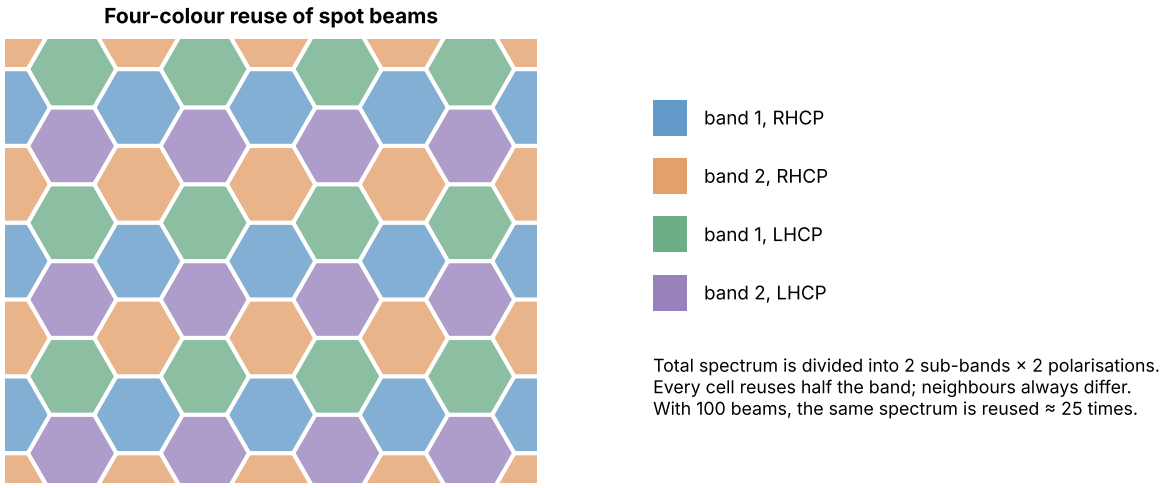



**FIGURE 17.11** Four-colour frequency reuse with spot beams. Neighbouring cells always use different sub-band and polarisation combinations.

### 17.6.3 High-Throughput Satellites

Applying this principle at Ka-band produced **high-throughput satellites (HTS)**. Eutelsat's KA-SAT (2010) had 82 spot beams and 90 Gbit/s of capacity; Viasat's ViaSat-1 (2011) offered 140 Gbit/s — more than all other satellites over North America combined at the time. Capacity per satellite has grown roughly a hundred-thousand-fold since Early Bird (Figure 17.12). Hughes' Jupiter 3 (EchoStar XXIV, 2023) provides about 500 Gbit/s. Viasat's **ViaSat-3** class was designed for more than 1 Tbit/s per satellite. But the first satellite's huge deployable reflector failed to deploy fully in 2023, sharply reducing its capacity. This is a reminder of the risks of large deployable structures (Chapter 5). The second ViaSat-3 was launched in late 2025.

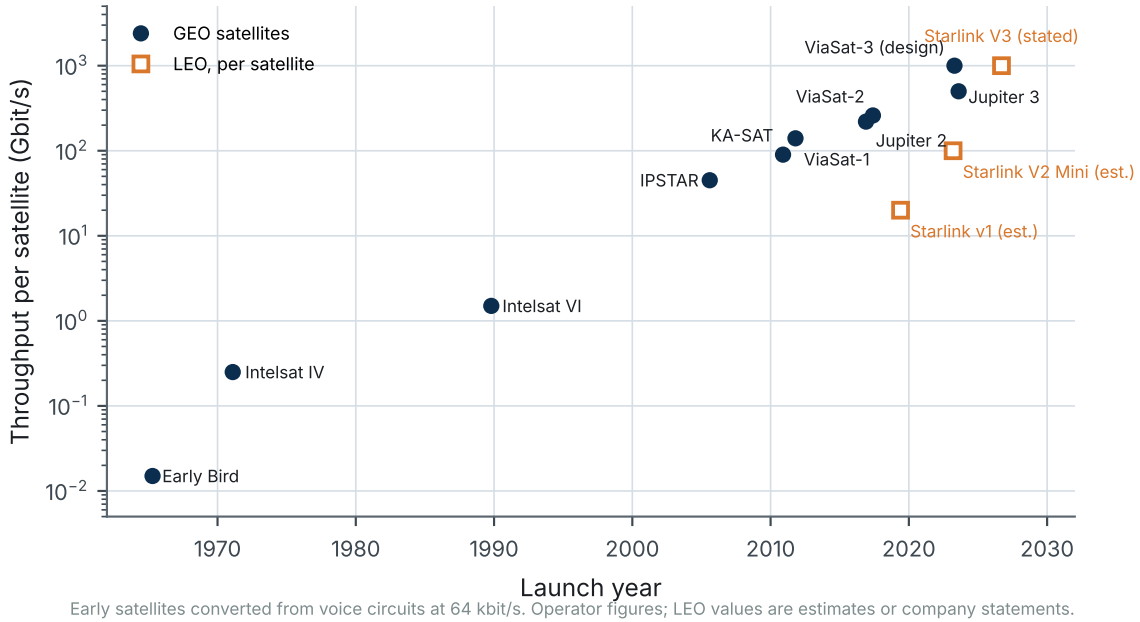



**FIGURE 17.12** Throughput per satellite from Early Bird (1965) to modern HTS in GEO and to individual LEO broadband satellites.

### 17.6.4 Software-Defined Satellites

A GEO satellite must serve markets for 15 years that cannot be predicted when it is ordered. **Software-defined** or **flexible** satellites include Eutelsat Quantum (2021), Airbus OneSat, Thales Alenia Space's Space Inspire and Boeing's 702X. They use digital processors and active phased-array antennas, so that coverage, beam shapes, frequencies and power can be reconfigured in orbit. This lets them follow demand, or even follow a moving ship or aircraft. Standardised designs also shorten manufacturing times from about three years to about 18 months. At the other end of the scale, start-ups such as **Astranis** build small (about 400 kg) "MicroGEO" satellites that serve a single customer or country with a dedicated software-defined payload.

### 17.6.5 The Geostationary Industry in Transition

The rise of LEO constellations has put pressure on traditional GEO operators. Orders for large commercial GEO satellites fell from 20–25 per year in the 2000s and early 2010s to around ten or fewer in the 2020s. Operators have also consolidated: Eutelsat merged with OneWeb (2023), Viasat acquired Inmarsat (2023), and SES completed its acquisition of Intelsat in July 2025. Most GEO operators now offer "multi-orbit" services combining their GEO capacity with LEO or MEO partners.

## 17.7 LEO Broadband Constellations

### 17.7.1 Earlier Attempts

LEO communication constellations are not new. Iridium (66 satellites, 1998), Globalstar (48 satellites, 1999) and Orbcomm (1990s) were built for mobile telephony and messaging, and Teledesic planned a constellation of 840 broadband satellites in the 1990s. Iridium and Globalstar went bankrupt shortly after entering service — undercut by cheaper terrestrial mobile networks — although both later recovered and replaced their constellations. What changed in the 2010s was the cost of building and launching satellites, the availability of cheap phased arrays, and soaring demand for broadband.

### 17.7.2 Starlink

SpaceX's **Starlink** has become by far the largest satellite system ever built (Figure 17.13, Figure 17.14). Its first 60 production-type satellites were launched in May 2019; by mid-October 2026 about 11 000 were in orbit, launched mostly by SpaceX's own reusable Falcon 9 rockets in batches of 20–30 satellites. Key features of its design include:

- **Flat-panel satellites** stacked in the fairing and released together. Each has phased-array antennas for user beams (Ku-band), parabolic or phased-array antennas for gateways (Ka- and E-band), a single large solar array and **Hall-effect thrusters**. The thrusters use krypton or argon for orbit raising, station-keeping and de-orbiting.
- **Low altitudes** — mostly 480–560 km, in "shells" at inclinations of 43°, 53°, 70° and 97.6° (Figure 17.15). At these altitudes, failed satellites re-enter within about five years by atmospheric drag, which limits long-term debris; in 2026 SpaceX began lowering many satellites towards about 480 km.

- **Laser inter-satellite links** on all recent satellites, forming an optical mesh network that carries traffic across oceans and to places without nearby gateways (Figure 17.16).
- **Rapid iteration**: successive versions (v0.9, v1.0, v1.5, V2 Mini) have increased capacity per satellite. The much larger **V3** satellites, designed for about 1 Tbit/s of downlink capacity each, require the Starship launch vehicle; the first V3 satellites were deployed on Starship's fourteenth flight in September 2026.
- **Direct to Cell** satellites with large antennas that connect to ordinary phones (see below).

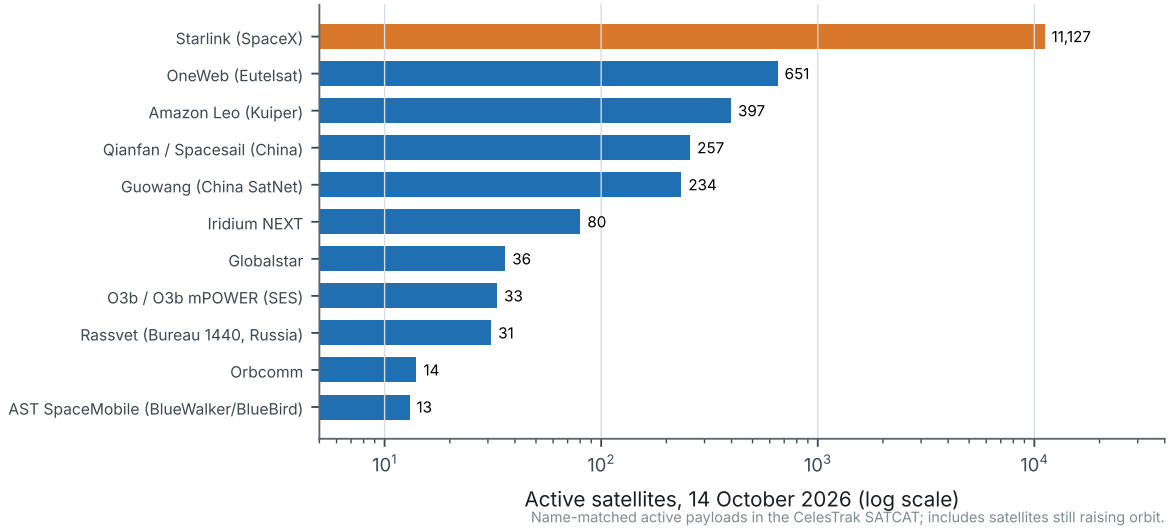



**FIGURE 17.13** Active satellites in the main broadband and mobile communication constellations in October 2026 (logarithmic scale).

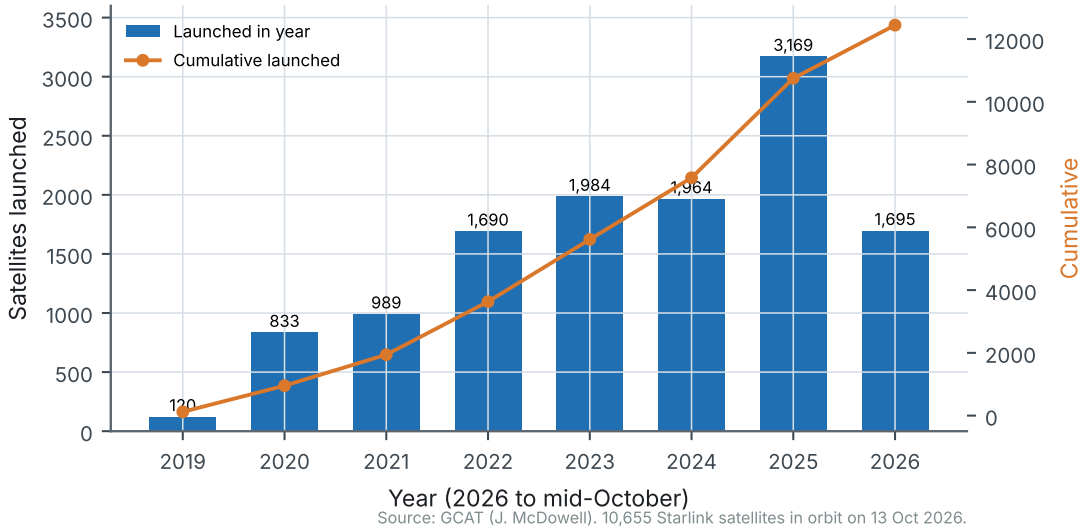
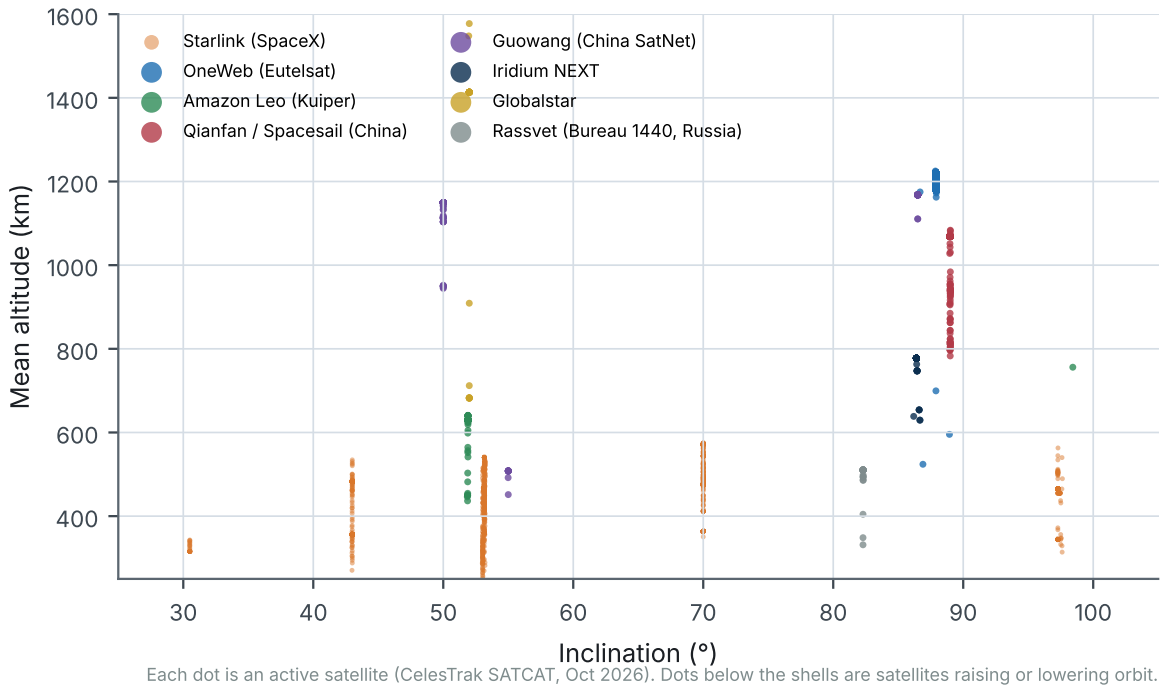



**FIGURE 17.14** Starlink satellites launched per year and cumulatively, 2019 to mid-October 2026.




Each dot is an active satellite (CelesTrak SATCAT, Oct 2026). Dots below the shells are satellites raising or lowering orbit.

**FIGURE 17.15** Orbital shells of the main LEO communication constellations, plotted from the October 2026 satellite catalogue. Each constellation uses a few distinct combinations of altitude and inclination.

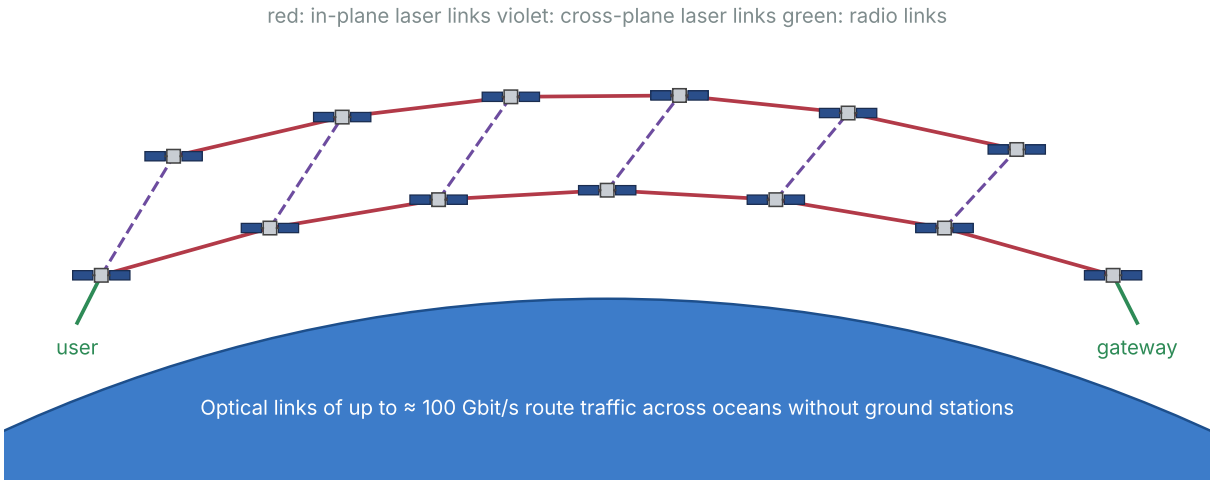



**FIGURE 17.16** Laser inter-satellite links create an optical mesh network in space, routing traffic between distant users and gateways.

## 17.7.3 Other Constellations

- **OneWeb** (Eutelsat): about 650 satellites at 1 200 km in near-polar orbits (87.9°), built with Airbus. Rather than selling directly to consumers, OneWeb serves telecom operators, governments, enterprises and mobility providers. A second generation is being procured.
- **Amazon Leo** (formerly Project Kuiper): authorised for 3 236 satellites in shells at 590–630 km. Amazon began full-scale launches in April 2025, using Atlas V, Falcon 9, Vulcan and other rockets, and renamed the system in November 2025. About 400 satellites were in orbit by October 2026; launch-vehicle shortages, including the loss of a New Glenn rocket in a pad explosion in May 2026, slowed deployment.

- **China's Guowang** (China Satellite Network Group, about 13 000 satellites planned) and **Qianfan** (Spacesail, about 14 000 planned) began deploying in 2024; each had more than 200 satellites in orbit by October 2026.
- **Telesat Lightspeed** (Canada): about 200 satellites built by MDA in polar and inclined orbits at about 1 000–1 300 km, serving enterprise and government customers, with launches from 2026.
- **IRIS²** (EU): a multi-orbit secure-connectivity constellation of roughly 300 satellites in LEO and MEO for governmental and commercial services, with first launches expected around 2029–2030.
- **Rassvet** (Bureau 1440, Russia) began launching in 2025.

**TABLE 17.2** Major non-geostationary communication constellations (approximate).

| System | Operator | Satellites (planned / active Oct 2026) | Altitude | Frequency bands | Market |
|---|---|---|---|---|---|
| Starlink | SpaceX (USA) | ≈ 12 000 authorised (Gen 1), up to ≈ 30 000 Gen 2 proposed / ≈ 11 000 | 340–570 km | Ku, Ka, E; PCS/AWS for D2C | Consumer, enterprise, mobility, government, D2C |
| OneWeb | Eutelsat (France/UK) | 648 Gen 1 / ≈ 650 | 1 200 km | Ku, Ka | Wholesale, enterprise, government |
| Amazon Leo | Amazon (USA) | 3 236 / ≈ 400 | 590–630 km | Ka | Consumer, enterprise, government |
| Guowang | China SatNet | ≈ 13 000 / ≈ 230 | 500–1 150 km | Ku, Ka, Q/V | National broadband |
| Qianfan (Spacesail) | Shanghai SSST | ≈ 14 000 / ≈ 260 | ≈ 800–1 100 km | Ku, Ka | National and international broadband |
| Lightspeed | Telesat (Canada) | 198 / first launches 2026 | ≈ 1 000–1 300 km | Ka | Enterprise, government |
| Iridium NEXT | Iridium (USA) | 66 + spares / ≈ 80 | 780 km | L, Ka | Mobile voice, IoT, safety |
| O3b mPOWER | SES (Luxembourg) | 13 / 13 | 8 062 km (MEO) | Ka | Enterprise, cruise, government |
| IRIS² | EU | ≈ 300 (LEO + MEO) / — | LEO and MEO | Ka and others | Government and commercial |

### 17.7.4 Challenges

Mega-constellations raise new problems. Their satellites are visible in astronomical images as streaks, and their radio emissions can interfere with radio telescopes. Operators have darkened satellites, adjusted orientations and coordinated with observatories. Even so, the effect on wide-field astronomy is significant (Chapter 38). The sheer number of satellites makes **collision avoidance** a continuous activity. Starlink satellites perform tens of thousands of automated avoidance manoeuvres per year. The number of satellites also raises concerns about the long-

term sustainability of LEO (Chapter 21). The re-entry of thousands of satellites per year is also adding aluminium oxide to the upper atmosphere, a possible environmental effect now being studied.

### 17.7.5 The Economics of a Constellation

A LEO broadband system is a capital-intensive bet. Building, launching and operating several thousand satellites, plus gateways and user terminals, costs on the order of $10–30 billion before the system covers its costs, and satellites must be replaced every five to seven years. Three factors dominate the economics:

1. **Cost per satellite and per launch.** Starlink, OneWeb and Amazon all built factories producing several satellites per day on assembly lines. This mass production has reduced satellite costs to a few hundred thousand to a few million dollars each. Reusable launchers have also cut launch costs per kilogram by a factor of five to ten (Chapter 34).
2. **Capacity per satellite.** Because most of the fleet flies over oceans and empty land at any moment, revenue comes only from the fraction over paying customers. Larger satellites with more capacity improve this ratio dramatically, which is why SpaceX's V3 satellites depend on Starship.
3. **Terminal cost.** A phased-array terminal that costs more to make than the customer pays for it must be subsidised. Driving terminal costs down from thousands to a few hundred dollars was essential to reaching consumers.

Vertical integration — one company building the satellites, the rockets, the terminals and the network — gave SpaceX a decisive cost advantage. Governments increasingly view sovereign broadband constellations (China's Guowang and Qianfan, the EU's IRIS$^2$) as strategic infrastructure, not just commercial ventures.

### 17.7.6 Satellite IoT and Machine-to-Machine Services

A separate class of constellations serves the **Internet of Things (IoT)**: tracking containers, livestock, wildlife, pipelines, ships and environmental sensors in remote areas with very small, low-power devices. Systems include Orbcomm, Iridium's short-burst data, Kinéis (France, 25 nanosatellites launched in 2024, continuing the Argos system used for wildlife tracking since 1978), Astrocast, Myriota and Swarm (acquired by SpaceX). Some use standard LoRa or NB-IoT protocols so that cheap terrestrial chips can talk to satellites. Data rates are tiny — bytes to kilobytes per day — but the number of devices could reach tens of millions.

## 17.8 Mobile Satellite Services and Direct-to-Device

### 17.8.1 Classic Mobile Satellite Systems

Mobile satellite services use L- and S-band frequencies, which suit small, non-directional antennas. **Inmarsat** (founded in 1979 for maritime safety, now part of Viasat) provides voice, data and safety services to ships and aircraft from GEO, including the Global Maritime Distress and Safety System. **Iridium**'s 66 cross-linked LEO satellites at 780 km in six near-polar planes cover

the entire Earth, including the poles. They serve handheld phones, IoT devices and aircraft tracking. The second-generation **Iridium NEXT** constellation was completed in 2019. **Globalstar** uses bent-pipe LEO satellites and, since 2022, provides emergency messaging for Apple iPhones.

### 17.8.2 Direct-to-Device

The newest frontier is connecting **unmodified smartphones** directly to satellites, using the phones' own cellular frequencies. The challenge is the link budget (Figure 17.9): a phone has an antenna gain of about 0 dBi and transmits only about 0.2 W, compared with a broadband terminal's 30–40 dBi. The missing gain must be provided by the satellite — with very large antennas — and by modest data rates.

- **SpaceX** launched its first Direct to Cell satellites in January 2024, and by 2025 several hundred were in orbit. T-Mobile's **T-Satellite** text and data service launched commercially in the USA in July 2025. Partners in Canada, Japan, Australia, Switzerland, Latin America and elsewhere followed. In 2025 SpaceX agreed to acquire mobile-satellite spectrum from EchoStar to expand the service.
- **AST SpaceMobile** builds satellites with enormous phased-array antennas: about 64 $m^2$ on its first five BlueBirds (2024) and about 223 $m^2$ on the second-generation BlueBirds. These are the largest commercial communication arrays ever flown. The satellites are designed to deliver broadband speeds of up to about 120 Mbit/s per cell to ordinary phones, in partnership with AT&T, Verizon, Vodafone and others. Thirteen BlueBirds had been launched by August 2026.
- **Lynk Global**, **Skylo** (using existing GEO satellites for messaging) and Chinese providers (Tiantong) offer other approaches. The 3GPP standards now include satellite access, so future phones and networks will treat satellites as just another cell.

## 17.9 Gateways, Teleports and Networks

Satellites are only part of a communication system. **Gateways** (also called teleports or ground stations) connect the satellite network to the terrestrial internet and telephone networks. A GEO HTS may need 10–50 gateways spread over its coverage area, each with 7–9 m Ka-band antennas and fibre connections, located far enough apart that rain fades are uncorrelated. A LEO constellation needs many more — Starlink operates well over a hundred gateway sites worldwide, each with several tracking antennas — although laser links reduce the need for gateways in remote areas. Network operations centres control the satellites, allocate capacity, manage subscribers and monitor quality of service, and increasingly the whole system is managed as software, with virtualised network functions running in commercial clouds.

Satellite networks are also integrating with terrestrial networks. Mobile operators use satellite **backhaul** to connect cell towers in remote areas. Airlines and shipping companies combine LEO, GEO and cellular links on the same terminal. The 5G non-terrestrial-network standards also allow a satellite to act as part of a mobile operator's radio-access network. The long-term vision is a seamless network in which users neither know nor care whether their data travel through fibre, a cell tower or a satellite.

**TABLE 17.3** Selected high-throughput satellites (approximate design capacities).

| Satellite (operator) | Launch | Orbit | Throughput | Notes |
|---|---|---|---|---|

| Satellite (operator) | Launch | Orbit | Throughput | Notes |
|---|---|---|---|---|
| KA-SAT (Eutelsat) | 2010 | GEO 9° E | ≈ 90 Gbit/s | 82 Ka-band spot beams over Europe |
| ViaSat-1 (Viasat) | 2011 | GEO | ≈ 140 Gbit/s | First HTS above 100 Gbit/s |
| Intelsat EpicNG series | 2016– | GEO | 25–60 Gbit/s each | Ku/Ka multi-spot, open architecture |
| ViaSat-2 (Viasat) | 2017 | GEO | ≈ 260 Gbit/s | Antenna deployment problem reduced capacity |
| Jupiter 3 / EchoStar XXIV (Hughes) | 2023 | GEO 95° W | ≈ 500 Gbit/s | One of the heaviest commercial GEO satellites (≈ 9 t) |
| ViaSat-3 F1, F2 (Viasat) | 2023, 2025 | GEO | designed > 1 Tbit/s each | F1 reflector partially deployed |
| O3b mPOWER (SES) | 2022– | MEO | multi-Tbit/s system | Up to ≈ 5 000 beams per satellite |

## 17.10 Optical Communications and Data Relay

### 17.10.1 Laser Links

Optical communications offer far higher data rates than radio with much smaller antennas and no spectrum licensing (Chapter 7). Inter-satellite laser links are now routine in LEO constellations: Starlink operates thousands of laser terminals carrying up to about 100 Gbit/s per link. The **European Data Relay System (EDRS)** has relayed Sentinel data from LEO to Europe via GEO at 1.8 Gbit/s since 2016. NASA's **Laser Communications Relay Demonstration (LCRD)** (2021) and the **ILLUMA-T** terminal on the ISS (2023) demonstrated an end-to-end laser relay from the space station through GEO to the ground. The TBIRD CubeSat (2022) downlinked at 200 Gbit/s. The main limitation of space-to-ground laser links is cloud cover, which is overcome with networks of optical ground stations in dry locations.

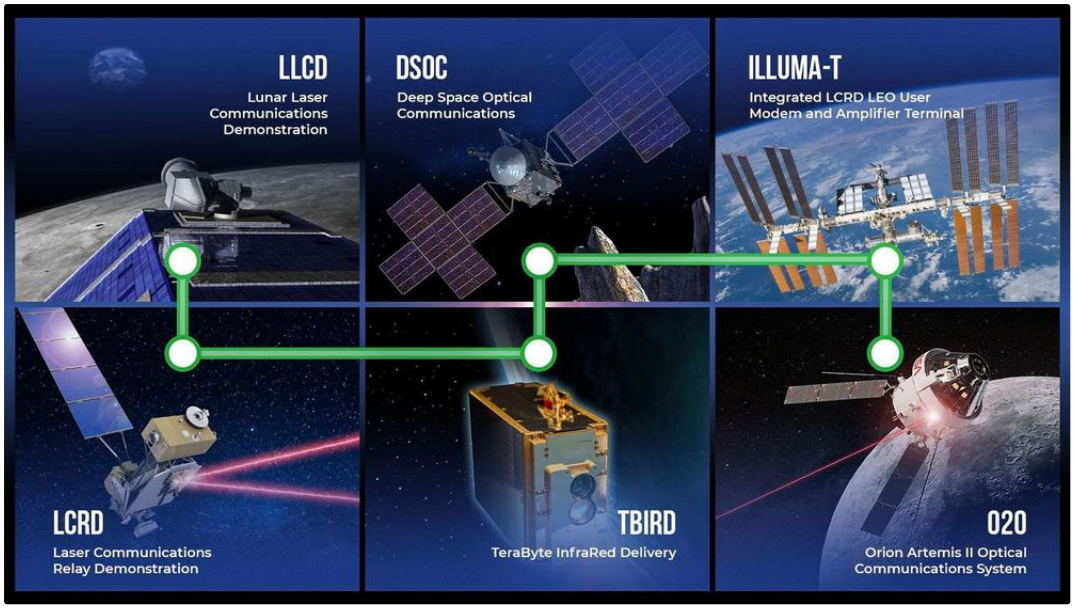



**FIGURE 17.17** NASA's optical communications demonstrations: the Lunar Laser Communications Demonstration (2013), Deep Space Optical Communications on Psyche (2023), ILLUMA-T on the ISS, LCRD in geostationary orbit, TBIRD and the Orion Artemis II optical system. Image: NASA/JSC.

## 17.10.2 Data Relay Satellites

Low-orbiting spacecraft see a given ground station for only about 10 minutes per pass. **Data relay satellites** in GEO provide near-continuous contact. NASA's **Tracking and Data Relay Satellite System (TDRSS)** has operated since 1983. It has supported the Space Shuttle, the ISS, the Hubble Space Telescope and many science missions. TDRS-M, launched in 2017 (the chapter's opening image), was the last of the series. NASA plans to retire TDRS in the 2030s and buy relay services from commercial providers instead, and in 2022 it awarded development agreements to several companies for this purpose. China operates the Tianlian relay satellites and Russia the Luch series; ESA's EDRS uses laser links.

### EXAMPLE: HOW MANY SATELLITES DOES A LEO BROADBAND SYSTEM NEED?

*Coverage.* A satellite at 550 km serving users at elevations above 25° covers a circle of Earth-central half-angle

$$\lambda = \arccos\left(\frac{R_E \cos\varepsilon}{R_E + h}\right) - \varepsilon = \arccos\left(\frac{6371\cos 25^\circ}{6921}\right) - 25^\circ \approx 8.4^\circ$$

so each footprint covers about 2.8 million km² — about 0.54 % of the Earth's surface. Continuous single coverage of the whole Earth therefore needs at least about 190 satellites, and in practice, with overlap and uneven distribution, 300–600.

*Capacity.* Suppose 10 million subscribers in the busiest regions each need an average of 3 Mbit/s at the evening peak, after statistical multiplexing: a total demand of 30 Tbit/s. Only satellites above populated areas carry this traffic, perhaps 10–15 % of the fleet at any moment. With 100 Gbit/s per satellite, about 300 satellites must be over the busy regions — implying a fleet of 2 000–3 000 satellites. Doubling the subscriber base or the demand per user doubles the fleet unless capacity per satellite grows.

This is why broadband constellations are sized by **capacity**, not coverage, and why capacity per satellite (and the launch vehicle needed to carry larger satellites) is the key to their economics.

### AI IN SATELLITE COMMUNICATIONS

Modern satellite networks are too complex to manage by hand. Machine learning predicts traffic demand and rain fades, allocates beams, frequencies and power dynamically on software-defined satellites, routes traffic through laser meshes, detects interference and jamming, and plans collision-avoidance manoeuvres for thousands of satellites. Chapter 49 discusses AI for satellite operations and networks.

### KEY TAKEAWAYS

- Communication satellites evolved from passive balloons (Echo, 1960) through GEO trunking and broadcasting to high-throughput GEO satellites and LEO broadband constellations.
- Orbit choice trades coverage and simplicity (GEO) against latency and terminal size (LEO); MEO and HEO fill specific niches.
- Higher frequencies give more bandwidth but suffer rain fade; systems use adaptive coding and modulation, power control and site diversity.
- The ITU allocates spectrum and coordinates orbital positions; EPFD limits protect GEO from LEO interference; milestone rules require constellations to be deployed on time.
- Capacity comes from frequency reuse with many spot beams; HTS increased GEO capacity per satellite to 0.5–1 Tbit/s.
- Starlink (about 11 000 satellites in October 2026 and about 12 million customers) has transformed the market; OneWeb, Amazon Leo, Guowang, Qianfan and others follow.
- Direct-to-device services connect ordinary phones by using very large antennas in orbit and 3GPP non-terrestrial-network standards.
- Laser links are now routine between satellites; data-relay satellites keep low-orbit spacecraft in near-continuous contact.

### REVIEW QUESTIONS

1. Trace the evolution of communication satellites from Echo to LEO broadband constellations and give one key innovation at each stage.
2. Calculate the one-way propagation delay of a link through a geostationary satellite (two legs of 38 000 km) and through a 550 km LEO satellite (two legs of 1 000 km).
3. Why does rain fade increase with frequency? Describe three techniques used to keep Ka-band links available.
4. What are EPFD limits and ITU milestone rules, and why do they matter for new constellations?
5. Compare bent-pipe and regenerative transponders.
6. Explain how frequency reuse with many spot beams increases the capacity of a high-throughput satellite.
7. Why do direct-to-device services need very large antennas in orbit?
8. How do laser inter-satellite links and data-relay satellites change the design of a LEO constellation's ground segment?

CHAPTER

# 18 Global Navigation Satellite Systems

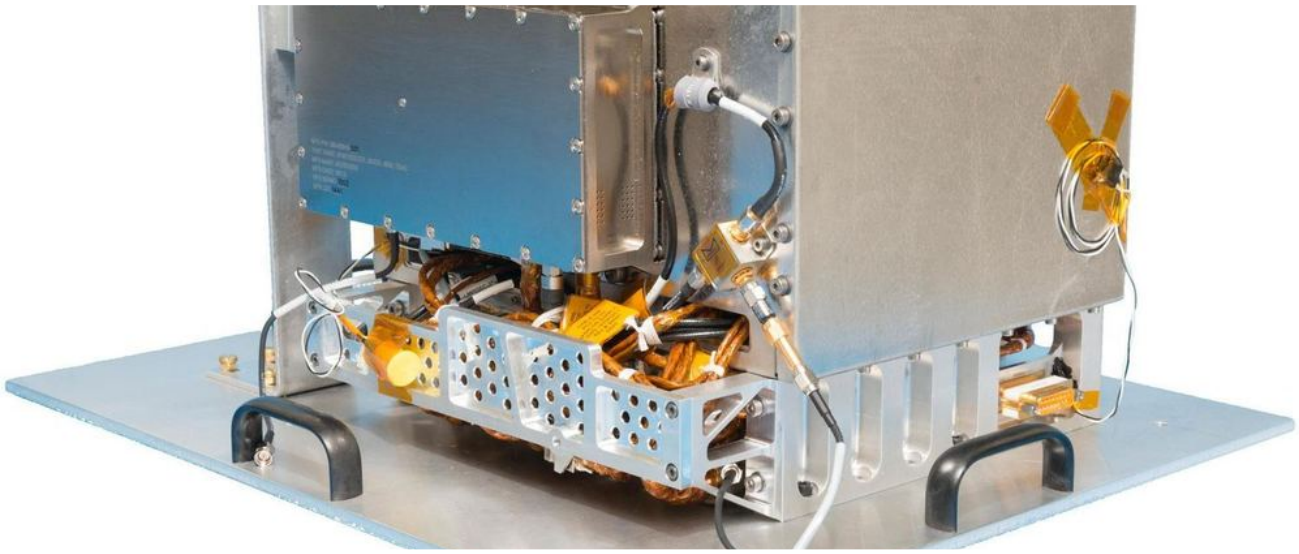


Image: NASA/JPL (NASA/JPL-Caltech)

*Global navigation satellite systems (GNSS) — GPS, GLONASS, Galileo and BeiDou, with regional systems in Japan and India — broadcast precisely timed radio signals from atomic clocks in orbit, allowing billions of receivers to determine their position, velocity and time anywhere on Earth. This chapter explains how satellite navigation works: one-way ranging, pseudoranges and the navigation solution; spread-spectrum signals and receivers; atomic clocks and relativity; the error budget and satellite geometry; and the augmentation techniques that improve accuracy from metres to centimetres. It describes the four global systems as they stand in 2026, the enormous range of applications that depend on them — including the timing of telecom networks, power grids and financial markets — and their vulnerability to interference, jamming and spoofing, which has grown into a serious problem. It ends with the next generation of satellites and with navigation in low Earth orbit and on the Moon.*

IN THIS CHAPTER

18.1 Positioning, Navigation and Timing
18.2 How Satellite Navigation Works
18.3 The Systems in 2026
18.4 Accuracy
18.5 Augmentation: From Metres to Centimetres
18.6 Applications
18.7 Vulnerabilities: Interference, Jamming and Spoofing
18.8 Future Systems

# 18.1 Positioning, Navigation and Timing

Satellite navigation is one of the most pervasive technologies ever created. In 2026 there are roughly seven billion GNSS receivers in use — in phones, vehicles, aircraft, ships, tractors, drones, watches, survey instruments and the timing equipment of telecommunications networks, power grids and stock exchanges. Most users think of GNSS as a way to find their position, but its three outputs — **positioning, navigation and timing (PNT)** — are equally important. Precise time from GNSS synchronises mobile-phone base stations, time-stamps financial transactions and aligns measurements across electricity grids. A study for the US government estimated that GPS alone generated about $1.4 trillion in economic benefits in the USA. This figure covers the years between its opening to civil use in the 1980s and 2017.

## 18.1.1 A Short History

The idea grew from the space race. When Sputnik was launched in 1957, scientists at the Johns Hopkins Applied Physics Laboratory found that they could determine its orbit from the Doppler shift of its radio signal. They then realised that, conversely, a receiver could determine its own position from the Doppler shift of a satellite in a known orbit. The resulting US Navy **Transit** system (first satellite 1960, operational 1964) allowed submarines to fix their position every hour or so. Experiments with atomic clocks in orbit (Timation) and the Air Force's 621B programme led in 1973 to the decision to build the **NAVSTAR Global Positioning System**. The first GPS satellite was launched in 1978 and the system was declared fully operational in 1995. The Soviet Union developed the similar **GLONASS** from 1982.

GPS was opened to civil users in the 1980s, but its civil signal was deliberately degraded by "Selective Availability" to about 100 m accuracy. This degradation was switched off in May 2000, instantly improving accuracy tenfold and unleashing the consumer navigation industry. Other powers then built their own systems for strategic independence: China's **BeiDou** (regional from 2012, global in 2020), Europe's civilian **Galileo** (initial services in 2016), Japan's regional **QZSS** and India's regional **NavIC** (Figure 18.1).

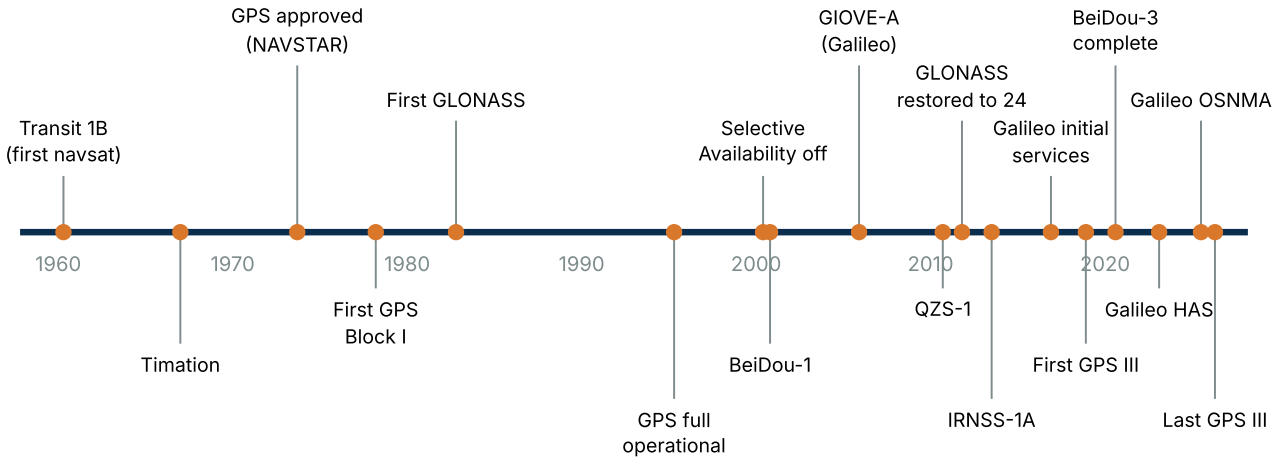



**FIGURE 18.1** Milestones in satellite navigation.

## 18.2 How Satellite Navigation Works

### 18.2.1 One-Way Ranging and Pseudoranges

Each GNSS satellite carries atomic clocks and continuously broadcasts radio signals that contain its precise time and its position (ephemeris). A receiver measures when each signal arrives; the difference between the time of reception and the time of transmission, multiplied by the speed of light, is the distance (range) to the satellite. With ranges to three satellites at known positions, the receiver's position could be found by **trilateration** — the intersection of three spheres (Figure 18.2).

The difficulty is that the receiver's own clock — a cheap quartz oscillator — is not synchronised with the satellites' atomic clocks. An error of just one microsecond would cause a range error of 300 m. The measured ranges, which include this unknown clock offset, are therefore called **pseudoranges**:

$$\rho_i = \sqrt{(x_i - x)^2 + (y_i - y)^2 + (z_i - z)^2} + c\,\delta t + \varepsilon_i$$

where $(x_i, y_i, z_i)$ is the position of satellite $i$, $(x, y, z)$ the receiver position, $\delta t$ the receiver clock offset and $\varepsilon_i$ the remaining errors. Because the clock offset is common to all measurements, it can be solved as a fourth unknown: **four satellites** are needed to determine three position coordinates and time. In practice receivers track 10–40 satellites from several constellations and solve the over-determined system by iterative least squares or, for continuous navigation, by a Kalman filter. As a by-product, every GNSS receiver becomes a clock synchronised to the system time within tens of nanoseconds. Velocity is found in a similar way from the Doppler shifts of the signals.

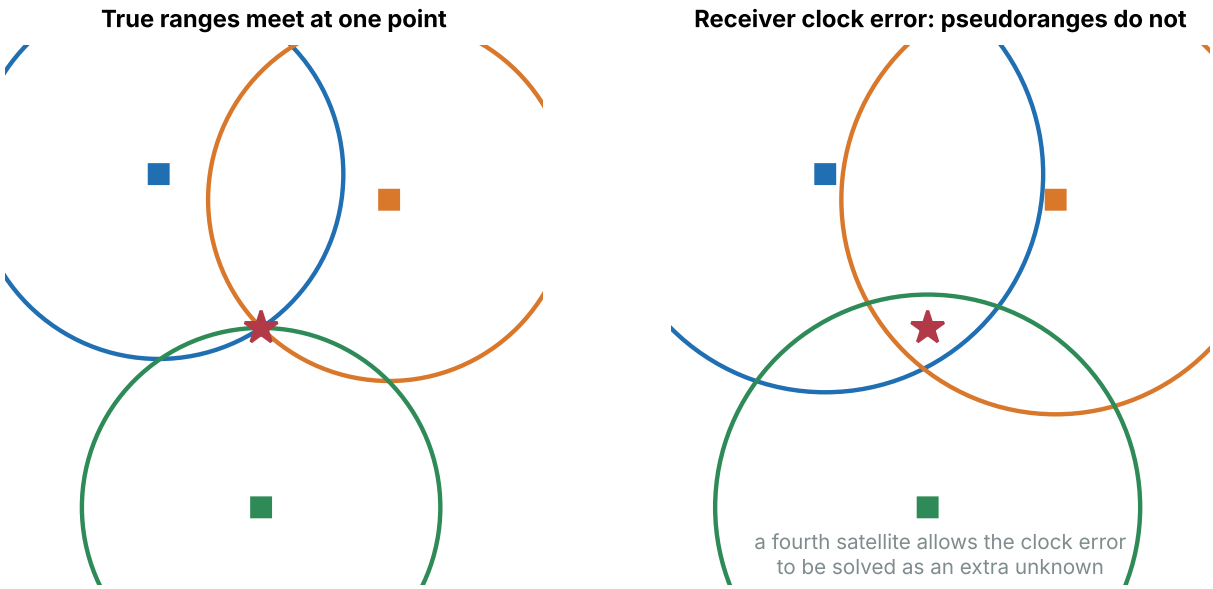
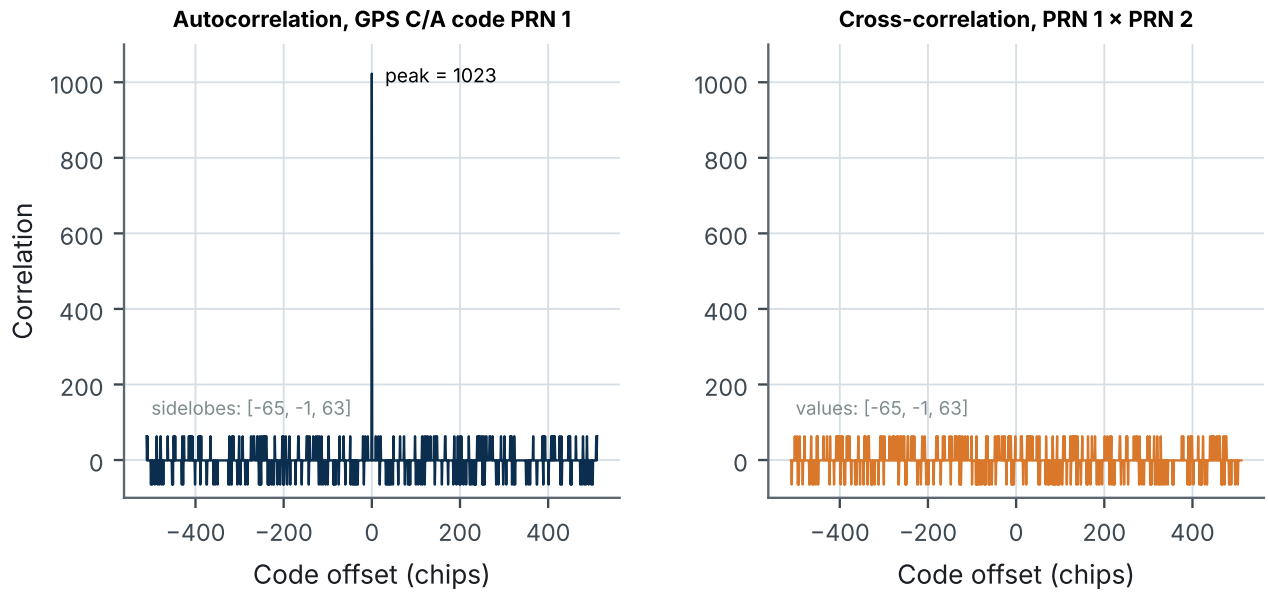



**FIGURE 18.2** Trilateration. With perfect clocks, ranges from three satellites intersect at the user's position (left). A receiver clock error changes all ranges by the same amount, so the circles no longer meet (right) — the error is solved for using a fourth satellite.

### 18.2.2 Spread-Spectrum Signals

GNSS signals are extraordinarily weak. A GPS L1 signal arrives at the Earth's surface with a minimum power of about −128.5 dBm ($1.4 \times 10^{-16}$ W). That is roughly the brightness of a 25 W light bulb seen from 20 000 km, and well **below** the thermal noise in the receiver's bandwidth. Receivers can nevertheless detect them because the signals use **direct-sequence spread spectrum**: the navigation data are multiplied by a fast **pseudo-random noise (PRN) code** that spreads the signal over a wide bandwidth. The receiver generates a replica of each satellite's code and correlates it with the incoming signal; only when the replica is aligned with the received code does a strong correlation peak appear (Figure 18.3). Correlating over 1–20 ms or more provides a **processing gain** of 30–45 dB that lifts the signal out of the noise.

Each satellite uses a different code chosen to have low **cross-correlation** with the others, so all satellites can transmit on the same frequency without interfering — **code division multiple access (CDMA)**. The GPS **coarse/acquisition (C/A) code**, for example, is a Gold code of 1 023 chips transmitted at 1.023 million chips per second, repeating every millisecond. The timing of the correlation peak, refined by tracking loops to a small fraction of a chip, gives the pseudorange. The phase of the carrier wave (wavelength 19 cm at L1) can be measured even more precisely, to about a millimetre.




Computed with the IS-GPS-200 G1/G2 Gold-code generators (1 023 chips, 1.023 Mchip/s).

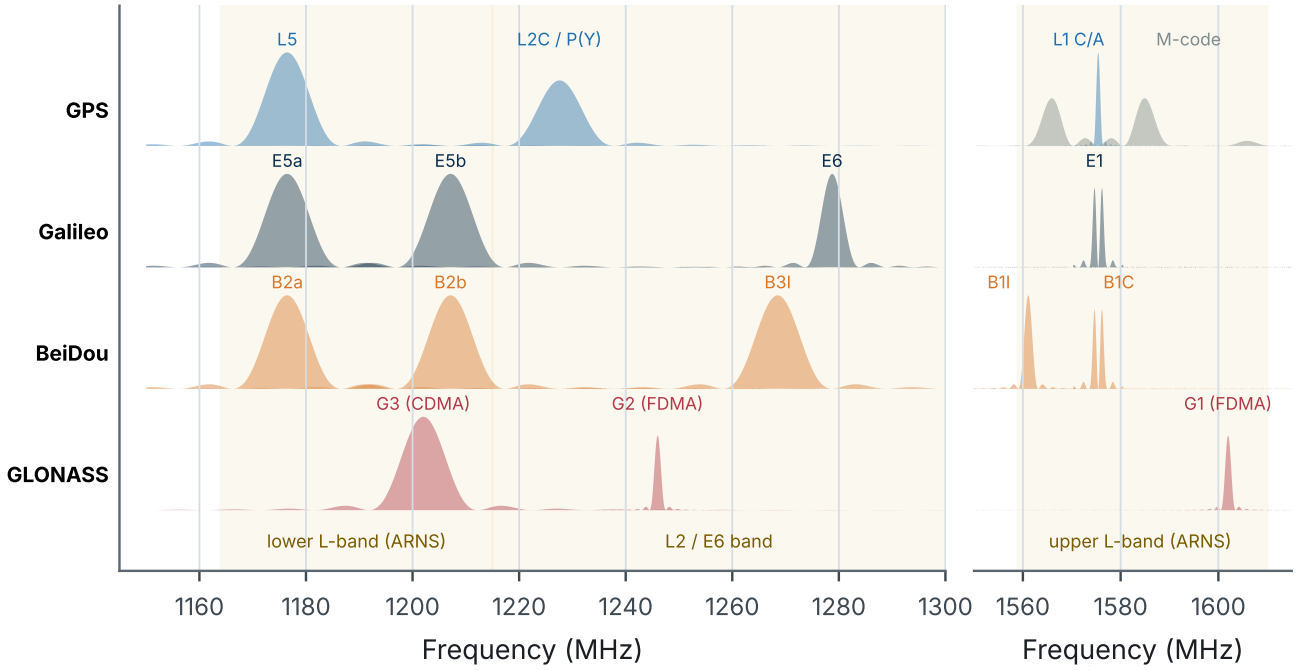
**FIGURE 18.3** Correlation properties of the GPS C/A code, computed from the generators in the GPS interface specification. The autocorrelation has a single sharp peak of 1 023 at zero offset; sidelobes and cross-correlations with other satellites' codes take only the values −65, −1 and +63.

### 18.2.3 Receivers

A GNSS receiver consists of an antenna (typically right-hand circularly polarised), a radio-frequency front end that amplifies, filters and down-converts the signals and digitises them, and a baseband processor with many **correlator channels**. **Acquisition** searches over all possible code delays and Doppler shifts for each satellite. Modern receivers perform this two-dimensional search in parallel using FFTs. It takes from about a second (with assistance data from the mobile network) to a minute (cold start). **Tracking** then follows each signal with a delay-locked loop for the code and a phase- or frequency-locked loop for the carrier, while the navigation message is decoded. A smartphone GNSS chip, which tracks four constellations on two frequencies, costs about a dollar and consumes tens of milliwatts.

### 18.2.4 Signals of the Four Global Systems

All four global systems now broadcast multiple civil signals in the radio-navigation bands around 1.2 and 1.575 GHz, which are protected for aeronautical safety use (Figure 18.4). Newer signals share common centre frequencies: 1575.42 MHz (GPS L1C, Galileo E1, BeiDou B1C) and 1176.45 MHz (GPS L5, Galileo E5a, BeiDou B2a). This allows a single dual-frequency receiver to use all of them. Some use **binary offset carrier (BOC)** modulation, which splits the signal's power away from the band centre, improving tracking precision and reducing interference with older signals.




Schematic power spectra (normalised). Shared centre frequencies (1575.42, 1176.45 MHz) simplify multi-GNSS receivers.

**FIGURE 18.4** Simplified frequency plan of the four global navigation systems.

**TABLE 18.1** Principal signals of the global navigation satellite systems (simplified).

| System | Signal | Centre frequency (MHz) | Modulation | Use |
|---|---|---|---|---|
| GPS | L1 C/A | 1575.42 | BPSK(1) | Legacy civil signal, used by almost every receiver |
| GPS | L1C | 1575.42 | TMBOC(6,1,1/11) | Modern civil, interoperable with Galileo E1 and BeiDou B1C (GPS III) |
| GPS | L2C | 1227.60 | BPSK(1) time-multiplexed | Second civil frequency (since 2005) |
| GPS | L5 | 1176.45 | BPSK(10) | Safety-of-life civil signal (since 2010) |
| GPS | P(Y), M-code | 1575.42, 1227.60 | BPSK(10), BOC(10,5) | Military, encrypted |
| Galileo | E1 OS | 1575.42 | CBOC(6,1,1/11) | Open service; carries OSNMA authentication |
| Galileo | E5a, E5b | 1176.45, 1207.14 | AltBOC(15,10) combined | Open service; E5b also for integrity |
| Galileo | E6 | 1278.75 | BPSK(5) | High-Accuracy Service; Public Regulated Service (encrypted) |
| BeiDou | B1I, B1C | 1561.098, 1575.42 | BPSK(2), MBOC | Open service |
| BeiDou | B2a, B2b | 1176.45, 1207.14 | QPSK(10) | Open service; B2b carries PPP corrections |
| BeiDou | B3I | 1268.52 | BPSK(10) | Open and authorised services |

| System | Signal | Centre frequency (MHz) | Modulation | Use |
|---|---|---|---|---|
| GLONASS | L1OF, L2OF | 1602 + k·0.5625, 1246 + k·0.4375 | FDMA, BPSK(0.511) | Legacy open signals, one frequency channel per satellite pair |
| GLONASS | L3OC | 1202.025 | CDMA | New CDMA signal (GLONASS-K) |

# 18.3 The Systems in 2026

## 18.3.1 Space Segments

**TABLE 18.2** The world's satellite navigation systems in 2026.

| System | Operator | Nominal constellation | MEO orbit | Other orbits | Clocks | Status (Oct 2026) |
|---|---|---|---|---|---|---|
| GPS (USA) | US Space Force | 24 slots (≈ 31 operational) | 20 180 km, 55°, 6 planes | — | Rubidium, caesium | Operational since 1995; ten GPS III launched 2018–2026; GPS IIIF from ≈ 2027 |
| GLONASS (Russia) | Roscosmos / Ministry of Defence | 24 | 19 100 km, 64.8°, 3 planes | — | Caesium, rubidium | Operational; transition to GLONASS-K/K2 with CDMA signals |
| Galileo (EU) | EU / EUSPA, ESA | 24 + spares | 23 222 km, 56°, 3 planes | — | Passive hydrogen maser, rubidium | Initial services since 2016; ≈ 30 satellites usable after the December 2025 launch; second generation from 2026–27 |
| BeiDou-3 (China) | China Satellite Navigation Office | 24 MEO + 3 IGSO + 3 GEO | 21 528 km, 55°, 3 planes | GEO and inclined GEO over Asia | Rubidium, hydrogen maser | Global service since 2020; legacy BeiDou-2 satellites still in use |
| QZSS (Japan) | Cabinet Office | 7 (growing to 11) | — | Quasi-zenith (inclined, elliptical GEO) and GEO | Rubidium | Six satellites by October 2026 |
| NavIC (India) | ISRO | 7 | — | GEO and IGSO over India | Rubidium | Several clock failures; NVS-series replacements, one (NVS-02) stranded in transfer orbit in 2025 |

The global systems use medium Earth orbits at 19 000–23 000 km (Figure 18.5). This is a compromise between coverage (each satellite sees about 38 % of the Earth) and the number of satellites needed. About 24 satellites give at least four visible everywhere at all times. GLONASS uses a higher inclination (64.8°) for better coverage of Russia's northern latitudes. BeiDou and the regional systems add satellites in geostationary and **inclined geosynchronous (IGSO)** orbits. These trace figure-of-eight ground tracks and remain over their service regions. Japan's quasi-zenith satellites are designed to stay at high elevation over Japan, so that their signals are not blocked by tall buildings and mountains.

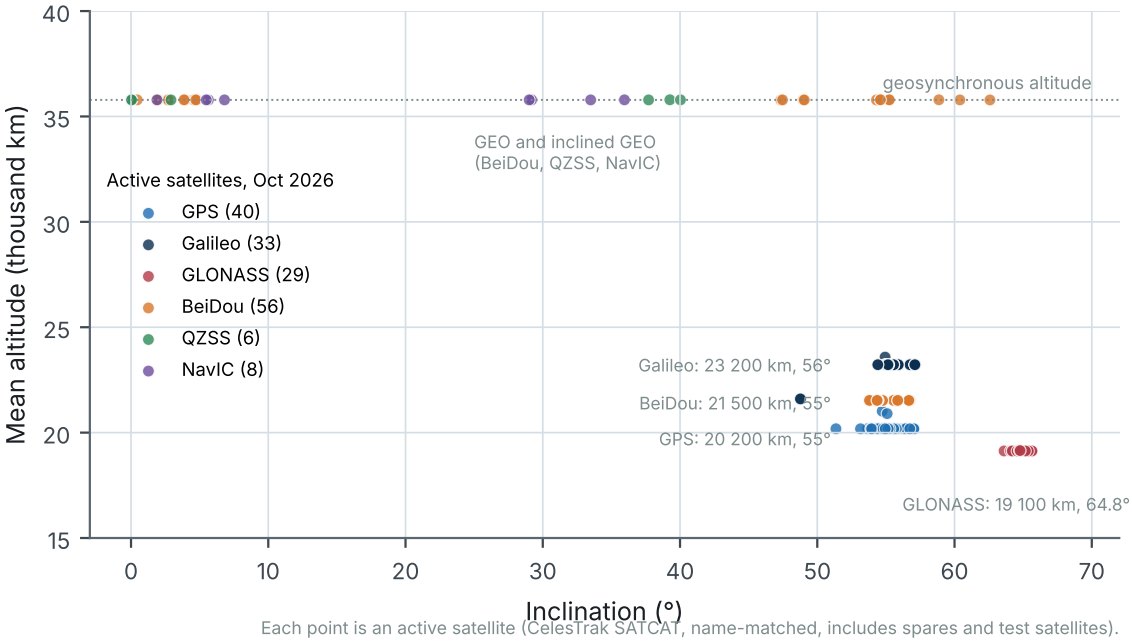



**FIGURE 18.5** Altitude and inclination of active navigation satellites in October 2026. Each global system occupies a distinct MEO shell; BeiDou, QZSS and NavIC also use geostationary and inclined geosynchronous orbits.

**FIGURE 18.6** A Delta II rocket ready to launch the 19th NAVSTAR GPS satellite from Cape Canaveral in 1993. Delta II rockets launched most GPS satellites from 1989 to 2009. Image: NASA/MSFC.

### 18.3.2 Atomic Clocks and Time

The precision of GNSS rests on atomic clocks. GPS satellites carry rubidium and caesium frequency standards. Galileo satellites carry two rubidium clocks and two **passive hydrogen masers**. These masers are the most stable clocks flying in navigation satellites, with a stability of better than one part in $10^{14}$ over a day. BeiDou-3 also carries hydrogen masers. A clock error of 1 ns produces a range error of 30 cm, so the clocks must be extremely stable between updates from the ground. Each system maintains its own **system time** (GPS Time, Galileo System Time, BeiDou Time, GLONASS Time), steered to Coordinated Universal Time (UTC). The offsets between them are broadcast, so that receivers can combine all systems. NASA's **Deep Space Atomic Clock** (the opening image of this chapter), a mercury-ion clock flown in 2019–2021, demonstrated stability good enough for spacecraft to navigate autonomously in deep space.

### 18.3.3 Relativity in Daily Life

GNSS is the most familiar practical application of Einstein's theories of relativity (Figure 18.7). A clock in a GPS orbit runs **faster** than a clock on the ground by about 45.7 µs per day. This is because it sits higher in the Earth's gravitational field (general relativity). It also runs **slower** by about 7.2 µs per day because of its orbital speed (special relativity). The net effect is +38.5 µs per day — which, left uncorrected, would cause range errors growing by about 11 km per day. GPS satellite clocks are therefore deliberately set to run slow by 4.465 parts in $10^{10}$ before launch, and receivers apply a further small correction for orbital eccentricity.

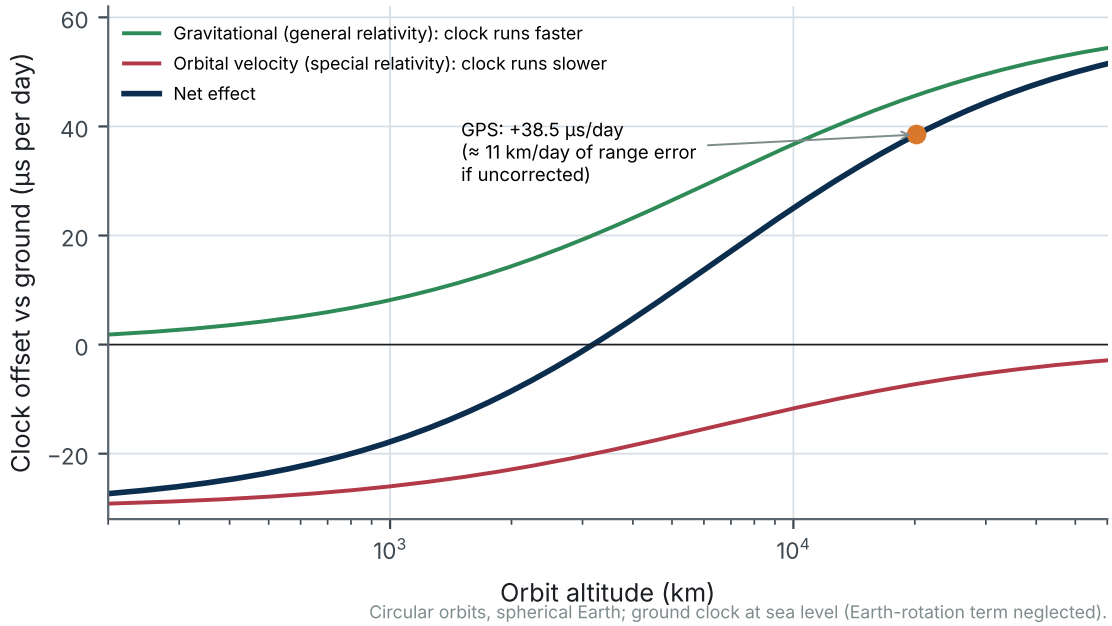



**FIGURE 18.7** Relativistic clock-rate offsets of orbiting clocks relative to a clock on the ground. Above about 3 000 km the gravitational effect dominates and orbiting clocks gain time.

### 18.3.4 Control Segments

Each system has a control segment of monitor stations around the world that track the satellites, and master control stations that compute orbit and clock predictions. The segment's uplink antennas upload these predictions to the satellites for broadcast. The GPS master control station at Schriever Space Force Base in Colorado updates each satellite's navigation message at least

daily. The modernised GPS ground system, **OCX**, is needed to exploit the new civil and military signals fully. It suffered years of delays and cost growth. Galileo's ground segment includes control centres in Germany and Italy and a worldwide network of sensor stations.

## 18.4 Accuracy

### 18.4.1 The Error Budget

The accuracy of a GNSS position depends on the errors in each pseudorange — summarised as the **user equivalent range error (UERE)** — and on the geometry of the satellites. The main error sources are (Figure 18.8):

- **Satellite clock and orbit errors**: the broadcast predictions are not perfect. The GPS signal-in-space range error has improved to about 0.5 m (RMS) in recent years, and Galileo's to about 0.2 m.
- **Ionospheric delay**: free electrons in the ionosphere (50–1 000 km) delay the signal. The delay is proportional to the **total electron content (TEC)** along the path and inversely proportional to the square of the frequency (Figure 18.9). It is typically 2–15 m at L1, and much more during solar maxima and geomagnetic storms (Chapter 22). Single-frequency receivers remove only about half of it using a broadcast model; **dual-frequency** receivers eliminate the first-order effect almost completely, which is the main reason for the new civil signals.
- **Tropospheric delay**: the neutral atmosphere delays signals by about 2.4 m at the zenith (more at low elevations); models remove most of it, leaving errors of decimetres.
- **Multipath**: reflections from buildings, the ground and vehicles distort the correlation peak — the dominant error in cities.
- **Receiver noise**, typically decimetres for code measurements and millimetres for carrier phase.

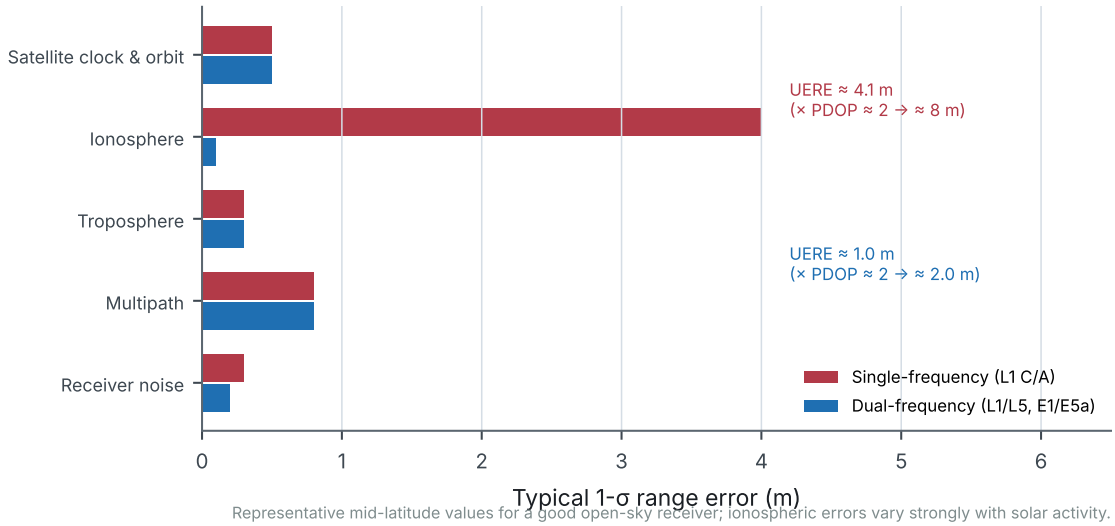



**FIGURE 18.8** Typical pseudorange error budget for single- and dual-frequency receivers. Dual-frequency operation removes the ionospheric error, the largest error source for single-frequency users.

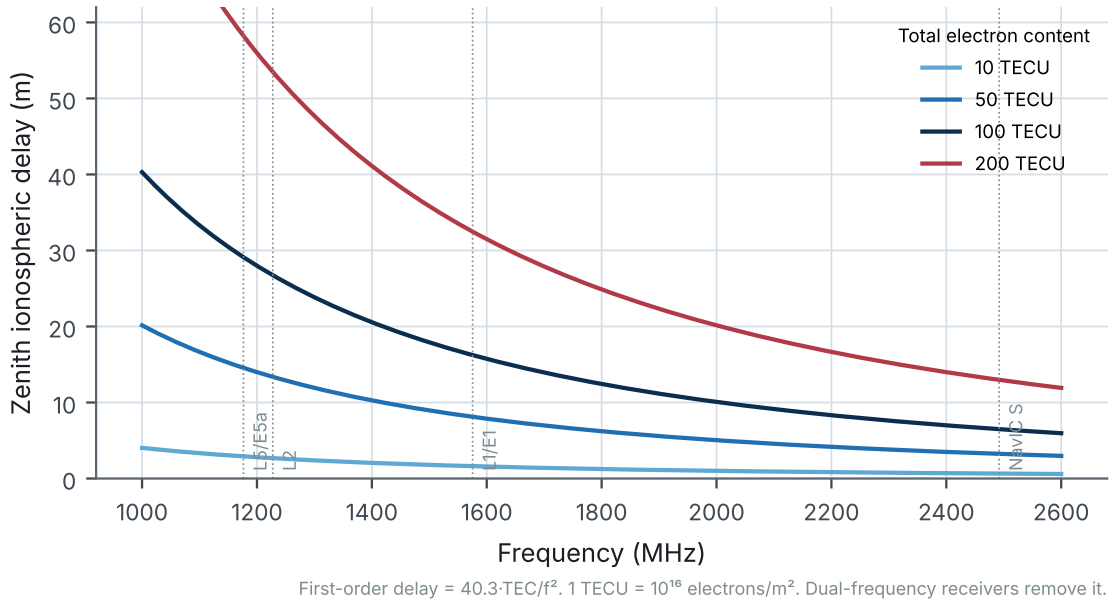



**FIGURE 18.9** Ionospheric delay versus frequency for different total electron contents. The delay at L1 can exceed 30 m in a strongly ionised ionosphere.

## 18.4.2 Satellite Geometry and DOP

The same range errors produce larger position errors when the satellites are poorly spread across the sky (Figure 18.10). This amplification is expressed by the **dilution of precision (DOP)**, computed from the geometry matrix of the least-squares solution. Position error ≈ PDOP × UERE (with HDOP and VDOP for horizontal and vertical components, and TDOP for time). Vertical errors are typically 1.5–2 times larger than horizontal errors because all satellites are above the horizon.

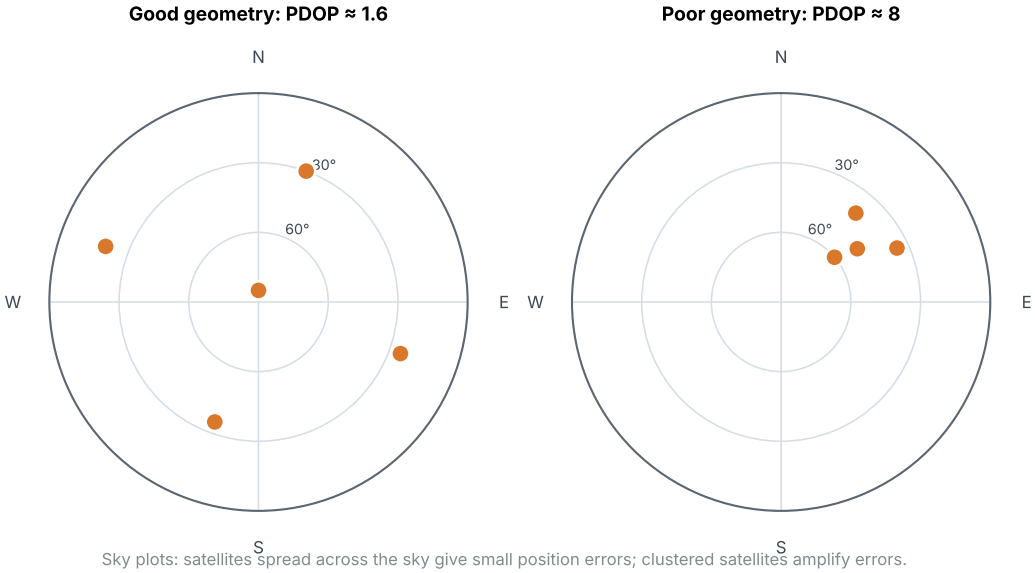
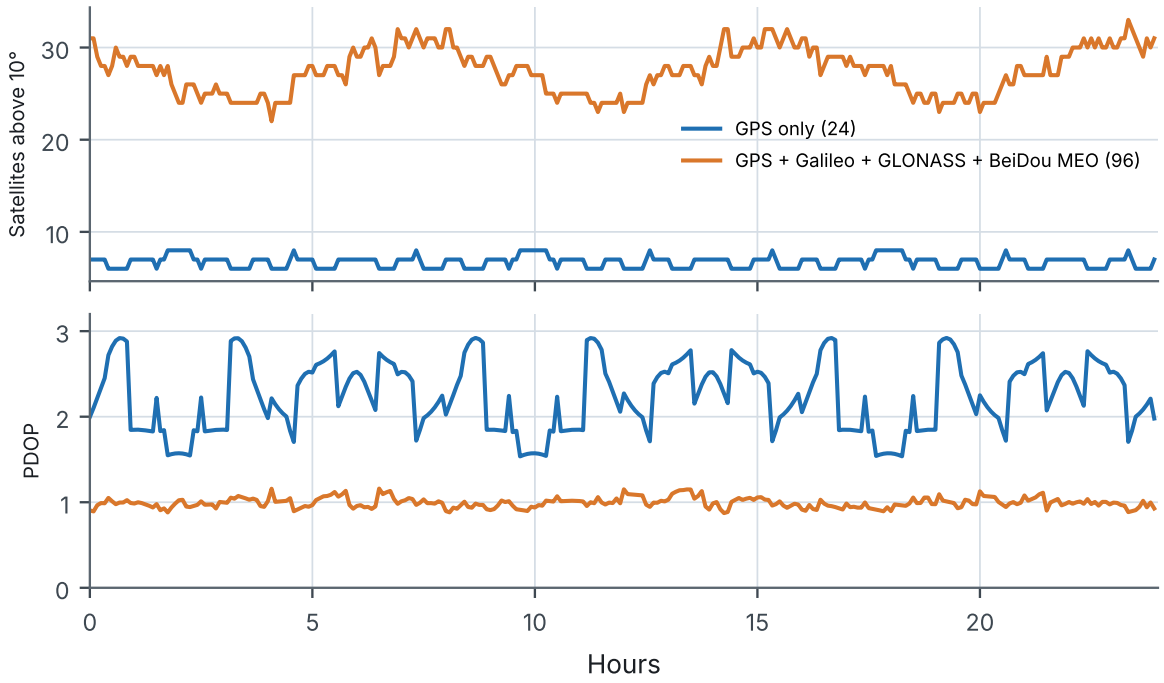



**FIGURE 18.10** Satellite sky plots illustrating good and poor geometry. Clustered satellites give a large dilution of precision.

Using several constellations together dramatically improves geometry and availability (Figure 18.11): a multi-GNSS receiver typically sees 30–40 satellites rather than 8–12, and PDOP falls from about 2 to about 1. This matters most where much of the sky is blocked — in city “urban canyons”, forests and mountains.




Simulated nominal constellations (Walker patterns) seen from 40° N, 0° E, 10° elevation mask.

**FIGURE 18.11** Simulated number of visible satellites and PDOP over 24 hours for GPS alone and for four global constellations together.

A typical smartphone achieves 3–5 m accuracy in open sky (dual-frequency phones, introduced in 2018, do better) and worse in cities. A single-frequency GPS receiver meets the official GPS performance standard of a few metres. A dual-frequency, multi-constellation receiver achieves about 1–2 m.

## 18.5 Augmentation: From Metres to Centimetres

Many applications need more accuracy or guaranteed **integrity** — assurance that the position is not wrong without warning. **Augmentation** systems provide corrections and integrity information (Figure 18.12).

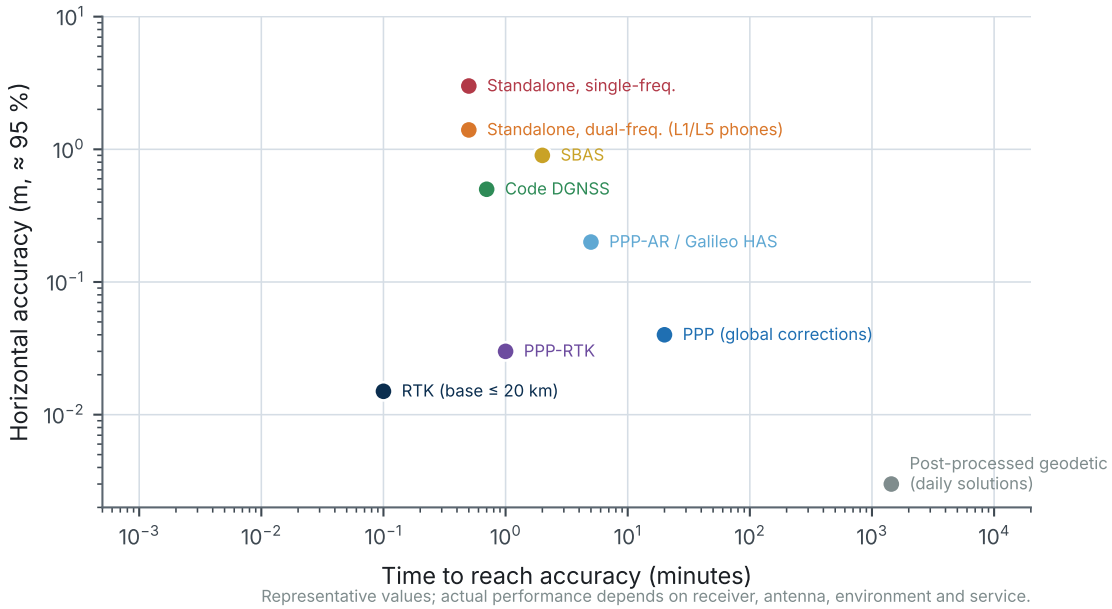
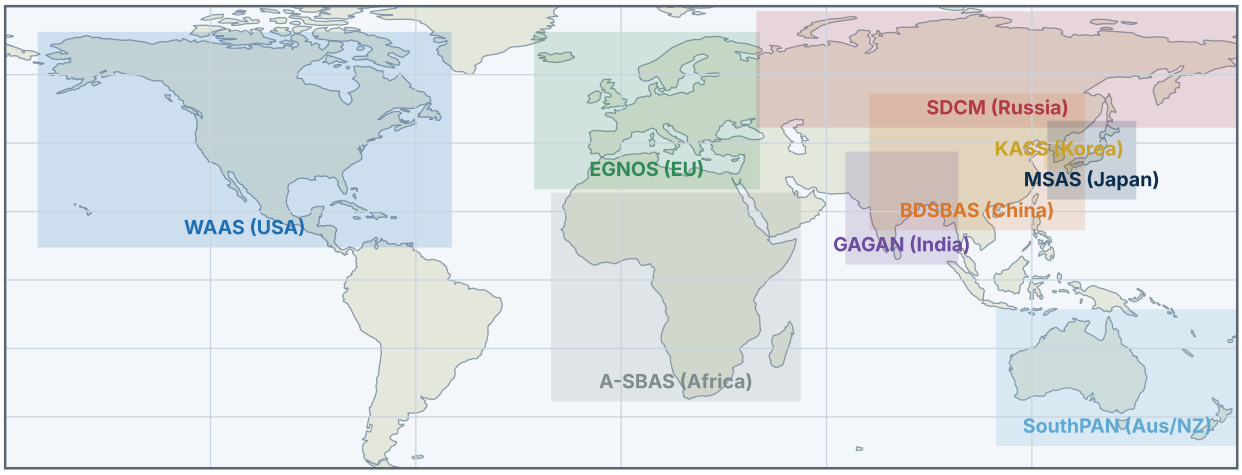



**FIGURE 18.12** Typical accuracy versus time to reach it for different GNSS positioning techniques.

- **Differential GNSS (DGNSS)**: a reference station at a known position computes the errors in each pseudorange and broadcasts corrections; nearby users, who experience nearly the same errors, apply them to obtain sub-metre accuracy.
- **Satellite-based augmentation systems (SBAS)** use networks of reference stations across a continent to compute corrections for satellite orbits, clocks and the ionosphere, plus integrity messages. They broadcast them from geostationary satellites on the GPS L1 frequency (Figure 18.13). The US **WAAS** (2003), Europe's **EGNOS** (2009–2011), Japan's MSAS, India's GAGAN and others allow aircraft to fly precision approaches with vertical guidance (LPV) to thousands of airports without ground-based landing systems. The next generation, **dual-frequency multi-constellation (DFMC) SBAS**, will use L1/L5 and Galileo signals.
- **Ground-based augmentation systems (GBAS)** provide corrections and integrity within about 40 km of an airport for Category I (and in future Category II/III) landings.
- **Real-time kinematic (RTK)** positioning uses carrier-phase measurements from a nearby base station (Figure 18.14). Once the integer number of carrier cycles (the **ambiguity**) has been resolved, the rover's position is known to 1–2 cm within seconds. Networks of continuously operating reference stations (CORS) provide RTK corrections across whole countries for surveying, construction and precision agriculture.
- **Precise point positioning (PPP)** uses precise satellite orbit and clock products computed from global networks (such as those of the International GNSS Service). It achieves centimetre accuracy with a single receiver anywhere, after a convergence time of 10–30 minutes. **PPP-RTK** adds regional atmospheric corrections to shorten convergence. Satellite systems now broadcast PPP corrections free of charge: Galileo's **High Accuracy Service (HAS)** since 2023, QZSS's CLAS and MADOCA, and BeiDou's PPP-B2b.




Approximate service regions of satellite-based augmentation systems (SBAS) operating or in development.

**FIGURE 18.13** Approximate service regions of satellite-based augmentation systems.

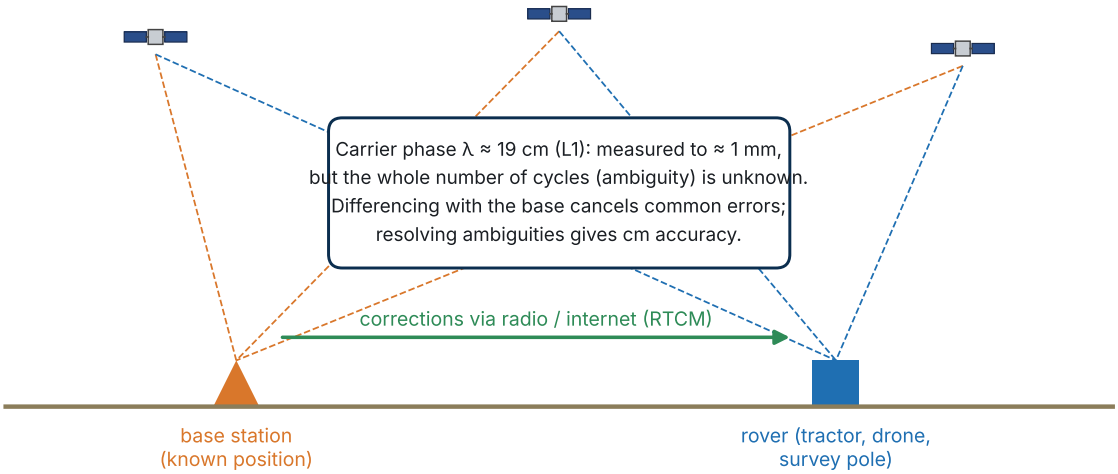



**FIGURE 18.14** Real-time kinematic positioning. A base station at a known position sends its carrier-phase measurements to the rover, which can then resolve the carrier-cycle ambiguities and position itself to about a centimetre.

## 18.6 Applications

**TABLE 18.3** Major applications of GNSS.

| Sector | Applications | Typical requirement |
|---|---|---|
| Consumer and location-based services | Smartphones, maps, ride-hailing, fitness, games | 3–5 m, low power, fast fix |
| Road transport | Navigation, fleet management, emergency call (eCall), tolling, connected and automated vehicles | 1–5 m; lane-level 0.5 m for automation, with integrity |
| Aviation | Performance-based navigation, precision approaches (SBAS, GBAS), ADS-B surveillance | Certified accuracy and integrity |
| Maritime | Navigation, port approaches, AIS, dynamic positioning of drilling ships | 1–10 m; decimetres for dynamic positioning |
| Rail | Train location, signalling (ERTMS), asset management | Safety-critical integrity |
| Agriculture | Auto-steering, variable-rate application, yield mapping | 2 cm (RTK) |

| Sector | Applications | Typical requirement |
|---|---|---|
| Surveying, construction, mapping | Cadastral surveys, machine control, GIS, mobile mapping | mm–cm |
| Timing and synchronisation | Telecom networks (5G needs ±1.5 µs), power-grid phasor measurements, financial time-stamping, data centres, broadcasting | 10 ns–1 µs |
| Drones and robotics | Autonomous flight, delivery, inspection | cm–m, high integrity |
| Search and rescue | Cospas-Sarsat MEOSAR distress beacons; Galileo Return Link Service confirms receipt | < 100 m, minutes |
| Science | Geodesy and plate tectonics (mm/year), sea level and ice, seismology, ionospheric and atmospheric sounding, reflectometry | mm, ns |
| Space | Orbit determination of LEO satellites, launch-vehicle tracking, GEO and lunar navigation | cm–m |
| Defence | Navigation, weapon guidance, timing of military networks | Encrypted, jam-resistant signals |

Scientific uses are remarkable. Networks of thousands of geodetic GNSS stations measure the motion of tectonic plates at millimetres per year. They also measure the elastic rebound of the crust as ice sheets melt, and the ground deformation of earthquakes in real time, which can improve tsunami warnings. The same signals measure the atmosphere: water vapour from tropospheric delay, electron content from ionospheric delay (Figure 18.15) and temperature profiles from radio occultation (Chapter 16).

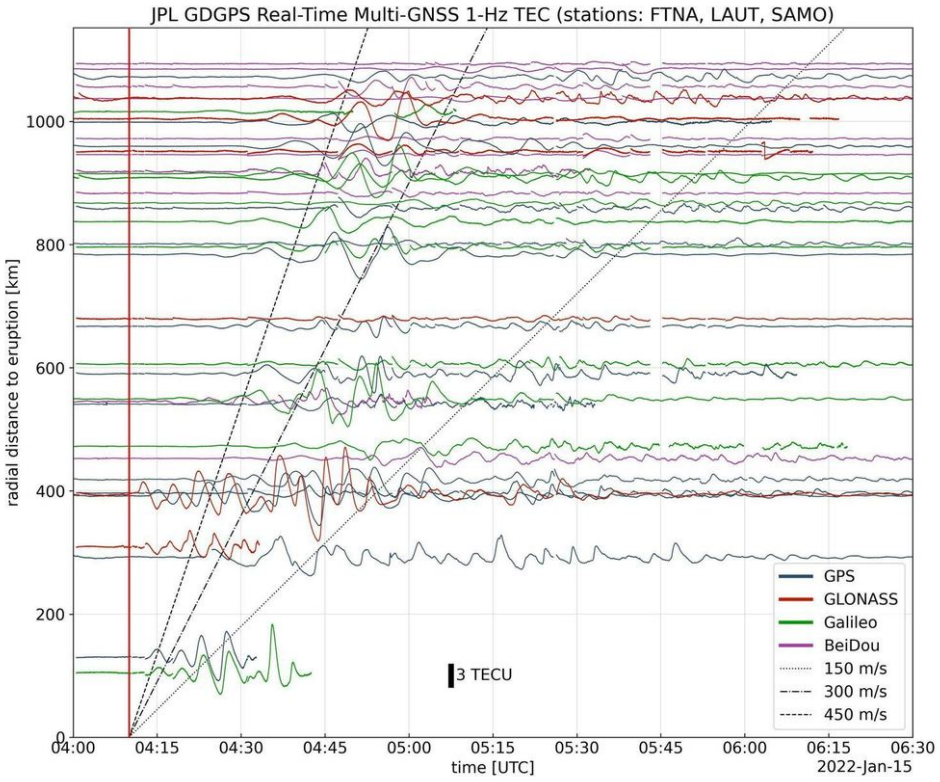



**FIGURE 18.15** Ionospheric total electron content measured in real time by NASA's Global Differential GPS network after the eruption of the Hunga Tonga–Hunga Ha'apai volcano on 15 January 2022. Waves travelling outward from the eruption appear as diagonal stripes, measured simultaneously with GPS, GLONASS, Galileo and BeiDou signals. Image: NASA/JPL (NASA/JPL-Caltech/GDGPS).

## 18.7 Vulnerabilities: Interference, Jamming and Spoofing

The weakness of GNSS signals makes them easy to disrupt. **Jamming** — transmitting noise in the GNSS bands — can deny service over large areas with modest power (Figure 18.16). A 1 W jammer can prevent a typical receiver from tracking GPS C/A signals at line-of-sight distances of more than 10 km. Military jammers of hundreds of watts affect aircraft hundreds of kilometres away. Cheap illegal "privacy jammers" in vehicles regularly disrupt airports and port operations. **Spoofing** — transmitting counterfeit GNSS signals — is more insidious: a spoofed receiver reports a false position or time without any warning.

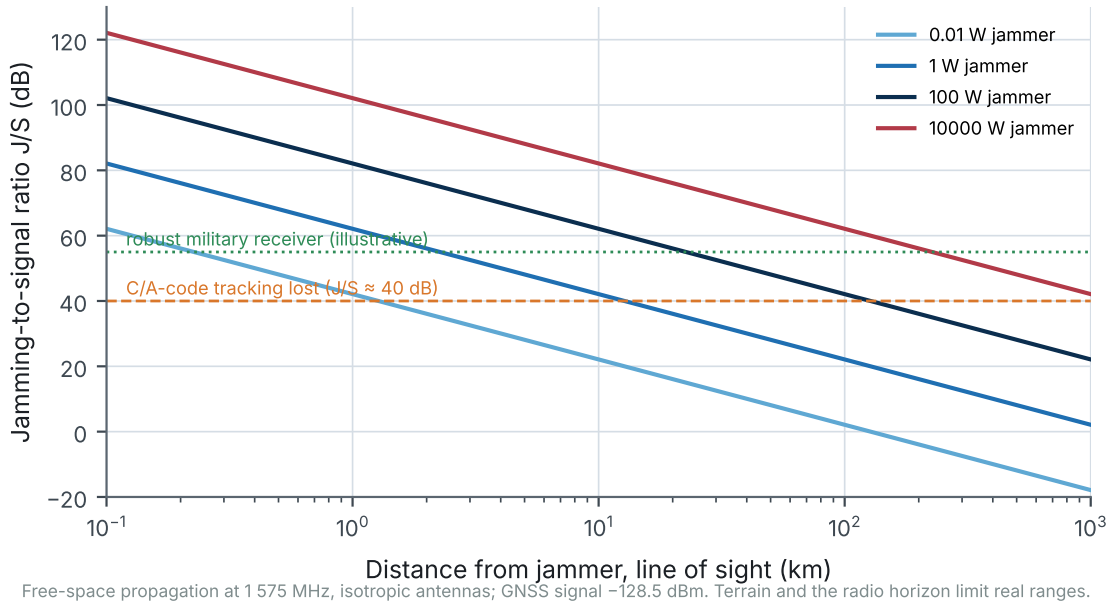



**FIGURE 18.16** Jamming-to-signal ratio versus distance from jammers of different power. Even very low-power jammers disrupt receivers at kilometres; high-power military jammers reach the radio horizon.

Since 2017, interference has grown into a serious international problem. Ships in the Black Sea reported spoofed positions placing them at inland airports; aircraft over the Middle East, the Baltic region and elsewhere routinely encounter jamming and spoofing linked to military conflicts. In Lithuania alone, more than 1 000 aircraft per month reported interference in the first half of 2025. The number of spoofing incidents in the region rose to several thousand per month in mid-2025, with transmitters located in the Kaliningrad region. Aviation authorities (ICAO, EASA, the FAA) have issued safety bulletins, and airlines have had to return to traditional radio navigation aids.

Countermeasures operate at several levels:

- **Robust receivers**: multi-frequency, multi-constellation tracking; interference detection and filtering; controlled-reception-pattern antennas (CRPAs) that steer nulls towards jammers; integration with inertial sensors that bridge outages.
- **Signal authentication**: Galileo's **Open Service Navigation Message Authentication (OSNMA)**, operational since 2025, digitally signs the navigation data so that receivers can detect forged messages. GPS plans similar protection (CHIMERA), and encrypted military signals (GPS M-code, Galileo PRS) resist spoofing.
- **Complementary PNT**: terrestrial systems (eLoran, mobile-network timing, fibre time distribution), signals from LEO satellites, and new technologies such as quantum inertial sensors and magnetic-anomaly navigation. The aim is that critical infrastructure does not depend on GNSS alone. US Executive Order 13905 (2020) requires critical-infrastructure operators to manage their PNT risks.

## 18.8 Future Systems

### 18.8.1 Modernisation

All the global systems are modernising. The final GPS III satellite (SV10) was launched in April 2026. The next generation, **GPS IIIF**, from about 2027, will add a regional high-power military signal, laser retroreflectors for precise orbit determination, and search-and-rescue payloads. Europe's **second-generation Galileo** satellites, built by Airbus and Thales Alenia Space, introduce electric propulsion, fully digital reconfigurable payloads, improved clocks and inter-satellite links, with first launches expected in 2026–27. China plans a next-generation BeiDou around 2035 with LEO augmentation, and GLONASS is moving to CDMA signals with GLONASS-K2. Japan is expanding QZSS to seven and later eleven satellites. India is replacing NavIC satellites with the NVS series, which carry indigenous atomic clocks and an L1 signal for wider receiver compatibility.

### 18.8.2 LEO PNT

Satellites in low Earth orbit are about 20 times closer than MEO navigation satellites. So their signals arrive about 30 dB stronger, which makes them much harder to jam. Their faster motion across the sky also allows rapid convergence of precise solutions. **Iridium**'s Satellite Time and Location service already provides timing and positioning that works indoors. Start-ups such as **Xona Space Systems** (which launched its first production-class Pulsar satellite in 2025) are building dedicated LEO PNT constellations. ESA's **LEO-PNT** in-orbit demonstrators (Celeste) are testing the concept. Researchers have also shown that even signals from broadband constellations such as Starlink can be used opportunistically for navigation. LEO PNT is expected to complement rather than replace MEO GNSS.

### 18.8.3 Navigation Beyond the Earth

GNSS signals are increasingly used above the constellations. In geostationary and high orbits, receivers use the weak signals that spill past the Earth's limb and the antenna side lobes — the **space service volume**. The **Lunar GNSS Receiver Experiment (LuGRE)**, an Italian–NASA receiver, flew on Firefly's Blue Ghost lander. In March 2025 it became the first to track GPS and Galileo signals on the Moon's surface and compute a navigation fix, some 380 000 km from Earth (Figure 18.17). For the coming era of lunar exploration, NASA, ESA (Moonlight), JAXA and others are developing **LunaNet**, a framework for lunar communications and navigation with dedicated lunar navigation satellites and a lunar time standard.

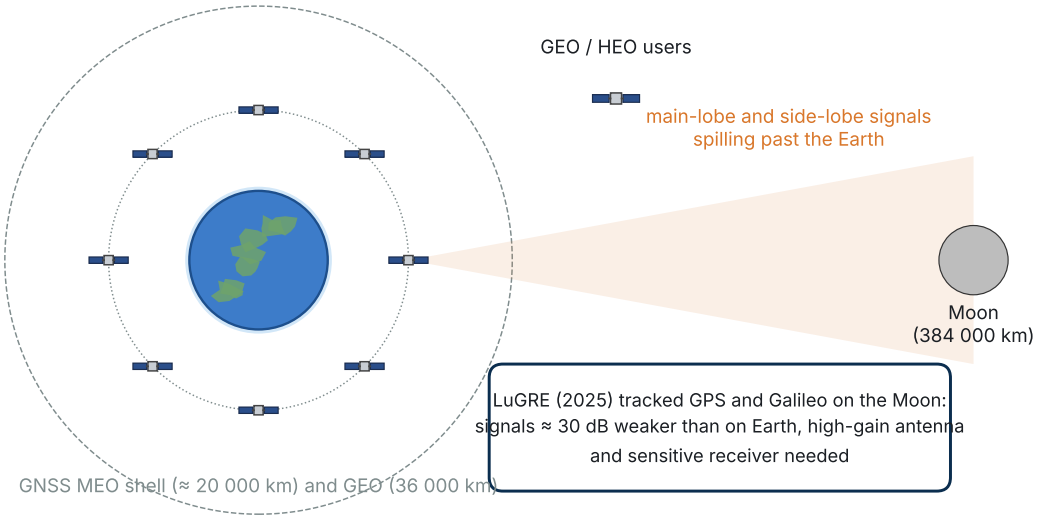



**FIGURE 18.17** GNSS signals beyond the constellations. Users in high orbits and at the Moon receive signals that spill past the Earth's limb.

**FIGURE 18.18** Firefly's Blue Ghost Mission 1 lander was launched on 15 January 2025 on a Falcon 9 rocket. It carried LuGRE, which made the first GNSS navigation fix on the Moon in March 2025. Image: NASA/KSC (NASA/Kim Shiflett).

**FIGURE 18.19** The Gravity Probe B spacecraft, launched in 2004 on a Delta II, confirmed predictions of general relativity — the same physics that every GNSS satellite clock must account for. Image: NASA/MSFC.

### EXAMPLE: WHAT ACCURACY DO GNSS CLOCKS AND ORBITS NEED?

1. *Clock stability.* Suppose the satellite clock contribution to the range error must stay below 0.3 m (1 ns) for 12 hours between uploads. Then the clock must keep time to 1 ns in 43 200 s. This is a fractional frequency stability of about $1 \times 10^{-9} / 43\,200 \approx 2 \times 10^{-14}$. Only hydrogen masers and the best rubidium clocks achieve this; the ground segment also estimates and broadcasts each clock's drift.
2. *Orbit accuracy.* An orbit error mainly affects the range along the line of sight. To contribute less than 0.3 m, the broadcast orbit must be known to about 0.3 m radially — even though the satellite moves at 3.9 km/s.
3. *Position accuracy.* With UERE = 1.0 m (dual-frequency) and PDOP = 1.5 (multi-GNSS), the 1-σ position error is about 1.5 m. With RTK (UERE ≈ 1 cm for carrier phase after ambiguity resolution), it falls to about 1.5 cm.
4. *Timing.* A static receiver at a known position needs only one satellite to solve for time; averaging many satellites gives timing accuracy of 10–20 ns relative to UTC — sufficient for 5G networks.

### AI IN SATELLITE NAVIGATION

Machine learning helps receivers detect and classify interference and spoofing, mitigate multipath in cities by learning which signals are reflected, fuse GNSS with cameras, lidar and inertial sensors in autonomous vehicles, and forecast ionospheric conditions. On the ground, AI models improve orbit and clock predictions and monitor the health of the constellation. Chapters 47 and 49 discuss these applications.

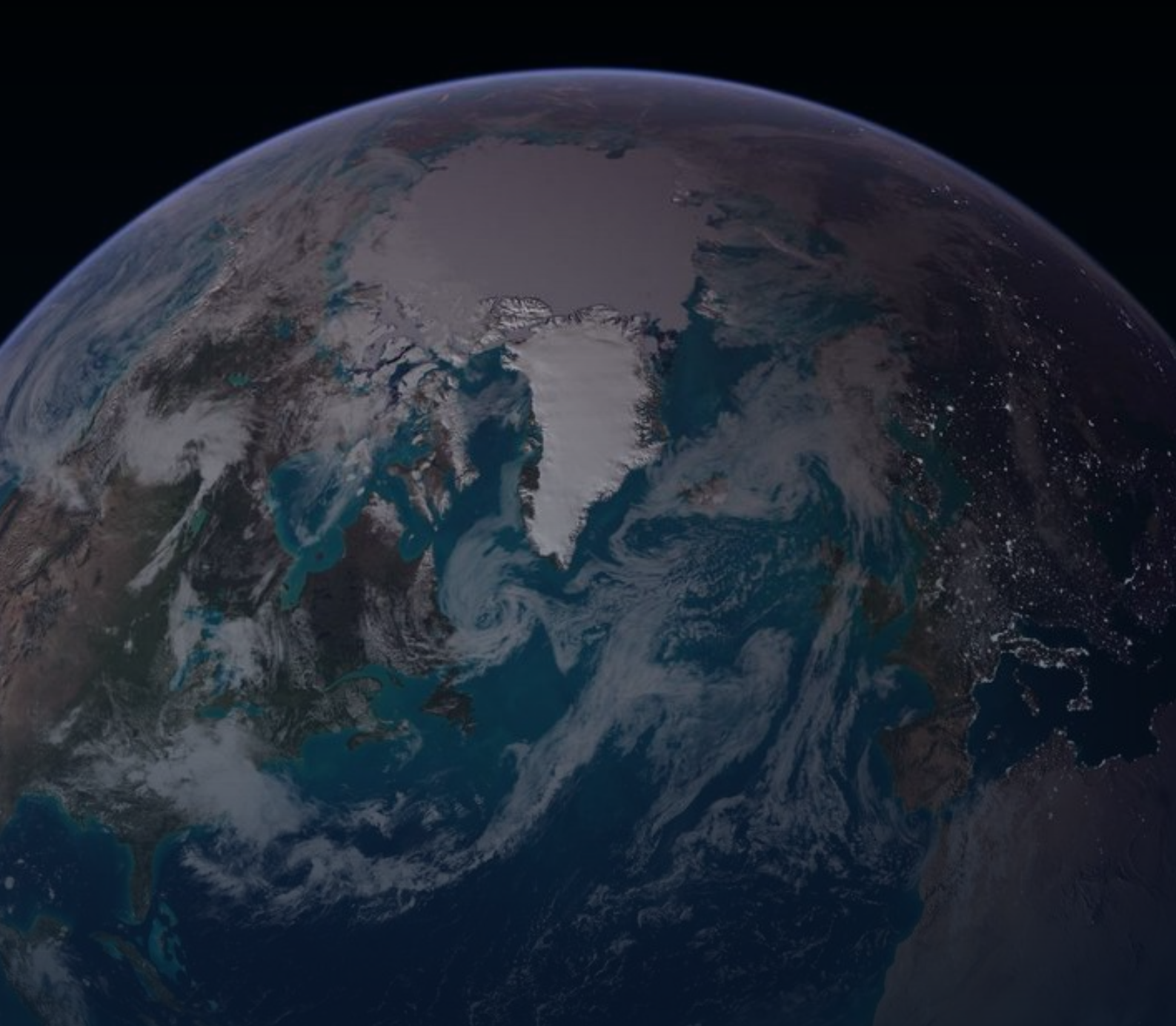
### KEY TAKEAWAYS

- GNSS provides positioning, navigation and timing worldwide; about seven billion receivers depend on it, and so do telecom, power and financial networks.
- Receivers measure pseudoranges to at least four satellites and solve for position and receiver clock offset; carrier phase allows centimetre precision.
- Signals are weaker than thermal noise but are recovered through spread-spectrum correlation; codes such as the GPS C/A Gold code allow all satellites to share frequencies.
- GPS, GLONASS, Galileo and BeiDou each operate about 24–30 MEO satellites with atomic clocks; QZSS and NavIC serve regions. Relativity changes GPS clock rates by +38.5 µs per day.
- Accuracy depends on the error budget (ionosphere, clocks, orbits, troposphere, multipath) and satellite geometry (DOP); dual-frequency, multi-GNSS receivers achieve 1–2 m.
- SBAS, GBAS, RTK and PPP improve accuracy to centimetres and provide integrity for aviation and other safety-critical uses.
- Jamming and spoofing have become widespread; authentication (Galileo OSNMA), robust receivers and complementary PNT (including LEO PNT) are the responses.
- GNSS is now used at the Moon: LuGRE made the first lunar GNSS fix in 2025.

### REVIEW QUESTIONS

1. Why does a GNSS receiver need signals from at least four satellites? What are the four unknowns?
2. The GPS C/A code has a chipping rate of 1.023 Mchip/s and carries data at 50 bit/s. What processing gain does despreading provide, in decibels?
3. GPS clocks gain about 38.5 µs per day relative to the ground because of relativity. What ranging error would accumulate in one day if this were not corrected?

4. A receiver sees satellites with a PDOP of 1.8, and the user equivalent range error is 1.2 m. Estimate the position error.
5. List the main sources in the GNSS error budget and explain how dual-frequency receivers remove the largest of them.
6. Compare SBAS, RTK and PPP in accuracy, coverage and what they need from the user.
7. How do jamming and spoofing work, and how does Galileo OSNMA help against spoofing?
8. Why are low-Earth-orbit PNT systems being developed, and what did LuGRE demonstrate at the Moon?



PART VI

# Military, Intelligence and Space Security

Space is a strategic domain. This Part describes, using openly published information, how signals intelligence and missile-warning satellites work, and how radars and telescopes track tens of thousands of objects to keep orbits safe from collisions and debris.


19 National Security Space and Signals Intelligence

20 Missile Early Warning Satellites

21 Space Surveillance, Orbital Debris and Space Situational Awareness


Image: NASA/GSFC

CHAPTER

# 19 National Security Space and Signals Intelligence

Image: NASA/JSC

*Space was a military and intelligence domain from the very beginning: the first reconnaissance satellites returned film from orbit in 1960, and today armed forces depend on space for intelligence, warning, navigation, communications and weather. This chapter describes, from public sources, the mission areas of national-security space and the organisations that conduct them; the principles of imagery and signals intelligence (SIGINT, ELINT and COMINT) from orbit, including the physics of intercepting and geolocating radio emitters; protected military satellite communications; the shift from a few exquisite satellites to proliferated constellations; the growing role of commercial space in conflict; and the counterspace threats — from anti-satellite missiles to jamming and cyber attacks — that have made space a contested domain, together with the legal and diplomatic efforts to limit them.*

IN THIS CHAPTER

19.1 Space as a National Security Domain
19.2 Imagery Intelligence from Space
19.3 Signals Intelligence from Space
19.4 Military Satellite Communications
19.5 Military Weather, Launch and the Wider Enterprise
19.6 From Exquisite to Proliferated Architectures
19.7 Counterspace: A Contested Domain
19.8 Law, Norms and Stability

# 19.1 Space as a National Security Domain

## 19.1.1 Mission Areas

Military and intelligence organisations use space for the same reasons as civilians: global coverage, a vantage point above any territory, and the ability to relay signals over the horizon. But they have demands for resilience, security and timeliness that civil systems rarely face. National-security space is usually divided into the mission areas shown in Figure 19.1:

- **Intelligence, surveillance and reconnaissance (ISR)**: imagery (IMINT), signals intelligence (SIGINT), and measurement and signature intelligence (MASINT).
- **Missile warning and tracking**: infrared satellites that detect missile launches and nuclear detonations (Chapter 20).
- **Positioning, navigation and timing**: GPS and its military signals (Chapter 18).
- **Satellite communications (SATCOM)**: protected, wideband and narrowband links to forces worldwide.
- **Environmental monitoring**: military weather satellites.
- **Space domain awareness**: tracking and characterising objects and activities in space (Chapter 21).
- **Space control**: protecting one's own space systems and, if necessary, denying an adversary's use of space.
- **Launch and ranges**: assured access to space.

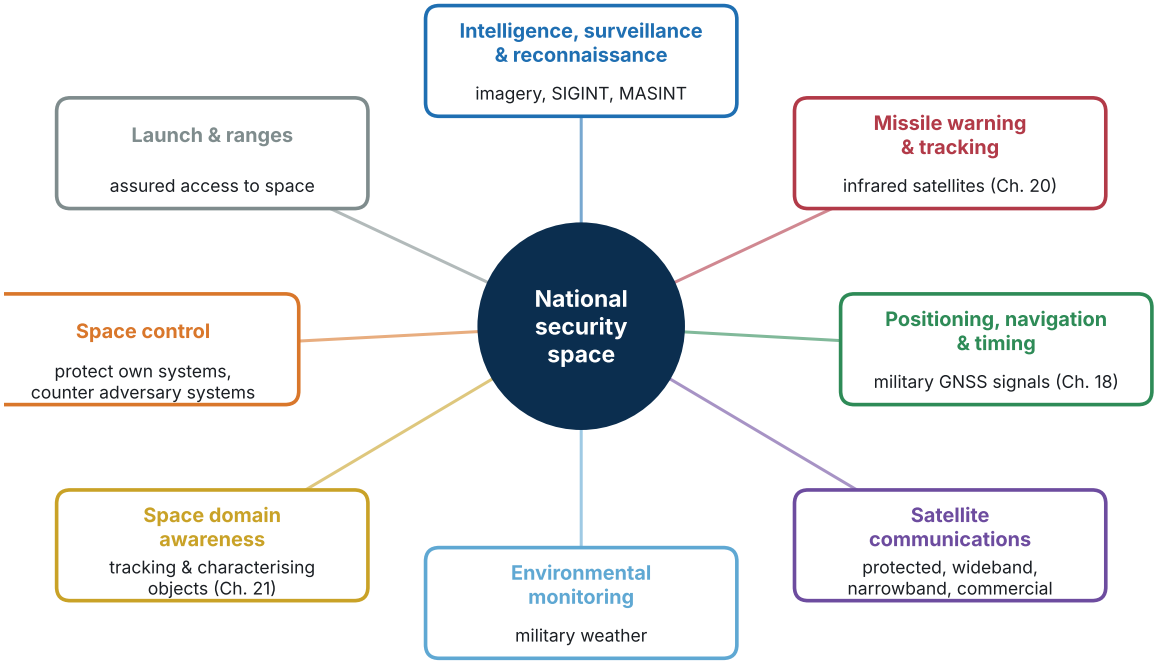



**FIGURE 19.1** Mission areas of national-security space.

## 19.1.2 History

The first US reconnaissance satellites flew within three years of Sputnik (Figure 19.2). The Naval Research Laboratory's **GRAB** satellite, launched in June 1960 under cover of a scientific mission, intercepted Soviet air-defence radar signals — the world's first SIGINT satellite. Two months later, the **Corona** programme achieved the first recovery of exposed film from orbit, dropped in a capsule and caught in mid-air by an aircraft. A single early Corona mission photographed more of the Soviet Union than all previous U-2 flights combined, settling the debate about the "missile gap". Corona flew from 1960 to 1972, its resolution improving from about 12 m to under 2 m; its more than 800 000 images were declassified in 1995 and remain a valuable record of mid-20th-century landscapes. The Soviet Union's **Zenit** satellites, derived from the Vostok crewed spacecraft, followed from 1962.

Film-return systems gave way to electro-optical imaging satellites, which transmit images digitally, in the mid-1970s. The **Vela** satellites (from 1963) watched for nuclear tests in space and the atmosphere, and the **Defense Support Program** (from 1970) for ballistic-missile launches. The 1991 Gulf War, in which GPS, satellite communications, missile warning and reconnaissance supported coalition forces on an unprecedented scale, is often called the first "space war". The very existence of the US **National Reconnaissance Office (NRO)**, founded in 1961, was declassified only in 1992.

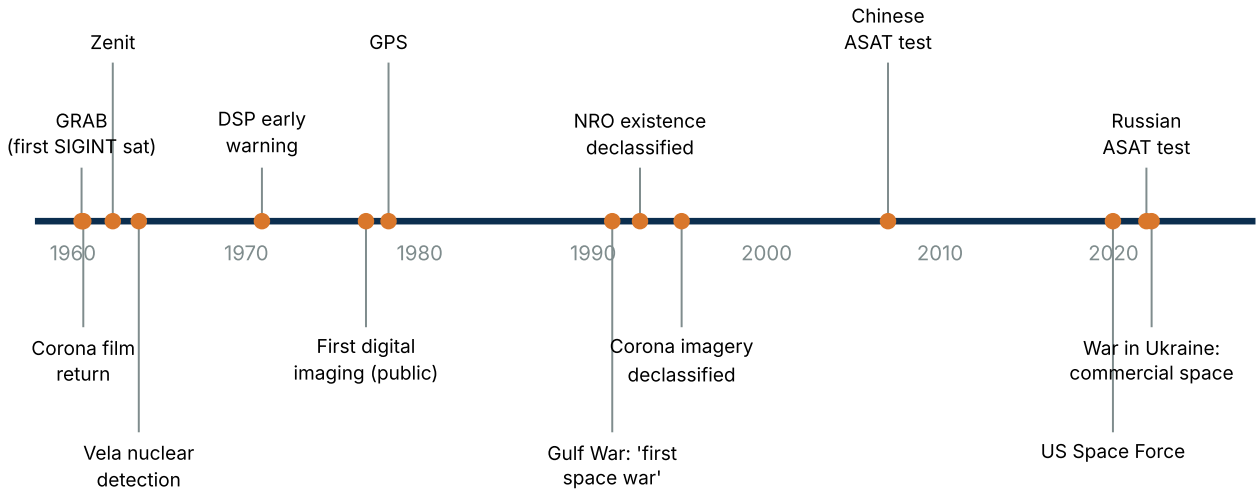



**FIGURE 19.2** Milestones in national-security space.

### 19.1.3 Organisations

Many countries have reorganised their armed forces to treat space as a domain of operations alongside land, sea, air and cyberspace (Table 19.1). NATO recognised space as an operational domain in 2019.

**TABLE 19.1** Selected military space organisations (public information).

| Country | Military space organisation | Intelligence / other agencies | Notes |
|---|---|---|---|
| USA | US Space Force (2019); US Space Command (re-established 2019); Space Development Agency (2019) | National Reconnaissance Office; National Geospatial-Intelligence Agency | Largest military space budget; extensive use of commercial services |
| China | PLA Aerospace Force (2024, formed from the Strategic Support Force) | — | Yaogan, Gaofen, Shijian and TJS series; rapid growth |
| Russia | Aerospace Forces (VKS), Space Forces | GRU and others | Persona, Bars-M, Lotos-S/Liana, Tundra |
| France | Space Command (CDE, 2019) | DGSE, DRM | CSO imaging satellites; CERES SIGINT trio |
| United Kingdom | UK Space Command (2021) | — | Skynet SATCOM; partnerships and commercial data |
| Germany, Italy, Spain | National space commands | — | SARah, COSMO-SkyMed, PAZ radar satellites |
| Japan | Air and Space Self-Defense Force (renamed) | Cabinet Satellite Intelligence Center | Information Gathering Satellites (IGS) |
| India | Defence Space Agency (2019) | — | RISAT, EMISAT (ELINT), dedicated GSAT |
| Israel | Israeli Air Force / Ministry of Defence | — | Ofek imaging satellites |

## 19.2 Imagery Intelligence from Space

Military imaging satellites use the same physics as civil Earth observation (Chapters 13–15): high-resolution optical and infrared telescopes, and synthetic aperture radars for all-weather, day-and-night coverage. The most capable systems are believed to use mirrors of about 2.4 m — the same size as the Hubble Space Telescope — giving resolutions of roughly 10–15 cm from low orbit (Chapter 14, Figure 14.3). Governments seldom publish the specifications of such systems. What distinguishes military imaging is less the sensor than the **system**: priority tasking, rapid processing, secure dissemination and the fusion of many sources into intelligence products.

### 19.2.1 Revisit and Constellation Design

For many military purposes the critical parameter is not resolution but **revisit** — how often a target can be observed. A single agile imaging satellite in a sun-synchronous orbit can view a mid-latitude target only about once every two to three days in daylight; doubling the number of satellites roughly halves the wait (Figure 19.3). Radar satellites, which can image at night as well, have about twice as many opportunities. To watch a target several times per day, a constellation of a dozen or more satellites is needed. An alternative is satellites in mid-inclination orbits, which pass over mid-latitudes more often and at varying local times. This is why both governments and commercial operators have moved towards larger constellations of smaller satellites.

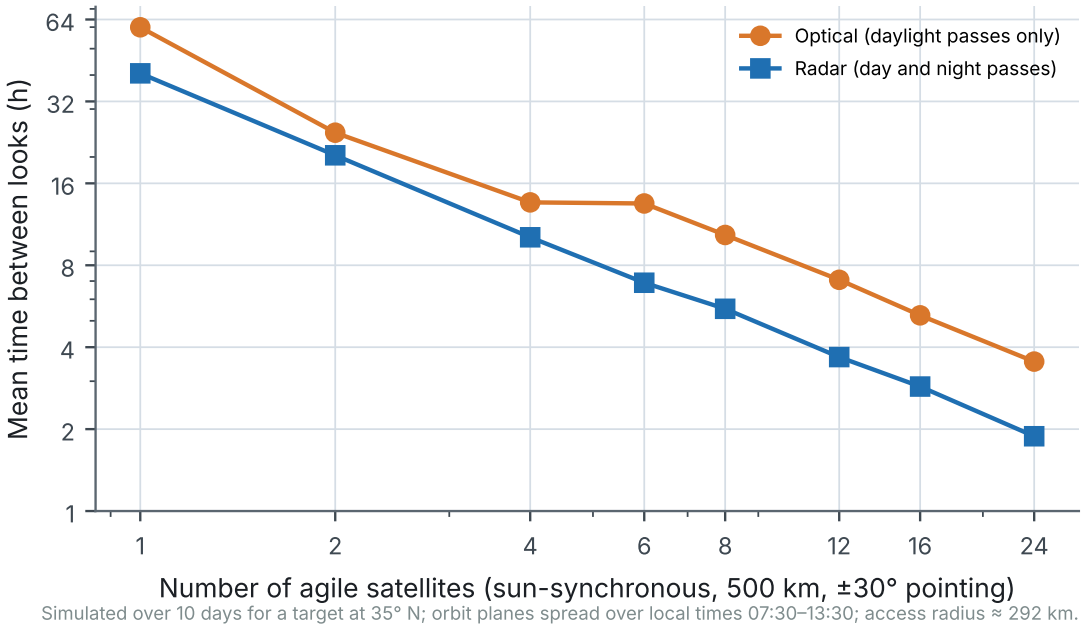



**FIGURE 19.3** Simulated mean time between looks at a target at 35° N as a function of the number of agile imaging satellites.

**TABLE 19.2** Selected government imaging and reconnaissance systems (public information).

| Country | System (public name) | Type | Notes |
|---|---|---|---|
| USA | NRO electro-optical and radar satellites | Optical, SAR | Specifications classified; large-aperture optical systems since 1976 |
| USA | NRO proliferated architecture | Optical and other sensors | Launched from 2024 with SpaceX and Northrop Grumman |

| Country | System (public name) | Type | Notes |
|---|---|---|---|
| France | CSO (Composante Spatiale Optique) | Optical | Three satellites (2018, 2020, 2025); shared with partners |
| Germany | SAR-Lupe, SARah | SAR | SARah constellation launched 2022–2023 |
| Italy | COSMO-SkyMed (1st and 2nd generation) | SAR | Dual-use military and civil |
| Japan | Information Gathering Satellites (IGS) | Optical, SAR | Programme started after the 1998 North Korean missile overflight |
| Israel | Ofek, TecSAR | Optical, SAR | Launched westward from Israel on Shavit rockets |
| China | Yaogan, Gaofen series | Optical, SAR, ELINT | About 170 Yaogan satellites in orbit; some in triplets |
| Russia | Persona, Bars-M, Kondor | Optical, SAR | Smaller fleet than the USA or China |
| India | Cartosat, RISAT, EMISAT | Optical, SAR, ELINT | Dual-use national programmes |

### 19.2.2 The Intelligence Cycle

Intelligence agencies describe their work as a cycle. It starts with **planning and direction** (what do decision-makers need to know?) and **collection** (tasking satellites and other sensors). Next comes **processing and exploitation** (turning raw data into information — locating, identifying and measuring objects). It ends with **analysis and production** (combining sources into assessments) and **dissemination**. Satellites contribute mainly to collection, but the bottlenecks have increasingly moved to processing and analysis. Modern constellations generate far more images than human analysts can examine. This is why AI-based object detection and change detection have become essential (Part XIII).

### 19.2.3 Commercial Imagery in National Security

Commercial imagery now complements government systems on a large scale. The USA buys imagery under the NRO's Electro-Optical Commercial Layer contracts and from commercial SAR providers, and many smaller nations rely entirely on commercial or allied data. During the war in Ukraine, governments have widely used commercial optical and radar imagery from companies such as Vantor (formerly Maxar), Planet, Airbus, ICEYE, Capella and BlackSky. The media have also published this imagery, transforming public understanding of military events. Commercial data are unclassified and shareable with allies and the public, which has given them diplomatic as well as operational value.

## 19.3 Signals Intelligence from Space

### 19.3.1 Disciplines

**Signals intelligence (SIGINT)** is the collection and analysis of electromagnetic signals emitted by an adversary's systems (Table 19.3). It is traditionally divided into:

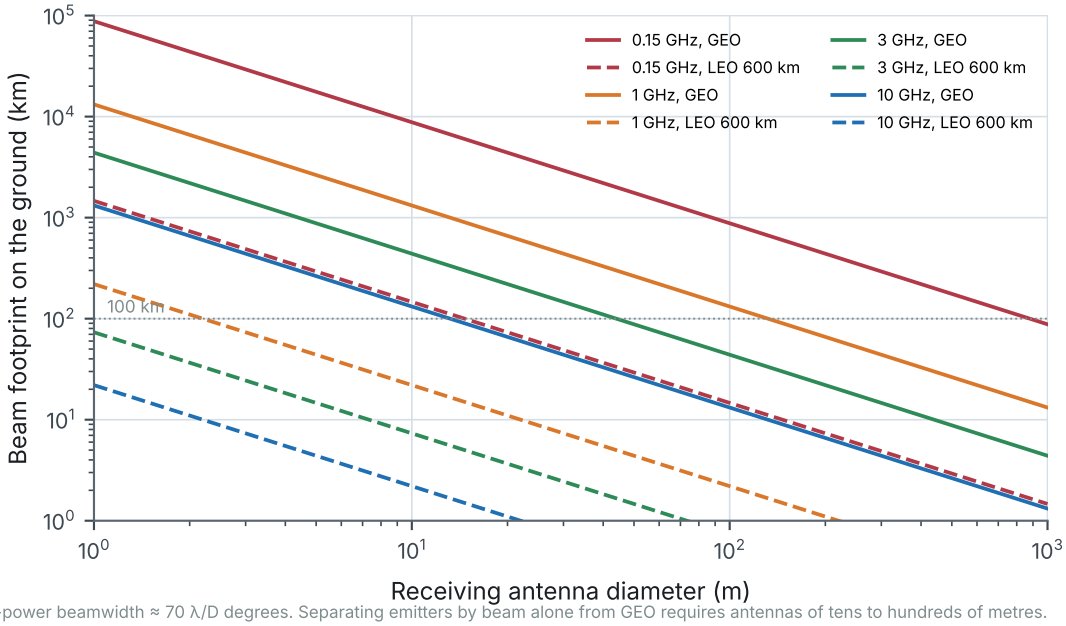
**TABLE 19.3** Signals intelligence disciplines.

| Discipline | Target | Typical frequencies | Information obtained |
|---|---|---|---|
| ELINT (electronic intelligence) | Radars and other non-communication emitters | ≈ 0.1–40 GHz | Location, type and operating mode of radars; air-defence order of battle |
| COMINT (communications intelligence) | Voice, data and messaging | HF to Ka-band | Content (if decrypted), traffic patterns, location of communicators |
| FISINT (foreign instrumentation signals intelligence) | Telemetry from missiles, aircraft and spacecraft | VHF to S-band | Performance of weapons under test |
| Commercial RF geolocation | Ships with AIS switched off, satellite phones, push-to-talk radios, GNSS jammers | VHF to Ku-band | Location and track of emitters for maritime and security customers |

## 19.3.2 Intercept Physics

A SIGINT satellite is a sensitive radio receiver with a large antenna. Intercepting radar signals is relatively easy because radars are powerful — often hundreds of kilowatts to megawatts of peak power — and even their side lobes are strong. Intercepting low-power communications from space is harder. The link budget (Chapter 7) shows that a 5 W VHF handheld radio received at 600 km arrives at about −95 dBm, which is easily detectable. At geostationary distance, however, it arrives about 36 dB weaker, so a significant receiving antenna is needed.

The bigger challenge is to **separate and locate** emitters. A receiving antenna's beamwidth is about $70\lambda/D$ degrees, so at low frequencies even large antennas have broad beams (Figure 19.4). From geostationary orbit, a 10 m antenna at 1 GHz has a beam about 1 300 km wide. At VHF, a beam narrow enough to isolate a single city would need an antenna several hundred metres across. Public reports describe very large deployable reflectors on some geostationary SIGINT satellites, and complementary geolocation techniques are used.



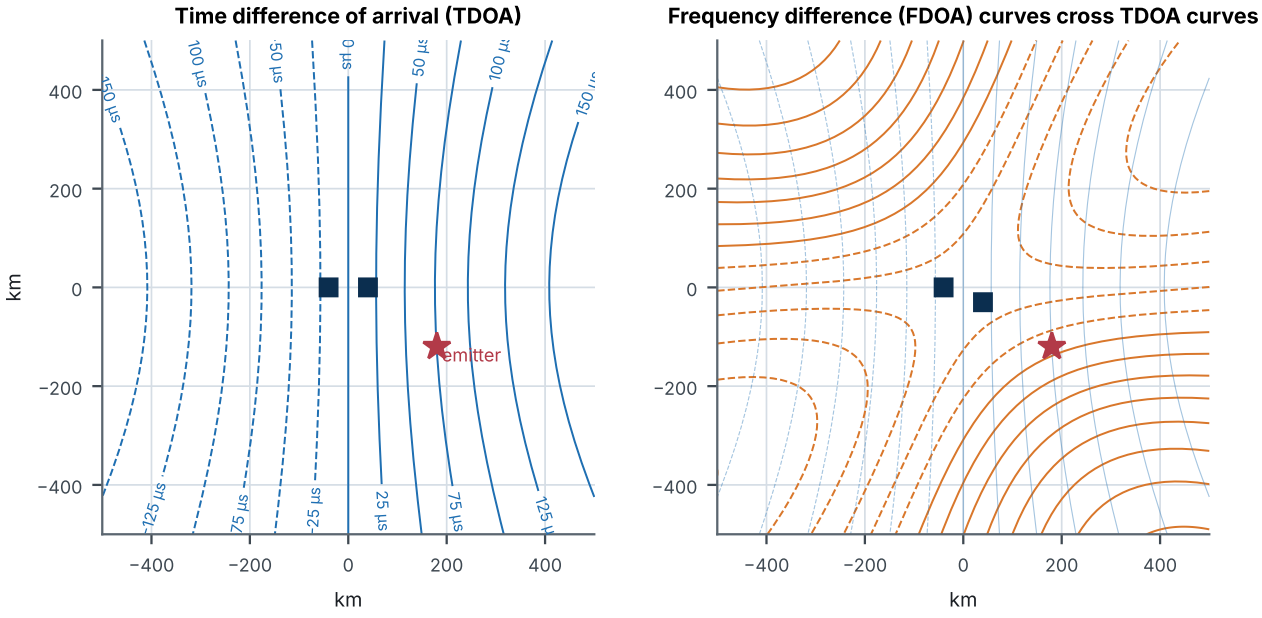

Half-power beamwidth ≈ 70 λ/D degrees. Separating emitters by beam alone from GEO requires antennas of tens to hundreds of metres.

**FIGURE 19.4** Ground footprint of a receiving antenna beam as a function of antenna diameter, for four frequencies, from geostationary and low orbit. Isolating emitters by beam alone requires very large antennas, especially at low frequencies.

### 19.3.3 Geolocation Techniques

Emitters are located by measuring how their signals arrive at one or more receivers:

- **Angle of arrival (AOA)**: an interferometer measures the phase difference of a signal between two or more antennas separated by a baseline $d$; the phase difference $2\pi d \sin\theta/\lambda$ gives the arrival direction $\theta$ (Figure 19.6). Long baselines give precise but ambiguous angles; combinations of baselines resolve the ambiguity.
- **Time difference of arrival (TDOA)**: two satellites receive the same signal at slightly different times; each measured time difference places the emitter on a hyperbolic curve on the Earth's surface (Figure 19.5). Three satellites give two independent TDOAs whose curves intersect at the emitter.
- **Frequency difference of arrival (FDOA)**: because the satellites move, they receive slightly different Doppler shifts. The frequency difference defines another family of curves that cross the TDOA curves. So a pair of satellites measuring both can fix an emitter.
- **Single-satellite Doppler**: like the old Transit system in reverse, one satellite can locate a continuously transmitting emitter from the Doppler curve observed during a pass.




Two satellites 80 km apart at 600 km altitude (squares: sub-satellite points). Each measurement places the emitter on a curve; combining TDOA and FDOA (or three satellites) fixes its position.

**FIGURE 19.5** Emitter geolocation from a pair of satellites. Lines of constant time difference of arrival (left) are hyperbolas on the ground; lines of constant frequency difference (right) cross them, so the two measurements together fix the emitter.

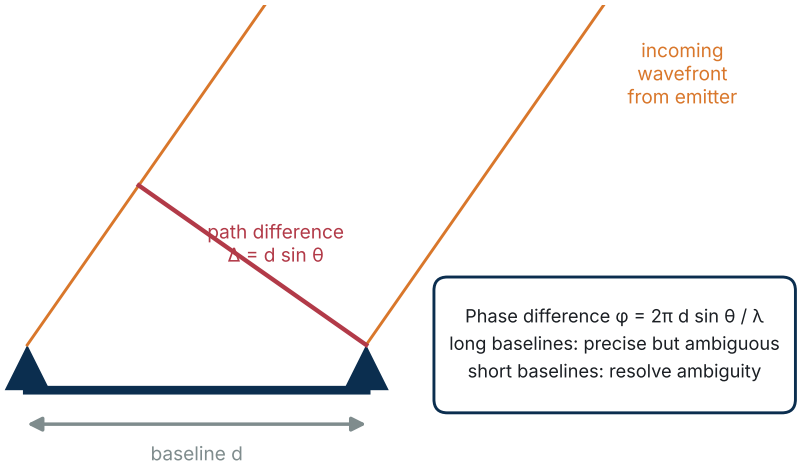



**FIGURE 19.6** Angle-of-arrival measurement with a two-element interferometer.

These principles explain the distinctive configurations of SIGINT satellites reported in public sources. Ocean-surveillance systems such as the historic US "White Cloud"/NOSS series and China's Yaogan triplets fly **clusters** of two or three satellites tens of kilometres apart to measure TDOA from ship radars and communications. France's **CERES** system (three satellites launched in 2021) and its ELISA demonstrators locate radars for the French armed forces. Russia's Lotos-S/Liana system and India's EMISAT perform similar roles.

### 19.3.4 Commercial RF Geolocation

Since 2018, commercial companies have brought these techniques to an open market. **HawkEye 360** (USA) flies clusters of three microsatellites that detect and geolocate signals from marine VHF radios, AIS transponders, satellite phones, radars and GNSS jammers; by 2026 it had about 40 satellites in orbit. France's **Unseenlabs** locates ships from their radio emissions using single satellites, and Luxembourg's **Kleos Space** and others offer similar services. Their customers include coast guards fighting illegal fishing and smuggling — finding "dark" vessels that switch off their AIS transponders — and defence agencies.

### 19.3.5 MASINT

**Measurement and signature intelligence (MASINT)** uses the physical signatures of objects and events. These include infrared signatures of missile plumes and industrial activity, spectral signatures of materials (hyperspectral imaging) and radar signatures. They also include the light flash, X-rays and electromagnetic pulse of nuclear detonations. Nuclear-detonation detection sensors have flown on GPS satellites since the 1980s as part of the US Nuclear Detonation Detection System, continuing the role of the Vela satellites in verifying the Limited Test Ban Treaty.

## 19.4 Military Satellite Communications

Military forces need communications that are global, secure and resistant to jamming and interception. Military SATCOM is usually divided into three tiers (Table 19.4):

**TABLE 19.4** Military satellite communication tiers.

| Tier | Purpose | Bands | Example systems |
|---|---|---|---|

| Tier | Purpose | Bands | Example systems |
|---|---|---|---|
| Protected / strategic | Survivable, jam-resistant, low-probability-of-intercept links for nuclear command and control and critical operations | EHF (44 GHz uplink / 20 GHz downlink) | US AEHF (six satellites, 2010–2020) and planned ESS; Milstar (legacy) |
| Wideband | High-capacity links for headquarters, ships, aircraft and drones | X, Ka | US WGS; UK Skynet 5/6; France Syracuse 4; Italy Sicral 2; Spain SpainSat NG |
| Narrowband / tactical | Voice and data to small mobile terminals | UHF | US MUOS (five satellites, 2012–2016); UHF Follow-On |
| Commercial and hybrid | Additional capacity, resilience and Arctic coverage | Ku, Ka, L | Commercial GEO and LEO services, Starshield, Space Norway ASBM |

**Jam resistance** is achieved mainly by spreading signals over very wide bandwidths — by direct-sequence spreading or rapid **frequency hopping** — so that a jammer must spread its power across the whole band. The protection, called **processing gain**, is $10\log(W/R)$, where $W$ is the spread bandwidth and $R$ the data rate (Figure 19.7). EHF systems hop over about 2 GHz, giving more than 60 dB of protection to low-rate strategic messages. Other techniques include **nulling antennas** that steer deep nulls towards jammers, on-board processing that demodulates signals and rejects interference, and narrow spot beams. **Low probability of intercept and detection (LPI/LPD)** waveforms hide transmissions below the noise.

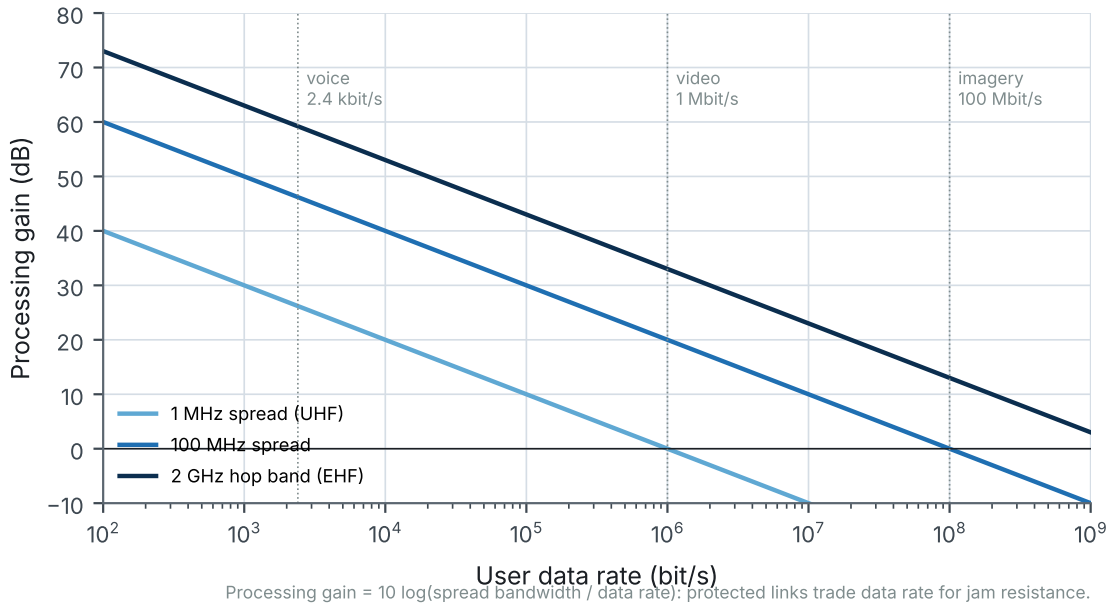



**FIGURE 19.7** Processing gain of spread-spectrum military links versus user data rate. High data rates leave little margin against jamming.

The war in Ukraine demonstrated the value — and vulnerability — of commercial communications in conflict. Thousands of Starlink terminals provided resilient connectivity to Ukrainian forces and civil infrastructure. Meanwhile, on 24 February 2022, the first day of the invasion, a cyber attack on the ground network of Viasat's KA-SAT system disabled tens of thousands of terminals across Europe.

# 19.5 Military Weather, Launch and the Wider Enterprise

## 19.5.1 Military Weather

Weather affects every military operation — cloud cover over a target, winds for parachute drops, sea state for landings, and space weather for communications and GPS. The US Defense Meteorological Satellite Program (DMSP), operating since the 1960s, pioneered night-time visible imaging and microwave sounding; its last satellites are reaching the end of their lives. Its replacements include the **Weather System Follow-on – Microwave (WSF-M)**. The first WSF-M was launched in April 2024 to measure ocean-surface winds and tropical-cyclone structure. The replacements also include small electro-optical/infrared weather satellites for cloud imagery over regions where geostationary coverage is poor. Military users also rely heavily on civil systems such as Metop and JPSS (Chapter 16).

## 19.5.2 National Security Launch

Assured access to space is a national-security requirement in itself. The USA certifies launch providers for national-security payloads under the **National Security Space Launch (NSSL)** programme. In the 2020s, launches were shared between SpaceX (Falcon 9 and Falcon Heavy) and United Launch Alliance (Vulcan Centaur, which replaced Atlas V and Delta IV). From Phase 3 (2025 onward), they were also shared with Blue Origin's New Glenn, with a separate "lane" for new entrants on lower-risk missions. **Tactically responsive space** demonstrations, such as Victus Nox in 2023, aim to show that satellites can be replaced quickly in a crisis. In Victus Nox, a satellite was launched 27 hours after the order was given. China, Russia, India and Europe maintain their own launchers for sovereign access (Chapter 36).

## 19.5.3 Who Operates Military Satellites?

Figure 19.8 summarises the publicly catalogued military satellites of the main space powers by orbit. The USA operates the largest number — boosted by proliferated LEO constellations — followed by China, whose Yaogan series alone counts about 170 satellites, and Russia. Many military satellites operate in LEO for imaging and SIGINT; communications, missile-warning and some SIGINT satellites are in GEO and highly elliptical orbits; and navigation satellites are in MEO.

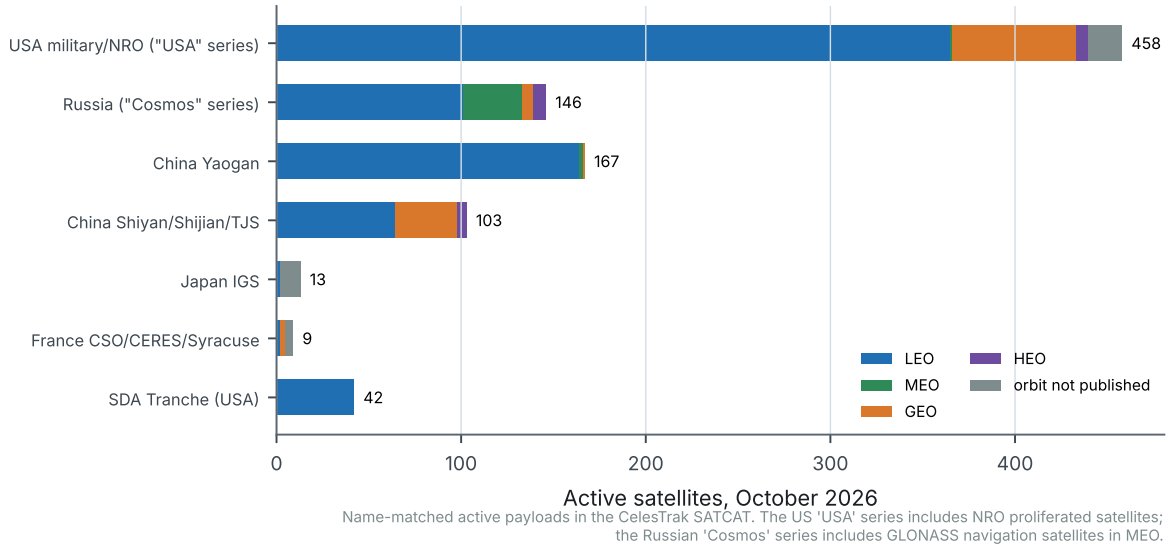



**FIGURE 19.8** Active military satellites of selected series by orbit regime in October 2026 (public catalogue).

### 19.5.4 Allies and Commercial Partners

No country can afford all space capabilities alone. The **Combined Space Operations (CSpO)** initiative brings together the space commands of Australia, Canada, France, Germany, Italy, Japan, New Zealand, Norway, the UK and the USA to coordinate operations and share data. NATO adopted its first space policy in 2019 and established a Space Centre at its air command in Germany. Commercial companies increasingly act as partners: providing imagery, RF geolocation, communications, launch and space-tracking data, and in some cases building dedicated government constellations. This raises new questions — such as whether commercial satellites supporting military operations become legitimate targets, and how governments protect companies that operate in conflict zones.

## 19.6 From Exquisite to Proliferated Architectures

For most of the space age, national-security satellites were few, large, expensive and long-lived — "exquisite" systems that took a decade to develop and cost billions of dollars each. Such systems are extremely capable but also tempting targets: losing one can remove a capability for years. Since about 2019, the USA and other countries have shifted towards **resilience** through:

- **Proliferation**: many cheaper satellites, so that losing a few does not matter;
- **Disaggregation**: separating functions (for example strategic and tactical missile warning) onto different satellites;
- **Diversification**: using several orbits, frequencies and providers, including commercial ones;
- **Rapid replenishment**: replacing satellites every few years in "tranches" that incorporate new technology.

The US **Space Development Agency (SDA)**, founded in 2019, is building the **Proliferated Warfighter Space Architecture (PWSA)** in low Earth orbit at about 1 000 km (Figure 19.9). Its **Transport Layer** forms a mesh network of optical inter-satellite links and radio links (including the Link 16 tactical data link) to deliver targeting data directly to forces. Its **Tracking Layer** carries infrared sensors to detect and track missiles, including hypersonic glide vehicles

(Chapter 20). About two dozen Tranche 0 demonstration satellites launched in 2023, and the first operational Tranche 1 satellites launched in September 2025. Tranche 1 comprises 154 satellites (126 transport and 28 tracking), with further tranches planned every two years. Separately, the NRO has been deploying a proliferated constellation of imaging satellites, built with SpaceX and Northrop Grumman, since 2024.

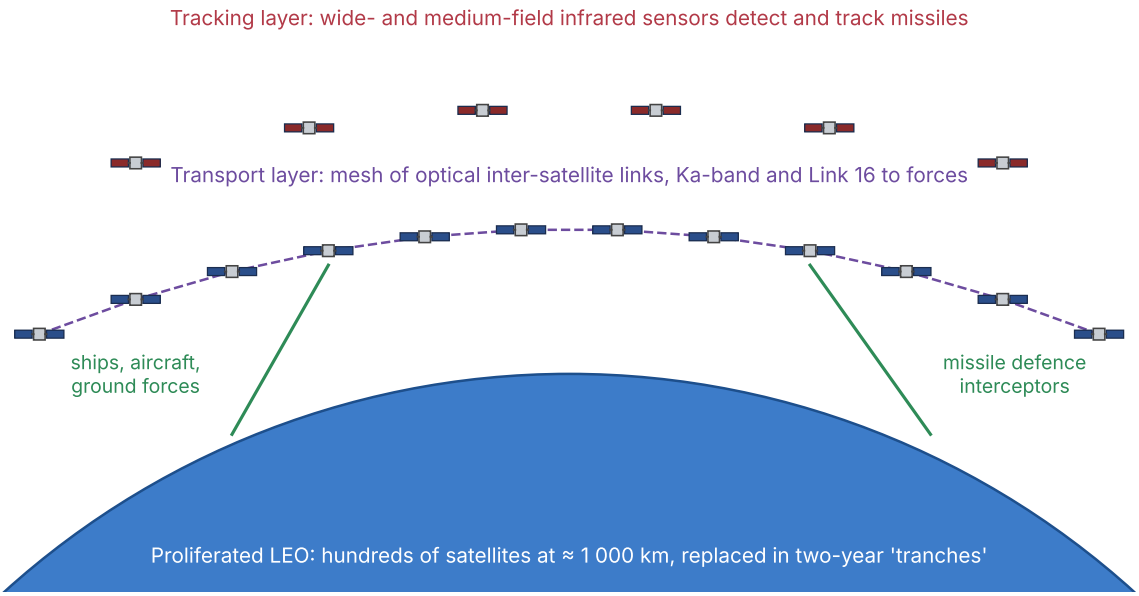



**FIGURE 19.9** The Proliferated Warfighter Space Architecture: a transport layer of laser-linked satellites delivers data from a tracking layer of infrared sensors to forces and missile-defence systems.

In 2025 the US government launched **Golden Dome**, an ambitious homeland missile-defence initiative that includes space-based sensors and, controversially, **space-based interceptors**. In April 2026 the Space Force awarded prototype agreements worth up to $3.2 billion to twelve companies to demonstrate space-based interceptors by 2028; independent analysts have questioned the technical feasibility and cost of the overall programme, which official estimates put at about $175–185 billion.

Other powers are following similar paths: China has launched large numbers of Yaogan and other military satellites and is building proliferated LEO constellations, and Europe's IRIS$^2$ constellation will include secure governmental communications.

### 19.6.1 Spaceplanes and Experimental Platforms

The US Space Force's **X-37B** Orbital Test Vehicle, a small reusable uncrewed spaceplane built by Boeing (Chapter 10), has flown eight missions since 2010, some lasting more than two years. Its seventh mission (2023–2025) demonstrated aerobraking from a highly elliptical orbit, and its eighth was launched in August 2025. China has flown a comparable reusable experimental spacecraft several times since 2020. Such platforms test new technologies and allow sensors to be returned to Earth for inspection.

# 19.7 Counterspace: A Contested Domain

## 19.7.1 Categories of Threat

As military dependence on space grew, so did interest in denying an adversary's use of it. Public assessments by organisations such as the Secure World Foundation and the Center for Strategic and International Studies group counterspace capabilities into four categories (Figure 19.10):

- **Kinetic physical**: direct-ascent anti-satellite (DA-ASAT) missiles that destroy a satellite by collision; co-orbital weapons that approach and attack a target; attacks on ground stations.
- **Non-kinetic physical**: lasers that dazzle or damage optical sensors, high-power microwaves and — most dangerously — nuclear detonations in space, whose radiation would damage large numbers of satellites indiscriminately.
- **Electronic**: jamming of uplinks and downlinks and spoofing of GNSS and other signals — the most widely used and most easily reversible form of counterspace.
- **Cyber**: attacks on the data, ground systems or satellites themselves.

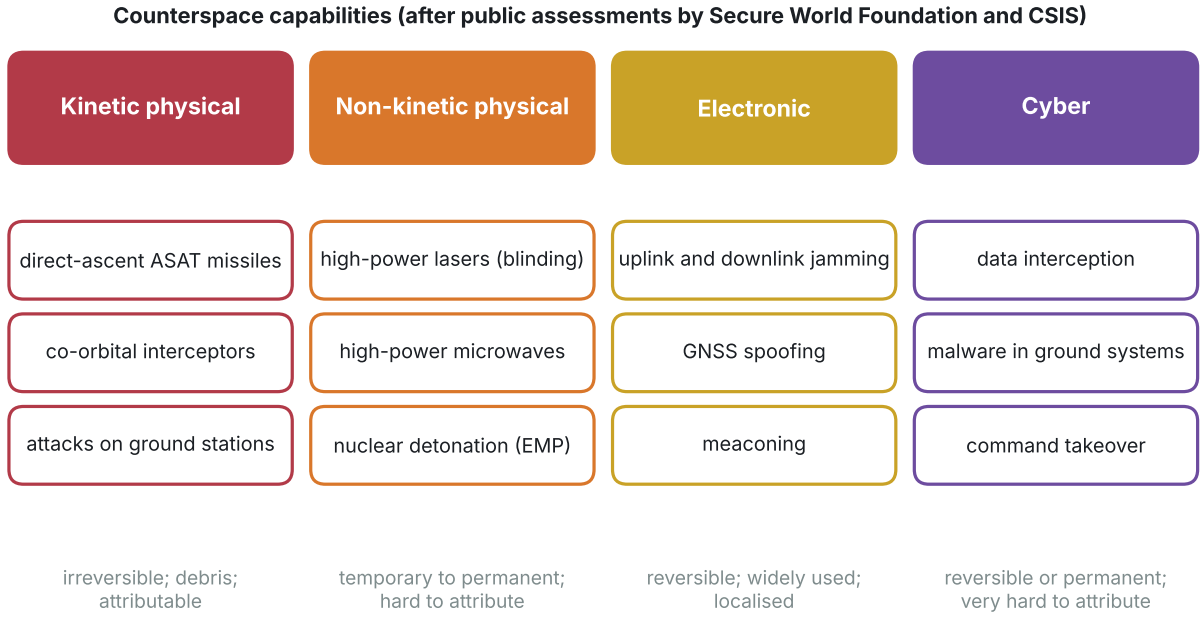



**FIGURE 19.10** Categories of counterspace capability.

## 19.7.2 Anti-Satellite Tests and Debris

Five destructive anti-satellite tests have created long-lived debris (Figure 19.11; Table 19.5). The worst was China's January 2007 destruction of its Fengyun-1C weather satellite at about 865 km. It produced more than 3 500 catalogued fragments — the largest debris-generating event in history. More than 2 300 were still in orbit in October 2026, threatening satellites in sun-synchronous orbits for decades. Russia's November 2021 test against the defunct Cosmos 1408 at about 480 km produced about 1 800 catalogued fragments and forced the ISS crew to shelter in their spacecraft; most have since re-entered. Tests at lower altitudes included the US destruction of the failing USA-193 satellite in 2008 at about 247 km and India's "Mission Shakti" in 2019 at about 283 km. These created debris that decayed within months to a few years.

**TABLE 19.5** Destructive direct-ascent anti-satellite tests (debris counts from the CelesTrak SATCAT).

| Year | Country | Target | Intercept altitude | Catalogued debris | Still in orbit (Oct 2026) |
|---|---|---|---|---|---|
| 1985 | USA | Solwind P78-1 (air-launched ASM-135) | ≈ 525 km | 286 | 0 |
| 2007 | China | Fengyun-1C (SC-19 missile) | ≈ 865 km | 3 534 | 2 314 |
| 2008 | USA | USA-193 (SM-3 from a ship) | ≈ 247 km | 174 | 0 |
| 2019 | India | Microsat-R (PDV Mk-II) | ≈ 283 km | 129 | 0 |
| 2021 | Russia | Cosmos 1408 (Nudol) | ≈ 480 km | 1 806 | 4 |

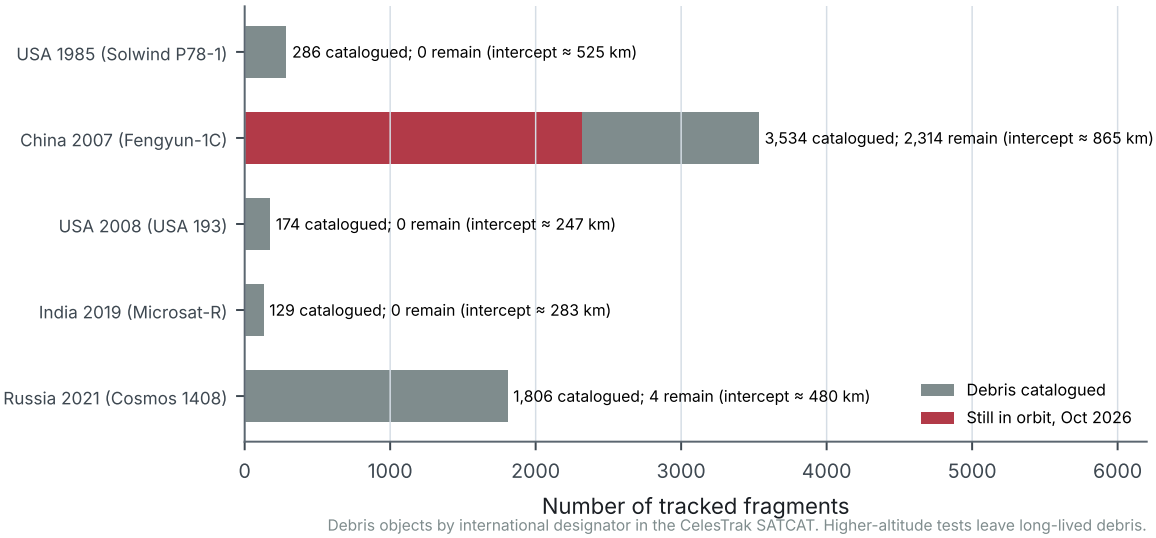
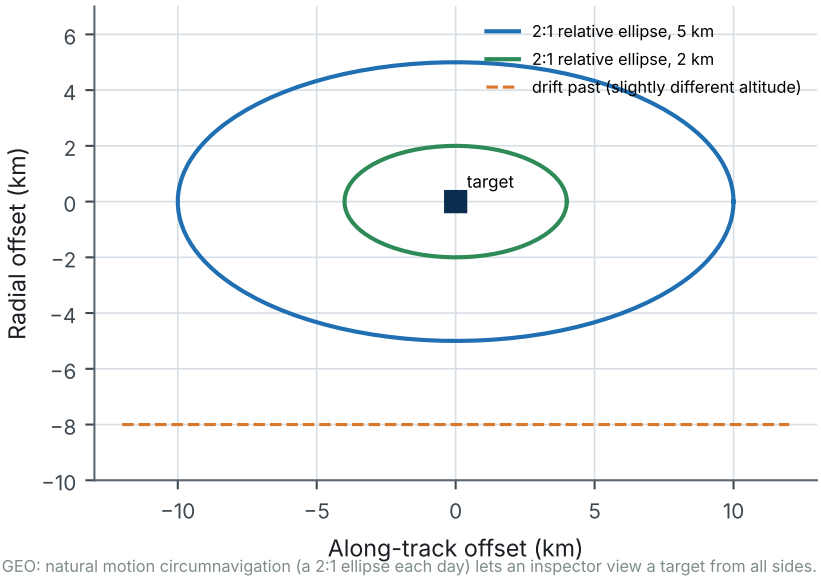



**FIGURE 19.11** Tracked debris from destructive anti-satellite tests. Debris from the high-altitude 2007 test remains in orbit; debris from lower-altitude tests has re-entered.

In April 2022 the USA announced a unilateral commitment not to conduct destructive direct-ascent ASAT missile tests. In December 2022 the UN General Assembly adopted a resolution calling on all states to make the same commitment. Dozens of countries have since joined the pledge.

### 19.7.3 Rendezvous and Proximity Operations

Satellites that can approach other satellites — for inspection, servicing or, potentially, attack — are a growing concern because the same technology serves peaceful and hostile purposes. The USA operates the Geosynchronous Space Situational Awareness Program (GSSAP) satellites to inspect objects in GEO. Russian satellites such as Cosmos 2542 and Luch/Olymp have caused public concern. Cosmos 2542 released a sub-satellite that manoeuvred near a US reconnaissance satellite in 2020, and Luch/Olymp has parked near Western communication satellites. China's Shijian-21 towed a dead BeiDou satellite to a graveyard orbit in 2022, demonstrating capture technology. Orbital mechanics shapes these operations: an inspector at a slightly different orbit can perform a natural circumnavigation of its target once per orbit without using propellant (Figure 19.12).




Clohessy–Wiltshire relative motion in GEO: natural motion circumnavigation (a 2:1 ellipse each day) lets an inspector view a target from all sides.

**FIGURE 19.12** Relative motion of an inspector satellite around a target in geostationary orbit (Clohessy–Wiltshire equations). A suitably offset orbit produces a daily 2:1 ellipse around the target.

### 19.7.4 Jamming, Spoofing and Cyber Attacks

Electronic and cyber attacks are now routine in conflicts. GNSS jamming and spoofing affect aviation and shipping across large regions (Chapter 18), and satellite television and communications are jammed. The 2022 KA-SAT attack showed how a cyber attack on a satellite operator's ground network can have international effects. Because these attacks are reversible, hard to attribute and often below the threshold of armed conflict, they are difficult to deter.

### 19.7.5 Nuclear Weapons in Space

The 1962 US Starfish Prime test, a 1.4-megaton nuclear explosion at 400 km, created artificial radiation belts that damaged or destroyed about a third of the satellites then in orbit (Chapter 22). The Outer Space Treaty (1967) prohibits placing nuclear weapons or other weapons of mass destruction in orbit. In 2024 US officials publicly expressed concern that Russia was developing a nuclear anti-satellite capability; a UN Security Council resolution reaffirming the prohibition was vetoed by Russia in April 2024.

## 19.8 Law, Norms and Stability

International law applies in space. The **Outer Space Treaty** declares space the province of all humankind and prohibits national appropriation and weapons of mass destruction in orbit. It also requires states to authorise and supervise their nationals' space activities. The UN Charter, including the right of self-defence, and the law of armed conflict also apply. However, there is no treaty banning conventional weapons in space or ground-based anti-satellite weapons. Diplomatic efforts include:

- Russian–Chinese proposals for a treaty on the Prevention of the Placement of Weapons in Outer Space (PPWT), criticised by Western states for lacking verification and ignoring ground-based ASATs;
- the UK-led UN Open-Ended Working Group on **reducing space threats through norms, rules and principles of responsible behaviour** (2022–23);

- the destructive DA-ASAT test moratorium;
- expert efforts to clarify how existing law applies, such as the Woomera Manual and the MILAMOS project.

Transparency measures include announcing manoeuvres, sharing space-tracking data, and establishing communication channels between space commands. They aim to reduce the risk of misunderstanding in a crowded orbital environment, where an accidental collision could be mistaken for an attack.

**EXAMPLE: GEOLOCATING A SHIP RADAR WITH TWO SATELLITES**

Two SIGINT satellites fly 80 km apart at 600 km altitude. A ship's navigation radar transmits pulses at 9.4 GHz.

1. The satellites time-tag each received pulse with GNSS-disciplined clocks. A time difference of 100 µs means the ship is about 30 km farther from one satellite than the other; the ship lies on a hyperbola-like curve on the sea surface (Figure 19.5, left).
2. A timing precision of 50 ns corresponds to a range-difference precision of 15 m; geometry stretches the position error along the curve, typically to a few hundred metres.
3. The satellites also measure the Doppler shift of the pulses: their slightly different velocity vectors give a frequency difference of a few kilohertz that places the ship on a second family of curves (Figure 19.5, right).
4. The intersection gives the ship's position. Repeating the measurement on successive pulses, or on the next pass, gives its course and speed — even if it has switched off its AIS transponder.

**AI IN NATIONAL-SECURITY SPACE**

AI is used to detect objects and changes in imagery, classify radio emitters from their signal characteristics, fuse data from many sensors, detect anomalous satellite behaviour, plan collections across large constellations and support battle management. These uses raise questions about reliability, adversarial manipulation and human control, discussed in Chapters 50 and 53.

**KEY TAKEAWAYS**

- National-security space includes ISR, missile warning, PNT, SATCOM, weather, space domain awareness, space control and launch.
- Reconnaissance satellites date from 1960 (Corona film return, GRAB SIGINT); imagery is now digital and complemented by large commercial constellations.
- SIGINT satellites intercept radar (ELINT), communications (COMINT) and telemetry (FISINT); emitters are located by angle of arrival, TDOA and FDOA, often with satellite clusters. Commercial RF geolocation now serves maritime security.
- Military SATCOM uses wide spreading and frequency hopping for jam resistance; commercial services have become important but are also targets.
- Architectures are shifting from a few exquisite satellites to proliferated, frequently replenished constellations (SDA PWSA) for resilience.
- Five destructive ASAT tests have created thousands of debris fragments; the 2007 Chinese test left debris that will persist for decades. Jamming, spoofing and cyber attacks are now routine in conflicts.
- International law applies in space, but there is no ban on conventional space weapons; norms such as the DA-ASAT test moratorium aim to improve stability.

### REVIEW QUESTIONS

1. List the mission areas of national-security space and give one example system for each.
2. Why was Corona so important, and how did it work?
3. Distinguish ELINT, COMINT and FISINT.
4. Two satellites measure a time difference of arrival of 2.0 μs for the same radar pulse. What path-length difference does this represent, and what locus on the Earth's surface does it define?
5. How does FDOA geolocation work, and why do commercial RF-geolocation companies fly satellites in clusters?
6. Explain why national-security architectures are moving from a few exquisite satellites to proliferated constellations.
7. Why are destructive anti-satellite tests especially harmful to all space users? Use the 2007 test as an example.
8. What legal principles apply to military activities in space, and what is the purpose of the moratorium on destructive direct-ascent ASAT tests?

# 20 CHAPTER Missile Early Warning Satellites

Image: NASA/JSC

*The fiery exhaust of a rocket is one of the brightest infrared sources on Earth. Since 1970, satellites watching for that signature have given the first warning of ballistic-missile launches anywhere in the world, underpinning nuclear deterrence and, increasingly, missile defence. This chapter explains, from public sources, how early-warning satellites work: the infrared physics of rocket plumes and the atmosphere, scanning and staring sensors, orbits that cover the globe and the poles, the processing that separates launches from sunlight and clouds, and the timelines of ballistic-missile flight. It traces the history from MIDAS and the Defense Support Program to SBIRS and Next-Generation OPIR, describes Russian, Chinese and European systems, and explains why hypersonic glide vehicles are driving a shift to proliferated low-orbit tracking constellations.*

IN THIS CHAPTER

20.1 Why Warning from Space?
20.2 History
20.3 The Physics of Seeing Rockets
20.4 Sensors and Satellites
20.5 Missile Flight and Warning Timelines
20.6 Hypersonic Glide Vehicles and the Tracking Layer
20.7 Systems in Operation
20.8 Beyond Missiles: Other Infrared Events
20.9 Stability and Arms Control

## 20.1 Why Warning from Space?

A ballistic missile launched from one continent can reach another in about 30 minutes; a missile launched from a submarine near the coast can arrive in 10–15 minutes. Ground-based radars, limited by the curvature of the Earth, can see such missiles only after they rise above the horizon

— typically several minutes into flight and thousands of kilometres from the launch point. A satellite high above the Earth, by contrast, can see a launch almost anywhere within seconds, because the rocket's exhaust plume is an intense source of infrared radiation.

Early-warning satellites therefore serve four purposes:

1. **Strategic warning**: detecting missile attacks on a nation's territory and confirming that an attack is real, a cornerstone of nuclear deterrence and crisis stability.
2. **Missile defence**: providing the earliest possible track of a missile to cue radars and interceptors, and increasingly providing tracks precise enough to guide interceptors directly.
3. **Battlespace awareness**: detecting and characterising other infrared events — tactical missile launches, explosions, aircraft afterburners, large fires.
4. **Technical intelligence**: measuring the performance of foreign missiles during tests.

The same satellites, together with sensors on GPS satellites, also detect nuclear detonations.

**FIGURE 20.1** The Defense Support Program satellite DSP Flight 16 on its Inertial Upper Stage in the payload bay of Space Shuttle Atlantis before deployment in November 1991. DSP satellites provided US missile warning for almost five decades. Image: NASA/JSC.

## 20.2 History

The US **MIDAS** (Missile Defense Alarm System) programme launched twelve experimental satellites between 1960 and 1966 (Figure 20.2). Many failed, and early sensors were overwhelmed by false alarms from sunlight reflected by clouds, but MIDAS proved that rocket launches could be detected from orbit. Its successor, the **Defense Support Program (DSP)**, launched its first satellite in 1970; 23 DSP satellites were launched until 2007, and some operated for more than 20 years. DSP gained public fame in the 1991 Gulf War, when it detected Iraqi Scud launches and warned Coalition forces and Israel.

The Soviet Union developed the **Oko** system from the 1970s, using satellites in highly elliptical Molniya orbits. These viewed US missile fields obliquely, against the dark background of space near the Earth's limb, to avoid confusion with sunlit clouds. On 26 September 1983 an Oko satellite falsely reported the launch of several US missiles — sunlight reflected off high-altitude clouds had triggered the alarm. The duty officer, Lieutenant Colonel Stanislav Petrov, judged it to be a false alarm and did not pass it up the chain of command. This episode is often cited as an example of the dangers of automated warning systems and the importance of human judgement.

The US **Space-Based Infrared System (SBIRS)** replaced DSP: sensors hosted on classified satellites in highly elliptical orbits from 2006–2008, and six SBIRS GEO satellites launched between 2011 and 2022. Russia replaced Oko with the **EKS (Tundra/Kupol)** system from 2015, and China, France and others have developed their own capabilities.

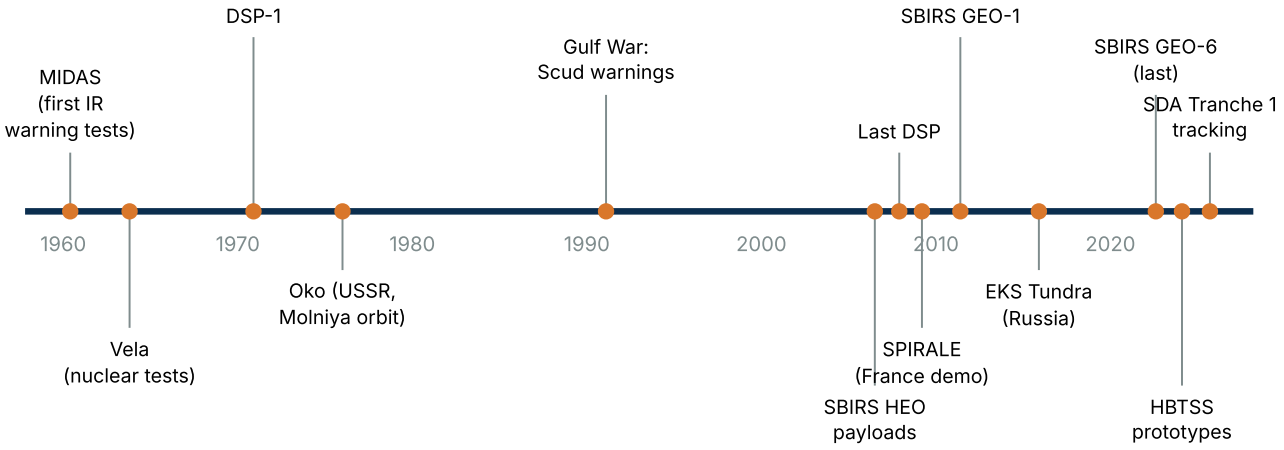
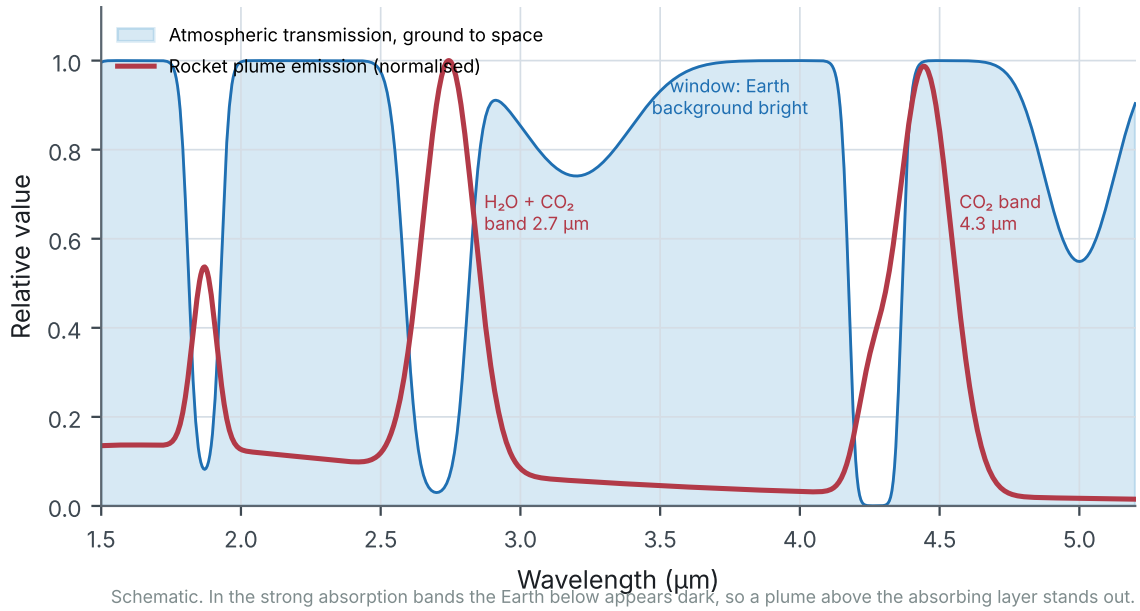



**FIGURE 20.2** Milestones in space-based missile warning.

## 20.3 The Physics of Seeing Rockets

### 20.3.1 Plume Emission

A rocket plume consists of hot combustion gases — mainly water vapour, carbon dioxide, carbon monoxide and, for solid-propellant motors, aluminium oxide particles — at temperatures of 1 500–3 000 K. Hot gases emit strongly in the vibrational bands of their molecules, especially near **2.7 µm** ($H_2O$ and $CO_2$) and **4.3 µm** ($CO_2$), while hot particles emit a continuum like a grey body (Figure 20.3). Where the plume mixes with the atmosphere, unburned fuel continues to burn ("afterburning"), increasing its size and brightness. A large booster plume radiates megawatts per steradian in the infrared.




Schematic. In the strong absorption bands the Earth below appears dark, so a plume above the absorbing layer stands out.

**FIGURE 20.3** Schematic emission of a rocket plume compared with the transmission of the atmosphere. The strongest plume emission falls in molecular bands where the lower atmosphere is opaque.

## 20.3.2 The Atmosphere as a Filter

The atmosphere is opaque in the same bands where the plume is brightest, because the same molecules — water vapour and carbon dioxide — absorb there. This is the key to early-warning sensor design. Suppose a sensor looks at the Earth in a strong absorption band such as 2.7 µm. The radiation from the ground, clouds and sunlit surfaces is then absorbed before it reaches space, so the Earth appears dark and uniform. A rocket plume is invisible while it is low in the atmosphere. The absorbing layer extends a few kilometres up for the water-vapour band, and higher in the more strongly absorbing parts of the $CO_2$ band. Once the plume climbs above this layer, it suddenly appears as a bright point against a dark background (Figure 20.4). This **"above-the-horizon of absorption"** approach suppresses clutter from clouds and sun glints; sensors working in more transparent bands see launches earlier but must cope with a bright, structured background.

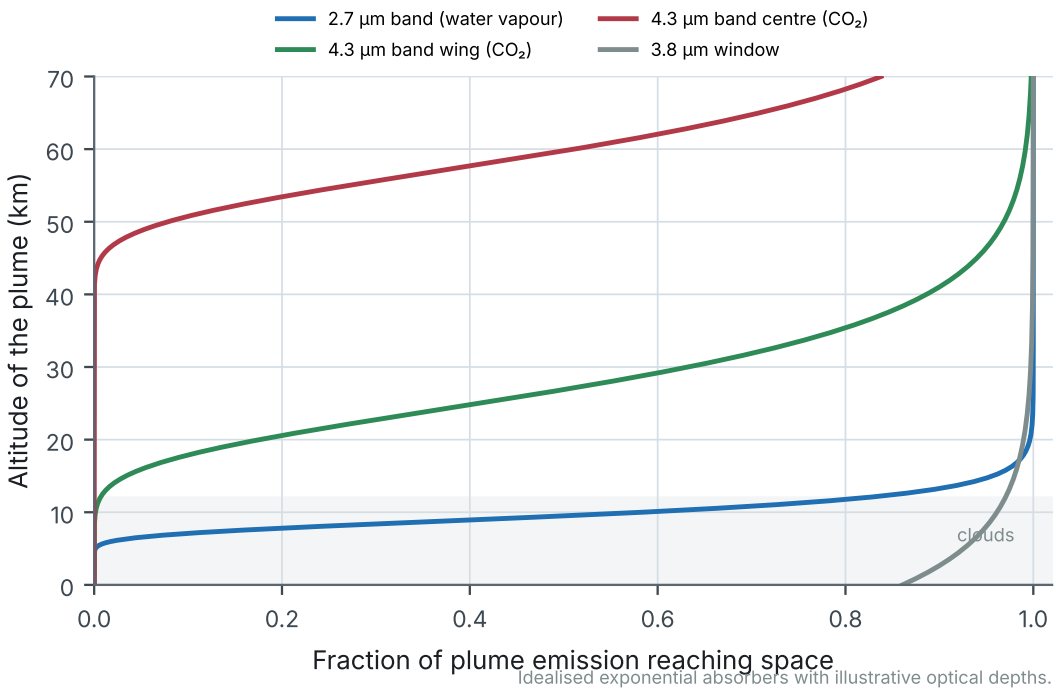



**FIGURE 20.4** Fraction of plume radiation escaping to space as a function of the plume's altitude in different spectral bands. In strongly absorbing bands, a plume becomes visible only after it climbs above the absorbing layer, but the background below is suppressed.

### 20.3.3 How Far Can Sensors See?

The irradiance at the sensor falls with the square of the distance (Figure 20.5). A bright ICBM booster seen from geostationary orbit, 36 000–40 000 km away, delivers an irradiance far above the sensitivity of modern infrared sensors. Dimmer targets include short-range missiles, the hot bodies of re-entry vehicles after burnout, and hypersonic glide vehicles. These are much harder to see from GEO, and they favour sensors in lower orbits, which are 30–40 times closer.

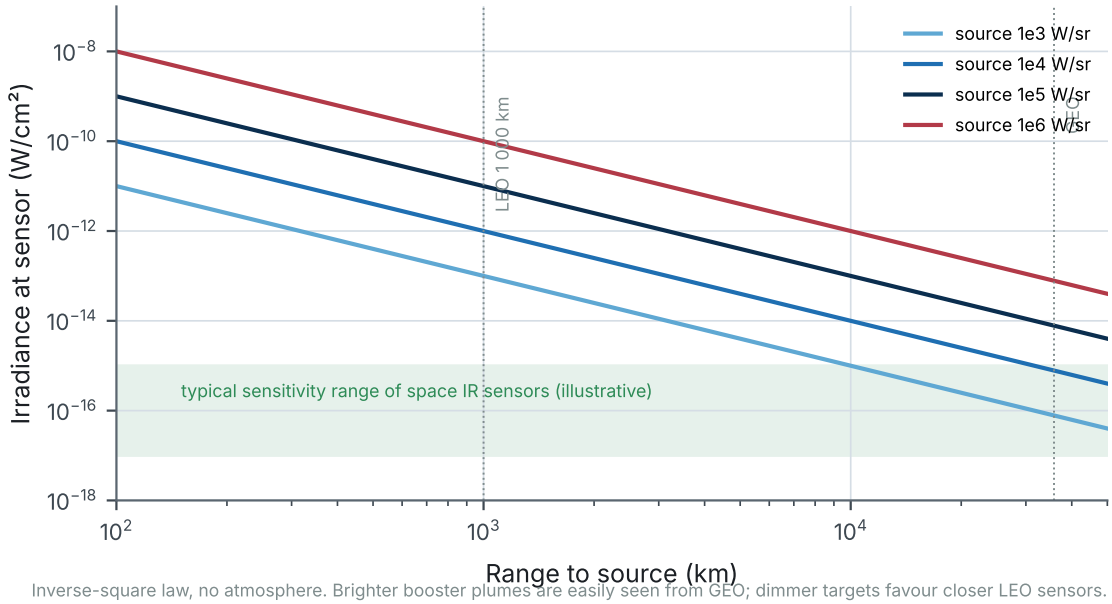



**FIGURE 20.5** Irradiance from infrared sources of different intensity as a function of range, compared with an illustrative sensitivity range for space infrared sensors.

# 20.4 Sensors and Satellites

## 20.4.1 Scanning and Staring

Two basic sensor architectures are used (Figure 20.6):

- **Scanning sensors**, as on DSP, sweep a linear array of detectors across the scene. DSP satellites rotated about six times per minute. A telescope tilted slightly off the spin axis swept a line of several thousand lead-sulphide (and later also mercury-cadmium-telluride) detectors around the Earth's disc. Each point was revisited about every 10 seconds. Scanning is efficient with small detector arrays but cannot follow a fast event continuously.
- **Staring sensors**, made possible by large two-dimensional infrared focal-plane arrays, watch an area continuously, measuring the brightness of every pixel many times per second. They detect dimmer and shorter events and measure the time history of a plume precisely. SBIRS GEO combined a **scanning sensor** covering the whole Earth with a **staring sensor** that could be pointed at areas of interest; Next-Gen OPIR uses large staring arrays.

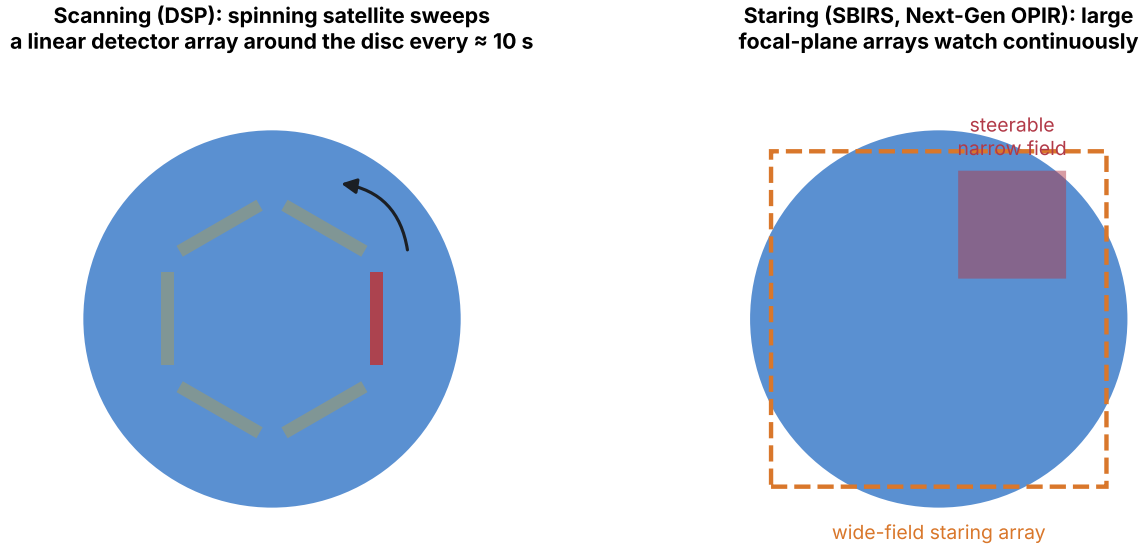



**FIGURE 20.6** Scanning (DSP) and staring (SBIRS, Next-Gen OPIR) sensor architectures.

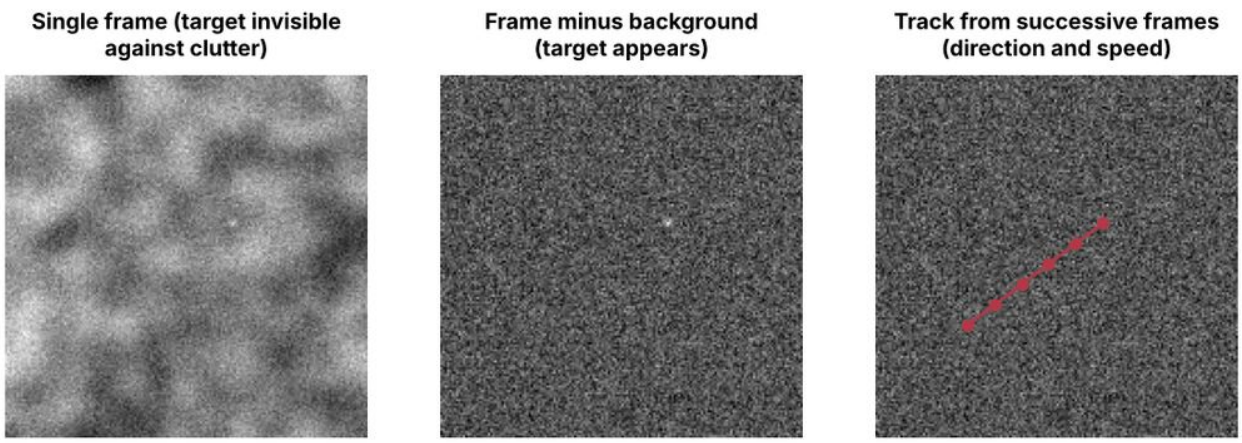
**FIGURE 20.7** A DSP satellite being prepared at Cape Canaveral in 1991 for launch on the Space Shuttle. The large telescope tube housing the infrared sensor projects from the spacecraft; the satellite spun around its axis to scan the Earth. Image: NASA/JSC.

### 20.4.2 Detection and Clutter Rejection

Turning detector signals into reliable warnings is a demanding signal-processing problem. The background — sunlight reflected by clouds and water (sun glint), hot surfaces, fires, volcanoes, lightning and even high-altitude aircraft — produces vastly more "events" than missile launches. On-board and ground processing:

- estimate and subtract the slowly varying background in each pixel;
- detect pixels whose brightness rises above a threshold (constant-false-alarm-rate detection);
- associate detections in successive frames into **tracks** with consistent motion (Figure 20.8);
- compare the track's intensity history and multi-band "colour" with the signatures expected from missiles;
- fuse data from two or more satellites to compute the three-dimensional trajectory by stereo triangulation, estimate the launch point and predict the impact area.

Within a minute or two of launch, a missile-warning centre (for the USA, at Buckley Space Force Base in Colorado) can report the launch, the missile type and its predicted impact region.




Simulated infrared frames: a faint moving source is detected by background subtraction and temporal tracking.

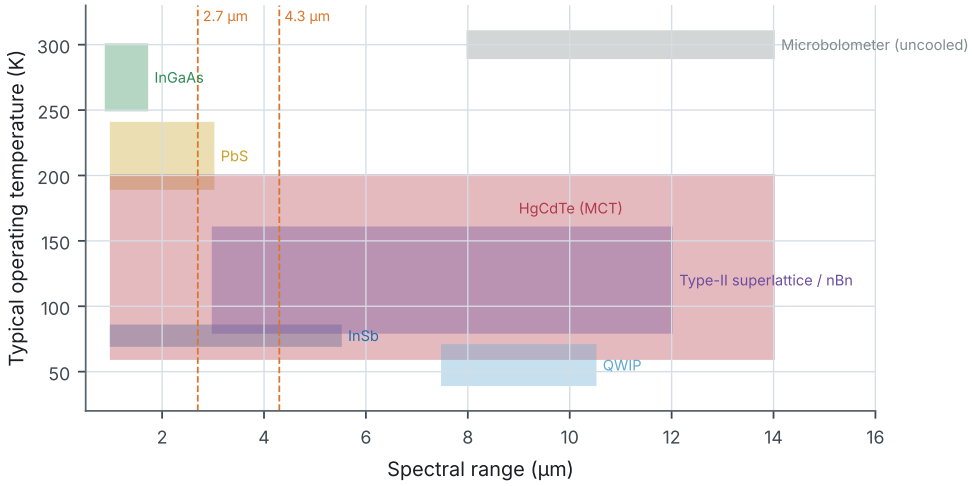
**FIGURE 20.8** Simulated detection of a faint moving infrared source: invisible in a single frame against structured clutter, it emerges after background subtraction and is confirmed by tracking it over successive frames.

### 20.4.3 Infrared Detectors

The heart of every early-warning sensor is its infrared **focal-plane array**: a two-dimensional array of photodetectors bonded to a silicon read-out integrated circuit that amplifies and digitises each pixel's signal. Detector materials are chosen for the spectral band (Figure 20.9):

- **Lead sulphide (PbS)**, used on early DSP sensors, responds to the 2.7 µm band at moderate cooling.
- **Indium antimonide (InSb)** covers 1–5.5 µm and must be cooled to about 77 K.
- **Mercury cadmium telluride (HgCdTe, "MCT")** can be tuned by its composition to cut off anywhere from about 1.7 to 14 µm; it is the workhorse for space infrared sensors, from SBIRS to astronomical telescopes.
- **Type-II strained-layer superlattices** and **nBn barrier detectors** are newer III–V semiconductor technologies that can operate at higher temperatures with lower dark current, easing cooling requirements.

Cooling is provided by passive radiators or mechanical cryocoolers (Chapter 5). Arrays with millions of pixels, read out hundreds of times per second, are now available — enabling wide-field staring sensors that were impossible in the DSP era. Radiation hardening is essential: energetic particles striking the array create false "spikes" that processing must reject, especially in highly elliptical orbits that pass through the radiation belts.




Approximate ranges. Longer-wavelength photon detectors must be cooled to suppress dark current; missile-warning sensors use SWIR and MWIR bands.

**FIGURE 20.9** Approximate spectral ranges and operating temperatures of infrared detector technologies.

**TABLE 20.1** Infrared detector materials.

| Material | Spectral range | Operating temperature | Use in missile warning |
|---|---|---|---|
| PbS | 1–3 µm | ≈ 190–240 K | Early DSP sensors (2.7 µm band) |
| InSb | 1–5.5 µm | ≈ 77 K | MWIR sensors |
| HgCdTe | 1–14 µm (tunable) | ≈ 60–200 K | SBIRS, Next-Gen OPIR, LEO tracking sensors |
| Type-II superlattice / nBn | 3–12 µm | ≈ 80–160 K | Newer MWIR and LWIR sensors; higher operating temperature |
| QWIP | 7–11 µm | ≈ 40–70 K | LWIR (e.g. Landsat TIRS) — not typical for warning |

### 20.4.4 From Detection to Decision

Missile-warning data flow from the satellites through relay and ground stations to mission-control centres. There, data from several satellites are fused into tracks with estimated launch points and impact areas. The data then flow to warning and command centres (in North America, the North American Aerospace Defense Command, NORAD) and to missile-defence systems and allied forces (Figure 20.10). The whole chain must be fast, highly reliable and protected against cyber attack and jamming. It is tested continuously with real launches, including the world's routine space launches, which look exactly like missile launches during boost.

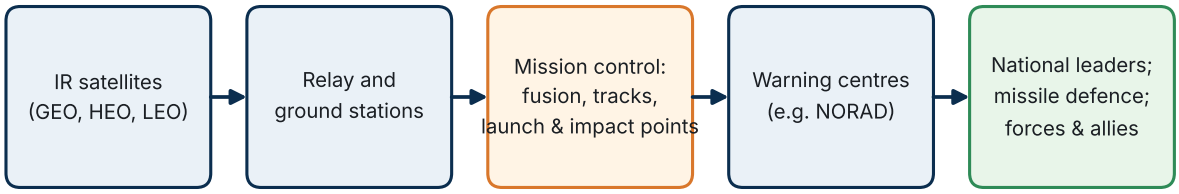



**FIGURE 20.10** The missile-warning data chain from satellites to decision-makers.

### 20.4.5 Orbits

Geostationary satellites are the backbone of missile warning: three or four can watch almost the whole Earth between about 70° N and 70° S continuously. They see poorly at very high latitudes, however — exactly where the shortest routes for ICBMs between North America and Eurasia and many submarine patrol areas lie. Sensors in **highly elliptical orbits** (Molniya- or Tundra-type), which spend many hours near apogee over the Arctic, fill this gap (Figure 20.11). Russia's Oko and EKS systems rely mainly on such orbits; the USA has used sensors hosted on satellites in highly elliptical orbits since 2006 and plans dedicated Next-Gen OPIR Polar satellites.

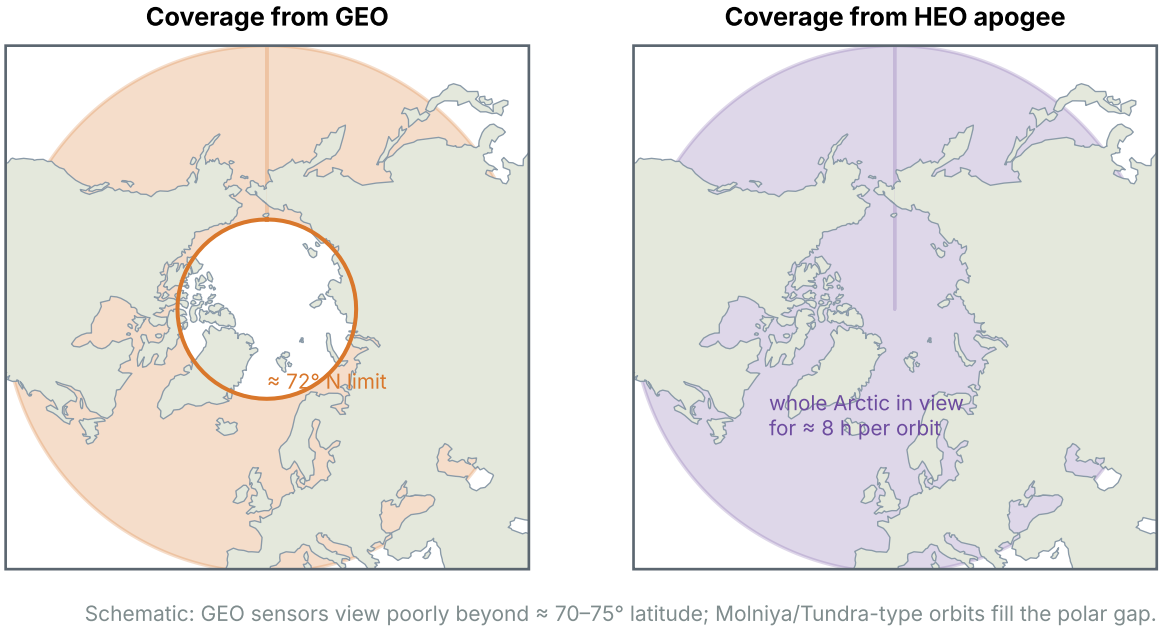



**FIGURE 20.11** Coverage of the Arctic from geostationary orbit and from the apogee of a highly elliptical orbit (schematic).

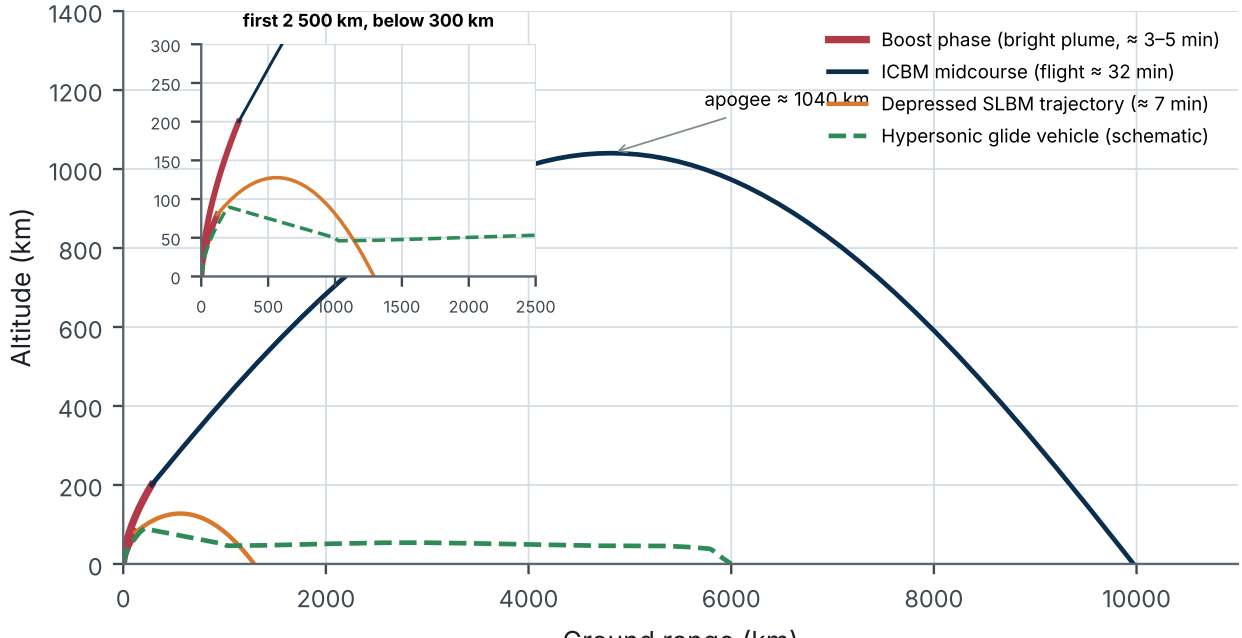
## 20.5 Missile Flight and Warning Timelines

### 20.5.1 Phases of Ballistic-Missile Flight

A ballistic missile's flight has three phases (Figure 20.12):

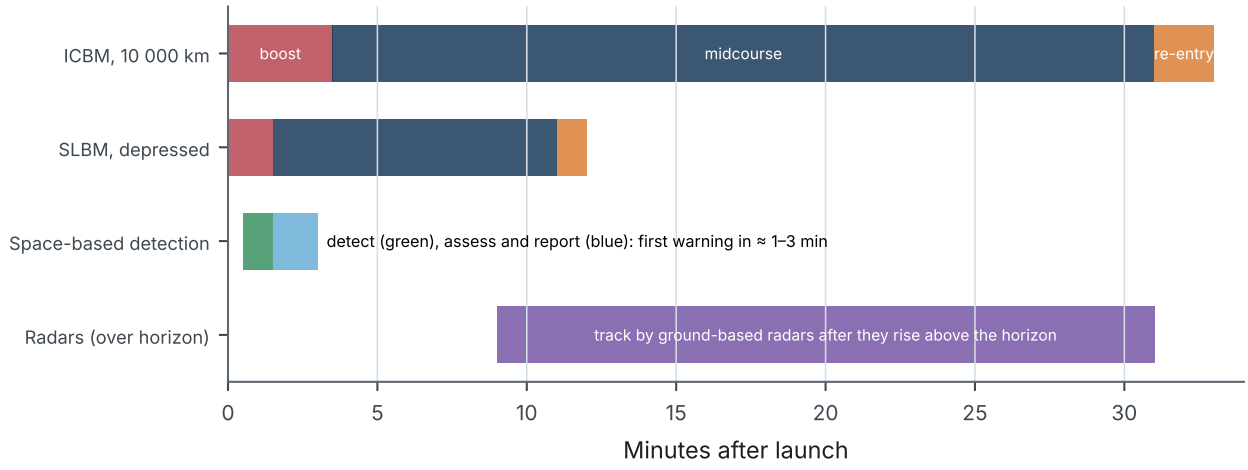
- **Boost**: the rocket engines burn for about 1–5 minutes, accelerating the payload to about 7 km/s for an intercontinental missile. This is when the plume is brightest and space-based infrared sensors see it best.
- **Midcourse**: after burnout, the warhead and any decoys coast above the atmosphere on an elliptical arc. For a minimum-energy ICBM trajectory over 10 000 km, they reach an apogee of about 1 000–1 300 km and take 20–25 minutes. Without a plume, the objects are cold and dim; they are tracked mainly by radars and by sensitive sensors observing their thermal emission against cold space.
- **Terminal**: re-entry vehicles plunge back through the atmosphere in the final minute or so.

Submarine-launched missiles can fly **depressed trajectories** — lower and faster — reducing warning time to about 10 minutes. These timelines (Figure 20.13) leave national leaders only minutes to assess an attack, which is why fast, reliable warning from space is so important for crisis stability.




Two-body trajectories after burnout (ICBM: 7.0 km/s at 17°; SLBM: 4.0 km/s at 12°). Glide vehicles fly low and manoeuvre, so they are harder to see from GEO and require low-orbit tracking sensors.

**FIGURE 20.12** Computed trajectories of an intercontinental ballistic missile and a depressed-trajectory submarine-launched missile, with a schematic hypersonic glide vehicle for comparison. The inset expands the first 2 500 km.




Illustrative. Space-based infrared sensors provide the first warning, within about a minute of launch.

**FIGURE 20.13** Illustrative timeline of a ballistic-missile attack and its detection. Space-based infrared sensors provide the first warning within minutes, long before the missile rises above the radar horizon.

**TABLE 20.2** Phases of ballistic-missile flight and the sensors that observe them.

| Phase | Duration (ICBM) | Main observables | Primary sensors |
|---|---|---|---|
| Boost | 1–5 min | Bright plume (2.7, 4.3 µm bands) | GEO and HEO infrared satellites; LEO tracking satellites |
| Post-boost / midcourse | 20–25 min | Warm warhead, decoys, debris against cold space | Ground- and sea-based radars; LEO infrared tracking with discrimination |
| Terminal | ≈ 1 min | Re-entry heating, radar signature | Terminal-defence radars |

### 20.5.2 From Warning to Defence

Early-warning satellites provide the first **cue** to a missile-defence system. Their tracks tell ground-based radars, such as the US Upgraded Early Warning Radars and the Long Range Discrimination Radar in Alaska, where to look, and they initiate the engagement planning for interceptors. Modern missile-defence architectures increasingly aim for **fire-control-quality** tracks from space — precise enough to launch and guide an interceptor before ground radars can see the target — and for **discrimination** between warheads and decoys.

## 20.6 Hypersonic Glide Vehicles and the Tracking Layer

**Hypersonic glide vehicles (HGVs)**, now deployed by Russia (Avangard) and China (DF-17), are boosted by rockets. They then glide through the upper atmosphere at 30–60 km altitude and speeds above Mach 5, manoeuvring over thousands of kilometres (Figure 20.12). They challenge existing warning systems in three ways. First, their low altitude keeps them below the horizon of ground radars until late. Second, their glide phase is much dimmer than a rocket plume, although the vehicle is heated by atmospheric friction. Third, their manoeuvres make their target uncertain.

Tracking HGVs requires sensors that are closer and that can follow dim, hot objects against the Earth background through their whole flight. This is the motivation for **proliferated low-Earth-orbit tracking constellations** (Figure 20.14):

- The US Space Development Agency's **Tracking Layer** (Chapter 19) carries wide-field-of-view infrared sensors on satellites at about 1 000 km; Tranche 1 includes 28 tracking satellites, with more in later tranches.
- The Missile Defense Agency's **Hypersonic and Ballistic Tracking Space Sensor (HBTSS)** — two prototypes launched in February 2024 — demonstrates medium-field-of-view sensors that can produce fire-control-quality tracks.
- The US Space Force is also deploying missile-warning and tracking satellites in **medium Earth orbit** ("Epoch 1"), for a layered architecture that is harder to defeat.

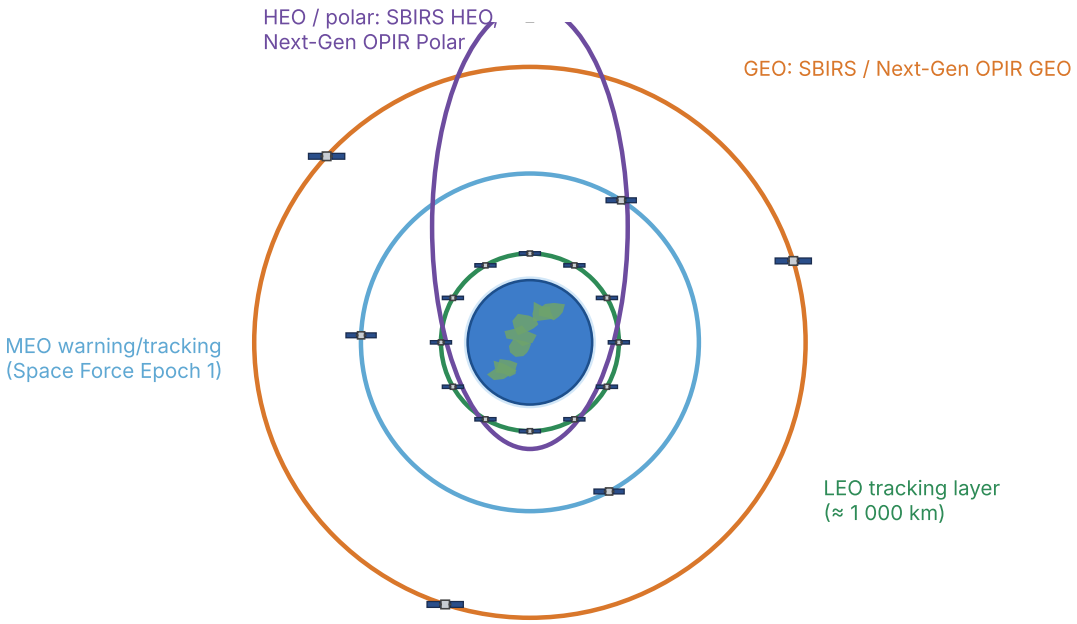



**FIGURE 20.14** The layered missile-warning and tracking architecture being built by the USA: geostationary and polar satellites for strategic warning, plus medium- and low-orbit constellations for resilience and tracking.

## 20.7 Systems in Operation

**TABLE 20.3** Space-based missile-warning systems (public information).

| Country | System | Orbit | Status (2026) |
|---|---|---|---|
| USA | Defense Support Program (DSP) | GEO | 23 launched 1970–2007; last satellites retired or in reserve |
| USA | SBIRS GEO-1 to GEO-6 | GEO | Operational; launched 2011–2022 |
| USA | SBIRS HEO payloads | HEO (hosted) | Operational since 2006–2008; replacement payloads added |
| USA | Next-Gen OPIR GEO (2 satellites) | GEO | First satellite awaiting launch in late 2026 after delays |
| USA | Next-Gen OPIR Polar (2 satellites) | HEO | Planned ≈ 2028 |
| USA | SDA Tracking Layer; MDA HBTSS | LEO | Prototypes 2023–2024; Tranche 1 deploying from 2025–26 |

| Country | System | Orbit | Status (2026) |
|---|---|---|---|
| USA | Resilient MW/MT (“Epoch 1”) | MEO | First satellites planned from about 2026–27 |
| Russia | EKS “Kupol” (Tundra satellites) | HEO (Tundra) | About six satellites launched 2015–2022 |
| China | Early-warning satellites (TJS/Huoyan series, reported) | GEO | Operational (limited public information) |
| France / Europe | SPIRALE (demonstrators, 2009); TWISTER and ODIN’S EYE studies | LEO, GEO | European system under study |

### 20.7.1 SBIRS and Next-Gen OPIR

Each **SBIRS GEO** satellite, built by Lockheed Martin around the A2100 bus, weighs about 4 500 kg at launch. It carries a scanning sensor that covers a wide area continuously and a staring sensor that can be pointed at regions of interest. Both operate in several short- and mid-wave infrared bands. The ground segment processes data from SBIRS, the remaining DSP satellites and hosted sensors together.

The **Next-Generation Overhead Persistent Infrared (Next-Gen OPIR)** programme replaces SBIRS with two GEO satellites (Lockheed Martin, with sensor payloads from RTX and a Northrop Grumman–Ball team) and two polar satellites (Northrop Grumman). Its sensors are designed to be more sensitive to dimmer targets and more resistant to jamming and attack. Like many large defence programmes, it has suffered delays: the first GEO launch, originally planned for 2025, slipped into late 2026.

### 20.7.2 Other Nations

Russia’s **EKS** (Integrated Space System) uses Tundra satellites in highly elliptical orbits with a “Kupol” (dome) sensor, reportedly able to detect not only ICBM launches but also smaller missiles, and to relay combat commands. China is reported to operate early-warning satellites in GEO and has cooperated with Russia on missile-warning systems. France flew the two **SPIRALE** microsatellites in 2009 to measure infrared backgrounds for a future system. The European Union is funding studies (ODIN’S EYE) and the PESCO **TWISTER** project for a European space-based early-warning capability linked to interceptors.

## 20.8 Beyond Missiles: Other Infrared Events

The same sensors see many other events. US officials have stated that space-based infrared sensors detected the missile that destroyed Malaysia Airlines flight MH17 over Ukraine in 2014. They have also stated that the sensors gave advance warning to troops in Iraq of the Iranian ballistic-missile attack in January 2020, allowing them to shelter. Large ballistic-missile attacks on Israel in 2024 and on Ukraine throughout the war were detected and tracked from space, cueing the defences. Researchers have also used declassified infrared data to study bolides — small asteroids exploding in the atmosphere — such as the Chelyabinsk event of 2013. US government sensors report such events publicly to NASA’s Center for Near-Earth Object Studies.

**FIGURE 20.15** The plume of Starship's sixth flight test seen from the International Space Station on 19 November 2024. Rocket plumes are among the brightest infrared sources on Earth. Image: NASA/JSC (NASA/Don Pettit).

### 20.8.1 Nuclear Detonation Detection

Nuclear explosions in the atmosphere or space produce a characteristic double-peaked flash of light, bursts of X-rays, gamma rays and neutrons, and an electromagnetic pulse. The **Vela** satellites (1963–1984) carried detectors for these signatures to verify the 1963 Limited Test Ban Treaty. Their successor, the **US Nuclear Detonation Detection System**, uses optical "bhangmeters", X-ray, gamma-ray, neutron and electromagnetic-pulse sensors carried on GPS and DSP/SBIRS satellites. Because GPS satellites are distributed around the Earth, at least four always see any point, allowing a detonation to be located by timing.

## 20.9 Stability and Arms Control

Reliable early warning is stabilising. If leaders know they will receive unambiguous warning of an attack, they need not adopt hair-trigger postures. They are also less likely to mistake an accident or a space launch for an attack. Conversely, attacks on early-warning satellites could be extremely escalatory, since they might be interpreted as the prelude to a nuclear strike. The **Ballistic Missile Launch Notification Agreement** (1988) between the USA and the Soviet Union, and later agreements, require advance notice of certain launches. Plans for a joint US–Russian data-exchange centre, agreed in 1998, were never implemented. The growing use of the same satellites for both strategic warning and conventional missile defence — "entanglement" — is a subject of concern among arms-control experts.

### EXAMPLE: WARNING TIME FOR AN ICBM ATTACK

Using the computed trajectory of Figure 20.12 (burnout at 7.0 km/s and 17° flight-path angle at 200 km altitude):

1. *Boost*: about 3 minutes; the plume becomes visible to a 2.7 μm sensor once it climbs above about 8–10 km, about 40–60 s after launch.
2. *Detection and assessment*: two or more satellites confirm the launch and compute an initial trajectory within about 1–2 minutes of the plume becoming visible.
3. *Flight*: the warhead coasts for about 29 minutes after burnout, reaching an apogee of about 1 040 km and impacting about 10 000 km from the launch site, 32 minutes after launch.
4. *Warning*: subtracting about 3 minutes for detection and reporting leaves roughly 25–28 minutes of warning. A ground-based radar 4 000 km from the target, by contrast, would see the warhead rise above its horizon only about halfway through its flight — giving about half as much warning.

**FIGURE 20.16** The exhaust plumes of a Falcon 9 rocket illuminated by sunlight before dawn during the launch of the CRS-35 cargo mission on 13 October 2026, seen from Florida. Twilight launches show the vast extent of rocket plumes at high altitude. Image: NASA/KSC (NASA/John Kraus).

### AI IN MISSILE WARNING

Machine learning helps reject clutter and false alarms, classify infrared events from their spectral and temporal signatures, fuse data from many satellites and radars into tracks, and discriminate warheads from decoys. Because errors could have grave consequences, these uses demand rigorous verification and human oversight (Chapters 50 and 53).

### KEY TAKEAWAYS

- Space-based infrared sensors detect ballistic-missile launches within about a minute, providing strategic warning and cueing missile defences.
- Rocket plumes emit strongly at 2.7 and 4.3 µm; sensors operating in these atmospheric absorption bands see a dark Earth background and a bright plume once it climbs above the absorbing layer.
- Scanning sensors (DSP) gave way to staring arrays (SBIRS, Next-Gen OPIR); processing rejects clutter from clouds, sun glint and fires and builds three-dimensional tracks from several satellites.
- GEO satellites provide global coverage to about 70° latitude; highly elliptical orbits cover the Arctic.
- ICBMs fly for about 30 minutes, submarine-launched missiles on depressed trajectories for about 10; space-based warning roughly doubles the warning time compared with ground radars alone.
- Hypersonic glide vehicles are driving a shift to proliferated LEO and MEO tracking constellations.
- Early-warning satellites also detect nuclear detonations, bolides and other infrared events; their reliability is central to strategic stability.

### REVIEW QUESTIONS

1. Why do missile-warning sensors observe in the 2.7 µm and 4.3 µm absorption bands rather than in atmospheric windows?
2. What angle does the Earth subtend from geostationary orbit? How many 10 µrad pixels would a staring sensor need across that angle?
3. Compare scanning and staring infrared sensors for missile warning.
4. Describe three sources of clutter and how processing rejects them.
5. Why do early-warning constellations combine geostationary satellites with highly elliptical orbits?
6. Outline the phases of an ICBM's flight and how much warning space-based sensors provide in each case for land-based and submarine-launched missiles.
7. Why do hypersonic glide vehicles require a new tracking layer in LEO or MEO?
8. Why is the reliability of early-warning satellites important for strategic stability?

CHAPTER

# 21 Space Surveillance, Orbital Debris and Space Situational Awareness

Image: NASA/JSC

*Some 35 000 catalogued objects — working satellites, dead satellites, spent rocket stages and fragments — orbit the Earth, together with an estimated million pieces of debris too small to track but large enough to destroy a satellite. Space situational awareness (SSA) is the knowledge of where these objects are, where they are going and what they are doing. This chapter uses the public satellite catalogue to show how the orbital population has grown and where it is concentrated, explains the hazards of hypervelocity impacts and the threat of a collisional cascade, and describes the radars, telescopes and analytical methods used to track objects, predict close approaches and decide when to manoeuvre. It then covers debris mitigation rules — from the 25-year guideline to the new five-year rule — active debris removal, re-entry risks and the emerging governance of space traffic.*

IN THIS CHAPTER

21.1 A Crowded Environment
21.2 Sources of Debris
21.3 Why Debris Is Dangerous
21.4 Tracking Objects in Space
21.5 Conjunction Assessment and Collision Avoidance
21.6 Debris Mitigation
21.7 Active Debris Removal and Remediation
21.8 Re-entries
21.9 Space Traffic Coordination

# 21.1 A Crowded Environment

## 21.1.1 Space Situational Awareness and Space Domain Awareness

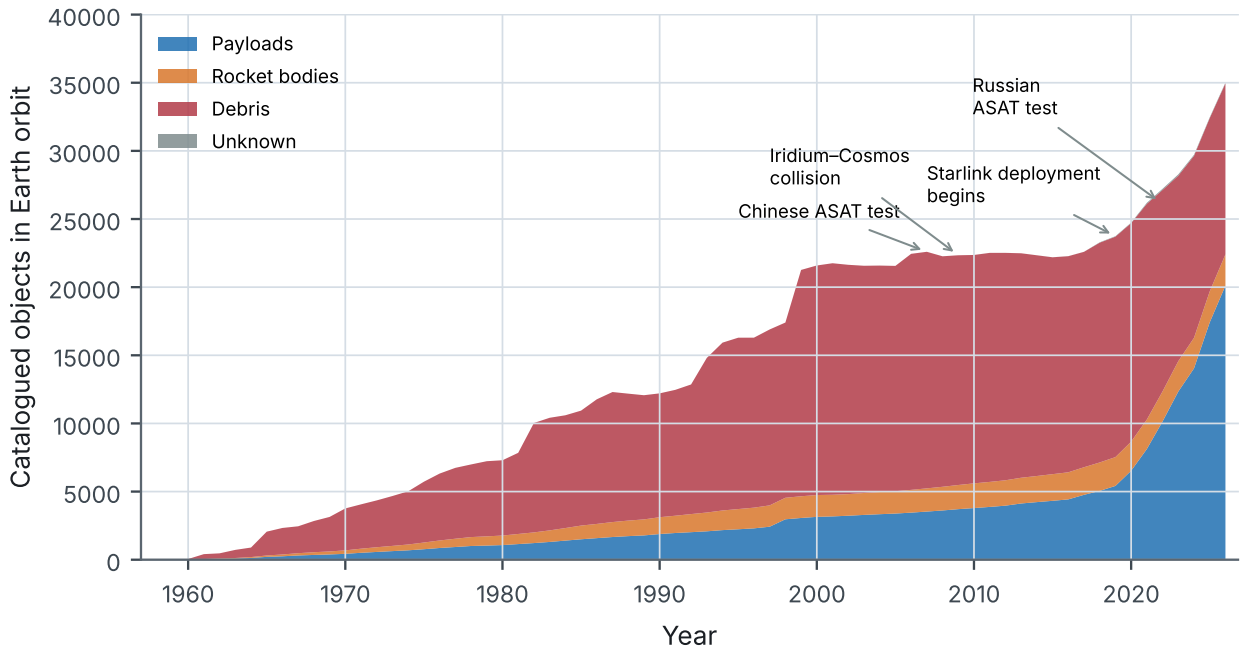
**Space situational awareness (SSA)** is the knowledge of the population of objects in space and of the space environment needed to operate safely. It covers where each object is, how its orbit is changing, what it is, and whether it threatens another. Military organisations increasingly use the broader term **space domain awareness (SDA)**, which adds understanding of the capabilities and intentions of other actors (Chapter 19). SSA has three main components:

- **Space surveillance and tracking (SST)** of objects in orbit;
- **Space weather** monitoring (Chapters 22 and 23);
- **Near-Earth object** monitoring — asteroids and comets that could hit the Earth (Chapter 41).

This chapter concentrates on the first: the population of artificial objects and the debris problem.

## 21.1.2 The Numbers

In October 2026 the publicly available catalogue maintained by the US Space Force and redistributed by CelesTrak contained about **35 000** objects in Earth orbit. These were some 20 000 payloads (about 17 000 of them active), 12 500 catalogued debris fragments and 2 300 rocket bodies (Figure 21.1). The total had more than doubled in a decade, driven mostly by the growth of large constellations. Surveillance networks track many more objects than are published — ESA estimates about 40 000 tracked objects — and statistical models estimate far larger populations of smaller debris (Table 21.1).




Computed from launch and decay dates in the CelesTrak SATCAT (publicly catalogued objects only); 35,012 in orbit on 14 Oct 2026.

**FIGURE 21.1** Catalogued objects in Earth orbit by type, 1957–2026, computed from the public satellite catalogue. Two destructive events (2007 and 2009) caused sudden jumps; since 2019 large constellations have dominated growth.

**TABLE 21.1** Estimated population of debris by size (ESA Space Environment Report 2025, model estimates).

| Size | Estimated number | Can it be tracked? | Effect of impact on a satellite |
|---|---|---|---|
| > 10 cm | > 50 000 | Yes (most, in LEO) | Catastrophic break-up; satellites manoeuvre to avoid |
| 1–10 cm | ≈ 1.2 million | Mostly not | Can destroy a satellite; cannot be avoided or shielded against easily |
| 1 mm – 1 cm | ≈ 130 million | No | Damage to sensitive surfaces; penetration of unshielded components |
| < 1 mm | Many billions | No | Erosion, pitting of windows and solar cells |

### 21.1.3 Where the Objects Are

Objects are not evenly distributed (Figure 21.2). The spatial density in LEO peaks sharply at 450–570 km, where most Starlink satellites fly. It also shows a broader band of old debris and rocket bodies at 700–1 000 km. This is the region of the 2007 Chinese ASAT test, the 2009 Iridium–Cosmos collision and many older break-ups. Debris at these higher altitudes will remain for decades to centuries because the atmosphere is too thin to remove it quickly. Other concentrations are found in the geostationary ring, where about 1 000 objects, active and dead, crowd a narrow belt, and in the MEO navigation orbits.

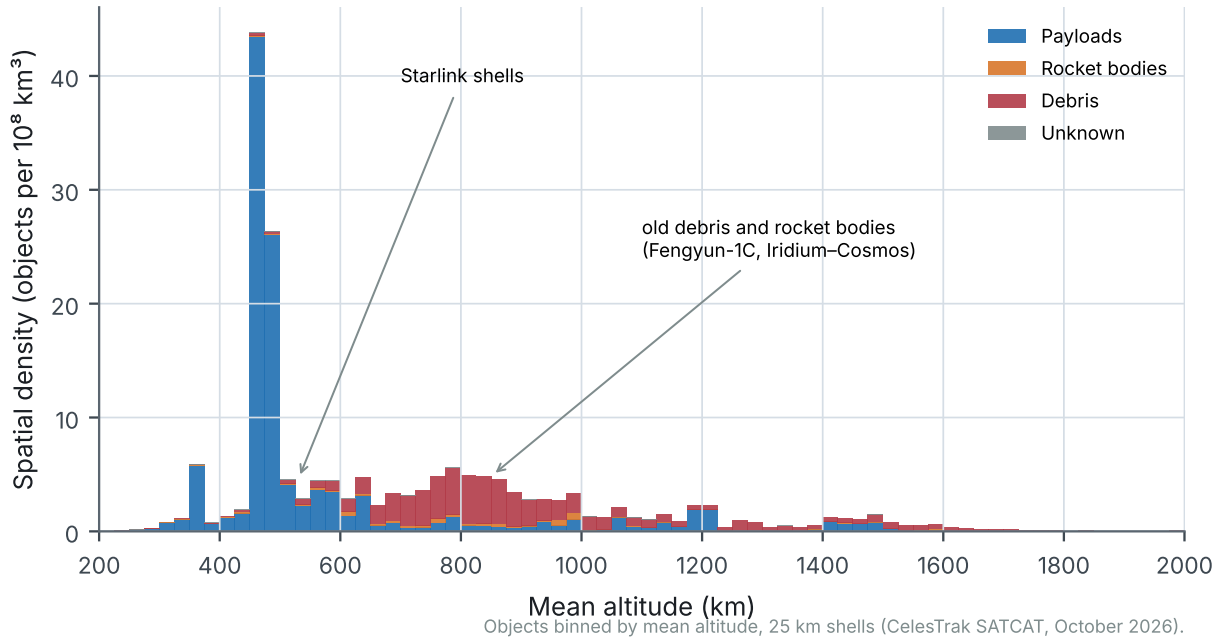



**FIGURE 21.2** Spatial density of catalogued objects in low Earth orbit by altitude and type (October 2026).

Plotting every object by altitude and inclination (Figure 21.3) reveals the structure of the population. It shows vertical streaks of satellites in the same orbital planes, the dense band of sun-synchronous orbits at 97–99°, and the shells of Starlink and OneWeb. It also shows clouds of debris spread in altitude around the orbits of the satellites and stages that broke up.

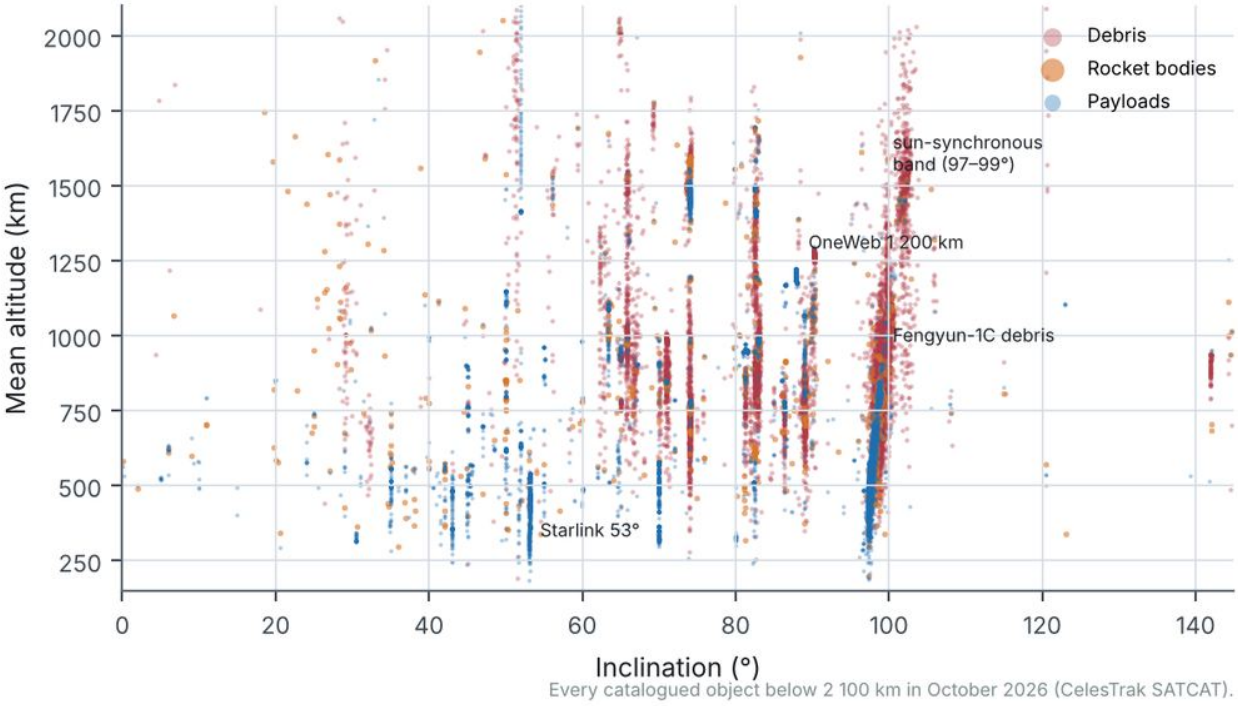



**FIGURE 21.3** Altitude and inclination of every catalogued object below 2 100 km in October 2026.

Responsibility for the population is concentrated (Figure 21.4). Objects attributed to the USA — including Starlink — are the most numerous, followed by Russia and China. Russia and China account for most of the catalogued debris, through old explosions, the 2007 and 2021 ASAT tests and recent upper-stage break-ups. Under the Outer Space Treaty and the Liability Convention, the launching state remains responsible for its objects, including debris, indefinitely.

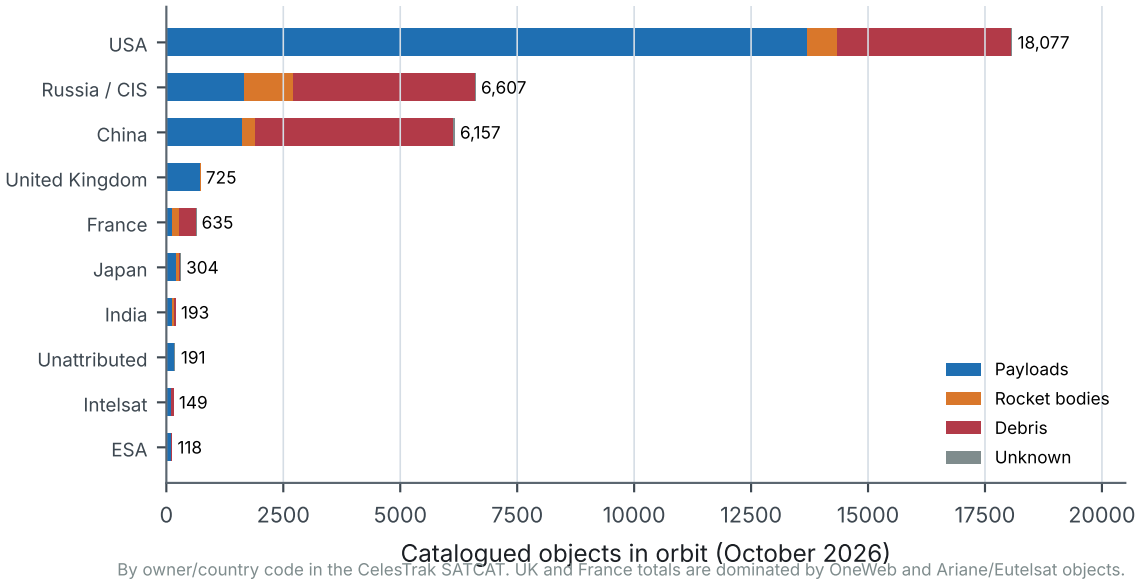



**FIGURE 21.4** Catalogued objects in orbit by owner and type (October 2026).

## 21.2 Sources of Debris

### 21.2.1 Fragmentations

Most catalogued debris comes from a few hundred **fragmentation events** (Figure 21.5):

- **Explosions** of rocket upper stages and satellites, caused by residual propellant, pressurised tanks or batteries — the most common cause historically. Stages that remain in orbit with leftover propellant can explode decades later, which is why "passivation" (venting propellant and discharging batteries) is now required.
- **Deliberate destruction**, above all the anti-satellite tests of 2007 (Fengyun-1C) and 2021 (Cosmos 1408) (Chapter 19).
- **Collisions**. On 10 February 2009 the active Iridium 33 communications satellite collided with the defunct Russian Cosmos 2251 at about 790 km and a relative speed of 11.7 km/s. It was the first accidental collision between two intact satellites, and it produced more than 2 300 catalogued fragments. Smaller collisions have occurred before and since; in 2021, for example, the Chinese satellite Yunhai 1-02 broke up after being struck by a fragment of debris from a 1996 launch.
- **Anomalous events**: the sudden break-up of old weather satellites (NOAA-16, NOAA-17, DMSP) and, in 2022 and 2024, of Long March 6A upper stages, each producing hundreds of fragments.

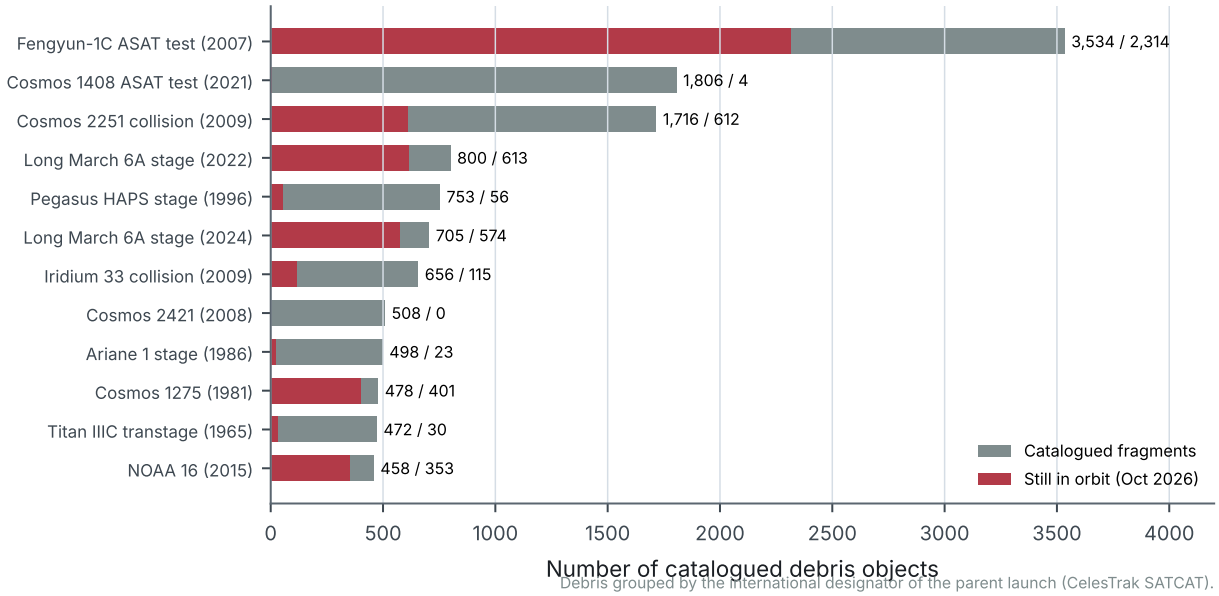



**FIGURE 21.5** The largest fragmentation events in the public catalogue: total catalogued fragments and those still in orbit in October 2026.

### 21.2.2 Other Sources

Smaller debris also comes from **mission-related objects** (lens covers, bolts, adapters released during deployment) and from **solid-rocket-motor** slag and aluminium-oxide dust. Other sources are **paint flakes** and degraded insulation shed by ageing spacecraft (Chapter 5 shows a damaged insulation blanket). There are also droplets of sodium–potassium coolant leaked by Soviet nuclear-powered radar satellites of the 1980s. Natural **micrometeoroids** — dust from comets and asteroids — add a background flux that dominates for sizes below about 0.1 mm.

## 21.3 Why Debris Is Dangerous

### 21.3.1 Hypervelocity Impacts

Objects in LEO collide at an average relative speed of about 10 km/s — ten times faster than a rifle bullet. At such speeds, even small particles carry enormous energy (Figure 21.6). A 1 cm aluminium sphere at 10 km/s has a kinetic energy of about 70 kJ, comparable to a small car crash. Both projectile and target are partly vaporised. In practice:

- particles smaller than about 1 mm cause pitting and erosion; windows of the Space Shuttle were replaced after almost every flight because of tiny impacts;
- particles of 1 mm to 1 cm can penetrate unshielded walls, tanks and cables, but **shielding** works against them;
- objects of 1–10 cm are too small to track reliably but too large to shield against — they represent the greatest unmanaged risk;
- objects larger than about 10 cm are tracked, and spacecraft can manoeuvre to avoid them.

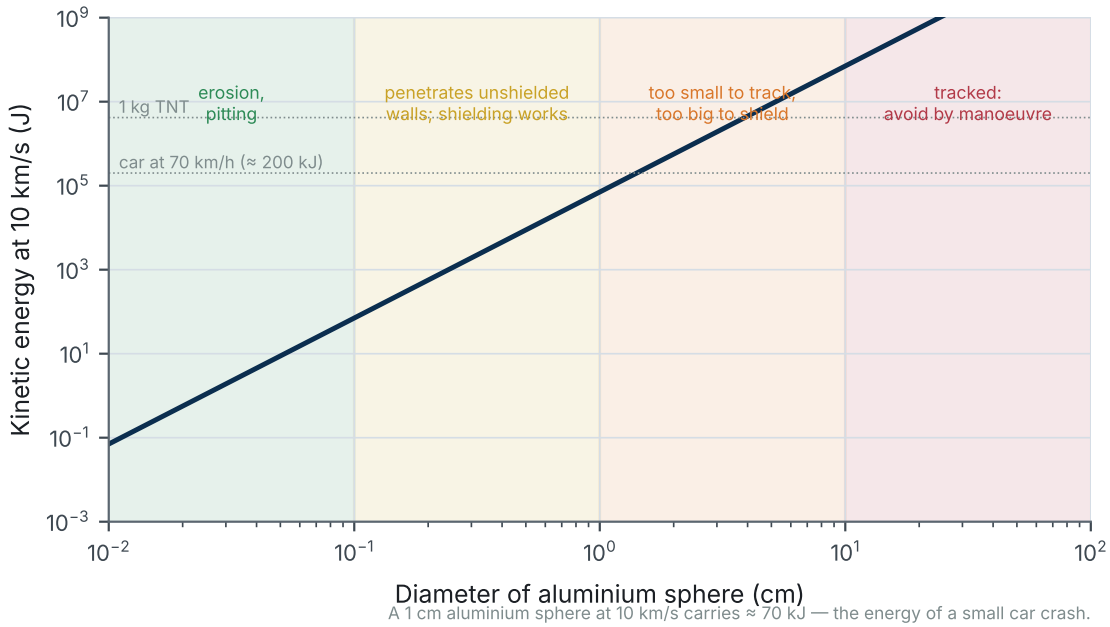



**FIGURE 21.6** Kinetic energy of an aluminium sphere at 10 km/s as a function of its diameter, with the effects of impacts in each size range.

## 21.3.2 Shielding

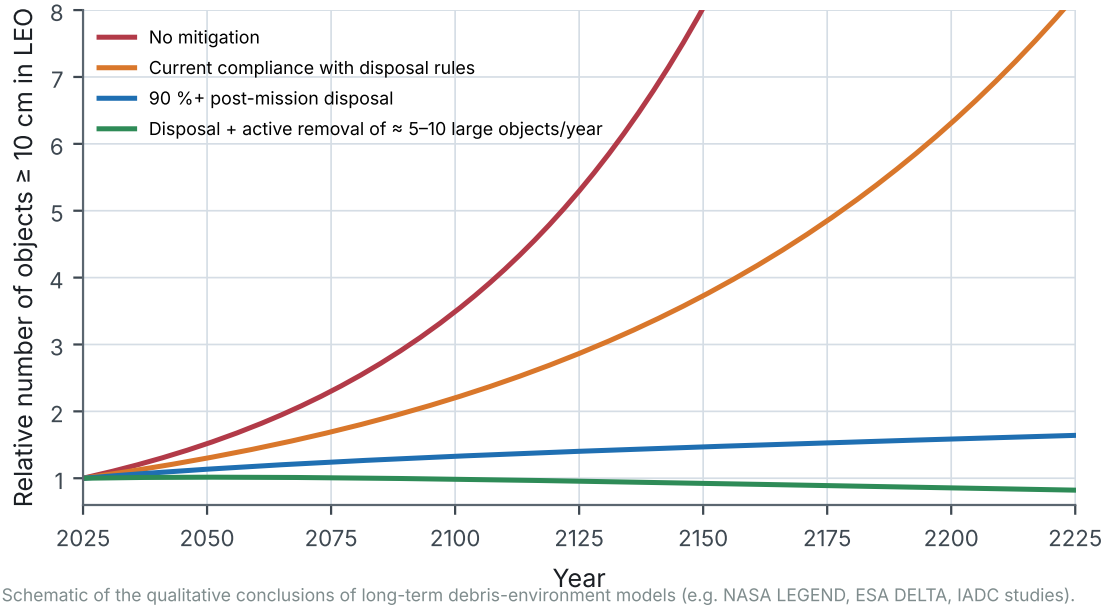
In 1947, the astronomer Fred Whipple proposed protecting spacecraft from meteoroids with a thin outer "bumper" separated from the main wall. A projectile striking a **Whipple shield** shatters and partly vaporises on the bumper. The resulting cloud of fragments spreads out over the gap and strikes the main wall over a wide area, where it can be stopped. **Stuffed Whipple shields**, with intermediate layers of Kevlar and Nextel ceramic fabric, protect the crewed modules of the ISS against particles up to about 1 cm. Shields are designed and tested with light-gas guns that fire small projectiles at up to about 7–8 km/s (Figure 21.7), complemented by computer simulations. Recovered hardware, such as Hubble's Wide Field and Planetary Camera 2 returned in 2009 (Figure 21.8), provides real data on the impact environment.

**FIGURE 21.7** A target plate after a hypervelocity impact test at 6.8 km/s at NASA's White Sands Test Facility, during tests of materials for the Orion spacecraft. Image: NASA/JSC.

**FIGURE 21.8** Engineers of NASA's Orbital Debris Program Office and Hypervelocity Impact Technology team examine impact craters on Hubble's Wide Field and Planetary Camera 2 after its return in 2009, after 16 years in orbit. Image: NASA/JSC.

### 21.3.3 The Kessler Syndrome

In 1978 Donald Kessler and Burton Cour-Palais of NASA predicted what would happen as the number of objects in orbit grew. Collisions between them would eventually produce debris faster than the atmosphere removes it. This would lead to a self-sustaining **collisional cascade** — now called the **Kessler syndrome**. Long-term models of the debris environment include NASA's LEGEND, ESA's DELTA and the studies of the Inter-Agency Space Debris Coordination Committee. They agree that some LEO altitude bands, especially around 700–1 000 km, may already be above this critical density. Even with no further launches, collisions would slowly increase the number of fragments. The models also show that the outcome depends strongly on behaviour (Figure 21.9). If spacecraft are reliably removed from orbit at the end of their missions, growth can be held in check. Removing a few large derelict objects per year can also stabilise the environment.



Schematic of the qualitative conclusions of long-term debris-environment models (e.g. NASA LEGEND, ESA DELTA, IADC studies).

**FIGURE 21.9** Schematic of the qualitative conclusions of long-term debris-environment models under different mitigation scenarios.

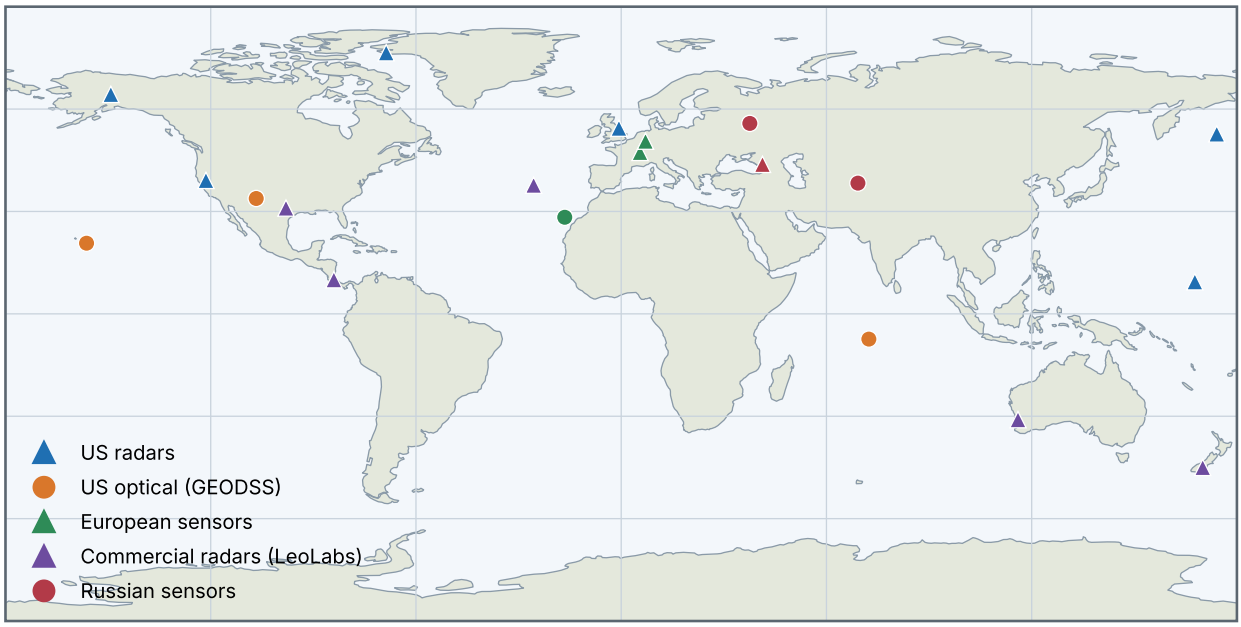
## 21.4 Tracking Objects in Space

### 21.4.1 Sensors

Objects in orbit are observed by networks of radars, optical telescopes and, increasingly, sensors in space (Figure 21.10; Table 21.2).

- **Radars** dominate in LEO because they work day and night in all weather and measure range and range-rate directly. Large phased-array radars include the US PAVE PAWS and Upgraded Early Warning Radars (originally built for missile warning, Chapter 20). Others are Cobra Dane in Alaska and the S-band **Space Fence** on Kwajalein Atoll (operational since 2020). Such radars can detect objects down to about 5–10 cm in LEO and track thousands of objects per hour. Tracking and imaging radars such as Germany's TIRA can produce radar images of satellites.

- **Optical telescopes** observe sunlit objects against a dark sky and are best for high orbits, where radar echoes are extremely weak. This is because the radar signal falls as the fourth power of range, but reflected sunlight only as the square (Figure 21.11). The US Ground-based Electro-Optical Deep Space Surveillance (GEODSS) telescopes, commercial networks and European and Russian systems such as ISON monitor the geostationary belt, detecting objects of about 0.5–1 m at GEO.
- **Space-based sensors** avoid weather and day–night limitations. The US Space-Based Space Surveillance satellite (since 2010) observes GEO from LEO. The Geosynchronous Space Situational Awareness Program (GSSAP) satellites — six launched in 2014–2022 and two more in 2026 — patrol near the GEO belt to inspect objects.
- **Commercial providers** now operate their own networks. These include LeoLabs' phased-array radars in several countries and optical networks such as those of ExoAnalytic Solutions and Slingshot Aerospace. Some companies also fly cameras on satellites to observe other objects.




Selected ground-based space-surveillance sensors (approximate locations, public information). ▲ radar, ● optical.

**FIGURE 21.10** Selected ground-based space-surveillance sensors around the world (public information).

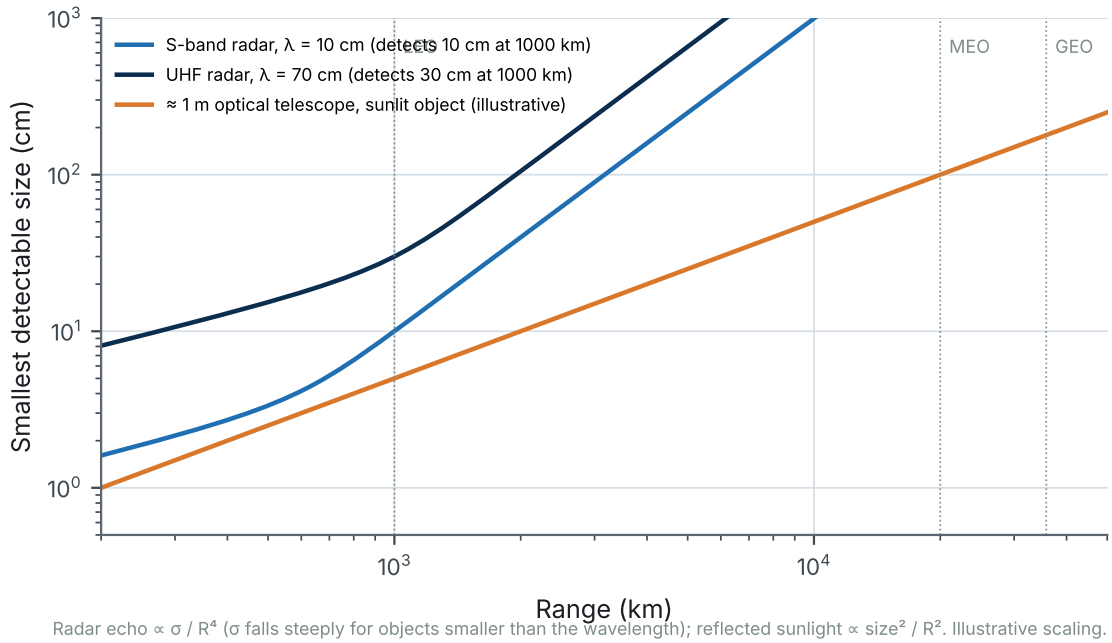



**FIGURE 21.11** Approximate smallest detectable object size as a function of range for radars and optical telescopes. Radars excel in LEO; telescopes are needed for GEO.

**TABLE 21.2** Selected space-surveillance sensors.

| Sensor | Operator | Type | Main role |
|---|---|---|---|
| Space Fence (Kwajalein) | US Space Force | S-band phased-array radar | Uncued detection of small LEO objects |
| Cobra Dane, PAVE PAWS/UEWR | US Space Force | UHF/L-band phased-array radars | Missile warning and LEO surveillance |
| GEODSS (Socorro, Maui, Diego Garcia) | US Space Force | 1 m optical telescopes | Deep-space (GEO) surveillance |
| GSSAP, SBSS | US Space Force | Space-based optical | GEO inspection and surveillance |
| GRAVES, TIRA, EU SST network | France, Germany, EU | Bistatic radar, tracking/imaging radar, optical | European surveillance and tracking |
| LeoLabs network | LeoLabs (commercial) | UHF/S-band phased-array radars | Commercial LEO tracking and conjunction services |
| ISON | Russian Academy of Sciences | Optical network | GEO and HEO objects |
| Okno, Krona | Russia | Optical and radar complexes | Military space surveillance |

## 21.4.2 The Catalogue

The observations are combined into a **catalogue** of orbits. The US Space Force (through its space defence squadrons) maintains the most complete catalogue, publishing data on most unclassified objects through the Space-Track website, where each object has a NORAD catalogue number. Orbits are published as **two-line element sets (TLEs)**, compatible with the SGP4 analytical propagator (Chapter 3). For precise applications, the military also maintains a "special perturbations" catalogue, computed by numerical integration with detailed force models. In addition, operators exchange precise ephemerides. Keeping the catalogue current requires

associating every new observation with the right object. This is difficult after break-ups or when many satellites are deployed together. It also requires modelling atmospheric drag, which changes with solar activity and is the largest source of prediction error in LEO (Chapter 22).

**Characterisation** goes beyond orbits. Brightness variations (**light curves**) reveal whether an object is tumbling and how fast, and radar cross-sections and radar images reveal its size and shape. Spectroscopy can identify surface materials, and observing manoeuvres reveals whether a satellite is active.

## 21.5 Conjunction Assessment and Collision Avoidance

### 21.5.1 Predicting Close Approaches

Every day, screening software compares the predicted trajectories of all catalogued objects to find **conjunctions** — close approaches — within the next several days. For each, the screening centre issues a **conjunction data message (CDM)** to the operators involved, giving the predicted time of closest approach, the miss distance and the uncertainty (covariance) of both objects' positions. Large constellations receive thousands of such messages per week.

The key quantity is the **probability of collision** ($P_c$), computed in the **encounter plane** perpendicular to the relative velocity (Figure 21.12). LEO encounters last only a fraction of a second at 10–15 km/s, so relative motion can be treated as a straight line. $P_c$ is then the integral of the two-dimensional probability density of the miss vector over a circle. The circle's radius is the combined size of the two objects (Figure 21.13). Counter-intuitively, very uncertain predictions give *low* probabilities, because the probability is spread over a large area ("dilution"); good tracking data are needed to recognise a real threat.

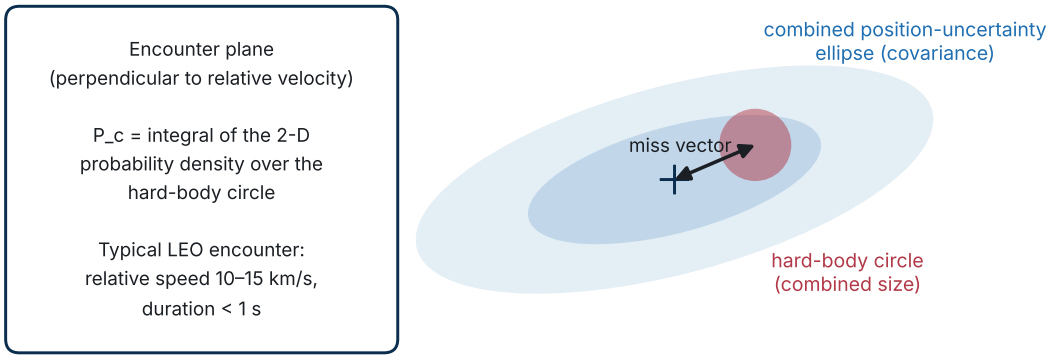



**FIGURE 21.12** Geometry of a conjunction in the encounter plane.

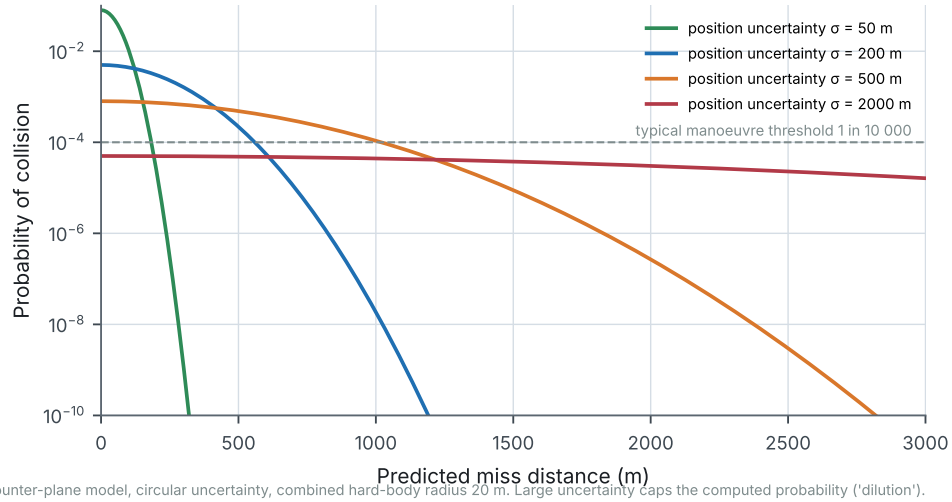



**FIGURE 21.13** Probability of collision versus predicted miss distance for different levels of position uncertainty.

### 21.5.2 Deciding to Manoeuvre

Operators typically plan an avoidance manoeuvre when $P_c$ exceeds about 1 in 10 000 (NASA's threshold for robotic missions), sometimes using more conservative values for crewed vehicles. Each manoeuvre costs propellant, interrupts the mission and may create new conjunctions, so operators weigh the risk carefully; the decision is often made only one or two days before the encounter, when uncertainties have shrunk. The ISS has performed dozens of debris-avoidance manoeuvres since 1999, and its crew has occasionally sheltered in their return vehicles when a threat was detected too late to manoeuvre. Large constellations automate the process: Starlink satellites perform tens of thousands of autonomous avoidance manoeuvres per year using screening data, and share their ephemerides publicly. When both objects can manoeuvre, operators must coordinate so that they do not both move the wrong way.

## 21.6 Debris Mitigation

### 21.6.1 Guidelines and Rules

International efforts to limit debris began in the 1990s. The **Inter-Agency Space Debris Coordination Committee (IADC)** was founded in 1993 by the major space agencies. It issued mitigation guidelines in 2002, which were endorsed by the UN Committee on the Peaceful Uses of Outer Space (COPUOS) in 2007. The UN adopted wider **Guidelines for the Long-term Sustainability of Outer Space Activities** in 2019 (Table 21.3). The main principles are:

1. **Limit debris released** during normal operations.
2. **Minimise the potential for break-ups**: passivate spacecraft and stages at the end of life by venting propellant and pressurants and discharging batteries.
3. **Post-mission disposal**: remove spacecraft and stages from the protected LEO region within a limited time — traditionally **25 years**. Do this by lowering their orbits so that drag brings them down, by controlled re-entry, or by boosting them to a disposal orbit. Move GEO satellites at least about 235 km above the geostationary ring (plus a term depending on their area-to-mass ratio) into a "graveyard" orbit.
4. **Prevent on-orbit collisions** through conjunction assessment and manoeuvre capability.

**TABLE 21.3** Selected debris-mitigation guidelines and rules.

| Instrument | Body | Year | Key provisions |
|---|---|---|---|
| IADC Space Debris Mitigation Guidelines | IADC | 2002 (updated) | Passivation, 25-year LEO disposal, GEO graveyard |
| Space Debris Mitigation Guidelines | UN COPUOS | 2007 | Endorsed IADC principles for all states |
| ISO 24113 | International Organization for Standardization | 2010 (updated) | Engineering requirements for debris mitigation |
| LTS Guidelines | UN COPUOS | 2019 | 21 guidelines on safety, sustainability and cooperation |
| Five-year rule | US FCC | adopted 2022, in force 2024 | LEO satellites below 2 000 km must be disposed of within 5 years of mission end |
| Zero Debris approach / Charter | ESA and partners | 2023 | Aim to stop generating debris by 2030; ≥ 99 % disposal success |
| French Space Operations Act | France | 2008 | Legally binding technical regulations, including disposal |

The 25-year guideline was set when LEO held a few thousand objects; with tens of thousands of satellites, it is now widely considered too lax. The US Federal Communications Commission's **five-year rule**, in force since 2024 for satellites it licenses, reflects the tightening of expectations. So does ESA's **Zero Debris** approach, which aims to stop creating debris by 2030. Figure 21.14 shows why altitude matters. Satellites below about 500 km re-enter naturally within a few years. Those above about 700 km may stay up for a century or more unless they carry propulsion to lower their orbits.

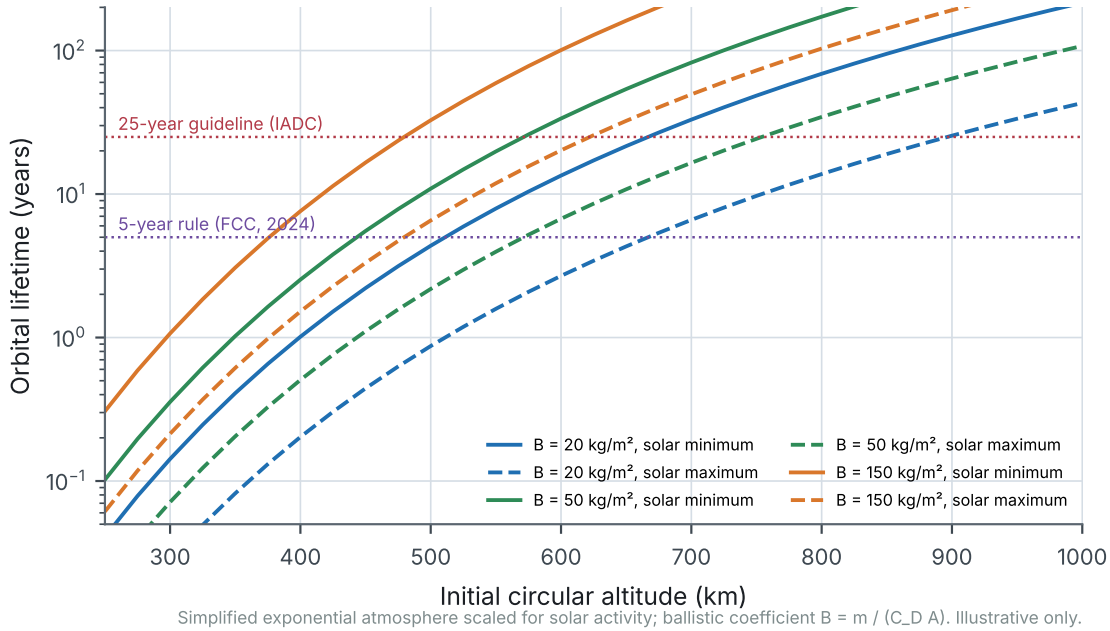



**FIGURE 21.14** Approximate orbital lifetime versus initial altitude for different ballistic coefficients and solar activity levels.

### 21.6.2 Compliance

Compliance with disposal guidelines has improved but remains imperfect. ESA's annual space-environment reports show that most new LEO payloads now comply, largely because so many fly in low orbits that decay quickly. However, many rocket bodies are still abandoned in long-lived orbits, and a few percent of large-constellation satellites fail before they can de-orbit themselves. A failure rate of even 1–2 % in a constellation of 10 000 satellites leaves 100–200 uncontrolled objects.

## 21.7 Active Debris Removal and Remediation

The most dangerous objects for the long term are the few thousand large, intact derelicts — especially rocket bodies of 1–9 tonnes in crowded orbits. So **active debris removal (ADR)** focuses on capturing them and de-orbiting them. The challenges are considerable: the targets were not designed to be captured, may be tumbling, and have no docking fixtures, and someone must pay for the service.

- Japan's **Astroscale** demonstrated magnetic capture of a cooperative target with **ELSA-d** (2021). With **ADRAS-J** (2024), it approached a discarded H-IIA upper stage to about 15 m and inspected it in detail — the first close approach to a real piece of debris. Its follow-on, **ADRAS-J2**, planned for launch around 2027–28, is designed to capture and remove such a stage.
- ESA's **ClearSpace-1** mission, now planned for around 2028–29, will capture and de-orbit the PROBA-1 satellite (launched in 2001).
- Other concepts include nets, harpoons, robotic arms, drag sails and tethers, ground- or space-based lasers that nudge small debris, and life-extension servicing that keeps valuable satellites working (Chapter 40).

Studies suggest that removing about five to ten of the most dangerous large objects per year, combined with high compliance with disposal rules, could stabilise the LEO environment.

## 21.8 Re-entries

What goes up comes down. About 2 000 catalogued objects now re-enter each year (Figure 21.15), most of them small objects and constellation satellites that burn up completely. Larger objects pose a risk to people on the ground: dense components such as titanium tanks, steel parts and optics can survive re-entry. Guidelines limit the casualty risk of an uncontrolled re-entry to 1 in 10 000. Larger spacecraft and stages must perform a **controlled re-entry** into a remote ocean area — the South Pacific "spacecraft cemetery" near Point Nemo — or be designed to **demise** completely. Notable uncontrolled re-entries include Skylab (1979), whose debris fell on Western Australia, and several 20-tonne Long March 5B core stages (2020–2022). Another was a fragment of an ISS battery pallet that struck a house in Florida in March 2024, prompting claims for damages under the Liability Convention. Scientists are also studying whether the aluminium oxide produced by thousands of burning satellites each year could affect the upper atmosphere and ozone layer.

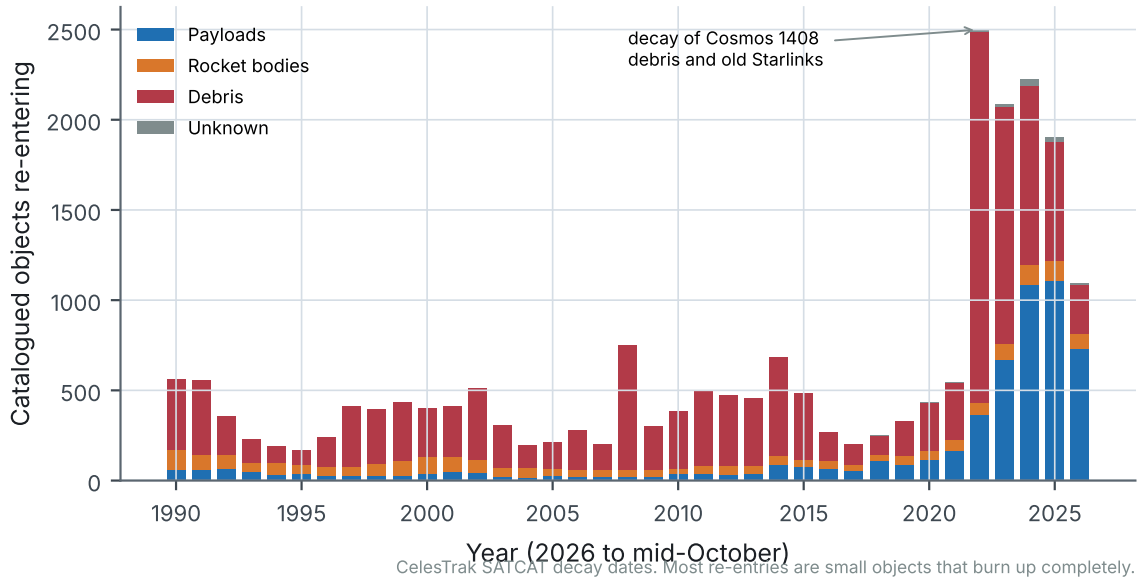



**FIGURE 21.15** Catalogued objects re-entering the atmosphere each year by type. Re-entries have increased sharply with large constellations and the decay of debris from the 2021 ASAT test.

## 21.9 Space Traffic Coordination

There is no international space traffic management system comparable to air traffic control. Each country licenses its own operators, and coordination relies on the sharing of data and good practice. Steps toward more organised **space traffic coordination** include:

- the US Office of Space Commerce's **Traffic Coordination System for Space (TraCSS)**, which is taking over civil conjunction services from the military. It was in pre-production with operators in 2026, though its funding has been contested;
- the **EU Space Surveillance and Tracking** partnership, which provides collision-avoidance services to hundreds of European and international satellites;
- industry initiatives such as the **Space Data Association** and the **Space Sustainability Rating**, which scores missions on their debris-mitigation and data-sharing practices;
- research on automated, interoperable coordination between operators — increasingly using AI (Part XIII).

### EXAMPLE: SHOULD WE MANOEUVRE?

A conjunction data message predicts that an Earth-observation satellite will pass a defunct rocket body at a miss distance of 400 m in 36 hours. The combined hard-body radius is 8 m, and the position uncertainty in the encounter plane is about σ = 300 m in each direction.

1. Using the circular-uncertainty approximation,
   $P_c \approx \frac{R^2}{2\sigma^2}\exp\left(-\frac{d^2}{2\sigma^2}\right) = \frac{64}{180\,000}\exp(-0.89) \approx 1.5 \times 10^{-4}$.
2. This exceeds the $1 \times 10^{-4}$ threshold, so the operator plans a manoeuvre but waits for the next update.
3. Twelve hours later, new tracking reduces σ to 100 m and the predicted miss distance to 150 m:
   $P_c \approx \frac{64}{20\,000}\exp(-1.13) \approx 1.0 \times 10^{-3}$ — a real threat.
4. A small along-track burn of a few centimetres per second, executed half an orbit before the encounter, changes the timing of arrival and increases the miss distance to about 2 km, reducing $P_c$ below $10^{-7}$. The operator confirms with the screening service that the new orbit creates no new conjunctions.

### AI FOR SPACE SITUATIONAL AWARENESS

Machine learning is used to associate observations with catalogue objects, detect manoeuvres and break-ups, characterise objects from light curves and radar data, improve drag and orbit predictions, prioritise conjunction alerts and recommend avoidance manoeuvres. As constellations grow, automated and AI-assisted coordination between operators will become essential (Chapter 50).

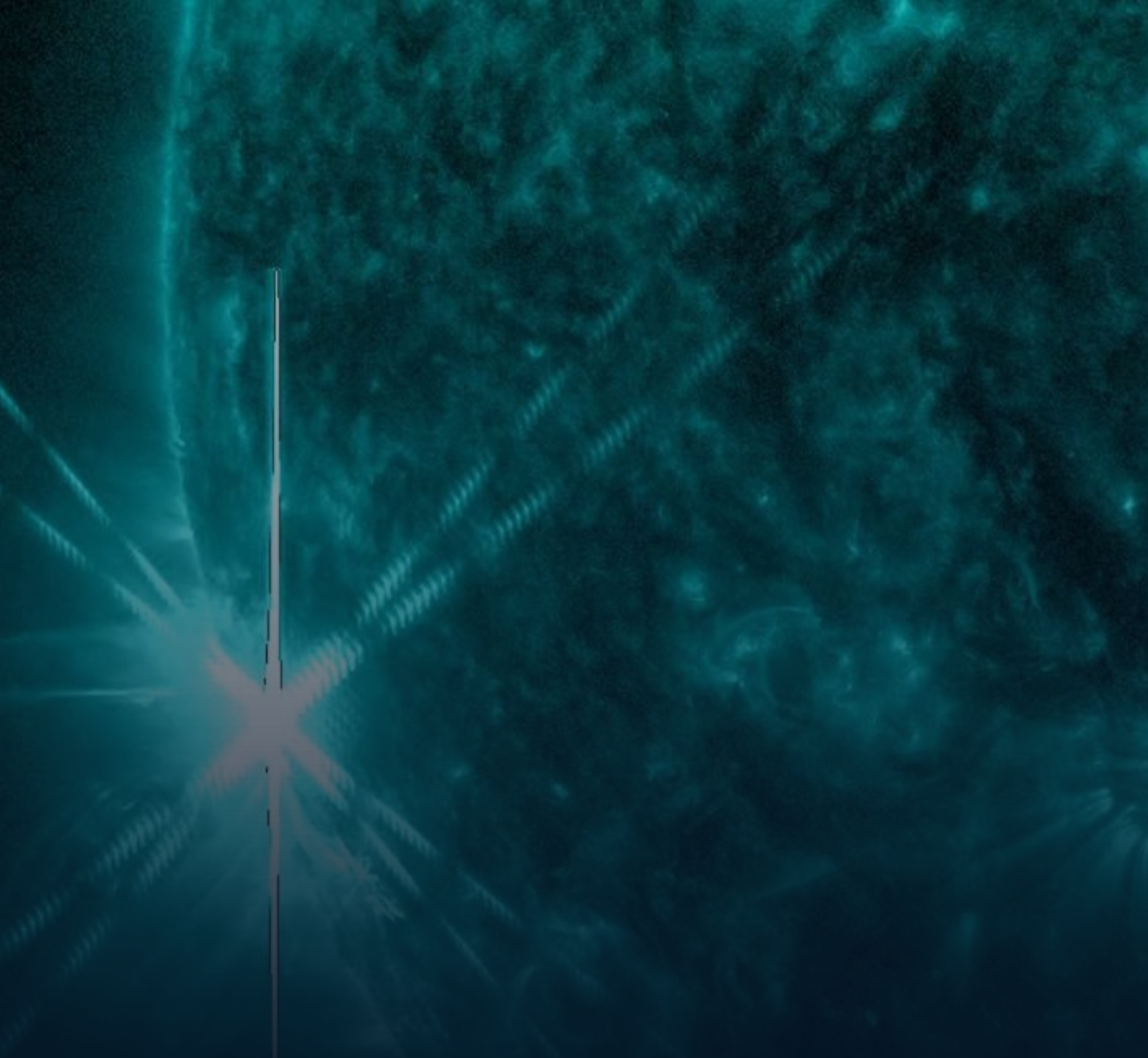
### KEY TAKEAWAYS

- About 35 000 objects are catalogued in Earth orbit (October 2026), of which about 17 000 are active payloads; models estimate more than a million debris objects larger than 1 cm.
- Debris comes mainly from explosions, deliberate destruction and collisions; the 2007 Chinese ASAT test and the 2009 Iridium–Cosmos collision are the largest sources still in orbit.
- At 10 km/s, a 1 cm fragment carries the energy of a small car crash; objects of 1–10 cm can neither be tracked reliably nor shielded against.
- Some LEO altitude bands may already exceed the critical density for a collisional cascade (Kessler syndrome); reliable post-mission disposal and removal of large derelicts can stabilise the environment.
- Radars track LEO objects down to about 5–10 cm; telescopes monitor GEO; catalogues are maintained by military, civil and commercial organisations.
- Conjunction assessment uses the probability of collision in the encounter plane; operators typically manoeuvre above about 1 in 10 000.
- Mitigation rules are tightening — from the 25-year guideline to the FCC five-year rule and ESA's Zero Debris approach; active removal missions such as ADRAS-J2 and ClearSpace-1 are preparing to remove debris.

### REVIEW QUESTIONS

1. How many objects are catalogued in Earth orbit in 2026, how many are active payloads, and how many debris objects larger than 1 cm are estimated to exist?

2. Calculate the kinetic energy of a 1 cm aluminium sphere (density 2 700 kg/m$^3$) at 10 km/s, and compare it with a 1-tonne car at 12 m/s.
3. Which events created the largest amounts of debris still in orbit?
4. Why are objects of 1–10 cm in LEO the most dangerous class for operational satellites?
5. What is the Kessler syndrome? Why do studies conclude that removing large derelict objects is more effective than removing small fragments?
6. How is the probability of collision for a conjunction calculated, and what thresholds do operators use to decide on a manoeuvre?
7. Compare the 25-year guideline, the FCC five-year rule and ESA's Zero Debris approach.
8. Describe two active debris-removal techniques and the missions planned to demonstrate them.



PART VII

# Space Environmental Science

The Sun drives a violent 'weather' system that reaches every satellite and power grid on Earth. Part VII explains solar activity, geomagnetic storms, the magnetosphere, radiation belts and the ionosphere — and the missions and instruments that study them.


22 Space Weather

23 Magnetospheric and Ionospheric Physics


Image: NASA/GSFC

# 22 CHAPTER Space Weather

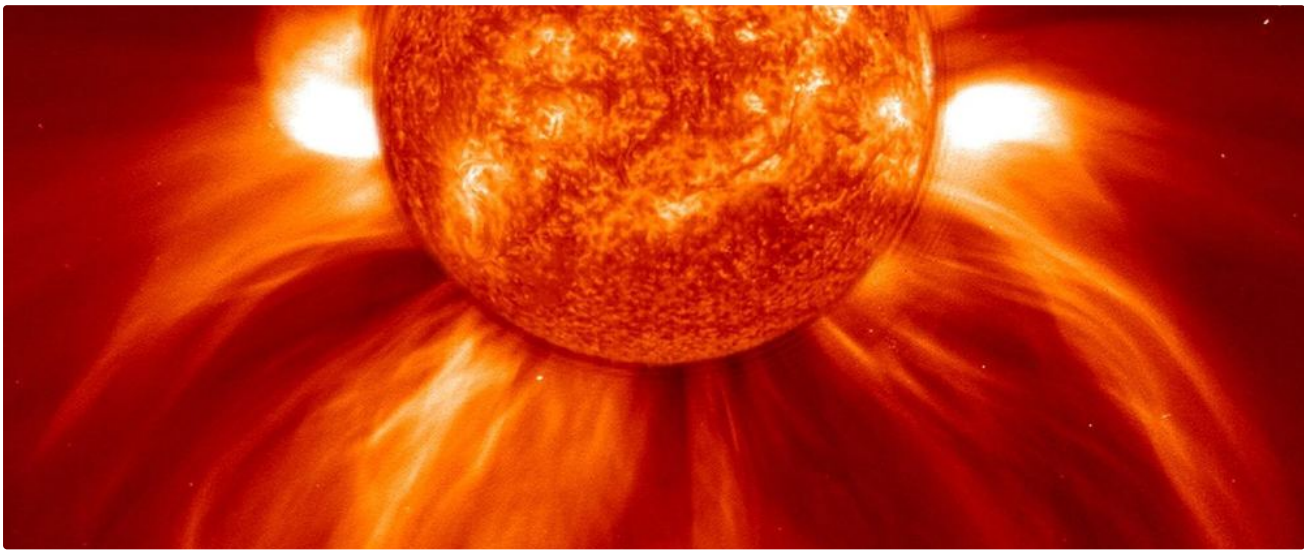


Image: NASA/GSFC

*The Sun is a variable star whose eruptions — flares, coronal mass ejections and storms of energetic particles — can disrupt satellites, navigation, radio communications, aviation and power grids on Earth and endanger astronauts. This chapter explains the physics of space weather from the solar dynamo and the 11-year sunspot cycle to the arrival of a storm at Earth, using real records of the sunspot number since 1749 and of geomagnetic activity since 1932. It describes the great historical storms, the May 2024 "Gannon" storm — the strongest in two decades — and the effects of space weather on technology, with engineering mitigations for each. It then covers the observatories, from SOHO to the new L1 monitors launched in 2025, and the models and services that forecast space weather today.*

IN THIS CHAPTER

22.1 What Is Space Weather?
22.2 The Sun: A Variable Star
22.3 Solar Eruptions and the Solar Wind
22.4 Space Weather at the Earth
22.5 Effects on Technology and People
22.6 Observing and Forecasting Space Weather
22.7 Designing for Space Weather

## 22.1 What Is Space Weather?

**Space weather** describes the conditions in space — on the Sun, in the solar wind, and in the Earth's magnetosphere, ionosphere and thermosphere — that can affect technology and human life. Just as terrestrial weather has its storms, space weather has its eruptions. There are **solar flares** that flood the sunlit side of the Earth with X-rays in minutes, and **solar energetic particle**

**(SEP) events** that fill near-Earth space with high-energy protons. There are also **coronal mass ejections (CMEs)** — billion-tonne clouds of magnetised plasma that can trigger **geomagnetic storms** when they reach the Earth one to four days later.

Modern society is increasingly exposed. Satellites can be damaged or lost, and GNSS positioning degrades. High-frequency radio used by aviation and maritime users blacks out, and airline passengers and crew receive extra radiation on polar routes. Long power lines and pipelines carry damaging induced currents. Several governments, including those of the UK and the USA, list severe space weather among their national risks. Insurance studies have estimated that a storm comparable to the largest on record could cause economic losses of hundreds of billions to trillions of dollars. These losses would come mainly through prolonged power outages.

## 22.2 The Sun: A Variable Star

### 22.2.1 Structure

The Sun is a sphere of hot plasma 1.39 million km across, powered by nuclear fusion in its core at about 15 million K (Chapter 2). Energy travels outward by radiation through the **radiative zone** and then by convection through the **convection zone**, the outer 30 % of the radius. The visible surface, the **photosphere**, has a temperature of about 5 800 K. Above it lie the thin **chromosphere** and the tenuous **corona**, which is mysteriously heated to 1–2 million K and continuously expands into space as the **solar wind**. NASA's **Parker Solar Probe**, which on 24 December 2024 passed just 6.1 million km from the Sun's surface at a record 192 km/s, is investigating how the corona is heated and the wind accelerated.

### 22.2.2 Magnetic Activity and the Solar Cycle

The Sun's activity is driven by its **magnetic field**, generated by a dynamo. In this dynamo, differential rotation (the equator rotates in about 25 days, the poles in about 35) and convection stretch, twist and amplify the field. Where bundles of strong magnetic field emerge through the photosphere, they inhibit convection and create cooler, darker **sunspots** (Figure 22.3), surrounded by active regions where flares and CMEs originate.

The number of sunspots rises and falls in a cycle of about **11 years** (Figure 22.1), during which the Sun's global magnetic field reverses; the full magnetic cycle therefore lasts about 22 years. Systematic sunspot counts began in the 18th century; the cycle that began in 1755 is numbered Cycle 1. Cycles vary greatly in strength: Cycle 19, peaking in 1958, was the strongest recorded, while from about 1645 to 1715 — the **Maunder Minimum** — sunspots almost disappeared.

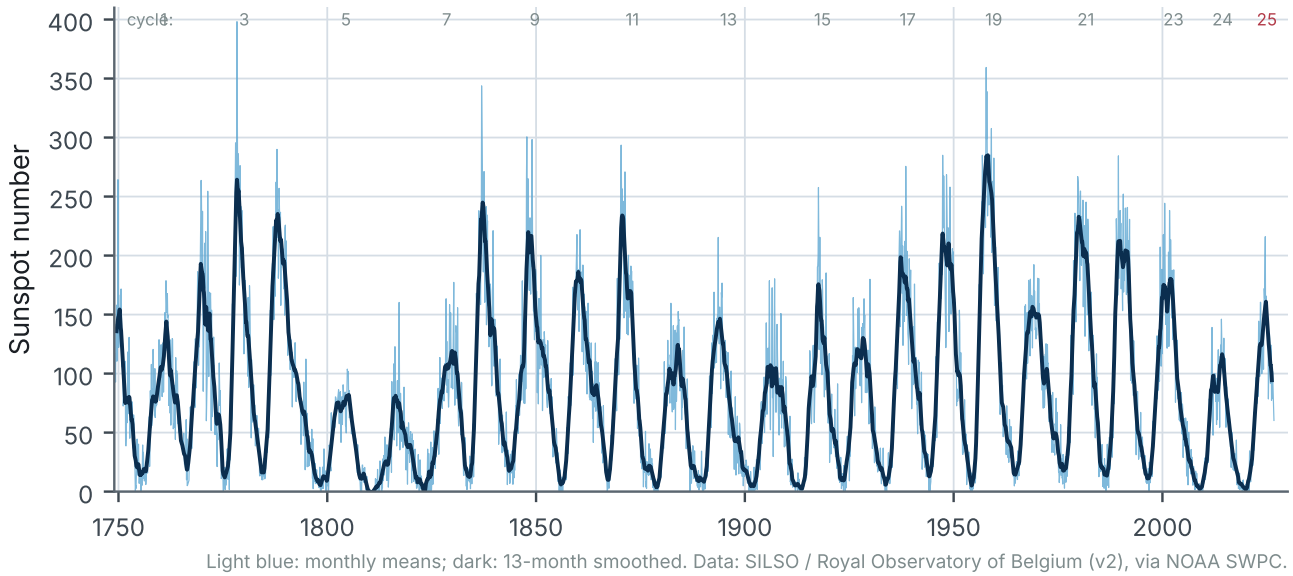



**FIGURE 22.1** The sunspot number since 1749. Cycle numbers are shown at the top. The strength and length of cycles vary considerably.

**Solar Cycle 25** began in December 2019. A NOAA–NASA expert panel predicted a weak cycle peaking at a smoothed sunspot number of about 115, similar to Cycle 24. Instead, activity rose faster and higher, reaching a smoothed maximum of about 161 in October 2024 (Figure 22.2). The declining phase, now under way, is historically when many of the largest geomagnetic storms occur, driven both by CMEs and by high-speed solar-wind streams from coronal holes.

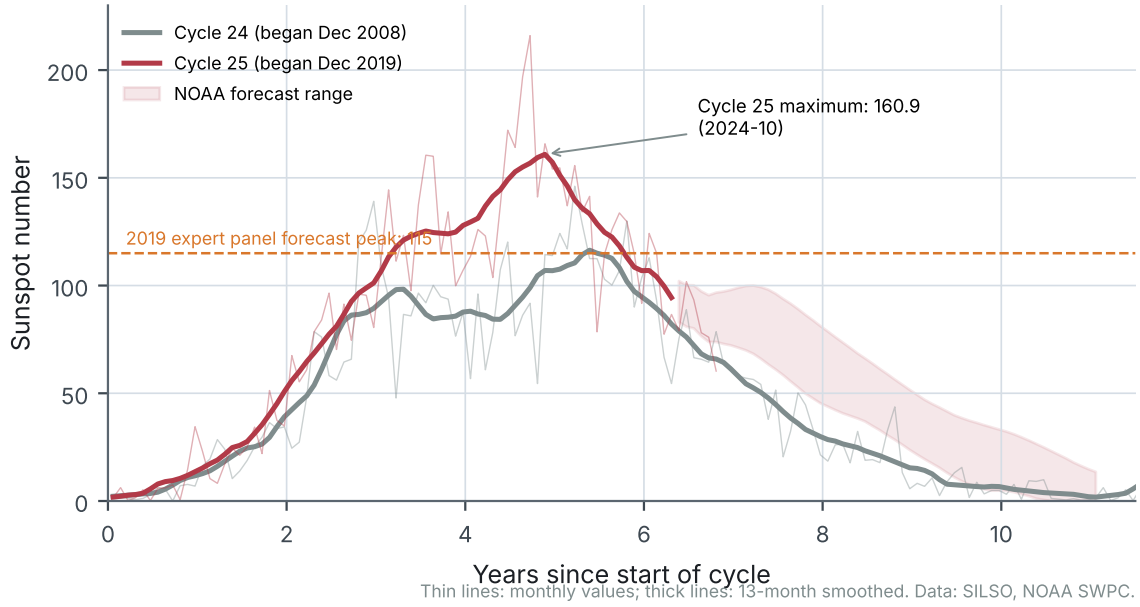



**FIGURE 22.2** Solar Cycle 25 compared with Cycle 24 and with the 2019 panel forecast.

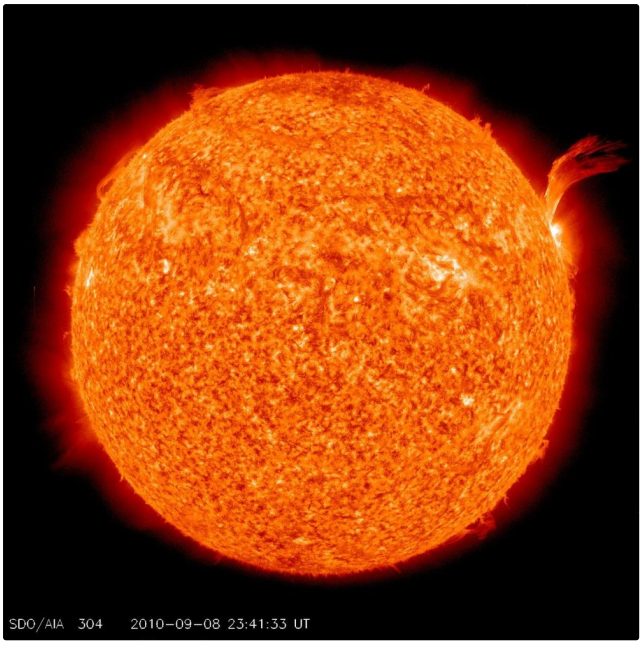



**FIGURE 22.3** The Sun seen by NASA's Solar Dynamics Observatory in extreme ultraviolet light on 8 September 2010, as a C3-class flare erupted from an active region near the edge of the disk. Active regions sit above sunspot groups, where strong magnetic fields break through the surface. Image: NASA/GSFC.

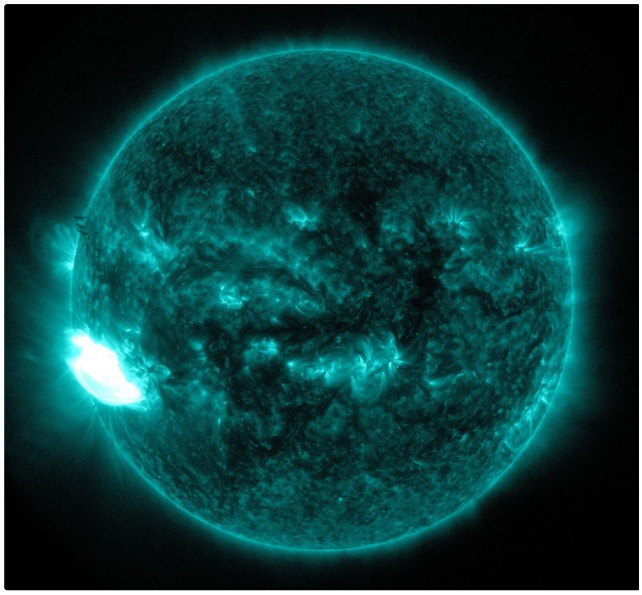


**FIGURE 22.4** An X-class solar flare (bright region at left) on 19 October 2014, observed by SDO in extreme ultraviolet light. Image: NASA/GSFC.

## 22.3 Solar Eruptions and the Solar Wind

### 22.3.1 Solar Flares

A **solar flare** is a sudden release of magnetic energy in an active region — up to about $10^{25}$ J in minutes to hours. It heats plasma to tens of millions of kelvin and accelerates particles. Flares emit across the spectrum from radio waves to gamma rays; their X-ray and extreme-ultraviolet radiation reaches the Earth in about 8 minutes and immediately increases ionisation in the dayside ionosphere. Flares are classified by their peak soft X-ray flux at 0.1–0.8 nm, measured by GOES satellites. The classes are A, B, C, M and X, each ten times stronger than the last. An X1 flare corresponds to $10^{-4}$ W/m$^2$. The largest flare of the space age, on 4 November 2003, saturated the detectors and has been estimated at about X40.

### 22.3.2 Coronal Mass Ejections

A **coronal mass ejection** expels $10^{11}$–$10^{13}$ kg of magnetised plasma from the corona at speeds from about 300 to more than 3 000 km/s (Figure 22.5). CMEs are observed with **coronagraphs**, which block the bright solar disc to reveal the faint corona. When a fast CME is directed towards the Earth, it typically arrives in one to three days; the fastest have crossed the 150 million km in under 18 hours. The geomagnetic impact of a CME depends above all on the direction of its internal magnetic field. A strong **southward** field component ($B_z$ negative) connects efficiently with the Earth's northward field at the dayside magnetopause. This opens the magnetosphere to the solar wind's energy (Chapter 23).

**FIGURE 22.5** A coronal mass ejection erupting from the Sun's limb on 17–18 June 2015, imaged by SDO in extreme ultraviolet light. Image: NASA/GSFC.

### 22.3.3 Solar Energetic Particles

Flares and the shocks driven by fast CMEs accelerate protons, electrons and heavier ions to energies of millions to billions of electronvolts. These **solar energetic particles** travel along the interplanetary magnetic field and can reach the Earth within tens of minutes. They raise the flux of energetic protons near the Earth by many orders of magnitude for hours to days. The most energetic events produce **ground-level enhancements** detected by neutron monitors at the Earth's surface. SEPs are the main acute radiation hazard for astronauts beyond low Earth orbit and a major cause of single-event effects in spacecraft electronics.

### 22.3.4 The Solar Wind

Between eruptions, the corona continuously emits the **solar wind**: about 5 protons and electrons per cubic centimetre at 1 AU. It flows at 300–500 km/s ("slow wind") or 600–800 km/s ("fast wind" from **coronal holes**), carrying the interplanetary magnetic field of about 5 nT. Where fast wind catches up with slow wind, **corotating interaction regions** form; because coronal holes can persist for months, they produce geomagnetic disturbances that recur every 27 days as the Sun rotates.

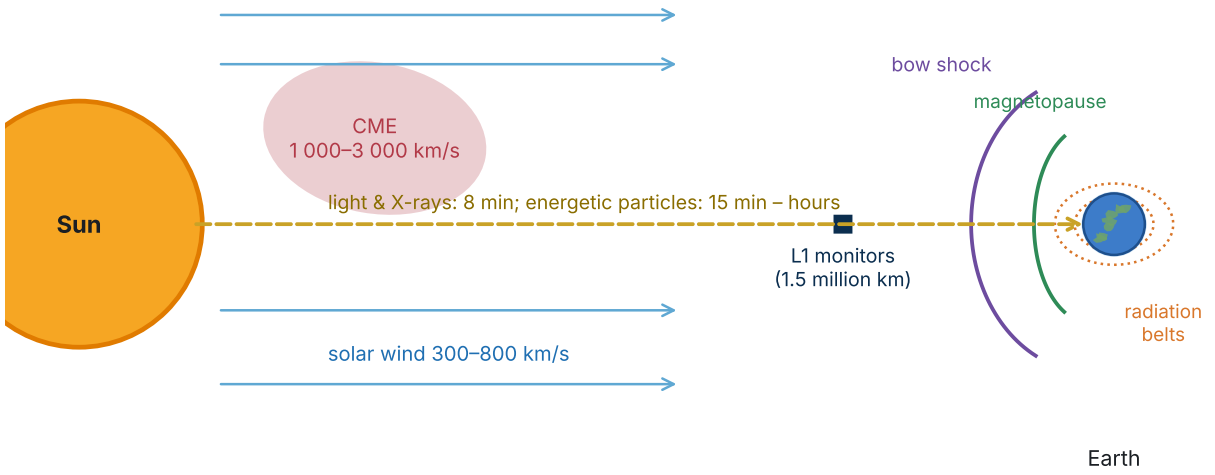



**FIGURE 22.6** The chain of space weather from the Sun to the Earth (schematic, not to scale).

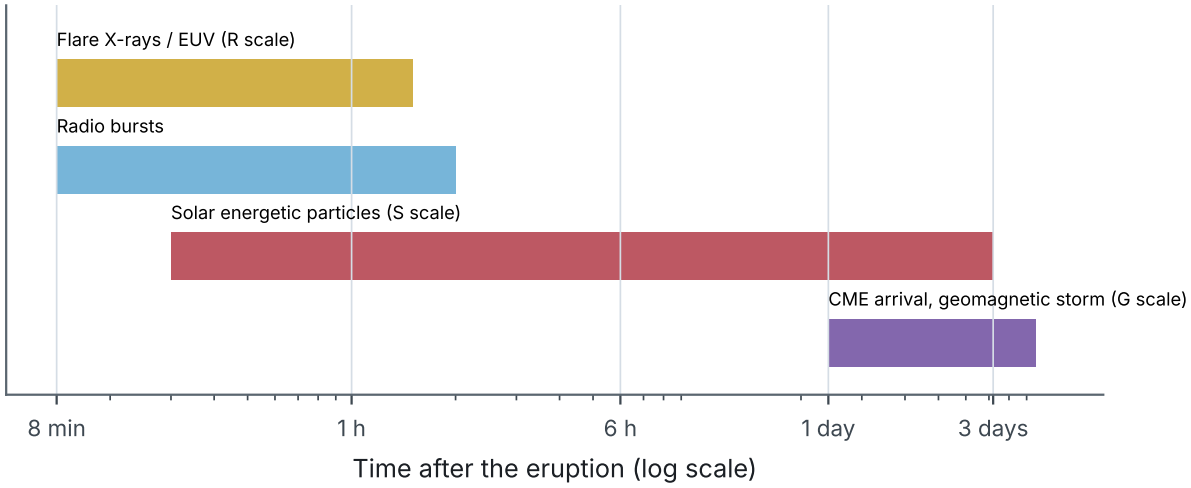



**FIGURE 22.7** Typical timing of the effects of a major solar eruption at the Earth.

## 22.4 Space Weather at the Earth

### 22.4.1 Geomagnetic Storms and Indices

When a CME or fast stream with a southward magnetic field reaches the Earth, energy pours into the magnetosphere, driving a **geomagnetic storm**. The ring current around the Earth intensifies and the auroral ovals expand towards the equator. Strong currents flow in the polar ionosphere, and the thermosphere heats and expands. Storms are measured by **geomagnetic indices** derived from networks of ground magnetometers:

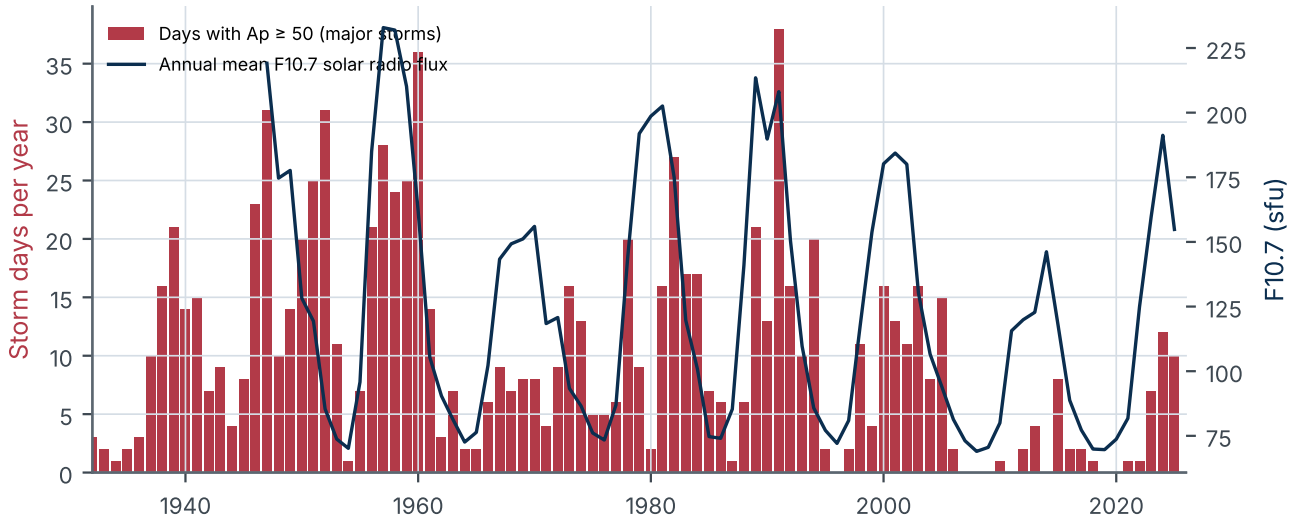
- **Kp** (0 to 9, quasi-logarithmic) measures global disturbance in three-hour intervals; **Ap** is its daily linear equivalent.
- **Dst** (in nanotesla) measures the depression of the horizontal magnetic field at low latitudes due to the ring current; major storms reach −200 to −600 nT.
- **AE** measures the intensity of the auroral electrojets.

NOAA's **space weather scales** (Table 22.1) translate these measurements into five levels of severity for geomagnetic storms (G), solar radiation storms (S) and radio blackouts (R).

**TABLE 22.1** NOAA space weather scales (pfu = particle flux units, particles $cm^{-2}$ $s^{-1}$ $sr^{-1}$).

| Level | Geomagnetic storm (G) | Solar radiation storm (S) | Radio blackout (R) | Approximate frequency per 11-year cycle |
|---|---|---|---|---|
| 1 Minor | Kp 5 | ≥ 10 pfu (≥ 10 MeV protons) | M1 flare | G1: ≈ 1 700; S1: ≈ 50; R1: ≈ 2 000 |
| 2 Moderate | Kp 6 | ≥ $10^2$ pfu | M5 | G2: ≈ 600; S2: ≈ 25; R2: ≈ 350 |
| 3 Strong | Kp 7 | ≥ $10^3$ pfu | X1 | G3: ≈ 200; S3: ≈ 10; R3: ≈ 175 |
| 4 Severe | Kp 8 (including 9−) | ≥ $10^4$ pfu | X10 | G4: ≈ 100; S4: ≈ 3; R4: ≈ 8 |
| 5 Extreme | Kp 9 | ≥ $10^5$ pfu | X20 | G5: ≈ 4; S5: < 1; R5: < 1 |

The long record of geomagnetic activity (Figure 22.8) shows that major storms are concentrated around solar maximum and in the declining phase of each cycle, but that severe storms can occur at almost any time.



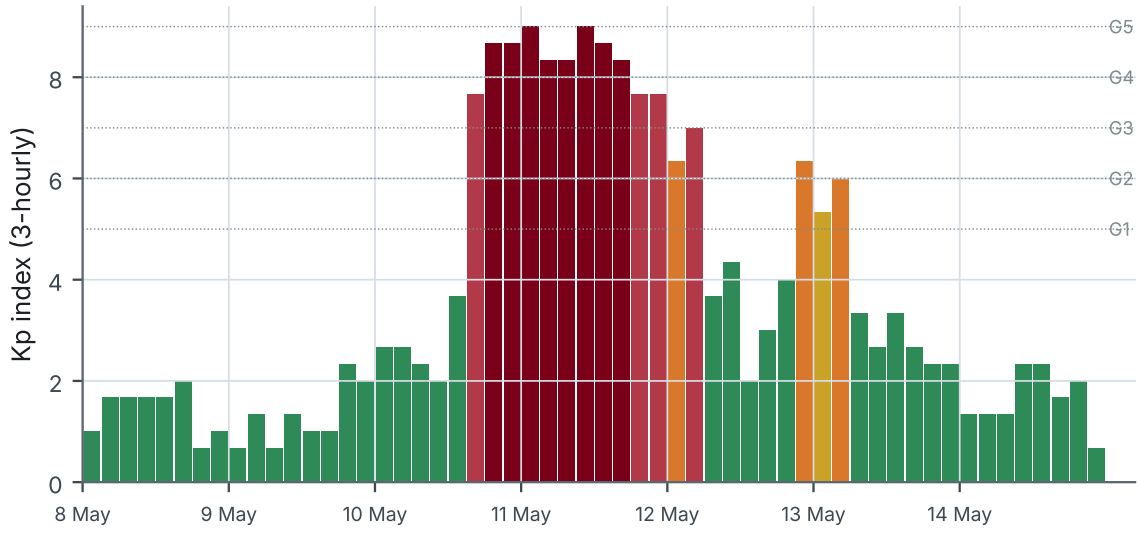

Data: GFZ Potsdam Kp/Ap index and F10.7 flux (Penticton/Ottawa). Storms are most frequent in the declining phase of each cycle.

**FIGURE 22.8** Number of days per year with major geomagnetic storms (Ap ≥ 50) and the annual mean solar radio flux, 1932–2025.

## 22.4.2 The May 2024 "Gannon" Storm

From 8 May 2024, a cluster of sunspots (region 13664) produced a series of X-class flares and at least six Earth-directed CMEs. They arrived on 10 May and combined to produce the first **G5 (extreme)** geomagnetic storm since 2003 (Figure 22.9), with Kp reaching 9 and Dst falling to about −410 nT. Aurorae were seen as far south as Florida, northern India and Namibia, and photographed by millions of people with smartphones. The storm, named after the solar physicist Jennifer Gannon, caused GNSS-guided tractors to lose accuracy during the planting season in North America. It also degraded HF communications and increased drag on thousands

of satellites, causing mass manoeuvres and temporary loss of tracking. In power grids it induced currents, which operators managed without major failures. Further severe storms followed in October 2024, November 2025 and January 2026.




Data: GFZ Potsdam definitive Kp. The 10–11 May 2024 'Gannon storm' reached Kp 9 (G5), the strongest since 2003.

**FIGURE 22.9** Three-hourly Kp index during the May 2024 geomagnetic storm (8–14 May).

## 22.4.3 Great Storms of the Past

**TABLE 22.2** Notable space weather events.

| Date | Event | Notable effects |
|---|---|---|
| 1–2 Sept 1859 | Carrington event | First observed white-light flare; telegraph systems failed and caught fire; aurora seen near the equator; Dst estimated −850 to −1 050 nT |
| 13–15 May 1921 | “New York Railroad” storm | Fires in telegraph and railway signalling stations; possibly comparable to Carrington |
| 4 Aug 1972 | Solar particle event and storm | Between Apollo 16 and 17 — a lethal dose risk for astronauts on the Moon; detonated sea mines off Vietnam |
| 13 Mar 1989 | Quebec storm | Hydro-Québec grid collapsed in 92 s, leaving 6 million people without power for 9 hours; tracking of thousands of satellites lost |
| 29–31 Oct 2003 | Halloween storms | X17 and X10 flares (and an estimated X40 on 4 Nov); satellite anomalies and a lost Japanese satellite (ADEOS-II); power outage in Malmö, Sweden; aircraft rerouted; WAAS vertical guidance unavailable for ≈ 30 hours |
| 23 Jul 2012 | Near miss | Extreme CME observed by STEREO-A missed the Earth by about nine days of solar rotation |
| 3–4 Feb 2022 | Minor storm, major loss | Increased drag prevented 38 of 49 newly launched Starlink satellites from raising their orbits |

| Date | Event | Notable effects |
|---|---|---|
| 10–11 May 2024 | Gannon storm | First G5 since 2003; worldwide aurorae; GNSS and satellite drag effects |

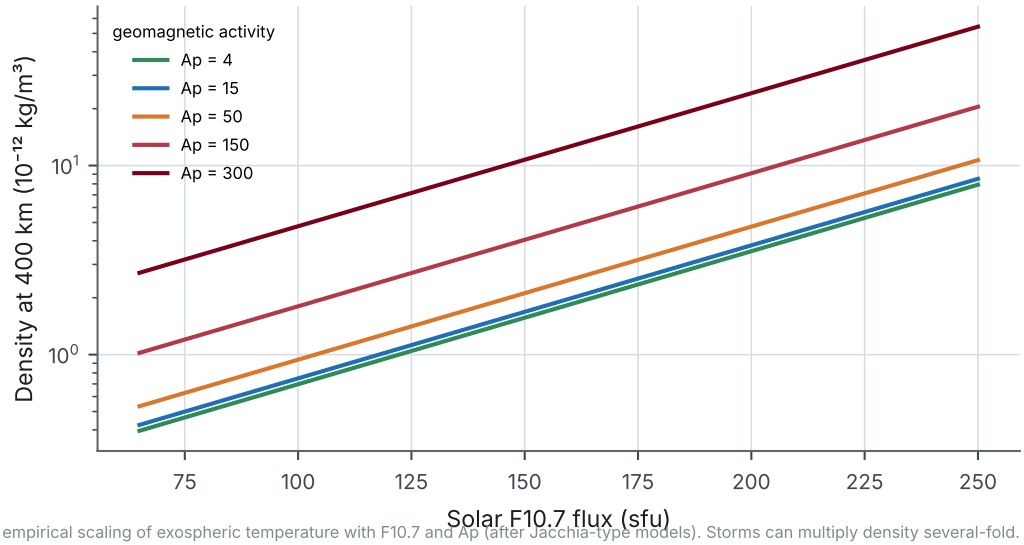
Even larger events have occurred. Tree rings and ice cores record sudden increases in radiocarbon and beryllium-10 in 774–775 and 993–994 CE (**Miyake events**). These are attributed to solar particle storms perhaps ten times stronger than any observed in the space age.

## 22.5 Effects on Technology and People

### 22.5.1 Satellites

Space weather affects spacecraft in several ways (Chapter 2):

- **Surface charging**: during storms, hot (keV) electrons injected into the inner magnetosphere charge spacecraft surfaces to thousands of volts, especially in the night-side GEO region. Discharges ("arcs") can damage solar arrays and electronics or cause phantom commands.
- **Internal (deep dielectric) charging**: high-energy (MeV) electrons of the outer radiation belt, which can increase dramatically for days after a storm, penetrate shielding and accumulate in cable insulation and circuit boards until they discharge. Internal charging is blamed for the failures of Canada's Anik E1 and E2 communications satellites in January 1994. It is also blamed for the failure of Galaxy 15 in 2010, which drifted out of control for eight months as a "zombie satellite" still relaying signals.
- **Single-event effects**: solar energetic particles and galactic cosmic rays cause bit flips, latch-ups and burnouts in electronics; star trackers become blinded by particle hits during SEP events.
- **Total ionising dose and displacement damage**: SEP events can deliver in days the dose expected over years, degrading solar cells and sensors.
- **Atmospheric drag**: heating of the thermosphere by storms increases density at satellite altitudes several-fold (Figure 22.10). In February 2022, a minor storm prevented 38 of 49 Starlink satellites, released at a low parking altitude, from raising their orbits; they re-entered within days. During major storms, thousands of objects change their orbits unpredictably, degrading the space catalogue and conjunction assessment (Chapter 21).



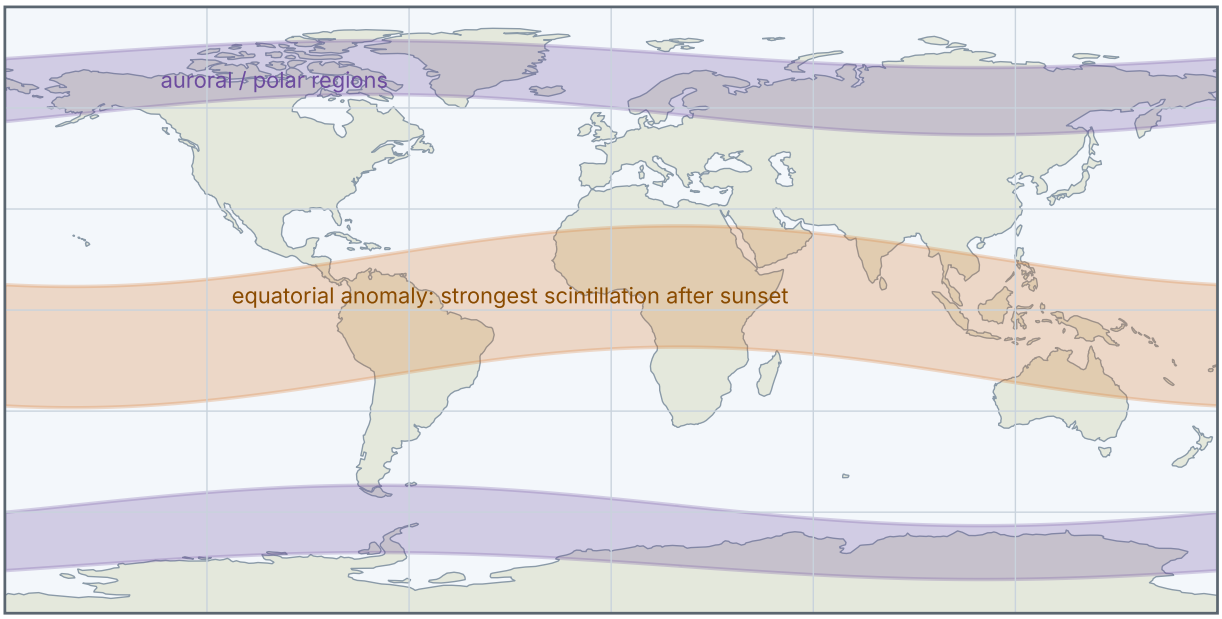

Illustrative empirical scaling of exospheric temperature with F10.7 and Ap (after Jacchia-type models). Storms can multiply density several-fold.

**FIGURE 22.10** Illustrative density of the thermosphere at 400 km as a function of solar EUV activity (F10.7) and geomagnetic activity (Ap). Both solar maximum and storms greatly increase satellite drag.

### 22.5.2 GNSS, Radio and Aviation

Disturbances of the ionosphere affect radio signals passing through it (Chapter 23):

- **GNSS errors**: increases and steep gradients of total electron content during storms produce positioning errors of tens of metres for single-frequency users. They also degrade augmentation systems. During the Halloween 2003 storms, WAAS could not provide vertical guidance for about 30 hours.
- **Scintillation**: small-scale irregularities in the ionosphere cause rapid fluctuations of signal amplitude and phase, which can make receivers lose lock. This is common after sunset near the geomagnetic equator and in the auroral and polar regions (Figure 22.11).
- **HF radio blackouts**: flare X-rays increase ionisation in the D region, absorbing HF signals on the dayside for minutes to hours. Energetic protons cause **polar cap absorption** lasting days, forcing airlines to reroute polar flights that rely on HF communications.
- **Radiation exposure**: during SEP events, aircrew and passengers on high-latitude, high-altitude flights receive increased radiation doses; aviation authorities issue advisories, and some flights are lowered or rerouted.




Schematic regions of frequent ionospheric scintillation affecting GNSS and satellite communications (after NOAA SWPC).

**FIGURE 22.11** Regions where ionospheric scintillation most often affects GNSS and satellite communications (schematic).

### 22.5.3 Power Grids and Other Ground Systems

Rapid changes in the Earth's magnetic field during storms (large $dB/dt$) induce **geoelectric fields** in the conducting ground — typically 1–10 V/km in severe storms, more over resistive rock (Figure 22.12). These fields drive **geomagnetically induced currents (GICs)** through long conductors grounded at both ends: high-voltage power lines, pipelines, railways and submarine cables. In power grids, GICs of tens to hundreds of amperes flow through transformer windings into the ground, saturating transformer cores, generating harmonics, causing overheating and tripping protective relays. The March 1989 storm collapsed the Hydro-Québec grid in about 90 seconds and damaged a large transformer at the Salem nuclear plant in New Jersey.

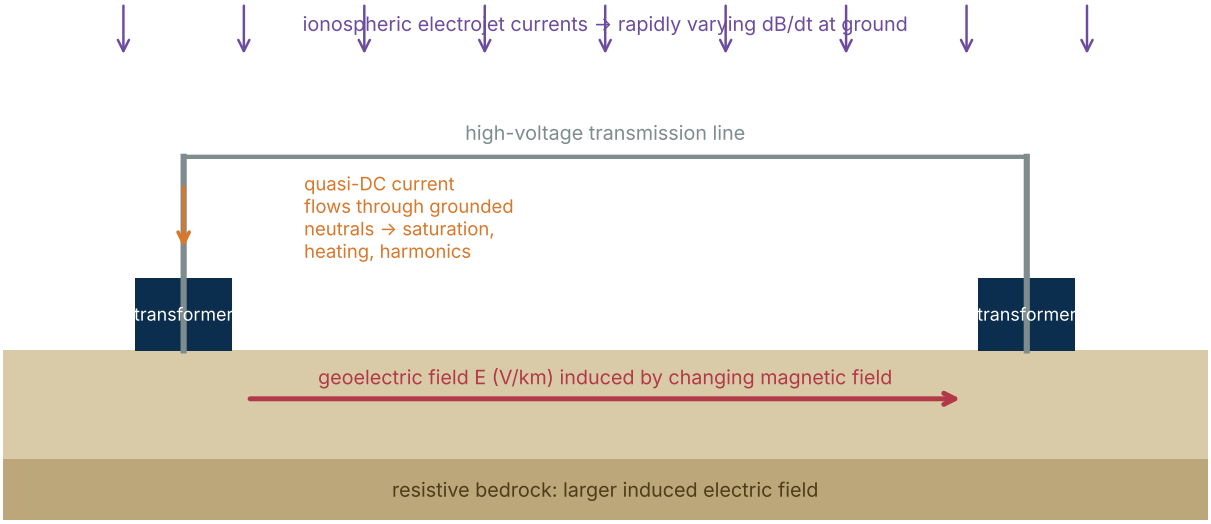



**FIGURE 22.12** Geomagnetically induced currents. Ionospheric currents cause rapid magnetic variations that induce an electric field in the ground and drive quasi-DC currents through grounded transformers.

Grid operators now receive storm warnings and run models of GIC flows. They can reduce loading on vulnerable transformers or reconfigure the network. Regulators in North America and Europe require utilities to assess their vulnerability to benchmark extreme storms.

### 22.5.4 Astronauts

Outside the protection of the Earth's magnetosphere, astronauts are exposed to SEP events that could deliver dangerous doses in hours. Had the August 1972 event occurred during an Apollo moonwalk, the astronauts could have suffered acute radiation sickness. Crewed spacecraft therefore provide **storm shelters** — areas surrounded by water, food, equipment and propellant — and missions rely on real-time warnings (Chapter 30). Artemis II's Orion spacecraft, flown around the Moon in April 2026, carried radiation sensors and a shelter procedure.

## 22.6 Observing and Forecasting Space Weather

### 22.6.1 Observatories

Space weather is monitored by a fleet of solar and heliospheric spacecraft and by ground networks (Table 22.3):

**TABLE 22.3** Space weather observatories (selected).

| Mission | Agencies | Launch | Location | Role |
|---|---|---|---|---|
| SOHO | ESA / NASA | 1995 | L1 | LASCO coronagraph: the main CME monitor for 30 years |
| ACE | NASA | 1997 | L1 | Solar-wind particles and magnetic field |
| STEREO-A | NASA | 2006 | Heliocentric orbit | Views of the Sun and CMEs from the side |
| SDO | NASA | 2010 | Geosynchronous | High-resolution imaging of the Sun in EUV and magnetic field |
| DSCOVR | NOAA / NASA | 2015 | L1 | Operational solar-wind monitor |
| GOES-R series | NOAA | 2016– | GEO | X-ray flux (flare classes), EUV imaging, particles, magnetometer; CCOR-1 operational coronagraph on GOES-19 |
| Parker Solar Probe | NASA | 2018 | Solar orbit to 6.1 million km | Origin of the solar wind |
| Solar Orbiter | ESA / NASA | 2020 | Inclined solar orbit | Close-up and polar views of the Sun |
| Aditya-L1 | ISRO | 2023 | L1 | India's first solar observatory |
| Proba-3 | ESA | 2024 | High elliptical orbit | Formation-flying coronagraph (two spacecraft 150 m apart) |

| Mission | Agencies | Launch | Location | Role |
|---|---|---|---|---|
| PUNCH | NASA | 2025 | LEO (sun-synchronous) | Wide-field imaging of the corona and solar wind |
| SWFO-L1, IMAP | NOAA, NASA | 2025 | L1 | Next operational monitor with coronagraph; interstellar boundary and solar-wind science |
| Vigil | ESA | ≈ 2031 | L5 | Side view of the Sun–Earth line for earlier CME warning |

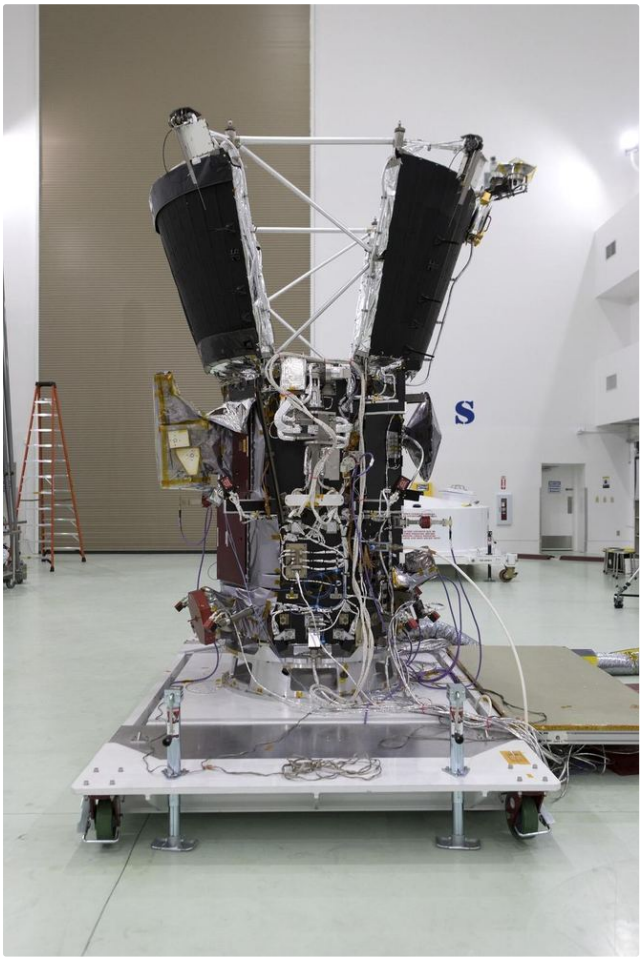
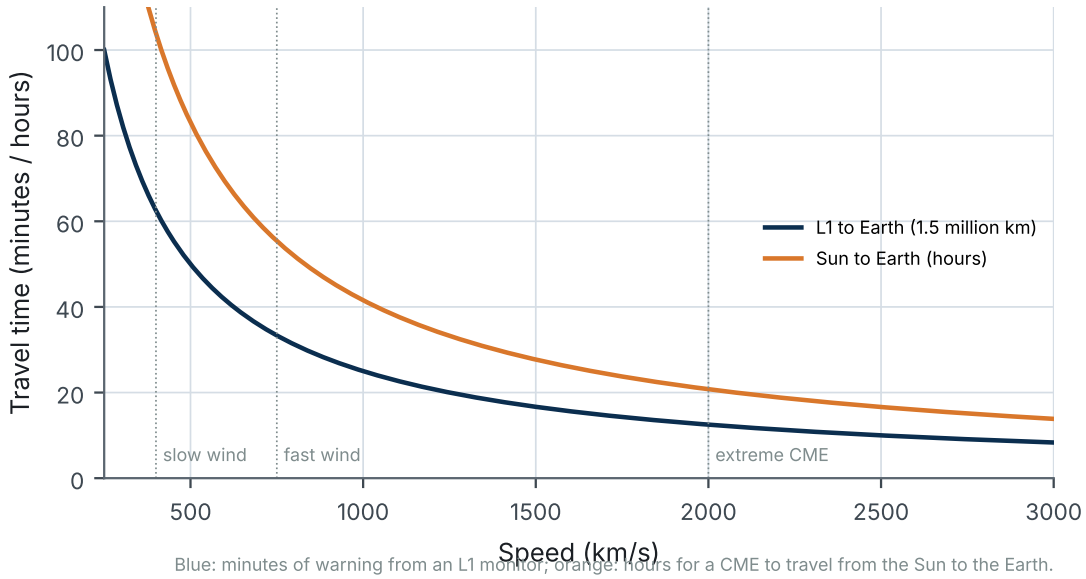


**FIGURE 22.13** NASA's Parker Solar Probe during testing in 2018. Its carbon-composite heat shield allows it to fly closer to the Sun than any previous spacecraft. Image: NASA/KSC (NASA/Kim Shiflett).

**FIGURE 22.14** One of the four PUNCH microsatellites during solar-array deployment tests in January 2025. Together they image the outer corona and the solar wind as a single "virtual instrument". Image: NASA/KSC (USSF 30th Space Wing/Alex Valdez).

The **L1 Lagrange point**, 1.5 million km sunward of the Earth, is the key location: a spacecraft there measures the solar wind, including the critical direction of its magnetic field, before it reaches the Earth — and an extreme CME as little as 15 minutes — this provides only short-term warning (Figure 22.15). Ground-based magnetometers, ionosondes, GNSS networks, neutron monitors and radio telescopes complete the observing system.




Blue: minutes of warning from an L1 monitor, orange: hours for a CME to travel from the Sun to the Earth.

**FIGURE 22.15** Travel time from L1 to the Earth (minutes) and from the Sun to the Earth (hours) as a function of solar-wind or CME speed.

## 22.6.2 Forecasting

Operational space-weather centres include NOAA's **Space Weather Prediction Center** in Boulder, the UK Met Office Space Weather Operations Centre and ESA's Space Weather Service Network. Others are the regional warning centres of the International Space Environment Service. These centres issue forecasts, watches, warnings and alerts. Their tools include:

- **CME arrival forecasts**: coronagraph images are used to fit a CME's speed and direction, and a heliospheric model such as **WSA–Enlil** propagates it to the Earth. Arrival-time errors are typically ±10 hours.
- **Geomagnetic forecasts**: once a CME reaches L1, measured solar-wind conditions drive magnetospheric models (such as NOAA's Geospace model) that predict Kp, regional magnetic disturbances and GIC risk tens of minutes ahead.
- **Flare and SEP forecasts**: based mainly on the magnetic complexity of active regions; these remain probabilistic and of limited skill.
- **Ionosphere and thermosphere models** for GNSS users and satellite operators.

The greatest limitation is that the magnetic orientation of a CME, which determines its geomagnetic impact, cannot yet be measured until it reaches L1. Missions viewing the Sun from the side (L5), new methods of measuring coronal magnetic fields and machine-learning models are among the approaches to improve forecasts.

## 22.6.3 Space Weather Beyond the Earth

Space weather affects all planets. During the May 2024 storms, NASA's MAVEN spacecraft observed global aurorae on Mars, and its Curiosity rover measured a radiation spike on the surface (Figure 22.16). Mars lacks a global magnetic field, so it is fully exposed to SEPs — a key concern for future crews.

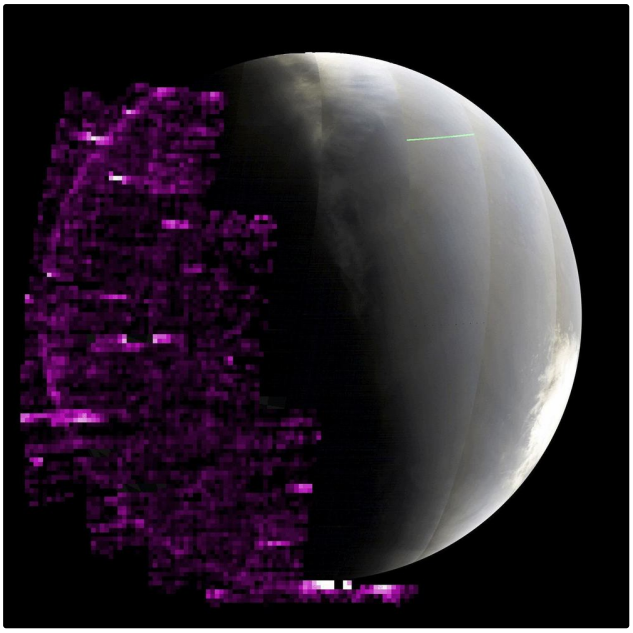


**FIGURE 22.16** Ultraviolet aurorae across the night side of Mars (purple) observed by NASA's MAVEN orbiter during the solar storm of May 2024. Image: NASA/JPL (NASA/University of Colorado/LASP).

## 22.7 Designing for Space Weather

Engineers design spacecraft and infrastructure for specified space-weather environments:

- **Environment specifications** such as ECSS-E-ST-10-04 and NASA and ISO standards define radiation-belt fluxes, SEP fluences, charging environments and thermospheric density ranges, often including a "worst case" or 1-in-100-year event.
- **Spacecraft design**: radiation-tolerant components and error-correcting memories; shielding of sensitive parts; grounding and conductive coatings to prevent charging; derating of solar arrays for degradation; propellant margins for higher drag; autonomous safe modes.
- **Operations**: delaying launches, EVAs or critical manoeuvres during storms; placing satellites in safe configurations; increasing tracking and conjunction screening.
- **Ground infrastructure**: GIC-blocking devices and transformer designs for power grids; GNSS receivers with multi-frequency and integrity monitoring; alternative timing sources.

### EXAMPLE: ESTIMATING A GEOMAGNETICALLY INDUCED CURRENT

During a severe storm, a ground magnetometer in northern Europe records magnetic variations of about 1 000 nT with a period of about 10 minutes ($\omega = 2\pi/600\ \text{s} \approx 0.0105\ \text{s}^{-1}$). For a plane wave over uniform ground of conductivity $\sigma = 10^{-3}$ S/m, the induced electric field is

$$E \approx B\sqrt{\frac{\omega}{\mu_0 \sigma}} = 10^{-6}\ \text{T} \times \sqrt{\frac{0.0105}{1.26 \times 10^{-6} \times 10^{-3}}} \approx 2.9 \times 10^{-3}\ \text{V/m} \approx 3\ \text{V/km}.$$

Over more resistive rock, or during faster variations, the field can be several times larger.

Taking E = 5 V/km along a 200 km high-voltage line, the driving voltage is 1 000 V. The loop consists of the line, two transformer windings and two substation earthings. If its total resistance is about 5 Ω, a quasi-DC current of about 200 A flows through each transformer neutral. This is enough to drive a large transformer into half-cycle saturation, causing heating, reactive-power demand and harmonics. Grid operators therefore monitor neutral currents and may reduce transformer loading during severe storms.

### AI IN SPACE WEATHER

Machine learning is used to predict flares from magnetograms, estimate CME arrival times, forecast geomagnetic indices from solar-wind data, nowcast GIC and ionospheric scintillation, and model thermospheric density for satellite drag. NASA and NOAA have explored deep-learning models that predict regional geomagnetic disturbances about 30 minutes ahead from L1 data. Chapter 47 discusses these applications.

### KEY TAKEAWAYS

- Space weather comprises solar flares (X-rays in 8 minutes), energetic particles (minutes to hours) and CMEs (1–3 days) that disturb the Earth's space environment.
- The Sun's magnetic activity follows an ~11-year sunspot cycle; Cycle 25 peaked at a smoothed sunspot number of about 161 in October 2024, well above forecast.
- Geomagnetic storms are measured by Kp, Ap and Dst and classified by NOAA's G scale; the May 2024 Gannon storm was the first G5 since 2003.
- Effects include satellite charging, radiation damage and drag (38 Starlink satellites lost in 2022), GNSS errors and scintillation, HF blackouts, aviation and astronaut radiation, and geomagnetically induced currents in power grids (Quebec, 1989).
- L1 monitors give 15–60 minutes of warning of a storm's severity; coronagraphs give 1–3 days of warning of CME arrival but not of its magnetic orientation.
- New observatories launched in 2024–2025 (Proba-3, PUNCH, SWFO-L1, IMAP) and a planned L5 mission (Vigil) aim to improve forecasts.

### REVIEW QUESTIONS

1. Order flare X-rays, solar energetic particles and coronal mass ejections by how quickly they reach the Earth, and explain the difference.
2. A CME travels from the Sun to the Earth at an average speed of 1 500 km/s. How long does it take to arrive?
3. A solar-wind monitor at L1 is 1.5 million km upstream of the Earth. How much warning does it give for solar wind travelling at 500 km/s?
4. Explain the Kp, Ap and Dst indices and the NOAA G scale. What made the May 2024 storm notable?
5. Why did 38 Starlink satellites re-enter shortly after launch in February 2022?

6. How do geomagnetic storms induce currents in power grids, and which grids are most at risk?
7. Why can coronagraphs predict when a CME will arrive but not how strong the resulting storm will be?
8. Describe three design measures that make a satellite tolerant of space weather.

CHAPTER

# 23 Magnetospheric and Ionospheric Physics

Image: NASA/JSC

*Every satellite in Earth orbit flies through geospace — the region dominated by the Earth's magnetic field and by electrically charged gas. This chapter explains the physics of this environment: the geomagnetic field and the magnetosphere it carves out of the solar wind; how charged particles gyrate, bounce and drift to form the Van Allen radiation belts; how magnetic reconnection couples the solar wind to the magnetosphere and drives storms, substorms and the aurora; and how solar radiation creates the ionosphere, whose layers reflect radio waves and delay navigation signals. Computed models — field lines, particle periods, ionospheric profiles and a synthetic ionogram — illustrate the key ideas, and the chapter describes the observatories, models and missions, from Explorer 1 to SMILE (launched in 2026), that have revealed geospace.*

IN THIS CHAPTER

23.1 Geospace
23.2 The Geomagnetic Field
23.3 The Magnetosphere
23.4 Charged Particles in Magnetic Fields
23.5 Coupling the Solar Wind to the Magnetosphere
23.6 The Aurora
23.7 The Upper Atmosphere
23.8 The Ionosphere
23.9 Measuring and Modelling Geospace

## 23.1 Geospace

Above about 60 km, the Earth's atmosphere becomes partly ionised by solar radiation. Above about 1 000 km, the environment is dominated by **plasma** — a gas of free electrons and ions. This plasma is controlled by electric and magnetic fields rather than by collisions. This region, from

the ionosphere out to the boundary with the solar wind, is called **geospace**. It contains the radiation belts that damage satellites (Chapter 2) and the ionosphere that affects every radio signal to and from space (Chapters 7 and 18). It also contains the thermosphere, whose drag determines satellite lifetimes (Chapters 11 and 21). Understanding its physics is the basis of space weather forecasting (Chapter 22).

## 23.2 The Geomagnetic Field

### 23.2.1 A Tilted Dipole

The Earth's magnetic field is generated by a dynamo in the liquid iron outer core. At the surface it is approximately that of a **magnetic dipole** tilted by about 10° from the rotation axis. Its field strength is about 30 μT at the equator and 60 μT near the poles. In a dipole, the field strength falls with distance as $r^{-3}$, and field lines follow

$$r = L\cos^2\lambda$$

where λ is magnetic latitude and $L$ is the distance, in Earth radii ($R_E$), at which the field line crosses the magnetic equator (Figure 23.1). The **L-shell** parameter is the most convenient way to label regions of the magnetosphere. The inner radiation belt lies at L ≈ 1.2–2, GNSS satellites at L ≈ 4–4.5, and geostationary satellites at L ≈ 6.6.

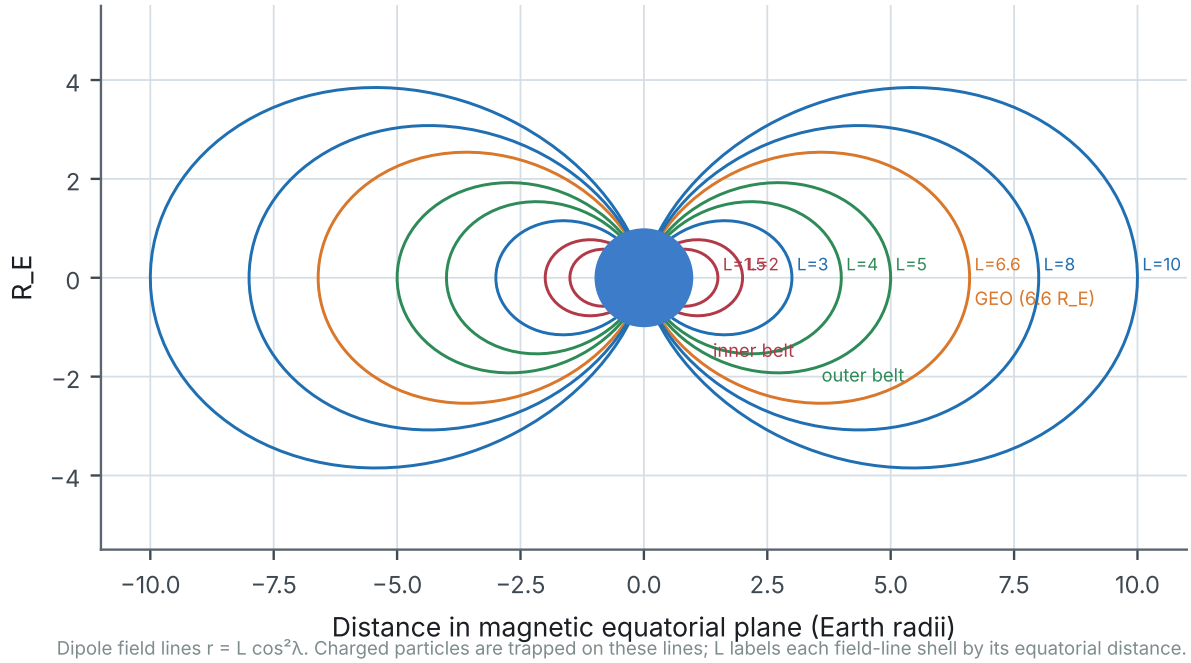



**FIGURE 23.1** Dipole magnetic field lines labelled by L-shell. Trapped particles spiral along these lines; geostationary satellites are on the L = 6.6 shell.

### 23.2.2 A Changing Field

The field is not static. The north magnetic pole has drifted from northern Canada towards Siberia at up to 50 km per year. The dipole moment has weakened by about 9 % since the mid-19th century. Over geological time the field has reversed many times — most recently about 780 000 years ago. A region of weak field over South America and the South Atlantic, the **South Atlantic Anomaly**, allows the inner radiation belt to dip to altitudes of 200–800 km. LEO satellites —

including the ISS and the Hubble Space Telescope — pass through it several times a day. They experience most of their radiation dose and single-event upsets there. The field is monitored by ground observatories and by ESA's three **Swarm** satellites (2013). It is modelled by the International Geomagnetic Reference Field (IGRF) and the World Magnetic Model. These models are updated every five years and used in every smartphone compass and aircraft heading system.

## 23.3 The Magnetosphere

### 23.3.1 Structure

The supersonic solar wind cannot penetrate the Earth's magnetic field. Instead, the field carves out a cavity — the **magnetosphere** — shaped like a comet, compressed on the dayside and stretched into a long tail on the nightside (Figure 23.2):

- The **bow shock**, about 13 $R_E$ upstream, where the solar wind slows abruptly from supersonic to subsonic speed;
- the **magnetosheath** of shocked, turbulent solar-wind plasma;
- the **magnetopause**, the boundary where the pressure of the solar wind balances the magnetic pressure of the Earth's field — typically about 10 $R_E$ from the Earth on the sunward side;
- the polar **cusps**, funnel-shaped regions through which solar-wind plasma can reach the upper atmosphere;
- the **magnetotail**, extending more than 200 $R_E$ downstream, consisting of two **lobes** of nearly empty, open field lines separated by the hot **plasma sheet**;
- the inner magnetosphere: the cold, dense **plasmasphere** (an extension of the ionosphere to about 4–5 $R_E$), the **radiation belts** and the **ring current** of energetic ions that flows around the Earth during storms.

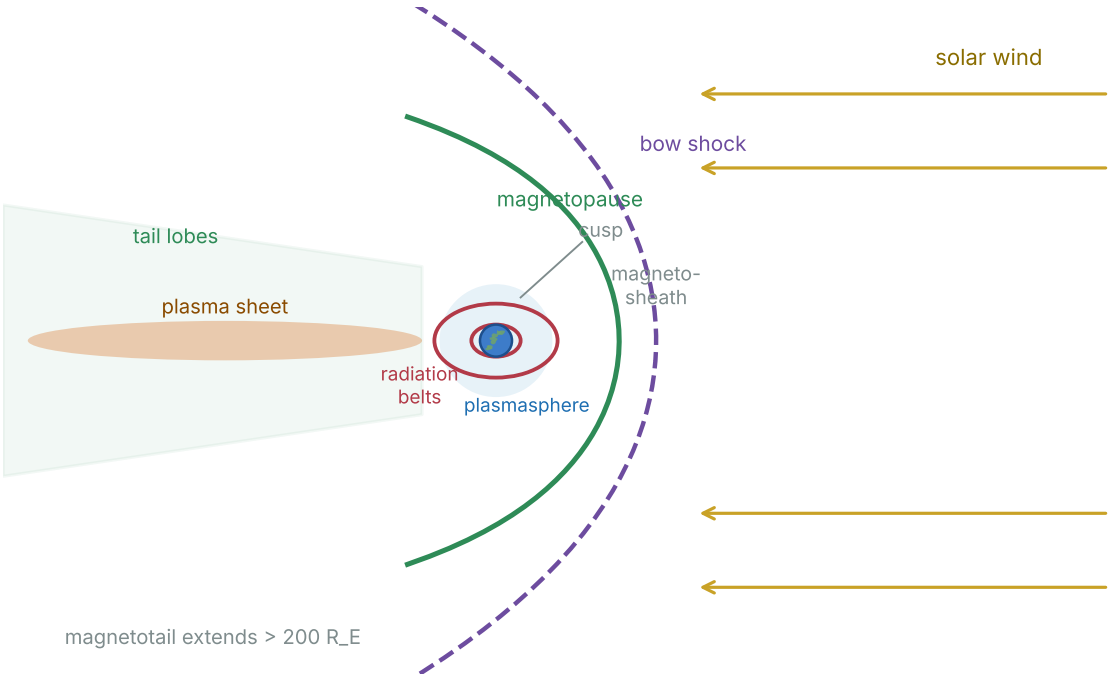



**FIGURE 23.2** Structure of the Earth's magnetosphere (schematic, not to scale). The Sun is to the right.

### 23.3.2 Where Is the Magnetopause?

The distance to the magnetopause follows from pressure balance. The solar wind's dynamic pressure $P = \rho v^2$ must equal the magnetic pressure of the compressed dipole field $B^2/2\mu_0$; allowing for a factor of about 2 compression by the boundary currents, the standoff distance is

$$r_{mp} \approx \left( \frac{2B_0^2}{\mu_0 P} \right)^{1/6} R_E$$

with $B_0 \approx 31$ μT the equatorial surface field. For a typical pressure of 2 nPa this gives $r_{mp} \approx 9.6\,R_E$; because of the sixth root, even large pressure increases move it only modestly (Figure 23.3). During extreme storms, high pressure combined with erosion by a southward interplanetary field pushes the magnetopause inside geostationary orbit, exposing GEO satellites directly to the solar wind and the magnetosheath.

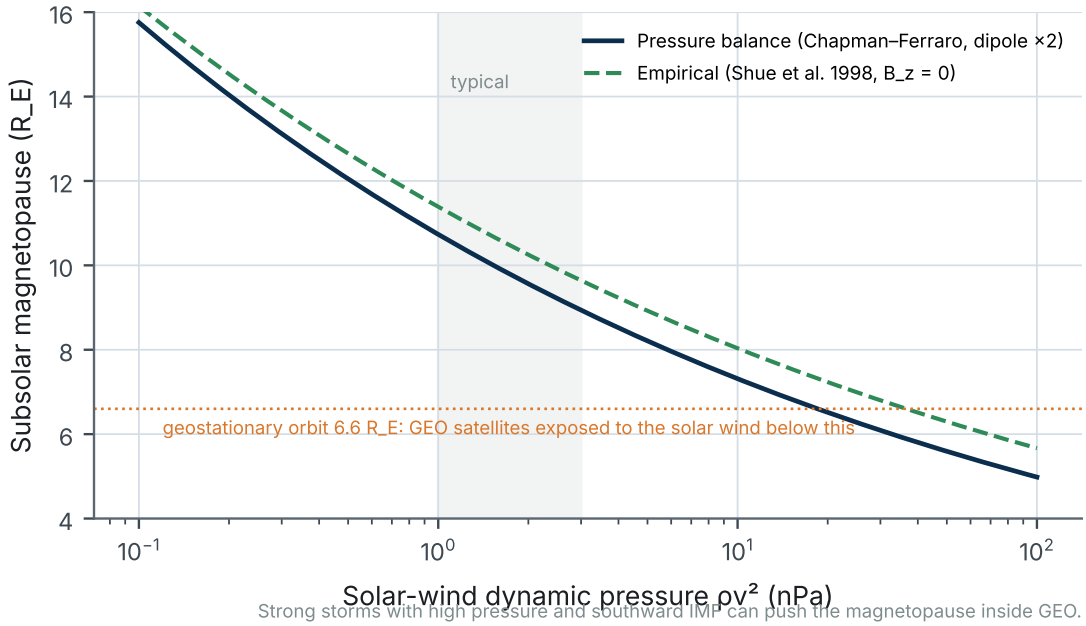



**FIGURE 23.3** Subsolar magnetopause distance versus solar-wind dynamic pressure, from pressure balance and from an empirical model.

## 23.4 Charged Particles in Magnetic Fields

### 23.4.1 Three Periodic Motions

A charged particle in a magnetic field performs three superimposed periodic motions (Figure 23.4):

1. **Gyration** around the field line, with the **gyrofrequency** $f_g = qB/(2\pi\gamma m)$ — tens of kilohertz for electrons and a few hertz for protons in the inner magnetosphere.
2. **Bounce** along the field line between **mirror points**. As a particle moves towards stronger field near the poles, its velocity component perpendicular to the field grows at the expense of the parallel component (conserving its magnetic moment) until it reverses direction. Particles whose **pitch angle** is too small mirror below about 100 km altitude, where they are lost by collisions with the atmosphere — they are said to be in the **loss cone**.

3. **Drift** around the Earth, caused by the gradient and curvature of the field: electrons drift eastward and ions westward, forming the ring current.

The three periods differ by orders of magnitude (Figure 23.5): gyration in microseconds to fractions of a second, bounce in fractions of a second to seconds, drift in minutes to hours. Each motion is associated with an **adiabatic invariant** that is conserved when the fields change slowly compared with that period. Violating the invariants — through waves, sudden compressions or electric fields — accelerates, transports and removes particles.

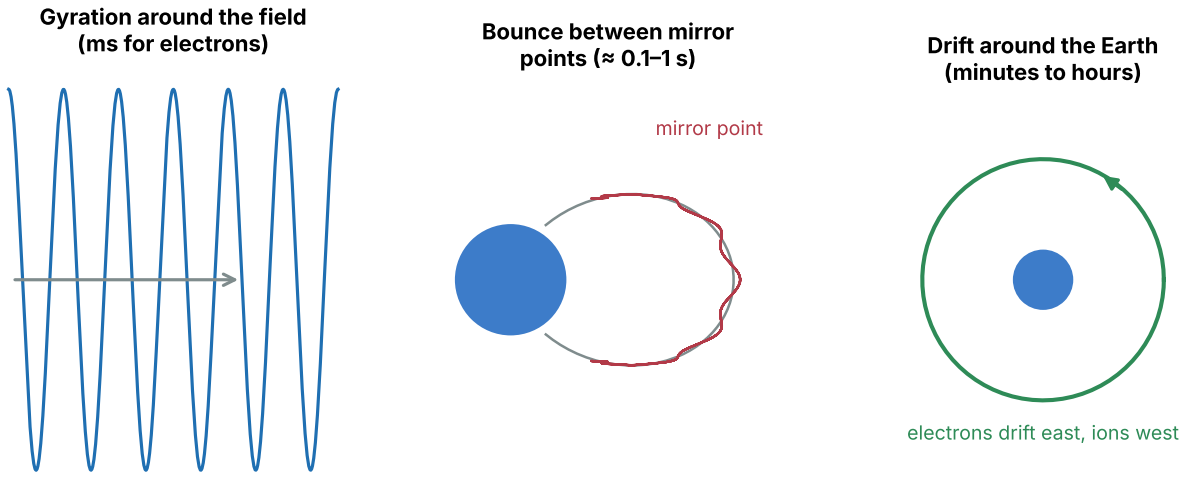
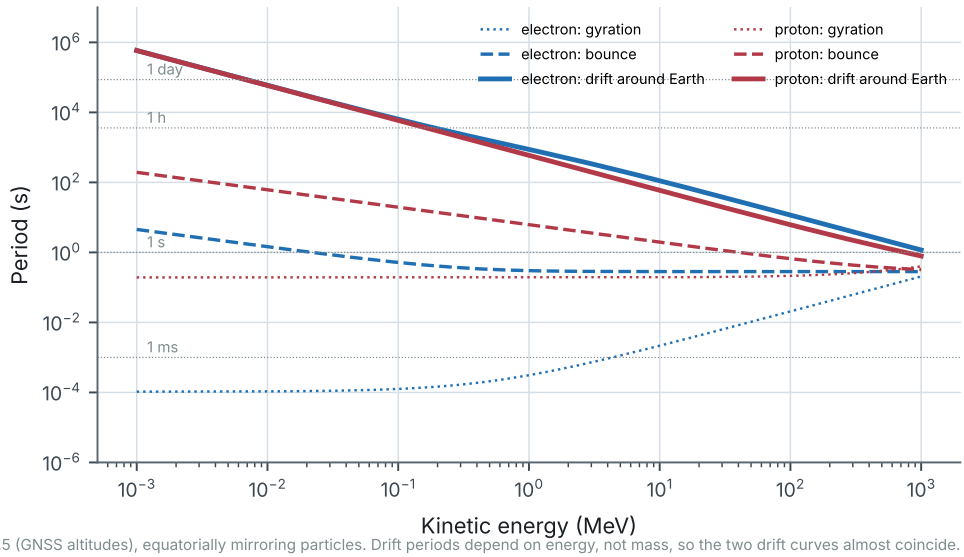



**FIGURE 23.4** The three periodic motions of trapped particles: gyration, bounce between mirror points and drift around the Earth.




Dipole field at L = 4.5 (GNSS altitudes), equatorially mirroring particles. Drift periods depend on energy, not mass, so the two drift curves almost coincide.

**FIGURE 23.5** Gyration, bounce and drift periods of electrons and protons as a function of energy at L = 4.5.

## 23.4.2 The Van Allen Radiation Belts

The first US satellite, **Explorer 1** (1958), carried a Geiger counter designed by James Van Allen of the University of Iowa. It recorded puzzling drop-outs — later understood as saturation by intense radiation — revealing that the Earth is surrounded by belts of trapped energetic particles. The **radiation belts** (Figure 23.6) consist of:

- An **inner belt** at L ≈ 1.2–2, dominated by protons of tens to hundreds of MeV, created mainly by the decay of neutrons produced by cosmic rays hitting the atmosphere (cosmic-ray albedo neutron decay). It is very stable.
- An **outer belt** at L ≈ 3–7, dominated by electrons of hundreds of keV to several MeV, which varies by orders of magnitude within hours during storms, as wave–particle interactions accelerate and scatter electrons.
- A **slot region** at L ≈ 2–3, kept relatively empty by scattering of electrons into the loss cone by plasmaspheric waves.

NASA's twin **Van Allen Probes** (2012–2019) transformed understanding of the belts. Within days of launch they discovered a transient **third belt** (Figure 23.7). They also showed that electrons are accelerated locally by "chorus" waves inside the belts, and found a sharp barrier at about 11 000 km that ultra-relativistic electrons cannot cross. Engineering models of the trapped radiation environment, such as AE8/AP8 and their successors AE9/AP9/SPM, are used to design spacecraft (Chapter 2).

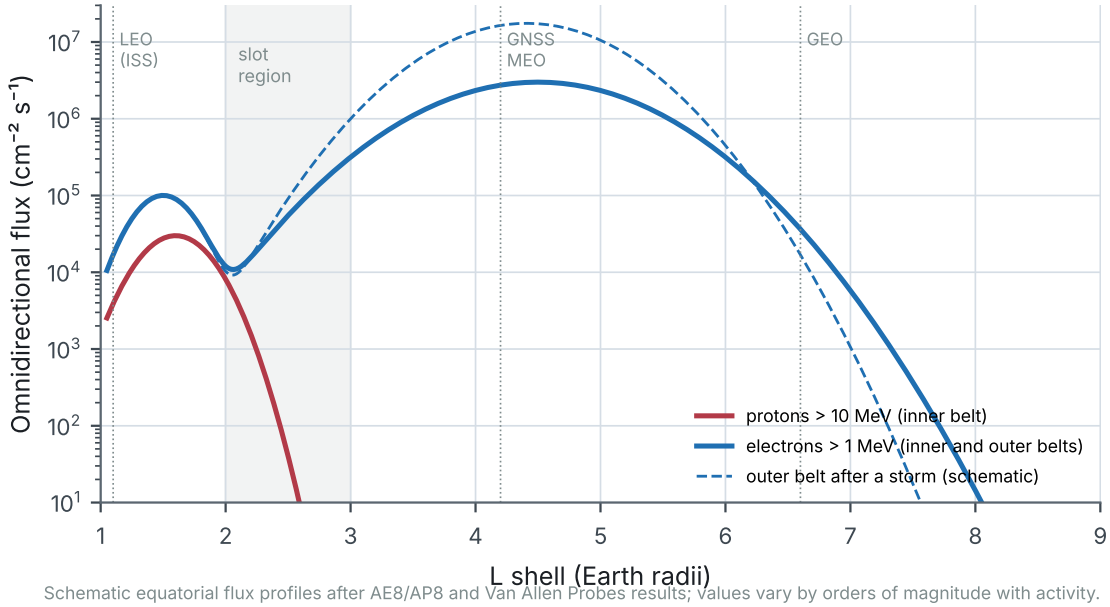



**FIGURE 23.6** Schematic radial profiles of trapped protons and electrons in the radiation belts. The outer electron belt varies greatly with geomagnetic activity.

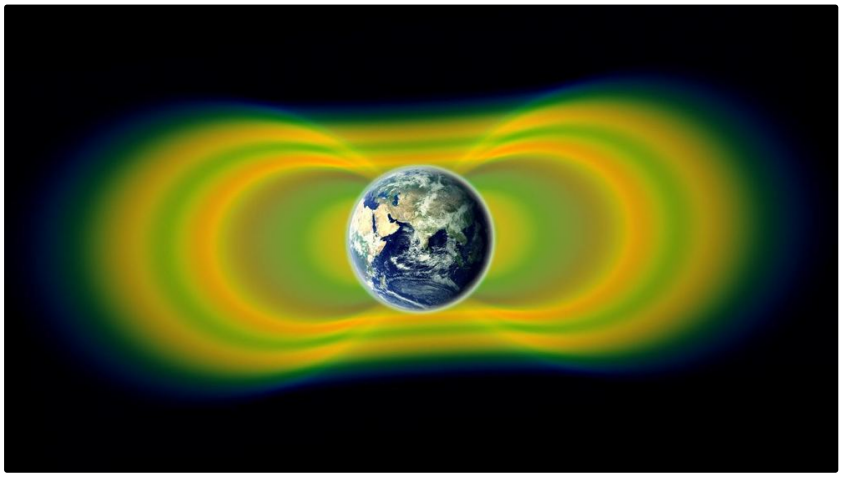


**FIGURE 23.7** The Van Allen Probes discovered in September 2012 that a third, transient radiation belt (in the middle) can form between the inner and outer belts. Visualisation by NASA. Image: NASA/GSFC.

The belts can also be created artificially. The US **Starfish Prime** nuclear test in July 1962, at 400 km altitude, injected high-energy electrons that formed an artificial belt lasting years. It damaged or destroyed about a third of the satellites then in orbit — one reason why nuclear explosions in space are now banned (Chapter 19).

## 23.5 Coupling the Solar Wind to the Magnetosphere

### 23.5.1 Magnetic Reconnection

How does solar-wind energy enter a magnetosphere that the solar wind cannot penetrate? The answer, proposed by James Dungey in 1961, is **magnetic reconnection**. When the interplanetary magnetic field points southward — opposite to the Earth's field at the dayside magnetopause — field lines from the two regions can break and reconnect. Terrestrial field lines then become connected to the solar wind (Figure 23.8). These "open" field lines are swept back over the poles by the solar wind and pile up in the tail lobes. There they reconnect again in the plasma sheet, and the closed field lines return to the dayside. This **Dungey cycle** drives a large-scale circulation of plasma — two convection cells in the polar ionosphere — and transfers energy, mass and momentum from the solar wind into the magnetosphere. When the interplanetary field is northward, reconnection occurs at high latitudes instead and the magnetosphere is relatively quiet.

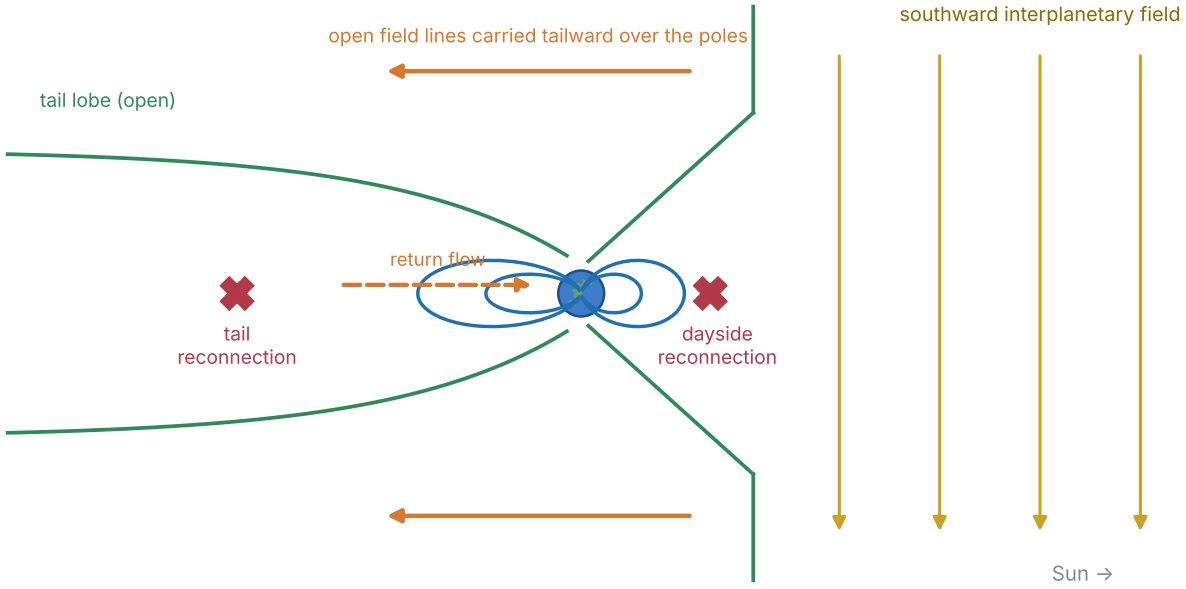



**FIGURE 23.8** The Dungey cycle. Reconnection with a southward interplanetary magnetic field opens dayside field lines, which are carried over the poles into the tail, where they reconnect and return.

Reconnection converts magnetic energy into particle energy in tiny regions, a few kilometres across for electrons, where the plasma decouples from the magnetic field. NASA's four **Magnetospheric Multiscale (MMS)** spacecraft were launched in 2015 and fly in a tetrahedron only a few kilometres across. They were the first to fly through these electron diffusion regions and measure the physics of reconnection directly (Figure 23.9). The same physics also operates in solar flares, fusion reactors and astrophysical plasmas.

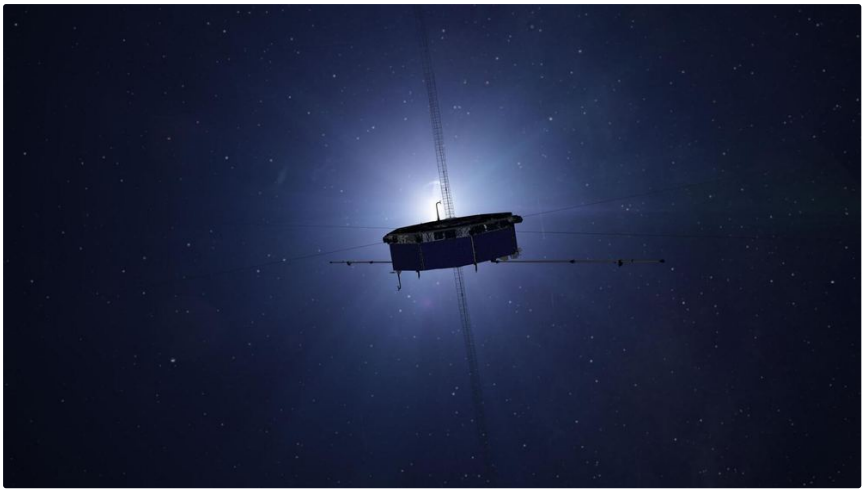


**FIGURE 23.9** Artist's impression of NASA's four Magnetospheric Multiscale spacecraft flying in formation through a region of magnetic reconnection. Image: NASA/GSFC.

### 23.5.2 Substorms and Storms

Energy stored in the stretched tail is released intermittently in **substorms**. Over about an hour, the tail field becomes increasingly stretched (growth phase). Then reconnection in the near-Earth tail suddenly releases energy (expansion phase), and the tail field snaps back towards a dipole. Particles are injected into the inner magnetosphere, and the aurora brightens and surges poleward and westward. Substorms occur several times a day even in moderately disturbed conditions; NASA's five **THEMIS** satellites (2007) were designed to determine where and how they start.

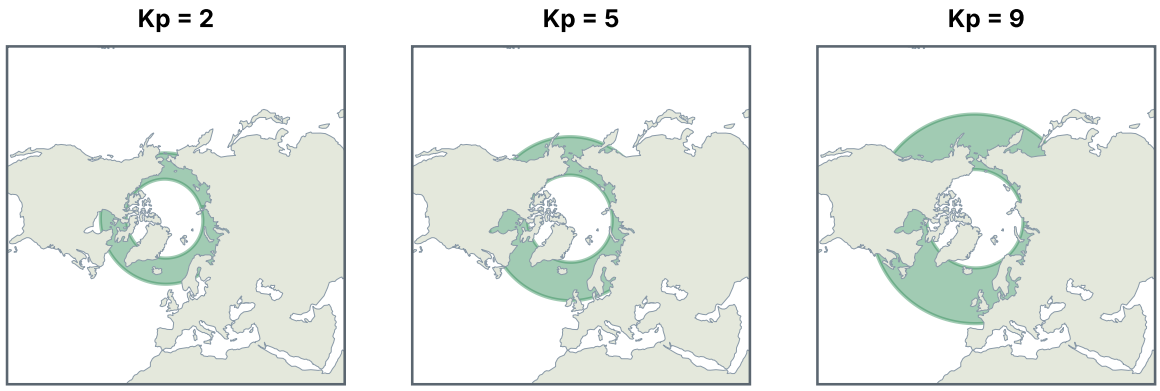
A **geomagnetic storm** is a longer, larger disturbance lasting one to several days, driven by a sustained period of strong southward interplanetary field — usually from a CME or a corotating interaction region (Chapter 22). During a storm, ions injected from the tail build up the **ring current**, whose magnetic field depresses the horizontal field at low latitudes (measured by the Dst index). The radiation belts are also reconfigured, and large currents flow along field lines into the polar ionosphere (**field-aligned** or Birkeland currents). These currents close through the auroral electrojets.

## 23.6 The Aurora

The aurora is produced when electrons with energies of about 1–10 keV, accelerated in the magnetosphere above the ionosphere, plunge along field lines into the upper atmosphere. There they excite oxygen atoms and nitrogen molecules, which emit light as they return to lower energy states:

- **Green** light at 557.7 nm from atomic oxygen, at about 100–250 km — the most common auroral colour;
- **Red** light at 630.0 nm from atomic oxygen, at higher altitudes (above about 200 km), where collisions are rare enough for this slow transition to occur;
- **Blue and violet** light from ionised molecular nitrogen (391.4 and 427.8 nm) at lower altitudes, often seen at the bottom edge of active displays.

Aurorae occur in two **auroral ovals** centred on the geomagnetic poles, typically between about 65° and 75° magnetic latitude. During storms the ovals expand towards the equator (Figure 23.10): in the May 2024 storm, aurorae were seen at magnetic latitudes below 30°. Astronauts on the ISS, whose orbit reaches 51.6° latitude, regularly fly through the auroral zones (Figure 23.11). Space-based ultraviolet imagers have photographed the entire oval from above since the 1980s.




Schematic auroral oval (geographic coordinates, approximate). The oval expands towards the equator as activity increases.

**FIGURE 23.10** Schematic auroral ovals for quiet (Kp 2), moderate (Kp 5) and extreme (Kp 9) geomagnetic activity.

**FIGURE 23.11** The aurora australis above the Indian Ocean, photographed from the International Space Station on 12 June 2025 between Australia and Antarctica. Image: NASA/JSC.

## 23.7 The Upper Atmosphere

Above about 90 km, the neutral atmosphere changes character (Figure 23.12). In the **thermosphere**, absorption of solar extreme-ultraviolet radiation heats the gas to 600–2 000 K, depending on solar activity. Molecules are dissociated, so **atomic oxygen** becomes the dominant species between about 200 and 600 km. This atomic oxygen causes material erosion on spacecraft (Chapter 2). Above about 600–1 000 km, in the **exosphere**, helium and then hydrogen dominate and atoms move on ballistic paths, sometimes escaping to space. Because the thermosphere

expands when heated, its density at a given altitude varies by a factor of ten or more over the solar cycle. Its density also increases several-fold during storms. This is the main reason for uncertainty in predicting satellite orbits (Chapters 21 and 22). NASA's TIMED (2001), GOLD (2018) and ICON (2019–2022) missions have studied how the thermosphere responds both to the Sun from above and to atmospheric waves from below.

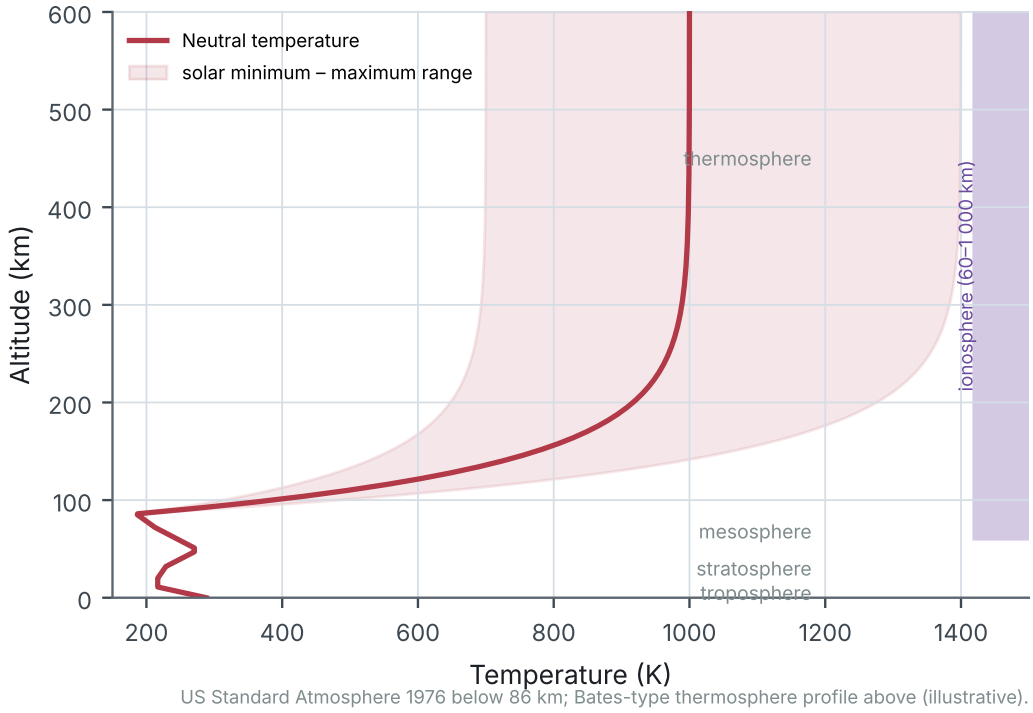



**FIGURE 23.12** Temperature of the neutral atmosphere with altitude: US Standard Atmosphere below 86 km and a thermospheric profile above, with the range between solar minimum and maximum.

## 23.8 The Ionosphere

### 23.8.1 Formation and Layers

Solar extreme-ultraviolet and X-ray radiation ionises the upper atmosphere, creating the **ionosphere** between about 60 and 1 000 km. Its electron density at each altitude results from a balance between production, which is greatest where the radiation is absorbed (a Chapman layer), and loss by recombination and transport. The ionosphere has distinct layers (Figure 23.13):

- the **D region** (60–90 km), present only in daytime, weakly ionised but dense enough in neutral gas to absorb HF radio waves;
- the **E region** (90–150 km), with a peak near 110 km and occasional thin, intense layers of metallic ions ("sporadic E");
- the **F1 layer** (150–220 km), present in daytime;
- the **F2 layer** (220–500 km), the densest, with peak densities of about $10^{11}$–$10^{12}$ electrons per $m^3$, which persists at night because recombination is slow at high altitudes.

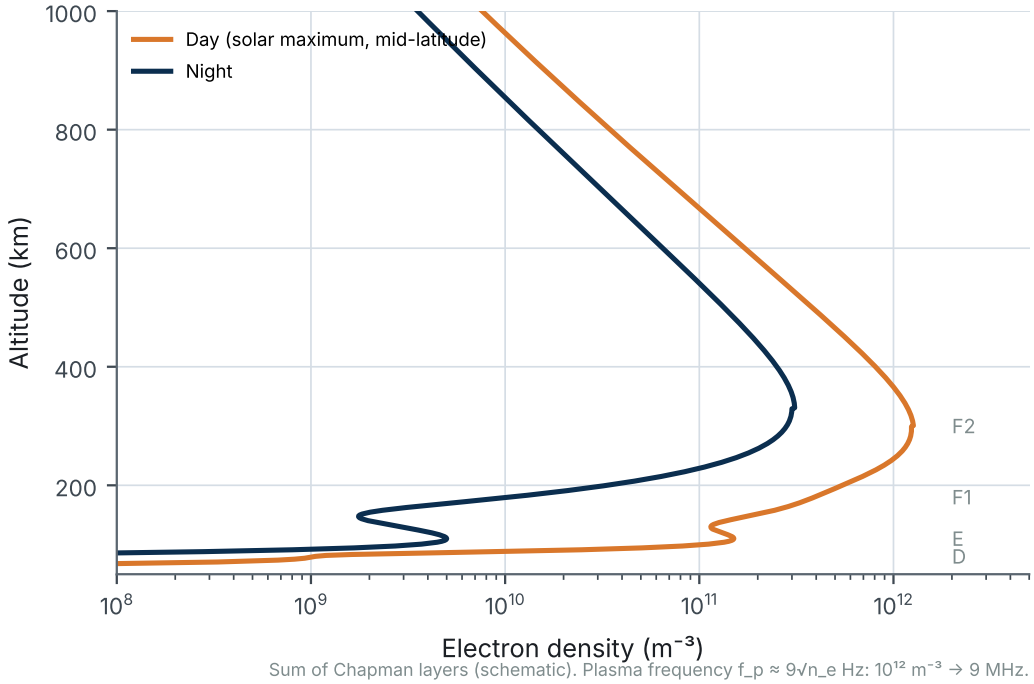



**FIGURE 23.13** Electron density profiles of the ionosphere by day and by night (schematic Chapman-layer model).

## 23.8.2 Variations and Irregularities

The ionosphere varies with time of day, season, solar cycle (peak densities are several times higher at solar maximum) and geomagnetic activity. Near the magnetic equator, the eastward electric field of the daytime ionosphere lifts plasma upward ("fountain effect"). The plasma then slides down the field lines to form two crests of high density about 15° north and south of the magnetic equator. This is the **equatorial ionization anomaly**, where total electron content reaches its global maximum (Figure 23.14). After sunset, the bottom of the equatorial F region can become unstable, forming **plasma bubbles** — depletions that rise through the ionosphere and fill with irregularities of all scales, causing strong **scintillation** of satellite signals. At high latitudes, polar-cap patches, auroral precipitation and travelling ionospheric disturbances create other irregularities.

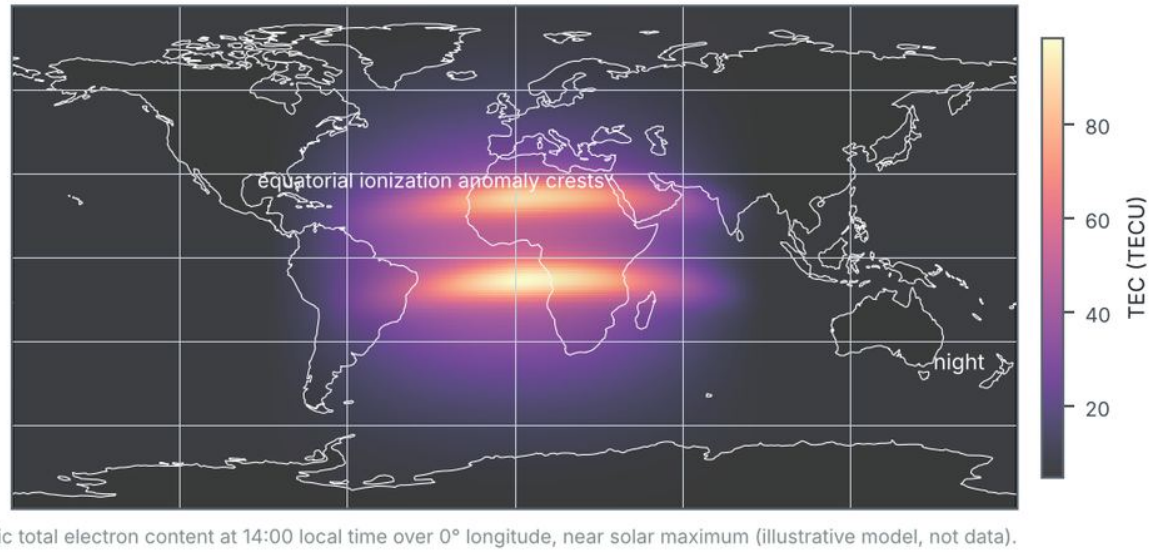



**FIGURE 23.14** Schematic global total electron content in the afternoon near solar maximum, showing the crests of the equatorial ionization anomaly.

### 23.8.3 Radio Propagation

The ionosphere's effect on radio waves depends on the **plasma frequency**

$$f_p \approx 8.98\sqrt{n_e}\ \text{Hz}$$

where $n_e$ is the electron density in $\text{m}^{-3}$. Radio waves below the local plasma frequency cannot propagate and are reflected; waves above it pass through but are delayed and refracted.

- **HF skywave propagation**: waves at 3–30 MHz are reflected by the ionosphere, allowing over-the-horizon communication and broadcasting worldwide. The highest frequency reflected at vertical incidence is the **critical frequency** foF2 (5–15 MHz); at oblique incidence the **maximum usable frequency** is higher by roughly the secant of the angle of incidence.
- **Ionosondes** sweep a radio frequency upward and measure the delay of the echo, producing an **ionogram** of virtual height versus frequency (Figure 23.15), from which the electron-density profile is reconstructed. Networks of ionosondes, and over-the-horizon radars, rely on the same physics.
- **Satellite signals** at frequencies far above the plasma frequency pass through, but they are delayed. The delay is proportional to the total electron content divided by the frequency squared (40.3 TEC/f$^2$ metres; Chapter 18). Their polarisation is also rotated by the magnetic field (**Faraday rotation**), which affects low-frequency radars and radio astronomy.
- **Scintillation**, measured by the S4 index (amplitude) and σϕ (phase), disrupts GNSS and satellite communications at L-band and below, mainly near the equator after sunset and at high latitudes.

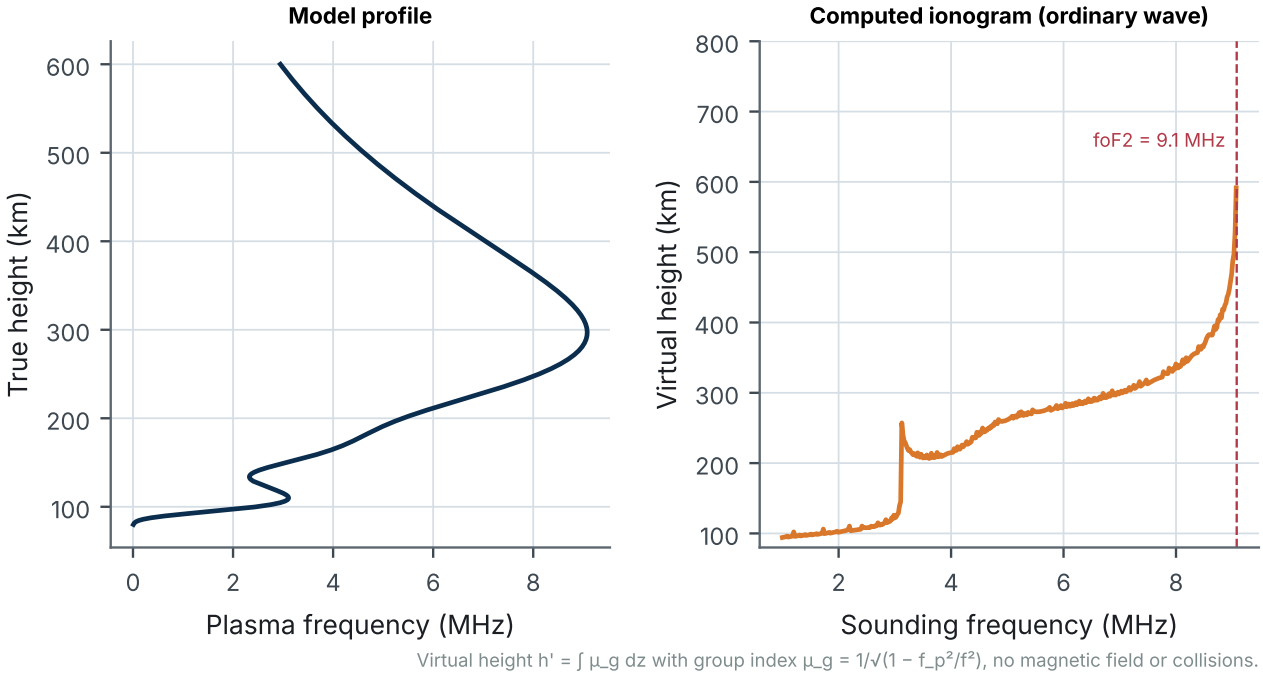



**FIGURE 23.15** A model electron-density profile (left) and the ionogram computed from it (right). Virtual heights rise steeply near the critical frequencies of the E and F2 layers, where the signal is slowed most.

# 23.9 Measuring and Modelling Geospace

## 23.9.1 Observations

Geospace is studied with a combination of ground and space instruments:

- **Ground**: magnetometer networks (which provide the Kp, Dst and AE indices), ionosondes, and coherent-scatter radars (SuperDARN, which maps ionospheric convection over both polar regions). There are also **incoherent-scatter radars** (such as those at Millstone Hill, Jicamarca and the European EISCAT system, being replaced by the volumetric EISCAT_3D). Other instruments are GNSS receiver networks that map total electron content globally, all-sky cameras and riometers.
- **Space**: missions from Explorer 1 onward (Figure 23.16). They include ISEE, Dynamics Explorer, Geotail, Polar and Wind, ESA's four-spacecraft **Cluster** (2000–2024), THEMIS, the Van Allen Probes, Swarm, MMS, Japan's Arase (2016), GOLD and ICON. Radio occultation also provides ionospheric profiles (Chapter 16). NASA's **TRACERS** twin satellites (launched July 2025) study reconnection in the cusp. The ESA–Chinese Academy of Sciences mission **SMILE** was launched on 19 May 2026. It carries a soft X-ray imager that, for the first time, images the magnetopause and cusps from outside, together with an ultraviolet auroral imager.

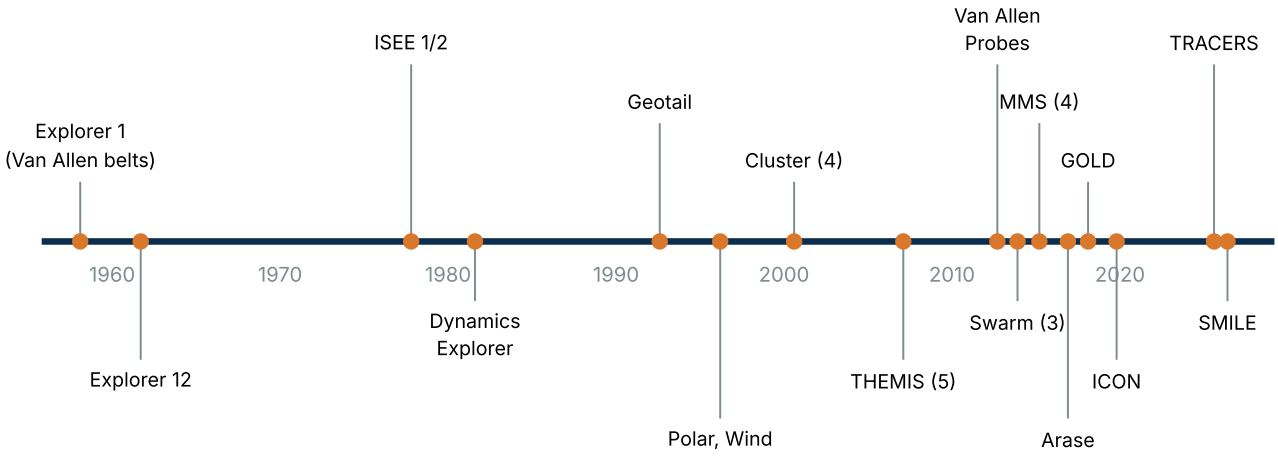



**FIGURE 23.16** Selected magnetospheric and ionospheric missions.

## 23.9.2 Models

Engineers and forecasters use a family of standard models:

**TABLE 23.1** Standard models of the geospace environment.

| Model | Describes | Typical use |
|---|---|---|
| IGRF, World Magnetic Model | Main geomagnetic field | Navigation, attitude determination, L-shell computation |
| Tsyganenko and other magnetospheric field models | External field from currents | Particle tracing, mapping field lines during storms |
| AE9/AP9/SPM (and legacy AE8/AP8) | Trapped radiation | Radiation-dose and single-event analysis in spacecraft design |

| Model | Describes | Typical use |
|---|---|---|
| IRI (International Reference Ionosphere) | Electron density, temperatures | Radio propagation, GNSS corrections, research |
| Klobuchar, NeQuick | Ionospheric delay | Broadcast corrections for GPS and Galileo single-frequency users |
| NRLMSIS 2.0, JB2008, DTM | Thermospheric density and composition | Orbit determination and lifetime prediction |
| Space Weather Modeling Framework, Geospace model | Coupled Sun-to-Earth physics | Operational forecasting |

### 23.9.3 Other Planets

Magnetospheres are common in the Solar System (Table 23.2). Jupiter's is the largest structure in the Solar System after the heliosphere — its magnetopause lies 50–100 Jovian radii from the planet. Its intense radiation belts, fed by sulphur and oxygen from Io's volcanoes, constrain the design of every Jupiter mission (Juno, JUICE, Europa Clipper). Mercury's small magnetosphere is so compressed that the solar wind sometimes reaches its surface. Mars and Venus have no global fields, so the solar wind interacts directly with their upper atmospheres and strips them away, as measured by MAVEN.

**TABLE 23.2** Magnetospheres of the planets (approximate).

| Planet | Global field (relative to Earth's dipole moment) | Typical magnetopause distance | Notes |
|---|---|---|---|
| Mercury | ≈ 0.0006 | ≈ 1.5 planetary radii | Studied by MESSENGER; BepiColombo arrives in 2026 |
| Venus | None (induced magnetosphere) | — | Ionosphere interacts directly with the solar wind |
| Earth | 1 | ≈ 10 $R_E$ | Radiation belts, aurora |
| Mars | None (crustal fields) | — | Atmospheric escape measured by MAVEN |
| Jupiter | ≈ 20 000 | 50–100 $R_J$ | Intense radiation belts; Io plasma torus |
| Saturn | ≈ 580 | ≈ 20 $R_S$ | Enceladus supplies water-group ions |
| Uranus, Neptune | ≈ 50, ≈ 25 | ≈ 20 planetary radii | Strongly tilted, offset fields |

### EXAMPLE: HF COMMUNICATION ACROSS THE ATLANTIC

An airline crew over the North Atlantic must contact a ground station 3 000 km away by HF radio. At night the F2-layer peak is at about 300 km with foF2 = 5 MHz.

1. For a single-hop path of 3 000 km, the ray reflects at the midpoint, 1 500 km from each end. Allowing for the Earth's curvature, the elevation angle at the ground is only a few degrees and the angle of incidence on the layer is about 75°, so the secant factor is about 3.5–4.
2. The maximum usable frequency is therefore about 5 × 3.7 ≈ 18 MHz. Operators choose a working frequency about 15 % below this, around 15 MHz.
3. After sunrise, foF2 rises to about 10 MHz and the MUF to more than 30 MHz, but the D region also forms and absorbs lower frequencies, so the crew must move to a higher band.
4. During a polar-cap absorption event after a solar proton event (Chapter 22), D-region absorption at high latitudes can block HF entirely for days; aircraft then use satellite communications or reroute to lower latitudes.

### AI IN MAGNETOSPHERIC AND IONOSPHERIC PHYSICS

Neural networks trained on decades of satellite and ground data now predict geomagnetic indices, map radiation-belt electron fluxes, forecast ionospheric total electron content and scintillation, and reconstruct three-dimensional ionospheric structure from sparse GNSS observations. Physics-informed machine learning combines these data-driven models with the conservation laws of plasma physics (Chapters 44 and 47).

### KEY TAKEAWAYS

- The Earth's field is approximately a tilted dipole; field lines are labelled by L-shell. The South Atlantic Anomaly brings the inner belt down to LEO altitudes.
- The magnetosphere is bounded by the magnetopause (about 10 R_E sunward) and bow shock and has a long tail; the standoff distance follows from pressure balance and scales as P^(−1/6).
- Trapped particles gyrate, bounce and drift; the radiation belts comprise a stable inner proton belt and a highly variable outer electron belt separated by a slot.
- Magnetic reconnection with a southward interplanetary field drives the Dungey cycle, substorms and storms; MMS measured reconnection directly.
- The aurora is produced by 1–10 keV electrons exciting oxygen (green, red) and nitrogen (blue, violet) in the auroral ovals, which expand during storms.
- The ionosphere's D, E, F1 and F2 layers reflect HF radio below the plasma frequency (f_p ≈ 9√n_e Hz) and delay and scintillate satellite signals.
- Geospace is monitored by ground networks and missions from Explorer 1 to SMILE (2026), and described by standard models used throughout space engineering.

### REVIEW QUESTIONS

1. The Earth's equatorial surface field is about 30 μT. Estimate the field at the equator of the L = 6.6 shell (geostationary distance) using the dipole law.
2. The magnetopause stand-off distance scales as the solar-wind dynamic pressure to the power −1/6. If the pressure rises eightfold from its typical value, how far does a 10 $R_E$ magnetopause move inwards?
3. Describe the three periodic motions of a charged particle trapped in the Earth's magnetic field.
4. Why is the outer radiation belt much more variable than the inner belt? What is the slot region?

5. What is the South Atlantic Anomaly, and why does it matter for satellites and astronauts in low orbit?
6. Explain how magnetic reconnection with a southward interplanetary magnetic field drives geomagnetic activity.
7. Why are aurorae mostly green and red? At what altitudes is each colour produced?
8. Calculate the plasma frequency for an electron density of $10^{12}\ \mathrm{m}^{-3}$. What happens to HF radio waves below and above this frequency?

PART VIII

# Deep Space Exploration

From the Moon to interstellar space, robotic explorers have visited every planet. Part VIII covers interplanetary mission design, orbiters, flyby probes, landers and entry systems, rovers and helicopters, and missions that bring samples of other worlds back to Earth.


24 Deep-Space Mission Design
25 Planetary Orbiters
26 Space Probes and Flyby Missions
27 Landers and Entry, Descent & Landing
28 Planetary Rovers and Aerial Explorers
29 Sample-Return Missions


Image: NASA/JPL (NASA/JPL-Caltech/SwRI/MSSS)

# 24 CHAPTER Deep-Space Mission Design

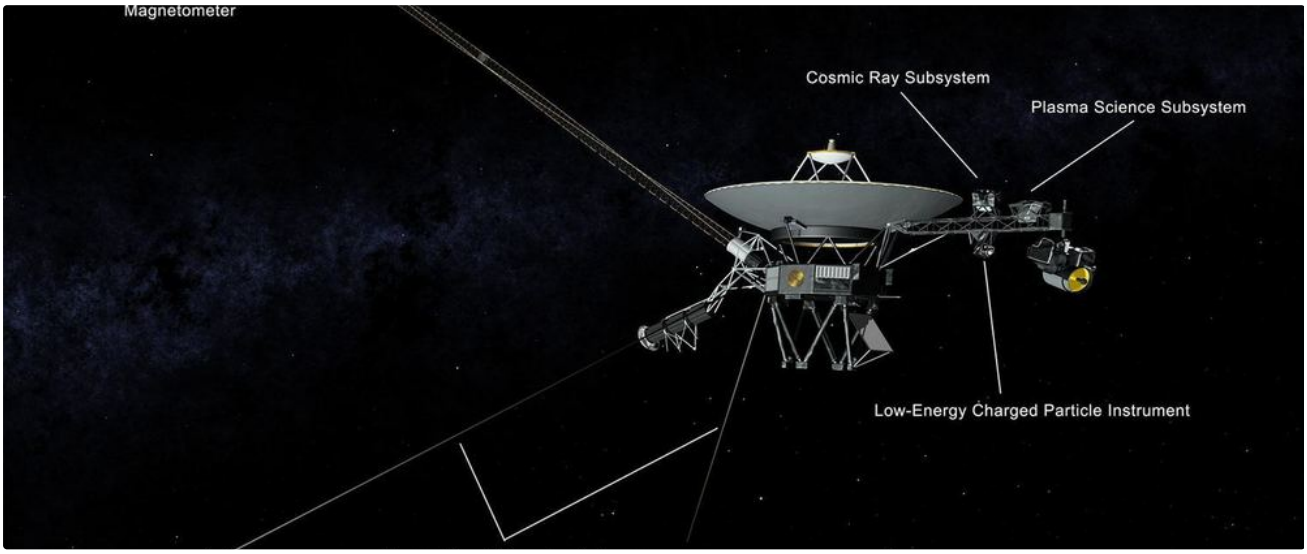



Image: NASA/JPL (NASA/JPL-Caltech)

*Sending a spacecraft to another world is an exercise in patience and precision. Launch windows open only every few years; journeys last months to decades; signals take minutes to hours to arrive; sunlight fades; radiation, dust and extreme temperatures test every component. This chapter explains how deep-space missions are designed: interplanetary trajectories and launch windows — with launch opportunities to Mars computed for 2026–2041 — gravity assists, low-thrust electric propulsion, orbit insertion and aerobraking, cislunar and libration-point orbits, deep-space navigation and communications, power and thermal design across the Solar System, planetary protection, and the life cycle and cost of a flagship mission. It ends with the fleet of deep-space missions active in 2026.*

**IN THIS CHAPTER**

24.1 The Challenge of Deep Space
24.2 Interplanetary Trajectories
24.3 Gravity Assists
24.4 Low-Thrust Trajectories
24.5 Arrival: Orbit Insertion, Aerobraking and Aerocapture
24.6 Cislunar Space and Libration Points
24.7 Navigating in Deep Space
24.8 Communications, Power and Thermal Design
24.9 Planetary Protection
24.10 From Idea to Arrival: The Mission Life Cycle
24.11 The Deep-Space Fleet in 2026

## 24.1 The Challenge of Deep Space

Deep-space missions — to the Moon, planets, asteroids, comets, the Sun and beyond — face constraints that Earth-orbiting satellites rarely meet:

- **Energy and time**: escaping the Earth and changing orbit around the Sun require velocity changes of several kilometres per second. Journeys last from days (to the Moon) to more than a decade (to the outer planets).
- **Launch windows** dictated by planetary alignment, which for Mars open only once every 26 months.
- **Distance**: radio signals take up to hours to arrive, so spacecraft must operate autonomously and data rates are limited.
- **Power and temperature**: sunlight falls with the square of the distance from the Sun, and spacecraft experience extremes from the heat near Mercury or the Sun to the cold of the outer Solar System.
- **Environment**: intense radiation at Jupiter, dust near comets, unknown surfaces at landing sites.
- **No second chance**: a missed orbit insertion or a failed landing usually ends the mission.

Missions are commonly classified as **flybys**, **orbiters**, **landers**, **rovers** and **sample-return** missions (Chapters 25–29), and grouped by agencies into cost classes. NASA's **Discovery** programme (cost cap around $500–600 million) and **New Frontiers** programme (around $1 billion) compete missions among scientists, while **flagship** missions such as Cassini, Perseverance and Europa Clipper cost several billion dollars. ESA's Science Programme distinguishes small, medium and large missions; China's **Tianwen** and **Chang'e** series, JAXA's sample-return missions and ISRO's Mars and lunar missions have made deep-space exploration a truly international enterprise. The two Voyager spacecraft, launched in 1977, are still returning data from interstellar space; on 18 November 2026 Voyager 1 reaches one light-day — about 26 billion km — from the Earth.

## 24.2 Interplanetary Trajectories

### 24.2.1 Escape and Hyperbolic Excess Speed

Trajectory design uses the **patched-conic** approximation (Chapter 3): near each body, the spacecraft follows a conic section governed by that body alone; between bodies, it follows an ellipse around the Sun. The energy with which a spacecraft leaves the Earth is described by its **hyperbolic excess speed** $v_\infty$ — its speed relative to the Earth after escape. It can also be described by the **characteristic energy** $C_3 = v_\infty^2$. This is the standard way launch-vehicle performance is quoted for interplanetary missions. The velocity change needed to depart from a circular parking orbit of radius $r$ is

$$\Delta v = \sqrt{v_{esc}^2 + v_\infty^2} - v_{circ}, \qquad v_{esc} = \sqrt{\frac{2\mu}{r}},\; v_{circ} = \sqrt{\frac{\mu}{r}}$$

which, by the **Oberth effect**, is much less than $v_\infty$ added to the escape speed separately: a burn deep in a gravity well, at high speed, gains the most energy.

### 24.2.2 Hohmann Transfers and the Cost of Getting There

The minimum-energy transfer between two circular coplanar orbits is the **Hohmann transfer** (Chapter 3), an ellipse tangent to both orbits (Figure 24.1). Figure 24.2 compares the launch energies and flight times. Venus and Mars are relatively cheap (C3 of about 6–9 km$^2$/s$^2$). Mercury is surprisingly expensive — a spacecraft must shed the Earth's orbital speed to fall towards the Sun. The outer planets require C3 values of 80–140 km$^2$/s$^2$ and flight times of years to decades. Only a few launch vehicles can deliver useful masses at such energies, which is why outer-planet missions rely on gravity assists.

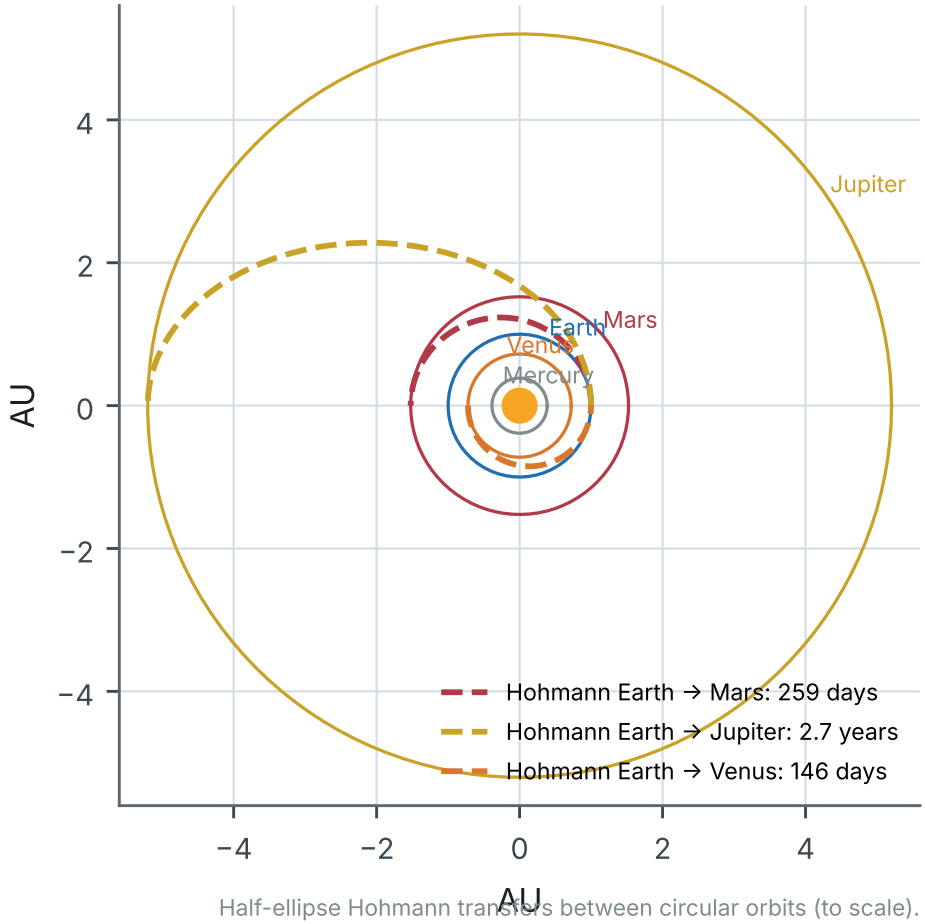
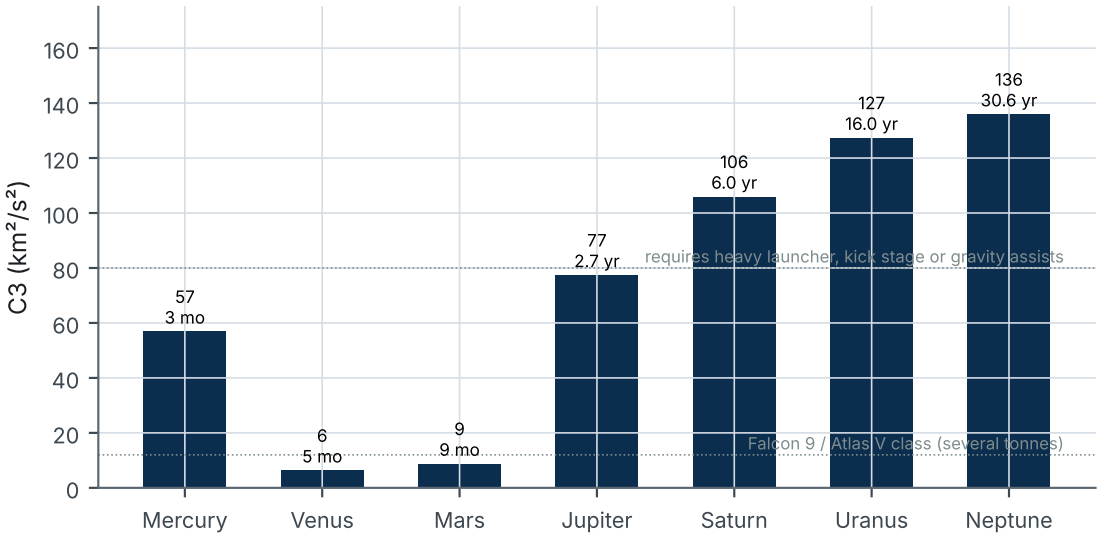



**FIGURE 24.1** Hohmann transfer ellipses from the Earth to Venus, Mars and Jupiter, drawn to scale.




Circular coplanar orbits; labels give C3 (km²/s²) and transfer time. Real missions use gravity assists to reach the outer planets with lower C3.

**FIGURE 24.2** Launch energy (C3) and flight time for Hohmann transfers from the Earth to each planet.

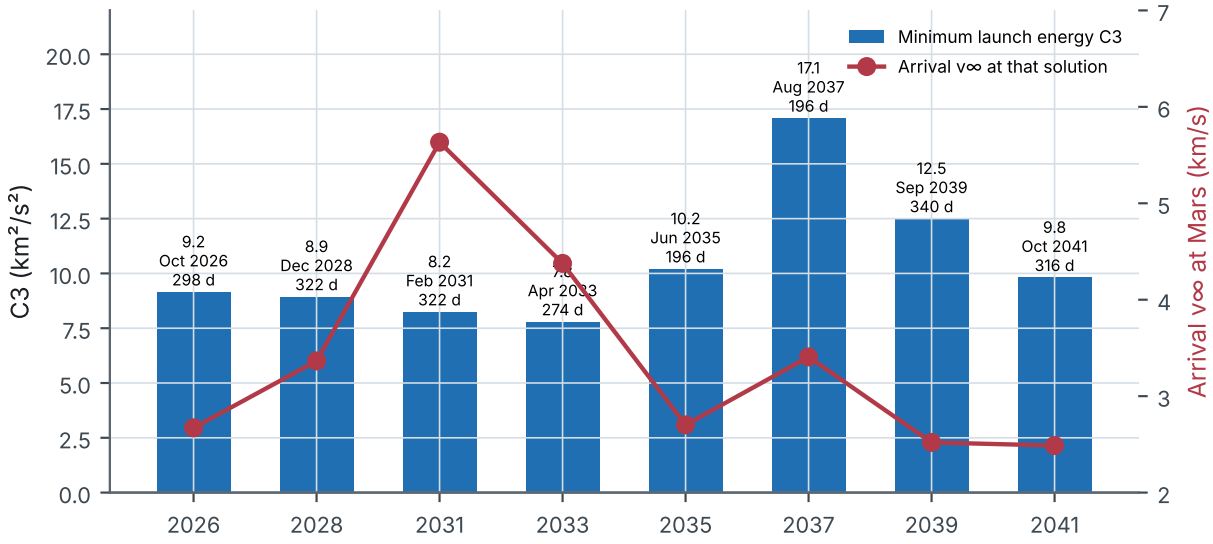
### 24.2.3 Launch Windows

A transfer works only if the target planet arrives at the end of the transfer at the right moment, so launches must take place when the planets are in the right relative positions. These **launch opportunities** recur at the **synodic period** of the two planets: about 584 days for Venus, 780 days (26 months) for Mars, and about 13 months for Jupiter and the more distant planets (Table 24.1). Each opportunity is a **launch period** of a few weeks, with a daily **launch window** of minutes to hours determined by the launch site and parking orbit.

**TABLE 24.1** Synodic periods and transfer characteristics.

| Planet | Synodic period with Earth | Hohmann transfer time | Typical C3 (km²/s²) |
|---|---|---|---|
| Mercury | 116 days | 3.5 months | ≈ 57 (direct; real missions use Venus and Mercury flybys) |
| Venus | 584 days (1.6 yr) | 4.8 months | ≈ 6–10 |
| Mars | 780 days (2.1 yr) | 8.5 months | ≈ 8–17 depending on opportunity |
| Jupiter | 399 days (1.1 yr) | 2.7 years | ≈ 77–85 |
| Saturn | 378 days (1.04 yr) | 6.0 years | ≈ 105–110 |

Because planetary orbits are elliptical and inclined, the real transfers are found by solving **Lambert's problem** (Chapter 3) for many combinations of launch and arrival dates. The results are plotted as a "porkchop plot" (Figure 3.16 in Chapter 3). Repeating this computation for each Mars opportunity (Figure 24.3) shows that their quality varies over a cycle of about 15–17 years. This is because Mars's eccentric orbit brings it closer to or farther from the Earth at opposition. The 2033 opportunity requires the least energy in the next decade, while 2037 is among the most demanding. Transfers shorter than half an orbit are called **Type I**, longer ones **Type II**.



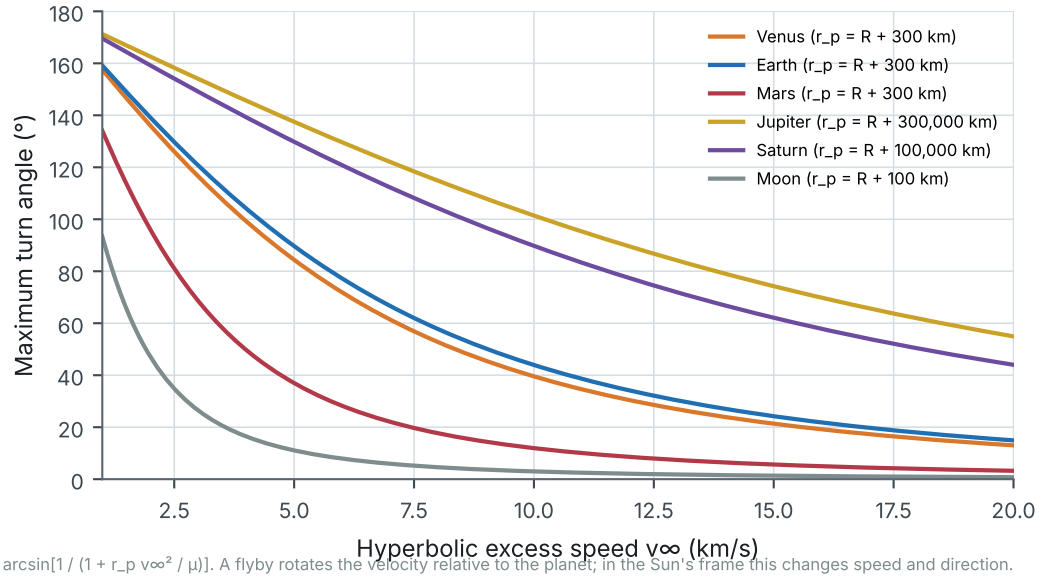

Computed for this manual: single-revolution Lambert solutions minimising C3 in each window (labels: C3, launch month, flight time).

**FIGURE 24.3** Minimum launch energy and the corresponding arrival speed for each Mars launch opportunity from 2026 to 2041, computed with a Lambert solver (flight times limited to 130–340 days).

## 24.3 Gravity Assists

### 24.3.1 Borrowing Momentum from Planets

A spacecraft flying past a planet enters and leaves the planet's sphere of influence with the same speed relative to the planet. However, its direction is turned through an angle that depends on how close it passes and how fast it is moving (Figure 24.4). Seen from the Sun, adding the planet's orbital velocity to this rotated relative velocity changes the spacecraft's heliocentric speed and direction. This is a **gravity assist**, which can add or remove several kilometres per second without using propellant (Chapter 3, Figure 3.17). Massive planets such as Jupiter can turn a spacecraft by large angles even at high speeds; small bodies like Mars or the Moon give only modest changes.




δ = 2 arcsin[1 / (1 + r_p v∞² / μ)]. A flyby rotates the velocity relative to the planet; in the Sun's frame this changes speed and direction.

**FIGURE 24.4** Maximum turn angle of a spacecraft's velocity during a flyby, as a function of the approach speed v∞, for different bodies.

### 24.3.2 Gravity Assists in Practice

Gravity assists have made the exploration of the Solar System possible:

- **Mariner 10** (1973) used Venus to reach Mercury — the first gravity assist.
- **Voyager 1 and 2** (1977) exploited a rare alignment of the outer planets, recurring only every 175 years, to tour Jupiter, Saturn, Uranus and Neptune.
- **Galileo** (1989) and **Cassini** (1997) used Venus and Earth flybys (VEEGA and VVEJGA trajectories) because their launch vehicles could not send them directly to Jupiter or Saturn.
- **MESSENGER** (2004) used six flybys to slow down enough to orbit Mercury. **BepiColombo** (2018) used nine — one of Earth, two of Venus and six of Mercury — before entering orbit around Mercury in November 2026.
- **Parker Solar Probe** used seven Venus flybys to lower its perihelion to 6.1 million km.
- ESA's **JUICE** (2023) performed the first-ever lunar–Earth double flyby in 2024 and a Venus flyby in 2025. Further Earth flybys follow in 2026 and 2029, before it arrives at Jupiter in 2031. NASA's **Europa Clipper** (2024) flew past Mars in March 2025 and returns to Earth in December 2026 on its way to Jupiter in 2030 (Figure 24.15).

Once in orbit around a giant planet, missions use repeated flybys of its moons to reshape their orbits. Cassini made more than 120 flybys of Titan, and Europa Clipper will make about 49 flybys of Europa. Designing such **tours** is a combinatorial problem solved with tools such as Tisserand graphs and global optimisation.

## 24.4 Low-Thrust Trajectories

**Electric propulsion** (Chapter 6) produces only millinewtons to a fraction of a newton of thrust. However, its exhaust velocities are 5–10 times higher than those of chemical rockets. It can therefore provide total velocity changes of 10 km/s or more with modest propellant (Figure 24.5). Instead of short impulsive burns, the engine operates for months or years, spiralling the spacecraft gradually from one orbit to another.

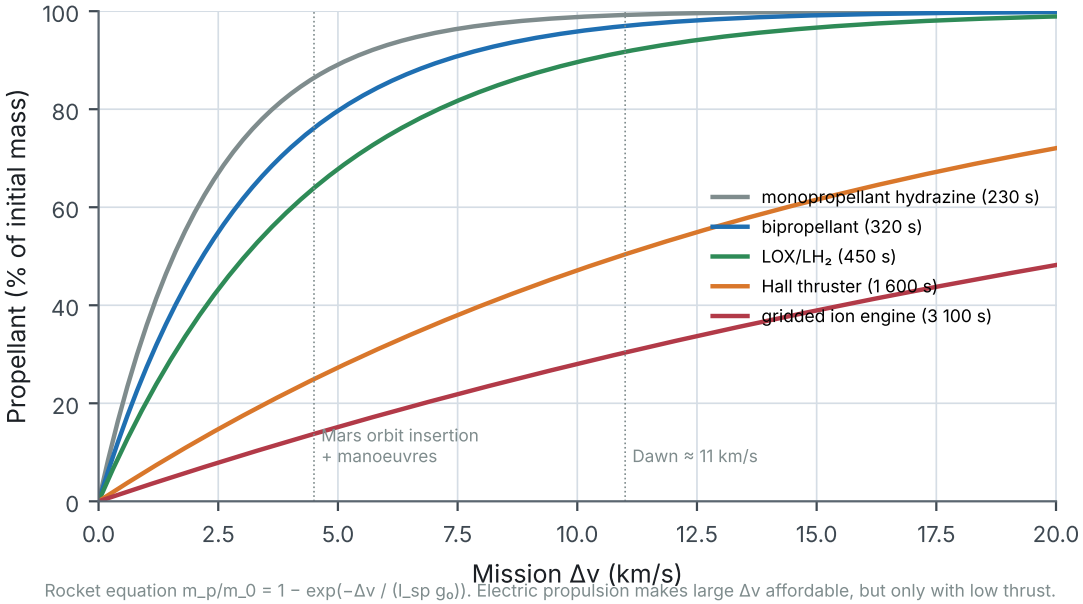



**FIGURE 24.5** Propellant mass fraction versus mission Δv for chemical and electric propulsion.

NASA's **Deep Space 1** (1998) demonstrated ion propulsion, and JAXA's **Hayabusa** (2003) used it for the first asteroid sample return. NASA's **Dawn** (2007) used three ion engines to orbit two different bodies — Vesta and Ceres. It achieved a record total Δv of about 11 km/s (Figure 24.6). ESA–JAXA's BepiColombo, NASA's Psyche (launched in 2023 with Hall-effect thrusters, flying past Mars in May 2026 on its way to the metal-rich asteroid Psyche in 2029) and many smaller missions use electric propulsion. Low-thrust trajectories are designed with numerical optimisation methods that balance thrust direction and timing against propellant use and flight time.

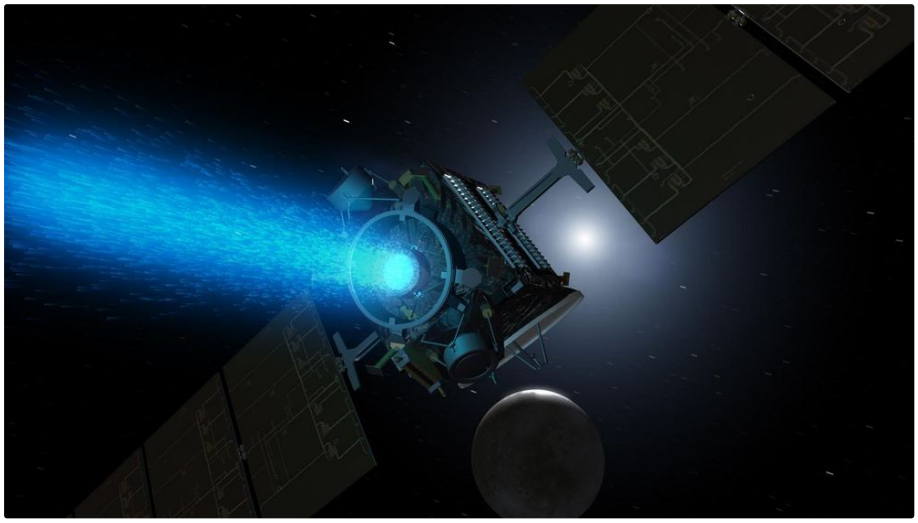


**FIGURE 24.6** Artist's impression of NASA's Dawn spacecraft approaching Ceres, its xenon ion engine glowing blue. Dawn's ion engines provided about 11 km/s of velocity change over its mission. Image: NASA/JPL (NASA/JPL-Caltech).

## 24.5 Arrival: Orbit Insertion, Aerobraking and Aerocapture

A spacecraft arrives at its target on a hyperbolic trajectory and must slow down to be captured. **Orbit insertion** burns of 0.5–1.5 km/s are typical at Mars and Venus, and can exceed 2 km/s at Mercury or for low orbits at the giant planets. They must be executed entirely autonomously at precisely the right moment, often during a communication blackout behind the planet. Failures have ended several missions. Mars Observer (1993) was lost just before insertion. Mars Climate Orbiter (1999) flew too deep into the atmosphere because one team used imperial units and another metric units in navigation software. This is the most famous units error in engineering history.

To save propellant, many Mars and Venus orbiters first enter a large elliptical orbit and then use **aerobraking**. Dipping repeatedly into the upper atmosphere at periapsis, they lose a little energy on each pass until the orbit is circularised. Mars Global Surveyor, Mars Odyssey, Mars Reconnaissance Orbiter, ESA's Trace Gas Orbiter and Venus Express all aerobraked; it takes months and careful monitoring of atmospheric density. **Aerocapture** is a single pass deep through the atmosphere inside a heat shield to go directly from a hyperbolic to an elliptical orbit. It could save even more mass, especially at Uranus and Neptune, but has not yet been flown.

## 24.6 Cislunar Space and Libration Points

Missions near the Moon and the Sun–Earth Lagrange points use the dynamics of the three-body problem (Chapter 3). The Sun–Earth **L1** and **L2** points, 1.5 million km from the Earth, host solar observatories (SOHO, DSCOVR, Aditya-L1) and astronomical telescopes (Gaia, JWST, Euclid, Roman). These fly in large **halo** or **Lissajous** orbits around the unstable equilibrium points, which require small station-keeping manoeuvres every few weeks.

Around the Moon, new families of orbits have become important (Figure 24.7). The **near-rectilinear halo orbit (NRHO)** is a highly elongated halo orbit around the Earth–Moon L2 point with a period of about 6.5 days. It was selected for the planned Gateway station and tested by the small CAPSTONE spacecraft in 2022. It is stable, almost always in view of the Earth, and accessible from both the Earth and the lunar surface (though Gateway itself was paused in 2026). Artemis I's Orion flew in a **distant retrograde orbit (DRO)**, a very stable orbit around the Moon. **Low-energy (ballistic) transfers** loop out to about 1.5 million km and exploit the Sun's gravity. They take three to four months instead of three days but save hundreds of metres per second of Δv. They were used by Korea's Danuri, CAPSTONE, Japan's SLIM and ispace's Hakuto-R commercial landers.

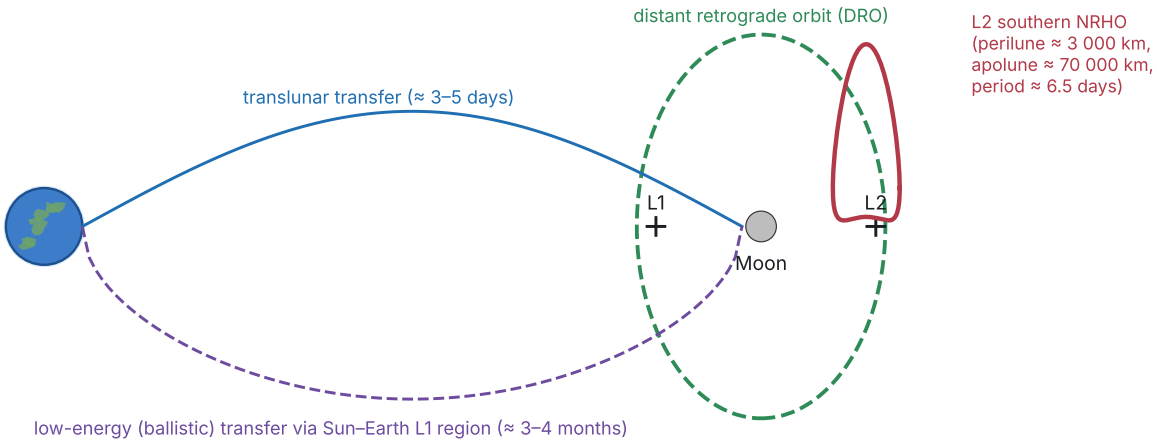



**FIGURE 24.7** Orbits and transfers in cislunar space (schematic, not to scale).

## 24.7 Navigating in Deep Space

### 24.7.1 Radiometric Tracking

Deep-space navigation is mostly done from the ground. Large antennas of NASA's **Deep Space Network (DSN)** — complexes at Goldstone (California), Madrid and Canberra, spaced about 120° apart in longitude so that every spacecraft is always in view of one — measure:

- **range**, by timing a coded signal sent to the spacecraft and returned by its transponder, to an accuracy of about 1 m;
- **range-rate**, from the Doppler shift of the carrier, to about 0.05 mm/s;

- **angular position**, by **delta differential one-way ranging (ΔDOR)** (Figure 24.8): two widely separated stations observe the spacecraft and then a quasar close to it in the sky. The difference in arrival times gives the spacecraft's direction with an accuracy of about one nanoradian — about 200 m at the distance of Mars.

Navigators combine these measurements with dynamical models in an **orbit determination** filter. The models include the gravity of all relevant bodies, solar radiation pressure, thruster firings and even the thermal recoil of radiation from the spacecraft. Navigators also design **trajectory correction manoeuvres** to keep the spacecraft on course. Errors at a target are expressed in the **B-plane** (Figure 24.9), the plane through the target body perpendicular to the incoming asymptote. Delivering a Mars lander to its entry corridor requires B-plane accuracy of a few kilometres.

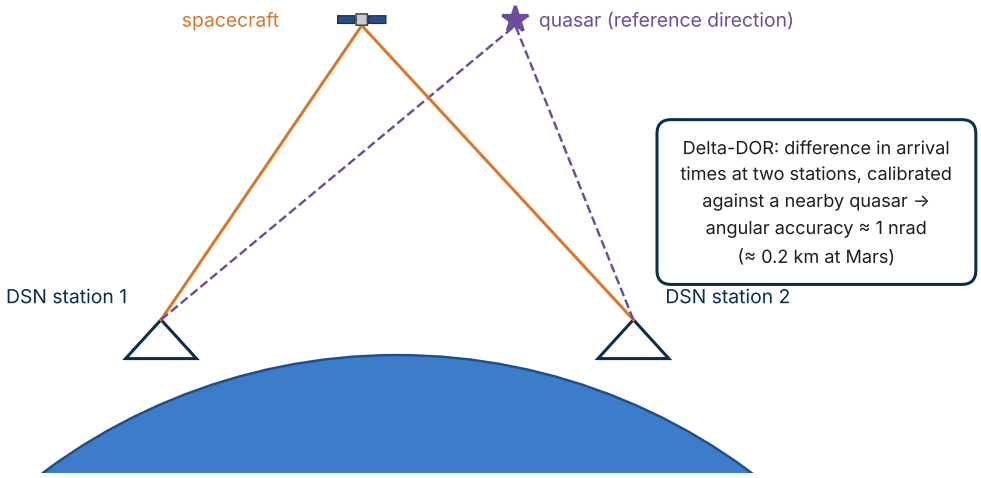



**FIGURE 24.8** Delta-DOR: comparing the arrival of a spacecraft's signal at two distant stations with that of a nearby quasar gives a very precise direction.

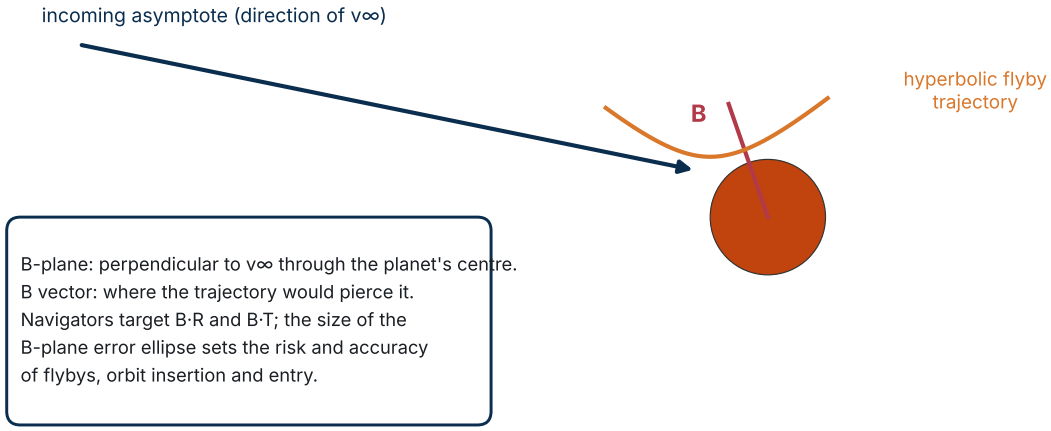



**FIGURE 24.9** The B-plane used to target planetary flybys, orbit insertion and atmospheric entry.

## 24.7.2 Optical and Autonomous Navigation

Near a target, cameras provide **optical navigation**: images of a moon or asteroid against background stars give its direction relative to the spacecraft with high precision. Autonomous navigation systems close the loop on board. NASA's **AutoNav** (Deep Space 1, Deep Impact,

Stardust) and the **SMART Nav** system, which guided DART into Dimorphos in 2022, identified their targets in images. They steered without help from the ground in the final hours. Landing systems use **terrain-relative navigation** (Chapter 27). Experimental techniques include navigation using X-ray pulsars as natural clocks (demonstrated by NASA's SEXTANT on the ISS) and onboard atomic clocks that enable one-way ranging (Chapter 18).

## 24.8 Communications, Power and Thermal Design

### 24.8.1 Talking Across the Solar System

Light takes 1.3 seconds to reach the Moon, between 3 and 22 minutes to reach Mars, and almost a day to reach Voyager 1 (Figure 24.10). Spacecraft therefore carry out pre-programmed sequences autonomously and protect themselves with fault-protection software that can put them in a safe mode and wait for instructions. Data rates fall with the square of distance (Chapter 7): a Mars orbiter can return several megabits per second, while Voyager 1 sends about 160 bit/s. The DSN's 70 m antennas (Figure 24.11) and arrays of 34 m antennas, ESA's ESTRACK 35 m deep-space antennas, and Chinese, Indian and Japanese deep-space networks support all missions. NASA's Deep Space Optical Communications experiment on Psyche demonstrated laser data rates of hundreds of megabits per second from tens of millions of kilometres.

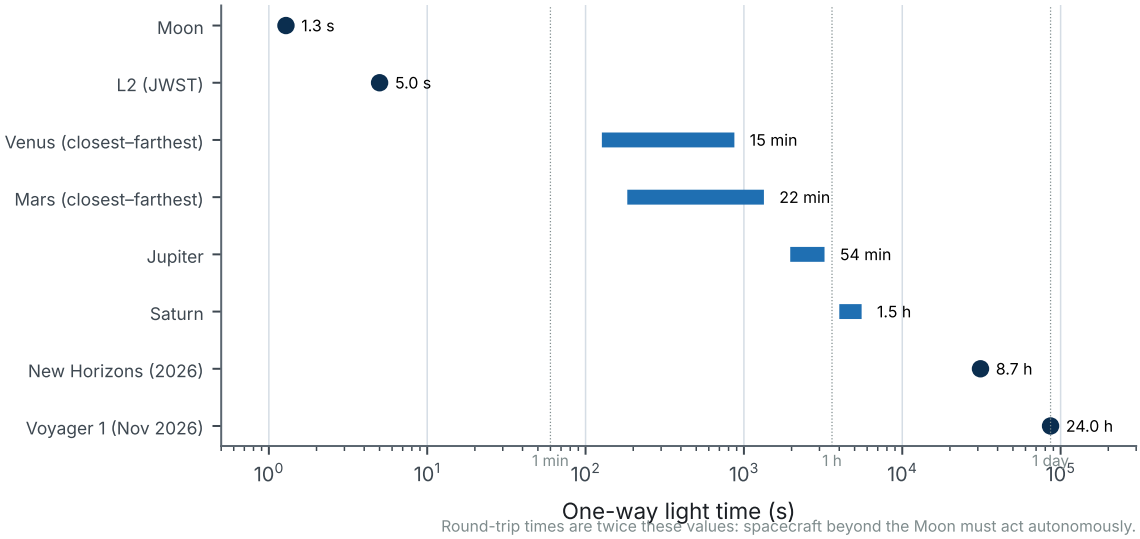



**FIGURE 24.10** One-way light time to destinations across the Solar System.

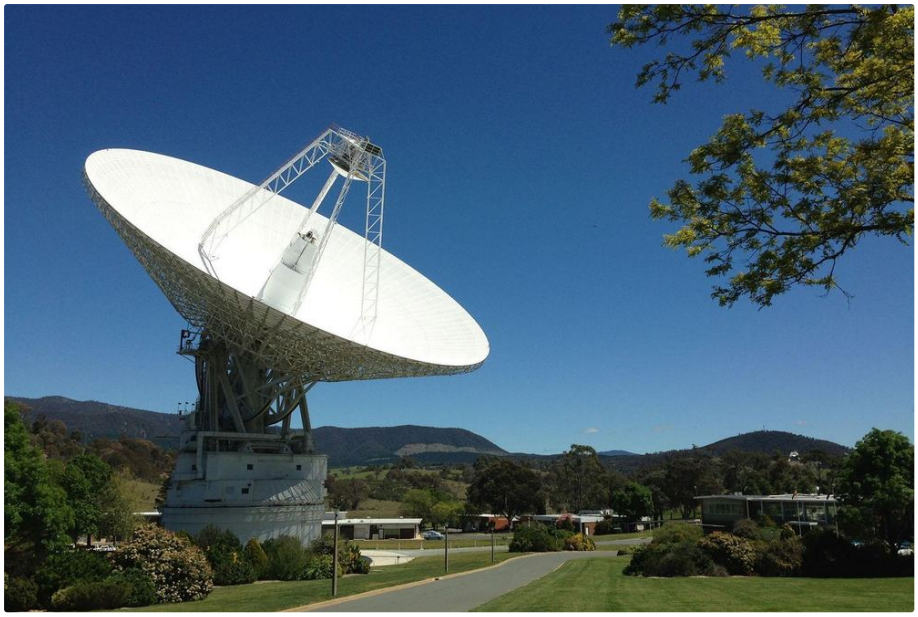


**FIGURE 24.11** The 70 m antenna DSS-43 at the Canberra Deep Space Communication Complex in Australia — the only antenna able to send commands to Voyager 2, which is below the Earth's equatorial plane. Image: NASA/JPL (NASA/Canberra Deep Space Communication Complex).

## 24.8.2 Power

At Jupiter, sunlight is 27 times weaker than at the Earth; at Saturn, 90 times (Figure 24.12). Missions as far as Jupiter can use very large solar arrays combined with efficient cells that work at low intensity and low temperature. Juno's three arrays span about 20 m and produce about 500 W at Jupiter, and Europa Clipper's span 30.5 m. ESA's JUICE carries 85 m² of cells, and NASA's Lucy, with two 7.3 m circular arrays, reaches 5.7 AU. Beyond about Jupiter, missions need **radioisotope power**: radioisotope thermoelectric generators (RTGs) convert the heat of plutonium-238 decay into about 100–300 W of electricity for decades. Voyager, Galileo, Cassini, New Horizons, Curiosity and Perseverance all use RTGs; the supply of plutonium-238 — whose US production restarted in the 2010s — limits how many such missions can fly.

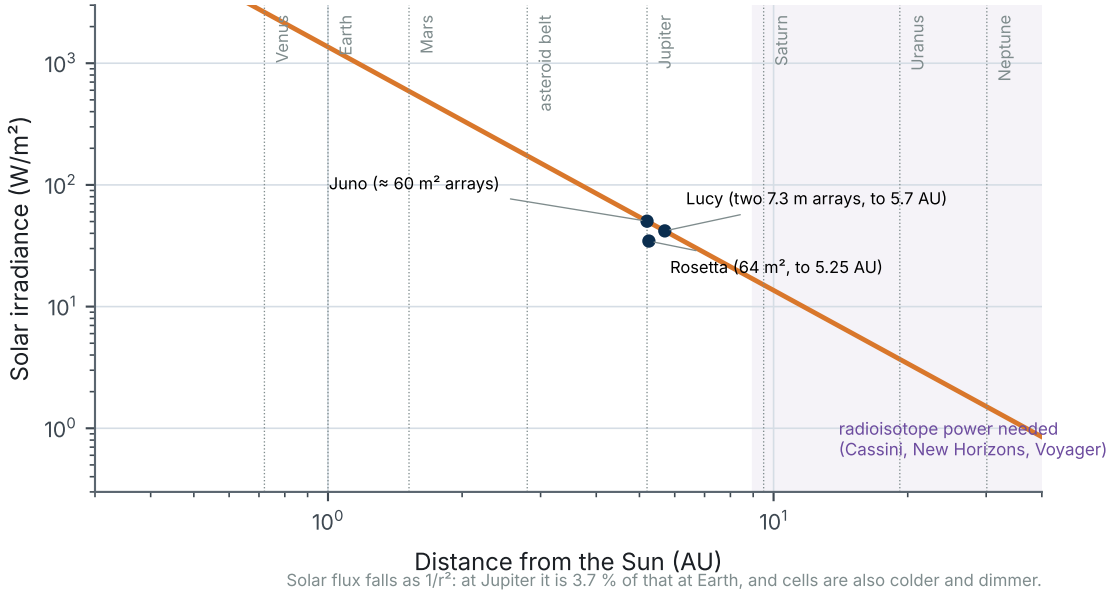



**FIGURE 24.12** Solar irradiance as a function of distance from the Sun. Large solar arrays have powered missions out to Jupiter; beyond, radioisotope power is required.

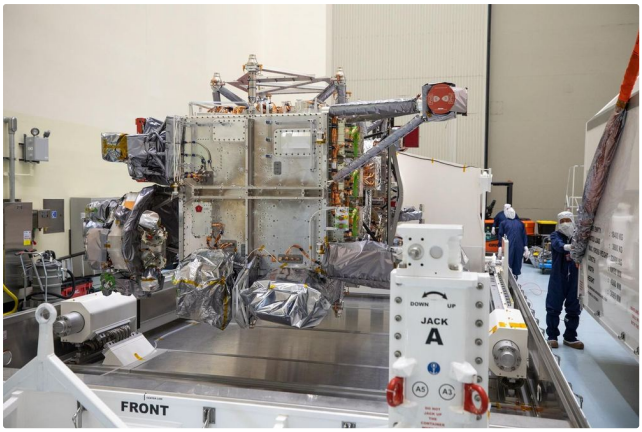



**FIGURE 24.13** Europa Clipper — NASA's largest planetary spacecraft, with solar arrays spanning 30.5 m — being unpacked at Kennedy Space Center in May 2024 before its launch in October 2024. Image: NASA/KSC (NASA/Kim Shiflett).

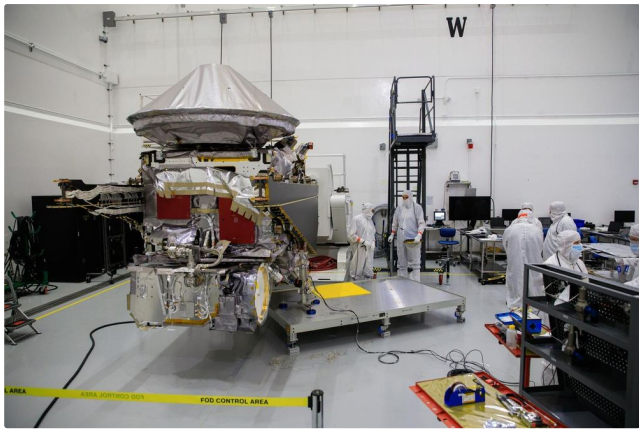



**FIGURE 24.14** One of Lucy's two 7.3 m circular solar arrays being deployed in a test in August 2021. Lucy is visiting Trojan asteroids at the distance of Jupiter's orbit using solar power. Image: NASA/KSC (NASA/Glenn Benson).

### 24.8.3 Thermal Extremes and Radiation

Thermal design must cope with radiation intensities from many times the Earth's value to almost nothing in the outer Solar System. At the hot extreme, Parker Solar Probe's heat shield reaches about 1 400 °C. At the cold extreme, spacecraft rely on radioisotope heater units and electrical heaters to keep propellant and batteries warm. At Jupiter, radiation is so intense that Juno and Europa Clipper protect their electronics inside **radiation vaults** with thick titanium or aluminium walls, and their orbits are designed to limit time in the worst regions.

## 24.9 Planetary Protection

Spacecraft must not contaminate other worlds with terrestrial organisms that could confuse the search for life or alter habitable environments, and samples returned to Earth must not pose a risk to the biosphere. The **COSPAR planetary protection policy**, implementing Article IX of the Outer Space Treaty, assigns missions to categories (Table 24.2).

**TABLE 24.2** COSPAR planetary-protection categories (simplified).

| Category | Mission type and target | Requirements |
|---|---|---|
| I | Any mission to bodies of no interest for chemical evolution or life (e.g. Mercury, undifferentiated asteroids) | None |
| II | Missions to bodies of significant interest but with remote chance of contamination (Venus, the Moon, comets, Jupiter, Saturn, Titan) | Documentation and impact-avoidance analysis |
| III | Flybys and orbiters of bodies with a significant chance of contamination (Mars, Europa, Enceladus) | Limits on impact probability or bioburden |
| IV | Landers and probes to these bodies | Strict cleaning and sterilisation; IVc for special regions (e.g. Martian ice or brines) |

| Category | Mission type and target | Requirements |
|---|---|---|
| V | Sample return to Earth: “unrestricted” (Moon, asteroid Ryugu) or “restricted” (Mars, Europa) | Restricted: break the chain of contact; containment and biohazard testing |

The Viking landers of 1976 were heat-sterilised as complete spacecraft. Today’s Mars landers are cleaned to limits of a few hundred thousand bacterial spores. At the end of their missions near Jupiter and Saturn, Galileo and Cassini were deliberately plunged into the planets to avoid any possibility of crashing on Europa or Enceladus.

## 24.10 From Idea to Arrival: The Mission Life Cycle

Deep-space missions follow the project life cycle of Chapter 4, but over unusually long timescales. Europa Clipper illustrates this (Figure 24.15). Studies began more than a decade before its launch in October 2024, and its cruise includes Mars (2025) and Earth (2026) gravity assists. Its science phase at Jupiter, beginning in 2030, will last about four years. Its life-cycle cost is about $5 billion.

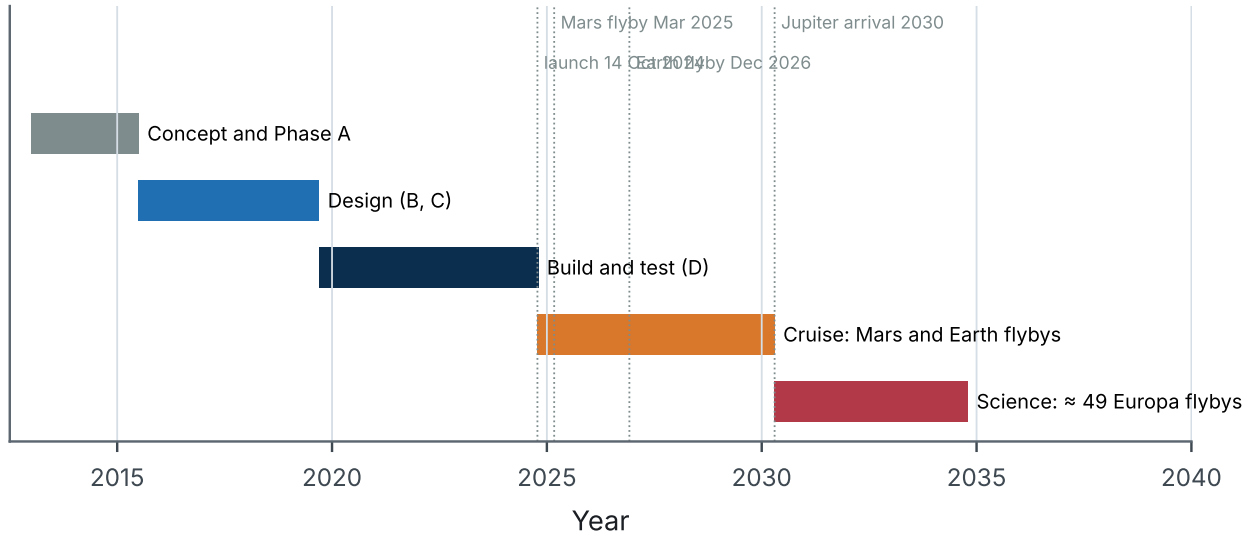



**FIGURE 24.15** The life cycle of NASA’s Europa Clipper mission, from concept studies to the end of its prime science phase.

## 24.11 The Deep-Space Fleet in 2026

**TABLE 24.3** Selected deep-space missions in operation or en route in October 2026.

| Destination | Active or en-route missions (selected) |
|---|---|
| Moon | LRO (USA), Chandrayaan-2 orbiter (India), Danuri (Korea), Chang’e-4 and Queqiao relays (China), commercial and Artemis missions (Chapters 25–27) |
| Sun and heliosphere | Parker Solar Probe, Solar Orbiter, SOHO, IMAP, Aditya-L1, PUNCH |
| Mercury | BepiColombo (ESA/JAXA), Mercury orbit insertion November 2026 |

| Destination | Active or en-route missions (selected) |
|---|---|
| Mars | Mars Odyssey, MRO, MAVEN, Mars Express, ExoMars TGO, Tianwen-1 orbiter, Hope; Curiosity and Perseverance rovers; ESCAPADE twin smallsats en route; MMX launching to Phobos in October 2026 |
| Jupiter | Juno (in orbit since 2016; extended operations uncertain after 2025); JUICE (arrival 2031) and Europa Clipper (arrival 2030) en route |
| Asteroids and comets | Lucy (Trojans), Psyche (arrival 2029), Hera (Didymos–Dimorphos, arrival November 2026), OSIRIS-APEX (Apophis, 2029), Hayabusa2 extended mission (2031), Tianwen-2 (Kamoʻoalewa, sample return ≈ 2027) |
| Outer Solar System and beyond | New Horizons (Kuiper Belt), Voyager 1 and 2 (interstellar space) |

### EXAMPLE: THE ΔV BUDGET OF A 2026 MARS ORBITER

From Figure 24.3, the best 2026 opportunity requires C3 ≈ 9.2 km²/s² (v∞ ≈ 3.0 km/s) with launch on about 29 October 2026 and arrival with v∞ ≈ 2.7 km/s.

1. *Departure* from a 300 km parking orbit ($r$ = 6 671 km): $v_{circ}$ = 7.73 km/s and $v_{esc}$ = 10.93 km/s. Then $v_p = \sqrt{10.93^2 + 9.2} = 11.34$ km/s, so Δv ≈ 3.61 km/s. This is normally provided by the launch vehicle's upper stage.
2. *Trajectory corrections* during cruise: typically 20–50 m/s.
3. *Mars orbit insertion* at 250 km altitude ($r$ = 3 640 km, μ = 42 828 km³/s²): the hyperbolic periapsis speed is $\sqrt{2\mu/r + v_\infty^2} = \sqrt{23.53 + 7.13} = 5.54$ km/s. To capture into an elliptical orbit with a period of one Martian day (a = 20 450 km), the periapsis speed must be $\sqrt{\mu(2/r - 1/a)} = 4.63$ km/s, so the insertion burn is about 0.91 km/s.
4. *Aerobraking* then lowers the apoapsis to a low circular science orbit over several months, saving about 1.2 km/s compared with doing the same with the engine.

With a bipropellant engine of 320 s specific impulse, the 0.95 km/s of spacecraft-provided Δv requires propellant equal to about 26 % of the spacecraft's mass at launch.

### AI IN DEEP-SPACE MISSION DESIGN

Machine learning and global optimisation are accelerating trajectory design (searching enormous spaces of gravity-assist sequences and low-thrust profiles), autonomous navigation and fault protection, scheduling of DSN antennas, and the analysis of returned data. Future spacecraft will need far greater on-board autonomy to explore distant worlds with hours of communication delay (Chapters 48 and 51).

### KEY TAKEAWAYS

- Deep-space missions are constrained by energy, launch windows, distance, power, environment and the need for autonomy.
- Departure energy is expressed as $C3 = v_\infty^2$; Hohmann transfers to Venus and Mars need C3 ≈ 6–10 $km^2/s^2$, to the outer planets 80–140 $km^2/s^2$.
- Launch opportunities recur at synodic periods (26 months for Mars); their quality varies with orbital geometry — 2033 is the best Mars window of the next decade.
- Gravity assists and low-thrust electric propulsion make otherwise impossible missions feasible; BepiColombo needed nine flybys to reach Mercury, Dawn achieved 11 km/s of Δv with ion engines.
- Orbit insertion, aerobraking and (future) aerocapture capture spacecraft at their targets; NRHOs, DROs and low-energy transfers are reshaping cislunar operations.
- Navigation relies on DSN range, Doppler and ΔDOR plus optical and autonomous navigation; light time ranges from seconds (Moon) to a day (Voyager 1).
- Solar power works to about Jupiter; beyond, RTGs are needed. Radiation vaults protect electronics at Jupiter.
- Planetary protection categories govern cleanliness and sample return; flagship missions take 15–20 years from concept to science.

### REVIEW QUESTIONS

1. List six constraints that make deep-space missions harder than Earth-orbit missions.
2. Calculate the Δv needed to depart from a 200 km circular parking orbit with a C3 of 10 $km^2/s^2$.
3. Why do Mars launch opportunities recur every 26 months, and why are some windows better than others?
4. Explain how a gravity assist can increase a spacecraft's speed relative to the Sun without violating energy conservation.
5. How did Dawn achieve a total Δv of about 11 km/s, and why would that have been impossible with chemical propulsion?
6. Compare orbit insertion by propulsive burn, aerobraking and aerocapture.
7. Describe range, Doppler and ΔDOR tracking, and the information each provides.
8. Why is solar power practical out to about Jupiter but not beyond? What do missions to the outer Solar System use instead?

# 25 CHAPTER Planetary Orbiters

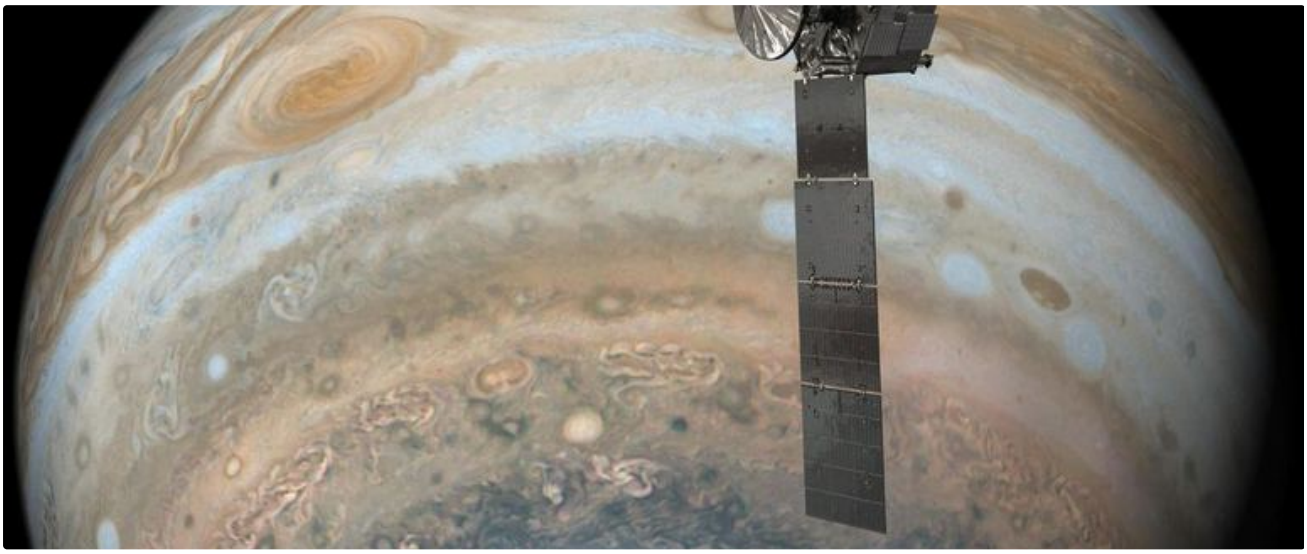


Image: NASA/JPL (NASA/JPL-Caltech)

*Orbiters are the workhorses of planetary exploration. Circling a world for years, they map its whole surface, probe beneath it with radar, weigh its interior through its gravity field, sample its atmosphere and magnetic environment, and relay the data of landers and rovers back to Earth. This chapter explains how orbits are designed around bodies as different as the Moon, Mars, Venus, Mercury, the giant planets and small asteroids, describes the instruments that orbiters carry, and reviews the orbiter missions — from Luna 10 and Mariner 9 to Juno, Cassini and BepiColombo — that have transformed our knowledge of the Solar System, including those operating in 2026.*

IN THIS CHAPTER

25.1 Why Orbit?
25.2 Designing Orbits Around Other Worlds
25.3 Getting Into Orbit
25.4 What Orbiters Carry
25.5 Orbiters World by World
25.6 Ending an Orbiter Mission

## 25.1 Why Orbit?

A flyby gives a few hours of close observation; an orbiter can observe a world for years, through changing seasons and lighting, at resolutions limited only by its instruments and its orbit. Orbiters provide:

- **global mapping** of surface composition, topography, geology and temperature;
- **long-term monitoring** of atmospheres, weather, seasonal ice caps and active processes such as dust storms, volcanism and geysers;
- **interior structure**, through precise measurements of the gravity and magnetic fields;

- **reconnaissance** of landing sites and safety;
- **communication relays** for surface missions.

**TABLE 25.1** First orbiters of Solar System bodies.

| Body | First orbiter | Year | Notes |
|---|---|---|---|
| Moon | Luna 10 (USSR) | 1966 | First spacecraft to orbit another body |
| Mars | Mariner 9 (USA) | 1971 | First to orbit another planet; mapped Mars after a global dust storm |
| Venus | Venera 9 (USSR) | 1975 | Orbiter of the first lander mission to return surface images |
| Jupiter | Galileo (USA) | 1995 | With an atmospheric entry probe |
| Asteroid | NEAR Shoemaker (USA) | 2000 | Orbited and then landed on Eros |
| Saturn | Cassini (USA/Europe/Italy) | 2004 | Carried ESA's Huygens Titan probe |
| Mercury | MESSENGER (USA) | 2011 | Found water ice in permanently shadowed polar craters |
| Dwarf planets | Dawn (USA) | 2011, 2015 | Orbited Vesta and then Ceres |
| Comet | Rosetta (Europe) | 2014 | Deployed the Philae lander on 67P/Churyumov–Gerasimenko |

## 25.2 Designing Orbits Around Other Worlds

### 25.2.1 Gravity Fields and Perturbations

The basic orbital mechanics are the same everywhere (Chapter 3), but each body's mass, size, rotation, shape and gravity field produce different orbits (Figure 25.1). A low orbit around the Moon takes about two hours; around Mars about 1 hour 50 minutes; around Earth about 90 minutes.

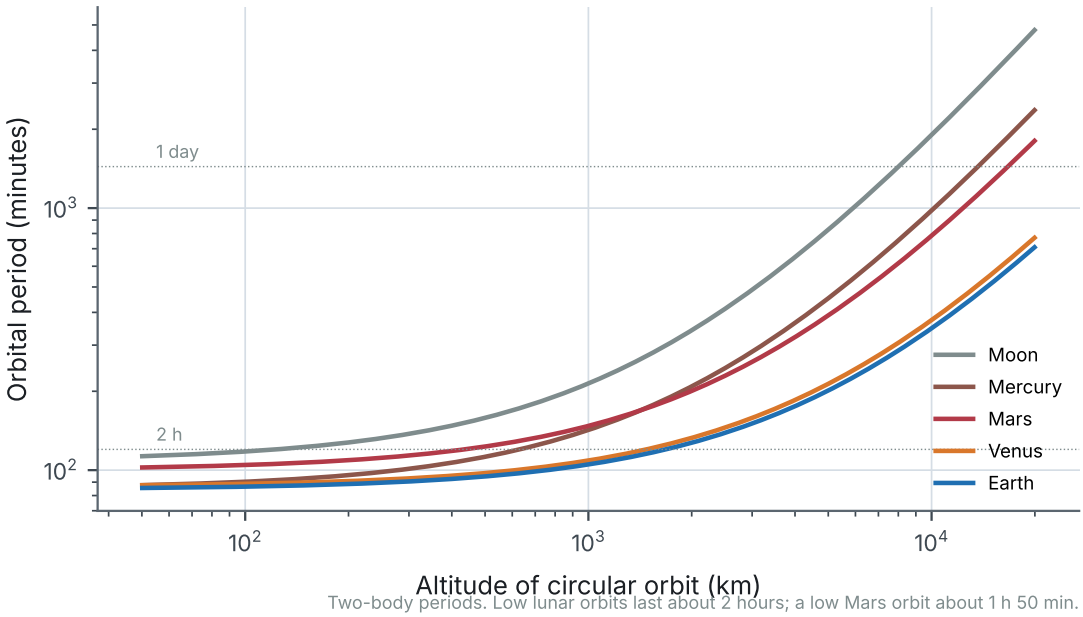



**FIGURE 25.1** Orbital period as a function of altitude around the Moon, Mercury, Mars, Venus and the Earth.

Irregularities in the gravity field strongly affect low orbits:

- The **Moon** has concentrations of mass under its large impact basins (**mascons**), discovered by tracking Lunar Orbiter spacecraft in 1968. They perturb low orbits so strongly that most low lunar orbits crash within months unless maintained. Only certain **frozen orbits**, at inclinations of about 27°, 50°, 76° and 86°, are naturally stable for long periods. NASA's **GRAIL** twin spacecraft (2012) measured the lunar gravity field with unprecedented precision by tracking the distance between them, revealing the structure of the lunar crust.
- **Mars** has a large equatorial bulge ($J_2$ about twice the Earth's), which allows **sun-synchronous orbits** that pass over each point at the same local time (Figure 25.2). Because Mars's year is longer, the required precession is slower, and its sun-synchronous orbits are nearly polar (about 93°). Most Mars orbiters since Mars Global Surveyor use them.
- **Venus** rotates once every 243 days (backwards) and has almost no bulge, so sun-synchronous orbits are impossible. A near-polar orbit lets the planet rotate slowly beneath it. NASA's **Magellan** (1990–1994) mapped 98 % of the surface with radar over three Venusian days (Figure 25.13).
- **Small bodies** — asteroids and comet nuclei — have such weak, irregular gravity that orbits are slow (hours to weeks) and strongly perturbed by solar radiation pressure and outgassing. OSIRIS-REx orbited the 500 m asteroid Bennu at about 1 km — the smallest body ever orbited.

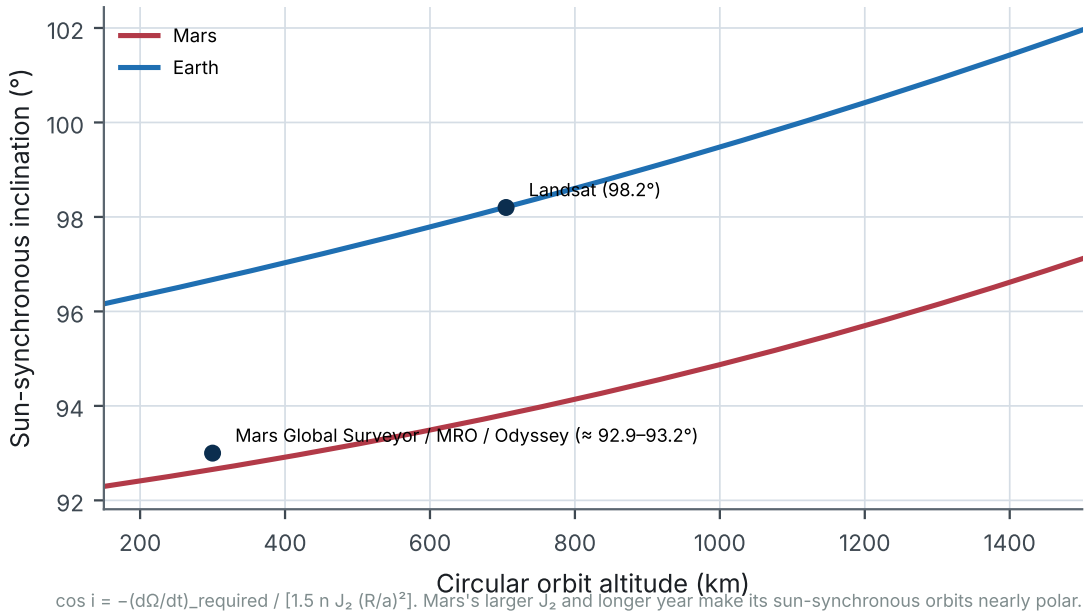



**FIGURE 25.2** Sun-synchronous inclination as a function of altitude for Mars and the Earth.

## 25.2.2 Thermal and Radiation Constraints

Orbits are also shaped by the environment. At **Mercury**, sunlight is up to 10 times more intense than at the Earth, and the dayside surface reaches 430 °C, radiating heat back at the spacecraft. MESSENGER flew a highly elliptical orbit to limit time over the hot surface, shaded behind a ceramic-cloth sunshade. BepiColombo's orbiters use high-temperature multilayer insulation and radiators that view away from the planet. At **Jupiter**, the radiation belts dominate orbit design (Chapter 23): Juno's elongated polar orbit (Figure 25.3) dives beneath the belts at each perijove, minimising the dose per orbit.

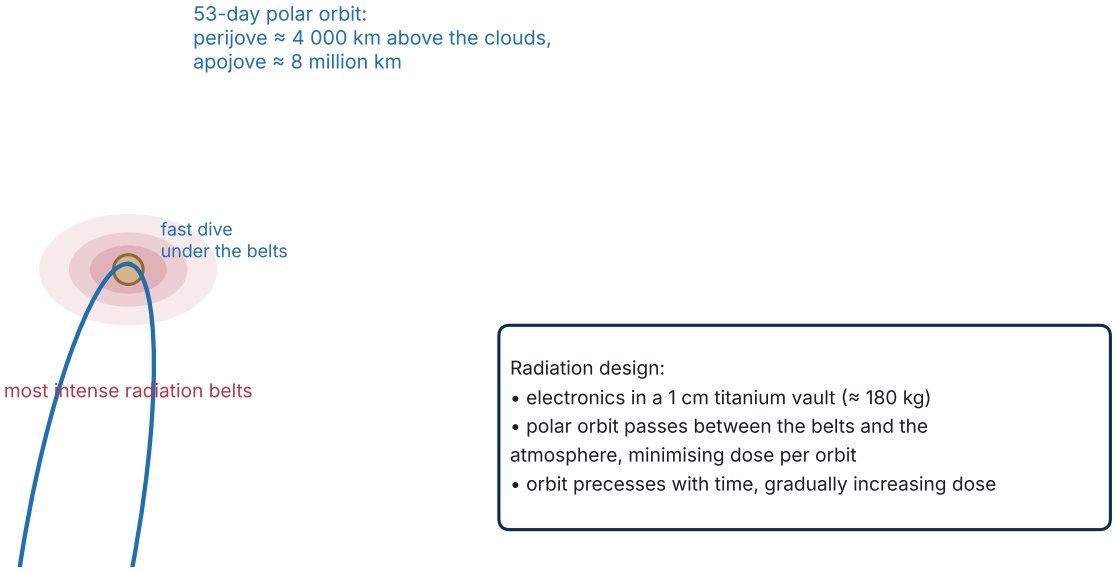



**FIGURE 25.3** Juno's 53-day polar orbit passes between Jupiter's atmosphere and its most intense radiation belts at each close approach (schematic).

### 25.2.3 Moon Tours

At the giant planets, orbiters use repeated flybys of the large moons to change their orbits (Chapter 24). Cassini's 13-year tour of the Saturn system included more than 120 Titan flybys and 23 close Enceladus flybys. It ended with the **Grand Finale** of 2017 — 22 orbits through the 2 000 km gap between the rings and the planet's atmosphere. This was followed by a deliberate plunge into Saturn to protect Enceladus and Titan from contamination (Figure 25.4, Figure 25.5).

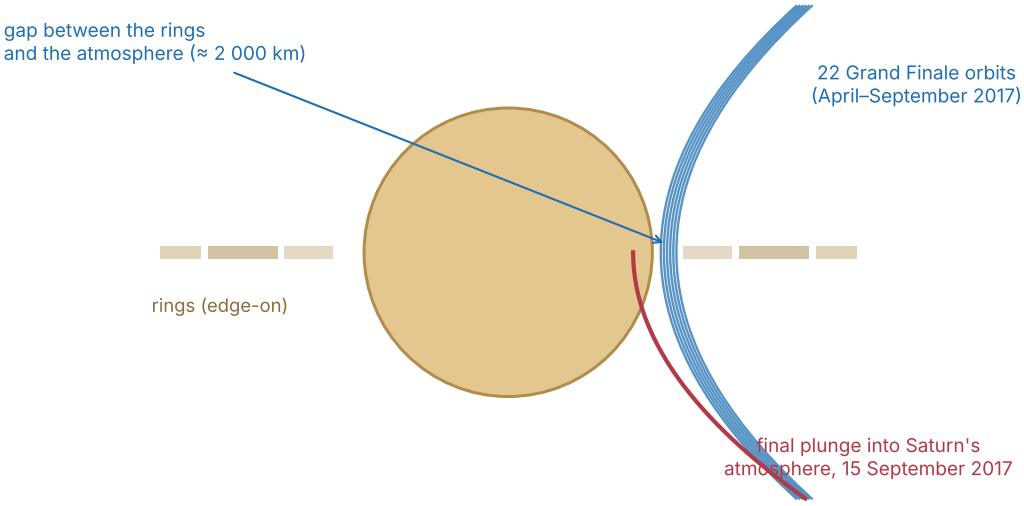



**FIGURE 25.4** Cassini's Grand Finale: 22 orbits through the gap between Saturn's rings and atmosphere, followed by the final plunge (schematic).

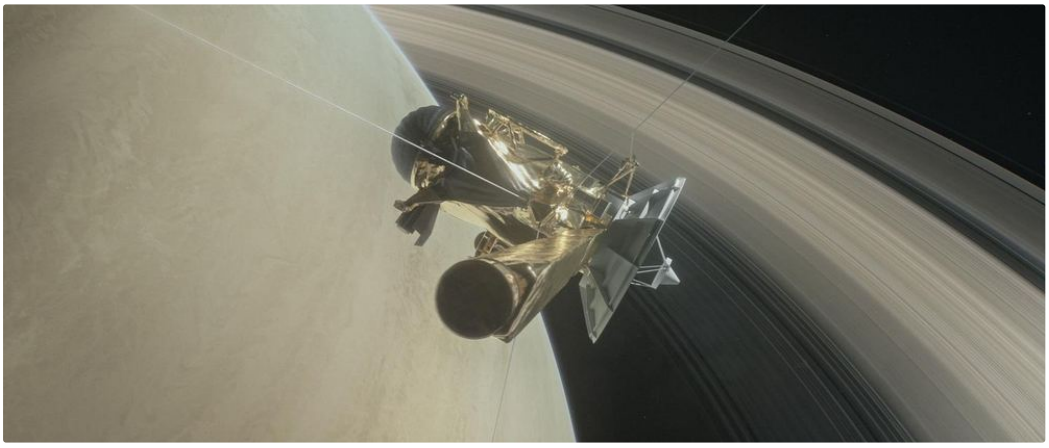


**FIGURE 25.5** Artist's impression of Cassini diving between Saturn and its rings during the Grand Finale in 2017. Image: NASA/JPL (NASA/JPL-Caltech).

## 25.3 Getting Into Orbit

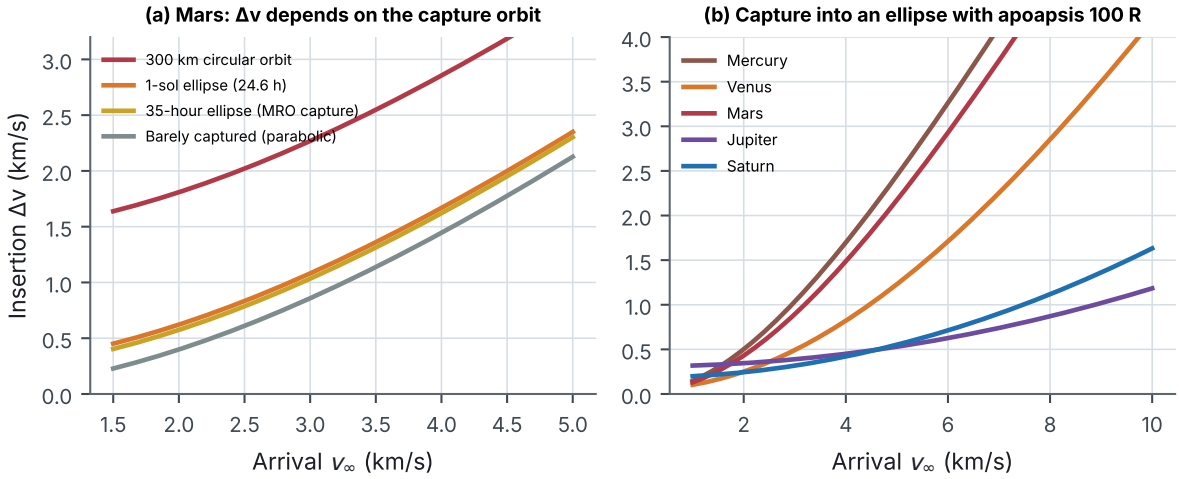
### 25.3.1 The Orbit-Insertion Burn

A spacecraft arrives at a planet on a hyperbola, moving faster than the local escape speed. To be captured, it must fire its main engine near closest approach and slow down by

$$\Delta v_{\text{insertion}} = \sqrt{v_\infty^2 + \frac{2\mu}{r_p}} - v_{\text{capture}},$$

where $v_\infty$ is the arrival speed relative to the planet (Chapter 24), $r_p$ is the periapsis radius and $v_{\text{capture}}$ is the periapsis speed on the desired capture orbit. Two facts dominate the design (Figure 25.6):

- **Capturing into a long ellipse is much cheaper than capturing into a low circular orbit.** At Mars, the difference is more than 1 km/s. Most orbiters therefore first enter a long, elliptical orbit and lower it later.
- **Deep gravity wells make capture cheap.** Because the burn is made where the spacecraft is moving fastest (the Oberth effect), Jupiter and Saturn can be entered for a few hundred metres per second. Juno's Jupiter orbit insertion in 2016 needed about 542 m/s (a 35-minute burn). Cassini's Saturn insertion in 2004 needed about 626 m/s (96 minutes). Mercury, with a shallow well deep inside the Sun's, is the hardest planet to orbit, which is why MESSENGER and BepiColombo needed six or more planetary flybys to reduce their arrival speed.




Periapsis 300 km in (a); 400 km (1.1 R for the giants) in (b). A deep gravity well makes capture cheap: the burn is made where the spacecraft is fastest (the Oberth effect).

**FIGURE 25.6** Orbit-insertion Δv. (a) At Mars, capturing into a long ellipse rather than a low circular orbit saves more than 1 km/s. (b) Capture into a long ellipse costs least at the giant planets with their deep gravity wells.

Orbit insertion is a **critical event**: it happens at a fixed time that cannot be postponed, often while the spacecraft is behind the planet and out of contact. It must therefore be fully autonomous, with fault protection that keeps the burn going through most failures. Losses have shaped today's practice. **Mars Observer** fell silent in 1993 just before insertion, probably after a propellant line ruptured while its tanks were being pressurised. **Mars Climate Orbiter** was lost in 1999 because ground software reported thruster impulse in pound-force seconds while the navigation software expected newton-seconds. The error accumulated over the cruise. The spacecraft passed about 57 km above the surface instead of the planned 140–150 km. Japan's **Akatsuki** failed to enter Venus orbit in 2010 when its main engine was damaged. Engineers kept it alive for five years in a solar orbit and inserted it in December 2015 using only its small attitude thrusters. This was one of the great recoveries in spaceflight.

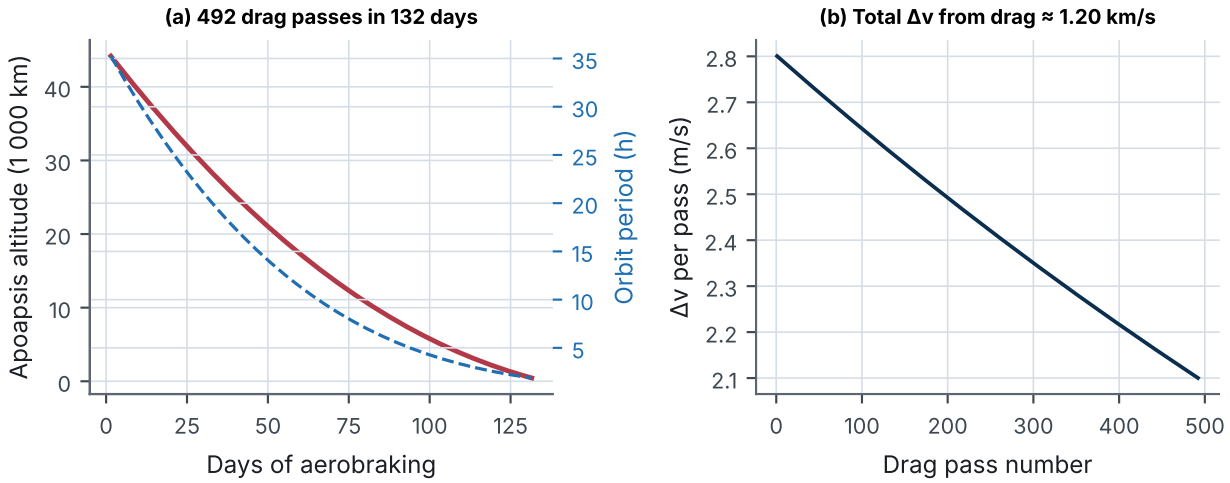
### 25.3.2 Aerobraking and Aerocapture

After capture, the orbit must be lowered from a long ellipse to the low science orbit. Doing this with the engine at Mars would cost another 1.2 km/s, so orbiters at Mars and Venus use **aerobraking**. The spacecraft lowers its periapsis to about 100–130 km. There, each pass through

the thin upper atmosphere removes a few metres per second of velocity, gradually shrinking the apoapsis (Figure 25.7). The solar arrays act as a drag brake. Ground teams (and increasingly onboard software) manage the **aerobraking corridor**. Small burns at apoapsis raise or lower the periapsis to keep dynamic pressure and heating within limits. This is needed because the density of the upper atmosphere can change by 30 % or more from one orbit to the next during dust storms. Aerobraking has three phases. In the **walk-in**, the periapsis is lowered step by step into the atmosphere. The **main phase** has hundreds of drag passes. In the **walk-out**, the periapsis is raised out of the atmosphere when the target apoapsis is reached.

**TABLE 25.2** Aerobraking campaigns.

| Mission | Planet | Years | Duration and notes |
|---|---|---|---|
| Magellan | Venus | 1993 | First aerobraking, used to circularise the orbit for gravity mapping |
| Mars Global Surveyor | Mars | 1997–1999 | About 16 months; slowed by a damaged solar-array hinge |
| Mars Odyssey | Mars | 2001–2002 | About 3 months |
| Mars Reconnaissance Orbiter | Mars | 2006 | 426 drag passes in about 5 months; from 35-hour to 2-hour orbit |
| Venus Express | Venus | 2014 | Experimental campaign at the end of the mission |
| ExoMars Trace Gas Orbiter | Mars | 2017–2018 | About 11 months, interrupted by solar conjunction |




Model: periapsis 100 km, density held at $4.5\times10^{-8}$ kg/m³ by corridor control, scale height 10 km, ballistic coefficient 18 kg/m², drag applied impulsively at periapsis.

**FIGURE 25.7** Aerobraking at Mars (illustrative model of an MRO-like orbiter). (a) Hundreds of drag passes shrink the orbit from a 35-hour ellipse to the 2-hour science orbit. (b) Each pass removes only a few metres per second; together they provide about 1.2 km/s — the equivalent of hundreds of kilograms of propellant.

**Aerocapture** goes further: the spacecraft, protected by an aeroshell, makes a single deep pass through the atmosphere to slow from hyperbolic speed directly into orbit, using lift to steer to the right exit conditions. It has never been flown, but studies show it could reduce the mass of orbiters bound for Uranus, Neptune or Titan by a third or more. It is a candidate technology for a future Uranus orbiter (Chapter 42).

**EXAMPLE: CAPTURING AT MARS**

An orbiter of 2 000 kg arrives at Mars from the 2026 launch window with $v_\infty$ = 2.67 km/s (Chapter 24). Its periapsis is 300 km (radius 3 696 km); its engine has a specific impulse of 320 s.

1. Speed at periapsis on the arrival hyperbola: $\sqrt{2.67^2 + 2\times 42\,828/3\,696}$ = 5.50 km/s.
2. **Option A — direct capture into a 300 km circular orbit** (speed $\sqrt{\mu/r}$ = 3.40 km/s): Δv = 2.10 km/s. By the rocket equation, $1 - e^{-2\,100/(320\times 9.81)}$ = 49 % of the arrival mass — 976 kg — would be propellant.
3. **Option B — capture into a 35-hour ellipse** (semi-major axis 25 825 km, apoapsis altitude about 44 600 km, periapsis speed 4.64 km/s): Δv = 0.87 km/s. This uses 24 % of the mass — 482 kg.
4. Lowering the 35-hour orbit to the 300 km circle with the engine would cost another 1.23 km/s. By aerobraking, it costs only a few tens of metres per second for corridor control and the final periapsis raise.
5. Option B plus aerobraking saves about 450 kg — almost a quarter of the arrival mass — at the cost of about 5 months of intensive operations before the science mission starts.

## 25.4 What Orbiters Carry

### 25.4.1 Instruments

Planetary orbiters carry many of the instrument types described in Chapter 12, adapted to their targets (Table 25.3).

**TABLE 25.3** Typical planetary orbiter instruments.

| Instrument type | Measures | Examples |
|---|---|---|
| High-resolution cameras | Morphology, geology, landing-site hazards | HiRISE (MRO, 25–30 cm/pixel), LROC (LRO, 0.5 m), CaSSIS (TGO), HiRIC (Tianwen-1) |
| Context and stereo cameras | Regional mapping, topography | CTX (MRO, 6 m), HRSC (Mars Express), Kaguya Terrain Camera |
| Visible–infrared imaging spectrometers | Minerals, ices, hydrated materials | CRISM (MRO), OMEGA (Mars Express), M3 (Chandrayaan-1), MISE (Europa Clipper) |
| Thermal infrared radiometers | Surface temperature, thermal inertia, rock abundance | THEMIS (Odyssey), Diviner (LRO) |
| Gamma-ray and neutron spectrometers | Elemental composition, subsurface hydrogen (ice) | Mars Odyssey GRS, Lunar Prospector, MESSENGER |
| Radar sounders | Subsurface layers, ice, Europa's ice shell | MARSIS (Mars Express), SHARAD (MRO), REASON (Europa Clipper), RIME (JUICE) |
| Laser and radar altimeters | Topography, shape, tides | MOLA (MGS), LOLA (LRO), BELA (BepiColombo), GALA (JUICE) |
| Radio science / gravity | Interior structure, atmospheres by radio occultation | GRAIL, Juno, all orbiters' telecom systems |
| Magnetometers, particles and fields | Magnetic fields, plasma, induced fields from oceans | Galileo, Juno, MAVEN, Europa Clipper |
| Atmospheric spectrometers | Trace gases, winds, escape | TGO's NOMAD and ACS, MAVEN's IUVS, Juno's MWR |

The **High Resolution Imaging Science Experiment (HiRISE)** on NASA's Mars Reconnaissance Orbiter, with a 0.5 m telescope, resolves objects about a metre across from 300 km. It has photographed landers descending under their parachutes (Chapter 27 shows Perseverance on its way down), tracked the movement of dunes and avalanches, and certified the safety of every NASA landing site since 2008.

**Radar sounders** operating at a few megahertz penetrate hundreds of metres to kilometres into ice and dry rock (Figure 25.8). On Mars, they have mapped the layered polar caps, buried glaciers at mid-latitudes and the thickness of ice deposits — essential information for future human missions seeking water.

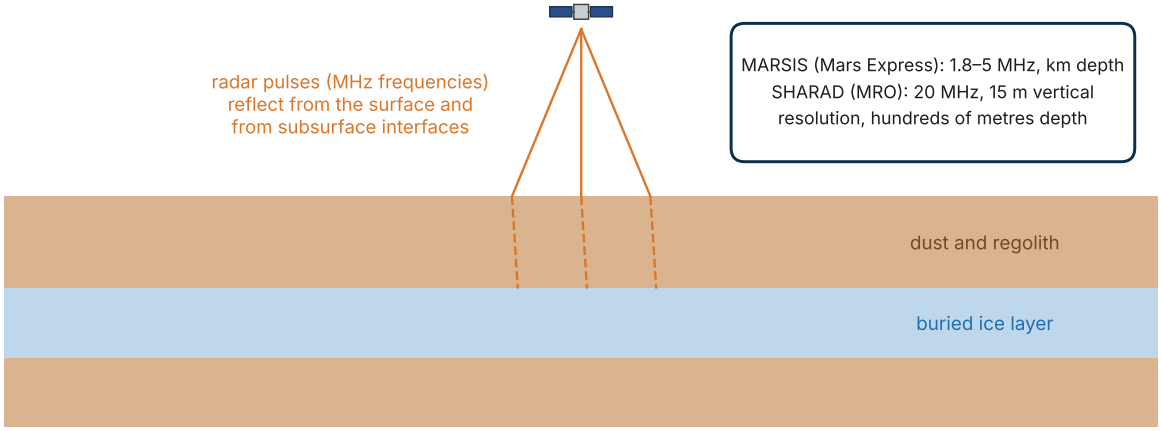



**FIGURE 25.8** Principle of orbital radar sounding of subsurface layers.

## 25.4.2 Relays for Surface Missions

Landers and rovers have small antennas and limited power, and they can talk directly to the Earth only at low data rates. Orbiters overhead act as **relays** (Figure 25.9). A rover transmits to an orbiter at UHF during a pass of about ten minutes, at up to about 2 Mbit/s. The orbiter then forwards the data to the Deep Space Network over its high-gain antenna. Most of the data from NASA's Mars rovers have been relayed this way, mainly by Mars Odyssey and MRO, with MAVEN and ESA's Trace Gas Orbiter. Chinese missions use Tianwen-1's orbiter and, on the Moon's far side, the Queqiao relay satellites. Ageing relay orbiters — Odyssey was launched in 2001 and MRO in 2005 — are a growing concern for future Mars surface missions.

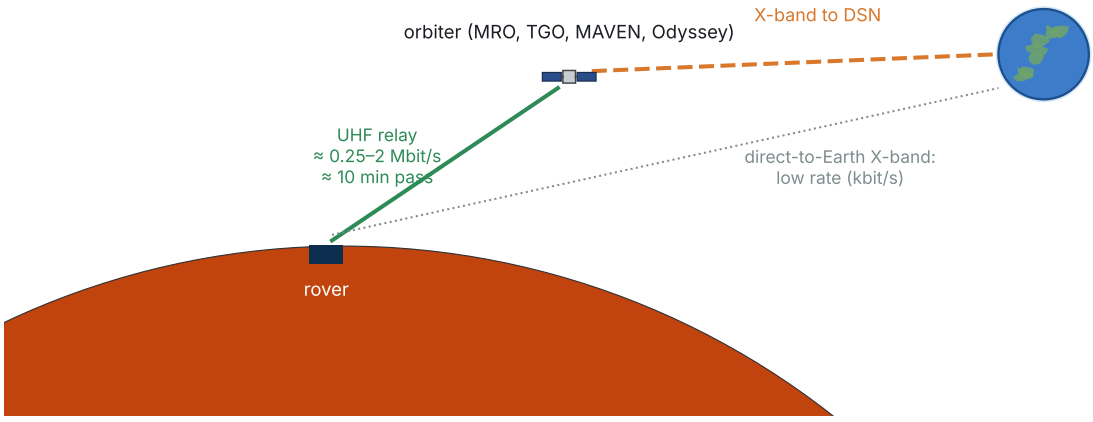



**FIGURE 25.9** Relay of data from a Mars rover via an orbiter.

## 25.5 Orbiters World by World

### 25.5.1 The Moon

The Moon has been orbited by more spacecraft than any other body. After Luna 10 (1966), NASA's five **Lunar Orbiters** (1966–67) photographed potential Apollo landing sites and 99 % of the surface. A long gap followed until **Clementine** (1994) and **Lunar Prospector** (1998), whose neutron spectrometer detected excess hydrogen at the poles — possibly water ice in permanently shadowed craters. The 2000s brought ESA's ion-propelled **SMART-1**, Japan's **Kaguya**, China's **Chang'e-1** and **Chang'e-2**, and India's **Chandrayaan-1**, whose NASA-supplied Moon Mineralogy Mapper detected hydroxyl and water molecules on the sunlit surface in 2009.

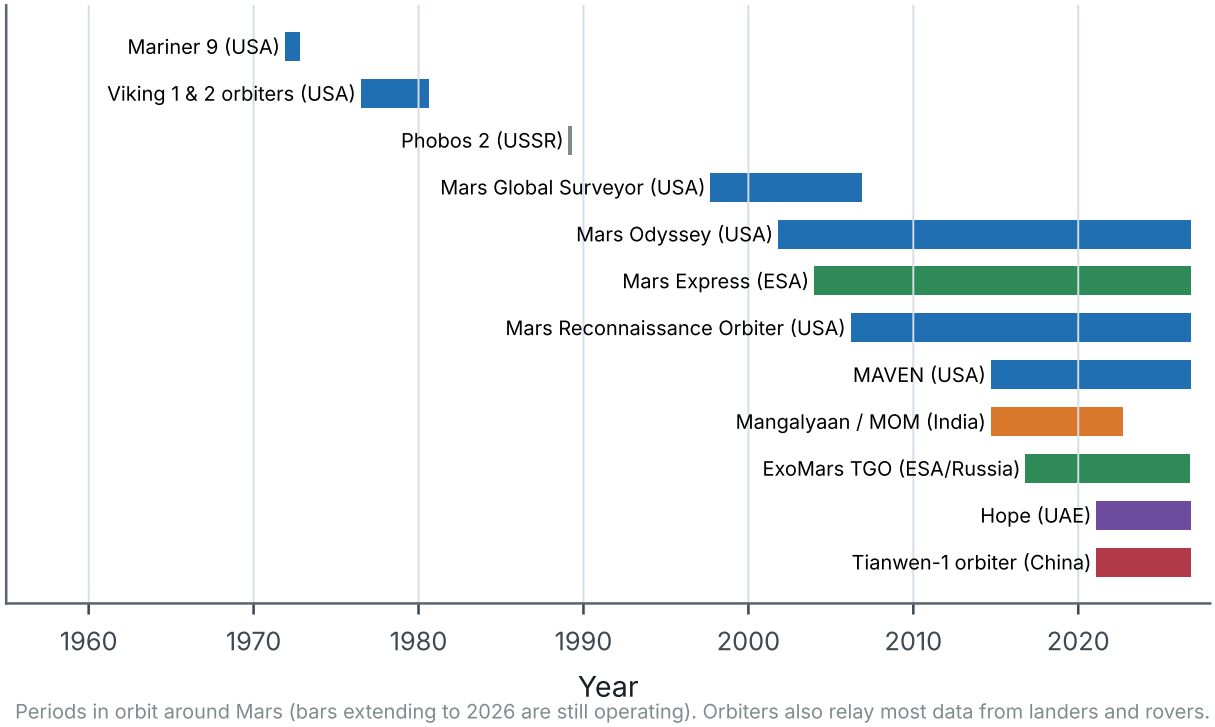
NASA's **Lunar Reconnaissance Orbiter (LRO)**, launched in 2009 and still operating, has mapped the Moon in unprecedented detail. Its cameras resolve the Apollo landers and astronauts' tracks, and its laser altimeter has produced a global topographic map. Its Diviner radiometer found the coldest places measured in the Solar System — about 25 K in polar craters (Figure 25.10). Its companion **LCROSS** deliberately crashed a rocket stage into the crater Cabeus in 2009 and detected water in the ejecta. Today, India's **Chandrayaan-2** orbiter, Korea's **Danuri** and China's Queqiao relays orbit the Moon, preparing for the Artemis and Chinese crewed landings (Chapters 27, 31). Danuri carries NASA's ShadowCam, which images inside shadowed craters by faint reflected light.

**FIGURE 25.10** Earthrise as seen by NASA's Lunar Reconnaissance Orbiter in 2015, with the crater Compton on the Moon's far side in the foreground. Image: NASA/GSFC.

### 25.5.2 Mars

Mars has had a continuous orbital presence since 1997 (Figure 25.11). **Mariner 9** (1971) revealed giant volcanoes, canyons and dry river valleys; the **Viking orbiters** (1976) mapped the planet; **Mars Global Surveyor** (1997–2006) measured its topography with a laser altimeter (MOLA) and discovered remnant crustal magnetism. **Mars Odyssey** (2001) found hydrogen-rich ground — ice — beneath the surface at high latitudes. ESA's **Mars Express** (2003) and NASA's **Mars**

**Reconnaissance Orbiter** (2006) discovered clays and sulphates formed in water, revealing a warmer, wetter early Mars. They also imaged active processes from avalanches to seasonal carbon-dioxide jets. NASA's **MAVEN** (2014) showed how the solar wind has stripped Mars's atmosphere. ESA–Roscosmos's **Trace Gas Orbiter** (2016) placed strict limits on atmospheric methane, and India's **Mangalyaan** (2014–2022) was the first Asian Mars orbiter. The UAE's **Hope** (2021) studies the atmosphere from a high orbit, and China's **Tianwen-1** (2021) delivered the Zhurong rover. NASA's twin **ESCAPADE** smallsats, launched in 2025, will study the Martian magnetosphere from 2027.




Periods in orbit around Mars (bars extending to 2026 are still operating). Orbiters also relay most data from landers and rovers.

**FIGURE 25.11** Orbiters at Mars since 1971. Six orbiters from four agencies and the UAE were operating in 2026.

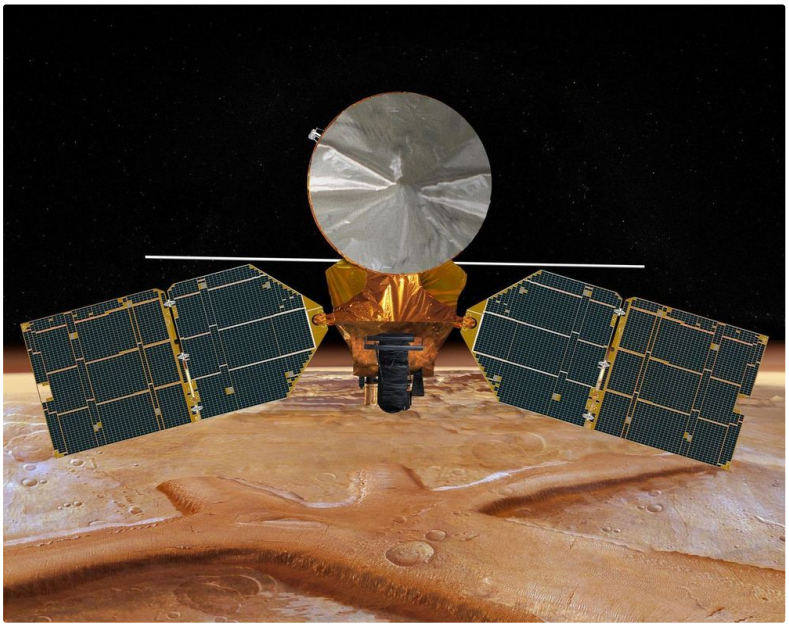


**FIGURE 25.12** Artist's concept of NASA's Mars Reconnaissance Orbiter, with its 3 m high-gain antenna and large solar arrays. Since 2006 it has returned more data than all other Mars missions combined. Image: NASA/JPL (NASA/JPL).

### 25.5.3 Venus

Venus's thick clouds hide its surface from optical cameras, so orbiters use radar. The Soviet **Venera 15 and 16** (1983) and NASA's **Magellan** (1990–1994) mapped it, revealing a young volcanic surface (Figure 25.13). ESA's **Venus Express** (2006–2014) and Japan's **Akatsuki** (2015–2024) studied its atmosphere, including the "super-rotation" in which the cloud tops circle the planet in four days. Three new orbiters were planned — NASA's **VERITAS**, ESA's **EnVision** and ISRO's **Shukrayaan** — together with NASA's DAVINCI descent probe, although US budget pressures in 2025–26 made the NASA missions' future uncertain.

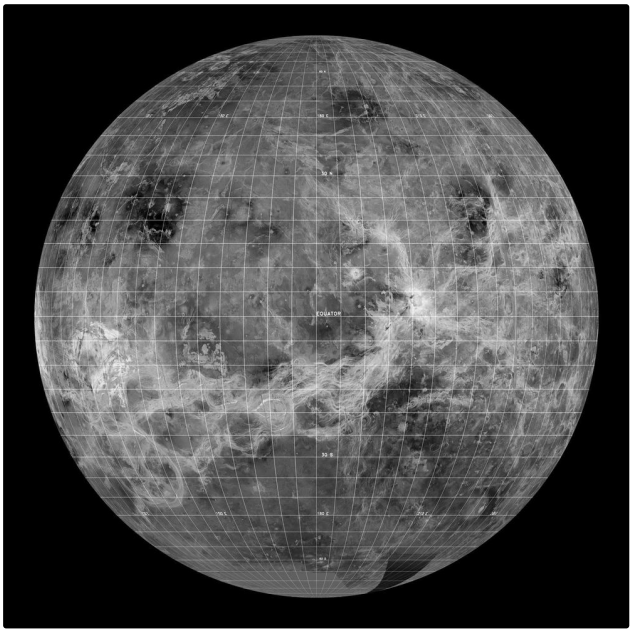


**FIGURE 25.13** A global radar view of Venus from Magellan data, centred on 180° longitude. Venus's surface is dominated by volcanic plains and highlands. Image: NASA/JPL (NASA/JPL).

### 25.5.4 Mercury

NASA's **MESSENGER** (2011–2015) was the first Mercury orbiter. It found that Mercury's surface is unexpectedly rich in volatile elements such as sulphur and potassium, and that its magnetic field is offset to the north. It also found that permanently shadowed craters near its poles contain water ice — on the planet closest to the Sun (Figure 25.14). The ESA–JAXA **BepiColombo** mission, launched in 2018, comprises two orbiters: ESA's Mercury Planetary Orbiter and JAXA's Mio, carried by a transfer module with electric propulsion. After a thruster power problem in 2024 delayed its arrival by about a year, it enters orbit around Mercury in November 2026.

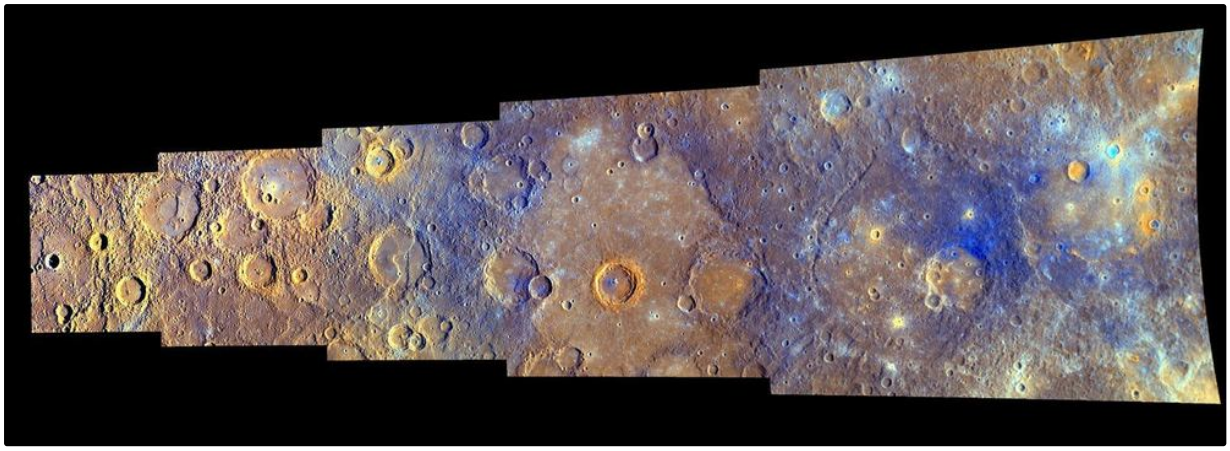


**FIGURE 25.14** MESSENGER colour mosaic of Mercury's surface. Enhanced colours reveal variations in composition between volcanic plains and older terrain. Image: NASA/JPL.

### 25.5.5 Jupiter

**Galileo** (1995–2003) dropped a probe into Jupiter's atmosphere and found evidence from magnetic-field measurements that Europa, Ganymede and Callisto have subsurface oceans. **Juno**, in orbit since 2016, studies Jupiter's interior from a polar orbit (Figure 25.3). It found that the planet's core is "dilute" — mixed with the surrounding envelope. It also found that its zonal winds extend about 3 000 km deep, and that its poles are covered by clusters of giant cyclones. Its JunoCam has produced spectacular images (Figure 25.15). Two new missions are on their way. NASA's **Europa Clipper** will make about 49 close flybys of Europa from 2030 to assess its habitability. ESA's **JUICE** will tour the Galilean moons from 2031 before entering orbit around Ganymede in 2034 — the first spacecraft to orbit a moon of another planet.

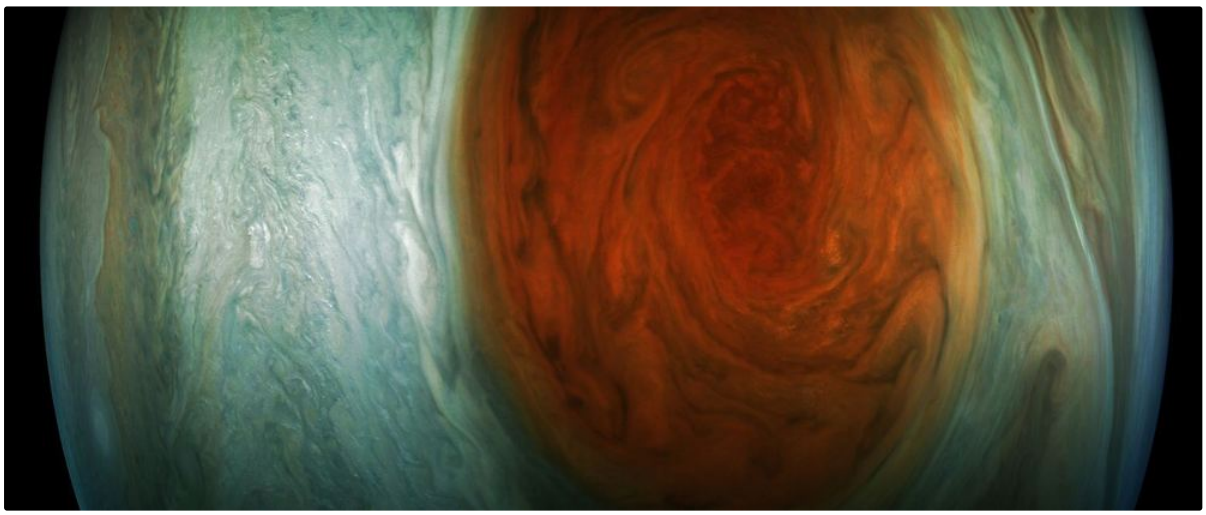


**FIGURE 25.15** Jupiter's Great Red Spot imaged by JunoCam on Juno in July 2017 (enhanced colour). The storm, larger than the Earth, extends about 300 km below the clouds. Image: NASA/JPL.

### 25.5.6 Saturn

**Cassini–Huygens** (2004–2017), a collaboration of NASA, ESA and the Italian Space Agency, was one of the most successful planetary missions. It delivered the Huygens probe to the surface of Titan (Chapter 27) and found hydrocarbon lakes and seas near Titan's poles. In 2005 it discovered

water-vapour and ice plumes erupting from the south pole of Enceladus (Figure 25.16). The plumes are fed by a subsurface ocean that contains salts, silica and organic molecules, making Enceladus one of the most promising places to search for life.

**FIGURE 25.16** Plumes of ice and water vapour rising from the south polar region of Enceladus, back-lit by the Sun, imaged by Cassini. Image: NASA/JPL (NASA/JPL-Caltech/Space Science Institute).

### 25.5.7 Small Bodies

Orbiting small bodies demands precise navigation in very weak gravity. NASA's **NEAR Shoemaker** orbited Eros for a year and landed on it in 2001. **Dawn** orbited Vesta (2011–12) and Ceres (2015–18), finding bright salt deposits on Ceres. ESA's **Rosetta** accompanied comet 67P/Churyumov–Gerasimenko for two years (2014–2016) as it passed perihelion. The sample-return missions Hayabusa2 and OSIRIS-REx also mapped their asteroids from orbit before collecting samples (Chapter 29). ESA's **Hera** will arrive at the binary asteroid Didymos–Dimorphos in November 2026, a month earlier than first planned. It will measure the effects of NASA's DART impact of 2022. NASA's **Psyche** will orbit the metal-rich asteroid Psyche from 2029.

**TABLE 25.4** Planetary orbiters operating or arriving in 2026 (selected).

| Mission | Agency | Target | Arrival | Status (Oct 2026) |
|---|---|---|---|---|
| LRO | NASA | Moon | 2009 | Operating |
| Chandrayaan-2 orbiter | ISRO | Moon | 2019 | Operating |
| Danuri (KPLO) | KARI | Moon | 2022 | Operating |
| Mars Odyssey, MRO, MAVEN | NASA | Mars | 2001, 2006, 2014 | Operating (relay and science) |
| Mars Express, ExoMars TGO | ESA (TGO with Roscosmos until 2022) | Mars | 2003, 2016 | Operating |
| Hope | UAE | Mars | 2021 | Operating |
| Tianwen-1 orbiter | CNSA | Mars | 2021 | Operating |
| Juno | NASA | Jupiter | 2016 | Extended operations uncertain after 2025 |

| Mission | Agency | Target | Arrival | Status (Oct 2026) |
|---|---|---|---|---|
| BepiColombo (MPO, Mio) | ESA, JAXA | Mercury | Nov 2026 | Orbit insertion |
| Hera | ESA | Didymos–Dimorphos | Nov 2026 | Final approach |
| Europa Clipper, JUICE | NASA, ESA | Jupiter system | 2030, 2031 | En route |
| Psyche | NASA | Asteroid Psyche | 2029 | En route |

## 25.6 Ending an Orbiter Mission

Orbiters usually end when they run out of propellant for attitude control, when a critical component fails, or when funding stops. How they end is governed by **planetary protection** (Chapter 24): an orbiter must not contaminate a world that might host life, or that future missions will search for life.

- **Disposal into the planet.** Galileo was deliberately plunged into Jupiter in 2003 and Cassini into Saturn in 2017, so that they could never crash into Europa, Enceladus or Titan. Juno's mission plan also ends with a plunge into Jupiter.
- **Long-lived parking orbits.** Mars orbiters that have not been sterilised must stay in orbit for at least 20 years after launch (with 99 % probability) and 50 years (with 95 % probability). When Dawn ran out of hydrazine in 2018, it was left in an orbit around Ceres that is expected to remain stable for decades.
- **Controlled impacts.** Where contamination is not a concern, orbiters often end with an impact that is itself an experiment. LCROSS, GRAIL's twin spacecraft (2012) and LADEE (2014) impacted the Moon, and MESSENGER crashed into Mercury in 2015. Rosetta made a slow descent onto comet 67P in 2016, taking images until the final seconds.
- **Unplanned endings.** Mars Global Surveyor was lost in 2006 after a software error pointed a solar array wrongly and exposed its battery to the Sun. India's Mangalyaan fell silent in 2022, probably after a long eclipse drained its battery. Contact with Akatsuki was lost in 2024.

### EXAMPLE: CHOOSING A MARS MAPPING ORBIT

A new Mars orbiter must image the surface at the same local time each day (to keep shadows consistent) from an altitude of about 300 km.

1. Mars's $J_2 = 1.96 \times 10^{-3}$, radius 3 396 km and $\mu = 42\,828$ km$^3$/s$^2$ give, at 300 km, a mean motion $n = \sqrt{\mu/a^3} = 9.2 \times 10^{-4}$ rad/s.
2. The orbit plane must precess 360° in one Martian year (687 days): $\dot{\Omega} = 1.06 \times 10^{-7}$ rad/s.
3. From $\dot{\Omega} = -\tfrac{3}{2} n J_2 (R/a)^2 \cos i$, $\cos i = -0.052$, so $i \approx 93.0^\circ$ (Figure 25.2).
4. The ground track shifts westward by about 28° of longitude per orbit (Mars rotates 360° in 24.6 hours while the orbit takes about 1.9 hours). A camera with a 6 km swath therefore needs thousands of orbits — several Earth years — to map the planet completely, as MRO's Context Camera did.
5. Planetary protection (Category III) requires the orbit to remain above the atmosphere for at least 20 years after launch, which a 300 km orbit easily satisfies if the spacecraft is clean or later raised.

### AI FOR PLANETARY ORBITERS

Planetary scientists use machine learning to detect craters, landslides, dust devils and new impacts in millions of orbital images, to map minerals from hyperspectral data, and to prioritise downlink of the most interesting observations. On board, autonomous instruments can detect transient events — such as plumes, storms or impacts — and decide what to observe and return (Chapter 51).

### KEY TAKEAWAYS

- Orbiters provide global, long-duration mapping, interior measurements and relays for surface missions.
- Orbit design depends on each body. Lunar mascons allow only a few frozen low orbits, and Mars's bulge allows nearly polar sun-synchronous orbits. Venus and Mercury pose thermal challenges, Jupiter's radiation shapes Juno's orbit, and small bodies have weak, irregular gravity.
- Instruments include high-resolution cameras (HiRISE, about 30 cm/pixel), imaging spectrometers, radar sounders, altimeters, gravity science and particle and field sensors.
- Key discoveries include water ice at the lunar and Mercurian poles, Mars's watery past and atmospheric loss, Venus's volcanic surface, Jupiter's dilute core, and Enceladus's ocean plumes.
- In 2026, orbiters operate at the Moon, Mars and Jupiter; BepiColombo and Hera arrive at Mercury and Didymos, and Europa Clipper and JUICE are en route to Jupiter.

### REVIEW QUESTIONS

1. Give four kinds of science that an orbiter can do better than a flyby or a lander.
2. Why are only a few low lunar orbits stable for long periods? What are frozen orbits?
3. A spacecraft arrives at Mars with $v_\infty$ = 2.65 km/s. Calculate the Δv to enter a circular orbit 400 km above the surface in one burn at periapsis.
4. Why do many Mars orbiters first capture into a long elliptical orbit and then aerobrake? What are the risks of aerobraking?
5. Explain why a radiation vault is needed at Jupiter and how Juno's orbit was designed to limit radiation dose.
6. How do Mars orbiters act as relays for surface missions, and why does this increase the data returned by rovers?
7. Describe three major discoveries made by planetary orbiters.
8. Why are orbiters around Jupiter and Saturn deliberately crashed into the planet at the end of their missions?

CHAPTER

# 26 Space Probes and Flyby Missions

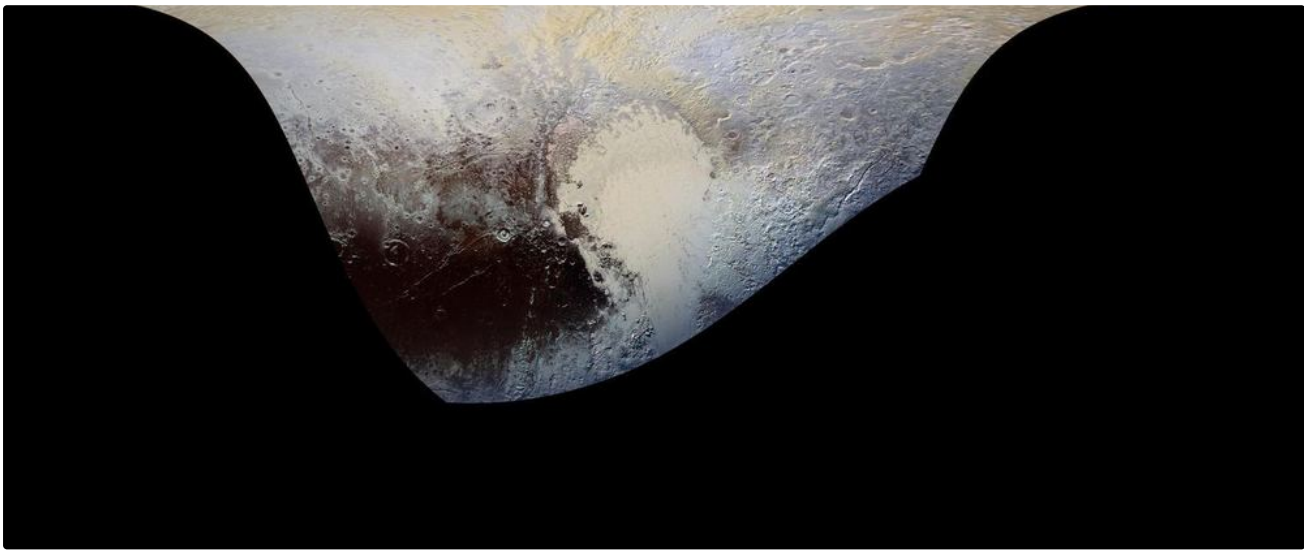


Image: NASA/JPL

*A flyby gives a few hours of close-up science after years of travel, yet flybys revealed most of the Solar System to us — the volcanoes of Io, the rings of Uranus, the geysers of Triton, the nitrogen glaciers of Pluto. This chapter explains the different kinds of space probes — flyby spacecraft, atmospheric entry probes, impactors and interstellar travellers — and the engineering behind them, from encounter navigation and imaging at 14 km/s to radioisotope power and data links across 25 billion kilometres. It follows the Pioneers, Voyagers and New Horizons to the edge of the heliosphere, describes how DART moved an asteroid, and reviews the probes active in 2026.*

IN THIS CHAPTER

26.1 Kinds of Probes
26.2 Sixty Years of Flybys
26.3 The Grand Tour: Pioneer and Voyager
26.4 Engineering a Flyby
26.5 Power and Communications Far from the Sun
26.6 Atmospheric Entry Probes
26.7 Impactors and Planetary Defence
26.8 Comets, Asteroids and Interstellar Visitors
26.9 Into Interstellar Space
26.10 Probes in 2026

A **space probe** is an uncrewed spacecraft that leaves Earth orbit to explore the Solar System. In a broad sense every planetary mission is a probe, but this chapter concentrates on missions that do not stay in orbit or land. These are spacecraft that **fly past** their targets, **plunge into** atmospheres, **crash into** surfaces on purpose, or travel **out of the Solar System**. Orbiters are described in Chapter 25, landers in Chapter 27, rovers in Chapter 28 and sample-return missions in Chapter 29.

## 26.1 Kinds of Probes

**TABLE 26.1** The main kinds of probes discussed in this chapter.

| Type | What it does | Advantages | Limitations | Examples |
|---|---|---|---|---|
| Flyby | Passes a target once, observing for hours to days | Cheapest way to reach a new world; can visit several targets | Brief, one-sided view; high speed blurs images | Mariner 4, Voyager 1 and 2, New Horizons, Lucy |
| Atmospheric entry probe | Enters an atmosphere, measuring it during descent | Direct measurement of composition, temperature, winds | Short life (minutes to hours); extreme entry heating | Pioneer Venus, Galileo probe, Huygens |
| Impactor | Strikes a surface deliberately | Excavates subsurface material; tests asteroid deflection | Single event; must be observed by another spacecraft or telescope | Deep Impact, LCROSS, DART |
| Heliospheric and interstellar | Travels outward to sample the solar wind and interstellar medium | The only direct measurements of the Sun's outer boundary | Power and communications limits; decades-long missions | Pioneer 10 and 11, Voyager 1 and 2 |
| Close-solar | Dives towards the Sun | In-situ measurements of the corona | Intense heat; needs many gravity assists | Helios, Parker Solar Probe, Solar Orbiter (Chapter 22) |

Flybys are the natural first step in exploring a world. A flyby needs no braking propellant and can be combined with gravity assists to reach several targets in one mission (Chapter 24). They are followed, where budgets allow, by orbiters for global mapping and landers for surface measurements. For the most distant worlds — Uranus, Neptune, Pluto and the Kuiper Belt — flybys remain our only close-up views.

## 26.2 Sixty Years of Flybys

The first spacecraft to escape Earth was the Soviet **Luna 1** in January 1959. It missed the Moon by about 6 000 km and became the first artificial object in orbit around the Sun. The same year **Luna 3** photographed the Moon's far side. The planetary era began with NASA's **Mariner 2**, which flew past Venus in December 1962 and showed that its surface was hot enough to melt lead. **Mariner 4** returned the first close-up pictures of another planet — 21 images of a cratered, Moon-like Mars — in July 1965 (Figure 26.1).

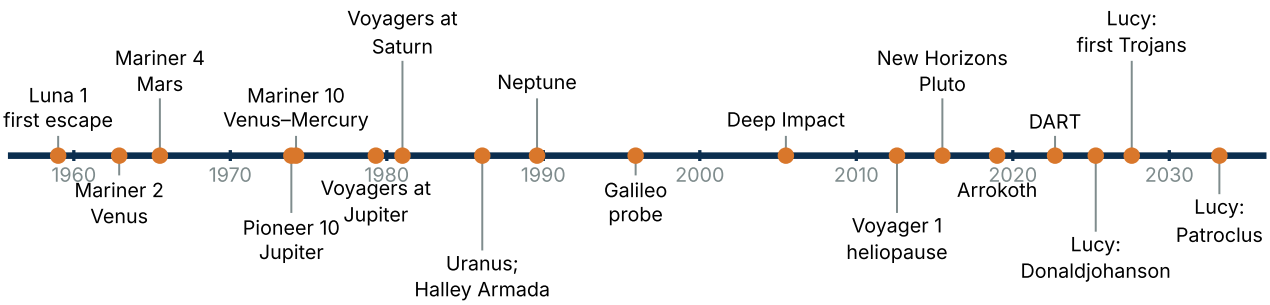



**FIGURE 26.1** Milestones in flyby and probe exploration, 1959–2033.

**Mariner 10** (1973–75) was the first spacecraft to use a gravity assist: a Venus flyby bent its path towards Mercury, which it passed three times. **Pioneer 10** became the first spacecraft to cross the asteroid belt and fly past Jupiter (December 1973); **Pioneer 11** followed it past Jupiter and then made the first Saturn flyby in 1979. In 1986 an international fleet — the **Halley Armada** of ESA's Giotto, the Soviet Vega 1 and 2, and Japan's Suisei and Sakigake — met Halley's Comet. Giotto passed 596 km from the nucleus and returned the first images of a comet's core.

**TABLE 26.2** Firsts in flyby and probe exploration.

| Year | Spacecraft | Achievement |
|---|---|---|
| 1959 | Luna 1 (USSR) | First spacecraft to escape Earth and orbit the Sun |
| 1959 | Luna 3 (USSR) | First images of the Moon's far side |
| 1962 | Mariner 2 (USA) | First successful planetary flyby (Venus) |
| 1965 | Mariner 4 (USA) | First close-up images of Mars |
| 1967 | Venera 4 (USSR) | First data from inside another planet's atmosphere (Venus) |
| 1973 | Pioneer 10 (USA) | First crossing of the asteroid belt; first Jupiter flyby |
| 1974 | Mariner 10 (USA) | First gravity assist; first Mercury flyby |
| 1979 | Pioneer 11 (USA) | First Saturn flyby |
| 1985 | ICE (USA) | First comet encounter (Giacobini–Zinner) |
| 1986 | Giotto (ESA) | First close images of a comet nucleus (Halley) |
| 1986, 1989 | Voyager 2 (USA) | Only flybys of Uranus and Neptune |
| 1991, 1993 | Galileo (USA) | First asteroid flybys (Gaspra, Ida) and first asteroid moon (Dactyl) |
| 1995 | Galileo probe (USA) | First entry probe into a giant planet |
| 2005 | Deep Impact (USA) | First impact experiment on a comet |
| 2012 | Voyager 1 (USA) | First spacecraft in interstellar space |
| 2015 | New Horizons (USA) | First Pluto flyby |
| 2019 | New Horizons (USA) | Most distant object explored (Arrokoth, 6.6 billion km) |
| 2022 | DART (USA) | First change of an asteroid's orbit by a human-made object |

## 26.3 The Grand Tour: Pioneer and Voyager

### 26.3.1 Pioneer 10 and 11

The two Pioneers were simple, spin-stabilised spacecraft of about 260 kg, powered by four SNAP-19 radioisotope generators and communicating through a 2.74 m dish (Figure 26.2). Their main task was to find out whether a spacecraft could survive the asteroid belt and Jupiter's radiation belts. They could — just: Pioneer 10's instruments recorded radiation doses close to their limits. Both spacecraft carry a gold-anodised plaque showing a man and a woman and the Sun's location relative to 14 pulsars. Pioneer 11 fell silent in 1995 and Pioneer 10 in 2003, at a distance of 80 AU.

**FIGURE 26.2** Pioneer 10 being prepared for launch in 1972. The 2.74 m antenna dominates the spacecraft; the radioisotope generators are on booms to keep their radiation away from the instruments. Image: NASA/ARC.

### 26.3.2 Voyager 1 and 2

In the late 1970s the four giant planets lined up so that a single spacecraft could visit all of them, using each one's gravity to reach the next. This alignment occurs about once every 175 years. NASA's two **Voyagers**, launched in August and September 1977, made use of it (Figure 26.3). Voyager 1 flew past Jupiter (March 1979) and Saturn (November 1980); it was then deliberately sent close to Saturn's moon Titan, which bent its path northward out of the ecliptic. Voyager 2 visited Jupiter (July 1979), Saturn (August 1981), Uranus (January 1986) and Neptune (August 1989) — still the only spacecraft to have seen the two ice giants close up.

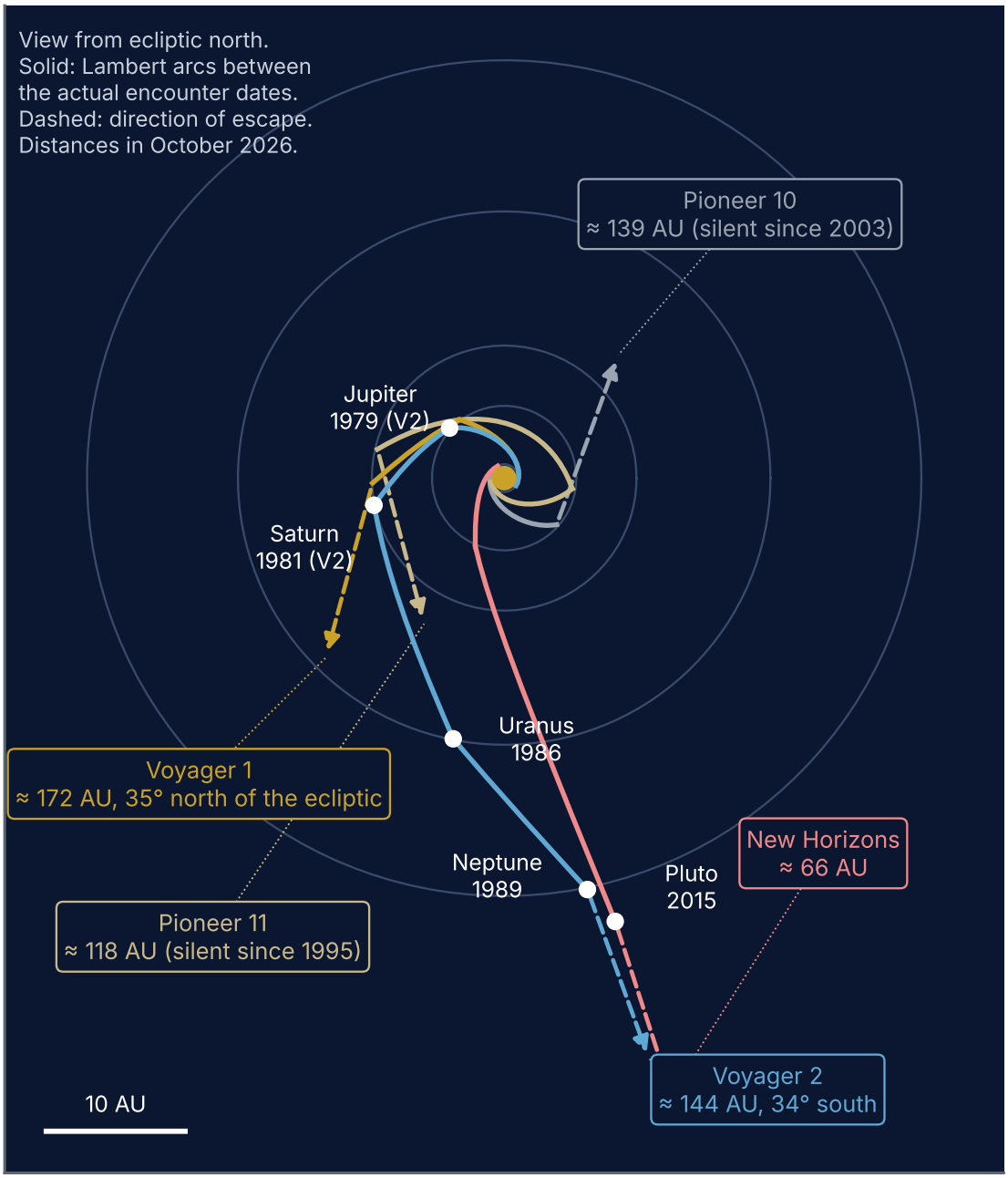



**FIGURE 26.3** The trajectories of the five spacecraft that are leaving the Solar System, computed from the planets' positions on the actual encounter dates.

Each Voyager weighed 825 kg at launch and carried ten science instruments, a 3.66 m high-gain antenna and three radioisotope generators producing 470 W. It also had a scan platform that pointed the cameras and spectrometers without turning the whole spacecraft (Figure 26.4). Their three onboard computers had a combined memory of about 69 kilobytes, and data were recorded on an eight-track digital tape recorder.

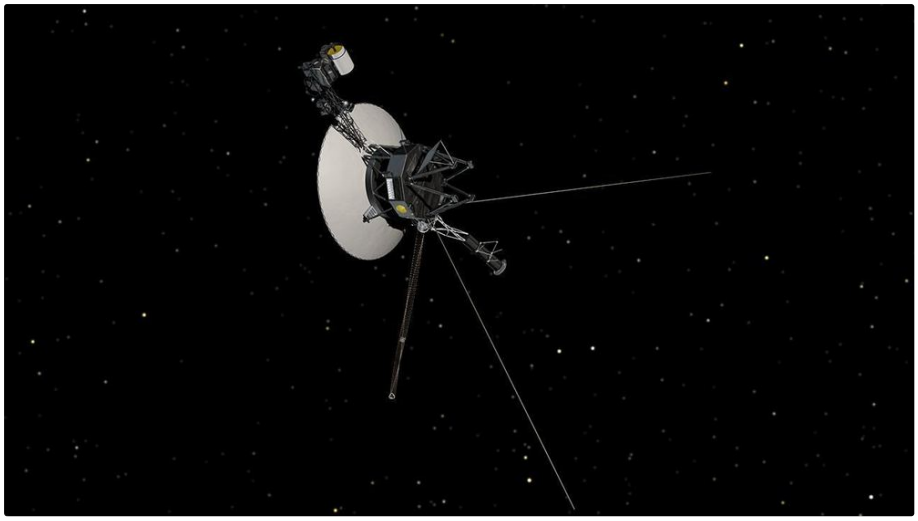


**FIGURE 26.4** Artist's concept of a Voyager spacecraft. The 3.66 m antenna points to Earth; the magnetometer boom (13 m) and the RTG boom extend to the sides. Image: NASA/JPL (NASA/JPL-Caltech).

The Voyagers transformed planetary science:

- **Jupiter:** active volcanoes on Io — the first seen beyond Earth — a thin ring, and the cracked, icy surface of Europa, hinting at an ocean beneath.
- **Saturn:** thousands of ringlets and the "spokes" in the rings, shepherd moons that confine narrow rings, and a thick nitrogen atmosphere on Titan.
- **Uranus:** a magnetic field tilted by 59° to the rotation axis, ten new moons and the bizarre, patchwork surface of Miranda.
- **Neptune:** the Great Dark Spot, the fastest winds in the Solar System (about 2 000 km/h), and nitrogen geysers on its large moon Triton. Triton orbits backwards and is probably a captured Kuiper Belt object (Figure 26.5).

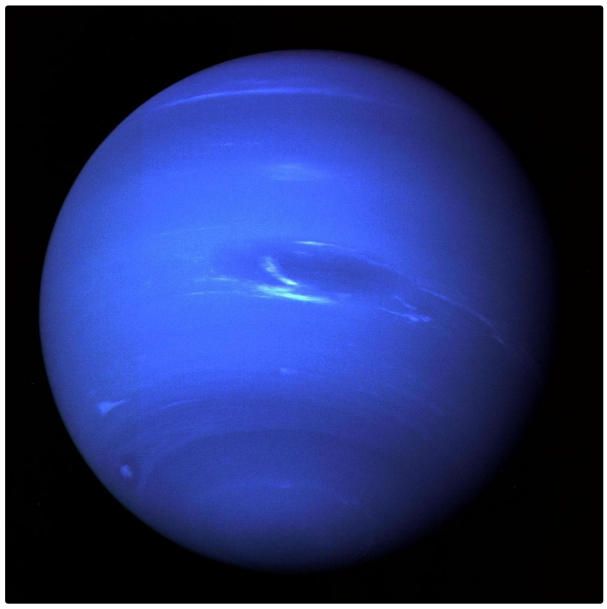


**FIGURE 26.5** Neptune imaged by Voyager 2 in August 1989, showing the Great Dark Spot (centre) and bright methane-ice clouds. Image: NASA/JPL (Voyager 2).

On 14 February 1990, at a distance of 6 billion km, Voyager 1 turned its cameras back towards the Sun and photographed the planets for a "family portrait". In one frame Earth appears as a pale blue dot less than a pixel across (Figure 26.6). The cameras were then switched off to save power

and memory; they have not been used since.

**FIGURE 26.6** The "Pale Blue Dot": Earth (in the band of scattered sunlight, right of centre) seen by Voyager 1 from 6 billion km in 1990, reprocessed in 2020. Image: NASA/JPL (NASA/JPL-Caltech).

Each Voyager carries a **Golden Record** in case either spacecraft is ever found by another civilisation. It is a gold-plated copper disc with sounds and images of Earth, greetings in 55 languages and 90 minutes of music.

## 26.4 Engineering a Flyby

### 26.4.1 Arriving at the Right Place at the Right Time

A flyby is aimed in the **B-plane**, the plane through the target's centre perpendicular to the incoming asymptote of the hyperbolic trajectory (Chapter 24). The aim point in this plane determines the closest-approach distance and the bending of the trajectory; the arrival time determines which side of the target is lit and which moons can be observed. Errors grow as the spacecraft approaches: a small velocity error made weeks before encounter can move the aim point by thousands of kilometres. Navigators therefore combine radio tracking (Doppler, ranging and delta-DOR) with **optical navigation** — images of the target against the stars taken by the spacecraft's own cameras. They also make a series of trajectory correction manoeuvres in the final weeks. For the Pluto flyby, the uncertainty in Pluto's own position, known only from 85 years of telescopic observations, was larger than that of the spacecraft. Optical navigation reduced it to a few hundred kilometres before the encounter. New Horizons arrived within about 90 seconds of the planned time.

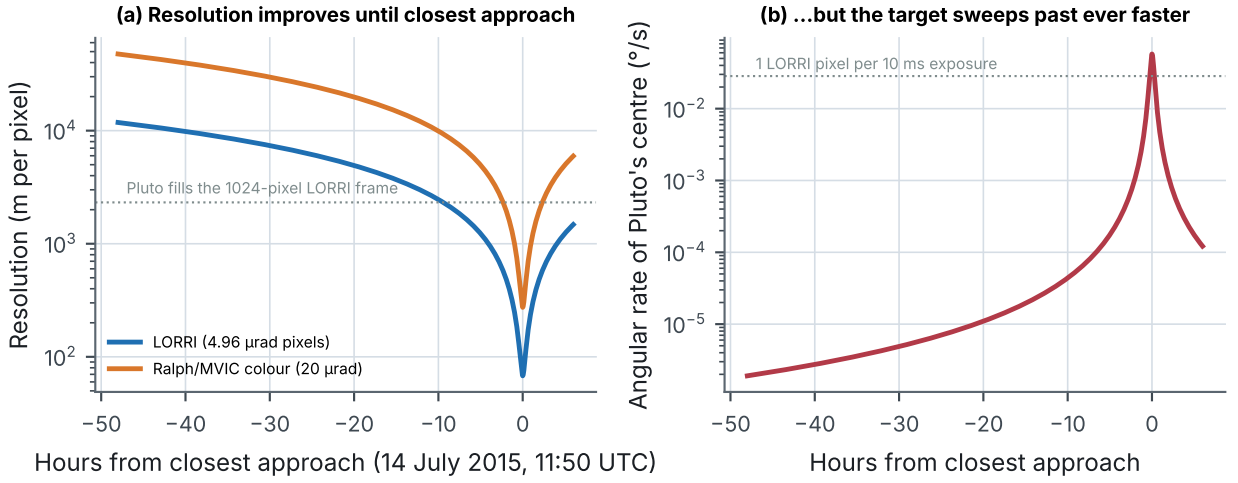
### 26.4.2 Sequences and Fault Protection

The most important hours of a flyby must run without help from the ground: at Pluto, a radio signal took 4.5 hours each way. The encounter is therefore executed by a **command sequence** loaded days in advance, which points the spacecraft and instruments, sets exposure times and records the data. **Fault protection** software monitors the spacecraft and, if a problem occurs,

puts it into a safe state — but during an encounter, entering safe mode could mean losing the unrepeatable observations. Flyby sequences therefore include special "critical sequence" fault protection that keeps the observations going through most faults. Ten days before the Pluto flyby, on 4 July 2015, New Horizons went into safe mode. Its main computer was overloaded while compressing data and loading the encounter sequence at the same time. The team recovered the spacecraft in three days and lost only minor science.

### 26.4.3 Imaging at High Speed

A flyby spacecraft sees its target grow from a dot to a world in a few days, and then sweep past in hours. The resolution — the size of a camera pixel projected on the target — improves in proportion to the distance, but the target's angular rate rises sharply near closest approach (Figure 26.7). New Horizons passed Pluto at 13.78 km/s. At closest approach, 12 500 km above the surface, Pluto's centre moved across the sky at 0.058° per second. A feature would cross one pixel of the LORRI telescopic camera in about 5 milliseconds. Exposures had to be short, or the spacecraft had to turn to follow the target, which it did with its thrusters.




Flyby speed 13.78 km/s; closest approach 12 500 km above Pluto's surface. Data were stored on board and returned over the next 15 months.

**FIGURE 26.7** New Horizons at Pluto. (a) The resolution of the LORRI camera and the colour imager improved from kilometres to about 70 m per pixel at closest approach. (b) Pluto's angular rate rose more than ten-thousandfold in the last two days.

Voyager mounted its cameras on a **scan platform** that could point independently of the spacecraft, so the high-gain antenna could stay pointed at Earth. New Horizons, to save mass and complexity, had no scan platform: the whole spacecraft turned to point its instruments, so it could not transmit while observing. It recorded about 6.25 gigabytes during the Pluto encounter on two solid-state recorders and then spent 15 months sending it to Earth at 1–4 kilobits per second.

**EXAMPLE: IMAGE SMEAR DURING A FLYBY**

At closest approach to Pluto, New Horizons was 13 690 km from Pluto's centre, moving at 13.78 km/s.

1. Angular rate of the target as seen from the spacecraft: $\omega = v/d$ = 13.78 / 13 690 = $1.0 \times 10^{-3}$ rad/s (0.058°/s).
2. LORRI's pixels subtend 4.96 µrad. The time for the image to move by one pixel is $4.96 \times 10^{-6}$ / $1.0 \times 10^{-3} \approx 5$ ms.
3. Sunlight at 33 AU is 1/1 100 as bright as at Earth, so long exposures are tempting. But an exposure of 50 ms would smear the image over 10 pixels unless the spacecraft rotated to track the target.
4. A day earlier, at 1.2 million km, Pluto was almost straight ahead. Its angular rate, $v\,b/r^2$ with $b$ = 13 690 km, was only about $10^{-7}$ rad/s and smear was negligible — but each pixel covered 6 km.

The design of a flyby encounter is therefore a trade between distance (resolution), speed of tracking (smear) and the time needed to observe each target.

### 26.4.4 Pluto and Arrokoth: New Horizons

**New Horizons** was launched on 19 January 2006 on an Atlas V with an additional solid-fuel upper stage, leaving Earth faster than any spacecraft before it (about 16.3 km/s relative to Earth). A Jupiter flyby in February 2007 shortened its journey by three years. Most of the 9.5-year cruise was spent in hibernation, with the spacecraft spinning slowly and sending a weekly beacon tone.

At Pluto on 14 July 2015, it found a geologically active world. **Sputnik Planitia** is a 1 000 km plain of nitrogen ice slowly convecting in cells tens of kilometres across. It also found mountains of water ice several kilometres high, a blue, layered haze and evidence of a possible ocean beneath the ice. Its large moon **Charon** has a vast canyon system and a reddish polar cap made of material from Pluto's escaping atmosphere (Figure 26.8).

**FIGURE 26.8** Charon in enhanced colour from New Horizons. The reddish polar region (top) is coloured by organic material formed from methane that escaped from Pluto. Image: NASA/JPL.

On 1 January 2019, New Horizons flew 3 500 km past **Arrokoth**, a 36 km Kuiper Belt object 6.6 billion km from the Sun — the most distant object ever explored. Arrokoth turned out to be a "contact binary" of two flattened lobes that merged gently at walking speed, supporting the

theory that planetesimals formed by the gentle collapse of clouds of pebbles. New Horizons continues to study the Kuiper Belt, the heliosphere and the faint cosmic optical background, which can only be measured from far beyond the zodiacal dust of the inner Solar System.

## 26.5 Power and Communications Far from the Sun

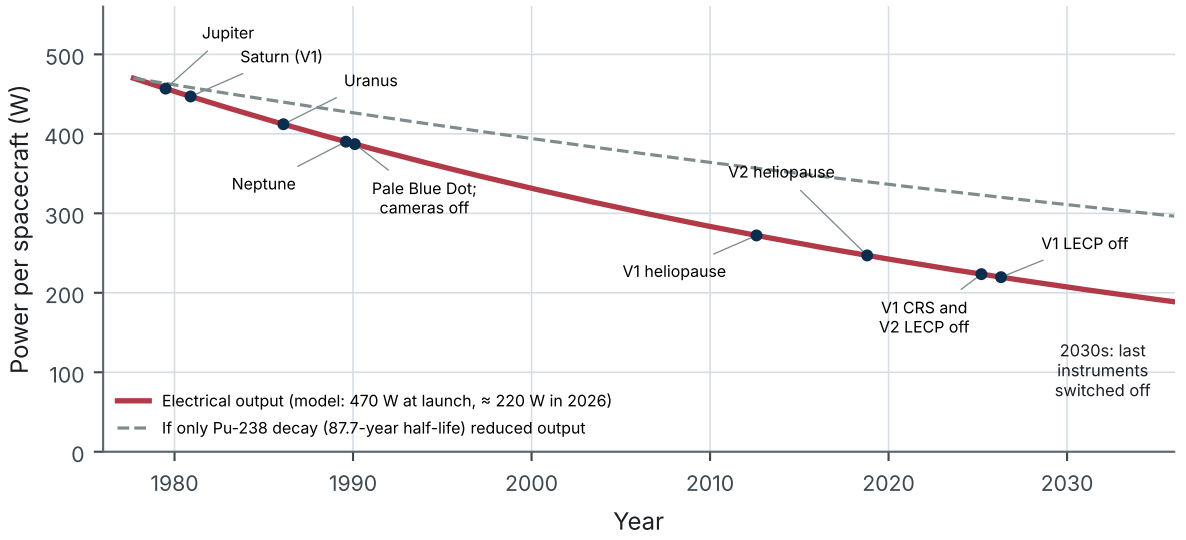
### 26.5.1 Radioisotope Power

Sunlight weakens with the square of the distance: at Jupiter it is 1/27 as strong as at Earth, at Saturn 1/90 and at Pluto about 1/1 100. Solar arrays work as far as Jupiter — Juno, JUICE, Europa Clipper and Lucy all use very large arrays. Beyond that, every probe has relied on **radioisotope thermoelectric generators** (RTGs), which convert heat from the decay of plutonium-238 into electricity (Chapter 5).

**TABLE 26.3** Radioisotope thermoelectric generators used on US deep-space probes.

| Generator | Electrical power at launch | Mass | Missions |
|---|---|---|---|
| SNAP-19 | ≈ 40 W | 14 kg | Pioneer 10 and 11 (four each), Viking landers |
| MHW-RTG | ≈ 158 W | 38 kg | Voyager 1 and 2 (three each), LES-8 and 9 |
| GPHS-RTG | ≈ 290 W | 56 kg | Ulysses, Galileo, Cassini, New Horizons |
| MMRTG | ≈ 110 W | 45 kg | Curiosity, Perseverance, Dragonfly |
| Next-Generation RTG | ≈ 250 W (design) | — | In development for missions of the 2030s |

An RTG's output declines for two reasons: the plutonium decays with a half-life of 87.7 years, and the thermocouples degrade as they age. For the Voyagers, the combined effect is a loss of about 4 W per year (Figure 26.9). From 470 W at launch, each Voyager had about 220 W in 2026. To keep the spacecraft operating, engineers have switched off the cameras (1990), the scan platforms, heaters and, one by one, the science instruments. Voyager 1's cosmic ray instrument was switched off in February 2025 and its low-energy charged particle instrument in April 2026. Voyager 2's low-energy charged particle instrument was switched off in March 2025. In 2026 Voyager 1 retained only its magnetometer and plasma wave instruments. The team plans to keep at least one instrument running on each spacecraft into the 2030s.



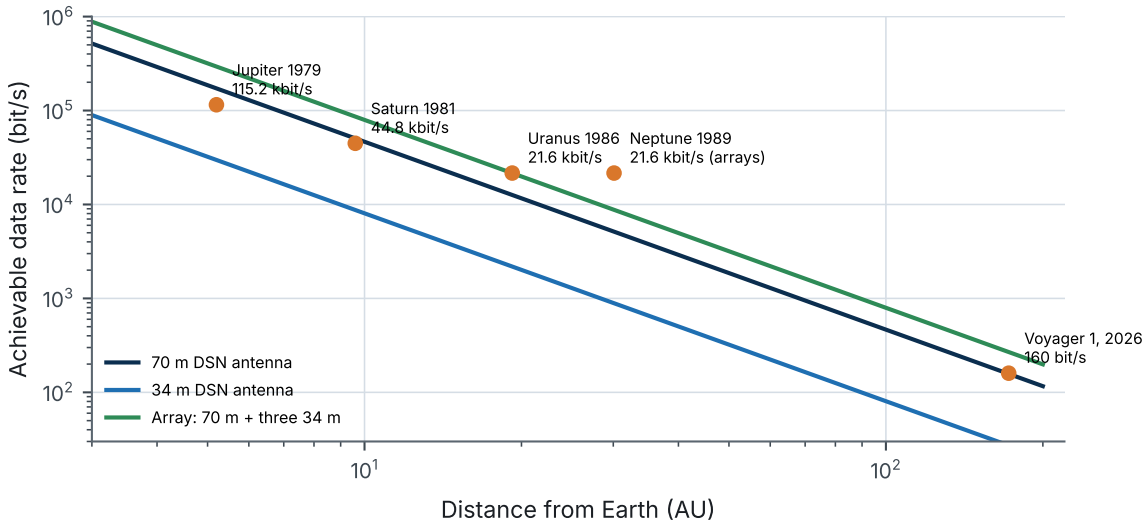

Three MHW-RTGs per spacecraft. Output also falls as the thermocouples age; the decline is now about 4 W per year.

**FIGURE 26.9** Electrical power available on each Voyager. Ageing thermocouples make the decline faster than plutonium decay alone; instruments are switched off as power falls.

Keeping 49-year-old spacecraft alive has required remarkable engineering. In November 2023, Voyager 1 began sending meaningless data. Engineers traced the fault to one failed memory chip in its flight data system. In April 2024, they moved the affected code to other parts of the memory — a software patch sent across 24 billion km. In 2025 they revived Voyager 1's primary roll thrusters, which had been considered dead since 2004, as a backup to its degrading secondary thrusters, whose fuel lines are slowly clogging with silicon dioxide.

### 26.5.2 Talking across the Solar System

Radio signals weaken with the square of the distance as well, so a probe's data rate falls as it travels outward (Figure 26.10). At Jupiter in 1979, each Voyager returned up to 115 kilobits per second. At Neptune, ten years later, it achieved 21.6 kbit/s only because NASA had enlarged its 64 m antennas to 70 m and combined them electronically with other radio telescopes. These included the Very Large Array and Australia's Parkes dish. In 2026, Voyager 1 transmits at 160 bits per second; its signal, from a 21 W transmitter, delivers only about $3 \times 10^{-19}$ W to a 70 m antenna on Earth.




Model: Voyager X-band link — 21.3 W, 3.66 m antenna, coded Eb/N0 ≈ 2.5 dB plus 3 dB margin.
At Jupiter and Saturn the DSN dishes were 64 m; for Uranus and Neptune they were enlarged and arrayed with other radio telescopes.

**FIGURE 26.10** Data rate versus distance for a Voyager-class X-band link. The orange points are rates actually achieved.

Because Voyager 2 is heading south, far below the ecliptic, it can be seen only from the southern hemisphere. Only the 70 m antenna at Canberra (DSS-43) has an S-band transmitter able to command it. When that antenna was upgraded between March and October 2020, Voyager 2 could receive no commands. In August 2023 an incorrect command turned Voyager 2's antenna 2° away from Earth. The team regained contact by "shouting" — transmitting at high power from Canberra in the hope that some of the signal would reach the spacecraft, which it did.

### EXAMPLE: VOYAGER 1'S LINK BUDGET IN 2026

Voyager 1 is 172 AU ($2.57 \times 10^{13}$ m) from Earth, transmitting 21.3 W at 8.415 GHz (wavelength 3.56 cm) through a 3.66 m antenna to a 70 m Deep Space Network antenna.

1. Transmit power: $10 \log_{10} 21.3 = 13.3$ dBW.
2. Antenna gains $G = \eta(\pi D/\lambda)^2$: transmit ($\eta = 0.55$) 47.6 dBi; receive ($\eta = 0.60$) 73.6 dBi.
3. Free-space loss $(4\pi d/\lambda)^2 = 319.2$ dB.
4. Received power: $13.3 + 47.6 + 73.6 - 319.2 = -184.7$ dBW, or $3.4 \times 10^{-19}$ W.
5. Noise density for a 22 K receiving system: $kT = -228.6 + 13.4 = -215.2$ dBW/Hz. Carrier-to-noise density: 30.5 dB-Hz.
6. At 160 bit/s (22.0 dB-Hz), the energy per bit relative to noise is $E_b/N_0 = 8.5$ dB. Voyager's concatenated coding needs about 2.5 dB, leaving about 6 dB of margin for pointing losses, weather and the noise of the sky.
7. The one-way light time is 23.8 hours: a command sent on Monday morning is answered on Wednesday morning. On 18 November 2026, Voyager 1 becomes the first spacecraft one light-day from Earth.

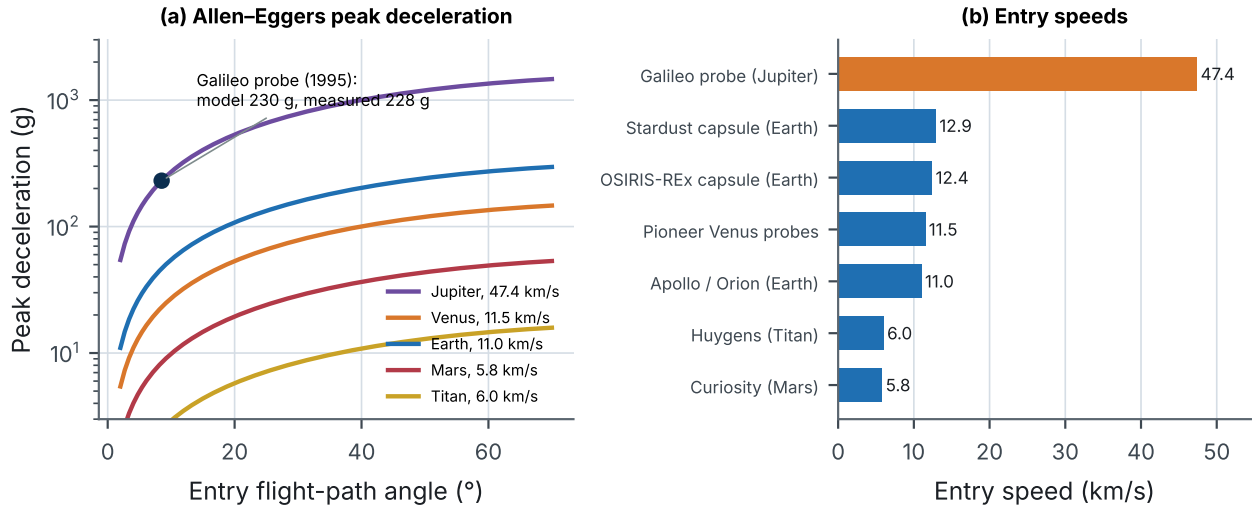
## 26.6 Atmospheric Entry Probes

An entry probe is released from a carrier spacecraft weeks before arrival, coasts to the planet and enters its atmosphere at high speed, protected by a heat shield. Once it has slowed, it deploys a parachute and descends, measuring temperature, pressure, composition, clouds and winds and

relaying its data to the carrier spacecraft overhead.

The Soviet **Venera 4** returned the first data from inside the atmosphere of Venus in 1967 before being crushed by the pressure. NASA's **Pioneer Venus Multiprobe** dropped one large and three small probes into Venus in December 1978; one of them survived impact and transmitted from the surface for 67 minutes. In 1985 the Soviet–French **Vega** missions placed two balloons in the Venus atmosphere at 54 km altitude, where they floated for two days, tracked by radio telescopes around the world.

**The Galileo probe** remains the only probe to have entered a giant planet. Released five months before arrival, it entered Jupiter's atmosphere on 7 December 1995 at 47.4 km/s — the fastest atmospheric entry ever — and decelerated at up to 228 *g*. Its carbon-phenolic heat shield, 152 kg of the probe's 339 kg, lost about 80 kg to ablation. The probe transmitted for 58 minutes and descended about 150 km below the cloud tops, to a pressure of 22 bar. It found less water than expected, because it had entered a dry "hot spot". It also found winds of over 600 km/h and heavy noble gases more enriched than models predicted, which changed theories of how Jupiter formed. ESA's **Huygens** probe, which descended through Titan's atmosphere and landed on its surface in 2005, is described in Chapter 27.




Peak deceleration $a = v^2 \sin\gamma / (2\,e\,H)$ for a ballistic entry into an exponential atmosphere of scale height H; it does not depend on the vehicle's mass or shape.

**FIGURE 26.11** Entry physics. (a) Peak deceleration depends only on entry speed, flight-path angle and the atmosphere's scale height; the Galileo probe's measured 228 g matches this simple model. (b) The Galileo probe entered at four times the speed of a lunar return capsule.

The heating rate at the stagnation point of an entry vehicle rises roughly as $\rho^{1/2} v^3$. The Galileo probe's entry was therefore far more severe than any Earth return. Much of its heating came as radiation from the extremely hot gas of the shock layer. For entries into Earth's atmosphere, see Chapters 31 and 29.

**EXAMPLE: THE GALILEO PROBE'S DECELERATION**

For a ballistic entry at speed $v_e$ and flight-path angle γ into an exponential atmosphere of scale height $H$, the peak deceleration (Allen and Eggers, 1958) is

$$a_{\max} = \frac{v_e^2 \sin\gamma}{2\,e\,H}.$$

For the Galileo probe, $v_e$ = 47.4 km/s, γ = 8.5° and $H \approx 27$ km in Jupiter's upper atmosphere:

$a_{\max} = (47\,400)^2 \times 0.148 / (2 \times 2.718 \times 27\,000) = 2\,260$ m/s$^2 \approx 230\,g$,

within 1 % of the measured 228 *g*. Steeper entries give higher decelerations but shorter, less intense heat pulses; shallower entries risk skipping out of the atmosphere. Probe designers choose γ to keep both deceleration and total heat load within limits.

New entry probes are planned but not yet secure. NASA's **DAVINCI** would descend through the atmosphere of Venus in the 2030s, and the US planetary science decadal survey (2022) chose a **Uranus Orbiter and Probe** as the highest-priority new flagship mission. In 2026, budget pressures left the timing of both uncertain.

## 26.7 Impactors and Planetary Defence

### 26.7.1 Deep Impact

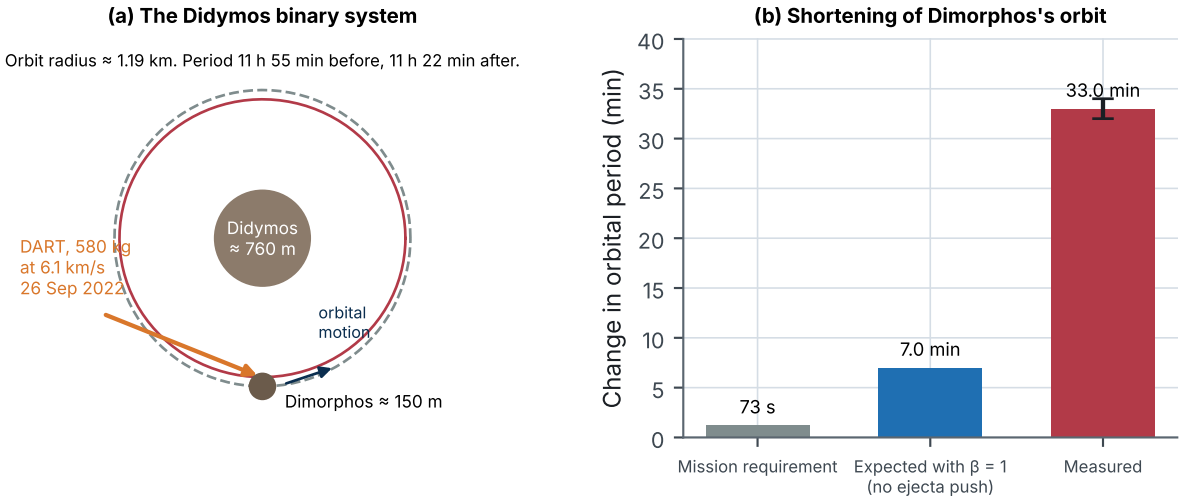
Some questions can be answered only by hitting a target. NASA's **Deep Impact** released a 372 kg copper-rich impactor into the path of comet Tempel 1, which struck the nucleus at 10.2 km/s on 4 July 2005 (Figure 26.12). In its final two hours the impactor steered itself using **AutoNav**, an autonomous navigation system that tracked the brightest part of the nucleus. The flyby spacecraft, 500 km away, observed the plume of ejecta, which revealed that the nucleus is covered in fine dust as loose as talcum powder. In 2011, NASA's Stardust-NExT spacecraft flew past Tempel 1 and imaged the 150 m crater.

**FIGURE 26.12** The flash and ejecta of Deep Impact's collision with comet Tempel 1 on 4 July 2005, imaged by the flyby spacecraft 67 seconds after impact. Image: NASA/JPL (NASA/JPL-Caltech/UMD).

### 26.7.2 DART: Moving an Asteroid

Asteroid impacts are rare but potentially catastrophic. An object 140 m across, striking land, could devastate a whole metropolitan region. The 10 km asteroid that ended the age of the dinosaurs released billions of times the energy of the Hiroshima bomb. The simplest way to deflect an asteroid that is discovered years before impact is a **kinetic impactor** — a spacecraft that strikes it to change its velocity slightly, so that years later it misses Earth.

NASA's **Double Asteroid Redirection Test (DART)** tested this idea on **Dimorphos**, a 150 m moon of the asteroid Didymos (Figure 26.13, Figure 26.14). A binary asteroid was chosen because a change in the moon's 12-hour orbit around its primary could be measured from Earth within weeks. A change in an asteroid's orbit around the Sun, by contrast, would take years to detect. DART struck Dimorphos at 6.1 km/s on 26 September 2022, steering itself during the last four hours with an autonomous navigation system (SMART Nav) that identified Dimorphos only an hour before impact. The Italian cubesat **LICIACube**, released 15 days earlier, flew past three minutes after the impact and imaged the ejecta plume.




Ejecta thrown back from the impact site multiplied the push: momentum enhancement β ≈ 3.6 (2.2–4.9, depending on Dimorphos's density, which Hera will measure).

**FIGURE 26.13** DART's impact on Dimorphos. (a) The Didymos system. (b) The orbit period shortened by 33 minutes, far more than DART's momentum alone could explain.

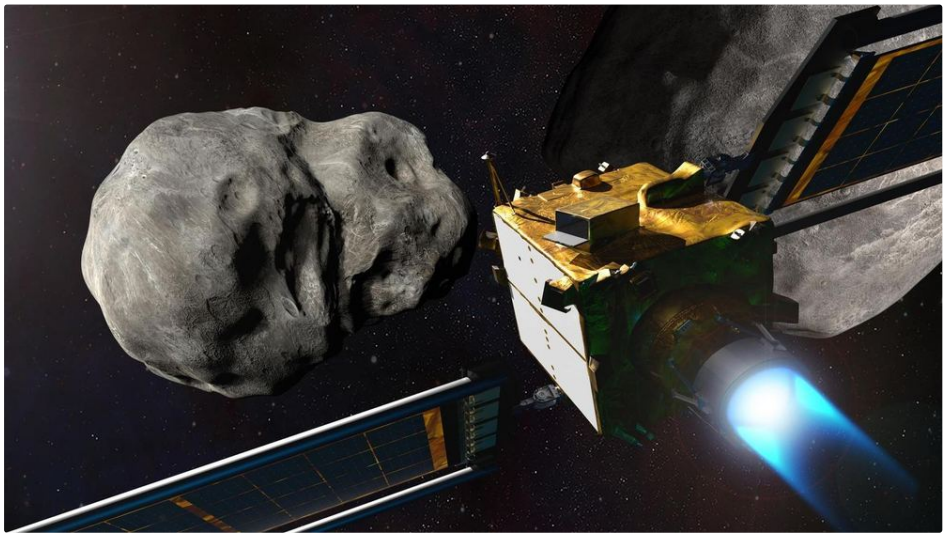


**FIGURE 26.14** Artist's illustration of DART approaching the binary asteroid Didymos (left) and its moon Dimorphos. Image: NASA/JPL (NASA/Johns Hopkins APL/Steve Gribben).

Telescope observations showed that Dimorphos's orbital period had shortened by 33.0 ± 1.0 minutes, from 11 h 55 min — more than 25 times the mission's minimum success criterion of 73 seconds. More than a million kilograms of rock were thrown off, forming a tail thousands of kilometres long; the recoil from this ejecta pushed Dimorphos several times harder than the spacecraft did. The ratio of the total momentum change to the momentum DART delivered, the **momentum enhancement factor β**, is about 3.6 (between 2.2 and 4.9, depending on Dimorphos's unknown density). In 2026, stellar occultation and radar measurements showed that the impact had also changed the orbit of the whole Didymos system around the Sun, by about 12 micrometres per second.

**EXAMPLE: HOW HARD DID DART PUSH?**

Dimorphos orbits 1.19 km from the centre of Didymos with a period $T$ = 11.92 h, so its orbital speed is $v = 2\pi \times 1\,190$ m / 42 900 s = 0.174 m/s.

1. For a nearly circular orbit, a small along-track velocity change $\Delta v$ changes the period by $\Delta T/T \approx 3\,\Delta v/v$.
2. The measured $\Delta T = -33$ min gives $\Delta T/T = -0.046$, so $\Delta v \approx 0.046 \times 0.174 / 3 = 2.7$ mm/s.
3. Dimorphos has a volume of about $1.8 \times 10^6$ m$^3$. For a density of 2 400 kg/m$^3$, its mass is $4.3 \times 10^9$ kg. DART's momentum (580 kg × 6 145 m/s) could change its velocity by at most $3.6 \times 10^6 / 4.3 \times 10^9 = 0.83$ mm/s.
4. The observed 2.7 mm/s is more than three times larger: the ejecta, thrown backwards out of the crater, provided most of the push.

Because the result depends on Dimorphos's mass, ESA's Hera mission (arriving in November 2026) will measure it, along with the shape of the crater, so that future deflection missions can be designed with confidence.

### 26.7.3 Finding the Threats First

A deflection mission needs years of warning, so planetary defence begins with discovering near-Earth objects (NEOs). US law set the goal of finding 90 % of NEOs larger than 140 m; by 2026 well under half of them had been found. Ground surveys such as Catalina, Pan-STARRS, ATLAS and the Vera C. Rubin Observatory, which began its survey in 2025, will be joined by NASA's **NEO Surveyor**, an infrared telescope to be stationed at the Sun–Earth L1 point, with launch targeted for September 2027. The value of these surveys was shown by asteroid **2024 YR4**, about 60 m across, discovered in December 2024. In February 2025 its calculated chance of striking Earth in 2032 briefly reached 3 %, the highest ever recorded for an object of its size. Further observations then ruled out an Earth impact (a small chance of hitting the Moon remains).

On 13 April 2029, the 340 m asteroid **Apophis** will pass about 32 000 km above Earth's surface — closer than geostationary satellites — and be visible to the naked eye. It will be studied by NASA's OSIRIS-APEX (the extended OSIRIS-REx spacecraft, Chapter 29) and by ESA's **Ramses**. Ramses was approved in 2025 for launch in April 2028 on a Japanese H3 rocket, and will observe how Earth's gravity shakes and reshapes the asteroid. China plans its own impactor-and-observer test later in the decade.

## 26.8 Comets, Asteroids and Interstellar Visitors

Flybys of small bodies have revealed their bewildering variety. Galileo's flyby of **Ida** in 1993 discovered **Dactyl**, the first known asteroid moon. **Deep Space 1** (1998–2001) tested ion propulsion and autonomous navigation and flew past comet Borrelly. **Stardust** collected dust from comet Wild 2 (Chapter 29). Rosetta flew past the asteroids Šteins and Lutetia on its way to its comet.

NASA's **Lucy**, launched in October 2021, is touring the **Trojan asteroids** — primitive bodies sharing Jupiter's orbit at its L4 and L5 Lagrange points (Chapter 3). On its way it flew past the main-belt asteroid **Dinkinesh** in November 2023, discovering that it has a moon, Selam, which is itself a contact binary, and past **Donaldjohanson** in April 2025. Lucy will fly past the Trojans Eurybates and its moon Queta (August 2027), Polymele (September 2027), Leucus (April 2028) and Orus (November 2028), and finally the binary pair Patroclus–Menoetius in March 2033.

Three objects from outside the Solar System have been discovered passing through it. The first was **'Oumuamua** (2017), an elongated body that accelerated slightly as it left, and the second was the comet **2I/Borisov** (2019). The third, **3I/ATLAS**, was discovered on 1 July 2025. It passed perihelion at 1.36 AU at the end of October 2025, moving at about 58 km/s relative to the Sun. When 3I/ATLAS was hidden behind the Sun as seen from Earth, spacecraft across the Solar System observed it. Mars orbiters observed it as it passed Mars in early October 2025, and the solar observatories SOHO and STEREO did so near perihelion. Later came ESA's JUICE and NASA's SPHEREx, which found a coma rich in carbon dioxide and a strong increase in water after perihelion. ESA's **Comet Interceptor**, to be launched in 2028 or 2029, will wait at the Sun–Earth L2 point for a newly discovered comet — or interstellar object — and then intercept it with three spacecraft.

## 26.9 Into Interstellar Space

The Sun blows a supersonic wind of charged particles, the solar wind, that carves a bubble in the interstellar medium — the **heliosphere** (Chapter 22). About 90 AU from the Sun the wind passes through a **termination shock**, where it slows abruptly; beyond it, in the **heliosheath**, the slowed, heated wind is deflected towards a long tail. The **heliopause** is the boundary where the solar wind meets the interstellar plasma (Figure 26.15).

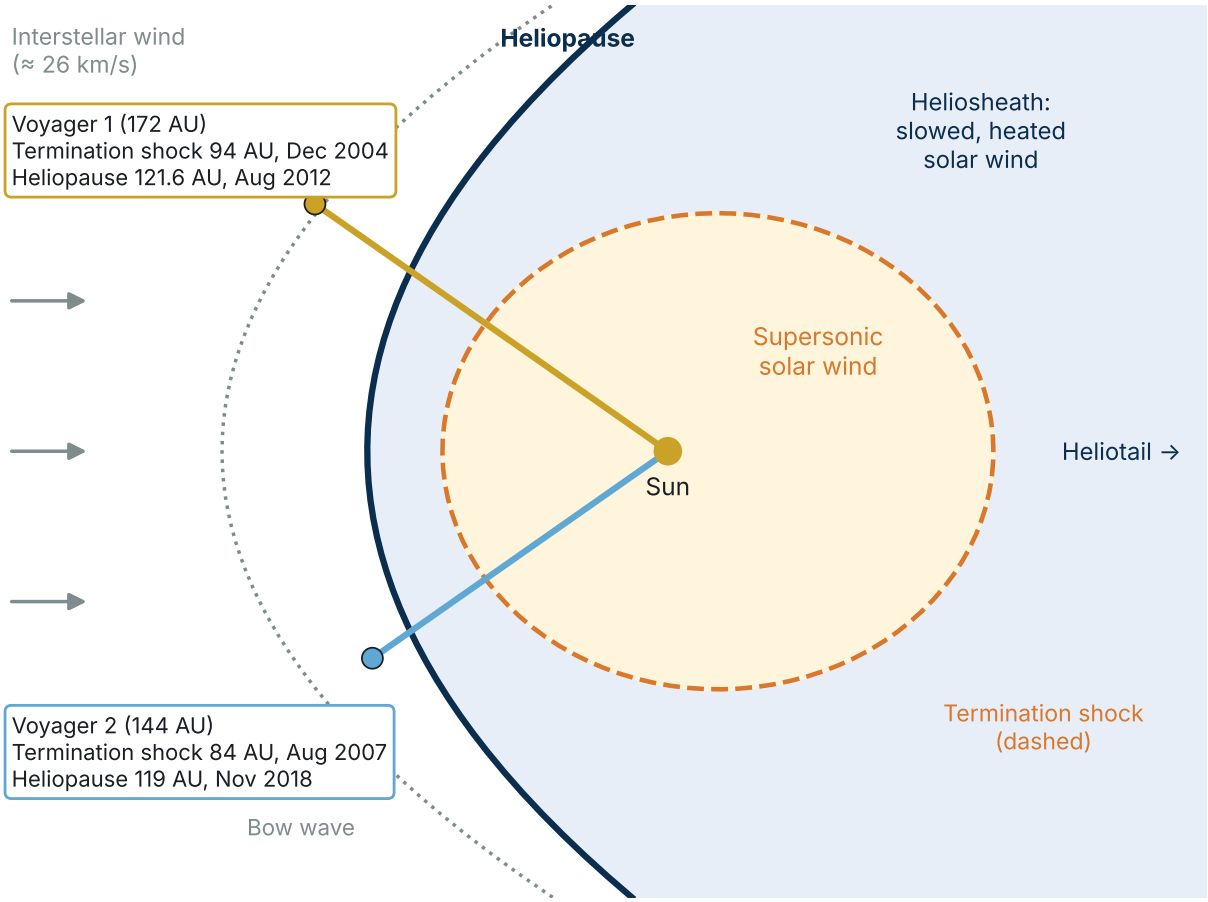



**FIGURE 26.15** The heliosphere and the Voyagers' crossings of its boundaries (schematic, not to scale).

Voyager 1 crossed the termination shock in December 2004 at 94 AU and the heliopause on 25 August 2012 at 121.6 AU, becoming the first human-made object in interstellar space. Voyager 2 followed in November 2018 at 119 AU. The crossings were identified by a sudden fall in solar-wind particles and a jump in galactic cosmic rays. Because Voyager 1's plasma instrument had failed decades earlier, they were also identified by its plasma wave instrument. This detected the plasma's natural oscillation frequency and revealed a density about 40 times higher than in the heliosheath. Surprisingly, the direction of the magnetic field barely changed across the heliopause. The Voyagers now measure the interstellar medium directly, including the faint "hum" of plasma waves, and show that the Sun's influence extends well beyond the heliopause.

The **Pioneer anomaly** — a tiny unexplained sunward acceleration of Pioneers 10 and 11, about $8.7 \times 10^{-10}$ m/s$^2$ — was once proposed as evidence of new physics. In 2012, a careful analysis of recovered tracking data and a thermal model of the spacecraft explained the anomaly. It was caused by the uneven emission of heat from the RTGs and electronics, which acts as a minute thrust.

No spacecraft is yet on its way to explore the interstellar medium deliberately. Concept studies of an **Interstellar Probe** assume a launch in the 2030s with a super-heavy launcher and a close gravity assist from Jupiter. They aim at speeds of about 7 AU per year — twice the Voyagers' — to reach several hundred AU within a working lifetime. Light-sail concepts that could reach other stars are discussed in Chapter 42.

## 26.10 Probes in 2026

**TABLE 26.4** Probes and flyby missions in 2026.

| Spacecraft | Status in October 2026 |
|---|---|
| Voyager 1 | 172 AU; one light-day from Earth on 18 November 2026; magnetometer and plasma wave instruments operating |
| Voyager 2 | About 144 AU, heading south; magnetometer, plasma wave and cosmic ray instruments operating early in 2026 |
| New Horizons | About 66 AU; Kuiper Belt and heliosphere observations; extended mission until it leaves the Kuiper Belt (≈ 2028–29) |
| Lucy | Cruising to the L4 Trojans; Eurybates flyby August 2027 |
| Psyche | Mars flyby May 2026; arrival at asteroid Psyche August 2029 |
| Europa Clipper | Earth flyby December 2026; Jupiter arrival April 2030 |
| JUICE | Earth flyby on 28 September 2026 added 3.5 km/s; one more Earth flyby (January 2029); Jupiter in July 2031 |
| Hera | Final approach; arrival at Didymos in November 2026 |
| BepiColombo | Mercury orbit insertion November 2026 (Chapter 25) |
| Parker Solar Probe | Closest solar passes at 6.1 million km and 192 km/s (Chapter 22) |
| In development | NEO Surveyor (2027), Ramses (2028), Comet Interceptor (2028–29), Dragonfly (2028, Chapter 28), Tianwen-4 to Jupiter and Uranus (≈ 2029–30) |

### AUTONOMY AND AI ON FLYBY MISSIONS

Flybys and impacts happen too fast and too far away for ground control, so they were among the first uses of onboard autonomy. **AutoNav**, developed for Deep Space 1 (1999–2001), navigated by images of asteroids against the stars and steered Deep Impact's impactor into Tempel 1. DART's **SMART Nav** found Dimorphos in its camera images an hour before impact and guided the spacecraft into the centre of the 150 m target. Lucy uses an autonomous **terminal tracking** system that keeps an asteroid centred in its instruments' fields of view during the fastest part of each flyby. Machine-learning tools are now used on the ground to plan encounter sequences, to detect new moons, rings and plumes in flyby images, and to search survey data for near-Earth asteroids and interstellar objects (Chapter 51).

### KEY TAKEAWAYS

- Probes include flyby spacecraft, atmospheric entry probes, impactors and heliospheric explorers. Flybys are the cheapest way to reach a new world and remain our only close-up views of Uranus, Neptune, Pluto and the Kuiper Belt.
- Flyby engineering requires precise targeting in the B-plane, autonomous encounter sequences protected against faults, and imaging strategies that trade resolution against smear.
- Beyond Jupiter, probes rely on RTGs, whose output declines by a few watts each year; data rates fall with the square of distance, and Voyager 1 now returns 160 bit/s from 172 AU.
- Entry probes face extreme heating and deceleration — up to 228 g for the Galileo probe at Jupiter — that can be estimated from simple physics.
- DART showed that a kinetic impactor can change an asteroid's orbit: Dimorphos's period shortened by 33 minutes, with ejecta multiplying the push about 3.6 times. Hera arrives in November 2026 to measure the result.
- The Voyagers crossed the heliopause in 2012 and 2018 and are now sampling interstellar space; Voyager 1 reaches one light-day from Earth in November 2026.

### REVIEW QUESTIONS

1. Why are flybys the cheapest way to reach a new world, and what are their scientific limitations?
2. What is B-plane targeting, and why must a flyby be navigated so precisely?
3. Voyager 1 returns 160 bit/s from about 172 AU. With the same antennas and transmitter, what rate would it have had at 86 AU?
4. Plutonium-238 has a half-life of 87.7 years. What fraction of its heat output remains after 49 years? Why has the electrical output of the Voyager RTGs fallen faster than this?
5. Why did the Galileo probe experience a peak deceleration of about 228 g? What does this imply for the design of entry probes?
6. What did DART demonstrate, and what is the momentum-enhancement factor β? Why did β exceed 1?
7. How do flyby sequences protect against faults when the spacecraft is too far away for real-time ground control?
8. What are the heliopause and interstellar space, and what have the Voyagers measured there?

CHAPTER

# 27 Landers and Entry, Descent & Landing

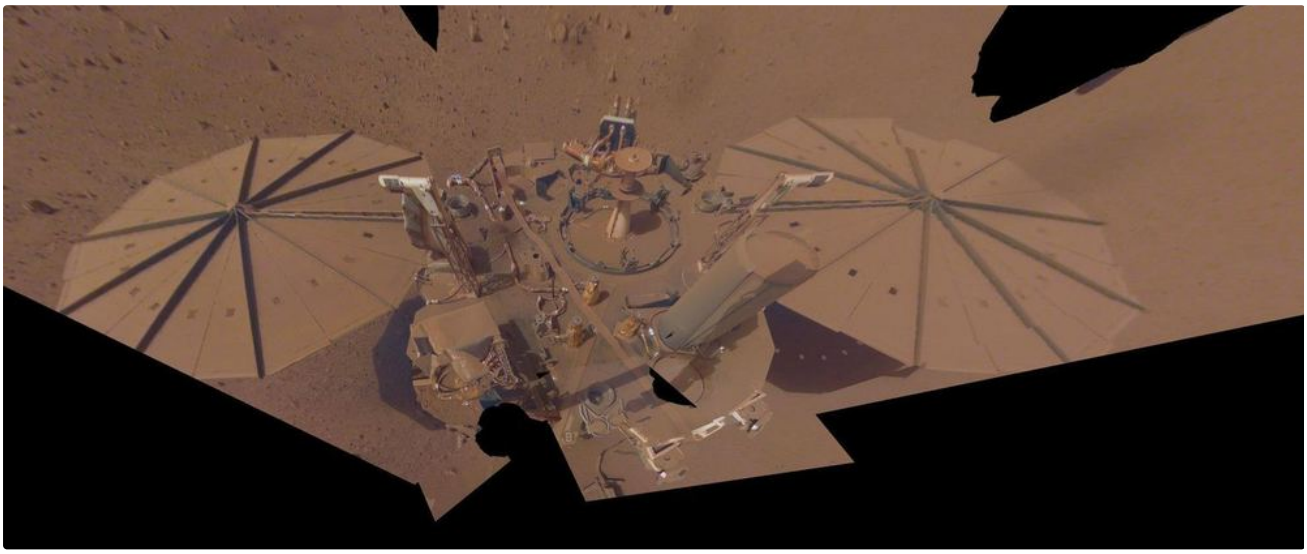


Image: NASA/JPL (NASA/JPL-Caltech)

*Landing is the most dangerous minute of any planetary mission. A lander must shed kilometres per second of speed, find a safe spot it has never seen, and touch down gently enough to survive — all on its own, minutes or hours of light-time from Earth. This chapter explains how this is done on airless worlds like the Moon, on Mars with its treacherously thin atmosphere, in the crushing heat of Venus, under the haze of Titan and in the feeble gravity of comets and asteroids. It describes powered descent, heat shields, parachutes, airbags and sky cranes, terrain-relative navigation and hazard avoidance, the record of lunar landings from Luna 9 to the commercial landers of 2024–25, and what landers have discovered on the surface.*

IN THIS CHAPTER

27.1 Why Land?
27.2 The Landing Problem
27.3 Landing on the Moon
27.4 Entry, Descent and Landing at Mars
27.5 Venus, Titan and Small Bodies
27.6 Surviving on the Surface

**Entry, descent and landing (EDL)** is the sequence of events that takes a spacecraft from its approach trajectory or orbit to a safe rest on a planetary surface. It is the phase in which most planetary missions have failed. Of the 34 robotic attempts to land softly on the Moon up to 2025, 12 crashed and 4 more landed damaged or on their sides. Only a little over half of all attempts to land on Mars have succeeded. This chapter covers stationary **landers**; rovers, helicopters and balloons are described in Chapter 28, and the return of samples to Earth in Chapter 29.

## 27.1 Why Land?

Orbiters map whole worlds, but some measurements can only be made by touching the surface (Table 27.1). A lander can analyse the chemistry and mineralogy of rocks and soil in contact, listen for quakes with a seismometer and measure the flow of heat from the interior. It can also record the weather day after day, search for organic molecules and signs of life, and collect samples for return to Earth. Landers also prove technologies needed for later crewed missions and for building permanent bases.

**TABLE 27.1** Science that requires a lander.

| Measurement | Instrument | Examples |
|---|---|---|
| Interior structure | Seismometer | Apollo passive seismic experiments (Moon, 1969–77); InSight SEIS (Mars, 2018–22) |
| Heat flow from the interior | Heat-flow probe | Apollo 15 and 17; InSight HP³; Blue Ghost LISTER (2025) |
| Surface chemistry and mineralogy | X-ray, gamma-ray and laser spectrometers, microscopes | Venera 13 (Venus); Phoenix and Perseverance (Mars); Chandrayaan-3 (Moon) |
| Volatiles and organics | Gas chromatograph–mass spectrometer, ovens | Viking (Mars); Phoenix; Philae (comet 67P) |
| Atmosphere and weather | Pressure, temperature, wind and dust sensors | Viking, InSight, Huygens (Titan) |
| Ground truth for orbital data | Cameras, spectrometers | Every lander |

## 27.2 The Landing Problem

A spacecraft arriving at a planet carries an enormous amount of kinetic energy per kilogram. Coming down from a low lunar orbit at 1.7 km/s, each kilogram carries 1.4 MJ — as much as 30 g of TNT. Entering the Martian atmosphere at 5.4 km/s, it carries 15 MJ; returning to Earth from the Moon at 11 km/s, 60 MJ. All of this energy must be removed before touchdown, together with the potential energy of the altitude. There are only three ways to do it:

- **Aerodynamic drag** — heat shields and parachutes convert kinetic energy into heat in the atmosphere. This is free in terms of propellant but requires an atmosphere and protection from heating.
- **Propulsion** — rocket engines fired against the direction of motion. This works anywhere but is expensive: every metre per second costs propellant, and slowing down against gravity adds **gravity losses**.
- **Impact attenuation** — airbags, crushable structures or shock-absorbing legs absorb the last few metres per second at touchdown.

Every lander combines these in proportions set by its destination (Table 27.2).

**TABLE 27.2** Landing on different worlds.

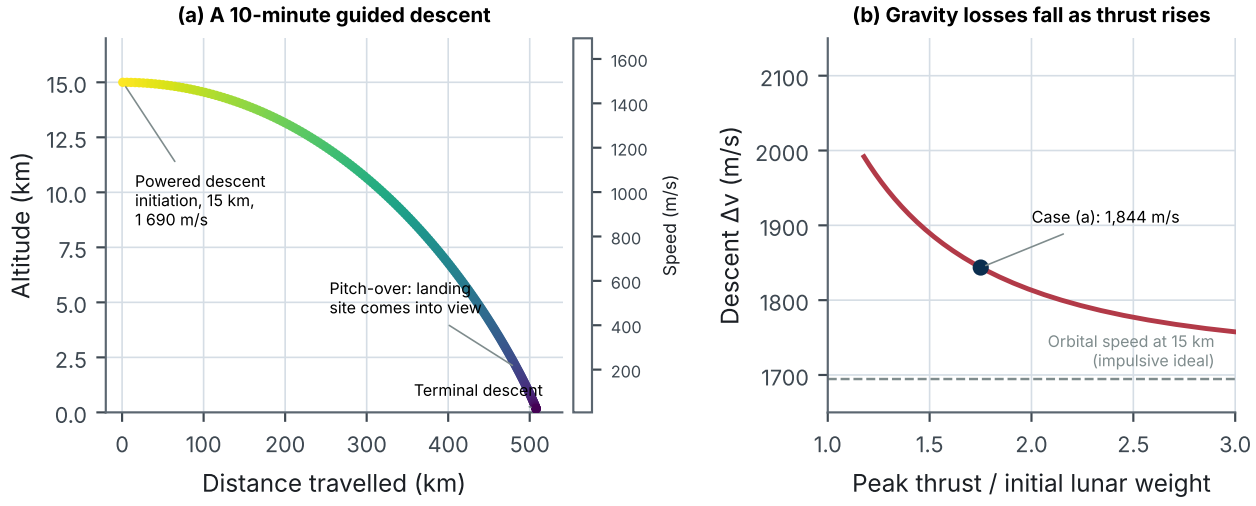
| Destination | Surface gravity | Atmosphere | How landers slow down |
|---|---|---|---|
| Moon, Mercury | 1.6 / 3.7 m/s² | None | Entirely propulsive, from orbital speed |

| Destination | Surface gravity | Atmosphere | How landers slow down |
|---|---|---|---|
| Mars | 3.7 m/s² | Thin (≈ 0.6 % of Earth's surface pressure) | Heat shield removes ≈ 90 % of speed; parachute; propulsion or airbags for the last 80–100 m/s |
| Venus | 8.9 m/s² | Very dense (92 bar at the surface) | Heat shield, parachute high up, then free fall: the dense air limits the landing speed to a few metres per second |
| Titan | 1.35 m/s² | Dense and extended (1.5 bar) | Heat shield and parachutes all the way to the surface |
| Asteroids, comets | $10^{-5}$–$10^{-3}$ m/s² | None | Small thrusters; the challenge is to avoid bouncing off |

# 27.3 Landing on the Moon

## 27.3.1 Powered Descent

With no atmosphere, a lunar lander must cancel its entire orbital velocity with its engines. The Apollo Lunar Module (LM) set the pattern still used today. It first lowered its orbit to a periapsis of about 15 km, 480 km before the landing site. At **powered descent initiation**, its throttleable descent engine fired against the direction of motion. During the **braking phase**, lasting about eight minutes, the engine removed most of the 1 690 m/s orbital speed while the lander gradually lost height. At the **pitch-over**, about 2 km up, the LM rotated to a more upright attitude so the crew — or, today, the cameras — could see the landing site. In the **approach phase** the site was checked and, if necessary, changed; and in the **terminal descent** the lander came straight down at a metre per second or so (Figure 27.1).




Simulation: Apollo-class lander (15.1 t, Isp 311 s) from a 15 × 110 km orbit to 150 m altitude above the site, using an optimal-guidance law of the kind flown by Apollo.

**FIGURE 27.1** Lunar powered descent. (a) A simulated Apollo-class descent from 15 km altitude. (b) Descent Δv versus the peak thrust-to-weight ratio: a weak engine must fight gravity for longer and wastes more propellant.

If the lander could stop instantly, it would need a Δv equal to its orbital speed. In reality it must also hold itself up against gravity during the descent, and the total Δv is 1.8–2.1 km/s, including margins for hovering and diverting to a safer site. The higher the thrust-to-weight ratio, the

shorter the burn and the smaller these **gravity losses** — but bigger engines are heavier and must be throttled deeply to hover.

> **EXAMPLE: PROPELLANT FOR A LUNAR DESCENT**
>
> An Apollo-class lander of 15 100 kg begins powered descent at 15 km. Its engine has a specific impulse of 311 s ($v_e$ = 3 050 m/s).
>
> 1. The guided descent of Figure 27.1 needs Δv = 1 844 m/s to reach 150 m above the site. Adding about 200 m/s for the final descent, hovering and redesignation margin gives about 2 040 m/s.
> 2. Rocket equation: $m_p = m_0\,(1 - e^{-\Delta v / v_e})$ = 15 100 × (1 − e^{−2 040/3 050}) = 7 360 kg.
> 3. Nearly half the lander's mass is descent propellant. The real LM descent stage carried about 8 200 kg, the rest being reserve. Apollo 11 touched down with less than a minute of usable propellant left before the abort limit, after Neil Armstrong flew beyond a boulder field.

### 27.3.2 Navigation and Hazard Avoidance

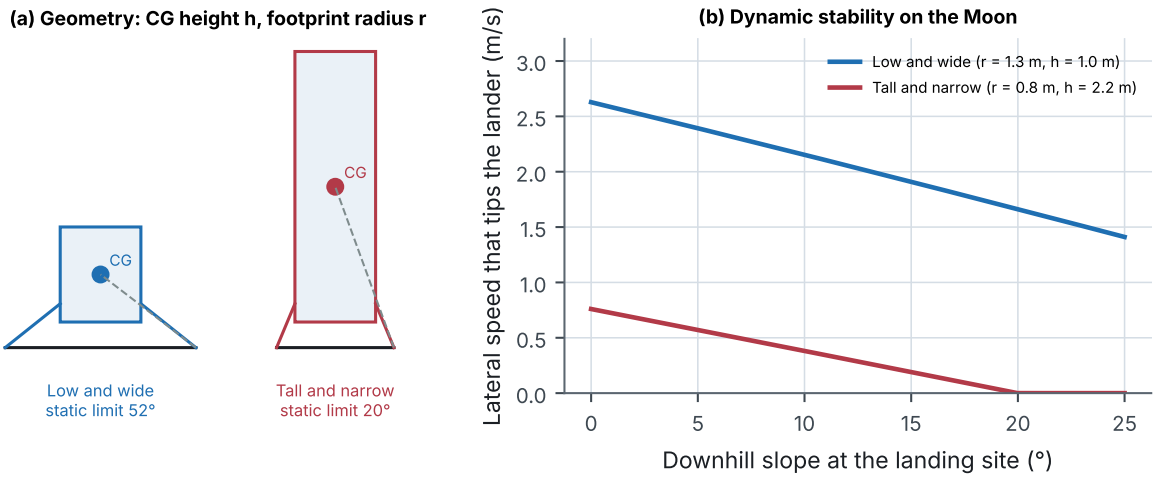
A lander must know its altitude and velocity precisely. Apollo used a **landing radar** that measured altitude and, by the Doppler shift of four beams, velocity. Modern landers add **laser rangefinders** and **Doppler lidar** — NASA's Navigation Doppler Lidar measures velocity to a few centimetres per second. They also add cameras for **terrain-relative navigation (TRN)**. TRN matches images of craters and other landmarks with an onboard map to determine position to tens of metres.

The final hazard is the landing site itself: a boulder or a slope of more than 10–15° can tip a lander over. **Hazard detection and avoidance** systems image the terrain below with lidar or cameras during the last few hundred metres, build a map of slopes and rocks, and steer to a safe spot. China's Chang'e 3 lander hovered 100 m above the Moon in 2013 to choose its landing point, and every Chang'e lander since has done the same. Japan's **SLIM** (Smart Lander for Investigating Moon) demonstrated vision-based "pinpoint" landing in January 2024. It touched down about 55 m from its target, within its 100 m goal. It did so even though one of its two main engines lost a nozzle about 50 m above the surface and the lander came to rest on its nose.

### 27.3.3 Touchdown

Landing legs must absorb the remaining vertical speed, typically 1–3 m/s, and keep the lander upright on slopes and rocks. Apollo's legs contained **crushable aluminium honeycomb** cartridges that absorbed energy by deforming permanently (Figure 27.2). A lander tips over when its centre of gravity (CG) passes outside the line joining its downhill feet. On level ground, the static limit is the angle whose tangent is the footprint radius divided by the CG height. But a sideways velocity at touchdown makes the lander rotate about its downhill feet. A tall, narrow lander can be tipped by less than 1 m/s of lateral speed on a modest slope (Figure 27.3). Both Intuitive Machines landers to date (2024 and 2025) — tall vehicles 4.3 m high — ended on their sides.

**FIGURE 27.2** Apollo 11's Buzz Aldrin beside a leg of the Lunar Module Eagle in July 1969. The footpad and the crushable struts above it absorbed the landing shock. Image: NASA/JSC.




Lander pivots about its downhill feet; it tips if the kinetic energy of rotation exceeds the work needed to raise the CG over the pivot. Radius of gyration 0.8 m; lunar gravity.

**FIGURE 27.3** Tip-over stability. (a) A low, wide lander tolerates much steeper slopes than a tall, narrow one. (b) The lateral speed at touchdown that would tip each lander over, as a function of slope.

The engine exhaust also interacts with the surface. Plumes blast away dust and rocks at high speed, erode shallow craters and can damage nearby hardware. When Apollo 12 landed about 160 m from the robotic Surveyor 3 lander in 1969, the astronauts found Surveyor's surfaces sandblasted by the LM's plume (Figure 27.4). Plume–surface interaction is a major concern for the large Artemis landers and for future lunar bases.

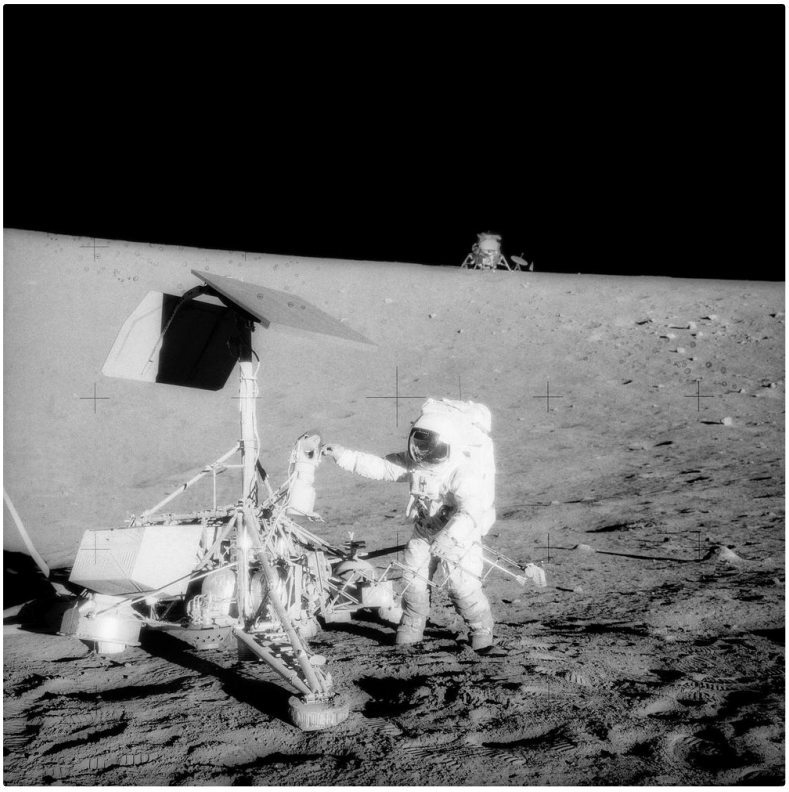
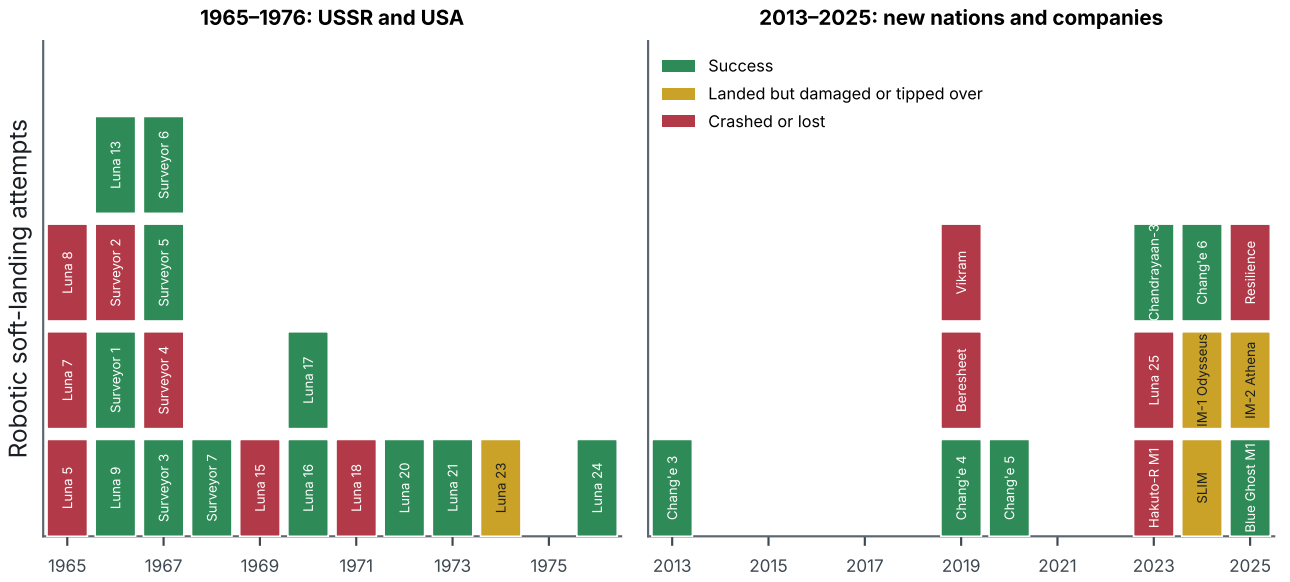


**FIGURE 27.4** Apollo 12's Pete Conrad inspects Surveyor 3, which had landed 31 months earlier; the Lunar Module Intrepid stands on the horizon. Parts brought back to Earth showed sandblasting by the LM's plume. Image: NASA/JSC.

### 27.3.4 Sixty Years of Lunar Landings

The Soviet **Luna 9** made the first soft landing on another world in February 1966, using an inflated airbag-like ball ejected just before impact. NASA's **Surveyor 1** followed four months later with a fully propulsive landing. Between 1966 and 1976, five of seven Surveyors and seven Soviet Luna landers succeeded, including three sample-return missions and two Lunokhod rovers, and six Apollo crews landed. Then, for 37 years, no spacecraft landed on the Moon — until China's **Chang'e 3** in 2013 (Figure 27.5).




Excludes the six crewed Apollo landings (1969–72, all successful) and missions lost before reaching the Moon. No lunar landing was attempted in 2026 up to October.

**FIGURE 27.5** Robotic lunar soft-landing attempts by year and outcome.

The 2010s and 2020s opened the Moon to new nations and to private companies. China has landed four times without failure, including the first landings on the far side (Chang'e 4, 2019) and the first far-side sample return (Chang'e 6, 2024). India's **Chandrayaan-3** became the first lander near the south pole region, at 69° S, in August 2023. Through NASA's **Commercial Lunar Payload Services (CLPS)**, companies now deliver NASA instruments under fixed-price contracts. Firefly's **Blue Ghost Mission 1** became the first fully successful commercial landing on 2 March 2025 in Mare Crisium. It operated for a full lunar day of 14 Earth days.

The failures are equally instructive (Table 27.3). Most were caused not by hardware breaking but by **navigation and software**: sensors that were not enabled, data rejected as faulty, or unexpected terrain that confused altitude estimates.

**TABLE 27.3** Lessons from recent lunar landing failures and anomalies.

| Mission | Date | What happened | Lesson |
|---|---|---|---|
| Beresheet (SpaceIL) | Apr 2019 | An inertial measurement unit fault led to a computer reset that shut down the main engine during descent | Fault responses must not remove thrust during descent |
| Vikram (ISRO) | Sep 2019 | Higher-than-expected thrust in the braking phase; the guidance could not recover | Robust guidance margins; Chandrayaan-3 widened them and succeeded |
| Hakuto-R M1 (ispace) | Apr 2023 | Passing over a crater rim, the altitude estimate jumped; software rejected correct radar data, and the lander ran out of propellant 5 km up | Test navigation with realistic terrain |
| Luna 25 (Roscosmos) | Aug 2023 | An orbit-lowering burn lasted 127 s instead of 84 s; the lander crashed | Independent burn-cutoff protection |
| SLIM (JAXA) | Jan 2024 | Nozzle lost 50 m up; landed on its nose but operated | Precision landing works; design for off-nominal touchdown |

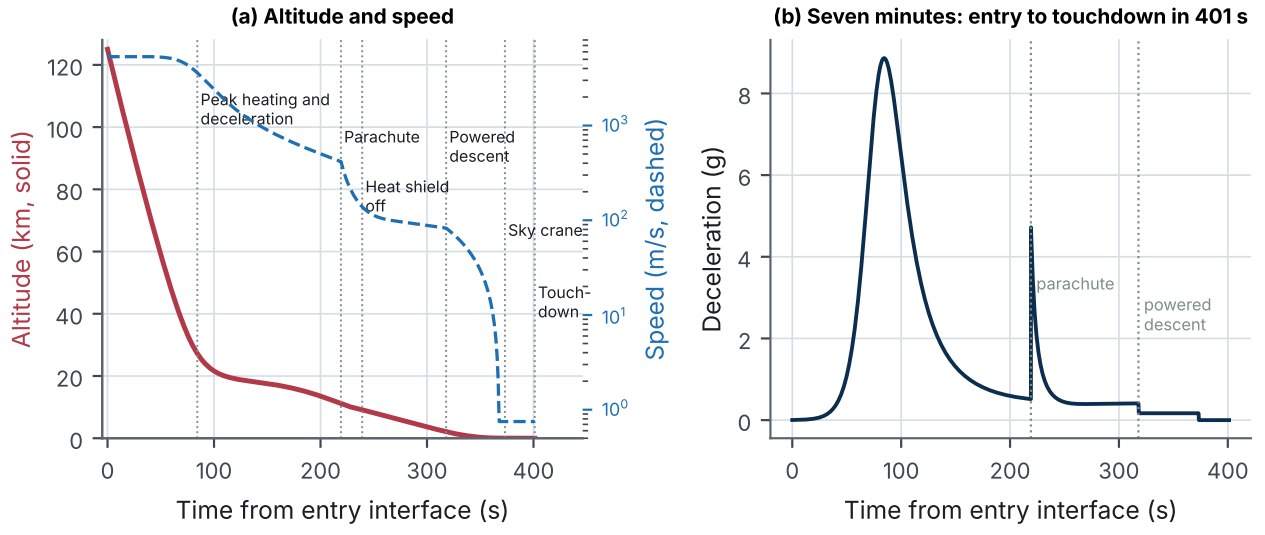
| Mission | Date | What happened | Lesson |
|---|---|---|---|
| IM-1 Odysseus (Intuitive Machines) | Feb 2024 | Laser rangefinders unusable because a safety switch was not enabled before launch; the team switched to a NASA lidar experiment; a leg caught at touchdown and the lander tipped | Pre-launch checks; end-to-end sensor testing |
| IM-2 Athena (Intuitive Machines) | Mar 2025 | Altimeter data degraded over dark polar terrain; landed on its side in a crater | Sensors for low-Sun polar lighting |
| Resilience (ispace) | Jun 2025 | Laser rangefinder measurements came late; the lander could not slow down in time | Sensor qualification for the descent environment |

In 2026, no lunar landing had been attempted by October. Astrobotic's Griffin-1 was due to launch no earlier than November. Blue Origin's Blue Moon MK1, Intuitive Machines' IM-3, Firefly's far-side Blue Ghost Mission 2 and China's Chang'e 7 (which missed its August 2026 launch window) were all expected in 2027.

## 27.4 Entry, Descent and Landing at Mars

### 27.4.1 The Thin-Atmosphere Problem

Mars is the hardest place to land in the inner Solar System. Its atmosphere is thick enough to heat an entering spacecraft to well over 1 000 °C. But it is also very thin — less than 1 % of Earth's surface pressure. A heat shield and parachute alone therefore cannot slow a spacecraft to a safe landing speed. Under the same parachute, a spacecraft descends about five times faster on Mars than it would on Earth, and the parachute must open at supersonic speed. Each Mars lander therefore uses a chain of devices, each taking over when the previous one has done all it can: the famous **"seven minutes of terror"** (Figure 27.6).




Simulation of a Perseverance-class entry: 3.4 t, 4.5 m aeroshell, lift-to-drag 0.24, entry at 5.3 km/s and −15.5°, exponential atmosphere (scale height 11.1 km).

**FIGURE 27.6** Entry, descent and landing at Mars, simulated for a Perseverance-class spacecraft. (a) Altitude and speed with the main events. (b) Deceleration: the heat shield removes about 90 % of the speed in the first two minutes.

### 27.4.2 The EDL Sequence

The sequence flown by NASA's Perseverance rover on 18 February 2021 illustrates the state of the art:

1. **Cruise-stage separation and entry.** Ten minutes before entry, the cruise stage that carried the spacecraft from Earth is jettisoned. The 4.5 m aeroshell enters the atmosphere at about 125 km altitude at 5.4 km/s.
2. **Guided entry.** The capsule flies at an angle of attack that gives it a little lift (lift-to-drag ratio about 0.24). By rolling to point this lift up, down or sideways with small thrusters, the onboard guidance corrects for variations in the atmosphere and steers towards the target, like the Apollo capsules returning to Earth. Peak heating and a deceleration of about 10 $g$ occur about 80 seconds after entry; the heat shield's surface reaches about 1 300 °C.
3. **Parachute.** About four minutes after entry, at 9–11 km altitude and close to Mach 1.7, a 21.5 m disk-gap-band parachute is fired out by a mortar. Perseverance used a **range trigger**, opening the parachute when its position estimate showed it was the right distance from the target, rather than at a fixed speed.
4. **Heat-shield separation.** Twenty seconds later the heat shield is released, exposing the radar and cameras.
5. **Terrain-relative navigation.** From about 4 km altitude, Perseverance's **Lander Vision System** photographed the ground, matched the images with an onboard map made from orbital images, and computed its position to within about 40 m. It then chose the safest reachable spot from a pre-loaded hazard map.
6. **Back-shell separation and powered descent.** At about 2.1 km altitude and 90 m/s, the descent stage with the rover beneath it drops away from the back shell and parachute, and eight throttleable engines take over (Figure 27.7).
7. **Sky crane.** About 20 m above the surface, the descent stage lowers the rover on three nylon bridles 6.4 m long. When the rover's wheels touch down at 0.75 m/s, the bridles are cut and the descent stage flies away to crash at a safe distance (Figure 27.9).

The whole sequence is autonomous: with 11 minutes of light-time, the landing was already over when the first signal of entry reached Earth.

**FIGURE 27.7** Perseverance's powered descent vehicle (descent stage, top, with the rover beneath) being stacked at Kennedy Space Center in 2020. Image: NASA/KSC (NASA JPL/Christian Mangano).

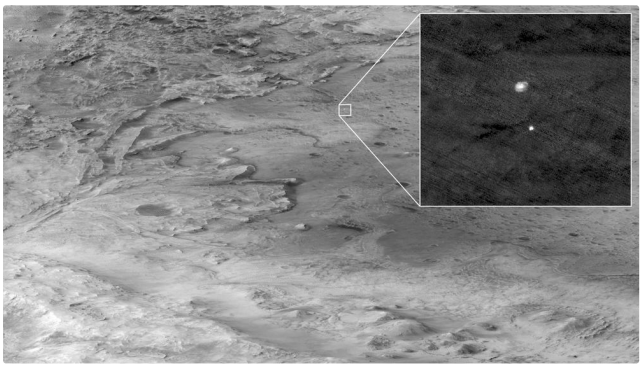


**FIGURE 27.8** Perseverance under its parachute, photographed from orbit by MRO's HiRISE camera during the descent on 18 February 2021. Image: NASA/JPL (NASA/JPL-Caltech/University of Arizona).

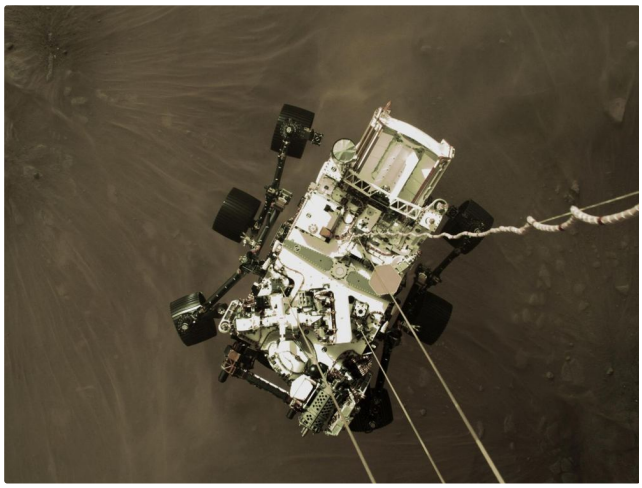


**FIGURE 27.9** The sky crane manoeuvre, seen from the descent stage: Perseverance hangs on its bridles a few metres above the surface. Image: NASA/JPL (NASA/JPL-Caltech).

### EXAMPLE: WHY A MARS PARACHUTE IS NOT ENOUGH

A 3 000 kg spacecraft hangs under a 21.5 m parachute with drag coefficient 0.6 (drag area $C_D A$ = 218 m²). At terminal velocity, drag equals weight: $v_t = \sqrt{2mg/(\rho C_D A)}$.

- **Mars**, 2 km above the surface: ρ ≈ 0.017 kg/m³, $g$ = 3.71 m/s²: $v_t = \sqrt{2\times3\,000\times3.71/(0.017\times218)}$ ≈ **78 m/s** (280 km/h).
- **Earth**, at the same altitude: ρ ≈ 1.0 kg/m³, $g$ = 9.81 m/s²: $v_t \approx$ **16 m/s**.

Even a parachute of 50 m diameter would leave the spacecraft falling at about 30 m/s on Mars. The final 80–100 m/s must be removed by rockets, or the impact must be absorbed by airbags. Mars Pathfinder and the Mars Exploration Rovers used airbags, bouncing up to 15 m high before rolling to a stop.

### 27.4.3 Landing Architectures

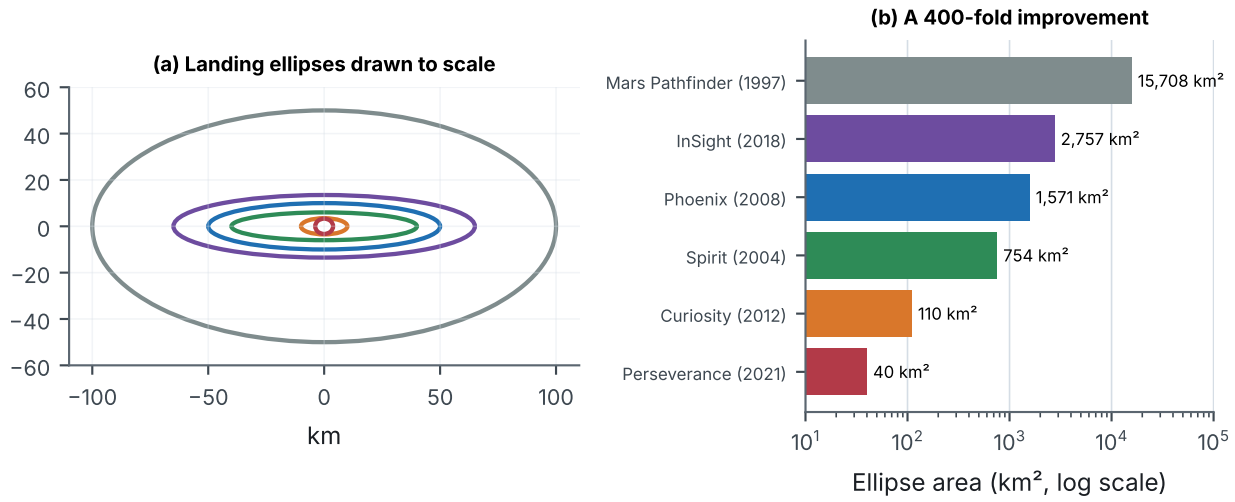
**TABLE 27.4** Successful and attempted Mars landing systems (selected).

| Mission | Year | Final landing system | Notes |
|---|---|---|---|
| Viking 1 and 2 (USA) | 1976 | Throttled rockets and three legs | Biology experiments; operated until 1980 and 1982 |
| Mars Pathfinder (USA) | 1997 | Airbags | Carried the first Mars rover, Sojourner |
| Spirit and Opportunity (USA) | 2004 | Airbags | Opportunity operated for 14 years |
| Phoenix (USA) | 2008 | Pulsed rockets and legs | Arctic plains, 68° N; dug into ground ice |
| Curiosity (USA) | 2012 | Sky crane | First guided lifting entry at Mars; 899 kg rover |
| Schiaparelli (ESA) | 2016 | Crushable structure | Failed: navigation error, see below |
| InSight (USA) | 2018 | Pulsed rockets and legs | Seismometer and heat-flow probe |
| Perseverance (USA) | 2021 | Sky crane | First terrain-relative navigation at Mars; 1 025 kg rover |
| Tianwen-1 / Zhurong (China) | 2021 | Rockets and legs, with hover and hazard avoidance | First successful landing on Mars at the first attempt by a new nation |

Airbags are simple and tolerant of rocks and slopes but limited to landed masses of a few hundred kilograms. Legged landers, as on Phoenix and InSight, can carry heavier instruments but need flat, rock-free sites. The **sky crane**, introduced for the 899 kg Curiosity, keeps the rocket engines high above the surface, avoiding dust and craters, and puts the rover directly on its wheels, ready to drive. Landing heavier payloads — such as the tonnes of cargo needed for human missions — will require new techniques, such as inflatable heat shields and **supersonic retropropulsion**. In retropropulsion, rocket engines are fired into the oncoming supersonic flow. SpaceX's Falcon 9 first stages use this technique when returning through Earth's upper atmosphere, and Starship is designed to land on Mars the same way.

### 27.4.4 Shrinking the Landing Ellipse

Because the atmosphere and the spacecraft's approach are never known exactly, a lander can only be targeted to land somewhere within a **landing ellipse**, and every point inside it must be safe. Better approach navigation, guided entry, the range trigger and terrain-relative navigation have shrunk Mars landing ellipses about 400 times in area in 25 years (Figure 27.10). This has opened scientifically rich but rugged sites such as Perseverance's Jezero crater, an ancient river delta.




Guided lifting entry (Curiosity) and terrain-relative navigation (Perseverance) shrank the ellipses; Japan's SLIM landed within 100 m of its lunar target in 2024.

**FIGURE 27.10** Mars landing ellipses. Smaller ellipses allow landings in complex, scientifically interesting terrain.

### 27.4.5 Failures at Mars

The Soviet **Mars 3** made the first soft landing on Mars in December 1971 but fell silent after about 20 seconds, possibly because of a dust storm. NASA's **Mars Polar Lander** was lost in 1999, almost certainly because the jolt of its legs deploying was misread by software as touchdown, shutting off the engines 40 m above the ground. Britain's **Beagle 2** landed in 2003 but was not heard from; in 2015 HiRISE images showed it on the surface with its solar panels only partly unfolded, blocking its antenna. ESA's **Schiaparelli** test lander crashed in October 2016 when its inertial measurement unit briefly saturated during parachute descent. The navigation software then calculated a negative altitude and released the parachute. It fired the braking thrusters for only three seconds while the lander was still 3.7 km up.

### 27.4.6 What Mars Landers Found

The two **Viking** landers of 1976 returned the first images from the surface (Figure 27.11) and ran three biology experiments. Their results — chemically active soil but no organic molecules — are still debated. The later discovery of perchlorates in the soil by **Phoenix** suggests that heating during the analysis destroyed any organics. Phoenix scraped through the soil at 68° N and found water ice a few centimetres down (Figure 27.12).

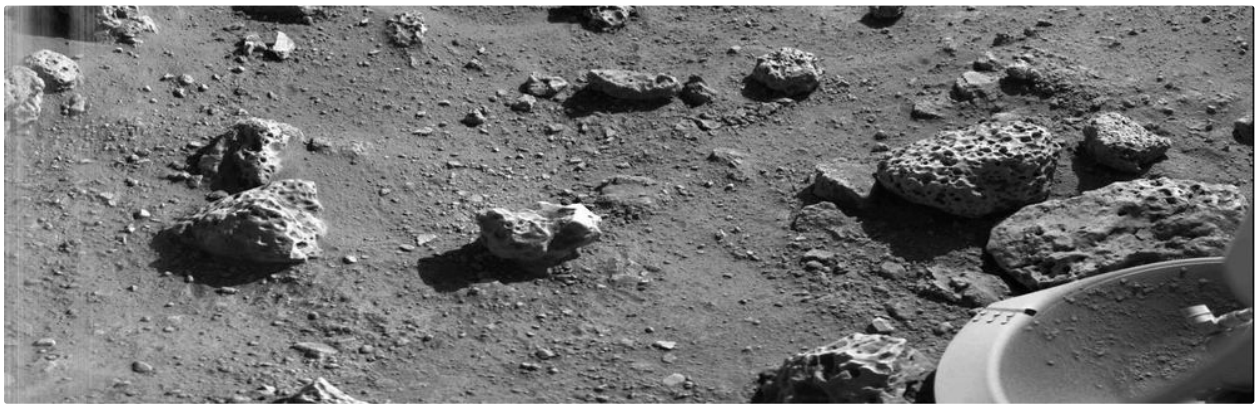


**FIGURE 27.11** The first photograph taken by Viking Lander 2 at Utopia Planitia in September 1976, showing rocks and part of the lander's footpad (lower right). Image: NASA/JPL (NASA/JPL).

**FIGURE 27.12** Phoenix's "Dodo-Goldilocks" trench in June 2008. Lumps of bright material in the lower left (top image) had vanished four days later (bottom), proving that they were water ice that sublimated. Image: NASA/JPL (NASA/JPL-Caltech/University of Arizona/Texas A&M University).

NASA's **InSight** (2018–2022) placed a seismometer on the Martian surface with its robotic arm (Figure 27.13) and recorded 1 319 marsquakes. These included one of magnitude 4.7 in May 2022 and the shaking from meteorite impacts whose fresh craters were then found by orbiters. Seismic waves passing through the planet revealed its internal structure (Figure 27.14). It has a crust 24–72 km thick and a large liquid iron-rich core surrounded by a layer of molten rock. According to a study published in 2025, it may also have a solid inner core. InSight's heat-flow probe, a self-hammering "mole", could not penetrate the unexpectedly cohesive soil. Dust slowly covered the solar panels (see the chapter-opening image) and the mission ended in December 2022.

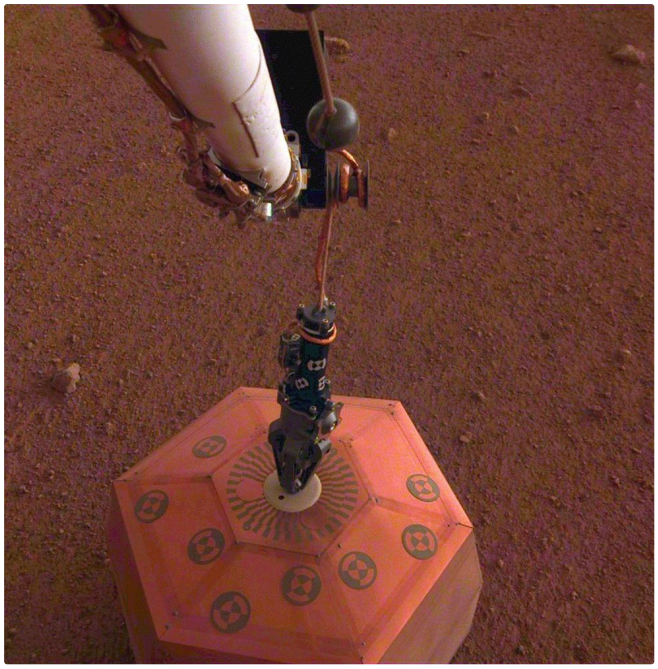


**FIGURE 27.13** InSight's robotic arm placing the SEIS seismometer on the surface of Mars in December 2018. Image: NASA/JPL (NASA/JPL-Caltech).

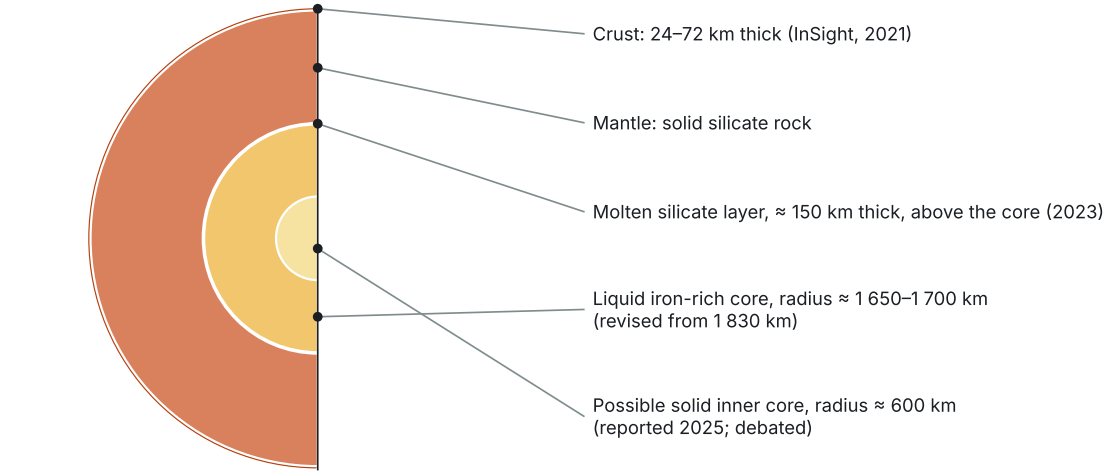



**FIGURE 27.14** The interior of Mars revealed by InSight's seismometer.

## 27.5 Venus, Titan and Small Bodies

### 27.5.1 Venus: Landing in an Oven

The surface of Venus is at 465 °C and 92 bar — the pressure 900 m deep in Earth's oceans — under clouds of sulphuric acid. The Soviet Union nevertheless landed ten spacecraft there between 1970 and 1985. **Venera 7** (1970) transmitted for 23 minutes from the surface, the first data ever received from the surface of another planet. **Venera 9** (1975) returned the first pictures. **Venera 13** (1982) returned colour panoramas, drilled a sample and analysed it with an X-ray fluorescence spectrometer, surviving for 127 minutes.

Venus landers are built like deep-sea submersibles: a spherical titanium pressure vessel, thick insulation, and a heat sink of lithium nitrate or similar phase-change material, with the electronics pre-chilled before entry. They are doomed from the moment they land, and the race is to return data before the interior overheats. Because the atmosphere is so dense, the parachute is released at about 50 km altitude. The lander falls the rest of the way with only a drag plate, reaching the surface at about 8 m/s. New electronics based on silicon carbide, which NASA operated for 60 days in a Venus simulation chamber in 2021, may one day allow long-lived Venus landers.

### 27.5.2 Titan: Huygens

ESA's **Huygens** probe, carried to Saturn by Cassini, entered Titan's atmosphere on 14 January 2005. Titan's dense, extended atmosphere and low gravity made the landing gentle. Huygens descended under parachutes for 2 hours 27 minutes, imaging river channels and shorelines below. It landed at about 4.5 m/s on damp ground strewn with rounded pebbles of water ice (Figure 27.15). It recorded a surface temperature of −179 °C and a pressure of 1.5 bar. It transmitted from the surface for 72 minutes until Cassini set below the horizon. Its landing is still the most distant ever made. NASA's Dragonfly rotorcraft will return to Titan in 2034 (Chapter 28).

**FIGURE 27.15** The surface of Titan photographed by Huygens after landing in 2005. The "pebbles" in the foreground, 10–15 cm across, are made of water ice. Image: NASA/JPL (NASA/JPL/ESA/University of Arizona).

### 27.5.3 Comets and Asteroids: Avoiding the Bounce

Landing on a small body is the opposite problem: gravity is so weak that a lander risks bouncing back into space. The escape speed from the 900 m asteroid Ryugu is about 0.37 m/s, a slow walking pace. NASA's **NEAR Shoemaker** orbiter, not designed as a lander, touched down on Eros at the end of its mission in 2001 and kept operating for two weeks. ESA's **Philae** landed on comet 67P/Churyumov–Gerasimenko on 12 November 2014, but its harpoons failed to fire and its hold-down thruster did not work. It bounced about a kilometre high and came to rest two hours later in the shadow of a cliff. There it operated on its batteries for about 64 hours before falling silent.

JAXA's Hayabusa2 deployed the first hopping rovers, **MINERVA-II-1A and 1B**, and the German–French **MASCOT** lander on Ryugu in 2018; sample-collecting "touch-and-go" landings are described in Chapter 29.

**EXAMPLE: HOW SLOWLY MUST YOU LAND ON A COMET?**

Comet 67P has a mass of about $1.0 \times 10^{13}$ kg and an effective radius of about 2 km.

1. Escape speed: $v_{esc} = \sqrt{2GM/r} = \sqrt{2 \times 6.67 \times 10^{-11} \times 10^{13} / 2\,000}$ = 0.82 m/s.
2. Philae touched down at about 1 m/s. Without anchoring, any rebound faster than 0.82 m/s would have escaped the comet entirely. Philae's damping legs absorbed most of the energy, and it rebounded at about 0.4 m/s. This was slow enough to fall back, but only after a two-hour flight.
3. Surface gravity is about $GM/r^2$ = 1.7 × $10^{-4}$ m/s², so the 100 kg Philae weighed only about 0.02 N — the weight of a 2 g object on Earth. This is why it carried harpoons, ice screws and a thruster to press it down: without them, almost any disturbance could launch it back into space.

## 27.6 Surviving on the Surface

Landing is only the beginning. A lander must then survive the local environment:

- **Night and cold.** A lunar night lasts 14 Earth days, and the surface cools to about −170 °C. Solar-powered landers usually die at sunset unless they carry radioisotope heater units, as China's Chang'e 3 and 4 landers do; SLIM, not designed to survive, revived after three lunar nights before failing in 2024. Martian nights reach −90 °C, and winter is the end of most solar-powered polar landers, as it was for Phoenix.
- **Dust.** Lunar dust is sharp and electrostatically charged, sticking to radiators, seals and optics; Martian dust settles on solar panels, steadily reducing power, as it ended InSight's mission.
- **Communications.** A lander on the Moon's near side can talk directly to Earth, but one on the far side, which never faces Earth, needs a relay satellite such as China's Queqiao and Queqiao-2. Mars landers relay most of their data through orbiters (Chapter 25).

**AI AND AUTONOMY FOR LANDING**

Landing is the most autonomous phase of any mission, and modern landers rely on computer vision. Terrain-relative navigation systems detect craters or match image patches with onboard maps. Hazard-detection systems turn lidar or stereo images into maps of slope and roughness and select a safe spot within seconds. SLIM's vision-based navigation demonstrated 100 m landing accuracy. On the ground, deep-learning models trained on orbital images detect boulders and craters to map hazards at candidate landing sites — for the Artemis south-pole sites, they have catalogued millions of boulders from LRO images. Future landers will combine these into fully autonomous "land anywhere" systems (Chapters 48 and 51).

### KEY TAKEAWAYS

- Landing removes huge kinetic energy — 1.4 MJ/kg from lunar orbit, 15 MJ/kg at Mars entry — using drag, propulsion and impact attenuation in proportions set by the destination.
- Airless bodies require fully propulsive descent; gravity losses add 10–20 % to the Δv, and the final phase depends on accurate altimetry, terrain-relative navigation and hazard avoidance.
- Tall, narrow landers are vulnerable to tipping over; most recent lunar failures and anomalies were caused by navigation sensors and software rather than hardware breakdowns.
- Mars's thin atmosphere requires a chain of heat shield, supersonic parachute and propulsive or airbag landing; guided entry and terrain-relative navigation have shrunk landing ellipses 400-fold.
- Venus landers survive for hours at 465 °C and 92 bar; Huygens landed on Titan under parachutes; small-body landers must avoid bouncing off in micro-gravity.
- Landers have revealed Mars's interior, water ice at its high latitudes and the surfaces of Venus and Titan; in 2026 a new wave of commercial and national lunar landers was preparing for 2027.

### REVIEW QUESTIONS

1. Calculate the specific kinetic energy that must be removed when landing from lunar orbit (1.68 km/s) and at Mars entry (5.5 km/s).
2. A lunar lander needs a Δv of 2.0 km/s with engines of 320 s specific impulse. What fraction of its initial mass must be propellant?
3. Why do airless bodies require fully propulsive descent, and why do gravity losses add 10–20 % to the Δv?
4. Why have recent commercial lunar landers tipped over? What design and navigation lessons follow?
5. Describe the Mars EDL sequence from entry to touchdown for a sky-crane landing.
6. How have guided entry and terrain-relative navigation reduced the size of Mars landing ellipses?
7. How did the Venera landers survive on the surface of Venus, and for how long?
8. Why is landing on a comet or small asteroid difficult? What happened to Philae?

CHAPTER

# 28 Planetary Rovers and Aerial Explorers

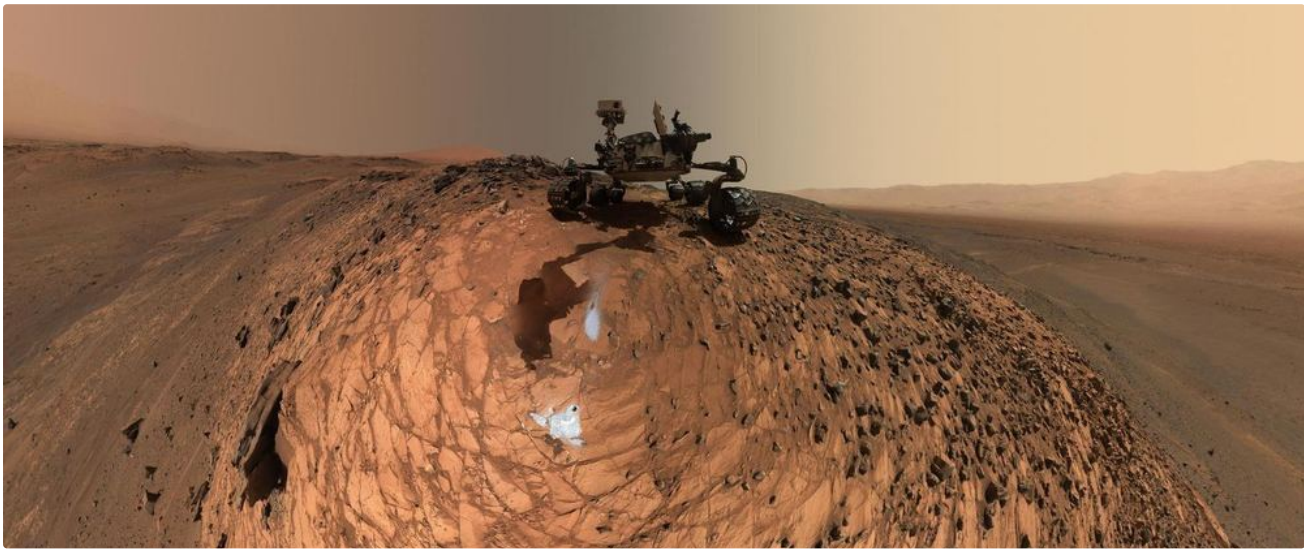


Image: NASA/JPL (NASA/JPL-Caltech/MSSS)

*A lander sees one spot; a rover can follow the clues. Since Lunokhod 1 rolled onto the Moon in 1970, rovers have crossed tens of kilometres of the Moon and Mars, climbed mountains, drilled rocks, cached samples for return to Earth and found the strongest evidence yet that Mars was once habitable. In 2021 a 1.8 kg helicopter made the first powered flight on another planet, and a nuclear-powered rotorcraft is being built to fly through the dense air of Titan. This chapter explains how rovers move over soft ground, stay warm and powered, navigate by themselves, and carry laboratories across other worlds; it reviews every rover mission to date and those planned for the next decade.*

**IN THIS CHAPTER**

28.1 Sixty Years of Rovers
28.2 Engineering a Rover
28.3 What Rovers Have Found
28.4 Aerial Explorers
28.5 Rovers at the Lunar Poles
28.6 The Next Generation

A **rover** is a mobile laboratory. Mobility lets one mission visit many sites, follow geological layers like the pages of a book, and reach targets first spotted from orbit. The price is complexity: a rover must cross unknown terrain, mostly on its own, with its power, temperature and communications all depending on where it stops. This chapter describes robotic rovers on the Moon and Mars, the vehicles that carried Apollo astronauts, and the newest class of explorer — aircraft that fly in other worlds' atmospheres.

# 28.1 Sixty Years of Rovers

## 28.1.1 The First Rovers: Lunokhod and the Apollo Lunar Roving Vehicle

The first rover on another world was the Soviet **Lunokhod 1**, landed by Luna 17 on 17 November 1970. The 756 kg, eight-wheeled vehicle was driven from Earth by a five-person crew who watched slow-scan television pictures arriving with a delay of several seconds; a polonium-210 heater kept it alive through the lunar nights. It operated for 11 lunar days and drove 10.5 km. **Lunokhod 2** (1973) drove 39 km in four months — the distance record for 41 years. Their landing sites have been photographed by NASA's Lunar Reconnaissance Orbiter (Figure 28.1).

**FIGURE 28.1** The Lunokhod 2 rover (arrow), as seen by NASA's Lunar Reconnaissance Orbiter, at the end of its tracks in the crater Le Monnier, where it stopped in 1973. Image: NASA/JPL (NASA/GSFC/Arizona State University).

The last three Apollo missions (1971–72) carried the **Lunar Roving Vehicle (LRV)**, a 210 kg, battery-powered, four-wheel-drive car that folded into the side of the Lunar Module. With wire-mesh wheels and a top speed of about 13 km/h, the LRVs carried astronauts and their samples a combined 90 km and let them explore up to 7.6 km from the lander (Figure 28.2).

**FIGURE 28.2** Apollo 15's Jim Irwin with the first Lunar Roving Vehicle at Hadley–Apennine in July 1971. Mount Hadley rises in the background. Image: NASA/JSC.

### 28.1.2 Mars Rovers

NASA's rovers have grown with each generation (Figure 28.3):

- **Sojourner** (1997), a 10.6 kg, solar-powered microrover carried by Mars Pathfinder, explored the rocks around the lander for 83 sols (Martian days) and travelled about 100 m.
- **Spirit and Opportunity** (2004), 185 kg each, were designed for 90 sols and 600 m of driving. Opportunity found layered rocks and spherules of hematite ("blueberries") formed in acidic groundwater. It operated for 14 years until a planet-wide dust storm in 2018 cut its power. Its 45.16 km is still the record for distance driven on another world. Spirit found almost pure silica deposited by ancient hot springs before it became stuck in soft soil in 2009.
- **Curiosity** (2012), 899 kg and powered by a radioisotope generator, landed in Gale crater and showed that it once held a long-lived lake whose water could have supported microbes. It has found organic molecules preserved in mudstones and seasonal changes in atmospheric methane, and has climbed more than 1.3 km up the layered Mount Sharp, driving more than 37 km by 2026.
- **Perseverance** (2021), 1 025 kg, explores Jezero crater, which contained a lake fed by a river delta 3.5 billion years ago. It is the first rover to **collect and seal samples for return to Earth**. By early 2026 it had filled 33 of its 43 sample tubes and left ten of them in a depot on the crater floor in 2022–23. It had also climbed out over the crater rim and driven more than 42 km. Its MOXIE experiment made 122 g of oxygen from the Martian carbon dioxide atmosphere.

**FIGURE 28.3** Three generations of NASA Mars rovers at JPL: a flight spare of Sojourner (front), a Mars Exploration Rover test vehicle (left) and a Curiosity test rover (right). Image: NASA/JPL (NASA/JPL-Caltech).

China's **Zhurong** rover (240 kg) was landed by Tianwen-1 in May 2021 in Utopia Planitia. It drove 1.9 km and found evidence of water-altered minerals and buried layers with its ground-penetrating radar. In May 2022 it entered hibernation under a coating of dust and did not wake.

### 28.1.3 Rovers on the Moon Since 2013

China's **Yutu** (2013) drove 114 m before its mobility failed. **Yutu-2**, landed by Chang'e 4 on the far side in January 2019, became the longest-lived lunar rover. Its ground-penetrating radar mapped layers of debris tens of metres deep. It was still active, though nearly immobile, in 2025 after driving about 1.6 km. India's 26 kg **Pragyan**, carried by Chandrayaan-3 (2023), confirmed sulphur in the south polar regolith with its laser-induced breakdown spectrometer before going to sleep at the end of the lunar day. Japan's SLIM carried two tiny rovers in 2024: LEV-1, a hopping rover, and LEV-2 (SORA-Q), a 250 g sphere that unfolded into a two-wheeled rover and photographed SLIM lying on its nose. The UAE's first rover, **Rashid**, was lost when the Hakuto-R lander crashed in 2023.

**TABLE 28.1** Planetary rovers to 2026.

| Rover | Operator | Body | Years | Mass | Power | Distance |
|---|---|---|---|---|---|---|
| Lunokhod 1, 2 | USSR | Moon | 1970–71, 1973 | 756 kg, 836 kg | Solar; Po-210 heater | 10.5 km, 39 km |
| Lunar Roving Vehicle (×3) | NASA | Moon | 1971–72 | 210 kg | Batteries | 90 km in total |
| Sojourner | NASA | Mars | 1997 | 10.6 kg | Solar | ≈ 0.1 km |
| Spirit, Opportunity | NASA | Mars | 2004–10, 2004–18 | 185 kg | Solar | 7.7 km, 45.2 km |
| Curiosity | NASA | Mars | 2012– | 899 kg | MMRTG | > 37 km |
| Yutu, Yutu-2 | CNSA | Moon | 2013–16, 2019– | 140 kg | Solar, radioisotope heaters | 0.1 km, 1.6 km |

| Rover | Operator | Body | Years | Mass | Power | Distance |
|---|---|---|---|---|---|---|
| Perseverance | NASA | Mars | 2021– | 1 025 kg | MMRTG | > 42 km |
| Zhurong | CNSA | Mars | 2021–22 | 240 kg | Solar | 1.9 km |
| Pragyan | ISRO | Moon | 2023 | 26 kg | Solar | ≈ 0.1 km |

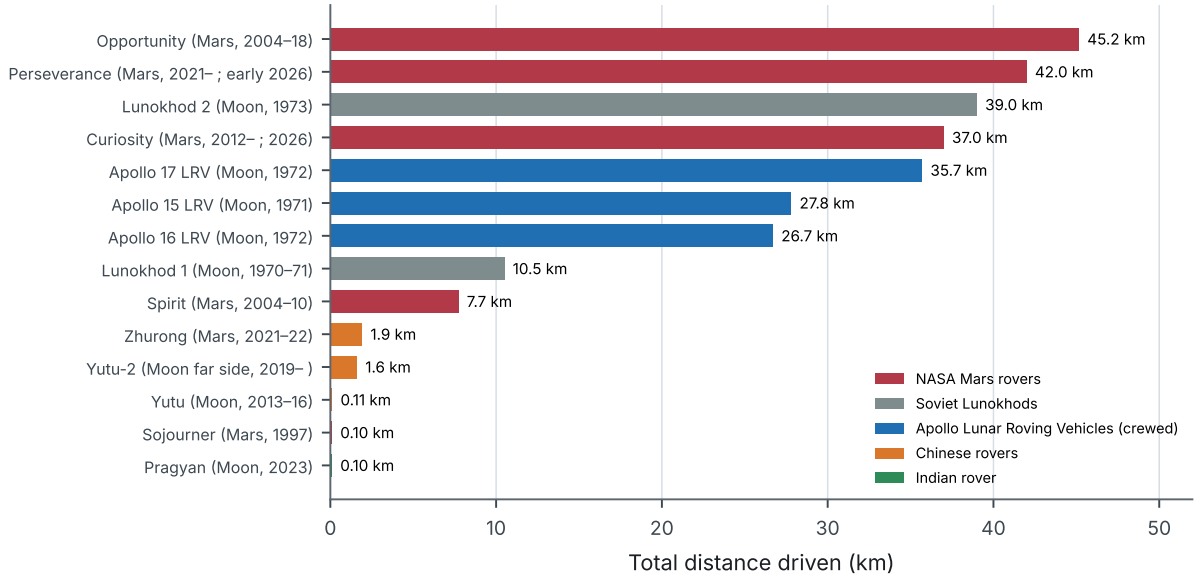



**FIGURE 28.4** The distances driven by rovers on the Moon and Mars.

## 28.2 Engineering a Rover

### 28.2.1 Mobility

All NASA Mars rovers since Sojourner have used the **rocker-bogie** suspension (Figure 28.5). Each side has a large “rocker” link carrying the rear wheel (or front, depending on direction) and a smaller “bogie” carrying the other two wheels. The two rockers are connected through a differential, so the body pitches only half as much as either side. The system has no springs: all six wheels stay in contact with the ground, sharing the load and providing traction, and the rover can climb obstacles almost as high as the wheel diameter. Each wheel has its own drive motor, and the four corner wheels can be steered so the rover can turn in place.

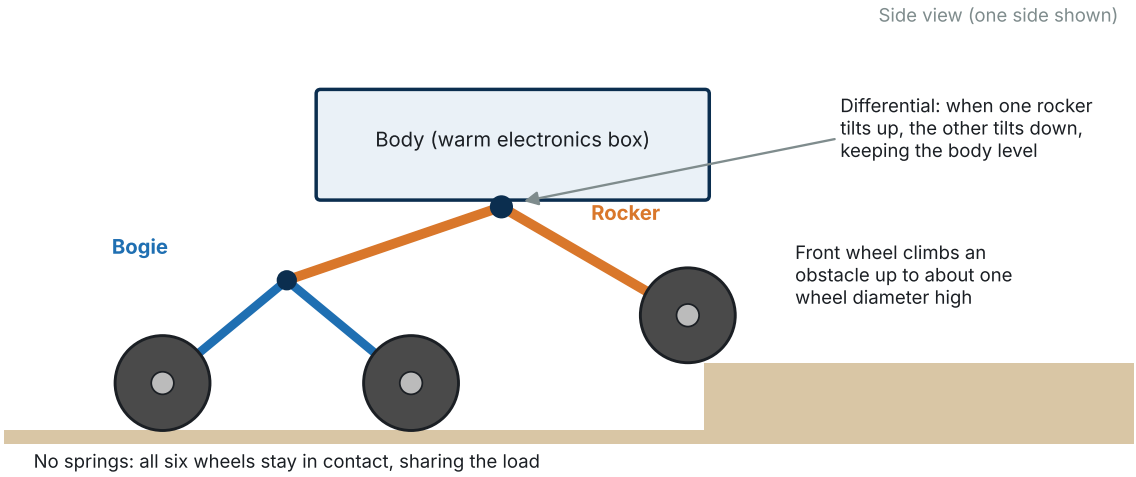
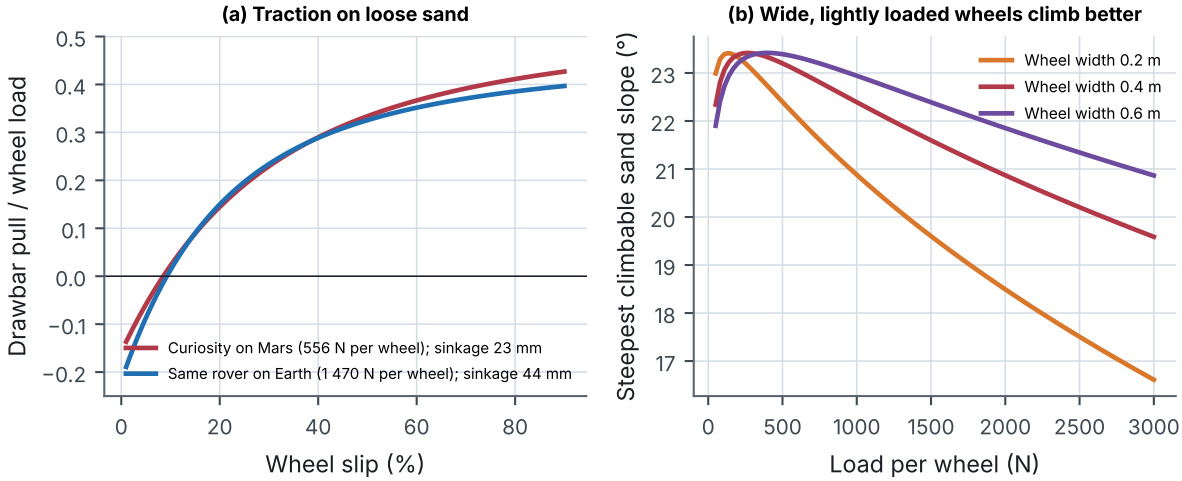



**FIGURE 28.5** The rocker-bogie suspension used on NASA's Mars rovers.

The greatest mobility hazard is soft ground. A wheel sinks until the soil beneath it can support its load. It must then bulldoze and compact soil in front of it — the **compaction resistance** — while its grousers (tread cleats) shear the soil to generate thrust. The science of this, **terramechanics**, developed by M. G. Bekker and J. Y. Wong for military vehicles, predicts the net force available to pull the rover forward, the **drawbar pull**, as a function of wheel slip (Figure 28.6). On loose sand, even a well-designed rover can climb only about 20–25°; Spirit and Opportunity both became trapped in sand, Opportunity escaping after five weeks of careful driving in 2005, Spirit never.




Bekker–Wong terramechanics for a rigid wheel of 0.26 m radius on cohesionless sand (n = 1, kc = 1.4 kN/m², kφ = 820 kN/m³, c = 0.17 kPa, φ = 35°). Representative values.

**FIGURE 28.6** Wheel–soil interaction. (a) Drawbar pull rises with slip; the same rover on Earth sinks deeper, which is why engineers test Mars rovers with lightened copies. (b) Wider wheels and lighter loads let a rover climb steeper sandy slopes.

**EXAMPLE: TESTING A MARS ROVER ON EARTH**

Curiosity has a mass of 899 kg. On Mars each of its six wheels carries 899 × 3.71 / 6 = 556 N; on Earth, 899 × 9.81 / 6 = 1 470 N.

1. With the soil model of Figure 28.6, the wheels sink about 23 mm on Mars but 44 mm on Earth, and the compaction resistance rises from 16 % to 22 % of the wheel load.
2. A rover tested at full mass on Earth would therefore perform worse than on Mars — misleading the engineers.
3. JPL instead tests mobility with a lightened rover, nicknamed “Scarecrow”, that has the same wheels and suspension. It weighs on Earth what Curiosity weighs on Mars (about 340 kg mass), with no computer or cameras on board.

Wheels must also survive rocks. Curiosity’s 0.5 m aluminium wheels, with skins only 0.75 mm thick, were punctured and torn by sharp, wind-carved rocks from 2013 (Figure 28.7). The team responded by planning routes over softer ground and, in 2017, uploading a **traction-control** algorithm that adjusts each wheel’s speed to reduce the forces on it. Perseverance’s wheels were made thicker and narrower, with more grousers.

**FIGURE 28.7** Damage to one of Curiosity’s aluminium wheels after about 4.8 km of driving, in December 2013. Image: NASA/JPL (NASA/JPL-Caltech/MSSS).

## 28.2.2 Power and Temperature

A rover must keep its electronics and batteries above about −40 °C through nights that, on Mars, reach −90 °C. Solar-powered rovers house their electronics in an insulated **warm electronics box** heated by electric heaters and radioisotope heater units (RHUs); Spirit and Opportunity carried eight RHUs each. Dust gradually covers solar arrays, reducing power by a fraction of a per cent per sol. Opportunity’s arrays were repeatedly cleaned by gusts of wind and dust devils, which is why it lasted 14 years. Zhurong’s power, by contrast, was not restored. Curiosity and Perseverance use **multi-mission radioisotope thermoelectric generators** (MMRTGs). These provide about 110 W of electricity at launch — around 2.5 kWh per sol, day and night. Their waste heat warms the rover through a pumped fluid loop. Batteries supply the peaks when the rover drives, drills or runs its laboratories.

### 28.2.3 Communications and Operations

Rovers can talk directly to Earth through high-gain X-band antennas, but most of their data — hundreds of megabits per sol — is relayed by orbiters passing overhead (Chapter 25). Lunar far-side rovers depend on relay satellites such as Queqiao. Because of the light-time (4–24 minutes each way at Mars) and the limited contact windows, rovers cannot be joysticked. Instead, the operations team plans each sol's activities — drives, images, arm placements, drilling — and uplinks them as a command sequence in the Martian morning. The rover executes them and sends back the results by the afternoon relay pass. The team then plans the next sol. A rover team works on "Mars time" during the first months of a mission, with its working day shifting 40 minutes later each Earth day.

### 28.2.4 Autonomous Navigation

To drive further than the operators can see, rovers navigate themselves (Figure 28.8). Stereo cameras build a three-dimensional map of the terrain ahead, and software assesses each patch for slope, rocks and steps relative to the rover's dimensions. The software chooses the path that makes most progress towards the goal set by the team. Meanwhile, **visual odometry** — tracking features between successive images — measures how far the rover has really moved, correcting for wheel slip. Opportunity's patterned tracks record this process (Figure 28.9). Spirit and Opportunity had to stop to think after each short step. Perseverance's faster computer and dedicated vision processor let it plan "while driving". It can cover more than 100 m per hour across favourable terrain — several times faster than Curiosity.

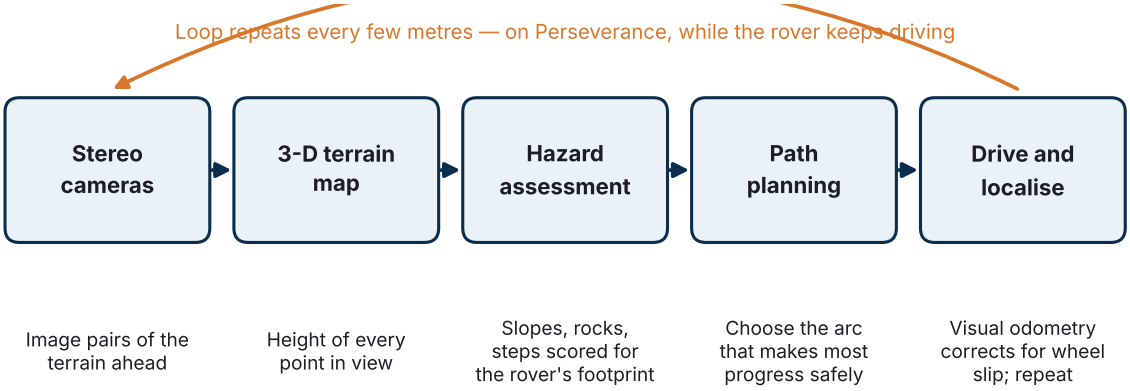



**FIGURE 28.8** The autonomous navigation loop of a Mars rover.

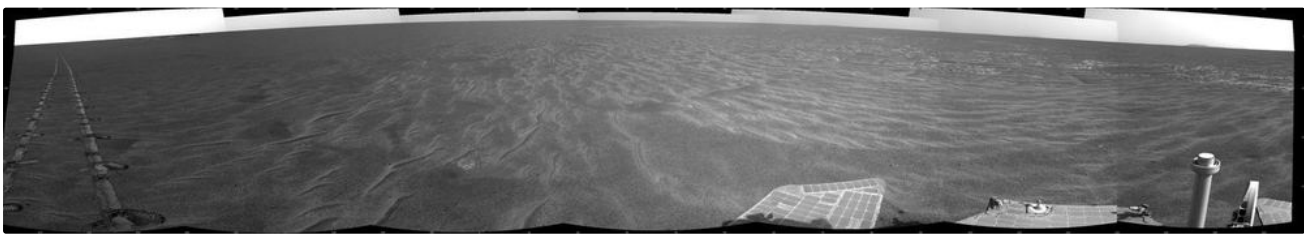


**FIGURE 28.9** Opportunity's tracks on Meridiani Planum in 2011. The pattern of short arcs records the rover stopping to check for hazards autonomously. Image: NASA/JPL (NASA/JPL-Caltech).

### 28.2.5 A Laboratory on Wheels

Rover science payloads combine remote-sensing instruments on the mast, contact instruments on a robotic arm, and laboratories inside the body (Table 28.2).

**TABLE 28.2** Science instruments of Curiosity and Perseverance.

| Location | Instrument type | Curiosity | Perseverance |
|---|---|---|---|
| Mast | Colour stereo cameras with zoom | Mastcam | Mastcam-Z |
| Mast | Laser-induced breakdown spectroscopy (zaps rocks from up to 7 m) | ChemCam | SuperCam (also Raman and infrared spectroscopy, microphone) |
| Arm | Hand lens imager | MAHLI | WATSON |
| Arm | X-ray spectrometer for chemistry | APXS | PIXL (maps elements at 0.1 mm scale) |
| Arm | Ultraviolet Raman and fluorescence for organics | — | SHERLOC |
| Arm | Rock drill | Powder drill | Coring drill producing intact cores |
| Body | Mineralogy by X-ray diffraction | CheMin | — |
| Body | Organic and isotope analysis by gas chromatography–mass spectrometry | SAM | — |
| Body | Sample sealing and caching | — | 43 tubes |
| Body or deck | Weather, radiation, ground-penetrating radar, oxygen production | REMS, RAD, DAN | MEDA, RIMFAX, MOXIE |

## 28.3 What Rovers Have Found

The Mars rovers have turned the question “was Mars ever habitable?” into a resounding “yes”. Opportunity found that Meridiani Planum was once soaked in acidic, salty groundwater, and Spirit found the deposits of hot springs. Curiosity found that Gale crater held fresh, neutral water for long periods, with all the chemical ingredients needed by microbes. Perseverance has studied the ancient delta and the volcanic rocks of Jezero’s floor. In July 2024, it drilled a rock nicknamed **Cheyava Falls** with millimetre-sized “leopard spots” — pale patches with dark rims rich in iron phosphate and iron sulphide minerals (Figure 28.10). On Earth, such reactions are often driven by microbes. In September 2025 the team reported in *Nature* that the sample, named Sapphire Canyon, contains a **potential biosignature** — a feature that might have a biological origin but could also have formed without life. Only analysis in laboratories on Earth can settle the question.

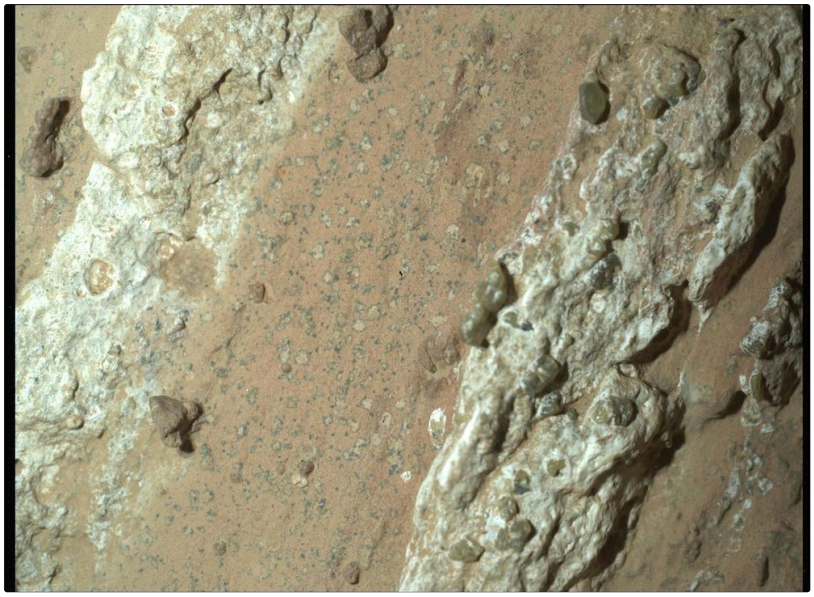


**FIGURE 28.10** The “leopard spots” in the Cheyava Falls rock, imaged by Perseverance’s WATSON camera in July 2024. The sample drilled here may preserve a potential biosignature. Image: NASA/JPL (NASA/JPL-Caltech/MSSS).

That question is now tied to the fate of **Mars Sample Return**. The NASA–ESA plan to collect Perseverance’s tubes with a lander, a small rocket and an Earth-return orbiter grew to an estimated cost of about US$11 billion with return no earlier than 2040. In 2025–26 the US Congress provided no money for the existing programme, leaving the samples on Mars with no retrieval plan. Meanwhile, China’s **Tianwen-3** prepares to launch in 2028 and return samples from a different site by 2031 (Chapter 29).

## 28.4 Aerial Explorers

### 28.4.1 Ingenuity: First Flight on Another World

On 19 April 2021, NASA’s **Ingenuity** helicopter, carried to Mars beneath Perseverance, rose 3 m above Jezero crater, hovered and landed in a flight lasting 39 seconds. This was the first powered, controlled flight of an aircraft on another planet (Figure 28.11). Flying on Mars is hard: the atmosphere has 1.4 % of the density of Earth’s at sea level. Ingenuity had a mass of 1.8 kg and two counter-rotating rotors 1.2 m across. They spun at about 2 500 rpm — five times faster than a terrestrial helicopter’s rotor. This speed was limited by the need to keep the blade tips below the speed of sound, which is lower on Mars than on Earth (Figure 28.12).

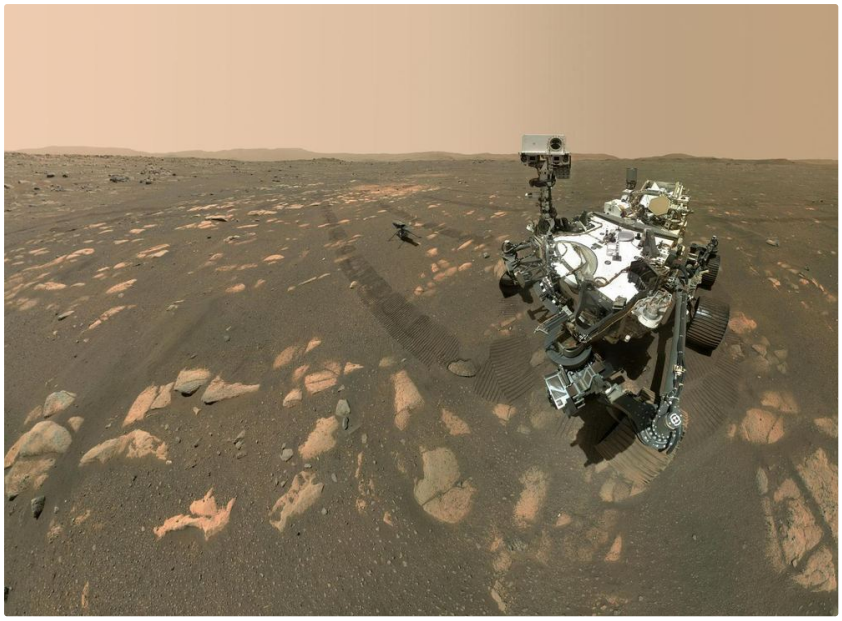
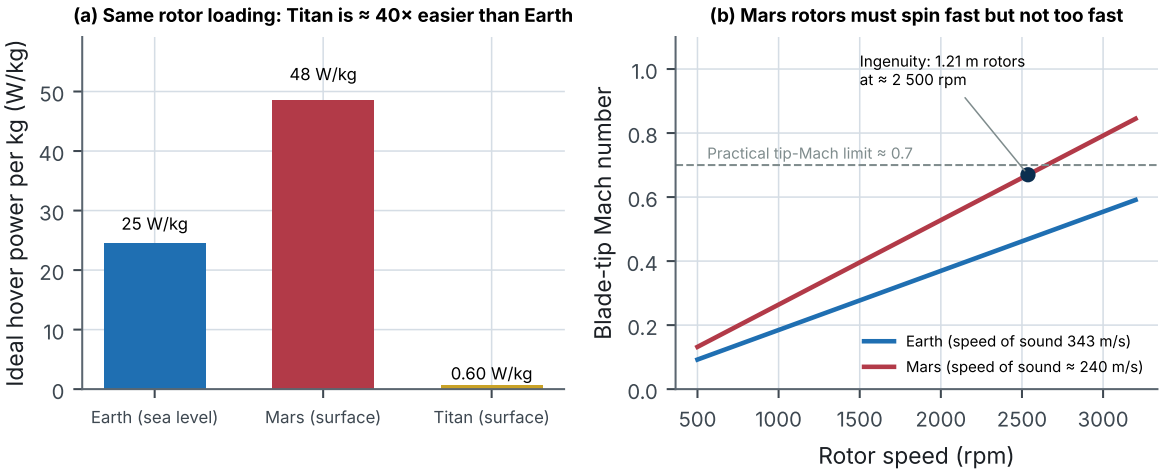


**FIGURE 28.11** Perseverance's selfie with Ingenuity in April 2021, shortly before the helicopter's first flight. Image: NASA/JPL (NASA/JPL-Caltech/MSSS).




Momentum theory: ideal hover power per unit mass = g^1.5 √(disk loading / 2ρ), here for Ingenuity's disk loading (1.8 kg on a 1.21 m disk). Real power is roughly twice the ideal.

**FIGURE 28.12** The physics of flying on other worlds. (a) Hover power per kilogram for the same rotor loading. (b) Rotor tip Mach number: Mars's lower speed of sound limits rotor speed.

Planned as a five-flight technology demonstration, Ingenuity flew 72 times over nearly three years, covering about 17 km in a total of 129 minutes. It reached altitudes of 24 m and speeds of 10 m/s, and scouted routes for Perseverance. It navigated by tracking features on the ground with a downward-looking camera. On its 72nd flight, on 18 January 2024, over featureless sand ripples, the navigation system lost track of the ground and the helicopter landed hard, breaking its rotor blades (Figure 28.13). It remains on Mars as a weather station.

**FIGURE 28.13** Ingenuity's own navigation camera reveals the shadow of a rotor blade with its tip missing after its final flight in January 2024. Image: NASA/JPL (NASA/JPL-Caltech).

**EXAMPLE: HOVERING ON MARS**

Ingenuity weighs $T = 1.8 \times 3.71$ = 6.7 N on Mars. Its rotor disk area is $A = \pi \times 0.605^2$ = 1.15 m$^2$, and the air density at Jezero is about 0.017 kg/m$^3$.

1. Momentum theory gives the ideal hover power $P = T^{3/2}/\sqrt{2\rho A}$ = 6.7^1.5 / √(2 × 0.017 × 1.15) = 17.3 / 0.198 ≈ **88 W**.
2. The same helicopter on Earth would weigh 17.7 N and need only $17.7^{1.5}/\sqrt{2 \times 1.225 \times 1.15}$ = 74.5 / 1.68 ≈ 44 W: half the power, because the dense air makes the rotor far more effective.
3. Real rotors have a figure of merit (ideal/actual power) of about 0.5–0.6, and the coaxial rotors interfere with each other. Ingenuity drew about 350 W in flight from six lithium-ion cells, so each flight was limited to about 2–3 minutes.

### 28.4.2 Balloons

The only balloons flown on another planet were the two **Vega** balloons of 1985, released into the atmosphere of Venus at about 54 km altitude. They drifted there for two days, tracked by a worldwide network of radio telescopes that measured the winds. Venus's cloud layer, where temperature and pressure are similar to Earth's surface, is attractive for long-lived balloons; NASA and others have tested variable-altitude "aerobot" prototypes that could rise and sink through the clouds.

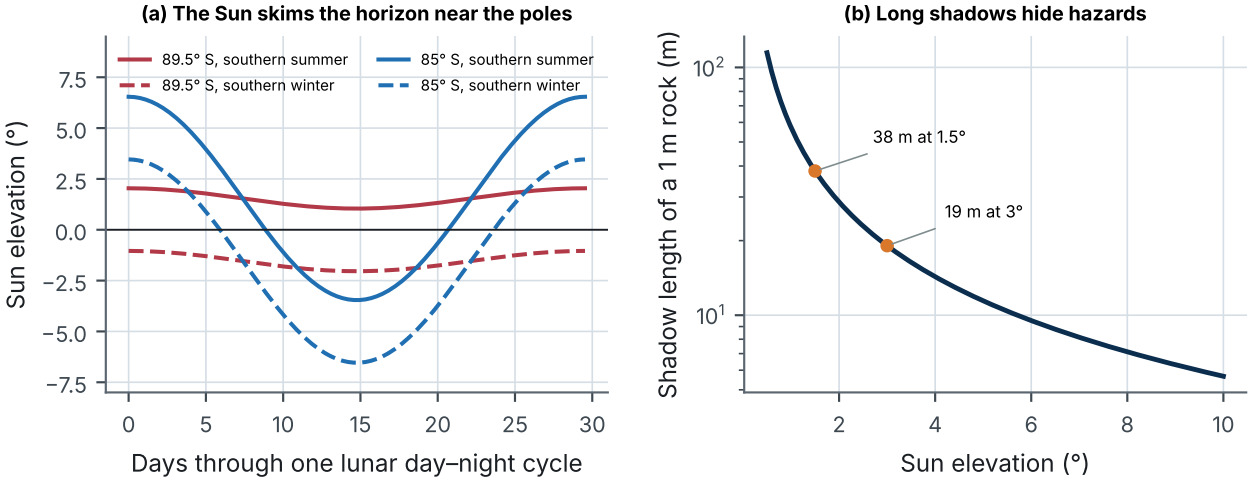
### 28.4.3 Dragonfly: Flying on Titan

Titan is the easiest place in the Solar System to fly. Its atmosphere is four times denser than Earth's and its gravity one-seventh, so hovering takes about 40 times less power per kilogram than on Earth (Figure 28.12). NASA's **Dragonfly**, a car-sized rotorcraft lander with eight rotors and an MMRTG, is scheduled for launch on a Falcon Heavy in July 2028 and arrival in 2034. It will land first in the Shangri-La dune fields. It will then fly from site to site — eventually more than 175 km in total — towards the 80 km Selk impact crater. There, liquid water and organic molecules may once have mixed. Its mass spectrometer and drill will analyse the surface for prebiotic chemistry. Dragonfly will recharge its batteries from the MMRTG between flights, making flights of several kilometres at a time, roughly once per Titan day (16 Earth days).

## 28.5 Rovers at the Lunar Poles

The next generation of lunar rovers is heading for the south pole, where water ice is thought to be trapped in **permanently shadowed regions**. These are crater floors that have not seen sunlight for billions of years and stay below −200 °C. The Moon's spin axis is tilted only 1.54° to the plane of its orbit around the Sun. As a result, the Sun near the poles never rises more than a few degrees above the horizon. It circles the sky once a month, casting shadows tens of times longer than the objects that cast them (Figure 28.14). This creates unique engineering problems:

- **Power and thermal.** Solar arrays must be vertical, and they must turn to follow the Sun around the horizon. A rover that drives into a shadow loses its power and starts to freeze within hours; NASA's VIPER was planned to travel between "safe havens" of sunlight, entering shadowed craters for short excursions only.
- **Navigation.** Low sun angles hide rocks and holes in long shadows and dazzle cameras pointed towards the Sun. Polar rovers carry their own headlights, and stereo vision and lidar must work in near-total darkness.
- **Communications.** Earth, too, hangs low on the horizon and is often hidden by terrain, so rovers need relay satellites or carefully planned routes to stay in contact.




The Moon's spin axis is tilted only 1.54° to the ecliptic, so near the poles the Sun circles the horizon and crater floors can stay dark for billions of years.

**FIGURE 28.14** Lighting at the lunar south pole. (a) Sun elevation over one lunar day–night cycle at two latitudes. (b) Shadow length of a 1 m boulder at low Sun angles.

### EXAMPLE: HOW LONG ARE THE SHADOWS AT THE POLE?

At 89.5° S during southern summer, the Sun stays about 2° above a flat horizon.

1. A boulder of height $h$ casts a shadow of length $h / \tan e$, where $e$ is the Sun's elevation: for $e = 2°$, a 1 m boulder casts a shadow 29 m long.
2. A crater only 10 m deep, with walls rising 5° from its floor, is entirely in shadow when the Sun is lower than 5° — which, at this latitude, is always.
3. On polar summits a few kilometres high, the Sun may be visible for 80–90 % of the year. This is why these "peaks of near-eternal light" are favoured sites for future lunar bases and their solar power stations.

## 28.6 The Next Generation

**TABLE 28.3** Rovers and aerial explorers in development in 2026.

| Mission | Body | Planned | Features |
|---|---|---|---|
| VIPER (NASA) | Moon, south pole | Late 2027, on Blue Origin Blue Moon MK1 | Cancelled in 2024 and revived in 2025; prospects for water ice with a 1 m drill and a neutron spectrometer; can raise and lower each wheel to free itself |
| CADRE (NASA) | Moon | 2027, on IM-3 | Three suitcase-sized rovers that explore cooperatively as a team |
| Chang'e 7 (CNSA) | Moon, south pole | 2027 | Rover and a hopping probe to search shadowed craters for ice |
| Lunar Terrain Vehicle (NASA) | Moon | From about 2028 | Unpressurised rover for Artemis crews, also driven remotely between missions; competitors Astrolab, Lunar Outpost and Intuitive Machines |
| Lunar Cruiser (JAXA/Toyota) | Moon | Early 2030s | Pressurised rover in which two astronauts can live and drive for weeks |
| IDEFIX (CNES/DLR) on MMX (JAXA) | Phobos | Launch Oct 2026; landing 2029 | 25 kg rover, first on a Martian moon |
| Rosalind Franklin (ESA) | Mars, Oxia Planum | Launch 2028; landing 2030 | Drill to 2 m depth to reach organics shielded from radiation; NASA provides radioisotope heaters and landing engines |
| Dragonfly (NASA) | Titan | Launch 2028; arrival 2034 | Rotorcraft lander |

**FIGURE 28.15** A mobility test model of NASA's VIPER lunar rover on a tilting bed of simulated regolith at NASA Glenn Research Center. Image: NASA/GRC (NASA/GRC/Bridget Caswell).

### AI ON ROVERS

Rovers are among the most autonomous robots ever built. Besides autonomous navigation, Curiosity and Perseverance use **AEGIS**, software that selects rocks in navigation-camera images and aims the laser spectrometer at them without waiting for the ground. Perseverance's PIXL uses **adaptive sampling**, deciding on board where to dwell longer on interesting mineral grains. An onboard scheduler rearranges activities when tasks run faster or slower than planned. On the ground, machine-learning classifiers of terrain type (sand, bedrock, sharp rocks) trained on orbital and rover images help plan routes that spare the wheels. Future rovers will combine these into "science autonomy" — recognising what is scientifically interesting and acting on it (Chapters 48 and 51).

### KEY TAKEAWAYS

- Rovers have explored the Moon since 1970 and Mars since 1997; Opportunity holds the distance record (45.2 km), with Perseverance (> 42 km) and Curiosity (> 37 km) still driving in 2026.
- Rocker-bogie suspensions, terramechanics, wheel design and traction control determine where rovers can go; soft sand and sharp rocks are the main hazards.
- Solar rovers are limited by dust and night-time cold; MMRTG-powered rovers have steady power day and night.
- Light-time forces sol-by-sol planning and growing autonomy: hazard-avoiding navigation, visual odometry, autonomous targeting.
- Rovers established that Mars was once habitable; Perseverance has cached samples, one with a potential biosignature, whose return to Earth is uncertain.
- Ingenuity proved that aircraft can fly on Mars; Dragonfly will fly on Titan, where flight is easiest of all.

### REVIEW QUESTIONS

1. Compare the Lunokhod rovers, the Apollo Lunar Roving Vehicle and the Mars rovers in purpose, mobility and power.
2. Explain how a rocker-bogie suspension works and why it suits rough terrain.
3. What are the advantages and disadvantages of solar power and MMRTG power for Mars rovers?
4. Why must rover operations be planned sol by sol? How have autonomous navigation and visual odometry increased driving distances?
5. What evidence have rovers found that Mars was once habitable?
6. For an ideal rotor, hover power scales as $(mg)^{3/2}/\sqrt{(\rho A)}$. Compare the hover power of the same vehicle on Titan ($\rho = 5.4$ kg/m$^3$, $g = 1.35$ m/s$^2$) with that on Earth ($\rho = 1.225$ kg/m$^3$, $g = 9.81$ m/s$^2$).
7. Why was Ingenuity's first flight a major engineering achievement given Mars's thin atmosphere?
8. What special challenges do rovers face at the lunar poles?

# 29 CHAPTER Sample-Return Missions

Image: NASA (NASA/Keegan Barber)

*The most powerful instruments for studying another world are not on spacecraft but in laboratories on Earth. Sample-return missions bring pieces of the Moon, asteroids, a comet and even the solar wind home, where they can be dated, dissected atom by atom and preserved for scientists not yet born. This chapter explains how samples are collected — by drill, scoop, gas jet, bullet or aerogel — how they are launched off other worlds and carried back through Earth's atmosphere at up to 13 km/s, and how they are kept pristine. It reviews every sample-return mission from Apollo to OSIRIS-REx and Chang'e 6, what their samples have revealed, and the missions under way in 2026, including China's Tianwen-2 at the asteroid Kamoʻoalewa and the uncertain future of Mars Sample Return.*

IN THIS CHAPTER

29.1 Sample-Return Missions So Far
29.2 Engineering a Sample Return
29.3 Mars Sample Return and the Next Decade

Flight instruments must be small, light, low-powered and able to survive launch. Laboratories on Earth have none of these limits. They can measure the age of a rock to within a few million years from the decay of radioactive isotopes, and image individual atoms with transmission electron microscopes. They can also analyse organic molecules at concentrations of parts per billion, and probe grains with synchrotron X-rays. Above all, samples can be **kept**: techniques that do not yet exist can be applied decades later. In 2022, NASA opened a tube of lunar soil that Apollo 17 astronauts had sealed in a vacuum on the Moon in 1972, so that it could be studied with instruments invented since.

**TABLE 29.1** What laboratories on Earth can do that flight instruments cannot.

| Capability | Spacecraft instrument | Laboratory on Earth |
|---|---|---|

| Capability | Spacecraft instrument | Laboratory on Earth |
|---|---|---|
| Radiometric age | Rough estimates only (one experimental dating on Curiosity) | Precise ages from many isotope systems (U–Pb, Rb–Sr, Ar–Ar, Sm–Nd) |
| Spatial resolution | ≈ 0.1 mm (PIXL on Perseverance) | Below 1 nm (electron microscopes, atom probes) |
| Organic analysis | A few target molecules | Thousands of compounds; chirality; isotopes of individual molecules |
| Number of techniques | A few, fixed at launch | Hundreds, improving every year |
| Re-analysis | Impossible once the mission ends | Samples curated for future generations |

## 29.1 Sample-Return Missions So Far

Nine missions or programmes have returned extraterrestrial material to Earth (Figure 29.1). Their returns range from 382 kg of lunar rock to less than a milligram of comet dust and a fraction of a milligram of solar-wind atoms.

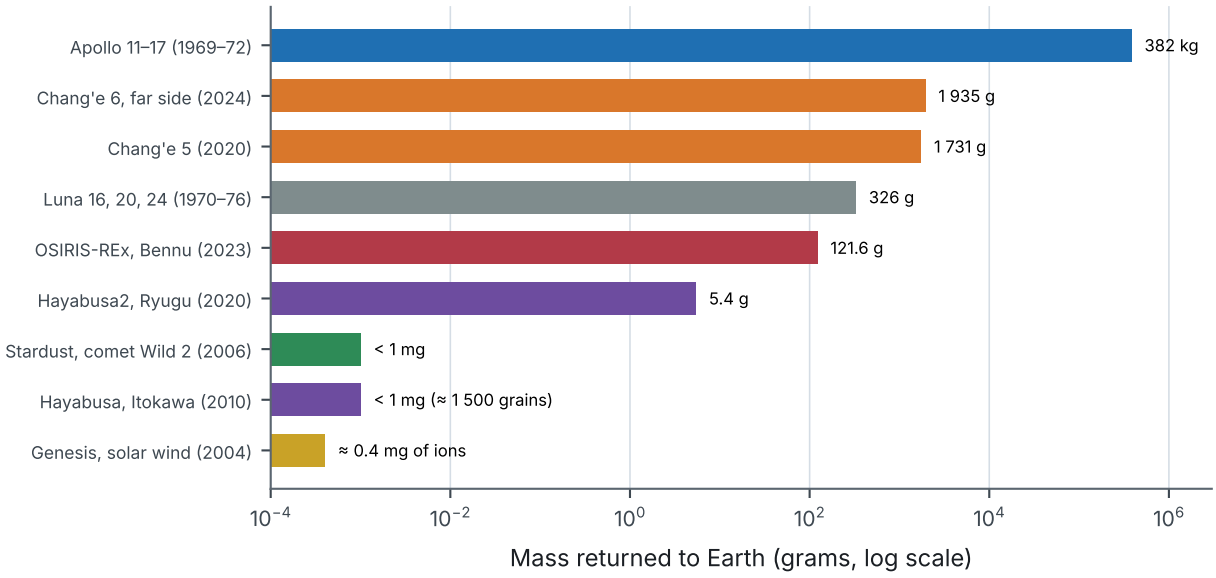



**FIGURE 29.1** Mass of extraterrestrial material returned by each mission (log scale).

### 29.1.1 The Moon

The six **Apollo** landings (1969–72) brought back 382 kg of rocks, soil and cores — 2 196 samples in all — collected by astronauts trained in field geology (Figure 29.2). Three robotic Soviet landers, **Luna 16, 20 and 24** (1970–76), returned a total of 326 g, including a 2 m drill core. Their analyses showed that the Moon formed hot, with a global magma ocean whose floating crystals created the bright highland crust. They also supported the theory that it formed from debris thrown out when a Mars-sized body collided with the young Earth. Lunar samples also calibrated the **crater-counting chronology** used to estimate the ages of surfaces throughout the Solar System.

**FIGURE 29.2** Apollo 17 geologist-astronaut Harrison Schmitt beside a large split boulder at Taurus–Littrow in December 1972. Image: NASA/JSC.

After 44 years, China resumed lunar sample return. **Chang'e 5** (November–December 2020) landed in Oceanus Procellarum and drilled and scooped 1 731 g of soil. It launched the soil into lunar orbit in an ascent stage that docked automatically with the orbiter — the first robotic rendezvous in lunar orbit. The sample was then returned to Inner Mongolia using a "skip" re-entry that bounced off the upper atmosphere to reduce heating. Its basalts are only about 2.0 billion years old, showing that the Moon was volcanically active far later than Apollo samples had suggested. **Chang'e 6** (May–June 2024) returned 1 935 g from the Apollo crater inside the South Pole–Aitken basin, the first samples from the Moon's far side, relaying its signals through the Queqiao-2 satellite. They include basalts about 2.8 billion years old and fragments from volcanism 4.2 billion years ago, and suggest that the far-side mantle was drier than the near side's.

### 29.1.2 A Comet and the Sun

NASA's **Stardust** flew 236 km from the nucleus of comet Wild 2 in January 2004 at 6.1 km/s. It captured thousands of dust particles in **aerogel** — a silica foam less than 1 % as dense as glass. The aerogel slowed them over a few millimetres without vaporising them (Figure 29.4). Its capsule re-entered Earth's atmosphere in January 2006 at 12.9 km/s, the fastest of any human-made object, protected by the first PICA (phenolic-impregnated carbon ablator) heat shield. The comet dust contained the amino acid glycine and, surprisingly, high-temperature minerals that could only have formed close to the young Sun. This was proof that material was mixed across the whole protoplanetary disk before the comets formed.

**FIGURE 29.3** The Stardust collector: blocks of aerogel in an aluminium grid, like an ice-cube tray, before launch. Image: NASA/JPL (NASA/JPL).



**FIGURE 29.4** Carrot-shaped tracks of comet particles that struck the aerogel at 6.1 km/s; the particles themselves lie at the ends of the tracks. Image: NASA/JPL (NASA/JPL).

NASA's **Genesis** spent 27 months at the Sun–Earth L1 point (2001–04) exposing wafers of ultra-pure silicon, sapphire, gold and diamond to the solar wind, collecting about 0.4 mg of solar atoms. On its return in September 2004 its parachute failed to open — an acceleration sensor had been installed upside down — and the capsule hit the Utah desert at more than 300 km/h. Many collectors survived, and their analysis showed that the Sun contains a higher proportion of the oxygen isotope $^{16}O$ than Earth, Moon and Mars, a clue to how the planets formed.

### 29.1.3 Asteroids

Japan's **Hayabusa** survived failed reaction wheels, a fuel leak, seven weeks without contact and failing ion engines to return about 1 500 tiny grains of the asteroid Itokawa in June 2010. They proved that the most common type of asteroid (S-type) is the source of the most common meteorites (ordinary chondrites), linking telescope spectra with laboratory samples.

**Hayabusa2** reached the 900 m carbon-rich asteroid **Ryugu** in June 2018. It released two hopping rovers and the MASCOT lander, and touched down to collect surface grains in February 2019. It then fired a 2 kg copper projectile into the asteroid at 2 km/s to blast an artificial crater. In July 2019 it touched down beside the crater to collect material excavated from below the surface. Its capsule landed at Woomera, Australia, in December 2020 with 5.4 g of Ryugu. The samples are among the most primitive materials ever studied. They are chemically similar to the rare CI

chondrite meteorites and to the composition of the Sun, but altered by liquid water. They contain more than 20 amino acids and the RNA nucleobase uracil. The spacecraft continues: it flew past the asteroid Torifune (2001 CC21) in July 2026 and will rendezvous with the small, fast-spinning asteroid 1998 KY26 in 2031.

NASA's **OSIRIS-REx** reached the 490 m asteroid **Bennu** in December 2018 and found a surface far rougher than expected, covered in boulders (Figure 29.5), and occasionally ejecting small particles into space. On 20 October 2020 it descended to the site called Nightingale and made a **touch-and-go (TAG)** contact lasting about six seconds. Its sampler head released a burst of nitrogen gas that blew loose material into a collection chamber. The head sank almost half a metre into the surface, which turned out to behave like a pit of plastic balls. Had the spacecraft not fired its thrusters to back away, it would have sunk further into the asteroid. So much material was collected that rocks wedged the sampler's flap open and particles leaked out, and the team stowed the sample early.

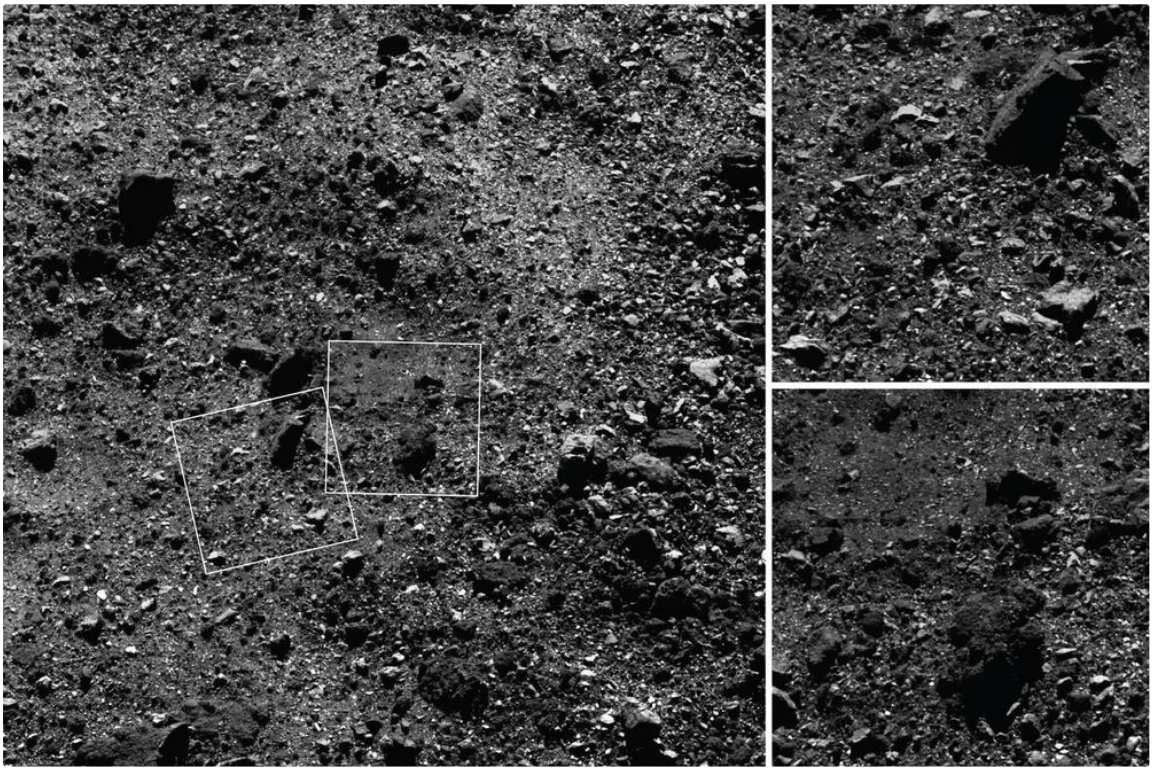


**FIGURE 29.5** Bennu's northern hemisphere seen by OSIRIS-REx in 2019. The asteroid's surface is covered in boulders, making a safe sampling site hard to find. Image: NASA/GSFC.

The capsule landed in Utah on 24 September 2023 (Figure 29.6) with 121.6 g — twice the mission requirement. Two of the 35 fasteners on the sampler head resisted removal, and the curation team had to design new tools before they could open it fully in January 2024 (Figure 29.7). In January 2025 scientists reported that the Bennu samples contain 14 of the 20 amino acids used by life on Earth and all five nucleobases of DNA and RNA. The samples also contain abundant ammonia and a complete sequence of eleven salt minerals that crystallised as salty water evaporated on Bennu's parent body. The spacecraft, renamed **OSIRIS-APEX**, will reach the asteroid Apophis just after its close pass by Earth in April 2029.

**FIGURE 29.6** The OSIRIS-REx sample return capsule shortly after landing in the Utah desert on 24 September 2023. Image: NASA (NASA/Keegan Barber).

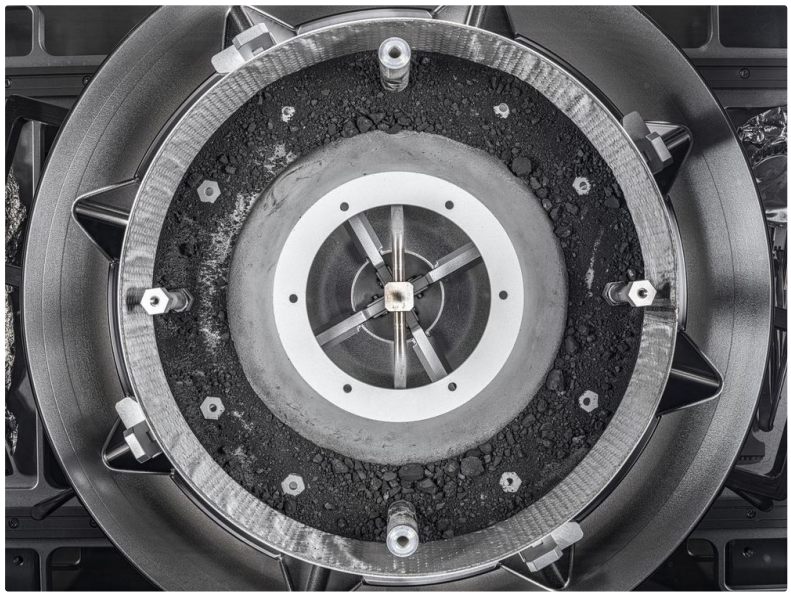


**FIGURE 29.7** The OSIRIS-REx sampler head (TAGSAM) opened in the curation laboratory at NASA's Johnson Space Center in January 2024, with dark Bennu material visible around its rim. Image: NASA/JSC (Erika Blumenfeld & Joseph Aebers).

**TABLE 29.2** Sample-return missions completed by 2026.

| Mission | Agency | Target | Launch | Return | Sample | Collection method |
|---|---|---|---|---|---|---|
| Apollo 11–17 | NASA | Moon | 1969–72 | 1969–72 | 382 kg | Astronauts |
| Luna 16, 20, 24 | USSR | Moon | 1970–76 | 1970–76 | 326 g | Drill |
| Stardust | NASA | Comet Wild 2 | 1999 | 2006 | < 1 mg | Aerogel during flyby |
| Genesis | NASA | Solar wind | 2001 | 2004 | ≈ 0.4 mg | Collector wafers |
| Hayabusa | JAXA | Asteroid Itokawa | 2003 | 2010 | ≈ 1 500 grains | Touchdown (projectile failed to fire) |

| Mission | Agency | Target | Launch | Return | Sample | Collection method |
|---|---|---|---|---|---|---|
| Hayabusa2 | JAXA | Asteroid Ryugu | 2014 | 2020 | 5.4 g | Touchdown with projectile, two sites |
| OSIRIS-REx | NASA | Asteroid Bennu | 2016 | 2023 | 121.6 g | Touch-and-go with nitrogen gas |
| Chang'e 5 | CNSA | Moon | 2020 | 2020 | 1 731 g | Drill and scoop |
| Chang'e 6 | CNSA | Moon, far side | 2024 | 2024 | 1 935 g | Drill and scoop |

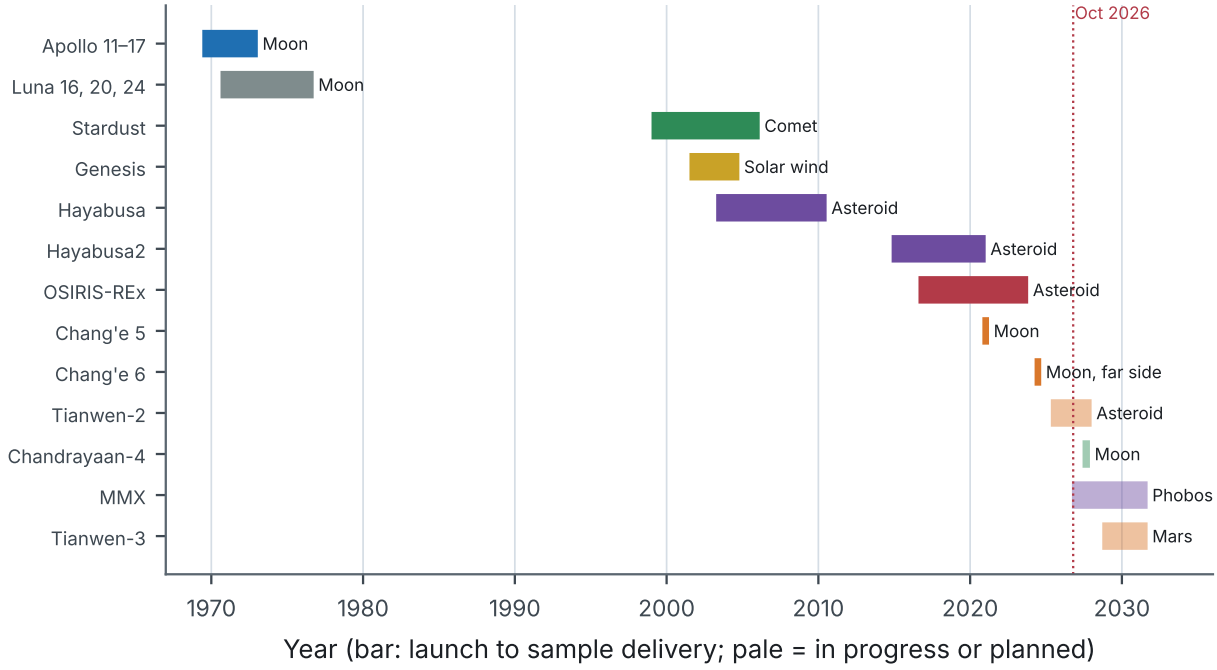



**FIGURE 29.8** Sample-return missions from launch to delivery, including those under way or planned.

## 29.2 Engineering a Sample Return

A sample-return mission must do everything a lander does — and then reverse it: collect material, leave the surface, travel home and survive a high-speed entry into Earth's atmosphere, all without contaminating the sample.

### 29.2.1 Collecting

The method depends on the target (Figure 29.9). On the Moon, drills and scoops on robotic arms collect regolith and cores. On small asteroids with almost no gravity, a spacecraft cannot stand on the surface and push, so **touch-and-go** techniques are used. Hayabusa's sampler horn touches the surface for about a second while a 5 g tantalum bullet fired at 300 m/s throws fragments up into a catcher. OSIRIS-REx's TAGSAM used nitrogen gas to stir the regolith into its collector. Stardust captured dust during a fast flyby in aerogel, and Genesis simply exposed collectors to the solar wind.

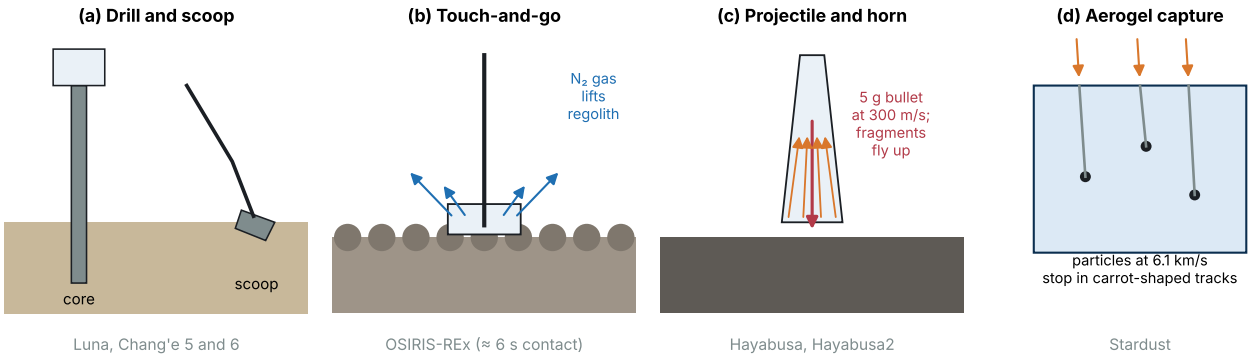
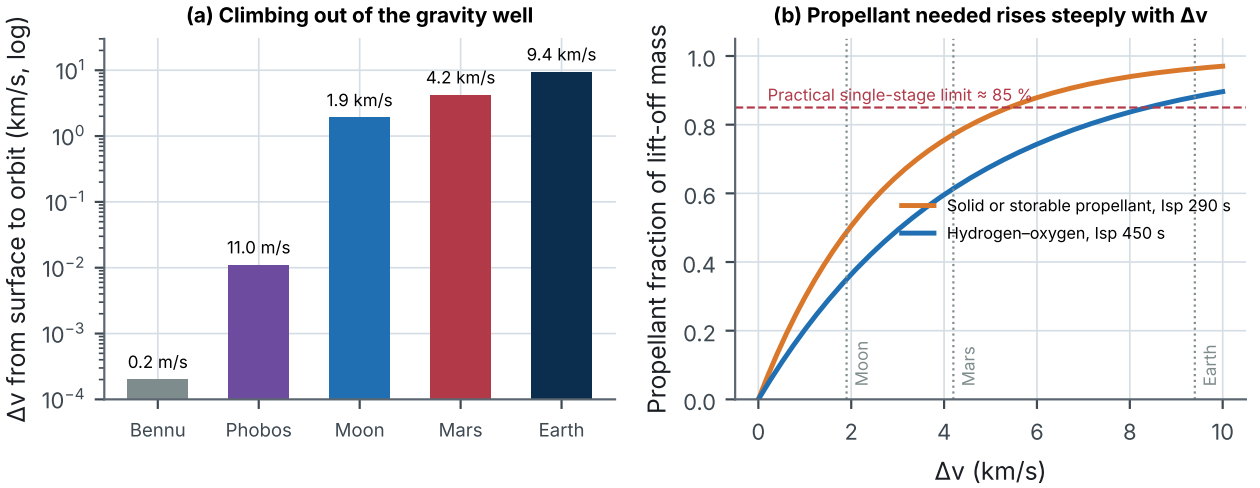



**FIGURE 29.9** Sample-collection techniques.

Missions must also confirm what they have collected. OSIRIS-REx planned to measure its sample by spinning the spacecraft and measuring the change in its moment of inertia. When it found it had collected too much, it instead photographed the sampler head and stowed it quickly to minimise losses.

## 29.2.2 Leaving the Surface

Lifting a sample back into orbit requires a rocket — a small one on the Moon and a substantial one on Mars. On a small asteroid almost nothing is needed, as a gentle push exceeds the escape speed (Figure 29.10). Chang'e 5 and 6 used ascent stages that docked with the orbiter in lunar orbit and transferred the sample container, the same technique Apollo used with crews. For Mars, a **Mars Ascent Vehicle** must reach about 4.2 km/s, including losses, while surviving months on the surface at Martian temperatures. NASA's design was a two-stage solid rocket of about 400 kg lifting a 16 kg sample container — the first launch from another planet.




Δv includes typical gravity and drag losses. Phobos and Bennu: escape speeds; a sample can be lifted off with a gentle push.

**FIGURE 29.10** Leaving the surface. (a) Ascent Δv to orbit, from a gentle push on an asteroid to 4.2 km/s on Mars. (b) Propellant fraction needed for two classes of propellant.

**EXAMPLE: SIZING A MARS ASCENT VEHICLE**

The ascent vehicle must deliver Δv = 4.2 km/s using solid motors with $I_{sp}$ = 290 s ($v_e$ = 2 844 m/s). Assume each stage's structure, nozzle and casing weigh 12 % of the stage's loaded mass (structural ratio ε = 0.12).

1. **Single stage.** Mass ratio $R = e^{4\,200/2\,844}$ = 4.38, so 77 % of the lift-off mass is propellant. The payload fraction is $(1/R - \varepsilon)/(1 - \varepsilon)$ = (0.228 − 0.12)/0.88 = **12.3 %**.
2. **Two equal stages,** each providing 2.1 km/s: $R$ = 2.09 per stage; payload fraction per stage (0.478 − 0.12)/0.88 = 0.407; overall $0.407^2$ = **16.5 %**.
3. Staging increases the payload by about a third and leaves margin for the real penalties. These are guidance, thermal protection, the launch structure and the need to fire after months in the cold. In NASA's design they reduced the payload fraction to about 4 %.
4. For comparison, a lunar ascent needs only 1.9 km/s: with storable propellants ($I_{sp} \approx$ 315 s), propellant is 46 % of the ascent stage's mass.

### 29.2.3 Coming Home

From the Moon, the return trip takes a few days; from asteroids and Mars, a year or more. Shortly before arrival the spacecraft releases a small **sample-return capsule** on a trajectory that intercepts Earth directly, and then diverts itself to fly past. The capsule enters at 11–13 km/s, much faster than a spacecraft returning from Earth orbit at 7.8 km/s. Because the heating rate rises roughly as the cube of the speed, a capsule entering at 12.4 km/s (OSIRIS-REx) experiences about four times the peak heat flux of one returning from low Earth orbit. Capsules use ablative heat shields of PICA or carbon phenolic and deploy parachutes at a few kilometres' altitude. They land in remote, closely controlled areas — the US Air Force's Utah Test and Training Range, the Woomera Range in South Australia, or Inner Mongolia in China. There, recovery teams wait with helicopters and clean containers.

### 29.2.4 Keeping Samples Pristine

A sample is only as valuable as its purity. Contamination by terrestrial organics, water or even fingerprints can ruin the search for prebiotic chemistry. Spacecraft that touch samples are built from a short list of approved, well-characterised materials, and **witness plates** fly alongside the samples to record any contamination. Capsules are purged with dry nitrogen immediately after landing, and the samples are opened and processed in glove boxes filled with pure nitrogen. This is done at dedicated **curation facilities**, such as NASA's at the Johnson Space Center (Figure 29.11) and JAXA's at Sagamihara. Fractions are shared among agencies: NASA received 10 % of the Ryugu sample, and JAXA and the Canadian Space Agency received portions of Bennu.

**FIGURE 29.11** Scientists in clean-room suits examining asteroid Ryugu particles at NASA's Johnson Space Center, part of the sample exchanged with JAXA. Image: NASA/JSC (ROBERT MARKOWITZ NASA-JSC).

### 29.2.5 Planetary Protection for Returned Samples

International planetary protection policy places samples from the Moon, comets and most asteroids in the **unrestricted Earth return** category: they are not thought to pose any biological risk. Samples from Mars, Europa or Enceladus — places that might harbour life — fall into **restricted Earth return** (Category V). The mission must "break the chain of contact" with the target body, so that nothing that has touched Mars can reach Earth's environment outside a sealed container. The samples must be opened in a high-containment receiving facility comparable to a biosafety-level-4 laboratory until they are shown to be safe. The capsule designed for Mars Sample Return was to land without a parachute. A parachute that might fail was considered a greater risk than a deliberate impact onto a crushable, energy-absorbing structure designed to keep the sample container intact (Figure 29.12).

**FIGURE 29.12** Artist's concept of the Earth Entry System designed for Mars Sample Return approaching Earth. Image: NASA/JPL (NASA/ESA/JPL-Caltech/GSFC).

## 29.3 Mars Sample Return and the Next Decade

**Mars Sample Return (MSR)** was planned as a NASA–ESA campaign to bring Perseverance's sealed tubes to Earth (Figure 29.13). A Sample Retrieval Lander would receive the tubes and launch them into Mars orbit on the Mars Ascent Vehicle; ESA's Earth Return Orbiter would capture the container and carry it home. As the design matured, the estimated cost rose to about US$ 8–11 billion and the return date slipped to the late 2030s or 2040. NASA studied cheaper alternatives in 2024–25, but the US budget for fiscal year 2026 provided no money for the existing programme. In October 2026 the samples, including one with a potential biosignature (Chapter 28), remain on Mars without a funded plan to retrieve them.

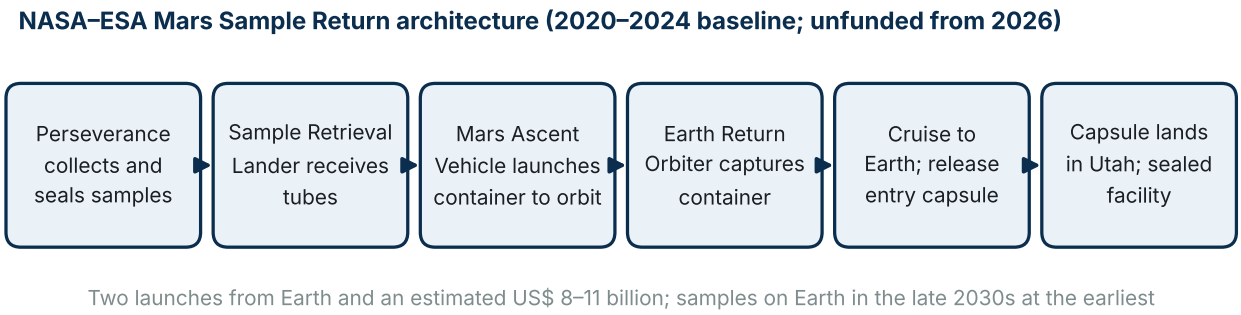



**FIGURE 29.13** The NASA–ESA Mars Sample Return architecture and China's simpler Tianwen-3 approach.

China plans to reach the goal first. **Tianwen-3**, to be launched on two Long March 5 rockets in 2028, will land, drill and scoop at its own site, launch the samples to a waiting orbiter and return them to Earth in 2031. Other sample-return missions under way or planned include:

- **Tianwen-2** (CNSA): launched in May 2025, it arrived at the small near-Earth asteroid **Kamoʻoalewa**, a quasi-satellite of Earth that spins once every 28 minutes and may be a fragment of the Moon, in mid-2026. It is mapping the asteroid before sampling, and is due to return its capsule to Earth in late 2027 before continuing to the main-belt comet 311P.
- **MMX** (JAXA): scheduled for launch in October 2026 on an H3 rocket, it will orbit Mars in 2027. It will land on Phobos with the French–German IDEFIX rover in 2029 and collect at least 10 g of Phobos regolith with a coring sampler and pneumatic sampler. It will return the samples in 2031.
- **Chandrayaan-4** (ISRO): approved in 2024, it will be launched in two parts on two LVM-3 rockets. Its five modules will dock in Earth and lunar orbit to return about 3 kg from near the south pole, around 2027–28.

- **Artemis**: astronauts landing near the lunar south pole from the late 2020s will collect samples from the oldest and most volatile-rich regions of the Moon (Chapters 31 and 33).

### AI IN SAMPLE-RETURN MISSIONS

Touch-and-go sampling depends on autonomous navigation: OSIRIS-REx used **Natural Feature Tracking**, comparing images taken during its descent with an onboard catalogue of Bennu's boulders. This let it update its trajectory and abort automatically if it was heading for a hazard. Hayabusa2 navigated using target markers it dropped on the surface. On Earth, machine learning segments X-ray computed-tomography scans of thousands of particles to classify minerals and find rare grains without destroying them. It also helps curators track every fragment as samples are subdivided among hundreds of laboratories (Chapter 51).

### KEY TAKEAWAYS

- Laboratories on Earth can date, image and analyse samples far beyond the capability of flight instruments, and curated samples serve future generations.
- Nine missions or programmes have returned extraterrestrial material: Apollo, Luna, Stardust, Genesis, Hayabusa, Hayabusa2, OSIRIS-REx, Chang'e 5 and Chang'e 6.
- Collection techniques are matched to gravity: drills and scoops on the Moon, touch-and-go with gas or projectiles on asteroids, aerogel during a comet flyby.
- Leaving the surface costs 1.9 km/s from the Moon and 4.2 km/s from Mars; return capsules enter Earth's atmosphere at 11–13 km/s and need ablative heat shields.
- Samples revealed the Moon's violent origin and late volcanism, the mixing of the early Solar System, and abundant prebiotic molecules on carbon-rich asteroids.
- Mars samples require "restricted Earth return". NASA's Mars Sample Return lost its funding in 2026, while China's Tianwen-3 aims to return Martian samples in 2031; Tianwen-2, MMX and Chandrayaan-4 are under way or in preparation.

### REVIEW QUESTIONS

1. Give three measurements that a laboratory on Earth can make on a returned sample but a flight instrument cannot, and explain why curation for future generations matters.
2. Why did Stardust use aerogel, and what did its comet particles reveal about mixing in the early Solar System?
3. Compare the sampling techniques of Hayabusa2 and OSIRIS-REx. Why are touch-and-go methods used on small asteroids?
4. Using the rocket equation, estimate the propellant fraction for a lunar ascent stage ($\Delta v = 1.9$ km/s, $I_{sp} = 315$ s). How does this compare with a Mars ascent?
5. A capsule returns from an asteroid at 12.4 km/s. By roughly what factor is its peak heating higher than that of a capsule returning from low Earth orbit at 7.8 km/s?
6. What is "restricted Earth return", and why did the Mars Sample Return capsule design omit a parachute?
7. What did the Chang'e 5 and Chang'e 6 samples reveal about the Moon's volcanic history?
8. List the organic molecules and minerals reported in the Bennu samples, and explain why they are significant for the origin of life.

PART IX

# Human Spaceflight and Habitation

Keeping people alive and productive in space is the hardest engineering problem of all. Part IX explains the human body in space, life support, crewed spacecraft, space stations and the spacesuits that let astronauts work outside.


30 Human Spaceflight Fundamentals and Life Support

31 Crewed Spacecraft

32 Space Stations and Habitats

33 Extravehicular Activity Systems


Image: NASA/JSC

CHAPTER

# 30 Human Spaceflight Fundamentals and Life Support

Image: NASA/JSC

*A human being is the most demanding payload a spacecraft can carry. Every day each crew member consumes almost a kilogram of oxygen and several kilograms of water and food, and produces carbon dioxide, heat and waste that must be removed. Weightlessness weakens bones, muscles, the heart and even the eyes, while cosmic radiation slowly raises the risk of cancer. This chapter explains how the body adapts to spaceflight and how crews are protected: the radiation environment and dose limits, the design of cabin atmospheres, the metabolic budget, and the environmental control and life support systems (ECLSS) that recycle air and water on the International Space Station. It closes with food, medicine, psychology, the history of fatal accidents and the safety standards that govern crewed vehicles in 2026.*

**IN THIS CHAPTER**

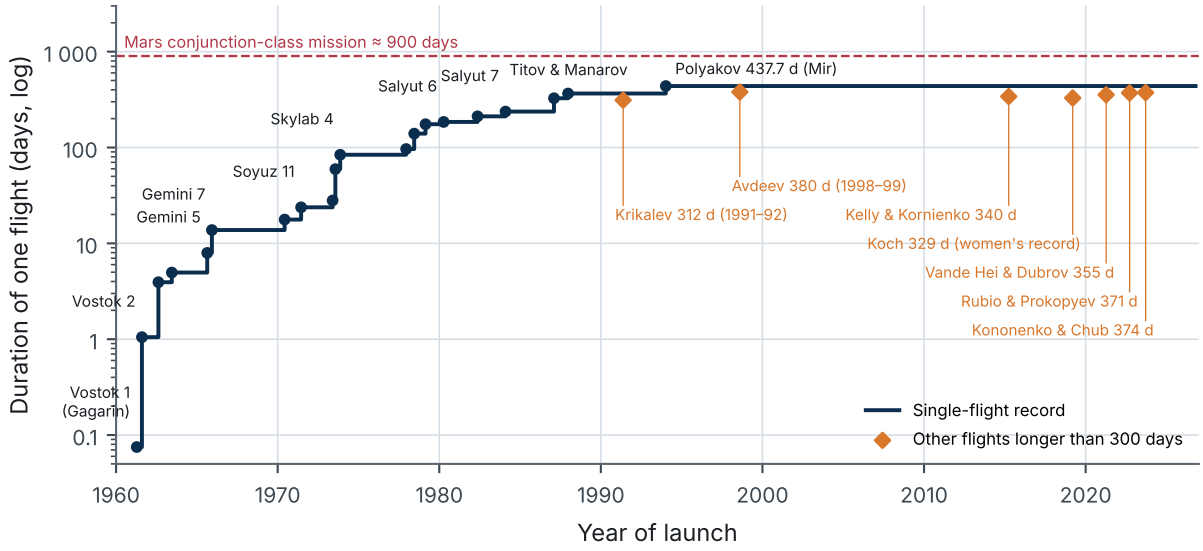
30.1 Humans in Space: How Long and How Far
30.2 The Human Body in Weightlessness
30.3 Radiation: The Long-Term Hazard
30.4 Designing the Cabin Atmosphere
30.5 The Human Metabolic Budget
30.6 Environmental Control and Life Support Systems
30.7 Living and Working in Space
30.8 Crew Safety and Risk

On 12 April 1961, Yuri Gagarin completed one orbit of the Earth in Vostok 1 and landed 108 minutes after launch. By mid-2026 about 780 people from 55 countries had flown above 100 km — a crew small enough to fit into a single airliner, but one that has lived continuously in orbit since November 2000. People are sent into space because they bring what machines still lack: judgement in unforeseen situations, dexterity for repairs and assembly, and a field scientist's eye. Five Shuttle crews serviced and upgraded the Hubble Space Telescope, and Apollo astronauts chose the most telling rocks on the Moon.

The price is steep. A crewed spacecraft must provide a pressurised, breathable, temperature-controlled bubble. It must remove carbon dioxide, heat, moisture and waste, supply water and food, and protect against radiation and fire. It must keep the crew healthy and bring them home, all with far higher reliability than an uncrewed spacecraft. This chapter covers the biology and the life support engineering; the vehicles themselves are described in Chapter 31, space stations and habitats in Chapter 32, and spacesuits in Chapter 33.

## 30.1 Humans in Space: How Long and How Far

Human spaceflight has progressed in steps of duration (Figure 30.1). Gemini 7 showed in 1965 that people could survive the two weeks needed for a lunar mission. Skylab and the Soviet Salyut and Mir stations pushed single flights from one month to over a year. The record of **437.7 days**, set by the Russian physician Valeri Polyakov on Mir in 1994–95, still stands in 2026. The longest flights on the International Space Station (ISS) are those of Oleg Kononenko and Nikolai Chub (374 days, 2023–24). Next come Sergey Prokopyev, Dmitri Petelin and Frank Rubio (371 days, 2022–23), whose flight is the US record. It was extended when their Soyuz MS-22 sprang a coolant leak. In June 2024 Kononenko became the first person to spend 1 000 days in space, finishing his fifth flight with 1 111 days in total. Peggy Whitson holds the US and women's cumulative record of 695 days.



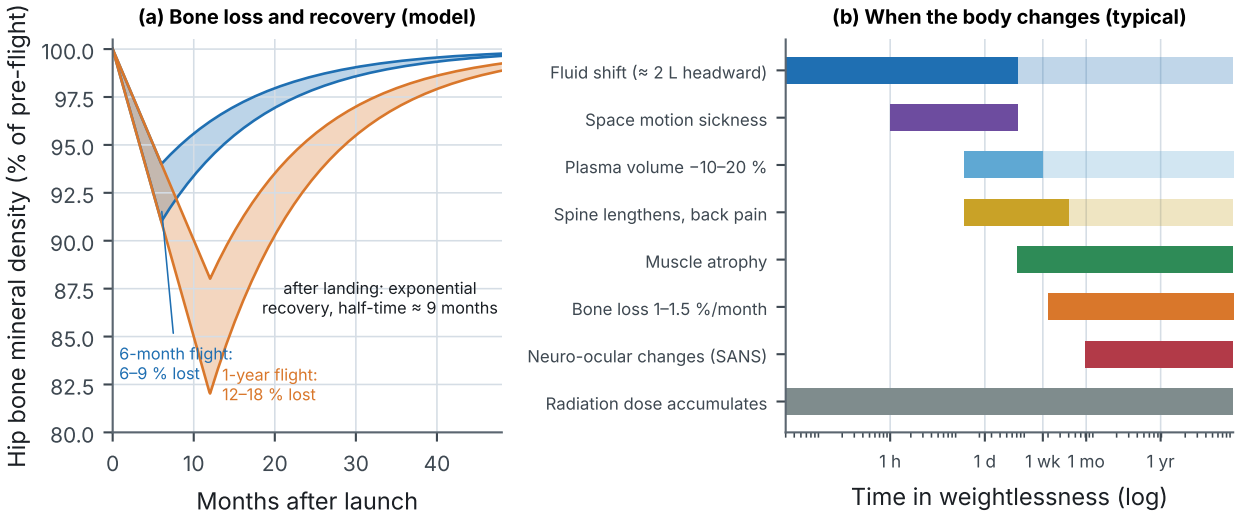

Record progression from mission durations; Kononenko holds the cumulative record, 1 111 days over five flights (2024).

**FIGURE 30.1** Single-flight duration record from Vostok 1 to Polyakov's 437.7 days on Mir, with the other flights that have lasted longer than 300 days. A crewed Mars mission of the conjunction class would last about twice as long as the record.

A round trip to Mars would last 500–1 000 days with no resupply and no quick return. Everything in this chapter — from bone loss to recycling — becomes harder as missions grow longer and further from Earth.

# 30.2 The Human Body in Weightlessness

In orbit the crew falls freely with the spacecraft, so nothing presses the body against a floor (Chapter 2). Tissues tuned to 1 g respond within minutes to months (Figure 30.2). Most changes are adaptations rather than illnesses, but they become problems on return to gravity — on Earth, the Moon or Mars.




(a) Linear loss of 1–1.5 %/month in flight, exponential recovery with 9-month half-time (Sibonga et al. 2007); modern resistive exercise reduces the loss. (b) Dark: onset/main phase; pale: new steady state.

**FIGURE 30.2** How the body changes in weightlessness. (a) A simple model of bone loss in the hip during 6- and 12-month flights and of recovery after landing. (b) Typical onset and duration of the main changes.

## 30.2.1 Fluids, Heart and Eyes

On Earth, gravity pulls blood and tissue fluid towards the legs. In weightlessness about two litres shift towards the chest and head within hours: faces become puffy, legs thin and noses feel blocked. The body treats the extra fluid in the chest as an overload and excretes it, so blood plasma volume falls by roughly 10–20 % within days, and the reflexes that control blood pressure weaken. After landing many astronauts suffer **orthostatic intolerance** — dizziness or fainting when standing up — so crews drink salt water before entry, wear compression garments and are often helped out of the capsule.

A more puzzling effect, first described in 2011, concerns the eyes. **Spaceflight-associated neuro-ocular syndrome (SANS)** is a set of changes to the eye. The optic disc swells where the nerve enters the eye, and folds form in the choroid layer behind the retina. The back of the eyeball flattens, and vision shifts towards long-sightedness. Imaging studies suggest that most long-duration crew members show some of these signs; in one prospective ISS study about two-thirds had optic-disc swelling by the 150th day of flight. The likely cause is the headward fluid shift, which keeps pressure in the veins of the head slightly raised all the time. Crews monitor their eyes with optical coherence tomography and ultrasound (Figure 30.4), and lower-body negative-pressure devices that draw fluid back towards the legs are being tested. Some changes persist for years, making SANS a leading health concern for Mars missions.

### 30.2.2 Balance and Motion Sickness

The inner ear's otolith organs sense linear acceleration and the direction of gravity. In free fall they send signals that conflict with what the eyes see. More than half of all space travellers suffer **space motion sickness** — nausea, vomiting, headache and lethargy — during the first two or three days. Spacewalks are not scheduled during this period, because vomiting inside a helmet could be fatal. The brain relearns to rely on vision and touch, but must readapt on return: astronauts walk unsteadily for days after landing — a serious matter for a crew stepping onto Mars after months in transit. Astronauts also grow up to about 3 % taller as the discs between the vertebrae expand, which often causes back pain.

### 30.2.3 Bone and Muscle

Bone is constantly rebuilt in response to the loads placed on it. Without body weight, the hip and lumbar spine lose mineral at **1–1.5 % per month** — about as much in one month as an elderly person loses in a year. The calcium released raises the risk of kidney stones. After landing, bone is rebuilt only slowly. A study of 45 crew members on 4–6-month flights found that it took about nine months to recover half of the loss. It also found that recovery was largely complete only after about three years. Figure 30.2 (a) shows what this implies for a one-year mission without effective exercise: a loss of 12–18 % in the hip, with years needed to recover.

Muscles that keep us upright — the calves, thighs and back — waste fastest. Even with exercise, ISS crews in a 2009 study lost about 13 % of calf muscle volume and almost a third of calf muscle power over six months. The main countermeasure is exercise, about two hours every day, on three machines:

- the **Advanced Resistive Exercise Device (ARED)**, which uses vacuum cylinders to provide up to about 2.7 kN (600 lbf) of resistance for squats, dead-lifts and heel raises (Figure 30.3);
- the **T2 treadmill**, on which the astronaut is held down by a harness and bungee cords;
- a **cycle ergometer** for aerobic exercise (CEVIS, replaced around 2023 by the smaller FERGO).

Since ARED arrived in late 2008, crews who also eat enough and take vitamin D have lost markedly less bone than earlier crews, and drugs that slow bone breakdown (bisphosphonates) have been tested in flight. Exercise has an engineering cost: more oxygen, heat and humidity, vibration that must be isolated from the structure, and equipment that a small Mars transit vehicle can barely accommodate.

**FIGURE 30.3** NASA astronaut Nick Hague exercising on the Advanced Resistive Exercise Device in the ISS Tranquility module, October 2024. Image: NASA/JSC.

**FIGURE 30.4** NASA astronaut Jonny Kim, assisted by JAXA astronaut Takuya Onishi, performing an eye ultrasound scan on the ISS in May 2025 to look for signs of SANS. Image: NASA/JSC.

### 30.2.4 Immune System, Sleep and the Twins Study

Spaceflight also disturbs the immune system: dormant herpes viruses reactivate and are shed in the saliva of many crew members, and skin rashes are common. Sleep is short — about six hours a night on average for Shuttle and ISS crews, despite 8.5 hours being scheduled — disturbed by noise, a sunrise every 92 minutes and workload.

The most detailed picture of these changes came from NASA's **Twins Study** (2015–16). Scott Kelly spent 340 days on the ISS while his identical twin Mark, also an astronaut, stayed on Earth as a genetic control (Figure 30.5). In flight, Scott's telomeres — the protective ends of the chromosomes — unexpectedly lengthened, then shortened rapidly after his return. Thousands of genes changed their activity, most returning to normal within six months. His cognitive test performance also dropped after landing. A 2024 package of 44 papers on the three-day private Inspiration4 flight found similar molecular changes even on a short mission, most of which faded within months. Spaceflight is a whole-body stress that people tolerate well for months, but whose long-term effects are still being measured.

Table 30.1 summarises the main effects and their countermeasures.

**TABLE 30.1** Main effects of weightlessness on the human body and their countermeasures.

| Body system | Effect in weightlessness | Typical size | Countermeasures |
|---|---|---|---|
| Fluids, heart | Headward fluid shift; fainting on return | ≈ 2 L shifted; plasma −10–20 % | Fluid loading before entry, compression garments |
| Balance | Motion sickness; unsteady after landing | Over half of crew, first 2–3 days | Medication, post-flight rehabilitation |
| Bone | Loss from hip and spine; kidney stones | 1–1.5 % per month | ARED exercise, diet, vitamin D, drugs |
| Muscle | Atrophy of postural muscles | Calf power ≈ −30 % in 6 months | Treadmill, ARED, ergometer, ≈ 2 h/day |
| Eyes (SANS) | Optic-disc swelling, choroidal folds | Signs in most long-duration crew | Monitoring; negative-pressure devices studied |
| Immune, sleep | Virus reactivation; ≈ 6 h of sleep | Many crew members | Lighting schedules, sleep stations |

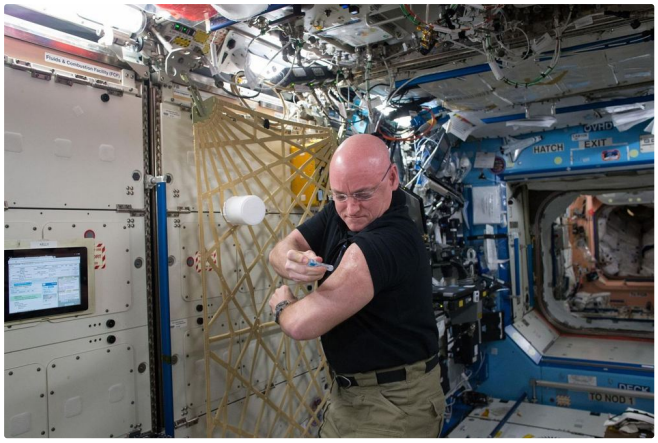



**FIGURE 30.5** Scott Kelly giving himself a flu vaccination aboard the ISS in September 2015, part of the Twins Study investigation of how the immune system responds in flight. Image: NASA/JSC (Scott Kelly).

## 30.3 Radiation: The Long-Term Hazard

Radiation is the one hazard that grows steadily with every day outside the Earth's protection. Chapter 2 describes the sources and Chapter 22 the solar storms; for a crew they mean three things:

- **Galactic cosmic rays (GCRs)**: atomic nuclei at close to the speed of light, arriving continuously and almost impossible to stop. The rare heavy nuclei such as iron (**HZE** particles, for high charge Z and energy E) leave dense tracks of damage through cells.
- **Solar particle events (SPEs)**: rare bursts of protons that are easier to shield, but can deliver a dangerous dose within hours to an unprotected crew.
- **Trapped particles** in the Van Allen belts, which dominate the dose in low Earth orbit and during transits of the belts.

### 30.3.1 Dose, Risk and Limits

The energy deposited per kilogram of tissue is the **absorbed dose**, in grays (1 Gy = 1 J/kg). Weighting it for the greater damage done by heavy ions and for the sensitivity of each organ gives the **effective dose**, in sieverts (Sv). Doses above about 1 Gy within hours cause acute radiation sickness. At the lower doses of spaceflight, the concern is an increased lifetime risk of cancer, and possible damage to the heart and central nervous system.

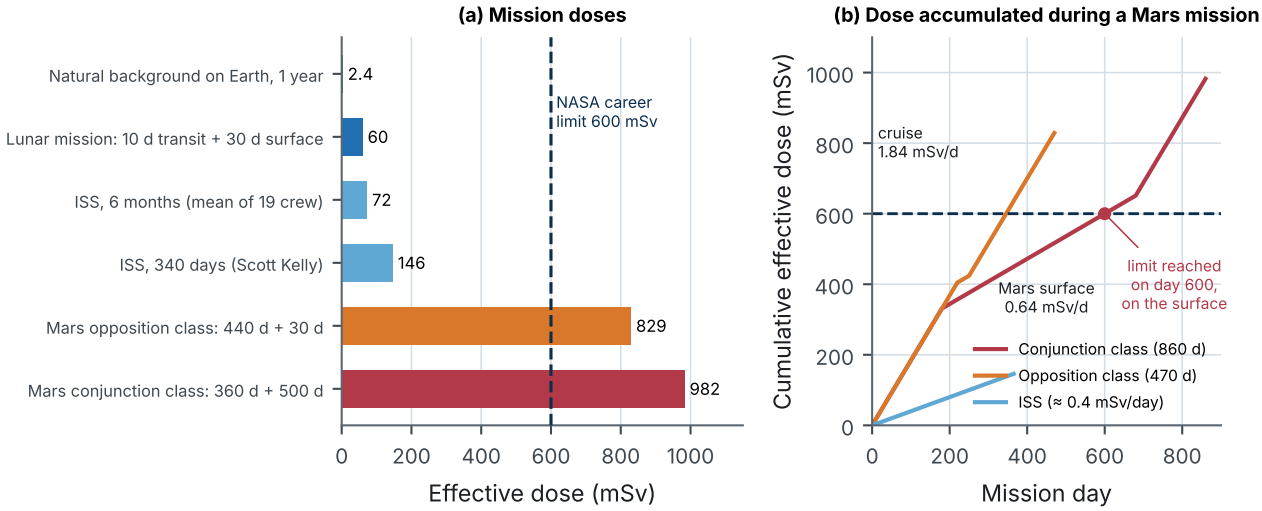
Table 30.2 lists measured dose rates. A six-month ISS mission gives about 70–160 mSv, depending on the solar cycle; Scott Kelly received 146 mSv in 340 days. Outside the magnetosphere the rate jumps. The Radiation Assessment Detector on the Curiosity rover measured 1.84 mSv per day during its cruise to Mars in 2011–12. On the surface, where the planet blocks half the sky, it measured 0.64 mSv per day. On the Moon, China's Chang'e 4 lander measured about 1.4 mSv per day.

In 2022 NASA adopted a single **career limit of 600 mSv** effective dose for all astronauts, regardless of age or sex. It corresponds to a calculated mean 3 % risk of dying from radiation-induced cancer for the most sensitive group (a 35-year-old woman). It replaced earlier limits that allowed older men a higher dose than young women. For comparison, international recommendations limit radiation workers on Earth to an average of 20 mSv per year. NASA also

requires shielding to keep the dose from a large SPE — modelled on the October 1989 event — below 250 mSv. Figure 30.6 compares mission doses with the limit. Low-Earth-orbit and lunar missions fit comfortably, but a Mars mission does not.

**TABLE 30.2** Measured radiation dose rates for crews.

| Environment | Dose rate | Notes |
|---|---|---|
| Earth's surface | ≈ 0.007 mSv/day | 2.4 mSv per year world average |
| ISS (400 km, 51.6°) | 0.4–0.9 mSv/day | Mostly trapped protons and GCRs |
| Lunar surface | ≈ 1.4 mSv/day | Chang'e 4, 2019 (solar minimum) |
| Interplanetary cruise | ≈ 1.8 mSv/day | Curiosity RAD, 2011–12 |
| Mars surface | ≈ 0.64 mSv/day | Curiosity RAD, 2012–13 |
| Large SPE in deep space | Up to hundreds of mSv in a day | Depends strongly on shielding |




Rates: MSL/RAD cruise 1.84 mSv/day and Mars surface 0.64 mSv/day (2012–13); lunar surface 1.37 mSv/day (Chang'e 4 LND); ISS from Cucinotta et al. 2008 and the Twins Study. No large solar particle event included.

**FIGURE 30.6** Radiation doses for crews. (a) Estimated effective doses for different missions compared with NASA's 600 mSv career limit. (b) Dose accumulated during conjunction-class and opposition-class Mars missions: the limit is reached during the surface stay.

## 30.3.2 Shielding and Storm Shelters

Shielding is specified by its **areal density** — mass per unit area, in g/cm² — because what matters is how much material a particle must cross. A spacecraft wall provides only a few g/cm²; racks, water and stowage inside add more. For SPE protons, shielding works well: the lower-energy protons stop, and NASA estimates that typical aluminium or polyethylene shielding halves the average SPE dose. For GCRs it barely helps — only about 7 %. This is because the most energetic nuclei pass through any practical thickness and break up into showers of **secondary** particles, including neutrons, that add their own dose. Hydrogen-rich materials such as polyethylene and water work better than aluminium because they fragment incoming ions without producing many secondaries.

The practical strategy is to accept the GCR dose, limit mission duration, and provide a **storm shelter** for SPEs. This is a small volume, surrounded by water, food and stowage, where the crew retreats for the hours to days of an event. Warnings come from the space-weather forecasters described in Chapter 22. During Artemis I in 2022 two female-torso phantoms, Helga and Zohar, flew in Orion's passenger seats carrying thousands of dosimeters (Figure 30.7). Dose rates

differed by a factor of four between the least and most shielded positions. Turning the spacecraft through 90° halved the dose rate in the inner belt. A well-placed shelter could keep the dose from a very large SPE below about 150 mSv. On Artemis II in April 2026 a planned in-flight shelter demonstration was shortened to prepare for entry. Building the shelter from stowage, already practised on the ground, was skipped. Only the ventilation of an astronaut sheltering in a locker was tested.

**FIGURE 30.7** The radiation phantoms Helga (below) and Zohar (above) installed in Orion's passenger seats before Artemis I in August 2022. Zohar wears the AstroRad radiation-protection vest. Image: NASA/KSC (NASA/Frank Michaux).

**EXAMPLE: RADIATION DOSE FOR A MARS MISSION**

A conjunction-class mission spends 180 days travelling to Mars, 500 days on the surface and 180 days returning. Use the Curiosity RAD dose rates: 1.84 mSv/day in cruise and 0.64 mSv/day on the surface. Ignore SPEs, which the storm shelter is designed to handle.

1. Cruise dose: 360 days × 1.84 mSv/day = **662 mSv** — already above the 600 mSv career limit.
2. Surface dose: 500 days × 0.64 mSv/day = **320 mSv**.
3. Total: 982 mSv, **164 %** of the limit. Following the dose day by day (Figure 30.6 b), the limit is reached on mission day 600, after 420 of the 500 days on the surface.
4. An opposition-class mission (220 days out, 30 days on Mars, 220 days back) gives 1.84 × 440 + 0.64 × 30 = **829 mSv**. It is shorter, but still over the limit because more time is spent in deep space.
5. To stay within 600 mSv with a 500-day surface stay, the total transit time would have to be below (600 − 320)/1.84 = **152 days** — 76 days each way, beyond chemical propulsion. Thicker shielding hardly helps against GCRs; the options are faster propulsion (Chapter 42), flying near solar maximum when GCRs are weaker, or a deliberate decision to accept a higher risk.

## 30.4 Designing the Cabin Atmosphere

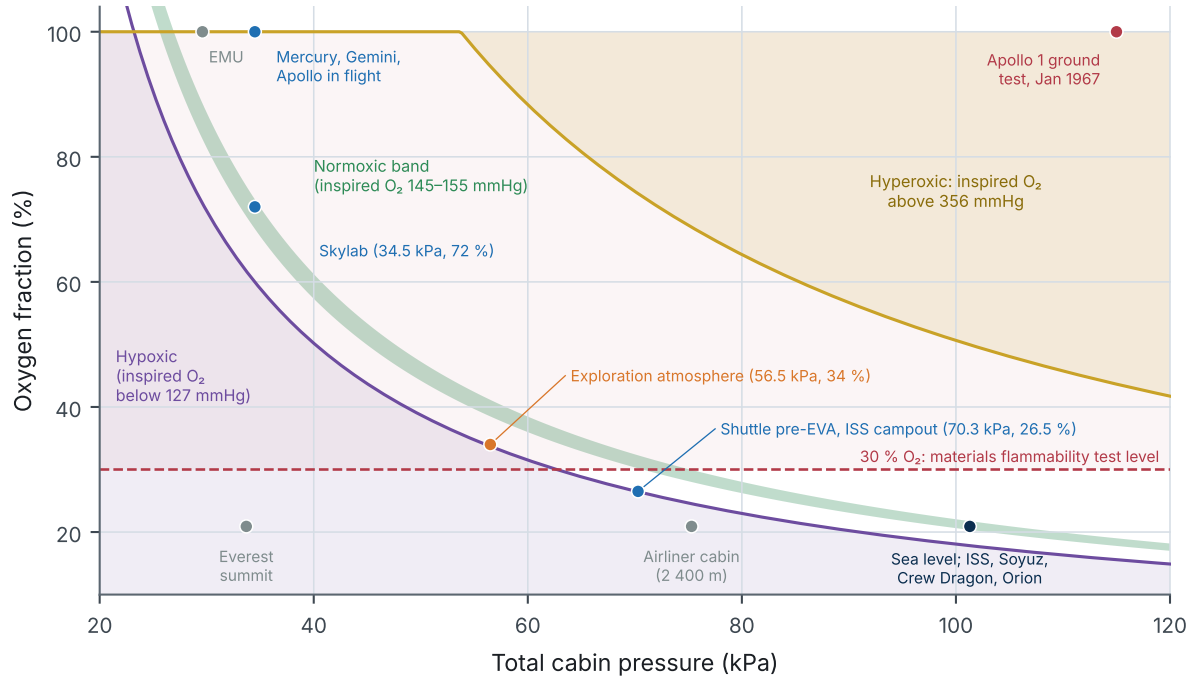
Engineers choose the cabin's total pressure and oxygen fraction, balancing four hazards: **hypoxia** (too little oxygen), **oxygen toxicity** (too much), **fire**, and **decompression sickness** when moving to a spacesuit at lower pressure. Lower pressure also reduces hull loads and leakage.

### 30.4.1 Oxygen and Hypoxia

What the lungs care about is the partial pressure of oxygen in the air reaching them. Air entering the lungs is saturated with water vapour at body temperature, which takes up 47 mmHg (6.3 kPa) regardless of altitude, so the **inspired oxygen partial pressure** is

$$P_{IO_2} = (P_{total} - 6.3\ \text{kPa}) \times F_{O_2}$$

where $F_{O_2}$ is the oxygen fraction. At sea level this gives 149 mmHg (19.9 kPa). NASA's standard for crewed spacecraft (NASA-STD-3001, Volume 2) treats 145–155 mmHg as normal and limits mild hypoxia to no lower than 127 mmHg. It sets 356 mmHg as the highest value tolerable indefinitely. This is because breathing very high oxygen pressures damages the lungs and, at extreme levels, causes seizures. On the summit of Everest the inspired value is only 43 mmHg. Figure 30.8 maps these limits onto the two design variables and shows the choices made for real spacecraft.




Inspired $O_2$ partial pressure = (total pressure − 47 mmHg water vapour) × $O_2$ fraction; limits from NASA-STD-3001 Vol. 2 (Rev F, 2026). Lower total pressure means less nitrogen and shorter pre-EVA oxygen prebreathe.

**FIGURE 30.8** The cabin atmosphere design space. Curves of constant inspired oxygen pressure show the hypoxic, normal and hyperoxic regions; the dashed line marks the 30 % oxygen level at which spacecraft materials are commonly tested for flammability.

### 30.4.2 Fire

Materials burn faster as the oxygen fraction rises, and in weightlessness, with no buoyant convection, flames become spherical and slow but can persist where on Earth they would go out. The danger was established tragically on 27 January 1967, when a fire swept through the Apollo 1 command module during a ground test, killing Gus Grissom, Ed White and Roger Chaffee. The cabin held pure oxygen at about 115 kPa and contained flammable nylon netting and Velcro. A spark from wiring started a fire that spread in seconds, and the inward-opening hatch could not be opened against the cabin pressure. NASA changed the launch atmosphere to 60 % oxygen and 40 % nitrogen (vented to pure oxygen at 34.5 kPa during ascent), replaced flammable materials, and fitted a quick-opening hatch.

Today, materials for US crewed vehicles must pass an upward flame-spread test, commonly run at 30 % oxygen and 70.3 kPa, the worst case expected in flight. NASA's Saffire experiments (2016–24) burned large samples inside departing Cygnus cargo ships, providing the first data on how big fires grow in weightlessness.

### 30.4.3 Decompression Sickness

Nitrogen dissolved in body tissues forms bubbles if the surrounding pressure falls quickly — **decompression sickness (DCS)**, the divers' "bends". Spacesuits work at low pressure to stay flexible: the US suit at 29.6 kPa of pure oxygen, the Russian Orlan at about 40 kPa (Chapter 33). Consider a crew member moving from a sea-level cabin, with 80 kPa of nitrogen in the tissues, into a suit at 29.6 kPa. The ratio of tissue nitrogen pressure to suit pressure would be 2.7 — far too high. Astronauts therefore **prebreathe** pure oxygen for hours before a spacewalk, sometimes with exercise or a night at reduced pressure (the "campout").

A lower-pressure cabin with more oxygen shortens prebreathe dramatically. NASA's **exploration atmosphere** — 56.5 kPa (8.2 psi) with 34 % oxygen — leaves about 37 kPa of nitrogen in the tissues, a ratio of only 1.26 to the suit. So about 15 minutes of in-suit prebreathe would be enough for frequent surface spacewalks. Its inspired oxygen pressure, 128 mmHg, is like living at about 1 200 m altitude. The penalty is fire: a 2026 NASA assessment found that most spacecraft subsystems would need material changes at 34 % oxygen, and far fewer at 70.3 kPa (10.2 psi). As of 2026 the atmosphere for lunar landers and surface habitats is still being evaluated. Table 30.3 compares the choices.

**TABLE 30.3** Cabin atmospheres of crewed spacecraft. The last column is the tissue nitrogen pressure after long exposure divided by the suit pressure; higher values need longer oxygen prebreathe.

| Vehicle or regime | Total pressure | $O_2$ fraction | $O_2$ partial pressure | $N_2$ ratio to 29.6 kPa suit |
|---|---|---|---|---|
| Mercury, Gemini, Apollo (in flight) | 34–38 kPa | 100 % | 34–38 kPa | ≈ 0 |
| Skylab (1973–74) | 34.5 kPa | ≈ 72 % | 24.8 kPa | 0.33 |
| ISS, Soyuz, Shuttle, Crew Dragon, Orion | 101.3 kPa | 21 % | 21.3 kPa | 2.70 |
| Shuttle pre-EVA; ISS campout | 70.3 kPa | 26.5 % | 18.6 kPa | 1.75 |
| Exploration atmosphere (studied) | 56.5 kPa | 34 % | 19.2 kPa | 1.26 |

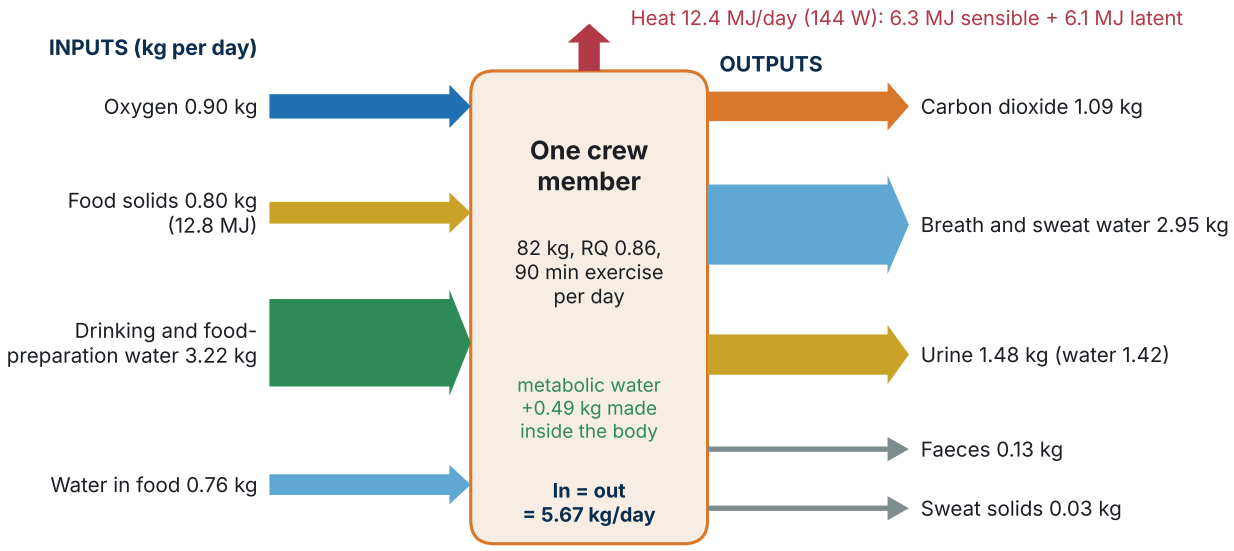
Early US spacecraft used pure oxygen at about 35 kPa to save mass: thinner walls, simpler life support, no nitrogen tanks. Soviet spacecraft used sea-level air, a choice the ISS inherited at the cost of long prebreathe.

## 30.5 The Human Metabolic Budget

A crew member burns food with oxygen. NASA's *Baseline Values and Assumptions Document* (BVAD) defines a reference astronaut of 82 kg who exercises for 90 minutes a day (Table 30.4). The ratio of carbon dioxide produced to oxygen consumed — the **respiratory quotient** — is about 0.86, so

more carbon dioxide is exhaled, by mass, than oxygen inhaled. The body also makes about half a kilogram of water a day from food and loses about three kilograms as vapour that the cabin air system must condense.

Two lessons stand out from Figure 30.9. Water dominates — about 4.5 kg of it passes through a person every day, four-fifths of the total — which is why water recycling pays off fastest. And although oxygen is a small mass, it must be supplied, and carbon dioxide removed, continuously: a few hours of failure can be dangerous.




Arrow width proportional to mass per day. Food as supplied with packaging: 1.8–2.4 kg/day.

**FIGURE 30.9** Daily mass balance of one crew member. Inputs and outputs balance at 5.67 kg per day; most of the mass flowing through a person is water.

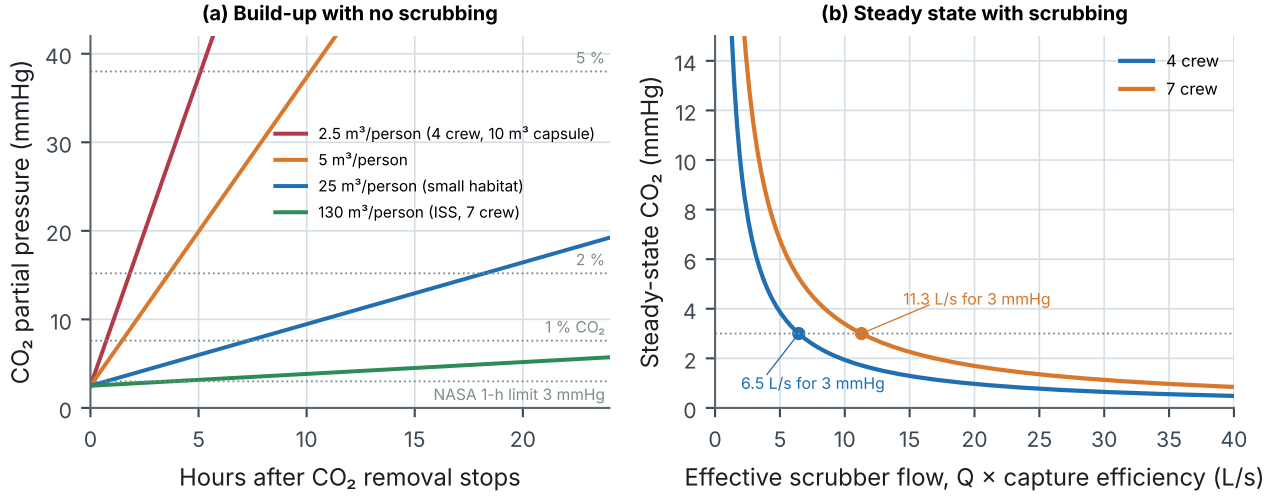
**TABLE 30.4** The metabolic budget of one crew member (NASA BVAD, 2022).

| Item | kg per crew member per day | Notes |
|---|---|---|
| Oxygen consumed | 0.90 | Older design value 0.84 |
| Carbon dioxide produced | 1.09 | Older design value 1.00 |
| Food solids (dry) | 0.80 | 12.8 MJ of energy |
| Food as supplied on ISS | 2.39 | Including water and 0.43 kg packaging |
| Drinking and food-preparation water | 3.22 | Plus 0.76 kg water in food |
| Breath and sweat water | 2.95 | Removed as humidity condensate |
| Urine | 1.48 | 1.42 kg water |
| Faeces | 0.13 | 0.10 kg water |
| Metabolic heat | 12.4 MJ | 144 W average; half as latent heat |

## 30.5.1 Carbon Dioxide

Carbon dioxide is toxic well before it displaces oxygen. Concentrations of a few per cent cause headaches, rapid breathing and impaired thinking, and above about 5–10 % they become incapacitating. ISS crews reported headaches at partial pressures of about 3–5 mmHg. Analysis showed that the odds of a headache doubled with every 1 mmHg increase. NASA-STD-3001 now limits the one-hour average to 3 mmHg (0.4 kPa).

Small cabins leave little margin (Figure 30.10 a). If the scrubber stops at 2.5 mmHg in a capsule with 2.5 $m^3$ of air per person, the level climbs by 7 mmHg per hour. It reaches 1 % in about 45 minutes and 2 % in under two hours. On the ISS, the pressurised volume of about 900 $m^3$ gives some 130 $m^3$ per person for a crew of seven. There, the same rise takes about a day and a half. With scrubbing, the steady-state level is inversely proportional to the air flow the scrubber cleans (panel b).




Ideal gas at 22 °C, well-mixed cabin, 1.0 kg $CO_2$ per person per day. (a) $dp/dt = \dot{n}RT/V$; (b) $p = \dot{n}RT/(Q\eta)$.

**FIGURE 30.10** Carbon dioxide in a closed cabin. (a) Rise in partial pressure after carbon dioxide removal stops, for different free gas volumes per crew member. (b) Steady-state partial pressure as a function of the effective scrubber flow.

### EXAMPLE: SIZING A LITHIUM HYDROXIDE SCRUBBER

A four-person capsule must remove carbon dioxide for a five-day free flight with non-regenerable lithium hydroxide (LiOH). Each crew member produces 1.0 kg of $CO_2$ per day.

1. $CO_2$ to remove: $4 \times 5 \times 1.0 =$ **20 kg**.
2. Chemistry: $2\,LiOH + CO_2 \rightarrow Li_2CO_3 + H_2O$. Two moles of LiOH ($2 \times 23.95$ g) absorb one mole of $CO_2$ (44.01 g). So the theoretical capacity is 0.92 kg of $CO_2$ per kg of LiOH, and the minimum LiOH mass is $20/0.92 =$ **21.8 kg**.
3. Real beds are never fully used, because the gas breaks through before the last granules react. Shuttle practice was a 7 kg canister for four crew-days, or 1.75 kg per crew-day: **35 kg**, five canisters.
4. By-products: the reaction releases $18/44 \times 20 =$ **8.2 kg of water** that the humidity system must remove, and about 2.1 MJ of heat per kilogram of $CO_2$ — some 25 W per crew member.
5. Air flow: to hold 3 mmHg (0.16 mol of $CO_2$ per $m^3$ at 22 °C) the bed must remove 91 mol/day = $1.05 \times 10^{-3}$ mol/s. If it captures 90 % per pass, the fan must move $1.05 \times 10^{-3} / (0.16 \times 0.9) \approx$ **7.2 L/s** through it (Figure 30.10 b).

## 30.6 Environmental Control and Life Support Systems

The **environmental control and life support system (ECLSS)** controls pressure and oxygen, and removes carbon dioxide and trace contaminants (**air revitalisation**). It also controls temperature and humidity, supplies and recovers water, manages waste, and detects and suppresses fire.

Figure 30.11 shows how these are linked on the ISS, the most complete system ever flown, and Table 30.5 compares the technologies.

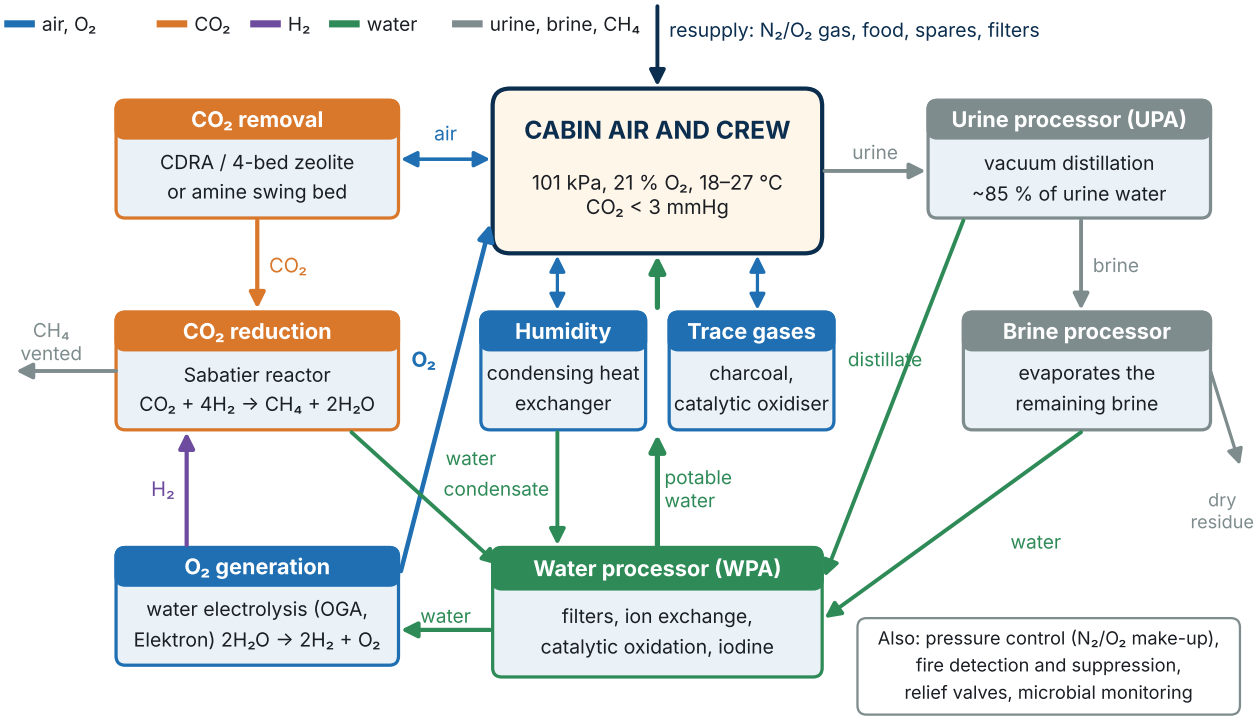



**FIGURE 30.11** Simplified ECLSS flow diagram of the ISS US segment. Air is scrubbed of $CO_2$, which a Sabatier reactor combines with hydrogen from the oxygen generator to make water; urine, humidity condensate and recovered brine water are purified for drinking and for electrolysis.

## 30.6.1 Removing Carbon Dioxide

Short missions use **lithium hydroxide** canisters that are thrown away when full: Apollo, the Shuttle and Crew Dragon all used them. The most famous are those of Apollo 13. After the oxygen-tank explosion in April 1970 the three astronauts sheltered in the lunar module, whose round canisters were sized for two people for two days; the command module's spare canisters were square. Engineers in Houston devised a way to connect them with plastic bags, cardboard and tape, and the crew built the "mailbox" from instructions read up to them (Figure 30.12). Soyuz uses potassium superoxide, which absorbs carbon dioxide and releases oxygen, with LiOH as a supplement.

Long missions need **regenerable** systems. The ISS Carbon Dioxide Removal Assembly (**CDRA**) passes cabin air through beds of zeolite — a mineral with molecular-sized pores — that adsorb carbon dioxide. A full bed is heated and exposed to vacuum to drive the gas out while a second bed takes over. Desiccant beds first remove water, which would block the zeolite, and return it to the cabin. The Russian **Vozdukh** works on the same swing principle with three beds of an amine sorbent. A redesigned Four Bed $CO_2$ Scrubber, delivered in 2021, now serves as a primary system. Orion uses an **amine swing bed**, regenerated by exposure to vacuum every few minutes — compact, but it vents some water overboard.

### 30.6.2 Making Oxygen

Short missions carry oxygen as high-pressure gas or cryogenic liquid. The ISS makes its own by **electrolysis** of water: the US Oxygen Generation Assembly (Figure 30.13) and the Russian Elektron split water into oxygen for the cabin and hydrogen. Each kilogram of oxygen requires 1.13 kg of water and, even for an ideal cell, about 5 kWh of electrical energy; real electrolysers need more. Backup oxygen comes from tanks delivered by cargo vehicles and from **solid-fuel oxygen generators** — "candles" of lithium perchlorate that release oxygen when burned. A candle on Mir caught fire in February 1997, filling the station with smoke.

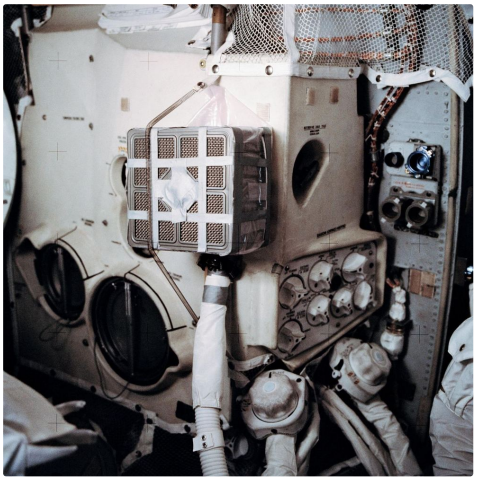


**FIGURE 30.12** The "mailbox" built by the Apollo 13 crew to use the command module's square lithium hydroxide canisters in the lunar module, April 1970. Image: NASA/JSC.

### 30.6.3 Closing the Oxygen Loop

Electrolysis consumes water, and carbon dioxide removed from the cabin still contains oxygen. Since 2010 the ISS has recovered part of it with a **Sabatier reactor**, which combines carbon dioxide with hydrogen from the oxygen generator over a heated catalyst:

$$\mathrm{CO_2 + 4\,H_2 \rightarrow CH_4 + 2\,H_2O}$$

The water returns to electrolysis. The methane is vented, taking half of the hydrogen with it, and because there is not enough hydrogen to convert all the carbon dioxide, the Sabatier closes only about 50 % of the oxygen loop. Future systems may recover hydrogen from the methane or use the **Bosch reaction** ($\mathrm{CO_2 + 2\,H_2 \rightarrow C + 2\,H_2O}$), which leaves solid carbon instead.

**FIGURE 30.13** NASA astronaut Nichole Ayers servicing components of the Oxygen Generation System rack in the ISS Destiny laboratory, May 2025. Image: NASA/JSC.

**FIGURE 30.14** NASA astronaut Jessica Watkins servicing the Water Recovery System rack in the Tranquility module, June 2022. Image: NASA/JSC.

### 30.6.4 Trace Contaminants, Temperature and Humidity

People, materials and equipment release hundreds of trace gases — ammonia, alcohols, aldehydes, siloxanes, carbon monoxide — that would accumulate in a sealed cabin. Activated charcoal adsorbs most of them and a heated catalytic oxidiser burns the rest, while filters remove dust and microbes. Fans keep the air moving, because without convection exhaled carbon dioxide would collect around a sleeping astronaut's head. A **condensing heat exchanger**, cooled by the internal water loop, keeps the cabin at about 18–27 °C. It also removes the 3 kg per person per day of water vapour, which joins the water recovery system as condensate.

### 30.6.5 Recovering Water

The ISS **Water Recovery System** (Figure 30.14) has two parts. The **Urine Processor Assembly** distils pretreated urine in a rotating drum at low pressure, recovering about 85 % of its water. A newer **Brine Processor Assembly** extracts most of the rest from the concentrated brine. The **Water Processor Assembly** treats the distillate and humidity condensate with filters and ion-exchange beds, oxidises the remaining organic compounds in a catalytic reactor, and adds iodine against microbes. In June 2023 NASA announced that the system had reached its goal of recovering **98 %** of the water, up from 93–94 % before the brine processor. The Russian segment recycles humidity condensate and uses silver ions as a biocide.

### 30.6.6 Waste, Hygiene and Fire

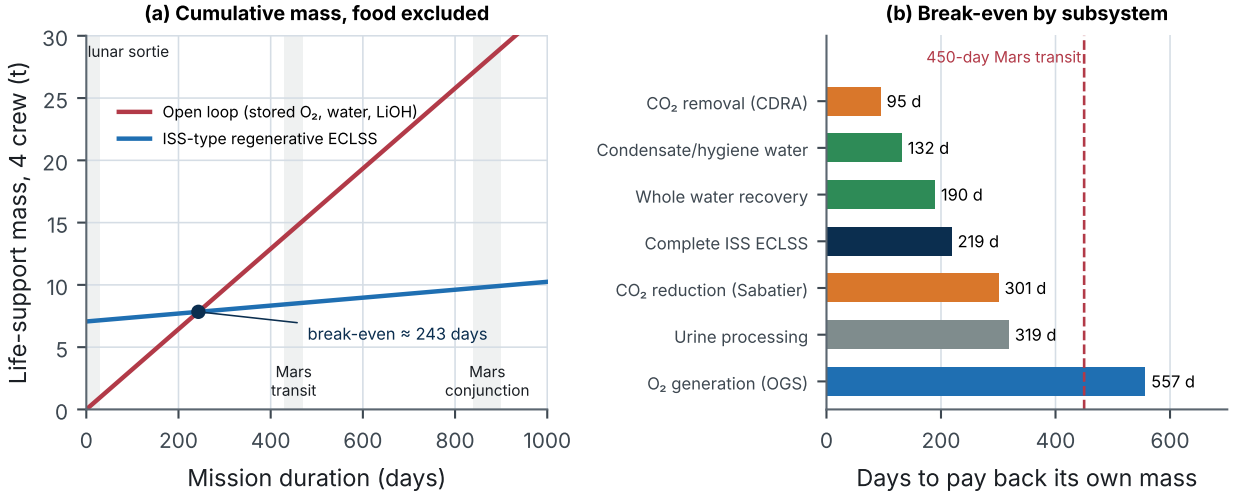
Toilets use air flow instead of gravity to carry waste away; the compact Universal Waste Management System, installed on the ISS in 2020, also flies on Orion. Faeces are stored and burned up in departing cargo vehicles, and crews wash with wet towels and rinseless shampoo. Smoke detectors sit in the ventilation ducts. From 2015 the US segment's carbon dioxide extinguishers were replaced by water-mist units, because carbon dioxide would create an atmosphere the crew's emergency masks cannot filter. In a fire the crew stops the ventilation, removes power, dons oxygen masks and, if necessary, retreats to the docked vehicles.

## 30.6.7 Open and Closed Loops

Recycling hardware is heavy, consumes power and needs spares, while stored consumables cost mass every day, so the choice depends on mission duration (Figure 30.15). An analysis looked at ISS hardware for a four-person, 450-day Mars transit. It found that the complete regenerative system, with three sets of spares, has a mass of 7 062 kg and replaces about 32 kg of supplies per day. It pays for itself after about 220 days — about 240 days if the residual resupply of oxygen and filters is counted — and halves the life support mass of a 450-day transit. Carbon dioxide removal pays off within about three months, but the oxygen generator, with its heavy spares, takes 557 days.

**TABLE 30.5** ECLSS technologies of crewed spacecraft.

| Function | Open loop | ISS US segment | ISS Russian segment | Capsules |
|---|---|---|---|---|
| $CO_2$ removal | LiOH canisters | CDRA, Four Bed $CO_2$ Scrubber (zeolite) | Vozdukh (amine beds) | LiOH (Dragon); amine swing bed (Orion); $KO_2$ (Soyuz) |
| $O_2$ supply | Gas or liquid tanks | Electrolysis (OGA) | Electrolysis (Elektron); candles | Tanks |
| $CO_2$ reduction | None, $CO_2$ discarded | Sabatier (since 2010) | None | None |
| Water | Stored, or fuel-cell water (Apollo, Shuttle) | Urine and brine processors, WPA: 98 % | Condensate recovery | Stored |
| Trace gases | Charcoal | Charcoal and catalytic oxidiser | Similar system | Charcoal, filters |




ISS hardware with three spare sets (7 062 kg) replacing 32.2 kg/day of supplies for 4 crew (Jones, ICES-2017-85); (a) adds 3.2 kg/day residual resupply ($O_2$ not recovered by Sabatier, filters). Tankage neglected.

**FIGURE 30.15** Open versus closed life support. (a) Cumulative mass of life support supplies for four crew with open-loop storage and with an ISS-type regenerative system. (b) Days needed for each ISS subsystem to save its own mass, including three sets of spares.

The ultimate step is a **bioregenerative** system, in which plants and microbes recycle air, water and food. The Soviet BIOS-3 facility in Krasnoyarsk supported three people for 180 days in 1972–73, with wheat and vegetables providing part of the food. China's Lunar Palace 1 ran for 370 days in 2017–18 with rotating crews of volunteers and reported about 98 % closure of the material cycle. ESA's MELiSSA programme is testing a loop of bacterial and plant compartments in Barcelona. Because one person's oxygen and food would need several tens of square metres of lit

crops, plants will long supplement packaged food rather than replace physico-chemical systems. NASA's Veggie and Advanced Plant Habitat chambers on the ISS have grown lettuce, mustard greens, radishes and chile peppers (Figure 30.16).

**FIGURE 30.16** Chile peppers growing in the Advanced Plant Habitat on the ISS in November 2021, during Expedition 66. Image: NASA/JSC (Mark Vande Hei).

**EXAMPLE: CONSUMABLES FOR A 30-DAY LUNAR SURFACE MISSION**

Three astronauts spend 30 days on the Moon with open-loop life support. Use the BVAD values: oxygen 0.89 kg, water 3.2 kg and food 1.83 kg (packaged) per crew member per day. For LiOH, use the Shuttle rate of 1.75 kg per crew-day, scaled up for a $CO_2$ output of 1.085 kg. This assumes that the canister rating corresponds to the older 1.0 kg design value.

1. Crew-days: $3 \times 30 =$ **90**.
2. Oxygen: $90 \times 0.89 =$ **80 kg**.
3. Water (drinking, food preparation, minimal hygiene): $90 \times 3.2 =$ **288 kg**.
4. Food: $90 \times 1.83 =$ **165 kg**.
5. LiOH: $90 \times 1.75 \times 1.085 =$ **171 kg**.
6. Total: **704 kg**, or about **775 kg** with a 10 % margin — before tanks, cabin leakage and gas lost through the airlock on each spacewalk.

An ISS-type regenerative system would weigh several tonnes, so open loop is clearly right for 30 days. Water dominates, so a condensate recycler would be the first system worth adding for longer stays.

## 30.7 Living and Working in Space

### 30.7.1 Food

Space food must be safe, nutritious, light and appetising for months. ISS menus offer some 200 items: thermostabilised pouches, freeze-dried dishes rehydrated with hot water, and irradiated meat. There are also natural-form foods such as nuts and tortillas (which do not make crumbs), and powdered drinks. Salt and pepper come as liquids. Meals are a social event (Figure 30.17). The ISS food supply has grown to 2.39 kg per person per day including packaging, which makes up almost a fifth of it. For Mars, food must keep its nutrients for up to five years, longer than current foods do.

**FIGURE 30.17** Expedition 31 crew members sharing a meal near the galley of the Zvezda service module, May 2012. Image: NASA/JSC.

**FIGURE 30.18** The four volunteers of NASA's CHAPEA Mission 2 inside the 3D-printed simulated Mars habitat at Johnson Space Center, November 2025. Image: NASA/JSC.

## 30.7.2 Habitability and Medical Care

Each ISS crew member has a sleeping station the size of a telephone booth, with a sleeping bag tied to the wall and a door to shut out the noise of fans and pumps. Habitable volume, noise, lighting, privacy and window views all affect performance and morale on long missions (Chapter 32).

ISS medical care relies on two trained **crew medical officers**, a medical kit with ultrasound scanner, defibrillator and drugs, and flight surgeons on the ground; serious illness means returning home in a docked capsule. In January 2026 NASA's Crew-11 returned about a month early, the first time an ISS mission was ended for medical reasons; the affected crew member was reported stable. A Mars crew, with radio delays of up to 22 minutes each way and months from home, must diagnose and treat emergencies themselves.

## 30.7.3 Psychology and Isolation

Isolation, confinement, monotony and separation from family are among the hardest problems of long missions. Agencies select crews for stability and teamwork and provide psychological support. **Analogue missions** on Earth test this: Mars-500 confined six men in a mock spacecraft in Moscow for 520 days in 2010–11, and Hawaii's HI-SEAS ran missions of up to a year on Mauna Loa. NASA's **CHAPEA** (Crew Health and Performance Exploration Analog) placed four volunteers in a 160 $m^2$ 3D-printed habitat at Johnson Space Center for 378 days in 2023–24. They faced simulated equipment failures, resource limits and communication delays. The second CHAPEA crew entered in October 2025 and is due to emerge at the end of October 2026 (Figure 30.18).

# 30.8 Crew Safety and Risk

## 30.8.1 Lessons Paid For in Lives

Eighteen people have died in four accidents on orbital missions, and three more in the Apollo 1 ground test (Table 30.6). Each accident changed how crewed spacecraft are designed and managed. The Soyuz 11 crew died during re-entry in 1971 when a valve opened as the modules separated and the cabin lost its air. They were not wearing pressure suits, and every Soyuz crew since has worn the Sokol suit for launch and landing. Both Shuttle losses were preceded by warning signs — eroded booster seals, foam strikes on earlier flights — that had come to be accepted as normal, a pattern known as the "normalisation of deviance".

Many more crews have survived serious failures, such as the Apollo 13 explosion (1970). Launch escape systems saved the crews of the Soyuz T-10-1 pad fire (1983) and the Soyuz MS-10 booster failure (2018). Crews also survived the Mir fire and collision (1997) and the Starliner thruster problems of 2024, after which its crew came home on a Crew Dragon. They show why crewed vehicles need **abort** capability in every phase, **safe havens**, and margins for the unexpected.

## 30.8.2 Human-Rating Requirements

NASA's requirements are set out in its human-rating process (NPR 8705.2C, with technical requirements in NASA-STD-8719.29 since 2023) and in **NASA-STD-3001**. Volume 1 of the standard covers crew health, including radiation limits, and Volume 2 (Revision F, July 2026) covers habitability and the cabin environment. One key principle is that the system must tolerate at least one failure without catastrophe, unless a component is designed for minimum risk (like structures and pressure vessels). The actual level of failure tolerance (one, two or more) is set by safety analysis. Other principles are that the crew can abort and escape, and that the crew can monitor and control the vehicle manually.

Risk is also set as a number. NASA's Commercial Crew Program required that the probability of **loss of crew** on a mission of up to 210 days to the ISS be no worse than 1 in 270. For ascent and entry, it had to be no worse than 1 in 500. NASA later estimated that the first Shuttle flight in 1981 carried a risk of about 1 in 12, and the mature Shuttle about 1 in 90 — consistent with two losses in 135 flights. Reliability and risk analysis (Chapter 4) identifies the dominant contributors, but experience shows that the largest risks are often the ones nobody modelled.

**TABLE 30.6** Fatal accidents in human spaceflight.

| Accident | Date | Crew lost | Cause | Main consequences |
|---|---|---|---|---|
| Apollo 1 (ground test) | 27 Jan 1967 | 3 | Fire in pure oxygen at 115 kPa; inward-opening hatch | Materials, 60/40 launch atmosphere, quick-opening hatch |
| Soyuz 1 | 24 Apr 1967 | 1 | Main parachute failed to deploy; reserve tangled | Parachute system redesign |
| Soyuz 11 | 30 Jun 1971 | 3 | Valve opened at module separation; cabin depressurised | Pressure suits for launch and entry |
| Challenger (STS-51-L) | 28 Jan 1986 | 7 | Solid-booster joint seal failed in cold weather | Booster redesign, crew escape system, management reform |

| Accident | Date | Crew lost | Cause | Main consequences |
|---|---|---|---|---|
| Columbia (STS-107) | 1 Feb 2003 | 7 | Foam strike broke the wing's leading edge; vehicle broke up on entry | Inspection and repair in orbit, Shuttle retirement in 2011 |

### AI FOR CREW HEALTH AND LIFE SUPPORT

A Mars crew will have to handle medical emergencies without real-time help from the ground. In 2025 NASA and Google tested a prototype clinical decision-support assistant for long-duration crews. In simulated cases it reached the correct diagnosis 74–88 % of the time, depending on the condition. Similar tools could guide an astronaut through an ultrasound scan. Life support is a natural target for machine learning too. ECLSS telemetry — pressures, flows, temperatures, valve positions — can train models that detect developing faults before they become failures. In 2025 researchers demonstrated supervised anomaly detection and diagnosis on data from the European Columbus module's ECLSS. Such **predictive maintenance** would let a crew far from Earth replace the right part before it fails and carry fewer spares (Chapter 46).

### KEY TAKEAWAYS

- About 780 people have flown in space; the longest single flight is 437.7 days and the longest cumulative time 1 111 days, while a Mars mission would last up to 900 days.
- Weightlessness shifts fluids headward and causes motion sickness, eye changes (SANS), bone loss of 1–1.5 % per month and muscle atrophy; about two hours of daily exercise is the main countermeasure.
- Deep-space crews receive about 1.8 mSv per day from cosmic rays; a Mars mission would exceed NASA's 600 mSv career limit. Storm shelters handle solar particle events, but shielding barely reduces the cosmic-ray dose.
- Cabin atmospheres balance hypoxia, fire and decompression sickness: the ISS uses sea-level air, while NASA has studied 56.5 kPa with 34 % oxygen for lunar surface operations.
- Each crew member consumes about 0.9 kg of oxygen, 3–4 kg of water and 1.8–2.4 kg of packaged food per day and produces 1.1 kg of carbon dioxide and 144 W of heat.
- The ISS ECLSS removes $CO_2$ with zeolite beds, makes oxygen by electrolysis, recovers half of the oxygen with a Sabatier reactor and recycles 98 % of its water. It pays back its mass after about 7–8 months, so open loops suit short missions and recycling suits long ones.
- Four accidents on orbital missions and the Apollo 1 fire killed 21 people; Commercial Crew vehicles must meet a loss-of-crew risk no worse than 1 in 270 per mission.

### REVIEW QUESTIONS

1. Why do astronauts suffer orthostatic intolerance after landing? Give two countermeasures.
2. A crew member loses hip bone at 1.2 % per month for 9 months. Estimate the loss at landing and how much is recovered a year later (9-month half-time).
3. Why does shielding reduce the dose from solar particle events much more than that from galactic cosmic rays? Why is polyethylene better than aluminium?
4. Using the dose rates in Table 30.2, calculate the dose for a 180-day lunar surface mission (with 10 days of transit) and compare it with the 600 mSv career limit.
5. Calculate the inspired oxygen partial pressure, in mmHg, for a cabin at 70.3 kPa with 26.5 % oxygen. Is it within NASA's normoxic range?
6. Why is a low-pressure, oxygen-rich cabin attractive for a lunar lander that supports frequent spacewalks, and what is its main disadvantage?

7. A four-person crew sits in a capsule with 10 $m^3$ of free air when its scrubber fails at a $CO_2$ partial pressure of 2.5 mmHg. How long until the partial pressure reaches 15 mmHg?
8. How do the CDRA, the oxygen generator and the Sabatier reactor work together, and why does the Sabatier close only about half of the oxygen loop?
9. Using Figure 30.15, explain why Apollo carried stored oxygen and water while the ISS recycles them. Which subsystem pays back its mass fastest?
10. What did the Apollo 1, Soyuz 11 and Columbia accidents each teach designers of crewed spacecraft?

# 31 CHAPTER Crewed Spacecraft

Image: NASA/JSC

*A crewed spacecraft is a small, self-contained world that must keep people alive, carry them away from an exploding rocket within seconds, and bring them home through a fireball at up to 11 km/s. This chapter traces the vehicles that have carried people since 1961 — from Vostok and Mercury through Apollo, Soyuz and the Space Shuttle to Shenzhou, Crew Dragon, Starliner and Orion, which carried four astronauts around the Moon in April 2026 — and the vehicles now being built in India and China and for lunar landings. It explains how they are engineered: pressure vessels, windows, seats and acceleration limits, avionics, propulsion, docking systems, launch abort, entry and parachutes, and how their safety is estimated, demonstrated and certified.*

IN THIS CHAPTER

31.1 Sixty-Five Years of Crewed Spacecraft
31.2 Anatomy of a Crewed Spacecraft
31.3 Launch Abort
31.4 Coming Home: Entry, Descent and Landing
31.5 Reliability and Safety

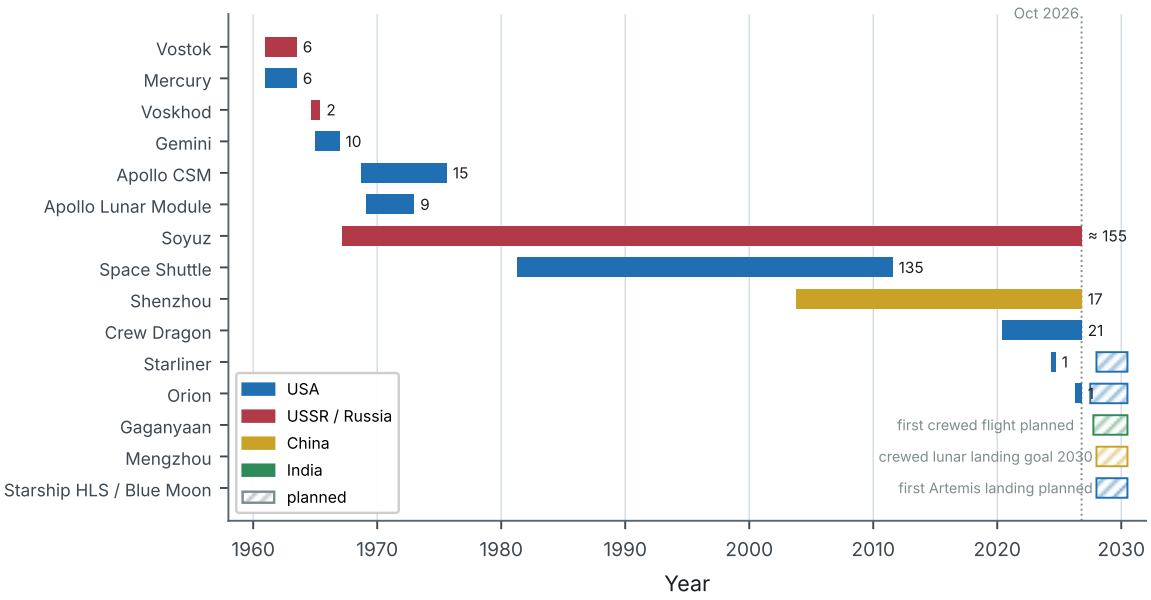
A **crewed spacecraft** carries people into space and back. Unlike a robotic spacecraft, its payload cannot be replaced and tolerates only a narrow range of pressure, temperature, acceleration and air composition. Three requirements set it apart. It must keep its crew alive (life support, Chapter 30), and it must be able to escape from a failing launch vehicle at any moment of the ascent. It must also bring its crew back to Earth gently enough for them to walk away.

Since Yuri Gagarin's flight in April 1961, fewer than twenty types of vehicle have carried people into space. Only four were in service in October 2026: Russia's Soyuz, China's Shenzhou, SpaceX's Crew Dragon and NASA's Orion. Boeing's Starliner was grounded after its troubled first crewed

flight, and India's Gaganyaan and China's Mengzhou were preparing for their first orbital flights. This chapter describes these vehicles and the engineering behind them; space stations are covered in Chapter 32, spacesuits in Chapter 33 and launch vehicles in Chapters 34 and 36.

## 31.1 Sixty-Five Years of Crewed Spacecraft

Figure 31.1 shows every orbital crewed spacecraft and how long it served, and Table 31.1 compares them. Two waves stand out: the 1960s race to the Moon, and a second wave from 2020 driven by commercial contracts and by China.




Numbers: crewed launches of each vehicle to October 2026 (Soyuz includes the 1975 and 1983 launch aborts; Apollo CSM includes Skylab and ASTP). Planned dates as announced in 2026.

**FIGURE 31.1** Crewed spacecraft from 1961 to 2026 and those planned for the late 2020s. The numbers give the crewed launches of each type. Soyuz has flown for almost sixty years; the Space Shuttle flew 135 times in 30 years.

### 31.1.1 The First Generation: Vostok, Mercury, Voskhod and Gemini

The Soviet **Vostok** was a 2.3 m sphere, which needed no attitude control during its ballistic 8–9 g entry. It landed too hard for safety, so the cosmonaut ejected at about 7 km and came down under a separate parachute. Six Vostoks flew in 1961–63, carrying Gagarin and Valentina Tereshkova, the first woman in space.

NASA's **Mercury** capsule weighed only about 1.4 t so that it could ride the Redstone and Atlas missiles of the day. It set the pattern of every American capsule since: a cone with a broad, blunt heat shield at the base (Chapter 10), and a crew lying on contoured couches facing it. Mercury flew twice suborbitally in 1961, starting with Alan Shepard on 5 May, and four times in orbit from John Glenn's flight on 20 February 1962.

**Voskhod** was a Vostok without its ejection seat — and so without any means of escape early in the ascent. It carried three cosmonauts in 1964 and two in 1965, when Alexei Leonov made the first spacewalk. **Gemini** carried two astronauts and was the first spacecraft built to be *flown*: thrusters, an onboard computer and radar let it change orbit, rendezvous and dock. Between

March 1965 and November 1966 ten Gemini crews demonstrated the techniques Apollo needed — rendezvous (Figure 31.2), docking, spacewalks, fuel-cell power, guided lifting entry and flights of up to 14 days.

**FIGURE 31.2** Gemini 7 photographed from Gemini 6A during the first space rendezvous, 15 December 1965, with the two spacecraft about 11 m apart. The white section is the adapter module carrying propellant, fuel cells and radiators. Image: NASA/JSC.

### 31.1.2 Apollo: The Command, Service and Lunar Modules

Apollo divided a lunar mission between three modules. The **command module (CM)**, a 3.9 m cone of 5.6 t, carried three astronauts and alone returned. The **service module** carried fuel cells, oxygen tanks, radiators and a single 91 kN engine, which had to work to bring the crew home from lunar orbit and was therefore pressure-fed, with redundant valves. The **Lunar Module (LM)** was the first spacecraft designed to fly only in vacuum. It had a descent stage with a throttleable engine and an ascent stage that carried two astronauts back to the CM in lunar orbit (Figure 31.3; landing is described in Chapter 27).

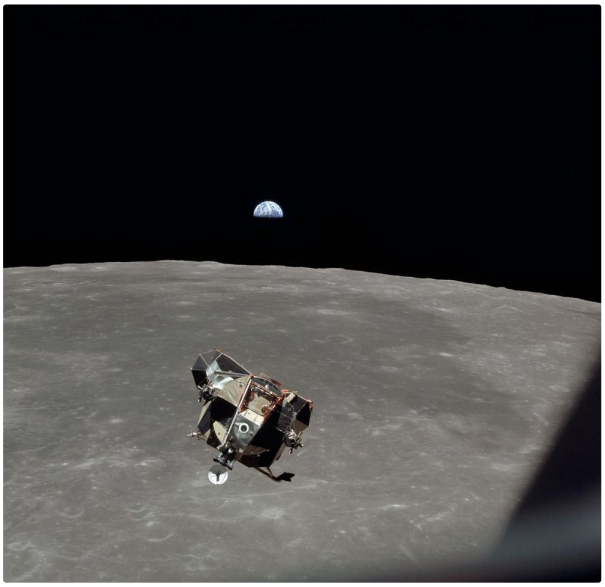


**FIGURE 31.3** The ascent stage of Apollo 11's Lunar Module Eagle returns from the surface to rendezvous with the command module Columbia in lunar orbit, July 1969. Image: NASA/KSC.

The command and service module flew 15 crewed missions: 11 Apollo flights from 1968 to 1972, three Skylab crews and the Apollo–Soyuz Test Project of 1975. When an oxygen tank in Apollo 13's service module exploded in April 1970, the crew survived for almost four days in the Lunar Module, using it as a lifeboat.

### 31.1.3 Soyuz and the Space Shuttle

**Soyuz**, first flown with a crew in April 1967, is the longest-serving crewed spacecraft. A spherical **orbital module** serves as living space and airlock, the crew launch and land in the bell-shaped **descent module**, and an unpressurised **service module** carries the engine, solar arrays and radiators. Only the 2.2 m descent module returns, which keeps the whole spacecraft to about 7 t. Two early accidents (Soyuz 1 and Soyuz 11, Table 31.5) killed four cosmonauts. Upgraded through the T, TM, TMA and MS versions, Soyuz had made about 155 crewed launches by October 2026, the latest, Soyuz MS-29, on 14 July 2026.

The **Space Shuttle** (1981–2011) was a reusable winged orbiter carrying up to eight people and a 24 t payload and landing on a runway. Its two-deck crew module had a pressurised volume of about 66 $m^3$, six times that of the Apollo command module. The Shuttle flew 135 times, assembled most of the International Space Station (ISS) and serviced Hubble five times, but two orbiters and 14 astronauts were lost (Chapter 10).

### 31.1.4 Shenzhou and China's Programme

China's **Shenzhou** follows the Soyuz layout but is larger, with a 2.5 m descent module, and on its early flights the orbital module stayed in orbit as a satellite after the crew returned. Yang Liwei made China's first spaceflight on Shenzhou 5 on 15 October 2003, and since 2021 Shenzhou has ferried crews of three to the Tiangong space station, usually for six months.

Shenzhou also showed why a station needs a spare lifeboat. In November 2025, shortly before its crew was due home, a crack — probably from the impact of a debris particle smaller than a millimetre — was found in a window of Shenzhou-20. The crew returned in the newer Shenzhou-21 instead. China therefore launched an uncrewed **Shenzhou-22** from its standby rocket on 25 November to give the Shenzhou-21 crew a ride home. They landed in it on 29 May 2026 after a Chinese-record 210 days. Russia had met the same problem in December 2022, when a coolant leak attributed to a micrometeoroid emptied the radiator loop of Soyuz MS-22 and an uncrewed Soyuz MS-23 was sent as a replacement.

### 31.1.5 The Commercial Era: Crew Dragon and Starliner

After the Shuttle retired, the United States relied on Soyuz for nine years while NASA's **Commercial Crew Program** paid two companies to develop vehicles that they own and operate under fixed-price contracts. SpaceX's **Crew Dragon** launched Doug Hurley and Bob Behnken to the ISS on 30 May 2020. By October 2026 Dragon had flown 21 crewed missions. They were the Demo-2 test flight, 13 NASA crew rotations, four private Axiom missions to the ISS and three free-flying private missions. The free-flying missions included Polaris Dawn's spacewalk in 2024 and Fram2's polar orbit in 2025. In January 2026 Dragon made the first medical evacuation from the ISS, bringing the Crew-11 astronauts home early. Five capsules are in service; Endeavour has flown six times.

Boeing's **CST-100 Starliner** has struggled. Its first uncrewed test in 2019 failed to reach the ISS because of software errors, and the second succeeded in 2022. The **Crew Flight Test** launched on 5 June 2024 with Butch Wilmore and Suni Williams. During the approach to the ISS, five of the 28 small thrusters on the service module failed, and helium leaks appeared in the propulsion system (Figure 31.4). The thrusters, packed in enclosed "doghouse" pods, had overheated during repeated firings, deforming the Teflon seals in their valves. NASA brought Starliner home uncrewed in September 2024 — it landed safely — and returned its crew on a Crew Dragon in March 2025, after 286 days in space instead of eight. In February 2026 NASA reclassified the flight as a Type A mishap, its most serious category. As of October 2026, an uncrewed cargo flight, **Starliner-1**, with new thruster valves and seals was planned for December 2026–January 2027, and the first crew rotation flight for 2028.

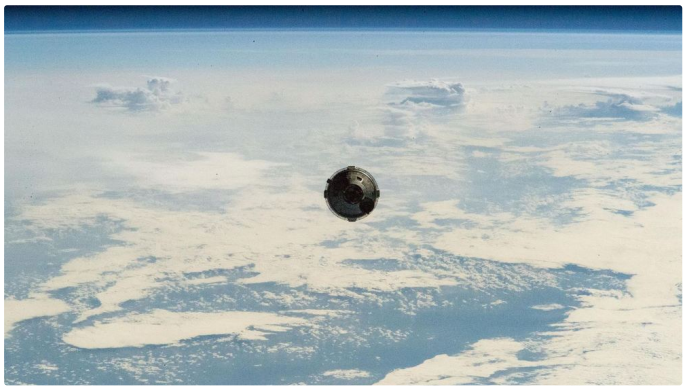


**FIGURE 31.4** Boeing's Starliner approaching the ISS on 6 June 2024 during the Crew Flight Test, with Butch Wilmore and Suni Williams aboard. Thruster failures during this approach led NASA to return the spacecraft without its crew. Image: NASA/JSC.

### 31.1.6 Orion and Artemis II

NASA's **Orion** is the first new deep-space crew vehicle since Apollo. Its 5.0 m crew module, built by Lockheed Martin, carries four astronauts for up to 21 days. The **European Service Module**, built by Airbus for ESA, provides propulsion, about 11 kW from four solar wings and life-support consumables. After the uncrewed Artemis I flight around the Moon in 2022, **Artemis II** was launched on 1 April 2026 by the Space Launch System. Reid Wiseman, Victor Glover, Christina Koch and the Canadian Jeremy Hansen flew Orion *Integrity* on a free-return trajectory around the Moon (Figure 31.5). They passed about 6 500 km above the far side and reached 406 771 km from Earth — farther than any humans before them. They splashed down off San Diego on 10 April after an entry at about 11 km/s. Problems were minor, such as a frozen urine vent line.

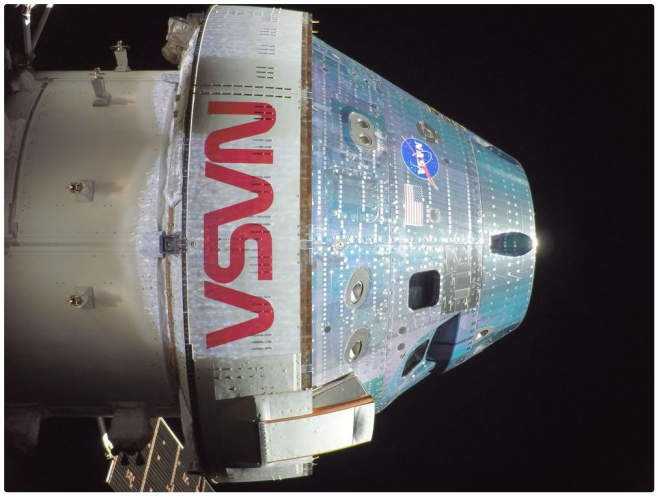



**FIGURE 31.5** Orion photographed itself with a camera on a solar array wing on the second day of Artemis II, 3 April 2026. The crew module, with its windows, sits on the European Service Module. Image: NASA/JSC (NASA).

### 31.1.7 Coming Next: Gaganyaan, Mengzhou and the Lunar Landers

India's **Gaganyaan** is a three-person spacecraft of about 8 t (crew module plus service module), launched by the human-rated LVM3, for flights of up to seven days at about 400 km. As of October 2026 the first uncrewed orbital flight, G1, carrying the humanoid robot Vyommitra, had not flown. ISRO aimed for the end of 2026, though mid-2026 reports pointed to 2027. A crewed flight was not expected before 2027–28.

China's **Mengzhou** is a reusable successor to Shenzhou for Earth orbit (up to seven crew) and the Moon (three crew). After abort tests in 2025 and 2026, its first uncrewed orbital flight is expected around 2027. For lunar missions, one Long March 10 will launch Mengzhou and another the **Lanyue** lander. The lander completed a landing and take-off test in August 2025 while suspended from a tower rig that simulated lunar gravity. China's goal is a crewed landing by 2030.

The Artemis lunar landers are the largest crewed spacecraft yet designed. SpaceX's **Starship HLS** is a version of the 50 m Starship upper stage, with a crew cabin near the top reached by an elevator. It must be refuelled in orbit by a series of tanker flights. This had not yet been demonstrated when Starship first reached orbit in September 2026. Blue Origin's **Blue Moon MK2** is a 16 m, 45 t-class lander burning liquid hydrogen and oxygen. In 2026 NASA turned Artemis III into a crewed test in Earth orbit, no earlier than late 2027, in which Orion is to dock with one or both landers. The first landing moved to Artemis IV, no earlier than 2028.

**TABLE 31.1** Crewed spacecraft compared (status in October 2026). Masses at launch without escape systems; free, habitable volume is roughly half the pressurised volume (5.9 $m^3$ in the Apollo CM, 8.9 $m^3$ in Orion).

| Vehicle | Crew | Mass (t) | Pressurised volume ($m^3$) | Launcher | Landing | First crewed flight; status |
|---|---|---|---|---|---|---|
| Vostok (USSR) | 1 | 4.7 | — | Vostok | Land; crew ejects | 1961; retired 1963 |
| Mercury (USA) | 1 | 1.4 | 2.8 | Redstone, Atlas | Water | 1961; retired 1963 |
| Gemini (USA) | 2 | 3.2–3.8 | ≈ 2.3 | Titan II | Water | 1965; retired 1966 |
| Apollo CSM (USA) | 3 | 28.8 (CM 5.6) | 10.4 | Saturn IB, V | Water | 1968; retired 1975 |
| Apollo LM (USA) | 2 | 15–16 | 6.7 | Saturn V | Moon | 1969; retired 1972 |

| Vehicle | Crew | Mass (t) | Pressurised volume (m³) | Launcher | Landing | First crewed flight; status |
|---|---|---|---|---|---|---|
| Soyuz MS (Russia) | 3 | 7.1 | 8.5 (two modules) | Soyuz-2.1a | Land, retro-rockets | 1967; flying |
| Space Shuttle (USA) | 7–8 | ≈ 110 (orbiter) | 66 | Shuttle stack | Runway | 1981; retired 2011 |
| Shenzhou (China) | 3 | 7.8 | 14 (two modules) | Long March 2F | Land, retro-rockets | 2003; flying |
| Crew Dragon (USA) | 4 | ≈ 12.5 | 9.3 | Falcon 9 | Water | 2020; flying |
| Starliner (USA) | 4 | ≈ 13 | 11 | Atlas V | Land, airbags | 2024; grounded |
| Orion (USA, Europe) | 4 | 26.5 (with ESM) | 19.6 | SLS | Water | 2026; next flight 2027 |
| Gaganyaan (India) | 3 | ≈ 8.2 | — | LVM3 | Water | Uncrewed G1 pending |
| Mengzhou (China) | 3–7 | 14 (LEO), 21.6 (Moon) | 13 | Long March 10A, 10 | Land, airbags | Uncrewed flight ≈ 2027 |

## 31.2 Anatomy of a Crewed Spacecraft

Most crewed spacecraft are **capsules** with a three-part layout (Figure 31.6). The **crew module**, the only part that returns, is a pressure vessel wrapped in a heat shield and thermal protection, containing the crew, the controls and the parachutes. The **service module** carries everything not needed for entry — main engine and most of the propellant, solar arrays, radiators and consumables — and burns up in the atmosphere. The **launch abort system** sits on top for the first minutes of flight.

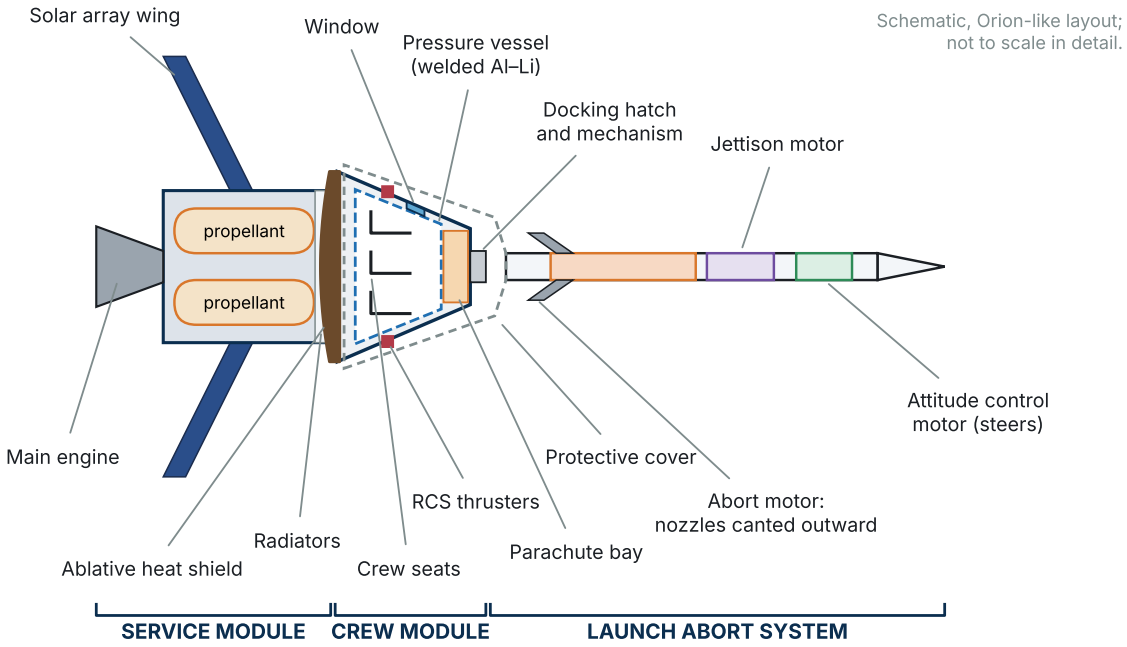



**FIGURE 31.6** Anatomy of a capsule stack, based on Orion's layout. Only the crew module returns to Earth; the service module provides propulsion, power and heat rejection during the mission, and the abort system protects the crew during ascent.

### 31.2.1 Pressure Vessel and Atmosphere

The heart of the crew module is the **pressure vessel**, usually welded from aluminium or aluminium–lithium alloy. At sea-level pressure (101 kPa) every square metre of wall carries about 10 t, so the vessel is built from cones, cylinders and domes that carry pressure efficiently as tension (Chapter 5). It is designed for 1.5–2 times its operating pressure, proof-tested, and shielded from debris by the heat shield and backshell around it.

The cabin atmosphere has large consequences. Mercury, Gemini and Apollo flew pure oxygen at about 34–38 kPa, which kept the structure light. But on 27 January 1967 a fire broke out in the Apollo 1 command module, filled with pure oxygen above sea-level pressure for a ground test. The fire killed its three astronauts, whose inward-opening hatch could not be opened against the cabin pressure. Apollo was redesigned with a quick-opening hatch, non-flammable materials and a nitrogen–oxygen mixture on the pad. Soyuz, the Shuttle, Shenzhou, Dragon and Orion all fly an Earth-like nitrogen–oxygen atmosphere at about 101 kPa, maintained by the systems described in Chapter 30.

### 31.2.2 Windows

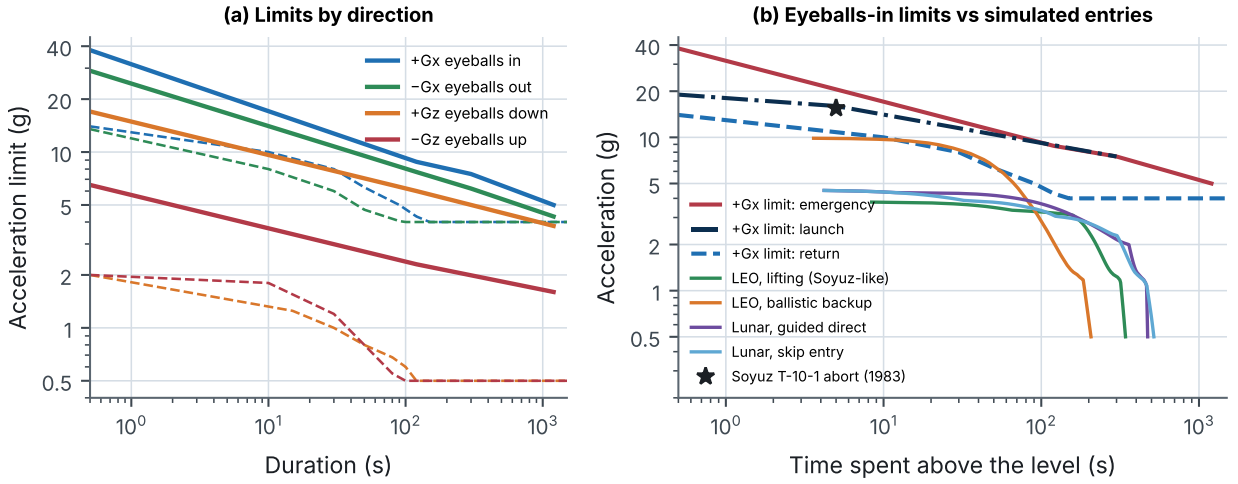
Windows are weak points in the pressure shell, but crews need them for docking and for seeing the horizon. They are typically stacked as three panes: an outer pane of fused silica that takes entry heating and debris impacts, and two inner pressure panes, either of which can hold cabin pressure on its own. Because glass fails by the slow growth of tiny flaws, windows have large safety factors. The crack in Shenzhou-20's window ran right through the outer, protective pane, but the two pressure panes behind it held — exactly what the redundancy is for.

### 31.2.3 Seats and Acceleration Limits

Capsule crews sit in contoured seats facing the heat shield, because the body tolerates acceleration very differently along its three axes (Figure 31.7). In the **eyeballs-in** direction (+Gx, chest to back) heart and brain are at the same level, so trained crews can withstand 15 g for several seconds and 4–5 g for many minutes. In the **eyeballs-down** direction (+Gz, head to foot) blood drains from the brain and, without protection, vision greys out at about 4 g. NASA's human-system standard, NASA-STD-3001, sets limits for each axis and duration, lower for crews deconditioned by months in weightlessness and higher for aborts and emergency entries.

In Figure 31.7 (b), a lifting entry from low Earth orbit (LEO) stays inside the nominal return limit. A ballistic backup entry, at almost 10 g, crosses it but stays within the emergency limit. This is why Soyuz and Shenzhou use ballistic entry only as a backup.

Seats spread the load over as much of the body as possible. Soyuz cosmonauts lie in individually moulded seat liners on seats that stroke on shock absorbers at touchdown; Dragon's four carbon-fibre seats sit side by side beneath the display panel (Figure 31.8).




Limits: NASA-STD-3001 Vol. 2 Rev. A, Figs 2–5 (sustained, > 0.5 s); in (a) solid = abort/emergency entry, dashed = nominal return. (b) Time each simulated entry of the entry-simulation figure spends above a given load; T-10-1: reported 14–17 g for about 5 s.

**FIGURE 31.7** Human tolerance of sustained acceleration. (a) NASA-STD-3001 limits for four directions: eyeballs-in is by far the most tolerable. (b) Eyeballs-in limits compared with the time spent above each acceleration level in the simulated entries of Figure 31.13, and with the 1983 Soyuz T-10-1 pad abort.

**FIGURE 31.8** The NASA SpaceX Crew-10 astronauts in their seats inside a Dragon capsule during a pre-flight equipment test at Kennedy Space Center, February 2025. The commander and pilot take the two central seats, facing the touchscreen display panel (out of view). Image: NASA/KSC (SpaceX).

### 31.2.4 Displays and Avionics

**Avionics** — the guidance, navigation and control computers, sensors and displays — show the largest generational change. The Apollo Guidance Computer had 2 048 words of erasable memory and was operated through a numeric keypad. The Shuttle had five computers, four running identical software and a fifth independently written backup software. Dragon's crew fly with three large touchscreens and a row of physical buttons for critical actions such as an emergency deorbit, parachute deployment or fire suppression. The vehicle normally rendezvouses and docks on its own. Orion uses airliner-derived flight computers with independent backup flight software, and its crew demonstrated manual piloting on Artemis II.

The governing principle is **fault tolerance**: no single failure — and for many functions no two — may cause loss of the crew. It is achieved with redundant, physically separated strings of equipment, voting computers that outvote a faulty member, dissimilar backups that cannot share a software error, and manual control of critical functions.

### 31.2.5 Propulsion, Power and Thermal Control

Crewed spacecraft carry a **main engine** for large manoeuvres (Orion's service module uses a refurbished 27 kN Space Shuttle orbital manoeuvring engine), **reaction-control thrusters** for attitude control and docking and, in pusher designs, **abort engines**. Almost all burn storable **hypergolic** propellants, which ignite on contact and can wait for months in orbit (Chapter 35). Dragon has 16 Draco thrusters of 400 N and 8 SuperDraco abort engines of about 71 kN, all fed from the same tanks. Thrusters are a recurring source of trouble. On Gemini 8 in 1966 a stuck thruster spun the spacecraft at about one revolution per second until Neil Armstrong regained control. Starliner's 2024 failures came from heat building up during long series of pulses. Qualification tests must therefore reproduce realistic firing sequences.

Gemini and Apollo introduced **fuel cells**, which combine hydrogen and oxygen to make electricity — on Apollo and the Shuttle also the crew's drinking water. Today's vehicles use **solar arrays** with batteries for eclipses and entry. Crew and electronics produce hundreds of watts to several kilowatts of heat, carried by pumped fluid loops to **radiators** on the service module or, on Dragon, the trunk (Chapter 5). The Soyuz MS-22 leak showed how exposed these external loops are.

### 31.2.6 Docking Systems

Docking joins two spacecraft so that crews can move between them through a pressurised tunnel. Early systems used a **probe** on the active vehicle and a cone-shaped **drogue** on the passive one, as Soyuz and the Russian ISS segment still do. **Androgynous** systems, with identical rings on both sides, let any two vehicles dock and leave the passage clear without removing a probe. The International Docking System Standard (IDSS) of 2010 defines a common interface for vehicles of different agencies and companies (Table 31.2). Modern systems first make a **soft capture**, with latches on a compliant, actuator-controlled ring absorbing the contact energy at closing speeds of about 0.1 m/s or less. They then make a **hard capture**, in which hooks pull the structures together and seal them.

**TABLE 31.2** Docking systems for crewed spacecraft.

| System | Type | Used by | Notes |
|---|---|---|---|
| Apollo probe and drogue | Probe on CSM, drogue on LM | Apollo, Skylab | Removed to open the tunnel |
| SSVP | Probe and drogue | Soyuz, Progress, Russian ISS modules | In use since 1971; 0.8 m hatch |
| APAS-75 / -89 / -95 | Androgynous, peripheral | Apollo–Soyuz, Mir, Shuttle | Shuttle docked to Mir and ISS |
| Chinese mechanism | APAS-derived | Shenzhou, Tianzhou, Tiangong | Automated 2011, manual 2012 |
| IDSS (NDS, SpaceX system) | Androgynous, low-impact | Dragon, Starliner, Orion; ISS adapters | Common standard; 0.8 m passage |

# 31.3 Launch Abort

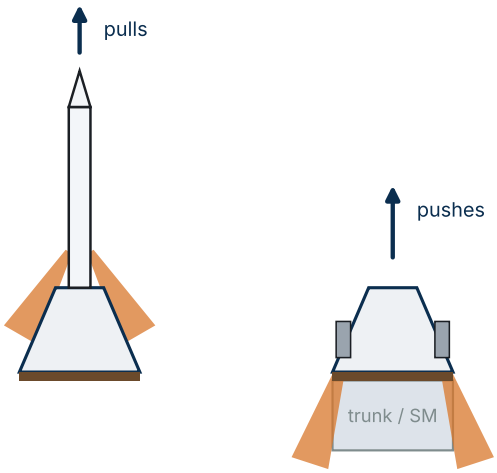
## 31.3.1 Why Crews Need to Escape

A launch vehicle is a controlled explosion of hundreds of tonnes of propellant. If it fails, the crew may have only seconds before blast overpressure, a fireball, flying fragments or, at altitude, aerodynamic forces reach them. The **launch abort system** must detect the failure, carry the crew module far enough away fast enough, stabilise it and hand it over to the normal parachute landing.

Vostok and Gemini used **ejection seats**, which work only at low altitude and speed. When Gemini 6A's engines shut down 1.2 s after ignition on 12 December 1965, Wally Schirra judged that the rocket had not lifted off. He chose not to eject, saving the mission and avoiding a risky ejection from a pure-oxygen cabin. Apollo and almost every crewed vehicle since — the Space Shuttle being the main exception — instead remove the whole crew module from the rocket.

## 31.3.2 Towers and Pushers

There are two ways to do this (Figure 31.9). A **tower**, or tractor, system mounts solid rocket motors above the capsule, with nozzles canted outwards so that the plumes clear it. It *pulls* the capsule away with a short, violent burn and is jettisoned once the rocket is high enough. Mercury, Apollo, Soyuz, Shenzhou, Orion, Gaganyaan and Mengzhou use towers. A **pusher** system builds liquid-propellant engines into the capsule or service module, like Dragon's SuperDracos and Starliner's four 178 kN RS-88 engines, and *pushes* it away.




(a) Tower (tractor) Mercury, Apollo, Soyuz, Shenzhou, Orion, Gaganyaan, Mengzhou

(b) Pusher (integrated engines) Crew Dragon, Starliner

| | Tower | Pusher |
|---|---|---|
| **Thrust-to-weight** | ≈ 7–13 | ≈ 4–6 |
| **Burn time** | 1–5 s | ≈ 5–10 s |
| **Available** | pad → jettison | pad → orbit |
| **Mass to orbit** | jettisoned | carried |
| **Re-use** | lost each flight | with capsule |
| **Propellant** | solid | hypergolic |
| **Risk** | jettison event | propellant on board |

(c) Trade-offs (typical values)

**FIGURE 31.9** Launch escape systems. (a) A tower pulls the capsule away; (b) a pusher uses engines built into the capsule or service module. (c) Typical trade-offs between them.

Towers give very high thrust for their mass from simple solid motors, but they are dead weight after jettison and are discarded on every flight. Pushers are available all the way to orbit, and their propellant can be used for manoeuvring. However, they carry tonnes of toxic propellant next to the crew for the whole mission, and they give a slower escape.

### 31.3.3 Abort Modes

The abort method changes as the rocket climbs. On the pad, the escape motor must lift the capsule high enough to open its parachutes; through **maximum dynamic pressure** (max-Q) it must also overcome drag. After the tower is jettisoned, two to three minutes after launch, the capsule separates with its own engines (on Soyuz, with motors on the payload fairing) and makes a ballistic or lifting entry downrange. Close to orbital speed, it may be better to continue to a lower orbit (**abort to orbit**).

### 31.3.4 Aborts in Flight

Table 31.3 lists the aborts that occurred during crewed launches or, for Soyuz T-10-1, with the crew aboard on the pad; every crew survived. (The Shuttle also had five engine shutdowns on the pad, which needed no escape.) The two Soyuz in-flight aborts showed that the backup motors and ballistic entry worked — at the cost of high accelerations.

**TABLE 31.3** Launch aborts with crews aboard.

| Flight | Date | Phase | What happened | Outcome |
|---|---|---|---|---|
| Soyuz 18a | 5 Apr 1975 | Staging | Stages failed to separate; steep entry from 192 km at about 21 g | Crew injured; landed in the Altai mountains |
| Soyuz T-10-1 | 26 Sep 1983 | On the pad | Booster fire; tower fired seconds before the explosion; 14–17 g for 5 s | Crew unhurt; landed 4 km away |
| STS-51F | 29 Jul 1985 | Ascent, 5 min 45 s | Main engine shut down by faulty sensors | Abort to orbit; mission completed |
| Soyuz MS-10 | 11 Oct 2018 | Booster separation, ≈ 2 min | Strap-on struck the core (sensor pin deformed in assembly); fairing motors pulled the spacecraft clear; ballistic entry at 6–7 g | Crew landed 19 min 41 s after launch |

### 31.3.5 Abort Tests and Abort Sizing

Abort systems are proven with uncrewed test flights. Orion's Pad Abort-1 in May 2010 accelerated a test capsule to about 200 m/s (445 mph) in three seconds. Ascent Abort-2 in July 2019 fired the abort motor at about 9.5 km and Mach 1.1 (Figure 31.10). SpaceX tested Dragon from the pad in 2015 and, in January 2020, near max-Q from a Falcon 9 whose engines were shut down at the abort command. Starliner's pad abort test in November 2019 succeeded although one of three main parachutes failed because of an improperly connected link (Figure 31.11). India tested Gaganyaan's system in flight in 2023, and China tested Mengzhou in 2025 and 2026.

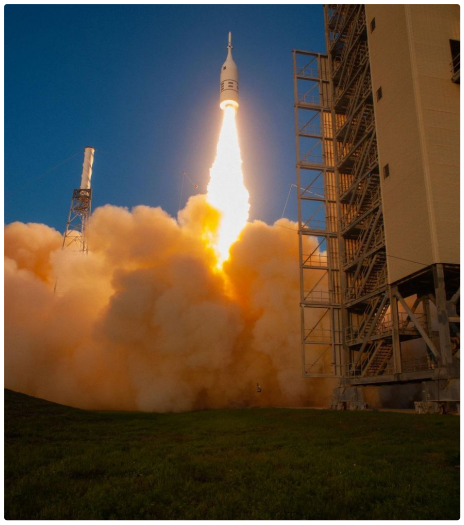
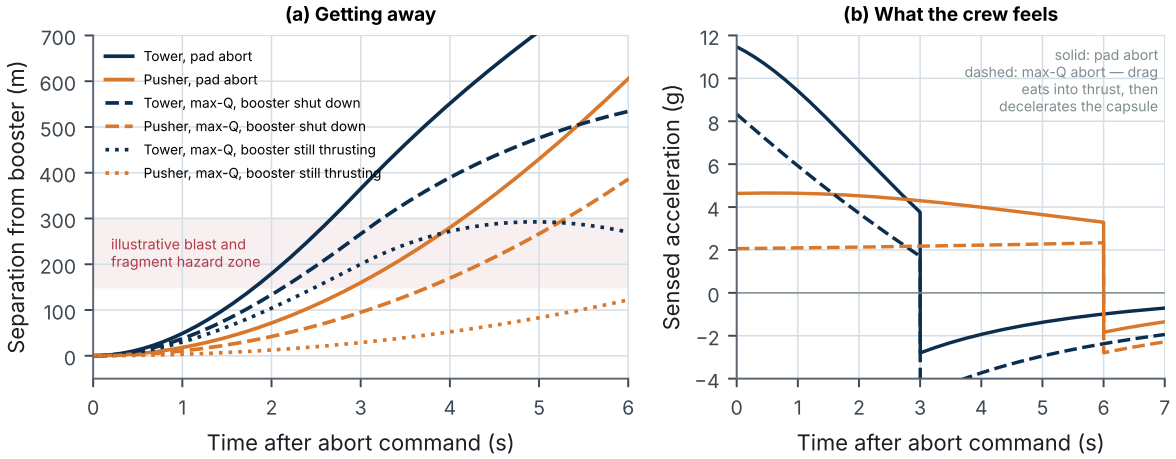


**FIGURE 31.10** Orion's Ascent Abort-2 test lifts off from Cape Canaveral on 2 July 2019. Less than a minute later the abort motor fired at about 9.5 km altitude, pulling the test crew module off the booster.
Image: NASA/KSC (NASA/Tony Gray and Kevin O'Conne).

**FIGURE 31.11** Starliner's pad abort test at White Sands, 4 November 2019: four abort engines and several manoeuvring thrusters push the spacecraft off the stand with a combined thrust of about 710 kN.
Image: NASA/KSC (NASA JSC/Boeing).

How much thrust is needed? That depends on how far the crew module must get from the booster, and how quickly; Figure 31.12 simulates both types of system.




1-D model along the flight path. Tower: 16 t abort vehicle, 1.8 MN regressive thrust for 3 s. Pusher: 12.5 t, 8 × 71 kN for 6 s. Max-Q: 420 m/s at 11 km (q ≈ 31 kPa), C_D ≈ 0.8; booster 300 t.

**FIGURE 31.12** Simulated separation after an abort. (a) Distance from the booster for a tower and a pusher, from the pad and at maximum dynamic pressure. If the booster keeps thrusting, even a tower struggles to escape. (b) The acceleration the crew feels: at max-Q, drag first reduces the net push and then decelerates the capsule once the motor burns out.

**EXAMPLE: SIZING AN ABORT MOTOR**

Suppose the crew module must be 300 m from the booster within 3 s of the abort command, and the abort vehicle (crew module plus tower) has a mass of 16 t.

1. Required relative acceleration: $a_{rel} = 2s/t^2 = 2{\times}300/3^2$ = 66.7 m/s² (6.8 g).
2. **Pad abort.** The booster stands still and the capsule climbs against gravity: $T/W = (a_{rel} + g)/g$ = **7.8**, a thrust of 1.22 MN.
3. **Max-Q, booster shut down.** At 420 m/s and 11 km (ρ = 0.356 kg/m³), $q = \frac{1}{2}\rho v^2$ = 31 kPa. With $C_D A$ = 0.8 × 19.6 = 15.7 m², the drag is 0.49 MN (3.1 g). Gravity acts equally on capsule and booster and cancels in their relative motion: $T = m\,a_{rel} + D$ = 1.07 + 0.49 = 1.55 MN, **T/W = 9.9**.
4. **Max-Q, booster still thrusting** at 1.5 g: $T = m(a_{rel} + 1.5g) + D$ = 1.79 MN, **T/W = 11.4** — close to the 1.8 MN (400 000 lbf) of Orion's abort motor.
5. A Dragon-class pusher (8 × 71 kN on 12.5 t, T/W = 4.6) needs about 4.1 s to open 300 m from the pad and 5.3 s at max-Q (Figure 31.12). Pushers therefore depend on shutting the booster's engines down the instant an abort begins. A solid booster cannot be shut down, which favours towers on rockets with solid first stages.

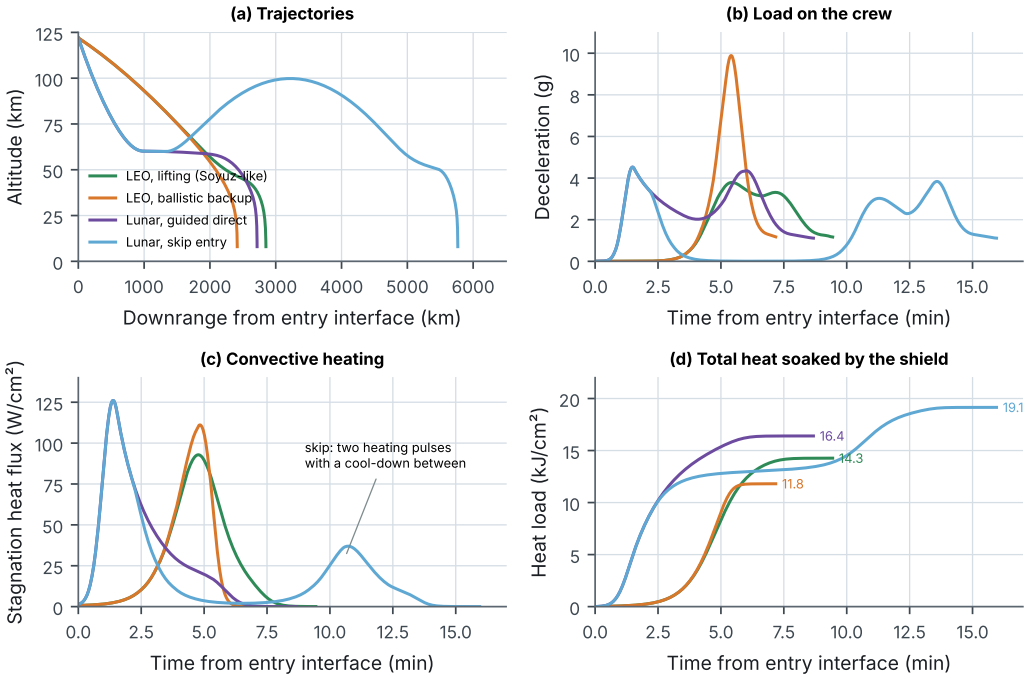
## 31.4 Coming Home: Entry, Descent and Landing

### 31.4.1 Capsule Shapes and Lift

A capsule returning from orbit must dissipate about 30 MJ of kinetic energy per kilogram, and from the Moon about 60 MJ (Chapter 10). Placing the centre of mass slightly off the axis makes the capsule fly at an angle of attack of 15–25°. It then produces a **lift-to-drag ratio** (L/D) of 0.25–0.4 — about 0.37 for Apollo and 0.25–0.3 for Soyuz, Shenzhou and Orion. By rolling to point this lift up, down or sideways (**bank-angle modulation**), the guidance controls both deceleration and landing point; Artemis II splashed down 0.95 km from its predicted point. A broad cone like Orion's has a low **ballistic coefficient** $\beta = m/(C_D A)$, so it decelerates high in thin air. But it must stay stable from hypersonic to subsonic speeds, and it carries uprighting bags in case it floats apex-down.

### 31.4.2 Ballistic, Lifting and Skip Entries

Figure 31.13 compares four ways of coming home. A **lifting entry from LEO** (Soyuz-like, at 7.6 km/s and −1.35°) peaks at 3.8 g. The **ballistic backup** of the same capsule is flown when the guidance fails, with the capsule spinning slowly to cancel its lift. It peaks near 10 g in the model, compared with the roughly 8 g reported on the ballistic Soyuz entries of 2007 and 2008. Returning from the Moon at 11 km/s, a capsule must enter more steeply (about −6°) so as not to skip out; with lift and guidance the model peaks at 4.5 g. A **skip entry** dips into the atmosphere, climbs back to about 100 km and enters again. This roughly doubles the downrange distance, so that the capsule can reach a fixed landing zone from a wider range of return trajectories.




Model: 3-DOF point mass, US-76 density, Sutton–Graves convective heating only (radiative heating, significant above 10 km/s, omitted). LEO: 7.6 km/s, −1.35°, β = 610 kg/m², L/D 0.3, bank 55°. Lunar: 11.0 km/s, −6.0°, β = 350 kg/m², L/D 0.27, simple bank-angle guidance. Entry interface 122 km. Illustrative, not flight trajectories.

**FIGURE 31.13** Simulated capsule entries. (a) Trajectories; (b) deceleration; (c) convective heat flux at the stagnation point; (d) total heat absorbed per unit area. Lift cuts the peak load from about 10 g to about 4 g; a lunar return raises the peak heating, and a skip entry splits it into two pulses and increases the total heat load.

### EXAMPLE: WHY LUNAR RETURN NEEDS LIFT

The Allen–Eggers formula (Chapter 10) gives the peak deceleration of a ballistic body entering on a straight path at speed $V_E$ and angle $\gamma_E$, with scale height $H$ = 7.2 km:

$$n_{max} = \frac{V_E^2 \sin|\gamma_E|}{2\,e\,H\,g_0}$$

1. **LEO return**, 7.6 km/s at −1.35°: $n_{max}$ = 7 600² × 0.0236 / (2 × 2.718 × 7 200 × 9.81) = **3.5 g**.
2. **Lunar return**, 11.0 km/s at −6.0°: $n_{max}$ = 11 000² × 0.105 / 383 900 = **33 g** — a factor (11.0/7.6)² = 2.1 from speed and 4.4 from the steeper angle.
3. **Simulation** (Figure 31.13). The LEO ballistic case actually peaks at 9.9 g: on such a shallow path gravity bends the trajectory down, steepening it to about −5.5° by the time of peak deceleration. The lunar ballistic case at −6.0° peaks at 9.7 g, because above orbital speed the curving path lifts the capsule — but entering just 0.5° steeper raises it to 15.5 g.
4. **Lesson.** The formula shows the scaling — deceleration grows with the square of speed and the steepness of entry — but real entries must be integrated numerically. A lunar-return capsule is so sensitive to its entry angle that it needs lift and closed-loop guidance; with L/D = 0.27 the simulated peak falls to 4.5 g.

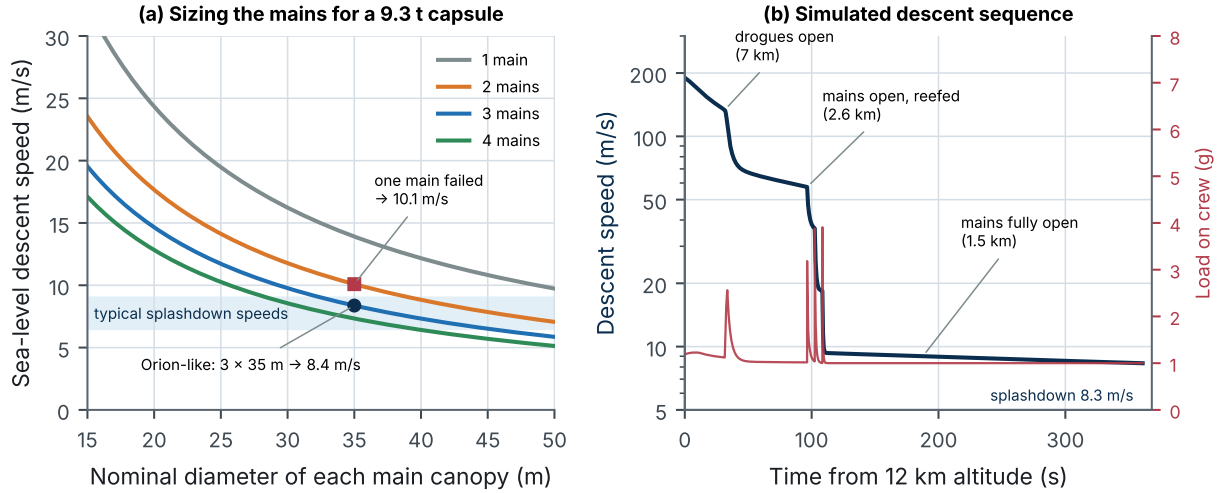
## 31.4.3 Heat Shields

A crewed heat shield cannot be tested at full scale before flight. Nearly all capsules use **ablators**, which char and erode, carrying heat away in the gases they release (Chapter 10). Apollo's **Avcoat**, an epoxy-novolac resin with silica fibres filling a fibreglass honeycomb, was revived for Orion, whose 5 m shield is made of 186 bonded blocks. Dragon uses SpaceX's **PICA-X**, a low-density carbon–phenolic ablator derived from NASA's PICA. Soyuz and Shenzhou jettison their heat shields shortly before landing.

After **Artemis I**, engineers found that charred Avcoat had broken away from more than a hundred places on Orion's heat shield. NASA traced the cause to the skip entry. As the heating fell between the two pulses — the dip in Figure 31.13 (c) — the Avcoat kept generating gas inside. But it formed permeable char too slowly to let the gas escape, so pressure built up and cracked off pieces of char. Rather than replace the Artemis II shield, NASA flew a modified entry with a shorter maximum downrange distance, reducing the time spent in the troublesome heating regime. Artemis II's shield, whose surface exceeded 2 200 °C, showed only nine spots of char loss against 103, and the Artemis III shield uses Avcoat of more uniform permeability.

### 31.4.4 Parachutes

Below about 10 km a capsule falls at 100–200 m/s, and parachutes take over in stages that keep opening loads within the strength of the structure and the tolerance of the crew. Pilot parachutes or mortars deploy **drogues**, which stabilise the capsule and slow it to 50–90 m/s. The drogues then pull out the **mains**. Their **reefing** lines hold each canopy partly closed until pyrotechnic cutters release them a few seconds later (Figure 31.14 b). Redundancy is essential: Apollo 15 landed safely in 1971 although one of its three mains collapsed, and modern systems are designed to land safely with one main failed. Soyuz has a single large main parachute and a complete reserve.




v = √(2mg / ρ C_D S); ringsail C_D ≈ 0.8 on nominal area, cluster efficiency 0.90–0.95. (b) drogues 2 × 7 m from 12 km at 190 m/s (vertical), mains reefed to 6 % then 25 % of area.

**FIGURE 31.14** Parachute design. (a) Sea-level descent speed versus main-canopy diameter for a 9.3 t capsule with one to four mains; Orion's three 35 m mains give about 8 m/s, and about 10 m/s with one failed. (b) A simulated descent: drogues at 7 km, reefed mains at 2.6 km, full mains below 1.5 km.

At terminal velocity, drag equals weight, so the descent speed is $v = \sqrt{2mg/(\rho C_D S)}$ for total canopy area $S$: halving the landing speed needs four times as much parachute.

**EXAMPLE: SIZING THE MAIN PARACHUTES**

A 9.3 t crew module must splash down at no more than 10 m/s at sea level (ρ = 1.225 kg/m$^3$) even if one of its three mains fails. Ringsail canopies have $C_D \approx 0.80$ on their nominal area, and two canopies in a cluster lose about 5 % of their drag through interference.

1. Drag area needed: $S = 2mg/(\rho v^2 C_D)$ = 2 × 9 300 × 9.81 / (1.225 × 10$^2$ × 0.80) = 1 860 m$^2$.
2. Two working canopies: each needs 1 860 / (2 × 0.95) = 980 m$^2$, a nominal diameter $D_0 = \sqrt{4 \times 980/\pi}$ = **35.3 m** — the size of Orion's mains.
3. With all three open (cluster efficiency 0.92), $v$ = **8.3 m/s** (19 mph), and the capsule meets the water with about 320 kJ of kinetic energy.
4. **Why reef?** The mains open at 2.6 km (ρ = 0.94 kg/m$^3$) and about 57 m/s, where $q$ = 1.55 kPa. Opened fully at once, three canopies would decelerate the capsule at about 36 g; reefed to 6 % of their area, about 2 g. The peak in the simulated sequence is 3.9 g.

### 31.4.5 Water or Land?

Water landings are gentle but need ships and divers, and seawater makes reuse harder. American capsules from Mercury to Orion have splashed down (Figure 31.15); since mid-2025 Dragon has usually returned to the Pacific off California, where its trunk, jettisoned after the deorbit burn, falls into the ocean.

**FIGURE 31.15** Orion Integrity descends under its three main parachutes before splashdown in the Pacific off San Diego on 10 April 2026, ending Artemis II. Image: NASA/KSC (NASA/Josh Valcarcel).

Landing on land makes recovery and reuse easier but needs a soft touchdown. Soyuz and Shenzhou fire **soft-landing rockets**, six small solid motors triggered by a gamma-ray altimeter about a metre above the ground, which cut the speed from about 7 m/s to about 1.5 m/s (Figure 31.16). Starliner inflates six **airbags** beneath its jettisoned heat shield and lands at White Sands, New Mexico; Mengzhou will also use airbags (Table 31.4).

**FIGURE 31.16** Soyuz MS-08 lands near Zhezkazgan, Kazakhstan, on 4 October 2018. The flash beneath the descent module is the soft-landing rockets firing just before touchdown. Image: NASA (NASA/Bill Ingalls).

**TABLE 31.4** Descent and landing systems of capsules.

| Vehicle | Drogues | Mains | Speed under mains | Final attenuation |
|---|---|---|---|---|
| Apollo CM | 2 × 5.0 m | 3 × 25.5 m ringsail | ≈ 10 m/s (22 mph) | Water |
| Soyuz | 1 (≈ 24 m²) after pilots | 1 × ≈ 1 000 m², plus reserve | ≈ 7 m/s | Soft-landing rockets → ≈ 1.5 m/s |
| Crew Dragon | 2, at ≈ 5.5 km | 4, at ≈ 1.8 km | ≈ 7 m/s (15–16 mph) | Water |
| Starliner | 2 | 3 | — | Six airbags, on land |
| Orion | 2 × 7 m | 3 × 35 m ringsail | < 9 m/s (< 20 mph) | Water |

At sea, Dragon's recovery ship lifts the capsule onto its deck within about an hour, and the crew are helped out while their balance readjusts to gravity (Figure 31.17). The Artemis II crew instead stepped onto a raft and were flown to their ship by helicopter. The winged alternative — the Shuttle's runway landing after a gliding entry with up to 2 000 km of cross-range — gave the gentlest deceleration of all. The cost was carrying wings, landing gear and a fragile thermal protection system to orbit and back (Chapter 10).

**FIGURE 31.17** The recovery team opens the hatch of Dragon Endeavour on the SpaceX recovery ship after the Crew-11 splashdown off California on 15 January 2026. The discolouration is from entry heating. Image: NASA (NASA/Bill Ingalls).

# 31.5 Reliability and Safety

## 31.5.1 What Has Gone Wrong

Four accidents in flight have killed 18 people in orbital spacecraft, and the Apollo 1 fire killed three on the ground (Table 31.5). Each occurred where the crew could not escape: during entry and landing (Soyuz 1, Soyuz 11, Columbia), or during ascent in a vehicle without an escape system (Challenger).

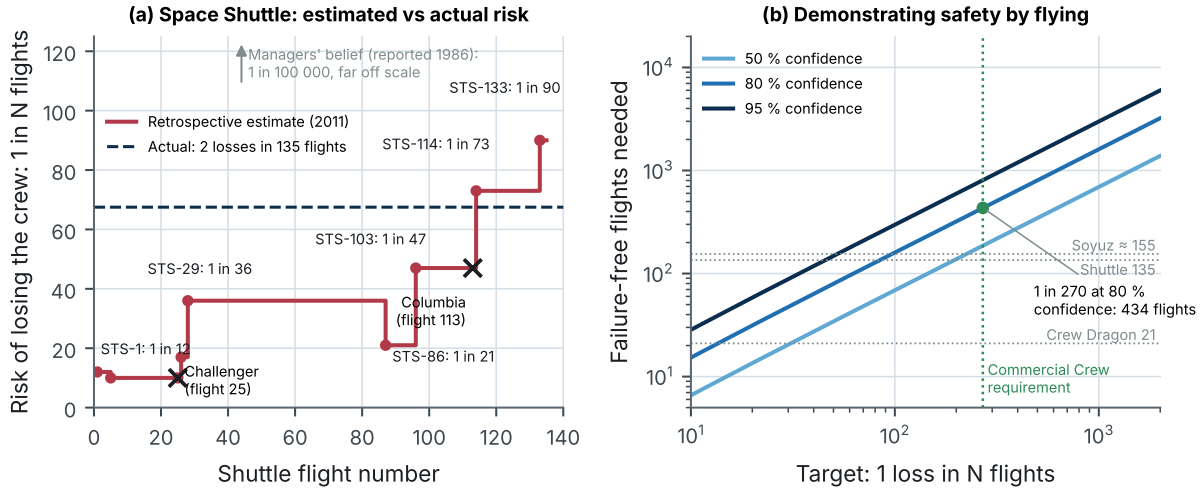
**TABLE 31.5** Fatal accidents in orbital crewed spacecraft.

| Accident | Date | Phase | Cause | Main changes afterwards |
|---|---|---|---|---|
| Apollo 1 | 27 Jan 1967 | Ground test | Fire in pure oxygen; inward-opening hatch; 3 killed | $N_2$–$O_2$ on the pad, new hatch, materials control |
| Soyuz 1 | 24 Apr 1967 | Landing | Main parachute failed, reserve tangled; 1 killed | Redesigned parachutes |
| Soyuz 11 | 30 Jun 1971 | Entry | Valve opened at module separation; cabin depressurised; 3 killed | Pressure suits for launch and landing |
| Challenger (STS-51L) | 28 Jan 1986 | Ascent | Booster joint seal failed in cold weather; 7 killed | Booster redesign, bail-out system |
| Columbia (STS-107) | 1 Feb 2003 | Entry | Wing holed by foam at launch; 7 killed | In-orbit inspection; Shuttle retirement |

## 31.5.2 Estimating Risk

How safe is a spacecraft? The best answer comes from **probabilistic risk assessment (PRA)**. This is a model of the vehicle built from fault trees and event sequences, using failure rates from tests and experience. It estimates the probability of **loss of crew (LOC)** per flight. Investigating the Challenger accident in 1986, the physicist Richard Feynman found that NASA managers believed the chance of losing a Shuttle was about 1 in 100 000, while engineers estimated about 1 in 100. A 2011 NASA study recomputed the risk as it really was through the programme (Figure 31.18 a). It was about 1 in 12 on the first flight and 1 in 10 at the time of Challenger. It improved to about 1 in 90 by the end as hazards were found and fixed. The actual record — 2 losses in 135 flights, or 1 in 68 — agreed with this retrospective estimate, not with the managers' belief.

Figure 31.18 (b) shows why flight experience alone cannot prove a vehicle safe. After $n$ flights without loss, a vehicle whose true risk is $p$ per flight would show that record with probability $(1-p)^n$. Claiming with confidence $C$ that the risk is no worse than $p$ requires $n = \ln(1-C)/\ln(1-p)$ successful flights. Showing 1 in 270 at 80 % confidence needs 434 flights — more than any crewed vehicle has made. Safety must be built in by design, analysis and test.




(a) Hamlin et al., NASA Shuttle risk progression (2011), loss-of-crew-and-vehicle odds at selected flights. (b) n = ln(1 − C) / ln(1 − 1/N): flights with no loss needed to show risk ≤ 1/N.

**FIGURE 31.18** Measuring safety. (a) Retrospective estimates of the Space Shuttle's loss-of-crew risk at selected flights, compared with the actual loss rate. (b) The number of consecutive successful flights needed to demonstrate a risk target with a given statistical confidence.

### 31.5.3 Human-Rating

NASA certifies a spacecraft for crews (**human-rating**) under NPR 8705.2, *Human-Rating Requirements for Space Systems*, together with the technical standards of NASA-STD-3001. Among other things, the system must:

- tolerate failures, so that no single failure — and for many functions no combination of two — causes a catastrophic event;
- provide abort and crew survival capability in every phase where it is feasible;
- give the crew insight into, and manual control of, critical functions, within human limits for acceleration, noise, vibration, atmosphere and workload;
- be verified by test and analysis, including flight tests of the integrated system before crews fly.

For Commercial Crew, NASA required a loss-of-crew risk no worse than **1 in 270** over a 210-day ISS mission, with no more than 1 in 500 from ascent and entry. This is three to five times safer than the Shuttle at its best. In 2019 the US Government Accountability Office found that debris strikes during the long docked period dominated the risk. Starliner showed the limits of certification: thruster failures had also occurred on its 2022 uncrewed flight, but their cause was not fully understood before a crew flew.

#### AI IN CREWED SPACECRAFT

Crewed vehicles already rely on autonomy. Soviet and Russian spacecraft have docked automatically using radio systems since 1967 (Igla, then Kurs from 1986). Crew Dragon flies its whole rendezvous and docking on its own with lidar, thermal and visible cameras while the crew monitors. Machine learning is being added cautiously, where it can be verified. Uses include vision-based pose estimation for docking and inspection, and anomaly detection that compares thousands of telemetry channels with learned normal behaviour. Such detection could flag slow degradations such as Starliner's thruster heating. There are also crew assistants, such as the voice assistant demonstrated on Artemis I. On Mars missions, light-time delays will make onboard fault diagnosis essential. Chapters 49 and 51 discuss these applications and how they can be qualified for human spaceflight.

### KEY TAKEAWAYS

- In October 2026 Soyuz, Shenzhou, Crew Dragon and Orion were in service, Starliner was grounded, and Gaganyaan and Mengzhou were preparing for uncrewed test flights.
- A capsule stack has a crew module (pressure vessel, heat shield, parachutes), a service module and a launch abort system; only the crew module returns.
- Crews tolerate acceleration best eyeballs-in: about 15 g for seconds and 4–5 g for minutes. Lift ($L/D \approx 0.3$) cuts entry loads from about 10 g to about 4 g and lets capsules steer to a precise landing.
- Abort systems must open hundreds of metres of separation within seconds: towers give $T/W \approx 7$–$13$; pushers ($T/W \approx 4$–$6$) work all the way to orbit but rely on shutting the booster down.
- Lunar return doubles the energy per kilogram and demands a precise entry angle, guided lift and robust heat shields.
- Parachute descent speed falls only with the square root of canopy area; systems are designed to land safely with one main parachute failed.
- Flight experience alone cannot demonstrate safety: the Shuttle's 1-in-68 record matched retrospective risk models, not its managers' 1-in-100 000 belief.

### REVIEW QUESTIONS

1. Why do capsule crews lie facing the heat shield? Use Figure 31.7 to compare +Gx and +Gz limits for 60 s.
2. List the three elements of a capsule stack and their functions. Which are reused on Crew Dragon?
3. Compare tower and pusher abort systems. Why might a rocket with a solid-fuel first stage favour a tower?
4. An abort vehicle of 10 t must open 200 m of separation from a stationary booster within 2.5 s. What thrust is required?
5. Use the Allen–Eggers formula to estimate the peak deceleration of a ballistic capsule entering at 11 km/s and −6.5°. Why is the real value lower, and why do lunar-return capsules still need lift?
6. How does a skip entry differ from a direct entry, and how did the skip lead to char loss on Orion's Artemis I heat shield?
7. A 6 t capsule must splash down at no more than 9 m/s with one of its two mains failed. What nominal canopy diameter is needed ($C_D = 0.8$)?
8. Why do Soyuz and Shenzhou need soft-landing rockets, while Dragon and Orion do not?
9. What do the Shenzhou-20 window damage (2025) and the Soyuz MS-22 leak (2022) teach about lifeboats for space stations?
10. Why can flight experience alone not prove a crewed vehicle safe? How many successful flights would show a risk of 1 in 500 with 90 % confidence?

# 32 CHAPTER Space Stations and Habitats

Image: NASA/JSC

*A space station is a spacecraft that people live in for months or years, supplied from Earth and maintained in orbit. This chapter traces stations from Salyut 1 and Skylab through Mir to the International Space Station and China's Tiangong, then takes the ISS apart system by system — power, thermal control, attitude control, reboost, communications and robotics — and explains the engineering that every station must solve: choosing an orbit against drag, sizing arrays and radiators, shielding against micrometeoroids and debris, providing enough habitable volume, and keeping cargo flowing. It covers expandable habitats and artificial gravity, the plan to deorbit the ISS around 2030, and the commercial and national stations competing to replace it as of October 2026.*

IN THIS CHAPTER

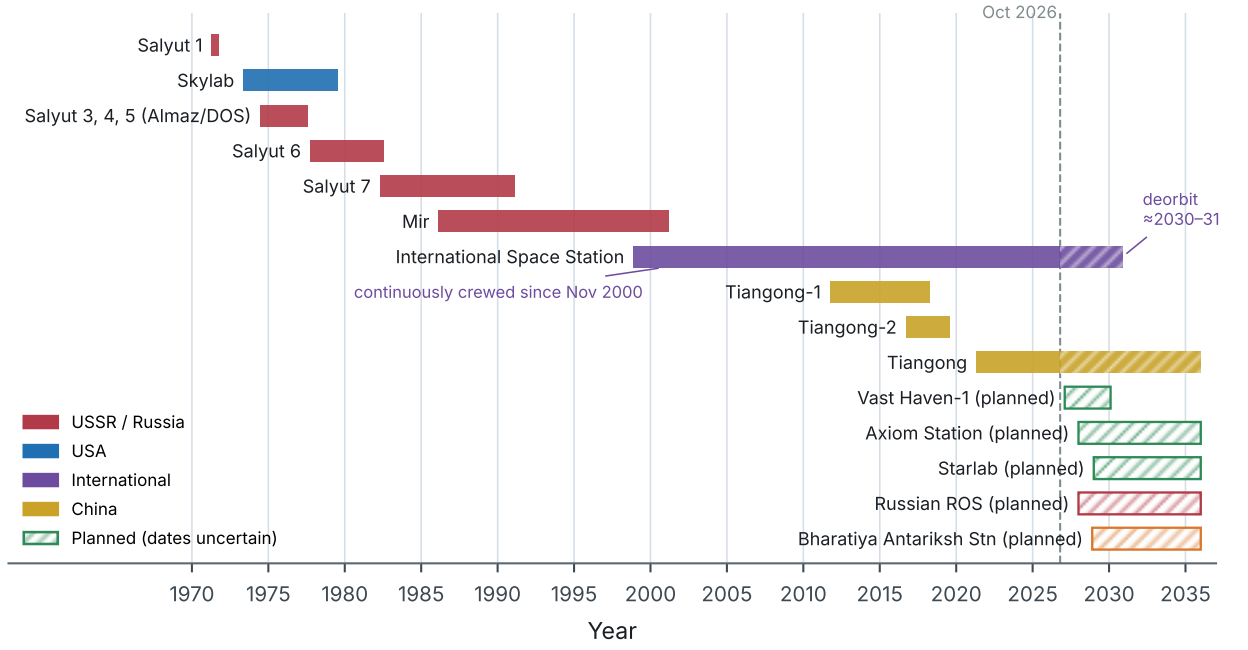
32.1 Fifty-Five Years of Space Stations
32.2 The International Space Station in Detail
32.3 Engineering a Station
32.4 Supplying a Station
32.5 Expandable Habitats and Artificial Gravity
32.6 After the ISS

A satellite can be designed once and left alone. A space station cannot: it is a house, a laboratory, a port and a factory in one, occupied by people who must be kept healthy and brought home safely. It must be supplied with tonnes of food, water, gas and spare parts every year, and boosted against atmospheric drag. It must also be repaired when things break, and protected against impacts from particles moving at ten times the speed of a rifle bullet. In return it offers something nothing else can — months of continuous human presence in weightlessness, for research, technology testing and preparation for journeys to the Moon and Mars.

People have lived aboard the **International Space Station (ISS)** without a break since November 2000, and Chinese crews have occupied the **Tiangong** station continuously since mid-2022. This chapter explains how stations work and how they are designed; the human body and life support are covered in Chapter 30, crewed spacecraft and docking systems in Chapter 31, and spacewalks in Chapter 33.

## 32.1 Fifty-Five Years of Space Stations

Each station since 1971 has taught how to keep people in orbit longer, with less help from the ground (Figure 32.1, Table 32.1).




Bars show time in orbit (launch to re-entry). Planned dates as reported in October 2026; Salyut 2 (failed 1973) omitted.

**FIGURE 32.1** Space stations from Salyut 1 to the stations planned for the late 2020s. Each bar runs from launch to re-entry; hatched bars are planned, with uncertain dates.

### 32.1.1 Salyut and Skylab

The Soviet Union launched the first station, **Salyut 1**, on 19 April 1971: a single 18-tonne cylinder with about 100 m$^3$ of pressurised volume. Its first crew, on Soyuz 10, docked but could not enter. The second, aboard Soyuz 11, spent more than three weeks on board. But on 30 June 1971 Georgy Dobrovolsky, Vladislav Volkov and Viktor Patsayev died during their return, when a pressure-equalisation valve was jolted open as the descent module separated. The valve vented the module's air to space. They are the only people to have died above the atmosphere, and since then Soyuz crews have worn pressure suits for launch and re-entry.

The United States converted a Saturn V third stage into **Skylab**, launched on 14 May 1973. It was a 77-tonne station with 361 m$^3$ of volume, still the most spacious station ever launched in one piece (Figure 32.2). Sixty-three seconds into the launch its micrometeoroid shield tore away; one solar wing was lost and debris jammed the other. The first crew, Pete Conrad, Joe Kerwin and Paul Weitz, put up a parasol sunshade to cool the overheating workshop and cut free the jammed wing during a spacewalk — the first major in-orbit repair. Three crews spent 28, 59 and 84 days on

board, operating the Apollo Telescope Mount solar observatory. High solar activity then expanded the upper atmosphere, and Skylab re-entered on 11 July 1979 over the Indian Ocean and Western Australia, before the Space Shuttle could boost it.

**Salyut 6** (1977–82) and **Salyut 7** (1982–91) introduced the feature that made long stays possible: a second docking port. Uncrewed **Progress** freighters, first flown in January 1978, could deliver food, water and propellant while a crew was on board, and short "visiting" crews could exchange the Soyuz lifeboat. Stays grew to 237 days in 1984. In 1985 Vladimir Dzhanibekov and Viktor Savinykh docked manually with a dead, frozen Salyut 7 and brought it back to life — a demonstration that crews can repair what automation cannot.

### 32.1.2 Mir

**Mir** grew from a core module launched on 19 February 1986 into a 130-tonne complex of seven modules by 1996 (Figure 32.3), the first station assembled in orbit. It was occupied for 4 592 days, including almost ten years without a break (1989–99), and in 1994–95 the physician Valeri Polyakov stayed 437 days, still the record for a single spaceflight. Between 1995 and 1998 nine Space Shuttle missions docked with Mir, and seven NASA astronauts lived on board — the rehearsal for the ISS. Mir also taught hard lessons. In February 1997 a burning oxygen-generating canister filled it with smoke. In June 1997 a remotely piloted Progress struck and punctured the Spektr module, which the crew had to seal off, losing much of the station's power. Mir was deliberately de-orbited into the South Pacific on 23 March 2001.

**TABLE 32.1** Space stations that have been crewed. Masses exclude visiting vehicles.

| Station | Operator | In orbit | Mass (t) | Pressurised volume (m³) | Notable for |
|---|---|---|---|---|---|
| Salyut 1 | USSR | 1971 | 18.4 | ≈ 100 | First station |
| Skylab | USA | 1973–79 | 77 | 361 | Largest module |
| Salyut 6, 7 | USSR | 1977–91 | 19–20 | ≈ 90 | Progress resupply |
| Mir | USSR, Russia | 1986–2001 | ≈ 130 | ≈ 350 | First modular station |
| ISS | 5 agencies | 1998– | ≈ 420 | 1 005 | Crewed since 2000 |
| Tiangong-1, -2 | China | 2011–19 | 8.5 | ≈ 15 | Docking, refuelling tests |
| Tiangong | China | 2021– | ≈ 70 | ≈ 340 | Crewed since 2022 |

**FIGURE 32.2** Skylab photographed by the departing Skylab 4 crew in February 1974. The gold twin-pole sunshade, erected by the second crew over the first crew's parasol, covers the workshop, which lost one of its two large solar wings during launch; the windmill-shaped arrays belong to the Apollo Telescope Mount. Image: NASA/JSC.

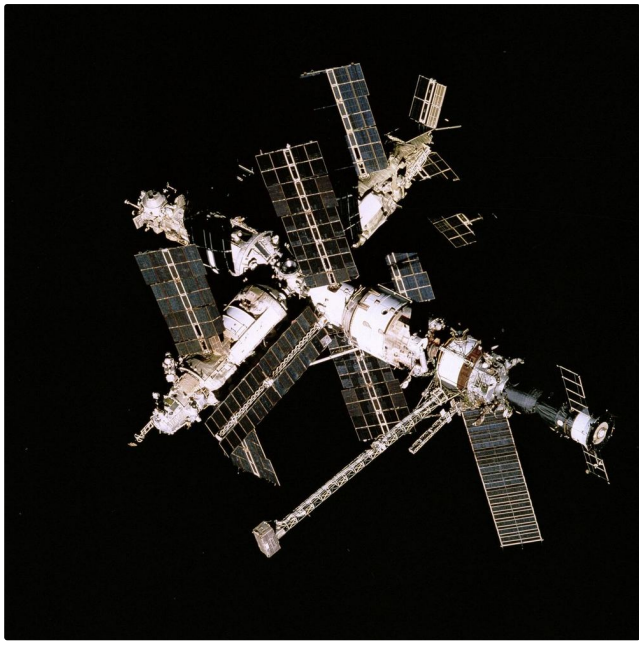


**FIGURE 32.3** Mir photographed from the approaching Space Shuttle Atlantis on 29 June 1995, during the first Shuttle–Mir docking mission. The modules radiate from the docking node at the front of the core. Image: NASA/JSC.

### 32.1.3 The International Space Station

The ISS combined the American Freedom station design with the planned Mir-2. Five agencies — NASA, Roscosmos, the European Space Agency (ESA), the Japan Aerospace Exploration Agency (JAXA) and the Canadian Space Agency (CSA) — representing 15 countries are partners. The Russian-built, US-funded Zarya module was launched on 20 November 1998 and the US Unity node two weeks later; the first resident crew, Expedition 1, arrived on 2 November 2000. Assembly took 42 flights — 37 Space Shuttle missions and five Russian launches — and the last major Russian module, the Prichal docking node, arrived in November 2021. On 2 November 2025 the station marked 25 years of continuous occupation, with more than 290 people from 26 countries having visited.

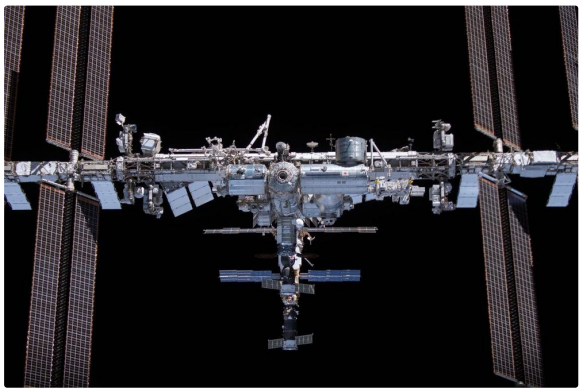


**FIGURE 32.4** The ISS seen from Crew Dragon Endeavour after undocking in November 2021: the pressurised modules run down the centre from the 109 m truss and its solar-array wings to the Russian segment. Image: NASA/JSC.

### 32.1.4 China's Tiangong

China followed a step-by-step path. **Tiangong-1** (2011), an 8.5-tonne laboratory, received the uncrewed Shenzhou 8 and two crews in 2012 and 2013, then re-entered uncontrolled in April 2018. **Tiangong-2** (2016) hosted a 30-day mission and tested in-orbit refuelling by the Tianzhou-1 freighter in 2017. The modular **Tiangong** station began with the 22.5-tonne Tianhe core module on 29 April 2021. The Wentian and Mengtian laboratory modules (July and October 2022) completed a T-shaped station of about 70 tonnes — roughly 90–100 tonnes with docked spacecraft. It has about 340 $m^3$ of pressurised volume. It orbits at about 390 km and 41.5° inclination, normally carries three astronauts (six during crew handovers), and was the first crewed station to use **Hall-effect electric thrusters** for reboost. In November 2025, when the Shenzhou-20 return capsule was found to have a cracked window, China launched the uncrewed Shenzhou-22 as a replacement lifeboat within weeks.

## 32.2 The International Space Station in Detail

The ISS is 109 m from end to end and has a mass of about 420 tonnes. It offers 1 005 $m^3$ of pressurised volume — about as much as a Boeing 747 — of which 388 $m^3$ is habitable. It circles Earth every 93 minutes, about 15.5 times a day, at about 415–420 km altitude and 51.6° inclination, passing over more than 90 % of the world's population.

### 32.2.1 Modules and Segments

The station has two halves (Figure 32.5). The **Russian Orbital Segment** includes the Zvezda service module, which provides propulsion, crew quarters and life support for the Russian crew. It also includes Zarya, which was the first element; the docking and research modules Poisk, Rassvet, Nauka and Prichal; and ports for Soyuz and Progress. The **US Orbital Segment** includes the nodes Unity, Harmony and Tranquility. Its laboratories are the US Destiny laboratory, ESA's Columbus and JAXA's Kibo. It also has the Quest and commercial Bishop airlocks, the Leonardo storage module, the seven-window Cupola and the expandable BEAM. Modules connect through standard ports: Russian probe-and-drogue docking ports, and **Common Berthing Mechanisms (CBMs)** with 1.27 m square hatches wide enough to pass equipment racks. There are also **International Docking Adapters (IDAs)** for Crew and Cargo Dragon and Starliner.

### 32.2.2 Power

Electrical power comes from eight **solar-array wings**, each about 35 m long, mounted in pairs on rotating joints at the ends of the truss so that they can track the Sun. Their silicon cells, designed for 15 years, have degraded; by about 2020 the arrays generated roughly 160 kW in sunlight. Since 2021 NASA has installed **ISS Roll-Out Solar Arrays (iROSAs)**, flexible blankets that unroll from a compact canister and each produce more than 20 kW. They are mounted in front of the old wings, partly shading them (Figure 32.6). Six iROSAs installed by 2023 raised the output to about 215 kW, and the final pair arrived in October 2026 aboard the CRS-35 Cargo Dragon for installation by spacewalk. Part of the power charges 24 lithium-ion batteries (which replaced 48 nickel–hydrogen batteries in 2017–21) for the eclipse. So the power available to the station's systems and

experiments over a whole orbit is much lower than the sunlit output. NASA's fact sheet quotes 75–90 kW. The US segment distributes power at 160 V direct current, converted to 124 V for users; the Russian segment uses 28 V.

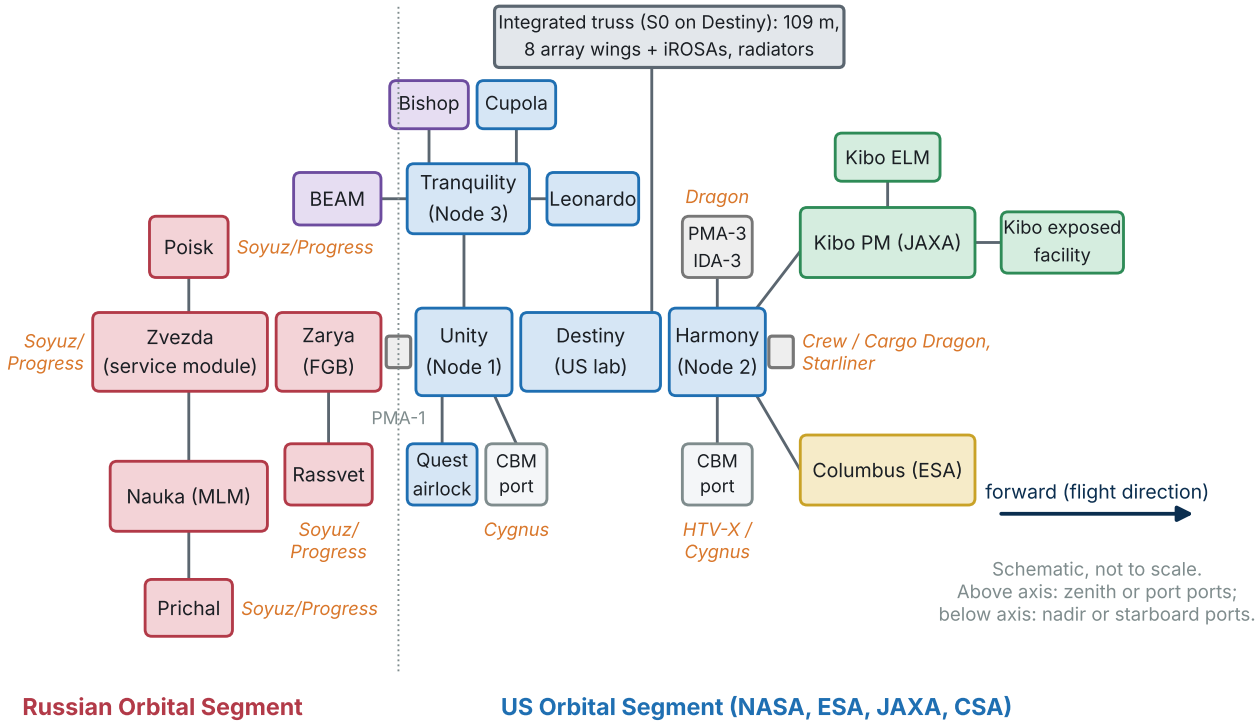



**FIGURE 32.5** Schematic connection diagram of the ISS pressurised modules and the ports used by visiting vehicles in 2026.

### 32.2.3 Thermal Control

Every watt the station uses ends up as heat that must be radiated to space. Inside the US segment, two water loops — a low-temperature loop at about 4 °C and a moderate-temperature loop at about 17 °C — collect heat from cold plates and air heat exchangers. Water is safe in the cabin but would freeze outside, so heat exchangers pass the heat to the **External Active Thermal Control System**. This consists of two independent loops of liquid ammonia that circulate through six radiator assemblies on the truss. There are three assemblies on each side; each has eight panels and rejects about 11.7 kW, 70 kW in all. The radiators rotate to keep edge-on to the Sun. Separate radiators cool the power electronics at the array roots. Ammonia leaks and pump failures have required several repair spacewalks.

### 32.2.4 Attitude Control and Reboost

The station normally flies with one axis along its velocity and another pointing at Earth, turning once per orbit. Small torques from gravity gradient, drag and venting would slowly twist it out of this attitude, so four **control moment gyroscopes (CMGs)** on the Z1 truss absorb them. Each is a 100 kg wheel spinning at 6 600 rpm with an angular momentum of 4 760 N·m·s; tilting the wheel's axis on motorised gimbals produces a torque on the station without using propellant. When the CMGs approach saturation, or for large manoeuvres, Russian thrusters take over. The station is flown slightly tilted, in a **torque-equilibrium attitude** where aerodynamic and gravity-gradient torques cancel on average.

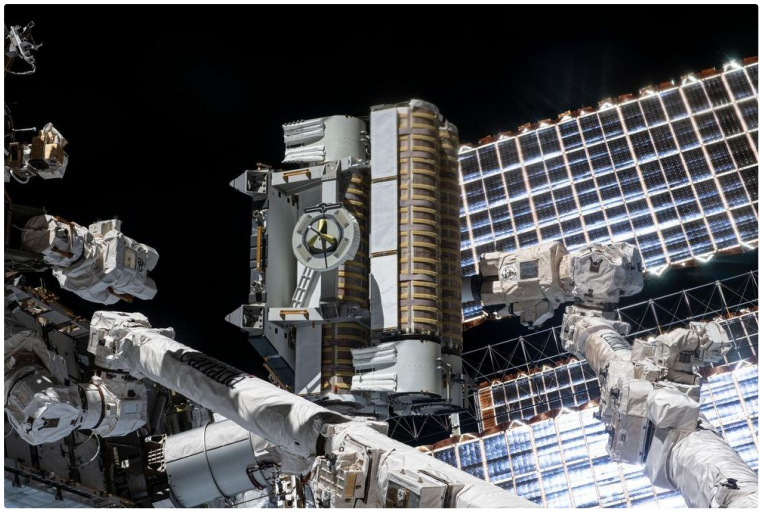


**FIGURE 32.6** Spacewalkers work with the first ISS Roll-Out Solar Array (iROSA) on the station's truss in June 2021, during the Expedition 65 spacewalks that installed it. Each iROSA unrolls like a carpet over an older, gold-coloured array wing. Image: NASA/JSC.

Drag lowers the station by roughly 1–5 km per month, depending on solar activity. It is periodically **reboosted** — raised by a few kilometres — by the engines of Zvezda and docked Progress freighters. Since November 2024 Cargo Dragon has also reboosted it, and one mission (CRS-33, 2025–26) carried a propulsion kit in its trunk. ESA's ATV and some Cygnus missions provided reboosts in the past. Thrusters also perform **debris-avoidance manoeuvres** when a tracked object's collision probability exceeds about 1 in 10 000: about 40 between 1999 and 2024. In July 2021, shortly after Nauka docked, its thrusters fired unexpectedly and turned the whole station about one and a half revolutions before controllers regained control.

### 32.2.5 Communications, Data and Robotics

Voice, telemetry and commands travel over S-band, and high-rate data and video over Ku-band. Both go through NASA's **Tracking and Data Relay Satellites (TDRS)** in geostationary orbit, which give almost continuous coverage (Chapter 7). The Russian segment uses ground stations and Luch relay satellites. More than 50 computers run about 1.5 million lines of flight software and monitor about 350 000 sensors.

**Canadarm2**, a 17.6 m arm with seven joints and a latching "hand" at each end, can move end over end between grapple fixtures and ride a mobile base along the truss. It assembled much of the station, captures visiting cargo vehicles and moves spacewalkers. Its two-armed attachment **Dextre** handles smaller replaceable units (Figure 32.7), and Kibo and Nauka carry their own arms. After a wrist joint failed in May 2026, spacewalkers replaced it with a spare on 30 June 2026.

**FIGURE 32.7** Canadarm2 with Dextre attached, extending from the Harmony module above the Sahara in August 2025. Image: NASA/JSC.

### 32.2.6 Crew, Research and Cost

The station normally hosts seven people — four launched on Crew Dragon and three on Soyuz, with seats swapped between the vehicles so that each segment always has trained crew. Nearly 3 000 investigations from more than 108 countries have used its racks, freezers, glove boxes, furnaces and external platforms, from protein crystal growth and combustion to Earth observation and the Alpha Magnetic Spectrometer. Total spending by all partners is often estimated at more than US$ 150 billion. NASA's Inspector General put NASA's own cost at about US$ 3 billion per year for operations and transport. Including research and related activities, it is about US$ 4.1 billion per year — about 16 % of NASA's budget.

**TABLE 32.2** International Space Station key figures (2026).

| Parameter | Value |
|---|---|
| Mass; dimensions | ≈ 420 t; 109 m × 73 m |
| Pressurised / habitable volume | 1 005 m³ / 388 m³ |
| Orbit | ≈ 415–420 km, 51.6°, 93 min |
| Crew | 7 (more during crew handovers) |
| Power | 8 wings + 6 iROSAs (≈ 215 kW in sunlight); 24 Li-ion batteries |
| Heat rejection | Two ammonia loops; 6 radiators, ≈ 70 kW |
| Attitude control | 4 CMGs (4 760 N·m·s each) + Russian thrusters |
| Communications | S- and Ku-band via TDRS; UHF for spacewalks and approaching vehicles |
| Robotics | Canadarm2 (17.6 m) with Dextre; Kibo and European arms |
| Docking ports | Up to 8 vehicles at once (Soyuz, Progress, Dragon, Cygnus, HTV-X) |

### 32.2.7 An Ageing Station

The oldest ISS modules have been in orbit since 1998. Since 2019 air has been leaking from the **PrK**, a small tunnel between Zvezda's main compartment and its aft docking port, through hairline cracks. In April 2024 the leak reached about 1.7 kg of air per day; NASA rated it its highest-level safety risk, and the hatch is normally kept closed. NASA and Roscosmos have not

agreed on the root cause. The tunnel was reported to be holding pressure in January 2026, but the leak returned in May. On 5 June 2026 five crew members briefly sheltered in their Crew Dragon while Roscosmos began, then paused, a structural repair. Press reports later said Roscosmos would stop pressurising the tunnel. Such ageing problems are a main reason for retiring the station around 2030.

## 32.3 Engineering a Station

### 32.3.1 Choosing the Orbit

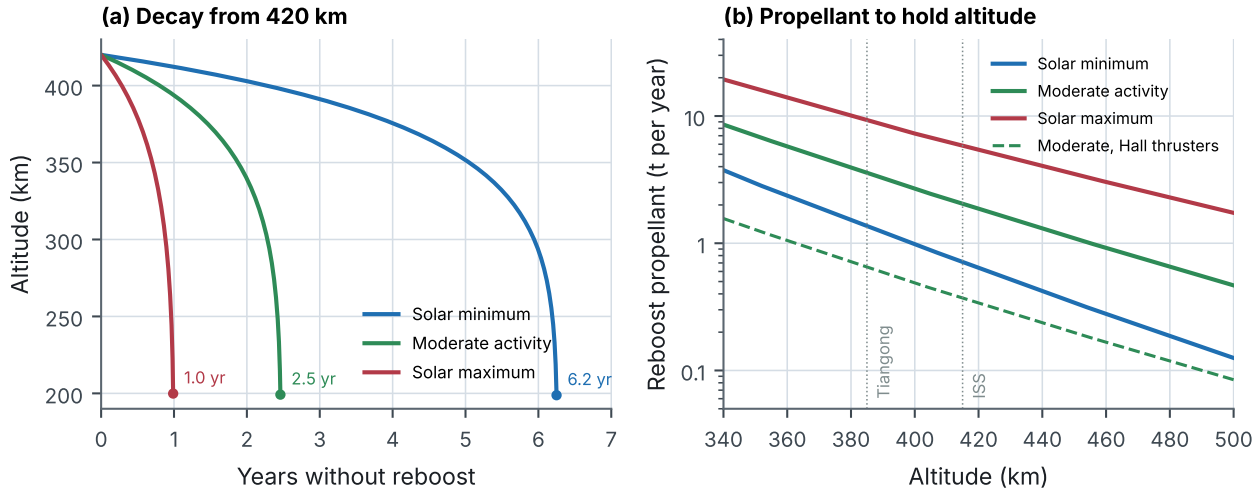
A station's **inclination** — the angle between its orbit and the equator — is usually set by its launch sites. A rocket launched due east reaches an inclination equal to the latitude of its launch site, and higher inclinations cost payload. The ISS flies at 51.6° because Russian vehicles from Baikonur (46° N) cannot launch due east without dropping stages on China; American vehicles from Florida (28.5° N) pay a payload penalty to reach it. Tiangong's 41.5° suits the Jiuquan launch site.

**Altitude** is a compromise. Lower is cheaper to reach — every kilometre of altitude costs payload for each crew and cargo flight — but the air is denser, so the station must spend more propellant fighting drag (Table 32.3). Higher reduces drag but increases radiation exposure, as orbits approach the inner Van Allen belt, and the density of orbital debris, and makes Earth-observation instruments less sharp. During the Shuttle era the ISS flew as low as 330–350 km so that the Shuttle could deliver the heaviest modules; since 2011 it has been kept near 400–420 km.

Drag decays the orbit as described in Chapter 2:

$$\frac{da}{dt} = -\frac{\rho \sqrt{\mu a}}{B}, \qquad B = \frac{m}{C_D A}$$

where $\rho$ is the air density, $a$ the orbit radius, $\mu$ Earth's gravitational parameter and $B$ the ballistic coefficient. Holding altitude requires a velocity increment each year equal to the drag deceleration integrated over time, $\Delta v = \int \rho v^2 / (2B)\, dt$. Because thermospheric density changes by a factor of about ten over the 11-year solar cycle, the reboost budget does too (Figure 32.8). Without reboosts, an ISS-class station released at 420 km would re-enter within about one year at solar maximum and about six years at solar minimum.




Model: da/dt = −ρ√(μa)/B with B = m/($C_D$A) = 150 kg/m² (450 t, $C_D$A ≈ 3 000 m²); MSIS-like density table of Chapter 2; chemical $I_{sp}$ = 290 s, Hall $I_{sp}$ = 1 600 s.

**FIGURE 32.8** (a) Altitude of an ISS-class station left without reboost, starting from 420 km. (b) Propellant needed each year to hold a constant altitude with chemical thrusters, and with Hall-effect electric thrusters at moderate solar activity.

**TABLE 32.3** Drag penalty of altitude for a 450 t station with B = 150 kg/m², computed with the model of Figure 32.8 (specific impulse 290 s).

| Altitude (km) | Decay, moderate activity (km/month) | Δv per year: min / moderate / max (m/s) | Propellant per year, moderate (t) | Life without reboost, moderate (years) |
|---|---|---|---|---|
| 350 | 6.5 | 19 / 45 / 106 | 7.0 | 0.6 |
| 380 | 3.7 | 10 / 25 / 65 | 3.9 | 1.1 |
| 400 | 2.5 | 6 / 17 / 46 | 2.7 | 1.7 |
| 420 | 1.7 | 4 / 12 / 34 | 1.9 | 2.5 |
| 450 | 1.0 | 2 / 7 / 22 | 1.1 | 4.3 |
| 500 | 0.4 | 1 / 3 / 11 | 0.5 | 11 |

### EXAMPLE: ANNUAL REBOOST FOR AN ISS-CLASS STATION

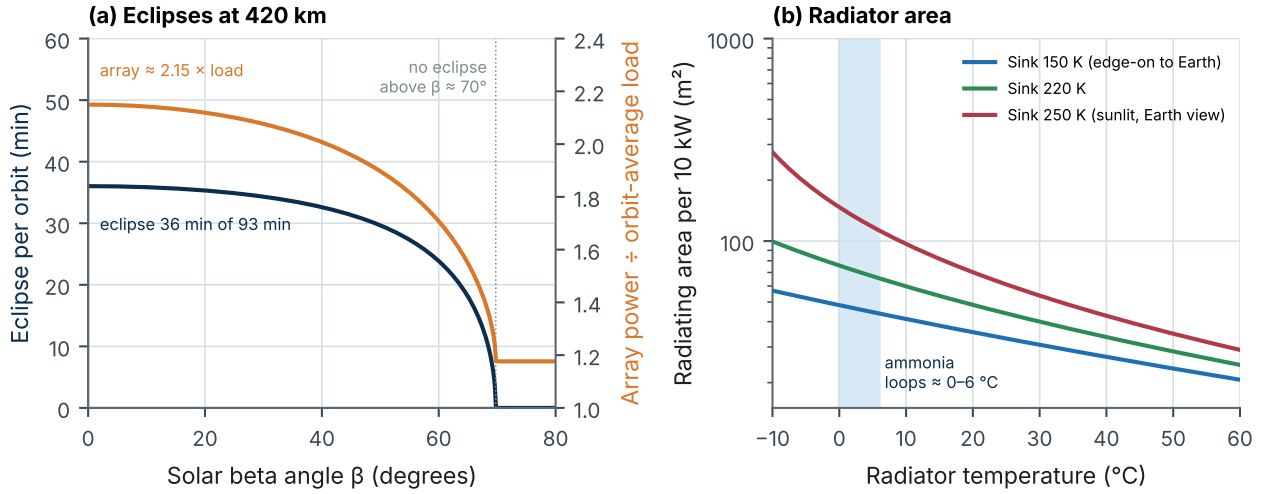
Take a station of 450 t, including docked vehicles, with an average drag area $C_DA \approx 3\,000$ m², so that $B = 150$ kg/m². It flies at 410 km in moderate solar activity ($\rho \approx 2.3 \times 10^{-12}$ kg/m³).

1. Orbital speed: $v = \sqrt{\mu/a} = \sqrt{3.986 \times 10^{14} / 6.788 \times 10^{6}} = 7\,663$ m/s.
2. Drag deceleration: $\rho v^2/(2B) = 2.3 \times 10^{-12} \times 7\,663^2 / 300 = 4.5 \times 10^{-7}$ m/s² — about 50 billionths of g.
3. Over a year ($3.16 \times 10^7$ s) this accumulates to **Δv ≈ 14 m/s**, and the orbit would sink by about 68 m per day (2.1 km per month) if not corrected.
4. With thrusters of $I_{sp} = 290$ s ($v_e = 2\,844$ m/s), the propellant is $m(1 - e^{-\Delta v/v_e}) = 450\,000 \times (1 - e^{-14.1/2\,844}) \approx$ **2.2 t per year**.
5. At solar maximum the density is nearly three times higher: Δv ≈ 40 m/s and **6.3 t per year**. Hall thrusters ($I_{sp} \approx 1\,600$ s) would need only about 0.4 t at moderate activity.

The reported ISS average of about 7 t per year is higher because it also covers attitude control and debris avoidance.

### 32.3.2 Power and Thermal Budgets

The angle between the orbit plane and the direction of the Sun is the **beta angle (β)**. It changes continuously as the orbit plane precesses and Earth moves round the Sun. For the ISS it swings between about −75° and +75° over roughly two months. At β = 0 the station spends 36 of every 93 minutes in shadow; above β ≈ 70° it is in permanent sunlight (Figure 32.9). The arrays must supply both the load and the battery charge for the next eclipse. Because storing energy in a battery and recovering it wastes about a third of it, the arrays must produce about 2.2 times the orbit-average load at low β. At high β, constant sunlight overheats some surfaces, so the attitude may be changed and visiting-vehicle traffic restricted.




(a) Path efficiencies 0.65 (via battery) and 0.85 (direct). (b) $q = \varepsilon \eta \sigma(T^4 - T_{sink}^4)$, ε = 0.85, fin efficiency η = 0.85; one radiating face.

**FIGURE 32.9** (a) Eclipse duration at 420 km and the ratio of array output to orbit-average load needed to recharge batteries, as functions of beta angle. (b) Radiator area needed to reject 10 kW as a function of radiator temperature for three effective sink temperatures.

The radiators must reject the whole electrical load plus the heat produced by the crew (about 100–150 W per person). Radiated flux follows the Stefan–Boltzmann law, $q = \varepsilon\sigma(T^4 - T_{sink}^4)$. So a radiator near 0 °C — cold enough to cool the cabin through heat exchangers — emits only about 200–250 W/m² even with a cold view of space. It emits much less when it sees the warm Earth or sunlight. That is why the ISS radiators are among its largest structures. Power, heat rejection and drag are coupled: larger arrays bring more heat to reject, larger radiators and more drag.

### 32.3.3 Protecting Against Micrometeoroids and Debris

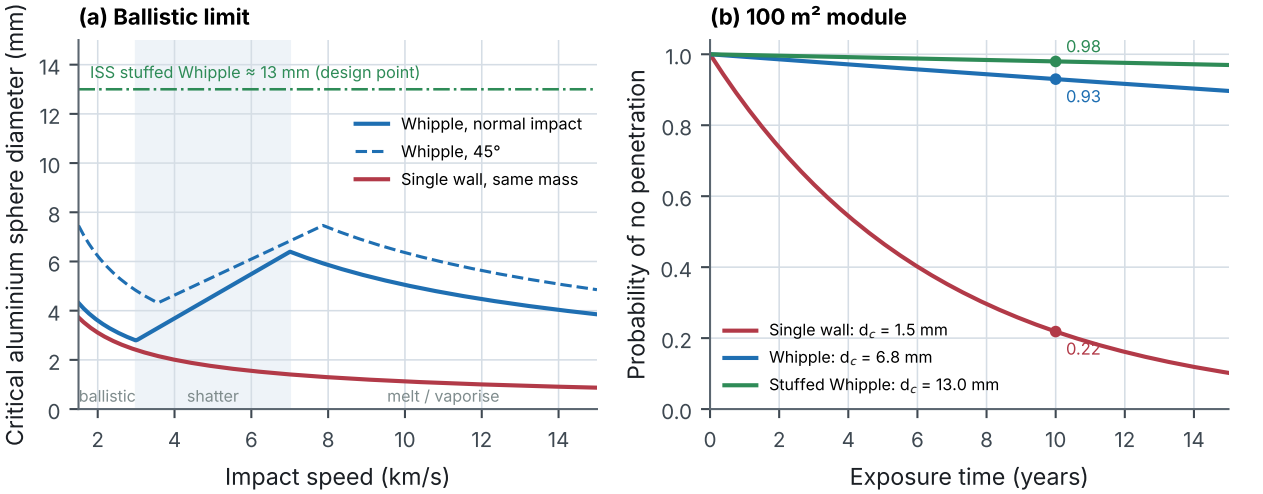
A station is a large target: the ISS presents more than 1 000 m² of pressurised-module surface to the particle environment described in Chapters 2 and 21. Its crew modules are protected by **Whipple shields**. In these, a thin aluminium bumper spaced in front of the pressure wall breaks up an incoming particle into a spreading cloud of fragments. Modules are also protected by **stuffed Whipple shields**, with layers of Nextel ceramic cloth and Kevlar between bumper and wall. The ISS uses more than 100 different shield configurations, the heaviest on forward-facing surfaces, which see the most debris impacts.

The performance of a shield is described by its **ballistic limit**: the diameter $d_c$ of the largest aluminium sphere it can stop, as a function of speed and impact angle (Figure 32.10). At low speeds (below about 3 km/s) a projectile stays intact and the shield behaves like a thick plate.

Between about 3 and 7 km/s the projectile shatters on the bumper, and the shield becomes *more* effective with speed. Above 7 km/s the fragments melt and vaporise, and $d_c$ falls slowly with speed. NASA's ballistic limit equation for the high-speed regime is

$$d_c = 3.918\, t_w^{2/3}\, \rho_p^{-1/3}\, \rho_b^{-1/9}\, (V\cos\theta)^{-2/3}\, S^{1/3} \left(\frac{\sigma}{70}\right)^{1/3}$$

in cm, with wall thickness $t_w$ and spacing $S$ in cm, projectile and bumper densities $\rho_p$, $\rho_b$ in g/cm$^3$, speed $V$ in km/s, impact angle $\theta$ and wall yield strength $\sigma$ in ksi. Doubling the spacing increases $d_c$ by 26 %; a single wall of the same mass stops particles four to five times smaller. The ISS stuffed Whipple shields are designed to stop particles of about 1.3 cm at typical impact conditions. Larger, catalogued objects are avoided by manoeuvring, but objects between about 1 and 10 cm can be neither shielded against nor tracked (Chapter 21).




Shield: 1.6 mm Al bumper, 10 cm gap, 3.2 mm Al 2219 wall (single wall 4.8 mm). Christiansen/Cour-Palais equations; flux from Chapter 2 model; (b) at 9 km/s, 45°.

**FIGURE 32.10** (a) Ballistic limit of a Whipple shield and of a single aluminium wall of equal mass, from NASA's empirical equations. (b) Probability that a 100 m² module suffers no penetration, for the three wall designs, using the illustrative flux model of Chapter 2.

### EXAMPLE: PENETRATION RISK FOR A MODULE OVER TEN YEARS

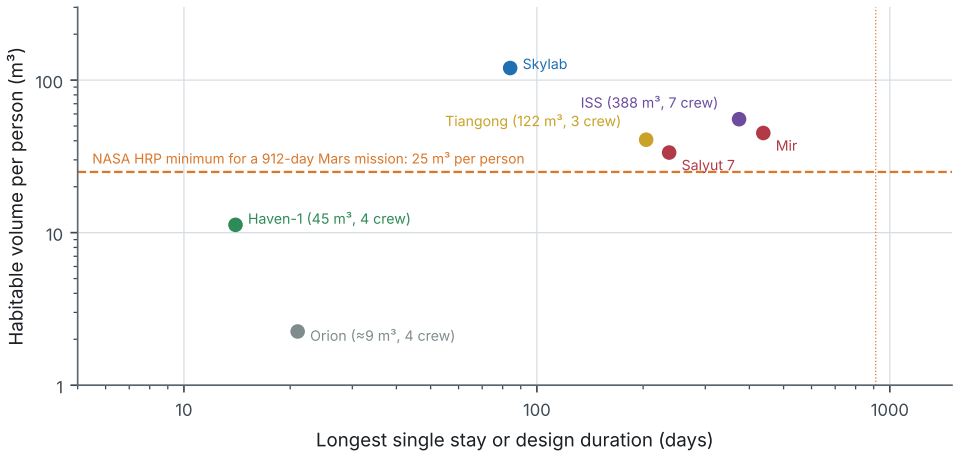
A laboratory module exposes 100 m² of effective area to impacts. Typical debris strikes at 9 km/s and 45°.

1. **Shield.** A Whipple shield with a 1.6 mm aluminium bumper, a 10 cm gap and a 3.2 mm aluminium-alloy wall has $d_c \approx 6.8$ mm at these conditions. A single 4.8 mm wall of the same mass has $d_c \approx 1.5$ mm. The ISS stuffed Whipple stops about 13 mm.
2. **Flux.** The cumulative flux of particles larger than $d_c$ is read from the illustrative ISS-altitude curve of Chapter 2, which follows the shape of NASA's ORDEM debris model. It is about $1.5 \times 10^{-3}$ per m² per year above 1.5 mm, $7 \times 10^{-5}$ above 6.8 mm and $2 \times 10^{-5}$ above 13 mm.
3. **Expected penetrations** in ten years: $N = FAT$ = 1.5, 0.073 and 0.021 respectively.
4. Impacts are random, so the **probability of no penetration** is $P = e^{-N}$: **0.22** for the single wall, **0.93** for the Whipple shield and **0.98** for the stuffed Whipple.
5. Ten such modules multiply $N$ by ten: PNP falls to about 0.81 even with the best shields. Not every penetration is catastrophic: a small hole leaks slowly enough to be found and sealed. NASA therefore also estimates the risk of evacuation or crew loss with tools such as BUMPER and MSCSurv.

The ISS has suffered damage. Impacts chip its windows, which is why the Cupola has external shutters. A debris strike punched a 5 mm hole in Canadarm2's thermal blanket in 2021. In December 2022 the coolant radiator of a docked Soyuz, MS-22, was punctured, probably by a micrometeoroid. The capsule had to return uncrewed, and a replacement Soyuz was launched. After Russia destroyed its Kosmos 1408 satellite in an anti-satellite test on 15 November 2021, the crew sheltered in their docked spacecraft while the station passed through the debris cloud.

### 32.3.4 Habitable Volume and Layout

Volume is measured in three ways. **Pressurised volume** is everything inside the hull. The **habitable volume** excludes the space taken by racks and fixed equipment. The **net habitable volume** is what remains free for people to move in once stowage and loose equipment are counted. In 2014 a NASA Human Research Program panel of experts recommended a minimum net habitable volume of 25 $m^3$ per person for a Mars mission lasting about 912 days. Its comparison listed about 120 $m^3$ per person for Skylab, 45 for Mir and 34 for Salyut. By comparison, today's ISS offers about 55 $m^3$ (388 $m^3$ for seven people) (Figure 32.11). The ISS crew quarters — private sleeping cabins — are about 2.1 $m^3$ each, the size of a telephone box.




Skylab, Salyut, Mir per-person values from the NASA HRP consensus report (2014); others computed from habitable volume ÷ nominal crew. Durations: record stays or design values.

**FIGURE 32.11** Habitable volume per person against the longest stay on several stations and spacecraft, with the minimum recommended by NASA's Human Research Program for a long Mars mission.

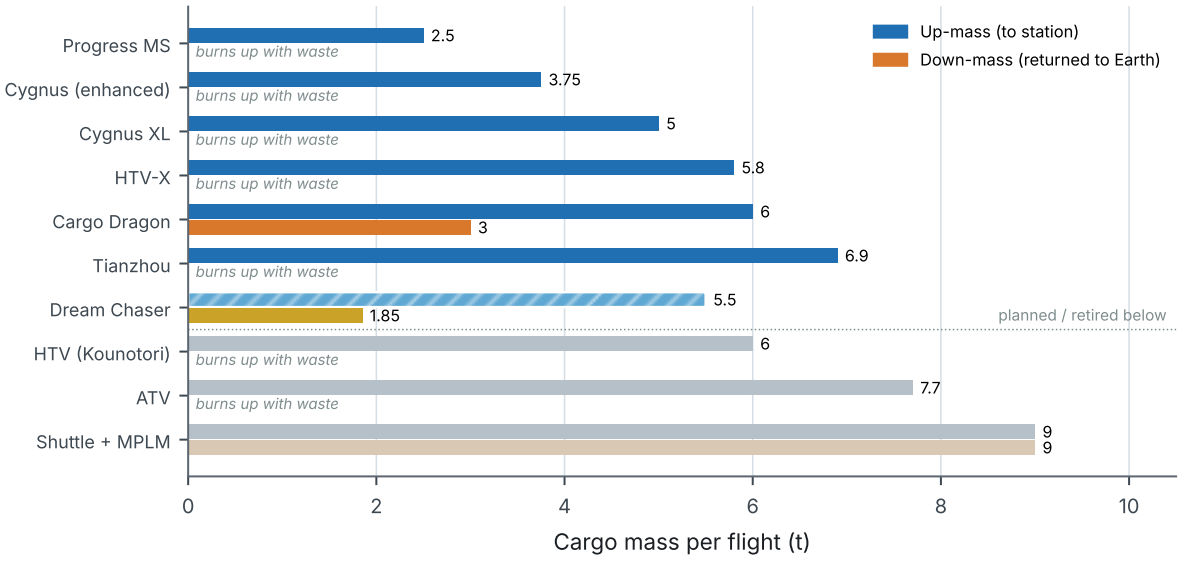
In weightlessness there is no natural up or down. Even so, designers give every module a consistent "floor" and "ceiling", with lighting overhead and labels the right way up. This helps people orient faster and make fewer mistakes. Handrails and foot restraints replace chairs, and fan and pump noise is a persistent habitability problem. Windows matter for morale as much as for operations: the Cupola, installed in 2010, gives a panoramic view used for robotics and photography (Figure 32.12).

**FIGURE 32.12** ESA astronaut Sophie Adenot and NASA astronaut Jack Hathaway in the Cupola in May 2026, monitoring the automated approach of a Cargo Dragon. Image: NASA/JSC.

## 32.4 Supplying a Station

### 32.4.1 Cargo Vehicles

Even recycling about 98 % of its water (Chapter 30), a crew of seven consumes several tonnes of food, gases, clothing and hygiene items a year, and the station also needs propellant, spares and experiments. Cargo vehicles differ in how much they carry up (**up-mass**) and whether they can carry unpressurised cargo such as pumps and batteries outside. They also differ in whether they can refuel and reboost the station, and whether anything comes back (**down-mass**) (Figure 32.13, Table 32.4). Most freighters are loaded with rubbish before they leave and burn up on re-entry. Apart from small items carried in crew capsules, only Dragon returns experiments and failed hardware to Earth. Dream Chaser could do so if NASA later orders ISS flights.




Maximum or typical capacities from operator data; Progress includes propellant and water; ATV up to 7.7 t including refuelling propellant. Dream Chaser not yet flown.

**FIGURE 32.13** Cargo capacity per flight of station supply vehicles. Most freighters burn up with the station's waste; Cargo Dragon is currently the only vehicle that returns cargo.

JAXA's **HTV-X**, successor to the HTV, made its first flight on an H3 rocket on 26 October 2025; it was captured by Canadarm2 on 29 October (Figure 32.14). Northrop Grumman's larger **Cygnus XL**, about a third roomier than its predecessor, first flew in September 2025 (Figure 32.15). Sierra

Space's winged **Dream Chaser** has repeatedly slipped. In September 2025 NASA removed its guaranteed ISS flights. Its first mission is a free-flying demonstration on a Vulcan rocket, planned for late 2026 at the earliest.

**TABLE 32.4** Station cargo vehicles (October 2026).

| Vehicle | Operator | Since | Arrival | Up / down (t) | Notes |
|---|---|---|---|---|---|
| Progress MS | Roscosmos | 1978 (MS: 2015) | Automatic docking | ≈ 2.5 / 0 | Refuels and reboosts Russian segment |
| Cygnus / Cygnus XL | Northrop Grumman | 2013 / 2025 | Berthed by Canadarm2 | 3.75–5 / 0 | XL first flew Sept 2025 |
| Cargo Dragon | SpaceX | 2012 (Dragon 2: 2020) | Automatic docking | ≈ 6 / 3 | Ocean splashdown; reboost kit |
| HTV-X | JAXA | Oct 2025 | Berthed by Canadarm2 | ≈ 5.8 / 0 | Can fly up to 1.5 years after leaving |
| Tianzhou | CMSA | 2017 | Automatic docking | ≈ 6–7 / 0 | Supplies Tiangong; refuels it |
| Dream Chaser | Sierra Space | Not yet flown | (Berthing planned) | ≈ 5.5 / 1.85 | Winged; runway landing; free-flyer demo first |
| ATV; HTV | ESA; JAXA | 2008–14; 2009–20 | Docking; berthing | 7.7; 6 / 0 | Retired |

**FIGURE 32.14** JAXA's first HTV-X in the grip of Canadarm2 on 29 October 2025, carrying about 5.8 t of supplies. Image: NASA/JSC.

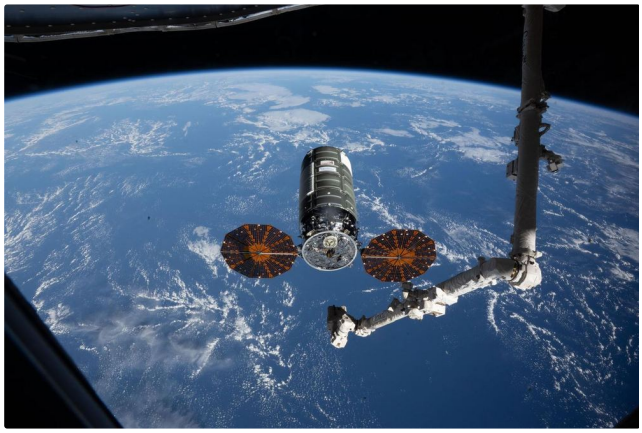


**FIGURE 32.15** A Cygnus XL freighter waiting at its capture point, 12 m from the station, as Canadarm2 prepares to grapple it in April 2026. Image: NASA/JSC.

## 32.4.2 Docking and Berthing

Vehicles join a station in two ways. In **docking**, the visiting vehicle flies itself into contact. Soyuz and Progress use the Russian Kurs radar system and probe-and-drogue ports, while Dragon uses its own sensors and the androgynous International Docking System Standard (Chapter 31). In **berthing**, the vehicle flies to a capture point 10–12 m below the station and holds position with its thrusters disabled. While it holds, an astronaut grapples it with Canadarm2 and installs it on a CBM port, where powered bolts pull the two rings together. Berthing needs no precision docking system on the visitor and gives a wide hatch for racks, but requires the arm and a trained crew.

Approaches follow strict rules. There is an **approach ellipsoid** of 4 × 2 × 2 km around the station, and a **keep-out sphere** of 200 m radius. A vehicle may enter the sphere only after a "go" from mission control, along a defined corridor and with an automatic abort capability. Solar arrays are usually turned edge-on to protect them from thruster plumes.

## 32.5 Expandable Habitats and Artificial Gravity

### 32.5.1 Expandable Modules

A rigid module can be no wider than the rocket fairing that launches it — about 4.5 m for most modules so far. An **expandable** (or inflatable) module is launched folded and then pressurised in orbit, giving several times more volume per launch. Its wall is built from layers of fabric. Inner gas bladders (usually redundant) hold the air, and a **restraint layer** of woven Kevlar or Vectran straps carries the pressure load. Many spaced layers of ceramic cloth and foam act as a multi-shock micrometeoroid shield. Then come multilayer thermal insulation and an outer cover against atomic oxygen.

The restraint layer must resist a hoop load per unit length of $N = p r$. For a module 8.2 m across at one atmosphere (101 kPa), that is about 415 kN per metre of wall — the weight of 42 tonnes hanging from each metre. The fabric must withstand about four times that. NASA's TransHab technology of the 1990s passed to Bigelow Aerospace, which flew the uncrewed Genesis I and II prototypes in 2006–07. The **Bigelow Expandable Activity Module (BEAM)**, about 1.4 t, was berthed to the ISS in April 2016 and expanded on 28 May 2016 to about 16 m³ (Figure 32.16). Planned as a two-year test, it is still attached a decade later and used for storage; its ownership passed to NASA in 2021 after Bigelow ceased operations. Sierra Space's **LIFE** habitat is about 8 m in diameter, with about 285 m³ inside. In full-scale tests in December 2023 and June 2024 it burst at 77 psi and 74 psi (530 and 510 kPa). These pressures are 22–27 % above the 60.8 psi (four times operating pressure) that NASA recommends.

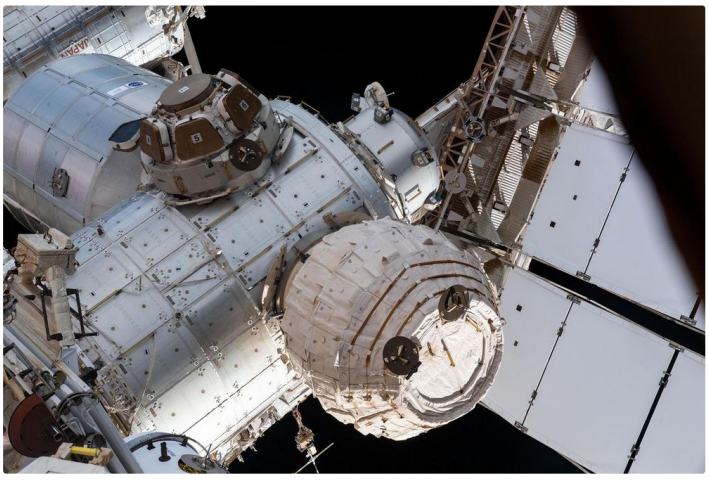


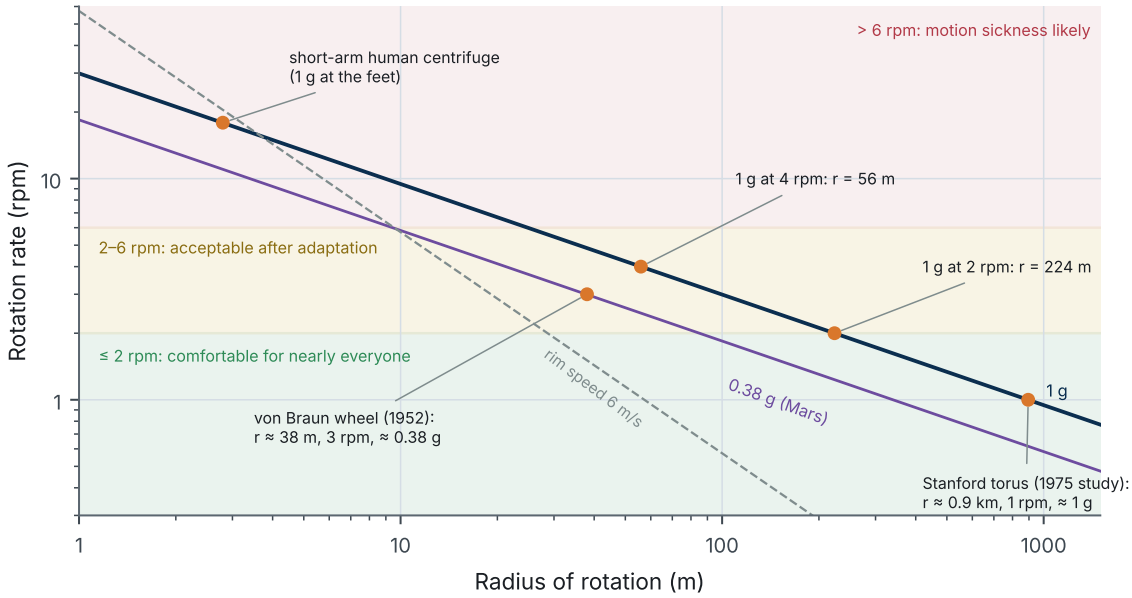
**FIGURE 32.16** BEAM (the fabric-covered module, lower centre), attached to the Tranquility node next to the shuttered Cupola and the Bishop airlock, photographed from the Russian segment in August 2022. Image: NASA/JSC.

## 32.5.2 Artificial Gravity

Weightlessness weakens bones, muscles and the heart and affects the eyes (Chapter 30). Exercise reduces but does not prevent these effects, and a rotating habitat could replace weight with centripetal acceleration:

$$a = \omega^2 r, \qquad a_{Cor} = 2\,\omega\, v_{rel}$$

where $\omega$ is the rotation rate, $r$ the radius and $v_{rel}$ the speed of a person moving inside the habitat. The **Coriolis acceleration** $a_{Cor}$ makes moving objects veer sideways, and turning the head in a rotating room confuses the inner ear, causing motion sickness. Experiments in rotating rooms suggest that almost everyone is comfortable at up to about 2 rpm, and that people can adapt to about 4–6 rpm (Figure 32.17). A low rotation rate requires a large radius: for 1 g at 2 rpm, the radius is 224 m. A **gravity gradient** between head and feet is a further issue for small radii.




ω = √(a/r). Comfort bands follow limits commonly quoted in the artificial-gravity literature (compiled by T. W. Hall); individual tolerance varies.

**FIGURE 32.17** Rotation rate against radius for 1 g and Mars-level artificial gravity, with commonly quoted comfort limits and some historical concepts.

### EXAMPLE: A ROTATING HABITAT FOR 1 G AT 2 RPM

1. Rotation rate: ω = 2 × 2π/60 = 0.209 rad/s.
2. Radius for 1 g: $r = g/\omega^2$ = 9.81/0.209² = **224 m** — a structure almost half a kilometre across. Rim speed: $\omega r$ = 47 m/s.
3. Coriolis acceleration on a crew member walking at 1 m/s: 2 × 0.209 × 1 = 0.42 m/s² — about 4 % of their weight, pushing sideways.
4. Walking 1 m/s in the direction of rotation raises apparent weight by 4.3 %; walking the other way lowers it by 4.2 %.
5. Gravity gradient over a 1.8 m person: 1.8/224 = **0.8 %** — negligible. At 4 rpm the radius shrinks to 56 m, but the Coriolis acceleration doubles and the gradient reaches 3 %.

Mars gravity (0.38 g) at 2 rpm needs 85 m. A cheaper option is to spin two masses — a habitat and a spent stage or counterweight — connected by a tether.

No crewed station has yet used artificial gravity. In 1966 Gemini 11 spun itself and an Agena stage on a 30 m tether, producing a tiny artificial gravity. On the ISS, JAXA's Multiple Artificial-gravity Research System centrifuge has kept mice at controlled artificial-gravity levels for comparison with weightless animals, and short-arm human centrifuges have been tested on the ground in bed-rest studies.

## 32.6 After the ISS

### 32.6.1 Retiring and Deorbiting the ISS

The partners plan to operate the ISS until 2030; Roscosmos said in April 2026 that it would begin rounding off its ISS work in 2028 and expected the deorbit to be completed in 2030. A 420-tonne station cannot be left to re-enter uncontrolled: tens of tonnes of dense parts — truss segments, docking rings, tanks — would survive and fall anywhere under its 51.6° ground track. It must instead be steered into a remote area of the South Pacific.

In June 2024 NASA awarded SpaceX a contract worth up to US$ 843 million for the **US Deorbit Vehicle (USDV)**, a heavily modified Dragon with 46 Draco thrusters and about 16 t of propellant; with launch, NASA puts the total at about US$ 1.5 billion. Launched about 18 months before the end, the USDV will dock at the forward port while the station sinks. The last crew will leave at about 330 km, some six months before re-entry. At about 225 km the USDV will begin a series of burns. The last burn, about four days later, aims the debris at the South Pacific — in early 2031 on current plans. Lowering the perigee from a 225 km circular orbit to about 50 km requires about 52 m/s. For 450 t, this is about 8 t of propellant at $I_{sp}$ = 300 s. Much of the remaining propellant is needed to control the station's attitude as aerodynamic torques grow during the final orbits.

### 32.6.2 Commercial Stations

NASA wants to buy services on commercial stations rather than own one. Its **Commercial LEO Destinations (CLD)** programme gave Space Act Agreements worth US$ 415.6 million in December 2021. They went to Blue Origin, Nanoracks (now part of Voyager, for Starlab) and Northrop Grumman (which later joined Starlab). Axiom Space separately holds a contract to attach modules to the ISS. The programme was restructured repeatedly in 2025–26. An August 2025 directive lowered the minimum capability NASA would require. A proposal in March 2026 to attach a government-owned core module to the ISS was dropped in June. On 9 October 2026 NASA issued its final request for proposals for firm-fixed-price contracts covering development, certification and services. Proposals are due on 8 December 2026, and two or more awards are expected in spring 2027. Because a commercial station must be certified and crewed before the ISS is deorbited, and schedules keep slipping, a gap in continuous US presence in orbit is possible (Table 32.5).

**Vast**'s **Haven-1** would host four-person crews arriving by Crew Dragon for about two weeks at a time; Vast proposes a larger multi-module **Haven-2** for the NASA competition. **Axiom Space** has flown private crews to the ISS since 2022. In December 2024 it changed its plan so that its first module, a power and thermal module, will attach to the ISS. This module will later depart to join a habitat module as a free-flying station. **Starlab** is a single 8 m-diameter station to be launched fully outfitted on one Starship.

### 32.6.3 National Stations: China, Russia and India

If the ISS ends on schedule, China's Tiangong will be the only government station in low Earth orbit, and its planned six-module expansion would double it. Russia's **Russian Orbital Station (ROS)** was redesigned in December 2025 to use the ISS's 51.6° orbit rather than a near-polar one, with its first modules attached to the ISS from about 2028. India's **Bharatiya Antariksh Station (BAS)** builds on the Gaganyaan crew vehicle and the SpaDeX docking experiment; its first module, of about 10 t, is to be launched on an LVM-3.

**TABLE 32.5** Successors and new stations (status as of October 2026).

| Station | Developer | First launch (planned) | Configuration | Status, October 2026 |
|---|---|---|---|---|
| Haven-1 | Vast | NET Q1 2027 | 14 t, 45 m³; 4 crew for ≈ 2 weeks | Integration and tests; Haven Demo flew Nov 2025 |
| Axiom Station | Axiom Space | ≈ 2028 | Power module first on ISS, then joins Hab-1 to fly free | Modules in production |
| Starlab | Voyager, Airbus and partners | NET 2029, one Starship launch | 8 m diameter, ≈ 450 m³, 4 crew | Critical design review Feb 2026 |
| Orbital Reef | Blue Origin, Sierra Space | Not announced | Multi-module, LIFE habitats | Few public milestones |
| ROS | Roscosmos | 2028 | First modules attached to ISS, 51.6° | Redesigned Dec 2025 |
| Bharatiya Antariksh Station | ISRO | BAS-01 by 2028 | Five modules by 2035 | First module approved 2024 |
| Tiangong expansion | CMSA | Not dated | Six modules, ≈ 180 t | Plan shown 2023, confirmed 2026 |

### 32.6.4 Gateway and Surface Habitats

The **Lunar Gateway**, a small station planned for a near-rectilinear halo orbit around the Moon, was to begin with a Power and Propulsion Element (PPE) and a Habitation and Logistics Outpost (HALO). The FY2026 budget request proposed cancelling it, and on 24 March 2026 NASA paused Gateway "in its current form" in favour of a lunar surface base. By August 2026 the HALO hardware was being repurposed for lunar-surface demonstration missions. The PPE was reported to be under study for other uses. Its future as a station is uncertain.

On the surface, NASA's Artemis Base Camp concept foresaw a pressurised rover and a fixed habitat near the lunar south pole; its Moon Base programme (March 2026) starts with demonstrations of surviving the lunar night. Surface habitats gain gravity, local materials and a planet that blocks half the sky, but must cope with abrasive dust, temperature swings of almost 300 °C on the Moon and no magnetic field. A 1 m layer of regolith (density about 1.5 g/cm³) provides about 150 g/cm² of shielding — enough against solar particle events, though several metres are needed to remove most of the cosmic-ray dose. Proposals include covering inflatable habitats with bagged regolith, sintering or 3D-printing regolith into walls, and using lava tubes (Chapters 30 and 41).

### AI IN STATION OPERATIONS

Station operations are moving towards autonomy. The ISS already uses automated fault detection and procedure execution. NASA's **Astrobee** — three cube-shaped free-flying robots that navigate Kibo with cameras and fans (Figure 32.18) — has been used since 2019 to test autonomous inventory, inspection and mapping. DLR's **CIMON**, an AI-powered floating assistant with voice interaction, flew from 2018, and JAXA's Int-Ball camera drone films crew activities. Machine-learning models are being trained on years of station telemetry to predict failures of pumps, fans and CMGs before they happen. Commercial stations plan to operate with very small ground teams and no permanent crew. Chapter 51 discusses AI for human spaceflight in detail.

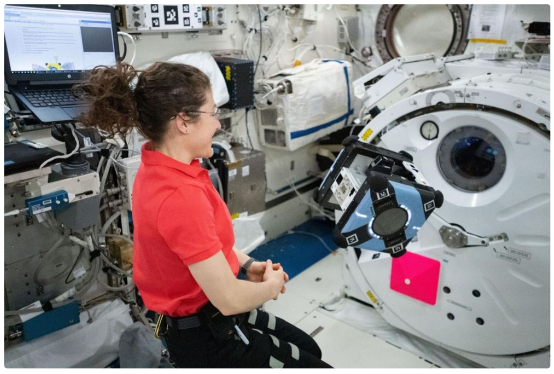


**FIGURE 32.18** NASA astronaut Christina Koch testing the mobility of an Astrobee free-flying robot in the Kibo laboratory in July 2019. Image: NASA/JSC (Christina Koch).

### KEY TAKEAWAYS

- Stations evolved from single modules (Salyut 1, Skylab) through resupply by freighters (Salyut 6) and modular assembly (Mir). This led to the 420 t ISS, crewed continuously since November 2000, and China's Tiangong, crewed continuously since 2022.
- A station is a coupled system: solar arrays sized for eclipses, radiators sized for the full load, CMGs for attitude, and propellant for reboost. Reboost needs about 4–40 m/s per year at ISS altitude, depending on the solar cycle.
- Orbit selection trades launch cost against drag: lowering the orbit from 420 to 350 km almost quadruples the reboost propellant, and solar maximum roughly triples it compared with average conditions.
- Whipple and stuffed Whipple shields stop particles up to about 1.3 cm; combining ballistic-limit equations with a flux model gives the probability of no penetration, which falls with area and time.
- Logistics depend on several freighters, most of them destroyed on re-entry; Cargo Dragon is the only vehicle returning cargo, and HTV-X and Cygnus XL entered service in 2025.
- Expandable habitats offer several times more volume per launch; artificial gravity at comfortable rotation rates needs radii of hundreds of metres.
- The ISS is to be deorbited by a SpaceX-built vehicle around 2030–31; commercial stations (Haven-1, Axiom, Starlab), Russia's ROS, India's BAS and an expanded Tiangong are planned, while Gateway has been paused.

### REVIEW QUESTIONS

1. What design feature introduced on Salyut 6 made long-duration stations possible, and why?
2. Why did Skylab re-enter earlier than NASA had planned? Link your answer to the solar cycle.

3. Explain how a control moment gyroscope produces torque without propellant, and why the ISS still needs thrusters for attitude control.
4. Using Table 32.3, estimate how much more propellant per year a station at 380 km needs than one at 420 km, at moderate solar activity. What are the advantages of the lower altitude?
5. At $\beta = 0$, an ISS-like station is in eclipse for 36 of 93 minutes. With path efficiencies of 0.65 through the battery and 0.85 directly, what array output is needed for a 100 kW orbit-average load?
6. Why is a Whipple shield more effective at 6 km/s than at 3 km/s? Sketch its ballistic-limit curve.
7. A module of 60 $m^2$ has a shield that stops particles up to 8 mm, and the flux above 8 mm is $5 \times 10^{-5}$ per $m^2$ per year. What is the probability of no penetration over 15 years?
8. Compare docking and berthing. Why do Cygnus and HTV-X berth rather than dock?
9. What radius is needed for 0.38 g at 3 rpm? What Coriolis acceleration acts on a crew member walking at 1.2 m/s?
10. List the main candidates to succeed the ISS and the status of each as of October 2026. What is the risk of a “gap” in continuous human presence in orbit?

# 33 CHAPTER Extravehicular Activity Systems

Image: NASA/JSC

*A spacesuit is a one-person spacecraft. It must hold a breathable atmosphere against vacuum, remove carbon dioxide and up to several hundred watts of body heat, shield its wearer from sunlight, cold and micrometeoroids, and still let a gloved hand turn a bolt for seven hours. This chapter follows sixty years of spacewalks — from Leonov's ballooning suit in 1965 through Apollo, the Space Shuttle, Hubble and the 284 spacewalks at the International Space Station to China's record nine-hour EVA and the first commercial spacewalk in 2024. It explains the engineering of pressure garments, joints, gloves and portable life support systems, the physiology of decompression sickness and prebreathe, how spacewalks are trained and run, and what suits for the Moon and Mars must do differently.*

IN THIS CHAPTER

33.1 Sixty Years of Spacewalks
33.2 Anatomy of a Spacesuit
33.3 The Portable Life Support System
33.4 Decompression Sickness and Prebreathe
33.5 Working Outside
33.6 Spacewalking on Other Worlds

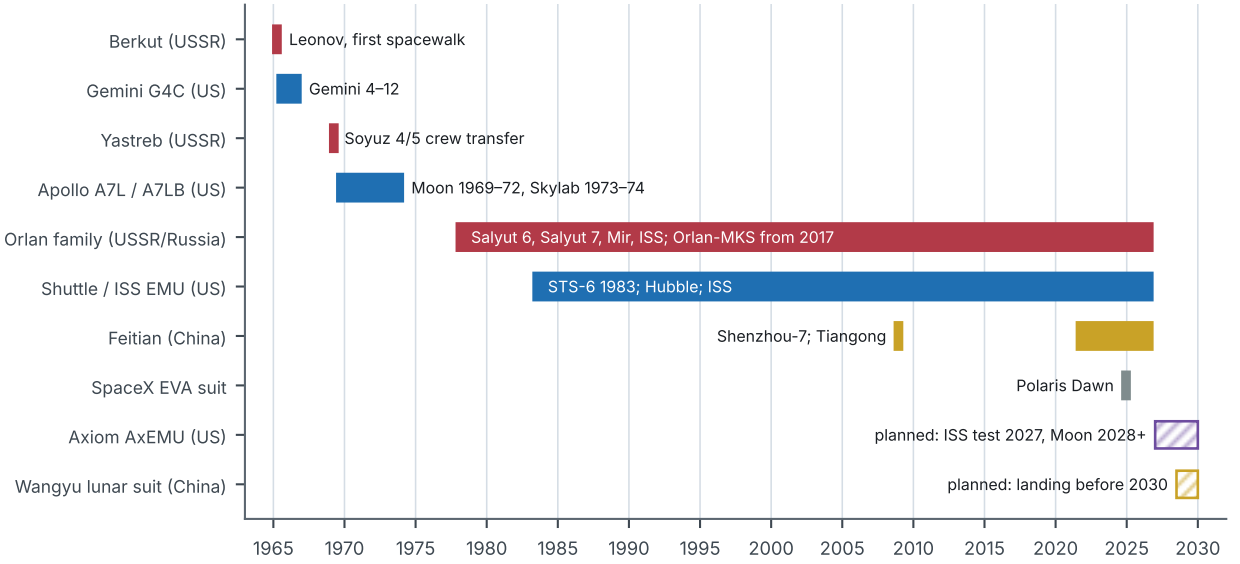
**Extravehicular activity (EVA)** — a spacewalk — is any time a crew member leaves the protection of a pressurised vehicle and works in a spacesuit in vacuum or on another world's surface. A spacesuit used this way is the smallest crewed spacecraft there is. Wrapped closely around one person, it must provide pressure, oxygen, carbon dioxide removal, cooling, power and communications. It must protect against radiation, temperature extremes and micrometeoroids, and give enough mobility for hands to do precise work for a full working day.

People go outside because some jobs still need hands and eyes on the spot. Spacewalkers built the International Space Station (ISS), upgraded the Hubble Space Telescope five times, saved Skylab and collected 382 kg of rock on the Moon. By October 2026 there had been 284 spacewalks

at the ISS alone, and the longest single EVA had reached 9 h 06 min. This chapter covers the history, the suit and its backpack, decompression, operations and the suits being built for the Moon and Mars. Life support in general is covered in Chapter 30, crewed vehicles in Chapter 31, stations and airlocks in Chapter 32, and robotic servicing in Chapter 40.

## 33.1 Sixty Years of Spacewalks

The history of EVA is a history of learning how hard it is. Figure 33.1 shows the suits that have been used outside and their years of service, and Table 33.1 compares the main ones.




Bars: years each suit type was used for spacewalks (hatched: planned, status October 2026). Red: Soviet/Russian; blue: NASA; gold: China.

**FIGURE 33.1** Spacesuits used for spacewalks since 1965, with the years in which each type was used outside. The Orlan and the Shuttle/ISS EMU have each served for more than four decades; the Axiom lunar suit and China's Wangyu are in development.

### 33.1.1 First Steps Outside: Voskhod and Gemini

On 18 March 1965 Alexei Leonov left Voskhod 2 through an inflatable airlock and floated for about 12 minutes at the end of a tether. His Berkut suit, pressurised to 0.4 atmospheres, ballooned so much in vacuum that his feet and hands pulled away from the boots and gloves, and he could not fit back into the airlock. He switched the suit to its reserve setting of about 0.27 atmospheres, accepting a risk of decompression sickness, and struggled back inside. His memoir's account of going in head first is more dramatic than his post-flight report. The first lesson of EVA: a pressurised suit is a balloon whose shape must be controlled.

Ed White followed on Gemini 4 on 3 June 1965 (Figure 33.2) and spent about 20 minutes outside; his hand-held gas gun ran dry after about three minutes, leaving him to pull on his tether. Most of the Gemini spacewalks that followed proved far harder than expected. On Gemini 9A in June 1966 Eugene Cernan spent most of a two-hour spacewalk struggling to reach and put on a manoeuvring backpack stowed on the spacecraft's adapter. With nothing to brace against, every push moved his body rather than the task; his heart rate reportedly reached about 180 beats per minute and his visor fogged until he could barely see. Richard Gordon on Gemini 11 was similarly exhausted.

**FIGURE 33.2** Ed White during the first American spacewalk, Gemini 4, 3 June 1965. The gold-coated visor protects his eyes from sunlight; the umbilical supplies oxygen and a tether; he holds the hand-held manoeuvring unit. Image: NASA/JSC.

The solution came on Gemini 12 in November 1966. Buzz Aldrin trained underwater beforehand, and his spacecraft carried handrails, foot restraints ("golden slippers"), waist tethers and rest periods written into the plan. He worked for about 5.5 hours over three EVAs without overexertion. Restraints, neutral-buoyancy training and paced timelines are still the foundation of EVA practice.

### 33.1.2 Apollo and Skylab

The Apollo **A7L** suit, built by ILC Dover, worked at 25.8 kPa (3.75 psi) of pure oxygen. For the Moon it was combined with a **portable life support system (PLSS)** backpack. This supplied oxygen, scrubbed carbon dioxide with lithium hydroxide and cooled the astronaut by sublimating water to vacuum — the architecture of every backpack since. On the Moon, twelve astronauts spent about 80 hours outside in 14 EVAs between July 1969 and December 1972 (see the section on other worlds below).

In May 1973 the Skylab space station reached orbit with its micrometeoroid shield torn away and one solar wing missing and the other jammed. On 7 June Pete Conrad and Joe Kerwin, in Apollo-type suits fed by umbilicals, cut the restraining strap and freed the remaining wing in a 3 h 25 min EVA. It was the first demonstration that spacewalkers could rescue a crippled spacecraft.

### 33.1.3 The Shuttle, the MMU and Hubble

The Space Shuttle brought the **Extravehicular Mobility Unit (EMU)**, built by Hamilton Standard and ILC Dover, first used by Story Musgrave and Don Peterson on STS-6 in April 1983. Unlike the custom-tailored Apollo suits, the EMU is assembled from standard-sized parts, so a fleet of parts can fit many astronauts. Its PLSS is mounted on the hard upper torso, and it works at 29.6 kPa (4.3 psi).

On 7 February 1984 Bruce McCandless flew the **Manned Manoeuvring Unit (MMU)**, a nitrogen-gas jet backpack with 24 thrusters, up to about 98 m from Challenger with no tether at all (Figure 33.3). MMUs were used on two more flights in 1984, to help repair the Solar Maximum Mission satellite and to capture two stranded communications satellites. They were then retired, because the Shuttle's arm and tethered crews could do the same work with less risk.

The **Hubble Space Telescope** turned EVA into a precision trade. Hubble was designed to be serviced: its instruments slide out on rails, its electronics boxes have captive bolts, and handrails and foot-restraint sockets cover its exterior. Five servicing missions between 1993 and 2009 made 23 spacewalks totalling 166 hours, replacing instruments, gyroscopes, batteries and solar arrays and fixing the telescope's flawed optics (Figure 33.4). On the final mission astronauts repaired two instruments never designed to be opened, removing more than a hundred tiny screws.

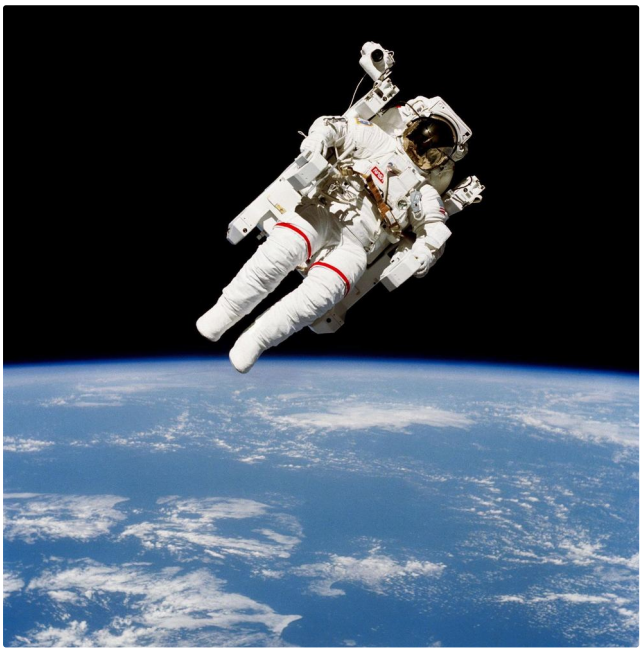


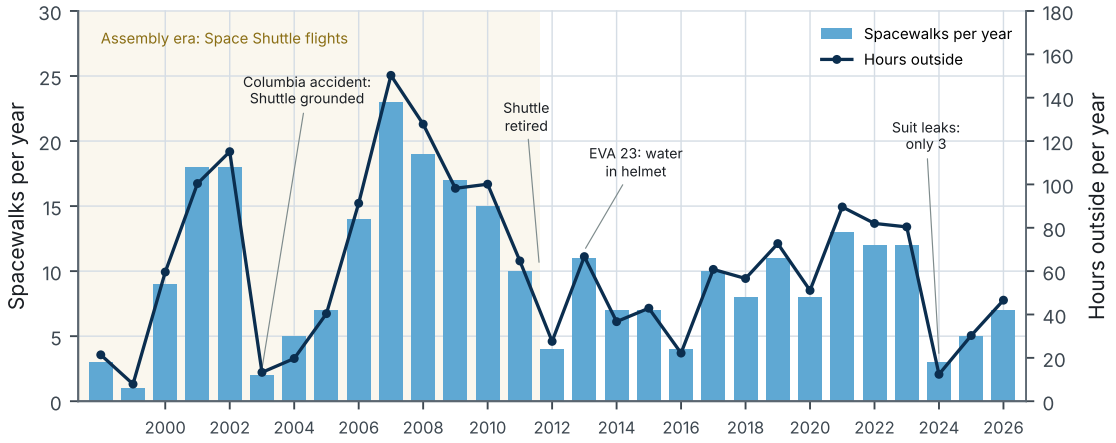
**FIGURE 33.3** Bruce McCandless flying the Manned Manoeuvring Unit a few metres from the Space Shuttle Challenger on 7 February 1984, the first untethered spacewalk. Image: NASA/JSC.

**FIGURE 33.4** John Grunsfeld and Andrew Feustel at work on Hubble during the final servicing mission, STS-125, on 14 May 2009. One rides a foot restraint on the Shuttle's robotic arm. Image: NASA/JSC.

### 33.1.4 Building and Keeping the Space Station

The ISS was assembled in orbit between 1998 and 2011, and spacewalks connected every module, truss segment and cable that robots could not. Figure 33.5 shows the result: from December 1998 to September 2026, there were 284 spacewalks totalling about 1 790 hours; NASA's itemised list shows a peak of 23 in 2007, when Shuttle assembly flights and station crews worked outside

almost monthly. Since the Shuttle's retirement, spacewalks have mostly been maintenance — replacing pumps, batteries and failed electronics, installing new roll-out solar arrays — at a rate of about eight a year. The longest ISS EVA, by Jim Voss and Susan Helms in March 2001, lasted 8 h 56 min.




Data: NASA itemised list of ISS spacewalks (US and Russian), accessed October 2026, corrected for duplicates: 283 of NASA's total of 284, ≈1 790 h, December 1998 – September 2026.

**FIGURE 33.5** Spacewalks at the International Space Station per year and hours spent outside, December 1998 to September 2026. The 2024 dip followed water leaks and other faults in the ageing US suits; US spacewalks resumed in 2025–26.

In 2024 only three spacewalks took place: one US EVA in June ended after 31 minutes when water leaked from a service-and-cooling umbilical, and NASA paused others while suit faults were investigated. Seven spacewalks followed in 2026 up to September, including a seven-hour repair of a wrist joint of the station's Canadarm2 robotic arm.

### 33.1.5 Russia, China and the First Commercial Spacewalk

The Soviet **Orlan** suits, first used outside Salyut 6 in December 1977, are semi-rigid. They have a single-piece aluminium torso with a rear hatch formed by the hinged backpack. So a cosmonaut climbs in from behind in a few minutes without help. The current **Orlan-MKS**, in use on the ISS since 2017, works at 40 kPa and has about 7 hours of autonomy. It has an onboard computer and automatic thermal control, and is rated for 15–20 spacewalks over five to six years (Figure 33.6). Russia delivered new Orlan-MKS suits to the ISS in 2025 and 2026.

China's **Feitian** suit was derived from the Orlan and, according to Chinese state media, also works at about 40 kPa. Zhai Zhigang wore the first version for China's first spacewalk, about 20 minutes long, from Shenzhou-7 on 27 September 2008. The second generation, in use at the Tiangong station since July 2021, supports up to 8 hours outside; Chinese sources report that its design life has been extended from 15 to 20 spacewalks. On 17 December 2024 Cai Xuzhe and Song Lingdong of Shenzhou-19 stayed outside for 9 h 06 min, the longest spacewalk ever. By September 2026 the second-generation suits had supported 27 spacewalks at Tiangong, many of them to fit debris shields to the station.

**FIGURE 33.6** Cosmonaut Alexey Zubritsky in an Orlan suit during a Russian spacewalk on 28 October 2025. The blue stripes distinguish the two cosmonauts; the bulky backpack is also the suit's entry hatch. Image: NASA/JSC.

The first commercial spacewalk took place on 12 September 2024 on the privately funded Polaris Dawn mission. Crew Dragon has no airlock, so after about two days of prebreathe with the cabin at reduced pressure, all four crew members suited up and the whole cabin was vented to vacuum. Jared Isaacman and SpaceX engineer Sarah Gillis in turn leaned out of the hatch to test the mobility of the new SpaceX EVA suit. The suit worked at about 34 kPa (5 psi) and drew oxygen, power and data through 3.7 m umbilicals rather than a backpack. From depressurisation to repressurisation the EVA lasted 1 h 46 min. No second commercial spacewalk had flown by October 2026.

**TABLE 33.1** Spacesuits for EVA. Masses are Earth masses of suit plus life support; the AxEMU is derived from NASA's xEMU, designed for pressures from 29.6 to 56.5 kPa.

| Suit | Operator | Pressure | Mass (approx.) | Time outside | First EVA |
|---|---|---|---|---|---|
| A7L / A7LB + PLSS | NASA (Apollo) | 25.8 kPa | 82–96 kg | 4 h; 8 h from Apollo 15 | 1969 |
| Shuttle/ISS EMU | NASA | 29.6 kPa | ≈ 145 kg (ISS) | 7 h + 1 h reserve | 1983 |
| Orlan-MKS | Roscosmos | 40 kPa | ≈ 110 kg | ≈ 7 h | 2017 (Orlan 1977) |
| Feitian (2nd gen.) | CMSA | ≈ 40 kPa (reported) | 120–130 kg | up to 8 h | 2021 (1st gen. 2008) |
| SpaceX EVA suit | SpaceX | ≈ 34 kPa | not published | umbilical, no backpack | 2024 |
| Axiom AxEMU | Axiom / NASA | variable (xEMU heritage) | not published | ≥ 8 h (goal) | planned 2027–28 |

## 33.2 Anatomy of a Spacesuit

A spacesuit has two main parts. The **pressure garment** holds the atmosphere around the body, and the life support system keeps that atmosphere breathable and cool. Figure 33.7 shows the main elements of the US EMU and the layers of its wall.

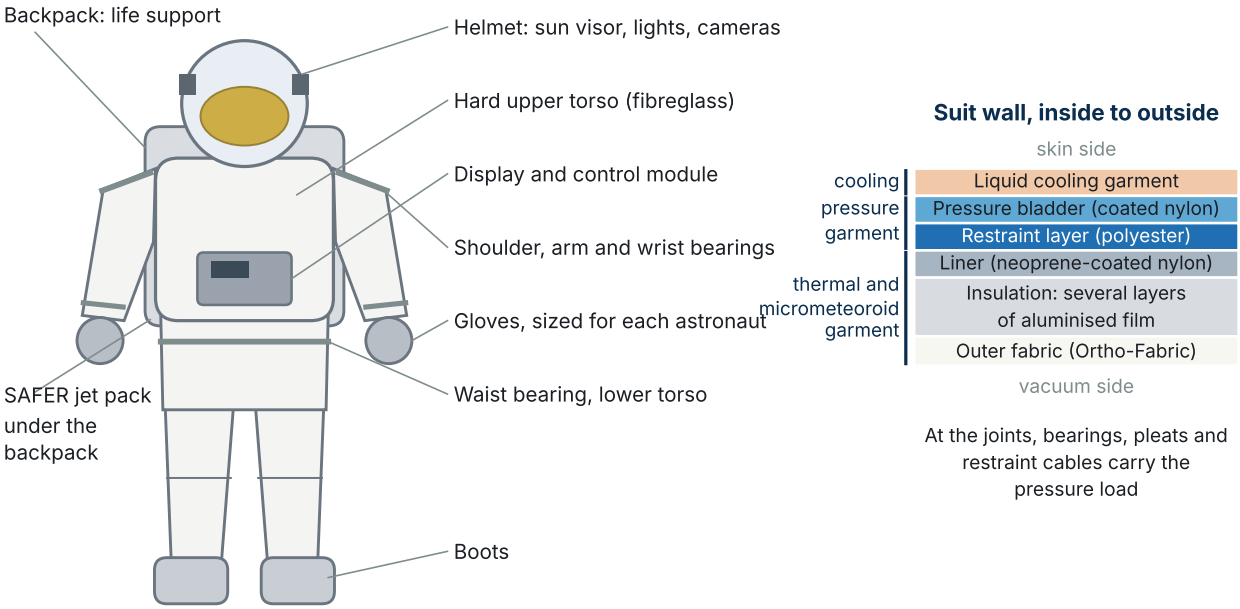



**FIGURE 33.7** Main elements of the US Extravehicular Mobility Unit and, right, the layers of the suit wall from the skin outwards. The bladder holds the gas, the restraint carries the pressure load, and the thermal and micrometeoroid garment protects both.

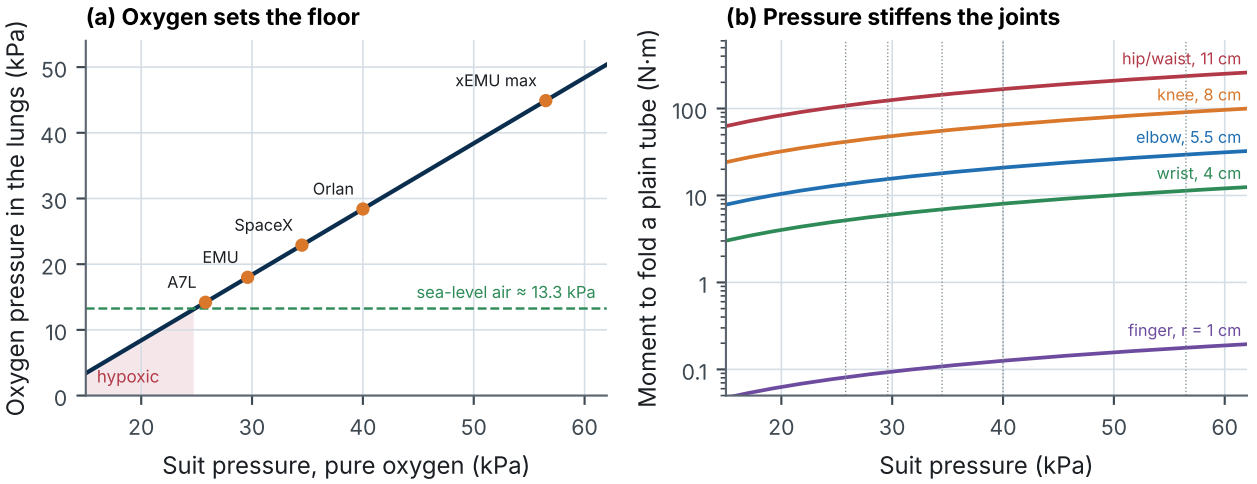
### 33.2.1 Why Suits Work at Low Pressure

The suit pressure is the most important number in its design. It must be high enough to supply oxygen: with pure oxygen, the oxygen pressure in the lungs is the suit pressure minus about 6.3 kPa of water vapour and 5.3 kPa of carbon dioxide. Matching the 13.3 kPa of a person breathing sea-level air needs about 25 kPa — almost exactly the Apollo suit pressure (Figure 33.8 a). Below about 6.3 kPa (the **Armstrong limit**) body fluids boil at body temperature, so even an emergency suit pressure must stay well above it.

Every kilopascal above the minimum makes the suit stiffer. An inflated fabric tube resists bending because bending it reduces its volume, and the gas must be compressed to make room. A simple model gives the moment needed to fold an unjointed tube of radius $r$ at pressure $p$:

$$M \approx \pi p r^3$$

For an elbow-sized tube, 5.5 cm in radius, this is about 15 N·m at 29.6 kPa and 30 N·m at 56.5 kPa; for a knee it is three times larger (Figure 33.8 b). An astronaut flexing an arm hundreds of times a day against such moments would tire quickly. Low pressure is therefore the first defence of mobility, and clever joints are the second. The price of low pressure is decompression sickness, discussed below.




(a) Alveolar gas equation with 100 % $O_2$, water vapour 6.3 kPa, $CO_2$ 5.3 kPa. (b) Collapse moment of an inflated fabric tube, $M = \pi p r^3$; real joints use pleats, bearings and constant-volume shapes to cut this by a large factor. Dotted lines: suit pressures of panel (a).

**FIGURE 33.8** Why suits are pressurised to 26–57 kPa. (a) Oxygen pressure in the lungs for a pure-oxygen suit: about 25 kPa matches breathing sea-level air. (b) Moment needed to fold a plain inflated tube of different radii, which grows in proportion to pressure.

### 33.2.2 Layers

The wall of a suit has three functional groups (Figure 33.7). Next to the skin is a **liquid cooling and ventilation garment (LCVG)**. In the EMU it is a stretch-mesh undergarment threaded with about 90 m of thin plastic tubing. It carries cool water over the body and collects ventilation gas from the limbs. Over it the **pressure garment** has two layers. One is a gas-tight **bladder** of urethane-coated nylon. The other is a **restraint layer** of strong polyester fabric that carries the pressure load and fixes the suit's shape, so the bladder never stretches. Outside is the **thermal micrometeoroid garment (TMG)**: a neoprene-coated nylon liner, several layers of aluminised film acting as multilayer insulation (Chapter 5), and an outer shell of Ortho-Fabric. This fabric is a weave of Gore-Tex, Kevlar and Nomex that resists abrasion, fire and small particles. The suit's surroundings range from about +120 °C in sunlight to −120 °C in shadow.

### 33.2.3 Joints and Mobility

The aim of joint design is a **constant-volume joint**: one whose internal volume, and therefore the work done against the gas, stays nearly the same as it bends. Three techniques are used. **Convolutes** are bellows-like folds that fill on the inside of the bend as the outside opens. **Flat-pattern joints** use sewn gores and restraint cables or webbing along the axis of the joint, so the fabric on one side shortens as the other lengthens. **Rotary bearings** — sealed rings at the shoulder, upper arm, wrist, waist and hip — let a limb rotate with almost no change in volume. Hard parts carry loads. The EMU's **hard upper torso** is a fibreglass shell to which the arms, helmet, lower torso and backpack attach. The Orlan, Feitian and AxEMU extend this to a rear-entry hard torso.

Fit matters as much as design: in a suit that is too large the joints of suit and body no longer coincide; one too small presses on shoulders and knees. In a small NASA survey, 14 of 22 EVA astronauts reported EVA-related shoulder injuries, most of them during training in the pool, where the hard torso restricts shoulder movement, especially when working upside down.

### 33.2.4 Gloves and Hands

Gloves are the hardest part of a suit to design: the hand has many small joints and weak muscles, and a glove must be pressurised, insulated and tough enough for sharp edges. EMU gloves are custom-sized from casts or scans of each astronaut's hands. They have a bladder and restraint with finger joints and a wrist bearing, heated fingertips for cold work, and rubberised palms. Even so, grasping with a pressurised glove takes much more effort than with a bare hand and fatigues the forearms within hours. In one NASA study, 770 suited training events produced 352 reported symptoms, 47 % of them in the hands. The most common was **fingernail delamination**, the loss of nails, attributed to moisture in the gloves and loads on the fingertips. Gloves are also vulnerable to damage: after a cut glove ended a spacewalk early in 2007, crews began photographing their gloves during spacewalks, and gloves are inspected after every EVA (Figure 33.9).

**FIGURE 33.9** Astronauts Mike Foreman (left) and Rick Linnehan in the ISS Quest airlock after a spacewalk in March 2008, still wearing their EMUs. Gloves are photographed and inspected for cuts after each spacewalk. Image: NASA/JSC.

### 33.2.5 Helmet and Visors

The EMU helmet is a clear polycarbonate bubble. Over it fits the **extravehicular visor assembly**, with a clear protective visor and a gold-coated sun visor that reflects visible light and infrared, plus adjustable eyeshades. Lights and video cameras are mounted on the sides. Inside are a microphone and earphones in a fabric "Snoopy cap", a drink bag of about 0.95 L and a foam block against which the astronaut can press the nose to clear the ears. The SpaceX EVA suit added a head-up display and a copper and indium tin oxide visor coating.

## 33.3 The Portable Life Support System

The **portable life support system (PLSS)** turns a pressure garment into a spacecraft. Figure 33.10 shows its two loops. In the **ventilation loop**, a fan blows oxygen into the helmet at the face. The gas sweeps over the head and body, collecting carbon dioxide, moisture and heat. It is drawn back from the wrists and ankles through a canister that removes carbon dioxide and odours. It then passes through the sublimator, which cools it and condenses moisture, and a water

separator, before returning to the helmet. In the **cooling-water loop**, a pump circulates water through the LCVG tubes and the sublimator. The EMU's fan, pump and separator are a single rotating machine.

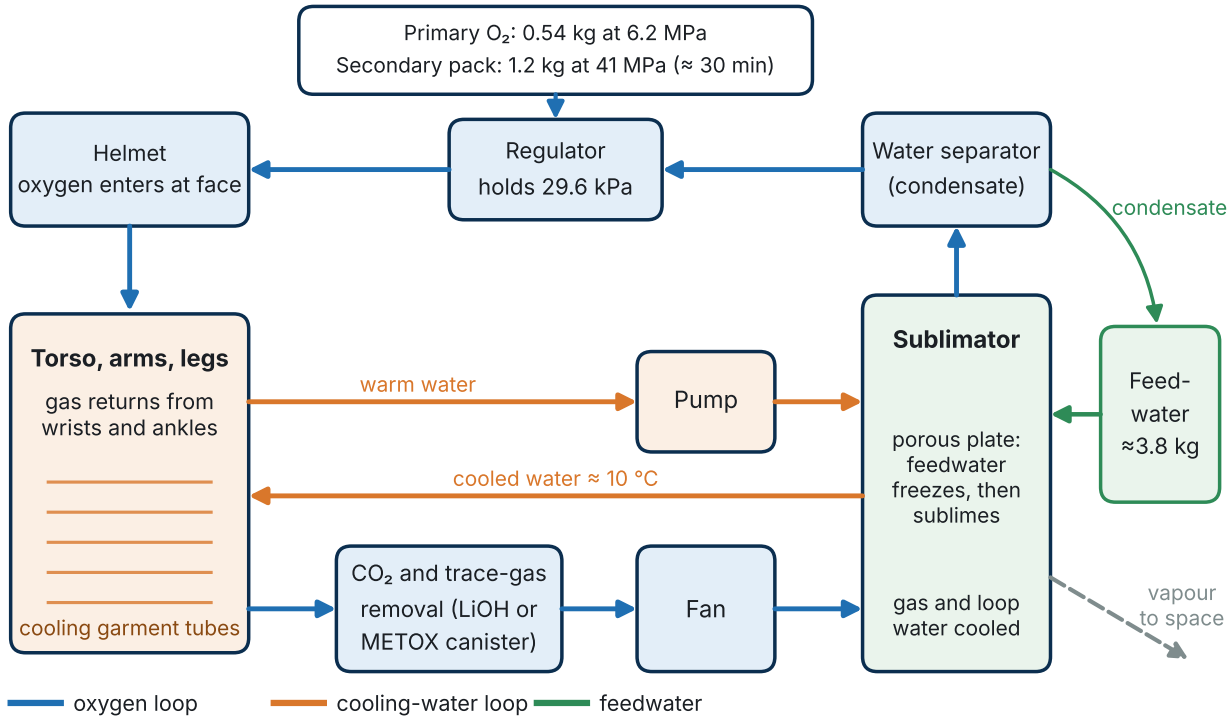



**FIGURE 33.10** Simplified flow diagram of the EMU portable life support system. Oxygen is recirculated through a carbon dioxide canister and a sublimator; cooling water circulates through the garment; feedwater is lost to space as vapour.

## 33.3.1 Oxygen and Pressure

The suit is a closed loop, so oxygen is needed only to replace what the astronaut consumes and what leaks out. An astronaut working at 250 W consumes about 63 g of oxygen per hour. The EMU's primary tanks hold about 0.54 kg at 6.2 MPa; a regulator keeps the suit at 29.6 kPa. A **secondary oxygen pack** of about 1.2 kg at 41 MPa can maintain pressure for about 30 minutes in **purge mode**. If the fan fails, it flushes oxygen straight through the suit to remove carbon dioxide and heat. This is the "get-back" time: EVA plans keep the astronaut within about 30 minutes of the airlock.

## 33.3.2 Removing Carbon Dioxide

Apollo backpacks and early Shuttle EMUs absorbed carbon dioxide in disposable **lithium hydroxide (LiOH)** cartridges (the chemistry is described in Chapter 30). The ISS EMU uses regenerable **metal oxide (METOX)** canisters based on silver oxide, which absorb about 0.54 kg of carbon dioxide. They are then regenerated by baking in an oven on the station, which takes more than 1 kW for up to 14 hours. The next generation, used in NASA's xEMU prototype and the AxEMU, uses a **rapid cycle amine (RCA)** swing bed. Two small beds of amine sorbent alternate every few minutes. One absorbs carbon dioxide and water while the other is exposed to vacuum to release them. Because the beds regenerate continuously, $CO_2$ capacity no longer limits EVA duration (Table 33.2).

**EXAMPLE: CARBON DIOXIDE REMOVAL FOR A 7-HOUR EVA**

An astronaut works for 7 hours at an average metabolic rate of 250 W, with a respiratory quotient ($CO_2$ produced ÷ $O_2$ consumed) of 0.85.

1. Oxygen: at RQ 0.85 each litre of oxygen releases about 20.4 kJ (Weir equation). So 250 W needs 250 × 3 600 / 20 400 = **44.1 L/h** of $O_2$ (at 0 °C, 1 atm), or 63 g/h.
2. Carbon dioxide: 0.85 × 44.1 = 37.5 L/h = **73.6 g/h**; over 7 hours **515 g**, or 589 g if a 1-hour reserve is added.
3. Lithium hydroxide absorbs at most 0.92 kg of $CO_2$ per kg (Chapter 30), so 589 g needs at least **0.64 kg**; real beds, never fully used, need about twice that. The Apollo 15–17 backpacks carried 1.42 kg.
4. A METOX canister with 0.54 kg capacity lasts 0.54/0.0736 = **7.3 h** at 250 W, but only 6.3 h at 293 W — one reason heavy work shortens EVAs (Figure 33.11 b).
5. Ventilation: the EMU fan moves about 0.17 $m^3$/min (6 cubic feet per minute) of gas at 29.6 kPa, or 0.034 mol/s. Carbon dioxide is added at $4.6 \times 10^{-4}$ mol/s, so the returning gas carries about 1.4 % more $CO_2$ than the gas entering the helmet — a rise of **0.40 kPa (3 mmHg)**. Fresh gas is therefore aimed at the face, where exhaled $CO_2$ is concentrated.

### 33.3.3 Rejecting Heat

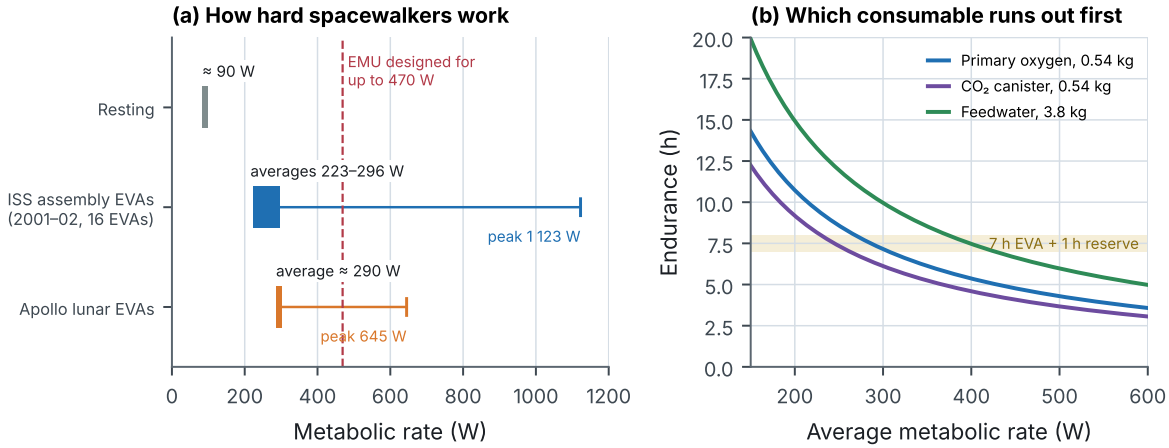
The astronaut's metabolic heat, typically 200–300 W and sometimes much more, must be removed without air to carry it away. A radiator alone would be impractical (see the example below), so the PLSS throws away water. In a **sublimator**, feedwater flows to one side of a porous metal plate whose other side faces vacuum. Water in the pores freezes, and the ice sublimates — turns directly to vapour — absorbing 2.83 MJ per kilogram. The process regulates itself: more heat sublimates ice faster. NASA's newer **spacesuit water membrane evaporator (SWME)**, used in the xEMU design, evaporates water through hydrophobic hollow fibres and is less sensitive to contamination, which has blocked sublimator plates in the past.

**EXAMPLE: WATER FOR A SUBLIMATOR**

The PLSS must reject 300 W for a 7-hour EVA using a sublimator (latent heat 2.83 MJ/kg).

1. Water rate: 300 / $2.83 \times 10^6$ = $1.06 \times 10^{-4}$ kg/s = **0.38 kg/h**.
2. For 7 hours: **2.7 kg**. The EMU carries about 3.8 kg of feedwater, enough for 10 hours at 300 W but only 6.4 hours at 470 W (1 600 Btu/h), the highest rate for which the EMU was designed.
3. Apollo's extended backpacks carried 5.2 kg of water for 8-hour lunar EVAs. This gives an average capacity of $5.2 \times 2.83 \times 10^6$ / 28 800 ≈ **510 W** — more than the EMU, for the heavier work of lunar surface EVAs.
4. A suit radiator would need 300 / ($0.9 \times 5.67 \times 10^{-8} \times 280^4$) ≈ **0.96 $m^2$** of surface viewing only cold space. That is roughly half the outer area of a suit, much of which faces the Sun, the Earth or the spacecraft at any moment.

Figure 33.11 puts these numbers together. Average metabolic rates on 16 ISS assembly spacewalks in 2001–02 were 223–296 W, with peaks above 1 000 W; Apollo lunar EVAs averaged about 290 W. At such rates the $CO_2$ canister runs out first, then the oxygen, and the water last; the first two give roughly the 7 hours plus reserve for which the EMU is sized. A 2024 NASA review concluded that EVAs of 6.5 hours are reasonable while average metabolic rates stay below about 290 W (1 000 Btu/h).




(a) ISS assembly EVAs on STS-104 to STS-113 (Katuntsev, Osipov and Gernhardt 2004, via NASA) and Apollo data. (b) Weir equation, RQ 0.85, 20.4 kJ per litre $O_2$; all metabolic heat rejected by sublimation (2.83 MJ/kg); suit leakage and heat leaks ignored. Capacities approximate those of the ISS EMU.

**FIGURE 33.11** Metabolic rates and consumables. (a) Measured averages and peaks for ISS assembly and Apollo lunar spacewalks. (b) Endurance set by each consumable for EMU-like capacities as a function of average metabolic rate.

**TABLE 33.2** Portable life support systems compared. xEMU and AxEMU values are design goals.

| Function | Apollo PLSS (1969–72) | Shuttle/ISS EMU | xEMU / AxEMU (2020s) |
|---|---|---|---|
| Oxygen storage | 9.9 MPa tank (Apollo 15–17) | 0.54 kg at 6.2 MPa + 1.2 kg secondary | high-pressure tanks, variable-pressure regulator |
| $CO_2$ removal | LiOH, 1.42 kg | LiOH or METOX (regenerable in oven) | rapid cycle amine, vented to vacuum |
| Cooling | sublimator, 5.2 kg water | sublimator, 3.8 kg water | water membrane evaporator |
| Emergency supply | oxygen purge system, ≥ 30 min | secondary oxygen pack, ≈ 30 min | secondary oxygen supply |
| Time outside | 4 h, later 8 h | 7 h + 1 h reserve | ≥ 8 h (goal) |

## 33.3.4 Power, Communications and Displays

A rechargeable battery powers the fan, pump, radios, sensors, lights and cameras. The EMU's **display and control module** on the chest carries the oxygen and cooling controls and a small text display; astronauts read its labels through a wrist mirror, and checklists are worn on the cuff. A caution-and-warning system monitors pressure, oxygen, carbon dioxide, battery and water, and suit data are radioed to the ground with voice and video. Newer suits add colour and head-up displays and record health data such as heart rate.

## 33.3.5 SAFER: the Self-Rescue Jet Pack

A broken tether could leave an astronaut drifting away from the station with no one able to follow. The **Simplified Aid For EVA Rescue (SAFER)** is a small nitrogen-jet backpack, about 38 kg, that latches under the EMU's PLSS. It has 24 small thrusters, a hand controller that folds out from its lower edge and an automatic attitude hold that stops tumbling at the push of a button. Its 1.4 kg of nitrogen gives a velocity change of only about 3 m/s — enough to stop, turn and fly back a short distance, but not to explore. SAFER was first tested on STS-64 in September 1994 (Figure 33.12), is worn on every US spacewalk at the ISS, and has never had to be used in an emergency.

**FIGURE 33.12** Astronaut Mark Lee testing SAFER above the Earth during the STS-64 mission in September 1994, the first untethered US spacewalk in ten years. Image: NASA/MSFC.

## 33.4 Decompression Sickness and Prebreathe

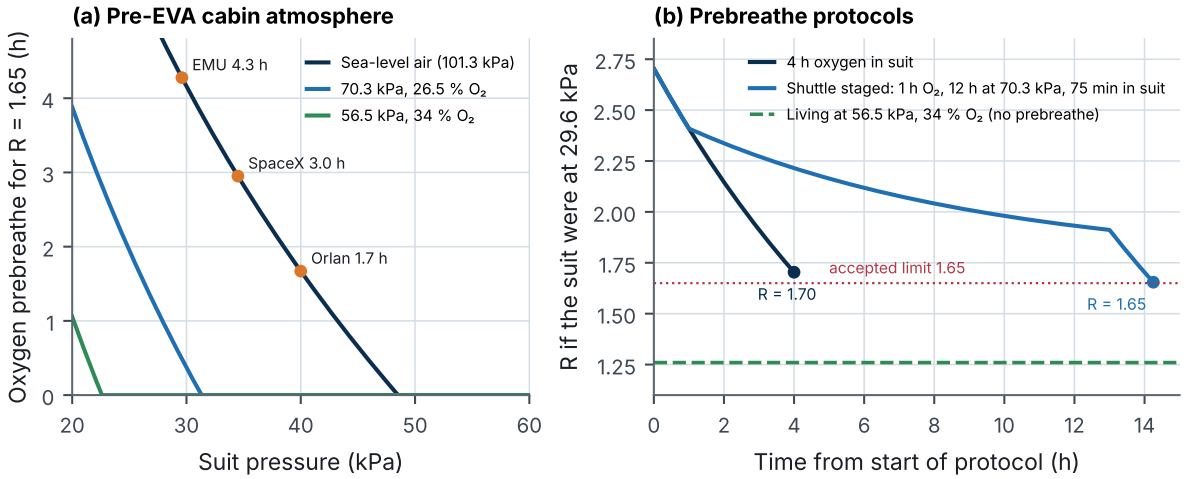
### 33.4.1 Nitrogen in the Tissues

The ISS cabin holds sea-level air, so the body's tissues are saturated with nitrogen at about 80 kPa. If an astronaut moves straight to a suit at 29.6 kPa, the nitrogen becomes supersaturated and can form bubbles — **decompression sickness (DCS)**, introduced in Chapter 30. Mild DCS is pain in the joints; serious ("Type II") DCS affects the brain, spinal cord or lungs. NASA uses the **tissue ratio** $R$ — tissue nitrogen pressure divided by suit pressure — as an index of risk:

$$R = \frac{P_{N_2,\,\text{tissue}}}{P_{\text{suit}}}, \qquad P_{N_2,\,\text{tissue}}(t) = P_{N_2,\,\text{amb}} + \left(P_{N_2,0} - P_{N_2,\,\text{amb}}\right) 2^{-t/t_{1/2}}$$

The tissue relaxes towards the nitrogen pressure it breathes with a half-time $t_{1/2}$ of 360 minutes, the value that best matched NASA and US Air Force chamber tests. Breathing pure oxygen drops the inspired nitrogen to almost zero, so nitrogen leaves the tissues. For the EMU at 29.6 kPa, protocols with $R \leq 1.65$ have historically been accepted. NASA's standard requires that any protocol be shown in ground trials to give less than a 4 % probability of DCS with 95 % confidence.

Without prebreathe, R is 2.70 for the EMU, 2.32 for a 34.5 kPa suit and 2.00 for the 40 kPa Orlan. Figure 33.13 a shows the oxygen prebreathe needed to reach R = 1.65. It is 4.3 hours from sea-level air for the EMU, 3.0 hours at 34.5 kPa and 1.7 hours for the Orlan. Above 48.5 kPa none is needed. Russian practice accepts a shorter prebreathe, and so a higher R: cosmonauts breathe oxygen in the Orlan for only about 30 minutes before going out. A low-pressure, oxygen-rich cabin changes the picture completely. Living at 70.3 kPa with 26.5 % oxygen, the "campout" atmosphere, leaves only 52 kPa of nitrogen in the tissues. NASA's 56.5 kPa, 34 % oxygen **exploration atmosphere** leaves 37 kPa, R = 1.26. By this criterion it needs no prebreathe at all, although NASA plans about 15 minutes of in-suit oxygen as a margin (Chapter 30).




Single 360-min tissue (NASA R-value model): tissue $N_2$ relaxes exponentially towards the inspired $N_2$ pressure; R = tissue $N_2$ / suit pressure. Initial tissue $N_2$ = 0.79 × 101.3 kPa. Real protocols were validated in chamber trials; exercise during prebreathe, not modelled here, speeds nitrogen removal.

**FIGURE 33.13** Decompression and prebreathe in the 360-minute tissue model. (a) Oxygen prebreathe needed to reach R = 1.65 as a function of suit pressure, for three cabin atmospheres. (b) R during two Shuttle-era protocols, compared with living in the exploration atmosphere.

## 33.4.2 Prebreathe Protocols

Four hours in a suit breathing oxygen wastes crew time, so shorter protocols were developed (Table 33.3). The Shuttle's **staged decompression** lowered the whole cabin to 70.3 kPa for at least 12 hours before EVA days; in practice crews spent an average of more than 40 hours at this pressure. The ISS cannot lower its whole cabin, so other approaches are used. In the **campout**, the EVA crew sleep in the closed Quest airlock at 70.3 kPa for up to 8 h 40 min before the EVA. In **exercise prebreathe**, the astronaut breathes oxygen while pedalling vigorously; exercise increases blood flow and speeds nitrogen removal, cutting the required time to about two hours. The **in-suit light exercise (ISLE)** protocol replaces hard exercise with light movement in the suit after 40 minutes on a mask and a 20-minute partial depressurisation. NASA has never had a Type II DCS case in spaceflight, and no DCS was reported in more than 140 Shuttle EVAs.

**TABLE 33.3** Ways to prepare for a spacewalk. Durations are approximate and protocols have been revised over time.

| Protocol | Main steps | Typical duration | Used on |
|---|---|---|---|
| In-suit prebreathe | $O_2$ in suit at 101 kPa | 4 h | ISS, contingency |
| Staged decompression | 1 h $O_2$; ≥ 12 h cabin at 70.3 kPa; 75 min $O_2$ in suit | 1–2 days of cabin time | Space Shuttle |
| Campout | night in airlock at 70.3 kPa (≤ 8 h 40 min); $O_2$ in suit | overnight + ≈ 1 h | ISS |
| Exercise prebreathe | $O_2$ on mask with cycle exercise; in-suit prebreathe | ≈ 2 h of $O_2$ | ISS |
| ISLE | 40 min $O_2$ mask; 20 min depress to 70.3 kPa; ≈ 100 min in suit with light exercise | ≈ 2.7 h | ISS (from 2011) |
| Orlan | $O_2$ in suit at reduced pressure | ≈ 30 min | Russian segment |
| Polaris Dawn | ≈ 2 days with cabin at ≈ 59 kPa | ≈ 45 h | Crew Dragon, 2024 |

**EXAMPLE: R-VALUE FOR THE SHUTTLE STAGED PROTOCOL**

Start from tissue nitrogen in equilibrium with sea-level air: 0.79 × 101.3 = 80.0 kPa. Use $t_{1/2}$ = 360 min and a suit at 29.6 kPa.

1. No prebreathe: R = 80.0/29.6 = **2.70** — unacceptable.
2. 60 min of pure oxygen at 101.3 kPa: $80.0 \times 2^{-60/360}$ = **71.3 kPa**.
3. 12 h in a cabin at 70.3 kPa with 26.5 % $O_2$, where inspired nitrogen is 0.735 × 70.3 = 51.7 kPa: $51.7 + (71.3 - 51.7) \times 2^{-720/360}$ = **56.6 kPa**.
4. 75 min of oxygen in the suit: $56.6 \times 2^{-75/360}$ = **49.0 kPa**, so R = 49.0/29.6 = **1.65** — exactly the accepted limit.
5. By comparison, 4 hours of in-suit oxygen from sea-level air gives $80.0 \times 2^{-240/360}$ = 50.4 kPa, R = **1.70** (Figure 33.13 b); the real 4-hour protocol was validated by chamber tests rather than by R alone.

### 33.4.3 The Pressure–Mobility Trade

The choice of suit pressure is a three-way trade between oxygen supply, mobility and decompression (Figure 33.8 and Figure 33.13). A higher-pressure suit needs no prebreathe but is stiffer, heavier and leakier. The answer adopted for NASA's xEMU, and inherited by the AxEMU, is a **variable-pressure suit** that can run at 29.6 kPa for mobility or up to 56.5 kPa to shorten prebreathe or treat suspected DCS. For frequent lunar or Martian surface EVAs, a reduced-pressure, oxygen-enriched habitat combined with a suit at 29.6–34 kPa is the preferred direction, limited mainly by the fire risk of oxygen-rich cabins (Chapter 30).

## 33.5 Working Outside

### 33.5.1 Tools, Tethers and Restraints

The first rule of EVA is never to let go. Each astronaut is clipped to the structure by a retractable **safety tether** of about 26 m on a reel, and moves hand over hand along handrails, attaching a **waist tether** at each worksite. Every tool and part is tethered too; lost objects become orbital debris (Chapter 21). For two-handed work the astronaut locks the boots into an **articulating portable foot restraint**, which fits sockets on the structure or the end of the robotic arm. The station's Canadarm2 (Chapter 32) can carry an astronaut and a large unit, such as a battery or pump, from one worksite to another, as the Shuttle arm did at Hubble (Figure 33.4).

Tools are designed for bulky gloves. The **pistol grip tool**, a battery-powered torque-limited wrench, does most bolt work; bolts are captive, connectors have large tabs, and replaceable units have guide pins and handles placed for a gloved hand (Figure 33.14).

### 33.5.2 Training in Water

The best simulation of weightlessness on Earth is neutral buoyancy. NASA's **Neutral Buoyancy Laboratory (NBL)** in Houston, built in the mid-1990s, is a pool 62 m long, 31 m wide and 12 m deep. It holds 23.5 million litres of water and full-size mockups of ISS modules. Suited astronauts are weighted until they neither float nor sink and rehearse every task with support divers, for

several hours per hour of planned EVA (Figure 33.15). Water drag and gravity inside the suit make the pool imperfect, so it is complemented by virtual reality, air-bearing floors and, for lunar suits, reduced-gravity rigs. Russia and China have similar hydrolabs.

**FIGURE 33.14** NASA astronaut Chris Williams during a 7 h 20 min spacewalk on 30 June 2026 to replace a faulty wrist joint of the Canadarm2 robotic arm. Image: NASA/JSC.

**FIGURE 33.15** Astronauts Loral O'Hara and Stan Love during the first two-suit test of Axiom's lunar spacesuit at the Neutral Buoyancy Laboratory in September 2025. Image: NASA/JSC (NASA).

### 33.5.3 Timelines and Workload

A typical ISS spacewalk lasts 6–7 hours, plus about as much again for suit checks, prebreathe, depressurisation and post-EVA servicing. The timeline is planned to the minute, with "get-ahead" tasks and pre-agreed stopping points. Flight controllers estimate metabolic rate from oxygen use and cooling-water temperatures, because consumables and heat limit the day (Figure 33.11). NASA measures **EVA efficiency** as EVA time divided by the crew time spent preparing for it and cleaning up afterwards; this was only about 0.4 for ISS protocols and 0.5 for the Shuttle's.

### 33.5.4 When Things Go Wrong

Serious EVA incidents have been few, but each taught a lesson (Table 33.4). The most dangerous at the ISS came on 16 July 2013, during Luca Parmitano's EVA 23. About 44 minutes in he reported water at the back of his head. It spread over his eyes, nose and ears. He found his way back to the airlock largely by feel before the EVA ended at 92 minutes. Between 1 and 1.5 litres of water had entered the helmet from the ventilation loop. The investigation traced it to contaminants blocking holes in the fan/pump/separator, and found that a smaller leak on the previous EVA had been misdiagnosed as a leaking drink bag. NASA added absorbent pads and a snorkel to the helmet and tightened the handling of suit anomalies.

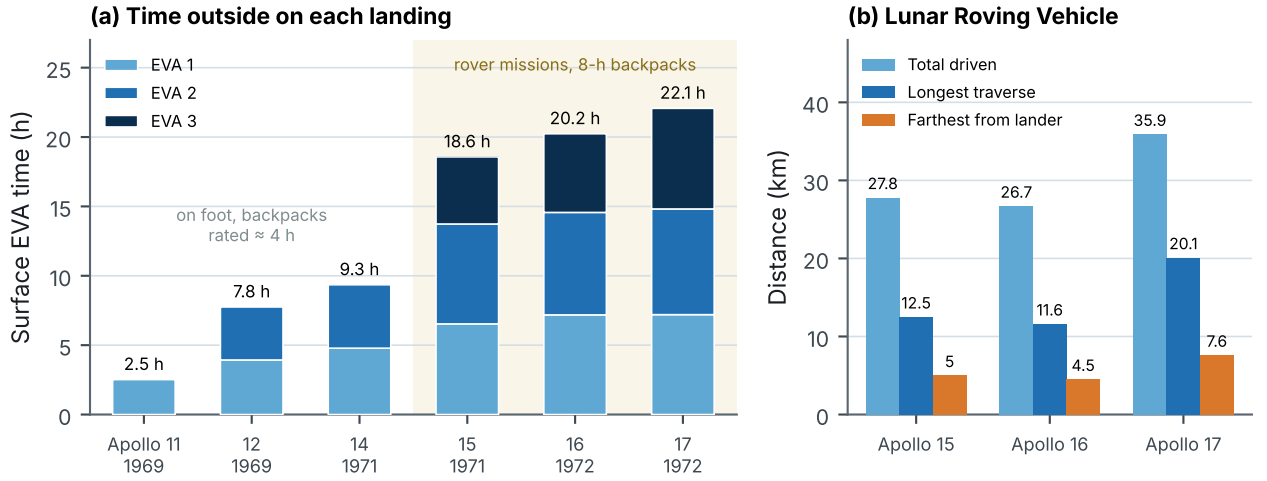
**TABLE 33.4** Notable spacewalk incidents and what they taught.

| Year | Event | Lesson |
|---|---|---|
| 1965 | Leonov's suit balloons; he lowers pressure to re-enter the airlock | control of suit shape; airlock sizing |
| 1973 | Skylab solar wing freed with improvised tools | EVA as a contingency capability |
| 2007 | Torn ISS solar array repaired from the end of a boom extension | improvised tools; rehearse with ground teams |
| 2013 | EVA 23: about 1.5 L of water in Parmitano's helmet | treat suit anomalies as life-threatening |
| 2024 | ISS EVA ends after 31 min: water leak in suit umbilical | ageing suits limit operations |

# 33.6 Spacewalking on Other Worlds

## 33.6.1 Lessons from Apollo

Gravity on the Moon is one sixth of Earth's, so the 82–96 kg Apollo suit and backpack felt like only 14–16 kg, but the suit's stiff legs and the low traction made normal walking awkward. Astronauts found that a **loping** gait, bounding from foot to foot, or a two-footed hop, was faster and less tiring, and falls were common but usually harmless. Apollo 11–14 crews walked, with backpacks rated for about 4 hours. Apollo 15–17 carried the **Lunar Roving Vehicle (LRV)** and 8-hour backpacks. Figure 33.16 shows how surface time and range grew. The Apollo 17 crew spent 22 h 04 min outside over three EVAs and drove 35.9 km, up to 7.6 km from their lander.




Data: NASA mission reports; NSSDC Lunar Roving Vehicle summary. Total ≈ 80 h of lunar surface EVA by 12 astronauts.

**FIGURE 33.16** Apollo surface EVAs. (a) Time outside per landing, by EVA. (b) Distances driven with the Lunar Roving Vehicle on Apollo 15–17; the farthest point from the lander was limited by the need to be able to walk back if the rover failed.

The rover was a safety system as well as a vehicle. Traverses were planned so that, if the rover broke down at the farthest point, both astronauts could walk back to the lander on the consumables left in their backpacks. This **walkback limit** is the surface equivalent of the 30-minute get-back time in orbit, and it will constrain Artemis traverses too until pressurised rovers are available.

Lunar dust was the biggest surprise. The regolith is made of sharp, glassy fragments never rounded by wind or water, and it clings electrostatically to everything. After three days on the Moon, Apollo 17 suits were grey to the knees, their bearings were harder to turn and their outer fabric was abraded. Dust in the cabin irritated the crew's eyes and lungs (Figure 33.17). Gloves wore fastest, and astronauts' hands and fingers were sore after hours of gripping tools.

## 33.6.2 Artemis Suits

NASA decided in 2022 to buy lunar and station suits as a service rather than build them, under its **Exploration Extravehicular Activity Services (xEVAS)** contracts. Collins Aerospace withdrew in 2024, leaving Axiom Space as the sole provider. Axiom's **AxEMU** is derived from NASA's xEMU prototype (Figure 33.18). It is a rear-entry suit with a hard upper torso and interchangeable parts intended to fit at least 90 % of the US male and female population. It has mobility joints at the

hips and knees that let the wearer kneel and squat, and an xEMU-style backpack. Its goals include EVAs of at least 8 hours and at least two hours in the permanently shadowed craters near the lunar south pole, where temperatures fall below −200 °C.

By early 2026 Axiom had completed its own design review and more than 850 hours of crewed pressurised tests. Qualification was planned to run to mid-2027, with a first suit flying in Earth orbit in 2027. In April 2026 NASA's Office of Inspector General warned that the schedule had little or no margin for the first landing, now planned for Artemis IV in 2028. It also warned that delays in line with recent spaceflight programmes (about 8.7 years on average from contract to test flight) could push suit demonstrations to 2031. In September 2026 NASA and Axiom were reported to be studying a simpler, lighter suit configuration for the first landing. China, meanwhile, unveiled its **Wangyu** lunar suit in 2024 for a crewed landing planned before 2030.

**FIGURE 33.17** Gene Cernan in the lunar module after the second Apollo 17 EVA in December 1972, his suit coated with lunar dust. Image: NASA/JSC.

**FIGURE 33.18** The flight design of Axiom Space's AxEMU lunar spacesuit, shown in 2024 with its white outer layer. Image: NASA/JSC (Axiom Space).

### 33.6.3 Suits for Mars

Mars adds problems that the Moon does not (Table 33.5).

**TABLE 33.5** How the environment changes spacesuit design.

| Factor | ISS orbit | Moon | Mars |
|---|---|---|---|
| Gravity | ≈ 0 | 0.17 g | 0.38 g |
| Ambient pressure | vacuum | vacuum | ≈ 0.6 kPa $CO_2$ |
| Heat rejection | sublimator | sublimator | evaporator or radiator; sublimator unreliable |
| Dust | none | sharp, electrostatic | fine, windblown, perchlorates |
| Return to safety | ≈ 30 min to airlock | walkback limit | walkback limit; no quick return to Earth |
| Spacewalks | ≈ 8 a year | a few per early landing | many over a long stay |

Its gravity, 0.38 g, makes suit mass more than twice as burdensome as on the Moon, so a Mars suit must be much lighter than the AxEMU. Its thin carbon dioxide atmosphere, about 0.6 kPa, carries heat by convection, so insulation that works in vacuum works less well. The surface pressure is close to the triple point of water, 0.61 kPa, so a sublimator will not work reliably, and evaporative coolers lose efficiency. An amine scrubber that regenerates by exposure to "vacuum" must vent against an atmosphere that is itself 95 % carbon dioxide, so it needs heating or a different sorbent. Martian dust is finer than lunar dust, is lifted by wind and contains perchlorates, which are toxic. Suits must also be maintained and repaired on site for months.

### 33.6.4 Suitports and Rear-Entry Suits

Every trip through an airlock costs gas and time and brings dust inside. A **suitport** avoids this: the suit is stored outside the habitat or rover with its rear hatch sealed against a port in the wall, and the crew member climbs in through the back and detaches. NASA tested the idea on its Space Exploration Vehicle rover prototype and the Z-1 prototype suit in the early 2010s, and rear-entry suits such as the Orlan and AxEMU make it practical. Suitports would allow quick EVAs from pressurised rovers on long traverses, if the seals survive dust and repeated use.

#### AI FOR SPACEWALKS

AI is starting to support EVA in three ways. *Suit health monitoring:* models trained on suit telemetry can watch pressures, temperatures, fan speed and carbon dioxide. They can flag early signs of faults like the water intrusion of EVA 23, or a rising metabolic rate, faster than controllers watching many screens. *EVA planning and support:* optimisation tools can schedule tasks against consumables, lighting and walkback limits. Voice assistants in the helmet can read procedures, log progress and answer questions. This helps when the light-time delay to Mars — up to 22 minutes one way — makes ground support slow. *Robotic assistants:* free-flying robots and dexterous manipulators can fetch tools, hold lights and inspect worksites, and may eventually do routine external maintenance so that people go outside only when judgement and dexterity are needed. Chapters 48 and 51 describe these systems and how they can be verified for human spaceflight.

#### KEY TAKEAWAYS

- A spacesuit is a one-person spacecraft: a pressure garment (bladder, restraint, thermal and micrometeoroid layers) plus a portable life support system that supplies oxygen, removes carbon dioxide and rejects heat.
- Suits run at 26–57 kPa: about 25 kPa of pure oxygen matches sea-level breathing, and lower pressure means more mobility, because the moment to bend a pressurised joint grows as $pr^3$.
- Constant-volume joints, bearings, hard torsos and custom gloves make work possible; hands and shoulders still suffer most.
- The PLSS of every suit since Apollo recirculates oxygen through a $CO_2$ scrubber and cools the astronaut by sublimating or evaporating water: about 0.38 kg of water per hour for 300 W.
- Moving from sea-level air to a 29.6 kPa suit gives a tissue ratio R = 2.70; oxygen prebreathe, exercise or a reduced-pressure campout brings it to about 1.65. Variable-pressure suits and low-pressure, oxygen-rich habitats can cut prebreathe to minutes or remove it altogether.
- EVA success depends on operations: tethers, restraints, neutral-buoyancy training, paced timelines and a 30-minute get-back or walkback reserve.
- In October 2026 the ISS still relied on suits designed in the 1970s; Axiom's AxEMU was in qualification for a first flight in 2027 and a landing planned for 2028, with little schedule margin.

### REVIEW QUESTIONS

1. What went wrong on Leonov's spacewalk and on Gemini 9A, and which design and operational changes followed?
2. Use the alveolar gas equation to show why a pure-oxygen suit must be pressurised to at least about 25 kPa. What is the oxygen pressure in the lungs at 40 kPa?
3. Estimate the moment needed to bend an unjointed knee section (radius 8 cm) of a suit at 29.6 kPa and at 56.5 kPa. How do constant-volume joints reduce it?
4. Describe the functions of the bladder, restraint layer and thermal micrometeoroid garment. Why must the restraint, not the bladder, carry the load?
5. An astronaut works for 6 hours at 350 W. Estimate the oxygen consumed, the carbon dioxide produced (RQ 0.85) and the sublimator water needed.
6. Why does a sublimator regulate itself? Why would it be unreliable on the surface of Mars?
7. Calculate R for an astronaut who breathes oxygen for 2 hours at 101.3 kPa before entering a 29.6 kPa suit ($t_{1/2}$ = 360 min). How much longer would be needed to reach R = 1.65?
8. Compare the campout and exercise prebreathe protocols. Why can the ISS not use the Shuttle's staged decompression?
9. What caused the water intrusion on EVA 23, and why was the earlier warning missed? What does this suggest about anomaly handling?
10. List three differences between designing a suit for the ISS and one for Mars, and explain how a suitport would help a Martian rover crew.

PART X

# Launch Systems and Infrastructure

Everything in space started on a launch pad. Part X explains launch vehicle engineering and rocket propulsion, surveys the world's launch vehicles including the new generation of reusable rockets, and describes spaceports and the ground systems that support them.


34 Launch Vehicle Engineering
35 Rocket Propulsion
36 Launch Vehicles of the World
37 Spaceports and Ground Infrastructure


Image: NASA (NASA/Joel Kowsky)

# 34 CHAPTER Launch Vehicle Engineering

Image: NASA/KSC (NASA/Kim Shiflett)

*A launch vehicle must accelerate its payload to nearly 8 km/s in under ten minutes while carrying all of its own propellant and surviving aerodynamic loads, noise, vibration and heat — reliably and cheaply. This chapter explains how engineers meet that challenge. It builds the velocity budget and shows how launch latitude changes it, derives the rocket equation and the logic of staging, follows a simulated two-stage kerosene vehicle from the pad to orbit, and describes tanks, pressurisation, guidance and flight termination. It then turns to the environments a payload must survive, the payload penalty of landing a booster, the economics of launch, and what seventy years of launch statistics reveal about reliability.*

IN THIS CHAPTER

34.1 Getting to Orbit
34.2 Staging
34.3 The Ascent Trajectory
34.4 Structures and Materials
34.5 Avionics, Guidance, Navigation and Control
34.6 The Payload's Ride
34.7 Reusability
34.8 Reliability and Testing

A rocket standing on the pad is mostly propellant. A Falcon 9 weighs about 549 t at lift-off, almost 490 t of it liquid oxygen and kerosene; the satellites on top make up only 3–4 % of the total. Within about eight minutes it must lift its payload above the atmosphere and accelerate it to 7.8 km/s. Launch is the shortest phase of most missions and the most violent: the payload meets its highest loads, loudest noise and strongest vibration in its first minutes of flight.

To keep the numbers concrete, this chapter follows a **reference vehicle**. It is a planar trajectory simulation of a two-stage kerosene rocket with Falcon 9-class masses and engines, flown from Cape Canaveral to a 200 km orbit (Figure 34.5). The engines themselves are the subject of Chapter

35, the world's launch vehicles of Chapter 36, and launch sites and ground systems of Chapter 37; the orbital mechanics used here is developed in Chapter 3.

# 34.1 Getting to Orbit

## 34.1.1 The Velocity Budget

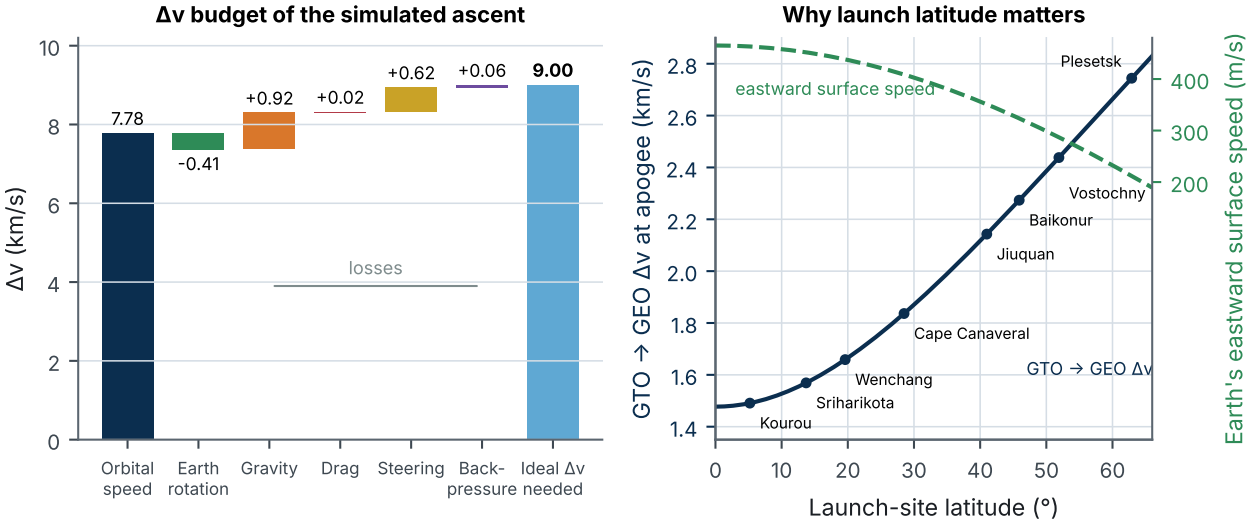
A satellite in a circular orbit 200 km above Earth moves at 7.78 km/s. Each kilogram carries about 30 MJ of kinetic energy and almost 2 MJ of potential energy. The launch vehicle supplies it by throwing propellant backwards, and the measure of what it can do is the **velocity change** Δv it can give itself (Figure 34.1).

### THE ROCKET EQUATION

A rocket of mass $m$ ejects propellant at speed $v_e$ relative to itself. When it expels a small mass $-dm$, conservation of momentum gives $m\,dv = -v_e\,dm$. Integrating from the initial mass $m_0$ to the final mass $m_f$:

$$\Delta v = v_e \ln \frac{m_0}{m_f} = I_{sp}\, g_0 \ln \frac{m_0}{m_f}$$

where $I_{sp}$ is the specific impulse (Chapter 35) and $g_0$ = 9.806 65 m/s². The ratio $m_0/m_f$ is the **mass ratio**. The equation, published by Konstantin Tsiolkovsky in 1903, gives the velocity change in empty space, with no gravity or air.




Left: losses integrated along the simulated ascent of the 4-panel ascent figure (ideal Δv = Σ $v_e$ ln($m_0$/$m_f$), vacuum Isp).
Right: GTO perigee 200 km, inclination = site latitude; the apogee burn circularises and removes the inclination. Surface speed = 465 m/s × cos(latitude).

**FIGURE 34.1** Left: Δv budget of the simulated reference ascent to a 200 km circular orbit from Cape Canaveral. Earth's rotation reduces the requirement by 0.41 km/s; gravity and steering losses add about 1.5 km/s. Right: how launch-site latitude changes the eastward surface speed and the Δv a satellite needs to turn a geostationary transfer orbit into a geostationary one.

Four **losses** separate this ideal Δv from the final orbital speed:

- **Gravity loss**, $\int g \sin\gamma\, dt$, where γ is the flight-path angle above the horizontal: while the rocket climbs, part of its thrust only holds it up. A rocket that hovered would burn all its propellant and gain nothing.
- **Drag loss**, $\int D/m\, dt$, the momentum given to the air.

- **Steering loss**, $\int (T/m)(1-\cos\alpha)\,dt$, when the thrust $T$ points at an angle α away from the velocity.
- **Back-pressure loss**: a nozzle produces less thrust in the atmosphere than in a vacuum (Chapter 35).

Earth's rotation helps, because the pad already moves east:

$$\Delta v_{ideal} = v_{orbit} - v_{rotation} + \Delta v_{gravity} + \Delta v_{drag} + \Delta v_{steering} + \Delta v_{pressure}$$

Figure 34.1 shows the terms for the reference vehicle, integrated along its simulated trajectory: it needs an ideal Δv of 9.00 km/s. Gravity loss (0.92 km/s) is the largest. Steering loss (0.62 km/s) is almost as large. At ignition the second stage's thrust is less than its weight, so it must point its nose above the flight path to avoid falling back while it gains speed. It trades gravity loss for steering loss. Drag costs only 0.02 km/s, because a large, dense vehicle has so much mass per unit of frontal area; small launchers lose more, often 0.1–0.2 km/s. Without the 0.41 km/s gift of Earth's rotation the reference vehicle would need 9.4 km/s, consistent with the usual rule of thumb of 9.3–9.6 km/s to low Earth orbit (LEO).

### 34.1.2 Latitude and Inclination

The ground at latitude ϕ moves east at 465 m/s × cos ϕ. A rocket launched on azimuth $Az$ (measured from north) into an orbit of inclination $i$ must satisfy sin $Az$ = cos $i$ / cos ϕ. The useful part of the rotation it inherits works out to 465 m/s × cos $i$. It depends only on the orbit, not the site. A launch due east gives the lowest inclination reachable directly, $i$ = ϕ. A launch to the International Space Station (51.6°) gains 289 m/s; a polar orbit gains nothing; a sun-synchronous orbit, inclined about 97–98°, is slightly retrograde and costs about 60 m/s extra.

Reaching an inclination *lower* than the site latitude needs a **dog-leg** (steering sideways during ascent) or a plane change in orbit, both expensive: at orbital speed each degree costs 2$v$ sin 0.5° ≈ 136 m/s. Latitude therefore matters most for geostationary missions (Table 34.1). A satellite released into a geostationary transfer orbit (GTO) inclined at the site latitude must remove that inclination at apogee, where it moves slowly and plane changes are cheapest, while circularising. From Kourou (5.2° N) this costs 1.49 km/s; from Cape Canaveral (28.5° N), 1.84 km/s; from Baikonur (45.9° N), 2.27 km/s. For a satellite with a 320 s apogee engine, the 0.35 km/s difference between Kourou and the Cape raises the propellant share of its mass from 38 % to 44 %. Alternatives are a "supersynchronous" transfer, whose higher apogee makes the plane change cheaper, or direct injection into geostationary orbit by the upper stage.

**TABLE 34.1** Latitude effects at selected launch sites. GTO → GEO Δv assumes a 200 km perigee and an inclination equal to the site latitude.

| Launch site | Latitude | Eastward surface speed | GTO → GEO Δv |
|---|---|---|---|
| Kourou (Guiana Space Centre) | 5.2° N | 463 m/s | 1.49 km/s |
| Sriharikota (India) | 13.7° N | 452 m/s | 1.57 km/s |
| Wenchang (China) | 19.6° N | 438 m/s | 1.66 km/s |
| Cape Canaveral / Kennedy | 28.5° N | 409 m/s | 1.84 km/s |
| Mahia (New Zealand) | 39.3° S | 360 m/s | 2.10 km/s |
| Baikonur (Kazakhstan) | 45.9° N | 324 m/s | 2.27 km/s |
| Plesetsk (Russia) | 62.9° N | 212 m/s | 2.74 km/s |

**EXAMPLE: THE ΔV BUDGET TO A 500 KM, 53° ORBIT FROM CAPE CANAVERAL**

1. Launch azimuth: $\sin Az = \cos 53° / \cos 28.5° = 0.602 / 0.879 = 0.685$, so $Az = 43°$ — the vehicle flies north-east along the US coast.
2. Earth-rotation gain: $465 \times \cos 53° = 280$ m/s, instead of 409 m/s for a due-east launch.
3. Insertion into a 200 km parking orbit (7 784 m/s), then a Hohmann transfer to 500 km: 86 m/s to raise apogee plus 85 m/s to circularise (Chapter 3).
4. Ascent losses from Figure 34.1: 0.92 + 0.62 + 0.02 + 0.06 = 1.62 km/s.
5. Ideal Δv = 7 784 − 280 + 1 620 + 172 ≈ 9 300 m/s. This is 0.30 km/s more than the due-east 200 km orbit — about 2.3 t of droneship payload in the model of Figure 34.14. This is one reason constellations are often deployed low and raise their own orbits with electric propulsion.

## 34.2 Staging

### 34.2.1 Why One Stage Is Not Enough

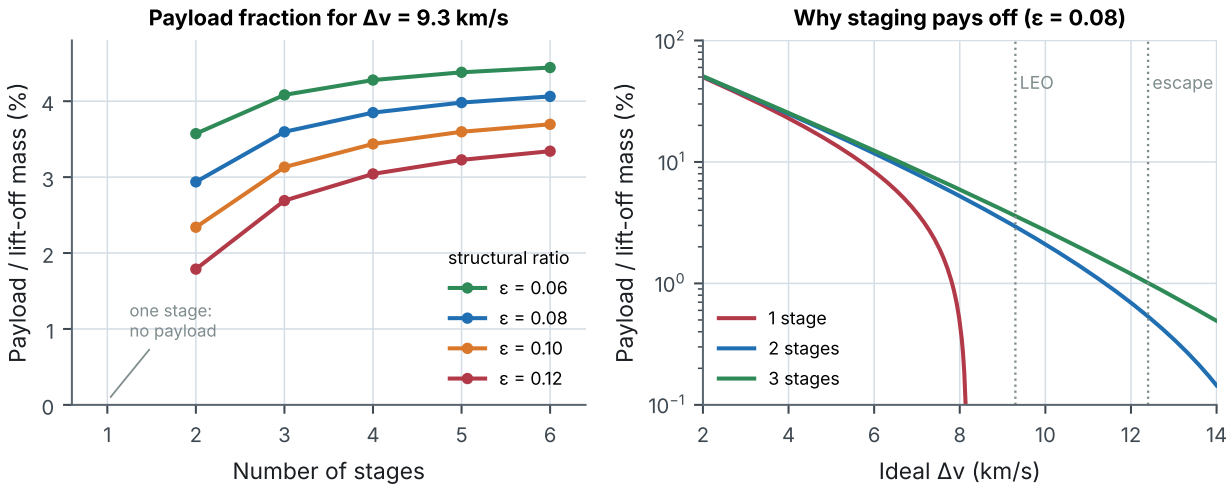
Every stage carries structure — tanks, engines, plumbing, avionics — that becomes dead weight once its propellant is gone. It is described by the **structural ratio** $\varepsilon = m_s/(m_s + m_p)$, the structure's share of a loaded stage; good modern stages achieve $\varepsilon \approx 0.05$–$0.10$. A single stage with an average specific impulse of 330 s that must deliver 9.3 km/s needs a mass ratio of $e^{9300/3236} = 17.7$: only 5.6 % of the lift-off mass may remain at burnout. If structure alone takes 8 %, the rocket cannot reach orbit even empty. This is the fundamental difficulty of **single-stage-to-orbit** vehicles, studied for decades but never flown to orbit (Chapter 10).

### 34.2.2 The Rocket Equation for Several Stages

Staging throws structure away as soon as it is empty. Each stage $k$ contributes $\Delta v_k = v_{e,k} \ln(m_{0,k}/m_{f,k})$, where the "payload" of stage $k$ is everything above it, and the contributions add. For $n$ identical stages the optimum is an equal split of Δv, and the overall payload fraction is

$$\frac{m_{payload}}{m_0} = \left( \frac{e^{-\Delta v/(n v_e)} - \varepsilon}{1 - \varepsilon} \right)^n$$

Figure 34.2 evaluates this formula. Going from one stage to two turns an impossible vehicle into one that delivers 3 % of its lift-off mass. A third stage adds about a fifth more. Beyond that the gains are small, while every extra stage adds engines, a separation event, an in-flight ignition and cost. Most modern launchers therefore have two stages. Real stages are not identical. Hydrogen upper stages (Isp near 450 s) take a larger share of the Δv. A reusable first stage is given a *smaller* share, so that it separates slowly and close to the launch site.




Tsiolkovsky equation for identical stages (Isp = 330 s for every stage; structural ratio ε = structure /(structure + propellant)); equal Δv split, which is optimal for identical stages.

**FIGURE 34.2** Left: payload fraction for a 9.3 km/s mission as a function of the number of identical stages and their structural ratio. Right: payload fraction versus total Δv for one, two and three stages with ε = 0.08; a single stage runs out of capability just above 8 km/s.

**EXAMPLE: TWO STAGES VERSUS ONE**

Take Isp = 330 s ($v_e$ = 3 236 m/s), Δv = 9.3 km/s and a lift-off mass of 550 t.

1. One stage, ε = 0.08: the final-to-initial mass ratio is $e^{-9300/3236}$ = 0.0565, less than ε, so no payload is possible. Even with an exceptionally light structure (ε = 0.05) the payload fraction is (0.0565 − 0.05)/0.95 = 0.7 %, or 3.8 t.
2. Two equal stages, ε = 0.08: each provides 4.65 km/s, so $e^{-4650/3236}$ = 0.238. Each stage's payload ratio is (0.238 − 0.08)/0.92 = 0.171, and the overall fraction is $0.171^2$ = 2.9 %, or **16 t**.
3. With ε = 0.06 this rises to 3.6 % (19.6 t); with ε = 0.10 it falls to 2.3 % (12.9 t). Each percentage point of structural ratio is worth about 1.7 t of payload, a tenth of the total, which is why engineers fight for every kilogram of tank wall.

## 34.2.3 Serial, Parallel and Cross-Feed

In **serial** staging the stages sit on top of one another and fire in sequence, as in the Saturn V (Figure 34.3) and Falcon 9. In **parallel** staging, boosters strapped around a core ignite on the ground and drop away first. The R-7 of 1957, ancestor of today's Soyuz, uses four, and Ariane 6 uses two or four solid boosters. The Space Launch System (SLS) uses two, and Falcon Heavy two Falcon 9 first stages. Parallel staging avoids lighting engines in flight, but the core must carry side loads. The Space Shuttle's orbiter engines burned propellant from an external tank that was discarded just short of orbit (Figure 34.4). **Cross-feed** means pumping propellant from the boosters into the core, so that its tanks are still full when they separate. It would give serial-staging performance with parallel hardware. It was proposed for Falcon Heavy but has not flown.

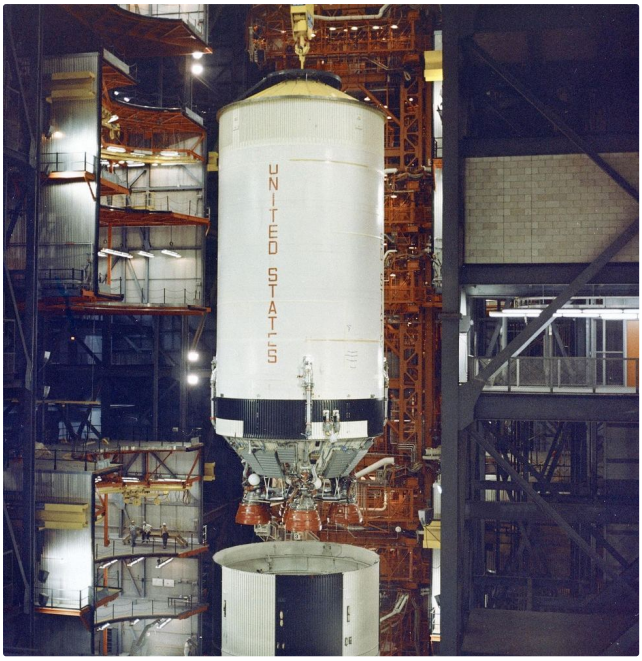



**FIGURE 34.3** The second stage (S-II) of the Saturn V for Apollo 6 being lowered onto the first stage in the Vehicle Assembly Building in 1967. Serial staging stacks each stage on top of the previous one. Image: NASA/MSFC.

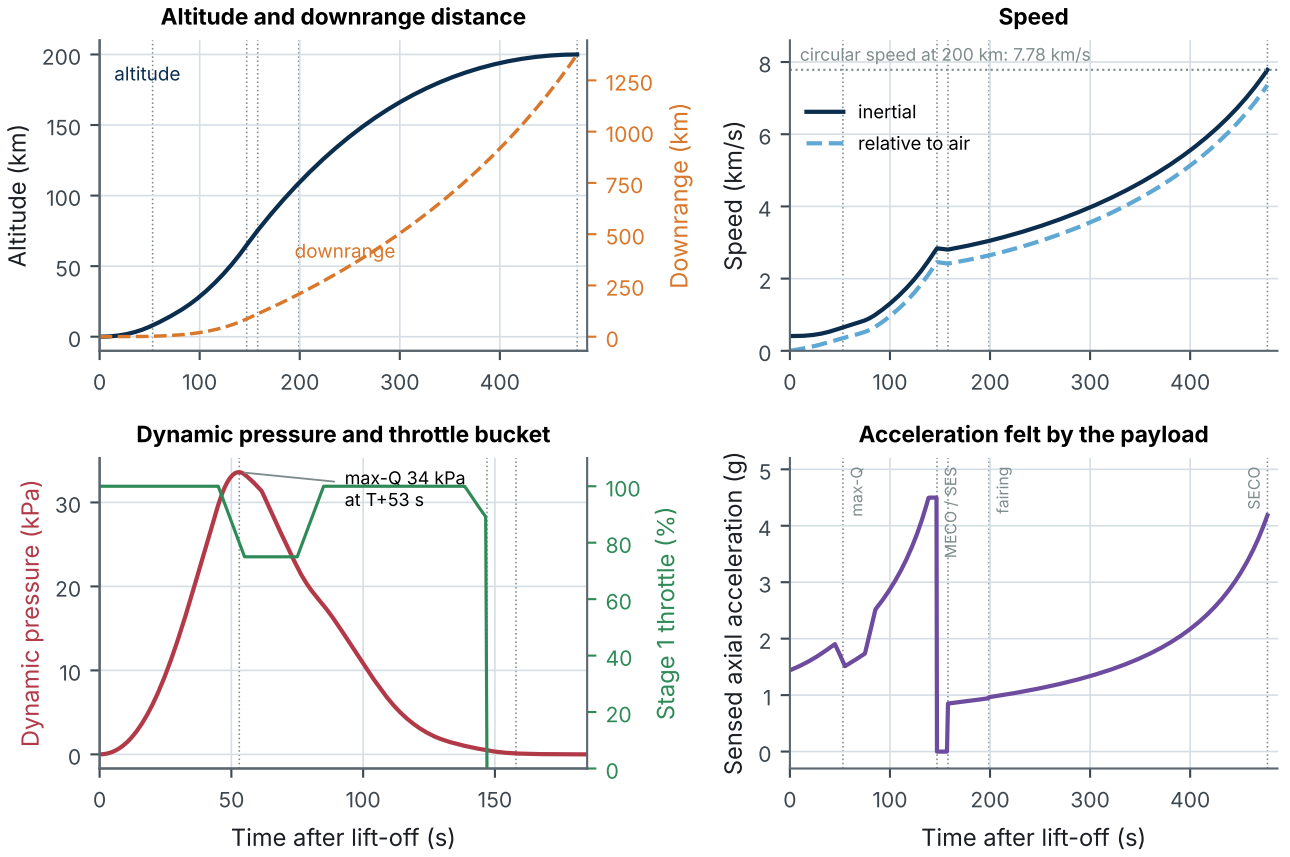
**FIGURE 34.4** The Space Shuttle's external tank falling away after separation in March 1989 (STS-29), photographed from Discovery. The tank fed the orbiter's main engines and was discarded just short of orbit. Image: NASA/JSC.

## 34.3 The Ascent Trajectory

### 34.3.1 Lift-Off, Pitch-Over and the Gravity Turn

A liquid-fuelled vehicle ignites its engines a few seconds before release and is held down while its computers confirm that every engine is running normally; if not, it shuts down safely on the pad. At release its thrust-to-weight ratio is typically 1.2–1.5 (1.44 for the reference vehicle). It rises vertically to clear the tower, rolls to its launch azimuth, then makes a small **pitch-over** — 4.8° between 8 and 18 s in the simulation. In a **gravity turn** the vehicle keeps its nose along the airflow, at nearly zero **angle of attack** α. Gravity gradually bends the path towards the horizontal. This keeps the aerodynamic side force and the bending of the long, thin structure small. The pitch-over sets the whole first-stage trajectory: too small and the rocket climbs too steeply; too large and it stays too long in thick air. The simulation chooses it to maximise payload.

**Dynamic pressure**, $q = \frac{1}{2}\rho v^2$, rises as the vehicle accelerates and falls as the air thins; its peak, **max-Q**, is typically 25–40 kPa. In the simulation it reaches 34 kPa at T+53 s, 8 km up, just after Mach 1; the Falcon 9 user's guide lists max-Q at T+67–74 s. The engines are throttled through a **throttle bucket** (75 % in the simulation) to limit it. What matters most for the structure is the product **qα**: a gust or **wind shear** at high dynamic pressure forces a non-zero angle of attack and bends the vehicle. Balloons and wind profilers measure the upper winds before each launch, and the trajectory is "wind-biased" and the loads recomputed to confirm that they stay within limits.




Planar simulation, Falcon 9-class vehicle (stage 1: 25.6 t dry, 395.7 t propellant, 7.6–8.2 MN, Isp 283–312 s; stage 2: 3.9 t dry, 92.7 t, 981 kN, 348 s; fairing 1.9 t), 19.1 t payload to a 200 km circular orbit from 28.5° N with 22 t of first-stage propellant kept for a droneship landing; US 1976 atmosphere, rotating Earth, 4.5 g limit.

**FIGURE 34.5** Simulated ascent of the reference vehicle to a 200 km orbit with a droneship booster recovery. Altitude climbs steeply while speed builds slowly; max-Q is limited by throttling the first stage; the sensed acceleration rises as propellant burns off, is capped at 4.5 g, and drops to zero during the coast between staging and second-stage ignition.

As propellant burns off, acceleration grows (Figure 34.5) from 1.4 g towards 4–5 g, where vehicles cap it by throttling or shutting down engines.

## 34.3.2 Staging, Fairing Jettison and the Upper Stage

At **main-engine cut-off** (MECO), 147 s after lift-off, the reference first stage is 65 km up and moving at 2.5 km/s relative to the air. Pneumatic pushers separate the stages, and the second stage coasts briefly while its propellant settles before igniting. The **payload fairing** is jettisoned once the aerodynamic heating it would let through is negligible. Falcon releases it when the **free-molecular heating** rate, $\tfrac{1}{2}\rho v^3$, falls below 1 135 W/m$^2$. In the simulation this happens at T+199 s and about 109 km.

The upper stage then flies above the atmosphere under closed-loop guidance. Its thrust-to-weight ratio is only 0.85 at ignition, so it pitches up to hold altitude while it accelerates; by **second-engine cut-off** (SECO) at T+477 s its acceleration has climbed above 4 g. For higher orbits the stage usually stops in a low **parking orbit**, coasts, and restarts to raise the apogee (Table 34.2). Restarting after a long coast requires settling the propellant with small thrusters and keeping engine and propellants within temperature limits. Afterwards a responsible stage performs a **deorbit burn** or vents its propellant and pressurant to “passivate” itself, so that it cannot explode and add to the debris population (Chapter 21).

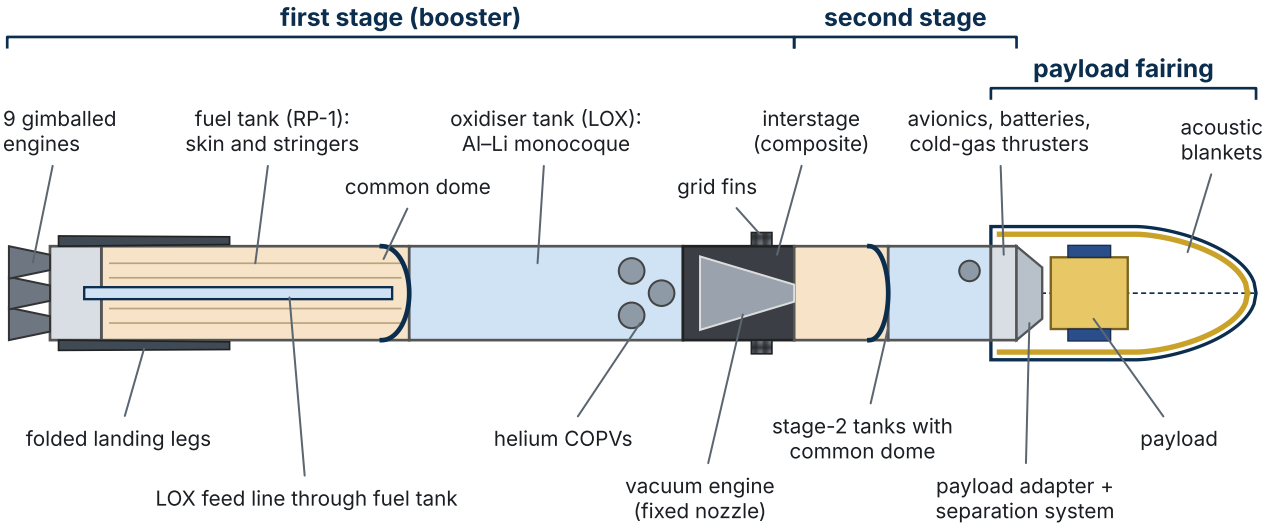
**TABLE 34.2** Sample Falcon 9 flight timelines from the Falcon User's Guide (May 2025). Every mission's profile is different.

| Event | LEO mission | GTO mission |
|---|---|---|
| Lift-off | T+0 | T+0 |
| Maximum dynamic pressure | T+67 s | T+74 s |
| First-stage main-engine cut-off | T+145 s | T+147 s |
| Stage separation | T+148 s | T+151 s |
| Second-stage ignition (1) | T+156 s | T+158 s |
| Fairing separation | T+195 s | T+222 s |
| Second-stage cut-off (1) | T+514 s | T+484 s |
| Second-stage ignition (2) | T+3 086 s | T+1 636 s |
| Second-stage cut-off (2) | T+3 090 s | T+1 696 s |
| Spacecraft separation | T+3 390 s | T+1 996 s |

# 34.4 Structures and Materials

## 34.4.1 Tanks, Domes and Stiffeners

A launch vehicle is essentially a stack of thin-walled pressure vessels with engines at one end and a payload at the other (Figure 34.6). A first stage 3.7 m across has tank walls only a few millimetres thick, and structure is typically 5–8 % of its loaded mass.




Schematic of a generic two-stage kerosene vehicle, loosely based on Falcon 9 (diameter exaggerated about ×1.5; length ≈ 70 m).

**FIGURE 34.6** Anatomy of a two-stage kerosene launch vehicle. The first stage carries its oxygen tank above its fuel tank, with a common dome between them and a feed line running through the fuel tank; the composite interstage houses the second stage's vacuum engine; the fairing, lined with acoustic blankets, encloses the payload on its adapter.

**Aluminium alloys** have been the standard since the 1950s (Table 34.3). The Saturn V's first stage, the original Shuttle external tank and the SLS core stage use alloy 2219, which welds well and stays tough at cryogenic temperatures. **Aluminium–lithium** alloys are lighter and stiffer. The Shuttle's Super Lightweight Tank, first flown on STS-91 in June 1998, used alloy 2195, about 30 % stronger and 5 % less dense than 2219. It weighed 3 400 kg less than its predecessor. Falcon 9's

tanks are aluminium–lithium joined by **friction-stir welding**, in which a rotating tool stirs the metal together below its melting point. Al–Li is harder to weld and less tolerant of flaws, and SLS returned to 2219.

**Stainless steel**, used in the balloon tanks of Atlas and Centaur, has returned in SpaceX's Starship. It is heavier but cheap, easy to weld, stronger when cold, and keeps its strength at several hundred degrees Celsius — an advantage for a stage that must survive re-entry. **Carbon-fibre composites** promise the lightest tanks. Rocket Lab's Electron uses them throughout, and its larger Neutron is built around composite tanks. However, voids found in a composite tank dome helped push Neutron's first launch into 2027. Composite tanks must resist micro-cracking, leakage and, for oxygen, ignition.

**TABLE 34.3** Tank and structure materials. Strengths are typical room-temperature tensile values; the metals are stronger at cryogenic temperature, and composite values depend strongly on the lay-up.

| Material | Density (g/cm³) | Typical strength (MPa) | Strength / density vs 2219 | Examples |
|---|---|---|---|---|
| Aluminium 2219 | 2.84 | 420–460 | 1.0 | Saturn V S-IC, SLS core stage |
| Aluminium–lithium (e.g. 2195) | 2.71 | 550–600 | ≈ 1.4 | Shuttle SLWT (2195), Falcon 9 |
| Stainless steel (300 series) | 7.9 | 500–600 (more when cold) | ≈ 0.5 | Atlas, Centaur, Starship |
| Carbon-fibre/epoxy laminate | 1.55–1.6 | 600–1 000 | ≈ 3 | Electron, Neutron, interstages |

A **common bulkhead** — a single dome separating the two propellants — saves the length and mass of a second dome. Falcon 9 uses one, as did both upper stages of the Saturn V, where the S-II's bulkhead had to be insulated to stop −183 °C oxygen freezing against −253 °C hydrogen. Walls in compression must resist **buckling**, so they are stiffened. Some use stringers attached to a thin skin (**skin-and-stringer**, as in Falcon 9's fuel tank). Others use ribs machined out of thick plate in a triangular **isogrid** or rectangular **orthogrid** pattern (Figure 34.7), which removes most of the metal but leaves a light, damage-tolerant structure without fasteners.

**FIGURE 34.7** A completed liquid-hydrogen tank barrel for the SLS core stage lifted off the Vertical Weld Center at NASA's Michoud Assembly Facility in February 2025. The orthogrid stiffening machined into the inner surface is visible. Image: NASA/MAF (NASA/Evan Deroche).

**FIGURE 34.8** The SLS liquid-hydrogen tank structural test article being placed in a test stand at NASA Marshall in January 2019. In December 2019 this test article was loaded to more than 260 % of its expected flight loads before it buckled and ruptured. Image: NASA/MSFC (Tyler Martin).

### 34.4.2 Pressurisation

Tanks are pressurised to push propellant into the turbopumps without cavitation and to stiffen the walls, so that internal pressure carries part of the compressive and bending loads. **Balloon tanks** such as those of Atlas had no stiffeners at all and had to stay pressurised even on the ground. The pressurant is usually helium, stored at very high pressure in **composite-overwrapped pressure vessels** (COPVs). These are often submerged in the cold liquid oxygen, where the gas is denser. Falcon heats the helium before it enters the tanks, so that less is needed. Failures of these systems destroyed two Falcon 9s (Table 34.6). **Autogenous** pressurisation avoids helium altogether by feeding warmed propellant gas back into the tanks, as Starship does with methane and oxygen.

### 34.4.3 Loads, Vibration and POGO

**Axial compression** peaks at the highest acceleration, just before first-stage cut-off; **bending** peaks around max-qα and in ground winds on the pad; internal pressure sizes the tank walls; and **acoustic** loads shake panels and equipment. A dangerous coupling is **POGO**, named after the pogo stick: a longitudinal vibration changes the pressure at the pump inlets, which changes the thrust, which reinforces the vibration. On Apollo 13 in April 1970, POGO oscillations shut down the centre engine of the Saturn V's second stage about two minutes early. The cure is to detune the feed system with gas-filled **accumulators** in the propellant lines; baffles damp propellant **slosh**.

Structures are designed with **factors of safety**, typically 1.1 on yield and 1.25–1.4 on ultimate strength, and proven by test, sometimes to destruction (Figure 34.8). Wind tunnels still provide the aerodynamic loads. In 2025 NASA's Ames Research Center tested a 1.2 % scale model of the Super Heavy Version 3 booster. The tests measured the forces on it during its return through the atmosphere (Figure 34.10).

### 34.4.4 Fairings and Adapters

The **payload fairing** is a two-piece shell, usually carbon-fibre composite, lined with acoustic blankets; Falcon's is 5.2 m in diameter and 13.2 m tall. Some fairings are split by pyrotechnic **frangible joints**, which fracture a grooved structural seam (Figure 34.9). Falcon's standard fairing releases mechanical latches and pushes the halves apart pneumatically. This system can be tested before flight and produces little shock or debris. Below the payload, a **payload adapter** carries loads into the upper stage and holds the **separation system**. This is typically a **clamp band** of standard diameter (937, 1 194 or 1 666 mm on Falcon). It is released by cutting its bolts, and springs push the spacecraft away.

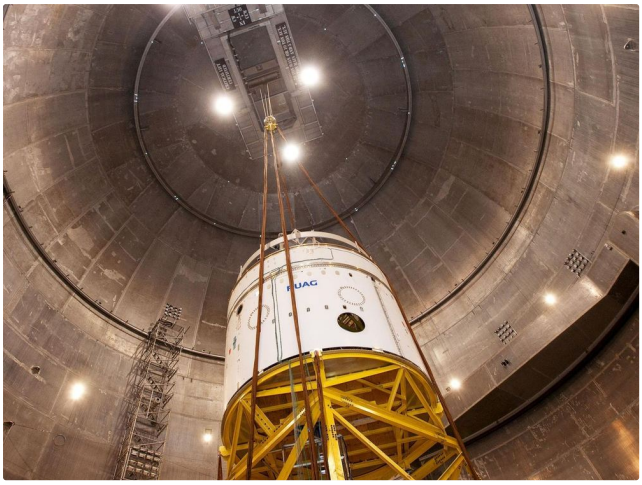


**FIGURE 34.9** An Ariane 5 payload fairing in NASA's Space Power Facility at Plum Brook, Ohio, in 2012, before a full-scale test of a new horizontal pyrotechnic separation system that blew the two halves apart. Image: NASA/GRC.

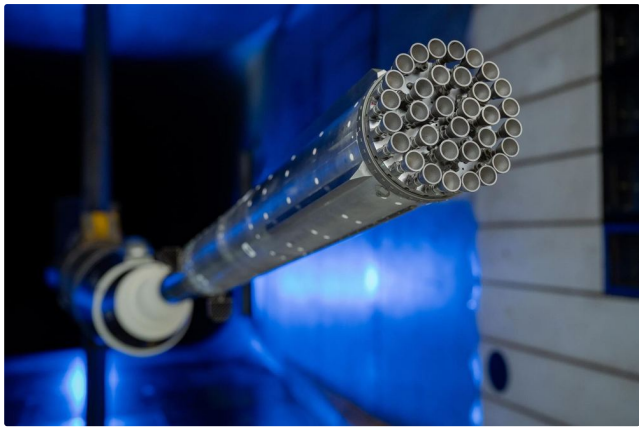


**FIGURE 34.10** A 1.2 % scale model of the Super Heavy Version 3 booster in a wind tunnel at NASA's Ames Research Center in 2025, used to measure the aerodynamic forces during re-entry. Image: NASA/MSFC (NASA/Brandon Torres).

## 34.5 Avionics, Guidance, Navigation and Control

### 34.5.1 Navigation and Guidance

**Navigation** rests on an **inertial measurement unit** (IMU): gyroscopes, typically ring-laser or fibre-optic, measure rotation, and accelerometers measure specific force; integrating them gives attitude, velocity and position without outside help. Its errors grow with time, so it is blended with satellite navigation (Chapter 18) in a **Kalman filter**.

**Guidance** decides where to steer. In the dense atmosphere most vehicles fly a pre-computed, **open-loop** pitch programme that keeps α near zero. Above it, **closed-loop** guidance recomputes, many times a second, the steering needed to reach the target orbit from the current state. The Saturn V's Iterative Guidance Mode and the Shuttle's **Powered Explicit Guidance** both rest on one result. In uniform gravity, the fuel-optimal steering law is the **linear-tangent law**, in which the tangent of the pitch angle varies linearly with time. The reference simulation uses this law for its second stage, solving for the two coefficients that place it on a 200 km circular orbit.

### 34.5.2 Control and Redundancy

Liquid engines are mounted on **gimbals** and swivelled by a few degrees for **thrust vector control** (TVC). Actuators are hydraulic or, increasingly, electromechanical. Falcon's first-stage actuators use high-pressure kerosene from the engine as hydraulic fluid, removing a separate hydraulic system. The solid first stage of Vega-C and Starship's Raptors use electric actuators. Single-engine stages control roll with small thrusters, such as Falcon's nitrogen cold-gas jets. Most rockets are aerodynamically unstable — their centre of pressure lies ahead of the centre of gravity — so the control loop must actively hold them straight without reacting to structural vibration or propellant slosh. Returning boosters steer with **grid fins**, lattice-like surfaces that work at hypersonic speed and fold flat during ascent.

Avionics are **fault tolerant**: Falcon uses a three-string architecture that votes between triplicated computers and sensors. Several engines give propulsion redundancy — Falcon 9 can shut down a misbehaving first-stage engine and burn the others longer (Chapter 35). A study by The Aerospace Corporation is cited in the Falcon user's guide. It found that 91 % of launch failures in the two decades before 2001 were caused by engines, avionics or stage separation. These are the areas where redundancy and simplicity pay off most.

### 34.5.3 Range Safety and Flight Termination

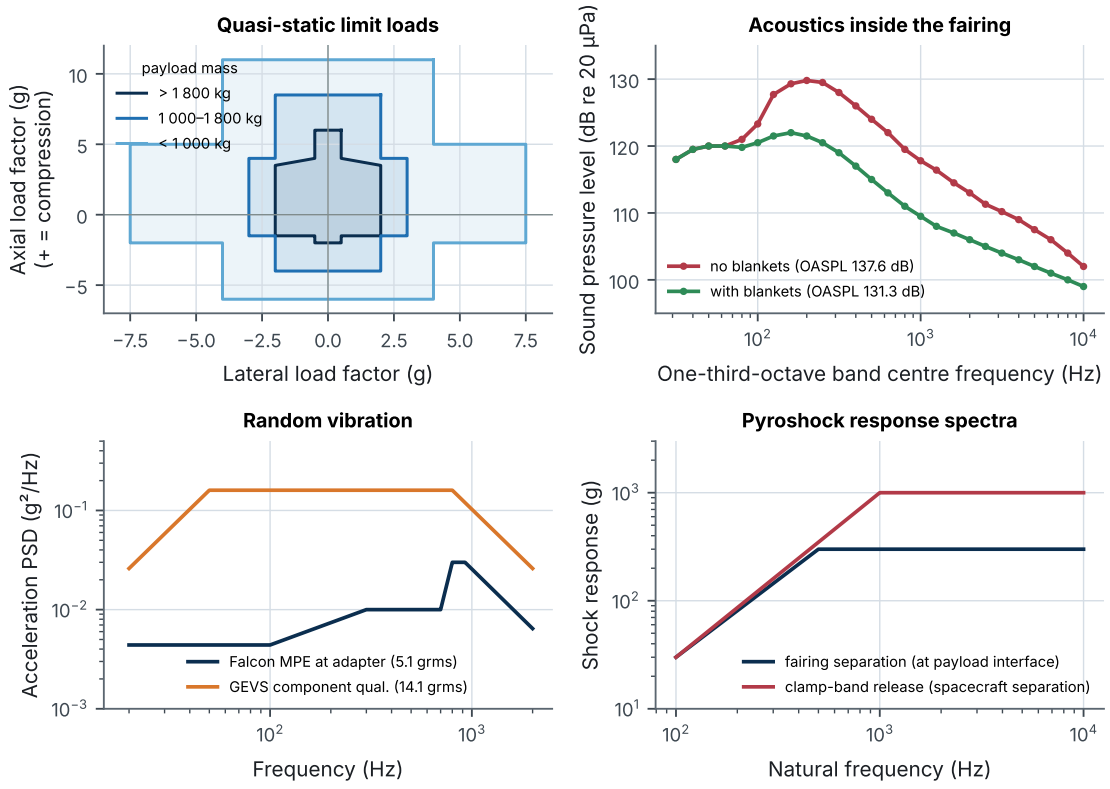
A rocket that strays threatens people on the ground. Ranges compute its **instantaneous impact point** — where it would land if thrust stopped — in real time. Every vehicle carries a **flight termination system**. Its charges cut open the tanks to disperse the propellant before it can reach populated areas. For decades a range safety officer watched radar and telemetry and sent the destruct command by radio. An **autonomous flight safety system** runs the same rules on board, using the vehicle's own GNSS and inertial navigation; its first operational use was on Falcon 9's CRS-10 launch on 19 February 2017. It removes tracking radars and ground crews and allows launches in quick succession. In the United States the Federal Aviation Administration licenses commercial launches under 14 CFR Part 450, which caps the collective risk to the public at an expected $1 \times 10^{-4}$ casualties per launch.

## 34.6 The Payload's Ride

### 34.6.1 Environments

The **launch vehicle user's guide** defines the environments a satellite must survive and the interfaces it must meet. Figure 34.11 and Table 34.4 summarise those of Falcon 9, taken from SpaceX's guide.

- **Quasi-static loads** combine steady acceleration and low-frequency vibration into load factors: up to +6 g axially and ±2 g sideways for satellites heavier than 1 800 kg, more for lighter ones.
- **Sine vibration** (5–100 Hz) comes from engine transients and structural modes; to avoid coupling, Falcon asks for payload fundamental frequencies above 25 Hz axially and 10 Hz laterally.
- **Acoustic** noise inside the fairing, from engine exhaust at lift-off and turbulent airflow in transonic flight, reaches 131 dB overall with acoustic blankets and 138 dB without. Solar arrays and antennas suffer most.
- **Random vibration** through the adapter (5.1 grms) matters mainly for small components. Fairing and spacecraft separation cause **shock** of up to 1 000 g at kilohertz frequencies. This can break crystals, relays and solder joints near the source.
- The payload must also tolerate heating, rapid depressurisation and contamination; it is processed in clean rooms (ISO class 8 on Falcon) and encapsulated before transport to the pad (Figure 34.12).




Data: SpaceX Falcon User's Guide (May 2025) — limit loads, P95/50 acoustic MPE for the standard fairing (Eastern Range, 60 % fill), random-vibration MPE at the top of the payload adapter, shock response spectra; GEVS = NASA GSFC-STD-7000B generalised qualification level for components up to 22.7 kg. Qualification adds 25 % (loads) or 3 dB (acoustics, vibration).

**FIGURE 34.11** Payload environments of Falcon 9, from the Falcon User's Guide (May 2025): quasi-static limit-load envelopes for three payload-mass classes; acoustic spectra inside the fairing with and without blankets; random vibration at the payload adapter compared with NASA's generalised component test level; and shock response spectra for fairing and spacecraft separation.

**TABLE 34.4** Falcon 9 payload environments and the corresponding qualification tests (Falcon User's Guide, May 2025). MPE = maximum predicted environment.

| Environment | Falcon 9 maximum predicted value | Typical verification test |
|---|---|---|
| Axial quasi-static load (> 1 800 kg) | +6.0 g compression, −2.0 g tension | Static load or sine burst, 1.25 × limit |
| Lateral quasi-static load (> 1 800 kg) | ±2.0 g | As above |
| Sine vibration | 0.5–1.0 g, 5–100 Hz | Sine sweep, 1.25 × limit |
| Acoustic (with blankets) | 131.4 dB overall | Reverberant chamber, MPE + 3 dB |
| Random vibration at adapter | 5.13 grms, 20–2 000 Hz | Shaker, MPE + 3 dB |
| Separation shock (clamp band) | up to 1 000 g, 1–10 kHz | Separation or pyroshock test |
| Fairing jettison | heating < 1 135 W/m² | Thermal analysis |

Satellites are verified against these environments with margin. A **qualification** test, on a unit that will not fly, exceeds the predicted loads by 25 % or the acoustic and vibration levels by 3 dB. An **acceptance** test applies the predicted levels to flight hardware to reveal workmanship defects. Many programmes use **protoflight** testing, at qualification levels for a shorter time on the flight unit (Chapter 4). Late in development the launch provider runs a **coupled loads analysis**, combining finite-element models of rocket and satellite to predict the loads at every interface.

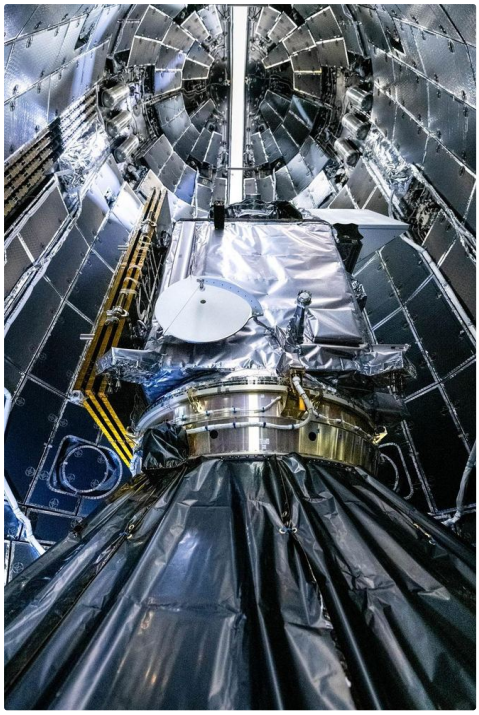
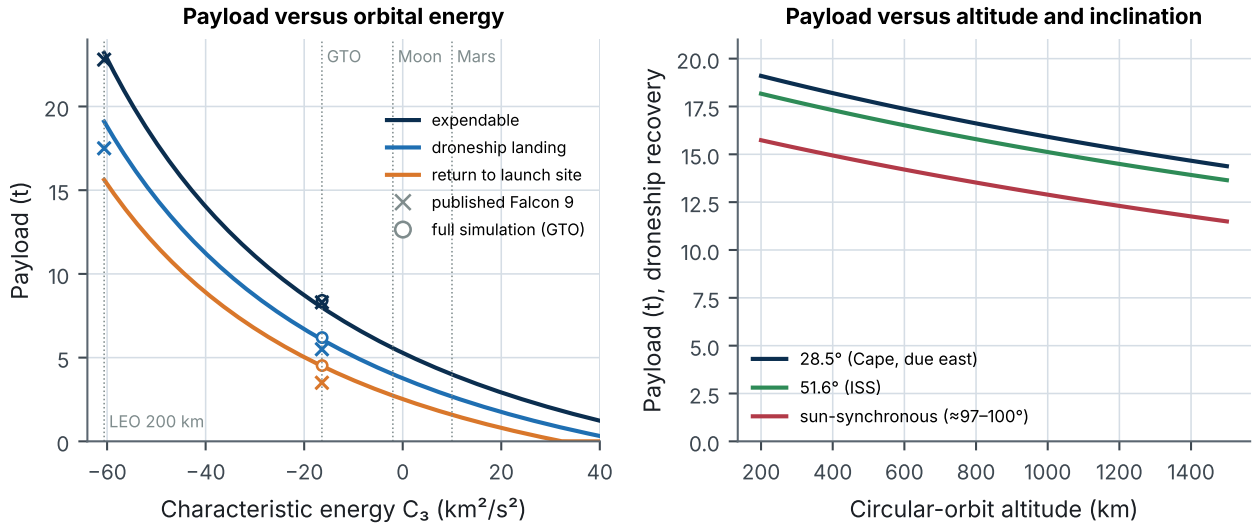


**FIGURE 34.12** NASA's PACE Earth-observation satellite being encapsulated in a Falcon 9 fairing in January 2024. The work is done in a clean room to protect the instruments from contamination. Image: NASA/KSC (NASA Goddard Katie Mellos).

**FIGURE 34.13** The Orion capsule that flew around the Moon on Artemis I, now an environmental test article, during vibro-acoustic testing at NASA's Armstrong Test Facility, Ohio, in 2024. Image: NASA/GRC (NASA/Sara Lowthian-Hanna).

### 34.6.2 Reading a Performance Curve

The other key part of a user's guide is the **performance curve** — payload against target orbit. Figure 34.14 shows the reference vehicle's, from a rocket-equation model calibrated to the trajectory simulation. For orbits reaching far from Earth the natural measure is the **characteristic energy** $C_3 = v_\infty^2$ (Chapter 24). It is negative for bound orbits (−60.6 km²/s² for a 200 km circular orbit, −16.4 km²/s² for GTO) and zero for escape. It is about 10 km²/s² for a typical trip to Mars. For low orbits, payload falls gently with altitude and with inclination. The model reproduces Falcon 9's published expendable capabilities well: 23.2 t to a 200 km orbit (published 22.8 t) and 8.4 t to GTO (8.3 t).




Rocket-equation model of the simulated vehicle; LEO requirement calibrated to the trajectory simulation (expendable 23.2 t, droneship 19.1 t, RTLS 15.6 t to 200 km); higher orbits add impulsive Δv from a 200 km parking orbit; inclination changes the Earth-rotation gain (465 m/s × cos i), no dog-legs; booster reserves held at their LEO values.

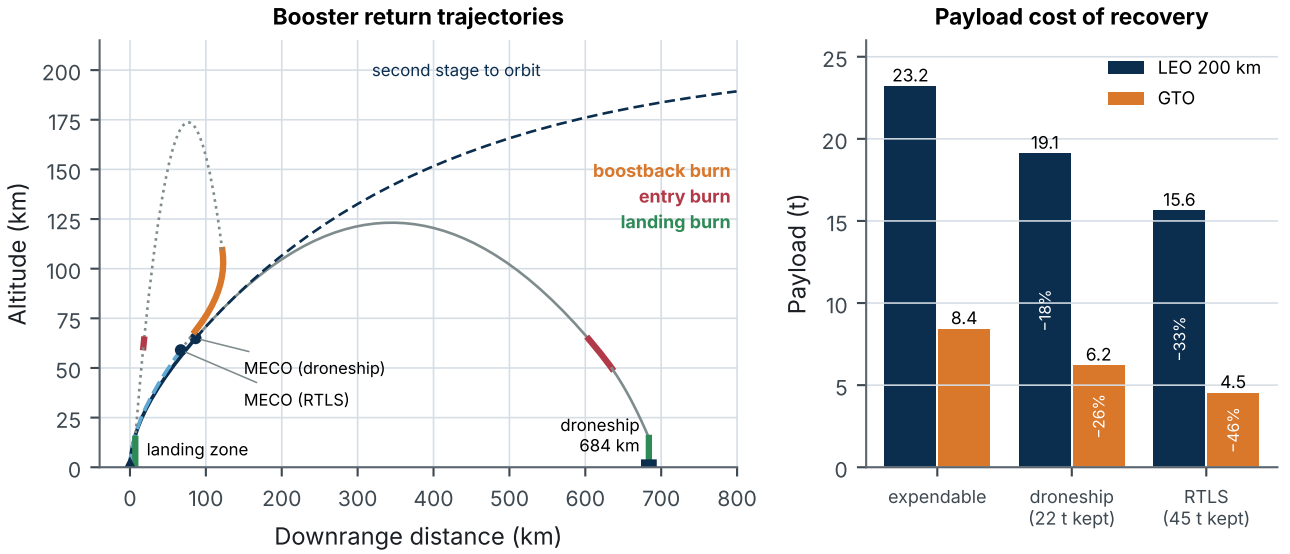
**FIGURE 34.14** Left: payload of the reference vehicle versus orbital energy for three booster-recovery modes, compared with published Falcon 9 values (crosses) and the full trajectory simulation (circles). Right: payload to circular orbits of different altitude and inclination with droneship recovery.

# 34.7 Reusability

## 34.7.1 Landing a Booster

For most of the space age every stage was thrown away. The Space Shuttle (Chapter 10) recovered its orbiter and boosters but needed months of refurbishment. The modern approach, pioneered by SpaceX's first Falcon 9 landing in December 2015 and first re-flight in 2017, flies the booster back under its own power to a vertical landing on legs. It must keep propellant for up to three burns. A **boostback** burn reverses its course, and an **entry burn** slows it before the dense atmosphere, limiting heating and loads. A **landing burn** is timed so that velocity reaches zero just as the legs touch down, because even at minimum throttle one engine lifts more than the empty stage weighs. In a **return to launch site** (RTLS) the booster cancels its downrange velocity and flies back to a landing zone near the pad. In a **droneship** landing it continues on its ballistic arc to an autonomous barge hundreds of kilometres downrange. This usually needs no boostback burn.

Figure 34.15 shows both for the reference vehicle. Every kilogram kept in the booster is a kilogram not used to accelerate the upper stage, so the first stage cuts off earlier and slower. In the simulation a droneship landing keeps 22 t of propellant and costs 18 % of the LEO payload; RTLS keeps 45 t and costs 33 %. GTO missions need a faster staging speed and so more braking: the penalties grow to 26 % and 46 %. Falcon 9's published figures show larger penalties (22.8 and 17.5 t to LEO; 8.3, 5.5 and 3.5 t to GTO), partly because the model ignores the legs, grid fins and performance reserves.




Same model as the ascent figure. Booster: 25.6 t dry, 3 engines for boostback (15° above horizontal) and entry burns (below 65 km, to 1.3 km/s), one engine at 75 % for landing; kept propellant = burns + 1.5 t residuals, iterated to consistency; payload re-optimised for each case. GTO = 200 × 35 786 km.

**FIGURE 34.15** Left: simulated booster return trajectories for a droneship landing about 680 km downrange and for a return to the launch site, with the boostback (orange), entry (red) and landing (green) burns highlighted. Right: payload delivered to LEO and GTO in each recovery mode.

### EXAMPLE: THE PROPELLANT BILL FOR LANDING

The reference booster has a dry mass of 25.6 t, plus 1.5 t of unusable residual propellant. Its exhaust speed is 3.06 km/s in vacuum and 2.78 km/s at sea level. Work backwards from touchdown with $m_{before} = m_{after}\, e^{\Delta v / v_e}$:

1. **Droneship.** Landing burn, 0.44 km/s at sea level: 27.1 t × $e^{0.44/2.78}$ = 31.8 t. Entry burn, 1.23 km/s: 31.8 × $e^{1.23/3.06}$ = 47.5 t. Propellant kept at MECO: 47.5 − 25.6 = **21.9 t** (simulation: 22.0 t).
2. **RTLS.** At MECO the stage is moving downrange at 1.87 km/s. Landing burn 0.38 km/s: 27.1 × $e^{0.38/2.78}$ = 31.1 t; entry burn 0.21 km/s: 33.3 t; boostback burn 2.32 km/s: 33.3 × $e^{2.32/3.06}$ = 71.0 t. Propellant kept: **45.4 t**, more than twice as much.
3. Payload: in the simulation each tonne kept in the booster costs about 0.17–0.19 t of LEO payload, roughly one tonne for every five to six held back. This is because the upper stage still provides most of the Δv. RTLS suits light payloads, where saving the time and cost of a droneship voyage matters more than the lost mass.

## 34.7.2 Records, Catches and Refurbishment

Falcon has turned landing into routine. SpaceX made its 600th Falcon booster landing on 19 April 2026, and the fleet leader, B1067, flew for the 37th time on 25 August 2026. By February 2025 SpaceX had also re-flown fairing halves, recovered from the sea, on 307 missions. Between flights engines, heat shielding, grid fins and legs are inspected and a few parts replaced; Falcon Heavy's side boosters routinely land side by side (Figure 34.16).

Others are following. Blue Origin's New Glenn landed its first booster in November 2025 and re-flew it in April 2026. China recovered its first orbital booster on 10 July 2026. On its maiden flight, the first stage of the new **Long March 10B** was caught at sea by hooks that snagged cables tensioned over a net on the ship *Linghangzhe*. In August 2026 LandSpace's stainless-steel **Zhuque-3** landed on legs at a desert site in Gansu province, although a fire broke out after touchdown. SpaceX's Super Heavy booster is caught by the launch tower's mechanical "chopstick" arms

instead of landing on legs. The first catch came on Starship Flight 5 on 13 October 2024 and was repeated on Flights 7 and 8 in 2025. A caught booster re-flew on Flight 9. On 28 September 2026 Starship reached orbit for the first time on Flight 14; catching the ship, needed to make the whole vehicle reusable, was planned for a later flight.

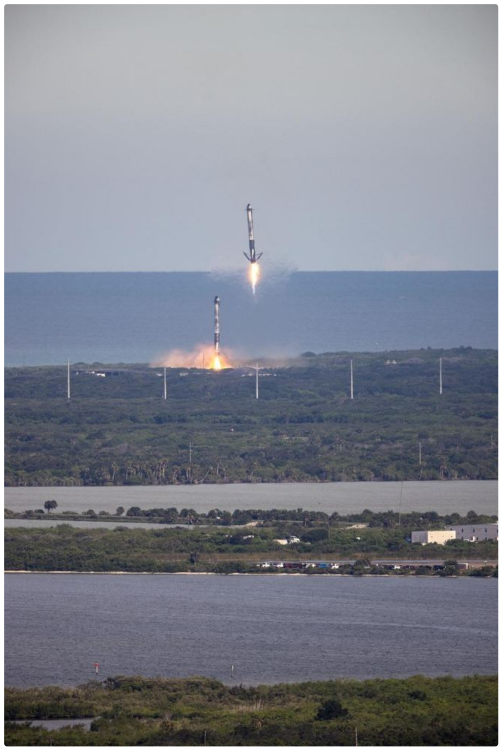


**FIGURE 34.16** Falcon Heavy's two side boosters returning to Landing Zones 1 and 2 at Cape Canaveral after launching the GOES-U weather satellite in June 2024. Image: NASA/KSC (NASA/Amber Jean Notvest).

### 34.7.3 The Economics of Launch

Reuse matters because hardware dominates launch cost: flying a first stage ten or thirty times spreads the cost of most of a vehicle's engines and structure. It pays only at a high flight rate, because factories, recovery ships and launch teams are fixed costs. Falcon 9 flew 165 times in 2025, half of all orbital launch attempts worldwide. Starlink satellites made up 3 169 of the 4 511 payloads launched that year — an in-house customer that keeps the fleet busy.

Table 34.5 compares launch prices per kilogram; list prices are not costs, and rockets rarely fly full. A 2018 NASA analysis found that vehicles introduced between 1970 and 2000 averaged about $18 500 per kilogram to LEO (2018 dollars), the Space Shuttle $61 700, and Falcon 9 at its advertised price $2 700 — less than a twentieth of the Shuttle's figure. Prices have since risen with inflation and demand: in 2026 SpaceX listed a dedicated Falcon 9 at $74 million and rideshare at $350 000 for 50 kg to sun-synchronous orbit. Cheaper access has transformed demand: the mass launched to orbit grew from about 400 t in 2015 to about 3 500 t in 2025.

**TABLE 34.5** Launch prices per kilogram to low Earth orbit. Historical values from Jones (NASA, 2018); 2026 values from SpaceX's published prices in current dollars; per-kilogram values assume a full vehicle.

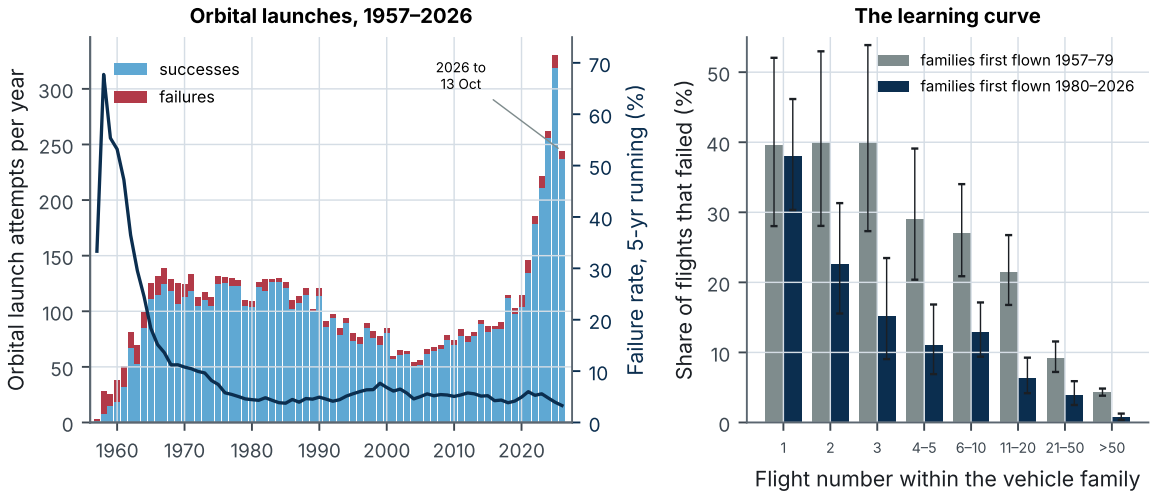
| Vehicle | First flight | Price per launch | Payload to LEO | Price per kg |
|---|---|---|---|---|
| Vanguard | 1957 | — | very small | ≈ $895 000 (2018 $) |
| Saturn V | 1967 | $728 M (2018 $) | 140 t | $5 200 (2018 $) |
| Space Shuttle | 1981 | $1 697 M (2018 $) | 27.5 t | $61 700 (2018 $) |

| Vehicle | First flight | Price per launch | Payload to LEO | Price per kg |
|---|---|---|---|---|
| Ariane 5 G | 1996 | — | — | $13 100 (2018 $) |
| Falcon 9 (2018 list) | 2010 | $62 M | 22.8 t | $2 700 |
| Falcon Heavy (2018 list) | 2018 | $90 M | 63.8 t | $1 400 |
| Falcon 9 (2026 list, booster landed) | 2010 | $74 M | ≈ 17.5 t | ≈ $4 200 |
| Falcon 9 rideshare (2026) | — | $350 000 per 50 kg | — | $7 000 |

# 34.8 Reliability and Testing

## 34.8.1 How Often Do Rockets Fail?

Of 330 orbital launch attempts in 2025, 11 failed — 3.3 %. Figure 34.17 puts this in context using Jonathan McDowell's General Catalog of Artificial Space Objects (GCAT). In the late 1950s a third or more of all launches failed. The five-year average fell below 10 % in the early 1970s. It has stayed at roughly 3–8 % since the mid-1970s, even as the launch rate tripled after 2020. The right-hand panel shows that **reliability is learned**. Across the 100 vehicle families first flown since 1980, 38 % of maiden flights failed, 23 % of second flights and 15 % of third flights. From the 21st to the 50th flight the rate was 4 %, and beyond the 50th, 0.7 %. New vehicles meet combinations of conditions that no ground test fully reproduces. Mature vehicles are very reliable: by October 2026 Falcon 9 had made 695 orbital attempts with 2 failures (99.7 %), and Soyuz more than 1 180 with 97.4 % success (Appendix E).




Data: GCAT (J. McDowell, launch list to 13 Oct 2026); partial failures counted as failures. Right: 143 vehicle families; error bars are 90 % Jeffreys intervals.

**FIGURE 34.17** Left: orbital launch attempts per year since 1957, split into successes and failures, with the five-year running failure rate. Right: the share of flights that failed as a function of flight number within each vehicle family, for families first flown before and after 1980.

A few flights say little about a new vehicle. With $k$ failures in $n$ flights a reasonable estimate of failure probability is $(k + ½)/(n + 1)$, the mean of the Jeffreys distribution used for the error bars of Figure 34.17. A rocket with ten successes in ten flights thus has an estimated failure probability of about 5 %. The upper end of its 90 % interval lies near 17 %. Only a hundred clean flights bring it below 2 %.

### 34.8.2 Why Rockets Fail

Propulsion causes the largest share of failures; Chapter 35 describes the engine failures that shaped design practice. Table 34.6 lists failures elsewhere in the vehicle — software, sensors, structures, separation and weather — and their lessons.

**TABLE 34.6** Launch failures outside the main engines and their lessons.

| Event | What failed | Lesson |
|---|---|---|
| Atlas-Centaur AC-67, Mar 1987 | Launched into cloud; lightning altered guidance-computer memory | Stricter weather launch-commit criteria |
| Ariane 5 flight 501, Jun 1996 | Reused Ariane 4 inertial-reference software overflowed | Re-validate reused software for the new flight |
| Taurus XL, Feb 2009 and Mar 2011 | Fairing did not separate; traced to falsified tests of aluminium parts | Verify the supply chain |
| Proton-M, Jul 2013 | Angular-rate sensors installed upside down | Error-proof mounting; inspection |
| Falcon 9 CRS-7, Jun 2015 | Strut holding a helium COPV in the upper-stage oxygen tank broke | Test critical small parts individually |
| Falcon 9 AMOS-6, Sep 2016 | Oxygen trapped in a COPV overwrap ignited during fuelling on the pad | Understand cryogenic loading; change procedures |
| H3 flight 8, Dec 2025 | Bonded composite payload adapter, delaminated in manufacture, failed under fairing-separation shock; satellite lost | Shock exposes hidden flaws; bolted adapter adopted |

### 34.8.3 Investigations and Testing

After a failure the operator leads a **mishap investigation** — in the United States under FAA oversight — and the vehicle is grounded until the cause is understood and the fix accepted. The evidence is mainly telemetry, analysed with fault trees that work through every possible cause. Falcon 9's Starlink 9-3 failure of 11 July 2024 shows the process at its fastest. An oxygen leak was traced to a pressure-sensor line cracked by vibration fatigue and a loose clamp; it chilled the engine's ignition system and caused a hard start at relight (Chapter 35). SpaceX removed the line, inspected clamps across the fleet, and the FAA cleared the vehicle on 25 July. Investigations take longer when the cause is less obvious. On New Glenn's third flight, in April 2026, one upper-stage engine fell short of full thrust, stranding its satellite in too low an orbit. Five weeks later the FAA accepted Blue Origin's finding that a cryogenic leak had frozen a hydraulic line. Japan's H3 needed six months: its December 2025 failure was traced to a manufacturing flaw in its bonded carbon-fibre payload adapter that grew under the shock of fairing separation. With a bolted adapter, H3 flew successfully again in June and August 2026.

The guiding rule of testing is **test like you fly**. Components are qualified on test stands, stages are fired whole (Chapter 35) and complete vehicles perform a **static fire** on the pad. Structures are tested to failure to learn their true margins. Two philosophies coexist. The traditional approach spends years on analysis and ground testing to make the first flight as safe as possible. The **iterative**, hardware-rich approach builds cheaper test articles, flies early and expects to lose some — SpaceX's approach with Starship, which lost several upper stages in 2025 before reaching orbit in 2026. Both converge on the lesson of Figure 34.17: reliability is demonstrated by flying.

### AI IN LAUNCH-VEHICLE ENGINEERING

Convex optimisation already flies: the G-FOLD algorithm, developed at NASA's Jet Propulsion Laboratory, computes fuel-optimal landing trajectories in milliseconds with guaranteed convergence, and SpaceX has described landing Falcon 9 boosters with automatically generated convex-optimisation code. Machine learning is being added around such algorithms: anomaly detectors scan thousands of telemetry channels for patterns that precede failures, and learned surrogates of loads and engine behaviour speed up design optimisation. Flight-critical software still rests on verifiable algorithms (Chapters 52 and 53).

### KEY TAKEAWAYS

- Reaching a 200 km orbit takes about 9.0–9.6 km/s of ideal Δv. This is 7.8 km/s of orbital speed plus about 1.5–2 km/s of gravity, steering, drag and back-pressure losses. Earth's rotation subtracts up to 0.46 km/s (465 m/s × cos *i*).
- Launch latitude matters most for geostationary missions: removing a 28.5° inclination at apogee costs 0.35 km/s more than removing 5°.
- With realistic structures one stage cannot reach orbit with useful payload; two stages capture most of the benefit of staging.
- A gravity turn keeps the angle of attack near zero; engines throttle through max-Q (typically 25–40 kPa) and acceleration is capped at about 4–5 g.
- Tanks are thin pressurised shells of aluminium–lithium, stainless steel or composite; payloads must survive 130–140 dB of noise and loads of several g.
- Landing a booster costs payload — about a fifth for a droneship landing and a third for a return to the launch site — but enables dozens of re-flights.
- About 40 % of maiden flights fail; reliability grows with flight experience to 99 % or better for mature vehicles.

### REVIEW QUESTIONS

1. List the four losses in a launch Δv budget and explain why steering loss can rival gravity loss for a vehicle with a low-thrust upper stage.
2. Show that the Earth-rotation gain for a launch into an orbit of inclination $i$ is 465 m/s × cos $i$, whatever the site latitude. What is it for a 97.5° sun-synchronous orbit?
3. A 4 000 kg geostationary satellite with a 320 s apogee engine is launched from Kourou or from Cape Canaveral. Estimate the difference in the propellant it needs.
4. For stages with Isp 340 s and ε = 0.07, compute the payload fraction of one, two and three identical stages for Δv = 9.4 km/s.
5. Why do launch vehicles fly a gravity turn? What is max-qα, and how is it managed on launch day?
6. Explain the purpose of a throttle bucket and why acceleration rises during each stage's burn.
7. Compare aluminium–lithium, stainless steel and composite tanks. Why might a reusable stage use steel?
8. What is POGO, and how is it suppressed?
9. A satellite's lowest lateral natural frequency is 8 Hz. Why might the launch provider object, and what analysis would settle the question?
10. Estimate the propellant a 25 t booster must keep for a 3 km/s return-to-launch-site profile with Isp 300 s, and explain why the payload penalty is much smaller than the propellant kept.

# 35 CHAPTER Rocket Propulsion

Image: NASA/SSC (Danny Nowlin)

*A rocket engine is a machine for throwing mass backwards as fast as possible. This chapter explains how engines and motors produce thrust and how performance is measured — specific impulse, characteristic velocity and thrust coefficient — and how a converging–diverging nozzle turns hot gas into a supersonic jet whose performance depends on altitude. It compares liquid, solid and hybrid propellants, explains the cycles that feed liquid engines, from pressure-fed thrusters to full-flow staged combustion and battery-driven pumps, and describes turbopumps, injectors, combustion instability, cooling, throttling and 3-D printing. It covers solid-motor ballistics, hybrid physics and engine testing, using the engines of 2026 as examples.*

IN THIS CHAPTER

35.1 How a Rocket Makes Thrust
35.2 Nozzle Flow
35.3 Propellants
35.4 Liquid-Propellant Engines
35.5 Engines of 2026
35.6 Solid Rocket Motors
35.7 Hybrid Rockets
35.8 Testing and Qualification
35.9 Beyond Chemical Propulsion

Every launch vehicle and almost every spacecraft moves by one principle. It carries its own propellant and releases the propellant's chemical energy as heat. It then expels the hot products backwards through a nozzle at several kilometres per second. The reaction to that stream of mass is **thrust**. Engines burning liquid propellants are called **rocket engines**; those with propellant cast as a solid block are **rocket motors**.

A Super Heavy booster lights 33 Raptor engines that together produce about 80 MN — more than eight thousand tonnes-force. By contrast, a Rutherford engine on Rocket Lab's Electron weighs 35 kg and produces 25 kN. Both obey the same few equations. Chapter 34 shows how engines are combined into stages and derives the rocket equation, Chapter 36 surveys the world's launch vehicles, and Chapter 6 covers spacecraft thrusters and electric propulsion.

## 35.1 How a Rocket Makes Thrust

### 35.1.1 The Thrust Equation

Draw an imaginary boundary around an engine. Propellant enters at almost zero velocity and leaves through the nozzle exit at velocity $v_e$ with mass flow rate $\dot{m}$. Newton's second law says the force on the engine equals the rate at which momentum leaves the boundary, plus any unbalanced pressure force across the exit plane:

$$F = \dot{m}\, v_e + (p_e - p_a)\, A_e$$

The first term is the **momentum thrust**. The second, the **pressure thrust**, is positive if the gas at the exit plane (pressure $p_e$, area $A_e$) is at a higher pressure than the surrounding air ($p_a$). It is negative if the pressure is lower. In vacuum $p_a = 0$, so every engine produces more thrust in space than at sea level — 8 % more for the Merlin 1D, 23 % for the RS-25. Both terms together define the **effective exhaust velocity** $c = F/\dot{m}$, which therefore changes with altitude.

### 35.1.2 Specific Impulse

Dividing the effective exhaust velocity by standard gravity $g_0$ = 9.806 65 m/s² gives the **specific impulse** in seconds:

$$I_{sp} = \frac{F}{\dot{m}\, g_0} = \frac{c}{g_0}$$

Specific impulse is the number of seconds that one kilogram of propellant can produce one kilogram-force of thrust — the "fuel economy" of a rocket. The rocket equation, $\Delta v = I_{sp}\, g_0 \ln(m_0/m_f)$ (Chapter 34), shows that the velocity a stage can deliver grows in direct proportion to it. Typical vacuum values are 250–290 s for solid motors, 310–340 s for kerosene, about 340–380 s for methane and 420–465 s for hydrogen engines. An engine's **thrust-to-weight ratio** measures how compact it is: about 35–60 for hydrogen engines and 70–185 for kerosene and methane engines.

### 35.1.3 Characteristic Velocity and Thrust Coefficient

Two further parameters separate the work of the combustion chamber from that of the nozzle. When the flow is choked at the nozzle throat (explained below), the mass flow is fixed by the chamber pressure $p_c$ and the throat area $A_t$:

$$\dot{m} = \frac{p_c A_t}{c^*}, \qquad F = C_F\, p_c A_t, \qquad c = c^* C_F$$

The **characteristic velocity** $c^*$ ("c-star") depends only on the combustion gas — its temperature $T_c$, mean molar mass $\mathcal{M}$ and ratio of specific heats γ. It is 1 500–1 600 m/s for solid propellants, about 1 800 m/s for kerosene or methane with oxygen and 2 300–2 400 m/s for hydrogen–oxygen.

The ratio of measured to theoretical $c^*$, the **c* efficiency** (typically 95–99 %), shows how well the injector mixes and burns the propellants. The **thrust coefficient** $C_F$, typically 1.3–2.0, measures how much the nozzle amplifies the force that the chamber pressure would exert on the throat alone; it depends only on γ, the **expansion ratio** ε = $A_e/A_t$ and the ambient pressure.

## 35.2 Nozzle Flow

### 35.2.1 The Ideal Rocket

The nozzle turns heat into directed motion. For the **ideal rocket** — steady, one-dimensional flow of a perfect gas with constant γ, no friction and no heat loss — the energy equation gives the exit velocity:

$$v_e = \sqrt{\frac{2\gamma}{\gamma-1}\,\frac{R_u T_c}{\mathcal{M}}\left[1-\left(\frac{p_e}{p_c}\right)^{(\gamma-1)/\gamma}\right]}$$

where $R_u$ = 8 314 J/(kmol·K). Exhaust velocity therefore rises with $\sqrt{T_c/\mathcal{M}}$: light exhaust matters as much as a hot flame. Hydrogen–oxygen burns slightly cooler than kerosene–oxygen, but its exhaust (mean molar mass about 13.5 kg/kmol) is much lighter than kerosene's (about 23.5). With γ = 1.2 and a pressure ratio of 1 000 the formula gives 4 200 m/s for hydrogen and 3 270 m/s for kerosene. Because the bracket approaches 1 slowly, higher pressure ratios add little velocity.

### 35.2.2 Area, Mach Number and the Throat

Subsonic gas accelerates in a narrowing duct, and supersonic gas in a widening one. So a rocket nozzle converges to a **throat**, where the flow reaches exactly Mach 1, and then diverges — the **de Laval nozzle**. For γ = 1.2 the throat pressure is 56 % and the throat temperature 91 % of the chamber values, so the throat carries the highest heat load. Downstream the Mach number climbs to 3–5 (Figure 35.1), following the **area–Mach relation**:

$$\frac{A}{A_t} = \frac{1}{M}\left[\frac{2}{\gamma+1}\left(1+\frac{\gamma-1}{2}M^2\right)\right]^{\frac{\gamma+1}{2(\gamma-1)}}$$

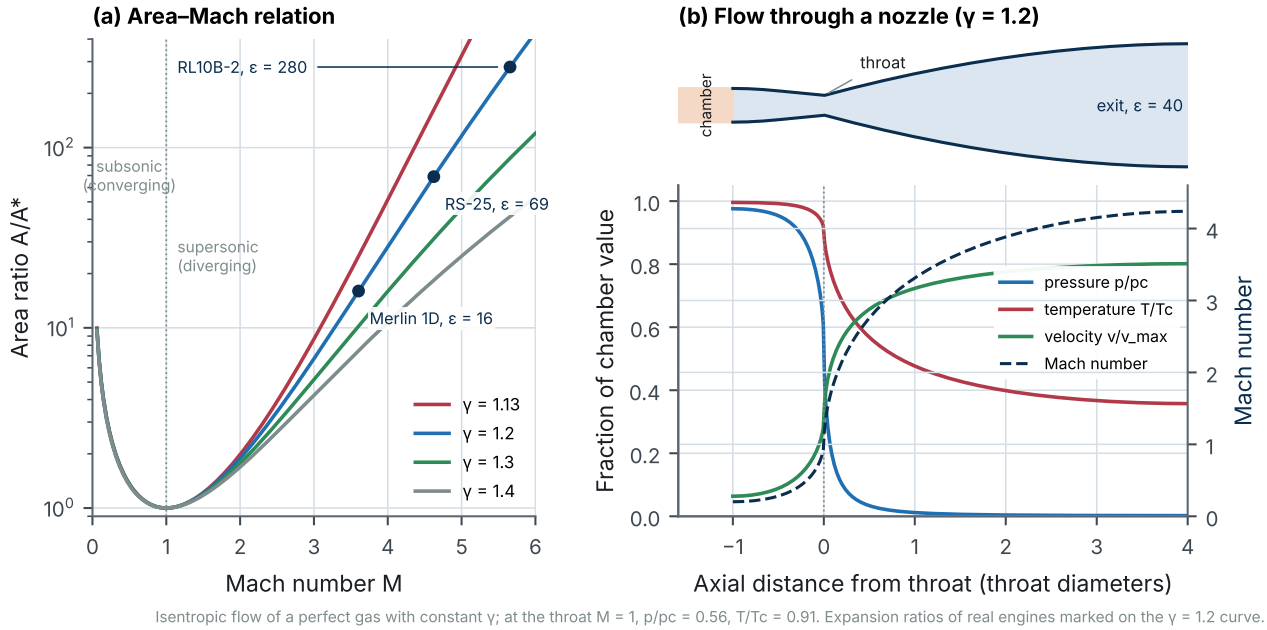


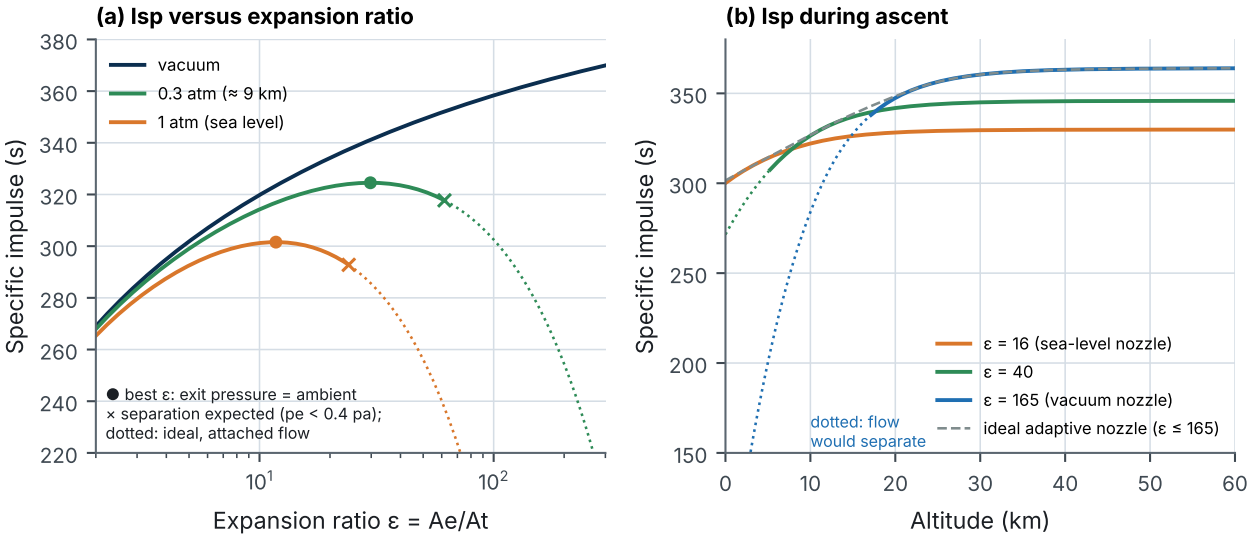

**FIGURE 35.1** The de Laval nozzle. (a) Area ratio needed to reach each Mach number for several values of γ, with the expansion ratios of three real engines. (b) Pressure, temperature, velocity and Mach number along a nozzle with ε = 40: pressure falls by almost half before the throat, and most of the velocity is gained within a few throat diameters downstream of it.

Once the throat is **choked**, no disturbance can travel upstream, so the mass flow depends only on chamber conditions and throat area; designers size an engine by choosing $p_c$ and $A_t$ first. A simple 15° conical nozzle loses 1.7 % of thrust because the exhaust leaves at an angle (divergence factor $(1+\cos\alpha)/2$ = 0.983). Most engines use a contoured **bell nozzle**, which gives the same performance at about 80 % of the cone's length.

### 35.2.3 Expansion Ratio and Altitude

A nozzle gives its best thrust when exit pressure equals ambient pressure. If $p_e > p_a$ it is **under-expanded** and the plume balloons outwards after the exit. If $p_e < p_a$ it is **over-expanded**: the air squeezes the jet into the diamond-shaped shock pattern seen in sea-level engine tests, and the pressure term subtracts thrust. Below roughly 40 % of ambient pressure the flow **separates** from the wall inside the nozzle, and asymmetric separation produces side loads that can break it. A first-stage nozzle is therefore a compromise: it must work at sea level but spends most of its burn at altitude (Figure 35.2).

With the same chamber, a nozzle of ε = 16 gives 300 s at sea level and 330 s in vacuum. A vacuum nozzle of ε = 165, like the Merlin Vacuum's, gives 364 s in space but would separate violently at sea level. Vehicles solve the problem with stages: first-stage engines use moderate expansion ratios (Merlin 1D 16, Raptor 34, RD-180 37), upper-stage engines large ones (Vinci 240, RL10B-2 280). The RL10B-2 and Vinci also have **extendible nozzles** that slide out after staging. There are also **dual-bell** and **aerospike** nozzles, which adapt to the surrounding pressure. They have been tested many times, but none has yet powered an orbital launch.




Ideal rocket, γ = 1.2, chamber pressure 10 MPa, c* = 1 800 m/s (typical of LOX/RP-1 or LOX/$CH_4$); US Standard Atmosphere; Summerfield separation criterion pe ≈ 0.4 pa.

**FIGURE 35.2** Why nozzles are sized for altitude (chamber pressure 10 MPa). (a) Specific impulse versus expansion ratio for three ambient pressures; the best ratio grows from about 12 at sea level to infinity in vacuum. (b) Isp during ascent for a sea-level nozzle (ε = 16), an intermediate one and a vacuum nozzle (ε = 165); the dashed curve is an ideal nozzle that adapts to the surroundings.

**FIGURE 35.3** The 27 Merlin engines of a Falcon Heavy at lift-off with NASA's Nancy Grace Roman Space Telescope on 30 August 2026. The sea-level engines are slightly over-expanded at lift-off. Image: NASA/KSC (SpaceX).

Real engines fall short of the ideal through incomplete mixing, boundary-layer losses, exhaust divergence and chemistry that lags the falling temperature. Published vacuum Isp can be compared with ideal values computed for each engine's own conditions. This gives about 92 % for the Merlin 1D, 94 % for the RD-180, 95 % for Raptor, 97 % for the RL10B-2 and 98 % for the RS-25. Solid motors lose a little more because alumina droplets lag behind the gas.

**EXAMPLE: SPECIFIC IMPULSE OF THE RS-25 FROM C* AND THE THRUST COEFFICIENT**

The RS-25 burns hydrogen and oxygen at a mixture ratio of 6.0, with $p_c$ = 20.6 MPa and ε = 69. Its rated vacuum thrust is 2 279 kN.

1. **Characteristic velocity.** An equilibrium calculation gives a theoretical $c^*$ of 2 322 m/s; with a c* efficiency of 99 %, $c^*$ = 2 299 m/s.
2. **Exit pressure.** With γ = 1.2 the area–Mach relation gives $M_e$ = 4.6 for ε = 69 and $p_e/p_c$ = 0.001 05, so $p_e$ = 21.6 kPa.
3. **Thrust coefficient.** In vacuum $C_F$ = 1.855 (momentum) + 0.072 (pressure term $p_e\varepsilon/p_c$) = 1.927. At sea level subtract $p_a\varepsilon/p_c$ = 0.339: $C_F$ = 1.587.
4. **Specific impulse.** $I_{sp} = c^*C_F/g_0$ = 2 299 × 1.927 / 9.807 = **452 s** in vacuum (published: 452.3 s) and 372 s at sea level (published: 366 s).
5. **Throat size.** $A_t = F/(C_F p_c)$ = 2 279 000 / (1.927 × 20.6 × $10^6$) = 0.0574 $m^2$, a throat 27 cm across. The mass flow $p_c A_t/c^*$ is **514 kg/s** — a tonne of propellant every two seconds.

**EXAMPLE: SIZING A FIRST-STAGE NOZZLE**

A kerosene engine must give 850 kN at sea level with $p_c$ = 10 MPa, γ = 1.2 and $c^*$ = 1 746 m/s (97 % of 1 800 m/s). To keep the nozzle short and avoid separation, choose an exit pressure of 70 kPa.

1. **Expansion ratio.** $p_c/p_e$ = 143; the isentropic relations give $M_e$ = 3.6 and **ε = 15.6**.
2. **Thrust coefficient.** Momentum term 1.685; pressure term (70 − 101.3) kPa × 15.6 / 10 MPa = −0.049; so $C_F$ = 1.636 at sea level. Since $p_e/p_a$ = 0.69 > 0.4, the flow stays attached.
3. **Throat and exit.** $A_t = F/(C_F p_c)$ = 0.0520 $m^2$ (diameter 25.7 cm); $A_e = \varepsilon A_t$ = 0.810 $m^2$ (diameter **1.02 m**).
4. **Performance.** $I_{sp}$ = 291 s at sea level and 320 s in vacuum, where thrust rises to 932 kN; the mass flow is 298 kg/s. The Merlin 1D (ε = 16, 845/914 kN, 282/311 s) is close, with slightly lower real efficiency.

# 35.3 Propellants

## 35.3.1 Choosing a Combination

A good combination releases much energy per kilogram, produces light exhaust, is dense, storable, safe and cheap, and can cool the chamber. None is best at everything (Table 35.1, Figure 35.4).

**Liquid hydrogen with liquid oxygen** (LOX/$LH_2$) gives the highest Isp of any practical chemical combination. But hydrogen is a cryogenic liquid at 20 K with a density of only 71 kg/$m^3$. Its tanks are huge and must be insulated against boil-off, and the hydrogen can embrittle metals. Hydrogen powers upper stages and core stages that burn from the ground up (RS-25, Vulcain 2.1, LE-9, YF-77). Engines run fuel-rich, at O/F 5–6 instead of the stoichiometric 7.9, because lighter exhaust and smaller hydrogen tanks more than repay the lost energy.

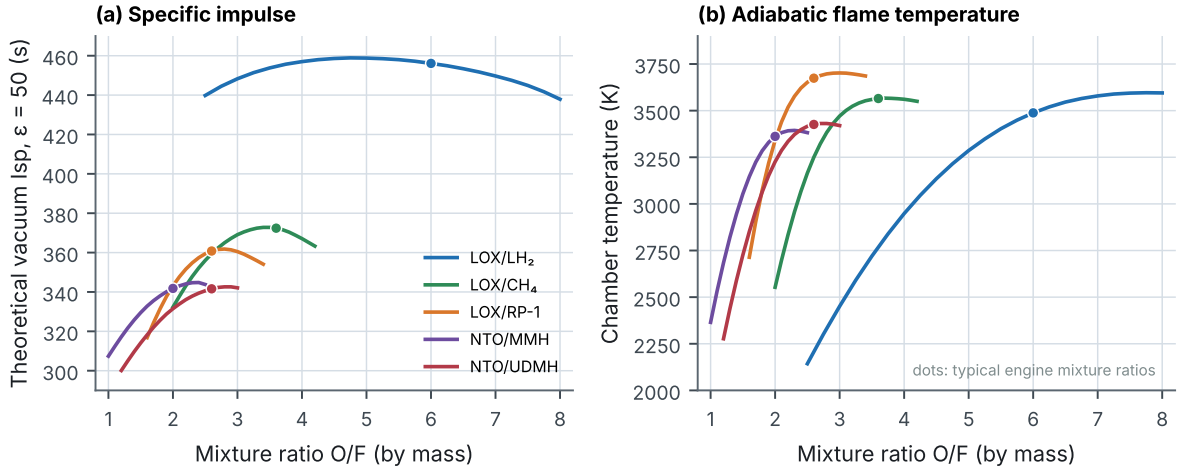
**Kerosene with oxygen** (LOX/RP-1, a highly refined kerosene) is dense and cheap, and the fuel is stored at room temperature. Its drawbacks are lower Isp and **coking** — carbon deposits in cooling channels and soot on turbines — which complicates reuse. SpaceX chills Falcon 9's oxygen well below its boiling point and its kerosene below room temperature to pack more propellant into

the tanks. **Methane with oxygen** ("methalox") sits between the two: better Isp than kerosene, far denser than hydrogen and little coking. Its boiling point (112 K) is close to that of oxygen (90 K). So the tanks can share a bulkhead, and each propellant can pressurise its own tank with its own vapour. Methane can also be made on Mars. China's Zhuque-2 was the first methane-fuelled rocket to reach orbit, on 12 July 2023, followed by, among others, Vulcan (January 2024), New Glenn (January 2025), China's Zhuque-3 (December 2025) and, in September 2026, Starship.

**Storable hypergolic propellants** — nitrogen tetroxide (NTO) with monomethylhydrazine (MMH), unsymmetrical dimethylhydrazine (UDMH) or UH25 (UDMH with 25 % hydrazine hydrate) — ignite on contact and stay liquid at room temperature for years. They power spacecraft (Chapter 6), some upper stages and India's Vikas engine, but are highly toxic.

**TABLE 35.1** Propellant performance. Liquid values are theoretical (shifting equilibrium, chamber pressure 6.9 MPa, sea-level Isp with optimum expansion, vacuum Isp with ε = 50), computed for this book; solid values are delivered figures of the Shuttle booster and GEM 63XL.

| Propellants | O/F | $T_c$ (K) | $c^*$ (m/s) | $I_{sp}$ SL / vac (s) | Bulk density (kg/m³) | Storage |
|---|---|---|---|---|---|---|
| $LOX/LH_2$ | 6.0 | 3 490 | 2 305 | 382 / 456 | 361 | Both cryogenic (90 K, 20 K) |
| $LOX/CH_4$ | 3.6 | 3 565 | 1 823 | 307 / 372 | 833 | Both cryogenic (90 K, 112 K) |
| LOX/RP-1 | 2.6 | 3 674 | 1 793 | 300 / 361 | 1 025 | Fuel storable |
| NTO/MMH | 2.0 | 3 362 | 1 755 | 288 / 342 | 1 186 | Storable, hypergolic |
| NTO/UDMH | 2.6 | 3 426 | 1 726 | 286 / 342 | 1 174 | Storable, hypergolic |
| APCP solid | — | — | ≈ 1 550 | 242 / 268–280 | ≈ 1 800 | Cast; storable for years |

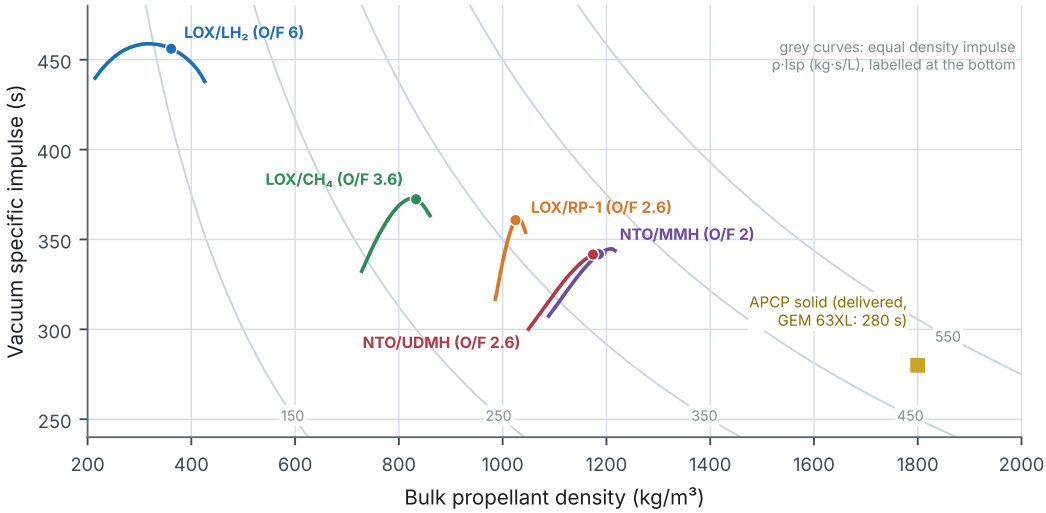



Shifting equilibrium, chamber pressure 6.9 MPa (1 000 psia), computed with an equilibrium code and NASA thermodynamic data; liquid propellants at their normal storage temperatures.

**FIGURE 35.4** Theoretical performance of liquid bipropellants versus mixture ratio. (a) Vacuum Isp peaks on the fuel-rich side of stoichiometric because lighter exhaust compensates for less heat. (b) Chamber temperature peaks near stoichiometric; running fuel-rich also keeps walls and turbines cooler. Dots mark mixture ratios typical of real engines.

### 35.3.2 Density versus Specific Impulse

A stage's tanks, structure and drag grow with the *volume* of its propellant, not its mass. For a fixed tank volume the total impulse is proportional to the **density impulse** $\rho I_{sp}$, where ρ is the bulk density of the combination. By this measure kerosene (370 kg·s/L) beats hydrogen (165 kg·s/L) by more than a factor of two, and solids (about 500 kg·s/L) do better still (Figure 35.5). This is why most first stages and strap-on boosters burn kerosene, methane or solid propellant, while hydrogen is favoured for upper stages, where every second of Isp is multiplied through the rocket equation.




Lines: theoretical Isp (ε = 50, 6.9 MPa, shifting equilibrium) as the mixture ratio varies; dots: typical mixture ratios. Bulk density ρ = (1 + O/F)/(O/F/ρo + 1/ρf).

**FIGURE 35.5** The density–impulse trade. Each line shows how theoretical vacuum Isp and bulk density change as the mixture ratio varies; dots mark mixture ratios typical of real engines, and grey curves join combinations with equal density impulse.

## 35.4 Liquid-Propellant Engines

### 35.4.1 Feeding the Chamber

Propellants must reach the injector at a pressure 20–30 % above chamber pressure, so that the injector's pressure drop isolates the feed system from chamber fluctuations. The simplest method is to pressurise the tanks with helium (**pressure-fed**). But tank wall thickness grows with pressure ($t = pr/\sigma$ for a cylinder of radius $r$ and allowable stress σ). A 3 m diameter aluminium-alloy tank needs walls 1.5 mm thick at 0.3 MPa but 15 mm at 3 MPa. Pressure-fed engines are therefore limited to chamber pressures of about 0.7–2 MPa and to spacecraft and small stages. Large stages keep their tanks at a few bar and raise the pressure with **turbopumps** — pumps driven by a turbine — and the way the turbine is powered defines the **engine cycle** (Figure 35.6, Table 35.2).

### 35.4.2 Engine Cycles

A **gas generator** burns a few per cent of the propellants very fuel-rich (below about 1 000 K) to drive the turbine and dumps the exhaust overboard: simple, but it costs several seconds of Isp. In an **expander** cycle the fuel, usually hydrogen, absorbs heat in the cooling jacket and then drives the turbine before being injected. Nothing burns in the turbine, so failures tend to be benign and

restarts easy. But the heat available limits the size of closed expanders such as the RL10 (flying since 1963) and Vinci. The **expander-bleed** variant dumps its turbine flow and allows larger engines such as Japan's 1.47 MN LE-9.

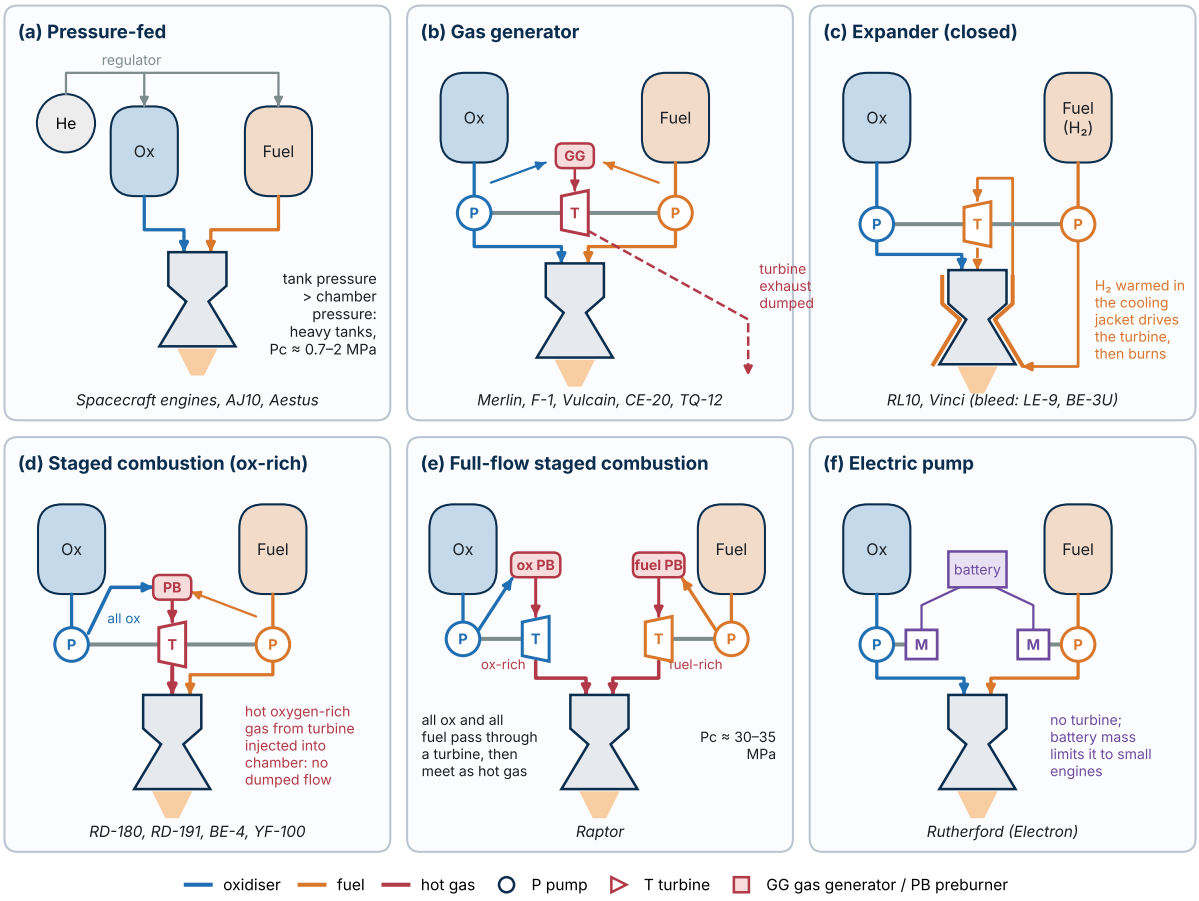



**FIGURE 35.6** The main liquid-engine cycles. Blue lines carry oxidiser, orange fuel, red hot gas. Gas-generator engines dump their turbine exhaust; expander engines drive the turbine with fuel heated in the cooling jacket; staged-combustion engines send all turbine exhaust into the main chamber; electric-pump engines use battery-powered motors.

In **staged combustion**, a **preburner** burns all of one propellant with a little of the other, and the hot gas drives the turbine and then flows into the main chamber, so nothing is wasted. The Soviet Union pioneered the **oxidiser-rich** version — the RD-253 (1965), NK-33 and the RD-170 family, including the RD-180 and RD-191. It needs special alloys because hot oxygen attacks most metals. The YF-100, BE-4 and Archimedes use it too. The RS-25 uses the **fuel-rich** version, with one preburner for each turbopump. **Full-flow staged combustion** (FFSC) passes *all* of both propellants through turbines. A fuel-rich preburner drives the fuel pump and an oxygen-rich one the oxygen pump. This gives cooler turbines and no need for a seal between fuel-rich gas and oxygen on one shaft. After Soviet and US ground demonstrators, SpaceX's Raptor, first flown in 2019, became the only FFSC engine to fly; Raptor 3 runs at about 33 MPa, the highest chamber pressure of any flying engine.

Rocket Lab's Rutherford replaces the turbine with two brushless electric motors of about 37 kW each, spinning near 40 000 rpm on lithium-polymer batteries. The pumps are simple and easy to control, but battery mass grows with power and burn time, so the **electric-pump** cycle suits only small engines; Electron's second stage even jettisons spent batteries in flight.

**TABLE 35.2** Liquid-engine cycles compared.

| Cycle | What drives the turbine | Typical $p_c$ (MPa) | Strengths | Weaknesses | Examples |
|---|---|---|---|---|---|
| Pressure-fed | No turbine: tank pressure | 0.7–2 | Simplest; easy restarts | Heavy tanks; low Isp | Spacecraft engines, AJ10, Aestus |
| Gas generator | Small fuel-rich combustor; exhaust dumped | 5–12 | Simple, easy to develop | Wastes a few per cent of flow | Merlin, F-1, Vulcain 2.1, CE-20 |
| Expander (closed / bleed) | Fuel heated in the cooling jacket | 4–6 / ≈ 10 | Benign failures; restartable | Heat pickup limits size | RL10, Vinci; LE-9, BE-3U |
| Staged combustion | Preburner gas, then main chamber | 10–27 | High Isp and pressure | Complex; hot oxygen (ox-rich) | RD-180, BE-4, YF-100; RS-25 |
| Full-flow staged comb. | Two preburners; all flow through turbines | 30–33 | Cooler turbines; no interpropellant seal | Most complex | Raptor |
| Electric pump | Battery and electric motors | Low (not published) | Simple; precise control | Battery mass | Rutherford |

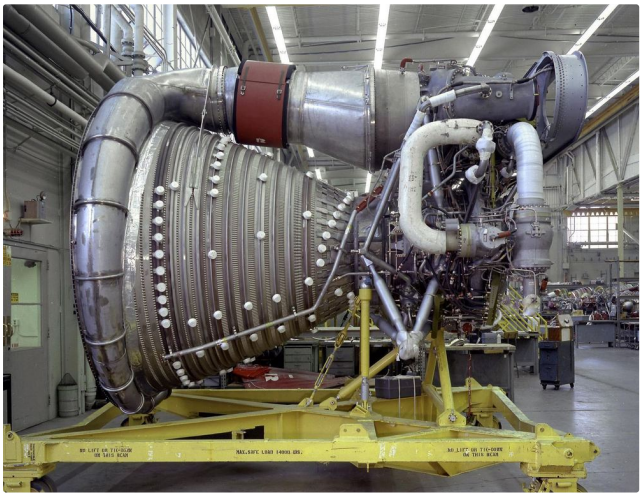


**FIGURE 35.7** A complete F-1 engine, 5.8 m tall. Five of these gas-generator engines powered each Saturn V first stage; the turbopump and turbine exhaust manifold are visible on the side of the tube-wall chamber. Image: NASA/MSFC.

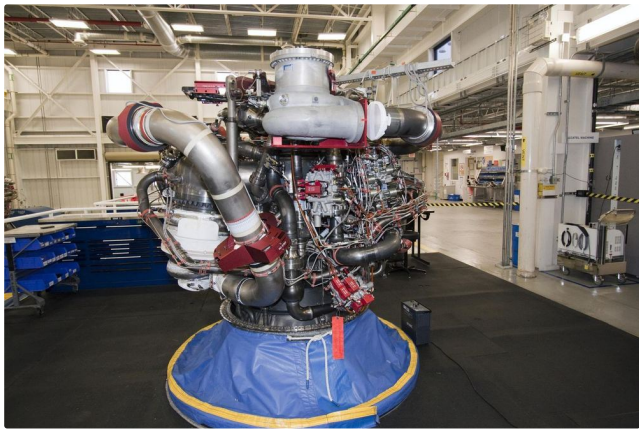


**FIGURE 35.8** Space Shuttle main engine No. 1 being fitted with a new turbopump at the Kennedy Space Center in June 2010. The staged-combustion powerhead sits above the nozzle, which is covered in blue. Image: NASA/KSC (NASA_Jack Pfaller).

### 35.4.3 Turbopumps

Pump power is $P = \dot{m}\,\Delta p/(\rho\,\eta)$, where $\Delta p$ is the pressure rise and η the efficiency, so light hydrogen is expensive to pump. Raising the RS-25's 73 kg/s of hydrogen (71 kg/m$^3$) by about 40 MPa at 75 % efficiency needs about 55 MW. This is close to the 53 MW (71 140 hp) rating of its high-pressure fuel turbopump, which spins at about 35 000 rpm (Figure 35.8). At the pump inlet the danger is **cavitation**: if the local pressure falls below the propellant's vapour pressure, bubbles form and collapse, destroying performance and eroding blades. The margin is the **net positive suction head** (NPSH), the excess of inlet pressure over vapour pressure expressed as a height of liquid. Designers provide it with tank pressure, subcooled propellants, **inducers** (helical impellers that tolerate some cavitation) and, on the RS-25, low-pressure boost pumps.

### 35.4.4 Injectors and Combustion Instability

The **injector** atomises and mixes the propellants through hundreds or thousands of holes. Kerosene engines often use **impinging** jets. Hydrogen engines use **coaxial** elements, in which fast hydrogen gas shears a central stream of liquid oxygen (the RS-25 main injector has 600). Russian engines favour **swirl** elements, and Merlin, like the Apollo lunar descent engine, uses a single central **pintle** that throttles well. A good injector releases 96–99 % of the theoretical energy.

The chamber is also an organ pipe. If heat release couples with a pressure wave, the wave grows until it destroys the engine. Low-frequency instability ("chugging") couples with the feed system — or with the whole vehicle's structure, when it is called **pogo** — while high-frequency acoustic instability ("screech") is the most destructive. In a chamber about 1 m across, where sound travels at about 1 100 m/s, the first transverse (tangential) mode lies near 650 Hz. Cures include **baffles** that divide the injector face, acoustic cavities that absorb the dangerous frequencies, and changes to the injection pattern.

> **TAMING THE F-1**
>
> The F-1, which produced 6.8 MN at sea level, remains the most powerful single-chamber liquid engine ever flown. In June 1962 one destroyed itself on the test stand through high-frequency instability. Rocketdyne and NASA spent more than three years and about 2 000 full-scale tests on injector and baffle variants, setting off small charges inside the running chamber to trigger instability. The final design, with copper baffles and a modified injection pattern, damped such disturbances within about 45 ms. No F-1 failed in flight: 65 engines powered 13 Saturn V launches.

### 35.4.5 Cooling

Near the throat of a high-pressure engine the heat flux exceeds 100 MW/m$^2$, about a thousand times that of a kitchen hot-plate, and the gas is hotter than any structural metal can survive. In **regenerative cooling** the fuel flows through channels in the wall before injection, returning the heat to the chamber. Modern chambers use copper alloys with milled channels, while the RS-25 nozzle is made of 1 080 brazed tubes (Figure 35.9). **Film cooling** protects the wall with a layer of fuel or, on Vulcain 2, re-injected turbine exhaust, while **ablative** walls char and erode slowly. Nozzle extensions can also be **radiation-cooled**: made of niobium alloy or carbon–carbon (Merlin Vacuum, RL10B-2), they glow and radiate their heat to space. Finally, **dump cooling** passes a little fuel through the wall of a nozzle extension and out of its exit.

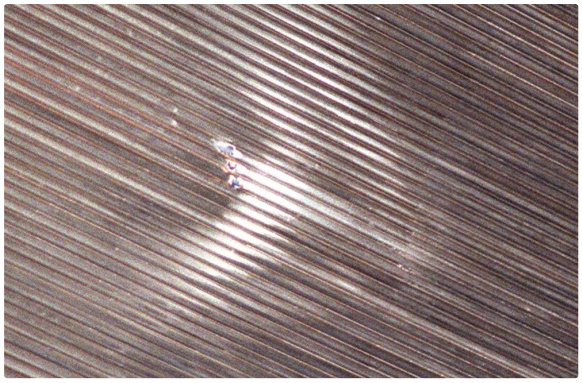


**FIGURE 35.9** Coolant tubes inside main engine No. 3 of Space Shuttle Columbia after STS-93 in July 1999. A pin ejected from the injector at ignition punctured three tubes; the resulting hydrogen leak made the engines shut down 0.16 s early, leaving Columbia about 5 m/s short. Image: NASA/KSC.

### 35.4.6 Ignition, Throttling and Restart

Merlin is ignited by **TEA–TEB**, a mixture of triethylaluminium and triethylborane that bursts into green flame on contact with oxygen; each start consumes a dose, so the supply carried limits restarts. The RS-25, RL10 and Raptor use electrically ignited **torch igniters**, and hypergolic engines need no igniter at all. Restarting in space also needs settled propellant (Chapter 6), chill-down of pumps and lines and careful sequencing; if too much propellant accumulates before ignition, a **hard start** can damage the engine. **Throttling** limits loads near maximum dynamic pressure and makes landing possible; it requires enough injector pressure drop at low thrust to avoid instability. The RS-25 throttles between 67 % and 109 % of rated thrust, the BE-4 between 40 and 100 % and the RD-191 between 27 and 105 %.

### 35.4.7 Additive Manufacturing

Rocket engines suit **additive manufacturing** (metal 3-D printing): they are built in small numbers from costly alloys with complex internal passages. Laser powder-bed fusion now produces injectors, chambers with integral cooling channels (including NASA's GRCop copper alloys), manifolds and pump housings in one piece instead of dozens of brazed parts. Rutherford is mostly printed. Raptor still uses printed parts, and Raptor 3 moved much of its plumbing into the engine's main structure. Up to half of Prometheus by mass is printed, part of its aim to cost a tenth as much as Vulcain 2. Porosity, rough passages and uncertain fatigue life mean printed parts still need extensive testing.

**FIGURE 35.10** Engineers at NASA's Marshall Space Flight Center prepare a 3-D printed turbopump for a test in 2016. At full power it pumped 2.3 $m^3$ of liquid methane per minute, enough for an engine of about 156 kN. Image: NASA/MSFC (EMMETT GIVEN).

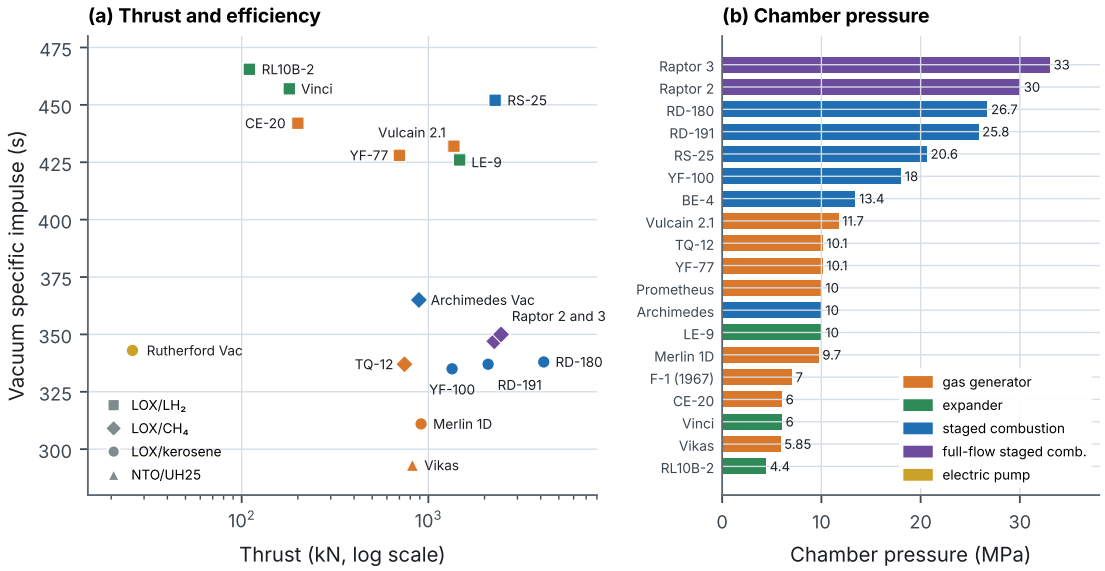
## 35.5 Engines of 2026

Table 35.3 and Figure 35.11 compare the main engines of 2026. Chamber pressures have risen from about 7 MPa for the F-1 to 33 MPa for Raptor 3, making engines smaller for their thrust. Methane has become the fuel of new reusable boosters in the United States, China and Europe. The newest kerosene and methane engines have thrust-to-weight ratios of 140–185, against about 54 for the RS-25, which shows the weight penalty of pumping hydrogen.

Several engines changed status in 2025–26. Raptor 3 first flew on Starship Flight 12 on 22 May 2026. Blue Origin is raising the BE-4 to about 2.85 MN with subcooled propellants, but a New Glenn booster exploded in a pre-launch hot fire on 28 May 2026 (Table 35.5). India hot-fired the CE-20 at an uprated 22 tonnes-force in 2026, and Prometheus awaits hop tests on Europe's Themis demonstrator. Rocket Lab moved the first launch of Neutron, with its Archimedes engines, to 2027 after problems with a composite tank.

**TABLE 35.3** Launch-vehicle engines in 2026. GG = gas generator, SC = staged combustion, FFSC = full-flow staged combustion. Raptor Isp: vacuum value of the sea-level engine; Rutherford and Archimedes vacuum figures: vacuum versions; arrows: uprating announced in 2025. T/W from sea-level thrust where available, otherwise vacuum thrust; "—" not published. CN China, EU Europe, IN India, JP Japan, NZ New Zealand, RU Russia. Sources: manufacturers and agencies, checked October 2026.

| Engine | Cycle | Propellants | Thrust SL / vac (kN) | $I_{sp}$ vac (s) | $p_c$ (MPa) | T/W |
|---|---|---|---|---|---|---|
| Merlin 1D (US) | GG | LOX/RP-1 | 845 / 914 | 311 | 9.7 | ≈ 180 |
| Raptor 2 (US) | FFSC | LOX/$CH_4$ | 2 256 / — | ≈ 347 | 30 | 141 |
| Raptor 3 (US) | FFSC | LOX/$CH_4$ | 2 452 / — | ≈ 350 | 33 | 164 |
| RS-25 (US) | SC fuel-rich | LOX/$LH_2$ | 1 859 / 2 279 | 452 | 20.6 | 54 |
| BE-4 (US) | SC ox-rich | LOX/$CH_4$ | 2 450 → 2 850 | — | 13.4 | — |
| BE-3U (US) | Exp. bleed | LOX/$LH_2$ | — / 711 → 890 | — | — | — |
| RL10B-2 (US) | Expander | LOX/$LH_2$ | — / 110 | 465.5 | 4.4 | ≈ 37 |
| RD-180 (RU) | SC ox-rich | LOX/RP-1 | 3 830 / 4 150 | 338 | 26.7 | 71 |
| RD-191 (RU) | SC ox-rich | LOX/RP-1 | 1 920 / 2 090 | 337 | 25.8 | 85 |
| Vulcain 2.1 (EU) | GG | LOX/$LH_2$ | — / ≈ 1 370 | ≈ 432 | 11.7 | — |
| Vinci (EU) | Expander | LOX/$LH_2$ | — / 180 | 457 | 6.0 | ≈ 33 |
| Prometheus (EU) | GG | LOX/$CH_4$ | ≈ 980 | — | 10 | — |
| YF-100 (CN) | SC ox-rich | LOX/RP-1 | 1 200 / 1 340 | 335 | 18 | — |
| YF-77 (CN) | GG | LOX/$LH_2$ | 518 / 700 | 428 | 10.1 | — |
| TQ-12 (CN) | GG | LOX/$CH_4$ | 658 / 745 | 337 | 10.1 | — |
| LE-9 (JP) | Exp. bleed | LOX/$LH_2$ | — / 1 471 | 426 | 10 | ≈ 62 |
| Rutherford (NZ) | Electric pump | LOX/RP-1 | 25 / 26 | 343 | — | ≈ 73 |
| Archimedes (US) | SC ox-rich | LOX/$CH_4$ | 730 / 890 | ≈ 365 | 10 | — |
| Vikas (IN) | GG | NTO/UH25 | 737 / 821 | 293 | 5.85 | — |
| CE-20 (IN) | GG | LOX/$LH_2$ | — / ≈ 200 | 442 | 6.0 | ≈ 35 |




Thrust: vacuum where published (sea level for Raptor). Raptor Isp: vacuum value of the sea-level engine. Colours give the cycle in both panels. Sources: manufacturers and agencies (2024–26).

**FIGURE 35.11** Launch-vehicle engines of 2026. (a) Thrust and vacuum specific impulse; marker shape gives the propellant, colour the cycle. (b) Chamber pressure: most staged-combustion engines run two to three times higher than gas-generator engines, although the methane-fuelled BE-4 and Archimedes use moderate pressures.

# 35.6 Solid Rocket Motors

## 35.6.1 Anatomy

A solid motor is a pressure vessel filled with propellant. The **case** — steel or, increasingly, carbon-fibre composite — is lined with rubbery insulation, and the propellant **grain** is cast inside it around a shaped mandrel that leaves an internal channel. An **igniter** fires hot gas down the channel, and the burning surface advances until it reaches the insulation. The ablative **nozzle**, with a graphite or carbon–carbon throat, is often partly submerged in the case and mounted on a flexible bearing for steering.

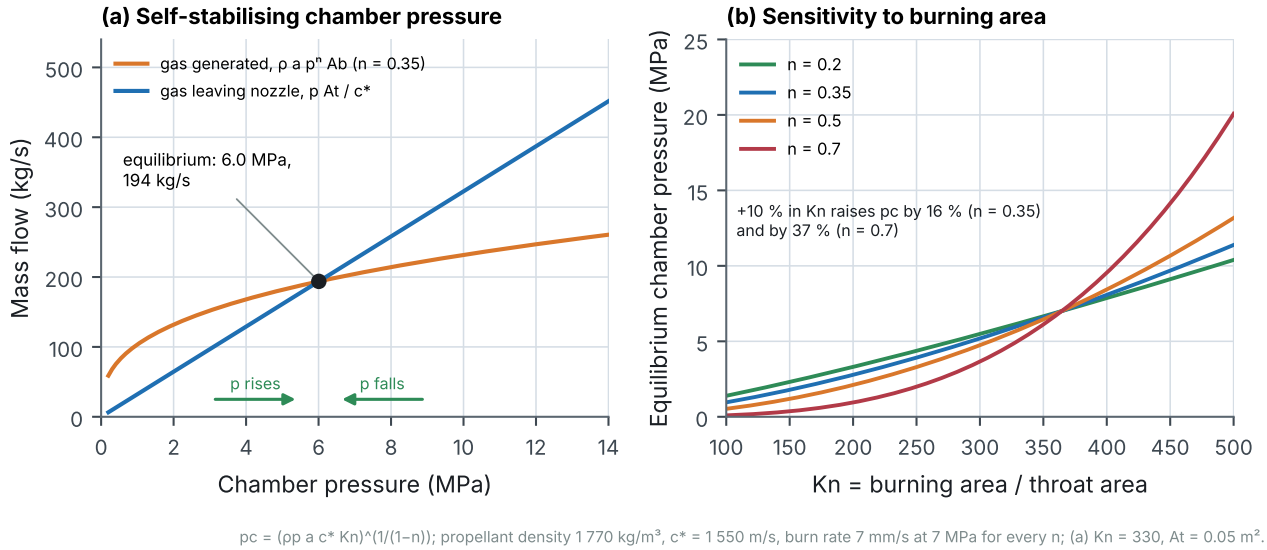
Large motors use **ammonium perchlorate composite propellant** (APCP): ammonium perchlorate oxidiser and aluminium powder in a rubbery binder. The Shuttle boosters used 69.8 % ammonium perchlorate, 16 % aluminium, 12 % PBAN binder, 2 % epoxy curing agent and 0.2 % iron oxide catalyst; newer motors use an HTPB binder. Solids are dense, store for years and give enormous thrust from a simple structure. But they have lower Isp, exhaust containing hydrogen chloride and no shut-down once lit. There is also the danger that a crack or void over-pressurises the case.

## 35.6.2 Burn Rate and Chamber Pressure

The burning surface recedes at a rate that depends mainly on pressure, following **Saint-Robert's law** $r = a\,p_c^{\,n}$, where $a$ depends on the propellant and its temperature and the exponent $n$ is typically 0.3–0.5. Gas is produced at $\rho_p r A_b$ ($\rho_p$ the propellant density, $A_b$ the burning area) and leaves through the nozzle at $p_c A_t / c^*$. Equating the two gives the chamber pressure:

$$p_c = \left(a\,\rho_p\,c^*\,K_n\right)^{1/(1-n)}, \qquad K_n = A_b / A_t$$

The equilibrium is stable only if $n < 1$ (Figure 35.12): a small rise in pressure then increases the nozzle outflow more than the gas production, so the pressure falls back. The exponent $1/(1-n)$ also amplifies any change in burning area. With $n$ = 0.35, a 10 % increase in $K_n$ — from a crack, a debond or the grain geometry — raises the pressure by 16 %; with $n$ = 0.7, by 37 %. Designers therefore choose low-exponent propellants and inspect grains for voids by X-ray tomography. Burn rate also rises with grain temperature.




pc = (ρp a c* Kn)^(1/(1−n)); propellant density 1 770 kg/m³, c* = 1 550 m/s, burn rate 7 mm/s at 7 MPa for every n; (a) Kn = 330, At = 0.05 m².

**FIGURE 35.12** Solid-motor ballistics. (a) Gas generated by the grain and gas discharged through the nozzle balance at the operating point, which is stable because n < 1. (b) Equilibrium pressure versus the ratio of burning area to throat area; a high pressure exponent makes the motor sensitive to small changes in burning area.

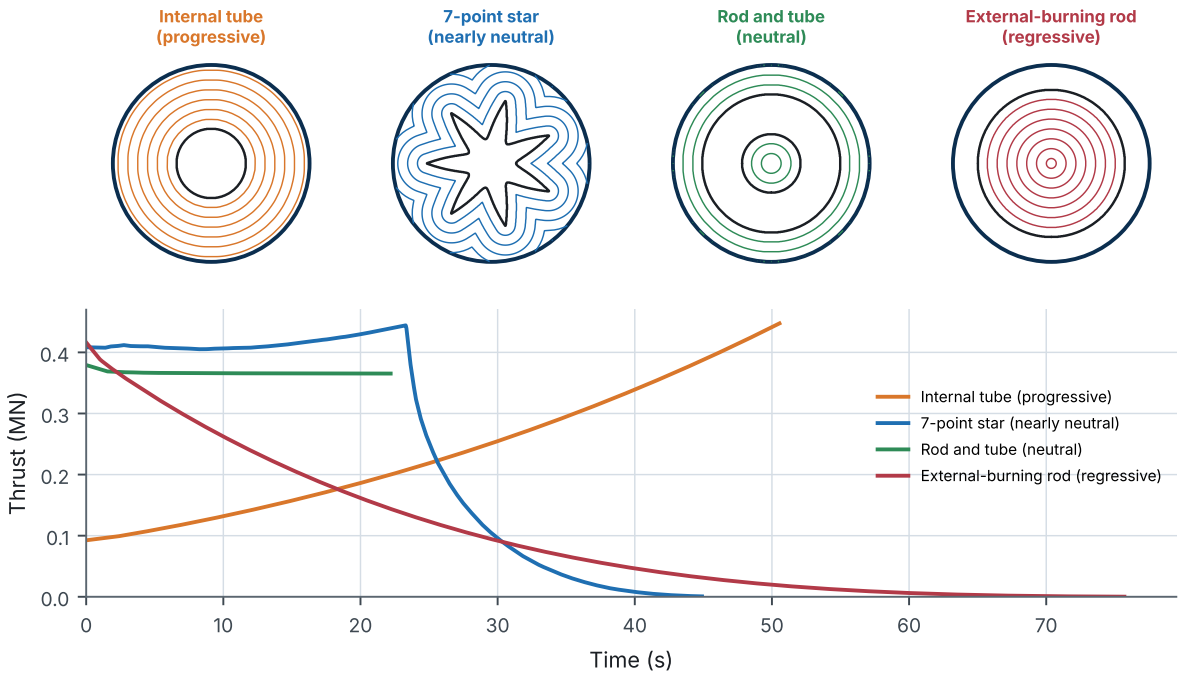
**EXAMPLE: EQUILIBRIUM PRESSURE AND THRUST OF A SOLID MOTOR**

A motor has $A_t$ = 0.10 m² and $A_b$ = 32 m² ($K_n$ = 320). The propellant has $\rho_p$ = 1 770 kg/m³, $c^*$ = 1 550 m/s and $n$ = 0.35, and burns at 7 mm/s at 7 MPa, so $a = 0.007/(7\times10^6)^{0.35}$ = 2.81 × 10⁻⁵ in SI units.

1. **Pressure.** $a\rho_p c^* K_n$ = 2.81 × 10⁻⁵ × 1 770 × 1 550 × 320 = 24 700; raised to the power 1/(1 − 0.35) = 1.538, this gives $p_c$ = **5.73 MPa**.
2. **Burn rate and mass flow.** $r = a p_c^n$ = 6.53 mm/s; $\dot m = \rho_p r A_b$ = 370 kg/s, equal to $p_c A_t / c^*$, as it must be.
3. **Thrust.** With a sea-level $C_F$ of 1.55, $F = C_F p_c A_t$ = **0.89 MN** and $I_{sp}$ = 245 s.
4. **Burn time and margin.** A neutral grain with a 0.40 m web burns for 0.40/0.006 53 ≈ **61 s**; a crack adding 10 % burning area would raise $p_c$ to 6.63 MPa.

## 35.6.3 Grain Geometry

A solid motor cannot be throttled, so its thrust history is designed into the grain. The burning surface advances at the same rate everywhere; if its area grows, thrust rises (**progressive**), if constant it is **neutral**, if it shrinks **regressive** (Figure 35.13). An internal tube burns progressively; stars, wagon wheels and slotted grains are nearly neutral; combined shapes give high lift-off thrust followed by a lower sustain level. The Shuttle boosters' forward segment had an 11-point star that burned away early, cutting thrust by about a third through the period of maximum aerodynamic pressure.




Grain 1 m in diameter, 4 m long, ends inhibited; burning surface traced by a level-set (distance-field) burn-back, contours every 5 cm of web; r = a p^0.35 (7 mm/s at 7 MPa), throat sized for 6 MPa at mean area, CF = 1.6.

**FIGURE 35.13** Grain geometry sets the thrust history. Top: cross-sections with burn-back contours every 5 cm of web, computed by tracing the advancing burning surface. Bottom: thrust versus time for each grain with the same size and propellant. The internal tube rises, the external-burning rod falls, and the star leaves slivers that burn out slowly.

## 35.6.4 Segmented and Monolithic Boosters

The largest motors are built from **segments** joined at the launch site by **field joints** sealed with O-rings. On 28 January 1986, at about 2 °C, cold O-rings in the aft field joint of Challenger's right booster failed to seal. Hot gas escaped and breached the external tank. The vehicle broke up 73 s after lift-off, killing all seven crew. The redesigned joint added a capture feature, a third O-ring and heaters.

NASA's Space Launch System uses five-segment boosters, each producing about 16 MN — together more than 75 % of lift-off thrust — for 126 s. The first composite-cased Booster Obsolescence and Life Extension (BOLE) motor ran for more than two minutes on 26 June 2025, but its nozzle suffered an anomaly near the end of the burn (Figure 35.14). Europe uses **monolithic** carbon-fibre motors: the P120C serves Ariane 6 and Vega-C, and the larger P160C first flew on Ariane 6 in June 2026. Vulcan's GEM 63XL boosters lost a nozzle in October 2024 and suffered an apparent burn-through in February 2026. After that, the US Space Force paused national-security launches on Vulcan. An improved motor was static-fired in April 2026, and Vulcan had not flown again by mid-October 2026 (Table 35.4). A large solid motor cannot be switched off; range safety destroys it by splitting the case.

**TABLE 35.4** Large solid rocket motors. Values are rounded; "—" = not published.

| Motor (vehicle) | Propellant (t) | Max thrust (kN) | $I_{sp}$ vac (s) | Burn (s) | Case |
|---|---|---|---|---|---|
| SLS five-segment booster | ≈ 726 | ≈ 16 000 | — | 126 | Steel, 5 segments |
| Shuttle SRB (1981–2011) | ≈ 500 | ≈ 14 700 | 268 | 123 | D6AC steel, 4 segments |
| S200 (LVM3) | ≈ 205 | 5 150 | 274.5 | ≈ 130 | Maraging steel, 3 segments |

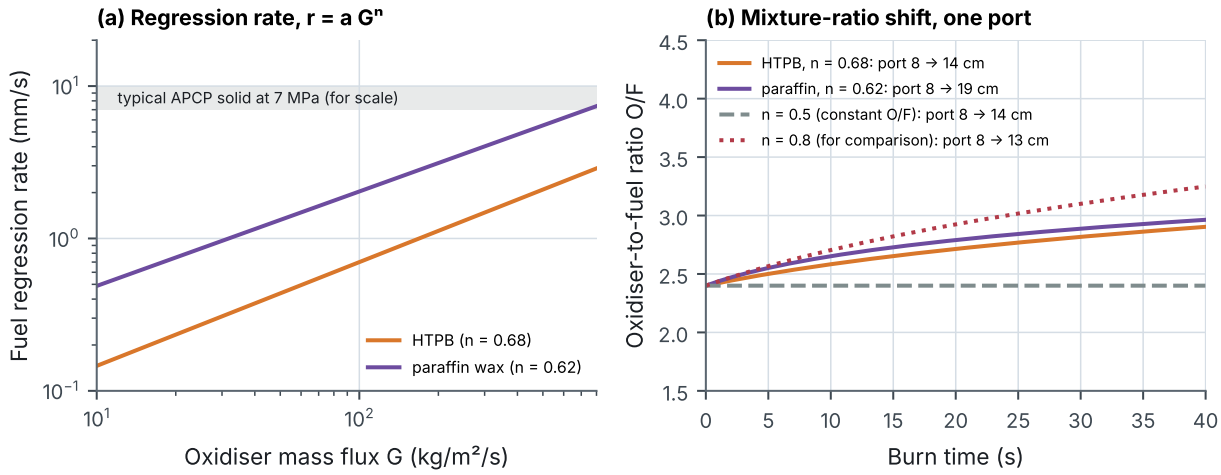
| Motor (vehicle) | Propellant (t) | Max thrust (kN) | $I_{sp}$ vac (s) | Burn (s) | Case |
|---|---|---|---|---|---|
| P160C (Ariane 6, 2026) | 156 | ≈ 4 780 | — | — | Carbon fibre, monolithic |
| P120C (Ariane 6, Vega-C) | ≈ 142 | ≈ 4 500 | — | ≈ 135 | Carbon fibre, monolithic |
| GEM 63XL (Vulcan) | 47.9 | 2 061 | 280 | 87 | Carbon fibre, monolithic |

**FIGURE 35.14** Northrop Grumman and NASA fire the BOLE five-segment demonstration motor horizontally at Promontory, Utah, on 26 June 2025. Image: NASA/MSFC (Northrop Grumman/Dylan Baker).

## 35.7 Hybrid Rockets

A **hybrid rocket** stores a liquid or gaseous oxidiser — nitrous oxide, oxygen or hydrogen peroxide — in a tank. Its solid fuel, usually a rubber such as HTPB or a wax, sits in the chamber. The oxidiser flows down a port in the fuel grain; fuel vaporises from the surface and burns in a flame within the boundary layer. With oxidiser and fuel kept apart, hybrids are safe to store, can be throttled and restarted with the oxidiser valve, and tolerate cracks in the grain.

Because the flame is fed by heat transferred through the boundary layer, the fuel **regression rate** depends not on pressure but on the oxidiser mass flux $G = \dot{m}_{ox}/A_{port}$: $r = a\,G^{\,n}$, with $n \approx 0.5$–$0.8$. HTPB regresses at only about 0.5–2 mm/s, several times slower than a solid propellant burns (Figure 35.15), so hybrids need long grains or several ports, which waste volume and leave unburnt slivers. Paraffin-based fuels, which form a thin liquid layer that sheds droplets into the flow, regress about three times faster. As the port widens the oxidiser flux falls, so even with constant oxidiser flow the **mixture ratio shifts**. O/F varies as $r_{port}^{\,2n-1}$, drifting upwards by about 20 % in a 40 s burn for typical exponents.




Laboratory correlations for gaseous oxygen. (b) Constant oxidiser flow 10 kg/s, single circular port of 8 cm initial radius, length set for O/F = 2.4 at ignition; O/F varies as r^(2n−1).

**FIGURE 35.15** Hybrid rocket physics. (a) Regression rate versus oxidiser mass flux for HTPB and paraffin fuels, compared with a typical solid propellant. (b) Mixture ratio during a 40 s burn with constant oxidiser flow through one circular port; only for n = 0.5 would it stay constant.

Hybrids have flown on SpaceShipOne and SpaceShipTwo, whose RocketMotorTwo burned nitrous oxide with a rubber and later a polyamide fuel (Chapter 9). They have also flown on Nammo's Nucleus sounding rocket (hydrogen peroxide and HTPB), which reached about 107 km in 2018. Australia's Gilmour Space built its Eris orbital launcher around Sirius hybrids burning hydrogen peroxide with a 3-D printed fuel grain, four of about 115 kN each on the first stage. Eris's first flight failed about 14 s after lift-off on 30 July 2025; a second is expected around the end of 2026.

**FIGURE 35.16** A hybrid rocket motor developed by Sierra Nevada Corporation for the Dream Chaser spaceplane, firing near San Diego in 2010. It was fired three times in one day to demonstrate restarts. Image: NASA/KSC.

## 35.8 Testing and Qualification

### 35.8.1 Test Stands and Campaigns

Engines are developed by testing: combustion stability, turbopump dynamics and chamber life cannot yet be predicted well enough to skip hot fire. A programme moves from component tests — injector elements, preburners, a turbopump on a "powerpack" rig — to complete engines, stages and often the whole vehicle on the pad. Development engines are run beyond their limits.

NASA's **Stennis Space Center** in Mississippi tested every Space Shuttle main engine and tests the RS-25s for SLS. Its Fred Haise (A-1) stand certified the new production RS-25 in 2023, running it at up to 113 % power. A January 2026 hot fire cleared engine 2063 for Artemis IV, and the B-2 stand fired the complete SLS core stage in 2021 (shown under the chapter title). SpaceX's **McGregor** site in Texas has about a dozen stands on which Merlin and Raptor engines are fired before delivery. At the German Aerospace Center's **Lampoldshausen** site the P5 stand tests Vulcain 2.1, P4.1 tests Vinci under simulated altitude conditions, and P5.2 fires the complete Ariane 6 upper stage. Upper-stage engines with large nozzles are fired in **altitude-simulation** facilities, vacuum chambers in which the exhaust, passing through a diffuser, helps hold the low pressure (Figure 35.17, Figure 35.18).

**FIGURE 35.17** A Merlin engine mounted on a test stand at SpaceX's McGregor facility in Texas, where engines are hot-fired before they fly. Image: NASA/KSC (SpaceX).

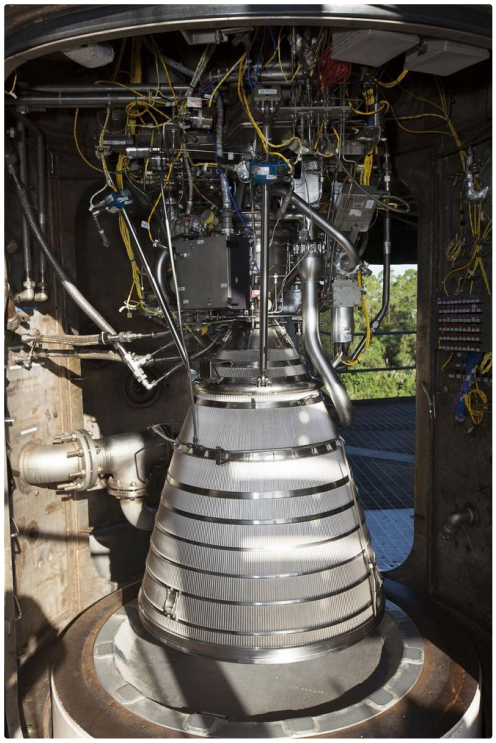


**FIGURE 35.18** An RL10 in the vacuum test chamber at Aerojet Rocketdyne's West Palm Beach facility, Florida, in 2016. It was one of two engines built for the Centaur upper stage of the Atlas V that would launch Starliner's crewed flight test. Image: NASA/KSC (NASA/Dimitri Gerondidakis).

**Qualification** shows that a design meets its requirements with margin. Engines are run for several times their mission duration, at the corners of the allowed ranges of thrust, mixture ratio and inlet conditions, and through many starts. The RS-25 has accumulated more than 1.1 million seconds of hot fire. Every flight engine then passes an **acceptance test** to catch defects.

## 35.8.2 Engine-Out and Failure Modes

Clustering allows engine-out survival. Falcon 9 lost an engine on the CRS-1 mission in October 2012 and still delivered Dragon to orbit. On Starship Flight 12 one of 33 Raptor 3s shut down early during ascent, yet the vehicle still reached its planned trajectory. Controllers shut an engine

down when a **redline** — a limit on pressure, temperature, speed or vibration — is crossed. Many vehicles are held down for a few seconds after ignition, so that a fault can stop the launch (Table 35.5).

**TABLE 35.5** Propulsion failures that changed engineering practice.

| Event | What happened | Consequence or lesson |
|---|---|---|
| F-1 development, 1962–65 | High-frequency combustion instability destroyed engines on the stand | Baffles, injector redesign, bomb tests |
| Challenger, 28 Jan 1986 | Cold O-rings let hot gas escape from a booster field joint | Joint redesign; temperature limits; decision-making reform |
| STS-93, 23 Jul 1999 | Injector pin punctured three nozzle coolant tubes | Hydrogen leak; main engines shut down early |
| Falcon 9 Starlink 9-3, Jul 2024 | Oxygen leak chilled the Merlin Vacuum; hard start at relight | Satellites left in a 135 km perigee orbit and lost |
| Vulcan, Oct 2024 and Feb 2026 | GEM 63XL nozzle lost; later an apparent burn-through | National-security flights paused; nozzle improved |
| New Glenn, 28 May 2026 | Booster exploded during pre-launch hot fire | Traced to the main oxygen valve of one BE-4 |

## 35.9 Beyond Chemical Propulsion

Chemical rockets are limited by the energy stored in their propellants: hydrogen and oxygen release about 13 MJ per kilogram, which in practice caps vacuum Isp near 470 s. Missions that need large velocity changes over months use **electric propulsion**, which accelerates ions or plasma to 15–50 km/s at tiny thrust (Chapter 6). Nuclear thermal rockets, solar sails and other advanced concepts are covered in Chapter 42.

### AI IN ROCKET PROPULSION

Combustion instability suits machine learning because it is hard to predict from first principles but leaves signatures in pressure data. Researchers at DLR and the University of Cambridge trained Bayesian neural networks on pressure signals from a hydrogen–oxygen research combustor. The networks forecast the onset of instability about half a second ahead — early enough for a controller to react. Engine teams also use machine-learning surrogates of slow simulations and anomaly detection on hot-fire data. **Computational engineering** tools generate whole engine geometries from requirements: in December 2024 the Dubai company LEAP 71 hot-fired a 5 kN kerosene–oxygen aerospike designed largely by its software and printed in copper. Chapter 52 discusses AI in design and manufacturing; certification still rests on hot fire.

### KEY TAKEAWAYS

- Thrust is $\dot{m}v_e + (p_e - p_a)A_e$; specific impulse $F/(\dot{m}g_0)$ measures propellant efficiency, while $c^*$ and $C_F$ separate the performance of the chamber from that of the nozzle.
- A de Laval nozzle chokes at the throat, so $\dot{m} = p_c A_t / c^*$; exhaust velocity grows with $\sqrt{T_c/\mathcal{M}}$, which is why hydrogen engines reach 420–465 s.
- Nozzles are sized for altitude: sea-level engines use expansion ratios of about 15–40 to avoid flow separation, vacuum engines 60–280.
- Propellants trade Isp against density, storability and handling; dense propellants suit first stages, hydrogen upper stages, and methane has become the fuel of new reusable boosters.
- Engine cycles differ in how they drive the turbopumps: gas generators are simple, expanders benign, staged combustion efficient; full-flow Raptor 3 reaches 33 MPa.
- A solid motor settles at $p_c = (a\rho_p c^* K_n)^{1/(1-n)}$; grain shape sets its thrust history and a low exponent keeps it tolerant of defects. Hybrids are safe and throttleable but regress slowly and shift mixture ratio.
- Engines are proven by hot fire, from components to whole vehicles, with margins found by exceeding limits.

### REVIEW QUESTIONS

1. Write the thrust equation and explain why an engine gives more thrust in vacuum than at sea level. By how much does the RS-25's thrust rise from sea level to vacuum?
2. A combustion gas has $c^*$ = 1 800 m/s and the nozzle a vacuum $C_F$ of 1.85. What is the vacuum Isp? For $p_c$ = 10 MPa and a thrust of 1 MN, what throat diameter is needed?
3. Using the exit-velocity formula, explain why hydrogen–oxygen gives a higher Isp than kerosene–oxygen although its flame is cooler.
4. What happens to the flow in an over-expanded nozzle, and why do first-stage engines not use the large nozzles of upper-stage engines?
5. Using density impulse, explain why most first stages burn kerosene, methane or solid propellant while many upper stages burn hydrogen.
6. Compare the gas-generator, expander and staged-combustion cycles. Why is the closed expander limited to engines of a few hundred kilonewtons?
7. What is NPSH, and how do designers prevent cavitation at a pump inlet?
8. A solid motor with n = 0.4 runs at 6 MPa. A debond increases the burning area by 15 %. Estimate the new chamber pressure and explain why a high pressure exponent is dangerous.
9. Why does the mixture ratio of a hybrid drift during the burn, and how do paraffin fuels help with the low regression rate of HTPB?

# 36 CHAPTER Launch Vehicles of the World

Image: NASA/KSC (NASA/Frank Michaux)

*In 2025 the world attempted 330 orbital launches, more than in any previous year, on some forty different families of rockets flown by providers in nine countries and Europe. This chapter surveys those vehicles. It traces the launch vehicle from the R-7 missile of 1957 through the Saturn V, the Space Shuttle, Energia, Proton, Ariane, Long March, H-II and PSLV, then describes the fleet of October 2026 country by country — from Falcon 9 and Starship to Ariane 6, Soyuz, the Long March family, China's commercial start-ups, H3, LVM3 and the newest small launchers. Launch data from 1957 to 2026 show who launches, how much mass reaches orbit, how reliable each class of rocket is and how fast cadence has grown. The chapter ends with the trends reshaping launch: reusability, rideshare, cadence, the heavy-lift race, the small-launcher shake-out and the demand for sovereign access to space.*

**IN THIS CHAPTER**

36.1 A Short History of the Launch Vehicle
36.2 The World Fleet in 2026
36.3 Launch by the Numbers
36.4 Trends Reshaping Launch

Between October 1957 and October 2026 the world made about 7 400 orbital launch attempts. For most of that time the annual total stayed between 50 and 140; then, after 2018, it took off. In 2025 there were 330 attempts, of which 318 succeeded, and by 13 October 2026 another 244 had been made. Half of all launches in 2025 used a single type of rocket, SpaceX's Falcon 9, and most of those carried the company's own Starlink satellites. At the same time more countries and companies than ever were building rockets: China alone flew sixteen different families of launch vehicles in 2025.

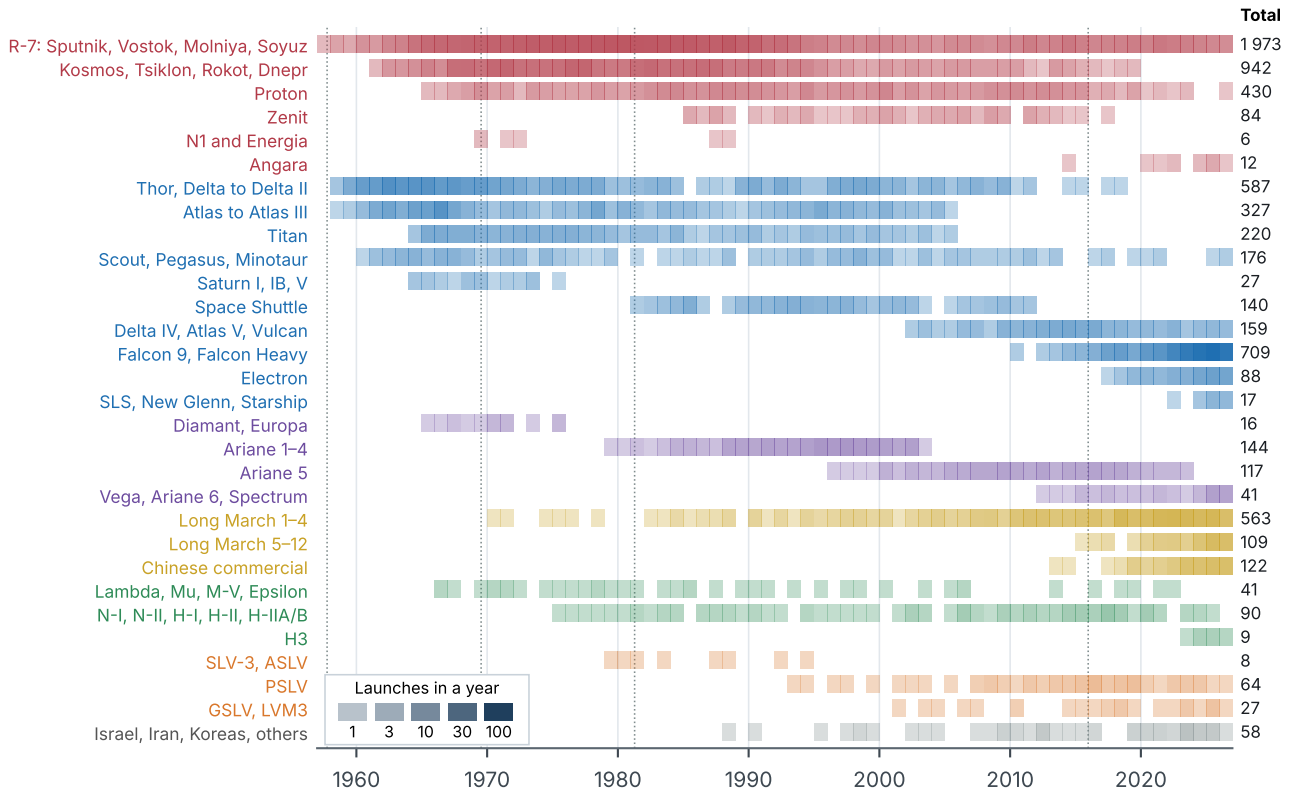
This chapter is a field guide to that fleet. It does not repeat the engineering of rockets: the rocket equation, staging, structures and guidance are the subject of Chapter 34, and engines of Chapter 35. Instead it uses that engineering to compare vehicles. It looks at how much they lift, what they burn, whether they can be reused, how often they fly and fail, and what they cost. Launch sites and ground systems are covered in Chapter 37, and suborbital rockets in Chapter 9. The launch statistics here come from Jonathan McDowell's General Catalog of Artificial Space Objects (GCAT) as of 13 October 2026. Appendix E lists every launch of 2025 and 2026 and the all-time totals by country, vehicle family and launch site.

## 36.1 A Short History of the Launch Vehicle

### 36.1.1 Missiles Become Launchers

The first launch vehicles were ballistic missiles with a satellite in place of a warhead. Sergei Korolev's **R-7** was built to throw a nuclear warhead of about 5 t across continents. It had a central core and four strap-on boosters, all burning kerosene and liquid oxygen and all ignited on the ground. A modified R-7 launched Sputnik on 4 October 1957, and a version with an upper stage carried Yuri Gagarin in 1961. Its descendants — Vostok, Voskhod, Molniya and Soyuz — have flown about 1 970 times, more than any other rocket family (Figure 36.1). Crews still fly to the International Space Station on a Soyuz-2, a modernised R-7 with digital avionics.

The United States followed a different route. The civilian Vanguard exploded on its first orbital attempt in December 1957, and the Army's Juno I, derived from the Redstone missile, put Explorer 1 into orbit on 31 January 1958. Three military missiles then became the American workhorses for half a century. The **Thor** intermediate-range missile grew, stage by stage, into the **Delta** family; the last Delta II flew in 2018. The **Atlas** missile, with its balloon tanks held rigid by internal pressure, launched the orbital Mercury flights. Topped by the hydrogen-fuelled **Centaur** stage (first successful flight in 1963), it also launched the Surveyor Moon landers and many planetary probes. The **Titan II** missile carried the Gemini crews, and with large solid boosters became the Titan III and IV, which launched heavy military satellites until 2005.




Each cell is one year; shading shows the number of orbital launch attempts (log scale). Dotted lines: Sputnik 1957, Apollo 11 1969, first Shuttle 1981, first Falcon 9 landing 2015. Data: GCAT (J. McDowell, CC-BY 4.0), launches to 13 Oct 2026.

**FIGURE 36.1** Orbital launch activity of the main launch-vehicle families, 1957–2026. Each cell is one year, shaded by the number of launch attempts; the totals at the right count all attempts since the first flight. Families born as missiles in the 1950s dominated until the 1990s; since 2015 the activity has shifted to Falcon 9, the new Long March rockets and Chinese commercial vehicles.

## 36.1.2 The Moon Race and the Super-Heavies

The race to the Moon produced the largest rockets of the century. NASA's **Saturn V** stood 110.6 m tall and weighed almost 3 000 t at lift-off. Its five F-1 engines (Chapter 35) gave 34 MN of thrust. The three stages could place about 140 t in low Earth orbit (LEO), including the third stage and the Apollo spacecraft bound for the Moon. All thirteen Saturn Vs launched between 1967 and 1973 reached space (Figure 36.2); the last lifted the Skylab station. The Soviet answer, the **N1**, had thirty engines in its first stage. Its first stage was never fired on a test stand as a complete stage. All four launches between 1969 and 1972 failed in that stage — through engine fires, a turbopump explosion and control problems — before the programme was cancelled. The next Soviet super-heavy, **Energia**, flew only twice, in 1987 and 1988, the second time carrying the uncrewed Buran shuttle; its four strap-on boosters were essentially first stages of the Zenit rocket.

## 36.1.3 Shuttles and Workhorses

The **Space Shuttle** (Chapter 10) promised cheap, routine access by reusing its orbiter and solid boosters. It flew 135 missions between 1981 and 2011 and carried most of the International Space Station into orbit (Figure 36.3). But refurbishment was slow and costly, and two orbiters were lost with their crews, *Challenger* in 1986 and *Columbia* in 2003. After *Challenger* the United States stopped launching commercial satellites on the Shuttle; the business moved back to expendable rockets, revived the Delta, Atlas and Titan lines and gave Europe's new Ariane its opening.

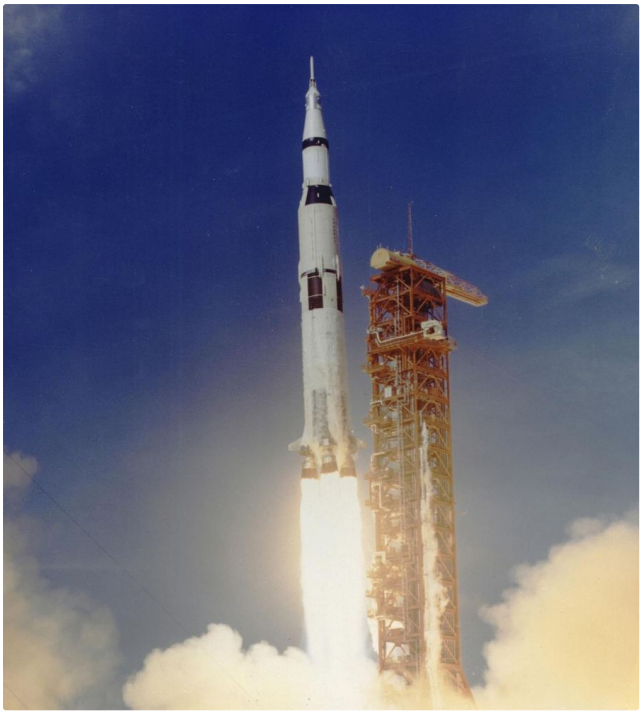


**FIGURE 36.2** A Saturn V lifting off from Kennedy Space Center with Apollo 11 on 16 July 1969. Each of its five F-1 engines burned about 2.6 t of kerosene and oxygen per second. Image: NASA/MSFC.

**FIGURE 36.3** The exhaust plume of Space Shuttle Atlantis climbing away from Pad 39A on STS-135, the last Shuttle mission, on 8 July 2011, seen from NASA's Shuttle Training Aircraft. Image: NASA (NASA/Dick Clark).

In the Soviet Union, two families did most of the work. The **Proton**, first launched in 1965, burns storable nitrogen tetroxide and unsymmetrical dimethylhydrazine (UDMH) — toxic but able to sit in the tanks for weeks. It lifted the Salyut and Mir stations, the Russian modules of the ISS and many geostationary satellites. About 430 have flown. The **Soyuz-U** flew 787 times between 1973 and 2017 (counting its PVB variant), at up to 47 launches a year. In the early 1980s the USSR launched about 100 rockets a year — 108 in 1982 — mostly carrying short-lived reconnaissance satellites that returned their film to Earth.

### 36.1.4 New Launching Nations

France became the third nation to launch its own satellite, with the Diamant rocket in 1965. Japan followed in February 1970 with the small solid Lambda-4S, China in April 1970 with the Long March 1, the United Kingdom in 1971 with Black Arrow and India in 1980 with the SLV-3. Israel (1988), Iran (2009), North Korea (2012) and South Korea (2013, with a Russian first stage; 2022 with its own Nuri) joined later. Table 36.1 lists the landmark vehicles.

After the failure of the multinational Europa rocket, Europe built **Ariane 1**, first launched on 24 December 1979. Its successor Ariane 4 (1988–2003) took about half of the commercial geostationary market. **Ariane 5** (1996–2023) flew 117 times, carrying two large satellites at once to geostationary transfer orbit (GTO). In its most famous flight it carried the James Webb Space Telescope (Figure 36.4). China's Long March family began with the Long March 1. Next came the

Long March 2, which returned film capsules and, as the 2F, has launched every Shenzhou spacecraft since 1999 and every crew since 2003. Then came the hydrogen-fuelled Long March 3 for geostationary missions. Japan developed both a solid-rocket line for science (Mu, M-V and Epsilon) and liquid rockets. Its liquid rockets were first built under licence from the Delta, followed by the all-Japanese **H-II** (1994) and **H-IIA**. The H-IIA flew 50 times from 2001 to 2025 with a single failure (Figure 36.5). India's four-stage **PSLV** (1993) launched Chandrayaan-1 and the Mars Orbiter Mission, and its LVM3 (originally GSLV Mk III) carried Chandrayaan-2 and -3.

**FIGURE 36.4** An Ariane 5 lifts off from Kourou with the James Webb Space Telescope on 25 December 2021. The two solid boosters provided about 90 % of the lift-off thrust. Image: NASA (NASA/Bill Ingalls).

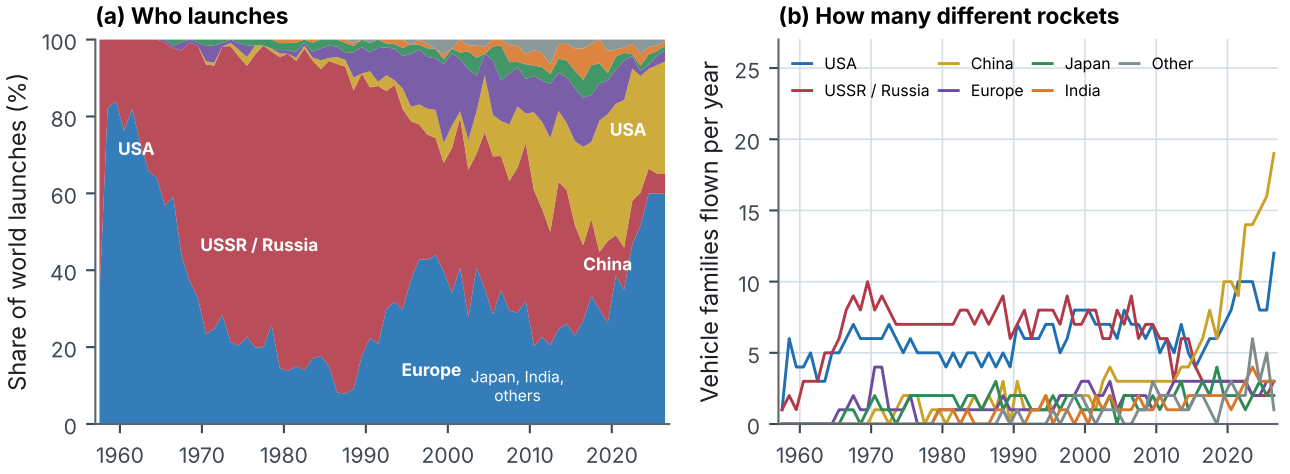
**FIGURE 36.5** An H-IIA launches the NASA–JAXA Global Precipitation Measurement Core Observatory from Tanegashima on 28 February 2014. The H-IIA retired in June 2025 after 50 flights. Image: NASA (NASA/Bill Ingalls).

**TABLE 36.1** Landmark launch vehicles. Flight counts are orbital launch attempts in GCAT to October 2026; payloads are approximate maxima to low orbit.

| Vehicle | Country | First flight | Max. LEO payload | Flights (failures) | Why it matters |
|---|---|---|---|---|---|
| R-7 / Soyuz family | USSR/Russia | 1957 | 1.3–8 t | ≈ 1 970 | First satellite and first human; still flying |
| Thor–Delta | USA | 1958 | to 6 t (Delta II) | ≈ 590 | Commercial workhorse; ended 2018 |
| Saturn V | USA | 1967 | ≈ 140 t | 13 (0) | Apollo Moon landings; Skylab |
| Proton | USSR/Russia | 1965 | 20–23 t | ≈ 430 | Space stations; storable propellants |
| N1 | USSR | 1969 | 95 t (design) | 4 (4) | Lunar rocket that never worked |
| Space Shuttle | USA | 1981 | 27.5 t | 135 missions (1) | Partly reusable; built the ISS |
| Energia | USSR | 1987 | 100 t | 2 (1) | Launched Buran; boosters became Zenit |

| Vehicle | Country | First flight | Max. LEO payload | Flights (failures) | Why it matters |
|---|---|---|---|---|---|
| Ariane 4 / 5 | Europe | 1988 / 1996 | 7.6 / 21 t | 116 / 117 | Commercial GTO market; Webb |
| Long March 2–4 | China | 1974–88 | 3.5–11.5 t | ≈ 530 | Basis of China's programme, crews since 2003 |
| H-II / H-IIA | Japan | 1994 / 2001 | 10 t | 7 / 50 | Independent Japanese hydrogen rockets |
| PSLV | India | 1993 | 3.8 t | 64 (4) | Low-cost rideshare; Moon and Mars probes |

The balance of power shifted twice (Figure 36.6). Until the mid-1960s the United States made most launches. From 1970 to the end of the 1980s the Soviet Union made 70–85 % of them. After its collapse the totals fell, and Europe, China, Japan and India took a growing share. In 2025 the United States and China made nearly nine launches in ten. Diversity has grown too: in 2026 Chinese providers flew 19 different vehicle families and American providers about a dozen, against two or three in Russia, Europe and Japan.

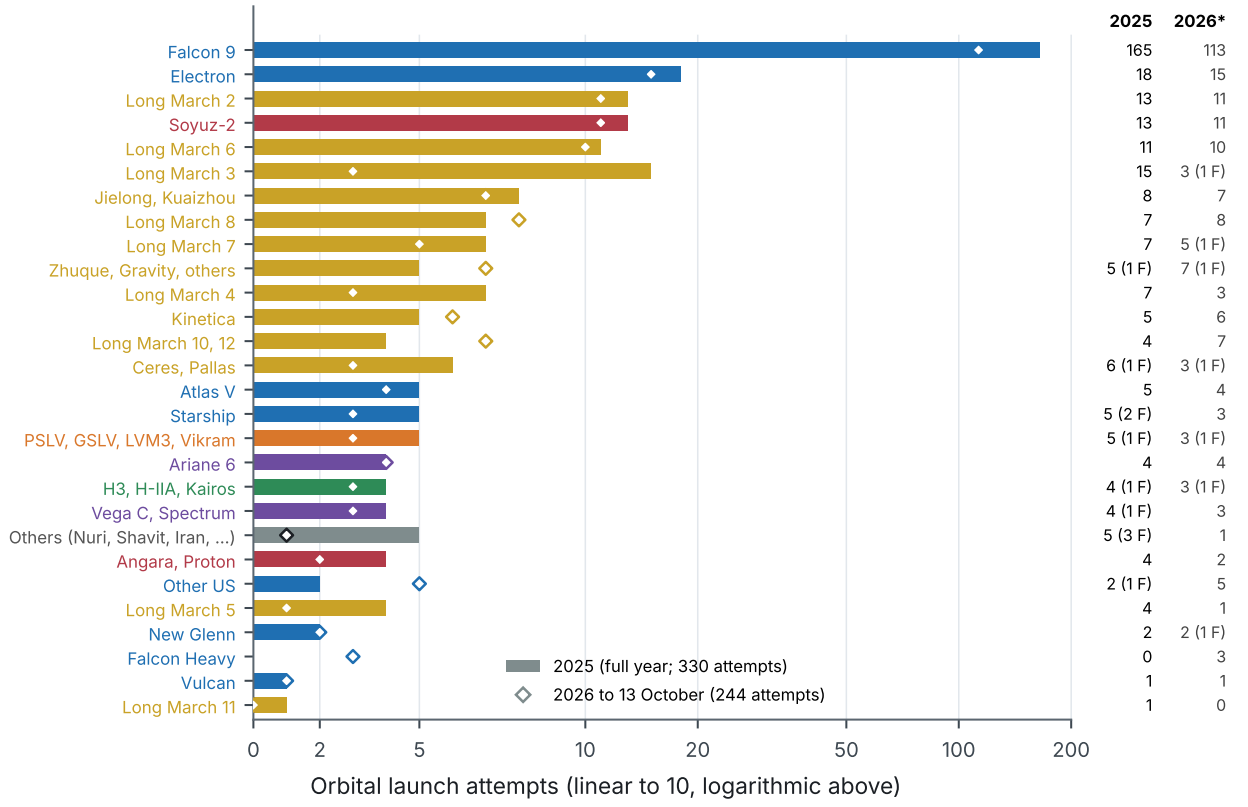



Country of the launch provider (Rocket Lab counted as USA). A family groups the variants of one vehicle (Long March split by number). 2026 to 13 Oct. Data: GCAT (J. McDowell, CC-BY 4.0), launches to 13 Oct 2026.

**FIGURE 36.6** Left: share of world orbital launch attempts by the country of the launch provider, 1957–2026. Right: the number of different vehicle families each country flew per year. China's commercial boom since 2020 has produced the most varied fleet any nation has operated.

## 36.2 The World Fleet in 2026

In 2025, 330 orbital launch attempts were made: 198 by US providers (including Rocket Lab's launches from New Zealand), 93 Chinese, 17 Russian, 8 European, 5 Indian, 4 Japanese and 5 by other countries. Eleven failed. Figure 36.7 breaks these down by vehicle family, together with the 244 attempts made in 2026 up to 13 October; Table 36.2 compares the main medium and heavy vehicles, and Figure 36.8 and Figure 36.9 place them against the rockets of the past.



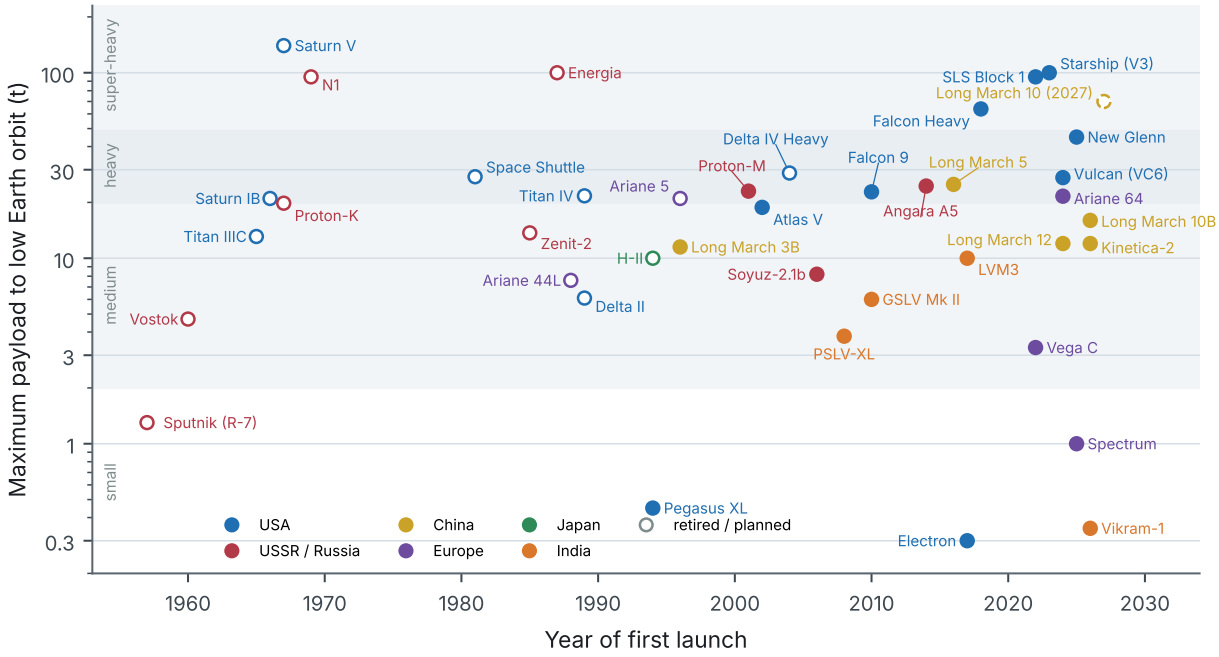

Colour = country of the launch provider (blue USA, red Russia, gold China, violet Europe, green Japan, orange India). F = failures. *To 13 Oct. Data: GCAT (J. McDowell, CC-BY 4.0), launches to 13 Oct 2026.

**FIGURE 36.7** Orbital launch attempts in 2025 (bars) and in 2026 up to 13 October (diamonds) by vehicle family, with failures (F). Falcon 9 made half of all launches in 2025; the Long March rockets, split here by series, made 69.

**TABLE 36.2** Main medium and heavy launch vehicles in 2026. Payloads are approximate published maxima (LEO at low inclination; GTO with 1 800 m/s to geostationary orbit); counts are orbital attempts from GCAT (*2026 to 13 October); prices are published list prices or estimates where available.

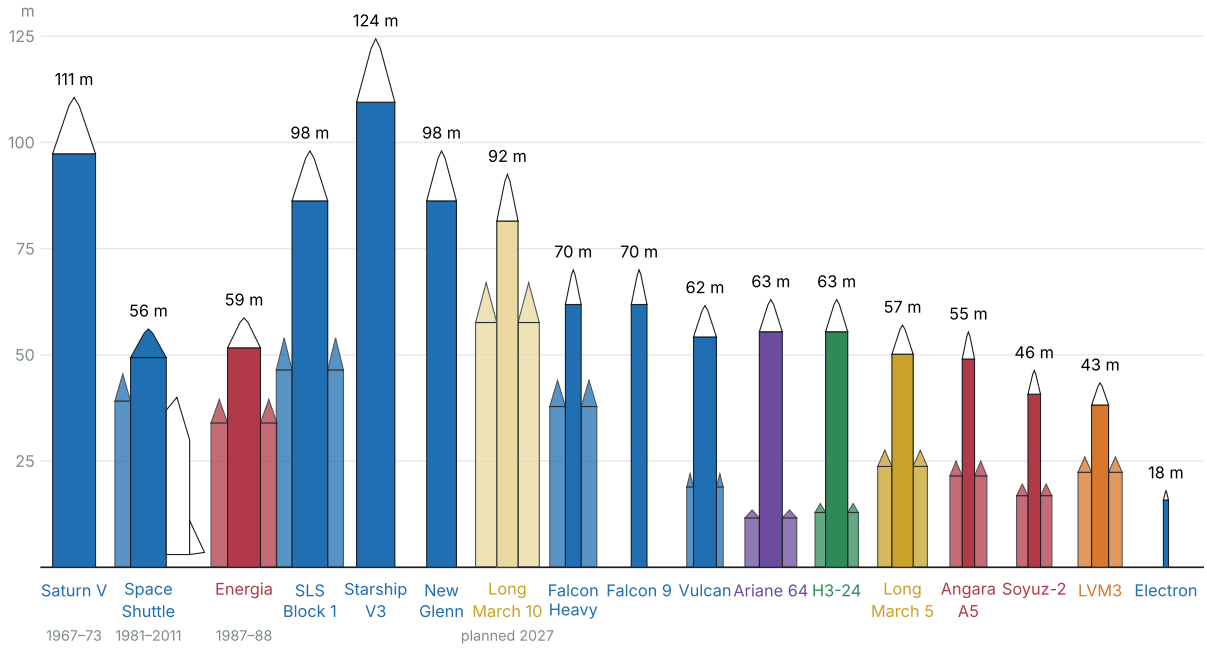
| Vehicle (first flight) | LEO / GTO (t) | Height | Stages and propellants | Reuse | Launches 2025 / 2026* | Price (public) |
|---|---|---|---|---|---|---|
| **Falcon 9** (2010), USA | 22.8 / 8.3 (17.5 / 5.5 reused) | 70 m | 2; LOX/kerosene | Booster, fairing | 165 / 113 | $74 M |
| **Falcon Heavy** (2018), USA | 63.8 / 26.7 | 70 m | 2 + 2 side cores; LOX/kerosene | Boosters | 0 / 3 | ≈ $97 M (reported) |
| **Starship** (2023; V3 2026), USA | ≈ 100 (reusable) | 124 m | 2; LOX/methane | Full (planned) | 5 / 3 | — |
| **SLS Block 1** (2022), USA | 95 (27 to the Moon) | 98 m | Core + 2 solids + upper stage; LOX/$LH_2$ | None | 0 / 1 | ≈ $2.2 billion (NASA OIG est.) |
| **Vulcan Centaur** (2024), USA | 27.2 / 14.4 (VC6) | 62 m | 2 + 0–6 solids; LOX/methane, LOX/$LH_2$ | None | 1 / 1 | — |
| **New Glenn** (2025), USA | 45 / 13 | 98 m | 2; LOX/methane, LOX/$LH_2$ | Booster | 2 / 1 | — |
| **Atlas V** (2002), USA | 18.8 / 8.9 (551) | 58–62 m | 2 + 0–5 solids; LOX/kerosene, LOX/$LH_2$ | None | 5 / 4 | Retiring |
| **Ariane 62 / 64** (2024), Europe | 10.4 / 4.5; 21.7 / 11.5 | 63 m | 2 + 2 or 4 solids; LOX/$LH_2$ | None | 4 / 4 | €100 M / €115 M (est.) |
| **Vega C** (2022), Europe | 2.3 to SSO | 35 m | 3 solid + 1 storable | None | 3 / 2 | — |
| **Soyuz-2** (2006), Russia | 7–8.7 / 3.1 | 46 m | 4 boosters + core + 3rd stage (+ Fregat); LOX/kerosene | None | 13 / 11 | — |
| **Angara A5** (2014), Russia | 24.5 / 5.4 | 55 m | 5 modules + 2nd + upper stage; LOX/kerosene | None | 1 / 0 | — |

| Vehicle (first flight) | LEO / GTO (t) | Height | Stages and propellants | Reuse | Launches 2025 / 2026* | Price (public) |
|---|---|---|---|---|---|---|
| **Proton-M** (2001), Russia | 23 / 6.4 | 58 m | 3 + upper; $N_2O_4$/UDMH | None | 0 / 1 | Retiring |
| **Long March 2C/2D** (1982/92), China | 3.5–3.9 | 41–42 m | 2; $N_2O_4$/UDMH | None | 10 / 9 | — |
| **Long March 3B** (1996), China | 11.5 / 5.5 | 56 m | 3 + 4 boosters; storable, LOX/$LH_2$ | None | 13 / 3 | — |
| **Long March 5 / 5B** (2016), China | 25 / 14 | 57 m | 2 + 4 boosters; LOX/$LH_2$, LOX/kerosene | None | 4 / 1 | — |
| **Long March 6A / 7A / 8A / 12** (2020–24), China | 8–12 / up to 7 | 50–60 m | 2–3 (+ boosters); LOX/kerosene (+ $LH_2$) | None (12A/B: tests) | 25 / 22 | — |
| **Long March 10B** (2026), China | 16 | 70 m | 2; LOX/kerosene, LOX/methane | Booster (caught at sea) | — / 1 | — |
| **H3** (2023), Japan | 6.5 GTO (H3-24); 4 SSO (H3-30) | 57–63 m | 2 + 0–4 solids; LOX/$LH_2$ | None | 3 / 2 | ≈ ¥5 billion target (H3-30) |
| **LVM3** (2017), India | 10 / 4.2 | 43 m | 2 solids + 2; storable, LOX/$LH_2$ | None | 2 / 0 | — |
| **PSLV / GSLV Mk II** (1993/2001), India | 3.8 / 1.3; 6 / 2.3 | 44 / 52 m | 4 / 3 stages; solid, storable, LOX/$LH_2$ | None | 3 / 2 | — |




Approximate published maximum LEO payloads (expendable configuration, low inclination; definitions vary between sources). Filled: flying in 2026.

**FIGURE 36.8** Maximum payload to low Earth orbit of major launch vehicles against the year of their first launch. The super-heavies of the 1960s–80s were retired; the 2020s have brought a second generation — SLS, Starship and, from 2027, China's Long March 10 — alongside reusable heavies such as Falcon Heavy and New Glenn.




Heights and diameters to a common scale (simplified outlines; side boosters shown in pairs). Grey dates: retired or planned vehicles.

**FIGURE 36.9** Launch vehicles of the past and present drawn to the same scale. Starship V3, at 124 m, is the tallest rocket ever flown; Electron, at 18 m, is the smallest in regular service.

## 36.2.1 United States

**SpaceX** dominates the world's launch market. Falcon 9 made 165 of the 330 attempts in 2025 and 113 more by mid-October 2026. Its first stage lands on a droneship or a landing zone after almost every flight (Chapter 34): the fleet leader flew for the 37th time in August 2026. SpaceX lists a Falcon 9 at $74 million in 2026. **Falcon Heavy**, three Falcon 9 cores strapped together, did not fly in 2025 but launched three times in 2026, including NASA's Nancy Grace Roman Space Telescope on 30 August. **Starship**, designed for full reuse, is the largest rocket ever built. The Version 3 vehicle stands 124.4 m tall and weighs about 5 000 t. It has 33 Raptor engines on its Super Heavy booster and six on the ship. After three ships were lost in the first half of 2025, the last two flights of Version 2 succeeded. On Flight 14, on 28 September 2026, a ship reached orbit for the first time and deployed 26 Starlink V3 satellites. It returned to splash down in the Pacific three hours later. Catching the ship on the launch tower, the step needed for full reuse, was the next goal, expected on Flight 15. NASA's Artemis lunar landings depend on refuelling Starships in orbit, which has yet to be demonstrated.

NASA's **Space Launch System** (SLS) is the government's own super-heavy rocket. It uses Space Shuttle engines and five-segment solid boosters and is thrown away on every flight. After the uncrewed Artemis I in 2022 it launched the four astronauts of Artemis II around the Moon on 1 April 2026 (Figure 36.10). In February 2026 NASA dropped plans for the larger Exploration Upper Stage, and in March it chose a Centaur V upper stage for later SLS rockets. NASA's Inspector General has estimated that each Artemis launch costs more than $4 billion, including the Orion spacecraft and ground systems.

**FIGURE 36.10** The SLS rocket lifts off from Pad 39B with the four Artemis II astronauts aboard Orion on 1 April 2026. The two solid boosters provide more than three-quarters of the lift-off thrust. Image: NASA/MSFC (NASA/Michael DeMocker).

**United Launch Alliance** (ULA), owned equally by Boeing and Lockheed Martin, is replacing its Atlas V and the retired Delta IV with **Vulcan Centaur**. Vulcan's first stage burns liquefied natural gas in two Blue Origin BE-4 engines. Atlas V flew 110 times from 2002 without losing a payload; its last Amazon Leo launch, on 2 July 2026, left six rockets reserved for Boeing's Starliner capsule. Vulcan has had a difficult start. On its second flight in October 2024 the nozzle fell off one of its solid boosters. When a booster anomaly recurred on the USSF-87 mission in February 2026, the Space Force paused national-security launches on Vulcan. ULA had planned 16–18 Vulcan launches in 2026; by mid-October it had made one, and return to flight was reported for late October.

**Blue Origin's New Glenn** is a 98 m, 7 m diameter rocket with a reusable first stage powered by seven BE-4 engines. It reached orbit on its first flight in January 2025. It landed its booster on a ship on its second flight, which launched NASA's ESCAPADE Mars probes in November 2025 (Figure 36.12). It re-flew that booster in April 2026 — but the upper stage under-performed and left AST SpaceMobile's BlueBird 7 satellite in a useless orbit. On 28 May 2026 a New Glenn exploded during a hot-fire test before its fourth flight, wrecking the transporter-erector at Launch Complex 36. Blue Origin traced the cause to an oxygen valve on one BE-4 and aimed to fly again before the end of 2026.

**FIGURE 36.11** Vulcan Centaur lifts off from Cape Canaveral on its first flight, carrying Astrobotic's Peregrine lunar lander, on 8 January 2024. Image: NASA/KSC (NASA/Isaac Watson).

**FIGURE 36.12** New Glenn lifts off from Launch Complex 36 with NASA's twin ESCAPADE Mars probes on 13 November 2025. Its first stage then made Blue Origin's first booster landing on a ship at sea. Image: NASA/KSC (Blue Origin).

Smaller American rockets serve satellites below about a tonne. **Rocket Lab's Electron** is an 18 m carbon-composite rocket with electric-pump-fed engines (Figure 36.13). It has made 88 orbital launches with four failures and flew 18 times in 2025, more than any other single rocket type except Falcon 9. Its average revenue per launch was $9.1 million in mid-2026. Rocket Lab's reusable medium-lift **Neutron** slipped to 2027 after a tank failed in testing. **Firefly Alpha** returned to flight in March 2026 after a failure in April 2025 and the loss of a booster in a ground test; Firefly and Northrop Grumman are developing the larger Eclipse for 2027. Northrop Grumman's solid **Minotaur IV** flew once in each of 2025 and 2026. The air-launched **Pegasus XL**, dropped from an L-1011 airliner, launched a robotic servicer in July 2026. The servicer is intended to raise the orbit of NASA's ageing Swift observatory. Relativity's Terran R and Stoke Space's Nova had not flown by October 2026.

**FIGURE 36.13** A Rocket Lab Electron lifts off from Māhia, New Zealand, on 26 May 2023 with two of NASA's TROPICS storm-observing CubeSats. Image: NASA/KSC (Rocket Lab).

## 36.2.2 Europe

**Ariane 6** replaced Ariane 5 in July 2024. It keeps a hydrogen-fuelled core stage with a Vulcain 2.1 engine, adds a restartable Vinci upper stage (Chapter 35) and carries either two (Ariane 62) or four (Ariane 64) P120C solid boosters. By October 2026 it had made nine flights: the first in 2024,

four Ariane 62s in 2025, then the first Ariane 64 on 12 February 2026, which placed 32 Amazon Leo satellites in orbit. In June 2026 the first Ariane 64 with the larger P160C boosters carried 36 of them, and Amazon raised its order from 18 to 24 Ariane 64 launches. **Vega C** is a four-stage rocket whose solid first stage is the same P120C motor used as Ariane 6's booster. It returned from a 2022 failure in December 2024 and is now marketed by its builder Avio. It launched the European–Chinese SMILE mission in May 2026.

Europe's dependence on so few rockets became painful in 2023–24. Ariane 5 retired, Vega C was grounded and Soyuz launches from Kourou ended after Russia's invasion of Ukraine. ESA had to buy Falcon 9 launches for Euclid, Galileo satellites and Hera. One response is the **European Launcher Challenge**, to which ESA's member states committed €902 million at their November 2025 ministerial meeting. In August 2026 ESA signed conditional contracts with Isar Aerospace (€197.8 million), Rocket Factory Augsburg (€186.9 million) and PLD Space (€158.9 million). Each is required to reach orbit by the end of 2027. Isar's **Spectrum** failed 30 s into its first flight in March 2025 but reached orbit from Andøya, Norway, on 5 September 2026 — the first satellite delivery by a European commercial launch company. RFA One and PLD's Miura 5 were preparing for first flights, MaiaSpace was developing a partly reusable rocket, and Orbex had closed down.

### 36.2.3 Russia

Russia launched only 17 rockets in 2025 and 13 by mid-October 2026. The workhorse is **Soyuz-2** (Figure 36.14), which launches crews and Progress cargo ships to the ISS from Baikonur in Kazakhstan, military satellites from Plesetsk and other missions from Vostochny in the Far East. When the Soyuz MS-28 crew launched in November 2025, a service platform fell into the flame trench of Site 31, Russia's only pad for crewed flights; it was repaired by March 2026. The modular **Angara** family, built around a common kerosene core, is meant to replace Proton and give Russia heavy launches from its own territory, but has flown only a dozen times since 2014. **Proton-M** made one launch in 2026. The new **Soyuz-5** medium rocket, with a single RD-171MV engine on its first stage, made a suborbital test flight from Baikonur on 30 April 2026.

**FIGURE 36.14** A Soyuz-2.1a launches the Soyuz MS-26 crew to the ISS from Baikonur on 11 September 2024. Its four strap-on boosters, which separate in the classic "Korolev cross", trace back to the R-7 of 1957. Image: NASA (NASA/Victor Zelentsov).

### 36.2.4 China

China launched 93 rockets in 2025 and 71 by mid-October 2026, more than half of them Long Marches built by the state-owned China Aerospace Science and Technology Corporation (CASC). The older Long March 2, 3 and 4 burn storable $N_2O_4$/UDMH. A new generation burns kerosene and oxygen (Long March 6, 7, 8 and 12) or hydrogen (the Long March 5 heavy-lift rocket and upper stages). Launches increasingly carry China's two broadband mega-constellations, Guowang and Qianfan, from the new commercial launch site next to Wenchang on Hainan Island. For crewed lunar missions CASC is developing the **Long March 10**: a 92.5 m, three-core rocket with 21 kerosene engines, able to send 27 t towards the Moon, due to fly in 2027. A single-core version, the Long March 10A, made a suborbital test with an abort of the Mengzhou crew capsule in February 2026. The commercial **Long March 10B** reached orbit on its first flight on 10 July 2026. Its first stage was caught at sea by hooks snagging cables stretched over a ship — China's first recovery of an orbital booster.

Since 2015 Chinese policy has opened launch to private and semi-private companies, and their rockets made about a quarter of China's launches in 2025. LandSpace's **Zhuque-2** became the first methane-fuelled rocket to reach orbit in 2023. The company's stainless-steel **Zhuque-3** reached orbit in December 2025. In August 2026 it landed its first stage on legs at a desert site in Gansu province, although a fire then broke out at its base. Galactic Energy's solid **Ceres-1** has flown 23 times, but its liquid Ceres-2 failed on its debut in January 2026 before the company's larger **Pallas-1** succeeded in September. CAS Space flies the solid **Kinetica-1** and debuted the 12 t **Kinetica-2** in March 2026. Orienspace launches the solid **Gravity-1** from a ship at sea. Space Pioneer's **Tianlong-3**, designed as a Falcon 9 rival, failed half a minute after lift-off on its first flight in April 2026. CASC's own reusable test vehicle, the Long March 12A, reached orbit in December 2025 but its booster crashed near the landing pad.

### 36.2.5 Japan, India and Other Nations

**Japan** relies on **H3**, a two-stage hydrogen rocket with the expander-bleed LE-9 engine. After a failed first flight in 2023 it succeeded five times, but on its seventh launch (flight F8) in December 2025 the QZS-5 navigation satellite was lost. JAXA's investigation concluded that the most likely cause was bonded joints in the honeycomb satellite adapter that had come apart during manufacture: the adapter buckled and broke piping on the second-stage hydrogen tank. H3 returned to flight in June 2026 with the first H3-30, which has no solid boosters. It launched the QZS-7 satellite in August and was scheduled to launch the Martian Moons eXploration (MMX) mission on 20 October 2026. The H-IIA retired in June 2025. The small solid Epsilon S has been delayed by two second-stage motor explosions in ground tests and is now planned with the older Epsilon second stage by March 2027. The private **Kairos** of Space One failed on all three of its attempts, most recently in March 2026.

**India** suffered two consecutive failures of the PSLV, in May 2025 and January 2026, both in its solid third stage, and had not flown again by October 2026. The **GSLV Mk II** launched the NASA–ISRO NISAR radar satellite in July 2025. The **LVM3** — renamed HLVM3 in its human-rated form — is being prepared for the first uncrewed Gaganyaan flight. That flight is expected at the end of 2026 or in 2027. The small SSLV has flown three times. On 18 July 2026 Skyroot Aerospace's **Vikram-1**, a three-stage solid rocket with a liquid kick stage, reached orbit on its first attempt, the first private Indian rocket to do so.

**South Korea's** three-stage kerosene **Nuri** made its fifth flight on 7 October 2026, delivering five satellites and nine CubeSats — its fourth success in five attempts. By contrast, the private Innospace's Hanbit-Nano failed shortly after lift-off from Brazil in December 2025. **Israel** launches its Ofeq reconnaissance satellites westward over the Mediterranean on the solid **Shavit-2**, accepting a retrograde orbit to keep spent stages away from populated land; the latest flew in September 2025. **Iran** launched satellites on the Simorgh and the solid Qaem-100 in 2024, but its Zoljanah failed in September 2025, and **North Korea** has not attempted an orbital launch since a failure in May 2024. **Australia's** first orbital rocket, Gilmour Space's hybrid-propellant Eris, failed 14 s after lift-off in July 2025; a second attempt was expected at the end of 2026. Table 36.3 summarises the small launchers.

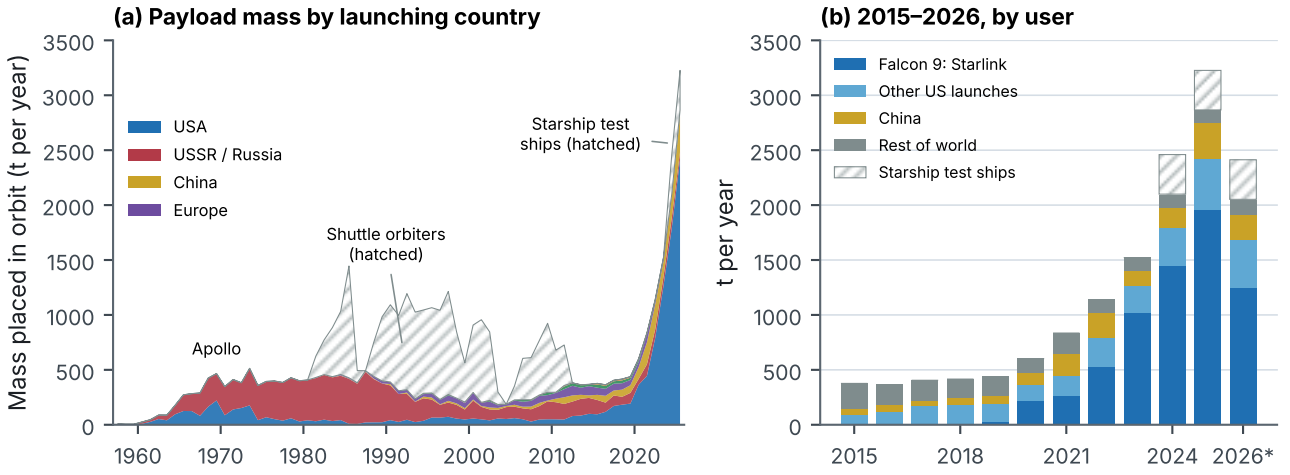
**TABLE 36.3** Small launchers (under about 2 t to LEO) of the 2020s. Attempts from GCAT to 13 October 2026 (a Kuaizhou-1A pad explosion in 2025 not counted); payloads are approximate published values.

| Vehicle | Country | LEO payload | Attempts (failures) | Status, October 2026 |
|---|---|---|---|---|
| Electron | USA/New Zealand | 0.3 t | 88 (4) | 15 launches in 2026 to date |
| Kuaizhou-1A / -11 | China | 0.3 / 1.5 t | 30 (2) / 7 (1) | Active (solid) |
| Ceres-1 | China | 0.4 t | 23 (2) | Active; land and sea launch |
| Kinetica-1 | China | 2 t | 16 (1) | Active (solid) |
| Jielong-3 | China | 1.6 t to SSO | 12 (0) | Active; sea launch |
| Firefly Alpha | USA | 1 t | 7 (3) | Returned to flight March 2026 |
| Spectrum | Germany | 1 t | 2 (1) | Reached orbit September 2026 |
| SSLV | India | 0.5 t | 3 (1) | No launch since 2024 |
| Vikram-1 | India | 0.35 t | 1 (0) | Reached orbit July 2026 |
| Kairos | Japan | 0.25 t | 3 (3) | Three failures |
| Eris | Australia | 0.3 t | 1 (1) | Second flight expected late 2026 |
| Astra Rocket 3, LauncherOne | USA | 0.05–0.5 t | 7 (5), 6 (2) | Retired in 2022–23 |

## 36.3 Launch by the Numbers

### 36.3.1 Mass to Orbit

Launch counts say little about what is lifted. Figure 36.15 sums the payload mass of every successful launch. The Moon race lifted the world total to almost 500 t a year. From 1981 to 2011 the Space Shuttle added several hundred tonnes a year. However, GCAT counts each orbiter — about 100 t that came back to Earth — as payload. Real growth began after 2018. In 2025 launches delivered about 2 870 t of satellites, cargo and crew vehicles, plus 360 t of Starship test ships. Including payloads lost in failures, the GCAT total is about 3 500 t, nearly nine times that of 2015. Falcon 9 carried 2 330 t of the 2 870 t — 81 % — and Starlink satellites alone 1 960 t. China lifted 330 t, about 11 %.

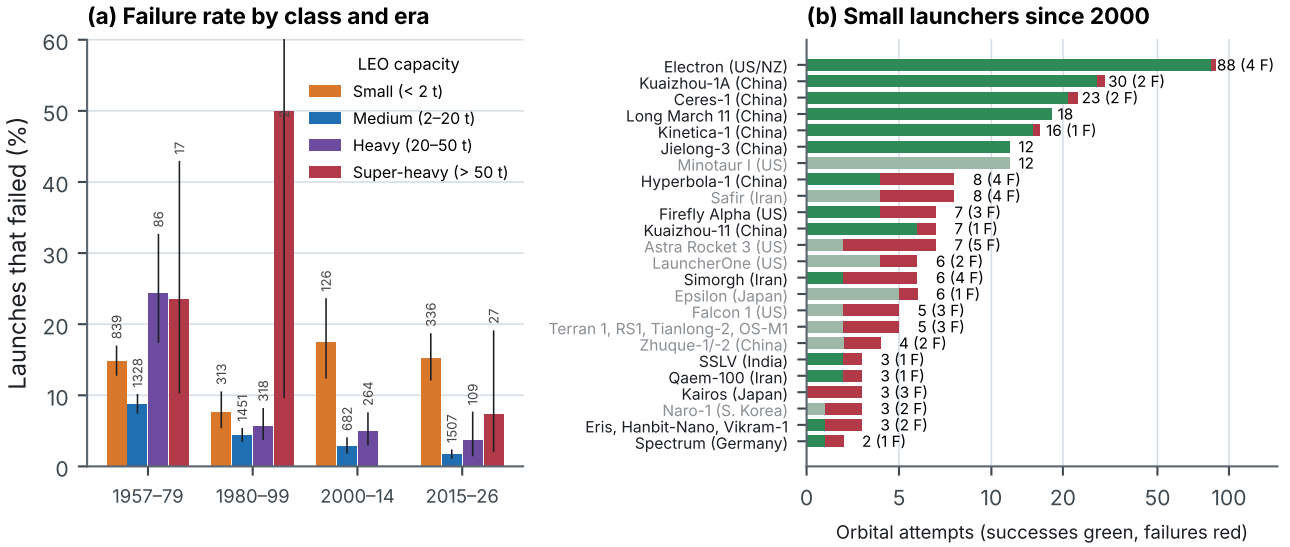



Sum of GCAT payload masses on successful launches. GCAT counts reusable orbiters (Shuttle, Buran, about 100 t each) and Starship test ships (120 t each) as payload: shown hatched. *2026 to 13 Oct. Data: GCAT (J. McDowell, CC-BY 4.0), launches to 13 Oct 2026.

**FIGURE 36.15** Left: payload mass placed in orbit each year by the country of the launch provider, 1957–2025. Right: 2015–2026 by user. GCAT counts reusable orbiters and Starship test ships (hatched) as payload; without them, 2025 saw about 2 870 t of satellites and spacecraft delivered, two-thirds of it Starlink.

## 36.3.2 Reliability by Class

Figure 36.16 adds two views to the learning curve of Chapter 34. Grouped by size, small launchers have always failed most often: 15 % of the 336 small-launcher attempts since 2015 failed, against 1.7 % of 1 507 medium launches and 3.7 % of 109 heavy launches. Part of the reason is that many small rockets are first or early flights of new designs by new companies, and few of those companies survive long enough to climb the learning curve. The right-hand panel shows the shake-out. Electron, the Chinese solid rockets and Jielong-3 have built long records. Astra and Virgin Orbit stopped flying after a handful of launches, and Kairos has yet to reach orbit.

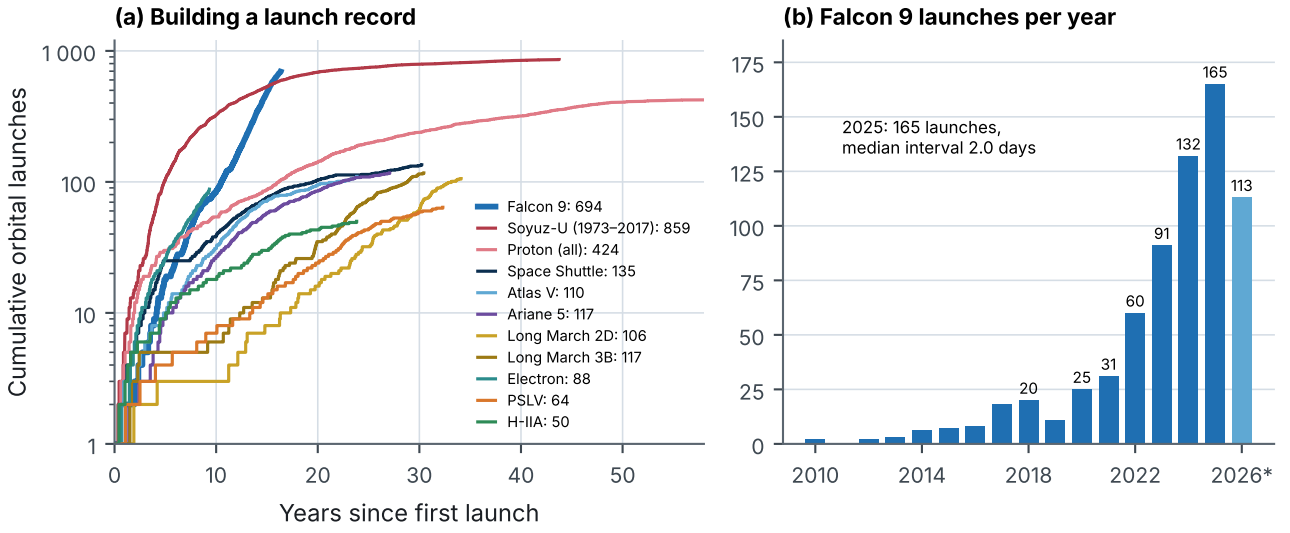



(a) Classes by approximate maximum LEO payload (Falcon 9 as medium); bars: failed share; whiskers: 90 % Jeffreys interval; numbers: attempts; pad aborts and explosions excluded. (b) Grey labels: retired or company closed. Data: GCAT (J. McDowell, CC-BY 4.0), launches to 13 Oct 2026.

**FIGURE 36.16** (a) Share of orbital launch attempts that failed, by vehicle size class and era. (b) Successes and failures of small launchers first flown since 2000.

### 36.3.3 Cadence

The most striking change of the 2020s is cadence — how often a rocket flies (Figure 36.17). Rockets used to need weeks or months between flights from one pad, because each was assembled and checked for weeks before launch. Falcon 9 flew 165 times in 2025, a median of 2.0 days between launches. The Soviet Soyuz-U had flown almost as often in the 1970s — it reached 500 launches in 14 years, Falcon 9 in 15 — but only by building a new rocket every time. Falcon 9 reached its 100th launch in late 2020, ten years after its first, and its 600th early in 2026, with most flights on re-used boosters.




Pad aborts excluded. *2026 to 13 Oct. Data: GCAT (J. McDowell, CC-BY 4.0), launches to 13 Oct 2026.

**FIGURE 36.17** (a) Cumulative launches against years since each vehicle's first flight (pad aborts excluded). (b) Falcon 9 launches per year; the fleet averaged one launch every 2.2 days in 2025.

Table 36.4 lists the first flights of new launch vehicles and major new versions in 2025–26. There were 19 debuts in 21 months, of which 7 failed (37 %). This is in line with the 38 % maiden-flight failure rate found in Chapter 34.

**TABLE 36.4** First orbital flights of new vehicles and major new versions, 2025–26 (dates UTC or local as reported).

| Date | Vehicle | Country | Outcome |
|---|---|---|---|
| 16 Jan 2025 | New Glenn | USA | Orbit; booster lost before landing |
| 16 Jan 2025 | Starship V2 | USA | Ship lost during ascent |
| 11 Feb 2025 | Long March 8A | China | Success |
| 30 Mar 2025 | Spectrum | Germany | Lost control at about 30 s |
| 30 Jul 2025 | Eris | Australia | Failed at about 14 s |
| 18 Sep 2025 | Zoljanah | Iran | Failed to reach orbit |
| 3 Dec 2025 | Zhuque-3 | China | Orbit; landing attempt failed |
| 22 Dec 2025 | Hanbit-Nano | South Korea | Failed shortly after lift-off |
| 23 Dec 2025 | Long March 12A | China | Orbit; landing attempt failed |
| 17 Jan 2026 | Ceres-2 | China | Failed to reach orbit |
| 12 Feb 2026 | Ariane 64 | Europe | Success, 32 Amazon Leo satellites |
| 30 Mar 2026 | Kinetica-2 | China | Success |
| 3 Apr 2026 | Tianlong-3 | China | Failed at about 33 s |

| Date | Vehicle | Country | Outcome |
|---|---|---|---|
| 22 May 2026 | Starship V3 | USA | Ship splashed down as planned; booster lost |
| 1 Jun 2026 | Long March 12B | China | Success |
| 12 Jun 2026 | H3-30 | Japan | Success (no solid boosters) |
| 10 Jul 2026 | Long March 10B | China | Success; first stage caught at sea |
| 18 Jul 2026 | Vikram-1 | India | Success |
| 1 Sep 2026 | Pallas-1 | China | Success |

## 36.4 Trends Reshaping Launch

### 36.4.1 Reusability

Before December 2015 no orbital booster had ever landed. By 2026 Falcon boosters had landed more than 650 times. Three other organisations had recovered orbital first stages: Blue Origin (on a ship), CASC (caught by cables at sea) and LandSpace (on legs, though a fire broke out after landing). Others are following — Rocket Lab's Neutron, Europe's Themis demonstrator, Japan's and India's experimental vehicles — and SpaceX aims to recover both stages of Starship. The economics are discussed in Chapter 34: reuse pays only at a high flight rate, which is why the leaders are operators with their own constellations to launch.

### 36.4.2 Rideshare

A **rideshare** mission carries dozens of small satellites from many customers to a common orbit. India's PSLV made it famous by launching 104 satellites at once in 2017. SpaceX's **Transporter** missions to sun-synchronous orbit (Transporter-1 carried 143 in January 2021; Transporter-18 flew on 1 October 2026) and **Bandwagon** missions to mid-inclination orbits now dominate. The published price is $350 000 for 50 kg. Rideshare undercuts dedicated small launchers on price; its drawbacks are that the satellite goes where and when the main mission goes, which **orbital transfer vehicles** (Chapter 11) partly overcome.

### 36.4.3 The Heavy-Lift Race

Between Energia's last flight in 1988 and Falcon Heavy's first in 2018, no rocket able to place more than 30 t in orbit flew. Now four are competing (Figure 36.8). Falcon Heavy (64 t expendable) and New Glenn (45 t with a reusable booster) are in service. SLS (95 t) is expensive and has flown only twice, in 2022 and 2026. Starship aims at 100 t or more with full reuse and orbital refuelling. China's Long March 10 (70 t to LEO, 27 t towards the Moon) is due in 2027 to support crewed lunar landings planned by 2030, and Russia has proposed a heavy Angara A5V. Heavy-lift rockets matter for crewed exploration, space stations and large telescopes, but much of the demand that pays for them is the launch of constellation satellites in large batches.

### 36.4.4 The Small-Launcher Shake-Out

Around 2018 more than a hundred small-launcher projects were under way worldwide. Few survived: rideshare took much of their market, and the cost of a dedicated small launch per kilogram is high (see the example below). Survivors offer what rideshare cannot — a specific orbit

at a specific time, for constellation replenishment, defence "responsive launch" or sovereign needs. Many are growing into medium rockets: Rocket Lab is moving to Neutron, Galactic Energy from Ceres-1 to Pallas-1 and Firefly to Eclipse.

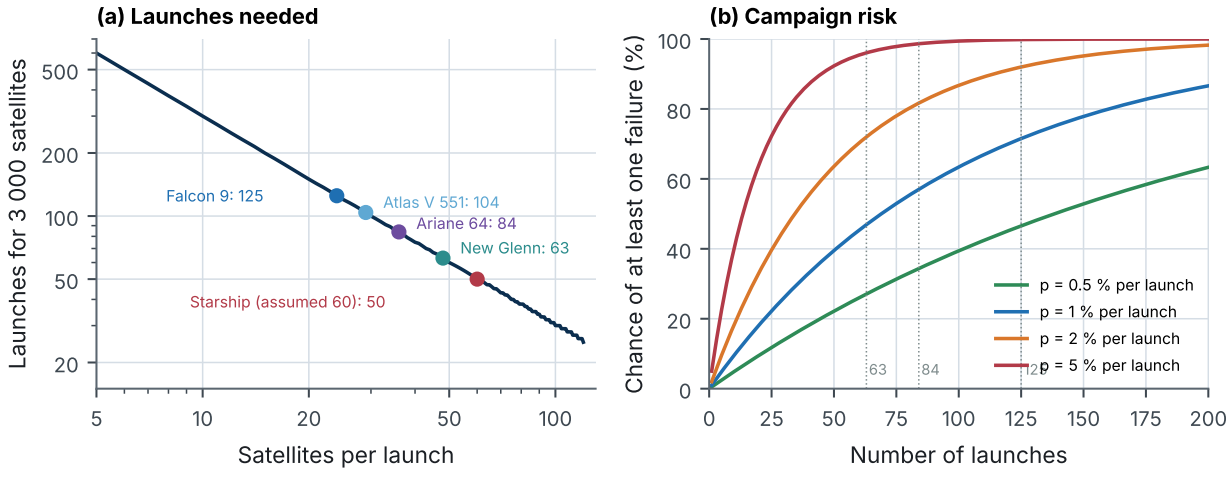
### 36.4.5 Sovereign Access

Governments value independent access to space as a matter of security. Europe's crisis of 2023–24, Russia's loss of Western customers after 2022, India's PSLV failures and Japan's H3 failure all show how fragile a fleet of one or two rockets is. The policy response is to fund several competing launchers. Examples are the European Launcher Challenge, Japan's support for private rockets, India's opening to Skyroot and others, and the US Space Force's practice of buying launches from several providers. The response also includes building new spaceports in Norway, Sweden, the United Kingdom and Australia (Chapter 37).

**EXAMPLE: PRICE PER KILOGRAM AND PAYLOAD FRACTION**

Compare three vehicles at their published prices, assuming each flies full to LEO. **Payload fraction** is payload mass divided by lift-off mass.

1. **Falcon 9** with booster landing: \$74 M for 17.5 t → $74 \times 10^6 / 17\,500 \approx$ **\$4 230/kg**. Lift-off mass 549 t → payload fraction 17.5/549 = **3.2 %**. Expending the booster raises the payload to 22.8 t (4.2 %), but the booster is then lost.
2. **Ariane 64**: estimated €115 M for 21.65 t → **€5 310/kg**, about \$6 100/kg at an assumed \$1.15 per euro. Lift-off mass 860 t → **2.5 %**. The hydrogen core gives high specific impulse, but the four solid boosters add mass.
3. **Electron**: \$9.1 M (average revenue per launch, 2026) for 0.3 t → **\$30 300/kg**; lift-off mass 13 t → **2.3 %**. Small rockets cost more per kilogram because avionics, range services and launch crews cost almost as much for a small rocket as for a big one.
4. Real missions are rarely full. A 150 kg satellite flying alone on Electron pays about \$60 000/kg, while large batches of constellation satellites fill a rocket almost to capacity. The comparison explains why customers who can accept a shared ride choose rideshare at \$7 000/kg, and why a dedicated small launch is bought for schedule and orbit, not price.




(a) Launches = ceil(3 000 / satellites per launch); per-launch counts from Amazon Leo missions (Starship value assumed). (b) 1 − (1 − p)^n for independent launches.

**FIGURE 36.18** (a) Launches needed to deploy 3 000 satellites as a function of how many each launch carries, with per-launch counts from Amazon Leo missions. (b) The chance that at least one launch fails during a campaign of n launches, for four per-launch failure probabilities.

### EXAMPLE: LAUNCHING A 3 000-SATELLITE CONSTELLATION

A broadband operator must deploy 3 000 satellites of about 600 kg — 1 800 t in all. How many launches are needed, and what is the risk (Figure 36.18)?

1. Satellites per launch are limited by the fairing volume and dispensers as much as by mass. Amazon Leo's missions carried 24 satellites on Falcon 9, 27–29 on Atlas V and 32–36 on Ariane 64; New Glenn was to carry 48.
2. Launches needed = $\lceil 3\,000 / k \rceil$: **125** on Falcon 9, **104** on Atlas V, **84** on Ariane 64, **63** on New Glenn, and about **50** on a Starship carrying 60.
3. At four launches a month, the Falcon 9 campaign takes 31 months, the New Glenn campaign 16 months. Real campaigns use several vehicles at once, because no single provider has enough capacity.
4. Risk: with independent launches each failing with probability $p$, the chance of at least one failure is $1 - (1 - p)^n$. For $p = 1$ % and 125 launches it is $1 - 0.99^{125} =$ **72 %**. For 63 launches on a new rocket with $p = 2$ %, it is $1 - 0.98^{63} =$ **72 %** as well.
5. The expected number of satellites lost is $N\,p$ — 30 at $p = 1$ %, whichever rocket is used. Bigger rockets reduce the number of launches but put more satellites at risk on each, so operators spread a constellation across several vehicles and keep spares in orbit.

### AI IN LAUNCH OPERATIONS AND RANGE SAFETY

Machine-learning models are being developed to forecast the weather that most often delays launches — lightning, upper-level winds and anvil clouds. They are also meant to help evaluate launch-commit criteria. Other models scan thousands of telemetry channels during countdowns for anomalies that a human console operator might miss. Range safety, once a human officer with a destruct button, is now largely automated: autonomous flight-safety systems on Falcon 9, Electron and many new rockets decide on board whether to terminate a flight. With hundreds of launches a year, optimisation algorithms schedule pads, ranges, airspace closures and recovery ships, and estimate the risk that falling stages or debris pose to people and aircraft. Because these decisions affect public safety, the algorithms on board stay deterministic and verifiable, while learning-based tools support the humans who certify them; Chapter 52 discusses AI across launch and its assurance.

### KEY TAKEAWAYS

- About 7 400 orbital launches have been attempted since 1957; the record year, 2025, saw 330 attempts and 11 failures, 60 % by US and 28 % by Chinese providers.
- Most early launchers were converted missiles: the R-7 (Soyuz) family has flown about 1 970 times and still carries crews. Saturn V, N1, Energia and the Shuttle were the super-heavies of the twentieth century.
- Falcon 9 dominates: half of all launches and 81 % of the satellite mass delivered in 2025, with a median of two days between flights and routinely re-used boosters.
- China flies the most varied fleet — 16–19 vehicle families a year — and in 2026 recovered orbital boosters for the first time; its Long March 10 is due in 2027.
- Europe (Ariane 6, Vega C, Spectrum), Japan (H3), India (LVM3, Vikram-1) and Russia (Soyuz-2, Angara) maintain sovereign access, but setbacks in 2025–26 show how fragile small fleets are.
- Small launchers fail far more often than larger ones (15 % since 2015) and many have disappeared; dedicated small launches cost several times more per kilogram than rideshare.
- Reusability, rideshare, high cadence and a new generation of super-heavies (Starship, SLS, Long March 10) are the trends shaping launch in the late 2020s.

### REVIEW QUESTIONS

1. Why were most of the first launch vehicles converted ballistic missiles? Give two examples from the USSR and two from the United States, and name what each became.
2. Compare the N1 and the Saturn V. Why did the N1 fail, and what lesson does the comparison hold for vehicles with many engines?
3. From Figure 36.6, describe how the shares of world launches changed between 1965, 1985, 2005 and 2025, and explain the main causes.
4. Using Table 36.2, compare Falcon 9, Ariane 64 and Long March 5 in payload to LEO and GTO, propellants and reusability. Why do hydrogen-fuelled vehicles often still need solid boosters?
5. GCAT reports about 3 500 t launched in 2025, yet only about 2 870 t of satellites and spacecraft were delivered. Explain the difference.
6. In 2015–2026, 15 % of small-launcher attempts failed against 1.7 % of medium launches. Give three reasons for this difference.
7. Falcon 9 and Soyuz-U both reached 500 launches in about 15 years. What made Falcon 9's cadence fundamentally different?
8. A rideshare costs $7 000/kg and a dedicated Electron launch about $30 000/kg when full. Under what circumstances would a satellite operator still choose the dedicated launch?
9. Estimate the number of launches needed to deploy 1 500 satellites if each launch carries 40, and the probability of at least one failure if each launch has a 1.5 % chance of failing.
10. What is "sovereign access to space"? Using examples from 2023–2026, explain why governments fund more than one launch vehicle.

CHAPTER

# 37 Spaceports and Ground Infrastructure

Image: NASA/KSC

*Every orbital rocket leaves from a few hectares of concrete surrounded by thousands of square kilometres of cleared sky and sea. This chapter explains how launch sites are chosen — latitude, reachable inclinations, safe corridors over the ocean, weather and logistics — and surveys the world's spaceports, from Florida, which hosted a third of all launches in 2025, to new sites in Norway, Australia and India. It then looks inside a launch complex: flame trenches, water deluge, umbilical towers, propellant farms and the three ways of assembling a rocket. It follows a launch campaign through the range, the countdown and the weather rules, and ends with noise, environmental licensing, and launching from aircraft and ships.*

IN THIS CHAPTER

37.1 Choosing a Launch Site
37.2 The World's Spaceports
37.3 Inside a Launch Complex
37.4 The Range: Tracking, Safety and Recovery
37.5 The Launch Campaign and Countdown
37.6 Environment, Community and Regulation
37.7 Launching from the Air and the Sea

A rocket spends weeks at its launch site and minutes flying from it. A modern **spaceport** combines an assembly hall and a chemical plant storing hundreds of tonnes of liquids at −253 °C to −183 °C. It has a deluge system that pours more than a million litres of water in half a minute, and a tracking network watching thousands of kilometres of ocean. It also includes a marine and aviation operation that clears the rocket's path and recovers what comes back.

In 2025 the world made 330 orbital launch attempts. More than half — 175 — left from just two places in the United States: Florida's Space Coast (109) and Vandenberg in California (66). China flew 93 times from four centres and from ships; Rocket Lab's private site on New Zealand's Mahia Peninsula launched 17 times, more than Russia's Baikonur and Plesetsk together. This chapter explains why the sites are where they are and what happens on them; the vehicles are described in Chapters 34 and 36.

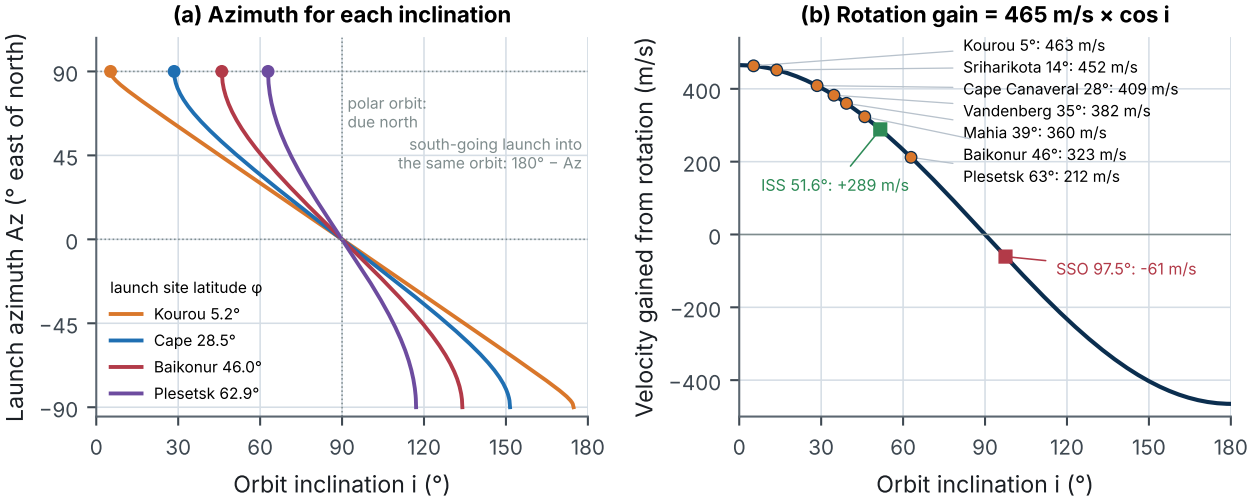
## 37.1 Choosing a Launch Site

### 37.1.1 Latitude, Azimuth and Inclination

Chapter 34 showed that a rocket launched from latitude $\phi$ on **azimuth** $Az$ (the compass direction of flight, measured clockwise from north) enters an orbit of inclination $i$ given by

$$\sin Az = \frac{\cos i}{\cos \varphi}$$

Because $|\sin Az|$ cannot exceed 1, a direct launch can reach only inclinations between $\phi$ and $180° - \phi$. Kourou (5.2° N) can reach almost any orbit. Plesetsk (62.9° N) can reach nothing below 62.9° without an expensive turn. Each allowed inclination can be reached flying north-east or south-east (north-west or south-west for a retrograde orbit), because the equation has two solutions, $Az$ and $180° - Az$ (Figure 37.1). A launch due east gives the lowest inclination, $i = \phi$; a polar orbit needs a launch due north or south; a sun-synchronous orbit (97–98°) a launch slightly west of north or south.




sin Az = cos i / cos φ; surface speed 465.1 m/s × cos φ. Dots in (b): the most each site can gain, launching due east into i = φ. Range-safety limits on azimuth ignored.

**FIGURE 37.1** (a) Launch azimuth needed to reach each orbit inclination from four sites (north-going solution; the south-going launch into the same orbit flies at 180° − Az). (b) The velocity a rocket gains from Earth's rotation depends only on the inclination of the orbit, 465 m/s × cos i; the dots mark the largest gain each site can obtain, by launching due east.

Earth's rotation carries the pad east at 465 m/s × cos $\phi$: 463 m/s at Kourou, 409 m/s at Cape Canaveral, 212 m/s at Plesetsk. Only the component along the orbit helps, and it equals 465 m/s × cos $i$ — the same for every site that can reach that inclination; the component across the orbit

must be cancelled. For the International Space Station's 51.6° orbit latitude therefore matters little. It matters greatly for orbits inclined *less* than the site's latitude, above all the geostationary orbit, because turning the orbital plane after launch is expensive.

**EXAMPLE: AN ISS ORBIT AND A GEOSTATIONARY TRANSFER FROM KOUROU AND FROM BAIKONUR**

Kourou lies at 5.24° N; Baikonur at 45.96° N. Assume a 200 km parking orbit (7 784 m/s).

1. **ISS orbit (51.64°).** From Kourou, $\sin Az = \cos 51.64° / \cos 5.24° = 0.6206 / 0.9958 = 0.6232$, so $Az = 38.6°$. From Baikonur, $\sin Az = 0.6206 / 0.6952 = 0.8927$, so $Az = 63.2°$.
2. **Rotation gain.** Kourou's surface moves at 463 m/s, Baikonur's at 323 m/s. The speed the rocket must add is the vector difference between orbital velocity and ground velocity, $\sqrt{v^2 - 2vv_r \sin Az + v_r^2}$. This gives a gain of 280 m/s at Kourou and 287 m/s at Baikonur: the equatorial site's extra rotation points across the orbit and is wasted.
3. **Geostationary transfer.** Launching due east, Kourou reaches a transfer orbit inclined about 5°. Baikonur's drop zones keep its launches at 51.5° or more, so a satellite would have to remove 51.6° of inclination at apogee, where it moves at 1.60 km/s and must reach 3.07 km/s. The combined burn is $\Delta v = \sqrt{v_1^2 + v_2^2 - 2v_1 v_2 \cos \Delta i}$ = 1.49 km/s from Kourou against 2.43 km/s from Baikonur.
4. **Propellant.** With a 320 s apogee engine ($v_e$ = 3.14 km/s), the propellant share of the satellite's mass is $1 - e^{-\Delta v / v_e}$: 38 % from Kourou and 54 % from Baikonur. This is why Proton's Breeze-M upper stage performs most of the plane change itself in a long, multi-burn ascent, and why Europe built its spaceport near the equator.

### 37.1.2 What the Rotation Is Worth

Of the roughly 9.3 km/s of ideal Δv needed to reach low orbit (Chapter 34), typically 0.2–0.45 km/s comes free from the spin of the Earth. The next example puts a price on it.

**EXAMPLE: THE PAYLOAD VALUE OF EARTH'S ROTATION**

Take an expendable two-stage kerosene rocket similar to Falcon 9: first stage 411 t of propellant, 25.6 t dry, average specific impulse 296 s; second stage 107.5 t of propellant, 4.0 t dry, 348 s.

1. Solving the rocket equation stage by stage for an ideal Δv of 9 300 m/s gives a payload of **21.9 t**.
2. Repeating for 9 400 m/s gives 21.1 t: each extra 100 m/s costs about 0.82 t, or 3.8 % of the payload.
3. A due-east launch from Kourou (463 m/s) rather than Cape Canaveral (409 m/s) gains 54 m/s, about 0.45 t; rather than Plesetsk (212 m/s), 251 m/s, or **2.2 t** — 10 % of the payload.
4. A sun-synchronous orbit (97.5°) is slightly retrograde: the rocket loses 61 m/s to rotation wherever it launches, about 0.5 t.

### 37.1.3 Safety Corridors and Overflight

Geometry says which orbits a site *could* reach; safety says which it *may*. Rockets drop spent stages and fairings along their path, and a failing vehicle can scatter debris wherever its **instantaneous impact point** — where it would land if the engines stopped — happens to be. Ranges therefore allow only azimuths whose early flight and drop zones lie over the sea or empty land (Figure 37.2). Cape Canaveral launches between north-east, along the US coast to inclinations of about

57°, and east-south-east over the Atlantic. Vandenberg launches south over the Pacific, and Kourou has open ocean to the north and east. Starship leaves Starbase over the Gulf of Mexico and passes between Florida and Cuba.

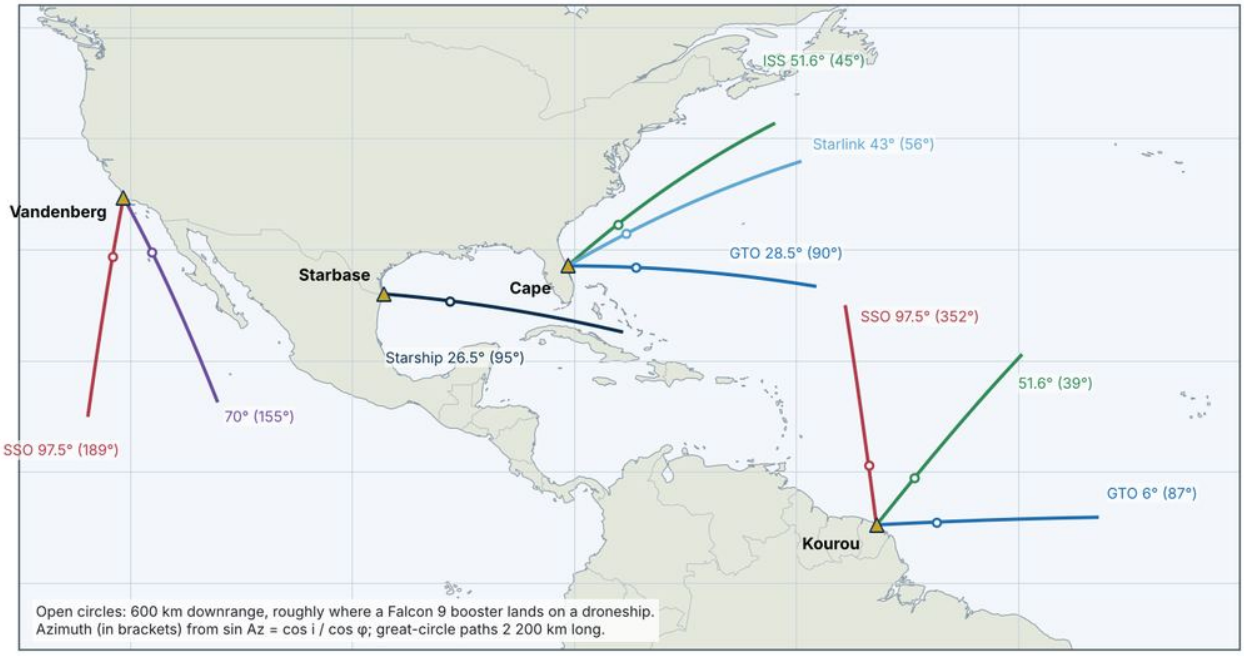
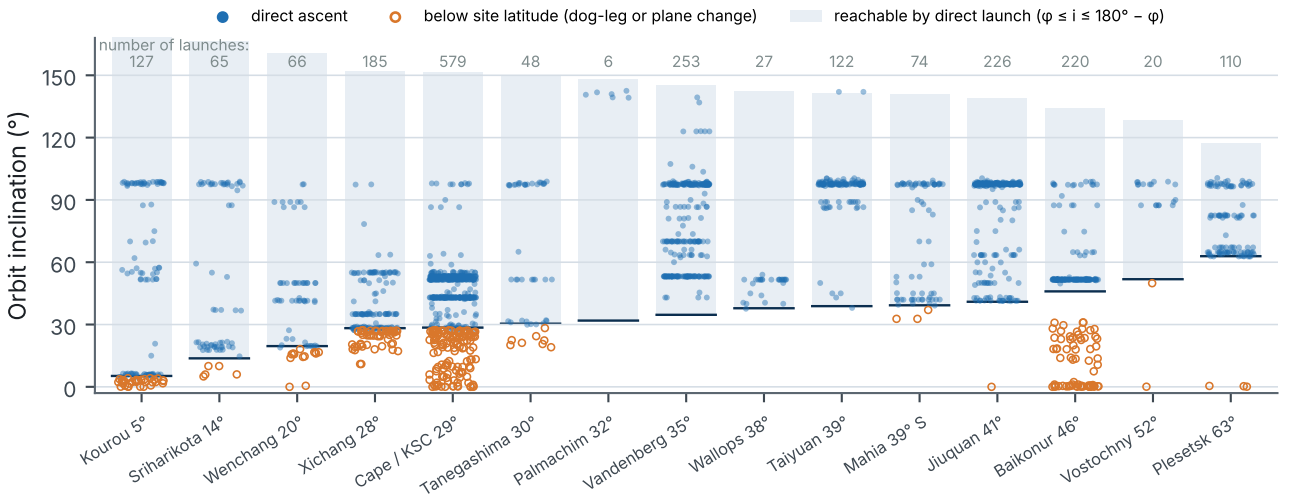



**FIGURE 37.2** Initial launch corridors from four spaceports in the Americas: great-circle paths for the first 2 200 km on the azimuth needed for each inclination.

The corridors leave a fingerprint in the orbits each site actually uses (Figure 37.3). Baikonur, deep inside Kazakhstan, drops its stages in designated zones on the steppe; partly to keep them out of China, it launches no lower than about 51.5°, which is why the ISS orbits at 51.6°. Palmachim, on Israel's coast, launches westward over the Mediterranean to avoid overflying its neighbours, so every satellite launched from Israel flies a retrograde orbit inclined 139–143°. Sriharikota's polar launches must turn ("dog-leg") to avoid Sri Lanka, one reason India is building a second spaceport farther south. Florida had not launched into a polar orbit since February 1969 until a Falcon 9 flew south along the coast on 30 August 2020. This "polar corridor" has since carried the PACE satellite and, in 2025, the first crewed polar flight, Fram2.




Successful orbital launches 2010 – 13 Oct 2026 (GCAT, J. McDowell). Inclination of the final rocket stage (or first payload). Sites ordered by latitude.

**FIGURE 37.3** Inclinations reached by successful orbital launches from 15 spaceports, 2010–2026, against the band each site's latitude allows (shaded). Clusters reveal the corridors: 51.6° at Baikonur, 43° and 53° (Starlink) at the Cape, 97–98° at Vandenberg, Taiyuan and Jiuquan, 139–143° at Palmachim. Open circles, below the site's latitude, are mostly geostationary missions whose upper stages removed inclination in orbit.

### 37.1.4 Weather, Logistics and Politics

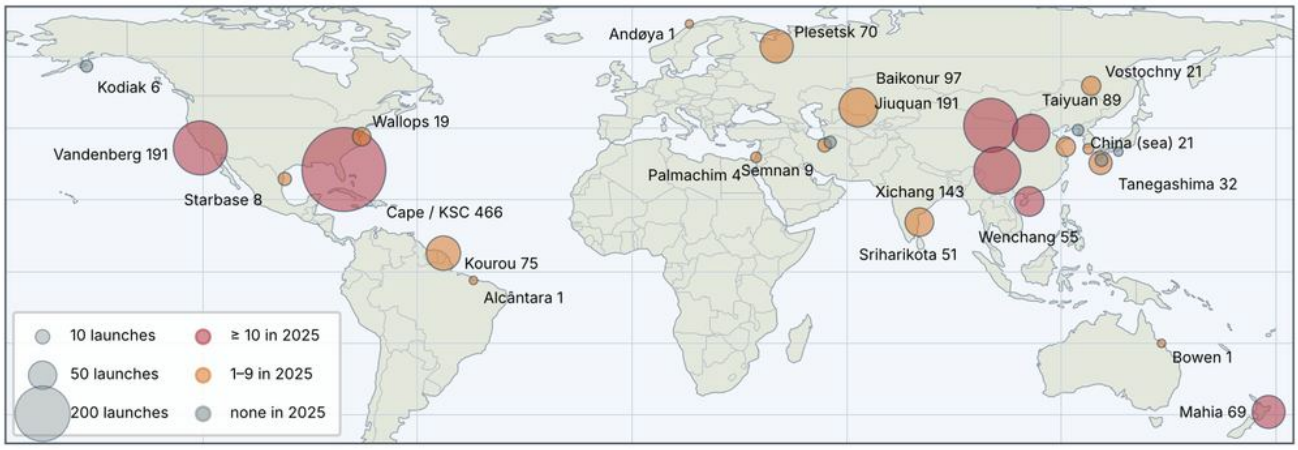
A launch site also needs tolerable weather (Florida's thunderstorms are discussed below), ports and roads for stages up to 9 m in diameter, power, fresh water, a workforce and an airport. Large stages are often built far away: the SLS core stage reaches Florida by barge from New Orleans, Ariane 6 stages cross the Atlantic by ship and Soyuz rockets reach Baikonur by rail. SpaceX builds Starship beside its pads at Starbase, a company town incorporated as a city in May 2025.

Politics shapes the map as much as physics. Russia leases Baikonur from Kazakhstan for $115 million a year until 2050 and built Vostochny, in its own Far East, to depend less on it. China placed Jiuquan, Taiyuan and Xichang far inland for security, which is why their spent stages fall on land. And many countries want a "sovereign" launch site even if it flies only a few times a year.

## 37.2 The World's Spaceports

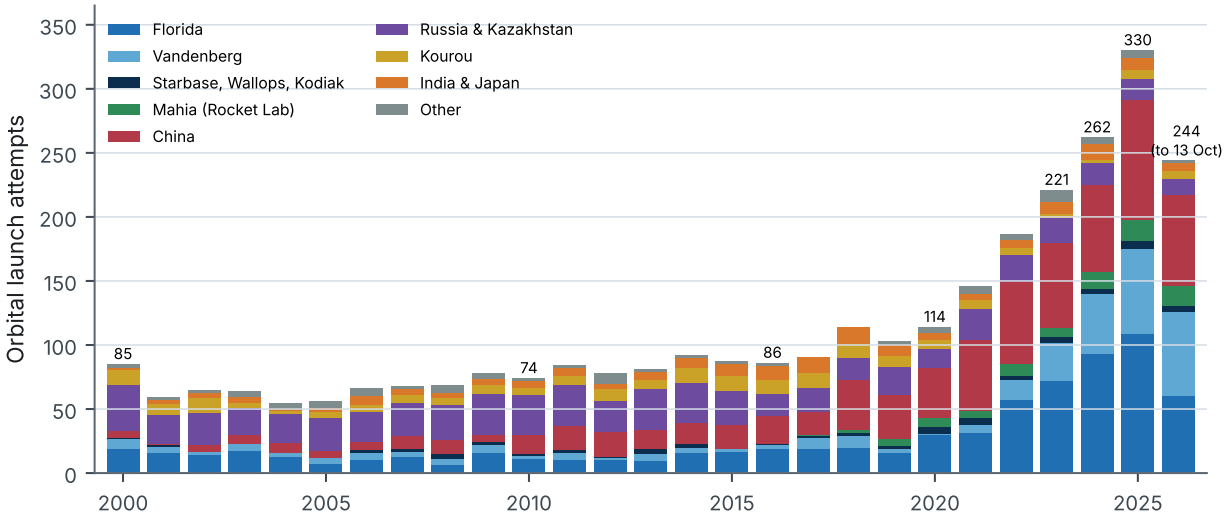
### 37.2.1 Where Rockets Fly From

About 40 sites, counting airfields used for air launch, have sent rockets towards orbit; 22 did so in 2025 (Figure 37.4). Over the whole space age the busiest has been Plesetsk, with 1 688 launches, mostly Soviet military satellites, followed by Baikonur (1 560) and Cape Canaveral with Kennedy (1 401). Since 2016 the world launch rate has nearly quadrupled, and the growth came almost entirely from Florida, Vandenberg and China (Figure 37.5).




Orbital launch attempts 2016–2025 by spaceport (GCAT, J. McDowell; Wenchang includes the Hainan commercial site; Chinese sea launches plotted at Haiyang).

**FIGURE 37.4** Orbital launch attempts from each spaceport, 2016–2025 (bubble area), coloured by activity in 2025. Florida, Vandenberg and China's four centres dominate; Rocket Lab's Mahia site in New Zealand outflew every Russian site.




GCAT (J. McDowell), orbital and deep-space launch attempts including failures and Starship test flights; 2026 to 13 October.

**FIGURE 37.5** Orbital launch attempts per year by spaceport, 2000–2026. Florida and Vandenberg together hosted 175 of the 330 attempts in 2025; China's launch rate has risen sixfold since 2010.

Table 37.1 summarises the busiest spaceports and Table 37.2 the smaller and newer ones.

**TABLE 37.1** The world's busiest spaceports. "2025" is the number of orbital launch attempts that year (GCAT, J. McDowell).

| Spaceport | Latitude | Main pads and vehicles (2026) | 2025 | Notes |
|---|---|---|---|---|
| Cape Canaveral SFS & Kennedy Space Center (USA) | 28.5° N | SLC-40, LC-39A (Falcon 9, Falcon Heavy); SLC-41 (Atlas V, Vulcan); LC-36 (New Glenn); LC-39B (SLS) | 109 | Busiest site; Starship pads at LC-39A and SLC-37 under construction |
| Vandenberg SFB (USA) | 34.7° N | SLC-4E (Falcon 9); SLC-6 (Falcon, being rebuilt); SLC-2W (Firefly Alpha); SLC-8 (Minotaur) | 66 | Polar and sun-synchronous launches south over the Pacific |
| Starbase, Texas (USA) | 26.0° N | Pads 1 and 2 (Starship) | 5 | Up to 25 launches a year approved in 2025 |
| Jiuquan (China) | 41.0° N | LC-43/91 (Long March 2F, crews); LC-43/94 (Long March 2C/2D, 4B/4C); commercial pads (Kuaizhou, Ceres-1, Zhuque, Long March 12B) | 34 | China's crewed launch site, in the Gobi Desert |

| Spaceport | Latitude | Main pads and vehicles (2026) | 2025 | Notes |
|---|---|---|---|---|
| Wenchang (China) | 19.6° N | LC-101 (Long March 5), LC-201 (Long March 7/7A), LC-301 (Long March 10); commercial Pads 1–2 (Long March 8, 12, 10B) | 21 | Coastal; heavy lift, station cargo and lunar missions |
| Xichang (China) | 28.2° N | LC-2, LC-3 (Long March 3A/3B/3C) | 19 | Geostationary and BeiDou launches; stages fall inland |
| Mahia, LC-1 (New Zealand) | 39.3° S | LC-1A, LC-1B (Electron) | 17 | Private; up to 120 launch opportunities a year |
| Taiyuan (China) | 38.8° N | LC-9, LC-9A, LC-16 (Long March 2, 4, 6, 6A) | 12 | Sun-synchronous and polar orbits |
| Plesetsk (Russia) | 62.9° N | Site 43 (Soyuz-2), Site 35 (Angara) | 9 | Military; high inclinations only |
| Kourou, Guiana Space Centre (France/ESA) | 5.2° N | ELA-4 (Ariane 6), ZLV (Vega-C); ELM-Diamant (micro-launchers, planned) | 7 | Near-equatorial; ocean to the north and east |
| Chinese sea launches | ≈ 35° N | Ships from Haiyang, Shandong (Long March 11, Jielong-3, Ceres-1S, Gravity-1) | 7 | 28 launches since 2019, all successful |
| Baikonur (Kazakhstan, leased by Russia) | 46.0° N | Site 31 (Soyuz crew and cargo); Site 81 (Proton); Site 45 (Soyuz-5) | 6 | Site 31 damaged in November 2025, repaired by March 2026 |
| Sriharikota, SDSC-SHAR (India) | 13.7° N | First and Second Launch Pads (PSLV, GSLV, LVM3, SSLV) | 5 | Third Launch Pad approved January 2025 |
| Tanegashima (Japan) | 30.4° N | Yoshinobu LP-1, LP-2 (H3) | 4 | H-IIA retired June 2025 |

### 37.2.2 The Americas

**Florida** is two neighbouring sites run as one range. NASA's Kennedy Space Center owns Launch Complex 39, built for the Saturn V. Pad 39A is leased to SpaceX for Falcon 9 and Falcon Heavy and is being equipped for Starship. Pad 39B was rebuilt for the Space Launch System, which flew Artemis II from it on 1 April 2026. Cape Canaveral Space Force Station hosts SpaceX's busiest pad, SLC-40, which has also launched crews since 2024, and United Launch Alliance's SLC-41 for Atlas V and Vulcan. It also hosts Blue Origin's LC-36 for New Glenn, where a booster exploded during a hot-fire test in May 2026. Other pads are being rebuilt for newcomers such as Relativity and Stoke. In late 2025 the Air Force approved SLC-37 for up to 76 Starship launches a year.

**Vandenberg**, on the Californian coast, is America's polar spaceport; its 66 launches in 2025 were almost all Falcon 9s from SLC-4E. In October 2025 the Air Force approved up to 100 Falcon launches a year from SLC-4 and a rebuilt SLC-6. The California Coastal Commission, worried about sonic booms and wildlife, objected, and the dispute went to court. **Wallops**, in Virginia, hosts Antares and Minotaur pads and Rocket Lab's LC-2 for Electron and LC-3 for Neutron, whose debut has slipped to 2027. At **Starbase**, in southern Texas, Starship's second pad, with a water-cooled flame trench, flew all three Starship flights of 2026, including the first orbital flight on 28 September.

Europe's spaceport is the **Guiana Space Centre** at Kourou, run by the French space agency CNES with ESA. Ariane 6 flies from ELA-4 and Vega-C from the Vega pad; the Soyuz pad fell idle when Russia withdrew in 2022, and the old Diamant site is being rebuilt for European micro-launchers. Brazil's **Alcântara**, at 2.3° S, hosted its first orbital attempt in more than 20 years in December 2025, when Innospace's HANBIT-Nano failed about 30 s after lift-off.

### 37.2.3 Russia and Kazakhstan

**Baikonur**, on the Kazakh steppe, launched Sputnik and Gagarin and is still Russia's only site for crews and ISS cargo. Soyuz rockets are assembled horizontally, rolled to the pad by train and raised by a hydraulic erector (Figure 37.6). On 27 November 2025, during the launch of Soyuz MS-28, the mobile service cabin under the rocket at Site 31, not properly secured after retraction, fell into the flame trench. With Gagarin's original pad retired in 2019, Russia briefly had no pad for crewed flights. More than 150 workers rebuilt the cabin, installing 19 m structural members of about 17 t. The pad was cleared for a Progress launch in March 2026. Site 45 flew the first Soyuz-5 on a suborbital test on 30 April 2026.

**FIGURE 37.6** A Soyuz rocket rolled out by train to Site 31 at Baikonur, September 2023. Assembled horizontally, it travels on the transporter-erector that will raise it at the pad. Image: NASA (NASA/Bill Ingalls).

**FIGURE 37.7** The erector raising a Soyuz into the launch table at Site 31, September 2024. The four support arms of the pad will hold the rocket by its waist until lift-off. Image: NASA (NASA/Bill Ingalls).

**Plesetsk**, 800 km north of Moscow, launches military Soyuz-2 and Angara rockets; **Vostochny**, in the Far East, has launched Soyuz-2 since 2016 and Angara A5 since 2024, but only twice in 2025.

### 37.2.4 China

China has four state launch centres and a growing commercial sector. **Jiuquan**, in the Gobi Desert, launches Shenzhou crews and, increasingly, commercial rockets from new pads nearby — among them LandSpace's Zhuque-3, whose booster touched down on its legs in August 2026, although a fire followed. **Taiyuan** specialises in sun-synchronous orbits and **Xichang**, in Sichuan, in geostationary and BeiDou missions. Both lie inland: spent stages carrying toxic hypergolic residue fall in mountain valleys, and residents downrange are warned to shelter or leave.

**Wenchang**, on Hainan Island, is the coastal site for Long March 5 and 7 and the new crew-capable Long March 10. The new rocket's first stage flew a suborbital test from the new pad LC-301 in February 2026. Beside it, the **Hainan commercial launch site**, China's first commercial spaceport, opened its two pads in 2024–25. It flew Long March 10B's maiden flight on 10 July 2026 and was building Pads 3 and 4 for more than 60 launches a year. China also launches solid rockets from **ships** sailing from Haiyang in Shandong (see the last section).

### 37.2.5 India, Japan, Korea and the Middle East

India's **Satish Dhawan Space Centre** on Sriharikota Island has two pads; in January 2025 the government approved a third (₹3 985 crore, about four years to build) for the Next Generation Launch Vehicle. **Kulasekarapattinam**, at 8.4° N in Tamil Nadu, is being built for SSLV and private rockets, with a first launch planned for 2027.

Japan launches H3 from the Yoshinobu complex at **Tanegashima**, where rockets are stacked in an assembly building and moved to the pad on a mobile launcher (Figure 37.8). The H-IIA made its final flight in June 2025; H3 failed in December 2025 and returned to flight in mid-2026. The solid Epsilon S, delayed by two motor explosions in ground tests, is due to fly from **Uchinoura** by March 2027. South Korea's **Naro Space Center** launched Nuri successfully in November 2025, the first time with Hanwha Aerospace building the rocket, and again in October 2026.

**FIGURE 37.8** An H-IIA rocket carrying the Global Precipitation Measurement Core Observatory rolls out on its mobile launcher to Launch Pad 1 at Tanegashima in February 2014. The red-and-white masts are lightning towers. Image: NASA (NASA/Bill Ingalls).

Israel launches Shavit rockets from **Palmachim** air base; Iran uses the **Imam Khomeini Spaceport** near Semnan and the military site at **Shahroud**; North Korea's **Sohae** last launched, unsuccessfully, in May 2024.

### 37.2.6 New and Small Spaceports

Small launchers have prompted a wave of new sites, many at high latitudes where polar orbits can be reached over empty sea (Table 37.2). Isar Aerospace's Spectrum failed 30 s into its first flight from **Andøya**, Norway, in March 2025 but reached orbit on 5 September 2026 — the first orbital launch from continental Western Europe.

**TABLE 37.2** Selected new and smaller spaceports.

| Spaceport | Latitude | Operator and vehicle | Status (October 2026) |
|---|---|---|---|

| Spaceport | Latitude | Operator and vehicle | Status (October 2026) |
|---|---|---|---|
| Andøya (Norway) | 69.3° N | Isar Aerospace Spectrum | Failed March 2025; reached orbit 5 September 2026 |
| SaxaVord, Shetland (UK) | 60.8° N | RFA One | Awaiting first launch; rocket de-stacked July 2026 |
| Esrange (Sweden) | 67.9° N | Firefly Alpha (LC-3C) | First orbital launch targeted 2028 |
| Kulasekarapattinam (India) | 8.4° N | ISRO SSLV, private rockets | Under construction; first launch 2027 |
| Bowen (Australia) | 20.0° S | Gilmour Space Eris | First flight failed after 14 s (July 2025); second planned late 2026 |
| Alcântara (Brazil) | 2.3° S | Innospace HANBIT-Nano | First flight failed (December 2025) |
| Spaceport Kii (Japan) | 33.5° N | Space One Kairos | Three failures (2024–26) |
| Spaceport America (USA) | 33.0° N | Virgin Galactic (suborbital) | Delta-class spaceplane being prepared for flight tests |

# 37.3 Inside a Launch Complex

## 37.3.1 The Pad: Mounts, Trenches and Deflectors

At the heart of every pad is the **launch mount**, a steel table that carries the rocket through **hold-down clamps** (Figure 37.9). Liquid-fuelled rockets ignite a few seconds before release while computers confirm that every engine has reached full thrust; if one misbehaves, everything shuts down with the vehicle still clamped — a **hold-down abort**.

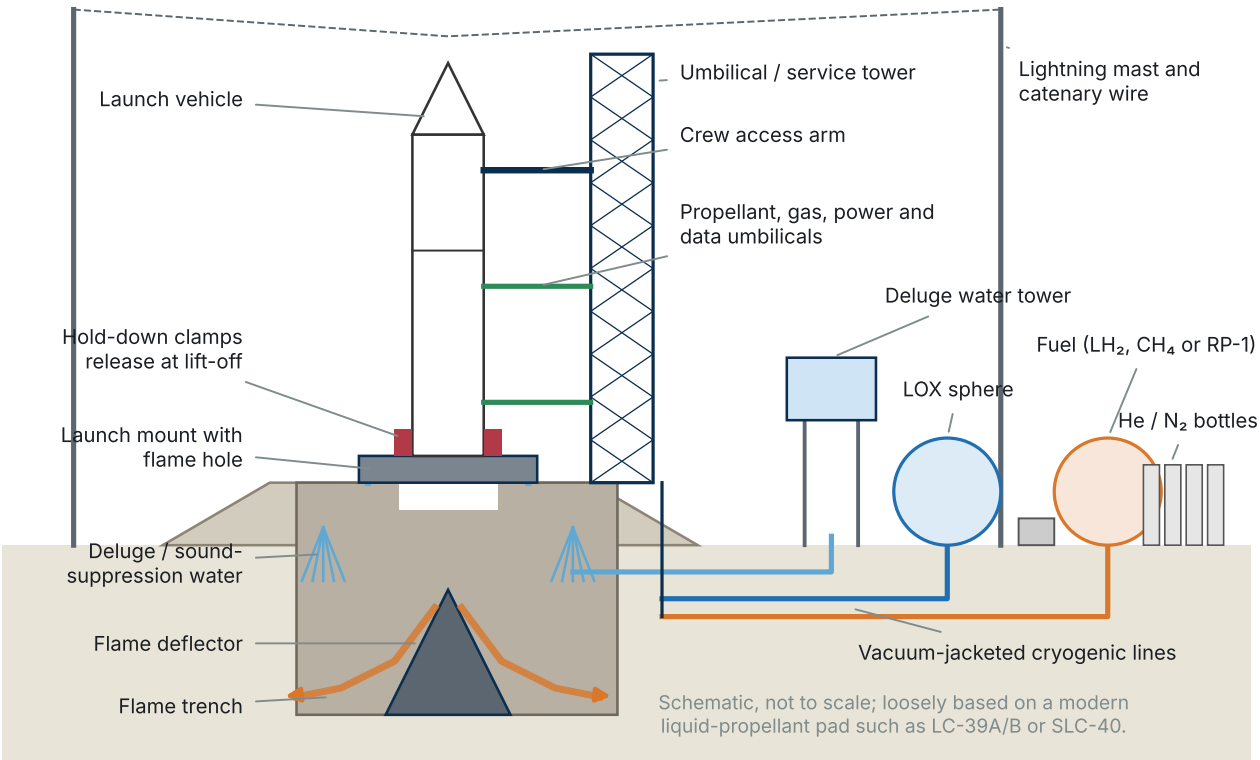



**FIGURE 37.9** Main elements of a liquid-propellant launch pad. The exhaust passes through the flame hole into a trench, where a deflector turns it sideways; water floods the mount and trench at ignition; umbilicals deliver propellant, gas, power and data until lift-off; lightning masts and propellant storage stand at a safe distance.

The exhaust, at 2–4 km/s and up to about 3 000 °C, passes through a hole in the mount into a **flame trench**, a concrete channel where a wedge-shaped **flame deflector** turns it sideways. At Pad 39B the trench is more than 130 m long and lined with about 96 000 heat-resistant bricks. The deflector, completed in 2018, is built from some 150 steel plates of up to 1.8 t each (Figure 37.10). Without a trench the jet strikes the ground directly: on Starship's first flight in April 2023, Super Heavy's 33 engines dug a crater under the mount and threw concrete across the site. SpaceX then fitted a water-cooled steel plate under the mount, and its second Starbase pad has a full trench.

**FIGURE 37.10** The new main flame deflector in the flame trench at Launch Complex 39B in May 2018, built to divert the exhaust of the Space Launch System sideways along the trench. Image: NASA/KSC (NASA/Kim Shiflett).

### 37.3.2 Water: Deluge and Sound Suppression

Ignition creates a sharp pressure pulse, the **ignition overpressure**, which reflects from the trench onto the rocket. On the first Space Shuttle flight in 1981 it caused most of the damage to the orbiter's heat shield, on which 16 tiles were lost and 148 damaged. Lift-off also radiates about 200 dB of sound power, which shakes the payload (Chapter 34). The cure is water sprayed into the exhaust and over the mount, which absorbs energy as it is shattered into droplets and turned to steam. At Pad 39B a 400 000-gallon (1.5 million litre) tower empties in under 30 s, at a peak of about 70 $m^3/s$ (Figure 37.11). The Shuttle's 300 000 gallons, released just before main-engine ignition, emptied in 41 s. Ariane 6 receives about a quarter of an Olympic pool in 20 s. Most of the white cloud seen at lift-off is this water turned to steam.

**FIGURE 37.11** A test of the Pad 39B sound-suppression system in September 2019: water pours over the SLS mobile launcher and through its flame hole into the trench. Image: NASA/KSC (NASA/Kim Shiflett).

### 37.3.3 Umbilicals, Towers and Catch Arms

Until lift-off the rocket is fed through **umbilicals** — hoses and cables for propellant top-up, pressurant gas, conditioned air, power and data. They are attached by quick-disconnect plates that separate in the last seconds or as the vehicle rises. Upper-stage umbilicals hang from swing arms on a **service tower**, which for crewed launches also carries a **crew access arm** and an escape system; tail umbilicals at the base retract into protective housings at lift-off. These interfaces are a common cause of scrubs: Artemis I was delayed from August to November 2022 partly by hydrogen leaks at a quick-disconnect on its tail service mast.

SpaceX's Starship towers add a new function. Their two "chopstick" arms stack the booster and ship on the mount and, since Flight 5 in October 2024, catch the returning booster in mid-air. This makes the tower a crane, umbilical tower and landing pad in one.

### 37.3.4 Propellants and Gases

A pad is also a cryogenic plant (Table 37.3). NASA's newest hydrogen sphere at Pad 39B, finished in 2022, holds about 4 700 $m^3$ of usable liquid hydrogen — the largest such tank in the world. Its glass-bubble insulation cuts boil-off by 40–50 % compared with the perlite of its 1960s neighbour. Transfer lines are **vacuum-jacketed** to limit heat leaks.

**TABLE 37.3** Commodities stored and handled at a typical launch site.

| Commodity | Storage | Density (kg/$m^3$) | Used for | Main hazard |
|---|---|---|---|---|
| Liquid oxygen (LOX) | −183 °C; Falcon 9 subcools to about −207 °C | 1 141 (more when subcooled) | Oxidiser for most liquid rockets | Intense fire with any fuel; cold burns |
| Liquid hydrogen ($LH_2$) | −253 °C | 71 | Fuel (SLS, Ariane 6, H3, Centaur) | Invisible flame; leaks; embrittlement |
| Liquid methane ($LCH_4$) | −162 °C | 422 | Fuel (Starship, Vulcan, New Glenn, Zhuque) | Flammable vapour cloud |
| RP-1 kerosene | Ambient; Falcon 9 chills to about −7 °C | ≈ 810 | Fuel (Falcon, Soyuz, Electron, Long March 6/7) | Fire |

| Commodity | Storage | Density (kg/m³) | Used for | Main hazard |
|---|---|---|---|---|
| Hydrazine, MMH, $N_2O_4$ | Ambient, sealed | 870–1 450 | Spacecraft thrusters; some upper stages | Toxic, carcinogenic; ignite on contact |
| Helium, nitrogen | Gas at 20–40 MPa | — | Pressurant; purges | Asphyxiation; stored energy |

Cold propellant is denser: Falcon 9 chills its oxygen to about −207 °C, 24 °C below its boiling point, and its kerosene to about −7 °C. Because the propellant then warms in the tanks, it is loaded only in the last 35 minutes, with the crew already aboard on Dragon flights — a "load and go" procedure NASA approved in 2018. Spacecraft propellants are **hypergolic** (they ignite on contact) and toxic. Technicians in protective suits load them days or weeks earlier in a payload processing facility (Figure 37.12). This is done before the spacecraft is sealed in its fairing (Chapter 34).

**FIGURE 37.12** Fuelling technicians in self-contained atmospheric protective ensemble (SCAPE) suits take a sample of monomethylhydrazine before loading it into NASA's Solar Dynamics Observatory at a payload processing facility near Kennedy Space Center in January 2010. Image: NASA/KSC.

### 37.3.5 Assembling the Rocket

Rockets are assembled in one of three ways (Figure 37.13). **Vertical assembly and transfer** stacks the vehicle on a mobile launcher inside a tall building, such as NASA's 160 m Vehicle Assembly Building, and carries it to the pad. The crawler-transporter weighs about 3 000 t, can carry 8 200 t and creeps the 6.8 km to Pad 39B at 1.6 km/h. It burns some 390 litres of diesel per kilometre, while hydraulic jacks keep its load level on the ramp (Figure 37.14). A second, taller mobile launcher for SLS Block 1B was abandoned in 2026 after its cost grew from $383 million to about $1.8 billion. Japan, China and United Launch Alliance use the same principle with platforms on rails or tow vehicles.

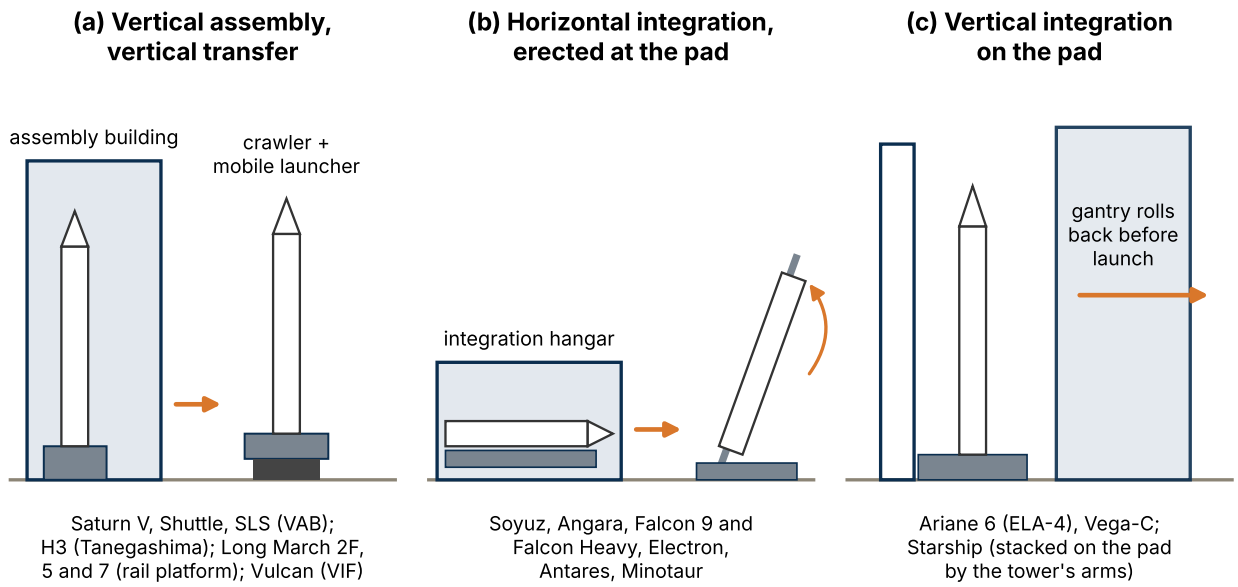



**FIGURE 37.13** Three ways to assemble a launch vehicle: (a) stacked vertically in an assembly building and carried to the pad on a mobile launcher; (b) integrated horizontally in a hangar and raised at the pad by a transporter-erector; (c) assembled vertically on the pad inside a mobile gantry that rolls away before launch.

**Horizontal integration**, the Soviet tradition followed by SpaceX and Rocket Lab, joins the stages lying down in a simple hangar; a transporter-erector then raises the rocket at the pad (Figure 37.6). The payload is usually mated horizontally too, which some national-security satellites cannot tolerate, so in 2020 SpaceX proposed a mobile service tower at Pad 39A for vertical integration. **Vertical integration on the pad** builds the rocket in place inside a mobile gantry: Ariane 6's, at ELA-4, is 90 m tall, weighs about 8 200 t and rolls 120 m away before launch.

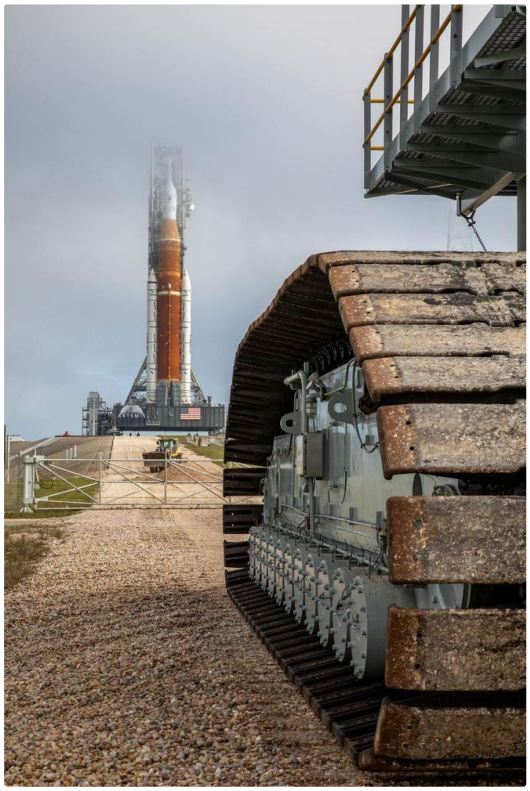


**FIGURE 37.14** The crawler-transporter that carried NASA's Space Launch System to Launch Pad 39B in March 2022, with the rocket standing on its mobile launcher at the pad in the background. The crawler rides on four double-tracked trucks, each driven by four 375 hp electric motors. Image: NASA/KSC (NASA/Kim Shiflett).

## 37.4 The Range: Tracking, Safety and Recovery

### 37.4.1 Instrumentation and Telemetry

The **range** is everything beyond the pad that keeps a launch safe and observed: radars, telemetry antennas, optical trackers, weather instruments and a control centre. The US Space Force runs the Eastern Range from Florida (Space Launch Delta 45) and the Western Range from Vandenberg (Space Launch Delta 30); NASA runs Wallops and CNES Kourou, each with downrange tracking stations. During ascent the range receives telemetry and compares the instantaneous impact point with pre-set limits. Since 2017 most American launches have carried an **autonomous flight safety system** that makes any destruct decision on board (Chapter 34), so ranges depend less on tracking radars and can support launches in quick succession.

### 37.4.2 Hazard Areas in the Sky and at Sea

Before each launch the range defines **hazard areas** where falling stages, fairings or failure debris could land. Aircraft are kept out by notices issued through the Federal Aviation Administration (FAA), ships by navigational warnings, and a single boat inside the zone can halt the count — the range is "fouled". If a vehicle breaks up unexpectedly, the FAA can activate a **debris response area**. When Starship Flight 7 broke up over the Caribbean on 16 January 2025, dozens of airliners were held or diverted. Debris was reported on the Turks and Caicos Islands.

### 37.4.3 Recovery Fleets and Landing Zones

Reusable rockets have turned ranges into recovery operations. SpaceX has autonomous **droneships**, one in the Atlantic and one in the Pacific since a third was reassigned to Starship in April 2026. They wait about 600 km downrange for Falcon boosters (Figure 37.2), and other ships collect fairing halves. Boosters that fly back land on concrete **landing zones**. At Cape Canaveral, Landing Zone 1 was retired when its lease ended in July 2025. LZ-40 beside SLC-40 took its first booster on 13 February 2026. Dragon capsules have splashed down off California since 2025, after pieces of their jettisoned trunks reached land in Australia and North Carolina. Blue Origin lands New Glenn boosters on the ship *Jacklyn*, and in July 2026 China caught a Long March 10B first stage in a net of cables on the ship *Linghangzhe*.

## 37.5 The Launch Campaign and Countdown

### 37.5.1 From Arrival to Lift-off

A campaign begins weeks before the countdown. A large satellite typically arrives four to eight weeks before launch, is tested, fuelled, mated to its adapter and encapsulated. The rocket's stages are checked, joined and often test-fired on the pad — a **static fire** — or put through a **wet dress rehearsal**, a countdown with propellant loading that stops before ignition. Falcon 9 is usually rolled out within a day of launch, Soyuz two days before, while SLS spends weeks at the pad.

### 37.5.2 The Countdown

The countdown is a scripted sequence (Table 37.4). The **launch director** polls each team at set points ("go/no-go"), and built-in **holds** give time to catch up. In the last minutes an automatic sequencer takes over, ready to stop if any measurement drifts out of limits.

**TABLE 37.4** Falcon 9 countdown milestones for a Dragon cargo mission from LC-39A (CRS-29, NASA, November 2023; times approximate).

| Time (min:s) | Event |
|---|---|
| −38:00 | Launch director verifies "go" for propellant load |
| −35:00 | RP-1 loading begins; first-stage LOX loading begins |
| −16:00 | Second-stage LOX loading begins |
| −07:00 | Pre-launch engine chill begins |
| −05:00 | Dragon transfers to internal power |
| −01:00 | Flight computer begins final checks; tanks pressurised for flight |
| −00:45 | Launch director verifies "go" for launch |
| −00:03 | Engine controller commands ignition sequence |
| 00:00 | Lift-off |

The **launch window** may last hours for a sun-synchronous satellite, or be **instantaneous** for a mission that must meet the ISS, leaving no time to resolve a problem.

### 37.5.3 Launch Commit Criteria

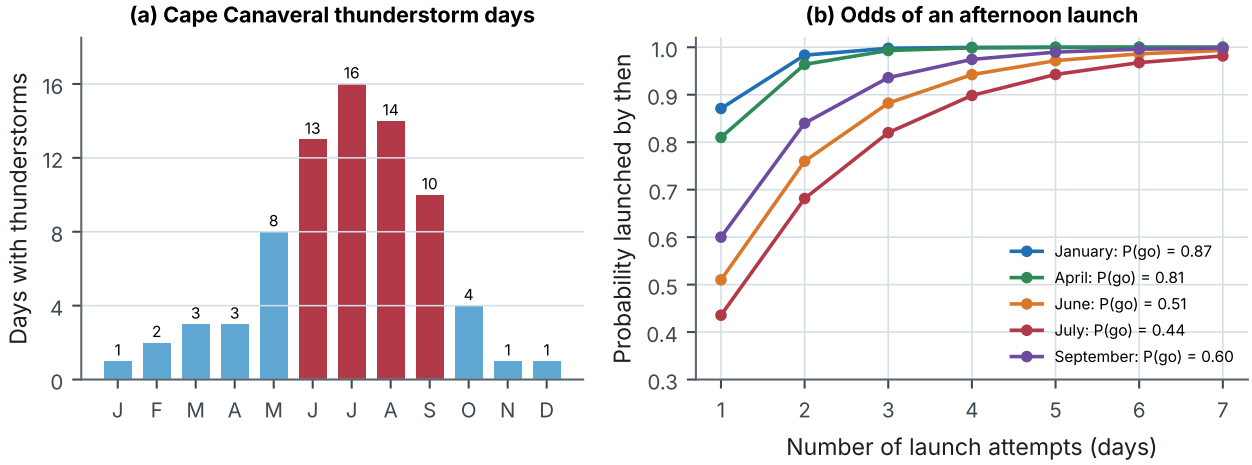
Weather causes about half of all scrubs at Cape Canaveral: in 2020 the 45th Weather Squadron reported that 49 % of scrubs were due to weather, which affected 24 % of all countdowns. The rules are written as **launch commit criteria** (Table 37.5). Wind limits protect the vehicle on the pad and in the jet stream, and cloud and lightning rules protect it from **triggered lightning**. A rocket and its long, ionised plume can trigger a strike in a field too weak to produce lightning on its own. This happened to Apollo 12 in 1969 and to Atlas-Centaur 67, destroyed in 1987. Electric-field mills around the pads measure the field continuously.

**TABLE 37.5** Weather launch commit criteria for a crewed Falcon 9 from Florida (NASA fact sheet, abridged).

| Parameter | Do not launch if… |
|---|---|
| Ground wind | Sustained wind at 49 m (162 ft) on the pad exceeds 13.4 m/s (30 mph) |
| Upper-level winds | Wind shear aloft could cause control problems |
| Lightning | Lightning within 18.5 km (10 nautical miles) of the pad or flight path in the last 30 min, unless specific conditions are met |
| Anvil clouds | Within 18.5 km of an attached anvil (with exceptions) or of any detached anvil |
| Cumulus clouds | Within 18.5 km of cumulus with tops colder than freezing, unless distance rules are met |
| Thick clouds | Flight through a layer more than 1.4 km thick extending into freezing temperatures |
| Electric field | Field mills within 9.3 km of the pad read more than ±1 500 V/m (or ±1 000 V/m under some conditions) in the last 15 min |
| Abort weather | Conditions at any of 50+ downrange sites would exceed Dragon's abort splashdown limits |

Florida is America's thunderstorm capital. In 1950–80 the Cape averaged 76 thunderstorm days a year, 80 % of them from May to September, peaking at 16 in July (Figure 37.15). Summer storms build in the afternoon, when sea breezes from both coasts collide. A 2024 study looked at 18 years

of field-mill records. It found that 88 % of successful launches in 2005–2020 lifted off in hours when the climatological chance of violating the electric-field rule was 15 % or less.




(a) NASA TM-103498 (1950–52, 1957–80). (b) Assumes an afternoon window is lost on any thunderstorm day, a 10 % chance of a non-weather scrub, and independent days.

**FIGURE 37.15** (a) Mean number of days with thunderstorms per month at Cape Canaveral, 1950–52 and 1957–80. (b) The probability that an afternoon launch has flown after a given number of daily attempts, if any thunderstorm day forbids launch and there is a 10 % chance of a non-weather scrub each day.

**FIGURE 37.16** Lightning near Launch Complex 39B on 10 June 2022, with the Artemis I rocket on the pad. The pad's three 180 m masts and their wires carry strikes to ground away from the vehicle. Image: NASA/KSC (NASA).

**EXAMPLE: THE ODDS OF A SUMMER SCRUB**

A satellite must launch in a two-hour window starting at 14:00 local time from Cape Canaveral. Use the thunderstorm climatology of Figure 37.15 as the chance that lightning rules forbid an afternoon window, and assume a 10 % chance that a technical problem scrubs any attempt, independent of the weather.

1. **July:** 16 thunderstorm days in 31, so the chance of good weather is $1 - 0.516 = 0.484$. The chance of launching on a given attempt is $0.484 \times 0.90 =$ **0.44**.
2. The number of attempts follows a geometric distribution, with mean $1/0.44 =$ **2.3 attempts**. The chance of having launched within three days is $1 - (1 - 0.44)^3 = 0.82$, so about one campaign in five will still be waiting after three days.
3. **January:** 1 thunderstorm day in 31 gives $0.97 \times 0.90 = 0.87$ per attempt, a mean of 1.15 attempts, and a 99.8 % chance of launching within three days. But winter cold fronts bring strong upper-level winds instead.
4. Over the year, 76 thunderstorm days in 365 is a 21 % chance, close to the 24 % of countdowns that weather affects. A morning window, when storms are rare, is the best cure.

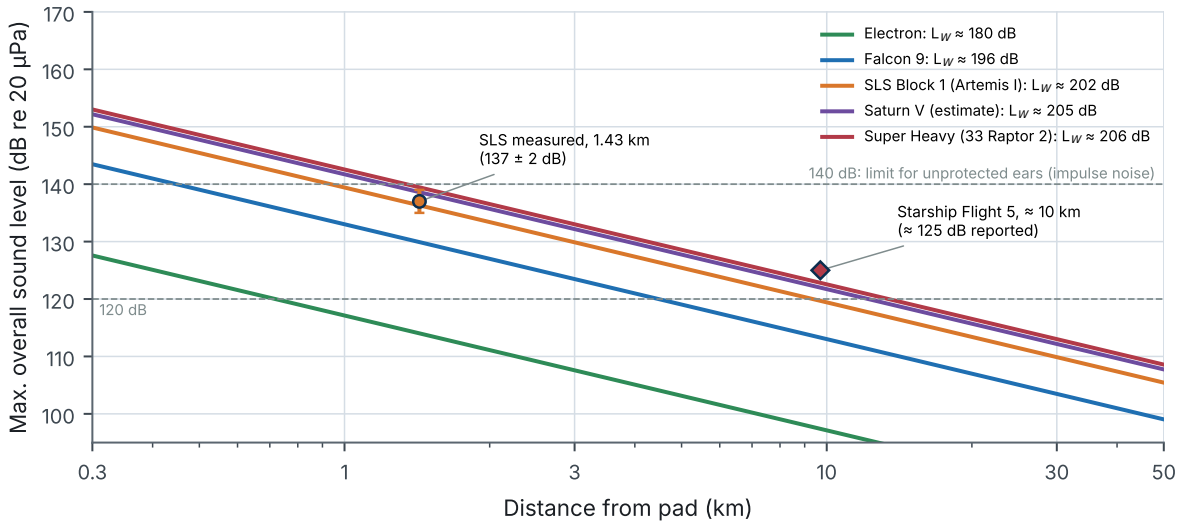
### 37.5.4 Scrubs, Holds and Recycles

When a criterion is violated the count **holds**; if the problem cannot be cleared within the window, the launch is **scrubbed**. The **turnaround** to the next attempt depends on draining and replacing propellant and fixing the fault. Falcon 9 often tries again 24 hours later, and a hydrogen-fuelled vehicle may need two days or more. A mission tied to a planetary window may wait weeks.

## 37.6 Environment, Community and Regulation

### 37.6.1 Noise and Sonic Booms

A rocket's **sound power** is a small fraction $\eta$, the **acoustic efficiency**, of the mechanical power of its jet, $\tfrac{1}{2} \times$ thrust $\times$ exhaust speed. Measurements 1.4–1.8 km from the Artemis I launch gave SLS a sound power level of 202.4 dB (re 1 pW). The measured efficiency was 0.33 %, below the 0.5 % long assumed. Falcon 9 radiates about 196 dB and the Saturn V about 205 dB. Treating the rocket as a source above hard ground gives the levels of Figure 37.17. The model reproduces SLS's measured 137 dB at 1.43 km and predicts 123 dB at 10 km from Super Heavy. This is close to the 125 dB recorded by Brigham Young University researchers about 10 km from Starbase on Starship Flight 5. The 120 dB contour lies about 4.5 km from a Falcon 9, 9 km from SLS and 13 km from Super Heavy.




$L_p = L_W + 5$ dB (directivity) − 10 $\log_{10}(2\pi r^2)$: hemispherical spreading over hard ground, no atmospheric absorption. $L_W = 10 \log_{10}(\eta \cdot ½ \cdot F \cdot v_e$ / 1 pW) with $\eta$ = 0.3 %, or measured/estimated values (Kellison & Gee 2023, 2024). Super Heavy: 33 × 2.26 MN, $v_e \approx 3.2$ km/s.

**FIGURE 37.17** Maximum overall sound level against distance for five launch vehicles, from their sound power assuming hemispherical spreading over hard ground and a 5 dB directivity gain. Points are measurements at the Artemis I launch and during Starship Flight 5.

The model ignores absorption by the air, small at the low frequencies (10–100 Hz) where rocket noise is strongest, and weather, which can focus sound far from the pad. Returning boosters add **sonic booms**: at Starbase the boom of the descending Super Heavy was the strongest pressure event of the flight, and booms lie at the centre of California's objections to more Vandenberg launches.

## 37.6.2 Debris, Wildlife and Water

Spaceports often share land with wildlife because their safety zones keep people out. Kennedy Space Center shares its land with the Merritt Island National Wildlife Refuge, and Starbase lies among protected tidal flats. Vandenberg's coast hosts nesting shorebirds and seals. Operators control lighting that disorients nesting sea turtles, monitor deluge water (which picks up hydrochloric acid from solid motors) and survey debris thrown out at lift-off. Downrange, falling stages have contaminated drop zones in Kazakhstan and China with the toxic fuel UDMH.

## 37.6.3 Licensing and Environmental Review

In the United States the FAA licenses launches, re-entries and spaceport operators. Its performance-based rule, **14 CFR Part 450**, adopted in 2021, became the only licensing route in March 2026, when the older licences expired. It caps the expected casualties from each launch at $1 \times 10^{-4}$ (Chapter 34). Federal decisions also need review under the **National Environmental Policy Act** — an environmental assessment where impacts are expected to be modest, or a full environmental impact statement that can take years. The May 2025 approval of 25 Starship launches a year from Starbase followed an assessment; SLC-37 and the Vandenberg increase needed full statements. Executive Order 14335 of August 2025 told the FAA to revise Part 450 and agencies to use simpler "categorical exclusions" for routine launch actions.

Elsewhere, the UK Civil Aviation Authority licenses spaceports and launches under the Space Industry Act 2018. France's 2008 Space Operations Act gives CNES responsibility for safety at Kourou. Australia, New Zealand and Japan passed launch laws in 2016–18, and India authorises

private launches through its regulator IN-SPACe. All implement the Outer Space Treaty principle that a state is liable for damage caused by objects it launches (Chapter 1).

## 37.7 Launching from the Air and the Sea

A rocket launched from an aircraft or a ship can choose its latitude and azimuth, avoid overflying land and escape a fixed site's weather. Air launch also saves a little Δv by starting above the densest air (Table 37.6). Northrop Grumman's **Pegasus** has flown 46 times since 1990, latterly from the L-1011 *Stargazer* at about 12 km (Figure 37.18). Its latest flight, on 3 July 2026 from Kwajalein, carried a servicer to raise the orbit of NASA's Swift observatory. Virgin Orbit's **LauncherOne**, dropped from a Boeing 747, reached orbit four times in six attempts. After a failure from Spaceport Cornwall on 9 January 2023 — the first orbital attempt from the UK — the company closed.

**FIGURE 37.18** A Pegasus XL rocket carrying NASA's NuSTAR telescope, attached beneath the L-1011 carrier aircraft Stargazer in June 2012. The aircraft carries the rocket to about 12 km and releases it over the ocean. Image: NASA/JPL (NASA).

**Sea Launch**, an American–Russian–Ukrainian–Norwegian venture, sailed a converted oil platform from Long Beach to the equator and launched Zenit-3SL rockets 36 times from 1999 to 2014; in 2007 one exploded on the platform. China is now the leading sea launcher. A Long March 11 flew from a barge in the Yellow Sea in June 2019. Since then, solid rockets such as Jielong-3, Ceres-1S and Gravity-1 have flown 28 times from ships based at Haiyang and Yangjiang. Solid rockets need little ground support; cryogenic propellants are far harder to handle on a moving deck.

**TABLE 37.6** Air and sea launch systems (flights from GCAT, J. McDowell).

| System | Platform | Flights | Period | Status (2026) |
|---|---|---|---|---|
| Pegasus (Northrop Grumman) | L-1011 *Stargazer* (earlier NB-52) | 46, of which 3 failed and 2 partly failed | 1990–2026 | Flown occasionally |
| LauncherOne (Virgin Orbit) | Boeing 747 *Cosmic Girl* | 6, of which 4 succeeded | 2020–23 | Company closed 2023 |
| Sea Launch (Zenit-3SL) | *Odyssey* platform at the equator, 154° W | 36, of which 3 failed | 1999–2014 | Ended; platform moved to Russia |

| System | Platform | Flights | Period | Status (2026) |
|---|---|---|---|---|
| Chinese sea launches | Barges and ships in the Yellow and South China Seas | 28, all successful | 2019–26 | Expanding |
| Russian submarine launches | Shtil′ and Volna from submarines in the Barents Sea | 3, of which 1 failed | 1998–2006 | Ended |

### AI ON THE LAUNCH RANGE

As launch rates climb, ranges are turning to software (Chapter 52). Scheduling several launches a day is an optimisation problem — windows, hazard areas, airspace closures and shared tracking and recovery assets — for which algorithms can propose conflict-free timetables. Weather is a second target: the 45th Weather Squadron's lightning forecasts still rely on statistical tools. But the Air Force Institute of Technology has trained machine-learning models on a decade of weather-balloon data to automate them. A model from NASA Marshall and the University of Alabama in Huntsville detected 84 % of lightning 15–30 minutes ahead with a 22 % false-alarm rate (Chapter 47). Autonomous flight safety systems remain deliberately simple, verifiable rules; machine learning for debris-risk estimates or faster airspace release must meet the same safety standards.

### KEY TAKEAWAYS

- A site at latitude $\phi$ can launch directly into inclinations from $\phi$ to $180° - \phi$, with $\sin Az = \cos i / \cos \phi$. Rotation adds 465 m/s × $\cos i$; latitude matters most for geostationary missions.
- Safety corridors decide which azimuths are allowed, explaining Baikonur's 51.6°, Palmachim's retrograde orbits and Florida's polar corridor of 2020.
- In 2025 Florida (109) and Vandenberg (66) hosted more than half of the world's 330 orbital attempts; China flew 93 times, and Rocket Lab's Mahia site alone matched all 17 Russian launches.
- A pad combines a flame trench and deflector, a deluge of up to 70 $m^3/s$, umbilical towers, lightning masts and a cryogenic farm. Rockets are stacked vertically and carried out, integrated horizontally and erected, or built on the pad.
- The range tracks the flight, clears hazard areas in the sky and at sea, and recovers boosters, fairings and capsules.
- Weather causes about half of Florida's scrubs; on a July afternoon the chance of launching on a given day can fall below one in two.
- Noise, sonic booms, debris and wildlife make licensing and environmental review a pacing item for new launch sites.

### REVIEW QUESTIONS

1. Derive the range of inclinations reachable by a direct launch from latitude $\phi$. Which inclinations can Vostochny (51.8° N) reach, and in which two directions can it launch into a 70° orbit?
2. Why does a launch into the ISS orbit gain almost the same velocity from Earth's rotation at Kourou as at Baikonur, while a geostationary mission gains far more at Kourou?
3. Using the worked example's sensitivity (0.82 t per 100 m/s), estimate how much payload a sun-synchronous mission loses to Earth's rotation, and how much a due-east launch from Kourou gains over one from Cape Canaveral.
4. Explain the reasons for Palmachim's westward launches, Sriharikota's dog-leg and Baikonur's minimum inclination of 51.6°.

5. List the main elements of a launch pad and the purpose of each. Why do pads need water at ignition, and what happened when Starship first launched without a flame trench or deluge?
6. Compare vertical assembly and transfer, horizontal integration and vertical integration on the pad. Give one advantage and one drawback of each, with an example vehicle.
7. What is triggered lightning, and which launch commit criteria are designed to prevent it?
8. A launch window opens at 15:00 in August at Cape Canaveral. Using Figure 37.15 and a 10 % technical-scrub probability, estimate the expected number of attempts and the chance of launching within two days.
9. Using the noise model, estimate the distance at which a Falcon Heavy (three Falcon 9 cores) would produce 120 dB. Why might real levels differ?
10. What are the advantages and limitations of air and sea launch? Why are most sea launches today made by solid-fuelled rockets?

PART XI

# Space Observatories

Above the atmosphere, telescopes see the whole electromagnetic spectrum without blur. Part XI covers the engineering of space telescopes and the great observatories, from Hubble and Webb to the X-ray, gamma-ray, gravitational-wave and survey missions of today and tomorrow.


38 Space Telescope Engineering

39 Great Observatories and Deep-Space Telescopes


Image: NASA/STScI (Webb) (NASA ESA CSA STScI)

# 38 CHAPTER Space Telescope Engineering

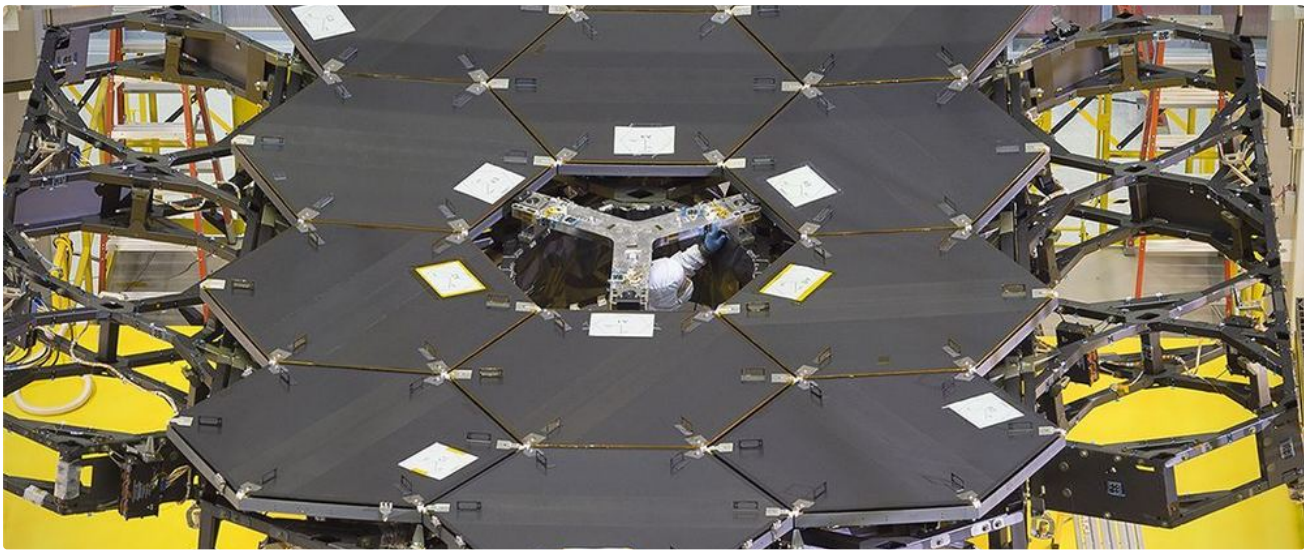


Image: NASA/GSFC

*A space telescope is a precision optical instrument that must survive launch, unfold or cool itself in orbit, hold its aim to a few thousandths of an arcsecond and send home data from as far as 1.5 million kilometres away. This chapter explains why astronomers go to space at all — atmospheric absorption, blurring and glow — and then the engineering that follows: diffraction-limited optics, segmented and grazing-incidence mirrors, detectors from CCDs to microcalorimeters, sunshields and cryocoolers, fine guidance and jitter control, the choice of orbit, servicing and testing, and the data chain from downlink to public archive. Worked examples size the resolution, photon budget and thermal shielding of real observatories.*

IN THIS CHAPTER

38.1 Why Go to Space?
38.2 Optics: Collecting and Focusing Light
38.3 Detectors
38.4 Keeping Cold and Clean
38.5 Pointing and Stability
38.6 Where to Put a Telescope
38.7 Building, Testing and Servicing
38.8 From Photons to Archives

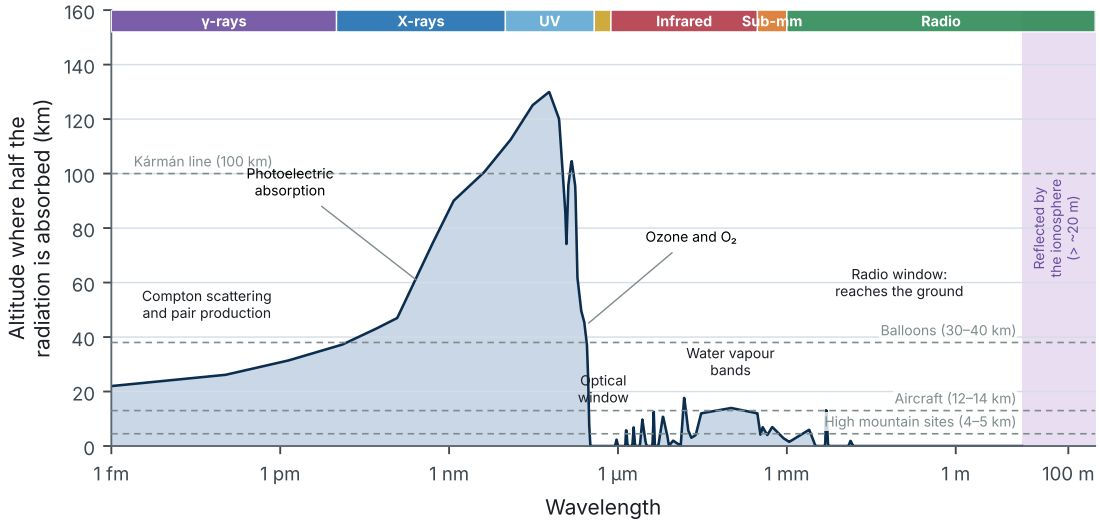
The first view of the Sun from above the ozone layer came in October 1946, when a spectrograph on a captured V-2 rocket recorded its ultraviolet spectrum. Eighty years later, the James Webb Space Telescope (JWST) observes galaxies as they were less than 300 million years after the Big Bang. Its 6.5 m segmented mirror is held below 50 K behind a sunshield the size of a tennis court. The 36-year-old Hubble Space Telescope still works in low Earth orbit. The Nancy Grace Roman Space Telescope, launched on 30 August 2026, is on its way to the Sun–Earth L2 point.

Every one of these observatories is a compromise between physics and engineering. A larger mirror collects more light and sees finer detail, but must fit inside a rocket fairing. A colder telescope sees fainter infrared sources, but needs sunshields or refrigerators. A more distant orbit is thermally quiet, but cannot be visited for repairs and has a weaker radio link. This chapter explains these trade-offs. Optical payloads and radiometry in general are introduced in Chapter 12; the history and science of individual observatories are covered in Chapter 39, and in-space servicing and assembly in Chapter 40.

# 38.1 Why Go to Space?

## 38.1.1 The Atmosphere as a Filter

Earth's atmosphere is transparent in only two broad windows: visible light with a few near-infrared bands, and radio waves from about 1 cm to 20 m (Figure 38.1). Gamma rays and hard X-rays are scattered and absorbed by nuclei and electrons, so that even balloons at 35–40 km lose much of the signal. Soft X-rays and extreme ultraviolet light are absorbed by nitrogen and oxygen at 80–150 km, which is why X-ray astronomy beyond the Sun began on a sounding rocket in 1962 (Chapter 9). Ozone absorbs ultraviolet light below about 300 nm. Water vapour and carbon dioxide absorb most of the infrared, leaving narrow windows at high, dry sites. The ionosphere reflects radio waves longer than about 20 m. Table 38.1 lists the lowest platform from which each band can be observed.




Schematic: approximate half-absorption altitudes after Giacconi et al. (1968) and standard texts; X/γ values from air attenuation coefficients and the standard atmosphere.

**FIGURE 38.1** Altitude at which half of the incoming radiation from space has been absorbed, as a function of wavelength. Only visible light and radio waves reach the ground; X-ray, gamma-ray, ultraviolet and most infrared astronomy must be done from balloons, rockets or spacecraft.

## 38.1.2 Seeing, Diffraction and Sky Brightness

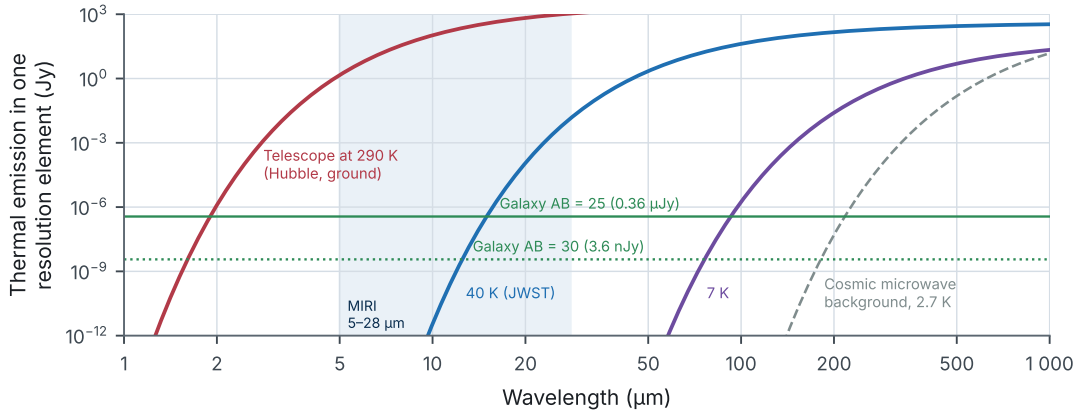
Even where the air is transparent, it is not still. Turbulent cells of air at slightly different temperatures act as moving lenses and smear a star into a flickering blob — astronomical **seeing**. At the best mountain sites the blur is about 0.4–1 arcsecond (″; 1″ = 4.85 µrad), whatever the telescope's size. A 2.4 m telescope in space is limited instead by **diffraction** to about 0.05″ in

visible light. Ground telescopes now partly correct seeing with **adaptive optics**. But the correction works best in the infrared, over small fields near a bright guide star. It never matches the stability of a space image, which can stay the same for years.

The atmosphere also **glows**: emission lines from hydroxyl radicals near 90 km make the near-infrared night sky bright. In space the darkest background is **zodiacal light**, sunlight scattered and re-emitted by interplanetary dust. Its visible brightness is about 22–23 magnitudes per square arcsecond — somewhat darker than the darkest ground sites. In the near infrared it is hundreds of times darker than the sky seen from the ground. It is this background that sets the limiting magnitude of Hubble and JWST.

### 38.1.3 The Infrared Problem: Telescopes That Glow

Everything warm emits infrared radiation. A telescope at room temperature radiates like a blackbody peaking near 10 µm, multiplied by its mirrors' emissivity of a few per cent. Figure 38.2 shows the consequence. At 10 µm a 290 K telescope puts a few hundred million times more power into each resolution element than a galaxy of AB magnitude 25. The only cure is cold. At 40 K, JWST's own emission is fainter than even a 30th-magnitude galaxy below about 12 µm and dominates its background only at the longest wavelengths of its Mid-Infrared Instrument (MIRI). Herschel's mirror, at about 85 K, was limited by its own glow, and future far-infrared missions aim for a few kelvin. This single fact drives the sunshields, orbits and cryogenics described later.




Model: emissivity 0.05 × Planck function, integrated over a beam of radius λ/D for a 6.5 m aperture; 1 Jy = $10^{-26}$ W $m^{-2}$ $Hz^{-1}$.

**FIGURE 38.2** Thermal emission from a telescope (mirror emissivity 0.05) at three temperatures, compared with faint galaxies of AB magnitude 25 and 30. Cooling to 40 K removes the problem below about 12–15 µm; only a few-kelvin telescope is dark in the far infrared.

**TABLE 38.1** Where each band of the electromagnetic spectrum can be observed from.

| Band | Wavelength / energy | Main absorber | Lowest useful platform | Space telescopes |
|---|---|---|---|---|
| Gamma rays | > 100 keV | Nuclei, electrons | Balloon (partly), satellite | Fermi, Swift BAT |
| X-rays | 0.1–100 keV | Photoelectric absorption | Rocket, satellite | Chandra, XMM-Newton, XRISM |
| Ultraviolet | 10–320 nm | $O_2$, $O_3$ | Rocket, satellite | Hubble, GALEX |
| Visible | 0.32–0.9 µm | (none) | Ground (blurred) | Hubble, Gaia, Euclid |
| Near/mid infrared | 0.9–30 µm | $H_2O$, $CO_2$ | Mountain (windows only) | JWST, Spitzer, Roman |

| Band | Wavelength / energy | Main absorber | Lowest useful platform | Space telescopes |
|---|---|---|---|---|
| Far infrared | 30–300 μm | $H_2O$ | Aircraft, balloon | Herschel, planned PRIMA |
| Sub-mm and radio | 0.3 mm–20 m | $H_2O$ (sub-mm) | Ground, high dry sites | Planck (microwave background) |

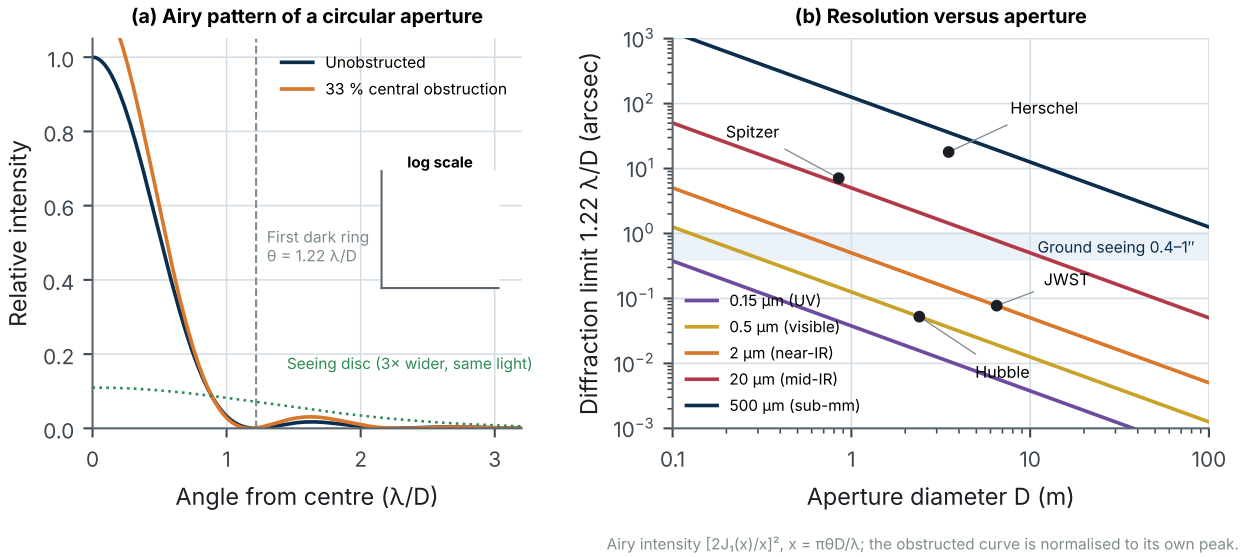
# 38.2 Optics: Collecting and Focusing Light

## 38.2.1 The Diffraction Limit

A telescope **collects** light in proportion to its area and **resolves** detail in proportion to its diameter. Because light is a wave, a circular aperture of diameter $D$ focuses a point source into an **Airy pattern** — a bright core surrounded by faint rings — whose first dark ring lies at

$$\theta = 1.22\,\frac{\lambda}{D}$$

where $\lambda$ is the wavelength (Figure 38.3). Two stars closer than about this angle merge into one (the **Rayleigh criterion**). The core holds 84 % of the light; a central obstruction, such as a Cassegrain secondary mirror, moves light into the rings, which matters for high-contrast imaging. To exploit the diffraction limit, the detector must have at least two pixels across $\lambda/D$ (**Nyquist sampling**). Hubble's cameras, with 40–50 mas (milliarcsecond) pixels, are undersampled about twofold in the visible (Nyquist at 0.5 μm needs 21 mas). They recover detail by combining several offset ("dithered") exposures.




Airy intensity $[2J_1(x)/x]^2$, $x = \pi\theta D/\lambda$; the obstructed curve is normalised to its own peak.

**FIGURE 38.3** (a) The Airy pattern of a circular aperture and the effect of a central obstruction, compared with a seeing-limited image of the same total flux. (b) Diffraction-limited resolution versus aperture at five wavelengths; ground-based seeing is shown for comparison.

**EXAMPLE: HOW SHARP IS A 6.5 M TELESCOPE?**

JWST has an effective diameter of 6.5 m. What is its resolution at 2 μm, and how does it compare with Hubble and with the ground?

1. Diffraction limit: $\theta = 1.22\times2\times10^{-6}/6.5$ = 3.75 × $10^{-7}$ rad. Multiplying by 206 265″/rad gives 0.077″ = 77 mas.
2. Nyquist sampling needs pixels of $\lambda/(2D)$ = 32 mas — almost exactly the 31 mas pixels of JWST's NIRCam short-wavelength channel.
3. Hubble at 0.5 μm: $\theta$ = 1.22 × 0.5 × $10^{-6}$ / 2.4 = 52 mas. JWST at 2 μm is slightly coarser than Hubble in the visible, but at 0.7 μm it reaches 27 mas.
4. Ground seeing of 0.8″ is ten times worse. At the Moon's distance (384 400 km), 77 mas corresponds to 144 m; JWST could separate two car headlights 1.5 m apart at about 4 000 km.

### 38.2.2 Telescope Designs

Almost all space telescopes are reflectors: mirrors work from the ultraviolet to the far infrared, have no chromatic aberration and can be supported from behind. The main designs (illustrated in Chapter 12) are the **Cassegrain**, with a concave primary and a convex secondary that folds the light back through a hole in the primary. Hubble is an f/24 **Ritchey–Chrétien** Cassegrain, whose two hyperbolic mirrors remove coma over a wide field. The other main design is the **three-mirror anastigmat (TMA)**, whose third curved mirror corrects astigmatism and field curvature to give a wide, flat field. JWST, Euclid (a compact Korsch variant) and Roman are TMAs; Roman's 0.28 square-degree field is about 200 times that of Hubble's infrared camera. **Off-axis** designs tilt the mirrors so that the secondary and its struts do not block the beam. This gives a clean Airy pattern without diffraction spikes. That is why the planned Habitable Worlds Observatory (HWO) is studying an off-axis aperture for coronagraphy.

### 38.2.3 Mirrors: Materials and Lightweighting

A mirror must keep its shape to a small fraction of the wavelength — typically 10–30 nm rms for a visible or near-infrared telescope. It must do so while being launched at several g, cooled or heated by tens of kelvin and held without distortion in weightlessness. The material is chosen for low thermal expansion, high stiffness per unit mass and good thermal conductivity (Table 38.2).

**TABLE 38.2** Typical room-temperature properties of space mirror materials.

| Material | Density (g/cm³) | Stiffness (GPa) | Expansion at 20 °C (ppm/K) | Conductivity (W/m·K) | Used on |
|---|---|---|---|---|---|
| ULE glass | 2.2 | 68 | ≈ 0 ± 0.03 | 1.3 | Hubble, Roman |
| Zerodur glass-ceramic | 2.5 | 90 | ≈ 0 ± 0.02 | 1.5 | Chandra |
| Beryllium | 1.85 | 300 | 11 (≈ 0 below 100 K) | 216 | JWST, Spitzer |
| Silicon carbide | 3.2 | 420 | 2.2 | 180 | Herschel, Gaia, Euclid |
| Aluminium alloy | 2.7 | 69 | 23 | 167 | Small and cryogenic telescopes |

Glass mirrors are **lightweighted** by fusing thin face sheets to a honeycomb core. Hubble's 2.4 m primary (Figure 38.4) weighs about 828 kg instead of roughly 3 600 kg for a solid blank; Roman's mirror of the same size weighs only 186 kg. **Beryllium** is as stiff as steel but lighter than aluminium, and its expansion almost vanishes below 100 K. That is why JWST and Spitzer use it. Each JWST segment was measured at cryogenic temperature and re-polished warm with the inverse of its cold distortion. **Silicon carbide** combines stiffness and conductivity: Herschel's 3.5 m primary was brazed from 12 SiC petals and machined from 720 kg down to 240 kg.

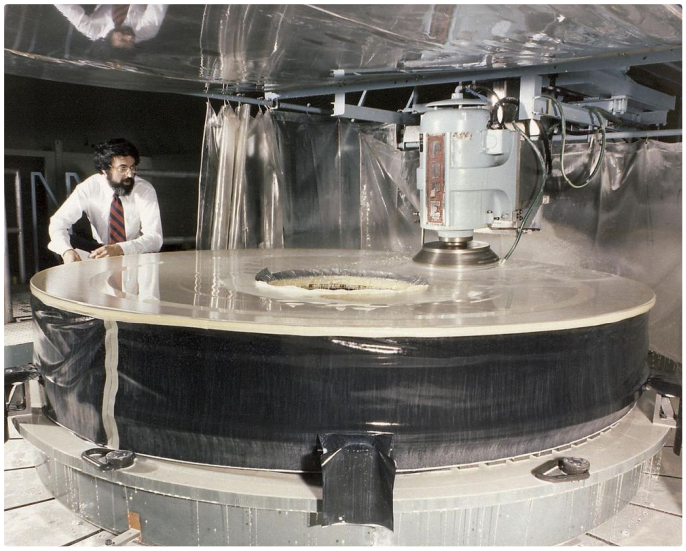


**FIGURE 38.4** Hubble's 2.4 m primary mirror being ground at Perkin-Elmer in Danbury, Connecticut, around 1979. Its ultra-low-expansion glass face sheets are fused to a honeycomb core to cut its mass to about 828 kg. Image: NASA/MSFC.

Hubble also taught the most expensive lesson in optical testing. Its primary was polished very precisely — to the wrong shape. A lens in the **null corrector**, the instrument used to check the mirror's figure, was misplaced by about 1.3 mm, so the mirror's edge was about 2 µm too flat. The resulting **spherical aberration** spread most of a star's light into a wide halo. Two cross-checks with other instruments had hinted at the error and been dismissed. The lesson — test flight hardware end-to-end, with independent methods ("test as you fly") — has shaped every programme since.

### 38.2.4 Segmented and Deployable Mirrors

Fairing diameter limits a monolithic mirror to about 4–5 m today. JWST's 6.5 m primary is therefore made of 18 hexagonal beryllium segments, each about 1.32 m across flats and weighing about 20 kg. They are coated with about 100 nm of gold, which reflects some 99 % of infrared light. Two wings of three segments fold back for launch, and the secondary mirror is held in front of the primary by a deployable tripod.

Each segment sits on six actuators (a **hexapod**) that set its position and tilt, plus a seventh that changes its curvature, moving in steps of roughly ten nanometres (Figure 38.5). Commissioning in February–April 2022 followed a **wavefront sensing and control** sequence. First, each segment formed its own image of a bright star, and the 18 images were tilted one by one until they overlapped. Then **coarse phasing** with a dispersed Hartmann sensor in NIRCam measured and

removed height (piston) differences between segments. Finally, in **fine phasing**, NIRCam took deliberately defocused images through weak lenses, and **phase-retrieval** software computed the wavefront error across the whole pupil for the actuators to correct.

The result beat the requirement. The wavefront error at NIRCam is typically 60–80 nm rms, against a prediction of under 100 nm. The segments drift by less than 10 nm between checks every two days, with small corrections every month or two. One surprise was a micrometeoroid that hit segment C3 in May 2022, larger than pre-launch models predicted. It raised that segment's error from about 56 to more than 250 nm rms, partly corrected to 178 nm. This added about 9 nm to the telescope total. JWST now limits the time it spends looking along its direction of orbital motion, where impacts are more energetic.

**FIGURE 38.5** A JWST engineering-model mirror segment with its gold coating. The actuators that position the segment and adjust its curvature are mounted behind it. Image: NASA/GSFC.

### 38.2.5 Grazing Incidence, Coded Masks and Pair Telescopes

X-rays penetrate a normal mirror; they reflect only at **grazing incidence**, at angles below about 1°, and the critical angle shrinks as photon energy rises. A **Wolter type I** telescope therefore uses a ring-shaped paraboloid followed by a hyperboloid, and nests several pairs to increase the collecting area (Figure 38.6). Chandra has four nested pairs of iridium-coated Zerodur shells, the largest 1.23 m across, with a 10 m focal length (Figure 38.7). Its 1 484 kg mirror assembly resolves better than 0.5″, still the sharpest of any X-ray telescope. XMM-Newton trades sharpness for area with three modules of 58 thin gold-coated nickel shells each. NuSTAR reaches 79 keV with 133 shells per optic coated with **multilayers** — alternating heavy and light layers that reflect like a crystal. The optics sit at the end of a 10 m deployable mast.

At higher energies, imaging uses shadows. Swift's Burst Alert Telescope places a **coded mask** of about 52 000 lead tiles one metre above 32 768 cadmium zinc telluride (CZT) crystals. Each source casts a shifted shadow of the mask. Software decodes the overlapping shadows to locate gamma-ray bursts within seconds. Above about 20 MeV, where a gamma ray mostly turns into matter, each photon is tracked individually. In a **pair-conversion telescope** such as Fermi's Large Area Telescope, a photon turns into an electron–positron pair in a thin tungsten foil. Silicon strip detectors track the pair, and a caesium iodide calorimeter measures its energy.

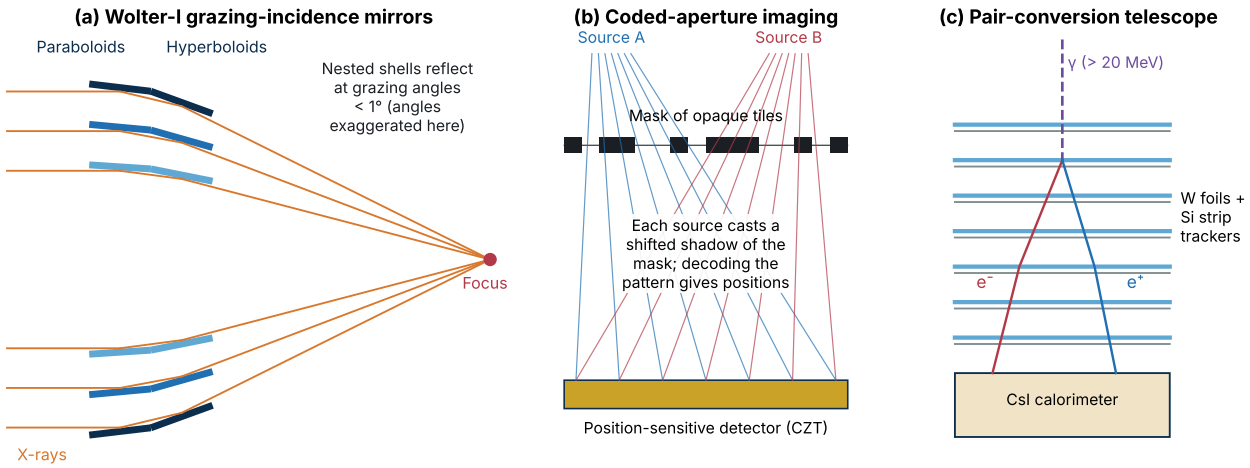



**FIGURE 38.6** Optics for high-energy astronomy: (a) nested Wolter-I grazing-incidence mirrors focus X-rays after two reflections; (b) a coded mask casts a shadow pattern that encodes the direction of each source; (c) a pair-conversion telescope tracks the electron and positron created by a gamma ray.

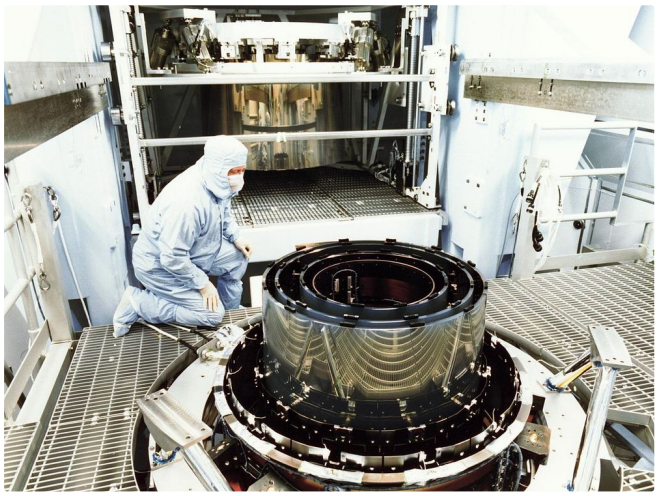


**FIGURE 38.7** The nested mirror shells of Chandra's High Resolution Mirror Assembly being assembled at Eastman Kodak in Rochester, New York. Image: NASA/MSFC.

## 38.2.6 Blocking the Star: Coronagraphs and Starshades

An Earth-like planet is extremely faint next to its star. With a geometric albedo of 0.2 its brightness relative to the star is $A_g(R_p/a)^2 \approx 3.6 \times 10^{-10}$ at full phase and about $1.2 \times 10^{-10}$ when half-lit. At 10 parsecs it lies only 0.1″ from the star — about six times $\lambda/D$ for a 6 m telescope in visible light. Seeing it requires suppressing starlight ten-billion-fold close to the diffraction limit.

A **coronagraph** blocks the star's image with a small focal-plane mask, then removes the diffracted light with a **Lyot stop** in a re-imaged pupil. JWST's coronagraphs reach contrasts of order $10^{-5}$, enough for young, self-luminous giant planets in the infrared. Roman's Coronagraph Instrument is the first in space to correct wavefront errors actively with two **deformable mirrors**, and aims to show contrasts of $10^{-7}$ or better. Reaching $10^{-10}$ needs a telescope roughly ten times more stable than Roman, with a wavefront steady to picometres over hours, and a coronagraph about a hundred times deeper. This is the central technology challenge for HWO, whose project office aims to mature it before a mission concept review at the end of the decade. However, NASA's FY2027 budget request (not enacted by October 2026) proposed cutting HWO's funding from US$150 million to US$5 million.

The alternative is a **starshade**: a flower-shaped screen tens of metres across, flying tens of thousands of kilometres in front of the telescope, whose petals diffract starlight away while the planet's light passes by (Figure 38.8). Laboratory masks have reached $10^{-10}$ contrast, but a starshade must travel for days to weeks between target stars. NASA closed out its starshade technology activity in April 2025, and HWO's baseline is a coronagraph.

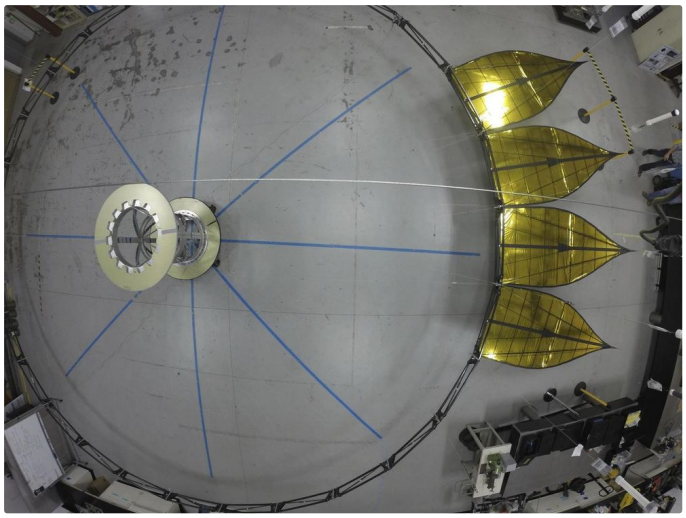


**FIGURE 38.8** A half-scale starshade with four petals deployed at NASA's Jet Propulsion Laboratory in 2014. The precisely shaped petal edges diffract starlight away from the telescope. Image: NASA/JPL (NASA/JPL-Caltech/Princeton).

## 38.3 Detectors

### 38.3.1 From Photons to Electrons

A detector converts photons into electrical charge and measures it. Its key properties are **quantum efficiency (QE)** — the fraction of photons that produce a detected electron — **read noise**, **dark current** (charge generated thermally in darkness), format and radiation tolerance (Table 38.3).

**Charge-coupled devices (CCDs)** store photoelectrons in potential wells and shift them pixel by pixel to one low-noise amplifier, reaching QE above 90 % and read noise of 2–4 electrons. Their weakness in space is **radiation damage**: proton-induced traps capture charge as it is shifted, smearing images. Hubble's CCDs lose transfer efficiency every year, and its pipeline corrects each pixel with a model of the traps. **CMOS** sensors put an amplifier in every pixel, read out faster, need less power and tolerate radiation better.

**TABLE 38.3** Detectors used in space astronomy.

| Detector | Range | Temperature | Typical format | Examples |
|---|---|---|---|---|
| Silicon CCD | 0.2–1.0 µm | 150–190 K | 4k × 4k | Hubble, Gaia (106 CCDs), Euclid VIS (36) |
| Silicon CMOS | 0.3–1.0 µm | 200–300 K | Up to 4k × 4k | Star trackers, guide cameras, small telescopes |
| HgCdTe (H2RG, H4RG) | 0.6–5.3 µm | 35–100 K | 2k × 2k, 4k × 4k | JWST, Euclid NISP, Roman WFI |

| Detector | Range | Temperature | Typical format | Examples |
|---|---|---|---|---|
| Si:As impurity band | 5–28 µm | < 7 K | 1k × 1k | JWST MIRI, Spitzer, WISE |
| Bolometers | 60 µm–3 mm | 0.1–0.3 K | $10^2$–$10^3$ pixels | Herschel SPIRE, Planck HFI |
| X-ray CCD | 0.2–12 keV | ≈ 150 K | 1k × 1k | Chandra ACIS, XMM-Newton EPIC |
| Microcalorimeter | 0.3–12 keV | 0.05 K | 6 × 6 | XRISM Resolve |
| CZT | 3–150 keV | ≈ 290 K | $10^4$ crystals | Swift BAT, NuSTAR |

Infrared photons have too little energy to free electrons in silicon. **Mercury cadmium telluride (HgCdTe)**, whose band gap is set by its composition, is bonded pixel by pixel with indium bumps to a silicon readout circuit. Teledyne's **H2RG** (2048 × 2048 pixels of 18 µm) serves JWST's near-infrared instruments and Euclid; Roman uses 18 **H4RG-10** arrays (4096 × 4096 pixels of 10 µm), about 300 million pixels (Figure 38.9). These arrays are read non-destructively many times "up the ramp" during an exposure, which averages down read noise and lets software remove cosmic-ray hits as sudden jumps. Beyond 5 µm, MIRI uses arsenic-doped silicon **blocked-impurity-band** detectors.

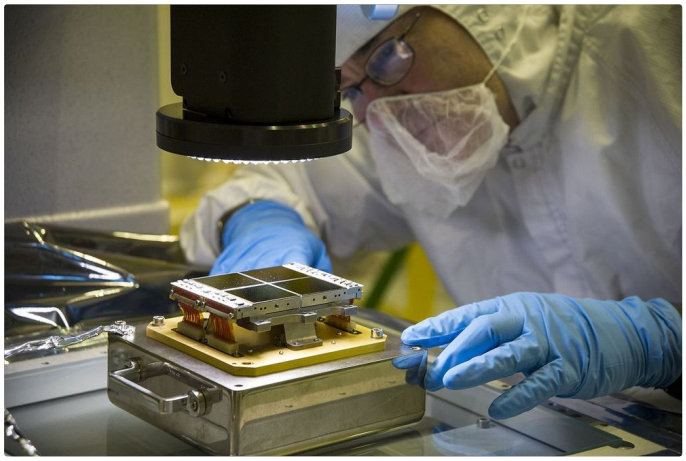


**FIGURE 38.9** A replacement focal-plane assembly for JWST's Near Infrared Camera being aligned under a three-dimensional microscope at NASA Goddard. Each assembly holds HgCdTe detector arrays on a precisely flat mount. Image: NASA/GSFC.

## 38.3.2 Bolometers and Microcalorimeters

At far-infrared and millimetre wavelengths, detectors measure the **heat** deposited by radiation. A **bolometer** is a tiny absorber on a weak thermal link to a cold bath, with a sensitive thermometer — a doped semiconductor or a **transition-edge sensor** held on the edge of superconductivity. Planck cooled its bolometers to 0.1 K and mapped the cosmic microwave background to millionths of a kelvin. **Kinetic inductance detectors**, superconducting resonators read by the thousand on one wire, are being developed for future far-infrared missions.

An X-ray **microcalorimeter** measures the heat of single photons. XRISM's Resolve, launched in 2023, has a 6 × 6 array at 50 mK, cooled by adiabatic demagnetisation refrigerators. The tiny temperature rise from each photon gives its energy to about 4.5 eV at 6 keV — some 30 times finer than an X-ray CCD. This reveals the motions of hot gas in galaxy clusters. Its protective gate valve

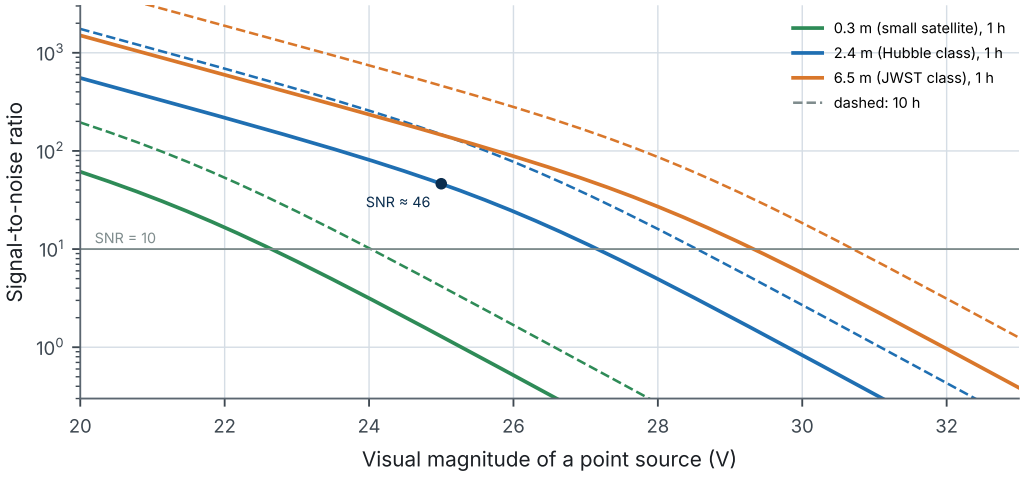
has failed to open, limiting it to energies above about 1.7 keV. Hard X-rays and gamma rays are absorbed in thick, dense semiconductors such as **CZT**, which work at room temperature, or in scintillating crystals.

### 38.3.3 Noise and the Photon Budget

Whether a source is detectable depends on how many photons arrive compared with the noise. The **signal-to-noise ratio (SNR)** of a point source measured in an aperture of $n_{pix}$ pixels is

$$\mathrm{SNR} = \frac{S}{\sqrt{S + n_{pix}\left(B + D\,t + n_{read}\,\sigma_{read}^{2}\right)}}$$

where $S$ is the number of source electrons, $B$ the sky electrons per pixel, $D$ the dark current, $t$ the exposure time and $\sigma_{read}$ the read noise for each of $n_{read}$ readouts. Faint sources are **background-limited**: SNR grows only as $\sqrt{t}$, so a tenfold longer exposure gains 1.25 magnitudes. A larger telescope helps twice — it collects more light, and its smaller image includes less sky. In one hour, a 2.4 m telescope reaches SNR = 10 at V ≈ 27.2 and a 6.5 m telescope at V ≈ 29.3 (Figure 38.10).




Model: V band (550 nm, 88 nm wide), 30 % throughput, 33 % obstruction, diffraction-limited aperture r = 1.22 λ/D holding 80 % of the light, zodiacal sky 22.7 mag/arcsec², Nyquist pixels, dark 0.005 e⁻/s, read noise 3 e⁻ per 1 000 s sub-exposure.

**FIGURE 38.10** Signal-to-noise ratio versus magnitude for 0.3 m, 2.4 m and 6.5 m diffraction-limited space telescopes observing in the visible for 1 hour (solid) and 10 hours (dashed). The dot marks the worked example.

#### EXAMPLE: HOW MANY PHOTONS FROM A 25TH-MAGNITUDE STAR?

A V = 25 star is observed for one hour with a 2.4 m telescope (33 % central obstruction) through a V filter (550 nm, 88 nm wide) with 30 % overall throughput.

1. A V = 0 star delivers $3.63 \times 10^{-11}$ W m$^{-2}$ nm$^{-1}$. Each 550 nm photon carries $hc/\lambda = 3.61 \times 10^{-19}$ J, so the flux is $1.0 \times 10^{8}$ photons s$^{-1}$ m$^{-2}$ nm$^{-1}$.
2. V = 25 is $10^{-10}$ times fainter: 0.88 photons s$^{-1}$ m$^{-2}$ across the 88 nm band.
3. Collecting area $(\pi/4) \times 2.4^{2} \times (1 - 0.33^{2}) = 4.03$ m$^{2}$ gives 3.6 photons/s; with 30 % throughput, 1.07 electrons/s, or 3 850 per hour, of which 80 % (3 080) fall in the measuring aperture.
4. Noise: zodiacal sky (22.7 mag/arcsec$^{2}$) adds 335 electrons, dark current 337 and read noise 673 (19 pixels, four readouts of 3 electrons). Total noise = √(3 080 + 335 + 337 + 673) = 66.5 electrons.
5. SNR = 3 080 / 66.5 ≈ **46**. A star some 40 million times fainter than the faintest visible to the naked eye (V ≈ 6) yields about one detected photon per second on a Hubble-sized telescope.

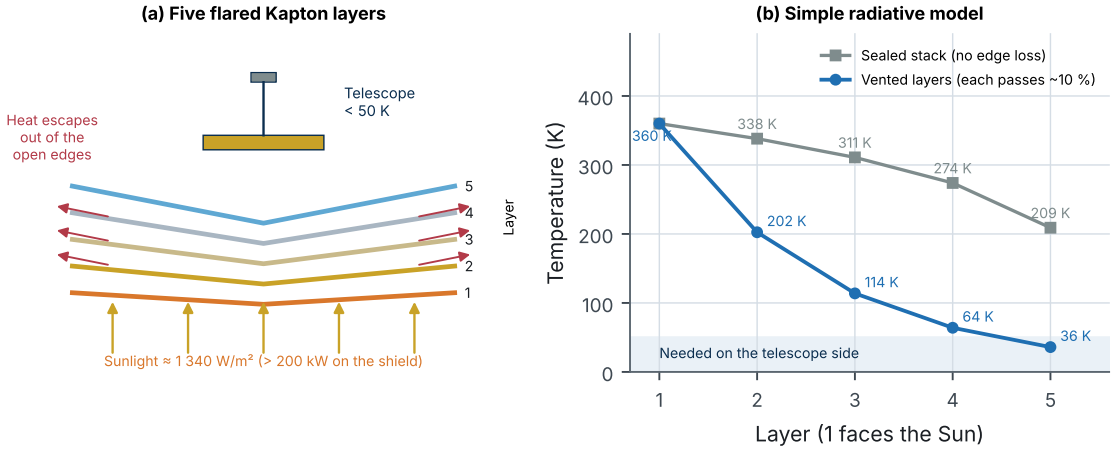
# 38.4 Keeping Cold and Clean

## 38.4.1 Passive Cooling and Sunshields

The simplest way to cool a telescope is to let it radiate to deep space while shading it from the Sun, Earth and Moon. Radiation scales with the fourth power of temperature, so a 40 K surface can reject only 0.15 W per square metre even with perfect emissivity. Every milliwatt leaking from the warm spacecraft must therefore be intercepted.

JWST's **sunshield** is the extreme example (Figure 38.11). Five membranes of Kapton E, the first 50 µm thick and the rest 25 µm, span about 21 m by 14 m. They are coated with about 100 nm of aluminium and, on the two Sun-facing layers, doped silicon. More than 200 kW of sunlight falls on the first layer, which reaches up to about 383 K in NASA's thermal models. Only about 23 mW reaches the telescope side, where the last layer is near 36–40 K. The layers never touch, and they are **flared** — closer at the centre and wider apart at the edges — so that heat radiated into each gap escapes sideways to space (Figure 38.12).

**FIGURE 38.11** JWST's full-size sunshield test unit stacked and unfurled in a cleanroom. The five membranes, each thinner than a human hair, are separated by gaps that let heat escape out of the edges. Image: NASA/GSFC.




Model: layer 1 at 360 K; sealed: four gaps with ε = 0.05 faces, layer 5 radiating to space; vented: flux falls ×10 per layer (reported for JWST). NASA quotes ≈383 K maximum and ≈36 K.

**FIGURE 38.12** (a) The five flared sunshield layers between the Sun and the telescope. (b) Layer temperatures in two simple radiative models: a sealed stack, in which heat can only pass through the layers, and vented layers, each of which passes on about a tenth of the heat it receives.

**EXAMPLE: WHY THE SUNSHIELD GAPS MUST BE OPEN**

At L2, sunlight delivers $S$ = 1 361 / 1.01² = 1 334 W/m², so the 200 kW on JWST's first layer implies about 150 m² facing the Sun.

1. A plate absorbing all this sunlight and radiating from one face reaches $(S/\sigma)^{1/4}$ = 392 K — close to the first layer's 383 K maximum.
2. **Sealed stack.** If the five layers formed a closed sandwich with low-emissivity faces (ε = 0.05), each of the four gaps would have a radiative resistance of $(2/\varepsilon - 1)$ = 39. With the first layer at 360 K and the last at 40 K, the heat flow is $q = \sigma(360^4 - 40^4)/(4\times39)$ = 6.1 W/m², about 900 W in all.
3. To radiate 6.1 W/m², the cold layer would have to warm to at least 102 K even with perfect emissivity — far too warm for a mid-infrared telescope.
4. **Vented layers.** The real leak, 23 mW over 150 m², is 0.15 mW/m² — about 40 000 times less. Suppose each vented layer passes on a tenth of the heat it receives. Then four steps reduce the radiated flux by $10^{-4}$ and the temperature by $(10^{-4})^{1/4}$ = 0.1: from 360 K to 36 K, as Figure 38.12 shows.

## 38.4.2 Cryocoolers and Cryogens

Passive cooling stops at about 30–40 K for a large telescope near Earth. Early infrared missions carried **cryostats** of liquid helium that slowly boiled away. Spitzer's lasted from 2003 to 2009, after which it continued a "warm" mission at about 30 K. Herschel's 2 300 litres of superfluid helium ran out in April 2013, ending its science. Spitzer pioneered the **warm launch**: only the instruments sat inside the cryostat, and the telescope cooled in space by radiating from its black outer shell.

Mechanical **cryocoolers** do not run out. JWST's MIRI detectors must stay below 7 K. Its cooler (Figure 38.13) uses a three-stage **pulse-tube** precooler, driven by a compressor in the warm spacecraft bus, to chill helium gas to about 14 K. The helium then flows about 10 m to a **Joule–Thomson** stage beside MIRI. There, expansion through an orifice less than 1 mm wide cools it to about 6 K. The only moving parts are horizontally opposed pistons, balanced so as not to shake the telescope. For 0.1 K and below, Planck used a dilution refrigerator and XRISM uses magnetic refrigeration.

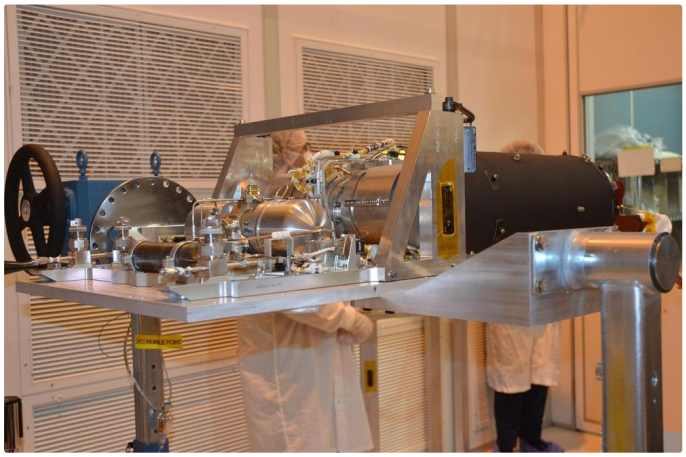


**FIGURE 38.13** The flight cryocooler for JWST's Mid-Infrared Instrument after testing in 2016, before it was shipped to Northrop Grumman for installation on the spacecraft. It cools MIRI's detectors to below 7 K, about 30 K colder than the rest of the telescope. Image: NASA/JPL (NASA/JPL-Caltech).

### 38.4.3 Stray Light and Contamination

**Stray light** is sunlight or earthshine scattered off baffles and struts, light from bright stars outside the field, or glow from warm structures. It is controlled with baffles, black coatings and field stops. These are designed by tracing millions of rays through a model of the telescope (Chapter 12). Mirror microroughness of a few nanometres scatters light into a halo that hides faint objects near bright ones.

**Contamination** is the other enemy. Molecules **outgassing** from adhesives, cables and insulation condense on cold optics; a film a few nanometres thick absorbs far-ultraviolet light. Space materials are screened for outgassing (typically below 1 % total mass loss and 0.1 % condensable material), baked in vacuum and assembled in cleanrooms under dry-nitrogen purge. Water is the hardest contaminant to eliminate. After **Euclid**'s launch in July 2023 its throughput fell steadily as water released by insulation condensed as ice on its cold mirrors. In March 2024 ESA warmed a mirror behind the main optics from about 126 K to 160 K for under two hours, and repeated the procedure in June. The visible camera regained about 15 % more light, and throughput has been stable since (Figure 38.14).

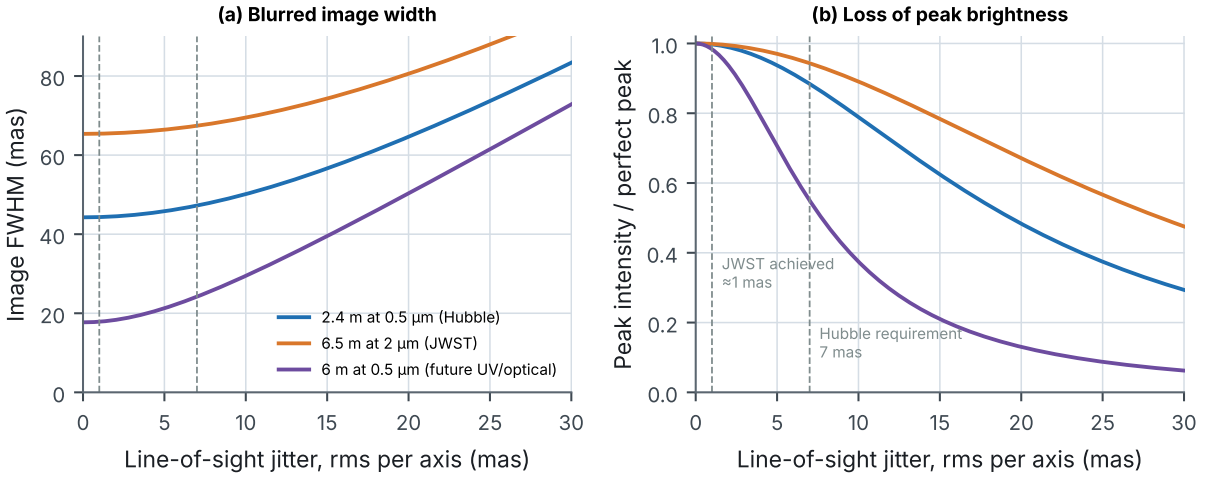
**FIGURE 38.14** ESA's Euclid being secured to the adapter of its Falcon 9 launcher in June 2023. The white cylinder houses the 1.2 m silicon-carbide telescope; the black solar panels also shade it from the Sun. Image: NASA/JPL (ESA).

## 38.5 Pointing and Stability

### 38.5.1 Sensing and Control

Coarse pointing uses sun sensors, star trackers and gyroscopes, as on any spacecraft (Chapter 6). The final step is a **fine guidance sensor (FGS)** that measures a guide star's position in the telescope's own focal plane, so that flexing between a separate star tracker and the instrument does not matter. Hubble's three interferometric FGS and four **reaction wheels** hold a target to about 7 mas rms over 24 hours. JWST's FGS reads guide stars 16 times a second and drives a **fine steering mirror**; in orbit its jitter is about 1 mas per axis, several times better than required.

Jitter blurs the image and lowers its peak brightness (Figure 38.15). A useful rule is that it should stay below about a tenth of $\lambda/D$: 4–5 mas for Hubble at 0.5 µm, and about 2 mas for a future 6 m ultraviolet–optical telescope. Coronagraphy is stricter still, because drift moves starlight off the mask; HWO studies discuss sub-milliarcsecond pointing together with picometre wavefront stability.




Gaussian jitter convolved with the diffraction core: $\mathrm{FWHM}^2 = (1.03\,\lambda/D)^2 + (2.355\,\sigma)^2$; peak ratio $\approx 1/[1 + (\pi^2/2)(\sigma D/\lambda)^2]$.

**FIGURE 38.15** Effect of line-of-sight jitter on (a) image width and (b) peak brightness for three telescopes. Hubble's 7 mas requirement and JWST's measured jitter of about 1 mas are marked.

Most jitter comes from inside the spacecraft. **Reaction wheels** spinning at thousands of revolutions per minute have tiny imbalances that shake the structure; cryocooler compressors, **thermal snaps** and flexing solar arrays add more. Hubble's first solar arrays vibrated every time the telescope crossed from shadow into sunlight, until stiffer arrays were fitted in 1993 and 2002. Engineers balance wheels and choose speeds that avoid structural resonances. They mount wheels or whole telescopes on soft **isolators** (JWST has a passive isolator between its bus and telescope). They also stiffen and damp the structure, and close fast optical loops with steering mirrors.

## 38.5.2 Hubble's Gyroscopes

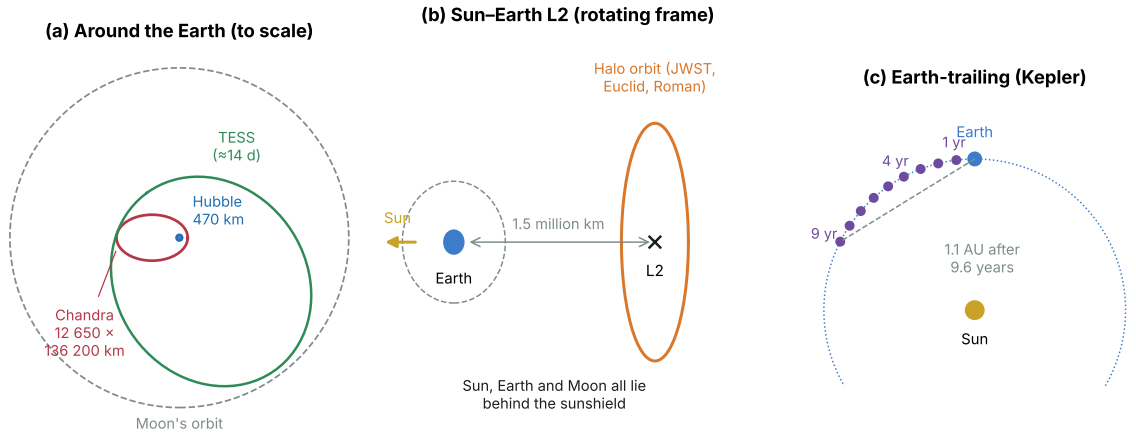
Hubble's gyroscopes sense rotation through a spinning wheel floating in fluid, connected by thin wires called **flex leads** that corrode; by 2023, 8 of the 22 gyros flown had failed. Hubble carries six and normally used three. Servicing Mission 4 replaced all six in 2009 — the last time this was possible. After a gyro repeatedly gave faulty readings in 2024, NASA switched Hubble to **one-gyro mode** in June 2024, keeping its other working gyro in reserve. Magnetometers, sun sensors and star trackers fill the gaps during slews. NASA expected about 12 % lower efficiency (74 instead of 85 orbits of observing a week) and slower target acquisition. It also expected less sky available at any time and no tracking of targets closer than Mars. Image quality is unchanged. NASA estimated a better than 70 % chance of at least one gyro working through 2035; by late 2025 the observing success rate was back above 90 %, with no safe-mode events since the switch.

## 38.5.3 Scanning Telescopes

Not every telescope stares. **Gaia** spun once every six hours, sweeping two fields of view across the sky. As it spun, its 106 CCDs moved charge in step with the star images — **time-delay integration**, as in pushbroom Earth imagers (Chapter 14). Over 10.5 years it measured positions of nearly two billion stars to tens of microarcseconds. Planck spun once a minute to map the microwave background, and WISE and SPHEREx scanned the whole sky. Scanning simplifies attitude control but demands precise knowledge of the spin axis and a readout synchronised to the motion.

## 38.6 Where to Put a Telescope

The orbit sets a telescope's thermal environment, observing efficiency, data rate and whether it can be serviced (Figure 38.16, Table 38.4).




(a) to scale; (b) schematic; (c) circular-orbit approximation of Kepler's 372.5-day heliocentric orbit (lag ≈ 7° per year), shown in a frame rotating with the Earth.

**FIGURE 38.16** Telescope orbits: (a) low Earth orbit (Hubble), Chandra's highly elliptical orbit and TESS's lunar-resonant orbit, to scale; (b) a halo orbit around Sun–Earth L2 (schematic); (c) the slow drift of an Earth-trailing telescope such as Kepler, in a frame rotating with the Earth.

**Low Earth orbit** is cheap to reach and was the only orbit the Space Shuttle could service, so Hubble flies at about 470 km. But Earth hides most targets for part of every 95-minute orbit. The telescope crosses the **South Atlantic Anomaly**, where trapped protons swamp the detectors, and it passes from sunlight to shadow about 15 times a day. Heaters hold Hubble's mirrors at about 21 °C. **Highly elliptical orbits** give long, uninterrupted views: Chandra's 64-hour orbit, now about 12 650 × 136 200 km, keeps it above the radiation belts for most of each orbit. TESS uses a stable 2:1 resonance with the Moon.

**Sun–Earth L2**, 1.5 million km beyond Earth on the anti-Sun side, has become the standard home of large infrared and survey telescopes: Herschel, Planck, Gaia, JWST, Euclid and, from December 2026, Roman. The Sun, Earth and Moon always lie in roughly the same direction, so one sunshield blocks all three, and the thermal environment is almost constant. L2 is unstable, so telescopes fly **halo orbits** around it with small station-keeping burns every few weeks; JWST's accurate launch left propellant for more than 20 years. **Earth-trailing** heliocentric orbits, used by Spitzer and Kepler, need no station-keeping, but the telescope drifts away — Kepler fell behind by about 7° a year — so its radio link weakens with time.

**TABLE 38.4** Trade-offs between orbits for space telescopes.

| Orbit | Examples | Thermal | Observing efficiency | Downlink | Servicing |
|---|---|---|---|---|---|
| LEO | Hubble, Swift, Fermi | Cycles every 95 min | ≈ 50 % | Relay satellites, ground passes | Possible |
| Highly elliptical | Chandra, XMM-Newton, TESS | Long stable arcs | High; day-long exposures | Ground stations | No |
| Sun–Earth L2 | JWST, Euclid, Gaia, Roman | Nearly constant | Very high | Deep-space antennas | Not designed for it |
| Earth-trailing | Spitzer, Kepler | Very stable, cold | Very high | Weakens with distance | No |

# 38.7 Building, Testing and Servicing

## 38.7.1 Test as You Fly

A telescope that cannot be repaired is tested on the ground at the temperatures and in the configuration it will fly. Complete observatories are shaken on vibration tables, exposed to the acoustic roar of launch and tested in **thermal-vacuum** chambers. JWST's telescope and instruments spent about 100 days in 2017 inside Chamber A at NASA's Johnson Space Center — a 36 m tall chamber built for Apollo. They were cooled by liquid-nitrogen and helium shrouds while optical tests confirmed that the segments could be aligned at operating temperature (Figure 38.17). Gravity sag, which cannot be removed on the ground, is predicted with finite-element models and checked by testing mirrors face-up and face-down. JWST's deployments were rehearsed many times; after launch on 25 December 2021, its roughly 50 major deployments took two weeks and all worked.

## 38.7.2 Hubble Servicing Missions

Hubble was designed to be serviced, with handholds, standard bolts and **orbital replacement units** — instruments, batteries, gyroscopes, computers — that astronauts could exchange during spacewalks from the Space Shuttle (Chapter 33). Five missions between 1993 and 2009 corrected its optics, replaced all of its original instruments and extended its life by decades (Table 38.5).

**TABLE 38.5** The five Hubble servicing missions.

| Mission | Date | Shuttle | Main work |
|---|---|---|---|
| SM1 | Dec 1993 | Endeavour (STS-61) | COSTAR corrective optics; WFPC2; new solar arrays and gyros |
| SM2 | Feb 1997 | Discovery (STS-82) | STIS spectrograph; NICMOS infrared camera; solid-state recorder |
| SM3A | Dec 1999 | Discovery (STS-103) | All six gyros after Hubble entered safe mode; new computer |
| SM3B | Mar 2002 | Columbia (STS-109) | Advanced Camera for Surveys; rigid solar arrays; NICMOS cryocooler |
| SM4 | May 2009 | Atlantis (STS-125) | WFC3 and COS; repair of ACS and STIS; six gyros; batteries |

On the first mission astronauts installed COSTAR — small deployable mirrors that refocused light for three instruments — and replaced the Wide Field/Planetary Camera with WFPC2, whose internal optics compensated for the mirror error (Figure 38.18). Without shuttle visits, Hubble's orbit is slowly decaying, from more than 600 km at its launch in 1990 to about 470 km in 2026. NASA has studied but not ordered a commercial reboost; without one, STScI's median forecast for re-entry is late 2033.

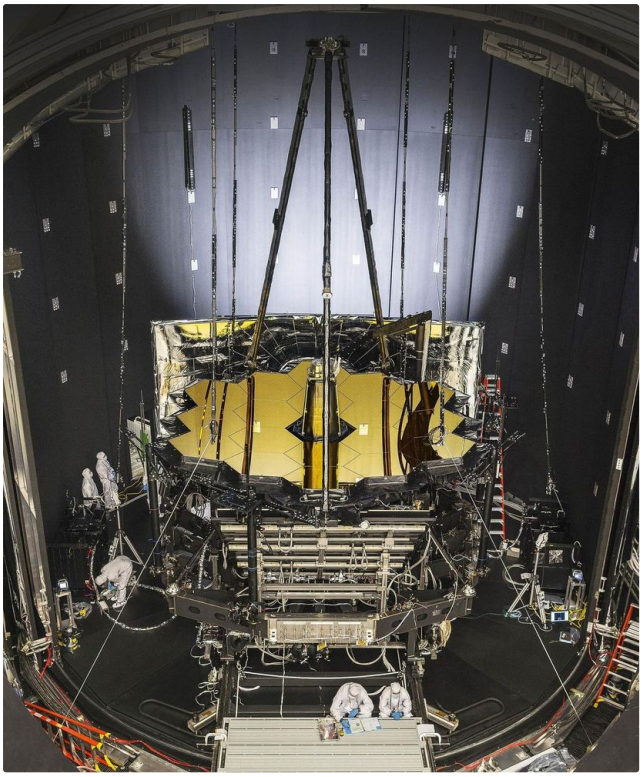


**FIGURE 38.17** JWST's telescope and science instruments inside Chamber A at NASA's Johnson Space Center in 2017, before cryogenic vacuum testing that simulated the cold of space. Image: NASA/GSFC.

**FIGURE 38.18** Astronaut Jeffrey Hoffman holds the original Wide Field/Planetary Camera in Endeavour's payload bay during the first Hubble servicing mission in December 1993, after replacing it with WFPC2. Image: NASA/MSFC.

### 38.7.3 Assembling the Next Generation in Space

JWST shows how far folding can go: a 6.5 m aperture inside a 5 m fairing. Apertures of 10–20 m may need to be assembled in orbit from separately launched segments, by robots or astronauts, and aligned with the same wavefront sensing used on JWST. HWO studies include standard interfaces and an orbit a future servicing vehicle could reach, so that instruments could be upgraded as Hubble's were (Chapter 40).

## 38.8 From Photons to Archives

### 38.8.1 Downlink and Storage

Data volume grows with detector size, and the downlink must keep pace (Table 38.6). Hubble sends about 18 GB of science data a week through NASA's Tracking and Data Relay Satellites. JWST stores data on a 68 GB solid-state recorder. It sends up to about 57 GB a day to the Deep Space Network's 34 m antennas on a 25.9 GHz Ka-band link. At up to 28 Mbit/s, this takes about

4.5 hours of transmission. Euclid returns about 100 GB of compressed data a day in a four-hour pass. Roman will send about 1.4 TB a day at up to 500 Mbit/s to stations in New Mexico, Australia and Japan — more than Hubble returns in a year.

**TABLE 38.6** Five major space telescopes and their data rates.

| Telescope | Launched | Aperture | Band | Orbit | Data per day |
|---|---|---|---|---|---|
| Hubble | 1990 | 2.4 m | 0.115–1.7 µm | LEO, 470 km | ≈ 2.6 GB |
| Chandra | 1999 | 1.2 m (outer shell) | 0.1–10 keV | 12 650 × 136 200 km | Small (photon lists) |
| JWST | 2021 | 6.5 m | 0.6–28.5 µm | L2 halo | ≈ 57 GB (28 Mbit/s) |
| Euclid | 2023 | 1.2 m | 0.55–2.0 µm | L2 halo | ≈ 100 GB (≈ 75 Mbit/s) |
| Roman | 2026 | 2.4 m | 0.48–2.3 µm | L2 halo | ≈ 1.4 TB (≤ 500 Mbit/s) |

## 38.8.2 Pipelines and Archives

Raw data are useless until calibrated. Each observatory runs a **pipeline**. For JWST, stage 1 fits the slope of each pixel's ramp and removes cosmic rays, and stage 2 applies flat fields, flux calibration and sky coordinates. Stage 3 combines dithered exposures into mosaics, spectra and catalogues. Data are reprocessed as calibrations improve. The **Mikulski Archive for Space Telescopes (MAST)** at the Space Telescope Science Institute holds data from about 20 NASA missions, including Hubble, JWST and TESS, and soon Roman. It passed 3 petabytes in 2017 and has grown rapidly since. Copies are kept in commercial clouds so that researchers can analyse data where they are stored. Most observations become public within a year, and archival research now produces a large share of published papers — the archive, not the telescope, is the observatory's lasting product.

### AI IN SPACE TELESCOPE OPERATIONS

Machine learning appears throughout a space telescope's life (Chapter 51). Neural networks trained on simulated defocused star images can estimate wavefront errors directly, complementing iterative phase retrieval. Scheduling has used artificial intelligence since Hubble's SPIKE planner, built at STScI in the late 1980s around constraint-based search; modern schedulers juggle thousands of observations and their constraints. In data reduction, deep-learning models flag cosmic rays and detector artefacts, classify galaxy shapes and find rare objects among billions. Euclid's first quick data release combined neural networks with citizen scientists to search for strong gravitational lenses. The challenge is trust: every automated decision must be validated against physics and be reproducible from the archive (Chapter 53).

### KEY TAKEAWAYS

- The atmosphere absorbs gamma rays, X-rays, ultraviolet and most infrared light, blurs images and glows; space offers the whole spectrum, diffraction-limited images and a dark, stable sky.
- Resolution is set by $1.22\,\lambda/D$ and sensitivity by collecting area and background. JWST resolves 77 mas at 2 µm, and a 2.4 m telescope reaches $V \approx 27$ at SNR = 10 in an hour.
- Large apertures use lightweighted glass, beryllium or silicon carbide, and segmented mirrors aligned in orbit to tens of nanometres; X-ray telescopes need nested grazing-incidence mirrors, gamma rays coded masks or pair-conversion trackers.
- Infrared telescopes must be cold: JWST's edge-vented, five-layer sunshield cuts 200 kW of sunlight to about 23 mW, and a cryocooler takes MIRI below 7 K.
- Fine guidance sensors, quiet wheels, isolation and steering mirrors hold pointing to milliarcseconds; Hubble has operated on one gyroscope since 2024.
- Sun–Earth L2 gives large telescopes a cold, stable home at the cost of servicing; Hubble's five servicing missions showed the value of designing for repair.
- Data rates have grown from gigabytes a week (Hubble) to terabytes a day (Roman); pipelines and archives matter as much as telescopes.

### REVIEW QUESTIONS

1. Why must X-ray and far-infrared astronomy be done above the atmosphere? What does a space telescope gain in the visible?
2. Calculate the diffraction limit of a 1.2 m telescope at 0.8 µm and at 2 µm in milliarcseconds, and the pixel size that Nyquist-samples each.
3. Using Figure 38.2, explain why MIRI is limited by the telescope's own emission at its longest wavelengths. What would cooling the telescope to 10 K gain?
4. Compare beryllium, ULE and SiC mirrors for a warm UV and a 40 K infrared telescope.
5. Describe the steps used to align JWST's 18 segments in orbit. Why could the alignment not be completed on the ground?
6. Why do X-ray telescopes use grazing incidence and nested mirror shells?
7. A 6 m telescope observes an Earth twin at 15 pc at 0.5 µm. How many multiples of $\lambda/D$ separate planet and star, and what contrast is needed?
8. Repeat the photon-budget example for a 6.5 m telescope. Why does the SNR improve by more than the ratio of diameters?
9. Use the sealed-stack calculation to explain why JWST's sunshield gaps are open.
10. Compare LEO, a highly elliptical orbit and Sun–Earth L2 for a new ultraviolet–optical telescope that should be serviceable.

CHAPTER

# 39 Great Observatories and Deep-Space Telescopes

Image: NASA/JPL (NASA)

*Most of the light in the Universe never reaches the ground. Space telescopes recover the gamma rays, X-rays, ultraviolet and infrared that the atmosphere absorbs, see sharper than turbulent air allows and measure brightness steadily enough to find Earth-sized planets. This chapter tells the story of NASA's four Great Observatories — Hubble, Compton, Chandra and Spitzer — and of the James Webb Space Telescope, which by 2026 had seen galaxies 280 million years after the Big Bang and analysed the air of distant worlds. It explains redshift, look-back time and the several meanings of cosmic distance, reviews the survey, exoplanet, high-energy and cosmic-microwave missions, and describes the fleet operating in October 2026 and the observatories planned for the 2030s and 2040s.*

**IN THIS CHAPTER**

39.1 Why Observe from Space
39.2 NASA's Great Observatories
39.3 The James Webb Space Telescope
39.4 Reading the Distant Universe: Redshift, Time and Distance
39.5 Surveys and Exoplanet Hunters
39.6 High-Energy and Time-Domain Astronomy
39.7 Microwaves, Radio and Gravitational Waves
39.8 The Fleet in 2026 and the Road Ahead

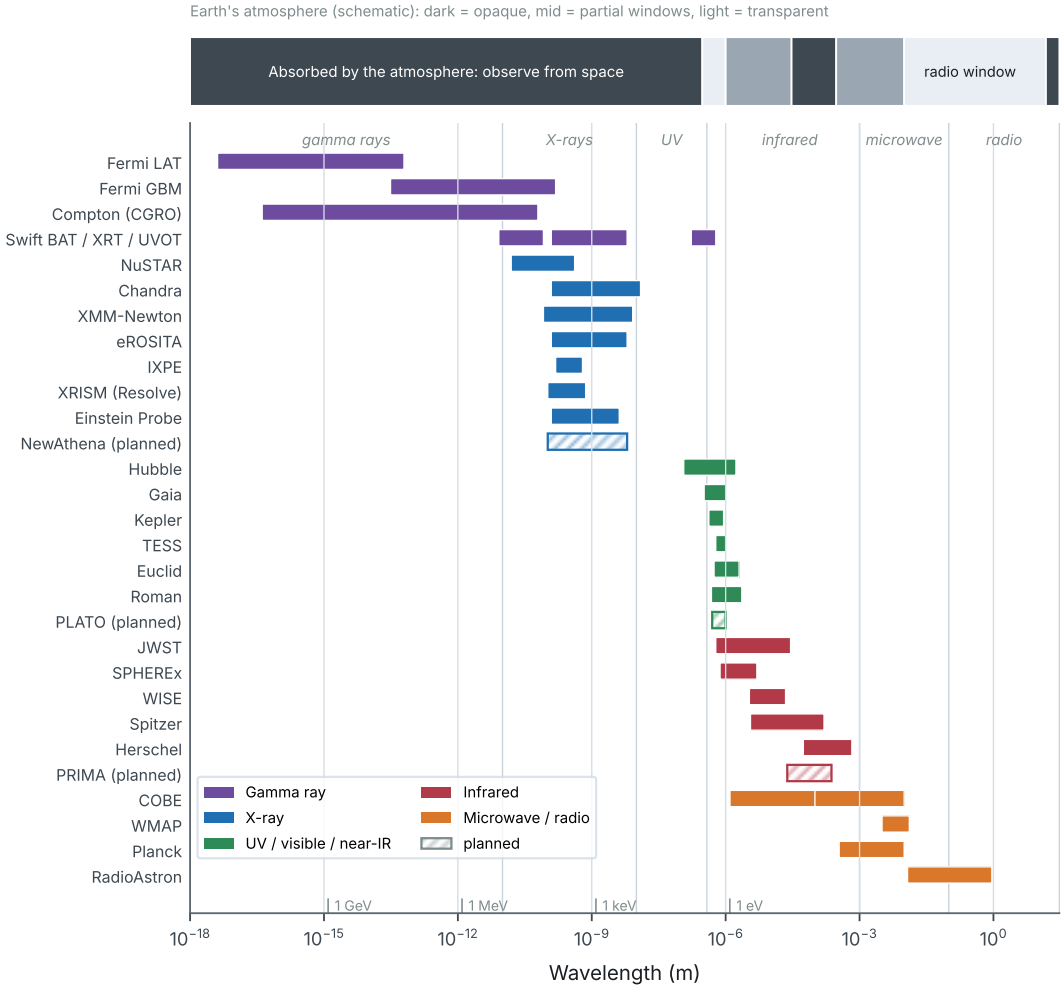
An astronomer on the ground looks out through a thick, warm, moving filter. Earth's atmosphere absorbs all gamma rays and X-rays, almost all ultraviolet light and most of the infrared. Water vapour glows brightly in the infrared. Turbulence smears every star into a shimmering blob about one arcsecond (1″, 1/3 600 of a degree) across. A telescope above the atmosphere escapes all three problems. It can observe at any wavelength, reach the sharpness set by diffraction, and measure a star's brightness to a few parts per million for weeks at a time.

This chapter covers the missions, capabilities and results of space observatories, from NASA's **Great Observatories** to the fleet of October 2026. How they are engineered — mirrors, detectors, cryogenics and pointing — is the subject of Chapter 38.

# 39.1 Why Observe from Space

## 39.1.1 The Atmospheric Filter

Figure 39.1 shows which parts of the electromagnetic spectrum reach the ground. Only two broad **windows** are open: the optical window from about 0.3 to 1 µm, with partial windows in the near-infrared, and the radio window from about 1 cm to 15 m. Gamma rays and X-rays are absorbed high in the atmosphere, ultraviolet by ozone and infrared by water vapour and carbon dioxide; the ionosphere reflects long radio waves. Balloons and sounding rockets (Chapters 8 and 9) sample some bands briefly; only satellites observe for years.




Ranges are the nominal bands of each mission's instruments (energy E = hc/λ). Hubble: current instruments; XRISM Resolve above 1.7 keV while its gate valve stays closed.

**FIGURE 39.1** Wavelength coverage of the main space observatories, past, present and planned. The strip at the top shows schematically where Earth's atmosphere is opaque, partly transparent or clear: gamma-ray, X-ray, ultraviolet and far-infrared astronomy is possible only from space.

## 39.1.2 Sharpness, Darkness and Stability

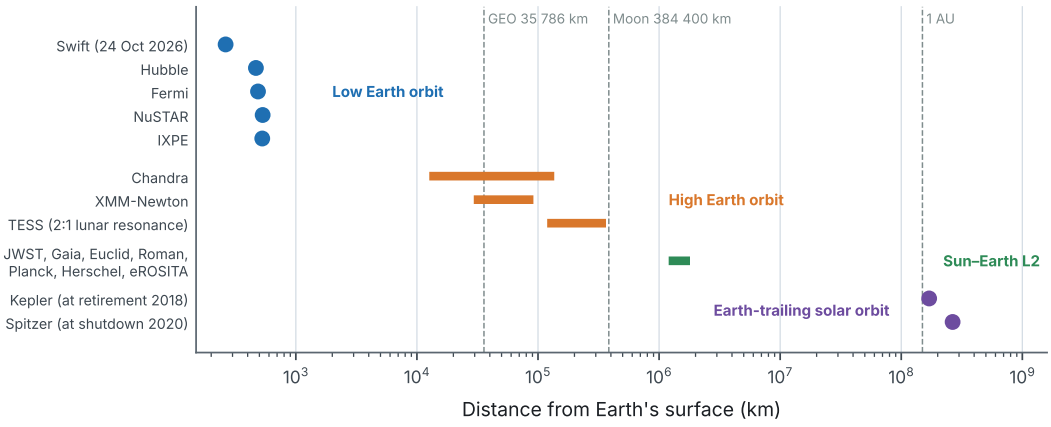
Even in the open windows, space wins three ways.

- **Resolution.** A perfect telescope of diameter $D$ observing at wavelength $\lambda$ cannot separate points closer than about $\theta = 1.22\,\lambda/D$ radians, the **diffraction limit**. For Hubble's 2.4 m mirror at 0.5 µm this is 0.05″, twenty times sharper than typical ground-based seeing. Adaptive optics now correct turbulence on large ground telescopes in the near-infrared, but only over small fields.
- **Background.** Airglow in the upper atmosphere makes the night sky glow, and in the infrared the atmosphere and the telescope itself shine like a warm oven. A telescope cooled to 40 K in deep space has an infrared background many orders of magnitude lower. This is why JWST, with a third of the collecting area of the largest ground telescopes, is far more sensitive than any of them beyond about 2 µm.
- **Stability.** Starlight twinkles (scintillation). Above the atmosphere a star's brightness can be measured continuously to tens of parts per million, the precision needed to see an Earth-sized planet cross a Sun-like star.

The price is high. A space telescope must fit inside a fairing about 5 m across, survive launch and work without adjustment by hand. It can rarely be repaired. Its cost per square metre of collecting area is hundreds of times that of a ground telescope. The two approaches are complementary: giant ground telescopes such as the 39 m Extremely Large Telescope under construction in Chile collect more light, while space telescopes provide wavelengths, sharpness and stability that the ground cannot.

### 39.1.3 Where Space Telescopes Fly

The orbit trades access, thermal environment and observing efficiency (Figure 39.2). **Low Earth orbit (LEO)**, used by Hubble, Fermi, NuSTAR and IXPE, is cheap to reach and shielded by Earth's magnetic field. But Earth hides most targets for part of every 95-minute orbit and warms the spacecraft. X-ray telescopes such as Chandra use **highly elliptical orbits** (HEO) reaching a third of the way to the Moon, above the radiation belts for most of the time. The cold, quiet observatories — WMAP, Planck, Herschel, Gaia, JWST, Euclid and Roman — use **halo or Lissajous orbits around the Sun–Earth L2 point**, 1.5 million km beyond Earth (Chapter 3). There the Sun, Earth and Moon always lie on one side, and one sunshield can block all three. Kepler and Spitzer used **Earth-trailing solar orbits**, drifting slowly away; when they were switched off they were 170 and 266 million km from Earth.




Bars: perigee–apogee (Earth orbits, CelesTrak elements Oct 2026) or approximate range of distance for L2 halo orbits. Dots: distance at end of mission (JPL Horizons).

**FIGURE 39.2** Where astronomical observatories fly, from the low orbits of Hubble and the X-ray and gamma-ray missions to the Sun–Earth L2 point and Earth-trailing solar orbits.

## 39.2 NASA's Great Observatories

In the 1980s NASA planned four large observatories, each covering a different part of the spectrum. They were the **Hubble Space Telescope** (visible and ultraviolet), the **Compton Gamma Ray Observatory**, the **Chandra X-ray Observatory** and the **Spitzer Space Telescope** (infrared). Three were launched by the Space Shuttle. The four never flew all at once — Compton came down three years before Spitzer was launched — but two remain in service in 2026 (Table 39.1).

**TABLE 39.1** NASA's four Great Observatories.

| Observatory | Band | Telescope | Launch | Orbit | End | Signature results |
|---|---|---|---|---|---|---|
| Hubble | 0.115–1.7 µm | 2.4 m mirror | 24 Apr 1990, STS-31 | LEO, 28.5° | Operating | Deep fields; expansion rate; dark energy |
| Compton (CGRO) | 20 keV–30 GeV | 4 instruments, 17 t | 5 Apr 1991, STS-37 | LEO, 28.5° | Deorbited 4 Jun 2000 | Gamma-ray bursts are cosmic; blazars |
| Chandra | 0.1–10 keV | 1.2 m nested mirrors, 0.5″ | 23 Jul 1999, STS-93 | ≈ 64-hour HEO | Operating | Dark matter in Bullet Cluster; black holes |
| Spitzer | 3.6–160 µm | 0.85 m, helium-cooled | 25 Aug 2003, Delta II | Earth-trailing | 30 Jan 2020 | TRAPPIST-1 planets; first exoplanet light |

### 39.2.1 Hubble Space Telescope

Hubble is a 2.4 m Ritchey–Chrétien reflector (Chapter 12) with a mass of about 11 t. Space Shuttle Discovery released it on 25 April 1990 at an altitude of about 615 km (Figure 39.3). Within weeks it was clear that the images were blurred: the primary mirror had been polished very precisely to the wrong shape. A 1.3 mm spacing error in the optical test device used during polishing (the null corrector) had made the mirror about 2.2 µm too flat at its edge. This caused **spherical aberration** — light from the edge and the centre of the mirror came to focus at different points.

**FIGURE 39.3** Hubble held by Discovery's robotic arm on 25 April 1990, during deployment on mission STS-31, with one solar array unrolled and the second still stowed. Image: NASA/JSC.

Because the error was known exactly, it could be corrected exactly, like prescribing spectacles. The first servicing mission (STS-61, December 1993) installed a new camera with the correction built in and **COSTAR**, small corrective mirrors on deployable arms that fed the other instruments. Astronauts visited Hubble five times between 1993 and 2009, replacing gyroscopes, batteries, solar arrays and every original instrument (Chapters 33 and 40). The last visit installed Wide Field Camera 3 and the Cosmic Origins Spectrograph, which still work alongside two older instruments.

Hubble measured the expansion rate of the Universe, the **Hubble constant**, to about 10 % (72 ± 8 km/s per megaparsec) in 2001. Its observations of distant supernovae helped show in 1998 that the expansion is accelerating — the evidence for **dark energy**. Its **Deep Fields** (1995, 2004), long exposures of apparently empty sky, revealed thousands of galaxies and traced how they grew. It found that almost every large galaxy has a central supermassive black hole and made the first detection of an exoplanet's atmosphere (2001). Its images, such as the "Pillars of Creation" (Figure 39.6), are among the best-known scientific pictures ever made.

**Hubble in 2026.** On 19 September 2026 Hubble completed its 200 000th orbit, after more than 1.7 million observations. Of the six gyroscopes installed in 2009, two still work, and since June 2024 Hubble has used only one, keeping the other as a spare. This **one-gyro mode** halves the sky it can reach at any moment and initially cut observations by about a tenth, but STScI reported no safe-mode events in the two years that followed. Drag is slowly lowering the orbit, now about 470 km: the Space Telescope Science Institute (STScI) puts the chance of re-entry before 2032 below 10 %, with a median date in late 2033. NASA declined a privately funded Crew Dragon reboost in June 2024, citing risks to the telescope. But it said in 2026 that it would consider a reboost if operating costs fell. STScI has meanwhile been told to plan for budgets about a fifth lower.

### 39.2.2 Compton Gamma Ray Observatory

At about 17 t, the **Compton Gamma Ray Observatory** (CGRO) was the heaviest astronomical observatory flown up to that time (Figure 39.4). Its four instruments covered photon energies from 20 keV to 30 GeV. The **Burst and Transient Source Experiment** (BATSE) recorded 2 704 **gamma-ray bursts** — brief, intense flashes of gamma rays. It showed that they come equally from all directions rather than concentrating along the Milky Way. Bursts must therefore occur at cosmological distances, which makes them the most powerful explosions known. The **EGRET** telescope showed that the high-energy sky is dominated by **blazars**, galaxies whose central black holes fire jets almost straight at us. After a gyroscope failed in 1999, NASA steered the observatory to a controlled re-entry over the Pacific on 4 June 2000.

### 39.2.3 Chandra X-ray Observatory

X-rays penetrate a mirror at normal incidence and can be focused only by **grazing incidence**, glancing off a polished surface at about a degree, as a stone skips across water. Chandra's mirror is four nested pairs of nearly cylindrical iridium-coated shells, the largest 1.2 m across, with a 10 m focal length (Chapter 38); its 0.5″ resolution is still the sharpest of any X-ray telescope. Columbia deployed Chandra on 23 July 1999, on the first Shuttle mission commanded by a woman, Eileen Collins (Figure 39.5). An Inertial Upper Stage then lifted it into an orbit that now ranges from 12 650 to 136 200 km above Earth, with a period of 63.5 hours. With its upper stage, Chandra was the heaviest payload ever launched by the Shuttle (22.8 t).

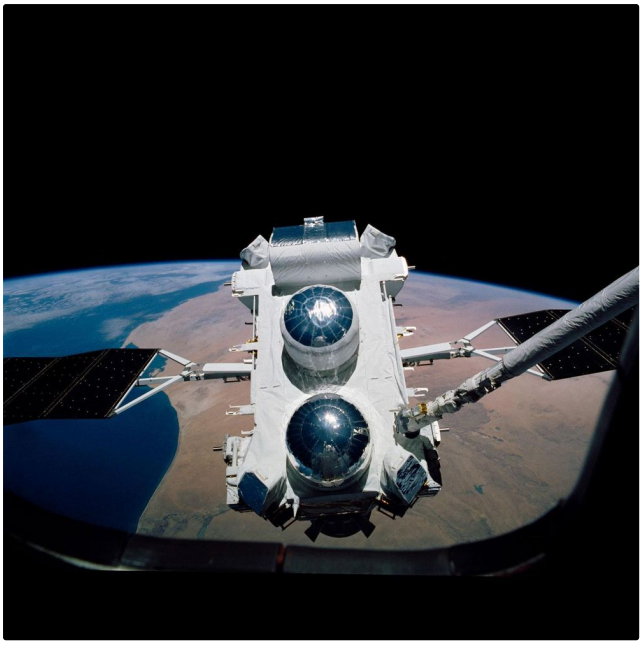


**FIGURE 39.4** The Compton Gamma Ray Observatory held by Atlantis's robotic arm during checkout on mission STS-37 in April 1991. The large cylinders are parts of the COMPTEL and EGRET telescopes. Image: NASA/JSC.

**FIGURE 39.5** Chandra, with its Inertial Upper Stage, in Columbia's payload bay shortly before being tilted up for release on 23 July 1999. Image: NASA/JSC.

Chandra sees the hot Universe: million-degree gas in galaxy clusters, the remnants of exploded stars and matter falling into black holes. In 2006 its map of hot gas in the colliding **Bullet Cluster** was compared with the mass mapped by gravitational lensing. It showed most of the mass passing through the collision while the gas was slowed — direct evidence for **dark matter**. With JWST in 2023 it detected X-rays from **UHZ1**, a growing black hole seen less than 500 million years after the Big Bang. Figure 39.7 shows how Chandra combines with other observatories. Each wavelength reveals a different part of the same object.

**FIGURE 39.6** The "Pillars of Creation" in the Eagle Nebula, re-imaged by Hubble in near-infrared light in 2014. Infrared light passes through much of the dust, revealing stars behind and inside the columns of cold gas. Image: NASA/GSFC.

**FIGURE 39.7** The Crab Nebula, the remnant of a supernova seen in 1054, combining radio (Very Large Array), infrared (Spitzer), visible (Hubble), ultraviolet (XMM-Newton) and X-ray (Chandra) data. The X-rays (purple) come from the most energetic particles near the central pulsar. Image: NASA/GSFC.

Chandra has also become a symbol of budget pressure. NASA's requests for 2025 and 2026 proposed winding it down to save about US$ 70 million a year. Congress refused, and the 2026 appropriation (signed in January 2026) gave astrophysics US$ 1.595 billion and Chandra at least US$ 63 million, enough for a "minimum-cost mission". The 2027 request would cut astrophysics by about 65 %, to US$ 552 million, ending Chandra and Fermi. A House committee rejected the cut, but no 2027 budget had been enacted in October 2026. A stopgap continuing resolution held NASA at its 2026 level until 11 December.

### 39.2.4 Spitzer Space Telescope

Spitzer pioneered a design that JWST later followed: it was **launched warm**. Earlier infrared telescopes such as IRAS (1983) and ISO (1995) sat inside large tanks of liquid helium. Spitzer kept only its instruments in a small cryostat. Its 0.85 m beryllium telescope cooled by radiating to space, helped by vented helium vapour, in an orbit far from the warm Earth (Figure 39.8). The helium lasted until May 2009. Spitzer then worked for ten more years in a "warm mission", using the two shortest-wavelength channels of its camera. These still functioned at the telescope's new temperature of roughly 30 K.

Spitzer made the first detections of light emitted by exoplanets (2005), found a huge, faint ring around Saturn, and in 2017 helped show that the cool dwarf star **TRAPPIST-1** has seven Earth-sized planets. As it drifted away its antenna had to point ever closer to the Sun to send data home, and NASA switched it off on 30 January 2020.

**FIGURE 39.8** Spitzer at Cape Canaveral being readied for launch in 2003. The telescope tube sits above the spacecraft bus, shaded by the solar panel on the right. Image: NASA/JPL (NASA).

## 39.3 The James Webb Space Telescope

JWST was conceived in the 1990s as Hubble's infrared successor, because the ultraviolet and visible light of the first galaxies is stretched by cosmic expansion into the infrared. Built by NASA with the European and Canadian space agencies, it is the largest and most complex scientific spacecraft ever launched (Table 39.2). It took about 25 years to develop and cost about US$ 10 billion, including the first five years of operations.

### 39.3.1 Launch and Commissioning

JWST's mirror and sunshield were launched folded (Figure 39.9). Its deployment involved about 344 **single-point failures** — items whose failure would have ended the mission. In the two weeks after launch the sunshield unfolded and tensioned, and the mirror wings locked into place. JWST reached L2 on 24 January 2022. Engineers spent three months aligning the 18 segments to within tens of nanometres so that they act as one mirror. The accurate Ariane 5 launch left propellant for about 20 years instead of the 10 planned.

The first images were released on 12 July 2022. The first deep field (Figure 39.10), a 12.5-hour exposure of the galaxy cluster SMACS 0723, reached infrared depths beyond those of Hubble's deepest fields, which had taken weeks. Image quality and background beat requirements. One surprise was **micrometeoroid** damage: an impact in May 2022 slightly distorted one segment, and JWST now limits the time it points towards its direction of orbital motion, where impacts are most energetic.

**FIGURE 39.9** The last of JWST's 18 primary-mirror segments being lowered into place by a robotic arm at NASA's Goddard Space Flight Center in February 2016, completing the mirror. Image: NASA/GSFC.

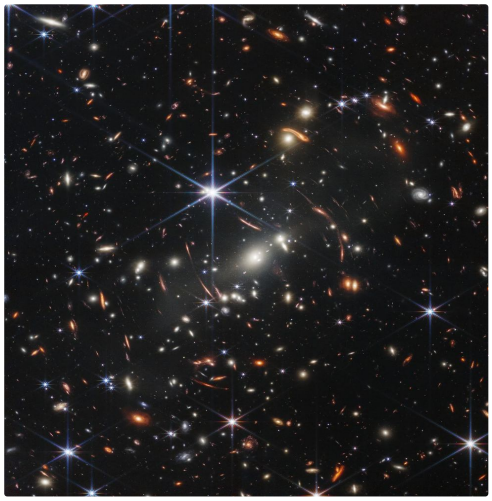


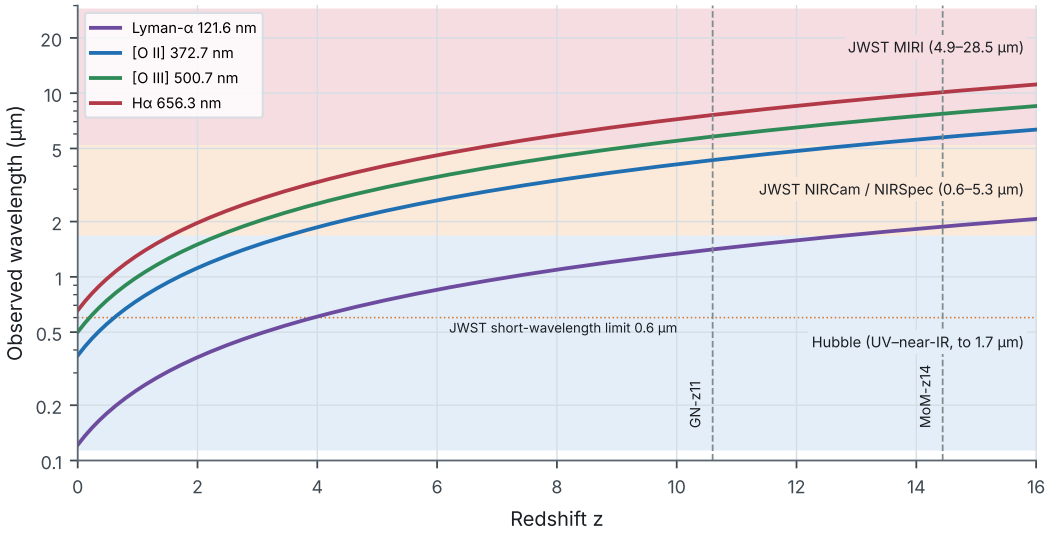
**FIGURE 39.10** JWST's first deep field (2022): the cluster SMACS 0723 acts as a gravitational lens, stretching background galaxies into arcs. Image: NASA/STScI (NASA ESA CSA STScI).

**TABLE 39.2** The James Webb Space Telescope at a glance.

| Parameter | Value |
|---|---|
| Primary mirror | 6.5 m; 18 hexagonal segments of gold-coated beryllium; 25.4 m² collecting area |
| Wavelength range | 0.6–28.5 µm (visible red to mid-infrared) |
| Instruments | NIRCam (camera), NIRSpec (spectrograph), MIRI (mid-infrared), NIRISS and Fine Guidance Sensor |
| Sunshield | Five Kapton layers, 21.2 × 14.2 m; Sun side above 350 K, cold side ≈ 40 K |
| Operating temperature | Telescope ≈ 40 K; MIRI 7 K (cryocooler) |
| Mass and orbit | About 6.2 t; halo orbit around Sun–Earth L2 |
| Launch | 25 December 2021, Ariane 5, Kourou |
| Lifetime | Propellant for about 20 years |

### 39.3.2 The First Galaxies

Cosmic expansion stretches light from a distant galaxy by a factor $(1+z)$, where $z$ is the **redshift**. Neutral hydrogen absorbs light bluer than the Lyman-α line at 121.6 nm, so a young galaxy vanishes abruptly below $121.6\,(1+z)$ nm. Figure 39.11 shows that for $z > 13$ this **Lyman break** lies beyond 1.7 µm, the red limit of Hubble's cameras. Hubble cannot see such galaxies at all. Hubble's most distant galaxy, GN-z11 (2016), lies at $z = 10.6$.




λ(observed) = λ(emitted) × (1 + z). Shading: wavelength range of the instruments; Earth's atmosphere blocks most light beyond 2.5 µm and below 0.3 µm.

**FIGURE 39.11** Observed wavelength of four bright spectral lines as a function of redshift, compared with the ranges of Hubble and JWST's instruments. Above a redshift of about 13, even ultraviolet Lyman-α light arrives at wavelengths longer than Hubble can detect.

JWST has repeatedly broken the distance record, confirming each redshift with a spectrum. In 2024 the JADES survey confirmed **JADES-GS-z14-0** at $z \approx 14.3$ (refined to 14.18 by the ALMA radio telescope). In 2025 the MoM survey measured **MoM-z14** at $z = 14.44$ (published in January 2026), seen just 280 million years after the Big Bang. As of October 2026 no galaxy had been spectroscopically confirmed at a greater distance. The surprise is less the distance than the brightness. JWST finds more luminous galaxies above $z = 10$ than most models predicted. This suggests that early star formation was more efficient than assumed — an active research problem rather than a contradiction of standard cosmology.

JWST has also found black holes earlier and larger than expected. Besides UHZ1, they include a feeding black hole in GN-z11 and many compact, very red **"little red dots"**. These appear to hide black holes of millions of solar masses in dense gas. One of them (CAPERS-LRD-z9, reported in 2025) is seen about 500 million years after the Big Bang. Whether such black holes began as collapsing giant gas clouds or grew unusually fast from the remnants of the first stars is one of JWST's central questions.

### 39.3.3 Exoplanet Atmospheres

When a planet passes in front of its star, a little starlight filters through its atmosphere. Molecules absorb at characteristic wavelengths, so the planet appears slightly larger at those wavelengths: this **transmission spectrum** reveals the atmosphere's composition. In its first months JWST

made the first clear detection of **carbon dioxide** in an exoplanet atmosphere, that of the hot Saturn WASP-39 b, and found sulphur dioxide made by starlight-driven chemistry. Heat measurements of the rocky planets TRAPPIST-1 b and c suggest little or no thick atmosphere.

**K2-18 b**, about two and a half times Earth's radius and in the habitable zone of a red dwarf, shows the limits of the method. JWST detected methane and carbon dioxide there in 2023, and in April 2025 a team reported evidence at about three standard deviations for dimethyl sulphide, a gas made on Earth mainly by marine life. Independent reanalyses found that different but reasonable choices of data reduction and atmospheric model remove the signal, and in October 2026 the claim remains unconfirmed. Features from small planets amount to tens of parts per million, close to instrumental limits, so a biosignature claim demands several independent lines of evidence.

### 39.3.4 JWST in 2026

JWST attracts more demand than any observatory before it: for its fifth cycle, which began on 1 July 2026, astronomers submitted 2 930 proposals requesting about twelve times the time available. It has confirmed that Hubble's measurements of the variable stars calibrating the distance ladder were not biased by crowding. Yet calibrations still give Hubble constants of about 70–73 km/s/Mpc, against 67–68 km/s/Mpc from the cosmic microwave background — the unexplained **Hubble tension**. Its budget is under pressure too: the 2027 request proposed reduced operating funds for JWST and Hubble.

## 39.4 Reading the Distant Universe: Redshift, Time and Distance

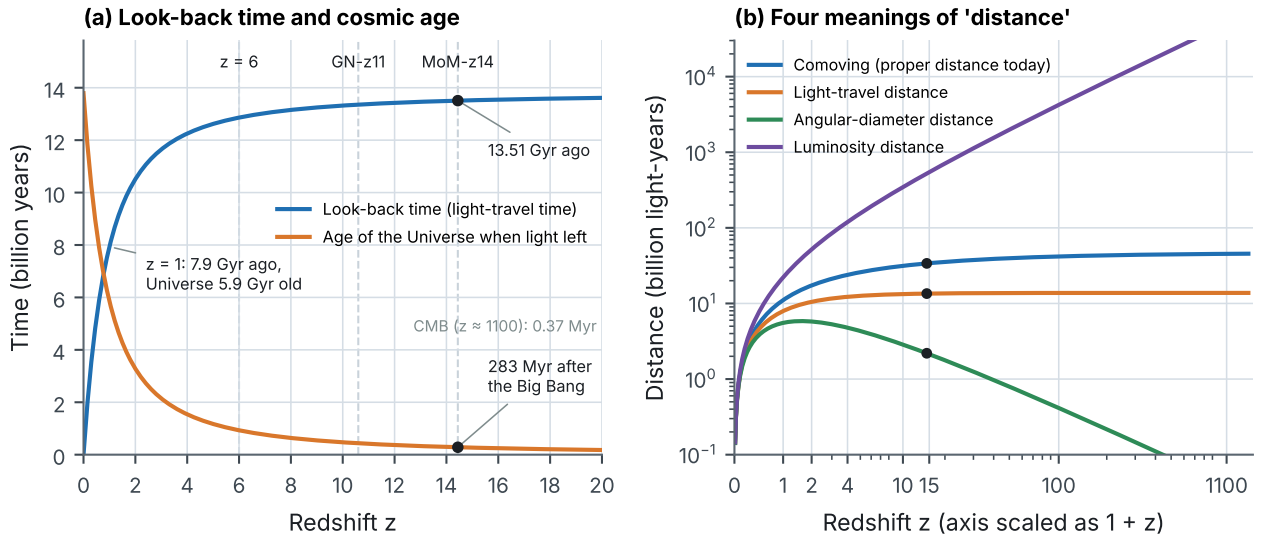
Looking far away means looking back in time, but in an expanding Universe "how far" has several meanings. The standard model is **flat ΛCDM**: a Universe dominated today by dark energy, Λ, and cold dark matter. With Hubble constant $H_0$ = 67.7 km/s/Mpc and matter density $\Omega_m$ = 0.31, it gives a present age of 13.79 billion years. The age of the Universe when light was emitted at redshift $z$ is

$$t(z) = \int_z^\infty \frac{dz'}{(1+z')\,H(z')}, \qquad H(z) = H_0\sqrt{\Omega_m(1+z)^3 + \Omega_r(1+z)^4 + \Omega_\Lambda},$$

where $\Omega_r \approx 9 \times 10^{-5}$ is the small contribution of radiation and $\Omega_\Lambda = 1 - \Omega_m - \Omega_r$. The **look-back time** is the present age minus $t(z)$ (Figure 39.12, Table 39.3). Four distances are in common use:

- **Light-travel distance**: look-back time × speed of light — the figure press releases quote, though no ruler could measure it.
- **Comoving distance**: the distance to the object *today*, if the expansion could be frozen and a ruler laid out. It exceeds the light-travel distance because space expanded while the light was in transit; the edge of the observable Universe is about 46 billion light-years away.
- **Angular-diameter distance**: relates an object's true size to the angle it covers. It peaks near $z \approx 1.6$ and then *decreases*: very distant galaxies look larger than galaxies of the same size at intermediate distance, because we see them as they were when they were closer to us.

- **Luminosity distance**: relates an object's power to the flux received, as used with "standard candles" such as type Ia supernovae. It exceeds the comoving distance by a factor $(1+z)$, because photons lose energy and arrive less often.




Flat ΛCDM: $H_0$ = 67.7 km/s/Mpc, Ωm = 0.31, Ωr = 9.1 × $10^{-5}$, ΩΛ = 1 − Ωm − Ωr. Present age 13.79 Gyr. Dots: MoM-z14.

**FIGURE 39.12** (a) Look-back time and age of the Universe when the light was emitted, as functions of redshift. (b) Four measures of distance. Light from MoM-z14 travelled for 13.5 billion years, but the galaxy is now about 34 billion light-years away.

### EXAMPLE: LOOK-BACK TIME TO REDSHIFT 14

A galaxy's spectrum shows the Lyman break at 1.824 µm. How long ago did its light leave, and how old was the Universe then?

1. **Redshift.** $1+z = 1.824/0.1216 = 15.0$, so $z = 14.0$. The break lies beyond Hubble's 1.7 µm limit; only JWST can measure it.
2. **Quick estimate.** At this redshift matter dominates, $H(z) \approx H_0\sqrt{\Omega_m}(1+z)^{3/2}$, and the integral gives $t \approx \frac{2}{3}\,(1/H_0)/[\sqrt{\Omega_m}(1+z)^{3/2}]$. With $1/H_0$ = 14.44 Gyr: $t \approx \frac{2}{3} \times 14.44/(0.557 \times 58.1)$ = **0.298 Gyr**.
3. **Exact integral.** Numerical integration including dark energy and radiation gives $t$ = **296 million years**, within 1 % of the estimate — dark energy was negligible then.
4. **Look-back time.** 13.79 − 0.30 = **13.50 billion years**: the light has travelled for 97.9 % of cosmic history.

**TABLE 39.3** Time and distance in flat ΛCDM ($H_0$ = 67.7 km/s/Mpc, $\Omega_m$ = 0.31); 1 Gly = $10^9$ light-years.

| Redshift $z$ | Example | Look-back time (Gyr) | Age then (Myr) | Comoving distance (Gly) | Angular-diameter distance (Gly) |
|---|---|---|---|---|---|
| 1 | Star-formation peak | 7.94 | 5 860 | 11.1 | 5.54 |
| 2 | Quasar era | 10.51 | 3 280 | 17.3 | 5.77 |
| 6 | Reionisation ends | 12.86 | 930 | 27.5 | 3.93 |
| 10.6 | GN-z11 | 13.36 | 436 | 31.9 | 2.75 |
| 14.44 | MoM-z14 | 13.51 | 283 | 33.9 | 2.19 |
| 1 100 | CMB | 13.79 | 0.37 | 45.3 | 0.041 |

**EXAMPLE: HOW FAR, AND HOW BIG?**

For the same galaxy at $z = 14.0$, compare the distances and decide whether JWST can resolve it.

1. **Light-travel distance**: 13.5 billion light-years (Gly), the number usually quoted.
2. **Comoving distance today**: integrating $c/H(z)$ from 0 to 14 gives 10 330 Mpc = **33.7 Gly**, 2.5 times the light-travel distance.
3. **Distance when the light left**: comoving distance ÷ $(1+z)$ = 33.7/15.0 = **2.25 Gly**, only a sixth of the light-travel distance. In a flat universe this equals the angular-diameter distance.
4. **Angular size**: a galaxy 500 parsecs (1 630 light-years) across subtends $500\ \mathrm{pc}/688.6\ \mathrm{Mpc} = 7.3 \times 10^{-7}$ rad = **0.15″**.
5. **Resolution**: JWST's diffraction limit at 2 µm is $1.22 \times 2 \times 10^{-6}/6.5$ = $3.75 \times 10^{-7}$ rad = **0.077″**, half the galaxy's size, so JWST resolves it into about two resolution elements across. Hubble at 1.6 µm (0.17″) could not, even if the light were in its band.
6. **Luminosity distance**: $(1+z)$ × comoving distance = **505 Gly**, so the flux is $15^2$ = 225 times lower than an inverse-square estimate using the comoving distance.

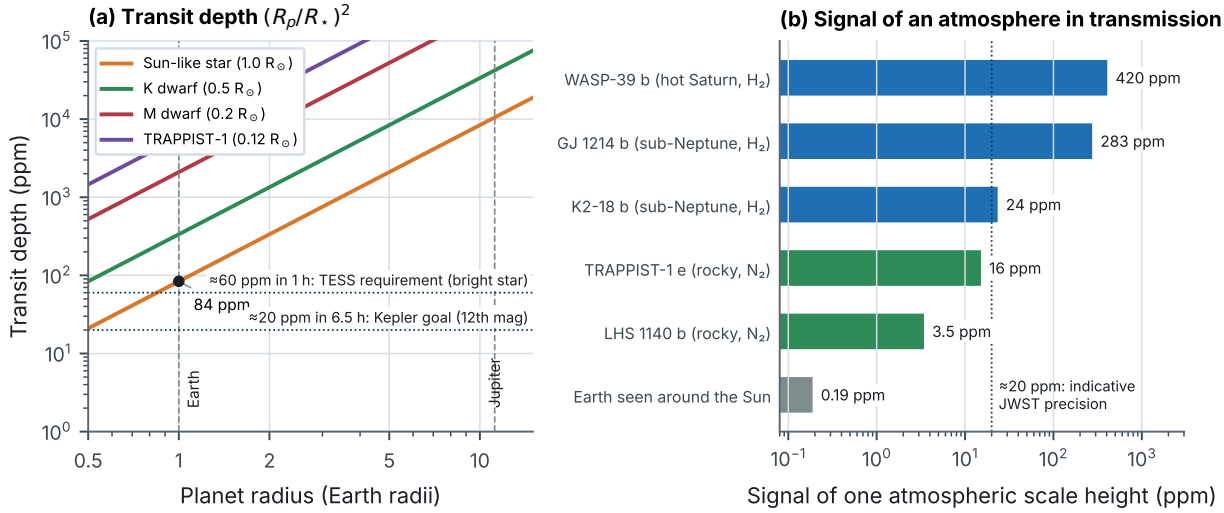
## 39.5 Surveys and Exoplanet Hunters

**Survey missions** observe large parts of the sky systematically to build catalogues, measure the shape of the Universe or find rare events. Their power depends less on aperture than on **étendue**, the product of collecting area and field of view, which sets how fast they map the sky (Figure 39.18 b).

### 39.5.1 Finding Planets: Kepler and TESS

When a planet of radius $R_p$ crosses a star of radius $R_\star$, the star dims by the **transit depth** $\delta = (R_p/R_\star)^2$ (Figure 39.13 a). This is 84 parts per million (ppm) for Earth crossing the Sun, and about 1 % for Jupiter. Only a fraction $R_\star/a$ of systems, for a planet at distance $a$, are aligned so that transits occur — 0.47 % for an Earth–Sun analogue. So a transit survey must monitor tens of thousands of stars or more.

NASA's **Kepler** (2009–18) used a 0.95 m Schmidt telescope and a 95-megapixel camera covering 115 square degrees (Figure 39.15) to measure about 150 000 stars in Cygnus and Lyra every 30 minutes for four years. When a second reaction wheel failed in 2013, engineers balanced the spacecraft against sunlight pressure, and the **K2** mission observed fields along the ecliptic until the propellant ran out in October 2018. Kepler showed that planets are the rule rather than the exception. It showed that the commonest planets in the Galaxy are between Earth and Neptune in size — a type absent from the Solar System. It also showed that a substantial fraction of Sun-like stars (estimates range from about one in ten to one in two) have an Earth-sized planet in the habitable zone.




Scale height H = kT/(μ m_H g), T = equilibrium temperature; signal = 2H·Rp/R*²; strong bands span 2–5 H. Data: NASA Exoplanet Archive; μ = 2.3 ($H_2$) or 28–29 ($N_2$-rich) assumed.

**FIGURE 39.13** (a) Transit depth versus planet radius for four types of star, with the photometric precision of Kepler and TESS. (b) Signal of one atmospheric scale height in transmission for selected planets: hydrogen-rich planets give tens to hundreds of ppm, rocky planets with nitrogen-rich air from a fraction of a ppm (Earth around the Sun) to about 15 ppm (TRAPPIST-1 e around a tiny star).

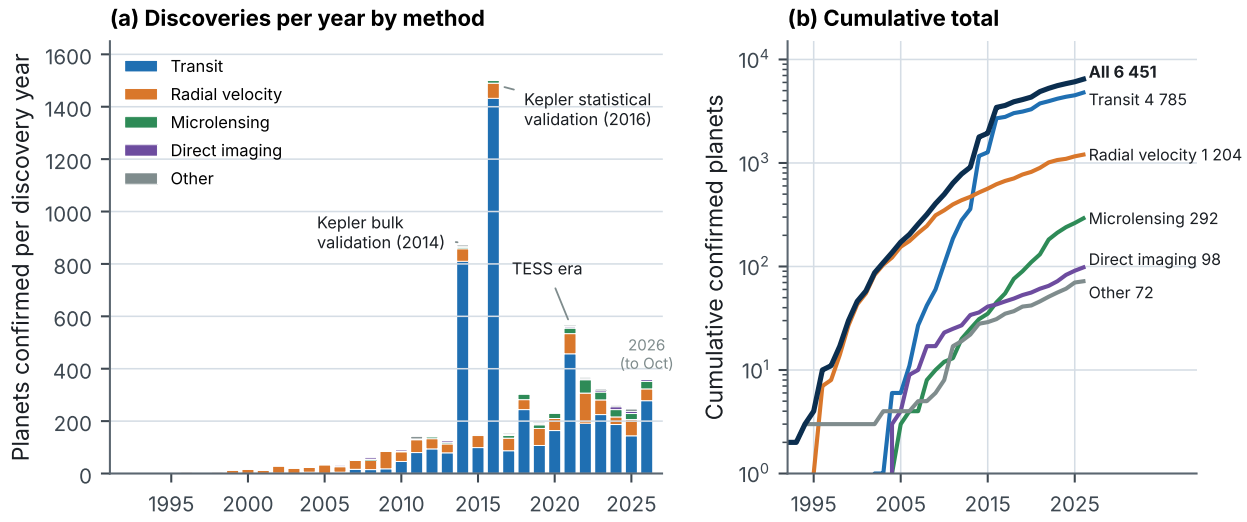
The **Transiting Exoplanet Survey Satellite (TESS)**, launched in 2018, has four 10.5 cm cameras covering a 24° × 96° strip of sky, which it observes for 27 days before moving on. It has covered more than 98 % of the sky, concentrating on bright, nearby stars whose planets can be weighed from the ground and studied by JWST. By October 2026 it had flagged 8 151 "TESS Objects of Interest" — possible planets, some later shown to be false alarms. Of its discoveries, 1 012 were confirmed planets. It continues in its third extended mission. ESA's **CHEOPS** (2019), a 32 cm telescope, measures the radii of known planets precisely.

### EXAMPLE: CAN TESS FIND AN EARTH TWIN?

An Earth-sized planet orbits a Sun-like star (1.0 $R_\odot$, 1.0 $M_\odot$) once a year. For a bright star TESS reaches about 60 ppm of noise in one hour.

1. **Depth**: $\delta = (6\,371/695\,700)^2 = 8.4 \times 10^{-5}$ = **84 ppm**.
2. **Duration** of a central transit: $T \approx (P/\pi)\,(R_\star/a)$ = (8 766 h/π) × (695 700/149.6 million) = **13.0 hours**.
3. **Noise over one transit**: 60 ppm/√13 = 16.6 ppm, giving a signal-to-noise ratio of 84/16.6 = **5.0**.
4. **Detection** conventionally needs a ratio of about 7.1, i.e. $(7.1/5.0)^2 \approx$ **2 transits** — a continuous baseline of more than a year. TESS watches most stars for only 27 days, so it essentially cannot find Earth twins, and only 0.47 % of such systems transit at all. That is the job of PLATO, which will stare at the same fields for two years or more.
5. **Around a small red dwarf** (0.2 $R_\odot$), the same planet in the habitable zone (about 0.07 AU) orbits in 14 days with a depth of **2 100 ppm**. Even with three times the noise from the fainter star, one 1.5-hour transit gives a ratio of about 14, and TESS sees two transits per sector. This is why most known small temperate planets — and those JWST can study — orbit red dwarfs.

By October 2026 the NASA Exoplanet Archive listed **6 451 confirmed planets** (Figure 39.14), with more than 8 000 candidates awaiting confirmation. Three-quarters (4 785) were found by transits, and over half (3 337) by Kepler and K2. Table 39.4 summarises the main space missions.




Data: NASA Exoplanet Archive, Planetary Systems Composite table, queried 19 Oct 2026 (6 451 planets). 'Other' = timing, astrometry, brightness modulation, disk kinematics.

**FIGURE 39.14** Exoplanet discoveries by year and method (NASA Exoplanet Archive, October 2026). The peaks in 2014 and 2016 are statistical validations of large batches of Kepler candidates.

**TABLE 39.4** Space missions for finding and characterising exoplanets.

| Mission | Years | Method | Telescope / field | Role and status (Oct 2026) |
|---|---|---|---|---|
| Kepler / K2 (NASA) | 2009–18 | Transit | 0.95 m, 115 deg² | Planet statistics; 3 337 planets |
| TESS (NASA) | 2018– | Transit | 4 × 10.5 cm, 2 304 deg² | Bright nearby stars; 1 012 planets |
| JWST (NASA/ESA/CSA) | 2021– | Spectroscopy, imaging | 6.5 m | Atmospheres of known planets |
| Roman (NASA) | 2026– | Microlensing, coronagraph | 2.4 m, 0.28 deg² | Cold planets; >1 000 expected |
| PLATO (ESA) | 2027– | Transit | 26 × 12 cm | Earth analogues around Sun-like stars |
| Ariel (ESA) | 2029– | Spectroscopy | 1.1 × 0.7 m | Atmospheres of about 1 000 planets |
| HWO (NASA) | 2040s | Direct imaging | ≈ 6 m | Search for life on ~25 Earth-like planets |

## 39.5.2 Gaia: Mapping the Galaxy

ESA's **Gaia** (2013–25) measured the positions, distances and motions of nearly two billion stars from L2. Two telescopes 106.5° apart scanned the sky as the spacecraft rotated, imaging onto nearly a billion pixels. As a result, angles between stars were measured to ten to a few tens of micro-arcseconds — roughly the size of a coin on the Moon seen from Earth. Its **parallaxes**, the annual shifts of stars caused by Earth's orbit, give distances to over a billion stars. Its data also revealed the debris of a dwarf galaxy that merged with the Milky Way about 10 billion years ago. Science operations ended on 15 January 2025; the fourth data release, from 5.5 years of data and more than 500 TB, is scheduled for December 2026.

### 39.5.3 Mapping the Dark Universe: Euclid, SPHEREx, Roman and Xuntian

**Euclid** (ESA, July 2023) is a 1.2 m telescope with a 0.53-square-degree field — over a hundred times that of Hubble's main camera. It is mapping a third of the sky in six years to measure the shapes of more than a billion galaxies. Their tiny, coherent distortions by intervening matter (**weak gravitational lensing**) and the clustering of galaxies reveal how dark matter is distributed and how dark energy has driven the expansion. Its first quick release (March 2025) covered 63 square degrees and 26 million galaxies. The first major data release (DR1) begins in November 2026 with images and catalogues of about 1 900 square degrees. The weak-lensing and clustering products needed for cosmology follow in mid-2027. NASA's **SPHEREx** (March 2025) splits light into 102 colours from 0.75 to 5 μm with a 20 cm telescope. It completed the first of four planned all-sky spectral maps in December 2025. It seeks imprints of **cosmic inflation** — the extremely rapid expansion thought to have occurred in the first instant after the Big Bang — and surveys interstellar ices.

The **Nancy Grace Roman Space Telescope** (NASA, about US$ 4 billion) pairs a 2.4 m mirror, donated by the US National Reconnaissance Office, with a 300-megapixel camera. The camera's field is about 200 times that of Hubble's infrared camera. Launched on a Falcon Heavy on 30 August 2026, ahead of schedule (Figure 39.16), it was being commissioned on its way to L2 in October 2026, with first science images expected in early 2027. Over five years it will map dark energy. It will find more than a thousand planets towards the Galactic centre by **gravitational microlensing** (the brief brightening of a background star as a planet passes in front of it). It will also test a coronagraph required to detect planets $10^{7}$ times fainter than their star (ground tests reached better than $5 \times 10^{-8}$). This is a step towards the $10^{10}$ needed for Earth twins.

**FIGURE 39.15** The Kepler spacecraft in a clean room at Ball Aerospace in Boulder, Colorado, in 2008. The white dust cover on the sunshade was jettisoned after launch. Image: NASA/JPL (NASA/JPL-Caltech/Ball Aerospace).

**FIGURE 39.16** The Nancy Grace Roman Space Telescope being encapsulated in its Falcon Heavy payload fairing at Kennedy Space Center on 21 August 2026, nine days before launch. Image: NASA/KSC (NASA/Sydney Rohde (Rocz)).

China's 2 m **Xuntian** survey telescope (CSST, 1.1-square-degree field) is to be launched in 2027 into the orbit of the Tiangong station, where it can dock for servicing. ESA's **PLATO**, with 26 small cameras, is due for launch on Ariane 6 in early 2027 (ESA's mission page gives April). Its aim is to find Earth-sized planets in the habitable zones of Sun-like stars. **Ariel** (2029) will measure the atmospheres of about a thousand exoplanets.

## 39.6 High-Energy and Time-Domain Astronomy

X-ray and gamma-ray observatories study the most violent events in the Universe, many lasting seconds to days. So high-energy astronomy is a **time-domain** discipline: wide-field monitors detect a burst and narrow-field telescopes swing onto it within minutes.

- **Fermi** (NASA, 2008) carries the Large Area Telescope (20 MeV to over 300 GeV), which surveys the whole gamma-ray sky every three hours and has catalogued about 7 000 sources. It combines this with a burst monitor (8 keV–40 MeV). On 17 August 2017 it detected a short gamma-ray burst 1.7 seconds after LIGO and Virgo recorded gravitational waves from merging neutron stars, opening **multi-messenger astronomy**.
- **Swift** (NASA, 2004) could turn to a new burst in about a minute and observe it with X-ray and ultraviolet/optical telescopes; it observed more than 2 000 gamma-ray bursts. The high solar activity of 2024–25 made its orbit decay much faster than predicted. NASA paid Katalyst Space US$ 30 million for a robotic spacecraft to grab Swift and raise its orbit. Launched on a Pegasus XL on 3 July 2026, it suffered attitude-control problems, and the attempt was abandoned on 19 August. Below 275 km drag prevented steady pointing: Swift made its last observation on 20 October 2026, NASA ended the mission on 23 October, and re-entry is expected within weeks.
- **NuSTAR** (NASA, 2012) was the first telescope to focus hard X-rays (3–79 keV), using a 10 m mast unfolded in orbit for the long focal length grazing-incidence optics need. **IXPE** (NASA–Italy, 2021) measures X-ray **polarisation**, revealing the geometry of magnetic fields in pulsars and black-hole jets.
- **XRISM** (JAXA with NASA and ESA, 2023) carries **Resolve**, a microcalorimeter. It measures each X-ray photon's energy from the tiny temperature rise it causes in a detector at 0.05 K. Its spectral resolution is 20–30 times finer than that of X-ray CCDs. Its protective gate valve has failed to open, blocking X-rays below about 1.7 keV, but it works superbly at higher energies and operations have been extended to 2030.
- **eROSITA**, the German telescope on Russia's Spektr-RG (2019), completed four all-sky X-ray surveys before being idled in February 2022 after Russia's invasion of Ukraine; its July 2026 release lists about two million sources.
- **Einstein Probe** (China with ESA, Germany and France, January 2024) uses "lobster-eye" optics, modelled on crustacean eyes, to watch 3 600 square degrees at once in soft X-rays. It has found many new transients. The Chinese–French **SVOM** (June 2024) detects gamma-ray bursts and follows up its alerts. ESA's **XMM-Newton** (1999) remains Europe's X-ray spectroscopy workhorse.

## 39.7 Microwaves, Radio and Gravitational Waves

The **cosmic microwave background (CMB)** is light released 370 000 years after the Big Bang, when the Universe became transparent, now stretched 1 100-fold into microwaves. NASA's **COBE** (1989) showed that it has a perfect blackbody spectrum at 2.725 K. It also detected temperature variations of a few parts in 100 000 — the seeds of all later structure. This work was recognised by the 2006 Nobel Prize in Physics. **WMAP** (2001–10) mapped those variations over the whole sky from L2, and ESA's **Planck** (2009–13) mapped them in nine frequency bands with about three times better resolution. Planck's final results (2018) gave an age of 13.8 billion years, a composition of about 5 % ordinary matter, 27 % dark matter and 68 % dark energy, and $H_0 = 67.4 \pm 0.5$ km/s/Mpc.

In radio astronomy space extends the size of an interferometer. Russia's **RadioAstron** (2011–19) linked a 10 m orbiting antenna, reaching about 350 000 km from Earth, with ground radio telescopes. It achieved resolutions of about ten micro-arcseconds.

The next frontier is not light at all. The **Laser Interferometer Space Antenna (LISA)** was adopted by ESA in January 2024 with NASA participation and is planned for launch on Ariane 6 in 2035. It will fly three spacecraft in a triangle 2.5 million km on a side and use lasers to measure picometre changes in their separations. It will detect low-frequency **gravitational waves** from merging supermassive black holes and compact binary stars, which ground-based detectors such as LIGO cannot hear.

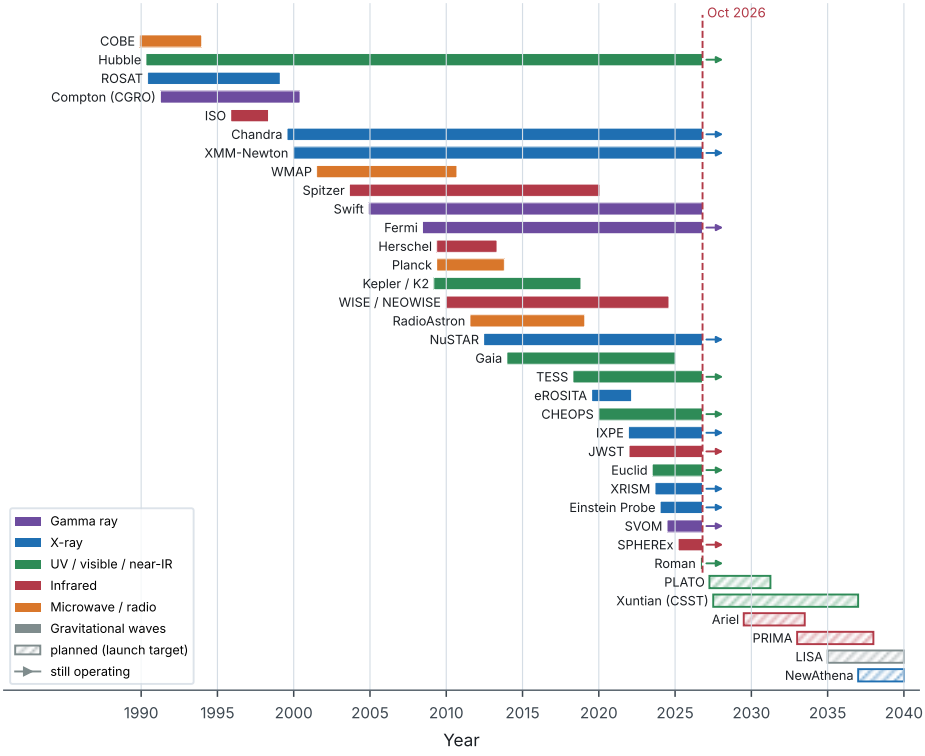
## 39.8 The Fleet in 2026 and the Road Ahead

Figure 39.17 shows how missions have succeeded one another since 1990, and Table 39.5 lists the main observatories operating or approved in October 2026. Missions outlive their plans — Hubble was designed for 15 years and Chandra for five. Europe, China and Japan now lead many missions. Apertures grow slowly, limited by fairings and cost (Figure 39.18 a), so progress comes largely from wider fields, better detectors and colder telescopes.

**TABLE 39.5** Major space observatories operating or approved in October 2026 (launch years of future missions are targets).

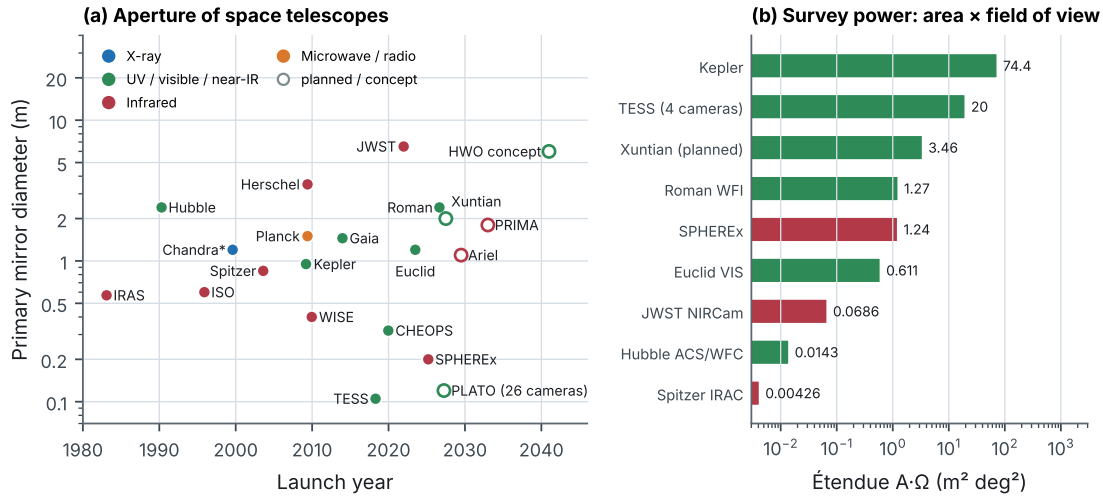
| Mission | Lead | Band | Aperture | Orbit | Launch | Status |
|---|---|---|---|---|---|---|
| Hubble | NASA/ESA | 0.115–1.7 µm | 2.4 m | LEO 470 km | 1990 | One-gyro mode |
| Chandra | NASA | 0.1–10 keV | 1.2 m | HEO | 1999 | Minimum-cost |
| Swift | NASA | γ, X-ray, UV | 0.3 m | LEO 265 km | 2004 | Ended Oct 2026 |
| Fermi | NASA | 8 keV–300 GeV | Detectors | LEO 490 km | 2008 | Budget threat |
| TESS | NASA | 0.6–1.0 µm | 4 × 10.5 cm | HEO | 2018 | Extended |
| JWST | NASA/ESA/CSA | 0.6–28.5 µm | 6.5 m | L2 | 2021 | Cycle 5 |
| Euclid | ESA | 0.55–2.0 µm | 1.2 m | L2 | 2023 | Surveying |
| XRISM | JAXA/NASA | 1.7–12 keV | 2 × 0.45 m | LEO 525 km | 2023 | Operating |
| Einstein Probe | CAS/ESA | 0.3–10 keV | Lobster-eye | LEO 600 km | 2024 | Operating |
| SPHEREx | NASA | 0.75–5.0 µm | 0.2 m | LEO 650 km | 2025 | Surveying |

| Mission | Lead | Band | Aperture | Orbit | Launch | Status |
|---|---|---|---|---|---|---|
| Roman | NASA | 0.48–2.3 µm | 2.4 m | L2 | 2026 | Commissioning |
| PLATO | ESA | 0.5–1.0 µm | 26 × 0.12 m | L2 | 2027 | Final testing |
| Xuntian | China | 0.25–1.0 µm | 2 m | LEO | 2027 | Building |
| Ariel | ESA | 0.5–7.8 µm | 1.1 × 0.7 m | L2 | 2029 | Building |
| PRIMA | NASA | 24–235 µm | 1.8 m | L2 | 2033 | Selected |
| LISA | ESA/NASA | Grav. waves | 3 × 0.3 m | Solar orbit | 2035 | Adopted |
| NewAthena | ESA | 0.2–12 keV | ≈ 2.5 m | L-point | 2037 | Study |
| HWO | NASA | UV–near-IR | ≈ 6 m | L2 | 2040s | Concept |




Bars run from launch to end of science operations. Planned bars start at the launch target (Oct 2026) and show nominal mission length; Swift ended science on 20 Oct 2026 ahead of re-entry.

**FIGURE 39.17** Timeline of major space observatories, 1990–2040. Bars show operating life from launch; hatched bars are planned missions starting at their target launch dates.




*Chandra: outer diameter of nested grazing-incidence mirrors. HWO: ~6 m concept, 2040s. Étendue uses geometric aperture area (JWST 25.4 m²) and the field of the main survey camera(s).

**FIGURE 39.18** (a) Primary-mirror diameter of space telescopes against launch year. (b) Survey power (étendue): collecting area multiplied by the field of view of the main camera. Wide-field survey telescopes map the sky thousands of times faster than Hubble or JWST.

## 39.8.1 New Missions and the Habitable Worlds Observatory

In September 2026 NASA selected **PRIMA**, a cryogenically cooled 1.8 m far-infrared telescope, as the first mission in a new **Probe** class with a cost cap of US$ 1.2 billion. Due for launch in 2033, it will study how galaxies, stars and planets form, filling the gap left by Spitzer and Herschel. The UVEX ultraviolet explorer (about 2030) and Israel's ULTRASAT (2027) will survey the ultraviolet sky, and ESA's **NewAthena** X-ray observatory is planned for adoption in 2027 and launch in the late 2030s.

The 2021 US astronomy decadal survey recommended as NASA's next great observatory a telescope of about 6 m to image Earth-like planets around Sun-like stars and search their spectra for oxygen, water and methane. This **Habitable Worlds Observatory (HWO)** needs a coronagraph suppressing starlight by $10^{10}$. An Earth seen from 30 light-years is ten billion times fainter than its star and only 0.1″ from it. It also needs a wavefront stable to picometres. NASA aims for launch in the early 2040s, but funding is uncertain: Congress provided about US$ 150 million for 2026, while the 2027 request proposed almost nothing.

Astronomers thus have more capability in space than ever, yet budget requests propose ending productive missions that cost tens of millions a year to run after costing billions to build.

### AI IN SPACE ASTRONOMY

Machine learning classifies **transients**: ground surveys such as the Vera C. Rubin Observatory, which released its first images in 2025, are expected to issue millions of alerts per night. Classifiers in "alert brokers" decide within minutes which supernovae, tidal-disruption events or X-ray flares deserve follow-up by satellites such as Einstein Probe. In **exoplanet vetting**, neural networks trained on Kepler light curves separate true transits from eclipsing binaries and instrumental artefacts. One found the eighth planet of Kepler-90 in 2017, and NASA's ExoMiner validated 301 new Kepler planets in 2021. **Galaxy morphology** is classified by deep networks trained on volunteers' labels from the Galaxy Zoo project, now applied to Euclid images. All such outputs must be validated, because a subtle bias can mimic a discovery (Chapter 51).

### KEY TAKEAWAYS

- Space telescopes gain wavelengths the atmosphere blocks, diffraction-limited sharpness and photometric stability, at hundreds of times the ground's cost per square metre.
- Of NASA's Great Observatories, Hubble (one-gyro mode, median re-entry late 2033) and Chandra (minimum-cost operations) still worked in October 2026; Compton and Spitzer ended in 2000 and 2020.
- JWST has seen galaxies to $z = 14.44$, 280 million years after the Big Bang, found surprisingly early black holes and measured exoplanet atmospheres; the K2-18 b biosignature claim is unconfirmed.
- Light from $z = 14$ left 13.5 billion years ago from a source now 34 billion light-years away, which was 2.25 billion light-years away when the light left: cosmic distances must be defined carefully.
- A transiting Earth dims a Sun-like star by 84 ppm for 13 hours. Of 6 451 confirmed exoplanets (October 2026), three-quarters were found by transits; PLATO will hunt true Earth analogues.
- Survey telescopes gain power from wide fields; high-energy missions form a rapid-response network.
- PLATO and Xuntian (2027), Ariel (2029), PRIMA (2033), LISA (2035), NewAthena and the Habitable Worlds Observatory lie ahead, but budget threats to operating missions are serious.

### REVIEW QUESTIONS

1. Give three reasons for putting a telescope in space, with an example of each.
2. What caused Hubble's spherical aberration, and why could it be corrected completely?
3. Compare LEO, HEO and an L2 halo orbit for an infrared telescope and for an X-ray telescope.
4. Why must X-ray telescopes use grazing-incidence mirrors, and with what consequences?
5. Why can JWST, but not Hubble, see galaxies beyond $z \approx 13$?
6. Using Table 39.3, give the look-back time, age at emission and comoving distance for $z = 6$. Why does the comoving distance exceed the light-travel distance?
7. Find the transit depth and duration for a 3.9 Earth-radius planet in a 30-day orbit around a Sun-like star.
8. Why is the K2-18 b dimethyl-sulphide claim unconfirmed, and what would establish a biosignature?
9. Define étendue, and explain why SPHEREx, with a 20 cm telescope, maps the sky faster than JWST.
10. Discuss the case for and against funding old observatories such as Hubble and Chandra when new missions need money.

PART XII

# Emerging Frontiers

The next era of spaceflight will service, refuel and build things in orbit, use resources from the Moon and asteroids, and move with new kinds of propulsion. Part XII explains the engineering and the economics behind these frontiers.


40 In-Space Servicing, Assembly and Manufacturing

41 Space Resources and Asteroid Mining

42 Advanced Propulsion


Image: NASA/JSC (Terry Virts)

CHAPTER

# 40 In-Space Servicing, Assembly and Manufacturing

Image: NASA/JSC

*Almost every satellite ever launched has been used once and thrown away, retired with working instruments because its propellant ran out or a single part failed. In-space servicing, assembly and manufacturing (ISAM) aims to change that by inspecting, refuelling, repairing, upgrading, moving and disposing of spacecraft in orbit, building structures too large for any rocket fairing, and making products that can only be made in weightlessness. This chapter traces servicing from the Skylab and Hubble spacewalks to the commercial robots docking with communication satellites in 2020–26, explains the engineering of rendezvous, capture and refuelling, and estimates the economics of life extension, debris removal and orbital propellant depots, with the state of every major programme as of October 2026.*

**IN THIS CHAPTER**

40.1 What ISAM Means and Why It Matters
40.2 Sixty Years of Servicing
40.3 Rendezvous, Proximity Operations and Capture
40.4 The Economics of Life Extension and Debris Removal
40.5 Assembling Large Structures in Orbit
40.6 Manufacturing in Space
40.7 Refuelling for Exploration

A geostationary communication satellite costs several hundred million dollars to build, launch and insure, and is designed to work for about 15 years. When it retires, its transponders usually still work: what it has run out of is the few hundred kilograms of propellant needed to stay in its slot. A science observatory can be crippled by one failed gyroscope; a broken solar-array hinge can end a mission worth billions. On Earth, no one would scrap a car because its fuel tank was empty or a tyre was flat. In orbit, until recently, there was no alternative.

**In-space servicing, assembly and manufacturing (ISAM)** is the set of technologies that treat spacecraft as maintainable, upgradeable and buildable in place. The term was adopted in the United States' *ISAM National Strategy* of April 2022; NASA previously used **OSAM** (on-orbit servicing, assembly and manufacturing), and European documents speak of **in-orbit servicing (IOS)**. The idea is old — astronauts repaired Skylab in 1973 — but for forty years it depended on the Space Shuttle and its crews. What changed after 2019 is that robotic servicers began to dock with commercial satellites as a paid service. In the same period China demonstrated similar capabilities in geostationary orbit, and refuelling became central to plans for returning people to the Moon.

This chapter follows the three strands of ISAM in turn. Servicing extends and repairs existing spacecraft; assembly builds structures larger than any launcher can carry; manufacturing makes things — parts, fibres, crystals, tissues — in orbit. Rendezvous and proximity operations, the common foundation of all three, are treated in detail. The orbital mechanics of relative motion is introduced in Chapter 3, and the security implications of satellites that approach other satellites in Chapter 19.

## 40.1 What ISAM Means and Why It Matters

Table 40.1 lists the main ISAM activities, from the simplest — looking at another spacecraft from close range — to the most ambitious. They share a ladder of difficulty. **Inspection** needs only safe proximity operations. **Relocation** and **life extension** need a physical connection that can carry thrust loads. **Refuelling** adds leak-tight fluid connections; **repair and upgrade** add dexterous robotics; **assembly** and **manufacturing** require all of these to be repeatable, hundreds of times, without people present.

**TABLE 40.1** The main ISAM activities, in rough order of increasing difficulty.

| Activity | What it does | Typical client | Status (Oct 2026) |
|---|---|---|---|
| Inspection | Close-range imaging of a satellite or debris object | Any; damaged or unknown objects | Operational (commercial and military) |
| Relocation and disposal | Tow to a new slot, graveyard orbit or re-entry | Dead or retiring satellites; debris | Demonstrated in GEO (2020, 2022) |
| Life extension | Attach a vehicle or pod that takes over attitude and orbit control | GEO satellites short of propellant | Commercial since 2020 |
| Refuelling | Transfer propellant through a valve | Prepared clients (port); later unprepared | Demonstrated 2007; commercial demos 2026–27 |
| Repair and upgrade | Replace units, free stuck mechanisms, add payloads | Prepared or high-value clients | Crewed (Hubble); robotic in development |
| Assembly | Join modules, trusses or mirror segments | Stations, antennas, telescopes | Crewed (ISS); robotic demos planned |
| Manufacturing | Print, draw, crystallise or grow material in orbit | Parts, fibres, drugs, semiconductors | Demonstrations on ISS and free-flyers |

The arguments for ISAM are economic, scientific and environmental:

- **Life extension.** A satellite that keeps earning for five more years defers the cost of a replacement worth hundreds of millions of dollars.

- **Repair and upgrade.** Hubble was launched with a flawed mirror and 1980s electronics; five servicing missions turned it into one of the most productive scientific instruments ever built (Chapter 38).
- **Debris removal.** The objects that threaten long-term sustainability in low Earth orbit are a few thousand large derelicts (Chapter 21); only something that can rendezvous with and grip them can remove them.
- **Size.** A rocket fairing limits a monolithic structure to about 5 m (or 9 m for Starship). Larger telescopes, antennas and solar power stations must be folded or built in orbit.
- **New products.** Weightlessness suppresses convection and sedimentation, enabling purer glasses and crystals; and refuelling in orbit multiplies what one rocket can send beyond it.

## 40.2 Sixty Years of Servicing

The history of servicing has two halves: four decades in which astronauts did the work, and a commercial robotic era that began in 2020 (Figure 40.1).

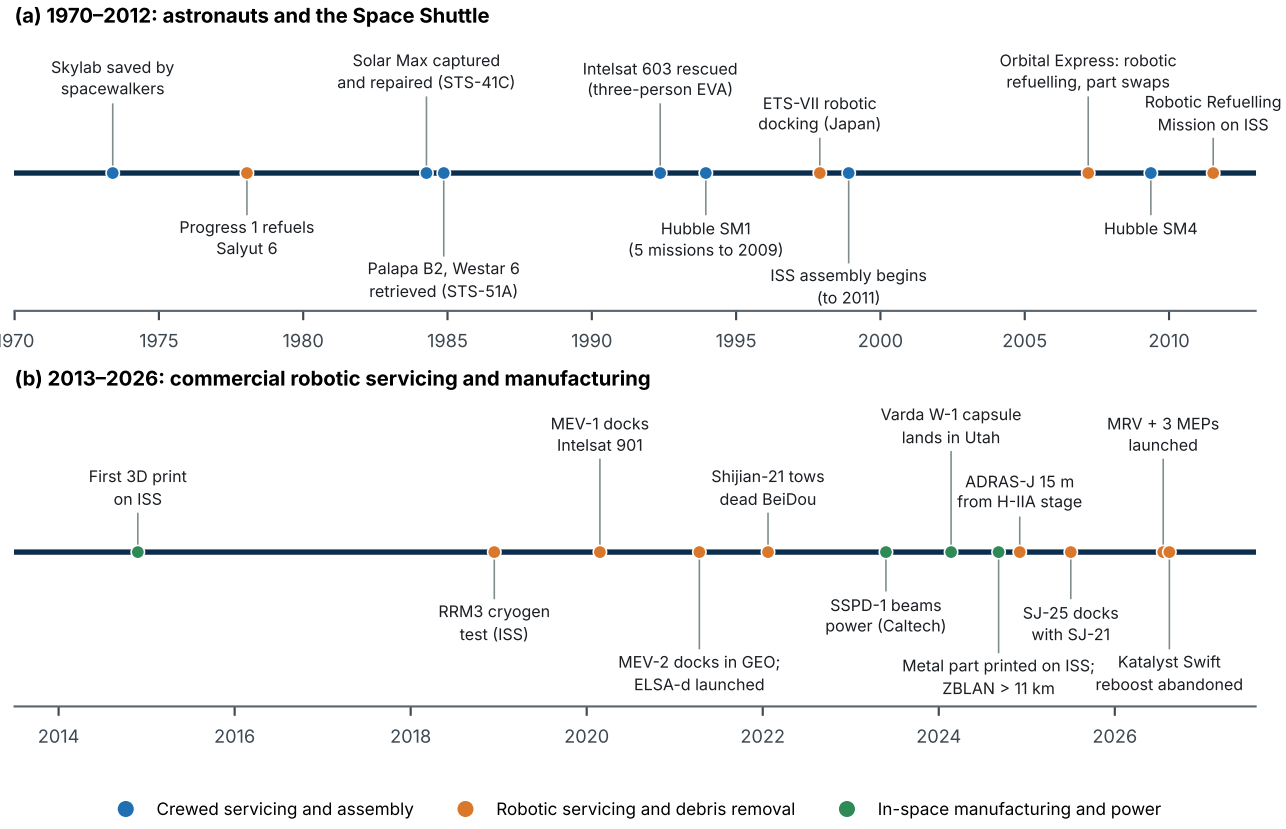



**FIGURE 40.1** Milestones of in-space servicing, assembly and manufacturing. (a) Until 2012 nearly all servicing was done by astronauts working from the Space Shuttle. (b) Since 2014 the field has shifted to commercial robotic servicers, debris-removal demonstrators and manufacturing experiments.

### 40.2.1 Astronauts as Servicers

The first repair was improvised. When Skylab reached orbit in May 1973, its micrometeoroid shield had been torn away and a solar wing was jammed. Its first crew deployed a sunshade through a small airlock and cut the wing free on a spacewalk (Chapter 33). The Space Shuttle then made servicing a planned activity. The **Solar Maximum Mission** satellite had lost its fine pointing in 1980 when fuses in its attitude-control system failed. It was built with a grapple

fixture and replaceable modules precisely so that the Shuttle could fix it. In April 1984 George Nelson flew to it with the Manned Manoeuvring Unit (Figure 40.2) to attach a docking device to a pin on its side. The device failed to latch — an unexpected small fitting was in the way — and Nelson's attempt to steady the satellite by hand set it tumbling. Controllers at NASA Goddard stabilised it overnight, Challenger's robotic arm grappled it, and Nelson and James van Hoften replaced the attitude-control module and repaired an instrument's electronics. Solar Max returned to work until it re-entered in 1989.

**FIGURE 40.2** George Nelson, flying the Manned Manoeuvring Unit, approaches the ailing Solar Maximum Mission satellite during STS-41C in April 1984. His attempt to dock by hand failed; the Shuttle's arm captured the satellite instead. Image: NASA/JSC.

**FIGURE 40.3** Richard Hieb, Thomas Akers and Pierre Thuot hold the 4.2-tonne Intelsat VI (603) satellite above Endeavour's payload bay on 13 May 1992, minutes after the only three-person capture by hand. Image: NASA/JSC.

Later in 1984, the crew of STS-51A retrieved two communication satellites, Palapa B2 and Westar 6, stranded in useless orbits by failed kick motors. They brought them back to Earth, and both were refurbished and launched again. The most dramatic rescue came in 1992. Intelsat 603 had been stranded in low orbit in 1990, when a wiring error prevented it from separating from its Titan III second stage together with its perigee motor. On Endeavour's first flight, STS-49, two attempts to attach a capture bar failed because the satellite rotated away at the slightest touch. On the third spacewalk three astronauts simply grabbed it by hand (Figure 40.3). This was the only three-person spacewalk ever made. They fitted a new perigee motor, and the satellite reached geostationary orbit and served for many years.

**Hubble** was designed for servicing from the start, with handrails, standard bolts and drawer-like **orbital replacement units**. Five Shuttle missions between 1993 and 2009 (Chapter 38) replaced every instrument, gyroscope set and battery and corrected the flawed optics. Each began with the Shuttle's arm grasping one of two grapple fixtures on the telescope and berthing it in the payload bay (Figure 40.4). The **International Space Station** was the culmination. Assembled from 1998 to 2011 by Shuttle and Russian launches, it remains a servicing laboratory. There, crews and the Canadarm2 and Dextre robots replace pumps, batteries and solar arrays (Chapter 32).

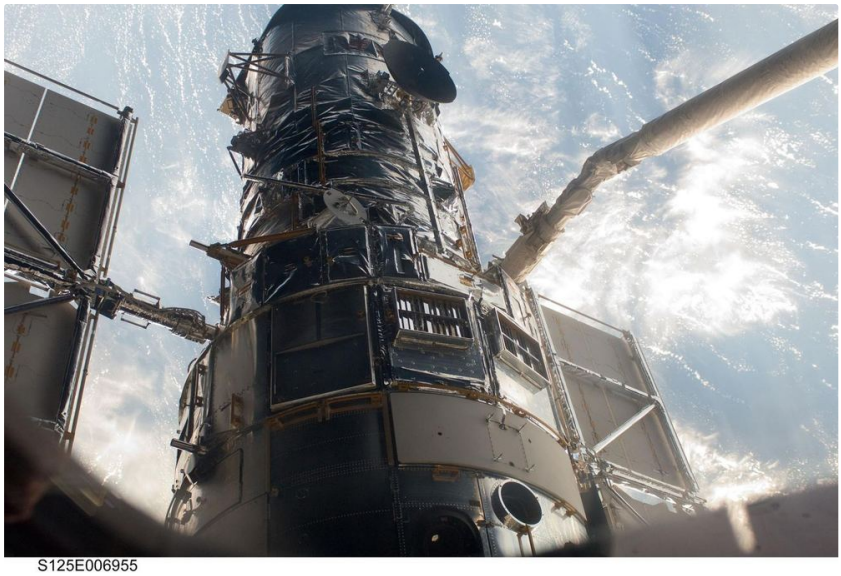



**FIGURE 40.4** Hubble held by Atlantis's robotic arm on 13 May 2009 at the start of the fifth and final servicing mission, STS-125. The grapple fixture and handrails were designed in from the start. Image: NASA/JSC.

The lesson of the Shuttle era is mixed. Servicing worked spectacularly on clients designed for it — Solar Max and Hubble — and was heroic but risky on clients that were not. And it was expensive: each Hubble mission required a Shuttle flight costing well over a billion dollars at full cost, so it was never economical for commercial satellites.

### 40.2.2 Robotic Pioneers

Robotic servicing began with fuel. In January 1978 the uncrewed **Progress 1** docked with the Salyut 6 station and pumped propellant into its tanks — the first transfer of propellant between spacecraft. Progress and China's Tianzhou freighters still refuel the ISS and Tiangong. Japan's **ETS-VII** (1997–99) repeated the automatic docking of two uncrewed satellites, first achieved by the Soviet Kosmos 186 and 188 in 1967. It was also the first uncrewed satellite to test a robotic arm in orbit. NASA's **DART** demonstrator of 2005 (not the later asteroid impactor) showed the dangers: errors in its navigation software made it collide gently with its target. In 2007 DARPA's **Orbital Express** — the Boeing-built ASTRO servicer and Ball's NextSat client — demonstrated nearly everything a servicer needs. This included autonomous rendezvous, capture with a robotic arm, transfer of hydrazine in both directions, and the exchange of a battery and a flight computer.

NASA then tackled a harder problem: refuelling a satellite never designed for it, whose fill-and-drain valves are covered by caps, lock wire and thermal blankets meant to be closed once, on the ground. The **Robotic Refuelling Mission (RRM)** was carried to the ISS on the last Shuttle flight in 2011 (Figure 40.5). It gave Dextre a set of specialised tools that cut lock wire, unscrewed caps and, in 2013, transferred liquid ethanol through a standard valve. RRM3 (2018–19) aimed to store and transfer liquid methane, a cryogenic propellant, but a failed cryocooler forced its methane to be vented.

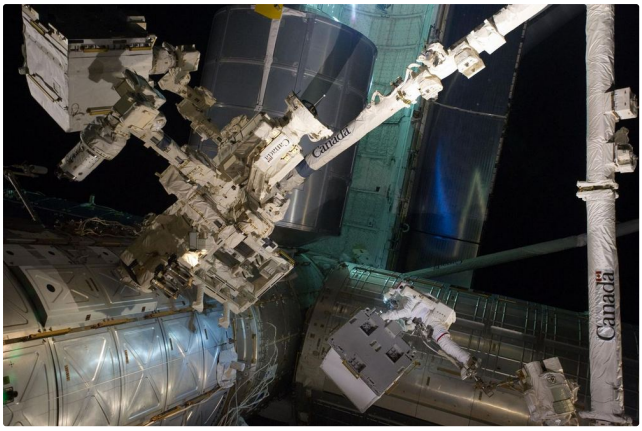



**FIGURE 40.5** Astronaut Mike Fossum, riding Canadarm2, carries the Robotic Refuelling Mission module during the final Shuttle mission in July 2011. Dextre later used its tools to open and refuel simulated satellite valves. Image: NASA/JSC.

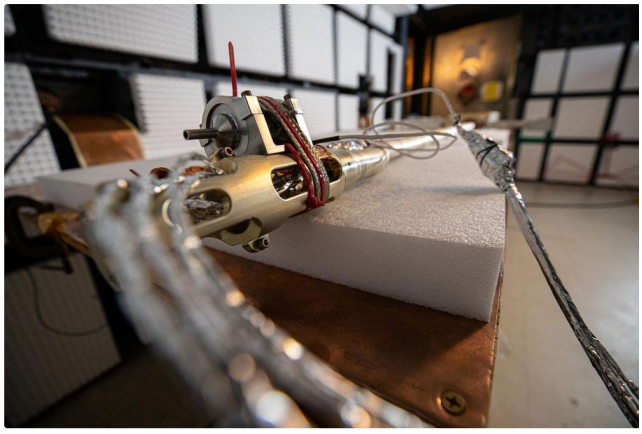


**FIGURE 40.6** The hose management assembly built for NASA's Restore-L (later OSAM-1) refuelling servicer, in electromagnetic testing at Kennedy Space Center in 2020. It extended and retracted the propellant hose like a tape measure. Image: NASA/KSC (NASA/Cory Huston).

RRM led to **Restore-L**, renamed **OSAM-1**: a servicer that would grasp the Landsat 7 satellite and refuel it. It would also use a separate arm called SPIDER to assemble a 3 m antenna and manufacture a 10 m beam (Figure 40.6). Its cost estimate grew from $626–753 million to more than $2 billion, and in March 2024 NASA cancelled it, citing "continued technical, cost, and schedule challenges" and "a broader community evolution away from refueling unprepared spacecraft". About $1.5 billion had been appropriated. A companion mission to 3D-print beams in orbit, OSAM-2 (Archinaut One), had been closed out in 2023 before flight.

### 40.2.3 Commercial Life Extension

The first commercial servicer avoided refuelling altogether. Northrop Grumman's **Mission Extension Vehicle (MEV)** is about 2.3 t at launch and built for a 15-year life. It docks with a client, then provides all of the combined vehicle's attitude and orbit control with its own electric thrusters. It needs nothing special on the client. It inserts a probe into the satellite's liquid apogee engine (the large rocket motor used once to reach geostationary orbit) and expands a tip inside the nozzle. It then pulls itself in until stanchions press against the launch-adapter ring (see Figure 40.8).

**MEV-1** docked with Intelsat 901, which had been parked in the graveyard orbit about 300 km above GEO, on 25 February 2020. This was the first docking of two commercial satellites, and MEV-1 returned Intelsat 901 to service in April 2020. After five years it moved Intelsat 901 back to the graveyard orbit, undocked on 9 April 2025 and went on to extend the life of Optus D3, from which it separated again in mid-2026. **MEV-2** docked with Intelsat 10-02 directly in its operational slot in April 2021; its planned five-year stay was extended in 2024 by several more years. Intelsat reportedly paid about $13 million per year for the service.

Northrop's second generation launched on a Falcon 9 on 21 July 2026: the **Mission Robotic Vehicle (MRV)**, together with three **Mission Extension Pods (MEPs)**. The MRV carries the robotic arms and tools of DARPA's Robotic Servicing of Geosynchronous Satellites (RSGS) programme. Each MEP is a small propulsion module that the MRV will attach to a client's launch-adapter ring. The pod then keeps the satellite on station for up to about eight years while the MRV moves on.

The MRV needs about a year to raise itself to geostationary orbit with electric propulsion; early customers include Optus and SES (which absorbed Intelsat in 2025). After checkout, RSGS operations are to pass to the US Space Force.

### 40.2.4 China's Geostationary Servicers

China has developed comparable capabilities, with less publicity. **Shijian-21**, launched in October 2021, docked with the dead BeiDou-2 G2 navigation satellite in December 2021. In January 2022 it towed the satellite about 3 000 km above the geostationary belt — far higher than the usual 300 km graveyard — before returning to GEO. **Shijian-25**, launched in January 2025, approached Shijian-21 and the two appeared merged in commercial optical tracking from about 2 July 2025, separating again by late November. The pairing was widely reported as the first refuelling in geostationary orbit, but China has not described what was transferred. Because the same rendezvous, grappling and towing capabilities could disable another nation's satellite, these missions are watched closely (Chapter 19).

### 40.2.5 Debris Removal and Inspection

Removing debris is harder than servicing a satellite: the target was never meant to be approached, may be tumbling, has no lights, reflectors or radio, and nobody pays it to cooperate. Japan's **Astroscale** has worked up to it step by step (Table 40.2). **ELSA-d** (launched March 2021) released and recaptured its own small client with a magnet in August 2021, but thruster failures in 2022 ended its more ambitious tests. **ADRAS-J** was launched on an Electron rocket in February 2024 under a JAXA contract. It made the first close inspection of a large piece of real debris: an 11 m, 3 t H-IIA upper stage left in orbit in 2009. It flew around the stage at 50 m in mid-2024, once aborting automatically when the target's attitude did not match predictions. In late November 2024 it closed to about 15 m from the planned capture point. **ADRAS-J2**, under a JAXA contract worth about ¥12 billion (about $80 million), is to capture the same stage with a robotic arm and lower it for re-entry. A launch on Isar Aerospace's Spectrum rocket in Japan's fiscal year 2027 (up to April 2028) was contracted in September 2026.

**TABLE 40.2** Robotic servicing and debris-removal missions (selected).

| Mission | Operator | Launch | Client | Result (as of Oct 2026) |
|---|---|---|---|---|
| Orbital Express | DARPA (Boeing, Ball) | 2007 | Own client (NextSat) | Capture, hydrazine transfer, battery and computer swap |
| MEV-1 | Northrop Grumman | 2019 | Intelsat 901, then Optus D3 | Intelsat 901 2020–25; Optus D3 2025–26 |
| MEV-2 | Northrop Grumman | 2020 | Intelsat 10-02 (GEO) | Docked in slot 2021; stay extended |
| ELSA-d | Astroscale | 2021 | Own client (17 kg) | Magnetic capture 2021; ended 2024 |
| Shijian-21 | China | 2021 | Dead BeiDou-2 G2 | Towed 3 000 km above GEO, 2022 |
| ADRAS-J | Astroscale / JAXA | 2024 | H-IIA upper stage | Inspected down to 15 m |
| Shijian-25 | China | 2025 | Shijian-21 | Docked 2025; refuelling reported, unconfirmed |

| Mission | Operator | Launch | Client | Result (as of Oct 2026) |
|---|---|---|---|---|
| Katalyst LINK | Katalyst / NASA | 2026 | Swift observatory | Lost attitude control; reboost abandoned |
| MRV + 3 MEPs | Northrop Grumman | 2026 | GEO comsats | Raising orbit to GEO |
| Otter | Starfish Space | 2026 | Defunct US satellites | Commissioning; inspection for NASA |
| ADRAS-J2 | Astroscale / JAXA | FY2027 | H-IIA upper stage | Capture and de-orbit planned |
| ClearSpace-1 | ESA (OHB, ClearSpace) | ≈ 2029 | PROBA-1 (95 kg) | Capture with four arms planned |

Europe's **ClearSpace-1**, led by OHB with the Swiss company ClearSpace, will capture ESA's 95 kg PROBA-1 satellite, launched in 2001, with four arms and de-orbit it; ESA lists the launch for 2029. In the USA, **Starfish Space** launched its first full-size Otter servicer on a SpaceX rideshare on 1 October 2026. For NASA it is to approach up to four defunct US satellites in early 2027, without touching them, ahead of planned life-extension work for Intelsat in GEO and de-orbit services for the Space Development Agency. Its smaller Otter Pup 2 demonstrator, in orbit since June 2025, had not yet docked with a client.

A reminder of how unforgiving the field is came in 2026. NASA's Swift gamma-ray observatory, launched in 2004, was decaying towards re-entry. In September 2025 NASA gave the start-up **Katalyst Space** a $30 million contract to boost it. This was a first attempt to reboost a satellite not designed for servicing, on a schedule of months rather than years. Its 425 kg LINK spacecraft reached orbit on the final Pegasus XL launch on 3 July 2026, but three weeks later two of its three reaction wheels failed and it lost attitude control. It came within 12–15 km of Swift before the reboost was abandoned in August; LINK re-entered on 25 September, and Swift is expected to re-enter around the end of 2026.

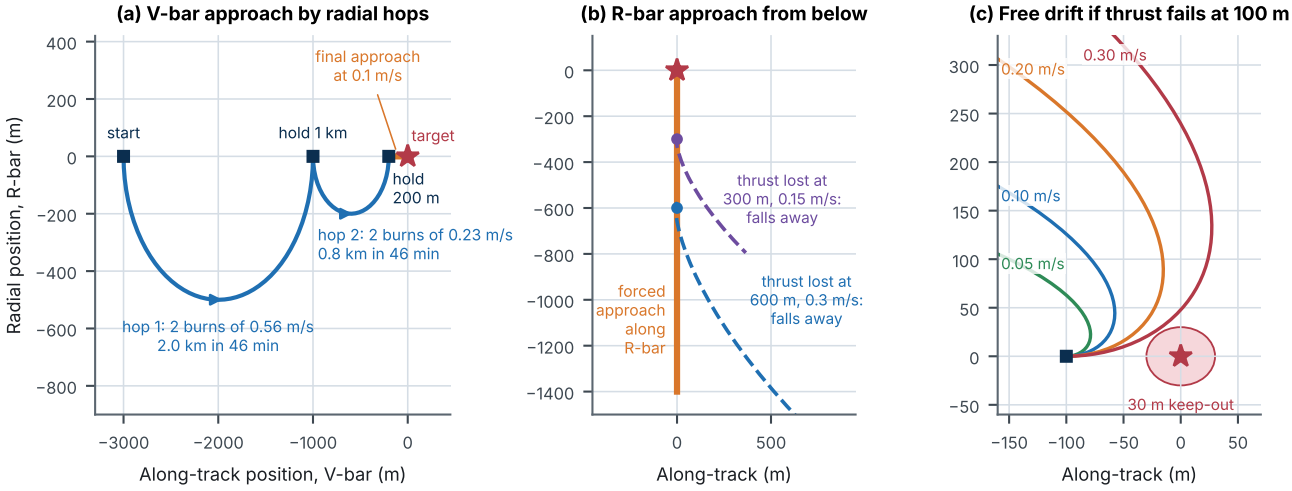
## 40.3 Rendezvous, Proximity Operations and Capture

Every ISAM activity starts with **rendezvous and proximity operations (RPO)**: bringing two spacecraft from hundreds of kilometres apart to contact, safely. The far-range phases — launch into a slightly lower or higher orbit, phasing, and orbit matching — use the methods of Chapter 3. Within a few kilometres the relative motion is described by the Clohessy–Wiltshire equations in the target's local frame. In this frame $x$ is radial (along the **R-bar**, the line through the target and Earth's centre; positive away from Earth) and $y$ is along the target's velocity (the **V-bar**). Their solutions are counter-intuitive: a forward push raises a spacecraft and makes it drift backwards, and a spacecraft below the target drifts forwards.

### 40.3.1 Approach Strategies

Operators approach along one of two straight lines through the target (Figure 40.7). On the **V-bar** a chaser can rest at any distance ahead of or behind the target, because it then shares the target's orbit. This makes the V-bar ideal for **hold points**, where the approach pauses while controllers check sensors and give the go-ahead. The chaser moves between hold points by **radial hops**. A

small burn towards Earth makes it dip below the target's orbit, travel forwards and return to the V-bar half an orbit later. There an equal and opposite burn stops it. The ISS's European ATV and Japanese HTV, and today's Cygnus and Dragon, all use variants of these patterns.




Clohessy–Wiltshire solutions for a target in a 420 km circular orbit (period 93 min). Radial axis up = away from Earth; target flies to the right. Axes not to equal scale.

**FIGURE 40.7** Approach strategies computed from the Clohessy–Wiltshire equations for a target in a 420 km orbit. (a) A V-bar approach: two radial hops carry the chaser from 3 km to a hold point 200 m behind the target, followed by a slow straight-line approach. (b) An approach up the R-bar from below: if thrust is lost, orbital mechanics makes the chaser fall away. (c) If thrusters fail during a V-bar approach from 100 m, a slow chaser stops short of the target while a fast one sweeps past it — the basis of "passively safe" approach speeds.

The **R-bar** approach climbs towards the target from directly below (or descends from above). Here orbital mechanics provides **natural braking**. A chaser below the target is in a lower orbit, so the tidal term in the equations pulls it down and away. To hold position on the R-bar it must thrust continuously towards the target. If the thrust stops, it falls away and drifts ahead, as in panel (b). The Shuttle approached the ISS this way, and so do servicers wanting to reach a client's underside. The final metres are flown along a narrow **approach corridor** at a few centimetres per second, with the docking interface pointing at the chaser.

The key design rule is **passive safety**: at every point, if the chaser's thrusters stopped working, its free drift must not hit the target. Panel (c) shows why approach speeds must be low. A chaser closing at 0.1 m/s from 100 m behind on the V-bar, with its thrusters suddenly dead, coasts forward only 42 m. Orbital mechanics then lifts it and carries it back. At 0.3 m/s it sweeps over the target within 34 m. Approach paths are surrounded by **keep-out spheres** that the chaser may enter only after explicit approval.

**EXAMPLE: HOPPING ALONG THE V-BAR**

A servicer must move from a hold point 1 km behind a client in a 420 km orbit to a point just behind it. The mean motion is $n = \sqrt{\mu/r^3} = 1.126 \times 10^{-3}$ rad/s, so the orbital period is 93.0 min.

1. **Radial hop.** For a radial burn $\Delta v$, the Clohessy–Wiltshire solution gives $x = -(\Delta v/n)\sin nt$ and $y = (2\Delta v/n)(1 - \cos nt)$. After half an orbit the chaser is back on the V-bar, $4\Delta v/n$ further forward. For a 1 000 m hop, $\Delta v = nL/4 = 1.126 \times 10^{-3} \times 1\,000 / 4 =$ **0.28 m/s**; an equal burn stops it, so the hop costs 0.56 m/s and takes 46.5 min. The path dips 250 m below the V-bar.
2. **Straight-line approach.** Flying the same 1 000 m straight along the V-bar requires continuous radial thrust to cancel the Coriolis term, $a_x = 2n\dot{y}$. Integrated over the approach this costs $2nL =$ **2.25 m/s** whatever the speed — four times the hop's cost, which is why straight lines are kept for the last few hundred metres.
3. **Passive safety.** A chaser drifting forward at speed $v$ along the V-bar travels at most $0.478\ v/n$ before reversing. In LEO at 0.1 m/s that is 42 m. In GEO, where $n = 7.29 \times 10^{-5}$ rad/s, the same speed would carry it 655 m. To stop short of a client 100 m away, a GEO servicer must approach at less than 1.5 cm/s. Hops are cheap there (0.036 m/s per kilometre) but each one takes 12 hours.

### 40.3.2 Sensors and Navigation

A servicer measures the client's relative position and attitude — its **pose** — with a sequence of sensors that hand over as the range shrinks. Relative GNSS or ground tracking works at tens of kilometres. Visible and infrared cameras see the client first as a point and then as a shape. Below a few hundred metres, lidar or time-of-flight cameras measure range and three-dimensional shape directly. Cooperative clients such as the ISS carry retroreflectors and radio beacons. Unprepared clients do not, so the servicer must estimate pose from images alone, using the client's known geometry. It must do this under harsh lighting that changes every few minutes. Thermal infrared cameras help in eclipse and against glinting surfaces. The guidance software runs on board because ground loops are too slow and too intermittent. It must recognise when measurements stop making sense and abort — as ADRAS-J did when the rocket stage's attitude departed from predictions.

### 40.3.3 Capturing a Client

How a servicer grips its client depends on whether the client was **prepared** — fitted before launch with a grapple fixture, docking plate or refuelling port — or **unprepared**. Figure 40.8 compares the main options. Prepared interfaces are simple and reliable: the Shuttle's and ISS's arms snare a grapple pin with three wires, and Astroscale's clients carry a ferromagnetic plate. Most satellites in orbit, though, were never prepared. So the commercial servicers have learned to grip features that almost every satellite has anyway. These are the **launch-adapter ring** and its bolt holes, by which the satellite was clamped to its rocket, and the nozzle of the **liquid apogee engine** that raised it to GEO. Both are strong enough to carry towing loads and are located on the satellite's axis.

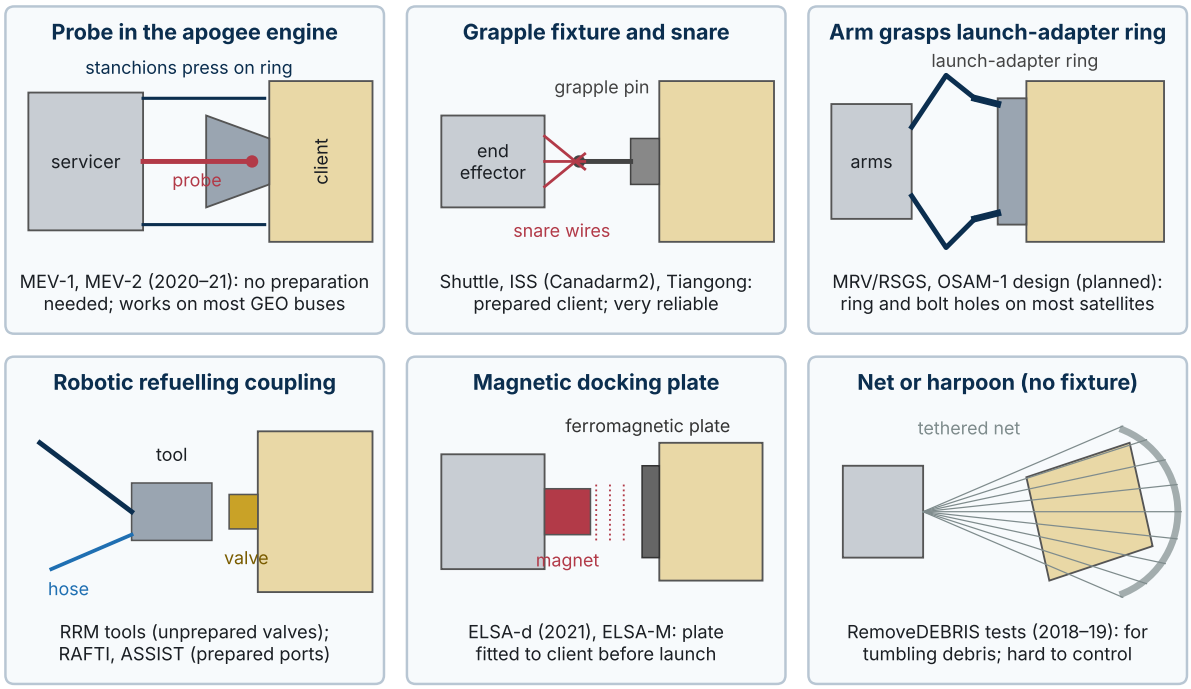
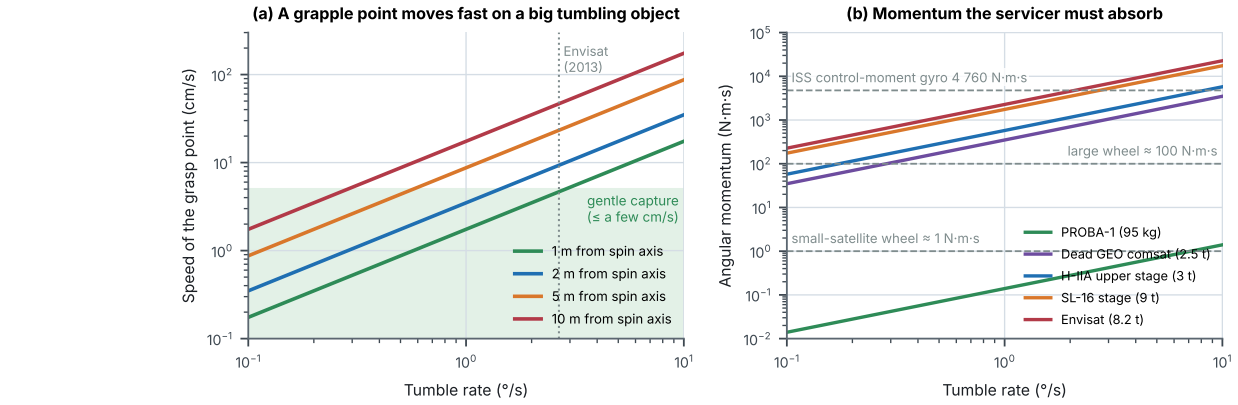



**FIGURE 40.8** Six ways for a servicer to grip a client. The apogee-engine probe, grapple fixture and magnetic plate have all worked in orbit, and refuelling couplings were tested by Orbital Express and on the ISS; prepared interfaces (grapple fixture, magnetic plate, refuelling port) need equipment fitted before launch, while the apogee-engine probe and robotic ring grasp (planned for the MRV) work on most satellites already in orbit. Nets and harpoons, tested by RemoveDEBRIS in 2018–19, can catch tumbling debris but give poor control afterwards.

**Tumbling** is the main obstacle to capturing debris. Without attitude control, objects are spun up over years by solar radiation pressure, gravity-gradient torques and eddy currents induced by Earth's magnetic field. Envisat, the 8.2 t ESA satellite that failed in 2012, was measured by laser ranging spinning once every 135 s in 2013 — about 2.7° per second. A grasp point 5 m from the spin axis of such an object moves at about 23 cm/s (Figure 40.9), far too fast to seize gently. So a servicer must either fly in synchrony with the rotation, matching the spin while hovering, or slow the target down first. It can slow the target, for example, by eddy-current braking, plume impingement or repeated gentle contacts. After capture, the stack's angular momentum must be absorbed. An H-IIA upper stage (moment of inertia about $3.3 \times 10^4$ kg·m$^2$) tumbling at 1° per second carries about 580 N·m·s. This is about six times what a large reaction wheel (about 100 N·m·s) can store, so servicers use thrusters to detumble the combined vehicle.




v = ωr; H = Iω about the largest principal axis. Moments of inertia approximated from mass and dimensions (stages as uniform cylinders; Envisat ≈ 1.3 × 10⁸ kg·m²). Envisat rate from laser ranging.

**FIGURE 40.9** Capturing a tumbling object. (a) The speed of a grasp point grows with tumble rate and with its distance from the spin axis; only slowly rotating targets can be seized directly. (b) The angular momentum of large debris objects at typical tumble rates exceeds the storage of reaction wheels, so thrusters must stop the rotation after capture.

### 40.3.4 Robotic Arms and Tools

Servicing robots descend from the Shuttle's arm and from the ISS's **Canadarm2** and **Dextre** (Chapter 32). But they must work without astronauts nearby, often with a time delay of a quarter of a second or more through geostationary relays. The arms that FREND and RSGS developed for GEO servicing have seven joints, like a human arm, so that they can reach around obstacles, and carry force–torque sensors at the wrist. Their controllers use **compliance** (also called impedance control). Instead of driving the hand rigidly to a position, they make it behave like a spring and damper. Contact forces then stay small even if the client is not exactly where the cameras think it is. Free-flying arms also disturb their own base: every arm motion rotates the servicer slightly in the opposite direction, and the control system must account for this coupling.

Tools turn arms into technicians. NASA's RRM tools were a wire cutter, a multifunction tool that unscrews caps, a nozzle tool that mates with standard fill-and-drain valves, and a camera tool for looking inside blanketed areas. They showed that even lock-wired valves could be opened robotically. Teleoperation from the ground remains normal for complex tasks, with operators watching stereo video and a simulation that predicts the arm's motion through the time delay; routine motions are increasingly automated.

### 40.3.5 Refuelling Interfaces and Standards

Refuelling an unprepared satellite is possible but slow, because the robot must reproduce work done by hand on the ground. The industry is therefore converging on **refuelling ports** fitted before launch (Table 40.3), which open, seal and verify the connection in one robotic operation, together with standard **grapple fixtures** for capture. Agreeing on such interfaces — and on how close servicers may approach other operators' satellites — is the work of industry groups. One is **CONFERS** (Consortium for Execution of Rendezvous and Servicing Operations), set up with DARPA support in 2017. Its principles fed into the international standard ISO 24330:2022 on rendezvous, proximity operations and on-orbit servicing.

**TABLE 40.3** Capture and refuelling interfaces and standards.

| Interface | Developer | Type | Notes |
|---|---|---|---|
| Standard fill-and-drain valve | Satellite industry | Unprepared | Opened robotically by RRM tools; slow, multi-step |
| Probe and ring docking | Northrop Grumman (MEV, MEP) | Unprepared | No fluid transfer; servicer takes over propulsion |
| RAFTI port | Orbit Fab | Prepared | Accepted by the US Space Force (2024) as a refuelling interface |
| GRIP | Orbit Fab | Servicer side | Docks with RAFTI-equipped clients; flight-qualified 2025 |
| ASSIST | GMV / ESA | Prepared | Docking and refuelling; servicer tool plus passive client fixture |
| Magnetic docking plate | Astroscale | Prepared | For capture and de-orbit at end of life |
| ISO 24330:2022 | ISO (CONFERS input) | Standard | Programmatic principles and practices for RPO and servicing |

Refuelling demonstrations on prepared clients were scheduled for 2026–27. Astroscale US's 300 kg APS-R servicer is to transfer hydrazine to one of the Space Force's Tetra-5 satellites in GEO. Orbit Fab planned to fly its GRIP system to service the same satellites. As of October 2026, neither had been reported as launched.

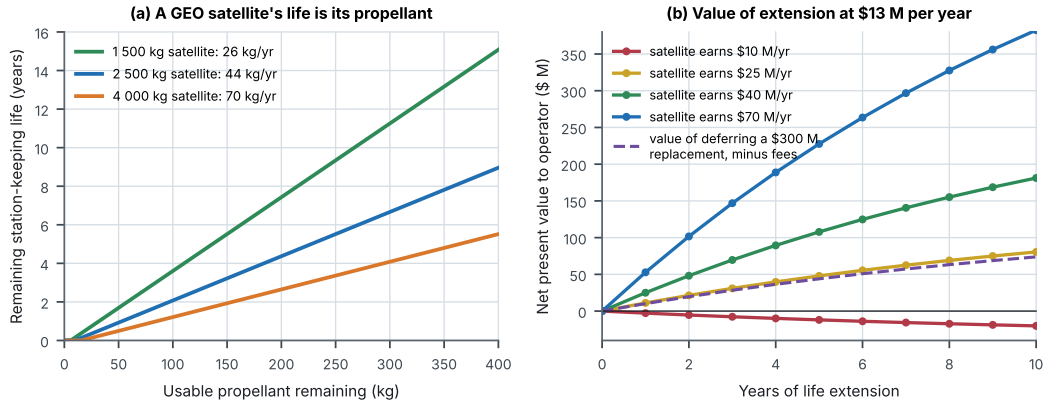
### 40.3.6 Safety, Law and Trust

Servicing raises questions that ordinary satellites do not. Under the Outer Space Treaty, a state keeps **jurisdiction and control** over an object it registered — including debris — indefinitely, so removing another nation's dead satellite or rocket stage needs that state's consent. Liability for damage in orbit depends on proving fault, which is hard when two spacecraft touch on purpose. Servicers must convince their clients' insurers and regulators that they will not turn a working satellite into a tumbling wreck. MEV-1 made the first docking with Intelsat 901 in the graveyard orbit, where an accident would not have endangered the GEO belt. And because the same technology can inspect, disable or move another country's satellite, transparency about intentions and approaches — what CONFERS calls being a "responsible actor" — matters as much as engineering (Chapter 19).

## 40.4 The Economics of Life Extension and Debris Removal

### 40.4.1 What a Satellite-Year Is Worth

A geostationary satellite drifts north–south under lunar and solar gravity and east–west under Earth's slightly elliptical equator, and needs about 50 m/s of Δv per year to stay in its slot. With chemical thrusters of 290 s specific impulse, a 2 500 kg satellite therefore burns about 44 kg of propellant a year. It must also keep about 10 kg in reserve to raise itself to the graveyard orbit at the end (Figure 40.10). With 100 kg of usable propellant left, it has about two years to live — and because propellant gauging is uncertain, operators often retire satellites with a margin that a servicer could have used.




(a) Δv 50 m/s per year, chemical Isp 290 s, 11 m/s reserved for disposal 300 km above GEO. (b) Fee ≈ $13 M/yr (reported MEV price); discount rate 8 %; net revenue and replacement cost illustrative.

**FIGURE 40.10** Economics of life extension. (a) Remaining life of a GEO satellite as a function of its usable propellant. (b) Net present value to an operator of extending a satellite's life at the reported MEV price of about $13 million a year: worthwhile if the satellite still earns well, or if extension defers an expensive replacement.

Whether extension pays depends on what the satellite still earns. A satellite with 40 transponders leased at about $1 million each per year earns $40 million a year; paying $13 million a year for five more years gives the operator a present value of about $108 million at an 8 % discount rate. Even if the satellite would otherwise be replaced immediately, deferring a $300 million replacement by five years is worth about $96 million in present value, $44 million more than the fees. Conversely, a satellite earning less than its servicing fee is not worth extending. The market is limited by the number of high-value satellites approaching the end of their propellant at any time — some tens per year. It is also limited by competition from cheaper new satellites and from LEO constellations.

### EXAMPLE: A TUG IN GEOSTATIONARY ORBIT

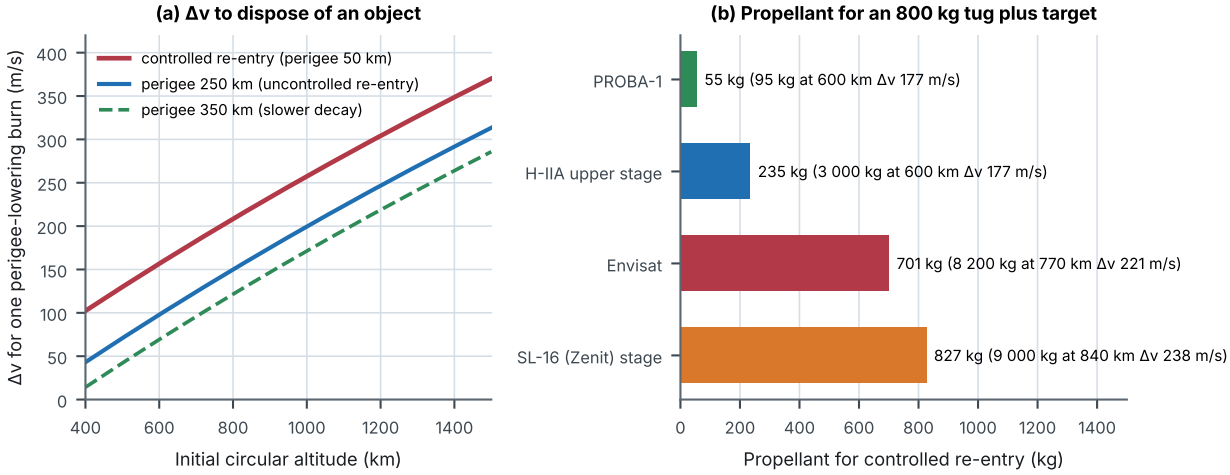
A servicer of 2 330 kg (MEV class) docks with a 2 500 kg satellite whose tanks are empty. The combined mass is 4 830 kg.

1. **Disposal.** Raising the stack 300 km above GEO (r = 42 164 km, v = 3.075 km/s) by a Hohmann transfer takes two burns of 5.44 m/s, 10.9 m/s in total. With chemical thrusters (Isp 300 s, $v_e$ = 2 942 m/s) this uses $m_p = m_0(1 - e^{-\Delta v/v_e}) = 4\,830 \times (1 - e^{-10.9/2\,942}) = \mathbf{17.8\ kg}$; with Hall-effect thrusters of Isp 1 700 s, only **3.2 kg**.
2. **Station keeping.** Keeping the stack in its slot for five years needs 5 × 50 = 250 m/s. That is **72 kg** of xenon at 1 700 s, but **393 kg** of chemical propellant at 300 s — almost the entire life-extension budget. This is why servicers use electric propulsion.
3. **Relocation.** To drift along the GEO arc at 1° per day, the orbit's semi-major axis must change by $\Delta a = \frac{2}{3}\,(\dot{\lambda}/n)\,a$ = 78 km, costing 2.8 m/s to start and 2.8 m/s to stop. Moving 90° to a new client takes three months and about 6 m/s. A single 15-year servicer can therefore serve several clients in turn, as MEV-1 has done.

## 40.4.2 Paying to Remove Debris

Debris removal has a harder business case: nobody earns revenue from a dead rocket stage. The Δv to dispose of an object grows with altitude (Figure 40.11). A controlled re-entry lowers the perigee to about 50 km so that the object falls into an empty ocean area. It needs about 157 m/s from 600 km and 218 m/s from 840 km, plus about 20 m/s for rendezvous and attitude control. For a tug of 800 kg, de-orbiting the 3 t H-IIA stage needs about 235 kg of propellant, and a 9 t Soviet-

era SL-16 stage about 830 kg, before any margin for a second attempt. One tug can therefore remove one large object, or perhaps a few small satellites — which is why proposals for multi-target removal favour attaching cheap de-orbit kits to several objects in the same orbital plane.




Single impulsive burn from a circular orbit. (b) Isp 300 s; +20 m/s for rendezvous and attitude margin; target masses and altitudes approximate.

**FIGURE 40.11** Disposing of debris. (a) Δv to lower the perigee of an object from a circular orbit, for a controlled re-entry or to accelerate natural decay. (b) Propellant needed by an 800 kg tug to de-orbit four candidate objects with a controlled re-entry.

Current removal missions are demonstrations funded by space agencies (Table 40.4): ADRAS-J2 costs about ¥12 billion for one object, and ESA's 2020 contract for ClearSpace-1 was worth €86 million. Studies suggest that removing five to ten of the most dangerous large objects per year would stabilise the LEO population (Chapter 21), which at today's prices means several hundred million dollars a year. Possible funding models include government procurement as a public good and removal fees built into launch licences. Another is a requirement that new satellites carry a docking plate, so that a cheap commercial tug can remove them if they fail.

**TABLE 40.4** Prices and budgets of selected servicing and debris-removal missions.

| Mission or service | Customer | Value | Notes |
|---|---|---|---|
| MEV life extension | Intelsat | ≈ $13 M per year (reported) | Five-year term per client |
| OSAM-1 (cancelled 2024) | NASA | ≈ $1.5 B appropriated; > $2 B estimate | Refuelling and assembly demonstration |
| ADRAS-J2 | JAXA | ≈ ¥12 B (≈ $80 M) | Capture and de-orbit one rocket stage |
| ClearSpace-1 | ESA | €86 M (2020 contract) | Capture and de-orbit PROBA-1 |
| Swift reboost (LINK) | NASA | $30 M | Abandoned 2026 |
| Otter inspection (SSPICY) | NASA | $15 M | Up to four defunct US satellites (2027) |
| Otter de-orbit services | Space Development Agency | $52.5 M | Proliferated LEO constellation |

## 40.5 Assembling Large Structures in Orbit

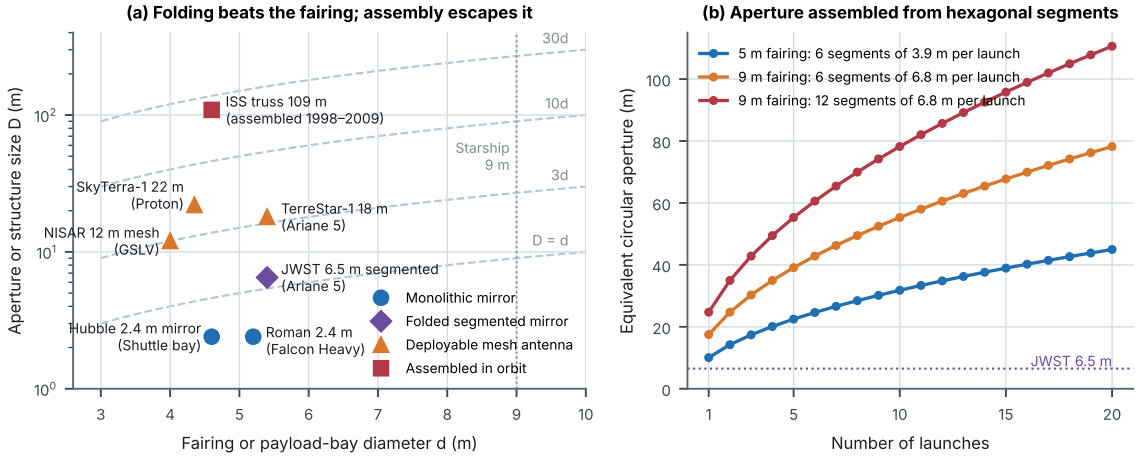
### 40.5.1 Lessons from the ISS

The ISS is the largest structure ever assembled in space: about 420 t, with a 109 m truss carrying eight solar wings, each 35 m long. It was built from components launched on more than 40 flights between 1998 and 2011. The Shuttle's arm lifted truss segments out of the payload bay and handed them to Canadarm2, which positioned them for bolting; spacewalkers then connected the power, data and coolant lines that robots could not (Figure 40.12). Hundreds of hours of spacewalks went into assembly (Chapter 33). The station's design shows what makes assembly possible. It uses standard mechanical interfaces (the segment-to-segment attach system and the common berthing mechanism). Its structures were tested together on the ground before launch wherever possible, and its connectors were designed for gloved hands or robot end-effectors.

**FIGURE 40.12** The Starboard One (S1) truss segment newly installed on the ISS in October 2002, during the STS-112 assembly mission; Canadarm2 is visible at left. Each truss segment was berthed by robotic arms and connected by spacewalkers. Image: NASA/JSC.

### 40.5.2 Beyond the Fairing

Every launch vehicle limits the size of what it carries to the diameter of its fairing — about 5 m for most large rockets, 7 m for New Glenn and 9 m for Starship. Designers escape this limit in three ways (Figure 40.13). A **monolithic** mirror must fit inside the fairing: Hubble's and Roman's are 2.4 m. **Folding** gains a factor of up to about five. JWST's 6.5 m segmented mirror was folded into a 5.4 m fairing. The gold-plated mesh reflectors of mobile communication satellites such as TerreStar-1 (18 m) and SkyTerra-1 (22 m), and NISAR's 12 m radar antenna, unfurl like umbrellas. Beyond that, structures must be **assembled**, as the ISS was.




(a) Nominal fairing/bay outer diameters. (b) Hexagons with flat-to-flat width 0.85 × usable fairing diameter (4.6 m or 8 m); D = √(4A/π); gaps, edge losses and support structure ignored.

**FIGURE 40.13** Aperture size versus launcher. (a) Monolithic, folded and mesh structures stay within a few fairing diameters; only assembly escapes the limit. (b) Equivalent aperture of a telescope assembled from hexagonal segments, versus number of launches, for illustrative packing assumptions.

Assembly trades launch volume for complexity. In the illustrative case of panel (b), a 20 m telescope needs about four launches of segments in 5 m fairings. It needs two in a 9 m fairing, or one if twelve segments can be stacked inside it. The challenges lie in precision. Mirror segments must be aligned to a small fraction of a wavelength — tens of nanometres for JWST, far finer for a telescope searching for Earth-like planets. So they need actuators and wavefront sensing like JWST's (Chapter 38), and the supporting truss must be stiff and thermally stable. Robotic assembly has been demonstrated on Earth and in small experiments; NASA's in-orbit demonstrations — OSAM-1's SPIDER antenna assembly and OSAM-2's printed beams — were cancelled before flight. NASA's early studies of the Habitable Worlds Observatory assume that it will be serviceable, with standard interfaces and an orbit reachable by servicers, so that instruments could be replaced as Hubble's were. Modular spacecraft with standard mechanical, electrical and data interfaces between their blocks — sometimes called "Lego-like" — are a related idea. A damaged module could be swapped and a new payload added without replacing the whole satellite.

## 40.5.3 Space-Based Solar Power

The most ambitious assembly concept is a **space solar power station**, collecting sunlight continuously in GEO and beaming it to Earth as microwaves. The physics sets its scale. To deliver 1 GW with about 10 % end-to-end efficiency from sunlight at 1.36 kW/m$^2$ requires a collecting area of about 7 km$^2$. That means a structure of thousands of tonnes even at 1 kg/m$^2$ — far beyond any foldable spacecraft. Caltech's **SSPD-1** demonstrator, launched in January 2023, showed the first steps. Its MAPLE experiment, a lightweight array of microwave transmitters steered electronically, beamed power to receivers in orbit. It also sent a detectable beam to a receiver on the roof of a laboratory at Caltech. Commercial viability would require launch costs, mass per unit area and robotic assembly far beyond today's.

# 40.6 Manufacturing in Space

## 40.6.1 Why Make Things in Orbit?

Manufacturing in orbit offers three things that are hard to get on Earth. The first is **weightlessness**, which suppresses buoyant convection and the settling of heavier components, so that crystals grow more evenly and mixtures do not separate. The second is **containerless processing**, in which a liquid can solidify without touching walls that would contaminate or seed it. The third is an unlimited **high vacuum**. Against these stand the costs of launch and return, a few cubic metres of space on a station, and the need to automate almost everything. In-space manufacturing therefore targets two niches: products so valuable per kilogram that they justify the round trip (Table 40.5), and parts made in space for use in space, which save the cost of launching spares.

**TABLE 40.5** Candidate products for in-space manufacturing.

| Product | Why microgravity helps | Status (2026) | Example |
|---|---|---|---|
| Spare parts (polymer) | Print on demand instead of stocking spares | ISS printers since 2014 | Made In Space (Redwire) printers |
| Metal parts | Strong tools and parts for long missions | First metal print 2024 | ESA/Airbus printer in Columbus |
| ZBLAN optical fibre | Less crystallisation, so lower optical loss | > 11 km drawn on ISS (2024) | Flawless Photonics |
| Protein and drug crystals | Larger, more uniform crystals | Routine ISS research; capsule returns | Merck, Varda (ritonavir) |
| Bioprinted tissue | Soft tissue holds shape without scaffolds | Meniscus printed 2023 | Redwire BioFabrication Facility |
| Semiconductor crystals | Purer crystals grown from gas or melt | Furnace operated in orbit 2025 | Space Forge ForgeStar-1 |
| Large structures | Unlimited length; no launch loads | Studies; demos cancelled | OSAM-2 beam printing |

## 40.6.2 Printing in Orbit

The first 3D printer in space, built by Made In Space for NASA, reached the ISS in 2014 and printed its first part — a faceplate for its own extruder — in November (Figure 40.14). Its test coupons, compared with identical ones printed on the ground, showed that extrusion works normally in weightlessness. Later in 2014 it printed a ratchet wrench from a file sent up from the ground. A commercial printer followed in 2016, and the company, now part of Redwire, printed hundreds of parts. Metal is harder: molten metal must be controlled without gravity, and powder beds cannot be used because loose powder floats. ESA and Airbus's printer, installed in the Columbus module in 2024 (Figure 40.15), melts stainless-steel wire with a laser. It produced its first part that summer. Samples were returned to Earth for testing against parts printed on the ground.

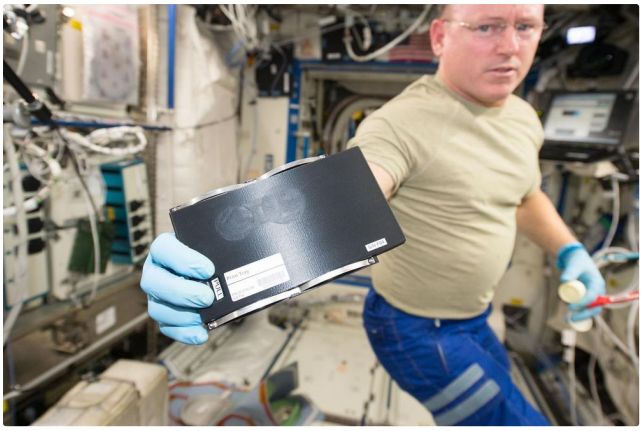


**FIGURE 40.14** Barry Wilmore holds a test coupon made by the first 3D printer in space, aboard the ISS in December 2014. Comparing such coupons with identical ones made on the ground showed that printing works in weightlessness. Image: NASA/JSC (Barry Wilmore).

**FIGURE 40.15** Jeanette Epps configures the ESA metal 3D printer in the Columbus module in August 2024. It melts stainless-steel wire with a laser inside a sealed chamber. Image: NASA/JSC.

### 40.6.3 Fibres, Crystals and Tissue

**ZBLAN** is a fluoride glass (named after its zirconium, barium, lanthanum, aluminium and sodium components) that in theory transmits infrared light far better than silica. But on Earth tiny crystals form as it is drawn into fibre, and they scatter light. In weightlessness, without convection and sedimentation, fewer crystals form. In 2024 the start-up Flawless Photonics drew more than 11 km of ZBLAN fibre on the ISS, the longest run yet; whether its quality justifies the cost of production in orbit is still being assessed.

**Protein crystals** grown in weightlessness are often larger and more regular. This helps determine molecular structures by X-ray diffraction and can improve drug formulation. Experiments on the ISS helped Merck develop a more uniform crystalline suspension of its cancer drug pembrolizumab. **Bioprinting** benefits because soft tissues printed from cells and gels do not collapse under their own weight. Redwire's BioFabrication Facility on the ISS printed a human knee meniscus in 2023, a first step towards printing tissues for transplantation.

### 40.6.4 Free-Flying Factories

Stations offer crew help but little throughput. **Varda Space Industries** builds small free-flying factories: a satellite bus carries a process module and a re-entry capsule, which returns its product at almost orbital speed and lands under a parachute. W-1 grew crystals of the HIV drug ritonavir. After months waiting for permission to land, it came down in Utah in February 2024. This was the first re-entry licensed under the US FAA's newer Part 450 rules, and the first time the FAA had licensed a commercial spacecraft to land on US soil. Its successors land at the Koonibba Test Range in South Australia: W-2 in February 2025 and W-3 in May 2025. W-5 followed in January 2026 with the first heat shield made entirely of Varda's own carbon ablator, and W-6 in May 2026. Each capsule is also a **hypersonic test bed**, carrying re-entry experiments for the US Air Force Research Laboratory and, on W-6, NASA — an important source of revenue alongside pharmaceutical work. In the UK, **Space Forge**'s ForgeStar-1, launched in June 2025, ignited a plasma at about 1 000 °C in its miniature furnace in December 2025. This was a step towards growing semiconductor crystals in orbit and returning them on later missions.

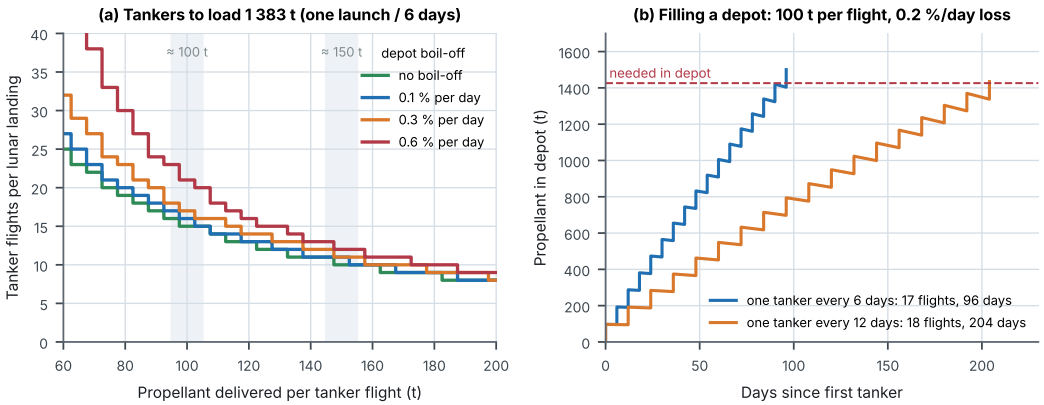
The economics of all these products depends on the price of the round trip. At a rideshare price of a few thousand dollars per kilogram to orbit, plus a capsule and recovery, only very valuable products are plausible candidates. They must be worth hundreds of thousands of dollars per kilogram — drug crystals, seed crystals and specialised fibre. Lower launch costs and reusable capsules would widen the range.

## 40.7 Refuelling for Exploration

Refuelling in orbit matters most for exploration. A rocket that arrives in low Earth orbit has used most of its propellant. If it can be refilled there, it can carry far more payload to the Moon or Mars than its launch vehicle alone could send. Both Artemis landers depend on this. SpaceX's Starship HLS is filled in LEO from a depot ship loaded by a series of tanker launches. Blue Origin's Blue Moon Mk2 is to be refuelled by a separate transporter vehicle.

Cryogenic propellants make this hard. Liquid oxygen boils at −183 °C and methane at −162 °C, so heat leaking into a tank boils them away continuously — **boil-off** — unless the tank is shaded, insulated or actively cooled. In weightlessness the liquid floats as blobs and does not sit over the outlet, so transfer needs a small settling acceleration or capillary devices. Transfer lines must first be chilled, wasting some propellant as gas. And the amount in a tank is hard to measure without gravity. RRM3's methane experiment failed before it could test these effects. In March 2024, on Starship's third flight, SpaceX transferred liquid oxygen between tanks inside one ship. The first ship-to-ship transfer, between two Starships docked in orbit, was planned for no earlier than late 2026; Starship first reached orbit on 28 September 2026. NASA restructured Artemis in 2026 so that Artemis III, in 2027, will rehearse rendezvous and docking with the landers in Earth orbit, with the first landing on Artemis IV.

How many tanker flights a landing needs depends on assumptions that SpaceX has not published in detail. Public estimates range from about 10 (SpaceX) to the "high teens" (NASA officials). Figure 40.16 shows how the number depends on how much each tanker delivers and how fast the depot loses propellant.




Assumed lander: 120 t dry + 15 t payload Δv 8.9 km/s after leaving LEO (to NRHO landing ascent) Isp 375 s → 1 383 t of propellant. 97 % efficiency for each transfer. Illustrative; not SpaceX figures.

**FIGURE 40.16** Propellant transfer for a lunar lander. (a) Tanker flights needed to load the lander, versus propellant delivered per tanker, for different depot boil-off rates. (b) Propellant in the depot during a campaign with one tanker every 6 or 12 days: slower launch rates lose more to boil-off.

### EXAMPLE: HOW MANY TANKERS FOR A LUNAR LANDING?

Assume (these are not SpaceX figures) a lander of 120 t dry mass carrying 15 t, with Raptor engines of Isp 375 s ($v_e$ = 3 677 m/s). After leaving LEO it needs about 3.6 km/s to reach near-rectilinear halo orbit (NRHO), 2.65 km/s to land and 2.65 km/s to return to NRHO: 8.9 km/s in total.

1. Mass ratio: $e^{8\,900/3\,677}$ = 11.25, so the lander must leave LEO with 135 × 11.25 = 1 518 t, of which **1 383 t** is propellant.
2. If each tanker delivers 100 t, and each transfer (tanker to depot, depot to lander) loses 3 %, the lander needs 1 383 / (0.97 × 0.97 × 100) = 14.7. This rounds up to **15 tanker flights** with no boil-off.
3. With boil-off of 0.2 % per day in the depot and one tanker every 6 days, the campaign takes **17 flights over 96 days**; at one every 12 days, 18 flights over 204 days. Add the depot and the lander themselves, and a single landing needs about 19–20 launches.
4. If each tanker could deliver 150 t, the count falls to about **11**. The answer is most sensitive to tanker payload and launch rate — which is why rapid reuse is central to the plan.

### AI AND AUTONOMY FOR SERVICING

Servicing is one of the most demanding applications of space autonomy. Approaching an unprepared, tumbling object requires the servicer to estimate its pose from camera images in real time. Convolutional neural networks trained on synthetic images now estimate pose from a single image. They have been tested in international satellite pose-estimation competitions run by ESA and Stanford. But the gap between rendered training images and real lighting remains the main difficulty. Onboard guidance must decide when to abort — ADRAS-J's autonomous abort in 2024 is an example — and learning-based planners are being studied for grasping tumbling targets and for scheduling multi-client missions. On the ground, machine-learning models monitor the health of the servicer and client and flag anomalies such as unexpected attitude behaviour (Chapters 45, 48 and 49).

### KEY TAKEAWAYS

- ISAM spans inspection, relocation, life extension, refuelling, repair, assembly and manufacturing; each step up the ladder adds harder requirements for capture, fluid transfer and dexterity.
- Astronauts serviced Solar Max, Intelsat 603 and Hubble and assembled the ISS. Since 2020 commercial robots have docked with satellites as a paid service (MEV-1, MEV-2; MRV and MEPs launched July 2026). China has towed and probably refuelled satellites in GEO.
- Rendezvous uses V-bar hops and R-bar approaches designed to be passively safe. Final approaches are slow — centimetres per second, slower still in GEO. Capture of unprepared or tumbling clients relies on launch-adapter rings, engine nozzles and careful detumbling.
- Life extension pays when a satellite earns more than the roughly $13 million per year fee or when it defers a replacement; debris removal remains a government-funded demonstration market costing about $80–100 million per object.
- Assembly escapes the fairing limit that caps monolithic and folded structures at a few fairing diameters; NASA's robotic assembly demonstrations were cancelled, but the ISS proves the principle.
- Manufacturing in orbit has moved from ISS printers to free-flying factories with return capsules; refuelling in orbit, still undemonstrated between two vehicles, is the key to Artemis landers that need 10–20 tanker flights each.

### REVIEW QUESTIONS

1. List the main ISAM activities in order of difficulty and explain what each additional step requires of the servicer and the client.
2. Why did servicing Solar Max and Hubble succeed while the first attempts to capture Intelsat 603 failed? What does this imply for designing future satellites?
3. How does the Mission Extension Vehicle attach to a client that was never designed for servicing, and why does it use electric propulsion?
4. Using the Clohessy–Wiltshire equations, calculate the $\Delta v$ for a 2 km radial hop along the V-bar in a 500 km orbit, and the time it takes.
5. Explain "passive safety" in rendezvous. Why must approaches in GEO be slower than in LEO?
6. A dead satellite with a moment of inertia of $5 \times 10^4$ kg·m$^2$ tumbles at 2° per second. What is its angular momentum, and how could a servicer remove it after capture?
7. A GEO satellite of 3 000 kg has 80 kg of usable propellant. Estimate its remaining life with chemical thrusters of 290 s, keeping 11 m/s for disposal.
8. Why do debris-removal missions struggle to find a business case, and what funding models have been proposed?
9. Explain why folding structures cannot grow beyond a few fairing diameters, and outline the engineering challenges of assembling a 20 m telescope in orbit.
10. Explain the difficulties of transferring cryogenic propellants in orbit and estimate how the number of tanker flights for a lunar landing changes if each tanker delivers 150 t instead of 100 t.

CHAPTER

# 41 Space Resources and Asteroid Mining

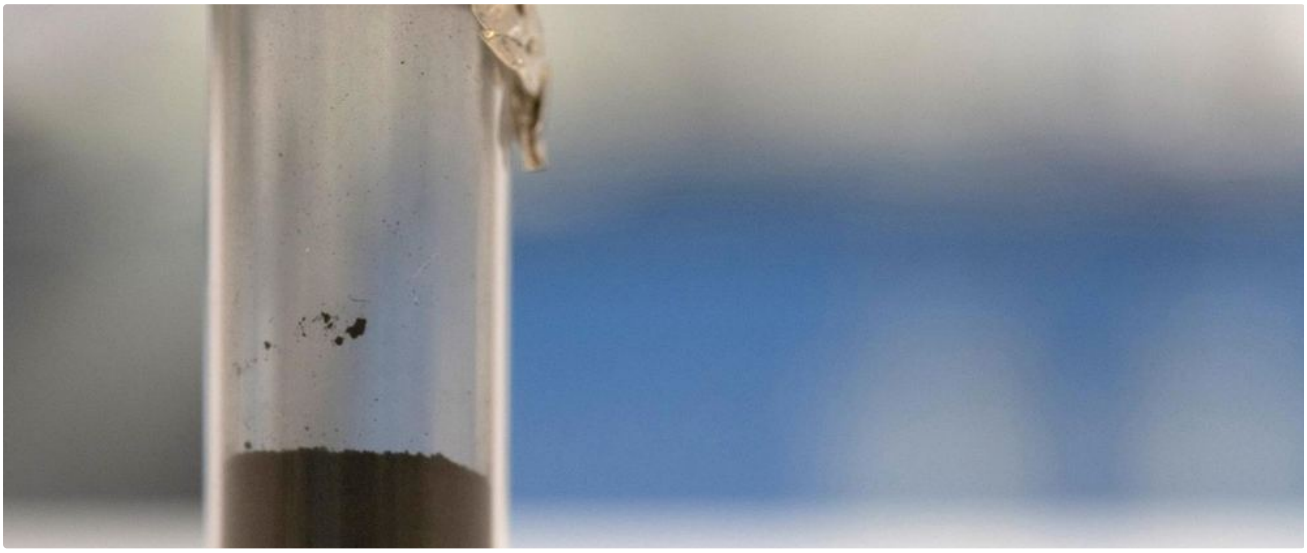


Image: NASA/GSFC

*Every kilogram of propellant, water or shielding used beyond Earth orbit must be lifted out of a deep gravity well at great cost. Space resources — ice and oxygen-rich rock on the Moon, carbon dioxide and ice on Mars, water and metals in asteroids — promise to cut that cost by making what is needed where it is needed. This chapter quantifies the "gear ratio" that makes local production attractive, surveys what is known about each resource, and explains the processes that turn regolith and ice into water, oxygen, propellant, metals and building material, with their energy costs. It then reviews the prospecting missions from LCROSS to VIPER, the rise and fall of asteroid-mining companies, the economics of selling space metals on Earth, and the laws that decide who may own what is extracted.*

**IN THIS CHAPTER**

41.1 Why Use Space Resources?
41.2 What Is Out There
41.3 Turning Rock and Ice into Products
41.4 Prospecting: Finding the Ore
41.5 The Business of Space Mining
41.6 Law and Policy

Polar explorers hauled everything on sledges until they learned to hunt and to build shelters from snow. Spaceflight is still in the sledge-hauling era: every Apollo crew carried all its oxygen, water and propellant from Earth. Because of the exponential rocket equation the penalty is severe — with the assumptions used in this chapter, each kilogram placed on the lunar surface requires about 130 kg of rocket on the launch pad.

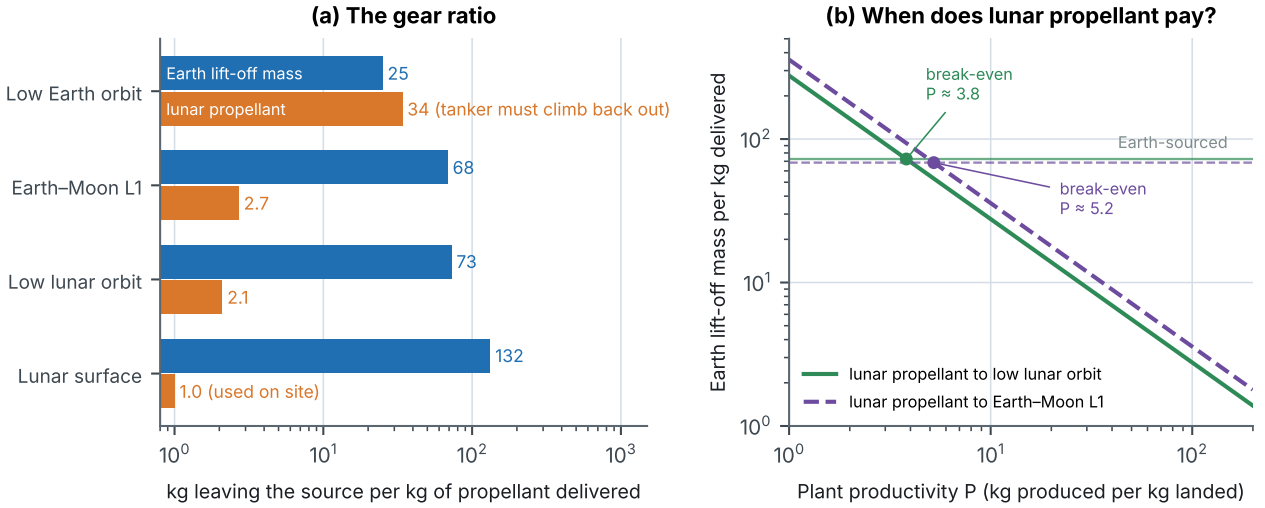
**In situ resource utilisation (ISRU)** — from the Latin *in situ*, "in place" — means producing useful materials from what is found at the destination. The most valuable products are mundane. One is **oxygen**, about four-fifths of the propellant mass of hydrogen–oxygen and methane–oxygen rockets. Another is **water**, for drinking, growing food and radiation shielding. A third is **regolith**, the loose rock and dust covering the Moon, Mars and asteroids, for building landing pads, roads and shelters. Further off lie metals for building in space, and precious metals and helium-3 for Earth.

This chapter covers the arithmetic that makes space resources attractive, the resources themselves, the chemistry and energy cost of extracting them, prospecting, the mining companies and the law. Landers and rovers are covered in Chapters 27 and 28, propellant transfer in orbit in Chapter 40.

## 41.1 Why Use Space Resources?

### 41.1.1 The Gear Ratio

The rocket equation (Chapter 34), $\Delta v = v_e \ln(m_0/m_f)$, makes the mass needed grow exponentially with the velocity change. Reaching low Earth orbit (LEO) takes about 9.4 km/s including losses, so even a large launcher delivers only about 4 % of its lift-off mass to orbit. From LEO, a further 4.0 km/s reaches low lunar orbit (LLO) and 5.9 km/s the lunar surface. The ratio of mass leaving Earth to mass delivered is the **gear ratio** of a destination (Figure 41.1 a).




Launcher: 4 % of lift-off mass reaches LEO. Earth-sourced: expendable LOX/$LH_2$ stages, Isp 450 s, dry mass 10 % of propellant; Δv LEO→L1 3.8, →LLO 4.0, →lunar surface 5.9 km/s. Moon-sourced: reusable LOX/$LH_2$ tanker (10 t dry, 30 t loaded) keeps propellant to return; Δv surface→LLO 1.9, →L1 2.5, →LEO 2.8 km/s (aerobraking; 5.9 km/s back). Plant and tanker mass in (b) landed from Earth.

**FIGURE 41.1** The gear ratio. (a) Mass leaving its source per kilogram of propellant delivered: Earth lift-off mass for Earth-launched propellant, and lunar propellant for a reusable lunar tanker. (b) Earth lift-off mass per kilogram delivered to low lunar orbit or L1, against the productivity of a lunar plant.

With hydrogen–oxygen stages of specific impulse 450 s whose empty mass is 10 % of their propellant, each kilogram delivered to LLO needs 2.9 kg in LEO and about 73 kg at lift-off. Each kilogram on the lunar surface needs 5.3 kg in LEO and 132 kg at lift-off. Propellant made on the Moon starts in a far shallower gravity well — escape velocity 2.4 km/s against Earth's 11.2 km/s,

with no atmosphere. A reusable tanker that carries lunar propellant to LLO, keeping enough to fly back down, burns only about 1.1 kg for every kilogram it delivers: a gear ratio of 2.1 instead of 73. Delivery to the Earth–Moon L1 point, a proposed depot site, costs 2.7.

The comparison reverses in LEO. A tanker can brake into LEO in the atmosphere (aerobraking), needing only about 2.8 km/s, but then needs the full 5.9 km/s to return. Each kilogram delivered costs 34 kg of lunar propellant — worse than launching from Earth — or about 5 if the tanker refuels with lunar propellant at L1 on the way back. Lunar propellant is most valuable close to the Moon.

### 41.1.2 When Does Local Production Pay?

A lunar plant is not free: its excavators, reactors, power system and tanker must be landed from Earth at 132 kg of lift-off mass per kilogram. The figure of merit is the plant's **productivity** $P$, the mass of product made over its life divided by its landed mass. Lunar propellant delivered to LLO beats Earth-launched propellant once $P$ exceeds about 3.8, and at L1 once it exceeds about 5.2 (Figure 41.1 b). The question is whether a lunar plant can survive abrasive dust, temperatures from 40 K to 390 K and years without maintenance for long enough.

**EXAMPLE: THE LEVERAGE OF A LUNAR-ORBIT TANKER**

A reusable tanker with an empty mass of 10 t carries 30 t of liquid oxygen and hydrogen from a polar plant to a depot in low lunar orbit and returns. Its engines have $I_{sp}$ = 450 s ($v_e$ = 4 413 m/s); each leg needs Δv = 1.9 km/s.

1. Mass ratio per leg: $R = e^{1\,900/4\,413}$ = 1.538.
2. Ascent propellant (10 t of tanker plus 30 t of cargo): 40 × (1.538 − 1) = 21.5 t. To return, the empty tanker keeps 10 × 0.538 = 5.4 t, so it delivers 30 − 5.4 = **24.6 t** per trip.
3. Lunar production per trip: 21.5 + 30 = 51.5 t — a gear ratio of 51.5 / 24.6 = **2.09**.
4. Delivering 24.6 t from Earth instead would need 24.6 × 2.90 = 71.5 t in LEO, or about **1 790 t** of launch vehicle at lift-off.
5. Suppose the plant, its power supply and the tanker land 40 t from Earth (5 290 t at lift-off) and produce 100 t of propellant a year for 10 years ($P$ = 25). They deliver 1 000 / 2.09 = 478 t to LLO, at 5 290 / 478 = **11 kg** of Earth lift-off mass per kilogram, against 73 kg for Earth-launched propellant — a sixfold gain. At $P$ = 3.8 the advantage vanishes.

The same logic applies to Mars, where an ascent needs about 4–5 km/s (Chapter 29), and to other products: a tonne of regolith piled over a habitat as shielding need not be launched.

## 41.2 What Is Out There

The resources differ in concentration by a factor of a billion — from oxygen, nearly half of every rock, to helium-3, a few parts per billion of lunar soil. Figure 41.4 compares the elemental make-up of typical feedstocks, and Table 41.1, at the end of this section, summarises the resources.

### 41.2.1 Lunar Regolith

The Moon is covered by **regolith**: rock fragments, glass and dust produced by four billion years of meteorite impacts. It is about 4–5 m deep on the dark volcanic plains (**maria**) and 10–15 m in the older **highlands**. It is made of oxides: oxygen is 41–45 % of the mass, bound to silicon, aluminium, iron, calcium, magnesium and titanium. Highland soils are rich in aluminium and calcium. Mare soils carry up to about 20 % FeO. In the "high-titanium" basalts sampled by Apollo 11 and 17, they carry 10 % or more of the iron–titanium oxide **ilmenite** ($FeTiO_3$), the favourite feedstock for oxygen extraction (Figure 41.2). Volcanic glass beads such as Apollo 17's orange soil (Figure 41.3) are also rich in these oxides.

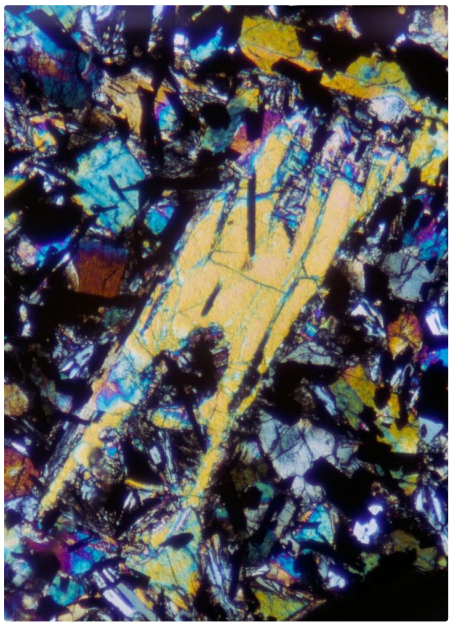
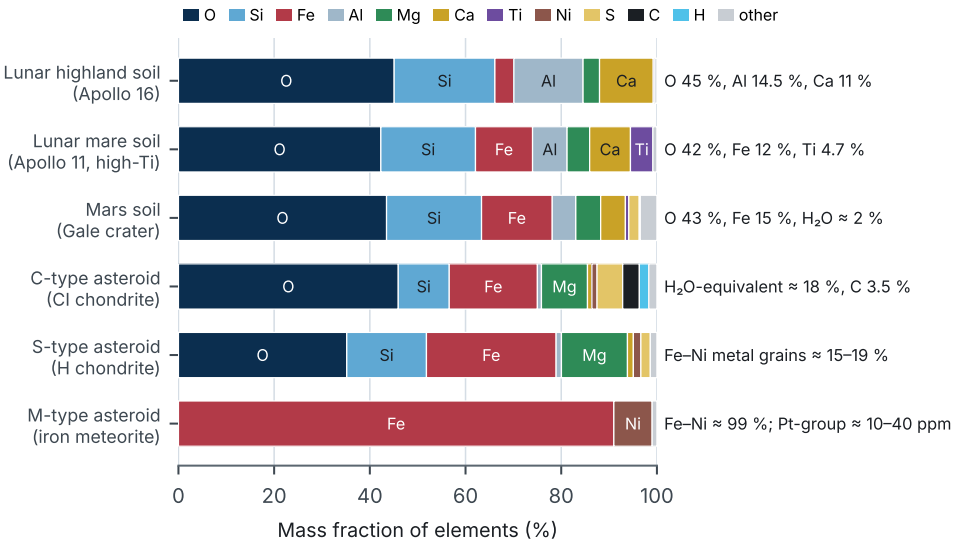


**FIGURE 41.2** Photomicrograph of a thin slice of Apollo 11 basalt 10022. Black grains are ilmenite, light blue and white grains are plagioclase and the brightly coloured grains are pyroxene. Image: NASA/JSC.

**FIGURE 41.3** The orange soil discovered by Apollo 17 at Shorty crater in December 1972 — tiny beads of volcanic glass rich in titanium and iron. Image: NASA/JSC.



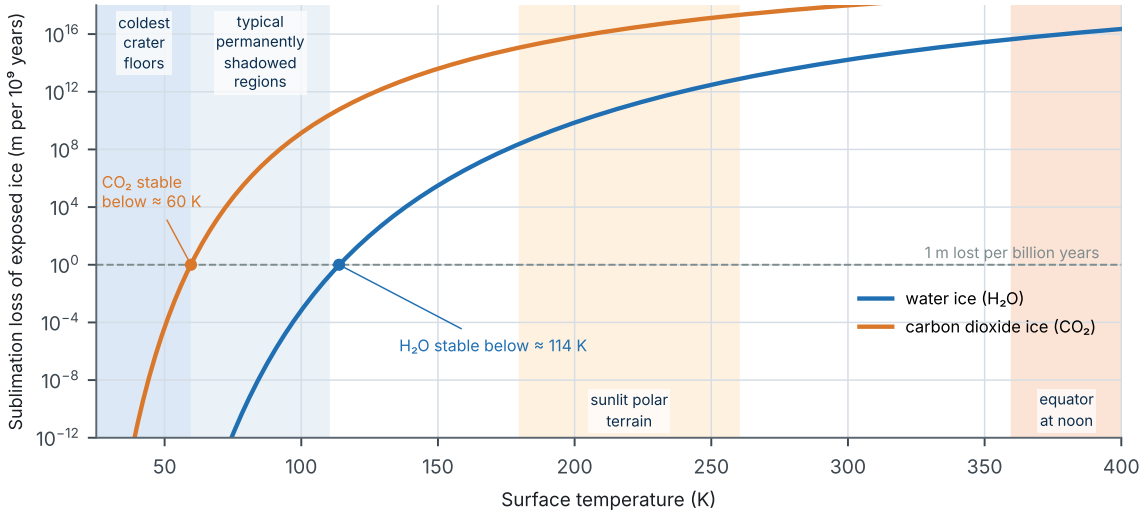

Element fractions from average oxide analyses (lunar soils 64501-type and 10084; MSL Rocknest with ≈ 2 % $H_2O$), CI (Lodders 2003), H-chondrite and iron-meteorite averages. Representative values only.

**FIGURE 41.4** Elemental composition by mass of lunar highland and mare soils, Martian soil and the meteorite types matching C-, S- and M-type asteroids.

Regolith also holds **solar-wind** atoms implanted into grain surfaces, hydrogen at tens of parts per million (ppm). On the western near side, the **Procellarum KREEP Terrane** is enriched in potassium (K), rare-earth elements (REE) and phosphorus (P). But its rare-earth concentrations, though hundreds of times those of primitive meteorites, are far below those of rich ores on Earth.

### 41.2.2 Ice at the Poles

The Moon's spin axis is tilted only 1.54° to the ecliptic, so crater floors near the poles never see the Sun (Chapter 28). In these **permanently shadowed regions (PSRs)** the Diviner radiometer on NASA's Lunar Reconnaissance Orbiter (LRO) measures temperatures below 40 K, and about 25 K in the coldest places. Water molecules from comets, asteroids, volcanic gases or the solar wind hop across the surface until they land in such a **cold trap** and freeze. Figure 41.5 shows why. Exposed ice at 114 K loses about a metre per billion years by sublimation, but at 150 K it loses hundreds of kilometres. Carbon dioxide ice needs temperatures below about 60 K.




Free sublimation into vacuum (Hertz–Knudsen) with ice vapour pressure from Murphy & Koop (2005) and $CO_2$ from Clausius–Clapeyron; a regolith cover slows the loss further.

**FIGURE 41.5** Sublimation loss of exposed water and carbon dioxide ice in vacuum. Water ice survives for billions of years only below about 110–115 K, on permanently shadowed crater floors; $CO_2$ ice only in the coldest.

The evidence accumulated over three decades:

- **Lunar Prospector** (1998) found excess hydrogen at both poles with a neutron spectrometer: hydrogen nuclei slow the neutrons leaking from the surface.
- NASA's Moon Mineralogy Mapper on India's **Chandrayaan-1** (2008–09) detected hydroxyl and water bound in minerals across much of the surface; a 2018 reanalysis found the infrared signature of water ice exposed in some PSRs.
- **LCROSS** (9 October 2009) crashed a spent 2.3 t Centaur upper stage into Cabeus crater and flew through the plume. It measured 5.6 ± 2.9 % water by mass in the ejecta, together with hydrogen sulphide, ammonia, sulphur dioxide, carbon dioxide and methanol — useful, but also contaminants that a water plant must remove.
- **LRO** (since 2009) mapped cold traps and found ultraviolet hints of patchy frost. **ShadowCam** on Korea's Danuri orbiter images PSRs by light scattered from sunlit terrain. In 2026 it found no widespread surface ice above about 20–30 % by mass — only a few patches 20–50 m across with probably more than 10 %.

- **Chandrayaan-3**'s ChaSTE probe measured 82 °C on a Sun-facing slope at 69° S in 2023, implying that steep pole-facing slopes at high latitudes may keep ice near the surface even outside PSRs.

Cold traps, including countless "micro cold traps" a few metres across, cover about 40 000 km$^2$, 60 % of it in the south. Their ice content is uncertain by orders of magnitude. If the top metre of all of them held LCROSS-like 5.6 % ice at a regolith density of 1 500 kg/m$^3$, the total would be about 3 billion tonnes. That is enough for a thousand tonnes of propellant a year for millions of years. The real deposits are probably patchy and buried, which is why prospecting on the ground matters.

### 41.2.3 Helium-3: A Realistic Assessment

Helium-3, a light isotope carried by the solar wind, is often cited as a reason to mine the Moon because its fusion with deuterium releases few neutrons. The arithmetic is daunting. Lunar regolith contains on average about 4 parts per billion (ppb) of helium-3, and 10–20 ppb in the richest titanium-bearing mare soils, where ilmenite retains it best. A litre of the gas weighs 0.135 g, so at 10 ppb about 13 t of regolith must be dug and heated to several hundred degrees Celsius per litre — 34 t at 4 ppb. Deuterium–helium-3 fusion also needs plasma temperatures several times higher than the deuterium–tritium reaction, which no power plant has yet used to produce net electricity.

The near-term market is **cryogenics**: dilution refrigerators that cool quantum computers to millikelvin temperatures need helium-3. Terrestrial supply, mostly from the decay of tritium, is tiny, and it sells for about US$ 2 500–3 000 per litre — some US$ 20 million per kilogram. The start-up **Interlune** has agreements to supply 3 litres to the US Department of Energy by April 2029 and up to 10 000 litres a year in 2028–37 to the cryostat maker Bluefors. At 10 ppb, 10 000 litres a year means processing 135 000 t of regolith a year, 15 t every hour: a credible niche product, not an energy source.

### 41.2.4 Mars

Mars offers two resources almost everywhere: an atmosphere that is 95 % carbon dioxide at a mean surface pressure of about 0.6 kPa, and water. The polar caps hold enough ice to cover the planet about 20 m deep, but lie far from the low latitudes where landing is easiest. Closer to hand, Phoenix scraped into ice centimetres below the soil at 68° N in 2008. Orbital radar has mapped buried mid-latitude glaciers. Fresh craters expose ice within a metre of the surface as far from the poles as about 40° (Figure 41.6). Even soil in Gale crater releases about 2 % water when heated.

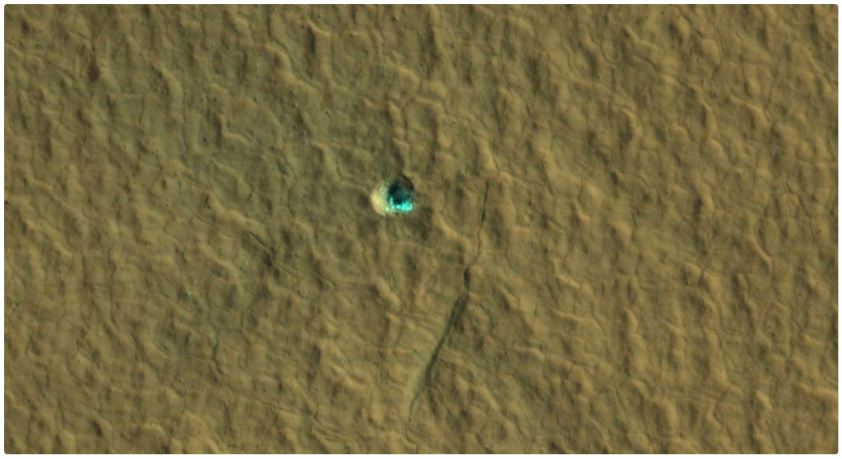


**FIGURE 41.6** A small, fresh impact crater on Mars that has exposed bright subsurface ice, surrounded by polygon-patterned ground, imaged by the HiRISE camera on Mars Reconnaissance Orbiter. Craters like this are used to map shallow ice near candidate landing sites. Image: NASA/JPL (NASA/JPL-Caltech/University of Arizona).

### 41.2.5 Asteroids

Asteroids are classified by their reflectance spectra; three classes matter most:

- **C-type** (carbonaceous) asteroids are dark, dominate the outer main belt and match carbonaceous chondrites. The most primitive, CI chondrites, hold water equivalent to 13–20 % of their mass, mostly in clays, plus 3–4 % carbon. Samples from Ryugu and Bennu are similar clay- and carbon-rich material, though Ryugu's grains hold only about 7 % water (Chapter 29).
- **S-type** (stony) asteroids dominate the inner belt and the known near-Earth population. They match ordinary chondrites, which in the H group contain 15–19 % iron–nickel metal; NEAR Shoemaker confirmed such a composition at the 34 km Eros in 2000–01 (Figure 41.7).
- **M-type** asteroids have spectra and radar echoes suggestive of metal. Iron meteorites, their likely counterparts, are about 90 % iron and 5–10 % nickel. They hold 10–40 ppm of **platinum-group metals** (platinum, palladium, rhodium, iridium, osmium, ruthenium). This is several times the grade of the richest ores on Earth, which hold a few grams per tonne.

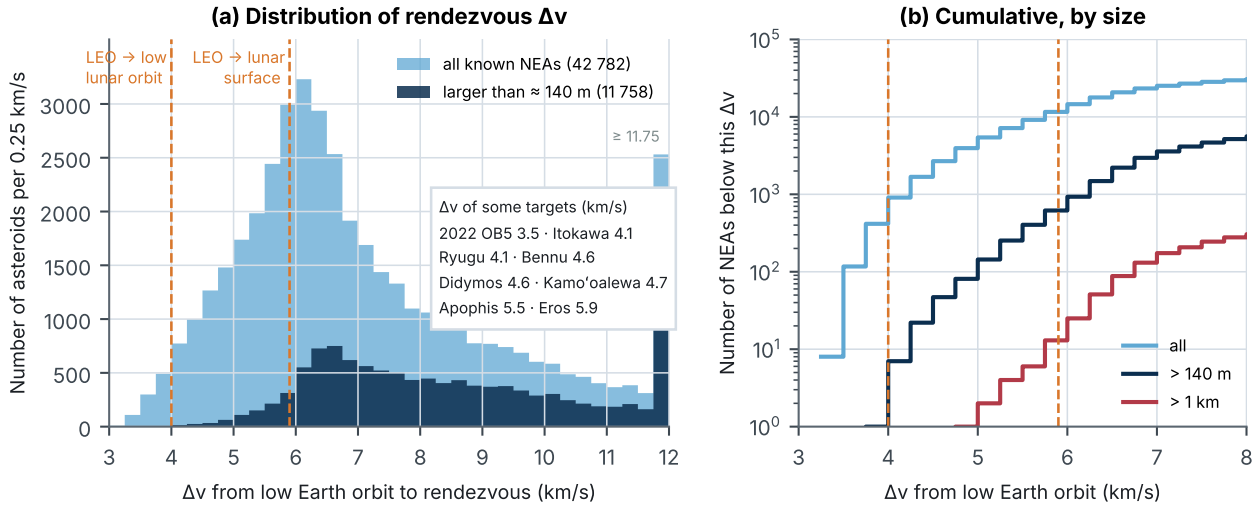
The largest M-type, **16 Psyche**, about 220 km across, was thought to be the bare iron core of a protoplanet. Its density of 3 400–4 100 kg/m$^3$ suggests instead perhaps 30–60 % metal by volume, mixed with rock. NASA's Psyche spacecraft (Figure 41.8), launched in October 2023 and driven by Hall-effect thrusters, flew past Mars in May 2026 and will arrive in 2029. Headline claims that Psyche is "worth" quintillions of dollars are meaningless: such supply would collapse metal prices.

**FIGURE 41.8** NASA's Psyche spacecraft in a clean room near Kennedy Space Center in June 2023, during final assembly before its launch to the metal-rich asteroid 16 Psyche. Image: NASA/KSC (NASA/Frank Michaux).

**FIGURE 41.7** Mosaic of the northern hemisphere of the S-type asteroid Eros, 34 km long, imaged by NEAR Shoemaker in 2000. Image: NASA/JPL (NASA/JPL/JHUAPL).

For mining, the important asteroids are the **near-Earth asteroids (NEAs)**, whose orbits come within 1.3 astronomical units of the Sun. On 19 October 2026 the JPL Small-Body Database listed 42 782, including 875 larger than about 1 km and 11 758 larger than about 140 m; new surveys will multiply the number (Chapter 26).

Figure 41.9 estimates the Δv to reach each one, in the spirit of Eugene Shoemaker and Eleanor Helin (1978). It uses an idealised two-burn transfer from a 400 km orbit to the asteroid's farthest point from the Sun. About 900 known NEAs need less Δv than a trip to low lunar orbit (4.0 km/s), and about 14 600 less than 6 km/s, close to the cost of landing on the Moon. The catch is size: only 7 of the easiest are larger than 140 m, and only about 25 objects larger than 1 km lie below 6 km/s. Their Earth-like orbits also mean that launch opportunities recur only every few years or decades.




42 782 NEAs from the JPL Small-Body Database (19 Oct 2026). Idealised two-impulse transfer from a 400 km orbit to the asteroid's aphelion, plane change split optimally; phasing and launch windows ignored. Size from absolute magnitude H (≤ 22 ≈ 140 m, ≤ 17.75 ≈ 1 km, albedo 0.14). Real missions usually need more Δv; Earth and lunar flybys can reduce it.

**FIGURE 41.9** Accessibility of the 42 782 known near-Earth asteroids. (a) Estimated Δv from low Earth orbit to rendezvous. (b) Number of NEAs below a given Δv, by size. Dashed lines: Δv from LEO to low lunar orbit and to the lunar surface.

**TABLE 41.1** The main resources beyond Earth.

| Resource | Where | Typical concentration | Main products | Key uncertainty |
|---|---|---|---|---|
| Polar water ice | Lunar cold traps (≈ 40 000 km²) | 0 to > 10 %; LCROSS 5.6 % | Water, $H_2$, $O_2$ | Amount, depth, patchiness |
| Oxides in regolith | Whole Moon | O 41–45 %, Fe ≤ 15 %, Al ≤ 15 % | $O_2$, Fe, Si, Al, Ti | Energy per kg |
| Solar-wind volatiles | Lunar regolith, esp. Ti-rich mare | H tens of ppm; $^3$He ≈ 4–20 ppb | $H_2O$, $^3$He | Very low grade |
| Atmospheric $CO_2$ | All of Mars | 95 % of a 0.6 kPa atmosphere | $O_2$, CO, $CH_4$ (with $H_2$) | Power, dust |
| Water on Mars | Poles, buried ice, soil | ≈ 2 % in soil to near-pure ice | Water, $H_2$ | Depth and access |
| Carbonaceous asteroids | Outer main belt; some NEAs | Up to ≈ 18 % water-equivalent, C 3–4 % | Water, carbon, organics | Access, anchoring |
| Metallic asteroids | Rare; Psyche; a few NEAs | Fe–Ni ≈ 90 %+; Pt-group 10–40 ppm | Metals | Composition, market |

## 41.3 Turning Rock and Ice into Products

Three process chains dominate ISRU planning (Figure 41.10): water from lunar polar ice; oxygen and metals from ordinary regolith; and oxygen and methane from Martian carbon dioxide and ice. Table 41.2 compares the main processes.

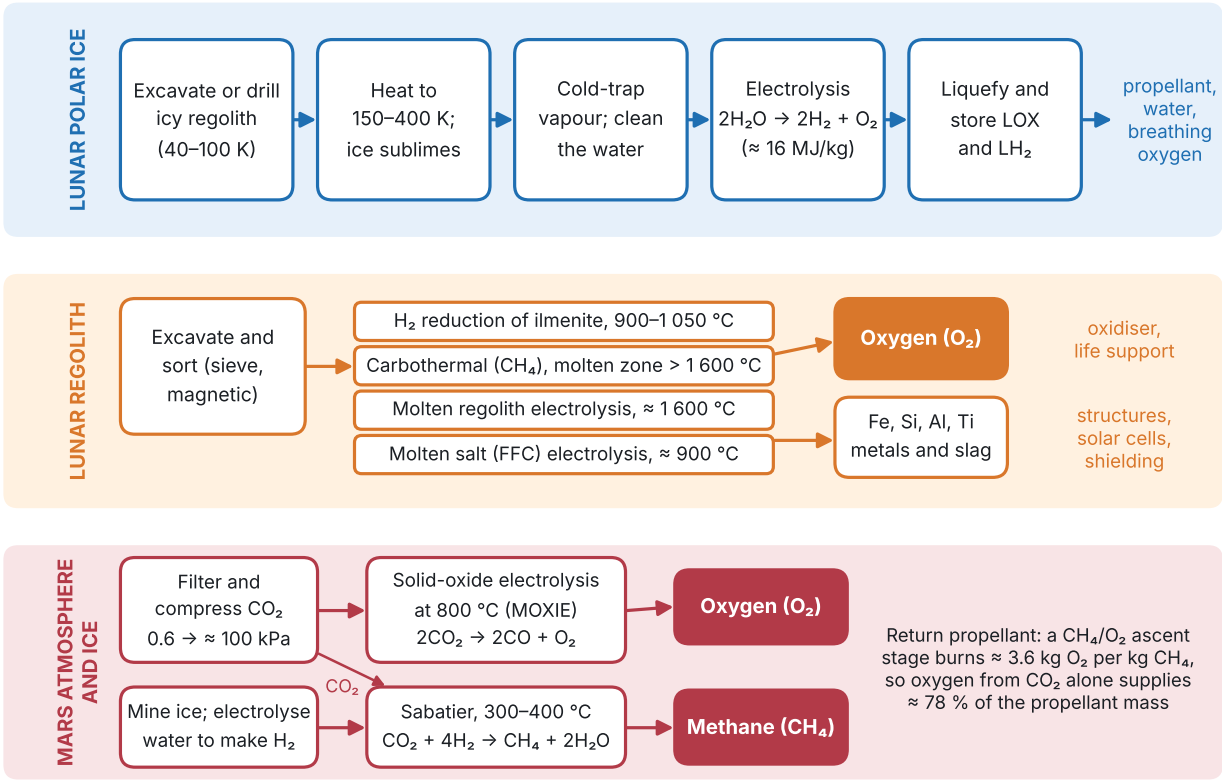



**FIGURE 41.10** The three main ISRU process chains: lunar polar ice, ordinary lunar regolith, and the Martian atmosphere and ice.

**TABLE 41.2** Main ISRU processes. Maximum oxygen yields are theoretical, for typical lunar soils.

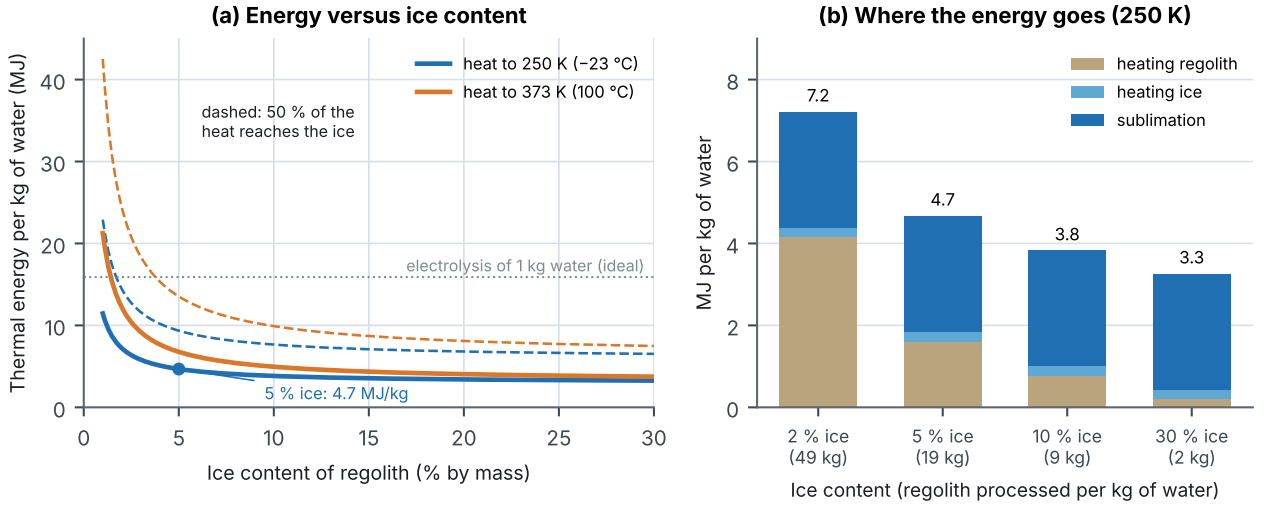
| Process | Feedstock | Temperature | Products | Max. $O_2$ (kg/t) | Status (Oct 2026) |
|---|---|---|---|---|---|
| Ice extraction + electrolysis | Icy polar regolith | −120 to +100 °C | $H_2O$, $H_2$, $O_2$ | Depends on ice | Lab tests; prospecting from 2027 |
| $H_2$ reduction of ilmenite | Mare soil, volcanic glass | 900–1 050 °C | $O_2$ (via $H_2O$), Fe, $TiO_2$ | 9–34 | Field tests in Hawaii (2008–10) |
| Carbothermal reduction | Any regolith | > 1 600 °C (melt zone) | $O_2$ (via $H_2O$), slag | 250–290 | Vacuum demonstration 2023 |
| Molten regolith electrolysis | Any regolith | ≈ 1 600 °C | $O_2$, Fe, Si, Al | 420–450 | Lab; Blue Alchemist |
| Molten salt (FFC) electrolysis | Any regolith | ≈ 900–950 °C | $O_2$, metal alloy powder | 400–430 | Lab (ESA, Metalysis) |
| Solid-oxide $CO_2$ electrolysis | Mars atmosphere | ≈ 800 °C | $O_2$, CO | — | Flown: MOXIE, 122 g $O_2$ |
| Sabatier reaction | $CO_2$ + $H_2$ | 300–400 °C | $CH_4$, $H_2O$ | — | Used on ISS from 2010 |

## 41.3.1 Water from Icy Regolith

The simplest chain digs polar regolith, heats it to drive off the ice, freezes the vapour on a cold surface, purifies it and splits it. Machines must dig soil at 40–100 K, where ice-cemented regolith may be as hard as weak rock, in darkness. Prototypes such as NASA's **RASSOR** and **IPEx** use counter-rotating bucket drums whose digging forces cancel, so a light robot can excavate in one-

sixth gravity without being pushed backwards (Figure 41.14). Alternatives heat the ground in place, with heated drills or — in one "thermal mining" proposal — mirrors on a crater rim that reflect sunlight onto icy ground beneath a dome.

The thermal energy has two main terms: warming the regolith that comes with the ice, and the latent heat of sublimation (2.83 MJ/kg). Figure 41.11 shows that at 30 % ice sublimation dominates and about 3.3 MJ is needed per kilogram of water; at 2 % ice, 49 kg of rock must be warmed per kilogram of water and the total more than doubles. Real plants lose heat and vapour, so the practical figure may be twice the ideal.




Regolith starts at 40 K; specific heat from Hayne et al. (2017), ice $c \approx 7.7 \cdot T$ J/(kg·K), sublimation 2.83 MJ/kg. Solid lines assume all heat reaches the feedstock and all vapour is captured.

**FIGURE 41.11** Thermal energy to extract water from icy regolith. (a) Energy per kilogram of water versus ice content; dashed lines assume half the heat reaches the feedstock. (b) Breakdown at 250 K: at low ice content, most energy heats rock.

**Electrolysis**, $2\mathrm{H_2O} \rightarrow 2\mathrm{H_2} + \mathrm{O_2}$, needs at least 15.9 MJ per kilogram of liquid water — more than the extraction. It yields hydrogen and oxygen in the mass ratio 1 : 8, while engines burn them at about 1 : 6, leaving surplus oxygen for life support. Oxygen liquefies at 90 K, which radiators can reach near the poles; hydrogen needs 20 K and active refrigeration (Chapter 40).

**EXAMPLE: POWER FOR A POLAR WATER PLANT**

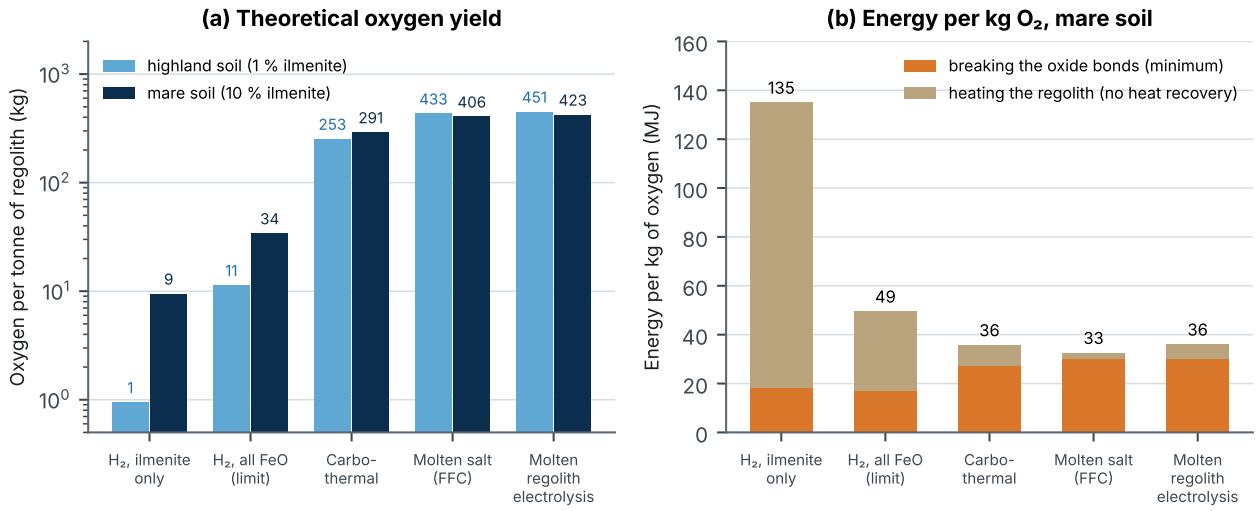
A plant mines regolith containing 5 % ice by mass and must produce 100 t of water a year. The regolith starts at 40 K and is warmed to 250 K.

1. Regolith processed: 19 kg per kilogram of water, so **1 900 t a year**, or 5.2 t a day.
2. Ideal thermal energy (Figure 41.11): warming the regolith 1.61 MJ, warming the ice 0.24 MJ, sublimation 2.83 MJ — **4.7 MJ** per kilogram of water; 9.4 MJ/kg if only half the heat reaches the feedstock.
3. Electrolysis at 70 % efficiency: 15.9 / 0.7 = 22.7 MJ/kg. Total about **32 MJ** (8.9 kWh) per kilogram of water.
4. Annual energy: 100 000 kg × 32 MJ = 3.2 TJ. Divided by the $3.16 \times 10^7$ s in a year, this is an average of **about 100 kW**, day and night, before liquefaction and losses. That is the output that NASA's 2025 directive set for a lunar fission reactor around 2030. However, the 2026 draft procurement for the first unit, Lunar Reactor-1, asks for at least 20 kWe.
5. Products: 11.2 t of hydrogen and 88.8 t of oxygen. At a mixture ratio of 6 : 1 the hydrogen supports 78 t of propellant, leaving 22 t of oxygen for breathing or an oxygen-rich engine.

### 41.3.2 Oxygen from Ordinary Regolith

Ice exists only at the poles, but oxygen is everywhere. Freeing it from the strong bonds of silicates and oxides takes at least the enthalpy of formation of the oxides. This is 18 MJ per kilogram of oxygen for ilmenite and about 30 MJ/kg for complete reduction of typical soil. The soil must also be heated to 900–1 600 °C. Figure 41.12 compares four processes.

- **Hydrogen reduction.** Hot hydrogen removes the oxygen bound to iron, $\mathrm{FeTiO_3 + H_2 \rightarrow Fe + TiO_2 + H_2O}$, at 900–1 050 °C; the water is electrolysed and the hydrogen recycled. Only iron oxide reacts, so the yield is about 1 % of the soil mass unless the ilmenite is first concentrated magnetically or electrostatically. NASA field tests on Mauna Kea, Hawaii, in 2008 and 2010 ran small hydrogen-reduction and carbothermal plants.
- **Carbothermal reduction.** Methane passed over molten regolith above about 1 600 °C reduces silicates too, giving carbon monoxide and hydrogen that are converted back into methane and water; yields reach 25–30 % of the regolith mass. In April 2023 a NASA demonstration with a laser-heated Sierra Space reactor extracted oxygen from simulant this way in vacuum for the first time.
- **Molten regolith electrolysis (MRE).** As in aluminium smelting, a current through regolith melted at about 1 600 °C releases oxygen at an inert anode, while iron and silicon collect at the cathode. MRE frees almost all the oxygen but needs anodes that survive hot, corrosive melts. Blue Origin's **Blue Alchemist**, partly funded by a US$ 34.7 million NASA award, has made solar cells from simulant this way and passed its critical design review in 2025.
- **Molten salt (FFC Cambridge) electrolysis.** A regolith pellet forms the cathode in molten calcium chloride at about 950 °C; oxygen ions migrate to the anode, leaving metal alloy powder. Researchers working with ESA and Metalysis extracted 96 % of the oxygen from a simulant in 2020; inert anodes are again the key development.




Typical Apollo 16 and Apollo 11 soil oxides; 90 % of ilmenite reduced; reaction energy = enthalpy of formation of the reduced oxides (298 K). Process temperatures 1 300 K ($H_2$), 1 200 K (FFC), 1 900 K (molten); fusion 0.48 MJ/kg. Real plants add electrical and thermal losses, and can recover much of the sensible heat.

**FIGURE 41.12** Oxygen from lunar regolith. (a) Theoretical yield per tonne of highland and mare soil (log scale). (b) Minimum energy per kilogram of oxygen from mare soil, without heat recovery.

**EXAMPLE: OXYGEN FROM ILMENITE**

A hydrogen-reduction plant processes high-titanium mare soil containing 10 % ilmenite by mass and reduces 90 % of it at 1 300 K.

1. Ilmenite per tonne: 100 kg, or 100 000 / 151.7 = 659 mol of $FeTiO_3$, of which 593 mol is reduced.
2. Each mole gives one mole of water and, after electrolysis, half a mole of $O_2$: 593 × 16.0 g = **9.5 kg of oxygen per tonne** (0.95 %). The 1.2 kg of hydrogen is recycled.
3. Chemical energy: 288 kJ per mole of oxygen atoms, or 171 MJ per tonne — **18 MJ per kilogram of $O_2$**.
4. Heating a tonne of regolith from 250 K to 1 300 K takes about 1 110 MJ, or 117 MJ per kilogram of $O_2$ — six times the chemistry. Recovering 75 % of the heat from the spent soil cuts the total from 135 to 47 MJ/kg; concentrating the ilmenite to 50 % first cuts the heating term five times.
5. To make 10 t of oxygen a year the plant must dig and process about **1 050 t of soil**, or 2.9 t a day. This is why low-yield processes need beneficiation, and why ice is attractive where it exists.

## 41.3.3 Oxygen and Methane on Mars

On Mars the feedstock comes to the plant. NASA's **Mars Oxygen In-Situ Resource Utilisation Experiment (MOXIE)** was a 17 kg unit inside Perseverance (Figure 41.13). It compressed Martian air from about 0.6 kPa to near 100 kPa and passed it over solid-oxide electrolysis cells at 800 °C. There a current split carbon dioxide: $\mathbf{2CO_2 \rightarrow 2CO + O_2}$. In 16 runs between April 2021 and August 2023, by day and night and in every season, it made 122 g of oxygen. It produced up to 12 g an hour, at 98 % purity or better.

NASA estimates that a crewed Mars ascent vehicle would need about 25 t of oxygen. Made over the 14 months before a crew arrives, that needs 2.4 kg an hour — 200 times MOXIE's best rate — and tens of kilowatts. Methane comes from the **Sabatier reaction**, $\mathbf{CO_2 + 4H_2 \rightarrow CH_4 + 2H_2O}$, over a nickel or ruthenium catalyst at 300–400 °C (used on the ISS from 2010), with hydrogen from

Martian ice or brought from Earth. A methane–oxygen engine burns about 3.6 kg of oxygen per kilogram of methane, so oxygen from the air alone supplies 78 % of the propellant — one reason SpaceX's Starship burns methane.

**FIGURE 41.13** Engineers lower MOXIE into the body of the Perseverance rover at JPL before launch. The experiment made the first oxygen produced on another planet. Image: NASA/JPL (NASA/JPL-Caltech).

### 41.3.4 Building with Regolith

By mass, the largest use of lunar material may be the least glamorous. This means landing pads that stop rocket exhaust blasting regolith across a base, **berms** (walls of piled soil) against ejecta, and roads. It also means a few metres of soil over habitats, against radiation and a day–night temperature swing of nearly 300 K. In June 2025 NASA's RASSOR robot built a berm about 0.9 m high from simulant at Kennedy Space Center (Figure 41.15).

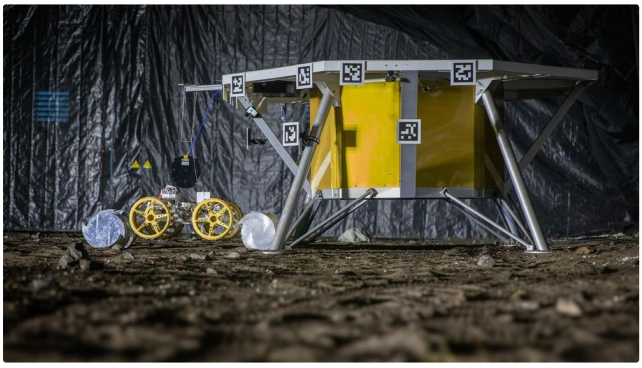


**FIGURE 41.14** NASA's ISRU Pilot Excavator (IPEx), an excavator and dump truck in one, during a simulated lunar mission in a regolith testbed at Kennedy Space Center in August 2024. Image: NASA/KSC (NASA/Frank Micheaux).

**FIGURE 41.15** RASSOR, with counter-rotating bucket drums at each end, building a berm of simulated lunar regolith at Kennedy Space Center in June 2025. Image: NASA/KSC (NASA/Frank Michaux).

**Sintering** fuses grains with focused sunlight, lasers or microwaves; **melting** makes glass and cast basalt; **binders** such as sulphur (melting at 115 °C) make concrete without water; and **3D printing** builds walls layer by layer. ICON, which printed NASA's simulated Mars habitat for the CHAPEA missions (Chapter 30), is developing laser vitrification of lunar soil under a US$ 57.2 million NASA contract running to 2028. China plans to test solar-concentrator brick-making on its Chang'e 8 lander around 2028.

### 41.3.5 Power: The Hidden Requirement

Every process is limited by power: 100 t of water a year needs about 100 kW continuously. Ridges near the south pole, such as the rim of Shackleton crater, are sunlit for 80–90 % of the year. But inside the cold traps, and through the 14-day lunar night elsewhere, nuclear power is the practical choice. NASA and the US Department of Energy renewed their Fission Surface Power partnership in January 2026. They aim for a 100 kW reactor ready to launch by about 2030 and to run for at least 10 years. Its waste heat could warm regolith.

## 41.4 Prospecting: Finding the Ore

Miners distinguish a **resource**, known to exist, from a **reserve**, which can be extracted at a profit with proven technology; no space resource is yet a reserve. Orbital data show where hydrogen and cold traps are, but not the form, depth and purity of the ice; these must be measured on the ground (Table 41.3).

**TABLE 41.3** Lunar prospecting missions.

| Mission | Agency | Year | Method | Result or status (Oct 2026) |
|---|---|---|---|---|
| Lunar Prospector | NASA | 1998–99 | Neutron spectrometer | Hydrogen excess at both poles |
| Chandrayaan-1 | ISRO, NASA | 2008–09 | Infrared spectra, radar | Surface OH/$H_2O$; ice in some PSRs |
| LCROSS | NASA | 2009 | Impactor and shepherding probe | 5.6 ± 2.9 % water in Cabeus |
| LRO; ShadowCam on Danuri | NASA; KARI | 2009–; 2022– | Thermal, UV, neutron mapping; PSR imaging | Cold-trap maps; no extensive surface ice |
| Lunar Trailblazer | NASA | 2025 | Infrared and thermal mapping | Lost the day after launch |
| PRIME-1 (on IM-2) | NASA | 2025 | 1 m drill, mass spectrometer | Lander tipped over; no drilling |
| VIPER | NASA | late 2027 | Rover, 1 m drill, spectrometers | Revived 2025; on Blue Moon MK1 |
| Chang'e 7 | CNSA | 2027 | Orbiter, lander, rover, hopper | Missed 2026 launch window |
| PROSPECT (on IM-4) | ESA | 2027 | 1 m drill and gas laboratory | In preparation |
| LUPEX (Chandrayaan-5) | ISRO, JAXA | ≈ 2028 | Lander and rover with drill | In development |
| Chang'e 8 | CNSA | ≈ 2028 | ISRU and construction tests | In development |

### 41.4.1 Bad Luck in 2025

Two missions launched on one Falcon 9 in February 2025 showed how hard the work is. NASA's US$ 94 million **Lunar Trailblazer** carried infrared and thermal mappers to map the form and abundance of polar water (Figure 41.16). Contact was lost the day after launch, probably because misaligned solar arrays let its batteries drain. The IM-2 lander *Athena* carried NASA's **PRIME-1**: the TRIDENT drill, to bring up regolith from 1 m depth in 10 cm steps, and the MSolo mass spectrometer to analyse the released gases (Figure 41.17). *Athena* landed on 6 March 2025 at 84.8° S near Mons Mouton but came to rest on its side in a crater. It operated for only about 12 hours instead of 10 days. TRIDENT never drilled, and MSolo mostly recorded gases vented by the lander itself.

**FIGURE 41.16** NASA's Lunar Trailblazer in a clean room at Lockheed Martin in 2023, after the Lunar Thermal Mapper (black box, upper right) was installed. The orbiter was lost the day after its launch in February 2025. Image: NASA/JPL (Lockheed Martin Space).

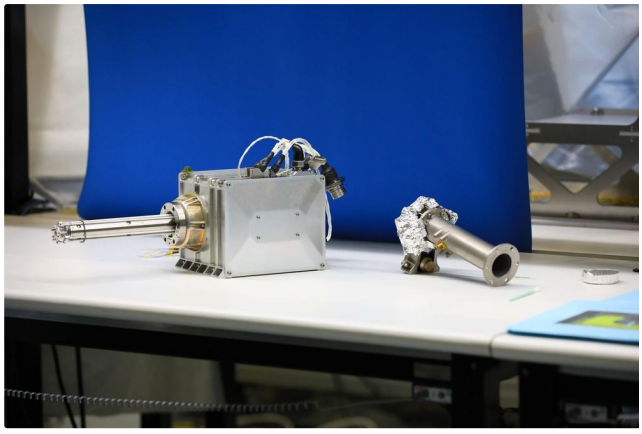


**FIGURE 41.17** Parts of the MSolo mass spectrometer, a modified commercial instrument, at Kennedy Space Center in 2020. MSolo flew on the IM-2 lander as part of PRIME-1. Image: NASA/KSC (NASA/Glenn Benson).

### 41.4.2 Next Steps on the Ground

NASA's half-tonne **VIPER** rover carries a neutron spectrometer, a near-infrared spectrometer, MSolo and a TRIDENT drill to map polar ice (Chapter 28). Cancelled in July 2024 over cost growth, it was revived in September 2025 with a task order worth up to US$ 190 million. Under this order Blue Origin is to land it near Mons Mouton in late 2027 on the second Blue Moon MK1. That flight depends on the first MK1 flight, delayed to early 2027 after a New Glenn booster exploded during a static-fire test in May 2026.

China's **Chang'e 7** — orbiter, lander, rover and a small "hopper" designed to jump into a shadowed crater and analyse its water — missed its August 2026 launch window and is expected in 2027. ESA's **PROSPECT** drill and gas laboratory, first developed for Russia's Luna 27, will fly on Intuitive Machines' IM-4 in 2027. The ISRO–JAXA **LUPEX** (Chandrayaan-5) lander and rover is planned for around 2028, as is China's **Chang'e 8**, which will test resource use and construction.

The stakes rose in 2026, when NASA paused the Gateway station and announced a permanent base near the south pole, starting with frequent robotic landings from 2027. Whether the base can live partly off the land depends on what these drills find.

## 41.5 The Business of Space Mining

### 41.5.1 The First Wave

Asteroid mining became a business in April 2012, when **Planetary Resources** was unveiled with backing from Google's Larry Page and Eric Schmidt and the film director James Cameron. It planned to find water-rich NEAs with small telescopes and sell the water as propellant in space. **Deep Space Industries (DSI)** followed in 2013, and Luxembourg's space-resources initiative of 2016 invested in Planetary Resources and agreed to co-fund DSI. With no customers for propellant in space, investors lost patience: Planetary Resources, after two test satellites, was bought by the blockchain company ConsenSys in October 2018, and DSI by Bradford Space in January 2019.

### 41.5.2 The Second Wave

A second generation started with smaller, cheaper missions (Table 41.4).

- **AstroForge** (founded 2022) aims to mine platinum-group metals from metallic NEAs. Brokkr-1, launched in April 2023 to test refining asteroid-like material in orbit, could not be commanded reliably. Odin, a 100 kg, US$ 3.5 million spacecraft that rode with IM-2 in February 2025 to fly past the possibly metallic asteroid 2022 OB5, was declared lost in March. DeepSpace-2, designed to land on 2022 OB5, was completed in June 2026 and is to ride on IM-3 in 2027.
- **Karman+** (Denver) raised US$ 20 million in 2025 to rendezvous with an NEA and extract about 1 kg of material around 2027, aiming to sell water and regolith to servicing tugs.
- **TransAstra** proposes "optical mining", using concentrated sunlight to drive volatiles out of an asteroid enclosed in a bag, and reported testing a 1 m capture bag inside the ISS's Bishop airlock in October 2025.
- **Interlune** plans to harvest lunar helium-3 (see above).

**TABLE 41.4** Space-mining companies.

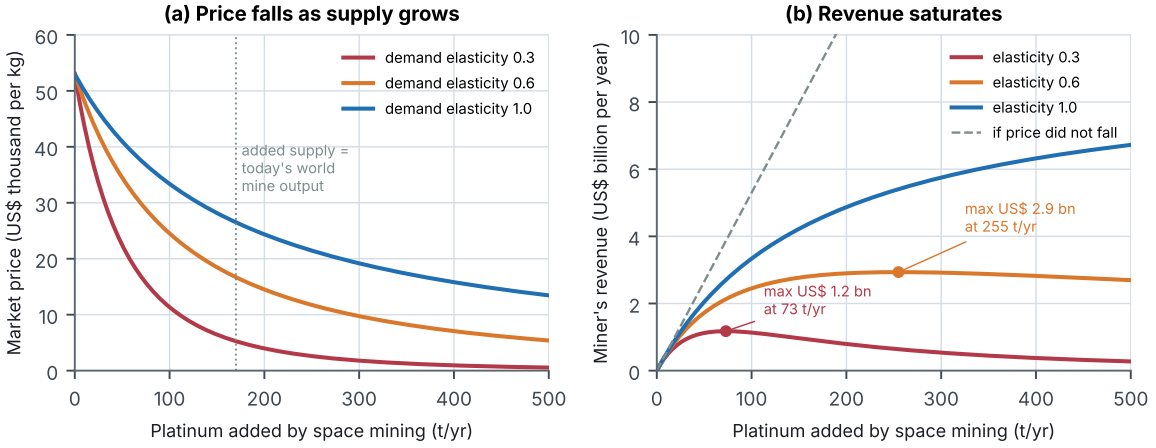
| Company | Founded | Target | Missions flown | Outcome or status (Oct 2026) |
|---|---|---|---|---|
| Planetary Resources | 2012 (public) | Water from NEAs | Arkyd-3R (2015), Arkyd-6 (2018) | Acquired by ConsenSys, 2018 |
| Deep Space Industries | 2013 | Water from NEAs | None of its own | Acquired by Bradford Space, 2019 |
| AstroForge | 2022 | Platinum-group metals | Brokkr-1 (2023), Odin (2025) | Both failed; DeepSpace-2 due 2027 |
| Karman+ | early 2020s | Water, regolith for use in space | None yet | Demonstration targeted 2027 |
| TransAstra | 2015 | Volatiles by optical mining | Capture-bag test on ISS (2025) | Development |
| Interlune | 2020 | Lunar helium-3 | None yet | Contracts for delivery from 2028–29 |

### 41.5.3 Who Would Buy?

A space resource can be sold in two markets. **In space**, the competitor is material launched from Earth: propellant in lunar orbit is worth what it costs to deliver from Earth, thousands of dollars per kilogram, but that market barely exists yet. **On Earth**, the competitor is terrestrial mining, and the product must be worth enough per kilogram to pay for its return.

Platinum is the usual example. World mine supply is about 170 t a year, and the price is volatile: in 2026 it reached a record of about US$ 2 900 per troy ounce in January, fell to about US$ 1 540 in July and stood near US$ 1 650 (about US$ 53 000 per kilogram) in mid-October. Figure 41.18 uses a simple constant-elasticity demand model, in which a 1 % increase in supply lowers the price by $1/\varepsilon$ %, where $\varepsilon$ is the **price elasticity of demand**. If demand is inelastic ($\varepsilon = 0.3$), the miner's revenue peaks at only about US$ 1.2 billion a year, at 73 t a year; at $\varepsilon = 0.6$ the peak is US$ 2.9 billion at 255 t a year. A single 100 m metallic asteroid of bulk density 5 000 kg/m$^3$ with 20 ppm of

platinum-group metals would hold about 50 t of them. If it were all platinum and sold in one year, that would be enough to cut the price by between about a quarter and more than a half. Space mining for Earth markets caps its own revenue.




Illustrative constant-elasticity demand $P = P_0(1 + X/S_0)^{(-1/\varepsilon)}$ with $P_0$ = US$ 53 000/kg (≈ US$ 1 650 per troy ounce, mid-October 2026; the price ranged from about US$ 1 540 to 2 920 per ounce during 2026) and $S_0$ = 170 t/yr (world mine supply, WPIC 2025).

**FIGURE 41.18** Why space metals cannot be valued at today's prices. (a) Platinum price as space-mined supply is added, for three demand elasticities. (b) The miner's annual revenue, which peaks when demand is inelastic.

The most plausible early business is therefore service to government exploration — data, delivery, demonstrations and, eventually, propellant sold in lunar orbit.

## 41.6 Law and Policy

### 41.6.1 Treaties

The **Outer Space Treaty** of 1967, ratified by more than 110 states, is the foundation of space law (Chapters 1 and 19). Article I guarantees the freedom to explore and use outer space. Article II declares that outer space, "including the Moon and other celestial bodies, is not subject to national appropriation by claim of sovereignty, by means of use or occupation, or by any other means". Article VI makes states responsible for authorising and supervising their companies. The treaty is silent on whether materials taken from a celestial body can be owned, though nobody objected when Apollo and Luna brought samples home.

The 1979 **Moon Agreement** declared the Moon and its natural resources "the common heritage of mankind" and called for an international regime to govern their exploitation once it became feasible. It entered into force in 1984 but was never joined by the United States, Russia or China. Since Saudi Arabia left in January 2024 it has 17 parties. Several of them, such as Australia and the Netherlands, have also signed the Artemis Accords.

### 41.6.2 National Laws and the Artemis Accords

Since 2015 several states have let their companies own extracted resources while disclaiming sovereignty over the body itself — like owning the fish but not the sea (Table 41.5). The United States led with the **Commercial Space Launch Competitiveness Act** of November 2015, followed by **Luxembourg** (2017), the **United Arab Emirates** (a 2019 space law and 2023 space-resources

regulations) and **Japan** (2021). Under Japan's law the government licensed ispace in 2022 and again in 2024 to transfer ownership of lunar regolith to NASA — the first licences issued under the law.

In a symbolic programme of December 2020, NASA agreed to buy small regolith samples, collected and photographed on the Moon and left in place. The sellers were Lunar Outpost for US$ 1, ispace Japan and ispace Europe for US$ 5 000 each and Masten Space Systems for US$ 15 000. By October 2026 no sale had been completed: ispace's landers crashed in April 2023 and June 2025, Masten went bankrupt in 2022, and Lunar Outpost's rover was aboard the toppled IM-2 lander.

**TABLE 41.5** Law and policy on space resources.

| Instrument | Year | Parties or scope | Position on resources |
|---|---|---|---|
| Outer Space Treaty | 1967 | > 110 states | No national appropriation (Art. II); silent on extracted resources |
| Moon Agreement | 1979 | 17 states | "Common heritage of mankind"; future international regime |
| US Commercial Space Launch Competitiveness Act | 2015 | United States | Citizens may own and sell recovered resources |
| Luxembourg space-resources law | 2017 | Luxembourg | "Space resources are capable of being appropriated" |
| UAE space law and regulations | 2019, 2023 | United Arab Emirates | Licensing of resource activities |
| Japan Space Resources Act | 2021 | Japan | Licensed operators own what they extract |
| Artemis Accords | 2020– | 76 signatories (Sept 2026) | Extraction is not in itself appropriation; safety zones |
| COPUOS working group principles | 2021– | UN Legal Subcommittee | Non-binding principles in draft (2026) |

The **Artemis Accords**, launched by the United States in October 2020 with eight founding signatories, had 76 signatories by late September 2026. Section 10 affirms that "the extraction of space resources does not inherently constitute national appropriation under Article II", and Section 11 introduces **safety zones** around operations to avoid harmful interference. China and Russia, leaders of the rival International Lunar Research Station, have not signed.

### 41.6.3 The United Nations Working Group

In 2021 the Legal Subcommittee of the UN **Committee on the Peaceful Uses of Outer Space (COPUOS)** set up a working group. This Working Group on Legal Aspects of Space Resource Activities is chaired by the Australian law professor Steven Freeland. In September 2026 it circulated draft non-binding "Initial Recommended Principles": extraction in accordance with the Outer Space Treaty would not by itself be national appropriation, and benefit sharing would be voluntary. Open questions remain about how large and lasting a safety zone may be, and who may work in the few square kilometres of the best cold traps. Another is how to protect pristine ice — a record of water in the inner Solar System — from contamination.

### AI FOR PROSPECTING AND MINING

Machine-learning classifiers map minerals and hydration from orbital spectra and fuse them with temperature and topography to rank candidate ice sites. A physics-based deep-learning denoiser has turned noisy LRO images of shadowed craters into boulder maps for planning rover routes. Networks also classify the spectra of tens of thousands of asteroids observed by Gaia. On the surface, delays and darkness demand autonomy. It is needed for perception under headlights, digging that adapts to unknown soil, and fleets of excavators and haulers. It is also needed for fault detection in plants that must run unattended for years (Chapters 46, 48 and 51).

### KEY TAKEAWAYS

- The gear ratio drives ISRU. A kilogram on the lunar surface costs about 130 kg at Earth lift-off, and one in low lunar orbit about 73 kg. But lunar propellant delivered to low lunar orbit costs only about 2.1 kg of lunar production.
- A lunar plant pays only if it makes several times its own landed mass (3.8 times for delivery to low lunar orbit) before it fails.
- Oxygen is everywhere (41–45 % of lunar regolith; Mars's $CO_2$ atmosphere); water ice is trapped in polar cold traps below about 110 K, in uncertain and probably patchy amounts.
- Water extraction needs 3–7 MJ of heat per kilogram plus at least 15.9 MJ/kg for electrolysis; oxygen from regolith needs 18–30 MJ/kg plus heat. Power is the limiting resource.
- MOXIE made 122 g of oxygen on Mars, but a crewed ascent needs 200 times its rate. Prospecting suffered losses in 2025; VIPER, Chang'e 7 and PROSPECT are due in 2027.
- Selling space metals on Earth collapses their price; the first asteroid miners failed, and the second wave has yet to succeed in space.
- The Outer Space Treaty bans national appropriation but is silent on extracted resources; national laws and the 76-signatory Artemis Accords allow ownership.

### REVIEW QUESTIONS

1. Explain the "gear ratio" of a destination. Using the rocket equation with $I_{sp}$ = 450 s, show why a kilogram in low lunar orbit costs about 2.9 kg in low Earth orbit.
2. Why does lunar propellant cost more to deliver to low Earth orbit than to low lunar orbit?
3. A 15 t lunar plant makes 20 t of propellant a year. How long must it run to pass the break-even productivity of 3.8 for delivery to low lunar orbit?
4. Using Figure 41.5, explain why water ice survives only in permanently shadowed craters.
5. Estimate the energy to extract 1 kg of water from regolith containing 2 % ice, and compare it with the energy to electrolyse the water.
6. Compare hydrogen reduction of ilmenite with molten regolith electrolysis.
7. MOXIE produced up to 12 g of oxygen an hour. How long would one MOXIE take to make the 25 t needed for a crewed ascent, and how many would be needed to make it in 14 months?
8. How much regolith must be processed for 1 litre of helium-3 at 4 ppb? Why is helium-3 more likely to be sold for cryogenics than for fusion?
9. Using the model of Figure 41.18, explain why the revenue from selling asteroid platinum on Earth reaches a maximum. Why might water sold in lunar orbit be a better business?
10. How do the Outer Space Treaty, the Moon Agreement and the Artemis Accords treat extracted resources?

# 42 CHAPTER Advanced Propulsion

Image: NASA/GRC

*Chemical rockets have reached the limit set by the energy stored in their propellants, so travelling faster or farther needs another source of energy or a way to avoid carrying propellant at all. This chapter shows, with worked numbers, why exhaust velocity and power decide every advanced concept. It covers high-power Hall and ion thrusters and the power-limited trajectories they fly; nuclear thermal rockets from NERVA to the cancelled DRACO; space fission reactors and NASA's SR-1 Freedom nuclear-electric mission planned for 2028; solar, electric and laser sails, including Breakthrough Starshot; fusion and antimatter; scramjets and rotating detonation engines; and tethers. It ends by explaining why "reactionless" drives and warp drives fail the test of physics.*

IN THIS CHAPTER

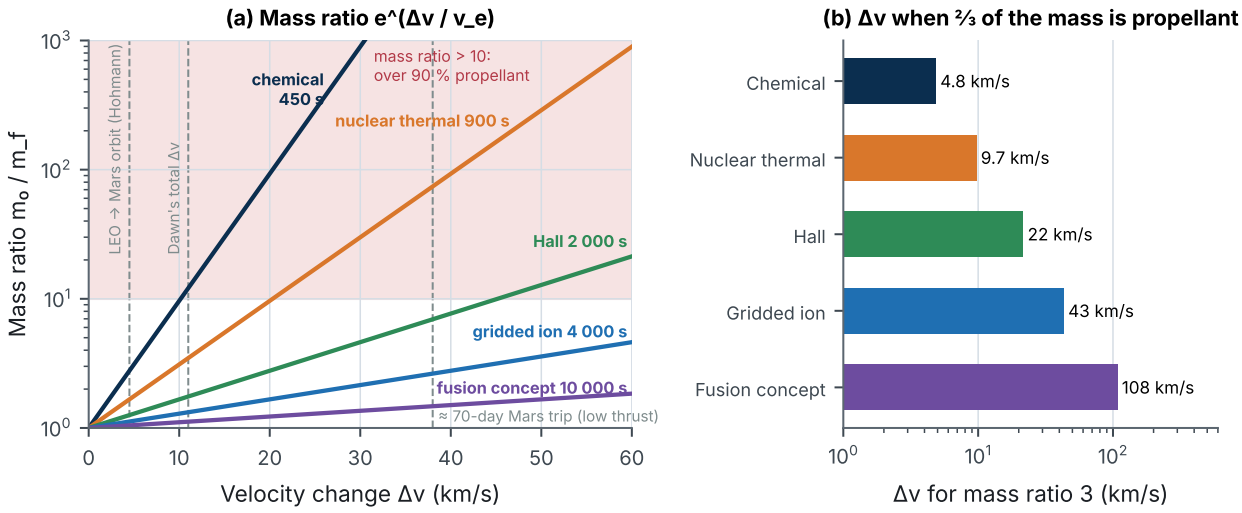
42.1 The Tyranny of the Rocket Equation
42.2 Advanced Electric Propulsion
42.3 Nuclear Thermal Propulsion
42.4 Nuclear Electric Propulsion and Space Reactors
42.5 Sailing on Light
42.6 Beamed Energy and the Interstellar Challenge
42.7 Breathing Air and Detonating Fuel
42.8 Propellantless and Exotic Ideas

Every spacecraft that has left Earth has done so on chemical rockets, whose energy is fixed by chemistry. The best practical combination, hydrogen and oxygen, gives an exhaust velocity of about 4.4 km/s (Chapter 35). The rocket equation turns that limit into an exponential penalty for every extra kilometre per second of velocity change. A crewed round trip to Mars, a fast mission to the outer planets or a probe to another star needs one of two things. One is a much faster exhaust; the other is a way of pushing a spacecraft without carrying its reaction mass along.

Advanced propulsion splits into two families. The first raises the exhaust velocity with energy from somewhere other than the propellant — a nuclear reactor, solar arrays, fusion or antimatter. The second dispenses with propellant and pushes on sunlight, laser beams, the solar wind, a planet's magnetic field or the atmosphere. Some fly by the thousand, others have never left the laboratory, and a few contradict the laws of physics. This chapter sorts them by exhaust velocity, thrust, power per kilogram and maturity.

## 42.1 The Tyranny of the Rocket Equation

The **rocket equation**, $\Delta v = v_e \ln(m_0/m_f)$, says that the velocity change Δv depends on the exhaust velocity $v_e$ and on the **mass ratio**. This is the starting mass $m_0$ divided by the mass $m_f$ left when the propellant is gone. Because the mass ratio is $e^{\Delta v/v_e}$, it grows exponentially once Δv exceeds the exhaust velocity (Figure 42.1). A hydrogen–oxygen stage needs a mass ratio of 2.75 to go from low Earth orbit (LEO) into Mars orbit on the cheapest trajectory (4.5 km/s). But it would need 12 for the 11 km/s that the ion-propelled Dawn accumulated (Chapter 24). Doubling the exhaust velocity takes the square root of the mass ratio; multiplying it by ten turns an impossible mission into a routine one.




Ideal rocket equation, v_e = g₀ I_sp. Reference Δv values: Hohmann transfer from 400 km LEO to a 1-sol Mars orbit (this chapter's model); Dawn ≈ 11 km/s; 70-day trip from the power-limited model in this chapter.

**FIGURE 42.1** The rocket equation for five exhaust velocities. (a) Mass ratio against velocity change. (b) The velocity change each technology gives when two-thirds of the vehicle is propellant.

### 42.1.1 Energy, Power and Thrust

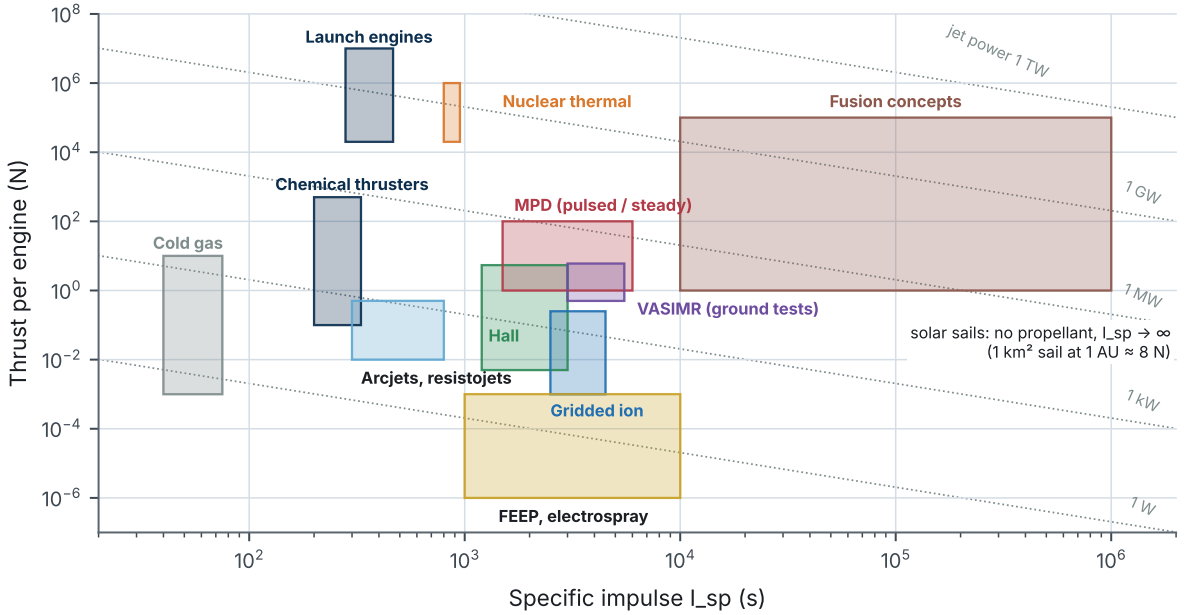
A faster exhaust has a price: the power it carries away, the **jet power**, is

$$P_{jet} = \tfrac{1}{2}\dot{m}v_e^2 = \tfrac{1}{2}Fv_e,$$

so, for a given thrust $F$, doubling the exhaust velocity doubles the power needed. A chemical rocket carries its power in its propellant — each F-1 engine of the Saturn V had a jet power near 9 GW. But an electric thruster must be fed by solar arrays or a reactor, and that power plant has mass. A thruster that turns a fraction η of its electrical input $P$ into jet power gives

$$F = \frac{2\eta P}{v_e} = \frac{2\eta P}{g_0 I_{sp}},$$

so a 12 kW thruster at 60 % efficiency and 2 600 s produces only about 0.56 N (Figure 42.2).




Indicative ranges of flown or ground-tested devices (fusion: design studies only).
Dotted lines: jet power $P = F\, g_0\, I_{sp} / 2$ — high thrust at high $I_{sp}$ needs enormous power.

**FIGURE 42.2** Thrust and specific impulse of propulsion technologies. Dotted lines join points of equal jet power: more thrust at higher specific impulse demands power that only large reactors, fusion or beamed energy could supply.

Advanced propulsion is therefore a contest of energy sources and their **specific power** α, the power per kilogram of power and propulsion system. For a mission that thrusts for a time $t$, the best exhaust velocity of a power-limited rocket is of the order of the **characteristic velocity** $v_{ch} = \sqrt{2\eta\alpha t}$. A faster exhaust would need a heavier power plant than the propellant it saves. With α = 50 W/kg, η = 0.6 and one year of thrusting, $v_{ch} \approx 43$ km/s (Isp ≈ 4 400 s). With α = 10 W/kg it falls to 19 km/s (Isp ≈ 2 000 s), the territory of Hall thrusters. Table 42.1 summarises the options. Technology readiness levels (TRLs, Chapter 4) run from 1 (principle observed) to 9 (proven in operation).

**TABLE 42.1** Advanced propulsion at a glance (October 2026). TRLs are approximate, judged from flight and test history.

| Technology | Isp (s) | Thrust | Energy source | TRL (approx.) | Examples |
|---|---|---|---|---|---|
| Chemical (reference) | 300–465 | N to MN | Propellant | 9 | RL10, Raptor |
| Nuclear thermal | 850–950 | 10–1 000 kN | Fission reactor | 5–6 | NERVA XE-Prime, RD-0410 |
| Hall thruster | 1 200–3 000 | mN to 5 N | Electric | 9 (≤ 5 kW), 6–7 (12 kW) | Starlink, Psyche, AEPS |
| Gridded ion | 2 500–4 500 | 1–250 mN | Electric | 9 | Dawn, BepiColombo, NEXT-C |
| MPD, VASIMR | 1 500–6 000 | 1–100 N | Electric, 0.1–10 MW | 4–5 | VX-200, laboratory MPDs |
| FEEP, electrospray | 1 000–10 000 | µN to mN | Electric, watts | 9 | LISA Pathfinder, Enpulsion |
| Solar sail | no propellant | ≈ 8 µN/m² at 1 AU | Sunlight | 6–7 | IKAROS, LightSail 2, ACS3 |

| Technology | Isp (s) | Thrust | Energy source | TRL (approx.) | Examples |
|---|---|---|---|---|---|
| Electric or magnetic sail | no propellant | μN–mN (studies) | Solar wind | 2–3 | ESTCube-1 (not deployed) |
| Laser sail | no propellant | ≈ 670 N per 100 GW | Ground laser | 2 | Breakthrough Starshot |
| Fusion | $10^4$–$10^6$ | N–kN (studies) | Fusion reactions | 1–3 | Daedalus, Direct Fusion Drive |
| Electrodynamic tether | no propellant | 0.1–1 N | Orbital energy, solar power | 5–6 | PMG, TSS-1R |
| Scramjet | 1 000–4 000 | kN | Fuel and air | 6 | X-43A, X-51A |

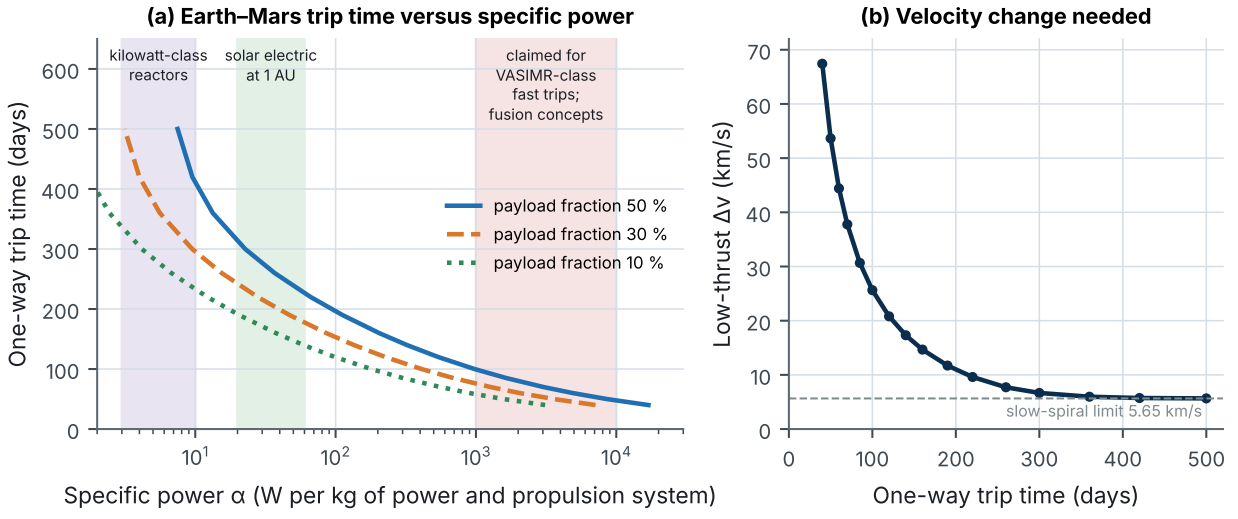
## 42.2 Advanced Electric Propulsion

Electric propulsion is already mass-produced — thousands of Hall thrusters fly on Starlink satellites, and ion and Hall engines have carried Dawn, BepiColombo and Psyche (Chapters 6 and 24). The frontier is power: tens to hundreds of kilowatts per spacecraft.

### 42.2.1 Power-Limited Trajectories

A power-limited spacecraft thrusts for months (Figure 42.3), and the ideal strategy is to vary the specific impulse along the way: more thrust near departure and arrival, higher Isp in between. For such a variable-Isp rocket the propellant consumed depends only on the integral $J = \int a^2\,dt$ of the squared thrust acceleration. With the power plant sized optimally, the fraction of the initial mass left for payload and structure is

$$\frac{m_{payload}}{m_0} = \left(1 - \sqrt{\frac{J}{2\eta\alpha}}\right)^2 .$$




Optimal variable-I_sp power-limited rendezvous, Earth escape to Mars capture (coplanar circular orbits, planetary spirals excluded); J = ∫a²dt by direct optimisation; payload fraction = (1 − √(J/2ηα))², η = 0.6. Shaded bands are indicative.

**FIGURE 42.3** Power-limited Earth–Mars transfers computed for this chapter. (a) Specific power needed for a given trip time when 50 %, 30 % or 10 % of the departing mass arrives as payload and structure. (b) The low-thrust velocity change. Escape and capture spirals at the planets are excluded.

$J$ rises roughly as the inverse cube of trip time. The figure applies this to a one-way trip from Earth escape to Mars capture, optimised numerically for each trip time. A slow spiral needs 5.7 km/s; a 100-day crossing needs 26 km/s and a 40-day crossing 67 km/s. If half the departing mass is to arrive as payload and structure, a 300-day trip needs $\alpha \approx 23$ W/kg and a 190-day trip about 110 W/kg. A 100-day trip needs about 1 kW/kg and a 40-day trip about 17 kW/kg. Solar-electric systems deliver of the order of tens of watts per kilogram at 1 AU; crewed crossings in a month or two would need power systems hundreds of times lighter.

Solar arrays are light and proven, but their output falls with the square of distance from the Sun — at Jupiter to 1/27 of its value near Earth. A reactor gives the same power anywhere, but small reactors manage only a few watts per kilogram and large ones are dominated by their radiators. Solar-electric propulsion suits the inner Solar System; nuclear-electric propulsion the outer planets and heavy cargo.

### 42.2.2 High-Power Hall Thrusters

Hall thrusters (Chapter 6) used to wear out as ions eroded their ceramic channel walls. **Magnetic shielding**, developed at JPL in the 2010s, shapes the field to keep energetic ions off the walls, extending design lifetimes to tens of thousands of hours. NASA Glenn and Aerojet Rocketdyne (now L3Harris) used it in the 12 kW **Advanced Electric Propulsion System (AEPS)**, which gives about 0.6 N at about 2 600 s. Three AEPS flight thrusters passed acceptance tests in 2025 (Figure 42.4) and were delivered for the **Power and Propulsion Element (PPE)** built for NASA's Gateway station. When NASA paused Gateway in March 2026, it reassigned the PPE as the bus of the nuclear-electric SR-1 Freedom mission described below; AEPS had by then completed environmental qualification, with a life test in preparation. In the laboratory, the three-channel X3 reached 102 kW in 2017 (Table 42.2).

### 42.2.3 Gridded Ion Engines

Among gridded ion engines, NASA's 6.9 kW **NEXT-C** reaches an Isp of about 4 190 s with 0.24 N of thrust, and a laboratory NEXT engine ran for 51 184 hours between 2005 and 2014. NEXT-C flew on the DART asteroid-deflection mission as a technology demonstration that the mission did not depend on. When it was first fired, in December 2021, it ran for about two hours. But an anomaly in the spacecraft's power electronics led the team to suspend further use (Chapter 26). The faint blue glow of the ion beam is the technology's signature (Figure 42.5).

### 42.2.4 Magnetoplasmadynamic Thrusters and VASIMR

Above about 100 kW, electromagnetic devices take over. A **magnetoplasmadynamic (MPD)** thruster drives thousands of amperes through a plasma between a central cathode and a surrounding anode, and the magnetic field pushes the plasma out by the Lorentz force $\vec{j} \times \vec{B}$. MPD thrusters handle megawatts in a compact device at 1 500–6 000 s. But their electrodes erode, and they are efficient only at very high power. A small pulsed version was carried by Japan's Space Flyer Unit in 1995.

The **Variable Specific Impulse Magnetoplasma Rocket (VASIMR)** of Ad Astra Rocket Company has no electrodes. Radio waves ionise argon, a second antenna heats the ions at their cyclotron frequency, and a magnetic nozzle turns their gyration into exhaust. The VX-200 prototype

reached 200 kW in 2009 and was measured at 72 % efficiency, about 4 900 s and 5.7 N; in July 2021 the VX-200SS ran for 88 hours at 80 kW. VASIMR has not flown; the fast crewed missions linked to it would need tens of megawatts at specific powers far beyond today's.

**FIGURE 42.4** Engineers at NASA Glenn with the third 12 kW AEPS flight thruster during its acceptance-test campaign in 2025. Image: NASA/GRC (NASA/GRC/Jef Janis).

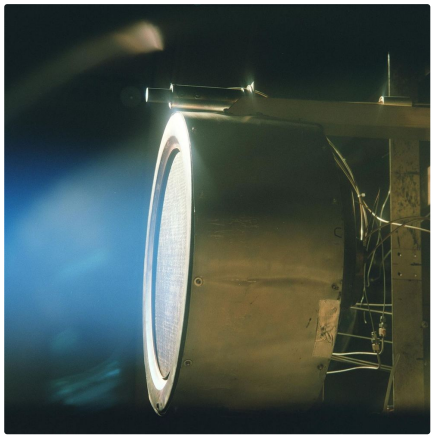


**FIGURE 42.5** A prototype of the xenon ion engine flown on Deep Space 1, seen through a port of a JPL vacuum chamber during a long-duration test; the faint blue glow is the ion beam. Image: NASA/JPL (NASA/JPL).

### 42.2.5 Field-Emission, Electrospray and Air-Breathing Thrusters

At the other extreme, **field-emission electric propulsion (FEEP)** pulls ions out of a liquid metal such as indium with about 10 kV on a sharp emitter, and **electrospray** thrusters do the same with ionic liquids. They give micronewtons to about a millinewton at 2 000–6 000 s or more. Colloid thrusters provided drag-free control for NASA's experiment on ESA's LISA Pathfinder (2015–17). Enpulsion had launched more than 200 indium FEEP systems by 2024.

For very low orbits (Chapter 14), **air-breathing electric propulsion** would collect the thin air as propellant. In 2018 an ESA-led team fired a Sitael thruster on a nitrogen–oxygen mixture representing the atmosphere at about 200 km, the first such test.

**TABLE 42.2** Electric thrusters at the frontier of power, efficiency and precision.

| Thruster | Type | Power | Isp (s) | Thrust | Status (Oct 2026) |
|---|---|---|---|---|---|
| AEPS (NASA, L3Harris) | Magnetically shielded Hall | 12 kW | ≈ 2 600 | ≈ 0.6 N | Three flight units delivered 2025 |
| X3 (Univ. of Michigan) | Hall, three channels | 102 kW | 1 800–2 650 | 5.4 N | Laboratory record, 2017 |
| NEXT-C (NASA, Aerojet) | Gridded ion | 6.9 kW | ≈ 4 190 | 0.24 N | Flown on DART; ≈ 2 h in 2021 |
| VX-200 / VX-200SS (Ad Astra) | Helicon and ion-cyclotron heating | 200 kW / 80 kW | ≈ 4 900 | ≈ 5.7 N | Laboratory; 88-hour run, 2021 |
| Applied-field MPD | Electromagnetic | 0.1–1 MW | 1 500–6 000 | 1–100 N | Laboratory; small pulsed units flown |
| Indium FEEP (Enpulsion) | Field emission | tens of W | 2 000–6 000 | ≤ ≈ 1 mN | More than 200 launched |

# 42.3 Nuclear Thermal Propulsion

A **nuclear thermal rocket (NTR)** replaces the combustion chamber with a fission reactor (Figure 42.6). Liquid hydrogen cools the nozzle walls, then flows through hundreds of channels in the hot fuel and leaves at 2 500–2 800 K. The reactor adds energy without adding mass, so the propellant can be the lightest gas of all. Exhaust velocity scales as $\sqrt{T/\mathcal{M}}$, where $\mathcal{M}$ is the exhaust's molar mass; hydrogen at 2 g/mol gives about twice the exhaust velocity of a hydrogen–oxygen engine, whose exhaust is mostly steam. The result is 850–950 s at thrusts of tens to hundreds of kilonewtons.

## 42.3.1 Reactor Physics in Brief

A uranium-235 nucleus that absorbs a neutron splits, releasing about 200 MeV and two or three neutrons; when on average exactly one of these causes another fission, the reactor is **critical**. Slow ("thermal") neutrons cause fission far more readily than fast ones, so most rocket reactors include a **moderator** — graphite, zirconium hydride or beryllium — that slows them by collisions. **Control drums** around the core, cylinders with a neutron absorber such as boron carbide on one face, rotate to absorb more or fewer neutrons. A **shield** between reactor and tank stops neutrons and gamma rays; elsewhere the spacecraft relies on distance.

The fuel must survive hot hydrogen, which attacks graphite: Rover/NERVA used uranium carbide particles in graphite coated with niobium or zirconium carbide, and today's programmes study carbides and "cermets" of uranium dioxide in tungsten. The 1960s reactors used highly enriched uranium (over 90 % U-235). Current designs use **high-assay low-enriched uranium (HALEU)**, just under 20 %. It is unusable for weapons without further enrichment and far cheaper to secure, at the cost of a larger core.

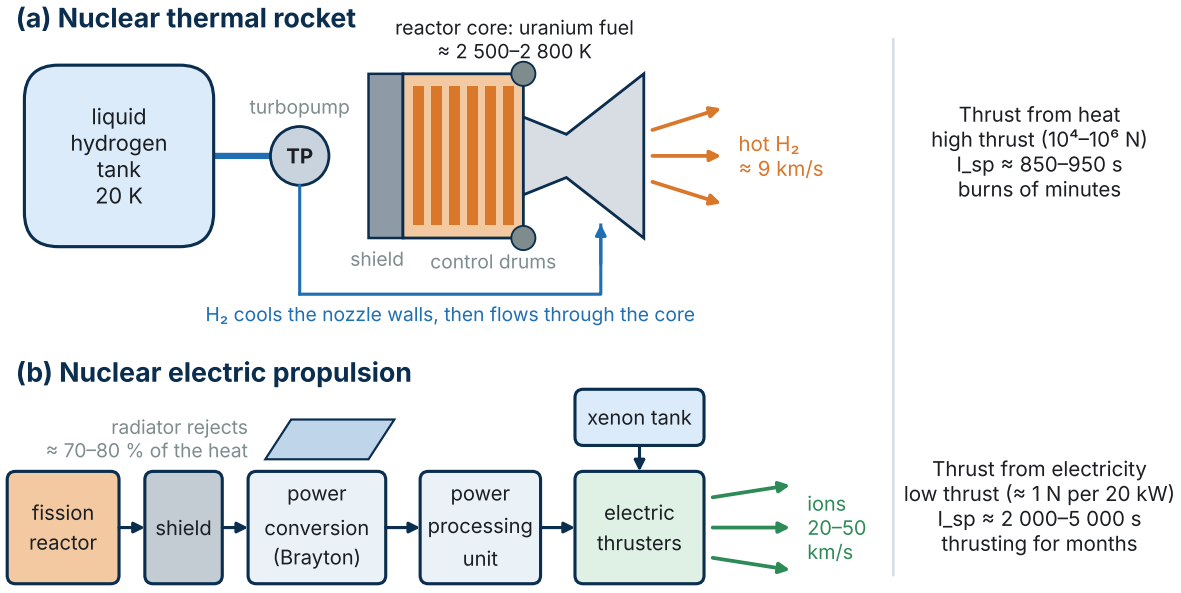



**FIGURE 42.6** Principles of nuclear propulsion. (a) In a nuclear thermal rocket, hydrogen cools the nozzle and is then heated in the reactor core. (b) In nuclear electric propulsion, reactor heat is partly converted into electricity for ion or Hall thrusters; radiators reject the rest.

### 42.3.2 Rover and NERVA

The United States tested about twenty nuclear rocket reactors between 1959 and 1972 in **Project Rover** and the **NERVA** (Nuclear Engine for Rocket Vehicle Application) programme at Jackass Flats, Nevada (Table 42.3). Phoebus-2A ran at 4 082 MW in 1968, still the most powerful nuclear rocket reactor ever operated, and the small Pewee delivered hydrogen at 2 550 K. The XE-Prime engine of 1969, fired downwards in a test stand, was started 28 times and produced about 247 kN. The programme, which had cost about US$ 1.4 billion, was cancelled in January 1973 (Figure 42.7); the Soviet RD-0410 was ground-tested at Semipalatinsk but never flew. Those tests vented exhaust carrying fission products into the open air; a new ground test must capture and filter it, one reason modern programmes are expensive.

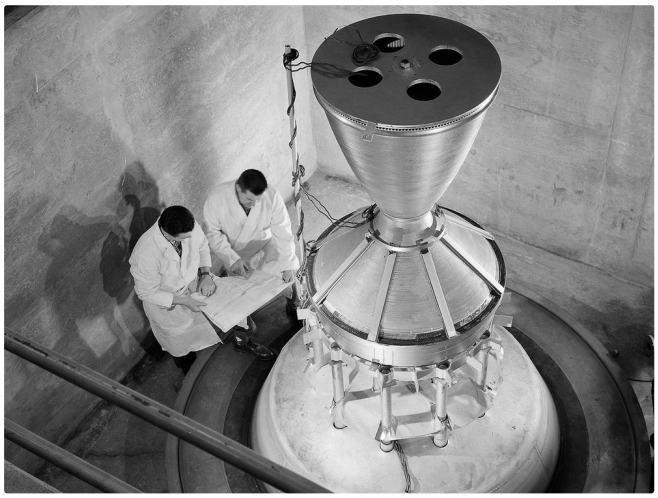


**FIGURE 42.7** Technicians at NASA's Lewis (now Glenn) Research Center manufacture a nozzle for the Kiwi B-1-B nuclear rocket engine in a vacuum oven in 1964. Image: NASA/GRC.

**TABLE 42.3** Milestones of nuclear thermal propulsion.

| Reactor or engine | Year | Reactor power | Milestone |
|---|---|---|---|
| Kiwi-B4E | 1964 | 937 MW | Full-power run and restart |
| Phoebus-2A | 1968 | 4 082 MW | Most powerful nuclear rocket reactor |
| Pewee | 1968 | 503 MW | Exit gas 2 550 K, Isp ≈ 850–900 s |
| NERVA XE-Prime | 1969 | ≈ 1 140 MW | 247 kN, 28 starts, flight-like engine |
| RD-0410 (USSR) | 1970s–80s | 196 MW | 35 kN, 910 s, ground tests only |
| DRACO (DARPA, NASA) | 2023–25 | — | HALEU flight demonstration; cancelled 2025 |

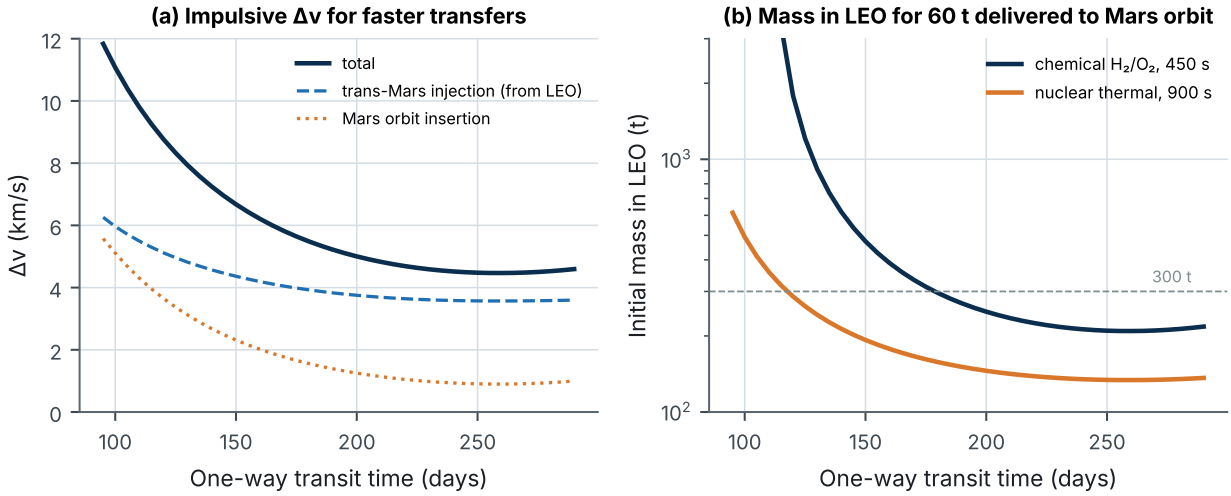
### 42.3.3 DRACO and After

In 2023 NASA joined DARPA's **DRACO** (Demonstration Rocket for Agile Cislunar Operations), which chose Lockheed Martin to build the spacecraft and BWX Technologies the HALEU reactor, aiming for an in-orbit test as early as 2027. NASA's budget request for 2026 then provided nothing for nuclear thermal or nuclear electric propulsion. In mid-2025 DARPA closed DRACO, citing the experiment's cost, the difficulty of ground testing and the falling cost of conventional launch. Congress nevertheless set aside at least US$ 110 million for nuclear thermal propulsion in NASA's

2026 appropriation, which the administrator told the Senate in April 2026 NASA would spend. Russian and Chinese work, mostly on nuclear-electric systems, is hard to judge from public sources.

### 42.3.4 Faster to Mars

The prize for crewed missions is time. Figure 42.8 compares single-stage chemical and nuclear thermal vehicles delivering 60 t into a high Mars orbit. Faster transfers need more Δv at both ends: 4.5 km/s for the 259-day minimum-energy transfer, 5.5 km/s for 180 days and 8.8 km/s for 120 days. With 300 t in LEO, the chemical vehicle can make the crossing in about 180 days and the nuclear one in about 120. Shorter transits matter because interplanetary crews receive about 1.8 mSv of cosmic radiation per day (Chapter 30).




Coplanar circular orbits, optimal phasing; 400 km LEO to 250 × 33 793 km Mars orbit. Stage inert mass: chemical 12 % of propellant; nuclear 20 % of propellant + 10 t of engines and shield. No aerocapture; one-way.

**FIGURE 42.8** Faster Mars transfers computed for this chapter. (a) Impulsive Δv from a 400 km Earth orbit to a 1-sol (250 × 33 793 km) Mars orbit against transit time. (b) Initial mass in low Earth orbit for chemical and nuclear thermal stages delivering 60 t.

**EXAMPLE: SIX MONTHS TO MARS — CHEMICAL OR NUCLEAR?**

A transfer vehicle must deliver 60 t (habitat and lander) from a 400 km LEO into a 1-sol Mars orbit in 180 days. The trajectory model gives a trans-Mars injection of 3.93 km/s and an orbit insertion of 1.56 km/s: Δv = 5.49 km/s.

1. **Chemical (Isp 450 s, $v_e$ = 4.41 km/s).** Mass ratio $R = e^{5.49/4.41}$ = 3.47. If tanks, engines and structure weigh 12 % of the propellant, the final mass is $m_f = 60/[1 - 0.12(R - 1)]$ = 85.3 t and the initial mass $m_0 = R\,m_f$ = **296 t**, of which 211 t is propellant.
2. **Nuclear thermal (Isp 900 s, $v_e$ = 8.83 km/s).** $R = e^{5.49/8.83}$ = 1.86. Bulky hydrogen tanks weigh 20 % of the propellant, plus 10 t of engines and shielding: $m_f = (60 + 10)/(1 - 0.20{\times}0.86)$ = 84.6 t and $m_0$ = **158 t**, with 73 t of propellant.
3. The nuclear vehicle needs about half the mass in LEO. Spending the same 300 t on it instead shortens the trip to about **118 days**, cutting the crew's dose in transit from about 0.32 Sv to 0.21 Sv.
4. The price is hydrogen's low density (71 kg/m³): 73 t fills about 1 000 m³, to be kept at 20 K for months (Chapter 40 discusses boil-off).

## 42.4 Nuclear Electric Propulsion and Space Reactors

Nuclear electric propulsion (NEP) separates energy from reaction mass completely: reactor heat is partly converted into electricity for ion or Hall thrusters (Figure 42.6 b). Its weak point is waste heat. A closed Brayton converter, in which a helium–xenon gas loop drives a turbine and alternator, is about 20–30 % efficient, so a 1 MWe system at 25 % must reject about 3 MW. A radiator at 500 K with an emissivity of 0.9 sheds only about 3.2 kW per square metre from each face, so it would need some 470 $m^2$, nearly two tennis courts.

### 42.4.1 Reactors in Space So Far

The United States has flown one reactor: **SNAP-10A**, launched in April 1965, which produced about 500 W of electricity for 43 days before an electrical failure shut it down. The Soviet Union flew 33: 31 **BES-5 Buk** reactors of about 3 kWe powered radar ocean-reconnaissance satellites (RORSATs) between 1970 and 1988, and two **TOPAZ** thermionic reactors of about 5 kWe flew in 1987 (Table 42.4). RORSATs boosted their reactors into storage orbits about 900 km high after use, but in January 1978 Kosmos 954 failed to and scattered radioactive debris across northern Canada. The accident shaped the United Nations principles on nuclear power sources in space (1992), including the rule that a reactor must not go critical before reaching its operating orbit or interplanetary trajectory.

### 42.4.2 Kilopower and Fission Surface Power

NASA's modern work began with **Kilopower**, a family of 1–10 kWe concepts with a solid cast uranium core, passive sodium heat pipes and Stirling engines. The **KRUSTY** prototype (Kilopower Reactor Using Stirling Technology) was tested in Nevada from November 2017 to March 2018. The tests ended with a 28-hour full-power run that simulated start-up, failed heat pipes and failed Stirling engines. This was the first test of a new US space-reactor design in decades. In 2022 NASA and the Department of Energy funded three industry designs for a 40 kWe **Fission Surface Power** system for the Moon (Figure 42.9).

Policy then accelerated. A NASA directive in August 2025 called for a lunar reactor of at least 100 kWe ready for launch around 2030. In April 2026 a White House memorandum (NSTM-3) launched a National Initiative for American Space Nuclear Power. It called for reactors in orbit as early as 2028 and on the Moon by 2030. The first lunar unit, **Lunar Reactor-1**, is specified in an August 2026 draft solicitation at no less than 20 kWe — a smaller first step than the 2025 goal. A landing is intended in 2030 to power a lunar base through the 14-day night (Chapter 41).

### 42.4.3 SR-1 Freedom

The most ambitious near-term project is **Space Reactor-1 (SR-1) Freedom**, announced by NASA Administrator Jared Isaacman in March 2026 as the first spacecraft to use a fission reactor for propulsion beyond Earth orbit. A reactor designed by Idaho National Laboratory, fuelled with HALEU uranium dioxide and cooled by heat pipes, will produce at least 20 kWe through a closed Brayton converter supplied by Sierra Lobo. It sits behind a boron-carbide shield at the end of a boom with radiator fins, and will be switched on only after leaving Earth orbit. It will then drive the xenon Hall thrusters of the repurposed Gateway PPE through an electric-propulsion system of up to 48 kW. The PPE was built by Lanteris Space Systems, part of Intuitive Machines since

January 2026. SR-1 is due to launch in December 2028 and reach Mars about a year later with three Ingenuity-derived helicopters (the SkyFall mission) to scout landing sites. Its reported cost is about US$ 2.1 billion without them. The schedule is tight — the reactor was still being designed in October 2026 — but SR-1 would be the first US reactor launched in more than 60 years. Russia's "Zeus" nuclear-electric tug, once proposed for a first flight around 2030, has shown few public signs of hardware progress since 2023.

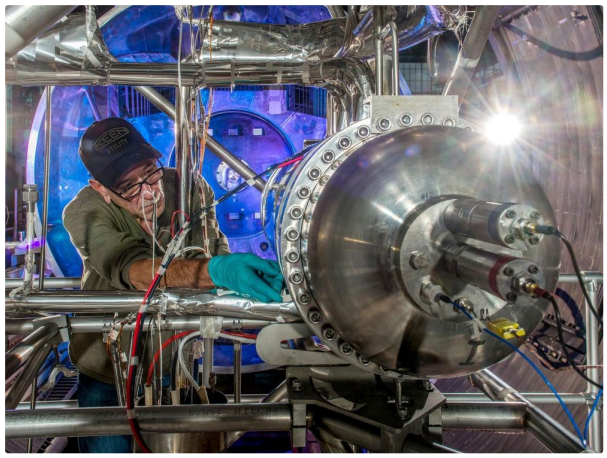


**FIGURE 42.9** A 10 kW Stirling power-conversion unit for NASA's Fission Surface Power Technology Demonstration Unit at Glenn Research Center; four such units would turn 180 kW of reactor heat into 40 kW of electricity. Image: NASA/GRC.

**TABLE 42.4** Space fission power systems, flown and planned.

| System | Country | Years | Electrical power | Conversion | Status |
|---|---|---|---|---|---|
| SNAP-10A | USA | 1965 | ≈ 0.5 kW | Thermoelectric | Flown; 43 days |
| BES-5 Buk | USSR | 1970–88 | ≈ 3 kW | Thermoelectric | 31 flown on RORSATs |
| TOPAZ | USSR | 1987 | ≈ 5 kW | Thermionic | 2 flown |
| KRUSTY (Kilopower) | USA | 2017–18 | 1 kW class | Stirling | Ground test; 28 h at full power |
| SR-1 Freedom | USA | 2028 (planned) | ≥ 20 kW | Closed Brayton | In development; drives Hall thrusters |
| Lunar Reactor-1 | USA | 2030 (planned) | ≥ 20 kW | — | Draft solicitation, August 2026 |

## 42.5 Sailing on Light

### 42.5.1 Radiation Pressure

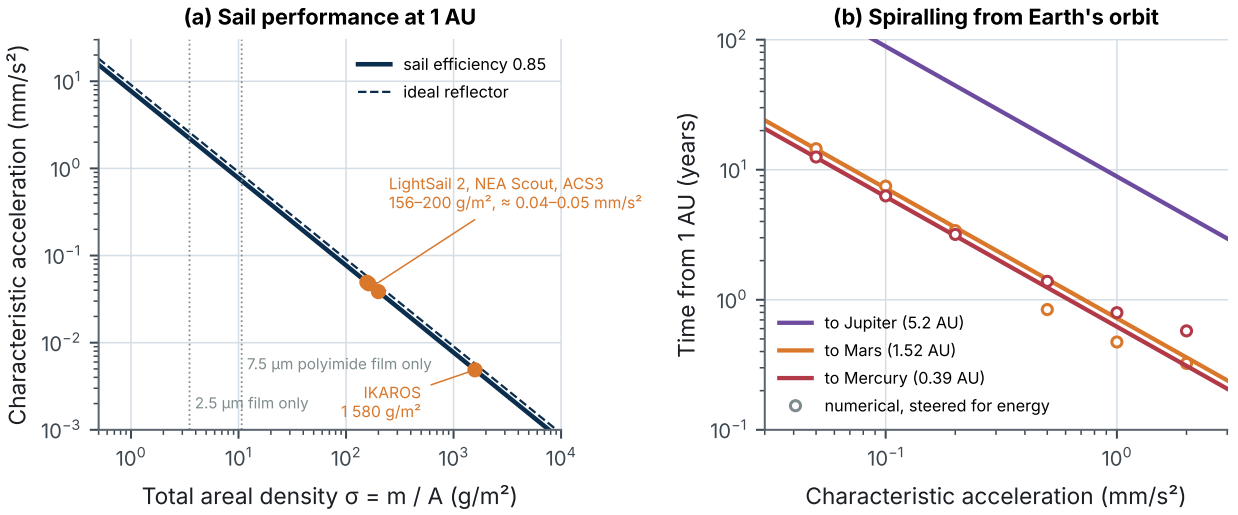
Light carries momentum: a beam of power $P$ pushes on an absorbing surface with a force $P/c$, and on a perfect mirror facing it with $2P/c$. At 1 AU sunlight delivers 1 361 W/m², a pressure of 4.54 μN/m² on an absorber and 9.08 μN/m² on a perfect mirror. A flat sail of area $A$, tilted at a **cone angle** α to the sunlight and with an overall efficiency η (about 0.85, allowing for absorption and wrinkles), feels

$$F = \frac{2\eta P_{\odot} A}{c}\cos^2\alpha.$$

Dividing by mass gives the **characteristic acceleration** $a_c$ at normal incidence, which depends only on the **areal density** σ = m/A of the whole spacecraft (Figure 42.10 a, Table 42.5). Sail acceleration and solar gravity both fall with the square of distance, so their ratio — the **lightness number** β — is the same everywhere. Tilting the sail by about 35° gives the largest push along the orbit, raising it in an outward spiral; tilting it the other way spirals inwards. For a slow spiral from radius $r_0$ to $r$ the time is about

$$T \approx \frac{\sqrt{\mu_\odot / r_0}\;\left|(r/r_0)^{3/2} - 1\right|}{3 \times 0.385\, a_c},$$

where 0.385 is $\cos^2\alpha \sin\alpha$ at the optimum angle (Figure 42.10 b).




a_c = 2ηP/(cσ), P/c = 4.54 µN/m² at 1 AU; σ uses total spacecraft mass. (b) Lines: quasi-circular spiral, cone angle 35.3°, T = √(μ/r₀) |(r/r₀)^1.5 − 1| / (3 × 0.385 a_c). Dots: numerical flat-sail trajectories steered to change orbital energy fastest — time to first reach the distance, without matching the planet's velocity.

**FIGURE 42.10** Solar-sail performance computed for this chapter. (a) Characteristic acceleration at 1 AU against the areal density of the whole spacecraft, with the sails flown so far. (b) Time to spiral from Earth's orbit to the distances of Mars, Mercury and Jupiter: lines from the spiral formula, dots from numerical trajectories.

**TABLE 42.5** Solar-sail missions. Masses are for the whole spacecraft.

| Mission | Operator | Launch | Sail area | Mass | Outcome |
|---|---|---|---|---|---|
| IKAROS | JAXA | 2010 | 196 m² | ≈ 310 kg | Venus flyby |
| NanoSail-D2 | NASA | 2010 | 10 m² | ≈ 4 kg | Deorbit test |
| LightSail 2 | Planetary Soc. | 2019 | 32 m² | ≈ 5 kg | Sailed in LEO |
| NEA Scout | NASA | 2022 | 86 m² | 14 kg | Never contacted |
| ACS3 | NASA | 2024 | 80 m² | 16 kg | Deployed; boom bent |
| Solar Cruiser | NASA | — | ≈ 1 670 m² | — | Cancelled 2023 |

## 42.5.2 Sails Flown So Far

JAXA's **IKAROS**, launched with the Venus orbiter Akatsuki in 2010, was the first interplanetary solar sail. It unfolded a 14 m square of 7.5 µm polyimide film by spinning, carried thin-film solar cells and steered with liquid-crystal panels of switchable reflectivity; JAXA measured a force of 1.12 mN. NASA's **NanoSail-D2** and The Planetary Society's **LightSail 2** followed in Earth orbit; LightSail 2 raised its apogee by about 2 km in its first days of sailing. NASA's 86 m² **NEA Scout**, launched on Artemis I in 2022, was never heard from (Figure 42.11), and the nearly 1 700 m² **Solar Cruiser** was not confirmed for flight and was closed out in 2023. The **Advanced Composite Solar**

**Sail System (ACS3)**, a 16 kg, 12-unit CubeSat, deployed an 80 m² sail on four 7 m carbon-fibre composite booms in August 2024 (Figure 42.12). One boom was later reported bent and the sail was not actively steered, but the deployment proved booms that larger sails will need.

**FIGURE 42.11** Engineers at NASA's Marshall Space Flight Center deploy a flight-like engineering unit of the 86 m² NEA Scout solar sail, first unfurled there in August 2016. Image: NASA/MSFC (EMMETT GIVEN/CHARLES BEASON/ JON).

**FIGURE 42.12** The ACS3 sail seen by one of the spacecraft's four wide-angle cameras during deployment in orbit on 28 August 2024: reflective quadrants and a composite boom, curved by the lens. Image: NASA (NASA).

### 42.5.3 Electric and Magnetic Sails

The solar wind, streaming outwards at about 400 km/s, carries far less momentum than sunlight but can be intercepted by structures much lighter than film. Pekka Janhunen's **electric sail** (2004) would spin out thin tethers tens of kilometres long, charged to about 20 kV, whose electric fields repel solar-wind protons over a width far greater than the wires. Because the electron sheaths around the wires widen as the solar wind thins, the thrust falls only as 1/r rather than 1/r². Estonia's ESTCube-1 (2013) and Finland's Aalto-1 (2017) carried tether experiments, but neither managed to unreel its tether. A **magnetic sail** would deflect the solar wind with a large superconducting loop or a plasma-inflated magnetic bubble; none has been tested in flight.

#### EXAMPLE: SAILING FROM EARTH'S ORBIT TO MARS'S

A sail of 1 600 m² of 2.5 µm aluminised film (about 3.5 g/m², 5.6 kg) is held by 6 kg of composite booms and carries a 40 kg spacecraft: 51.6 kg in all.

1. **Areal density:** σ = 51.6 kg/1 600 m² = 32.2 g/m².
2. **Force at 1 AU:** $F = 2 \times 0.85 \times 4.54\ \mu\text{N/m}^2 \times 1\,600\ \text{m}^2 = \mathbf{12.3\ mN}$, so $a_c = 12.3\ \text{mN}/51.6\ \text{kg} = \mathbf{0.239\ mm/s^2}$ and β = 0.239/5.93 = 0.040.
3. **Δv budget:** at the optimum cone angle the push along the orbit is 0.385 × 0.239 = 0.092 mm/s², or 2.9 km/s per year at 1 AU, falling as the sail moves out.
4. **Spiral time:** with $\sqrt{\mu_\odot / r_0} = 29.8$ km/s and $(1.524)^{3/2} - 1 = 0.881$, $T \approx 29\,800 \times 0.881/(3 \times 0.385 \times 0.000239)\ \text{s} = 9.5 \times 10^7\ \text{s} \approx \mathbf{3.0\ years}$ to Mars's distance. The numerical trajectory gives 3.3 years, against 0.7 years for a chemical Hohmann transfer.
5. By the same formula, a one-year spiral would need $a_c \approx 0.72\ \text{mm/s}^2$, or σ ≈ 11 g/m² for the whole spacecraft — beyond any sail yet built. Sails win where the Δv is enormous, such as stations hovering sunward of L1, where no propellant supply would last.

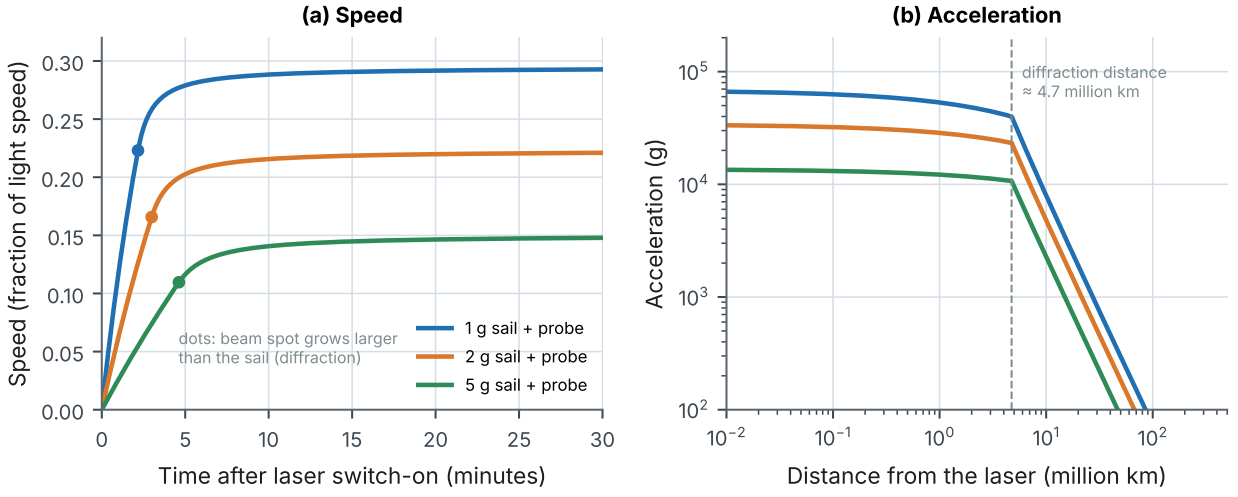
## 42.6 Beamed Energy and the Interstellar Challenge

The nearest star system, Alpha Centauri, is 4.37 light-years — about 276 000 AU — away. Voyager 1, the fastest object leaving the Solar System at about 17 km/s (3.6 AU per year), would need some 77 000 years to cover that distance. Reaching another star within a lifetime needs a tenth to a fifth of the speed of light.

### 42.6.1 Breakthrough Starshot

In 2016 the Breakthrough Initiatives announced **Starshot**, with US$ 100 million for research into leaving the engine on the ground. A phased array of about ten million lasers totalling up to 100 GW, spread over about a square kilometre, would focus on a sail a few metres across carrying a gram-scale “StarChip” with cameras, a laser transmitter and a radioisotope power source. Accelerating at about 10 000 g for some minutes, the probe would reach 15–20 % of the speed of light, pass Alpha Centauri about 20 years later and send images home by laser. A cost model by Kevin Parkin put the ground system at about US$ 8 billion, with an optimum 4.1 m sail accelerated for about nine minutes.

The physics is simple and severe (Figure 42.13). A perfectly reflecting sail struck by 100 GW feels $2P/c$ = 667 N; on a 2 g probe that is 34 000 g. But a laser beam spreads by diffraction. An array of diameter $D$ focuses light of wavelength λ into a spot about $2.44\,\lambda L/D$ across at distance $L$. So a 3 km array keeps a 4.1 m sail fully lit only to about 4.7 million km, some 12 times the distance to the Moon. Beyond that the captured power falls with the square of distance and the relativistic Doppler shift weakens the push, so most of the speed is gained in the first three minutes.




100 GW at 1.064 µm from a 3 km phased array, 4.1 m perfectly reflecting sail; F = (2P/c)(1−β)/(1+β) × captured fraction (1, or $(L_o/L)^2$ beyond the diffraction distance $L_o$ = D d/2.44λ); relativistic motion; light-travel delay neglected.

**FIGURE 42.13** Starshot-class laser sails computed for this chapter: 100 GW from a 3 km array at 1.064 µm, a 4.1 m perfectly reflecting sail and total masses of 1, 2 and 5 g. (a) Speed against time. (b) Acceleration against distance from the laser.

The engineering is far harder. It needs a sail under a micrometre thick that reflects almost all of an intense beam and rides it stably, and ten million lasers phased through a turbulent atmosphere. The probe must also survive dust impacts at 60 000 km/s and send data home across 4 light-years. Starshot’s first phase judged these problems to have credible conceptual solutions, but the programme was paused in 2024 to seek broader support.

**EXAMPLE: THE NUMBERS BEHIND A LASER SAIL**

A 2 g probe (1 g sail, 1 g chip) with a 4.1 m perfectly reflecting sail is pushed by 100 GW of 1.064 µm laser light from a 3 km array.

1. **Force and acceleration:** $F = 2P/c = 2 \times 10^{11}$ W/3.00 × $10^8$ m/s = **667 N**, a pressure of 51 Pa on the 13.2 m² sail; $a = F/m = 3.3 \times 10^5$ m/s² ≈ **34 000 g**.
2. **Without diffraction**, reaching 0.2c ($6.0 \times 10^7$ m/s) would take $v/a$ = 180 s over $v^2/2a$ = 5.4 million km.
3. **Diffraction distance:** $L_0 = Dd/2.44\lambda = 3\,000 \times 4.1/(2.44 \times 1.064 \times 10^{-6})$ m = $4.7 \times 10^9$ m. The simulation in Figure 42.13 reaches 0.17c there after 179 s and 0.22c after 10 minutes, 31 million km out.
4. **Energy:** the laser emits 100 GW × 600 s = 60 TJ; the probe's kinetic energy at 0.2c, $(\gamma - 1)mc^2$, is 3.7 TJ — an efficiency of about 6 %.
5. **Heating:** absorbing just $10^{-5}$ of the beam, the sail would have to shed 1 MW from 13 m², so it must be an almost perfect dielectric mirror. At 0.2c, Alpha Centauri is about 22 years away.

## 42.6.2 Fusion and Antimatter

Fusion releases about four times more energy per kilogram than fission, and its charged products could in principle be directed out as exhaust by magnetic fields (Figure 42.14). The British Interplanetary Society's **Project Daedalus** (1973–78) designed an uncrewed two-stage probe of about 54 000 t. Most of this was deuterium–helium-3 pellets, ignited by electron beams 250 times a second. The probe would carry a 450 t payload to Barnard's Star, 5.9 light-years away, at 12 % of the speed of light in about 50 years. Earlier, **Project Orion** (1958–65) studied a ship pushed by small nuclear explosions, until the 1963 Partial Test Ban Treaty and lack of funding ended it.

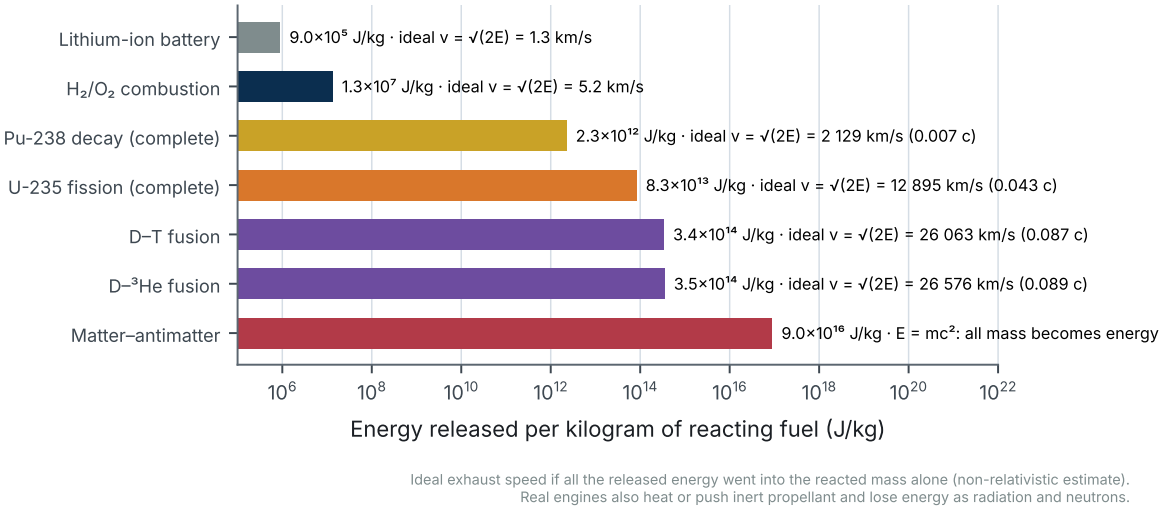



**FIGURE 42.14** Energy released per kilogram of reacting fuel and the corresponding ideal exhaust speed, computed from reaction energies. Each step from chemistry to fission, fusion and annihilation multiplies the energy by orders of magnitude.

No fusion device has yet produced more energy than its whole system consumes. The US National Ignition Facility first achieved ignition in December 2022, releasing 3.15 MJ from a capsule heated by 2.05 MJ of laser light, but its lasers drew about 300 MJ from the grid. Compact

propulsion concepts such as the **Direct Fusion Drive**, based on the Princeton field-reversed configuration, are estimated to give 2.5–5 N per megawatt of fusion power at about 10 000 s. They remain small laboratory experiments.

Matter–antimatter annihilation converts all of the mass into energy: 1 g of antimatter meeting 1 g of matter releases $1.8 \times 10^{14}$ J, as much as a 43-kiloton explosion. But antimatter is made in particle accelerators one particle at a time, at minute efficiency. CERN makes about $10^{12}$ antiprotons a day, and all it has ever produced amounts to less than 10 nanograms; at that rate a gram would take more than a billion years. Antimatter rockets remain thought experiments.

## 42.7 Breathing Air and Detonating Fuel

### 42.7.1 Scramjets and Combined Cycles

A hydrogen–oxygen rocket carries about 6 kg of oxygen for every kilogram of hydrogen. Air-breathing engines avoid that mass and reach fuel-specific impulses of 1 000–4 000 s, but only inside the atmosphere (Chapter 10). Above about Mach 5 the air must stay supersonic through the combustor, the regime of the **scramjet**. NASA's X-43A (Figure 42.15) reached Mach 9.6 for about 10 seconds in November 2004, and the US Air Force's X-51A flew for 210 s at up to Mach 5.1 in 2013. Combined-cycle engines switch between modes. The British SABRE design pre-cooled incoming air from about 1 000 °C to −150 °C in a fraction of a second. This let it run as an air-breathing rocket up to about Mach 5 before switching to pure rocket mode. Its developer, Reaction Engines, tested the pre-cooler at Mach 5 conditions in 2019 but entered administration in October 2024, and its intellectual property was being sold in 2025.

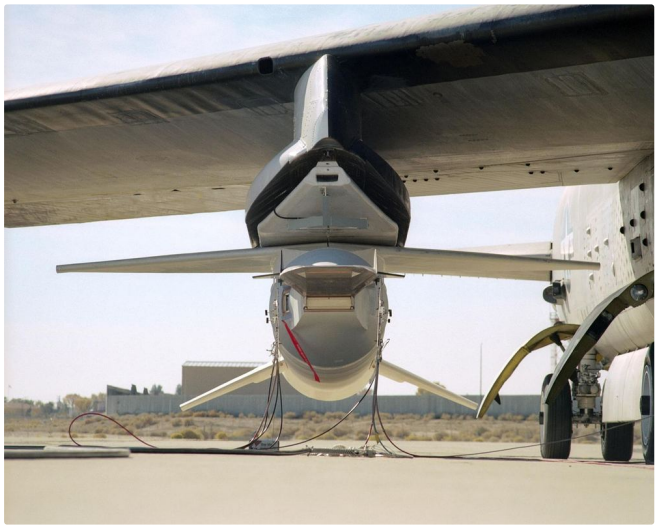


**FIGURE 42.15** The X-43A scramjet research vehicle — the small wedge at the nose of its modified Pegasus booster — under the wing of NASA's B-52B on 15 November 2004, the day before its record Mach 9.6 flight. Image: NASA/AFRC (NASA/Tony Landis).

### 42.7.2 Rotating Detonation Engines

Conventional rocket engines burn propellant at almost constant pressure. In a **rotating detonation engine (RDE)**, propellants injected into an annular channel are consumed by one or more detonation waves — shock waves coupled to combustion. The waves race around the annulus at about 2 km/s, thousands of times a second. Detonation raises the pressure as it burns,

so analyses predict a higher specific impulse for the same propellants, and the thin reaction zone allows a much shorter chamber. In July 2021 Nagoya University and JAXA flew an RDE in space for the first time on the S-520-31 sounding rocket, producing about 500 N for 6 seconds. At NASA's Marshall Space Flight Center a 3D-printed copper-alloy rotating detonation rocket engine (RDRE) produced about 18 kN (4 000 lbf) in 2022. In September 2023 it fired for 251 seconds (Figure 42.16) at more than 25 kN (5 800 lbf). In May 2025 the US company Venus Aerospace reported a short flight of an RDRE-powered test vehicle from Spaceport America. Stabilising the waves and cooling the walls are the main challenges.

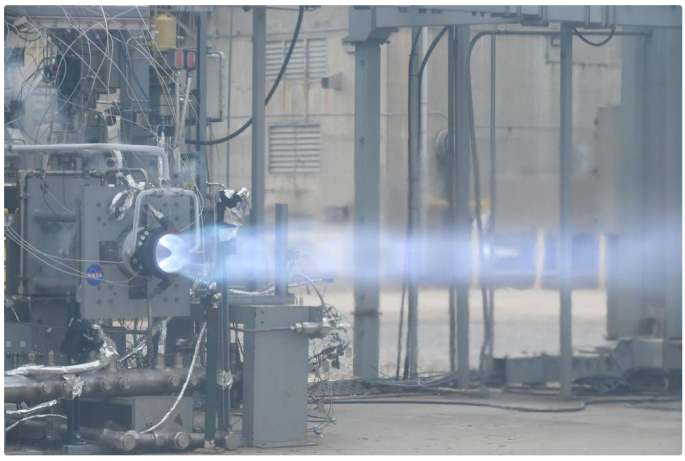


**FIGURE 42.16** A 251-second hot fire of a full-scale, 3D-printed rotating detonation rocket engine at NASA's Marshall Space Flight Center in 2023, producing more than 25 kN (5 800 lbf) of thrust. Image: NASA/MSFC.

## 42.8 Propellantless and Exotic Ideas

### 42.8.1 Tethers

A long conducting wire moving through Earth's magnetic field acts as a generator. Its **motional EMF** is $\varepsilon = vBL$: about 2 kV for a 10 km tether in low orbit, where v = 7.6 km/s and B ≈ 30 µT (Figure 42.17). If the circuit is closed through the ionosphere by plasma contactors, a current flows and the magnetic force $I\vec{L} \times \vec{B}$ opposes the motion. A current of 1 A gives about 0.3 N of drag, enough to deorbit dead satellites without propellant (Chapter 21). Driving the current the other way with solar power turns the tether into a motor that raises the orbit. NASA's Plasma Motor Generator (1993) tested both modes with a 500 m tether. On the TSS-1R mission in February 1996, Space Shuttle Columbia deployed an Italian satellite on a conducting tether to 19.7 km. The system generated about 3 500 V and nearly 0.5 A before arcing burned through the tether and it broke.

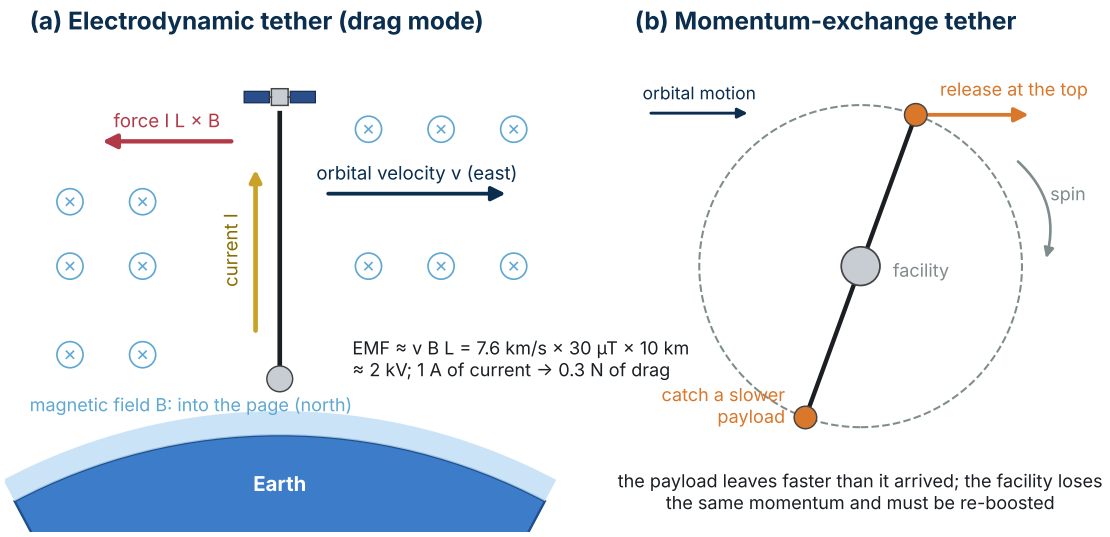



**FIGURE 42.17** Tethers. (a) An electrodynamic tether in Earth's magnetic field: the induced current feels a force opposing the motion (or, driven by solar power, along it). (b) A rotating tether catches a slow payload at the bottom of its swing and releases it faster at the top, giving up momentum.

Non-conducting tethers exchange momentum mechanically. In 1993 the SEDS-1 experiment lowered a 26 kg payload on a 20 km tether below a Delta II stage and cut it loose into a re-entry trajectory. In 2007 the student-built YES2 deployed 31.7 km of tether — the longest structure ever deployed in space. The ultimate tether, a **space elevator** from the equator to beyond geostationary orbit, fails on materials. Even with a cable ten times thicker at geostationary altitude than at the ground, it would need a breaking length of about 2 100 km under Earth's gravity. The best commercial fibres offer about 400 km.

Two propellant-free techniques already serve missions: gravity assists (Chapter 24) and atmospheric braking. **Aerocapture** is a single deep pass through a planet's atmosphere inside an aeroshell to brake straight into orbit. It has not yet flown, but could cut the mass of an orbiter for Uranus, Neptune or Titan by a third or more (Chapter 25).

### 42.8.2 Reactionless Drives and Warp Drives

Every working propulsion system conserves momentum: something — exhaust, photons, the solar wind, a planet's magnetic field — must carry momentum the other way. The **EmDrive**, a closed, tapered microwave cavity proposed around 2001, was claimed to produce thrust without exhaust; a NASA Eagleworks paper in 2016 reported about 1.2 mN per kilowatt. Tests at TU Dresden with new thrust balances and suspensions reproduced the apparent thrust and then made it disappear. It came from thermal expansion of the mountings and from currents in the power cables interacting with Earth's magnetic field. The 2021 results bounded any anomalous force below that of a photon rocket of the same power: 3.3 μN per kilowatt, the push of simply shining the microwaves into space. This bound is hundreds to thousands of times below the earlier claims. "Mach-effect" thrusters tested the same way also failed to show real thrust.

In 1994 Miguel Alcubierre showed that Einstein's equations allow, mathematically, a "warp bubble" that contracts space ahead of a ship and expands it behind, carrying it faster than light without local motion. It needs negative energy density — "exotic matter" unknown in bulk — and the first estimates of the energy required exceeded the mass-energy of the observable universe.

Later designs cut the energy enormously but not to zero, and a bubble could not be steered from inside, because its front edge would lie beyond the reach of any signal from the ship. Warp drives remain an exercise in general relativity, not a propulsion technology.

### AI IN ADVANCED PROPULSION

Plasma thrusters are hard to model: electrons in a Hall thruster cross the magnetic field far faster than classical collisions allow, and no first-principles theory yet predicts this "anomalous" transport. Researchers have used Bayesian inference and machine learning on plasma measurements to build data-driven transport models and fast design surrogates. In fusion, reinforcement learning has controlled the shape of a tokamak plasma at EPFL's TCV device (2022). Neural networks also approximate optimal low-thrust control for fast trajectory searches, and Hall-thruster fleets supply telemetry for anomaly detection (Chapters 48, 49 and 52).

### KEY TAKEAWAYS

- Mass ratio grows as $e^{\Delta v/v_e}$, and jet power $\tfrac{1}{2}Fv_e$ makes faster exhaust cost power.
- Electric propulsion is scaling up (12 kW AEPS built, 100 kW run), but 100-day Mars trips need about 1 kW/kg.
- Nuclear thermal rockets (850–950 s) roughly halve the LEO mass of a six-month Mars transfer; DRACO was cancelled in 2025.
- Space fission power is returning: SR-1 Freedom (≥ 20 kWe, nuclear-electric) is planned for December 2028, Lunar Reactor-1 for 2030.
- Sails flown so far reach at most about 0.05 mm/s$^2$; a one-year trip to Mars's distance needs about 11 g/m$^2$.
- A 100 GW laser could push a gram-scale sail to about 0.2c in minutes, but the engineering is unsolved and Starshot is paused.
- Scramjets, detonation engines and tethers serve niches; reactionless and warp drives fail the tests of physics.

### REVIEW QUESTIONS

1. Compare mass ratios for Δv = 10 km/s at $v_e$ = 4.4, 8.8 and 40 km/s. Why not raise $v_e$ without limit?
2. A 50 kW Hall-thruster system works at 60 % efficiency and 2 500 s. What is its thrust, and how long would it take to give a 20 t spacecraft 5 km/s?
3. Define specific power α. Using Figure 42.3, explain why a 40-day crewed trip to Mars is unrealistic with present technology.
4. Why does a nuclear thermal rocket reach twice the Isp of a hydrogen–oxygen engine? What do its moderator, control drums and shield do?
5. Why was DRACO cancelled, and why is ground-testing an NTR harder now than in the 1960s?
6. A 1 MWe NEP system has 25 % efficient converters and radiators at 500 K (emissivity 0.9). Estimate the radiator area.
7. A 4 kg CubeSat carries a 40 m$^2$ sail. Calculate its characteristic acceleration at 1 AU (η = 0.85) and the time to spiral out to 1.5 AU.
8. Why does a laser sail's push collapse beyond a certain distance, and how does doubling the array diameter change it?
9. How does a rotating detonation engine work? Describe two tests made since 2021.
10. Why do careful experiments rule out the EmDrive? What thrust must any device radiating 1 kW of electromagnetic power produce?

PART XIII

# Artificial Intelligence in the Space Ecosystem

Artificial intelligence now runs on satellites, plans rover drives, forecasts the weather, screens collision warnings and designs rocket parts. Part XIII is an in-depth guide in eleven chapters: the foundations of machine learning and deep learning, the data and hardware that make AI work in orbit, its applications in Earth observation, weather, autonomy, operations, security, science, exploration, design and business — and the assurance and governance needed to trust it.


43 AI Foundations: Machine Learning for Space Engineers

44 Deep Learning, Foundation Models and Generative AI

45 Data, Computing Hardware and Edge AI in Orbit

46 AI for Earth Observation

47 AI for Weather, Climate and Space Weather

48 AI for Autonomy, Navigation and Robotics

49 AI for Mission Operations, Communications and Networks

50 AI for Space Situational Awareness, Defence and Security

51 AI for Space Science, Planetary Exploration and Human Spaceflight

52 AI in Spacecraft Design, Manufacturing, Launch and the Space Business

53 Trustworthy AI for Space: Assurance, Security, Governance and Outlook


Image: NASA/ARC (Dominic Hart)

CHAPTER

# 43 AI Foundations: Machine Learning for Space Engineers

Image: NASA (BRANDON TORRES)

*Artificial intelligence now screens clouds on board satellites, flags anomalies in telemetry, maps the planet's forests and plans rover drives on Mars. This chapter gives space engineers the foundations needed to judge, specify and build such systems. It separates symbolic AI — rules, search and planning — from machine learning, traces how each wave of AI reached space from the expert systems of the 1980s to the foundation models of the 2020s, and maps the main problem types and learning paradigms onto space examples. It then explains, with equations and computed examples, how models learn by gradient descent, why they overfit, how classical methods work, how to measure performance honestly for rare events, and how spatial and temporal autocorrelation can make a useless model look excellent.*

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A modern Earth-observation satellite can collect more data in a day than its operators could inspect in a lifetime. NASA's Earth science archive alone held 178.7 PB at the end of September 2025. It was growing by about 160 TB a day, and a large mission may record thousands of telemetry channels every second for decades. Nobody can write explicit rules for every pattern hidden in such data. **Machine learning** — programs that improve their performance by learning from examples — has therefore become as much a part of the space engineer's toolkit as orbital mechanics or link budgets.

Machine learning is also easy to misuse. A model can score 99 % accuracy and be useless; it can pass every validation test and fail on the first image from a new region. This chapter builds the foundations needed to avoid these traps: what artificial intelligence (AI) is and is not, how machines learn, which classical methods suit which problems, and how to measure performance honestly. The ten chapters that follow apply these ideas across the space ecosystem.

## 43.1 What AI Is — and Is Not

**Artificial intelligence** is the field of building computer systems that perform tasks which normally require human intelligence: perceiving, reasoning, planning, deciding and learning. The term was coined for a workshop at Dartmouth College in 1956. It implies neither consciousness nor understanding, and it is not one technique but a family of them (Figure 43.1).

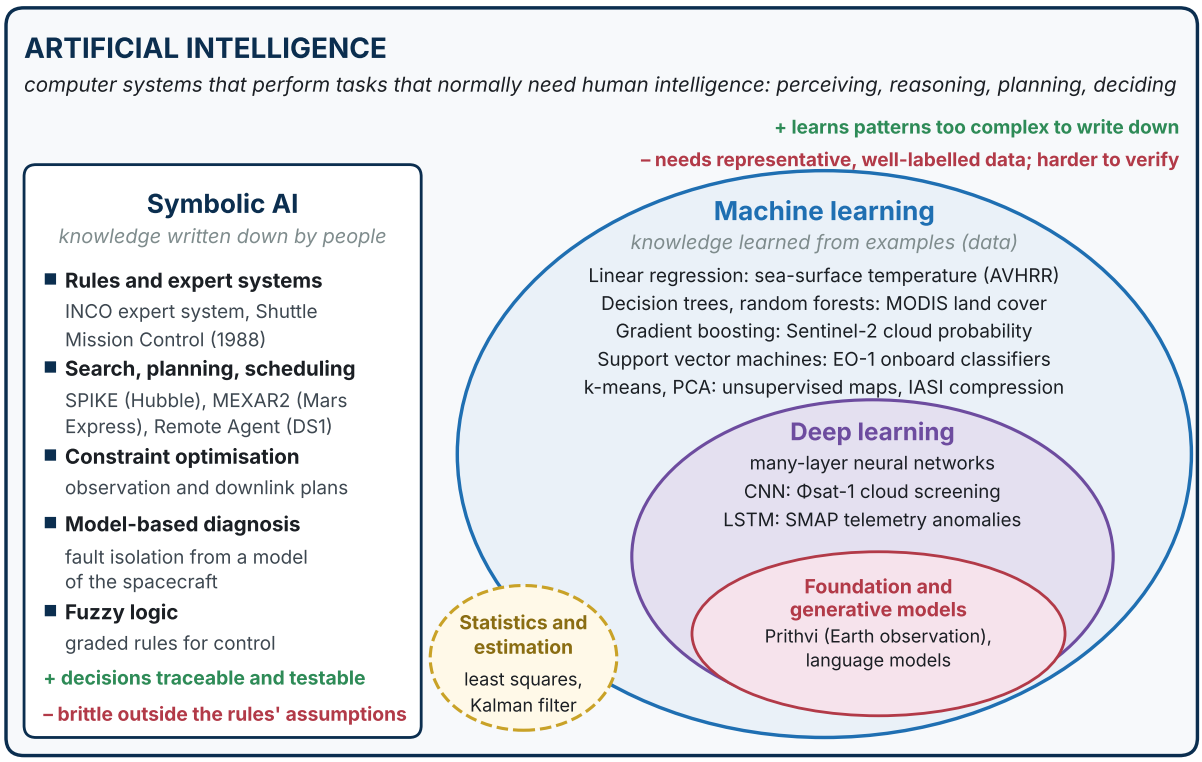



**FIGURE 43.1** How the families of AI relate. Symbolic AI encodes knowledge written down by people; machine learning learns it from data; deep learning uses many-layer neural networks; foundation models are large networks pre-trained on huge unlabelled data sets. Statistics and estimation theory overlap machine learning. All examples are space systems discussed in this chapter.

### 43.1.1 Symbolic and Learned AI

**Symbolic AI** represents knowledge explicitly, in a form that people write and read: rules (“if the bus voltage is below 26 V and the battery is discharging, shed heater loads”), facts, models and search procedures. Its main branches are **expert systems**, which apply hundreds or thousands of if–then rules elicited from specialists, and **search and planning**, which find a sequence of actions that reaches a goal within constraints. **Constraint optimisation** builds schedules that satisfy many conditions at once. **Model-based diagnosis** explains symptoms by reasoning over a model of the hardware. In **fuzzy logic**, rules hold to a degree between true and false.

**Machine learning (ML)** takes the opposite approach: the engineer provides examples and an algorithm finds a function that fits them. A cloud detector is not told that "clouds are bright in the visible and moderately bright at 1.6 μm". Instead, it is shown thousands of pixels labelled "cloud" or "clear" and works out a boundary between them. **Deep learning** uses neural networks with many layers, which learn their own features from raw images, spectra or time series (Chapter 44). **Foundation models** are very large deep networks pre-trained on enormous unlabelled data sets and then adapted to many tasks.

Table 43.1 compares the two traditions. The boundary is not sharp: a rover's planner may call a learned terrain classifier, and a learned model may be wrapped in hand-written safety rules. Many of the strongest space systems combine both.

**TABLE 43.1** Symbolic AI and machine learning compared.

| Aspect | Symbolic AI | Machine learning |
|---|---|---|
| Source of knowledge | Experts write rules, models, constraints | Algorithm infers patterns from examples |
| Typical methods | Expert systems, planners, schedulers, fuzzy logic | Regression, trees, forests, SVMs, clustering, neural networks |
| Space examples | Shuttle INCO expert system, Hubble's SPIKE, Mars Express MEXAR2 | Sea-surface temperature regression, MODIS land cover, onboard cloud screening |
| Strengths | Transparent, predictable, verifiable against requirements | Handles patterns too complex to write down; improves with data |
| Weaknesses | Brittle outside its assumptions; costly to maintain | Needs representative labelled data; can fail silently on new inputs |

### 43.1.2 Why Symbolic Methods Still Matter in Space

Spaceflight puts a premium on **verifiability**. Flight software must be shown, before launch, to do what its requirements say in every mode the mission can enter (Chapter 6). A rule base or planner can be reviewed line by line and sometimes proved correct with formal methods. A neural network's behaviour is known only where it has been tested, and engineers must argue statistically about everything in between (Chapter 53).

Safety-critical layers — fault protection, safe modes, attitude control — are therefore still built from explicit logic and classical control. Learned models are used where their errors are tolerable or can be caught: choosing images to downlink, flagging telemetry for a human, proposing plans that a validated planner checks. On 8 and 10 December 2025 Perseverance drove 210 m and 246 m along waypoints generated by a generative AI model. Every command was first checked on a digital twin of the rover that tracked more than 500 000 telemetry variables.

### 43.1.3 Four Waves of AI in Space

AI reached space in waves, each riding on progress elsewhere in computing (Figure 43.2).

**Expert systems and schedulers (1980s–1990s).** In 1985 NASA's Johnson Space Center began developing **CLIPS**, a rule engine written in C because the LISP-based tools of the day rarely left the laboratory; it is still maintained. In 1988, during STS-26, two expert systems ran in Mission Control for the first time. One covered the Shuttle's communications and instrumentation and was used by the INCO flight controller; the other covered its main engines. Both ran beside

conventional displays so that controllers could compare them (Figure 43.3). Hubble's long-range observing plan has been built since shortly after its launch in April 1990 with **SPIKE**, a scheduler from the Space Telescope Science Institute (Figure 43.4). SPIKE expresses each constraint — Sun and Moon avoidance, target visibility, guide-star availability — as a "suitability" between 0 and 1 over time and searches for the schedule with the highest total suitability. In May 1999 the Remote Agent experiment took control of Deep Space 1 (Chapter 48).

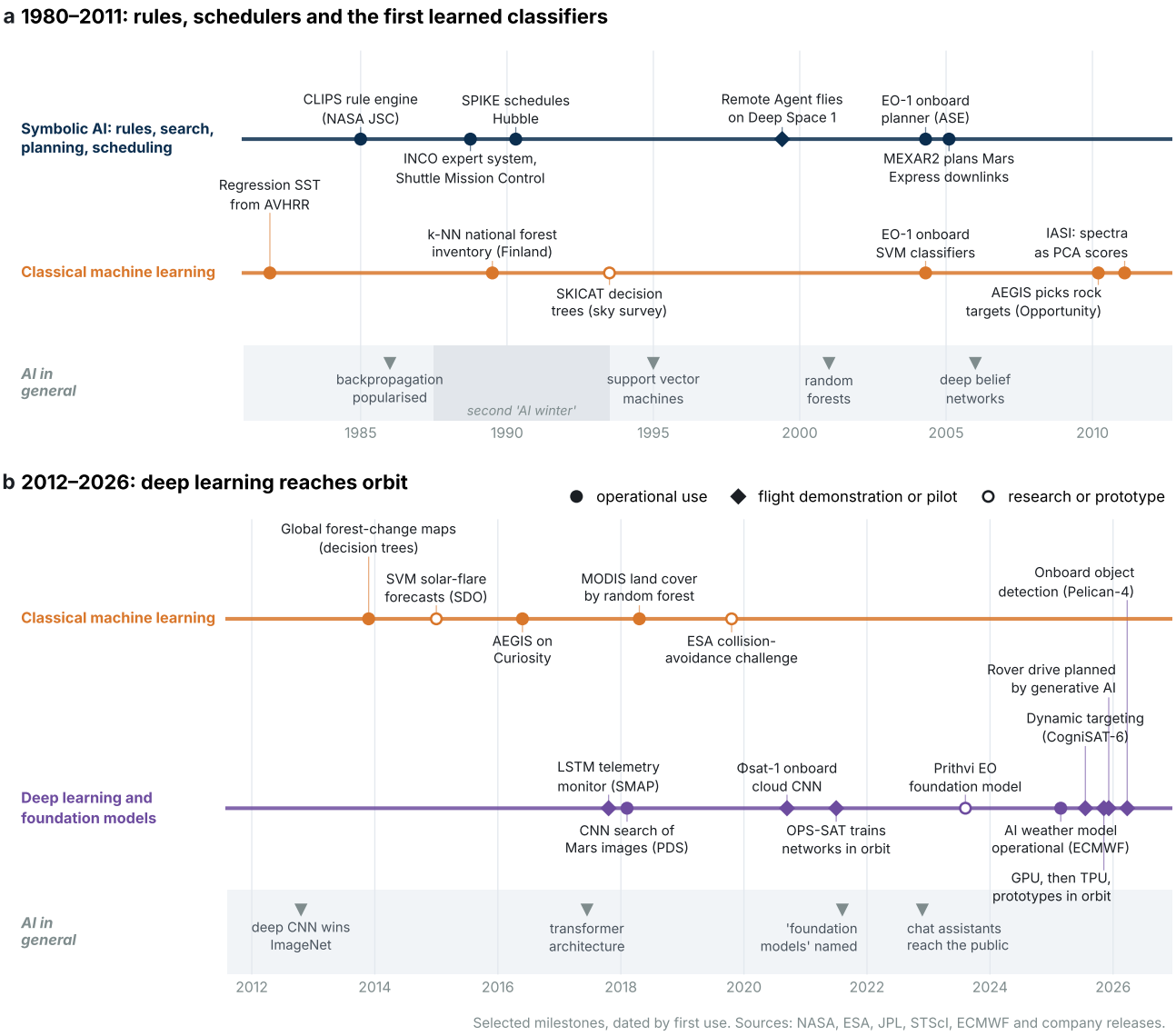



**FIGURE 43.2** Selected milestones of artificial intelligence in space, 1980–2026, sorted into symbolic AI, classical machine learning and deep learning, with general milestones of AI along the bottom. Filled circles mark operational use, diamonds flight demonstrations or pilots, open circles research or prototypes.

**FIGURE 43.3** Flight controller John Muratore at the INCO console in June 1988, during a simulation for STS-26. Beside him an operator works at the Real Time Data System, an expert-system workstation that monitored Shuttle telemetry. Image: NASA/JSC.

**FIGURE 43.4** Hubble above Discovery's payload bay before its release in April 1990 (STS-31). Its long-term observing schedule has been built ever since with SPIKE, a constraint-based AI scheduler. Image: NASA/JSC.

**Classical machine learning (1980s–2010s).** Statistical learning arrived quietly, often without the AI label. From the early 1980s NOAA computed sea-surface temperature from AVHRR's thermal channels with coefficients found by regression against drifting buoys. In the early 1990s JPL and Caltech's SKICAT used decision trees to classify faint stars and galaxies on the digitised Palomar sky survey. It reached about 94 % accuracy on objects too faint to classify by eye. From 2003–04 Earth Observing-1 ran onboard classifiers and a planner that retasked it automatically, and from 2010 the AEGIS software let Opportunity and later Curiosity choose their own rock targets.

**Deep learning (2012–).** In 2012 a deep convolutional network, AlexNet, cut the error rate of the ImageNet image-recognition contest from 26.2 % to 15.3 %. It showed that networks trained on graphics processors (GPUs) could learn better visual features than any designed by hand. In 2017 NASA's SMAP team began piloting a recurrent neural network that flags telemetry anomalies, and in 2018 JPL used networks pre-trained on everyday photographs to make NASA's Mars images searchable by content. Launched in September 2020, ESA's **Φsat-1** demonstrated a deep network screening its images for cloud on board, on a low-power Intel Movidius Myriad 2 vision processor.

**Foundation models and generative AI (2020s).** The **transformer** architecture (2017) made it practical to train networks with billions of parameters on unlabelled data; "foundation model" was coined in 2021. NASA and IBM released the Prithvi geospatial foundation model in 2023; its second version (December 2024) has 300 or 600 million parameters, pre-trained on about 4.2 million samples of harmonised Landsat and Sentinel-2 imagery. In February 2025 ECMWF made a machine-learned global weather model operational (Chapter 47). In December 2025 a satellite carrying a data-centre GPU trained a small language model in orbit. Planet's Pelican-4 detected aircraft in its own images over Alice Springs on 25 March 2026. On 1 October 2026 Google and Planet launched a prototype carrying four tensor processing units for a year-long test (Chapter 45). Most of these are **demonstrations**; operational systems still rest mainly on classical methods and carefully bounded deep networks.

## 43.2 Problems, Data and Ways of Learning

### 43.2.1 Ten Kinds of Problem

Before choosing an algorithm, an engineer must say what output is wanted. Table 43.2 maps ten problem types onto space examples.

**TABLE 43.2** Machine-learning problem types and space examples.

| Problem type | Output | Space example | Typical methods | Typical metric |
|---|---|---|---|---|
| Classification | A class for each input | Cloud or clear pixel; land-cover type | Boosted trees, forests, CNNs | Precision, recall, F1 |
| Regression | A continuous value | Sea-surface temperature; forest biomass | Linear regression, forests, Gaussian processes | RMSE, MAE, bias |
| Semantic segmentation | A class for every pixel, keeping shapes | Flood extent, burn scars, buildings | Convolutional encoder–decoders | Intersection over union |
| Object detection | Boxes and classes | Ships in radar images, aircraft | CNN detectors | Mean average precision |
| Forecasting | Future values or event probabilities | Solar flares, geomagnetic storms, weather | Trees, SVMs, recurrent and graph networks | Skill scores, Brier score |
| Anomaly detection | "Normal" or "unusual" | Telemetry faults, radio interference | Prediction errors, isolation forests | Event precision and recall |
| Clustering | Groups found without labels | Spectral classes, solar-wind types | k-means, Gaussian mixtures | Purity, Rand index |
| Dimensionality reduction | A few variables summarising many | 8 461-channel IASI spectra sent as about 300 numbers | PCA, autoencoders | Variance explained |
| Ranking | An ordering by value | Which rock to zap, which image to downlink first | Scoring models | Precision of top items |
| Decision and control | Actions | Scheduling, rover driving, pointing | Planners, reinforcement learning | Reward, mission return |

Anomaly detection and ranking show ML supporting people (Figure 43.5, Figure 43.6). JPL's telemetry monitor for SMAP and Curiosity, described in 2018, used a recurrent network to predict each of 82 telemetry channels from its recent history. It flagged values that departed from

the prediction by more than an automatically set threshold. On 105 labelled anomaly sequences it reached a precision of 87.5 % and a recall of 80 %. Even so, false positives kept it from becoming central to telemetry review. Curiosity's AEGIS ranks candidate rocks in navigation-camera images by criteria set by scientists and sends the best to the ChemCam laser. Over 2.5 km of driving it picked the preferred material more than 93 % of the time, against about 24 % expected without it.

**FIGURE 43.5** NASA's SMAP satellite being weighed before launch in November 2014. Since 2017 its operations team has piloted a neural-network system that flags anomalies in its telemetry. Image: NASA/KSC.

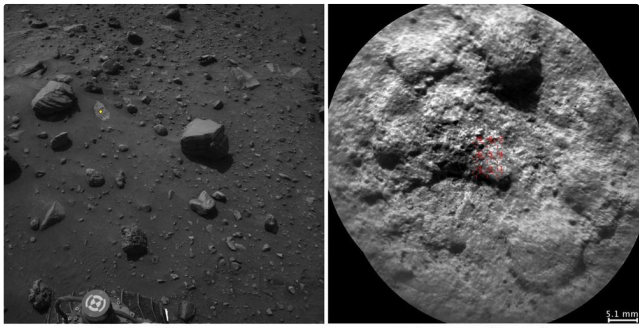



**FIGURE 43.6** Ranking on Mars: after a 13 m drive on 14 July 2016, Curiosity's AEGIS software picked the bright outcrop marked in this navigation-camera image (left); within 30 minutes ChemCam had fired its laser at nine points on it (right). Image: NASA/JPL (NASA/JPL-Caltech/LANL/CNES/IRAP/LPGNantes/CNRS/IAS).

### 43.2.2 Data, Features and Labels

Every ML problem starts with **data**: examples described by **features** $\boldsymbol{x}$ (reflectances, telemetry values, orbital elements) and usually a **label** $\boldsymbol{y}$ (the right answer). Good features encode physics. Clouds and snow are both bright at visible wavelengths, but snow absorbs strongly near 1.6 µm while water clouds do not (Chapter 13). A classifier given only visible bands confuses them (Figure 43.7), while one given a short-wave infrared band can separate them. This chapter's examples use this two-band problem, with synthetic pixels from eight surface and cloud types.

Labels are usually the scarcest ingredient. They come from experts drawing polygons, field campaigns and in-situ sensors, volunteers, simulations and operations logs, and they are never perfect. In the EO-1 study described below, the classifiers correctly found small lakes that the expert had missed — and were scored as wrong. Label errors set a floor on the achievable error and give flexible models something false to memorise.

### 43.2.3 Ways of Learning

The learning paradigms of Table 43.3 differ in the supervision they need, and are often combined: a network may be pre-trained without labels and fine-tuned with a few thousand.

**TABLE 43.3** Learning paradigms.

| Paradigm | Labels needed | How it learns | Space example |
|---|---|---|---|
| Supervised | A label for every example | Fits a mapping from features to labels | EO-1 onboard classifiers; cloud masks |
| Unsupervised | None | Finds clusters or low-dimensional structure | Spectral clustering; PCA of IASI spectra |

| Paradigm | Labels needed | How it learns | Space example |
|---|---|---|---|
| Self-supervised | None — data supply their own targets | Predicts hidden parts of the input | Prithvi, pre-trained on 4.2 million image samples |
| Semi-supervised | A few labels, many unlabelled examples | Extends labels to similar data | Mapping from a handful of field plots |
| Active learning | Chosen by the model | Asks for labels where most uncertain | Galaxy Zoo: 35–60 % fewer labelled galaxies |
| Transfer learning | Few, for the new task | Reuses a model trained on another task | Photo-trained networks fine-tuned on Mars images |
| Reinforcement learning | None; a reward instead | Learns actions by trial and error | Scheduling and guidance research (Chapter 48) |

Galaxy Zoo shows the value of combining people and machines. Launched in July 2007, it attracted about 150 000 volunteers who made some 50 million galaxy classifications in its first year. Later, Bayesian convolutional networks sent the galaxies they were least sure about to volunteers. This **active learning** cut the number of labelled galaxies needed by 35–60 %. **Reinforcement learning (RL)**, in which an agent learns by trial and error to maximise a reward, is covered in Chapter 48.

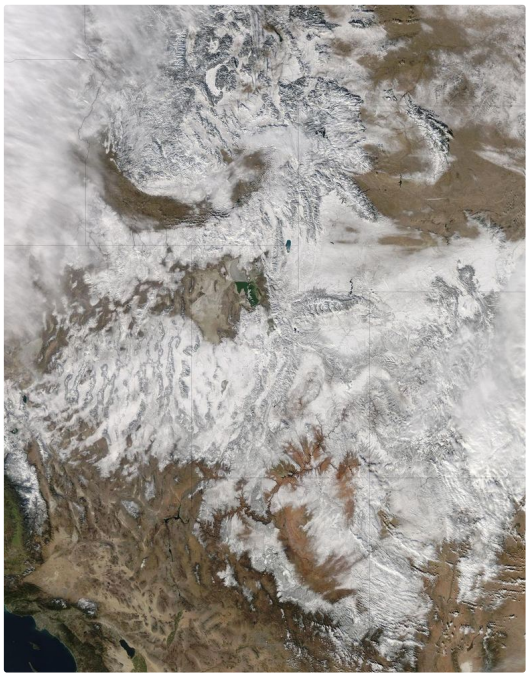


**FIGURE 43.7** Snow and cloud over the Rocky Mountains in a natural-colour MODIS image from Aqua, 19 December 2012. Both are bright in the visible; telling them apart needs a short-wave infrared band, in which snow is dark. Image: NASA/GSFC.

## 43.3 The Mathematics of Learning

### 43.3.1 Models, Losses and Empirical Risk

A **model** is a function $f_\theta(x)$ with adjustable **parameters** $\theta$ — two coefficients in a straight-line fit, millions of weights in a neural network. A **loss function** $\ell(f_\theta(x), y)$ measures how wrong one prediction is. For regression the usual choice is the squared error $(f_\theta(x) - y)^2$. For classification the model outputs a probability $p$ that the label is 1, and the loss is the **cross-entropy**

$$\ell = -\,[\,y \ln p + (1-y)\ln(1-p)\,],$$

which is zero for a confident correct answer and grows without limit for a confident wrong one. Learning means choosing the parameters that minimise the average loss over the $N$ training examples, the **empirical risk**:

$$\hat{R}(\theta) = \frac{1}{N}\sum_{i=1}^{N} \ell\big(f_\theta(x_i), y_i\big).$$

What we really want is a small loss on future data, the **expected risk**; the difference between the two — the **generalisation gap** — is the central difficulty of machine learning.

### 43.3.2 Gradient Descent

For a few models, such as ordinary least squares, the minimum has a closed form; in general it is found iteratively by **gradient descent**. The gradient $\nabla_\theta \hat{R}$ points uphill on the loss surface, so each step moves the parameters a little way downhill (Figure 43.8):

$$\theta_{t+1} = \theta_t - \eta\, \nabla_\theta \hat{R}(\theta_t),$$

where the **learning rate** $\eta$ sets the step size. Computing the full gradient over millions of examples at every step is wasteful, so **stochastic gradient descent (SGD)** estimates it from a small random **mini-batch** $B$:

$$\theta_{t+1} = \theta_t - \frac{\eta}{|B|}\sum_{i \in B} \nabla_\theta\, \ell\big(f_\theta(x_i), y_i\big).$$

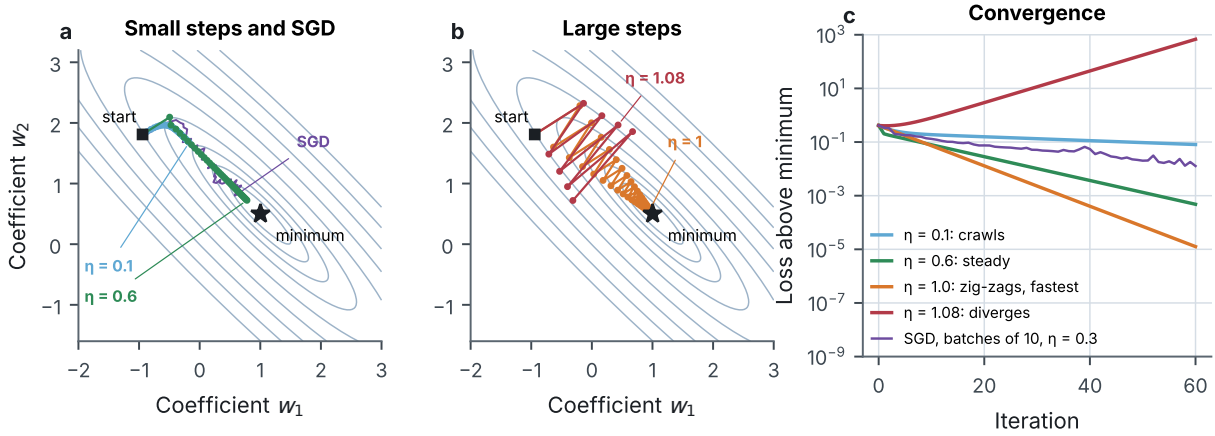



**FIGURE 43.8** Gradient descent on the least-squares loss of a linear model with correlated predictors (r = 0.9). (a) Small steps crawl; stochastic gradient descent (mini-batches of 10) wanders near the minimum. (b) Steps near the stability limit zig-zag; η = 1.08 diverges. (c) Excess loss after 60 iterations: 1.3 × $10^{-5}$ for η = 1.0, 5 × $10^{-4}$ for η = 0.6, 0.08 for η = 0.1.

Each step is noisy but cheap. Practical optimisers such as **Adam** add momentum (an average of recent gradients) and scale each parameter's step by the size of its recent gradients.

The learning rate is the most important setting. Figure 43.8 shows gradient descent on the least-squares loss of a linear model with two strongly correlated predictors, which stretch the loss surface into a narrow valley. Near a minimum any loss surface behaves like a quadratic with curvatures given by the eigenvalues $\lambda$ of its Hessian (here 0.083 and 1.92), and gradient descent is stable only if $\eta < 2/\lambda_{\max}$ = 1.04. A small step crawls along the valley; a step near the limit zig-zags

yet converges fastest; a slightly larger one diverges. The ratio $\kappa = \lambda_{\max}/\lambda_{\min}$, here 23, is the **condition number**: the larger it is, the slower the convergence, which is why inputs are routinely **standardised** to zero mean and unit variance.

**EXAMPLE: ONE GRADIENT-DESCENT STEP BY HAND**

A logistic-regression cloud detector computes $p = \sigma(w_1 x_1 + w_2 x_2 + b)$, where $x_1$ and $x_2$ are the visible and 1.6 µm reflectances and $\sigma(z) = 1/(1+e^{-z})$. It starts with all parameters zero and sees one cloudy pixel, $x = (0.70, 0.45)$, $y = 1$, with learning rate $\eta = 0.5$.

1. **Prediction.** $z = 0$, so $p = 0.5$ and the cross-entropy loss is $-\ln 0.5 = \mathbf{0.693}$.
2. **Gradient.** For logistic regression $\partial \ell/\partial w_j = (p - y)\, x_j$ and $\partial \ell/\partial b = p - y$. Here $p - y = -0.5$, so the gradients are $(-0.35, -0.225)$ for the weights and $-0.5$ for the bias.
3. **Update.** $w \leftarrow w - \eta \nabla_w \ell = (0.175, 0.1125)$ and $b \leftarrow 0.25$.
4. **Check.** Now $z = 0.175 \times 0.70 + 0.1125 \times 0.45 + 0.25 = 0.423$, $p = 0.604$ and the loss has fallen to **0.504**. But a clear snow pixel, $x = (0.80, 0.10)$, now also gets $p = 0.599$: the model has learned "bright means cloud". Only more examples — including snow — and a model able to bend its boundary can fix this.

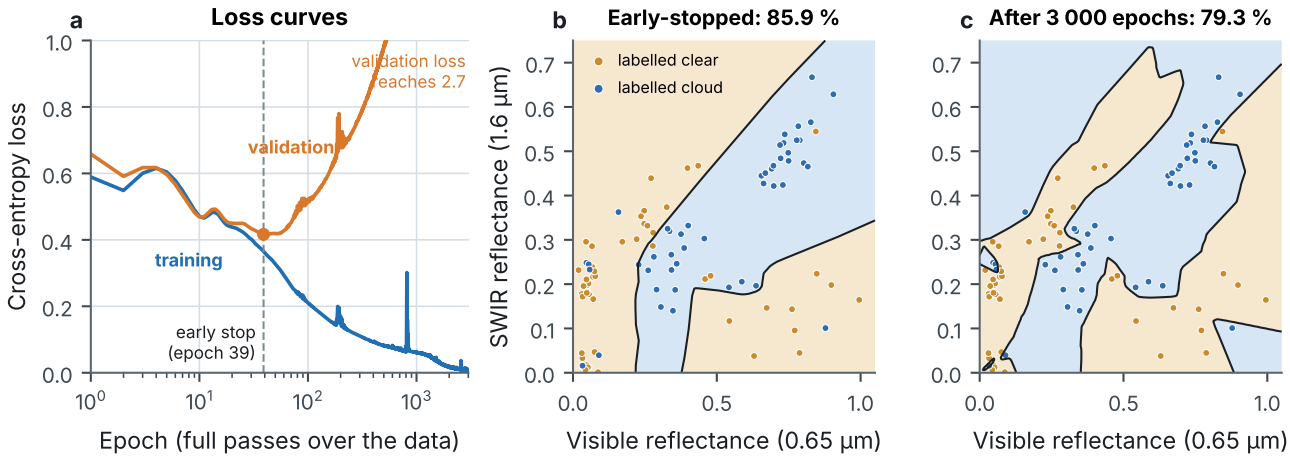
### 43.3.3 Neural Networks and Backpropagation

A **neural network** builds a flexible function by stacking simple layers. Each layer multiplies its input vector by a weight matrix, adds a bias and applies a non-linear **activation function**; the popular **ReLU** (rectified linear unit) replaces negative values by zero. A **multilayer perceptron (MLP)** with two hidden layers computes

$$h_1 = \mathrm{ReLU}(W_1 x + b_1), \qquad h_2 = \mathrm{ReLU}(W_2 h_1 + b_2), \qquad p = \sigma(w_3^{\top} h_2 + b_3).$$

Training needs the gradient of the loss with respect to every weight, computed by **backpropagation**. A forward pass stores each layer's output. A backward pass then applies the chain rule layer by layer, starting from $\partial \ell/\partial z = p - y$ at the output and multiplying by each layer's weights and activation slopes on the way back. The backward pass costs only about twice the forward pass, which is why networks with billions of parameters can be trained (Chapter 44).

Figure 43.9 shows such a network, written from scratch in numpy, learning the cloud-or-clear problem from only 100 pixels, one label in ten deliberately wrong. At first training and validation losses fall together. Then the network starts to fit the mislabelled pixels: training loss heads towards zero, while the loss on 2 000 independent validation pixels climbs from 0.42 to 2.7. This is **overfitting**, and the simplest cure is **early stopping** — keeping the weights from the epoch with the lowest validation loss, here epoch 39. The early-stopped network classifies 85.9 % of 20 000 test pixels correctly; trained to the end, it scores 99 % on its own training data but only 79.3 % on the test pixels.




Numpy network 2–64–64–1 (ReLU, Adam, step 0.01) trained on 100 synthetic pixels with 10 % wrong labels; test set 20 000 pixels.

**FIGURE 43.9** A numpy neural network (two hidden layers of 64 ReLU units, about 4 400 parameters) trained by backpropagation on 100 synthetic cloud and clear pixels, one label in ten wrong. (a) Validation loss is lowest at epoch 39, then rises as the network memorises label errors. (b) The early-stopped network: 85.9 % of 20 000 test pixels correct. (c) After 3 000 epochs the boundary wraps around mislabelled pixels: 79.3 %.

### 43.3.4 Regularisation, Bias and Variance

Any technique that discourages a model from fitting noise is **regularisation**. Early stopping is one; another adds a penalty on large weights, $\hat{R}(\theta) + \lambda\|\theta\|^2$ (**weight decay**, or **ridge** regression for linear models); the penalty $\lambda \sum_j |\theta_j|$ (**lasso**) drives unimportant weights to zero. Networks are also regularised by **dropout**, which randomly switches off units in training, and **data augmentation** with rotated or noisier copies of images; trees, by limiting depth.

The **bias–variance trade-off** explains why regularisation works. For squared loss, the expected error on new data splits into three parts:

$$\mathbb{E}\big[(y - \hat{f}(x))^2\big] = \underbrace{\big(\mathbb{E}[\hat{f}(x)] - f(x)\big)^2}_{\text{bias}^2} + \underbrace{\operatorname{Var}\big[\hat{f}(x)\big]}_{\text{variance}} + \sigma^2 .$$

**Bias** comes from assumptions too simple for the problem; **variance** is how much the fitted model changes between training sets; $\sigma^2$ is the **irreducible error** from noise in data and labels. Simple models underfit (high bias); flexible models overfit (high variance). Figure 43.10 (a) shows the trade-off for decision trees on 1 000 pixels: the cross-validated error is lowest, 15.1 %, at a depth of 5. Panel (b) shows **learning curves** — test error against training-set size. Logistic regression levels off at about 31 % whatever the amount of data, because a straight line cannot separate snow from cloud. A random forest and a k-nearest-neighbour classifier reach 12.3 % and 12.1 % with 20 000 pixels. That is close to the irreducible 11.9 % set by overlapping classes and 10 % label noise. Learning curves show whether more labels or a better model is needed.

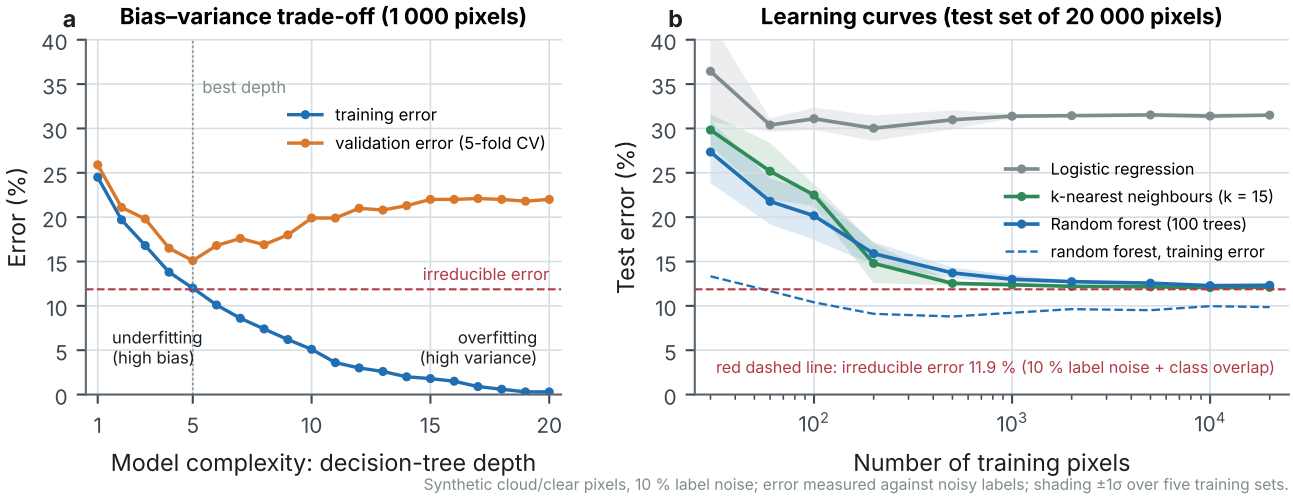



**FIGURE 43.10** (a) The bias–variance trade-off for decision trees fitted to 1 000 synthetic cloud and clear pixels: cross-validated error is lowest (15.1 %) at depth 5 and rises to about 22 % for deep trees. (b) Learning curves: flexible models approach the irreducible error of 11.9 % as training data grow; logistic regression stays near 31 % (high bias).

### 43.3.5 Training, Validation and Test Sets

Labelled data are split three ways. The **training set** fits the parameters. The **validation set** is used to choose **hyper-parameters** — settings that are not learned, such as the learning rate, network size or tree depth — and to decide when to stop. The **test set** is locked away and used once, at the end; every look at it that influences a design decision makes the final estimate optimistic. When data are scarce, **k-fold cross-validation** trains the model $k$ times, each time testing on a different fifth or tenth of the data, and averages the scores, with the test set still untouched.

### 43.3.6 Leakage: When Space Data Cheat

**Data leakage** occurs when information about the test data reaches the model during training, so that the test score no longer measures performance on genuinely new inputs. The most dangerous leak in space applications is **autocorrelation**. Neighbouring pixels are alike, and so are consecutive telemetry samples or successive days of solar activity. With a random split, almost every test sample has a near-twin in the training set. The model can then score well by remembering places or times, rather than by learning physics that would carry it to a new region or season.

Figure 43.11 makes the effect concrete. A random forest learns a synthetic target made of a genuine function of one predictor plus a stronger smooth spatial signal (correlation length 12 km) that no predictor explains. Five further predictors are smooth random fields unrelated to the target — but they let the forest recognise *where* a sample is. Random five-fold cross-validation reports an excellent $R^2$ of 0.80. Cross-validation with **spatial blocks** larger than the correlation length reports about 0.25. Tested in an unseen region, the model scores −0.39 — worse than predicting the regional average — while the true relationship explains only 17 % of the variance there. The forest had learned a map, not a law of nature.

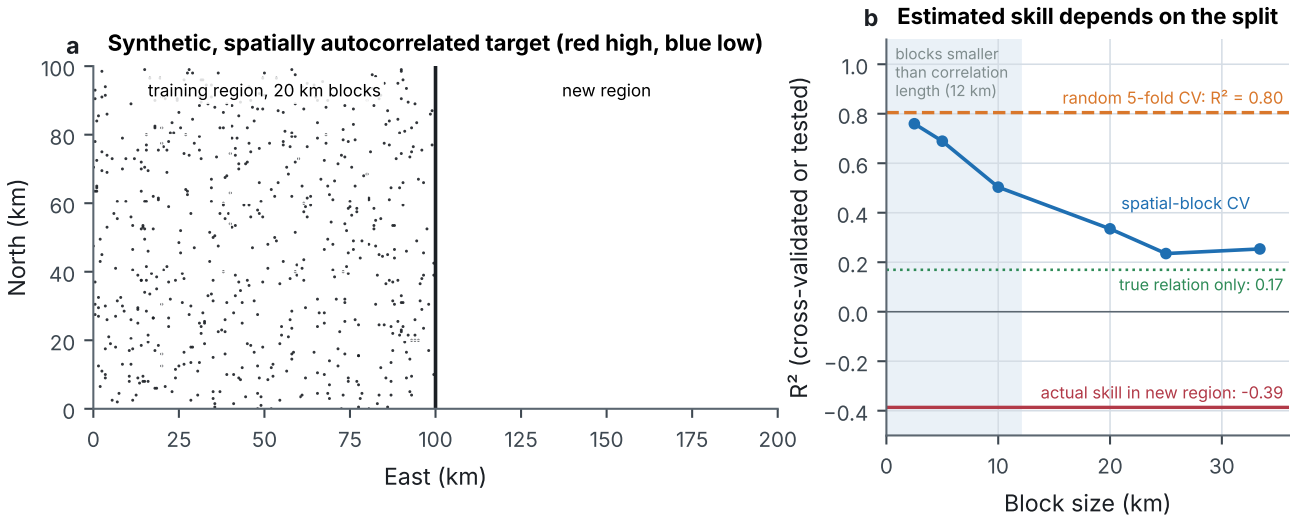



**FIGURE 43.11** Leakage through spatial autocorrelation, computed with a random forest on a synthetic 200 km × 100 km field. (a) The target, with training samples in the western half and test samples in an unseen eastern region. (b) Random five-fold cross-validation reports $R^2$ = 0.80 and spatial-block cross-validation about 0.25; the actual skill in the new region is −0.39, and even the true relationship explains only 17 % of the variance there.

**DATA BOX — SPATIAL VALIDATION OF BIOMASS MAPS**

- In 2020 Ploton and colleagues rebuilt a typical mapping workflow: a random forest predicting forest biomass in central Africa from 9 MODIS and 27 environmental predictors, trained on an inventory of 11.8 million trees.
- With random 10-fold cross-validation the model appeared to explain more than half of the variation ($R^2$ = 0.53); with spatial cross-validation, only 14 % ($R^2$ = 0.14). For test plots more than about 100 km from any training plot, its predictive power was close to zero; biomass was spatially autocorrelated up to about 120 km.
- Predictors with no real relationship to the target, the authors concluded, can appear powerful simply because they vary smoothly in space.

The cure is to split the data the way the model will be used. Maps that must be predicted far from training sites need **spatial blocks**, or **buffers** that remove training samples near each test sample, larger than the range of autocorrelation (estimated from a variogram of the residuals). Forecasts need **forward chaining**: train on the past, test on later periods only. Products from several satellites or instruments need **grouped** splits that keep each orbit, pass, sensor or mission within one fold.

## 43.4 Classical Machine Learning

Deep networks dominate image and language tasks, but most operational space products still rest on the classical methods of Table 43.4. They train quickly on modest data and hardware and are easier to verify.

**TABLE 43.4** Classical machine-learning methods and their space uses.

| Method | Key idea | Strengths | Limitations | Space use (status) |
|---|---|---|---|---|
| Linear regression | Weighted sum fitted by least squares | Fast, interpretable | Linear only | AVHRR sea-surface temperature (operational since 1980s) |
| Logistic regression | Sigmoid of a weighted sum | Probabilities; baseline | Linear boundary | Baselines for flare and fire detection |
| k-nearest neighbours | Vote of the $k$ closest examples | No training | Slow; few features | Finnish forest inventory (operational since about 1990) |
| Decision tree | Sequence of threshold tests | Readable | Unstable; overfits | SKICAT (1990s); MODIS land cover (2000s) |
| Random forest | Average of randomised trees | Robust, little tuning | Large; opaque | MODIS land cover since Collection 6 |
| Gradient boosting | Trees added to correct errors | Top accuracy on tables | Needs tuning | Sentinel-2 cloud probability (operational) |
| Support vector machine | Maximum-margin boundary | Good with few samples | Slow on big data | EO-1 onboard classifiers (2004–17) |
| k-means | Nearest-centroid clustering | Fast, simple | Choice of $k$ | Unsupervised image classification |
| PCA | Main directions of variance | Compression, denoising | Linear only | IASI spectra as PC scores (since 2011) |
| Gaussian process | Prior over functions | Honest error bars | Cost grows as $N^3$ | Stellar-activity models (research) |

### 43.4.1 Linear and Logistic Regression

**Linear regression** predicts $\hat{y} = w^{\top}x + b$ and chooses the weights that minimise the squared error, with the closed-form solution $w = (X^{\top}X)^{-1}X^{\top}y$. NOAA's **multichannel sea-surface temperature** algorithm exploits the fact that water vapour absorbs more at 12 µm than at 11 µm. Temperature is a weighted combination of the 11 µm brightness temperature and the difference between the channels. The coefficients are found by regression against simultaneous readings from drifting buoys. In 1985 its root-mean-square difference from ship and buoy data was about 0.5 °C (Chapter 16).

**Logistic regression** turns the weighted sum into a probability, $p = \sigma(w^{\top}x + b)$, and is trained by minimising cross-entropy, as in the worked example. Its decision boundary, $p = 0.5$, is a straight line (a plane, in more dimensions). That makes it a good **baseline** — every new model should be compared with a simple one. It also explains its failure in Figure 43.12: no straight line can put snow on the clear side and cloud on the other.

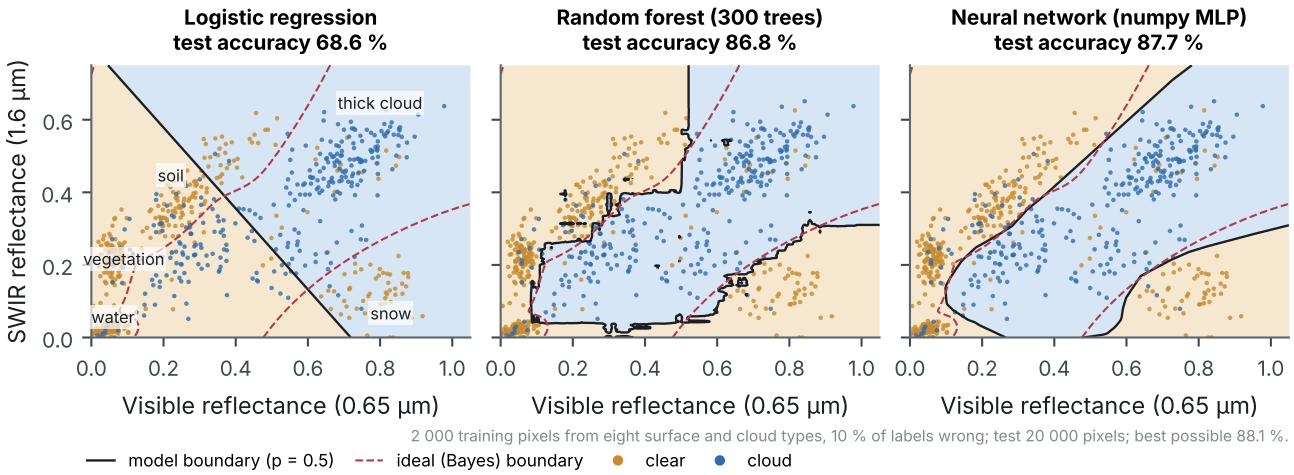



**FIGURE 43.12** Decision boundaries of three classifiers trained on the same 2 000 synthetic pixels from eight surface and cloud types, 10 % of labels wrong. Logistic regression labels 96 % of snow pixels as cloud (test accuracy 68.6 %); a random forest (86.8 %) and the early-stopped numpy network (87.7 %) approach the best possible 88.1 %, whose ideal boundary is dashed.

## 43.4.2 Nearest Neighbours

The **k-nearest-neighbour (k-NN)** method stores the training data and, for each new input, takes a majority vote (classification) or an average (regression) of the $k$ closest examples. It needs no training and adapts to any boundary shape, but prediction is slow for large data sets and distances lose meaning when there are many features — the "curse of dimensionality". Since about 1990 Finland's multi-source national forest inventory has used k-NN to give every satellite pixel the timber volume, species and age of the field plots whose spectra it most resembles.

## 43.4.3 Decision Trees, Random Forests and Gradient Boosting

A **decision tree** asks a sequence of threshold questions, such as "is the 1.6 µm reflectance above 0.3?". Each question is chosen to make the resulting groups as pure as possible, measured for example by the **Gini impurity** $G = 1 - \sum_k p_k^2$, where $p_k$ is the fraction of class $k$ in a group. Trees are readable and fast — the best tree JPL built for EO-1 needed at most 24 comparisons per pixel — but deep trees memorise their training data and are unstable.

A **random forest** averages hundreds of trees, each grown on a random bootstrap sample of the data and choosing each split from a random subset of features. The trees make different errors, so their average has much lower variance. NASA's MODIS land-cover product switched from a C4.5 decision tree in Collection 5 to a random forest in Collection 6, classifying the globe each year since 2001 at 500 m resolution. Hansen and colleagues' 2013 global forest-change maps applied decision trees to about 654 000 Landsat scenes, using one million CPU-core hours in Google Earth Engine.

**Gradient boosting** builds trees in sequence. Each new small tree $h_m$ is fitted to the errors (strictly, the negative gradient of the loss) left by the ensemble so far. It is added with a small weight $\nu$, so $F_m(x) = F_{m-1}(x) + \nu\, h_m(x)$. Boosted trees, in libraries such as XGBoost and LightGBM, are usually the most accurate models for tabular data. Sentinel Hub's **s2cloudless** detector is a LightGBM model working on ten Sentinel-2 bands. It underlies the cloud-probability layer distributed with Sentinel-2 imagery in Google Earth Engine, although it gives high probabilities over very bright surfaces such as snow.

### 43.4.4 Support Vector Machines

A **support vector machine (SVM)** chooses, among all boundaries that separate two classes, the one with the widest **margin** — the largest gap to the nearest training points, the support vectors. For a linear boundary $w^{\top}x + b = 0$ the margin is $2/\|w\|$, so the SVM minimises

$$\tfrac{1}{2}\|w\|^2 + C\sum_i \max\left(0,\; 1 - y_i(w^{\top}x_i + b)\right),$$

with labels coded $y_i = \pm 1$; the second term, the **hinge loss**, penalises points on the wrong side of the margin. A **kernel** such as the Gaussian $K(x, x') = \exp(-\gamma\|x - x'\|^2)$ lets the same mathematics draw curved boundaries. In 2015 Bobra and Couvidat used a Gaussian-kernel SVM to forecast M- and X-class flares within 24 hours (Chapter 47). It worked on magnetic-field features of solar active regions measured by the Solar Dynamics Observatory, and reached a true skill statistic of 0.76 on 303 flaring and 5 000 non-flaring examples.

#### EO-1: LEARNED CLASSIFIERS IN COMMAND

NASA's Earth Observing-1, launched in 2000 as a technology demonstrator, became from 2003–04 the testbed of the **Autonomous Sciencecraft Experiment**. This onboard software analysed images from the Hyperion spectrometer, detected floods, eruptions and changes in snow and ice, and replanned observations without waiting for the ground (Figure 43.13). The payload computer was a Mongoose V of about 8 MIPS with 256 MB of memory, and only 12 of Hyperion's bands could be read on board. JPL compared four pixel classifiers for land, water, snow, ice and cloud: one built by an expert, the best of thousands of thresholded band ratios, a decision tree and a **linear SVM** on 11 bands. The SVM needed just 11 multiplications and 11 additions per class for each pixel. On 50 expert-labelled test images (about 91 million pixels) the SVM classified 81.3 % of pixels correctly, the expert classifier 82.9 % and the tree 80.9 %. The averages hid a weakness: ice made up only 2 % of the test pixels, and the SVM found just 47.8 % of them. The expert and SVM classifiers ran on board — the SVMs for more than a dozen years, until the mission ended in 2017. The team reported a hundred-fold increase in science returned per megabyte downlinked.

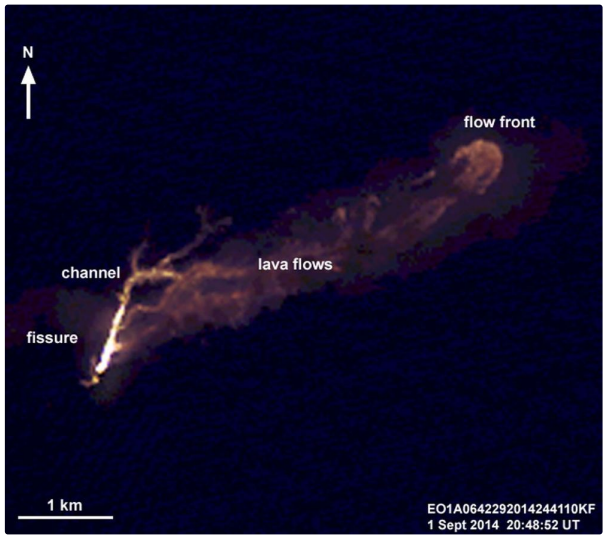
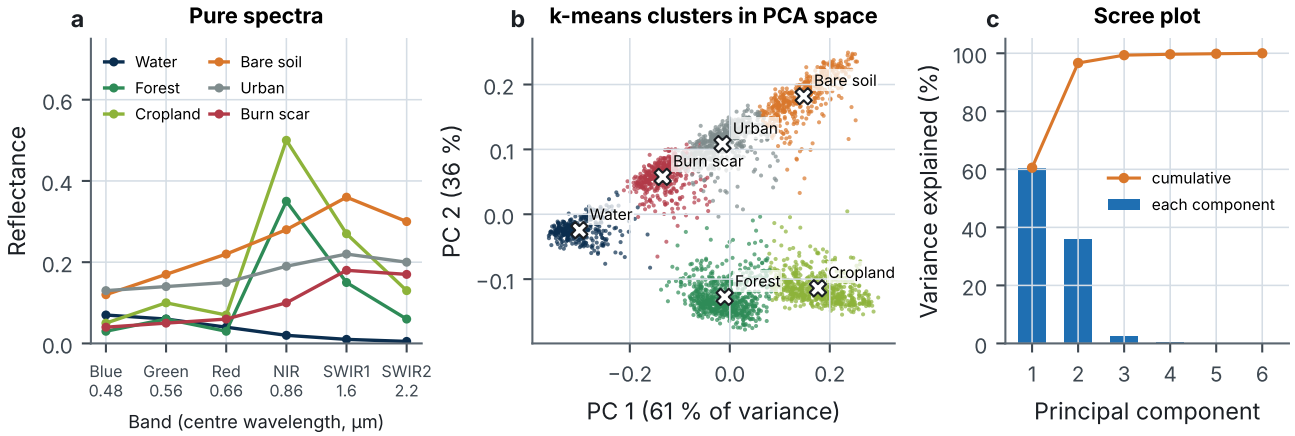



**FIGURE 43.13** The Holuhraun eruption in Iceland on 1 September 2014, imaged by EO-1 in false colour to emphasise the hottest vents and lava flows. Detecting such thermal events on board, and retasking the satellite to observe them again, was one of the jobs of EO-1's autonomy software. Image: NASA/JPL (NASA/JPL/EO-1 Mission/GSFC/Ashley Davies).

### 43.4.5 Clustering and Dimensionality Reduction

**k-means** clustering divides unlabelled data into $k$ groups. It starts from $k$ trial centres, assigns each sample to the nearest centre, moves each centre to the mean of its samples, and repeats, minimising the total squared distance $\sum_i \|x_i - \mu_{c(i)}\|^2$. Its variant ISODATA, which splits and merges clusters, has been used for unsupervised image classification since the 1970s; the analyst names each spectral cluster afterwards (Chapter 13). The number of clusters must be chosen, for example from an "elbow" in the total distance as $k$ grows.

**Principal component analysis (PCA)** finds new axes, the eigenvectors of the covariance matrix $C = \frac{1}{N}\sum_i (x_i - \bar{x})(x_i - \bar{x})^{\top}$, ordered by the variance $\lambda_k$ along each. Keeping only the first few components compresses the data and removes noise. In Figure 43.14 two components hold 96.6 % of the variance of six-band pixels, and k-means, never shown a label, recovers the six cover types well (adjusted Rand index 0.89, where 1 is perfect agreement). The IASI sounder on the Metop satellites measures 8 461 spectral channels. Since 2011 EUMETSAT has also distributed each spectrum as about 300 principal-component scores — a 28-fold reduction. Users reconstruct the radiances from them with much of the instrument noise removed.




3 000 synthetic six-band pixels (shaded mixtures dominated by one cover, plus noise); k-means with k = 6: adjusted Rand index 0.89.

**FIGURE 43.14** Unsupervised learning on 3 000 synthetic six-band pixels, each a shaded, noisy mixture dominated by one cover type. (a) The pure spectra. (b) PCA compresses six bands into two components holding 96.6 % of the variance; k-means with six clusters, run without labels, finds groups (crosses: centres) matching the cover types (adjusted Rand index 0.89). (c) Variance explained by each component.

### 43.4.6 Gaussian Processes

A **Gaussian process (GP)** treats the unknown function itself as random, with a **kernel** $k(x, x')$ saying how strongly the function values at two inputs are correlated — typically more strongly the closer they are. Conditioning on $N$ noisy observations $y$ gives, at a new input $x_*$, a predictive mean and variance

$$\mu_* = k_*^{\top}(K + \sigma_n^2 I)^{-1} y, \qquad \sigma_*^2 = k(x_*, x_*) - k_*^{\top}(K + \sigma_n^2 I)^{-1} k_*,$$

where $K$ holds the kernel values between training inputs, $k_*$ those between them and $x_*$, and $\sigma_n^2$ is the noise variance. The variance widens away from the data, giving honest error bars; the cost of the $N \times N$ matrix inversion grows as $N^3$. In geostatistics the same mathematics is called **kriging**. GPs are a standard tool for separating planetary signals from stellar activity in radial-

velocity planet searches. In 2014 a GP model of activity in the star CoRoT-7 gave masses of 4.73 ± 0.95 and 13.56 ± 1.08 Earth masses for its two planets (Chapter 51). They also underlie **Bayesian optimisation** of hyper-parameters.

### 43.4.7 The Kalman Filter: Estimation Meets Learning

The **Kalman filter** (Chapter 6) estimates the state of a dynamic system — position and velocity, or attitude and gyro bias — from noisy measurements. Its **prediction** step propagates the estimate and its covariance with a dynamic model, $\hat{\boldsymbol{x}}^- = \boldsymbol{F}\hat{\boldsymbol{x}}$ and $\boldsymbol{P}^- = \boldsymbol{F}\boldsymbol{P}\boldsymbol{F}^\top + \boldsymbol{Q}$, where $\boldsymbol{Q}$ is the process noise. Its **update** blends in a measurement $\boldsymbol{z}$:

$$\boldsymbol{K} = \boldsymbol{P}^- \boldsymbol{H}^\top (\boldsymbol{H}\boldsymbol{P}^- \boldsymbol{H}^\top + \boldsymbol{R})^{-1}, \qquad \hat{\boldsymbol{x}} = \hat{\boldsymbol{x}}^- + \boldsymbol{K}(\boldsymbol{z} - \boldsymbol{H}\hat{\boldsymbol{x}}^-), \qquad \boldsymbol{P} = (\boldsymbol{I} - \boldsymbol{K}\boldsymbol{H})\boldsymbol{P}^-,$$

where $\boldsymbol{H}$ maps the state to the measurement, $\boldsymbol{R}$ is the measurement noise and the **gain** $\boldsymbol{K}$ weights new information by the relative confidence in prediction and measurement. This is exact **Bayesian inference** for linear systems with Gaussian noise: prior (prediction) times likelihood (measurement) gives the posterior (update). Stanley Schmidt's group at NASA Ames adapted Kalman's 1960 theory to the non-linear problem of navigating Apollo to the Moon, and the resulting **extended Kalman filter** flew in the Apollo guidance computer (Figure 43.15).

**FIGURE 43.15** Jim Lovell at the guidance and navigation station of Apollo 8 in December 1968, in a frame from 16 mm film. Star sightings made here were combined with a dynamic model in a recursive estimator of the kind adapted for Apollo at NASA Ames. Image: NASA/JSC.

Estimation theory and machine learning are two dialects of one language. Least-squares orbit determination (Chapter 3) is regression, and a Kalman filter can be seen as Gaussian-process regression in time. A network trained with weight decay is a maximum *a posteriori* estimate with a Gaussian prior on its weights. Estimation starts from physics with few parameters; learning starts from data with many. The best systems use physics where it is known and learning where it is not.

# 43.5 Measuring Performance

## 43.5.1 From the Confusion Matrix to F1

For a two-class detector — fire or no fire, anomaly or normal — every prediction falls into one cell of the **confusion matrix**. A **true positive** (TP) is an event correctly flagged and a **false positive** (FP) is a false alarm. A **false negative** (FN) is a miss and a **true negative** (TN) is a correct rejection. In map accuracy assessment (Chapter 13) precision is called user's accuracy and recall producer's accuracy. Table 43.5 lists the scores built from these counts. **Precision** is the fraction of alarms that are real, **recall** (probability of detection) the fraction of events found, and the **F1 score** their harmonic mean, which is high only if both are.

**TABLE 43.5** Evaluation metrics.

| Metric | Definition | Use it when | Caution |
|---|---|---|---|
| Accuracy | (TP + TN) / N | Classes are balanced | Meaningless for rare events |
| Precision | TP / (TP + FP) | False alarms are costly | Falls as events become rarer |
| Recall | TP / (TP + FN) | Misses are costly | Ignores false alarms |
| F1 | 2 × precision × recall / (precision + recall) | One number for a rare class | Depends on threshold |
| ROC AUC | Area under recall versus false-alarm rate | Comparing rankings | Optimistic for rare events |
| Average precision | Area under precision versus recall | Rare events | Depends on prevalence |
| RMSE, MAE | √(mean error²); mean absolute error | Regression | RMSE stresses large errors |
| CSI | TP / (TP + FP + FN) | Event forecasts | Depends on event rate |
| TSS | Recall − false-alarm rate | Flare and storm forecasts | Can be high with poor precision |
| Brier score | Mean of (probability − outcome)² | Probability forecasts | Compare with climatology |

**EXAMPLE: SCORING A WILDFIRE DETECTOR**

A thermal-infrared fire detector is tested on a scene of 1 000 000 land pixels containing 500 fire pixels, checked by an independent high-resolution sensor (Figure 43.16). It raises 620 alarms: 410 at real fires and 210 elsewhere.

1. **Counts.** TP = 410, FP = 210, FN = 500 − 410 = 90 and TN = 1 000 000 − 500 − 210 = 999 290.
2. **Precision** = 410 / 620 = **0.661**: one alarm in three is false. **Recall** = 410 / 500 = **0.820**. **F1** = 2 × 0.661 × 0.820 / (0.661 + 0.820) = **0.732**.
3. **False-alarm rate** = 210 / 999 500 = $2.1 \times 10^{-4}$; CSI = 410 / 710 = 0.577; TSS = 0.820 − 0.0002 = 0.820.
4. **Accuracy** = (410 + 999 290) / 1 000 000 = **99.97 %** — but a "detector" that never raises an alarm scores 99.95 %. Accuracy cannot tell them apart; precision and recall can.
5. **Sensitivity.** If a season of sun glint doubles the false alarms to 420, precision falls to 0.494 and F1 to 0.617, while accuracy barely changes (99.95 %).

**FIGURE 43.16** Smoke from lightning-caused wildfires in northern California on 4 August 2015, in a natural-colour image from MODIS on Terra. Red outlines mark actively burning areas detected by MODIS's thermal bands — a detection problem in which fire pixels are a tiny fraction of the scene. Image: NASA/GSFC.

## 43.5.2 ROC and Precision–Recall Curves

Most classifiers output a score, and an alarm is raised above a chosen **threshold**; lowering it finds more events but raises more false alarms. The **receiver operating characteristic (ROC)** curve plots recall against false-alarm rate for every threshold; its area (AUC) is the probability that a random event scores higher than a random non-event. The **precision–recall (PR)** curve plots precision against recall; its area is the **average precision** (Figure 43.17).

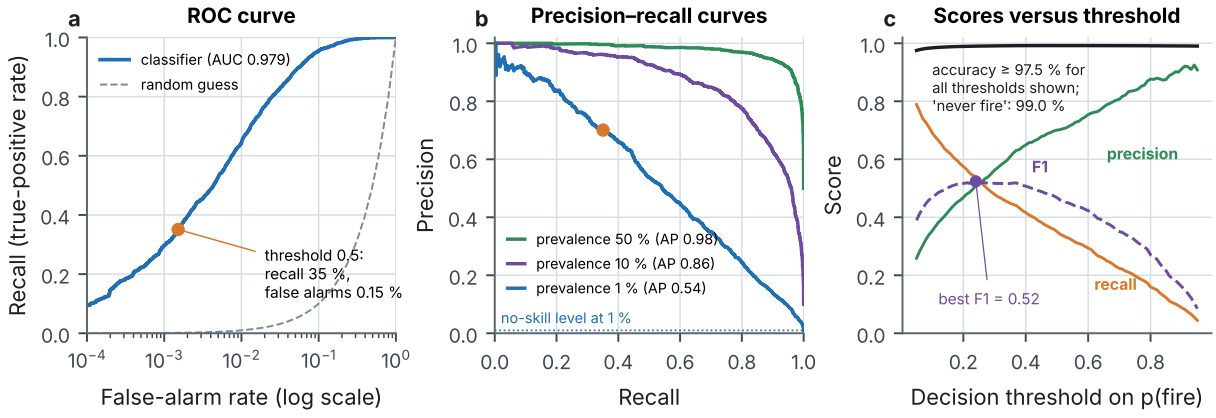



**FIGURE 43.17** Why accuracy misleads for rare events: a logistic-regression detector on a synthetic fire task (2 001 fires among 200 000 test pixels; 5 % of clear pixels fire-like). (a) The ROC area is 0.979, yet at threshold 0.5 only 35 % of fires are found. (b) Average precision is 0.54 at 1 % prevalence, 0.86 at 10 % and 0.98 at 50 %. (c) Accuracy stays above 97.5 % at all thresholds from 0.05 to 0.95; F1 peaks at 0.52.

For rare events the two tell very different stories. A logistic-regression detector for a synthetic fire task with 1 % prevalence has a near-perfect-looking ROC area of 0.979. Yet at the default threshold of 0.5 it finds only 35 % of the fires, and its average precision is just 0.54. On samples in which fires make up 10 % or 50 % of the pixels, the same detector reaches average precisions of 0.86 and 0.98 — its ROC curve does not change at all. Because the false-alarm rate divides by the

huge number of non-events, the ROC curve hides the fact that even a small rate then produces as many false alarms as detections. Accuracy is worse still: it stays above 97.5 % at every threshold from 0.05 to 0.95, and a detector that never fires scores 99.0 % — the **accuracy paradox**. The threshold should be set from the real costs of misses and false alarms, not by a software default.

### 43.5.3 Detection, Segmentation and Regression Metrics

For segmentation and object detection the basic unit is the **intersection over union** $\mathrm{IoU} = |A \cap B| / |A \cup B|$ of predicted and true regions; a detection usually counts as correct if its IoU exceeds 0.5. **Mean average precision (mAP)** averages the average precision over classes, sometimes also over IoU thresholds of 0.5–0.95, so mAP figures are comparable only under the same definition. For regression the usual scores are the **root-mean-square error** $\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_i (\hat{y}_i - y_i)^2}$, which emphasises large errors, and the **mean absolute error** (MAE). Others are the mean **bias** and $R^2$, the fraction of variance explained. As Figure 43.11 showed, $R^2$ becomes negative when a model is worse than predicting the mean.

### 43.5.4 Forecast Skill and Calibration

Forecasts of events use further scores. The **critical success index**, CSI = TP / (TP + FP + FN), ignores the many correct "no event" forecasts. The **true skill statistic**, TSS = recall − false-alarm rate, ranges from −1 to 1 and does not depend on how often events occur. That is why it is widely recommended for comparing flare forecasts made on different samples. A **skill score**, $\mathrm{SS} = 1 - S/S_{\mathrm{ref}}$, compares a score $S$ (lower is better) with that of a reference such as climatology or persistence.

Probabilistic forecasts must also be **calibrated**: of all the occasions on which a flare model says "30 % chance", about 30 % should see a flare. The **Brier score**, $\mathrm{BS} = \frac{1}{N}\sum_i (p_i - o_i)^2$ with outcomes $o_i$ of 0 or 1, rewards both calibration and sharpness, and a **reliability diagram** plots observed frequency against forecast probability. Many classifiers are poorly calibrated out of the box. Modern deep networks tend to be over-confident, while boosted trees and SVMs distort probabilities in other ways. They can be recalibrated on validation data, for example by Platt scaling or isotonic regression. Calibration matters whenever probabilities feed decisions, such as collision-avoidance manoeuvres (Chapter 21).

### 43.5.5 Rare Events

Fires, anomalies, high-risk conjunctions and large flares are all rare, and a model that minimises average loss can do well by ignoring the rare class. Remedies include **re-weighting** the loss, **re-sampling** the classes, setting the threshold for the actual costs and evaluating with precision, recall and PR curves rather than accuracy. The EO-1 classifiers showed the problem: overall accuracy above 80 % concealed a recall below 50 % for ice, which made up 2 % of the pixels.

DATA BOX — RARE EVENTS: ESA'S COLLISION-AVOIDANCE CHALLENGE

- In 2019 ESA released 199 082 conjunction data messages for 15 321 close approaches involving its satellites between 2015 and 2019, each with 103 attributes. It asked competitors to predict the final collision risk two days before closest approach.
- Only 515 events (3.4 %) had a final risk of $10^{-6}$ or more, 131 (0.9 %) of $10^{-5}$ or more and 30 of $10^{-4}$ or more.
- 96 teams made 862 submissions over two months. A naive forecast — "the final risk equals the latest known risk" — scored 0.694 on the competition's loss; only 12 teams beat it, and the winners only modestly.
- The organisers concluded that naive forecasts are a surprisingly strong benchmark that machine learning can beat, but that the data's imbalance limited off-the-shelf methods (Chapter 50).

### 43.5.6 How Sure Are We?

A test score must be quoted with its uncertainty. If a detector produces $k$ false alarms among $n$ independent negative samples, the rate $\hat{p} = k/n$ has a standard error of about $\sqrt{\hat{p}(1-\hat{p})/n}$. An approximate 95 % confidence interval is $\hat{p} \pm 1.96$ standard errors. The **Wilson interval** is more accurate when $k$ is small. If no false alarms are seen in $n$ trials, the "rule of three" gives a 95 % upper limit of about $3/n$. Two models tested on the same samples are compared with a paired test such as McNemar's.

EXAMPLE: HOW BIG MUST THE TEST SET BE?

A requirement says that an onboard event detector must have a false-alarm rate of about 1 %, measured to within ±0.2 % at 95 % confidence.

1. **Sample size.** Solving $1.96\sqrt{p(1-p)/n} = E$ for $n$ gives $n = 1.96^2 \times 0.01 \times 0.99 / 0.002^2$ = **9 508 negative samples**.
2. **Check.** With 95 false alarms in 9 508 samples, the Wilson 95 % interval is 0.82–1.22 %, as required.
3. **Correlation.** Pixels are not independent. Suppose they come in clusters of 25 neighbouring pixels with an intra-cluster correlation of 0.2. The **design effect** is then 1 + (25 − 1) × 0.2 = 5.8, and about **55 000 pixels** — some 2 200 independent clusters — are needed.
4. **Rarer requirements.** Halving the interval needs four times the samples; a 0.1 % rate measured to ±0.02 % needs about 96 000 independent negatives.
5. **The positives too.** The 500 fire pixels of the wildfire example measure its recall of 0.82 only to within about ±0.034.

## 43.6 From Notebook to Operations

In operations a model becomes one component of a data system (Figure 43.18). It may run on the ground, in a processing chain or cloud service. After verification on flight-like hardware it may also run on the spacecraft itself, where it can decide what to downlink or whether to retask the sensor (Chapter 45). In JPL's Dynamic Targeting experiment on the CogniSAT-6 cubesat in July–August 2025, the satellite looked about 74–90 seconds ahead along its track. It analysed the image for cloud on board and then pointed its camera at the clearest scenes.

### PRACTICE: BUILDING AN ONBOARD CLOUD SCREEN FOR A NEW IMAGER

1. **Define the decision.** The screen decides which scenes are downlinked. State the cost of each error — a cloudy scene wastes link capacity, a discarded clear scene is lost — and the onboard power, memory and time limits.
2. **Gather data.** Use data processed as on board, not ground products, covering all seasons, latitudes, Sun angles and surfaces — especially snow, ice, deserts, glint and smoke.
3. **Label.** Have experts mark cloud, thin cloud, shadow and clear; label a sample twice to measure the label-error rate.
4. **Set a baseline.** Score a physics-based threshold test on visible and 1.6 µm bands; any learned model must beat it clearly.
5. **Split by space and time.** Assign whole scenes, regions and dates to each set; hold out a region and a season entirely, and lock the test set.
6. **Train and tune.** Start with boosted trees on bands and ratios, then try a small convolutional network; tune on validation data only.
7. **Evaluate.** Report precision and recall per surface type at the chosen threshold, with confidence intervals; inspect the worst errors by eye.
8. **Deploy and verify.** Convert the model to the flight processor's number format, check its outputs against the ground version, and run it in "shadow mode" — deciding but not acting — before it controls the downlink.
9. **Monitor.** Compare flagged fractions with ground cloud masks; retrain after recalibration or for new regions.

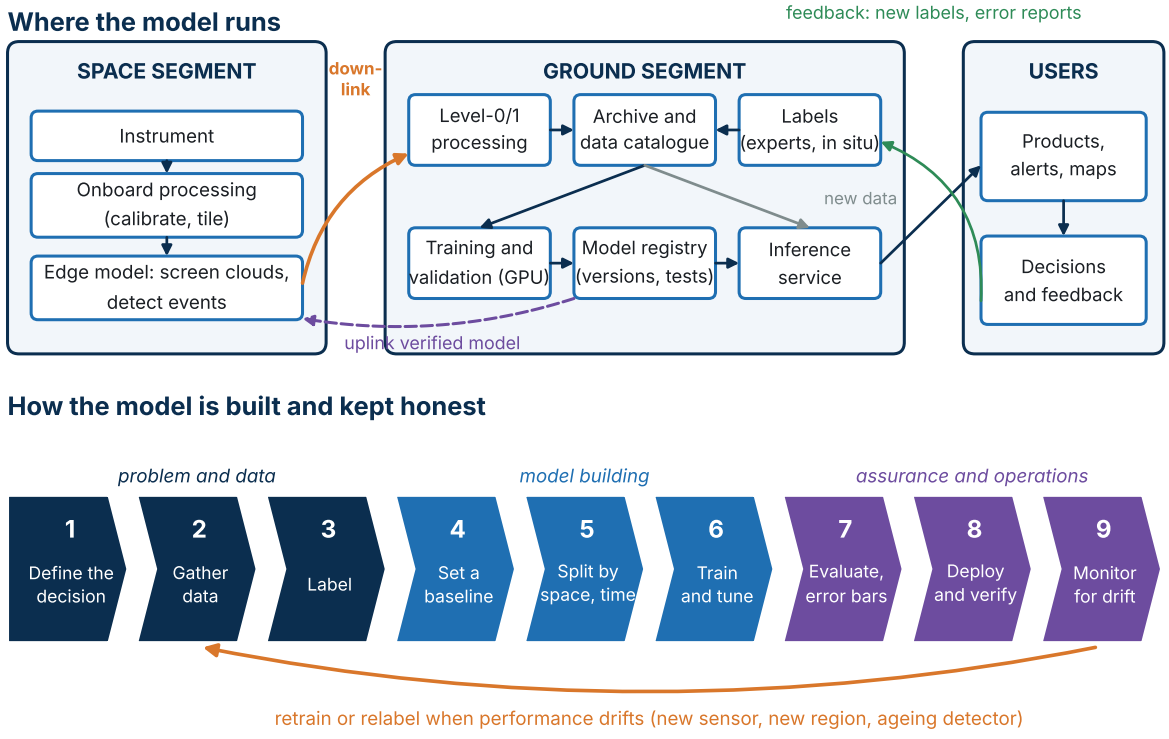



**FIGURE 43.18** Where machine learning sits in a space data system (top) and the nine-step workflow for building and operating a model (bottom). Models run on the ground or, once verified, are uplinked to the spacecraft; new labels and error reports from users feed retraining when performance drifts.

Operations bring **distribution shift**: instruments degrade and are recalibrated, new satellites have slightly different spectral responses, and models meet new regions and seasons. The remedy is monitoring inputs and outputs, comparing predictions with independent reference

data, and retraining when performance falls. A model must also be versioned and documented like any other flight or ground software, with its training data, test results and known limitations recorded (Chapter 53).

## 43.7 Road Map of Part XIII

The rest of Part XIII builds on these foundations (Table 43.6). Chapters 44 and 45 cover deep-learning methods, data and computers, and Chapters 46–52 apply them to each domain of the space ecosystem. Chapter 53 asks how learned systems can be verified, secured and governed well enough to be trusted.

**TABLE 43.6** Road map of Part XIII.

| Chapter | Topic | Main themes |
|---|---|---|
| 44 | Deep learning, foundation models and generative AI | CNNs, transformers, pre-trained and generative models |
| 45 | Data, computing hardware and edge AI in orbit | Data, labels, processors for onboard AI |
| 46 | AI for Earth observation | Mapping, change detection, disasters, image triage |
| 47 | AI for weather, climate and space weather | AI forecasts versus physics; flares and storms |
| 48 | AI for autonomy, navigation and robotics | Planners, AutoNav, learned guidance, RL |
| 49 | AI for mission operations, communications and networks | Anomaly detection, scheduling, cognitive radio |
| 50 | AI for space situational awareness, defence and security | Tracking, conjunctions, intelligence |
| 51 | AI for space science, planetary exploration and human spaceflight | Planet finding, planetary data, crew support |
| 52 | AI in spacecraft design, manufacturing, launch and the space business | Design, inspection, launch operations, markets |
| 53 | Trustworthy AI for space | Verification, security, governance, outlook |

### KEY TAKEAWAYS

- Symbolic AI writes knowledge down as rules and plans; machine learning learns it from examples. Space systems combine them, with verifiable logic guarding learned parts.
- AI reached space in waves — expert systems and schedulers, classical machine learning, deep learning, foundation models; most AI flying in 2026 is still demonstration.
- Learning minimises empirical risk by (stochastic) gradient descent, with learning rates below $2/\lambda_{\max}$; regularisation, early stopping and validation data control overfitting.
- Spatial and temporal autocorrelation leak information between random splits; split by block, region, time or mission, or a useless model can score $R^2 = 0.80$.
- Classical methods — regression, k-NN, trees, forests, boosting, SVMs, k-means, PCA, Gaussian processes, Kalman filters — still underpin most operational products.
- For rare events report precision, recall and skill scores with confidence intervals, not accuracy; a 1 % false-alarm rate needs about 9 500 independent negatives to measure to ±0.2 %.

### REVIEW QUESTIONS

1. Contrast symbolic AI and machine learning, with a space example of each. Why is safe mode still implemented with explicit logic?

2. Describe how Hubble's SPIKE scheduler represents constraints, and why a planner of this kind is easier to verify than a neural network.
3. Name the problem type and a suitable metric for: mapping flood extent from radar images; predicting the Kp index three hours ahead; choosing which of 50 images to downlink first; finding unusual behaviour in battery telemetry.
4. A loss surface has Hessian eigenvalues of 0.5 and 20. What is the largest stable learning rate for gradient descent, and what is the condition number? Why does standardising the inputs help?
5. Repeat the gradient-descent example for a clear vegetation pixel $\boldsymbol{x} = (0.06, 0.22)$, $\boldsymbol{y} = 0$, starting from the updated parameters. How does the prediction change?
6. Sketch training and validation loss against epochs for a network that overfits. What are early stopping, weight decay and data augmentation?
7. Why did random cross-validation give $R^2 = 0.80$ in Figure 43.11 when the model's skill in a new region was negative? How would you choose the block size for spatial cross-validation of a soil-moisture model?
8. How does a random forest reduce the variance of decision trees, and how does gradient boosting differ? Why did EO-1 use a linear SVM on board?
9. A conjunction-screening model raises 40 alerts in a year, of which 4 are real high-risk events, and misses 1. Compute its precision, recall and F1. Why is accuracy not useful here?
10. How many independent negatives are needed to measure a 2 % false-alarm rate to ±0.5 % at 95 % confidence? How does spatial correlation change this?

CHAPTER

# 44 Deep Learning, Foundation Models and Generative AI

Image: NASA/ARC (Dominic Hart)

*Deep neural networks now map floods from radar, find fresh craters on Mars, forecast the weather and help plan rover drives. This chapter explains how they work, from first principles and with small networks written and trained from scratch. It covers the building blocks of deep networks and the scaling laws that govern them; convolutional networks for images; recurrent networks, transformers and graph networks for telemetry, sequences and meshes; and generative models from autoencoders to diffusion. It then describes self-supervised foundation models for Earth, Sun and Moon, large language models and agents, physics-informed learning, uncertainty quantification and explainability. Verified examples show what each method has achieved in space as of October 2026, and where it can fail.*

**IN THIS CHAPTER**

44.1 Building Blocks of Deep Networks
44.2 Convolutional Networks for Images
44.3 Sequences, Transformers and Graphs
44.4 Generative Models
44.5 Foundation Models and Self-Supervised Learning
44.6 Large Language Models and Agents
44.7 Physics, Uncertainty and Explanation

Chapter 43 showed how a small neural network learns by gradient descent. **Deep learning** stacks many such layers — tens to hundreds — and trains them on large data sets. The network then learns its own features from raw pixels, spectra or telemetry, instead of relying on features chosen by an engineer. Since about 2012 this approach has overtaken older methods in image recognition, speech, translation and, since 2023, global weather forecasting.

This chapter explains the main families of deep networks, each in plain words with its key equations, then its space uses. Most figures come from small numpy networks trained for this book, so every number can be checked. Data and onboard processors are covered in Chapter 45, applications in Chapters 46–52 and assurance in Chapter 53.

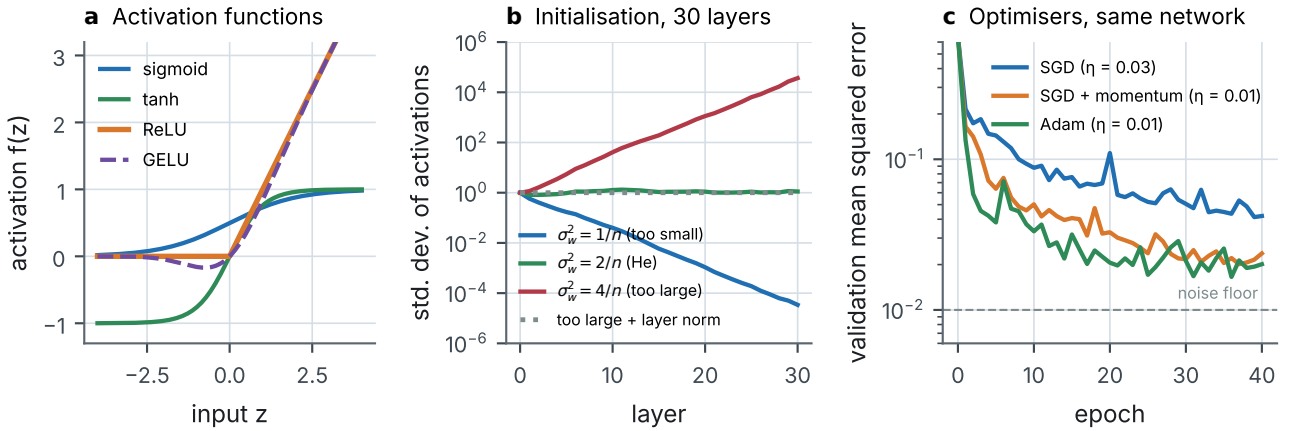
## 44.1 Building Blocks of Deep Networks

A deep network is a chain of simple, differentiable steps. Each step is cheap; the power comes from stacking many of them and fitting millions of weights to data. Almost every modern network uses the same few parts: linear layers, activations, careful initialisation, normalisation, residual connections and an adaptive optimiser.

### 44.1.1 Neurons, Activations and Layers

An artificial **neuron** computes a weighted sum of its inputs, adds a bias and passes the result through an **activation function** $f$: $y = f(\mathbf{w}^{\top}\mathbf{x} + b)$. A **layer** is many neurons side by side, so it computes $\mathbf{h} = f(W\mathbf{x} + \mathbf{b})$. Without the activation, any stack of layers would collapse into a single matrix multiplication and could only draw straight boundaries.

Figure 44.1 (a) shows the common activations. The **sigmoid** and **tanh** squash inputs into a fixed range; their slope is almost zero for large inputs, which slows learning. The **rectified linear unit (ReLU)**, $\max(0, z)$, has slope 1 for every positive input and is cheap to compute. **GELU** and **SiLU** are smooth versions of it used in transformers. For classification the last layer uses the **softmax**, $p_k = e^{z_k} / \sum_j e^{z_j}$, which turns scores into probabilities that sum to one.




(b) 512 random inputs through 30 ReLU layers of width 256. (c) Network 2-64-64-1 (ReLU, He init.) trained on 2 000 noisy samples of a smooth 2-D function, mini-batches of 32; best of four learning rates per optimiser on 2 000 validation samples; one seed.

**FIGURE 44.1** Building blocks, computed in numpy. (a) Common activations. (b) Activation spread through 30 ReLU layers: He initialisation (variance 2/n) keeps it near 1; 1/n and 4/n make it vanish or explode, unless layer normalisation is used. (c) One small network trained by three optimisers, each at its best learning rate.

### 44.1.2 Training: Backpropagation and Optimisers

Training uses **backpropagation**, which applies the chain rule backwards through the layers to find the gradient of the loss with respect to every weight (Chapter 43). Plain **stochastic gradient descent (SGD)** then steps each weight downhill. Two refinements make training much faster.

**Momentum** keeps a running sum of past gradients, $\mathbf{v}_t = \beta \mathbf{v}_{t-1} + \mathbf{g}_t$, and steps along it, $\theta_t = \theta_{t-1} - \eta \mathbf{v}_t$, with $\beta \approx 0.9$. Like a heavy ball, it rolls through small bumps and along narrow valleys. **Adam** also tracks the mean square of each weight's gradient and divides by its square root:

$$\mathbf{m}_t = \beta_1 \mathbf{m}_{t-1} + (1-\beta_1)\mathbf{g}_t, \quad \mathbf{v}_t = \beta_2 \mathbf{v}_{t-1} + (1-\beta_2)\mathbf{g}_t^2, \quad \theta_t = \theta_{t-1} - \eta \, \frac{\hat{\mathbf{m}}_t}{\sqrt{\hat{\mathbf{v}}_t} + \epsilon},$$

where $\hat{\mathbf{m}}_t$ and $\hat{\mathbf{v}}_t$ are corrected for their start at zero, $\beta_1$ = 0.9 and $\beta_2$ = 0.999. Each weight thus gets its own step size. In Figure 44.1 (c), a network with two hidden layers of 64 units learns a smooth two-dimensional function. After 10 epochs (passes through the data) the validation error is 0.088 with plain SGD, 0.050 with momentum and 0.033 with Adam. After 40 epochs Adam reaches 0.020, against a noise floor of 0.010. Large models also use a **learning-rate schedule**: a short warm-up, then a slow decay.

### 44.1.3 Initialisation, Normalisation and Regularisation

Deep networks are fragile at the start. If the initial weights are slightly too small, the signal shrinks at every layer; if slightly too large, it explodes. Figure 44.1 (b) sends random inputs through 30 ReLU layers. With weight variance $1/n$ ($n$ inputs per neuron) the spread of the activations falls to $3 \times 10^{-5}$ of its starting value. With $4/n$ it grows to $4 \times 10^{4}$. **He initialisation**, with variance $2/n$, keeps it near 1, because ReLU zeroes half of its inputs. Gradients flowing backwards suffer the same fate, which is why badly initialised deep networks do not learn.

**Normalisation layers** fix the scale during training as well. **Layer normalisation** rescales each sample's activations to zero mean and unit variance, then applies a learned scale $\gamma$ and shift $\beta$: $y = \gamma(x - \mu)/\sigma + \beta$. **Batch normalisation** does the same across a mini-batch. **Dropout** randomly zeroes 10–50 % of units during training, so the network cannot rely on any single one.

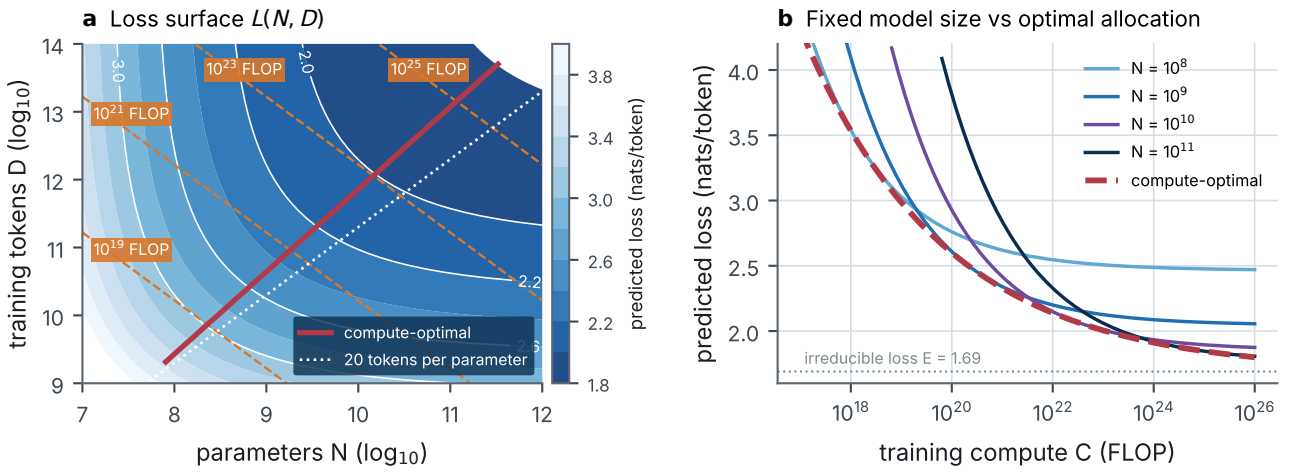
### 44.1.4 Scaling Laws

Since 2020, studies of language models have found that test loss falls smoothly, as a **power law**, as the number of parameters $N$, the number of training tokens $D$ and the compute $C$ grow. Hoffmann and colleagues (2022) fitted

$$L(N, D) = E + \frac{A}{N^{\alpha}} + \frac{B}{D^{\beta}}, \qquad C \approx 6ND,$$

with $E$ = 1.69, $A$ = 406.4, $B$ = 410.7, $\alpha$ = 0.34 and $\beta$ = 0.28. $E$ is the loss that no model can beat, set by the randomness of language itself. The cost rule $C \approx 6ND$ counts about six floating-point operations (FLOP) per parameter per training token: two for the forward pass and four for the backward pass.

The formula answers a practical question: for a fixed compute budget, how big should the model be? Figure 44.2 computes it. For $10^{22}$ FLOP the best choice is about 5 billion parameters trained on about 320 billion tokens, giving a loss of 2.14. A 1-billion-parameter model trained with the same compute on five times more data reaches only 2.20. With these constants the optimal model size grows as $C^{0.45}$ and the data as $C^{0.55}$, so data matter as much as parameters.




Computed from the published fit L = E + A/N^α + B/D^β (E = 1.69, A = 406.4, B = 410.7, α = 0.34, β = 0.28; Hoffmann et al. 2022) for language models; training compute C ≈ 6ND floating-point operations (dashed lines in a: equal compute). Constants differ for other data types and architectures.

**FIGURE 44.2** Scaling laws from the published fit of Hoffmann et al. (2022). (a) Predicted loss against model size and data; dashed lines join points of equal compute, and the red line marks the best split. (b) Loss against compute: each fixed model size flattens, while the compute-optimal curve keeps falling.

Two cautions apply. First, for budgets of $10^{20}$–$10^{24}$ FLOP these published constants imply 40–100 tokens per parameter. The paper's other two methods recommend about 20. A 2024 replication traced the difference to errors in this fit. Second, the constants are for mostly English text. Earth-observation and weather models show similar trends, but with their own constants.

## 44.2 Convolutional Networks for Images

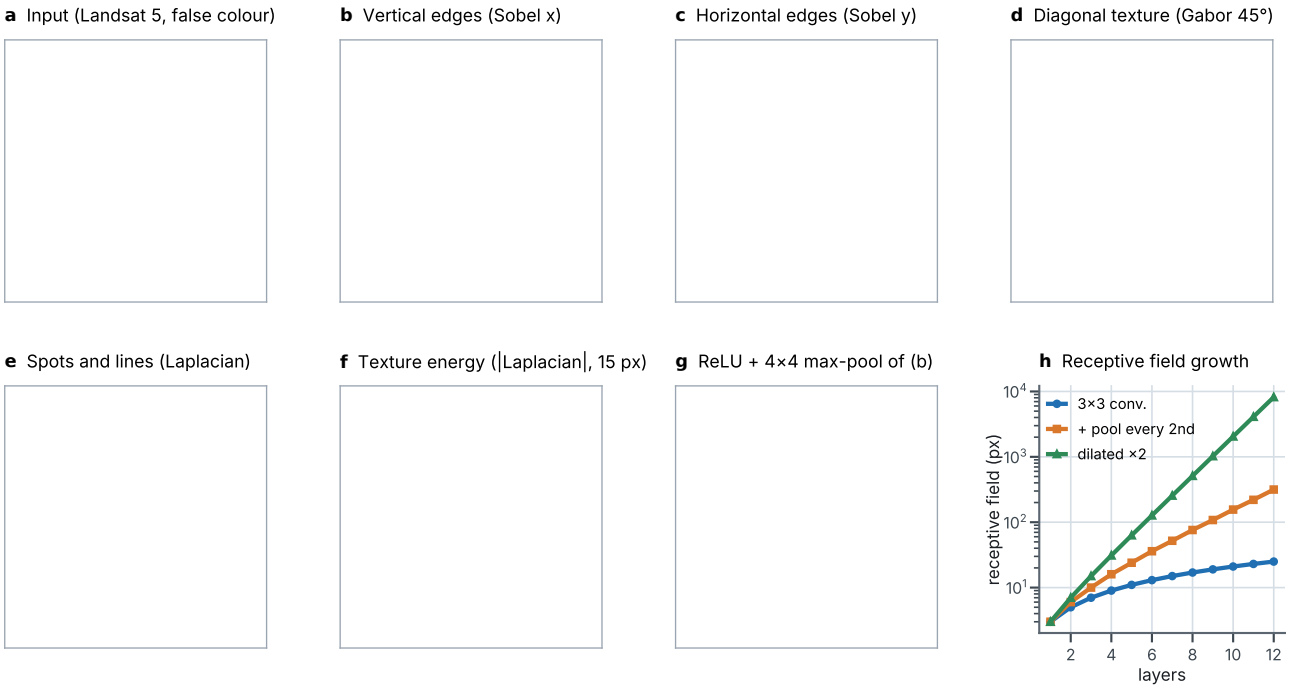
Images have structure: nearby pixels are related, and a coastline or a crater looks the same wherever it appears. A **convolutional neural network (CNN)** builds these facts into its design. It is the workhorse of image analysis in space, from cloud masks to crater catalogues.

### 44.2.1 Convolution, Pooling and Receptive Field

A **convolution layer** slides a small grid of weights, the **kernel**, over the image. At each position it multiplies the kernel by the pixels beneath and sums the result. Output channel $c'$ at pixel $(i, j)$ is

$$y_{i,j,c'} = b_{c'} + \sum_{c} \sum_{u=0}^{k-1} \sum_{v=0}^{k-1} w_{u,v,c,c'}\, x_{i+u,\, j+v,\, c}.$$

Here $c$ counts input channels and the kernel is $k \times k$ pixels. The same kernel is used everywhere — **weight sharing** — so a layer needs only $k^2 c\, c'$ weights, whatever the image size. Each output channel is a **feature map** that responds to one pattern. In Figure 44.3, hand-made kernels detect edges, texture and spots in a Landsat 5 image of Las Vegas. **Max-pooling** then keeps the largest value in each 4 × 4 block. A trained CNN learns similar filters in its first layer and combines them into shapes and objects in deeper layers.




Fixed (hand-designed) 3×3 and 9×9 kernels convolved with a 480 × 480-pixel crop (about 40 m per pixel) of a Landsat 5 image of Las Vegas, Nevada, 2 November 2011 (bands 7, 4, 2; NASA/USGS). Red and blue: positive and negative responses. A trained CNN learns similar filters in its first layer.

**FIGURE 44.3** Feature maps of a real satellite image: fixed kernels applied to a 480 × 480-pixel crop of a Landsat 5 false-colour image of Las Vegas (2 November 2011). (b–e) Edge, texture and spot filters; (f) texture energy; (g) ReLU and 4 × 4 max-pooling of (b); (h) receptive-field growth with depth.

The **receptive field** is the patch of input that can influence one output pixel. Twelve stacked 3 × 3 layers see only 25 × 25 pixels. Adding 2 × 2 pooling after every second layer widens this to 316 pixels. **Dilated (atrous) convolutions** space the kernel taps apart; doubling the spacing at each layer gives 8 191 pixels after 12 layers, without losing resolution (Figure 44.3 h). Context matters: telling a lake from a cloud shadow needs a view of the surroundings.

### 44.2.2 Residual Networks

Simply stacking more layers eventually makes networks harder to train. **Residual networks (ResNets)** (2015) fixed this with a shortcut: each block learns a correction $\boldsymbol{F}(\mathbf{x})$ and outputs $\mathbf{y} = \mathrm{ReLU}(\boldsymbol{F}(\mathbf{x}) + \mathbf{x})$ (Figure 44.4). An unneeded block can learn $\boldsymbol{F} \approx \mathbf{0}$ and pass its input straight through, and gradients flow back through the shortcuts undiminished. An ensemble of ResNets, the deepest with 152 layers, won the 2015 ImageNet competition with a top-5 error of 3.57 %. Residual connections are now in almost every deep network.

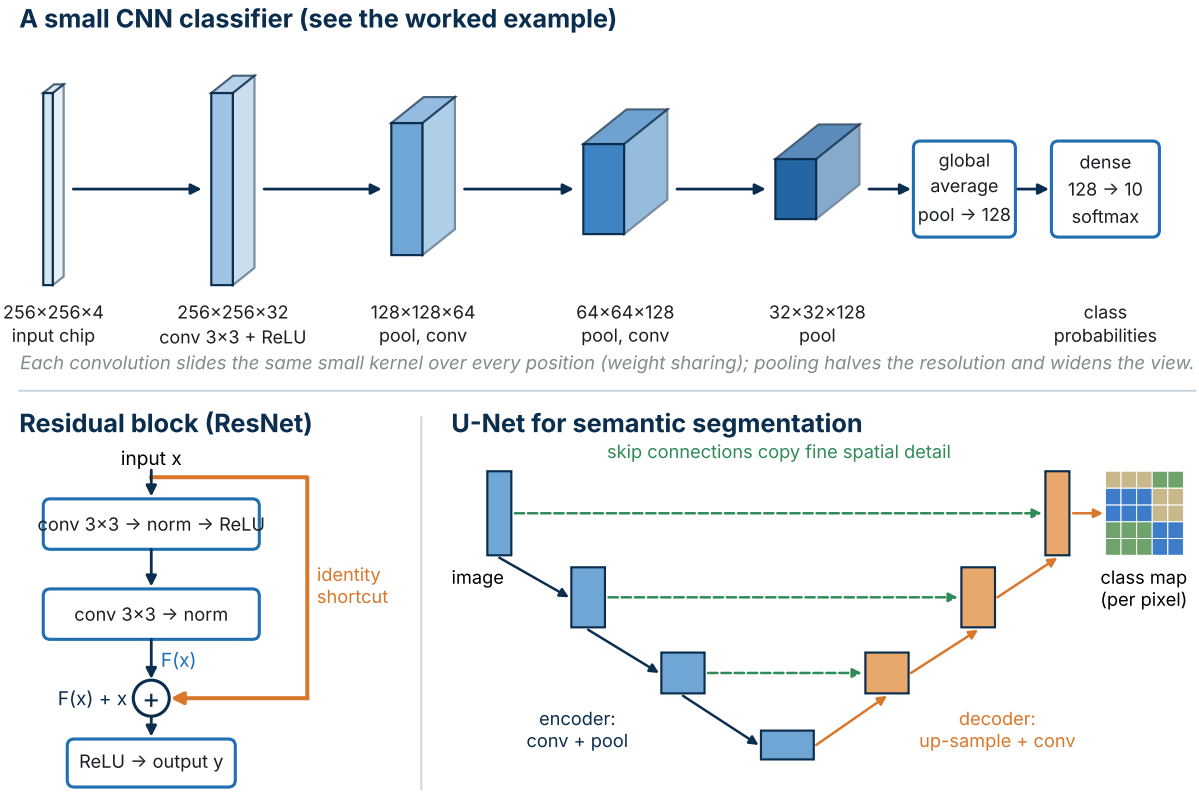



**FIGURE 44.4** Three convolutional architectures. Top: a small classifier for 256 × 256-pixel, four-band chips, with the data shape after each stage (see the worked example). Bottom: a residual block, and a U-Net whose skip connections copy fine detail from encoder to decoder.

### EXAMPLE: PARAMETERS AND OPERATIONS OF A SMALL CNN

The classifier at the top of Figure 44.4 takes 256 × 256-pixel chips with four bands. It has three 3 × 3 convolution layers with 32, 64 and 128 channels, with 2 × 2 pooling before the second and third. Global average pooling and a dense layer for 10 classes follow.

1. **One layer.** A convolution has $k^2 c_{in} c_{out} + c_{out}$ parameters and needs $HWk^2 c_{in} c_{out}$ multiply–accumulate operations (MACs). Layer 1: 9 × 4 × 32 + 32 = **1 184 parameters**, and 256 × 256 × 9 × 4 × 32 = 75.5 million MACs.
2. **Layers 2 and 3.** Layer 2 has 9 × 32 × 64 + 64 = 18 496 parameters at 128 × 128 pixels (302 million MACs). Layer 3 has 9 × 64 × 128 + 128 = 73 856 parameters at 64 × 64 pixels (also 302 million MACs).
3. **Head.** Pooling has no weights; the dense layer has 128 × 10 + 10 = 1 290.
4. **Totals.** The network has **94 826 parameters** and needs 679 million MACs, or **1.36 GFLOP** (two operations per MAC), per chip.
5. **Weight sharing.** A dense layer mapping the same input to the same 256 × 256 × 32 output would need $5.5 \times 10^{11}$ weights — about six million times more than the whole network.
6. **A whole scene.** A 10 000 × 10 000-pixel scene holds about 1 526 chips: about $2.1 \times 10^{12}$ operations, or some 2 seconds on a processor sustaining $10^{12}$ operations per second (1 TOPS; Chapter 45).

## 44.2.3 Object Detection

**Object detection** finds each object in an image and draws a box around it with a class label. **Two-stage detectors**, such as Faster R-CNN (2015), first propose regions that might contain an object, then classify and refine each one. They are accurate but slower. **One-stage detectors**, such as the YOLO ("you only look once", 2016) family, divide the image into a grid and predict boxes and

classes for every cell in a single pass. Overlapping boxes for the same object are then merged by **non-maximum suppression**. Ultralytics' YOLO26, released in January 2026, removes that step by predicting one box per object directly. Detection is scored by mean average precision (mAP, Chapter 43). Ships, aircraft and vehicles in satellite images are detected this way (Chapter 46), as are craters and boulders on planetary surfaces.

**LANDMARK: AI FINDS FRESH CRATERS ON MARS (2020)**

- Fresh impacts on Mars are found by comparing before-and-after images, a slow task for people. JPL researchers trained a classifier on 6 830 Context Camera (CTX) images from Mars Reconnaissance Orbiter.
- Running up to 750 copies in parallel on a JPL cluster, it scanned about 112 000 CTX images, at about 5 seconds per image against about 40 minutes for a human.
- Its first confirmed find was a cluster of craters up to about 4 m across in Noctis Fossae, formed between March 2010 and May 2012. HiRISE confirmed it on 26 August 2020 (Figure 44.5), and more than 20 further candidates were queued for follow-up.
- The network did not replace the scientists: it ranked the archive so that HiRISE looked where a discovery was likely.

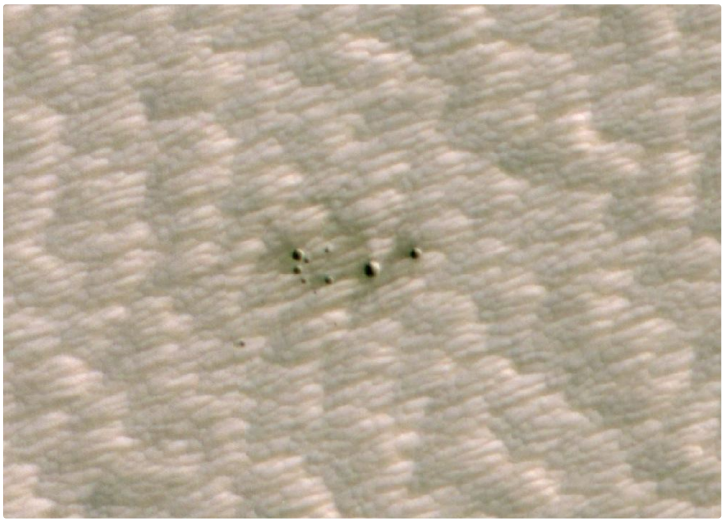


**FIGURE 44.5** The first crater cluster on Mars found by artificial intelligence, seen by HiRISE on Mars Reconnaissance Orbiter. A classifier flagged the dark spots in Context Camera images; HiRISE confirmed fresh craters made by one impactor that broke apart. Image: NASA/JPL (NASA/JPL-Caltech/University of Arizona).

### 44.2.4 Semantic Segmentation

**Semantic segmentation** assigns a class to every pixel: water or land, cloud or clear, sand or bedrock. The **U-Net** (2015) is an encoder–decoder (Figure 44.4): the encoder shrinks the image while widening the view, the decoder up-samples back to full resolution, and **skip connections** copy fine detail across. **DeepLab** (2014–18) instead uses dilated convolutions and a pyramid of dilation rates to capture context at several scales. Table 44.1 summarises the vision architectures and their space uses.

DeepMoon (2019) trained a U-Net on lunar elevation maps. It recovered 92 % of the craters in its test catalogue and nearly doubled the number found, though about 11 % of the new detections checked by hand were false. NASA JPL's **AI4Mars** project collected about 326 000 crowd-sourced

labels on about 35 000 rover images, marking soil, bedrock, sand and big rocks (Figure 44.6). A DeepLabv3 network trained on them classified more than 96 % of test pixels correctly. Terrain classifiers of this kind support Curiosity's drive planning on the ground (Chapter 48).

**TABLE 44.1** Vision architectures and typical space uses.

| Task | Architecture | Key idea | Space example |
|---|---|---|---|
| Scene classification | ResNet, EfficientNet, ViT | Deep features, one label per image | Land-use classes, image triage on board |
| Object detection | Faster R-CNN; YOLO family | Region proposals; single-pass grid | Ships, aircraft, craters, boulders |
| Semantic segmentation | U-Net, DeepLab | Encoder–decoder with skips; dilation | Floods, clouds, burn scars, Mars terrain |
| Instance segmentation | Mask R-CNN | Box plus per-object mask | Building footprints, individual fields |
| Super-resolution | Residual CNNs, GANs, diffusion | Learned up-sampling | Sharpening multispectral bands |

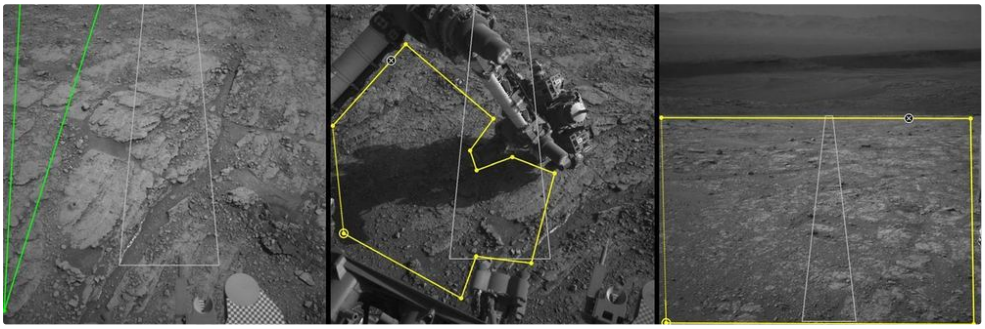


**FIGURE 44.6** The public AI4Mars labelling tool. Volunteers outlined soil, bedrock, sand and large rocks in Curiosity images to train terrain-segmentation networks. Image: NASA/JPL (NASA/JPL-Caltech).

## 44.3 Sequences, Transformers and Graphs

Telemetry, light curves and solar-wind data are **sequences**, in which the past helps predict the future. Weather on a sphere and satellite networks are better described as **graphs**. This section covers networks for both, including the now-dominant transformer.

### 44.3.1 Recurrent Networks, LSTM and Temporal Convolutions

A **recurrent neural network (RNN)** reads a sequence one step at a time and keeps a **hidden state** that summarises what it has seen: $\mathbf{h}_t = \tanh(W_h \mathbf{h}_{t-1} + W_x \mathbf{x}_t + \mathbf{b})$. Gradients are multiplied by one matrix per step backwards through time, so they shrink or explode over long sequences. Plain RNNs forget after a few tens of steps.

The **long short-term memory (LSTM)** network (1997) adds a separate **cell state** $\mathbf{c}_t$ and three gates, each a sigmoid between 0 and 1. The forget gate $\mathbf{f}_t$ decides what to keep, the input gate $\mathbf{i}_t$ what to add and the output gate $\mathbf{o}_t$ what to show:

$$\mathbf{c}_t = \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \tilde{\mathbf{c}}_t, \qquad \mathbf{h}_t = \mathbf{o}_t \odot \tanh(\mathbf{c}_t),$$

where $\odot$ means element-by-element multiplication. Because the cell state is updated by addition, information and gradients survive hundreds of steps. The **gated recurrent unit (GRU)** (2014) is a lighter version with two gates. In 2018 JPL researchers used LSTMs to predict telemetry from the

SMAP satellite and the Curiosity rover; large prediction errors flagged anomalies (Chapter 49).

A **temporal convolutional network (TCN)** applies one-dimensional convolutions that look only at the past (**causal** convolutions), with dilations 1, 2, 4, 8 and so on. With kernel size $k$ and $L$ layers, its receptive field is $1 + (k-1)(2^L - 1)$ samples: 2 047 samples for $k = 3$ and ten layers. Unlike an RNN, a TCN processes all time steps in parallel during training. Table 44.2 compares the options.

**TABLE 44.2** Sequence models compared.

| Model | How it remembers | Parallel training | Cost per sequence | Typical space use |
|---|---|---|---|---|
| RNN | Hidden state | No | Linear in length | Rarely used alone today |
| LSTM / GRU | Gated cell state | No | Linear | Telemetry prediction, anomaly detection |
| TCN | Dilated causal convolutions | Yes | Linear | Fast onboard forecasting, event detection |
| Transformer | Attention over all past steps | Yes | Quadratic in length | Long telemetry, solar time series, language |

### 44.3.2 Attention

**Attention** lets each element of a sequence look directly at every other element and weigh its importance. Each input vector is projected into a **query** $\mathbf{q}$ ("what am I looking for?"), a **key** $\mathbf{k}$ ("what do I contain?") and a **value** $\mathbf{v}$ ("what do I pass on?"). Stacking them into matrices $Q$, $K$ and $V$ gives **scaled dot-product attention**:

$$\mathrm{Attention}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^{\top}}{\sqrt{d_k}}\right)V.$$

Each softmax row holds weights that sum to one, so each output is a weighted average of the values. The weights are large where the query matches the key. Dividing by $\sqrt{d_k}$, where $d_k$ is the key length, keeps the softmax in a range with useful gradients. For forecasting, a **causal mask** sets the weights on future positions to zero.

In Figure 44.7, a single attention layer trained in numpy predicts the next sample of noisy periodic "telemetry". The period is 16 steps, but the model was not told this. After 4 000 training steps it looks back 15 and 31 steps, each with a mean weight of 0.21. These are the samples at the same phase one and two cycles before the predicted one. Averaging several noisy copies of the same phase cancels much of the noise. On new sequences its error is 0.331, against 0.551 for repeating the last value and 0.425 for copying the value one period earlier. The noise sets a floor of 0.30.

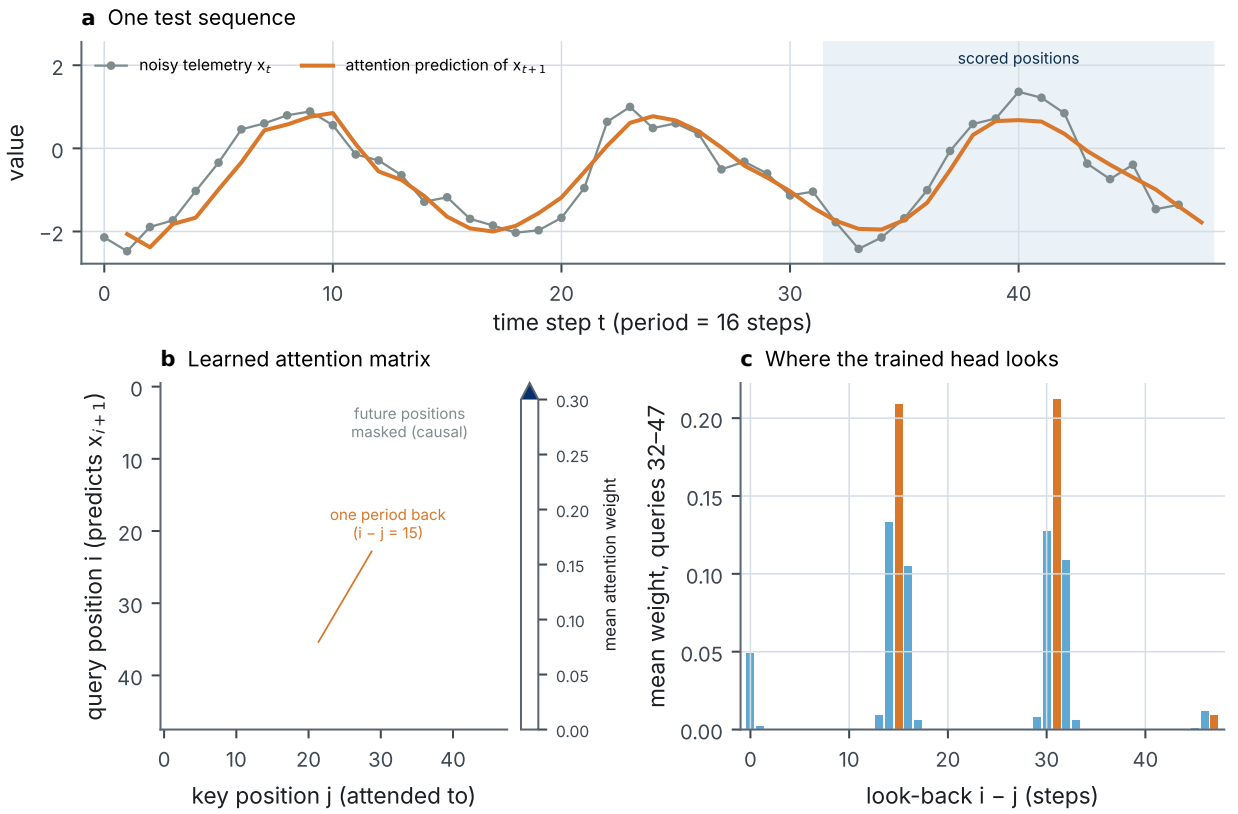



**FIGURE 44.7** A causal self-attention layer trained in numpy on noisy periodic signals. (a) One test sequence and its forecast. (b) Mean learned attention weights: bright diagonals one and two periods back. (c) Mean weight against look-back: 0.21 at 15 and at 31 steps. Test RMSE 0.331, against 0.551 for persistence.

### 44.3.3 The Transformer

The **transformer** (2017) is built from attention. Each layer has a **multi-head attention** block: several attention operations in parallel, so that different heads can track different relationships. A small two-layer network, applied to each position, follows. Both parts have residual connections and layer normalisation. A layer of width $d$ has about $12d^2$ weights: $4d^2$ for the attention projections and $8d^2$ for the position-wise network.

Attention ignores order, so each element's position is added to its input vector as a **positional encoding**. The original transformer used sine waves of different frequencies, $PE_{p,2i} = \sin(p/10\,000^{2i/d})$ and $PE_{p,2i+1} = \cos(p/10\,000^{2i/d})$; many newer models learn them.

The **vision transformer (ViT)** (2020) cuts an image into patches, typically 16 × 16 pixels, and treats them as a sequence of "words". The **Swin transformer** (2021) limits attention to windows of 7 × 7 patches and shifts the windows between layers so that information still spreads. Prithvi uses ViTs; Microsoft's Aurora weather model uses a three-dimensional Swin transformer.

For time series, transformers treat each time step, or a short patch of steps, as a token. They can relate events far apart, such as a solar active region seen again one rotation (27 days) later. But they are not always better: a 2023 study found that a simple linear model beat several transformer forecasters on standard benchmarks. Always compare with persistence, climatology and a linear model.

### 44.3.4 Why Attention Is Expensive

Attention compares every token with every other one, so its cost grows with the **square** of the sequence length $n$. Per layer, the projections and position-wise network cost about $24nd^2$ FLOP and the attention itself $4n^2d$. The first term dominates for short sequences, the second for long ones.

> **EXAMPLE: ATTENTION COST FOR AN IMAGE SPLIT INTO 16 × 16 PATCHES**
>
> Take a ViT-Base layer: width $d = 768$ and 12 heads.
>
> 1. **A 224 × 224 image** gives $(224/16)^2 =$ **196 patches**. The linear parts cost $24 \times 196 \times 768^2 = 2.8$ GFLOP and attention $4 \times 196^2 \times 768 = 0.12$ GFLOP: only 4 % of the layer. The attention matrices of all 12 heads, stored in 16-bit numbers, take about 1 MB.
> 2. **A 1 024 × 1 024 image** gives 4 096 patches. Linear parts: 58 GFLOP; attention: 51.5 GFLOP, now 47 % of the cost. The score matrices take $4\,096^2 \times 12 \times 2$ bytes = **403 MB per layer**.
> 3. **A 4 096 × 4 096 solar image** from SDO gives 65 536 patches. Attention costs 13 200 GFLOP per layer — 93 % of the total — and its score matrices would need **103 GB per layer**, more than most processors hold.
> 4. **Windows.** Restricting attention to 7 × 7-patch windows, as in Swin, cuts the attention cost to $4 \times 65\,536 \times 49 \times 768 = 9.9$ GFLOP, about 1 340 times less.
> 5. **Conclusion.** Full-resolution images and long records need windowed, sparse or memory-efficient attention. The Surya solar model, for example, combines spectral filtering with long- and short-range attention to handle 4 096 × 4 096-pixel SDO images.

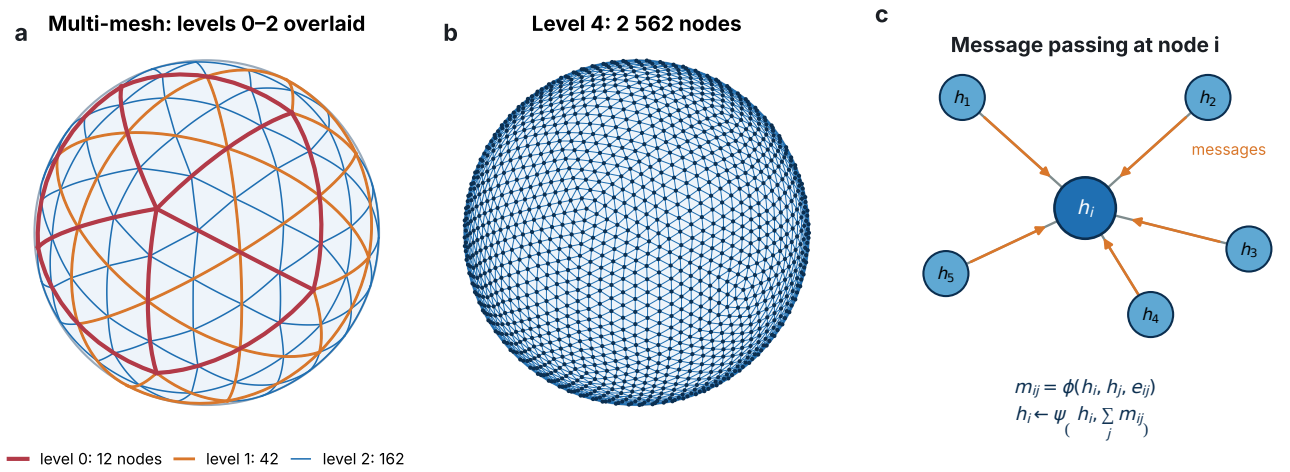
### 44.3.5 Graph Neural Networks

Many space problems are naturally **graphs**: nodes joined by edges. Weather on a sphere can be held on the points of a mesh, and a constellation is satellites joined by inter-satellite links. A **graph neural network (GNN)** learns on such structures by **message passing**. Each node $i$ holds a feature vector $\mathbf{h}_i$. In each layer, every neighbour $j$ sends a message computed by a small network $\phi$. The node then updates itself with another network $\psi$:

$$\mathbf{m}_{ij} = \phi(\mathbf{h}_i, \mathbf{h}_j, \mathbf{e}_{ij}), \qquad \mathbf{h}_i \leftarrow \psi\Big(\mathbf{h}_i, \sum_j \mathbf{m}_{ij}\Big),$$

where $\mathbf{e}_{ij}$ describes the edge, such as its length and direction. The same small networks are used at every node, so a GNN works on graphs of any size. After $L$ layers each node has heard from nodes up to $L$ edges away.

Google DeepMind's **GraphCast** (2023) maps the atmosphere from a 0.25° grid (about one million points, 37 pressure levels) onto an icosahedral mesh, passes messages 16 times and decodes the result back to the grid. The mesh is made by splitting each triangle of an icosahedron into four, six times over (Figure 44.8). The finest level has 40 962 nodes about 120 km apart. Its "multi-mesh" keeps the edges of all coarser levels too, so its 327 660 directed edges include links about 7 000 km long, and information crosses the globe in a few steps. With 36.7 million parameters, it makes a 10-day forecast in under a minute (Chapter 47).




Computed: a regular icosahedron refined by splitting each triangle into four; level 6 has 40 962 nodes (81 920 faces, mean edge 120 km) — the finest level of GraphCast's mesh. Level-0 edges are about 7 054 km long, so a few message-passing steps on the multi-mesh carry information across the globe.

**FIGURE 44.8** The icosahedral mesh of graph-network weather models, computed by repeated subdivision. (a) Levels 0–2 overlaid. (b) Level 4, 2 562 nodes. (c) Message passing at one node. Level 6 has 40 962 nodes with a mean spacing of 120 km.

ECMWF's operational AIFS model, running since 25 February 2025, also uses graph layers to encode and decode the grid, with a transformer in between. GNNs for routing traffic through constellations and for screening conjunctions are still mostly research (Chapters 49 and 50).

## 44.4 Generative Models

A **generative model** learns the probability distribution of its training data and can draw new samples from it: new images, new weather states, new designs. In space engineering this supports super-resolution, synthetic training data, filling gaps under clouds, ensemble forecasts and design exploration. Table 44.3 compares the main families.

**TABLE 44.3** Families of generative model.

| Family | How it generates | Strengths | Weaknesses | Space uses |
|---|---|---|---|---|
| Autoencoder, VAE | Decodes a point from a compact latent space | Compression, smooth latent space | Blurry samples | Anomaly detection, compression |
| GAN | Generator trained to fool a discriminator | Sharp images, fast sampling | Unstable training, misses modes | Super-resolution, SAR-to-optical translation |
| Diffusion | Removes noise step by step | High quality, covers all modes | Many steps, slow sampling | Ensemble weather (GenCast), downscaling |
| Flow matching | Follows a learned velocity field from noise to data | Straighter paths, fewer steps | Newer, less tested | Fast image and weather generation |
| Autoregressive | Predicts the next token or patch | Flexible, works for text | Errors accumulate | Language models, any-to-any EO models |

### 44.4.1 Autoencoders and Variational Autoencoders

An **autoencoder** squeezes its input through a narrow **latent** layer and then reconstructs it, so it must capture the main structure of the data in a few numbers. Inputs that it reconstructs badly are unusual, which makes autoencoders popular for telemetry anomaly detection (Chapter 49).

A **variational autoencoder (VAE)** (2013) makes the latent space probabilistic. Its encoder outputs a mean and spread for a latent vector $\mathbf{z}$. Training maximises a lower bound on the log-probability of the data.

$$\log p(\mathbf{x}) \;\geq\; \mathbb{E}_{q(\mathbf{z}|\mathbf{x})}\big[\log p(\mathbf{x}|\mathbf{z})\big] - \mathrm{KL}\big(q(\mathbf{z}|\mathbf{x}) \,\|\, p(\mathbf{z})\big).$$

The first term rewards good reconstructions. The second keeps the latent codes close to a simple Gaussian, so new samples can be made by decoding random $\mathbf{z}$. VAE samples tend to be blurry, but their compact latents are used inside many larger models.

### 44.4.2 Generative Adversarial Networks

A **generative adversarial network (GAN)** (2014) trains two networks against each other. A **generator** $G$ turns random noise $\mathbf{z}$ into a fake sample, and a **discriminator** $D$ tries to spot the fakes. They play the game:

$$\min_{G}\max_{D}\; \mathbb{E}_{\mathbf{x}}\big[\log D(\mathbf{x})\big] + \mathbb{E}_{\mathbf{z}}\big[\log\big(1 - D(G(\mathbf{z}))\big)\big].$$

GANs make sharp images quickly. They have been used to sharpen satellite images and to make radar images look optical. Their training is unstable, and they can suffer **mode collapse**, producing only a few kinds of output. A GAN can also invent plausible detail: a "super-resolved" building or crater must never be mistaken for a measurement.

### 44.4.3 Diffusion Models

**Diffusion models** learn to reverse a gradual noising process. The **forward process** adds a little Gaussian noise at each of $T$ steps, with variances $\beta_t$, until only noise remains. Its useful property is that the noisy sample at any step can be written in one go:

$$q(\mathbf{x}_t \mid \mathbf{x}_0) = \mathcal{N}\big(\sqrt{\bar{\alpha}_t}\,\mathbf{x}_0,\; (1-\bar{\alpha}_t)\,\mathbf{I}\big), \qquad \bar{\alpha}_t = \prod_{s=1}^{t}(1-\beta_s).$$

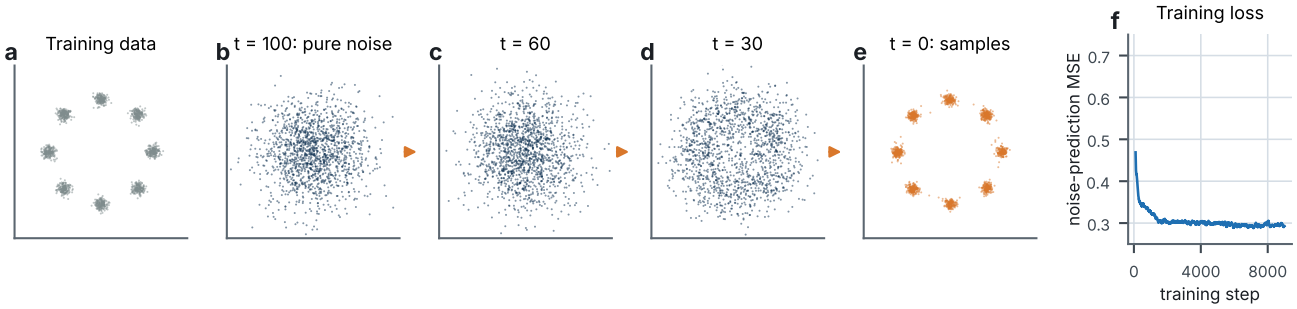
A network $\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)$ is trained to predict the noise that was added, with the simple loss $\mathbb{E}\,\|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)\|^2$. To generate, the **reverse process** starts from pure noise and removes a little predicted noise at each step:

$$\mathbf{x}_{t-1} = \frac{1}{\sqrt{1-\beta_t}}\left(\mathbf{x}_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\,\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)\right) + \sigma_t \mathbf{z},$$

where $\mathbf{z}$ is fresh noise and $\sigma_t$ a set variance. The noise predictor is, in effect, learning the direction towards more probable data at every noise level.

In Figure 44.9, a complete diffusion model written in numpy learns a ring of eight clusters. It uses 100 noise steps and a network with three hidden layers of 128 units, trained for 9 000 steps (a little over a minute on a desktop processor). The reverse process gathers 4 000 points of pure

noise onto the ring. Of the generated points, 98.5 % lie within three standard deviations of a cluster centre, against 98.7 % of real samples. Every cluster gets 10.9–14.0 % of the points, close to the ideal 12.5 %, so no mode has been lost.




Numpy DDPM: noise predictor 18-128-128-128-2 (SiLU), sinusoidal time embedding, T = 100 steps, linear β from $10^{-4}$ to 0.12, 9 000 Adam steps on a ring of eight Gaussians. 98 % of 4 000 generated samples lie within 3σ of a mode; every mode is populated (11–14 % each).

**FIGURE 44.9** A two-dimensional diffusion model trained in numpy. (a) Training data: eight clusters on a ring. (b–e) Samples during the reverse process, from pure noise at step 100 to step 0. (f) Training loss. 98.5 % of samples fall within 3σ of a cluster, and every cluster is populated.

**Flow matching** (2022) is a close relative. It draws a straight line from a noise sample $\mathbf{x}_0$ to a data sample $\mathbf{x}_1$, $\mathbf{x}_t = (1-t)\mathbf{x}_0 + t\mathbf{x}_1$, and trains a network to predict the velocity $\mathbf{x}_1 - \mathbf{x}_0$. Sampling integrates $d\mathbf{x}/dt = \mathbf{v}_\theta(\mathbf{x}, t)$, and the straighter paths need fewer steps.

### 44.4.4 Generative Models in Space

The most important space use so far is **ensemble weather forecasting**. Google DeepMind's **GenCast** (December 2024) is a diffusion model on the sphere that generates 15-day, 0.25° forecasts as ensembles of 50 or more members. On 2019 weather it beat ECMWF's ensemble on 97.2 % of 1 320 combinations of variable, level and lead time. One member takes about 8 minutes on one TPU v5 processor. For hurricanes (Figure 44.10), forecasters need the probability of each track, not one best guess (Chapter 47).

**FIGURE 44.10** Hurricane Milton, a Category 5 storm, in the Gulf of Mexico, seen from the International Space Station on 8 October 2024. Diffusion-based ensembles aim to give the probabilities of tracks and intensities of storms like this. Image: NASA/JSC.

Generative models also **downscale** coarse fields: NVIDIA's CorrDiff research model turns 25 km atmospheric data into 2 km detail over Taiwan. They create **synthetic training data**, such as rendered spacecraft images (Chapter 45), and propose structures in **generative design** (Chapter 52). TerraMind can generate a land-cover map or radar-like image from an optical one. In every case the output is a plausible sample, not a measurement.

## 44.5 Foundation Models and Self-Supervised Learning

A **foundation model** is a large network pre-trained on huge amounts of unlabelled data, then adapted to many tasks with few labels. In space, labelled flood maps or flare catalogues take experts months to build, while satellites collect petabytes of unlabelled data. Foundation models exploit that imbalance (Figure 44.11).

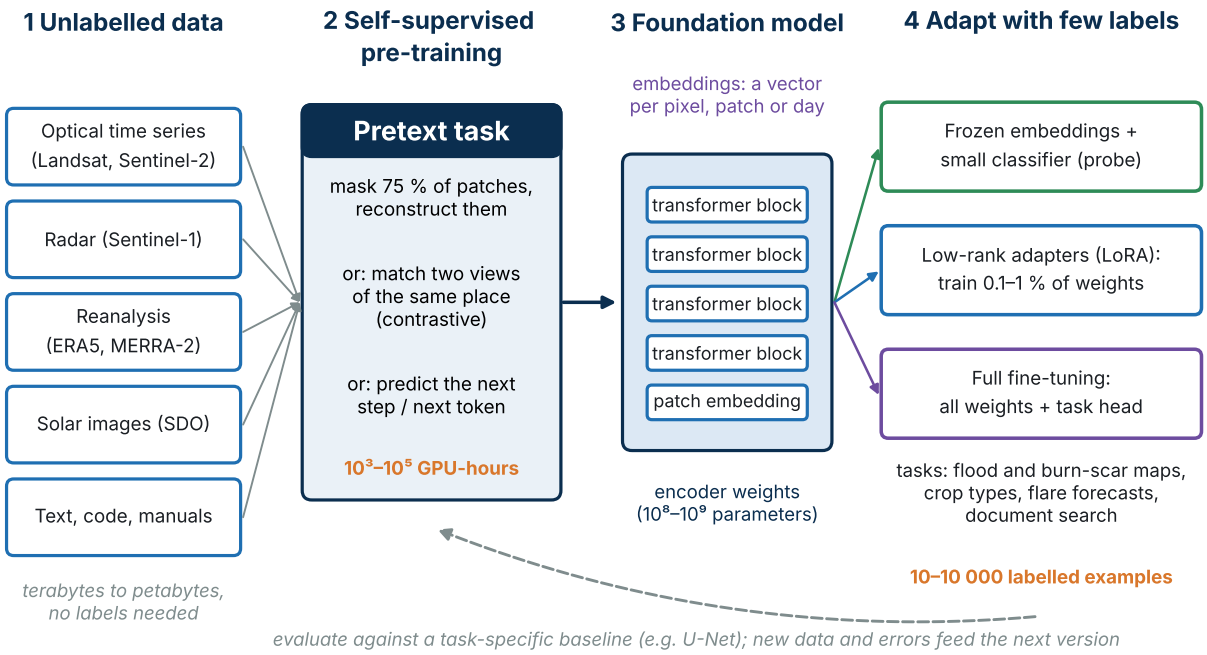



**FIGURE 44.11** How foundation models are built and used. Self-supervised pre-training on unlabelled archives produces an encoder that turns any input into embeddings. Small labelled data sets then adapt it to each task, and results are checked against a task-specific baseline.

### 44.5.1 Learning Without Labels

**Self-supervised learning** invents a training task from the data itself. In **masked autoencoding (MAE)** (2021) the image is cut into patches, typically 75 % of them are hidden, and the network must reconstruct the missing ones. To fill in a hidden field or coastline, it must learn what landscapes, seasons and clouds look like.

**Contrastive learning** teaches the network that two views of the same thing should map to nearby vectors. The views might be two crops of a scene, radar and optical images of the same place, or an image and its caption. Views of different things should map far apart. The common **InfoNCE** loss for a pair of embeddings $\mathbf{z}_i$ and $\mathbf{z}_j$ is

$$\ell_{ij} = -\log \frac{\exp\left(\mathrm{sim}(\mathbf{z}_i, \mathbf{z}_j)/\tau\right)}{\sum_{k \neq i} \exp\left(\mathrm{sim}(\mathbf{z}_i, \mathbf{z}_k)/\tau\right)},$$

where sim is the cosine similarity and $\tau$ a "temperature" setting. In 2021 an image–text model trained this way on 400 million captioned images learned to recognise objects it had never been labelled for.

The output of a pre-trained encoder is an **embedding**: a vector of tens to thousands of numbers that summarises a pixel, patch, image or day. Similar inputs have similar embeddings. Google DeepMind's **AlphaEarth Foundations** (July 2025) summarises a year of optical, radar, lidar and other data for every 10 m patch of land in 64 numbers. Annual layers for 2017–2025 are open in Google Earth Engine. Users map crops or change with a simple model on top. Google reports about 24 % lower error than the other approaches it tested.

### 44.5.2 Adapting a Foundation Model

There are three main ways to adapt a pre-trained model (Figure 44.11). A **linear probe** freezes the encoder and trains only a small classifier on its embeddings; it is cheap and hard to overfit. **Full fine-tuning** updates all weights with a new task head; it is usually the most accurate but needs memory for every weight's gradients and optimiser state.

**Parameter-efficient fine-tuning** sits in between. Its most popular form, **low-rank adaptation (LoRA)** (2021), freezes each large weight matrix $W_0$. It learns a small correction made of two thin matrices:

$$W = W_0 + BA, \qquad B \in \mathbb{R}^{d\times r},\ A \in \mathbb{R}^{r\times k},\ r \ll d, k.$$

Here $r$ is the rank. For a 1 024 × 1 024 matrix with $r$ = 8, the correction has 16 384 trainable numbers instead of 1 048 576 — 1.6 %. For a 175-billion-parameter language model, the LoRA authors cut the trainable parameters about 10 000-fold and the memory about threefold. **Few-shot learning** goes further, adapting from a handful of examples.

### 44.5.3 Earth and Space Foundation Models

Since 2023 many foundation models have been released for Earth observation (Table 44.4) and for the atmosphere, the Sun and the Moon (Table 44.5). Most are open. They differ in data, size and design, so their published scores are hard to compare.

**TABLE 44.4** Earth-observation foundation models (status October 2026). HLS: Harmonized Landsat Sentinel-2.

| Model | Developer, year | Pre-training data | Size (parameters) | Licence |
|---|---|---|---|---|
| SatMAE | Stanford, 2022 | fMoW RGB and Sentinel-2 images, with time | ViT-B, ViT-L | Public code and weights |
| Prithvi-EO-1.0 | NASA, IBM, 2023 | > 1 TB of HLS optical time series, contiguous US | 100 M | Apache-2.0 |
| SkySense | Ant Group, Wuhan Univ., 2024 | 21.5 M optical and SAR time series | 2.06 B | Non-commercial research |
| Clay v1.5 | Clay Foundation, 2024 | ~70 M chips: Sentinel-1/2, Landsat, NAIP, MODIS, others | 632 M (311 M encoder) | Apache-2.0 |
| Prithvi-EO-2.0 | NASA, IBM, Jülich, 2024 | 4.2 M global HLS time series, 2014–23 | 300 M, 600 M | Apache-2.0 |

| Model | Developer, year | Pre-training data | Size (parameters) | Licence |
|---|---|---|---|---|
| TerraMind | IBM, ESA Φ-lab, Jülich, 2025 | 9 M aligned samples, 9 modalities incl. SAR, DEM, text | Base, Large | Apache-2.0 |
| AlphaEarth Foundations | Google DeepMind, 2025 | Optical, radar, lidar, climate | 64-number embeddings | Embeddings CC-BY 4.0 |
| OlmoEarth | Allen Institute for AI, 2025 | ~10 TB optical, radar and map layers | 1.4 M–300 M | OlmoEarth Artifact License |

**TABLE 44.5** Foundation models for the atmosphere, Sun, Moon and science text (status October 2026).

| Model | Developer, year | Pre-training data | Size | Main uses | Licence |
|---|---|---|---|---|---|
| Prithvi WxC | NASA, IBM, 2024 | 160 MERRA-2 variables, 40 years | 2.3 B | Downscaling, gravity waves, forecasts | CDLA-Permissive-2.0 |
| Aurora | Microsoft, 2024 (Nature 2025) | > 1 million hours of geophysical data | 1.3 B | Weather, air quality, waves, cyclones | MIT |
| Surya | NASA, IBM, 2025 | 9 years of SDO images, ~218 TB | 366 M | Flares, solar wind, active regions | Apache-2.0 |
| Lunar Foundation Model | NASA, IBM, 2026 | ~2 M tile bundles from LRO, Kaguya, GRAIL and others | ViT-B size | Craters, ice prospecting | Apache-2.0 |
| INDUS | NASA, IBM, 2024 | 60 B tokens of science text | 125 M encoder, smaller variants | Search, retrieval | Apache-2.0 |
| EVE | ESA Φ-lab, Pi School, 2026 | Earth-science text, open base model | 24 B | Question answering with sources | Open release planned |

DATA BOX — PRITHVI-EO-2.0 IN NUMBERS

- **Data.** About 4.2 million samples from the global Harmonized Landsat Sentinel-2 archive (Figure 44.12), each 256 × 256 pixels at 30 m with four time steps, from 2014–2023.
- **Compute.** 400 epochs on the JUWELS Booster supercomputer in Jülich: about 21 000 GPU-hours on 80 A100 processors for the 300-million-parameter model, and about 58 000 GPU-hours on 240 processors for the 600-million version.
- **Gains.** About 8 % better than Prithvi-EO-1.0 on the GEO-Bench suite. Burn scars: 90.5 % mean intersection-over-union (mIoU), against 86.9 % for version 1.0; floods: 90.3 % against 88.3 %.
- **Few labels.** Landslide mapping with about 1 % of the training data, some 50 images: 67.0 % mIoU, against 59.7 % for a U-Net trained from scratch.

Foundation models do not always win. On the PANGAEA benchmark, the TerraMind authors report a mean mIoU of 59.6 % for their large model and 58.4 % for the base model. A U-Net trained for each task scored 57.6 %, and the first Prithvi model 51.0 %: only TerraMind beat the U-Net. Foundation models are most valuable when labels are scarce, tasks change often or data types must be combined.

The same recipe now reaches beyond Earth. **Surya** (August 2025) was pre-trained on about nine years of full-resolution images from the Solar Dynamics Observatory (Figure 44.13). Fine-tuned with LoRA, it segmented active regions with an intersection-over-union of 0.768 and, in preliminary tests, forecast flares about 16 % better than existing benchmarks. The **NASA-IBM**

**Lunar Foundation Model** (September 2026) cut the error of a polar-ice prospectivity map by about 22 % compared with a general vision model. Its developers state that it is not yet validated for operational decisions.

**FIGURE 44.12** Landsat 9 at Vandenberg Space Force Base on 16 August 2021, before encapsulation. Harmonised Landsat and Sentinel-2 data trained the Prithvi Earth-observation models. Image: NASA/KSC (NASA/Randy Beaudoin).

**FIGURE 44.13** The Solar Dynamics Observatory with its arrays deployed in pre-launch tests, 2009. About nine years of its images, some 218 TB, trained the Surya model. Image: NASA/KSC.

LANDMARK: THE FIRST FOUNDATION MODEL IN ORBIT (2025–26)

- Researchers from Adelaide University, ESA Φ-lab, Thales Alenia Space and the SmartSat Cooperative Research Centre compressed Prithvi-EO-2.0-300M for spacecraft processors.
- They used knowledge distillation: training a small “student” network to copy the large “teacher”. It cut the width from 1 024 to 256 and the file from 1.2 GB to 73 MB, about 16 times smaller. Accuracy on cloud detection stayed within about one percentage point, and biomass error within 3 %.
- The compact model ran on the IMAGIN-e payload on the International Space Station, on an ordinary 16-core ARM processor, using an uploaded test set. For South Australia’s Kanyini cubesat, the December 2025 preprint reported only ground tests on an engineering model. In May 2026 NASA reported that the model had also been uploaded to Kanyini. It called it the first geospatial foundation model deployed in orbit.
- Open weights spared the team from training its own model. Onboard hardware and compression are discussed in Chapter 45.

## 44.6 Large Language Models and Agents

A **large language model (LLM)** is a transformer trained to predict the next piece of text. Scaled to billions of parameters and trillions of words, this simple task yields systems that answer questions, summarise documents, write code and follow instructions. In 2025 such models even helped plan a Mars rover drive.

### 44.6.1 How Language Models Are Trained

Text is first split into **tokens**: common words, parts of words or single characters, typically about three-quarters of an English word each. A model with parameters $\theta$ learns to give a high probability to each actual next token. **Pre-training** minimises the average negative log-probability over a huge text corpus:

$$\mathcal{L}(\theta) = -\frac{1}{T}\sum_{t=1}^{T} \log p_\theta(x_t \mid x_1, \dots, x_{t-1}).$$

To predict well, the model must absorb grammar, facts and some reasoning patterns, but it only continues text. **Instruction tuning** fine-tunes it on examples of requests and good answers. In **reinforcement learning from human feedback (RLHF)**, people rank alternative answers, a **reward model** learns their preferences, and the language model is tuned to please it. In a 2022 study, raters preferred a 1.3-billion-parameter model tuned this way to an untuned one over 100 times larger.

The **context window** is the amount of text the model can consider at once. In 2026 leading commercial models accepted about a million tokens, well over a thousand pages, though they use long contexts less reliably than advertised. **Multimodal** models also accept images, diagrams or audio, often turning image patches into tokens with a ViT.

### 44.6.2 Retrieval, Tools and Agents

A language model's knowledge is frozen at training time and may be wrong. **Retrieval-augmented generation (RAG)** helps with both. Documents such as requirements, anomaly reports and standards are split into passages and turned into embeddings. For each question, the nearest passages are retrieved and placed in the prompt, and the model answers from them with citations. NASA and IBM's **INDUS** models (2024), trained on 60 billion tokens of science text, support this kind of search. ESA's **Earth Virtual Expert (EVE)** combines a 24-billion-parameter domain-tuned model with retrieval and a hallucination check; it had about 350 pilot users in March 2026.

**Tool use** lets a model call programs: a calculator, a database, an orbit propagator or a code interpreter. An **agent** wraps the model in a loop — plan, call tools, read the results, repeat — until the goal is reached. In December 2025 JPL engineers let commercial vision–language models plan waypoints for two Perseverance drives, of 210 m and 246 m. Every command was first checked on JPL's digital twin of the rover, which simulates over 500 000 telemetry variables (Figure 44.14).

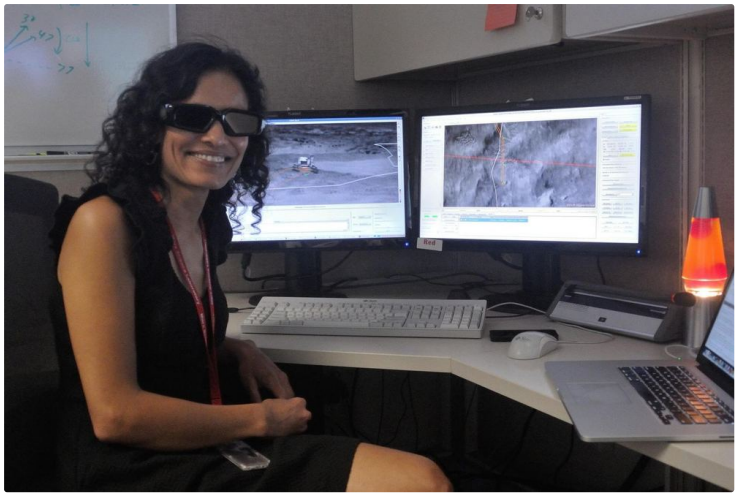


**FIGURE 44.14** A JPL rover driver plans a Curiosity drive, using 3D glasses to see the terrain's shape. In December 2025, vision–language models proposed waypoints for two Perseverance drives; people checked every command on a digital twin. Image: NASA/JPL (NASA/JPL-Caltech).

### 44.6.3 Hallucination

A language model generates fluent text whether or not it knows the answer. A confident, plausible but false statement is called a **hallucination**. Retrieval reduces it but does not remove it. A 2025 study of commercial legal-research tools built on RAG found that they still gave false or unsupported answers to 17–33 % of test questions.

In engineering this is a serious hazard: a wrong torque value or an invented requirement looks exactly like a correct one. Treat language-model output as a draft from an unverified source. Answers should cite sources that people can check, and safety-related changes need normal engineering review. Space uses are discussed in Chapters 49 (operations assistants), 52 (systems engineering) and 53 (assurance).

**EXAMPLE: MEMORY FOR A 7-BILLION-PARAMETER LANGUAGE MODEL**

A typical open model of this size has 32 transformer layers of width 4 096.

1. **Weights.** At 16 bits (2 bytes) per parameter: $7 \times 10^9 \times 2$ = **14 GB**. At 8 bits: **7 GB**. At 4 bits: **3.5 GB**, or about 3.7 GB with one 16-bit scale factor per 64 weights (4.25 bits per weight).
2. **Key–value cache.** During generation each layer stores a key and a value vector for every token: 2 × 32 × 4 096 × 2 bytes = 0.5 MB per token. A 4 096-token context needs **2.1 GB**; 32 768 tokens need **17 GB** — more than the 16-bit weights.
3. **Full fine-tuning.** Adam in mixed precision needs about 16 bytes per parameter for weights, gradients, two moments and a 32-bit master copy: about **112 GB** before activations.
4. **LoRA.** Rank-16 adapters on two projection matrices per layer add $32 \times 2 \times 16 \times (4\,096 + 4\,096)$ = **8.4 million** trainable parameters, 0.12 % of the model. Their optimiser state needs about 0.13 GB.
5. **Conclusion.** With 4-bit weights and LoRA, the model can be fine-tuned on one 24 GB processor and run on a laptop, but it is far too large for most flight computers (Chapter 45).

## 44.7 Physics, Uncertainty and Explanation

Space engineering needs more than a good average score. A model must respect physics, say how sure it is and give reasons that engineers can check.

### 44.7.1 Physics-Informed Networks and Neural Operators

A **physics-informed neural network (PINN)** (2019) adds the governing equations to the loss. If the physics is written as a differential equation $\mathcal{N}[u] = 0$, the network $u_\theta$ is trained on

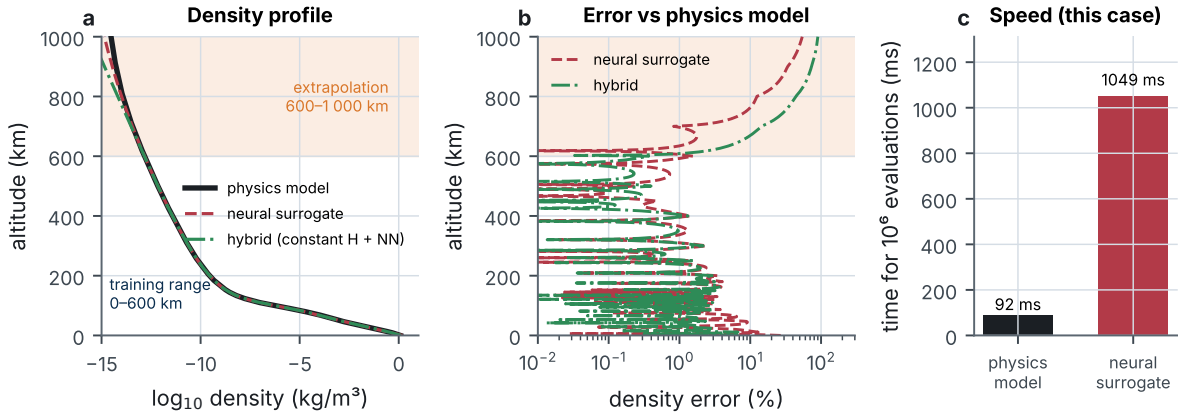
$$\mathcal{L} = \frac{1}{N_d}\sum_{i=1}^{N_d}\left|u_\theta(\mathbf{x}_i) - u_i\right|^2 + \lambda\,\frac{1}{N_r}\sum_{j=1}^{N_r}\left|\mathcal{N}[u_\theta](\mathbf{x}_j)\right|^2.$$

The first term fits the $N_d$ measurements. The second penalises the **residual** of the equations at $N_r$ points, where no data are needed; derivatives come from the same automatic differentiation used in backpropagation. Martin and Schaub's PINN gravity models learn the potential of an irregular asteroid such as Eros and take the acceleration as its gradient, which helps them extrapolate.

A **neural operator** learns a mapping between whole functions, such as from today's weather field to tomorrow's. The **Fourier neural operator (FNO)** (2020) applies a fast Fourier transform, multiplies the lowest frequencies by learned weights $R_\phi$ and transforms back: $v_{l+1} = \sigma\big(W v_l + \mathcal{F}^{-1}[R_\phi \cdot \mathcal{F}(v_l)]\big)$. Working with frequencies, it can be trained at one resolution and applied at another (Chapter 47).

A **surrogate** or **emulator** is a fast network trained to imitate a slow simulation, such as fluid dynamics or a climate model, enabling thousands of design trades (Chapter 52). **Differentiable simulators** compute gradients through the physics itself, so a trajectory or antenna shape can be optimised directly.

In Figure 44.15, a small network learns atmospheric density from 0 to 600 km from a standard-atmosphere model. Within that range its median error is 0.75 %, with a worst case of 27 % at the ground, the edge of the range. Above 600 km, where it has never seen data, the error grows to −12 % at 800 km and −54 % at 1 000 km. A hybrid that adds a learned correction to a constant-scale-height model fits better in range (median 0.57 %, worst case 6.9 %). But it extrapolates even worse, −90 % at 1 000 km, because the real scale height keeps rising as light helium takes over: a hybrid is only as good as its physics. Nor is speed automatic. Here the closed-form physics model is about ten times faster than the network, so surrogates pay off only when the model they replace is expensive.




Physics model: US Standard Atmosphere 1976 below 86 km and exponential segments (Vallado) to 1 000 km. Surrogates: numpy MLP 1-32-32-1 (tanh), 12 000 Adam steps on 601 altitudes, 0–600 km; hybrid = constant-scale-height fit to 500–600 km plus a learned correction. Timings: one CPU core, vectorised.

**FIGURE 44.15** A neural surrogate of atmospheric density, trained in numpy on 0–600 km only. (a) Density from the physics model and two surrogates. (b) Error: below 1 % on average in range, tens of per cent beyond it. (c) Time for a million evaluations: here the physics model is about ten times faster.

### 44.7.2 Uncertainty Quantification

Every prediction has two kinds of uncertainty. **Aleatoric** uncertainty is noise in the data themselves, such as a sensor's random error; more data do not reduce it. **Epistemic** uncertainty is the model's ignorance, largest where training data are sparse or absent; more data do reduce it. A useful model should report both.

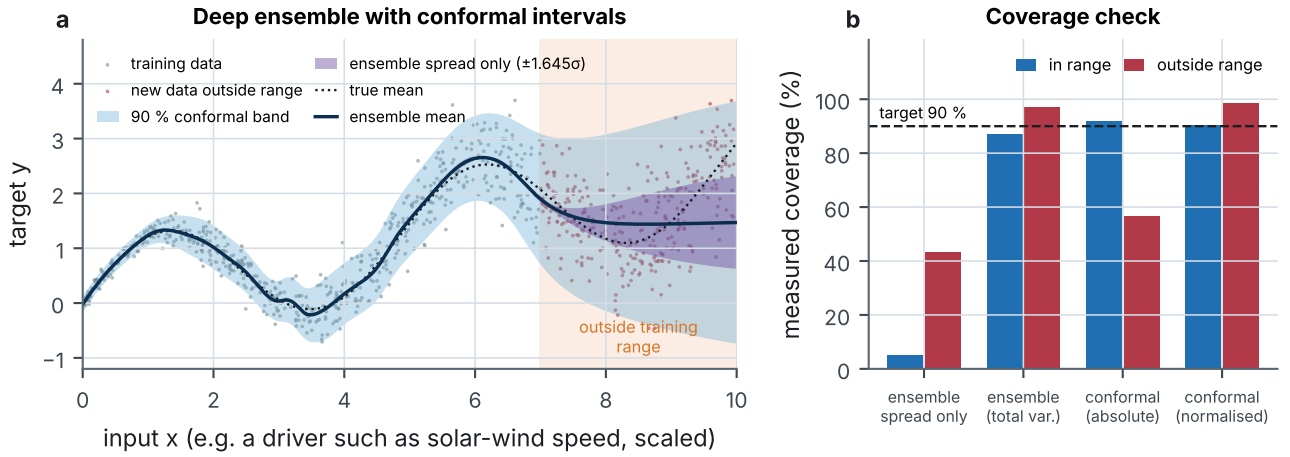
A network can learn aleatoric uncertainty by predicting a mean $\mu(x)$ and a variance $\sigma^2(x)$, trained with the Gaussian negative log-likelihood $\frac{1}{2}\log\sigma^2 + (y-\mu)^2/(2\sigma^2)$. For epistemic uncertainty, the methods in Table 44.6 compare several plausible models. A **deep ensemble** trains several networks from different random starts; their disagreement measures epistemic uncertainty. The total variance is the mean of the members' variances plus the variance of their means.

**TABLE 44.6** Ways to quantify the uncertainty of a neural network.

| Method | Captures | Extra cost | Guarantee | Notes |
|---|---|---|---|---|
| Mean–variance network | Aleatoric | None | None | Overconfident far from data |
| Deep ensemble | Both | 5–10 × training and running | None | Simple, strong baseline |
| Monte Carlo dropout | Mostly epistemic | Many passes at run time | None | Cheap to add; often underestimates |
| Bayesian neural network | Both | High | Only if the prior and approximation are right | Hard to scale |
| Conformal prediction | Total (calibrated) | A calibration set | Coverage if data are exchangeable | Wraps any model |

**Monte Carlo dropout** keeps dropout on at prediction time and treats many passes as an ensemble. **Bayesian neural networks** place probability distributions on the weights. **Conformal prediction** wraps any model in intervals with guaranteed coverage. On a separate calibration set of $n$ points it computes the scores $s_i = |y_i - \hat{y}_i|/\hat{\sigma}_i$. It takes $q$ as the $\lceil (n+1)(1-\alpha) \rceil$-th smallest score and reports $\hat{y} \pm q\hat{\sigma}$. If new data are **exchangeable** with the calibration data, meaning drawn from the same distribution, a fraction of at least $1-\alpha$ of the intervals will contain the truth.

In Figure 44.16, the noise of a regression problem grows across the input range. In range, the spread of five ensemble members alone covers the truth only 5 % of the time: they agree with each other but ignore the noise. The ensemble's total variance gives 87 % coverage at a nominal 90 %; conformal intervals give 91.7 % (absolute) and 90.4 % (normalised). Outside the range, the members disagree, with 40 times more epistemic spread, so the normalised intervals widen and still cover 98 %. The fixed-width band covers only 57 %: its guarantee assumed data like the calibration set.




Synthetic regression y = sin(1.3x) + 0.25x with noise s.d. rising from 0.08 to 0.50. Five numpy MLPs (1-64-64-2, tanh) trained on 600 points with the Gaussian negative log-likelihood; conformal quantiles from 400 calibration points; coverage measured on 4 000 in-range and 2 000 out-of-range test points.

**FIGURE 44.16** A deep ensemble of five numpy networks with conformal prediction. (a) Data, ensemble mean, member spread and a 90 % normalised conformal band. (b) Coverage of nominal 90 % intervals: member spread alone 5 %, total variance 87 %, conformal 91.7 % and 90.4 % in range; outside it, only variance-scaled intervals hold.

In space weather, Licata and Mehta (2022) trained a thermospheric-density model with calibrated uncertainty on US Space Force density data. A 2026 study of flare-intensity forecasts found that conformalised quantile regression gave the best coverage with the narrowest intervals. Calibrated uncertainty lets a conjunction analyst or forecaster decide whether to act (Chapters 47 and 50).

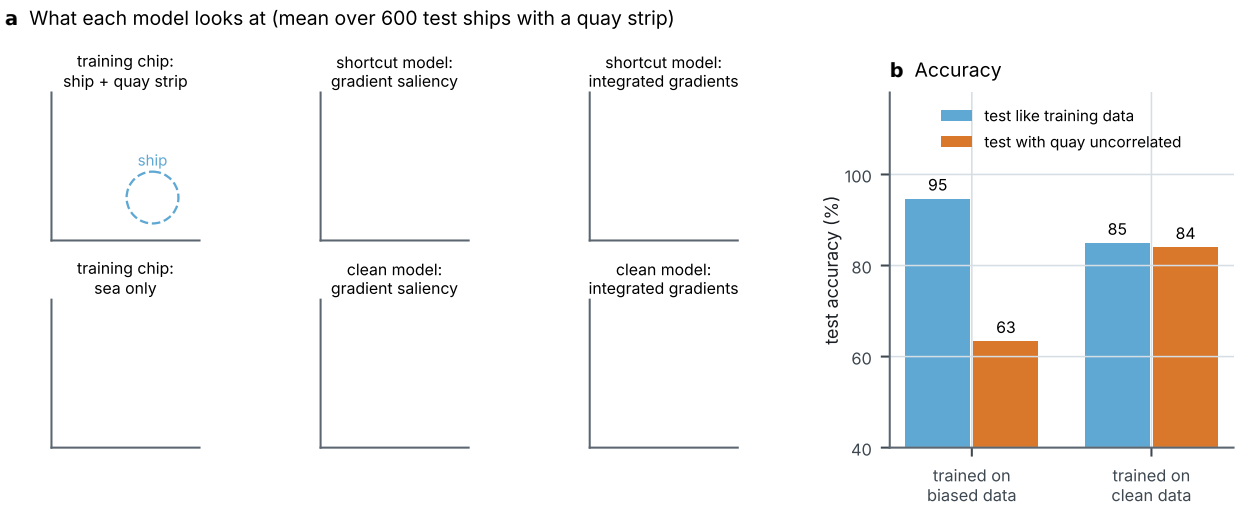
### 44.7.3 Explainability

**Explainable AI** tries to show why a model produced its output. A **saliency map** shows the gradient of the output with respect to each input pixel: where a small change would most alter the answer. **Grad-CAM** (2017) weights the feature maps $A^k$ of the last convolution layer by the average gradient of the class score $y^c$, $\alpha_k^c = \frac{1}{Z}\sum_{i,j} \partial y^c / \partial A_{ij}^k$, and shows $\mathrm{ReLU}\big(\sum_k \alpha_k^c A^k\big)$ as a heat map. **Integrated gradients** (2017) average the gradient along a straight path from a neutral baseline image $\mathbf{x}'$ to the input:

$$\mathrm{IG}_i(\mathbf{x}) = (x_i - x_i') \int_0^1 \frac{\partial F\big(\mathbf{x}' + a(\mathbf{x} - \mathbf{x}')\big)}{\partial x_i}\, da.$$

The attributions add up exactly to $F(\mathbf{x}) - F(\mathbf{x}')$. **SHAP** values (2017) share a prediction among the input features using Shapley values from game theory: each feature's average contribution over all subsets of the others.

Explanations are most useful for catching **shortcuts**, where a model learns an accidental cue instead of the real signal. In Figure 44.17, radar-like chips contain sea clutter and, in half of them, a ship. In the biased training set a bright quay strip along one edge accompanies 95 % of the ships. The trained network scores 95 % on similar test chips but 63 % when the strip appears at random, finding only 28 % of ships without one. Integrated gradients expose the shortcut: the strip, 8 % of the pixels, carries 68 % of the net evidence for "ship". When the strip is replaced by sea, detections fall from 98 % to 26 %. A network trained on clean data scores 84–85 % on both test sets, and its strip carries 1 % of the evidence.




Synthetic 24 × 24-pixel radar-like chips (gamma sea clutter; ships = bright elongated blobs). In the biased training set a bright quay strip on the left edge accompanies 95 % of ships. Numpy MLP 576-64-1 (ReLU), 60 Adam epochs on 6 000 chips. Gradient saliency: mean |∂logit/∂pixel|. Integrated gradients (red = evidence for ship, blue = against): 32 steps from a uniform clutter-mean baseline.

**FIGURE 44.17** Explaining a ship detector that learned a shortcut (synthetic data, numpy networks). (a) Training chips, and what each model uses, averaged over 600 test ships beside a quay strip: gradient saliency looks alike, but integrated gradients show the strip dominating the shortcut model. (b) Its accuracy falls from 95 % to 63 % on unbiased data.

The same figure shows the limits of explanation. The plain gradient maps of the two models look alike, and the strip carries only 7 % of the biased model's gradient: two methods, two answers. A 2018 study found that some popular saliency methods gave almost the same maps for trained and untrained networks. Test explanations, for example by removing the highlighted input and checking that the output changes.

Pandey and colleagues (2023) trained a convolutional network to forecast M-class and larger solar flares from full-disk magnetograms, with a true skill statistic of about 0.51. Guided Grad-CAM showed that it attended to active regions, even near the edge of the disk. Such an explanation supports a model but cannot prove it correct (Chapter 53).

### PRACTICE: FINE-TUNING AN EARTH-OBSERVATION FOUNDATION MODEL WITH FEW LABELS

1. **Define the task and the baseline.** For example, map flood water at 30 m within six hours of a pass. Choose the metric (water-class IoU) and a simple baseline, such as a water-index threshold or a small U-Net.
2. **Choose the model.** Pick an open model pre-trained on data like yours — Prithvi-EO-2.0 for HLS imagery, TerraMind for radar plus optical. Check its licence.
3. **Label a little, well.** Label 50–200 chips from many regions, seasons and event types. Have a second expert relabel 10 % of them to measure label noise.
4. **Split by place and time.** Hold out whole events and regions for validation and testing (Chapter 43). Never let chips from the same scene fall on both sides of a split.
5. **Start with a probe.** Freeze the encoder and train a small segmentation head on its embeddings. This takes minutes and sets a floor.
6. **Then adapt.** Add LoRA adapters (rank 8–16) or fine-tune all weights with a low learning rate ($10^{-5}$–$10^{-4}$), with early stopping on the validation set.
7. **Compare honestly.** Report the foundation model, the probe and the baseline on the locked test set, with confidence intervals. Keep the simplest model that meets the requirement.
8. **Check failure modes.** Inspect errors by surface type: cloud shadow, terrain shadow, urban areas, wet snow. Add labels where errors cluster.
9. **Calibrate and document.** Add conformal intervals, record data, version and limits in a model card, and monitor after deployment (Chapters 45 and 53).

### KEY TAKEAWAYS

- Deep networks stack linear layers and non-linear activations. They train reliably only with careful initialisation, normalisation, residual connections and optimisers such as Adam.
- Scaling laws predict how loss falls with parameters, data and compute; for a fixed budget, model size and data should grow together.
- CNNs exploit weight sharing in images: U-Nets and DeepLab segment, YOLO-type and two-stage networks detect. A small CNN may have under 100 000 weights yet need over a gigaflop per chip.
- Attention cost grows with the square of sequence length, so large images and long records need windowed or sparse attention. Graph networks pass messages on meshes.
- Diffusion models generate by removing noise step by step and underpin ensemble weather forecasts; generated outputs are samples, not measurements.
- Foundation models such as Prithvi, TerraMind, AlphaEarth, Aurora and Surya adapt to new tasks with few labels, but must still beat a simple baseline.
- Language models hallucinate: use retrieval with citations and human review. Check every model for extrapolation, calibration and shortcuts.

### REVIEW QUESTIONS

1. Why can a network without activation functions draw only straight decision boundaries? Why is ReLU preferred to the sigmoid in hidden layers?
2. Using Figure 44.1, explain what goes wrong when the weight variance in a 30-layer ReLU network is 1/n or 4/n, and how He initialisation and layer normalisation fix it.
3. Using $C \approx 6ND$, estimate the training compute of a 1-billion-parameter model trained on 20 billion tokens. Would extra compute be better spent on a bigger model or on more data?
4. Compute the parameters and multiply–accumulate operations of a 3 × 3 convolution with 64 input and 64 output channels on a 128 × 128 feature map. How would they change for a 5 × 5 kernel?

5. Which architecture would you use to count ships, to map flood extent and to outline individual buildings? Explain why.
6. Write the scaled dot-product attention equation and explain the roles of the queries, keys, values and the factor $\sqrt{d_k}$. In Figure 44.7, why does the trained head look back 15 and 31 steps rather than 16 and 32?
7. A transformer must process a 2 048 × 2 048 image in 16 × 16 patches. How many tokens is that, and how much memory do the 16-bit attention scores of 12 heads need per layer? Suggest two ways to reduce it.
8. Write the forward process of a diffusion model and state what the network learns to predict. Why is a super-resolved image not a measurement?
9. Describe masked autoencoding and LoRA. A team has 60 labelled landslide chips: outline how they should use a foundation model and how they would show it beats a U-Net.
10. In Figure 44.16, why does the member spread alone cover only 5 % of test points, and why does the fixed-width conformal band fail outside the training range? What does exchangeability mean for a space-weather model during a record storm?

CHAPTER

# 45 Data, Computing Hardware and Edge AI in Orbit

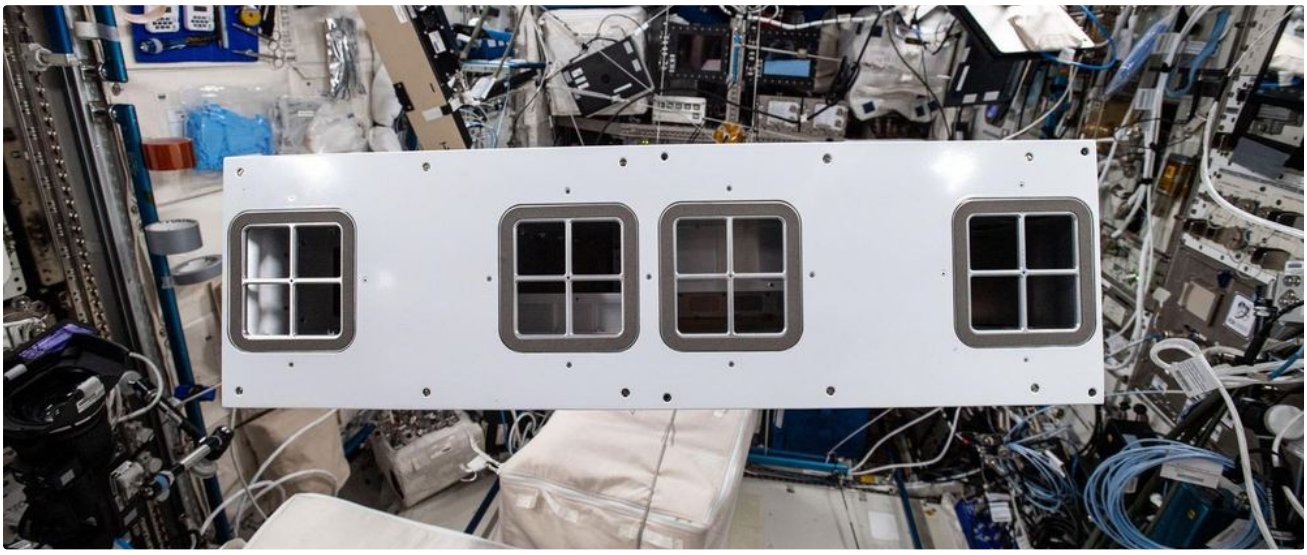


Image: NASA/JSC (Michael Hopkins)

*Every AI system in space rests on two foundations, data and computers. This chapter follows data from sensor to model. It covers petabyte archives, the kinds of space data, how labels are made, and the benchmarks and synthetic data used to train and test models. It explains the machine-learning operations (MLOps) practices that keep models honest in service. It then turns to hardware: why radiation flips bits in neural networks and how to protect them, which processors fly in 2026, and how compression makes models fit. It weighs processing on board against processing on the ground. It ends with updating models in orbit, federated learning across constellations, and the physics and economics of orbital data centres.*

IN THIS CHAPTER

45.1 Data for Space AI
45.2 Benchmarks, Competitions and Synthetic Data
45.3 MLOps: From Archive to Operations
45.4 Radiation and AI Hardware
45.5 Processors for AI in Orbit
45.6 Making Models Fit: Compression
45.7 On Board or on the Ground?
45.8 Updating AI in Orbit
45.9 Computing in Orbit: Orbital Data Centres

A modern Earth-observation satellite collects far more data than its radio link can send home. A model on the ground must wait for the data; a model in orbit must live with little power, intense radiation and a computer that cannot be repaired. Either way, it is only as good as the data it learned from. This chapter is about these two foundations of space AI: **data** and **computing hardware**.

The numbers are large. NASA's Earth science archive held 178.7 PB at the end of September 2025 and grew by about 160 TB a day. Yet in 2020 ESA's Φsat-1 screened its images with a deep neural network running on a vision chip of about 1 W. In 2025 a satellite carrying a data-centre graphics processor (GPU), of a kind that draws several hundred watts, trained a small language model in orbit. The hero image above shows HPE's Spaceborne Computer-2 on the International Space Station (ISS). Launched in 2021, it followed a first HPE unit of 2017 and added a GPU to commercial servers in orbit.

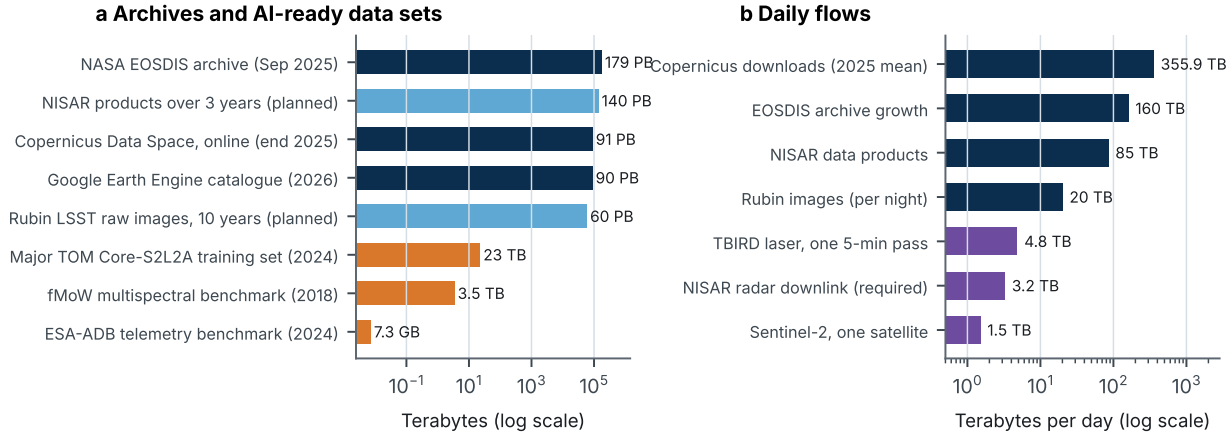
Chapters 43 and 44 explain the learning methods, and Chapters 46–53 apply them and ask how far to trust them. Radiation, flight computers and radio links are introduced in Chapters 2, 6 and 7.

## 45.1 Data for Space AI

The key idea of this section is simple: in space AI, data are plentiful but labels are scarce. Archives hold petabytes, yet a carefully labelled training set is often a few gigabytes.

### 45.1.1 How Much Data?

Archives dwarf the data sets that models are trained on (Figure 45.1). NASA's Earth Observing System Data and Information System (EOSDIS) is the largest, and 116.2 PB of it already sits in commercial cloud storage. The **Copernicus Data Space Ecosystem**, the European Union's archive, held about 91 PB online at the end of 2025, and its users downloaded 129.9 PB directly that year. **NISAR**, the NASA–ISRO radar satellite launched in July 2025, is expected to produce close to 85 TB of products a day and up to about 140 PB in three years. The **Vera C. Rubin Observatory** in Chile, which began its ten-year survey in 2026, records about 20 TB a night. It issues about 10 million alerts a night, each within about 60 seconds.




Sources: NASA Earthdata metrics (FY2025); Copernicus Data Space Ecosystem annual report (2025 data); Google Earth Engine; NASA/JPL NISAR; Rubin Observatory; Major TOM (ESA Phi-lab); fMoW; ESA-ADB (Kotowski et al. 2024); MIT Lincoln Laboratory TBIRD; Airbus (Sentinel-2). In (b) blue bars are ground-side flows, violet bars are what a single spacecraft sends down.

**FIGURE 45.1** Data volumes in the space ecosystem. (a) Archives hold tens to hundreds of petabytes; AI training and benchmark sets are thousands to millions of times smaller. (b) Users download and archives ingest (blue) far more than one spacecraft sends down (violet).

Telemetry — the housekeeping measurements a spacecraft sends about its own health — is smaller but richer in structure. ESA's anomaly-detection benchmark of 2024 holds 1.55 billion data points from two missions, yet only 7.32 GB compressed. Most telemetry has never been labelled.

### 45.1.2 Kinds of Data

Each form of space data needs different models and has different traps (Table 45.1). Optical images resemble photographs, so everyday computer-vision methods transfer well. Radar images do not: they are speckled, complex-valued and taken from the side (Chapter 15). Telemetry, light curves and solar data are time series. Text — anomaly reports, logs and procedures — is the newest data type for space AI (Chapter 49).

**TABLE 45.1** Main types of space data for AI.

| Data type | Examples | Structure | Typical AI tasks | Main traps |
|---|---|---|---|---|
| Optical multispectral | Landsat, Sentinel-2, Planet | 3–13 bands, 0.3–60 m pixels | Land cover, objects, change | Clouds, shadows, haze, sensor differences |
| Hyperspectral | EnMAP, PRISMA, Tanager-1 | 200+ bands per pixel | Minerals, methane, crops | Noisy bands, huge volumes |
| Synthetic aperture radar | Sentinel-1, NISAR, ICEYE | Complex amplitude and phase | Floods, ships, ice, ground motion | Speckle, side-looking geometry |
| Telemetry | Housekeeping channels | Multichannel time series | Anomaly detection, forecasting | Few labelled faults, mode changes |
| Spectra and light curves | Space telescopes, Rubin | 1-D signals over wavelength or time | Classification, transient alerts | Uneven sampling, rare classes |
| Text and logs | Anomaly reports, procedures | Free text | Search, summarising, assistants | Jargon, confidential content |

### 45.1.3 Labels: Where the Truth Comes From

A **label** is the answer a model should learn: “this pixel is water”, “this telemetry interval is an anomaly”. Labels are the most expensive part of most projects, and there are four main ways to obtain them.

**Expert labels** are drawn by trained analysts. They are the most reliable but slow and costly. The So2Sat LCZ42 data set took 15 experts six months, and they judged 85 % of the labels correct. Even experts disagree, so a good project labels a sample twice and measures the disagreement.

**Crowdsourced labels** come from many volunteers or paid workers. They are cheaper but less reliable, so several people label each item and their answers are combined by vote.

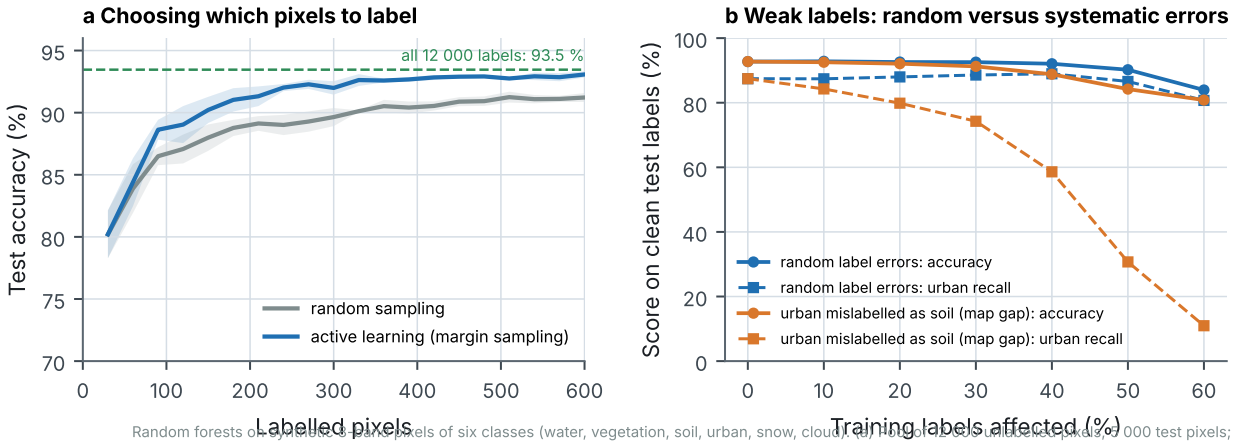
**Weak labels** come from existing sources that were not made for training, such as OpenStreetMap buildings and roads or national land-cover maps. They are cheap and cover huge areas, but they contain errors. Some errors are random. Others are **systematic**: a map may omit new buildings, so the model learns that buildings are bare soil.

**Active learning** lets the model choose which samples experts should label next. The simplest rule, **uncertainty sampling**, labels the samples the current model is least sure about. One measure is the **margin** between the two most likely classes,

$$m(x) = p_{(1)}(x) - p_{(2)}(x),$$

where $p_{(1)}$ and $p_{(2)}$ are the largest and second-largest predicted probabilities. A small margin means the sample lies near a decision boundary, where a new label teaches the most. Another measure is the **entropy** $H(x) = -\sum_k p_k(x) \ln p_k(x)$.

In Figure 45.2 a random forest classifies synthetic eight-band pixels. In panel (a), margin sampling with 210 labels reaches 91.3 % accuracy, while random sampling reaches only 91.2 % with 600 labels. Active learning needed about a third as many labels. In panel (b), random label errors barely matter: the forest still scores 90.2 % with half of all labels wrong, because the errors are spread over five other classes. A systematic map gap is far worse. When 40 % of urban pixels are labelled as soil, urban recall — the share of urban pixels found — falls from 87.4 % to 58.6 %. At 50 % it falls to 30.7 %.




Random forests on synthetic 8-band pixels of six classes (water, vegetation, soil, urban, snow, cloud). (a) Pool of 12 000 unlabelled pixels, 4 000 test pixels; start with 5 labels per class, add 30 per round; margin sampling labels the pixels whose two most likely classes are closest; mean and +/-1 s.d. of 4 runs. (b) 4 000 training pixels; 'random' replaces a label by another class at random; 'map gap' relabels urban pixels as soil, as when buildings are missing from a reference map; mean of 3 runs.

**FIGURE 45.2** Labelling strategies, computed with random forests on synthetic eight-band pixels of six classes. (a) Margin sampling (blue) reaches 91.3 % with 210 labels; random sampling (grey) is still below that with 600. (b) Random label errors (blue) hurt little; turning urban labels into soil (orange), as when a map misses buildings, stops the model finding urban areas.

The lesson is clear. Random noise can be averaged away; systematic omissions cannot. Weak labels should be checked against a small expert-labelled sample before they are trusted.

### 45.1.4 Analysis-Ready, AI-Ready and FAIR Data

**Analysis-ready data (ARD)** are calibrated, geolocated, corrected to surface reflectance and cloud-masked, so that users can analyse them directly (Chapter 13). **AI-ready data** go further. They are cut into consistent patches or time windows, paired with labels, split into training, validation and test sets, and documented. ESA's **Major TOM** collection (2024) is an example. It offers 2 245 886 Sentinel-2 patches of 1 068 × 1 068 pixels covering more than half of Earth's surface; the L2A part alone is about 23 TB.

Good data sets follow the **FAIR principles** of 2016. Data should be **Findable**, with persistent identifiers and searchable metadata, and **Accessible** through open protocols. They should be **Interoperable**, in common formats and vocabularies, and **Reusable**, with clear licences and records of their origin. In Earth observation, the SpatioTemporal Asset Catalog (STAC) and Cloud-Optimized GeoTIFF (COG) formats put these principles into practice.

**Licensing** matters as much as format. Copernicus and Landsat data are free and open for any use. Many benchmark sets allow only non-commercial use; CloudSEN12, for example, carries a CC-BY-NC 4.0 licence. A model may inherit such limits, so the licence of every training set must be recorded before a model is sold or flown.

## 45.2 Benchmarks, Competitions and Synthetic Data

A **benchmark** is a fixed data set with an agreed task and score, so that different methods can be compared fairly. Benchmarks drive progress, but they also mislead when models are tuned to them rather than to the real problem.

### 45.2.1 Benchmark Data Sets

Widely used benchmarks are listed in Table 45.2. Most Earth-observation sets come from the free Sentinel-1 and -2 data or from commercial imagery donated for research. SPEED+ serves spacecraft pose estimation, and the NASA and ESA telemetry sets serve anomaly detection.

**TABLE 45.2** Benchmark data sets for space AI.

| Benchmark | Year | Data | Size | Task |
|---|---|---|---|---|
| EuroSAT | 2019 | Sentinel-2, 13 bands | 27 000 patches | 10 land-cover classes |
| BigEarthNet (v2.0 2024) | 2019 | Sentinel-2 (+ Sentinel-1 in v2.0) | 590 326 patches (v2.0: 549 488 pairs) | Multi-label land cover |
| SEN12MS | 2019 | Sentinel-1, Sentinel-2, MODIS | 180 662 triplets | Land cover |
| So2Sat LCZ42 | 2020 | Sentinel-1 and -2 | 400 673 patch pairs | 17 local climate zones |
| SpaceNet 1–8 | 2016–22 | Commercial optical (0.3–4 m) and radar | SpaceNet 8: 850 km² | Buildings, roads, floods |
| xView | 2018 | WorldView-3, 0.3 m | > 1 million objects, 1 400 km² | 60 object classes |
| fMoW | 2018 | High-resolution multispectral | 1 047 691 images | 63 categories of land use |
| xBD | 2019 | Before and after disasters | 850 736 buildings, 19 disasters | Damage level |
| CloudSEN12 | 2022 | Sentinel-2 and -1 | 49 400 patches | Cloud and shadow masks |
| SPEED+ | 2021 | Rendered and laboratory images | 59 960 + 9 531 images | Spacecraft pose |
| SMAP/MSL (Telemanom) | 2018 | Spacecraft telemetry | 82 channels, 105 anomalies | Anomaly detection |
| ESA-ADB | 2024 | Telemetry of two missions | 1.55 billion points, 844 events | Anomaly detection |

Three traps recur. **Leakage**: neighbouring patches in training and test sets look alike, so random splits overstate accuracy (Chapter 43); BigEarthNet v2.0 therefore splits by geography. **Saturation**: once top scores pass 98 %, a benchmark no longer separates good methods from great ones. **Label quality**: errors in a benchmark's own labels cap the score any honest model can reach.

### 45.2.2 Competitions and Research Sprints

ESA's **Kelvins** platform has run open competitions on real mission problems since 2016. Topics have included Mars Express power, PROBA-V super-resolution, collision avoidance and spacecraft pose. In the **OPS-SAT case** (July–October 2022), teams built a small, quantisation-aware model from few labelled images to run on board OPS-SAT.

NASA's **Frontier Development Lab (FDL)** was created in 2016 by NASA, the SETI Institute and Trillium Technologies. It brings machine-learning researchers and space scientists together for research sprints of eight to ten weeks; a European version has run with ESA in Oxford. Sprints produce prototypes, and turning one into an operational system takes years.

### 45.2.3 Synthetic Data and the Sim-to-Real Gap

Some data cannot be collected in advance: nobody has thousands of labelled photographs of a particular satellite tumbling in orbit. Engineers therefore **render** synthetic images with computer graphics, or generate synthetic telemetry from a **digital twin** — a simulation detailed enough to stand in for the spacecraft.

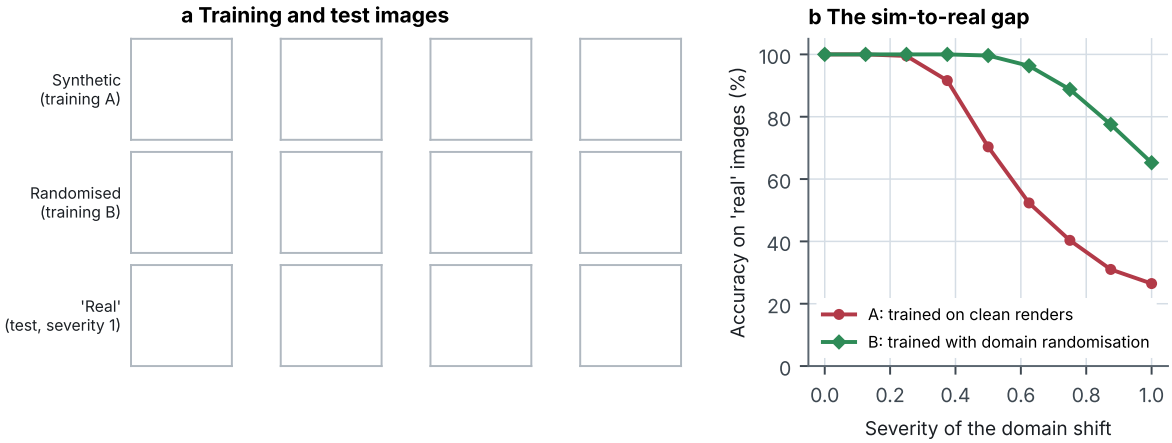
The danger is the **sim-to-real gap**: a model that excels on synthetic data fails on real data, because rendering never captures every detail of lighting, materials, optics and noise. SPEED+ measures this gap. A pose network trained on its rendered images scored 0.16 on synthetic validation images (lower is better). It scored 1.21 on laboratory images lit by a lightbox and 1.73 under a lamp that mimics the Sun — eight to eleven times worse.

**Domain randomisation** is the main defence. Instead of rendering the most realistic images possible, the engineer randomises everything that should not matter — lighting, background, texture, blur, noise, gain. The model then learns the one thing that stays constant, the object itself. Formally, training minimises the expected loss over the randomisation parameters $\xi$:

$$\min_{\theta} \; \mathbb{E}_{\xi \sim p(\xi)} \Big[ L\big( f_\theta(\mathrm{render}(s, \xi)),\, y(s) \big) \Big],$$

where $s$ is the scene (here, the spacecraft's pose), $y(s)$ its label and $f_\theta$ the network. With luck, the real world is just one more random draw.

A toy renderer shows the effect (Figure 45.3). A numpy network trained on clean renders identifies a spacecraft's orientation perfectly on similar images. On "real" images with a different Sun angle, an Earth background, blur and noise, its accuracy falls to 70.3 % at half severity and 26.5 % at full severity. The same network trained on randomised renders keeps 99.6 % and 65.2 %. Randomisation narrows the gap but does not close it. The winners of ESA's 2021 pose challenge also used **domain adaptation**, which trains the network so that its internal features look the same for synthetic and real images.




Toy renderer: 20 × 20-pixel images of a T-shaped spacecraft in 8 orientation classes (45 deg apart, +/-11 deg jitter). Numpy network 400-96-8 trained on 8 000 images per set. A: Sun fixed, black sky, no noise. B: random Sun angle, random gradient and blob backgrounds, blur, gain and noise. 'Real' test images (3 000 per point): Sun angle spread, textured Earth-limb background, blur, gain error and noise all grow with severity (0 = identical to A).

**FIGURE 45.3** The sim-to-real gap, computed with a numpy network on 20 × 20-pixel renders of a toy spacecraft in 8 orientations. (a) Clean training images (A), randomised training images (B) and “real” test images at full severity. (b) Accuracy on 3 000 “real” images as the shift grows; chance is 12.5 %.

## 45.3 MLOps: From Archive to Operations

**MLOps** (machine-learning operations) is the set of practices that take a model into reliable service and keep it there. The key idea is that a model is never finished. It is one stage in a loop of data, training, testing, deployment and monitoring that runs for the life of the mission.

### 45.3.1 The End-to-End Pipeline

The loop for an Earth-observation mission is drawn in Figure 45.4. Data flow from the sensor through the ground stations into processing and the archive. Labels are attached, and a frozen **snapshot** of training data is taken. Models are trained, registered and tested against validation gates. A model that passes is deployed on the ground or, after further checks, uplinked. Monitoring watches for drift and triggers retraining. The data shrink at each step, from terabytes a day to a model of megabytes.

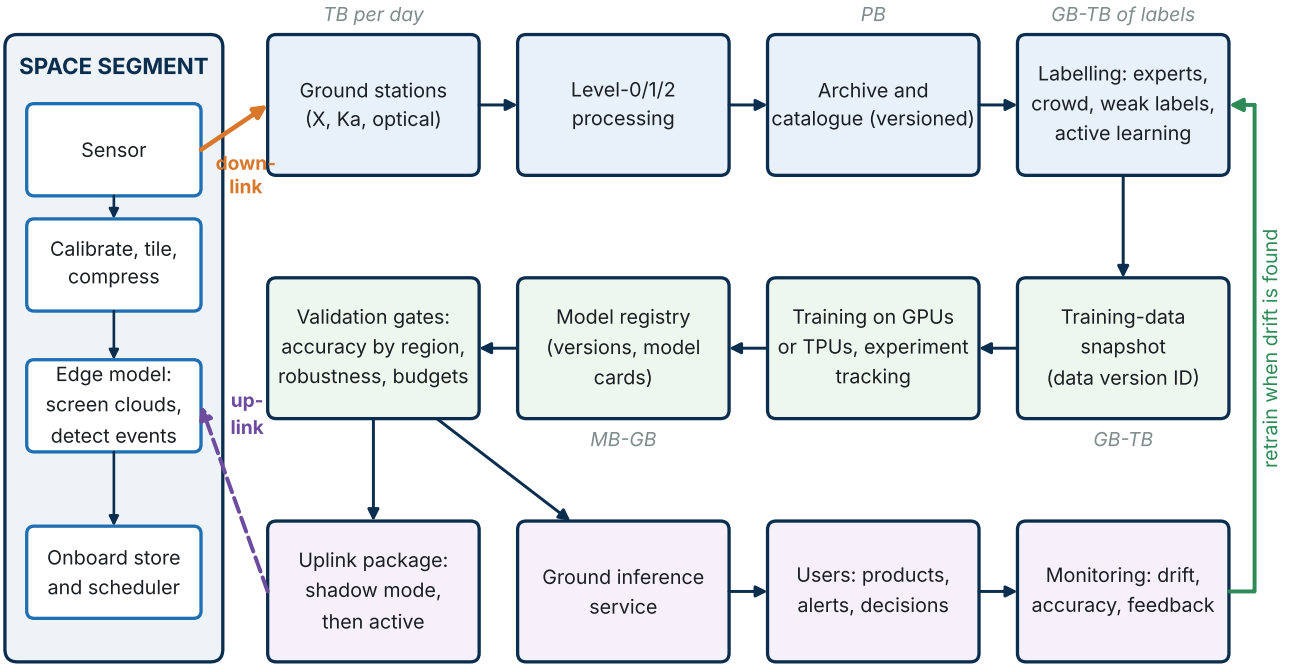



**FIGURE 45.4** An end-to-end MLOps pipeline from satellite to model to user. Volumes shrink from terabytes a day at the downlink to megabytes for the model. Validation gates come before deployment on the ground or in orbit; monitoring triggers retraining when the data drift.

## 45.3.2 Versions, Registries and Validation Gates

Four habits make a pipeline trustworthy.

1. **Version the data.** Each training snapshot gets a unique identifier, often a hash of its contents, so that any model can be traced to the exact data it saw.
2. **Track experiments.** Every training run records its code version, settings, random seeds and results, so that it can be reproduced.
3. **Register models.** A **model registry** stores each released model with its version, data identifier and test results. It adds a **model card**: a short note on what the model is for, how it was tested and where it fails.
4. **Gate releases.** A **validation gate** is a test the model must pass before release. It checks accuracy by region, season and surface type, and robustness to noise and calibration changes. It also checks memory, time and power on the flight processor, and **bit-exact** agreement with the ground reference.

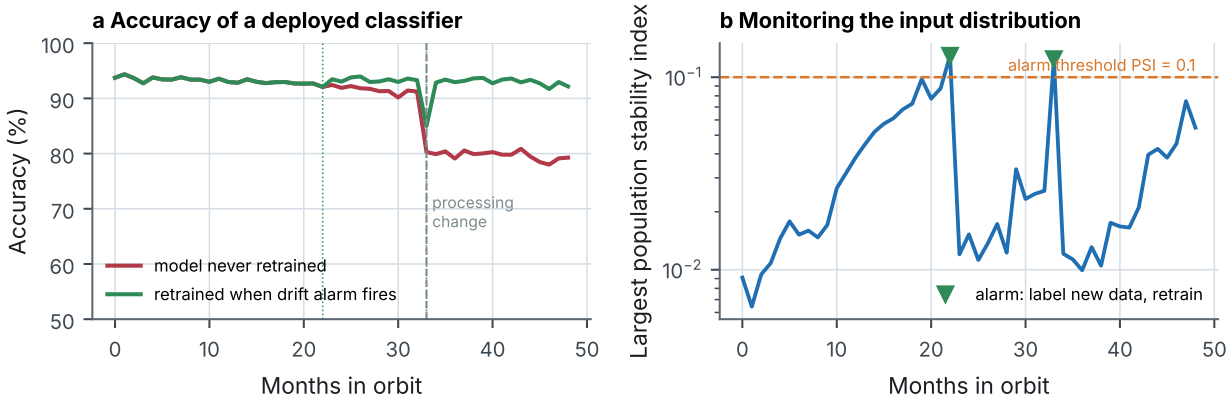
## 45.3.3 Monitoring for Drift

After launch the world changes. Detectors age, calibration is updated, and seasons bring snow and smoke. This is **drift**: the data no longer match the data the model learned from. Labels arrive slowly in operations, so drift is usually detected first from the inputs. A common statistic is the **population stability index (PSI)**. For each bin $i$ of a variable, it compares the share $p_i$ of training data with the share $q_i$ of current data.

$$\mathrm{PSI} = \sum_i (q_i - p_i)\,\ln\frac{q_i}{p_i}.$$

Values below 0.1 are usually taken as stable, and values above 0.25 signal a large shift.

In Figure 45.5 a deployed classifier watches a detector that loses 0.1–0.6 % of its gain per month. At month 33, a ground-processing change raises two short-wave infrared bands by 20 %. Without retraining, accuracy slides from 93.7 % to about 90–91 % by month 32, then drops to 80.3 %. With a PSI alarm at 0.1, the monitor fires at months 22 and 33. Each time the model is retrained on 2 000 newly labelled pixels and recovers to about 92–93 %.




Random forest (80 trees) trained on 4 000 synthetic 8-band pixels at launch; each month 3 000 new pixels are classified. The detector loses 0.1-0.6 % of its gain per month (fastest in the blue band); at month 33 a processing change raises the two short-wave infrared bands by 20 %. PSI: population stability index of each band against the training data (10 quantile bins); after an alarm the model is retrained on 2 000 newly labelled pixels, which become the new reference.

**FIGURE 45.5** Drift monitoring, computed with a random forest on synthetic pixels over 48 months. (a) A model never retrained (red) loses accuracy as the detector ages and after a processing change; one retrained at each alarm (green) recovers. (b) The largest population stability index across the bands, with the threshold and alarms.

### 45.3.4 Cloud Platforms and Ground Compute

Most training and much inference happens on the ground, close to the archives (Table 45.3). Users send their code to the data instead of downloading petabytes. Chapter 13 introduces these platforms; their status in October 2026 is summarised here.

**TABLE 45.3** Cloud platforms for Earth-observation AI.

| Platform | Operator | What it offers | Status, October 2026 |
|---|---|---|---|
| Google Earth Engine | Google | Catalogue of 80–90 PB and more than 1 000 data sets; analysis on Google's servers | Free for research; commercial use through Google Cloud |
| Copernicus Data Space Ecosystem | European Commission, ESA | About 91 PB online; all Sentinel data; processing services | Opened January 2023; replaced the old Open Access Hub (closed November 2023); 542 622 users at end of 2025 |
| NASA Earthdata Cloud | NASA, on Amazon Web Services | 116.2 PB of the 178.7 PB EOSDIS archive | Operational; serves NISAR products |
| AWS Open Data | Amazon | Free hosting of public data sets, including Landsat, Sentinel-2 and SpaceNet | Operational |
| Microsoft Planetary Computer | Microsoft | Open-data catalogue with a STAC search interface; paid "Pro" service for private data | Free computing hub retired June 2024; catalogue continues |

Training large models also needs clusters of GPUs or tensor processing units (TPUs), in the cloud or in agency supercomputers (Figure 45.6). A foundation model may occupy hundreds of accelerators for weeks, so compute time is budgeted like ground-station time.

**FIGURE 45.6** The Discover supercomputer at the NASA Center for Climate Simulation, Goddard Space Flight Center. Ground clusters and cloud services do most of the training for space AI. Image: NASA/GSFC.

## 45.4 Radiation and AI Hardware

The key idea of this section is that a neural network is a large block of numbers held in memory, and radiation changes numbers. Some changes are harmless; a few are catastrophic. Good design finds and protects the few.

### 45.4.1 How Radiation Disturbs Computers

Chapter 2 explains radiation and its effects; here is a short reminder. **Total ionising dose (TID)** slowly degrades transistors as charge builds up in their oxides; it is measured in krad(Si). **Displacement damage** knocks atoms out of the crystal lattice and mainly harms detectors and solar cells. **Single-event effects** are caused by one energetic particle. A **single-event upset (SEU)** flips a stored bit. A **single-event functional interrupt (SEFI)** hangs a processor until it is reset. A **single-event latch-up (SEL)** creates a short circuit that can destroy the chip unless the power is cut quickly.

The upset rate of a memory comes from folding its measured **cross-section** $\sigma(L)$, the sensitive area per bit, with the flux of particles at each linear energy transfer $L$.

$$R = \int \sigma(L)\,\frac{d\Phi}{dL}\,dL \quad \text{upsets per bit per second.}$$

Measured rates give a feel for the result. Static memory (SRAM) on two Chinese internet satellites at 1 050–1 425 km recorded about $7.6 \times 10^{-7}$ upsets per bit per day, mostly over the South Atlantic Anomaly. At that rate, a model of 3.4 million 8-bit weights (27.2 Mbit) would collect about 21 upsets a day; stored as 32-bit floating-point numbers, about 83.

### 45.4.2 Why Neural Networks Are Vulnerable

A network's **weights** are its learned parameters, stored in memory for the whole mission. Its **activations** are intermediate results, recomputed for every input. An upset in an activation spoils one result. An upset in a weight spoils every result until the memory is repaired.

Not all bits are equal. A 32-bit floating-point number has 1 sign bit, 8 exponent bits and 23 fraction bits. Flipping a low fraction bit changes a weight slightly. Flipping the highest exponent bit can multiply it by about $2^{128}$, or $10^{38}$, and one such weight turns the network's output into garbage. One flip changes an 8-bit integer weight by at most 128 quantisation steps, so its errors are bounded.

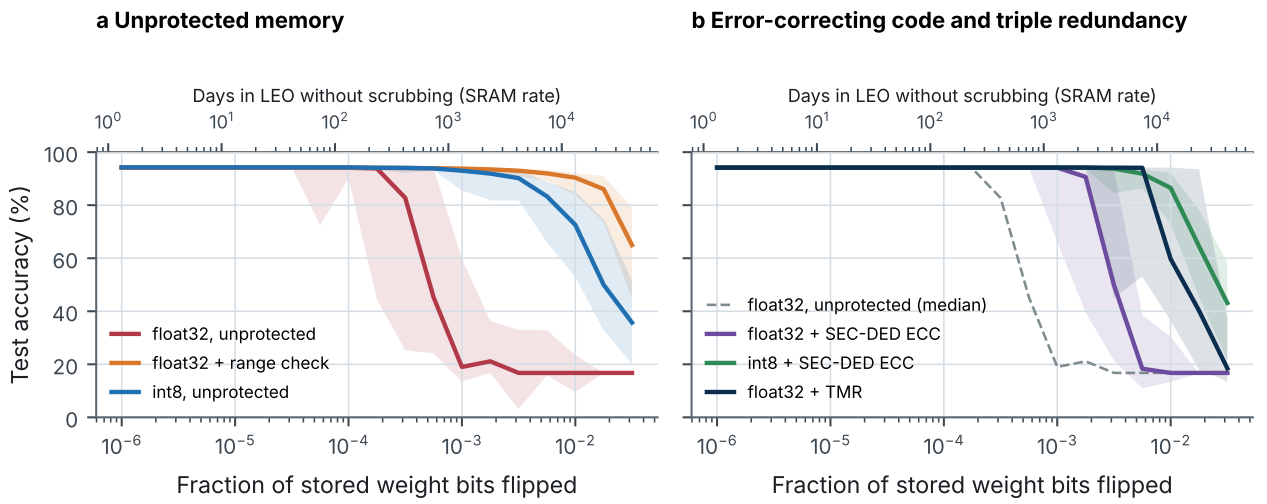
A flip in control logic, program code or a register can hang the processor or corrupt a whole layer. Errors that go unnoticed are called **silent data corruption (SDC)**. They are more dangerous than crashes, which at least announce themselves.

### 45.4.3 Mitigation

Engineers combine several defences, from hardware to algorithms.

- **Triple modular redundancy (TMR)** keeps three copies and takes a majority vote. If each copy works with probability $R$, the voted system works with probability $R_{\mathrm{TMR}} = 3R^2 - 2R^3$, better than $R$ whenever $R > 0.5$.
- **Error-correcting codes (ECC)** add check bits to each memory word. The common **SEC-DED** Hamming code adds 8 bits to each 64-bit word (12.5 % overhead). It corrects any single-bit error and detects any double error.
- **Scrubbing** reads and rewrites memory regularly, correcting single errors before a second one lands in the same word. With $n$ bits per word and upset probability $p$ per bit between scrubs, the chance of two errors in a word is about $\binom{n}{2} p^2$. Halving the scrub interval quarters it.
- **Watchdogs** restart a processor that stops responding, and current limiters cut power within microseconds of a latch-up.
- **Selective hardening** protects only the most sensitive parts: the exponent bits of weights, the first and last layers, or the control code. A cheap software version is a **range check** that clamps any weight larger than its known maximum.
- **Algorithm-based fault tolerance (ABFT)** checks the arithmetic. For a matrix product $C = AB$, the column sums must satisfy $\mathbf{1}^{\top} C = (\mathbf{1}^{\top} A)\, B$; checking this is cheap.
- **Redundant inference** runs the model twice, or on two processors, and compares the answers.

The fault-injection experiment in Figure 45.7 tests these defences on a small network of 4 992 weights. Random bit flips are injected into its stored weights. Unprotected float32 weights are the weakest. At a bit-error rate of $10^{-4}$, the median accuracy is still 94.1 %, but one trial in ten falls to 90.0 % or less. At $10^{-3}$ the median collapses to 19.0 %, because almost every trial hits a high exponent bit. A simple range check keeps 93.8 % at $10^{-3}$, and int8 storage keeps 93.0 % with no protection at all. SEC-DED ECC without scrubbing keeps the float32 median at full accuracy up to $10^{-3}$, but fails at $10^{-2}$, when many words collect two errors. Int8 with ECC keeps a median of 86.5 % at $10^{-2}$, and TMR keeps nine trials in ten at full accuracy up to $10^{-3}$. On the top axis, $10^{-4}$ equals about 130 days in LEO without scrubbing.




Numpy network 8-64-64-6 (4 992 weights) trained on 20 000 synthetic 8-band pixels of six classes; 2 000 test pixels; 40 random trials per point (line: median; band: 10th-90th percentile). Flips land uniformly on all stored weight bits (biases protected). Range check: weights non-finite or > 2x the largest original weight are set to zero. ECC: SEC-DED on 64-bit words, multi-bit words left uncorrected. Top axis: 7.6e-7 upsets per bit per day, measured in SRAM on LEO satellites.

**FIGURE 45.7** Fault injection into a numpy network (8–64–64–6) trained on synthetic pixels; 40 trials per point. (a) Unprotected float32 weights fail once a high exponent bit flips; a range check or int8 storage removes most of the damage. (b) SEC-DED ECC on 64-bit words (no scrubbing) and triple modular redundancy; grey dashes repeat unprotected float32.

The experiment points to a practical recipe. Store weights as integers. Protect them with ECC and scrub often. Add a range check. Keep a protected golden copy of the model to reload when the ECC reports an uncorrectable error.

### 45.4.4 Radiation Testing

No calculation replaces a test. **TID tests** expose parts to a cobalt-60 gamma source (ESCC 22900, MIL-STD-883 Method 1019). **Single-event tests** fire heavy ions or protons from an accelerator at a working device and count upsets, interrupts and latch-ups (ESCC 25100, JESD57). For AI processors, the best test runs the real model during exposure and checks every output, which measures silent data corruption directly.

Several AI processors have been tested this way. Intel's Myriad 2 vision chip passed a 36-hour beam test at CERN in late 2018 and two later tests without modification, which cleared it for Φsat-1. Heavy-ion tests at GANIL in France on NVIDIA Jetson Orin NX and Xavier NX modules found that about 80 % of events were correctable upsets, with no destructive latch-up. The HiMassSEE experiment on the ISS measured the environment that causes such events (Figure 45.8). Tested commercial chips have even flown on Mars (Figure 45.9).

**DATA BOX — RADIATION TESTING OF TPUS FOR PROJECT SUNCATCHER**

Google's Suncatcher study (November 2025) tested Trillium (v6e) TPUs with 67 MeV protons for an 81-satellite cluster at about 650 km.

- Dose behind 10 mm of aluminium: about 150 rad(Si) a year, 750 rad(Si) in five years.
- Memory irregularities began after 2 krad(Si); one chip had no hard failure up to 15 krad(Si).
- Silent data corruption: about one event per 14.4–20 rad, or one bad result in three million at one inference per second.
- Uncorrectable memory errors: about one per 44 rad; functional interrupts about one per 5 krad per chip.

**FIGURE 45.8** The HiMassSEE detector kit in the ISS's US Laboratory in August 2012. It measured how heavy shielding changes the radiation that causes single-event effects. Image: NASA/JSC.

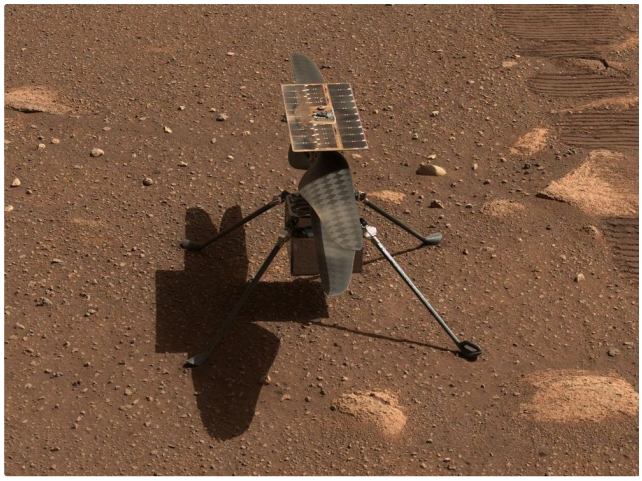


**FIGURE 45.9** Ingenuity on Mars, photographed by Perseverance's Mastcam-Z on 5 April 2021. Its navigation ran on a Snapdragon 801 smartphone processor. Image: NASA/JPL (NASA/JPL-Caltech/ASU).

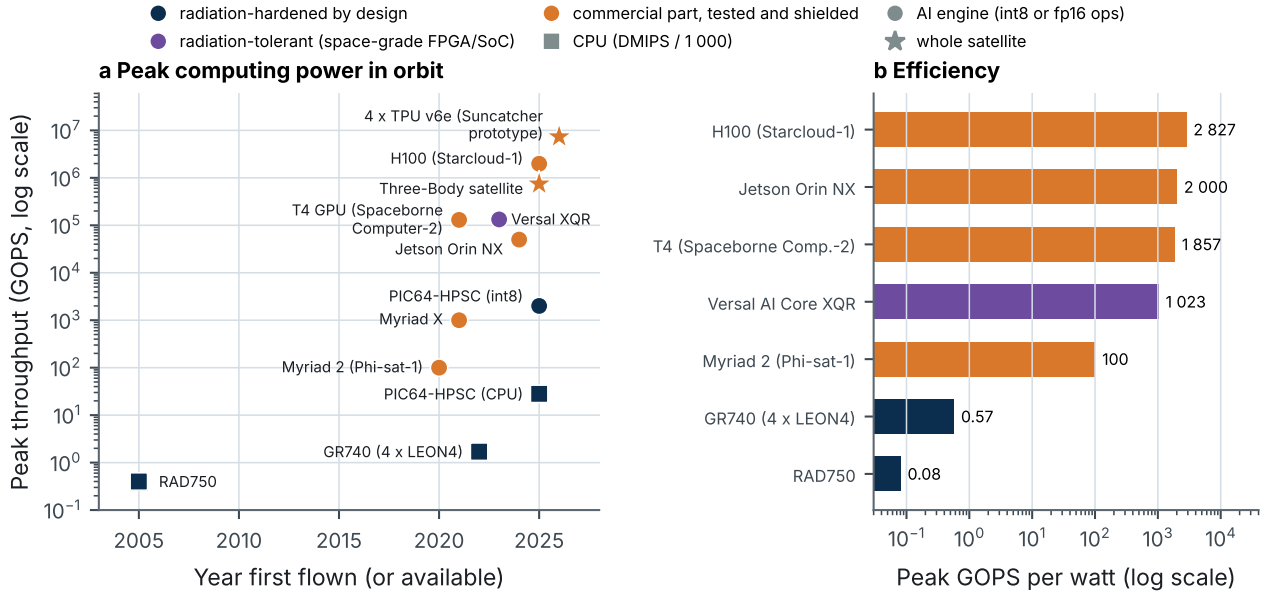
## 45.5 Processors for AI in Orbit

Space processors fall into three families. **Radiation-hardened** processors are designed and built to survive large doses and to be immune to latch-up. They are reliable but lag commercial chips by one to two decades. **Radiation-tolerant** parts, such as space-grade field-programmable gate arrays (FPGAs) and adaptive systems-on-chip, use commercial technology with careful design and testing. **Commercial** processors — vision chips, GPUs and TPUs — are screened, tested, shielded and supervised. Table 45.4 lists examples, and Figure 45.10 plots their peak throughput and efficiency.

**TABLE 45.4** Processors for AI in space (vendor peak figures). DMIPS: millions of Dhrystone instructions per second, a CPU benchmark; TOPS: trillions of int8 operations per second (dense).

| Processor | In space | Peak throughput | Power | Radiation approach | Example use |
|---|---|---|---|---|---|
| RAD750 (BAE Systems) | 2005 | 400 DMIPS | 5 W (CPU) | Hardened, 0.2–1 Mrad | Perseverance, JWST |
| GR740 (Gaisler), 4 × LEON4 | 2022 | 1 700 DMIPS | < 3 W (core) | Hardened, 300 krad | Rashid lunar rover |
| PIC64-HPSC (Microchip) | 2025 (samples) | 28 130 DMIPS; 2 TOPS | Not published | Hardened, 100 krad | In test; not flown |
| Versal XQRVC1902 (AMD) | 2023 (shipping) | 133 TOPS | Up to ~130 W | Tolerant; scrubbing | Payload processing |
| Myriad 2 (Intel) | 2020 | > 100 GFLOPS | About 1 W | Commercial; tested | Φsat-1 |
| Myriad X (Intel) | 2021 | 1 TOPS | As Myriad 2 | Commercial | SpaceCloud, CogniSAT-6 |
| Snapdragon 801 (Qualcomm) | 2021 | 4 × 2.26 GHz | Not published | Commercial; tested | Ingenuity |
| T4 GPU (NVIDIA) | 2021 | 130 TOPS | 70 W | Commercial; inside ISS | Spaceborne Computer-2 |

| Processor | In space | Peak throughput | Power | Radiation approach | Example use |
|---|---|---|---|---|---|
| SpaceCloud iX5, iX10 (Unibap) | 2021 (iX5) | iX10: CPU, GPU, Myriad X, FPGA | Tens of watts (iX10) | Commercial parts | D-Orbit ION carriers |
| Jetson Orin NX (NVIDIA) | 2024 | 50 TOPS (100 sparse) | 10–25 W | Commercial; tested | Aethero cubesat |
| H100 GPU (NVIDIA) | 2025 | ~2 000 TOPS | Up to 700 W | Commercial | Starcloud-1 |
| TPU v6e (Google) | 2026 | 1 836 TOPS | Not published | Commercial; tested | Suncatcher (4 chips) |
| Three-Body satellite (China) | 2025 | 744 TOPS | Not published | Not published | 12 satellites |




Vendor peak figures from the processor table: CPUs in millions of Dhrystone instructions per second (DMIPS / 1 000); AI engines in int8 operations (Myriad 2: fp16), dense, without sparsity. H100: SXM version (700 W); the variant flown was not disclosed. Versal power: about 130 W when fully used (commercial part). Parts without published power are omitted from (b). Real applications reach 10-50 % of peak.

**FIGURE 45.10** Processors for AI in orbit, from Table 45.4. (a) Peak throughput rose about seven orders of magnitude from the RAD750 to the H100. (b) Peak operations per watt: the best commercial AI engines reach 2 000–2 800 GOPS/W, the RAD750 0.08 — though CPU instructions and int8 operations are not directly comparable.

The comparison needs care. A CPU instruction does more varied work than an 8-bit multiply–add, vendors quote peaks, and real networks reach perhaps 10–50 % of them. Even so, the gap is real. The RAD750 delivers about 0.08 GOPS per watt; Myriad 2 about 100; a Jetson Orin NX about 2 000; an H100 about 2 800.

NASA's **High-Performance Spaceflight Computing (HPSC)** project aims to narrow the gap for hardened parts. Microchip's PIC64-HPSC, announced in July 2024, has eight 64-bit RISC-V cores with vector units. NASA's figures give it about 1 343 times the speed of the GR740 on its SpaceBench benchmark at 1 GHz. As of mid-2026, its chips were in radiation, thermal and shock testing; none had flown.

**FPGAs** are chips whose logic can be rewired after manufacture, which suits high-rate payload processing. AMD's space-grade Versal AI Core adds 400 small AI engines to an FPGA. It completed Class B qualification in December 2022 and has shipped since early 2023. A lidless version aiming at the stricter Class Y is sampling in 2026, with flight units planned for 2027.

Commercial chips dominate new demonstrations because they are thousands of times more efficient. Planet's Pelican-4, for example, used a Jetson Orin module to detect aircraft in its own images on 25 March 2026. Unibap's SpaceCloud computers have hosted customers' applications on D-Orbit's ION carriers since June 2021. The price is risk. The Ingenuity helicopter flew 72 times on Mars between April 2021 and January 2024 on a Snapdragon 801 phone processor (Figure 45.9). Its team accepted that the chip was less robust than flight parts, and two small microcontrollers ran the time-critical flight control.

> **ΦSAT-1: DEEP LEARNING SCREENS IMAGES IN ORBIT**
>
> Φsat-1 was an ESA cubesat. It was launched on 2/3 September 2020 on Vega flight VV16 into a sun-synchronous orbit at about 530 km. Its HyperScout-2 hyperspectral camera fed an Intel Myriad 2 vision chip of about 1 W. On it ran CloudScout, a small convolutional network developed with the University of Pisa. It kept images less than 70 % cloudy for downlink. Because no real images yet existed, it was trained on Sentinel-2 images processed to imitate the new camera. On test images it was 92 % accurate with 1 % false positives, taking about a third of a second per image at about 2 W. ESA estimated that discarding cloudy images could save about 30 % of the downlink. Φsat-1 showed that a commercial AI chip could survive and work in orbit.

## 45.6 Making Models Fit: Compression

The key idea of compression is that most networks are far larger and more precise than they need to be. Removing the excess saves memory, uplink time, energy and radiation exposure, often with almost no loss of accuracy.

### 45.6.1 Quantisation

**Quantisation** stores weights and activations as integers with fewer bits. In symmetric quantisation with $b$ bits, a scale factor

$$s = \frac{\max |w|}{2^{\,b-1} - 1}$$

maps each weight $w$ to the integer $q = \mathrm{round}(w/s)$, clipped to $\pm(2^{b-1} - 1)$. The network then uses $\hat{w} = s\,q$. The rounding error is roughly uniform between $\pm s/2$, so its variance is $s^2/12$. Each extra bit halves $s$ and improves the signal-to-noise ratio by about 6 dB. One scale per tensor is simplest; one scale per output neuron or channel fits uneven weights better.

**Post-training quantisation (PTQ)** converts a trained network directly. **Quantisation-aware training (QAT)** fine-tunes the network with quantised weights in the forward pass, so that it learns to tolerate rounding. Rounding has no useful gradient. QAT therefore uses the **straight-through estimator**, which passes the gradient through the rounding step as if it were absent.

These methods were applied to the 5 126-parameter network of Figure 45.7 (Figure 45.11). At 8 and 6 bits, PTQ loses nothing measurable: 94.2–94.3 % in every case. At 4 bits it loses about one point (93.2 % per tensor, 93.7 % per neuron). At 2 bits — weights of −1, 0 or +1 times a scale — PTQ collapses to 16.7 % per tensor and 51.5 % per neuron, while QAT still reaches 93.2 %. Model size falls from 20.0 kB to 5.9 kB at 8 bits and 2.3 kB at 2 bits; the 32-bit biases and scales set a floor.

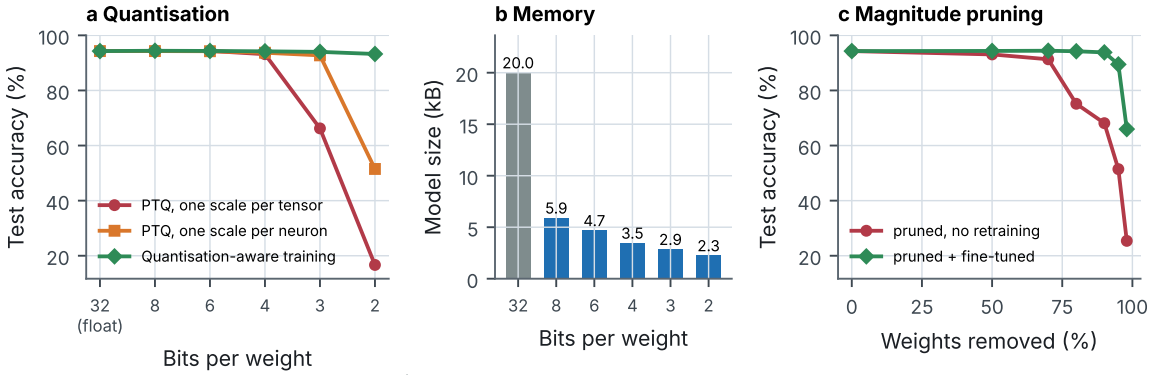
**EXAMPLE: SIZING AN ONBOARD CLOUD SCREEN**

A cloud screen the size of MobileNetV2 has 3.4 million parameters and needs about 0.30 billion multiply–accumulates (MACs) per 224 × 224-pixel tile. The processor draws 5 W with a peak of 1 TOPS in int8; assume 30 % of peak.

1. **Model size.** In float32: $3.4 \times 10^6 \times 4$ bytes = 13.6 MB. In int8: 3.4 MB, four times smaller.
2. **Uplink time.** At 256 kbit/s: $3.4 \times 8 \times 10^6 / 256\,000 \approx 106$ s for int8 (one pass), 425 s for float32.
3. **Work per tile.** Two operations per MAC give $0.6 \times 10^9$ operations; at $0.3 \times 10^{12}$ per second, a tile takes 2.0 ms.
4. **Energy per inference.** E = P t = 5 W × 2.0 ms = 10 mJ per tile.
5. **Per image.** A 10 000 × 10 000-pixel image holds about 2 000 tiles: about 4 s and 20 J. Screening 300 images a day uses about 1.7 Wh. (CloudScout on Φsat-1 took about 0.3 s at about 2 W, roughly 0.6 J per image.)
6. **Radiation.** At $7.6 \times 10^{-7}$ upsets per bit per day, the 27.2 Mbit of int8 weights collect about 21 upsets a day. With SEC-DED ECC and daily scrubbing, a double error in one word occurs only about once in four years.

## 45.6.2 Pruning, Distillation and Efficient Architectures

**Pruning** removes weights that matter little, usually the smallest in magnitude. Panel (c) of Figure 45.11 shows that removing 80 % of the weights without retraining drops accuracy to 75.2 %. After ten epochs of fine-tuning with the remaining weights, it recovers to 94.2 %. At 95 % pruning the fine-tuned network keeps 89.5 %. Pruning saves memory only if the hardware can skip zeros, either through sparse formats or by removing whole neurons or channels.




Same network and data as the bit-flip figure (5 126 parameters). Weights quantised symmetrically, biases and activations kept in 32-bit floating point. QAT: 12 epochs of fine-tuning with quantised weights in the forward pass (straight-through estimator). Pruning: global magnitude threshold, then 10 epochs of fine-tuning with the mask fixed. Sizes include biases and one scale per neuron.

**FIGURE 45.11** Compressing the network of Figure 45.7, tested on 8 000 synthetic pixels. (a) Post-training quantisation (PTQ) is harmless down to 6 bits; below 4 bits only quantisation-aware training (QAT) holds accuracy. (b) Model size. (c) Magnitude pruning works only with fine-tuning.

**Knowledge distillation** trains a small "student" network to imitate a large "teacher". The student learns from the teacher's soft probabilities, which carry more information than hard labels. Its loss has two terms:

$$L = (1-\alpha)\,\mathrm{CE}(y, p_s) + \alpha\, T^2\, \mathrm{KL}\big(p_t^{(T)} \,\|\, p_s^{(T)}\big),$$

Here CE is the cross-entropy with the true labels and KL the Kullback–Leibler divergence between the two networks' outputs. The "temperature" $T$ softens both distributions, and $\alpha$ weights the two terms.

**Efficient architectures** are designed small from the start. Mobile networks replace a standard convolution with a **depthwise separable** one: a per-channel spatial filter followed by a 1 × 1 convolution that mixes channels. For a $k \times k$ kernel and $N$ output channels, this cuts the work by a factor of about $1/N + 1/k^2$ — nearly nine times for 3 × 3 kernels. Chapter 44 describes these architectures.

## 45.7 On Board or on the Ground?

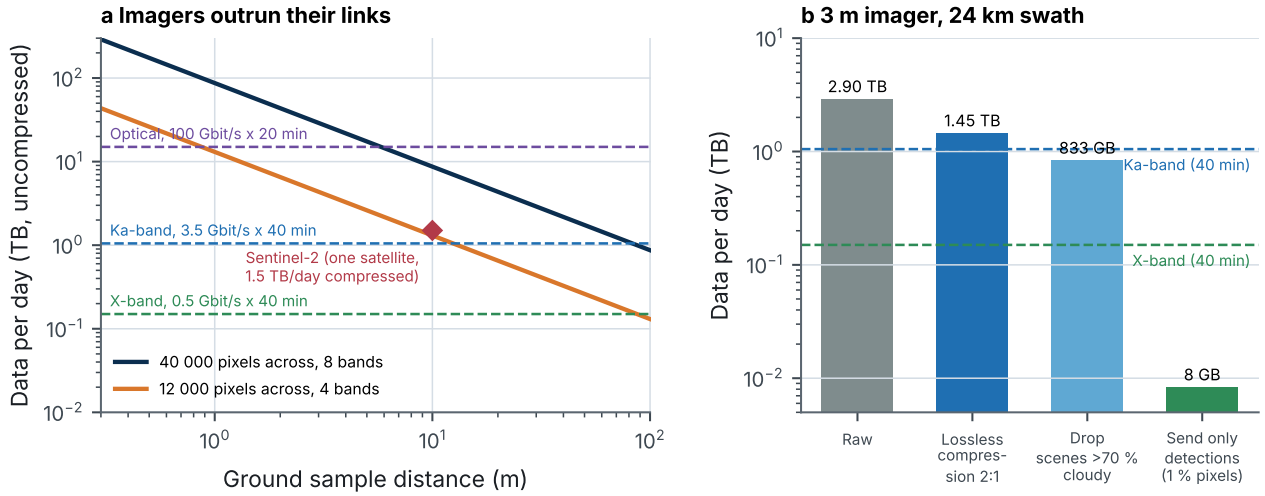
Every mission must decide where its models run. The key trade is between the cost of sending data down and the cost of processing them in orbit.

### 45.7.1 The Downlink Bottleneck

A pushbroom imager with $N$ pixels across its track, ground sample distance $g$ and ground speed $v_g$ scans $v_g/g$ lines per second. With $N_b$ bands of $b$ bits each, its data rate is

$$R = N \frac{v_g}{g} N_b b.$$

At 1 m resolution, 20 000 pixels across, 4 bands, 12 bits and a ground speed of 7 km/s, that is 6.7 Gbit/s. A link's daily capacity is its rate times the daily contact time with ground stations. The two are compared in Figure 45.12. An X-band link of 0.5 Gbit/s with 40 minutes of contact a day carries 0.15 TB. A Ka-band link of 3.5 Gbit/s carries 1.05 TB, and an optical link of 100 Gbit/s for 20 minutes carries 15 TB in clear weather. Applied to Sentinel-2's bands, the same model gives about 4.4 TB a day. That is consistent with the 1.5 TB a day each satellite sends after onboard compression of about 3:1.




Pushbroom model: data rate = pixels across track x lines per second (ground speed 7.0 km/s / GSD) x bands x 12 bits, imaging 30 % of the time. Link capacity = rate x daily contact time (assumed). Sentinel-2 check: the same model with its 10/20/60 m bands gives about 4.4 TB/day before compression. (b) Scene cloud fractions follow a U-shaped beta distribution with a mean of 55 % (land); scenes more than 70 % cloudy are discarded.

**FIGURE 45.12** Imager data versus link capacity. (a) Daily data from pushbroom imagers imaging 30 % of the time, against assumed daily capacities of X-band, Ka-band and optical links; Sentinel-2's reported 1.5 TB a day for comparison. (b) For a 3 m imager, compression, cloud screening and sending only detections cut 2.9 TB a day to 8 GB.

Optical links change the picture (Chapter 7). NASA's TBIRD cubesat sent 4.8 TB in a single pass of under five minutes in April 2023, at 200 Gbit/s. But clouds block lasers, so optical ground stations must be spread across regions with clear skies (Figure 45.14).

### 45.7.2 Cloud Screening

About two-thirds of Earth is covered by cloud at any moment. More than a decade of MODIS observations gives a global cloud fraction of about 67 %: about 55 % over land and 72 % over ocean (Figure 45.13). For an optical imager, most cloudy pixels are useless, so a model that discards them on board saves link capacity and ground storage. The risk is a **false negative** — a clear image wrongly discarded is lost forever.

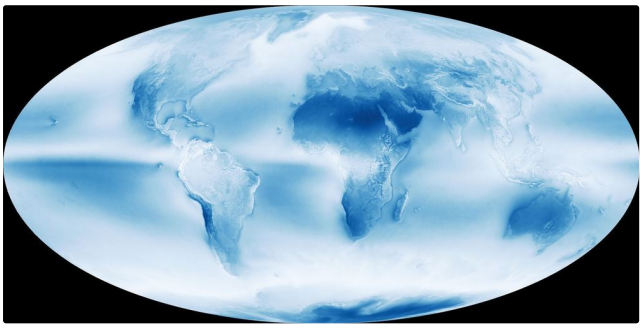


**FIGURE 45.13** Average cloud cover from MODIS. About 67 % of Earth is typically cloud-covered, most of all over the oceans. Image: NASA/GSFC.

**FIGURE 45.14** The 34 m antenna DSS-13 at Goldstone, California, with an optical terminal (below the centre of the dish) that receives laser signals from NASA's Psyche spacecraft. Clouds limit optical links. Image: NASA/JPL (NASA/JPL-Caltech).

**EXAMPLE: DOWNLINK SAVED BY ONBOARD CLOUD SCREENING**

A small satellite carries a 3 m imager with 8 000 pixels across (a 24 km swath). It has 4 bands of 12 bits, images 30 % of the time over land, and downlinks on X-band at 0.5 Gbit/s.

1. **Raw data.** 8 000 × (7 000 / 3) lines/s × 4 × 12 bits = 0.90 Gbit/s; over 30 % of a day this is 2.90 TB.
2. **Compression.** Lossless compression of 2:1 gives 1.45 TB a day. That needs 6.5 hours of X-band contact — far more than a small network of stations provides.
3. **Cloud statistics.** Assume scene cloud fractions follow a U-shaped beta distribution with a mean of 55 %, typical of land: many scenes are nearly clear and many nearly overcast. Then 42.6 % of scenes are more than 70 % cloudy.
4. **Screening.** Discarding those scenes leaves 0.83 TB a day and cuts the contact time to 3.7 hours. The discarded scenes held about 70 % of all cloudy pixels but only about 9 % of the clear ones.
5. **Detections only.** If the mission needs only objects such as ships, sending small image chips around detections (1 % of pixels) reduces the volume to about 8 GB a day — under three minutes of contact.

The screen saves about 40 % of the link but costs 9 % of the clear pixels. Whether that trade is right depends on the mission's users.

### 45.7.3 Latency, Power, Thermal and Verification Costs

Downlink is not the only factor (Table 45.5). Onboard processing can cut the delay from observation to alert from hours to minutes. It also lets a satellite retask itself, as JPL's dynamic-targeting experiment on CogniSAT-6 did in 2025 (Chapter 43). But every watt of computing must be generated, stored for eclipse and rejected as heat, and flight software must be verified to a higher standard.

**TABLE 45.5** Trade-offs between onboard and ground processing.

| Factor | Processing on board | Processing on the ground |
|---|---|---|
| Downlink | Products and alerts, megabytes to gigabytes | All raw data, terabytes |
| Latency | Seconds to minutes (with relays) | Hours, waiting for passes and processing |
| Computing | Limited; tolerant parts lag | Practically unlimited GPUs and TPUs |
| Power and heat | Adds 1–700 W to the bus | Cheap mains power |
| Updating models | Limited by uplink; must be re-verified | Any time |
| Re-analysis | Discarded data are lost | Full archive kept for future methods |
| Assurance | Flight-software standards; hard to fix | Normal software practice |

The usual answer is a mix: simple, well-tested models on board to screen, compress and alert, and heavier models on the ground.

## 45.8 Updating AI in Orbit

A model frozen at launch will age (see Figure 45.5). The key idea of this section is that space AI must be designed to be updated safely, like any other flight software.

### 45.8.1 Uploading New Models

Uploading a model needs four things. The flight software must accept new applications without a full reload. The uplink must carry the model, which 8-bit storage makes easier. Each upload must be authenticated. And a rollback path must restore the previous version if the new one fails.

Missions now demonstrate this routinely. ESA's **Φsat-2**, a 6U cubesat built by Open Cosmos and launched on 16 August 2024, runs its AI on Ubotica's CogniSAT processing board. Its applications run on the NanoSat MO Framework and can be installed and updated from the ground. At launch it carried four: image-to-map conversion, cloud detection, vessel detection and image compression. Two more — marine anomaly detection and wildfire detection — were uploaded after launch. Its operational phase from April 2025 to March 2026 was used to fine-tune the applications in orbit.

**OPS-SAT: A FLYING LABORATORY FOR SOFTWARE**

ESA's OPS-SAT, a 3U cubesat launched on 18 December 2019, was built to let outside teams test new software on a real spacecraft. ESA described its experimental computer as about ten times more powerful than those on its other spacecraft of the time. By its re-entry on 22–23 May 2024, 134 teams from 26 countries had run more than 284 experiments. Firsts included training machine-learning models on board and retraining them with live data. Its 2022 Kelvins competition asked teams to build small, quantised image classifiers designed to run on the satellite. OPS-SAT showed that frequent, safe software changes are possible when the spacecraft is designed for them.

### 45.8.2 Software-Defined Payloads and Split Computing

A **software-defined payload** is one whose function is set largely by software: a radio whose waveforms can change, or an imager whose processing chain can be replaced. Packaged as containers, its applications can serve several customers on one satellite.

**Split computing** divides a network between satellite and ground. The first layers run on board and turn the image into compact features, which are sent down to the rest of the network. Splitting where the features are smallest saves downlink without fitting the whole model on board.

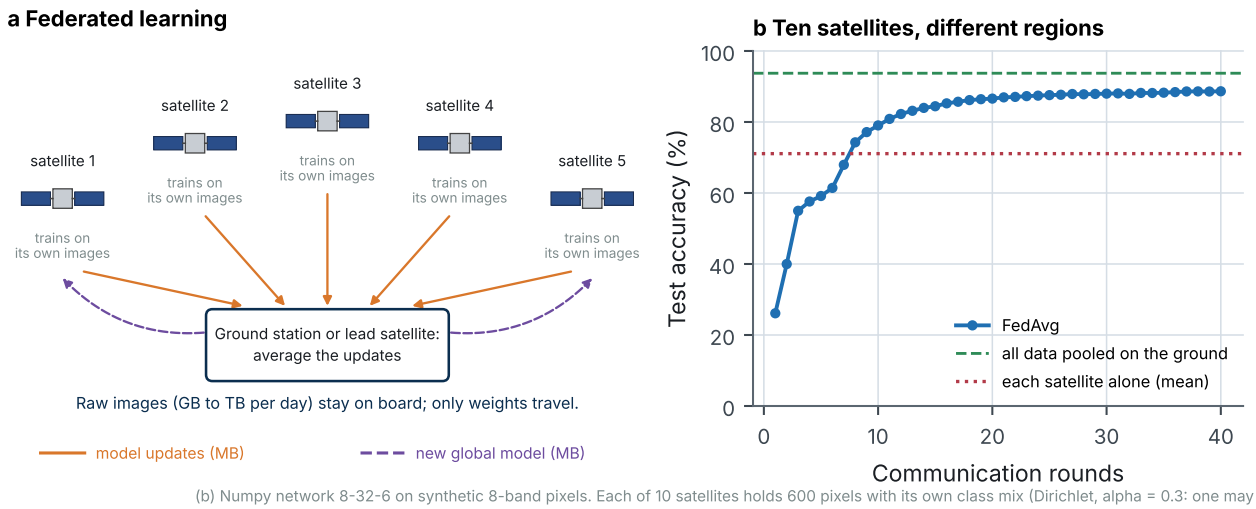
### 45.8.3 Federated Learning across Constellations

**Federated learning** trains one model across many devices without collecting their raw data. In a constellation, each satellite trains on its own images and sends only model updates, megabytes instead of terabytes. A ground station or lead satellite averages them. In the standard **FedAvg** algorithm, the new global weights are a weighted mean:

$$w_{t+1} = \sum_{k=1}^{K} \frac{n_k}{n} \, w_{t+1}^{(k)},$$

Here $w_{t+1}^{(k)}$ are the weights trained by satellite $k$ in round $t$ on its $n_k$ samples, and $n = \sum_k n_k$.

The simulation in Figure 45.15 has ten satellites, each seeing a different mix of surfaces, as if one flew mostly over ocean and another over desert. A satellite training alone reaches only 71.1 % on average on a balanced test set, and pooling all data on the ground gives 93.7 %. FedAvg reaches 79.0 % after 10 rounds and 88.6 % after 40 without moving any raw data. The remaining gap comes from the satellites' very different data, a known weakness of simple averaging.




(b) Numpy network 8-32-6 on synthetic 8-band pixels. Each of 10 satellites holds 600 pixels with its own class mix (Dirichlet, alpha = 0.3: one may see mostly ocean, another desert). Each round every satellite trains 2 epochs from the global model; updates are averaged by sample count. Test: 6 000 pixels with all classes equally likely.

**FIGURE 45.15** Federated learning across a constellation. (a) Satellites train on their own images and send only model updates, which are averaged and returned. (b) Computed with numpy networks on synthetic pixels: FedAvg (blue) approaches pooling all data on the ground (green) and far exceeds satellites training alone (red).

As of October 2026, federated learning across satellites remains **research**. Simulation studies have adapted it to intermittent contacts and to about 100 satellites, but no in-orbit demonstration across a constellation had been reported.

## 45.9 Computing in Orbit: Orbital Data Centres

The boldest idea in this chapter turns edge AI inside out: put data-centre computers into orbit, powered by sunlight and cooled by radiating heat to space. The physics allows it, but the engineering and economics are unproven. Table 45.6 lists the main projects as of October 2026.

### 45.9.1 Demonstrations and Proposals

**TABLE 45.6** Computing in orbit: demonstrations and proposals, October 2026.

| Project | Who | When | What | Status |
|---|---|---|---|---|
| Spaceborne Computer-2 | HPE, NASA | 2021; new unit 2024 | Servers and a T4 GPU on the ISS | Demonstration |
| Kepler optical relay | Kepler | January 2026 | 10 relay satellites with GPU modules | In service since August 2026 |
| Three-Body constellation | ADA Space, Zhejiang Lab | May 2025 | 12 satellites, laser links to 100 Gbit/s | Demonstration; 2 800 planned |
| Starcloud-1 | Starcloud | November 2025 | One H100 on a 60 kg satellite | Demonstration; next satellite 2027 |
| Suncatcher prototype | Google, Planet | October 2026 | Four TPUs; one-year test | Demonstration |
| Orbital data centres | SpaceX | FCC filing, January 2026 | Up to one million satellites | Proposal |

**Spaceborne Computer-2** showed the first benefit: less data to send. In a 2021 genomics test it reduced 182 GB of raw data to 13 MB of results, over a link limited to about two hours a week at up to 250 KB/s.

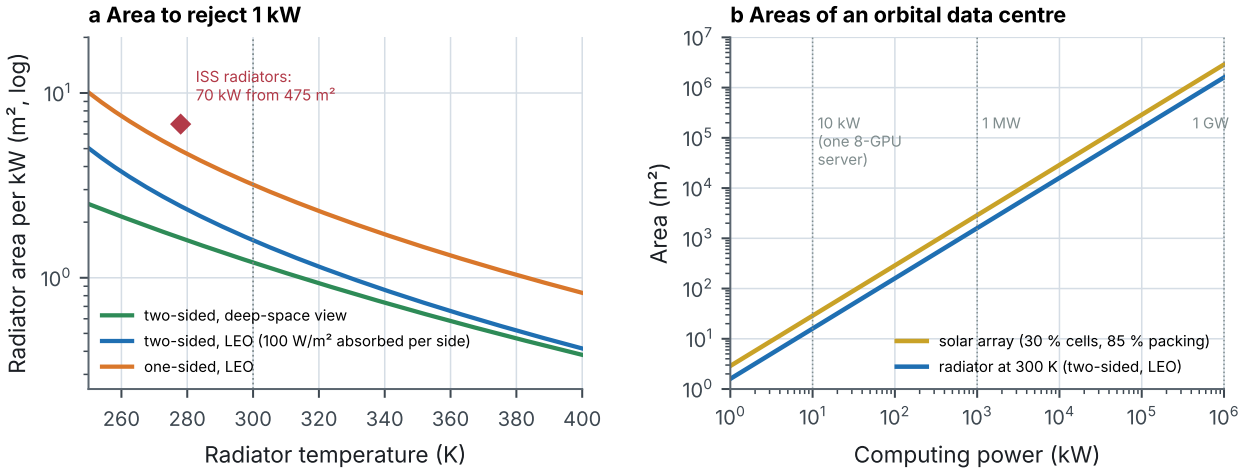
China's **Three-Body Computing Constellation** is the largest effort so far. Its first 12 satellites, launched on 14 May 2025, offer about 5 POPS (peta-operations per second) combined, linked by lasers. In February 2026 Chinese state media reported ten AI models deployed, two with 8 billion parameters. **Starcloud-1**, launched on 2 November 2025, carried one H100 GPU. In December 2025 the company reported training a small language model on the works of Shakespeare in orbit. Both are demonstrations; neither yet sells computing at scale.

## 45.9.2 Cooling: the Radiator Problem

In orbit, the only way to lose heat is to radiate it, and almost all the power a processor uses becomes heat. A radiator panel of area $A$, emissivity $\varepsilon$ and temperature $T$ rejects

$$P = \varepsilon \sigma A \left(T^4 - T_{\text{sink}}^4\right) - \alpha A q_{\text{abs}},$$

Here $\sigma$ is the Stefan–Boltzmann constant, $5.67 \times 10^{-8}$ W m$^{-2}$ K$^{-4}$. The flux $q_{\text{abs}}$ is the sunlight and Earth infrared that the panel absorbs (Chapter 5). At 300 K with $\varepsilon$ = 0.9, each side radiates 413 W/m$^2$. Because of the fourth power, area grows steeply as temperature falls (Figure 45.16). Hotter chips mean smaller radiators, but electronics age faster when hot.




Net rejection per side = emissivity (0.9) x Stefan-Boltzmann constant x T^4 minus absorbed environmental flux; fin efficiency, temperature drop from chip to radiator and design margin are ignored, which is why the real ISS radiators need about four times the ideal area. Solar array in continuous sunlight (dawn-dusk orbit), 1 361 W/m².

**FIGURE 45.16** Radiator physics for computing in orbit. (a) Area to reject 1 kW: 1.6 m² for a two-sided panel at 300 K in LEO; the ISS's radiators need about four times the ideal. (b) Radiator and solar-array areas grow with computing power; a gigawatt would need about 1.6 km² of radiator and 2.9 km² of solar array.

The ISS gives a reality check (Figure 45.17). Its external active thermal control system rejects up to 70 kW. It uses two radiator wings of three units each; every unit is 23.3 m × 3.4 m, for 475 m² of panel in all. That is 6.8 m² per kW, against an ideal of about 1.6 m² per kW for two-sided panels near 280 K.

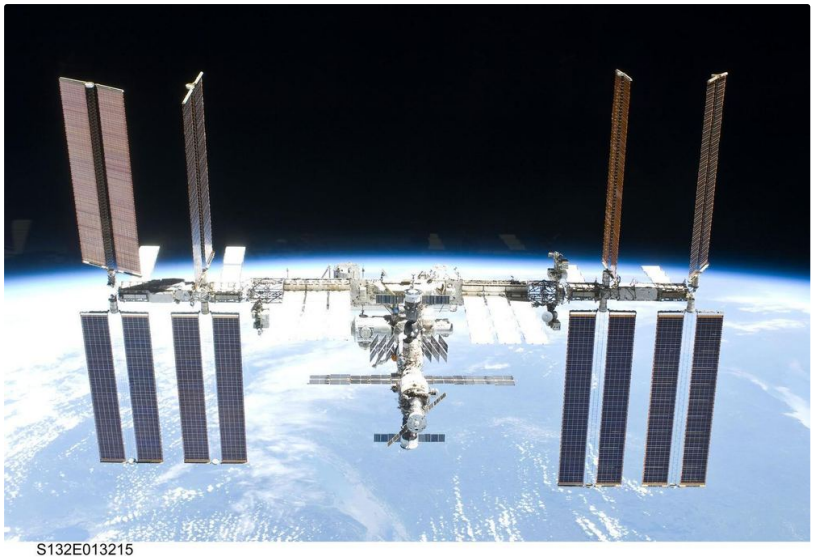


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**FIGURE 45.17** The ISS in May 2010 (STS-132). Between the solar-array wings, white radiator panels on the truss reject up to 70 kW — the heat of about seven eight-GPU AI servers. Image: NASA/JSC.

**EXAMPLE: A RADIATOR FOR 10 KW OF COMPUTING**

A 10 kW AI server — about one machine with eight data-centre GPUs — must reject its heat at 300 K in low Earth orbit. Assume a two-sided panel with emissivity 0.9, absorbing 100 W/m² from Earth and the Sun on each side.

1. **Ideal flux.** $\varepsilon\sigma T^4 = 0.9 \times 5.67 \times 10^{-8} \times 300^4 = 413$ W/m² per side.
2. **Ideal area.** With no absorbed flux, A = 10 000 / (2 × 413) = 12.1 m².
3. **With environment.** Net flux 413 − 100 = 313 W/m² per side, so A = 10 000 / (2 × 313) = 16.0 m².
4. **With margin.** Allow 30 % for fin efficiency, temperature drops and degradation: about 21 m².
5. **Mass.** ISS radiator units weigh about 14 kg/m² (1 123 kg for 79 m²), giving about 300 kg.
6. **Launch cost.** At about $3 600/kg, the radiator alone costs about $1.1 million to launch. The solar array for 10 kW needs another 29 m².

At 350 K the area before margin would fall to 7.5 m², if the chips tolerate the heat.

### 45.9.3 Economics

The economic case rests on three claims. Sunlight in a dawn–dusk orbit is almost continuous; Google estimates that panels in some orbits collect up to eight times the yearly energy of a panel at mid-latitudes. No land, water or grid connection is needed. And launch costs are falling.

Google's Suncatcher study puts numbers on the last point. At about $200/kg, possible by the mid-2030s, launched power would cost about $810 per kilowatt per year. Terrestrial data centres pay $570–3 000 per kilowatt per year for electricity. At a current price of about $3 600/kg, the estimate is $14 700 per kilowatt per year. The **ASCEND** study for the EU found orbital data centres feasible only if launchers became ten times less polluting over their life cycle.

The counter-arguments are strong. Radiators and solar arrays are large and heavy. Failed chips cannot be swapped. Radiation causes silent errors, as the Suncatcher tests showed. Data must still cross limited links to Earth, and crowded orbits add debris risk (Chapter 21). For now, orbital data centres are serious demonstrations and bold proposals, not an industry.

### PRACTICE: QUALIFYING AN AI PAYLOAD FOR A SMALL SATELLITE

Small satellites are often the first home of a new AI payload (Figure 45.18). A typical workflow:

1. **Define the function and its criticality.** State what the model decides and what happens if it is wrong; keep it out of safety-critical loops. The European handbook ECSS-E-HB-40-02A (November 2024) gives guidance.
2. **Budget resources.** Set limits for power, energy per orbit, heat, memory, mass and data volume. Estimate throughput at 20–30 % of the processor's peak.
3. **Choose hardware and protection.** Predict the dose and upset rate for the orbit (Chapter 2) with a design margin of at least two. Add a latch-up current limiter, a watchdog and a radiation-tolerant supervisor for any commercial chip.
4. **Test for radiation.** Run total-dose tests on parts from the flight lot. Run proton or heavy-ion tests with the real model executing, logging every upset, interrupt and wrong output.
5. **Freeze and quantise the model.** Compare the flight version bit for bit with the ground reference on the locked test set. Measure accuracy, time and energy on the flight hardware.
6. **Inject faults.** Flip bits in weights, activations and code. Choose ECC, scrub interval and range checks that hold accuracy at the predicted upset rate with margin.
7. **Test the environment.** Run vibration, thermal-vacuum and electromagnetic tests with the processor at full load in the hot case.
8. **Secure updates.** Package the model as an application with signed uploads, version records and a rollback path. Keep safe mode independent of the AI.
9. **Plan operations.** Commission in shadow mode — deciding but not acting — and downlink samples to check the decisions. Then monitor for drift.

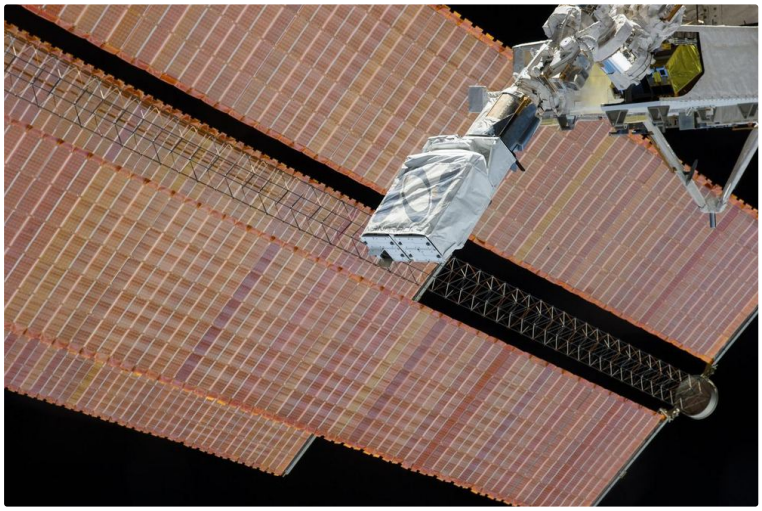


**FIGURE 45.18** Japan's Small Satellite Orbital Deployer, held by the Kibo robotic arm on 11 February 2014, shortly before it released NanoRacks cubesats from the ISS. Many AI payloads are first proven on small satellites like these. Image: NASA/JSC.

### KEY TAKEAWAYS

- Archives hold hundreds of petabytes, but labelled, AI-ready data are scarce. Active learning can cut labelling about threefold; systematic label errors harm far more than random ones.
- Benchmarks enable fair comparison but leak, saturate and contain label errors. Synthetic data need domain randomisation and adaptation to cross the sim-to-real gap.
- MLOps — versioned data, tracked experiments, model registries, validation gates and drift monitoring — keeps a model trustworthy for the life of a mission.
- Radiation flips stored bits. Float32 weights fail when an exponent bit flips; int8 storage, range checks, ECC with scrubbing and TMR restore robustness.
- Commercial AI chips are thousands of times more efficient than hardened CPUs but need testing, shielding and supervision; hardened parts such as PIC64-HPSC are closing the gap slowly.
- Quantisation to 8 bits costs almost nothing, and QAT, pruning and distillation go further. Onboard screening saves downlink but discards data for ever.
- Models in orbit must be updatable, as Φsat-2 and OPS-SAT showed. Federated learning and orbital data centres remain research and demonstration, limited by radiators, radiation and economics.

### REVIEW QUESTIONS

1. Compare expert, crowdsourced and weak labels. How does margin sampling choose what to label next?
2. In Figure 45.2, why does random label noise of 50 % reduce accuracy only slightly, while 50 % of urban pixels labelled as soil cuts urban recall to about 31 %?
3. What is the sim-to-real gap? Use the SPEED+ numbers to describe its size, and explain how domain randomisation and domain adaptation reduce it.
4. List the stages of an MLOps pipeline. What should a validation gate check before a model is uplinked?
5. Compute the PSI for a band where the training data fall into four bins in proportions 0.25 each and current data in proportions 0.10, 0.20, 0.30 and 0.40. Is retraining advised?
6. Why does a single flipped exponent bit in a float32 weight do so much more damage than a flipped bit in an int8 weight? How do ECC, scrubbing and range checks work together?
7. A model has 10 million int8 weights in SRAM with $7.6 \times 10^{-7}$ upsets per bit per day. How many upsets occur per day, and how often must memory be scrubbed so that a 72-bit ECC word rarely holds two errors?
8. Explain post-training quantisation and quantisation-aware training. Why did 2-bit PTQ fail in Figure 45.11 while 2-bit QAT kept 93.2 %?
9. A 0.5 m imager with 30 000 pixels across, 4 bands and 12 bits images 20 % of the time. Estimate its daily data volume and compare it with X-band and optical links. What onboard processing would you propose?
10. Estimate the radiator area needed to reject 100 kW at 330 K from a two-sided panel absorbing 100 W/m$^2$ per side. Discuss two technical and two economic obstacles to orbital data centres.

# 46 CHAPTER AI for Earth Observation

Image: NASA/GSFC

> *Earth-observation satellites now collect far more images than people could ever look at, so most pixels are first seen by a machine. This chapter explains, from first principles and with small models trained from scratch, how AI turns pixels into information. It covers land-cover maps, objects, change, crop calendars, radar speckle, ships, floods and methane plumes. It then follows verified deployments in agriculture, forests, cities, oceans, disasters, greenhouse gases and development, as of October 2026. It describes AI on board satellites and smart tasking, the commercial analytics business and geospatial foundation models in practice. It ends with the traps — domain shift, label noise, biased samples, wrong area estimates and privacy — and how good practice avoids them.*

**IN THIS CHAPTER**

46.1 From Pixels to Information
46.2 Land: Crops, Forests, Cities and Water
46.3 Oceans and Maritime Awareness
46.4 Disasters
46.5 Greenhouse Gases and Human Development
46.6 AI on Board and Smart Tasking
46.7 The EO Analytics Business
46.8 Pitfalls and Good Practice

Every day, Earth-observation (EO) satellites image the whole land surface several times over. Planet's fleet alone can collect about 200 million km$^2$ a day. Sentinel-1's radars see through cloud, day and night. Thermal, hyperspectral and very-high-resolution sensors add more. No team of people can inspect all of it. The images are only useful if software finds the floods, ships, fires, new buildings and methane leaks in them, quickly and reliably.

That software is increasingly **machine learning**: models whose behaviour is learned from examples rather than written as rules. This chapter shows how such models work on EO data, where they are used today, and where they fail. The hero image above shows centre-pivot fields

in Kansas in a Landsat false-colour composite. Crop type, health and harvest date can all be read from such images by a trained model.

The physics of remote sensing is covered in Chapters 13–16. The learning methods themselves are explained in Chapters 43 and 44, and data and onboard hardware in Chapter 45. Weather and climate models are left to Chapter 47, and military uses to Chapter 50. Here we join the two: what a learning model needs to know about EO data, and what EO users need to know about learning models.

## 46.1 From Pixels to Information

The key idea is that EO analysis is a short list of tasks — classify, segment, detect, compare, estimate — applied to images that are unlike everyday photographs. They have many spectral bands, a time dimension and an exact location on Earth. Each task has a matching family of models and a matching way to measure success (Table 46.1).

The pipeline in Figure 46.1 shows where the models sit. Raw data are calibrated and corrected into **analysis-ready data** (ARD) — pixels in physical units on a map grid (Chapter 13). Models turn ARD into maps, alerts and estimates. Users send back corrections and new labels, and some detections trigger new images of the same place. A growing share of the work also happens on the satellite itself.

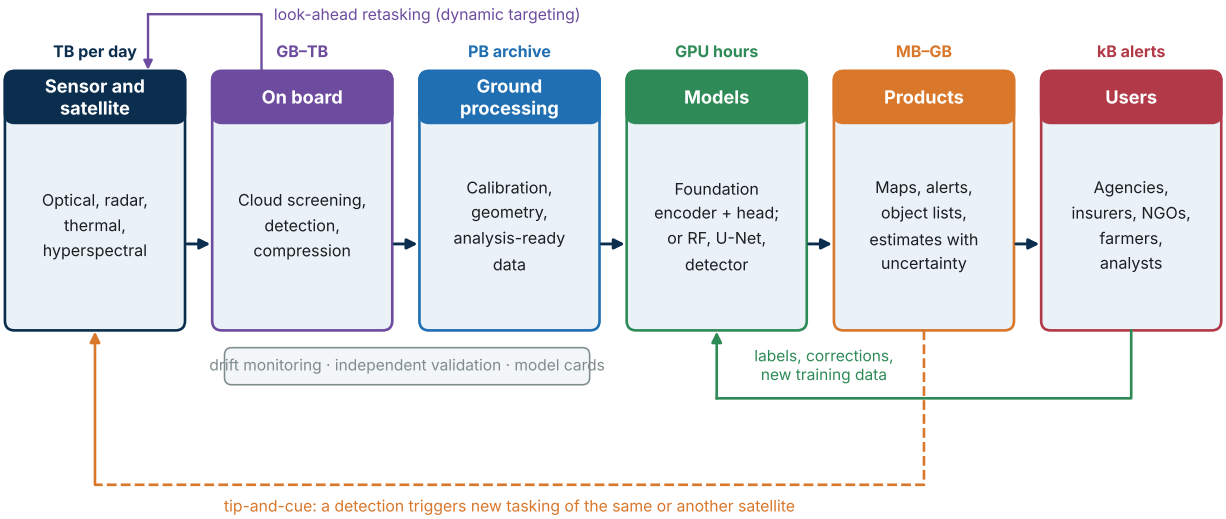



**FIGURE 46.1** The EO data pipeline with AI. Each stage shrinks the data: terabytes of pixels a day become megabytes of maps and kilobytes of alerts. Three loops close the system: users supply new labels, detections cue new collections (tip-and-cue), and an onboard look-ahead camera can retask the satellite in under 90 seconds.

### 46.1.1 What Makes EO Data Different

Four things set EO images apart from the photographs on which computer vision grew up. First, they have **many bands**: Sentinel-2 records 13, from blue to short-wave infrared, and imaging spectrometers record over 200. Second, they come in **time series**: the same field is seen every few days. Third, every pixel has a **location**, so neighbouring pixels are alike and nearby samples are not independent. Fourth, the **physics changes with the sensor**: radar measures surface roughness and moisture, not colour (Chapter 15).

Labels are scarce. A satellite can image a country in a day, but surveying land cover on the ground takes months. Many EO models are therefore trained on a few thousand labelled pixels, on labels made by other models, or on simulations (Chapter 45).

**TABLE 46.1** Common EO tasks, the models used for them and how they are scored. IoU is intersection over union, the overlap between mapped and true areas divided by their union.

| Task | Output | Typical models | Usual metrics | EO examples |
|---|---|---|---|---|
| Scene classification | One label per image chip | CNN, vision transformer | Accuracy, F1 | Cloudy or clear; land-use type |
| Semantic segmentation | A class for every pixel | Random forest, U-Net, segmentation transformer | Overall accuracy, IoU | Land cover, floods, burn scars |
| Object detection | Boxes with labels | Faster R-CNN, YOLO, CFAR + classifier | Precision, recall, mean average precision | Ships, aircraft, vehicles |
| Instance segmentation | An outline per object | Mask R-CNN | Mask average precision | Building footprints, fields |
| Change detection | Changed pixels or objects | Siamese CNN, bi-temporal classifier | F1 of the change class | Deforestation, damage |
| Time-series classification | A label per pixel or field | Random forest, temporal CNN, attention | Accuracy, F1 | Crop type, phenology |
| Regression and retrieval | A physical quantity | Gradient boosting, CNN, physics + learning | RMSE, bias | Biomass, methane emission rate |
| Image-to-image | A new image | U-Net, GAN, diffusion model | PSNR, SSIM, task skill | Super-resolution, despeckling, SAR to optical |

### 46.1.2 Land Cover: Pixels and Neighbourhoods

**Land-cover mapping** gives every pixel a class such as water, forest, cropland or built-up land. The classic approach treats each pixel on its own. Its features are the reflectances in each band, plus indices such as the **normalised difference vegetation index**,

$$\mathrm{NDVI} = \frac{\rho_{\mathrm{NIR}} - \rho_{\mathrm{red}}}{\rho_{\mathrm{NIR}} + \rho_{\mathrm{red}}},$$

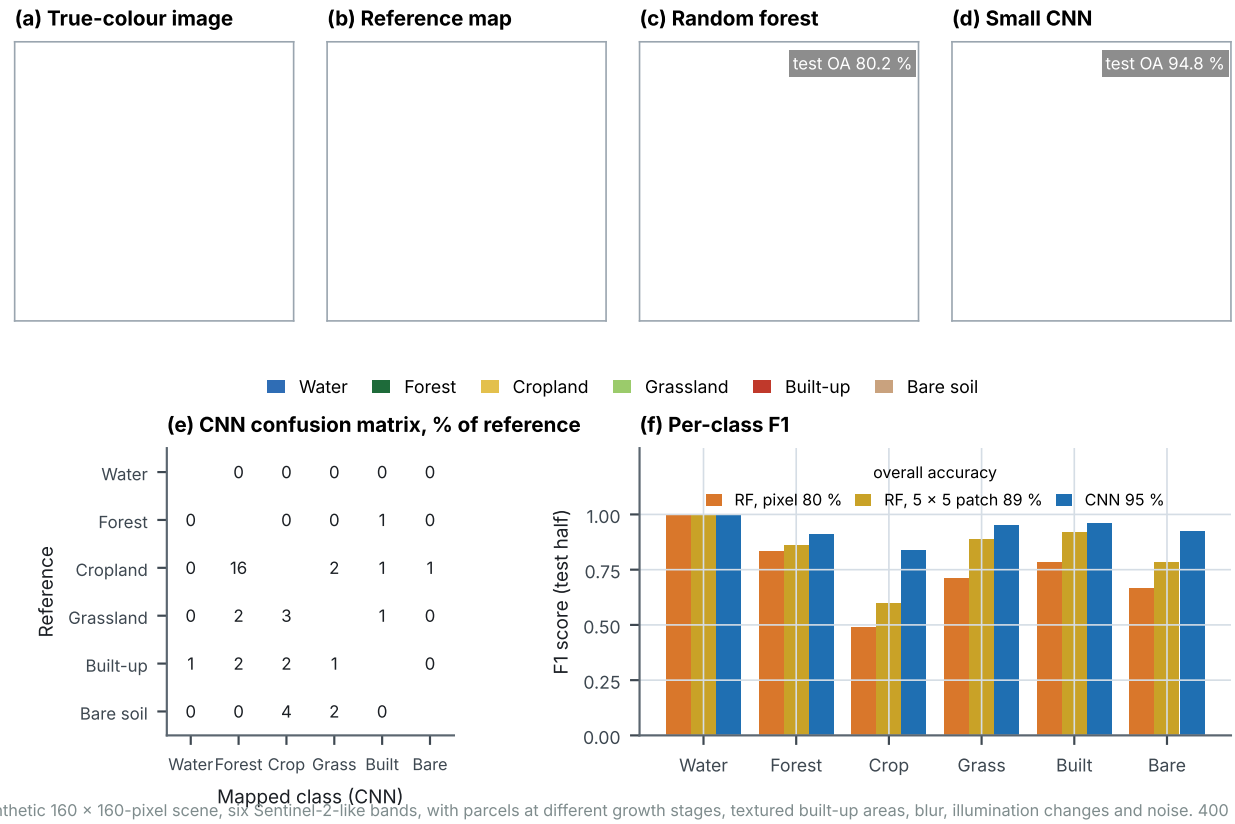
where $\rho$ is reflectance. Healthy plants reflect strongly in the near infrared (NIR) and absorb red light, so NDVI is high for vegetation and near zero for soil and water. A **random forest** — hundreds of decision trees, each trained on a random subset of the data, voting together (Chapter 43) — is still the workhorse for such pixel-based maps.

A pixel alone is often ambiguous. A dry field looks like bare soil, and a suburb is a mixture of roofs, roads and gardens. A **convolutional neural network** (CNN) looks at a neighbourhood instead. Each layer slides small filters over the image and sums weighted pixel values,

$$y_{ij} = \sigma\Big(\sum_{c}\sum_{u,v} w_{c,uv}\, x_{c,\,i+u,\,j+v} + b\Big),$$

where $\sigma$ is a simple non-linear function. The filters learn textures and shapes as well as colours (Chapter 44). For whole maps, **encoder–decoder** networks such as the U-Net segment large tiles in one pass.

In Figure 46.2 the two are compared on a synthetic six-class scene with Sentinel-2-like bands. Both models learned from 400 labelled pixels per class in the western half and were tested on all 12 800 pixels of the eastern half. The random forest on single pixels reached an overall accuracy of 80.2 %. Cropland was its weakest class, with an F1 score — the balance of precision and recall — of only 0.49, because fields at different growth stages look like grass or bare soil. A random forest given the same 5 × 5-pixel neighbourhoods reached 89.0 %. A small CNN with about 10 000 weights, written in numpy, reached 94.8 % and raised cropland F1 to 0.84. Context, not the network as such, did most of the work.




Synthetic 160 × 160-pixel scene, six Sentinel-2-like bands, with parcels at different growth stages, textured built-up areas, blur, illumination changes and noise. 400 labelled pixels per class from the western half train each model; all 12 800 pixels of the eastern half test it. CNN: one 5 × 5 × 64 convolution, ReLU, 1 × 1 output (10 054 weights), written in numpy.

**FIGURE 46.2** Land-cover mapping of a synthetic scene. (a)–(d) Image, reference and two maps; models learned from the western half and were tested on the eastern half. (e) The CNN's remaining confusion is mostly cropland labelled as forest. (f) Neighbourhood information raises F1 for every mixed or textured class.

The synthetic test is easier than reality: global maps reach overall accuracies of about 74–77 % (Table 46.2). But the ranking is typical. Pixel-based random forests remain strong baselines, and convolutional and transformer networks win where texture and shape matter.

### 46.1.3 Objects: Ships, Aircraft and Buildings

**Object detection** finds each object and draws a box around it. Two-stage detectors first propose candidate regions and then classify them; one-stage detectors such as the YOLO family do both in one pass, which suits fast processing (Chapter 44). **Instance segmentation** outlines each object, which is what building-footprint maps need.

EO objects are small and rotated. A 50 m ship is five pixels long at 10 m resolution, and a car is a few pixels at 0.3 m. Detectors for EO therefore use rotated boxes, image pyramids and tiles that overlap so that objects on tile edges are not cut in half. Large public sets such as xView (over a

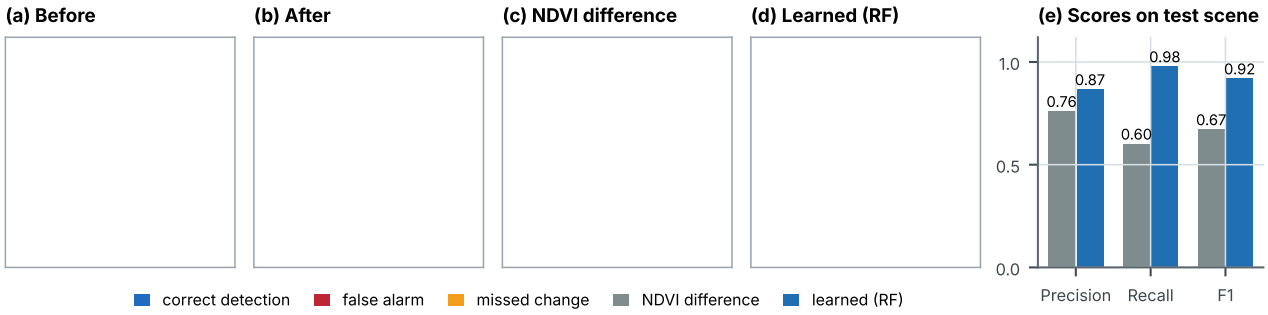
million objects in 0.3 m images) and xBD (850 736 buildings before and after 19 disasters) made this progress possible (Chapter 45).

### 46.1.4 Change Detection

Most EO questions are about **change**: what was cleared, flooded, built or burned. The simplest method subtracts an index between two dates and applies a threshold, for example flagging forest loss where $\Delta\mathrm{NDVI} = \mathrm{NDVI}_2 - \mathrm{NDVI}_1$ falls below −0.5. It is fast and easy to explain. But it reacts to every change in the index: harvests, seasons, haze, shadows and small misalignments between the images.

A **learned change detector** is trained on examples of the changes that matter and of those that do not. It sees both images, all bands and the local neighbourhood. Deep versions use **Siamese networks**, two copies of one encoder that process the two dates and compare their features.

The test in Figure 46.3 makes the difference concrete. The task is to find forest clearings in a synthetic scene that also contains harvested fields, drying grass, haze, a cloud shadow and a one-pixel misregistration. Both methods were tuned on four other scenes. The best NDVI threshold (−0.55) found only 60 % of the cleared pixels, with a precision of 0.76 and an F1 of 0.67. A random forest on 16 bi-temporal features found 98 %, with a precision of 0.87 and an F1 of 0.92. It had learned that a drop in NDVI over a field is a harvest, not deforestation.




Synthetic 128 × 128-pixel scene (false colour: NIR, red, green). Forest clearings are the only true change; harvests, drying grass, haze, a cloud shadow and 1-pixel misregistration are not. The NDVI threshold (-0.55) and the random forest (200 trees, 16 bi-temporal features) were both tuned on four other scenes.

**FIGURE 46.3** Change detection on a synthetic scene where only forest clearings count as change. (a, b) False-colour images before and after. (c) An NDVI-difference threshold misses partly cleared patches and flags harvests. (d) A random forest trained on other scenes finds almost all clearings. (e) Precision, recall and F1 on the test scene.

Dense time series go further. Algorithms such as CCDC and BFAST fit a seasonal curve to every pixel's history and flag a break when new data stop fitting it (Chapter 13). Operational forest alerts, described below, combine such statistics with machine learning.

### 46.1.5 Time Series and Crop Types

Crops are best recognised by their **calendar**. Winter wheat greens up in spring and is harvested in summer. Maize and soybean start later. Grassland drops after each mowing (Figure 46.4 a). A model that sees the whole season can tell them apart even when single images cannot.

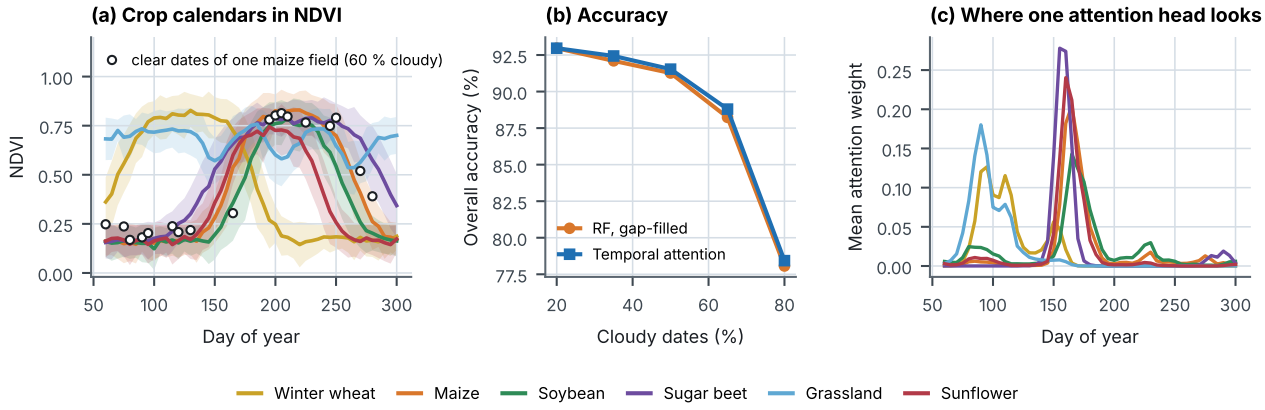
Clouds make the series irregular. One option is to fill the gaps by interpolation and give the filled series to a random forest. Another is **temporal attention**, which works directly on the clear dates. Each observation $t$ becomes a feature vector $\boldsymbol{h}_t$ that encodes its value and its day of year. A

learned query $q$ scores each date, and the model averages the dates with weights

$$a_t = \frac{\exp(q \cdot k_t / \sqrt{d})}{\sum_{s \in \text{clear}} \exp(q \cdot k_s / \sqrt{d})}, \qquad z = \sum_t a_t \, h_t,$$

where $k_t$ is a "key" computed from $h_t$ and $d$ its length. Cloudy dates simply drop out of the sum. The pooled vector $z$ is then classified. Garnot and colleagues' pixel-set encoder with temporal attention (2020) reached 94.2 % overall accuracy on the S2-Agri benchmark of French fields.

In Figure 46.4 both approaches are tested on synthetic NDVI series for six crops, with timing shifts and undetected thin cloud. A numpy attention network with six heads and about 5 000 weights matched a random forest on gap-filled series. Both reached 93.0 % with 20 % of dates cloudy and about 78 % with 80 % cloudy. Panel (c) shows what one head learned: it looks at early spring, which separates wheat and grass, and at early June, which separates the summer crops. On small, clean problems attention gives no free gain. Its advantages appear with large and varied training sets, many bands and irregular sampling across regions.




Synthetic fields: double-logistic NDVI curves (grass: three mowing cuts) with ±10-day timing shifts, amplitude spread and noise; 49 dates every 5 days; clouds remove dates at random and 8 % of 'clear' dates carry undetected thin cloud. Both models trained on 5 000 fields, tested on 3 000 per point. Attention network: six heads, 4 998 weights, numpy.

**FIGURE 46.4** Crop types from NDVI time series (synthetic data). (a) Mean seasonal curves ± one standard deviation, and the clear observations of one field. (b) Accuracy against the share of cloudy dates for a random forest on gap-filled series and a temporal-attention network. (c) Mean attention weights of one head for each crop.

### 46.1.6 Clouds, Sharper Images and Fusion

**Cloud and shadow masking** is the first step for any optical model, because cloud covers about two-thirds of Earth at any time. Rule-based masks such as Fmask test brightness, temperature and band ratios. Learned masks now do better. The s2cloudless detector uses gradient-boosted trees on ten Sentinel-2 bands, and CNNs trained on sets such as CloudSEN12 (49 400 patches) also find thin cloud and shadows. Missed thin cloud is a quiet source of error in every downstream product.

**Pan-sharpening** merges a sharp panchromatic band with coarser colour bands (Chapter 14). **Super-resolution** goes further and predicts detail finer than any band, for example 10 m Sentinel-2 images sharpened to 2.5 m. Networks trained on pairs of images from fine and coarse sensors learn plausible detail. But they can **hallucinate**: they may invent edges, cars or field boundaries that look real. Super-resolved images are useful for display and as model inputs, but they are not new measurements.

**Data fusion** combines sensors. Radar sees through cloud, optical data are easier to interpret, and the two together beat either alone for crops, floods and forests. Networks can even **translate** radar into optical-looking images. Such images help analysts read radar, but they must never be used as if they were observations.

### 46.1.7 Radar: Speckle and Ships

Radar images carry **speckle**, the grainy pattern made when echoes from many small scatterers in a pixel add with random phases (Chapter 15). For a uniform surface with true reflectivity $R$, the measured intensity is

$$I = R\,n, \qquad n \sim \Gamma(L, 1/L),$$

where $L$ is the number of independent looks averaged. A single-look image ($L = 1$) has a standard deviation equal to its mean. Smoothness is measured by the **equivalent number of looks**, $\mathrm{ENL} = \mu^2/\sigma^2$, over a uniform area.

Averaging neighbours raises ENL but blurs edges. The **Lee filter** (1980) adapts. It smooths where the image is uniform and keeps the original pixel where it is not:

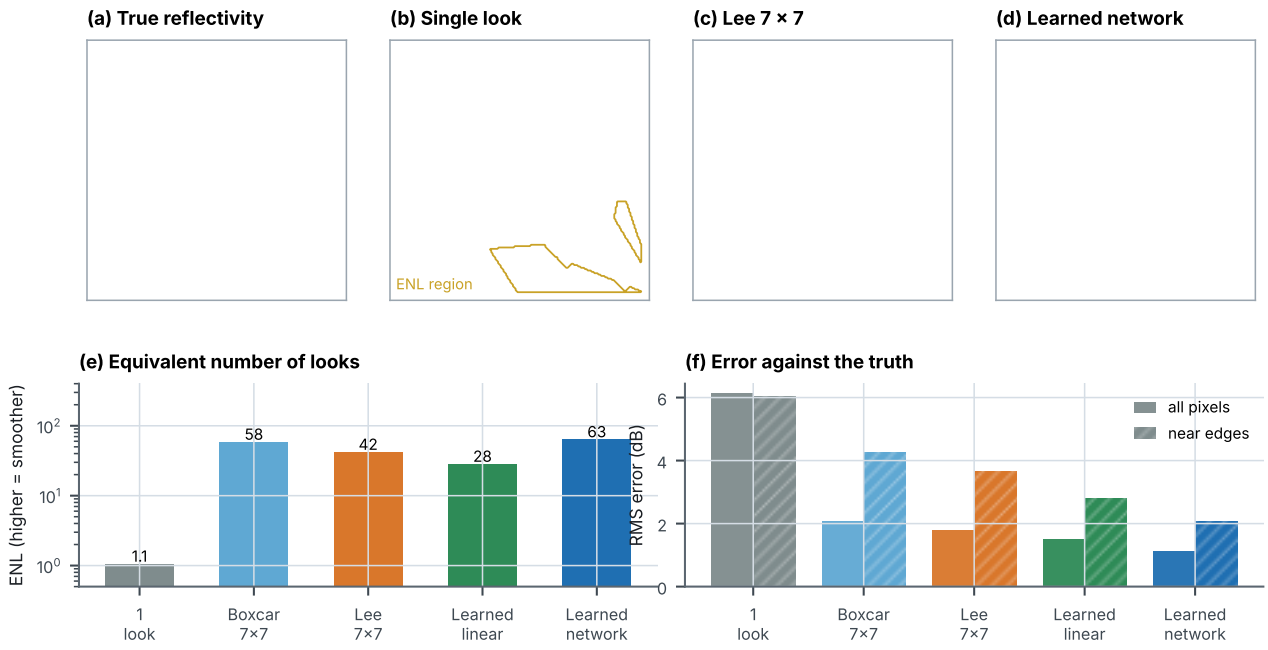
$$\hat{R} = \bar{I} + k\,(I - \bar{I}), \qquad k = \frac{1 - C_u^2/C_I^2}{1 + C_u^2}.$$

Here $\bar{I}$ is the local mean. $C_I$ and $C_u$ are the local and speckle-only coefficients of variation, the standard deviation divided by the mean. Learned filters instead train on pairs of speckled and clean images. Since real clean images do not exist, they use simulated speckle or several dates averaged together. Self-supervised methods train on speckled data alone.

In Figure 46.5 four filters are compared on a synthetic single-look scene. A 7 × 7 boxcar reached an ENL of 58 but blurred edges, with an error of 4.25 dB near them. The Lee filter reached an ENL of 42 and an overall error of 1.81 dB. A least-squares 9 × 9 linear filter, learned from data, cut the error to 1.49 dB. A small numpy network on the same 9 × 9 patches reached an ENL of 63 and an error of only 1.14 dB, and 2.08 dB near edges. The lesson: ENL alone rewards blurring, so always measure error near edges and small targets too.

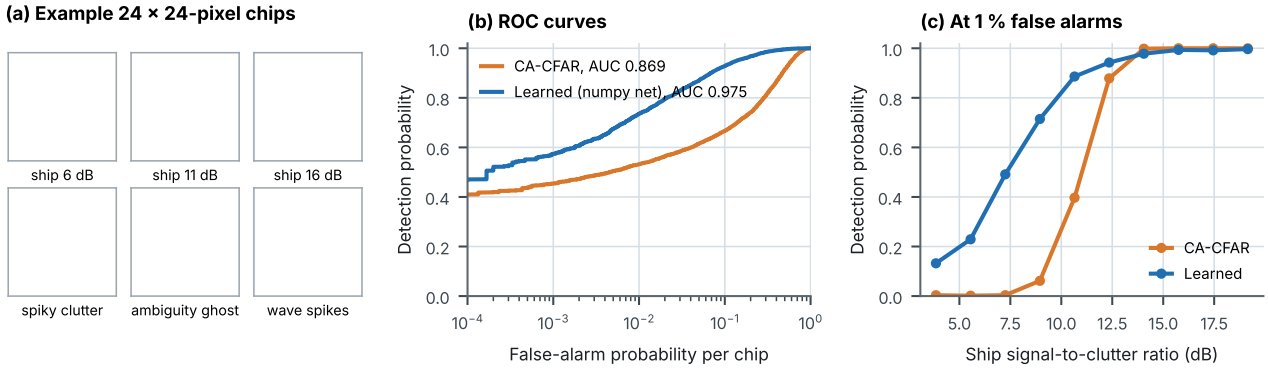
**Ship detection** in radar is the classic EO detection task. Ships are bright metal on a darker sea. The standard detector is the **constant false alarm rate** (CFAR) test. It estimates the local sea brightness $\hat{\mu}$ from a ring of pixels around each test cell and declares a target when the cell exceeds $T = \alpha\hat{\mu}$. For exponential clutter with a known mean, the false-alarm probability is $P_{fa} = e^{-\alpha}$, so $\alpha = -\ln P_{fa}$. Real sea clutter is spikier, so $\alpha$ must be larger.

CFAR flags anything bright. Breaking waves, azimuth “ghosts” of bright objects and small islands all cause false alarms. A learned classifier can look at shape. In Figure 46.6, chips of synthetic K-distributed sea clutter contain ships 2–7 pixels long or look-alikes. At a false-alarm rate of 1 % per chip, cell-averaging CFAR found 53 % of ships, and 286 of its 300 false alarms were ghosts or wave spikes. A small numpy network found 74 %. Its area under the receiver operating characteristic (ROC) curve, which sums up performance at all thresholds, rose from 0.869 to 0.975. Its advantage was largest for weak ships, 5–11 dB above the clutter. In practice the two are chained: CFAR proposes candidates and a network confirms them.




Synthetic 192 × 192-pixel scene: 34 fields between −17 and −5 dB, a −22 dB road and ten +8 dB point targets, multiplied by single-look exponential speckle. Learned filters see 9 × 9 log-intensity patches and were trained on 120 000 patches from three other scenes; the network (two hidden layers, 18 945 weights) is written in numpy.

**FIGURE 46.5** Speckle filtering of a synthetic single-look SAR scene. (a)–(d) Truth, speckled image, Lee filter and learned network (dB scale). (e) Equivalent number of looks in the outlined uniform region. (f) RMS error against the truth for all pixels and near edges.




Synthetic Sentinel-1-like chips: K-distributed sea clutter (shape 1.5), ships 2–7 pixels long at 3–20 dB above the clutter, and look-alikes (azimuth ghosts, breaking-wave spikes). Training: 12 000 chips with flips and rotations; test: 6 000 ships and 30 000 negatives. CA-CFAR: brightest 2 × 2 mean over the background ring mean.

**FIGURE 46.6** Ship detection in synthetic SAR chips. (a) Ships at three signal-to-clutter ratios and three look-alikes. (b) ROC curves: detection probability against false alarms per chip. (c) Detection probability against ship brightness at 1 % false alarms.

### 46.1.8 Spectra and Heights

**Hyperspectral images** record a full spectrum in each pixel (Chapter 14). Their pixels are often mixtures. The **linear mixing model** writes a pixel spectrum $\boldsymbol{x}$ as

$$\boldsymbol{x} = \sum_k a_k \boldsymbol{e}_k + \boldsymbol{\varepsilon}, \qquad a_k \geq 0, \quad \sum_k a_k = 1,$$

where $\boldsymbol{e}_k$ are pure "endmember" spectra, such as soil, grass and roof, and $a_k$ their area fractions. **Unmixing** solves for the fractions. Learned methods add non-linear mixing and use the spatial context. Classifiers for hyperspectral data must also cope with hundreds of correlated, noisy bands and few labels. Dimension reduction and foundation models pre-trained on unlabelled cubes help (Chapter 44).

**3-D reconstruction** turns stereo pairs, radar interferometry or lidar into heights. Learned stereo matching fills gaps in shadows and on uniform roofs. Networks also estimate building heights from single images. Microsoft's building data set includes height estimates for more than 170 million buildings.

## 46.2 Land: Crops, Forests, Cities and Water

On land, AI's main job is to make **wall-to-wall maps** that are updated often. Many are now free and global (Table 46.2).

**TABLE 46.2** Global land products made with machine learning (status October 2026).

| Product | Producer, first release | Input, resolution | Method | Reported quality or size |
|---|---|---|---|---|
| WorldCover | ESA, 2021 (2020 map) | Sentinel-1 and -2, 10 m, 11 classes | Supervised classifier on yearly band statistics | Overall accuracy 74.4 % (2020), 76.7 % (2021) |
| Dynamic World | Google, World Resources Institute, 2022 | Every Sentinel-2 image ≤ 35 % cloud, 10 m, 9 classes | Fully convolutional network; class probabilities | 73.8 % agreement with expert labels |
| WorldCereal | ESA consortium, 2023 (2021 maps) | Sentinel-1, -2, Landsat, 10 m | Gradient-boosted trees | Crop extent: user's 88.5 %, producer's 92.1 % |
| Global ML Building Footprints | Microsoft, 2022– | Bing Maps and Vexcel imagery, 2014–2025 | Segmentation network + polygon step | About 1.4 billion buildings; precision 92–97 % |
| Open Buildings v3 | Google, 2023 | 50 cm imagery; 58 million km² | Segmentation network | 1.8 billion buildings |
| Tree cover loss and alerts | Univ. of Maryland, Wageningen, Global Forest Watch | Landsat, Sentinel-2, Sentinel-1 | Decision trees, statistical change tests | Annual since 2000; weekly alerts |

### 46.2.1 Agriculture

**Crop-type maps** feed food-security outlooks, subsidy checks and insurance. In Europe, paying agencies use Sentinel time series to check farmers' declarations field by field. ESA's **WorldCereal** produced the first global, seasonal 10 m maps of temporary crops, maize, cereals and irrigation for 2021. Its crop-extent map had a user's accuracy of 88.5 % and a producer's accuracy of 92.1 %; crop types agreed with field data at least 80 % of the time. The maps were released in 2023 as open data and open code.

**Yield forecasting** is harder. Models combine vegetation indices through the season with weather, soil and past yields. Such models work well where long yield records exist and poorly where they do not, which is often where forecasts matter most. Field patterns like those in Figure 46.7 are easy for people to read but differ greatly between regions.

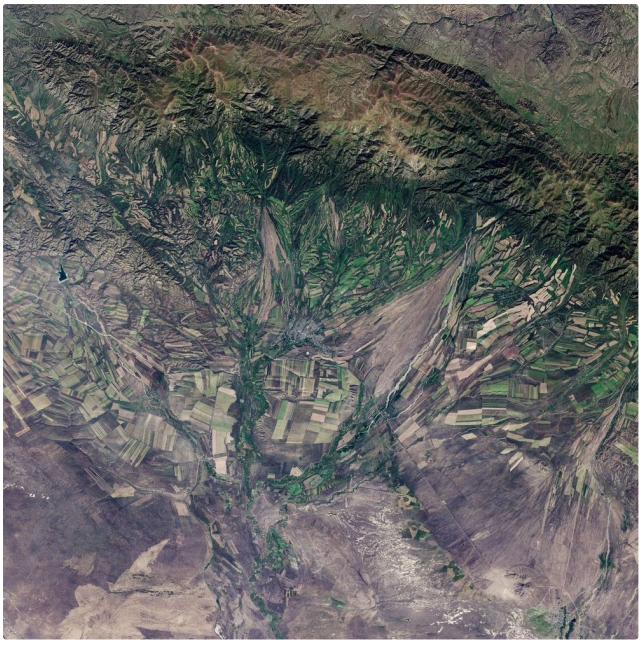


**FIGURE 46.7** Fields in eastern Kazakhstan seen by Landsat 8 on 9 September 2013. Some are harvested and bare, others still green — the kind of pattern a crop-type model learns from a season of images.
Image: NASA/GSFC.

**FIGURE 46.8** Forest loss in the Xingu basin, Brazil, from L-band radar: areas cleared before 1996 (black) and between 1996 and 2007 (red), from JERS-1 and ALOS-1 data. Radar alerts see through tropical cloud.
Image: NASA/JPL.

### 46.2.2 Forests and Deforestation Alerts

The University of Maryland's global forest-change maps processed about 654 000 Landsat scenes with decision trees in 2013. Updated every year, they are the basis of **Global Forest Watch**. Its 2025 data, released in April 2026, showed 4.3 million hectares of tropical primary forest lost. That was 36 % less than the record 6.7 million hectares of 2024, a year of extreme fires.

Annual maps are too slow for enforcement, so **alerts** were added. GLAD alerts flag new loss in each Landsat image at 30 m and later in Sentinel-2 at 10 m. The **RADD** alerts from Wageningen University use Sentinel-1 radar, which sees through the clouds that hide tropical forests for months (Figure 46.8). They cover about 50 tropical countries at 10 m, with a new radar image every 6–12 days. An alert is raised from a single new radar image. It is confirmed when the disturbance probability exceeds 97.5 % within 90 days. Global Forest Watch merges these streams into "integrated alerts" used by governments, companies checking supply chains and journalists.

### 46.2.3 Cities, Mines and Water

**Building footprints** are the base layer for population estimates, disaster planning and electrification. Microsoft has extracted about 1.4 billion footprints from aerial and satellite imagery, with regional precision of 92–97 % but recall as low as 71 % in Africa. Google's Open Buildings holds 1.8 billion detections across Africa, South and South-East Asia, Latin America and the Caribbean. Both are open and often fill gaps that community mapping has not reached.

**Mining** is a growing target. Amazon Mining Watch, from Earth Genome and partners, runs an ensemble of CNNs over Sentinel-2 to map gold mining in nine Amazon countries. Its August 2026 update reported about 99 % specificity and 90 % sensitivity, with maps updated every quarter

since 2025. **Water and wetlands** are mapped from long Landsat records and from radar, which separates open water, flooded vegetation and dry land.

## 46.3 Oceans and Maritime Awareness

At sea, the key question is "who is out there?" Ships are supposed to broadcast their identity and position through the **Automatic Identification System** (AIS). Many do not: their transponders are off, broken or not required. Satellite imagery finds these **"dark" vessels**, and machine learning makes it possible to do so for whole oceans.

> **MAPPING THE DARK FLEET (GLOBAL FISHING WATCH, 2024)**
>
> In January 2024, Paolo and colleagues at Global Fishing Watch published the first global map of industrial vessels and offshore structures in *Nature*. They processed about 2 PB of imagery from 2017–2021: 753 030 Sentinel-1 radar scenes for vessels and about 2.5 million Sentinel-2 scenes for infrastructure. A CFAR detector proposed objects. Three deep CNNs then confirmed vessels (about 97.5 % accurate), estimated their length and separated fishing from non-fishing vessels (90.5 % accurate). Detections were matched against 53 billion AIS messages.
>
> The study found 23.1 million vessel detections, about 63 300 vessels at sea at any moment. It concluded that 72–76 % of the world's industrial fishing vessels and 21–30 % of transport and energy vessels are not publicly tracked. It also counted about 28 000 offshore structures, with wind turbines more than doubling since 2017. Detection was above 70 % for 25 m vessels and above 90 % for vessels of 50 m or more. Smaller boats remain largely invisible to 10 m radar.

Before this, the **xView3** challenge (2021–22) had released close to 1 000 Sentinel-1 scenes labelled for dark-vessel detection. Models had to find vessels, estimate their length and say whether they were fishing. Commercial services now combine radar, optical, radio-frequency and AIS data. One example is Vantor's (formerly Maxar's) Maritime Sentry, which uses AIS-to-optical and radar-to-optical tipping. Fully automatic cueing between different companies' satellites is still mostly at the research stage.

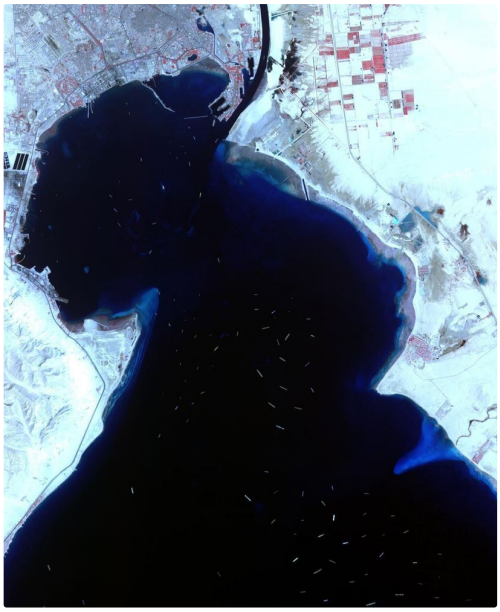


**FIGURE 46.9** More than 100 freighters waiting at the southern entrance of the Suez Canal on 4 April 2021, after the grounded Ever Given was freed, seen by ASTER on NASA's Terra satellite. Each bright dot is a ship — easy to count by eye here, but across whole oceans it is a job for detectors. Image: NASA/JPL.

Other maritime tasks follow the same pattern. **Oil spills** damp the small waves that scatter radar, so they appear as dark slicks. The European Maritime Safety Agency's CleanSeaNet service has screened radar images for spills since April 2007. It alerts national authorities within 30 minutes of the image. The difficulty is look-alikes: calm water, algae and rain cells also appear dark, so classifiers use shape, context and wind. **Sea ice** charts are still drawn largely by expert analysts from radar. ESA's AutoICE challenge (2023) showed that a U-Net combining Sentinel-1 with passive-microwave data could map ice concentration, type and floe size together (Chapter 47).

## 46.4 Disasters

In a disaster, speed matters more than perfection. The key idea is to deliver a good-enough map within hours of the first image, then improve it.

### 46.4.1 Flood Mapping

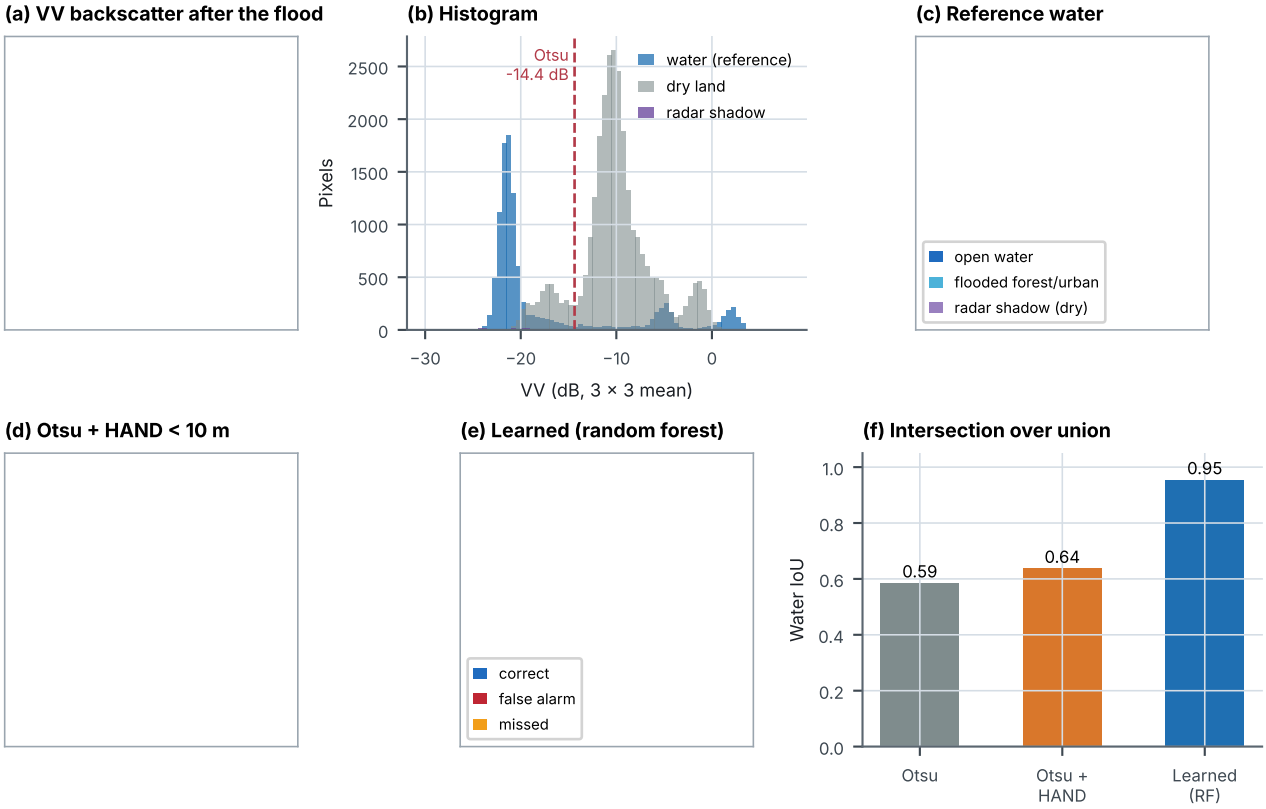
Radar is the main tool for floods because rain clouds hide the ground from optical sensors. Calm open water reflects radar away from the satellite, so it appears dark (Chapter 15). The traditional method picks a backscatter threshold, often with **Otsu's method**. It chooses the threshold $t$ that maximises the between-class variance of the histogram.

$$\sigma_B^2(t) = w_0(t)\, w_1(t)\, \left[\mu_0(t) - \mu_1(t)\right]^2$$

Here $w_0$ and $w_1$ are the fractions of pixels below and above $t$, and $\mu_0$ and $\mu_1$ their mean values. Errors come from dark land (smooth sand, tarmac, radar shadow), from flooded forests and towns that look *brighter* because of double bounce between water and trunks or walls, and from wind-roughened water.

The synthetic Sentinel-1-like scene of Figure 46.10 shows these effects. Otsu's threshold of −14.4 dB on VV backscatter (vertical polarisation sent and received) gave a water IoU of 0.59. Adding a **height above nearest drainage** (HAND) mask, which removes land more than 10 m

above the river, raised it to 0.64. It removed most radar-shadow false alarms but also some real ponds. A random forest that used both polarisations before and after the flood, plus HAND, reached 0.95. It found 1 836 of the 2 202 bright flooded-forest and flooded-urban pixels; the thresholds found almost none. Synthetic scenes flatter every method. Real scenes add wind, wet snow and errors in the labels themselves.




Synthetic 200 × 200-pixel Sentinel-1-like scene (VV and VH, before and after, about 4.4 looks) over a valley; floodwater fills land up to 2–5 m above the river (HAND, uneven), plus a few ponds; the HAND map given to the models carries ±1.5 m DEM errors. Look-alikes: smooth sand or tarmac, radar shadow on hills; flooded forest and buildings appear bright. Random forest: 150 trees, 8 features, trained on three other scenes.

**FIGURE 46.10** Flood mapping from a synthetic Sentinel-1-like scene. (a) Backscatter after the flood. (b) Histogram with Otsu's threshold; radar shadow is as dark as water. (c) Reference, including bright flooded forest and buildings. (d) Threshold plus HAND mask. (e) A random forest using both dates and polarisations. (f) Water intersection over union.

Operational systems already work this way. The Copernicus **Global Flood Monitoring** (GFM) service maps every new Sentinel-1 interferometric-wide-swath image automatically. It runs three independently developed algorithms, from LIST, DLR and TU Wien, and combines them by majority vote, with a likelihood layer for uncertainty. Since Sentinel-1A ended operations on 30 June 2026 after 12 years, GFM has used Sentinel-1C and -1D. The **Sen1Floods11** data set (2020) of 4 831 chips from 11 flood events showed a useful result: models trained on many automatically generated labels beat models trained on 446 hand-labelled chips.

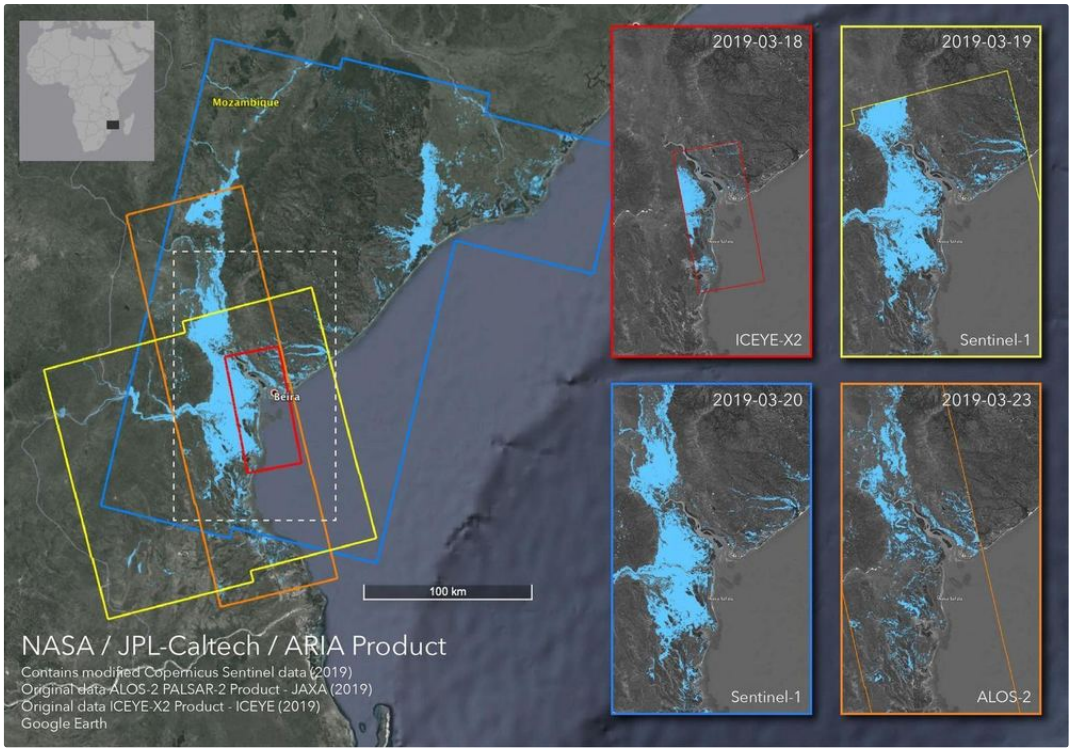



**FIGURE 46.11** Flood proxy maps of Mozambique after Cyclone Idai in March 2019, made by NASA JPL's ARIA team from ICEYE, Sentinel-1 and ALOS-2 radar images. Light blue marks likely floodwater; the panels track the flood from 18 to 23 March. Image: NASA/JPL (NASA/JPL-Caltech).

**EXAMPLE: HOW MUCH DOES ONE SENTINEL-1 PASS MAP?**

Sentinel-1 orbits at 693 km. In interferometric wide-swath mode its swath is 250 km. How much area does a 10-minute acquisition cover, and can a model keep up?

1. **Orbital speed.** $v = \sqrt{\mu/(R_E + h)} = \sqrt{398\,600/7\,064} = 7.51$ km/s.
2. **Ground speed.** $v_g = v\,R_E/(R_E + h) = 7.51 \times 6\,371/7\,064 = 6.77$ km/s.
3. **Area.** In 600 s the ground track advances 6.77 × 600 = 4 065 km. Times 250 km, that is about 1.02 million km$^2$ — roughly the area of Egypt.
4. **Pixels.** At 10 m spacing, $1.02 \times 10^{12}$ m$^2$ / 100 m$^2$ = $1.0 \times 10^{10}$ pixels per polarisation. Two polarisations at 2 bytes each are about 40 GB.
5. **Computing.** Cut into 512 × 512 tiles, that is about 38 800 tiles. At an assumed 50 billion operations per tile for a U-Net, the total is $1.9 \times 10^{15}$ operations. A graphics processor sustaining $30 \times 10^{12}$ operations a second needs about 65 s.

Inference is not the bottleneck. Downloading, calibrating and terrain-correcting the data take longer, and the satellite returns only every 6 days (12 days for each of the two satellites).

## 46.4.2 Rapid Mapping and Damage Assessment

Two coordination systems turn images into maps for responders. The **International Charter "Space and Major Disasters"**, in operation since 2000, lets disaster agencies request free imagery from member space agencies. It reached activation number 1 052 in August 2026; 2024 set a record of 85 activations. The **Copernicus Emergency Management Service** (CEMS) adds on-demand mapping. For the Myanmar earthquake of March 2025 it delivered 93 maps from 106 post-event images. Most rapid maps are still drawn or checked by analysts, with AI speeding up the steps.

**Building damage** is assessed by comparing images from before and after. Models trained on xBD grade each building on a four-level scale, from no damage to destroyed. After the February 2023 earthquakes in Türkiye, Microsoft's AI for Good Lab, Planet and the Turkish disaster agency AFAD used such a model on imagery from the first three days. They estimated that 3 849 buildings were damaged or destroyed in four cities. The World Food Programme and Google Research built the open **SKAI** tool. A UN evaluation over nine events found that it covered about seven times more area and gave first results about six times faster, in under a day. A later study comparing two companies' damage maps for the same disaster found that they disagreed in many places. Maps from different models should be treated as estimates, not facts.

Radar change detection gives a complementary view (Figure 46.12). It needs no cloud-free image and no labels, but it marks any change in the radar echo, not damage as such.

**FIGURE 46.12** Damage proxy map of eastern Puerto Rico after Hurricane Maria (September 2017), from Sentinel-1 images before and after landfall. Red and yellow pixels in the San Juan inset mark likely damage. Image: NASA/JPL (NASA/JPL-Caltech/ESA/Copernicus/Google).

**FIGURE 46.13** Thick smoke from the Grasshopper wildfire near Mount Hood, Oregon, photographed from the International Space Station on 31 July 2026. Constellations such as FireSat aim to detect fires far smaller than this one. Image: NASA/JSC.

## 46.4.3 Wildfires

Active fires are found by their heat at 3.9 µm (Chapter 14). New constellations add machine learning and speed (Figure 46.13). The German company **OroraTech** had 20 thermal satellites of its FOREST series in orbit by October 2026. Its satellites carry a commercial AI module that detects fires on board and sends compact alerts rather than images. The nonprofit Earth Fire Alliance's **FireSat**, built by Muon Space with Google, launched a prototype in March 2025. Its first three operational satellites followed in July 2026, on SpaceX's Transporter-17. They aim to detect fires as small as 5 m × 5 m, with at least twice-daily revisits of fire-prone regions. The stated goal is hourly global revisit by 2029. In both systems, learning helps separate real fires from hot roofs, sun glint and volcanoes.

These services are summarised in Table 46.3.

**TABLE 46.3** AI in disaster services.

| Service | Data | What AI does | Status (Oct 2026) |
|---|---|---|---|
| Copernicus GFM | Every Sentinel-1 IW image | Three-algorithm ensemble flood maps with likelihood | Operational, global |
| International Charter | Member agencies' satellites | Mostly analyst mapping; AI tools in some workflows | Operational; > 1 050 activations |

| Service | Data | What AI does | Status (Oct 2026) |
|---|---|---|---|
| Copernicus EMS rapid mapping | Commercial and Copernicus images | Analyst maps, partly automated | Operational, on demand |
| xBD-trained damage models (e.g. Microsoft, SKAI) | Sub-metre optical, before and after | Building-by-building damage grades | Used in many disasters; accuracy varies |
| FireSat | Six-channel infrared constellation | Fire detection from small hot spots | First 3 operational satellites launched 2026 |
| OroraTech FOREST | Thermal smallsats | Fire detection on board | Commercial service; 20 satellites |

## 46.5 Greenhouse Gases and Human Development

The key idea of this section is that machine learning now finds individual sources of methane from space, and estimates how much they emit. The same tools of image regression also map wealth and poverty.

### 46.5.1 Finding Methane Plumes

Methane traps about 80 times more heat than carbon dioxide over 20 years, but lasts only about a decade in the air. Stopping large leaks — "super-emitters" — is one of the quickest ways to slow warming. Methane absorbs sunlight near 1.65 and 2.3 µm. Imaging spectrometers detect it with a **matched filter**, which tests each pixel spectrum $\boldsymbol{x}$ against the expected absorption signature $\boldsymbol{t}$:

$$\hat{\alpha} = \frac{(\boldsymbol{x} - \boldsymbol{\mu})^{\top} \boldsymbol{\Sigma}^{-1} \boldsymbol{t}}{\boldsymbol{t}^{\top} \boldsymbol{\Sigma}^{-1} \boldsymbol{t}},$$

where $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ are the mean and covariance of the background spectra. The result $\hat{\alpha}$ is a map of methane enhancement. Bright surfaces, roads and some minerals produce false plumes, and checking every map by eye is slow.

Machine learning helps in two ways. CNNs separate real plumes from artefacts, and regression models estimate emission rates. Schuit and colleagues (2023) combined a CNN with a second classifier on TROPOMI data from Sentinel-5P. They found 2 974 plumes in 2021, with a mean source rate of about 44 t/h. In September 2026 Google and NASA JPL published MAPL-EMIT, a model for the EMIT spectrometer on the International Space Station. It was trained on 3.6 million physics-simulated plumes. It found about 50 % more plumes than human experts and more than 23 000 additional plumes worldwide.

**TABLE 46.4** Satellites used to find methane sources.

| Mission | Pixel size | Role | Status (Oct 2026) |
|---|---|---|---|
| TROPOMI (Sentinel-5P) | About 5.5 × 7 km | Daily global mapping; finds the largest plumes | Operating |
| GHGSat constellation | About 25 m | Commercial point-source monitoring | Operating; more than a dozen satellites |
| EMIT (ISS) | 60 m | Imaging spectrometer; mineral dust and methane | Operating |
| Tanager-1 (Carbon Mapper, Planet) | About 30 m | Methane and $CO_2$ point sources | Operating since 2024 |
| Sentinel-2, Landsat | 20–30 m | Multispectral; detect only very large plumes | Operating |

| Mission | Pixel size | Role | Status (Oct 2026) |
|---|---|---|---|
| MethaneSAT (EDF) | About 100 × 400 m | Area-wide emissions | Lost on 20 June 2025 |

DATA BOX — METHANE ALERTS IN NUMBERS

- **UNEP's Methane Alert and Response System (MARS)**, launched in November 2022, combines satellite data with AI-assisted analysis and notifies governments and operators. By July 2026 it had sent more than 5 000 alerts to 33 countries.
- **Response.** Only 13 % of alerts receive a reply from the operator or government, up from about 1 % in 2024. More than 40 verified mitigations have been recorded in 10 countries.
- **Tanager-1's first year** (September 2024 – September 2025): about 5 392 published methane plumes and 1 234 $CO_2$ plumes. Of the methane plumes, 2 661 came from oil and gas, 1 327 from waste and 1 211 from coal.
- **Loss of MethaneSAT.** Contact with MethaneSAT was lost on 20 June 2025, about 15 months after launch. It had been designed to map emissions across whole oil and gas basins.

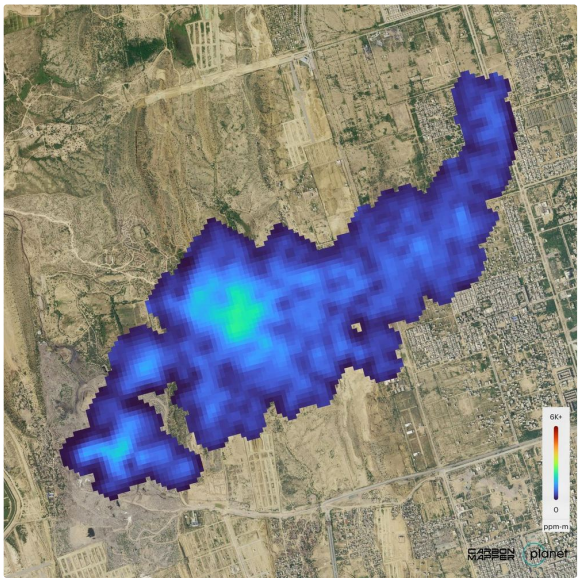



**FIGURE 46.14** The first methane plume seen by Tanager-1, on 19 September 2024 over Karachi, Pakistan: a plume about 4 km long from a landfill. Carbon Mapper's preliminary estimate was about 1 200 kg of methane an hour. Image: NASA/JPL (Carbon Mapper/Planet Labs PBC).

## 46.5.2 Estimating the Emission Rate

A plume shows how much methane is in the air, not how fast it is being released. The **integrated mass enhancement** (IME) method links the two. The IME is the total extra methane mass in the plume mask:

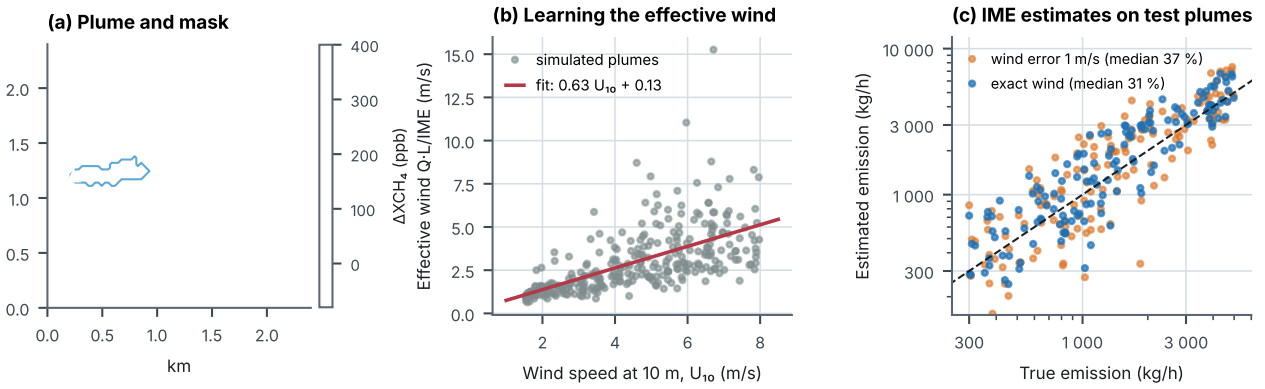
$$\mathrm{IME} = \sum_{i \in \mathrm{mask}} \Delta\Omega_i \, A_i ,$$

where $\Delta\Omega_i$ is the column enhancement in kg/m² and $A_i$ the pixel area. If the plume has length scale $L = \sqrt{A_{\mathrm{mask}}}$ and is carried by an effective wind $U_{\mathrm{eff}}$, the methane is renewed every $L/U_{\mathrm{eff}}$ seconds. The emission rate is then

$$Q = \frac{U_{\mathrm{eff}} \, \mathrm{IME}}{L} .$$

The effective wind is not the wind at 10 m, $U_{10}$. It is **learned** from simulated plumes, usually as a straight line $U_{\text{eff}} = \alpha U_{10} + \beta$. For Sentinel-2, published fits use $\alpha = 0.33$ and $\beta = 0.45$ m/s.

In Figure 46.15 this calibration is repeated with a simple plume model at 30 m pixels. A fit on 300 simulated plumes gave $U_{\text{eff}} = 0.63\,U_{10} + 0.13$ m/s; the coefficients depend on the sensor, the mask rule and the plume model. On 146 test plumes the median error was 31 % with the true wind, and 37 % when the wind had a 1 m/s error. A single estimate can be wrong by a factor of two, so repeated passes matter more than any one number.




Synthetic 30 m imagery (2.4 km × 2.4 km): puffy Gaussian plumes, 300–5 000 kg/h, winds 1.5–8 m/s, 50 ppb pixel noise plus correlated surface artefacts. Mask: smoothed image above 2σ, connected to the source. U_eff fitted on 300 plumes, tested on 146. Q = U_eff · IME / L, L = √(mask area).

**FIGURE 46.15** The IME method on synthetic 30 m methane images. (a) A 1 500 kg/h plume with its detection mask. (b) Effective wind needed to recover the true emission, and a straight-line fit. (c) Estimated against true emission for test plumes, with exact wind and with a 1 m/s wind error.

**EXAMPLE: A METHANE EMISSION RATE FROM A PLUME**

A Sentinel-2 image shows a plume from a gas well. The plume mask holds 1 250 pixels of 20 m. The mean column enhancement in the mask is 100 ppb (parts per billion of the total air column). The 10 m wind from a weather model is 3.5 m/s. Assume a dry air column of 101 325 Pa / 9.81 m/s² = 10 329 kg/m².

1. **Unit conversion.** The air column holds 10 329 / 0.028 97 = 356 500 mol/m². A 1 ppb enhancement is 356 500 × $10^{-9}$ × 0.016 04 kg/mol = 5.72 × $10^{-6}$ kg/m².
2. **Mask area and length.** A = 1 250 × 400 m² = 0.5 km², so L = √(5 × $10^5$) = 707 m.
3. **IME.** 100 × 5.72 × $10^{-6}$ kg/m² × 5 × $10^5$ m² = 286 kg.
4. **Effective wind.** $U_{\text{eff}}$ = 0.33 × 3.5 + 0.45 = 1.61 m/s.
5. **Emission rate.** Q = 1.61 × 286 / 707 = 0.649 kg/s ≈ 2 340 kg/h.
6. **Wind uncertainty.** With 2.5 or 4.5 m/s instead, Q is 1 860 or 2 820 kg/h, about ±20 %.

Sustained for a year, 2.3 t/h is about 20 500 t of methane. With a 20-year warming potential of about 80, that is about 1.6 million t of $CO_2$-equivalent — from one source.

## 46.5.3 Mapping Poverty

Surveys of household income are expensive and rare in many low-income countries. In 2016, Jean and colleagues showed in *Science* that a CNN could predict village wealth in five African countries from daytime satellite images. They trained it first to predict night-time lights, a cheap proxy for economic activity, and explained up to 75 % of the variation in surveyed wealth. In 2022 Chi and colleagues mapped relative wealth for all 135 low- and middle-income countries at 2.4 km resolution. They used satellite imagery with phone-network, connectivity and terrain

data. Trained on surveys from 56 countries, the estimates explained 56–70 % of the variation in held-out survey wealth. In 2020 the Government of Togo used such maps, with phone data, to direct emergency cash transfers. The maps are estimates with known errors. Used for targeting, they can wrongly exclude poor households, so they should be checked against local knowledge.

## 46.6 AI on Board and Smart Tasking

The key idea is to decide in orbit what is worth collecting and sending down, because downlink and imaging time are the scarce resources (Chapter 45). Two uses stand out: **screening** images on board, and **retasking** the satellite before it reaches a target.

### 46.6.1 Demonstrations so far

ESA's **Φsat-1** (2020) showed that a deep network on a vision chip drawing only a watt or two could discard cloudy images in orbit. Its successor, **Φsat-2** (2024), carried several uploadable apps, including vessel and wildfire detection (Chapter 45). The main EO demonstrations are listed in Table 46.5. Most are still demonstrations; few run as part of a paying service.

**TABLE 46.5** Onboard AI for Earth observation.

| Mission or system | Year | What the AI does on board | Status (Oct 2026) |
|---|---|---|---|
| Φsat-1 (ESA) | 2020 | Discards images more than 70 % cloudy | Completed demonstration |
| Satellogic NewSat with Palantir Edge AI | 2022 | Runs third-party detection models | Demonstration |
| Φsat-2 (ESA, Open Cosmos) | 2024 | Cloud, vessel, fire and anomaly apps; map generation; compression | Demonstration; apps updated in orbit |
| CogniSAT-6 with JPL dynamic targeting | 2025 | Looks ahead, skips cloudy targets | First flight test, July 2025 |
| OroraTech FOREST | 2022– | Fire detection and alert compression | Commercial service |
| Planet Pelican-4 | 2026 | Aircraft detection on raw images (about 80 % accurate) | Demonstration, March 2026 |
| Compressed Prithvi model on Kanyini and IMAGIN-e | 2026 | Geospatial foundation model run in orbit | Research demonstration |

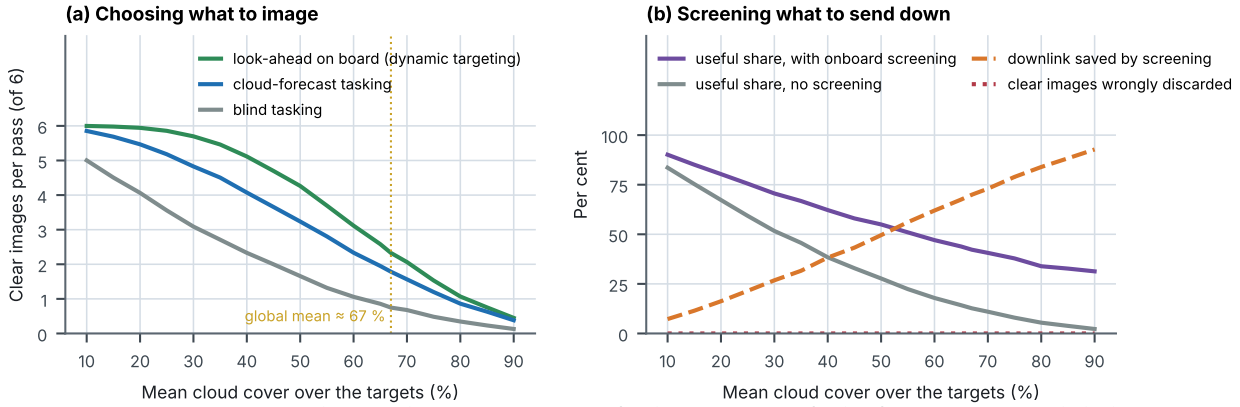
**DYNAMIC TARGETING ON COGNISAT-6 (2025)**

CogniSAT-6 is a 6U cubesat designed, built and operated by Open Cosmos. Its payload, from Ubotica, pairs a commercial AI processor with a hyperspectral camera from Simera Sense. It was launched in March 2024. In mid-July 2025, NASA JPL's **Dynamic Targeting** software flew on it for the first time. The satellite tilted 40–50° forward to image the ground about 500 km ahead. It analysed that image on board, judged whether the target was clear, and decided to image it or skip it — all in under 90 seconds and without human help. JPL's next steps are to hunt short-lived events such as storms, fires and eruptions, and to let a leading satellite pass targets to the ones behind it.

### 46.6.2 What Smart Tasking Is Worth

A satellite passing over a region can usually image only some of the possible targets. A Monte Carlo model in Figure 46.16 estimates the value of choosing well. A pass has 20 candidate targets and room for six images. An image is useful if it is less than 20 % cloudy. At a mean cloud cover of 67 %, blind tasking yields 0.75 clear images per pass. Tasking with a cloud forecast yields 1.78. A look-ahead camera on board yields 2.33, three times as many. At 30 % cloud the gains are smaller: 3.1, 4.8 and 5.7.

Screening after imaging is a different trade. It does not create clear images, but it stops cloudy ones from using the link. At 67 % cloud, keeping only images estimated under 50 % cloudy freed 70 % of the downlink. It raised the useful share of what was sent from 13 % to 42 %, and almost no clear image was lost.




Monte Carlo, 4 000 passes per point: 20 candidate targets per pass, room for 6 images; scene cloud fractions follow a U-shaped beta distribution, correlated along track (ρ = 0.6). An image is useful if under 20 % cloudy. Forecast skill: correlation 0.7 with the truth; look-ahead camera: 0.97. Screening keeps images estimated under 50 % cloudy (error 0.08).

**FIGURE 46.16** Clouds and onboard intelligence (Monte Carlo). (a) Clear images per pass from blind tasking, forecast-based tasking and an onboard look-ahead camera, against mean cloud cover. (b) Effect of onboard screening on the share of useful images sent down and on the downlink used.

**Tip-and-cue** extends the idea across a constellation. A wide-area sensor — a radar, a radio-frequency receiver or AIS — detects something, and a sharper sensor is tasked to look. A dark vessel found by radar can be imaged by an optical satellite on its next pass. Such chains are automated within single companies today. Between companies, and from satellite to satellite without a ground loop, they are mostly at the research stage.

## 46.7 The EO Analytics Business

Most commercial value lies not in pixels but in answers. Novaspace valued the EO data and services market at about $5 billion in 2023 and expected it to pass $8 billion by 2033, with services rising from $3.1 billion to $4.9 billion. Defence is one of the main drivers of growth. Planet, one of the largest commercial imaging operators, reported revenue of $307.7 million for the year to January 2026, up 26 %, with a backlog of over $900 million.

Three changes define the business in 2026 (Chapter 52). First, operators sell **subscriptions and analytics** rather than single images: alerts for new construction, vessel lists, crop statistics. Second, data are sold as **AI-ready products**: calibrated, co-registered time series with cloud masks and standard tiling. EarthDaily, for example, launched six satellites in May 2026 for a

constellation designed for broad-area change detection, with 22 spectral bands. Third, **foundation models** are moving into practice. Google's AlphaEarth embeddings summarise each 10 m pixel and year since 2017 as 64 numbers in Earth Engine. A small classifier on these embeddings can map a new class from relatively few labels. NASA and IBM's Prithvi models are fine-tuned for floods, burn scars and crops (Chapter 44).

**EXAMPLE: IMAGES PER DAY VERSUS ANALYSTS**

A constellation collects about 200 million km$^2$ a day in scenes of about 32.5 km × 19.6 km, as PlanetScope does. Suppose an analyst needs 5 minutes to check one scene for new buildings and works 7 hours a day.

1. **Scenes per day.** 32.5 × 19.6 = 637 km$^2$ per scene; 200 × 10$^6$ / 637 ≈ 314 000 scenes a day.
2. **Analyst capacity.** 7 × 60 / 5 = 84 scenes per analyst per day.
3. **Analysts needed.** 314 000 / 84 ≈ 3 740 analysts, every day, for one constellation and one question.
4. **With AI triage.** A change detector flags 2 % of scenes (6 280), and analysts also check a random 1 % of the rest (about 3 080) to watch for misses. That is about 9 360 scenes, or about 111 analysts.
5. **Pixels.** At 3 m, each scene holds about 71 million pixels per band, and the day about 2.2 × 10$^{13}$.

The model does not replace the analysts. It points their attention and lets them spend time on what matters. Its miss rate, measured on the random checks, sets how far its output can be trusted.

## 46.8 Pitfalls and Good Practice

EO models fail in predictable ways. The key idea is that an accuracy figure means little unless you know where, when and how it was measured (Table 46.6).

### 46.8.1 Domain Shift

A model trained in one region, season or sensor often fails in another. Soils, crops, building styles, sun angles and atmospheric haze all differ. In Figure 46.17 (a), a random forest trained on one synthetic region lost accuracy as the test region became drier and hazier: from 77.8 % to 53.2 % at the largest shift. Adding random spectral changes to the training data helped only slightly and cost accuracy at home. Adding 240 labelled pixels from the new region kept 67.5 %. A few local labels are usually worth more than clever tricks.

Shifts also come from the data chain. A change in processing, such as a new radiometric baseline, alters every pixel slightly (Chapter 45). Models should be monitored after deployment and checked whenever the data chain changes.

### 46.8.2 Labels, Imbalance and Validation

**Label noise** is common. Reference maps are out of date, photo-interpreters disagree, and labels from other models carry their errors along. **Class imbalance** is the rule: forest loss, ships and methane plumes are rare, so a model can look accurate while missing most of them (Chapter 43).

**Spatial autocorrelation** inflates accuracy. If training and test pixels come from the same fields, the test measures memory, not skill. Chapter 43 shows how random splits overstate accuracy compared with splits by region. Hold out whole regions, scenes or events.

**Geographic bias** follows from where labels exist. Many large labelled sets were built mainly from North America and Europe, so models often perform worst where data are needed most. Report accuracy by region, not just globally.

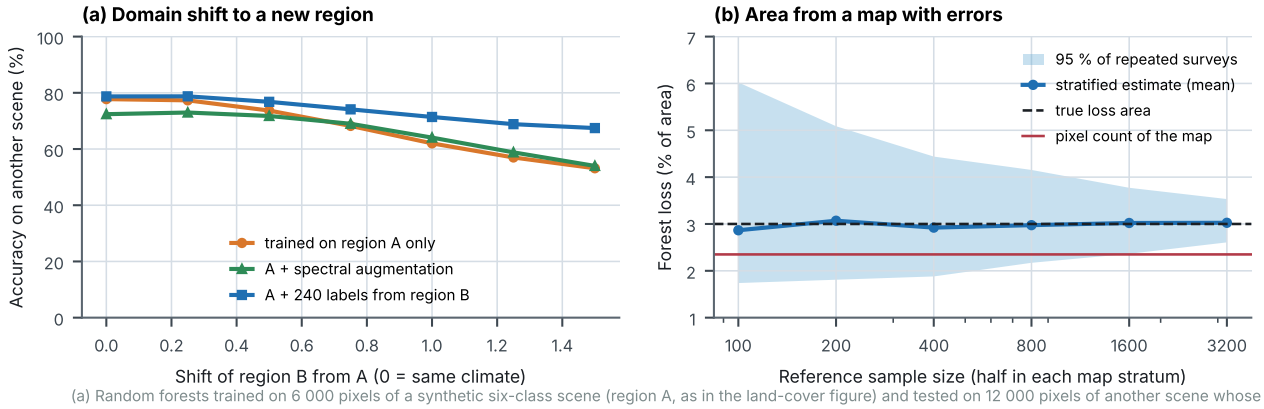
### 46.8.3 Getting the Area Right

A map is often used to report an **area**: hectares of forest lost, or square kilometres flooded. Counting mapped pixels is biased whenever omission and commission errors do not cancel. Good practice, set out by Olofsson and colleagues (2014), uses a **stratified random sample** of reference points, checked independently, inside each map class. The estimated proportion of true class $k$ is

$$\hat{p}_k = \sum_h W_h \frac{n_{hk}}{n_h}, \qquad \widehat{\mathrm{SE}}(\hat{p}_k) = \sqrt{\sum_h W_h^2 \frac{\hat{p}_{hk}(1-\hat{p}_{hk})}{n_h - 1}},$$

where $W_h$ is the mapped share of stratum $h$, $n_h$ the number of samples in it, $n_{hk}$ those found to be class $k$, and $\hat{p}_{hk} = n_{hk}/n_h$.

In Figure 46.17 (b), a map of a million pixels finds 65 % of 30 000 true loss pixels and adds 4 000 false ones. Counting gives 2.35 % loss, while the truth is 3.00 %. The stratified estimate is unbiased. With 3 200 reference points, 95 % of repeated surveys fell between 2.61 % and 3.53 %. With only 100 points, the reported confidence interval contained the truth in only 37 % of surveys. The few omission errors hidden in the large “no loss” stratum were usually missed by the sample. Small samples give confident-looking but wrong intervals.




(a) Random forests trained on 6 000 pixels of a synthetic six-class scene (region A, as in the land-cover figure) and tested on 12 000 pixels of another scene whose soils, vegetation and haze shift towards a drier climate (shift 1: soils up to 35 % brighter, plant NIR 25 % lower, more haze). (b) A $10^6$-pixel map finds 65 % of 30 000 loss pixels and adds 4 000 false ones; 400 simulated surveys per sample size, estimator of Olofsson et al. (2014).

**FIGURE 46.17** Two pitfalls, computed. (a) Accuracy of a land-cover random forest as the test region drifts away from the training region, with two remedies. (b) Forest-loss area from pixel counting and from stratified sampling (Olofsson et al. 2014), against reference sample size.

**TABLE 46.6** Common pitfalls of AI for Earth observation and how to avoid them.

| Pitfall | What goes wrong | Remedy |
|---|---|---|
| Domain shift | Accuracy falls in new regions, seasons or sensors | Train on varied data; add local labels; monitor drift |
| Label noise | Model learns the errors of its labels | Second interpreter on a sample; robust losses; clean the test set |
| Class imbalance | Rare events missed while accuracy looks high | Report recall, precision and PR curves; set thresholds for real prevalence |

| Pitfall | What goes wrong | Remedy |
|---|---|---|
| Spatial autocorrelation | Test scores overstated | Split by region, scene or event; use buffers |
| Geographic bias | Worst performance where data are scarcest | Collect labels in under-served regions; report by region |
| Biased area estimates | Pixel counts misreport change | Stratified reference sample; area-adjusted estimates with confidence intervals |
| Hallucinated detail | Generated images mistaken for data | Label synthetic outputs; never measure from them |

### 46.8.4 Explainability, Ethics and Privacy

**Explanations** help to catch shortcuts, such as a ship detector that keys on harbour walls rather than ships (Chapter 44). For maps, the most useful explanation is often simple: show the input images beside the output, with a per-pixel confidence.

**Privacy** matters more as resolution improves. At 30 cm, people cannot be recognised, but vehicles, crowds and the routines of a household can be. Combining imagery with other data can identify individuals. Licensing rules limit some collection; for example, the US rule on imagery of Israel was relaxed from 2 m to 0.4 m in 2020. Other ethical questions concern who controls alerts about people's land, and whether maps are used to help or to punish. Assurance and governance are treated in Chapter 53.

#### PRACTICE: BUILDING A FLOOD-MAPPING MODEL FROM SENTINEL-1

1. **Define the product.** Example: water extent at 20 m within 6 hours of each pass, scored by water-class IoU against hand-labelled chips, with a separate score for flooded vegetation.
2. **Get the data.** Use Sentinel-1 Ground Range Detected (GRD) images in VV and VH. Calibrate them to the backscatter coefficient $\sigma^0$ and terrain-correct them with the Copernicus digital elevation model (DEM). Add a pre-flood reference image from the same orbit, a permanent-water map and a HAND layer.
3. **Get labels.** Start from Sen1Floods11 and similar open sets. Add hand labels from your region, including urban and forest floods, and have a second person relabel 10 % to measure label noise.
4. **Set a baseline.** Run Otsu's threshold on VV (in dB) with the HAND and permanent-water masks. Record its IoU; every model must beat it.
5. **Train.** Fit a random forest on per-pixel features (VV, VH, their changes, HAND), then a small U-Net on 256 × 256 tiles. Use flips and rotations for augmentation, never mixing tiles from one event across splits.
6. **Validate by event.** Hold out whole floods from other continents and seasons. Report IoU, precision and recall per event, and for open water and flooded vegetation separately.
7. **Check failure cases.** Inspect wind-roughened lakes, wet snow, sand, radar shadow and dense cities, and add labels where errors cluster.
8. **Deploy and monitor.** Run on each new image automatically, publish a confidence layer, and compare with GFM and analyst maps after every major event. Retrain when the sensor or processing changes.

### KEY TAKEAWAYS

- EO analysis reduces to a few tasks — segmentation, detection, change, time series and retrieval — on images with many bands, a time axis and a location. Neighbourhood and time information often matter more than the choice of network.
- Simple baselines stay strong. Random forests, NDVI differences, Lee filters, CFAR and Otsu thresholds should be beaten, by a measured margin, before a deep model is adopted.
- Operational AI products now map land cover, buildings, crops and forest loss globally, track dark vessels, map floods from every Sentinel-1 image and find methane plumes from orbit.
- Emission rates, flood areas and loss areas are estimates with real uncertainty: the IME method's single-plume errors are often 30–40 %, and pixel counts of maps are biased.
- Onboard AI is moving from demonstration to service. Look-ahead tasking can triple the clear images collected in cloudy regions, and screening frees much of the downlink.
- Most failures come from the data: domain shift, label noise, imbalance, spatially correlated validation and geographic bias. Split by region, sample properly and report uncertainty.

### REVIEW QUESTIONS

1. Name the task type, a suitable model and a metric for: counting aircraft at airports; mapping burned area; classifying crop types from a season of images; estimating forest biomass.
2. In Figure 46.2, why did a random forest given 5 × 5 neighbourhoods gain almost nine points over the same forest on single pixels? Which classes gained most, and why?
3. Explain why an NDVI-difference threshold flags harvested fields as deforestation. What extra information did the learned detector of Figure 46.3 use?
4. Write the temporal-attention weights for a time series in which two of five dates are cloudy. Why does attention not need gap filling?
5. Derive the CFAR multiplier $\alpha$ for exponential clutter and $P_{fa} = 10^{-6}$. Why do azimuth ghosts and breaking waves still cause false alarms?
6. Why can the equivalent number of looks reward a bad speckle filter? Which other measures would you report?
7. A flooded forest appears brighter in radar than the same forest when dry. Explain why, and how a flood-mapping model can still classify it as water.
8. A methane plume has an IME of 500 kg over a mask of 1 km$^2$. The 10 m wind is 5 m/s. Using $U_{\text{eff}} = 0.33\,U_{10} + 0.45$ m/s, estimate the emission rate in kg/h, and its range if the wind is uncertain by ±1.5 m/s.
9. Using Figure 46.16, estimate the number of clear images per pass at 55 % mean cloud cover for each tasking strategy. When would onboard screening be a poor choice?
10. A forest-loss map shows 2 000 km$^2$ of loss. A stratified sample shows that 20 % of mapped loss is false and that 1 % of the mapped "no loss" area of 98 000 km$^2$ is real loss. Estimate the true loss area and explain why pixel counting failed.

CHAPTER

# 47 AI for Weather, Climate and Space Weather

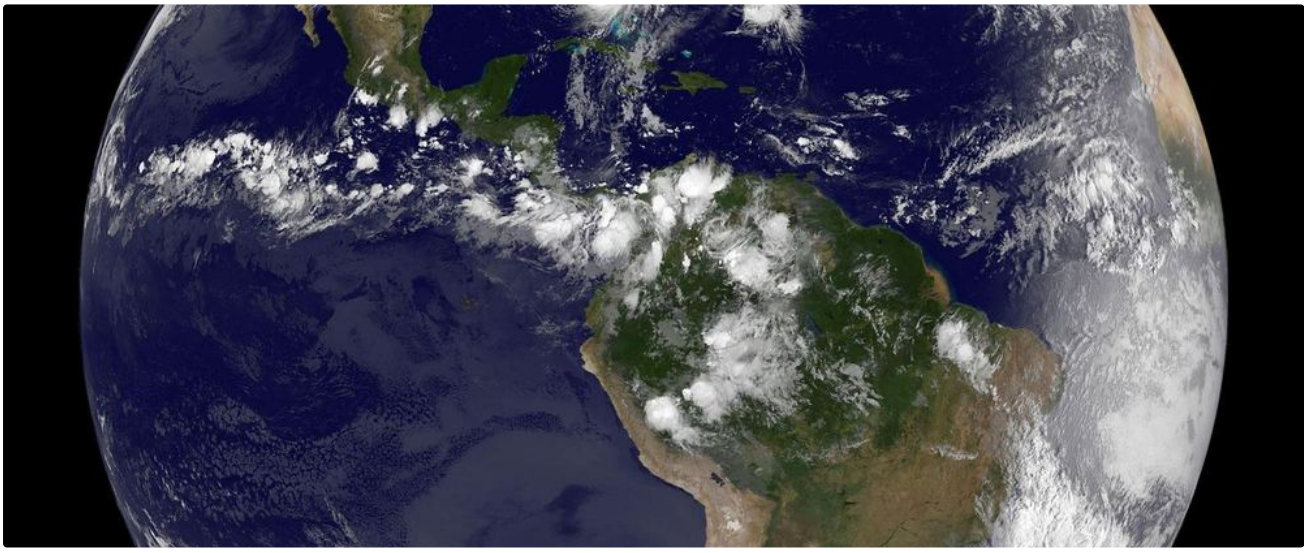


Image: NASA/GSFC

*Between 2022 and 2026 machine learning changed weather forecasting faster than any idea since the first computer forecasts. This chapter explains how. It recalls how physics models and data assimilation turn satellite data into forecasts, then follows the learned models from the first research papers to operational service at ECMWF, NOAA and other centres. It shows how they are built and trained, how their skill is measured fairly, and where they still fall short. It covers machine learning inside the observing chain, nowcasting, climate emulators, downscaling and digital twins. It ends with space weather: flares, coronal mass ejections, geomagnetic storms and the slow road to operations. Small models trained in Python make each idea concrete.*

IN THIS CHAPTER

47.1 From Physics to Learning
47.2 The Machine-Learning Weather Revolution, 2022–26
47.3 How Learned Weather Models Work
47.4 Measuring Skill Fairly
47.5 Machine Learning in the Observing Chain
47.6 Climate: Emulators, Downscaling and Digital Twins
47.7 Space Weather

Weather forecasting is one of the largest users of satellite data. It is also where machine learning has changed operational practice fastest. In 2022 a learned weather model was a research curiosity. By October 2026 learned models were running every day at the European Centre for Medium-Range Weather Forecasts (ECMWF), the US National Oceanic and Atmospheric Administration (NOAA), Germany's weather service and China's. A learned model used at the US National Hurricane Center slightly beat the official track forecasts of 2025 at some lead times.

These models did not appear from nowhere. They learned from decades of **reanalysis**, the best reconstruction of past weather, made by physics models that absorbed billions of satellite observations. They still start from analyses made the same way. The hero image above shows Hurricane Matthew nearing Florida on 7 October 2016, seen by NOAA's GOES-East satellite. Images like this one are the raw material of every learned forecast.

This chapter builds on Chapter 16 (weather satellites and forecasting), Chapters 22 and 23 (the space environment) and Chapters 43 and 44 (learning methods). Data and computing are covered in Chapter 45, and AI for Earth observation in Chapter 46.

## 47.1 From Physics to Learning

The key idea of this section is that a forecast needs two things: a good picture of the atmosphere now, and a good rule for how it changes. For 70 years both came from physics. Machine learning first replaced the rule, and is now starting on the picture.

### 47.1.1 The Forecasting Chain

A modern weather service is a pipeline (Figure 47.1). Observations come from satellites, weather balloons (radiosondes), aircraft, ships, buoys, radars and ground stations. They are checked, corrected for instrument biases and thinned. **Data assimilation** then combines them with a short-range forecast, the **background**, to make the **analysis**: the best estimate of the whole atmosphere at one moment. A **numerical weather prediction (NWP)** model then steps the analysis forward in time by solving the equations of fluid flow, heat and moisture. Finally, the output is corrected statistically and turned into products and warnings.

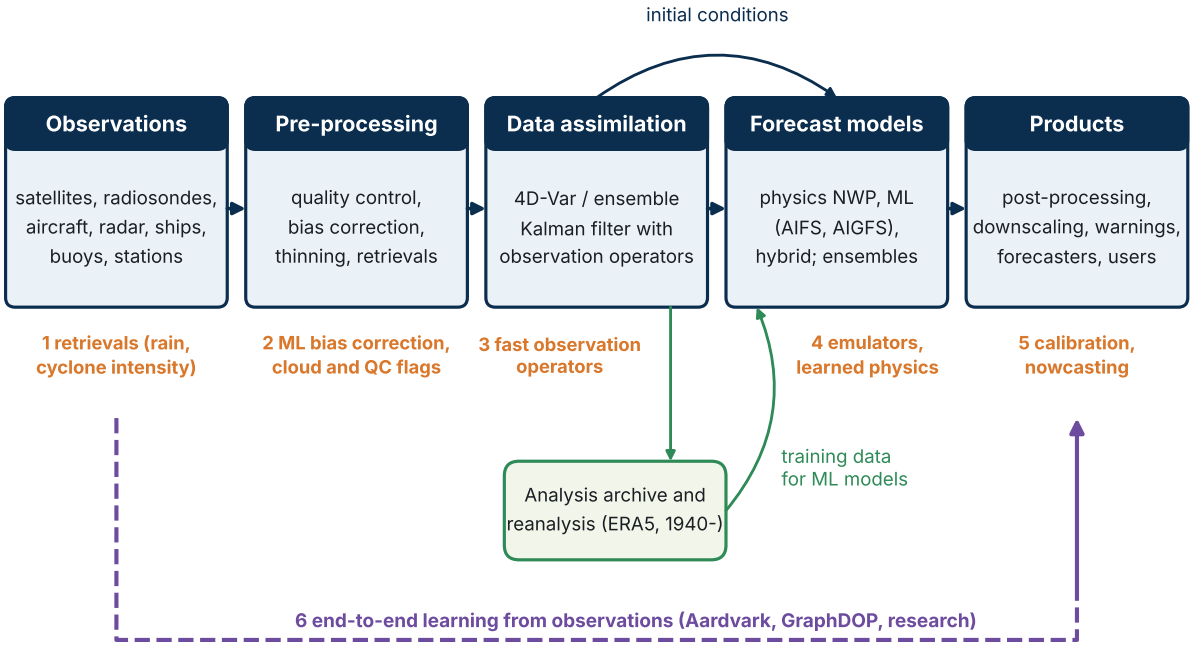



**FIGURE 47.1** The weather forecasting chain and the places where machine learning enters it. Learned models trained on reanalysis now run alongside physics models (4); end-to-end systems that learn directly from observations (6) are still research.

Variational data assimilation finds the atmospheric state $\mathbf{x}$ that best fits both the background $\mathbf{x}_b$ and the observations $\mathbf{y}$. It does this by minimising a cost function:

$$J(\mathbf{x}) = \tfrac{1}{2}(\mathbf{x}-\mathbf{x}_b)^{\mathsf{T}}\mathbf{B}^{-1}(\mathbf{x}-\mathbf{x}_b) + \tfrac{1}{2}\big(\mathbf{y}-H(\mathbf{x})\big)^{\mathsf{T}}\mathbf{R}^{-1}\big(\mathbf{y}-H(\mathbf{x})\big).$$

Here $\mathbf{B}$ and $\mathbf{R}$ are the error covariances of the background and of the observations, and $H$ is the **observation operator**. For a satellite sounder, $H$ is a radiative-transfer model that computes the radiances the instrument would see through the model atmosphere (Chapter 16). The four-dimensional version, **4D-Var**, fits observations spread over a 6–12-hour window. Every term in this equation is a place where learning can help, as the section on the observing chain shows.

### 47.1.2 Reanalysis: the Training Set

A **reanalysis** runs one fixed assimilation system over decades of archived observations. Because the system never changes, the record is consistent in time. ECMWF's **ERA5**, produced for the European Union's Copernicus Climate Change Service, covers 1940 to the present. It gives hourly fields on a 0.25° grid (about 31 km) with 137 levels from the surface to about 80 km, and many products on 37 pressure levels. Its successor, ERA6, is now in production at about 14 km. Its first decades are due for release towards the end of 2027.

Almost every learned global model has been trained on ERA5. A typical training set holds 1979–2017 or 1979–2018 at 6-hour intervals: about 57 000 global states, each with about a million grid points and tens to hundreds of variables. Later years are kept for testing. A learned model can only be as good as the reanalysis it imitates: it inherits ERA5's errors, its smoothing of small scales and its weak points, such as rainfall.

### 47.1.3 Chaos Sets the Limit

The atmosphere is **chaotic**: small errors grow until the forecast is no better than a guess from climate statistics. Errors in the large-scale flow double in about one to two days, so medium-range forecasts lose most of their skill after 10–15 days. A better model or analysis can push that limit outwards, but cannot remove it.

## 47.2 The Machine-Learning Weather Revolution, 2022–26

The key idea of this section is that within three years learned models went from first papers to running beside physics models in operations. They matched or beat the best physics models on most standard scores, at a tiny fraction of the computing cost.

### 47.2.1 Landmark Models

Table 47.1 lists the models that set the pace. **FourCastNet** (NVIDIA and partners, February 2022) produced a week-long global forecast at 0.25° in under two seconds. It matched ECMWF's physics model at short lead times for large-scale variables, but its skill fell behind at longer range. Still, it showed what was possible. **Pangu-Weather** (Huawei, published in *Nature* in July 2023) was the first learned model reported to beat ECMWF's deterministic forecast on its main scores. It used a transformer that treats pressure levels and surface fields as one three-dimensional volume.

**GraphCast** (Google DeepMind, *Science*, November 2023) was more accurate than ECMWF's high-resolution forecast (HRES) on 90 % of 1 380 verification targets. It made a 10-day forecast in under one minute on a single Google TPU v4 chip.

**TABLE 47.1** Landmark machine-learning global weather models. Ranges and grids are those of the first published versions.

| Model | Developer, year | Architecture | Grid and step | Range | Status, October 2026 |
|---|---|---|---|---|---|
| FourCastNet | NVIDIA and partners, 2022 | Fourier neural operator | 0.25°, 6 h | About 1 week | Research; later versions in NVIDIA Earth-2 |
| Pangu-Weather | Huawei, 2023 | 3D transformer | 0.25°, 13 levels | 7 days and more | Research; run experimentally at ECMWF until May 2026 |
| GraphCast | Google DeepMind, 2023 | Graph network on a multi-mesh | 0.25°, 37 levels, 6 h | 10 days | Open; basis of NOAA's AIGFS |
| FengWu | Shanghai AI Laboratory and partners, 2023 | Multi-modal transformer | 0.25°, 37 levels | Useful to 10.75 days | Research |
| FuXi | Fudan University, 2023 | Cascade of three transformers | 0.25°, 6 h | 15 days | Research |
| NeuralGCM | Google, 2024 | Physics core plus learned physics | 2.8°–0.7° | Days to decades | Research; open |
| GenCast | Google DeepMind, 2024 | Diffusion ensemble | 0.25°, 12 h | 15 days, 50 members | Research; followed by WeatherNext 2 (2025) |
| Aurora | Microsoft, 2025 | 1.3-billion-parameter foundation model | 0.25° and 0.1° | 10 days; air quality, waves | Open; run experimentally at ECMWF until May 2026 |
| AIFS Single and ENS | ECMWF, 2024–25 | Graph encoder and decoder, transformer processor | 28–31 km, 6 h | 15 days; ENS 51 members | Operational since February and July 2025 |
| Aardvark Weather | Cambridge, Turing Institute, Microsoft, ECMWF, 2025 | End to end from observations | 1.41°, 5 levels | 10 days | Research |

Two further steps followed. **GenCast** (Google DeepMind, *Nature*, December 2024) produced ensembles instead of single forecasts. It was more skilful than ECMWF's physics ensemble (ENS) on 97.2 % of 1 320 targets, and on 99.8 % beyond 36 hours. Each 15-day member took about 8 minutes on one TPU v5 chip. **NeuralGCM** (Google, 2024) kept a differentiable physics core for the large-scale flow and learned only small-scale processes such as clouds. It stays stable over decades, so it can also simulate climate.

Foundation models followed (Chapter 44). **Aurora** (Microsoft, *Nature*, May 2025) was pre-trained on over a million hours of weather and climate data, then fine-tuned for forecasts at 0.1°, air pollution, ocean waves and cyclone tracks. In November 2025 Google DeepMind released

**WeatherNext 2**. Its **functional generative network** injects random noise inside the network, so each run gives a different but realistic forecast. It makes hundreds of scenarios in under a minute on one TPU and drives the forecasts in Google's consumer products.

### 47.2.2 Into Operations

The decisive change came when weather centres put learned models into daily service (Table 47.2). "Operational" means run every day on a fixed schedule, monitored, and delivered to users with a service commitment. In every case the learned model runs **beside** the physics model, not instead of it.

ECMWF built its own model (see the box at the end of this section); NOAA started from an outside one. Through its Project EAGLE, it fine-tuned GraphCast on analyses from its own assimilation system. Of its three models of December 2025, **AIGFS** is a single forecast with better cyclone tracks but weaker intensity than the physics GFS. **AIGEFS** is a 31-member learned ensemble that needs about 9 % of the computing of the physics ensemble. **HGEFS** joins 31 physics and 31 learned members into one 62-member "grand ensemble" — the first such hybrid in operations, NOAA believes.

**TABLE 47.2** Machine-learning global models at national and international centres, October 2026.

| Centre | System | In service | How it was built | Notes |
|---|---|---|---|---|
| ECMWF (Europe) | AIFS Single | 25 Feb 2025 | Own graph–transformer model, Anemoi software | 28 km, 4 runs a day to 15 days; open data; v2 on 12 May 2026 added waves |
| ECMWF | AIFS ENS | 1 Jul 2025 | Trained on CRPS | 50 members plus control, 31 km |
| NOAA (USA) | AIGFS, AIGEFS, HGEFS | Dec 2025 | GraphCast fine-tuned on NOAA analyses | 16-day forecast in about 40 min with 0.3 % of GFS computing; 31-member AI ensemble; 62-member hybrid |
| DWD (Germany) | AICON | Runs since Sep 2025; fully operational 2 Mar 2026 | Graph network built with Anemoi | About 13 km, to 180 h |
| CMA (China) | Fengqing | 2025 | With Tsinghua University | Rain skill comparable to physics models to 10 days |
| Met Office (UK) | FastNet | Not yet | With the Alan Turing Institute | Research; similar RMSE to the Met Office global model (2026) |

### 47.2.3 Learning Directly from Observations

All the models above start from an analysis made by physics-based assimilation. The next step is to learn the whole chain. **Aardvark Weather** (Cambridge, the Alan Turing Institute, Microsoft Research and ECMWF; *Nature*, 2025) takes in raw data from scatterometers, sounders, geostationary imagers, stations, ships and radiosondes. It outputs forecasts on a 1.41° grid and for individual stations in about one second on four GPUs; ECMWF's physics system needs about 1 000

node-hours for assimilation and forecast. Using about 8 % of the observations that operational systems ingest, it beat NOAA's GFS on many variables but trailed ECMWF's HRES in the upper atmosphere and at short range.

ECMWF's **GraphDOP** (2024–25) learns from satellite and in-situ observations alone and was skilful to about five days in its first version. A Google DeepMind preprint of September 2026, **WeatherNext 3**, ingests low-latency geostationary images and predicts station temperatures anywhere. These systems are research, but they could free learned forecasting from its dependence on physics-based analyses.

LANDMARK: ECMWF'S AIFS GOES OPERATIONAL (2025–26)

- **25 February 2025.** ECMWF put its **Artificial Intelligence Forecasting System (AIFS)** into operations, the first data-driven global model in service at a major international centre. It runs at 28 km beside the 9 km physics model (IFS). ECMWF reported gains of up to 20 % on some scores, including cyclone tracks, and about 1 000 times less energy per forecast.
- **Design.** A graph network maps the input grid onto an internal grid of about 1°, a transformer steps it forward 6 hours, and a decoder maps it back. Pre-trained on four decades of ERA5 and fine-tuned on operational analyses, it trained in about a week on 64 GPUs. A 10-day forecast takes about 2.5 minutes on one GPU.
- **1 July 2025. AIFS ENS** followed: a 51-member ensemble trained on the continuous ranked probability score. ECMWF reported gains over the physics ensemble of up to 20 % for surface temperature.
- **12 May 2026.** Version 2 of both arrived with the physics model's new cycle, 50r1, and added ECMWF's first operational data-driven wave forecasts. Both were re-tuned on analyses from the new cycle. Started from the new analyses, version 1 had lost skill, with 2 m temperature errors up to 20 % larger.
- **Open by design.** AIFS forecasts are free open data (CC-BY-4.0). Its training software, Anemoi, is shared with national services; Germany's AICON is built on it.

## 47.3 How Learned Weather Models Work

The key idea of this section is that a learned weather model is a function that maps today's state to the state a few hours later. It is trained by comparing millions of its predictions with reanalysis, then applied again and again to reach the desired lead time.

### 47.3.1 The Forecast as a Learned Time Step

Let $\mathbf{X}_t$ be the global state at time $t$: temperature, wind, humidity and geopotential on every level, plus surface fields. A learned model $f_\theta$ with parameters $\theta$ predicts the change over one step $\Delta t$, usually 6 hours:

$$\hat{\mathbf{X}}_{t+\Delta t} = \mathbf{X}_t + f_\theta(\mathbf{X}_t, \mathbf{X}_{t-\Delta t}).$$

A 10-day forecast applies the model 40 times, each time feeding its own output back in. This is called an **autoregressive roll-out**. Predicting the change rather than the full state makes learning easier, because most of the atmosphere changes little in 6 hours.

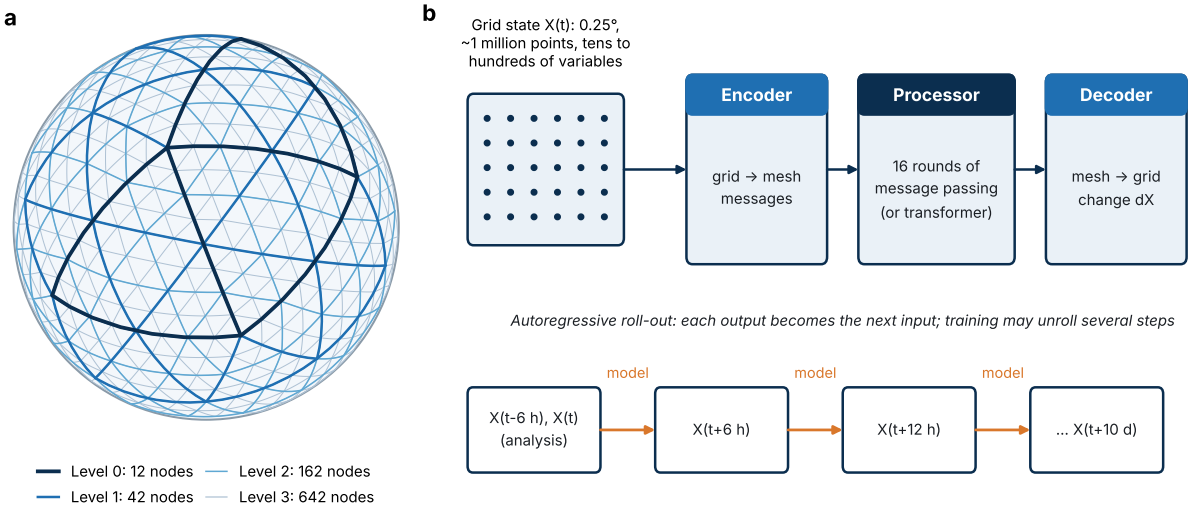
Training minimises a weighted mean squared error between the model output and reanalysis over many past dates:

$$\mathcal{L}(\theta) = \frac{1}{N} \sum_{\text{dates}} \sum_{j=1}^{J} \sum_{i,v,\ell} a_i \, w_v \, w_\ell \left( \hat{x}^{(j)}_{i,v,\ell} - x^{(j)}_{i,v,\ell} \right)^2 .$$

Here $i$ labels grid points, $v$ variables and $\ell$ levels. The **area weight** $a_i \propto \cos(\text{latitude})$ stops the crowded grid points near the poles from dominating. The weights $w_v$ and $w_\ell$ set how much each variable and level matter. The sum over $j$ covers several roll-out steps: fine-tuning on 2–12 steps teaches the model to correct its own errors, but also makes its forecasts smoother.

### 47.3.2 Graphs, Transformers and Meshes

The network inside $f_\theta$ must let information travel across the globe. **Graph neural networks** (Chapter 44) pass messages between the nodes of a mesh. GraphCast and AIFS use an **encode–process–decode** structure (Figure 47.2). An encoder maps the latitude–longitude grid onto an internal mesh, a processor exchanges messages along its edges many times, and a decoder maps the result back as the predicted change. GraphCast's **multi-mesh** keeps the edges of every refinement level of an icosahedron, from 12 nodes up to 40 962. Long edges carry information quickly across the globe; short edges resolve local detail.




(a) Computed by repeated subdivision of an icosahedron; a multi-mesh keeps the edges of every level, so messages travel both short and long distances (GraphCast uses levels 0-6, 40 962 nodes).

**FIGURE 47.2** (a) An icosahedral multi-mesh computed by repeated subdivision: each level splits every triangle into four, and the multi-mesh keeps the edges of all levels. (b) The encode–process–decode structure of a graph weather model and its autoregressive roll-out.

**Transformers** (Chapter 44), used by Pangu-Weather, FuXi and Aurora, let patches of the globe attend to each other, usually within local windows to save memory. **Neural operators**, such as FourCastNet's, learn in Fourier or spherical-harmonic space. Once trained on enough data, all three designs reach similar skill; they differ mainly in memory and speed.

### 47.3.3 Ensembles: Diffusion and CRPS Training

A single forecast trained on mean squared error has a known flaw. When the future is uncertain, the forecast that minimises squared error is the **average of all possible futures**. An average is smooth: a storm that may go left or right is forecast as a weak storm in the middle. The fix is to forecast a probability distribution, as physics ensembles do.

Two learned approaches are in use. **Diffusion models** (Chapter 44), as in GenCast, learn to turn random noise into a realistic weather state, step by step, conditioned on the current state. Each fresh noise sample gives another ensemble member. **CRPS training**, used by AIFS ENS and WeatherNext 2, adds noise inside the network and directly optimises the **continuous ranked probability score** of the ensemble.

$$\mathrm{CRPS}(F, y) = \mathbb{E}\,|X - y| - \tfrac{1}{2}\,\mathbb{E}\,|X - X'|$$

Here $X$ and $X'$ are independent members drawn from the forecast distribution $F$ and $y$ is what happened. The first term rewards members close to the truth. The second rewards spread between members, so the ensemble does not collapse to one forecast. For one deterministic forecast the CRPS reduces to the absolute error.

### 47.3.4 A Learned Weather Model in Miniature

The ideas above can be tried on a small chaotic system. The **Lorenz-96 model** is a ring of 40 variables that behave like a weather quantity around a latitude circle.

$$\frac{dx_k}{dt} = (x_{k+1} - x_{k-2})\,x_{k-1} - x_k + F, \qquad k = 1, \ldots, 40$$

The quadratic term moves "weather" around the ring, $-x_k$ damps it and $F = 8$ drives it, giving chaos. By a common convention, 0.05 time units count as 6 hours.

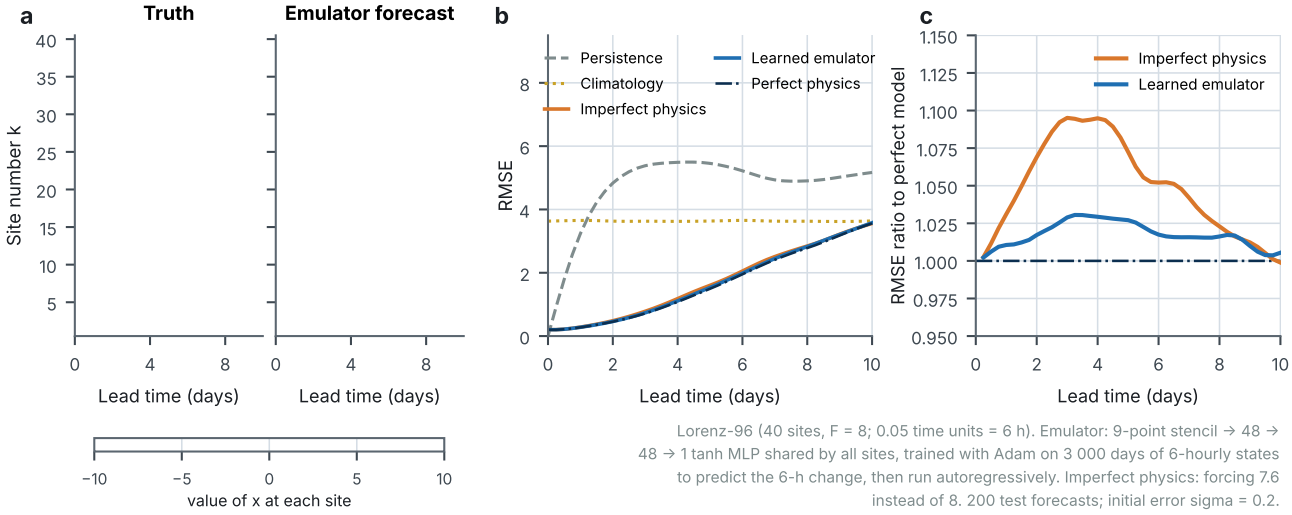
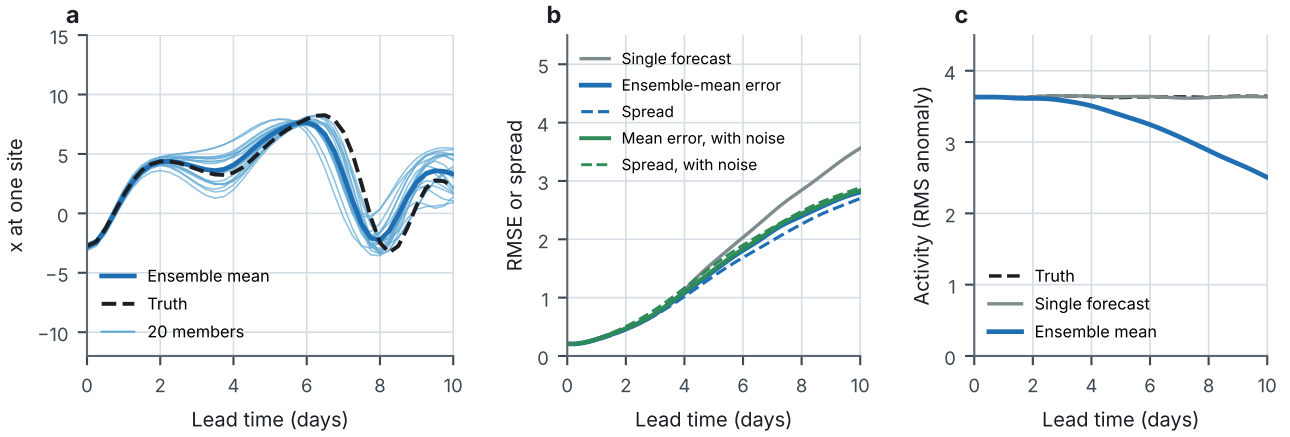



**FIGURE 47.3** A learned emulator of the chaotic Lorenz-96 system. (a) Truth and a 10-day emulator forecast as space–time diagrams. (b) Error growth for 200 forecasts: the emulator (blue) tracks the perfect physics model, beats a physics model with a 5 % forcing error (orange) and far outperforms persistence. (c) Error relative to the perfect model.

We generated 3 000 days of 6-hourly "reanalysis" and trained a small neural network to predict the 6-hour change at each site from the values at that site and its eight nearest neighbours. Like a graph model, the same network serves all sites. It has 2 881 parameters. We then ran 200 forecasts from initial states with small errors, as if from an imperfect analysis (Figure 47.3). The learned emulator's error grew almost exactly like that of the perfect physics model: 0.74 against 0.71 after 3 days and 1.53 against 1.49 after 5 days. It beat a physics model with a 5 % error in its forcing, which reached 0.78 and 1.60. All three reached half the climatological spread after 5.5–5.7 days. Persistence — forecasting no change — was worse than climatology after just over a day. Chaos, not the model, set the limit.

The same emulator shows how ensembles help (Figure 47.4). Twenty members started from perturbed analyses spread out as errors grew. After 10 days the ensemble-mean error was 2.81, against 3.56 for a single forecast. The **spread** — the scatter of the members — was 2.70. A spread close to the error of the ensemble mean means the ensemble is **well calibrated**: its uncertainty is honest. With perturbations only at the start, the spread was 4–7 % too small. Adding a little noise at each step, a crude stand-in for model error, brought spread and error to within 2–7 % of each other.



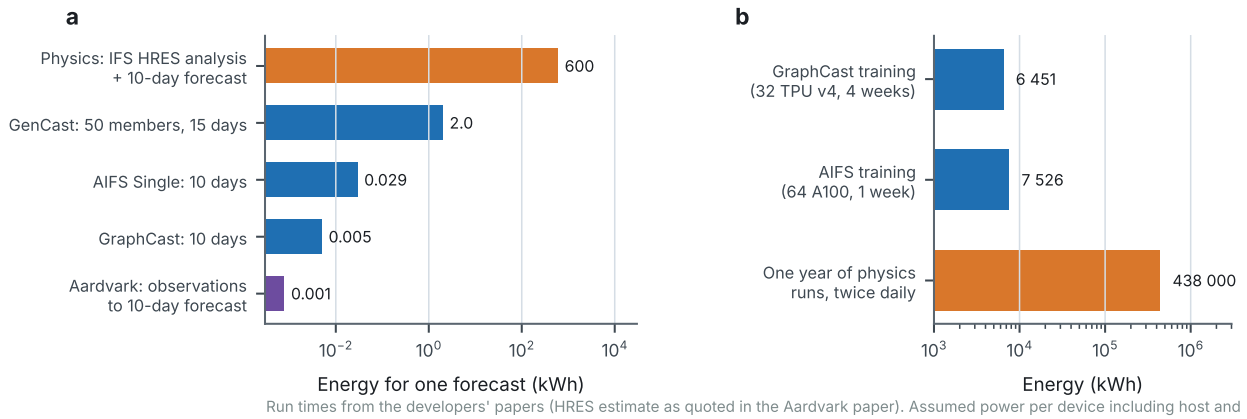

Lorenz-96 learned emulator of the previous figure; 150 start dates, 20 members from analysis perturbations (sigma = 0.2). Green: extra Gaussian noise (sigma = 0.06) added at every 6-h step to represent model error.

**FIGURE 47.4** An ensemble of the learned Lorenz-96 emulator. (a) Twenty members, their mean and the truth at one site. (b) The ensemble mean beats a single forecast, and its spread matches its error once a little model-error noise is added (green). (c) The ensemble mean loses "activity": it smooths towards climatology, as any forecast trained on squared error does.

Panel (c) shows the price of the mean. The root-mean-square anomaly of a single forecast, its **activity**, stays at 3.6, like the truth. The ensemble mean's activity falls to 2.5 by day 10. A deterministic model trained on squared error over long roll-outs drifts towards the same smooth mean. This is why learned forecasts look blurred at long range, and why the best systems now forecast ensembles.

### 47.3.5 Computing and Energy

Learned models are cheap to run and expensive to train (Figure 47.5). The physics system behind ECMWF's HRES needs roughly 1 000 node-hours of a supercomputer for one assimilation and 10-day forecast. AIFS needs 2.5 minutes of one GPU, and GraphCast under a minute of one TPU. Even a 50-member GenCast ensemble needs only about 7 TPU-hours. Training is the expensive part: about 21 500 TPU-hours for GraphCast and 10 750 GPU-hours for AIFS. Physics models need supercomputers (Figure 47.6); a trained learned model does not.




Run times from the developers' papers (HRES estimate as quoted in the Aardvark paper). Assumed power per device including host and cooling share: CPU node 0.6 kW, A100 GPU 0.7 kW, TPU 0.3 kW. ML forecasts except Aardvark still need an analysis from data assimilation.

**FIGURE 47.5** Energy for one forecast (a) and for training (b), on a logarithmic scale, from published run times and assumed power draws. A learned forecast uses thousands of times less energy, but it still needs an analysis made by the physics system.

### EXAMPLE: ENERGY OF ONE LEARNED AND ONE PHYSICS FORECAST

Compare one 10-day forecast from AIFS with one from a physics model. The physics run time is not published, so it is an assumption: 100 nodes for 1 hour, at 0.6 kW per node including cooling. Assume 0.7 kW for one GPU with its share of host and cooling.

1. **Physics forecast.** E = 100 nodes × 1 h × 0.6 kW = 60 kWh.
2. **Learned forecast.** AIFS takes 2.5 min = 0.0417 h on one GPU: E = 0.0417 h × 0.7 kW ≈ 0.029 kWh, or about 105 kJ.
3. **Ratio.** 60 / 0.029 ≈ 2 000. ECMWF quotes about 1 000; the exact factor depends on the assumptions, but three orders of magnitude is robust.
4. **Training.** AIFS used about 64 GPUs for one week: 64 × 168 h × 0.7 kW ≈ 7 500 kWh. That equals about 125 physics forecasts, or two months of twice-daily runs. Training pays for itself quickly, even if repeated every year.
5. **What is not saved.** The learned forecast still starts from an analysis made by 4D-Var with the physics model; in the Aardvark paper's estimate, assimilation plus forecast take about 1 000 node-hours. Until assimilation is also learned, the saving for the whole system is far smaller than in step 3.

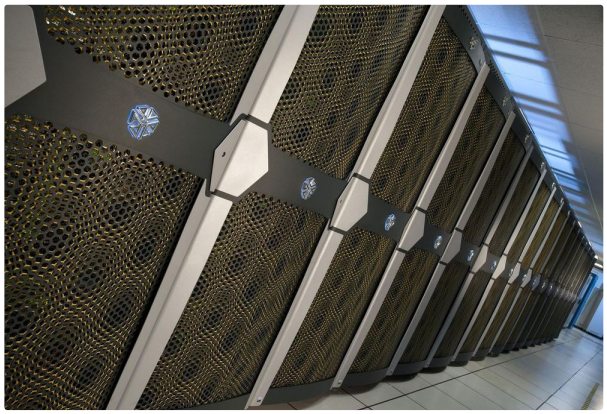


**FIGURE 47.6** The Pleiades supercomputer at NASA's Ames Research Center in 2008. Physics-based weather and climate models need machines like this; learned models run on a single graphics processor once trained. Image: NASA/ARC (Dominic Hart).

## 47.4 Measuring Skill Fairly

The key idea of this section is that a forecast is only as good as the test it passes. Learned models win most standard scores, but scores reward different things, and a smooth forecast can win a score while missing the storm.

### 47.4.1 Scores

Table 47.3 lists the main scores. **RMSE** penalises large errors and rewards smooth forecasts. The **anomaly correlation coefficient (ACC)** measures how well the pattern of departures from climate is forecast:

$$\mathrm{ACC} = \frac{\sum_i a_i (f_i - c_i)(o_i - c_i)}{\sqrt{\sum_i a_i (f_i - c_i)^2 \, \sum_i a_i (o_i - c_i)^2}},$$

where $f$, $o$ and $c$ are forecast, observation and climatology, and $a_i$ are area weights. By convention, a 500 hPa height forecast with ACC above 0.6 still has useful skill.

**TABLE 47.3** Common scores for weather forecasts.

| Score | What it measures | Typical use |
|---|---|---|
| RMSE | Size of errors; favours smooth forecasts | All variables and lead times |
| Anomaly correlation (ACC) | Pattern of departures from climate | 500 hPa height; useful above 0.6 |
| CRPS | Quality of a probability forecast | Ensembles |
| Spread–error ratio | Honesty of the ensemble spread | Ensembles; ideal value 1 |
| Track and intensity error | Distance to the centre; wind error | Tropical cyclones |
| Critical success index | Hits ÷ (hits + misses + false alarms) | Rain, nowcasting |
| Power spectrum | Energy at each spatial scale | Detecting blurring |

### 47.4.2 Where Learned Models Win

On the scores of the WeatherBench 2 benchmark and the centres' own verification, the best learned models beat the best physics models for most upper-air variables from day 1 to day 10. The gains are largest for large-scale patterns, such as 500 hPa height, and for tropical-cyclone tracks, which depend mainly on the steering flow. Learned forecasts also jump around less from run to run, as DWD found for AICON.

> **DATA BOX — AN AI MODEL AT THE NATIONAL HURRICANE CENTER, 2025**
>
> - In June 2025 Google opened Weather Lab, with an experimental cyclone model giving 50 scenarios out to 15 days. NHC forecasters saw it live all season, including Category 5 Melissa (Figure 47.7).
> - NHC's verification report (spring 2026) found that its ensemble mean "slightly outperformed" the official Atlantic track forecasts at 12–72 hours, with comparable intensity skill.
> - In the eastern Pacific it beat all other models, the official forecast and the consensus aids at 48–120 hours.
> - It was one input to human forecasters, whose own track forecasts beat their 2020–24 average.

**FIGURE 47.7** Category 5 Hurricane Melissa about 80 km south of Jamaica, photographed from the International Space Station on 28 October 2025. Learned models forecast the tracks of such storms well; their peak intensity is harder. Image: NASA/JSC.

### 47.4.3 Where They Fall Short

**Smoothing.** Models trained on squared error lose small-scale detail with lead time (Figure 47.4). Their power spectra fall below the real atmosphere's at scales under a few hundred kilometres, so fronts broaden and rain spreads into drizzle.

**Extremes.** In a study of Storm Ciarán (November 2023), four learned models placed the storm well but all underestimated its peak winds, which gusted to 111 knots (about 205 km/h) in Brittany. A 2025–26 study of record-breaking heat, cold and wind found that ECMWF's HRES beat GraphCast, Pangu-Weather and FuXi at nearly all lead times. The learned models underestimated how often and how far records were broken. A model trained on the past struggles with states it has never seen — a growing problem as the climate warms.

**Intensity.** Learned models tend to underestimate the peak winds of tropical cyclones. A 0.25° grid is too coarse to resolve an eyewall, and the training data are themselves smooth. NOAA's AIGFS 1.0 and its hybrid ensemble were both weaker on intensity than their physics counterparts.

**Physical consistency.** Nothing forces a learned model to conserve mass or energy, or to keep winds balanced with pressure. Outputs are mostly close to balance because the training data are, but small errors matter when the output drives other models. Learned models also lack many of the hundreds of fields, such as cloud ice or radiation fluxes, that physics models provide.

**Dependence on analyses.** A learned model inherits every error of its starting analysis; ECMWF's AIFS had to be re-tuned when the physics assimilation changed in 2026. A system of learned models alone would still need the satellites, the assimilation and the physics model that create its inputs.

**PRACTICE: EVALUATING A LEARNED WEATHER MODEL FAIRLY AGAINST A PHYSICS BASELINE**

1. **Fix the test period.** Use at least one full year, preferably two, that the model never saw in training or validation.
2. **Use the same starting point.** Start both models from the same analysis, or verify each against its own analysis and against observations. Verifying a model against the analysis it was trained to imitate flatters it.
3. **Match the resolution.** Compare on a common grid. Coarsening a high-resolution physics forecast removes its small-scale "double penalty" and makes the comparison fair.
4. **Score probabilistically.** Use CRPS and spread–error ratio for ensembles, RMSE and ACC for single forecasts. Report by region, season and lead time, with confidence intervals.
5. **Check the physics.** Plot power spectra, check mass and energy budgets, and look at balance between wind and pressure.
6. **Test the extremes.** Score storms, heatwaves and records separately, against station observations. Use event-based scores (CSI, hit rate) and tropical-cyclone track and intensity errors.
7. **Read real cases.** Have forecasters examine high-impact cases side by side. Track run-to-run consistency.
8. **Count the whole cost.** Include the assimilation that makes the analysis, the training and the retraining, not only the run time.

## 47.5 Machine Learning in the Observing Chain

The key idea of this section is that learning is useful long before the forecast model. It helps turn raw satellite signals into usable quantities, correct biases, speed up the physics inside assimilation, and forecast the next few hours.

### 47.5.1 Retrievals from Satellites

A **retrieval** estimates a geophysical quantity, such as rain rate, from what an instrument measures. Geostationary imagers see cloud tops every 5–15 minutes but not the rain beneath. Microwave instruments in low orbit see the rain but pass only a few times a day (Chapter 16). Neural networks have linked the two since the 1990s. PERSIANN ("precipitation estimation from remotely sensed information using artificial neural networks") maps infrared cloud-top temperatures to rain rate, trained on microwave and radar estimates. Multi-satellite products such as NASA's IMERG have used such estimates to fill gaps between microwave passes.

Deep networks now use several channels and the pattern of clouds. GREMLIN, from Colorado State University's CIRA, turns GOES imagery and lightning data into synthetic radar maps. In a one-week test, assimilating them gave better forecasts out to 12 hours over the mountainous western United States, where radar coverage is poor. It remained research in 2025.

**Tropical-cyclone intensity** from satellites still relies largely on the **Dvorak technique**: a trained analyst matches cloud patterns to an intensity scale. When only satellite estimates exist, the National Hurricane Center puts their uncertainty at 10–20 %. Convolutional networks trained on infrared and microwave images, with aircraft reconnaissance as ground truth, can make the estimate objective and repeatable. Most such networks are still research tools.

### 47.5.2 Bias Correction, Quality Control and Fast Observation Operators

Every satellite channel has biases that drift with orbit, temperature and age. Operational 4D-Var corrects them with **variational bias correction**, using a few predictors per channel. Neural networks can learn more complex, state-dependent biases and flag radiances contaminated by cloud, a task now done with hand-tuned thresholds. Both remain mostly research.

The observation operator $\boldsymbol{H}$ is costly: a radiative-transfer model runs for every observation in every iteration of 4D-Var. A network trained on that model can reproduce it many times faster, especially for cloudy and rainy scenes. It must also give accurate derivatives, because 4D-Var follows the gradient of $\boldsymbol{J}$. Such emulators are a step towards learned assimilation (Table 47.4).

**TABLE 47.4** Machine learning in the observing and assimilation chain.

| Task | Classical method | Learned approach | Status, 2026 |
|---|---|---|---|
| Rain from geostationary infrared | Cloud-top temperature thresholds | Neural networks since PERSIANN (1997); deep networks on many channels | Used in products; deep versions in research |
| Synthetic radar where radar is missing | None | Convolutional network on imager and lightning data (GREMLIN) | Research and demonstration |
| Tropical-cyclone intensity | Dvorak pattern matching | Convolutional networks on infrared and microwave images | Research |
| Radiance bias correction | Variational bias correction | Neural networks with state-dependent predictors | Research |
| Observation operator | Radiative-transfer models | Network emulators with accurate gradients | Research |
| Nowcasting rain, 0–2 h | Optical-flow extrapolation | Generative networks (DGMR, NowcastNet), MetNet-3 | MetNet-3 in Google products; others research |

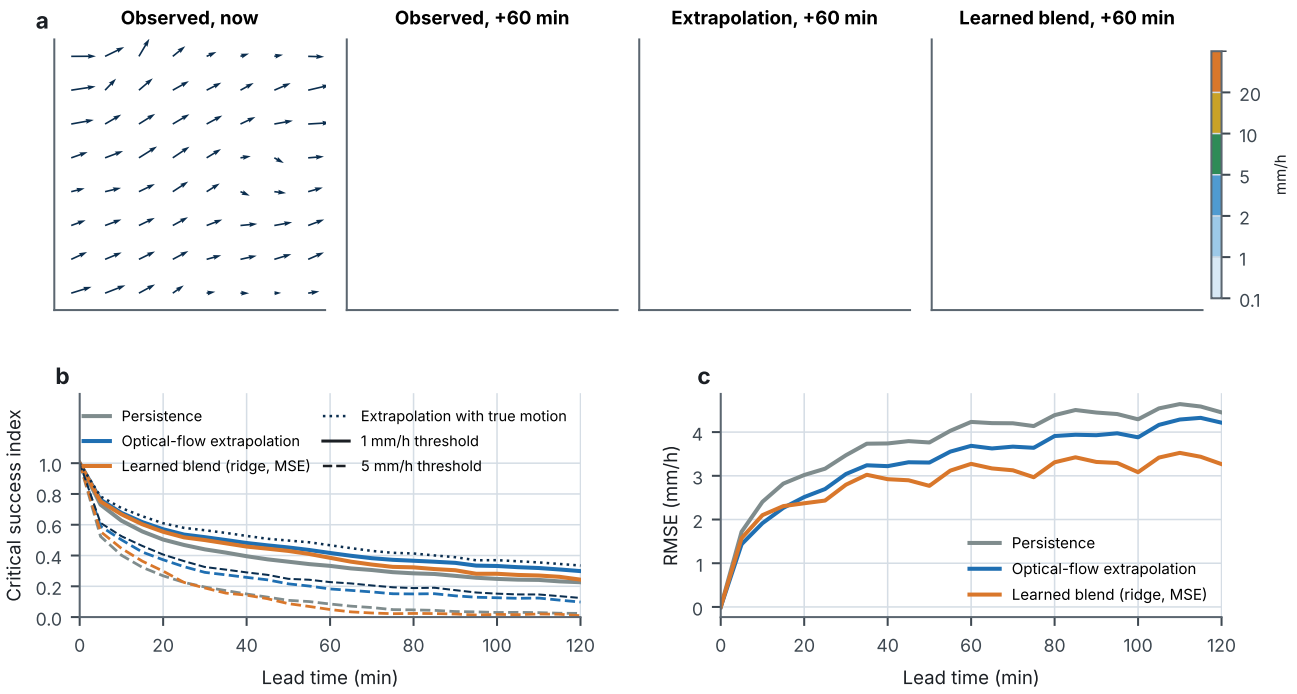
### 47.5.3 Nowcasting

A **nowcast** forecasts the next 0–6 hours, mainly for rain and storms. Physics models need time to "spin up" storms, so for the first hours the best tool has long been **extrapolation**: estimate how radar echoes move, then move the latest image along. The motion is found by **optical flow**, which assumes that rain is carried along unchanged.

$$\frac{\partial R}{\partial t} + \mathbf{u}\cdot\nabla R = 0, \qquad \hat{R}(\mathbf{r}, t + n\Delta t) = R(\mathbf{r} - n\,\mathbf{u}\,\Delta t,\; t)$$

The first equation defines the motion field $\mathbf{u}$, found by optimisation as the shift that best matches successive images. The second moves the latest image along $\mathbf{u}$. Systems based on this idea, such as STEPS and the open-source pySTEPS, run at many weather services.

Figure 47.8 tests it on a synthetic radar field. Rain cells of many sizes grow and decay, small cells faster than large ones, while the whole field moves at 39 km/h. We estimated the motion by block matching across three pairs of frames. At 60 minutes, extrapolation reached a critical success index (CSI) of 0.42 for rain above 1 mm/h, against 0.33 for persistence. Extrapolation with the true motion reached 0.47, so the motion estimate cost a little skill. For heavy rain above 5 mm/h, the CSI fell to 0.18, because small intense cells die quickly.




Synthetic radar: 384 km x 384 km, 2 km pixels, 5-min frames; power-law log-rain field whose small scales decorrelate faster, moving uniformly at 39 km/h. Motion from block matching of three frame pairs; learned blend = ridge regression of log rain on the extrapolated field smoothed at 4 scales, trained on earlier frames. 15 test times.

**FIGURE 47.8** Nowcasting a synthetic advecting rain field. (a) The field now with estimated motion vectors, the truth 60 minutes later, the optical-flow extrapolation and a learned blend. (b) Critical success index for rain above 1 mm/h (solid) and 5 mm/h (dashed). (c) RMSE: the learned blend has the lowest error but loses heavy rain.

We then trained a simple learned correction. A regression predicted future rain from the extrapolated field smoothed at four scales, by minimising squared error on earlier frames. It had the lowest RMSE at every lead time: 3.27 mm/h at 60 minutes against 3.69 for extrapolation. Yet it scored worse on the CSI: at 60 minutes only 0.05 for heavy rain, against 0.18 for extrapolation. Squared-error training spreads rain into a smooth, light field, so intense cores vanish. This is the nowcasting version of the blurring seen in Figure 47.4.

Deep-learning nowcasts face the same trap; the leading ones avoid it with generative training. **DGMR**, from DeepMind and the Met Office (*Nature*, 2021), makes sharp 90-minute forecasts from 20 minutes of radar. More than 50 Met Office meteorologists ranked it first in 89 % of cases. **NowcastNet** (Tsinghua University, *Nature*, 2023) joins a physical advection scheme to a generative network; 62 Chinese meteorologists ranked its 3-hour forecasts first in 71 % of cases. Google's **MetNet-3** (2023) forecasts rain, temperature and wind up to 24 hours ahead at 1–4 km from radar, satellite and station data. It runs in Google products for the United States and 27 European countries. NVIDIA's **StormScope** (January 2026) forecasts storms 0–6 hours ahead directly from geostationary images.

## 47.6 Climate: Emulators, Downscaling and Digital Twins

The key idea of this section is that climate questions differ from weather questions. They ask about statistics over decades under conditions that have never occurred. Learning helps where it can be checked against physics, and is dangerous where it must extrapolate.

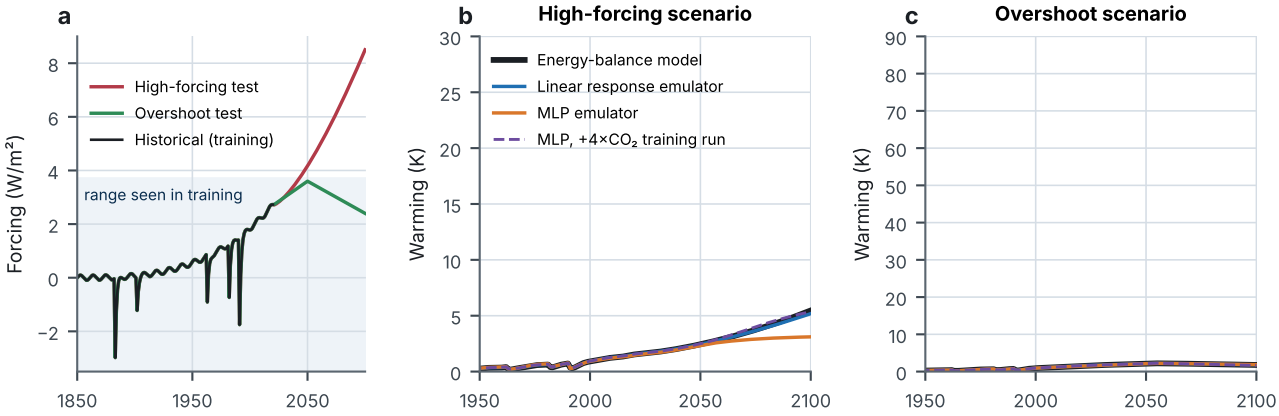
### 47.6.1 Climate Emulators

A **climate emulator** is a fast learned model that imitates an expensive climate model. **Full emulators** step the atmosphere forward like a weather model but stay stable for centuries. ACE2, from the Allen Institute for AI, was trained on ERA5 and physics-model output and runs about 1 500 simulated years per day on one GPU at 1°. A coupled atmosphere–ocean version, SamudrACE, followed in 2025. **Reduced emulators** map emission scenarios to global or regional warming, so thousands of scenarios can be explored.

The danger is **extrapolation**. A weather model sees the same kinds of state in training and in use. A climate model is asked about a world warmer than anything in its training data. Figure 47.9 shows what can happen. The "truth" is a two-layer **energy-balance model**: an upper ocean and atmosphere of heat capacity $C$, coupled to a deep ocean of heat capacity $C_d$.

$$C\frac{dT}{dt} = F - \lambda(T)\,T - \gamma(T - T_d), \qquad C_d\frac{dT_d}{dt} = \gamma(T - T_d)$$

Here $F$ is the radiative forcing, $\lambda$ the climate feedback, which weakens slightly as the planet warms, and $\gamma$ the exchange with the deep ocean. Its equilibrium warming for doubled $CO_2$ is 3.7 K. We trained two emulators on three runs with forcing up to 3.7 W/m$^2$: a historical run, a 1 % per year rise in $CO_2$ to doubling, and an abrupt doubling. Both emulators see the forcing smoothed over 1–400 years. One is linear, like a physical response function. The other is a small neural network (MLP).




Two-layer energy-balance model (C = 7.3 and 106 W yr m$^{-2}$ K$^{-1}$, feedback 1.13 - 0.035 T W m$^{-2}$ K$^{-1}$, exchange 0.73 W m$^{-2}$ K$^{-1}$) with red-noise variability. Emulators map forcing smoothed over 1-400 years to warming; trained on historical, 1 %/yr to 2×$CO_2$ and abrupt 2×$CO_2$ runs; MLP = mean of 5 networks.

**FIGURE 47.9** Emulating a two-layer energy-balance climate model. (a) Training forcing (black and shaded range) and two unseen test scenarios. (b) Under high forcing, outside the training range, the neural-network emulator (orange) fails, while the linear response emulator (blue) stays close. Adding an abrupt 4×$CO_2$ run to training (violet) fixes the network. (c) In an overshoot scenario inside the training range, all emulators work.

In an overshoot scenario that stayed within the training range, all emulators reproduced the truth, with RMSE of 0.06 K or less over 2021–2100. In a high-forcing scenario reaching 8.5 W/m$^2$ by 2100, the true warming was 5.47 K. The linear emulator gave 5.19 K: close, but low, because it cannot know that the feedback weakens. The neural network gave only 3.11 K. Its tanh units had saturated outside the range they were trained on. When an abrupt 4×$CO_2$ run was added to training, the network gave 5.36 K. The lesson applies to every learned climate model: **train on the conditions you will predict**, prefer structures that respect physics, and test outside the training range on purpose.

### 47.6.2 Downscaling

Global models run at 10–100 km; impacts happen at 1 km. **Downscaling** fills the gap. Statistical downscaling regresses local weather on large-scale predictors. **Generative downscaling** adds realistic small-scale detail, giving many possible fine fields for one coarse input. NVIDIA's CorrDiff, trained with data from Taiwan's Central Weather Administration, maps 25 km ERA5 fields to about 2 km: a network predicts the mean field and a diffusion model adds the detail. The output must be checked against stations, because a network can invent convincing but false detail.

### 47.6.3 Satellite Climate Records and Extremes

Learning from observations needs long, stable records. ESA's **Climate Change Initiative (CCI)** reprocesses decades of satellite data into over 100 consistent data sets, covering many of the 55 **essential climate variables** of the Global Climate Observing System. They are used to train and test learned models and to detect change. Statistical learning can even find the fingerprint of global warming in a single day's weather pattern. For **attribution** of extremes, learned models can generate thousands of plausible seasons to estimate how much more likely a heatwave has become. Their answers are only as good as their behaviour in the tails, which must be tested separately.

### 47.6.4 Digital Twins of the Earth

A **digital twin** is a model kept in step with observations that users can question. For example, how would this heatwave look in a world 2 °C warmer? The EU's **Destination Earth**, run by ECMWF, ESA and EUMETSAT, has a Climate Change Adaptation twin and a Weather-Induced Extremes twin, both at kilometre scale. Phase 2 added learned components for land, ocean, sea ice, waves and hydrology. Phase 3 (June 2026–June 2028) will couple them into a European learned Earth-system model. NVIDIA's **Earth-2** is a platform of open models (Table 47.5); in January 2026 it added a 15-day model and a storm nowcaster.

**TABLE 47.5** Learning in climate science: selected systems.

| System | Developer, year | What it does | Status, October 2026 |
|---|---|---|---|
| ClimaX | Microsoft and partners, 2023 | Foundation model pre-trained on climate-model output | Research |
| ACE2, SamudrACE | Allen Institute for AI and partners, 2024–25 | Emulates atmosphere (and ocean) at 1°, about 1 500 years a day on one GPU | Research; open |
| NeuralGCM | Google, 2024 | Hybrid physics–learned model, stable for decades | Research; open |
| CorrDiff | NVIDIA, 2024 | Generative downscaling, 25 km to about 2 km | Open; developed with Taiwan's weather agency |
| Destination Earth | EU: ECMWF, ESA, EUMETSAT, 2022– | Kilometre-scale climate and extremes twins with learned parts | Phase 3, June 2026–June 2028 |
| Earth-2 | NVIDIA, 2024– | Platform of open weather and climate models | New models released January 2026 |
| ESA CCI | ESA, 2010– | Over 100 satellite climate data sets | Operational archive |

## 47.7 Space Weather

The key idea of this section is that space weather has the same forecasting needs as the weather, with far fewer observations. That makes machine learning attractive and also risky. The Sun is watched by a handful of spacecraft, and big storms are rare.

### 47.7.1 Why Space Weather Is Hard to Learn

Space weather (Chapter 22) runs from the Sun to the ground. A flare or coronal mass ejection (CME) leaves the Sun. The solar wind carries it to the Earth in one to four days. It reaches the L1 monitors about an hour before the Earth. It then drives currents in the magnetosphere and ionosphere (Chapter 23). Each link has its own data and forecasting problem (Table 47.6).

**TABLE 47.6** Machine learning for space weather (selected tasks). TSS: true skill statistic.

| Task | Main inputs | Typical learned method | Reported skill | Status |
|---|---|---|---|---|
| Solar flares (24 h) | Magnetogram features or images | SVM, CNN, foundation models | TSS about 0.5–0.8 in research | Research; human-led forecasts in operations |
| CME arrival time | Coronagraph images, CME speed | Regression, CNN | Mean absolute error 6–12 h on research samples | Research |
| Solar-wind speed at L1 | Solar EUV images | CNN with LSTM | RMSE 72–82 km/s, 4 days ahead | Research |
| Ground magnetic disturbance | L1 solar wind | Recurrent and graph networks (DAGGER) | 30 min ahead, worldwide | Research demonstration |
| Kp and Dst indices | L1 solar wind, past index | Neural networks, LSTM, Gaussian processes | Kp ensemble up to 72 h (GFZ) | Some products |
| Ionosphere (TEC) | GNSS maps, solar indices | LSTM, ConvLSTM | Better than climatological models in studies | Research |
| Radiation belts | Particle data, solar wind | Neural networks | Electron flux forecasts in studies | Research |

Three things make learning hard. Data are few: the best magnetograms, from the Solar Dynamics Observatory (SDO), cover only since 2010, about one and a half solar cycles. Events are rare: there are only a few dozen great storms in the whole space age. And the key quantity for geomagnetic storms, the north–south component of the magnetic field in a CME ($B_z$), cannot be measured until the CME reaches L1.

### 47.7.2 Flare Forecasting from Magnetograms

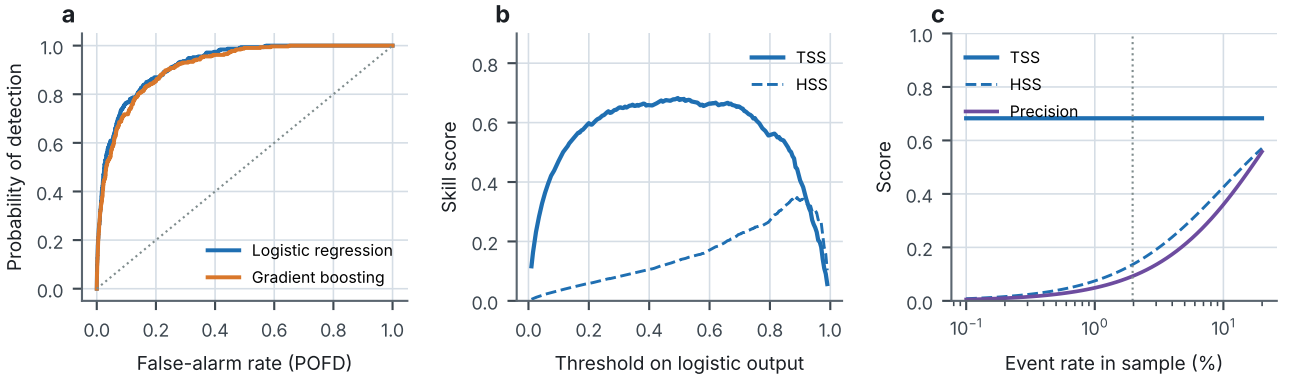
Large flares come from active regions with strong, twisted magnetic fields (Figure 47.12), mapped by SDO's Helioseismic and Magnetic Imager (HMI). Features computed from each map include the total unsigned flux, the flux near the **polarity inversion line** where the field changes sign, electric currents and shear. These features, or the images themselves, feed a classifier that gives the probability of an M- or X-class flare in the next 24 hours.

Flares are rare, so accuracy is useless as a score. A model that always says "no flare" can be 98 % accurate. Flare forecasts are compared with the **true skill statistic** (Chapter 43):

$$\mathrm{TSS} = \frac{\text{hits}}{\text{hits} + \text{misses}} - \frac{\text{false alarms}}{\text{false alarms} + \text{correct negatives}},$$

the probability of detection minus the probability of false detection. It ranges from −1 to 1 and does not depend on how common flares are in the sample. In 2015 Bobra and Couvidat reached TSS 0.76 with a support vector machine on HMI features, though only about 42 % of its flare warnings came true. A 2019 comparison of operational systems over 2016–17 found that most scored above "no skill", with TSS values roughly from 0 to 0.5. No method was best on every score.

Figure 47.10 shows the method on synthetic data. A hidden "free energy" drives five magnetogram-like features and the chance of a flare, with an unobserved trigger added. The flare rate is 2 %. A logistic regression reached a maximum TSS of 0.68, and gradient-boosted trees 0.66. Panel (c) shows why TSS is used. Keep the detection and false-alarm rates fixed: TSS stays constant, but precision rises from 2.5 % at a 0.5 % event rate to 56 % at 20 %. Scores like precision and the Heidke skill score (HSS) cannot be compared between solar maximum and solar minimum.




Synthetic region-days: five magnetogram-like features driven by a hidden free-energy variable; flare (M class or above within 24 h) drawn from a logistic law with an unobserved trigger term. First 60 % for training, last 40 % for testing. (c) Fixed detection and false-alarm rates of the logistic model.

**FIGURE 47.10** Flare forecasting on synthetic magnetogram features. (a) ROC curves for logistic regression and gradient boosting. (b) TSS and HSS against the decision threshold. (c) At fixed detection and false-alarm rates, TSS does not depend on the event rate; precision and HSS do.

### EXAMPLE: THE TRUE SKILL STATISTIC FROM A FLARE CONTINGENCY TABLE

The logistic model of Figure 47.10 was tested on 16 000 region-days, of which 315 had an M-class or larger flare. At its best threshold it gave 268 hits, 47 misses, 2 635 false alarms and 13 050 correct negatives.

1. **Probability of detection.** POD = 268 / (268 + 47) = 0.851.
2. **Probability of false detection.** POFD = 2 635 / (2 635 + 13 050) = 0.168.
3. **True skill statistic.** TSS = 0.851 − 0.168 = 0.683.
4. **Precision.** 268 / (268 + 2 635) = 0.092: only about 1 warning in 11 is followed by a flare.
5. **Accuracy.** (268 + 13 050) / 16 000 = 0.832. A forecast of "never" scores 0.980 but has TSS 0.
6. **Heidke skill score.** HSS = 2(ad − bc) / [(a + c)(c + d) + (a + b)(b + d)] = 0.136, with a–d the hits, false alarms, misses and correct negatives. False alarms keep it low.
7. **Meaning.** A high TSS shows that the model separates flaring from quiet regions, not that its warnings are usually right. Users who pay for false alarms should choose a higher threshold.

Since 2018 deep networks have learned directly from magnetogram and extreme-ultraviolet images, with results similar to feature-based methods on small samples. The NASA–IBM **Surya** foundation model (August 2025) was pre-trained on more than 200 TB of SDO data. Fine-tuned for flares, it reportedly beat existing benchmarks by about 16 % in preliminary tests (Chapter 44).

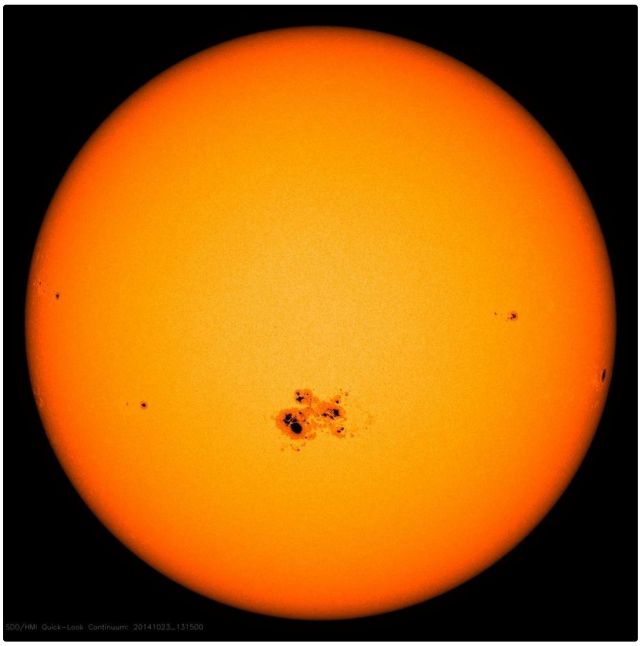


**FIGURE 47.11** The largest sunspot group of solar cycle 24, active region 12192, seen by SDO in October 2014. It produced six X-class flares, none of which launched a coronal mass ejection. Image: NASA/GSFC.

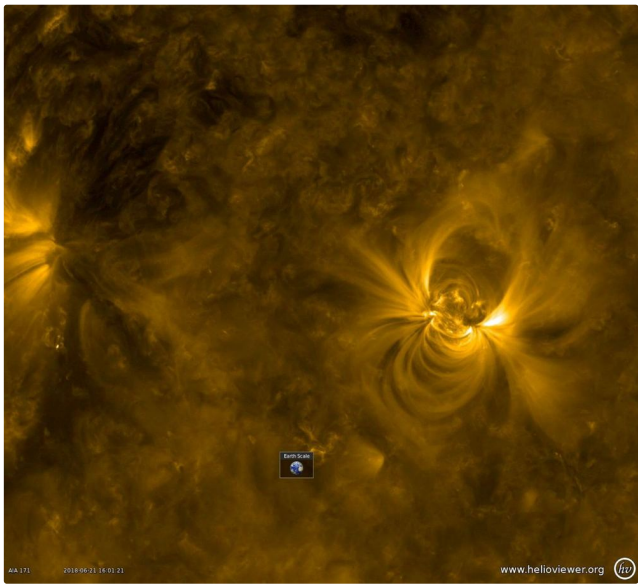



**FIGURE 47.12** Magnetic loops above a fast-growing active region in extreme-ultraviolet light, imaged by SDO in June 2018. Flare forecasts learn from the magnetic fields beneath such loops. Image: NASA/JPL (NASA/GSFC/Solar Dynamics Observatory).

Region 12192 (Figure 47.11) is a warning about labels. It produced 6 X-class, 22 M-class and 53 C-class flares in one disk passage, yet none of its X-class flares launched a CME. A model trained to predict flares would rate it highly. A model trained to predict geomagnetic storms should not.

### 47.7.3 CMEs and the Solar Wind

Forecasting when a CME will arrive is still hard. NASA's Community Coordinated Modeling Center collects real-time forecasts from many teams in its CME Scoreboard. A 2024 re-analysis found a mean absolute arrival-time error of 13.2 hours, a standard deviation of 17.4 hours and a bias of −2.5 hours (slightly early). That is only slightly better than in the Scoreboard's first five years, from 2013. Learned models report 6–12 hours on their own samples, such as 5.9 hours for a support vector machine on 182 CMEs and 9.75 hours for an ensemble on 363 CMEs. The samples differ, so the numbers cannot be compared directly.

Networks can also forecast the background solar wind. Coronal holes in extreme-ultraviolet images release fast wind that reaches the Earth 3–4 days later. An image network combined with a recurrent network forecast the wind speed at L1 four days ahead with an RMSE of 82 km/s; a later study reached 72 km/s. The physics-based WSA–Enlil model remains the operational standard.

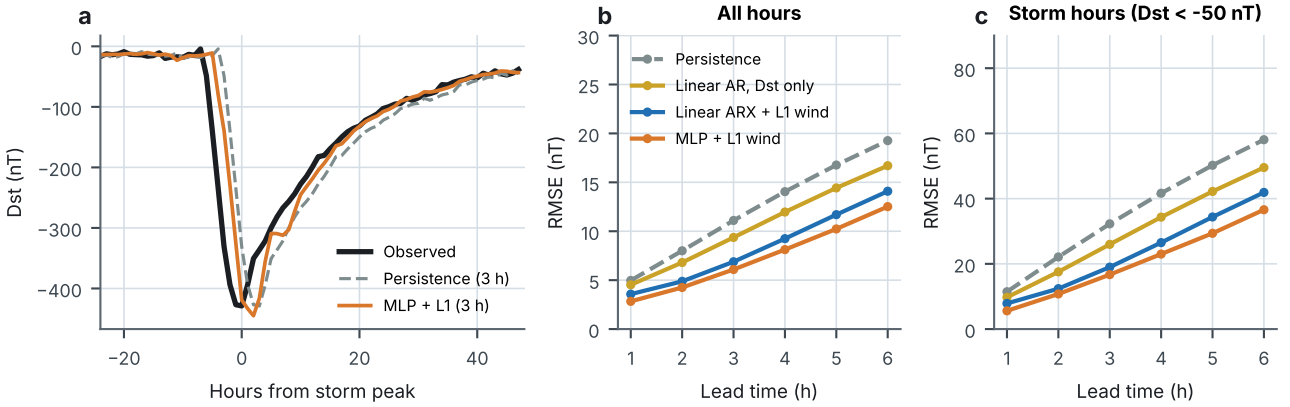
### 47.7.4 Geomagnetic Indices

Once the solar wind reaches L1, forecasts become much better, because the driver is measured. The **Dst index** measures the ring current that weakens the field at low latitudes during storms. **Kp** measures disturbance at mid-latitudes on a 0–9 scale (Chapter 22). A classic physics-based model links Dst to the solar-wind electric field $VB_s$, the solar-wind speed times the southward magnetic field:

$$\frac{d\,\mathrm{Dst}^*}{dt} = Q(VB_s) - \frac{\mathrm{Dst}^*}{\tau(VB_s)},$$

where $Q$ is the energy injection and $\tau$ a recovery time of several hours. Storms push the aurora towards the equator (Figure 47.14). Neural networks have forecast Dst and Kp since the early 1990s. Today's models include recurrent networks and Gaussian processes that also give uncertainty. GFZ in Potsdam runs a machine-learning Kp ensemble every hour, up to 72 hours ahead.

We built a synthetic test (Figure 47.13). Four years of hourly solar wind, with recurring fast streams and random CMEs, drove the Dst model above. Three years were used for training and the fourth for testing; the test year held a −429 nT superstorm, stronger than any in training. With only past Dst, a linear model reduced the 3-hour-ahead RMSE from 11.1 nT for persistence to 9.4 nT. Adding L1 solar-wind data cut it to 6.9 nT, and a neural network with the same inputs to 6.1 nT. In storm hours the network's skill score against persistence, $1 - \mathrm{MSE}/\mathrm{MSE}_{\mathrm{pers}}$, was 0.73. Most of the gain came from the L1 data; the network added a smaller step.




Four years of hourly synthetic solar wind (recurrent streams, CMEs, random Bz) drive the O'Brien-McPherron Burton-type Dst model with pressure correction and 2 nT noise. Years 1-3 train, year 4 tests; inputs: 6 hours of past Dst, and for ARX/MLP 7 hours of L1 VBs (one hour ahead of Earth) and dynamic pressure.

**FIGURE 47.13** Forecasting a Dst-like index from synthetic solar wind. (a) The largest test storm, beyond anything in training, with 3-hour forecasts. (b) RMSE over all test hours against lead time. (c) RMSE in storm hours. Upstream L1 data give most of the skill; a neural network adds more.

NASA's **DAGGER** model (2023), from the Frontier Development Lab, used solar-wind data from upstream spacecraft such as ACE and Wind to forecast the ground magnetic disturbance worldwide, 30 minutes ahead. It runs in under a second and updates every minute. Its authors noted that its average errors were close to persistence and that it underestimated the strongest disturbances, near 1 000 nT. Like most work in this field, it is a research demonstration.

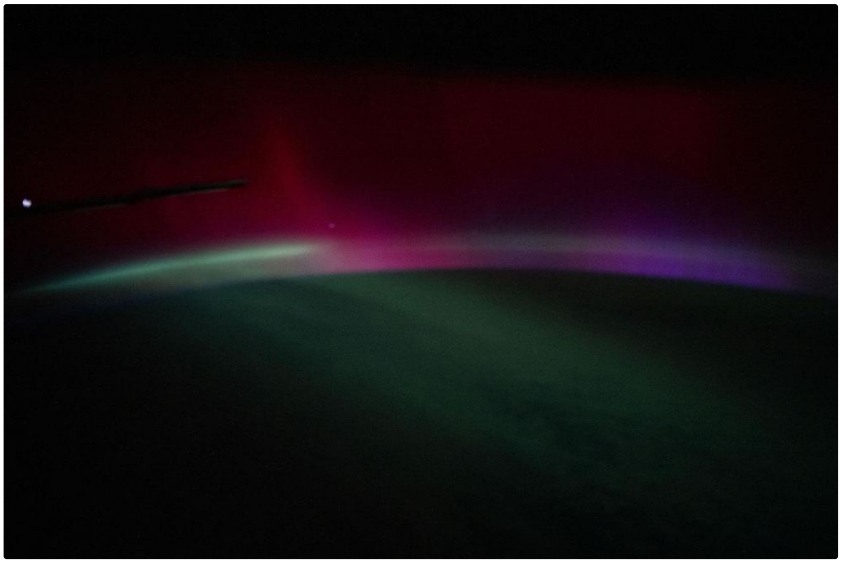


**FIGURE 47.14** An aurora with rare deep-red and purple colours over the Atlantic off Portugal, photographed from the International Space Station on 19 January 2026, during a geomagnetic storm that reached Kp 9−. Image: NASA/JSC.

**EXAMPLE: HOW MUCH WARNING CAN AN L1 MONITOR PLUS A LEARNED MODEL GIVE?**

NOAA's SOLAR-1 observatory sits near L1, about 1.5 million km sunward of the Earth. Assume 4 minutes to downlink, process the data, run a learned model (under 1 s) and issue an alert.

1. **Slow wind, 400 km/s.** Travel time $t = 1.5 \times 10^9$ m / $4 \times 10^5$ m/s = 3 750 s = 62.5 min. Warning ≈ 62.5 − 4 = 58.5 min.
2. **Fast CME, 750 km/s.** t = 2 000 s = 33.3 min; warning ≈ 29 min.
3. **Extreme CME, 1 500 km/s.** t = 1 000 s = 16.7 min; warning ≈ 13 min.
4. **What learning adds.** It cannot make the solar wind travel more slowly. It can turn the measured wind into a local forecast of ground disturbance or Kp within seconds, as DAGGER does for 30 minutes ahead at moderate speeds.
5. **Longer warnings.** A CME seen leaving the Sun gives one to four days of notice, but arrival errors of about 13 hours and the unknown $B_z$ make this a "watch", not a warning. ESA's planned Vigil spacecraft at L5 would see CMEs from the side and improve arrival times.

### 47.7.5 Ionosphere and Radiation Belts

The ionosphere delays GNSS signals in proportion to its **total electron content (TEC)** (Chapters 18 and 23). Global TEC maps from thousands of receivers form long image sequences, well suited to recurrent and convolutional networks. Studies report 1–24-hour forecasts that beat empirical models in quiet conditions but are weaker in storms, which are rare in training data. Neural networks driven by solar-wind and particle data also forecast radiation-belt electrons (Chapter 23) between the few spacecraft that measure them. Both remain mostly research.

### 47.7.6 From Research to Operations

Operational space-weather centres have been slower than weather centres to adopt learned models. NOAA's Space Weather Prediction Center (SWPC), the Met Office Space Weather Operations Centre and ESA's Space Weather Service Network rely mainly on human forecasters. Their tools are physics models, such as WSA–Enlil and NOAA's Geospace model, and empirical

rules. Learning appears in a few products, such as neural-network Kp forecasts. Three obstacles stand out: few events to validate against, the high cost of a missed storm, and the long, formal path from research to operations.

**LANDMARK: THREE DECADES OF NEURAL NETWORKS IN SPACE WEATHER**

- **Early 1990s.** A group at the Swedish Institute of Space Physics in Lund began using neural networks to predict geomagnetic indices from solar-wind data. By the early 2000s it ran real-time forecasts of Dst; its Kp and Dst forecasts are now distributed through ESA's space-weather service.
- **2010s.** SDO magnetograms enabled image-based flare forecasting. Research classifiers reached TSS near 0.8; operational comparisons showed far lower skill.
- **2023.** NASA's DAGGER forecast ground magnetic disturbances worldwide 30 minutes ahead in under a second.
- **2025.** NASA and IBM released the Surya foundation model openly. NOAA's SWFO-L1 launched with NASA's IMAP on 24 September 2025, reached L1 in January 2026 and was renamed SOLAR-1 (Figure 47.15).
- **2026.** SOLAR-1 was declared operational in mid-2026, after its CCOR-2 coronagraph in June. Operational use of learning was still limited to a few products; the bottleneck had moved from algorithms to validation and trust (Chapter 53).

**FIGURE 47.15** The Falcon 9 carrying NASA's IMAP, the Carruthers Geocorona Observatory and NOAA's SWFO-L1 (now SOLAR-1) at Launch Complex 39A on 22 September 2025, two days before launch. SOLAR-1 carries a coronagraph and solar-wind instruments for operational forecasts. Image: NASA/KSC (SpaceX).

Space-weather forecasting will probably follow the weather path, more slowly. New monitors at L1, and later L5, will supply data, and long archives with agreed benchmarks will allow fair tests. Hybrid models that keep the physics of the solar wind and learn the rest are the most likely to earn forecasters' trust.

### KEY TAKEAWAYS

- Learned weather models step the atmosphere forward 6 hours at a time and learn from ERA5 reanalysis. Since 2023 the best have matched or beaten physics models on most scores, using about a thousandth of the energy per forecast.
- ECMWF's AIFS (2025), NOAA's AIGFS, AIGEFS and HGEFS (December 2025), Germany's AICON and China's Fengqing run operationally — always beside physics models.
- Squared-error training smooths forecasts and weakens extremes; diffusion and CRPS-trained ensembles restore detail and honest uncertainty. Record extremes and cyclone intensity remain weak points.
- Learned models depend on physics-based analyses. End-to-end systems that learn from raw observations, such as Aardvark Weather and GraphDOP, are promising research.
- In the observing chain, learning helps with retrievals, bias correction, fast observation operators and nowcasting; nowcasts need generative training to keep heavy rain sharp.
- Climate emulators and downscaling are fast and useful, but learned models fail outside their training range. Digital twins such as Destination Earth combine physics and learning.
- Space weather has far less data and fewer events. Flare forecasts are compared by the true skill statistic, CME arrival errors are still about 13 hours, and L1 data plus learned models give tens of minutes of warning. Operational use is still limited.

### REVIEW QUESTIONS

1. Explain why almost all learned weather models still depend on data assimilation and satellites, even though they replace the physics model.
2. Write the autoregressive update of a learned weather model. Why does it predict the change over 6 hours rather than the full state, and why is the loss weighted by the cosine of latitude?
3. In Figure 47.3 the learned emulator beats a physics model with a 5 % forcing error, yet all forecasts reach half the climatological spread after about 5.5–5.7 days. What limits the forecast range?
4. Why does a deterministic model trained on squared error produce smooth forecasts at long lead times? Use Figure 47.4 to explain how ensembles and CRPS training address this.
5. Give two reasons why the energy saving of a learned forecast is smaller for the whole forecasting system than for the forecast step alone.
6. List four weaknesses of learned weather models and a test from the Practice box that would reveal each.
7. In Figure 47.8 the learned blend has the lowest RMSE but the worst CSI for heavy rain. Explain the difference and how DGMR avoided it.
8. Why did the neural-network climate emulator in Figure 47.9 fail in the high-forcing scenario while the linear emulator did not? What would you change in its training?
9. A flare model gives 40 hits, 10 misses, 300 false alarms and 9 650 correct negatives. Compute its POD, POFD, TSS and precision. Would a satellite operator be satisfied?
10. A CME travels from L1 to the Earth at 900 km/s. With 5 minutes of processing latency, how much warning does an L1 monitor give? Why can a CME seen leaving the Sun not give a reliable warning days ahead?

CHAPTER

# 48 AI for Autonomy, Navigation and Robotics

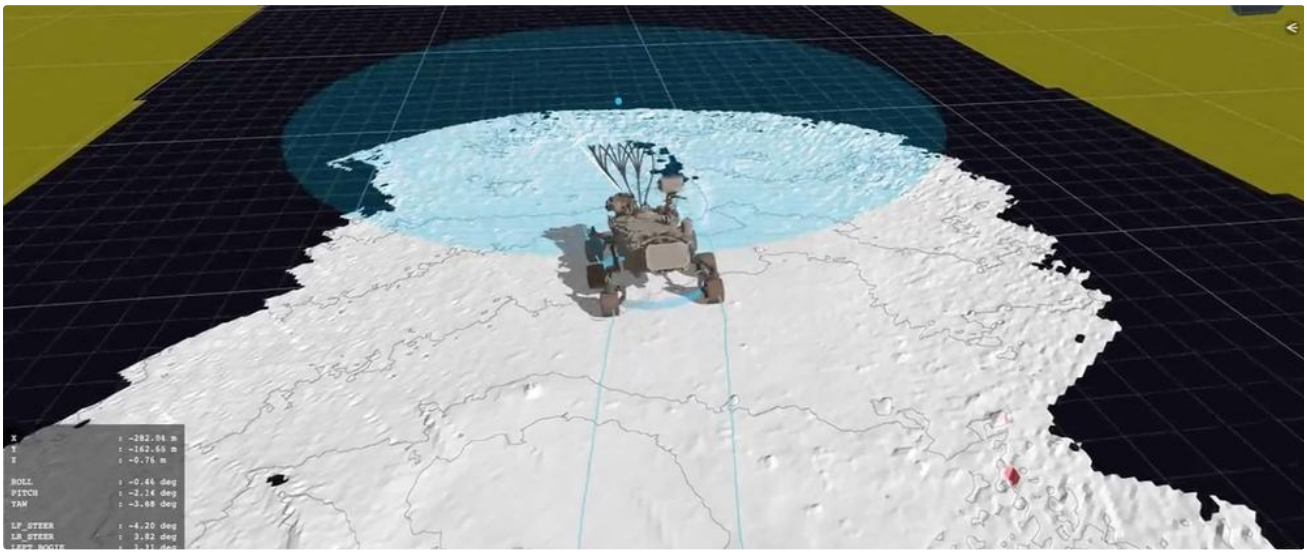


Image: NASA/JPL (NASA/JPL-Caltech)

*A spacecraft that waits for instructions from Earth cannot land on Mars, dodge debris every few minutes or grab a tumbling satellite. It must sense, decide and act on its own. This chapter explains how. It starts with why autonomy is needed and the ECSS autonomy levels, then looks inside an autonomous spacecraft: executives, fault protection, model-based diagnosis and onboard planners. It covers navigation by stars, craters and terrain maps, pose estimation for rendezvous, convex guidance and reinforcement learning, docking, rovers, robot arms, free-flyers, swarms and constellations. Small optimisation and learning experiments, run from scratch, show what each method does well and where it fails. Status is given as of October 2026.*

IN THIS CHAPTER

48.1 Why Spacecraft Need Autonomy
48.2 Inside an Autonomous Spacecraft
48.3 Onboard Planning and Scheduling
48.4 Navigation Without the Ground
48.5 Pose Estimation for Rendezvous and Servicing
48.6 Guidance and Control: Optimisation or Learning?
48.7 Rendezvous, Docking and Proximity Operations
48.8 Robots and Swarms

Every spacecraft is a robot. Most of the time it follows a script written on the ground. But some decisions cannot wait for the ground. A Mars lander has about seven minutes from the top of the atmosphere to the surface, while a radio signal takes up to 22 minutes to arrive. A rover that stops after every metre to ask for advice will never get anywhere. A fleet of ten thousand satellites meets debris many times an hour. In each case the spacecraft must sense its situation, decide what to do and act — on its own.

This chapter is about the software that makes those decisions. Much of it is not machine learning. Planners search through possible schedules, filters fuse sensor data, and optimisers compute fuel-optimal trajectories with mathematical guarantees. Learning enters where hand-written rules struggle: recognising craters, finding a satellite's corners in a camera image, or choosing a path across rough ground. The hero image above shows the idea. It is a replay of telemetry from Perseverance's first self-driving (AutoNav) drive: the rover's own three-dimensional map of the ground ahead and the path it chose.

Background is covered elsewhere: control and estimation in Chapter 6, deep-space navigation in Chapter 24, landing in Chapter 27, rovers in Chapter 28 and servicing in Chapter 40. Machine-learning basics are in Chapters 43 and 44, and onboard computers in Chapter 45.

## 48.1 Why Spacecraft Need Autonomy

The key idea is simple. A spacecraft needs autonomy whenever the ground cannot respond fast enough, often enough or cheaply enough. Four things set the limit: light-time, gaps in contact, the number of spacecraft and the speed of events.

### 48.1.1 Light-Time, Gaps and Scale

**Light-time** is the time a radio signal takes to travel one way. It is 0.12 s to geostationary orbit and 1.3 s to the Moon. To Mars it ranges from about 3 to 22 minutes, and to Jupiter from 33 to 54 minutes. Voyager 1 will be one light-day from Earth in November 2026. A question and its answer take twice as long.

**Contact gaps** often matter more. A low-orbit satellite passes over a ground station for about ten minutes at a time, a few times a day. A Mars rover talks to Earth mostly through orbiters that fly over a few times per sol (a Martian day of 24 h 39 min). Deep-space missions share NASA's Deep Space Network with dozens of others and may get a few hours of contact a day.

**Scale** is the newest driver. Over ten thousand Starlink satellites are in orbit, and other constellations are following (Chapter 17). No team of people can plan every pointing change, data dump and collision-avoidance manoeuvre for such a fleet by hand.

**Speed of events** sets the final limit. A failed thruster valve, a boulder in a landing zone or a closing piece of debris must be dealt with in seconds to hours. Autonomy also saves money, because fewer people are needed to watch routine operations.

### 48.1.2 Levels of Autonomy

The European standard on space segment operability, ECSS-E-ST-70-11C (2008, revised in October 2025), defines levels on three separate scales (Table 48.1). The **E** scale covers the execution of nominal operations. The **D** scale covers onboard storage of mission data, and the **F** scale covers what the spacecraft does after a failure. A mission states a level on each scale. A typical Earth-observation satellite might be E2, D2 and F1; a Mars rover is E3 or E4 for driving, with F2 for many faults.

**TABLE 48.1** Autonomy levels of ECSS-E-ST-70-11C for execution (E), data management (D) and failure management (F). The standard defines two data levels and two failure levels, not four.

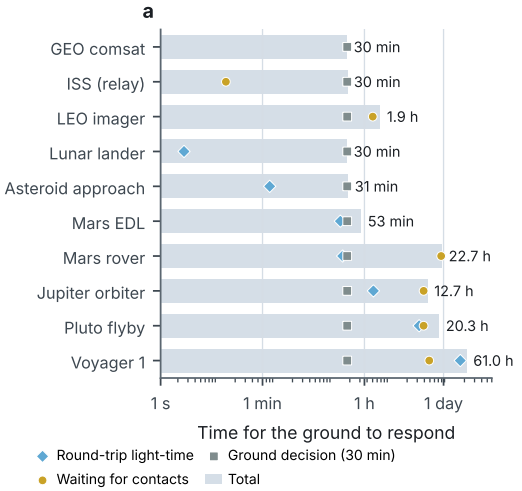
| Level | The spacecraft can... | Typical example |
|---|---|---|
| E1 | Run ground commands in real time; time-tagged safety commands | Early satellites |
| E2 | Run a pre-planned, ground-written timeline | Most Earth-observation satellites |
| E3 | React to events with onboard rules and procedures | Rover hazard avoidance |
| E4 | Re-plan on board from goals | EO-1; Perseverance's planner |
| D1 | Store essential data during a ground outage or failure | Event logs kept for the next pass |
| D2 | Store all mission data, independent of the ground | Memory sized for the longest gap |
| F1 | Detect and isolate a failure; go to a safe state | Sun-pointing safe mode |
| F2 | Recover and resume nominal operations | Deep-space probes using spare units |

### 48.1.3 How Much Autonomy Is Enough?

A simple model shows where each level is needed (Figure 48.1). Suppose an event happens on board. The ground learns of it, decides and replies. The time this takes is

$$T_{\text{ground}} = 2\tau + 2W + T_{\text{decide}}, \qquad W = \tfrac{1}{2}(1-f)^2 P,$$

where $\tau$ is the one-way light-time and $T_{\text{decide}}$ the time the team needs to decide. $W$ is the mean wait for a contact window, for a spacecraft in contact a fraction $f$ of a cycle of length $P$. The factor $(1-f)^2$ appears because an event falls outside contact with probability $1-f$ and then waits, on average, half the gap.




b

| Equipment fault | Landing guidance | Driving hazard | Science event | Plan change | Collision warning |
|---|---|---|---|---|---|
| F1 | – | – | – | E2 | E2 |
| F1 | – | – | – | E2 | E2 |
| F1 | – | – | E3 | E3 | E2 |
| **F2** | **E4** | – | – | – | – |
| **F2** | **E4** | – | E2 | – | – |
| **F2** | **E4** | – | – | – | – |
| **F2** | – | **E4** | **E4** | E3 | – |
| **F2** | – | – | E3 | E3 | E3 |
| **F2** | – | – | **E4** | E3 | – |
| **F2** | – | – | – | **E4** | – |

Model: response = 2 × one-way light-time + 2 × mean wait for a contact, (1 − f)²P/2 for contact fraction f and cycle P, + 30 min for a ground decision. (b) compares it with each event's time constant (fault, guidance 1 min; driving hazard 10 s; science event 2 h; plan change 6 h; conjunction 1 day): ratio < 0.3 ground in the loop (E2); 0.3–10 onboard reactions (E3, F1); ≥ 10 onboard re-planning (E4); F2 where ground recovery takes ≥ 6 h or in critical phases. Illustrative heuristic, not an ECSS rule.

**FIGURE 48.1** How much autonomy different missions need, from a simple model. (a) Time for the ground to respond to an event on board: round-trip light-time, waiting for contact windows and a 30-minute decision. (b) The autonomy level implied when that time is compared with how fast typical events unfold. Grey cells are events that do not apply.

The figure applies the model with a 30-minute decision time. A geostationary satellite or the space station gets an answer in about half an hour. A low-orbit imager with two polar ground stations waits about 1.9 hours, mostly for the next pass. A Mars rover waits about 23 hours, and

Voyager 1 about 61 hours. Where the ground is more than about ten times too slow, the spacecraft must re-plan for itself (E4). Equipment faults need at least a safe mode (F1) everywhere. Far from Earth, or during landing, they need full recovery on board (F2).

## 48.2 Inside an Autonomous Spacecraft

An autonomous spacecraft is not one clever program. It is a stack of simple, well-tested parts, each working on its own time scale. Learning is usually confined to the parts that interpret sensor data.

### 48.2.1 Sense, Plan, Act

Most designs follow a **three-layer architecture** (Figure 48.2). The **functional layer** runs fast control loops: attitude control, engine firing, camera exposure, and the perception software that turns images into estimates of position or hazards. The **executive layer** runs the plan. It starts activities, checks that resources and timing stay within limits, and switches modes. The **deliberative layer** thinks ahead. A planner turns goals into a timeline, science software decides what is worth observing, and navigation software chooses paths and manoeuvres.

Two functions cut across the stack. **Fault detection, isolation and recovery (FDIR)** watches for anything that goes wrong. A **runtime monitor** checks commands against simple safety rules before they reach the hardware. The ground does not disappear. It sends goals, models and rules instead of individual commands, and receives summaries and selected data instead of everything.

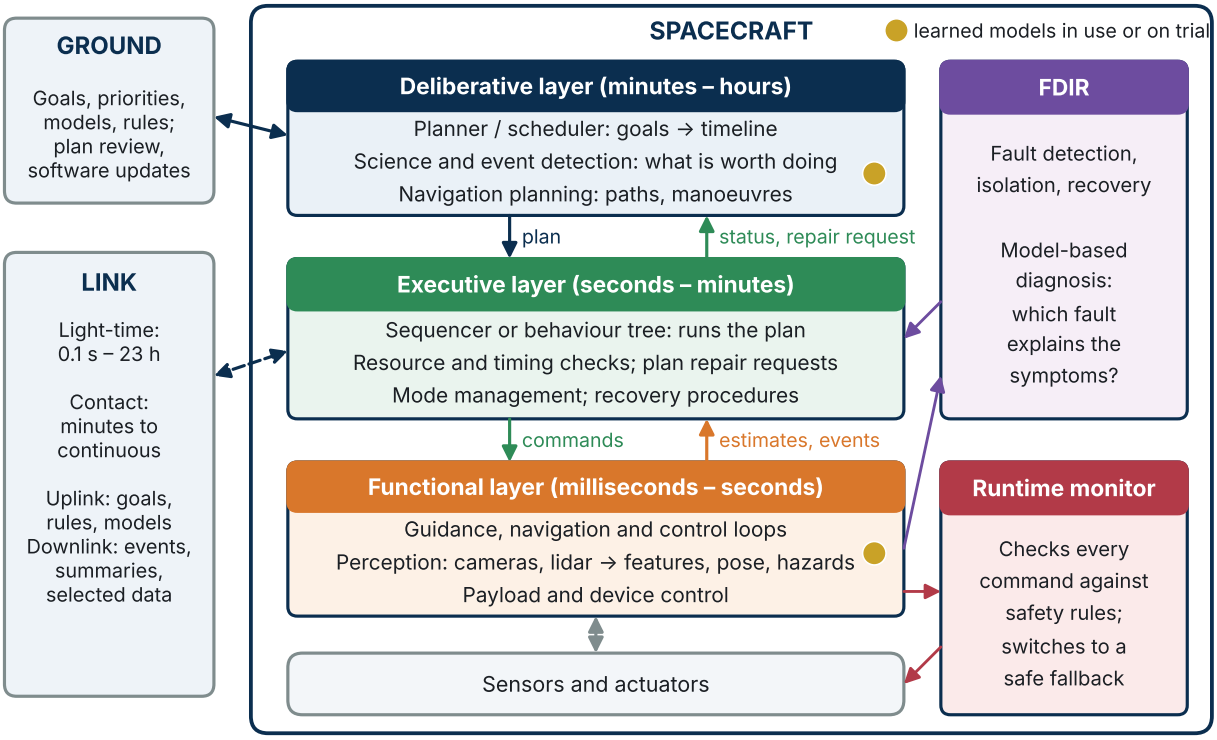



**FIGURE 48.2** Block diagram of an autonomous spacecraft. Faster layers sit lower in the stack; the ground sends goals and rules rather than commands. Gold dots mark where learned models are used or tested today: event detection and perception.

### 48.2.2 Executives, Behaviour Trees and Fault Protection

The **executive** turns a plan into action. Classic flight executives are timeline sequencers: time-tagged commands plus rules such as "if the battery falls below 30 %, stop the payload". **Behaviour trees**, taken from computer games and ground robotics, organise behaviours as a tree. Each node is "ticked" in turn and reports success, failure or running. A *sequence* node runs its children in order until one fails; a *fallback* node tries alternatives until one succeeds. Trees are easy to read and test, which is why they are increasingly proposed for space robots.

**FDIR** is the oldest form of autonomy. Monitors compare telemetry with limits. When a limit is crossed, a recovery action switches to a redundant unit or puts the spacecraft in **safe mode**: Sun-pointing, power-positive, listening for the ground. That is level F1. Level F2 goes further and resumes the mission, as deep-space probes must.

Fault protection can also cause failures, when the rules meet a situation nobody imagined. In October 2016 ESA's Schiaparelli lander saw its inertial unit saturate for about a second under the parachute. The navigation software then computed a negative altitude, released the parachute and fired the thrusters for only a few seconds. The lander fell from about 3.7 km. In April 2023 ispace's Hakuto-R lander crossed a crater rim during descent. The sudden jump in measured altitude made its software reject the good altimeter data as faulty. The lander kept descending slowly from about 5 km until its fuel ran out. Both lessons are the same: autonomy must be tested against the real terrain and sensors, not only the expected ones.

### 48.2.3 Model-Based Diagnosis

Rules say "if symptom A, then do B". They become unmanageable when faults combine. **Model-based diagnosis** takes another approach. Engineers write a model of each component and its possible modes — a valve can be *open*, *closed*, *stuck open* or *stuck closed* — and how the modes connect. The software then searches for the most probable set of modes that explains what the sensors see:

$$\hat{m} = \arg\max_{m} \; P(m \mid \text{observations}) \;\propto\; P(\text{observations} \mid m)\, P(m).$$

Faults are rare, so the prior $P(m)$ favours explanations with few failed parts. NASA's **Livingstone** engine, built at Ames Research Center in the 1990s, did this with logical models and fast search. It then planned the reconfiguration needed to work around the fault. Learned anomaly detectors on the ground (Chapter 49) now complement such models.

### 48.2.4 Twenty-Five Years of Flight Experience

Landmark flights of onboard autonomy are listed in Table 48.2. Three patterns stand out. Autonomy has grown from one-off experiments into everyday operations. Most of it is search, estimation and optimisation, not learning. And learned components have entered mainly through perception and science, such as EO-1's classifiers and rover targeting. The first in-orbit test of a learned attitude controller was reported only in 2025.

**TABLE 48.2** Landmark flights of onboard autonomy, from experiments to daily operations. "Operational" means used in normal mission operations.

| Year | Mission | Autonomous function | Status | Result |
|---|---|---|---|---|
| 1999 | Deep Space 1 | Remote Agent planning, execution, diagnosis | Flight experiment | 100 % of objectives |
| 1999–2001 | Deep Space 1 | AutoNav optical navigation | Flight demonstration | Comet Borrelly flyby, 2001 |
| 2004–17 | EO-1 | Onboard planning and event detection (ASE) | Operational | About $1 M a year saved |
| 2010– | Opportunity, Curiosity, Perseverance | AEGIS science targeting | Operational | 889 targets 2016–May 2026 |
| 2019– | Starlink | Autonomous collision avoidance | Operational | 207 152 manoeuvres in 6 months |
| 2020 | OSIRIS-REx | Natural Feature Tracking to sample site | Operational | Within about 1 m |
| 2021 | Perseverance | Lander Vision System | Operational | Touchdown 5 m from aim |
| 2021– | Perseverance | Enhanced AutoNav | Operational | > 90 % of 45 km |
| 2022 | DART | SMART Nav terminal guidance | Operational | Hit 160 m Dimorphos |
| 2023– | Perseverance | Onboard planner (from 5 Oct 2023) | Operational | Hundreds of plans |
| 2024 | SLIM | Crater-matching pinpoint landing | Flight demonstration | Under 10 m before final avoidance |
| 2025 | InnoCube | Learned attitude controller (LeLaR) | Flight experiment | First full manoeuvre 30 Oct 2025 |

#### REMOTE AGENT: AN AI IN CHARGE OF A SPACECRAFT (DEEP SPACE 1, MAY 1999)

NASA's Deep Space 1 tested twelve new technologies. One was the **Remote Agent**, the first AI software to control a spacecraft without ground commanding. It had three parts: a constraint-based planner and scheduler, a multi-threaded "Smart Executive", and Livingstone for diagnosis and recovery. The experiment ran during 17–21 May 1999, for about 35 hours in all.

A 2-day scenario started at 11:04 PDT on 17 May. Remote Agent planned and ran ion-engine thrusting and imaging. When a simulated fault left a camera stuck on, it tried to switch it off, then re-planned around the extra power use. Early on 18 May the part of the executive that issues commands stopped, so the ion engine was not turned off as planned; a backup command shut it down later. The cause was a race condition between two executive threads that had not appeared in extensive ground testing. About 70 % of the objectives had been met by then. There was no time to fix and re-test the software that week. The team instead wrote a new 6-hour scenario and ran it with the bug still on board. On 21 May that scenario completed the remaining objectives, so the experiment met 100 % of them. Its deadlock became a classic argument for formal verification of flight software.

## 48.3 Onboard Planning and Scheduling

A planner decides *what* to do; a scheduler decides *when*. In space the two are usually combined. The key idea is that a plan is a set of activities that must satisfy hard constraints — time windows, power, data volume and instrument states — while earning as much value as possible.

### 48.3.1 Plans, Constraints and Search

Written as mathematics, choosing observations is an integer programme. Each candidate observation $i$ has a value $v_i$, and $x_i$ is 1 if it is chosen and 0 if not.

$$\max_{x} \sum_{i} v_i x_i \quad \text{subject to} \quad Ax \le b, \qquad x_i \in \{0, 1\}.$$

Each row of $Ax \le b$ is one constraint. Two observations that overlap in time cannot both be chosen. The energy used in an orbit must stay within the battery budget, and data taken between downlinks must fit in memory. Such problems are **NP-hard**: no known algorithm solves every case quickly, and the worst-case effort grows exponentially with size. Flight planners therefore use three families of methods.

- **Greedy heuristics** sort candidates by a rule, such as value per unit of resource, and add each one that still fits. They are fast and predictable but can miss better combinations.
- **Local search** starts from a plan and keeps making small changes — add an observation, remove a clashing one — accepting changes that help, and sometimes ones that do not, to escape dead ends. JPL's **CASPER** planner uses a related idea called *iterative repair*: when reality departs from the plan, it fixes only the parts that conflict.
- **Exact optimisation**, such as mixed-integer linear programming (MILP) or constraint programming, proves that a plan is the best possible. It needs more computing and its run time is harder to bound.

### 48.3.2 A Toy Scheduling Problem

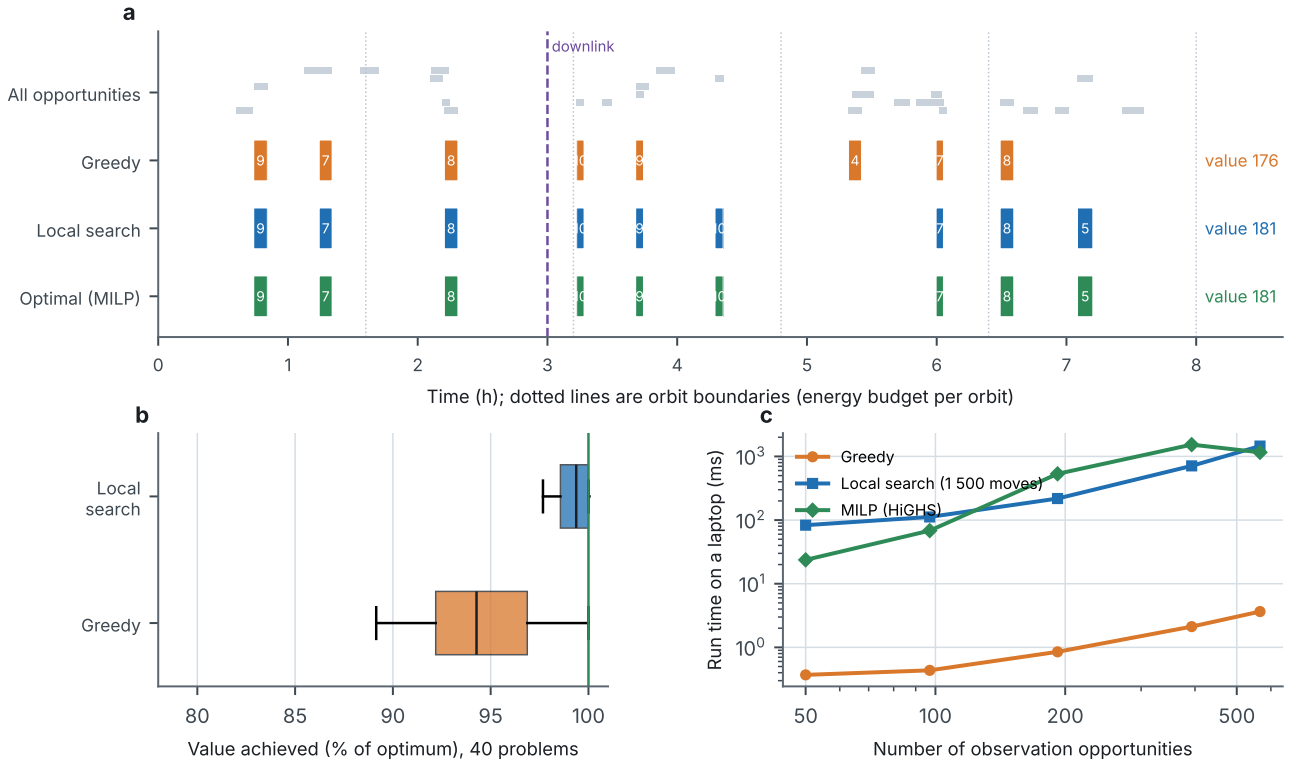
Figure 48.3 compares the three on a synthetic day for an imaging satellite. There are 45 targets, each visible one to three times, with values from 1 to 10. The satellite has one instrument with 2-minute slews, an energy budget per 96-minute orbit and a memory emptied at four downlink passes. A typical day has about 90 opportunities and just over 100 constraints.

Over 40 random days, the greedy rule reached 94.4 % of the optimal value on average and 89 % in the worst case, in under a millisecond on a laptop. Local search by **simulated annealing** reached 99.2 % on average in about a tenth of a second, and found the exact optimum on 15 of the 40 days. The exact MILP solver also took about a tenth of a second. As problems grew to about 570 opportunities, MILP took over a second and the greedy rule's shortfall stayed near 9 %. Flight computers are a hundred to a thousand times slower than a laptop. Flight planners therefore pair a fast heuristic, which always yields a safe plan, with improvement when time allows.

### 48.3.3 Planners in Flight Today

The most visible planner today runs on Perseverance. Its **onboard planner** (also called Simple Planner) was developed from 2016 and verified from October 2021, and entered operations on 5 October 2023. It is a priority-first, non-backtracking scheduler: it places activities in priority order and never undoes a placement, which keeps its run time bounded. It re-schedules when an activity ends early or late, or when energy differs from predictions, so the rover can use margin that ground planners had to hold back. In its first two years it ran hundreds of plans with very few anomalies.

Europe has tested autonomy mainly through small missions. ESA's OPS-SAT CubeSat (2019–2024) was an open flying laboratory for experiments in onboard planning and machine learning, among others. France's space agency CNES flew autonomous orbit control on the DEMETER microsatellite from 2005 and an autonomous guidance kit on PICARD in 2014. Its ASTERIA orbit-control software was later demonstrated on OPS-SAT. These steps matter because operators accept autonomy gradually, function by function, once each one has proved itself.




Synthetic day of 45 targets with 1–3 visibility windows each (3–10 min), values 1–10, one instrument with 2 min slews, payload energy ≤ 14 units per 96-min orbit, memory ≤ 40 units between four downlink passes. Exact solution by mixed-integer linear programming (scipy/HiGHS). Flight processors are 100–1 000 times slower than a laptop.

**FIGURE 48.3** Scheduling one day of observations for a synthetic imaging satellite. (a) Part of one day: all opportunities (top) and the plans chosen by a greedy rule, local search and an exact mixed-integer solver, with values. (b) Value achieved relative to the optimum over 40 random days. (c) Run time against problem size on a laptop.

THE AUTONOMOUS SCIENCECRAFT ON EO-1 (2004–2017)

NASA's Earth Observing-1 satellite, launched in 2000 to test new imagers, became the longest-running test of an onboard planner. Its **Autonomous Sciencecraft Experiment** (ASE) began operating on board in January 2004 and was declared validated in May 2004. In November 2004 it became EO-1's primary operations system. It had three parts. Onboard classifiers analysed Hyperion images for events such as eruptions, floods, fires and ice break-up. The CASPER planner re-planned observations around new goals. A robust executive ran the result.

The computer was modest: a 12 MHz Mongoose V processor of about 8 million instructions per second (MIPS), separate from the main flight computer for safety. ASE had only about 4 MIPS and 128 MB of memory to itself. CASPER's model was kept small so that it could build a plan in tens of minutes. By 2007 ASE had made over 11 000 autonomous data acquisitions and over 1 500 closed-loop re-tasks. Alerts from other satellites, such as volcano warnings, could trigger Hyperion images through ground "sensorwebs". EO-1's operations costs fell from $3.6 million in fiscal year 2005 to $1.6 million in 2006, and JPL attributed about $1 million a year of the saving to ASE. Science events captured per megabyte of downlink rose over a hundredfold. ASE remained the primary planning system until the end of the mission in 2017, after more than 60 000 autonomous observations.

## 48.4 Navigation Without the Ground

Navigation means knowing where you are and how fast you are moving. Most spacecraft are navigated from the ground by radio tracking (Chapter 24), which is accurate but slow. The key idea of autonomous navigation is to measure position against something the spacecraft can see — stars, asteroids, craters or terrain — and fuse those measurements in an onboard filter.

### 48.4.1 Optical Navigation in Deep Space

A camera measures a direction. If it sees a known asteroid against background stars, it measures the unit vector $\hat{\boldsymbol{u}} = (\mathbf{r}_a - \mathbf{r})/\|\mathbf{r}_a - \mathbf{r}\|$ from the spacecraft at $\mathbf{r}$ to the asteroid at $\mathbf{r}_a$. A Kalman filter (Chapter 6) combines such directions, taken over days, with the equations of motion to fix the orbit.

Deep Space 1's **AutoNav** was the first system to do this on its own, in 1999. It also steered the spacecraft through its flyby of comet Borrelly in September 2001. The same software kept Stardust's camera locked on comet Wild 2 during its 2004 flyby. In July 2005 it flew Deep Impact's impactor into comet Tempel 1, making the final targeting manoeuvres in the last two hours. DART's **SMART Nav** went further in September 2022. It took over about four hours before impact, picked out the 160 m moonlet Dimorphos from its larger partner only about an hour before impact, and steered into it at about 6 km/s. None of these systems used machine learning: they relied on image centroiding, geometry and filtering.

### 48.4.2 Terrain-Relative Navigation

Near a surface, the spacecraft can compare what it sees with a map. This is **terrain-relative navigation (TRN)**. It solves a problem that radio tracking cannot: an inertial unit alone may drift by kilometres during entry, while the safe spots in a rugged landing area may be only tens of metres across.

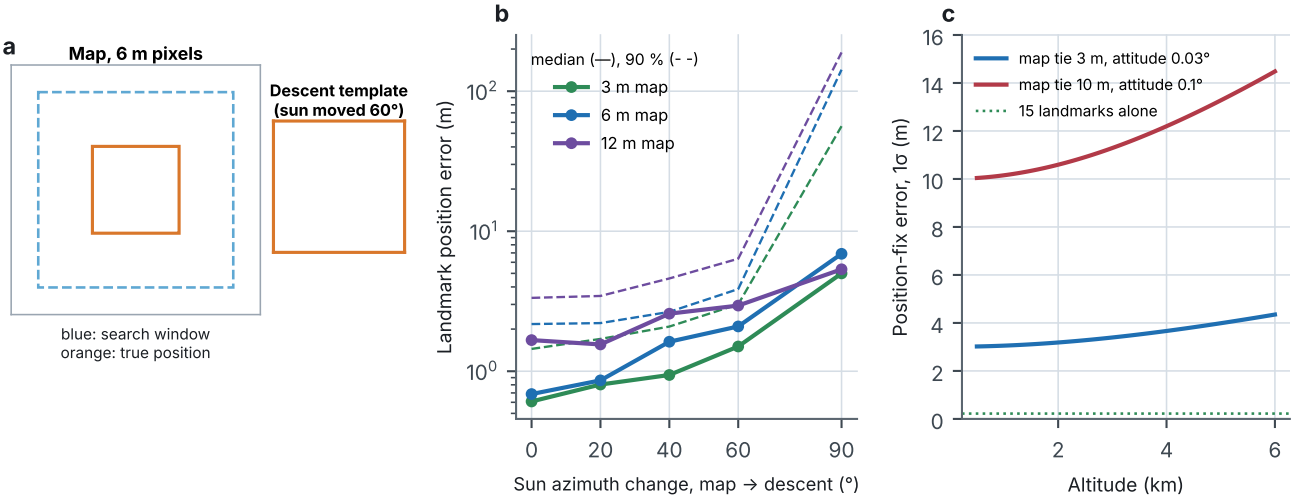
Perseverance's **Lander Vision System** (LVS) set the standard in February 2021. Its maps came from HiRISE orbital images, resampled to 12 m pixels for a coarse search and 6 m for a fine one. During descent, LVS first matched 15 landmarks across three images, correlating 128 × 128-pixel patches against large map windows in the frequency domain. It then matched up to 150 small landmarks per image and fused them with inertial data. This cut the position error from up to 3.2 km to the required 40 m or less. LVS then chose the safest reachable spot from an onboard hazard map. Perseverance touched down within 5 m of the point LVS aimed for.

The matching step uses **normalised cross-correlation**. For a template $t$ from the descent image and a map patch $m$ at offset $(i, j)$,

$$\rho(i,j) = \frac{\sum_k (t_k - \bar{t})(m_{k+ij} - \bar{m}_{ij})}{N\,\sigma_t\,\sigma_{m,ij}},$$

where the sums run over the $N$ pixels of the template. The offset with the highest $\rho$ is the match. Because $\rho$ removes the mean and scales by the spread, it tolerates changes in brightness and contrast. It does not tolerate changes in shadow direction, because shadows move when the Sun moves.

Figure 48.4 tests this on synthetic terrain. With the Sun in the same place for the map and the descent image, 32 × 32-pixel templates matched a 6 m map well. The median error was 0.7 m, about a tenth of a pixel. When the Sun had moved 60° in azimuth, the median grew to 2.1 m. At 90° a third of matches failed outright. Real systems therefore plan landing times to match the lighting of the map, and reject outliers. Panel (c) shows the bigger lesson: with many landmarks, the matching error is not the limit. Errors in tying the map to the inertial frame, and in knowing the camera's attitude, dominate the final fix.




Synthetic 3 km terrain (fractal relief, 1 200 craters, albedo noise) rendered at 25° sun elevation; 32 × 32-pixel descent templates matched by normalised cross-correlation in a ±20-pixel window, 40 trials per point. (c) Root-sum-square of landmark error/√N (6 m map, 20° sun change), map registration and attitude-knowledge error projected over the altitude.

**FIGURE 48.4** Terrain-relative navigation on synthetic terrain. (a) A 6 m map with a 32 × 32-pixel descent template and its search window. (b) Error of single landmark matches as the Sun's azimuth changes between map and descent; dashed lines show the 90th percentile. (c) A position-fix error budget against altitude: the map tie and attitude knowledge dominate.

**EXAMPLE: HOW ACCURATE CAN TERRAIN-RELATIVE NAVIGATION BE?**

A lander carries a 6 m-pixel map tied to the inertial frame with 3 m (1σ) error. Its attitude is known to 0.03°. At 4 km altitude it matches $N = 15$ landmarks, each with the median error of Figure 48.4 for a modest lighting change: 0.86 m, or 0.14 pixel.

1. **Landmarks.** Averaging $N$ independent matches gives $0.86/\sqrt{15} = \mathbf{0.22\ m}$.
2. **Attitude.** A pointing error $\delta\theta$ shifts the projected position by $h\,\delta\theta = 4\,000\ \text{m} \times 5.2 \times 10^{-4}\ \text{rad} = \mathbf{2.09\ m}$.
3. **Map tie.** The map itself may sit 3 m from where the navigation frame thinks it is: **3.0 m**.
4. **Total.** Independent errors add in root-sum-square: $\sqrt{0.22^2 + 2.09^2 + 3.0^2} = \mathbf{3.7\ m}$.

The answer is about ten times better than the 40 m requirement LVS had to meet, and matches its 5 m touchdown. Doubling the number of landmarks hardly helps. Halving the map-tie error helps most. With a 10 m map tie and 0.1° attitude knowledge, the same matcher gives 12 m.

Other landers use similar ideas. China's Chang'e landers have hovered about 100 m above the Moon since 2013 to choose a safe spot with lidar and cameras (Chapter 27). Japan's SLIM matched craters in its descent images with an onboard crater map in January 2024. All 14 of its image-matching fixes were judged accurate, and its position error before the final hazard avoidance was under 10 m. NASA's **SPLICE** project packages TRN, hazard-detection lidar, a Navigation Doppler Lidar and a descent computer for commercial landers. On Intuitive Machines' IM-1 in February 2024, a safety switch left the lander's own laser rangefinders disabled. A software patch to use NASA's Doppler lidar instead reportedly failed for want of a data flag, and Odysseus landed on its side. In June 2025 ispace's second lander crashed after its laser rangefinder reported late. In these recent failures the trouble lay in sensors and in the logic that judges their data, not in the navigation algorithms.

### 48.4.3 Crater Detection with Learning

Craters are the natural landmarks of airless worlds. A crater-based navigator detects craters in the descent image, matches the pattern with a crater catalogue and solves for position. Classical detectors look for the bright-and-dark pair that a bowl makes in low sunlight, or correlate the image with crater templates. Learned detectors instead train a network on many labelled examples (Chapter 44). The best-known example, **DeepMoon** (2019), trained a U-Net on lunar elevation maps. It recovered 92 % of the craters in a human-made catalogue, roughly doubled the number of detections, and had an estimated 11 % false-positive rate among new finds.

Figure 48.5 compares the two approaches on synthetic scenes. The template matcher was given an ideal crater rendered at the *true* Sun angle. The learned detector was a small numpy neural network with about 16 500 weights, trained in seconds on 29 689 patches from ten other scenes under random lighting. At a low Sun of 10°, both found craters equally well: average precision (the area under the precision–recall curve) was 0.56 and 0.62. As the Sun rose, shadows shrank and eroded craters faded. At 25° the template matcher dropped to 0.28, and at 45° to 0.09, while the network kept 0.57 and 0.34. At 65° both failed.

What matters for navigation is the **position fix**. Matching the 40 best detections to the catalogue with RANSAC gave errors of 0.2–0.8 m with templates at 10°. The network's fixes were slightly less precise, 0.5–1.7 m, because its sliding window is coarser. Template-based fixes locked onto wrong

matches, more than 100 m off, in one of three scenes at 25° and in all scenes from 45° upwards. The network's fixes stayed within 3 m to 45° and mostly within 10 m at 65°. The learned detector was not more accurate, but it was more *robust*, because it had seen many lighting conditions. Synthetic tests flatter both methods; real images add blur and craters missing from the catalogue.

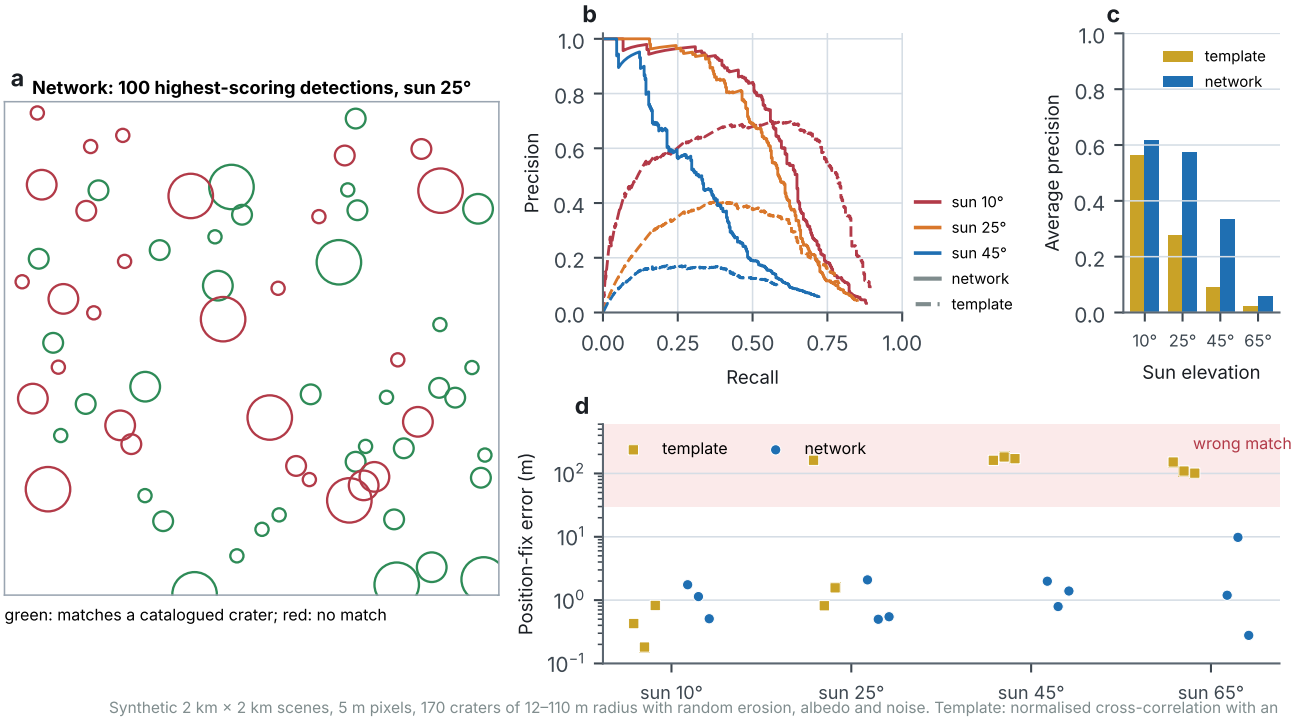
### 48.4.4 Small Bodies

Asteroids and comets are hard targets. Their gravity is weak and irregular, and maps must be built on arrival. **OSIRIS-REx** navigated to the Nightingale site on Bennu in October 2020 with **Natural Feature Tracking**. The spacecraft rendered the expected terrain from an onboard shape model and correlated it with its camera images. It touched down within about 1 m of its target. Japan's **Hayabusa2** dropped reflective target markers on Ryugu and homed in on them. Its second touchdown, in July 2019, was 60 cm from the aim point. ESA's Hera, due at Didymos in November 2026, relies on an autonomous "self-driving" navigation system around the pair.

Table 48.3 summarises the main onboard navigation systems.

**TABLE 48.3** Onboard navigation systems that have flown. None used machine learning in its flight loop.

| System | Mission | Sensor | Method | Reported accuracy |
|---|---|---|---|---|
| AutoNav | DS1, Stardust, Deep Impact | Camera | Asteroid and comet directions, filter | Comet impact, 2005 |
| SMART Nav | DART, 2022 | DRACO camera | Target centroiding, onboard guidance | Hit 160 m moonlet |
| Natural Feature Tracking | OSIRIS-REx, 2020 | Navigation camera | Rendered-template correlation | About 1 m |
| Target markers | Hayabusa2, 2019 | Camera, flash, rangefinder | Marker tracking | 60 cm |
| Lander Vision System | Perseverance, 2021 | Camera, inertial unit | Landmark correlation with maps | Within 5 m of aim |
| Hover and avoid | Chang'e 3–6, 2013–24 | Lidar, cameras | Hazard map at about 100 m | Safe landings |
| Crater matching | SLIM, 2024 | Camera | Crater detection and catalogue match | Under 10 m before avoidance |




Synthetic 2 km × 2 km scenes, 5 m pixels, 170 craters of 12–110 m radius with random erosion, albedo and noise. Template: normalised cross-correlation with an ideal crater rendered at the true sun angle. Network: 256–64–1 perceptron trained on patches from ten other scenes under random lighting. (d) Position fix by RANSAC matching of the 40 best detections to the crater catalogue, three test scenes per sun angle.

**FIGURE 48.5** Crater detection on synthetic lunar scenes with 5 m pixels. (a) The 100 highest-scoring detections of a small neural network at 25° Sun elevation; green circles match a catalogued crater. (b) Precision–recall curves for the network (solid) and template matching at the true Sun angle (dashed). (c) Average precision against Sun elevation. (d) Error of the position fix obtained by matching detections to the crater catalogue; the shaded band marks wrong matches.

## 48.5 Pose Estimation for Rendezvous and Servicing

To dock with, inspect or capture another spacecraft, a chaser must know the target's **pose**: its position and its attitude relative to the camera (Chapter 40). A cooperative target carries markers, reflectors or radio beacons. A non-cooperative one — a dead satellite or a rocket stage — does not.

Its pose must be estimated from images and the known shape alone, under harsh lighting that changes every few minutes.

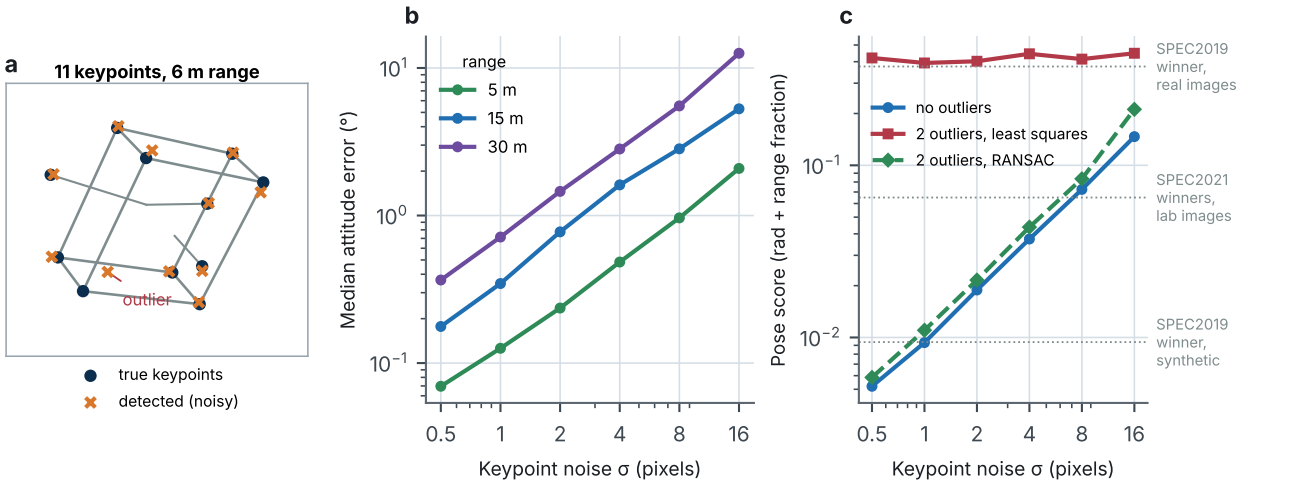
### 48.5.1 Keypoints and Perspective-n-Point

The most successful learned approach splits the task in two. First, a convolutional network finds the target in the image and predicts the pixel positions of a few **keypoints** — corners, antenna tips, nozzle centres — whose 3-D positions on the target are known. Second, a geometric solver recovers the pose. A point $\mathbf{X}_i$ on the target appears in the image at

$$\mathbf{u}_i = \pi\big(K(R\,\mathbf{X}_i + \mathbf{t})\big),$$

where $R$ and $\mathbf{t}$ are the rotation and translation, $K$ holds the camera's focal length and centre, and $\pi$ divides by depth. The **Perspective-n-Point (PnP)** solver finds the $R$ and $\mathbf{t}$ that minimise the squared reprojection error $\sum_i \|\mathbf{u}_i - \hat{\mathbf{u}}_i\|^2$. **RANSAC** (random sample consensus) protects it from wrong keypoints. It fits poses to many small random subsets, keeps the pose that most points agree with, and refits on those "inliers". This split is popular because the network does what networks are good at — recognising image patterns — and the geometry stays exact and checkable.

Keypoint errors become pose errors as shown in Figure 48.6, for a camera like the one used in the Stanford–ESA challenges and an 11-keypoint target. At 15 m range, keypoint noise of 1 pixel gave a median attitude error of 0.35° and a position error of 0.2 % of range. At 8 pixels the attitude error grew to about 3°. Doubling the range roughly doubled the error, because the target covers fewer pixels. Two badly misplaced keypoints out of 11 ruined a plain least-squares fit, giving a pose score of about 0.4. RANSAC restored almost the full accuracy.




Monte Carlo, 120 random attitudes per point; SPEED-like camera (focal length 3 003 pixels); 11 keypoints on a 0.8 m body with antennas; DLT and POSIT initialisation, Levenberg–Marquardt refinement, RANSAC on 5-point POSIT samples. (c) at 15 m; mean score as in the ESA–Stanford challenges, position error capped at 100 %.

**FIGURE 48.6** Pose from keypoints by Perspective-n-Point. (a) True and noisy keypoints on a satellite-like target, with one outlier. (b) Median attitude error against keypoint noise at three ranges. (c) Mean pose score at 15 m, with and without two outliers and with RANSAC; dotted lines mark the winning scores of the 2019 and 2021 challenges.

### 48.5.2 The ESA–Stanford Challenges and the Domain Gap

Two open competitions shaped this field. In the **Satellite Pose Estimation Challenge** of 2019, run by ESA's Advanced Concepts Team and Stanford's Space Rendezvous Laboratory, 48 teams estimated the pose of the PRISMA mission's Tango satellite. The data set held about 15 000

rendered images and 305 laboratory photographs of a mock-up. The score adds the attitude error in radians to the position error divided by range. The winner, from the University of Adelaide, used keypoints and PnP. It scored 0.0094 on rendered test images, about 0.4° and 3 cm. On real photographs its score was 0.375, forty times worse.

That jump is the **domain gap**, also called the sim-to-real gap: networks trained only on rendered images meet real images with different noise, reflections and lighting (Chapter 45). The 2021 challenge attacked it directly. Teams trained on 59 960 rendered images and were tested on 9 531 hardware-in-the-loop images taken with two kinds of lighting: a “lightbox” imitating Earth glow and a “sunlamp” imitating direct sunlight. With heavy data augmentation, style randomisation and adaptation methods, the winning scores were 0.073 (lightbox) and 0.059 (sunlamp). By Figure 48.6, such scores correspond to six to eight pixels of keypoint noise at 15 m, against about one pixel for the 2019 winner on rendered images.

### 48.5.3 Lidar and Sensor Fusion

Cameras fail in darkness and glare. **Lidar** measures range directly by timing laser pulses. Its point cloud is matched to the target’s 3-D model, often with the **iterative closest point** algorithm, which pairs each measured point with the nearest model point and solves for the best rigid motion. Lidar works at any Sun angle but has fewer pixels and costs more power. Crew Dragon, ATV and the commercial servicers combine laser sensors with cameras.

## 48.6 Guidance and Control: Optimisation or Learning?

Guidance decides the path; control makes the vehicle follow it. The key message of this section is that the most impressive autonomous guidance in flight today is **optimisation**, not learning. Learning is promising but much harder to verify.

### 48.6.1 Convex Guidance

A powered landing must reach a target with zero velocity, using as little propellant as possible, without exceeding thrust limits or flying into the ground. The thrust magnitude has a lower bound as well as an upper one, because rocket engines cannot be throttled to zero. That lower bound makes the problem non-convex — it can have many local optima — and hard to solve reliably on board.

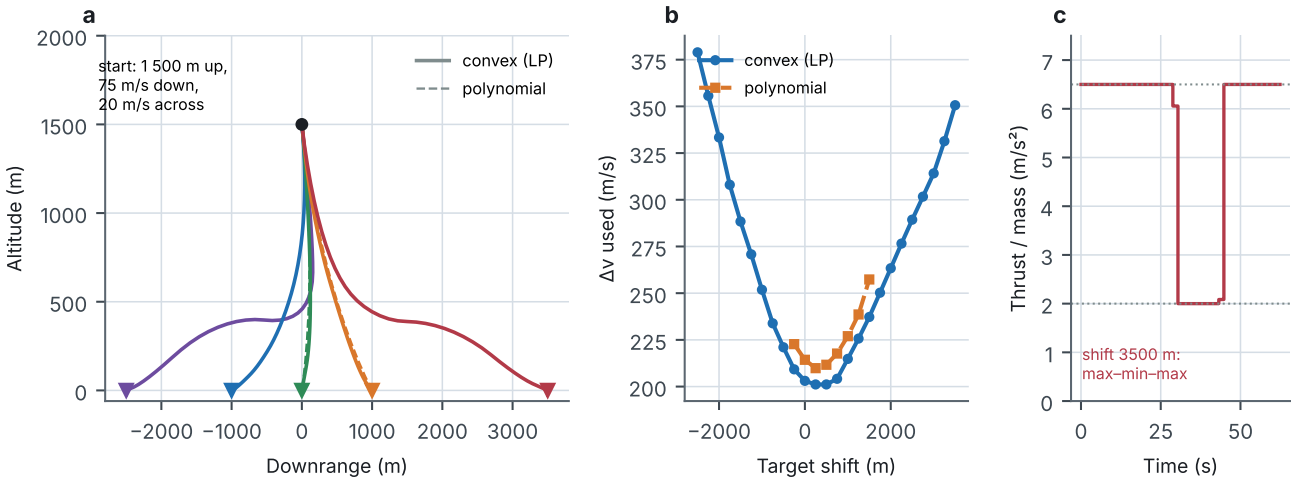
In 2007 Behçet Açıkmeşe and Scott Ploen at JPL showed a way out, called **lossless convexification**. They introduced a slack variable $\sigma$ for the thrust acceleration magnitude and wrote the constraints as

$$\|\mathbf{u}(t)\| \le \sigma(t), \qquad \rho_{\min} \le \sigma(t) \le \rho_{\max}, \qquad \min \int_0^{t_f} \sigma(t)\, dt.$$

This relaxed problem is convex, so a solver finds its global optimum reliably and quickly. They proved that at the optimum $\|\mathbf{u}\| = \sigma$, so the relaxation loses nothing. The resulting **G-FOLD** algorithm (guidance for fuel-optimal large diverts) was flown on Masten’s Xombie rocket in 2012–13. In July 2013 it diverted 750 m sideways at about 25 m/s. Its custom solver could compute a

trajectory in about 100 ms on a 1.4 GHz processor. SpaceX engineers have described using convex optimisation to land Falcon 9 boosters. This is not machine learning: nothing is learned from data, and every answer can be checked against the constraints.

A 2-D version for a Mars landing can be solved by linear programming, approximating the circle $\|\mathbf{u}\| \le \sigma$ by a 16-sided polygon (Figure 48.7). From 1 500 m altitude and 75 m/s descent, the convex method could reach targets from 2.5 km behind to 3.5 km ahead. A classical polynomial guidance law, with acceleration linear in time, could reach only from 250 m behind to 1.5 km ahead. Where both worked, the polynomial law used 4–8 % more Δv. The optimal thrust profile is "max–min–max": brake hard, coast at minimum thrust, then brake hard again.




2-D Mars landing (g = 3.71 m/s²), thrust acceleration 2–6.5 m/s², within 60° of vertical, 10° glide slope. Convex: 40-step discretisation, ‖u‖ ≤ σ by a 16-sided polygon, minimise Σσ Δt by linear programming (HiGHS) with a search over flight time. Polynomial: acceleration linear in time, best feasible flight time.

**FIGURE 48.7** Fuel-optimal powered descent solved by linear programming (lossless convexification) compared with polynomial guidance. (a) Trajectories to targets at different distances; dashed lines show polynomial guidance where it is feasible. (b) Δv against target shift: convex guidance reaches a three times wider span of targets. (c) The optimal thrust profile for the largest shift.

## 48.6.2 Reinforcement Learning

**Reinforcement learning (RL)** learns a control **policy** $\pi(a \mid s)$ — what action $a$ to take in state $s$ — by trial and error in a simulator. Each trial earns rewards $r_t$, and learning maximises the expected **return**

$$J(\pi) = \mathbb{E}\Big[\sum_{t=0}^{T} \gamma^t r_t\Big],$$

where $\gamma \le 1$ discounts distant rewards. Policy-gradient methods nudge the policy's parameters in the direction that raises $J$. Simpler **policy-search** methods such as the cross-entropy method try many random parameter sets, keep the best few and sample around them. RL's appeal is that it needs no explicit model of the optimal solution, only a simulator and a reward.

In Figure 48.8 a small neural-network policy, with 12 hidden units and 49 weights, learns to land on the Moon in one dimension. Training used the cross-entropy method: 60 generations, 64 candidate policies each and 24 simulated landings per candidate — 92 160 landings in about 7 seconds. From generation 8 every landing in a validation set was soft (below 1.5 m/s). The policy

then kept improving its fuel use, from 58 m/s of Δv to 31 m/s. A simple proportional–derivative (PD) controller that tracks a speed profile used 48 m/s. The learned policy had discovered a near-optimal "suicide burn": fall fast, then brake at full thrust at the last moment.

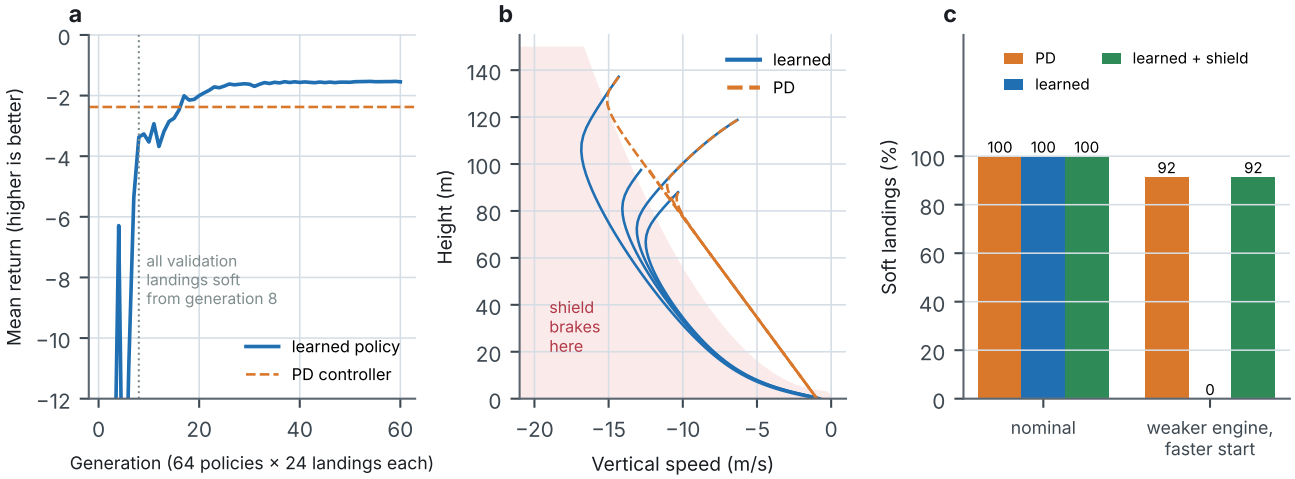
That efficiency is also its weakness. With 20 % less thrust than expected and faster starting speeds, the PD controller still landed softly 92 % of the time. The learned policy never did: its late braking left no margin. Research on landing, docking and attitude control shows the same pattern. RL policies match or beat classical controllers inside their training conditions but can fail abruptly outside them.

Real flights are beginning. In October 2025 the University of Würzburg's **InnoCube** CubeSat performed a full attitude manoeuvre with a controller trained by deep RL in simulation. Its team reported this LeLaR experiment as the first in-orbit demonstration of an AI-based attitude controller. On the ISS in February 2025, Stanford researchers used a neural network to give the Astrobee robot's trajectory optimiser a starting guess. Solver iterations fell by 50–60 %, and the optimiser still enforced every constraint. Learning that speeds up an optimiser, rather than replacing it, is the most likely route into flight.

### 48.6.3 Verifying Learned Controllers

Classical controllers come with proofs such as stability margins and worst-case bounds. A neural-network policy has none by default. It may exploit flaws in the simulator and behave strangely in states it never saw. Testing can show the presence of failures but not their absence.

**Runtime assurance** offers a practical answer. In the *simplex* architecture, an untrusted high-performance controller runs inside a "safety envelope". A simple monitor checks every command. If the state approaches the edge of the envelope, a verified backup controller takes over. In Figure 48.8 the monitor computes the stopping distance at 70 % of rated thrust and orders full braking when it approaches the height. With this **shield**, the learned policy landed softly 92 % of the time in the shifted test, as often as the PD controller. The price was fuel: the shield acted in every landing, raising the nominal Δv from 31 to 42 m/s. Assurance methods, from formal verification of networks to simplex designs, are covered in Chapter 53.




1-D lunar landing (g = 1.62 m/s², thrust acceleration 0–4 m/s², 0.2 s steps) from 80–150 m at 5–15 m/s descent; soft landing ≤ 1.5 m/s. Policy: 2–12–1 network trained by the cross-entropy method on return = −Δv/20 − 2 × excess touchdown speed. Shifted test: 20 % less thrust, start up to 20 m/s. Shield: full braking when the stopping distance at 70 % of rated thrust nears the height.

**FIGURE 48.8** Learning to land, in one dimension on the Moon. (a) Return of a neural-network policy trained by the cross-entropy method, with a PD controller for comparison. (b) Height against speed for four landings; in the shaded region a runtime monitor would order full braking. (c) Soft landings in nominal conditions and with a weaker engine and faster starts, for the PD controller, the learned policy and the learned policy with a shield.

Table 48.4 compares the three approaches.

**TABLE 48.4** Ways to compute guidance and control on board, from fixed laws to learned policies.

| Approach | How it decides | Guarantees | Flight status (2026) | Main risk |
|---|---|---|---|---|
| Classical (PD, LQR) | Fixed feedback law | Stability margins | Universal | Wastes fuel and envelope |
| Convex optimisation | Solves an optimisation on board | Global optimum; checkable | Rocket landing; G-FOLD tests | Solver time; model errors |
| Learned policy (RL) | Network trained in simulation | Statistical testing only | First flight tests (2025) | Fails outside training |
| Learned + runtime assurance | Network inside a monitored envelope | From verified monitor and backup | Research | Conservative envelope |
| Learned warm start | Network guesses, optimiser solves | Optimiser's guarantees kept | ISS test (2025) | A bad guess slows the solver |

## 48.7 Rendezvous, Docking and Proximity Operations

Automated docking is the oldest everyday autonomy in human spaceflight. The key idea is a strict sequence of hold points and approach corridors (Chapter 40), with the chaser flying itself and humans or the ground able to stop it at any time.

Soviet spacecraft have docked automatically since 1967, with the Igla system and, from 1986, the Kurs radio system still used by Soyuz and Progress. Europe's five **Automated Transfer Vehicles** (2008–2014) docked with the ISS using relative GPS, then laser rangefinders and a camera-based "videometer" reading retroreflectors (Figure 48.9). **Crew Dragon** docked itself for the first time in March 2019, using lidar and visible and thermal cameras. China has made automated rendezvous fast. Tianzhou-5 docked with the Tiangong station 2 hours 7 minutes after launch in November 2022, a world record. Shenzhou-21 brought its crew in about 3.5 hours in 2025.

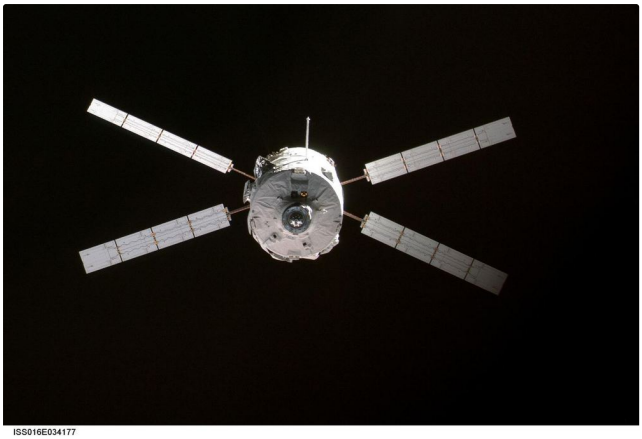



**FIGURE 48.9** ESA's first Automated Transfer Vehicle, Jules Verne, approaching the ISS during a rehearsal of its automated approach on 31 March 2008. Image: NASA/JSC.

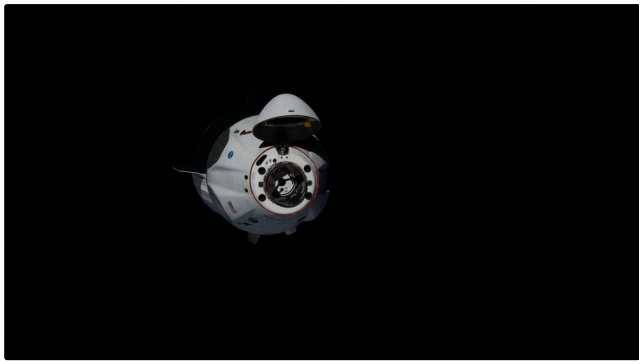


**FIGURE 48.10** Crew Dragon on its first crewed approach to the ISS on 31 May 2020, nose cone open to show the docking mechanism. Image: NASA/JSC (Chris Cassidy).

Servicing adds non-cooperative targets. DARPA's Orbital Express captured its partner autonomously in 2007, and Northrop Grumman's MEV-1 docked with Intelsat 901 in 2020. Astroscale's ADRAS-J approached a spent rocket stage to about 15 m in 2024. Learned pose estimation is the main entry point for AI here. The systems are compared in Table 48.5.

**TABLE 48.5** Automated rendezvous and docking systems. Most combine onboard control with human go/no-go decisions at hold points.

| Vehicle | First automated docking | Relative sensors | Who decides | Typical time to dock |
|---|---|---|---|---|
| Soyuz, Progress | 1967 (Igla); Kurs from 1986 | Radio (Kurs) | Onboard; crew backup (TORU) | 3 h – 2 days |
| ATV | 2008 | Relative GPS, laser, videometer | Onboard; ground can abort | About 10 days (ATV-4) |
| Crew Dragon | 2019 | Lidar, visible and thermal cameras | Onboard; crew monitors | About 1 day |
| Shenzhou, Tianzhou | 2011 | Radar, laser, cameras | Onboard; manual backup | 2 h 7 min (record) to 2 days |
| MEV-1, MEV-2 | 2020 | Cameras, infrared, lidar | Onboard; ground go/no-go | Months (slow orbit raising) |
| ADRAS-J | 2024 (approach only) | Cameras, lidar | Onboard; ground holds | Months |

## 48.8 Robots and Swarms

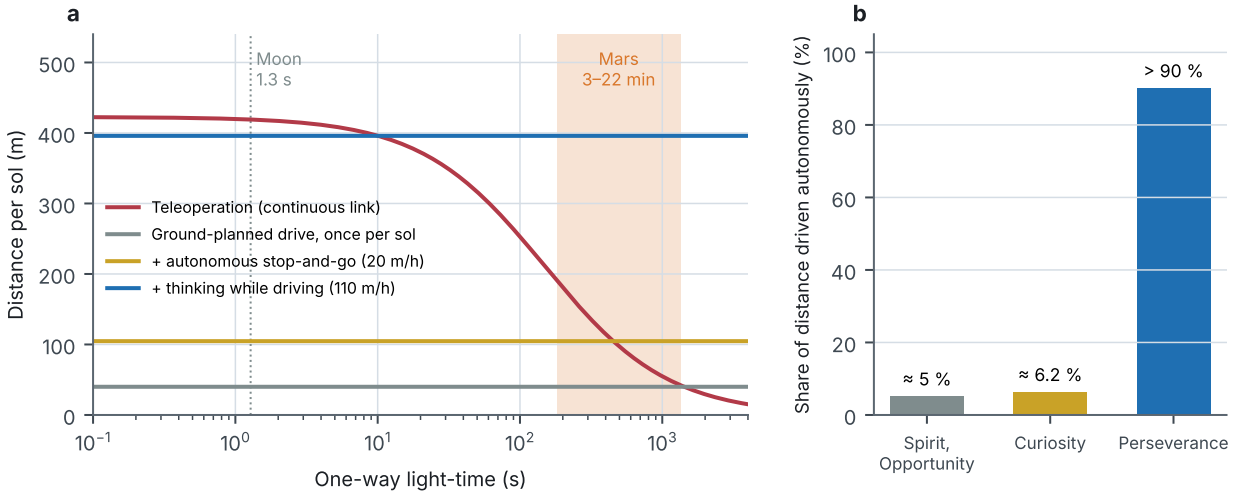
Robots are where autonomy is most visible. A robot's usefulness grows with the distance it can safely go, or the work it can safely do, between two contacts with its operators.

### 48.8.1 Rover Autonomy

A Mars rover drives in three ways (Chapter 28). In a **blind drive**, operators plan a path through terrain they have seen in stereo images, and the rover follows it with **visual odometry** — tracking features between images to measure its true motion. In **AutoNav**, the rover builds its own 3-D map, scores each patch for slope, rocks and steps, and chooses paths towards a goal.

Perseverance's **Enhanced AutoNav** adds "thinking while driving". A dedicated vision processor makes the maps while the wheels turn. An algorithm called Approximate Clearance Evaluation checks whether the rover's body can clear the rocks along each candidate path.

The benefit is large (Figure 48.11). Spirit and Opportunity used their early AutoNav for about 5 % of their driving, and Curiosity for about 6 %. Perseverance used AutoNav for more than 90 %. On 11 October 2026 it passed Opportunity's off-world driving record, reaching 45.22 km in about five and a half years. Opportunity needed more than fourteen years for the same distance. Perseverance can cover up to 412 m in a sol.




(a) Model: 3.5 h drive window per sol; top speed 4.2 cm/s; teleoperation in 10 m steps, each waiting a round trip plus 60 s of operator time; ground plans 40 m blind per sol; autonomous average speeds assumed. (b) NASA and IEEE Spectrum figures to 2024–26.

**FIGURE 48.11** Why rovers drive themselves. (a) Modelled distance per sol against one-way light-time for teleoperation with a continuous link, a single ground-planned drive per sol, and two kinds of onboard autonomy. (b) Share of distance driven autonomously by NASA's Mars rovers.

### EXAMPLE: HOW MUCH AUTONOMY DOES A MARS ROVER NEED?

Assume a one-way light-time of 20 minutes and one uplink of commands per sol. The rover can drive for 3.5 hours a day at a top speed of 4.2 cm/s.

1. **Teleoperation.** Even with a continuous link, each 10 m step takes 238 s of driving and 2 400 s of round-trip light-time. Add a minute to decide, and a step takes 2 698 s. In 3.5 hours that is 4.7 steps, or **47 m**.
2. **Ground planning.** With one uplink per sol, operators can plan only as far as their images show, about **40 m** of blind driving.
3. **Onboard autonomy.** At an average of 110 m/h while thinking while driving, the rest of the window adds about 356 m, giving about **400 m** per sol.
4. **Decisions.** If AutoNav assesses the terrain about once per metre, a 400 m drive needs about 400 hazard decisions. The ground can make one set of decisions per sol.
5. **Faults.** If a fault happens just after the morning uplink, the team may not hear of it for several hours, until the next relay pass. The rover must make itself safe (F1) and often continue (F2).

So driving needs event-driven autonomy (E3), resource-aware re-planning helps (E4), and fault handling must be F1–F2. To cross 10 km, ground planning alone would need about 250 driving sols; autonomy needs about 25.

Autonomy also chooses what to study. **AEGIS** (Autonomous Exploration for Gathering Increased Science) finds rocks in navigation-camera images after a drive and points a laser spectrometer at the best one, without waiting for the ground (Figure 48.12). It was first demonstrated on Opportunity in 2010. On Curiosity from 2016 it selected the scientists' preferred kind of target over 93 % of the time, against about 24 % expected without it. By May 2026 it had picked 596 targets on Curiosity and 293 on Perseverance.

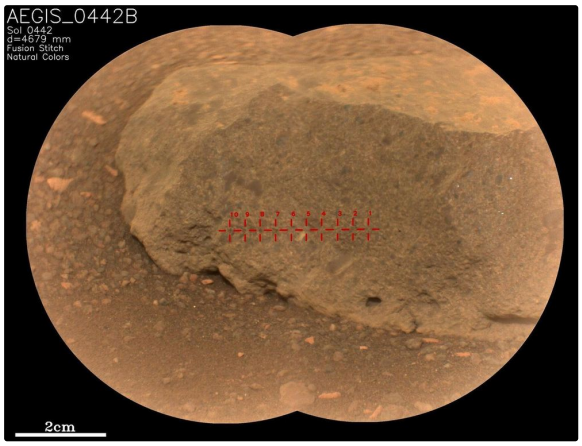



**FIGURE 48.12** A rock targeted by AEGIS on Perseverance on 18 May 2022, one of the first two chosen by the software for SuperCam without input from Earth. The red marks show the laser spots. Image: NASA/JPL (NASA/JPL-Caltech/LANL/CNES/IRAP).

The newest step moves AI into route planning on the ground. On 8 and 10 December 2025, Perseverance drove 210 m and 246 m along waypoints chosen by generative vision-language models. The models worked from orbital images and terrain-slope maps. Before upload, engineers checked the commands in JPL's digital twin of the rover, verifying more than 500 000 telemetry variables.

### 48.8.2 Arms, Free-Flyers and Humanoids

Robot arms in space are mostly supervised rather than autonomous. **Canadarm2** (2001) and **Dextre** (2008) on the ISS are driven by ground controllers or crew. Ground teams run pre-planned, automatic motion sequences for routine jobs, while astronauts usually fly the capture of cargo ships. Canada's **Canadarm3** was designed for the Gateway station. After NASA paused Gateway in March 2026, its builder MDA Space continued under its Canadian Space Agency contract, refocused on lunar-surface logistics, with launch no earlier than 2029.

Inside the ISS, three cube-shaped **Astrobee** free-flyers have operated since 2019. They navigate by matching camera images to a map of the station, move with electric fans and serve as a test bed for autonomy research (Figure 48.13). **Robonaut 2**, the first humanoid robot in space, worked on the ISS from 2011 to 2018. It was mostly teleoperated, testing whether a dexterous robot could use tools and switches made for people (Figure 48.14). The Japanese start-up **GITAI** tested an arm inside the commercial Bishop airlock in 2021. In March 2024 it mounted a second arm outside and performed assembly and servicing tasks, reaching technology readiness level 7.

### 48.8.3 Swarms, Formations and Constellations

Spacecraft working together must know where the others are and agree on who does what. The Swedish-led **PRISMA** mission (2010) demonstrated autonomous formation flying with two satellites, Mango and Tango — the Tango of the 2019 pose challenge. ESA's **Proba-3**, launched in December 2024, flies two satellites about 150 m apart. One casts a precise shadow on the other's coronagraph to create artificial solar eclipses. Holding the formation to about a millimetre needs onboard control, using GPS, star trackers, lasers and radio links. Its first eclipse images were taken in May 2025, and it was commissioned for autonomous formation flying in June 2025.

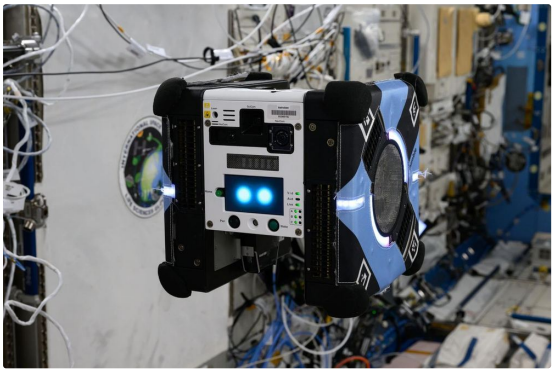


**FIGURE 48.13** An Astrobee free-flying robot during a Zero Robotics student programming demonstration on the ISS on 1 May 2024. Image: NASA/JSC.

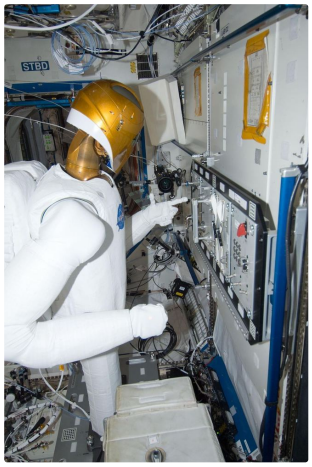


**FIGURE 48.14** Robonaut 2 in the ISS Destiny laboratory on 2 January 2013, operating valves on a task board under remote command from the ground. Image: NASA/JSC.

NASA's **Starling** mission (July 2023) flew four CubeSats to test swarm autonomy. Its Distributed Spacecraft Autonomy software let the satellites share observations of the ionosphere over crosslinks and plan follow-up measurements together. Its StarFOX experiment tracked swarm members and other objects with commercial star trackers. In a test with SpaceX reported in March 2025, a Starling satellite accepted responsibility for avoiding a close approach with a Starlink satellite through a shared screening service. It then planned and executed the avoidance manoeuvre on its own. On the Moon, the **CADRE** project will land three suitcase-sized rovers that explore and map as a team without moment-to-moment commands (Figure 48.15). It is due to fly on Intuitive Machines' IM-3, no earlier than early 2027.

**FIGURE 48.15** A development model of a CADRE rover on its first autonomous drive in JPL's Mars Yard in June 2023. Three such rovers are to map the Moon together as a team. Image: NASA/JPL (NASA/JPL-Caltech).

Table 48.6 gives the status of the robotic systems discussed.

**TABLE 48.6** Robots, swarms and formations with onboard autonomy. NET means no earlier than.

| System | Type | Autonomy | Status, October 2026 |
|---|---|---|---|
| Perseverance | Mars rover | Enhanced AutoNav, onboard planner, AEGIS | Operating; record 45.22 km; extended to about 2028 |
| Canadarm2, Dextre | ISS arms | Supervised; automated routine motions | Operating |
| Canadarm3 | Lunar arm system | Planned supervised autonomy | Refocused on lunar surface; NET 2029 |
| Astrobee | ISS free-flyers | Map-based navigation; research autonomy | Operating since 2019 |
| GITAI S2 | Arm on Bishop airlock | Autonomous and teleoperated tasks | Demonstrated 2024 |
| Starling | 4-CubeSat swarm | Distributed planning, star-tracker navigation | Primary mission done; extended tests 2024–25 |
| CADRE | 3 lunar rovers | Cooperative mapping | Awaiting IM-3, NET early 2027 |
| Proba-3 | 2-satellite formation | Millimetre formation control | Operating since 2025 |

At constellation scale, autonomy is already routine. Starlink satellites receive conjunction warnings and plan and execute avoidance manoeuvres themselves.

### STARLINK'S AUTONOMOUS COLLISION AVOIDANCE IN NUMBERS

- **Volume.** From 1 December 2025 to 31 May 2026, Starlink satellites made 207 152 propulsive collision-avoidance manoeuvres: 65 137 by first-generation and 142 015 by second-generation satellites. The previous six months (June–November 2025) saw 148 696.
- **Rate.** That is about 36 manoeuvres per satellite per year for the first generation and 46 for the second.
- **Threshold.** SpaceX manoeuvres when the probability of collision exceeds 3 in 10 million ($3 \times 10^{-7}$), plus miss-distance criteria. The common industry threshold is 1 in 10 000.
- **Misses.** The onboard system ran without system-wide outages. In 22 individual cases the targeted risk was not reduced, mostly because predictions of the other object were poor or arrived too late. Newly launched satellites not yet tracked and short Space-Track.org outages were among the causes.
- **New sensing.** In January 2026 SpaceX announced Stargaze, which uses about 30 000 Starlink star trackers to spot nearby objects. It aims to screen conjunctions in minutes rather than hours.

Source: SpaceX semi-annual report to the US Federal Communications Commission, mid-2026; SpaceX announcements.

### EXAMPLE: COLLISION-AVOIDANCE DECISIONS FOR A 10 000-SATELLITE CONSTELLATION

Assume Starlink's second-generation rate of 46 manoeuvres per satellite per year.

1. **Per year.** 10 000 × 46 = **460 000** manoeuvres.
2. **Per day and hour.** 460 000 / 365.25 = 1 259 a day, or 52 an hour — one every **69 s**, day and night.
3. **Human review.** If each decision took an analyst 10 minutes, that would be 210 person-hours a day. At about 1 800 working hours per person per year, it would need about **43 full-time analysts** for this task alone.

This is why the decision is automated: humans set the thresholds and handle the exceptions, while the satellites plan and execute routine manoeuvres. The weak point moves from deciding to *knowing*: most failures come from poor tracking data about the other object (Chapter 50).

### PRACTICE: DESIGNING AND VERIFYING AN AUTONOMY FUNCTION WITH A RUNTIME MONITOR

1. **State the decision.** Write down what the function decides, how often, and what a wrong decision costs. Example: "choose a safe landing point within 10 s, with at most one hazardous choice in 10 000 landings".
2. **Define the operating domain.** List the conditions it must handle — Sun angles, terrain, sensor noise, speeds — and which of them the training data cover.
3. **Build the simplest baseline.** Start with a classical method, such as a template, a PD law or a greedy rule. A learned method must beat it by a measured margin.
4. **Design the safety envelope.** Write simple, checkable rules the system must never break, such as stopping distance or keep-out zones, and a backup that always keeps it inside them.
5. **Add the monitor.** Make it independent of the learned component, and simple enough to verify exhaustively or formally.
6. **Test in layers.** Run millions of randomised simulations, then hardware-in-the-loop tests. Report failure rates with confidence intervals, and test outside the domain too.
7. **Measure the cost of safety.** Record how often the monitor intervenes and what that costs. Frequent interventions mean the learned part or the envelope needs work.
8. **Fly in shadow mode first.** Run the function on board without letting it act, compare its choices with the existing system, then give it authority step by step.

### KEY TAKEAWAYS

- Spacecraft need autonomy when the ground cannot respond fast enough, often enough or cheaply enough. Light-time, contact gaps, fleet size and event speed set the level; ECSS defines E1–E4 for execution, D1–D2 for data and F1–F2 for failures.
- Most autonomy in flight is search, estimation and optimisation, not learning: executives, fault protection, diagnosis, planners, optical navigation and convex guidance.
- Learning has entered through perception and science: crater detection, keypoint networks for pose and rover science targeting. Learned detectors are often more robust to lighting than templates, but not more precise.
- Terrain-relative navigation can reach metre-level accuracy. Map registration and attitude knowledge, not image matching, usually limit it. Recent landing failures came from sensors and the logic that judges their data.
- Reinforcement learning can find efficient policies but may fail abruptly outside its training conditions. Runtime monitors with verified backups, and learning that speeds up an optimiser, are the practical routes to flight.
- Autonomy is now routine at scale: Perseverance drives itself for over 90 % of its distance, and Starlink satellites made over 200 000 autonomous avoidance manoeuvres in six months.

### REVIEW QUESTIONS

1. Using $T_{\text{ground}} = 2\tau + 2W + T_{\text{decide}}$, estimate the ground response time for a Jupiter orbiter with 43 minutes of one-way light-time and 8 hours of contact a day. Which ECSS execution level does a 2-hour science event require?
2. Explain the difference between levels F1 and F2. Give one mission for which F1 is enough and one that needs F2, with reasons.
3. Describe the three layers of an autonomous spacecraft and their time scales. Where are learned models most often used, and why?
4. How does model-based diagnosis choose between explanations of a fault? Why does the prior probability favour explanations with few failed components?
5. In Figure 48.3, why does a greedy rule fall short of the optimum, and why might a flight planner still use one? What property of Perseverance's onboard planner keeps its run time bounded?
6. Why does normalised cross-correlation tolerate changes in brightness but not in Sun direction? Give two ways a landing system can cope.
7. Repeat the terrain-relative navigation example for a 12 m map, 150 landmarks, a 5 m map tie and 0.05° attitude knowledge at 3 km. Which term dominates?
8. In the 2019 pose-estimation challenge, the winning score was forty times worse on real images than on rendered ones. What is this gap called, and what methods narrowed it in 2021?
9. Why is G-FOLD not machine learning? What does lossless convexification achieve?
10. In Figure 48.8, why did the learned policy fail with a weaker engine, and how did a runtime monitor restore safety? What did the monitor cost?

CHAPTER

# 49 AI for Mission Operations, Communications and Networks

Image: NASA/JSC (ROBERT MARKOWITZ NASA-JSC)

*Every spacecraft is flown from the ground by people and software that watch its telemetry, plan its activities, schedule its contacts and send its commands. This chapter shows where machine learning and optimisation now help with that work, and where they do not yet. It covers anomaly detection in telemetry, from fixed limits to learned models, false-alarm budgets, prognostics of batteries and wheels, automation, digital twins and language-model assistants, and the human factors that decide whether they are trusted. It then turns to scheduling ground contacts and to communications — cognitive radio, signal recognition, adaptive coding, beam and spectrum management, interference, laser-mesh routing and 6G — and ends with the security of ground segments. Small models run in Python make each method concrete.*

IN THIS CHAPTER

49.1 Mission Operations Today
49.2 Watching Telemetry: Anomaly Detection
49.3 Prognostics and Health Management
49.4 Automating Operations
49.5 Scheduling Contacts and Ground Networks
49.6 AI in Space Communications
49.7 Networks in Orbit
49.8 Securing Ground Segments and Links

Spacecraft are flown from the ground. A flight control team watches the stream of **telemetry** (TM) — measurements that a spacecraft sends about its own health. It plans activities, books contacts with ground stations and sends **telecommands** (TC). The hero image above shows NASA's Artemis II flight control team in Houston on 10 April 2026, during the splashdown of the first crewed flight to the Moon since 1972. Rooms like this one have worked in much the same way since Apollo: skilled people, many screens and fixed alarm limits.

Two trends are changing this. The first is scale: one observatory produces tens of thousands of telemetry channels, and one operator flies more than ten thousand satellites. The second is practical machine learning, optimisation and language models. This chapter explains how they help to watch telemetry, predict failures, automate routine work, schedule ground networks, run radio links and route data. It separates what is operational, demonstrated or still research, as of October 2026.

The chapter builds on Chapters 7, 11 and 17 (ground segment, constellations and communication satellites) and on the learning methods of Chapters 43–45. Onboard autonomy is covered in Chapter 48, space situational awareness and defence in Chapter 50, and assurance of AI in Chapter 53.

## 49.1 Mission Operations Today

The key idea of this section is that mission operations are a loop: plan, command, monitor and analyse. Machine learning helps at every step, but the loop and the people in it remain.

### 49.1.1 What a Control Centre Does

A **mission operations centre** holds the people and software that fly a spacecraft (Chapter 7). Its core, the **mission control system** (MCS), turns raw telemetry frames into named, calibrated parameters, checks limits, raises alarms and sends commands. Four other functions surround it. **Flight dynamics** determines the orbit and plans manoeuvres. **Mission planning** turns requests into a timeline that respects power, thermal, data and visibility limits. The **ground network** provides antennas and contact time. **Engineering support** — subsystem experts on call — investigates anything unusual.

Work follows a daily cycle. Planners build tomorrow's timeline, and command sequences are checked on a simulator before uplink. During each contact, controllers watch live telemetry and alarms; afterwards, engineers study trends. When something goes wrong, an **anomaly report** records the symptoms, the cause and the fix. Years of such reports, telemetry and command logs are the raw material for the learning methods in this chapter.

### 49.1.2 How Much Telemetry?

Spacecraft telemetry is small in bytes but rich in structure. NASA's James Webb Space Telescope records telemetry at about 100 kbit/s. On the ground it is unpacked into about 36 000 named parameters, called **mnemonics**, with 4 000 more derived on the ground. Some are sampled every 0.064 s, others every 32 s or more. A 28-minute file holds about 10 million samples. EUMETSAT routinely reports about 46 000 housekeeping parameters from Sentinel-3. Table 49.1 gives more examples.

**TABLE 49.1** Examples of the scale of modern operations. Sources: SpaceOps 2025 papers (JWST, Sentinel-3), ESA-ADB paper, Planet (IAC 2018) and SpaceX's report to the US Federal Communications Commission.

| System | Scale | What it means for operations |
|---|---|---|
| James Webb Space Telescope | 36 000 telemetered + 4 000 derived points; ~100 kbit/s | Too many for people to watch; trending software is essential |
| Sentinel-3 (EUMETSAT) | ~46 000 housekeeping parameters routinely reported | Learned outlier detection is trialled on a subset |

| System | Scale | What it means for operations |
|---|---|---|
| ESA anomaly benchmark (2024) | 1.55 billion data points, 844 annotated events | Real anomalies are rare: 31–118 per mission |
| Planet Dove fleet (2018) | >200 satellites, >2 000 passes a day, 8 engineers | Operations run by exception; nominal passes automated |
| Starlink (June–November 2025) | 148 696 collision-avoidance manoeuvres in six months | Decisions too frequent for people to make one by one |

### 49.1.3 From One Satellite to Thousands

A traditional mission has a team for each spacecraft, with people on console around the clock. That model does not scale to constellations. In 2018 Planet flew more than 200 Dove imaging satellites with eight engineers in Berlin and San Francisco. By 2022 Planet reported that commanding each pass, scheduling contacts, reporting and delivering images were fully automated. Periodically run Python scripts handled the most common anomalies. This cut daily Dove monitoring from about four hours to 30 minutes and ended night shifts for the larger SkySats.

This style is called **management by exception**. Software flies the nominal mission; people handle what the software cannot. At the largest scale, decisions are taken by machines because there are too many for people. Starlink satellites made 148 696 collision-avoidance manoeuvres between June and November 2025, decided autonomously, without a person in the loop, when the predicted collision probability exceeded about 3 in 10 million (Chapters 21 and 48).

As Figure 49.1 shows, decision support now grows beside the data path from spacecraft to engineers. It includes monitoring models trained on the archive, planning solvers, automation scripts, assistants and digital twins. Monitoring and scheduling tools are operational in some centres. Language-model assistants are pilots, and natural-language commanding is research.

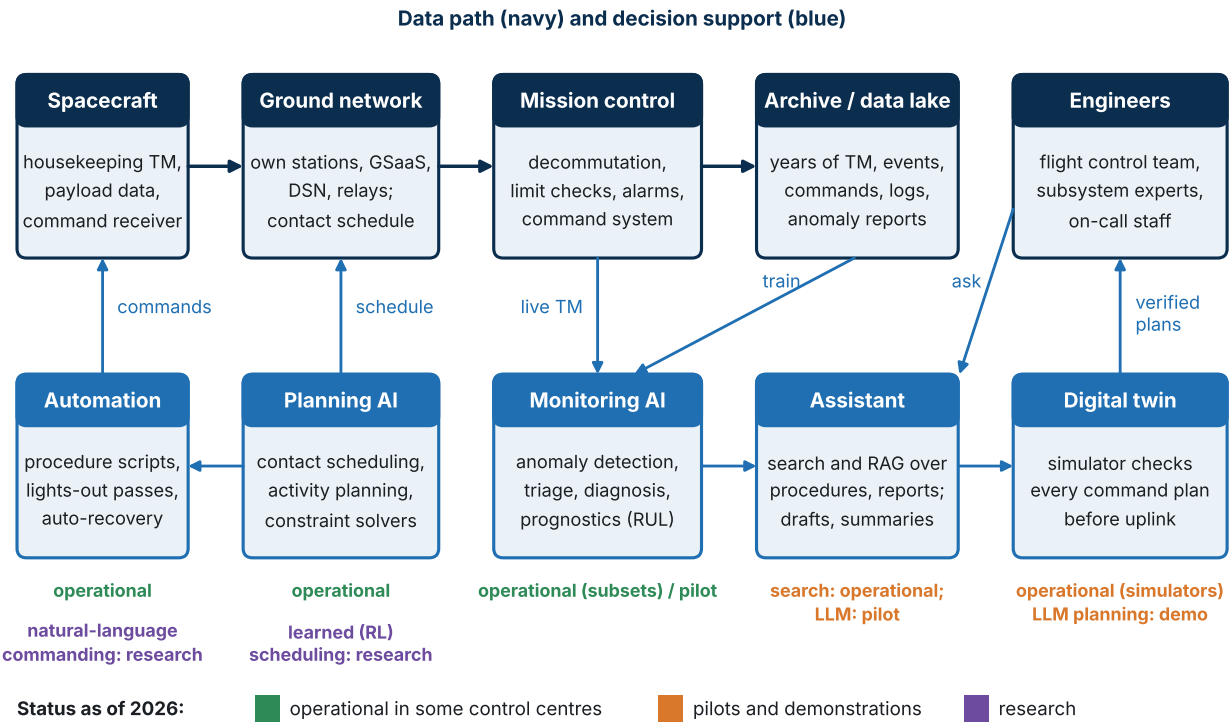



**FIGURE 49.1** Where learning and automation enter the ground segment. The data path (navy) runs from the spacecraft through the ground network and mission control system to the archive and the engineers. Decision-support functions (blue) are trained on the archive or fed by live telemetry; their status in 2026 is colour-coded.

## 49.2 Watching Telemetry: Anomaly Detection

The key idea of this section is that a monitor must find rare, varied faults among millions of normal samples, while raising few false alarms. Learned models of normal behaviour find faults that fixed limits miss, but only if their false alarms are controlled.

### 49.2.1 Limits and Their Limits

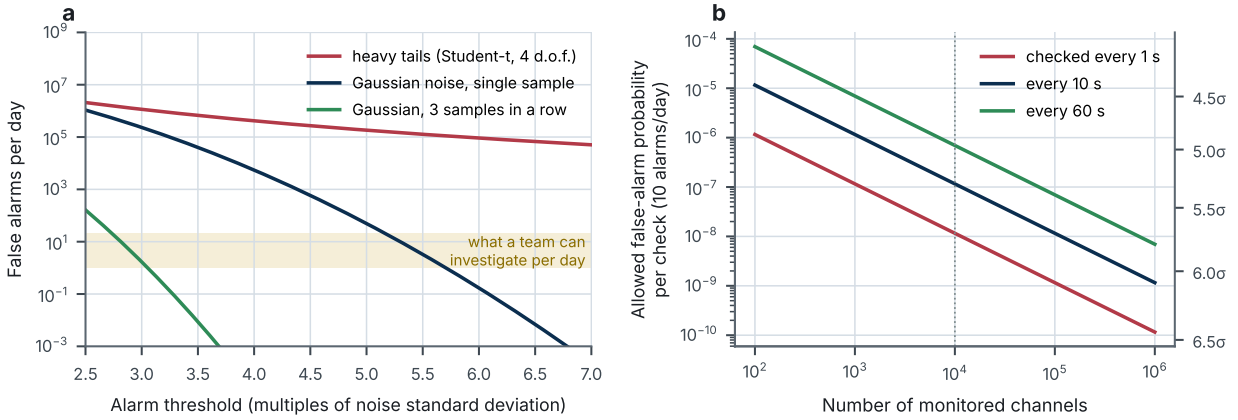
The classic monitor is the **limit check**. Each parameter has a lower and an upper limit, often a soft "warning" pair and a hard "alarm" pair. A value outside them raises an alarm. Limits can depend on the mode: a heater current is checked against one range when the heater is commanded on and another when it is off. Limit checks are simple, fast and easy to explain. They also catch most serious failures, because a failed unit usually drives some parameter far out of range.

Their weakness is everything else. A temperature can be normal for daylight but wrong in eclipse. A slow drift stays inside the limits for months, then crosses them too late. And limits set too tight produce false alarms, which cost attention and teach people to ignore alarms.

The false-alarm budget is simple arithmetic. Suppose $N$ channels are each checked $f$ times a second, and each check has a false-alarm probability $p$. The expected number of false alarms per day is then given below.

$$n_{\mathrm{FA}} = N\, f \cdot 86\,400 \cdot p.$$

For a threshold at $k$ standard deviations of Gaussian noise, the two-sided probability is $p = 2Q(k)$, where $Q$ is the tail probability of the standard normal distribution. In Figure 49.2 this is applied to 10 000 channels checked every 10 s. A 3σ threshold, common in textbook statistics, gives about 233 000 false alarms a day. Fewer than 10 a day needs about 5.3σ for perfectly Gaussian noise. Real telemetry noise has heavier tails, so empirical thresholds must be set from the data. Requiring several consecutive exceedances helps greatly when noise is independent, but much less when it is correlated in time.




Model: 10 000 channels, each checked against its threshold every 10 s (8.64 × 10⁷ checks a day), two-sided thresholds, independent noise. Real telemetry noise is often heavy-tailed and correlated in time, so the green curve is a best case.

**FIGURE 49.2** The false-alarm budget of a large telemetry set. (a) False alarms per day for 10 000 channels checked every 10 s, against the threshold in multiples of the noise standard deviation. (b) The per-check false-alarm probability allowed for 10 alarms a day, against the number of channels; the right axis gives the equivalent Gaussian threshold.

### EXAMPLE: FALSE ALARMS PER DAY FOR 10 000 CHANNELS

A science satellite has 10 000 monitored channels, each checked every 10 s. The team can investigate about 10 alarms a day.

1. **Checks per day.** 10 000 × 86 400 s / 10 s = $8.64 \times 10^7$ checks.
2. **At $p = 10^{-6}$ per check.** $8.64 \times 10^7 \times 10^{-6} \approx$ **86 false alarms a day** — about nine times the budget. Checking every second instead would give 864.
3. **Allowed probability.** $10 / 8.64 \times 10^7 =$ **$1.16 \times 10^{-7}$** per check. For Gaussian noise this is a two-sided threshold of **5.3σ**.
4. **At a 5σ threshold.** $p = 5.7 \times 10^{-7}$, giving about 50 false alarms a day.
5. **Persistence.** Requiring three exceedances in a row at 3σ gives $(2.7 \times 10^{-3})^3 \approx 2 \times 10^{-8}$ per check, or about 1.7 a day — if the noise is independent. Correlated noise raises this, so the rule must be tested on archived data.
6. **Lesson.** With thousands of channels, the per-channel false-alarm rate must be tiny. Any learned detector needs the same budget, set on real data.

## 49.2.2 Kinds of Anomaly

An **anomaly** is behaviour that differs from what is expected. Not every anomaly is a fault: a commanded manoeuvre or a calibration looks unusual the first time but is normal for the operators. ESA's benchmark calls these **rare nominal events**. Table 49.2 lists the main kinds of anomaly and the detectors that suit them.

**TABLE 49.2** Kinds of telemetry anomaly and the detectors that suit them.

| Kind | What it looks like | Example | Detectors that suit it |
|---|---|---|---|
| Point | One sample far from its neighbours | Bit flip, sensor glitch | Limits, prediction residuals |
| Contextual | Normal value at the wrong time or mode | Daylight temperature in eclipse | Models that use orbit phase and mode |
| Collective | Normal values in an abnormal sequence | Oscillating control loop | Sequence models, windowed features |
| Drift | Slow change of level or trend | Rising friction, sensor ageing | Trend tests, residuals against other channels |
| Multivariate | Each channel normal, their combination not | Current rises while load is unchanged | Cross-channel models, isolation forests |

### 49.2.3 Learning What Normal Looks Like

Learned monitors share one idea: model normal behaviour from history, then score how far new data departs from it. There are three main families.

**Prediction-based** detectors forecast each channel and watch the error. A model $f$ predicts the value $x_t$ from recent history and from context $\mathbf{c}_t$, such as orbit phase and commanded mode. The standardised residual is

$$z_t = \frac{\lvert x_t - f(x_{t-L}, \dots, x_{t-1}, \mathbf{c}_t) \rvert}{\sigma_{\text{res}}},$$

where $\sigma_{\text{res}}$ is the residual spread on normal training data. An alarm is raised when $z_t$ exceeds a threshold. JPL's **Telemanom** system, described in 2018, used a **long short-term memory** (LSTM) network for each of 82 channels of the SMAP satellite and the Curiosity rover (Chapter 44). It smoothed the errors and set thresholds automatically from their recent distribution. On 105 labelled anomalies it reached 87.5 % precision and 80 % recall (Chapter 43).

**Reconstruction-based** detectors squeeze the data through a narrow layer and rebuild it. An **autoencoder** with encoder $h$ and decoder $g$ is trained so that $g(h(\mathbf{x})) \approx \mathbf{x}$ for normal data. The anomaly score is the reconstruction error $\lVert \mathbf{x} - g(h(\mathbf{x})) \rVert^2$. Patterns the network never saw are rebuilt badly.

**Distance- and isolation-based** detectors look at where a sample sits among the training data. The k-nearest-neighbour score is the distance to the $k$-th closest normal sample. A one-class support vector machine draws a boundary around normal data. An **isolation forest** (2008) builds random trees that split the data at random values of random features. Unusual points are isolated after few splits. The score is

$$s(\mathbf{x}) = 2^{-\bar{h}(\mathbf{x})/c(n)},$$

where $\bar{h}$ is the average path length over the trees and $c(n)$ the average for $n$ normal points. Scores near 1 mean anomalous.

### 49.2.4 A Test on Synthetic Telemetry

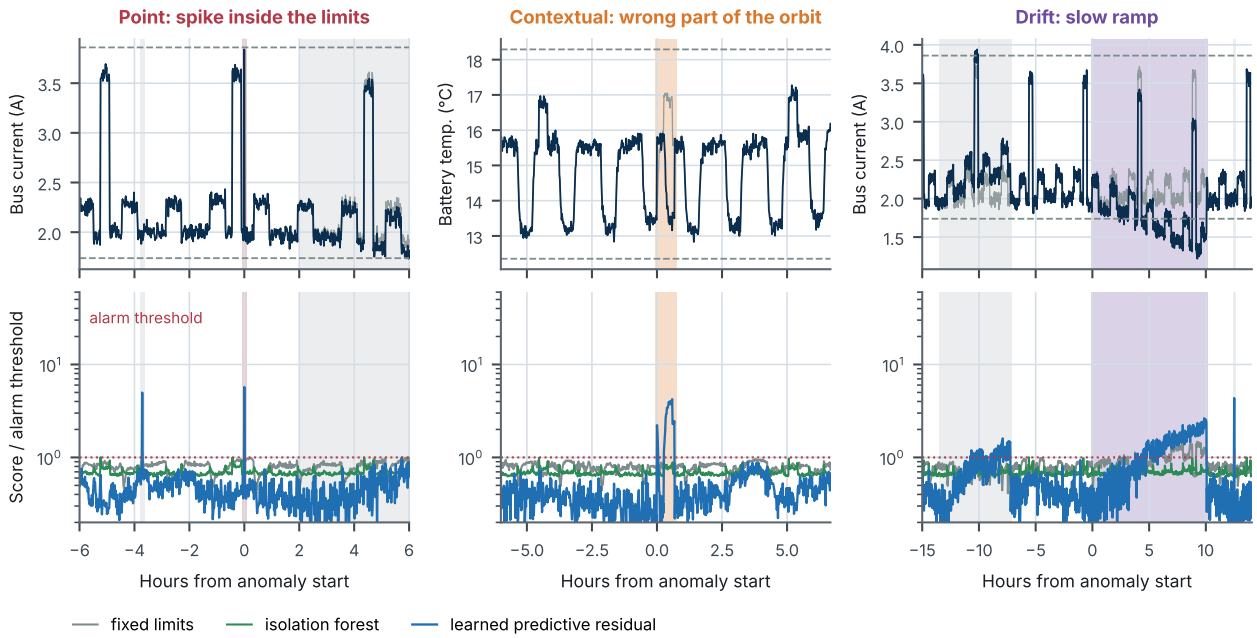
To see the strengths of each family, this chapter's script builds six housekeeping channels for a satellite in a 96-minute orbit with 36 % eclipse. Battery voltage, battery temperature, bus current, panel temperature, wheel speed and heater duty all follow the orbit, a commanded payload mode

and a 14-day seasonal cycle. Slow unmodelled variations are added. Fourteen days train each detector. Fourteen more days test it, with 14 point, 12 contextual and 8 drift anomalies injected (Figure 49.3).

Three detectors are compared. The **fixed limits** are the training minimum and maximum, widened by 5 % of the range. The **learned predictive detector** is a small neural network for each channel, written in numpy. It predicts the channel from the other five channels, the orbit phase and the mode, and scores the largest standardised residual. The **isolation forest** from scikit-learn sees the channel values, orbit phase and their changes. Each detector's threshold lets 0.1 % of normal samples alarm — about 1.4 false alarms a day.

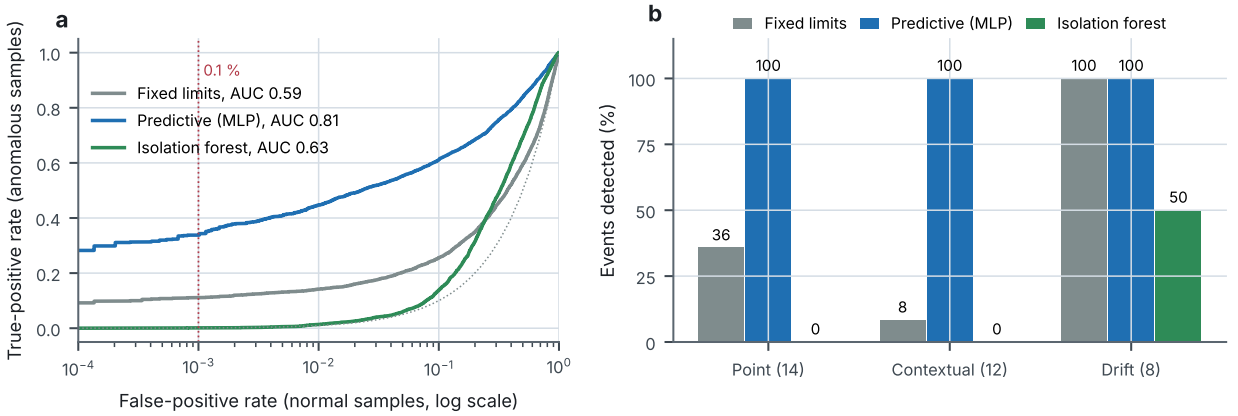
The results appear in Figure 49.4. The predictive network found all 34 injected events, and its sample-by-sample area under the ROC curve (AUC; Chapter 43) was 0.81. The fixed limits found all 8 drifts, but only after 45 % of each ramp, on average. They caught 5 of the 14 spikes — those that left the normal range — and 1 of the 12 contextual events. The isolation forest found only half the drifts and none of the spikes or contextual events.

Two lessons follow. First, context makes learned detectors useful: the network knew the orbit phase and the other channels, so a value copied from daylight into eclipse stood out. Second, a generic detector is not automatically better than limits. The isolation forest's scores saturate outside the training range, so a far-out spike looks no stranger than the many normal points near the edge. Real systems should keep limit checks and add learned detectors beside them. Real telemetry is also much harder than these synthetic data, as the next section shows.




Synthetic telemetry: six channels sampled once a minute, driven by a 96-min orbit (36 % eclipse), a commanded payload mode, a 14-day season and slow unmodelled variations. Grey: signal without the anomaly; dashed: fixed limits (training minimum and maximum). Thresholds are set so that 0.1 % of normal test samples alarm.

**FIGURE 49.3** Three injected anomalies and the detectors' responses. Top: a spike that stays inside the limits, a segment copied from the wrong part of the orbit, and a slow drift (navy; grey shows the signal without the anomaly; dashed lines are the fixed limits). Bottom: each detector's score divided by its alarm threshold.




Same data as the previous figure. (a) Sample-by-sample ROC curves. (b) Share of injected events with at least one alarm when each detector's threshold lets 0.1 % of normal samples (about 1.4 per day) through. Neural network: 2 hidden layers of 24 tanh units per channel, trained 12 epochs with Adam; isolation forest: 300 trees (scikit-learn).

**FIGURE 49.4** Detector performance on the synthetic test set. (a) Sample-by-sample receiver operating characteristic (ROC) curves on a logarithmic false-positive axis. (b) Share of injected events caught at a false-alarm rate of 0.1 % of normal samples.

## 49.2.5 Benchmarks and Real Results

Comparing detectors fairly needs real, labelled data. Until 2024 the main public sets were the SMAP and Curiosity channels released with Telemanom. ESA then published a much larger benchmark, built with operations engineers at its European Space Operations Centre (ESOC) in Darmstadt.

### DATA BOX — ESA'S ANOMALY-DETECTION BENCHMARK (ESA-ADB), 2024

- **Data.** Real telemetry from three ESA missions, two used in the benchmark; 1.55 billion data points, 7.32 GB compressed. Developed by ESA with KP Labs (Poland); first released in June 2024.
- **Mission 1.** 76 channels (58 monitored), 14 years, 200 annotated events: 118 anomalies, 78 rare nominal events and 4 data gaps.
- **Mission 2.** 100 channels (47 monitored), 3.5 years, 644 events: only 31 anomalies, but 613 rare nominal events such as manoeuvres and resets.
- **Scoring.** Event-wise precision first, because operators care about false alarms; then which channels and subsystems were flagged, and timing.
- **Results.** An adapted Telemanom was best for Mission 1, with an event-wise F0.5 score of 0.968 for anomalies alone. A simple windowed isolation forest was best on a small channel subset of Mission 2 (0.949). No algorithm found the nine real anomalies hidden among Mission 2's rare nominal events, and adding channels added false detections.
- **Conclusion of the authors.** Typical algorithms are not yet good enough for operators; new methods are needed. A July 2026 follow-up found supervised graph networks more accurate, and simple unsupervised methods much cheaper to run.

## 49.2.6 Operational Tools

Learned monitors have been used in real control centres for about two decades, usually beside limit checks and for selected parameters. Table 49.3 summarises the best-documented ones.

EUMETSAT's experience is typical. Its Sentinel-3 flight control team ran a k-nearest-neighbour tool and a neural network in parallel. The choice of training data mattered more than the algorithm. Novel-but-harmless detections were the main burden, and adding the timeline of

commanded operations cut them. Engineers also needed to see *why* a parameter was flagged, which favoured simple methods. The tools did find real anomalies, including micrometeoroid or debris impacts.

**TABLE 49.3** Learned telemetry monitors with published operational use or trials.

| Tool | Organisation | Method | Status |
|---|---|---|---|
| Inductive Monitoring System (IMS) | NASA Ames | Clusters of nominal data; distance to nearest cluster | Used for ISS gyroscopes from about 2005; about a dozen Mission Control systems by 2012 |
| Novelty Detection | ESA/ESOC | k-nearest-neighbour outlier score on orbit statistics | Developed at ESOC in the early 2010s; trialled by CNES in 2013 |
| NOSTRADAMUS | CNES with TeSA | One-class support vector machine | Prototype (2014), operational assessment |
| ATHMoS | DLR German Space Operations Center | Outlier detection; later neural prediction about 4.5 h ahead | Developed 2016–19 |
| Telemanom | NASA JPL | LSTM prediction and dynamic thresholds | Piloted on SMAP; false positives limited its use |
| CHART outlier tools | EUMETSAT | k-nearest neighbours; neural network | Routine on a subset of parameters; Sentinel-3 assessment (2025) |

LANDMARK: TWENTY YEARS OF LEARNED TELEMETRY MONITORING

- **1988.** Houston's Mission Control tested expert systems beside conventional Space Shuttle displays (Chapter 43).
- **About 2005.** ISS engineers adopted NASA Ames' Inductive Monitoring System, learned from months of normal data, to watch a failing gyroscope.
- **2010s.** ESA, CNES and DLR brought outlier detection into European control centres.
- **2018.** JPL's Telemanom showed that LSTM prediction with automatic thresholds could catch most labelled anomalies. In its SMAP pilot, false positives were the main obstacle to central use.
- **2024.** ESA released ESA-ADB, the first large, operator-labelled benchmark. It showed how far typical algorithms remain from operators' needs.
- **2025–26.** EUMETSAT reported its Sentinel-3 trial. ESOC's search assistant was in use on 14 missions, and the Aerospace Corporation and Google showed a proof-of-concept agent for triage in large constellations.

### 49.2.7 From Detection to Diagnosis

An alarm says that something is wrong. **Diagnosis** says what, and **root-cause analysis** says why. In a large constellation one fault often triggers dozens of alarms across subsystems. The first step is therefore to group alarms that share a time, a satellite and a likely cause. The diagnosis itself is still mostly human work (Figure 49.5).

Classical diagnosis uses fault trees, failure-mode tables and model-based reasoners that compare telemetry with a model of the system (Chapter 6). Learning adds three things. It can match a new alarm pattern to similar past anomalies. It can learn which alarms occur together and which comes first, a hint of cause. And it can gather the context — recent commands, space weather, ground-station problems — that an engineer would otherwise collect by hand.

In April 2026 the Aerospace Corporation and Google's public-sector arm showed a proof-of-concept tool for large constellations in low Earth orbit. It combines telemetry, payload and ground-station data in one view. It links an anomaly to recent events, such as payload activity and environmental conditions, to help engineers find the cause. The aim is to save the minutes on-call staff now spend correlating data by hand. It is decision support, not a replacement for engineers. A 2025 JPL study of similar "agentic" detection for Deep Space Network data systems reported preliminary results, not yet robust to false positives.

**FIGURE 49.5** Nzinga Tull, Hubble systems anomaly response manager, in the control room at NASA's Goddard Space Flight Center on 15 July 2021. A month earlier Hubble's payload computer had stopped, and the team switched to backup hardware to restore science. Diagnosis of this kind is still human work, supported by data tools. Image: NASA/GSFC (Rebecca Roth).

## 49.3 Prognostics and Health Management

The key idea of this section is that predicting *when* a part will fail is worth more than detecting that it has. Prognosis combines a physical degradation model with data, and must report its uncertainty honestly.

### 49.3.1 Remaining Useful Life

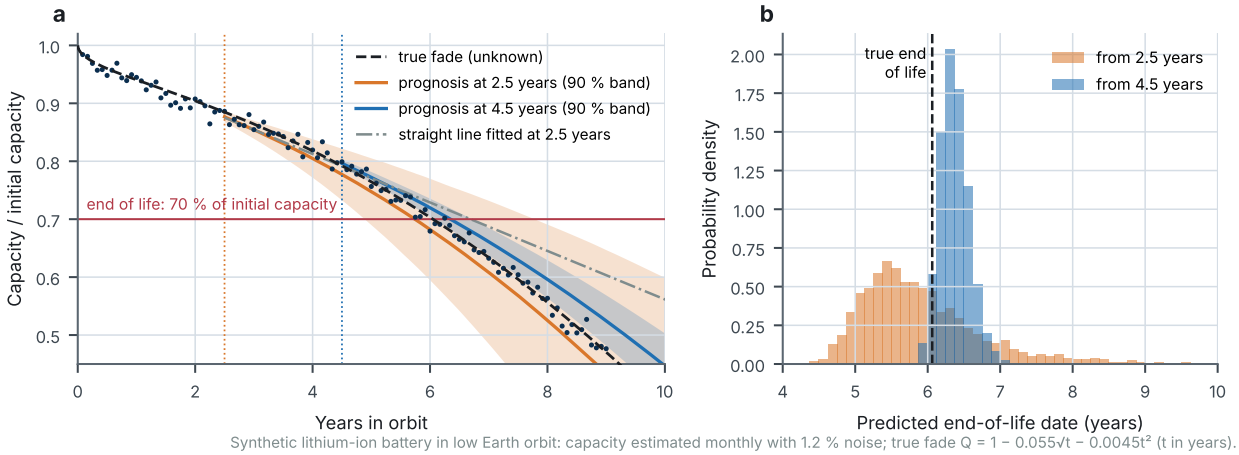
**Prognostics and health management** (PHM) estimates the **remaining useful life** (RUL) of a component: the time until it can no longer meet its requirement. Batteries, reaction wheels, gyroscopes, thrusters and solar arrays all wear out in ways that can be trended.

A lithium-ion battery in low Earth orbit charges and discharges about 15 times a day, some 5 500 cycles a year. Its capacity fades. Early fade is often proportional to the square root of time, as a surface layer grows on the electrode. Later, fade can accelerate. A simple model of the capacity fraction $Q$ is

$$Q(t) = 1 - a\sqrt{t} - b\,t^2,$$

with the coefficients $a$ and $b$ learned from the satellite's own data. Because $Q$ is linear in $a$ and $b$, **Bayesian linear regression** gives their joint uncertainty in closed form. Thousands of sampled curves then give a distribution for the date when $Q$ falls below the end-of-life threshold.

In Figure 49.6 this is applied to a synthetic battery with monthly capacity estimates and 1.2 % noise. The true end of life, at 70 % capacity, is at 6.06 years. A prognosis made after 2.5 years puts the end of life at 5.75 years (median), with a 90 % interval of 4.9–7.8 years. After 4.5 years of data the median is 6.35 years and the interval shrinks to 6.08–6.71 years. A straight line fitted at 2.5 years says 6.7 years, with no hint of its uncertainty. The model form matters: in tests with an extra free linear term, the early intervals often missed the true date.




Synthetic lithium-ion battery in low Earth orbit: capacity estimated monthly with 1.2 % noise; true fade $Q = 1 - 0.055\sqrt{t} - 0.0045t^2$ (t in years). Prognosis: Bayesian linear regression on $\sqrt{t}$ and $t^2$ (form assumed known from ground life tests) with a Gaussian prior; 4 000 posterior samples give the end-of-life distribution.

**FIGURE 49.6** Battery capacity-fade prognosis. (a) Monthly capacity estimates (dots), the true fade (dashed), prognoses made at 2.5 and 4.5 years with 90 % bands, and a straight-line fit. (b) The predicted end-of-life date from each prognosis.

Operators use exactly this kind of tool. Planet's PROTOS, built for the five RapidEye satellites, used a **Kalman filter** (Chapter 43) to estimate each battery's depth of discharge from charge counting and voltage. A fade model with measured temperatures predicted when each 15 Ah battery would reach 50 % capacity. A genetic algorithm then re-planned imaging, downlinks and heater use to slow the fade. Planet estimated a 15 % longer mission life for satellites already past their 7.25-year design life. On the ISS, trending told engineers when to replace ageing batteries, a job done by astronauts (Figure 49.7).

**FIGURE 49.7** NASA astronaut Andrew Morgan on the International Space Station's port truss on 11 October 2019, during a spacewalk to replace older nickel–hydrogen batteries with lithium-ion ones. Trending of battery capacity tells operators when such work is needed. Image: NASA/JSC.

### 49.3.2 Wheels, Gyroscopes and Other Wear-Out Parts

Reaction wheels and gyroscopes have ended or reshaped several missions. Kepler lost its second wheel in 2013, which ended its original mission. Dawn lost wheels in 2010 and 2012 and finished its mission on thrusters. Hubble has flown with a single gyroscope since June 2024, which costs about 12 % of its observing efficiency.

Wheel health shows in friction torque, motor current, temperature and speed noise; rising drag can precede failure by months. Learned models remove the normal dependence on speed and temperature, so that the remaining trend shows wear, and compare identical wheels across a fleet. Most published work still uses ground life tests or single satellites.

## 49.4 Automating Operations

The key idea of this section is that automation should take over routine, well-understood work, while people keep the judgement and the authority. The newest tools — digital twins and language-model assistants — are useful when their output is checked.

### 49.4.1 From Scripts to Lights-Out Operations

Automation in a control centre grows in steps (Table 49.4). At the first level, people type every command. At the highest, software runs the mission and people are called only when something unusual happens. Most modern constellations work at the third or fourth level. Most flagship science missions work at the second, with automation for routine passes and people on console for critical events.

**Lights-out operations** means that nobody is on console for routine passes. The ground system schedules contacts, uplinks commands, downlinks data and checks health, and pages an on-call engineer if a check fails. The key technology is not AI but reliable procedure automation: scripts tested on a simulator and kept under configuration control. Learning enters where rules are hard to write, such as deciding whether an unusual pattern is safe to ignore. The example below shows what automated triage can save.

**TABLE 49.4** Levels of automation in spacecraft operations.

| Level | What software does | What people do | Typical use |
|---|---|---|---|
| 1 Manual | Displays data, checks limits | Send every command, watch every pass | Early operations, critical events |
| 2 Assisted | Runs approved procedures step by step | Start procedures, confirm critical steps | Science missions, GEO satellites |
| 3 Automated passes | Runs nominal contacts and planning | Review reports in working hours | Small constellations, mature missions |
| 4 Management by exception | Detects, triages and fixes known anomalies | Handle new problems, on call | Large constellations (e.g. Planet) |
| 5 Autonomy | Decides on board, including safety actions | Set goals and policies, audit | Collision avoidance, deep space (Chapter 48) |

**EXAMPLE: OPERATOR WORKLOAD SAVED BY AUTOMATED TRIAGE**

A 300-satellite constellation raises about 1 200 alarms a day, mostly known, harmless patterns such as eclipse transitions. Manual triage takes about 3 minutes per alarm.

1. **Manual workload.** 1 200 × 3 min = 3 600 min = **60 hours a day**, or 7.5 eight-hour shifts.
2. **Grouping.** Merging correlated alarms, 3 per group on average, leaves 400 incidents.
3. **Classification.** A learned classifier closes 80 % as known patterns, leaving **80** for people.
4. **Review.** With more context, each takes 6 minutes: 80 × 6 = 480 min = **8.0 hours**.
5. **Audit and result.** Spot-checking 5 % of the 320 closed incidents at 2 minutes adds 32 min. The total is 8.5 hours instead of 60: a saving of **86 %**, from 7.5 shifts to about 1.1.
6. **Price.** If 5 real anomalies a day pass through and 1 % are wrongly closed, one is missed every 20 days. The audit and limit checks that bypass the classifier are there to catch it.

### 49.4.2 Ground Digital Twins

A **digital twin** is a model kept in step with one real system, using its own telemetry. In operations, the oldest form is the **operational simulator**. It runs the real flight software on emulated processors, with models of the rest of the spacecraft and its environment. Every command plan for a major mission is checked on such a simulator before uplink, and teams train on it for failures. Rover missions also keep physical twins on Earth (Figure 49.8).

Learning makes twins faster and closer to the real vehicle. Learned surrogates can replace slow thermal or power models, and parameters can be refitted to flight data as the spacecraft ages. Twins can then answer “what if” questions in seconds. In December 2025 JPL checked rover drives proposed by vision–language models on a digital twin of Perseverance before sending them to Mars. The twin simulates more than 500 000 telemetry variables (Chapters 44 and 48).

**FIGURE 49.8** OPTIMISM, the full-scale engineering twin of the Perseverance rover, at the Mars Yard garage at NASA’s Jet Propulsion Laboratory on 29 October 2021. Physical test beds like this one and software twins are used to check commands before they are sent to Mars. Image: NASA/JPL (NASA/JPL-Caltech).

### 49.4.3 Language-Model Assistants

Operations knowledge is scattered across procedures, design documents, test reports, anomaly reports, logs and the telemetry database. Finding the right page during an anomaly can take longer than fixing the fault. **Large language models** (LLMs; Chapter 44) can search, summarise

and draft from this material. The safest pattern is **retrieval-augmented generation** (RAG): the system retrieves the relevant passages and the model answers from them, citing its sources.

ESOC's **Operations Companion (OCAI)** is the best-documented example. A 2025 paper reports it in use on 14 ESOC missions, for search and for training new flight control teams. It combines keyword search, semantic search on embeddings and a knowledge graph of operations terms. It covers command and telemetry archives, procedures, anomaly reports, manuals and logs, and language-model answers with references were being added. Telespazio Germany planned to build it into its commercial mission control system, with constellr's HiVE constellation as pilot.

Start-ups are also building language-model features into mission control software. One, Quindar, raised 18 million US dollars in November 2025 for software that covers planning, flight dynamics and command and control. Earlier that year it described an "AI autopilot" with manual override. Published evidence of how well such assistants work in real operations is still thin. Table 49.5 lists the main risks and the usual defences.

**TABLE 49.5** Risks of language-model assistants in mission operations and common defences.

| Risk | Example in operations | Defence |
|---|---|---|
| Hallucination | Invented limit value or procedure step | Answers only from retrieved sources, with citations; people check |
| Stale or wrong sources | Superseded procedure retrieved | Configuration-controlled document store; version tags |
| Over-trust | Operator accepts a plausible diagnosis | Show evidence and confidence; training; audits |
| Unsafe action | Assistant drafts a hazardous command | No direct command access; validation and approval gates |
| Data leakage | Export-controlled data sent to an outside service | Models hosted on site or in accredited clouds; access control |
| Prompt injection | Malicious text in a log or document changes behaviour | Treat retrieved text as data; restrict tools (Chapter 53) |

### 49.4.4 Natural-Language Commanding

Telling a spacecraft what to do in plain words is still research. In one 2024 study, a language model flew spacecraft in a simulated pursuit–evasion game and ranked second in the challenge. Others propose models that turn goals into plans for an onboard planner.

The risks are clear. A spoken request is ambiguous, while a telecommand must be exact. A model can produce a command that is valid in form but wrong in intent: the right command sent to the wrong unit, or at the wrong time. Any practical system therefore needs strict layers. The model translates the request into a structured command. The command is checked against the command database and flight rules. It is simulated on the digital twin. A person approves it, and critical commands keep their usual two-step protection.

### 49.4.5 People and Automation

Automation changes the human job; it does not remove it. Decades of research in aviation and process control name three effects. **Automation bias** is the tendency to accept a machine's output without checking, or to miss what it fails to flag. **Trust calibration** is the match between how

much people trust a tool and how reliable it is. Both over-trust and under-trust are harmful. **Skill fade** is the loss of manual skill and system knowledge when people rarely practise them.

Together they create an irony first described in 1983: the more reliable the automation, the less prepared people are for the rare day it fails. Good practice follows. Show the evidence behind each alert, not just a score. Report each detector's measured precision. Keep people practising failures on the simulator. And hand over from software to person with enough context to act.

## 49.5 Scheduling Contacts and Ground Networks

The key idea of this section is that antenna time is a scarce resource shared by many spacecraft. Optimisation, not learning, does most of the work, but learning helps to predict demand and to speed up solvers.

### 49.5.1 The Deep Space Network

NASA's **Deep Space Network** (DSN) serves about 40 missions with antennas at Goldstone (Figure 49.9), Madrid and Canberra (Chapter 7). Demand routinely exceeds capacity, often by up to 40 %. JPL reported that the gap between requested and schedulable time grew from about 300 hours a week in 2011–12 to about 500 hours by late 2015.

Since July 2011 the DSN has been scheduled with **Service Scheduling Software** (S3). Each mission states its needs: hours, antennas and time windows. A scheduling engine turns them into tracks, using systematic search and then local repair of conflicts, and the missions negotiate the draft peer to peer. JPL has studied deep reinforcement learning and mixed-integer programming for the same problem (2021); both are research. The hardest part is agreeing what "fair" means.

### 49.5.2 Ground Stations as a Service

Small-satellite operators often buy antenna time instead of building stations. **Ground-station-as-a-service** (GSaaS) providers sell passes or minutes on shared networks through web interfaces (Chapter 7). Amazon Web Services opened AWS Ground Station in 2019. Microsoft retired its Azure Orbital Ground Station on 18 December 2024 and sold its ten 6 m antennas. KSAT lists more than 300 antennas at 28 locations, and in March 2022 its small-satellite network handled more than 52 000 passes.

On a shared network, each operator wants enough contacts with short gaps at low cost, and the provider wants busy antennas without conflicts. Automated interfaces with priorities and pre-emption rules make this work. Learning helps to forecast demand and to predict which passes will fail. Contacts through relay satellites are booked in the same way (Figure 49.10).

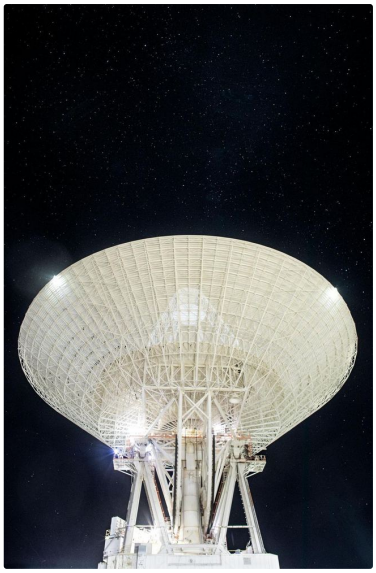


**FIGURE 49.9** The 70 m antenna DSS-14 at Goldstone, California, in September 2025. Its time is shared among dozens of missions through automated scheduling and negotiation. Image: NASA/JPL (NASA/JPL-Caltech).

**FIGURE 49.10** The relay satellite TDRS-L being prepared in January 2014, shortly before launch. Relay contacts are booked like ground-station passes. Image: NASA/KSC (NASA/KSC - Dimitri Gerondidakis ).

### 49.5.3 Scheduling a Constellation's Contacts

The problem can be stated exactly. A **pass** $p$ is a window when satellite $s(p)$ is visible from station $g(p)$, from $t_p^{\text{start}}$ to $t_p^{\text{end}}$. A binary variable $x_p$ is 1 if the pass is used. Each antenna can track one satellite at a time, with a set-up gap between passes. Each satellite can talk to one station at a time and has $D_s$ gigabytes waiting. The delivered data $y_s$ is limited by both:

$$\max \sum_{s} y_s + \lambda\, z \quad \text{subject to} \quad y_s \le D_s, \quad y_s \le \sum_{p:\, s(p)=s} R\,(t_p^{\text{end}} - t_p^{\text{start}})\, x_p, \quad z \le y_s / D_s \ \text{for all } s,$$

with the antenna and satellite conflict constraints. Here $R$ is the downlink rate and $z$ is the share of data delivered for the worst-served satellite, so $\lambda$ buys fairness. This is a **mixed-integer linear programme** (MILP). Standard solvers handle problems with thousands of passes in seconds.

In Figure 49.11 it is compared with two greedy rules for 36 imaging satellites in six polar planes and three single-antenna stations: Svalbard, Inuvik and Awarua. Over 24 hours there are 827 visible passes. Each satellite has 100–300 GB waiting, 7 040 GB in all, at 3 GB a minute. At Svalbard more than one satellite is in view 54 % of the time, so passes compete. Taking passes first-come, first-served delivers 69.8 % of the data and leaves nine satellites below half. Taking the longest passes first delivers 76.9 %. The MILP, solved in under a second, delivers 81.6 %, and every satellite gets at least 66 %.

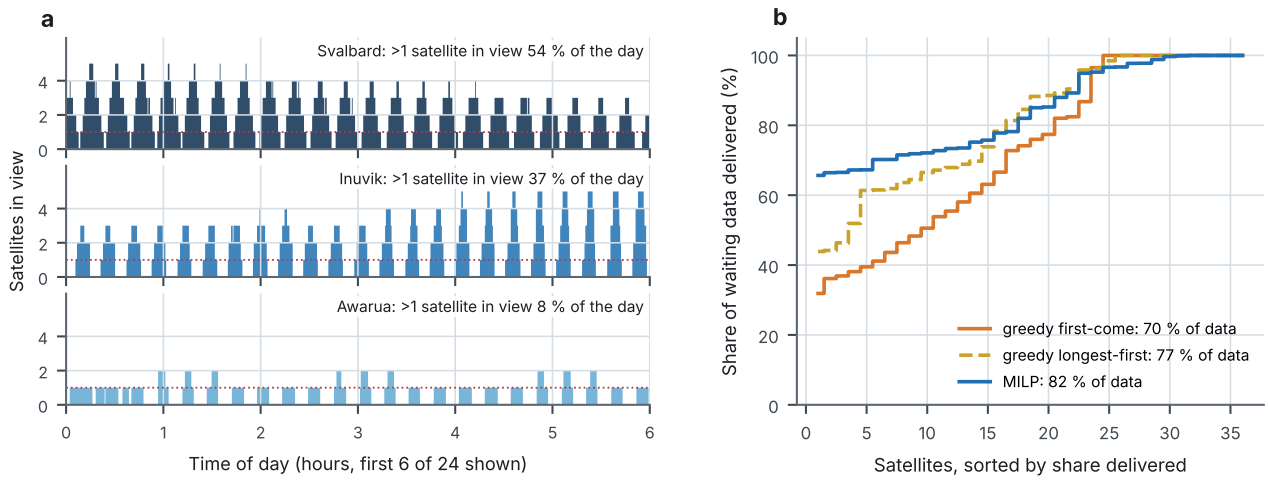



**FIGURE 49.11** Scheduling 36 satellites over three single-antenna stations for one day. (a) Number of satellites in view of each station during the first 6 hours; above the red line passes compete for one antenna. (b) Share of each satellite's waiting data delivered, sorted, for two greedy rules and the mixed-integer programme.

Real schedulers add slew times, uplinks, data generated through the day, station costs and weather, and re-plan as passes fail. Learning can predict inputs, such as data volumes and pass reliability, or learn heuristics that speed up a solver. The optimisation model still provides the constraints and the yardstick.

## 49.6 AI in Space Communications

The key idea of this section is that radios increasingly sense their environment and adapt to it. Learning helps to recognise signals, choose waveforms, share beams and spectrum, and find interference. Most of it is research or early deployment, but some is already inside commercial systems.

### 49.6.1 Cognitive Radio and Spectrum Sensing

A **cognitive radio** senses its radio environment and changes its frequency, power, waveform or timing to work better. The simplest sensing method is **energy detection**: compare the power in a band with the noise floor. More advanced methods recognise the type of signal present, which is where learning enters.

NASA's **SCaN Testbed** on the International Space Station (Figure 49.12), operated from 2012, was the main flight laboratory for such ideas. It carried three software-defined radios that could be reprogrammed from the ground. In a 2018 experiment, a cognitive engine on the ground used reinforcement learning to choose channels. When jammed, it told the satellite transmitter to switch to the channel expected to stay clear longest. The closed loop worked against several types of jammer, but the published summary gives no throughput figures.

**FIGURE 49.12** The SCaN Testbed, then called CoNNeCT, during a test at NASA's Glenn Research Center before its launch to the International Space Station. Its three software-defined radios were used from 2012 to try reconfigurable and cognitive communication in orbit. Image: NASA/GRC.

## 49.6.2 Recognising Signals

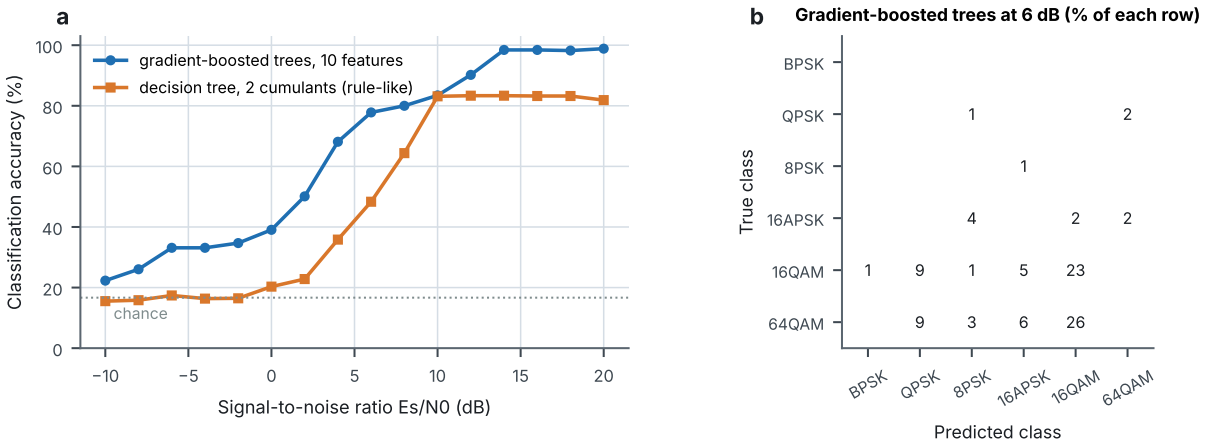
**Automatic modulation classification** identifies how a received signal is modulated — BPSK, QPSK, 16QAM and so on (Chapter 7). It is used in spectrum monitoring, interference hunting and signals intelligence (Chapter 19). The classical method uses **higher-order cumulants**: statistics of the complex baseband samples $y$ that differ between constellations but not with carrier phase. For example, with $M_{pq} = \mathbf{E}[y^{p-q}(y^*)^q]$,

$$C_{40} = M_{40} - 3M_{20}^2, \qquad C_{42} = M_{42} - |M_{20}|^2 - 2M_{21}^2.$$

Normalised by $M_{21}^2$, noise-free BPSK has $C_{42} = -2$, QPSK $-1$ and 16QAM $-0.68$. Noise pushes all of them toward zero.

Deep learning entered this field in 2016, when researchers fed raw I/Q samples to convolutional networks. Their open RadioML data set of 2018 holds 24 modulations and 2.56 million examples at signal-to-noise ratios (SNR) from −20 to +30 dB. Residual networks reached about 95 % accuracy at high SNR, and much less at low SNR.

The script behind Figure 49.13 repeats the classical approach on synthetic signals. Six modulations used in satellite links are generated, 512 symbols each, with random phase, a small frequency offset, unknown gain and noise. Ten cumulant and amplitude features feed two classifiers. A depth-3 decision tree on only $|C_{40}|$ and $C_{42}$, a rule-like expert classifier, levels off at 83 % because it cannot separate 16QAM from 64QAM. Gradient-boosted trees on all ten features reach 98–99 % above 14 dB, 78 % at 6 dB and 39 % at 0 dB, against 17 % for chance. At 6 dB the dense QAM constellations are still confused with each other.




Synthetic signals: 6 modulations, 512 symbols per example at one sample per symbol, random carrier phase, small frequency offset and unknown gain, white Gaussian noise; 160 training and 160 test examples per class and SNR. Features: normalised cumulants (C20, C40, C41, C42, C60, C63, M80) and amplitude statistics.

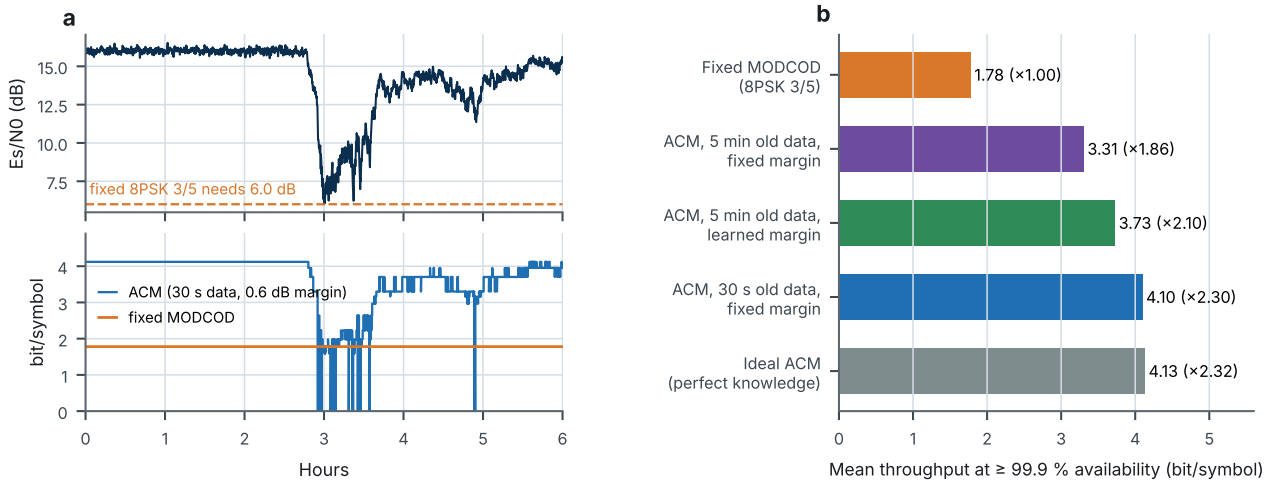
**FIGURE 49.13** Modulation classification of synthetic I/Q signals. (a) Accuracy against signal-to-noise ratio for a two-feature decision tree and for gradient-boosted trees on ten features. (b) Confusion matrix of the boosted trees at 6 dB.

### 49.6.3 Adaptive Coding and Modulation

Satellite links above 10 GHz fade in rain (Chapter 17). A link with **constant coding and modulation** (CCM) must use a robust waveform that works through the worst fade it is designed for. In clear sky, which is most of the time, it wastes capacity. **Adaptive coding and modulation** (ACM), standardised in DVB-S2 (2005) and DVB-S2X (2014), changes the modulation and code rate — the **MODCOD** — as the measured signal-to-noise ratio changes. The receiver reports its $E_s/N_0$ (energy per symbol over noise density). The transmitter then picks the most efficient MODCOD whose threshold is below the measured value minus a safety margin.

The simulation in Figure 49.14 covers one year of a Ka-band downlink with a clear-sky $E_s/N_0$ of 16 dB. Rain falls 5.4 % of the time, with fades from a standard log-normal model: 2.3 dB is exceeded 1 % of the time and 9.4 dB 0.1 % of the time. For 99.9 % availability, a fixed link must use 8PSK 3/5 at 1.78 bit/symbol. ACM driven by measurements 30 s old needs only a 0.6 dB margin and averages 4.10 bit/symbol — 2.3 times more — within 1 % of ideal ACM.

Learning does not help a fast loop, which already works almost ideally. It helps when decisions must be made well ahead, such as planning a pass or choosing beams and gateways minutes early. If the measurement is 5 minutes old, a fixed margin of 3.0 dB is needed and throughput falls to 3.31 bit/symbol. The script then learns a state-dependent margin from the first 90 days. It is a quantile of how much the fade can change, given the current fade and its trend. This raises throughput to 3.73 bit/symbol — 13 % more at the same availability — because the margin is small in clear sky and large in rain. Published research uses neural networks and weather radar or forecasts in the same role.




Synthetic Ka-band downlink, one year at 10 s steps: rain 5 % of the time with log-normal fades (Maseng–Bakken model; exceeded 1 % of the time: 2.3 dB, 0.1 %: 9.4 dB) plus scintillation; clear-sky Es/N0 16 dB; DVB-S2 thresholds (ETSI EN 302 307-1) plus 0.5 dB implementation margin. (b) Each ACM scheme uses the smallest safety margin that keeps unavailability at or below 0.1 %; the learned margin is a quantile of the fade change predicted from the current fade and its trend (trained on the first 90 days, scored on the rest).

**FIGURE 49.14** Adaptive coding and modulation over a synthetic Ka-band rain-fade year. (a) Six hours around the year's deepest fade: Es/N0 (top) and spectral efficiency of fixed and adaptive MODCODs (bottom). (b) Mean throughput at 99.9 % availability for fixed coding, three ACM schemes and ideal ACM.

### EXAMPLE: THROUGHPUT GAIN OF ADAPTIVE CODING AND MODULATION

Use the results of Figure 49.14 for a carrier of 250 million symbols per second (Mbaud). Ignore framing overhead.

1. **Fixed coding.** 8PSK 3/5 needs 5.5 dB plus a 0.5 dB implementation margin; it works 99.912 % of the time. Throughput: $250 \times 10^6 \times 1.78 \times 0.99912 \approx$ **445 Mbit/s**.
2. **ACM, 30 s loop.** Mean 4.096 bit/symbol: $250 \times 10^6 \times 4.096 \approx$ **1 024 Mbit/s**, a factor of **2.30**.
3. **Ideal ACM.** 4.130 bit/symbol, 1 033 Mbit/s: the 0.6 dB safety margin costs less than 1 %.
4. **Why so large?** In clear sky the link uses 32APSK 5/6 at 4.12 bit/symbol, which needs 14.3 dB plus margin. The fixed link must keep about 10 dB in reserve (16 dB available, 6.0 dB needed) for rare fades.
5. **Five-minute-old data.** A fixed 3.0 dB margin gives 3.305 bit/symbol (826 Mbit/s). The learned, fade-dependent margin gives 3.727 bit/symbol (932 Mbit/s): **13 % more** at the same 99.9 % availability.
6. **Caveat.** The gain depends on climate, frequency and clear-sky margin. Systems with a smaller clear-sky margin gain less.

## 49.6.4 Beams, Power and Spectrum in High-Throughput and LEO Systems

High-throughput satellites have hundreds of spot beams, and LEO broadband satellites steer many beams with phased arrays (Chapter 17). Demand is uneven in space and time. **Beam hopping** lights each beam only part of the time, in a pattern that follows demand. The first over-the-air demonstration, in ESA's BEHOP project in June 2018, used DVB-S2X "superframes" through a conventional Eutelsat satellite. Digital payloads, such as those on Eutelsat Quantum (launched 2021) and many newer satellites, make such flexibility routine.

Choosing which beams to light, with how much power and bandwidth, is a large combinatorial problem that changes from milliseconds to minutes. Operators use optimisation and heuristics. Since about 2018, many papers have trained **deep reinforcement learning** agents for beam

hopping, power control and channel allocation; in simulation they often beat simple heuristics. Hybrids, in which a network proposes a pattern and an optimiser repairs it, look most promising. Operators rarely publish what their resource managers use, so operational use is hard to confirm.

### 49.6.5 Interference: Detection and Geolocation

Interference is a daily problem for satellite operators. Causes include badly pointed terminals, cross-polarised carriers, terrestrial 4G and 5G signals and, increasingly, deliberate jamming (Chapter 18 for navigation signals). Monitoring systems scan the spectrum around the clock. Learning helps in three steps, as in ESA's SkyMon project, completed in 2021. A model first recognises known carriers from their spectral shape. A detector then flags spectra that depart from the expected mask, and a classifier names the type of interference. Researchers at KTH and the Swedish Space Corporation showed in 2019 that LSTM models trained on real spectrum data could find short and long interference events in time and frequency.

Finding *where* an interferer is uses physics. The uplink is received through the wanted satellite and a neighbour; the differences in arrival time and Doppler shift define curves on the Earth that cross at the transmitter (Chapter 19).

## 49.7 Networks in Orbit

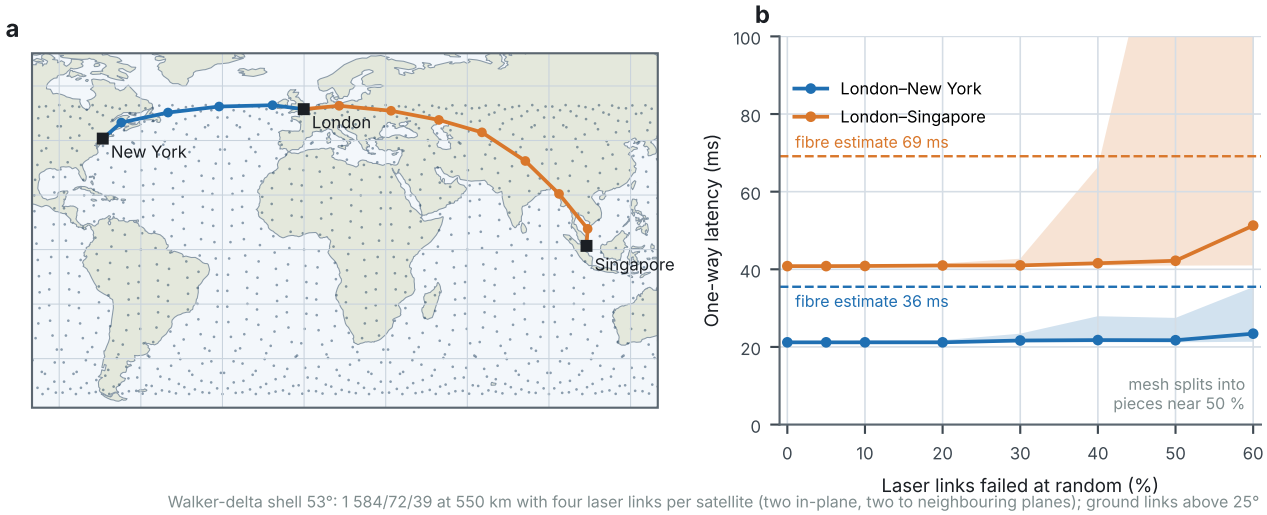
The key idea of this section is that a constellation with inter-satellite links is a network whose shape changes every second. Classical routing works well. Learning is studied for traffic prediction, failures and the joint control of space and ground networks.

### 49.7.1 Routing in Laser Meshes

Iridium has routed calls through radio crosslinks since its service began in 1998, with up to four links per satellite. Starlink uses lasers. In January 2024 SpaceX reported more than 9 000 lasers in orbit carrying more than 42 petabytes a day. Links ran at up to 100 Gbit/s over distances up to 5 400 km. The constellation has grown much larger since. The US Space Development Agency's Transport Layer also links its satellites optically (Chapter 19).

In a **Walker-delta** shell (Chapter 11), each satellite can link to two neighbours in its own plane and one in each neighbouring plane. This gives a grid that wraps around the Earth. Routing then becomes a **shortest-path** problem on a graph: satellites are nodes, links are edges weighted by distance (delay), and Dijkstra's algorithm finds the fastest path. Because the satellites move, the graph is recomputed every few seconds. Light in vacuum travels about 47 % faster than in optical fibre, so long space paths can beat fibre.

The script behind Figure 49.15 builds a 1 584-satellite shell at 550 km and 53°, with four laser links per satellite, using the networkx library. From London to New York the best path uses four satellites and three laser hops. The one-way delay is 21.2 ms, against about 35.5 ms for fibre that is 1.3 times longer than the great circle. London to Singapore takes 40.8 ms, against about 69 ms. When links fail at random, routing simply goes around them. With 30 % of links failed, the median delay hardly changes. Near 50 %, the grid starts to break into pieces, as theory predicts for a square grid, and some routes need long detours.




Walker-delta shell 53°: 1 584/72/39 at 550 km with four laser links per satellite (two in-plane, two to neighbouring planes); ground links above 25° elevation; snapshot geometry. Shortest paths by distance (Dijkstra, networkx); 30 random failure draws per point, band 5–95 %. Fibre: 1.3 × great-circle distance at c/1.468.

**FIGURE 49.15** Routing in a laser-linked Walker-delta constellation. (a) Snapshot of the 1 584 satellites (dots) and the shortest paths from London to New York and Singapore. (b) One-way delay against the share of laser links failed at random (median and 5–95 % band of 30 draws), with fibre estimates.

Shortest paths are easy; managing load is not. Traffic moves with the time of day, and gateways and laser terminals fail or lose sight of each other. Research uses learned traffic forecasts, reinforcement-learning routing and graph neural networks to spread load and pre-compute routes. The bar is high: a learned router must beat classical algorithms that are fast, predictable and easy to verify.

### 49.7.2 Network Digital Twins and Self-Driving Networks

A **network digital twin** is a running model of a whole network: satellites, links, gateways, traffic and the control software. Operators use it to test changes before applying them — a new routing policy, a software update, a gateway closure. Learned models make twins fast enough to run in a loop. One example is a graph neural network that predicts delay and loss for a traffic pattern far faster than a packet-level simulator.

The long-term aim is a **self-driving network**: operators state goals, such as “keep latency below 40 ms”, and software configures, watches and corrects the network. How much of today’s automation uses learning, rather than rules and optimisation, is rarely published.

### 49.7.3 Toward AI-Native 6G Non-Terrestrial Networks

Satellites are joining the mobile standards. The 3GPP body added **non-terrestrial networks** (NTN) in Release 17 (2022), so that phones and IoT devices can use satellites. Release 19 (2025) added **regenerative payloads**, which carry a full base station on the satellite. 6G studies started in Release 20, including learning-based features of the radio interface such as compressed channel-state feedback. First 6G specifications are expected around 2028–29. The ITU’s framework for 6G (IMT-2030), adopted in 2023, lists AI and ubiquitous connectivity among its aims.

“AI-native” means that learned models are built into the radio and network from the start, with standard ways to train, test and update them. For satellites this could cover beam management, handover prediction for fast-moving cells, Doppler and timing correction, and energy saving. All

are research or pre-standard work in 2026. Direct-to-device services from Starlink, AST SpaceMobile and others (Chapter 17) give the commercial pull. Table 49.6 summarises the status of AI across communications and networks.

**TABLE 49.6** Status of AI in space communications and networks, October 2026.

| Problem | Typical method | Status (2026) |
|---|---|---|
| Spectrum monitoring, carrier recognition | Learned classifiers on spectra | Commercial products and ESA projects |
| Modulation classification | Cumulant features; convolutional and residual networks | Mature research; used in monitoring and signals intelligence |
| Interference detection | LSTM and anomaly detection on spectra | Demonstrated on real data; early products |
| ACM and power control | Feedback loops; learned fade prediction | ACM operational; learning research |
| Beam hopping, resource allocation | Optimisation; deep reinforcement learning | Optimisation operational; RL research |
| Anti-jamming channel selection | Reinforcement learning | Flight experiment (SCaN Testbed, 2018) |
| Laser-mesh routing | Shortest paths; learned traffic forecasts, RL | Classical operational; learning research |
| 6G NTN air interface | AI/ML features under study in 3GPP | Standards work |

## 49.8 Securing Ground Segments and Links

The key idea of this section is that the ground segment is the easiest way into a space system. Machine learning helps defenders spot unusual behaviour, but attackers can target the learning itself (Chapter 53).

Ground systems are ordinary computer networks with antennas, modems and cloud links attached, and they face the same attacks as any business network. The best-known case came on 24 February 2022, the first day of Russia's full-scale invasion of Ukraine. Attackers abused the management network of Viasat's KA-SAT system to push a "wiper" program, later named AcidRain, to user modems. Tens of thousands of modems in Ukraine and across Europe stopped working. Spill-over cut remote monitoring of 5 800 Enercon wind turbines in central Europe. The European Union attributed the attack to Russia in May 2022.

Learned defences work at several layers:

- **Network intrusion detection.** Models of normal traffic between ground-segment computers flag unusual connections, volumes or timing. Like telemetry monitors, they must meet a strict false-alarm budget.
- **Command anomaly detection.** A model of normal command sequences flags commands that are valid but unusual: wrong time, wrong order, wrong source. This adds to the encryption and authentication of telecommands (CCSDS Space Data Link Security; Chapter 7).
- **Telemetry for security.** Telemetry anomaly detectors can also reveal tampering, such as an unexpected mode change.

- **Jamming and spoofing detection.** Learned classifiers on signal power, spectra and receiver statistics detect jamming of command links and spoofing of navigation signals (Chapter 18). Research has trained such detectors for intermittent jamming of telemetry and telecommand links.

The Aerospace Corporation's SPARTA framework (2022) catalogues attack techniques against spacecraft, including some shown only in laboratories, and helps defenders plan detections. Two warnings apply to learned defences. Attackers can evade them by mimicking normal behaviour or slowly poisoning training data. And the models, pipelines and assistants become new targets (Chapter 53).

### PRACTICE: DEPLOYING AN ANOMALY-DETECTION SYSTEM IN A CONTROL CENTRE

1. **Start from the operators' needs.** Agree which subsystems matter most, what a useful alert looks like and how many alerts a day the team can handle — for example, 10. This budget sets every threshold.
2. **Assemble the data.** Collect several years of telemetry, command logs, timelines and anomaly reports. Label known anomalies and rare nominal events, such as manoeuvres and calibrations, with the engineers who saw them.
3. **Keep the baseline.** Keep limit checks running unchanged. The learned system adds alerts; it does not remove existing protection.
4. **Choose simple models first.** Start with per-channel prediction residuals and a nearest-neighbour or isolation-forest score, using orbit phase and commanded modes as inputs. Try deep networks only if they clearly do better on held-out data.
5. **Test like an operator.** Use a time-ordered split with a full year of test data. Score detected events, alerts per day and warning time, not just sample accuracy (ESA-ADB is a good template).
6. **Run in shadow mode.** Run the system beside normal operations for several months without paging anyone. Review every alert with the engineers and record whether it was useful.
7. **Explain each alert.** Show the channels involved, the expected and measured values, and similar past cases. Engineers will not act on a bare score.
8. **Plan for change.** Mark planned events so that they are not flagged, retrain after configuration changes and season shifts, and track alert precision every month.
9. **Go live gradually.** Start with one subsystem during working hours, with a named owner and a way to suppress or tune noisy detectors. Write down what the system does not cover.

### KEY TAKEAWAYS

- Modern spacecraft produce tens of thousands of telemetry parameters (JWST about 40 000, Sentinel-3 about 46 000), and constellations fly hundreds to thousands of satellites. Operations are moving to automation and management by exception.
- The false-alarm budget drives every monitor: 10 000 channels checked every 10 s need fewer than about $1.2 \times 10^{-7}$ false alarms per check to stay below 10 a day.
- Learned detectors that use context — orbit phase, mode and other channels — catch contextual anomalies and drifts that fixed limits miss. Generic detectors can do worse than limits, and real benchmarks such as ESA-ADB remain hard.
- Prognosis needs a physical degradation model, honest uncertainty and data from the vehicle itself. Predictions tighten as data accumulate.
- Language-model assistants are useful for search and drafting when grounded in controlled documents. Natural-language commanding is research and needs strict validation, simulation and human approval.
- Contact scheduling is an optimisation problem: in this chapter's example a mixed-integer programme delivered 81.6 % of waiting data, against 69.8 % for first-come scheduling, and served every satellite fairly.
- In communications, ACM roughly doubles Ka-band throughput over fixed coding. Learning helps most where decisions are made well ahead. Laser meshes route around many failures with classical shortest-path algorithms. Ground segments remain the main cyber target.

### REVIEW QUESTIONS

1. A satellite has 4 000 channels sampled every 2 s. How many checks are made per day? What per-check false-alarm probability allows 5 false alarms a day, and what Gaussian threshold does that correspond to?
2. Explain the difference between point, contextual and drift anomalies, with one spacecraft example of each. Which kind can fixed limits never detect, and why?
3. Write the standardised residual used by a prediction-based detector. Why did the predictive network in Figure 49.3 detect a value copied from another part of the orbit, while the fixed limits did not?
4. Why did the isolation forest in Figure 49.4 miss spikes that left the normal range? What would you change in its inputs or in the system around it?
5. ESA-ADB scores detectors event by event and puts precision first. Why is this closer to operators' needs than sample-by-sample accuracy? What made Mission 2 hard?
6. In Figure 49.6, why is the 90 % interval for the end-of-life date much wider at 2.5 years than at 4.5 years? What would happen to the intervals if the model form were wrong?
7. List four risks of a language-model assistant in a control centre, and a defence against each. Why should such an assistant not send commands directly?
8. State the decision variables, constraints and objective of the contact-scheduling MILP. Why did first-come, first-served scheduling leave nine satellites below half their data?
9. A Ka-band link has a clear-sky Es/N0 of 16 dB and must tolerate a 9.4 dB fade. Using the DVB-S2 thresholds in this chapter, compare the spectral efficiency of fixed coding with that of ACM in clear sky. Why does a learned margin help more when decisions use 5-minute-old data than 30-second-old data?
10. In a laser mesh, why does the delay between London and New York change little when 30 % of links fail, yet rise sharply near 50 %? Give two ways in which learning could improve routing beyond shortest paths.

CHAPTER

# 50 AI for Space Situational Awareness, Defence and Security

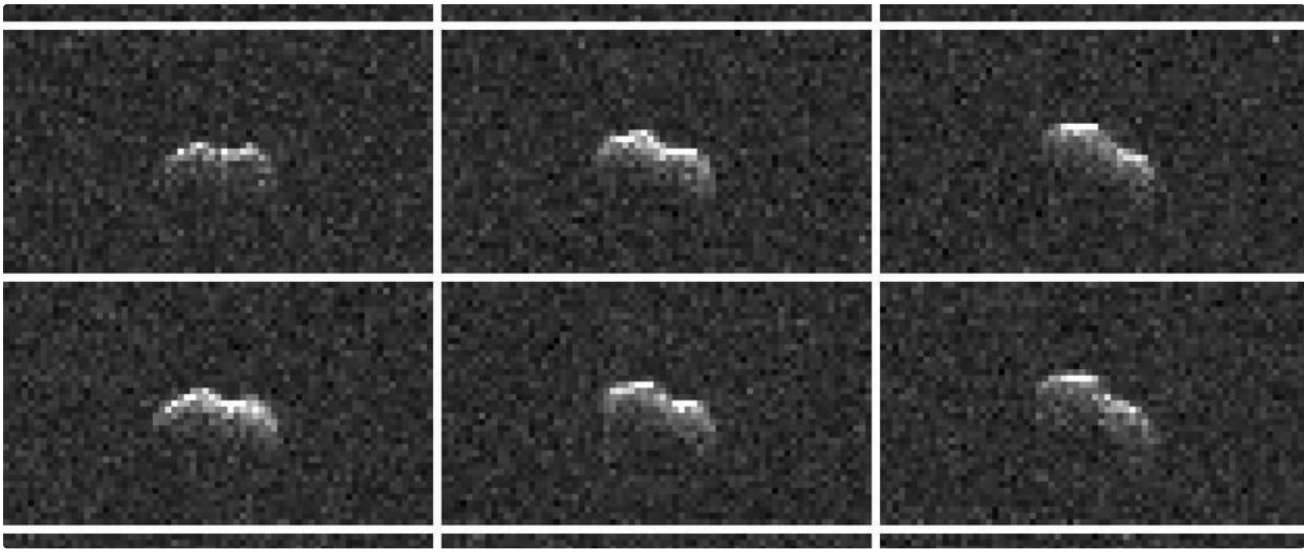


Image: NASA/JPL (NASA/JPL-Caltech and NSF/AUI/GBO)

*More than 35 000 objects are catalogued in Earth orbit, and the number of close approaches grows with the square of that number. This chapter shows where machine learning and optimisation help to keep track of it all, and where they do not. It covers streak detection, data association, orbit prediction with learned corrections, storm-time drag, conjunction screening and triage, manoeuvre detection, light-curve characterisation and re-entry. It then turns, from open sources only, to defence and intelligence uses — imagery exploitation, tip-and-cue, emitter recognition, missile tracking, battle management, proximity operations, deception and human–machine teaming — and ends with the law, policy and stability questions they raise. Small models run in Python make each method concrete.*

IN THIS CHAPTER

50.1 Space Situational Awareness at Machine Scale
50.2 Conjunction Assessment and Space Traffic
50.3 Watching Behaviour: Manoeuvres, Characterisation and Re-entry
50.4 AI in Defence and Intelligence Space Systems
50.5 Law, Ethics and Policy
50.6 Lessons from the Experiments

Space situational awareness (SSA) means knowing what is in orbit, where it is going and what it is doing. The hero image above shows an older side of the job. Engineers of NASA's Orbital Debris Program Office are recording impact craters on a Hubble camera that astronauts brought home

in 2009. Every crater is a small piece of evidence about the debris population. Today the evidence arrives as millions of radar and telescope observations a day. Software must turn them into orbits, warnings and decisions for more than fifteen thousand active satellites.

That software increasingly includes machine learning (ML) — models that learn patterns from examples rather than following hand-written rules. The same methods also serve defence and intelligence. They sift satellite images for change, fuse signals from many sensors and point the next satellite at the right place. This chapter explains both uses. It separates what is operational, what has been demonstrated and what is still research, as of October 2026. Defence topics are covered from open sources only and without operational detail.

The chapter builds on the engineering background of Chapter 21 (SSA and debris), Chapters 19 and 20 (national-security space and missile warning) and Chapter 22 (space weather). Learning methods are introduced in Chapters 43 and 44, data and onboard computing in Chapter 45 and imagery analysis in Chapter 46. Autonomy for rendezvous is covered in Chapter 48 and the assurance of AI in Chapter 53.

## 50.1 Space Situational Awareness at Machine Scale

The key idea of this section is that SSA is a pipeline, from raw sensor data to decisions. Physics does most of the work. Learning helps at a few points where the data are messy, the volume is large or the physics is uncertain.

### 50.1.1 From Sensors to Decisions

Figure 50.1 shows the chain. **Sensors** — radars, telescopes, space-based cameras and commercial networks (Chapter 21) — produce raw data. **Detection** finds objects in that data. **Association** decides which known object each detection belongs to. Detections that match nothing are called **uncorrelated tracks** (UCTs). **Orbit determination** fits an orbit to each object's observations and predicts where it will go. The results form the **catalogue**. In October 2026 the public catalogue held about 35 000 objects in orbit, of which about 17 000 were working satellites.

From the catalogue, **screening** finds every close approach in the next few days. **Risk triage** decides which of them matter. Operators then **decide** whether to manoeuvre and **coordinate** with each other. Alongside this runs **behaviour analysis**: spotting manoeuvres and break-ups and learning what each object is.

Most of this pipeline is classical. Orbit determination uses least-squares fits and Kalman filters (Chapter 43), and screening uses geometry. The US Space Force declared its new catalogue software, ATLAS, operational in September 2025. It is replacing a system that dates from the 1980s. Its Unified Data Library pools government and commercial sensor data for space domain awareness. The service's 2025 Data and AI Strategic Action Plan made the library a formal programme and set goals for adopting AI. Table 50.1 summarises where learning fits.

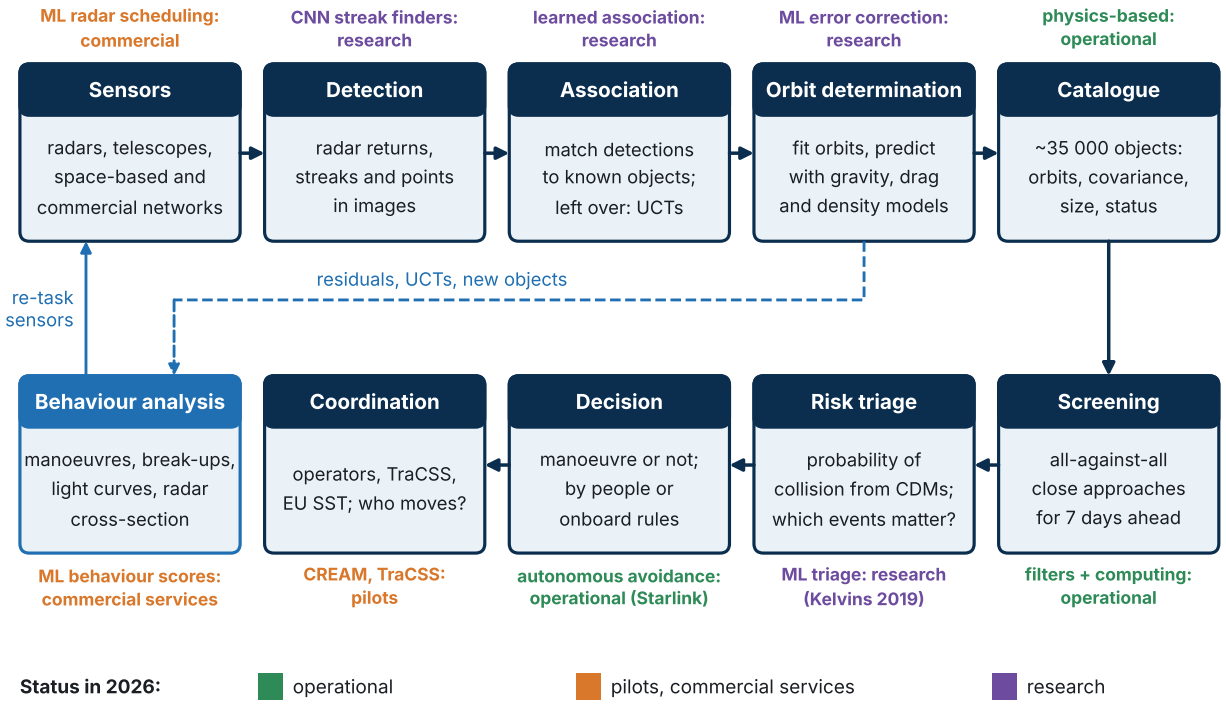



**FIGURE 50.1** The space situational awareness pipeline. The data path (top) turns sensor data into a catalogue; the services path (bottom) turns the catalogue into warnings and decisions. Tags give the status in 2026 of machine learning at each step.

**TABLE 50.1** Space situational awareness tasks, their classical methods and the places where machine learning helps.

| Task | Classical method | Where learning helps | Status (2026) |
|---|---|---|---|
| Detecting objects in images | Matched filters, stacking | CNN detectors for streaks and faint objects | Research; first pipelines |
| Associating detections | Gating and global assignment | Learned similarity after break-ups | Research |
| Predicting orbits | Numerical propagation, fitted drag | Learned corrections of prediction errors | Research |
| Thermospheric density | Empirical and data-assimilating models | Fast surrogates with uncertainty | Research |
| Conjunction triage | Probability of collision thresholds | Ranking events from CDM histories | Research; pilots |
| Manoeuvre detection | Residual tests, CUSUM | Classifiers on orbit histories | Commercial services |
| Characterisation | Expert reading of light curves | Classifiers of light curves | Research; commercial |
| Radar tasking | Fixed schedules and priorities | Learned detection probability | Commercial (vendor-reported) |

## 50.1.2 Finding Faint Streaks

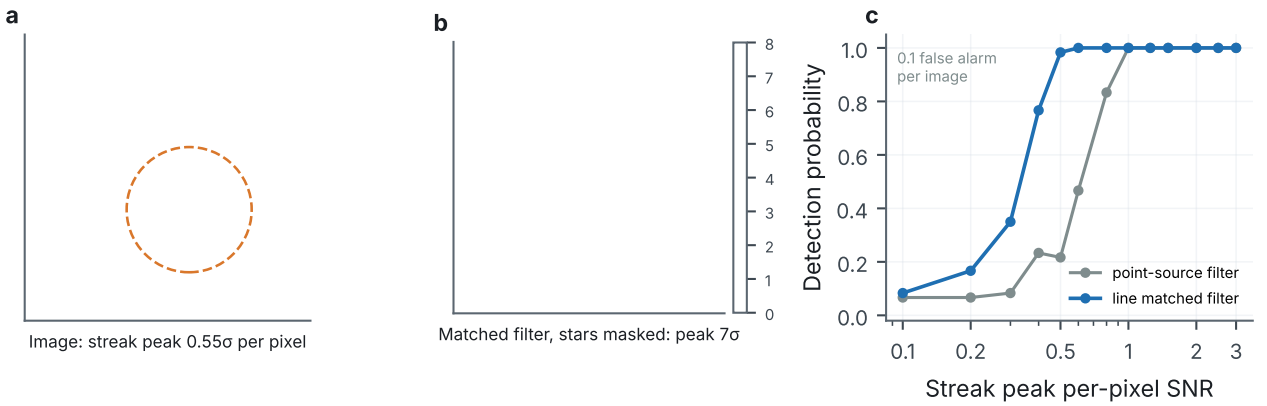
A satellite moves across the sky while a telescope tracks the stars. In a long exposure it therefore appears as a **streak**, not a point. A faint streak spreads its light over many pixels, so no single pixel stands out from the noise. The classic answer is the **matched filter**: correlate the image with a template shaped like the expected streak. For white noise this is the best linear detector.

Suppose the template $s_i$ matches the signal and the noise has standard deviation $\sigma$ per pixel. The filter output then has a signal-to-noise ratio of

$$\mathrm{SNR}_{\mathrm{MF}} = \frac{\sqrt{\sum_i s_i^2}}{\sigma} \approx \frac{A}{\sigma}\sqrt{\sqrt{\pi}\,\sigma_{\mathrm{psf}}\,L},$$

Here $A$ is the peak brightness along the streak, $L$ its length and $\sigma_{\mathrm{psf}}$ the width of the blur. A long streak therefore collects signal from many pixels. The catch is that the direction and length are unknown. The filter must be applied at many angles and positions, and each extra trial adds chances for a false alarm.

Figure 50.2 shows this on a synthetic star field. A 64-pixel streak whose peak is only 0.55 times the noise is invisible to the eye. After bright stars are masked, the matched filter finds it at 7.3σ. In the Monte Carlo test, at the same rate of 0.1 false alarm per frame, the matched filter detected half of the streaks at a peak signal of 0.34σ. A point-source detector needed 0.62σ, so the streak filter reached objects about 1.8 times fainter.




Synthetic 192 × 192 px frame, unit Gaussian noise, 90 stars (PSF σ = 1.3 px) and a 64 px streak. Stars masked above 6σ after PSF smoothing. (c) 60 trials per point on 128 px frames with stars masked; thresholds set from 240 empty frames for 0.1 false alarm per frame.

**FIGURE 50.2** Streak detection by matched filtering. (a) Synthetic frame with stars and a faint streak (circle). (b) Matched-filter output, maximum over 36 angles, after masking stars. (c) Detection probability against the streak's peak signal-to-noise ratio per pixel, for a point-source filter and a line matched filter at 0.1 false alarm per frame.

Learning enters in two ways. Convolutional networks (CNNs, Chapter 44) can be trained on real frames to find streaks of any shape while ignoring satellite trails, cosmic rays and stars. A 2026 study trained an object detector on 2 335 telescope images and reported 94 % precision and 97 % recall. Networks also help with **synthetic tracking**: shifting and adding many short frames along trial velocities, so that faint objects build up while noise averages out. Both are in research use and in some commercial pipelines. Matched filtering and stacking remain the operational workhorses.

Moving-object detection is older than satellites. Asteroid surveys find objects by their motion against the stars, as in Figure 50.3. The same software problems — faint signals, many trials and false alarms — run through both fields.

**FIGURE 50.3** Discovery images of asteroid 2018 LA, taken by the Catalina Sky Survey on 2 June 2018. The small object (circled) moves between frames against fixed stars. It entered the atmosphere over Botswana about eight hours later. Image: NASA/JPL (NASA/JPL-Caltech/CSS-Univ. of Arizona).

### 50.1.3 Association and Uncorrelated Tracks

Each new detection must be linked to a catalogue object. The usual measure is the **Mahalanobis distance** between the detection $\mathbf{z}$ and the predicted position $\hat{\mathbf{z}}$:

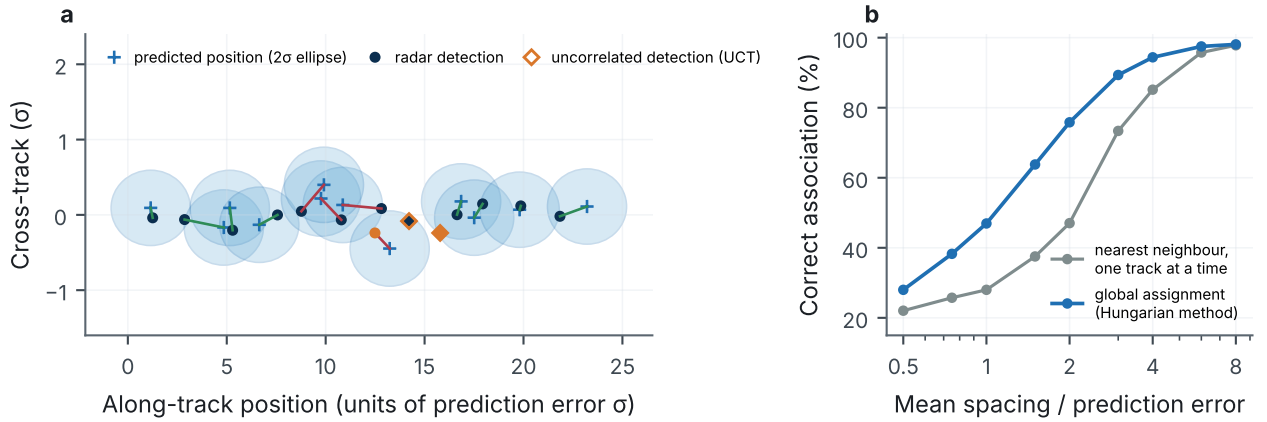
$$d^2 = (\mathbf{z} - \hat{\mathbf{z}})^{\mathsf{T}}\, \mathbf{S}^{-1}\, (\mathbf{z} - \hat{\mathbf{z}}),$$

where $\mathbf{S}$ is the combined covariance of prediction and measurement. It measures distance in units of uncertainty. A **gate** rejects pairs with $d^2$ above a threshold, such as 13.8 for 99.9 % confidence in two dimensions.

Association is easy when objects are far apart. It is hard after a launch that releases dozens of satellites together, or after a break-up. Then many predictions overlap and a greedy rule — give each object its nearest detection — makes chains of mistakes. **Global assignment** instead chooses all links at once to minimise the total cost $\sum_{ij} c_{ij} x_{ij}$, where $x_{ij} = 1$ links object $i$ to detection $j$. The Hungarian method solves this exactly.

Figure 50.4 tests both rules on a synthetic radar pass over a cluster of 24 objects. When the spacing was twice the prediction error, global assignment linked 76 % of objects correctly and the greedy rule 47 %. Both reach about 98 % when the objects are eight errors apart. Detections that match nothing become UCTs. They may be new fragments, unknown objects or simply wrong associations — so each UCT costs analyst time.

Learning helps by improving the inputs. Better predictions shrink the ellipses, and learned similarity — from brightness, radar cross-section or radio signatures — separates objects that geometry alone cannot. Research also uses multi-target trackers that keep several hypotheses alive until new data resolve them.




Synthetic radar pass over objects released together and spread along-track; prediction error σ along-track and 0.25σ cross-track; 92 % detection probability; two unknown objects per pass. Gate $\chi^2 < 13.8$ (99.9 %). (b) 24 objects, 300 trials per point.

**FIGURE 50.4** Data association after a cluster deployment. (a) One synthetic pass: predicted positions with 2σ ellipses, detections, global-assignment links (green right, red wrong) and unmatched detections (UCTs). (b) Share of objects correctly associated against the ratio of spacing to prediction error.

### 50.1.4 Orbit Prediction: Physics Plus Learning

In low Earth orbit (LEO), the largest error in an orbit prediction is usually drag. Drag lowers the semi-major axis $a$ at a rate $\dot{a} = -\rho B \sqrt{\mu a}$, where $\rho$ is air density, $B$ the ballistic coefficient ( $C_D A/m$) and $\mu$ Earth's gravitational parameter. A lower orbit is faster, so a density error makes the object run ahead of or behind its prediction. For a fractional density error $\varepsilon$, the along-track error grows with the square of time:

$$\Delta s(t) \approx \tfrac{3}{4}\, n\, \varepsilon\, |\dot{a}|\, t^2,$$

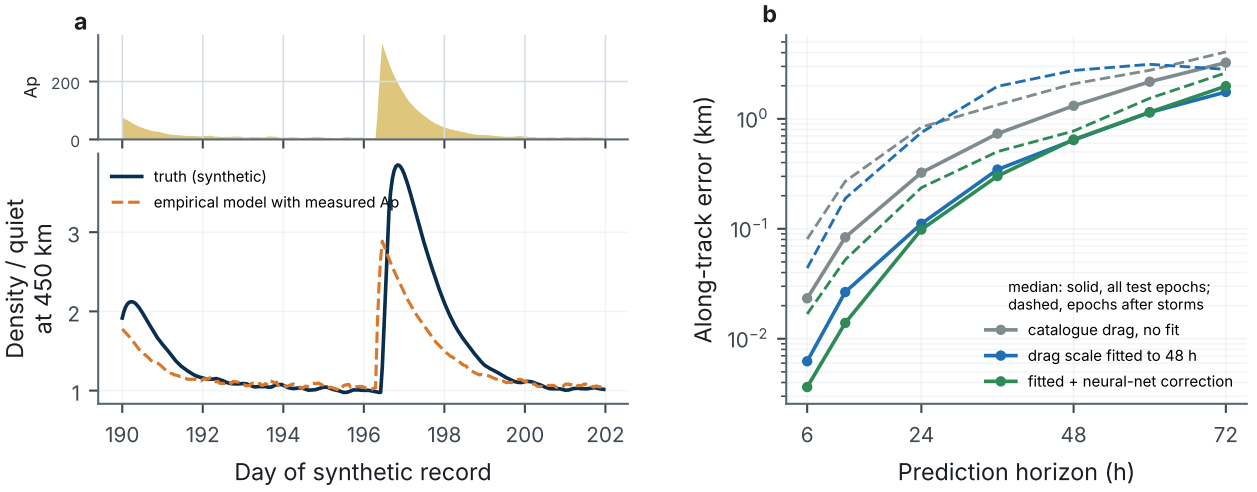
where $n$ is the mean motion. At 450 km, with $|\dot{a}|$ = 50 m/day and $n \approx 96.6$ rad/day, a 20 % density error gives about 0.7 km after one day and 6.5 km after three. This quadratic growth is why conjunction warnings beyond a few days are unreliable.

Operators fight it by fitting a drag scale factor to recent tracking data. Learning adds a **hybrid** step. A physics propagator makes the prediction. A learned model $g_\theta$ predicts the propagator's typical error, which is then removed.

$$\hat{\mathbf{x}}(t) = \mathbf{f}_{\text{phys}}(\mathbf{x}_0, t) + g_\theta(\text{features}).$$

The features include the forecast horizon, altitude, space-weather indices and the errors the propagator made in the recent past. Peng and Bai (2018) showed the idea with neural networks, support vector machines and Gaussian processes on a simulated catalogue. The physics keeps the prediction sensible; the learned part corrects patterns that the physics misses.

Figure 50.5 runs this on 160 synthetic objects over a 240-day record with storms. Fitting the drag scale to 48 hours of data nearly halved the median three-day error, from 3.2 km to 1.75 km. A small neural network on top did not help on average: its median three-day error was 1.98 km. But in the nine test epochs just after storms it cut the median one-day error from 0.75 km to 0.24 km. That sample is small, so the gain is only indicative. The lesson is general. A learned correction must be tested separately in the conditions that matter, because a gain in one regime can hide a loss in another.




160 synthetic objects at 350–650 km over a 240-day record with random storms. Truth: density with a lagged storm response; empirical model: weaker, unlagged response. Forecasts assume Ap persists. Network (2 × 32 tanh, numpy) trained on days 3–150, tested on days 160–236.

**FIGURE 50.5** Storm-time drag and a hybrid correction. (a) Synthetic Ap index (top) and density at 450 km relative to quiet conditions: truth with a lagged storm response and an empirical model with a weaker, unlagged response. (b) Median along-track error against forecast horizon for three predictors, for all test epochs (solid) and for epochs just after storms (dashed).

### 50.1.5 Density and Drag During Storms

Geomagnetic storms heat the thermosphere and can multiply density at satellite heights several times (Chapter 22). Empirical models such as NRLMSIS and JB2008 capture average behaviour. The US Space Force's High Accuracy Satellite Drag Model (HASDM) goes further: it adjusts density every few hours from the observed drag of dozens of calibration objects. Learning offers fast **surrogates** of such models. Licata, Mehta and colleagues compressed 20 years of HASDM density maps with principal component analysis and trained a network on the result. Their model reproduced the database with about 10 % error and gave calibrated uncertainty, even during storms. Such models are in research use; operational density still comes from physics and data assimilation.

#### THE MAY 2024 STORM AND THE CATALOGUE

- **Strength.** The storm of 10–11 May 2024 was the strongest in two decades, with the planetary ap index reaching 400.
- **Density.** Model density at 400 km rose to about six times its value 12 hours before.
- **Forecasts.** The 0–3-day ap forecasts underestimated the rise by 100–300.
- **Decay.** One Earth-observation satellite's decay rate jumped from 38 m to 180 m per day.
- **Operators.** Before the storm about 300 of nearly 10 000 active LEO satellites were manoeuvring. Afterwards thousands manoeuvred together — nearly half of all active LEO satellites, mostly Starlink.
- **Consequence.** Mass manoeuvres invalidate the conjunctions predicted for the coming week, so screening has to restart from new orbits.

Source: Parker and Linares, Journal of Spacecraft and Rockets (2024), from public two-line element data.

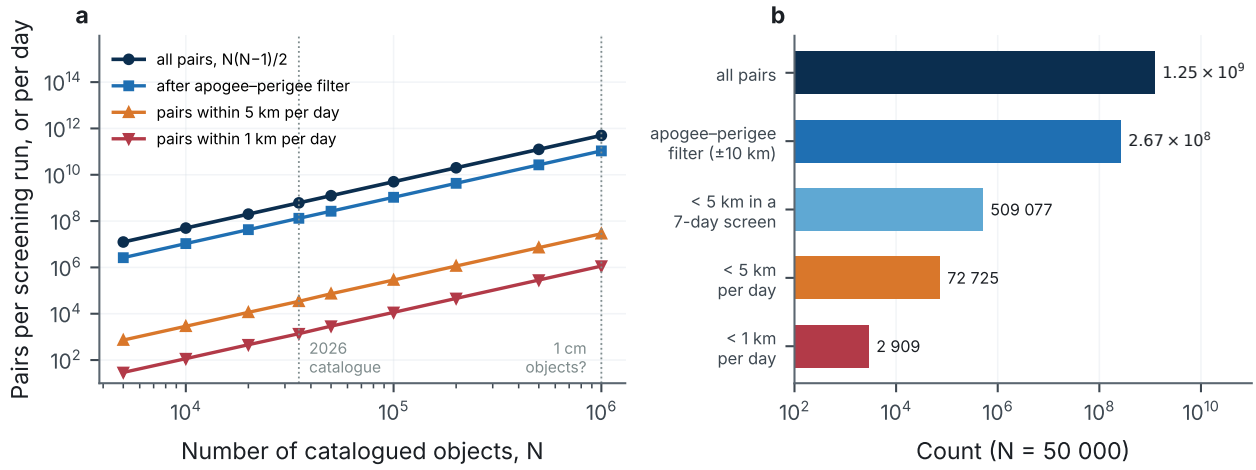
## 50.2 Conjunction Assessment and Space Traffic

The key idea of this section is a funnel. Billions of possible pairs shrink to thousands of close approaches a day and then to a handful of decisions. Computing handles the first steps easily. The hard part is deciding which few events deserve a manoeuvre.

### 50.2.1 Screening at Scale

A catalogue of $N$ objects has $N(N-1)/2$ pairs. Screening programs do not test every pair at every moment. They first apply cheap **filters**. The **apogee–perigee filter** drops pairs whose altitude ranges do not overlap. Orbit-path and time filters then drop pairs whose orbits never come close, or whose closest points are never reached at the same time.

Figure 50.6 shows the result for a synthetic catalogue shaped like today's. The apogee–perigee filter keeps about 21 % of pairs, because most objects crowd into LEO. The number of real close approaches is far smaller. A kinetic-gas estimate — treating objects as molecules in shells of the atmosphere — gives about 1 400 pairs within 1 km per day for 35 000 objects. Pairs inside one constellation are left out, because their orbits are designed to keep apart. Every count grows as $N^2$. Doubling the catalogue, for example by tracking smaller debris, quadruples the work.




Synthetic catalogue shaped like the 2026 public catalogue (78 % near-circular LEO with dense shells at 480–560 km, eccentric debris, MEO, GTO/HEO and GEO). Close approaches: kinetic-gas estimate with objects spread over their radial range in 5 km shells and a mean relative speed of 10 km/s.

**FIGURE 50.6** Conjunction-screening workload. (a) Pairs per screening run and close approaches per day against catalogue size, for a synthetic catalogue shaped like the 2026 public one. (b) The screening funnel for 50 000 objects.

**EXAMPLE: DAILY SCREENING WORKLOAD FOR A 50 000-OBJECT CATALOGUE**

The numbers come from the synthetic catalogue of Figure 50.6.

1. **Pairs.** $N(N-1)/2 = 50\,000 \times 49\,999/2 = 1.25 \times 10^9$ pairs.
2. **Filter.** The apogee–perigee filter (±10 km) keeps 21.3 %, or $2.67 \times 10^8$ pairs.
3. **Brute-force check.** Testing each surviving pair every minute for seven days takes $2.67 \times 10^8 \times 10\,080 = 2.7 \times 10^{12}$ distance checks. At $10^9$ checks per second, that is 45 minutes on one processor core, or 27 seconds on 100 cores. Computing is not the bottleneck.
4. **Close approaches.** The kinetic-gas estimate gives about 72 700 pairs within 5 km and 2 900 within 1 km each day — about a million a year within 1 km.
5. **People.** Suppose 1 % of the 1 km events needed an analyst for two hours. That would be 29 reviews and 58 hours a day — about 12 full-time analysts at 1 800 working hours a year.

This is why triage — choosing which events deserve people — matters more than raw computing speed.

### 50.2.2 Triage: Which Events Matter?

For each close approach, a **conjunction data message** (CDM) gives the predicted miss and the uncertainty of both objects (Chapter 21). The **probability of collision** $P_c$ is the chance that the true miss is smaller than the combined hard-body radius $R$. For a round uncertainty of standard deviation $\sigma$ in the encounter plane and $R \ll \sigma$,

$$P_c \approx \frac{R^2}{2\sigma^2} \exp\!\left(-\frac{d^2}{2\sigma^2}\right),$$

where $d$ is the predicted miss distance. For a fixed miss distance, $P_c$ peaks at $R^2/(e\,d^2)$ when $\sigma = d/\sqrt{2}$. With larger uncertainty, $P_c$ falls again. This is **dilution**: a poorly tracked object looks safe because nobody knows where it is.

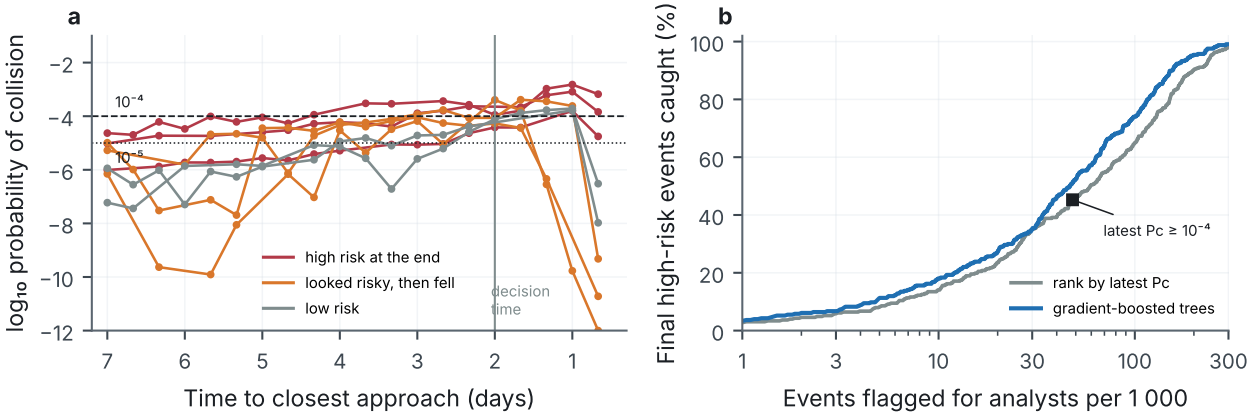
Each event produces a series of CDMs, typically one to three a day for about a week. The uncertainty shrinks as tracking accumulates, so $P_c$ jumps around (Figure 50.7). Operators usually decide about two days before the closest approach, the last moment for a cheap manoeuvre. The question for learning is whether the history of an event's CDMs predicts its final risk better than the latest value alone.

**THE KELVINS COLLISION-AVOIDANCE CHALLENGE (2019)**

In 2019 ESA ran a two-month machine-learning competition on its Kelvins platform. ESA's Space Debris Office released a database of 15 321 conjunction events and 199 082 CDMs, collected in 2015–2019 for its satellites. The data became public in October 2019. Each CDM had 103 attributes. Only 30 events had a final risk of $10^{-4}$ or more. Teams had to predict the final risk from CDMs available two days before closest approach.

Ninety-six teams made 862 submissions. The main result surprised many: the **naive forecast** — the latest known risk — was very hard to beat. Later tests with gradient-boosted trees beat it by about 1.3 % in the F2 score, a small but statistically clear gain. The organisers concluded that naive forecasts are an "unavoidable benchmark" for any future method (Uriot et al., 2022).

Figure 50.7 repeats the experiment on 40 000 synthetic events. Each has a true miss within 3 km, a hard-body radius of 15 m, shrinking and partly wrong covariances, and missing CDMs. An event counts as high risk if its final $P_c$ is at least $10^{-5}$. Ranking events by their latest $P_c$ at two days and flagging the top 40 per 1 000 caught 39 % of the high-risk ones. Gradient-boosted trees using the whole CDM history caught 46 %. The common rule "latest $P_c \geq 10^{-4}$" flagged 48 events per 1 000 and caught 45 %; the trees caught 51 % with the same number of alerts. The gain is real but modest, as in the Kelvins challenge. No method can recover information that the tracking data do not contain.




40000 synthetic events: true miss uniform within 3 km, hard-body radius 15 m, reported σ shrinking from km to tens of metres, covariance realism error 30 %, correlated estimation errors, 15 % of CDMs missing. High risk: final Pc ≥ $10^{-5}$. Trees trained on half, tested on the other half using CDMs up to 2 days before closest approach.

**FIGURE 50.7** Conjunction triage on synthetic CDM sequences. (a) Probability of collision against time to closest approach for example events. (b) Share of final high-risk events caught against the number of events flagged per 1 000, for ranking by the latest Pc and for gradient-boosted trees.

The last resort, when warning comes too late to manoeuvre, is to protect the people. On 24 March 2012 the six crew members of the International Space Station took shelter in their two Soyuz spacecraft as a debris fragment passed close by (Figure 50.8).

**FIGURE 50.8** Expedition 30 crew members in the ISS Zvezda module on 24 March 2012, preparing to shelter in their Soyuz spacecraft because a piece of debris could pass close to the station. Image: NASA/JSC (Andre Kuipers).

**EXAMPLE: PROBABILITY-OF-COLLISION THRESHOLDS AND MANOEUVRE NUMBERS**

A fleet of 1 000 satellites each has 300 close approaches a year with a miss distance inside $D$ = 5 km. Assume the true miss is spread evenly over that disc, the hard-body radius is $R$ = 5 m and the final uncertainty is $\sigma$ = 100 m.

1. **Risk without avoidance.** Each event has a collision chance of $R^2/D^2 = 10^{-6}$. The fleet expects $1\,000 \times 300 \times 10^{-6} = 0.3$ collisions a year.
2. **Trigger distance.** Setting $P_c$ equal to the threshold $P_{\text{th}}$ gives $d_{\text{th}} = \sigma\sqrt{2\ln(R^2/(2\sigma^2 P_{\text{th}}))}$. For $10^{-4}$ this is 225 m.
3. **Manoeuvres.** The share of events inside $d_{\text{th}}$ is $(d_{\text{th}}/D)^2 = 2.0 \times 10^{-3}$. That is 0.61 manoeuvres per satellite per year, or 606 for the fleet.
4. **Residual risk.** The risk left in events below the threshold is $2\sigma^2 P_{\text{th}}/R^2$ = 8 % of the total. A threshold of $10^{-4}$ thus removes 92 % of the risk; $10^{-5}$ removes 99.2 % for 1.16 manoeuvres per satellite per year.
5. **Poor tracking.** With $\sigma$ = 500 m the largest possible $P_c$ is $5 \times 10^{-5}$. A $10^{-4}$ threshold then never triggers, although the true risk is unchanged. A threshold of $3 \times 10^{-7}$ removes 99.4 % of the risk but needs 31 manoeuvres per satellite per year.

Better tracking reduces manoeuvres more than any threshold rule can (Table 50.2).

**TABLE 50.2** Manoeuvre numbers and risk removed for different collision-probability thresholds, from the worked example (300 events a year within 5 km, hard-body radius 5 m).

| Threshold $P_{\text{th}}$ | Trigger miss, σ = 100 m | Manoeuvres per satellite-year | Risk removed | Manoeuvres per satellite-year, σ = 500 m | Risk removed, σ = 500 m |
|---|---|---|---|---|---|
| $10^{-3}$ | 67 m | 0.05 | 20 % | never triggers | 0 % |
| $10^{-4}$ | 225 m | 0.61 | 92 % | never triggers | 0 % |
| $10^{-5}$ | 311 m | 1.16 | 99.2 % | 9.7 | 80 % |
| $3 \times 10^{-7}$ | 408 m | 2.0 | 99.98 % | 30.7 | 99.4 % |

## 50.2.3 Automated Decisions and Coordination

At the largest scale, decisions are already automatic. Starlink satellites receive warnings and plan and execute their own avoidance manoeuvres. They made more than 200 000 such manoeuvres in the six months to May 2026, at a threshold of $3 \times 10^{-7}$ (Chapter 48). This is rule-based autonomy rather than learning, and it works because the operator controls both the rules and the satellites. The weak point is data. In the few cases where a manoeuvre did not reduce the risk as planned, the cause was mostly a poor or late prediction of the other object.

Coordination between operators is the next step. ESA's **CREAM** project (Collision Risk Estimation and Automated Mitigation), begun in 2020, aims to cut operator workload, false alerts and response time. In August 2025 its ground prototype, built by GMV and Guardtime, could issue alerts, propose manoeuvres and support coordination. ESA was preparing a pilot phase and in-orbit demonstrations, first as a software payload on other satellites. Table 50.3 lists the main systems and their status.

In the USA, civil conjunction services are moving from the military to the Office of Space Commerce's **TraCSS**. In July 2026 it was still a pilot, with 62 users covering more than 11 000 satellites. Congress funded the office at $52.5 million for 2026, but the 2027 budget request did not

mention TraCSS. Europe's EU SST protected more than 700 satellites in 2026. It reported 15 639 close approaches in 2023, of which about 1 000 were high risk.

**TABLE 50.3** Selected conjunction, coordination and behaviour-analysis systems. Sources: operators' and agencies' public statements.

| System | Operator | What it does | Status (October 2026) |
|---|---|---|---|
| ATLAS and Space-Track | US Space Force | Catalogue processing; public data and CDMs | Operational (ATLAS from Sept. 2025) |
| TraCSS | US Office of Space Commerce | Civil conjunction service | Pilot: 62 users, >11 000 satellites (July 2026); funding contested |
| EU SST | EU Space Surveillance and Tracking | Collision avoidance, re-entry, fragmentation services | Operational; >700 satellites protected |
| CREAM | ESA with GMV and Guardtime | Automated risk assessment, manoeuvre design, coordination | Ground prototype; pilots and in-orbit demonstration in preparation |
| Starlink onboard avoidance | SpaceX | Autonomous screening response and manoeuvres | Operational |
| LeoLabs services | LeoLabs | Radar catalogue; ML manoeuvre, break-up and behaviour detection | Commercial (vendor-reported) |
| Agatha | Slingshot Aerospace for DARPA | Flags outlier satellites in constellations | Research and development (2023–24) |

## 50.3 Watching Behaviour: Manoeuvres, Characterisation and Re-entry

The key idea of this section is that a catalogue must know not only where objects are but how they behave. Has a satellite moved? Is it tumbling? Will it re-enter soon? Learning is most useful here, because behaviour leaves subtle patterns in long data histories.

### 50.3.1 Manoeuvre Detection

A manoeuvre is a sudden change in velocity. A burn of 1 cm/s along the track of a LEO satellite raises its semi-major axis by about 18 m. That is less than the noise in a single orbit estimate, and drag changes keep blurring the picture. Yet missed manoeuvres corrupt predictions, and unexplained ones may signal intent.

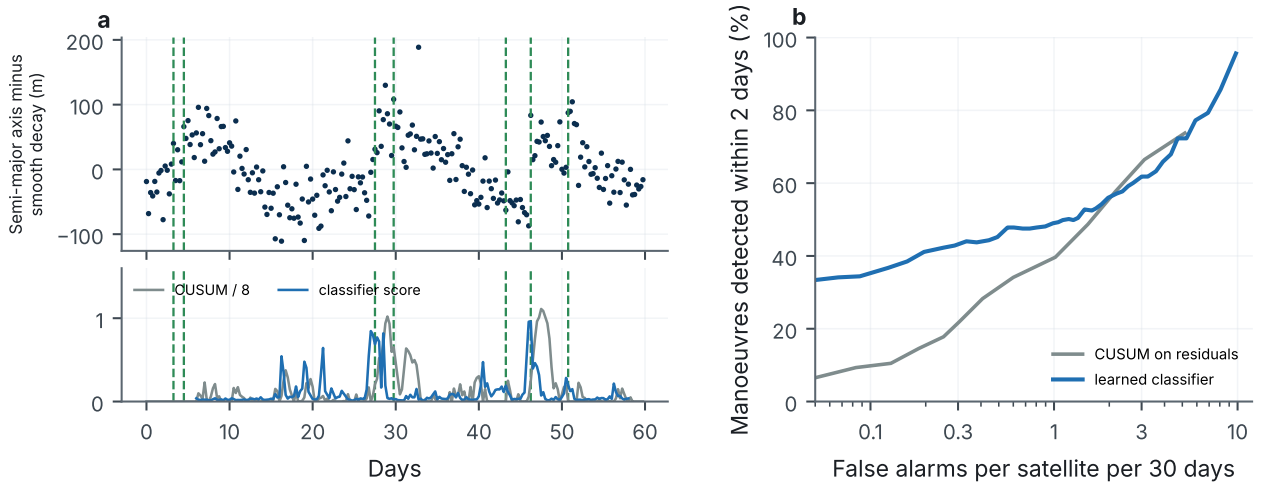
The classical tool is the **cumulative sum** (CUSUM) test. A short-term model predicts the next estimate. The standardised residual $r_k$ feeds two running sums,

$$S_k^{\pm} = \max\bigl(0,\; S_{k-1}^{\pm} \pm r_k - \kappa\bigr),$$

which raise an alarm when either exceeds a threshold $h$. The allowance $\kappa$ absorbs normal noise. A step change pushes one sum steadily upward, so CUSUM detects small persistent shifts that a single-point test would miss.

A learned classifier instead looks at a whole window of data around each moment. It learns what a burn looks like compared with a storm or an outlier. Figure 50.9 compares the two on 240 synthetic satellites. Each has 60 days of orbit estimates every six hours, drag that rises in storms

and burns of 15–150 m. At one false alarm per satellite per month, a tuned CUSUM found 39 % of burns within two days and gradient-boosted trees found 49 %. At 0.3 false alarms the gap widened to 22 % against 43 %. Both found more than 80 % of the largest burns (100–150 m). Burns below 50 m — two noise widths — stayed hard: CUSUM found 19 % of them and the trees 33 %.




240 synthetic LEO satellites, 60 days of semi-major-axis estimates every 6 h (noise 25 m, 2 % outliers ×5), drag decay 15–60 m/day rising in storms, burns of 15–150 m (about 1–8 cm/s; green dashed lines). Classifier: gradient-boosted trees on 8-day windows (6 days before, 2 after), trained on 120 satellites, tested on the other 120.

**FIGURE 50.9** Manoeuvre detection in noisy orbit estimates. (a) One synthetic satellite: semi-major axis minus its smooth decay, with burns (green dashed lines), the CUSUM statistic and the classifier score. (b) Share of burns detected within two days against false alarms per satellite per month.

Commercial providers now sell such detection. LeoLabs says it uses machine learning to detect manoeuvres in real time, score unusual “patterns of life” across its catalogue and spot new objects from deployments or break-ups. It also reports that learned radar scheduling lets it track objects 40 % more often. These are vendor claims without independent evaluation. Slingshot Aerospace’s **Agatha**, built for the US Defense Advanced Research Projects Agency (DARPA) from 2023, takes a different route. It was trained on simulated constellations and uses **inverse reinforcement learning** — inferring goals from observed behaviour — to flag satellites that act unlike their neighbours.

Behaviour analysis matters most in geostationary orbit, where inspector satellites manoeuvre near others. In mid-2025 China’s Shijian-21 and Shijian-25 appear to have docked in GEO, probably to test refuelling, and separated in late 2025. China did not confirm the operation. Commercial trackers such as COMSPOC and S2a followed it from the ground, and the US Space Force reportedly watched with its GSSAP inspector satellites.

**PRACTICE: BUILDING A MANOEUVRE-DETECTION PIPELINE**

1. **Define the need.** Decide which satellites to watch, the smallest burn that matters (for example 2 cm/s) and how quickly it must be found. Agree how many false alarms per day the analysts can handle.
2. **Gather data.** Collect orbit histories — two-line elements, ephemerides or your own orbit fits — with their uncertainties. Add space-weather indices, because storms mimic burns.
3. **Get labels.** Use operators' published manoeuvre histories, such as constellation ephemerides, or ask cooperating operators. Mark storms, deployments and orbit-raising phases separately.
4. **Build a baseline.** Fit a smooth decay model and run a CUSUM test on the residuals. Tune its window and allowance on training satellites only.
5. **Add a learned model.** Train a gradient-boosted classifier on windows of residuals and space-weather features. Split by satellite and by time, never at random, to avoid leakage (Chapter 43).
6. **Score like an operator.** Report the share of burns found within the required time at your false-alarm budget. Report results for small burns and storm periods separately.
7. **Estimate each burn.** For every alarm, fit the size, direction and time of the burn and its uncertainty. A flag without an estimate is of little use.
8. **Run in shadow mode.** Run beside existing tools for months. Review every alarm and missed burn with analysts.
9. **Monitor and retrain.** Track precision monthly. Retrain after new satellite types, new sensors or a change in solar activity.
10. **Share what you find.** Publish detected manoeuvres in a standard format, with their uncertainty, so that screening services and other operators can use them at once.

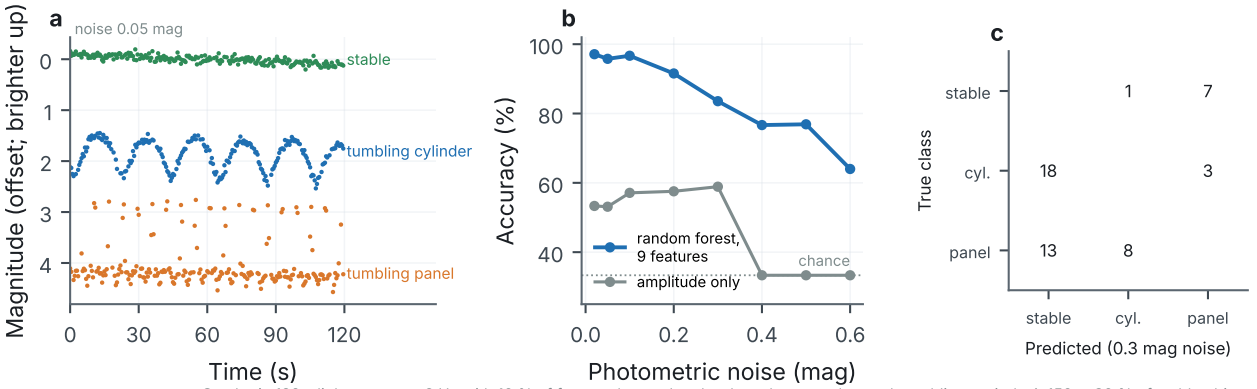
The same steps work for break-up detection and for change detection in imagery.

### 50.3.2 Characterising Objects from Light Curves

A **light curve** is an object's brightness over time. A stable, pointed satellite shines steadily. A tumbling rocket body brightens twice per turn as its long side faces the observer. A tumbling panel flashes when its flat surface briefly acts as a mirror. Period, amplitude and shape thus reveal what an object is and how it moves.

A good starting point is a set of physical **features**. The **Lomb–Scargle periodogram** finds the strongest period in unevenly sampled data. Skewness and the share of bright outliers measure flashes. The autocorrelation at the best period tests whether the pattern repeats. A classifier then learns from labelled examples.

Figure 50.10 tests this on synthetic two-minute light curves of three classes. A random forest on nine features was about 96–97 % accurate up to 0.1 magnitude of noise and 84 % at 0.3 magnitude. The most useful features were periodogram power, skewness and autocorrelation. A rule using amplitude alone managed only 53–59 %. Real studies report similar trends. Classifiers trained on light curves from Russia's Mini-MegaTORTORA wide-field camera reach about 75 % for three broad classes and 85–92 % for some finer tasks. Results differ with the class definitions, so the 2024 RoBo6 data set was released as a common benchmark.




Synthetic 120 s light curves at 2 Hz with 10 % of frames dropped and a slow phase-angle trend; tumbling periods 4–150 s; 30 % of stable objects show one glint. Features: spread, amplitude, Lomb–Scargle peak power and period, skewness, kurtosis, flash fraction, autocorrelation at the period. Trained on 2 400 curves (noise 0.02–0.6 mag); 450 test curves per noise level.

**FIGURE 50.10** Light-curve classification. (a) Synthetic light curves of a stable satellite, a tumbling cylinder and a tumbling panel. (b) Accuracy against photometric noise for a random forest on nine features and for a rule using amplitude alone. (c) Confusion matrix of the forest at 0.3 magnitude noise.

Radar adds a second view. **Radar cross-section** — how strongly an object reflects radio waves — varies with size, shape and attitude. Imaging radars such as Germany's TIRA can make pictures of satellites, which networks can help interpret. Characterisation feeds association, because two objects with different light curves are unlikely to be the same. It also feeds risk, because a tumbling derelict cannot manoeuvre.

### 50.3.3 Re-entry Prediction

About 2 000 catalogued objects re-enter each year (Chapter 21). Predicting when an uncontrolled object will re-enter is hard because the final days are dominated by drag at low altitude, where density is least certain. A common rule of thumb is an uncertainty of about 10–20 % of the remaining lifetime. A prediction made a day ahead is therefore uncertain by a few hours. That is more than a full orbit, or tens of thousands of kilometres along the track.

Learning has been tried here too. Salmaso, Trisolini and Colombo (2023) trained a sequence-to-sequence network on the altitude histories of more than 400 re-entered objects, adding the drag term, solar activity and area-to-mass ratio as inputs. It worked best for objects similar to its training set, a familiar limit of learned models. Controlled re-entries avoid the problem, as for ESA's ATV-3 in 2012 (Figure 50.11).

**FIGURE 50.11** The destructive re-entry of ESA's Automated Transfer Vehicle-3 (Edoardo Amaldi) over the South Pacific on 3 October 2012, photographed from the International Space Station. The re-entry was controlled, so the time and place were known in advance. Image: NASA/JSC.

## 50.4 AI in Defence and Intelligence Space Systems

The key idea of this section is that defence uses of AI in space are mostly about **attention**. Sensors now collect far more than people can look at. Machines filter, flag and cue; people check, judge and decide. This section uses open sources only. It describes what governments and companies have said publicly, without operational detail.

### 50.4.1 Imagery Exploitation at National Scale

Satellite imagery has shaped wars since the first reconnaissance satellites (Chapter 19). In 1991 Space Shuttle crews photographed the Kuwaiti oil fires set during the Gulf War (Figure 50.12). Today hundreds of commercial and government imaging satellites revisit most places several times a day. No analyst workforce could inspect all those images, so automated **object detection** and **change detection** (Chapter 46) now do the first pass.

**FIGURE 50.12** Smoke from burning oil wells north of Kuwait City, photographed from the Space Shuttle in April 1991 after the Gulf War. Space imagery has been part of conflict monitoring ever since. Image: NASA/JSC.

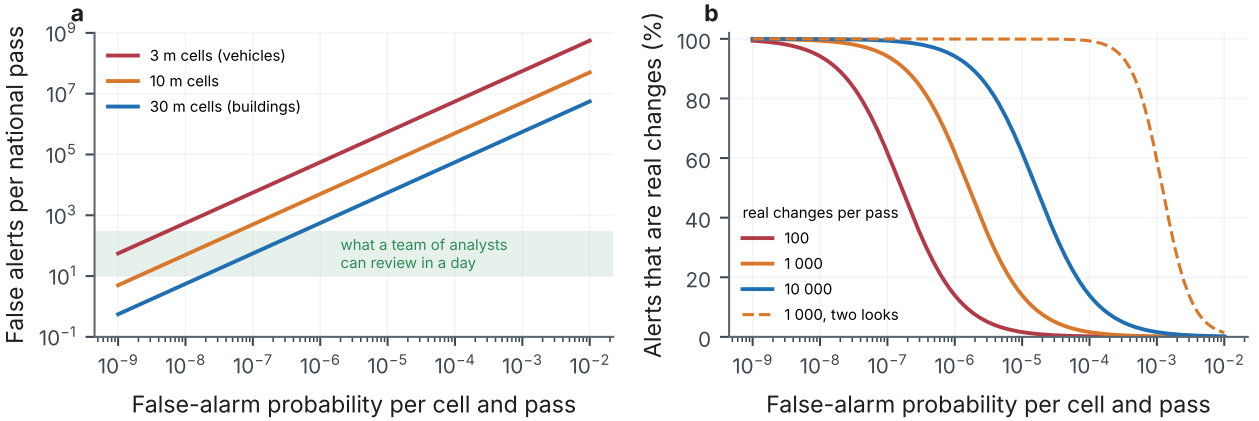
**FROM PROJECT MAVEN TO NGA MAVEN**

- **2017.** The US Department of Defense set up an Algorithmic Warfare team, known as Project Maven, to apply computer vision to drone video and other imagery. Its first algorithms were in use by December 2017.
- **2022.** Parts of Maven moved to the National Geospatial-Intelligence Agency (NGA), which handles imagery and maps.
- **November 2023.** NGA made NGA Maven a formal programme of record.
- **May 2025.** NGA's director reported more than 20 000 active users across more than 35 military tools, available to all services and combatant commands. Targeting timelines in one exercise fell "from hours to minutes". The ceiling of the related Maven Smart System contract rose to about $1.3 billion through 2029.
- **June 2025.** NGA began labelling intelligence products that are "100 percent machine-generated", with no human hand in the template or its release — the first of the 18 US intelligence agencies to do so routinely.

Sources: Breaking Defense and DefenseScoop reporting of NGA statements, 2025.

NGA also buys analytics from industry. Its Luno A contract (September 2024, up to $290 million, ten companies) buys computer vision and analytic services based on commercial imagery. Luno B (January 2025, up to $200 million, 13 companies) has funded orders such as change and movement detection and maritime analytics.

At national scale the main enemy is the false alarm. Clouds, shadows, seasons and small misregistrations all look like change. Figure 50.13 shows the arithmetic. A country of 500 000 km² divided into 30 m cells has $5.6 \times 10^8$ cells per look. Even a tiny per-cell false-alarm rate produces more alerts than any team can read.




Area 500 000 km² imaged once. (b) 30 m cells ($5.6 \times 10^8$ cells) and 90 % detection probability per look. Assumes false alarms independent between cells and looks; real clutter (clouds, shadows, seasons, misregistration) is correlated, so measured rates are needed.

**FIGURE 50.13** False alarms of automated change detection at national scale. (a) False alerts per pass over 500 000 km² against the per-cell false-alarm probability, for three cell sizes, with the range a team can review in a day. (b) Share of alerts that are real changes for 30 m cells, for different numbers of real changes per pass, and with confirmation by a second independent look.

**EXAMPLE: FALSE ALARMS IN AUTOMATED CHANGE DETECTION OVER A COUNTRY**

A detector screens 500 000 km$^2$ in 30 m cells after each pass. It finds 90 % of real changes and has a false-alarm probability of $10^{-4}$ per cell. There are 1 000 real changes per pass.

1. **Cells.** $500\,000\ \text{km}^2/(30\ \text{m})^2 = 5.56 \times 10^8$ cells.
2. **False alerts.** $5.56 \times 10^8 \times 10^{-4} = 55\,556$ per pass.
3. **Real alerts.** $0.9 \times 1\,000 = 900$.
4. **Precision.** Only $900/(900 + 55\,556) = 1.6\ \%$ of alerts are real.
5. **Target.** For half of the alerts to be real, the false-alarm rate must fall to $900/5.56 \times 10^8 = 1.6 \times 10^{-6}$ per cell.
6. **Second look.** Confirming each alert with an independent second image squares the false-alarm rate to $10^{-8}$. False alerts drop to about 6, while detection falls to 81 %. Precision becomes 99.3 %.

Step 6 assumes independent errors. Real clutter repeats from image to image, so the gain is smaller in practice. This is why measured false-alarm rates, not textbook ones, must drive the design.

## 50.4.2 Activity-Based Intelligence and Tip-and-Cue

**Activity-based intelligence** looks for patterns of activity rather than single objects: ships that meet at sea, vehicles that gather before an exercise, a radar that switches on at unusual times. It needs data from many sources over long periods. Machine learning helps to build a picture of normal behaviour and flag departures from it.

**Tip-and-cue** turns such flags into new collection (Figure 50.14). A wide-area sensor, such as a radar satellite or a radio-frequency receiver, notices something. Software then tasks a high-resolution satellite or another sensor to look again. The US National Reconnaissance Office (NRO) has described its **Sentient** research programme as an automated tipping-and-cueing system. In 2025 it told Congress that its new constellations and ground systems will "fuse data across collection platforms" to automate tipping and cueing. By mid-2026 the NRO had made 14 launches of its proliferated LEO constellation and said it had hundreds of satellites in orbit.

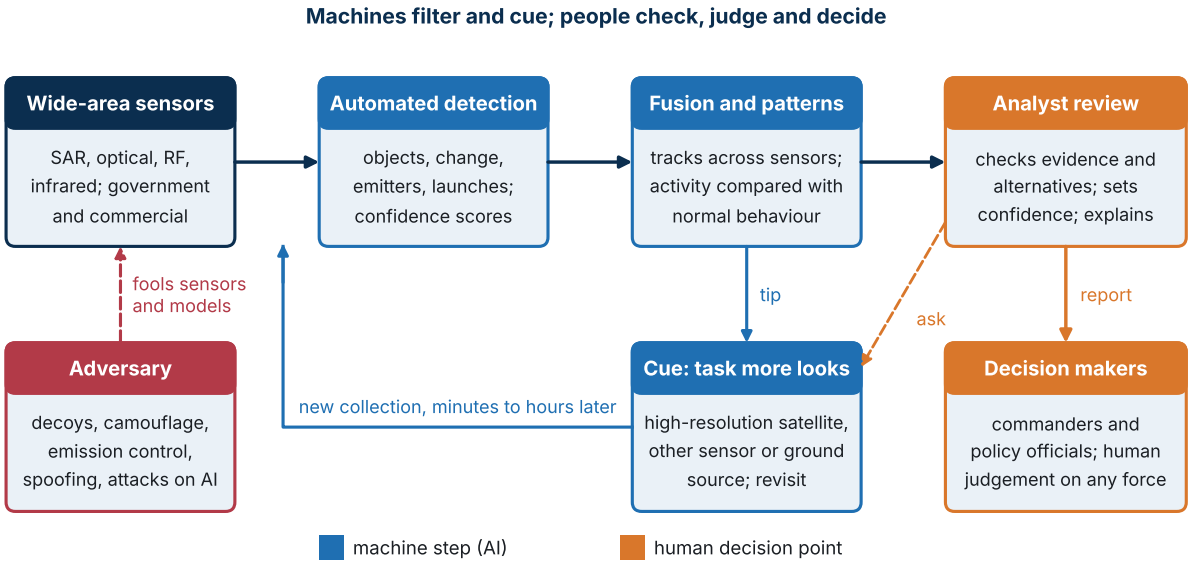



**FIGURE 50.14** Tip-and-cue across constellations. Machine steps (blue) detect, fuse and cue new collection; human decision points (orange) review evidence and make decisions. An adversary (red) tries to deceive both sensors and models.

Radar satellites are central to maritime monitoring because they see through cloud and darkness. In an early example from 1994, the Space Shuttle's imaging radar recorded a ship and its 28 km wake in the Atlantic (Figure 50.15).

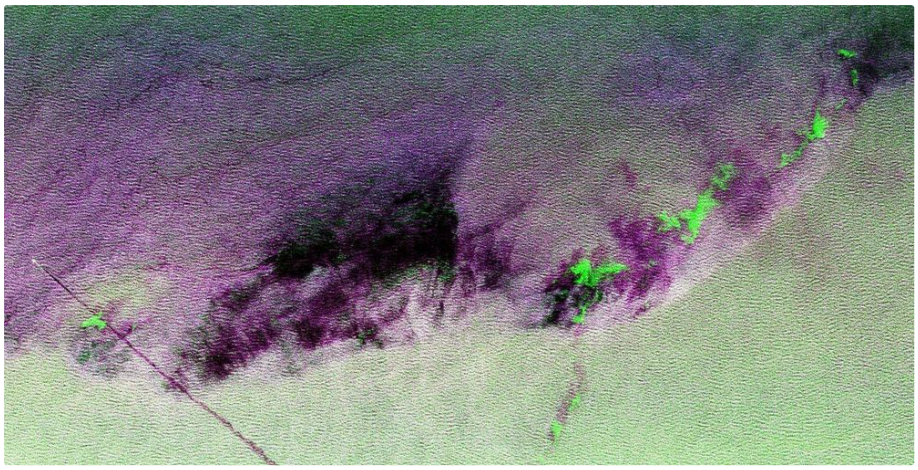


**FIGURE 50.15** A Spaceborne Imaging Radar-C/X-SAR image of the north-east Atlantic taken from Space Shuttle Endeavour on 11 April 1994. The bright spot at left is a ship and the line at lower left its wake, about 28 km long. Investigators thought that discharged oil made the wake last longer. The area shown is about 35 km × 65 km. Image: NASA/JPL (NASA/JPL).

### 50.4.3 Signals and Emitters

Radio receivers in orbit locate and classify emitters, such as radars, satellite phones and ship transponders (Chapter 19). Learning helps in two ways. **Signal classification** recognises the type of signal from its modulation (Chapter 49). **Specific emitter identification** goes further: it tells apart individual transmitters of the same type from tiny hardware imperfections, such as oscillator drift and amplifier distortion — a radio fingerprint.

Open research shows what is possible. In 2023 researchers at Oxford fingerprinted Iridium satellite transmitters from about 1.7 million recorded messages, using an autoencoder and a twin ("Siamese") network. Against messages replayed with a software-defined radio — a spoofing attack — the system had an equal error rate of 12 %. Commercial operators such as HawkEye 360 use clusters of satellites to geolocate signals and flag ships that switch off their identification beacons.

### 50.4.4 Missile Warning and Tracking

Missile-warning satellites stare at the Earth in the infrared and must find a rocket plume among clouds, sun glints and fires (Chapter 20). Classical processing subtracts the background, applies thresholds and confirms detections by tracking them over several frames. Learning is being studied for clutter suppression, event classification and **track fusion** — combining detections from several satellites and radars into one three-dimensional track. In September 2025 the US Space Force accepted the second delivery of its new FORGE missile-warning ground system for operations. Public material says little about the use of learning in operational chains. Errors here could have grave consequences, so changes demand long testing.

### 50.4.5 Battle Management and Golden Dome

Missile defence must turn tracks into decisions in minutes. **Battle management, command and control** (BMC2) software links sensors to interceptors and proposes which interceptor should engage which threat. The US **Golden Dome** programme, launched by executive order in January 2025, aims at a layered homeland defence that includes space sensors and possibly space-based interceptors.

Public facts are limited and changing. In April 2026 the Space Force gave 12 companies agreements worth up to $3.2 billion for space-based interceptor prototypes. Golden Dome's director, General Michael Guetlein, calls the command-and-control layer the programme's "glue layer". An industry consortium of nine firms ran a live demonstration of part of it in early 2026, and initial operational capability is planned for 2028. He put the cost at $185 billion through 2035. A Congressional Budget Office estimate in May 2026, for an illustrative architecture, put it at about $1.2 trillion over 20 years. In September 2026 Guetlein said that the military must make decisions "at machine speed". Senior leaders described a role for AI in helping human operators handle the flood of sensor data. How far humans stay in each decision has not been described in public detail.

### 50.4.6 Autonomy for Rendezvous and Proximity Operations

Inspector satellites, servicers and possible weapons all fly close to other spacecraft. Their guidance and navigation methods are covered in Chapter 48. The defence angle is speed and secrecy. In June 2026 the Space Force's **Victus Haze** exercise launched a Rocket Lab target spacecraft 16 hours and 42 minutes after notice. True Anomaly's Jackal, already in orbit, then approached it and imaged it from several sides. Its mission software planned the sortie and commanded the manoeuvres, and it finished in 61 hours of a 72-hour window. Such missions rely partly on onboard autonomy, because at close range there is little time to wait for commands from the ground. They also stress SSA, because observers must decide quickly whether an approach is inspection, servicing or a threat (Table 50.4).

**TABLE 50.4** Defence and intelligence uses of AI in space systems, from open sources.

| Application | Example (public) | What the machine does | Status (October 2026) |
|---|---|---|---|
| Imagery exploitation | NGA Maven; NGA Luno contracts | Detects objects and change; drafts reports | Operational at scale |
| Tip-and-cue | NRO Sentient; proliferated constellation | Flags events; tasks other sensors | Described as operational in part; details classified |
| Emitter identification | SatIQ research; commercial RF constellations | Recognises signal types and individual transmitters | Research; commercial geolocation operational |
| Missile-warning processing | FORGE; SDA Tracking Layer | Rejects clutter; builds and fuses tracks | Operational processing; learning mostly research |
| Battle management | Golden Dome C2 consortium | Pairs threats with interceptors; presents options | In development; demonstrations 2026 |
| Proximity operations | Victus Haze; GSSAP; Shijian-21/25 | Guides approach; classifies others' approaches | Demonstrations and operations |

### 50.4.7 Deception and Counter-AI

Every detector invites an attempt to fool it. Classic **deception** uses decoys, camouflage and emission control. In 2023–24 analysts of commercial imagery found aircraft outlines painted on Russian airfields, including Tu-95 bomber shapes at Engels. The UK Ministry of Defence noted in 2024 that helicopters were still landing on the painted fighter outlines, which undermined the deception. Commercial imagery separated the paintings from real aircraft easily.

AI adds new targets. An **adversarial patch** is a printed pattern designed to make a neural network miss an object. A 2020 study showed that a patch covering about 10 % of an aircraft hid it from a common detector in aerial images, although only digitally. Training data can be poisoned, and models can be probed until they reveal their blind spots. Defences include training on adversarial examples, combining different sensors — a painted shape has no radar or heat signature — and keeping people in the review loop. Chapter 53 covers the security of AI in more depth.

### 50.4.8 Human–Machine Teaming

Automation changes how people think. **Automation bias** is the tendency to accept a machine's suggestion without checking. Its opposite, **disuse**, follows when a system cries wolf too often. Good teaming gives each side what it does best. Machines sift volume and keep track of many objects. People bring context, scepticism and responsibility.

Three practices help. Systems should show their evidence and confidence, not just a verdict. Operators should train with systems that sometimes fail, so that they learn when to doubt them. And the design should state clearly who decides what. NGA's label for machine-generated products is one example of making the machine's role visible. Flight dynamics officers at NASA, who own trajectory decisions for crewed missions, offer another model of human authority supported by software (Figure 50.16).

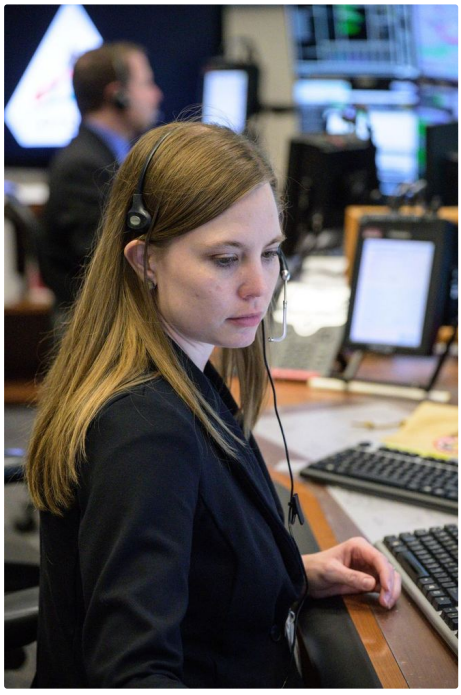


**FIGURE 50.16** Flight controller Julie Reed at the flight dynamics officer console in Mission Control, Houston, on 1 December 2022, as Orion left its distant retrograde orbit around the Moon during Artemis I. Software computes the trajectory options; a named person owns the decision. Image: NASA/JSC (ROBERT MARKOWITZ NASA-JSC).

## 50.5 Law, Ethics and Policy

The key idea of this section is that existing law applies to AI, but states disagree on whether new rules are needed. Space adds special worries about stability, because satellites are fragile and hard to defend.

### 50.5.1 Rules for Autonomy in Weapons

The main US policy is **Department of Defense Directive 3000.09**, "Autonomy in Weapon Systems", updated on 25 January 2023. It requires that systems let commanders and operators exercise "appropriate levels of human judgment over the use of force". Autonomous weapon systems need a senior-level review before formal development and again before fielding. The directive does not ban such weapons, and it does not require a human to approve every individual engagement. In June 2026 a presidential memorandum, NSPM-11, ordered an update of the directive within 90 days and a review every year. No revised text had been published by late October 2026.

### 50.5.2 International Norms

The USA launched the **Political Declaration on Responsible Military Use of Artificial Intelligence and Autonomy** in February 2023 and revised it in November 2023. It is a non-binding set of commitments: legal reviews, human accountability, testing, and safeguards such as the ability to disengage systems that behave unexpectedly. Almost 60 states had endorsed it by late 2024. Other forums run in parallel (Table 50.5).

**TABLE 50.5** Policy and legal instruments on military AI and autonomy.

| Instrument | Year | Type | Main content | Status (October 2026) |
|---|---|---|---|---|
| DoD Directive 3000.09 | 2012; updated 2023 | US policy | Human judgement over force; senior reviews | In force; update ordered June 2026 |
| Political Declaration | 2023 | Non-binding, ~60 states | Responsible military AI practices | Endorsements open |
| NATO AI strategy | 2021; revised 2024 | Alliance policy | Six principles of responsible use; testing | In force |
| REAIM summits | 2023, 2024, 2026 | Multi-stakeholder | Blueprint (Seoul); "Pathways to Action" (A Coruña) | About 35 states endorsed 2026 text (reported) |
| UN General Assembly resolutions | 2023, 2024, 2025 | Non-binding | Concern about lethal autonomous weapons | Resolution 80/57: 164 for, 6 against |
| CCW expert group (GGE) | 2016–2026 | Treaty forum, consensus | Elements of a possible instrument | Consensus report Sept. 2026; Review Conference Nov. 2026 |

Under the Convention on Certain Conventional Weapons (CCW), with 128 states parties, a group of governmental experts set up in 2016 has discussed **lethal autonomous weapon systems**. Its mandate ended in 2026. At its final session (31 August–4 September 2026) it agreed by consensus on elements that could form the basis of an instrument. The Seventh Review Conference, on 16–20 November 2026, will decide whether to open negotiations. The USA and Russia have opposed

new legally binding rules. Many other states and the International Committee of the Red Cross want a treaty. NATO's revised 2024 strategy keeps its six principles: lawfulness, responsibility and accountability, explainability and traceability, reliability, governability and bias mitigation.

### 50.5.3 Stability and Escalation in Space

AI could make space more stable or less. Better SSA helps: knowing quickly that a close approach is an accident, not an attack, reduces the risk of overreaction. Shared data and automated coordination make accidents less likely. But speed cuts both ways. If machines classify a manoeuvre as hostile within seconds, decision makers may feel pressure to respond before checking. Errors in one system can trigger responses in another, as in financial "flash crashes". And satellites that support nuclear command and early warning (Chapter 20) are especially sensitive. Misreading an attack on them could escalate a conflict quickly.

Common proposals include keeping human decision points for any action against a satellite, exchanging data on manoeuvres, testing systems against realistic deception, and agreeing norms against destructive anti-satellite tests (Chapter 19).

### 50.5.4 Export Controls and Dual Use

Almost everything in this chapter is **dual-use**: the same tracker serves a commercial operator and an air force. Space hardware falls under export controls such as the US International Traffic in Arms Regulations and Export Administration Regulations, and the multilateral Wassenaar Arrangement. Advanced AI chips have been controlled since 2022. A US rule of January 2025 that would also have limited exports of AI chips and model weights to most countries was rescinded in May 2025, and no replacement had been issued by mid-2026. Engineers should expect rules on chips, models and SSA data to keep changing and should check them early in any international project.

## 50.6 Lessons from the Experiments

Table 50.6 collects the results of this chapter's experiments. All used synthetic data with fixed random seeds, so they illustrate methods rather than measure real systems.

**TABLE 50.6** Summary of this chapter's synthetic experiments.

| Task and condition | Baseline | Learned or optimised method | Result |
|---|---|---|---|
| Streak detection | Point-source filter | Line matched filter | 1.8× fainter |
| Association, spacing 2σ | Greedy nearest neighbour | Global assignment | 76 % vs 47 % |
| Drag after storms, one day | Fitted drag scale | Plus neural network | 0.24 vs 0.75 km |
| Triage, 40 alerts per 1 000 | Latest Pc | Boosted trees | 46 % vs 39 % |
| Manoeuvres, 1 false alarm a month | CUSUM | Boosted trees | 49 % vs 39 % |
| Light curves, 0.3 mag noise | Amplitude rule | Random forest | 84 % vs 59 % |

Four lessons stand out. Physics and geometry carry most of the load. Learning gives real but moderate gains, mostly where data are noisy or patterns are subtle. Gains must be measured in the conditions that matter, such as storms or low false-alarm rates. And no model can create

information that the sensors did not collect.

### KEY TAKEAWAYS

- About 35 000 objects are catalogued in orbit (October 2026). The number of close approaches grows with the square of that number: 50 000 objects make about $1.25 \times 10^9$ pairs and thousands of approaches within 1 km a day.
- Most of the SSA pipeline is physics and geometry. Learning helps with faint detections, association, prediction errors, density during storms, triage, manoeuvre detection and characterisation — mostly in research or commercial services so far.
- Along-track errors grow with the square of time, so density errors dominate LEO predictions. Hybrid models that learn a correction to physics can help after storms but must be tested regime by regime.
- In conjunction triage the latest probability of collision is a strong baseline, as ESA's 2019 Kelvins challenge showed. Better tracking reduces manoeuvres more than any threshold rule, because poor tracking dilutes the probability.
- In defence, AI's main job is managing attention: NGA Maven had more than 20 000 users in 2025, and tip-and-cue links constellations. At national scale, false alarms set the limit — $10^{-4}$ per cell gives over 55 000 false alerts per pass over 500 000 km².
- Adversaries can deceive both sensors and models with decoys, camouflage and adversarial patterns. Fusing different sensors and keeping people in the review loop are the main defences.
- US policy requires "appropriate levels of human judgment over the use of force". International talks on autonomous weapons reached a consensus text in 2026, with the decision on negotiations due in November 2026.

### REVIEW QUESTIONS

1. A 40-pixel streak has a peak brightness of 0.5σ per pixel and a blur of $\sigma_{\mathrm{psf}}$ = 1.5 px. Estimate its matched-filter signal-to-noise ratio. Why does searching over many angles reduce the real gain?
2. Explain the Mahalanobis distance and the role of a gate. Why does global assignment outperform the greedy rule in Figure 50.4 when objects are close together?
3. Using $\Delta s \approx \frac{3}{4} n \varepsilon |\dot{a}| t^2$, estimate the along-track error after two days for a satellite at 450 km with $|\dot{a}|$ = 50 m/day and a 30 % density error.
4. In Figure 50.5 the neural-network correction helped after storms but not on average. Give two reasons, and say how you would test such a model before using it.
5. For a catalogue of 100 000 objects, how many pairs are there? By what factor does the number of daily close approaches change compared with 50 000 objects?
6. What is dilution in the probability of collision? Using the formula in this chapter, find the largest possible $P_c$ for a miss of 300 m and a hard-body radius of 10 m.
7. What did ESA's Kelvins challenge teach about machine learning for conjunction triage? Why is the latest risk such a strong baseline?
8. Compare CUSUM and a learned classifier for manoeuvre detection. Why do storms cause false alarms, and how can space-weather data reduce them?
9. A change detector screens 200 000 km² in 10 m cells with a false-alarm probability of $10^{-6}$ per cell. How many false alerts per pass does it make? What would confirmation by a second look achieve, and why is the real gain smaller?
10. Describe two ways AI in space systems could make a crisis more stable and two ways it could make it less stable. What human decision points would you keep?

CHAPTER

# 51 AI for Space Science, Planetary Exploration and Human Spaceflight

Image: NASA/JSC

*Space science now produces far more data than people can inspect. A survey telescope sends millions of alerts a night, and orbiters have imaged whole planets at metre scale. A crew on the way to Mars will be twenty minutes from help. This chapter shows how machine learning finds planets in light curves, classifies transients, maps galaxies and lenses, estimates redshifts and speeds up gravitational-wave analysis. It then turns to craters, boulders and minerals on other worlds, to science autonomy on rovers, and to AI for crew health, assistants, robots, smart habitats and Earth-independent operations. Small experiments, run from scratch, show what each method does and where it fails. Status is given as of October 2026.*

IN THIS CHAPTER

51.1 Why Science Needs Machine Learning
51.2 Exoplanets: Finding Small Signals in Big Data
51.3 The Time-Domain Sky
51.4 Galaxies, Lenses and Cosmology
51.5 The Sun: Features and Event Catalogues
51.6 Planetary Surfaces and Science Autonomy
51.7 AI for Human Spaceflight
51.8 Validating an AI Discovery

Science missions are built to collect data. For most of the space age, people could look at all of it. Astronomers inspected every photographic plate, and planetary geologists mapped every image by hand. That is no longer possible. Kepler watched about 150 000 stars every 30 minutes for four

years. The Vera C. Rubin Observatory now sends up to several million alerts each night. Orbiters around the Moon and Mars have returned millions of images sharp enough to show single boulders.

Machine learning (ML) helps in two ways. As a **filter and ranker**, it sorts a flood of data so that people look first at what most likely matters. As a **fast approximation** to slow physics, it gives in a second an answer that a full calculation needs hours to produce. Either way, scientists make the final decision. The hero image shows a third role: a robot that works beside a crew on the International Space Station (ISS).

This chapter covers astronomy, the Sun, planetary science and human spaceflight. Telescopes are described in Chapters 38 and 39, planetary missions in Chapters 24–29 and human spaceflight in Chapters 30–33. Learning methods are explained in Chapters 43 and 44, onboard computing in Chapter 45, and rover autonomy in Chapter 48.

## 51.1 Why Science Needs Machine Learning

The key idea is that data now grow faster than the number of people who can look at them. Learning methods turn huge archives into short lists that people can check.

### 51.1.1 The Data Deluge

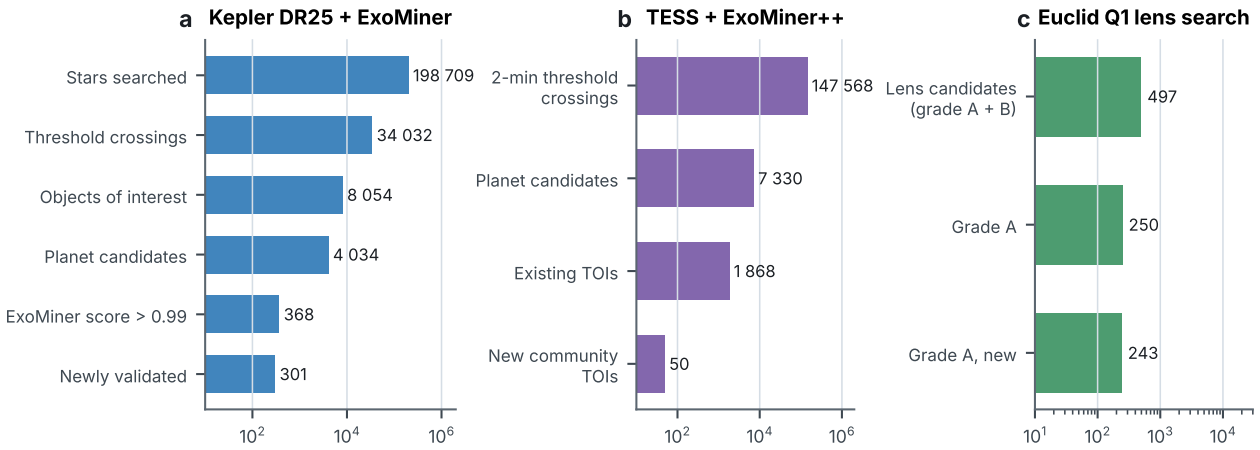
The data rates of some major science facilities are compared in Table 51.1. Light curves call for signal detection, images for object detection and segmentation. Alert streams need decisions within minutes, while archives can be mined for years.

**TABLE 51.1** Data rates of some major science facilities and the learning tasks they drive (status October 2026).

| Facility | Data rate or volume | Main learning tasks |
|---|---|---|
| Kepler (2009–18) | ≈ 150 000 stars every 30 min | Planet search and vetting |
| TESS (2018–) | Most of the sky in 27-day sectors | Vetting thousands of candidates |
| Gaia (2013–25) | > 500 TB for its fourth release (Dec 2026) | Classifying stars and their motions |
| Euclid (2023–) | ≈ 100 GB per day | Shapes, lenses, morphology |
| SDO (2010–) | About 1.5 TB per day | Solar features and event catalogues |
| Rubin LSST (2026–) | ≈ 10 TB per night; up to ~7 million alerts | Real/bogus, transient classes, redshifts |
| LIGO–Virgo–KAGRA | About 250 candidates in 2023–25 | Glitches, rapid parameter estimation |
| SKA (2030s) | ≈ 700 PB of products per year (estimate) | Imaging, radio transients |

### 51.1.2 Funnels, Not Oracles

A discovery usually passes through a **funnel**: a search finds many candidates, filters remove most of them, and people examine the rest. Three real funnels are shown in Figure 51.1. Kepler's final search of 198 709 stars produced 34 032 "threshold-crossing events" (TCEs): signals strong enough to pass the detection limit. Only 4 034 survived as planet candidates. A neural network then scored 1 922 unlabelled candidates, and 301 were validated as new planets.




Sources: Kepler DR25 catalogue (Thompson et al. 2018; 198 709 stars searched, 34 032 TCEs); ExoMiner (Valizadegan et al. 2022, 1 922 unlabelled KOIs scored); ExoMiner++ (Valizadegan et al. 2025); Euclid Q1 strong-lensing discovery engine (Euclid Collaboration 2025/26, 63 deg$^2$).

**FIGURE 51.1** Discovery funnels from published numbers. (a) Kepler's final search and the ExoMiner validation of 301 new planets. (b) The ExoMiner++ catalogue for two-minute TESS data. (c) The Euclid Q1 search for strong gravitational lenses. Each step removes most of the objects left by the step before.

Machine learning sits at the narrow middle of each funnel. It must keep nearly all real objects (high **recall**) while passing few false ones (high **precision**; Chapter 43). A model that ranks well saves years of human effort. A model that hides a class of objects can bias a whole field.

### 51.1.3 From Pixels to Alerts

A typical pipeline for a survey telescope is shown in Figure 51.2. The observatory side is automatic. Images are calibrated and a deep "template" image of the same field is subtracted. Anything that has changed leaves a residual. A first network rejects artefacts — cosmic rays, satellite trails and bad subtractions — that are far more common than real changes. Alerts then go to **brokers**: software services that add information from catalogues, classify each object and pass it to the science teams that asked for it.

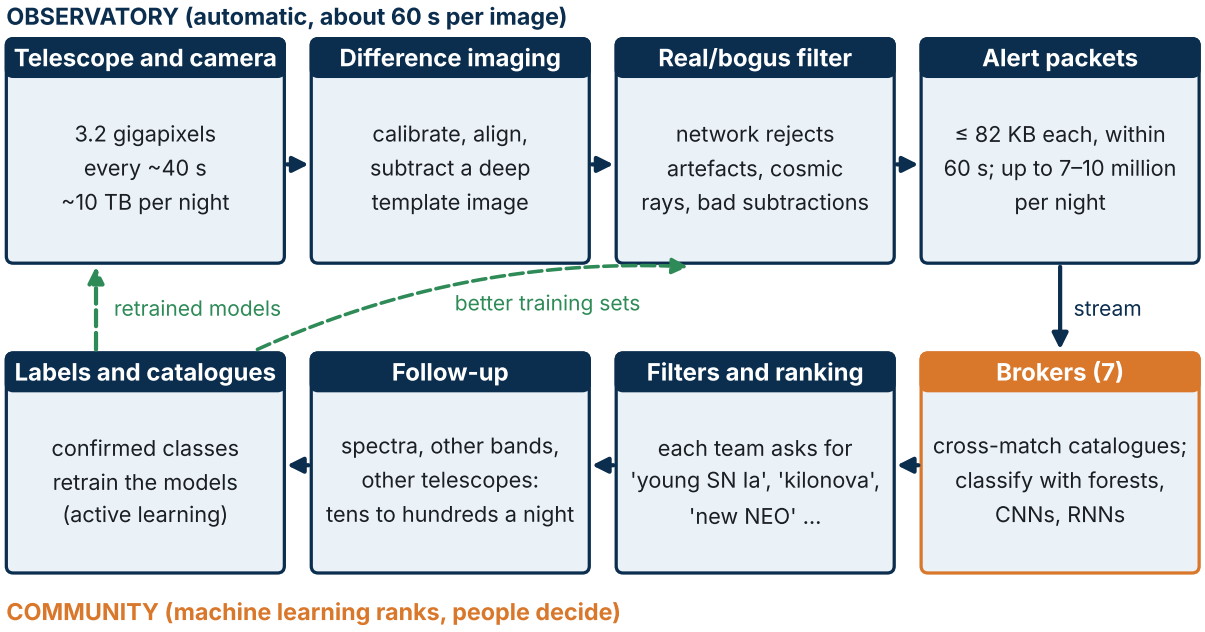



**FIGURE 51.2** A survey alert pipeline, modelled on the Vera C. Rubin Observatory. Learning appears at three points: rejecting artefacts, classifying alerts in the brokers, and ranking targets for follow-up. Confirmed classes return as labels and improve the next models.

The loop at the bottom matters as much as the arrows at the top. Every spectrum taken for follow-up becomes a new label, so the models learn from their own mistakes.

## 51.2 Exoplanets: Finding Small Signals in Big Data

Planet searches find far more false signals than planets. Learning is most useful in **vetting**: deciding which signals are real.

### 51.2.1 Searching Light Curves

When a planet crosses its star, the star dims by the transit depth $\delta = (R_p/R_\star)^2$ (Chapter 39). An Earth-sized planet in front of a Sun-like star gives 84 parts per million (ppm). Noise from the detector, from the star and from the spacecraft is usually larger than that for a single transit. The signal becomes visible only when many transits are added together.

The standard search is **box least squares (BLS)**. It tries every period $P$, every phase and a few transit durations. For each trial it folds the light curve and compares the mean flux inside a box with the mean outside. The result is a signal-to-noise ratio:

$$\mathrm{SNR} = \frac{\hat{\delta}}{\sigma}\sqrt{n_{\mathrm{in}}}.$$

Here $\hat{\delta}$ is the measured depth, $\sigma$ the noise per point and $n_{\mathrm{in}}$ the number of points in transit. A signal with SNR above a threshold, 7.1 for Kepler, becomes a TCE.

A search on simulated data is shown in Figure 51.3. Two years of photometry at 30-minute cadence contain starspots, gaps and white noise of 250 ppm per point. A planet 1.3 times Earth's radius, with a depth of 142 ppm, is hidden in it; one transit has an SNR of only 1.7. The light curve is first **detrended**: a two-day running median removes the starspot signal. The BLS search over 40 000 trial periods then finds a peak at 17.632 days (true value 17.63) with SNR 8.3 from 42 transits. The next-highest unrelated peak reaches 6.4.

The recovered box depth is 130 ppm, a little less than the 142 ppm injected. Limb darkening makes a real transit rounded rather than box-shaped, and detrending removes a little depth. Such losses are why pipelines fit a physical transit model after the search.

### 51.2.2 Vetting with Neural Networks

Most TCEs are not planets. In Kepler's final catalogue, 88 % of them did not become planet candidates. Some are **eclipsing binaries**: pairs of stars that eclipse each other. Their eclipses are often V-shaped, may show a shallow secondary eclipse half an orbit later, and may differ in depth between odd and even events. Others are instrumental artefacts, such as sudden drops in pixel sensitivity, or stellar variability that the search mistook for a dip.

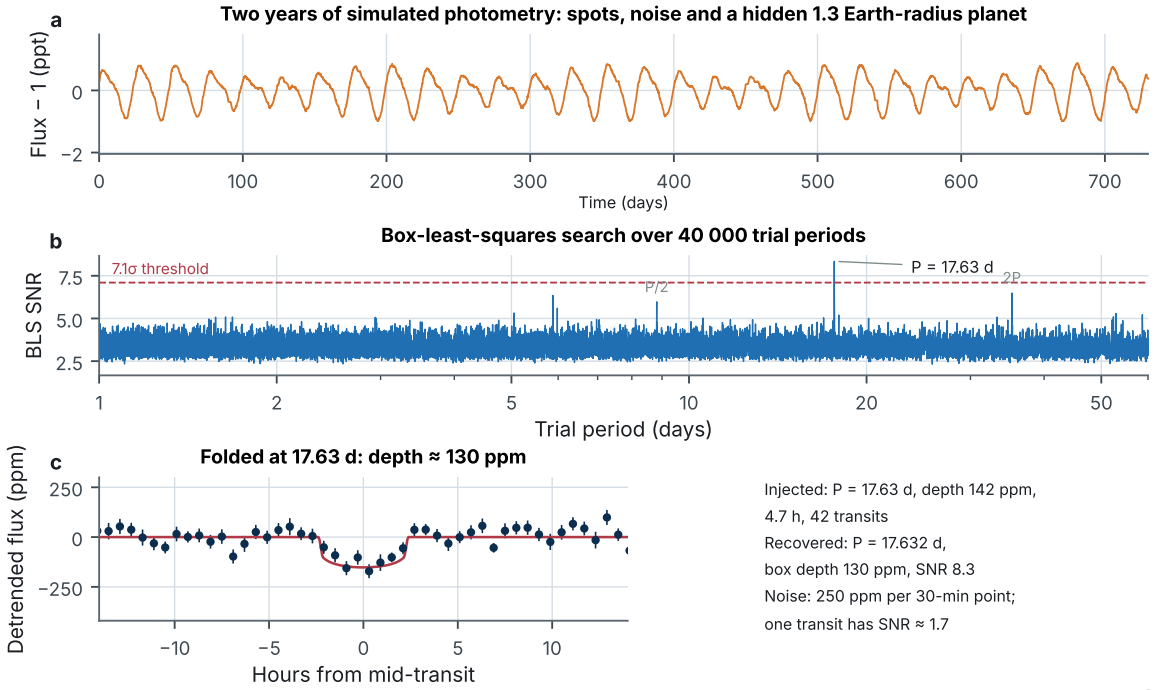



**FIGURE 51.3** A transit search on simulated data. (a) Two years of photometry with starspot modulation; the orange line is the trend removed before the search. (b) Box-least-squares signal-to-noise against trial period; the peak lies at the planet's period, with weaker peaks at half and twice the period. (c) Data folded at the recovered period, binned (dark points) and compared with the injected transit (red).

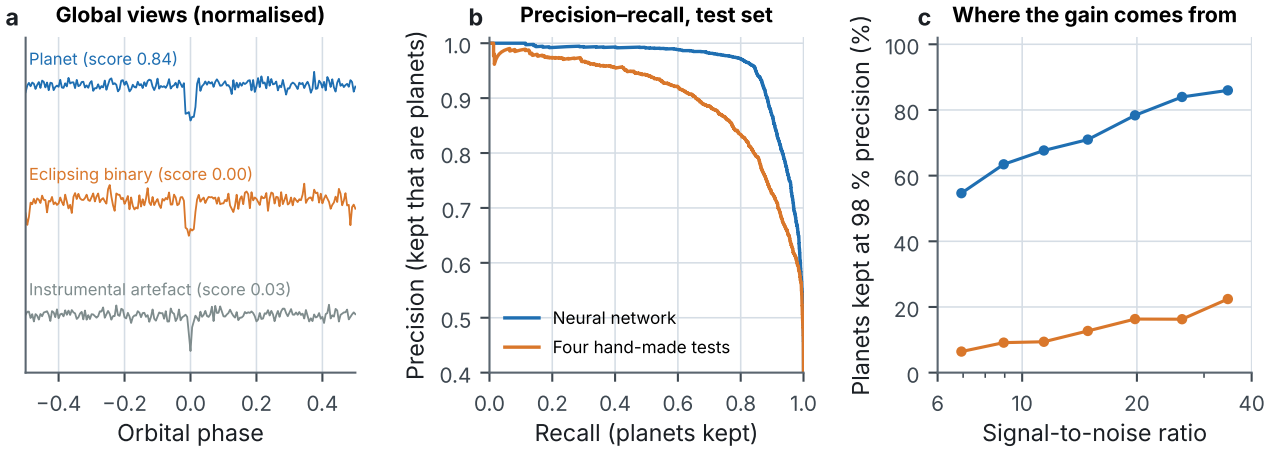
Kepler's team first vetted signals by eye, then with the **Robovetter**, a set of expert rules. Each rule tests one property, such as a secondary eclipse or a shift of the star's image during the dip. In 2017 a Google engineer and an astronomer at the University of Texas at Austin trained a **convolutional neural network** (Chapter 44) on about 15 000 labelled Kepler signals. Their network, AstroNet, looked at two "views" of each folded light curve: the whole orbit and a zoom on the transit. It reached about 96 % accuracy on test data.

### LANDMARK: KEPLER-90I, THE FIRST PLANET FOUND BY A NEURAL NETWORK (DECEMBER 2017)

- AstroNet was run on 670 stars that already had several known planet candidates. Weak signals around such stars are more likely to be real.
- It found Kepler-90i, a planet about 30 % larger than Earth with a 14.4-day orbit, and Kepler-80g. Kepler-90 became the first known star with eight planets, tying the Solar System.
- The team said openly that the network also produced many false positives. Each new planet was confirmed by further statistical tests and by checking the folded data by hand.
- The code was released openly in 2018, and learned vetting has since become a standard step in planet searches.

NASA's **ExoMiner** (2021) went further. Its network has separate branches for transit shape, secondary eclipse, odd–even depths and image centroids, so it can explain why a signal looks false. On Kepler test data it kept 93.6 % of planets at a precision of 99 %, against 76.3 % for the best earlier classifier. Applied to 1 922 unlabelled candidates, it gave 368 a score above 0.99, and 301 of these were validated as new planets. **ExoMiner++** (2025) adapted the method to TESS. It was trained on Kepler's well-labelled signals together with TESS data, whose labels are noisier. It flagged 7 330 of 147 568 two-minute-cadence TESS signals as planet candidates.

In Figure 51.4 the idea is repeated on 36 000 synthetic folded signals. They include planets, several kinds of eclipsing binary and artefacts, with SNR from 6 to 40. A small numpy neural network sees the raw global and local views. A logistic-regression vetter sees only four hand-made tests: secondary eclipse, V shape, asymmetry and depth just outside the transit. Both were trained on 24 000 signals and tested on 8 000.




Synthetic data: 36 000 folded signals (40 % planets, 35 % eclipsing binaries with V shapes, secondary eclipses or odd–even differences, 25 % artefacts), SNR 6–40. Network: 262 inputs (201-bin global and 61-bin local views), two ReLU layers (64, 32), dropout, Adam; trained on 24 000, tested on 8 000. Seed 510.

**FIGURE 51.4** Vetting synthetic transit signals. (a) Examples of a planet, an eclipsing binary and an artefact, with the network's planet score. (b) Precision against recall: the network keeps far more planets at high precision than a classifier built on four hand-made tests. (c) The fraction of planets kept at 98 % precision, by signal-to-noise ratio.

The network reached 92.0 % accuracy. At 95 % precision it kept 84.8 % of planets, against 43.7 % for the hand-made tests. At 99 % precision the gap widened: 58.7 % against 6.5 %. The network gained most at low SNR, where single tests are noisy but the whole shape still carries information. Diluted background binaries, with weak secondaries and U-shaped eclipses, stayed hard for both. In real data they are rejected by other evidence, such as shifts of the star's image or spectra from the ground.

**FIGURE 51.5** The Transiting Exoplanet Survey Satellite (TESS) being lifted onto a work stand at Kennedy Space Center in February 2018. Its four cameras produce thousands of candidate signals that learned vetting tools help to sort. Image: NASA/KSC (NASA/Ben Smegelsky).

### 51.2.3 Radial Velocities and Stellar Noise

Transits give a planet's size. Its mass comes from the **radial-velocity** method: the star's spectrum shifts as the planet pulls it back and forth. Earth makes the Sun wobble by only about 9 cm/s. Starspots and convection on the star's surface change the shapes of its spectral lines. They produce false velocity signals of around 1 m/s, ten times larger than the planet.

**Gaussian processes** (Chapter 43) are the usual tool: they model the star's activity as correlated noise, fitted together with the planet's orbit. In 2022 a team trained a convolutional network on the line shapes in more than 600 solar spectra from the HARPS-N solar telescope, taken over three years. It reduced the scatter of the solar velocities from 1.75 m/s to 1.04 m/s — a large step, but still ten times too much to find an Earth twin.

> **EXAMPLE: AN EARTH TWIN IN KEPLER DATA, AND THE FALSE-ALARM BUDGET**
>
> Kepler's noise for a bright, quiet Sun-like star was about 30 ppm over 6.5 hours. An Earth-sized planet in a one-year orbit gives a depth of 84 ppm and a central transit of about 13 hours.
>
> 1. **Noise over one transit.** Noise falls as the square root of time: $30 \times \sqrt{(6.5/13)} = \mathbf{21.2\ ppm}$. One transit gives SNR = 84/21.2 = **4.0**.
> 2. **Several transits.** Four transits give $4.0 \times \sqrt{4} = \mathbf{7.9}$, just above the 7.1 threshold. Three transits give **6.8**, which is below it. About 3.2 transits are needed, so an Earth twin is at the edge of a four-year mission.
> 3. **Why 7.1?** For pure Gaussian noise, the chance of a value above $7.1\sigma$ is $6.2 \times 10^{-13}$. Suppose each of the 198 709 stars needs about five million independent trials of period, phase and duration. That is $10^{12}$ tests, which gives about **0.6** false alarms. The threshold was set for exactly this reason.
> 4. **Reality.** Real noise is not Gaussian. The search produced 34 032 TCEs, and 88 % were not planet candidates.
> 5. **Human cost.** At five minutes per signal, vetting all TCEs by eye would take about 2 840 hours, or about one and a half working years. A neural network scores them all in minutes.
>
> The threshold controls false alarms from random noise; vetting controls false alarms from everything else. Learned vetting does not lower the threshold. It makes the work above the threshold affordable.

## 51.3 The Time-Domain Sky

The key idea of this section is speed. Transients fade in hours to weeks, so they must be classified while they can still be observed.

### 51.3.1 Rubin and the Alert Brokers

The **Vera C. Rubin Observatory** in Chile has an 8.4 m telescope and the world's largest astronomical camera, with 3 200 megapixels. Its Legacy Survey of Space and Time (LSST) images the whole southern sky every few nights for ten years. Each image is compared with a template within about a minute. Every change becomes an **alert**: a small packet with the measurement, image cut-outs and the object's history.

DATA BOX — RUBIN'S FIRST YEAR OF ALERTS

- **First look.** First images were released in June 2025. The first world-public alerts reached the brokers on the night of 24 February 2026, about 800 000 of them.
- **Survey.** The ten-year survey formally began in mid-2026. The camera takes an image about every 40 s and the observatory produces about 10 TB of data a night.
- **Alerts.** The design allows 10 000 alerts per image and about 1 000 images a night, or 10 million alerts. The current estimate is up to 7 million a night and about 20 billion over the survey.
- **Speed and size.** 98 % of alerts must leave within 60 s of readout; each packet is up to 82 KB.
- **Brokers.** Seven "full-stream" brokers receive every alert: ALeRCE, AMPEL, ANTARES, Babamul, Fink, Lasair and Pitt-Google. Two downstream services take subsets, one for Solar System objects.
- **Early science.** In about six weeks of early test surveys, Rubin found more than 11 000 new asteroids, including 33 near-Earth objects.

Brokers were rehearsed for years. In the 2018 **PLAsTiCC** challenge, more than 1 000 teams classified simulated Rubin light curves, with only about 7 850 labelled for training. The winner degraded bright training objects to look faint and distant, then used gradient-boosted trees. The lesson — make the training set look like the data — applies far beyond astronomy.

### 51.3.2 Classifying Transients

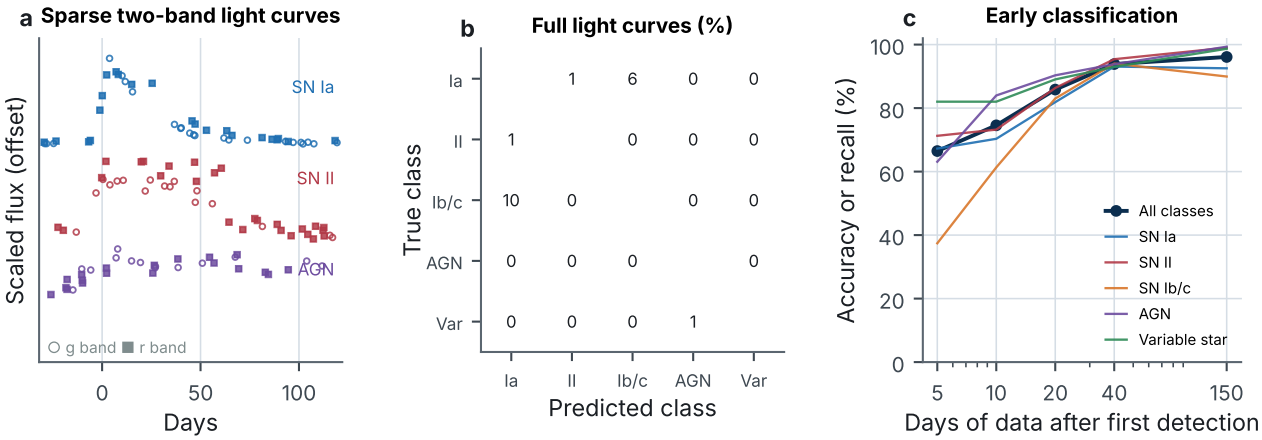
A survey light curve is sparse: each object is seen every few days, in different filters, with noisy fluxes. Classifiers work on the raw sequence or on features such as rise time, decline rate, peak colour and periodicity. A **random forest** (Chapter 43) takes a vote among many decision trees $T_b$, each trained on a random sample of the data.

$$\hat{y}(\mathbf{x}) = \operatorname{mode}\{T_1(\mathbf{x}), \dots, T_B(\mathbf{x})\}.$$

ALeRCE, one of the Rubin brokers, has long used such a forest for light curves, alongside a convolutional network that classifies image cut-outs.

A small version is shown in Figure 51.6. Synthetic two-band light curves of five classes were generated: type Ia supernovae, type II and type Ib/c supernovae, active galactic nuclei (AGN) and periodic variable stars. Each had 28–45 irregular epochs with noise. A random forest of 200 trees used 15 features and was trained on 6 000 objects.

With full light curves the forest classified 96.1 % of 3 000 test objects correctly. Most errors confused type Ia with type Ib/c supernovae, which also happens in real data. The useful number is the early one. Five days after the first detection, accuracy was only 66.5 %, and type Ib/c supernovae were recognised just 37 % of the time. After 20 days it reached 85.8 %. Astronomers want spectra *before* the peak, so brokers must rank objects while the classification is still uncertain. They therefore report probabilities, not labels.




Synthetic two-band light curves: SN Ia, II, Ib/c (Bazin-type profiles with class-dependent rise, decline, colour and secondary peak), AGN (damped random walk) and periodic variables; 28–45 irregular epochs with noise. Random forest (200 trees, 15 features); 6 000 training and 3 000 test objects. Seed 5152.

**FIGURE 51.6** Classifying synthetic transient light curves. (a) Examples of a type Ia supernova, a type II supernova and an active galactic nucleus. (b) Confusion matrix for full light curves. (c) Accuracy, and recall for each class, as more days of data arrive after the first detection.

**EXAMPLE: RUBIN'S ALERT STREAM AGAINST HUMAN ATTENTION**

Take 7 million alerts in a 10-hour night.

1. **Rate.** 7 000 000 / 36 000 s = **194 alerts per second**.
2. **People.** If a trained person needs 3 s to glance at one alert, a night's stream needs 5 830 hours of attention. That is about **730 people** working 8-hour shifts, every night for ten years.
3. **Data.** At up to 82 KB per alert, the stream is about 0.6 TB a night. Most alerts concern known variable stars and asteroids.
4. **Follow-up.** Suppose the world's telescopes can take 100 spectra of new transients a night. That is one alert in **70 000**.
5. **What machine learning must do.** It must remove artefacts, recognise known variables and asteroids, and rank the rest. A 1 % error in a filter applied to 7 million alerts is still 70 000 mistakes a night. Errors must be counted, not just accuracy.

No human team can read the stream. People decide which questions to ask the brokers, check the top of each ranked list, and take the spectra that create new labels.

### 51.3.3 Radio Bursts and Gravitational Waves

**Fast radio bursts (FRBs)** are millisecond flashes of radio waves from distant galaxies. In 2018 a convolutional network re-analysed five hours of Green Bank Telescope data from Breakthrough Listen. It found 72 new bursts from the repeating source FRB 121102, on top of the 21 found by the original search. The second CHIME/FRB catalogue (2026) lists 4 539 bursts from 3 641 sources. At this rate, sorting real bursts from radio interference must be automatic.

**Gravitational-wave** detectors measure changes in arm length smaller than a proton. The standard search is a **matched filter**. It correlates the data with a bank of theoretical waveforms $h$, weighting each frequency $f$ by the detector noise $S_n(f)$.

$$\rho^2 = 4 \int_0^\infty \frac{|\tilde{h}(f)|^2}{S_n(f)}\, df.$$

The fourth observing run of LIGO, Virgo and KAGRA ran from May 2023 to 18 November 2025 and produced about 250 candidate signals. Learning helps at three points. **Gravity Spy** combines volunteers and a network to classify detector "glitches", short noise bursts that can mimic signals. Research pipelines such as Aframe search for signals with neural networks in milliseconds, though they do not yet issue official alerts.

The biggest gain is in **parameter estimation**. Standard Bayesian sampling of a merger's masses, spins and sky position can take hours to days. **Simulation-based inference** instead trains a network on millions of simulated signals to output the posterior directly. In March 2025 a team reported DINGO-BNS, which infers all 17 parameters of a neutron-star merger in about one second. Its 90 % sky areas were typically about 30 % smaller than those of the standard rapid method. Such speed lets telescopes point at a kilonova before it fades.

Learning can also improve the instrument itself. In September 2025 researchers from Google DeepMind and LIGO reported **Deep Loop Shaping**, a reinforcement-learning controller (Chapter 48) for one mirror-alignment loop at LIGO Livingston. It cut control noise between 10 and 30 Hz by more than 30 times, and by up to 100 times in narrow bands. It was a demonstration on one loop, not yet part of routine observing.

## 51.4 Galaxies, Lenses and Cosmology

Learning turns images of billions of galaxies into measurements: shapes, distances and the rare objects among them.

### 51.4.1 Galaxy Shapes

**Galaxy Zoo**, launched in 2007, asked volunteers to classify galaxy images (Chapter 43). Its labels trained **Zoobot**, a family of networks that predict how volunteers would answer each question about a galaxy. In 2023 Zoobot measured detailed shapes for 8.67 million galaxies in the DESI Legacy Imaging Surveys. For Euclid, tests on simulated images suggest that about 800 million galaxies in the wide survey could be measured reliably.

In 2024 a network trained on labelled Hubble images was adapted to the sharper, redder images of the James Webb Space Telescope (JWST). It classified about 20 000 distant galaxies in the CEERS survey and agreed with visual classifications 80–90 % of the time. Moving a model between instruments with few new labels is called **domain adaptation** (Chapter 45).

### 51.4.2 Strong Lenses in Euclid

A massive galaxy bends the light of a galaxy behind it into arcs or rings. Such **strong lenses** weigh the dark matter in the lensing galaxy and test cosmology, but only roughly one galaxy in ten thousand shows one. Euclid's first quick release (March 2025) covered 63 square degrees. An ensemble of networks, including Zoobot, ranked the galaxies. Volunteers on the Space Warps project then inspected the best candidates, and experts graded and modelled them.

The result was a catalogue of 497 galaxy–galaxy lenses, 250 of them "grade A"; 243 of those were new. The same engine is expected to yield about 7 000 high-confidence (grade A or B) lenses in the first major data release, DR1. Its first part is due in November 2026 and the full release in mid-2027. The full survey should yield more than 100 000.

### 51.4.3 Photometric Redshifts

Most galaxies are too faint for a spectrum, so their distance must be estimated from colours in a few broad filters. This is a **photometric redshift**. Spectral features, such as the 4000 Å break and the Lyman break, move through the filters as redshift grows. Learning methods map colours directly to redshift, trained on galaxies whose spectroscopic redshifts are known. Quality is measured by the normalised scatter of $\Delta z = (z_{\mathrm{phot}} - z_{\mathrm{spec}})/(1 + z_{\mathrm{spec}})$:

$$\sigma_{\mathrm{NMAD}} = 1.4826 \times \mathrm{median}\,|\Delta z - \mathrm{median}(\Delta z)|,$$

and by the fraction of **outliers** with $|\Delta z| > 0.15$. Rubin's goals for its "gold" sample, brighter than magnitude 25.3 in the i band, are a scatter below 0.02 (requirement 0.05) and fewer than 10 % large (3σ) outliers.

In Figure 51.7, a random forest is tested on 60 000 synthetic galaxies observed in six Rubin-like filters. Trained on a representative sample of 30 000, it reached a scatter of 0.017 with 12.0 % outliers. Errors grew steeply for faint galaxies, from 0.007 at i ≈ 20 to 0.13 at i ≈ 25. Then the forest was trained only on galaxies brighter than i = 22.5 and nearer than z = 1.2, like a typical spectroscopic sample. The outlier fraction rose to 19.4 %, and to 61 % for the faintest galaxies. A model cannot learn what its training set never shows.

Synthetic galaxies: continuum, 4000 Å break and emission lines varying with type, dust; Lyman-α forest absorption; six LSST-like Gaussian filters with 10-year depths; i < 25.3. Random forest (120 trees) on i magnitude and five colours; 30 000 training (b: only the 13 929 with i < 22.5 and z < 1.2), 30 000 test galaxies; outliers |Δz|/(1+z) > 0.15. Seed 5151.

**FIGURE 51.7** Photometric redshifts from a random forest. (a) Photometric against true redshift when the training set represents the whole population. (b) The same when training uses only bright, nearby galaxies: faint and distant galaxies are pulled to the wrong redshifts. (c) Outlier fraction against brightness for both cases.

### 51.4.4 Simulation-Based Inference in Cosmology

Cosmology usually compresses a survey into a few summary statistics, such as the two-point correlation of galaxy positions, and fits them with a formula. Much information lies in the non-linear structure that such summaries miss. **Simulation-based inference (SBI)** uses Bayes' theorem,

$$p(\boldsymbol{\theta} \mid \mathbf{x}) \propto p(\mathbf{x} \mid \boldsymbol{\theta})\, p(\boldsymbol{\theta}),$$

without writing down the likelihood $p(\mathbf{x} \mid \boldsymbol{\theta})$. Instead, thousands of simulated universes with known parameters $\boldsymbol{\theta}$ train a network to estimate the posterior. The SimBIG project applied this to part of the BOSS galaxy survey. Using the full three-dimensional galaxy field, it constrained the

clustering amplitude $\sigma_8$ 2.65 times more tightly than a standard power-spectrum analysis of the same data. The method is only as good as its simulations: if they miss a physical effect, the answer is precise but wrong.

Learning also works inside telescope pipelines (Chapter 38), flagging cosmic rays and artefacts and separating overlapping objects. Landmark results in astronomy are listed in Table 51.2.

**TABLE 51.2** Landmark uses of machine learning in astronomy. CNN: convolutional neural network; RNN: recurrent neural network.

| Result | Year | Data | Method | Outcome | Status (Oct 2026) |
|---|---|---|---|---|---|
| AstroNet | 2017 | Kepler | 1-D CNN | Kepler-90i, Kepler-80g | Research, open code |
| ExoMiner | 2021 | Kepler | Explainable CNN | 301 validated planets | Planets in NASA archive |
| ExoMiner++ | 2025 | TESS | Kepler + TESS training | 7 330 candidates | Public catalogue |
| FRB 121102 search | 2018 | Green Bank | CNN | 72 new bursts | Research |
| Galaxy Zoo DESI | 2023 | DESI imaging | Zoobot CNN | 8.67 million galaxies | Public catalogue |
| Euclid lens engine | 2025 | Euclid Q1 | CNN ensemble + volunteers | 497 lenses | In Euclid pipeline |
| SimBIG | 2023–24 | BOSS | Simulation-based inference | $\sigma_8$ 2.65× tighter | Research |
| DINGO-BNS | 2025 | LIGO–Virgo | Neural posterior | 17 parameters in 1 s | Research |
| Deep Loop Shaping | 2025 | LIGO Livingston | Reinforcement learning | Control noise ÷ 30–100 | Demonstration |
| Rubin brokers | 2026 | LSST alerts | Forests, CNNs, RNNs | Real-time classes | Operational |

## 51.5 The Sun: Features and Event Catalogues

The key idea is that heliophysics needs *catalogues*: lists of every flare, sunspot group, filament and eruption, with times and positions. Forecasting space weather is covered in Chapter 47. Here the aim is to describe what happened.

The Solar Dynamics Observatory (SDO) sends about 1.5 TB of images a day. Around its launch in 2010, NASA funded a **Feature Finding Team** to build automatic detection modules. Their reports feed the **Heliophysics Event Knowledgebase** (HEK), which scientists search for, say, every filament that erupted near an active region. A 2016 study counted more than 280 000 event reports for seven kinds of solar phenomenon. Coronagraph catalogues such as CACTus detect coronal mass ejections automatically, and are compared with catalogues made by people.

The early modules used thresholds, edge detection and tracking. Since about 2018, convolutional networks have segmented coronal holes, filaments and active regions in several wavelengths at once; the Surya foundation model is described in Chapter 44. The challenge is consistency. A catalogue that changes its detector can create false trends over the 11-year solar cycle, so versions must be recorded and old data reprocessed.

# 51.6 Planetary Surfaces and Science Autonomy

Planetary images and spectra become maps — of craters, rocks and minerals — and those maps drive landing, driving and sampling decisions. Learning makes global maps possible. Physics decides what they mean.

## 51.6.1 Craters and Surface Ages

On airless bodies, craters accumulate at a known rate. Counting them dates a surface: more craters per square kilometre means an older surface. The method rests on two functions calibrated with Apollo and Luna samples (Chapter 29). The **production function** gives the shape of the size distribution, and the **chronology function** gives the density of craters larger than 1 km, $N_1$, after $T$ billion years. For the Moon, Neukum and colleagues (2001) give

$$N_1(T) = 5.44 \times 10^{-14}\left(e^{6.93\,T} - 1\right) + 8.38 \times 10^{-4}\,T \quad \text{per km}^2.$$

The first term describes the heavy bombardment of the early Moon; the second, the nearly constant rate since about 3 billion years ago.

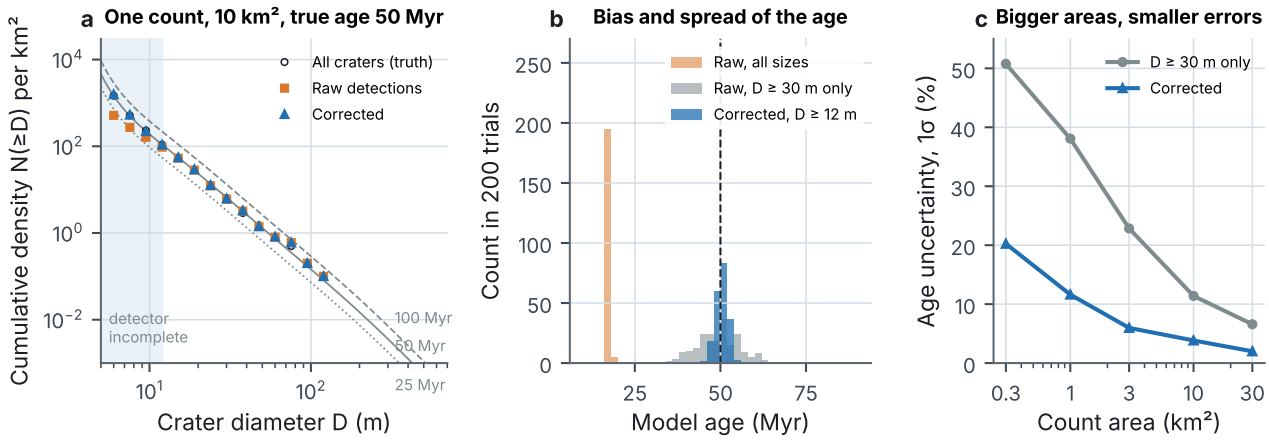
Counting small craters by hand is slow and inconsistent between people, so trained detectors now do it at planetary scale. In 2020 a team used transfer learning on Chang'e-1 and Chang'e-2 data to find 109 956 new lunar craters. In 2021 a detector trained on HiRISE images found about 90 million craters in the Context Camera mosaic of Mars. Rays of small secondary craters then traced a group of Martian meteorites to their source craters in the Tharsis region. Crater detection for navigation is covered in Chapter 48.

A detector does not see every crater. Small craters, only a few pixels across, are often missed, and shadows or boulders can be mistaken for craters. The fix is to measure **completeness** $c(D)$ and the false-positive density $f(D)$ by testing the detector on images with known craters. A model age is then fitted by **maximum likelihood**. Suppose $k_i$ craters are counted in size bin $i$ of an area $A$. The expected count $\lambda_i$ and the log-likelihood are

$$\lambda_i(T) = A\,\Delta N_i(T)\,c_i + A f_i, \qquad \ln L(T) = \sum_i \left[k_i \ln \lambda_i - \lambda_i\right],$$

Here $\Delta N_i$ is the number per unit area in bin $i$ from the production and chronology functions.

In Figure 51.8 this is applied to a synthetic 10 km$^2$ lunar surface aged 50 million years, imaged at 1.5 m per pixel. The synthetic detector finds half the craters 7 pixels across and 95 % of large ones. About 40 % of its detections between 6 and 10 m are false. Fitting the raw counts at all sizes gives an age of 17 million years, three times too young. Counting only craters of 30 m and larger gives an unbiased median of 49 million years, but with a spread of ±13 %. Correcting for completeness and false positives, and fitting from 12 m upwards, gives 51 million years with a spread of only ±3 %.




Lunar production and chronology functions of Neukum et al. (2001). Synthetic detector on 1.5 m pixels: recall 50 % at 7 px rising to 95 %, 7 % diameter error, false positives near the resolution limit. Completeness and false-positive rates measured by injection on a separate 30 km² labelled set; ages by Poisson maximum likelihood in 10 bins per decade. Seed 5153.

**FIGURE 51.8** Crater-count dating with an imperfect detector. (a) Cumulative crater densities for one synthetic count, with model curves for three ages; raw detections fall below the truth at small sizes. (b) Ages from 200 repeated counts: raw counts at all sizes are far too young; corrected counts are accurate and precise. (c) Age uncertainty against count area.

The lesson applies to every automated catalogue: a detector's errors must be measured and modelled, not just reduced. Small craters also reach **equilibrium** on old surfaces, where new craters erase old ones as fast as they form. Below that size, no detector can date the surface.

### 51.6.2 Boulders, Rockfalls and Landslides

Boulders tell geologists how deep the regolith is and how surfaces erode. They are also hazards for landers and rovers (Chapter 27). In 2020 a convolutional network searched images from the Narrow Angle Camera (NAC) of NASA's Lunar Reconnaissance Orbiter for boulders that had rolled downhill. It mapped 136 610 such rockfalls and showed that impacts trigger most of them. In 2025 a team mapped about 94 million boulders larger than about 4.5 m, using 635 000 NAC images between 60° S and 60° N.

Shadows hide some of the most valuable terrain. Permanently shadowed craters near the lunar poles may hold water ice (Chapter 41). In 2021 a deep-learning tool called HORUS removed noise from NAC images of these craters, lit only by faint light scattered from nearby slopes. It revealed features a few metres across at about 1.5 m per pixel. It was later used to examine candidate target regions for NASA's VIPER rover.

Such mapping projects are summarised in Table 51.3. Fresh impacts found by a classifier in Mars orbiter images are described in Chapter 44.

**TABLE 51.3** Planetary mapping with machine learning.

| Project | Body, data | Method | Result | Use |
|---|---|---|---|---|
| Lunar rockfalls (2020) | Moon, LRO NAC | CNN detector | 136 610 rockfalls | Erosion; hazards |
| Chang'e craters (2020) | Moon, Chang'e-1/2 | Transfer learning | 109 956 new craters | Crater ages |
| HORUS (2021) | Moon, NAC in shadow | Physics-guided denoising | ≈ 1.5 m images of shadowed craters | Ice prospecting, VIPER |
| Mars craters (2021) | Mars, CTX mosaic | CNN detector | ≈ 90 million craters | Meteorite source craters |

| Project | Body, data | Method | Result | Use |
|---|---|---|---|---|
| AI4Mars (2021) | Mars, rover images | Crowd labels + segmentation | 326 000 labels | Terrain classes for driving |
| Lunar boulders (2025) | Moon, 635 000 NAC images | CNN detector | ≈ 94 million boulders | Regolith depth; landing |

### 51.6.3 Minerals from Spectra

Minerals absorb light at characteristic wavelengths. Olivine has a broad band near 1.05 µm, pyroxenes absorb near 0.9 and 1.9 µm, and clays and carbonates have narrow bands between 2.2 and 2.5 µm (Chapter 13). Imaging spectrometers record hundreds of wavelengths for every pixel. CRISM, on NASA's Mars Reconnaissance Orbiter, did this at about 18 m per pixel from 2006 until it was retired in April 2023. Its maps of carbonates in Jezero crater helped select the landing site of Perseverance (Figure 51.9).

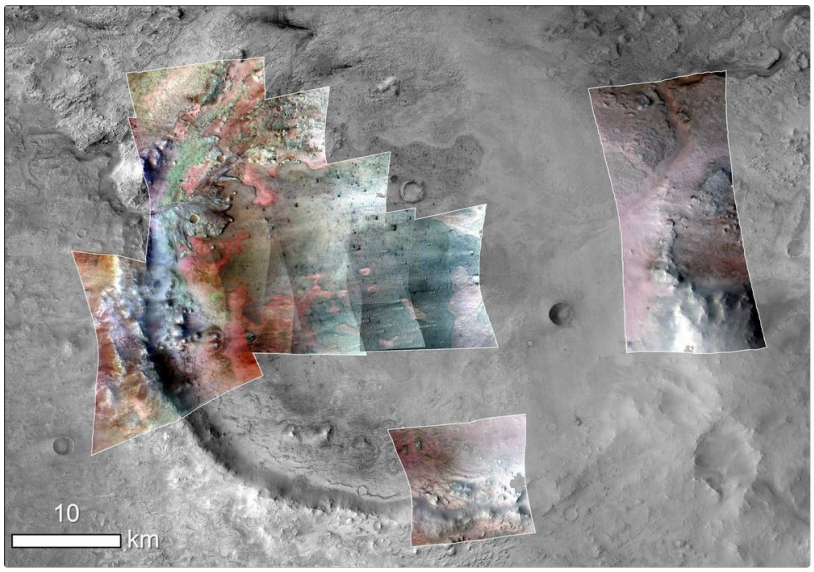



**FIGURE 51.9** Minerals in Jezero crater, mapped by the CRISM imaging spectrometer and the Context Camera on Mars Reconnaissance Orbiter. Green marks carbonates; red marks olivine-rich sand. Maps like this helped NASA choose Perseverance's landing site. Image: NASA/JPL (NASA/JPL-Caltech/MSSS/JHU-APL/Purdue/USGS).

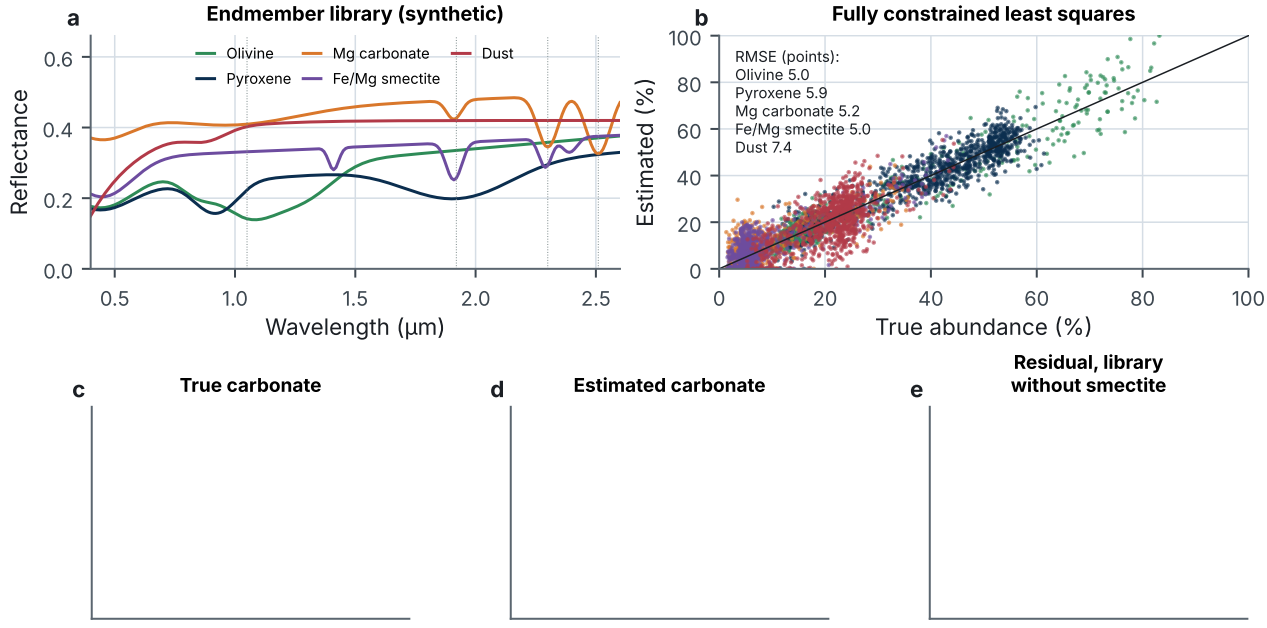
A pixel usually contains several minerals. In **linear spectral unmixing**, the spectrum $\mathbf{x}$ of a pixel is modelled as a weighted sum of pure "endmember" spectra, the columns of a matrix $\mathbf{E}$.

$$\mathbf{x} = \mathbf{E}\,\mathbf{a} + \mathbf{n}, \qquad a_k \geq 0, \quad \sum_k a_k = 1.$$

The abundances $\mathbf{a}$ are found by least squares with both constraints. This is called **fully constrained least squares (FCLS)**. When no spectral library is available, **non-negative matrix factorisation (NMF)** can estimate endmembers and abundances together, but its endmembers need not be pure minerals.

In Figure 51.10 both methods are applied to a synthetic scene of 3 456 pixels with 221 bands. It contains a carbonate- and clay-bearing fan, an olivine-rich channel, pyroxene and dust. The endmembers vary from pixel to pixel in band depth and brightness, and topographic shading changes the overall brightness. With the right library, FCLS recovered the abundances with

errors of 5.0–7.4 percentage points. Unconstrained least squares gave similar average errors, but 5.3 % of its abundances were negative, which is physically impossible. NMF found endmembers that differed from the true minerals by 8–19° in spectral angle.




Synthetic 48 × 72-pixel scene, 221 bands (0.40–2.60 µm), five endmembers with absorption bands at the positions of real minerals, linear (areal) mixing, noise SNR 150. FCLS = non-negative least squares with a sum-to-one row. Without the smectite spectrum, the residual map outlines where the library is incomplete. Seed 5154.

**FIGURE 51.10** Linear unmixing of a synthetic hyperspectral scene. (a) The five endmember spectra; dotted lines mark key absorption bands. (b) True against estimated abundances from fully constrained least squares. (c–d) True and estimated carbonate abundance. (e) Fit residual when the clay (smectite) spectrum is left out of the library: the error outlines where the clay is.

Panel (e) shows the most useful diagnostic. With smectite removed from the library, the fit residual rose sharply where smectite was present; its 99th percentile was 2.8 times the median. The residual map points to minerals the library lacks. In fine-grained real surfaces, light passes through several grains and mixing is non-linear. Networks trained on laboratory mixtures are an active research topic, but their results are still checked against band positions a spectroscopist can explain.

### 51.6.4 Landing Sites, Traverses and Resources

Before a landing, engineers must know where the ground is safe. JPL's **SPOC** (Soil Property and Object Classification) network classifies terrain in HiRISE images into types such as sand, bedrock and ridges. It was used to judge how easily a rover could drive at candidate landing sites for Perseverance. The **AI4Mars** project (2021) collected about 326 000 terrain labels on 35 000 rover images from volunteers, ten per image. A segmentation network trained on them reached more than 96 % accuracy on an expert-labelled test set (the labelling tool is shown in Chapter 44).

Resource prospecting combines these tools (Chapter 41). Maps of shadowed craters, boulders, slopes and temperature are merged to find ice that a rover could reach. The NASA–IBM Lunar Foundation Model (2026) is one example; Chapter 44 gives its results and limits.

### 51.6.5 Science Autonomy on Rovers and Probes

A rover or probe can also decide what to measure. The first steps were small. In 2006 the Mars Exploration Rovers received software that looked for dust devils and clouds in sequences of images. It first detected a dust devil on board Spirit in April 2007. On dust-devil sequences it cut the data volume by about 75 %, because only frames with moving features were sent home.

**AEGIS** (Chapter 48) picks rock targets for the laser spectrometers on Curiosity and Perseverance; by May 2026 it had chosen nearly 900. Perseverance's X-ray spectrometer **PIXL** goes further. Its hexapod mount places it close to a rock, and it scans a postage-stamp-sized patch with thousands of X-ray shots. Its **adaptive sampling** software watches the results. When it sees minerals of interest, such as carbonates or phosphates, it stops and collects data for longer (Figure 51.11). Each decision is made on Mars, without waiting for the ground.

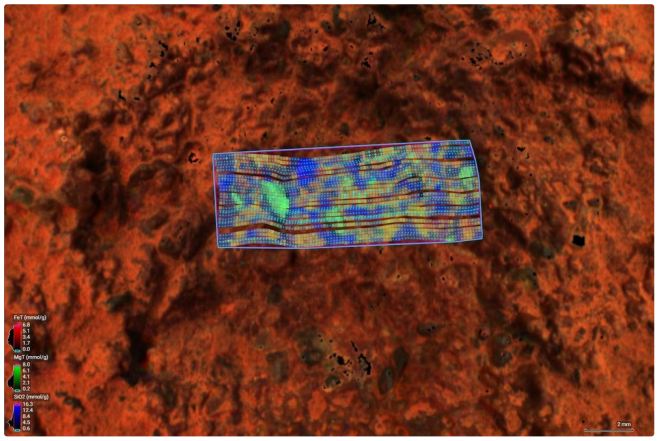


**FIGURE 51.11** A PIXL scan of the rock target "Thunderbolt Peak", recorded by Perseverance on 8 July 2024. Each dot marks a point hit by the X-ray beam; colours show the amounts of iron (red), magnesium (green) and silica (blue). PIXL's software can lengthen its measurements on board when it finds minerals of interest. Image: NASA/JPL (NASA/JPL-Caltech/DTU/QUT).

Future probes to icy moons may have to detect a plume or outburst and re-plan within hours. In sample curation (Chapter 29), learning could help find rare grains in images and keep records consistent between laboratories, without ever risking irreplaceable material. Onboard science autonomy is summarised in Table 51.4.

**TABLE 51.4** Onboard science autonomy on planetary missions.

| System | Mission | What it decides | Since | Status (Oct 2026) |
|---|---|---|---|---|
| Dust-devil and cloud detection | Spirit, Opportunity | Which frames to send home | 2006–07 | Missions ended |
| Autonomous Sciencecraft | EO-1 | Re-plans imaging after events (Chapter 48) | 2004 | Mission ended 2017 |
| AEGIS | Opportunity, Curiosity, Perseverance | Laser targets after a drive | 2010 | Operational |
| PIXL adaptive sampling | Perseverance | Where to dwell longer in a scan | 2021 | Operational |
| Onboard plume or outburst detection | Icy moons, comets | When to re-target | — | Research |

## 51.7 AI for Human Spaceflight

A crew far from Earth must become its own doctor, engineer and mission control. AI can help, but only if it works without a link to Earth and its advice can be checked.

### 51.7.1 Watching Crew Health

Wearable sensors measure the body continuously. Canada's **Bio-Monitor** has been used on the ISS since early 2019 (Figure 51.12). It is a stretchable shirt with a headband and a small recorder. It tracks heart activity, breathing, skin temperature, blood pressure and blood oxygen during work, exercise and sleep. On Artemis II in April 2026, all four astronauts wore wrist devices for the ARCHeR study. These logged movement, sleep and exposure to light around the Moon.

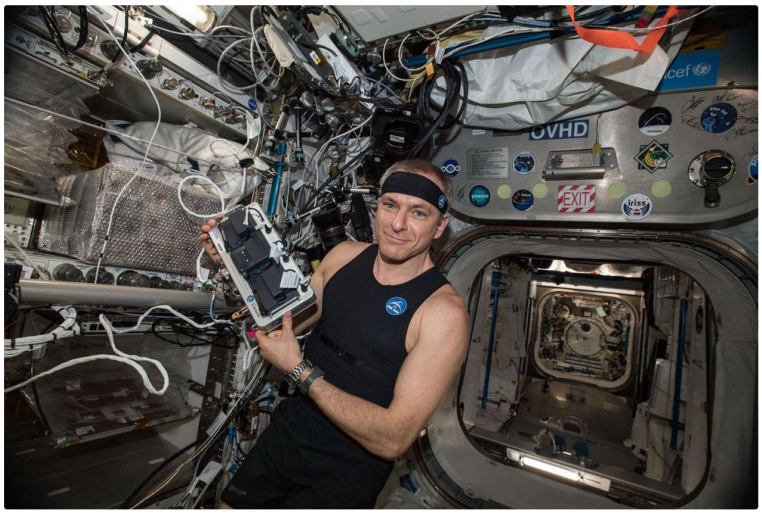



**FIGURE 51.12** Canadian astronaut David Saint-Jacques demonstrates the Bio-Monitor smart shirt on the ISS in January 2019. Its sensors record heart activity, breathing, blood pressure, blood oxygen and skin temperature. Image: NASA/JSC (David Saint-Jacques).

Such data are only useful if someone looks at them. Learning methods can build a **personal baseline** for each astronaut, the normal range of heart rate, temperature and sleep, and flag departures from it. A rising resting heart rate and temperature can precede the symptoms of an infection. The statistics are those of anomaly detection (Chapter 49), but a miss can cost a life.

### 51.7.2 Exploration Medicine

On the ISS, a sick astronaut can come home within hours. In January 2026, NASA's Crew-11 returned about a month early, the first time an ISS crew was brought home for medical reasons (Chapter 30). A Mars crew will have no such option for many months. Its medical kit, training and software must cover almost everything that can go wrong.

NASA plans medical kits with simulation. Its **Integrated Medical Model** runs Monte Carlo missions over about 100 medical conditions. Each simulated case uses drugs and supplies from the kit, and untreated conditions count as risk, so planners can choose the kit that lowers risk most for its mass. A newer tool, IMPACT, adds conditions that matter for exploration.

Decision support is the next step. In 2025 NASA and Google tested a prototype **Crew Medical Officer Digital Assistant**. It takes speech, text and images, works with little or no link to Earth and records each recommendation. Three physicians, one of them an astronaut, graded it with

the structured exams used for medical students. They rated its assessment and treatment plan 88 % likely to be correct for an ankle injury, 80 % for ear pain and 74 % for flank pain. NASA called these early results; the tool is not approved for use.

**Ultrasound** is the main imaging tool in space, but good images need skill. On the ISS, experts on the ground guide astronauts in real time, which will not work with a 40-minute round trip. ESA's ALISSE prototype (2024) uses a network trained on hundreds of thousands of labelled scans. It tells the user where to move the probe and how likely it is that the image shows the right organ. In tests on kidneys and bladders, nine out of ten images taken by untrained students were clinically acceptable.

**Behavioural health** may be the hardest part (Chapter 30). ARCHeR also studied stress, cognition and teamwork, and researchers are testing whether voice, sleep and activity patterns give early warning of fatigue or conflict. Such monitoring raises questions of privacy and trust, so crews must agree to it and control what is shared.

### 51.7.3 Assistants and Robots in the Cabin

Voice assistants that can read procedures, log work and answer questions have been tested for several years (Table 51.5). Language models now make them far more capable. Their main risk is the confident wrong answer, or **hallucination** (Chapter 44). For crew use, an assistant must quote its sources, such as the procedure, the step and the telemetry, so that the astronaut can check them.

LANDMARK: FROM CIMON TO LANGUAGE MODELS IN ORBIT (2018–2025)

- **CIMON** (Crew Interactive Mobile Companion), a floating sphere built by Airbus and IBM for the German space agency DLR, reached the ISS in July 2018. On 15 November 2018, in a 90-minute session, it recognised Alexander Gerst and talked with him. It returned to Earth in August 2019.
- **CIMON-2** arrived in December 2019, adding tone analysis to judge a speaker's mood. Its speech processing ran on ground servers.
- **Callisto** flew inside Orion on the uncrewed Artemis I mission in late 2022, testing an offline voice assistant and video calls (Figure 51.14).
- In **August 2024** a generative language model ran on HPE's Spaceborne Computer-2 on the ISS. The Space Llama project followed in April 2025, with an open-source model tuned for station science.
- In **July 2025** China delivered the **Wukong AI** model to its Tiangong station. Its onboard part helped crew members prepare for a spacewalk.
- Also in **July 2025**, in the ICHIBAN test, JAXA astronaut Takuya Onishi spoke to CIMON in the Columbus module. CIMON passed his commands to JAXA's Int-Ball2 camera drone in Kibo and showed its video (Figure 51.13). JAXA called it the first time that robots built by different organisations had worked together in orbit.

**FIGURE 51.13** JAXA astronaut Takuya Onishi sets up CIMON on the ISS on 18 July 2025, during the ICHIBAN tests in which it relayed his voice commands to the free-flying camera robot Int-Ball2. Image: NASA/JSC.

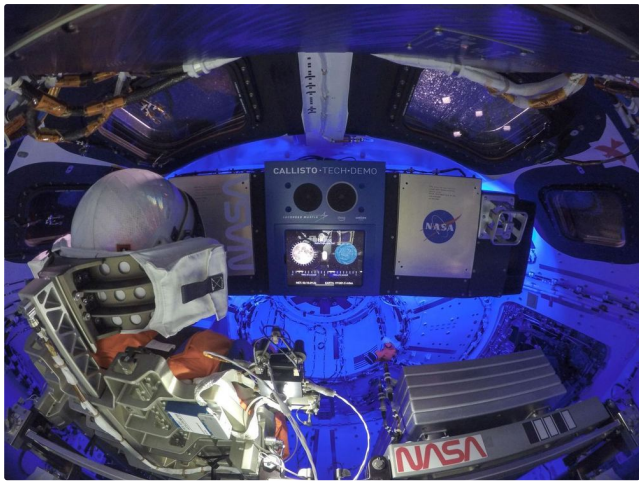



**FIGURE 51.14** The Callisto technology demonstration inside the Orion crew module on Artemis I, 18 November 2022. The payload tested an offline voice assistant and video calls for future crews. Image: NASA/JSC.

Robots are the physical side of the same idea. **Astrobee**, the free-flyer in the hero image, has tested autonomous inventory and inspection since 2019 (Chapter 48). JAXA's **Int-Ball2** films crew activities in Kibo. For a Mars habitat, the goal is a robot that keeps the station running while the crew sleeps or works outside, and before any crew arrives.

**TABLE 51.5** AI systems for crew support (status October 2026).

| System | Who, where | Function | Years | Status (Oct 2026) |
|---|---|---|---|---|
| Bio-Monitor | CSA, ISS | Wearable vital signs | 2019– | In use for research |
| ARCHeR wrist devices | NASA, Artemis II | Sleep, activity, light exposure | 2026 | Flown |
| CIMON, CIMON-2 | DLR, Airbus, IBM; ISS | Voice assistant, robot control | 2018– | Demonstrations; used in 2025 |
| Callisto | Lockheed Martin; Orion | Offline voice assistant | 2022 | Flown once (uncrewed) |
| Space Llama | Booz Allen, Meta, HPE; ISS | Language model on board | 2025– | Demonstration |
| Wukong AI | China; Tiangong | Language model, EVA preparation | 2025– | In use (reported) |
| Medical Officer Digital Assistant | NASA, Google | Diagnosis support | 2025 | Prototype, ground tests |
| ALISSE | ESA, GMV | Ultrasound guidance | 2024 | Prototype |
| Columbus ECLSS toolchain | Airbus, ESA | Anomaly detection, diagnosis | 2020– | Ground pilot |
| Astrobee, Int-Ball2 | NASA; JAXA | Free-flying robots | 2019–; 2023– | Operational test beds |

### 51.7.4 Smart Habitats and Predictive Maintenance

A habitat is a machine that keeps people alive (Chapters 30 and 32). Its fans, pumps, valves and filters wear out, and on a Mars mission every failed part must be fixed with a spare carried from Earth. Detecting wear early lets the crew plan repairs and can reduce the spares carried.

The European Columbus module on the ISS sends about 17 000 telemetry parameters. Its life-support system alone produces more than 1 000 signals sampled once a second. Since about 2020, Airbus has trained autoencoders on these data to learn normal behaviour. In 2025 researchers described a ground toolchain, integrated at Airbus, that detects anomalies, diagnoses their cause and recommends actions. In ESA's ORBIT-STAR project, Airbus built a demonstrator that reads Columbus sensor data and sends event messages to ground and crew. It was tested on a ground copy of the Columbus software; running it on board is the next step.

A synthetic test shows why learned models help (Figure 51.15). A fleet of 60 cabin fans ran for 1 100 days. Each wore out at a random time, and its vibration, current and temperature rose in its last weeks. But the fans also changed speed, and the cabin warmed and cooled. A fixed vibration limit therefore either alarmed whenever a fan ran fast, or alarmed late. **Gradient-boosted trees** — many small trees, each correcting the errors of the ones before — learned to predict "failure within 30 days" from the raw signals and their trends. They were trained on 40 fans and tested on 20 others. Allowing at most 0.5 false alarms per fan-year, the model gave a median warning of 29 days and warned of every failure. The fixed limit gave 15 days and missed 15 % of failures.

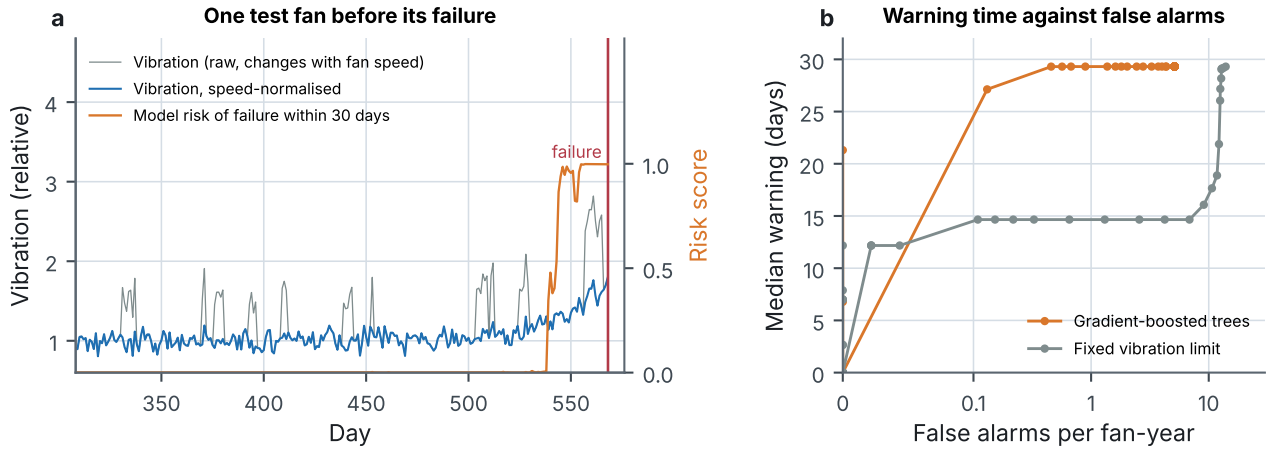
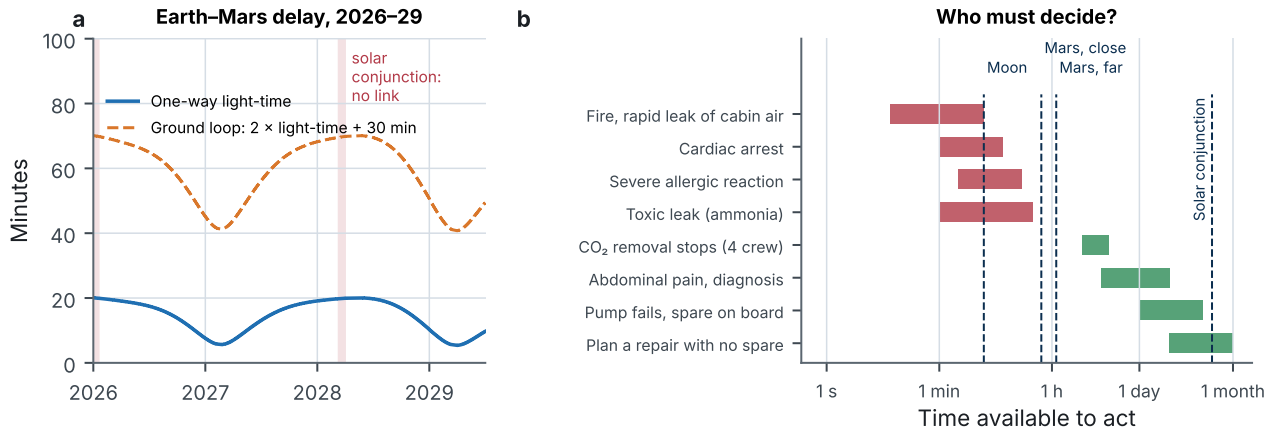



**FIGURE 51.15** Predictive maintenance of synthetic cabin fans. (a) One test fan before its failure: raw vibration rises whenever the fan runs fast; the speed-normalised signal and the model's risk score rise only as wear grows. (b) Median warning time against false alarms per fan-year, for gradient-boosted trees and a fixed vibration limit.

Real fleets are smaller and failures rarer, so models must share data across similar units and stations (Chapter 49).

### 51.7.5 Earth-Independent Operations

On the ISS, mission control answers within seconds. At Mars the one-way **light-time** — the time a signal takes to cross the distance $d$, $\tau = d/c$ — is 3 to 22 minutes. A ground answer takes at least $2\tau$ plus the time experts need to respond. For about two weeks near **solar conjunction**, when Mars is behind the Sun, there is no reliable link at all. In Figure 51.16 this is computed from the orbits of Earth and Mars for 2026–29. Mars passed behind the Sun on 9 January 2026 and will do so again around 20 March 2028, staying within 2° of it for about two and a half weeks (11–30 March 2028).




(a) Heliocentric Keplerian orbits of Earth and Mars (J2000 elements, coplanar approximation); shading where Mars is within 3° of the Sun. (b) Approximate times available before harm, compared with a ground loop of two light-times plus 30 minutes for experts to respond. Red: crew must act alone; green: ground advice can arrive in time, except during solar conjunction.

**FIGURE 51.16** Why a Mars crew must decide for itself. (a) One-way light-time between Earth and Mars in 2026–29 and the resulting ground loop, with solar conjunctions shaded. (b) Approximate time available to act in several emergencies and failures, compared with the ground loop for the Moon, for Mars near and far, and for solar conjunction.

NASA's analogue missions test this. The second one-year CHAPEA mission, with four volunteers in a 3-D-printed Mars habitat in Houston, ran from 19 October 2025 to its planned end on 31 October 2026, with simulated communication delays.

**EXAMPLE: AUTONOMY REQUIREMENTS FOR A MARS CREW WITH A 20-MINUTE DELAY**

Assume a one-way light-time of 20 minutes and 30 minutes for experts on Earth to respond.

1. **Ground loop.** $2 \times 20 + 30 =$ **70 minutes** for one question and answer. A diagnosis that needs three exchanges takes **3.5 hours**.
2. **Emergencies.** A fire, a fast air leak, cardiac arrest, a severe allergic reaction or a toxic leak must be handled within seconds to tens of minutes. The ground cannot help in time. Procedures, alarms and decision support must all be on board and work offline.
3. **Slower failures.** Four crew members exhale about 1.0 kg of $CO_2$ each per day. In 100 $m^3$ of air at 22 °C, the partial pressure rises by $\dot{m}RT/(MV) \approx$ **93 Pa per hour**. From 0.4 to 1.0 kPa takes **6.5 hours**, about 5.5 ground loops. The ground can advise, but onboard diagnosis should find the cause in the first hour.
4. **Conjunction.** For about 16–18 days no link is possible. A 900-day mission includes at least one conjunction. Any illness or failure in that window must be handled alone.
5. **Requirements.** The crew needs offline medical decision support, guided imaging, health monitoring with personal baselines, prognostics for critical spares, and a searchable record of every procedure. All must keep working if a computer fails, so simple manual procedures remain the backup.

The ground's role shifts from control to support. It reviews logs, updates models and plans the next days, while the crew and its software handle the present.

## 51.7.6 Augmented Reality for EVA and Maintenance

**Augmented reality (AR)** overlays instructions on what a person sees. In NASA's Project Sidekick, Microsoft HoloLens headsets reached the ISS in December 2015 (Figure 51.17). In "remote expert" mode, engineers on the ground saw what the astronaut saw and drew guidance into the view. In

"procedure" mode, the headset showed animated steps. With Mars delays, only the procedure mode is practical.

**FIGURE 51.17** NASA astronaut Scott Kelly wears a Microsoft HoloLens headset on the ISS on 20 February 2016, during Project Sidekick, a test of augmented reality for crew procedures. Image: NASA/JSC (Mikhail Kornienko).

Learning adds recognition: a headset that recognises a part, checks that the right bolt is being turned and records each step. For spacewalks (Chapter 33), helmet displays could show suit data, routes and procedures. Such systems must be verified like flight software, because a wrong overlay can be worse than none.

## 51.8 Validating an AI Discovery

A discovery made with machine learning must pass the same tests as any other, and some extra ones. A network's score is a ranking, not a probability that an object is real, until it is calibrated.

Three kinds of evidence matter. **Completeness** says how many real objects the method finds, measured by injecting fake signals into real data, as in the crater example. **Reliability** says how many of the claimed objects are real, measured with known false positives and with independent data. **Physical plausibility** asks whether the object makes sense, such as a transit shape that fits its star. Learned models can also exploit quirks of a data set, such as a camera artefact that happens to correlate with a class. This **leakage** (Chapter 43) is found only by careful checks.

Status needs the same care (Table 51.2, Table 51.4, Table 51.5). Most science results so far come from archives on the ground, with people checking the final list. Onboard science autonomy is operational only in narrow, well-tested forms, such as AEGIS and PIXL. AI for crew health and assistance is still at the stage of demonstrations and prototypes.

### PRACTICE: VALIDATING AN AI DISCOVERY BEFORE PUBLICATION

1. **Freeze the test set.** Set aside labelled data that the model never sees during development, split by object, region or time so that neighbours cannot leak between sets.
2. **Inject and recover.** Insert synthetic signals — transits, craters, lenses — into real data and run the full pipeline. Report completeness against signal strength and size.
3. **Beat a baseline.** Compare with the simplest established method, such as BLS with hand-made vetting, template matching or a $\chi^2$ fit. Report both on the same data.
4. **List the false-positive scenarios.** For each candidate, check every known impostor: eclipsing binaries, background stars, artefacts, shadows, unmodelled minerals. Use evidence the model did not see, such as centroids, spectra or other instruments.
5. **Calibrate the score.** Convert scores to probabilities with held-out data, then combine them with prior rates to estimate the chance that each object is a false positive.
6. **Explain and ablate.** Check which inputs drive a decision, and remove them one at a time. A planet score that depends on a detector edge is a warning.
7. **Confirm independently.** Seek follow-up with a different method or instrument, or ask an independent team to repeat the analysis.
8. **Publish everything needed to repeat it.** Release code, model weights, data versions, thresholds and the selection function, so that later studies can correct for what the method missed.

### KEY TAKEAWAYS

- Science data now outgrow human attention: Rubin can send up to about 7 million alerts a night, and Kepler's final search produced 34 032 signals. Machine learning acts as a filter and ranker in a discovery funnel; people make the final decision.
- Learned vetting finds planets that rules miss. AstroNet found Kepler-90i in 2017, and ExoMiner validated 301 Kepler planets. In this chapter's test, a small network kept 84.8 % of planets at 95 % precision, against 43.7 % for hand-made tests.
- Training sets must represent the data. Random-forest photometric redshifts trained only on bright, nearby galaxies had 19.4 % outliers instead of 12.0 %.
- Speed creates science: transients must be classified within days, and neutron-star mergers can now be characterised in about one second.
- Automated catalogues of craters, boulders and minerals now cover whole planets, but detectors must be calibrated. Ignoring completeness made a 50-million-year surface look 17 million years old.
- Onboard science autonomy is operational in narrow forms: AEGIS selects laser targets and PIXL lengthens its scans on Mars without waiting for Earth.
- A Mars crew faces a ground loop of over an hour and two weeks without contact at conjunction. Medical decision support, guided ultrasound, assistants and predictive maintenance are promising, but mostly prototypes in October 2026.

### REVIEW QUESTIONS

1. Explain the discovery funnel with Kepler's numbers. Why does a 7.1σ threshold still leave tens of thousands of false signals?
2. A planet gives a depth of 200 ppm and 3-hour transits. The noise is 300 ppm per 30-minute point. How many transits are needed for SNR 7.1?
3. In Figure 51.4, why does the neural network gain most at low signal-to-noise ratio? Which false positives remain hard, and how are they rejected in practice?

4. Rubin sends 7 million alerts in a 10-hour night. If a broker's filter has a 0.5 % false-positive rate for "young supernova", how many false candidates does it pass? Why should brokers report probabilities rather than labels?
5. Define $\sigma_{\mathrm{NMAD}}$ and the outlier fraction for photometric redshifts. Why did training on bright, nearby galaxies hurt faint galaxies most?
6. What does simulation-based inference avoid writing down? What is its main weakness?
7. Using the lunar chronology function, estimate $N_1$ for surfaces aged 0.05, 1 and 3.5 billion years. Why does the exponential term matter only for old surfaces?
8. Explain how completeness and false positives bias crater-count ages. How are they measured?
9. Write the linear mixing model with its constraints. How can the fit residual reveal a mineral missing from the library?
10. List the conditions that decide whether a Mars crew must act without help from Earth. Which AI tools would you fly first, and how would you verify them?

CHAPTER

# 52 AI in Spacecraft Design, Manufacturing, Launch and the Space Business

Image: NASA/MSFC (Fred Deaton)

*Long before a spacecraft flies, AI is at work in the engineering behind it. Algorithms now propose structures, search large design spaces with few simulations, check requirements, watch metal printers layer by layer and screen CT scans for flaws. On the launch range, learning helps with engine health, weather and scheduling. Flight-critical decisions stay with optimisation and fixed rules. In the business, models turn satellite data into market signals, price launch risk and size constellations. This chapter explains each method from first principles. Small experiments, run from scratch, measure what each method gains and where it fails. It closes with a safe way to bring AI tools into an engineering team. Status is given as of October 2026.*

IN THIS CHAPTER

52.1 AI Across the Life Cycle
52.2 Design: Letting Algorithms Propose Shapes
52.3 Exploring the Trade Space
52.4 AI for Systems Engineers
52.5 Manufacturing and Test
52.6 Launch: Where Learning Stays on the Ground
52.7 The Space Business
52.8 Bringing AI into an Engineering Team

A spacecraft is shaped by thousands of decisions taken years before launch. Each is made with incomplete information and a fixed budget. Artificial intelligence (AI) now helps with many of them. It does not replace the engineer. It proposes options, searches faster and checks more data than people can.

The hero image shows the starting point of much of this work: an engineer at NASA's Marshall Space Flight Center explaining rocket-engine hardware made by additive manufacturing (3-D printing). Printing frees the designer from the limits of machining. An algorithm can then draw shapes that no person would draw. Printing also creates new risks, such as hidden pores in the metal, which learned models help to find.

This chapter follows a spacecraft from design to the market. Systems engineering is covered in Chapter 4, structures in Chapter 5, launch vehicles in Chapters 34–36 and spaceports in Chapter 37. Learning methods are explained in Chapters 43 and 44, and computing hardware in Chapter 45. Trust, assurance and governance are the subject of Chapter 53.

## 52.1 AI Across the Life Cycle

The key idea of this chapter is simple. **AI proposes; physics, test and people verify.** A learned model can be wrong in ways that are hard to see. So nothing it produces enters the controlled design without passing the same checks as human work.

Figure 52.1 shows where AI enters the life cycle. Each stage has its own tools, but they share two features. First, every output passes a **verification gate**: a high-fidelity analysis, a test, an inspection or a review by a responsible engineer. Second, every stage writes data into a **digital thread** — one linked record of requirements, models, analyses, build records and test results.

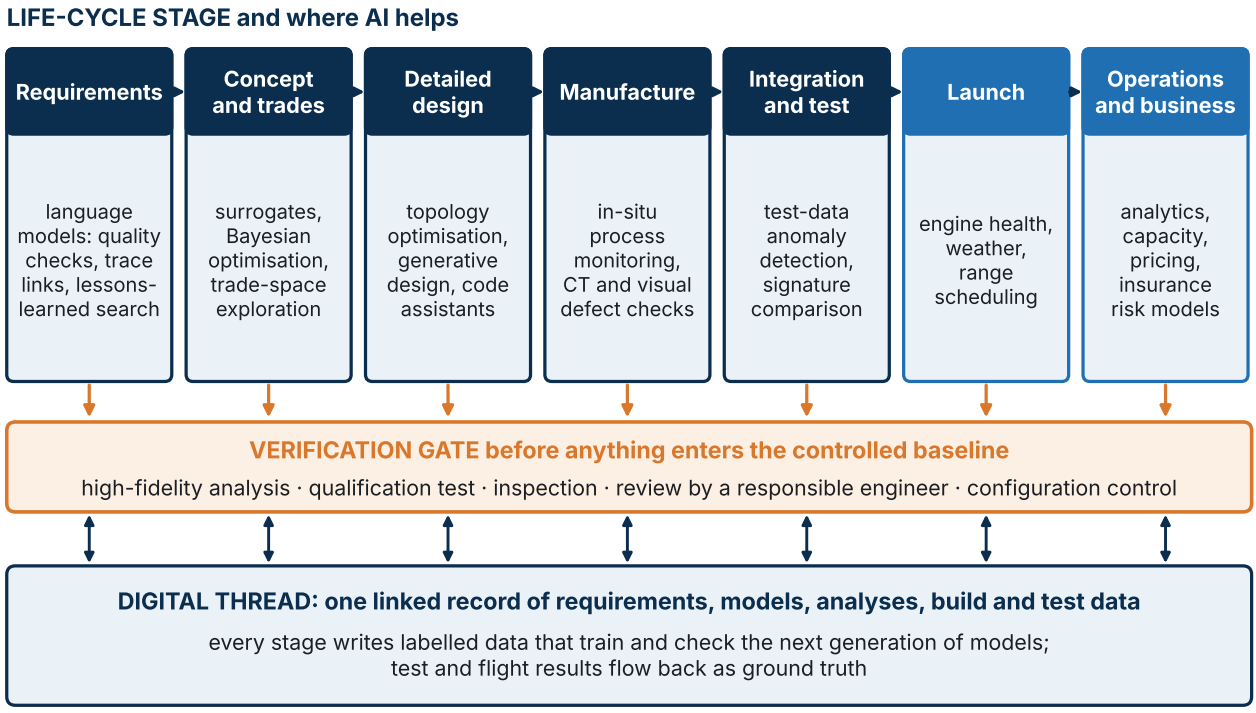



**FIGURE 52.1** Where AI helps in the life cycle of a spacecraft. Each stage has its own tools, but every output passes a verification gate before it enters the controlled design. A digital thread links all stages, so that test and flight results return as labelled data.

### 52.1.1 Where the Value Lies

Early choices fix most of the cost of a programme (Chapter 4), so improving the concept saves more than speeding up drawing. Yet early stages have the least data, and models there must work from physics and small samples. Later stages, such as inspection, have plenty of data but little freedom to change the design.

### 52.1.2 Operational, Demonstrated or Research?

Claims about AI range from routine use to press releases about prototypes. Table 52.1 sorts the main uses by status as of October 2026. "Operational" means used routinely on real programmes. "Demonstrated" means shown on real hardware or data but not yet routine. "Research" means published studies and prototypes.

**TABLE 52.1** Main uses of AI in design, manufacturing, launch and business, with an example and their status as of October 2026.

| Use | Main method | Example | Status (Oct 2026) |
|---|---|---|---|
| Structural brackets and mounts | Topology optimisation, generative design | NASA Goddard "evolved structures"; RUAG Sentinel-1 bracket | Operational |
| Antennas | Evolutionary search | NASA ST5 antennas (2006) | Flown |
| Whole engines from requirements | Rule-based computational models | LEAP 71 engines, 5–20 kN | Demonstrated (hot fire) |
| Fast simulation | Neural surrogates and operators | Commercial tools (PhysicsX, Neural Concept) | Early operational |
| Requirements and documents | Language models, retrieval | ESA SpaceQA, NASA drafts | Pilots and research |
| Printer monitoring | Image segmentation | ORNL Peregrine | Operational in labs and industry |
| CT and visual inspection | Classifiers | Factory inspection | Operational in industry |
| Engine health | Redlines; learned detectors on the ground | Shuttle AHMS (2006–11) | Redlines flown; learning on ground |
| Satellite-data analytics | Detection, regression | Retail, shipping, commodities | Operational |
| Insurance and pricing | Statistical risk models | Launch and in-orbit risk, parametric cover | Operational |

## 52.2 Design: Letting Algorithms Propose Shapes

The key idea of this section is that a computer can search shapes, not just check them. The engineer states the loads, the interfaces and the forbidden zones. The algorithm finds where material is needed.

### 52.2.1 Evolutionary Design

The first computer-designed hardware in space was a small wire antenna. NASA's Space Technology 5 (ST5) mission needed an X-band antenna with a wide beam for three small satellites. Around 2004 a team at NASA Ames used **evolutionary algorithms** to design it (Figure 52.2).

An evolutionary algorithm copies natural selection. It starts with a population of random designs, here wire shapes with a few bends. Each design is scored by a simulation; the score is called its **fitness**. The best designs are copied with small random changes (**mutation**) and mixed together (**crossover**). After many generations the population converges on good designs. The method needs no gradients, so it works with any simulator, but it may need tens of thousands of evaluations.

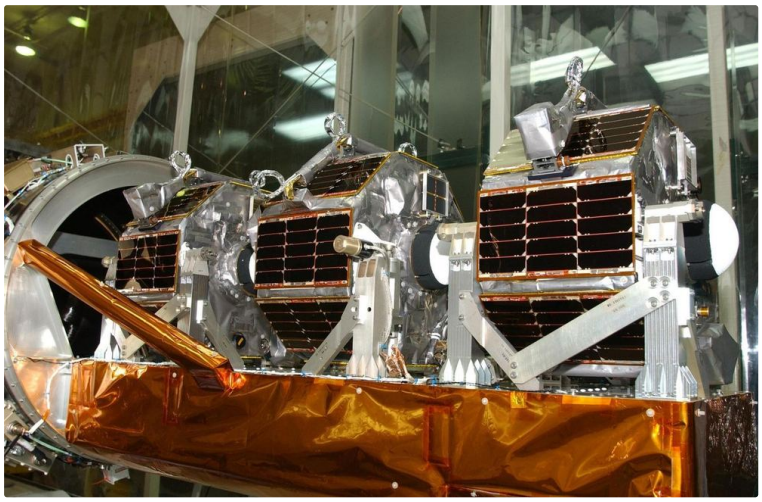


**FIGURE 52.2** The three ST5 microsatellites on their payload structure at Vandenberg Air Force Base in February 2006, a month before launch. Each carried an X-band antenna whose shape was found by an evolutionary algorithm. Image: NASA/KSC.

LANDMARK: THE EVOLVED ANTENNAS OF ST5 (2006)

- Jason Lohn of NASA Ames, Gregory Hornby of the University of California, Santa Cruz, and Derek Linden of JEM Engineering ran two evolutionary algorithms. One bent single wires; the other grew branching shapes.
- The winning antennas looked like bent paper clips. No human designer would have drawn them.
- Mid-project, a new launch vehicle put the mission into a much lower orbit, and the requirements changed. The team changed mainly the fitness function. Within a month of the change, a new antenna had been evolved and prototyped.
- On a spacecraft mock-up, two evolved antennas reached 93 % efficiency, against 38 % for two conventional quadrifilar helix antennas designed by a contractor. The evolved design also took about three person-months to design and build, against about five.
- The three satellites were launched on 22 March 2006 on a Pegasus XL. Each carried an evolved antenna, the first computer-evolved hardware in space.

### 52.2.2 Topology Optimisation

For structures, the most widely used method is **topology optimisation**. It decides where material should go inside a fixed design space. The space is divided into small finite elements. Each element $e$ gets a density $\rho_e$ between 0 (empty) and 1 (solid).

The popular **SIMP** method (solid isotropic material with penalisation) links the stiffness of an element to its density. The rule is

$$E(\rho_e) = E_{\min} + \rho_e^{\,p}\,(E_0 - E_{\min}), \qquad p \approx 3.$$

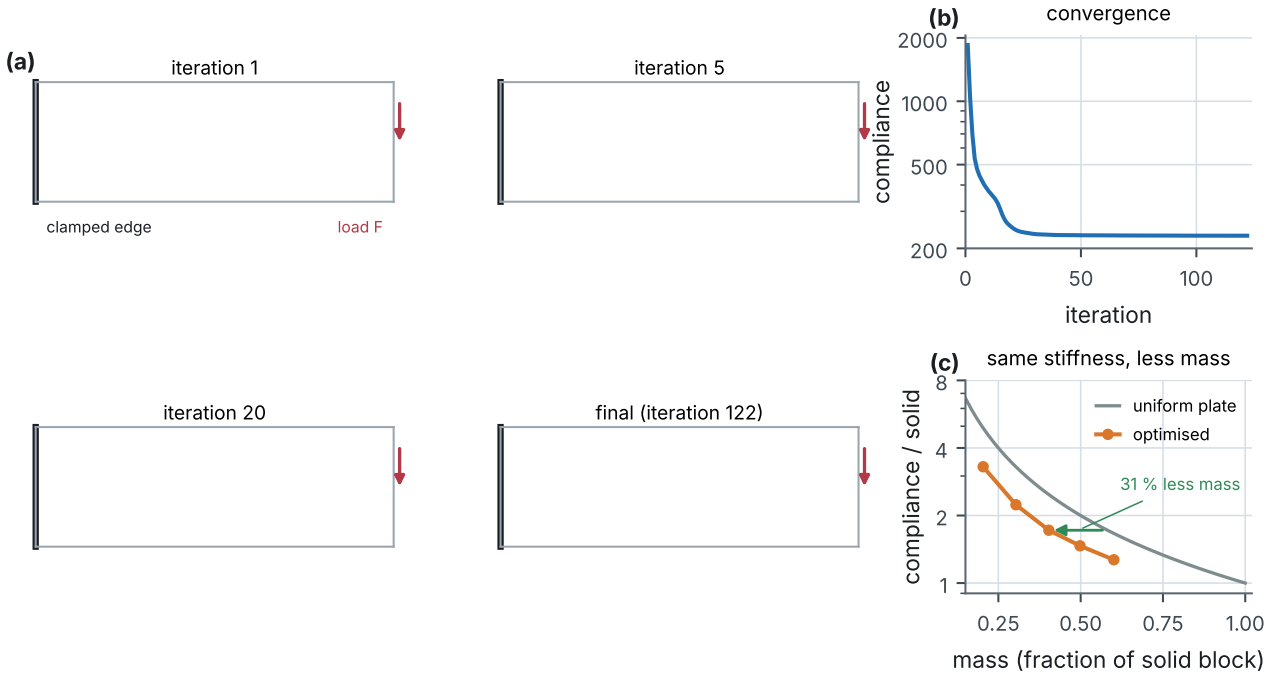
Here $E_0$ is the stiffness of solid material and $E_{\min}$ a tiny value that keeps the maths stable. The penalty $p$ makes half-dense elements poor value: they cost half the mass but give only an eighth of the stiffness. The optimiser therefore pushes elements towards 0 or 1.

The usual goal is the stiffest structure for a given mass. Stiffness is measured by its inverse, the **compliance** $c = \mathbf{F}^{\top}\mathbf{u}$: the work done by the load $\mathbf{F}$ through the displacements $\mathbf{u}$. The problem is

$$\min_{\rho} \; c(\rho) \quad \text{subject to} \quad \mathbf{K}(\rho)\,\mathbf{u} = \mathbf{F}, \qquad \sum_{e} \rho_e v_e \le V^{*},$$

where $\mathbf{K}$ is the stiffness matrix, $v_e$ the volume of an element and $V^{*}$ the allowed volume. The gradient has a simple form. For each element, $\partial c / \partial \rho_e = -p\,\rho_e^{p-1}(E_0 - E_{\min})\,\mathbf{u}_e^{\top}\mathbf{k}_0\mathbf{u}_e$, where $\mathbf{k}_0$ is the stiffness matrix of a solid element. Elements that store much strain energy gain material; the others lose it. A **filter** averages the densities over a small radius, which stops chequerboard patterns and sets a minimum member size.

The method is classical optimisation, not machine learning. It is still the engine under most "AI-designed" parts, so it belongs at the start of this chapter. Figure 52.3 shows it at work on a cantilever, clamped on the left and loaded at the right. The code is about a hundred lines of numpy. It follows the well-known "88-line" program published by Andreassen and colleagues in 2011.




SIMP method (penalty p = 3, density filter radius 1.6 elements, optimality-criteria update) on 120 × 40 four-node plane-stress elements, written in numpy/scipy after Andreassen et al. (2011). Compliance in (c) is recomputed for the black-and-white design (densities > 0.5 set to solid). The uniform plate is a plate of reduced thickness, whose stiffness scales with mass.

**FIGURE 52.3** Topology optimisation of a cantilever with 40 % of the material, run from scratch. (a) The density field after 1, 5, 20 and 122 iterations; a truss-like structure emerges. (b) The compliance falls by a factor of eight in the first 30 iterations. (c) For the same stiffness, optimised designs need 24–33 % less mass than a plate of uniform thickness.

The run took 122 iterations. Compliance fell from 1 851 to 230 units, and most of the gain came in the first 30 steps. The final design is a set of struts and ties, like a bridge truss.

How much mass does this save? Compare a plate of the same outline, thinned until it is as stiff as the optimised part. Rounded to solid or empty, the optimised design uses 40.3 % of the material of a solid block and is 1.72 times as compliant. An equally stiff uniform plate needs 58.2 %, so the saving is 30.6 % of the mass; across volume fractions from 0.2 to 0.6 it was 24–33 %.

**EXAMPLE: MASS SAVING FROM TOPOLOGY OPTIMISATION**

A small satellite carries 25 aluminium brackets machined as plates of uniform thickness, 12.0 kg in total. Each bracket is replaced by a topology-optimised part with the same stiffness, using the ratio from Figure 52.3.

1. **Mass ratio at equal stiffness.** The optimised design uses 40.33 % of the solid material; the equally stiff plate uses 58.15 %. The ratio is 0.4033 / 0.5815 = **0.694**.
2. **New mass.** 12.0 kg × 0.694 = **8.32 kg**. The saving is **3.68 kg** per satellite.
3. **Natural frequency.** The stiffness $k$ and the carried mass are unchanged, so $f = \frac{1}{2\pi}\sqrt{k/m}$ stays the same or rises slightly.
4. **Launch value.** At a reported rideshare price of about $7 000 per kilogram in 2026, 3.68 kg is worth about **$25 800** per satellite. For a batch of 100 satellites it is 368 kg.

The mass saving is real but modest per part. Its main value is often margin: kilograms freed for propellant, payload or growth late in the programme.

### 52.2.3 Generative Design in Practice

Commercial **generative design** tools wrap topology optimisation in a friendlier workflow. The engineer marks surfaces to keep, such as bolt holes, and zones to leave empty, such as an optical path. The tool returns many shapes, each checked for stress, frequency and how it can be made.

Industry uses such tools widely. RUAG Space, now Beyond Gravity, topology-optimised an antenna support bracket for Sentinel-1 and printed it in aluminium. It was half the mass of the original part and exceeded its stiffness requirement by more than 30 %; it was qualified for flight around 2016.

At NASA's Goddard Space Flight Center, research engineer Ryan McClelland has used such tools since about 2020 to design what he calls **evolved structures**. NASA reports that they can be up to two-thirds lighter than conventional parts, and that designs take an hour or two. They have been used on balloon telescopes, instruments and a mount for the Mars Sample Return programme, which lost its funding in 2026 (Chapter 29).

The best-documented case is a bracket for the **EXCITE** balloon telescope (Table 52.2 and Figure 52.4). The bracket holds a 1.35 kg tip/tilt mirror stage to the back of the telescope. It must survive 10 g vertically and 3 g sideways. Its first natural frequency must be above 100 Hz, and its target mass was 0.2 kg.

**TABLE 52.2** The EXCITE tip/tilt bracket: four human design iterations against generative designs that met the 0.2 kg target and the 100 Hz requirement. Data: R. McClelland, NASA Goddard (2023).

| Design | Time | Mass (kg) | 1st mode (Hz) | Stress (MPa) | Cost, lead time |
|---|---|---|---|---|---|
| Human, best of 4 | 2 days | 0.27 | 65 | 103 | No quote |
| Generative, milled | ≈ 1 hour | 0.20 | 147 | 14.8 | $1 000, 3 days |
| Generative, printed | ≈ 1 hour | 0.20 | 177 | 11.2 | $2 000, 3 weeks |

In four iterations, no human design that could be manufactured met the mass target. The one design that met both targets could be neither machined nor printed. The generative designs met both, with seven to nine times less peak stress. McClelland stresses that people must still review every result. Left alone, the algorithm can make members too thin to build or inspect.

**FIGURE 52.4** The EXCITE balloon-borne telescope hanging in a hangar at NASA's Columbia Scientific Balloon Facility in Fort Sumner, New Mexico, before its test flight on 31 August 2024. Several of its mounts were evolved structures designed with generative software. Image: NASA/GSFC.

### 52.2.4 Computational Engineering of Whole Systems

A rocket engine couples flow, combustion, heat transfer and structure. The Dubai company **LEAP 71** calls its approach **computational engineering**. Its software, Noyron, encodes engineering rules, physics and manufacturing limits as code, and generates a whole engine from a short specification. It is closer to an expert system than to a neural network: "AI-designed" can mean many things.

The results are measured by hot fire, not by claims. In June 2024 LEAP 71 fired a 5 kN kerosene–oxygen engine at Airborne Engineering in Westcott, UK. In December 2024 it fired a printed aerospike of the same thrust, and the measured pressures and temperatures were used to tune the model. In December 2025 the company reported two 20 kN methane–oxygen engines printed in a copper alloy. They went from specification to first firing in under three weeks. The bell-nozzle engine reached more than 93 % combustion efficiency. The aerospike reached its design chamber pressure of 50 bar, but start-up problems limited it to one firing. A 200 kN aerospike was printed in March 2026 for hot-fire tests later that year, and a 2 000 kN design has been shown.

### 52.2.5 Designing for the Printer

Printing changes what is cheap. In laser powder-bed fusion, a laser melts metal powder in layers about 30–60 µm thick. Complex internal channels cost little more than plain walls. But steep overhangs need supports, thin walls warp, and trapped powder and internal flaws must be found. Good design tools include these rules, so the optimiser proposes shapes that can be printed, cleaned and inspected.

## 52.3 Exploring the Trade Space

The key idea of this section is sample efficiency. A detailed simulation of a spacecraft may take hours. Learning methods help by choosing which simulations to run.

### 52.3.1 Multidisciplinary Design Optimisation

A spacecraft is a coupled system. A larger solar array gives more power but adds mass, drag and inertia. That changes the attitude control system, which changes power needs again. **Multidisciplinary design optimisation (MDO)** treats the whole design as one problem:

$$\min_{\mathbf{x}} f(\mathbf{x}, \mathbf{y}) \quad \text{subject to} \quad g_j(\mathbf{x}, \mathbf{y}) \le 0, \qquad \mathbf{y}_i = A_i(\mathbf{x}, \mathbf{y}_{j \neq i}).$$

Here $\mathbf{x}$ holds the design variables, such as orbit altitude. Each discipline $A_i$ — structures, thermal, power, orbit — computes outputs $\mathbf{y}_i$ that others need, and the constraints $g_j$ hold the requirements. Frameworks such as NASA's open-source OpenMDAO link the discipline codes and compute gradients. ESA's Concurrent Design Facility, open since 1998, does the same job with people and linked models.

When each evaluation is cheap, any optimiser works. When each takes hours, the number of runs becomes the limit. A surrogate model then stands in for the expensive analysis.

### 52.3.2 Surrogate Models

A **surrogate** (or emulator) is a fast approximation fitted to a modest number of expensive runs. Engineers have used polynomial "response surfaces" for decades. **Gaussian processes (GPs)** are now the standard choice for small data sets (Chapter 43). A GP predicts both a mean $\mu(\mathbf{x})$ and an uncertainty $\sigma(\mathbf{x})$. The uncertainty grows away from the runs already made. That is exactly what is needed to decide where to look next.

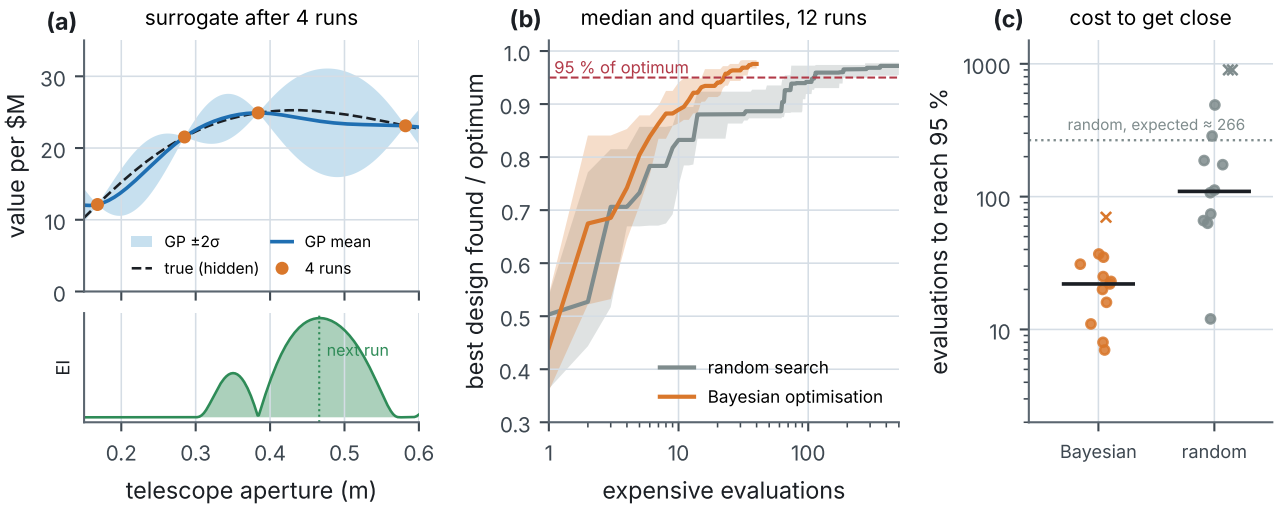
### 52.3.3 Bayesian Optimisation

**Bayesian optimisation (BO)** uses the GP to choose each new run. It scores every candidate design by an **acquisition function**. The most common is the **expected improvement** over the best value so far, $f^{*}$:

$$\mathrm{EI}(\mathbf{x}) = \left(\mu - f^{*}\right)\Phi(z) + \sigma\,\phi(z), \qquad z = \frac{\mu - f^{*}}{\sigma},$$

where $\Phi$ and $\phi$ are the cumulative and density functions of the standard normal distribution. The first term rewards designs that are predicted to be good. The second rewards designs that are uncertain and might be good. The run with the highest EI is made, the GP is refitted, and the loop repeats.

Figure 52.5 tests the method on a toy model of an Earth-observation smallsat. It stands in for an expensive simulation. There are four design variables: orbit altitude, telescope aperture, solar-array area and propellant mass. Lower orbits give sharper images but decay sooner unless propellant is carried. A larger aperture sharpens images but adds mass and heater power. The goal is image value per million dollars over the satellite's life. The best design lies at 428 km with a 0.44 m aperture, a 4.9 m$^2$ array and 16 kg of propellant.




Toy model of an Earth-observation smallsat standing in for an expensive simulation: four variables (altitude 400–800 km, aperture 0.15–0.6 m, array 1–6 m², propellant 0–60 kg); value = lifetime × images per day × value per image / cost. Gaussian process with a Matérn kernel (scikit-learn), expected-improvement (EI) acquisition, 6 Latin-hypercube starts, 40 evaluations; crosses = runs that did not reach 95 %. In (a) the other three variables are held at their optimum values.

**FIGURE 52.5** Bayesian optimisation against random search on a toy smallsat design model. (a) The Gaussian-process surrogate after four runs, along one variable, with its uncertainty band; expected improvement (EI) picks the next run where the model is both promising and uncertain. (b) Best design found against the number of expensive evaluations, over 12 independent runs. (c) Evaluations needed to come within 5 % of the optimum.

Only 0.38 % of random designs come within 5 % of the optimum. Random search therefore needs about 266 evaluations on average to find one. Bayesian optimisation needed a median of 22, and 11 of its 12 runs got there within 40. After 40 evaluations its median result was 97.6 % of the optimum, against 88.6 % for random search. Random search needed about 500 evaluations to reach 97 %.

### EXAMPLE: SIMULATIONS SAVED BY BAYESIAN OPTIMISATION

Suppose each evaluation in Figure 52.5 is a coupled thermal, structural and orbit simulation taking 8 hours on 256 processor cores.

1. **Runs to get within 5 % of the optimum.** Random search: about **266** on average. Bayesian optimisation: median **22**. The saving is 244 runs.
2. **Computer time.** 244 runs × 8 h = **1 952 hours**, or 81 days if run one after another.
3. **Core-hours.** 1 952 h × 256 cores ≈ **500 000 core-hours**. At an assumed $0.03 per core-hour, that is about **$15 000** of computing.
4. **Calendar time.** BO is sequential: 22 runs take 7.3 days. Random runs can be spread over many machines; with 10 in parallel, 266 runs take 8.9 days. Batch versions of BO choose several runs at once to close this gap.
5. **The real gain.** The computing cost is small. The calendar time and the engineer's attention are what matter, and BO gives a better design in the same time.

BO works best with up to about 20 variables and a smooth objective. With hundreds of variables, gradient-based MDO is usually better.

## 52.3.4 Neural Surrogates and Neural Operators

A GP becomes slow beyond a few thousand runs, because its cost grows with the cube of the number of points. For large simulation archives, **neural surrogates** take over. A neural network is trained on thousands of past simulations. It then predicts a new case in milliseconds. **Neural**

**operators**, such as the Fourier neural operator (Chapter 44), learn a mapping from one field to another. For example, they can map a wing shape and flow conditions to the whole pressure field around it.

Uses include aerodynamic databases for launchers and capsules, thermal maps of electronics boxes and fast stress checks inside optimisation loops. Investors have noticed: PhysicsX (London) raised $135 million in June 2025 and Neural Concept (Switzerland) $100 million in December 2025. Both sell learned simulation tools to aerospace and other industries.

Two cautions apply. First, a surrogate is only trustworthy inside the range of its training data; Chapter 44 shows how errors explode outside it. Second, the final design must be checked with the full, validated simulation and by test. A surrogate is a tool for search, not for certification.

## 52.4 AI for Systems Engineers

The key idea of this section is that language models are good at reading and drafting text, and poor at guaranteeing truth. They help most where a person checks every output.

### 52.4.1 Checking Requirements

A good requirement is necessary, clear, verifiable and states one thing (Chapter 4). Bad ones contain "weak" words such as *adequate*, *as appropriate* or *user-friendly*, open values such as "TBD", or two requirements joined by "and". Rule-based checkers have flagged such patterns since the 1990s. NASA's Automated Requirement Measurement tool, for example, counted weak phrases and options in specifications.

Language models go further. They can judge whether a requirement is testable, suggest a measurable rewrite and find conflicts between documents. Researchers at the University of Strathclyde trained **SpaceTransformers** (2021). These language models were further trained on space-systems text and tested on recognising concepts in requirements from the European ECSS standards. Domain training improved results over general models. Still, any rewrite is a suggestion. A wrong unit or a lost tolerance in a requirement can survive into hardware.

### 52.4.2 Searching Lessons Learned

Agencies hold decades of lessons in reports that few people read. **Retrieval-augmented generation (RAG)** combines a search engine with a language model (Chapter 44). The question is used to find relevant passages in a trusted archive, and the model answers from those passages with citations. Researchers working with ESA built **SpaceQA**, a question-answering system over reports from ESA's Concurrent Design Facility. NASA teams have proposed and prototyped similar tools for the NASA Lessons Learned system and for tracing requirements. One team building a NASA retrieval tool heard that some of its answers were wrong. On checking the sources, it found that the official guidance had changed, and the tool was right. The reverse happens too: a RAG system is only as current and as correct as its archive.

### 52.4.3 Models from Text

**Model-based systems engineering** keeps the design in a connected model rather than in documents (Chapter 4). Its language, SysML, gained a textual form in **SysML v2**, formally adopted by the Object Management Group in 2025. A textual language is something language models can write. Research teams have built systems that draft SysML v2 models from natural-language descriptions. They report that a language model alone often produces invalid syntax. Adding a grammar checker and retrieval of correct examples makes the drafts much better. These are research tools in 2026.

### 52.4.4 Code and Tests

Code assistants draft functions, unit tests and scripts. The evidence on productivity is mixed. In a controlled trial published by METR in July 2025, 16 experienced open-source developers completed 246 tasks. When AI tools were allowed, they took 19 % longer, although they believed they had been 20 % faster. Quality is also a concern. A 2022 study had a code assistant complete 89 security-relevant scenarios. About 40 % of the 1 689 programs it produced contained a vulnerability.

For flight software the rules do not change: NASA's NPR 7150.2 and ECSS-E-ST-40 still demand reviews, tests and traceability, whoever wrote the code. Assistants are most useful for ground tools, documentation and test cases. A person decides which tests are meaningful, and the tests themselves check the code.

### 52.4.5 Data Security and Export Control

Engineering data are often export-controlled under ITAR or EAR (US export-control rules) or commercially sensitive. Sending them to a public online service can break the law or a contract, so agencies and primes use approved, access-controlled deployments. NASA's digital-engineering team used commercial language models to draft internal procedures. This saved months of work, and every output was treated as a draft to verify.

Table 52.3 summarises the tools for systems engineers and the check that each one needs.

**TABLE 52.3** AI tools for systems engineers, the main risk of each and the check it needs.

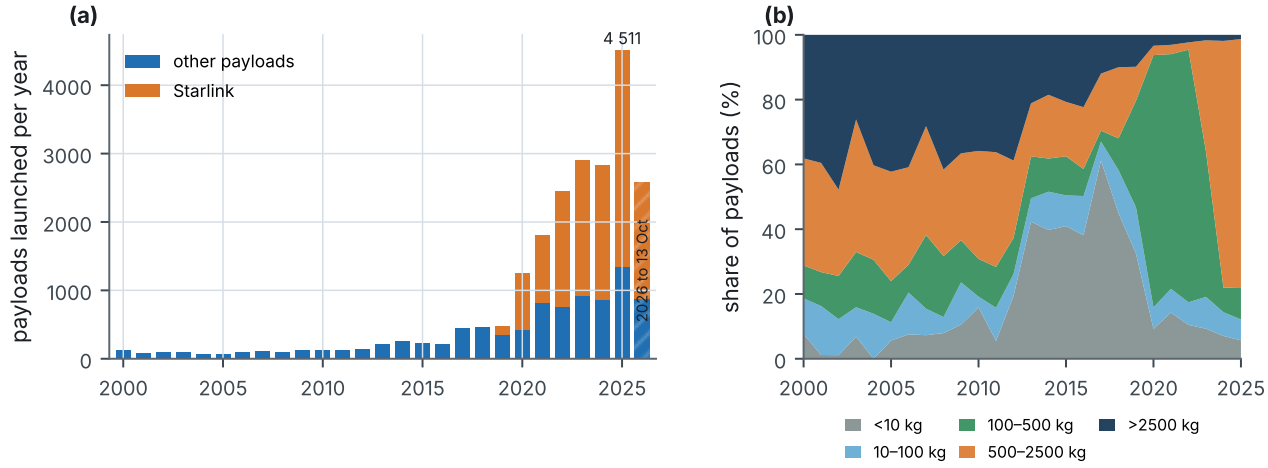
| Task | How AI helps | Main risk | Required check |
|---|---|---|---|
| Requirement quality | Flags vague, compound or untestable text; suggests rewrites | Changed meaning, lost tolerances | Owner approves every change |
| Lessons-learned search | Retrieves and summarises past reports with citations | Out-of-date or invented answers | Read the cited source |
| Traceability | Proposes links between requirements, design and tests | Missing or false links | Review by the systems engineer |
| SysML v2 drafting | Writes model text from descriptions | Invalid syntax, wrong semantics | Grammar check and model review |
| Code and tests | Drafts functions, test cases and scripts | Subtle bugs, security flaws | Standard review, static analysis, test |
| Report drafting | Turns data and notes into a draft | Plausible but wrong numbers | Check every number against its source |

# 52.5 Manufacturing and Test

The key idea of this section is that factories produce data at a scale people cannot inspect by eye. Learning sorts the data so that people look first at what most likely matters. The cost of a missed flaw sets how the sorting is tuned.

## 52.5.1 From One-Off to Mass Production

For most of the space age, satellites were built one at a time. That has changed (Figure 52.6). In 2025, 4 511 payloads were launched, nineteen times as many as in 2015. Starlink satellites made up 70 % of them. Most payloads launched in 2025 weighed 500–2 500 kg, because Starlink's current satellites fall in that class.




Data: GCAT (J. McDowell), payloads by launch year and mass class, updated 13 October 2026.

**FIGURE 52.6** The shift to mass production. (a) Payloads launched each year, with Starlink shown separately; 2026 is to 13 October. (b) The share of payloads by mass class: small satellites rose from 2013, then mass-produced constellation satellites of 500–2 500 kg took over.

Factories now build satellites at rates once reserved for aircraft parts (Table 52.4). SpaceX's filing for its 2026 share sale reported that its plant in Redmond, Washington, averaged about 70 Starlink satellites per week from December 2025 to April 2026. Amazon's factory in Kirkland, Washington, is designed to build up to five satellites per day at peak. Amazon says its test process there cut test time per satellite from months to days.

**TABLE 52.4** Satellite production rates reported for 2025–26.

| Producer | Satellites | Rate | Source and status |
|---|---|---|---|
| SpaceX, Redmond (USA) | Starlink | ≈ 70 per week (Dec 2025–Apr 2026) | SpaceX share-sale filing, 2026 |
| Amazon, Kirkland (USA) | Amazon Leo (formerly Kuiper) | Up to 5 per day at peak | Company, design capacity |
| ICEYE (Finland) | Radar satellites | ≈ 50 per year, 100 planned from 2027 | Company |
| Typical science mission | One spacecraft | One in 5–10 years | — |

At 70 satellites a week, no team can examine every weld, scan and test trace in detail. Statistical process control, automated inspection and anomaly detection become part of the production line. The same tools serve one-off science missions, where each part is precious and a missed flaw may end the mission.

### 52.5.2 Watching the Printer

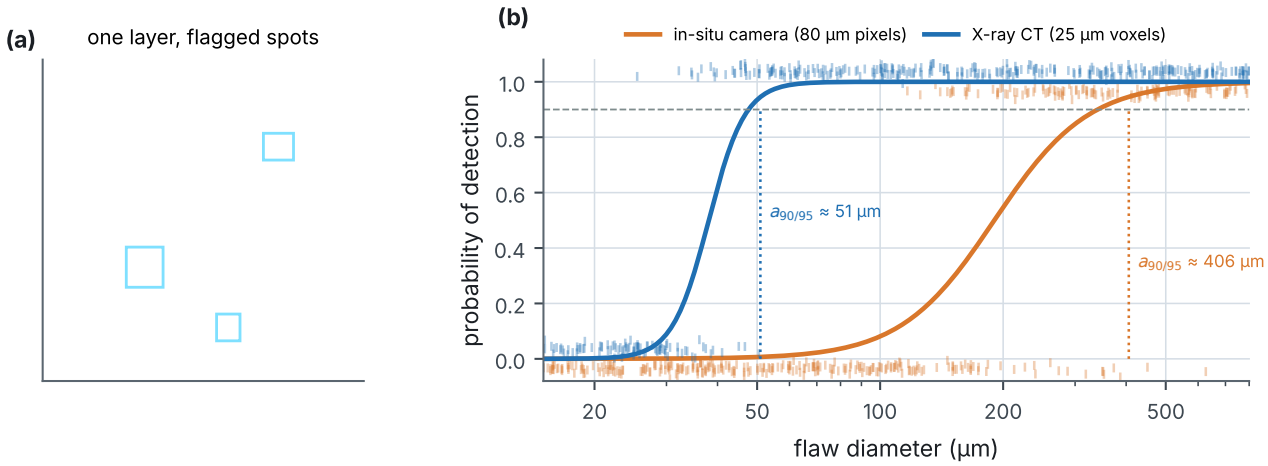
A metal printer builds a part from thousands of layers. Each layer is a chance for a flaw. **Lack-of-fusion** voids form where the laser did not melt enough powder. **Keyhole** pores form where it melted too deeply and trapped gas. Spatter, recoater streaks and warping cause other defects. Cameras, photodiodes and thermal sensors can watch every layer and every melt track. A build may produce tens of gigabytes of images.

Oak Ridge National Laboratory's **Peregrine** software is a leading example. It segments layer images with a convolutional network (Chapter 44) into classes such as powder, part, spatter and streaks, and raises alerts during the build. It was trained using X-ray computed tomography (CT) of finished parts as ground truth, and works with many makes of printer. By 2023, 15 US government laboratories used it, and 14 research licences had been granted to industry.

How small a flaw can such monitoring find? Non-destructive evaluation answers this with a **probability-of-detection (POD) curve**. Many flaws of known size are inspected, and each is recorded as a hit or a miss. A logistic curve in the logarithm of flaw size $a$ is fitted:

$$\mathrm{POD}(a) = \frac{1}{1 + \exp\!\left(-(\ln a - \mu)/\sigma\right)}.$$

The key number is $a_{90/95}$: the flaw size found 90 % of the time, with 95 % confidence. For fracture-critical parts, the structure must tolerate any flaw smaller than this size (Chapter 5).




Synthetic data. (a) Simulated melt-pool emission image of one 7.7 mm × 7.7 mm layer with scan-track texture and three dark flaws; boxes mark pixels more than 4σ below the local median. (b) 400 simulated flaws per sensor (ticks: hits at 1, misses at 0), contrast reduced by blur of one pixel, detection at 3.5σ; logistic hit/miss POD fit in log size (MIL-HDBK-1823A style), a90/95 = 95 % bootstrap bound on the size found 90 % of the time.

**FIGURE 52.7** Simulated in-situ monitoring of powder-bed fusion. (a) A melt-pool image of one layer; boxes mark spots darker than the local background. (b) Hit/miss probability-of-detection curves for an in-situ camera and for X-ray CT, with the 90/95 flaw size of each.

Figure 52.7 shows the idea with simulated data. A camera with 80 µm pixels blurs small flaws into the background. Its $a_{90/95}$ came out at about 406 µm. A CT scan with 25 µm voxels found flaws down to about 51 µm. These numbers are illustrative. Real values depend on the machine, the metal and the sensor, and must be measured for each process. The lesson holds, though. In-situ monitoring is a fast first screen and a record of the build. It does not replace CT for critical parts.

### 52.5.3 Seeing Inside: CT and Visual Inspection

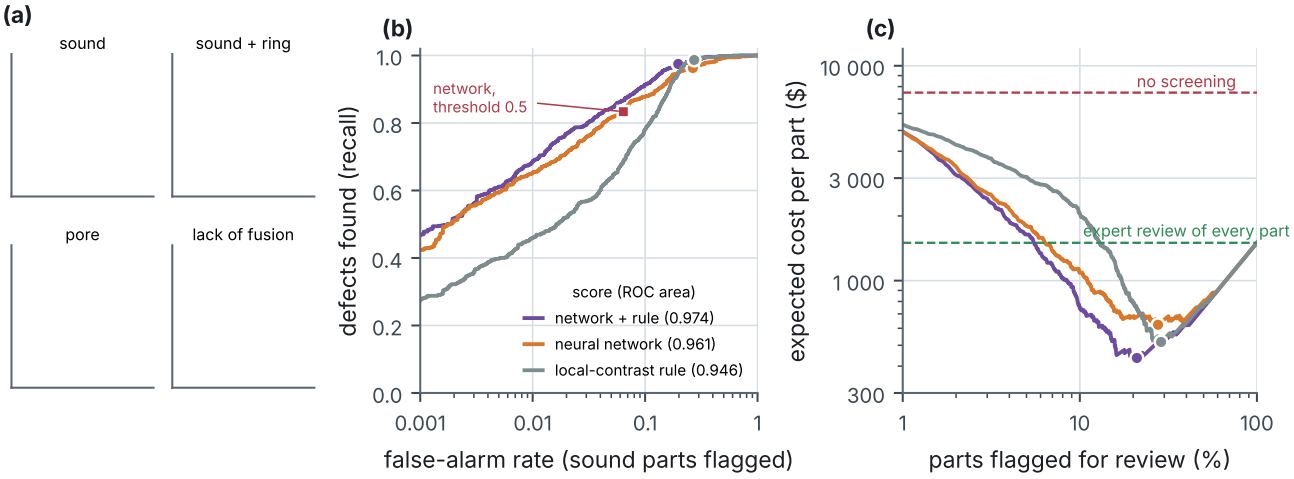
X-ray CT gives a 3-D density map of a part. A single scan holds billions of voxels, and an expert may need an hour to review it. A **classifier** can screen the scan and flag regions that look like pores or lack-of-fusion flaws. The expert then reviews only those regions.

Every screening model must choose a **threshold**. Above it a region is flagged; below it the part passes. The best threshold depends on what errors cost. Let a review cost $C_{\text{rev}}$ and a missed defect cost $C_{\text{esc}}$ (an "escape"). For a defect prevalence $\pi$, the expected cost per part is

$$\text{cost} = (\text{fraction flagged}) \times C_{\text{rev}} + \pi\,(1 - \text{recall})\,C_{\text{esc}},$$

where recall is the fraction of defects caught. If the model's probabilities are well calibrated, decision theory gives a simple rule: flag a part when $P(\text{defect}) > C_{\text{rev}}/(C_{\text{rev}} + C_{\text{esc}})$. With a review costing \$1 500 and an escape \$250 000, the optimal threshold is about 0.6 %, not the 50 % that software uses by default.

Figure 52.8 tests this on simulated CT patches. The patches contain pores and lack-of-fusion flaws. They also contain things that look like flaws but are not: ring and streak artefacts of the scanner, part edges and dense inclusions. Three screens were compared. The first is a classic **local-contrast rule**: smooth the image and flag the darkest spot relative to its surroundings. The second is a small neural network that sees the raw pixels. The third combines the two scores with a logistic regression.




Synthetic 24 × 24-pixel CT patches with build texture, beam hardening, part edges, ring and streak artefacts and noise; defects are pores and lack-of-fusion flaws. Network: two hidden layers (128, 32), scikit-learn, trained on 20 000 patches (30 % defective); tests on 24 000 patches at 3 % prevalence; the combined score is a logistic regression on both outputs. Costs: review \$1 500 per flagged part, escape \$250 000 per missed defect; dots = minimum-cost threshold chosen on a separate validation set.

**FIGURE 52.8** Screening synthetic X-ray CT patches. (a) Examples: sound material, a ring artefact, a pore and a lack-of-fusion flaw. (b) ROC curves of three screens; dots mark the minimum-cost operating points, and the square the network at its default threshold of 0.5. (c) Expected cost per part against the fraction of parts flagged; both baselines are far more expensive.

The combined screen had the largest area under the ROC curve, 0.974, against 0.961 for the network and 0.946 for the rule. But the ranking by cost was closer than the ranking by ROC area. At the lowest-cost threshold, chosen on a separate validation set, the combined screen cost \$521 per part. The rule cost \$543 and the network alone \$709. The network at its default threshold of 0.5 missed one defect in six and cost \$1 381 per part. A well-designed physical rule is a strong baseline; a learned model must beat it where it matters.

**EXAMPLE: COST–BENEFIT OF AN INSPECTION MODEL**

A factory prints 2 000 flight parts a year. About 3 % contain a defect. Reviewing a part's CT scan costs $1 500; an escaped defect costs $250 000 on average, through later test failures, rework and mission risk. Use the results in Figure 52.8.

1. **No screening.** 2 000 × 0.03 × $250 000 = **$15.0 million** a year of expected escape cost.
2. **Expert review of every part.** 2 000 × $1 500 = **$3.0 million** a year, assuming the experts miss nothing.
3. **Combined screen at its best threshold.** It flags 22.2 % of parts (444 reviews) and catches 97.5 % of defects. The cost is 2 000 × $521 = **$1.04 million**, a saving of **$1.96 million** a year against full review.
4. **Escapes.** 2 000 × 0.03 × (1 − 0.975) = **1.5 defective parts** a year still escape. With the network at its default threshold, about **10** would escape.
5. **Sensitivity.** If escapes cost ten times more, the best threshold moves lower and more parts are reviewed. If the model is retrained or the printer changes, the threshold must be checked again.

The saving comes from choosing the threshold with real costs, not from a higher ROC area. For fracture-critical parts, the residual escapes must also be covered by design margins and proof tests.

### 52.5.4 Composites and Electronics

Carbon-fibre parts are checked by **ultrasonic C-scans**, which map echoes from inside the laminate, and by **thermography**, which watches heat spread after a flash. Classifiers learn to flag the patterns of delaminations, porosity and wrinkles. Electronics boards pass through **automated optical inspection (AOI)**, and X-ray machines check hidden solder joints. Deep learning has cut the false calls that used to send many good boards for manual review. Throughout, the threshold is tuned to the cost of escapes, and each model is rechecked on new production lots.

### 52.5.5 Test Data: Vibration and Thermal Vacuum

Every flight unit goes through environmental tests (Chapter 4): vibration, acoustic, shock and thermal vacuum (Figure 52.9). A vibration test of a spacecraft may record hundreds of accelerometer channels. Before and after each full-level run, a **low-level sine sweep** measures the natural frequencies. If a main frequency shifts by more than a few per cent, something may have loosened or cracked. Engineers compare these "signatures" by eye, channel by channel.

Anomaly detection can make this comparison for every channel and flag the few that changed (Chapter 49 explains the methods). In thermal vacuum tests, models of expected temperatures flag a heater that switches too often or a sensor that drifts. Real test failures are rare, so most methods learn what "normal" looks like and flag departures. The test conductor makes the final call.

### 52.5.6 Supply-Chain Risk

A spacecraft contains parts from hundreds of suppliers, some with a single source. **Supply-chain risk analytics** combine purchase records, supplier finances, news, export rules and shipping data. They flag parts at risk of delay, counterfeiting or obsolescence, as a ranked list for buyers and engineers to act on.

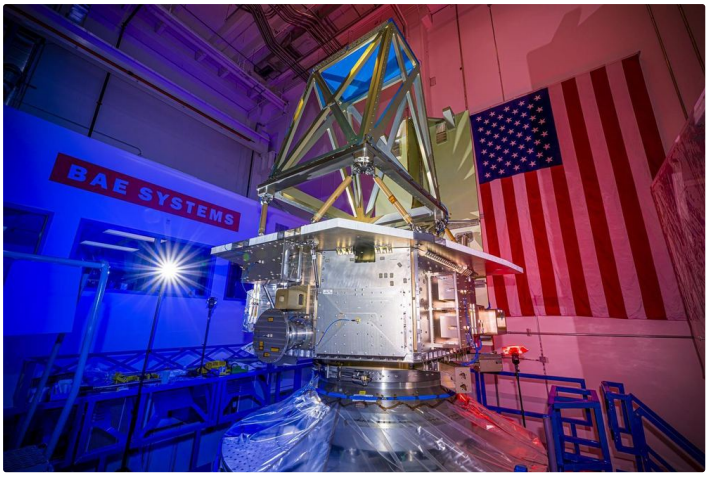



**FIGURE 52.9** The bus structure of NASA's NEO Surveyor on a shaker table at BAE Systems in Boulder, Colorado, during vibration testing in August 2025, with mass simulators standing in for the telescope and sunshade. Image: NASA/JPL (BAE Systems Space & Mission Systems).

## 52.6 Launch: Where Learning Stays on the Ground

The key idea of this section is that the decisions made on board a rocket in flight use optimisation and fixed rules that can be verified. Learning helps on the ground: in tests, forecasts, schedules and analysis.

### 52.6.1 Why Landing Guidance Is Optimisation

A booster must land on a small target at zero speed, within thrust limits. Chapter 48 shows how **convex optimisation** solves this problem. A convex problem has a single optimum, and modern solvers find it in a bounded number of steps. That gives three guarantees that a learned controller lacks. If a solution exists, it will be found. Every constraint will hold. The time needed can be bounded. Masten's tests of the G-FOLD algorithm in 2012–13 used this approach, and SpaceX engineers have described using it to land Falcon 9 boosters (Figure 52.10).

A neural-network policy might land well in simulation, but it offers no such guarantees outside its training conditions (Chapter 48). Learning can still help by giving the solver a good starting guess, which speeds it up while the solver keeps every constraint.

**FIGURE 52.10** A Falcon 9 first stage landing at Landing Zone 40, Cape Canaveral Space Force Station, on 1 October 2026, after launching NASA's SpaceX Crew-13 mission. SpaceX engineers have described its landing guidance as convex optimisation solved in real time. Image: NASA/KSC (NASA/John Kraus).

### 52.6.2 Engine and Vehicle Health

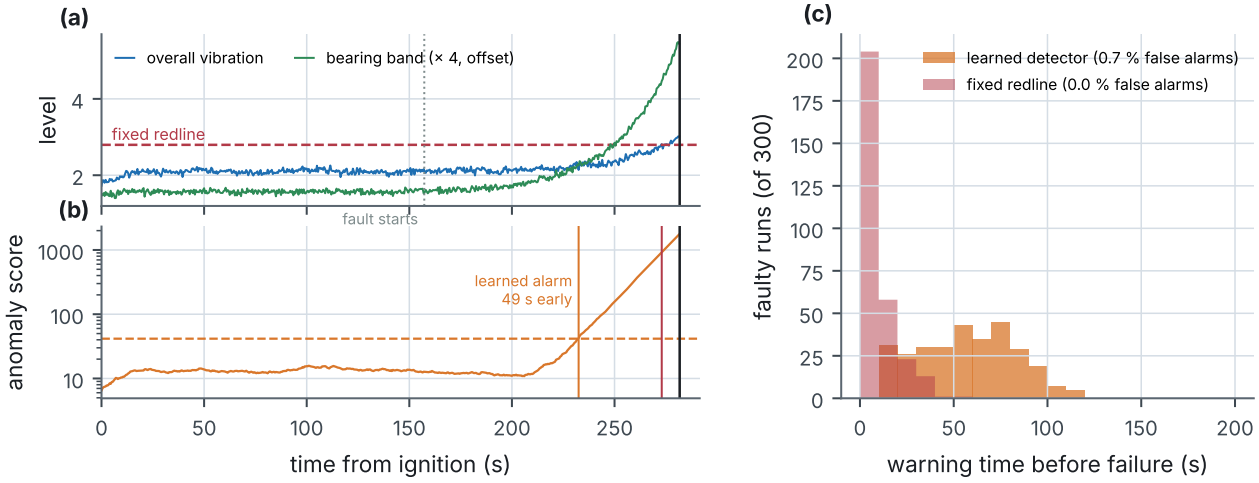
A turbopump that fails can destroy an engine in less than a second. **Redlines** — fixed limits on pressure, temperature or vibration — shut the engine down if any limit is crossed. They are simple, fast and verifiable. But a fixed limit must allow for every normal operating condition, so it is often set high and trips late.

A learned detector can find faults earlier by modelling how each sensor *should* behave at the current operating point. The residual — measured minus predicted — is small in normal running. Several residuals $\mathbf{e}$ are combined by the **Mahalanobis distance**,

$$d^2 = \mathbf{e}^\top \boldsymbol{\Sigma}^{-1} \mathbf{e},$$

where $\boldsymbol{\Sigma}$ is the covariance of the residuals in normal runs. This distance weights each residual by its usual spread and allows for correlations between channels.

Figure 52.11 compares the two approaches on 300 simulated hot fires with a growing bearing fault. The engine throttles between 80 % and 109 % of rated power. The fixed redline, set just above the highest normal vibration, gave a median warning of only 5 seconds; 19 % of failures came with no warning at all. The learned detector gave a median warning of 58 seconds, with false alarms in 0.7 % of normal runs. Earlier warning in test means less damage to expensive test hardware.




Synthetic 500 s hot fires with throttle changes between 80 % and 109 % power level; six channels (two synchronous vibration levels, bearing-band and broadband vibration, temperature, pressure). Learned detector: quadratic model of each channel against power level fitted on 40 normal runs, Mahalanobis distance of the residuals, smoothed (EWMA), threshold set on 200 validation runs. Black line: failure. Faults: bearing wear growing exponentially (time constant 8–50 s); 8 % of runs contain a brief sensor glitch.

**FIGURE 52.11** Health monitoring of simulated turbopump hot fires. (a) One run with a growing bearing fault: overall vibration stays under the fixed redline almost until failure. (b) The learned anomaly score crosses its threshold much earlier. (c) Warning time before failure over 300 faulty runs: the learned detector warns about a minute early, the redline only seconds early or not at all.

Why not fly the learned detector? A false shutdown in flight can lose a mission, and a learned model can fail on conditions it never saw. Learned detectors suit ground tests, such as the RS-25 firings in Figure 52.12, and analysis after flight. On board they are at most advisory, unless they meet the assurance standards of Chapter 53.

**LANDMARK: THE SHUTTLE'S ADVANCED HEALTH MANAGEMENT SYSTEM (2006–11)**

- The Space Shuttle main engine (SSME, also called RS-25) ran its high-pressure turbopumps at tens of thousands of revolutions per minute. A failed bearing or blade could destroy the engine.
- The Advanced Health Management System upgraded the engine controller with digital signal processors and new software. It watched turbopump vibration in real time against a **vibration redline**, and could shut down an engine if vibration grew too high.
- It first flew in monitor-only mode on STS-116 in December 2006 and was then made active.
- The logic was deterministic: fixed signal processing and limits, verified like any flight software.

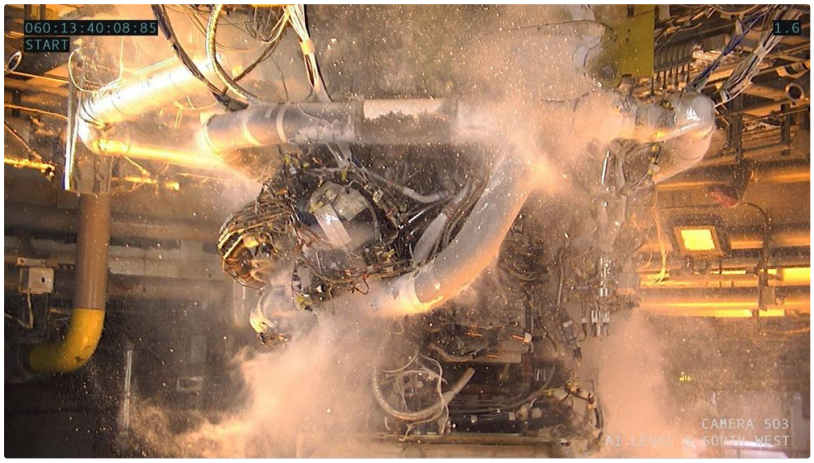



**FIGURE 52.12** A full-duration hot fire of an RS-25 engine on the Fred Haise Test Stand at NASA's Stennis Space Center on 29 February 2024, during certification of new production engines for SLS. Each test records hundreds of sensor channels. Image: NASA/SSC (NASA Stennis).

### 52.6.3 Weather, Lightning and the Range

Weather causes about half of all scrubs at Cape Canaveral (Chapter 37), and lightning rules are the hardest to meet. A model from NASA Marshall and the University of Alabama in Huntsville detected 84 % of lightning 15–30 minutes ahead, with a 22 % false-alarm rate. A study at the Air Force Institute of Technology trained models that could automate the 45th Weather Squadron's daily lightning forecast (Chapter 47). The launch-commit criteria themselves remain fixed rules.

Busy ranges also need scheduling. Florida's Space Coast hosted a record 109 launches in 2025 and planned for 100–120 in 2026. Each launch needs a pad, a window, hazard areas at sea and in the air, tracking assets and recovery ships. Choosing a conflict-free timetable is an optimisation problem. Learning helps predict the inputs, such as the chance of a scrub or of a ship entering a hazard area. Launch teams rehearse these decisions in simulations (Figure 52.13).

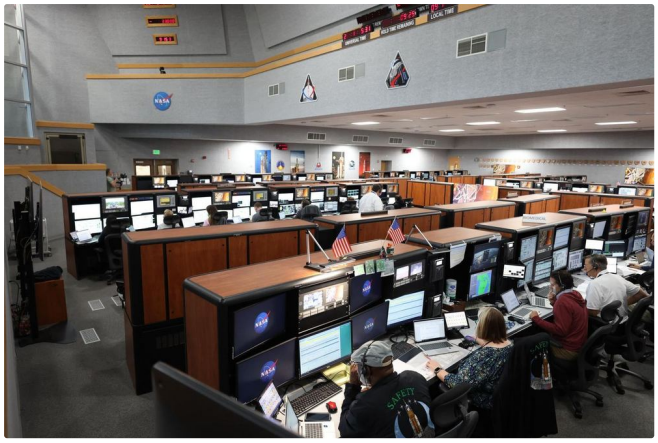


**FIGURE 52.13** The Artemis launch team in Firing Room 1 of the Launch Control Center at Kennedy Space Center during an Artemis II countdown simulation on 8 October 2025. Simulations like this test software, procedures and people against realistic failures. Image: NASA/KSC (NASA/Glenn Benson).

### 52.6.4 Flight Safety: Deterministic by Design

An **autonomous flight safety system (AFSS)** decides on board whether to end a flight that strays (Chapter 34). It uses the vehicle's own navigation and fixed rules, such as "terminate if the impact point crosses this line". Its first operational use was on Falcon 9 in February 2017. Here determinism is the point. Every rule can be reviewed, every path tested and every decision explained. Learning enters only upstream, for example in better models of falling debris used to draw the safety lines.

### 52.6.5 Trajectories and Mission Design

Trajectory optimisation relies on mature numerical methods (Chapter 24). Learning helps by screening: surrogates estimate the propellant for thousands of candidate transfers in seconds, and networks give the solver starting guesses. The final trajectory is always computed and checked with the full physics.

Table 52.5 summarises the division of labour.

**TABLE 52.5** Who decides what during a launch: verifiable logic on board, learning on the ground.

| Function | On board in flight | Role of learning |
|---|---|---|
| Landing and ascent guidance | Convex or classical optimisation | Starting guesses; offline design |
| Engine protection | Fixed redlines and logic | Ground-test anomaly detection; setting limits |
| Flight termination | Rule-based AFSS | Debris and risk models used to set rules |
| Launch weather | None (ground decision) | Lightning, wind and cloud forecasts |
| Range scheduling | None (ground decision) | Predicting scrubs, conflicts and delays |
| Trajectory design | Stored or optimised trajectory | Surrogates, warm starts, screening |

## 52.7 The Space Business

The key idea of this section is that the space business sells answers and manages risk. Learning turns satellite data into decisions. Statistics prices the risk of launch and operations. Optimisation sizes and prices the capacity of constellations.

### 52.7.1 Selling Answers from Orbit

Most commercial value lies not in pixels but in what they reveal (Chapter 46). Analytics companies count cars, ships, aircraft and containers. They estimate crop yields, measure oil in storage tanks from the shadows of their floating roofs, and track ships. Customers include traders, insurers, governments and supply-chain managers.

Satellite-tracked shipping became big business. In November 2024 the analytics company Kpler agreed to buy Spire's maritime business, which tracks ships from orbit, for about $241 million. The deal closed in 2025. The best-studied use is in finance (see the box).

> **DATA BOX: SATELLITE CAR-PARK COUNTS AND SHARE PRICES**
>
> A study published in 2025 in the *Journal of Financial and Quantitative Analysis* tested whether satellite data give investors an edge.
>
> - **Data:** about 4.7 million daily counts of cars in the car parks of 67 078 stores of 44 major US retailers, 2011–2017. The counts came from RS Metrics and Orbital Insight, which used image analysis to count cars.
> - **Signal:** an unusual change in how full the car parks were during a quarter predicted the retailer's earnings.
> - **Returns:** in the three days around earnings announcements, retailers with fuller car parks beat the market by 1.66 %. Those with emptier car parks fell behind it by 3.10 %. The gap was 4.76 %.
> - **Who gained:** mostly sophisticated investors who bought the data, above all short sellers betting on bad news. Buying by individual investors became less well informed, and shares became less liquid around the reports.
> - **Lesson:** satellite analytics carry real information. But unequal access to data can widen the gap between large and small investors rather than making prices fairer.

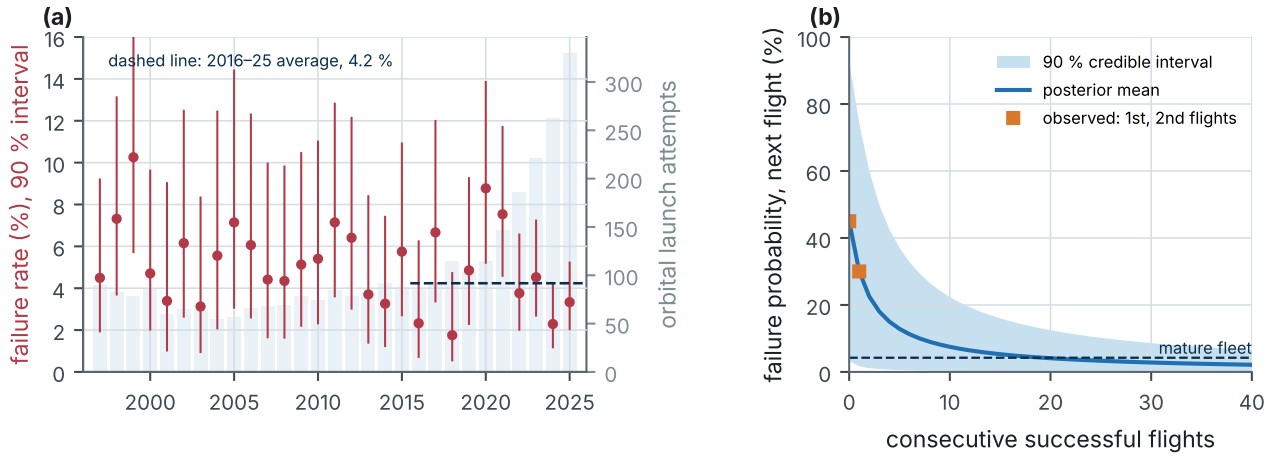
### 52.7.2 Insurance and Risk Models

Insurers cover launch failure and loss of a satellite in orbit. The market is small, with annual premiums of roughly $550–700 million in 2023–25, so one bad year can wipe out several good ones. On Earth, satellites also enable **parametric insurance**. It pays a fixed amount when a measured index crosses a trigger, such as wind speed or flood extent mapped by radar, with no claim assessment. Descartes Underwriting, for example, uses satellite imagery and machine learning to set triggers and pay quickly.

How does an insurer price a rocket that has flown only twice? The simplest honest model treats each flight as a coin toss with an unknown failure probability $p$. Prior knowledge is expressed as a **beta distribution**, $\mathrm{Beta}(\alpha, \beta)$. After $n$ flights with $k$ failures, the updated (posterior) estimate has mean

$$\hat{p} = \frac{\alpha + k}{\alpha + \beta + n}.$$

This is the **beta–binomial** model. It lets each flight shift the estimate by an amount that depends on how much was already known.




(a) Orbital launch attempts and failures by year from GCAT (J. McDowell, updated 13 October 2026); intervals from Jeffreys priors. (b) Beta(0.9, 1.1) prior with a mean of 45 %, the reported maiden-flight failure rate (Seradata); orange squares are the reported first- and second-flight failure rates since 1990.

**FIGURE 52.14** Launch risk from data. (a) Orbital launch failure rates by year from 1997 to 2025, with 90 % intervals, over the number of launch attempts. (b) A Bayesian estimate of the failure probability of a new rocket as it builds a record of successes, starting from the reported 45 % failure rate of maiden flights; one success brings it to 30 %, the reported rate for second flights.

Figure 52.14 applies the model. Across the world, 70 of 1 652 orbital launch attempts failed in 2016–25, a rate of 4.2 %. Maiden flights fail far more often: about 45 % according to the analysts Seradata, and second flights about 30 % since 1990. A prior of $\mathbf{Beta(0.9, 1.1)}$ has a mean of 45 %. After one success the model gives 30 %, matching the reported second-flight rate. After ten straight successes it gives 7.5 %, and after 20 it gives 4.1 % — the rate of the mature fleet. One failure in the first three flights keeps the estimate at 12.7 % even after ten more successes. Real underwriters add engineering judgement, the record of the company and the cost of the payload.

### DATA BOX: SPACE INSURANCE, 2023–25

- **2023:** one of the worst years on record. Claims were about $995 million against premiums of about $557 million. Two satellites caused most of the loss: ViaSat-3 Americas, whose large antenna failed to deploy (a claim of about $420 million), and Inmarsat-6 F2, lost to a power failure ($348 million). More than 85 % of losses came after the satellite had separated from the rocket, not from launch failures.
- **2024:** claims fell to about $211 million against premiums of about $580 million (Seradata). Insurers raised rates after 2023, and total working capacity shrank to about $550 million in 2024 and $502 million in 2025 (Gallagher).
- **2025:** a quiet year until late December. Then SpainSat NG-2, a Spanish military communications satellite insured for over $400 million, was damaged on its way to geostationary orbit. Its likely total loss raised the year's loss ratio from about 15 % to about 75 % (Gallagher).

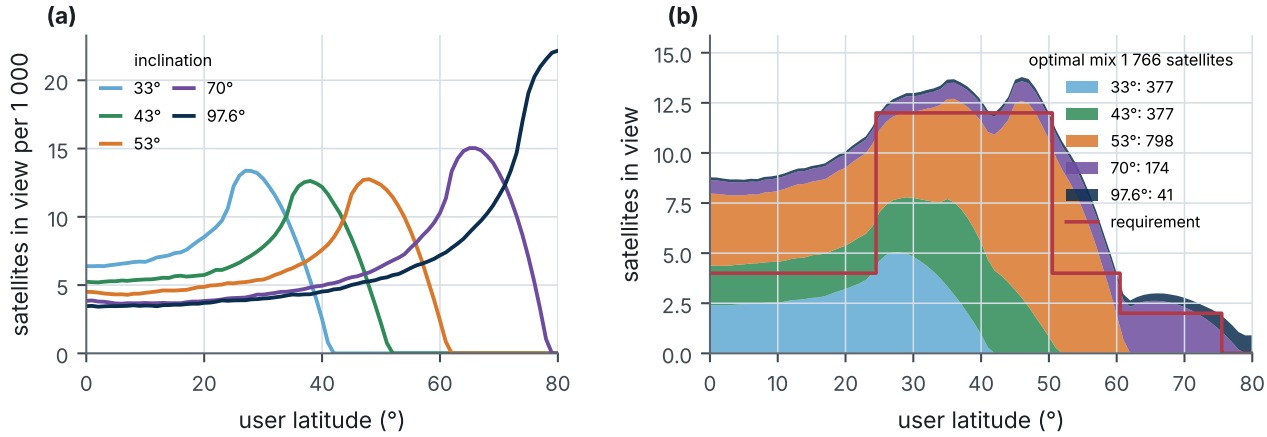
## 52.7.3 Constellation Design and Capacity

A broadband constellation sells capacity where its users are. Satellites in an inclined orbit spend most time near the edges of their latitude range. For a circular orbit of inclination $i$, the density of satellites per unit area at latitude $\varphi$ varies as

$$n(\varphi) \propto \frac{1}{\sqrt{\sin^2 i - \sin^2 \varphi}}, \qquad |\varphi| < i.$$

It is lowest at the equator and highest near latitude $\varphi = i$. Figure 52.15 (a) shows the number of satellites in view, above 25° elevation, for shells at 550 km. A shell at 53° inclination gives most coverage near 50° latitude, where much of the demand in North America and Europe lies. A polar shell is needed for high latitudes.

Choosing the mix of shells is an optimisation problem. In Figure 52.15 (b), a **linear programme** chooses how many satellites to put in each of five shells. It minimises the total while giving at least 12 satellites in view at 25–50° latitude, 4 elsewhere up to 60° and 2 up to 75°. The answer is 1 766 satellites in a mix of all five shells, with 798 at 53°. A single polar shell would need 3 095 satellites for the same requirement. Real operators solve much larger versions of this problem, with beams, spectrum, ground stations and launch costs. Learning helps with the input that matters most and is least certain: where demand will be.




Circular shells at 550 km with satellites spread evenly in orbit position and node (Monte Carlo); a satellite is in view above 25° elevation. (b) Linear programme: choose the number of satellites in each shell to minimise the total while meeting an illustrative requirement of 12 satellites in view at 25–50° latitude, 4 elsewhere up to 60° and 2 up to 75°.

**FIGURE 52.15** Constellation geometry and sizing. (a) Satellites in view above 25° elevation per 1 000 satellites, for shells at 550 km with different inclinations. (b) A linear programme chooses the number of satellites in each shell to meet an illustrative demand profile with the fewest satellites.

### 52.7.4 Customers, Pricing and Spectrum

Capacity in a constellation is fixed in each place, but demand is not. When a cell fills, the operator must refuse new users or let service slow for everyone. Starlink has done both. From 2024 it charged a one-off "congestion charge" of $100 in some US areas. By 2025–26 the charge was reported at $250 to more than $1 000 in busy areas, and some cities were shown as sold out. Demand models forecast sign-ups by cell, churn models flag users likely to leave, and pricing balances revenue against quality of service.

Spectrum is the other scarce resource. New systems must coordinate their frequencies through the ITU and national regulators and respect interference limits. Learning helps operators detect and locate interference and manage their beams (Chapter 49); the rules themselves are set by treaties.

### 52.7.5 AI Companies, Investment and New Demand

Money flowed into space in 2025–26, and much of it was linked to AI. The trackers differ, because they count different things, as Table 52.6 shows. Seraphim reported a record $12.4 billion of space-technology venture investment in 2025. Space Capital, which counts a wider range of

companies, reported a record $36 billion in the first quarter of 2026 alone. It named AI, geopolitics and computing in orbit as the drivers.

**TABLE 52.6** Money in the space business, 2025–26. Trackers use different definitions, so their figures cannot be compared directly.

| Item | Figure | Source and note |
|---|---|---|
| Global space economy, 2025 | $686 billion (+12 %) | Space Foundation, July 2026 |
| Space-tech venture investment, 2025 | $12.4 billion (record) | Seraphim |
| Investment, Q1 2026 | $36 billion, 148 companies (record) | Space Capital (wide definition) |
| Investment, Q1 2026 | $9.4 billion, 82 companies | Novaspace (narrower definition) |
| SpaceX IPO, June 2026 | ≈ $75 billion ($85.7 billion with extra shares) | Largest IPO on record |
| Space insurance premiums, 2025 | ≈ $675 million | Secondary sources |

The largest deal joined a space company and an AI company. In February 2026 SpaceX completed an all-share takeover of the AI company xAI. In June 2026 it sold shares to the public for about $75 billion at a valuation of about $1.77 trillion. It was the largest initial public offering (IPO) on record. Its filing reportedly pointed to orbital AI computing from as early as 2028. It also warned that the technology is unproven and may never be commercially viable. Google, with Planet, launched a prototype of its Suncatcher computing satellites in October 2026, and Starcloud flew a data-centre graphics processor in 2025. Chapter 45 discusses whether computing in orbit can compete with ground data centres.

AI creates demand for space in three ways. It needs data, such as images and weather records, to train models. It needs connectivity to reach ships, aircraft and remote sites. And, more speculatively, it may move computing itself into orbit. Each claim deserves the usual engineering test: what are the numbers, and has it flown?

## 52.8 Bringing AI into an Engineering Team

The key idea of this section is that AI tools are engineering tools. They need requirements, verification and configuration control like any other. Teams that treat them this way gain speed without losing safety.

Most failures of AI in engineering are not dramatic. A generated number is copied into a report unchecked. A surrogate is used outside its range. A model trained on last year's printer is used on this year's. Ordinary engineering discipline catches each of them. The practice box gives a short workflow; Chapter 53 covers formal assurance, security and governance.

### PRACTICE: INTRODUCING AN AI TOOL INTO A SPACECRAFT DESIGN TEAM SAFELY

1. **State the job and the error budget.** Write down what the tool will do and what a wrong answer would cost. Example: "suggest bracket shapes; a wrong shape costs one re-analysis." Avoid uses where errors would reach flight hardware unchecked.
2. **Check the data rules.** Confirm with your export-control and security officers which data the tool may see. Use an approved deployment, and never paste controlled data into public services.
3. **Find a baseline.** Measure how the task is done today, by people or by simple rules. A new tool must beat that baseline on your own data, not on the vendor's examples.
4. **Test on held-out cases.** Keep a set of past problems with known answers that the tool never saw during tuning. Measure errors, false alarms and misses. For classifiers, set the threshold from real costs.
5. **Keep a human gate.** Decide who reviews each output and how. Nothing enters the controlled baseline — drawings, requirements, code or test reports — without the usual analysis, test or review.
6. **Record provenance.** Log the tool, its version, its inputs and the reviewer for each output that is used. Store this in the digital thread, so that any result can be traced and repeated.
7. **Watch for drift.** Re-check performance when the process, the supplier, the software version or the design space changes. Retire tools that no longer pass.
8. **Share lessons.** Report both successes and failures to the team's lessons-learned system, so that the next project starts from evidence.

### KEY TAKEAWAYS

- AI proposes; physics, test and people verify. Every AI output must pass the same verification gate as human work before it enters a controlled design.
- Most "AI-designed" structures rest on topology optimisation. In a 2-D test, it saved about 31 % of mass at equal stiffness. Goddard's generative EXCITE bracket met mass and frequency targets that no buildable human design met in four iterations.
- Bayesian optimisation with a Gaussian-process surrogate found a near-optimal design in a median of 22 expensive runs, where random search needed about 266.
- Language models help with requirements, lessons learned, SysML v2 drafts and code. But they make plausible errors, and export-controlled data need approved tools.
- In manufacturing, the value of a screening model comes from choosing its threshold with real costs. In our test, the cost-optimal screen cost a third of full expert review, while the default threshold missed one defect in six.
- On rockets in flight, guidance, engine protection and flight safety use optimisation and fixed rules. Learning works on the ground: health monitoring in tests, weather, scheduling and design.
- In the business, AI turns satellite data into market signals, prices risk and sizes constellations. AI-linked investment and the SpaceX–xAI merger reshaped space finance in 2026.

### REVIEW QUESTIONS

1. Explain why the SIMP method raises density to a power $p \approx 3$. What would happen with $p = 1$?
2. A topology-optimised bracket uses 40 % of the material of a solid block and is 1.7 times as compliant. What fraction of the solid material would a uniform plate of equal stiffness need, and what is the mass saving?

3. How does an evolutionary algorithm differ from a gradient-based optimiser? Give one advantage and one disadvantage of each.
4. Write the expected-improvement formula and explain the role of each of its two terms. Why does Bayesian optimisation need fewer evaluations than random search?
5. Name three tasks in systems engineering where language models help. For each, describe the check that must follow.
6. What does $a_{90/95}$ mean on a probability-of-detection curve, and why is it needed for fracture-critical printed parts?
7. A review costs \$1 000 and an escaped defect \$400 000. With calibrated probabilities, what threshold minimises the expected cost? Why is the default of 0.5 a poor choice?
8. Give three reasons why landing guidance uses convex optimisation rather than a learned policy.
9. Using a $\mathrm{Beta}(0.9, 1.1)$ prior, estimate the failure probability of a new rocket after five consecutive successes. How does one early failure change the picture?
10. Explain why satellites in a 53° shell give more coverage at 45° latitude than at the equator. How would an operator use this in planning capacity?

CHAPTER

# 53 Trustworthy AI for Space: Assurance, Security, Governance and Outlook

Image: NASA/GSFC (NASA/IV&V)

*Machine learning now helps to forecast weather, screen images, plan operations and steer robots. Before a learned function may act on its own in space, engineers must show that it is safe, secure and lawful. That is hard: behaviour is learned from data rather than written down, and it can fail silently when conditions change. The chapter reviews the standards that appeared in 2024–26 and the main assurance methods — design domains, testing, formal verification, runtime monitors and assurance cases — with small experiments that show their strengths and limits. It covers attacks on learned systems and their defences, the fast-changing law of AI, the energy cost of computing and an outlook to 2035. It ends with a review of Part XIII.*

IN THIS CHAPTER

53.1 Why Learned Systems Are Hard to Assure
53.2 Standards and Guidance, 2024–26
53.3 Building the Evidence: Assurance Methods
53.4 Security: Attacks on Learned Systems
53.5 Governance, Law and Ethics
53.6 Outlook to 2035
53.7 Part XIII in Review

A spacecraft's flight software is among the most carefully checked code in the world. Every requirement is written down, every line is traced to a requirement, and every branch is tested. A neural network breaks this pattern. Its behaviour is not written by people. It is learned from examples and stored in millions of numbers that nobody can read. Yet Chapters 43–52 show

learned functions spreading into weather forecasting, image screening, operations, navigation and design. The question is no longer whether AI can help, but how far we can trust it and how we can show that trust to others.

This chapter is about that showing. **Assurance** means collecting evidence that a system is good enough for its job, and building an argument from that evidence that an independent reviewer can check. The hero image shows NASA's Katherine Johnson Independent Verification and Validation Facility in Fairmont, West Virginia. Since 1993, engineers there who are independent of each project have checked the software of NASA missions. Learned software needs the same scrutiny, with new tools.

The chapter explains why learned systems are hard to assure, surveys the standards and describes assurance methods, with small experiments run for this book. It then covers attacks, law and ethics as of October 2026, and an outlook to 2035. It closes Part XIII with a synthesis. Learning methods are explained in Chapters 43 and 44, and onboard hardware in Chapter 45.

## 53.1 Why Learned Systems Are Hard to Assure

The key idea of this section is that classical assurance checks code against a written specification. A learned model has neither readable code nor a complete specification, so the evidence must come from data, tests and protection around the model.

### 53.1.1 Behaviour Learned from Data

A learned model $f_\theta$ has parameters $\theta$ chosen to make the average loss small on $n$ training examples (Chapter 43). What matters in service is the loss on the inputs it will actually meet, drawn from the operational distribution $P_{\text{op}}$.

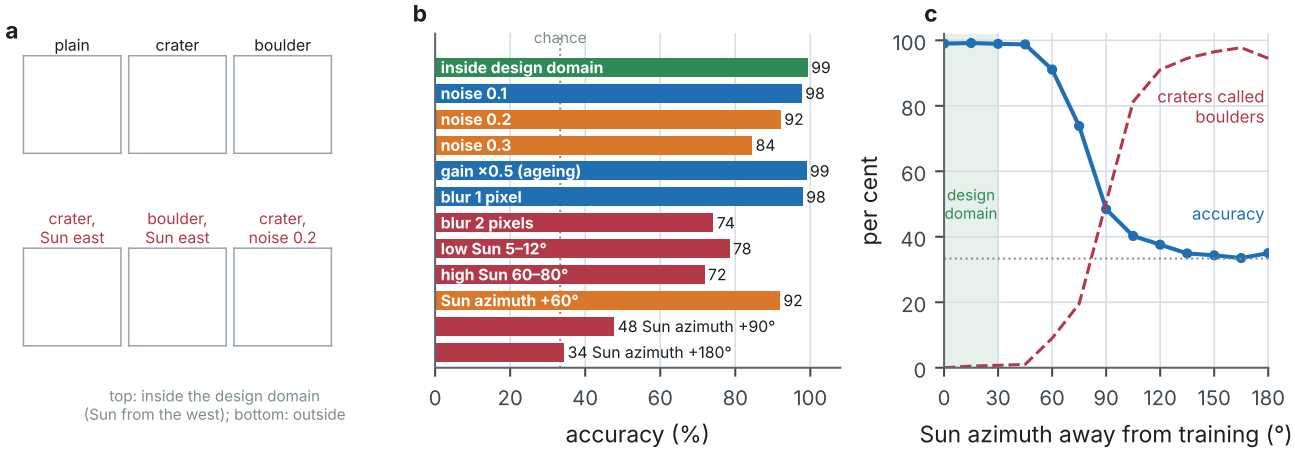
$$\hat{R}(\theta) = \frac{1}{n}\sum_{i=1}^{n} \ell\big(f_\theta(\mathbf{x}_i), y_i\big), \qquad R_{\text{op}}(\theta) = \mathbb{E}_{(\mathbf{x},y)\sim P_{\text{op}}}\, \ell\big(f_\theta(\mathbf{x}), y\big).$$

If training, test and operational data all come from the same distribution, the test error is a fair estimate of $R_{\text{op}}$. If they do not — a **distribution shift** — nothing in training limits the error. Four problems follow:

- **The specification gap.** Nobody can define "a crater" or "a safe landing site" completely in words. The labelled examples become the real specification, so the data must be checked as carefully as requirements.
- **Opacity.** The model is a large set of numbers. No single number maps to a requirement, and code-coverage measures used for normal software say little about what a network has learned.
- **Shift.** Light, seasons, detectors and targets change. A model can be excellent on its test set and poor in orbit, often without any sign that it is wrong.
- **Fragility.** Tiny, carefully chosen changes to an input can flip the output (see the section on security). Models with the same test accuracy can also behave very differently on unfamiliar inputs.

### 53.1.2 A Small Experiment in Distribution Shift

Figure 53.1 makes these points concrete. A small neural network learns to label 16 × 16-pixel surface patches as plain ground, crater or boulder. The patches are synthetic. Each is drawn from a height map and lit by the Sun using a simple shading law. In training the Sun always comes from the west, between 20° and 45° above the horizon. That set of conditions is the model's **operational design domain (ODD)**: the conditions in which it is meant to work.




Synthetic 16 × 16 patches rendered from height maps with Lambertian shading; network 256–64–32–3 (ReLU, numpy, Adam) trained on 9 000 patches with the Sun from the west (azimuth 150–210°, elevation 20–45°) and noise 0.02. Each bar or point: 1 200 new test patches. Seeds fixed.

**FIGURE 53.1** Distribution shift in a learned crater and boulder classifier. (a) Example patches; the bottom row lies outside the training conditions. (b) Accuracy under noise, a halved detector gain, blur, and changes of Sun elevation and direction. (c) Accuracy and the share of craters called boulders as the Sun moves away from the training direction.

Inside its domain the network is 99.2 % accurate. It shrugs off some changes that look large to a person. With noise of 0.2 (half the brightness of an average pixel), accuracy is still 92.1 %, and halving the detector gain changes almost nothing. Other changes are fatal. With the Sun 90° from the training direction, accuracy falls to 47.7 %. With the Sun from the east, it falls to 34.3 %, no better than guessing among three classes. More than 90 % of craters are then called boulders, because a crater lit from one side looks like a boulder lit from the other. Very low or very high Sun and strong blur also cost 21–27 points.

Three lessons carry over to real systems. The ODD must list every factor that changes the input, not just the obvious ones. Each factor must be tested across its range, including just outside the domain. And the failures are silent. With the Sun from the east, 95 % of the wrong answers came with more than 90 % confidence.

### 53.1.3 How Classical Flight Software Is Assured

Space and aviation software standards share one idea: the worse a failure's consequences, the more evidence is required. The European standards ECSS-E-ST-40C (software engineering) and ECSS-Q-ST-80C (software product assurance), both revised in April 2025, assign each function a **criticality category** from A to D. The category follows from the severity of the worst hazard it can cause. NASA's NPR 7150.2D classifies software from Class A (human-rated) to Class F (general computing). Civil aviation uses DO-178C (also published as ED-12C), with design assurance levels (DAL) A–E. Table 53.1 compares the schemes.

At the top levels, every requirement is traced to code and tests, and structural coverage shows that tests exercised every decision in the code. **MC/DC**, for example, requires each condition in a decision to be shown to change the outcome on its own. A network has almost no branches, so such measures are trivially met and prove nothing. The new ML guidance therefore adds evidence about data, learning and runtime behaviour on top of the classical process.

**TABLE 53.1** How space and aviation standards rank software by criticality, and where machine learning stands in October 2026. MC/DC: modified condition/decision coverage.

| Scheme | Levels | Top level means | What the top level demands | Status of ML |
|---|---|---|---|---|
| ECSS-Q-ST-80C Rev.2 (2025) | Categories A–D | Failure can cause loss of life or of the system | Full verification, independent review, strict coverage | ML handbook covers B–D only |
| NASA NPR 7150.2D (2022) | Classes A–F | Human-rated software | Full process, IV&V where required | Handbook topics added 2025 |
| DO-178C / ED-12C (2011) | DAL A–E | Catastrophic failure condition | 71 objectives; MC/DC structural coverage | No ML provisions; ED-324 in work |
| EASA AI concept papers | AI levels 1–3 | Level 3: advanced automation | Learning assurance plus classical assurance | Level 3 in consultation (2026) |

## 53.2 Standards and Guidance, 2024–26

The key idea of this section is that the first ML assurance standards now exist, but they are mostly guidance for lower criticality. No standard yet allows a learned function to be the only protection against a catastrophic failure.

Table 53.2 lists the main documents. Space has one dedicated document so far. **ECSS-E-HB-40-02A**, the European machine-learning handbook, was published on 15 November 2024. It covers whether to use ML at all, data management, training, model verification, system-level evaluation and a **safety cage** architecture (described below). It is guidance, not a binding standard, and covers criticality categories B–D; category A is out of scope for now. In 2025 NASA added topics on AI to its Software Engineering Handbook, under the existing NPR 7150.2 classes.

**TABLE 53.2** Standards and guidance for assuring AI, as of October 2026.

| Document | Body and date | Scope | Status, October 2026 |
|---|---|---|---|
| ECSS-E-HB-40-02A | ECSS, Nov 2024 | ML handbook: data, training, verification, safety cage | Guidance; categories B–D |
| ECSS-E-ST-40C Rev.1, ECSS-Q-ST-80C Rev.2 | ECSS, Apr 2025 | Space software engineering and assurance | In force |
| NPR 7150.2D; handbook topics 7.25, 8.25 | NASA, 2022; 2025 | Software classes; AI in engineering and assurance | In force; guidance |
| AI Roadmap 2.0; Concept Paper Issue 2 | EASA, 2023; 2024 | Learning assurance, AI levels 1–2 | Guidance; Issue 3 consulted Jun–Aug 2026 |
| NPA 2025-07 | EASA, Nov 2025 | First AI rules for aviation | Proposal; consultation closed 2026 |
| ED-324 / ARP6983 | EUROCAE WG-114, SAE G-34 | Process standard for aeronautical AI | Drafting |
| ISO/IEC 22989, 23053 | ISO/IEC, 2022 | AI concepts, terms, ML framework | Published |

| Document | Body and date | Scope | Status, October 2026 |
|---|---|---|---|
| ISO/IEC 23894; 42001 | ISO/IEC, 2023 | AI risk management; AI management system | Published; 42001 certifiable |
| ISO/IEC TR 24029-1; ISO/IEC 24029-2 | ISO/IEC, 2021; 2023 | Robustness of neural networks; formal methods | Published (report; standard) |
| ISO/IEC TR 5469 | ISO/IEC, Jan 2024 | Functional safety and AI | Published (report) |
| AI RMF 1.0; AI 600-1 | NIST, 2023; 2024 | Voluntary risk framework; generative-AI profile | Revision ordered 2025; none by Oct 2026 |
| UL 4600 | UL, 2020; 3rd ed. 2023 | Safety cases for autonomous products | Published |

Aviation is further ahead, and space engineers follow its lead. The European Union Aviation Safety Agency (EASA) defines three **AI levels**. Level 1 assists a human. Level 2 is human–AI teaming. Level 3 is "advanced automation", with little or no human oversight. EASA's Concept Paper Issue 2 (2024) gave guidance for Levels 1 and 2. In November 2025 EASA proposed its first binding AI rules, in Notice of Proposed Amendment (NPA) 2025-07. A proposed Issue 3 of the concept paper, published on 3 June 2026, extends the guidance to reinforcement learning and to Level 3. A joint EUROCAE–SAE group, set up in 2019, is drafting the process standard ED-324 (ARP6983 in the USA); it was unpublished in October 2026.

General standards come from the joint ISO and IEC committee on AI. ISO/IEC 42001 (December 2023) defines an AI management system that organisations can be certified against, as ISO 9001 does for quality. ISO/IEC TR 5469 (January 2024) covers AI inside safety functions, conventional safety functions guarding AI, and AI used to design safety functions. The voluntary AI Risk Management Framework of the US National Institute of Standards and Technology (NIST) has four functions: govern, map, measure and manage. A revision ordered in July 2025 had not appeared by October 2026. **UL 4600** is the best model for the argument itself. Instead of prescribing a process, it asks for a complete, checkable **safety case** for an autonomous product.

### LANDMARK: THE FIRST DESIGN-ASSURANCE STUDY FOR NEURAL NETWORKS (2020–21)

- In 2019–20 EASA and the Swiss company Daedalean studied how a neural network could be certified for a safety-related avionics task. The case study was a camera-based system guiding an aircraft to a runway.
- Their March 2020 report, *Concepts of Design Assurance for Neural Networks* (CoDANN), introduced **learning assurance**: evidence about the data, the learning process and generalisation, added to the usual software assurance.
- It proposed the **W-shaped process** (Figure 53.2): the classical V of design and verification, with a second V for data and learning inserted in the middle.
- A second study (May 2021) applied the ideas to visual traffic detection, and the US Federal Aviation Administration assessed them in 2022. These studies shaped the EASA concept papers and the ECSS handbook.

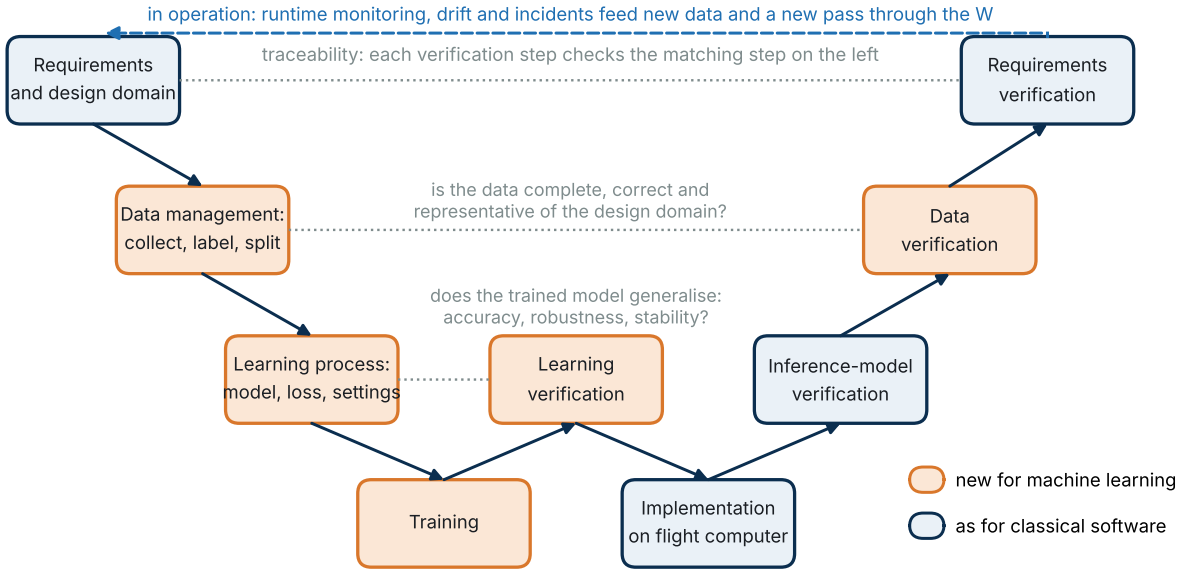



**FIGURE 53.2** A W-shaped learning-assurance life cycle, simplified after the EASA concept papers. Orange steps are new for machine learning; dotted lines link each verification step to the step it checks. In operation, monitoring and incidents feed new data and a new pass through the W.

## 53.3 Building the Evidence: Assurance Methods

The key idea of this section is that no single method is enough. Assurance combines a defined domain, good data, tests, small proofs and runtime protection, tied together in an argument.

### 53.3.1 Design Domain, Data and Requirements

Everything starts with the **ODD**. For a lunar hazard detector, it lists altitude, speed, Sun elevation and azimuth, terrain types, dust, camera settings and sensor health, each with a range. The data must then **cover** the domain: every combination that matters must appear often enough to train and test on. Coverage is never complete in many dimensions, so engineers sample key factors on grids, fill the rest randomly and record what is missing.

Data need their own paperwork. A **datasheet** for a dataset (proposed in 2018) records how the data were collected and labelled, who labelled them, how they were split and what is known to be missing. A **model card** (2019) records what a model is for, how it was tested, on which slices of the domain, and its known limits. Both are now part of the model registry in a good machine-learning operations (MLOps) pipeline (Chapter 45).

**Requirements for an ML component** go beyond “accuracy above 99 %”. They state performance for each slice of the domain, such as each Sun elevation band, and cover robustness, stability, calibrated confidence (Chapter 44), timing, memory and behaviour under radiation upsets (Chapter 45). Each must be testable and traced to evidence.

### 53.3.2 Robustness Testing

Robustness tests change inputs in controlled ways. **Corruption tests** add noise, blur, compression, stray light, dead pixels and changes of gain, as in Figure 53.1. **Metamorphic tests** use known relations instead of labels. For example, a crater shifted by two pixels must stay a

crater. **Adversarial tests** search for the worst input within a small budget (see the section on security below). Tests can show weaknesses but never prove their absence. Still, they are cheap and find most real problems early.

### 53.3.3 Formal Verification of Neural Networks

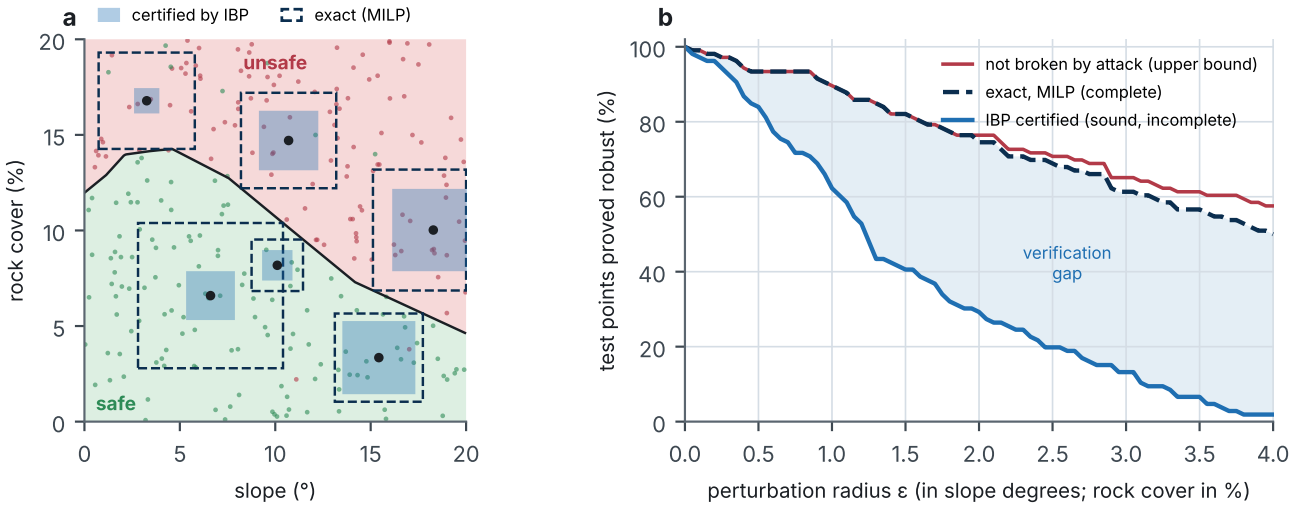
Formal verification aims to **prove** a property for every input in a region, not just for the inputs tested. The usual property is **local robustness**: for an input $\mathbf{x}_0$, no input within distance $\varepsilon$ changes the decision. In the $L_\infty$ norm the region is a box: each input value may move by up to $\varepsilon$.

The simplest sound method is **interval bound propagation (IBP)**. It pushes the whole box through the network layer by layer. Write the box as a centre $\mathbf{c}$ and a radius $\mathbf{r}$. A linear layer with weights $\mathbf{W}$ and bias $\mathbf{b}$ maps it to

$$\mathbf{c}' = \mathbf{W}\mathbf{c} + \mathbf{b}, \qquad \mathbf{r}' = |\mathbf{W}|\,\mathbf{r},$$

where $|\mathbf{W}|$ takes the absolute value of every weight. A ReLU unit, $\max(0, z)$, never decreases, so it simply clips both ends of each interval at zero. At the output, if the lowest possible score of the predicted class exceeds the highest possible score of every other class, the property is proved. IBP is fast, but its boxes grow at each layer. It can fail to prove a property that is true.

**Complete** methods give an exact answer: either a proof or a counter-example. They encode each ReLU exactly, either with logic in a satisfiability-modulo-theories (SMT) solver or with a binary variable in a **mixed-integer linear program (MILP)**. The Reluplex solver (2017) used this approach to verify properties of the neural networks proposed for the ACAS Xu aircraft collision-avoidance system. Today's leading tools combine fast bounds with branch-and-bound search. One of them, α,β-CROWN, won the annual international verification competition VNN-COMP every year from 2021 to 2025.




Synthetic landing-site data (1 500 training points, 3 % label noise); network 2–16–16–2 (ReLU, numpy). 106 correctly classified test points. Radii in the L∞ norm (a box). MILP: scipy milp with a big-M encoding of each ReLU; attack: projected gradient with 3 random starts.

**FIGURE 53.3** Proving robustness on a tiny network. (a) A learned "safe or unsafe" boundary for landing sites from slope and rock cover, with the largest box around six test points that interval bound propagation (IBP) can prove robust and the true largest box from an exact mixed-integer solver. (b) Share of test points proved robust against the box size.

Figure 53.3 compares the methods on a toy landing-site classifier with two inputs: slope and rock cover. The network has two hidden layers of 16 ReLU units and is 96.4 % accurate. For 106 correctly classified test points, IBP proved a median radius equal to 1.2° of slope. The exact MILP radius was 4.0°: IBP proved, on average, only 46 % of the true radius. A gradient attack found counter-examples only beyond the exact radius, as it must. At a radius of 1° (one degree of slope and one point of rock cover), IBP certified 62 % of points and the exact method 90 %.

The exact method took under a second per point here, but its cost can grow exponentially with network size. Complete verification works for small and medium networks, not for large vision or language models. And local robustness is not correctness: a network can be provably robust and still wrong. Formal methods therefore suit small, critical networks — controllers, monitors, surrogates — and specific properties such as "never command thrust above the limit".

### 53.3.4 Runtime Monitors and Safety Cages

If a model cannot be fully verified, it can be **contained**. The **simplex architecture** (Lui Sha, 2001) runs an unverified high-performance controller beside a simple verified one. A verified switch hands control to the simple one whenever the system nears the edge of a safe region. Chapter 48 shows a simplex "shield" on a learned lunar-landing controller. The ECSS handbook calls the general idea a **safety cage** (Figure 53.4). An input monitor checks that inputs lie inside the domain. An output monitor checks that outputs are plausible. A verified arbiter uses the learned result only if both checks pass.

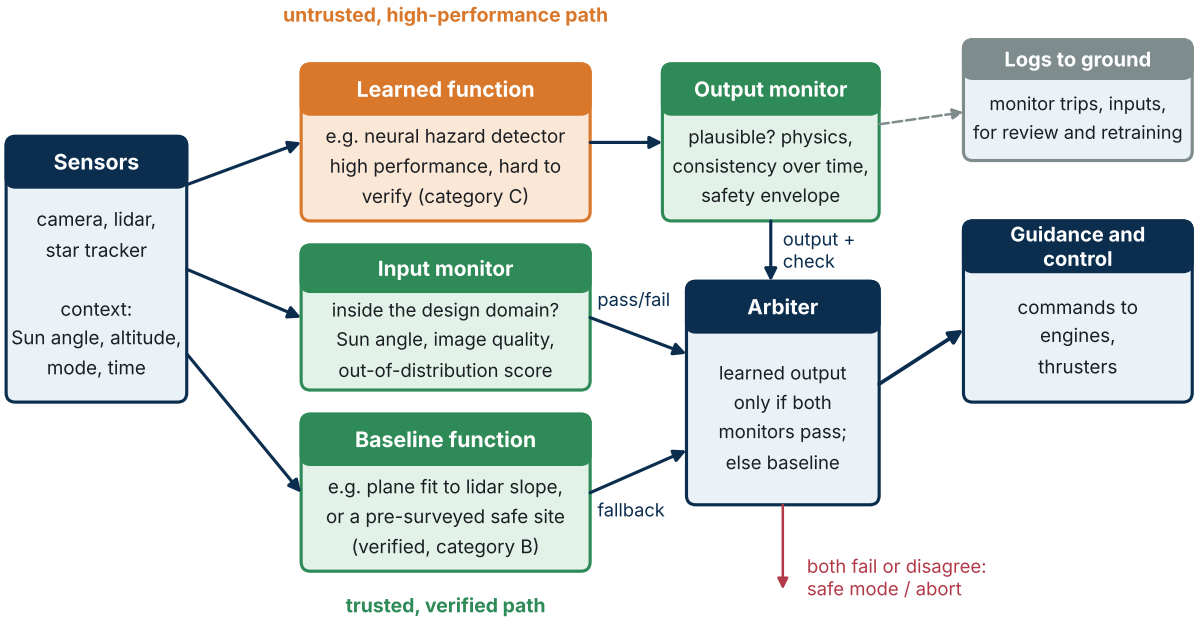


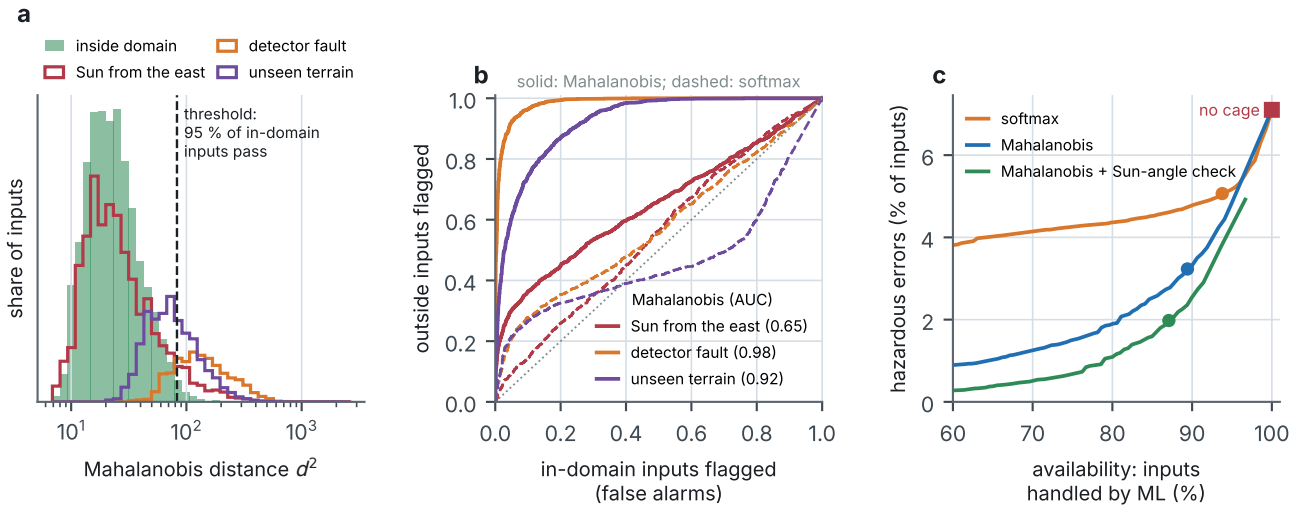

**FIGURE 53.4** A safety-cage (simplex) architecture for an onboard learned function. Only the arbiter, monitors and baseline need the highest assurance; the learned function can sit at a lower category.

The cage changes the assurance problem. The arbiter, monitors and baseline are small and conventional, so classical methods can verify them to a high category. The learned function then needs less assurance, because its failures are caught.

A common input monitor measures how far an input's internal features lie from the training data. With hidden features $\mathbf{h}$, and class means $\boldsymbol{\mu}_k$ and covariances $\boldsymbol{\Sigma}_k$ from training data, the score is the **Mahalanobis distance** to the nearest class.

$$d^2(\mathbf{x}) = \min_k (\mathbf{h} - \boldsymbol{\mu}_k)^\mathsf{T} \boldsymbol{\Sigma}_k^{-1} (\mathbf{h} - \boldsymbol{\mu}_k).$$

A large $d^2$ means the input looks unlike anything seen in training, so the monitor rejects it. The threshold sets the trade between false alarms and missed detections. A simpler baseline flags inputs for which the network's top **softmax** probability is low.




Same network and patches as the previous figure. Monitor: distance to the nearest class in the 96 hidden units, with class means and covariances from the training set. Each outside set: 1 800 patches. Stream in (c): 90 % inside the domain, 10 % outside (equal thirds); dots mark the threshold that passes 95 % of in-domain inputs; flagged inputs go to a safe fallback.

**FIGURE 53.5** Runtime monitoring of the patch classifier. (a) Mahalanobis distances for inputs inside the design domain and for three kinds of input outside it. (b) Receiver operating characteristic (ROC) curves for the Mahalanobis and softmax monitors. (c) Availability against hazardous errors for safety cages built from each monitor.

Figure 53.5 tests both monitors on the patch classifier. Three kinds of outside input were used: Sun from the east, a detector fault with bad columns and hot pixels, and an unseen terrain type (a long trough). With the threshold set so that only 5 % of in-domain inputs are rejected, the Mahalanobis monitor caught 92 % of faults and 60 % of unseen terrain. The softmax monitor caught 21 % and 22 %. Neither monitor saw the Sun problem well (30 % and 9 %). A crater lit from the east really does look like a familiar boulder, so the network's features look normal too. The cure is not a better image monitor but a **domain check on data the spacecraft already knows**: the Sun's direction comes from navigation. The design lesson: check the domain directly wherever possible, and use learned monitors only for what cannot be checked directly.

**Uncertainty-aware decisions** extend the idea. The model reports a calibrated probability or a conformal prediction set (Chapter 44), and the logic acts only when the uncertainty is small enough for what is at stake. Otherwise it asks the ground or falls back to a safe default.

**EXAMPLE: WHAT A SAFETY CAGE COSTS AND BUYS**

Take the patch classifier of Figure 53.5 and a stream of inputs. 90 % lie inside its domain and 10 % outside, split equally among Sun from the east, detector faults and unseen terrain. A **hazardous error** is an input on which the learned function acts and is wrong; a flagged input goes to a safe fallback.

1. **No cage.** In-domain errors: 0.90 × 1.1 % = 1.0 % of all inputs. Outside: 3.3 % × (65 % + 18 % + 100 %) = 6.1 %, since unseen terrain is always mislabelled. Total **7.1 %**, with 100 % availability.
2. **Mahalanobis cage.** Reject inputs with $d^2$ above the 95th percentile of in-domain scores. Availability falls to **89.4 %** and hazardous errors to **3.2 %**.
3. **Add the Sun-angle check.** All Sun-from-east inputs are now rejected. In-domain inputs keep 0.90 × 95 % = 85.5 % availability with 0.65 % errors. Faults that slip through add 0.3 % availability and almost no errors. Unseen terrain passes 40 % of the time, adding 1.3 % to both. Total: availability **87.1 %**, hazardous errors **2.0 %**.
4. **Trade.** The cage cut hazardous errors by a factor of 3.6 for a loss of 12.9 points of availability. Two-thirds of the remaining hazard is unseen terrain, so the next step is an output check, such as a slope from lidar, not a tighter threshold.

A cage turns an unbounded failure mode into a known rate of fallbacks. Whether the trade is worth it depends on what the fallback costs: a slower landing, a lost image, or an aborted manoeuvre.

**LANDMARK: A NEURAL NETWORK FLIES A SATELLITE UNDER A WATCHDOG (2026)**

- In July 2026 the US Air Force Research Laboratory (AFRL) reported a new flight test. A neural network had controlled a CubeSat's attitude for one orbit without human help.
- The network was trained by reinforcement learning (Chapter 48), then tested in simulation and in the laboratory before upload.
- Two runtime-assurance systems flew with it in **shadow mode**: an "AI watchdog" that detects unverified situations and would switch to backup systems, and a "guardrail" that would intervene near predefined safety limits. They observed and logged, but did not take control.
- AFRL gave no performance figures, and the flight was a demonstration. But it follows the pattern this chapter recommends: learned control, conventional limits, and evidence from shadow mode before monitors get authority.

### 53.3.5 Testing, Simulation and Independent Review

Statistical testing gives the most direct evidence: run the function on many independent inputs drawn from the domain and count failures. If $n$ tests show no failures, the failure probability $p$ is below a bound $p_U$ with confidence $C$ when $(1 - p_U)^n = 1 - C$. Rearranged,

$$n = \frac{\ln(1 - C)}{\ln(1 - p)} \approx \frac{-\ln(1 - C)}{p},$$

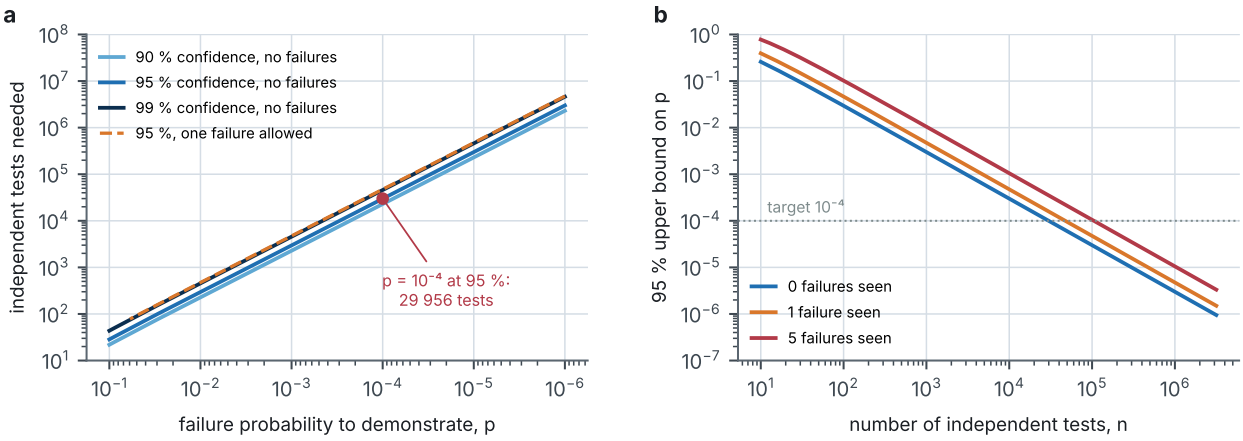
which at 95 % confidence gives the "rule of three": $n \approx 3/p$ (Figure 53.6).

### EXAMPLE: HOW MANY TESTS PROVE A FAILURE RATE BELOW $10^{-4}$?

A learned hazard detector must mislabel fewer than one landing-site image in 10 000, with 95 % confidence.

1. **No failures allowed.** $n = \ln(0.05)/\ln(1 - 10^{-4}) = 29\,956$ independent test images, close to $3/p = 30\,000$.
2. **Other confidence levels.** At 90 %, 23 025 images; at 99 %, 46 049.
3. **One failure seen.** If one test fails, the campaign must grow to **47 437** images to support the same claim (binomial distribution).
4. **A smaller campaign.** 10 000 images with no failure support only $p < 3.0 \times 10^{-4}$.
5. **The catch.** The images must be independent and drawn from the operational domain. A thousand frames from one descent are not a thousand independent tests. For aviation's catastrophic level of $10^{-9}$ per flight hour, the same rule needs $3 \times 10^{9}$ hours, about 342 000 years of testing.

Testing alone can support moderate failure rates. Rare-failure claims need architecture: a verified cage around the learned part, so that its failures are caught rather than merely counted.




Binomial statistics, assuming independent tests drawn from the operational distribution. (a) Smallest n with P(at most k failures | n, p) ≤ 1 − confidence. (b) One-sided Clopper–Pearson 95 % upper bound. For no failures, n ≈ 3/p at 95 % (the 'rule of three').

**FIGURE 53.6** What testing can show. (a) Independent tests needed to demonstrate a failure probability, for several confidence levels and with one failure allowed. (b) The 95 % upper bound on the failure probability after a test campaign, for 0, 1 or 5 failures seen.

**Simulation** supplies the volume. Landers, rovers and rendezvous systems are tested on millions of simulated scenarios, with **scenario coverage** measured over the ODD. But a learned model can exploit flaws of the simulator, and synthetic images differ from real ones (the sim-to-real gap; Chapter 45). The answer is a ladder of evidence: simulation, then hardware-in-the-loop tests with real cameras and computers, field and flight tests (Figure 53.7), and shadow-mode operation in space.

Three other kinds of evidence complete the picture:

- **Explainability as evidence.** Attribution maps (Chapter 44) show which inputs drove a decision and catch shortcuts, such as a ship detector that looks at the quay. They only support other evidence: methods can disagree, and a plausible map does not prove a correct model.

- **MLOps traceability.** Every deployed model must be traceable to its code, data version, settings and test results, so that it can be rebuilt and an anomaly traced to its cause (Chapter 45).
- **Independent verification and validation (IV&V).** A team that did not build the model repeats key tests, writes its own adversarial cases and reviews the argument. This matters even more for ML, because the developers' data choices are themselves a source of error.

**FIGURE 53.7** NASA's Morpheus prototype lander lifting off on free flight 15 at Kennedy Space Center in December 2014. During the 97-second flight its autonomous landing and hazard avoidance sensors surveyed a field of hazards, chose a safe site and guided the lander to it. Flight tests like this close the gap between simulation and reality. Image: NASA/KSC.

Table 53.3 summarises what each method can and cannot show.

**TABLE 53.3** Assurance methods for learned functions: what each shows and its limits.

| Method | What it shows | Main strength | Main limit | Use in space, 2026 |
|---|---|---|---|---|
| Design domain and data coverage | Data match the conditions of use | Basis for all other evidence | Never complete in many dimensions | ECSS and EASA guidance |
| Statistical testing | Bound on failure rate | Simple and convincing | About 3/p independent tests | Standard practice |
| Robustness tests | Stability to noise, blur, attacks | Finds weaknesses early | Cannot show their absence | Growing practice |
| Formal verification | Proof over an input region | A real guarantee | Small networks, local properties | Research; aviation benchmarks |
| Runtime monitors, safety cage | Catches outside inputs and bad outputs | Handles the unknown | Misses look-alikes; costs availability | Flight demos (shadow mode) |
| Explanations | Which inputs drove a decision | Reveals shortcuts | Not proof; methods disagree | Reviews and audits |
| Simulation, hardware in the loop | System behaviour in many scenarios | Volume and rare cases | Sim-to-real gap | Standard for autonomy |
| Traceability, IV&V | What was built, checked by others | Audit and repeatability | Discipline and cost | NASA IV&V since 1993 |

### 53.3.6 Putting It Together: the Assurance Case

The evidence is joined by an **assurance case**: a structured argument that a system is acceptably safe (or secure) for a given use in a given domain. **Goal Structuring Notation (GSN)** draws it as a tree (Figure 53.8). Each goal is a claim. A strategy explains how it is broken into sub-goals. Each

sub-goal ends in evidence. Context and assumptions are stated, so that a reviewer can attack them.

### PRACTICE: WRITING AN ASSURANCE CASE FOR AN ONBOARD ML FUNCTION

1. **State the claim and its limits.** Write the top goal in one sentence naming the function, the hazard and the domain. "The detector does not make landing less safe inside its domain" is checkable; "the AI is safe" is not.
2. **Define the domain.** List every factor that changes the input — light, geometry, terrain, sensor state, radiation — with ranges. Note which factors the spacecraft can measure directly.
3. **Fix the criticality.** Run a hazard analysis. Decide which parts must reach which category, and design a cage so that the learned part can sit at category C or D.
4. **Choose a strategy.** Argue over the life cycle (requirements, data, learning, implementation, operations) and over the runtime protection. Assign each sub-goal an owner.
5. **Collect evidence for each sub-goal.** Datasheet and coverage report; tests per domain slice with confidence bounds; robustness and adversarial tests; proofs for the monitor; bit-exact checks on the flight computer; radiation and fault-injection results.
6. **Attack your own case.** List "defeaters": ways each claim could be false. Unseen terrain, label errors, a monitor blind spot. Add evidence or change the design until each is answered.
7. **Get independent review.** A team that did not build the system reviews the case and data; record their findings and your answers.
8. **Keep it alive.** Every retraining, new domain or anomaly in orbit reopens the case. Version it with the model.

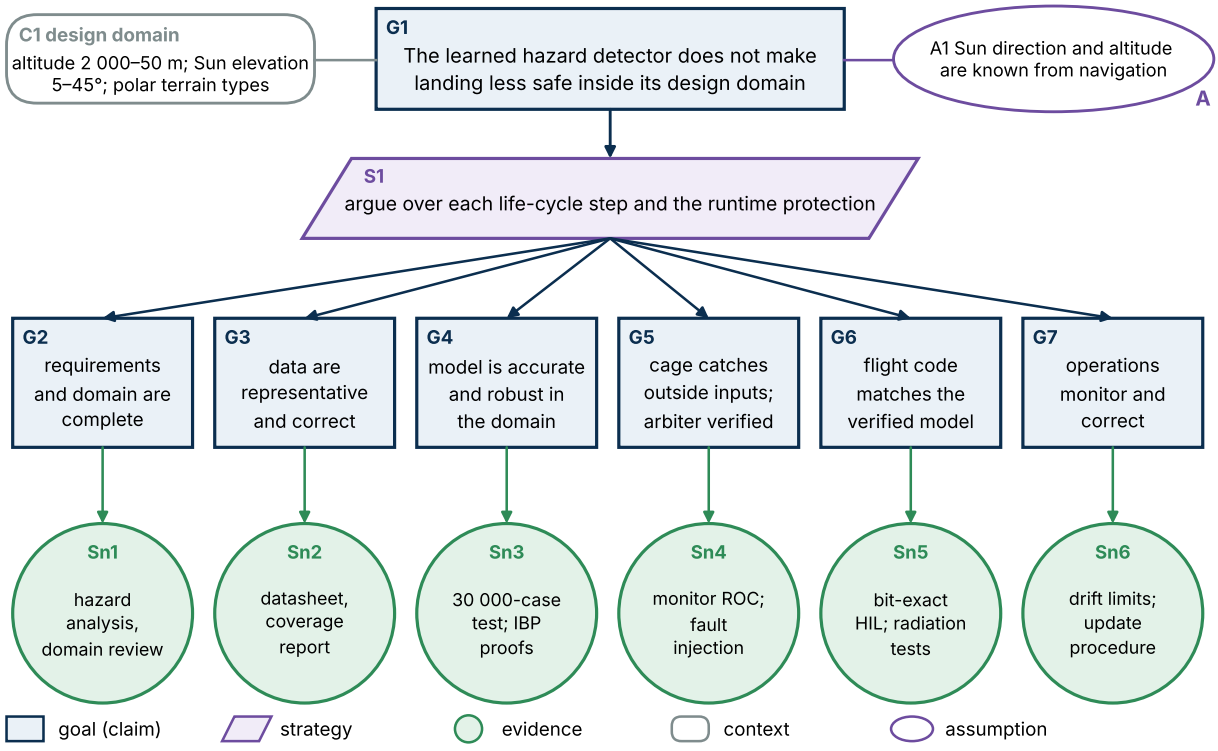



**FIGURE 53.8** An assurance case in Goal Structuring Notation for a learned hazard detector on a lunar lander (simplified). Each claim must end in evidence that a reviewer can inspect.

## 53.4 Security: Attacks on Learned Systems

The key idea of this section is that learning creates new ways to attack a system. An attacker can target the data, the model or the inputs, as well as the computers and links around them.

Learned systems add attacks at every stage of the life cycle (Table 53.4). NIST's taxonomy of adversarial machine learning (March 2025) groups them into evasion, poisoning and privacy attacks, plus supply-chain and prompt-injection attacks on generative AI.

**TABLE 53.4** Attacks on learned systems across the life cycle, with examples and defences.

| Attack | Stage | What the attacker needs | Example | Main defences |
|---|---|---|---|---|
| Evasion (adversarial examples) | Operation | Control of inputs | Patches hid aircraft from a detector (2020, Ch. 50) | Robust training, input checks, sensor fusion |
| Poisoning and backdoors | Training data | Write access to a data source | 0.01 % of a web-scale set for $60 (2023) | Provenance, hashes, data audits |
| Model extraction and inversion | Deployment | Query access | Copying a cloud analytics model | Rate limits, privacy-preserving training |
| Supply chain | Development | A shared model or library | About 100 malicious models on a public hub (2024) | Signed artefacts, safe file formats, scans |
| Prompt injection, leakage | Operation | Text the assistant will read | Instructions hidden in a log or document (Ch. 49) | Treat text as data; limit tools; human approval |
| Classic cyber attack | All | Network access | KA-SAT modems wiped (2022, Ch. 49) | Segmentation, encryption, NIST IR 8401 |

### 53.4.1 Adversarial Examples

An **adversarial example** is an input changed slightly, on purpose, so that a model gets it wrong. Such inputs were first described in 2013. The simplest way to make one is the **fast gradient sign method (FGSM)** of 2014. It moves every pixel by $\varepsilon$ in the direction that most increases the loss:

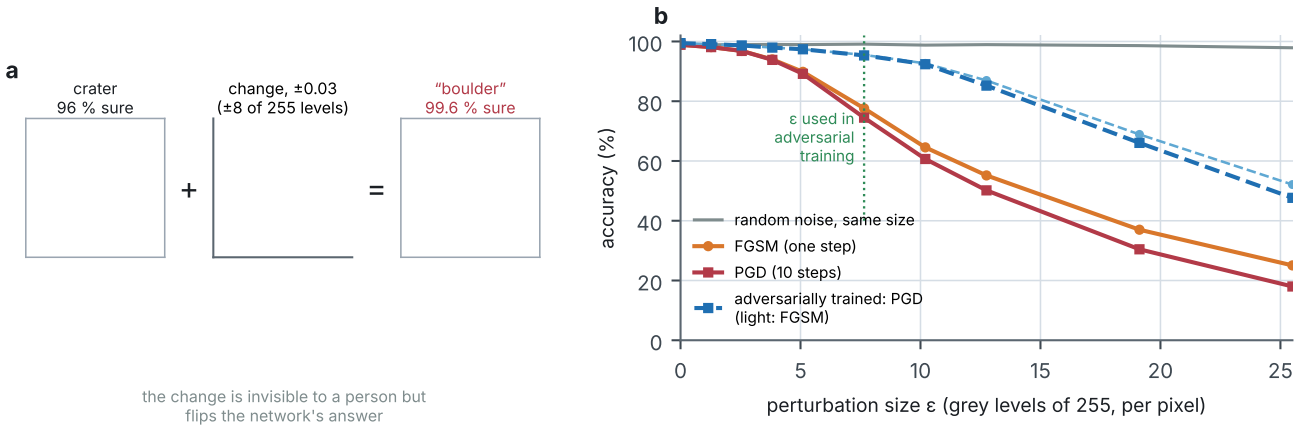
$$\mathbf{x}' = \mathrm{clip}_{[0,1]}\left(\mathbf{x} + \varepsilon\ \mathrm{sign}\,\nabla_{\mathbf{x}}\,\ell(f_\theta(\mathbf{x}), y)\right).$$

**Projected gradient descent (PGD)** repeats small steps of this kind and keeps the result within $\varepsilon$ of the original. It is the standard strong attack for testing.

Figure 53.9 attacks the patch classifier of Figure 53.1. Random noise of up to ±26 grey levels (of 255) hardly matters: accuracy never falls below 97.9 %. A crafted change of only ±8 levels cuts accuracy to 77.6 % with FGSM and 74.5 % with PGD. At ±13 levels PGD leaves 50.2 %, and at ±26 levels 18.0 %. The example crater, 96 % "crater" before the attack, becomes a 99.6 % "boulder", yet a person sees no difference. **Adversarial training**, which mixes attacked examples into training, helps: the retrained network keeps 95.3 % against PGD at ±8 levels and 85.2 % at ±13. It still falls to 47.7 % at ±26 levels.

How real is the threat in space? A digital attack needs access to the pixels — in the onboard computer, the ground pipeline or the data link — so normal cyber security is the first defence. **Physical** attacks need no such access. Printed patches, paint patterns and decoys can change

what a sensor sees (Chapter 50). Defences are layered: robust training, input checks, several sensors (optical and radar rarely share the same blind spot), and people reviewing important decisions.




Same network and patches as before (1 800 in-domain test patches). White-box attacks use the gradient of the loss with respect to the pixels; PGD = projected gradient descent, step ε/4, random start. The adversarially trained network saw FGSM examples with ε = 0.03 (about 8 grey levels) during training.

**FIGURE 53.9** Adversarial examples on the patch classifier. (a) A crater, an FGSM change of at most ±8 grey levels per pixel, and the result, which the network calls a boulder. (b) Accuracy against the size of the change for random noise, FGSM and PGD attacks, and for a network trained on adversarial examples.

### 53.4.2 Poisoning, Backdoors and the Supply Chain

**Data poisoning** inserts bad examples into training data. A **backdoor** is a poisoning that teaches the model a hidden trigger: the model behaves normally until a pattern appears, then outputs what the attacker wants. Large datasets built from the web are easy targets. In 2023 a team showed that, for about US$60, an attacker could have bought expired internet domains hosting images for two popular web-scale datasets. Through them, it could have controlled 0.01 % of a dataset of hundreds of millions of image–text pairs. Space datasets are smaller and better curated, but they are open and re-used, so labels, benchmarks and archives need provenance checks and cryptographic hashes.

**Pre-trained models** are a supply-chain risk like any third-party software. In February 2024 security researchers found about 100 models on a popular public model hub that ran hidden code when loaded, through an unsafe file format. Most were probably tests by other researchers, but one opened a remote connection to the user's computer. A downloaded model must be treated as untrusted code: loaded only in safe formats, scanned and signed.

A model offered as a service can also be **extracted** — copied by training on its answers — or probed to reveal its training data (**model inversion** and **membership inference**). Rate limits, coarse outputs and training that limits memorisation reduce the risk.

### 53.4.3 Language Models in Operations

Language-model assistants (Chapters 44 and 49) add two specific risks. In **prompt injection**, text that the assistant reads — a web page, an email, an anomaly report — contains instructions that it follows, even if the attacker never talks to the assistant. **Data leakage** happens when an assistant reveals confidential procedures, keys or personal data it has seen. The defences are architectural. Treat all retrieved text as data, never as commands. Give the assistant the fewest tools and permissions it needs. Keep a human approval step before any command reaches a spacecraft.

### 53.4.4 Securing AI-Enabled Space Systems

Learned components live inside ordinary space systems, so classical space cyber security comes first (Chapters 7 and 49). NIST's report IR 8401 (December 2022) applies its Cybersecurity Framework to the satellite ground segment. The Aerospace Corporation's **SPARTA** framework (first released in October 2022) catalogues attacks on spacecraft: in 2026 it listed 9 tactics and 85 techniques. In Europe, the NIS2 directive already covers operators of space ground infrastructure. The proposed EU Space Act of June 2025 would add space-specific cyber rules; it was still being negotiated in October 2026.

> DATA BOX — ATTACKING SATELLITES ON PURPOSE, 2023
>
> - **OPS-SAT (April 2023).** In an exercise run with ESA before the CYSAT conference in Paris, four researchers from Thales took control of the application environment of ESA's OPS-SAT test satellite. They inserted code that could, for example, mask areas in its images. ESA kept control of the satellite throughout.
> - **Moonlighter (June 2023).** The Aerospace Corporation and the US Space Systems Command built Moonlighter, a CubeSat designed to be hacked. It launched on 5 June 2023 on a cargo flight to the ISS.
> - **Hack-A-Sat 4 (August 2023).** At the DEF CON 31 conference in Las Vegas, five finalist teams attacked Moonlighter in orbit. An Italian team, mHACKeroni, won the US$50 000 first prize.
> - **Why it matters for AI.** The OPS-SAT attack could alter imagery at the source. A learned function downstream would analyse falsified data with full confidence, so integrity checks belong before the model.

A sound defence for an AI-enabled mission combines four layers:

- **Protect the pipeline**: signed data, models and code; separate networks for training, ground operations and commanding; secure model upload with rollback (Chapter 45).
- **Harden the model**: adversarial and corruption tests, robust training and monitoring of input statistics.
- **Contain the model** in a safety cage, so that a fooled model cannot command a hazardous action.
- **Keep people in the loop** for irreversible decisions, and train them to spot a model under attack.

## 53.5 Governance, Law and Ethics

The key idea of this section is that AI law is moving fast and differs between regions. In October 2026 it touches space mostly through data, ground services and products sold on Earth, and less through onboard functions.

### 53.5.1 The European Union AI Act

The **AI Act**, Regulation (EU) 2024/1689, is the first broad AI law. It entered into force on 1 August 2024 and sorts AI uses by risk. Unacceptable practices, such as social scoring, are banned. These bans, and a duty to build staff AI literacy, apply from 2 February 2025. **High-risk** systems — for example safety components of critical infrastructure, or AI in products under EU safety law such

as medical devices — need risk management, data governance, logging, human oversight, accuracy and robustness. They also need conformity assessment. Providers of **general-purpose AI models** have had transparency and copyright duties since 2 August 2025. Fines for banned practices reach €35 million or 7 % of world turnover. AI used only for military or national-security purposes is excluded.

In 2026 a "digital omnibus" amendment, agreed in May and published in July, delayed the rules for stand-alone high-risk systems to 2 December 2027, and for high-risk AI in regulated products to 2 August 2028. Transparency duties, such as labelling AI-generated content, still apply from 2 August 2026. Generative systems already on the market have until 2 December 2026 to add machine-readable marks to their output. For space engineers the main routes into the Act are data services used in critical infrastructure and, through EASA's rules, AI in aviation; NPA 2025-07 is partly a response to the Act. Most onboard spacecraft functions are not listed as high-risk, but customers increasingly ask for the same evidence.

### 53.5.2 The United States

US policy turned in 2025 from managing AI risk to accelerating adoption. Executive Order 14110 of 2023 was revoked on 20 January 2025. Executive Order 14179 (23 January 2025) and America's AI Action Plan (July 2025) stress innovation, infrastructure and exports. Memoranda of April 2025 tell federal agencies to appoint chief AI officers, list their AI uses and apply minimum practices to "high-impact" ones. Executive Order 14365 (December 2025) directs federal action against state AI laws judged "onerous". A national policy framework of March 2026 asked Congress to pre-empt conflicting state laws. As of October 2026 there is no general federal AI statute, and state AI laws continue to apply. Military AI rules are described in Chapter 50.

### 53.5.3 China, the United Kingdom and Others

**China** regulates specific services. Interim measures for generative AI services, with registration of algorithms, took effect in August 2023. Rules requiring visible and hidden labels on AI-generated content took effect on 1 September 2025. An amendment adding the first AI article to the Cybersecurity Law, adopted in October 2025, took effect on 1 January 2026. The **United Kingdom** has no general AI law. It relies on existing regulators applying shared principles. Its AI Safety Institute was renamed the AI Security Institute in February 2025.

### 53.5.4 International Agreements

The **Council of Europe Framework Convention on AI** is the first binding international treaty on AI and human rights. It opened for signature in Vilnius on 5 September 2024. The EU ratified it in May 2026, and by the latest treaty records no other party had. It needs five ratifications, including three by Council of Europe member states, to enter into force. At the United Nations, the **Global Digital Compact** (September 2024) led to resolution 79/325 of August 2025. It created an Independent International Scientific Panel on AI, whose 40 members were appointed in February 2026, and a **Global Dialogue on AI Governance**. The Dialogue held its first session in Geneva on 6–7 July 2026, where the Panel presented a preliminary report. A series of summits — Bletchley Park (2023), Seoul (2024), Paris (2025) and New Delhi (February 2026) — produced non-binding declarations. The New Delhi Declaration had 91 signatories, including the USA, the UK and China. Table 53.5 summarises the main instruments.

**TABLE 53.5** Main AI governance instruments as of October 2026. Military AI is covered in Chapter 50.

| Instrument | Jurisdiction | Type | Key dates | Status, October 2026 |
|---|---|---|---|---|
| AI Act, Reg. (EU) 2024/1689 | European Union | Binding, risk-based | In force Aug 2024; bans Feb 2025 | High-risk rules delayed to Dec 2027 and Aug 2028 |
| Executive orders, Action Plan, OMB memos | United States | Executive policy | Jan–Jul 2025; Dec 2025 | No general federal AI law; state laws apply |
| Generative-AI measures, labelling rules | China | Binding regulations | Aug 2023; Sep 2025 | In force; AI added to Cybersecurity Law Jan 2026 |
| Sector-based principles | United Kingdom | Regulators' guidance | 2023–25 | No general AI law |
| Framework Convention on AI | Council of Europe | Treaty | Opened Sep 2024 | Ratified by the EU; not yet in force |
| Global Digital Compact; Scientific Panel; Global Dialogue | United Nations | Non-binding | 2024; 2025; Jul 2026 | Panel report Jul 2026; next Dialogue May 2027 |
| New Delhi Declaration | 91 signatories | Non-binding | Feb 2026 | Signed |

### 53.5.5 Export Controls

Space hardware has long been export-controlled; the USA eased licensing of some spacecraft items to close allies in October 2024. Advanced AI chips have been controlled since 2022. A US rule of January 2025 that would have limited chip and model-weight exports worldwide was rescinded in May 2025. In January 2026 the USA moved some high-end chips to case-by-case licensing for China, combined with a 25 % import duty. Engineers buying processors for an AI payload, or sharing models and training data with foreign partners, should check controls early (Chapter 50).

### 53.5.6 Agency Principles and Human Oversight

NASA published a *Framework for the Ethical Use of Artificial Intelligence* in April 2021. It sets six principles: fair; human-centric and societally beneficial; explainable and transparent; accountable; secure and safe; and scientifically and technically robust. NASA appointed its first chief AI officer in May 2024. By October 2026 NASA's web pages referred to the same framework as one for "super intelligence". ESA works through the ECSS standards and EU law. Defence users follow principles such as the US Department of Defense's five of 2020: responsible, equitable, traceable, reliable and governable.

**FIGURE 53.10** The Artemis II flight control team in the White Flight Control Room at NASA's Johnson Space Center during the crew's lunar flyby on 6 April 2026. Every principle for trustworthy AI comes back to people like these, who must understand, question and, when necessary, overrule the tools they use. Image: NASA/JSC (ROBERT MARKOWITZ NASA-JSC).

All these documents call for **meaningful human oversight**. In practice that means operators who understand what a tool can and cannot do (Figure 53.10). They need the time and information to question it, and the authority to overrule it. It also means guarding against **automation bias**: the tendency to accept a machine's answer because it is usually right. Interfaces should show confidence and reasons, and training should include cases where the tool is wrong.

### 53.5.7 Privacy and Dual Use

Satellites now see the ground at 30 cm or better, and some revisit a site many times a day (Figure 53.11). One image reveals little about a person. But learned detectors can track vehicles, ships and crowds over months and link them to other data, building a "pattern of life" that can identify people. The results can then be personal data under laws such as the EU's General Data Protection Regulation. Good practice limits resolution and retention to what a task needs, aggregates results, and logs who queried what. Almost every tool in Part XIII is also **dual-use**: the same detector serves a port authority and an army (Chapter 50).

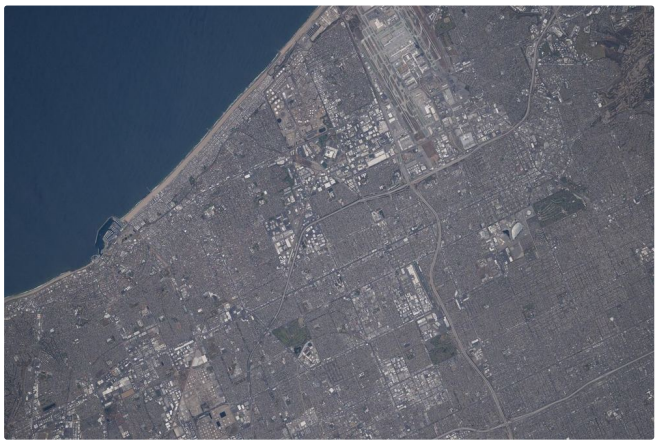


**FIGURE 53.11** Los Angeles International Airport (top) and the surrounding city, photographed from the International Space Station on 29 May 2026. Learned detectors can count aircraft, cars and ships in images like this; combined over time, such counts raise questions of privacy and dual use. Image: NASA/JSC.

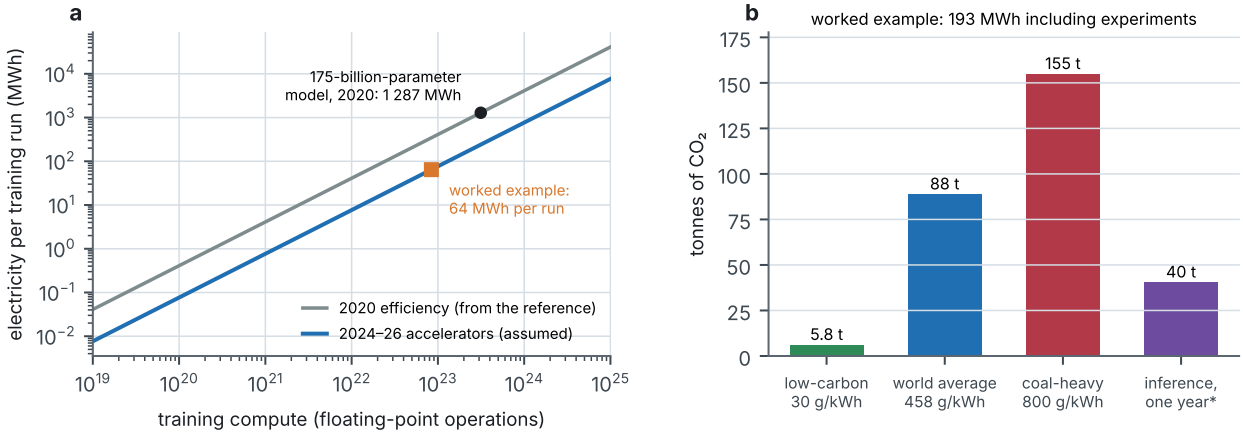
### 53.5.8 The Environmental Cost of Computing

The International Energy Agency estimated that data centres used about 415 TWh in 2024, about 1.5 % of the world's electricity. It expects this to roughly double, to about 945 TWh, by 2030, with AI the main driver (Figure 53.12). The energy for one training run can be estimated from the compute: a model with $N$ parameters trained on $D$ tokens needs about $C = 6ND$ floating-point operations (Chapter 44). Then

$$E \approx \frac{C}{F_{\text{peak}}\, u}\, P_{\text{acc}}\, k_{\text{host}}\, \text{PUE},$$

where $F_{\text{peak}}$ is the peak speed of one accelerator, $u$ the fraction of it actually used, and $P_{\text{acc}}$ its power. The factor $k_{\text{host}}$ adds host computers and networking, and the power usage effectiveness (PUE) adds cooling and other overheads (Figure 53.13).

**FIGURE 53.12** City lights from Tokyo to Osaka and Hiroshima, photographed from the International Space Station on 15 September 2025. Data centres for AI are a fast-growing part of the electricity demand that lights scenes like this. Image: NASA/JSC.




Compute C = 6 N D (N parameters, D training tokens). Assumed: 989 TFLOP/s peak per accelerator, 40 % utilisation, 0.7 kW board power × 1.3 for host and network × 1.2 data-centre overhead (PUE). Reference: Patterson et al. (2021). *1 million queries a day at 0.24 Wh each (a median reported by one large provider in 2025), world-average grid.

**FIGURE 53.13** Energy and emissions of training. (a) Electricity per training run against compute, for 2020-era efficiency (from a published estimate for a 175-billion-parameter model) and for assumed 2024–26 accelerators. (b) Emissions of the worked example on three electricity grids, and of one year of a popular inference service.

**EXAMPLE: ENERGY AND CARBON TO TRAIN A MID-SIZE MODEL**

Estimate the cost of training a 7-billion-parameter model on 2 trillion tokens — a mid-size language model, or a large Earth-observation foundation model (Chapter 44).

1. **Compute.** $C = 6 \times 7 \times 10^{9} \times 2 \times 10^{12} = 8.4 \times 10^{22}$ operations.
2. **Accelerator time.** Assume 989 × $10^{12}$ operations per second at peak and 40 % utilisation: $8.4 \times 10^{22} / (3.96 \times 10^{14})$ s = **59 000 accelerator-hours**, or 4.8 days on 512 accelerators.
3. **Power.** 0.7 kW per accelerator × 1.3 for hosts and network × PUE 1.2 = 1.09 kW. One run: 59 000 h × 1.09 kW = **64 MWh**.
4. **Real projects.** Failed runs, tuning and evaluation often triple the total: **193 MWh**.
5. **Carbon.** On a low-carbon grid (30 g $CO_2$/kWh): **5.8 t**. At the 2025 world average of 458 g/kWh: **88 t**. On a coal-heavy grid (800 g/kWh): **155 t**.
6. **Use.** One million queries a day at 0.24 Wh each — a median reported by one large provider in 2025 — need 88 MWh a year, or 40 t $CO_2$ at the world average. After about 2.2 years the answers have used more energy than the whole training effort. At ten million queries a day this takes under three months.

For comparison, a 175-billion-parameter model trained in 2020 was estimated at 1 287 MWh and 552 t $CO_2$; the assumed modern hardware uses about 5 times less energy per operation. The grid changes the carbon by a factor of 27.

Onboard inference uses only a few watts (Chapter 45); the large costs lie on the ground.

### 53.5.9 Workforce and Skills

Trustworthy AI needs people who understand both machine learning and safety-critical engineering, and they are rare. Data scientists seldom know hazard analysis, and many assurance engineers have never trained a model. Mixed teams and training on the new standards are closing the gap.

## 53.6 Outlook to 2035

The key idea of this section is to separate what has happened from what may happen. Projections are marked with a confidence level.

Figure 53.14 sets out one view of the next decade. First, the facts as of October 2026. Learned weather forecasts have been operational at the European Centre for Medium-Range Weather Forecasts (ECMWF) since February 2025 (Chapter 47). Foundation models for Earth observation and the Sun are openly available (Chapter 44). Onboard AI has flown on several small satellites (Chapters 45 and 46), and a neural network has flown a satellite's attitude for one orbit. Single operators already run autonomous collision avoidance across thousands of satellites (Chapter 48). No learned function has yet been certified as the sole protection against a catastrophic failure in space or aviation.

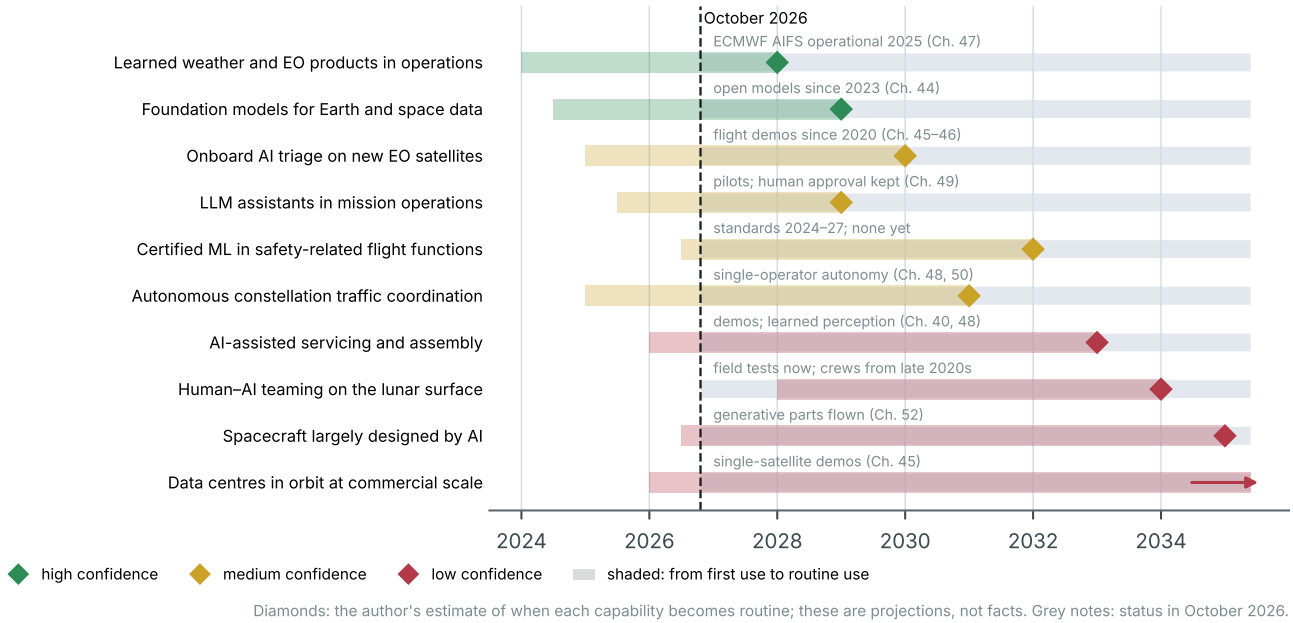



**FIGURE 53.14** An outlook to 2035. Shaded bars run from first use to the estimated year of routine use (diamonds), coloured by the author's confidence. Grey notes give the status in October 2026. These are projections, not facts.

The projections, with confidence:

- **Foundation models for Earth and space (high confidence).** Shared pre-trained models will be the normal starting point for EO, weather and heliophysics products by about 2030. Assuring a *base model* then becomes a shared task, like qualifying a processor.
- **Certified ML in safety-related functions (medium).** Aviation will probably certify Level 1–2 functions first, once ED-324 and EASA's rules are final. Space will follow at ECSS category B, inside safety cages, in the early 2030s.
- **Autonomous constellations (medium).** Coordination between operators — who moves when two satellites meet — needs shared rules and data more than better models (Chapters 21 and 50).
- **AI-designed spacecraft (low).** Algorithms already shape parts: evolved antennas flew on NASA's ST5 in 2006 (Chapter 52). A spacecraft designed largely by AI is unlikely before 2035, because every design must still be verified.
- **Human–AI teaming on the Moon (low).** Artemis astronauts are expected to land from about 2028 (Figure 53.15). Assistants for traverses, procedures and medical care will be tried first in analogue tests (Figure 53.16), and trusted only as far as they can explain themselves.
- **AI-assisted servicing and orbital computing (low).** Docking life-extension vehicles have served satellites commercially since 2020 (Chapter 40). Robotic servicing with learned perception and data centres in orbit are still at the demonstration stage (Chapters 40 and 45). Both depend more on hardware and economics than on AI.

**FIGURE 53.15** NASA astronauts Joe Acaba and Mark Vande Hei test Lunar Outpost's Pegasus lunar terrain vehicle prototype at Johnson Space Center in September 2026. NASA selected Lunar Outpost and Astrolab in May 2026 to build the first lunar terrain vehicles. Image: NASA/JSC (NASA/Bill Stafford).

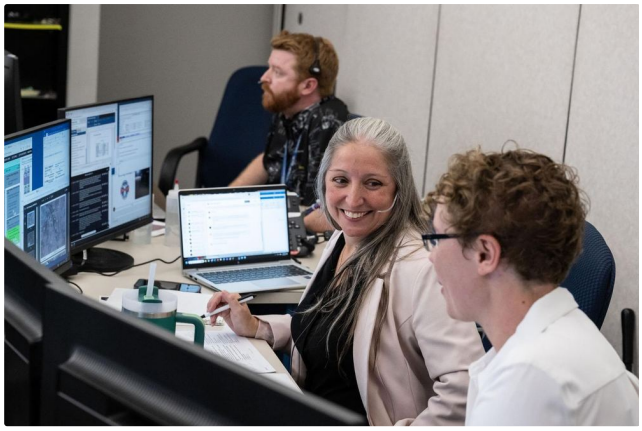


**FIGURE 53.16** Flight controllers at Johnson Space Center guiding simulated moonwalks in Arizona's San Francisco Volcanic Field during the JETT 5 field test in May 2024. Analogue tests like this are where decision-support tools for surface operations can be tried before flight. Image: NASA/JSC (James Blair - NASA - JSC).

Several problems remain open:

- **Assuring foundation models** trained by others on data nobody can fully inspect.
- **Monitors with guarantees**: detecting inputs outside the domain that look familiar, as in Figure 53.5.
- **Verification at scale** for convolutional and transformer networks, and for properties beyond local robustness.
- **Assurance of learning in orbit**, when a model updates itself from new data.
- **Agents that act**: language-model systems that use tools need limits that can be checked.
- **Common evidence formats**, so that a model qualified by one organisation can be re-used by another.

## 53.7 Part XIII in Review

Part XIII began with the foundations of machine learning and ends with trust. Table 53.6 gathers the main lesson of each chapter.

**TABLE 53.6** Part XIII in one table: the lesson of each chapter and the most mature use as of October 2026.

| Chapter | Topic | Central lesson | Most mature use, 2026 |
|---|---|---|---|
| 43 | AI foundations | Test on data split by place and time; report uncertainty | Classical ML in data products |
| 44 | Deep learning, foundation models | Pre-train once, adapt many times; check calibration | Open EO and weather models |
| 45 | Data, hardware, edge AI | Data and radiation, not algorithms, limit onboard AI | Cloud screening, triage demos |
| 46 | Earth observation | Domain shift and area estimation decide real accuracy | Land cover, floods, methane |
| 47 | Weather, climate, space weather | Learned forecasts rival physics but need its analyses | ECMWF AIFS operational |

| Chapter | Topic | Central lesson | Most mature use, 2026 |
|---|---|---|---|
| 48 | Autonomy and robotics | Learning helps perception; verified logic keeps control | Rover and avoidance autonomy |
| 49 | Operations and networks | Alert budgets, not accuracy, decide usefulness | Telemetry anomaly tools |
| 50 | SSA, defence, security | Physics carries most of the load; people keep decisions | Tracking, imagery triage |
| 51 | Science and human spaceflight | ML is a funnel; discoveries need independent checks | Planet vetting, alert brokers |
| 52 | Design, manufacturing, launch, business | AI proposes; physics, test and people verify | Generative parts, inspection |
| 53 | Trustworthy AI | Assure data, model and runtime; contain what cannot be proved | Handbooks; shadow-mode demos |

Three threads run through all eleven chapters. First, **physics and engineering still carry most of the load**: learning works best as a fast approximation, a filter or a perception layer inside a well-understood system. Second, **data decide everything** — how they were collected, labelled and split, and how far operation differs from training. Third, **trust is earned by evidence and architecture together**. A model is trusted when its domain is defined, its performance honestly measured, its failures contained and its behaviour visible to people. Space has always demanded this discipline; machine learning makes it more necessary, not less.

KEY TAKEAWAYS

- Learned behaviour comes from data, not a written specification, and fails silently outside its design domain. A crater classifier fell from 99.2 % to 34.3 % when the Sun came from the other side, while noise barely mattered.
- The first ML assurance documents exist — ECSS-E-HB-40-02A (2024), EASA's concept papers and NPA 2025-07, ISO/IEC 42001 and TR 5469 — but they cover lower criticality. No learned function is yet the sole protection against a catastrophic failure.
- Evidence combines a defined domain with data coverage, statistical testing (about 3/p failure-free tests: 29 956 for $10^{-4}$), robustness tests, formal proofs for small networks, traceability and independent review.
- A safety cage contains what cannot be proved. Here it cut hazardous errors from 7.1 % to 2.0 % for 12.9 points of availability; checking the domain directly (Sun angle) beat any learned monitor for that factor.
- Attacks target data, models and inputs: changes of ±8 grey levels cut accuracy to 74.5 %, and poisoning web-scale data can cost tens of dollars. Defences are layered: secure pipelines, robust training, cages and people.
- AI law diverges: the EU AI Act's high-risk rules now apply from December 2027 and August 2028; the USA has no general AI statute; China regulates services and labels; international agreements are mostly non-binding.
- Training a 7-billion-parameter model takes about 190 MWh including experiments; the grid changes its emissions 27-fold, and a popular service can use more energy answering queries than training.

### REVIEW QUESTIONS

1. Write the empirical and operational risks. Why does a low test error say nothing about performance after a distribution shift? Give two space examples.
2. In Figure 53.1, why is the classifier robust to noise and gain but not to the Sun's direction? What would you add to its design domain and its test plan?
3. Compare ECSS criticality categories, NASA software classes and DO-178C levels. Why do structural-coverage measures such as MC/DC say little about a neural network?
4. Describe the W-shaped learning-assurance process. Which steps are new compared with classical software, and what evidence does each produce?
5. Show how interval bound propagation passes a box through a linear layer and a ReLU. Why is it sound but incomplete? What did Figure 53.3 show about the size of the gap?
6. Draw a safety cage for a learned star-tracker or crater-navigation function. Which parts need the highest assurance category, and why?
7. Using the Mahalanobis monitor results, explain why some outside inputs are hard to detect. How does a direct domain check change the cage's trade between availability and hazard?
8. How many independent failure-free tests demonstrate a failure probability below $10^{-3}$ with 99 % confidence? Why can testing alone never support $10^{-9}$ per hour?
9. Explain FGSM and PGD. Why does random noise of the same size barely affect the classifier while a crafted change does? List four defences for a spacecraft image pipeline.
10. A company will train a 7-billion-parameter foundation model and sell flood maps from it to an EU water authority. Which legal, security and environmental questions should it answer first?

PART XIV

# Appendices and Reference Data

Constants, equations, planetary data, frequency bands, mission databases, a timeline, acronyms, a glossary, standards and sources, and image credits — the quick-reference tools an engineer keeps within reach.


- A Constants, Units and Time Systems
- B Key Equations
- C Solar System Data
- D Radio-Frequency Bands and Spectrum
- E Mission and Vehicle Databases
- F Timeline of Space Exploration and Engineering
- G Acronyms and Abbreviations
- H Glossary
- I Standards, Data Sources and Further Reading
- J Image Credits


Image: NASA/GSFC

APPENDIX

# A Constants, Units and Time Systems

*The physical and astronomical constants used throughout this manual, the SI units and prefixes, conversion factors for the non-SI units still common in aerospace, decibel conventions used in link budgets, and the time scales and epochs used to timestamp orbits, telemetry and observations. Values follow the 2019 SI, CODATA 2022 recommended values, the IAU 2012 and 2015 resolutions, and the WGS 84 Earth model.*

IN THIS CHAPTER

A.1 The SI and Its Defining Constants
A.2 Physical Constants
A.3 Astronomical and Earth Constants
A.4 Conversion Factors
A.5 Decibels
A.6 Time Scales and Epochs

This appendix collects the numbers an engineer reaches for most often. Exact values (by definition) are marked **exact**; other values are given to the precision useful in engineering work, with the last digits uncertain as in the source. Where two sources differ slightly — for example the Earth's gravitational parameter in WGS 84 and in the latest geodetic models — the value used throughout this manual is given, and the difference is far smaller than any engineering margin.

## A.1 The SI and Its Defining Constants

Since 20 May 2019 all seven SI base units are defined by fixing the numerical values of seven constants of nature (Table A.1). The kilogram is no longer a metal cylinder in Paris; it is defined through the Planck constant. For practical engineering nothing changed, but every unit can now in principle be realised anywhere — including in orbit.

**TABLE A.1** The seven defining constants of the SI (2019).

| Defining constant | Symbol | Value (exact) | Unit it defines |
|---|---|---|---|
| Caesium hyperfine transition frequency | $\Delta\nu_{Cs}$ | 9 192 631 770 Hz | second |
| Speed of light in vacuum | $c$ | 299 792 458 m/s | metre |
| Planck constant | $h$ | $6.626\,070\,15 \times 10^{-34}$ J s | kilogram |
| Elementary charge | $e$ | $1.602\,176\,634 \times 10^{-19}$ C | ampere |
| Boltzmann constant | $k$ | $1.380\,649 \times 10^{-23}$ J/K | kelvin |
| Avogadro constant | $N_A$ | $6.022\,140\,76 \times 10^{23}$ mol$^{-1}$ | mole |
| Luminous efficacy of 540 THz light | $K_{cd}$ | 683 lm/W | candela |

**TABLE A.2** SI prefixes. Ronna, quetta, ronto and quecto were added in 2022 — partly so that data volumes and planetary masses could be written compactly (the Earth's mass is about 6 Rg).

| Factor | Prefix | Symbol | Factor | Prefix | Symbol |
|---|---|---|---|---|---|
| $10^{30}$ | quetta | Q | $10^{-1}$ | deci | d |
| $10^{27}$ | ronna | R | $10^{-2}$ | centi | c |
| $10^{24}$ | yotta | Y | $10^{-3}$ | milli | m |
| $10^{21}$ | zetta | Z | $10^{-6}$ | micro | μ |
| $10^{18}$ | exa | E | $10^{-9}$ | nano | n |
| $10^{15}$ | peta | P | $10^{-12}$ | pico | p |
| $10^{12}$ | tera | T | $10^{-15}$ | femto | f |
| $10^{9}$ | giga | G | $10^{-18}$ | atto | a |
| $10^{6}$ | mega | M | $10^{-21}$ | zepto | z |
| $10^{3}$ | kilo | k | $10^{-24}$ | yocto | y |
| $10^{2}$ | hecto | h | $10^{-27}$ | ronto | r |
| $10^{1}$ | deca | da | $10^{-30}$ | quecto | q |

For data storage the binary prefixes are also standardised (IEC 80000-13): 1 KiB (kibibyte) = 1 024 bytes, 1 MiB = 1 048 576 bytes, 1 GiB = $2^{30}$ bytes, 1 TiB = $2^{40}$ bytes. Space-mission documents usually quote downlink rates in decimal units (1 Mbit/s = $10^{6}$ bit/s) and on-board memory in binary units, so a "256 GB" recorder may hold 256 GiB = 275 × $10^{9}$ bytes. Always check which convention a specification uses.

## A.2 Physical Constants

Table A.3 lists the physical constants used in this manual. The gravitational constant $G$ is known only to about 22 parts per million, which is why orbital mechanics always uses the product $GM$ (the *gravitational parameter* μ) — it can be measured from spacecraft tracking about a million times more precisely than $G$ or $M$ separately.

**TABLE A.3** Physical constants (CODATA 2022 recommended values).

| Constant | Symbol | Value | Notes |
|---|---|---|---|
| Newtonian constant of gravitation | $G$ | $6.674\,30 \times 10^{-11}$ m$^3$ kg$^{-1}$ s$^{-2}$ | ± 0.000 15 |
| Stefan–Boltzmann constant | σ | $5.670\,374\,419 \times 10^{-8}$ W m$^{-2}$ K$^{-4}$ | exact (derived) |
| Wien displacement constant | $b$ | $2.897\,771\,955 \times 10^{-3}$ m K | $\lambda_{peak} = b/T$ |
| Molar gas constant | $R$ | 8.314 462 618 J mol$^{-1}$ K$^{-1}$ | exact (derived) |
| Vacuum permittivity | $\varepsilon_0$ | $8.854\,187\,8188 \times 10^{-12}$ F/m | |
| Vacuum permeability | $\mu_0$ | $1.256\,637\,061\,27 \times 10^{-6}$ N/A$^2$ | $\approx 4\pi \times 10^{-7}$ |
| Impedance of free space | $Z_0$ | 376.730 Ω | $\mu_0 c$ |
| Electron mass | $m_e$ | $9.109\,383\,7139 \times 10^{-31}$ kg | 0.511 MeV/c$^2$ |
| Proton mass | $m_p$ | $1.672\,621\,925\,95 \times 10^{-27}$ kg | 938.272 MeV/c$^2$ |
| Atomic mass constant | $m_u$ | $1.660\,539\,068\,92 \times 10^{-27}$ kg | 1 Da |
| Electronvolt | eV | $1.602\,176\,634 \times 10^{-19}$ J | exact |

| Constant | Symbol | Value | Notes |
|---|---|---|---|
| Standard acceleration of gravity | $g_o$ | 9.806 65 m/s² | exact; used for $I_{sp}$ |
| Standard atmosphere | atm | 101 325 Pa | exact |
| Boltzmann constant in decibels | 10 log $k$ | −228.6 dBW K$^{-1}$ Hz$^{-1}$ | link budgets |

## A.3 Astronomical and Earth Constants

**TABLE A.4** Astronomical and Earth constants.

| Quantity | Value | Source / note |
|---|---|---|
| Astronomical unit (au) | 149 597 870 700 m | exact (IAU 2012) |
| Light-time for 1 au | 499.005 s (8 min 19 s) | |
| Light-year (Julian year × $c$) | 9.460 730 × 10$^{15}$ m = 63 241 au | exact |
| Parsec | 3.085 678 × 10$^{16}$ m = 3.262 ly = 206 265 au | exact via au |
| Sun, $GM_\odot$ | 1.327 124 400 41 × 10$^{20}$ m$^3$/s$^2$ | planetary ephemerides |
| Sun, mass | 1.988 4 × 10$^{30}$ kg | $GM/G$ |
| Sun, nominal radius | 695 700 km | IAU 2015 |
| Sun, nominal luminosity | 3.828 × 10$^{26}$ W | IAU 2015 |
| Sun, effective temperature | 5 772 K | IAU 2015 |
| Total solar irradiance at 1 au | ≈ 1 361 W/m² | varies ± 0.1 % with the cycle |
| Earth, $GM_\oplus$ | 3.986 004 418 × 10$^{14}$ m$^3$/s$^2$ | WGS 84 |
| Earth, mass | 5.972 2 × 10$^{24}$ kg | |
| Earth, equatorial radius $a$ | 6 378 137 m | WGS 84 |
| Earth, flattening $f$ | 1/298.257 223 563 | WGS 84 |
| Earth, polar radius $b$ | 6 356 752.3 m | WGS 84 |
| Earth, mean radius | 6 371.0 km | |
| Earth, $J_2$ zonal harmonic | 1.082 63 × 10$^{-3}$ | drives nodal regression |
| Earth, rotation rate $\omega_\oplus$ | 7.292 115 × 10$^{-5}$ rad/s | |
| Sidereal day | 86 164.091 s (23 h 56 min 4.09 s) | |
| Mean solar day | 86 400 s | |
| Tropical year | 365.242 19 days | seasons |
| Sidereal year | 365.256 36 days | |
| Obliquity of the ecliptic (J2000) | 23.439° | |
| Moon, $GM$ | 4 902.800 km$^3$/s$^2$ | |
| Moon, mean distance | 384 400 km (light-time 1.28 s) | varies 356 400–406 700 km |
| Moon, mean radius | 1 737.4 km | |

USEFUL ORBITAL NUMBERS FOR EARTH

**TABLE A.5** Circular-orbit period, speed and horizon distance (spherical Earth, *GM* from WGS 84). One degree of longitude along the geostationary arc is 736 km.

| Altitude | Period | Circular speed | Distance to horizon | Typical use |
|---|---|---|---|---|
| 200 km | 88.5 min | 7.78 km/s | 1 610 km | Lowest practical orbit, parking orbits |
| 400 km | 92.6 min | 7.67 km/s | 2 290 km | ISS, Tiangong |
| 550 km | 95.6 min | 7.59 km/s | 2 710 km | Starlink shells, EO satellites |
| 800 km | 100.9 min | 7.45 km/s | 3 290 km | Sun-synchronous weather and EO |
| 1 200 km | 109.4 min | 7.25 km/s | 4 090 km | OneWeb, Jason/Sentinel-6 |
| 20 200 km | 11 h 58 min | 3.87 km/s | 25 800 km | GPS (half a sidereal day) |
| 35 786 km | 23 h 56 min | 3.07 km/s | 41 680 km | Geostationary orbit (radius 42 164 km) |

Escape speed from the Earth's surface is 11.19 km/s; from a 400 km orbit it is 10.85 km/s, only 3.18 km/s more than circular speed.

## A.4 Conversion Factors

Aerospace still mixes SI with US customary units — rocket thrust in pounds-force, altitude in feet, pressures in psi, antenna gain in decibels. Table A.6 lists the conversions used most often; all are exact except where three or fewer significant figures are shown as a rounded value.

**TABLE A.6** Conversion factors.

| Quantity | Unit | SI value |
|---|---|---|
| Length | inch / foot | 0.025 4 m / 0.304 8 m (exact) |
| | statute mile / nautical mile | 1 609.344 m / 1 852 m (exact) |
| Mass | pound (lb) | 0.453 592 37 kg (exact) |
| | slug | 14.593 9 kg |
| | metric tonne (t) | 1 000 kg |
| Force | pound-force (lbf) | 4.448 222 N |
| | kilogram-force (kgf) | 9.806 65 N (exact) |
| Pressure | psi (lbf/in²) | 6 894.757 Pa |
| | bar / atm / torr | $10^5$ Pa / 101 325 Pa / 133.322 Pa |
| Energy | kilowatt-hour | 3.6 MJ (exact) |
| | British thermal unit (BTU) | 1 055.06 J |
| | thermochemical calorie | 4.184 J (exact) |
| | kilotonne of TNT | $4.184 \times 10^{12}$ J |
| Power | horsepower (mechanical) | 745.700 W |
| Speed | knot / mile per hour | 0.514 444 m/s / 0.447 04 m/s |
| | Mach 1 at sea level (15 °C) | 340.3 m/s |

| Quantity | Unit | SI value |
|---|---|---|
| Angle | degree / arcminute / arcsecond | 17.453 3 mrad / 290.888 µrad / 4.848 14 µrad |
| | milliarcsecond (mas) | 4.848 14 nrad |
| | steradian (sphere = 4π sr) | 1 sr = 3 282.8 deg² |
| Temperature | °C / °F / °R | $T_K = T_{°C} + 273.15$; $T_{°F} = 1.8\ T_{°C} + 32$; $T_{°R} = 1.8\ T_K$ |
| Radiation dose | rad / rem | 0.01 Gy / 0.01 Sv |
| Magnetic flux density | gauss (G) / gamma (γ) | $10^{-4}$ T / 1 nT |
| Specific impulse | second (s) | effective exhaust speed $v_e = I_{sp} \times g_0$ (300 s ≈ 2.94 km/s) |
| Thrust-specific fuel consumption | lb/(lbf h) | 28.33 mg/(N s) |
| Data rate | bit/s (decimal) | 1 Gbit/s = $10^9$ bit/s |

## A.5 Decibels

Radio engineers express ratios that span many orders of magnitude in decibels: a power ratio $P_2/P_1$ is $10 \log_{10}(P_2/P_1)$ dB. A factor of two is 3.01 dB, ten is 10 dB, a thousand is 30 dB, and a million is 60 dB. Multiplications in a link budget become additions. A suffix tells you the reference (Table A.7).

**TABLE A.7** Decibel conventions used in link budgets (Chapter 7).

| Unit | Reference | Example |
|---|---|---|
| dBW | 1 W | 20 W transmitter = 13.0 dBW |
| dBm | 1 mW | 1 W = 30 dBm; GPS signal at the ground ≈ −128.5 dBm |
| dBi | isotropic antenna | 1 m dish at 8.4 GHz with 60 % efficiency ≈ 36.7 dBi |
| dBic | isotropic, circular polarisation | common for GNSS and S-band patch antennas |
| dBK | 1 kelvin | 290 K system noise temperature = 24.6 dBK |
| dB/K | gain-to-noise temperature ratio $G/T$ | ground station 35 dB/K |
| dBHz | 1 Hz | 1 MHz bandwidth = 60 dBHz; $C/N_0$ in dBHz |
| dBW/m² | flux density | saturated flux density of a GEO transponder |
| dBsm | 1 m² (radar cross-section) | a 1U CubeSat ≈ −20 dBsm at high frequency |
| dBc | carrier power | spurious emission 60 dBc below the carrier |

## A.6 Time Scales and Epochs

A spacecraft clock, a ground station log and a planetary ephemeris may each use a different time scale. Mixing them up produces errors of a minute or more — enough to put a planetary encounter or a collision-avoidance manoeuvre in the wrong place. Table A.8 summarises the

scales in use.

**TABLE A.8** Time scales used in space operations.

| Time scale | What it is | Relation (2026) |
|---|---|---|
| TAI (International Atomic Time) | Weighted average of about 450 atomic clocks worldwide | Reference |
| UTC (Coordinated Universal Time) | Civil time; stays within 0.9 s of Earth rotation by inserting leap seconds | UTC = TAI − 37 s (since 1 January 2017) |
| UT1 | Time defined by the Earth's actual rotation angle | UT1 − UTC within ± 0.9 s, published by the IERS |
| GPS time | Continuous atomic time of the GPS system, started 6 January 1980 | GPS = TAI − 19 s = UTC + 18 s |
| Galileo System Time / BeiDou Time | Continuous GNSS time scales | GST = GPS time; BDT = UTC + 4 s |
| TT (Terrestrial Time) | Ideal time on the geoid, used for ephemerides | TT = TAI + 32.184 s = UTC + 69.184 s |
| TDB (Barycentric Dynamical Time) | Time at the Solar System barycentre, used by planetary ephemerides | TDB − TT within ± 1.7 ms (periodic) |
| Spacecraft clock (SCLK) | On-board counter; drifts | Correlated to UTC by the ground (SPICE kernels) |

**Leap seconds.** Because the Earth's rotation is irregular, 27 leap seconds were added to UTC between 1972 and 2016. None has been needed since, because the Earth has recently been spinning slightly faster. In 2022 the General Conference on Weights and Measures decided that the tolerance between UTC and UT1 will be relaxed by 2035, effectively ending leap seconds. Software that assumes every minute has 60 seconds, or that GPS time equals UTC, has caused real outages.

**Julian Date.** Astronomers count days continuously from noon on 1 January 4713 BC (Julian calendar). The Julian Date (JD) at 00:00 UTC on 1 January 2026 is 2 461 041.5. The Modified Julian Date is MJD = JD − 2 400 000.5, so a day starts at midnight and the numbers are shorter (MJD 61 041 for the same instant). The standard epoch **J2000.0** is JD 2 451 545.0 TT — 12:00 TT on 1 January 2000 (11:58:55.816 UTC). Orbital elements in two-line element sets use a two-digit year and day-of-year with fraction.

**GNSS weeks.** GPS broadcasts a week number that originally rolled over every 1 024 weeks (19.6 years); rollovers on 21 August 1999 and 6 April 2019 broke unprepared receivers. Modernised GPS messages use a 13-bit week number that will not roll over until 2137.

**Relativity matters.** A GPS satellite clock runs fast by about 38 µs per day compared with a clock on the ground — about +45.9 µs from the weaker gravity at 20 200 km and −7.2 µs from its orbital speed. Uncorrected, this would cause ranging errors growing by about 11 km per day; the clocks are deliberately set slightly slow before launch. A clock on the Moon runs fast by about 56 µs per day relative to Earth. In 2024 the US government asked NASA to establish a Coordinated Lunar Time, and the International Astronomical Union adopted resolutions calling for a lunar reference system and time scale, so that lunar navigation systems such as LunaNet and ESA's Moonlight can share a common clock.

**Mars time.** A Martian solar day (*sol*) lasts 24 h 39 min 35.244 s (88 775.244 s). Mars rover teams work on "Mars time" for the first weeks of a mission, shifting their working day by about 40 minutes every Earth day. A Mars year lasts 668.6 sols (687.0 Earth days).

**KEY TAKEAWAYS**

- Seven constants now define the SI exactly; $c$, $h$, $e$ and $k$ have no uncertainty.
- Use the gravitational parameter $\mu = GM$, not $G$ and $M$ separately — it is known far more precisely.
- $g_0 = 9.806\,65\ \text{m/s}^2$ exactly converts specific impulse in seconds to exhaust speed.
- In link budgets, add decibels; −228.6 dBW/(K Hz) is Boltzmann's constant.
- UTC = TAI − 37 s, GPS = UTC + 18 s, TT = TAI + 32.184 s in 2026 — always state which time scale a timestamp uses.

APPENDIX

# B Key Equations

> *The essential equations of space engineering gathered in one place — orbital mechanics and manoeuvres, rocket propulsion, atmospheric flight and entry, power, thermal and attitude control, communication links, remote sensing and radar, the space environment and astronomy — each with its symbols defined and a pointer to the chapter where it is derived and used.*

**IN THIS CHAPTER**


B.1 Orbital Mechanics
B.2 Rocket Propulsion
B.3 Atmospheric Flight, Drag and Entry
B.4 Power, Thermal and Attitude Control
B.5 Communication Links
B.6 Remote Sensing, Optics and Radar
B.7 Space Environment and Astronomy


This appendix is a formula sheet. Each table gives the equation, defines its symbols briefly and names the chapter where it is explained. SI units are assumed throughout unless stated: metres, kilograms, seconds, kelvin, watts and radians. μ = *GM* is the gravitational parameter of the central body (Appendix C), *r* a distance from its centre, $g_0 = 9.806\ 65$ m/s², σ the Stefan–Boltzmann constant and *k* the Boltzmann constant (Appendix A).

## B.1 Orbital Mechanics

**TABLE B.1** Orbital mechanics.

| Quantity | Equation | Symbols and notes |
|---|---|---|
| Vis-viva (speed anywhere on an orbit) | $v^2 = \mu\left(\frac{2}{r} - \frac{1}{a}\right)$ | *a* semi-major axis; Chapter 3 |
| Circular and escape speed | $v_c = \sqrt{\mu/r},\ v_{esc} = \sqrt{2\mu/r}$ | |
| Orbital period | $T = 2\pi\sqrt{a^3/\mu}$ | Kepler's third law |
| Specific orbital energy | $\varepsilon = \frac{v^2}{2} - \frac{\mu}{r} = -\frac{\mu}{2a}$ | negative for bound orbits |
| Specific angular momentum | $h = rv\cos\gamma = \sqrt{\mu a(1-e^2)}$ | γ flight-path angle |
| Orbit equation (conic) | $r = \frac{a(1-e^2)}{1+e\cos\nu}$ | *e* eccentricity, ν true anomaly |
| Periapsis and apoapsis | $r_p = a(1-e),\ r_a = a(1+e)$ | |
| Kepler's equation | $M = E - e\sin E,\ M = n(t-t_p),\ n = \sqrt{\mu/a^3}$ | *E* eccentric anomaly, *n* mean motion |
| True from eccentric anomaly | $\tan\frac{\nu}{2} = \sqrt{\frac{1+e}{1-e}}\tan\frac{E}{2}$ | |

| Quantity | Equation | Symbols and notes |
|---|---|---|
| Hohmann transfer $\Delta v$ | $\Delta v_1 = \sqrt{\frac{\mu}{r_1}}\left(\sqrt{\frac{2r_2}{r_1+r_2}}-1\right)$, $\Delta v_2 = \sqrt{\frac{\mu}{r_2}}\left(1-\sqrt{\frac{2r_1}{r_1+r_2}}\right)$ | circular, coplanar orbits |
| Hohmann transfer time | $t_H = \pi\sqrt{(r_1+r_2)^3/(8\mu)}$ | half the transfer-orbit period |
| Pure plane change | $\Delta v = 2v\sin(\Delta i/2)$ | cheapest at low speed (apoapsis) |
| Combined manoeuvre | $\Delta v^2 = v_1^2 + v_2^2 - 2v_1 v_2\cos\Delta i$ | GTO-to-GEO insertion |
| Nodal regression ($J_2$) | $\dot\Omega = -\frac{3}{2}nJ_2\left(\frac{R}{p}\right)^2\cos i$, $p = a(1-e^2)$ | Earth $J_2$ = 1.0826 × $10^{-3}$ |
| Sun-synchronous condition | $\dot\Omega = 360^\circ/365.2422\ \text{d} = 0.9856^\circ/\text{day}$ | gives $i$ ≈ 97–99° in LEO |
| Apsidal precession ($J_2$) | $\dot\omega = \frac{3}{4}nJ_2\left(\frac{R}{p}\right)^2(5\cos^2 i - 1)$ | zero at $i$ = 63.4° (Molniya) |
| Ground-track shift per orbit | $\Delta\lambda \approx T\,(\omega_\oplus - \dot\Omega)$ | westward, in radians |
| Sphere of influence | $r_{SOI} = a_p(m_p/M_\odot)^{2/5}$ | patched conics; Chapter 24 |
| Hyperbolic excess and C3 | $v_\infty^2 = C_3 = v^2 - \frac{2\mu}{r}$ | departure energy |
| $\Delta v$ to depart from a circular orbit | $\Delta v = \sqrt{v_\infty^2 + 2\mu/r} - \sqrt{\mu/r}$ | Oberth effect: burn deep in the well |
| Gravity-assist turn angle | $\sin\frac{\delta}{2} = \frac{1}{1 + r_p v_\infty^2/\mu}$ | velocity change $2v_\infty\sin(\delta/2)$ |
| Clohessy–Wiltshire (relative motion) | $\ddot x - 3n^2x - 2n\dot y = 0$, $\ddot y + 2n\dot x = 0$, $\ddot z + n^2 z = 0$ | $x$ radial, $y$ along-track; Chapters 21 and 40 |
| Synodic period | $\frac{1}{T_{syn}} = \left\lvert\frac{1}{T_1} - \frac{1}{T_2}\right\rvert$ | interval between launch windows |

## B.2 Rocket Propulsion

**TABLE B.2** Rocket propulsion.

| Quantity | Equation | Symbols and notes |
|---|---|---|
| Rocket equation (Tsiolkovsky) | $\Delta v = v_e\ln\frac{m_0}{m_f} = I_{sp}\,g_0\ln\frac{m_0}{m_f}$ | $m_0$ initial, $m_f$ final mass; Chapter 35 |
| Propellant for a given $\Delta v$ | $m_p = m_f\left(e^{\Delta v/v_e} - 1\right)$ | |
| Thrust | $F = \dot m v_e + (p_e - p_a)A_e$ | $p_e$ exit, $p_a$ ambient pressure, $A_e$ exit area |
| Specific impulse | $I_{sp} = \frac{F}{\dot m g_0}$ | seconds |
| Characteristic velocity | $c^* = \frac{p_c A_t}{\dot m}$ | combustion efficiency; $A_t$ throat area |
| Thrust coefficient | $C_F = \frac{F}{p_c A_t}$, $F = C_F\,p_c A_t$ | nozzle efficiency; 1.4–2.0 |
| Ideal exhaust velocity | $v_e = \sqrt{\frac{2\gamma}{\gamma-1}\frac{R_u T_c}{\mathcal{M}}\left[1 - \left(\frac{p_e}{p_c}\right)^{(\gamma-1)/\gamma}\right]}$ | γ ratio of specific heats, $\mathcal{M}$ molar mass, $T_c$ chamber temperature |

| Quantity | Equation | Symbols and notes |
|---|---|---|
| Nozzle area ratio | $\frac{A_e}{A_t} = \frac{1}{M_e}\left[\frac{2}{\gamma+1}\left(1+\frac{\gamma-1}{2}M_e^2\right)\right]^{\frac{\gamma+1}{2(\gamma-1)}}$ | $M_e$ exit Mach number |
| Staged rocket $\Delta v$ | $\Delta v = \sum_i v_{e,i} \ln \frac{m_{0,i}}{m_{f,i}}$ | Chapter 34 |
| Losses in ascent | $\Delta v_{total} = \Delta v_{orbit} + \Delta v_{gravity} + \Delta v_{drag} + \Delta v_{steering}$ | ≈ 9.3–9.6 km/s to LEO |
| Electric thruster power | $P_{jet} = \frac{F v_e}{2} = \frac{\dot{m} v_e^2}{2},\ F = \frac{2\eta P}{v_e}$ | η thruster efficiency; Chapters 6 and 42 |
| Burn time at constant thrust | $t_b = \frac{m_p v_e}{F}$ | |

## B.3 Atmospheric Flight, Drag and Entry

**TABLE B.3** Atmospheric flight and entry.

| Quantity | Equation | Symbols and notes |
|---|---|---|
| Exponential atmosphere | $\rho = \rho_0 e^{-h/H},\ H = \frac{R_u T}{\mathcal{M} g}$ | $H$ scale height (8.5 km on Earth); Chapter 2 |
| Dynamic pressure | $q = \frac{1}{2}\rho v^2$ | "max-Q" during ascent |
| Drag force | $D = \frac{1}{2}\rho v^2 C_D A$ | $C_D$ ≈ 2.2 for satellites in free-molecular flow |
| Ballistic coefficient | $\beta = \frac{m}{C_D A}$ | kg/m²; higher β → longer orbital life |
| Decay of a circular orbit | $\frac{da}{dt} = -\frac{\sqrt{\mu a}\,\rho}{\beta}$ | per orbit $\Delta a = -2\pi \rho a^2/\beta$; Chapter 21 |
| Peak deceleration (ballistic entry, Allen–Eggers) | $a_{max} = \frac{v_E^2 \sin\gamma_E}{2eH}$ | $v_E$, $\gamma_E$ entry speed and angle; $e$ = 2.718; Chapter 10 |
| Stagnation-point heat flux (Sutton–Graves) | $\dot{q} = k\sqrt{\rho/r_n}\,v^3$ | $k \approx 1.74 \times 10^{-4}$ (Earth), $1.90 \times 10^{-4}$ (Mars) in SI; $r_n$ nose radius |
| Terminal velocity | $v_t = \sqrt{\frac{2mg}{\rho C_D A}}$ | parachute sizing; Chapter 27 |
| Lift-to-drag ratio and cross-range | $L/D$; cross-range grows roughly with $(L/D)^2$ | capsule ≈ 0.3, Shuttle ≈ 1–4.5 |
| Mach number | $M = v/a,\ a = \sqrt{\gamma R_u T/\mathcal{M}}$ | 340 m/s at sea level |

## B.4 Power, Thermal and Attitude Control

**TABLE B.4** Power, thermal and attitude control.

| Quantity | Equation | Symbols and notes |
|---|---|---|
| Solar array power | $P = S\,\eta\,A\cos\theta \cdot L_d$ | $S$ ≈ 1 361 W/m² at 1 au (∝ $1/d^2$); η cell efficiency; $L_d$ degradation; Chapter 5 |
| Array sizing with eclipses | $P_{sa} = \frac{\frac{P_e T_e}{X_e} + \frac{P_d T_d}{X_d}}{T_d}$ | $e$ eclipse, $d$ daylight; $X$ path efficiencies |
| Battery capacity | $C = \frac{P_e T_e}{\mathrm{DoD} \cdot n \cdot \eta_b}$ | DoD depth of discharge |

| Quantity | Equation | Symbols and notes |
|---|---|---|
| Radiated power | $P = \varepsilon \sigma A T^4$ | ε emissivity |
| Equilibrium temperature of a plate | $T = \left( \frac{\alpha S + Q_{int}/A}{\varepsilon \sigma} \right)^{1/4}$ | α solar absorptance; one side lit, one radiating |
| Planetary equilibrium temperature | $T_{eq} = \left( \frac{S(1 - A_B)}{4\sigma} \right)^{1/4}$ | $A_B$ Bond albedo; Earth 255 K |
| Conduction | $Q = \frac{kA}{L} \Delta T$ | $k$ conductivity |
| Euler's equations | $I_x \dot{\omega}_x + (I_z - I_y)\omega_y \omega_z = T_x$ (cyclic) | principal axes; Chapter 6 |
| Gravity-gradient torque | $T_{gg} = \frac{3\mu}{2r^3} \lvert I_z - I_y \rvert \sin 2\theta$ | θ deviation from local vertical |
| Solar radiation pressure torque | $T_{srp} = \frac{S}{c} A_s (1 + q) \cos\phi \, (c_{ps} - c_g)$ | $q$ reflectance |
| Aerodynamic torque | $T_a = \frac{1}{2} \rho v^2 C_D A_s (c_{pa} - c_g)$ | dominant below ≈ 500 km |
| Magnetic torque | $T_m = DB$, $B \approx B_0 (R/r)^3 \sqrt{1 + 3\sin^2 \lambda}$ | $D$ residual dipole (A m²) |
| Wheel momentum to store | $H = T_d \cdot \frac{T_{orbit}}{4} \cdot 0.707$ | cyclic disturbance estimate |
| Slew time (bang-bang) | $t = 2\sqrt{\theta I / T}$ | θ slew angle, $T$ torque |

## B.5 Communication Links

**TABLE B.5** Communication links.

| Quantity | Equation | Symbols and notes |
|---|---|---|
| Free-space path loss | $L_{fs} = \left( \frac{4\pi d}{\lambda} \right)^2$; in dB: $92.45 + 20\log f_{GHz} + 20\log d_{km}$ | Chapter 7, Appendix D |
| Antenna gain | $G = \eta \left( \frac{\pi D}{\lambda} \right)^2$ | η ≈ 0.55–0.70 for dishes |
| Half-power beamwidth | $\theta_{3dB} \approx 70 \frac{\lambda}{D}$ degrees | |
| Effective isotropic radiated power | $\mathrm{EIRP} = P_t G_t / L_t$ | dBW |
| Received carrier-to-noise density | $\frac{C}{N_0} = \mathrm{EIRP} - L_{fs} - L_a + \frac{G}{T} - 10\log k$ | in dB; −10 log $k$ = +228.6 |
| Energy per bit to noise density | $\frac{E_b}{N_0} = \frac{C}{N_0} - 10\log R_b$ | $R_b$ bit rate |
| Link margin | $M = \left( \frac{E_b}{N_0} \right)_{received} - \left( \frac{E_b}{N_0} \right)_{required}$ | typically ≥ 3 dB |
| System noise temperature | $T_{sys} = T_{ant} + (L - 1)T_0 + L T_{rx}$ | $T_0$ = 290 K |
| Shannon capacity | $C = B \log_2 (1 + S/N)$ | upper limit on bit rate |
| Doppler shift | $\Delta f = f \frac{v_r}{c}$ | ± 50 kHz at 2.2 GHz from LEO |
| Slant range to a satellite | $d = \sqrt{(R + h)^2 - R^2 \cos^2 \varepsilon} - R \sin \varepsilon$ | ε elevation angle |

## B.6 Remote Sensing, Optics and Radar

**TABLE B.6** Remote sensing, optics and radar.

| Quantity | Equation | Symbols and notes |
|---|---|---|
| Planck's law | $B_\lambda(T) = \frac{2hc^2}{\lambda^5}\,\frac{1}{e^{hc/(\lambda kT)}-1}$ | spectral radiance; Chapter 13 |
| Wien's law | $\lambda_{max} = 2898\ \mu\mathrm{m\,K}/T$ | Sun 0.50 µm, Earth 10 µm |
| Stefan–Boltzmann law | $M = \varepsilon\sigma T^4$ | |
| Diffraction limit (Rayleigh) | $\theta = 1.22\,\lambda/D$ | Chapters 14 and 38 |
| Ground sample distance | $\mathrm{GSD} = \frac{H\,p}{f}$ | $p$ pixel pitch, $f$ focal length, $H$ altitude |
| Swath width (nadir) | $W \approx 2H\tan(\mathrm{FOV}/2)$ | |
| Signal-to-noise ratio of a detector | $\mathrm{SNR} = \frac{S}{\sqrt{S + n_{pix}(B + Dt + R^2)}}$ | $S$ signal electrons, $B$ background, $D$ dark current, $R$ read noise |
| NDVI | $\mathrm{NDVI} = \frac{\rho_{NIR} - \rho_{red}}{\rho_{NIR} + \rho_{red}}$ | vegetation index |
| Radar range resolution | $\delta_r = \frac{c}{2B}$ (slant), ground: $\delta_r/\sin\theta_i$ | $B$ chirp bandwidth; Chapter 15 |
| SAR azimuth resolution (stripmap) | $\delta_a \approx L_a/2$ | $L_a$ antenna length |
| Radar equation | $P_r = \frac{P_t G^2 \lambda^2 \sigma}{(4\pi)^3 R^4}$ | σ radar cross-section |
| Altimeter range | $R = \frac{c\,\Delta t}{2}$ | $\Delta t$ echo delay |

## B.7 Space Environment and Astronomy

**TABLE B.7** Space environment and astronomy.

| Quantity | Equation | Symbols and notes |
|---|---|---|
| Dipole magnetic field | $B = B_0\left(\frac{R_E}{r}\right)^3\sqrt{1+3\sin^2\lambda}$ | $B_0 \approx 30$ µT at Earth's equator; Chapter 23 |
| Dipole field line (L-shell) | $r = L R_E \cos^2\lambda$ | λ magnetic latitude |
| Gyrofrequency | $f_g = \frac{qB}{2\pi m}$ | electrons: 28 GHz per tesla |
| Plasma frequency | $f_p \approx 8.98\sqrt{n_e}$ Hz | $n_e$ in m$^{-3}$; sets ionospheric reflection |
| Ionospheric group delay | $\Delta\rho = \frac{40.3}{f^2}\,\mathrm{TEC}$ | metres, TEC in electrons/m$^2$; Chapter 18 |
| Dynamic pressure of the solar wind | $p = \rho_{sw} v_{sw}^2 = n m_p v^2$ | ≈ 1–2 nPa, sets magnetopause distance |
| Magnetopause stand-off | $\frac{r_{mp}}{R_E} \approx \left(\frac{2B_0^2}{\mu_0\, n m_p v^2}\right)^{1/6}$ | ≈ 10 $R_E$ |
| Apparent magnitude | $m_1 - m_2 = -2.5\log_{10}\frac{F_1}{F_2}$ | 5 magnitudes = factor 100 |
| Distance modulus | $m - M = 5\log_{10}\frac{d}{10\ \mathrm{pc}}$ | |

| Quantity | Equation | Symbols and notes |
|---|---|---|
| Transit depth | $\delta = (R_p/R_\star)^2$ | Earth–Sun: 84 ppm; Chapter 39 |
| Hubble–Lemaître law | $v = H_0 d$ | $H_0$ ≈ 67–73 km/s/Mpc |
| Redshift | $1 + z = \frac{\lambda_{obs}}{\lambda_{emit}}$ | |
| Radiation dose | $D = E/m$ (gray), $H = w_R D$ (sievert) | Chapter 30 |

**USING THESE EQUATIONS SAFELY**

Every equation here rests on assumptions — two bodies and point masses for the conic equations, ideal gas and frozen chemistry for the nozzle equations, an exponential atmosphere for the entry formulas, isotropic antennas for path loss. Use them to size, estimate and check; use validated tools (high-fidelity propagators, chemical-equilibrium codes, ray-tracing and finite-element models) for design decisions, and always check that an answer has the right units and order of magnitude.

APPENDIX

# C Solar System Data

*Reference data for mission design and planetary science — physical and orbital properties of the Sun, the planets, the Moon and Pluto; gravitational parameters, spheres of influence and idealised transfer requirements from Earth; the major moons; the small bodies visited by spacecraft; Lagrange points; and the US Standard Atmosphere. Values follow the NASA Planetary Fact Sheets, JPL Solar System Dynamics and IAU working-group data, rounded for engineering use.*

**IN THIS CHAPTER**


C.1 The Planets
C.2 Gravitational Parameters and Mission Design Data
C.3 Major Moons
C.4 Small Bodies Visited by Spacecraft
C.5 Lagrange Points
C.6 The US Standard Atmosphere 1976


The tables in this appendix give the values used in the chapters on orbital mechanics (Chapter 3), deep-space mission design (Chapter 24) and planetary exploration (Chapters 25–29). Masses and radii are known far more precisely than shown here; the values are rounded so that the tables stay readable. When designing a real trajectory, use the gravitational parameters and ephemerides distributed by JPL (the DE440 series) through the SPICE toolkit.

## C.1 The Planets

**TABLE C.1** Physical properties of the planets, the Moon and Pluto. A negative rotation period means retrograde rotation. Gravity and escape speed are at the equator of the 1-bar level for the giant planets. Source: NASA Planetary Fact Sheet.

| Body | Mass ($10^{24}$ kg) | Equatorial radius (km) | Mean density (kg/m$^3$) | Surface gravity (m/s$^2$) | Escape speed (km/s) | Sidereal rotation (h) | Obliquity (°) |
|---|---|---|---|---|---|---|---|
| Mercury | 0.330 | 2 440 | 5 429 | 3.7 | 4.3 | 1 407.6 | 0.03 |
| Venus | 4.87 | 6 052 | 5 243 | 8.9 | 10.4 | −5 832.5 | 177.4 |
| Earth | 5.97 | 6 378 | 5 514 | 9.8 | 11.2 | 23.93 | 23.4 |
| Moon | 0.073 | 1 738 | 3 344 | 1.6 | 2.4 | 655.7 | 6.7 |
| Mars | 0.642 | 3 396 | 3 934 | 3.7 | 5.0 | 24.62 | 25.2 |
| Jupiter | 1 898 | 71 492 | 1 326 | 23.1 | 59.5 | 9.93 | 3.1 |
| Saturn | 568 | 60 268 | 687 | 9.0 | 35.5 | 10.66 | 26.7 |
| Uranus | 86.8 | 25 559 | 1 270 | 8.7 | 21.3 | −17.24 | 97.8 |
| Neptune | 102 | 24 764 | 1 638 | 11.0 | 23.5 | 16.11 | 28.3 |
| Pluto | 0.013 0 | 1 188 | 1 854 | 0.62 | 1.2 | −153.3 | 122.5 |

**TABLE C.2** Orbital properties of the planets and Pluto (epoch J2000). Moon counts change as surveys find more small irregular moons; the 2026 values include the Jupiter and Saturn moons announced by the Minor Planet Center in March 2026.

| Body | Semi-major axis (au) | Eccentricity | Inclination to ecliptic (°) | Orbital period | Mean surface temperature | Surface pressure | Known moons (2026) |
|---|---|---|---|---|---|---|---|
| Mercury | 0.387 | 0.206 | 7.0 | 88.0 d | 167 °C (−180 to 430) | ≈ 0 | 0 |
| Venus | 0.723 | 0.007 | 3.4 | 224.7 d | 464 °C | 92 bar | 0 |
| Earth | 1.000 | 0.017 | 0.0 | 365.26 d | 15 °C | 1.014 bar | 1 |
| Mars | 1.524 | 0.093 | 1.8 | 687.0 d | −65 °C | 0.006 bar | 2 |
| Jupiter | 5.203 | 0.048 | 1.3 | 11.86 yr | −110 °C (1 bar) | — | 101 |
| Saturn | 9.555 | 0.054 | 2.5 | 29.45 yr | −140 °C (1 bar) | — | 285 |
| Uranus | 19.22 | 0.047 | 0.8 | 84.0 yr | −195 °C (1 bar) | — | 29 |
| Neptune | 30.11 | 0.009 | 1.8 | 164.8 yr | −200 °C (1 bar) | — | 16 |
| Pluto | 39.48 | 0.249 | 17.2 | 248 yr | −225 °C | ≈ 10 µbar | 5 |

The giant planets have no solid surface; their “surface” values refer to the level where the pressure is 1 bar. Saturn’s mean density is lower than that of water. Venus and Uranus rotate backwards; Uranus is tipped on its side, so each pole faces the Sun for 42 years at a time.

## C.2 Gravitational Parameters and Mission Design Data

Table C.3 gives the gravitational parameter $\mu = GM$ of each body, its $J_2$ oblateness coefficient, and the radius of its *sphere of influence* — the region inside which the planet’s gravity, rather than the Sun’s, is the better reference for a patched-conic trajectory (Chapter 24). The sphere-of-influence radius is $r_{\mathrm{SOI}} = a\,(m/M_\odot)^{2/5}$, where $a$ is the planet’s orbital radius.

**TABLE C.3** Gravitational parameters, oblateness and spheres of influence. Computed from JPL values; the Moon’s sphere of influence is with respect to the Earth.

| Body | $\mu = GM$ (km³/s²) | $J_2$ (× 10⁻⁶) | Sphere of influence ($10^3$ km) | in planet radii |
|---|---|---|---|---|
| Sun | 132 712 440 041 | 0.2 | — | — |
| Mercury | 22 032 | 50 | 112 | 46 |
| Venus | 324 859 | 4.5 | 616 | 102 |
| Earth | 398 600 | 1 082.6 | 925 | 145 |
| Moon | 4 902.8 | 203 | 66 (about Earth) | 38 |
| Mars | 42 828 | 1 960 | 577 | 170 |
| Jupiter | 126 686 534 | 14 736 | 48 200 | 674 |
| Saturn | 37 931 187 | 16 298 | 54 600 | 907 |
| Uranus | 5 793 939 | 3 343 | 51 800 | 2 028 |
| Neptune | 6 836 529 | 3 411 | 86 800 | 3 504 |
| Pluto | 869.6 | — | 3 150 | 2 648 |

Table C.4 gives the idealised *Hohmann transfer* from Earth to each planet, assuming circular, coplanar orbits. Real launch opportunities differ — Mars needs a C3 of 8–16 km²/s² depending on the year, and missions to the outer planets use gravity assists to cut both C3 and flight time (Chapter 24) — but the table shows the scale of each journey. *C3* is the square of the hyperbolic excess speed at departure; $v_\infty$ at arrival is the speed relative to the planet before its gravity accelerates the spacecraft.

**TABLE C.4** Idealised Hohmann transfers from Earth (circular coplanar orbits; Pluto at its mean distance). The synodic period is the interval between successive launch opportunities.

| Destination | Departure C3 (km²/s²) | Δv from 400 km LEO (km/s) | Arrival $v_\infty$ (km/s) | Transfer time | Synodic period (days) |
|---|---|---|---|---|---|
| Mercury | 56.7 | 5.54 | 9.61 | 105 d | 116 |
| Venus | 6.2 | 3.46 | 2.71 | 146 d | 584 |
| Mars | 8.7 | 3.57 | 2.65 | 259 d | 780 |
| Jupiter | 77.3 | 6.29 | 5.64 | 2.73 yr | 399 |
| Saturn | 105.9 | 7.28 | 5.44 | 6.06 yr | 378 |
| Uranus | 127.3 | 7.98 | 4.66 | 16.1 yr | 370 |
| Neptune | 135.8 | 8.25 | 4.05 | 30.7 yr | 367 |
| Pluto | 139.6 | 8.37 | 3.69 | 45.5 yr | 367 |

## C.3 Major Moons

**TABLE C.5** Properties of the major moons. Source: JPL Solar System Dynamics planetary satellite physical and mean orbital parameters.

| Moon | Planet | Mean radius (km) | μ (km³/s²) | Orbit radius ($10^3$ km) | Period (days) | Notes |
|---|---|---|---|---|---|---|
| Moon | Earth | 1 737.4 | 4 902.8 | 384.4 | 27.32 | Synchronous rotation; polar ice in shadowed craters |
| Phobos | Mars | 11.1 | $7.1 \times 10^{-4}$ | 9.38 | 0.319 | Target of JAXA's MMX sample return |
| Deimos | Mars | 6.2 | $1.0 \times 10^{-4}$ | 23.46 | 1.263 | |
| Io | Jupiter | 1 821.6 | 5 959.9 | 421.8 | 1.769 | Most volcanically active body known |
| Europa | Jupiter | 1 560.8 | 3 202.7 | 671.1 | 3.551 | Subsurface ocean; Europa Clipper arrives 2030 |
| Ganymede | Jupiter | 2 631.2 | 9 887.8 | 1 070.4 | 7.155 | Largest moon; own magnetic field; JUICE target |
| Callisto | Jupiter | 2 410.3 | 7 179.3 | 1 882.7 | 16.69 | Heavily cratered |

| Moon | Planet | Mean radius (km) | μ (km³/s²) | Orbit radius ($10^3$ km) | Period (days) | Notes |
|---|---|---|---|---|---|---|
| Mimas | Saturn | 198.2 | 2.50 | 185.5 | 0.942 | Possible young internal ocean |
| Enceladus | Saturn | 252.1 | 7.21 | 238.0 | 1.370 | South-polar water plumes |
| Tethys | Saturn | 531.1 | 41.2 | 294.7 | 1.888 | |
| Dione | Saturn | 561.4 | 73.1 | 377.4 | 2.737 | |
| Rhea | Saturn | 763.8 | 153.9 | 527.1 | 4.518 | |
| Titan | Saturn | 2 574.7 | 8 978.1 | 1 221.9 | 15.945 | 1.5 bar $N_2$ atmosphere; methane lakes; Dragonfly target |
| Iapetus | Saturn | 734.5 | 120.5 | 3 560.8 | 79.33 | Two-toned surface |
| Miranda | Uranus | 235.8 | 4.3 | 129.9 | 1.413 | Extreme cliffs |
| Ariel | Uranus | 578.9 | 83.5 | 190.9 | 2.520 | |
| Umbriel | Uranus | 584.7 | 85.1 | 266.0 | 4.144 | |
| Titania | Uranus | 788.4 | 228.2 | 436.3 | 8.706 | Largest moon of Uranus |
| Oberon | Uranus | 761.4 | 192.4 | 583.5 | 13.46 | |
| Triton | Neptune | 1 353.4 | 1 427.6 | 354.8 | 5.877 (retrograde) | Captured Kuiper-belt object; nitrogen geysers |
| Charon | Pluto | 606 | 106.1 | 19.6 | 6.387 | Mutually locked with Pluto |

## C.4 Small Bodies Visited by Spacecraft

**TABLE C.6** Small bodies visited, or soon to be visited, by spacecraft (Chapters 25–29).

| Body | Type | Size | Orbit | Spacecraft (year of visit) |
|---|---|---|---|---|
| 1 Ceres | Dwarf planet (C-type) | 939 km diameter | 2.77 au, 4.6 yr | Dawn orbiter (2015–18) |
| 4 Vesta | Differentiated asteroid | 525 km | 2.36 au, 3.6 yr | Dawn orbiter (2011–12) |
| 433 Eros | S-type NEA | 34 × 11 × 11 km | 1.46 au, 1.76 yr | NEAR Shoemaker; landed 2001 |
| 25143 Itokawa | S-type NEA | 535 × 294 × 209 m | 1.32 au | Hayabusa sample return (2005, returned 2010) |
| 162173 Ryugu | C-type NEA | 0.9 km | 1.19 au | Hayabusa2 sample return (2018–19, returned 2020) |
| 101955 Bennu | B-type NEA | 0.49 km | 1.13 au | OSIRIS-REx sample return (2018–21, returned 2023) |

| Body | Type | Size | Orbit | Spacecraft (year of visit) |
|---|---|---|---|---|
| 65803 Didymos / Dimorphos | S-type binary NEA | 0.78 km / 0.15 km | 1.64 au | DART impact (2022); Hera (arrives November 2026) |
| 152830 Dinkinesh | S-type main-belt | 0.79 km, with contact-binary moon | 2.19 au | Lucy flyby (2023) |
| 52246 Donaldjohanson | C-type main-belt | ≈ 8 km elongated | 2.38 au | Lucy flyby (April 2025) |
| 16 Psyche | M-type (metal-rich) | ≈ 220 km | 2.92 au | Psyche orbiter (arrival 2029) |
| 67P/Churyumov–Gerasimenko | Jupiter-family comet | 4.1 × 3.3 × 1.8 km (two lobes) | 3.46 au, 6.4 yr | Rosetta and Philae (2014–16) |
| 1P/Halley | Halley-type comet | 15 × 8 km | 17.8 au, 76 yr | Giotto, Vega 1/2, Suisei, Sakigake (1986) |
| 486958 Arrokoth | Cold classical KBO | 36 km contact binary | 44.6 au | New Horizons flyby (2019) |
| 99942 Apophis | S-type NEA | ≈ 340 m | 0.92 au | OSIRIS-APEX (arrives 2029, after the 13 April 2029 Earth flyby at ≈ 32 000 km altitude) |

## C.5 Lagrange Points

In the three-body problem a small spacecraft can stay fixed relative to two large bodies at five *Lagrange points* (Chapter 3). L1, L2 and L3 lie on the line through the two bodies and are unstable — a spacecraft must make small station-keeping burns, typically a few metres per second per year — while L4 and L5 lead and trail the smaller body by 60° and are stable.

**TABLE C.7** Lagrange-point locations computed from the circular restricted three-body problem.

| System | L1 (from smaller body) | L2 (from smaller body) | L3 | L4 / L5 | Spacecraft |
|---|---|---|---|---|---|
| Sun–Earth | 1.498 million km sunward | 1.508 million km anti-sunward | opposite side of the Sun | 60° ahead / behind Earth | L1: SOHO, ACE, DSCOVR, Aditya-L1, SWFO-L1, IMAP; L2: JWST, Gaia, Euclid, Roman |
| Earth–Moon | 58 000 km Earth-side of the Moon (326 400 km from Earth) | 64 500 km beyond the Moon (448 900 km from Earth) | 381 700 km from Earth, opposite the Moon | at lunar distance, 60° ahead / behind | L2: Queqiao-1 and -2 halo orbits; the Gateway's NRHO is related to the L1/L2 halo family |

## C.6 The US Standard Atmosphere 1976

The *US Standard Atmosphere 1976* is an idealised, steady-state model of the Earth's atmosphere for moderate solar activity. Below 86 km it is identical to the ICAO standard atmosphere used in aviation. Above about 150 km the real density can be ten times higher or lower than shown,

depending on solar and geomagnetic activity (Chapters 2 and 22) — use an empirical model such as NRLMSIS 2.1 or JB2008 for drag and lifetime work.

**TABLE C.8** US Standard Atmosphere 1976 (geometric altitude).

| Altitude (km) | Temperature (K) | Pressure (Pa) | Density (kg/m$^3$) | Layer |
|---|---|---|---|---|
| 0 | 288.15 | 101 325 | 1.225 | Troposphere |
| 5 | 255.68 | 54 048 | 0.736 | |
| 11 | 216.77 | 22 700 | 0.365 | Tropopause |
| 20 | 216.65 | 5 529 | 0.088 9 | Stratosphere |
| 32 | 228.49 | 889.1 | 0.013 6 | |
| 47 | 269.68 | 115.9 | $1.50 \times 10^{-3}$ | Stratopause region |
| 51 | 270.65 | 70.46 | $9.07 \times 10^{-4}$ | Mesosphere |
| 71 | 216.82 | 4.480 | $7.20 \times 10^{-5}$ | |
| 86 | 186.87 | 0.373 4 | $6.96 \times 10^{-6}$ | Mesopause |
| 100 | 195.08 | $3.20 \times 10^{-2}$ | $5.60 \times 10^{-7}$ | Kármán line, thermosphere |
| 150 | 634.39 | $4.54 \times 10^{-4}$ | $2.08 \times 10^{-9}$ | |
| 200 | 854.56 | $8.47 \times 10^{-5}$ | $2.54 \times 10^{-10}$ | |
| 300 | 976.01 | $8.77 \times 10^{-6}$ | $1.92 \times 10^{-11}$ | |
| 400 | 995.83 | $1.45 \times 10^{-6}$ | $2.80 \times 10^{-12}$ | ISS altitude |
| 500 | 999.24 | $3.02 \times 10^{-7}$ | $5.22 \times 10^{-13}$ | |
| 600 | 999.85 | $8.21 \times 10^{-8}$ | $1.14 \times 10^{-13}$ | |
| 800 | 999.99 | $1.70 \times 10^{-8}$ | $1.14 \times 10^{-14}$ | |
| 1 000 | 1 000.00 | $7.51 \times 10^{-9}$ | $3.56 \times 10^{-15}$ | Exosphere |

**TABLE C.9** Atmospheres relevant to entry, descent and aerial vehicles (Chapters 27 and 28).

| Planet / moon | Main gases | Surface pressure | Scale height (km) | Notes for entry and flight |
|---|---|---|---|---|
| Venus | 96.5 % $CO_2$, 3.5 % $N_2$ | 92 bar | 15.9 | Surface 464 °C; sulphuric-acid clouds 48–70 km |
| Earth | 78 % $N_2$, 21 % $O_2$ | 1.014 bar | 8.5 | |
| Mars | 95 % $CO_2$, 2.6 % $N_2$, 1.9 % Ar | ≈ 6 mbar (varies with season and height) | 11.1 | Thin: parachutes alone cannot land heavy payloads |
| Jupiter | 90 % $H_2$, 10 % He | — | 27 | Galileo probe entered at 47.4 km/s (1995) |
| Titan | ≈ 95 % $N_2$, ≈ 5 % $CH_4$ | 1.5 bar | ≈ 20–40 | Dense and cold (94 K): ideal for flight — Dragonfly |

KEY TAKEAWAYS

- Use μ = *GM* from JPL ephemerides for trajectory work; the tables here are rounded for estimates.
- The Hohmann table gives the scale of each journey: Mars and Venus need about 3.5 km/s from LEO, the outer planets 6–8 km/s, and Neptune more than 30 years without gravity assists.
- Launch opportunities repeat at the synodic period — every 26 months for Mars, every 19 months for Venus, about every 13 months for the outer planets.
- Atmospheric density above 150 km varies by a factor of ten with solar activity; standard atmospheres are only a starting point.

APPENDIX

# D Radio-Frequency Bands and Spectrum

*A quick reference to the radio spectrum as used in space — the ITU and IEEE band names, the frequencies allocated to telemetry and command, deep-space, Earth observation, satellite communication and navigation services, the bands used by radars and radiometers, how the atmosphere and rain attenuate each band, free-space path loss from LEO to the outer planets, optical-communication wavelengths, and the international rules that govern who may transmit where.*

**IN THIS CHAPTER**


D.1 Band Names
D.2 Telemetry, Command and Deep-Space Frequencies
D.3 Communication and Broadcasting Satellites
D.4 Navigation Signals
D.5 Remote-Sensing Frequencies
D.6 Propagation Through the Atmosphere
D.7 Free-Space Path Loss
D.8 Optical Communication Wavelengths
D.9 Who Decides: Spectrum Regulation


Every satellite depends on the radio spectrum — to be commanded, to send its data home, to navigate, or to sense the Earth. Spectrum is a finite natural resource shared by all nations, managed through the International Telecommunication Union (ITU) and its Radio Regulations. This appendix gathers the frequencies and rules referred to in Chapters 7, 13, 15, 16, 17, 18 and 24. Band edges are approximate and simplified; the exact allocation for any service depends on the ITU Region (Region 1: Europe, Africa, the Middle East and northern Asia; Region 2: the Americas; Region 3: the rest of Asia and Oceania) and on national rules.

## D.1 Band Names

Two naming systems are in use. The ITU names bands by decade of frequency (Table D.1); the radar "letter bands" standardised by the IEEE (Table D.2) are what satellite engineers use day to day. The letters began as wartime secrecy codes, which is why they follow no logical order.

**TABLE D.1** ITU frequency band designations.

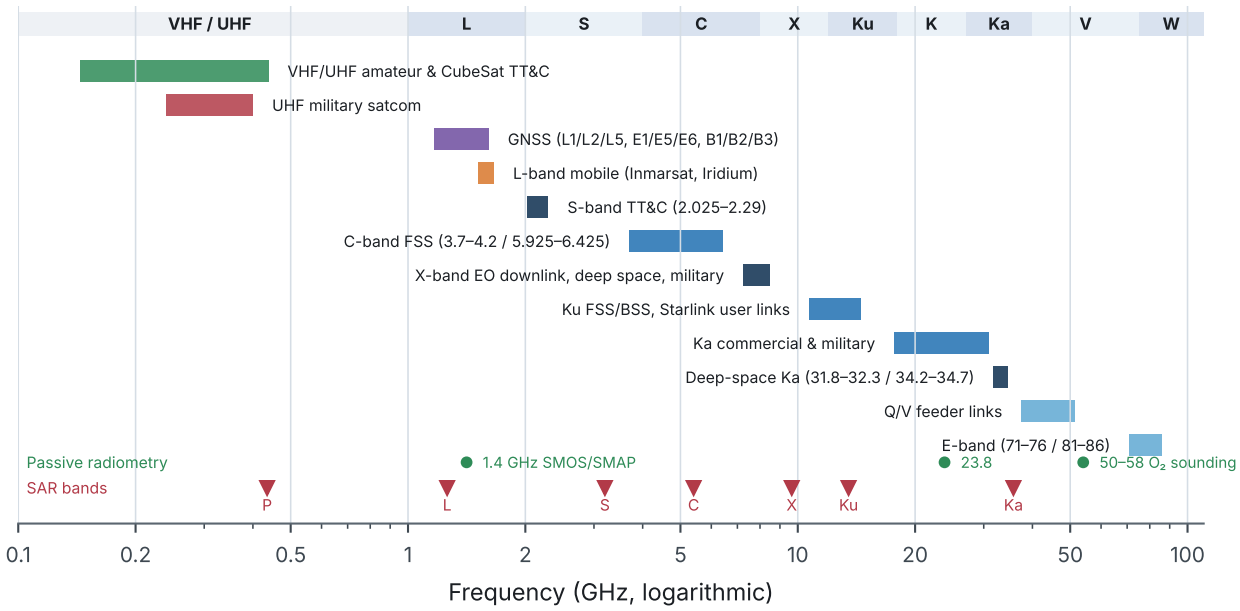
| ITU band no. | Name | Frequency | Wavelength | Typical space uses |
|---|---|---|---|---|
| 4 | VLF (very low) | 3–30 kHz | 100–10 km | Whistlers and plasma waves measured by science payloads |
| 5 | LF (low) | 30–300 kHz | 10–1 km | Ground navigation (Loran), not used for space links |
| 6 | MF (medium) | 0.3–3 MHz | 1 000–100 m | Reflected by the ionosphere; radio astronomy impossible from the ground |

| ITU band no. | Name | Frequency | Wavelength | Typical space uses |
|---|---|---|---|---|
| 7 | HF (high) | 3–30 MHz | 100–10 m | Ionospheric sounding; amateur satellites (rare) |
| 8 | VHF (very high) | 30–300 MHz | 10–1 m | CubeSat and amateur links (145 MHz), weather APT (137 MHz), AIS (162 MHz) |
| 9 | UHF (ultra high) | 300 MHz–3 GHz | 1 m–10 cm | CubeSat TT&C (435–438 MHz), military narrowband satcom, GNSS, mobile satcom, S-band TT&C |
| 10 | SHF (super high) | 3–30 GHz | 10–1 cm | C, X, Ku and Ka bands: most satellite communication, SAR, EO downlinks |
| 11 | EHF (extremely high) | 30–300 GHz | 10–1 mm | Ka, Q/V and E bands; protected military satcom (44 GHz); radiometry |
| 12 | THF (tremendously high) | 0.3–3 THz | 1–0.1 mm | Sub-millimetre astronomy and limb sounders |

**TABLE D.2** IEEE radar letter bands (IEEE Std 521) and their main space uses. Note that the satellite industry often calls 17.7–21.2 GHz "Ka-band downlink" although it lies in the IEEE K band.

| Letter band | Frequency (GHz) | Wavelength | Typical space uses |
|---|---|---|---|
| L | 1–2 | 30–15 cm | GNSS, mobile satcom (Inmarsat, Iridium), L-band SAR, weather-satellite direct broadcast |
| S | 2–4 | 15–7.5 cm | TT&C, deep space (legacy), mobile satcom, S-band SAR |
| C | 4–8 | 7.5–3.75 cm | Fixed satellite service (TV distribution), C-band SAR (Sentinel-1) |
| X | 8–12 | 3.75–2.5 cm | Deep space, EO data downlinks, military satcom, X-band SAR |
| Ku | 12–18 | 2.5–1.67 cm | Direct-to-home TV, broadband (Starlink user links), altimeters, ISS Ku-band |
| K | 18–27 | 1.67–1.11 cm | Ka-band downlinks (17.7–21.2 GHz), EESS data downlink (25.5–27 GHz) |
| Ka | 27–40 | 11.1–7.5 mm | Uplinks of high-throughput satellites, deep-space Ka-band, SWOT radar interferometer |
| V | 40–75 | 7.5–4 mm | Q/V-band feeder links, oxygen-band temperature sounding (50–60 GHz) |
| W | 75–110 | 4–2.7 mm | E-band feeder links (71–86 GHz), cloud radars (94 GHz) |

Figure D.1 places the main space services on a single logarithmic frequency axis.




Approximate band edges; exact allocations depend on ITU Region and service (ITU Radio Regulations, Article 5).

**FIGURE D.1** The radio spectrum used in space, from VHF to E-band. Coloured bars show approximate allocations for the main satellite services; the triangles mark the centre frequencies of the radar bands used by imaging radars, and the dots mark some of the passive bands that radiometers use and that are protected from transmissions.

## D.2 Telemetry, Command and Deep-Space Frequencies

Telemetry, tracking and command (TT&C) and science downlinks use bands allocated to the *space operation*, *space research* and *Earth exploration-satellite* (EESS) services. The ITU distinguishes *Category A* missions, within 2 million km of Earth, from *Category B* deep-space missions beyond that distance, which use separate, specially protected sub-bands because their signals arrive extremely weak (Table D.3).

**TABLE D.3** Frequencies for telemetry, command and science data (simplified).

| Use | Earth-to-space (uplink) | Space-to-Earth (downlink) | Examples |
|---|---|---|---|
| Amateur-satellite and CubeSats | 144–146 MHz; 435–438 MHz | same bands | University CubeSats, AMSAT |
| Near-Earth S-band TT&C | 2 025–2 110 MHz | 2 200–2 290 MHz | Almost all LEO and GEO satellites, ISS, launchers |
| Near-Earth X-band | 7 190–7 235 MHz | 8 450–8 500 MHz | Lunar and L1/L2 missions |
| EESS X-band data downlink | — | 8 025–8 400 MHz | Landsat, Sentinel, commercial EO |
| EESS / space research Ka-band | 22.55–23.15 GHz | 25.5–27.0 GHz | JPSS, NISAR, SWOT, Euclid, Roman, TDRS |
| Deep space S-band (Cat. B) | 2 110–2 120 MHz | 2 290–2 300 MHz | Legacy missions, Voyager (S/X) |
| Deep space X-band (Cat. B) | 7 145–7 190 MHz | 8 400–8 450 MHz | Workhorse band of the DSN, ESTRACK and CDSN |
| Deep space Ka-band (Cat. B) | 34.2–34.7 GHz | 31.8–32.3 GHz | Kepler; MRO demonstration; radio science on Cassini, Juno and BepiColombo |
| Mars and lunar proximity links | 435–450 MHz (to surface) | 390–405 MHz (from surface) | Electra relay radios on Mars orbiters, rovers and landers |

## D.3 Communication and Broadcasting Satellites

**TABLE D.4** Main satellite communication bands (simplified).

| Band | Downlink | Uplink | Services and users |
|---|---|---|---|
| UHF (military) | 240–270 MHz; MUOS 360–380 MHz | 290–320 MHz; MUOS 300–320 MHz | Narrowband tactical satcom (UFO legacy channels, MUOS) |
| L (mobile) | 1 518–1 559 MHz | 1 626.5–1 660.5 MHz | Inmarsat, Ligado/Viasat, Thuraya |
| L (Iridium) | 1 616–1 626.5 MHz | same (time-division) | Iridium voice, data and Certus |
| L/S (Globalstar) | 2 483.5–2 500 MHz | 1 610–1 618.725 MHz | Globalstar; Apple emergency SOS |
| S (2 GHz MSS) | 2 170–2 200 MHz | 1 980–2 010 MHz | EchoStar, EU 2 GHz MSS; spectrum SpaceX agreed to buy in 2025 for direct-to-device |
| Cellular (direct-to-device) | 600 MHz – 2.7 GHz | same | Terrestrial mobile bands reused from space: T-Mobile/Starlink, AST SpaceMobile |
| C | 3.7–4.2 GHz | 5.925–6.425 GHz | TV distribution, maritime, trunking; US 3.7–3.98 GHz cleared for 5G in 2020–23 |
| X (military) | 7.25–7.75 GHz | 7.9–8.4 GHz | WGS, Skynet, Syracuse, SATCOMBw |
| Ku | 10.7–12.75 GHz | 12.75–13.25, 13.75–14.5 GHz | Direct-to-home TV, VSAT, Starlink and OneWeb user links, aero and maritime |
| Ka (commercial) | 17.7–20.2 GHz | 27.5–30.0 GHz | High-throughput satellites (ViaSat-3, Jupiter), gateways, ESIM |
| Ka (government) | 20.2–21.2 GHz | 30.0–31.0 GHz | WGS, military high-throughput payloads |
| Q/V | 37.5–42.5 GHz | 47.2–50.2, 50.4–51.4 GHz | Gateway (feeder) links for new constellations |
| EHF (protected) | 20.2–21.2 GHz | 43.5–45.5 GHz | AEHF and its successor ESS — anti-jam, nuclear-survivable |
| E | 71–76 GHz | 81–86 GHz | Experimental gateway links |

## D.4 Navigation Signals

Global navigation satellite systems share three frequency regions: 1 164–1 215 MHz (lower L-band), 1 215–1 300 MHz, and 1 559–1 610 MHz (upper L-band). Systems deliberately share centre frequencies (1 575.42 MHz and 1 176.45 MHz) so that one antenna and receiver front end can use satellites from several constellations (Chapter 18).

**TABLE D.5** GNSS signals. GPS signals arrive at the ground with a minimum power of about −158.5 dBW — far below the noise.

| System | Signal | Centre frequency (MHz) | Notes |
|---|---|---|---|
| GPS | L1 C/A, L1C | 1 575.42 | 1.023 Mchip/s C/A code, used by every receiver |
| | L2C, P(Y) | 1 227.60 | Second civil frequency for ionospheric correction |

| System | Signal | Centre frequency (MHz) | Notes |
|---|---|---|---|
| | L5 | 1 176.45 | Safety-of-life signal, 10.23 Mchip/s |
| Galileo | E1 | 1 575.42 | Open Service; OSNMA authentication |
| | E5a / E5b | 1 176.45 / 1 207.14 | Wideband AltBOC signal |
| | E6 | 1 278.75 | High Accuracy Service (≈ 20 cm) |
| GLONASS | L1OF / L2OF | 1 602 + 0.5625 *k* / 1 246 + 0.437 5 *k* | FDMA: channel *k* from −7 to +6 |
| | L3OC | 1 202.025 | New CDMA signal |
| BeiDou | B1I / B1C | 1 561.098 / 1 575.42 | |
| | B2a / B2b | 1 176.45 / 1 207.14 | B2b carries PPP corrections |
| | B3I | 1 268.52 | |
| QZSS | L1, L2, L5, L6 | as GPS; L6 1 278.75 | Centimetre-level augmentation (CLAS) on L6 |
| NavIC | L5, S | 1 176.45 / 2 492.028 | L1 added on second-generation satellites |
| SBAS | L1, L5 | 1 575.42 / 1 176.45 | WAAS, EGNOS, MSAS, GAGAN, SouthPAN |

# D.5 Remote-Sensing Frequencies

## D.5.1 Active Sensors

Radar imagers and altimeters choose their band by trading resolution and antenna size against penetration of vegetation, soil and rain (Chapter 15). Longer wavelengths penetrate further into forest canopies and dry soil; shorter wavelengths give finer detail with smaller antennas.

**TABLE D.6** Radar bands used by imaging radars, altimeters and cloud and rain radars.

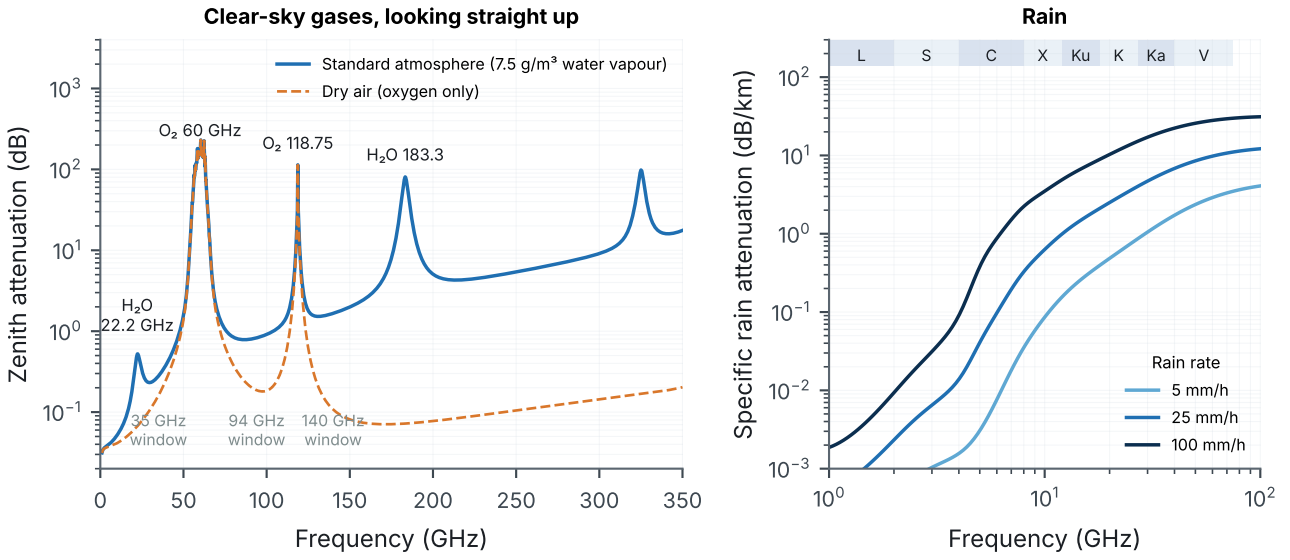
| Band | Centre frequency | Wavelength | What it sees | Missions |
|---|---|---|---|---|
| P | 435 MHz | 69 cm | Penetrates forest canopy to trunks and branches | ESA BIOMASS (launched April 2025) |
| L | 1.26 GHz | 24 cm | Forests, soil moisture, ground deformation, ice | ALOS-2/ALOS-4, SAOCOM, NISAR (L) |
| S | 3.2 GHz | 9.4 cm | Crops, snow, sea ice | NISAR (S), NovaSAR-1 |
| C | 5.405 GHz | 5.5 cm | All-round monitoring, sea ice, winds, floods | Sentinel-1, RADARSAT Constellation, Gaofen-3 |
| X | 9.65 GHz | 3.1 cm | High-resolution imaging, change detection | TerraSAR-X/TanDEM-X, COSMO-SkyMed, ICEYE, Capella, Umbra |
| Ku | 13.575 GHz | 2.2 cm | Sea-surface height, winds, rain | Sentinel-3 and Sentinel-6 altimeters, GPM (13.6 GHz) |
| Ka | 35.75 GHz | 8.4 mm | Water-surface topography, snow, light rain | SWOT KaRIn, SARAL/AltiKa, GPM (35.5 GHz) |
| W | 94.05 GHz | 3.2 mm | Cloud droplets and ice | CloudSat, EarthCARE |

### D.5.2 Passive Sensors

Microwave radiometers measure the natural thermal emission of the Earth's surface and atmosphere. They rely on frequencies set by physics — the oxygen absorption complex near 50–60 GHz for temperature profiles, the 22.235 GHz and 183.31 GHz water-vapour lines for humidity, the 1.4 GHz "window" for soil moisture and ocean salinity. Because those frequencies cannot be moved, the Radio Regulations protect them: in bands such as 1 400–1 427 MHz, 23.6–24.0 GHz and 50.2–50.4 GHz *all emissions are prohibited*. Unwanted emissions from neighbouring services remain a concern: the decision at the 2019 World Radiocommunication Conference on 5G emissions next to the 23.8 GHz water-vapour band was criticised by meteorologists as too permissive.

## D.6 Propagation Through the Atmosphere

The atmosphere is nearly transparent to radio below about 10 GHz, but above that gases and rain absorb increasingly (Figure D.2). Clear-sky absorption peaks at the oxygen lines (60 GHz and 118.75 GHz) and the water-vapour lines (22.2 GHz and 183.3 GHz); between them lie *windows* near 35, 94 and 140 GHz used by radars and radiometers. Rain is usually the larger problem for satellite links: at Ku-band a heavy shower (25 mm/h) attenuates about 1 dB per kilometre of rain path, at Ka-band (30 GHz) about 5 dB/km. Link designers therefore add a *rain margin* chosen from the statistics of rain at each ground station — for example, enough to keep the link closed for 99.9 % of an average year — and use adaptive coding and modulation, site diversity and uplink power control to ride out the worst storms. The ionosphere affects lower frequencies instead: it delays GNSS signals by up to tens of metres, rotates the polarisation of VHF and UHF signals (Faraday rotation) and causes scintillation near the equator and the poles (Chapter 23).




Gases: ITU-R P.676 line-by-line model, sea level, 1013 hPa, 15 °C. Rain: ITU-R P.838, 40° elevation, circular polarisation. Computed with ITU-Rpy.

**FIGURE D.2** Atmospheric attenuation of radio waves. Left: clear-sky attenuation by oxygen and water vapour for a path looking straight up from sea level, computed with the ITU-R P.676 line-by-line model. Right: specific attenuation by rain for three rain rates, from ITU-R P.838; the total rain loss is this value multiplied by the length of the path through rain, typically a few kilometres.

## D.7 Free-Space Path Loss

Most of the loss in any space link is simply geometric spreading. The *free-space path loss* between isotropic antennas is

$$L_{\text{fs}}\,(\text{dB}) = 92.45 + 20\log_{10} f_{\text{GHz}} + 20\log_{10} d_{\text{km}}$$

so every doubling of distance or frequency costs 6 dB. (The frequency dependence is offset by the higher gain of an antenna of a given size at higher frequency, which is why deep-space links moved from S- to X- to Ka-band.) Table D.7 gives typical values.

**TABLE D.7** Free-space path loss (dB) for typical space links.

| Link | Distance | 437 MHz | 2.2 GHz | 8.4 GHz | 20 GHz | 32 GHz |
|---|---|---|---|---|---|---|
| LEO, overhead | 550 km | 140.1 | 154.1 | 165.7 | 173.3 | 177.4 |
| LEO, low elevation | 2 000 km | 151.3 | 165.3 | 177.0 | 184.5 | 188.6 |
| GEO | 38 000 km | 176.9 | 190.9 | 202.5 | 210.1 | 214.1 |
| Moon | 384 400 km | 197.0 | 211.0 | 222.6 | 230.2 | 234.2 |
| Sun–Earth L2 | 1.5 million km | 208.8 | 222.8 | 234.5 | 242.0 | 246.1 |
| Mars, closest | 0.37 au | 240.2 | 254.2 | 265.9 | 273.4 | 277.5 |
| Mars, farthest | 2.67 au | 257.3 | 271.3 | 283.0 | 290.5 | 294.6 |
| Jupiter | 6.2 au | 264.6 | 278.6 | 290.3 | 297.8 | 301.9 |
| Voyager 1 (2026) | ≈ 170 au | 293.4 | 307.4 | 319.0 | 326.6 | 330.7 |

## D.8 Optical Communication Wavelengths

Laser links avoid spectrum licensing for the beam itself and offer far higher data rates from small terminals (Chapters 7 and 17). Most terminals work at the telecommunication wavelength of 1 550 nm, where fibre-optic components such as erbium-doped amplifiers are mature; some use 1 064 nm (Nd:YAG lasers). Examples include NASA's LCRD relay (1.2 Gbit/s, 2021), the TBIRD CubeSat (200 Gbit/s bursts, 2022–23), the DSOC experiment on Psyche (267 Mbit/s from 33 million km in 2023, and links from beyond the distance of Mars until 2025), the European Data Relay System (1.8 Gbit/s at 1 064 nm), the inter-satellite lasers of Starlink, and the optical terminals specified by the US Space Development Agency for its Proliferated Warfighter Space Architecture. Clouds block optical links, so ground networks use several geographically separated stations.

## D.9 Who Decides: Spectrum Regulation

**The ITU Radio Regulations** are an international treaty, revised every three to four years by a World Radiocommunication Conference (WRC). They allocate each band to one or more *services* (fixed-satellite, mobile-satellite, broadcasting-satellite, radionavigation-satellite, Earth exploration-satellite, space research, space operation, radio astronomy and others) with primary or secondary status. WRC-23 (Dubai, 2023) opened Ka-band to earth stations in motion

communicating with non-geostationary satellites and placed direct-to-device satellite connectivity in mobile bands and lunar-surface communications on the agenda of WRC-27, due in late 2027.

**Filing and coordination.** Before a satellite network can use a frequency, its national administration files it with the ITU Radiocommunication Bureau — first an advance publication or coordination request, then a notification for recording in the Master International Frequency Register. Coordination with earlier filings is negotiated between administrations; date priority ("first come, first served") governs most bands, while some broadcasting and fixed-satellite bands are governed by *plans* that guarantee each country access.

**Non-geostationary constellations.** Article 22 of the Radio Regulations sets *equivalent power flux-density* (EPFD) limits that protect geostationary networks from interference by non-geostationary constellations in shared Ku- and Ka-bands. Since WRC-19, a constellation must bring 10 % of its filed satellites into use within two years of the end of its seven-year regulatory period, 50 % within five years and 100 % within seven years, or its filing is reduced — a rule that discourages "paper satellites" (Chapter 17).

**National licensing.** Operators also need licences from their own regulator — in the United States the Federal Communications Commission (FCC) for commercial systems and the NTIA for federal ones; in the United Kingdom Ofcom — and landing rights in each country they serve.

KEY TAKEAWAYS

- Satellite engineers name bands by IEEE letters: L, S, C, X, Ku, K, Ka, V, W. The ITU allocates spectrum by service and Region.
- TT&C lives in S-band; deep space in X- and Ka-band sub-bands reserved for distances beyond 2 million km.
- GNSS signals cluster at 1 176, 1 207–1 278 and 1 575 MHz; shared centre frequencies enable multi-constellation receivers.
- Above 10 GHz rain fade dominates link margins; gas absorption peaks at 22, 60, 118 and 183 GHz.
- Free-space loss rises 6 dB per doubling of distance or frequency — from about 154 dB for LEO S-band to over 280 dB for Mars X-band.
- Spectrum is a treaty-governed shared resource: filings, coordination, EPFD limits and milestones shape every constellation.

APPENDIX

# E Mission and Vehicle Databases

*Reference tables of launch statistics, launch vehicles, launch sites and every orbital launch of 2025 and 2026, compiled from Jonathan McDowell's General Catalog of Artificial Space Objects (GCAT) and from the chapters of this manual.*

IN THIS CHAPTER

E.1 Orbital Launch Attempts by Year
E.2 Most-Launched Vehicle Families
E.3 Busiest Launch Sites
E.4 Orbital Launch Log: 2025
E.5 Orbital Launch Log: 2026, to 13 October

This appendix collects reference data that would interrupt the flow of the chapters. The launch statistics and launch logs are generated directly from the **General Catalog of Artificial Space Objects (GCAT)** maintained by Jonathan McDowell (planet4589.org/space/gcat, licensed CC-BY 4.0), as updated on 13 October 2026. An orbital launch attempt is counted when a vehicle intended to reach orbit lifts off, whether or not it succeeds; suborbital test flights (including most Starship flights before September 2026) are not included. Small differences from other published counts arise from the treatment of such tests, of launches from one country by another's vehicle, and of partial failures.

## E.1 Orbital Launch Attempts by Year

Orbital launch attempts (successful or not) by the country or region of the launching organisation, 1957 to 13 October 2026. Rocket Lab's launches from New Zealand count as US. Source: GCAT (J. McDowell, CC-BY 4.0).

**TABLE E.1** Orbital launch attempts per year, 1957–2026 (* 2026 to 13 October).

| Year | USA | USSR/Russia | China | Europe | Japan | India | Other | Total | Failures |
|---|---|---|---|---|---|---|---|---|---|
| 1957 | 1 | 2 | – | – | – | – | – | 3 | 1 |
| 1958 | 23 | 5 | – | – | – | – | – | 28 | 20 |
| 1959 | 21 | 4 | – | – | – | – | – | 25 | 10 |
| 1960 | 29 | 9 | – | – | – | – | – | 38 | 19 |
| 1961 | 41 | 9 | – | – | – | – | – | 50 | 18 |
| 1962 | 59 | 22 | – | – | – | – | – | 81 | 14 |
| 1963 | 46 | 24 | – | – | – | – | – | 70 | 17 |
| 1964 | 64 | 36 | – | – | – | – | – | 100 | 15 |
| 1965 | 71 | 53 | – | 1 | – | – | – | 125 | 14 |
| 1966 | 78 | 51 | – | 1 | 2 | – | – | 132 | 17 |
| 1967 | 61 | 74 | – | 3 | 1 | – | – | 139 | 14 |

| Year | USA | USSR/Russia | China | Europe | Japan | India | Other | Total | Failures |
|---|---|---|---|---|---|---|---|---|---|
| 1968 | 48 | 80 | – | 1 | – | – | – | 129 | 10 |
| 1969 | 41 | 82 | – | 1 | 1 | – | – | 125 | 18 |
| 1970 | 29 | 87 | 1 | 5 | 2 | – | – | 124 | 11 |
| 1971 | 33 | 91 | 1 | 6 | 2 | – | – | 133 | 15 |
| 1972 | 32 | 79 | – | 1 | 1 | – | – | 113 | 8 |
| 1973 | 25 | 90 | 1 | 1 | – | – | – | 117 | 7 |
| 1974 | 23 | 85 | 2 | 2 | 1 | – | – | 113 | 8 |
| 1975 | 30 | 93 | 3 | 4 | 2 | – | – | 132 | 7 |
| 1976 | 26 | 100 | 3 | – | 2 | – | – | 131 | 5 |
| 1977 | 26 | 102 | – | – | 2 | – | – | 130 | 7 |
| 1978 | 33 | 91 | 1 | – | 3 | – | – | 128 | 5 |
| 1979 | 16 | 89 | 1 | 1 | 2 | 1 | – | 110 | 5 |
| 1980 | 15 | 90 | – | 1 | 2 | 1 | – | 109 | 5 |
| 1981 | 19 | 100 | 1 | 2 | 3 | 1 | – | 126 | 4 |
| 1982 | 18 | 108 | 1 | 1 | 1 | – | – | 129 | 10 |
| 1983 | 22 | 100 | 1 | 2 | 3 | 1 | – | 129 | 2 |
| 1984 | 23 | 97 | 3 | 4 | 3 | – | – | 130 | 3 |
| 1985 | 19 | 100 | 1 | 4 | 2 | – | – | 126 | 5 |
| 1986 | 9 | 94 | 2 | 3 | 2 | – | – | 110 | 8 |
| 1987 | 9 | 97 | 2 | 2 | 3 | 1 | – | 114 | 6 |
| 1988 | 11 | 94 | 4 | 8 | 2 | 1 | 1 | 121 | 6 |
| 1989 | 18 | 75 | – | 7 | 2 | – | – | 102 | 1 |
| 1990 | 27 | 79 | 5 | 6 | 3 | – | 1 | 121 | 7 |
| 1991 | 19 | 61 | 1 | 8 | 2 | – | – | 91 | 5 |
| 1992 | 29 | 55 | 4 | 7 | 1 | 1 | – | 97 | 3 |
| 1993 | 27 | 48 | 1 | 7 | 1 | 1 | – | 85 | 6 |
| 1994 | 28 | 49 | 5 | 8 | 2 | 2 | – | 94 | 4 |
| 1995 | 30 | 33 | 3 | 11 | 2 | – | 1 | 80 | 7 |
| 1996 | 33 | 27 | 4 | 11 | 1 | 1 | – | 77 | 6 |
| 1997 | 38 | 29 | 6 | 12 | 2 | 1 | 1 | 89 | 4 |
| 1998 | 36 | 25 | 6 | 11 | 2 | – | 2 | 82 | 6 |
| 1999 | 31 | 22 | 4 | 16 | 1 | 1 | 3 | 78 | 8 |
| 2000 | 29 | 32 | 5 | 16 | 1 | – | 2 | 85 | 4 |
| 2001 | 24 | 23 | 1 | 8 | 1 | 2 | – | 59 | 2 |
| 2002 | 18 | 25 | 5 | 12 | 3 | 1 | 1 | 65 | 4 |
| 2003 | 26 | 19 | 7 | 6 | 3 | 2 | 1 | 64 | 2 |
| 2004 | 19 | 22 | 8 | 3 | – | 1 | 1 | 54 | 3 |
| 2005 | 16 | 23 | 6 | 8 | 2 | 1 | – | 56 | 4 |
| 2006 | 23 | 23 | 6 | 7 | 6 | 1 | – | 66 | 4 |
| 2007 | 20 | 23 | 10 | 9 | 2 | 3 | 1 | 68 | 3 |

| Year | USA | USSR/Russia | China | Europe | Japan | India | Other | Total | Failures |
|---|---|---|---|---|---|---|---|---|---|
| 2008 | 20 | 26 | 11 | 7 | 1 | 3 | 1 | 69 | 3 |
| 2009 | 25 | 32 | 6 | 7 | 3 | 2 | 3 | 78 | 4 |
| 2010 | 15 | 30 | 15 | 7 | 2 | 3 | 2 | 74 | 4 |
| 2011 | 19 | 28 | 19 | 9 | 3 | 3 | 3 | 84 | 6 |
| 2012 | 16 | 23 | 19 | 11 | 2 | 2 | 5 | 78 | 5 |
| 2013 | 20 | 31 | 15 | 8 | 3 | 3 | 1 | 81 | 3 |
| 2014 | 24 | 32 | 16 | 11 | 4 | 4 | 1 | 92 | 3 |
| 2015 | 20 | 25 | 19 | 12 | 4 | 5 | 2 | 87 | 5 |
| 2016 | 23 | 17 | 22 | 11 | 4 | 7 | 2 | 86 | 2 |
| 2017 | 30 | 18 | 18 | 11 | 7 | 5 | 1 | 90 | 6 |
| 2018 | 34 | 17 | 39 | 11 | 6 | 7 | – | 114 | 2 |
| 2019 | 27 | 22 | 34 | 9 | 2 | 6 | 3 | 103 | 5 |
| 2020 | 44 | 12 | 39 | 10 | 4 | 2 | 3 | 114 | 10 |
| 2021 | 51 | 16 | 56 | 15 | 3 | 2 | 3 | 146 | 11 |
| 2022 | 87 | 21 | 64 | 6 | 1 | 5 | 2 | 186 | 7 |
| 2023 | 114 | 19 | 67 | 3 | 3 | 7 | 8 | 221 | 10 |
| 2024 | 157 | 17 | 68 | 3 | 7 | 5 | 5 | 262 | 6 |
| 2025 | 198 | 17 | 93 | 8 | 4 | 5 | 5 | 330 | 11 |
| 2026* | 146 | 13 | 71 | 7 | 3 | 3 | 1 | 244 | 7 |

## E.2 Most-Launched Vehicle Families

**TABLE E.2** The thirty most-launched orbital vehicle families (to 13 October 2026).

| Family | First orbital attempt | Attempts | Failures | Success rate |
|---|---|---|---|---|
| Soyuz | 1966 | 1 181 | 31 | 97.4 % |
| Falcon 9 | 2010 | 695 | 2 | 99.7 % |
| Kosmos | 1961 | 622 | 45 | 92.8 % |
| Proton | 1967 | 425 | 43 | 89.9 % |
| Atlas | 1958 | 327 | 40 | 87.8 % |
| Molniya | 1960 | 320 | 33 | 89.7 % |
| Delta | 1972 | 301 | 9 | 97.0 % |
| Voskhod | 1963 | 299 | 13 | 95.7 % |
| Thor | 1958 | 288 | 38 | 86.8 % |
| Long March 2 | 1974 | 235 | 4 | 98.3 % |
| Titan | 1964 | 216 | 15 | 93.1 % |
| Long March 3 | 1984 | 177 | 8 | 95.5 % |
| Vostok | 1960 | 156 | 8 | 94.9 % |
| Ariane | 1979 | 144 | 7 | 95.1 % |
| Space Shuttle | 1981 | 140 | 1 | 99.3 % |
| Tsiklon-3 | 1977 | 122 | 7 | 94.3 % |

| Family | First orbital attempt | Attempts | Failures | Success rate |
|---|---|---|---|---|
| Long March 4 | 1988 | 119 | 3 | 97.5 % |
| Ariane 5 | 1996 | 117 | 3 | 97.4 % |
| Atlas V | 2002 | 110 | 0 | 100.0 % |
| Tsiklon-2 | 1969 | 105 | 1 | 99.0 % |
| Scout | 1960 | 99 | 11 | 88.9 % |
| Electron | 2017 | 88 | 4 | 95.5 % |
| PSLV | 1993 | 64 | 4 | 93.8 % |
| H-IIA | 2001 | 50 | 1 | 98.0 % |
| Pegasus | 1990 | 46 | 4 | 91.3 % |
| Long March 6 | 2015 | 44 | 0 | 100.0 % |
| Thorad | 1966 | 43 | 2 | 95.3 % |
| Kuaizhou | 2013 | 40 | 3 | 92.5 % |
| Zenit-2 | 1985 | 36 | 7 | 80.6 % |
| Zenit-3SL | 1999 | 36 | 3 | 91.7 % |

## E.3 Busiest Launch Sites

**TABLE E.3** The twenty-five busiest orbital launch sites of all time (to 13 October 2026).

| Launch site | Orbital launch attempts |
|---|---|
| Plesetsk | 1 688 |
| Baikonur | 1 560 |
| Cape Canaveral | 1 112 |
| Vandenberg | 893 |
| Kourou | 336 |
| Jiuquan | 299 |
| Kennedy Space Center | 287 |
| Xichang | 242 |
| Taiyuan | 161 |
| Sriharikota | 103 |
| Kapustin Yar | 101 |
| Tanegashima | 99 |
| Mahia (Rocket Lab LC-1) | 84 |
| Wenchang | 51 |
| Point Arguello | 43 |
| Pacific Ocean (Sea Launch) | 36 |
| Uchinoura (Kagoshima) | 35 |
| Wallops (MARS) | 31 |
| Wallops Island | 28 |
| Hainan commercial site (Wenchang) | 24 |
| Vostochny | 21 |

| Launch site | Orbital launch attempts |
|---|---|
| Yellow Sea (sea launch) | 20 |
| Semnan | 16 |
| Palmachim | 13 |
| Starbase | 11 |

## E.4 Orbital Launch Log: 2025

All 330 orbital launch attempts. × marks a failure to reach orbit. Times are UTC.

**TABLE E.4** Orbital launch attempts in 2025.

| No. | Date (UTC) | Vehicle | Site | Mission / payload | |
|---|---|---|---|---|---|
| 1 | Jan 4 | Falcon 9 | Canaveral | Thuraya 4-NGS | |
| 2 | Jan 6 | Long March 3B | Xichang | Shi Jian 25 | |
| 3 | Jan 6 | Falcon 9 | Canaveral | Starlink Group 6-71 | |
| 4 | Jan 8 | Falcon 9 | Kennedy | Starlink Group 12-11 | |
| 5 | Jan 10 | Falcon 9 | Vandenberg | NRO Starshield A6 | |
| 6 | Jan 10 | Falcon 9 | Canaveral | Starlink Group 12-12 | |
| 7 | Jan 13 | Jielong-3 | Yellow Sea | Weili Kongjian 01 zu | |
| 8 | Jan 13 | Falcon 9 | Canaveral | Starlink Group 12-4 | |
| 9 | Jan 14 | Falcon 9 | Vandenberg | Transporter-12 | |
| 10 | Jan 15 | Falcon 9 | Kennedy | BG-1/Hakuto-R M2 | |
| 11 | Jan 16 | New Glenn 7×2 | Canaveral | DS-1 | |
| 12 | Jan 16 | Starship V2 | Starbase | Ship 33 | × |
| 13 | Jan 17 | Long March 2D | Jiuquan | PRSC-EO1 | |
| 14 | Jan 20 | Gushenxing 1 | Jiuquan | Yunyao-1 37-40 | |
| 15 | Jan 21 | Falcon 9 | Kennedy | Starlink Group 13-1 | |
| 16 | Jan 21 | Falcon 9 | Vandenberg | Starlink Group 11-8 | |
| 17 | Jan 23 | Long March 6A | Taiyuan | Qianfan Jigui 06 | |
| 18 | Jan 23 | Long March 3B | Xichang | TJS 14 | |
| 19 | Jan 24 | Falcon 9 | Vandenberg | Starlink Group 11-6 | |
| 20 | Jan 27 | Falcon 9 | Canaveral | Starlink Group 12-7 | |
| 21 | Jan 29 | GSLV Mk II | Sriharikota | IRNSS-1K? | |
| 22 | Jan 30 | Falcon 9 | Kennedy | Spainsat NG-1 | |
| 23 | Feb 1 | Falcon 9 | Vandenberg | Starlink Group 11-4 | |
| 24 | Feb 2 | H3-22S | Tanegashima | QZS 6 | |
| 25 | Feb 4 | Falcon 9 | Canaveral | Starlink Group 12-3 | |
| 26 | Feb 4 | Falcon 9 | Kennedy | Maxar-3 | |
| 27 | Feb 5 | Soyuz-2-1V | Plesetsk | MKA-B 1 to 3 | |
| 28 | Feb 8 | Falcon 9 | Canaveral | Starlink Group 12-9 | |
| 29 | Feb 8 | Electron | Mahia | Kineis 2A-2E | |
| 30 | Feb 11 | Falcon 9 | Vandenberg | Starlink Group 11-10 | |
| 31 | Feb 11 | Long March 8A | Wenchang | WHDW 02 | |
| 32 | Feb 11 | Falcon 9 | Canaveral | Starlink Group 12-18 | |

| No. | Date (UTC) | Vehicle | Site | Mission / payload | |
|---|---|---|---|---|---|
| 33 | Feb 15 | Falcon 9 | Canaveral | Starlink Group 12-8 | |
| 34 | Feb 18 | Electron | Mahia | BlackSky Gen3 | |
| 35 | Feb 18 | Falcon 9 | Canaveral | Starlink Group 10-12 | |
| 36 | Feb 21 | Falcon 9 | Canaveral | Starlink Group 12-14 | |
| 37 | Feb 22 | Long March 3B | Xichang | ZX-10R | |
| 38 | Feb 23 | Falcon 9 | Vandenberg | Starlink Group 15-1 | |
| 39 | Feb 27 | Falcon 9 | Kennedy | IM-2/Odin/LTB/CGEO-1 | |
| 40 | Feb 27 | Falcon 9 | Canaveral | Starlink Group 12-13 | |
| 41 | Feb 27 | Long March 2C | Jiuquan | Superview 1-03/04 | |
| 42 | Feb 27 | Soyuz-2-1A | Baikonur | Progress 7K-TGM No. 460 | |
| 43 | Mar 1 | Kuaizhou-1A | Jiuquan | Unknown | |
| 44 | Mar 2 | Soyuz-2-1B | Plesetsk | Glonass-K2 No. 14L | |
| 45 | Mar 3 | Falcon 9 | Canaveral | Starlink Group 12-20 | |
| 46 | Mar 6 | Ariane 62 | Kourou | CSO 3 | |
| 47 | Mar 6 | Starship V2 | Starbase | Ship 34 | × |
| 48 | Mar 9 | Long March 3B | Xichang | TJS 15 | |
| 49 | Mar 11 | Long March 8 | Hainan (comm.) | Qianfan | |
| 50 | Mar 12 | Falcon 9 | Vandenberg | SPHEREx | |
| 51 | Mar 13 | Falcon 9 | Canaveral | Starlink Group 12-21 | |
| 52 | Mar 14 | Falcon 9 | Kennedy | Dragon C210.4 | |
| 53 | Mar 15 | Electron | Mahia | QPS-SAR-9 | |
| 54 | Mar 15 | Long March 2D | Jiuquan | Superview 3-02 | |
| 55 | Mar 15 | Falcon 9 | Vandenberg | Transporter-13 | |
| 56 | Mar 15 | Falcon 9 | Canaveral | Starlink Group 12-16 | |
| 57 | Mar 16 | Angara-1.2 | Plesetsk | Rodnik Blok 17 | |
| 58 | Mar 17 | Gushenxing 1 | Jiuquan | Yunyao-1 55-60 | |
| 59 | Mar 18 | Electron | Mahia | Kineis 4A-4E | |
| 60 | Mar 18 | Falcon 9 | Canaveral | Starlink Group 12-25 | |
| 61 | Mar 21 | Falcon 9 | Vandenberg | NRO Starshield C1 | |
| 62 | Mar 21 | Gushenxing 1 | Jiuquan | Yunyao-1 43-48 | |
| 63 | Mar 24 | Falcon 9 | Canaveral | INTRUDER? | |
| 64 | Mar 26 | Electron | Mahia | OTC Phase 1 | |
| 65 | Mar 26 | Long March 3B | Xichang | Tianlian 2-04 | |
| 66 | Mar 26 | Falcon 9 | Vandenberg | Starlink Group 11-7 | |
| 67 | Mar 29 | Long March 7A | Wenchang | TJS 16 | |
| 68 | Mar 30 | Spectrum | Andoya | Going Full Spectrum | × |
| 69 | Mar 31 | Falcon 9 | Canaveral | Starlink Group 6-80 | |
| 70 | Apr 1 | Falcon 9 | Kennedy | Dragon C207.4 | |
| 71 | Apr 1 | Long March 2D | Jiuquan | WHJSW 06 | |
| 72 | Apr 3 | Long March 6 | Taiyuan | Tianping 3A-02 | |
| 73 | Apr 4 | Falcon 9 | Vandenberg | Starlink Group 11-13 | |
| 74 | Apr 6 | Falcon 9 | Canaveral | Starlink Group 6-72 | |

| No. | Date (UTC) | Vehicle | Site | Mission / payload | |
|---|---|---|---|---|---|
| 75 | Apr 7 | Falcon 9 | Vandenberg | Starlink Group 11-11 | |
| 76 | Apr 8 | Soyuz-2-1A | Baikonur | Soyuz 11F732A48 No. 758 | |
| 77 | Apr 10 | Long March 3B | Xichang | TJS 17 | |
| 78 | Apr 12 | Falcon 9 | Vandenberg | NRO Starshield A7 | |
| 79 | Apr 13 | Falcon 9 | Kennedy | Starlink Group 12-17 | |
| 80 | Apr 14 | Falcon 9 | Canaveral | Starlink Group 6-73 | |
| 81 | Apr 16 | Minotaur IV | Vandenberg | NROL-174 Asterion | |
| 82 | Apr 18 | Long March 6A | Taiyuan | Shiyan 27 01-06 | |
| 83 | Apr 20 | Falcon 9 | Vandenberg | NRO Starshield A8 | |
| 84 | Apr 21 | Falcon 9 | Kennedy | Dragon C209.5 | |
| 85 | Apr 22 | Falcon 9 | Canaveral | Bandwagon-3 | |
| 86 | Apr 24 | Long March 2F | Jiuquan | Shenzhou 20 | |
| 87 | Apr 25 | Falcon 9 | Canaveral | Starlink Group 6-74 | |
| 88 | Apr 27 | Long March 3B | Xichang | Tianlian 2-05 | |
| 89 | Apr 28 | Falcon 9 | Canaveral | Starlink Group 12-23 | |
| 90 | Apr 28 | Long March 5B/YZ-2 | Wenchang | WHDW 03 | |
| 91 | Apr 28 | Falcon 9 | Vandenberg | Starlink Group 11-9 | |
| 92 | Apr 28 | Atlas V 551 | Canaveral | KA-01 | |
| 93 | Apr 29 | Falcon 9 | Kennedy | Starlink Group 12-10 | |
| 94 | Apr 29 | Vega C | Kourou | Biomass | |
| 95 | Apr 29 | Firefly Alpha | Vandenberg | LM-400 Tech Demo | × |
| 96 | May 2 | Falcon 9 | Canaveral | Starlink Group 6-75 | |
| 97 | May 4 | Falcon 9 | Kennedy | Starlink Group 6-84 | |
| 98 | May 7 | Falcon 9 | Canaveral | Starlink Group 6-93 | |
| 99 | May 10 | Falcon 9 | Vandenberg | Starlink Group 15-3 | |
| 100 | May 10 | Falcon 9 | Canaveral | Starlink Group 6-91 | |
| 101 | May 11 | Long March 6A | Taiyuan | Yaogan 40-02 | |
| 102 | May 12 | Long March 3C | Xichang | TJS 19 | |
| 103 | May 13 | Falcon 9 | Vandenberg | Starlink Group 15-4 | |
| 104 | May 13 | Falcon 9 | Kennedy | Starlink Group 6-83 | |
| 105 | May 14 | Long March 2D | Jiuquan | Taikong Jisuan WX | |
| 106 | May 14 | Falcon 9 | Canaveral | Starlink Group 6-67 | |
| 107 | May 16 | Falcon 9 | Vandenberg | Starlink Group 15-5 | |
| 108 | May 17 | Zhuque-2E | Jiuquan | Tianyi | |
| 109 | May 17 | Electron | Mahia | QPS-SAR-10 | |
| 110 | May 18 | PSLV-XL | Sriharikota | RISAT-1B | × |
| 111 | May 19 | Gushenxing 1 | Yellow Sea | Tianqi 16-18,20 | |
| 112 | May 20 | Long March 7A | Wenchang | ZX-3B | |
| 113 | May 21 | Falcon 9 | Canaveral | Starlink Group 12-15 | |
| 114 | May 21 | Lijian-1 | Jiuquan | Taijing | |
| 115 | May 23 | Soyuz-2-1B | Plesetsk | Nivelir-L No. 5? | |
| 116 | May 23 | Falcon 9 | Vandenberg | Starlink Group 11-16 | |

| No. | Date (UTC) | Vehicle | Site | Mission / payload |
|---|---|---|---|---|
| 117 | May 24 | Falcon 9 | Canaveral | Starlink Group 12-22 |
| 118 | May 27 | Falcon 9 | Vandenberg | Starlink Group 17-1 |
| 119 | May 27 | Starship V2 | Starbase | Ship 35 |
| 120 | May 28 | Falcon 9 | Kennedy | Starlink Group 10-32 |
| 121 | May 28 | Long March 3B | Xichang | Tianwen-2 |
| 122 | May 29 | Long March 4B | Jiuquan | SJ-26 |
| 123 | May 30 | Falcon 9 | Canaveral | GPS III SV08 |
| 124 | May 31 | Falcon 9 | Vandenberg | Starlink Group 11-18 |
| 125 | Jun 2 | Electron | Mahia | BlackSky Gen3-2 |
| 126 | Jun 3 | Falcon 9 | Canaveral | Starlink Group 12-19 |
| 127 | Jun 4 | Falcon 9 | Vandenberg | Starlink Group 11-22 |
| 128 | Jun 5 | Long March 6A | Taiyuan | WHDW 04 |
| 129 | Jun 7 | Falcon 9 | Canaveral | SXM-10 |
| 130 | Jun 8 | Falcon 9 | Vandenberg | Starlink Group 15-8 |
| 131 | Jun 10 | Falcon 9 | Canaveral | Starlink Group 12-24 |
| 132 | Jun 11 | Electron | Mahia | QPS-SAR-11 |
| 133 | Jun 13 | Falcon 9 | Vandenberg | Starlink Group 15-6 |
| 134 | Jun 13 | Falcon 9 | Canaveral | Starlink Group 12-26 |
| 135 | Jun 14 | Long March 2D | Jiuquan | ZH1-02 |
| 136 | Jun 17 | Falcon 9 | Vandenberg | Starlink Group 15-9 |
| 137 | Jun 18 | Falcon 9 | Canaveral | Starlink Group 10-18 |
| 138 | Jun 19 | Angara A5 | Plesetsk | 14F166A No. 1 |
| 139 | Jun 20 | Long March 3B | Xichang | ZX 9C |
| 140 | Jun 23 | Falcon 9 | Canaveral | Starlink Group 10-23 |
| 141 | Jun 23 | Atlas V 551 | Canaveral | KA-02 |
| 142 | Jun 23 | Falcon 9 | Vandenberg | Transporter-14 |
| 143 | Jun 25 | Falcon 9 | Kennedy | Dragon C213.1 |
| 144 | Jun 25 | Falcon 9 | Canaveral | Starlink Group 10-16 |
| 145 | Jun 26 | Electron | Mahia | Hawk 12/Kestrel 0A |
| 146 | Jun 28 | Falcon 9 | Canaveral | Starlink Group 10-34 |
| 147 | Jun 28 | Electron | Mahia | Lyra 4 |
| 148 | Jun 28 | H-IIA 202 | Tanegashima | GOSAT-GW |
| 149 | Jun 28 | Falcon 9 | Vandenberg | Starlink Group 15-7 |
| 150 | Jul 1 | Falcon 9 | Kennedy | MTG-S1 |
| 151 | Jul 2 | Falcon 9 | Canaveral | Starlink Group 10-25 |
| 152 | Jul 3 | Long March 4C | Xichang | SY28B-01 |
| 153 | Jul 3 | Soyuz-2-1A | Baikonur | Progress 7K-TGM No. 461 |
| 154 | Jul 8 | Falcon 9 | Canaveral | Starlink Group 10-28 |
| 155 | Jul 13 | Falcon 9 | Canaveral | Dror-1 |
| 156 | Jul 14 | Long March 7 | Wenchang | Tianzhou-9 |
| 157 | Jul 16 | Falcon 9 | Vandenberg | Starlink Group 15-2 |
| 158 | Jul 16 | Falcon 9 | Canaveral | Kuiper KF-01 |

| No. | Date (UTC) | Vehicle | Site | Mission / payload | |
|---|---|---|---|---|---|
| 159 | Jul 19 | Falcon 9 | Vandenberg | Starlink Group 17-3 | |
| 160 | Jul 22 | Falcon 9 | Canaveral | O3b mPOWER 9/10 | |
| 161 | Jul 23 | Falcon 9 | Vandenberg | TRACERS | |
| 162 | Jul 25 | Soyuz-2-1B | Vostochny | Soyuz Rideshare 15 | |
| 163 | Jul 26 | Vega C | Kourou | CO3D/Microcarb | |
| 164 | Jul 26 | Falcon 9 | Canaveral | Starlink Group 10-26 | |
| 165 | Jul 27 | Falcon 9 | Vandenberg | Starlink Group 17-2 | |
| 166 | Jul 27 | Long March 6A | Taiyuan | WHDW 05 | |
| 167 | Jul 29 | Shuang Quxian 1 | Jiuquan | Enshi Xidou Shanquan | |
| 168 | Jul 29 | Eris Block 1 | Bowen | Test | × |
| 169 | Jul 30 | Falcon 9 | Canaveral | Starlink Group 10-29 | |
| 170 | Jul 30 | Long March 8A | Hainan (comm.) | WHDW 06 | |
| 171 | Jul 30 | GSLV Mk II | Sriharikota | NISAR | |
| 172 | Jul 31 | Kuaizhou-1A | Xichang | PRSS | |
| 173 | Jul 31 | Falcon 9 | Vandenberg | Starlink Group 13-4 | |
| 174 | Aug 1 | Falcon 9 | Kennedy | Dragon C206.6 | |
| 175 | Aug 4 | Falcon 9 | Canaveral | Starlink Group 10-30 | |
| 176 | Aug 4 | Long March 12 | Hainan (comm.) | WHDW 07 | |
| 177 | Aug 5 | Electron | Mahia | QPS-SAR-12 | |
| 178 | Aug 8 | Jielong-3 | Yellow Sea | Gili 04 zu 01-11 | |
| 179 | Aug 11 | Falcon 9 | Canaveral | Kuiper KF-02 | |
| 180 | Aug 13 | Ariane 62 | Kourou | METOP-SG A1 | |
| 181 | Aug 13 | Vulcan Centaur VC4S | Canaveral | NTS-3 | |
| 182 | Aug 13 | Long March 5B/YZ-2 | Wenchang | WHDW 08 | |
| 183 | Aug 14 | Falcon 9 | Vandenberg | Starlink Group 17-4 | |
| 184 | Aug 14 | Falcon 9 | Canaveral | Starlink Group 10-20 | |
| 185 | Aug 15 | Zhuque-2E | Jiuquan | WHJSW 07? | × |
| 186 | Aug 17 | Long March 4C | Xichang | SY28B-02 | |
| 187 | Aug 17 | Long March 6A | Taiyuan | WHDW 09 | |
| 188 | Aug 18 | Falcon 9 | Vandenberg | Starlink Group 17-5 | |
| 189 | Aug 19 | Lijian-1 | Jiuquan | AIRSAT/DGS | |
| 190 | Aug 20 | Soyuz-2-1B | Baikonur | Bion-M No. 2 | |
| 191 | Aug 21 | Angara-1.2 | Plesetsk | OO MKA 3-6 | |
| 192 | Aug 22 | Falcon 9 | Kennedy | X-37B No. 1/USSF-36 | |
| 193 | Aug 22 | Falcon 9 | Vandenberg | Starlink Group 17-6 | |
| 194 | Aug 23 | Electron | Mahia | E-Space | |
| 195 | Aug 24 | Falcon 9 | Canaveral | Dragon C211.3 | |
| 196 | Aug 25 | Long March 8A | Hainan (comm.) | WHDW 10 | |
| 197 | Aug 26 | Falcon 9 | Vandenberg | NAOS | |
| 198 | Aug 26 | Starship V2 | Starbase | Ship 37 | |
| 199 | Aug 27 | Falcon 9 | Canaveral | Starlink Group 10-56 | |
| 200 | Aug 28 | Falcon 9 | Kennedy | Starlink Group 10-11 | |

| No. | Date (UTC) | Vehicle | Site | Mission / payload | |
|---|---|---|---|---|---|
| 201 | Aug 30 | Falcon 9 | Vandenberg | Starlink Group 17-7 | |
| 202 | Aug 31 | Falcon 9 | Canaveral | Starlink Group 10-14 | |
| 203 | Sep 2 | Shavit 2 | Palmachim | ʻOfeq-19 | |
| 204 | Sep 3 | Falcon 9 | Vandenberg | Starlink Group 17-8 | |
| 205 | Sep 3 | Falcon 9 | Canaveral | Starlink Group 10-22 | |
| 206 | Sep 5 | Long March 3C/YZ-1 | Xichang | SY29 | |
| 207 | Sep 5 | Gushenxing 1 | Jiuquan | Yunyao/Jiazhou | |
| 208 | Sep 5 | Falcon 9 | Kennedy | Starlink Group 10-57 | |
| 209 | Sep 6 | Long March 6A | Taiyuan | Yaogan 40-03 | |
| 210 | Sep 6 | Falcon 9 | Vandenberg | Starlink Group 17-9 | |
| 211 | Sep 8 | Jielong-3 | Yellow Sea | Gili 05 zu 01-11 | |
| 212 | Sep 9 | Long March 7A | Wenchang | Yaogan 45 | |
| 213 | Sep 10 | Falcon 9 | Vandenberg | Tranche 1 TL-B | |
| 214 | Sep 11 | Soyuz-2-1A | Baikonur | Progress 7K-TGM No. 462 | |
| 215 | Sep 12 | Falcon 9 | Canaveral | Nusantara Lima | |
| 216 | Sep 13 | Soyuz-2-1B | Plesetsk | Glonass/Mozhaets 6 | |
| 217 | Sep 13 | Falcon 9 | Vandenberg | Starlink Group 17-10 | |
| 218 | Sep 14 | Falcon 9 | Canaveral | Cygnus NG-23 | |
| 219 | Sep 16 | Long March 2C/YZ-1S | Jiuquan | WHJSW-07 | |
| 220 | Sep 18 | Falcon 9 | Canaveral | Starlink Group 10-61 | |
| 221 | Sep 18 | Zoljanah | Semnan | Test? | × |
| 222 | Sep 19 | Falcon 9 | Vandenberg | Starlink Group 17-12 | |
| 223 | Sep 21 | Falcon 9 | Canaveral | Starlink Group 10-27 | |
| 224 | Sep 22 | Falcon 9 | Vandenberg | NRO Starshield C2 | |
| 225 | Sep 24 | Jielong-3 | Yellow Sea | Gili 06 zu 01-11,SK1 | |
| 226 | Sep 24 | Falcon 9 | Kennedy | IMAP | |
| 227 | Sep 25 | Falcon 9 | Canaveral | Starlink Group 10-15 | |
| 228 | Sep 25 | Atlas V 551 | Canaveral | KA-03 | |
| 229 | Sep 26 | Falcon 9 | Vandenberg | Starlink Group 17-11 | |
| 230 | Sep 26 | Long March 4C | Jiuquan | FY3-08 | |
| 231 | Sep 27 | Long March 6A | Taiyuan | WHDW 11 | |
| 232 | Sep 29 | Falcon 9 | Vandenberg | Starlink Group 11-20 | |
| 233 | Sep 29 | Long March 2D | Xichang | Shiyan 30-01/02 | |
| 234 | Oct 3 | Falcon 9 | Vandenberg | Starlink Group 11-39 | |
| 235 | Oct 7 | Falcon 9 | Canaveral | Starlink Group 10-59 | |
| 236 | Oct 8 | Falcon 9 | Vandenberg | Starlink Group 11-17 | |
| 237 | Oct 11 | Yinli-1 | Yellow Sea | Jilin-1 Kuanfu 02B07 | |
| 238 | Oct 13 | Long March 2D | Jiuquan | Shiyan 31 | |
| 239 | Oct 13 | Starship V2 | Starbase | Ship 38 | |
| 240 | Oct 14 | Falcon 9 | Canaveral | Kuiper KF-03 | |
| 241 | Oct 14 | Electron | Mahia | Strix-5 | |
| 242 | Oct 15 | Falcon 9 | Vandenberg | Tranche 1 TL-C | |

| No. | Date (UTC) | Vehicle | Site | Mission / payload | |
|---|---|---|---|---|---|
| 243 | Oct 16 | Long March 8A | Hainan (comm.) | WHDW 12 | |
| 244 | Oct 16 | Falcon 9 | Canaveral | Starlink Group 10-52 | |
| 245 | Oct 17 | Long March 6A | Taiyuan | Qianfan Jigui 18 | |
| 246 | Oct 19 | Lijian-1 | Jiuquan | PRSC-HS1/AIRSAT | |
| 247 | Oct 19 | Falcon 9 | Canaveral | Starlink Group 10-17 | |
| 248 | Oct 19 | Falcon 9 | Vandenberg | Starlink Group 11-19 | |
| 249 | Oct 22 | Falcon 9 | Vandenberg | Starlink Group 11-5 | |
| 250 | Oct 23 | Long March 5 | Wenchang | TJS 20 | |
| 251 | Oct 24 | Falcon 9 | Canaveral | Spainsat NG-II | |
| 252 | Oct 25 | Falcon 9 | Vandenberg | Starlink Group 11-12 | |
| 253 | Oct 26 | H3-24W | Tanegashima | HTV-X1 | |
| 254 | Oct 26 | Long March 3B | Xichang | GF14-02 | |
| 255 | Oct 26 | Falcon 9 | Canaveral | Starlink Group 10-21 | |
| 256 | Oct 28 | Falcon 9 | Vandenberg | Starlink Group 11-21 | |
| 257 | Oct 29 | Falcon 9 | Canaveral | Starlink Group 10-37 | |
| 258 | Oct 31 | Long March 2F | Jiuquan | Shenzhou 21 | |
| 259 | Oct 31 | Falcon 9 | Vandenberg | Starlink Group 11-23 | |
| 260 | Nov 2 | Falcon 9 | Canaveral | Bandwagon-4 | |
| 261 | Nov 2 | LVM3 | Sriharikota | GSAT-7R | |
| 262 | Nov 3 | Long March 7A | Wenchang | Yaogan 46 | |
| 263 | Nov 4 | Ariane 62 | Kourou | Sentinel-1D | |
| 264 | Nov 5 | Electron | Mahia | QPS-SAR-14 | |
| 265 | Nov 6 | Falcon 9 | Canaveral | Starlink Group 6-81 | |
| 266 | Nov 6 | Falcon 9 | Vandenberg | Starlink Group 11-14 | |
| 267 | Nov 8 | Long March 11 | Yellow Sea | Shiyan 32 01-03 | |
| 268 | Nov 9 | Lijian-1 | Jiuquan | Chutian | |
| 269 | Nov 9 | Falcon 9 | Kennedy | Starlink Group 10-51 | |
| 270 | Nov 10 | Long March 12 | Hainan (comm.) | WHDW 13 | |
| 271 | Nov 10 | Gushenxing 1 | Jiuquan | Jixing Gaofen 04C | × |
| 272 | Nov 11 | Falcon 9 | Canaveral | Starlink Group 6-87 | |
| 273 | Nov 13 | New Glenn 7×2 | Canaveral | ESCAPADE | |
| 274 | Nov 14 | Atlas V 551 | Canaveral | Viasat-3 F2 | |
| 275 | Nov 15 | Falcon 9 | Kennedy | Starlink Group 6-89 | |
| 276 | Nov 15 | Falcon 9 | Canaveral | Starlink Group 6-85 | |
| 277 | Nov 17 | Falcon 9 | Vandenberg | Sentinel-6B | |
| 278 | Nov 19 | Falcon 9 | Canaveral | Starlink Group 6-94 | |
| 279 | Nov 19 | Long March 2C | Jiuquan | Shi Jian 30A-C | |
| 280 | Nov 20 | Electron | Mahia | BlackSky Gen3-3 | |
| 281 | Nov 21 | Falcon 9 | Kennedy | Starlink Group 6-78 | |
| 282 | Nov 21 | Long March 3B | Xichang | TJS 21 | |
| 283 | Nov 22 | Falcon 9 | Canaveral | Starlink Group 6-79 | |
| 284 | Nov 23 | Falcon 9 | Vandenberg | Starlink Group 11-30 | |

| No. | Date (UTC) | Vehicle | Site | Mission / payload | |
|---|---|---|---|---|---|
| 285 | Nov 25 | Long March 2F | Jiuquan | Shenzhou 22 | |
| 286 | Nov 25 | Angara-1.2 | Plesetsk | Rodnik Blok 18 | |
| 287 | Nov 26 | Nuri | Naro | CAS500-3 | |
| 288 | Nov 27 | Soyuz-2-1A | Baikonur | Soyuz 11F732A48 No. 753 | |
| 289 | Nov 28 | Falcon 9 | Vandenberg | Transporter-15 | |
| 290 | Nov 30 | Long March 7A | Wenchang | Shijian 28 | |
| 291 | Dec 1 | Falcon 9 | Kennedy | Starlink Group 6-86 | |
| 292 | Dec 1 | Vega C | Kourou | Kompsat-7 | |
| 293 | Dec 2 | Falcon 9 | Vandenberg | Starlink Group 15-10 | |
| 294 | Dec 2 | Falcon 9 | Canaveral | Starlink Group 6-95 | |
| 295 | Dec 3 | Zhuque-3 | Jiuquan | Dummy payload | |
| 296 | Dec 4 | Falcon 9 | Vandenberg | Starlink Group 11-25 | |
| 297 | Dec 5 | Kuaizhou-1A | Jiuquan | Jiaotong VDES A/B | |
| 298 | Dec 6 | Long March 8A | Hainan (comm.) | WHDW 14 | |
| 299 | Dec 7 | Falcon 9 | Vandenberg | Starlink Group 11-15 | |
| 300 | Dec 8 | Long March 6A | Taiyuan | WHDW 15 | |
| 301 | Dec 8 | Falcon 9 | Kennedy | Starlink Group 6-92 | |
| 302 | Dec 9 | Long March 4B | Jiuquan | Yaogan 47 | |
| 303 | Dec 9 | Long March 3B | Xichang | TJS 22 | |
| 304 | Dec 9 | Falcon 9 | Canaveral | INTRUDER? | |
| 305 | Dec 10 | Lijian-1 | Jiuquan | Jilin-1/UAE-813 | |
| 306 | Dec 10 | Falcon 9 | Vandenberg | Starlink Group 15-11 | |
| 307 | Dec 11 | Falcon 9 | Canaveral | Starlink Group 6-90 | |
| 308 | Dec 11 | Long March 12 | Hainan (comm.) | WHDW 16 | |
| 309 | Dec 13 | Kuaizhou-11 | Jiuquan | Dier-5/Xiwang-5-2 | |
| 310 | Dec 14 | Electron | Mahia | RAISE-4 | |
| 311 | Dec 14 | Falcon 9 | Vandenberg | Starlink Group 15-12 | |
| 312 | Dec 15 | Falcon 9 | Canaveral | Starlink Group 6-82 | |
| 313 | Dec 16 | Long March 4B | Taiyuan | ZY3-04 | |
| 314 | Dec 16 | Atlas V 551 | Canaveral | LA-04 | |
| 315 | Dec 17 | Ariane 62 | Kourou | Galileosat 33-34 | |
| 316 | Dec 17 | Falcon 9 | Kennedy | Starlink Group 6-99 | |
| 317 | Dec 17 | Falcon 9 | Vandenberg | Starlink Group 15-13 | |
| 318 | Dec 18 | Electron | Wallops MARS | DiskSat 1-4 | |
| 319 | Dec 20 | Long March 5 | Wenchang | TJS 23 | |
| 320 | Dec 21 | Electron | Mahia | QPS-SAR-15 | |
| 321 | Dec 22 | H3-22S | Tanegashima | QZS 5 | × |
| 322 | Dec 23 | HANBIT-NANO | Alcantara | Test | × |
| 323 | Dec 23 | Long March 12A | Jiuquan | Dummy payload | |
| 324 | Dec 24 | LVM3 | Sriharikota | BlueBird Block-2 FM1 | |
| 325 | Dec 25 | Soyuz-2-1A | Plesetsk | Obzor-R No. 1 | |
| 326 | Dec 25 | Long March 8A | Hainan (comm.) | WHDW 17 | |

| No. | Date (UTC) | Vehicle | Site | Mission / payload |
|---|---|---|---|---|
| 327 | Dec 26 | Long March 3B | Xichang | FY4-03 |
| 328 | Dec 28 | Soyuz-2-1B | Vostochny | Soyuz Rideshare 16 |
| 329 | Dec 30 | Long March 4B | Jiuquan | Tianhui-7 |
| 330 | Dec 30 | Long March 7A | Wenchang | Shijian 29A/B |

## E.5 Orbital Launch Log: 2026, to 13 October

All 244 orbital launch attempts. × marks a failure to reach orbit. Times are UTC.

**TABLE E.5** Orbital launch attempts in 2026, to 13 October.

| No. | Date (UTC) | Vehicle | Site | Mission / payload | |
|---|---|---|---|---|---|
| 1 | Jan 3 | Falcon 9 | Vandenberg | COSMO-Skymed SG FM3 | |
| 2 | Jan 4 | Falcon 9 | Canaveral | Starlink Group 6-88 | |
| 3 | Jan 9 | Falcon 9 | Canaveral | Starlink Group 6-96 | |
| 4 | Jan 11 | Falcon 9 | Vandenberg | Twilight | |
| 5 | Jan 12 | PSLV-DL | Sriharikota | PSLV Rideshare 20 | × |
| 6 | Jan 12 | Falcon 9 | Canaveral | Starlink Group 6-97 | |
| 7 | Jan 13 | Long March 6A | Taiyuan | YG50-01 | |
| 8 | Jan 13 | Long March 8A | Hainan (comm.) | WHDW 18 | |
| 9 | Jan 14 | Falcon 9 | Canaveral | Starlink Group 6-98 | |
| 10 | Jan 15 | Long March 2C | Jiuquan | Alsat-3A | |
| 11 | Jan 15 | Gushenxing 1 | Yellow Sea | Tianqi 37-40 | |
| 12 | Jan 16 | Long March 3B | Xichang | SJ-32 | × |
| 13 | Jan 17 | Gushenxing-2 | Jiuquan | Unk Sat 1-6 | × |
| 14 | Jan 17 | Falcon 9 | Vandenberg | NRO Starshield C3 | |
| 15 | Jan 18 | Falcon 9 | Canaveral | Starlink Group 6-100 | |
| 16 | Jan 19 | Long March 12 | Hainan (comm.) | WHDW 19 | |
| 17 | Jan 22 | Falcon 9 | Vandenberg | Starlink Group 17-30 | |
| 18 | Jan 22 | Electron | Mahia | Open Cosmos | |
| 19 | Jan 25 | Falcon 9 | Vandenberg | Starlink Group 17-20 | |
| 20 | Jan 28 | Falcon 9 | Canaveral | GPS III SV09 | |
| 21 | Jan 29 | Falcon 9 | Vandenberg | Starlink Group 17-19 | |
| 22 | Jan 30 | Electron | Mahia | NEONSAT-1A | |
| 23 | Jan 30 | Falcon 9 | Canaveral | Starlink Group 6-101 | |
| 24 | Jan 31 | Long March 2C | Jiuquan | Alsat-3B | |
| 25 | Feb 2 | Falcon 9 | Vandenberg | Starlink Group 17-32 | |
| 26 | Feb 5 | Soyuz-2-1B | Plesetsk | Unknown | |
| 27 | Feb 7 | Long March 2F | Jiuquan | CSSHQ | |
| 28 | Feb 7 | Falcon 9 | Vandenberg | Starlink Group 17-33 | |
| 29 | Feb 11 | Falcon 9 | Vandenberg | Starlink Group 17-34 | |
| 30 | Feb 12 | Jielong-3 | Yangjiang | PRSC-E02 | |
| 31 | Feb 12 | Proton-M/DM-3 | Baikonur | Elektro-L No. 5 | |
| 32 | Feb 12 | Vulcan Centaur VC4S | Canaveral | GSSAP | |

| No. | Date (UTC) | Vehicle | Site | Mission / payload | |
|---|---|---|---|---|---|
| 33 | Feb 12 | Ariane 64 | Kourou | Amazon LE-01 | |
| 34 | Feb 13 | Falcon 9 | Canaveral | Dragon C212.5 | |
| 35 | Feb 15 | Falcon 9 | Vandenberg | Starlink Group 17-13 | |
| 36 | Feb 16 | Falcon 9 | Canaveral | Starlink Group 6-103 | |
| 37 | Feb 20 | Falcon 9 | Canaveral | Starlink Group 10-36 | |
| 38 | Feb 21 | Falcon 9 | Vandenberg | Starlink Group 17-25 | |
| 39 | Feb 22 | Falcon 9 | Canaveral | Starlink Group 6-104 | |
| 40 | Feb 24 | Falcon 9 | Canaveral | Starlink Group 6-110 | |
| 41 | Feb 25 | Falcon 9 | Vandenberg | Starlink Group 17-26 | |
| 42 | Feb 27 | Falcon 9 | Canaveral | Starlink Group 6-108 | |
| 43 | Mar 1 | Falcon 9 | Vandenberg | Starlink Group 17-23 | |
| 44 | Mar 2 | Falcon 9 | Canaveral | Starlink Group 10-41 | |
| 45 | Mar 4 | Falcon 9 | Canaveral | Starlink Group 10-40 | |
| 46 | Mar 5 | Kairos | Kii | Tatara-1R | × |
| 47 | Mar 5 | Electron | Mahia | BlackSky Gen3-4 | |
| 48 | Mar 8 | Falcon 9 | Vandenberg | Starlink Group 17-18 | |
| 49 | Mar 10 | Falcon 9 | Canaveral | Echostar 25 | |
| 50 | Mar 12 | Firefly Alpha | Vandenberg | LM-400 Tech Demo | |
| 51 | Mar 12 | Long March 8A | Hainan (comm.) | WHDW 20 | |
| 52 | Mar 12 | Long March 2D | Xichang | Shiyan 30-03/04 | |
| 53 | Mar 13 | Falcon 9 | Vandenberg | Starlink Group 17-31 | |
| 54 | Mar 14 | Falcon 9 | Canaveral | Starlink Group 10-48 | |
| 55 | Mar 15 | Long March 6A | Taiyuan | YG50-02 | |
| 56 | Mar 16 | Kuaizhou-11 | Jiuquan | Juntian-1-04A/Dongp | |
| 57 | Mar 17 | Falcon 9 | Vandenberg | Starlink Group 17-24 | |
| 58 | Mar 17 | Falcon 9 | Canaveral | Starlink Group 10-46 | |
| 59 | Mar 19 | Falcon 9 | Canaveral | Starlink Group 10-33 | |
| 60 | Mar 20 | Electron | Mahia | StriX-6 | |
| 61 | Mar 20 | Falcon 9 | Vandenberg | Starlink Group 17-15 | |
| 62 | Mar 22 | Soyuz-2-1A | Baikonur | Progress 7K-TGM No. 463 | |
| 63 | Mar 22 | Falcon 9 | Canaveral | Starlink Group 10-62 | |
| 64 | Mar 22 | Jielong-3 | Yellow Sea | Weili Kongjian 02 zu | |
| 65 | Mar 23 | Soyuz-2-1B | Plesetsk | Rassvet-3 G1 | |
| 66 | Mar 25 | Long March 2D | Taiyuan | Superview 2-05/06 | |
| 67 | Mar 26 | Falcon 9 | Vandenberg | Starlink Group 17-17 | |
| 68 | Mar 27 | Long March 2C/YZ-1S | Jiuquan | Shiyan 33 | |
| 69 | Mar 28 | Electron | Mahia | Celeste | |
| 70 | Mar 30 | Lijian-2 | Jiuquan | XZC01/02 | |
| 71 | Mar 30 | Falcon 9 | Vandenberg | Transporter-16 | |
| 72 | Mar 30 | Falcon 9 | Canaveral | Starlink Group 10-44 | |
| 73 | Apr 1 | SLS Block 1 | Kennedy | Integrity | |
| 74 | Apr 2 | Falcon 9 | Canaveral | Starlink Group 10-58 | |

| No. | Date (UTC) | Vehicle | Site | Mission / payload | |
|---|---|---|---|---|---|
| 75 | Apr 3 | Tianlong-3 | Jiuquan | - | × |
| 76 | Apr 3 | Soyuz-2-1A | Plesetsk | Meridian No. 21L? | |
| 77 | Apr 4 | Atlas V 551 | Canaveral | LA-05 | |
| 78 | Apr 7 | Falcon 9 | Vandenberg | Starlink Group 17-35 | |
| 79 | Apr 7 | Minotaur IV | Vandenberg | STP-S29A | |
| 80 | Apr 7 | Long March 8 | Hainan (comm.) | Qianfan | |
| 81 | Apr 8 | Long March 6A | Taiyuan | WHDW 21 | |
| 82 | Apr 11 | Falcon 9 | Vandenberg | Starlink Group 17-21 | |
| 83 | Apr 11 | Jielong-3 | Yangjiang | WHJSW-8 | |
| 84 | Apr 11 | Falcon 9 | Canaveral | Cygnus NG-24 | |
| 85 | Apr 14 | Lijian-1 | Jiuquan | Jilin-1 GF07 | |
| 86 | Apr 14 | Falcon 9 | Canaveral | Starlink Group 10-24 | |
| 87 | Apr 15 | Falcon 9 | Vandenberg | Starlink Group 17-27 | |
| 88 | Apr 16 | Soyuz-2-1B | Plesetsk | Unknown | |
| 89 | Apr 17 | Long March 4C | Jiuquan | Daqi 2 | |
| 90 | Apr 19 | New Glenn 7×2 | Canaveral | SpaceMobile-007 | × |
| 91 | Apr 19 | Falcon 9 | Vandenberg | Starlink Group 17-22 | |
| 92 | Apr 21 | Falcon 9 | Canaveral | GPS III SV10 | |
| 93 | Apr 23 | Electron | Mahia | JAXA ISTD-4 | |
| 94 | Apr 23 | Falcon 9 | Vandenberg | Starlink Group 17-14 | |
| 95 | Apr 23 | Angara-1.2 | Plesetsk | OO MKA 7-10 | |
| 96 | Apr 24 | Long March 2D | Xichang | WHJSW 9 | |
| 97 | Apr 25 | Long March 6 | Taiyuan | PRSC-EO3 | |
| 98 | Apr 25 | Soyuz-2-1A | Baikonur | Progress 7K-TGM No. 464 | |
| 99 | Apr 26 | Falcon 9 | Vandenberg | Starlink Group 17-16 | |
| 100 | Apr 28 | Atlas V 551 | Canaveral | LA-06 | |
| 101 | Apr 29 | Falcon Heavy | Kennedy | Viasat 3F3 | |
| 102 | Apr 30 | Falcon 9 | Vandenberg | Starlink Group 17-36 | |
| 103 | Apr 30 | Ariane 64 | Kourou | Amazon LE-02 | |
| 104 | May 1 | Falcon 9 | Canaveral | Starlink Group 10-38 | |
| 105 | May 3 | Falcon 9 | Vandenberg | RideshareX-2026-100 | |
| 106 | May 6 | Falcon 9 | Vandenberg | Starlink Group 17-29 | |
| 107 | May 11 | Long March 7 | Wenchang | Tianzhou-10 | |
| 108 | May 12 | Falcon 9 | Vandenberg | NRO Starshield A9 | |
| 109 | May 12 | Long March 6A | Taiyuan | Qianfan Jigui 09 | |
| 110 | May 14 | Zhuque-2E | Jiuquan | Dummy sat | |
| 111 | May 15 | Lijian-1 | Jiuquan | Taijing-3 | |
| 112 | May 15 | Falcon 9 | Canaveral | Dragon C209.6 | |
| 113 | May 17 | Long March 8 | Hainan (comm.) | Qianfan | |
| 114 | May 19 | Vega C | Kourou | SMILE | |
| 115 | May 20 | Falcon 9 | Vandenberg | Starlink Group 17-42 | |
| 116 | May 21 | Falcon 9 | Canaveral | Starlink Group 10-31 | |

| No. | Date (UTC) | Vehicle | Site | Mission / payload |
|---|---|---|---|---|
| 117 | May 22 | Electron | Mahia | StriX-7 |
| 118 | May 22 | Starship V3 | Starbase | Ship 39 |
| 119 | May 24 | Long March 2F | Jiuquan | Shenzhou 23 |
| 120 | May 25 | Falcon 9 | Canaveral | Starlink Group 10-47 |
| 121 | May 26 | Falcon 9 | Vandenberg | Starlink Group 17-37 |
| 122 | May 26 | Long March 7A | Wenchang | TJS-24 |
| 123 | May 29 | New Glenn 7×2 | Canaveral | - |
| 124 | May 29 | Falcon 9 | Canaveral | Starlink Group 10-53 |
| 125 | May 29 | Atlas V 551 | Canaveral | LA-07 |
| 126 | May 30 | Falcon 9 | Vandenberg | Starlink Group 17-41 |
| 127 | May 30 | Long March 2D | Xichang | WHJSW 10 |
| 128 | Jun 1 | Long March 12B | Jiuquan | Qianfan |
| 129 | Jun 3 | Falcon 9 | Vandenberg | Starlink Group 17-47 |
| 130 | Jun 4 | Falcon 9 | Canaveral | Starlink Group 10-43 |
| 131 | Jun 4 | Long March 6A | Taiyuan | Qianfan Jigui 11 |
| 132 | Jun 5 | Long March 8 | Hainan (comm.) | Qianfan Jigui 12 |
| 133 | Jun 7 | Falcon 9 | Vandenberg | Starlink Group 17-43 |
| 134 | Jun 8 | Falcon 9 | Canaveral | Starlink Group 10-35 |
| 135 | Jun 9 | Zhuque-2E | Jiuquan | Qianfan DTC1 |
| 136 | Jun 11 | Long March 5 | Wenchang | TJS 25 |
| 137 | Jun 11 | HASTE | Wallops MARS | MACH-TB? |
| 138 | Jun 11 | Falcon 9 | Vandenberg | Starlink Group 17-44 |
| 139 | Jun 12 | H3-30S | Tanegashima | VEP-5 |
| 140 | Jun 12 | Falcon 9 | Canaveral | Starlink Group 10-54 |
| 141 | Jun 15 | Lijian-1 | Jiuquan | Jilin-1 GF07 |
| 142 | Jun 15 | Falcon 9 | Vandenberg | Starlink Group 17-54 |
| 143 | Jun 16 | Long March 3B | Xichang | SJ-31 (TJS-26?) |
| 144 | Jun 17 | Long March 12 | Hainan (comm.) | WHDW 22 |
| 145 | Jun 17 | Kuaizhou-11 | Jiuquan | Centispace 05 |
| 146 | Jun 17 | Falcon 9 | Canaveral | Bluebird 8 to 10 |
| 147 | Jun 17 | Ariane 64 | Kourou | Amazon LE-03 |
| 148 | Jun 19 | Falcon 9 | Vandenberg | NRO Starshield C4 |
| 149 | Jun 19 | Electron | Mahia | VICTUS HAZE Puma |
| 150 | Jun 21 | Falcon 9 | Vandenberg | Starlink Group 17-28 |
| 151 | Jun 23 | Long March 7A | Wenchang | TJS-26A |
| 152 | Jun 23 | Falcon 9 | Canaveral | Starfall Demo |
| 153 | Jun 25 | Falcon 9 | Vandenberg | Starlink Group 17-45 |
| 154 | Jun 26 | Electron | Mahia | StriX-8 |
| 155 | Jun 28 | Falcon 9 | Vandenberg | Starlink Group 17-40 |
| 156 | Jun 29 | Falcon 9 | Canaveral | SXM-11 |
| 157 | Jul 1 | Long March 4B | Jiuquan | HY-2E |
| 158 | Jul 2 | Falcon 9 | Vandenberg | Starlink Group 17-46 |

| No. | Date (UTC) | Vehicle | Site | Mission / payload | |
|---|---|---|---|---|---|
| 159 | Jul 2 | Atlas V 551 | Canaveral | LA-08 | |
| 160 | Jul 3 | Pegasus XL | Kwajalein | Swift Reboost | |
| 161 | Jul 4 | Long March 6A | Taiyuan | Qianfan Jigui 13 | |
| 162 | Jul 5 | Falcon 9 | Canaveral | Starlink Group 10-50 | |
| 163 | Jul 5 | Long March 8A | Hainan (comm.) | Qianfan Jigui 15 | |
| 164 | Jul 7 | Falcon 9 | Vandenberg | Transporter-17 | |
| 165 | Jul 9 | Falcon 9 | Canaveral | Starlink Group 10-42 | |
| 166 | Jul 10 | Long March 10B | Hainan (comm.) | CX-26 | |
| 167 | Jul 11 | Falcon 9 | Vandenberg | Starlink Group 17-48 | |
| 168 | Jul 14 | Falcon 9 | Vandenberg | Starlink Group 15-14 | |
| 169 | Jul 14 | Falcon 9 | Canaveral | Starlink Group 10-45 | |
| 170 | Jul 14 | Soyuz-2-1A | Baikonur | Soyuz 11F732A48 No. 759 | |
| 171 | Jul 16 | Falcon 9 | Vandenberg | Tranche 1 TL-E | |
| 172 | Jul 18 | Vikram-1 | Sriharikota | Solaras-S3/SCOPE | |
| 173 | Jul 19 | Soyuz-2-1B | Plesetsk | Rassvet-3 G2 | |
| 174 | Jul 21 | Falcon 9 | Vandenberg | Starlink Group 17-39 | |
| 175 | Jul 21 | Falcon 9 | Canaveral | MRV/MEP | |
| 176 | Jul 22 | Yinli-1 | East China Sea | Dongpo/ZDG-3 | |
| 177 | Jul 23 | Long March 3B | Xichang | Tianlian 2-06 | |
| 178 | Jul 23 | Lijian-1 | Jiuquan | Xiguang | |
| 179 | Jul 24 | Starship V3 | Starbase | Ship 40 | |
| 180 | Jul 25 | Falcon 9 | Vandenberg | Starlink Group 17-51 | |
| 181 | Jul 29 | Long March 7A | Wenchang | Tianlian 3-01 | |
| 182 | Jul 30 | Long March 6A | Taiyuan | TJS-27A/B | |
| 183 | Jul 30 | Falcon 9 | Canaveral | INTRUDER? | |
| 184 | Aug 1 | Falcon 9 | Vandenberg | Starlink Group 17-52 | |
| 185 | Aug 4 | Long March 8A | Hainan (comm.) | WHDW 23 | |
| 186 | Aug 4 | Falcon 9 | Vandenberg | Starlink Group 17-53 | |
| 187 | Aug 5 | Jielong-3 | Yellow Sea | Lampung/Samarqand | |
| 188 | Aug 5 | Falcon 9 | Canaveral | Bluebird 11 to 13 | |
| 189 | Aug 6 | Electron | Mahia | QPS-SAR-13 | |
| 190 | Aug 8 | Falcon 9 | Vandenberg | Starlink Group 17-38 | |
| 191 | Aug 10 | Long March 7A | Wenchang | Zhongxing-4B | × |
| 192 | Aug 10 | H3-22S | Tanegashima | QZS 7 | |
| 193 | Aug 11 | Falcon 9 | Canaveral | Starlink Group 10-19 | |
| 194 | Aug 12 | Falcon 9 | Vandenberg | Starlink Group 17-49 | |
| 195 | Aug 16 | Falcon 9 | Canaveral | Globalstar 2R-1 | |
| 196 | Aug 16 | Falcon 9 | Vandenberg | SBST R-1 | |
| 197 | Aug 16 | Long March 12 | Hainan (comm.) | WHDW 24 | |
| 198 | Aug 17 | Long March 2C | Taiyuan | SEO | |
| 199 | Aug 18 | Zhuque-3 | Jiuquan | Honghu-3 | |
| 200 | Aug 19 | Falcon 9 | Vandenberg | Starlink Group 17-50 | |

| No. | Date (UTC) | Vehicle | Site | Mission / payload |
|---|---|---|---|---|
| 201 | Aug 20 | Electron | Mahia | QPS-SAR-18 |
| 202 | Aug 21 | Falcon 9 | Canaveral | Starlink Group 10-39 |
| 203 | Aug 22 | Falcon 9 | Vandenberg | Starlink Group 15-20 |
| 204 | Aug 24 | Soyuz-2-1B | Plesetsk | Glonass |
| 205 | Aug 25 | Long March 6C | Taiyuan | ZK-14/15 |
| 206 | Aug 25 | Falcon 9 | Canaveral | Starlink Group 10-49 |
| 207 | Aug 26 | Falcon 9 | Vandenberg | Starlink Group 15-22 |
| 208 | Aug 27 | Ariane 62 | Kourou | MTG-I2 |
| 209 | Aug 30 | Falcon Heavy | Kennedy | NGRST |
| 210 | Sep 1 | Zhishenxing-1 | Jiuquan | Pallas-1 Test |
| 211 | Sep 2 | Falcon 9 | Vandenberg | Starlink Group 15-23 |
| 212 | Sep 2 | Electron | Mahia | StriX-9 |
| 213 | Sep 3 | GSLV Mk II | Sriharikota | GISAT-1A |
| 214 | Sep 5 | Spectrum | Andoya | Onward and Upward |
| 215 | Sep 6 | Falcon 9 | Vandenberg | Starlink Group 15-24 |
| 216 | Sep 10 | Long March 4B | Jiuquan | YG53/YG56 |
| 217 | Sep 10 | Falcon 9 | Vandenberg | SBST R-2 |
| 218 | Sep 11 | Electron | Mahia | BlackSky Gen3-5 |
| 219 | Sep 13 | Falcon 9 | Canaveral | O3b mPOWER 11/13 |
| 220 | Sep 15 | Vega C | Kourou | Sentinel-3C/Flex |
| 221 | Sep 15 | Zhuque-2E | Jiuquan | Qianfan Digui 19 |
| 222 | Sep 15 | Yinli-1 | East China Sea | Qianfan Digui 26 |
| 223 | Sep 16 | Soyuz-2-1B | Baikonur | Progress 7K-TGM No. 465 |
| 224 | Sep 17 | Long March 12 | Hainan (comm.) | WHDW 25 |
| 225 | Sep 17 | Falcon 9 | Vandenberg | SBST TH-1 |
| 226 | Sep 17 | Kuaizhou-11 | Jiuquan | Tianyi-51/52 |
| 227 | Sep 17 | Soyuz-2-1B | Plesetsk | Glonass |
| 228 | Sep 19 | Electron | Mahia | StriX-10 |
| 229 | Sep 19 | Long March 2D | Jiuquan | PIESAT 2-13 to 16 |
| 230 | Sep 20 | Falcon 9 | Vandenberg | Starlink Group 15-27 |
| 231 | Sep 20 | Lijian-1 | Jiuquan | Pengcheng |
| 232 | Sep 23 | Long March 8A | Hainan (comm.) | WHDW 26 |
| 233 | Sep 24 | Long March 6A | Taiyuan | Yaogan 40-04 |
| 234 | Sep 26 | Electron | Mahia | StriX-12 |
| 235 | Sep 26 | Falcon 9 | Vandenberg | SBST R-3 |
| 236 | Sep 28 | Starship V3 | Starbase | Ship 41 |
| 237 | Oct 1 | Falcon 9 | Canaveral | Dragon C213.2 |
| 238 | Oct 1 | Falcon 9 | Vandenberg | Transporter-18 |
| 239 | Oct 2 | Falcon Heavy | Kennedy | NROL-97 |
| 240 | Oct 7 | Nuri | Naro | NeonSat |
| 241 | Oct 9 | Long March 12 | Hainan (comm.) | WHDW 27 |
| 242 | Oct 10 | Falcon 9 | Vandenberg | Tranche 1 TL-A |

| No. | Date (UTC) | Vehicle | Site | Mission / payload |
|---|---|---|---|---|
| 243 | Oct 13 | Falcon 9 | Vandenberg | Starlink Group 15-25 |
| 244 | Oct 13 | Falcon 9 | Canaveral | Dragon C211.4 |

APPENDIX

# F Timeline of Space Exploration and Engineering

*A chronology of the milestones described in this manual — from the theoretical foundations of rocketry and the first liquid-fuelled rocket, through the Space Race, the Shuttle and station era, and the rise of commercial spaceflight, to the events of 2025 and 2026 — followed by the missions planned for 2026–2035 as of October 2026.*

**IN THIS CHAPTER**

F.1 Foundations (1903–1956)
F.2 The Space Race (1957–1975)
F.3 Shuttles, Stations and Space Science (1976–1999)
F.4 The New Century (2000–2019)
F.5 The Commercial Era (2020–2026)
F.6 Planned Missions (as of October 2026)

Dates are in Coordinated Universal Time (UTC) where the precise day matters; a launch late in the evening in the Americas can fall on the next UTC day. "First" claims refer to the first *successful* achievement. Planned dates in the final table are those announced by the responsible agencies or companies as of October 2026; experience suggests that many will slip.

## F.1 Foundations (1903–1956)

| Date | Event |
|---|---|
| 1903 | Konstantin Tsiolkovsky publishes *Exploration of Outer Space by Means of Reaction Devices*, deriving the rocket equation |
| 1919 | Robert Goddard publishes *A Method of Reaching Extreme Altitudes* |
| 1923 | Hermann Oberth publishes *Die Rakete zu den Planetenräumen* (The Rocket into Planetary Space) |
| 16 Mar 1926 | Goddard launches the first liquid-propellant rocket, at Auburn, Massachusetts — a 2.5 s flight to 12.5 m |
| 17 Aug 1933 | GIRD-09, the first Soviet rocket with a liquid oxidiser, flies near Moscow |
| 20 Jun 1944 | A German V-2 (A4) test rocket reaches about 175 km — the first human-made object in space |
| 24 Oct 1946 | A camera on a captured V-2 launched from White Sands takes the first photographs of Earth from space (105 km) |
| 24 Feb 1949 | The two-stage Bumper-WAC reaches 393 km |
| Jul–Aug 1955 | The United States and the Soviet Union announce plans to launch satellites for the International Geophysical Year |

## F.2 The Space Race (1957–1975)

| Date | Event |
|---|---|
| 21 Aug 1957 | First successful flight of the R-7, the world's first intercontinental ballistic missile |
| 4 Oct 1957 | **Sputnik 1**, the first artificial satellite, launched by an R-7 from Baikonur |

| Date | Event |
|---|---|
| 3 Nov 1957 | Sputnik 2 carries the dog Laika, the first animal in orbit |
| 1 Feb 1958 | Explorer 1, the first US satellite; its Geiger counter discovers the Van Allen radiation belts |
| 17 Mar 1958 | Vanguard 1, the first solar-powered satellite — still in orbit, the oldest human-made object in space |
| 1 Oct 1958 | NASA begins operations |
| 2 Jan 1959 | Luna 1, the first spacecraft to escape Earth's gravity, flies past the Moon |
| 14 Sep 1959 | Luna 2 strikes the Moon, the first spacecraft to reach another world |
| 7 Oct 1959 | Luna 3 photographs the far side of the Moon |
| 1 Apr 1960 | TIROS-1, the first weather satellite |
| 12 Aug 1960 | Echo 1, a 30 m passive reflecting balloon, bounces telephone and television signals across the United States |
| 18–19 Aug 1960 | Discoverer 14 (Corona): the first film capsule recovered from orbit |
| 12 Apr 1961 | **Yuri Gagarin** orbits the Earth in Vostok 1 — the first human in space |
| 5 May 1961 | Alan Shepard's suborbital flight in Freedom 7 |
| 20 Feb 1962 | John Glenn, the first American in orbit (Friendship 7) |
| 10 Jul 1962 | Telstar 1 relays the first live transatlantic television |
| 14 Dec 1962 | Mariner 2 flies past Venus — the first successful planetary encounter |
| 16 Jun 1963 | Valentina Tereshkova, the first woman in space (Vostok 6) |
| 26 Jul 1963 | Syncom 2, the first geosynchronous communication satellite (Syncom 3, geostationary, follows in 1964) |
| 18 Mar 1965 | Alexei Leonov makes the first spacewalk (Voskhod 2) |
| 6 Apr 1965 | Intelsat I "Early Bird", the first commercial communication satellite |
| 15 Jul 1965 | Mariner 4 returns the first close-up images of Mars |
| 26 Nov 1965 | France launches Astérix, becoming the third nation to orbit a satellite |
| 3 Feb 1966 | Luna 9 makes the first soft landing on the Moon |
| 16 Mar 1966 | Gemini 8 performs the first docking in orbit |
| 27 Jan 1967 | Apollo 1 fire kills three astronauts during a ground test |
| 10 Oct 1967 | The Outer Space Treaty enters into force |
| 21–27 Dec 1968 | Apollo 8: the first humans to orbit the Moon |
| 20 Jul 1969 | **Apollo 11**: Neil Armstrong and Buzz Aldrin land on the Moon |
| 11 Feb 1970 | Japan launches Ohsumi (fourth nation) |
| 24 Apr 1970 | China launches Dong Fang Hong 1 (fifth nation) |
| 17 Nov 1970 | Lunokhod 1, the first robotic rover on another world |
| 15 Dec 1970 | Venera 7 makes the first landing on Venus and transmits from the surface |
| 19 Apr 1971 | Salyut 1, the first space station |
| 28 Oct 1971 | The United Kingdom launches Prospero on Black Arrow |
| 14 Nov 1971 | Mariner 9, the first spacecraft to orbit another planet (Mars) |
| 2 Dec 1971 | Mars 3 makes the first soft landing on Mars; it transmits for about 20 seconds |
| 23 Jul 1972 | ERTS-1 (Landsat 1), the start of continuous civil land imaging |
| 7–19 Dec 1972 | Apollo 17, the last crewed lunar landing of the twentieth century |
| 14 May 1973 | Skylab, the first US space station |
| 4 Dec 1973 | Pioneer 10, the first spacecraft to fly past Jupiter |
| 17 Jul 1975 | Apollo and Soyuz dock in orbit |

## F.3 Shuttles, Stations and Space Science (1976–1999)

| Date | Event |
|---|---|
| 20 Jul 1976 | Viking 1 lands on Mars; Viking 2 follows on 3 September |
| 20 Aug / 5 Sep 1977 | Voyager 2 and Voyager 1 launched |
| 22 Feb 1978 | First Navstar GPS satellite launched |
| 27 Jun 1978 | Seasat, the first civil imaging radar satellite |
| 24 Dec 1979 | First flight of Ariane 1 |
| 18 Jul 1980 | India orbits Rohini with its SLV-3 rocket |
| 12 Apr 1981 | STS-1: first flight of the Space Shuttle Columbia |
| Apr 1984 | STS-41C crew captures and repairs the Solar Maximum Mission satellite |
| 28 Jan 1986 | Challenger breaks up 73 s after launch; seven astronauts killed |
| 20 Feb 1986 | Launch of the Mir core module |
| Mar 1986 | Giotto, Vega 1/2, Suisei and Sakigake fly past Halley's Comet |
| 15 Nov 1988 | Buran makes its only, uncrewed, orbital flight |
| 19 Sep 1988 | Israel launches Ofeq 1 |
| 25 Aug 1989 | Voyager 2 flies past Neptune |
| 18 Nov 1989 | COBE launched; it maps the cosmic microwave background |
| 24 Apr 1990 | Hubble Space Telescope launched by STS-31 |
| May 1992 | STS-49 crew rescues Intelsat 603 by hand |
| Dec 1993 | First Hubble servicing mission corrects its flawed optics |
| 17 Jul 1995 | GPS declared fully operational |
| 2 Dec 1995 | SOHO launched to the Sun–Earth L1 point |
| 7 Dec 1995 | The Galileo probe enters Jupiter's atmosphere; the orbiter begins eight years of study |
| 4 Jul 1997 | Mars Pathfinder lands; the Sojourner rover begins exploring |
| 20 Nov 1998 | Zarya, the first module of the International Space Station |
| Nov 1998 | Iridium begins the first global satellite telephone service |
| 23 Jul 1999 | Chandra X-ray Observatory launched |

## F.4 The New Century (2000–2019)

| Date | Event |
|---|---|
| 2 Nov 2000 | Expedition 1 arrives on the ISS — the start of continuous human presence in orbit |
| 12 Feb 2001 | NEAR Shoemaker lands on the asteroid Eros |
| 1 Feb 2003 | Columbia breaks up during re-entry; seven astronauts killed |
| 15 Oct 2003 | Yang Liwei flies Shenzhou 5; China becomes the third nation to launch a human |
| Jan 2004 | Mars Exploration Rovers Spirit and Opportunity land; Opportunity drives for 14 years |
| 21 Jun 2004 | SpaceShipOne makes the first privately funded crewed spaceflight; wins the Ansari X Prize in October |
| 1 Jul 2004 | Cassini enters orbit around Saturn |
| 14 Jan 2005 | Huygens lands on Titan, the most distant landing yet |
| 4 Jul 2005 | Deep Impact strikes Comet Tempel 1 |
| 19 Jan 2006 | New Horizons launched towards Pluto |
| 11 Jan 2007 | China's anti-satellite test destroys Fengyun-1C, creating more than 3 500 catalogued debris fragments |

| Date | Event |
|---|---|
| 25 May 2008 | Phoenix lands in Mars's northern arctic and touches water ice |
| 28 Sep 2008 | Falcon 1 becomes the first privately developed liquid-fuelled rocket to reach orbit |
| 10 Feb 2009 | Iridium 33 and Kosmos 2251 collide — the first accidental collision between intact satellites |
| 7 Mar 2009 | Kepler launched; it finds thousands of exoplanets |
| 9 Oct 2009 | LCROSS impact confirms water ice in a permanently shadowed lunar crater |
| 4 Jun 2010 | First flight of Falcon 9 |
| 13 Jun 2010 | Hayabusa returns the first samples from an asteroid (Itokawa) |
| Jun 2010 | IKAROS demonstrates solar-sail propulsion in interplanetary space |
| 18 Mar 2011 | MESSENGER, the first spacecraft to orbit Mercury |
| 21 Jul 2011 | Atlantis lands, ending the Space Shuttle programme after 135 flights |
| 25 May 2012 | SpaceX Dragon becomes the first commercial spacecraft to berth with the ISS |
| 6 Aug 2012 | Curiosity lands in Gale Crater using the sky crane |
| 25 Aug 2012 | Voyager 1 crosses the heliopause into interstellar space |
| 14 Dec 2013 | Chang'e 3 lands with the Yutu rover — the first lunar soft landing since 1976 |
| 19 Dec 2013 | Gaia launched |
| 6 Aug 2014 | Rosetta arrives at Comet 67P; Philae lands on 12 November |
| 24 Sep 2014 | India's Mangalyaan enters Mars orbit on the country's first attempt |
| 6 Mar 2015 | Dawn enters orbit around Ceres after orbiting Vesta |
| 14 Jul 2015 | New Horizons flies past Pluto |
| 22 Dec 2015 | A Falcon 9 first stage lands after an orbital launch for the first time |
| 5 Jul 2016 | Juno enters orbit around Jupiter |
| 15 Feb 2017 | PSLV-C37 deploys 104 satellites in one launch |
| 30 Mar 2017 | First reflight of an orbital-class booster (Falcon 9, SES-10) |
| 15 Sep 2017 | Cassini plunges into Saturn |
| 6 Feb 2018 | First flight of Falcon Heavy |
| 12 Aug 2018 | Parker Solar Probe launched |
| 1 Jan 2019 | New Horizons flies past Arrokoth, the most distant object explored |
| 3 Jan 2019 | Chang'e 4 makes the first landing on the far side of the Moon |
| 27 Mar 2019 | India's anti-satellite test (Mission Shakti) |
| 24 May 2019 | First 60 operational-design Starlink satellites launched |
| 9 Oct 2019 | Mission Extension Vehicle 1 launched; it docks with Intelsat 901 on 25 Feb 2020 |

## F.5 The Commercial Era (2020–2026)

| Date | Event |
|---|---|
| 30 May 2020 | Crew Dragon Demo-2: the first crewed flight of a commercially built orbital spacecraft |
| 31 Jul 2020 | BeiDou-3 global navigation system commissioned |
| 5 Dec 2020 | Hayabusa2 capsule lands in Australia with samples of Ryugu |
| 16 Dec 2020 | Chang'e 5 returns 1.73 kg of lunar samples |
| 18 Feb 2021 | Perseverance lands in Jezero Crater |
| 19 Apr 2021 | Ingenuity makes the first powered, controlled flight on another planet |
| 29 Apr 2021 | Tianhe, the core module of China's Tiangong station |

| Date | Event |
|---|---|
| 14 May 2021 | Tianwen-1 lands the Zhurong rover; China becomes the second nation to operate a rover on Mars |
| Jul 2021 | Virgin Galactic (11 July) and Blue Origin (20 July) fly their founders on suborbital flights |
| 16 Sep 2021 | Inspiration4, the first all-private orbital crew |
| 15 Nov 2021 | Russian anti-satellite test destroys Kosmos 1408, forcing the ISS crew to shelter |
| 25 Dec 2021 | James Webb Space Telescope launched by Ariane 5 |
| 12 Jul 2022 | First full-colour images from JWST |
| 26 Sep 2022 | DART strikes Dimorphos and shortens its orbit by 33 minutes — the first planetary-defence test |
| 16 Nov – 11 Dec 2022 | Artemis I: uncrewed Orion flies around the Moon on the first SLS launch |
| 20 Apr 2023 | First integrated flight of Starship |
| 23 Aug 2023 | Chandrayaan-3 lands near the lunar south pole; India becomes the fourth nation to soft-land on the Moon |
| 24 Sep 2023 | OSIRIS-REx capsule returns samples of Bennu |
| 19 Jan 2024 | Japan's SLIM lands within 100 m of its target |
| 22 Feb 2024 | Intuitive Machines' Odysseus: the first commercial lunar landing (it tips over) |
| 25 Jun 2024 | Chang'e 6 returns the first samples from the lunar far side |
| 9 Jul 2024 | First flight of Ariane 6 |
| 12 Sep 2024 | Polaris Dawn: the first commercial spacewalk |
| 13 Oct 2024 | Starship's Super Heavy booster is caught by the launch tower on its first attempt |
| 14 Oct 2024 | Europa Clipper launched |
| 16 Jan 2025 | First flight of New Glenn reaches orbit |
| 2 Mar 2025 | Firefly's Blue Ghost makes the first fully successful commercial lunar landing |
| 18 Mar 2025 | Crew-9 returns Butch Wilmore and Suni Williams after their Starliner test flight stretched to nine months |
| 27 Mar 2025 | Gaia passivated after mapping about two billion stars |
| 1 Apr 2025 | Fram2: the first crewed flight over the poles |
| 29 Apr 2025 | ESA's BIOMASS, the first P-band radar satellite |
| 28 May 2025 | China's Tianwen-2 launched to the near-Earth asteroid Kamoʻoalewa |
| 30 Jul 2025 | NISAR, the NASA–ISRO dual-band radar satellite, launched |
| Nov 2025 | A crack in Shenzhou-20's window, probably from debris, forces its crew to return in Shenzhou-21; an uncrewed Shenzhou-22 is launched as a replacement |
| 13 Nov 2025 | New Glenn launches NASA's ESCAPADE and lands its booster |
| 2025 | Record 330 orbital launch attempts worldwide; more than 4 500 payloads launched |
| 15 Jan 2026 | Crew-11 returns a month early — the first medical evacuation from the ISS |
| 1–10 Apr 2026 | **Artemis II**: Wiseman, Glover, Koch and Hansen fly around the Moon — the first crewed lunar flight since 1972 |
| 19 May 2026 | ESA–China SMILE magnetosphere mission launched on Vega-C |
| 28 May 2026 | A New Glenn rocket explodes during a static-fire test |
| Jun–Jul 2026 | Tianwen-2 arrives at Kamoʻoalewa |
| 30 Aug 2026 | Nancy Grace Roman Space Telescope launched by Falcon Heavy, about eight months early |
| 28 Sep 2026 | Starship reaches orbit for the first time on Flight 14 and deploys Starlink V3 satellites |
| 28 Sep 2026 | JUICE flies past Earth on its way to Jupiter |
| 8 Oct 2026 | Crew-12 returns after about 235 days on the ISS |

# F.6 Planned Missions (as of October 2026)

The following events were scheduled as of mid-October 2026. They are listed to show where the field is heading; check current sources before relying on any date.

| Planned date | Event |
|---|---|
| 20 Oct 2026 (JST) | JAXA's Martian Moons eXploration (MMX) launch on H3, to return samples of Phobos around 2031 |
| Nov 2026 | ESA's Hera arrives at Didymos and Dimorphos to study the aftermath of DART |
| 18 Nov 2026 | Voyager 1 becomes the first spacecraft one light-day from Earth |
| 21 Nov 2026 | BepiColombo enters orbit around Mercury; its two orbiters separate in December |
| Late 2026 | Gaganyaan G1, India's first uncrewed orbital test of its crew capsule |
| Late 2026 | Starship Flight 15 with a planned ship catch; uncrewed Starliner-1 cargo flight |
| 2027 | Artemis III: crewed rendezvous and docking test with commercial lunar landers in Earth orbit |
| 2027 | Chang'e-7 lunar south-pole mission (postponed from 2026); Blue Moon MK1 lunar lander |
| 2027 | First crewed Gaganyaan flight (ISRO) |
| 2027 | ESCAPADE arrives at Mars; MMX arrives in Mars orbit |
| Nov 2027 | Tianwen-2 sample capsule returns to Earth |
| 2028 | Ramses launched to Apophis (with DESTINY+ on H3); Space Rider first flight |
| Jul 2028 | Dragonfly rotorcraft lander launched to Titan (arrival 2034) |
| 13 Apr 2029 | Asteroid Apophis passes about 32 000 km above the Earth's surface; OSIRIS-APEX and Ramses study it |
| 2029 | Psyche arrives at the metal-rich asteroid 16 Psyche |
| By 2030 | China's planned first crewed lunar landing (Mengzhou spacecraft, Lanyue lander, Long March 10) |
| Apr 2030 | Europa Clipper enters orbit around Jupiter |
| 2030–31 | ISS operations end; the US Deorbit Vehicle steers the station into the South Pacific |
| 2031 | JUICE enters orbit around Jupiter (July); MMX returns Phobos samples |
| 2034 | Dragonfly arrives at Titan |
| ≈ 2035 | ESA's LISA gravitational-wave observatory and NewAthena X-ray observatory (launches planned 2035–37) |

**KEY TAKEAWAYS**

- It took 54 years from Tsiolkovsky's rocket equation (1903) to Sputnik (1957), and only 12 more to a human landing on the Moon.
- Most "firsts" of robotic planetary exploration came between 1959 and 1976; the decades since have brought orbiters, rovers and sample returns.
- Since 2015 reusable rockets and commercial spacecraft have transformed launch rates: 330 orbital launch attempts in 2025, compared with fewer than 100 a year in 2000–2017.
- 2026 brought the first crewed lunar flight in 54 years, the first orbital flight of Starship and the launch of the Roman Space Telescope.

APPENDIX

# G Acronyms and Abbreviations

*The acronyms and abbreviations used in this manual, with their meanings and the chapters in which they are defined or mainly used.*

**IN THIS CHAPTER**


G.1 A
G.2 B
G.3 C
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G.11 K
G.12 L
G.13 M
G.14 N
G.15 O
G.16 P
G.17 Q
G.18 R
G.19 S
G.20 T
G.21 U
G.22 V
G.23 W
G.24 X
G.25 Y
G.26 Z


Space engineering is notoriously rich in acronyms. This list gives every acronym that is spelled out in the chapters, together with common ones used without definition. The last column gives the chapters where each is defined or mainly used (letters denote appendices). Acronyms in other languages are followed by the English meaning where helpful. Units and chemical formulae are not listed; see Appendix A.

## G.1 A

| Acronym | Meaning | Chapters |
|---|---|---|
| **ABFT** | algorithm-based fault tolerance | 45 |
| **ABI** | Advanced Baseline Imager | 16 |
| **ACC** | anomaly correlation coefficient | 47 |
| **ACE** | Advanced Composition Explorer | |
| **ACM** | adaptive coding and modulation | 17, 49 |
| **ACS** | attitude control system | 9 |
| **ACS3** | Advanced Composite Solar Sail System | |
| **ADCS** | attitude determination and control subsystem | 6 |
| **ADR** | active debris removal | 21 |
| **ADRAS-J** | Active Debris Removal by Astroscale-Japan | |
| **ADS-B** | Automatic Dependent Surveillance–Broadcast (aircraft) | |

| Acronym | Meaning | Chapters |
|---|---|---|
| **AEGIS** | Autonomous Exploration for Gathering Increased Science | |
| **AEPS** | Advanced Electric Propulsion System | 42 |
| **AFRL** | Air Force Research Laboratory | 53 |
| **AFSS** | autonomous flight safety system | 52 |
| **AGN** | active galactic nuclei | 51 |
| **AHI** | Advanced Himawari Imager | 16 |
| **AI** | artificial intelligence | 43, 52 |
| **AIAA** | American Institute of Aeronautics and Astronautics | |
| **AICON** | DWD's AI-based global weather model (built with ECMWF's Anemoi software) | |
| **AIFS** | Artificial Intelligence Forecasting System | 16, 47 |
| **AIGEFS** | AI Global Ensemble Forecast System (NOAA) | |
| **AIGFS** | Artificial Intelligence Global Forecast System (NOAA) | |
| **AIS** | Automatic Identification System (ships) | 46 |
| **AIT** | assembly, integration and test | 4 |
| **AMSR2** | Advanced Microwave Scanning Radiometer 2 | |
| **AO** | atomic oxygen | 2 |
| **AOA** | angle of arrival | 19 |
| **AOD** | aerosol optical depth | 13 |
| **AOI** | automated optical inspection | 52 |
| **APCP** | ammonium perchlorate composite propellant | 35 |
| **API** | application programming interface | 14 |
| **APSK** | amplitude-phase shift keying | 7 |
| **AR** | augmented reality | 51 |
| **ARD** | analysis-ready data | 13, 45, 46 |
| **ARED** | Advanced Resistive Exercise Device | 30 |
| **ASAT** | anti-satellite (weapon) | |
| **ASCAT** | Advanced Scatterometer | |
| **ASE** | Autonomous Sciencecraft Experiment | 48 |
| **ASI** | Agenzia Spaziale Italiana (Italian Space Agency) | |
| **ASTER** | Advanced Spaceborne Thermal Emission and Reflection Radiometer | |
| **ATLID** | atmospheric lidar | 16 |
| **ATMS** | Advanced Technology Microwave Sounder | |
| **ATV** | Automated Transfer Vehicle | |
| **AUC** | area under the (ROC) curve | |
| **AVHRR** | Advanced Very High Resolution Radiometer | |

## G.2 B

| Acronym | Meaning | Chapters |
|---|---|---|
| **B/H** | base to the height | 14 |
| **BAS** | Bharatiya Antariksh Station | 32 |
| **BATSE** | Burst and Transient Source Experiment | 39 |

| Acronym | Meaning | Chapters |
|---|---|---|
| **BEAM** | Bigelow Expandable Activity Module | 32 |
| **BLE** | ballistic limit equation | 2 |
| **BLS** | box least squares | 51 |
| **BO** | bayesian optimisation | 52 |
| **BOC** | binary offset carrier | 18 |
| **BOLE** | Booster Obsolescence and Life Extension | 35 |
| **BPSK** | binary phase-shift keying | |
| **BRDF** | bidirectional reflectance distribution function | 13 |
| **BSS** | Broadcasting Satellite Service | 17 |
| **BTU** | British thermal unit | A |
| **BVAD** | Baseline Values and Assumptions Document | 30 |

## G.3 C

| Acronym | Meaning | Chapters |
|---|---|---|
| **C&DH** | command and data handling | 6 |
| **C/A** | coarse/acquisition | 18 |
| **CADRE** | Cooperative Autonomous Distributed Robotic Exploration | |
| **CAPSTONE** | Cislunar Autonomous Positioning System Technology Operations and Navigation Experiment | |
| **CAS** | Chinese Academy of Sciences | |
| **CASC** | China Aerospace Science and Technology Corporation | |
| **CASPER** | Continuous Activity Scheduling Planning Execution and Replanning | |
| **CBM** | Common Berthing Mechanism | 32 |
| **CCD** | charge-coupled device | 38 |
| **CCI** | Climate Change Initiative | 47 |
| **CCM** | constant coding and modulation | 49 |
| **CCSDS** | Consultative Committee for Space Data Systems | 7 |
| **CCW** | Certain Conventional Weapons | 50 |
| **CDM** | conjunction data message | 21, 50 |
| **CDMA** | code division multiple access | 18 |
| **CDR** | Critical Design Review | 4 |
| **CDRA** | Carbon Dioxide Removal Assembly | 30 |
| **CEMS** | Copernicus Emergency Management Service | 46 |
| **CEOS** | Committee on Earth Observation Satellites | 13 |
| **CEOS-ARD** | CEOS Analysis Ready Data | 13 |
| **CER** | cost-estimating relationship | 4 |
| **CERES** | Clouds and the Earth's Radiant Energy System | |
| **CFAR** | constant false alarm rate | 46 |
| **CFDP** | CCSDS File Delivery Protocol | 7, I |
| **CFR** | Code of Federal Regulations (USA) | |
| **CFRP** | carbon-fibre reinforced polymers | 5 |
| **CG** | centre of gravity | 27 |

| Acronym | Meaning | Chapters |
|---|---|---|
| **CGMS** | Coordination Group for Meteorological Satellites | 16 |
| **CGRO** | Compton Gamma Ray Observatory | 39 |
| **CIMON** | Crew Interactive MObile companioN (ISS assistant) | |
| **CLA** | coupled loads analysis | 5 |
| **CLD** | Commercial LEO Destinations | 32 |
| **CLPS** | Commercial Lunar Payload Services | 27 |
| **CM** | command module | 31 |
| **CMB** | cosmic microwave background | 39 |
| **CME** | coronal mass ejection | 22, 47 |
| **CMG** | control-moment gyroscope | 6, 14, 32 |
| **CMOS** | complementary metal–oxide–semiconductor | |
| **CMSA** | China Manned Space Agency | |
| **CNES** | Centre national d'études spatiales (French space agency) | |
| **CNN** | convolutional neural network | 44, 46 |
| **CNSA** | China National Space Administration | |
| **CoDANN** | Concepts of Design Assurance for Neural Networks | 53 |
| **COG** | Cloud-Optimized GeoTIFF | 13, 45 |
| **COMINT** | communications intelligence | |
| **CONFERS** | Consortium for Execution of Rendezvous and Servicing Operations | |
| **COPUOS** | Committee on the Peaceful Uses of Outer Space | 21, 41 |
| **COPV** | composite-overwrapped pressure vessel | 34 |
| **CORS** | continuously operating reference stations | 18 |
| **COSPAR** | Committee on Space Research | |
| **COSTAR** | Corrective Optics Space Telescope Axial Replacement | |
| **COTS** | commercial off-the-shelf | 2 |
| **CPU** | central processing unit | |
| **CRPA** | controlled-reception-pattern antenna | 18 |
| **CRPS** | continuous ranked probability score | |
| **CSA** | Canadian Space Agency | 32 |
| **CSBF** | Columbia Scientific Balloon Facility | 8 |
| **CSDA** | Commercial SmallSat Data Acquisition | 13 |
| **CSI** | critical success index | 47 |
| **CSpO** | Combined Space Operations | 19 |
| **CT** | computed tomography | 52 |
| **CUSUM** | cumulative sum | 50 |
| **CVCM** | collected volatile condensable material | 2 |
| **CW** | Clohessy–Wiltshire | |
| **CZT** | cadmium zinc telluride | 38 |

## G.4 D

| Acronym | Meaning | Chapters |
|---|---|---|
| **D2C** | direct-to-cell | |
| **DA-ASAT** | direct-ascent anti-satellite (weapon) | |
| **DAGGER** | Deep leArninG Geomagnetic pErtuRbation (NASA geomagnetic-disturbance model) | |
| **DAL** | design assurance levels | 53 |
| **DARPA** | Defense Advanced Research Projects Agency | 50 |
| **DART** | Double Asteroid Redirection Test | 26 |
| **DDD** | displacement damage dose | 2 |
| **DEM** | digital elevation model | 13, 46 |
| **DESI** | Dark Energy Spectroscopic Instrument | |
| **DET** | direct energy transfer | 5 |
| **DFMC** | dual-frequency multi-constellation | 18 |
| **DGMR** | Deep Generative Model of Rain (nowcasting) | |
| **DGNSS** | Differential GNSS | 18 |
| **DIAL** | differential absorption lidar | 12 |
| **DInSAR** | Differential InSAR | 15 |
| **DLR** | Deutsches Zentrum für Luft- und Raumfahrt (German Aerospace Center) | |
| **DMIPS** | Dhrystone million instructions per second | |
| **DMSP** | Defense Meteorological Satellite Program | 19 |
| **DNB** | Day/Night Band | 14 |
| **DO-178C** | Software Considerations in Airborne Systems and Equipment Certification (RTCA) | |
| **DoD** | depth of discharge | 5 |
| **DOP** | dilution of precision | 18 |
| **DR** | Decommissioning Review | 4 |
| **DRACO** | Demonstration Rocket for Agile Cislunar Operations | |
| **DRO** | distant retrograde orbit | 3, 24 |
| **DSCOVR** | Deep Space Climate Observatory | |
| **DSI** | Deep Space Industries | 41 |
| **DSM** | digital surface model | 14 |
| **DSN** | Deep Space Network | 7, 24, 49 |
| **DSOC** | Deep Space Optical Communications | |
| **DSP** | Defense Support Program | 20 |
| **DTN** | delay- and disruption-tolerant networking | |
| **DVB-S2** | Digital Video Broadcasting – Satellite, second generation | |
| **DVB-S2X** | DVB-S2 extensions | |

## G.5 E

| Acronym | Meaning | Chapters |
|---|---|---|
| **EASA** | European Union Aviation Safety Agency | |
| **ECC** | error-correcting codes | 45 |

| Acronym | Meaning | Chapters |
|---|---|---|
| **ECEF** | Earth-centred, Earth-fixed | 3 |
| **ECI** | earth-centred inertial | 3 |
| **ECLSS** | environmental control and life support systems | 30 |
| **ECMWF** | European Centre for Medium-Range Weather Forecasts | 16, 47, 53 |
| **ECOSTRESS** | ECOsystem Spaceborne Thermal Radiometer Experiment on Space Station | |
| **ECSS** | European Cooperation for Space Standardization | |
| **ECV** | Essential Climate Variable | 13 |
| **ED-324** | EUROCAE guidance on machine learning in aeronautical systems (with SAE ARP6983) | |
| **EDAC** | error detection and correction | 2, 6 |
| **EDL** | entry, descent and landing | 27 |
| **EDRS** | European Data Relay System | 14, 17 |
| **EEE** | electrical, electronic and electromechanical | 4 |
| **EESS** | Earth exploration-satellite service | |
| **EGMS** | European Ground Motion Service | 15 |
| **EGNOS** | European Geostationary Navigation Overlay Service | |
| **EHF** | extremely high frequency (30–300 GHz) | |
| **EIRP** | effective isotropic radiated power | 7 |
| **ELA** | Ensemble de Lancement Ariane (Ariane launch complex) | |
| **ELINT** | electronic intelligence | |
| **EM** | engineering model | 4 |
| **EMC** | electromagnetic compatibility | |
| **EMIT** | Earth Surface Mineral Dust Source Investigation | |
| **EMU** | Extravehicular Mobility Unit | 33 |
| **ENL** | equivalent number of looks (SAR speckle) | |
| **ENS** | ensemble | 47 |
| **EO** | earth observation | 1, 13, 14, 46 |
| **EO-1** | Earth Observing-1 | |
| **EO/IR** | electro-optical / infrared | |
| **EOL** | end of life | 5 |
| **EOSDIS** | Earth Observing System Data and Information System | 45 |
| **EP** | electric propulsion | 6 |
| **EPFD** | equivalent power-flux-density | 11, 17, D |
| **EPS** | electrical power subsystem | 5 |
| **ERA5** | ECMWF Reanalysis v5 | |
| **ESA** | European Space Agency | 32 |
| **ESA-ADB** | ESA's anomaly-detection benchmark | 49 |
| **ESCAPADE** | Escape and Plasma Acceleration and Dynamics Explorers | |
| **ESOC** | European Space Operations Centre | 49 |
| **ESTRACK** | ESA Tracking Station Network | |
| **EUMETSAT** | European Organisation for the Exploitation of Meteorological Satellites | |
| **EUSPA** | EU Agency for the Space Programme | |
| **EUV** | extreme ultraviolet | |

| Acronym | Meaning | Chapters |
|---|---|---|
| **EVA** | extravehicular activity | |
| **EVE** | Earth Virtual Expert | 44 |
| **EVI** | Enhanced Vegetation Index | 13 |

## G.6 F

| Acronym | Meaning | Chapters |
|---|---|---|
| **FAA** | Federal Aviation Administration | 37 |
| **FAI** | Fédération Aéronautique Internationale | 1 |
| **FCC** | Federal Communications Commission | 1, D |
| **FCI** | Flexible Combined Imager | 16 |
| **FCLS** | fully constrained least squares | 51 |
| **FDIR** | fault detection, isolation and recovery | 6, 48 |
| **FDL** | Frontier Development Lab | 45 |
| **FDOA** | frequency difference of arrival | 19 |
| **FEC** | forward error correction | 7 |
| **FEEP** | field-emission electric propulsion | 42 |
| **FEM** | finite-element method | 5 |
| **FFSC** | full-flow staged combustion | 35 |
| **FGS** | Fine Guidance Sensor | 38 |
| **FGSM** | fast gradient sign method | 53 |
| **FIRMS** | Fire Information for Resource Management System | 14 |
| **FISINT** | foreign instrumentation signals intelligence | |
| **FLOP** | floating-point operations | 44 |
| **FM** | flight model | 4 |
| **FMECA** | failure modes, effects and criticality analysis | 4 |
| **FN** | false negative | 43 |
| **FNO** | fourier neural operator | 44 |
| **FP** | false positive | 43 |
| **FPGA** | field-programmable gate array | 45 |
| **FRB** | fast radio burst | 51 |
| **FRP** | fire radiative power | 14 |
| **FRR** | Flight Readiness Review | 4 |
| **FSS** | Fixed Satellite Service | 17 |
| **FSW** | flight software | |
| **FTS** | fourier-transform spectrometers | 12 |

## G.7 G

| Acronym | Meaning | Chapters |
|---|---|---|
| **G-FOLD** | Guidance for Fuel-Optimal Large Diverts | |
| **GAN** | generative adversarial network | 44 |

| Acronym | Meaning | Chapters |
|---|---|---|
| **GBAS** | ground-based augmentation systems | 18 |
| **GCAT** | General Catalog of Artificial Space Objects (J. McDowell) | |
| **GCOS** | Global Climate Observing System | 13 |
| **GCP** | ground control point | 13 |
| **GCR** | galactic cosmic ray | 2, 30 |
| **GCRF** | Geocentric Celestial Reference Frame | 3 |
| **GEDI** | Global Ecosystem Dynamics Investigation | |
| **GEM** | Graphite Epoxy Motor (solid booster) | |
| **GEO** | Geostationary | 1, 3, 16, 17 |
| **GEODSS** | Ground-based Electro-Optical Deep Space Surveillance | 21 |
| **GEVS** | General Environmental Verification Standard | 2 |
| **GFLOP** | giga ($10^9$) floating-point operations | |
| **GFM** | Global Flood Monitoring | 46 |
| **GFS** | Global Forecast System (NOAA) | |
| **GIC** | geomagnetically induced current | 22 |
| **GIQE** | General Image Quality Equation | 13 |
| **GLM** | Geostationary Lightning Mapper | 16 |
| **GLONASS** | Globalnaya Navigatsionnaya Sputnikovaya Sistema (Russian GNSS) | |
| **GNC** | guidance, navigation and control | 1, 6 |
| **GNN** | graph neural network | 44 |
| **GNSS** | global navigation satellite systems | 18 |
| **GNSS-R** | GNSS reflectometry | 12 |
| **GOCE** | Gravity field and steady-state Ocean Circulation Explorer | |
| **GOES** | Geostationary Operational Environmental Satellite | |
| **GOSAT** | Greenhouse gases Observing Satellite (Japan) | |
| **GP** | gaussian process | 43, 52 |
| **GPM** | Global Precipitation Measurement | 16 |
| **GPS** | Global Positioning System | |
| **GPU** | graphics processing unit | |
| **GRACE** | Gravity Recovery and Climate Experiment | |
| **GRACE-FO** | GRACE Follow-On | |
| **GRAIL** | Gravity Recovery and Interior Laboratory | |
| **GRD** | Ground Range Detected | 46 |
| **GRU** | gated recurrent unit | 44 |
| **GSaaS** | Ground-station-as-a-service | 49 |
| **GSD** | ground sampling distance | 11, 12, 13 |
| **GSLV** | Geosynchronous Satellite Launch Vehicle | |
| **GSN** | Goal Structuring Notation | 53 |
| **GSSAP** | Geosynchronous Space Situational Awareness Program | 19, 21 |
| **GTO** | geostationary transfer orbit | 34, 36 |

## G.8 H

| Acronym | Meaning | Chapters |
|---|---|---|
| **HALEU** | high-assay low-enriched uranium | 42 |
| **HALO** | Habitation and Logistics Outpost | 32 |
| **HAND** | height above nearest drainage | 46 |
| **HAPS** | high-altitude platform stations | 1, 8 |
| **HAS** | High Accuracy Service | 18 |
| **HASDM** | High Accuracy Satellite Drag Model | 50 |
| **HBTSS** | Hypersonic and Ballistic Tracking Space Sensor | 20 |
| **HDRM** | hold-down and release mechanism | 5 |
| **HEK** | Heliophysics Event Knowledgebase | 51 |
| **HEO** | highly elliptical orbits | 1, 17, 39 |
| **HGA** | high-gain antenna | 7 |
| **HGEFS** | Hybrid Global Ensemble Forecast System (NOAA) | |
| **HGV** | hypersonic glide vehicle | 20 |
| **HiRISE** | High Resolution Imaging Science Experiment | 25 |
| **HLS** | Harmonized Landsat and Sentinel-2 | 14 |
| **HMI** | Helioseismic and Magnetic Imager | 47 |
| **HPE** | Hewlett Packard Enterprise | |
| **HPSC** | High-Performance Spaceflight Computing | 6, 45 |
| **HRES** | ECMWF high-resolution deterministic forecast | |
| **HSS** | heidke skill score | 47 |
| **HTPB** | hydroxyl-terminated polybutadiene (solid-propellant binder) | |
| **HTS** | high-throughput satellites | 17 |
| **HTV-X** | H-II Transfer Vehicle X | |
| **HWO** | Habitable Worlds Observatory | 38, 39 |

## G.9 I

| Acronym | Meaning | Chapters |
|---|---|---|
| **IADC** | Inter-Agency Space Debris Coordination Committee | |
| **IASI** | Infrared Atmospheric Sounding Interferometer | |
| **IAU** | International Astronomical Union | |
| **IBP** | interval bound propagation | 53 |
| **ICAO** | International Civil Aviation Organization | |
| **ICBM** | intercontinental ballistic missile | |
| **IDA** | International Docking Adapter | 32 |
| **IDSS** | International Docking System Standard | 31 |
| **IEEE** | Institute of Electrical and Electronics Engineers | |
| **IGRF** | International Geomagnetic Reference Field | 23 |
| **IGS** | Information Gathering Satellites | 19 |
| **IKAROS** | Interplanetary Kite-craft Accelerated by Radiation Of the Sun | |

| Acronym | Meaning | Chapters |
|---|---|---|
| **ILLUMA-T** | Integrated LCRD Low Earth Orbit User Modem and Amplifier Terminal | |
| **ILRS** | International Lunar Research Station | 1 |
| **IMAP** | Interstellar Mapping and Acceleration Probe | |
| **IME** | integrated mass enhancement | 46 |
| **IMS** | Inductive Monitoring System | 49 |
| **IMU** | inertial measurement unit | 34 |
| **INCO** | Instrumentation and Communications Officer (Mission Control) | |
| **INCOSE** | International Council on Systems Engineering | 4 |
| **IOS** | in-orbit servicing | 40 |
| **IoT** | Internet of Things | 17 |
| **IPCC** | Intergovernmental Panel on Climate Change | 16 |
| **IPO** | initial public offering | 52 |
| **ISAM** | in-space servicing, assembly and manufacturing | 40 |
| **ISLE** | in-suit light exercise | 33 |
| **ISO** | International Organization for Standardization | |
| **ISO/IEC** | International Organization for Standardization / International Electrotechnical Commission | |
| **ISR** | intelligence, surveillance and reconnaissance | 1, 8, 19 |
| **ISRO** | Indian Space Research Organisation | |
| **ISRU** | in situ resource utilisation | |
| **ISS** | International Space Station | 30, 31, 32, 33 … |
| **ITRF** | International Terrestrial Reference Frame | 3 |
| **ITU** | International Telecommunication Union | 7, 17, D |
| **ITU-R** | ITU Radiocommunication Sector | |
| **IV&V** | independent verification and validation | 4, 6, 53 |
| **IW** | Interferometric Wide-swath | 15 |
| **IXPE** | Imaging X-ray Polarimetry Explorer | |

## G.10 J

| Acronym | Meaning | Chapters |
|---|---|---|
| **J2000** | Standard astronomical epoch, 12:00 TT on 1 January 2000 | |
| **JAXA** | Japan Aerospace Exploration Agency | |
| **JB2008** | Jacchia–Bowman 2008 atmospheric density model | |
| **JD** | Julian Date | A |
| **JPL** | Jet Propulsion Laboratory (NASA/Caltech) | |
| **JPSS** | Joint Polar Satellite System | |
| **JUICE** | JUpiter ICy moons Explorer | |
| **JWST** | James Webb Space Telescope | 38, 51 |

## G.11 K

| Acronym | Meaning | Chapters |
|---|---|---|
| **KARI** | Korea Aerospace Research Institute | |
| **KaRIn** | ka-band radar interferometer | 16 |

## G.12 L

| Acronym | Meaning | Chapters |
|---|---|---|
| **L/D** | lift-to-drag ratio | |
| **LADEE** | Lunar Atmosphere and Dust Environment Explorer | |
| **LC** | Launch Complex | |
| **LCL** | latching current limiter | 5 |
| **LCRD** | Laser Communications Relay Demonstration | 17 |
| **LCROSS** | Lunar Crater Observation and Sensing Satellite | |
| **LCVG** | liquid cooling and ventilation garment | 33 |
| **LDB** | long-duration balloon | 8 |
| **LDPC** | low-density parity-check | 7 |
| **LDSD** | Low-Density Supersonic Decelerator | 8 |
| **LEO** | low Earth orbit | 1, 3, 16, 17 … |
| **LEOP** | launch and early orbit phase | 11 |
| **LET** | linear energy transfer | 2 |
| **LGA** | low-gain antenna | 7 |
| **LI** | Lightning Imager | 16 |
| **LIGO** | Laser Interferometer Gravitational-Wave Observatory | |
| **LISA** | Laser Interferometer Space Antenna | 39 |
| **LLM** | large language model | 44, 49 |
| **LLO** | low lunar orbit | 41 |
| **LM** | Lunar Module | 27, 31 |
| **LMI** | Lightning Mapping Imager | 16 |
| **LOC** | loss of crew | 31 |
| **LOLA** | Lunar Orbiter Laser Altimeter | 12 |
| **LORRI** | Long Range Reconnaissance Imager (New Horizons) | |
| **LOX** | liquid oxygen | 37 |
| **LQR** | linear-quadratic regulators | 6 |
| **LRO** | Lunar Reconnaissance Orbiter | 25, 41 |
| **LRR** | Launch Readiness Review | 4 |
| **LRV** | Lunar Roving Vehicle | 28, 33 |
| **LSST** | Legacy Survey of Space and Time | 51 |
| **LST** | land-surface temperature | 14 |
| **LSTM** | long short-term memory | 44, 49 |
| **LTE** | Long-Term Evolution (4G mobile standard) | |
| **LuGRE** | Lunar GNSS Receiver Experiment | 18 |

| Acronym | Meaning | Chapters |
|---|---|---|
| **LVM3** | Launch Vehicle Mark 3 | |
| **LVS** | Lander Vision System | 48 |
| **LWIR** | long-wave infrared | |

## G.13 M

| Acronym | Meaning | Chapters |
|---|---|---|
| **MAE** | mean absolute error | 43 |
| **MARS** | Methane Alert and Response System | 46 |
| **MASINT** | measurement and signature intelligence | 19 |
| **MAST** | Mikulski Archive for Space Telescopes | 38 |
| **MAVEN** | Mars Atmosphere and Volatile Evolution | |
| **MBSE** | model-based systems engineering | 4 |
| **MC/DC** | modified condition/decision coverage | |
| **MCR** | Mission Concept Review | 4 |
| **MCS** | mission control system | 49 |
| **MDO** | multidisciplinary design optimisation | 52 |
| **MECO** | main-engine cut-off | 34 |
| **MEMS** | microelectromechanical systems | |
| **MEO** | Medium Earth orbit | 1, 3, 17 |
| **MEP** | Mission Extension Pod | 40 |
| **MESSENGER** | MErcury Surface, Space ENvironment, GEochemistry and Ranging | |
| **METOX** | metal oxide | 33 |
| **MEV** | Mission Extension Vehicle | 40 |
| **MILP** | mixed-integer linear programming | 48, 49, 53 |
| **MIPS** | million instructions per second | 48 |
| **MIRI** | Mid-Infrared Instrument (JWST) | |
| **MISSE** | Materials International Space Station Experiment | 2 |
| **ML** | machine learning | 43, 50, 51 |
| **MLI** | multi-layer insulation | 5 |
| **MMH** | monomethylhydrazine | |
| **MMOD** | micrometeoroids and orbital debris | |
| **MMRTG** | multi-mission radioisotope thermoelectric generator | 28 |
| **MMS** | Magnetospheric Multiscale | |
| **MMU** | Manned Manoeuvring Unit | 33 |
| **MMX** | Martian Moons eXploration | |
| **MOC** | mission operations centre | 7 |
| **MODIS** | Moderate Resolution Imaging Spectroradiometer | |
| **MOLA** | Mars Orbiter Laser Altimeter | 12 |
| **MOXIE** | Mars Oxygen In-Situ Resource Utilization Experiment | |
| **MPC** | model predictive control | 6 |
| **MPD** | magnetoplasmadynamic (thruster) | |

| Acronym | Meaning | Chapters |
|---|---|---|
| **MPE** | maximum predicted environment | |
| **MRE** | molten regolith electrolysis | 41 |
| **MRL** | Manufacturing Readiness Levels | 4 |
| **MRO** | Mars Reconnaissance Orbiter | |
| **MRV** | Mission Robotic Vehicle | 40 |
| **MSI** | MultiSpectral Instrument (Sentinel-2) | |
| **MSR** | Mars Sample Return | 29 |
| **MSS** | Mobile Satellite Service | 17 |
| **MTF** | modulation transfer function | 12, 13 |
| **MTG** | Meteosat Third Generation | |
| **MTTF** | mean time to failure | |
| **MUOS** | Mobile User Objective System | |
| **MWIR** | mid-wave infrared | |

## G.14 N

| Acronym | Meaning | Chapters |
|---|---|---|
| **NAC** | Narrow Angle Camera | 51 |
| **NASA** | National Aeronautics and Space Administration (USA) | |
| **NBL** | Neutral Buoyancy Laboratory | 33 |
| **NBR** | Normalised Burn Ratio | 13 |
| **NDBI** | Normalised Difference Built-up Index | 13 |
| **NDMI** | Normalised Difference Moisture Index | 13 |
| **NDSI** | Normalised Difference Snow Index | 13 |
| **NDVI** | Normalised Difference Vegetation Index | 13 |
| **NDWI** | Normalised Difference Water Index | 13 |
| **NEA** | near-Earth asteroid | 41 |
| **NEO** | near-Earth object | 26 |
| **NEP** | nuclear electric propulsion | 42 |
| **NERVA** | Nuclear Engine for Rocket Vehicle Application | |
| **NESZ** | noise-equivalent sigma-zero | 15 |
| **NET** | no earlier than | |
| **NGA** | National Geospatial-Intelligence Agency | 50 |
| **NIIRS** | National Imagery Interpretability Rating Scale | 13 |
| **NIR** | near-infrared | |
| **NISAR** | NASA–ISRO Synthetic Aperture Radar | |
| **NIST** | National Institute of Standards and Technology (USA) | 53 |
| **NMF** | non-negative matrix factorisation | 51 |
| **NOAA** | National Oceanic and Atmospheric Administration (USA) | 47 |
| **NPA** | Notice of Proposed Amendment | 53 |
| **NPP** | (Suomi) National Polar-orbiting Partnership | |
| **NPR** | NASA Procedural Requirements | I |

| Acronym | Meaning | Chapters |
|---|---|---|
| **NPSH** | net positive suction head | 35 |
| **NRHO** | near-rectilinear halo orbit | 3, 24, 40 |
| **NRLMSIS** | Naval Research Laboratory Mass Spectrometer and Incoherent Scatter radar model | |
| **NRO** | National Reconnaissance Office | 19, 50 |
| **NSSL** | National Security Space Launch | 19 |
| **NTN** | non-terrestrial networks | 7, 17, 49 |
| **NTO** | nitrogen tetroxide | |
| **NTR** | nuclear thermal rocket | 42 |
| **NWP** | numerical weather prediction | 16, 47 |

## G.15 O

| Acronym | Meaning | Chapters |
|---|---|---|
| **O/F** | oxidiser-to-fuel mixture ratio | |
| **OBC** | on-board computer | |
| **OBDH** | on-board data handling | 6 |
| **OBIA** | object-based image analysis | 13 |
| **OCI** | ocean-colour instrument | 16 |
| **OCO-2** | Orbiting Carbon Observatory-2 | |
| **OD** | orbit determination | 3 |
| **ODD** | operational design domain | 53 |
| **OLCI** | Ocean and Land Colour Instrument (Sentinel-3) | |
| **OLI** | Operational Land Imager (Landsat) | |
| **OMM** | Orbit Mean-elements Message | 3 |
| **OPIR** | Overhead Persistent Infrared | |
| **OPS-SAT** | ESA's flying laboratory satellite for operations experiments (2019–24) | |
| **ORR** | Operational Readiness Review | 4 |
| **OSAM** | on-orbit servicing, assembly and manufacturing | |
| **OSIRIS-REx** | Origins, Spectral Interpretation, Resource Identification, Security–Regolith Explorer | |
| **OSNMA** | Open Service Navigation Message Authentication | 18 |
| **OTV** | orbital transfer vehicle | 11 |

## G.16 P

| Acronym | Meaning | Chapters |
|---|---|---|
| **PA** | product assurance | 4 |
| **PACE** | Plankton, Aerosol, Cloud, ocean Ecosystem | |
| **PCA** | principal component analysis | 43 |
| **PCEC** | Project Cost Estimating Capability | 4 |
| **PDGS** | payload data ground segment | 7 |
| **PDOP** | position dilution of precision | |
| **PDR** | Preliminary Design Review | 4 |

| Acronym | Meaning | Chapters |
|---|---|---|
| **PDU** | power distribution unit | 5 |
| **PGD** | projected gradient descent | 53 |
| **PHM** | prognostics and health management | 49 |
| **PICA** | phenolic impregnated carbon ablator | |
| **PINN** | physics-informed neural network | 44 |
| **PIXL** | Planetary Instrument for X-ray Lithochemistry (Perseverance) | |
| **PLAR** | Post-Launch Assessment Review | 4 |
| **PLATO** | PLAnetary Transits and Oscillations of stars | |
| **PLSS** | portable life support system | 33 |
| **PMD** | propellant management device | 6 |
| **PNP** | probability of no penetration | 2 |
| **PnP** | Perspective-n-Point | 48 |
| **PNT** | positioning, navigation and timing | 1, 18 |
| **POD** | probability-of-detection | 52 |
| **POGO** | longitudinal vibration from coupling of structure and propulsion (not an acronym) | |
| **PPE** | Power and Propulsion Element | 32, 42 |
| **PPP** | precise point positioning | 18 |
| **PRA** | probabilistic risk assessment | 31 |
| **PRF** | pulse repetition frequency | 15 |
| **PRN** | pseudo-random noise | 18 |
| **PSI** | persistent scatterer interferometry | 15, 45 |
| **PSLV** | Polar Satellite Launch Vehicle | |
| **PSR** | permanently shadowed region | 41 |
| **PTQ** | post-training quantisation | 45 |
| **PUE** | power usage effectiveness | 53 |
| **PUNCH** | Polarimeter to Unify the Corona and Heliosphere | |
| **PWSA** | Proliferated Warfighter Space Architecture | 19 |

## G.17 Q

| Acronym | Meaning | Chapters |
|---|---|---|
| **QAT** | quantisation-aware training | 45 |
| **QE** | quantum efficiency | 38 |
| **QM** | qualification model | 4 |
| **QPSK** | quadrature phase-shift keying | |
| **QWIP** | quantum-well infrared photodetector | 14 |
| **QZSS** | Quasi-Zenith Satellite System (Japan) | |

## G.18 R

| Acronym | Meaning | Chapters |
|---|---|---|
| **R-CNN** | region-based convolutional neural network | |

| Acronym | Meaning | Chapters |
|---|---|---|
| **RAAN** | right ascension of the ascending node | 3 |
| **RAG** | retrieval-augmented generation | 44, 49, 52 |
| **RANSAC** | random sample consensus | |
| **RCA** | rapid cycle amine | 33 |
| **RCC** | reinforced carbon-carbon | 10 |
| **RDE** | rotating detonation engine | 42 |
| **RDM** | radiation design margin | 2 |
| **RDRE** | rotating detonation rocket engine | 42 |
| **REE** | rare-earth elements | 41 |
| **ReLU** | rectified linear unit | 44 |
| **RF** | radio-frequency | 1 |
| **RHU** | radioisotope heater unit | 5, 28 |
| **RL** | reinforcement learning | 43, 48 |
| **RLHF** | reinforcement learning from human feedback | 44 |
| **RLV** | reusable launch vehicle | |
| **RMSE** | root-mean-square error | |
| **RNN** | recurrent neural network | 44 |
| **RO** | radio occultation | 12, 16 |
| **ROC** | receiver operating characteristic | 43, 46 |
| **RORSAT** | radar ocean-reconnaissance satellite | 42 |
| **ROS** | Russian Orbital Station | 32 |
| **ROSA** | Roll-Out Solar Array | 5 |
| **ROSES** | Research Opportunities in Space and Earth Sciences | 9 |
| **RP-1** | Rocket Propellant-1 (refined kerosene) | |
| **RPC** | rational polynomial coefficient | 13 |
| **RPO** | rendezvous and proximity operations | 40 |
| **RPS** | radioisotope power systems | 5 |
| **RRM** | Robotic Refuelling Mission | 40 |
| **RSGS** | Robotic Servicing of Geosynchronous Satellites | 40 |
| **RTG** | radioisotope thermoelectric generator | 5, 24, 26 |
| **RTK** | real-time kinematic | 18 |
| **RTLS** | return to launch site | 34 |
| **RUL** | remaining useful life | 49 |

## G.19 S

| Acronym | Meaning | Chapters |
|---|---|---|
| **SAA** | South Atlantic Anomaly | 2 |
| **SADA** | solar array drive assembly | 5 |
| **SAFER** | Simplified Aid For EVA Rescue | 33 |
| **SANS** | spaceflight-associated neuro-ocular syndrome | 30 |
| **SAR** | synthetic aperture radar | 12, 15 |

| Acronym | Meaning | Chapters |
|---|---|---|
| **SATCAT** | Satellite catalogue | |
| **SATCOM** | satellite communications | 19 |
| **SBAS** | small baseline subset | 15, 18 |
| **SBI** | simulation-based inference | 51 |
| **SBIRS** | Space-Based Infrared System | |
| **SDA** | space domain awareness | 11, 19, 21 |
| **SDC** | silent data corruption | 45 |
| **SDO** | Solar Dynamics Observatory | 47, 51 |
| **SEC-DED** | single-error correction, double-error detection | |
| **SECO** | second-engine cut-off | 34 |
| **SEE** | single-event effect | 2 |
| **SEFI** | single-event functional interrupt | 2, 45 |
| **SEL** | single-event latch-up | 2, 45 |
| **SEP** | solar energetic particle | 22 |
| **SEU** | single-event upset | 2, 45 |
| **SGD** | stochastic gradient descent | 43, 44 |
| **SGP4** | Simplified General Perturbations model 4 | |
| **SIGINT** | signals intelligence | 19 |
| **SIR** | System Integration Review | 4 |
| **SLC** | Space Launch Complex | |
| **SLIM** | Smart Lander for Investigating Moon | |
| **SLS** | Space Launch System | 34, 36 |
| **SLSTR** | Sea and Land Surface Temperature Radiometer (Sentinel-3) | |
| **SMAP** | Soil Moisture Active Passive | |
| **SMILE** | Solar wind Magnetosphere Ionosphere Link Explorer | |
| **SMOS** | Soil Moisture and Ocean Salinity | |
| **SMT** | satisfiability-modulo-theories | 53 |
| **SNR** | signal-to-noise ratio | 12, 13, 38, 49 |
| **SOHO** | Solar and Heliospheric Observatory | |
| **SOI** | sphere of influence | 3 |
| **SOLAR-1** | NOAA space-weather observatory at the Sun–Earth L1 point (formerly SWFO-L1) | |
| **SPB** | super-pressure balloon | 8 |
| **SPE** | solar particle event | 2, 30 |
| **SPF** | single points of failure | 4 |
| **SPICE** | Spacecraft, Planet, Instrument, C-matrix, Events (NAIF toolkit) | |
| **SPIKE** | Hubble Space Telescope long-range scheduling system | |
| **SR-1** | Space Reactor-1 | 42 |
| **SRP** | solar radiation pressure | 3 |
| **SRR** | System Requirements Review | 4 |
| **SRS** | shock response spectrum | 2 |
| **SRTM** | Shuttle Radar Topography Mission | 15 |
| **SSA** | space situational awareness | 21, 50 |

| Acronym | Meaning | Chapters |
|---|---|---|
| **SSLV** | Small Satellite Launch Vehicle | |
| **SSME** | Space Shuttle main engine | 52 |
| **SSO** | sun-synchronous orbits | 3 |
| **SST** | sea-surface temperature | 16, 21 |
| **SSTO** | Single-stage-to-orbit | 10 |
| **ST5** | Space Technology 5 | 52 |
| **STIS** | Space Telescope Imaging Spectrograph (Hubble) | |
| **STM** | structural-thermal model | 4 |
| **STScI** | Space Telescope Science Institute | 39 |
| **SVM** | support vector machine | 43 |
| **SVOM** | Space Variable Objects Monitor | |
| **SWFO-L1** | Space Weather Follow On–Lagrange 1 | |
| **SWIR** | short-wave infrared | |
| **SWME** | spacesuit water membrane evaporator | 33 |
| **SWOT** | Surface Water and Ocean Topography (mission) | |
| **SWPC** | Space Weather Prediction Center | 47 |

## G.20 T

| Acronym | Meaning | Chapters |
|---|---|---|
| **T/W** | thrust-to-weight ratio | |
| **TAEM** | Terminal Area Energy Management | 10 |
| **TAG** | touch-and-go | 29 |
| **TAI** | International Atomic Time | |
| **TBIRD** | TeraByte InfraRed Delivery | |
| **TCE** | threshold-crossing events” | 51 |
| **TCN** | temporal convolutional network | 44 |
| **TCS** | thermal control subsystem | 5 |
| **TDB** | Barycentric Dynamical Time | |
| **TDI** | time-delay integration | 12, 14 |
| **TDOA** | time difference of arrival | 19 |
| **TDRS** | Tracking and Data Relay Satellites | 32 |
| **TDRSS** | Tracking and Data Relay Satellite System | 17 |
| **TEC** | total electron content | 18, 47 |
| **TEMPO** | Tropospheric Emissions: Monitoring of Pollution | |
| **TESS** | Transiting Exoplanet Survey Satellite | 39 |
| **TGO** | Trace Gas Orbiter (ExoMars) | |
| **THEMIS** | Time History of Events and Macroscale Interactions during Substorms | |
| **TID** | total ionising dose | 2, 45 |
| **TIR** | thermal infrared | |
| **TIRS** | Thermal Infrared Sensor (Landsat) | |
| **TMA** | three-mirror anastigmat | 12, 38 |

| Acronym | Meaning | Chapters |
|---|---|---|
| **TMG** | thermal micrometeoroid garment | 33 |
| **TML** | total mass loss | 2 |
| **TMR** | triple modular redundancy | 2, 45 |
| **TN** | true negative | 43 |
| **TOA** | top-of-atmosphere | 13 |
| **TOMS** | Total Ozone Mapping Spectrometer | 16 |
| **TOPS** | tera ($10^{12}$) operations per second | |
| **TP** | true positive | 43 |
| **TPS** | thermal protection system | 10 |
| **TPU** | tensor processing unit | 45 |
| **TraCSS** | Traffic Coordination System for Space | 21 |
| **TRL** | Technology Readiness Level | 4, 42 |
| **TRMM** | Tropical Rainfall Measuring Mission | 16 |
| **TRN** | terrain-relative navigation | 27, 48 |
| **TROPOMI** | Tropospheric Monitoring Instrument (Sentinel-5P) | |
| **TSS** | true skill statistic | |
| **TT&C** | telemetry, tracking and command | 1, 5, D |
| **TVC** | thrust vector control | 34 |
| **TWTA** | travelling-wave-tube amplifier | 17 |

## G.21 U

| Acronym | Meaning | Chapters |
|---|---|---|
| **UAE** | United Arab Emirates | |
| **UDMH** | unsymmetrical dimethylhydrazine | |
| **UERE** | user equivalent range error | 18 |
| **UHF** | ultra high frequency (300 MHz–3 GHz) | |
| **ULA** | United Launch Alliance | 36 |
| **ULE** | ultra-low expansion (glass) | |
| **USAF** | United States Air Force | |
| **USDV** | US Deorbit Vehicle | 32 |
| **USGS** | United States Geological Survey | |
| **USSR** | Union of Soviet Socialist Republics | |
| **UT1** | Universal Time based on the Earth's rotation | |
| **UTC** | Coordinated Universal Time | |
| **UTM** | Universal Transverse Mercator | 13 |

## G.22 V

| Acronym | Meaning | Chapters |
|---|---|---|
| **VAE** | variational autoencoder | |
| **VASIMR** | Variable Specific Impulse Magnetoplasma Rocket | 42 |

| Acronym | Meaning | Chapters |
|---|---|---|
| **VCM** | variable coding and modulation | 7 |
| **VHF** | very high frequency (30–300 MHz) | |
| **VHR** | very high resolution | 14 |
| **VIIRS** | Visible Infrared Imaging Radiometer Suite | 14 |
| **VIPER** | Volatiles Investigating Polar Exploration Rover | |
| **ViT** | vision transformer | 44 |
| **VLEO** | very low Earth orbit | 2, 14 |
| **VNIR** | visible and near-infrared | |
| **VSAT** | very small aperture terminal | |

## G.23 W

| Acronym | Meaning | Chapters |
|---|---|---|
| **WAAS** | Wide Area Augmentation System | |
| **WGS** | Wideband Global SATCOM; also World Geodetic System | |
| **WISE** | Wide-field Infrared Survey Explorer | |
| **WMAP** | Wilkinson Microwave Anisotropy Probe | |
| **WMO** | World Meteorological Organization | 16 |
| **WRC** | World Radiocommunication Conference | D |

## G.24 X

| Acronym | Meaning | Chapters |
|---|---|---|
| **XMM** | X-ray Multi-Mirror (XMM-Newton) | |
| **XRISM** | X-Ray Imaging and Spectroscopy Mission | |

## G.25 Y

| Acronym | Meaning | Chapters |
|---|---|---|
| **YAG** | yttrium aluminium garnet (laser crystal) | |
| **YOLO** | “You Only Look Once” (single-stage object detector) | |

## G.26 Z

| Acronym | Meaning | Chapters |
|---|---|---|
| **ZBLAN** | heavy-metal fluoride glass ($ZrF_4$–$BaF_2$–$LaF_3$–$AlF_3$–NaF) | |

APPENDIX

# H Glossary

*Plain-language definitions of about 300 technical terms used in this manual, from ablator to zenith, covering orbital mechanics, spacecraft engineering, launch, remote sensing, communications, the space environment, exploration, human spaceflight and artificial intelligence, with pointers to the chapters where each is explained.*

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H.19 U–Z

Terms in *italics* within a definition have their own entry. The number in brackets is the chapter where the term is explained most fully; letters refer to appendices. Acronyms are listed in Appendix G.

## H.1 A

**Ablator.** A heat-shield material that protects a vehicle during atmospheric entry by charring, melting or vaporising, carrying heat away with the lost material. Examples are PICA and Avcoat (10, 27).

**Absorptance (α).** The fraction of incident radiation, usually sunlight, that a surface absorbs. With *emittance* it sets a surface's equilibrium temperature (5).

**Active debris removal.** Deliberately capturing and de-orbiting or re-orbiting a defunct satellite or rocket body with another spacecraft (21, 40).

**Adaptive coding and modulation.** Changing a link's modulation and error-correcting code from moment to moment to match the received signal quality, so throughput stays high in clear weather and the link survives rain fades (7).

**Aerobraking.** Using repeated passes through the upper atmosphere of a planet to lower an orbit gradually, saving propellant (25).

**Aerocapture.** Using a single deep pass through an atmosphere to slow a spacecraft from a hyperbolic approach into orbit (25).

**Airy disc.** The bright central spot of the diffraction pattern produced by a circular aperture; its angular radius, $1.22\,\lambda/D$, sets the resolution limit of a telescope (14, 38).

**Albedo.** The fraction of incoming sunlight that a body or surface reflects. Earth's average is about 0.3 (2, 13).

**Analysis-ready data.** Satellite data already corrected and formatted (for example to surface reflectance on a standard grid) so that users can analyse it without further processing (13, 46).

**Anomaly.** (1) In orbital mechanics, an angle giving a body's position along its orbit — true, eccentric or mean anomaly (3). (2) In operations, any unexpected behaviour of a spacecraft (49).

**Antenna gain.** How strongly an antenna concentrates power in its main direction compared with an isotropic antenna, usually in dBi (7).

**Aperture synthesis.** Combining signals from separated antennas or positions to obtain the resolution of a much larger aperture — the principle behind *synthetic aperture radar* and radio interferometers (15).

**Apoapsis / periapsis.** The farthest and nearest points of an orbit from the central body; called apogee and perigee for Earth, aphelion and perihelion for the Sun (3).

**Argument of periapsis (ω).** The angle, in the orbital plane, from the *ascending node* to periapsis (3).

**Artemis Accords.** Non-binding principles for civil exploration of the Moon and beyond, led by the United States; 76 nations had signed by late September 2026 (41).

**Artificial intelligence (AI).** Computer systems that perform tasks normally requiring human intelligence, such as perception, planning or language. Most current space applications use *machine learning* (43).

**Ascending node.** The point where an orbit crosses the equatorial plane going north. Its longitude is the *right ascension of the ascending node* (3).

**Astronomical unit (au).** The mean Earth–Sun distance, fixed at exactly 149 597 870 700 m (A).

**Atomic oxygen.** Single oxygen atoms, the main constituent of the atmosphere at 200–600 km, which erode polymers and some coatings on spacecraft (2).

**Attitude.** The orientation of a spacecraft in space. The attitude determination and control subsystem measures and controls it (6).

**Autonomy.** The ability of a spacecraft or ground system to make decisions and act without human intervention. ECSS defines levels E1 (ground control) to E4 (goal-oriented on-board operations) (48).

## H.2 B

**Ballistic coefficient (β).** Mass divided by the product of drag coefficient and reference area, $m/(C_D A)$. A high ballistic coefficient means the vehicle slows down less in the atmosphere (10, 21).

**Ballistic missile.** A missile that is powered only during the first part of its flight and then follows a free-fall trajectory, often reaching space (20).

**Bandwidth.** The range of frequencies a signal occupies or a system can pass, in hertz. It limits data rate and radar range resolution (7, 15).

**Barycentre.** The centre of mass of two or more bodies, around which they both orbit (3).

**Beamforming.** Steering and shaping an antenna's beam electronically by adjusting the phase and amplitude of many elements of a *phased array* (17).

**Bipropellant.** A rocket propellant system using separate fuel and oxidiser, such as hydrazine and nitrogen tetroxide or methane and liquid oxygen (6, 35).

**Blackbody.** An ideal object that absorbs all radiation falling on it and emits radiation according to *Planck's law* (13).

**Body-mounted array.** Solar cells fixed to a spacecraft's outer surfaces rather than on deployable wings (5).

**Boil-off.** The loss of cryogenic propellant by evaporation due to heat leaking into a tank (35, 40).

**Bond albedo.** The fraction of total incident solar power reflected by a planet in all directions and at all wavelengths (C).

**Bus (satellite bus).** The part of a satellite that provides structure, power, thermal control, attitude control, propulsion and communications to support the *payload* (12).

## H.3 C

**C3 (characteristic energy).** Twice the specific orbital energy of a departure hyperbola, equal to the square of the *hyperbolic excess speed*; a launch vehicle's capability is quoted as payload mass versus C3 (24).

**Cadence (revisit).** How often a satellite or constellation can observe the same place (11, 14).

**Capsule.** A blunt, wingless re-entry vehicle, such as Soyuz, Dragon or Orion (31).

**Catalogue (space object catalogue).** The database of tracked objects in orbit with their orbital elements, maintained by the US Space Force and others (21).

**Characteristic velocity (c*).** A measure of combustion performance in a rocket chamber: chamber pressure times throat area divided by mass flow rate (35).

**Chemical propulsion.** Rocket propulsion in which energy comes from chemical reaction of the propellants (35).

**Clean room.** A room in which airborne particles are controlled to protect sensitive hardware, classified by particle count (ISO class) (4).

**Coherent.** Describing waves that keep a fixed phase relationship. Radar and interferometric processing depend on coherent signals (15).

**Collision probability (Pc).** The estimated probability that two objects will collide during a close approach, used to decide whether to manoeuvre (21).

**Commissioning.** The period after launch in which a spacecraft's systems and instruments are switched on, checked and calibrated before routine operations (4).

**Conjunction.** A predicted close approach between two space objects (21).

**Constellation.** A group of satellites working together, arranged in planes and phases to provide coverage (11).

**Convolutional neural network (CNN).** A *neural network* that applies learned filters across an image; the standard tool for image classification and detection (44).

**Coronagraph.** An instrument that blocks the light of a star inside a telescope so faint nearby objects, such as planets, can be seen (38).

**Coronal mass ejection (CME).** A huge cloud of magnetised plasma thrown out from the Sun, which can cause a *geomagnetic storm* when it reaches Earth (22).

**Cosmic rays.** Very energetic charged particles, mostly protons and heavier nuclei, from outside the Solar System (*galactic cosmic rays*) or from the Sun (2, 22).

**Cross-range.** How far sideways from its original ground track a re-entry vehicle can steer, which depends on its *lift-to-drag ratio* (10).

**Cryocooler.** A mechanical refrigerator that cools detectors or instruments to cryogenic temperatures, as on JWST's MIRI instrument (38).

**Cryogenic propellant.** A propellant stored as a very cold liquid — liquid hydrogen (20 K), liquid oxygen (90 K) or liquid methane (112 K) (35).

**CubeSat.** A standard small satellite built from 10 cm units (1U); 3U, 6U and 12U are common sizes (11).

## H.4 D

**Data assimilation.** Combining observations with a numerical model to produce the best estimate of the state of a system, such as the atmosphere for a weather forecast (16, 47).

**Decibel (dB).** Ten times the base-10 logarithm of a power ratio (A, D).

**Deep learning.** *Machine learning* with neural networks of many layers (44).

**Deep Space Network.** NASA's three ground stations, at Goldstone, Madrid and Canberra, about 120° apart in longitude, which communicate with deep-space missions (7, 24).

**Delta-v (Δv).** The change in velocity a manoeuvre or mission requires; the currency of mission design (3).

**Depth of discharge.** The fraction of a battery's capacity used in a discharge cycle; lower depth gives longer life (5).

**Diffraction limit.** The finest angular detail an optical system can resolve, set by the wavelength divided by the aperture diameter (14, 38).

**Digital twin.** A continuously updated computer model of a physical system used to predict its behaviour and test operations (49, 52).

**Direct-to-device.** Satellite service to unmodified mobile phones, using terrestrial mobile spectrum (17).

**Dispersion (launch).** The spread of possible trajectories or impact points around the nominal, used in range safety (37).

**Distribution shift.** A change between the data on which an AI model was trained and the data it meets in operation, often degrading performance (43, 53).

**Docking / berthing.** Joining two spacecraft. In docking the active vehicle flies into contact; in berthing a robotic arm moves one vehicle onto the other's port (31, 32).

**Doppler shift.** The change in received frequency caused by relative motion between transmitter and receiver, used for navigation and radar (7, 18).

**Dose (radiation).** The energy deposited by radiation per kilogram of material (gray); weighted by biological effect it becomes equivalent dose (sievert) (2, 30).

**Downlink / uplink.** Transmission from space to ground / from ground to space (7).

**Drag.** The aerodynamic force opposing motion through an atmosphere; in low orbit it slowly lowers satellites (2, 21).

## H.5 E

**Earth observation.** Gathering information about the Earth's surface, oceans and atmosphere from space (13).

**Eccentricity (e).** A number describing the shape of an orbit: 0 for a circle, between 0 and 1 for an ellipse, 1 for a parabola and above 1 for a hyperbola (3).

**Ecliptic.** The plane of the Earth's orbit around the Sun (3).

**EIRP (effective isotropic radiated power).** The transmitter power multiplied by the antenna gain — the power an isotropic antenna would need to give the same signal in the main beam (7).

**Electric propulsion.** Propulsion that accelerates propellant electrically — ion, Hall-effect, arcjet or electrospray thrusters — giving high *specific impulse* at low thrust (6, 42).

**Electrolysis.** Splitting a compound with electricity, for example water into hydrogen and oxygen for propellant or breathing (30, 41).

**Emittance (ε).** How effectively a surface radiates heat compared with a *blackbody* (5).

**Engineering model.** A representative version of a spacecraft or unit built to verify design and test procedures, but not intended to fly (4).

**Entry, descent and landing (EDL).** The sequence that takes a spacecraft from the top of an atmosphere to the surface (27).

**Ephemeris.** A table or model of the positions of a body over time (3).

**Equatorial bulge (J2).** The flattening of the Earth, which makes orbital planes precess; described by the $J_2$ coefficient (3).

**Escape velocity.** The speed needed at a given distance to leave a body's gravity without further propulsion, $\sqrt{(2\mu/r)}$ (3).

**Exhaust velocity (effective).** The speed at which a rocket's exhaust leaves relative to the vehicle, equal to *specific impulse* × $g_0$ (35).

**Exoplanet.** A planet orbiting a star other than the Sun (39).

**Extravehicular activity (EVA).** Any activity by an astronaut outside a spacecraft — a spacewalk or moonwalk (33).

## H.6 F

**Fairing (payload fairing).** The aerodynamic nose cone that protects payloads during ascent and is jettisoned above the dense atmosphere (34).

**Fault detection, isolation and recovery (FDIR).** The on-board logic that detects faults, identifies the failed unit and switches to a safe configuration or a redundant unit (6, 48).

**Field of regard / field of view.** The region a sensor can be pointed at / the region it sees at one instant (13).

**Flyby.** A mission that passes a target body without orbiting or landing (26).

**Flight-path angle (γ).** The angle between a vehicle's velocity and the local horizontal (3, 10).

**Footprint.** The area on the ground covered by a satellite's antenna beam or sensor (11, 17).

**Formation flying.** Two or more spacecraft keeping a precise relative position, for example for interferometry (11).

**Foundation model.** A large *machine-learning* model trained on broad data, usually with *self-supervised learning*, that can be adapted to many tasks (44, 46).

**Free-space path loss.** The spreading loss of a radio signal with distance between isotropic antennas (7, D).

**Frozen orbit.** An orbit whose eccentricity and argument of perigee are chosen so that they stay nearly constant despite perturbations (11).

## H.7 G

**Gain-to-noise temperature ratio (G/T).** The figure of merit of a receiving system: antenna gain divided by system noise temperature (7).

**Gateway (feeder link).** A ground station that connects a communication satellite to terrestrial networks (17).

**Geomagnetic storm.** A major disturbance of the Earth's magnetosphere caused by solar-wind structures, measured by indices such as Kp and Dst (22).

**Geostationary orbit (GEO).** A circular orbit 35 786 km above the equator with a period of one *sidereal day*, in which a satellite appears fixed in the sky (11).

**Geosynchronous orbit.** Any orbit with a period of one sidereal day; geostationary orbit is the equatorial, circular case (11).

**Gimbal.** A pivoted mount that lets an engine or instrument rotate about one or two axes; engine gimbals steer rockets (34).

**Gravity assist (swing-by).** Using a close pass of a planet to change a spacecraft's speed and direction relative to the Sun without propellant (24).

**Gravity gradient.** The variation of gravity across a body, which produces torques and tidal forces (6).

**Gravity loss.** The part of a rocket's Δ*v* spent fighting gravity during ascent rather than gaining speed (34).

**Ground sample distance (GSD).** The distance on the ground between the centres of adjacent pixels in an image (13, 14).

**Ground segment.** All the ground facilities — antennas, control centres, data processing — that operate a space system (7).

**Ground track.** The path on the Earth's surface directly beneath an orbiting satellite (3).

## H.8 H

**Halo orbit.** A periodic three-dimensional orbit around a *Lagrange point*, used by JWST and many solar observatories (3, 38).

**Hall-effect thruster.** An *electric propulsion* device that traps electrons in a magnetic field to ionise and accelerate propellant, typically xenon or krypton (6, 42).

**Heat pipe.** A sealed tube containing a working fluid that moves heat by evaporation and condensation with very small temperature drop (5).

**Heliopause.** The boundary where the solar wind meets the interstellar medium, crossed by Voyager 1 in 2012 and Voyager 2 in 2018 (26).

**High-altitude platform station (HAPS).** An aircraft or airship that stays in the stratosphere, typically at about 20 km, for weeks or months (8).

**Highly elliptical orbit (HEO).** An orbit with high eccentricity, such as a Molniya or Tundra orbit, giving long dwell time over high latitudes (11, 20).

**Hohmann transfer.** The two-impulse transfer between coplanar circular orbits with the least $\Delta v$ (3, 24).

**Hydrazine.** $N_2H_4$, a storable propellant used alone over a catalyst (monopropellant) or with an oxidiser (6).

**Hyperbolic excess speed ($v_\infty$).** The speed a spacecraft has relative to a planet far outside its gravity well (24).

**Hypergolic.** Describing propellants that ignite spontaneously on contact, such as hydrazines with nitrogen tetroxide (35).

**Hyperspectral imaging.** Imaging in many (often hundreds of) narrow contiguous spectral bands, giving a spectrum for each pixel (13).

## H.9 I

**Inclination (i).** The angle between an orbital plane and the reference plane, usually the Earth's equator (3).

**Inertial measurement unit (IMU).** A package of gyroscopes and accelerometers that measures rotation and acceleration for navigation (6, 34).

**In situ resource utilisation (ISRU).** Producing useful materials, such as oxygen, water or propellant, from local resources on the Moon, Mars or asteroids (41).

**Integration and test.** Assembling a spacecraft and verifying it by functional, environmental and end-to-end tests (4).

**Interferometric SAR (InSAR).** Comparing the phase of two radar images to measure topography or surface movement to millimetres (15).

**Interplanetary trajectory.** A path between planets, usually designed with *patched conics* and refined numerically (24).

**Ionosphere.** The ionised upper atmosphere, from about 60 km to more than 1 000 km, which reflects, delays and scatters radio waves (23).

**Isotropic radiator.** An ideal antenna radiating equally in all directions; the reference for *antenna gain* (7).

## H.10 J–K

**Jitter.** Small, rapid unwanted motions of a pointing system, which blur images (6, 38).

**Kármán line.** The altitude of 100 km often used as the boundary of space (FAI); the US military and FAA use 50 miles (80 km) (1, 9).

**Kepler's laws.** Three laws of planetary motion: orbits are ellipses with the central body at one focus; a line from the central body sweeps equal areas in equal times; the square of the period is proportional to the cube of the semi-major axis (3).

**Kessler syndrome.** A cascade in which collisions between debris objects create more debris, making further collisions more likely (21).

**Kick stage.** A small upper stage that delivers payloads to their final orbits after separation from the main launch vehicle (34).

## H.11 L

**Lagrange point.** One of five points in a two-body system where a small object can remain stationary relative to the two bodies (3, C).

**Lander.** A spacecraft that makes a controlled landing on a surface (27).

**Laser communication (optical communication).** Using modulated laser light instead of radio waves to send data, allowing higher rates with smaller terminals (7).

**Latency.** The delay between sending and receiving a signal or a response — about 0.25 s each way via GEO, about 25–40 ms round trip via low-orbit broadband (7, 17).

**Launch window.** The period during which a launch can reach the desired orbit or trajectory (24, 34).

**Lift-to-drag ratio (L/D).** The ratio of aerodynamic lift to drag; it sets how much a re-entry vehicle can steer and how gently it decelerates (10).

**Light-time.** The time light takes to travel between two points — 1.3 s to the Moon, 3–22 minutes to Mars (7, 24).

**Link budget.** An accounting of all gains and losses between a transmitter and receiver, used to show that a communication link closes with margin (7).

**Liquid rocket engine.** An engine burning liquid propellants fed from tanks by pressure or pumps (35).

**Long-duration balloon.** A scientific balloon that flies for weeks, typically circling Antarctica in summer (8).

**Low Earth orbit (LEO).** Orbits from about 160 km to 2 000 km altitude (11).

**Lunar south pole.** The region around the Moon's south pole, of interest because permanently shadowed craters may hold water ice (27, 41).

## H.12 M

**Machine learning (ML).** Methods by which computers learn patterns from data instead of following explicitly programmed rules (43).

**Magnetometer.** An instrument that measures magnetic field strength and direction, used for attitude sensing and space science (6, 23).

**Magnetorquer.** An electromagnet that produces attitude control torques by interacting with the Earth's magnetic field (6).

**Magnetosphere.** The region around a planet dominated by its magnetic field, shaped by the solar wind into a long tail (23).

**Main-belt asteroid.** An asteroid orbiting between Mars and Jupiter (26, 41).

**Margin.** The difference between what a design can do and what is required, kept to cover uncertainty (4).

**Mass fraction (propellant mass fraction).** The fraction of a stage's mass that is propellant (34).

**Max-Q.** The point in ascent at which *dynamic pressure* is greatest (34).

**Medium Earth orbit (MEO).** Orbits between LEO and GEO, notably those of GNSS satellites at about 19 000–23 000 km (11, 18).

**Micrometeoroid.** A natural particle smaller than about 1 mm travelling at tens of km/s (2).

**Microgravity.** The condition of apparent weightlessness in free fall, with residual accelerations around a millionth of $g$ (9, 30).

**Mission control centre.** The facility from which a mission is commanded and monitored (7).

**Model compression.** Reducing an AI model's size and computational cost — by *quantisation*, pruning or distillation — so that it can run on limited hardware such as spacecraft processors (45).

**Molniya orbit.** A highly elliptical, 12-hour orbit with 63.4° inclination, giving long dwell over high northern latitudes (11).

**Monopropellant.** A single propellant that decomposes over a catalyst to produce thrust, usually hydrazine (6).

**Multi-layer insulation (MLI).** A blanket of many thin, reflective layers separated by spacers, which greatly reduces radiative heat transfer (5).

**Multispectral imaging.** Imaging in several (typically 4–15) spectral bands (13).

## H.13 N

**Near-Earth object (NEO).** An asteroid or comet whose orbit brings it within 1.3 au of the Sun (26).

**Near-rectilinear halo orbit (NRHO).** A highly elongated *halo orbit* around the Moon, planned for the Gateway station (32).

**Neural network.** A machine-learning model made of layers of simple connected units (“neurons”) whose connection weights are learned from data (43, 44).

**Nodal precession (regression of the nodes).** The slow rotation of an orbital plane about the Earth’s axis caused by the *equatorial bulge* (3).

**Noise temperature.** A way of expressing noise power as the temperature of an equivalent resistor (7).

**Nowcasting.** Forecasting weather for the next few hours, largely by extrapolating the latest observations (16, 47).

**Nuclear thermal propulsion.** Heating hydrogen in a nuclear reactor and expanding it through a nozzle, giving roughly twice the specific impulse of chemical rockets (42).

## H.14 O

**Oberth effect.** A burn performed at high speed, deep in a gravity well, produces a greater change in orbital energy than the same burn at low speed (3, 24).

**Occultation.** The passing of one body in front of another; radio and GNSS signals passing through an atmosphere during occultation are used to measure its profile (16, 26).

**On-board processing.** Processing data on the spacecraft, for example to filter clouds or detect events, rather than downlinking raw data (12, 45).

**Optical depth.** A measure of how much a medium attenuates light; an optical depth of 1 transmits 37 % (13).

**Orbit determination.** Estimating a spacecraft’s orbit from tracking measurements (3, 21).

**Orbital elements.** Six numbers that define an orbit and a position on it, such as semi-major axis, eccentricity, inclination, right ascension of the ascending node, argument of periapsis and true anomaly (3).

**Orbital period.** The time to complete one orbit (3).

**Outgassing.** The release of gas from materials in vacuum, which can contaminate optics and cold surfaces (2, 38).

**Overfitting.** When a machine-learning model learns the details and noise of its training data so closely that it performs poorly on new data (43).

## H.15 P

**Parking orbit.** A temporary orbit in which a spacecraft waits before a departure burn (24).

**Patched conics.** Approximating a trajectory as a sequence of two-body conic sections, each inside one body’s *sphere of influence* (24).

**Payload.** The part of a spacecraft or launch vehicle that performs the mission — instruments, transponders or passengers (12, 34).

**Perturbation.** Any force other than the central body’s point-mass gravity that alters an orbit — $J_2$, drag, third bodies, solar radiation pressure (3).

**Phased array.** An antenna made of many elements whose relative phases are controlled to steer the beam electronically (17).

**Physics-informed machine learning.** Machine-learning models constrained or guided by known physical laws (44).

**Pixel.** One element of a digital image or detector array (13).

**Planck’s law.** The equation giving the spectrum of radiation emitted by a *blackbody* at a given temperature (13, B).

**Planetary protection.** Measures to prevent biological contamination of other worlds by Earth organisms, and of Earth by returned samples (24, 29).

**Plasma.** An ionised gas of free electrons and ions; most of the matter in space is plasma (23).

**Polar orbit.** An orbit with inclination near 90°, passing over both poles (11).

**Pose estimation.** Determining the position and orientation of an object, such as a target spacecraft, from sensor data (48).

**Precession.** A slow rotation of the axis of a spinning body or of an orbital plane (3, 6).

**Pressure-fed / pump-fed.** Engine feed systems in which propellant is pushed into the chamber by tank pressure or by turbopumps (35).

**Primary mirror.** The main light-collecting mirror of a reflecting telescope (38).

**Proximity operations.** Manoeuvring close to another spacecraft for inspection, docking or servicing (40).

**Push-broom scanner.** An imager with a line of detectors across the ground track that builds an image as the satellite moves forward (13).

## H.16 Q–R

**Quantisation (AI).** Representing a model's weights and activations with fewer bits, for example 8-bit integers instead of 32-bit floating point (45).

**Quaternion.** A four-number representation of attitude that avoids the singularities of Euler angles (6).

**Radar cross-section (RCS).** The effective area that an object presents to a radar, in $m^2$ or dBsm (15, 21).

**Radiation belts (Van Allen belts).** Regions of trapped energetic electrons and protons in the Earth's magnetosphere (2, 23).

**Radiator.** A surface that rejects waste heat to space by thermal radiation (5).

**Radioisotope thermoelectric generator (RTG).** A power source that converts heat from radioactive decay of plutonium-238 into electricity (5, 26).

**Reaction wheel.** A spinning flywheel whose speed is changed by a motor to exchange angular momentum with the spacecraft and control its attitude (6).

**Re-entry.** The return of a vehicle into the atmosphere from space (10).

**Regolith.** The layer of loose, broken rock and dust covering a planetary surface (27, 41).

**Reinforcement learning.** Machine learning in which an agent learns a policy by trial and error, guided by rewards (43, 48).

**Relay satellite.** A satellite that forwards communications between other spacecraft and the ground, such as TDRS or Queqiao (7).

**Remote sensing.** Acquiring information about an object without physical contact, usually by measuring electromagnetic radiation (13).

**Rendezvous.** Bringing two spacecraft to the same orbit and position at the same time (31, 40).

**Resolution.** The ability to distinguish detail: spatial, spectral, radiometric or temporal (13).

**Retrograde orbit.** An orbit in the opposite direction to the central body's rotation (inclination above 90°) (3).

**Reusability.** Recovering and reflying launch vehicle stages, fairings or spacecraft (34, 36).

**Revisit time.** The time between successive observations of the same location (11).

**Right ascension of the ascending node (RAAN, Ω).** The angle from the vernal equinox direction to the *ascending node*, measured in the equatorial plane (3).

**Rover.** A mobile robotic vehicle that explores a planetary surface (28).

## H.17 S

**Safe mode.** A minimal, stable spacecraft configuration entered automatically after a fault, waiting for ground instructions (6, 48).

**Sample return.** A mission that collects material from another body and brings it to Earth (29).

**Scale height.** The altitude change over which atmospheric pressure or density falls by a factor of $e$ (2.718) (2).

**Scintillation.** Rapid fluctuations of a signal's amplitude and phase caused by irregularities in the medium it passes through, such as the ionosphere (23).

**Scramjet.** A supersonic-combustion ramjet, an air-breathing engine for hypersonic flight (42).

**Segmented mirror.** A large mirror made of smaller segments controlled to act as one surface, as on JWST (38).

**Self-supervised learning.** Machine learning in which a model learns from unlabelled data by predicting hidden parts of its input (44).

**Semi-major axis (a).** Half the longest diameter of an ellipse; it determines an orbit's period and energy (3).

**Sensor fusion.** Combining data from different sensors to obtain better estimates than any single sensor (6, 48).

**Servicing (on-orbit).** Inspecting, repairing, refuelling or upgrading a spacecraft in orbit (40).

**Sidereal day.** The Earth's rotation period relative to the stars, 23 h 56 min 4.1 s (3, A).

**Signal-to-noise ratio (SNR).** The ratio of signal power to noise power (7, 13).

**Signals intelligence (SIGINT).** Intelligence obtained by intercepting communications and electronic emissions (19).

**Single-event effect.** A malfunction caused by one energetic particle striking a sensitive part of a microcircuit — an upset (bit flip), latch-up or burnout (2).

**Small satellite.** A satellite of less than about 500 kg, including minisatellites, microsatellites, nanosatellites and *CubeSats* (11).

**Solar array.** An assembly of photovoltaic cells that converts sunlight into electricity (5).

**Solar cycle.** The roughly 11-year cycle of solar activity, measured by sunspot number (22).

**Solar flare.** A sudden release of energy in the Sun's atmosphere, producing X-rays and often energetic particles (22).

**Solar sail.** A large, thin reflective membrane propelled by the pressure of sunlight (42).

**Solar wind.** The continuous outflow of plasma from the Sun at 300–800 km/s (22).

**Sounding rocket.** A rocket that carries instruments on a suborbital flight, typically for 5–20 minutes above 100 km (9).

**Space situational awareness (SSA) / space domain awareness.** Knowledge of the location and behaviour of objects in space, and of the environment they operate in (21).

**Space weather.** Conditions in space, driven mainly by the Sun, that affect technology and human health (22).

**Specific impulse (Isp).** Thrust per unit weight flow of propellant, in seconds — a measure of propellant efficiency (35).

**Sphere of influence.** The region around a planet within which its gravity dominates for trajectory calculations (24, C).

**Spin stabilisation.** Keeping a spacecraft's attitude stable by spinning it (6).

**Staging.** Discarding empty tanks and engines during ascent so that the remaining vehicle is lighter (34).

**Star tracker.** A camera that identifies star patterns to determine a spacecraft's attitude to arc-second accuracy (6).

**Starshade.** A large, flower-shaped screen flown in front of a telescope to block a star's light for exoplanet imaging (38).

**Station-keeping.** Small manoeuvres that keep a satellite in its assigned orbit or slot (11).

**Stratosphere.** The atmospheric layer from about 11–15 km to 50 km, where temperature rises with height (2, 8).

**Suborbital flight.** A flight that reaches space but does not have enough speed to complete an orbit (9).

**Sun-synchronous orbit.** A near-polar orbit whose plane precesses once a year, so the satellite crosses each latitude at the same local solar time (11).

**Supervised learning.** Machine learning from examples labelled with the correct answer (43).

**Surrogate model (emulator).** A fast approximate model, often learned from data, that stands in for a slow physical simulation (44).

**Swath.** The width of the strip of ground imaged by a satellite on each pass (13).

**Synodic period.** The time between successive identical alignments of two orbiting bodies, such as Earth and Mars (24, C).

**Synthetic aperture radar (SAR).** An imaging radar that uses the motion of the platform to synthesise a long antenna, giving fine resolution day and night through clouds (15).

## H.18 T

**Telemetry.** Measurements sent from a spacecraft to the ground describing its health and status (7).

**Telecommand.** Instructions sent from the ground to a spacecraft (7).

**Terrain-relative navigation.** Navigating during descent by comparing camera images with onboard maps, used by Perseverance and SLIM to land precisely (27, 48).

**Thermal vacuum test.** Testing a spacecraft in a chamber that reproduces space vacuum and temperature extremes (4).

**Three-axis stabilisation.** Controlling a spacecraft's attitude actively about all three axes (6).

**Throttling.** Varying a rocket engine's thrust (35).

**Thrust-to-weight ratio.** Thrust divided by weight; it must exceed 1 for a rocket to lift off (34).

**Total electron content (TEC).** The number of free electrons in a column through the ionosphere, which determines GNSS signal delay (18, 23).

**Training data.** The examples from which a machine-learning model learns (43).

**Transfer orbit.** An intermediate orbit between two others, such as a geostationary transfer orbit (3).

**Transformer.** A neural-network architecture based on attention, used in large language models and increasingly for images and weather (44).

**Transit method.** Detecting an exoplanet by the small dip in starlight as it passes in front of its star (39).

**Transponder.** A unit on a communication satellite that receives, amplifies, frequency-shifts and retransmits signals (17).

**Troposphere.** The lowest layer of the atmosphere, up to 8–18 km, where weather occurs (2, 16).

**True anomaly (ν).** The angle from periapsis to a body's current position, measured at the central body (3).

**Turbopump.** A turbine-driven pump that feeds propellant into a rocket engine at high pressure (35).

**Two-line element set (TLE).** A standard text format for publishing the orbits of catalogued objects (3, 21).

# H.19 U–Z

**Uncertainty quantification.** Estimating how confident a model's predictions are (44).

**Unsupervised learning.** Machine learning that finds structure in unlabelled data, for example clustering (43).

**Upper stage.** The final stage(s) of a launch vehicle, which deliver the payload to orbit (34).

**Validation / verification.** Verification shows that a product meets its specified requirements ("built it right"); validation shows that it meets the user's real needs ("built the right thing") (4, 53).

**Vernal equinox.** The direction from the Earth to the Sun at the March equinox, used as the reference direction in celestial coordinates (3).

**Vis-viva equation.** The relation between speed, distance and semi-major axis in a two-body orbit (3, B).

**Wavefront sensing and control.** Measuring errors in the light waves reaching a telescope's focus and correcting them by moving mirror segments or deforming mirrors (38).

**Wet mass / dry mass.** A spacecraft's mass with and without propellant (4).

**Whipple shield.** A shield of a thin outer bumper spaced away from the main wall, which breaks up impacting particles (2).

**Window (atmospheric).** A wavelength band in which the atmosphere is relatively transparent (13, D).

**Zenith / nadir.** The point directly overhead / directly below (13).

APPENDIX

# Standards, Data Sources and Further Reading

*Where to go next — the engineering standards that govern spacecraft design, test, operations, debris mitigation and artificial intelligence; the open data services used to produce the figures and tables in this manual; and a selected reading list of textbooks, handbooks and reports for each Part of the book.*

IN THIS CHAPTER

I.1 Engineering Standards
I.2 Open Data Sources
I.3 Further Reading

This manual explains principles and gives typical numbers. Real projects are built to standards and checked against primary data. This appendix lists the most widely used standards, the open data sources behind the computed figures and tables, and a short list of recommended books and reports. Standards are revised frequently; always use the issue specified in your contract or agency requirements.

## I.1 Engineering Standards

### I.1.1 Who Writes Space Standards

**TABLE I.1** Main producers of space standards.

| Organisation | What it publishes | Access |
|---|---|---|
| ECSS — European Cooperation for Space Standardization | More than 100 standards, plus handbooks, for project management (M), engineering (E), product assurance (Q) and sustainability (U), used by ESA and European industry | Free at ecss.nl |
| NASA | NASA Procedural Requirements (NPR), Standards (NASA-STD) and Handbooks (NASA-HDBK) | Free at standards.nasa.gov and nodis3.gsfc.nasa.gov |
| US Space Force / Aerospace Corporation | Space and Missile Systems Center standards (SMC-S series), mission-assurance guides | Mostly public |
| CCSDS — Consultative Committee for Space Data Systems | Data-system and communication protocols (“Blue Books”) adopted by more than 30 agencies | Free at public.ccsds.org |
| ISO TC 20/SC 14 | International standards for space systems and operations (with ISO/IEC JTC 1/SC 42 for AI) | Purchase |
| ITU-R | Radio Regulations and Recommendations on propagation, interference and coordination | Free (Radio Regulations) at itu.int |

| Organisation | What it publishes | Access |
|---|---|---|
| IADC and UN COPUOS | Space debris mitigation and long-term sustainability guidelines | Free |
| AIAA, IEEE, SAE | Industry standards (e.g. AIAA S-series for pressure vessels and mass properties; IEEE 1149, SpaceWire under ECSS) | Purchase |

## I.1.2 Selected Standards by Topic

**TABLE I.2** Selected standards and guidelines cited in this manual.

| Topic | Standard | Title or subject | Chapter |
|---|---|---|---|
| Project management | NASA NPR 7120.5 | NASA Space Flight Program and Project Management Requirements | 4 |
| Systems engineering | NASA NPR 7123.1; NASA/SP-2016-6105 Rev 2 | Systems Engineering Processes and Requirements; NASA Systems Engineering Handbook | 4 |
| | ECSS-E-ST-10C | Space engineering — System engineering general requirements | 4 |
| | ISO/IEC/IEEE 15288 | Systems and software engineering — System life cycle processes | 4 |
| Risk classification | NASA NPR 8705.4 | Risk Classification for NASA Payloads (classes A–D) | 4 |
| Verification and test | ECSS-E-ST-10-03C | Space engineering — Testing | 4, 5 |
| | GSFC-STD-7000B (GEVS) | General Environmental Verification Standard for GSFC flight programs | 4, 5 |
| | SMC-S-016 | Test Requirements for Launch, Upper-Stage and Space Vehicles | 4, 34 |
| Space environment | ECSS-E-ST-10-04C; ECSS-E-ST-10-12C | Space environment; methods for the calculation of radiation received and its effects | 2, 22 |
| Structures | NASA-STD-5001; ECSS-E-ST-32C | Structural design and test factors of safety; structural general requirements | 5 |
| Thermal control | ECSS-E-ST-31C | Thermal control general requirements | 5 |
| Mechanisms | ECSS-E-ST-33-01C | Mechanisms | 5 |
| Electrical and power | ECSS-E-ST-20C | Electrical and electronic | 5 |
| Propulsion | ECSS-E-ST-35C | Propulsion general requirements | 6, 35 |
| Attitude and orbit control | ECSS-E-ST-60-30C | Satellite attitude and orbit control system requirements | 6 |
| Software | ECSS-E-ST-40C; ECSS-Q-ST-80C; NASA NPR 7150.2; RTCA DO-178C | Software engineering; software product assurance; NASA software requirements; airborne software (often used as a reference) | 6, 53 |
| EEE components | ECSS-Q-ST-60C; NASA EEE-INST-002 | Electrical, electronic and electromechanical components; parts selection and screening | 5, 12 |

| Topic | Standard | Title or subject | Chapter |
|---|---|---|---|
| Ground segment and operations | ECSS-E-ST-70C | Ground systems and operations | 7 |
| Space data links | CCSDS 131.0-B, 132.0-B, 231.0-B, 232.0-B, 732.0-B, 732.1-B | TM and TC synchronisation and coding; TM, TC, AOS and Unified Space Data Link protocols | 7 |
| Packets and files | CCSDS 133.0-B; 727.0-B; 734.2-B | Space Packet Protocol; CCSDS File Delivery Protocol (CFDP); Bundle Protocol (delay-tolerant networking) | 7 |
| RF and optical links | CCSDS 401.0-B; 141.0-B | Radio frequency and modulation systems; optical communications physical layer | 7 |
| Proximity links | CCSDS 211.0-B | Proximity-1 space link protocol (Mars relay) | 7, 28 |
| Data compression | CCSDS 121.0-B; 122.0-B; 123.0-B | Lossless data compression; image data compression; multispectral and hyperspectral compression | 12, 13 |
| Orbit and conjunction data | CCSDS 502.0-B; 508.0-B | Orbit Data Messages; Conjunction Data Message | 3, 21 |
| Radio spectrum | ITU Radio Regulations (2024 edition) | International allocation and coordination rules | 7, 17, App. D |
| Propagation | ITU-R P.618, P.676, P.837, P.838 | Earth–space propagation, gaseous attenuation, rain rate and rain attenuation | 7, App. D |
| CubeSats | CubeSat Design Specification Rev. 14.1 (Cal Poly); ISO 17770 | Mechanical and electrical interface requirements for CubeSats | 11 |
| Debris mitigation | ISO 24113:2023; ECSS-U-AS-10C; IADC-02-01; NASA-STD-8719.14; ESA ESSB-ST-U-007 | Debris mitigation requirements, including post-mission disposal and casualty-risk limits | 21 |
| Debris policy | UN COPUOS Space Debris Mitigation Guidelines (2007); Guidelines for the Long-term Sustainability of Outer Space Activities (2019); FCC five-year rule (in force 2024) | International and national debris guidelines and rules | 21 |
| Launch and re-entry safety | FAA 14 CFR Part 450; DAFMAN 91-710 | US commercial launch and re-entry licensing; range safety user requirements | 34, 37 |
| Human spaceflight | NASA-STD-3001 Vols 1 and 2; NASA Human Integration Design Handbook | Crew health and human-system design requirements | 30–33 |
| Planetary protection | COSPAR Policy on Planetary Protection; NASA NPR 8715.24 | Contamination control for missions to other worlds | 24, 29 |
| Machine learning in space software | ECSS-E-HB-40-02A (15 November 2024) | Machine learning qualification handbook | 45, 53 |
| AI management and risk | ISO/IEC 22989:2022; ISO/IEC 23053:2022; ISO/IEC 23894:2023; ISO/IEC 42001:2023; NIST AI RMF 1.0 (2023) | AI concepts; ML framework; AI risk management; AI management systems | 53 |
| AI regulation and aviation guidance | Regulation (EU) 2024/1689 (EU AI Act); EASA AI Concept Paper Issue 2 (2024) | Risk-based regulation of AI systems; guidance for machine-learning applications in aviation | 53 |

## I.2 Open Data Sources

Most figures in this manual that show real data were computed from the following open sources, which readers can use to reproduce and extend them.

**TABLE I.3** Open data sources.

| Source | Content | Used in |
|---|---|---|
| GCAT — General Catalog of Artificial Space Objects (J. McDowell, planet4589.org) | Every launch and catalogued object since 1957, with orbits, masses and owners (CC-BY 4.0) | Chapters 1, 11, 36, 37; Appendix E |
| CelesTrak (celestrak.org) and Space-Track.org | Orbital elements (TLE/OMM), satellite catalogue, debris counts, space weather indices | Chapters 3, 11, 21 |
| JPL Horizons and NAIF SPICE | Planetary, satellite and spacecraft ephemerides; mission geometry kernels | Chapters 3, 24–29; Appendix C |
| JPL Solar System Dynamics (ssd.jpl.nasa.gov) | Physical and orbital data for planets, moons, asteroids and comets; small-body database | Chapters 24–29, 41; Appendix C |
| NASA Planetary Fact Sheets (NSSDCA) | Planetary data and mission records | Appendix C |
| NOAA Space Weather Prediction Center; SILSO (Royal Observatory of Belgium) | Solar-wind data, geomagnetic indices, sunspot number | Chapters 22, 23 |
| NASA Exoplanet Archive | Confirmed exoplanets and their properties | Chapter 39 |
| Copernicus Data Space Ecosystem; USGS EarthExplorer | Sentinel and Landsat imagery and products | Chapters 13–16, 46 |
| ECMWF (ERA5 reanalysis, open forecast data) | Weather and climate data for numerical and machine-learning models | Chapters 16, 47 |
| ESA DISCOS and the annual ESA Space Environment Report | Debris and re-entry statistics | Chapter 21 |
| UCS Satellite Database (to 2023) | Active satellites by purpose and owner | Chapters 1, 11 |
| NASA Technical Reports Server (ntrs.nasa.gov) | NASA technical reports, conference papers and mission documents | All |
| eoPortal (ESA) | Mission and instrument descriptions | Chapters 12–16 |
| ITU Space Network List; ITU-Rpy software | Satellite network filings; ITU-R propagation models in Python | Chapter 17; Appendix D |
| NASA Image and Video Library (images.nasa.gov) | Public-domain photographs used throughout this manual | All; Appendix J |

## I.3 Further Reading

### I.3.1 Part I — Foundations

- J. R. Wertz, D. F. Everett and J. J. Puschell (eds.), *Space Mission Engineering: The New SMAD*, Microcosm Press, 2011 — the standard reference for mission and spacecraft sizing.
- P. Fortescue, G. Swinerd and J. Stark (eds.), *Spacecraft Systems Engineering*, 4th ed., Wiley, 2011.
- H. D. Curtis, *Orbital Mechanics for Engineering Students*, 4th ed., Butterworth-Heinemann, 2020.
- D. A. Vallado, *Fundamentals of Astrodynamics and Applications*, Microcosm Press (latest edition).
- R. R. Bate, D. D. Mueller and J. E. White, *Fundamentals of Astrodynamics*, Dover, 1971 (2nd ed. 2020).
- A. C. Tribble, *The Space Environment: Implications for Spacecraft Design*, revised ed., Princeton University Press, 2003.

- D. G. Gilmore (ed.), *Spacecraft Thermal Control Handbook, Vol. I*, 2nd ed., The Aerospace Press, 2002.
- F. L. Markley and J. L. Crassidis, *Fundamentals of Spacecraft Attitude Determination and Control*, Springer, 2014.
- NASA, *NASA Systems Engineering Handbook*, NASA/SP-2016-6105 Rev 2, 2016.
- NASA Ames Research Center, *State-of-the-Art of Small Spacecraft Technology* (updated annually).
- G. Maral, M. Bousquet and Z. Sun, *Satellite Communications Systems*, 6th ed., Wiley, 2020.


### I.3.2 Parts II–III — Near Space, Suborbital Flight and Satellites


- M. D. Griffin and J. R. French, *Space Vehicle Design*, 2nd ed., AIAA, 2004.
- NASA Sounding Rockets Program Office, *Sounding Rocket Program Handbook* (NASA Wallops Flight Facility).
- Cal Poly CubeSat Laboratory, *CubeSat Design Specification*, Rev. 14.1, 2022.


### I.3.3 Parts IV–V — Earth Observation, Communications and Navigation


- C. Elachi and J. van Zyl, *Introduction to the Physics and Techniques of Remote Sensing*, 3rd ed., Wiley, 2021.
- T. M. Lillesand, R. W. Kiefer and J. W. Chipman, *Remote Sensing and Image Interpretation*, 7th ed., Wiley, 2015.
- A. Moreira *et al.*, “A tutorial on synthetic aperture radar”, *IEEE Geoscience and Remote Sensing Magazine*, 1(1), 2013.
- T. Pratt and J. E. Allnutt, *Satellite Communications*, 3rd ed., Wiley, 2019.
- P. Misra and P. Enge, *Global Positioning System: Signals, Measurements, and Performance*, 2nd ed., Ganga-Jamuna Press, 2006.
- P. J. G. Teunissen and O. Montenbruck (eds.), *Springer Handbook of Global Navigation Satellite Systems*, Springer, 2017.


### I.3.4 Parts VI–VII — Security, Debris and the Space Environment


- H. Klinkrad, *Space Debris: Models and Risk Analysis*, Springer, 2006.
- ESA Space Debris Office, *ESA’s Annual Space Environment Report* (published each year).
- Secure World Foundation, *Global Counterspace Capabilities: An Open Source Assessment* (updated annually).
- M. G. Kivelson and C. T. Russell (eds.), *Introduction to Space Physics*, Cambridge University Press, 1995.
- D. Hastings and H. Garrett, *Spacecraft–Environment Interactions*, Cambridge University Press, 1996.


### I.3.5 Part VIII — Deep-Space Exploration


- S. Kemble, *Interplanetary Mission Analysis and Design*, Springer, 2006.
- A. J. Ball, J. R. C. Garry, R. D. Lorenz and V. V. Kerzhanovich, *Planetary Landers and Entry Probes*, Cambridge University Press, 2007.
- R. D. Braun and R. M. Manning, “Mars exploration entry, descent and landing challenges”, *Journal of Spacecraft and Rockets*, 44(2), 2007.

- A. A. Siddiqi, *Beyond Earth: A Chronicle of Deep Space Exploration, 1958–2016*, NASA SP-2018-4041, 2018.


### I.3.6 Parts IX–X — Human Spaceflight and Launch


- W. J. Larson and L. K. Pranke (eds.), *Human Spaceflight: Mission Analysis and Design*, McGraw-Hill, 2000.
- NASA, *Human Integration Design Handbook*, NASA/SP-2010-3407 (revised).
- G. P. Sutton and O. Biblarz, *Rocket Propulsion Elements*, 9th ed., Wiley, 2017.
- D. K. Huzel and D. H. Huang, *Modern Engineering for Design of Liquid-Propellant Rocket Engines*, AIAA, 1992.
- D. M. Goebel, I. Katz and I. G. Mikellides, *Fundamentals of Electric Propulsion*, 2nd ed., Wiley/JPL Space Science and Technology Series.
- R. W. Humble, G. N. Henry and W. J. Larson (eds.), *Space Propulsion Analysis and Design*, McGraw-Hill, 1995.


### I.3.7 Parts XI–XII — Observatories and Emerging Frontiers


- P. Y. Bely (ed.), *The Design and Construction of Large Optical Telescopes*, Springer, 2003.
- J. P. Gardner *et al.*, "The James Webb Space Telescope Mission", *Publications of the Astronomical Society of the Pacific*, 135, 068001, 2023.
- K. F. Long, *Deep Space Propulsion: A Roadmap to Interstellar Flight*, Springer, 2012.
- V. Badescu (ed.), *Moon: Prospective Energy and Material Resources*, Springer, 2012.
- F. von der Dunk and F. Tronchetti (eds.), *Handbook of Space Law*, Edward Elgar, 2015.


### I.3.8 Part XIII — Artificial Intelligence


- S. Russell and P. Norvig, *Artificial Intelligence: A Modern Approach*, 4th ed., Pearson, 2020.
- I. Goodfellow, Y. Bengio and A. Courville, *Deep Learning*, MIT Press, 2016.
- C. M. Bishop and H. Bishop, *Deep Learning: Foundations and Concepts*, Springer, 2024.
- G. Camps-Valls, D. Tuia, X. X. Zhu and M. Reichstein (eds.), *Deep Learning for the Earth Sciences*, Wiley, 2021.
- ECSS, *ECSS-E-HB-40-02A Machine Learning Handbook*, 15 November 2024.


> **KEEPING UP TO DATE**
>
> Space changes quickly. For current events, the most reliable sources are the agencies themselves (NASA, ESA, JAXA, CNSA, ISRO, Roscosmos press releases), company announcements checked against independent trackers, Jonathan McDowell's *Jonathan's Space Report*, Gunter's Space Page, the trade press (SpaceNews, SpacePolicyOnline, NASASpaceflight, Spaceflight Now) and peer-reviewed journals (*Acta Astronautica*, *Journal of Spacecraft and Rockets*, *Advances in Space Research*, *Remote Sensing of Environment*). Cross-check any number that matters against a primary source.

APPENDIX

# J Image Credits

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## J.1 Part Opening Images

**TABLE J.1** Part opening images.

| Part | NASA ID | Title | Date |
|---|---|---|---|
| I | iss028e007274 | Earth Observation | 2011-06-11 |
| II | GSFC_20171208_Archive_e000153 | Balloons on Ice: Launch # 2 takes flight in Antarctica | 2017-12-08 |
| III | KSC-20210811-PH-RNB01_0028 | Landsat 9 Spacecraft Mate to EPSA | 2021-08-11 |
| IV | GSFC_20171208_Archive_e001386 | Most Amazing High Definition Image of Earth - Blue Marble 2012 | 2017-12-08 |
| V | KSC-2013-1231 | KSC-2013-1231 | 2013-01-29 |
| VI | GSFC_20171208_Archive_e002131 | NASA Blue Marble 2007 West | 2017-12-08 |
| VII | GSFC_20171208_Archive_e001050 | Two X Flares in Quick Succession | 2017-12-08 |
| VIII | PIA21381 | Jupiter From Below (Enhanced Color) | 2017-02-08 |
| IX | iss065e145170 | iss065e145170 | 2021-06-25 |
| X | NHQ202211160030 | Artemis I Launch | 2022-11-16 |
| XI | carina_nebula | James Webb Space Telescope NIRCam Image of the “Cosmic Cliffs” in Cari | 2022-07-12 |
| XII | iss043e124426 | SpaceX (SpX)-6 Dragon spacecraft move to berth sequence | 2015-04-17 |
| XIII | ACD15-0062-004 | ACD15-0062-004 | 2015-04-10 |
| XIV | GSFC_20171208_Archive_e001282 | NASA Google+ Hangout: ‘Earthrise’ A New Visualization - 45th Anniversa | 2017-12-08 |

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**TABLE J.2** Images in Chapter 1.

| Image | NASA ID | Title | Date |
|---|---|---|---|
| Chapter opening | s134e010592 | View of ISS taken during STS-134 Flyaround | 2011-05-30 |

## J.3 Chapter 2: The Space Environment

**TABLE J.3** Images in Chapter 2.

| Image | NASA ID | Title | Date |
|---|---|---|---|
| Chapter opening | iss066e146457 | iss066e146457 | 2022-02-18 |
| misse | 0201585 | International Space Station (ISS) | 2001-08-17 |
| ldef | s32-541-008 | LDEF grappled by remote manipulator system (RMS) during STS-32 retriev | 1990-01-20 |
| tvac | PIA13936 | Juno Emerges from Thermal-Vac Testing | 2011-03-29 |

## J.4 Chapter 3: Orbital Mechanics and Astrodynamics

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|---|---|---|---|
| Chapter opening | art001e001712 | Flight Day 14: Orion's Solar Array Divides Earth and Moon | 2022-11-29 |

## J.5 Chapter 4: Spacecraft Systems Engineering

**TABLE J.5** Images in Chapter 4.

| Image | NASA ID | Title | Date |
|---|---|---|---|
| Chapter opening | GSFC_20171208_Archive_e000495 | James Webb Space Telescope Mirror Halfway Complete | 2017-12-08 |
| highbay | PIA23519 | High Bay 1 in JPL's Spacecraft Assembly Facility | 2019-12-23 |
| clipper | PIA25958 | Powerful Antenna Installed On Europa Clipper | 2023-08-15 |
| shake | GRC-2024-C-01582 | Sierra Space Dream Chaser Spaceplane at the Space Environments Complex | 2024-03-07 |

## J.6 Chapter 5: Spacecraft Subsystems I: Structures, Thermal Control, Power and Mechanisms

**TABLE J.6** Images in Chapter 5.

| Image | NASA ID | Title | Date |
|---|---|---|---|
| Chapter opening | PIA13925 | Juno Solar Panel Deployment Test | 2009-10-13 |

| Image | NASA ID | Title | Date |
|---|---|---|---|
| maven | GSFC_20171208_Archive_e001847 | MAVEN Mission Primary Structure Complete | 2017-12-08 |
| radiator | s114e5283 | P6 truss and radiator panels | 2005-07-28 |
| mli | iss015e10854 | Survey view of damage to a MLI Blanket on the exterior of the FGB take | 2007-06-06 |
| irosa | iss068e026401 | Expedition 68 EVA 82 | 2022-12-03 |
| mmrtg | PIA23306 | Mars 2020's MMRTG | 2019-07-24 |

## J.7 Chapter 6: Spacecraft Subsystems II: Attitude Control, Propulsion, Data Handling and Flight Software

**TABLE J.7** Images in Chapter 6.

| Image | NASA ID | Title | Date |
|---|---|---|---|
| Chapter opening | GRC-2003-C-00585 | GRC-2003-C-00585 | 2004-05-01 |
| startrackers | PIA26562 | Star Trackers for Europa Clipper | 2025-02-04 |
| wheels | PIA25495 | NASA's Europa Clipper Reaction Wheels Installed | 2022-11-23 |
| wheel_eva | S109E5396 | STS-109 MS Massimino and Newman replace Reaction Wheel assembly during | 2002-03-05 |

## J.8 Chapter 7: Space Communications and the Ground Segment

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| Image | NASA ID | Title | Date |
|---|---|---|---|
| Chapter opening | PIA13277 | Tracks of a Giant | 2010-08-25 |
| dsoc_ftl | PIA24569 | DSOC's Flight Laser Transceiver | 2022-04-06 |
| illumat | jsc2023e064875 | ILLUMA-T Investigation Preflight Imagery | 2023-10-19 |
| madrid | PIA26147 | Six Deep Space Network Antennas in Madrid Arrayed For the First Time | 2024-04-26 |
| dss53 | PIA25137 | The Deep Space Network's New DSS-53 at Night | 2022-03-16 |

## J.9 Chapter 8: High-Altitude Platforms and Space Balloons

**TABLE J.9** Images in Chapter 8.

| Image | NASA ID | Title | Date |
|---|---|---|---|
| Chapter opening | GSFC_20171208_Archive_e000115 | Alien crop circle? No, that's just NASA's newest balloon launch pad | 2017-12-08 |
| ldb | GSFC_20171208_Archive_e000799 | Balloon Program Wraps up in Antarctica, Heading to New Zealand | 2017-12-08 |

| Image | NASA ID | Title | Date |
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| part2 | GSFC_20171208_Archive_e000153 | Balloons on Ice: Launch # 2 takes flight in Antarctica | 2017-12-08 |
| helios | ED01-0230-1 | ED01-0230-1 | 2001-08-13 |
| er2 | EC01-0232-2 | Lockheed ER-2 #809 high altitude research aircraft in flight | 2001-08-01 |

## J.10 Chapter 9: Sounding Rockets and Suborbital Vehicles

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| Image | NASA ID | Title | Date |
|---|---|---|---|
| Chapter opening | GSFC_20171208_Archive_e000703 | NASA Successfully Launches Suborbital Rocket from Wallops with Student | 2017-12-08 |
| mtex | GSFC_20171208_Archive_e000831 | M-TeX and MIST Experiments Launched from Alaska | 2017-12-08 |
| evex | GSFC_20171208_Archive_e001503 | NASA EVEX Experiment Launches from the Marshall Islands | 2017-12-08 |
| clouds | GSFC_20171208_Archive_e000130 | NASA rocket to display artificial clouds in space | 2017-12-08 |
| x43 | ED04-0261-1 | NASA's B-52B launch aircraft takes off carrying the third X-43A hypers | 2004-09-27 |

## J.11 Chapter 10: Spaceplanes, Atmospheric Re-entry and Reusable Vehicles

**TABLE J.11** Images in Chapter 10.

| Image | NASA ID | Title | Date |
|---|---|---|---|
| Chapter opening | sts135-s-273 | Space Shuttle Atlantis after its Final Landing | 2011-06-21 |
| orionhs | KSC-20240203-EM1_CM_320_0-0001 1 | Artemis I Orion Heat Shield Inspection at Kennedy Space Center | 2023-02-03 |
| arcjet | jsc2024e078233 | Arc Jet Testing of Orion Heat Shield Avcoat | 2023-12-05 |
| sts1 | 8111969 | Space Shuttle Projects | 1981-04-12 |
| tiles | KSC-03pd3213 | KENNEDY SPACE CENTER, FLA. - In the Orbiter Processing Facility, KSC | 2003-12-09 |
| dreamchaser | AFRC2017-0016-016 | Dream Chaser Spacecraft Arrives at NASA Armstrong | 2017-01-25 |
| sts53 | sts053-s-080 | STS-53 Discovery, Orbiter Vehicle (OV) 103, lands on runway 22 at EAFB | 1992-12-09 |

## J.12 Chapter 11: Earth Orbits, Constellations and Small Satellites

**TABLE J.12** Images in Chapter 11.

| Image | NASA ID | Title | Date |
|---|---|---|---|
| Chapter opening | iss050e031340 | J-SSOD-6 Deployment | 2017-01-16 |
| marco | PIA20342 | MarCO CubeSat Engineers 2 | 2016-01-20 |
| nanoracks | iss069e004397 | Astronaut Frank Rubio installs the NanoRacks CubeSat Deployer | 2023-04-20 |
| starling | ACD23-0077-003 | Starling's four CubeSats in space (illustration) | 2023-07-12 |

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| Chapter opening | KSC-20171205-PH_MAS01_0031 | GOES-S Rotate to Vertical | 2017-12-05 |
| psp_mag | KSC-20180523-PH_LCH01_0084 | Parker Solar Probe Magnetometer Boom Deployment | 2018-05-23 |
| clippermag | PIA24786 | Europa Clipper Magnetometer (ECM) Boom | 2021-08-05 |
| goesfull | GSFC_20171208_Archive_e000007 | Eclipse from Space - GOES Full Disk | 2017-12-08 |

## J.14 Chapter 13: Remote Sensing Fundamentals

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| Image | NASA ID | Title | Date |
|---|---|---|---|
| Chapter opening | GSFC_20171208_Archive_e002144 | Guinea-Bissau | 2017-12-08 |
| aljowf | PIA23955 | Al Jowf, Saudi Arabia | 2020-07-06 |
| sf | PIA02606 | ASTER Images San Francisco Bay Area | 2000-04-26 |
| gardencity | GSFC_20171208_Archive_e001881 | Garden City, Kansas | 2017-12-08 |
| rimfire | GSFC_20171208_Archive_e001383 | Satellite View of Rim Fire On August 25, 2013 | 2017-12-08 |
| lakemead | PIA19731 | Lake Mead and Drought | 2015-07-20 |

## J.15 Chapter 14: Electro-Optical and Infrared Imaging Satellites

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| Image | NASA ID | Title | Date |
|---|---|---|---|
| Chapter opening | KSC-20210720-PH-RNB01_0022 | Landsat 9 Spacecraft (Observatory) Move from High Bay into Cell | 2021-07-20 |
| laheat | PIA23786 | Los Angeles Heat Wave | 2020-08-19 |
| applefire | PIA23798 | ECOSTRESS Gauges the Apple Fire's Temperature From Space | 2020-08-03 |
| nile | GSFC_20171208_Archive_e001586 | City Lights Illuminate the Nile | 2017-12-08 |
| methane | PIA26113 | EMIT Identifying Methane Plumes Around the Globe | 2023-11-17 |

| Image | NASA ID | Title | Date |
|---|---|---|---|
| eyja | PIA13098 | Iceland Volcano Puffs Ash as Lava Flow Cuts Through Eyjafjallajökull I | 2010-05-05 |

## J.16 Chapter 15: Synthetic Aperture Radar

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| Image | NASA ID | Title | Date |
|---|---|---|---|
| Chapter opening | PIA26612 | NISAR Team Deploys Radar Antenna Reflector | 2025-08-15 |
| seasat | PIA18137 | NASA Seasat Satellite Shows Massachusetts Coast | 2014-03-18 |
| sirc | s93-48551 | STS-59 payload SIR-C/X-SAR antenna view | 1993-10-30 |
| nisarfirst | PIA26616 | NASA-ISRO Radar Satellite Captures First Image of Mount Desert Island | 2025-09-25 |
| ndakota | PIA26615 | NASA-ISRO Radar Satellite Captures First Image of North Dakota Wetland | 2025-09-25 |
| srtm | PIA02723 | Radar Image, Wrapped Color as Height, Lanai and West Maui, Hawaii | 2000-02-19 |
| izmit | PIA00557 | Izmit, Turkey 1999 Earthquake Interferogram | 2001-03-30 |
| kilauea | PIA01710 | Space Radar Image of Kilauea, Hawaii | 1999-01-27 |

## J.17 Chapter 16: Weather and Environmental Satellites

**TABLE J.17** Images in Chapter 16.

| Image | NASA ID | Title | Date |
|---|---|---|---|
| Chapter opening | iss072e001649 | Hurricane Helene pictured from the space station | 2024-09-25 |
| goesu | KSC-20240624-PH-SPX01_0003 | GOES-U Rollout for Launch | 2024-06-24 |
| ozone | GSFC_20171208_Archive_e001962 | Snapshot of the Antarctic Ozone Hole 2010 | 2017-12-08 |
| gpm | GSFC_20171208_Archive_e001331 | GPM Core Observatory | 2017-12-08 |
| pace | KSC-20231204-PH-KLS01_0024 | PACE Spacecraft Processing | 2023-12-04 |
| swot | PIA25772 | SWOT Satellite's Sea Level 'First Light' | 2023-03-24 |

## J.18 Chapter 17: Communication Satellites

**TABLE J.18** Images in Chapter 17.

| Image | NASA ID | Title | Date |
|---|---|---|---|
| Chapter opening | KSC-20170808-PH_KLS01_0532 | TDRS-M Spacecraft Lift to Transport Vehicle | 2017-08-08 |
| echo | LRC-1958-B701_P-03603 | Inflation Tests of the Echo 1 Satellite in Weeksville, N.C. | 1958-08-13 |

| Image | NASA ID | Title | Date |
|---|---|---|---|
| syncom | 41d-32-0067 | View of the Syncom IV satellite in orbit over the earth | 1984-08-31 |
| intelsat | 9301572 | Space Shuttle Projects | 1992-05-13 |
| illumat | jsc2023e064876 | ILLUMA-T Investigation Preflight Imagery | 2023-10-19 |

## J.19 Chapter 18: Global Navigation Satellite Systems

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| Image | NASA ID | Title | Date |
|---|---|---|---|
| Chapter opening | PIA24573 | Deep Space Atomic Clock Hardware | 2021-06-30 |
| navstar | 9304431 | Launch Vehicles | 1993-03-29 |
| tonga | PIA24905 | Tonga Eruption Atmospheric Wave | 2022-01-27 |
| blueghost | KSC-20250114-PH-KLS01_0039 | CLPS Firefly Blue Ghost Mission 1 | 2025-01-14 |
| delta | 0400536 | Space Science | 2004-01-01 |

## J.20 Chapter 19: National Security Space and Signals Intelligence

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| Image | NASA ID | Title | Date |
|---|---|---|---|
| Chapter opening | iss071e439624 | An orbital sunrise colorfully illuminates the Earth's atmosphere | 2024-08-06 |

## J.21 Chapter 20: Missile Early Warning Satellites

**TABLE J.21** Images in Chapter 20.

| Image | NASA ID | Title | Date |
|---|---|---|---|
| Chapter opening | iss074e0431736 | The exhaust plume left behind by Artemis II is pictured from the Inter | 2026-04-01 |
| dspbay | S44-71-011 | STS-44 DSP / IUS spacecraft tilted to predeployment position in OV-104 | 1991-11-24 |
| dsp | s91-50773 | STS-44 DSP satellite and IUS during preflight processing at Cape Canav | 1991-10-19 |
| starship6 | iss072e220043 | The launch of the SpaceX Starship 6 rocket seen from the space station | 2024-11-19 |
| crs35 | NHQ20261013_admin_0004 | NASA's SpaceX CRS-35 Launch | 2026-10-13 |

## J.22 Chapter 21: Space Surveillance, Orbital Debris and Space Situational Awareness

**TABLE J.22** Images in Chapter 21.

| Image | NASA ID | Title | Date |
|---|---|---|---|
| Chapter opening | iss074e0540224 | A piece of unidentified space debris is pictured from the Internationa | 2026-04-27 |
| hvi | wstf0507e04048 | WSTF Hypervelocity Impact Test - CEV | 2007-05-31 |
| sts125 | jsc2010e054445 | STS-125 Orbital Debris | 2009-07-02 |

## J.23 Chapter 22: Space Weather

**TABLE J.23** Images in Chapter 22.

| Image | NASA ID | Title | Date |
|---|---|---|---|
| Chapter opening | GSFC_20171208_Archive_e002109 | Blasting CME | 2017-12-08 |
| sunspot | GSFC_20171208_Archive_e001978 | C3-class Solar Flare Erupts on Sept. 8, 2010 [Full Disk] | 2017-12-08 |
| xflare | GSFC_20171208_Archive_e000934 | NASA's SDO Observes an X-class Solar Flare | 2017-12-08 |
| cme | cme-blow-out_18529951894_o | CME Blow Out | 2015-06-25 |
| parker | KSC-20180419-PH_KLS01_0009 | Parker Solar Probe Antenna Deployment | 2018-04-19 |
| punch | KSC-20250121-PH-ANV01_0028 | PUNCH Satellites Solar Array Deployment Test | 2025-01-21 |
| maven | PIA26304 | MAVEN Detects Auroras During Solar Storm in 2024 | 2024-06-10 |

## J.24 Chapter 23: Magnetospheric and Ionospheric Physics

**TABLE J.24** Images in Chapter 23.

| Image | NASA ID | Title | Date |
|---|---|---|---|
| Chapter opening | iss072e159172 | The aurora borealis blankets the Earth | 2024-10-30 |
| belt3 | GSFC_20171208_Archive_e001541 | NASA's Van Allen Probes Discover a Surprise Circling Earth | 2017-12-08 |
| mms | GSFC_20171208_Archive_e001111 | Magnetospheric Multiscale (MMS) | 2017-12-08 |
| aurora | iss073e0247726 | The aurora australis arcs above a partly cloudy Indian Ocean | 2025-06-12 |

## J.25 Chapter 24: Deep-Space Mission Design

**TABLE J.25** Images in Chapter 24.

| Image | NASA ID | Title | Date |
|---|---|---|---|

| Image | NASA ID | Title | Date |
|---|---|---|---|
| Chapter opening | PIA22915 | Voyager 2 Spacecraft Instruments | 2018-12-10 |
| dawn | PIA18922 | Dawn Blue Glow Artist Concept | 2015-03-02 |
| dsn | PIA23682 | NASA's Deep Space Antenna Upgrade to Affect Voyager | 2020-03-04 |
| clipper | KSC-20240528-PH-KLS01_0036 | Europa Clipper Spacecraft Unboxing | 2024-05-28 |
| lucy | KSC-20210819-PH-GEB01_0026 | Lucy Solar Array Pop & Catch | 2021-08-19 |

## J.26 Chapter 25: Planetary Orbiters

**TABLE J.26** Images in Chapter 25.

| Image | NASA ID | Title | Date |
|---|---|---|---|
| Chapter opening | PIA21771 | Juno Over Jupiter's South Pole (Illustration) | 2017-07-11 |
| finale | PIA21439 | Cassini Grand Finale Dive Illustration | 2017-04-04 |
| earthrise | GSFC_20171208_Archive_e000496 | NASA Releases New High-Resolution Earthrise Image | 2017-12-08 |
| mro | PIA07245 | Mars Reconnaissance Orbiter, Front View Artist Concept | 2005-06-03 |
| venus | PIA00478 | Venus - Global View Centered at 180 degrees | 1996-11-26 |
| mercury | PIA13823 | MESSENGER Explores Mercury - In Color | 2011-01-25 |
| grs | PIA21773 | Jupiter's Great Red Spot (Enhanced Color) | 2017-07-12 |
| enceladus | PIA14599 | Dark Moon, Dramatic Plume | 2012-03-12 |

## J.27 Chapter 26: Space Probes and Flyby Missions

**TABLE J.27** Images in Chapter 26.

| Image | NASA ID | Title | Date |
|---|---|---|---|
| Chapter opening | PIA19956 | Pluto in Extended Color | 2015-09-24 |
| pioneer | ARC-1972-AC72-2135 | ARC-1972-AC72-2135 | 1972-02-26 |
| voyager | PIA17049 | Voyager in Space Artist Concept | 2013-09-12 |
| neptune | PIA01492 | PIA01492 | 1998-10-30 |
| paleblue | PIA23645 | Pale Blue Dot Revisited | 2020-02-12 |
| charon | PIA19968 | Charon in Enhanced Color | 2015-10-01 |
| tempel | PIA02137 | Tempel Alive with Light | 2005-07-05 |
| dart | PIA25329 | NASA's DART at the Didymos System | 2022-09-22 |

## J.28 Chapter 27: Landers and Entry, Descent & Landing

**TABLE J.28** Images in Chapter 27.

| Image | NASA ID | Title | Date |
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| Image | NASA ID | Title | Date |
|---|---|---|---|
| Chapter opening | PIA25287 | InSight's Final Selfie | 2022-05-23 |
| aldrin | as11-40-5902 | Astronaut Edwin Aldrin walks on lunar surface near leg of Lunar Module | 1969-07-20 |
| surveyor | AS12-48-7134 | Apollo 12 Mission image - Astronaut Alan L. Bean,lunar module pilot,an | 1969-11-20 |
| pdv | KSC-20200422-PH-JPL01_0005 | Mars 2020 Perseverance Powered Descent Vehicle (PDV) Stack Operations | 2020-04-22 |
| hirisedesc | PIA24270 | HiRISE Captured Perseverance During Descent to Mars | 2021-02-19 |
| skycrane | PIA24428 | High-Resolution Still Image of Perseverance's Landing | 2021-02-19 |
| viking | PIA00396 | Viking Lander 2 First Picture On The Surface Of Mars | 1996-12-12 |
| phoenix | PIA10910 | Ice on Mars - Now You See It | 2008-06-23 |
| seis | PIA22977 | Putting SEIS on the Ground | 2018-12-20 |
| huygens | PIA07232 | First Color View of Titan Surface | 2005-01-15 |

## J.29 Chapter 28: Planetary Rovers and Aerial Explorers

**TABLE J.29** Images in Chapter 28.

| Image | NASA ID | Title | Date |
|---|---|---|---|
| Chapter opening | PIA19807 | Curiosity Low-Angle Self-Portrait at Buckskin Drilling Site on Mount S | 2015-08-19 |
| lunokhod | PIA12967 | Soviet Union Lunar Rovers | 2010-03-17 |
| lrv | as15-86-11603 | Astronaut James Irwin works at Lunar Roving Vehicle during Apollo 15 E | 1971-07-31 |
| rovers3 | PIA11436 | Newest is Biggest: Three Generations of NASA Mars Rovers | 2008-11-19 |
| wheel | PIA17751 | Left-Front Wheel of Curiosity Rover, Approaching Three Miles | 2013-12-20 |
| opptracks | PIA14129 | Autonomous Hazard Checks Leave Patterned Rover Tracks on Mars | 2011-05-18 |
| leopard | PIA26368 | Perseverance Finds a Rock With 'Leopard Spots' | 2024-07-25 |
| ingenuity | PIA24542 | Perseverance's Selfie with Ingenuity | 2021-04-07 |
| blade | PIA26244 | Ingenuity's Navcam Reveals a Missing Rotor Blade | 2024-02-27 |
| viper | GRC-2019-C-09947 | Virtual Intelligent Planetary Exploration Rover, VIPER Mobili... | 2019-10-22 |

# J.30 Chapter 29: Sample-Return Missions

**TABLE J.30** Images in Chapter 29.

| Image | NASA ID | Title | Date |
|---|---|---|---|
| Chapter opening | NHQ202308300001 | OSIRIS-REx Sample Return Training | 2023-08-30 |
| schmitt | as17-140-21496 | Astronaut Harrison Schmitt standing next to boulder during third EVA | 1972-12-13 |
| collector | PIA03188 | Dust Collector with Aerogel | 2005-11-03 |
| aerogel | PIA03186 | Particle Tracks in Aerogel | 2005-11-03 |
| bennu | 2019-02-25_regolith_image_compilation | A Region of Bennu's Northern Hemisphere Close Up | 2019-03-14 |
| capsule | NHQ202309240002 | OSIRIS-REx Sample Return | 2023-09-24 |
| tagsam | jsc2024e006057 | OSIRIS-REx Asteroid Sample in Touch-and-Go-Sample-Acquisition-Mechanis | 2024-01-17 |
| ryugu | jsc2021e062383 | jsc2021e062383 | 2021-11-30 |
| ees | PIA25986 | Artist's Concept of the Earth Entry System for Mars Sample Return | 2023-08-03 |

# J.31 Chapter 30: Human Spaceflight Fundamentals and Life Support

**TABLE J.31** Images in Chapter 30.

| Image | NASA ID | Title | Date |
|---|---|---|---|
| Chapter opening | iss074e0381623 | ESA astronaut Sophie Adenot poses for a portrait inside the Internatio | 2026-03-12 |
| ared | iss072e126509 | Astronaut Nick Hague exercises on the advanced resistive exercise devi | 2024-10-29 |
| eyeus | iss073e0026012 | iss073e0026012 | 2025-05-06 |
| twins | iss045e028258 | Kelly during Twins Study Experiment Operations | 2015-09-24 |
| phantoms | KSC-20220808-PH-FMX01_0031 | Artemis I MARES Payloads - Install into Crew Module | 2022-08-08 |
| mailbox | as13-62-8929 | Interior view of "mail box" for purging carbon dioxide from Lunar Modu | 1970-04-14 |
| ogs | iss073e0118757 | NASA astronaut Nichole Ayers cleans and services life support componen | 2025-05-29 |
| wrs | iss067e099373 | iss067e099373 | 2022-06-01 |
| peppers | iss066e083264 | Chile Peppers growing inside the Advanced Plant Habitat in the Columbu | 2021-11-26 |
| meal | iss031e041675 | Expedition 31 Crewmembers Share a Meal in the SM | 2012-05-18 |
| chapea | jsc2026e403153 | CHAPEA Mission 2 Crew Poses in Habitat | 2025-11-24 |

## J.32 Chapter 31: Crewed Spacecraft

**TABLE J.32** Images in Chapter 31.

| Image | NASA ID | Title | Date |
|---|---|---|---|
| Chapter opening | iss073e0505071 | The SpaceX Dragon crew spacecraft approaches the International Space S | 2025-08-02 |
| gemini | S65-63220 | View of the Gemini 6 and Gemini 7 rendezvous | 1965-12-15 |
| lmA | as11-44-6642 | KSC-as11-44-6642 | 1969-07-21 |
| starliner | iss071e170979 | Boeing's Starliner Spacecraft Approaches the International Space Stati | 2024-06-06 |
| orionselfie | art002e004357 | Orion Snaps a Selfie During External Inspection | 2026-04-03 |
| ceit2 | KSC-20250217-PH-SPX01_0002 | SpaceX Crew-10 CEIT | 2025-02-17 |
| aa2D | KSC-20190702-PH_AWG07_0011 | Ascent Abort-2 Liftoff | 2019-07-02 |
| spaA | KSC-20191104-PH-JSC01_0002 | Boeing CST-100 Starliner Pad Abort Test | 2019-11-04 |
| a2splash | KSC-20260410-PH-JNV01_0002 | Artemis II Splashdown | 2026-04-10 |
| soyuzC | NHQ201810040001 | Expedition 56 Soyuz MS-08 Landing | 2018-10-04 |
| dsplash | NHQ202601150016 | NASA's SpaceX Crew-11 Splashdown | 2026-01-15 |

## J.33 Chapter 32: Space Stations and Habitats

**TABLE J.33** Images in Chapter 32.

| Image | NASA ID | Title | Date |
|---|---|---|---|
| Chapter opening | iss01-389-023 | Expedition One crew flyaround of ISS | 2001-02-24 |
| skylab | sl4-143-4706 | View of Skylab space station cluster in Earth orbit from CSM | 2008-08-18 |
| mir | sts071-744-017 | Mir space station | 1995-06-29 |
| iss2021 | iss066e080762 | iss066e080762 | 2021-11-08 |
| irosa | iss065e116519 | iss065e116519 | 2021-06-15 |
| dextre | iss073e0546366 | The Canadarm2 with its fine-tuned robotic hand, Dextre | 2025-08-27 |
| cupola | iss074e0604606 | Astronauts Sophie Adenot and Jack Hathaway look out a window in the cu | 2026-05-17 |
| htvx | iss073e0988464 | The new HTV-X1 in the grips of the Canadarm2 | 2025-10-29 |
| cygnusxl | iss074e0458241 | The Cygnus XL cargo spacecraft at its capture point 12 meters from the | 2026-04-13 |
| beamx2 | iss067e214089 | iss067e214089 | 2022-08-02 |
| astrobee | iss060e007162 | Mobility Test of the Free-Flying Astrobee Robotic Assistant | 2019-07-12 |

# J.34 Chapter 33: Extravehicular Activity Systems

**TABLE J.34** Images in Chapter 33.

| Image | NASA ID | Title | Date |
|---|---|---|---|
| Chapter opening | iss063e052528 | iss063e052528 | 2020-07-21 |
| white | s65-30427 | ASTRONAUT EDWARD H. WHITE II - GEMINI-TITAN (GT)-IV - ZERO GRAVITY - O | 2015-03-20 |
| mmu | s84-27017 | INFLIGHT (CREW ACTIVITY) - STS-11/41B | 1984-02-11 |
| hubble | s125e007221 | STS-125 MS5 Feustel and MS3 Grunsfeld during EVA1 | 2009-05-14 |
| orlan | iss073e0988196 | Cosmonaut Alexey Zubritsky is pictured in his Orlan spacesuit conducti | 2025-10-28 |
| gloves | s123e006438 | Whitson takes photo of EMU gloves after EVA 2 in the A/L during Joint | 2008-03-16 |
| safer | 9412846 | Space Shuttle Projects | 1994-09-16 |
| iss2026 | iss074e0788297 | NASA astronaut Chris Williams works outside the International Space St | 2026-06-30 |
| axnbl | jsc2025e077817 | NASA Astronauts Complete First Dual Suit Run Wearing AxEMU | 2025-09-24 |
| cernan | AS17-145-22224 | Astronaut Eugene Cernan inside the lunar module on lunar surface after | 1972-12-12 |
| axemu | axemu-prada-spacesuit-mockup-20241007-423 | Artemis III Lunar Spacesuit | 2024-10-07 |

# J.35 Chapter 34: Launch Vehicle Engineering

**TABLE J.35** Images in Chapter 34.

| Image | NASA ID | Title | Date |
|---|---|---|---|
| Chapter opening | KSC-20240928-PH-KLS03_0026 | NASA's SpaceX Crew-9 Liftoff | 2024-09-28 |
| sii | 6758331 | Saturn Apollo Program | 1967-01-01 |
| etsep | s29-72-059 | STS-29 Discovery, OV-103, external tank (ET) separation | 1989-03-13 |
| barrel2 | MAF_20250210_CS5_LH2_Barrel_out_VWC-04 | Artemis V Moon Rocket in Early Phase of Production | 2025-02-10 |
| lh2tank | MSFC_011519_SLS_LH2_Lift-7 | Space Launch System Liquid Hydrogen Tank Test Article Positioned in Te | 2019-03-05 |
| fairingtest | GRC-2012-C-01689 | Ariane 5 Payload Fairing Test | 2012-04-30 |
| windtunnel | ACD25-0156-012 | NASA, SpaceX Advance Wind Tunnel Tests at NASA Ames | 2025-07-31 |
| encap | GSFC_20240130_PACE_001980 | PACE Encapsulation | 2024-01-30 |

| Image | NASA ID | Title | Date |
|---|---|---|---|
| acoustic | GRC-2024-C-09957 | Orion Environmental Test Article in the Vibro Acoustic Lab at Armstron | 2024-09-11 |
| fhland2 | KSC-20240625-PH-AJN01_0191 | NASA's SpaceX GOES-U Booster Landing | 2024-06-25 |

## J.36 Chapter 35: Rocket Propulsion

**TABLE J.36** Images in Chapter 35.

| Image | NASA ID | Title | Date |
|---|---|---|---|
| Chapter opening | SSC20210116_GR8438 | SSC20210116_GR8438 | 2021-01-16 |
| fh27 | KSC-20260830-PH-SPX02_0005 | NASA's Nancy Grace Roman Space Telescope Liftoff, Launch Comple | 2026-08-30 |
| f1 | 6862832 | Saturn Apollo Program | 1968-01-01 |
| ssme | 2010-3891 | KSC-2010-3891 | 2010-06-25 |
| tubes | KSC-99pp0981 | KSC-99pp0981 | 1999-07-28 |
| amturbo | MSFC-1600485 | 3-D PRINTED TURBOPUMP TEST SET UP | 2016-03-29 |
| bole | C0180.00_01_41_23.Still002_Edit | NASA, Northrop Grumman Test Fire SLS BOLE Booster | 2025-06-26 |
| hybrid | KSC-2012-1013 | KSC-2012-1013 | 2010-09-21 |
| merlintest | KSC-20180102-PH_SPX01_0004 | Commerical Crew Program - SpaceX | 2018-01-02 |
| rl10 | KSC-20160812-PH_DNG01_0078 | Commerical Crew Program (CCP) Astronauts visit Aerojet Rocketdyn | 2016-08-12 |

## J.37 Chapter 36: Launch Vehicles of the World

**TABLE J.37** Images in Chapter 36.

| Image | NASA ID | Title | Date |
|---|---|---|---|
| Chapter opening | KSC-20220816-PH-FMX01_0067 | Artemis I Launch Rollout | 2022-08-16 |
| saturnv | 6901000 | Saturn Apollo Program | 1969-07-16 |
| shuttle | 201107080006HQ | STS-135 Atlantis Launch | 2011-07-08 |
| ariane5 | NHQ202112250015 | James Webb Space Telescope Launch | 2021-12-25 |
| h2a | 201402280004HQ | Global Precipitation Measurement (GPM) Mission | 2014-02-28 |
| sls | SLS_MAF_20260401_ArtemisIILaunch_05 | Artemis II launch | 2026-04-01 |
| vulcan | KSC-20240108-PH-ILW01_0013 | CLPS Peregrine Mission One (PM-1) Astrobotic/ULA Liftoff | 2024-01-08 |
| newglenn | KSC-20251113-PH-BLU01_0007 | NASA's ESCAPADE Launch on Blue Origin's NG-2 | 2025-11-13 |
| electron | KSC-20230530-PH-RKL02_0004 | TROPICS Rocket Launch | 2023-05-26 |
| soyuz | NHQ202409110047 | Expedition 72 Launch | 2024-09-11 |

# J.38 Chapter 37: Spaceports and Ground Infrastructure

**TABLE J.38** Images in Chapter 37.

| Image | NASA ID | Title | Date |
|---|---|---|---|
| Chapter opening | KSC-05pd2403 | KSC-05pd2403 | 2005-11-02 |
| soyuzrail | NHQ202309120007 | Expedition 70 Soyuz Rollout | 2023-09-12 |
| soyuzerect | NHQ202409080023 | Expedition 72 Soyuz Rollout | 2024-09-08 |
| h2a | 201402270008HQ | Global Precipitation Measurement (GPM) Mission | 2014-02-27 |
| deflector | KSC-20180516-PH_KLS01_0024 | Flame Deflector Complete at Launch Complex 39B | 2018-05-16 |
| wetflow | KSC-20190913-PH_KLS01_0030 | Nominal Wet Flow Test at Pad 39B | 2019-09-13 |
| scape | KSC-2010-1055 | KSC-2010-1055 | 2010-01-07 |
| crawler | KSC-20220318-PH-KLS01_0025 | Artemis I Arrives at Pad 39B | 2022-03-18 |
| lightning | KSC-20220610-PH-NAS01_0006 | Lightning Strikes at Pad 39B | 2022-06-10 |
| pegasus | PIA15633 | NuSTAR Hitches a Ride on the Stargazer | 2012-06-04 |

# J.39 Chapter 38: Space Telescope Engineering

**TABLE J.39** Images in Chapter 38.

| Image | NASA ID | Title | Date |
|---|---|---|---|
| Chapter opening | GSFC_20171208_Archive_e000440 | By the Dozen: NASA's James Webb Space Telescope Mirrors | 2017-12-08 |
| hstmirror | 7995584 | History of Hubble Space Telescope (HST) | 1979-03-01 |
| actuator | GSFC_20171208_Archive_e001917 | The Webb Telescope's Actuators: Curving Mirrors in Space | 2017-12-08 |
| hrma | 9704166 | History of Chandra X-Ray Observatory | 1997-01-01 |
| starshade | PIA20904 | Half-Scale Starshade | 2016-08-09 |
| fpa | GSFC_20171208_Archive_e001055 | Webb Instruments Perfected to Microscopic Levels | 2017-12-08 |
| sunshield | GSFC_20171208_Archive_e001023 | NASA's Webb Sunshield Stacks Up to Test | 2017-12-08 |
| cryocooler | PIA20688 | MIRI Cryocooler Packing | 2016-06-13 |
| euclid | PIA25783 | Euclid Spacecraft in Cleanroom | 2023-07-01 |
| chambera | GSFC_20171208_Archive_e000032 | NASA's Webb Telescope "chilling out" in Houston for the summer | 2017-12-08 |
| wfpc | 9400368 | History of Hubble Space Telescope (HST) | 1993-12-01 |

## J.40 Chapter 39: Great Observatories and Deep-Space Telescopes

**TABLE J.40** Images in Chapter 39.

| Image | NASA ID | Title | Date |
|---|---|---|---|
| Chapter opening | PIA18165 | Hubble Against Earth Horizon 1997 | 1997-10-10 |
| sts31 | s31-03-009 | STS-31 Hubble Space Telescope (HST) (SAs & HGAs deployed) is grappled | 1990-04-25 |
| gro | s37-99-056 | STS-37 Gamma Ray Observatory (GRO) grappled by RMS | 1991-04-07 |
| chandrabay | sts093-705-020 | View of the Chandra Observatory before its deployment from Columbia PL | 1999-07-23 |
| pillars | GSFC_20171208_Archive_e000842 | Hubble Goes High Def to Revisit the Iconic 'Pillars of Creation' | 2017-12-08 |
| crab | GSFC_20171208_Archive_e000053 | Observatories Combine to Crack Open the Crab Nebula | 2017-12-08 |
| spitzer | PIA23644 | Spitzer Space Telescope Ready for Launch | 2020-01-22 |
| jwstmirror | GSFC_20171208_Archive_e000422 | NASA's James Webb Space Telescope Primary Mirror Fully Assembled | 2017-12-08 |
| deepfield | webb_first_deep_field | Webb_first_deep_field_SMACS_0723 | 2022-07-12 |
| kepler | PIA11188 | NASA Kepler Spacecraft in a Clean Room | 2008-09-23 |
| romanfairing | KSC-20260821-PH-SER01_0002 | Roman Space Telescope Encapsulation | 2026-08-21 |

## J.41 Chapter 40: In-Space Servicing, Assembly and Manufacturing

**TABLE J.41** Images in Chapter 40.

| Image | NASA ID | Title | Date |
|---|---|---|---|
| Chapter opening | iss071e361950 | The Canadarm2 robotic arm above the Mozambique Channel | 2024-07-23 |
| solarmax | 41c-34-1380 | View of the damaged Solar Maximum Mission Satellite from the 41-C Chal | 1984-04-08 |
| intelsat | s49-91-026 | STS-49 crew captures INTELSAT VI above OV-105's payload bay (PLB) duri | 1992-05-13 |
| hstgrapple | s125e006955 | Hubble Space Telescope (HST) during rendezvous with STS-125 Shuttle At | 2009-05-13 |
| rrm | s135e007549 | Fossum transfers Robotics Refueling Mission during EVA 1 | 2011-07-12 |
| restorel | KSC-20200219-PH-CSH01_0001 | Restore-L Testing | 2020-02-19 |
| s1truss | sts112-709-033 | S1 truss installed on ISS | 2002-10-12 |
| printer | iss042e031282 | 3D Printer Coupon removal and stowage | 2014-12-09 |

| Image | NASA ID | Title | Date |
|---|---|---|---|
| metal | iss071e522127 | NASA astronaut Jeanette Epps configures the Metal 3D printer | 2024-08-21 |

## J.42 Chapter 41: Space Resources and Asteroid Mining

**TABLE J.42** Images in Chapter 41.

| Image | NASA ID | Title | Date |
|---|---|---|---|
| Chapter opening | Bennu_Sample2 | Vial with Sample of Asteroid Bennu | 2024-07-22 |
| ilmenite | S69-47900 | PHOTO MICROGRAPH - LUNAR SAMPE 10022 | 1969-08-28 |
| orange | as17-137-20989 | View of the orange soil which Apollo 17 crewmen found at Station 4 dur | 1972-12-12 |
| marsice2 | PIA26044 | Ice-Exposing Impact Crater Surrounded by Polygon Terrain | 2023-10-26 |
| eros | PIA02923 | Mosaic of Eros Northern Hemisphere | 2000-06-10 |
| psyche | KSC-20230626-PH-FMX01_0040 | Psyche Prelaunch Spacecraft Processing | 2023-06-26 |
| moxie | PIA24176 | Engineers Lower MOXIE into Perseverance | 2020-11-24 |
| ipex | KSC-20240830-PH-FMX01_0027 | IPEx Testbed Filming | 2024-08-30 |
| rassor | KSC-20250603-PH-FMX01_0086 | RASSOR Excavation Testing Swamp Works | 2025-06-03 |
| trailblazer | PIA25837 | Lunar Trailblazer Spacecraft Nears Completion | 2023-08-16 |
| msolo | KSC-20200925-PH-GEB01_0004 | MSolo Instrument Work | 2020-09-25 |

## J.43 Chapter 42: Advanced Propulsion

**TABLE J.43** Images in Chapter 42.

| Image | NASA ID | Title | Date |
|---|---|---|---|
| Chapter opening | GRC-2017-C-01056 | Hall Thruster | 2017-03-06 |
| aeps | grc-2025-c-07151 | Testing Gateway's AEPS Thruster at NASA Glenn (grc-2025-c-07151) | 2025-09-02 |
| ds1ion | PIA04238 | Deep Space 1 Ion Engine | 2002-12-21 |
| kiwi | GRC-1964-C-69680 | Technicians Manufacture a Nozzle for the Kiwi B-1-B Engine | 1964-05-23 |
| fsptdu | GRC-2015-C-06315 | Fission Surface Power Technology Demonstration Unit | 2016-11-09 |
| neascout | MSFC-1601350 | NEAR EARTH ASTERIOD SCOUT SOLAR SAIL | 2015-01-08 |
| acs3 | CamC_Seq109_2024-08-28_17-17-13Z_V2_S595171 | Advanced Composite Solar Sail System (ACS3) Onboard View of Sail/Boom | 2024-08-28 |

| Image | NASA ID | Title | Date |
|---|---|---|---|
| x43nose | EC04-0324-146 | The small size of the X-43A scramjet is evident in this nose-on view w | 2004-11-15 |
| rdre | 09-27-2023-HF-12 _062 | 3D-printed Rotating Detonation Rocket Engine Test | 2023-09-27 |

## J.44 Chapter 43: AI Foundations: Machine Learning for Space Engineers

**TABLE J.44** Images in Chapter 43.

| Image | NASA ID | Title | Date |
|---|---|---|---|
| Chapter opening | ACD24-0072-012 | NASA Advanced Supercomputing Modular Facility Hyperwall Visualiz | 2024-05-23 |
| inco | S88-38979 | JSC MCC Bldg 30 Instrumentation and Communications Officer (INCO) RTDS | 1988-06-02 |
| hubble | s31-04-027 | Hubble Space Telescope (HST) above OV-103's PLB during STS-31 deployme | 1990-04-25 |
| smap | KSC-314D-0492_0074 | SMAP During Weighing | 2014-11-07 |
| aegis | PIA20762 | Autonomous Selection of a Rover Laser Target on Mars | 2016-07-21 |
| rockies | GSFC_20171208_Archive_e001576 | Rocky Mountain Snow | 2017-12-08 |
| eo1 | PIA18796 | NASA Spacecraft Eyes Iceland Volcanic Eruption | 2014-09-03 |
| lovell | S69-35099 | ASTRONAUT LOVELL, JAMES A., JR. - APOLLO VIII (GUIDANCE & NAVIGATION [ | 1969-05-25 |
| fires | GSFC_20171208_Archive_e000668 | Series of Wildfires in Northern California Continue Blazing | 2017-12-08 |

## J.45 Chapter 44: Deep Learning, Foundation Models and Generative AI

**TABLE J.45** Images in Chapter 44.

| Image | NASA ID | Title | Date |
|---|---|---|---|
| Chapter opening | ACD15-0062-021 | ACD15-0062-021 | 2015-04-10 |
| marscraters | PIA24040 | AI Spots a Cluster of Mars Craters: HiRISE's view | 2020-10-01 |
| ai4mars | PIA23897 | Martian Terrain as Viewed in AI4Mars | 2020-06-12 |
| milton | iss072e029135 | Hurricane Milton is pictured in the Gulf of Mexico | 2024-10-08 |
| landsat9b | KSC-20210816-PH-RNB01_0090 | Landsat 9 Encapsulation | 2021-08-16 |
| sdo | KSC-2009-6419 | KSC-2009-6419 | 2009-11-16 |
| roverdrivers | PIA24724 | 3D Glasses Used for Rover Driving | 2021-07-01 |

# J.46 Chapter 45: Data, Computing Hardware and Edge AI in Orbit

**TABLE J.46** Images in Chapter 45.

| Image | NASA ID | Title | Date |
|---|---|---|---|
| Chapter opening | iss065e009485 | Spaceborne Computer-2 | 2021-04-29 |
| discover | GSFC_20171208_Archive_e002034 | Discover Supercomputer 1 | 2017-12-08 |
| himasssee | iss032e016954 | Spacecraft Single Event Enviroments at High Shielding Mass (HiMassSEE) | 2012-08-11 |
| ingenuity | PIA24547 | Mastcam-Z Gives Ingenuity a Close-up | 2021-04-06 |
| cloudyearth | cloudy-earth_16810027983_o | Cloudy Earth | 2015-05-08 |
| laserdownlink | PIA26148 | DSN's Experimental Hybrid Antenna Tracks DSOC's Laser Downlink | 2024-02-08 |
| issflyaround | s132e013215 | Flyaround view of ISS after undocking | 2010-05-23 |
| cubesats | iss038e044994 | NanoRacks CubeSat Deployment | 2014-02-11 |

# J.47 Chapter 46: AI for Earth Observation

**TABLE J.47** Images in Chapter 46.

| Image | NASA ID | Title | Date |
|---|---|---|---|
| Chapter opening | GSFC_20171208_Archive_e001699 | Landsat View: Pearl River Delta, China | 2017-12-08 |
| kazakh | GSFC_20171208_Archive_e001371 | Fall Harvest in Kazakhstan | 2017-12-08 |
| arcdefor | PIA26111 | Composite of Brazil's 'Arc of Deforestation' Shows Imagery NISAR Will | 2023-10-27 |
| suez | PIA24553 | Suez Canal Crisis | 2021-04-06 |
| idai | PIA23142 | Satellites Image Mozambique Flooding after Cyclone Idai | 2019-04-01 |
| mariadpm | PIA21964 | NASA Damage Map Aids FEMA's Hurricane Maria Rescue Operation in Puerto | 2017-09-26 |
| grasshopper | iss075e0001471 | Thick smoke emanates from the Grasshopper wildfire burning near Mount | 2026-07-31 |
| tanager | PIA26416 | Tanager-1 First Methane and Carbon Dioxide Plume Detections | 2024-10-10 |

# J.48 Chapter 47: AI for Weather, Climate and Space Weather

**TABLE J.48** Images in Chapter 47.

| Image | NASA ID | Title | Date |
|---|---|---|---|
| Chapter opening | GSFC_20171208_Archive_e000204 | Hurricane Matthew from Space | 2017-12-08 |
| pleiades | ARC-2008-ACD08-0272-005 | ARC-2008-ACD08-0272-005 | 2008-12-02 |

| Image | NASA ID | Title | Date |
|---|---|---|---|
| melissa | iss073e1044319 | Category 5 Hurricane Melissa is pictured in the Caribbean Sea | 2025-10-28 |
| sunspot | GSFC_20171208_Archive_e000923 | NASA's SDO Observes Largest Sunspot of the Solar Cycle | 2017-12-08 |
| activeregion | PIA22582 | Rambunctious Active Region | 2018-07-05 |
| aurora | iss074e0150531 | A wispy, multi-colored aurora created by a geomagnetic storm | 2026-01-19 |
| imap | KSC-20250922-PH-SPX01_0008 | NASA/SpaceX IMAP Vertical at LC 39A (Sunrise) | 2025-09-22 |

## J.49 Chapter 48: AI for Autonomy, Navigation and Robotics

**TABLE J.49** Images in Chapter 48.

| Image | NASA ID | Title | Date |
|---|---|---|---|
| Chapter opening | PIA24723 | Computer Simulation of Perseverance's First Autonav Drive | 2021-07-01 |
| atvjv | iss016e034177 | ATV during Demonstration Day 2 Rendezvous Test | 2008-03-31 |
| dm2 | iss063e021460 | Historic SpaceX DM-2 Approach and Docking to the ISS | 2020-05-31 |
| aegis | PIA25289 | Perseverance's SuperCam Uses AEGIS For the First Time | 2022-05-31 |
| astrobee | iss071e046284 | A free-flying Astrobee robot pictured aboard the International Space S | 2024-05-01 |
| robonaut | iss034e013990 | Robonaut 2 in the U.S. Laboratory | 2013-01-02 |
| cadre | PIA25665 | CADRE Rover Testing in JPL's Mars Yard | 2023-08-02 |

## J.50 Chapter 49: AI for Mission Operations, Communications and Networks

**TABLE J.50** Images in Chapter 49.

| Image | NASA ID | Title | Date |
|---|---|---|---|
| Chapter opening | jsc2026e022251 | Artemis II Flight Control Team During Splashdown | 2026-04-10 |
| hubbleops | GSFCRR2021071501 | NASA Begins Switch to Backup Spacecraft Hardware | 2021-07-15 |
| batteryeva | iss061e005514 | iss061e005514 | 2019-10-11 |
| optimism | PIA24527 | OPTIMISM Ready for Testing | 2021-11-17 |
| goldstone | PIA26716 | Goldstone's 70-Meter Antenna Under the Stars | 2026-01-27 |
| tdrs | KSC-2014-1075 | Shots of TDRS-L Spacecraft Prior to Media Day | 2014-01-03 |
| scantb | GRC-2011-C-03408 | GRC-2011-C-03408 | 2007-09-07 |

# J.51 Chapter 50: AI for Space Situational Awareness, Defence and Security

**TABLE J.51** Images in Chapter 50.

| Image | NASA ID | Title | Date |
|---|---|---|---|
| Chapter opening | PIA24561 | Goldstone Radar Observations of Asteroid 2001 FO32 | 2021-04-05 |
| 2018la | PIA22468 | Asteroid 2018 LA | 2018-06-03 |
| shelter | iss030e173969 | Expedition 30 crewmembers in the SM | 2012-03-24 |
| atv3 | iss033e009232 | ATV reentry | 2012-10-03 |
| kuwait | s37-152-092 | Kuwait Oil Fires, Kuwait | 1991-04-11 |
| sarship | PIA01799 | Space Radar Image of North Atlantic Ocean | 1999-04-15 |
| fdo | jsc2022e090753 | Flight Day 16: Mission Control during Distant Retrograde Departure | 2022-12-01 |

# J.52 Chapter 51: AI for Space Science, Planetary Exploration and Human Spaceflight

**TABLE J.52** Images in Chapter 51.

| Image | NASA ID | Title | Date |
|---|---|---|---|
| Chapter opening | iss069e010815 | Astronaut Sultan Alneyadi observes a free-flying Astrobee robotic assi | 2023-05-16 |
| tess | KSC-20180212-PH-JBS03_0030 | TESS Spacecraft Lift to Work Stand | 2018-02-12 |
| jezero | PIA23380 | Jezero Crater Minerals | 2019-11-12 |
| pixl | PIA26365 | Perseverance's PIXL Instrument Examines 'Thunderbolt Peak' | 2024-07-16 |
| biomonitor | iss058e003984 | Bio-Monitor | 2019-01-16 |
| cimon | iss073e0384102 | JAXA astronaut Takuya Onishi sets up the CIMON AI-powered robotic assi | 2025-07-18 |
| callisto | art001e000192 callisto | Inside Orion With Callisto | 2022-11-18 |
| hololens | iss046e043637 | HoloLens | 2016-02-20 |

# J.53 Chapter 52: AI in Spacecraft Design, Manufacturing, Launch and the Space Business

**TABLE J.53** Images in Chapter 52.

| Image | NASA ID | Title | Date |
|---|---|---|---|
| Chapter opening | MSFC-201900153 | NASA Deputy Administrator, Jim Morhard, Tours MSFC Additive Manu | 2019-03-25 |
| st5 | KSC-06pd0434 | KSC-06pd0434 | 2006-02-14 |
| excite | GSFC2024excite02 | GSFC2024excite02 | 2024-08-29 |

| Image | NASA ID | Title | Date |
|---|---|---|---|
| shaker | PIA26713 | NEO Surveyor's Spacecraft Bus on the 'Shaker Table' | 2026-01-14 |
| landing | NHQ20261001_admin_0012 | Falcon 9 Landing Following NASA's SpaceX Crew-13 Launch | 2026-10-01 |
| rs25 | SSC-20240229-s00309 | NASA Caps Off February with RS-25 Engine Certification Test | 2024-02-29 |
| lcc | KSC-20251008-PH-GEB01_0136 | Artemis II Terminal Count Simulation | 2025-10-18 |

## J.54 Chapter 53: Trustworthy AI for Space: Assurance, Security, Governance and Outlook

**TABLE J.54** Images in Chapter 53.

| Image | NASA ID | Title | Date |
|---|---|---|---|
| Chapter opening | ivv | NASA Renames Facility in Honor of 'Hidden Figure' Katherine Johnson | 2019-02-21 |
| morph3 | KSC-2014-4818 | KSC-2014-4818 | 2014-12-15 |
| artemisfcr | jsc2026e021924 | Artemis II Flight Control Team during Lunar Flyby | 2026-04-06 |
| lax | iss074e0622240 | Los Angeles International Airport on Southern California's Pacific coa | 2026-05-29 |
| cities | iss073e0842595 | City lights from Tokyo to Osaka and Hiroshima | 2025-09-15 |
| ltv | jsc2026e417568 | Crewed Lunar Terrain Vehicle Testing | 2026-09-15 |
| jett | jsc2024e035443 | JETT 5 - jsc2024e035443 | 2024-05-16 |

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